NISTIR XXXX Draft

Face Recognition Technology Evaluation (FRTE)

Part 1: Verification

Patrick Grother
Mei Ngan
Kayee Hanaoka
Joyce C. Yang
Austin Hom

Information Access Division
Information Technology Laboratory

This publication is available free of charge from:
https://www.nist.gov/programs-projects/face-recognition-vendor-test-frvt-ongoing

2026/11/06

U.S. Department of Commerce
Howard Lutnick, Secretary

National Institute of Standards and Technology
Arvind Raman, Under Secretary of Commerce for Standards and Technology and NIST Director

ACKNOWLEDGMENTS

The authors are grateful for the long-standing support and collaboration of the the Department of Homeland Security's Science & Technology Directorate (S&T) and the Office of Biometric Identity Management (OBIM).

Additionally, the authors are grateful to staff in the NIST Biometrics Research Laboratory for infrastructure supporting rapid evaluation of algorithms.

DISCLAIMER

Specific hardware and software products identified in this report were used in order to perform the evaluations described in this document. In no case does identification of any commercial product, trade name, or vendor, imply recommendation or endorsement by the National Institute of Standards and Technology, nor does it imply that the products and equipment identified are necessarily the best available for the purpose.

INSTITUTIONAL REVIEW BOARD

The National Institute of Standards and Technology's Research Protections Office reviewed the protocol for this project and determined it is not human subjects research as defined in Department of Commerce Regulations, 15 CFR 27, also known as the Common Rule for the Protection of Human Subjects (45 CFR 46, Subpart A).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

FRTE STATUS

This report is a draft NIST Interagency Report, and is open for comment. It is the thirty seventh edition of the report since the first was published in June 2017. Prior editions of this report are maintained on the FRTE website, and may contain useful information about older algorithms and datasets no longer used in FRTE.

FRTE remains open: All tracks of the FRTE are open to new algorithm submissions.

2026-11-06 changes since 2026-10-06:

- We have added results for first algorithms from six developers: Cvartel Inc, Elkosta Security Systems India, FacePlugin LLC, InfoDif Software and Information Technologies Inc, Johannes Gutenberg-Universitat Mainz, and Xailient.
- We have added results for new algorithms from thirteen returning developers: 20Face, AI Implementation Group, Chunghwa Telecom, Enqura Ltd, FPT Telecom JSC, ID3 Technology, Incode Technologies Inc, RealSense, Shanghai Jiao Tong University, Stellar Technologies, TAHAKOM, Via Technologies Inc, and Vivotek.

2026-10-06 changes since 2026-09-01:

- We have added results for first algorithms from eight developers: Akor IT Services LLC, IDBIO, Nmetics Private Ltd, Northern Point Technology Pvt Ltd, Toss, Universidade Federal de Goiás, Universal Ubiquitous AI Co Ltd, and Verismart
- We have added results for new algorithms from thirteen returning developers: Akiyama S.A, Cognitec Systems GmbH, Onfido, FacePhi, Innov8tif Solutions, Lishui Shuyan Huirong Technology Co, Metsakuur, Neurotechnology, Paravision, PNPSecure, STCON LLC, Toshiba, and Viettel Artificial Intelligence and Data Services Center

2026-09-01 changes since 2026-07-31:

- We have added results for first algorithms from eight developers: Intpixel Labs Private Ltd, Integrated Software and Technologies Ltd, Norwegian University of Science and Technology, OCR Studio L.L.C.-FZ, OORU Digital Private Ltd, Security Pitch Co Ltd, Softtech Yazilim Teknolojileri Arastirma Gelistirme ve, and UniverseAI.
- We have added results for new algorithms from thirteen returning developers: Advance.AI, InfoCert, Innovatrics, Iris ID Systems, Miaxis Biometrics, Regula Forensics, ROC, Sensetime Group, Sum and Substance Ltd, Verigram, Viante.AI, Vision-Box, and Vsoft.

2026-07-31 changes since 2026-06-15:

- We have added results for first algorithms from two developers: Kaizen Secure Voiz, and Veertec SAL (Offshore).
- We have added results for new algorithms from fourteen returning developers: CAKE by VPBank, Compal Electronics, Dermalog, Facehawk Ltd, FRP LLC, Intozi Tech Pvt Ltd, IQSEC, Nominder, RealSense, Sparktek International Pte Ltd, STCON LLC, TAHAKOM, Technology Control Company, Techainer, and Veridium.

2026-06-15 changes since 2026-05-08:

- We have added results for first algorithms from five developers: AIPRACTIX, Bynn Intelligence Inc, ScanalitiX, Stellar Technologies, and TrueSight Laboratories Tech LLC.
- We have added results for new algorithms from nine returning developers: Cyberlink Corp, Enqura Ltd, Hyundai Motor Company, ID3 Technology, IDENTY, Matrix Comsec Pvt Ltd, Sightropic, Viettel Artificial Intelligence and Data Services Center, and Visica AI Solutions.

2026-05-08 changes since 2026-03-30:

- We have added results for first algorithms from four developers: Ethiopian Labor Market Information System, Face Int, Lishui Shuyan Huirong Technology Co, and YuanLi Technology.
- We have added results for new algorithms from twenty-one returning developers: Advance.AI, BioID GmbH, Tripleize, Dralmey Solutions, FaceLocate, HyperVerge Inc, InfoCert, ioNetworks Inc, Innov8tif Solutions, Kneron Inc, Mobai, Onfido, PAPIL11 S.R.O., PT Technology Biometric Indonesia, Shanghai Jiao Tong University, TNG Digital Sdn Bhd, TOPPAN Security Co, Unissey, UZINFOCOM, Viante.AI, and Vsoft.

2026-03-30 changes since 2026-03-05:

- We have added results for first algorithms from three developers: Ice House, Shufti, and Smarg Technologies Pvt Ltd
- We have added results for new algorithms from four returning developers: AI Implementation Group, Daon, PNPSecure, and Sensetime Group

2026-03-05 changes since 2026-03-02:

- We have updated the missing results for VisionBox.

2026-03-02 changes since 2026-01-22:

- We have added results for first algorithms from two developers: FPT Telecom JSC, and Iris ID Systems
- We have added results for new algorithms from six returning developers: Biometrik Systems Private Limited, GTEL OTS JSC, ID3 Technology, NextMeta Labs, Nominder, and ROC.

2025-01-22 changes since 2025-12-19:

- We have added results for first algorithms from five developers: CAKE by VPBank, RealSense, Sum and Substance Ltd, Transmit Security Ltd, and Vaidio.
- We have added results for new algorithms from eleven returning developers: Beijing DeepSense Technologies, Compal Electronics, Candour Biometrics, Clearview AI Inc, IE SARSENOV, Miaxis Biometrics, Onfido, Paravision, Shanghai Jiao Tong University, STCON LLC, and Toshiba.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2025-12-19 changes since 2025-09-30:

- ▷ We have added results for first algorithms from one developers: Sparktek International Pte Ltd.
- ▷ We have added results for new algorithms from 13 returning developers: BioID GmbH, Chunghwa Telecom, Dermalog, Hyundai Motor Company, IDENTY, Innovatrics, InterLink labs, Matrix Comsec Pvt Ltd, Sightropic, Trust Stamp, Viettel Artificial Intelligence and Data Services Center, Viettel High Technology, and Vsoft.

2025-09-30 changes since 2025-08-13:

- ▷ We have added results for first algorithms from three developers: Biometrik Systems Private Limited, OOTO AI, and Toss.
- ▷ We have added results for new algorithms from four returning developers: Jaak IT, Online Mobile Services JSC, Regula Forensics, and Unissey

2025-08-13 changes since 2025-07-15:

- ▷ We have added results for first algorithms from four developers: BTS Digital, IE SARSENOV, UZINFOCOM, and Vsoft.
- ▷ We have added results for new algorithms from seventeen returning developers: 2060.io, Beijing DeepSense Technologies, ID R&D Inc, Innovatrics, IntelliVIX, Metsakuur, Mobai, NextMeta Labs, Neurotechnology, Onfido, QazSmartVision.AI, ROC, STCON LLC, Toshiba, VinBigData, Vision-Box, and Visica AI Solutions.

2025-07-15 changes since 2025-05-28:

- ▷ We have added results for first algorithms from three developers: Matrix Comsec Pvt Ltd, PNPSecure, and Urielsoft Co. Ltd.
- ▷ We have added results for new algorithms from eleven returning developers: Daon, Dermalog, Hyundai Motor Company, Incode Technologies Inc, Intel Research Group, Kloudspot Inc, Omnigarde Ltd, Paravision, Universidade de Coimbra, Viante.AI, and Vietnam Posts and Telecommunications Group.

2025-05-28 changes since 2025-04-25:

- ▷ We have added results for new algorithms from seven returning developers: BioID GmbH, Cubox, DiDi ChuXing Technology, InterLink labs, IDENTY, PAPIL11 S.R.O., and Trust Stamp.

2025-04-25 changes since 2025-03-18:

- ▷ We have added results for first algorithms from seven developers: ArVİs Teknoloji San.Tic.A.Ş., Compal Electronics, FCI Vietnam, GTEL OTS JSC, Shenzhen Polytechnic University, Sightropic, and TNG Digital Sdn Bhd.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for new algorithms from eleven returning developers: Awidit Systems, Cyberextruder, FRP LLC, Kasikorn Labs, NCS Pte Ltd, Nominder, Shanghai Jiao Tong University, STCON LLC, Tech5 SA, Veridas Digital Authentication Solutions S.L., and Viettel High Technology.

2025-03-18 changes since 2025-03-05:

- ▷ We have added results for first algorithms from one developers: FaiceTech Ltd.
- ▷ We have added results for new algorithms from ten returning developers: Beijing DeepSense Technologies, BioID Technologies SA, EI Networks Private Ltd, Hangzhuo AIlu Network Information Technology, Innovatrics, Private Identity LLC, Samsung S1 Corp, TigerIT Bangladesh Ltd, Via Technologies Inc, and Viettel Cyberspace Center.

2025-03-05 changes since 2025-01-30:

- ▷ We have added results for first algorithms from three developers: 2060.io, MB Bank, and Visica AI Solutions
- ▷ We have added results for new algorithms from twelve returning developers: AFR Engine, AI Implementation Group, Authme, Hyundai Motor Company, FacePhi, Inspur (Beijing) Electronic Information Industry Co, LightVision Inc, QazSmartVision.AI, ROC, Toshiba, Techainer, and VinBigData.

2025-01-30 changes since 2024-12-18:

- ▷ We have added results for first algorithms from four developers: Innov8tif Solutions, Maverick Research, PT Technology Biometric Indonesia, and TTI Group.
- ▷ We have added results for new algorithms from ten returning developers: Coretech Knowledge Inc, Intel Research Group, IQSEC, i2v Systems, PAPIL11 S.R.O., Paravision, Taiwan Mobile, Touchless ID, useB, and Viante.AI.

2024-12-18 changes since 2024-11-08:

- ▷ We have added results for first algorithms from six developers: BioID GmbH, InterLink labs, KYB-AI Ltd, Keyless Technologies, Pyramid Cyber Security + Forensic (P),and Syscake S.r.l..
- ▷ We have added results for new algorithms from sixteen returning developers: Alchera Inc, Awidit Systems, Cubox, DeCloak Intellegences, Beijing DeepSense Technologies, Innovatrics, Kasikorn Labs, Mitek Systems, Mukh Technologies, NCS Pte Ltd, NextMeta Labs, Nominder, PT Autentika Digital Indonesia, Sensetime Group, Shanghai Jiao Tong University, and STCON LLC.

2024-11-08 changes since 2024-10-07:

- ▷ We have added results for first algorithms from two developers: Hyderabad-Centre for Development of Advanced Computing, and TAHAKOM.
- ▷ We have added results for new algorithms from ten returning developers: Candour Biometrics, Cyberextruder, DICIO, First Credit Bureau Kazakhstan, Metsakuur, Mobbeel Solutions, Samsung S1 Corp, Vietnam Payment Solution, VinBigData, and Yuan High-Tech Development.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2024-10-07 changes since 2024-09-16:

- ▷ We have added results for first algorithms from three developers: Enqura, Kloudspot, and Penacomp Systemy Informatyczne.
- ▷ We have added results for new algorithms from fourteen returning developers: Chunghwa Telecom, Cyberlink, Dactionable Technologies, Daon, FPT Information System, Hangzhuo AIlu Network Information Technology, Kakao Bank, Panasonic R+D Center Singapore, Paravision, Private Identity, QazSmartVision.AI, Toshiba, Vietnam Posts and Telecommunications Group, and UNICC-Solution Architecture Section.

2024-09-16 changes since 2024-08-09:

- ▷ We have added results for first algorithms from one developer: Taiwan Mobile.
- ▷ We have added results for new algorithms from eleven returning developers: Cognitec Systems GmbH, Coretech Knowledge Inc, FaceOnLive Inc, Idemia, Intel Research Group, Luxand Inc, Ovision, PAPIL11 S.R.O., Unissey, useB, and Vivotek.

2024-08-09 changes since 2024-07-03:

- ▷ We have added results for first algorithms from seven developers: Affinsys AI Ltd, AI Implementation Group, Celloscope Limited, Hyundai Motor Company, ID R&D Inc, LightVision Inc and NextMeta Labs.
- ▷ We have added results for new algorithms from eleven returning developers: AFR Engine, Antheus Technologia, CTBC Bank, Innovatrics, Mitek Systems, Neurotechnology, Nominder, Seventh Sense Artificial Intelligence, Universidade de Coimbra, University of Surrey-CVSSP, Viettel Cyberspace Center, Vision-Box and Yoti.

2024-07-03 changes since 2024-05-30:

- ▷ We have include all algorithms evaluated since the ongoing FRTE evaluation started in 2017. Previously we listed only results for each developer's two most recent implementations. The main results page has been revised with a "by developer" tab which lists only the most accurate algorithm from each developer, where accuracy is defined by the lowest FNMR value for the visa-border dataset.
- ▷ We have added results for first algorithms from two developers: Taiwan Computer Vision and Telexper International.
- ▷ We have added results for new algorithms from fifteen returning developers: Alchera Inc, Beijing DeepSense Technologies, Cyberlink Corp, Dermalog, FacePhi, Facia.ai, FRP LLC, NCS Pte Ltd, Omnigarde Ltd, Online Mobile Services JSC, Qnap Security, ROC, STCON LLC, Videmo Intelligente Videoanalyse, and Viettel Group.

2024-05-30 changes since 2024-04-17:

- ▷ We have added results for first algorithms from three developers: ALTTEKGlobal, Fourthline, and IQSEC

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for new algorithms from thirteen returning developers: Candour Biometrics, Chunghwa Telecom, Clearview AI Inc, Cognitec Systems GmbH, Incode Technologies Inc, ioNetworks Inc, HyperVerge Inc, Mobbeel Solutions, Panasonic R+D Center Singapore, QazSmartVision.AI, Vietnam Posts and Telecommunications Group, VinBigData and Yuan High-Tech Development
- ▷ We have retired results for eleven algorithms per our policy to only list results for eleven algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-04-17 changes since 2024-03-26:

- ▷ We have added results for first algorithms from four developers: Ovision, Vietnam National Cyber Security Technology Corporation, Vivotek, and Vocalize.
- ▷ We have added results for new algorithms from four returning developers: Intel Research Group, Kakao Bank, Luxand, and Turkcell Technology.
- ▷ We have retired results for nine algorithms per our policy to only list results for four algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-03-26 changes since 2024-02-21:

- ▷ We have added results for first algorithms from six developers: FaceLocate, FPT Education, FPT Information System, Kogniza Technology, i2v Systems, and Seamfix.
- ▷ We have added results for new algorithms from eleven returning developers: AFR Engine, Aware, Beijing Hisign Technology, Cu-Face, Dactionable Technologies, First Credit Bureau Kazakhstan, Innovatrics, Nominder, Securif AI, Veridium, and Viettel Cyberspace Center.
- ▷ We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-02-21 changes since 2024-01-22:

- ▷ We discontinued the visa-visa benchmark in FRTE since Feburary 14th 2024. For highly performing algorithms, the only remaining false negative error in that set were in babies and in a few greyscale poor quality half-tone scans of paper photos.
- ▷ We have added results for first algorithms from four developers: Online Mobile Services JSC, Openedge Technologies, TNI Technoology, and Yoti.
- ▷ We have added results for new algorithms from nine returning developers: CMC Institute of Science and Technology, City and County of Honolulu, Dermalog, Euronovate SA, Maxvision Technology, Megvii/Face++, Neurotechnology, Toshiba, useB, and Universidade de Coimbra.
- ▷ We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2024-01-22 changes since 2023-12-15:

- ▷ We have added results for first algorithms from six developer: GPS Vietnam Trading, iCOMM Media & Tech, PAPIL11 S.R.O., QazSmartVision.AI, Sparsh CCTV, and VNIS Joint Stock.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have added results for new algorithms from eight returning developers: Cyberlink Corp, Intellivision, Hangzhuo AIlu Network Information Technology, Lebentech Biometrics, Momentum Digital, Regular Biometrics Solutions, ROC, and Via Technologies Inc.
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-12-15 changes since 2023-11-21:

- ▷ We have added results for first algorithms from five developer: CMC University, Element System Solutions Company, Fraud.com, Techainer, and Vietnam Payment Solutions.
- ▷ We have added results for new algorithms from thirteen returning developers: Accurascan, Adera Global, Alchera, Aware, Clearview AI, Cognitec Systems, Incode Technologies Inc, Inspur (Beijing) Electronic Information, IntelliVIX, ioNetworks, NHN Corp, PT Autentika Digital Indonesia, and Qnap Security.
- ▷ We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-11-21 changes since 2023-10-27:

- ▷ We have added results for new algorithms from ten returning developers: Coretech Knowledge Inc, Beijing DeepSense Technologies, General Interface Solutions Holding Ltd, Glory, Kakao Brain, Nominder, Smarvist Teknoloji, STCON LLC, Suprema AI Inc, and Yuan High-Tech Development
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-10-27 changes since 2023-09-29:

- ▷ We have added results for first algorithms from two developer: Authme and Dactionable Technologies.
- ▷ We have added results for new algorithms from twelve returning developers: Intel Research Group, Guangzhou Pixel Solutions, Kasikorn Labs, Kakao Bank, Luxand, Megvii/Face++, Panasonic R+D Center Singapore, Private Identity LLC, Recognito, Seventh Sense Artificial Intelligence, UNICC-Solution Architecture Section, and Universidade de Coimbra.
- ▷ We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-09-29 changes since 2023-09-09:

- ▷ We have added results for first algorithms from one developer: Intozi Tech.
- ▷ We have added results for new algorithms from seven returning developers: Euronovate, Maxvision Technology, Miaxis Biometrics, Mukh Technologies, Toshiba, University of Surrey-CVSSP, and useB.
- ▷ We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-09-09 changes since 2023-08-16:

- We have renamed and split FRVT into two sets of technology evaluations: The Face Recognition Technology Evaluation (FRTE) is the umbrella for the ongoing 1:1, 1:N and twins disambiguation tracks; the Face Analysis Technology Evaluation (FATE) holds the morph and attacked detection tracks, the quality assessment tracks, and the new age estimation tracks.
- We have added results for first algorithms from two developers: Facehawk and Regular Biometrics Solutions.
- We have added results for new algorithms from eight returning developers: CMC Institute of Science and Technology, Canon, Cu-Face, EI Networks, ICM Airport Technics, Securif AI, Securifai, Vietnam Posts and Telecommunications Group, and Vision Intelligence Center of Meituan
- We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-08-16 changes since 2023-07-21:

- We have added results for first algorithms from one developers: Facia.ai.
- We have added results for new algorithms from fifteen returning developers: Accurascan, Alchera, FOO, Hangzhuo AIlu Network Information Technology, HyperVerge, ioNetworks, Innovatrics, Nominder, Rank One Computing, Neurotechnology, Tech5, Tripleize, Unissey, UX Labs, and Verigram.
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-07-21 changes since 2023-06-16:

- We have added results for first algorithms from five developers: DeCloak Intellegences, Innominds Software SEZ India, Trust Stamp, Vcortex Labs, and Viettel Cyberspace Center
- We have added results for new algorithms from fourteen returning developers: AFR Engine, Alice Biometrics, Euronovate SA, IMDS Software, Idemia, IntelliVIX, Intellivision, NHN, Pangiam, STCON, Samtech InfoNet, Veridium, VinBigData, and Vision-Box
- We have retired results for eleven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We have stopped using wild images in FRVT since May 1st 2023. All results for that set have been removed from the website, and will be removed from future PDF reports.

2023-06-16 changes since 2023-04-20:

- We have added results for first algorithms from seven developers: EI Networks Private Ltd, General Interface Solutions Holding Ltd, Recognito, Serendipity Ltd, ST Engineering, and useB.
- We have added results for new algorithms from eighteen returning developers: Advance.AI, AYF Technology, Cyberlink Corp, Incode Technologies Inc, Intel Research Group, Intellibrain Technological Projects, Maxvision Technology, Megvii/Face++, Omnigarde Ltd, Paravision, Qnap Security, Seventh Sense Artificial Intelligence, Shanghai Jiao Tong University, Veridas Digital Authentication Solutions S.L., Universidade de Coimbra, Yuan High-Tech Development, UNICC-Solution Architecture Section, and Verihubs,verihubs-inteligensia.
- We have retired results for 15 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2023-04-20 changes since 2023-04-04:

- ▷ We have added results for first algorithms from one developers: IDENTY.
- ▷ We have added results for new algorithms from three returning developers: Metsakuur, Autentika Digital Indonesia, and Verigram.
- ▷ FRVT will re-open 2023-05-01.
- ▷ We have retired results for 3 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-04-04 changes since 2023-03-09:

- ▷ We have added results for first algorithms from six developers: Aratek Biometrics City and County of Honolulu, FOO, Intelligent Control Technology NCS, and Swsam Solutions.
- ▷ We have added results for new algorithms from nine returning developers: Cubox, Glory, InsightFace AI, Miaxis Biometrics, Mukh Technologies, Turkcell Technology, Samsung S1, Via Technologies, and Vision Intelligence Center of Meituan
- ▷ We have retired results for 7 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2023-03-09 changes since 2023-02-01:

- ▷ We have added results for first algorithms from eight developers: Biometric LLC (biometric.vision) KZ, Candour Biometrics, Fast Enterprises, KakaoBank, Mitek Systems, Nominder, Private Identity, and UNICC-Solution Architecture Section.
- ▷ We have added results for new algorithms from 22 returning developers: AFR Engine, Biocube Matrics, CMC Institute of Science and Technology, Cloudwalk - Moontime Smart Technology, Cyberlink Corp, Beijing DeepSense Technologies, First Credit Bureau Kazakhstan, Enface, Hangzhuo AIlu Network Information Technology, Herta Security, IMDS Software, Inspur (Beijing) Electronic Information Industry Co, Intellivision, MicroFocus, Neurotechnology, Pangiam, NSENSE Corp, STCON LLC, Touchless ID, University of Surrey-CVSSP, Vision-Box, and YooniK.
- ▷ We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have introduced new set of non-frontal portrait to border comparisons. The new images are described in section 2.3 and their use in seciton 3.2.

2023-02-01 changes since 2022-12-15:

- ▷ We have added results for first algorithms from four developers: CU-Face, Korea ID, Onfido, and TrueID-VNG.
- ▷ We have added results for new algorithms from 21 returning developers: Alchera, Armatura, Cogent-Thales, Dermalog, Didi ChuXing Global Face, Gorilla, Hyperverge, Innovatrics, Intel Research, IntelliVIX, Intema-LGL, Kasikorn Labs, Paravision, Rank One Computing, Sensetime Group, Suprema AI, Tech5, Unissey, U. Coimbra Visteam, Vixvizion (Imagus), and Yuan High-Tech Development.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ We have retired results for 20 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- ▷ We have introduced new set of non-frontal portrait to border comparisons. The new images are described in section 2.3 and their use in seciton 3.2.

2022-12-15 changes since 2022-11-06:

- ▷ We have added results for first algorithms from four developers: Miaxis Biometrics, PT Autentika Digital Indonesia, PT Qlue Performa Indonesia, and STCON.
- ▷ We have added results for new algorithms from 14 returning developers: Adera Global, Aiseemu Technology, Chunghwa Telecom,chtface, FRP, Griaule, Line Corporation, Maxvision Technology, Mukh Technologies, Papilon Savunma, Qnap Security, Realnetworks, Securif AI, SQIsoft, and Veridium.
- ▷ We have retired results for 10 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-11-06 changes since 2022-09-26:

- ▷ We have added results for first algorithms from six developers: AFR Engine, CMC Institute of Science and Technology, Saga Densan Center, Turkcell Technology, UXLabs, and Wise AI SDN BHD.
- ▷ We have added results for new algorithms from 14 returning developers: Coretech Knowledge, Cloudwalk - Moontime, Cloudmatrix, Deepglint, Guangzhou Pixel Solutions, Hangzhuo AIlu Network Information Technology, NEO Systems, One More Security, Palit Microsystems, Panasonic R+D Center Singapore, Samsung S1, Seventh Sense Artificial Intelligence, Touchless ID, and Veridas Digital Authentication Solutions S.L.
- ▷ We have retired results for 10 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-09-26 changes since 2022-08-30:

- ▷ We have added results for first algorithms from three developers: Codeline, First Credit Bureau Kazakhstan, and InfoCert.
- ▷ We have added results for new algorithms from 14 returning developers: Advancegroup, Armatura LLC, Beijing Hisign Technology, Cybercore, Cyberlink Corp, Herta Security, ICM Airport Technics, InsightFace AI, Metsakuur, NSENSE Corp, Samsung-SDS, Videmo Intelligente Videoanalyse, Vietnam Posts and Telecommunications Group, and Vision Intelligence Center of Meituan.
- ▷ We have retired results for 11 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-08-30 changes since 2022-07-29:

- ▷ We have added results for first algorithms from two developers: Aximetria, Intellibrain Technological Projects
- ▷ We have added results for new algorithms from twelve returning developers: Alchera Inc, Dermalog, Idemia, Incode Technologies Inc, Intellivision, Kasikorn Labs, Megvii/Face++, Techsign, TuringTech.vip, Universidade de Coimbra, Verijelais, Vixvizon

- We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-07-29 changes since 2022-06-27:

- We have added results for first algorithms from seven developers: FRP LLC (Hawaii), IMDS Software, Inspur (Beijing) Electronic Information Industry, Intema - LGL Group, PAPAGO, Qaz Biometric Systems, and VIDA-Digital Identity
- We have added results for new algorithms from nine returning developers: Cyberextruder, Glory, Maxvision Technology, Rank One Computing, Securif AI, Suprema AI, Suprema ID, Toshiba, and Yuan High-Tech Development.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-07-29 changes since 2022-06-27:

- We have added results for first algorithms from seven developers: FRP LLC (Hawaii), IMDS Software, Inspur (Beijing) Electronic Information Industry, Intema - LGL Group, PAPAGO, Qaz Biometric Systems, and VIDA-Digital Identity
- We have added results for new algorithms from nine returning developers: Cyberextruder, Glory, Maxvision Technology, Rank One Computing, Securif AI, Suprema AI, Suprema ID, Toshiba, and Yuan High-Tech Development.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-06-27 changes since 2022-06-03:

- We have added results for first algorithms from two developers: Krungthai Bank, and Smartbiometrik.
- We have added results for new algorithms from thirteen returning developers: Aiseemu, Corsight, Digidata, Griaule, Guangzhou Pixel Solutions, Hangzhuo AI Network Information Technology, Neurotechnology, Real Networks, Samsung S1, Sensetime Group, Smart Engines, Verihubs Inteligensia, and VinBigData.
- We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-06-03 changes since 2022-05-05:

- We have added results for first algorithms from seven developers: Jaak IT, Metsakuur, Palit Microsystems, Smarvist Teknoloji, and Touchless ID.
- We have added results for new algorithms from sixteen returning developers: Cyberlink, FaceOnLive, Kakao Enterprise, Line Corporation (Line Clova), Multi-Modality Intelligence, NEO Systems, and Unissey
- We have retired results for four algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- We have moved the results for the twenty human-difficult pairs used in the May 2018 paper *Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms* by Phillips et al. [1]. to the algorithm-specific report cards (example: PDF).
- Likewise, we have added figures showing impostor distribution shifts across demographics to the report card.

2022-05-05 changes since 2022-03-18:

- We have added results for first algorithms from seven developers: Accurascan, DICIO, FacePhi, Pangiam, University of Surrey-CVSSP, and Veridium.
- We have added results for new algorithms from sixteen returning developers: ACI Software, Canon Inc, Cloudwalk - Moontime Smart Technology, Cybercore,

2022-05-05 changes since 2022-03-18:

- We have added results for first algorithms from seven developers: Accurascan, DICIO, FacePhi, Pangiam, University of Surrey-CVSSP, and Veridium.
- We have added results for new algorithms from sixteen returning developers: ACI Software, Canon Inc, Cloudwalk - Moontime Smart Technology, Cybercore, Cyberextruder, Gemalto Cogent, HyperVerge Inc, KuKe3D Technology, Megvii/Face++, Mobbeel Solutions, Panasonic R+D Center Singapore, Qnap Security, Samsung-SDS, Vietnam Posts and Telecommunications Group, Viettel Group, and Vision Intelligence Center of Meituan.
- We have retired results for 12 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-03-18 changes since 2022-02-23:

- We have added support for the detection of multiple people in a single image (see Section 1.2). Specifically the API allows an algorithm to extract features from one or more faces it detects in an image. NIST scores such cases as a correct match when any detected face matches the reference photo, and as a false positive when either face matches a non-mated reference photo. The expected effect of doing this will be to improve reported false non-match rates, and to minimally elevate false match rates. This technique was only applied to images of type "border" and "kiosk".
- We have added results for first algorithms from four developers: IntelliVIX, Kasikorn Labs, Lebentech Biometrics, and Wicket.
- We have added results for new algorithms from 10 returning developers: Chunghwa Telecom, Cloudmatrix, Beijing DeepSense Technologies, FarBar Inc, Imagus Technology Pty, Intellivision, Maxvision Technology, NHN Corp, Seventh Sense Artificial Intelligence, and Verigram.
- We have retired results for 4 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-02-23 changes since 2022-01-24:

- We have added results for first algorithms from four developers: AFIS and Biometrics Consulting, Digidata, Graymatics, Hangzhuo AIlu Network Information Technology, KnowUTech LLC, Sukshi Technology Innovation, T4iSB, and TuringTech.vip

- We have added results for new algorithms from 18 returning developers: Cognitec Systems GmbH, GeoVision Inc, Glory, Herta Security, Intel Research Group, InsightFace AI, Kakao Enterprise, N-Tech Lab, Omnigarde Ltd, Papilon Savunma, Paravision, Realnetworks Inc, Reveal Media Ltd, Shenzhen Inst Adv Integrated Tech CAS, Suprema AI Inc, Toshiba, Universidade de Coimbra, and Yuan High-Tech Development
- We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2022-01-24 changes since 2022-01-20:

- We have added results for new algorithms from one returning developer: Vocord.

2022-01-20 changes since 2021-12-18:

- We have added results for first algorithms from four developers: Armatura, Beyne.AI, One More Security, and VinBigData
- We have added results for new algorithms from 19 returning developers: AuthenMetric, BOE Technology Group, Cybercore, Cyberlink, Dahua Technology, FaceTag Co, Innovatrics, Megvii, Mobbeel Solutions, Neurotechnology, Oz Forensics, Rank One Computing, Regula Forensics, Samsung S1, Securif AI, Sensetime Group, TigerIT Americas, Videmo Intelligente Videoanalyse, and YooniK.
- We have retired results for 14 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

: **2021-12-16** changes since 2021-11-22:

- We have added results for first algorithms from five developers: Alfabeta, Cloudmatrix, Euronovate SA, FaceOnLive Inc, and Mobipin Technology.
- We have added results for new algorithms from ten returning developers: ACI Software, ITMO University, NEO Systems, Guangzhou Pixel Solutions, Panasonic R+D Center Singapore, Qnap Security, Scanovate, Tevian, Unissey, and Vietnam Posts and Telecommunications Group.
- We have retired results for eight algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We have revamped the figure showing performance on 20 pairs of open-source images. It now color-codes false negatives and postives against a default threshold value.

2021-11-22 changes since 2021-10-28:

- We have added results to the website for kiosk-collected images where the design and geometry configuration mean that many images have considerable downward pitch angle. In some images, the face is partially cropped. Some images have other background faces.
- We have stopped using child exploitation images in FRVT, as we lost access to the imagery. All results for that set have been removed from the website, and will be removed from future PDF reports.
- We have added results for first algorithms from seven new developers: CUDO Communication, Daon, KuKe3D Technology, Mantra Softech India, Maxvision Technology, Multi-Modality Intelligence, and Samsung-SDS.

- We have added results for new algorithms from seven returning developers: Acer Incorporated, Cloudwalk-Moontime Smart Technology, Gorilla Technology, ID3 Technology, Incode Technologies, NSENSE Corp., and SQIsoft.
- We have retired results for six algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-10-28 changes since 2021-09-08:

- We have substantially revised the algorithm-specific report cards that are linked from the FRVT results page. (Example: HTML).
- We have added results for first algorithms from eight new developers: Beijing Mendaxia Technology, Beijing Hisign Technology, Biocube Matrics, Clearview AI, Reveal Media, Toppan ID Gate, Verigram, and Viettel High Technology.
- We have added results for new algorithms from thirty returning developers: 20Face, 3divi, Canon Inc Chunghwa Telecom, Corsight, Decatur Industries, Deepglint, Dermalog, FaceTag, Fiberhome Telecommunication Technologies, GeoVision, ICM Airport Technics, Imagus Technology, InsightFace AI, Kakao Enterprise, Kookmin University, Line Corporation, N-Tech Lab, NotionTag Technologies, Realnetworks, Suprema ID, Taiwan-Certificate Authority, Toshiba, Tripleize, Trueface.ai, Veridas Digital Authentication, Visidon, VisionLabs, YooniK, and Yuan High-Tech Development.
- We have retired results for twenty algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-09-08 changes since 2021-08-02:

- We have added results for first algorithms from seven new developers: Griaule, SQIsoft, Qnap Security, Techsign, Smart Engines, Verihubs, and Wuhan Tianyu Information Industry.
- We have added results for new algorithms from sixteen returning developers: ADVANCE.AI, AuthenMetric, CloudSmart Consulting, Code Everest Pvt, Cognitec Systems, Thales Gemalto Cogent, Intel Research Group, Omnigarde, Oz Forensics, Rank One Computing, Samsung S1 Corp, Securif AI, Tevian, TigerIT Americas, Universidade de Coimbra, and Vigilant Solutions
- We have retired results for eleven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-08-02 changes since 2021-06-25:

- We have added results for first algorithms from eight new developers: Bee the Data, Closeli Inc, Coretech Knowledge Inc, Deepsense (France), ioNetworks Inc, Kakao Pay Corp, Seventh Sense Artificial Intelligence, and SK Telecom.
- We have added results for new algorithms from fifteen returning developers: Alchera Inc, Adera Global PTE, Aware, Bresee Technology, Cyberlink Corp, Expasoft LLC, Fujitsu Research and Development Center, Gorilla Technology, Idemia, Neurotechnology, NEO Systems, NHN Corp, Paravision, Panasonic R+D Center Singapore, and Shenzhen University-Macau University of Science and Technology.
- We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2021-06-25 changes since 2021-05-21:

- ▷ We have added results for first algorithms from six new developers: Alice Biometrics, BOE Technology Group, Fincore, Neosecu, Sodec App, and Yuntu Data and Technology.
- ▷ We have added results for new algorithms from seven returning developers: Incode Technologies, HyperVerge, Mobbeel Solutions, Guangzhou Pixel Solutions, Remark Holdings, Sensetime, and Vietnam Posts and Telecommunications Group.
- ▷ We have retired results for four algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-05-21 changes since 2021-04-26:

- ▷ We have added results for first algorithms from five new developers: Ekin Smart City Technologies, Suprema ID, Tripleize, Taiwan-Certificate Authority, and Vision Intelligence Center of Meituan.
- ▷ We have added results for new algorithms from eight returning developers: ID3 Technology, Imagus Technology, Momentum Digital, N-Tech Lab, NSENSE, Shanghai Jiao Tong University, Vision-Box, and Yuan High-Tech Development
- ▷ We have retired results for seven algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-04-26 changes since 2021-04-16:

- ▷ We have added results for first algorithms from three new developers: Quantasoft, Rendip, and NEO Systems.
- ▷ We have added results for new algorithms from four returning developers: 3Divi, Realnetworks, Veridas Digital Authentication Solutions, and Universidade de Coimbra.
- ▷ We have retired results for three algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-04-16 changes since 2021-03-19:

- ▷ We have added results for first algorithms from six new developers: 20Face, Beijing DeepSense Technologies, BitCenter UK, Enface, FaceTag, InsightFace AI, Line Corporation, Lema Labs, Nanjing Kiwi Network Technology, Omnigarde, Regula Forensics, and Suprema.
- ▷ We have added results for new algorithms from ten returning developers: CloudSmart Consulting, Dermalog, GeoVision, Neurotechnology, Panasonic R+D Center Singapore, Samsung S1, Securif AI, Trueface.ai, Vigilant Solutions, and Visidon.
- ▷ We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-03-19 changes since 2021-03-05:

- ▷ We have added results for first algorithms from six new developers: Ajou University, AuthenMetric, Code Everest, Corsight, Papilon Savunma, and NHN Corp
- ▷ We have added results for new algorithms from seven returning developers: Alchera, Deepglint, Fiberhome Telecommunication Technologies, Kakao Enterprise, Kookmin University, Megvii/Face++, and NotionTag Technologies.
- ▷ We have updated many of the hyperlinked HTML report-cards to include seven figures on demographic dependence. Figures of this kind first appeared, and are documented in, the December 2019 document, NIST Interagency Report 8280 on demographic differentials in face recognition. The figures quantify false negative dependence on demographics using "visa-border" comparisons, and false positive dependence using comparisons of "application" photos that uniformly of quality and similar to visa photos.

2021-03-05 changes since 2021-01-19:

- ▷ We have added results for first algorithms from three new developers: IVA Cognitive, Mobbeel, and MoreDian Technology.
- ▷ We have added results for new algorithms from returning developers: Ability Enterprise - Andro Video, ACI Software, Adera Global, AnyVision, BioID Technologies, China Electronics Import-Export, Cognitec Systems, Fujitsu Research and Development Center, Glory, Guangzhou Pixel Solutions, Hengrui AI Technology, Incode Technologies, Intel Research, iQIYI, Mobai, Oz Forensics, Paravision, VisionLabs, and Xforward AI Technology.
- ▷ We have added a new "resources" tab to the main webpage. It includes sortable columns for data related to speed, model size, storage, and memory consumption.
- ▷ We have retired results for 13 algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2021-01-19 changes since 2020-12-18:

- ▷ This report adds results for first algorithms from four developers: Herta Security, Irex AI, Shenzhen University-Macau University of Science and Technology, and Vietnam Posts and Telecommunications Group. See Table 22 for more information.
- ▷ The report also includes results for thirteen developers who have previously submitted algorithms: Bresee Technology, Canon (previously Canon Information Technology (Beijing)), Cyberlink, CSA IntelliCloud Technology, Dahua Technology, ID3 Technology, Imagus Technology (Vixvizion), Moontime Smart Technology, N-Tech Lab, Thales Cogent, Veridas Digital Authentication Solutions, Vocord, and Yuan High-Tech Development.
- ▷ We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

2020-12-18 changes since 2020-10-09:

- ▷ This report adds results for first algorithms from ten developers: BitCenter UK, CloudSmart Consulting, Cubox, Institute of Computing Technology, Naver Corp, Minivision, NSENSE Corp, Viettel Group, Visage Technologies, and Xiamen University. See Table 22 for more information.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- The report also includes results for eighteen developers who have previously submitted algorithms: ADVANCE.AI, Awidit Systems, Chosun University, Dermalog, GeoVision, ICM Airport Technics, Idemia, Institute of Information Technologies, Kakao Enterprise, Neurotechnology, Panasonic R+D Center Singapore, Rank One Computing, Sensetime Group, Shanghai Jiao Tong University, TigerIT Americas LLC, Vigilant Solutions, Winsense, and YooniK
- We have retired results for twelve algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since September 18, 2020:

- This report adds results for first algorithms from five developers: Aigen, Cortica, Kookmin University, Securif AI and Vinai.
- The report also includes results for three developers who have previously submitted algorithms: Fujitsu Laboratories, Hengrui AI, and X-Forward AI.
- In the per-algorithm report-cards linked from tables and the main webpage, we have added a chart to showing reduction in error rates over the course of FRVT i.e. from 2017 onwards for all algorithms supplied by that developer. Similarly we have added a chart showing error rate reductions for our test of protective face mask verification.
- We plan to continue evaluating algorithms on various mask datasets. We hold that algorithms should be capable of detecting masks and verifying identity of all combinations of masked and unmasked faces. We have accordingly increased the amount to time allowed to extract those features from 1.0 to 1.5 seconds.

Changes since August 25, 2020:

- This report adds results for first algorithms from eight new developers. Akurat Satu Indonesia, Cybercore, Decatur Industries, Innef Labs, Satellite Innovation/Eocortex, Expasoft, and Mobai.
- The report includes results for seven developers who have previously submitted algorithms: 3Divi, BioID Technologies, Incode Technologies, Innovatrics, iSAP Solution, Synology, and Tevian.
- We have retired results for five algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since July 27, 2020:

- We have introduced per-algorithm report sheets. These are HTML documents linked from the accuracy tables in this report (i.e. Table 92) and on the FRVT 1:1 homepage. The sheets contain interactive graphics allowing, for example, mouseover exploration of FNMR(T) and FMR(T). Some of their content had previously appeared in this document.
- This report adds results for algorithms from six new developers. ACI Software, Bresee Technology, Fiberhome Telecommunication Technologies, Imageware Systems, Oz Forensics, and Pensees.
- The report includes results for thirteen developers who have previously submitted algorithms: Canon Information Technology (Beijing), Cyberlink, Dahua Technology, Gorilla Technology, ID3 Technology, Intel Research Group, iQIYI Inc, Momentum Digital, Netbridge Technology, Tech5 SA, Shenzhen AiMall Tech, Vigilant Solutions, and VisionLabs.
- We have retired results for nine algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.

Changes since May 18, 2020:

- The report is the first FRVT update since the pandemic closed it from March to June 2020.
- This report includes results for algorithms from nine new developers: GeoVision Inc, Su Zhou NaZhiTianDi Intelligent Technology, YooniK, AYF Technology, PXL Vision AG, Yuan High-Tech Development, Beihang University-ERCACAT, ICM Airport Technics, and Staqu Technologies
- This report includes results for algorithms from 15 returning developers Acer Incorporated, Antheus Technologia, Chosun University, Chunghwa Telecom, Idemia, Moontime Smart Technology, Neurotechnology, Guangzhou Pixel Solutions, Panasonic R+D Center Singapore, Rank One Computing, Scanovate, Shanghai Universiy - Shanghai Film Academy, Synesis, Trueface.ai, and Veridas Digital Authentication Solutions
- We have retired results for ten algorithms per our policy to only list results for two algorithms per developer. Results for retired algorithms appear in prior versions of this report in the archive.
- We separated timing and other resource consumption from the main participation table. The new Table 57 includes template generation durations for four kinds of images, not just mugshots.
- We have published a separate report, NIST Interagency Report 8311 on accuracy of pre-pandemic algorithms on subjects wearing face masks. We plan to track improvements in accuracy on masked images going forward. In particular, we invite submission of algorithms that can detect whether a person is wearing a mask, extract features from the full face or the exposed periocular region, and do appropriate comparison. We do not intend to evaluate algorithms that assume 100% of images will be of masked individuals.

Changes since March 25, 2020:

- The report is a maintenance release - it does not add any new algorithms, and FRVT has been closed to new algorithms since mid March 2020.
- We modified the primary accuracy summary, Table 92, as follows:
  - For visa images, the column for FNMR at FMR = 0.0001 has been removed. The visa images are so highly controlled that the error rates for the most accurate algorithms are dominated by false rejection of very young children and by the presence of a few noisy greyscale images. For now, two visa columns remain: FNMR at FMR= 10^{-6} and, for matched covariates, FNMR at FMR= 10^{-4}.
  - We have inserted a new column labelled "BORDER" giving accuracy for comparison of moderately poor webcam border-crossing photos that exhibit pose variations, poor compression, and low contrast due to strong background illumination. The accuracies are the worst from all cooperative image datasets used in FRVT.
- Accordingly, we updated the failure-to-template rates in Table 118.
- We withdrew a figure showing how false matches are concentrated in certain visa images used in cross-comparison, because it didn't attempt to include demographic information.

Changes since February 27, 2020:

- The report adds results algorithms from two new developers: Beijing Alleyes Technology, and the Chinese University of Hong Kong. Results for newly submitted algorithms from two other developers will appear in the next report.

- The report adds results for algorithms from thirteen returning developers: ASUSTek Computer, Aware, Cyberlink Corp, Gorilla Technology, Innovative Technology, Kakao Enterprise, Lomonosov Moscow State University, Panasonic R+D Center Singapore, Shenzhen AiMall Technology, Shenzhen Intellifusion Technologies, Synology, Tech5 SA, and Via Technologies.
- Per policy to only list results for two algorithms per developer, we have dropped results for algorithms from Aware, Cyberlink, Gorilla Technology, Kakao Enterprise, Lomonosov Moscow State University, Panasonic R+D Center Singapore, and Tech5 SA.

Changes since January 20, 2020:

- The report adds results for five new developers: Ability Enterprise (Andro Video), Chosun University, Fujitsu Research and Development Center, University of Coimbra, and Xforward AI Technology.
- The report adds results for algorithms from six returning developers: AlphaSSTG, Incode Technologies, Kneron. Shanghai Jiao Tong University, Vocord, and X-Laboratory.
- We have corrected template comparison timing numbers for algorithms submitted September 2019 to January 2020. The values reported previously were slower due to a software bug.
- We have dropped results for algorithms from Vocord and Incode per policy to only list results for two algorithms per developer.
- The FRVT 1:1 homepage has been updated with latest accuracy results.
- The FRVT 1:N homepage now includes an update to the September 2019 NIST Interagency Report 8271. The new report adds results for one-to-many search algorithms submitted to NIST from June 2019 to January 2020.

Changes since January 6, 2020:

- Section 2 has been updated to better describe the Visa and Border images. The caption for Table 92 has been updated to better relate the accuracy values to particular image comparisons.
- The report adds results for five new developers: Acer, Advance.AI, Expasoft, Netbridge Technology, and Videmo Intelligente Videoanalyse.
- The report adds results for algorithms from 7 returning developers: China Electronics Import-Export Corp, Intel Research Group, ITMO University, Neurotechnology, N-Tech Lab, Rokid, and VisionLabs.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: N-Tech Lab, Neurotechnology, ITMO, Visionlabs, and CEIEC.
- The FRVT homepage has been updated with latest accuracy results.

Changes since November 11, 2019:

- Table 57 has been updated to include runtime memory usage. This is the first time such a quantity has been reported. The value is the peak size of the resident set size logged during enrollment of single images.
- We have migrated summary results table to a new platform that supports sortable tables: https://pages.nist.gov/frvt/html/frvt11.html
- The report adds results for four new developers: Antheus Technologia, BioID Technologies SA, Canon Information Tech. (Beijing), Samsung S1 (listed in the tables as S1), and Taiwan AI Labs.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- The report adds results for algorithms from 13 returning developers: Anke Investments, Chunghwa Telecom, Deepglint, Institute of Information Technologies, iQIYI, Kneron, Ping An Technology, Paravision, KanKan Ai, Rokid Corporation, Shanghai Universiy - Shanghai Film Academy, Veridas Digital Authentication Solutions, and Videonetics Technology.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: remarkai-000, veridas-001, sensetime-001, iit-000, anke-003, and everai-002. Results for these are available in prior editions of this report linked from the FRVT page.
- We issued NIST Interagency Report 8280: FRVT Part 3: Demographics on 2019-12-19. It includes results for many of the algorithms covered by this report.

Changes since October 16, 2019:

- The report adds results for ten new developers: Ai-Union Technology, ASUSTek Computer, DiDi ChuXing Technology, Innovative Technology, Luxand, MVision, Pyramid Cyber Security + Forensic, Scanovate, Shenzhen AiMall Tech, and TUPU Technology.
- The report adds results for 12 returning developers: CTBC Bank Glory Gorilla Technology Guangzhou Pixel Solutions Imagus Technology Incode Technologies Lomonosov Moscow State University Rank One Computing Samtech InfoNet Shanghai Ulucu Electronics Technology Synesis, and Winsense.
- We have dropped results from this edition of the report per policy to only list results for two algorithms per developer: glory-000, gorilla-002, incode-003, rankone-006, and synesis-004.
- Results for five recently submitted algorithms will appear in the next report.

Changes since September 11, 2019:

- The report adds results for five new participants: Awidit Systems (Awiros), Momemtum Digital (Sertis), Trueface AI, Shanghai Jiao Tong University, and X-Laboratory.
- The reports adds results for five new algorithms from returning developers: Cyberlink, Hengrui AI Technology, Idemia, Panasonic R+D Singapore, and Tevian. This causes three algorithm, to be de-listed from the report per policy to list results for two algorithms per developer.

Changes since July 31 2019:

- The HTML table on the FRVT 1:1 homepage has been updated to include a column for cross-domain Visa-Border verification. Results for this new dataset appeared in the July 29 report under the name "CrossEV" - these are now renamed "Visa-Border".
- The FRVT 1:1 homepage lists algorithms according to lowest mean rank accuracy:
  Rank($\text{FNMR}_{\text{VISA}}$ at FMR = 0.000001) +
  Rank($\text{FNMR}_{\text{VISA-BORDER}}$ at FMR = 0.000001) +
  Rank($\text{FNMR}_{\text{MUGSHOT}}$ at FMR = 0.00001 after 14 years) +
  Rank($\text{FNMR}_{\text{WILD}}$ at FMR = 0.00001)
  This ordering rewards high accuracy across all datasets.
- The main results in Table 92 is now in landscape format to accomodate extra columns for the Visa-Border set, and mugshot comparisons after at least 12 years.
- The report adds results for nine new participants: Alpha SSTG, Intel Research, ULSee, Chungwa Telecon, iSAP Solution, Rokid, Shenzhen EI Networks, CSA Intellicloud, Shenzhen Intellifusion Technologies.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- The reports adds results for six new algorithms from returning developers: Innovatrics, Dahua Technology, Tech5 SA, Intellivision, Nodeflux and Imperial College, London. One algorithm, from Imperial has been retired, per policy to list results for two algorithms per developer.
- The cross-country false match rate heatmaps have been replotted to reveal more structure by listing countries by region instead of alphabetically.
- The next version of this report will be posted around October 18, 2019.

Changes since July 3 2019:

- The HTML table on the FRVT 1:1 homepage has been updated to list the 20 most accurate developers rather than algorithms, choosing the most accurate algorithm from each developer based on visa and mugshot results. Also, the algorithms are ordered in terms of lowest mean rank across mugshot, visa and wild datasets, rewarding broad accuracy over a good result on one particular dataset.
- This report includes results for a new dataset - see the column labelled "visa-border" in Table 5. It compares a new set of high quality visa-like portraits with a set webcam border-crossing photos that exhibit moderately poor pose variations and background illumination. The two new sets are described in sections 2.2 and 2.4. The comparisons are "cross-domain" in that the algorithm must compare "visa" and "wild" images. Results for other algorithms will be added in future reports as they become available.
- This report adds results for algorithms from 9 developers submitted in early July 2019. These are from 3DiVi, Camvi, EverAI-Paravision, Facesoft, Farbar (F8), Institute of Information Technologies, Shanghai U. Film Academy, Via Technologies, and Ulucu Electronics Tech. Six of these are new participants.
- Several other algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for September 5. That report will include results for new datasets.
- Older algorithms from Everai, Camvi and 3DiVi, have been retired, per the policy to list only two algorithms per developer.

Changes since June 20 2019:

- This report adds results for algorithms from 18 developers submitted in early June 2019. These are from CTBC Bank, Deep Glint, Thales Cogent, Ever AI Paravision, Gorilla Technology, Imagus, Incode, Kneron, N-Tech Lab, Neurotechnology, Notiontag Technologies, Star Hybrid, Videonetics, Vigilant Solutions, Winsense, Anke Investments, CEIEC, and DSK. Nine of these are new participants.
- Several other algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for August 1.
- Older algorithms from Everai, Thales Cogent, Gorilla Technology, Incode, Neurotechnology, N-Tech Lab and Vigilant Solutions have been retired, per the policy to list only two algorithms per developer.

Changes since April 2019:

- This report adds results for nine algorithms from nine developers submitted in early June 2019. These are from Tencent Deepsea, Hengrui, Kedacom, Moontime, Guangzhou Pixel, Rank One Computing, Synesis, Sensetime and Vocord.
- Another 23 algorithms have been submitted and are being evaluated. Results will be released in the next report, scheduled for July 3.
- Older algorithms for Rank One, Synesis, and Vocord have been retired, per the policy to list only two algorithms per developer.

Changes since February 2019:

- This report adds results for 49 algorithms from 42 developers submitted in early March 2019.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- This report omits results for algorithms that we retired. We retired for three reasons: 1. The developer submitted a new algorithm, and we only list two. 2. The algorithm needs a GPU, and we no longer allow GPU-based algorithms. 3. Inoperable algorithms.
- Previous results for retired algorithms are available in older editions of this report linked here.
- The mugshot database used from February 2017 to January 2019 has been replaced with an extract of the mugshot database documented in NIST Interagency Report 8238, November 2018. The new mugshot set is described in section 2.5 and is adopted because:
  - It has much better identity label integrity, so that false non-match rates are substantially lower than those reported in FRVT 1:1 reports to date - see Figure 296.
  - It includes images collected over a 17 year period such that ageing can be much better characterized - - see Figure 813.
- Using the new mugshot database, Figure 813 shows accuracy for four demographic groups identified in the biographic metadata that accompanies the data: black females, black males, white females and white males.
- The report added a figure (now moved to web) with results for the twenty human-difficult pairs used in the May 2018 paper *Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms* by Phillips et al. [1].
- The report uses an update to the wild image database that corrects some ground truth labels.
- Some results for the child exploitation database are not complete. They are typically updated less frequently than for other image sets.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Contents

ACKNOWLEDGMENTS 1

DISCLAIMER 1

INSTITUTIONAL REVIEW BOARD 1

1 METRICS 138
1.1 CORE ACCURACY . . . 138
1.2 MULTI-TEMPLATE SCORING METHODOLOGY . . . 138

2 DATASETS 139
2.1 VISA IMAGES . . . 139
2.2 APPLICATION IMAGES . . . 139
2.3 APPLICATION IMAGES WITH HEAD YAW . . . 139
2.4 BORDER CROSSING IMAGES . . . 140
2.5 MUGSHOT IMAGES . . . 140
2.6 KIOSK IMAGES . . . 140
2.7 WILD IMAGES . . . 141

3 RESULTS 141
3.1 TEST GOALS . . . 141
3.2 TEST DESIGN . . . 142
3.3 FAILURE TO ENROLL . . . 145
3.4 RECOGNITION ACCURACY . . . 172
3.5 GENUINE DISTRIBUTION STABILITY . . . 805
3.5.1 EFFECT OF BIRTH PLACE ON THE GENUINE DISTRIBUTION . . . 805
3.5.2 EFFECT OF AGEING . . . 896
3.5.3 EFFECT OF AGE ON GENUINE SUBJECTS . . . 985
3.6 IMPOSTOR DISTRIBUTION STABILITY . . . 1078
3.6.1 EFFECT OF BIRTH PLACE ON THE IMPOSTOR DISTRIBUTION . . . 1078
3.6.2 EFFECT OF AGE ON IMPOSTORS . . . 1081

List of Tables

1 PARTICIPANT INFORMATION . . . 44
2 PARTICIPANT INFORMATION . . . 45
3 PARTICIPANT INFORMATION . . . 46
4 PARTICIPANT INFORMATION . . . 47
5 PARTICIPANT INFORMATION . . . 48
6 PARTICIPANT INFORMATION . . . 49
7 PARTICIPANT INFORMATION . . . 50
8 PARTICIPANT INFORMATION . . . 51
9 PARTICIPANT INFORMATION . . . 52
10 PARTICIPANT INFORMATION . . . 53
11 PARTICIPANT INFORMATION . . . 54
12 PARTICIPANT INFORMATION . . . 55
13 PARTICIPANT INFORMATION . . . 56
14 PARTICIPANT INFORMATION . . . 57
15 PARTICIPANT INFORMATION . . . 58
16 PARTICIPANT INFORMATION . . . 59
17 PARTICIPANT INFORMATION . . . 60
18 PARTICIPANT INFORMATION . . . 61

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

19 PARTICIPANT INFORMATION . . . 62
20 PARTICIPANT INFORMATION . . . 63
21 PARTICIPANT INFORMATION . . . 64
22 PARTICIPANT INFORMATION . . . 65
23 ALGORITHM SUMMARY . . . 66
24 ALGORITHM SUMMARY . . . 67
25 ALGORITHM SUMMARY . . . 68
26 ALGORITHM SUMMARY . . . 69
27 ALGORITHM SUMMARY . . . 70
28 ALGORITHM SUMMARY . . . 71
29 ALGORITHM SUMMARY . . . 72
30 ALGORITHM SUMMARY . . . 73
31 ALGORITHM SUMMARY . . . 74
32 ALGORITHM SUMMARY . . . 75
33 ALGORITHM SUMMARY . . . 76
34 ALGORITHM SUMMARY . . . 77
35 ALGORITHM SUMMARY . . . 78
36 ALGORITHM SUMMARY . . . 79
37 ALGORITHM SUMMARY . . . 80
38 ALGORITHM SUMMARY . . . 81
39 ALGORITHM SUMMARY . . . 82
40 ALGORITHM SUMMARY . . . 83
41 ALGORITHM SUMMARY . . . 84
42 ALGORITHM SUMMARY . . . 85
43 ALGORITHM SUMMARY . . . 86
44 ALGORITHM SUMMARY . . . 87
45 ALGORITHM SUMMARY . . . 88
46 ALGORITHM SUMMARY . . . 89
47 ALGORITHM SUMMARY . . . 90
48 ALGORITHM SUMMARY . . . 91
49 ALGORITHM SUMMARY . . . 92
50 ALGORITHM SUMMARY . . . 93
51 ALGORITHM SUMMARY . . . 94
52 ALGORITHM SUMMARY . . . 95
53 ALGORITHM SUMMARY . . . 96
54 ALGORITHM SUMMARY . . . 97
55 ALGORITHM SUMMARY . . . 98
56 ALGORITHM SUMMARY . . . 99
57 ALGORITHM SUMMARY . . . 100
58 FALSE NON-MATCH RATE . . . 101
59 FALSE NON-MATCH RATE . . . 102
60 FALSE NON-MATCH RATE . . . 103
61 FALSE NON-MATCH RATE . . . 104
62 FALSE NON-MATCH RATE . . . 105
63 FALSE NON-MATCH RATE . . . 106
64 FALSE NON-MATCH RATE . . . 107
65 FALSE NON-MATCH RATE . . . 108
66 FALSE NON-MATCH RATE . . . 109
67 FALSE NON-MATCH RATE . . . 110
68 FALSE NON-MATCH RATE . . . 111
69 FALSE NON-MATCH RATE . . . 112
70 FALSE NON-MATCH RATE . . . 113
71 FALSE NON-MATCH RATE . . . 114
72 FALSE NON-MATCH RATE . . . 115
73 FALSE NON-MATCH RATE . . . 116
74 FALSE NON-MATCH RATE . . . 117

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

75 FALSE NON-MATCH RATE . . . 118
76 FALSE NON-MATCH RATE . . . 119
77 FALSE NON-MATCH RATE . . . 120
78 FALSE NON-MATCH RATE . . . 121
79 FALSE NON-MATCH RATE . . . 122
80 FALSE NON-MATCH RATE . . . 123
81 FALSE NON-MATCH RATE . . . 124
82 FALSE NON-MATCH RATE . . . 125
83 FALSE NON-MATCH RATE . . . 126
84 FALSE NON-MATCH RATE . . . 127
85 FALSE NON-MATCH RATE . . . 128
86 FALSE NON-MATCH RATE . . . 129
87 FALSE NON-MATCH RATE . . . 130
88 FALSE NON-MATCH RATE . . . 131
89 FALSE NON-MATCH RATE . . . 132
90 FALSE NON-MATCH RATE . . . 133
91 FALSE NON-MATCH RATE . . . 134
92 FALSE NON-MATCH RATE . . . 135
93 FAILURE TO ENROL RATES . . . 146
94 FAILURE TO ENROL RATES . . . 147
95 FAILURE TO ENROL RATES . . . 148
96 FAILURE TO ENROL RATES . . . 149
97 FAILURE TO ENROL RATES . . . 150
98 FAILURE TO ENROL RATES . . . 151
99 FAILURE TO ENROL RATES . . . 152
100 FAILURE TO ENROL RATES . . . 153
101 FAILURE TO ENROL RATES . . . 154
102 FAILURE TO ENROL RATES . . . 155
103 FAILURE TO ENROL RATES . . . 156
104 FAILURE TO ENROL RATES . . . 157
105 FAILURE TO ENROL RATES . . . 158
106 FAILURE TO ENROL RATES . . . 159
107 FAILURE TO ENROL RATES . . . 160
108 FAILURE TO ENROL RATES . . . 161
109 FAILURE TO ENROL RATES . . . 162
110 FAILURE TO ENROL RATES . . . 163
111 FAILURE TO ENROL RATES . . . 164
112 FAILURE TO ENROL RATES . . . 165
113 FAILURE TO ENROL RATES . . . 166
114 FAILURE TO ENROL RATES . . . 167
115 FAILURE TO ENROL RATES . . . 168
116 FAILURE TO ENROL RATES . . . 169
117 FAILURE TO ENROL RATES . . . 170
118 FAILURE TO ENROL RATES . . . 171

List of Figures

1 PERFORMANCE SUMMARY: FNMR VS. TEMPLATE SIZE TRADEOFF . . . 136
2 PERFORMANCE SUMMARY: FNMR VS. TEMPLATE TIME TRADEOFF . . . 137
3 EXAMPLE IMAGES . . . 141
(A) VISA . . . 141
(B) MUGSHOT . . . 141
(C) WILD . . . 141
(D) BORDER . . . 141
4 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 173

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

5 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 174
6 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 175
7 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 176
8 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 177
9 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 178
10 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 179
11 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 180
12 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 181
13 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 182
14 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 183
15 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 184
16 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 185
17 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 186
18 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 187
19 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 188
20 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 189
21 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 190
22 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 191
23 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 192
24 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 193
25 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 194
26 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 195
27 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 196
28 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 197
29 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 198
30 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 199
31 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 200
32 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 201
33 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 202
34 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 203
35 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 204
36 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 205
37 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 206
38 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 207
39 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 208
40 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 209
41 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 210
42 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 211
43 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 212
44 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 213
45 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 214
46 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 215
47 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 216
48 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 217
49 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 218
50 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 219
51 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 220
52 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 221
53 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 222
54 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 223
55 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 224
56 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 225
57 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 226
58 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 227
59 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 228
60 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 229
61 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 230

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

62 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 231
63 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 232
64 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 233
65 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 234
66 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 235
67 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 236
68 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 237
69 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 238
70 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 239
71 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 240
72 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 241
73 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 242
74 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 243
75 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 244
76 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 245
77 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 246
78 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 247
79 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 248
80 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 249
81 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 250
82 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 251
83 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 252
84 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 253
85 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 254
86 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 255
87 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 256
88 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 257
89 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 258
90 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 259
91 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 260
92 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 261
93 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 262
94 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 263
95 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 264
96 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 265
97 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 266
98 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 267
99 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 268
100 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 269
101 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 270
102 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 271
103 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 272
104 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 273
105 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 274
106 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 275
107 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 276
108 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 277
109 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 278
110 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 279
111 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 280
112 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 281
113 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 282
114 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 283
115 PERFORMANCE ON 20 HUMAN-DIFFICULT PAIRS . . . 284
116 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 285
117 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 286
118 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 287

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

119 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 288
120 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 289
121 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 290
122 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 291
123 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 292
124 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 293
125 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 294
126 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 295
127 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 296
128 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 297
129 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 298
130 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 299
131 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 300
132 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 301
133 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 302
134 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 303
135 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 304
136 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 305
137 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 306
138 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 307
139 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 308
140 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 309
141 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 310
142 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 311
143 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 312
144 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 313
145 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 314
146 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 315
147 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 316
148 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 317
149 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 318
150 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 319
151 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 320
152 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 321
153 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 322
154 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 323
155 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 324
156 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 325
157 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 326
158 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 327
159 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 328
160 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 329
161 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 330
162 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 331
163 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 332
164 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 333
165 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 334
166 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 335
167 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 336
168 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 337
169 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 338
170 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 339
171 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 340
172 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 341
173 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 342
174 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 343
175 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 344

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

176 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 345
177 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 346
178 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 347
179 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 348
180 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 349
181 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 350
182 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 351
183 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 352
184 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 353
185 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 354
186 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 355
187 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 356
188 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 357
189 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 358
190 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 359
191 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 360
192 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 361
193 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 362
194 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 363
195 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 364
196 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 365
197 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 366
198 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 367
199 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 368
200 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 369
201 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 370
202 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 371
203 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 372
204 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 373
205 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 374
206 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 375
207 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 376
208 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 377
209 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 378
210 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 379
211 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 380
212 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 381
213 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 382
214 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 383
215 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 384
216 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 385
217 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 386
218 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 387
219 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 388
220 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 389
221 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 390
222 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 391
223 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 392
224 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 393
225 ERROR TRADEOFF CHARACTERISTIC: VISA IMAGES . . . 394
226 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 395
227 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 396
228 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 397
229 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 398
230 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 399
231 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 400
232 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 401

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

233 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 402
234 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 403
235 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 404
236 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 405
237 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 406
238 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 407
239 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 408
240 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 409
241 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 410
242 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 411
243 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 412
244 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 413
245 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 414
246 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 415
247 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 416
248 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 417
249 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 418
250 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 419
251 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 420
252 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 421
253 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 422
254 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 423
255 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 424
256 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 425
257 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 426
258 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 427
259 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 428
260 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 429
261 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 430
262 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 431
263 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 432
264 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 433
265 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 434
266 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 435
267 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 436
268 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 437
269 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 438
270 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 439
271 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 440
272 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 441
273 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 442
274 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 443
275 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 444
276 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 445
277 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 446
278 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 447
279 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 448
280 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 449
281 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 450
282 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 451
283 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 452
284 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 453
285 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 454
286 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 455
287 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 456
288 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 457

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

289 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 458
290 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 459
291 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 460
292 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 461
293 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 462
294 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 463
295 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 464
296 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 465
297 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 466
298 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 467
299 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 468
300 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 469
301 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 470
302 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 471
303 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 472
304 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 473
305 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 474
306 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 475
307 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 476
308 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 477
309 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 478
310 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 479
311 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 480
312 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 481
313 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 482
314 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 483
315 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 484
316 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 485
317 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 486
318 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 487
319 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 488
320 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 489
321 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 490
322 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 491
323 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 492
324 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 493
325 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 494
326 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 495
327 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 496
328 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 497
329 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 498
330 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 499
331 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 500
332 ERROR TRADEOFF CHARACTERISTIC: WILD IMAGES . . . 501
333 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 502
334 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 503
335 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 504
336 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 505
337 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 506
338 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 507
339 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 508
340 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 509
341 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 510
342 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 511
343 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 512
344 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 513
345 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 514

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

346 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 515
347 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 516
348 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 517
349 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 518
350 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 519
351 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 520
352 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 521
353 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 522
354 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 523
355 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 524
356 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 525
357 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 526
358 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 527
359 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 528
360 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 529
361 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 530
362 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 531
363 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 532
364 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 533
365 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 534
366 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 535
367 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 536
368 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 537
369 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 538
370 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 539
371 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 540
372 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 541
373 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 542
374 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 543
375 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 544
376 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 545
377 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 546
378 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 547
379 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 548
380 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 549
381 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 550
382 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 551
383 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 552
384 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 553
385 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 554
386 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 555
387 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 556
388 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 557
389 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 558
390 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 559
391 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 560
392 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 561
393 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 562
394 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 563
395 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 564
396 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 565
397 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 566
398 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 567
399 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 568
400 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 569
401 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 570
402 FALSE MATCH RATES WITHIN AND ACROSS DEMOGRAPHIC GROUPS . . . 571

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

403 SEX AND RACE EFFECTS: MUGSHOT IMAGES 572
404 SEX AND RACE EFFECTS: MUGSHOT IMAGES 573
405 SEX AND RACE EFFECTS: MUGSHOT IMAGES 574
406 SEX AND RACE EFFECTS: MUGSHOT IMAGES 575
407 SEX AND RACE EFFECTS: MUGSHOT IMAGES 576
408 SEX AND RACE EFFECTS: MUGSHOT IMAGES 577
409 SEX AND RACE EFFECTS: MUGSHOT IMAGES 578
410 SEX AND RACE EFFECTS: MUGSHOT IMAGES 579
411 SEX AND RACE EFFECTS: MUGSHOT IMAGES 580
412 SEX AND RACE EFFECTS: MUGSHOT IMAGES 581
413 SEX AND RACE EFFECTS: MUGSHOT IMAGES 582
414 SEX AND RACE EFFECTS: MUGSHOT IMAGES 583
415 SEX AND RACE EFFECTS: MUGSHOT IMAGES 584
416 SEX AND RACE EFFECTS: MUGSHOT IMAGES 585
417 SEX AND RACE EFFECTS: MUGSHOT IMAGES 586
418 SEX AND RACE EFFECTS: MUGSHOT IMAGES 587
419 SEX AND RACE EFFECTS: MUGSHOT IMAGES 588
420 SEX AND RACE EFFECTS: MUGSHOT IMAGES 589
421 SEX AND RACE EFFECTS: MUGSHOT IMAGES 590
422 SEX AND RACE EFFECTS: MUGSHOT IMAGES 591
423 SEX AND RACE EFFECTS: MUGSHOT IMAGES 592
424 SEX AND RACE EFFECTS: MUGSHOT IMAGES 593
425 SEX AND RACE EFFECTS: MUGSHOT IMAGES 594
426 SEX AND RACE EFFECTS: MUGSHOT IMAGES 595
427 SEX AND RACE EFFECTS: MUGSHOT IMAGES 596
428 SEX AND RACE EFFECTS: MUGSHOT IMAGES 597
429 SEX AND RACE EFFECTS: MUGSHOT IMAGES 598
430 SEX AND RACE EFFECTS: MUGSHOT IMAGES 599
431 SEX AND RACE EFFECTS: MUGSHOT IMAGES 600
432 SEX AND RACE EFFECTS: MUGSHOT IMAGES 601
433 SEX AND RACE EFFECTS: MUGSHOT IMAGES 602
434 SEX AND RACE EFFECTS: MUGSHOT IMAGES 603
435 SEX AND RACE EFFECTS: MUGSHOT IMAGES 604
436 SEX AND RACE EFFECTS: MUGSHOT IMAGES 605
437 SEX AND RACE EFFECTS: MUGSHOT IMAGES 606
438 SEX AND RACE EFFECTS: MUGSHOT IMAGES 607
439 SEX AND RACE EFFECTS: MUGSHOT IMAGES 608
440 SEX AND RACE EFFECTS: MUGSHOT IMAGES 609
441 SEX AND RACE EFFECTS: MUGSHOT IMAGES 610
442 SEX AND RACE EFFECTS: MUGSHOT IMAGES 611
443 SEX AND RACE EFFECTS: MUGSHOT IMAGES 612
444 SEX AND RACE EFFECTS: MUGSHOT IMAGES 613
445 SEX AND RACE EFFECTS: MUGSHOT IMAGES 614
446 SEX AND RACE EFFECTS: MUGSHOT IMAGES 615
447 SEX AND RACE EFFECTS: MUGSHOT IMAGES 616
448 SEX AND RACE EFFECTS: MUGSHOT IMAGES 617
449 SEX AND RACE EFFECTS: MUGSHOT IMAGES 618
450 SEX AND RACE EFFECTS: MUGSHOT IMAGES 619
451 SEX AND RACE EFFECTS: MUGSHOT IMAGES 620
452 SEX AND RACE EFFECTS: MUGSHOT IMAGES 621
453 SEX AND RACE EFFECTS: MUGSHOT IMAGES 622
454 SEX AND RACE EFFECTS: MUGSHOT IMAGES 623
455 SEX AND RACE EFFECTS: MUGSHOT IMAGES 624
456 SEX AND RACE EFFECTS: MUGSHOT IMAGES 625
457 SEX AND RACE EFFECTS: MUGSHOT IMAGES 626
458 SEX AND RACE EFFECTS: MUGSHOT IMAGES 627
459 SEX AND RACE EFFECTS: MUGSHOT IMAGES 628

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

460 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 629
461 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 630
462 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 631
463 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 632
464 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 633
465 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 634
466 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 635
467 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 636
468 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 637
469 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 638
470 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 639
471 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 640
472 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 641
473 SEX AND RACE EFFECTS: MUGSHOT IMAGES . . . 642
474 SEX EFFECTS: VISA IMAGES . . . 643
475 SEX EFFECTS: VISA IMAGES . . . 644
476 SEX EFFECTS: VISA IMAGES . . . 645
477 SEX EFFECTS: VISA IMAGES . . . 646
478 SEX EFFECTS: VISA IMAGES . . . 647
479 SEX EFFECTS: VISA IMAGES . . . 648
480 SEX EFFECTS: VISA IMAGES . . . 649
481 SEX EFFECTS: VISA IMAGES . . . 650
482 SEX EFFECTS: VISA IMAGES . . . 651
483 SEX EFFECTS: VISA IMAGES . . . 652
484 SEX EFFECTS: VISA IMAGES . . . 653
485 SEX EFFECTS: VISA IMAGES . . . 654
486 SEX EFFECTS: VISA IMAGES . . . 655
487 SEX EFFECTS: VISA IMAGES . . . 656
488 SEX EFFECTS: VISA IMAGES . . . 657
489 SEX EFFECTS: VISA IMAGES . . . 658
490 SEX EFFECTS: VISA IMAGES . . . 659
491 SEX EFFECTS: VISA IMAGES . . . 660
492 SEX EFFECTS: VISA IMAGES . . . 661
493 SEX EFFECTS: VISA IMAGES . . . 662
494 SEX EFFECTS: VISA IMAGES . . . 663
495 SEX EFFECTS: VISA IMAGES . . . 664
496 SEX EFFECTS: VISA IMAGES . . . 665
497 SEX EFFECTS: VISA IMAGES . . . 666
498 SEX EFFECTS: VISA IMAGES . . . 667
499 SEX EFFECTS: VISA IMAGES . . . 668
500 SEX EFFECTS: VISA IMAGES . . . 669
501 SEX EFFECTS: VISA IMAGES . . . 670
502 SEX EFFECTS: VISA IMAGES . . . 671
503 SEX EFFECTS: VISA IMAGES . . . 672
504 SEX EFFECTS: VISA IMAGES . . . 673
505 SEX EFFECTS: VISA IMAGES . . . 674
506 SEX EFFECTS: VISA IMAGES . . . 675
507 SEX EFFECTS: VISA IMAGES . . . 676
508 SEX EFFECTS: VISA IMAGES . . . 677
509 SEX EFFECTS: VISA IMAGES . . . 678
510 SEX EFFECTS: VISA IMAGES . . . 679
511 SEX EFFECTS: VISA IMAGES . . . 680
512 SEX EFFECTS: VISA IMAGES . . . 681
513 SEX EFFECTS: VISA IMAGES . . . 682
514 SEX EFFECTS: VISA IMAGES . . . 683
515 SEX EFFECTS: VISA IMAGES . . . 684
516 SEX EFFECTS: VISA IMAGES . . . 685

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

517 SEX EFFECTS: VISA IMAGES . . . 686
518 SEX EFFECTS: VISA IMAGES . . . 687
519 SEX EFFECTS: VISA IMAGES . . . 688
520 SEX EFFECTS: VISA IMAGES . . . 689
521 SEX EFFECTS: VISA IMAGES . . . 690
522 SEX EFFECTS: VISA IMAGES . . . 691
523 SEX EFFECTS: VISA IMAGES . . . 692
524 SEX EFFECTS: VISA IMAGES . . . 693
525 SEX EFFECTS: VISA IMAGES . . . 694
526 SEX EFFECTS: VISA IMAGES . . . 695
527 SEX EFFECTS: VISA IMAGES . . . 696
528 SEX EFFECTS: VISA IMAGES . . . 697
529 SEX EFFECTS: VISA IMAGES . . . 698
530 SEX EFFECTS: VISA IMAGES . . . 699
531 SEX EFFECTS: VISA IMAGES . . . 700
532 SEX EFFECTS: VISA IMAGES . . . 701
533 SEX EFFECTS: VISA IMAGES . . . 702
534 SEX EFFECTS: VISA IMAGES . . . 703
535 SEX EFFECTS: VISA IMAGES . . . 704
536 SEX EFFECTS: VISA IMAGES . . . 705
537 SEX EFFECTS: VISA IMAGES . . . 706
538 SEX EFFECTS: VISA IMAGES . . . 707
539 SEX EFFECTS: VISA IMAGES . . . 708
540 SEX EFFECTS: VISA IMAGES . . . 709
541 SEX EFFECTS: VISA IMAGES . . . 710
542 SEX EFFECTS: VISA IMAGES . . . 711
543 SEX EFFECTS: VISA IMAGES . . . 712
544 SEX EFFECTS: VISA IMAGES . . . 713
545 SEX EFFECTS: VISA IMAGES . . . 714
546 SEX EFFECTS: VISA IMAGES . . . 715
547 SEX EFFECTS: VISA IMAGES . . . 716
548 SEX EFFECTS: VISA IMAGES . . . 717
549 SEX EFFECTS: VISA IMAGES . . . 718
550 SEX EFFECTS: VISA IMAGES . . . 719
551 SEX EFFECTS: VISA IMAGES . . . 720
552 SEX EFFECTS: VISA IMAGES . . . 721
553 SEX EFFECTS: VISA IMAGES . . . 722
554 SEX EFFECTS: VISA IMAGES . . . 723
555 SEX EFFECTS: VISA IMAGES . . . 724
556 SEX EFFECTS: VISA IMAGES . . . 725
557 SEX EFFECTS: VISA IMAGES . . . 726
558 SEX EFFECTS: VISA IMAGES . . . 727
559 SEX EFFECTS: VISA IMAGES . . . 728
560 SEX EFFECTS: VISA IMAGES . . . 729
561 SEX EFFECTS: VISA IMAGES . . . 730
562 SEX EFFECTS: VISA IMAGES . . . 731
563 SEX EFFECTS: VISA IMAGES . . . 732
564 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 733
565 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 734
566 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 735
567 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 736
568 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 737
569 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 738
570 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 739
571 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 740
572 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 741
573 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 742

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

574 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 743
575 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 744
576 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 745
577 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 746
578 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 747
579 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 748
580 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 749
581 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 750
582 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 751
583 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 752
584 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 753
585 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 754
586 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 755
587 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 756
588 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 757
589 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 758
590 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 759
591 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 760
592 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 761
593 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 762
594 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 763
595 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 764
596 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 765
597 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 766
598 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 767
599 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 768
600 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 769
601 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 770
602 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 771
603 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 772
604 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 773
605 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 774
606 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 775
607 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 776
608 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 777
609 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 778
610 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 779
611 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 780
612 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 781
613 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 782
614 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 783
615 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 784
616 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 785
617 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 786
618 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 787
619 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 788
620 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 789
621 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 790
622 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 791
623 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 792
624 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 793
625 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 794
626 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 795
627 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 796
628 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 797
629 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 798
630 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES 799

631 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 800
632 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 801
633 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 802
634 FALSE MATCH RATE CALIBRATION: MUGSHOT IMAGES . . . 803
635 FALSE MATCH RATE CALIBRATION: VISA IMAGES . . . 804
636 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 806
637 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 807
638 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 808
639 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 809
640 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 810
641 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 811
642 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 812
643 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 813
644 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 814
645 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 815
646 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 816
647 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 817
648 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 818
649 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 819
650 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 820
651 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 821
652 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 822
653 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 823
654 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 824
655 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 825
656 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 826
657 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 827
658 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 828
659 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 829
660 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 830
661 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 831
662 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 832
663 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 833
664 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 834
665 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 835
666 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 836
667 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 837
668 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 838
669 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 839
670 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 840
671 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 841
672 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 842
673 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 843
674 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 844
675 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 845
676 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 846
677 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 847
678 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 848
679 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 849
680 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 850
681 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 851
682 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 852
683 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 853
684 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 854
685 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 855
686 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 856
687 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 857

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

688 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 858
689 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 859
690 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 860
691 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 861
692 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 862
693 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 863
694 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 864
695 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 865
696 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 866
697 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 867
698 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 868
699 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 869
700 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 870
701 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 871
702 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 872
703 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 873
704 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 874
705 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 875
706 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 876
707 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 877
708 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 878
709 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 879
710 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 880
711 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 881
712 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 882
713 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 883
714 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 884
715 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 885
716 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 886
717 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 887
718 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 888
719 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 889
720 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 890
721 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 891
722 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 892
723 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 893
724 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 894
725 EFFECT OF COUNTRY OF BIRTH ON FNMR . . . 895
726 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 897
727 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 898
728 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 899
729 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 900
730 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 901
731 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 902
732 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 903
733 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 904
734 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 905
735 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 906
736 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 907
737 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 908
738 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 909
739 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 910
740 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 911
741 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 912
742 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 913
743 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 914
744 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 915

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

745 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 916
746 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 917
747 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 918
748 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 919
749 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 920
750 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 921
751 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 922
752 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 923
753 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 924
754 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 925
755 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 926
756 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 927
757 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 928
758 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 929
759 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 930
760 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 931
761 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 932
762 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 933
763 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 934
764 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 935
765 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 936
766 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 937
767 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 938
768 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 939
769 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 940
770 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 941
771 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 942
772 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 943
773 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 944
774 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 945
775 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 946
776 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 947
777 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 948
778 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 949
779 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 950
780 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 951
781 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 952
782 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 953
783 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 954
784 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 955
785 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 956
786 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 957
787 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 958
788 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 959
789 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 960
790 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 961
791 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 962
792 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 963
793 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 964
794 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 965
795 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 966
796 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 967
797 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 968
798 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 969
799 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 970
800 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 971
801 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES 972

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

802 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 973
803 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 974
804 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 975
805 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 976
806 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 977
807 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 978
808 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 979
809 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 980
810 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 981
811 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 982
812 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 983
813 ERROR TRADEOFF CHARACTERISTIC: MUGSHOT IMAGES . . . 984
814 EFFECT OF SUBJECT AGE ON FNMR . . . 986
815 EFFECT OF SUBJECT AGE ON FNMR . . . 987
816 EFFECT OF SUBJECT AGE ON FNMR . . . 988
817 EFFECT OF SUBJECT AGE ON FNMR . . . 989
818 EFFECT OF SUBJECT AGE ON FNMR . . . 990
819 EFFECT OF SUBJECT AGE ON FNMR . . . 991
820 EFFECT OF SUBJECT AGE ON FNMR . . . 992
821 EFFECT OF SUBJECT AGE ON FNMR . . . 993
822 EFFECT OF SUBJECT AGE ON FNMR . . . 994
823 EFFECT OF SUBJECT AGE ON FNMR . . . 995
824 EFFECT OF SUBJECT AGE ON FNMR . . . 996
825 EFFECT OF SUBJECT AGE ON FNMR . . . 997
826 EFFECT OF SUBJECT AGE ON FNMR . . . 998
827 EFFECT OF SUBJECT AGE ON FNMR . . . 999
828 EFFECT OF SUBJECT AGE ON FNMR . . . 1000
829 EFFECT OF SUBJECT AGE ON FNMR . . . 1001
830 EFFECT OF SUBJECT AGE ON FNMR . . . 1002
831 EFFECT OF SUBJECT AGE ON FNMR . . . 1003
832 EFFECT OF SUBJECT AGE ON FNMR . . . 1004
833 EFFECT OF SUBJECT AGE ON FNMR . . . 1005
834 EFFECT OF SUBJECT AGE ON FNMR . . . 1006
835 EFFECT OF SUBJECT AGE ON FNMR . . . 1007
836 EFFECT OF SUBJECT AGE ON FNMR . . . 1008
837 EFFECT OF SUBJECT AGE ON FNMR . . . 1009
838 EFFECT OF SUBJECT AGE ON FNMR . . . 1010
839 EFFECT OF SUBJECT AGE ON FNMR . . . 1011
840 EFFECT OF SUBJECT AGE ON FNMR . . . 1012
841 EFFECT OF SUBJECT AGE ON FNMR . . . 1013
842 EFFECT OF SUBJECT AGE ON FNMR . . . 1014
843 EFFECT OF SUBJECT AGE ON FNMR . . . 1015
844 EFFECT OF SUBJECT AGE ON FNMR . . . 1016
845 EFFECT OF SUBJECT AGE ON FNMR . . . 1017
846 EFFECT OF SUBJECT AGE ON FNMR . . . 1018
847 EFFECT OF SUBJECT AGE ON FNMR . . . 1019
848 EFFECT OF SUBJECT AGE ON FNMR . . . 1020
849 EFFECT OF SUBJECT AGE ON FNMR . . . 1021
850 EFFECT OF SUBJECT AGE ON FNMR . . . 1022
851 EFFECT OF SUBJECT AGE ON FNMR . . . 1023
852 EFFECT OF SUBJECT AGE ON FNMR . . . 1024
853 EFFECT OF SUBJECT AGE ON FNMR . . . 1025

854 EFFECT OF SUBJECT AGE ON FNMR 1026
855 EFFECT OF SUBJECT AGE ON FNMR 1027
856 EFFECT OF SUBJECT AGE ON FNMR 1028
857 EFFECT OF SUBJECT AGE ON FNMR 1029
858 EFFECT OF SUBJECT AGE ON FNMR 1030
859 EFFECT OF SUBJECT AGE ON FNMR 1031
860 EFFECT OF SUBJECT AGE ON FNMR 1032
861 EFFECT OF SUBJECT AGE ON FNMR 1033
862 EFFECT OF SUBJECT AGE ON FNMR 1034
863 EFFECT OF SUBJECT AGE ON FNMR 1035
864 EFFECT OF SUBJECT AGE ON FNMR 1036
865 EFFECT OF SUBJECT AGE ON FNMR 1037
866 EFFECT OF SUBJECT AGE ON FNMR 1038
867 EFFECT OF SUBJECT AGE ON FNMR 1039
868 EFFECT OF SUBJECT AGE ON FNMR 1040
869 EFFECT OF SUBJECT AGE ON FNMR 1041
870 EFFECT OF SUBJECT AGE ON FNMR 1042
871 EFFECT OF SUBJECT AGE ON FNMR 1043
872 EFFECT OF SUBJECT AGE ON FNMR 1044
873 EFFECT OF SUBJECT AGE ON FNMR 1045
874 EFFECT OF SUBJECT AGE ON FNMR 1046
875 EFFECT OF SUBJECT AGE ON FNMR 1047
876 EFFECT OF SUBJECT AGE ON FNMR 1048
877 EFFECT OF SUBJECT AGE ON FNMR 1049
878 EFFECT OF SUBJECT AGE ON FNMR 1050
879 EFFECT OF SUBJECT AGE ON FNMR 1051
880 EFFECT OF SUBJECT AGE ON FNMR 1052
881 EFFECT OF SUBJECT AGE ON FNMR 1053
882 EFFECT OF SUBJECT AGE ON FNMR 1054
883 EFFECT OF SUBJECT AGE ON FNMR 1055
884 EFFECT OF SUBJECT AGE ON FNMR 1056
885 EFFECT OF SUBJECT AGE ON FNMR 1057
886 EFFECT OF SUBJECT AGE ON FNMR 1058
887 EFFECT OF SUBJECT AGE ON FNMR 1059
888 EFFECT OF SUBJECT AGE ON FNMR 1060
889 EFFECT OF SUBJECT AGE ON FNMR 1061
890 EFFECT OF SUBJECT AGE ON FNMR 1062
891 EFFECT OF SUBJECT AGE ON FNMR 1063
892 EFFECT OF SUBJECT AGE ON FNMR 1064
893 EFFECT OF SUBJECT AGE ON FNMR 1065
894 EFFECT OF SUBJECT AGE ON FNMR 1066
895 EFFECT OF SUBJECT AGE ON FNMR 1067
896 EFFECT OF SUBJECT AGE ON FNMR 1068
897 EFFECT OF SUBJECT AGE ON FNMR 1069
898 EFFECT OF SUBJECT AGE ON FNMR 1070
899 EFFECT OF SUBJECT AGE ON FNMR 1071
900 EFFECT OF SUBJECT AGE ON FNMR 1072
901 EFFECT OF SUBJECT AGE ON FNMR 1073
902 EFFECT OF SUBJECT AGE ON FNMR 1074
903 EFFECT OF SUBJECT AGE ON FNMR 1075
904 EFFECT OF SUBJECT AGE ON FNMR 1076

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

905 IMPOSTOR COUNTS FOR CROSS COUNTRY FMR CALCULATIONS 1080

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1	KR	Urielsoft Co. Ltd	urielsoft-000	000	2025-06-30
2	EE	2060.io	2060io-001	001	2025-01-22
3	EE	2060.io	2060io-002	002	2025-07-23
4	NL	20Face	20face-000	000	2021-04-12
5	NL	20Face	20face-001	001	2021-09-29
6	NL	20Face	20face-002	002	2026-10-15
7	US	3Divi	3divi-000	000	2017-03-16
8	US	3Divi	3divi-001	001	2017-06-22
9	US	3Divi	3divi-002	002	2017-10-20
10	US	3Divi	3divi-003	003	2018-10-09
11	US	3Divi	3divi-004	004	2019-07-22
12	US	3Divi	3divi-005	005	2020-08-28
13	US	3Divi	3divi-006	006	2021-04-14
14	US	3Divi	3divi-007	007	2021-09-27
15	TH	ACI Software	acisw-003	003	2020-08-03
16	TH	ACI Software	acisw-006	006	2021-02-25
17	TH	ACI Software	acisw-007	007	2021-11-15
18	TH	ACI Software	acisw-008	008	2022-03-22
19	US	AFIS and Biometrics Consulting	afisbiometrics-000	000	2022-01-27
20	US	AFR Engine	afrengine-000	000	2022-09-29
21	US	AFR Engine	afrengine-001	001	2023-01-26
22	US	AFR Engine	afrengine-002	002	2023-06-27
23	US	AFR Engine	afrengine-003	003	2024-03-15
24	US	AFR Engine	afrengine-004	004	2024-07-22
25	US	AFR Engine	afrengine-005	005	2025-02-18
26	UZ	AI Implementation Group	aigroup-000	000	2024-07-17
27	UZ	AI Implementation Group	aigroup-001	001	2025-02-18
28	UZ	AI Implementation Group	aigroup-002	002	2026-03-12
29	UZ	AI Implementation Group	aigroup-003	003	2026-10-06
30	NV	AIPRACTIX	aipractix-000	000	2026-05-12
31	VN	ALTTEKGlobal	alttekglobal-000	000	2024-04-12
32	TW	ASUSTek Computer Inc	asusaics-000	000	2019-10-24
33	TW	ASUSTek Computer Inc	asusaics-001	001	2020-02-25
34	CN	AYF Technology	ayftech-001	001	2020-07-06
35	CN	AYF Technology	ayftech-003	003	2023-05-15
36	TW	Ability Enterprise - Andro Video	androvideo-000	000	2021-01-25
37	IN	Accurascan	accurascan-001	001	2023-0-15
38	IN	Accurascan	accurascan-002	002	2023-08-01
39	IN	Accurascan	accurascan-003	003	2023-12-06
40	TW	Acer Incorporated	acer-000	000	2020-01-08
41	TW	Acer Incorporated	acer-001	001	2020-06-30
42	TW	Acer Incorporated	acer-002	002	2021-11-10
43	SG	Adera Global PTE	adera-001	001	2019-06-17
44	SG	Adera Global PTE	adera-002	002	2021-02-16
45	SG	Adera Global PTE	adera-003	003	2021-07-12
46	SG	Adera Global PTE	adera-004	004	2022-11-14
47	SG	Adera Global PTE	adera-005	005	2023-11-17
48	SG	Advance.AI	advance-002	002	2019-12-19
49	SG	Advance.AI	advance-003	003	2021-08-05
50	SG	Advance.AI	advance-004	004	2022-09-06
51	SG	Advance.AI	advance-005	005	2023-05-25
52	SG	Advance.AI	advance-006	006	2026-04-03
53	SG	Advance.AI	advance-007	007	2026-08-07
54	AE	Affinsys AI Ltd	affinsys-ai-000	000	2024-07-11
55	TH	Ai First	aifirst-001	001	2019-11-21
56	TW	AiUnion Technology	aiunionface-000	000	2019-10-22
57	TH	Aigen	aigen-001	001	2020-10-06
58	TH	Aigen	aigen-002	002	2021-03-15
59	CN	Aiseemu Technology	aiseemu-001	001	2022-06-16
60	CN	Aiseemu Technology	aiseemu-002	002	2022-11-18
61	KR	Ajou University	ajou-001	001	2021-03-08
62	br	Akiyama S.A	akiyama-003	003	2026-08-24
63	AE	Akor IT Services LLC	akor-000	000	2026-09-16
64	ID	Akurat Satu Indonesia	ptakuratsatu-000	000	2020-09-11
65	KR	Alchera Inc	alchera-000	000	2019-03-01
66	KR	Alchera Inc	alchera-001	001	2019-03-01
67	KR	Alchera Inc	alchera-002	002	2021-03-05
68	KR	Alchera Inc	alchera-003	003	2021-07-13
69	KR	Alchera Inc	alchera-004	004	2022-08-12
70	KR	Alchera Inc	alchera-005	005	2023-01-04

Table 1: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
71	KR	Alchera Inc	alchera-006	006	2023-07-18
72	KR	Alchera Inc	alchera-007	007	2023-11-22
73	KR	Alchera Inc	alchera-008	008	2024-06-17
74	KR	Alchera Inc	alchera-009	009	2024-11-14
75	ID	Alfabeta	alfabeta-001	001	2021-12-02
76	ES	Alice Biometrics	alice-000	000	2021-06-15
77	ES	Alice Biometrics	alice-001	001	2023-07-05
78	RU	Alivia / Innovation Sys	isystems-000	000	2018-01-11
79	RU	Alivia / Innovation Sys	isystems-001	001	2018-06-12
80	RU	Alivia / Innovation Sys	isystems-002	002	2018-10-18
81	IN	AllGoVision	allgovision-000	000	2019-03-01
82	CN	AlphaSSTG	alphaface-001	001	2019-09-03
83	CN	AlphaSSTG	alphaface-002	002	2020-02-20
84	GB	Amplified Group	amplifiedgroup-001	001	2019-03-01
85	CN	Anke Investments	anke-000	000	2018-06-26
86	CN	Anke Investments	anke-001	001	2018-06-26
87	CN	Anke Investments	anke-002	002	2018-10-19
88	CN	Anke Investments	anke-003	003	2019-02-27
89	CN	Anke Investments	anke-004	004	2019-06-27
90	CN	Anke Investments	anke-005	005	2019-11-21
91	BR	Antheus Technologia	akiyama-000	000	2019-12-05
92	BR	Antheus Technologia	akiyama-001	001	2020-06-25
93	BR	Antheus Technologia	akiyama-002	002	2024-07-15
94	GB	AnyVision	anyvision-001	001	2018-01-31
95	GB	AnyVision	anyvision-002	002	2018-01-31
96	GB	AnyVision	anyvision-004	004	2018-06-15
97	GB	AnyVision	anyvision-005	005	2021-02-03
98	TR	ArVİs Teknoloji San.Tic.A.Ş.	arvis-000	000	2025-04-08
99	CN	Aratek Biometrics Co Ltd	aratek-001	001	2023-03-27
100	US	Armatura LLC	armatura-001	001	2022-01-04
101	US	Armatura LLC	armatura-002	002	2022-09-16
102	US	Armatura LLC	armatura-003	003	2023-01-13
103	CN	AuthenMetric	authenmetric-002	002	2021-03-10
104	CN	AuthenMetric	authenmetric-003	003	2021-08-09
105	CN	AuthenMetric	authenmetric-004	004	2022-01-03
106	TW	Authme	authme-001	001	2023-10-17
107	TW	Authme	authme-002	002	2025-02-10
108	US	Aware	aware-000	000	2017-10-27
109	US	Aware	aware-001	001	2017-10-27
110	US	Aware	aware-002	002	2018-06-15
111	US	Aware	aware-003	003	2018-10-19
112	US	Aware	aware-004	004	2019-03-01
113	US	Aware	aware-005	005	2020-02-27
114	US	Aware	aware-006	006	2021-07-03
115	US	Aware	aware-007	007	2023-11-15
116	US	Aware	aware-008	008	2024-03-15
117	IN	Awidit Systems	awiros-001	001	2019-09-23
118	IN	Awidit Systems	awiros-002	002	2020-10-28
119	IN	Awidit Systems	awiros-003	003	2024-11-01
120	IN	Awidit Systems	awiros-004	004	2025-03-31
121	CH	Aximetria	aximetria-001	001	2022-08-10
122	JP	Ayonix	ayonix-000	000	2017-06-22
123	CN	BOE Technology Group	boetech-001	001	2021-06-22
124	CN	BOE Technology Group	boetech-002	002	2021-12-21
125	KZ	BTS Digital	btsdigital-001	001	2025-07-31
126	ES	Bee the Data	beethedata-000	000	2021-07-26
127	CN	Beihang University-ERCACAT	ercacat-001	001	2020-07-06
128	CN	Beijing Alleyes Technology	alleyes-000	000	2020-03-09
129	CN	Beijing DeepSense Technologies	deepsense-000	000	2021-03-19
130	CN	Beijing DeepSense Technologies	deepsense-001	001	2022-03-11
131	CN	Beijing DeepSense Technologies	deepsense-002	002	2023-02-22
132	CN	Beijing DeepSense Technologies	deepsense-003	003	2023-10-23
133	CN	Beijing DeepSense Technologies	deepsense-004	004	2024-06-21
134	CN	Beijing DeepSense Technologies	deepsense-005	005	2024-10-28
135	CN	Beijing DeepSense Technologies	deepsense-006	006	2025-03-07
136	CN	Beijing DeepSense Technologies	deepsense-007	007	2025-07-08
137	CN	Beijing DeepSense Technologies	deepsense-008	008	2026-01-13
138	CN	Beijing Hisign Technology	hisign-001	001	2021-09-24
139	CN	Beijing Hisign Technology	hisign-002	002	2022-09-09
140	CN	Beijing Hisign Technology	hisign-003	003	2024-03-15

Table 2: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
141	CN	Beijing Mendaxia Technology	mendaxiatech-000	000	2021-09-15
142	CN	Beijing Vion Technology Inc	vion-000	000	2018-10-19
143	DE	Betaface	betaface-000	000	2018-05-29
144	DE	Betaface	betaface-002	002	2018-05-29
145	KZ	Beyne.AI	beyneai-000	000	2022-01-03
146	DE	BioID GmbH	bioid-001	001	2024-11-13
147	DE	BioID GmbH	bioid-002	002	2025-04-25
148	DE	BioID GmbH	bioid-003	003	2025-11-26
149	DE	BioID GmbH	bioid-004	004	2026-04-13
150	CH	BioID Technologies SA	bioidtechswiss-000	000	2019-11-15
151	CH	BioID Technologies SA	bioidtechswiss-001	001	2020-08-28
152	CH	BioID Technologies SA	bioidtechswiss-002	002	2021-02-17
153	CH	BioID Technologies SA	bioidtechswiss-003	003	2025-03-12
154	IN	Biocube Matrics	biocube-001	001	2021-09-08
155	IN	Biocube Matrics	biocube-002	002	2023-02-09
156	KZ	Biometric LLC	biometric-vision-000	000	2023-01-25
157	IN	Biometrik Systems Private Limited	biometrik-001	001	2025-09-18
158	IN	Biometrik Systems Private Limited	biometrik-002	002	2026-01-20
159	GB	BitCenter UK	farfaces-001	001	2021-04-09
160	CN	Bitmain	bm-001	001	2018-10-17
161	CN	Bresee Technology	bresee-000	000	2020-08-07
162	CN	Bresee Technology	bresee-001	001	2020-12-30
163	CN	Bresee Technology	bresee-002	002	2021-06-30
164	US	Bynn Intelligence Inc	bynn-000	000	2026-05-29
165	VN	CAKE by VPBank	cakevn-001	001	2026-01-12
166	VN	CAKE by VPBank	cakevn-002	002	2026-06-21
167	VN	CMC Institute of Science and Technology	cist-001	001	2022-10-20
168	VN	CMC Institute of Science and Technology	cist-002	002	2023-02-09
169	VN	CMC Institute of Science and Technology	cist-003	003	2023-08-14
170	VN	CMC Institute of Science and Technology	cist-004	004	2024-02-15
171	VN	CMC Univerisity	cmcuni-001	001	2023-12-05
172	CN	CSA IntelliCloud Technology	intellicloudai-001	001	2019-08-13
173	CN	CSA IntelliCloud Technology	intellicloudai-002	002	2020-12-17
174	TW	CTBC Bank	ctbcbank-000	000	2019-06-28
175	TW	CTBC Bank	ctbcbank-001	001	2019-10-28
176	TW	CTBC Bank	ctbcbank-002	002	2024-07-10
177	KR	CUDO Communication	cudocommunication-001	001	2021-10-20
178	US	Camvi Technologies	camvi-001	001	2017-09-13
179	US	Camvi Technologies	camvi-002	002	2018-10-19
180	US	Camvi Technologies	camvi-003	003	2019-03-01
181	US	Camvi Technologies	camvi-004	004	2019-07-12
182	FI	Candour Biometrics	candour-001	001	2023-02-10
183	FI	Candour Biometrics	candour-002	002	2024-03-07
184	FI	Candour Biometrics	candour-003	003	2024-10-23
185	FI	Candour Biometrics	candour-004	004	2026-01-08
186	CN	Canon Inc	canon-002	002	2020-12-29
187	JP	Canon Inc	canon-003	003	2021-09-15
188	JP	Canon Inc	canon-004	004	2022-04-25
189	JP	Canon Inc	canon-005	005	2023-09-05
190	CN	Canon Inc	cib-000	000	2019-12-11
191	CN	Canon Inc	cib-001	001	2020-08-05
192	BD	Celloscope Limited	celloscope-001	001	2024-07-15
193	CN	China Electronics Import-Export Corp	ceiec-001	001	2019-03-01
194	CN	China Electronics Import-Export Corp	ceiec-002	002	2019-06-12
195	CN	China Electronics Import-Export Corp	ceiec-003	003	2020-01-06
196	CN	China Electronics Import-Export Corp	ceiec-004	004	2021-01-18
197	CN	China University of Petroleum	upc-001	001	2019-06-05
198	CN	Chinese University of Hong Kong	cuhkee-001	001	2020-03-18
199	KR	Chosun University	chosun-000	000	2020-02-12
200	KR	Chosun University	chosun-001	001	2020-07-01
201	KR	Chosun University	chosun-002	002	2020-11-25
202	TW	Chunghwa Telecom	chtface-001	001	2019-08-06
203	TW	Chunghwa Telecom	chtface-002	002	2019-12-07
204	TW	Chunghwa Telecom	chtface-003	003	2020-06-24
205	TW	Chunghwa Telecom	chtface-004	004	2021-10-08
206	TW	Chunghwa Telecom	chtface-005	005	2022-03-09
207	TW	Chunghwa Telecom	chtface-006	006	2022-11-03
208	TW	Chunghwa Telecom	chtface-007	007	2024-04-23
209	TW	Chunghwa Telecom	chtface-008	008	2024-09-27
210	TW	Chunghwa Telecom	chtface-009	009	2025-10-21

Table 3: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
211	US	City and County of Honolulu	cchonolulu-000	000	2023-03-27
212	US	City and County of Honolulu	cchonolulu-001	001	2024-01-18
213	US	Clearview AI Inc	clearviewai-000	000	2021-09-22
214	US	Clearview AI Inc	clearviewai-001	001	2023-12-05
215	US	Clearview AI Inc	clearviewai-002	002	2024-05-22
216	US	Clearview AI Inc	clearviewai-003	003	2026-01-09
217	CN	Closeli Inc	closeli-001	001	2021-07-15
218	US	CloudSmart Consulting LLC	csc-001	001	2020-11-20
219	US	CloudSmart Consulting LLC	csc-002	002	2021-03-24
220	US	CloudSmart Consulting LLC	csc-003	003	2021-08-26
221	TW	Cloudmatrix	cloudmatrix-000	000	2021-10-22
222	TW	Cloudmatrix	cloudmatrix-001	001	2022-02-16
223	TW	Cloudmatrix	cloudmatrix-002	002	2022-10-17
224	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-000	000	2019-03-01
225	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-001	001	2019-06-04
226	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-002	002	2019-10-08
227	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-003	003	2020-09-25
228	CN	Cloudwalk - Hengrui AI Technology	cloudwalk-hr-004	004	2021-02-10
229	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-000	000	2019-06-03
230	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-002	002	2020-07-02
231	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-003	003	2020-12-22
232	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-004	004	2021-11-09
233	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-005	005	2022-03-29
234	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-006	006	2022-10-20
235	CN	Cloudwalk - Moontime Smart Technology	cloudwalk-mt-007	007	2023-02-21
236	IN	Code Everest Pvt	facex-001	001	2021-03-08
237	IN	Code Everest Pvt	facex-002	002	2021-08-24
238	KR	Codeline	codeline-000	000	2022-09-13
239	DE	Cognitec Systems GmbH	cognitec-000	000	2018-10-19
240	DE	Cognitec Systems GmbH	cognitec-001	001	2019-03-01
241	DE	Cognitec Systems GmbH	cognitec-002	002	2021-02-24
242	DE	Cognitec Systems GmbH	cognitec-003	003	2021-07-30
243	DE	Cognitec Systems GmbH	cognitec-004	004	2022-02-10
244	DE	Cognitec Systems GmbH	cognitec-005	005	2023-12-01
245	DE	Cognitec Systems GmbH	cognitec-006	006	2024-04-29
246	DE	Cognitec Systems GmbH	cognitec-007	007	2024-08-29
247	DE	Cognitec Systems GmbH	cognitec-008	008	2026-09-24
248	TW	Compal Electronics	compal-000	000	2025-03-20
249	TW	Compal Electronics	compal-001	001	2025-12-08
250	TW	Compal Electronics	compal-002	002	2026-07-09
251	TW	Coretech Knowledge Inc	coretech-000	000	2021-07-12
252	TW	Coretech Knowledge Inc	coretech-001	001	2022-09-29
253	TW	Coretech Knowledge Inc	coretech-002	002	2023-11-02
254	TW	Coretech Knowledge Inc	coretech-003	003	2024-08-07
255	TW	Coretech Knowledge Inc	coretech-004	004	2024-12-23
256	IL	Corsight	corsight-001	001	2021-03-11
257	IL	Corsight	corsight-002	002	2021-09-01
258	IL	Corsight	corsight-003	003	2022-06-09
259	IL	Cortica	cor-001	001	2020-09-24
260	TW	Cu-Face	cu-face-002	002	2023-01-05
261	TW	Cu-Face	cu-face-003	003	2023-08-28
262	TW	Cu-Face	cu-face-004	004	2024-03-13
263	KR	Cubox	cubox-001	001	2020-12-07
264	KR	Cubox	cubox-002	002	2021-08-24
265	KR	Cubox	cubox-003	003	2023-03-07
266	KR	Cubox	cubox-004	004	2024-12-05
267	KR	Cubox	cubox-005	005	2025-05-14
268	US	Cvartel Inc	cvartel-001	001	2026-10-13
269	JP	Cybercore	cybercore-000	000	2020-08-26
270	JP	Cybercore	cybercore-001	001	2021-12-15
271	JP	Cybercore	cybercore-002	002	2022-04-25
272	JP	Cybercore	cybercore-003	003	2022-08-31
273	US	Cyberextruder	cyberextruder-001	001	2017-08-02
274	US	Cyberextruder	cyberextruder-002	002	2018-01-30
275	US	Cyberextruder	cyberextruder-003	003	2022-03-16
276	US	Cyberextruder	cyberextruder-004	004	2022-07-20
277	US	Cyberextruder	cyberextruder-005	005	2024-10-01
278	US	Cyberextruder	cyberextruder-006	006	2025-03-19
279	TW	Cyberlink Corp	cyberlink-000	000	2019-03-01
280	TW	Cyberlink Corp	cyberlink-001	001	2019-03-01

Table 4: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
281	TW	Cyberlink Corp	cyberlink-002	002	2019-06-12
282	TW	Cyberlink Corp	cyberlink-003	003	2019-10-07
283	TW	Cyberlink Corp	cyberlink-004	004	2020-02-27
284	TW	Cyberlink Corp	cyberlink-005	005	2020-07-31
285	TW	Cyberlink Corp	cyberlink-006	006	2021-01-08
286	TW	Cyberlink Corp	cyberlink-007	007	2021-07-16
287	TW	Cyberlink Corp	cyberlink-008	008	2022-01-07
288	TW	Cyberlink Corp	cyberlink-009	009	2022-05-12
289	TW	Cyberlink Corp	cyberlink-010	010	2022-09-16
290	TW	Cyberlink Corp	cyberlink-011	011	2023-01-30
291	TW	Cyberlink Corp	cyberlink-012	012	2023-06-02
292	TW	Cyberlink Corp	cyberlink-013	013	2023-12-15
293	TW	Cyberlink Corp	cyberlink-014	014	2024-05-23
294	TW	Cyberlink Corp	cyberlink-015	015	2024-09-26
295	TW	Cyberlink Corp	cyberlink-016	016	2026-06-02
296	MX	DICIO	dicio-001	001	2022-03-22
297	MX	DICIO	dicio-002	002	2024-10-03
298	CN	DSK	dsk-000	000	2019-06-28
299	VN	Dactionable Technologies	datech-000	000	2023-10-16
300	VN	Dactionable Technologies	datech-001	001	2024-02-21
301	VN	Dactionable Technologies	datech-002	002	2024-09-11
302	CN	Dahua Technology	dahua-000	000	2018-10-19
303	CN	Dahua Technology	dahua-001	001	2018-10-19
304	CN	Dahua Technology	dahua-002	002	2019-03-01
305	CN	Dahua Technology	dahua-003	003	2019-08-14
306	CN	Dahua Technology	dahua-004	004	2019-12-18
307	CN	Dahua Technology	dahua-005	005	2020-08-13
308	CN	Dahua Technology	dahua-006	006	2020-12-30
309	CN	Dahua Technology	dahua-007	007	2021-12-20
310	IE	Daon	daon-000	000	2021-11-03
311	IE	Daon	daon-001	001	2024-09-26
312	IE	Daon	daon-002	002	2025-07-01
313	IE	Daon	daon-003	003	2026-03-12
314	TW	DeCloak Intellegences	decloakface-001	001	2023-06-13
315	TW	DeCloak Intellegences	decloakface-002	002	2024-10-28
316	US	Decatur Industries Inc	decatur-000	000	2020-08-18
317	US	Decatur Industries Inc	decatur-001	001	2021-09-27
318	CN	Deepglint	deepglint-001	001	2019-06-21
319	CN	Deepglint	deepglint-002	002	2019-11-15
320	CN	Deepglint	deepglint-003	003	2021-03-03
321	CN	Deepglint	deepglint-004	004	2021-09-17
322	CN	Deepglint	deepglint-005	005	2022-10-17
323	FR	Deepsense	dps-000	000	2021-07-16
324	DE	Dermalog	dermalog-001	001	2017-02-22
325	DE	Dermalog	dermalog-002	002	2017-02-22
326	DE	Dermalog	dermalog-003	003	2017-07-10
327	DE	Dermalog	dermalog-004	004	2017-10-26
328	DE	Dermalog	dermalog-005	005	2018-02-02
329	DE	Dermalog	dermalog-006	006	2018-10-18
330	DE	Dermalog	dermalog-007	007	2020-11-25
331	DE	Dermalog	dermalog-008	008	2021-03-25
332	DE	Dermalog	dermalog-009	009	2021-10-06
333	DE	Dermalog	dermalog-010	010	2022-07-25
334	DE	Dermalog	dermalog-011	011	2022-12-12
335	DE	Dermalog	dermalog-012	012	2024-01-19
336	DE	Dermalog	dermalog-013	013	2024-06-18
337	DE	Dermalog	dermalog-014	014	2025-06-17
338	DE	Dermalog	dermalog-015	015	2025-11-26
339	DE	Dermalog	dermalog-016	016	2026-06-25
340	CN	DiDi ChuXing Technology	didiglobalface-001	001	2019-10-23
341	CN	DiDi ChuXing Technology	didiglobalface-002	002	2023-01-09
342	CN	DiDi ChuXing Technology	didiglobalface-003	003	2025-05-01
343	IN	Digidata	digidata-000	000	2022-01-27
344	IN	Digidata	digidata-001	001	2022-06-10
345	GB	Digital Barriers	digitalbarriers-000	000	2017-05-31
346	GB	Digital Barriers	digitalbarriers-001	001	2017-07-20
347	GB	Digital Barriers	digitalbarriers-002	002	2019-03-01
348	UY	Dralmey Solutions	dralmey-001	001	2026-03-27
349	IN	EI Networks Private Ltd	einetworksindia-000	000	2023-05-03
350	IN	EI Networks Private Ltd	einetworksindia-001	001	2023-09-05

Table 5: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
351	IN	EI Networks Private Ltd	einetworksindia-002	002	2025-02-21
352	TR	Ekin Smart City Technologies	ekin-002	002	2021-05-04
353	TH	Element System Solutions Company Ltd	element-000	000	2023-11-14
354	IN	Elkosta Security Systems India	essi-000	000	2026-09-25
355	AE	Enface	enface-000	000	2021-04-09
356	AE	Enface	enface-001	001	2021-12-17
357	AE	Enface	enface-002	002	2023-02-27
358	GB	Enqura Ltd	enqura-001	001	2024-09-25
359	GB	Enqura Ltd	enqura-002	002	2026-05-19
360	GB	Enqura Ltd	enqura-003	003	2026-10-07
361	ET	Ethiopian Labor Market Information System	ethiopianlmis-000	000	2026-03-18
362	CH	Euronovate SA	euronovate-001	001	2021-11-15
363	CH	Euronovate SA	euronovate-002	002	2023-06-28
364	CH	Euronovate SA	euronovate-003	003	2023-09-19
365	CH	Euronovate SA	euronovate-004	004	2024-01-19
366	RU	Expasoft LLC	expasoft-000	000	2020-01-06
367	RU	Expasoft LLC	expasoft-001	001	2020-09-03
368	RU	Expasoft LLC	expasoft-002	002	2021-07-26
369	VN	FCI Vietnam	fci-000	000	2025-04-01
370	LB	FOO	foomobi-001	001	2023-03-13
371	LB	FOO	foomobi-002	002	2023-08-07
372	VN	FPT Education	fedu-001	001	2024-03-15
373	VN	FPT Information System	fpt-000	000	2024-02-26
374	VN	FPT Information System	fpt-001	001	2024-09-16
375	VN	FPT Telecom JSC	fpttelecom-000	000	2026-02-03
376	VN	FPT Telecom JSC	fpttelecom-001	001	2026-10-20
377	US	FRP LLC	frpkauai-001	001	2022-07-18
378	US	FRP LLC	frpkauai-002	002	2022-11-21
379	US	FRP LLC	frpkauai-003	003	2024-05-30
380	US	FRP LLC	frpkauai-004	004	2025-03-25
381	US	FRP LLC	frpkauai-005	005	2026-06-22
382	IL	Face Int	faceint-001	001	2026-03-27
383	CA	FaceLocate	facelocate-001	001	2024-02-22
384	CA	FaceLocate	facelocate-002	002	2026-03-25
385	DE	FaceOnLive Inc	faceonlive-001	001	2021-11-23
386	DE	FaceOnLive Inc	faceonlive-002	002	2022-04-11
387	DE	FaceOnLive Inc	faceonlive-003	003	2024-08-12
388	ES	FacePhi	facephi-000	000	2022-04-06
389	ES	FacePhi	facephi-001	001	2024-05-24
390	ES	FacePhi	facephi-002	002	2025-02-11
391	US	FacePlugin LLC	faceplugin-000	000	2026-10-15
392	GB	FaceSoft	facesoft-000	000	2019-07-10
393	KR	FaceTag Co	facetag-000	000	2021-03-22
394	KR	FaceTag Co	facetag-001	001	2021-08-17
395	KR	FaceTag Co	facetag-002	002	2022-01-06
396	GB	Facehawk Ltd	facehawk-000	000	2023-08-22
397	GB	Facehawk Ltd	facehawk-001	001	2026-06-17
398	GB	Facia.ai	facia-001	001	2023-07-21
399	GB	Facia.ai	facia-002	002	2024-06-21
400	GB	FaiceTech Ltd	faicetech-000	000	2025-03-11
401	TW	FarBar Inc	f8-001	001	2019-07-11
402	TW	FarBar Inc	f8-002	002	2022-03-02
403	US	Fast Enterprises	fastenterprises-000	000	2023-03-01
404	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-002	002	2020-08-10
405	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-003	003	2021-03-12
406	CN	Fiberhome Telecommunication Technologies	fiberhome-nanjing-004	004	2021-09-14
407	GB	Fincore Ltd	fincore-000	000	2021-06-07
408	KZ	First Credit Bureau Kazakhstan	firstcreditkz-001	001	2022-08-22
409	KZ	First Credit Bureau Kazakhstan	firstcreditkz-002	002	2023-02-21
410	KZ	First Credit Bureau Kazakhstan	firstcreditkz-003	003	2024-02-20
411	KZ	First Credit Bureau Kazakhstan	firstcreditkz-004	004	2024-10-02
412	NL	Fourthline	fourthlinecom-001	001	2024-04-15
413	GB	Fraud.com	fraudcom-000	000	2023-12-01
414	CN	Fudan University	fdu-000	000	2017-11-22
415	CN	Fudan University	fdu-001	001	2018-01-11
416	CN	Fujitsu Research and Development Center	fujitsulab-000	000	2020-02-04
417	CN	Fujitsu Research and Development Center	fujitsulab-001	001	2020-09-30
418	CN	Fujitsu Research and Development Center	fujitsulab-002	002	2021-02-24
419	CN	Fujitsu Research and Development Center	fujitsulab-003	003	2021-07-12
420	VN	GPS Vietnam Trading	gpstechvn-000	000	2024-01-02

Table 6: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
421	VN	GTEL OTS JSC	gtelots-000	000	2025-04-14
422	VN	GTEL OTS JSC	gtelots-001	001	2026-02-02
423	US	Gemalto Cogent	cogent-000	000	2018-01-12
424	US	Gemalto Cogent	cogent-001	001	2018-05-29
425	US	Gemalto Cogent	cogent-002	002	2018-10-19
426	US	Gemalto Cogent	cogent-003	003	2019-03-01
427	US	Gemalto Cogent	cogent-004	004	2019-06-14
428	US	Gemalto Cogent	cogent-005	005	2020-12-29
429	US	Gemalto Cogent	cogent-006	006	2021-07-28
430	US	Gemalto Cogent	cogent-007	007	2022-04-11
431	US	Gemalto Cogent	cogent-008	008	2023-01-03
432	TW	General Interface Solutions Holding Ltd	gistouch-000	000	2023-05-08
433	TW	General Interface Solutions Holding Ltd	gistouch-001	001	2023-11-09
434	TW	GeoVision Inc	geo-000	000	2020-06-29
435	TW	GeoVision Inc	geo-001	001	2020-10-30
436	TW	GeoVision Inc	geo-002	002	2021-04-01
437	TW	GeoVision Inc	geo-003	003	2021-09-15
438	TW	GeoVision Inc	geo-004	004	2022-02-10
439	JP	Glory	glory-000	000	2018-06-06
440	JP	Glory	glory-001	001	2018-06-08
441	JP	Glory	glory-002	002	2019-11-12
442	JP	Glory	glory-003	003	2021-01-15
443	JP	Glory	glory-004	004	2022-02-08
444	JP	Glory	glory-005	005	2022-07-08
445	JP	Glory	glory-006	006	2023-03-23
446	JP	Glory	glory-007	007	2023-11-10
447	TW	Gorilla Technology	gorilla-000	000	2018-02-13
448	TW	Gorilla Technology	gorilla-001	001	2018-05-25
449	TW	Gorilla Technology	gorilla-002	002	2018-10-17
450	TW	Gorilla Technology	gorilla-003	003	2019-06-19
451	TW	Gorilla Technology	gorilla-004	004	2019-11-04
452	TW	Gorilla Technology	gorilla-005	005	2020-03-11
453	TW	Gorilla Technology	gorilla-006	006	2020-07-31
454	TW	Gorilla Technology	gorilla-007	007	2021-06-28
455	TW	Gorilla Technology	gorilla-008	008	2021-11-08
456	TW	Gorilla Technology	gorilla-009	009	2022-12-14
457	US	Graymatics	graymatics-001	001	2022-01-13
458	US	Griaule	griaule-000	000	2021-08-20
459	US	Griaule	griaule-001	001	2022-05-31
460	US	Griaule	griaule-002	002	2022-12-02
461	CN	Guangzhou Pixel Solutions	pixelall-002	002	2019-06-06
462	CN	Guangzhou Pixel Solutions	pixelall-003	003	2019-10-15
463	CN	Guangzhou Pixel Solutions	pixelall-004	004	2020-07-02
464	CN	Guangzhou Pixel Solutions	pixelall-005	005	2021-02-05
465	CN	Guangzhou Pixel Solutions	pixelall-006	006	2021-06-17
466	CN	Guangzhou Pixel Solutions	pixelall-007	007	2021-12-01
467	CN	Guangzhou Pixel Solutions	pixelall-008	008	2022-06-16
468	CN	Guangzhou Pixel Solutions	pixelall-009	009	2022-10-26
469	CN	Guangzhou Pixel Solutions	pixelall-010	010	2023-10-18
470	CN	Hangzhuo AIlu Network Information Technology	hzailu-001	001	2022-01-27
471	CN	Hangzhuo AIlu Network Information Technology	hzailu-002	002	2022-06-02
472	CN	Hangzhuo AIlu Network Information Technology	hzailu-003	003	2022-10-11
473	CN	Hangzhuo AIlu Network Information Technology	hzailu-004	004	2023-02-27
474	CN	Hangzhuo AIlu Network Information Technology	hzailu-005	005	2023-08-10
475	CN	Hangzhuo AIlu Network Information Technology	hzailu-006	006	2023-12-28
476	CN	Hangzhuo AIlu Network Information Technology	hzailu-007	007	2024-09-06
477	CN	Hangzhuo AIlu Network Information Technology	hzailu-008	008	2025-02-27
478	ES	Herta Security	hertasecurity-000	000	2021-01-05
479	ES	Herta Security	hertasecurity-001	001	2022-01-18
480	ES	Herta Security	hertasecurity-002	002	2022-09-02
481	ES	Herta Security	hertasecurity-003	003	2023-01-27
482	CN	Hikvision Research Institute	hik-001	001	2019-03-01
483	IN	Hyderabad-Centre for Development of Advanced Computing	cdachyd-001	001	2024-10-01
484	IN	HyperVerge Inc	hyperverge-001	001	2020-12-13
485	IN	HyperVerge Inc	hyperverge-002	002	2021-05-27
486	IN	HyperVerge Inc	hyperverge-003	003	2022-04-11
487	IN	HyperVerge Inc	hyperverge-004	004	2022-12-14
488	IN	HyperVerge Inc	hyperverge-005	005	2023-08-08
489	IN	HyperVerge Inc	hyperverge-006	006	2024-05-02
490	IN	HyperVerge Inc	hyperverge-007	007	2026-04-10

Table 7: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
491	KR	Hyundai Motor Company	hyundai-roboticslab-000	000	2024-07-12
492	KR	Hyundai Motor Company	hyundai-roboticslab-001	001	2025-02-03
493	KR	Hyundai Motor Company	hyundai-roboticslab-002	002	2025-06-03
494	KR	Hyundai Motor Company	hyundai-roboticslab-003	003	2025-11-14
495	KR	Hyundai Motor Company	hyundai-roboticslab-004	004	2026-06-04
496	AU	ICM Airport Technics	icm-002	002	2020-11-13
497	AU	ICM Airport Technics	icm-003	003	2021-09-06
498	AU	ICM Airport Technics	icm-004	004	2022-09-07
499	AU	ICM Airport Technics	icm-005	005	2023-09-05
500	US	ID R&D Inc	idrnd-000	000	2024-07-26
501	US	ID R&D Inc	idrnd-001	001	2025-07-31
502	FR	ID3 Technology	id3-001	001	2017-08-04
503	FR	ID3 Technology	id3-002	002	2017-08-04
504	FR	ID3 Technology	id3-003	003	2018-10-05
505	FR	ID3 Technology	id3-004	004	2019-03-01
506	FR	ID3 Technology	id3-005	005	2020-08-04
507	FR	ID3 Technology	id3-006	006	2020-12-17
508	FR	ID3 Technology	id3-007	007	2021-05-17
509	FR	ID3 Technology	id3-008	008	2021-11-10
510	FR	ID3 Technology	id3-009	009	2026-02-05
511	FR	ID3 Technology	id3-010	010	2026-06-05
512	FR	ID3 Technology	id3-011	011	2026-10-05
513	US	IDBIO	idbio-000	000	2026-09-08
514	GB	IDENTY	identy-000	000	2023-04-04
515	GB	IDENTY	identy-001	001	2025-05-21
516	GB	IDENTY	identy-002	002	2025-11-14
517	GB	IDENTY	identy-003	003	2026-05-15
518	KZ	IE SARSENOV	asr-000	000	2025-07-30
519	KZ	IE SARSENOV	asr-001	001	2025-12-01
520	CA	IMDS Software	imds-software-001	001	2022-07-06
521	CA	IMDS Software	imds-software-002	002	2023-02-10
522	CA	IMDS Software	imds-software-003	003	2023-06-20
523	MX	IQSEC	iqsec-001	001	2024-04-12
524	MX	IQSEC	iqsec-002	002	2025-01-08
525	MX	IQSEC	iqsec-003	003	2026-07-20
526	RU	ITMO University	itmo-001	001	2017-06-12
527	RU	ITMO University	itmo-002	002	2017-08-07
528	RU	ITMO University	itmo-003	003	2018-01-23
529	RU	ITMO University	itmo-004	004	2018-06-13
530	RU	ITMO University	itmo-005	005	2018-10-19
531	RU	ITMO University	itmo-006	006	2019-03-01
532	RU	ITMO University	itmo-007	007	2020-01-06
533	RU	ITMO University	itmo-008	008	2021-11-19
534	RU	IVA Cognitive	ivacognitive-001	001	2021-01-29
535	FR	Idemia	idemia-003	003	2018-10-19
536	FR	Idemia	idemia-004	004	2019-03-01
537	FR	Idemia	idemia-005	005	2019-10-11
538	FR	Idemia	idemia-006	006	2020-07-06
539	FR	Idemia	idemia-007	007	2020-12-04
540	FR	Idemia	idemia-008	008	2021-07-07
541	FR	Idemia	idemia-009	009	2022-07-27
542	FR	Idemia	idemia-010	010	2023-06-30
543	FR	Idemia	idemia-011	011	2024-08-06
544	FR	Idemia	morpho-000	000	2017-07-11
545	FR	Idemia	morpho-001	001	2017-07-10
546	FR	Idemia	morpho-002	002	2017-12-28
547	US	Imageware Systems	iws-000	000	2020-08-12
548	GB	Imperial College London	imperial-000	000	2019-03-01
549	GB	Imperial College London	imperial-001	001	2019-03-01
550	GB	Imperial College London	imperial-002	002	2019-08-28
551	US	Incode Technologies Inc	incode-000	000	2018-06-01
552	US	Incode Technologies Inc	incode-001	001	2018-06-13
553	US	Incode Technologies Inc	incode-002	002	2018-10-18
554	US	Incode Technologies Inc	incode-003	003	2019-03-01
555	US	Incode Technologies Inc	incode-004	004	2019-06-12
556	US	Incode Technologies Inc	incode-005	005	2019-10-17
557	US	Incode Technologies Inc	incode-006	006	2020-02-20
558	US	Incode Technologies Inc	incode-007	007	2020-08-25
559	US	Incode Technologies Inc	incode-008	008	2021-01-19
560	US	Incode Technologies Inc	incode-009	009	2021-06-22

Table 8: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
561	US	Incode Technologies Inc	incode-010	010	2021-10-22
562	US	Incode Technologies Inc	incode-011	011	2022-08-10
563	US	Incode Technologies Inc	incode-012	012	2023-05-05
564	US	Incode Technologies Inc	incode-013	013	2023-12-08
565	US	Incode Technologies Inc	incode-014	014	2024-04-08
566	US	Incode Technologies Inc	incode-015	015	2025-06-12
567	US	Incode Technologies Inc	incode-016	016	2026-09-30
568	IT	InfoCert	infocert-001	001	2022-09-08
569	IT	InfoCert	infocert-002	002	2026-03-18
570	IT	InfoCert	infocert-003	003	2026-07-20
571	TR	InfoDif Software and Information Technologies Inc	infodif-001	001	2026-10-14
572	IN	Innef Labs	innefulabs-000	000	2020-09-04
573	IN	Innominds Software SEZ India Pvt	innominds-001	001	2023-06-23
574	MY	Innov8tif Solutions	innov8tif-000	000	2024-12-10
575	MY	Innov8tif Solutions	innov8tif-001	001	2026-04-22
576	MY	Innov8tif Solutions	innov8tif-002	002	2026-09-07
577	GB	Innovative Technology	innovativetechnologyltd-001	001	2019-10-22
578	GB	Innovative Technology	innovativetechnologyltd-002	002	2020-02-26
579	SK	Innovatrics	innovatrics-000	000	2017-07-25
580	SK	Innovatrics	innovatrics-001	001	2017-07-25
581	SK	Innovatrics	innovatrics-002	002	2018-02-05
582	SK	Innovatrics	innovatrics-003	003	2018-06-14
583	SK	Innovatrics	innovatrics-004	004	2018-10-19
584	SK	Innovatrics	innovatrics-005	005	2019-03-01
585	SK	Innovatrics	innovatrics-006	006	2019-08-13
586	SK	Innovatrics	innovatrics-007	007	2020-08-19
587	SK	Innovatrics	innovatrics-008	008	2021-12-15
588	SK	Innovatrics	innovatrics-009	009	2022-01-19
589	SK	Innovatrics	innovatrics-010	010	2023-08-03
590	SK	Innovatrics	innovatrics-011	011	2024-03-08
591	SK	Innovatrics	innovatrics-012	012	2024-07-10
592	SK	Innovatrics	innovatrics-013	013	2024-11-13
593	SK	Innovatrics	innovatrics-014	014	2025-03-13
594	SK	Innovatrics	innovatrics-015	015	2025-07-24
595	SK	Innovatrics	innovatrics-016	016	2025-11-28
596	SK	Innovatrics	innovatrics-017	017	2026-07-30
597	CN	InsightFace AI	insightface-000	000	2021-03-17
598	CN	InsightFace AI	insightface-001	001	2021-09-27
599	CN	InsightFace AI	insightface-002	002	2022-01-31
600	CN	InsightFace AI	insightface-003	003	2022-08-23
601	CN	InsightFace AI	insightface-004	004	2023-03-31
602	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-000	000	2022-07-19
603	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-001	001	2023-02-24
604	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-002	002	2023-12-08
605	CN	Inspur (Beijing) Electronic Information Industry Co	inspur-003	003	2025-02-11
606	CN	Institute of Computing Technology	icthtc-000	000	2020-11-29
607	RU	Institute of Information Technologies	iit-000	000	2019-03-01
608	RU	Institute of Information Technologies	iit-001	001	2019-07-05
609	RU	Institute of Information Technologies	iit-002	002	2019-12-04
610	RU	Institute of Information Technologies	iit-003	003	2020-12-01
611	BD	Integrated Software and Technologies Ltd	istl-001	001	2026-08-07
612	IS	Intel Research Group	intelresearch-000	000	2019-07-08
613	IS	Intel Research Group	intelresearch-001	001	2020-01-14
614	IS	Intel Research Group	intelresearch-002	002	2020-07-24
615	IS	Intel Research Group	intelresearch-003	003	2021-01-18
616	IS	Intel Research Group	intelresearch-004	004	2021-08-24
617	IS	Intel Research Group	intelresearch-005	005	2022-02-13
618	IS	Intel Research Group	intelresearch-006	006	2022-12-19
619	IS	Intel Research Group	intelresearch-007	007	2023-05-30
620	IS	Intel Research Group	intelresearch-008	008	2023-10-03
621	IS	Intel Research Group	intelresearch-009	009	2024-03-26
622	IS	Intel Research Group	intelresearch-010	010	2024-08-20
623	IS	Intel Research Group	intelresearch-011	011	2024-12-24
624	IS	Intel Research Group	intelresearch-012	012	2025-07-03
625	KR	IntelliVIX	intellivix-001	001	2022-02-25
626	KR	IntelliVIX	intellivix-002	002	2022-07-14
627	KR	IntelliVIX	intellivix-003	003	2022-12-12
628	KR	IntelliVIX	intellivix-004	004	2023-06-21
629	KR	IntelliVIX	intellivix-005	005	2023-11-14
630	KR	IntelliVIX	intellivix-006	006	2025-07-31

Table 9: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
631	AE	Intellibrain Technological Projects	g42-intellibrain-001	001	2022-07-27
632	AE	Intellibrain Technological Projects	g42-intellibrain-002	002	2023-05-10
633	CN	Intelligent Control Technology Co Ltd - IGearx	igearx-face-000	000	2023-03-28
634	US	Intellivision	intellivision-001	001	2017-10-10
635	US	Intellivision	intellivision-002	002	2019-08-23
636	US	Intellivision	intellivision-003	003	2022-03-07
637	US	Intellivision	intellivision-004	004	2022-07-28
638	US	Intellivision	intellivision-005	005	2023-02-06
639	US	Intellivision	intellivision-006	006	2023-07-14
640	US	Intellivision	intellivision-007	007	2023-12-19
641	LU	Intema-LGL Group	intema-000	000	2022-07-15
642	LU	Intema-LGL Group	intema-001	001	2023-01-11
643	US	InterLink labs	interlinklabs-001	001	2024-11-08
644	US	InterLink labs	interlinklabs-002	002	2025-04-21
645	US	InterLink labs	interlinklabs-003	003	2025-11-18
646	IN	Intozi Tech Pvt Ltd	intozi-001	001	2023-09-25
647	IN	Intozi Tech Pvt Ltd	intozi-002	002	2026-06-21
648	IN	Intpixel Labs Private Ltd	vogic-000	000	2026-07-30
649	US	IrexAI	irex-000	000	2020-12-17
650	US	Iris ID Systems	irisid-001	001	2026-01-30
651	US	Iris ID Systems	irisid-002	002	2026-08-05
652	IL	Is It You	isityou-000	000	2017-06-26
653	MX	Jaak IT	jaakit-001	001	2022-05-20
654	MX	Jaak IT	jaakit-002	002	2025-09-11
655	DE	Johannes Gutenberg-Universitat Mainz	jgumainz-000	000	2026-10-27
656	GB	KYB-AI Ltd	kbyai-000	000	2024-11-29
657	IN	Kaizen Secure Voiz	kaizenvoiz-001	001	2026-06-08
658	KR	Kakao Bank	kakaobank-000	000	2023-02-27
659	KR	Kakao Bank	kakaobank-001	001	2023-10-10
660	KR	Kakao Bank	kakaobank-002	002	2024-04-01
661	KR	Kakao Bank	kakaobank-003	003	2024-09-30
662	KR	Kakao Brain	kakao-001	001	2019-03-01
663	KR	Kakao Brain	kakao-002	002	2019-06-19
664	KR	Kakao Brain	kakao-003	003	2020-02-26
665	KR	Kakao Brain	kakao-004	004	2020-10-28
666	KR	Kakao Brain	kakao-005	005	2021-03-09
667	KR	Kakao Brain	kakao-007	007	2022-01-12
668	KR	Kakao Brain	kakao-008	008	2022-05-12
669	KR	Kakao Brain	kakao-009	009	2023-11-07
670	KR	Kakao Pay Corp	kakaopay-001	001	2021-07-06
671	TH	Kasikorn Labs	kasikornlabs-000	000	2022-03-02
672	TH	Kasikorn Labs	kasikornlabs-001	001	2022-07-26
673	TH	Kasikorn Labs	kasikornlabs-002	002	2022-12-13
674	TH	Kasikorn Labs	kasikornlabs-003	003	2023-10-05
675	TH	Kasikorn Labs	kasikornlabs-004	004	2024-11-04
676	TH	Kasikorn Labs	kasikornlabs-005	005	2025-04-04
677	SG	Kedacom International Pte	kedacom-000	000	2019-06-03
678	GB	Keyless Technologies	keyless-000	000	2024-11-21
679	US	Kloudspot Inc	kloudspot-001	001	2024-09-03
680	US	Kloudspot Inc	kloudspot-002	002	2025-06-16
681	US	Kneron Inc	kneron-003	003	2019-07-01
682	US	Kneron Inc	kneron-005	005	2020-02-21
683	US	Kneron Inc	kneron-006	006	2026-04-10
684	US	KnowUTech LLC	knowutech-000	000	2022-02-13
685	US	Kogniza Technology	kogniza-001	001	2024-03-01
686	KR	Kookmin University	kookmin-001	001	2020-09-28
687	KR	Kookmin University	kookmin-002	002	2021-03-05
688	KR	Korea Identification Inc	koreaid-001	001	2022-12-12
689	TH	Krungthai	krungthai-002	002	2022-06-21
690	CN	KuKe3D Technology	kuke3d-001	001	2021-10-28
691	CN	KuKe3D Technology	kuke3d-002	002	2022-04-14
692	MX	Lebentech Biometrics	lebentech-000	000	2022-02-16
693	MX	Lebentech Biometrics	lebentech-001	001	2023-12-12
694	IN	Lema Labs	lemalabs-001	001	2021-04-13
695	KR	LightVision Inc	lightvision-000	000	2024-06-27
696	KR	LightVision Inc	lightvision-001	001	2025-02-18
697	JP	Line Corporation	lineclova-000	000	2021-03-31
698	JP	Line Corporation	lineclova-001	001	2021-09-26
699	JP	Line Corporation	lineclova-002	002	2022-05-18
700	JP	Line Corporation	lineclova-003	003	2022-11-28

Table 10: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
701	CN	Lishui Shuyan Huirong Technology Co	lssytech-000	000	2026-03-18
702	CN	Lishui Shuyan Huirong Technology Co	lssytech-001	001	2026-09-16
703	RU	Lomonosov Moscow State University	intsysmsu-000	000	2019-06-18
704	RU	Lomonosov Moscow State University	intsysmsu-001	001	2019-10-22
705	RU	Lomonosov Moscow State University	intsysmsu-002	002	2020-03-12
706	IN	Lookman Electroplast Industries	lookman-002	002	2018-06-13
707	IN	Lookman Electroplast Industries	lookman-004	004	2019-06-03
708	US	Luxand Inc	luxand-000	000	2019-11-07
709	US	Luxand Inc	luxand-001	001	2023-10-17
710	US	Luxand Inc	luxand-003	003	2024-03-25
711	US	Luxand Inc	luxand-004	004	2024-08-01
712	VN	MB Bank	mbbankvn-000	000	2025-02-12
713	RU	MVision	mvision-001	001	2019-11-12
714	IN	Mantra Softech India	mantra-000	000	2021-10-28
715	ID	Matrix Comsec Pvt Ltd	matrixcomsec-000	000	2025-06-11
716	ID	Matrix Comsec Pvt Ltd	matrixcomsec-001	001	2025-11-26
717	ID	Matrix Comsec Pvt Ltd	matrixcomsec-002	002	2026-05-21
718	AE	Maverick Research	maverickresearch-000	000	2024-12-26
719	CN	Maxvision Technology	maxvision-000	000	2021-10-27
720	CN	Maxvision Technology	maxvision-001	001	2022-03-03
721	CN	Maxvision Technology	maxvision-002	002	2022-07-12
722	CN	Maxvision Technology	maxvision-003	003	2022-11-14
723	CN	Maxvision Technology	maxvision-004	004	2023-05-08
724	CN	Maxvision Technology	maxvision-005	005	2023-09-21
725	CN	Maxvision Technology	maxvision-006	006	2024-01-19
726	CN	Megvii/Face++	megvii-000	000	2018-02-13
727	CN	Megvii/Face++	megvii-001	001	2018-06-15
728	CN	Megvii/Face++	megvii-002	002	2018-10-19
729	CN	Megvii/Face++	megvii-003	003	2021-03-08
730	CN	Megvii/Face++	megvii-004	004	2021-11-19
731	CN	Megvii/Face++	megvii-005	005	2022-03-28
732	CN	Megvii/Face++	megvii-006	006	2022-08-08
733	CN	Megvii/Face++	megvii-007	007	2023-05-17
734	CN	Megvii/Face++	megvii-008	008	2023-10-12
735	CN	Megvii/Face++	megvii-009	009	2024-02-12
736	KR	Metsakuur	metsakuurcompany-001	001	2022-05-12
737	KR	Metsakuur	metsakuurcompany-002	002	2022-09-14
738	KR	Metsakuur	metsakuurcompany-003	003	2023-04-04
739	KR	Metsakuur	metsakuurcompany-004	004	2024-10-24
740	KR	Metsakuur	metsakuurcompany-005	005	2025-07-31
741	KR	Metsakuur	metsakuurcompany-006	006	2026-09-17
742	CN	Miaxis Biometrics	miaxis-001	001	2022-11-15
743	CN	Miaxis Biometrics	miaxis-002	002	2023-03-22
744	CN	Miaxis Biometrics	miaxis-003	003	2023-09-19
745	CN	Miaxis Biometrics	miaxis-004	004	2025-12-09
746	CN	Miaxis Biometrics	miaxis-005	005	2026-07-24
747	GB	MicroFocus	microfocus-001	001	2018-06-13
748	GB	MicroFocus	microfocus-002	002	2018-10-17
749	GB	MicroFocus	microfocus-003	003	2023-02-23
750	GB	Microfocus	microfocus-000	000	2018-06-13
751	CN	Minivision	minivision-000	000	2020-10-28
752	GB	Mitek Systems	mitek-000	000	2023-01-27
753	GB	Mitek Systems	mitek-001	001	2024-07-02
754	GB	Mitek Systems	mitek-002	002	2024-11-19
755	NO	Mobai	mobai-000	000	2020-08-26
756	NO	Mobai	mobai-001	001	2021-02-17
757	NO	Mobai	mobai-002	002	2025-07-07
758	NO	Mobai	mobai-003	003	2026-04-13
759	ES	Mobbeel Solutions	mobbl-000	000	2021-01-28
760	ES	Mobbeel Solutions	mobbl-001	001	2021-06-16
761	ES	Mobbeel Solutions	mobbl-002	002	2021-12-16
762	ES	Mobbeel Solutions	mobbl-003	003	2022-04-19
763	ES	Mobbeel Solutions	mobbl-004	004	2024-05-20
764	ES	Mobbeel Solutions	mobbl-005	005	2024-10-15
765	KR	Mobipin Technology	mobipintech-000	000	2021-11-23
766	TH	Momentum Digital	sertis-000	000	2019-10-07
767	TH	Momentum Digital	sertis-001	001	2020-07-30
768	TH	Momentum Digital	sertis-002	002	2021-05-13
769	TH	Momentum Digital	sertis-003	003	2023-12-27
770	CN	MoreDian Technology	moredian-000	000	2021-02-24

Table 11: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
771	US	Mukh Technologies	mukh-001	001	2022-03-22
772	US	Mukh Technologies	mukh-002	002	2022-11-01
773	US	Mukh Technologies	mukh-003	003	2023-03-15
774	US	Mukh Technologies	mukh-004	004	2023-09-13
775	US	Mukh Technologies	mukh-005	005	2024-12-05
776	CN	Multi-Modality Intelligence	multimodality-000	000	2021-10-19
777	CN	Multi-Modality Intelligence	multimodality-001	001	2022-05-16
778	RU	N-Tech Lab	ntechlab-000	000	2017-03-13
779	RU	N-Tech Lab	ntechlab-001	001	2017-05-10
780	RU	N-Tech Lab	ntechlab-002	002	2017-08-23
781	RU	N-Tech Lab	ntechlab-003	003	2017-12-29
782	RU	N-Tech Lab	ntechlab-004	004	2018-06-14
783	RU	N-Tech Lab	ntechlab-005	005	2018-10-19
784	RU	N-Tech Lab	ntechlab-006	006	2019-03-01
785	RU	N-Tech Lab	ntechlab-007	007	2019-06-25
786	RU	N-Tech Lab	ntechlab-008	008	2020-01-06
787	RU	N-Tech Lab	ntechlab-009	009	2020-12-30
788	RU	N-Tech Lab	ntechlab-010	010	2021-04-30
789	RU	N-Tech Lab	ntechlab-011	011	2021-09-13
790	RU	N-Tech Lab	ntechlab-012	012	2022-01-20
791	SG	NCS Pte Ltd	ncssg-001	001	2023-03-24
792	SG	NCS Pte Ltd	ncssg-002	002	2024-06-25
793	SG	NCS Pte Ltd	ncssg-003	003	2024-10-30
794	SG	NCS Pte Ltd	ncssg-004	004	2025-04-03
795	CA	NEO Systems	neosystems-001	001	2021-03-02
796	CA	NEO Systems	neosystems-002	002	2021-07-03
797	CA	NEO Systems	neosystems-003	003	2021-11-11
798	CA	NEO Systems	neosystems-004	004	2022-05-02
799	KR	NHN Corp	nhn-001	001	2021-03-15
800	KR	NHN Corp	nhn-002	002	2021-07-15
801	KR	NHN Corp	nhn-003	003	2022-02-22
802	KR	NHN Corp	nhn-004	004	2023-06-26
803	KR	NHN Corp	nhn-005	005	2023-12-06
804	KR	NSENSE Corp	nsensecorp-001	001	2020-10-20
805	KR	NSENSE Corp	nsensecorp-002	002	2021-05-06
806	KR	NSENSE Corp	nsensecorp-003	003	2021-10-29
807	KR	NSENSE Corp	nsensecorp-004	004	2022-09-08
808	KR	NSENSE Corp	nsensecorp-005	005	2023-02-08
809	CN	Nanjing Kiwi Network Technology	kiwitech-000	000	2021-03-19
810	KR	Naver Corp	clova-000	000	2020-10-21
811	KR	Neosecu Co	openface-001	001	2021-06-15
812	TW	Netbridge Technology Incoporation	netbridgetech-001	001	2020-01-08
813	TW	Netbridge Technology Incoporation	netbridgetech-002	002	2020-08-11
814	LT	Neurotechnology	neurotechnology-000	000	2017-03-22
815	LT	Neurotechnology	neurotechnology-001	001	2017-08-07
816	LT	Neurotechnology	neurotechnology-002	002	2017-11-02
817	LT	Neurotechnology	neurotechnology-003	003	2018-02-02
818	LT	Neurotechnology	neurotechnology-004	004	2018-10-19
819	LT	Neurotechnology	neurotechnology-005	005	2019-03-01
820	LT	Neurotechnology	neurotechnology-006	006	2019-06-26
821	LT	Neurotechnology	neurotechnology-008	008	2020-01-08
822	LT	Neurotechnology	neurotechnology-009	009	2020-07-07
823	LT	Neurotechnology	neurotechnology-010	010	2020-11-26
824	LT	Neurotechnology	neurotechnology-011	011	2021-03-26
825	LT	Neurotechnology	neurotechnology-012	012	2021-07-26
826	LT	Neurotechnology	neurotechnology-013	013	2022-01-07
827	LT	Neurotechnology	neurotechnology-015	015	2022-06-07
828	LT	Neurotechnology	neurotechnology-016	016	2023-02-01
829	LT	Neurotechnology	neurotechnology-017	017	2023-07-28
830	LT	Neurotechnology	neurotechnology-018	018	2024-02-09
831	LT	Neurotechnology	neurotechnology-019	019	2024-07-31
832	LT	Neurotechnology	neurotechnology-020	020	2025-08-01
833	LT	Neurotechnology	neurotechnology-021	021	2026-09-17
834	IN	NextMeta Labs	nextmetalabs-001	001	2024-07-02
835	IN	NextMeta Labs	nextmetalabs-002	002	2024-12-04
836	IN	NextMeta Labs	nextmetalabs-003	003	2025-07-08
837	IN	NextMeta Labs	nextmetalabs-004	004	2026-01-27
838	IN	Nmetics Private Ltd	nmetics-001	001	2026-09-16
839	US	Noblis	noblis-000	000	2017-07-05
840	ID	Nodeflux	nodeflux-000	000	2019-03-01

Table 12: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
841	ID	Nodeflux	nodeflux-001	001	2019-03-01
842	ID	Nodeflux	nodeflux-002	002	2019-08-13
843	LT	Nominder	nominder-000	000	2023-02-06
844	LT	Nominder	nominder-001	001	2023-07-19
845	LT	Nominder	nominder-002	002	2023-11-04
846	LT	Nominder	nominder-003	003	2024-03-06
847	LT	Nominder	nominder-004	004	2024-07-11
848	LT	Nominder	nominder-005	005	2024-11-12
849	LT	Nominder	nominder-006	006	2025-04-07
850	LT	Nominder	nominder-007	007	2026-02-10
851	LT	Nominder	nominder-008	008	2026-07-20
852	SG	Northern Point Technology Pvt Ltd	npt-000	000	2026-09-10
853	NO	Norwegian University of Science and Technology	ntnu-000	000	2026-08-18
854	IN	NotionTag Technologies Private Limited	notiontag-000	000	2019-06-12
855	IN	NotionTag Technologies Private Limited	notiontag-001	001	2021-03-04
856	IN	NotionTag Technologies Private Limited	notiontag-002	002	2021-09-17
857	AE	OCR Studio L.L.C.-FZ	ocrstudio-001	001	2026-07-24
858	US	OOTO AI	ooto-001	001	2025-09-11
859	US	Omnigarde Ltd	omnigarde-000	000	2021-04-05
860	US	Omnigarde Ltd	omnigarde-001	001	2021-08-23
861	US	Omnigarde Ltd	omnigarde-002	002	2022-01-19
862	US	Omnigarde Ltd	omnigarde-003	003	2023-05-10
863	US	Omnigarde Ltd	omnigarde-004	004	2024-06-24
864	US	Omnigarde Ltd	omnigarde-005	005	2025-06-02
865	KR	One More Security	omface-000	000	2021-12-15
866	KR	One More Security	omface-001	001	2022-10-21
867	GB	Onfido	onfido-000	000	2022-12-13
868	GB	Onfido	onfido-001	001	2025-07-17
869	GB	Onfido	onfido-002	002	2025-12-16
870	GB	Onfido	onfido-003	003	2026-04-22
871	GB	Onfido	onfido-004	004	2026-08-28
872	VN	Online Mobile Services JSC	momovn-001	001	2024-01-26
873	VN	Online Mobile Services JSC	momovn-002	002	2024-06-04
874	VN	Online Mobile Services JSC	momovn-003	003	2025-09-19
875	IN	Ooru Digital Private Ltd	oorudigital-000	000	2026-07-22
876	IN	Openedge Technologies Pvt	openedge-000	000	2024-01-29
877	AE	Ovision	ovisionllc-001	001	2024-04-03
878	AE	Ovision	ovisionllc-002	002	2024-08-05
879	RU	Oz Forensics LLC	oz-001	001	2020-07-29
880	RU	Oz Forensics LLC	oz-002	002	2021-01-18
881	RU	Oz Forensics LLC	oz-003	003	2021-08-09
882	RU	Oz Forensics LLC	oz-004	004	2021-12-13
883	TW	PAPAGO Inc	papago-001	001	2022-07-19
884	CZ	PAPIL11 S.R.O.	papil11-000	000	2024-01-04
885	CZ	PAPIL11 S.R.O.	papil11-001	001	2024-08-06
886	CZ	PAPIL11 S.R.O.	papil11-002	002	2025-01-02
887	CZ	PAPIL11 S.R.O.	papil11-003	003	2025-04-29
888	CZ	PAPIL11 S.R.O.	papil11-004	004	2026-03-31
889	KR	PNPSecure	pnpsecure-000	000	2025-06-05
890	KR	PNPSecure	pnpsecure-001	001	2026-03-03
891	KR	PNPSecure	pnpsecure-002	002	2026-09-10
892	ID	PT Autentika Digital Indonesia	autentika-000	000	2022-12-05
893	ID	PT Autentika Digital Indonesia	autentika-001	001	2023-04-05
894	ID	PT Autentika Digital Indonesia	autentika-002	002	2023-11-21
895	ID	PT Autentika Digital Indonesia	autentika-003	003	2024-11-12
896	ID	PT Ice House	icehouse-000	000	2026-02-27
897	ID	PT Qlue Performa Indonesia	qluevision-001	001	2022-11-15
898	ID	PT Technology Biometric Indonesia	techbiometricid-001	001	2025-01-02
899	ID	PT Technology Biometric Indonesia	techbiometricid-002	002	2026-04-09
900	CH	PXL Vision AG	pxl-001	001	2020-06-30
901	TW	Palit Microsystems	palit-000	000	2022-05-16
902	TW	Palit Microsystems	palit-001	001	2022-09-26
903	SG	Panasonic R+D Center Singapore	psl-000	000	2018-06-18
904	SG	Panasonic R+D Center Singapore	psl-001	001	2018-10-12
905	SG	Panasonic R+D Center Singapore	psl-002	002	2019-02-28
906	SG	Panasonic R+D Center Singapore	psl-003	003	2019-10-01
907	SG	Panasonic R+D Center Singapore	psl-004	004	2020-03-03
908	SG	Panasonic R+D Center Singapore	psl-005	005	2020-07-06
909	SG	Panasonic R+D Center Singapore	psl-006	006	2020-11-13
910	SG	Panasonic R+D Center Singapore	psl-007	007	2021-03-19

Table 13: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
911	SG	Panasonic R+D Center Singapore	psl-008	008	2021-07-21
912	SG	Panasonic R+D Center Singapore	psl-009	009	2021-12-08
913	SG	Panasonic R+D Center Singapore	psl-010	010	2022-04-19
914	SG	Panasonic R+D Center Singapore	psl-011	011	2022-10-06
915	SG	Panasonic R+D Center Singapore	psl-012	012	2023-10-13
916	SG	Panasonic R+D Center Singapore	psl-013	013	2024-04-18
917	SG	Panasonic R+D Center Singapore	psl-014	014	2024-09-11
918	US	Pangiam	pangiam-000	000	2022-04-04
919	US	Pangiam	pangiam-001	001	2023-02-21
920	US	Pangiam	pangiam-002	002	2023-06-21
921	TR	Papilon Savunma	papsav1923-001	001	2021-03-10
922	TR	Papilon Savunma	papsav1923-002	002	2022-01-20
923	TR	Papilon Savunma	papsav1923-003	003	2022-11-25
924	US	Paravision	everai-000	000	2018-10-19
925	US	Paravision	everai-001	001	2018-10-19
926	US	Paravision	everai-002	002	2019-03-01
927	US	Paravision	paravision	003	2019-07-01
928	US	Paravision	paravision-004	004	2019-12-11
929	US	Paravision	paravision-006	006	2021-02-01
930	US	Paravision	paravision-008	008	2021-06-30
931	US	Paravision	paravision-010	010	2022-02-02
932	US	Paravision	paravision-011	011	2022-12-12
933	US	Paravision	paravision-013	013	2023-05-08
934	US	Paravision	paravision-015	015	2024-09-09
935	US	Paravision	paravision-017	017	2025-01-13
936	US	Paravision	paravision-018	018	2025-06-12
937	US	Paravision	paravision-020	020	2025-12-11
938	US	Paravision	paravision-022	022	2026-09-24
939	SG	Pensees Pte	pensees-001	001	2020-08-17
940	PL	Pentacomp Systemy Informatyczne	pentacomp-000	000	2024-09-26
941	CN	Ping An Technology	pa-002	002	2017-09-25
942	CN	Ping An Technology	pa-003	003	2017-05-25
943	US	Private Identity LLC	privid-001	001	2023-02-06
944	US	Private Identity LLC	privid-002	002	2023-10-03
945	US	Private Identity LLC	privid-003	003	2024-09-24
946	US	Private Identity LLC	privid-004	004	2025-03-05
947	IN	Pyramid Cyber Security + Forensic (P)	pyramid-000	000	2019-11-04
948	IN	Pyramid Cyber Security + Forensic (P)	pyramid-001	001	2024-11-26
949	KZ	Qaz Biometric Systems	qazbs-000	000	2022-06-22
950	KZ	QazSmartVision.AI	qazsmartvisionai-000	000	2024-01-11
951	KZ	QazSmartVision.AI	qazsmartvisionai-001	001	2024-05-21
952	KZ	QazSmartVision.AI	qazsmartvisionai-002	002	2024-09-25
953	KZ	QazSmartVision.AI	qazsmartvisionai-003	003	2025-01-30
954	KZ	QazSmartVision.AI	qazsmartvisionai-004	004	2025-07-18
955	TW	Qnap Security	qnap-000	000	2021-08-09
956	TW	Qnap Security	qnap-001	001	2021-12-09
957	TW	Qnap Security	qnap-002	002	2022-04-15
958	TW	Qnap Security	qnap-003	003	2022-12-09
959	TW	Qnap Security	qnap-004	004	2023-05-05
960	TW	Qnap Security	qnap-005	005	2023-11-16
961	TW	Qnap Security	qnap-006	006	2024-06-14
962	CZ	Quantasoft	quantasoft-003	003	2021-04-19
963	US	ROC	rankone-000	000	2017-03-21
964	US	ROC	rankone-001	001	2017-04-12
965	US	ROC	rankone-002	002	2017-08-18
966	US	ROC	rankone-003	003	2017-11-28
967	US	ROC	rankone-004	004	2018-06-12
968	US	ROC	rankone-005	005	2018-10-09
969	US	ROC	rankone-006	006	2019-02-27
970	US	ROC	rankone-007	007	2019-06-03
971	US	ROC	rankone-008	008	2019-11-12
972	US	ROC	rankone-009	009	2020-06-26
973	US	ROC	rankone-010	010	2020-11-05
974	US	ROC	rankone-011	011	2021-08-27
975	US	ROC	rankone-012	012	2021-12-27
976	US	ROC	rankone-013	013	2022-07-09
977	US	ROC	rankone-014	014	2022-12-21
978	US	ROC	rankone-015	015	2023-07-20
979	US	ROC	roc-016	016	2023-12-19
980	US	ROC	roc-017	017	2024-06-24

Table 14: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
981	US	ROC	roc-018	018	2025-01-30
982	US	ROC	roc-019	019	2025-07-21
983	US	ROC	roc-020	020	2026-02-06
984	US	ROC	roc-021	021	2026-08-05
985	US	RealSense	realsense-001	001	2025-12-01
986	US	RealSense	realsense-002	002	2026-06-09
987	US	RealSense	realsense-003	003	2026-10-19
988	US	Realnetworks Inc	realnetworks-000	000	2018-05-30
989	US	Realnetworks Inc	realnetworks-001	001	2018-10-19
990	US	Realnetworks Inc	realnetworks-002	002	2019-02-28
991	US	Realnetworks Inc	realnetworks-003	003	2019-06-12
992	US	Realnetworks Inc	realnetworks-004	004	2021-04-15
993	US	Realnetworks Inc	realnetworks-005	005	2021-09-27
994	US	Realnetworks Inc	realnetworks-006	006	2022-02-09
995	US	Realnetworks Inc	realnetworks-007	007	2022-06-14
996	US	Realnetworks Inc	realnetworks-008	008	2022-11-10
997	AE	Recognito	recognito-000	000	2023-05-24
998	AE	Recognito	recognito-001	001	2023-09-27
999	US	Regula Forensics	regula-000	000	2021-04-13
1000	US	Regula Forensics	regula-001	001	2021-12-14
1001	US	Regula Forensics	regula-002	002	2025-09-15
1002	US	Regula Forensics	regula-003	003	2026-08-20
1003	AM	Regular Biometrics Solutions	rebs-000	000	2023-08-22
1004	AM	Regular Biometrics Solutions	rebs-001	001	2023-12-22
1005	CN	Remark Holdings	remarkai-000	000	2019-03-01
1006	CN	Remark Holdings	remarkai-001	001	2019-03-01
1007	CN	Remark Holdings	remarkai-002	002	2019-11-21
1008	CN	Remark Holdings	remarkai-003	003	2021-06-22
1009	SG	Rendip	rendip-000	000	2021-04-19
1010	GB	Reveal Media Ltd	revealmedia-005	005	2021-09-24
1011	GB	Reveal Media Ltd	revealmedia-006	006	2022-01-26
1012	CN	Rokid Corporation	rokid-000	000	2019-08-01
1013	CN	Rokid Corporation	rokid-001	001	2019-12-13
1014	KR	SK Telecom	sktelecom-000	000	2021-07-09
1015	KR	SQIsoft	sqisoft-001	001	2021-07-27
1016	KR	SQIsoft	sqisoft-002	002	2021-11-03
1017	KR	SQIsoft	sqisoft-003	003	2022-10-26
1018	SG	ST Engineering	stengg-000	000	2023-06-06
1019	SA	STCON LLC	stcon-000	000	2022-11-02
1020	SA	STCON LLC	stcon-001	001	2023-03-03
1021	SA	STCON LLC	stcon-002	002	2023-07-05
1022	SA	STCON LLC	stcon-003	003	2023-11-07
1023	SA	STCON LLC	stcon-004	004	2024-06-18
1024	SA	STCON LLC	stcon-005	005	2024-11-13
1025	SA	STCON LLC	stcon-006	006	2025-03-17
1026	SA	STCON LLC	stcon-007	007	2025-07-18
1027	SA	STCON LLC	stcon-008	008	2025-12-30
1028	SA	STCON LLC	stcon-009	009	2026-05-01
1029	SA	STCON LLC	stcon-010	010	2026-09-11
1030	DE	Saffe	saffe-001	001	2018-10-19
1031	DE	Saffe	saffe-002	002	2019-03-01
1032	JP	Saga Densan Center Co Ltd	sdc-000	000	2022-10-18
1033	KR	Samsung S1 Corp	s1-001	001	2019-12-06
1034	KR	Samsung S1 Corp	s1-002	002	2021-03-24
1035	KR	Samsung S1 Corp	s1-003	003	2021-08-24
1036	KR	Samsung S1 Corp	s1-004	004	2022-01-04
1037	KR	Samsung S1 Corp	s1-005	005	2022-06-17
1038	KR	Samsung S1 Corp	s1-006	006	2022-10-17
1039	KR	Samsung S1 Corp	s1-007	007	2023-03-20
1040	KR	Samsung S1 Corp	s1-008	008	2024-10-16
1041	KR	Samsung S1 Corp	s1-009	009	2025-02-25
1042	KR	Samsung-SDS	samsungsds-000	000	2021-10-28
1043	KR	Samsung-SDS	samsungsds-001	001	2022-04-18
1044	KR	Samsung-SDS	samsungsds-002	002	2022-09-16
1045	IN	Samtech InfoNet Limited	samtech-000	000	2017-05-02
1046	IN	Samtech InfoNet Limited	samtech-001	001	2019-10-15
1047	IN	Samtech InfoNet Limited	samtech-002	002	2023-07-06
1048	RU	Satellite Innovation/Eocortex	eocortex-000	000	2020-08-26
1049	IN	ScanalitiX	scanalitix-001	001	2026-05-20
1050	IL	Scanovate	scanovate-001	001	2019-11-12

Table 15: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1051	IL	Scanovate	scanovate-002	002	2020-06-26
1052	IL	Scanovate	scanovate-003	003	2021-11-15
1053	NG	Seamfix	seamfix-001	001	2024-03-08
1054	RO	Securif AI	securifai-001	001	2020-10-06
1055	RO	Securif AI	securifai-002	002	2021-03-19
1056	RO	Securif AI	securifai-003	003	2021-08-03
1057	RO	Securif AI	securifai-004	004	2021-12-21
1058	RO	Securif AI	securifai-005	005	2022-05-16
1059	RO	Securif AI	securifai-006	006	2022-11-14
1060	RO	Securif AI	securifai-007	007	2023-08-17
1061	RO	Securif AI	securifai-008	008	2024-02-26
1062	TH	Security Pitch Co Ltd	securitypitch-000	000	2026-08-19
1063	CN	Sensetime Group	sensetime-001	001	2018-10-19
1064	CN	Sensetime Group	sensetime-002	002	2018-10-19
1065	CN	Sensetime Group	sensetime-003	003	2019-06-04
1066	CN	Sensetime Group	sensetime-004	004	2020-11-20
1067	CN	Sensetime Group	sensetime-005	005	2021-05-24
1068	CN	Sensetime Group	sensetime-006	006	2021-12-28
1069	CN	Sensetime Group	sensetime-007	007	2022-06-17
1070	CN	Sensetime Group	sensetime-008	008	2023-01-04
1071	CN	Sensetime Group	sensetime-010	010	2024-12-03
1072	CN	Sensetime Group	sensetime-011	011	2026-03-17
1073	CN	Sensetime Group	sensetime-012	012	2026-08-10
1074	UZ	Serendipity Ltd	serendipity-000	000	2023-05-31
1075	SG	Seventh Sense Artificial Intelligence	seventhsense-000	000	2021-06-29
1076	SG	Seventh Sense Artificial Intelligence	seventhsense-001	001	2022-03-04
1077	SG	Seventh Sense Artificial Intelligence	seventhsense-002	002	2022-10-17
1078	SG	Seventh Sense Artificial Intelligence	seventhsense-003	003	2023-05-31
1079	SG	Seventh Sense Artificial Intelligence	seventhsense-005	005	2023-10-05
1080	SG	Seventh Sense Artificial Intelligence	seventhsense-006	006	2024-07-09
1081	US	Shaman Software	shaman-000	000	2017-12-05
1082	US	Shaman Software	shaman-001	001	2018-01-13
1083	CN	Shanghai Jiao Tong University	sjtu-001	001	2019-09-27
1084	CN	Shanghai Jiao Tong University	sjtu-002	002	2020-02-12
1085	CN	Shanghai Jiao Tong University	sjtu-003	003	2020-11-02
1086	CN	Shanghai Jiao Tong University	sjtu-004	004	2021-05-13
1087	CN	Shanghai Jiao Tong University	sjtu-005	005	2023-05-05
1088	CN	Shanghai Jiao Tong University	sjtu-006	006	2024-12-06
1089	CN	Shanghai Jiao Tong University	sjtu-007	007	2025-04-16
1090	CN	Shanghai Jiao Tong University	sjtu-008	008	2025-12-11
1091	CN	Shanghai Jiao Tong University	sjtu-009	009	2026-04-13
1092	CN	Shanghai Jiao Tong University	sjtu-010	010	2026-10-19
1093	CN	Shanghai Ulucu Electronics Technology	uluface-002	002	2019-07-10
1094	CN	Shanghai Ulucu Electronics Technology	uluface-003	003	2019-11-12
1095	CN	Shanghai University - Shanghai Film Academy	shu-001	001	2019-06-17
1096	CN	Shanghai University - Shanghai Film Academy	shu-002	002	2019-12-10
1097	CN	Shanghai University - Shanghai Film Academy	shu-003	003	2020-06-24
1098	CN	Shanghai Yitu Technology	yitu-000	000	2017-05-23
1099	CN	Shanghai Yitu Technology	yitu-001	001	2018-06-12
1100	CN	Shanghai Yitu Technology	yitu-002	002	2018-10-19
1101	CN	Shanghai Yitu Technology	yitu-003	003	2019-03-01
1102	CN	Shenzhen AiMall Tech	aimall-001	001	2019-11-12
1103	CN	Shenzhen AiMall Tech	aimall-002	002	2020-03-12
1104	CN	Shenzhen AiMall Tech	aimall-003	003	2020-08-12
1105	CN	Shenzhen EI Networks	einetworks-000	000	2019-08-13
1106	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-002	002	2018-06-13
1107	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-004	004	2019-03-01
1108	CN	Shenzhen Inst Adv Integrated Tech CAS	siat-005	005	2022-02-08
1109	CN	Shenzhen Intellifusion Technologies	intellifusion-001	001	2019-08-22
1110	CN	Shenzhen Intellifusion Technologies	intellifusion-002	002	2020-03-18
1111	CN	Shenzhen Polytechnic University	szpu-000	000	2025-03-28
1112	CN	Shenzhen University-Macau University of Science and Technology	sztu-000	000	2020-12-17
1113	CN	Shenzhen University-Macau University of Science and Technology	sztu-001	001	2021-07-13
1114	GB	Shufti	shufti-001	001	2026-03-17
1115	QA	Sightropic	sightropic-000	000	2025-03-26
1116	QA	Sightropic	sightropic-001	001	2025-11-25
1117	QA	Sightropic	sightropic-002	002	2026-06-04
1118	IN	Smarg Technologies Pvt Ltd	smargtech-001	001	2026-03-17
1119	RU	Smart Engines	smartengines-000	000	2021-08-25
1120	RU	Smart Engines	smartengines-001	001	2022-05-31

Table 16: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1121	ES	Smartbiometrik	smartbiometrik-001	001	2022-05-16
1122	TR	Smarvist Teknoloji	smartvist-000	000	2022-05-10
1123	TR	Smarvist Teknoloji	smartvist-001	001	2023-11-02
1124	DE	Smilart	smilart-001	001	2018-02-06
1125	DE	Smilart	smilart-002	002	2018-02-06
1126	DE	Smilart	smilart-003	003	2019-03-01
1127	TR	Sodec App Inc	sodec-000	000	2021-06-02
1128	TR	Softtech Yazilim Teknolojileri Arastirma Gelistirme ve	softtech-000	000	2026-07-30
1129	CN	Sparktek International Pte Ltd	sparktek-000	000	2025-11-24
1130	CN	Sparktek International Pte Ltd	sparktek-001	001	2026-06-23
1131	IN	Sparsh CCTV	sparsh-001	001	2024-01-02
1132	IN	Staqu Technologies	staqu-000	000	2020-07-15
1133	CN	Star Hybrid Limited	starhybrid-001	001	2019-06-19
1134	US	Stellar Technologies	stellartecs-001	001	2026-05-25
1135	US	Stellar Technologies	stellartecs-003	003	2026-10-22
1136	CN	Su Zhou NaZhiTianDi intelligent technology	nazhiai-000	000	2020-06-25
1137	IN	Sukshi Technology Innovation	sukshi-000	000	2022-02-13
1138	GB	Sum and Substance Ltd	sumsub-000	000	2025-12-15
1139	GB	Sum and Substance Ltd	sumsub-001	001	2026-07-27
1140	KR	Suprema AI Inc	suprema-000	000	2021-03-31
1141	KR	Suprema AI Inc	suprema-001	001	2021-09-23
1142	KR	Suprema AI Inc	suprema-002	002	2022-02-11
1143	KR	Suprema AI Inc	suprema-003	003	2022-07-20
1144	KR	Suprema AI Inc	suprema-004	004	2023-01-09
1145	KR	Suprema AI Inc	suprema-005	005	2023-10-25
1146	KR	Suprema ID Inc	supremaid-001	001	2021-05-04
1147	KR	Suprema ID Inc	supremaid-002	002	2022-06-24
1148	GB	Swsam Solutions	swsam-001	001	2023-03-13
1149	RU	Synesis	synesis-000	000	2018-02-09
1150	RU	Synesis	synesis-001	001	2018-05-31
1151	RU	Synesis	synesis-003	003	2018-10-19
1152	RU	Synesis	synesis-004	004	2019-03-01
1153	RU	Synesis	synesis-005	005	2019-06-06
1154	RU	Synesis	synesis-006	006	2019-10-10
1155	RU	Synesis	synesis-007	007	2020-06-24
1156	TW	Synology Inc	synology-000	000	2019-10-23
1157	TW	Synology Inc	synology-001	001	2020-02-26
1158	TW	Synology Inc	synology-002	002	2020-08-20
1159	IT	Syscake S.r.l.	syscake-000	000	2024-11-26
1160	BR	T4iSB	t4isb-000	000	2022-01-28
1161	SA	TAHAKOM	tahakom-001	001	2024-10-17
1162	SA	TAHAKOM	tahakom-002	002	2026-06-15
1163	SA	TAHAKOM	tahakom-003	003	2026-10-15
1164	MY	TNG Digital Sdn Bhd	tngd-001	001	2025-03-17
1165	MY	TNG Digital Sdn Bhd	tngd-002	002	2026-03-18
1166	VN	TNI Tech Co	tnitech-000	000	2024-02-05
1167	AE	TOPPAN Security Co	toppansecurity-001	001	2026-03-30
1168	VN	TTI Group	ttigroup-000	000	2024-12-16
1169	CN	TUPU Technology	tuputech-000	000	2019-10-11
1170	TW	Taiwan AI Labs	ailabs-001	001	2019-12-18
1171	TW	Taiwan Computer Vision	tcv-001	001	2024-05-30
1172	TW	Taiwan Mobile	taiwanmobile-001	001	2024-08-06
1173	TW	Taiwan Mobile	taiwanmobile-002	002	2024-12-11
1174	TW	Taiwan-Certificate Authority Incorporation	twface-000	000	2021-05-14
1175	TW	Taiwan-Certificate Authority Incorporation	twface-001	001	2021-09-14
1176	CH	Tech5 SA	tech5-000	000	2018-10-09
1177	CH	Tech5 SA	tech5-001	001	2018-10-19
1178	CH	Tech5 SA	tech5-002	002	2019-03-01
1179	CH	Tech5 SA	tech5-003	003	2019-08-19
1180	CH	Tech5 SA	tech5-004	004	2020-03-09
1181	CH	Tech5 SA	tech5-005	005	2020-07-24
1182	CH	Tech5 SA	tech5-007	007	2022-12-30
1183	CH	Tech5 SA	tech5-008	008	2023-08-11
1184	CH	Tech5 SA	tech5-009	009	2025-04-17
1185	VN	Techainer	techainer-001	001	2023-11-30
1186	VN	Techainer	techainer-002	002	2025-02-14
1187	VN	Techainer	techainer-003	003	2026-06-08
1188	SA	Technology Control Company	tcc-000	000	2026-02-10
1189	SA	Technology Control Company	tcc-001	001	2026-07-13
1190	TR	Techsign	techsign-000	000	2021-08-25

Table 17: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1191	TR	Techsign	techsign-001	001	2022-07-01
1192	TW	Telexper International	telexper-000	000	2024-06-21
1193	CN	Tencent Deepsea Lab	deepsea-001	001	2019-06-03
1194	RU	Tevian	tevian-000	000	2017-12-10
1195	RU	Tevian	tevian-001	001	2018-01-19
1196	RU	Tevian	tevian-002	002	2018-06-14
1197	RU	Tevian	tevian-003	003	2018-10-19
1198	RU	Tevian	tevian-004	004	2019-03-01
1199	RU	Tevian	tevian-005	005	2019-09-21
1200	RU	Tevian	tevian-006	006	2020-09-11
1201	RU	Tevian	tevian-007	007	2021-08-06
1202	RU	Tevian	tevian-008	008	2021-12-06
1203	US	TigerIT Bangladesh Ltd	tiger-001	001	2018-01-23
1204	US	TigerIT Bangladesh Ltd	tiger-002	002	2018-06-13
1205	US	TigerIT Bangladesh Ltd	tiger-003	003	2018-10-16
1206	US	TigerIT Bangladesh Ltd	tiger-004	004	2020-12-01
1207	US	TigerIT Bangladesh Ltd	tiger-005	005	2021-07-29
1208	US	TigerIT Bangladesh Ltd	tiger-006	006	2021-12-13
1209	US	TigerIT Bangladesh Ltd	tiger-007	007	2025-03-05
1210	RU	Tinkoff Bank	tinkoff-001	001	2021-05-13
1211	CN	TongYi Transportation Technology	tongyi-004	004	2018-10-19
1212	CN	TongYi Transportation Technology	tongyi-005	005	2019-06-12
1213	CN	TongYi Transportation Technology	tongyitrans-001	001	2017-04-01
1214	CN	TongYi Transportation Technology	tongyitrans-002	002	2017-07-15
1215	CN	TongYi Transportation Technology	tongyitrans-003	003	2018-05-15
1216	TW	Toppan ID Gate	toppanidgate-000	000	2021-09-28
1217	JP	Toshiba	toshiba-000	000	2018-01-11
1218	JP	Toshiba	toshiba-001	001	2018-01-11
1219	JP	Toshiba	toshiba-002	002	2018-10-19
1220	JP	Toshiba	toshiba-003	003	2019-03-01
1221	JP	Toshiba	toshiba-004	004	2021-09-27
1222	JP	Toshiba	toshiba-005	005	2022-02-09
1223	JP	Toshiba	toshiba-006	006	2022-06-29
1224	JP	Toshiba	toshiba-007	007	2023-09-13
1225	JP	Toshiba	toshiba-008	008	2024-02-01
1226	JP	Toshiba	toshiba-009	009	2024-09-16
1227	JP	Toshiba	toshiba-010	010	2025-02-03
1228	JP	Toshiba	toshiba-011	011	2025-07-08
1229	JP	Toshiba	toshiba-012	012	2026-01-09
1230	JP	Toshiba	toshiba-013	013	2026-09-16
1231	KR	Toss	toss-000	000	2025-09-22
1232	KR	Toss	toss-001	001	2026-08-28
1233	ES	Touchless ID	touchlessid-000	000	2022-05-02
1234	ES	Touchless ID	touchlessid-001	001	2022-09-21
1235	ES	Touchless ID	touchlessid-002	002	2023-01-23
1236	ES	Touchless ID	touchlessid-003	003	2023-07-12
1237	ES	Touchless ID	touchlessid-004	004	2024-12-27
1238	Il	Transmit Security Ltd	transmitsecurity-001	001	2025-12-04
1239	JP	Tripleize	aize-001	001	2021-04-23
1240	JP	Tripleize	aize-002	002	2021-10-08
1241	JP	Tripleize	aize-003	003	2023-08-10
1242	JP	Tripleize	aize-006	006	2026-04-08
1243	VN	TrueID-VNG	trueidvng-001	001	2023-01-05
1244	AE	TrueSight Laboratories Tech LLC	tsight-000	000	2026-05-22
1245	US	Trueface.ai	trueface-000	000	2019-10-08
1246	US	Trueface.ai	trueface-001	001	2020-07-20
1247	US	Trueface.ai	trueface-002	002	2021-03-29
1248	US	Trueface.ai	trueface-003	003	2021-09-30
1249	GB	Trust Stamp	truststamp-001	001	2023-07-14
1250	GB	Trust Stamp	truststamp-002	002	2025-05-23
1251	GB	Trust Stamp	truststamp-003	003	2025-11-19
1252	CN	TuringTech.vip	turingtechvip-001	001	2022-02-03
1253	CN	TuringTech.vip	turingtechvip-002	002	2022-07-27
1254	TR	Turkcell Technology	turkcell-000	000	2022-10-11
1255	TR	Turkcell Technology	turkcell-001	001	2023-03-15
1256	TR	Turkcell Technology	turkcell-002	002	2024-03-18
1257	CN	ULSee Inc	ulsee-001	001	2019-07-31
1258	UN	UNICC-Solution Architecture Section	unicc-001	001	2023-02-17
1259	UN	UNICC-Solution Architecture Section	unicc-002	002	2023-06-07
1260	UN	UNICC-Solution Architecture Section	unicc-003	003	2023-10-16

Table 18: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1261	UN	UNICC-Solution Architecture Section	unicc-004	004	2024-09-23
1262	TW	UXLabs	uxlabs-001	001	2022-09-19
1263	TW	UXLabs	uxlabs-003	003	2023-07-28
1264	UZ	UZINFOCOM	uzinfocom-000	000	2025-07-28
1265	UZ	UZINFOCOM	uzinfocom-001	001	2026-04-08
1266	HU	Ultinous	ultinous-000	000	2017-12-18
1267	HU	Ultinous	ultinous-001	001	2018-02-02
1268	FR	Unissey	unissey-001	001	2021-11-29
1269	FR	Unissey	unissey-002	002	2022-04-29
1270	FR	Unissey	unissey-003	003	2022-12-19
1271	FR	Unissey	unissey-004	004	2023-07-19
1272	FR	Unissey	unissey-005	005	2024-08-06
1273	FR	Unissey	unissey-006	006	2025-09-11
1274	FR	Unissey	unissey-007	007	2026-03-25
1275	CN	Universal Ubiquitous AI Co Ltd	uniubi-000	000	2026-09-24
1276	SG	UniverseAI	universeai-000	000	2026-08-11
1277	BR	Universidade Federal de Goiás	ceiaufgbr-000	000	2026-08-06
1278	PT	Universidade de Coimbra	visteam-000	000	2020-01-14
1279	PT	Universidade de Coimbra	visteam-001	001	2021-03-16
1280	PT	Universidade de Coimbra	visteam-002	002	2021-08-20
1281	PT	Universidade de Coimbra	visteam-003	003	2022-01-31
1282	PT	Universidade de Coimbra	visteam-004	004	2022-08-01
1283	PT	Universidade de Coimbra	visteam-005	005	2023-01-04
1284	PT	Universidade de Coimbra	visteam-006	006	2023-05-25
1285	PT	Universidade de Coimbra	visteam-007	007	2023-10-18
1286	PT	Universidade de Coimbra	visteam-008	008	2024-02-06
1287	PT	Universidade de Coimbra	visteam-009	009	2024-07-26
1288	PT	Universidade de Coimbra	visteam-010	010	2025-05-28
1289	GB	University of Surrey-CVSSP	surrey-cvssp-000	000	2022-03-25
1290	GB	University of Surrey-CVSSP	surrey-cvssp-001	001	2022-09-22
1291	GB	University of Surrey-CVSSP	surrey-cvssp-002	002	2023-02-16
1292	GB	University of Surrey-CVSSP	surrey-cvssp-003	003	2023-09-13
1293	GB	University of Surrey-CVSSP	surrey-cvssp-004	004	2024-07-11
1294	US	VCognition	vcog-002	002	2017-06-12
1295	VN	VNIS Joint Stock	vnis-000	000	2024-01-08
1296	US	Vaidio	vaidio-001	001	2025-12-10
1297	US	Vcortex Labs	vcortex-001	001	2023-06-14
1298	LB	Veertec SAL (Offshore)	veertec-001	001	2026-07-07
1299	ES	Veridas Digital Authentication Solutions S.L.	veridas-001	001	2019-03-01
1300	ES	Veridas Digital Authentication Solutions S.L.	veridas-002	002	2019-03-01
1301	ES	Veridas Digital Authentication Solutions S.L.	veridas-003	003	2019-11-27
1302	ES	Veridas Digital Authentication Solutions S.L.	veridas-004	004	2020-07-21
1303	ES	Veridas Digital Authentication Solutions S.L.	veridas-005	005	2020-12-15
1304	ES	Veridas Digital Authentication Solutions S.L.	veridas-006	006	2021-04-15
1305	ES	Veridas Digital Authentication Solutions S.L.	veridas-007	007	2021-09-02
1306	ES	Veridas Digital Authentication Solutions S.L.	veridas-008	008	2022-10-17
1307	ES	Veridas Digital Authentication Solutions S.L.	veridas-009	009	2023-05-08
1308	ES	Veridas Digital Authentication Solutions S.L.	veridas-010	010	2025-03-18
1309	GB	Veridium	veridium-000	000	2022-03-28
1310	GB	Veridium	veridium-001	001	2022-11-03
1311	GB	Veridium	veridium-002	002	2023-06-30
1312	GB	Veridium	veridium-003	003	2024-02-21
1313	GB	Veridium	veridium-004	004	2026-07-07
1314	KZ	Verigram	verigram-000	000	2021-09-06
1315	KZ	Verigram	verigram-001	001	2022-03-09
1316	KZ	Verigram	verigram-002	002	2023-04-03
1317	KZ	Verigram	verigram-003	003	2023-08-09
1318	KZ	Verigram	verigram-004	004	2026-08-10
1319	ID	Verihubs	verihubs-inteligensia-000	000	2021-07-27
1320	ID	Verihubs	verihubs-inteligensia-001	001	2022-06-16
1321	ID	Verihubs	verihubs-inteligensia-002	002	2023-06-07
1322	ID	Verijelas	verijelas-000	000	2022-08-01
1323	IN	Verismart	verismart-001	001	2026-08-25
1324	TW	Via Technologies Inc	via-000	000	2019-07-08
1325	TW	Via Technologies Inc	via-001	001	2020-01-08
1326	TW	Via Technologies Inc	via-004	004	2023-02-23
1327	TW	Via Technologies Inc	via-005	005	2023-12-26
1328	TW	Via Technologies Inc	via-006	006	2025-02-24
1329	TW	Via Technologies Inc	via-007	007	2026-09-28
1330	AE	Viante.AI	viante-000	000	2023-06-09

Table 19: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1331	AE	Viante.AI	viante-001	001	2024-12-30
1332	AE	Viante.AI	viante-002	002	2025-06-17
1333	AE	Viante.AI	viante-003	003	2026-03-27
1334	AE	Viante.AI	viante-004	004	2026-08-05
1335	DE	Videmo Intelligente Videoanalyse	videmo-000	000	2019-12-19
1336	DE	Videmo Intelligente Videoanalyse	videmo-001	001	2021-12-22
1337	DE	Videmo Intelligente Videoanalyse	videmo-002	002	2022-08-31
1338	DE	Videmo Intelligente Videoanalyse	videmo-003	003	2024-06-18
1339	IN	Videonetics Technology Pvt	videonetics-001	001	2019-06-19
1340	IN	Videonetics Technology Pvt	videonetics-002	002	2019-11-21
1341	VN	Vietnam National Cyber Security Technology Corporation	ncsgroup-000	000	2024-03-18
1342	VN	Vietnam Payment Solution	vnpay-000	000	2023-12-05
1343	VN	Vietnam Payment Solution	vnpay-001	001	2024-10-15
1344	VN	Vietnam Posts and Telecommunications Group	vnpt-001	001	2021-01-08
1345	VN	Vietnam Posts and Telecommunications Group	vnpt-002	002	2021-06-08
1346	VN	Vietnam Posts and Telecommunications Group	vnpt-003	003	2021-12-01
1347	VN	Vietnam Posts and Telecommunications Group	vnpt-004	004	2022-04-15
1348	VN	Vietnam Posts and Telecommunications Group	vnpt-005	005	2022-08-24
1349	VN	Vietnam Posts and Telecommunications Group	vnpt-006	006	2023-08-18
1350	VN	Vietnam Posts and Telecommunications Group	vnpt-007	007	2024-04-29
1351	VN	Vietnam Posts and Telecommunications Group	vnpt-008	008	2024-09-20
1352	VN	Vietnam Posts and Telecommunications Group	vnpt-009	009	2025-06-04
1353	VN	Viettel Artificial Intelligence	viettelai-000	000	2023-06-26
1354	VN	Viettel Artificial Intelligence	viettelai-001	001	2023-11-20
1355	VN	Viettel Artificial Intelligence	viettelai-002	002	2024-07-17
1356	VN	Viettel Artificial Intelligence	viettelai-003	003	2025-02-27
1357	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-004	004	2025-11-26
1358	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-005	005	2026-05-11
1359	VN	Viettel Artificial Intelligence and Data Services Center	viettelai-006	006	2026-09-11
1360	VN	Viettel Group	vts-000	000	2020-11-04
1361	VN	Viettel Group	vts-001	001	2022-04-20
1362	VN	Viettel Group	vts-002	002	2024-06-04
1363	VN	Viettel High Technology	viettelhightech-000	000	2021-08-04
1364	VN	Viettel High Technology	viettelhightech-001	001	2025-03-18
1365	VN	Viettel High Technology	viettelhightech-002	002	2025-10-15
1366	US	Vigilant Solutions	vigilantsolutions-000	000	2017-03-30
1367	US	Vigilant Solutions	vigilantsolutions-001	001	2017-06-13
1368	US	Vigilant Solutions	vigilantsolutions-002	002	2017-09-28
1369	US	Vigilant Solutions	vigilantsolutions-003	003	2018-01-23
1370	US	Vigilant Solutions	vigilantsolutions-004	004	2018-06-14
1371	US	Vigilant Solutions	vigilantsolutions-005	005	2018-10-11
1372	US	Vigilant Solutions	vigilantsolutions-006	006	2019-03-01
1373	US	Vigilant Solutions	vigilantsolutions-007	007	2019-06-27
1374	US	Vigilant Solutions	vigilantsolutions-008	008	2020-08-03
1375	US	Vigilant Solutions	vigilantsolutions-009	009	2020-12-07
1376	US	Vigilant Solutions	vigilantsolutions-010	010	2021-04-07
1377	US	Vigilant Solutions	vigilantsolutions-011	011	2021-08-07
1378	VN	VinAI Research VietNam	vinai-000	000	2020-09-24
1379	VN	VinBigData	vinbigdata-001	001	2022-01-06
1380	VN	VinBigData	vinbigdata-002	002	2022-06-07
1381	VN	VinBigData	vinbigdata-003	003	2023-06-23
1382	VN	VinBigData	vinbigdata-004	004	2024-05-08
1383	VN	VinBigData	vinbigdata-005	005	2024-10-16
1384	VN	VinBigData	vinbigdata-006	006	2025-02-19
1385	VN	VinBigData	vinbigdata-007	007	2025-07-24
1386	SE	Visage Technologies	visage-000	000	2020-12-09
1387	BR	Visica AI Solutions	visica-001	001	2025-02-14
1388	BR	Visica AI Solutions	visica-002	002	2025-07-15
1389	BR	Visica AI Solutions	visica-003	003	2026-05-06
1390	FI	Visidon	vd-000	000	2018-06-13
1391	FI	Visidon	vd-001	001	2019-02-26
1392	FI	Visidon	vd-002	002	2021-04-12
1393	FI	Visidon	vd-003	003	2021-10-12
1394	CN	Vision Intelligence Center of Meituan	meituan-000	000	2021-05-14
1395	CN	Vision Intelligence Center of Meituan	meituan-001	001	2022-03-25
1396	CN	Vision Intelligence Center of Meituan	meituan-002	002	2022-09-14
1397	CN	Vision Intelligence Center of Meituan	meituan-003	003	2023-03-06
1398	CN	Vision Intelligence Center of Meituan	meituan-004	004	2023-08-16
1399	PT	Vision-Box	visionbox-000	000	2019-02-26
1400	PT	Vision-Box	visionbox-001	001	2019-03-01

Table 20: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1401	PT	Vision-Box	visionbox-002	002	2021-04-29
1402	PT	Vision-Box	visionbox-003	003	2023-02-01
1403	PT	Vision-Box	visionbox-004	004	2023-06-30
1404	PT	Vision-Box	visionbox-005	005	2024-07-29
1405	PT	Vision-Box	visionbox-006	006	2025-07-29
1406	PT	Vision-Box	visionbox-007	007	2026-02-03
1407	PT	Vision-Box	visionbox-008	008	2026-07-27
1408	RU	VisionLabs	visionlabs-001	001	2017-06-12
1409	RU	VisionLabs	visionlabs-002	002	2017-09-08
1410	RU	VisionLabs	visionlabs-003	003	2018-02-02
1411	RU	VisionLabs	visionlabs-004	004	2018-06-15
1412	RU	VisionLabs	visionlabs-005	005	2018-10-19
1413	RU	VisionLabs	visionlabs-006	006	2019-03-01
1414	RU	VisionLabs	visionlabs-007	007	2019-06-12
1415	RU	VisionLabs	visionlabs-008	008	2020-01-06
1416	RU	VisionLabs	visionlabs-009	009	2020-07-27
1417	RU	VisionLabs	visionlabs-010	010	2021-01-25
1418	RU	VisionLabs	visionlabs-011	011	2021-10-13
1419	TW	Vivotek	vivotek-001	001	2024-04-03
1420	TW	Vivotek	vivotek-002	002	2024-08-15
1421	TW	Vivotek	vivotek-003	003	2026-10-22
1422	AU	Vixvizon	vixvizion-000	000	2019-06-19
1423	AU	Vixvizon	vixvizion-001	001	2019-10-22
1424	AU	Vixvizon	vixvizion-002	002	2020-12-31
1425	AU	Vixvizon	vixvizion-003	003	2021-05-18
1426	AU	Vixvizon	vixvizion-004	004	2021-09-20
1427	AU	Vixvizon	vixvizion-005	005	2022-03-03
1428	AU	Vixvizon	vixvizion-006	006	2022-08-11
1429	AU	Vixvizon	vixvizion-007	007	2023-01-17
1430	US	Vocalize	vocalize-001	001	2024-03-27
1431	RU	Vocord	vocord-001	001	2017-04-21
1432	RU	Vocord	vocord-002	002	2017-06-07
1433	RU	Vocord	vocord-003	003	2018-01-23
1434	RU	Vocord	vocord-004	004	2018-06-15
1435	RU	Vocord	vocord-005	005	2018-10-19
1436	RU	Vocord	vocord-006	006	2019-03-01
1437	RU	Vocord	vocord-007	007	2019-06-06
1438	RU	Vocord	vocord-008	008	2020-01-31
1439	RU	Vocord	vocord-009	009	2020-12-28
1440	RU	Vocord	vocord-010	010	2021-12-20
1441	BR	Vsoft	vsoft-001	001	2025-08-01
1442	BR	Vsoft	vsoft-002	002	2025-11-28
1443	BR	Vsoft	vsoft-003	003	2026-03-30
1444	BR	Vsoft	vsoft-004	004	2026-07-31
1445	US	Wicket	wicket-000	000	2022-02-14
1446	CN	Winsense	winsense-000	000	2019-06-17
1447	CN	Winsense	winsense-001	001	2019-10-16
1448	CN	Winsense	winsense-002	002	2020-11-20
1449	MY	Wise AI SDN BHD	wiseai-001	001	2022-10-25
1450	CN	Wuhan Tianyu Information Industry	wuhantianyu-001	001	2021-08-05
1451	CN	X-Laboratory	x-laboratory-000	000	2019-09-03
1452	CN	X-Laboratory	x-laboratory-001	001	2020-01-21
1453	AU	Xailient	xailient-001	001	2026-10-22
1454	CN	Xforward AI Technology	xforwardai-000	000	2020-02-06
1455	CN	Xforward AI Technology	xforwardai-001	001	2020-09-25
1456	CN	Xforward AI Technology	xforwardai-002	002	2021-02-10
1457	CN	Xiamen Meiya Pico Information	meiya-001	001	2019-03-01
1458	CN	Xiamen University	xm-000	000	2020-10-19
1459	PT	YooniK	yoonik-000	000	2020-06-24
1460	PT	YooniK	yoonik-001	001	2020-10-26
1461	PT	YooniK	yoonik-002	002	2021-09-06
1462	PT	YooniK	yoonik-003	003	2022-01-06
1463	PT	YooniK	yoonik-004	004	2023-02-10
1464	GB	Yoti	yoti-001	001	2024-02-13
1465	GB	Yoti	yoti-002	002	2024-07-03
1466	TW	Yuan High-Tech Development	yuan-000	000	2020-06-30
1467	TW	Yuan High-Tech Development	yuan-001	001	2021-01-08
1468	TW	Yuan High-Tech Development	yuan-002	002	2021-05-17
1469	TW	Yuan High-Tech Development	yuan-003	003	2021-09-17
1470	TW	Yuan High-Tech Development	yuan-004	004	2022-01-14

Table 21: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Location	Developer Name	Short Name	Seq. Num.	Validation Date
1471	TW	Yuan High-Tech Development	yuan-005	005	2022-06-22
1472	TW	Yuan High-Tech Development	yuan-006	006	2022-12-14
1473	TW	Yuan High-Tech Development	yuan-007	007	2023-05-11
1474	TW	Yuan High-Tech Development	yuan-008	008	2023-11-06
1475	TW	Yuan High-Tech Development	yuan-009	009	2024-04-24
1476	TW	Yuan High-Tech Development	yuan-010	010	2024-10-22
1477	CN	YuanLi Technology	yuanli-001	001	2026-03-23
1478	CN	Yuntu Data and Technology	ytu-000	000	2021-06-16
1479	CN	Zhuhai Yisheng Electronics Technology	yisheng-000	000	2017-08-17
1480	CN	Zhuhai Yisheng Electronics Technology	yisheng-001	001	2017-08-17
1481	CN	Zhuhai Yisheng Electronics Technology	yisheng-002	002	2018-02-01
1482	CN	Zhuhai Yisheng Electronics Technology	yisheng-004	004	2018-06-12
1483	IN	i2v Systems	i2v-001	001	2024-03-11
1484	IN	i2v Systems	i2v-002	002	2025-01-15
1485	VN	iCOMM Media and Tech	icomm-000	000	2023-12-26
1486	CN	iQIYI Inc	iqface-000	000	2019-06-04
1487	CN	iQIYI Inc	iqface-001	001	2019-12-11
1488	CN	iQIYI Inc	iqface-003	003	2021-02-23
1489	TW	iSAP Solution Corporation	isap-001	001	2019-08-07
1490	TW	iSAP Solution Corporation	isap-002	002	2020-09-01
1491	TW	ioNetworks Inc	ionetworks-000	000	2021-07-20
1492	TW	ioNetworks Inc	ionetworks-001	001	2023-08-01
1493	TW	ioNetworks Inc	ionetworks-002	002	2023-12-08
1494	TW	ioNetworks Inc	ionetworks-003	003	2024-05-09
1495	TW	ioNetworks Inc	ionetworks-004	004	2026-04-22
1496	KR	useB	useb-000	000	2023-05-08
1497	KR	useB	useb-001	001	2023-09-12
1498	KR	useB	useb-002	002	2024-01-19
1499	KR	useB	useb-003	003	2024-08-09
1500	KR	useB	useb-004	004	2024-12-23

Table 22: Summary of participant information included in this report.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1	2060io-001	567279	43224	252 629	1486 10488 ± 0	1054 958 ± 22	778 983 ± 38	786 1045 ± 205	1109 1724 ± 310	1136 3474 ± 1170	1159 7105 ± 42	1155 7071 ± 47
2	2060io-002	579607	42980	669 1559	1487 10488 ± 0	358 398 ± 0	246 418 ± 1	308 525 ± 1	510 786 ± 1	-	1155 7000 ± 91	1149 6965 ± 85
3	20face-000	117155	324083	620 1417	551 2048 ± 0	167 232 ± 1	99 223 ± 1	79 226 ± 4	64 222 ± 1	46 224 ± 1	1410 44880 ± 134	1409 44462 ± 163
4	20face-001	226824	324119	649 1512	1273 4096 ± 0	211 279 ± 2	120 266 ± 1	93 266 ± 1	82 267 ± 1	63 267 ± 0	1103 5553 ± 54	1101 5541 ± 65
5	20face-002	262323	67761	211 559	597 2048 ± 0	255 314 ± 0	149 298 ± 0	116 304 ± 2	101 308 ± 2	83 336 ± 3	322 755 ± 4	338 775 ± 8
6	3divi-000	169360	139460	-	142 512 ± 0	219 285 ± 52	-	-	-	-	114 378 ± 20	116 375 ± 19
7	3divi-001	190867	139473	-	1239 4096 ± 0	203 274 ± 47	-	-	-	-	247 636 ± 19	259 634 ± 16
8	3divi-002	190867	175117	-	1330 4096 ± 0	209 279 ± 48	-	-	-	-	282 692 ± 22	307 707 ± 26
9	3divi-003	191636	55903	-	1289 4096 ± 0	681 650 ± 90	-	-	-	-	236 627 ± 11	245 623 ± 32
10	3divi-004	263670	50794	-	660 2048 ± 0	1083 984 ± 131	-	-	-	-	341 794 ± 35	354 801 ± 40
11	3divi-005	264098	53870	-	422 2048 ± 0	1093 993 ± 83	897 1136 ± 88	1005 1309 ± 137	953 1348 ± 131	993 1671 ± 166	338 790 ± 20	345 791 ± 23
12	3divi-006	273866	52656	559 1252	656 2048 ± 0	690 654 ± 1	471 651 ± 0	439 660 ± 1	407 678 ± 2	395 759 ± 13	333 775 ± 19	336 770 ± 22
13	3divi-007	483115	24723	572 1288	571 2048 ± 0	634 615 ± 1	427 616 ± 1	392 623 ± 1	370 644 ± 1	369 727 ± 5	295 707 ± 31	312 712 ± 25
14	accurascan-001	14057	24437	417 944	122 512 ± 0	143 204 ± 27	94 215 ± 17	76 220 ± 16	72 237 ± 18	88 339 ± 19	1283 15634 ± 54	1285 15647 ± 77
15	accurascan-002	14057	24437	419 945	95 512 ± 0	101 173 ± 23	72 188 ± 14	62 191 ± 14	56 208 ± 16	74 317 ± 14	1244 12255 ± 69	1246 12231 ± 72
16	accurascan-003	998397	27018	781 1996	669 2048 ± 0	723 676 ± 1	501 677 ± 7	486 700 ± 8	477 751 ± 16	536 950 ± 47	171 540 ± 10	174 536 ± 10
17	acer-000	109735	88323	-	830 2048 ± 0	160 222 ± 0	105 233 ± 4	84 238 ± 4	80 262 ± 18	99 356 ± 46	441 1065 ± 40	468 1109 ± 35
18	acer-001	36650	66086	140 417	99 512 ± 0	140 199 ± 0	106 237 ± 28	81 229 ± 26	74 242 ± 37	59 259 ± 21	821 2453 ± 44	826 2461 ± 62
19	acer-002	43922	624858	73 192	953 2048 ± 0	117 184 ± 0	69 184 ± 0	54 185 ± 0	45 185 ± 0	32 186 ± 0	951 3370 ± 47	956 3350 ± 54
20	acisw-003	282029	35664	-	1494 18467 ± 8	166 232 ± 1	122 267 ± 22	273 488 ± 28	660 990 ± 24	1123 2977 ± 129	1496 847908 ± 16757	1496 851850 ± 17018
21	acisw-006	282029	36107	-	1493 18465 ± 8	158 219 ± 0	103 227 ± 0	198 410 ± 1	548 838 ± 1	1099 2532 ± 10	1492 548137 ± 16513	1492 549586 ± 9238
22	acisw-007	267619	36111	91 287	764 2048 ± 0	218 283 ± 0	141 293 ± 3	202 414 ± 0	173 404 ± 0	178 484 ± 1	529 1316 ± 22	533 1297 ± 23
23	acisw-008	171703	39359	497 1102	537 2048 ± 0	362 400 ± 1	195 362 ± 28	165 369 ± 9	100 300 ± 2	84 336 ± 5	531 1327 ± 19	542 1337 ± 32
24	adera-001	0	79272	-	1190 2560 ± 0	45 97 ± 0	-	-	-	-	622 1604 ± 71	635 1649 ± 56
25	adera-002	0	749797	-	1433 5120 ± 0	1382 1394 ± 11	1072 1381 ± 1	1063 1393 ± 1	1002 1403 ± 1	907 1464 ± 2	756 2163 ± 32	754 2158 ± 28
26	adera-003	0	749778	407 917	1432 5120 ± 0	1367 1381 ± 12	1075 1385 ± 1	1066 1394 ± 1	996 1401 ± 1	909 1469 ± 1	753 2148 ± 34	749 2130 ± 32
27	adera-004	0	959123	724 1748	1441 6144 ± 0	1289 1246 ± 1	956 1204 ± 1	939 1230 ± 2	844 1207 ± 2	759 1254 ± 1	684 1840 ± 34	685 1828 ± 31
28	adera-005	0	959113	718 1734	1439 6144 ± 0	1004 914 ± 1	682 879 ± 3	634 880 ± 2	582 886 ± 2	510 910 ± 2	700 1896 ± 14	696 1874 ± 20
29	advance-002	257173	20434	-	396 2048 ± 0	891 811 ± 2	619 803 ± 2	479 696 ± 2	429 699 ± 4	359 718 ± 1	407 987 ± 10	413 988 ± 45
30	advance-003	258867	78699	194 518	743 2048 ± 0	583 586 ± 0	-	355 583 ± 0	316 588 ± 0	256 591 ± 1	683 1813 ± 17	676 1788 ± 26
31	advance-004	803133	954494	281 681	775 2048 ± 0	1168 1099 ± 20	871 1107 ± 15	824 1093 ± 21	746 1103 ± 21	679 1138 ± 21	710 1935 ± 35	714 1936 ± 32
32	advance-005	277607	126218	248 624	773 2048 ± 0	561 573 ± 1	386 574 ± 2	345 573 ± 2	302 574 ± 2	266 601 ± 2	604 1517 ± 16	605 1518 ± 26
33	advance-006	472567	136481	233 605	877 2048 ± 0	715 673 ± 0	488 670 ± 0	456 673 ± 0	414 684 ± 0	387 745 ± 0	352 808 ± 4	361 808 ± 1
34	advance-007	1059702	136627	728 1768	864 2048 ± 0	1270 1224 ± 0	975 1228 ± 0	943 1234 ± 0	875 1236 ± 0	773 1277 ± 0	344 796 ± 2	352 800 ± 1
35	affinsys-ai-000	38016	46159	45 142	101 512 ± 0	15 48 ± 0	12 52 ± 0	26 106 ± 0	23 110 ± 0	37 194 ± 0	723 2007 ± 25	725 2006 ± 28
36	afisbiometrics-000	545886	32882	486 1085	146 512 ± 0	1263 1219 ± 1	896 1135 ± 1	863 1137 ± 2	783 1137 ± 1	686 1147 ± 1	550 1400 ± 29	548 1357 ± 32
37	afrengine-000	151875	382842	63 178	1274 4096 ± 0	54 107 ± 0	35 112 ± 0	104 284 ± 2	428 697 ± 2	1134 3299 ± 17	1431 54329 ± 140	1432 56195 ± 256
38	afrengine-001	410712	391289	444 1015	1327 4096 ± 0	1069 970 ± 0	777 982 ± 1	720 979 ± 2	665 996 ± 0	562 990 ± 4	1430 54240 ± 169	1431 56125 ± 179
39	afrengine-002	579618	239125	447 1018	703 2048 ± 0	938 851 ± 3	838 1065 ± 16	736 993 ± 7	561 855 ± 5	477 857 ± 9	511 1259 ± 13	524 1270 ± 23
40	afrengine-003	579563	239117	445 1017	378 2048 ± 0	1161 1088 ± 1	706 911 ± 15	810 1067 ± 5	542 833 ± 3	561 989 ± 2	525 1294 ± 48	530 1284 ± 29
41	afrengine-004	579587	239117	446 1018	940 2048 ± 0	964 875 ± 2	743 951 ± 119	759 1018 ± 5	590 895 ± 1	600 1043 ± 6	509 1254 ± 17	520 1258 ± 23
42	afrengine-005	694908	239155	448 1018	791 2048 ± 0	916 831 ± 10	765 972 ± 52	597 834 ± 2	586 891 ± 9	626 1065 ± 1	518 1276 ± 19	526 1273 ± 20
43	aifirst-001	224157	808777	171 485	849 2048 ± 0	588 587 ± 2	381 568 ± 2	357 584 ± 3	326 601 ± 6	391 755 ± 5	458 1099 ± 14	462 1087 ± 45
44	aigen-001	256958	595227	517 1154	473 2048 ± 0	1417 1448 ± 9	1127 1451 ± 8	1162 1759 ± 6	1188 2594 ± 4	1165 5691 ± 44	1004 3772 ± 57	1002 3736 ± 56

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 23: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
45	aigen-002	205300	1316138	398 896	906 2048 ± 0	581 586 ± 24	395 582 ± 4	665 920 ± 4	1115 1758 ± 5	1162 5427 ± 17	988 3678 ± 44	990 3646 ± 48
46	aigroup-000	357973	390046	133 398	382 2048 ± 0	298 351 ± 0	172 337 ± 1	151 353 ± 1	132 356 ± 1	116 385 ± 1	1205 9603 ± 216	1207 9689 ± 373
47	aigroup-001	298974	390068	223 587	780 2048 ± 0	512 535 ± 0	346 531 ± 0	313 533 ± 0	280 544 ± 5	259 595 ± 0	1199 9526 ± 170	1202 9562 ± 213
48	aigroup-002	271124	396396	395 888	1159 2078 ± 0	204 275 ± 0	130 276 ± 0	103 283 ± 2	110 318 ± 5	140 419 ± 0	1450 66379 ± 111	1450 66387 ± 124
49	aigroup-003	903149	92606	954 3170	1370 4104 ± 0	794 722 ± 0	549 720 ± 0	504 723 ± 0	473 746 ± 0	446 822 ± 3	964 3476 ± 25	975 3492 ± 14
50	ailabs-001	1054663	338989	558 1252	713 2048 ± 0	704 664 ± 4	597 774 ± 50	872 1145 ± 12	1153 1972 ± 74	1159 5205 ± 272	1479 104034 ± 661	1479 103415 ± 7722
51	aimall-001	128820	57257	-	239 1024 ± 0	271 326 ± 1	-	-	-	-	1032 4127 ± 72	1016 3909 ± 166
52	aimall-002	370156	25210	675 1576	642 2048 ± 0	849 776 ± 4	725 927 ± 27	681 940 ± 21	633 955 ± 34	575 1003 ± 75	1462 72811 ± 7399	1458 71216 ± 6286
53	aimall-003	504324	171935	763 1913	233 1024 ± 0	699 662 ± 1	568 740 ± 51	535 752 ± 62	469 741 ± 46	437 807 ± 47	1389 34565 ± 93	1391 34598 ± 118
54	aipractix-000	419503	113764	647 1504	1382 4112 ± 0	539 554 ± 0	367 553 ± 0	335 556 ± 0	286 548 ± 0	249 581 ± 0	773 2243 ± 37	779 2257 ± 40
55	aiseemu-001	0	1005354	901 2701	1288 4096 ± 0	1096 1001 ± 1	806 1017 ± 0	756 1014 ± 5	692 1022 ± 2	618 1059 ± 4	1072 4864 ± 25	1075 4855 ± 32
56	aiseemu-002	0	1216980	972 3453	1295 4096 ± 0	1324 1298 ± 5	1031 1303 ± 4	1008 1313 ± 2	945 1329 ± 0	829 1348 ± 2	1077 4917 ± 37	1077 4916 ± 37
57	aiunionface-000	241642	840295	135 402	567 2048 ± 0	670 637 ± 13	583 754 ± 41	765 1025 ± 28	820 1179 ± 29	985 1639 ± 47	445 1072 ± 19	457 1080 ± 47
58	aize-001	268456	168970	660 1546	353 2048 ± 0	410 437 ± 10	268 440 ± 8	321 542 ± 17	481 756 ± 27	968 1583 ± 53	712 1937 ± 22	705 1919 ± 23
59	aize-002	257106	182517	222 586	970 2048 ± 0	441 467 ± 1	300 479 ± 1	539 756 ± 1	1056 1477 ± 1	1152 4617 ± 41	197 597 ± 16	215 598 ± 14
60	aize-003	257130	33045	206 546	891 2048 ± 0	121 187 ± 0	73 188 ± 0	51 182 ± 0	44 183 ± 0	38 195 ± 0	229 621 ± 37	207 592 ± 1
61	aize-006	276515	27909	179 497	1144 2060 ± 0	480 501 ± 0	325 504 ± 0	291 505 ± 0	251 508 ± 0	272 612 ± 0	973 3515 ± 35	976 3496 ± 40
62	ajou-001	363257	31734	151 442	566 2048 ± 0	507 530 ± 0	351 536 ± 0	316 535 ± 0	287 549 ± 0	248 577 ± 0	198 597 ± 19	212 596 ± 13
63	akiyama-000	119453	41994	32 116	178 520 ± 0	55 109 ± 1	71 187 ± 1	60 189 ± 1	49 195 ± 1	50 236 ± 2	1150 6901 ± 268	1147 6936 ± 103
64	akiyama-001	119453	41962	33 118	176 520 ± 0	62 120 ± 1	119 265 ± 13	257 468 ± 22	861 1223 ± 27	1102 2660 ± 87	1133 6218 ± 47	1128 6216 ± 45
65	akiyama-002	271192	189635	384 875	532 2048 ± 0	1172 1104 ± 1	899 1138 ± 1	867 1140 ± 1	787 1142 ± 0	740 1240 ± 1	356 817 ± 5	365 824 ± 16
66	akiyama-003	0	784249	547 1211	1387 4124 ± 0	559 570 ± 0	382 568 ± 0	341 569 ± 0	303 574 ± 0	331 685 ± 0	1022 3981 ± 133	1021 3933 ± 120
67	akor-000	250711	95296	646 1497	300 1280 ± 0	810 738 ± 0	564 739 ± 0	523 739 ± 0	474 747 ± 0	430 795 ± 2	59 246 ± 5	80 283 ± 10
68	alchera-000	258450	18848	-	698 2048 ± 0	589 587 ± 13	533 706 ± 48	1058 1389 ± 58	1191 2722 ± 123	1182 9011 ± 455	932 3189 ± 32	920 3031 ± 142
69	alchera-001	174013	18848	-	432 2048 ± 0	653 627 ± 11	-	-	-	-	948 3342 ± 81	947 3243 ± 47
70	alchera-002	405409	22275	-	406 2048 ± 0	1066 968 ± 1	768 976 ± 2	719 979 ± 1	657 988 ± 1	590 1025 ± 2	968 3488 ± 63	966 3430 ± 63
71	alchera-003	487718	24613	610 1376	763 2048 ± 0	942 854 ± 3	663 862 ± 2	625 870 ± 1	579 882 ± 2	517 918 ± 1	958 3426 ± 57	959 3383 ± 53
72	alchera-004	1001019	388616	565 1271	744 2048 ± 0	1074 975 ± 0	747 955 ± 0	705 960 ± 0	658 989 ± 0	688 1152 ± 1	976 3529 ± 54	983 3530 ± 63
73	alchera-005	1001019	388616	566 1272	745 2048 ± 0	1067 969 ± 1	781 987 ± 3	726 985 ± 3	668 998 ± 0	697 1162 ± 2	965 3481 ± 59	964 3422 ± 57
74	alchera-006	1000967	493456	624 1426	862 2048 ± 0	498 522 ± 2	354 540 ± 3	326 544 ± 2	299 569 ± 3	379 739 ± 3	771 2233 ± 38	777 2253 ± 77
75	alchera-007	1000948	493456	630 1430	888 2048 ± 0	504 530 ± 2	370 556 ± 2	339 566 ± 2	307 581 ± 2	397 762 ± 4	761 2178 ± 35	757 2169 ± 44
76	alchera-008	1002024	493460	631 1430	863 2048 ± 0	590 589 ± 3	377 563 ± 4	385 614 ± 3	338 614 ± 1	439 810 ± 3	1085 5129 ± 35	1230 11226 ± 120
77	alchera-009	2086479	454616	915 2763	756 2048 ± 0	1439 1486 ± 4	1147 1507 ± 4	1095 1438 ± 9	1070 1508 ± 11	1041 2008 ± 4	783 2284 ± 44	783 2278 ± 69
78	alfabeta-001	128232	21780	18 74	123 512 ± 0	199 271 ± 0	131 276 ± 0	246 459 ± 2	581 886 ± 2	1100 2547 ± 9	141 470 ± 25	145 458 ± 20
79	alice-000	1741293	19355	962 3329	1279 4096 ± 0	1046 950 ± 2	731 933 ± 1	692 949 ± 1	688 1011 ± 3	764 1264 ± 8	1273 14975 ± 201	1275 14890 ± 229
80	alice-001	1428001	17074	287 690	809 2048 ± 0	1156 1080 ± 1	842 1075 ± 0	828 1095 ± 1	788 1143 ± 1	889 1435 ± 1	1325 21885 ± 278	1330 21921 ± 265
81	alleyes-000	507636	997090	378 857	907 2048 ± 0	857 784 ± 1	761 970 ± 61	715 974 ± 62	623 943 ± 69	613 1057 ± 23	526 1298 ± 34	535 1303 ± 51
82	allgovision-000	172509	155862	214 561	705 2048 ± 0	342 384 ± 8	226 395 ± 17	200 413 ± 14	223 471 ± 14	352 710 ± 21	1369 29903 ± 406	1370 29735 ± 194
83	alphaface-001	259849	81636	197 527	414 2048 ± 0	629 612 ± 1	420 613 ± 3	380 612 ± 1	344 619 ± 1	298 640 ± 2	416 1008 ± 10	422 1002 ± 19
84	alphaface-002	768995	70692	632 1434	555 2048 ± 0	654 628 ± 2	571 746 ± 19	532 751 ± 18	501 779 ± 22	449 828 ± 40	395 945 ± 25	400 935 ± 17
85	alttekglobal-000	649718	3060239	276 673	639 2048 ± 0	1148 1066 ± 2	917 1158 ± 2	903 1184 ± 1	810 1168 ± 1	720 1204 ± 2	540 1370 ± 23	553 1377 ± 26
86	amplifiedgroup-001	0	47053	20 81	212 866 ± 2	40 93 ± 0	-	-	-	-	1439 57803 ± 4210	1433 56365 ± 1196
87	androvideo-000	174847	585063	137 405	459 2048 ± 0	207 277 ± 0	137 285 ± 0	122 314 ± 0	144 372 ± 1	281 620 ± 0	889 2860 ± 28	888 2847 ± 22
88	anke-000	323402	39031	-	1156 2072 ± 0	551 566 ± 12	-	-	-	-	182 567 ± 20	177 545 ± 29

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 24: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
89	anke-001	642061	37364	-	1155 2072 ± 0	526 541 ± 4	-	-	-	-	172 540 ± 16	172 530 ± 20
90	anke-002	798686	28364	-	1157 2072 ± 0	385 419 ± 30	-	-	-	-	157 501 ± 17	158 492 ± 17
91	anke-003	340160	198638	-	1109 2056 ± 0	889 811 ± 23	-	-	-	-	124 425 ± 28	135 437 ± 32
92	anke-004	349388	410776	290 706	1112 2056 ± 0	649 625 ± 1	445 627 ± 2	419 635 ± 3	380 653 ± 2	556 982 ± 8	241 633 ± 22	255 632 ± 34
93	anke-005	328553	429160	509 1134	1114 2056 ± 0	592 590 ± 2	406 594 ± 5	371 601 ± 3	365 638 ± 4	445 821 ± 24	277 685 ± 19	293 687 ± 26
94	anyvision-001	662659	518032	-	268 1024 ± 0	177 244 ± 0	-	-	-	-	1467 74285 ± 212	1466 74145 ± 200
95	anyvision-002	662659	520039	-	242 1024 ± 0	182 248 ± 0	-	-	-	-	1466 74069 ± 188	1465 74019 ± 198
96	anyvision-004	401001	630797	498 1102	267 1024 ± 0	302 355 ± 1	-	-	-	-	698 1891 ± 51	687 1829 ± 85
97	anyvision-005	190979	116595	462 1048	249 1024 ± 0	1084 985 ± 1	792 997 ± 1	750 1004 ± 1	664 995 ± 1	567 995 ± 1	313 733 ± 14	326 733 ± 16
98	aratek-001	254521	15993	805 2135	638 2048 ± 0	186 251 ± 1	129 275 ± 0	91 252 ± 0	87 278 ± 0	67 286 ± 0	666 1776 ± 20	665 1744 ± 38
99	armatura-001	0	374608	978 3518	856 2048 ± 0	745 688 ± 1	513 689 ± 1	477 693 ± 1	440 708 ± 3	392 756 ± 13	68 270 ± 17	76 268 ± 11
100	armatura-002	0	1258644	551 1222	1438 6144 ± 0	1432 1476 ± 3	1132 1458 ± 5	1137 1535 ± 5	1069 1505 ± 12	975 1605 ± 26	623 1605 ± 26	623 1589 ± 25
101	armatura-003	0	836082	674 1572	1447 6144 ± 0	1122 1028 ± 1	820 1032 ± 1	767 1027 ± 0	699 1036 ± 1	598 1041 ± 3	1428 51850 ± 56	1428 51835 ± 48
102	arvis-000	529004	705484	386 878	235 1024 ± 0	148 209 ± 5	95 215 ± 1	140 342 ± 4	382 655 ± 13	1066 2158 ± 55	966 3484 ± 41	969 3461 ± 48
103	asr-000	461347	236001	422 951	1100 2056 ± 0	418 450 ± 0	290 462 ± 1	239 454 ± 0	203 453 ± 1	181 486 ± 0	504 1226 ± 14	518 1251 ± 23
104	asr-001	461347	236016	421 949	1090 2056 ± 0	771 701 ± 0	534 706 ± 0	489 704 ± 0	441 709 ± 0	433 800 ± 0	716 1951 ± 28	722 1982 ± 23
105	asusaics-000	257418	245320	235 605	583 2048 ± 0	463 484 ± 13	328 506 ± 21	610 850 ± 26	1122 1789 ± 61	1168 6305 ± 188	1094 5455 ± 78	1095 5422 ± 112
106	asusaics-001	257418	245330	229 595	1232 4096 ± 0	930 842 ± 17	800 1008 ± 20	1053 1377 ± 28	1185 2423 ± 90	1176 7284 ± 277	1192 8618 ± 42	1192 8638 ± 136
107	autentika-000	266093	3200425	764 1922	965 2048 ± 0	538 553 ± 1	413 605 ± 1	377 609 ± 2	330 608 ± 1	280 618 ± 2	1463 72833 ± 577	1459 71829 ± 541
108	autentika-001	0	3198534	695 1650	609 2048 ± 0	739 683 ± 2	707 912 ± 4	654 908 ± 3	604 919 ± 3	501 893 ± 4	1459 71804 ± 600	1457 70747 ± 582
109	autentika-002	0	3198136	701 1676	368 2048 ± 0	329 374 ± 1	330 511 ± 1	310 528 ± 1	272 534 ± 1	230 546 ± 1	1418 46941 ± 1172	1414 46351 ± 822
110	autentika-003	0	3198136	694 1648	711 2048 ± 0	408 437 ± 7	429 617 ± 14	356 584 ± 3	333 611 ± 8	275 614 ± 2	1420 47827 ± 1020	1416 46728 ± 916
111	authenmetric-002	449944	91489	-	392 2048 ± 0	1031 942 ± 1	742 950 ± 1	706 960 ± 1	636 960 ± 1	565 991 ± 2	647 1712 ± 20	657 1719 ± 19
112	authenmetric-003	293599	39492	381 866	760 2048 ± 0	1091 992 ± 1	798 1006 ± 1	749 1003 ± 2	674 1002 ± 1	596 1036 ± 1	661 1757 ± 19	666 1755 ± 19
113	authenmetric-004	381165	39492	548 1214	409 2048 ± 0	997 910 ± 1	704 909 ± 1	657 915 ± 1	605 921 ± 2	537 950 ± 1	652 1724 ± 14	646 1691 ± 29
114	authme-001	1015124	37791	800 2097	1240 4096 ± 0	1043 949 ± 1	760 969 ± 7	704 960 ± 6	637 961 ± 1	545 966 ± 4	882 2836 ± 34	884 2827 ± 54
115	authme-002	1016682	37819	795 2079	1241 4096 ± 0	1056 959 ± 1	753 959 ± 7	741 995 ± 5	643 966 ± 5	559 985 ± 6	877 2796 ± 14	875 2775 ± 34
116	aware-000	240240	27639	-	326 1572 ± 0	694 655 ± 26	-	-	-	-	1027 4030 ± 83	913 2984 ± 152
117	aware-001	240240	27639	-	328 1572 ± 0	695 656 ± 26	-	-	-	-	898 2902 ± 51	883 2810 ± 111
118	aware-002	271312	36134	-	327 1572 ± 0	574 582 ± 14	-	-	-	-	353 810 ± 32	362 812 ± 28
119	aware-003	377729	27975	-	1217 3108 ± 0	855 783 ± 10	-	-	-	-	546 1392 ± 42	539 1334 ± 80
120	aware-004	427829	28219	-	1167 2084 ± 0	986 900 ± 10	869 1105 ± 10	936 1226 ± 25	1093 1585 ± 72	1121 2937 ± 166	520 1279 ± 50	531 1287 ± 100
121	aware-005	300017	26320	563 1265	329 1572 ± 0	979 886 ± 23	826 1038 ± 21	850 1121 ± 22	948 1337 ± 58	1068 2195 ± 144	582 1475 ± 63	568 1427 ± 115
122	aware-006	298543	14124	416 944	79 352 ± 0	1206 1148 ± 3	908 1146 ± 2	907 1190 ± 2	924 1306 ± 20	1004 1754 ± 84	846 2598 ± 42	845 2559 ± 60
123	aware-007	135312	18711	130 388	93 480 ± 0	783 713 ± 1	544 716 ± 1	513 731 ± 3	508 785 ± 8	571 999 ± 30	434 1058 ± 23	463 1091 ± 37
124	aware-008	135312	18715	129 387	92 480 ± 0	823 751 ± 1	562 737 ± 0	524 739 ± 3	538 825 ± 9	578 1007 ± 27	468 1120 ± 25	489 1152 ± 35
125	awiros-001	15499	87480	23 88	139 512 ± 0	44 97 ± 6	27 98 ± 4	34 138 ± 6	67 225 ± 7	238 556 ± 8	449 1079 ± 44	445 1050 ± 45
126	awiros-002	289016	203723	209 556	836 2048 ± 0	453 479 ± 0	321 500 ± 0	315 534 ± 0	343 618 ± 0	533 946 ± 1	718 1966 ± 31	718 1957 ± 25
127	awiros-003	292302	598824	428 976	921 2048 ± 0	398 428 ± 2	270 442 ± 1	241 454 ± 2	196 442 ± 1	154 444 ± 1	841 2558 ± 20	846 2580 ± 42
128	awiros-004	489913	598874	577 1295	977 2048 ± 0	481 502 ± 0	335 517 ± 0	303 519 ± 3	261 518 ± 1	203 512 ± 0	834 2527 ± 17	840 2537 ± 20
129	aximetria-001	408902	487912	871 2539	715 2048 ± 0	1108 1013 ± 1	810 1023 ± 21	771 1029 ± 5	670 999 ± 2	643 1091 ± 5	1045 4401 ± 94	1049 4490 ± 80
130	ayftech-001	195423	43580	309 731	108 512 ± 0	374 408 ± 23	297 476 ± 52	573 814 ± 108	1126 1827 ± 384	1161 5412 ± 1029	225 615 ± 16	381 885 ± 44
131	ayftech-003	526164	127257	690 1640	821 2048 ± 0	1098 1003 ± 5	807 1017 ± 6	753 1006 ± 6	685 1009 ± 7	622 1063 ± 5	528 1301 ± 20	534 1300 ± 25
132	ayonix-000	58505	5252	15 69	280 1036 ± 0	2 18 ± 2	-	-	-	-	231 621 ± 23	240 620 ± 26

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 25: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
133	beethedata-000	227849	1087592	210 557	564 2048 ± 0	438 465 ± 0	293 467 ± 0	258 468 ± 0	222 467 ± 0	169 467 ± 0	747 2121 ± 34	747 2110 ± 38
134	betaface-000	0	1060599	-	1174 2116 ± 0	923 837 ± 9	-	-	-	-	602 1514 ± 28	607 1520 ± 66
135	betaface-002	0	1060599	-	1173 2116 ± 0	952 870 ± 13	-	-	-	-	581 1474 ± 50	596 1488 ± 41
136	beyneai-000	256958	591433	783 1998	852 2048 ± 0	420 451 ± 8	278 449 ± 1	547 767 ± 7	1096 1603 ± 25	1154 4669 ± 124	996 3730 ± 57	992 3668 ± 54
137	biocube-001	25030	6192987	157 457	1328 4096 ± 0	215 282 ± 22	140 292 ± 24	306 521 ± 57	411 684 ± 59	779 1282 ± 68	1324 21787 ± 96	1328 21812 ± 109
138	biocube-002	69898	10651580	163 468	1318 4096 ± 0	911 826 ± 4	647 838 ± 4	894 1175 ± 4	1194 2933 ± 1074	1183 9144 ± 1090	1336 23074 ± 94	1338 23036 ± 99
139	bioid-001	361549	144	251 629	1468 7680 ± 0	124 190 ± 1	81 192 ± 1	105 290 ± 2	290 556 ± 7	947 1528 ± 25	1082 5051 ± 35	1084 5042 ± 19
140	bioid-002	364015	149	618 1415	1469 7680 ± 0	122 187 ± 2	75 190 ± 3	109 295 ± 4	294 560 ± 13	982 1630 ± 43	1083 5052 ± 34	1083 5031 ± 23
141	bioid-003	530171	156	668 1558	1489 15368 ± 0	573 582 ± 2	401 587 ± 2	493 706 ± 3	663 994 ± 7	1060 2123 ± 33	1138 6396 ± 25	1134 6377 ± 29
142	bioid-004	530301	156	672 1568	1490 15368 ± 0	610 603 ± 1	418 611 ± 1	426 640 ± 1	453 726 ± 2	615 1058 ± 3	1152 6948 ± 25	1148 6948 ± 19
143	bioidtechswiss-000	758466	119955	-	36 256 ± 0	658 630 ± 2	-	-	-	-	1388 34416 ± 137	1390 34403 ± 126
144	bioidtechswiss-001	1178769	120811	636 1455	96 512 ± 0	1064 966 ± 4	1007 1270 ± 270	995 1294 ± 96	1006 1409 ± 157	1012 1793 ± 79	847 2610 ± 25	849 2624 ± 32
145	bioidtechswiss-002	744786	114842	434 990	124 512 ± 0	1008 917 ± 2	727 930 ± 2	695 952 ± 2	625 947 ± 3	616 1058 ± 11	760 2177 ± 29	758 2170 ± 31
146	bioidtechswiss-003	518317	10378	1036 4730	963 2048 ± 0	1154 1075 ± 1	902 1139 ± 7	1202 2207 ± 9	1141 1891 ± 11	1157 4787 ± 21	1423 50218 ± 204	1424 50184 ± 163
147	biometric-vision-000	445487	72132	405 915	816 2048 ± 0	922 836 ± 3	640 831 ± 3	623 869 ± 5	632 955 ± 8	841 1370 ± 19	1265 14437 ± 161	1266 14325 ± 167
148	biometrik-001	23995	68818	28 101	140 512 ± 0	1414 1446 ± 3	1124 1448 ± 3	1106 1449 ± 3	1041 1456 ± 3	923 1485 ± 3	417 1016 ± 5	417 999 ± 17
149	biometrik-002	171539	84430	117 353	786 2048 ± 0	1444 1499 ± 3	1144 1499 ± 3	1129 1501 ± 3	1067 1503 ± 3	932 1507 ± 2	532 1328 ± 19	538 1321 ± 19
150	bm-001	287734	38076	50 148	1 64 ± 0	415 444 ± 88	-	-	-	-	697 1887 ± 31	698 1877 ± 26
151	boetech-001	261376	88710	987 3658	575 2048 ± 0	200 271 ± 1	124 268 ± 1	95 273 ± 0	90 286 ± 1	75 318 ± 1	1453 68519 ± 1921	1452 67648 ± 822
152	boetech-002	294347	88710	790 2057	534 2048 ± 0	246 305 ± 4	146 296 ± 1	112 302 ± 1	107 313 ± 1	96 348 ± 2	1455 68921 ± 2137	1456 69473 ± 2104
153	bresee-000	287880	22872	-	354 2048 ± 0	1332 1309 ± 1	1211 2126 ± 1	1020 1330 ± 0	934 1312 ± 0	823 1333 ± 0	1411 45317 ± 228	1421 48256 ± 425
154	bresee-001	287880	23227	552 1223	437 2048 ± 0	1269 1223 ± 3	962 1216 ± 1	1022 1331 ± 1	867 1227 ± 1	838 1360 ± 1	1397 37240 ± 655	1400 37167 ± 584
155	bresee-002	313627	30902	771 1951	439 2048 ± 0	815 743 ± 4	904 1143 ± 2	873 1146 ± 2	793 1148 ± 2	707 1176 ± 2	668 1778 ± 22	672 1775 ± 23
156	btsdigital-001	362015	234409	423 955	955 2048 ± 0	800 728 ± 2	543 713 ± 1	509 726 ± 4	449 716 ± 2	378 738 ± 0	925 3123 ± 22	930 3134 ± 37
157	bynn-000	271083	37637	134 400	982 2049 ± 0	1477 1834 ± 0	1183 1835 ± 0	1174 1840 ± 0	1128 1838 ± 0	1022 1875 ± 0	380 899 ± 7	393 915 ± 1
158	cakevn-001	271412	587471	312 739	889 2048 ± 0	586 587 ± 0	397 584 ± 0	358 585 ± 0	319 589 ± 0	314 664 ± 0	1470 78258 ± 72	1468 78329 ± 90
159	cakevn-002	1015434	73945	553 1228	885 2048 ± 0	989 905 ± 0	701 905 ± 0	652 907 ± 0	597 910 ± 0	512 916 ± 0	923 3115 ± 28	926 3086 ± 11
160	camvi-001	118759	21678	-	263 1024 ± 0	113 181 ± 6	-	-	-	-	145 481 ± 16	156 487 ± 23
161	camvi-002	236278	225285	311 737	223 1024 ± 0	726 677 ± 7	552 726 ± 36	624 869 ± 28	773 1129 ± 43	1111 2785 ± 113	221 612 ± 26	222 603 ± 20
162	camvi-003	285657	225320	-	237 1024 ± 0	821 750 ± 3	-	-	-	-	184 571 ± 23	188 565 ± 26
163	camvi-004	280733	615819	408 919	644 2048 ± 0	834 759 ± 10	662 861 ± 17	728 986 ± 34	910 1279 ± 51	1118 2891 ± 158	396 948 ± 40	406 963 ± 31
164	candour-001	150086	97632	169 477	649 2048 ± 0	1386 1400 ± 1	1088 1396 ± 1	1078 1408 ± 1	1005 1407 ± 1	908 1464 ± 1	1221 10725 ± 133	1222 10712 ± 127
165	candour-002	149560	65228	132 392	778 2048 ± 0	80 144 ± 0	44 143 ± 0	36 145 ± 0	35 152 ± 1	41 206 ± 0	1176 7798 ± 56	1176 7781 ± 51
166	candour-003	339247	65570	359 833	1206 3072 ± 0	337 380 ± 0	224 392 ± 0	173 378 ± 0	163 393 ± 1	157 445 ± 0	1387 34039 ± 45	1389 34020 ± 44
167	candour-004	1192219	65226	866 2490	1231 4096 ± 0	1500 2243 ± 0	1213 2247 ± 0	1203 2223 ± 0	1178 2270 ± 1	1082 2330 ± 0	863 2714 ± 20	872 2755 ± 23
168	canon-002	446491	130232	-	1269 4096 ± 0	1330 1308 ± 2	1041 1315 ± 1	1017 1326 ± 2	951 1345 ± 1	902 1452 ± 1	1132 6211 ± 25	1126 6194 ± 25
169	canon-003	2550850	101378	1055 5487	1449 6180 ± 0	1300 1263 ± 3	1002 1263 ± 1	986 1283 ± 1	938 1320 ± 1	920 1482 ± 2	1066 4783 ± 17	1067 4780 ± 19
170	canon-004	2399160	114188	1078 7037	1451 6200 ± 0	1042 948 ± 4	746 955 ± 3	703 959 ± 3	647 977 ± 3	625 1064 ± 2	1163 7172 ± 63	1160 7169 ± 51
171	canon-005	2521818	80309	1073 6700	1399 4148 ± 0	1268 1223 ± 21	968 1221 ± 2	937 1230 ± 4	871 1234 ± 5	821 1329 ± 3	1128 6142 ± 99	1129 6226 ± 146
172	cchonolulu-000	727461	37284	139 416	120 512 ± 0	67 123 ± 13	41 126 ± 10	33 129 ± 10	30 134 ± 10	45 223 ± 10	653 1726 ± 35	647 1697 ± 35
173	cchonolulu-001	733744	37284	143 426	654 2048 ± 0	94 162 ± 21	82 197 ± 23	297 513 ± 96	758 1118 ± 267	1064 2142 ± 584	1046 4418 ± 37	1040 4361 ± 35
174	cdachyd-001	238593	1941345	40 135	132 512 ± 0	1458 1642 ± 2	1173 1749 ± 1	1215 6467 ± 3	1213 17252 ± 10	1204 58681 ± 45	1233 11594 ± 85	1238 11639 ± 130
175	ceiaufgbr-000	254585	44874	156 453	962 2048 ± 0	184 250 ± 0	114 251 ± 0	90 252 ± 0	77 251 ± 0	58 253 ± 0	349 803 ± 4	358 804 ± 4
176	ceiec-001	159618	46136	-	269 1024 ± 0	254 314 ± 3	-	-	-	-	1334 22831 ± 108	1337 22813 ± 120

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 26: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
177	ceiec-002	269063	90975	-	408 2048 ± 0	626 612 ± 17	-	-	-	-	762 2188 ± 57	790 2301 ± 56
178	ceiec-003	260371	88707	147 430	633 2048 ± 0	899 817 ± 4	687 883 ± 57	644 897 ± 60	592 899 ± 72	531 944 ± 72	777 2256 ± 38	772 2241 ± 54
179	ceiec-004	263476	67011	138 408	584 2048 ± 0	1115 1024 ± 1	813 1027 ± 1	768 1027 ± 1	697 1030 ± 1	609 1055 ± 1	686 1844 ± 26	688 1836 ± 20
180	celloscope-001	38016	59158	42 138	100 512 ± 0	16 49 ± 0	11 52 ± 0	56 188 ± 0	254 509 ± 1	-	869 2733 ± 35	868 2730 ± 35
181	chosun-000	167093	694	-	752 2048 ± 0	414 441 ± 1	-	-	-	-	405 983 ± 20	411 983 ± 29
182	chosun-001	765615	707	176 491	696 2048 ± 0	854 783 ± 2	632 826 ± 4	1155 1662 ± 13	1202 3679 ± 67	1197 11694 ± 243	412 998 ± 25	434 1035 ± 11
183	chosun-002	234001	31875	154 451	694 2048 ± 0	181 248 ± 3	126 273 ± 3	1127 1495 ± 14	1208 7920 ± 90	1205 80302 ± 1349	232 623 ± 17	258 634 ± 13
184	chtface-001	94088	369529	-	812 2048 ± 0	154 212 ± 10	-	-	-	-	776 2256 ± 26	774 2252 ± 91
185	chtface-002	363153	369529	-	680 2048 ± 0	578 584 ± 14	541 712 ± 41	780 1038 ± 42	1132 1861 ± 75	1153 4661 ± 232	780 2264 ± 26	771 2234 ± 103
186	chtface-003	363153	369529	-	463 2048 ± 0	601 594 ± 16	548 720 ± 33	789 1050 ± 41	1139 1884 ± 90	1163 5606 ± 334	746 2110 ± 37	767 2219 ± 65
187	chtface-004	409656	311027	626 1428	446 2048 ± 0	277 332 ± 0	161 323 ± 1	132 329 ± 1	121 335 ± 1	112 377 ± 1	654 1727 ± 17	658 1720 ± 16
188	chtface-005	408364	311100	617 1415	447 2048 ± 0	266 322 ± 0	155 316 ± 1	127 325 ± 2	116 324 ± 1	136 411 ± 2	703 1907 ± 19	701 1898 ± 23
189	chtface-006	733645	610439	849 2424	485 2048 ± 0	501 522 ± 1	332 514 ± 1	317 536 ± 2	295 561 ± 1	340 693 ± 2	729 2034 ± 41	736 2049 ± 29
190	chtface-007	733528	610443	843 2393	509 2048 ± 0	225 292 ± 0	139 290 ± 0	117 307 ± 1	119 328 ± 1	190 501 ± 1	615 1570 ± 53	619 1561 ± 34
191	chtface-008	810179	583222	1089 8766	488 2048 ± 0	1134 1043 ± 8	808 1019 ± 9	763 1022 ± 14	703 1039 ± 16	627 1066 ± 23	600 1513 ± 14	614 1535 ± 44
192	chtface-009	810179	342213	1093 9371	449 2048 ± 0	1422 1455 ± 0	1134 1468 ± 16	1130 1501 ± 26	1065 1501 ± 0	955 1544 ± 15	766 2213 ± 13	769 2226 ± 29
193	cib-000	340288	69515	-	424 2048 ± 0	1092 993 ± 40	-	-	-	-	1344 24340 ± 60	1355 25972 ± 97
194	cib-001	436253	133766	-	968 2048 ± 0	683 651 ± 2	537 707 ± 13	499 716 ± 15	457 728 ± 3	442 820 ± 5	1005 3783 ± 38	1005 3765 ± 37
195	cist-001	0	300551	220 582	855 2048 ± 0	1071 972 ± 0	772 977 ± 1	723 981 ± 0	653 983 ± 0	621 1061 ± 0	905 2947 ± 20	907 2940 ± 22
196	cist-002	0	1186073	715 1730	1223 3200 ± 0	1077 976 ± 0	787 992 ± 0	817 1085 ± 1	909 1279 ± 0	1055 2101 ± 1	1130 6165 ± 22	1127 6207 ± 38
197	cist-003	0	1267350	902 2706	1437 5248 ± 0	1320 1291 ± 1	1030 1300 ± 1	1037 1352 ± 3	917 1287 ± 2	805 1313 ± 1	1177 7838 ± 63	1177 7823 ± 40
198	cist-004	0	1603990	877 2563	1436 5248 ± 0	1287 1243 ± 1	1008 1271 ± 1	969 1264 ± 3	885 1252 ± 1	781 1287 ± 1	937 3216 ± 25	943 3220 ± 32
199	clearviewai-000	342491	211852	909 2745	920 2048 ± 0	1389 1402 ± 1	1090 1403 ± 1	1085 1412 ± 1	1014 1420 ± 1	875 1418 ± 1	620 1592 ± 37	618 1561 ± 37
200	clearviewai-001	613532	511807	599 1353	748 2048 ± 0	669 635 ± 2	463 642 ± 1	438 659 ± 1	391 664 ± 1	324 674 ± 1	360 827 ± 10	368 829 ± 20
201	clearviewai-002	713008	509504	665 1556	897 2048 ± 0	828 755 ± 2	596 773 ± 2	543 765 ± 3	506 783 ± 2	424 790 ± 1	369 864 ± 34	370 842 ± 29
202	clearviewai-003	712800	540093	591 1344	375 2048 ± 0	1176 1109 ± 0	855 1091 ± 0	820 1088 ± 0	739 1090 ± 0	670 1126 ± 0	878 2812 ± 45	885 2831 ± 50
203	closeli-001	420342	9851	327 773	1315 4096 ± 0	927 839 ± 1	650 843 ± 1	602 841 ± 1	552 845 ± 1	482 865 ± 1	1093 5404 ± 17	1094 5400 ± 25
204	cloudmatrix-000	309939	542141	-	772 2048 ± 0	827 754 ± 10	576 750 ± 2	536 754 ± 4	490 764 ± 1	427 793 ± 2	1421 49192 ± 206	1422 49275 ± 176
205	cloudmatrix-001	10390	542121	12 65	939 2048 ± 0	58 114 ± 1	36 117 ± 0	30 118 ± 0	27 123 ± 1	27 169 ± 1	1424 50263 ± 212	1425 50243 ± 237
206	cloudmatrix-002	256635	693318	459 1036	708 2048 ± 0	356 395 ± 1	231 398 ± 1	188 399 ± 1	170 402 ± 1	150 437 ± 20	1422 49578 ± 120	1423 49602 ± 180
207	cloudwalk-hr-000	284600	105489	-	1130 2057 ± 0	605 600 ± 2	-	-	-	-	1292 16747 ± 238	1294 16627 ± 220
208	cloudwalk-hr-001	346156	104483	-	1128 2057 ± 0	707 665 ± 3	-	-	-	-	1299 17816 ± 260	1304 17878 ± 464
209	cloudwalk-hr-002	390059	327494	-	1133 2057 ± 0	994 908 ± 3	-	-	-	-	1332 22530 ± 416	1326 21651 ± 533
210	cloudwalk-hr-003	383739	144263	431 983	1126 2057 ± 0	617 606 ± 0	402 588 ± 0	365 594 ± 0	335 612 ± 1	231 547 ± 2	1154 6982 ± 80	1151 6972 ± 84
211	cloudwalk-hr-004	502916	520169	614 1399	988 2049 ± 0	960 873 ± 1	680 877 ± 1	631 876 ± 1	574 879 ± 1	508 902 ± 3	1236 11652 ± 127	1236 11608 ± 123
212	cloudwalk-mt-000	372169	282036	-	995 2049 ± 0	795 724 ± 12	-	-	-	-	639 1678 ± 47	627 1614 ± 85
213	cloudwalk-mt-002	290753	145340	-	983 2049 ± 0	562 573 ± 1	545 717 ± 78	485 700 ± 66	475 749 ± 96	411 770 ± 80	1164 7205 ± 204	1161 7211 ± 244
214	cloudwalk-mt-003	490365	494959	-	990 2049 ± 0	1015 923 ± 1	711 918 ± 1	674 926 ± 1	607 925 ± 1	528 936 ± 1	1235 11620 ± 179	1239 11661 ± 128
215	cloudwalk-mt-004	1384602	512628	-	427 2048 ± 0	1017 923 ± 2	713 919 ± 1	661 918 ± 0	603 919 ± 0	522 927 ± 1	1237 11744 ± 170	1237 11631 ± 126
216	cloudwalk-mt-005	846026	573253	983 3615	716 2048 ± 0	1230 1179 ± 3	951 1200 ± 3	924 1209 ± 3	865 1226 ± 5	735 1229 ± 3	1252 12525 ± 225	1251 12394 ± 152
217	cloudwalk-mt-006	563322	480071	976 3499	976 2048 ± 0	1374 1385 ± 0	1081 1392 ± 1	1068 1398 ± 1	994 1397 ± 4	895 1444 ± 2	950 3364 ± 96	954 3324 ± 83
218	cloudwalk-mt-007	563322	480071	977 3499	547 2048 ± 0	1366 1379 ± 2	1109 1425 ± 7	1091 1427 ± 6	1010 1417 ± 2	894 1443 ± 7	910 3000 ± 68	910 2971 ± 74
219	clova-000	198420	6824	-	922 2048 ± 0	407 437 ± 0	256 431 ± 0	221 435 ± 0	201 452 ± 2	200 508 ± 7	671 1794 ± 16	678 1795 ± 19
220	cmcuni-001	0	604227	527 1177	813 2048 ± 0	1215 1161 ± 1	941 1187 ± 1	911 1194 ± 5	836 1200 ± 1	724 1207 ± 1	929 3153 ± 22	934 3158 ± 26

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 27: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
221	codeline-000	361659	138388	542 1194	930 2048 ± 0	1419 1453 ± 0	1131 1456 ± 2	1111 1456 ± 0	1043 1457 ± 0	921 1483 ± 1	757 2171 ± 69	763 2194 ± 84
222	cogent-000	633812	18848	-	86 439 ± 0	489 511 ± 7	-	-	-	-	1200 9541 ± 54	1199 9500 ± 51
223	cogent-001	530508	18869	-	90 455 ± 0	542 557 ± 7	-	-	-	-	1218 10477 ± 278	1220 10392 ± 75
224	cogent-002	696959	18881	-	339 1979 ± 0	1027 941 ± 0	-	-	-	-	1264 14394 ± 134	1268 14385 ± 176
225	cogent-003	698290	40430	-	222 973 ± 0	1052 952 ± 0	-	-	-	-	1250 12496 ± 75	1241 11822 ± 163
226	cogent-004	722919	389164	-	340 1983 ± 0	1086 987 ± 50	-	-	-	-	1282 15536 ± 75	1287 15964 ± 708
227	cogent-005	1876796	75276	-	1189 2523 ± 0	1265 1221 ± 2	980 1236 ± 1	988 1289 ± 2	1015 1420 ± 4	974 1602 ± 5	1351 24854 ± 69	1350 24858 ± 71
228	cogent-006	1078167	58108	666 1557	289 1062 ± 0	843 768 ± 0	607 789 ± 1	594 831 ± 2	611 930 ± 1	549 971 ± 1	678 1802 ± 17	680 1797 ± 23
229	cogent-007	621565	72316	760 1888	197 550 ± 0	1342 1329 ± 2	1055 1333 ± 5	1027 1337 ± 4	957 1353 ± 5	859 1390 ± 4	103 355 ± 8	113 367 ± 14
230	cogent-008	856817	73587	815 2176	198 550 ± 0	1396 1412 ± 1	1107 1419 ± 2	1090 1426 ± 3	1027 1437 ± 3	917 1476 ± 1	127 436 ± 14	136 441 ± 23
231	cognitec-000	474759	27371	-	1014 2052 ± 0	161 224 ± 1	-	-	-	-	1008 3835 ± 108	1007 3782 ± 83
232	cognitec-001	476809	27487	-	1020 2052 ± 0	226 293 ± 17	-	-	-	-	1036 4253 ± 59	1034 4102 ± 167
233	cognitec-002	394088	62354	-	1036 2052 ± 0	129 192 ± 6	96 219 ± 6	83 233 ± 8	73 241 ± 6	73 314 ± 10	943 3250 ± 41	946 3241 ± 48
234	cognitec-003	471458	62502	347 817	1028 2052 ± 0	321 366 ± 9	235 403 ± 9	197 408 ± 9	184 424 ± 9	201 509 ± 13	957 3417 ± 51	967 3433 ± 53
235	cognitec-004	705645	62678	218 580	1049 2052 ± 0	434 463 ± 9	316 497 ± 9	290 504 ± 10	263 521 ± 10	286 631 ± 12	916 3028 ± 197	924 3059 ± 238
236	cognitec-005	326804	198231	350 819	1041 2052 ± 0	251 311 ± 10	198 364 ± 12	166 370 ± 10	156 390 ± 10	188 498 ± 16	794 2332 ± 48	797 2330 ± 40
237	cognitec-006	368383	40008	365 844	928 2048 ± 0	284 335 ± 7	214 381 ± 8	184 390 ± 7	165 396 ± 8	163 455 ± 11	792 2316 ± 60	775 2252 ± 50
238	cognitec-007	368401	40132	366 844	502 2048 ± 0	430 458 ± 20	409 601 ± 17	441 660 ± 17	528 815 ± 23	897 1445 ± 45	733 2051 ± 26	734 2045 ± 34
239	cognitec-008	407825	39616	403 914	648 2048 ± 0	519 538 ± 20	436 622 ± 23	407 631 ± 23	401 672 ± 24	373 735 ± 26	1071 4854 ± 124	1085 5052 ± 287
240	compal-000	1003818	186200	826 2242	232 1024 ± 0	1295 1255 ± 1	1023 1293 ± 2	998 1294 ± 2	921 1300 ± 2	811 1322 ± 4	1385 33116 ± 349	1387 33182 ± 252
241	compal-001	619796	186054	966 3395	985 2049 ± 0	946 861 ± 0	673 871 ± 0	612 851 ± 0	562 857 ± 0	497 887 ± 0	1484 149573 ± 724	1484 149664 ± 582
242	compal-002	620531	185671	950 3134	986 2049 ± 0	967 877 ± 0	676 873 ± 1	642 892 ± 1	575 879 ± 0	489 874 ± 0	514 1264 ± 13	525 1272 ± 16
243	cor-001	1194948	11240	557 1249	1142 2060 ± 0	765 699 ± 3	664 863 ± 76	620 865 ± 80	568 872 ± 89	539 952 ± 39	1489 270145 ± 2259	1489 282686 ± 11788
244	coretech-000	186423	43964	102 318	97 512 ± 0	609 602 ± 15	476 659 ± 12	865 1139 ± 24	794 1149 ± 25	700 1165 ± 23	91 333 ± 14	93 321 ± 13
245	coretech-001	235361	305490	650 1523	601 2048 ± 0	747 688 ± 7	519 695 ± 7	626 870 ± 17	576 879 ± 15	491 877 ± 15	233 625 ± 25	267 641 ± 25
246	coretech-002	339881	296194	960 3300	802 2048 ± 0	431 459 ± 1	280 451 ± 1	611 851 ± 1	570 875 ± 2	555 980 ± 2	819 2443 ± 17	817 2436 ± 13
247	coretech-003	339885	296194	754 1861	519 2048 ± 0	896 815 ± 2	648 838 ± 3	977 1273 ± 6	912 1279 ± 8	842 1373 ± 6	812 2416 ± 13	813 2415 ± 19
248	coretech-004	339885	296194	747 1828	935 2048 ± 0	394 427 ± 1	272 443 ± 2	607 847 ± 4	564 858 ± 2	532 945 ± 4	815 2436 ± 35	816 2432 ± 20
249	corsight-001	1437763	31525	-	1145 2064 ± 0	1319 1291 ± 3	1016 1285 ± 1	993 1293 ± 1	922 1303 ± 2	849 1379 ± 3	1488 249340 ± 1713	1488 248929 ± 1909
250	corsight-002	1474921	32093	761 1892	1165 2080 ± 0	1318 1290 ± 1	1018 1287 ± 1	989 1290 ± 1	926 1307 ± 2	858 1388 ± 4	1352 24953 ± 637	1346 24263 ± 578
251	corsight-003	1413063	32198	691 1642	1164 2080 ± 0	1252 1202 ± 2	943 1190 ± 5	918 1199 ± 3	874 1236 ± 3	831 1349 ± 7	1365 28754 ± 434	1367 28279 ± 446
252	csc-001	0	240698	-	188 544 ± 0	240 302 ± 1	151 303 ± 0	114 304 ± 0	103 309 ± 1	91 341 ± 4	102 354 ± 8	108 344 ± 11
253	csc-002	0	519768	4 30	189 544 ± 0	450 473 ± 0	315 494 ± 0	265 481 ± 1	237 490 ± 1	208 514 ± 5	110 367 ± 11	115 371 ± 10
254	csc-003	0	400435	5 31	194 544 ± 0	475 499 ± 0	322 500 ± 1	286 502 ± 0	252 508 ± 1	220 535 ± 4	117 393 ± 8	126 397 ± 7
255	ctbcbank-000	257208	599238	217 570	828 2048 ± 0	556 568 ± 43	414 606 ± 38	472 690 ± 53	443 711 ± 50	455 831 ± 51	978 3551 ± 87	1070 4805 ± 209
256	ctbcbank-001	275511	599238	231 603	738 2048 ± 0	688 652 ± 35	604 781 ± 30	630 875 ± 43	591 898 ± 51	593 1030 ± 47	1014 3926 ± 45	1017 3924 ± 56
257	ctbcbank-002	550523	253310	485 1084	113 512 ± 0	760 694 ± 0	536 707 ± 0	510 727 ± 0	472 744 ± 0	393 758 ± 0	1061 4656 ± 146	1098 5486 ± 179
258	cu-face-002	812008	38655	298 723	1338 4096 ± 0	1138 1054 ± 1	836 1059 ± 0	801 1060 ± 0	716 1063 ± 1	630 1070 ± 0	1438 57287 ± 1750	1437 57027 ± 945
259	cu-face-003	1015124	37791	802 2116	1259 4096 ± 0	1037 945 ± 1	748 955 ± 3	701 957 ± 2	628 949 ± 3	543 963 ± 2	874 2769 ± 19	878 2782 ± 41
260	cu-face-004	1016682	37791	796 2083	1257 4096 ± 0	1113 1022 ± 1	818 1031 ± 1	745 998 ± 3	725 1070 ± 1	587 1021 ± 2	891 2884 ± 38	891 2864 ± 48
261	cubox-001	369627	75427	265 652	972 2048 ± 0	990 907 ± 1	699 902 ± 1	649 903 ± 0	601 917 ± 0	525 931 ± 0	543 1379 ± 37	562 1417 ± 38
262	cubox-002	542254	90975	775 1966	403 2048 ± 0	1012 921 ± 1	717 921 ± 1	669 922 ± 1	614 933 ± 1	576 1003 ± 1	724 2008 ± 72	721 1969 ± 57
263	cubox-003	1694397	209684	975 3491	625 2048 ± 0	1232 1179 ± 0	942 1189 ± 0	917 1198 ± 0	845 1208 ± 0	765 1264 ± 1	786 2289 ± 112	786 2297 ± 68
264	cubox-004	1304512	209634	911 2747	595 2048 ± 0	1059 963 ± 1	780 986 ± 1	735 992 ± 0	684 1008 ± 0	635 1087 ± 0	583 1476 ± 62	581 1460 ± 42

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 28: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
265	cubox-005	225237	209600	611 1387	854 2048 ± 0	315 362 ± 0	206 371 ± 0	169 373 ± 0	159 390 ± 0	161 453 ± 0	659 1748 ± 25	655 1718 ± 40
266	cudocommunication-001	385258	341277	483 1080	570 2048 ± 0	1018 925 ± 1	720 923 ± 1	677 928 ± 1	612 932 ± 0	544 964 ± 1	836 2534 ± 20	841 2537 ± 20
267	cuhkee-001	787853	74917	869 2515	1010 2052 ± 0	1078 977 ± 31	-	-	-	-	865 2719 ± 60	879 2783 ± 56
268	cvartel-001	1310489	39129	891 2651	1046 2052 ± 0	1200 1143 ± 2	1019 1287 ± 23	1110 1453 ± 2	1097 1610 ± 4	1067 2167 ± 9	168 537 ± 26	179 547 ± 19
269	cybercore-000	86008	55441	-	156 512 ± 0	692 655 ± 3	514 689 ± 71	433 649 ± 6	376 648 ± 8	327 680 ± 6	1271 14800 ± 75	1286 15757 ± 782
270	cybercore-001	166096	7791	-	632 2048 ± 0	465 487 ± 0	309 486 ± 0	272 488 ± 0	236 487 ± 0	192 502 ± 0	1429 52119 ± 111	1429 52127 ± 111
271	cybercore-002	166096	7374	878 2565	346 2048 ± 0	467 489 ± 1	320 500 ± 4	285 500 ± 1	244 499 ± 2	218 528 ± 1	1248 12389 ± 123	1250 12352 ± 112
272	cybercore-003	289176	7969	1019 4311	1293 4096 ± 0	931 844 ± 2	657 855 ± 4	618 864 ± 4	566 862 ± 4	493 878 ± 2	1110 5744 ± 41	1112 5737 ± 31
273	cyberextruder-001	121211	13629	-	37 256 ± 0	984 893 ± 25	-	-	-	-	450 1083 ± 16	456 1079 ± 19
274	cyberextruder-002	168909	13924	-	361 2048 ± 0	511 532 ± 6	-	-	-	-	680 1803 ± 14	673 1779 ± 22
275	cyberextruder-003	253300	12354	152 444	148 512 ± 0	352 390 ± 1	220 388 ± 1	187 393 ± 1	168 399 ± 1	146 435 ± 1	41 198 ± 4	39 189 ± 8
276	cyberextruder-004	169301	12354	98 302	5 128 ± 0	145 206 ± 0	90 208 ± 0	71 209 ± 0	70 229 ± 0	56 249 ± 1	24 145 ± 14	27 155 ± 14
277	cyberextruder-005	269610	45367	236 606	135 512 ± 0	99 168 ± 2	56 169 ± 1	68 205 ± 3	96 295 ± 5	465 835 ± 9	23 142 ± 8	24 144 ± 10
278	cyberextruder-006	184816	45435	128 382	28 256 ± 0	37 91 ± 1	23 89 ± 0	21 97 ± 0	17 95 ± 0	20 130 ± 1	9 98 ± 7	9 101 ± 11
279	cyberlink-000	221801	91440	-	1009 2052 ± 0	288 338 ± 6	-	-	-	-	744 2102 ± 40	745 2093 ± 32
280	cyberlink-001	222009	91440	-	1033 2052 ± 0	391 425 ± 29	-	-	-	-	732 2051 ± 32	739 2060 ± 31
281	cyberlink-002	222311	95086	-	1056 2052 ± 0	696 656 ± 22	-	-	-	-	779 2264 ± 71	853 2649 ± 195
282	cyberlink-003	470949	104369	-	1027 2052 ± 0	390 423 ± 1	-	-	-	-	802 2366 ± 38	803 2373 ± 45
283	cyberlink-004	340894	104662	-	1392 4140 ± 0	780 712 ± 1	-	-	-	-	993 3693 ± 51	1014 3898 ± 71
284	cyberlink-005	341680	111358	-	1393 4140 ± 0	790 721 ± 1	581 752 ± 2	537 754 ± 2	480 755 ± 1	425 791 ± 3	989 3680 ± 37	1031 4021 ± 97
285	cyberlink-006	341666	102456	-	1453 6212 ± 0	753 690 ± 1	527 702 ± 0	488 703 ± 0	444 712 ± 0	384 741 ± 0	69 270 ± 13	77 271 ± 13
286	cyberlink-007	380046	102446	-	1454 6212 ± 0	796 725 ± 1	558 732 ± 1	518 734 ± 1	466 736 ± 1	407 767 ± 1	82 304 ± 19	88 304 ± 16
287	cyberlink-008	380047	102470	-	1455 6212 ± 0	801 729 ± 1	551 725 ± 0	511 727 ± 0	463 732 ± 0	396 760 ± 0	65 263 ± 17	70 255 ± 13
288	cyberlink-009	588443	102201	708 1707	1401 4164 ± 0	1372 1384 ± 2	1086 1395 ± 2	1067 1398 ± 2	995 1401 ± 2	903 1456 ± 2	79 299 ± 17	87 304 ± 16
289	cyberlink-010	1590818	102180	989 3678	1480 8260 ± 0	1302 1265 ± 2	1039 1314 ± 5	996 1294 ± 2	906 1273 ± 2	798 1305 ± 2	143 476 ± 23	151 472 ± 14
290	cyberlink-011	1397496	102171	940 3029	1457 6212 ± 0	1245 1196 ± 3	945 1193 ± 2	914 1195 ± 3	854 1215 ± 6	737 1231 ± 5	1003 3768 ± 75	1001 3735 ± 57
291	cyberlink-012	1200318	102281	918 2789	1456 6212 ± 0	607 601 ± 1	434 621 ± 5	379 610 ± 2	337 614 ± 2	305 649 ± 5	798 2343 ± 26	804 2377 ± 34
292	cyberlink-013	893519	102285	773 1964	1452 6212 ± 0	611 603 ± 1	410 602 ± 2	375 606 ± 2	336 613 ± 2	300 642 ± 1	85 307 ± 7	89 306 ± 5
293	cyberlink-014	1190911	102304	925 2876	1482 8260 ± 0	789 720 ± 1	561 736 ± 5	520 735 ± 10	461 731 ± 3	398 762 ± 3	113 378 ± 8	117 377 ± 13
294	cyberlink-015	1190911	102309	924 2875	1481 8260 ± 0	803 730 ± 1	569 741 ± 6	519 735 ± 12	462 732 ± 2	419 778 ± 1	831 2497 ± 26	835 2493 ± 51
295	cyberlink-016	1503336	105850	981 3591	1485 10308 ± 0	1478 1846 ± 1	1194 1908 ± 1	1178 1863 ± 1	1134 1864 ± 0	1058 2115 ± 3	452 1085 ± 4	450 1069 ± 19
296	dahua-000	283642	171917	-	393 2048 ± 0	317 363 ± 7	-	-	-	-	112 373 ± 16	111 364 ± 17
297	dahua-001	283642	171917	-	868 2048 ± 0	316 363 ± 6	-	-	-	-	108 367 ± 10	110 354 ± 16
298	dahua-002	526452	171518	-	440 2048 ± 0	655 628 ± 7	-	-	-	-	135 461 ± 23	140 454 ± 20
299	dahua-003	605337	141332	-	523 2048 ± 0	513 535 ± 2	-	-	-	-	255 648 ± 20	252 631 ± 34
300	dahua-004	832455	141070	-	495 2048 ± 0	807 735 ± 3	-	-	-	-	311 730 ± 25	308 707 ± 44
301	dahua-005	1586899	169478	-	1324 4096 ± 0	1398 1418 ± 34	-	-	-	-	398 957 ± 23	408 969 ± 19
302	dahua-006	831641	119261	1047 5068	493 2048 ± 0	1385 1398 ± 2	1089 1397 ± 1	1075 1404 ± 1	1000 1402 ± 1	863 1402 ± 1	60 249 ± 13	65 250 ± 11
303	dahua-007	1578737	119418	1080 7237	1297 4096 ± 0	1380 1393 ± 2	1070 1373 ± 1	1054 1378 ± 1	978 1378 ± 1	850 1379 ± 2	109 367 ± 102	132 434 ± 108
304	daon-000	280726	2307	785 2010	1149 2065 ± 0	548 562 ± 3	393 581 ± 5	557 791 ± 9	546 838 ± 15	610 1055 ± 32	1285 16052 ± 88	1288 16041 ± 85
305	daon-001	296035	69405	225 587	978 2048 ± 0	736 682 ± 0	481 661 ± 0	470 688 ± 0	410 683 ± 0	308 657 ± 0	348 803 ± 23	363 815 ± 22
306	daon-002	295360	66546	305 731	345 2048 ± 0	838 765 ± 0	591 767 ± 1	548 768 ± 6	499 776 ± 1	412 770 ± 0	363 840 ± 17	374 858 ± 26
307	daon-003	931820	66767	731 1776	1242 4096 ± 0	1497 2048 ± 0	1206 2046 ± 0	1196 2050 ± 0	1163 2069 ± 0	1047 2047 ± 0	763 2194 ± 59	765 2203 ± 28
308	datech-000	617605	139575	1057 5599	1253 4096 ± 0	563 573 ± 32	396 583 ± 44	364 593 ± 37	1212 14123 ± 9564	1200 15514 ± 2343	1178 7857 ± 29	1178 7852 ± 25

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 29: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
309	datech-001	835404	139615	1079 7179	1237 4096 ± 0	1111 1017 ± 83	832 1057 ± 65	805 1062 ± 88	751 1109 ± 81	793 1299 ± 92	1181 7944 ± 31	1182 7981 ± 56
310	datech-002	854184	139615	1083 7446	1236 4096 ± 0	1159 1086 ± 115	893 1130 ± 119	842 1113 ± 123	842 1204 ± 107	873 1413 ± 145	1179 7860 ± 20	1179 7857 ± 26
311	decatur-000	350495	171271	401 907	1352 4100 ± 0	1114 1024 ± 2	-	-	-	-	1230 11439 ± 80	1232 11418 ± 112
312	decatur-001	342866	253734	648 1508	1017 2052 ± 0	1171 1103 ± 2	837 1064 ± 2	806 1063 ± 2	719 1067 ± 2	634 1084 ± 2	215 610 ± 19	220 602 ± 8
313	decloakface-001	29693	170719	123 368	758 2048 ± 0	135 196 ± 58	85 201 ± 52	80 229 ± 53	98 298 ± 60	219 532 ± 61	1060 4638 ± 19	1060 4646 ± 22
314	decloakface-002	1692	143986	82 260	859 2048 ± 0	70 127 ± 0	43 135 ± 0	44 157 ± 0	63 218 ± 0	165 458 ± 2	1043 4380 ± 31	1044 4383 ± 45
315	deepglint-001	569802	206788	-	1268 4096 ± 0	792 721 ± 4	-	-	-	-	990 3680 ± 35	981 3517 ± 182
316	deepglint-002	459642	272878	-	1243 4096 ± 0	727 677 ± 2	633 826 ± 74	608 848 ± 42	553 849 ± 55	496 886 ± 27	1257 13633 ± 87	1256 12905 ± 440
317	deepglint-003	838065	262081	-	1444 6144 ± 0	1214 1159 ± 1	907 1145 ± 1	874 1148 ± 1	792 1148 ± 1	698 1163 ± 1	1295 17227 ± 41	1297 17210 ± 51
318	deepglint-004	1073382	261571	948 3116	549 2048 ± 0	1427 1470 ± 1	1138 1474 ± 1	1124 1485 ± 1	1053 1474 ± 1	927 1492 ± 2	1120 5961 ± 34	1118 5955 ± 29
319	deepglint-005	960326	213877	934 2991	344 2048 ± 0	1394 1408 ± 1	1115 1431 ± 2	1089 1424 ± 3	1018 1424 ± 3	898 1446 ± 2	1146 6765 ± 38	1143 6765 ± 40
320	deepsea-001	147497	336250	118 358	229 1024 ± 0	660 630 ± 7	579 752 ± 37	528 746 ± 30	455 727 ± 32	443 820 ± 32	553 1401 ± 37	584 1467 ± 50
321	deepsense-000	357113	936618	-	580 2048 ± 0	706 664 ± 3	466 645 ± 1	440 660 ± 2	417 687 ± 2	438 808 ± 3	144 480 ± 22	147 459 ± 34
322	deepsense-001	73173	1288355	1050 5314	143 512 ± 0	1199 1142 ± 2	921 1164 ± 3	902 1183 ± 3	838 1201 ± 3	812 1323 ± 2	800 2356 ± 35	799 2354 ± 42
323	deepsense-002	73173	1266131	1048 5280	214 896 ± 0	1349 1350 ± 1	1029 1299 ± 8	1014 1324 ± 0	954 1348 ± 0	916 1474 ± 4	969 3505 ± 32	980 3515 ± 40
324	deepsense-003	73173	1359497	1069 6555	216 901 ± 0	919 832 ± 2	735 940 ± 2	605 845 ± 4	652 983 ± 1	553 976 ± 7	673 1797 ± 25	679 1795 ± 19
325	deepsense-004	560774	1052093	1082 7310	219 916 ± 0	1297 1259 ± 2	1026 1294 ± 9	971 1268 ± 7	894 1261 ± 8	824 1334 ± 7	695 1884 ± 19	699 1884 ± 25
326	deepsense-005	560774	1052093	1081 7309	218 916 ± 0	1405 1430 ± 1	1003 1264 ± 1	952 1246 ± 1	888 1254 ± 1	833 1353 ± 4	707 1920 ± 26	712 1932 ± 40
327	deepsense-006	643809	1052098	1084 7719	217 916 ± 0	1348 1348 ± 2	1059 1347 ± 2	904 1184 ± 4	972 1369 ± 3	912 1473 ± 4	690 1860 ± 13	692 1853 ± 22
328	deepsense-007	643809	1070446	1085 7769	221 920 ± 0	1218 1163 ± 1	928 1173 ± 3	906 1190 ± 2	846 1209 ± 3	801 1310 ± 4	687 1847 ± 11	694 1860 ± 28
329	deepsense-008	643809	1070340	1086 7770	220 920 ± 0	1453 1571 ± 1	1157 1572 ± 0	1152 1610 ± 1	1110 1729 ± 1	1072 2216 ± 1	1050 4469 ± 109	1050 4498 ± 68
330	dermalog-001	0	287736	-	283 1043 ± 0	53 106 ± 1	-	-	-	-	1329 22160 ± 166	1333 22131 ± 131
331	dermalog-002	0	287738	-	284 1043 ± 0	30 81 ± 0	-	-	-	-	1330 22169 ± 114	1332 22105 ± 146
332	dermalog-003	0	309430	-	282 1043 ± 0	63 121 ± 22	-	-	-	-	1335 22957 ± 93	1336 22808 ± 131
333	dermalog-004	0	245770	-	9 128 ± 0	86 149 ± 5	-	-	-	-	151 490 ± 26	150 471 ± 26
334	dermalog-005	0	317687	-	11 128 ± 0	72 130 ± 11	-	-	-	-	156 499 ± 22	161 500 ± 22
335	dermalog-006	0	452387	-	6 128 ± 0	509 532 ± 12	-	-	-	-	160 506 ± 23	146 459 ± 23
336	dermalog-007	0	503332	-	105 512 ± 0	171 239 ± 9	-	-	-	-	377 887 ± 23	297 690 ± 25
337	dermalog-008	0	937895	-	149 512 ± 0	367 404 ± 2	239 410 ± 3	215 424 ± 5	187 430 ± 5	176 477 ± 5	139 468 ± 31	99 328 ± 13
338	dermalog-009	0	319363	607 1369	141 512 ± 0	296 349 ± 0	185 351 ± 0	149 352 ± 0	134 357 ± 0	120 389 ± 0	149 487 ± 34	121 385 ± 29
339	dermalog-010	0	525908	994 3789	153 512 ± 0	668 635 ± 0	462 640 ± 1	423 639 ± 4	374 647 ± 3	338 691 ± 5	131 444 ± 13	106 341 ± 26
340	dermalog-011	0	278395	296 716	7 128 ± 0	291 343 ± 0	177 345 ± 0	143 347 ± 0	128 351 ± 0	105 363 ± 0	80 299 ± 19	68 253 ± 14
341	dermalog-012	0	293431	299 723	4 128 ± 0	114 181 ± 0	62 179 ± 0	52 182 ± 0	46 186 ± 0	55 242 ± 0	27 163 ± 10	43 196 ± 16
342	dermalog-013	0	293658	1017 4263	8 128 ± 0	116 183 ± 2	64 180 ± 2	50 181 ± 1	43 183 ± 1	39 205 ± 1	35 186 ± 8	52 221 ± 7
343	dermalog-014	0	756422	545 1201	226 1024 ± 0	144 204 ± 0	83 197 ± 0	65 198 ± 1	53 199 ± 0	53 241 ± 0	123 424 ± 17	118 377 ± 17
344	dermalog-015	0	909348	859 2452	234 1024 ± 0	1488 1929 ± 1	1192 1891 ± 0	1190 1955 ± 1	1144 1899 ± 1	1027 1904 ± 1	106 360 ± 25	74 267 ± 14
345	dermalog-016	0	910481	858 2452	228 1024 ± 0	1486 1913 ± 2	1193 1908 ± 1	1182 1918 ± 1	1145 1917 ± 1	1029 1927 ± 1	165 523 ± 23	144 457 ± 25
346	dicio-001	61751	119517	22 84	175 520 ± 0	520 538 ± 0	376 563 ± 10	658 915 ± 3	1124 1800 ± 7	1160 5286 ± 30	880 2818 ± 20	881 2807 ± 31
347	dicio-002	571566	85868	575 1290	1143 2060 ± 0	934 846 ± 1	587 761 ± 0	551 775 ± 1	507 784 ± 1	456 831 ± 0	1134 6242 ± 74	1132 6256 ± 80
348	didiglobalface-001	259849	70680	196 527	954 2048 ± 0	624 612 ± 1	459 633 ± 3	417 634 ± 3	377 650 ± 15	318 666 ± 4	403 973 ± 20	412 988 ± 20
349	didiglobalface-002	260054	161508	343 812	559 2048 ± 0	646 622 ± 1	458 633 ± 1	428 642 ± 2	387 659 ± 4	368 726 ± 15	178 560 ± 10	189 567 ± 13
350	didiglobalface-003	1001649	83986	567 1274	377 2048 ± 0	812 740 ± 2	565 740 ± 2	546 767 ± 5	500 779 ± 4	474 853 ± 17	612 1544 ± 13	615 1539 ± 17
351	digidata-000	133370	30249	81 259	413 2048 ± 0	313 361 ± 0	190 360 ± 0	158 361 ± 0	137 363 ± 0	113 380 ± 0	742 2084 ± 37	731 2039 ± 42
352	digidata-001	254564	33036	121 367	480 2048 ± 0	544 559 ± 1	374 561 ± 1	336 562 ± 1	297 564 ± 1	267 602 ± 1	1216 10308 ± 102	1217 10301 ± 121

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 30: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
353	digitalbarriers-000	157794	427472	-	1072 2056 ± 0	52 104 ± 0	-	-	-	-	1254 13232 ± 166	1258 13226 ± 146
354	digitalbarriers-001	236915	427537	-	1105 2056 ± 0	229 294 ± 1	-	-	-	-	1247 12311 ± 164	1249 12347 ± 197
355	digitalbarriers-002	83002	598577	767 1930	1124 2056 ± 0	147 209 ± 11	112 250 ± 19	199 411 ± 37	521 808 ± 72	1073 2236 ± 123	1255 13409 ± 228	1259 13267 ± 206
356	dps-000	0	2211812	616 1405	1309 4096 ± 0	949 868 ± 2	694 893 ± 6	1102 1445 ± 9	1192 2910 ± 38	1185 9345 ± 17	580 1473 ± 37	589 1479 ± 37
357	dralmey-001	571556	82266	574 1290	1137 2060 ± 0	1239 1186 ± 0	920 1162 ± 0	888 1170 ± 0	821 1180 ± 0	743 1243 ± 0	1261 13996 ± 56	1264 13971 ± 60
358	dsk-000	11967	782905	79 252	137 512 ± 0	243 304 ± 47	159 317 ± 33	747 1001 ± 96	1189 2660 ± 170	1193 10451 ± 832	1162 7152 ± 115	1157 7134 ± 111
359	einetworks-000	372608	219883	388 880	1098 2056 ± 0	678 645 ± 3	-	-	-	-	1075 4876 ± 66	1086 5156 ± 77
360	einetworksindia-000	372375	479472	588 1322	1086 2056 ± 0	335 378 ± 1	215 382 ± 1	179 383 ± 1	167 397 ± 0	164 456 ± 3	1278 15171 ± 41	1279 15138 ± 62
361	einetworksindia-001	562398	481792	740 1808	1119 2056 ± 0	454 479 ± 0	304 481 ± 1	269 485 ± 1	242 494 ± 1	227 541 ± 1	1276 15147 ± 63	1278 15134 ± 74
362	einetworksindia-002	1214466	143768	863 2462	750 2048 ± 0	1160 1087 ± 1	846 1082 ± 1	808 1065 ± 0	742 1098 ± 3	687 1150 ± 2	1258 13785 ± 34	1261 13767 ± 56
363	ekin-002	51434	278	43 139	1211 3072 ± 0	1238 1186 ± 13	935 1180 ± 12	898 1181 ± 11	834 1191 ± 11	725 1207 ± 8	1039 4294 ± 80	1105 5569 ± 112
364	element-000	721679	24895	570 1287	604 2048 ± 0	360 399 ± 1	242 417 ± 1	216 428 ± 9	231 478 ± 14	334 688 ± 45	187 583 ± 10	198 585 ± 13
365	enface-000	369598	153781	275 671	244 1024 ± 0	540 555 ± 4	372 558 ± 4	454 669 ± 6	656 987 ± 15	1084 2349 ± 54	1158 7059 ± 62	1153 6980 ± 65
366	enface-001	370710	173609	274 669	255 1024 ± 0	536 550 ± 4	369 555 ± 3	453 668 ± 7	650 981 ± 15	1094 2416 ± 59	1145 6734 ± 68	1144 6766 ± 69
367	enface-002	858356	98741	1091 9004	256 1024 ± 0	1119 1026 ± 1	824 1038 ± 3	797 1057 ± 7	763 1123 ± 4	825 1338 ± 8	607 1529 ± 23	610 1530 ± 35
368	enqura-001	0	6136892	560 1263	643 2048 ± 0	159 221 ± 5	97 220 ± 3	85 239 ± 3	79 262 ± 3	68 288 ± 3	1 0 ± 0	1 0 ± 0
369	enqura-002	724175	1441517	382 870	1132 2057 ± 0	427 456 ± 1	285 457 ± 1	271 486 ± 1	345 619 ± 1	948 1532 ± 3	1367 29132 ± 232	1369 28946 ± 171
370	enqura-003	1014924	630597	585 1310	1127 2057 ± 0	1164 1097 ± 0	860 1093 ± 0	905 1189 ± 0	850 1213 ± 0	775 1279 ± 0	1377 31750 ± 263	1385 32710 ± 154
371	eocortex-000	255937	59432	75 224	685 2048 ± 0	245 305 ± 22	174 341 ± 25	228 440 ± 47	219 464 ± 45	205 513 ± 44	390 923 ± 11	395 918 ± 11
372	ercacat-001	811623	58012	920 2816	1013 2052 ± 0	1137 1052 ± 3	-	-	-	-	839 2551 ± 62	836 2501 ± 81
373	essi-000	170837	16264	100 310	901 2048 ± 0	123 188 ± 0	70 186 ± 0	94 272 ± 0	105 311 ± 0	77 320 ± 0	626 1613 ± 31	626 1612 ± 35
374	ethiopianlmis-000	309182	549917	170 481	688 2048 ± 0	973 882 ± 2	689 887 ± 1	583 824 ± 2	547 838 ± 2	509 903 ± 1	810 2400 ± 34	810 2407 ± 28
375	euronovate-001	0	1774966	593 1346	297 1177 ± 0	1128 1034 ± 2	922 1165 ± 3	882 1160 ± 3	817 1177 ± 3	705 1172 ± 2	1472 81294 ± 591	1470 81631 ± 931
376	euronovate-002	0	5476674	349 818	1418 4249 ± 0	227 293 ± 0	145 296 ± 0	111 298 ± 0	102 309 ± 0	89 339 ± 0	1383 32673 ± 391	1383 32587 ± 391
377	euronovate-003	0	5637394	541 1193	1184 2201 ± 0	237 300 ± 0	150 300 ± 0	113 303 ± 0	104 310 ± 0	90 340 ± 0	1399 37908 ± 968	1403 38285 ± 938
378	euronovate-004	0	1335993	463 1049	1183 2201 ± 0	222 290 ± 0	142 294 ± 0	108 294 ± 0	95 295 ± 0	69 294 ± 0	1413 46290 ± 490	1413 46199 ± 484
379	everai-000	229700	144802	-	554 2048 ± 0	425 455 ± 1	-	-	-	-	116 385 ± 10	120 382 ± 13
380	everai-001	449149	144802	-	593 2048 ± 0	772 701 ± 1	-	-	-	-	118 395 ± 11	127 404 ± 23
381	everai-002	561727	281965	-	1272 4096 ± 0	832 758 ± 0	-	-	-	-	252 644 ± 14	246 624 ± 35
382	everai-paravision-003	539802	118876	-	1305 4096 ± 0	720 674 ± 4	-	-	-	-	287 699 ± 20	314 713 ± 47
383	expasoft-000	15341	240451	-	617 2048 ± 0	22 68 ± 0	15 71 ± 0	10 75 ± 0	10 71 ± 0	7 72 ± 1	669 1779 ± 26	667 1757 ± 97
384	expasoft-001	39057	983064	46 142	389 2048 ± 0	23 70 ± 0	16 74 ± 0	12 77 ± 0	11 73 ± 0	9 74 ± 0	637 1660 ± 35	641 1676 ± 48
385	expasoft-002	38760	59825	60 170	840 2048 ± 0	8 34 ± 0	5 34 ± 0	5 34 ± 0	4 34 ± 0	2 34 ± 0	1195 8870 ± 78	1196 8838 ± 77
386	f8-001	272977	19668	568 1276	912 2048 ± 0	906 822 ± 39	-	-	-	-	1281 15262 ± 139	1284 15277 ± 212
387	f8-002	28278	215616	21 84	938 2048 ± 0	11 39 ± 0	7 41 ± 0	11 75 ± 0	52 197 ± 1	348 702 ± 1	1270 14765 ± 131	1273 14790 ± 133
388	facehawk-000	120009	7737	39 129	209 776 ± 1	359 399 ± 0	234 402 ± 0	201 414 ± 1	160 391 ± 2	151 440 ± 3	1225 10931 ± 155	1225 10915 ± 149
389	facehawk-001	173511	35863	113 343	1154 2070 ± 0	314 362 ± 0	197 363 ± 0	167 371 ± 0	148 375 ± 0	148 437 ± 0	576 1465 ± 19	586 1475 ± 11
390	faceint-001	110744	649612	930 2903	895 2048 ± 0	133 195 ± 0	78 191 ± 0	64 195 ± 0	55 203 ± 3	48 232 ± 10	565 1434 ± 11	569 1431 ± 19
391	facelocate-001	4739	11106	855 2449	19 168 ± 0	7 34 ± 1	6 38 ± 0	14 84 ± 1	24 112 ± 4	40 206 ± 13	1202 9565 ± 91	1206 9629 ± 131
392	facelocate-002	15198	486568	86 271	109 512 ± 0	156 214 ± 1	107 239 ± 0	209 422 ± 2	215 463 ± 5	245 571 ± 15	218 611 ± 8	242 620 ± 17
393	faceonlive-001	0	71529	97 300	1106 2056 ± 0	110 179 ± 0	61 179 ± 0	61 190 ± 0	61 217 ± 0	93 343 ± 1	439 1064 ± 37	433 1033 ± 35
394	faceonlive-002	155220	141019	436 997	418 2048 ± 0	856 783 ± 1	614 797 ± 2	560 794 ± 2	522 809 ± 3	505 901 ± 2	1259 13798 ± 197	1260 13743 ± 127
395	faceonlive-003	518321	560984	1033 4691	250 1024 ± 0	1127 1032 ± 4	856 1091 ± 3	1188 1942 ± 6	1108 1723 ± 5	1148 4168 ± 3	1286 16149 ± 41	1289 16135 ± 54
396	facephi-000	148904	5219	1102 20248	539 2048 ± 0	953 871 ± 2	683 881 ± 3	635 880 ± 4	585 888 ± 4	535 949 ± 12	1030 4067 ± 53	1033 4047 ± 53

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 31: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
397	facephi-001	138133	282874	594 1347	147 512 ± 0	220 285 ± 0	125 272 ± 2	107 291 ± 5	97 296 ± 3	183 489 ± 5	933 3194 ± 41	938 3184 ± 54
398	facephi-002	1031362	112529	1066 6074	880 2048 ± 0	1285 1240 ± 5	991 1251 ± 5	961 1253 ± 21	920 1298 ± 5	999 1713 ± 19	1241 11961 ± 81	1244 11970 ± 83
399	faceplugin-000	482373	20827	549 1217	285 1048 ± 0	816 744 ± 2	559 733 ± 2	604 844 ± 1	795 1149 ± 1	1125 3003 ± 3	997 3735 ± 533	996 3694 ± 495
400	facesoft-000	370120	10612	335 796	586 2048 ± 0	722 675 ± 18	487 669 ± 3	467 686 ± 3	403 675 ± 5	332 687 ± 2	772 2239 ± 28	782 2277 ± 96
401	facetag-000	1232331	4022	424 965	205 684 ± 0	303 355 ± 17	205 369 ± 8	730 989 ± 33	1184 2408 ± 91	1178 7930 ± 316	1460 72003 ± 625	1460 71912 ± 612
402	facetag-001	1258247	4022	-	1347 4100 ± 0	1090 991 ± 3	791 995 ± 3	760 1018 ± 3	723 1069 ± 5	780 1284 ± 8	946 3323 ± 28	952 3287 ± 34
403	facetag-002	819806	4021	291 706	589 2048 ± 0	530 544 ± 1	360 544 ± 0	323 542 ± 0	282 545 ± 0	236 554 ± 0	655 1730 ± 25	662 1733 ± 25
404	facex-001	305074	930372	963 3336	735 2048 ± 0	389 422 ± 4	261 434 ± 4	305 520 ± 7	467 737 ± 13	992 1670 ± 27	692 1871 ± 23	690 1846 ± 29
405	facex-002	305074	928334	939 3011	486 2048 ± 0	393 426 ± 5	254 429 ± 4	299 516 ± 8	459 730 ± 12	1002 1738 ± 36	239 631 ± 25	237 614 ± 19
406	facia-001	296496	745403	396 889	919 2048 ± 0	1167 1099 ± 1	874 1109 ± 0	843 1113 ± 0	768 1125 ± 3	749 1250 ± 1	1446 62245 ± 397	1443 62249 ± 496
407	facia-002	585227	694738	762 1907	1003 2052 ± 0	1345 1347 ± 4	1058 1337 ± 3	1133 1509 ± 7	1146 1919 ± 11	1142 3982 ± 34	1326 21963 ± 400	1329 21869 ± 318
408	faicetech-000	247687	34022	596 1349	370 2048 ± 0	1234 1179 ± 1	949 1199 ± 1	909 1193 ± 1	840 1202 ± 0	742 1240 ± 4	195 594 ± 4	209 593 ± 1
409	farfaces-001	346494	44581	83 262	102 512 ± 0	1233 1179 ± 1	934 1180 ± 1	897 1180 ± 0	827 1185 ± 1	727 1209 ± 2	368 855 ± 25	375 860 ± 31
410	fastenterprises-000	273365	235899	452 1030	612 2048 ± 0	700 662 ± 1	491 671 ± 0	450 664 ± 0	402 673 ± 3	333 688 ± 5	1006 3806 ± 96	1382 32064 ± 185
411	fci-000	1486985	454513	949 3123	394 2048 ± 0	1425 1464 ± 12	1143 1494 ± 9	1112 1457 ± 10	1048 1462 ± 11	945 1525 ± 5	791 2309 ± 8	793 2314 ± 14
412	fdu-000	202296	1307	-	184 534 ± 0	33 84 ± 0	-	-	-	-	649 1713 ± 28	653 1715 ± 20
413	fdu-001	345583	1329	-	332 1808 ± 0	73 131 ± 0	-	-	-	-	524 1288 ± 26	528 1281 ± 22
414	fedu-001	466605	496375	667 1557	501 2048 ± 0	595 592 ± 1	439 623 ± 1	395 624 ± 0	352 623 ± 0	301 647 ± 0	801 2362 ± 13	800 2361 ± 22
415	fiberhome-nanjing-002	207398	596827	-	251 1024 ± 0	1261 1217 ± 2	1025 1294 ± 30	1007 1312 ± 12	1055 1477 ± 14	818 1326 ± 11	617 1582 ± 57	616 1560 ± 51
416	fiberhome-nanjing-003	352895	1482309	379 859	557 2048 ± 0	1194 1136 ± 7	894 1134 ± 4	861 1132 ± 3	785 1139 ± 3	690 1154 ± 5	457 1097 ± 38	460 1083 ± 42
417	fiberhome-nanjing-004	443779	1482313	481 1077	1267 4096 ± 0	1337 1321 ± 5	1034 1304 ± 3	1004 1307 ± 2	929 1308 ± 3	815 1326 ± 5	519 1276 ± 40	522 1265 ± 38
418	fincore-000	256615	19409	201 536	790 2048 ± 0	487 508 ± 3	326 505 ± 0	292 508 ± 1	257 513 ± 2	221 535 ± 1	663 1765 ± 31	669 1763 ± 22
419	firstcreditkz-001	553811	24803	502 1113	441 2048 ± 0	884 808 ± 0	793 997 ± 0	803 1061 ± 1	815 1174 ± 1	1008 1774 ± 54	382 904 ± 20	385 903 ± 23
420	firstcreditkz-002	977538	24879	770 1951	911 2048 ± 0	1301 1265 ± 1	1011 1275 ± 4	1002 1299 ± 7	950 1341 ± 17	938 1515 ± 40	326 761 ± 28	337 774 ± 32
421	firstcreditkz-003	1145884	24907	809 2144	347 2048 ± 0	1241 1188 ± 79	1142 1492 ± 77	1208 2566 ± 293	1205 4939 ± 888	1199 14753 ± 3518	227 616 ± 10	234 613 ± 16
422	firstcreditkz-004	1205746	38973	884 2587	682 2048 ± 0	1144 1060 ± 5	979 1236 ± 4	1030 1343 ± 6	1113 1752 ± 15	1141 3765 ± 42	167 525 ± 16	167 521 ± 14
423	foomobi-001	0	219961	68 189	908 2048 ± 0	307 357 ± 0	219 388 ± 2	864 1138 ± 1	1196 2974 ± 10	1190 10101 ± 10	1419 47409 ± 453	1419 47413 ± 421
424	foomobi-002	0	219961	67 189	945 2048 ± 0	236 299 ± 0	162 323 ± 0	700 957 ± 1	1186 2496 ± 2	1180 8549 ± 7	1372 30257 ± 724	1371 30010 ± 320
425	fourthlinecom-001	0	437655	844 2410	732 2048 ± 0	1360 1372 ± 5	997 1257 ± 4	985 1283 ± 2	878 1243 ± 3	745 1244 ± 4	809 2391 ± 17	809 2403 ± 26
426	fpt-000	19013	170633	74 219	677 2048 ± 0	61 118 ± 0	38 122 ± 1	31 122 ± 0	26 120 ± 0	31 178 ± 2	1010 3871 ± 31	1012 3883 ± 29
427	fpt-001	256529	142347	230 600	550 2048 ± 0	608 602 ± 0	415 606 ± 0	393 623 ± 1	349 621 ± 0	294 637 ± 0	1016 3934 ± 10	1019 3931 ± 14
428	fpttelecom-000	266955	273536	172 485	788 2048 ± 0	1455 1587 ± 0	1163 1589 ± 0	1151 1593 ± 0	1094 1591 ± 0	973 1599 ± 0	1180 7881 ± 40	1180 7901 ± 48
429	fpttelecom-001	262561	24300	261 642	757 2048 ± 0	758 692 ± 0	478 660 ± 1	484 700 ± 1	435 705 ± 1	345 701 ± 0	727 2029 ± 11	729 2024 ± 35
430	fraudcom-000	354020	561861	425 966	1023 2052 ± 0	546 561 ± 2	353 539 ± 2	320 541 ± 1	292 557 ± 5	258 594 ± 6	778 2258 ± 10	778 2256 ± 29
431	frpkauai-001	507771	24807	1013 4143	936 2048 ± 0	748 689 ± 1	517 691 ± 0	481 697 ± 2	446 714 ± 6	416 775 ± 31	320 752 ± 29	334 764 ± 23
432	frpkauai-002	519141	24803	503 1113	448 2048 ± 0	878 799 ± 0	782 987 ± 0	787 1046 ± 1	803 1163 ± 2	1007 1769 ± 4	384 907 ± 20	382 886 ± 28
433	frpkauai-003	1135621	24939	817 2184	865 2048 ± 0	1041 948 ± 3	875 1110 ± 2	935 1225 ± 5	973 1369 ± 16	1040 2007 ± 34	166 524 ± 14	171 529 ± 19
434	frpkauai-004	513748	3521	556 1248	74 296 ± 0	263 317 ± 1	156 316 ± 3	125 317 ± 1	115 322 ± 1	106 370 ± 9	1174 7700 ± 81	1173 7659 ± 78
435	frpkauai-005	1250936	39141	874 2544	1011 2052 ± 0	1456 1592 ± 2	1182 1814 ± 2	1181 1910 ± 6	1167 2094 ± 12	1120 2906 ± 39	492 1181 ± 71	499 1182 ± 53
436	fujitsulab-000	0	205894	-	130 512 ± 0	386 419 ± 1	-	-	-	-	984 3613 ± 37	988 3621 ± 29
437	fujitsulab-001	0	548917	-	1373 4104 ± 0	1325 1300 ± 5	1022 1291 ± 2	1000 1295 ± 2	923 1304 ± 2	853 1382 ± 2	887 2853 ± 16	887 2847 ± 14
438	fujitsulab-002	0	1088887	703 1685	1369 4104 ± 0	1281 1237 ± 2	969 1222 ± 2	944 1236 ± 1	884 1251 ± 2	819 1327 ± 2	883 2836 ± 25	882 2809 ± 44
439	fujitsulab-003	662263	318209	1072 6691	1375 4104 ± 0	1049 951 ± 20	736 941 ± 19	693 952 ± 19	645 971 ± 20	601 1045 ± 21	888 2855 ± 16	889 2849 ± 19
440	g42-intellibrain-001	1031335	235521	1105 25154	67 269 ± 0	1076 976 ± 6	767 975 ± 1	743 997 ± 2	721 1068 ± 3	839 1362 ± 8	1116 5878 ± 96	1116 5865 ± 71

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 32: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
441	g42-intellibrain-002	1043920	132389	1101 19747	68 269 ± 0	1006 915 ± 2	715 920 ± 2	689 946 ± 2	669 998 ± 4	733 1224 ± 8	1268 14703 ± 56	1272 14658 ± 97
442	geo-000	114483	304983	-	876 2048 ± 0	937 851 ± 1	-	-	-	-	1483 137544 ± 6225	1483 137925 ± 6548
443	geo-001	258516	70163	-	365 2048 ± 0	1155 1079 ± 0	843 1076 ± 0	813 1076 ± 0	732 1078 ± 0	647 1102 ± 0	931 3163 ± 47	935 3158 ± 38
444	geo-002	369903	98667	449 1021	355 2048 ± 0	865 791 ± 1	610 793 ± 0	561 794 ± 0	513 795 ± 1	434 803 ± 1	956 3407 ± 45	965 3422 ± 65
445	geo-003	371712	102175	-	870 2048 ± 0	1312 1283 ± 1	1021 1290 ± 1	987 1285 ± 1	919 1292 ± 1	796 1302 ± 1	411 997 ± 13	420 1001 ± 20
446	geo-004	168980	107714	571 1287	808 2048 ± 0	1303 1268 ± 1	1012 1279 ± 1	978 1274 ± 0	892 1259 ± 1	788 1296 ± 1	422 1023 ± 20	431 1028 ± 22
447	gistouch-000	155338	444582	205 541	818 2048 ± 0	270 325 ± 1	166 326 ± 1	138 337 ± 1	131 354 ± 1	186 495 ± 1	1198 9505 ± 604	1197 9443 ± 388
448	gistouch-001	426635	444545	696 1651	896 2048 ± 0	1021 930 ± 3	714 919 ± 3	680 937 ± 4	644 966 ± 2	654 1113 ± 1	1188 8475 ± 133	1189 8460 ± 173
449	glory-000	0	98176	-	85 418 ± 0	96 165 ± 2	-	-	-	-	1156 7003 ± 84	1152 6978 ± 71
450	glory-001	0	144786	-	331 1726 ± 0	354 393 ± 2	-	-	-	-	1206 9607 ± 128	1201 9539 ± 182
451	glory-002	0	385177	-	1171 2106 ± 0	600 594 ± 3	566 740 ± 3	691 948 ± 3	1172 2168 ± 6	35 191 ± 15	1147 6787 ± 85	1137 6551 ± 249
452	glory-003	0	536910	-	1414 4234 ± 0	466 489 ± 0	379 565 ± 0	516 732 ± 0	1138 1876 ± 2	1181 8941 ± 20	1122 6020 ± 90	1122 6003 ± 72
453	glory-004	0	999639	822 2200	1406 4182 ± 0	746 688 ± 0	585 759 ± 1	683 941 ± 1	1169 2134 ± 4	1186 9360 ± 47	1079 4982 ± 66	1080 4990 ± 63
454	glory-005	0	999999	856 2450	1409 4182 ± 0	773 703 ± 1	608 789 ± 0	714 972 ± 1	1174 2200 ± 25	1188 9679 ± 22	1089 5224 ± 93	1088 5176 ± 81
455	glory-006	0	1004784	852 2438	1407 4182 ± 0	774 704 ± 1	599 774 ± 0	711 968 ± 2	1180 2286 ± 18	1192 10381 ± 151	1114 5853 ± 77	1115 5819 ± 74
456	glory-007	0	1004846	853 2442	1408 4182 ± 0	346 385 ± 0	255 431 ± 0	349 575 ± 14	1049 1464 ± 24	1172 6782 ± 62	970 3506 ± 51	973 3489 ± 66
457	gorilla-000	98290	1122916	-	1412 4204 ± 0	188 256 ± 14	-	-	-	-	1012 3875 ± 29	1011 3872 ± 51
458	gorilla-001	93768	1101838	-	1178 2156 ± 0	92 160 ± 14	-	-	-	-	959 3429 ± 145	953 3288 ± 51
459	gorilla-002	93869	1281839	-	296 1132 ± 0	265 322 ± 14	-	-	-	-	864 2715 ± 68	847 2585 ± 84
460	gorilla-003	94409	1281823	-	295 1132 ± 0	283 334 ± 25	-	-	-	-	884 2840 ± 42	892 2865 ± 87
461	gorilla-004	186952	1274131	-	1182 2192 ± 0	349 389 ± 3	-	-	-	-	873 2768 ± 44	870 2745 ± 44
462	gorilla-005	100684	1297614	-	1181 2192 ± 0	373 407 ± 3	265 437 ± 2	233 448 ± 4	227 473 ± 4	254 589 ± 15	856 2678 ± 42	874 2770 ± 112
463	gorilla-006	172743	1318812	-	1416 4240 ± 0	424 454 ± 3	307 484 ± 3	281 497 ± 5	279 543 ± 25	523 928 ± 60	1002 3755 ± 38	1003 3737 ± 44
464	gorilla-007	441058	708166	702 1683	1458 6288 ± 0	596 592 ± 1	404 592 ± 1	372 603 ± 1	353 625 ± 2	363 722 ± 9	991 3686 ± 37	998 3709 ± 36
465	gorilla-008	450175	707000	743 1812	1483 8338 ± 0	603 595 ± 1	403 590 ± 0	370 600 ± 1	348 621 ± 2	361 720 ± 9	1055 4530 ± 44	1053 4524 ± 38
466	gorilla-009	329584	297395	584 1304	1417 4242 ± 0	985 899 ± 2	719 922 ± 1	647 901 ± 3	606 924 ± 4	595 1032 ± 12	787 2294 ± 74	788 2301 ± 66
467	gpstechvn-000	108365	345975	889 2646	729 2048 ± 0	231 296 ± 1	147 297 ± 1	115 304 ± 0	108 316 ± 0	114 382 ± 0	818 2442 ± 11	820 2440 ± 16
468	graymatics-001	13095	70406	37 128	1326 4096 ± 0	126 191 ± 1	86 203 ± 1	363 592 ± 5	1104 1698 ± 9	1174 7150 ± 34	1404 39874 ± 309	1406 39762 ± 295
469	griaule-000	0	598214	467 1055	1032 2052 ± 0	381 416 ± 6	249 425 ± 7	549 770 ± 14	1112 1749 ± 43	1170 6406 ± 189	1023 3987 ± 42	1023 3938 ± 38
470	griaule-001	0	412061	564 1271	1018 2052 ± 0	1219 1164 ± 1	864 1096 ± 5	831 1099 ± 4	782 1136 ± 2	935 1509 ± 2	1019 3948 ± 23	1026 3957 ± 32
471	griaule-002	0	1320474	756 1870	1004 2052 ± 0	908 822 ± 1	721 924 ± 4	651 907 ± 1	702 1038 ± 21	885 1430 ± 9	1024 4005 ± 32	1030 4012 ± 31
472	gtelots-000	252631	59244	208 552	129 512 ± 0	859 785 ± 1	612 794 ± 0	569 810 ± 0	533 823 ± 2	490 875 ± 0	1327 22029 ± 130	1331 21988 ± 93
473	gtelots-001	442881	58709	426 966	1180 2176 ± 0	1121 1027 ± 0	814 1028 ± 0	779 1037 ± 0	722 1069 ± 0	701 1167 ± 1	154 493 ± 31	159 496 ± 38
474	hertasecurity-000	0	780014	-	33 256 ± 0	47 99 ± 0	26 98 ± 0	24 100 ± 0	22 107 ± 0	21 139 ± 0	300 710 ± 31	280 667 ± 28
475	hertasecurity-001	0	944427	536 1188	107 512 ± 0	294 346 ± 0	178 345 ± 0	145 349 ± 0	130 354 ± 0	119 388 ± 0	665 1770 ± 45	659 1726 ± 48
476	hertasecurity-002	0	944582	530 1182	151 512 ± 0	462 484 ± 7	299 478 ± 3	263 480 ± 3	243 495 ± 3	213 520 ± 3	784 2289 ± 40	781 2267 ± 48
477	hertasecurity-003	0	944583	529 1181	98 512 ± 0	471 495 ± 7	301 480 ± 3	270 486 ± 3	239 491 ± 3	214 520 ± 4	1172 7619 ± 65	1172 7597 ± 94
478	hik-001	667866	9290	1070 6597	308 1408 ± 0	686 651 ± 0	486 667 ± 8	462 677 ± 16	415 686 ± 13	376 737 ± 12	150 488 ± 19	153 477 ± 22
479	hisign-001	732412	167488	670 1560	1162 2080 ± 0	1328 1306 ± 1	1044 1320 ± 1	1010 1315 ± 1	933 1312 ± 1	814 1325 ± 1	45 201 ± 10	37 185 ± 13
480	hisign-002	1014906	102652	804 2131	1163 2080 ± 0	875 797 ± 0	616 800 ± 5	564 800 ± 0	516 801 ± 0	435 803 ± 1	56 232 ± 11	46 207 ± 11
481	hisign-003	2232829	156367	682 1599	1161 2080 ± 0	1132 1040 ± 5	847 1083 ± 4	818 1085 ± 4	738 1088 ± 4	644 1093 ± 4	28 167 ± 8	25 150 ± 10
482	hyperverge-001	260819	88624	-	619 2048 ± 0	737 682 ± 20	521 695 ± 17	915 1196 ± 37	1183 2400 ± 68	1175 7178 ± 204	1124 6026 ± 40	1120 5984 ± 38
483	hyperverge-002	2951900	198832	777 1975	262 1024 ± 0	1025 938 ± 1	734 939 ± 1	684 941 ± 1	624 945 ± 1	552 975 ± 1	1123 6023 ± 37	1119 5966 ± 40
484	hyperverge-003	1167779	282156	896 2687	236 1024 ± 0	1433 1477 ± 2	1146 1503 ± 3	1135 1520 ± 3	1075 1525 ± 4	964 1565 ± 3	181 566 ± 11	185 561 ± 8

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 33: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
485	hyperverge-004	4924393	282156	1000 3908	937 2048 ± 0	1428 1471 ± 2	1117 1434 ± 1	1104 1446 ± 6	1033 1445 ± 3	926 1491 ± 3	410 996 ± 10	418 1000 ± 19
486	hyperverge-005	3413303	284050	895 2686	252 1024 ± 0	750 689 ± 0	516 690 ± 0	474 693 ± 0	432 702 ± 6	389 752 ± 5	136 465 ± 2	149 464 ± 10
487	hyperverge-006	3901500	284050	793 2070	253 1024 ± 0	842 768 ± 1	595 770 ± 2	545 766 ± 1	497 773 ± 1	451 828 ± 6	137 466 ± 2	148 463 ± 4
488	hyperverge-007	2223041	169168	813 2159	260 1024 ± 0	1340 1327 ± 1	1028 1296 ± 0	983 1280 ± 0	916 1286 ± 0	828 1348 ± 0	264 659 ± 19	291 684 ± 26
489	hyundai-roboticslab-000	181134	191546	516 1149	810 2048 ± 0	331 376 ± 3	202 368 ± 1	163 367 ± 2	146 374 ± 0	132 408 ± 1	879 2815 ± 47	880 2790 ± 50
490	hyundai-roboticslab-001	361330	106869	352 820	674 2048 ± 0	309 358 ± 1	192 361 ± 1	159 362 ± 2	147 374 ± 3	124 394 ± 0	924 3115 ± 34	927 3117 ± 45
491	hyundai-roboticslab-002	181140	106752	348 817	546 2048 ± 0	295 347 ± 0	188 355 ± 0	153 357 ± 0	136 362 ± 3	118 387 ± 0	940 3243 ± 111	948 3245 ± 114
492	hyundai-roboticslab-003	484277	106656	787 2027	460 2048 ± 0	1473 1800 ± 0	1180 1792 ± 0	1169 1802 ± 0	1140 1889 ± 0	1014 1810 ± 0	1194 8841 ± 42	1195 8819 ± 54
493	hyundai-roboticslab-004	484277	106656	786 2026	574 2048 ± 0	1481 1882 ± 1	1187 1842 ± 2	1173 1839 ± 0	1127 1837 ± 0	1020 1871 ± 0	1224 10743 ± 201	1224 10767 ± 219
494	hzailu-001	0	372018	-	1123 2056 ± 0	1189 1126 ± 1	890 1128 ± 1	859 1130 ± 1	778 1132 ± 1	693 1159 ± 1	379 894 ± 19	384 899 ± 22
495	hzailu-002	1515880	74047	1035 4722	1122 2056 ± 0	1208 1150 ± 5	889 1127 ± 6	857 1129 ± 7	784 1137 ± 7	704 1172 ± 3	448 1079 ± 53	452 1070 ± 31
496	hzailu-003	1923030	222185	1042 4858	1216 3080 ± 0	1377 1389 ± 5	1054 1331 ± 7	1025 1334 ± 2	955 1349 ± 6	881 1424 ± 8	586 1483 ± 35	583 1464 ± 31
497	hzailu-004	1938475	275639	1006 4023	1215 3080 ± 0	1442 1495 ± 8	1141 1490 ± 6	1126 1493 ± 8	1066 1502 ± 8	967 1583 ± 17	538 1361 ± 32	563 1419 ± 44
498	hzailu-005	1573726	275643	959 3292	1434 5128 ± 0	741 684 ± 10	497 673 ± 7	390 622 ± 1	409 682 ± 6	356 715 ± 2	689 1853 ± 51	686 1829 ± 34
499	hzailu-006	2996729	275643	1065 6070	1465 7176 ± 0	1443 1498 ± 11	1140 1483 ± 15	1098 1439 ± 8	1031 1441 ± 10	962 1564 ± 40	814 2434 ± 19	824 2452 ± 32
500	hzailu-007	2906424	275643	1059 5897	1467 7176 ± 0	1404 1430 ± 19	1125 1448 ± 10	1019 1328 ± 3	1068 1504 ± 17	888 1434 ± 4	828 2481 ± 28	832 2488 ± 34
501	hzailu-008	2906977	275643	1060 5898	1466 7176 ± 0	1418 1452 ± 13	1150 1557 ± 10	1138 1545 ± 12	1081 1561 ± 20	949 1533 ± 16	830 2492 ± 62	833 2489 ± 66
502	i2v-001	276820	26600	346 815	747 2048 ± 0	791 721 ± 0	555 729 ± 0	514 731 ± 0	460 731 ± 0	403 764 ± 0	1232 11523 ± 176	1234 11533 ± 142
503	i2v-002	339732	122488	712 1721	767 2048 ± 0	1411 1438 ± 1	1139 1476 ± 1	1099 1442 ± 1	1058 1480 ± 9	950 1533 ± 1	263 656 ± 35	223 604 ± 28
504	icehouse-000	287028	178880	394 887	771 2048 ± 0	656 629 ± 1	432 620 ± 1	432 648 ± 1	419 690 ± 4	542 961 ± 11	388 910 ± 11	401 938 ± 16
505	icm-002	621586	903	-	905 2048 ± 0	1126 1031 ± 7	-	-	-	-	1343 24052 ± 118	1344 24049 ± 124
506	icm-003	1513988	940	192 516	379 2048 ± 0	733 681 ± 6	492 672 ± 4	498 714 ± 11	545 837 ± 41	851 1381 ± 131	1345 24351 ± 161	1345 24227 ± 146
507	icm-004	2012129	1089	466 1052	572 2048 ± 0	387 419 ± 6	237 407 ± 6	240 454 ± 15	328 603 ± 51	946 1527 ± 235	1269 14730 ± 154	1271 14521 ± 152
508	icm-005	877530	1097	504 1116	472 2048 ± 0	119 185 ± 3	67 181 ± 3	69 208 ± 9	113 321 ± 36	585 1014 ± 185	1214 10245 ± 65	1216 10245 ± 69
509	icomm-000	807818	139431	1074 6728	1320 4096 ± 0	614 605 ± 26	417 610 ± 30	374 605 ± 27	332 610 ± 23	299 641 ± 32	1189 8507 ± 25	1190 8516 ± 26
510	icthtc-000	172459	1471004	521 1165	721 2048 ± 0	287 338 ± 11	173 338 ± 9	223 437 ± 16	436 705 ± 24	1000 1719 ± 44	1092 5284 ± 63	1093 5290 ± 54
511	id3-001	225574	34433	-	174 520 ± 0	169 238 ± 19	-	-	-	-	435 1058 ± 32	444 1049 ± 28
512	id3-002	225574	34433	-	173 520 ± 0	459 482 ± 34	-	-	-	-	460 1100 ± 59	441 1048 ± 32
513	id3-003	265951	49731	-	60 264 ± 0	258 316 ± 19	-	-	-	-	533 1330 ± 25	546 1354 ± 28
514	id3-004	171526	49725	-	66 264 ± 0	524 541 ± 11	-	-	-	-	471 1135 ± 23	493 1156 ± 32
515	id3-005	153439	22125	-	61 264 ± 0	457 479 ± 1	-	-	-	-	1126 6072 ± 35	1125 6077 ± 38
516	id3-006	210116	7706	432 984	177 520 ± 0	740 683 ± 0	852 1088 ± 1	908 1192 ± 1	847 1209 ± 1	769 1270 ± 1	1102 5547 ± 34	1104 5563 ± 34
517	id3-007	185030	7728	-	63 264 ± 0	1109 1016 ± 1	900 1139 ± 1	1212 3911 ± 2	1210 10752 ± 12	1206 135182 ± 1912	1098 5500 ± 29	1097 5486 ± 35
518	id3-008	242416	8151	471 1059	58 264 ± 0	901 819 ± 0	960 1209 ± 2	1001 1297 ± 2	946 1329 ± 1	886 1433 ± 1	1108 5658 ± 44	1109 5624 ± 40
519	id3-009	128638	9510	465 1051	64 264 ± 0	496 520 ± 1	336 518 ± 1	298 515 ± 2	262 519 ± 1	207 514 ± 2	272 682 ± 34	283 677 ± 44
520	id3-010	128638	9510	469 1056	59 264 ± 0	515 535 ± 2	350 535 ± 2	461 677 ± 2	791 1147 ± 4	1116 2889 ± 5	258 651 ± 34	243 621 ± 16
521	id3-011	128638	9562	470 1056	62 264 ± 0	518 536 ± 2	343 526 ± 2	442 661 ± 2	776 1132 ± 2	1119 2899 ± 2	260 654 ± 20	281 668 ± 11
522	idbio-000	2	325054	336 796	180 522 ± 0	933 845 ± 3	652 847 ± 2	606 847 ± 2	556 853 ± 2	516 918 ± 1	1057 4553 ± 22	1058 4623 ± 16
523	idemia-003	427244	106432	-	80 352 ± 0	324 368 ± 6	-	-	-	-	1142 6654 ± 75	1073 4835 ± 90
524	idemia-004	406924	106436	-	81 352 ± 0	247 306 ± 5	-	-	-	-	1105 5592 ± 518	1100 5533 ± 426
525	idemia-005	509824	116761	-	201 588 ± 0	491 514 ± 15	-	-	-	-	1143 6657 ± 54	1140 6616 ± 53
526	idemia-006	570566	115633	-	204 668 ± 0	731 679 ± 4	724 926 ± 54	512 728 ± 7	486 760 ± 29	453 830 ± 31	1088 5223 ± 80	1089 5193 ± 72
527	idemia-007	353242	67485	-	91 468 ± 0	343 384 ± 0	221 389 ± 0	186 393 ± 1	174 405 ± 2	152 441 ± 8	939 3243 ± 63	939 3202 ± 63
528	idemia-008	374017	69922	540 1192	77 348 ± 0	428 457 ± 1	289 461 ± 0	253 466 ± 1	230 476 ± 2	204 513 ± 10	919 3080 ± 41	921 3046 ± 56

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 34: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
529	idemia-009	1066728	70572	900 2700	203 636 ± 0	1254 1207 ± 1	965 1218 ± 1	932 1222 ± 2	860 1222 ± 3	776 1280 ± 10	1109 5664 ± 84	1107 5597 ± 90
530	idemia-010	996690	66244	837 2306	84 412 ± 0	604 596 ± 1	456 632 ± 22	394 624 ± 18	357 633 ± 7	360 719 ± 5	645 1699 ± 37	649 1703 ± 38
531	idemia-011	472694	65098	967 3401	202 612 ± 0	897 816 ± 25	656 853 ± 9	437 658 ± 2	509 785 ± 3	371 732 ± 8	844 2583 ± 29	850 2627 ± 48
532	identy-000	271397	14217	166 474	588 2048 ± 0	751 689 ± 0	529 702 ± 0	497 712 ± 3	483 757 ± 0	548 970 ± 1	262 655 ± 25	263 637 ± 23
533	identy-001	385977	17850	535 1187	560 2048 ± 0	848 772 ± 1	649 841 ± 1	796 1056 ± 1	1085 1567 ± 5	1145 4081 ± 9	589 1484 ± 16	590 1480 ± 17
534	identy-002	1136161	40480	845 2414	1207 3072 ± 0	1446 1548 ± 0	1161 1584 ± 0	1141 1549 ± 0	1114 1757 ± 1	1075 2246 ± 1	913 3020 ± 41	919 3007 ± 19
535	identy-003	1245276	74623	635 1444	1210 3072 ± 0	1465 1730 ± 1	1172 1739 ± 0	1168 1794 ± 1	1155 1996 ± 3	1109 2775 ± 11	495 1204 ± 11	506 1198 ± 5
536	idrnd-000	1069136	69522	965 3357	1371 4104 ± 0	887 810 ± 1	638 828 ± 11	592 830 ± 10	573 877 ± 5	476 857 ± 9	1231 11504 ± 47	1233 11494 ± 62
537	idrnd-001	1080396	107151	971 3437	1372 4104 ± 0	599 594 ± 1	424 615 ± 6	387 615 ± 6	341 616 ± 6	297 640 ± 8	662 1757 ± 8	668 1759 ± 16
538	igearx-face-000	665623	298245	393 885	390 2048 ± 0	945 860 ± 1	658 855 ± 4	603 844 ± 0	578 881 ± 0	521 924 ± 3	630 1628 ± 45	637 1657 ± 41
539	iit-000	237317	33898	-	265 1024 ± 0	136 197 ± 8	-	-	-	-	609 1537 ± 81	529 1282 ± 20
540	iit-001	269176	52055	-	477 2048 ± 0	767 699 ± 4	-	-	-	-	436 1060 ± 48	453 1074 ± 54
541	iit-002	259579	52070	307 731	438 2048 ± 0	492 514 ± 1	345 531 ± 2	328 547 ± 1	309 583 ± 1	372 733 ± 2	423 1023 ± 7	423 1011 ± 66
542	iit-003	261288	53791	353 821	918 2048 ± 0	458 482 ± 0	313 493 ± 0	294 509 ± 0	277 541 ± 0	312 661 ± 0	90 324 ± 17	96 326 ± 8
543	imds-software-001	373399	352623	330 785	799 2048 ± 0	437 465 ± 1	752 958 ± 6	860 1131 ± 5	779 1134 ± 2	663 1119 ± 10	1239 11885 ± 120	1240 11779 ± 174
544	imds-software-002	373632	352627	356 833	576 2048 ± 0	445 469 ± 1	745 954 ± 2	827 1094 ± 5	717 1064 ± 4	689 1153 ± 4	1227 11100 ± 155	1228 11090 ± 145
545	imds-software-003	373648	352628	324 769	664 2048 ± 0	262 317 ± 5	473 654 ± 4	559 793 ± 6	496 770 ± 4	472 848 ± 2	1141 6624 ± 66	1138 6603 ± 77
546	imperial-000	370120	10623	337 796	679 2048 ± 0	709 669 ± 1	499 675 ± 3	465 683 ± 17	405 676 ± 2	335 689 ± 2	749 2130 ± 32	737 2052 ± 100
547	imperial-001	370260	10623	-	944 2048 ± 0	712 671 ± 0	-	-	-	-	743 2090 ± 28	740 2062 ± 74
548	imperial-002	472327	16134	746 1826	453 2048 ± 0	557 569 ± 1	394 581 ± 15	348 575 ± 5	304 576 ± 2	252 588 ± 3	782 2278 ± 90	750 2131 ± 44
549	incode-000	23697	18014	-	248 1024 ± 0	134 196 ± 9	-	-	-	-	1086 5175 ± 81	1076 4907 ± 77
550	incode-001	155416	18020	-	759 2048 ± 0	476 499 ± 9	-	-	-	-	1129 6154 ± 96	1124 6045 ± 182
551	incode-002	73239	32435	-	761 2048 ± 0	213 281 ± 15	-	-	-	-	611 1544 ± 38	595 1488 ± 69
552	incode-003	170632	42857	-	1286 4096 ± 0	344 384 ± 11	-	-	-	-	709 1928 ± 44	697 1876 ± 81
553	incode-004	260224	51685	-	690 2048 ± 0	455 479 ± 23	-	-	-	-	704 1913 ± 60	821 2443 ± 114
554	incode-005	256242	51670	-	833 2048 ± 0	552 566 ± 1	-	-	-	-	644 1695 ± 38	638 1663 ± 85
555	incode-006	266095	63518	-	528 2048 ± 0	449 472 ± 1	-	-	-	-	670 1788 ± 41	681 1798 ± 59
556	incode-007	266103	63524	-	684 2048 ± 0	446 470 ± 1	333 516 ± 0	300 518 ± 1	269 530 ± 2	268 603 ± 4	675 1799 ± 35	677 1789 ± 59
557	incode-008	266103	21014	-	884 2048 ± 0	478 500 ± 0	312 493 ± 0	280 496 ± 0	250 506 ± 1	222 537 ± 0	444 1070 ± 28	459 1081 ± 34
558	incode-009	266103	21014	-	857 2048 ± 0	482 503 ± 0	311 490 ± 1	282 498 ± 0	249 505 ± 0	223 537 ± 0	461 1102 ± 28	472 1113 ± 29
559	incode-010	627808	21014	873 2543	544 2048 ± 0	1235 1180 ± 2	930 1178 ± 1	900 1182 ± 1	825 1184 ± 1	730 1221 ± 1	487 1164 ± 32	482 1144 ± 32
560	incode-011	477280	21781	642 1483	603 2048 ± 0	958 872 ± 0	678 875 ± 0	636 881 ± 1	588 892 ± 1	529 939 ± 0	466 1117 ± 31	470 1109 ± 37
561	incode-012	477371	21786	640 1479	607 2048 ± 0	497 521 ± 0	329 506 ± 0	287 503 ± 0	258 514 ± 0	246 571 ± 0	817 2441 ± 25	828 2466 ± 37
562	incode-013	483720	41434	693 1647	513 2048 ± 0	514 535 ± 1	365 548 ± 1	333 555 ± 4	283 545 ± 1	279 618 ± 1	482 1152 ± 37	492 1154 ± 41
563	incode-014	2365157	41516	1076 6927	1474 8196 ± 0	1146 1064 ± 4	853 1089 ± 2	833 1100 ± 5	750 1107 ± 7	711 1190 ± 3	629 1627 ± 42	631 1640 ± 59
564	incode-015	1808779	70449	1068 6430	1448 6148 ± 0	1142 1056 ± 2	936 1182 ± 12	816 1081 ± 2	736 1084 ± 1	710 1187 ± 2	542 1373 ± 22	544 1352 ± 35
565	incode-016	2321212	105692	1098 12294	1475 8196 ± 0	1495 2027 ± 1	1204 2036 ± 0	1193 2040 ± 0	1160 2051 ± 1	1050 2068 ± 0	755 2162 ± 51	760 2177 ± 93
566	infocert-001	1204340	38972	644 1490	529 2048 ± 0	962 874 ± 1	691 891 ± 1	790 1050 ± 5	1052 1473 ± 2	1130 3174 ± 8	1084 5055 ± 108	1082 5008 ± 100
567	infocert-002	186823	19501	257 641	672 2048 ± 0	814 742 ± 0	563 738 ± 0	521 738 ± 0	468 738 ± 0	381 739 ± 0	901 2917 ± 114	905 2930 ± 50
568	infocert-003	186823	17115	193 517	599 2048 ± 0	835 763 ± 0	588 764 ± 0	541 764 ± 0	489 763 ± 0	405 766 ± 0	420 1022 ± 7	428 1023 ± 19
569	infodif-001	113229	2541	30 113	1477 8200 ± 0	38 92 ± 0	21 84 ± 0	18 89 ± 0	15 90 ± 0	22 140 ± 1	47 203 ± 1	64 245 ± 8
570	innefulabs-000	370588	162172	149 439	975 2048 ± 0	1102 1006 ± 3	811 1025 ± 3	773 1030 ± 4	705 1041 ± 2	677 1135 ± 3	1112 5782 ± 41	1113 5741 ± 45
571	innominds-001	102207	777689	506 1124	351 2048 ± 0	281 333 ± 21	184 350 ± 19	135 335 ± 16	125 339 ± 17	94 346 ± 18	829 2482 ± 23	834 2491 ± 29
572	innov8tif-000	170383	2163199	645 1494	1262 4096 ± 0	285 335 ± 23	182 348 ± 21	655 912 ± 64	1190 2661 ± 142	1195 10665 ± 528	192 591 ± 11	197 582 ± 11

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 35: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
573	innov8tif-001	118502	1241515	186 508	1310 4096 ± 0	963 875 ± 83	708 915 ± 73	1207 2360 ± 185	1206 6410 ± 390	1201 24097 ± 1477	332 775 ± 31	339 778 ± 14
574	innov8tif-002	258255	1108662	263 648	1147 2064 ± 0	435 464 ± 17	295 471 ± 15	344 572 ± 23	306 581 ± 24	284 628 ± 23	1013 3917 ± 72	1018 3924 ± 72
575	innovativetechnologyltd-001	177232	335757	111 341	899 2048 ± 0	401 433 ± 7	276 446 ± 8	226 439 ± 4	202 452 ± 4	179 485 ± 7	694 1877 ± 42	708 1924 ± 97
576	innovativetechnologyltd-002	173939	372324	402 912	727 2048 ± 0	697 661 ± 2	553 726 ± 4	721 981 ± 27	666 997 ± 40	406 766 ± 3	685 1841 ± 50	693 1857 ± 59
577	innovatrics-000	0	143500	-	15 146 ± 0	566 578 ± 5	-	-	-	-	1078 4964 ± 63	1061 4665 ± 262
578	innovatrics-001	0	143500	-	69 284 ± 0	677 645 ± 5	-	-	-	-	1099 5506 ± 131	1079 4975 ± 308
579	innovatrics-002	0	249195	-	183 530 ± 0	88 153 ± 0	-	-	-	-	1263 14384 ± 260	1270 14497 ± 207
580	innovatrics-003	0	538170	-	182 530 ± 0	185 250 ± 0	-	-	-	-	1284 15709 ± 134	1282 15193 ± 292
581	innovatrics-004	0	496988	-	293 1076 ± 0	353 391 ± 0	-	-	-	-	1190 8573 ± 274	1181 7929 ± 244
582	innovatrics-005	0	664175	-	292 1076 ± 0	565 577 ± 1	-	-	-	-	1293 16880 ± 194	1298 17232 ± 385
583	innovatrics-006	0	466269	-	187 538 ± 0	903 820 ± 5	615 799 ± 4	566 805 ± 3	514 796 ± 9	498 890 ± 15	1115 5855 ± 204	1091 5266 ± 118
584	innovatrics-007	0	493269	-	291 1064 ± 0	1438 1485 ± 7	1178 1785 ± 184	1199 2078 ± 24	1168 2123 ± 15	1070 2210 ± 42	1121 5978 ± 88	1111 5690 ± 102
585	innovatrics-008	307323	59842	628 1429	186 538 ± 0	850 778 ± 6	592 767 ± 3	550 770 ± 3	519 803 ± 3	473 853 ± 10	915 3021 ± 66	858 2673 ± 88
586	innovatrics-009	624485	105187	769 1947	1391 4136 ± 0	1184 1116 ± 1	872 1107 ± 5	837 1104 ± 5	753 1110 ± 5	685 1146 ± 6	1081 5051 ± 54	1064 4733 ± 102
587	innovatrics-010	604731	111234	710 1715	1390 4136 ± 0	620 611 ± 3	454 632 ± 2	505 723 ± 1	458 730 ± 1	408 768 ± 6	1031 4077 ± 42	1010 3857 ± 65
588	innovatrics-011	643997	50017	722 1745	1411 4184 ± 0	825 754 ± 2	593 767 ± 3	544 765 ± 3	505 783 ± 4	484 867 ± 3	1033 4135 ± 57	1022 3937 ± 91
589	innovatrics-012	0	680831	734 1789	1427 4721 ± 0	872 797 ± 3	645 837 ± 3	562 795 ± 1	594 904 ± 4	573 1001 ± 3	1471 80879 ± 981	1469 80521 ± 655
590	innovatrics-013	0	1665828	1004 3931	1413 4212 ± 0	1244 1195 ± 2	977 1230 ± 11	933 1223 ± 4	869 1233 ± 5	820 1329 ± 3	1475 85968 ± 990	1473 85796 ± 836
591	innovatrics-014	0	2870004	1067 6138	1460 6303 ± 0	1112 1022 ± 6	817 1031 ± 2	791 1051 ± 1	709 1052 ± 5	699 1163 ± 11	1442 60311 ± 644	1439 60186 ± 499
592	innovatrics-015	0	5116879	1099 15251	1461 6327 ± 0	1350 1350 ± 5	1093 1405 ± 4	1047 1371 ± 3	1003 1405 ± 21	930 1502 ± 17	1449 65175 ± 1027	1445 64294 ± 1390
593	innovatrics-016	0	3130540	1077 6982	1459 6303 ± 0	1499 2084 ± 0	1210 2097 ± 0	1200 2103 ± 1	1171 2139 ± 0	1071 2215 ± 0	1384 32998 ± 309	1386 32967 ± 351
594	innovatrics-017	2456649	28156	1049 5307	1415 4237 ± 0	1451 1569 ± 0	1160 1579 ± 0	1147 1570 ± 0	1088 1574 ± 0	969 1584 ± 0	1386 33161 ± 158	1388 33197 ± 143
595	insightface-000	806953	16606	-	1317 4096 ± 0	1104 1009 ± 1	809 1019 ± 2	758 1017 ± 2	691 1020 ± 2	594 1032 ± 2	667 1778 ± 31	670 1773 ± 35
596	insightface-001	776777	16606	996 3858	717 2048 ± 0	1356 1366 ± 2	1067 1368 ± 3	1049 1372 ± 3	977 1375 ± 5	855 1386 ± 4	467 1119 ± 29	467 1108 ± 34
597	insightface-002	800572	16606	-	913 2048 ± 0	1383 1396 ± 2	1077 1389 ± 4	1074 1403 ± 3	999 1402 ± 2	872 1413 ± 3	488 1169 ± 40	474 1118 ± 38
598	insightface-003	1016917	26668	664 1550	450 2048 ± 0	1153 1073 ± 0	858 1092 ± 2	811 1070 ± 1	735 1082 ± 1	646 1101 ± 1	201 597 ± 16	211 595 ± 17
599	insightface-004	928816	26668	738 1800	971 2048 ± 0	1323 1298 ± 1	1063 1356 ± 2	1009 1313 ± 1	932 1311 ± 0	830 1349 ± 1	180 565 ± 7	184 559 ± 5
600	inspur-000	364844	91926	342 808	1281 4096 ± 0	1357 1367 ± 1	1053 1331 ± 2	1045 1368 ± 2	1051 1465 ± 1	1019 1861 ± 3	1210 9831 ± 37	1210 9860 ± 40
601	inspur-001	364862	91934	358 833	801 2048 ± 0	1375 1385 ± 1	1065 1361 ± 4	1071 1401 ± 2	1064 1495 ± 1	1026 1891 ± 1	1107 5641 ± 42	1108 5622 ± 40
602	inspur-002	447948	91934	441 1009	798 2048 ± 0	1140 1056 ± 5	831 1056 ± 7	830 1096 ± 7	1119 1776 ± 23	896 1444 ± 12	1051 4504 ± 32	1051 4515 ± 35
603	inspur-003	444594	91934	455 1033	706 2048 ± 0	1272 1225 ± 2	950 1199 ± 3	948 1241 ± 1	935 1316 ± 1	987 1645 ± 5	1054 4525 ± 16	1052 4523 ± 26
604	intellicloudai-001	220831	868246	267 655	531 2048 ± 0	443 468 ± 2	283 456 ± 1	252 466 ± 3	241 492 ± 1	287 632 ± 2	433 1056 ± 4	446 1051 ± 72
605	intellicloudai-002	259047	58559	980 3585	1366 4100 ± 0	935 847 ± 1	653 847 ± 2	609 849 ± 1	557 853 ± 1	492 878 ± 4	357 822 ± 28	364 818 ± 23
606	intellifusion-001	271872	289387	321 762	443 2048 ± 0	837 764 ± 38	598 774 ± 39	563 797 ± 42	518 803 ± 34	436 805 ± 33	463 1112 ± 28	475 1128 ± 41
607	intellifusion-002	762731	385841	415 941	1325 4096 ± 0	1045 950 ± 2	865 1096 ± 42	821 1088 ± 33	809 1168 ± 31	702 1171 ± 10	648 1713 ± 57	639 1665 ± 87
608	intellivision-001	43692	11649	-	1121 2056 ± 0	21 62 ± 2	-	-	-	-	842 2573 ± 91	842 2544 ± 38
609	intellivision-002	43692	14505	-	1073 2056 ± 0	267 322 ± 1	187 355 ± 2	168 372 ± 1	181 422 ± 2	263 600 ± 1	1256 13525 ± 134	1255 12782 ± 278
610	intellivision-003	64023	133748	338 800	1117 2056 ± 0	369 407 ± 3	232 398 ± 2	207 418 ± 2	200 450 ± 1	255 591 ± 4	1226 11069 ± 56	1227 11066 ± 75
611	intellivision-004	117727	131310	191 515	1079 2056 ± 0	275 330 ± 0	168 330 ± 0	144 347 ± 0	152 382 ± 0	206 514 ± 0	1228 11197 ± 63	1229 11165 ± 72
612	intellivision-005	372608	257259	285 686	1096 2056 ± 0	640 618 ± 0	437 623 ± 0	415 634 ± 0	385 659 ± 0	413 771 ± 0	1313 19244 ± 68	1315 19270 ± 77
613	intellivision-006	287736	262742	297 723	1080 2056 ± 0	155 214 ± 0	93 214 ± 0	77 221 ± 0	85 272 ± 0	100 357 ± 1	1277 15153 ± 32	1280 15142 ± 38
614	intellivision-007	192454	262742	141 419	1116 2056 ± 0	84 148 ± 0	51 157 ± 0	46 166 ± 0	42 180 ± 0	62 265 ± 0	1274 15100 ± 50	1276 15095 ± 45
615	intellivix-001	256654	111858	369 846	867 2048 ± 0	332 378 ± 1	210 379 ± 1	177 381 ± 1	154 384 ± 1	141 421 ± 3	459 1100 ± 16	469 1109 ± 22
616	intellivix-002	361566	116162	525 1175	490 2048 ± 0	1053 956 ± 0	741 947 ± 6	717 976 ± 0	655 984 ± 4	639 1089 ± 1	1371 30096 ± 128	1376 31287 ± 140

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 36: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
617	intellivix-003	234409	116167	582 1301	825 2048 ± 0	991 908 ± 0	709 916 ± 1	679 930 ± 0	638 961 ± 1	674 1129 ± 3	1370 30025 ± 137	1375 31190 ± 131
618	intellivix-004	361668	128097	539 1189	510 2048 ± 0	396 427 ± 0	266 437 ± 0	231 444 ± 0	212 461 ± 0	243 565 ± 2	597 1498 ± 37	588 1478 ± 16
619	intellivix-005	774405	128093	990 3692	373 2048 ± 0	718 673 ± 1	465 643 ± 3	435 654 ± 3	395 666 ± 3	410 769 ± 5	585 1481 ± 65	575 1451 ± 26
620	intellivix-006	792822	128093	992 3753	367 2048 ± 0	708 665 ± 1	475 657 ± 2	447 662 ± 2	406 677 ± 2	423 784 ± 1	591 1486 ± 13	592 1482 ± 16
621	intelresearch-000	388229	307985	-	585 2048 ± 0	988 902 ± 6	-	-	-	-	1069 4800 ± 152	1056 4561 ± 97
622	intelresearch-001	353997	272602	-	831 2048 ± 0	734 682 ± 4	-	-	-	-	979 3553 ± 57	970 3462 ± 161
623	intelresearch-002	452850	86454	-	398 2048 ± 0	777 707 ± 2	609 790 ± 33	555 788 ± 26	541 831 ± 29	480 862 ± 22	1034 4204 ± 91	1035 4153 ± 93
624	intelresearch-003	401343	85085	526 1177	966 2048 ± 0	1276 1232 ± 3	982 1237 ± 2	950 1242 ± 2	897 1263 ± 2	813 1324 ± 3	1048 4443 ± 75	1042 4374 ± 77
625	intelresearch-004	646918	85290	755 1866	628 2048 ± 0	1336 1319 ± 2	1046 1322 ± 3	1021 1330 ± 3	952 1345 ± 3	870 1411 ± 5	1063 4696 ± 63	1063 4692 ± 66
626	intelresearch-005	398137	85290	520 1164	736 2048 ± 0	1341 1328 ± 1	1056 1334 ± 2	1031 1344 ± 2	958 1356 ± 2	880 1423 ± 4	1053 4524 ± 87	1046 4461 ± 74
627	intelresearch-006	445223	101126	478 1069	646 2048 ± 0	1010 918 ± 1	684 881 ± 0	641 892 ± 0	599 913 ± 1	580 1008 ± 3	1127 6137 ± 410	1123 6024 ± 109
628	intelresearch-007	523218	101150	544 1197	957 2048 ± 0	593 590 ± 1	398 585 ± 1	359 588 ± 1	329 603 ± 4	320 668 ± 6	999 3738 ± 42	1004 3752 ± 53
629	intelresearch-008	523218	101214	543 1196	426 2048 ± 0	667 634 ± 1	448 630 ± 1	410 632 ± 1	358 635 ± 2	354 711 ± 4	987 3661 ± 34	989 3641 ± 44
630	intelresearch-009	290599	120716	656 1535	723 2048 ± 0	545 560 ± 1	482 664 ± 3	421 638 ± 3	408 678 ± 13	339 692 ± 4	1007 3829 ± 207	993 3679 ± 71
631	intelresearch-010	307886	120767	697 1655	500 2048 ± 0	1211 1153 ± 1	916 1158 ± 2	883 1160 ± 2	806 1166 ± 3	739 1237 ± 5	1009 3850 ± 26	1009 3822 ± 75
632	intelresearch-011	317134	120757	726 1758	934 2048 ± 0	1279 1236 ± 4	966 1219 ± 3	941 1231 ± 1	876 1237 ± 3	803 1312 ± 6	1011 3873 ± 47	1013 3890 ± 57
633	intelresearch-012	317135	120715	725 1758	454 2048 ± 0	1311 1280 ± 2	996 1256 ± 4	973 1270 ± 3	915 1284 ± 3	822 1331 ± 5	1001 3744 ± 32	1000 3722 ± 44
634	intema-000	1532392	19488	489 1088	159 513 ± 0	1106 1010 ± 0	796 1001 ± 4	739 994 ± 0	662 993 ± 5	611 1056 ± 1	387 910 ± 29	390 906 ± 32
635	intema-001	1122562	19536	637 1458	161 513 ± 0	1353 1354 ± 1	1043 1318 ± 5	1026 1336 ± 4	944 1328 ± 2	848 1375 ± 0	404 977 ± 31	410 980 ± 31
636	interlinklabs-001	155897	1013105	314 746	687 2048 ± 0	272 328 ± 3	179 346 ± 8	141 343 ± 7	129 352 ± 3	139 418 ± 11	568 1441 ± 10	574 1450 ± 14
637	interlinklabs-002	286341	1013147	605 1364	1255 4096 ± 0	411 438 ± 1	263 435 ± 4	222 435 ± 5	195 439 ± 0	196 505 ± 2	859 2699 ± 35	860 2678 ± 25
638	interlinklabs-003	528962	951057	662 1549	1296 4096 ± 0	805 734 ± 0	577 750 ± 0	529 746 ± 0	493 769 ± 0	526 932 ± 0	972 3513 ± 50	977 3504 ± 26
639	intozi-001	21863	74210	44 140	371 2048 ± 0	35 85 ± 0	22 86 ± 0	16 86 ± 0	13 87 ± 0	11 87 ± 0	1203 9579 ± 252	1200 9530 ± 167
640	intozi-002	172046	95119	367 844	372 2048 ± 0	1452 1571 ± 0	1162 1585 ± 1	1150 1580 ± 0	1092 1585 ± 0	990 1665 ± 1	351 805 ± 14	355 803 ± 4
641	intsysmsu-000	650193	171481	-	352 2048 ± 0	516 535 ± 20	-	-	-	-	217 610 ± 22	235 613 ± 31
642	intsysmsu-001	384409	172480	333 789	652 2048 ± 0	632 614 ± 2	423 615 ± 2	427 642 ± 2	476 750 ± 3	695 1159 ± 4	230 621 ± 8	229 611 ± 31
643	intsysmsu-002	765921	172298	332 786	266 1024 ± 0	598 593 ± 1	611 793 ± 2	587 827 ± 1	571 875 ± 104	787 1293 ± 3	173 549 ± 25	180 548 ± 29
644	ionetworks-000	287609	51236	116 352	415 2048 ± 0	399 430 ± 0	262 435 ± 0	220 433 ± 0	188 432 ± 0	155 444 ± 0	1151 6913 ± 102	1158 7150 ± 160
645	ionetworks-001	39860	333205	95 298	452 2048 ± 0	412 439 ± 2	257 433 ± 2	249 461 ± 2	204 454 ± 2	182 488 ± 2	1136 6335 ± 103	1133 6358 ± 91
646	ionetworks-002	337801	73038	723 1746	751 2048 ± 0	749 689 ± 1	494 672 ± 5	452 665 ± 3	398 669 ± 3	315 664 ± 6	1149 6884 ± 154	1145 6808 ± 102
647	ionetworks-003	309918	68642	686 1631	1001 2050 ± 0	784 713 ± 0	505 681 ± 1	451 665 ± 0	397 668 ± 0	322 671 ± 0	1219 10649 ± 225	1221 10658 ± 106
648	ionetworks-004	302266	135028	745 1813	1002 2050 ± 0	1075 975 ± 0	705 910 ± 0	666 921 ± 0	600 916 ± 0	515 917 ± 1	1362 27437 ± 130	1364 27472 ± 114
649	iqface-000	268819	596337	289 704	1428 4750 ± 32	522 538 ± 26	314 494 ± 2	325 543 ± 3	464 734 ± 4	861 1393 ± 4	1495 636433 ± 38446	1495 632654 ± 85615
650	iqface-001	1145590	760941	-	1429 4752 ± 35	612 603 ± 1	-	-	-	-	1493 568761 ± 3659	1493 570335 ± 5122
651	iqface-003	370803	963398	354 821	1430 4763 ± 37	503 529 ± 1	347 532 ± 2	368 599 ± 8	554 850 ± 2	995 1694 ± 2	1494 575924 ± 2601	1494 576653 ± 2051
652	iqsec-001	172239	41022	374 850	462 2048 ± 0	189 258 ± 6	115 254 ± 4	262 474 ± 9	764 1124 ± 24	1147 4155 ± 119	1347 24586 ± 143	1353 25631 ± 157
653	iqsec-002	172239	41022	376 852	886 2048 ± 0	174 243 ± 6	117 259 ± 4	289 504 ± 10	826 1184 ± 25	1146 4081 ± 124	1348 24688 ± 149	1354 25737 ± 142
654	iqsec-003	293643	15245	247 622	1386 4116 ± 0	961 873 ± 0	674 872 ± 0	627 872 ± 0	577 880 ± 0	524 929 ± 0	1047 4441 ± 20	1047 4462 ± 41
655	irex-000	741899	47419	798 2090	1213 3080 ± 0	940 852 ± 2	655 850 ± 1	629 874 ± 2	621 939 ± 1	748 1249 ± 5	44 201 ± 11	47 208 ± 8
656	irisid-001	480885	100974	437 1000	992 2049 ± 0	1063 966 ± 0	764 971 ± 0	769 1028 ± 0	802 1163 ± 0	989 1662 ± 1	699 1892 ± 13	732 2040 ± 17
657	irisid-002	735508	101002	590 1334	991 2049 ± 0	1466 1732 ± 3	1171 1730 ± 2	1161 1752 ± 2	1147 1923 ± 2	1093 2410 ± 2	1166 7367 ± 16	1166 7410 ± 48
658	isap-001	99049	204201	1 18	1278 4096 ± 0	1 0 ± 0	-	-	-	-	134 459 ± 17	143 456 ± 11
659	isap-002	256765	49931	92 288	665 2048 ± 0	844 769 ± 3	812 1027 ± 2	632 877 ± 2	487 761 ± 1	511 912 ± 2	917 3045 ± 94	911 2973 ± 66
660	isityou-000	48010	36621	29 110	1495 19200 ± 0	57 113 ± 5	-	-	-	-	1487 237517 ± 1318	1487 237374 ± 1279

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 37: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
661	istl-001	271118	27325	215 561	1141 2060 ± 0	602 595 ± 0	408 598 ± 0	367 598 ± 0	327 603 ± 0	269 605 ± 0	739 2073 ± 19	742 2072 ± 34
662	isystems-000	209270	713870	-	227 1024 ± 0	89 154 ± 4	-	-	-	-	119 398 ± 14	125 396 ± 16
663	isystems-001	274621	639268	494 1091	527 2048 ± 0	223 291 ± 9	-	-	-	-	175 557 ± 16	187 564 ± 22
664	isystems-002	358984	803389	679 1595	670 2048 ± 0	907 822 ± 8	-	-	-	-	318 749 ± 31	253 632 ± 28
665	itmo-001	1923215	62191	-	1500 37997 ± 0	1055 959 ± 18	-	-	-	-	1366 28901 ± 1492	1365 27517 ± 186
666	itmo-002	1923215	62191	-	1400 4162 ± 0	621 611 ± 17	-	-	-	-	1169 7423 ± 96	1169 7451 ± 94
667	itmo-003	579961	82630	-	1172 2114 ± 0	28 79 ± 2	-	-	-	-	1212 10038 ± 180	1213 10030 ± 134
668	itmo-004	487023	79012	-	1170 2100 ± 0	404 436 ± 0	-	-	-	-	1182 8219 ± 100	1186 8214 ± 130
669	itmo-005	482155	81168	-	1404 4173 ± 0	833 759 ± 1	-	-	-	-	1253 13214 ± 164	1254 12576 ± 257
670	itmo-006	599187	96762	-	1177 2121 ± 0	893 814 ± 1	641 831 ± 26	591 830 ± 17	540 830 ± 3	538 952 ± 38	1355 26154 ± 148	1356 26217 ± 260
671	itmo-007	415979	245376	820 2199	548 2048 ± 0	813 741 ± 2	-	-	-	-	838 2551 ± 50	839 2529 ± 80
672	itmo-008	726866	318238	609 1376	1256 4096 ± 0	1145 1060 ± 1	833 1058 ± 1	799 1059 ± 1	727 1072 ± 4	650 1104 ± 1	981 3578 ± 25	987 3580 ± 28
673	ivacognitive-001	256958	62791	427 975	843 2048 ± 0	1321 1292 ± 3	1020 1289 ± 4	991 1292 ± 4	918 1292 ± 3	809 1321 ± 4	1035 4228 ± 41	1036 4226 ± 41
674	iws-000	30875	3063	19 77	144 512 ± 0	206 277 ± 5	136 283 ± 1	277 494 ± 3	654 984 ± 3	1124 2987 ± 39	413 999 ± 40	415 992 ± 22
675	jaakit-001	99024	24754	78 251	114 512 ± 0	26 76 ± 0	17 77 ± 0	13 79 ± 0	12 81 ± 0	13 93 ± 0	826 2466 ± 57	827 2465 ± 66
676	jaakit-002	179256	75037	87 277	634 2048 ± 0	831 758 ± 0	586 760 ± 0	673 925 ± 0	936 1318 ± 1	1112 2799 ± 2	394 943 ± 7	405 963 ± 5
677	jgumainz-000	256539	21867	165 472	1069 2056 ± 0	1306 1272 ± 3	1004 1265 ± 2	1016 1325 ± 6	1062 1491 ± 15	1069 2199 ± 53	290 703 ± 2	299 697 ± 17
678	kaizenvoiz-001	254811	74894	131 388	883 2048 ± 0	300 354 ± 0	191 360 ± 0	293 508 ± 0	-	-	960 3439 ± 19	978 3506 ± 31
679	kakao-001	107616	285676	-	254 1024 ± 0	336 379 ± 1	-	-	-	-	391 930 ± 22	402 948 ± 38
680	kakao-002	479406	412822	-	543 2048 ± 0	818 747 ± 6	-	-	-	-	651 1720 ± 62	654 1715 ± 83
681	kakao-003	414379	113944	-	445 2048 ± 0	969 878 ± 3	-	-	-	-	748 2128 ± 34	752 2134 ± 60
682	kakao-004	414411	135270	-	411 2048 ± 0	943 856 ± 1	659 855 ± 0	615 857 ± 0	563 858 ± 0	479 860 ± 1	717 1957 ± 28	717 1953 ± 37
683	kakao-005	414316	152216	-	1030 2052 ± 0	1150 1068 ± 1	841 1073 ± 1	814 1079 ± 0	729 1077 ± 1	640 1089 ± 1	737 2067 ± 26	733 2043 ± 34
684	kakao-007	526993	129545	953 3150	837 2048 ± 0	1051 952 ± 1	756 961 ± 1	702 958 ± 1	640 962 ± 1	547 968 ± 1	432 1056 ± 16	439 1047 ± 28
685	kakao-008	734583	104820	999 3879	455 2048 ± 0	1193 1135 ± 3	910 1148 ± 3	875 1150 ± 3	797 1156 ± 1	706 1175 ± 1	315 736 ± 23	323 727 ± 22
686	kakao-009	766989	104820	1008 4042	861 2048 ± 0	597 593 ± 1	411 603 ± 4	384 613 ± 2	342 617 ± 2	292 636 ± 4	133 457 ± 20	138 448 ± 17
687	kakaobank-000	570796	98479	897 2690	1306 4096 ± 0	1277 1234 ± 1	993 1254 ± 7	955 1248 ± 3	986 1389 ± 3	1016 1843 ± 11	1068 4790 ± 35	1069 4803 ± 35
688	kakaobank-001	780621	98829	819 2189	484 2048 ± 0	579 584 ± 0	416 608 ± 1	376 607 ± 5	412 684 ± 1	519 922 ± 2	1361 27092 ± 174	1363 27065 ± 158
689	kakaobank-002	780621	98833	862 2459	457 2048 ± 0	1100 1004 ± 1	803 1010 ± 0	764 1025 ± 5	731 1078 ± 1	806 1313 ± 1	1359 26972 ± 118	1361 27022 ± 139
690	kakaobank-003	1409862	98833	1015 4186	753 2048 ± 0	1243 1194 ± 2	973 1224 ± 6	940 1231 ± 2	899 1269 ± 5	952 1535 ± 5	1360 26986 ± 183	1362 27026 ± 151
691	kakaopay-001	397864	179869	284 685	1313 4096 ± 0	417 448 ± 0	356 542 ± 0	324 542 ± 0	278 542 ± 0	233 553 ± 0	242 633 ± 22	250 630 ± 22
692	kasikornlabs-000	256471	61000	266 655	728 2048 ± 0	992 908 ± 36	681 878 ± 22	712 969 ± 39	824 1184 ± 54	1089 2382 ± 145	1376 31669 ± 188	1377 31714 ± 182
693	kasikornlabs-001	256471	61037	279 676	782 2048 ± 0	999 912 ± 38	668 868 ± 10	752 1005 ± 50	816 1176 ± 44	1090 2387 ± 147	1373 30759 ± 198	1373 30867 ± 174
694	kasikornlabs-002	256431	61063	241 613	526 2048 ± 0	1009 917 ± 35	703 907 ± 13	708 963 ± 13	939 1320 ± 45	1101 2629 ± 178	1375 31025 ± 180	1374 31054 ± 186
695	kasikornlabs-003	256442	61063	317 751	943 2048 ± 0	687 652 ± 5	474 656 ± 5	538 755 ± 11	682 1007 ± 27	1074 2240 ± 168	1342 24022 ± 121	1343 23998 ± 97
696	kasikornlabs-004	466064	61063	461 1045	582 2048 ± 0	1047 950 ± 7	762 970 ± 4	774 1031 ± 11	975 1372 ± 30	1086 2368 ± 156	1339 23867 ± 56	1341 23919 ± 91
697	kasikornlabs-005	466202	61063	451 1028	874 2048 ± 0	1023 936 ± 8	751 956 ± 14	798 1057 ± 14	881 1243 ± 28	1078 2285 ± 145	1341 23900 ± 44	1342 23923 ± 56
698	kbyai-000	502373	34357	848 2417	75 296 ± 0	1251 1202 ± 3	1212 2167 ± 3	1206 2293 ± 9	1187 2586 ± 13	1140 3689 ± 20	1298 17785 ± 51	1302 17801 ± 125
699	kedacom-000	245292	37401	1104 23574	72 292 ± 0	484 506 ± 3	363 547 ± 10	386 614 ± 9	318 588 ± 10	316 665 ± 24	274 684 ± 14	287 682 ± 16
700	keyless-000	551302	27472	519 1157	594 2048 ± 0	635 615 ± 0	560 736 ± 5	1201 2122 ± 2	1166 2086 ± 1	1056 2110 ± 4	385 909 ± 8	389 905 ± 16
701	kiwitech-000	369711	21375	362 841	391 2048 ± 0	594 591 ± 0	405 594 ± 0	366 595 ± 1	325 596 ± 0	271 609 ± 0	660 1755 ± 20	663 1734 ± 16
702	kloudspot-001	261662	887420	390 882	681 2048 ± 0	274 330 ± 1	160 319 ± 6	128 325 ± 1	109 318 ± 0	76 318 ± 0	1101 5539 ± 48	1103 5555 ± 66
703	kloudspot-002	635738	460698	658 1540	1332 4096 ± 0	1190 1128 ± 1	891 1128 ± 2	853 1123 ± 0	800 1158 ± 2	675 1132 ± 0	1095 5456 ± 154	1096 5460 ± 108
704	kneron-003	58366	1747	66 188	405 2048 ± 0	214 281 ± 3	134 280 ± 1	123 315 ± 13	139 365 ± 7	732 1224 ± 30	1091 5237 ± 63	1092 5274 ± 99

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 38: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
705	kneron-005	375374	13633	158 457	533 2048 ± 0	495 518 ± 2	339 522 ± 4	334 556 ± 5	484 757 ± 19	1005 1760 ± 25	708 1922 ± 11	709 1926 ± 20
706	kneron-006	590352	63741	615 1402	827 2048 ± 0	568 579 ± 0	392 580 ± 0	369 599 ± 0	454 726 ± 1	774 1278 ± 4	774 2247 ± 22	776 2252 ± 28
707	knowutech-000	808045	32886	586 1312	309 1536 ± 0	1401 1419 ± 2	1069 1372 ± 1	1052 1377 ± 1	979 1382 ± 2	856 1386 ± 2	1000 3743 ± 31	995 3693 ± 38
708	kogniza-001	0	394421	573 1288	310 1536 ± 0	1315 1285 ± 2	1033 1304 ± 4	984 1280 ± 2	908 1278 ± 2	816 1326 ± 2	907 2968 ± 37	909 2971 ± 20
709	kookmin-001	234001	31875	-	423 2048 ± 0	59 114 ± 1	33 110 ± 1	29 116 ± 1	29 128 ± 1	29 172 ± 1	237 629 ± 35	239 616 ± 11
710	kookmin-002	371771	30734	355 827	397 2048 ± 0	1131 1038 ± 2	827 1047 ± 1	785 1045 ± 1	715 1061 ± 1	659 1116 ± 1	250 638 ± 19	261 636 ± 20
711	koreaid-001	256261	20152	742 1810	873 2048 ± 0	341 384 ± 2	222 390 ± 1	230 444 ± 2	291 556 ± 6	429 795 ± 5	246 636 ± 11	260 636 ± 10
712	krungthai-002	2360957	15033	534 1185	431 2048 ± 0	248 308 ± 0	154 314 ± 5	119 309 ± 0	111 319 ± 0	104 362 ± 0	911 3014 ± 20	912 2980 ± 22
713	kuke3d-001	403462	68786	199 530	1246 4096 ± 0	894 814 ± 2	622 811 ± 2	572 814 ± 2	524 814 ± 1	464 834 ± 1	1139 6412 ± 57	1135 6413 ± 51
714	kuke3d-002	270544	1227855	351 820	428 2048 ± 0	483 504 ± 3	324 504 ± 1	295 511 ± 1	265 523 ± 2	251 585 ± 1	904 2943 ± 22	908 2966 ± 38
715	lebentech-000	0	10360	2 26	117 512 ± 0	3 22 ± 0	1 22 ± 0	1 22 ± 0	1 23 ± 0	1 23 ± 0	347 801 ± 42	366 825 ± 51
716	lebentech-001	0	144456	671 1567	366 2048 ± 0	104 174 ± 0	60 175 ± 0	66 200 ± 1	68 227 ± 9	129 397 ± 20	148 485 ± 17	152 473 ± 28
717	lemalabs-001	748400	198794	910 2745	636 2048 ± 0	888 810 ± 0	624 812 ± 0	571 813 ± 0	530 819 ± 0	469 844 ± 1	1240 11930 ± 35	1243 11913 ± 37
718	lightvision-000	182569	62181	411 930	834 2048 ± 0	377 411 ± 1	213 381 ± 0	175 379 ± 1	155 386 ± 3	127 396 ± 2	588 1484 ± 13	587 1477 ± 13
719	lightvision-001	266840	62181	683 1604	666 2048 ± 0	505 530 ± 2	340 523 ± 2	322 542 ± 1	285 548 ± 10	224 537 ± 1	572 1456 ± 10	579 1457 ± 14
720	lineclova-000	264443	407003	-	561 2048 ± 0	584 586 ± 0	419 612 ± 0	378 609 ± 1	334 611 ± 0	278 618 ± 1	870 2753 ± 19	871 2745 ± 23
721	lineclova-001	944355	407058	825 2233	890 2048 ± 0	920 833 ± 10	639 830 ± 3	588 828 ± 4	549 838 ± 8	460 833 ± 4	858 2696 ± 23	859 2677 ± 35
722	lineclova-002	475779	406756	597 1352	740 2048 ± 0	1314 1284 ± 1	1010 1275 ± 2	979 1275 ± 1	905 1273 ± 2	777 1281 ± 2	872 2765 ± 10	873 2767 ± 31
723	lineclova-003	585149	410482	714 1725	471 2048 ± 0	1413 1444 ± 1	1120 1438 ± 1	1096 1439 ± 2	1029 1440 ± 1	899 1446 ± 2	893 2890 ± 23	898 2899 ± 29
724	lookman-002	138200	25410	1100 16518	195 548 ± 0	102 173 ± 1	-	-	-	-	216 610 ± 19	232 612 ± 22
725	lookman-004	244775	37401	1103 23548	196 548 ± 0	486 507 ± 5	361 545 ± 12	383 613 ± 12	321 590 ± 11	307 656 ± 16	370 871 ± 29	378 878 ± 29
726	lssytech-000	1316125	197944	1031 4622	1212 3076 ± 0	1212 1154 ± 0	885 1124 ± 0	856 1129 ± 0	772 1128 ± 0	673 1128 ± 0	314 733 ± 17	320 722 ± 13
727	lssytech-001	3239801	232221	1096 10421	1410 4184 ± 0	1420 1454 ± 1	1126 1450 ± 0	1136 1521 ± 2	1036 1452 ± 0	906 1462 ± 0	506 1237 ± 19	500 1183 ± 13
728	luxand-000	0	57908	606 1366	281 1040 ± 0	372 407 ± 23	258 433 ± 11	232 444 ± 14	220 464 ± 14	241 562 ± 25	361 828 ± 28	367 828 ± 32
729	luxand-001	85194	72189	254 636	627 2048 ± 0	400 431 ± 0	273 443 ± 0	229 443 ± 2	198 448 ± 0	211 519 ± 0	564 1433 ± 16	571 1433 ± 16
730	luxand-003	127425	72189	372 847	556 2048 ± 0	775 706 ± 4	522 697 ± 1	482 698 ± 3	425 694 ± 0	421 782 ± 3	570 1443 ± 17	570 1431 ± 17
731	luxand-004	127425	72189	373 848	491 2048 ± 0	725 677 ± 0	507 682 ± 0	473 691 ± 0	437 705 ± 0	404 766 ± 0	569 1441 ± 10	572 1440 ± 11
732	mantra-000	471458	62566	315 749	1047 2052 ± 0	379 413 ± 18	310 487 ± 19	275 494 ± 18	255 511 ± 18	262 598 ± 19	927 3151 ± 51	929 3127 ± 63
733	matrixcomsec-000	114103	177320	85 264	474 2048 ± 0	41 94 ± 0	25 94 ± 0	20 95 ± 0	40 173 ± 1	23 140 ± 1	226 616 ± 22	251 630 ± 28
734	matrixcomsec-001	179657	168204	512 1141	387 2048 ± 0	97 166 ± 0	55 168 ± 0	47 169 ± 0	39 171 ± 0	34 191 ± 0	67 268 ± 53	73 261 ± 62
735	matrixcomsec-002	93740	168203	244 618	951 2048 ± 0	83 147 ± 0	47 146 ± 0	39 147 ± 0	32 148 ± 0	30 178 ± 0	339 791 ± 13	344 790 ± 16
736	maverickresearch-000	256262	20268	107 331	902 2048 ± 0	1094 995 ± 1	794 999 ± 1	746 999 ± 2	667 997 ± 0	589 1023 ± 2	1320 20268 ± 108	1322 20254 ± 121
737	maxvision-000	133114	56426	-	115 512 ± 0	310 359 ± 0	189 356 ± 0	157 359 ± 0	133 356 ± 0	107 370 ± 1	823 2461 ± 20	823 2452 ± 17
738	maxvision-001	256146	61793	941 3037	350 2048 ± 0	205 275 ± 3	128 274 ± 2	96 277 ± 4	88 280 ± 4	79 325 ± 3	301 714 ± 13	316 717 ± 13
739	maxvision-002	171894	60623	727 1765	637 2048 ± 0	100 172 ± 0	58 171 ± 0	49 172 ± 0	41 174 ± 0	44 221 ± 0	309 725 ± 5	321 725 ± 5
740	maxvision-003	234062	61252	857 2452	916 2048 ± 0	451 474 ± 0	294 468 ± 0	259 471 ± 0	228 475 ± 0	212 519 ± 0	827 2467 ± 28	831 2488 ± 23
741	maxvision-004	394121	61583	956 3217	348 2048 ± 0	676 644 ± 1	468 648 ± 0	430 647 ± 1	363 637 ± 1	342 695 ± 1	701 1897 ± 10	703 1903 ± 14
742	maxvision-005	309326	61739	807 2138	545 2048 ± 0	1103 1007 ± 1	987 1245 ± 5	757 1017 ± 1	704 1040 ± 4	792 1299 ± 3	715 1950 ± 38	716 1948 ± 34
743	maxvision-006	315987	61739	955 3172	651 2048 ± 0	1290 1247 ± 3	958 1208 ± 3	926 1211 ± 5	852 1215 ± 1	767 1268 ± 4	713 1938 ± 17	710 1929 ± 35
744	mbbankvn-000	472410	139431	1016 4227	356 2048 ± 0	440 466 ± 17	302 480 ± 26	266 481 ± 27	232 479 ± 21	191 501 ± 29	1044 4388 ± 14	1043 4380 ± 17
745	megvii-000	1358915	16014	-	658 2048 ± 0	858 785 ± 0	-	-	-	-	1316 19816 ± 158	1318 19805 ± 174
746	megvii-001	1361523	16486	-	785 2048 ± 0	529 543 ± 0	-	-	-	-	1090 5228 ± 32	1090 5252 ± 60
747	megvii-002	1809564	16491	-	1356 4100 ± 0	675 644 ± 0	-	-	-	-	1426 50630 ± 183	1420 47591 ± 716
748	megvii-003	4430290	42790	-	1287 4096 ± 0	1255 1210 ± 1	971 1223 ± 0	1041 1356 ± 4	1091 1582 ± 7	1105 2727 ± 23	1485 225342 ± 3574	1486 225413 ± 6344

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 39: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
749	megvii-004	3962505	44019	-	1339 4097 ± 0	1316 1287 ± 1	1068 1369 ± 2	1006 1310 ± 2	981 1384 ± 3	891 1436 ± 5	1416 46801 ± 204	1417 46832 ± 207
750	megvii-005	1378009	44038	1007 4025	998 2049 ± 0	1335 1319 ± 5	989 1247 ± 6	947 1240 ± 2	882 1245 ± 2	791 1298 ± 3	1381 32025 ± 121	1381 32008 ± 114
751	megvii-006	1554938	44038	1020 4346	984 2049 ± 0	1317 1287 ± 3	1017 1286 ± 0	1065 1393 ± 5	937 1319 ± 1	836 1360 ± 1	1379 31845 ± 100	1379 31872 ± 118
752	megvii-007	1366028	44035	713 1723	989 2049 ± 0	1205 1148 ± 1	911 1153 ± 0	892 1172 ± 4	814 1172 ± 2	784 1288 ± 2	1319 20191 ± 1753	1320 20191 ± 1620
753	megvii-008	1644071	44035	797 2086	994 2049 ± 0	1333 1313 ± 1	1092 1404 ± 4	1044 1363 ± 2	965 1362 ± 3	877 1420 ± 2	1317 20059 ± 1598	1324 20448 ± 1633
754	megvii-009	1644071	44039	832 2294	996 2049 ± 0	1338 1322 ± 1	1052 1328 ± 1	1029 1339 ± 3	1165 2075 ± 3	865 1404 ± 4	1318 20081 ± 1621	1323 20260 ± 1577
755	meituan-000	259514	333178	-	641 2048 ± 0	406 436 ± 4	269 441 ± 1	401 626 ± 5	743 1098 ± 15	1128 3126 ± 53	249 638 ± 17	257 633 ± 16
756	meituan-001	615387	333249	501 1113	824 2048 ± 0	1110 1017 ± 4	801 1008 ± 3	755 1010 ± 2	687 1010 ± 3	583 1011 ± 4	259 654 ± 10	277 658 ± 14
757	meituan-002	686111	244091	821 2200	1329 4096 ± 0	1136 1052 ± 0	851 1086 ± 1	807 1064 ± 2	714 1060 ± 5	623 1063 ± 1	438 1064 ± 10	451 1070 ± 16
758	meituan-003	860010	244091	917 2772	1258 4096 ± 0	1440 1487 ± 1	1149 1518 ± 0	1119 1469 ± 6	1057 1478 ± 0	911 1471 ± 7	462 1112 ± 16	464 1097 ± 20
759	meituan-004	850507	244091	913 2759	1260 4096 ± 0	871 796 ± 1	630 822 ± 0	581 820 ± 11	527 815 ± 11	447 824 ± 0	476 1149 ± 10	487 1148 ± 5
760	meiya-001	280055	264913	184 507	987 2049 ± 0	645 622 ± 12	-	-	-	-	1184 8356 ± 615	1184 8134 ± 97
761	mendaxiatech-000	1941475	45484	1024 4397	1340 4097 ± 0	1286 1243 ± 2	994 1255 ± 1	1050 1373 ± 2	1095 1598 ± 3	1103 2689 ± 8	1417 46906 ± 275	1418 46872 ± 217
762	metsakuurcompany-001	445177	1091558	676 1588	1085 2056 ± 0	567 578 ± 1	400 587 ± 3	360 590 ± 1	386 659 ± 1	475 854 ± 1	1191 8600 ± 192	1185 8155 ± 298
763	metsakuurcompany-002	0	957558	430 980	1115 2056 ± 0	1080 980 ± 1	773 978 ± 1	716 976 ± 2	680 1005 ± 1	648 1103 ± 2	1193 8766 ± 326	1194 8786 ± 324
764	metsakuurcompany-003	0	957562	440 1004	1092 2056 ± 0	1061 965 ± 2	763 970 ± 5	800 1060 ± 12	678 1005 ± 6	666 1121 ± 1	1183 8234 ± 145	1183 8111 ± 280
765	metsakuurcompany-004	0	1035754	468 1056	1111 2056 ± 0	1038 946 ± 10	661 861 ± 6	633 877 ± 3	609 926 ± 4	563 991 ± 5	1135 6281 ± 160	1131 6232 ± 191
766	metsakuurcompany-005	0	305952	273 661	667 2048 ± 0	448 472 ± 0	298 477 ± 0	276 494 ± 1	267 527 ± 1	418 778 ± 2	953 3376 ± 22	960 3383 ± 40
767	metsakuurcompany-006	0	532431	443 1010	754 2048 ± 0	680 648 ± 0	472 651 ± 0	431 648 ± 0	373 647 ± 0	306 649 ± 0	408 993 ± 2	442 1049 ± 8
768	miaxis-001	0	215019	105 325	119 512 ± 0	210 279 ± 0	133 278 ± 0	97 278 ± 1	89 285 ± 0	70 297 ± 0	371 872 ± 14	380 881 ± 22
769	miaxis-002	0	216484	106 328	118 512 ± 0	224 292 ± 0	138 290 ± 1	110 297 ± 0	99 298 ± 0	102 359 ± 1	442 1065 ± 48	436 1040 ± 54
770	miaxis-003	0	439726	232 604	783 2048 ± 0	666 634 ± 0	440 624 ± 2	382 613 ± 0	266 524 ± 0	317 665 ± 0	735 2053 ± 20	735 2048 ± 28
771	miaxis-004	729303	89655	532 1183	993 2049 ± 0	1491 1941 ± 0	1199 1947 ± 0	1187 1940 ± 0	1200 3153 ± 0	1037 1963 ± 0	1478 103338 ± 44	1478 103361 ± 65
772	miaxis-005	738575	172996	1025 4416	999 2049 ± 0	1468 1741 ± 1	1176 1776 ± 13	1165 1771 ± 4	1118 1774 ± 1	1028 1925 ± 5	998 3738 ± 17	1006 3774 ± 45
773	microfocus-000	103483	27242	-	41 256 ± 0	194 263 ± 18	-	-	-	-	46 202 ± 5	45 203 ± 7
774	microfocus-001	104524	27242	-	38 256 ± 0	195 264 ± 18	-	-	-	-	50 215 ± 8	50 217 ± 10
775	microfocus-002	96288	27362	62 176	31 256 ± 0	190 259 ± 18	-	-	-	-	93 337 ± 34	56 230 ± 25
776	microfocus-003	169603	27689	185 507	40 256 ± 0	468 490 ± 19	334 517 ± 30	533 752 ± 55	770 1126 ± 121	1104 2713 ± 345	20 121 ± 5	19 117 ± 5
777	minivision-000	836697	16597	1018 4263	1316 4096 ± 0	1129 1035 ± 1	822 1033 ± 2	777 1035 ± 1	700 1037 ± 1	617 1059 ± 2	825 2466 ± 26	825 2460 ± 25
778	mitek-000	105584	44643	88 277	240 1024 ± 0	187 256 ± 21	118 265 ± 10	171 375 ± 24	281 545 ± 42	645 1098 ± 124	885 2846 ± 88	876 2777 ± 74
779	mitek-001	519928	69522	732 1777	1076 2056 ± 0	328 371 ± 0	229 397 ± 0	196 407 ± 4	164 394 ± 5	137 416 ± 2	1144 6682 ± 28	1142 6694 ± 48
780	mitek-002	534528	68069	730 1773	1101 2056 ± 0	318 365 ± 0	207 372 ± 0	162 366 ± 2	149 377 ± 2	126 394 ± 1	415 1005 ± 19	404 963 ± 23
781	mobai-000	365451	80573	331 786	1443 6144 ± 0	840 766 ± 8	670 869 ± 6	921 1205 ± 31	1136 1867 ± 45	1138 3549 ± 190	1290 16458 ± 333	1293 16423 ± 1473
782	mobai-001	265297	60164	200 534	610 2048 ± 0	625 612 ± 3	422 614 ± 3	469 687 ± 9	583 886 ± 31	997 1707 ± 103	544 1386 ± 25	552 1377 ± 26
783	mobai-002	254811	33242	641 1482	691 2048 ± 0	304 356 ± 0	201 366 ± 0	156 359 ± 4	141 367 ± 2	144 428 ± 0	1185 8387 ± 126	1187 8381 ± 105
784	mobai-003	1046772	427612	979 3555	1331 4096 ± 0	1461 1691 ± 7	1170 1698 ± 3	1157 1693 ± 1	1105 1705 ± 2	1009 1786 ± 1	1480 105417 ± 231	1480 105475 ± 169
785	mobbl-000	182052	58727	-	342 2048 ± 0	191 261 ± 16	121 267 ± 22	170 375 ± 92	381 655 ± 273	1048 2059 ± 1129	1243 12061 ± 142	1245 12050 ± 133
786	mobbl-001	231160	58706	76 226	835 2048 ± 0	115 183 ± 32	68 184 ± 25	152 354 ± 76	535 823 ± 396	1110 2781 ± 1166	1238 11832 ± 109	1242 11851 ± 88
787	mobbl-002	242920	60119	-	420 2048 ± 0	701 663 ± 6	477 660 ± 5	446 662 ± 5	389 663 ± 5	326 676 ± 5	1234 11616 ± 78	1235 11588 ± 97
788	mobbl-003	172248	60960	89 282	497 2048 ± 0	705 664 ± 6	480 661 ± 5	449 663 ± 5	394 665 ± 6	337 691 ± 5	1251 12506 ± 111	1252 12509 ± 100
789	mobbl-004	432204	970955	383 872	520 2048 ± 0	221 288 ± 0	135 281 ± 0	100 281 ± 1	92 288 ± 2	71 312 ± 2	947 3327 ± 22	958 3369 ± 48
790	mobbl-005	352570	983522	429 978	960 2048 ± 0	1166 1098 ± 1	887 1125 ± 2	852 1123 ± 2	808 1167 ± 2	834 1354 ± 1	1097 5498 ± 62	1099 5492 ± 53
791	mobipintech-000	370514	303291	523 1173	712 2048 ± 0	1288 1245 ± 1	978 1234 ± 1	968 1264 ± 1	964 1360 ± 1	998 1707 ± 1	1267 14506 ± 214	1269 14433 ± 197
792	momovn-001	246938	1013076	325 769	675 2048 ± 0	168 237 ± 1	123 268 ± 7	118 308 ± 10	94 293 ± 5	122 390 ± 3	567 1438 ± 13	573 1443 ± 13

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 40: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
793	momovn-002	320036	1013104	316 750	671 2048 ± 0	176 243 ± 3	110 248 ± 0	87 245 ± 1	78 253 ± 0	81 328 ± 2	571 1454 ± 8	577 1456 ± 16
794	momovn-003	528962	951056	661 1548	1300 4096 ± 0	841 766 ± 1	601 777 ± 0	553 783 ± 0	525 815 ± 1	602 1045 ± 1	1021 3979 ± 23	1028 3984 ± 29
795	moredian-000	525259	21374	458 1035	587 2048 ± 0	762 694 ± 0	523 698 ± 0	483 699 ± 0	430 700 ± 0	355 713 ± 1	679 1803 ± 11	674 1779 ± 23
796	morpho-000	100806	107129	-	3 116 ± 0	56 109 ± 1	-	-	-	-	409 993 ± 31	419 1000 ± 34
797	morpho-001	100806	368573	-	2 116 ± 0	12 41 ± 1	-	-	-	-	424 1030 ± 38	437 1042 ± 37
798	morpho-002	383021	110995	-	78 352 ± 0	378 411 ± 7	-	-	-	-	985 3620 ± 94	830 2481 ± 65
799	mukh-001	866223	451194	692 1647	246 1024 ± 0	1363 1375 ± 17	1078 1390 ± 12	1076 1406 ± 8	990 1394 ± 10	837 1360 ± 11	126 433 ± 14	134 435 ± 14
800	mukh-002	693809	454936	484 1081	569 2048 ± 0	1065 968 ± 1	718 921 ± 12	699 957 ± 2	630 954 ± 6	540 953 ± 5	220 612 ± 13	231 611 ± 17
801	mukh-003	556405	180996	752 1853	487 2048 ± 0	1482 1896 ± 3	1201 1948 ± 1	1192 2003 ± 0	1154 1981 ± 0	1043 2016 ± 0	199 597 ± 13	225 604 ± 16
802	mukh-004	405118	181085	598 1353	542 2048 ± 0	543 558 ± 1	380 567 ± 1	342 569 ± 1	300 569 ± 3	257 593 ± 1	292 704 ± 1	305 703 ± 2
803	mukh-005	555239	264759	592 1346	518 2048 ± 0	405 436 ± 3	308 486 ± 6	279 495 ± 7	268 527 ± 16	261 597 ± 25	293 705 ± 1	306 706 ± 2
804	multimodality-000	0	503924	619 1417	436 2048 ± 0	384 416 ± 0	248 420 ± 0	210 423 ± 0	185 427 ± 0	168 463 ± 0	366 848 ± 25	353 800 ± 28
805	multimodality-001	185719	545045	612 1390	1294 4096 ± 0	1242 1190 ± 2	925 1169 ± 2	886 1165 ± 2	807 1167 ± 2	708 1177 ± 2	561 1424 ± 35	557 1384 ± 42
806	mvision-001	227502	149531	300 723	128 512 ± 0	755 691 ± 21	528 702 ± 19	480 697 ± 24	439 708 ± 29	353 710 ± 27	469 1123 ± 40	490 1154 ± 38
807	nazhiai-000	547484	16141	904 2716	515 2048 ± 0	738 683 ± 3	511 687 ± 2	598 835 ± 27	551 840 ± 31	463 834 ± 34	770 2230 ± 34	751 2133 ± 81
808	ncsgroup-000	95110	345861	460 1041	468 2048 ± 0	91 157 ± 0	49 153 ± 0	41 151 ± 0	36 157 ± 0	47 231 ± 0	811 2407 ± 16	812 2411 ± 22
809	ncssg-001	148743	490151	370 847	931 2048 ± 0	1024 936 ± 2	739 946 ± 3	687 944 ± 2	619 938 ± 5	550 972 ± 2	982 3579 ± 50	968 3459 ± 34
810	ncssg-002	148859	415290	420 945	364 2048 ± 0	350 389 ± 1	227 397 ± 0	189 401 ± 1	175 407 ± 0	170 469 ± 0	1451 66390 ± 246	1449 66363 ± 179
811	ncssg-003	148876	414848	320 761	591 2048 ± 0	383 416 ± 1	267 437 ± 2	224 437 ± 1	197 443 ± 2	187 497 ± 1	1349 24709 ± 68	1348 24768 ± 134
812	ncssg-004	521184	483011	831 2272	647 2048 ± 0	744 687 ± 1	512 687 ± 0	492 706 ± 5	423 693 ± 1	370 731 ± 21	1350 24815 ± 117	1349 24799 ± 117
813	neosystems-001	575295	349959	-	434 2048 ± 0	1195 1137 ± 4	867 1098 ± 1	1164 1767 ± 4	1116 1769 ± 3	1006 1765 ± 4	1305 18557 ± 189	1310 18640 ± 192
814	neosystems-002	599441	349942	-	811 2048 ± 0	1192 1135 ± 2	1190 1855 ± 3	1204 2258 ± 5	1175 2238 ± 3	1076 2247 ± 3	1306 18752 ± 167	1309 18610 ± 213
815	neosystems-003	599442	349942	-	466 2048 ± 0	1202 1143 ± 2	1184 1836 ± 7	1205 2260 ± 3	1179 2273 ± 6	1077 2273 ± 3	1311 19130 ± 223	1314 19167 ± 186
816	neosystems-004	243546	352623	198 529	793 2048 ± 0	269 324 ± 0	540 711 ± 3	585 827 ± 7	559 854 ± 2	513 916 ± 2	1266 14437 ± 176	1267 14355 ± 173
817	netbridgetech-001	133108	205875	187 508	1254 4096 ± 0	34 85 ± 1	20 83 ± 0	15 84 ± 0	16 92 ± 0	16 113 ± 4	1196 9280 ± 74	1198 9446 ± 512
818	netbridgetech-002	257687	49931	96 299	697 2048 ± 0	926 838 ± 6	646 838 ± 2	600 839 ± 1	550 839 ± 3	478 859 ± 3	894 2893 ± 65	922 3050 ± 123
819	neurotechnology-000	62129	33905	-	1464 7148 ± 0	622 611 ± 48	-	-	-	-	1468 74288 ± 2194	1463 72879 ± 2640
820	neurotechnology-001	280771	35716	-	1203 2718 ± 0	972 881 ± 46	-	-	-	-	1458 69356 ± 684	1454 69140 ± 579
821	neurotechnology-002	280771	36036	-	1202 2718 ± 0	572 581 ± 6	-	-	-	-	1454 68738 ± 748	1453 68905 ± 993
822	neurotechnology-003	339962	36125	-	1435 5214 ± 0	571 580 ± 5	-	-	-	-	1464 73589 ± 2548	1462 72837 ± 1546
823	neurotechnology-004	293384	54399	-	26 256 ± 0	363 401 ± 0	-	-	-	-	52 219 ± 8	58 231 ± 13
824	neurotechnology-005	270450	58088	-	30 256 ± 0	361 399 ± 0	-	-	-	-	57 238 ± 10	62 237 ± 7
825	neurotechnology-006	525541	54522	-	145 512 ± 0	728 678 ± 56	-	-	-	-	162 513 ± 14	173 535 ± 26
826	neurotechnology-008	119718	53462	-	34 256 ± 0	289 339 ± 0	-	-	-	-	138 467 ± 19	155 486 ± 26
827	neurotechnology-009	137515	53213	-	25 256 ± 0	348 388 ± 1	-	-	-	-	164 513 ± 10	164 512 ± 16
828	neurotechnology-010	362144	50226	-	166 514 ± 0	870 796 ± 1	613 796 ± 1	575 815 ± 3	555 853 ± 1	577 1006 ± 11	19 120 ± 8	17 115 ± 7
829	neurotechnology-011	364137	51141	-	164 514 ± 0	877 798 ± 1	617 802 ± 1	586 827 ± 3	569 873 ± 2	619 1059 ± 15	16 114 ± 11	16 114 ± 8
830	neurotechnology-012	147830	51395	-	29 256 ± 0	345 384 ± 0	218 387 ± 0	195 404 ± 1	192 435 ± 1	250 583 ± 7	18 119 ± 7	18 116 ± 7
831	neurotechnology-013	474749	85552	1041 4825	165 514 ± 0	1095 1000 ± 1	799 1006 ± 2	761 1022 ± 2	711 1053 ± 2	712 1195 ± 8	14 109 ± 4	13 110 ± 4
832	neurotechnology-015	474782	86045	1038 4819	169 515 ± 0	1123 1028 ± 3	821 1033 ± 3	792 1055 ± 4	741 1097 ± 4	797 1304 ± 18	21 130 ± 2	21 130 ± 4
833	neurotechnology-016	661450	89972	1010 4074	167 515 ± 0	1343 1339 ± 3	1061 1351 ± 2	1046 1369 ± 4	987 1390 ± 4	934 1508 ± 21	32 179 ± 20	30 171 ± 20
834	neurotechnology-017	371650	98242	842 2385	27 256 ± 0	488 508 ± 2	331 513 ± 0	304 519 ± 1	271 533 ± 1	282 622 ± 15	4 80 ± 1	4 79 ± 1
835	neurotechnology-018	698669	98271	1021 4367	168 515 ± 0	797 725 ± 1	573 747 ± 1	530 747 ± 4	494 769 ± 2	461 833 ± 22	6 85 ± 2	7 87 ± 7
836	neurotechnology-019	365649	103237	839 2321	35 256 ± 0	650 625 ± 1	444 625 ± 0	414 633 ± 3	413 684 ± 2	385 742 ± 17	2 71 ± 2	2 72 ± 4

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 41: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
837	neurotechnology-020	807529	103241	1046 5010	157 513 ± 0	1250 1201 ± 4	946 1196 ± 5	913 1195 ± 8	863 1224 ± 7	844 1373 ± 22	11 102 ± 1	8 101 ± 1
838	neurotechnology-021	372239	147289	741 1810	32 256 ± 0	1384 1396 ± 0	1084 1394 ± 0	1083 1410 ± 5	1028 1438 ± 0	956 1545 ± 41	25 147 ± 1	26 152 ± 4
839	nextmetalabs-001	361607	4729276	250 627	875 2048 ± 0	1001 914 ± 0	726 928 ± 0	678 928 ± 0	626 948 ± 0	518 919 ± 0	1074 4876 ± 185	1066 4774 ± 72
840	nextmetalabs-002	571398	5046464	505 1123	451 2048 ± 0	1298 1259 ± 2	976 1229 ± 1	960 1253 ± 0	890 1257 ± 14	789 1296 ± 1	1067 4786 ± 103	1065 4753 ± 97
841	nextmetalabs-003	1105548	5046426	836 2301	678 2048 ± 0	1262 1218 ± 1	963 1217 ± 2	934 1225 ± 6	843 1207 ± 2	757 1253 ± 1	1073 4874 ± 57	1072 4833 ± 85
842	nextmetalabs-004	1105548	4174065	589 1332	349 2048 ± 0	1490 1940 ± 1	1202 1949 ± 0	1183 1919 ± 0	1148 1925 ± 1	1042 2011 ± 2	1249 12441 ± 88	1253 12539 ± 154
843	nhn-001	336391	817674	-	1284 4096 ± 0	1120 1027 ± 3	816 1029 ± 1	772 1029 ± 1	706 1044 ± 1	642 1090 ± 1	1435 56650 ± 260	1436 56639 ± 210
844	nhn-002	363471	817674	280 679	1285 4096 ± 0	1197 1141 ± 3	898 1138 ± 2	869 1141 ± 2	796 1151 ± 6	719 1203 ± 2	1434 56608 ± 579	1435 56549 ± 606
845	nhn-003	933665	432730	625 1427	1266 4096 ± 0	1273 1229 ± 2	1000 1261 ± 1	967 1263 ± 3	911 1279 ± 2	846 1375 ± 3	1425 50560 ± 105	1426 50592 ± 142
846	nhn-004	931648	432753	923 2864	1251 4096 ± 0	628 612 ± 0	455 632 ± 0	420 636 ± 0	404 676 ± 6	444 820 ± 0	1296 17323 ± 99	1299 17334 ± 133
847	nhn-005	1278729	432776	926 2890	1271 4096 ± 0	890 811 ± 0	489 671 ± 13	434 652 ± 0	452 726 ± 12	499 890 ± 3	1300 18123 ± 400	1305 18214 ± 250
848	nmetics-001	1304721	145561	811 2157	894 2048 ± 0	1449 1559 ± 0	1154 1564 ± 0	1146 1570 ± 1	1082 1561 ± 1	984 1634 ± 1	558 1407 ± 115	549 1357 ± 174
849	noblis-000	713573	253204	-	288 1061 ± 0	103 174 ± 20	-	-	-	-	535 1343 ± 45	551 1373 ± 56
850	nodeflux-000	347079	797842	-	553 2048 ± 0	527 542 ± 1	-	-	-	-	945 3283 ± 51	951 3285 ± 56
851	nodeflux-001	262553	797842	-	456 2048 ± 0	180 247 ± 1	-	-	-	-	938 3242 ± 81	950 3255 ± 93
852	nodeflux-002	774668	690213	161 466	909 2048 ± 0	778 708 ± 4	539 709 ± 4	500 716 ± 5	448 716 ± 7	375 736 ± 3	963 3475 ± 62	963 3408 ± 143
853	nominder-000	547519	102693	720 1741	46 258 ± 0	770 701 ± 0	525 701 ± 2	503 721 ± 2	451 720 ± 3	431 796 ± 1	86 314 ± 5	54 226 ± 45
854	nominder-001	332018	102774	522 1172	47 258 ± 0	120 186 ± 0	74 189 ± 1	58 188 ± 0	54 202 ± 1	64 271 ± 1	55 226 ± 1	32 172 ± 41
855	nominder-002	249049	116931	751 1852	42 258 ± 0	368 405 ± 1	250 425 ± 2	214 424 ± 1	190 434 ± 0	197 505 ± 0	10 101 ± 11	10 102 ± 14
856	nominder-003	249964	116931	847 2417	49 258 ± 0	721 675 ± 2	509 683 ± 5	459 677 ± 2	431 701 ± 3	417 775 ± 3	7 87 ± 4	5 84 ± 4
857	nominder-004	249964	114221	846 2416	44 258 ± 0	829 755 ± 2	578 751 ± 4	526 740 ± 3	492 768 ± 3	470 846 ± 1	5 81 ± 1	6 86 ± 8
858	nominder-005	289014	114221	899 2699	43 258 ± 0	1224 1170 ± 4	947 1196 ± 5	884 1161 ± 6	858 1221 ± 8	826 1343 ± 7	8 94 ± 14	20 117 ± 7
859	nominder-006	304867	114225	893 2660	45 258 ± 0	809 736 ± 2	572 746 ± 3	540 759 ± 3	488 762 ± 2	485 869 ± 5	13 105 ± 5	12 108 ± 10
860	nominder-007	249970	101106	412 932	48 258 ± 0	1157 1085 ± 0	840 1071 ± 0	832 1099 ± 1	757 1117 ± 2	738 1235 ± 5	29 171 ± 5	40 190 ± 7
861	nominder-008	484231	102193	704 1688	163 514 ± 0	1493 1982 ± 1	1203 1953 ± 0	1185 1937 ± 1	1152 1954 ± 3	1051 2074 ± 6	30 172 ± 2	42 191 ± 2
862	notiontag-000	92753	406791	-	200 584 ± 0	533 548 ± 64	364 548 ± 35	1107 1450 ± 99	1203 3771 ± 251	1198 13146 ± 792	1409 44672 ± 269	1410 44593 ± 358
863	notiontag-001	92753	427967	260 642	199 584 ± 0	1020 929 ± 35	857 1092 ± 39	1211 3709 ± 81	1209 10233 ± 180	1203 31181 ± 521	1406 43636 ± 286	1407 43724 ± 330
864	notiontag-002	271987	967207	932 2965	1175 2120 ± 0	423 453 ± 2	281 453 ± 3	238 453 ± 3	206 458 ± 2	172 471 ± 3	1321 20278 ± 194	1321 20195 ± 186
865	npt-000	466008	22512	357 833	1146 2064 ± 0	1072 974 ± 0	790 994 ± 0	738 994 ± 0	659 990 ± 1	579 1007 ± 0	714 1947 ± 29	723 1983 ± 34
866	nsensecorp-001	187421	258593	-	755 2048 ± 0	286 336 ± 0	171 335 ± 0	136 336 ± 0	120 335 ± 0	86 337 ± 0	1414 46605 ± 93	1415 46613 ± 112
867	nsensecorp-002	187421	122407	-	444 2048 ± 0	282 333 ± 0	169 333 ± 0	137 337 ± 0	124 338 ± 0	98 351 ± 0	1412 45965 ± 213	1412 45988 ± 158
868	nsensecorp-003	199895	117041	292 710	789 2048 ± 0	698 661 ± 0	483 664 ± 0	445 662 ± 1	384 659 ± 1	310 659 ± 0	1408 44658 ± 51	1411 44654 ± 72
869	nsensecorp-004	513276	139178	700 1675	476 2048 ± 0	1408 1433 ± 0	1122 1445 ± 7	1108 1450 ± 3	1061 1487 ± 5	-	807 2388 ± 42	806 2385 ± 63
870	nsensecorp-005	411845	138437	554 1240	645 2048 ± 0	1421 1455 ± 0	1129 1453 ± 0	1105 1449 ± 0	1044 1457 ± 2	925 1490 ± 4	750 2131 ± 51	756 2165 ± 66
871	ntechlab-000	191530	22085	-	1204 2906 ± 1	208 278 ± 13	-	-	-	-	1374 30787 ± 142	1372 30846 ± 77
872	ntechlab-001	691296	22848	-	1463 6744 ± 1	587 587 ± 11	-	-	-	-	1452 67692 ± 833	1451 67486 ± 244
873	ntechlab-002	894169	25023	-	1462 6744 ± 1	276 330 ± 11	-	-	-	-	1473 82508 ± 213	1471 82524 ± 220
874	ntechlab-003	1422738	24649	-	1205 2906 ± 1	71 128 ± 3	-	-	-	-	1391 35327 ± 117	1394 35371 ± 123
875	ntechlab-004	3752378	115782	-	1225 3482 ± 1	900 818 ± 11	-	-	-	-	1402 38929 ± 81	1405 38916 ± 111
876	ntechlab-005	1726214	116068	-	338 1940 ± 0	782 712 ± 1	-	-	-	-	375 882 ± 42	379 880 ± 29
877	ntechlab-006	7901590	43009	-	1200 2600 ± 0	819 749 ± 1	-	-	-	-	431 1055 ± 93	372 844 ± 48
878	ntechlab-007	2509686	52474	-	1224 3348 ± 0	867 792 ± 3	-	-	-	-	497 1209 ± 59	523 1267 ± 65
879	ntechlab-008	1138002	52483	-	305 1300 ± 0	541 556 ± 1	-	-	-	-	456 1095 ± 45	443 1049 ± 51
880	ntechlab-009	1748043	43730	-	337 1940 ± 0	1183 1115 ± 2	878 1114 ± 1	847 1119 ± 1	774 1130 ± 2	716 1202 ± 4	223 614 ± 20	210 595 ± 28

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 42: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
881	ntechlab-010	698591	217167	-	298 1280 ± 0	1229 1177 ± 2	933 1180 ± 2	916 1197 ± 2	864 1224 ± 1	817 1326 ± 3	121 405 ± 13	128 416 ± 31
882	ntechlab-011	786933	209458	1075 6867	301 1280 ± 0	1207 1148 ± 2	903 1142 ± 1	881 1159 ± 1	828 1185 ± 1	785 1290 ± 3	31 179 ± 11	33 173 ± 11
883	ntechlab-012	570796	212350	1053 5450	1194 2560 ± 0	1331 1309 ± 1	1047 1323 ± 1	1023 1331 ± 1	962 1360 ± 1	905 1460 ± 3	49 211 ± 8	49 211 ± 7
884	ntnu-000	293085	433109	698 1665	1491 16896 ± 0	1412 1440 ± 0	1121 1441 ± 0	1093 1430 ± 0	1037 1452 ± 0	913 1473 ± 1	674 1799 ± 53	645 1690 ± 42
885	ocrstudio-001	9010	57372	13 66	24 240 ± 0	32 83 ± 0	19 83 ± 0	17 88 ± 0	21 104 ± 0	33 190 ± 2	1222 10725 ± 62	1226 11030 ± 130
886	omface-000	45945	844976	52 154	261 1024 ± 0	118 185 ± 1	88 206 ± 2	67 203 ± 1	48 195 ± 1	36 193 ± 1	146 481 ± 42	142 456 ± 20
887	omface-001	146370	1799745	49 145	264 1024 ± 0	132 194 ± 2	98 222 ± 2	72 209 ± 0	60 216 ± 1	49 233 ± 1	1304 18369 ± 19	1307 18366 ± 32
888	omnigarde-000	264057	32882	-	238 1024 ± 0	1035 944 ± 0	690 887 ± 0	640 888 ± 1	589 892 ± 0	507 902 ± 0	854 2671 ± 35	848 2620 ± 29
889	omnigarde-001	200523	32882	162 466	155 512 ± 0	1026 941 ± 0	686 883 ± 1	639 886 ± 1	587 891 ± 1	504 898 ± 0	555 1405 ± 31	555 1379 ± 26
890	omnigarde-002	368860	32882	319 760	241 1024 ± 0	1326 1303 ± 1	988 1246 ± 1	956 1249 ± 1	886 1253 ± 1	763 1261 ± 1	868 2727 ± 34	863 2686 ± 32
891	omnigarde-003	427751	35153	318 756	171 517 ± 0	717 673 ± 0	450 631 ± 1	413 633 ± 1	364 637 ± 1	302 647 ± 1	513 1259 ± 22	516 1244 ± 16
892	omnigarde-004	995019	35158	757 1873	172 517 ± 0	1248 1201 ± 1	1001 1261 ± 1	923 1207 ± 1	853 1215 ± 0	734 1229 ± 0	515 1274 ± 22	514 1241 ± 26
893	omnigarde-005	1316067	35158	875 2545	181 525 ± 0	1223 1170 ± 1	918 1159 ± 0	887 1169 ± 7	830 1186 ± 1	713 1199 ± 1	592 1490 ± 40	591 1481 ± 38
894	onfido-000	273478	959781	404 914	469 2048 ± 0	432 459 ± 17	279 451 ± 15	236 451 ± 14	213 462 ± 15	198 505 ± 18	628 1617 ± 50	630 1637 ± 53
895	onfido-001	344394	1021960	912 2756	376 2048 ± 0	146 208 ± 4	89 208 ± 4	74 213 ± 4	65 223 ± 3	57 250 ± 4	373 874 ± 28	373 852 ± 20
896	onfido-002	852775	252949	439 1003	1235 4096 ± 0	1247 1199 ± 5	953 1202 ± 5	919 1203 ± 5	849 1213 ± 5	750 1250 ± 5	850 2625 ± 60	851 2628 ± 84
897	onfido-003	852775	252949	442 1010	1247 4096 ± 0	1253 1203 ± 5	954 1202 ± 5	922 1205 ± 5	851 1214 ± 5	755 1252 ± 5	824 2463 ± 14	856 2669 ± 37
898	onfido-004	855472	1085274	1012 4116	1440 6144 ± 0	1457 1635 ± 5	1165 1639 ± 6	1153 1630 ± 6	1099 1650 ± 5	994 1681 ± 6	857 2686 ± 112	893 2869 ± 50
899	oorudigital-000	1027828	75456	238 609	1150 2066 ± 0	703 664 ± 0	484 664 ± 0	476 693 ± 0	482 756 ± 0	628 1066 ± 8	624 1606 ± 13	636 1652 ± 23
900	ooto-001	382081	95138	766 1923	1209 3072 ± 0	1036 944 ± 0	744 953 ± 1	744 997 ± 0	634 957 ± 0	570 999 ± 0	1080 4993 ± 17	1081 5004 ± 50
901	openedge-000	447048	272235	608 1376	1065 2056 ± 0	323 367 ± 0	236 404 ± 1	148 352 ± 1	127 350 ± 0	117 385 ± 0	1275 15122 ± 41	1277 15110 ± 48
902	openface-001	0	40111	27 100	525 2048 ± 0	85 148 ± 1	50 154 ± 0	161 365 ± 3	178 409 ± 9	276 616 ± 31	214 608 ± 14	226 604 ± 13
903	ovisionllc-001	192059	252531	476 1066	224 1024 ± 0	106 175 ± 0	59 175 ± 0	53 183 ± 0	50 196 ± 0	54 241 ± 0	892 2889 ± 38	900 2901 ± 47
904	ovisionllc-002	247778	96584	639 1478	245 1024 ± 0	197 266 ± 0	116 256 ± 0	88 250 ± 0	81 263 ± 1	72 313 ± 0	871 2763 ± 72	866 2727 ± 29
905	oz-001	303723	238311	-	1388 4125 ± 0	1204 1147 ± 3	937 1182 ± 3	975 1273 ± 4	1098 1617 ± 7	1117 2890 ± 19	1486 228011 ± 5455	1485 220746 ± 5572
906	oz-002	716022	170261	-	1148 2065 ± 0	1147 1064 ± 3	926 1171 ± 3	1210 2953 ± 6	1207 7352 ± 13	1202 26658 ± 29	1482 131108 ± 1408	1482 126758 ± 913
907	oz-003	484147	519652	1097 11575	1059 2053 ± 0	1362 1375 ± 12	1076 1388 ± 3	1166 1773 ± 16	1159 2039 ± 6	1131 3209 ± 5	1465 73905 ± 456	1464 73892 ± 444
908	oz-004	373982	1075452	1087 7845	1058 2053 ± 0	918 832 ± 7	672 871 ± 6	646 899 ± 10	734 1078 ± 12	978 1608 ± 10	1444 61654 ± 418	1442 61749 ± 450
909	pa-002	442564	308954	-	1492 18436 ± 0	74 132 ± 7	-	-	-	-	920 3088 ± 32	923 3051 ± 28
910	pa-003	714102	1464254	-	959 2048 ± 0	915 831 ± 31	-	-	-	-	400 962 ± 23	391 914 ± 38
911	palit-000	428754	144958	602 1360	1275 4096 ± 0	560 570 ± 1	390 578 ± 1	351 576 ± 3	311 583 ± 1	274 614 ± 1	768 2227 ± 16	770 2226 ± 16
912	palit-001	173886	145564	224 587	797 2048 ± 0	162 227 ± 0	100 224 ± 1	78 224 ± 1	69 229 ± 3	61 262 ± 2	477 1150 ± 16	478 1135 ± 23
913	pangiam-000	464252	24512	1005 3963	598 2048 ± 0	652 627 ± 5	430 618 ± 4	388 615 ± 3	346 620 ± 3	295 639 ± 3	17 118 ± 7	15 113 ± 7
914	pangiam-001	1015455	37259	673 1569	866 2048 ± 0	1070 972 ± 1	776 982 ± 3	722 981 ± 2	661 990 ± 7	607 1052 ± 2	22 136 ± 11	22 139 ± 13
915	pangiam-002	486035	39834	546 1205	917 2048 ± 0	257 316 ± 0	157 316 ± 0	124 317 ± 0	112 319 ± 1	80 326 ± 2	15 111 ± 13	11 103 ± 2
916	papago-001	669274	52817	840 2341	410 2048 ± 0	1305 1272 ± 6	1027 1296 ± 7	999 1295 ± 6	913 1281 ± 3	827 1345 ± 3	1280 15236 ± 169	1281 15184 ± 142
917	papil11-000	1146378	24895	818 2184	602 2048 ± 0	1030 942 ± 3	873 1108 ± 2	920 1204 ± 6	971 1369 ± 10	1044 2033 ± 35	219 612 ± 7	236 613 ± 11
918	papil11-001	1221825	38957	886 2618	860 2048 ± 0	1048 951 ± 2	895 1135 ± 2	930 1220 ± 4	983 1386 ± 10	1079 2292 ± 69	159 506 ± 22	166 519 ± 20
919	papil11-002	1208976	38985	872 2543	910 2048 ± 0	1007 916 ± 2	862 1094 ± 2	885 1163 ± 1	927 1307 ± 4	1063 2140 ± 10	177 560 ± 22	193 577 ± 22
920	papil11-003	1263123	39005	892 2657	541 2048 ± 0	1130 1038 ± 1	970 1222 ± 1	992 1293 ± 2	1024 1431 ± 3	1080 2298 ± 8	161 508 ± 10	162 501 ± 10
921	papil11-004	560851	18192	499 1104	73 296 ± 0	500 522 ± 0	338 522 ± 0	307 524 ± 0	270 531 ± 0	237 556 ± 2	1337 23204 ± 63	1339 23224 ± 68
922	papsav1923-001	279210	52652	167 474	629 2048 ± 0	651 626 ± 1	446 628 ± 1	406 630 ± 1	375 648 ± 2	386 744 ± 3	308 725 ± 25	325 731 ± 28
923	papsav1923-002	491185	24727	511 1139	1057 2052 ± 0	868 792 ± 1	774 978 ± 1	782 1042 ± 1	799 1158 ± 1	986 1641 ± 19	496 1209 ± 29	508 1206 ± 38
924	papsav1923-003	515576	24803	500 1113	592 2048 ± 0	873 797 ± 0	783 987 ± 1	784 1043 ± 1	818 1178 ± 1	1013 1809 ± 7	381 903 ± 26	388 905 ± 34

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 43: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
925	paravision-004	556670	145440	-	1291 4096 ± 0	912 829 ± 2	644 834 ± 6	595 832 ± 2	543 833 ± 4	462 833 ± 2	316 737 ± 31	317 718 ± 38
926	paravision-006	542423	202234	-	1314 4096 ± 0	768 700 ± 1	531 704 ± 0	491 705 ± 0	434 704 ± 0	349 705 ± 0	88 321 ± 16	90 315 ± 14
927	paravision-008	542190	204400	613 1399	1261 4096 ± 0	766 699 ± 0	524 700 ± 0	487 701 ± 0	433 702 ± 1	346 702 ± 0	94 337 ± 17	101 330 ± 13
928	paravision-010	688291	205854	810 2150	1355 4100 ± 0	665 634 ± 0	461 635 ± 0	418 635 ± 0	359 635 ± 0	291 635 ± 1	616 1577 ± 35	621 1571 ± 32
929	paravision-011	781138	95589	850 2424	1351 4100 ± 0	939 852 ± 0	671 871 ± 1	617 858 ± 1	560 854 ± 0	488 873 ± 1	625 1608 ± 35	628 1625 ± 32
930	paravision-013	1537125	95602	1032 4648	1354 4100 ± 0	1088 991 ± 9	788 993 ± 5	734 992 ± 4	675 1002 ± 5	574 1002 ± 7	584 1481 ± 26	597 1490 ± 25
931	paravision-015	1584003	164658	1039 4820	1346 4100 ± 0	1359 1371 ± 12	1159 1577 ± 1	1128 1500 ± 4	1034 1449 ± 1	942 1521 ± 7	563 1432 ± 11	576 1455 ± 32
932	paravision-017	1584003	164639	1040 4820	1343 4100 ± 0	1322 1295 ± 11	909 1147 ± 4	878 1158 ± 8	805 1166 ± 7	682 1140 ± 12	574 1461 ± 10	580 1459 ± 16
933	paravision-018	1596372	164639	1043 4909	1359 4100 ± 0	1431 1474 ± 11	1135 1472 ± 12	1122 1475 ± 11	1045 1459 ± 11	943 1523 ± 16	608 1532 ± 68	606 1520 ± 40
934	paravision-020	1646513	155644	974 3487	1362 4100 ± 0	1496 2043 ± 0	1205 2043 ± 0	1194 2048 ± 0	1158 2038 ± 0	1054 2100 ± 0	745 2110 ± 10	746 2110 ± 17
935	paravision-022	1646513	155659	973 3460	1348 4100 ± 0	1498 2070 ± 0	1209 2057 ± 2	1198 2056 ± 0	1162 2068 ± 1	1053 2098 ± 0	736 2063 ± 72	761 2181 ± 38
936	pensees-001	1619431	408932	765 1922	1476 8200 ± 0	1174 1108 ± 3	1123 1448 ± 17	1097 1439 ± 10	1050 1464 ± 5	958 1546 ± 9	928 3151 ± 34	931 3143 ± 25
937	pentacomp-000	261914	192048	270 658	676 2048 ± 0	422 452 ± 1	274 444 ± 0	245 458 ± 4	205 456 ± 5	167 461 ± 3	451 1085 ± 25	458 1081 ± 38
938	pixelall-002	0	169030	-	1193 2560 ± 0	127 191 ± 1	-	-	-	-	502 1223 ± 56	511 1230 ± 47
939	pixelall-003	0	707030	-	1195 2560 ± 0	764 699 ± 8	-	-	-	-	489 1174 ± 28	479 1139 ± 68
940	pixelall-004	0	550919	-	1191 2560 ± 0	421 451 ± 11	277 449 ± 3	237 451 ± 1	208 459 ± 2	193 502 ± 4	472 1139 ± 65	503 1186 ± 31
941	pixelall-005	0	1001355	-	1431 5120 ± 0	1182 1112 ± 3	879 1115 ± 1	849 1120 ± 1	765 1124 ± 2	683 1143 ± 2	512 1259 ± 29	515 1243 ± 23
942	pixelall-006	0	746305	-	1192 2560 ± 0	1116 1024 ± 3	815 1028 ± 2	775 1033 ± 1	698 1032 ± 1	608 1054 ± 2	321 754 ± 14	319 722 ± 10
943	pixelall-007	0	444912	-	720 2048 ± 0	1118 1026 ± 4	823 1038 ± 2	822 1089 ± 2	737 1087 ± 2	667 1124 ± 2	297 708 ± 14	303 701 ± 19
944	pixelall-008	0	992249	721 1742	1471 8192 ± 0	1429 1471 ± 3	1094 1405 ± 4	1081 1409 ± 4	1008 1413 ± 3	882 1426 ± 4	676 1799 ± 50	684 1807 ± 48
945	pixelall-009	0	1009114	716 1732	1472 8192 ± 0	1437 1484 ± 3	1087 1395 ± 3	1069 1400 ± 4	988 1391 ± 3	887 1433 ± 3	688 1848 ± 13	689 1842 ± 19
946	pixelall-010	0	1469903	898 2693	1470 8192 ± 0	1210 1153 ± 3	849 1084 ± 4	836 1103 ± 4	740 1093 ± 3	661 1118 ± 4	702 1906 ± 13	702 1898 ± 29
947	pnpsecure-000	9726	146266	24 93	563 2048 ± 0	9 35 ± 0	8 42 ± 0	37 145 ± 0	177 409 ± 1	963 1565 ± 5	706 1918 ± 11	704 1916 ± 16
948	pnpsecure-001	1015928	59243	791 2060	887 2048 ± 0	719 674 ± 0	526 701 ± 0	638 885 ± 0	1038 1452 ± 1	1133 3286 ± 3	849 2621 ± 19	854 2649 ± 41
949	pnpsecure-002	1015393	146387	808 2139	739 2048 ± 0	974 883 ± 0	669 869 ± 0	621 867 ± 0	608 926 ± 0	582 1009 ± 0	852 2640 ± 11	865 2711 ± 11
950	privid-001	0	76008	48 143	1497 30720 ± 0	419 450 ± 0	282 454 ± 0	254 467 ± 1	226 472 ± 1	194 504 ± 1	1294 17041 ± 74	1296 17053 ± 100
951	privid-002	0	219733	454 1031	540 2048 ± 0	1299 1261 ± 3	913 1156 ± 4	877 1155 ± 2	801 1162 ± 3	795 1302 ± 4	279 686 ± 22	286 680 ± 20
952	privid-003	0	188341	472 1062	620 2048 ± 0	1406 1430 ± 2	1119 1437 ± 6	1115 1459 ± 7	1035 1451 ± 7	929 1499 ± 18	310 725 ± 7	318 720 ± 13
953	privid-004	0	992936	583 1304	608 2048 ± 0	1327 1303 ± 1	1013 1282 ± 2	1015 1324 ± 3	966 1363 ± 0	857 1387 ± 0	303 717 ± 29	301 699 ± 19
954	psl-000	260619	1537815	-	631 2048 ± 0	657 629 ± 11	-	-	-	-	72 280 ± 17	79 280 ± 17
955	psl-001	382035	1461947	-	1102 2056 ± 0	860 785 ± 16	-	-	-	-	78 298 ± 13	84 292 ± 14
956	psl-002	804934	1462048	-	1038 2052 ± 0	982 888 ± 9	-	-	-	-	619 1590 ± 48	477 1133 ± 78
957	psl-003	1159643	1434859	-	1176 2120 ± 0	947 865 ± 3	-	-	-	-	429 1052 ± 14	429 1025 ± 51
958	psl-004	771727	1560145	-	1218 3144 ± 0	928 839 ± 3	-	-	-	-	270 680 ± 34	282 670 ± 22
959	psl-005	774031	1560153	-	1220 3144 ± 0	1014 922 ± 37	-	-	-	-	289 703 ± 32	309 708 ± 26
960	psl-006	1211577	524525	-	1221 3144 ± 0	1410 1438 ± 6	1116 1434 ± 4	1094 1434 ± 4	1025 1434 ± 5	892 1437 ± 5	75 291 ± 34	75 268 ± 25
961	psl-007	954351	524521	-	1219 3144 ± 0	1395 1408 ± 5	1104 1417 ± 3	1087 1418 ± 3	1012 1419 ± 2	878 1422 ± 3	66 265 ± 22	71 258 ± 17
962	psl-008	954351	524525	-	1222 3144 ± 0	1397 1412 ± 4	1102 1415 ± 3	1086 1416 ± 2	1011 1418 ± 2	876 1418 ± 2	64 259 ± 22	67 252 ± 22
963	psl-009	411027	411504	-	1402 4168 ± 0	1369 1382 ± 2	1071 1381 ± 1	1056 1383 ± 1	980 1383 ± 2	854 1385 ± 1	87 316 ± 14	83 289 ± 14
964	psl-010	411027	591157	1052 5362	1403 4168 ± 0	1391 1403 ± 9	1082 1393 ± 3	1062 1392 ± 3	991 1395 ± 3	862 1396 ± 3	101 354 ± 53	100 329 ± 29
965	psl-011	814579	606050	1045 4959	1479 8248 ± 0	1339 1324 ± 2	1049 1323 ± 8	1018 1326 ± 8	943 1324 ± 8	810 1322 ± 4	640 1680 ± 37	644 1688 ± 40
966	psl-012	1318147	622004	1030 4535	1394 4144 ± 0	1180 1111 ± 2	883 1122 ± 2	839 1104 ± 2	760 1119 ± 1	660 1117 ± 1	188 583 ± 17	203 589 ± 17
967	psl-013	1318151	622036	1027 4525	1395 4144 ± 0	1188 1119 ± 1	868 1105 ± 1	841 1110 ± 1	747 1104 ± 1	671 1126 ± 1	437 1062 ± 62	465 1098 ± 51
968	psl-014	915965	622044	1029 4525	1396 4144 ± 0	1175 1109 ± 1	901 1139 ± 1	838 1104 ± 1	759 1118 ± 1	657 1114 ± 1	470 1132 ± 105	473 1114 ± 37

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 44: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
969	ptakuratsatu-000	0	585434	595 1347	185 538 ± 0	965 875 ± 3	665 863 ± 48	676 928 ± 9	635 958 ± 17	629 1066 ± 26	1118 5900 ± 103	1110 5687 ± 167
970	pxl-001	110116	78231	55 168	152 512 ± 0	49 101 ± 5	32 104 ± 5	59 189 ± 12	176 408 ± 27	910 1470 ± 144	1106 5598 ± 45	1106 5590 ± 68
971	pyramid-000	372608	219883	339 804	1094 2056 ± 0	576 583 ± 2	-	-	-	-	1161 7147 ± 59	1171 7586 ± 425
972	pyramid-001	1040862	143739	801 2098	475 2048 ± 0	886 809 ± 0	621 806 ± 0	574 815 ± 4	544 837 ± 0	494 881 ± 2	1245 12265 ± 48	1247 12276 ± 51
973	qazbs-000	362015	805258	456 1034	483 2048 ± 0	1329 1307 ± 1	986 1243 ± 0	954 1248 ± 9	887 1253 ± 1	768 1270 ± 0	1087 5181 ± 62	1087 5167 ± 93
974	qazsmartvisionai-000	560570	99515	841 2371	126 512 ± 0	759 694 ± 1	538 708 ± 7	501 717 ± 5	442 711 ± 5	366 723 ± 7	342 795 ± 29	340 781 ± 34
975	qazsmartvisionai-001	1417181	102251	933 2987	190 544 ± 0	1170 1103 ± 20	892 1129 ± 3	855 1124 ± 5	789 1145 ± 7	676 1134 ± 16	340 793 ± 29	359 804 ± 32
976	qazsmartvisionai-002	1365926	102252	928 2895	193 544 ± 0	1149 1068 ± 6	886 1125 ± 4	845 1114 ± 6	718 1066 ± 1	655 1113 ± 15	346 800 ± 31	348 798 ± 34
977	qazsmartvisionai-003	1365926	102265	927 2894	191 544 ± 0	1141 1056 ± 5	844 1078 ± 12	815 1079 ± 10	733 1078 ± 10	637 1088 ± 18	358 822 ± 28	356 803 ± 32
978	qazsmartvisionai-004	1564226	102265	958 3281	192 544 ± 0	1309 1280 ± 13	1036 1310 ± 14	1028 1338 ± 10	941 1323 ± 13	658 1115 ± 2	323 758 ± 38	331 758 ± 31
979	qluevision-001	173605	205230	126 376	1473 8192 ± 0	163 229 ± 1	104 230 ± 1	82 231 ± 1	71 233 ± 1	52 239 ± 1	952 3374 ± 38	957 3365 ± 41
980	qnap-000	186731	15598	-	435 2048 ± 0	799 726 ± 9	284 457 ± 1	244 458 ± 0	218 464 ± 1	177 482 ± 2	265 660 ± 25	275 654 ± 29
981	qnap-001	196210	13399	90 287	923 2048 ± 0	631 614 ± 1	425 615 ± 1	402 627 ± 1	350 623 ± 1	288 634 ± 2	256 649 ± 11	270 648 ± 14
982	qnap-002	346963	33284	389 882	402 2048 ± 0	905 821 ± 1	631 824 ± 1	584 824 ± 1	539 826 ± 1	458 832 ± 1	76 293 ± 13	81 287 ± 17
983	qnap-003	245476	61427	341 807	464 2048 ± 0	347 387 ± 0	225 393 ± 0	185 393 ± 0	162 393 ± 1	130 400 ± 2	273 683 ± 20	272 651 ± 17
984	qnap-004	371274	61422	438 1002	433 2048 ± 0	319 365 ± 0	199 365 ± 1	164 367 ± 0	142 368 ± 0	110 375 ± 3	271 681 ± 7	288 682 ± 13
985	qnap-005	977818	61429	921 2828	430 2048 ± 0	1068 969 ± 5	785 988 ± 8	732 992 ± 6	672 1000 ± 7	581 1008 ± 6	243 633 ± 1	256 632 ± 13
986	qnap-006	1244083	61533	964 3351	1007 2052 ± 0	1117 1025 ± 5	829 1051 ± 6	781 1041 ± 6	713 1055 ± 8	614 1057 ± 10	281 688 ± 29	279 664 ± 10
987	quantasoft-003	370518	211354	473 1063	803 2048 ± 0	662 632 ± 2	460 634 ± 0	408 632 ± 0	355 631 ± 1	289 634 ± 0	42 201 ± 7	44 203 ± 8
988	rankone-000	0	96896	-	14 144 ± 0	31 82 ± 9	-	-	-	-	1405 39932 ± 468	1193 8722 ± 171
989	rankone-001	0	97092	-	21 208 ± 0	18 56 ± 4	-	-	-	-	1461 72754 ± 658	109 345 ± 29
990	rankone-002	0	98585	-	22 224 ± 0	25 75 ± 1	-	-	-	-	1456 69113 ± 3802	114 369 ± 26
991	rankone-003	0	93554	-	23 228 ± 0	10 37 ± 1	-	-	-	-	89 322 ± 20	94 324 ± 19
992	rankone-004	0	135727	-	12 133 ± 0	27 78 ± 1	-	-	-	-	96 340 ± 19	85 298 ± 17
993	rankone-005	0	104048	-	13 133 ± 0	24 71 ± 2	-	-	-	-	120 403 ± 10	103 337 ± 19
994	rankone-006	0	133045	-	17 165 ± 0	149 210 ± 1	-	-	-	-	130 443 ± 26	124 395 ± 22
995	rankone-007	0	136942	-	16 165 ± 0	179 245 ± 5	-	-	-	-	280 688 ± 20	219 601 ± 16
996	rankone-008	2	141234	-	18 165 ± 0	202 272 ± 3	-	-	-	-	319 750 ± 20	233 613 ± 23
997	rankone-009	0	107688	-	51 260 ± 0	111 179 ± 4	-	-	-	-	299 710 ± 32	181 552 ± 25
998	rankone-010	0	138435	-	57 261 ± 0	130 193 ± 1	-	-	-	-	73 282 ± 13	60 234 ± 16
999	rankone-011	0	179209	-	52 261 ± 0	554 567 ± 1	371 557 ± 1	340 567 ± 1	314 586 ± 1	328 682 ± 3	74 283 ± 14	51 220 ± 19
1000	rankone-012	0	264182	-	53 261 ± 0	550 564 ± 3	368 554 ± 1	337 564 ± 1	313 586 ± 1	343 695 ± 1	70 273 ± 17	57 231 ± 14
1001	rankone-013	0	228729	51 150	54 261 ± 0	752 690 ± 5	493 672 ± 1	496 712 ± 1	503 780 ± 1	662 1118 ± 3	104 356 ± 23	86 304 ± 23
1002	rankone-014	0	243130	54 164	55 261 ± 0	769 701 ± 1	532 705 ± 0	515 732 ± 1	515 800 ± 1	656 1113 ± 1	83 306 ± 16	66 251 ± 13
1003	rankone-015	0	289078	70 190	56 261 ± 0	532 547 ± 5	358 543 ± 6	352 578 ± 2	354 629 ± 7	502 894 ± 9	43 201 ± 19	31 172 ± 16
1004	realnetworks-000	99044	107893	-	1342 4100 ± 0	81 144 ± 2	-	-	-	-	822 2458 ± 26	829 2467 ± 35
1005	realnetworks-001	99044	107893	-	1345 4100 ± 0	82 144 ± 2	-	-	-	-	832 2500 ± 47	814 2420 ± 34
1006	realnetworks-002	95328	107088	-	335 1848 ± 0	183 250 ± 2	108 242 ± 4	102 282 ± 5	151 381 ± 10	414 774 ± 15	523 1285 ± 17	517 1247 ± 42
1007	realnetworks-003	95334	104498	-	334 1848 ± 0	108 177 ± 10	-	-	-	-	603 1516 ± 29	608 1522 ± 60
1008	realnetworks-004	172335	913988	-	1120 2056 ± 0	273 330 ± 4	170 333 ± 3	192 402 ± 7	312 585 ± 15	864 1402 ± 51	498 1210 ± 29	507 1202 ± 17
1009	realnetworks-005	172253	56755	-	1081 2056 ± 0	151 211 ± 4	87 205 ± 3	106 290 ± 6	259 515 ± 17	802 1312 ± 78	499 1213 ± 17	509 1207 ± 16
1010	realnetworks-006	466225	56771	681 1596	1070 2056 ± 0	672 638 ± 4	449 630 ± 3	455 672 ± 5	438 706 ± 5	415 774 ± 5	140 469 ± 19	154 478 ± 25
1011	realnetworks-007	570797	101527	951 3145	1099 2056 ± 0	1346 1348 ± 2	1064 1358 ± 11	1043 1363 ± 10	982 1386 ± 9	939 1517 ± 6	176 559 ± 31	175 539 ± 35
1012	realnetworks-008	73346	75421	136 402	1071 2056 ± 0	233 296 ± 3	143 294 ± 3	150 353 ± 4	135 361 ± 5	180 485 ± 5	170 539 ± 31	176 543 ± 29

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 45: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1013	realsense-001	363245	116558	780 1995	1055 2052 ± 0	1480 1867 ± 0	1189 1845 ± 0	1177 1854 ± 0	1137 1875 ± 1	1033 1952 ± 2	1287 16252 ± 121	1290 16305 ± 100
1014	realsense-002	406105	116558	750 1852	714 2048 ± 0	1489 1939 ± 0	1196 1923 ± 0	1186 1939 ± 0	1150 1940 ± 1	1046 2042 ± 2	1288 16369 ± 206	1291 16347 ± 167
1015	realsense-003	665083	124487	687 1637	357 2048 ± 0	1403 1422 ± 1	1050 1326 ± 2	981 1278 ± 1	1063 1492 ± 2	954 1540 ± 3	1187 8429 ± 100	1191 8560 ± 127
1016	rebs-000	1375350	62664	931 2934	1245 4096 ± 0	647 623 ± 0	457 633 ± 10	398 626 ± 10	340 616 ± 1	277 617 ± 0	478 1150 ± 1	488 1150 ± 2
1017	rebs-001	2038561	62673	883 2582	1363 4100 ± 0	880 802 ± 1	623 812 ± 8	590 829 ± 1	520 807 ± 2	440 811 ± 4	479 1151 ± 8	483 1145 ± 4
1018	recognito-000	549308	247223	1061 5903	1197 2564 ± 0	1158 1086 ± 2	866 1096 ± 1	834 1102 ± 2	786 1142 ± 3	729 1219 ± 7	37 193 ± 8	41 190 ± 16
1019	recognito-001	517547	62334	995 3833	1198 2564 ± 0	1152 1072 ± 1	854 1089 ± 1	829 1095 ± 4	754 1112 ± 2	722 1205 ± 4	33 183 ± 10	35 183 ± 7
1020	regula-000	262444	29384	239 610	949 2048 ± 0	1240 1187 ± 1	888 1126 ± 1	858 1129 ± 0	777 1132 ± 1	694 1159 ± 1	153 491 ± 16	160 500 ± 22
1021	regula-001	256075	25980	433 987	878 2048 ± 0	1313 1284 ± 1	967 1220 ± 1	931 1222 ± 1	866 1226 ± 1	760 1255 ± 1	107 361 ± 10	107 342 ± 25
1022	regula-002	355812	42119	475 1066	1053 2052 ± 0	1057 962 ± 0	712 918 ± 0	660 917 ± 0	613 932 ± 0	572 1000 ± 1	494 1194 ± 14	512 1233 ± 53
1023	regula-003	495453	49195	627 1428	1125 2056 ± 0	1177 1110 ± 0	828 1048 ± 0	794 1055 ± 0	730 1077 ± 1	680 1140 ± 1	1469 77846 ± 428	1467 77865 ± 437
1024	remarkai-000	240152	868276	-	517 2048 ± 0	914 829 ± 7	-	-	-	-	372 873 ± 4	369 835 ± 35
1025	remarkai-001	241857	868314	303 730	1016 2052 ± 0	917 831 ± 6	654 849 ± 18	793 1055 ± 25	835 1198 ± 34	941 1519 ± 38	505 1229 ± 20	360 805 ± 56
1026	remarkai-002	224157	808777	-	964 2048 ± 0	1044 950 ± 6	732 935 ± 4	696 953 ± 3	642 964 ± 4	632 1073 ± 4	464 1115 ± 25	449 1068 ± 54
1027	remarkai-003	280516	58559	1001 3909	1341 4100 ± 0	1085 986 ± 1	789 993 ± 1	733 992 ± 1	671 999 ± 3	586 1019 ± 2	336 787 ± 20	346 793 ± 22
1028	rendip-000	0	437653	310 735	815 2048 ± 0	436 464 ± 2	286 458 ± 0	260 473 ± 0	234 483 ± 1	239 556 ± 4	185 576 ± 13	191 573 ± 11
1029	revealmedia-005	293933	202465	323 767	1361 4100 ± 0	397 428 ± 0	252 428 ± 0	218 430 ± 0	189 433 ± 0	153 442 ± 0	726 2023 ± 38	726 2009 ± 26
1030	revealmedia-006	293933	200912	313 745	1019 2052 ± 0	340 381 ± 0	211 381 ± 0	178 382 ± 0	153 384 ± 0	125 394 ± 0	234 626 ± 35	218 600 ± 2
1031	roc-016	0	674869	71 191	82 392 ± 0	547 562 ± 1	359 543 ± 2	332 553 ± 2	296 564 ± 2	283 626 ± 2	38 194 ± 8	28 161 ± 26
1032	roc-017	0	565239	93 291	83 396 ± 0	685 651 ± 2	479 660 ± 1	457 675 ± 1	399 669 ± 2	374 736 ± 2	40 197 ± 4	34 178 ± 7
1033	roc-018	0	932665	159 459	210 792 ± 0	1274 1231 ± 3	974 1227 ± 4	962 1253 ± 2	880 1243 ± 4	800 1308 ± 3	51 217 ± 10	91 316 ± 14
1034	roc-019	0	944816	119 360	88 444 ± 0	1246 1196 ± 4	944 1191 ± 5	929 1218 ± 6	868 1232 ± 4	770 1272 ± 5	84 306 ± 7	72 259 ± 8
1035	roc-020	0	1095173	124 375	87 444 ± 0	1483 1906 ± 0	1198 1936 ± 0	1191 1985 ± 0	1157 2030 ± 1	1057 2112 ± 1	211 602 ± 53	186 564 ± 22
1036	roc-021	0	1004172	125 375	89 448 ± 0	1467 1738 ± 0	1177 1781 ± 0	1176 1853 ± 0	1133 1863 ± 0	1036 1960 ± 23	179 561 ± 11	178 545 ± 11
1037	rokid-000	258612	396624	550 1218	1082 2056 ± 0	531 546 ± 3	357 542 ± 2	327 545 ± 1	264 522 ± 3	242 563 ± 4	962 3457 ± 62	971 3463 ± 77
1038	rokid-001	641223	413733	479 1071	1140 2060 ± 0	998 911 ± 2	697 901 ± 5	645 899 ± 2	593 900 ± 3	506 901 ± 3	949 3345 ± 50	955 3346 ± 149
1039	s1-001	435491	844340	-	1169 2092 ± 0	615 605 ± 24	438 623 ± 20	663 920 ± 39	1084 1567 ± 92	-	562 1428 ± 34	561 1415 ± 85
1040	s1-002	520163	95479	-	1442 6144 ± 0	1296 1257 ± 1	999 1260 ± 1	965 1261 ± 1	896 1262 ± 1	772 1273 ± 1	1052 4513 ± 25	1048 4479 ± 25
1041	s1-003	145509	95446	-	1250 4096 ± 0	1040 947 ± 0	754 959 ± 0	694 952 ± 0	629 952 ± 1	541 955 ± 1	986 3657 ± 19	991 3652 ± 16
1042	s1-004	246514	202623	-	956 2048 ± 0	895 815 ± 0	626 818 ± 1	579 818 ± 1	531 820 ± 1	450 828 ± 1	941 3245 ± 100	936 3161 ± 88
1043	s1-005	482369	95685	515 1146	360 2048 ± 0	1097 1001 ± 0	797 1002 ± 0	751 1004 ± 0	683 1008 ± 0	592 1029 ± 2	235 626 ± 74	202 589 ± 14
1044	s1-006	482372	95681	514 1146	363 2048 ± 0	1050 951 ± 0	749 956 ± 0	698 957 ± 0	639 962 ± 0	557 983 ± 0	284 696 ± 23	298 696 ± 29
1045	s1-007	482385	59657	513 1144	709 2048 ± 0	1087 988 ± 1	757 965 ± 0	710 966 ± 0	646 972 ± 0	564 991 ± 0	254 648 ± 26	276 658 ± 25
1046	s1-008	786695	102436	879 2573	850 2048 ± 0	879 800 ± 5	625 814 ± 3	580 818 ± 10	523 809 ± 8	467 841 ± 3	257 651 ± 32	269 647 ± 34
1047	s1-009	1457134	102436	1026 4435	429 2048 ± 0	1259 1213 ± 9	972 1224 ± 13	938 1230 ± 4	883 1251 ± 3	762 1258 ± 11	240 632 ± 13	244 622 ± 2
1048	saffe-001	85973	62488	57 168	302 1280 ± 0	212 281 ± 1	-	-	-	-	516 1274 ± 19	527 1277 ± 26
1049	saffe-002	260622	28285	377 855	623 2048 ± 0	898 817 ± 11	620 805 ± 15	568 809 ± 19	526 815 ± 29	441 813 ± 23	304 717 ± 7	315 714 ± 29
1050	samsungsds-000	0	307431	-	804 2048 ± 0	259 316 ± 0	165 326 ± 5	130 328 ± 4	117 327 ± 1	92 343 ± 0	1338 23722 ± 295	1340 23874 ± 305
1051	samsungsds-001	1189592	147444	1003 3922	1301 4096 ± 0	1196 1140 ± 3	906 1145 ± 4	1032 1344 ± 5	969 1366 ± 5	937 1514 ± 7	1427 51559 ± 773	1427 51721 ± 1003
1052	samsungsds-002	1040732	147475	961 3321	1074 2056 ± 0	1186 1118 ± 1	929 1175 ± 12	1048 1372 ± 6	942 1324 ± 2	924 1489 ± 4	1394 35803 ± 266	1397 36181 ± 674
1053	samtech-000	109774	12807	-	1060 2056 ± 0	193 262 ± 2	-	-	-	-	1056 4550 ± 26	1054 4541 ± 28
1054	samtech-001	288082	219883	234 605	1108 2056 ± 0	230 294 ± 3	-	-	-	-	1173 7694 ± 59	1174 7678 ± 91
1055	samtech-002	192220	474779	295 714	1097 2056 ± 0	153 212 ± 0	92 212 ± 0	75 215 ± 0	66 225 ± 0	66 285 ± 1	1279 15231 ± 50	1283 15215 ± 59
1056	scanalitix-001	270730	152135	227 591	1039 2052 ± 0	1399 1418 ± 1	1108 1421 ± 1	1072 1401 ± 1	1032 1442 ± 1	933 1507 ± 3	1363 27691 ± 59	1366 27723 ± 65

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 46: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1057	scanovate-001	257083	328532	-	[776]2048 ± 0	[564]577 ± 24	[388]577 ± 21	[412]632 ± 27	[495]770 ± 28	[866]1404 ± 32	[1242]12054 ± 699	[1262]13795 ± 705
1058	scanovate-002	256986	457227	[375]850	[872]2048 ± 0	[763]696 ± 32	[542]713 ± 33	[522]738 ± 28	[502]779 ± 32	[703]1172 ± 53	[914]3021 ± 38	[928]3120 ± 163
1059	scanovate-003	135585	89469	[340]805	[506]2048 ± 0	[580]585 ± 1	[421]613 ± 12	[361]591 ± 1	[331]610 ± 2	[329]684 ± 1	[902]2926 ± 22	[903]2925 ± 20
1060	sdc-000	256814	481583	[329]784	[795]2048 ± 0	[1000]913 ± 14	[702]906 ± 9	[870]1142 ± 19	[1117]1774 ± 45	[1155]4719 ± 222	[1382]32645 ± 93	[1384]32653 ± 112
1061	seamfix-001	429267	86391	[104]320	[558]2048 ± 0	[787]718 ± 8	[535]707 ± 25	[690]947 ± 18	[1073]1521 ± 55	[1149]4179 ± 226	[1117]5897 ± 118	[1117]5867 ± 123
1062	securifai-001	120291	12456	-	[1368]4104 ± 0	[152]211 ± 1	[91]211 ± 1	[73]211 ± 1	[59]211 ± 1	[43]211 ± 1	[641]1681 ± 29	[648]1701 ± 25
1063	securifai-002	198824	13496	-	[277]1032 ± 0	[68]123 ± 0	[39]123 ± 0	[32]123 ± 0	[28]123 ± 0	[19]123 ± 0	[224]615 ± 22	[214]597 ± 19
1064	securifai-003	303794	13512	-	[1376]4104 ± 0	[534]549 ± 7	[366]550 ± 7	[329]549 ± 7	[284]546 ± 6	[229]546 ± 6	[650]1714 ± 26	[651]1713 ± 37
1065	securifai-004	282177	12027	[255]636	[404]2048 ± 0	[951]869 ± 1	[667]867 ± 1	[622]867 ± 1	[567]867 ± 1	[483]865 ± 1	[646]1711 ± 19	[650]1705 ± 29
1066	securifai-005	252532	81777	[195]526	[481]2048 ± 0	[1364]1377 ± 2	[1062]1355 ± 1	[1039]1353 ± 0	[960]1357 ± 0	[835]1356 ± 0	[693]1873 ± 25	[691]1847 ± 35
1067	securifai-006	452474	81856	[326]771	[1292]4096 ± 0	[1162]1090 ± 2	[850]1086 ± 3	[825]1093 ± 1	[748]1104 ± 10	[641]1090 ± 2	[954]3376 ± 42	[962]3399 ± 40
1068	securifai-007	94589	88169	[221]583	[832]2048 ± 0	[479]500 ± 0	[327]506 ± 0	[284]499 ± 1	[246]501 ± 1	[195]505 ± 1	[587]1484 ± 11	[593]1483 ± 13
1069	securifai-008	111759	84658	[293]711	[879]2048 ± 0	[673]641 ± 4	[467]647 ± 3	[448]663 ± 3	[372]645 ± 3	[304]649 ± 1	[596]1498 ± 11	[594]1487 ± 13
1070	securitypitch-000	271112	45153	[212]559	[1151]2066 ± 0	[820]750 ± 0	[575]749 ± 0	[531]750 ± 0	[479]751 ± 0	[394]758 ± 0	[634]1644 ± 14	[640]1669 ± 20
1071	sensetime-001	531783	7203	-	[1025]2052 ± 0	[798]725 ± 3	-	-	-	-	[837]2546 ± 102	[802]2371 ± 45
1072	sensetime-002	531783	7203	-	[1012]2052 ± 0	[874]797 ± 3	-	-	-	-	[862]2713 ± 90	[789]2301 ± 25
1073	sensetime-003	787853	74950	-	[1029]2052 ± 0	[993]908 ± 4	-	-	-	-	[835]2527 ± 65	[917]3004 ± 174
1074	sensetime-004	954663	30733	-	[272]1028 ± 0	[1409]1437 ± 15	-	-	-	-	[507]1239 ± 31	[498]1171 ± 22
1075	sensetime-005	765353	37673	-	[270]1028 ± 0	[1355]1361 ± 27	[1032]1304 ± 1	[1012]1319 ± 1	[963]1360 ± 1	[936]1514 ± 1	[503]1223 ± 28	[501]1184 ± 29
1076	sensetime-006	765353	37673	[8]33	[276]1028 ± 0	[1352]1352 ± 17	[1037]1311 ± 1	[1013]1323 ± 1	[959]1357 ± 1	[944]1523 ± 2	[490]1179 ± 28	[494]1157 ± 29
1077	sensetime-007	765353	37533	[1063]5950	[275]1028 ± 0	[1376]1386 ± 41	[1048]1323 ± 2	[1034]1347 ± 2	[970]1366 ± 2	[972]1593 ± 8	[573]1460 ± 29	[566]1425 ± 26
1078	sensetime-008	1176483	60067	[1058]5882	[274]1028 ± 0	[1435]1479 ± 31	[1118]1436 ± 4	[1123]1482 ± 4	[1074]1525 ± 5	[991]1669 ± 2	[522]1283 ± 51	[513]1240 ± 47
1079	sensetime-010	1175952	62384	[1071]6671	[273]1028 ± 0	[1228]1176 ± 36	[905]1144 ± 2	[876]1153 ± 9	[813]1172 ± 6	[736]1229 ± 7	[100]351 ± 1	[95]324 ± 5
1080	sensetime-011	1176053	63554	[1062]5909	[287]1060 ± 0	[1445]1531 ± 1	[1145]1502 ± 0	[1139]1547 ± 0	[1071]1508 ± 0	[970]1586 ± 0	[158]501 ± 1	[163]503 ± 2
1081	sensetime-012	1252842	63563	[1064]6047	[290]1064 ± 0	[1464]1711 ± 0	[1166]1651 ± 0	[1154]1656 ± 0	[1100]1651 ± 0	[1003]1753 ± 0	[212]603 ± 77	[274]652 ± 4
1082	serendipity-000	0	346443	[160]462	[478]2048 ± 0	[987]900 ± 8	[698]901 ± 3	[667]922 ± 2	[617]936 ± 1	[560]987 ± 2	[210]602 ± 1	[228]607 ± 10
1083	sertis-000	265572	68770	[145]427	[839]2048 ± 0	[826]754 ± 0	[584]759 ± 0	[542]764 ± 0	[485]760 ± 0	[400]763 ± 0	[594]1497 ± 29	[622]1582 ± 38
1084	sertis-001	292591	68831	-	[979]2048 ± 0	[664]634 ± 1	-	-	-	-	[820]2449 ± 23	[862]2685 ± 41
1085	sertis-002	460790	68929	[271]660	[980]2048 ± 0	[1237]1181 ± 1	[931]1178 ± 0	[901]1183 ± 0	[832]1187 ± 0	[731]1221 ± 0	[453]1086 ± 32	[454]1076 ± 31
1086	sertis-003	0	451507	[774]1965	[893]2048 ± 0	[413]441 ± 0	[264]436 ± 2	[250]461 ± 2	[216]463 ± 1	[174]473 ± 4	[389]915 ± 10	[394]917 ± 8
1087	seventhsense-000	369850	1561668	-	[1044]2052 ± 0	[1292]1250 ± 3	[998]1257 ± 1	[964]1261 ± 1	[893]1259 ± 1	[771]1272 ± 2	[677]1800 ± 35	[675]1787 ± 32
1088	seventhsense-001	369850	3183365	[345]812	[1034]2052 ± 0	[1294]1255 ± 2	[1024]1294 ± 15	[980]1277 ± 3	[907]1275 ± 2	[783]1288 ± 3	[711]1936 ± 26	[715]1943 ± 34
1089	seventhsense-002	452197	1567903	[418]944	[1026]2052 ± 0	[1293]1252 ± 1	[1009]1271 ± 1	[972]1269 ± 1	[904]1272 ± 1	[786]1290 ± 1	[751]2131 ± 45	[748]2123 ± 45
1090	seventhsense-003	603091	453847	[492]1090	[1051]2052 ± 0	[1099]1003 ± 2	[771]977 ± 0	[589]828 ± 1	[529]817 ± 1	[448]826 ± 2	[605]1523 ± 25	[611]1530 ± 22
1091	seventhsense-005	4000865	453775	[882]2581	[1365]4100 ± 0	[1187]1119 ± 1	[882]1120 ± 1	[862]1134 ± 4	[775]1131 ± 2	[684]1144 ± 1	[876]2787 ± 28	[877]2777 ± 22
1092	seventhsense-006	1444389	451757	[880]2573	[1353]4100 ± 0	[1216]1162 ± 1	[919]1161 ± 2	[879]1158 ± 2	[798]1156 ± 0	[709]1182 ± 2	[867]2724 ± 16	[867]2728 ± 16
1093	shaman-000	0	120033	[183]507	[1334]4096 ± 0	[689]653 ± 16	-	-	-	-	[115]380 ± 25	[119]379 ± 31
1094	shaman-001	0	174446	[189]511	[1308]4096 ± 0	[228]294 ± 2	-	-	-	-	[245]635 ± 19	[137]441 ± 25
1095	shu-001	329513	62157	-	[407]2048 ± 0	[627]612 ± 5	-	-	-	-	[848]2619 ± 19	[914]2987 ± 143
1096	shu-002	731250	148309	[397]890	[1238]4096 ± 0	[824]751 ± 2	[594]769 ± 4	[668]922 ± 4	[1023]1431 ± 9	[1137]3489 ± 47	[1500]2930763 ± 47355	[1500]2929759 ± 39149
1097	shu-003	428774	146940	[188]511	[826]2048 ± 0	[902]820 ± 6	[637]828 ± 3	[685]941 ± 9	[928]1308 ± 15	[1126]3045 ± 44	[833]2506 ± 26	[837]2512 ± 38
1098	shufti-001	442529	2108442	[768]1941	[1042]2052 ± 0	[743]687 ± 0	[518]694 ± 0	[582]824 ± 1	[855]1216 ± 4	[1115]2875 ± 1	[926]3133 ± 19	[933]3153 ± 54
1099	siat-002	486842	7738	[851]2434	[1021]2052 ± 0	[570]579 ± 0	-	-	-	-	[330]769 ± 13	[330]750 ± 13
1100	siat-004	940063	6984	-	[1349]4100 ± 0	[711]670 ± 0	[490]671 ± 7	[475]693 ± 10	[471]742 ± 10	[527]935 ± 17	[1025]4013 ± 45	[1008]3782 ± 173

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 47: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1101	siat-005	380936	16935	580 1299	573 2048 ± 0	365 403 ± 0	233 400 ± 0	190 401 ± 0	172 403 ± 1	143 422 ± 7	186 577 ± 13	195 580 ± 17
1102	sightropic-000	355144	205017	371 847	470 2048 ± 0	494 515 ± 0	342 524 ± 0	301 519 ± 0	260 516 ± 0	228 544 ± 0	1213 10207 ± 35	1214 10214 ± 59
1103	sightropic-001	765429	490992	1044 4914	1067 2056 ± 0	1191 1134 ± 0	1045 1321 ± 6	866 1140 ± 1	781 1136 ± 2	696 1162 ± 6	1322 21280 ± 192	1325 21306 ± 176
1104	sightropic-002	611800	467645	772 1954	1045 2052 ± 0	1039 946 ± 0	740 947 ± 0	688 946 ± 0	631 955 ± 0	554 978 ± 0	1476 89034 ± 130	1475 89056 ± 133
1105	sjtu-001	347115	148322	-	565 2048 ± 0	682 650 ± 2	-	-	-	-	1499 2243804 ± 12751	1499 2249915 ± 19380
1106	sjtu-002	446215	148306	-	737 2048 ± 0	921 835 ± 2	-	-	-	-	1498 2211198 ± 26052	1498 2210428 ± 21877
1107	sjtu-003	480795	148243	203 539	653 2048 ± 0	904 821 ± 2	627 820 ± 2	670 923 ± 3	837 1201 ± 3	1087 2373 ± 9	614 1560 ± 20	617 1560 ± 14
1108	sjtu-004	1953267	241108	908 2741	1424 4608 ± 0	1280 1236 ± 2	959 1209 ± 2	994 1294 ± 4	1078 1554 ± 5	1107 2738 ± 8	918 3057 ± 14	925 3070 ± 20
1109	sjtu-005	1953267	241151	906 2734	1423 4608 ± 0	606 601 ± 1	428 617 ± 1	506 724 ± 2	771 1127 ± 3	1114 2855 ± 11	896 2898 ± 4	899 2900 ± 4
1110	sjtu-006	2685387	241394	942 3071	1425 4608 ± 0	684 651 ± 1	496 672 ± 4	554 788 ± 9	649 979 ± 2	633 1074 ± 7	893 2897 ± 4	897 2896 ± 4
1111	sjtu-007	1126310	244296	914 2759	1422 4608 ± 0	1002 914 ± 12	750 956 ± 12	899 1181 ± 37	1106 1707 ± 169	1059 2118 ± 191	900 2903 ± 14	901 2902 ± 11
1112	sjtu-008	1451928	245324	938 2997	1420 4608 ± 0	1476 1832 ± 1	1185 1836 ± 2	1172 1835 ± 0	1129 1839 ± 1	1024 1877 ± 1	1058 4621 ± 10	1057 4619 ± 25
1113	sjtu-009	894841	245301	936 2997	1426 4608 ± 0	1475 1832 ± 0	1188 1845 ± 1	1175 1846 ± 1	1130 1846 ± 1	1021 1874 ± 2	992 3688 ± 17	997 3694 ± 11
1114	sjtu-010	894838	403745	937 2997	1421 4608 ± 0	1278 1234 ± 1	924 1167 ± 1	889 1170 ± 1	822 1181 ± 1	753 1252 ± 2	995 3716 ± 16	999 3712 ± 23
1115	sktelecom-000	527132	298496	587 1321	312 1536 ± 0	1179 1110 ± 1	876 1113 ± 1	844 1114 ± 1	761 1120 ± 1	691 1155 ± 1	1358 26583 ± 128	1357 26508 ± 126
1116	smargtech-001	164944	16729	282 683	774 2048 ± 0	1460 1664 ± 6	1167 1659 ± 3	1158 1717 ± 11	1131 1847 ± 31	1088 2375 ± 105	961 3448 ± 10	974 3492 ± 17
1117	smartbiometrik-001	30875	92620	16 72	121 512 ± 0	644 620 ± 7	443 625 ± 7	425 640 ± 4	456 728 ± 6	604 1047 ± 8	288 703 ± 31	310 710 ± 40
1118	smartengines-000	1711	3025	10 50	70 288 ± 0	98 168 ± 7	63 180 ± 1	57 188 ± 3	62 217 ± 3	65 275 ± 1	39 197 ± 5	29 167 ± 11
1119	smartengines-001	7095	4601	9 47	71 288 ± 0	280 333 ± 89	238 408 ± 1	212 423 ± 1	209 460 ± 2	235 553 ± 5	26 153 ± 11	23 143 ± 13
1120	smartvist-000	5959	134084	59 169	116 512 ± 0	20 59 ± 0	14 56 ± 0	9 56 ± 0	8 58 ± 0	12 90 ± 1	566 1435 ± 31	565 1422 ± 48
1121	smartvist-001	349219	25644	400 904	359 2048 ± 0	326 370 ± 0	208 374 ± 0	172 377 ± 1	183 422 ± 0	156 445 ± 0	881 2835 ± 99	886 2836 ± 80
1122	smilart-001	67339	45737	-	125 512 ± 0	164 230 ± 11	-	-	-	-	541 1371 ± 29	547 1356 ± 57
1123	smilart-002	111826	87805	84 263	247 1024 ± 0	107 176 ± 16	-	-	-	-	1307 18784 ± 136	1311 18795 ± 151
1124	smilart-003	67339	91670	72 192	106 512 ± 0	112 180 ± 12	66 181 ± 10	121 313 ± 22	392 665 ± 49	1081 2299 ± 196	548 1395 ± 74	430 1027 ± 66
1125	sodec-000	836592	13142	929 2902	1319 4096 ± 0	1133 1041 ± 2	819 1032 ± 1	778 1035 ± 1	701 1037 ± 2	620 1061 ± 2	672 1794 ± 37	671 1775 ± 23
1126	softtech-000	656559	88676	508 1127	844 2048 ± 0	1135 1051 ± 0	830 1055 ± 0	795 1055 ± 0	712 1053 ± 0	624 1064 ± 0	618 1585 ± 29	624 1600 ± 40
1127	sparktek-000	2050067	115114	1094 9564	1333 4096 ± 0	1459 1656 ± 0	1168 1661 ± 0	1156 1671 ± 0	1101 1657 ± 0	988 1662 ± 0	155 495 ± 11	194 577 ± 32
1128	sparktek-001	2095581	116978	1095 10113	1405 4180 ± 0	1474 1806 ± 10	1181 1799 ± 0	1171 1818 ± 0	1125 1820 ± 0	1015 1818 ± 0	355 816 ± 13	357 804 ± 10
1129	sparsh-001	0	366290	1028 4525	536 2048 ± 0	732 680 ± 2	469 648 ± 1	422 638 ± 0	369 642 ± 0	357 717 ± 0	1204 9600 ± 275	1203 9601 ± 290
1130	sqisoft-001	278968	386291	286 689	1068 2056 ± 0	452 477 ± 5	1060 1348 ± 18	1038 1353 ± 26	949 1340 ± 14	860 1393 ± 28	345 797 ± 22	342 788 ± 22
1131	sqisoft-002	278039	386291	278 675	1066 2056 ± 0	439 466 ± 8	292 466 ± 2	255 468 ± 11	211 461 ± 6	173 472 ± 4	324 758 ± 11	332 760 ± 23
1132	sqisoft-003	362737	607964	344 812	1103 2056 ± 0	671 638 ± 2	498 674 ± 7	502 718 ± 17	393 665 ± 6	362 720 ± 6	364 844 ± 11	371 844 ± 23
1133	staqu-000	879661	624676	474 1064	1276 4096 ± 0	892 813 ± 25	-	-	-	-	908 2979 ± 31	918 3007 ± 75
1134	starhybrid-001	100509	289356	368 845	700 2048 ± 0	308 358 ± 82	186 355 ± 49	174 379 ± 58	169 401 ± 79	123 393 ± 67	447 1075 ± 51	455 1078 ± 53
1135	stcon-000	408095	49619	510 1137	243 1024 ± 0	639 617 ± 1	453 632 ± 4	416 634 ± 1	371 645 ± 2	325 676 ± 6	128 437 ± 10	133 434 ± 11
1136	stcon-001	322871	64463	380 860	231 1024 ± 0	1081 980 ± 1	786 991 ± 1	742 995 ± 5	676 1003 ± 6	599 1042 ± 4	92 336 ± 14	102 334 ± 14
1137	stcon-002	1255490	62667	799 2095	1299 4096 ± 0	577 583 ± 0	384 570 ± 2	347 574 ± 2	308 582 ± 1	285 630 ± 0	473 1143 ± 2	480 1142 ± 4
1138	stcon-003	2119710	62673	997 3862	1358 4100 ± 0	909 823 ± 0	629 821 ± 1	593 831 ± 3	534 823 ± 1	466 838 ± 2	480 1151 ± 5	495 1157 ± 5
1139	stcon-004	2831397	64930	993 3781	1054 2052 ± 0	1361 1375 ± 1	1074 1384 ± 4	1036 1348 ± 1	967 1365 ± 10	845 1375 ± 8	191 591 ± 4	201 589 ± 1
1140	stcon-005	3126319	50632	969 3424	1350 4100 ± 0	1388 1401 ± 2	1103 1415 ± 8	1070 1400 ± 10	1026 1435 ± 1	852 1382 ± 1	491 1181 ± 31	497 1162 ± 5
1141	stcon-006	3126319	50636	970 3426	1360 4100 ± 0	1371 1384 ± 1	1110 1426 ± 3	1084 1410 ± 1	1013 1419 ± 21	871 1412 ± 7	475 1148 ± 1	485 1148 ± 1
1142	stcon-007	3126319	50636	968 3413	1344 4100 ± 0	1407 1431 ± 1	1111 1427 ± 9	1131 1503 ± 8	1004 1406 ± 7	915 1473 ± 1	486 1163 ± 10	502 1185 ± 32
1143	stcon-008	2124613	56551	876 2552	1357 4100 ± 0	1447 1558 ± 0	1153 1564 ± 0	1144 1560 ± 0	1083 1565 ± 0	961 1561 ± 0	595 1497 ± 14	601 1505 ± 11
1144	stcon-009	2186212	56547	885 2599	1380 4108 ± 0	1450 1560 ± 0	1152 1561 ± 0	1145 1568 ± 2	1089 1580 ± 0	960 1555 ± 0	590 1485 ± 2	603 1515 ± 5

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 48: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1145	stcon-010	2227081	56547	888 2644	1381 4108 ± 0	1448 1558 ± 0	1158 1573 ± 0	1148 1574 ± 0	1087 1568 ± 0	965 1572 ± 0	599 1504 ± 2	609 1523 ± 5
1146	stellartecs-001	1431086	33554	782 1997	1302 4096 ± 0	1351 1351 ± 0	1101 1414 ± 0	1080 1409 ± 2	1030 1441 ± 0	976 1605 ± 1	720 1987 ± 8	724 1986 ± 31
1147	stellartecs-003	1431086	33571	779 1978	1227 4096 ± 0	1019 926 ± 1	755 961 ± 2	707 963 ± 1	648 978 ± 1	651 1106 ± 1	610 1543 ± 17	613 1535 ± 19
1148	stengg-000	361104	34674	301 724	663 2048 ± 0	885 809 ± 1	716 921 ± 2	748 1002 ± 1	512 794 ± 1	1062 2130 ± 5	722 2005 ± 22	727 2011 ± 34
1149	sukshi-000	94035	688738	127 382	1499 32768 ± 0	370 407 ± 11	240 413 ± 8	288 504 ± 8	418 689 ± 11	966 1574 ± 28	1209 9817 ± 50	1208 9787 ± 62
1150	sumsub-000	1120840	252	480 1074	467 2048 ± 0	1492 1947 ± 1	1200 1947 ± 0	1189 1947 ± 1	1151 1950 ± 1	1035 1958 ± 0	1457 69114 ± 201	1455 69212 ± 246
1151	sumsub-001	848331	252	531 1183	374 2048 ± 0	1487 1916 ± 0	1197 1934 ± 0	1184 1928 ± 0	-	1039 1978 ± 0	393 939 ± 5	398 926 ± 13
1152	suprema-000	246761	38507	-	846 2048 ± 0	845 771 ± 2	602 778 ± 1	619 864 ± 2	752 1109 ± 2	1065 2150 ± 4	643 1690 ± 17	643 1688 ± 13
1153	suprema-001	373423	41460	-	622 2048 ± 0	863 788 ± 1	634 826 ± 2	656 914 ± 2	790 1146 ± 7	1096 2443 ± 4	935 3212 ± 16	944 3220 ± 22
1154	suprema-002	373808	41473	717 1733	496 2048 ± 0	862 787 ± 3	643 833 ± 3	672 924 ± 4	829 1185 ± 6	1098 2479 ± 3	944 3255 ± 17	949 3253 ± 14
1155	suprema-003	498231	116054	555 1244	958 2048 ± 0	1416 1448 ± 1	1105 1417 ± 4	1088 1418 ± 3	1016 1421 ± 4	901 1451 ± 5	764 2201 ± 10	764 2198 ± 13
1156	suprema-004	1430475	116085	945 3096	1336 4096 ± 0	1434 1478 ± 2	1137 1472 ± 2	1118 1469 ± 1	1054 1476 ± 1	928 1496 ± 1	934 3201 ± 14	940 3202 ± 22
1157	suprema-005	1430177	116089	944 3093	1337 4096 ± 0	869 794 ± 1	666 863 ± 1	567 808 ± 5	580 883 ± 1	454 830 ± 5	1020 3950 ± 7	1025 3948 ± 4
1158	supremaid-001	258193	23479	204 541	784 2048 ± 0	456 479 ± 1	303 481 ± 0	264 481 ± 0	238 490 ± 0	216 522 ± 0	291 704 ± 19	273 652 ± 19
1159	supremaid-002	256273	23899	109 334	847 2048 ± 0	461 483 ± 0	323 501 ± 0	274 488 ± 0	248 503 ± 0	244 565 ± 0	721 1990 ± 19	707 1923 ± 29
1160	surrey-cvssp-000	158030	70795	387 879	508 2048 ± 0	1198 1141 ± 3	914 1157 ± 3	880 1158 ± 4	804 1163 ± 3	746 1245 ± 3	1201 9557 ± 143	1204 9602 ± 186
1161	surrey-cvssp-001	900280	76392	707 1705	579 2048 ± 0	1266 1221 ± 1	984 1238 ± 2	946 1240 ± 0	879 1243 ± 0	761 1257 ± 0	1310 18970 ± 161	1313 18999 ± 176
1162	surrey-cvssp-002	903493	87776	711 1720	659 2048 ± 0	1258 1213 ± 1	961 1211 ± 1	982 1279 ± 0	914 1282 ± 0	794 1300 ± 0	1211 9906 ± 120	1211 9911 ± 167
1163	surrey-cvssp-003	1013099	109557	824 2208	458 2048 ± 0	864 789 ± 1	651 846 ± 1	614 854 ± 1	870 1234 ± 2	1132 3283 ± 3	1165 7358 ± 130	1163 7336 ± 93
1164	surrey-cvssp-004	1142444	108961	823 2203	779 2048 ± 0	817 747 ± 1	1156 1568 ± 2	1214 4350 ± 12	1211 11245 ± 7	-	1175 7794 ± 109	1175 7738 ± 118
1165	swsam-001	265887	95072	259 642	482 2048 ± 0	1143 1058 ± 34	845 1079 ± 31	823 1091 ± 35	728 1076 ± 9	652 1109 ± 45	899 2902 ± 31	896 2892 ± 32
1166	synesis-000	340264	76158	-	150 512 ± 0	60 117 ± 0	-	-	-	-	331 775 ± 206	190 568 ± 48
1167	synesis-001	340264	78195	-	110 512 ± 0	65 122 ± 0	-	-	-	-	362 828 ± 182	300 698 ± 44
1168	synesis-003	256748	155901	-	1312 4096 ± 0	36 90 ± 0	-	-	-	-	1160 7118 ± 121	1159 7162 ± 100
1169	synesis-004	270628	16252	-	882 2048 ± 0	806 735 ± 15	-	-	-	-	122 424 ± 14	131 430 ± 22
1170	synesis-005	146509	17726	-	881 2048 ± 0	150 211 ± 9	-	-	-	-	203 599 ± 23	196 581 ± 32
1171	synesis-006	731941	21817	638 1472	1374 4104 ± 0	535 549 ± 1	362 546 ± 1	330 552 ± 1	293 558 ± 2	296 639 ± 28	285 697 ± 32	294 688 ± 31
1172	synesis-007	1442961	24145	854 2443	1214 3080 ± 0	1260 1215 ± 5	1006 1268 ± 30	1003 1306 ± 67	931 1311 ± 58	879 1423 ± 52	275 684 ± 32	292 686 ± 25
1173	synology-000	221021	25809	155 453	794 2048 ± 0	371 407 ± 14	241 415 ± 14	478 694 ± 31	992 1396 ± 58	1151 4568 ± 211	1315 19720 ± 203	1316 19767 ± 379
1174	synology-001	256895	25804	-	869 2048 ± 0	1029 942 ± 21	-	-	-	-	554 1402 ± 29	661 1730 ± 220
1175	synology-002	256713	25943	175 488	381 2048 ± 0	981 886 ± 4	692 892 ± 3	664 920 ± 2	673 1000 ± 5	807 1317 ± 12	577 1466 ± 32	599 1496 ± 45
1176	syscake-000	50022	12609	65 184	10 128 ± 0	175 243 ± 1	109 245 ± 0	86 241 ± 1	83 269 ± 0	101 357 ± 1	34 185 ± 1	36 185 ± 1
1177	szpu-000	283522	15225	894 2660	417 2048 ± 0	125 190 ± 1	80 192 ± 1	92 259 ± 6	193 438 ± 11	721 1204 ± 32	1026 4026 ± 57	1032 4021 ± 56
1178	sztu-000	338637	15871	579 1298	858 2048 ± 0	508 531 ± 0	348 532 ± 0	314 533 ± 0	275 537 ± 0	232 548 ± 0	189 585 ± 11	206 592 ± 13
1179	sztu-001	338650	15871	581 1299	358 2048 ± 0	517 535 ± 0	352 537 ± 0	319 538 ± 0	276 540 ± 0	234 553 ± 0	205 599 ± 10	217 598 ± 10
1180	t4isb-000	234227	115237	114 345	822 2048 ± 0	1101 1006 ± 5	795 1001 ± 1	754 1006 ± 1	686 1009 ± 1	588 1022 ± 2	983 3586 ± 34	984 3534 ± 34
1181	tahakom-001	198869	724103	219 581	362 2048 ± 0	239 302 ± 2	132 277 ± 3	160 364 ± 7	143 368 ± 4	121 389 ± 4	806 2387 ± 28	805 2381 ± 22
1182	tahakom-002	609724	37452	533 1184	1252 4096 ± 0	1275 1231 ± 1	990 1250 ± 1	942 1234 ± 0	856 1218 ± 0	766 1265 ± 0	601 1513 ± 2	604 1515 ± 2
1183	tahakom-003	1015539	68142	812 2157	683 2048 ± 0	659 630 ± 1	470 649 ± 1	444 661 ± 0	368 641 ± 1	321 669 ± 2	190 589 ± 1	241 620 ± 8
1184	taiwanmobile-001	170523	1947147	216 570	1196 2560 ± 0	569 579 ± 8	600 775 ± 16	464 683 ± 4	450 718 ± 8	420 781 ± 6	1400 38086 ± 62	1402 38109 ± 66
1185	taiwanmobile-002	170523	1969728	94 295	535 2048 ± 0	90 156 ± 0	52 162 ± 0	101 281 ± 0	288 549 ± 3	1038 1978 ± 21	1432 55746 ± 469	1430 55598 ± 410
1186	tcc-000	54057	1188952	101 318	257 1024 ± 0	330 375 ± 93	196 363 ± 76	671 924 ± 279	1181 2339 ± 806	1184 9265 ± 3411	430 1053 ± 38	448 1064 ± 23
1187	tcc-001	989049	1238700	262 644	596 2048 ± 0	375 408 ± 0	243 417 ± 0	204 415 ± 0	179 413 ± 0	159 450 ± 0	455 1090 ± 5	466 1105 ± 29
1188	tcv-001	171624	816225	120 363	973 2048 ± 0	46 98 ± 0	28 98 ± 0	23 99 ± 0	20 100 ± 0	18 122 ± 0	443 1070 ± 31	438 1045 ± 17

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 49: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1189	tech5-000	1451185	470097	-	225 1024 ± 0	474 498 ± 4	-	-	-	-	328 763 ± 23	335 766 ± 28
1190	tech5-001	636346	470101	-	1311 4096 ± 0	499 522 ± 5	-	-	-	-	454 1087 ± 34	350 799 ± 44
1191	tech5-002	1150887	111744	-	299 1280 ± 0	852 780 ± 10	-	-	-	-	557 1406 ± 120	440 1048 ± 57
1192	tech5-003	1427464	118901	-	311 1536 ± 0	978 885 ± 20	-	-	-	-	481 1151 ± 37	476 1128 ± 32
1193	tech5-004	2410272	118858	-	76 321 ± 0	957 872 ± 2	880 1117 ± 164	846 1114 ± 182	780 1134 ± 179	569 999 ± 44	200 597 ± 13	208 592 ± 16
1194	tech5-005	1178769	120517	623 1426	154 512 ± 0	1304 1272 ± 109	825 1038 ± 63	788 1046 ± 39	766 1124 ± 38	832 1351 ± 44	843 2573 ± 37	843 2545 ± 32
1195	tech5-007	0	340324	890 2647	134 512 ± 0	1354 1360 ± 0	1066 1366 ± 0	1051 1376 ± 0	976 1373 ± 0	893 1438 ± 6	169 538 ± 19	165 516 ± 22
1196	tech5-008	0	365877	1056 5567	492 2048 ± 0	693 655 ± 3	441 624 ± 11	397 625 ± 11	356 631 ± 11	351 710 ± 12	1167 7371 ± 81	1164 7367 ± 99
1197	tech5-009	0	361983	860 2453	766 2048 ± 0	1125 1029 ± 25	784 988 ± 21	770 1028 ± 23	681 1006 ± 20	653 1111 ± 22	1171 7478 ± 131	1170 7464 ± 120
1198	techainer-001	198738	703687	778 1977	848 2048 ± 0	338 381 ± 9	223 391 ± 5	193 402 ± 5	161 391 ± 1	166 461 ± 5	1223 10726 ± 189	1223 10714 ± 243
1199	techainer-002	286340	1013105	576 1293	1303 4096 ± 0	395 427 ± 2	253 429 ± 4	213 424 ± 1	194 438 ± 11	199 506 ± 6	579 1472 ± 10	582 1462 ± 16
1200	techainer-003	557482	1098186	663 1550	1248 4096 ± 0	836 763 ± 0	589 765 ± 0	556 789 ± 0	517 803 ± 0	591 1025 ± 2	1018 3940 ± 17	1027 3978 ± 37
1201	techbiometricid-001	172553	886937	144 427	807 2048 ± 0	157 218 ± 8	102 226 ± 6	242 455 ± 19	694 1023 ± 36	1139 3669 ± 176	1441 59682 ± 512	1441 61123 ± 662
1202	techbiometricid-002	601291	465337	150 439	969 2048 ± 0	528 543 ± 5	355 541 ± 4	331 552 ± 5	315 586 ± 5	402 764 ± 20	1289 16395 ± 120	1292 16374 ± 124
1203	techsign-000	0	1101622	776 1969	692 2048 ± 0	320 366 ± 1	230 398 ± 1	891 1172 ± 3	1199 3065 ± 18	1194 10460 ± 65	1064 4758 ± 112	1068 4789 ± 93
1204	techsign-001	0	586983	759 1881	961 2048 ± 0	847 772 ± 35	606 788 ± 23	565 802 ± 42	627 949 ± 10	868 1409 ± 26	194 592 ± 11	205 592 ± 13
1205	telexper-000	266638	74277	249 625	400 2048 ± 0	585 586 ± 1	387 576 ± 0	362 592 ± 1	366 639 ± 0	495 886 ± 0	1104 5559 ± 54	1102 5553 ± 47
1206	tevian-000	455746	16296	-	929 2048 ± 0	173 242 ± 14	-	-	-	-	125 430 ± 38	129 427 ± 23
1207	tevian-001	682319	16344	-	903 2048 ± 0	178 245 ± 13	-	-	-	-	99 350 ± 22	98 327 ± 13
1208	tevian-002	724437	16352	-	997 2049 ± 0	235 298 ± 15	-	-	-	-	111 369 ± 16	123 386 ± 25
1209	tevian-003	791725	16348	-	1000 2049 ± 0	366 404 ± 15	-	-	-	-	98 350 ± 11	105 338 ± 25
1210	tevian-004	863474	16544	-	924 2048 ± 0	485 506 ± 30	-	-	-	-	142 474 ± 31	97 326 ± 20
1211	tevian-005	921043	16556	-	781 2048 ± 0	663 633 ± 21	495 672 ± 25	578 818 ± 37	756 1117 ± 64	1085 2364 ± 121	183 568 ± 22	227 607 ± 35
1212	tevian-006	692492	19339	-	419 2048 ± 0	623 611 ± 1	485 666 ± 41	443 661 ± 32	400 672 ± 37	367 723 ± 31	193 591 ± 19	192 573 ± 28
1213	tevian-007	779934	19523	709 1712	279 1032 ± 0	575 583 ± 1	391 579 ± 0	354 580 ± 0	317 588 ± 1	293 636 ± 0	1076 4894 ± 65	1074 4841 ± 83
1214	tevian-008	847177	19519	982 3610	278 1032 ± 0	976 884 ± 2	700 903 ± 1	648 903 ± 1	598 911 ± 1	534 946 ± 1	1070 4828 ± 40	1071 4811 ± 41
1215	tiger-001	171041	102515	-	1484 8632 ± 0	252 312 ± 12	-	-	-	-	1477 98339 ± 34	1477 98327 ± 32
1216	tiger-002	341638	178194	-	1104 2056 ± 0	355 393 ± 20	-	-	-	-	752 2135 ± 29	753 2137 ± 38
1217	tiger-003	426164	560292	-	1113 2056 ± 0	429 458 ± 21	-	-	-	-	728 2031 ± 35	730 2029 ± 38
1218	tiger-004	342866	253683	-	1048 2052 ± 0	876 798 ± 2	-	-	-	-	196 595 ± 5	216 598 ± 7
1219	tiger-005	342866	253734	653 1532	1024 2052 ± 0	1165 1097 ± 2	839 1065 ± 2	809 1066 ± 2	720 1067 ± 3	638 1088 ± 3	228 620 ± 19	238 615 ± 16
1220	tiger-006	421186	394688	294 714	1050 2052 ± 0	1379 1392 ± 16	1099 1411 ± 10	1101 1444 ± 10	1076 1531 ± 11	1017 1848 ± 10	682 1810 ± 20	683 1801 ± 13
1221	tiger-007	516470	192057	272 661	1015 2052 ± 0	426 455 ± 1	291 464 ± 2	251 464 ± 3	225 472 ± 4	225 537 ± 1	425 1033 ± 8	435 1037 ± 19
1222	tinkoff-001	274660	389272	243 616	915 2048 ± 0	1227 1176 ± 3	932 1179 ± 3	896 1178 ± 3	811 1169 ± 2	718 1203 ± 3	1042 4361 ± 74	1041 4364 ± 75
1223	tngd-001	73167	519258	213 560	1035 2052 ± 0	105 175 ± 0	76 190 ± 0	194 404 ± 0	210 460 ± 1	145 432 ± 0	781 2274 ± 29	780 2265 ± 20
1224	tngd-002	134027	35925	364 843	577 2048 ± 0	761 694 ± 0	554 728 ± 0	1035 1348 ± 0	1195 2974 ± 1	1196 10990 ± 7	418 1016 ± 4	425 1019 ± 4
1225	tnitech-000	234032	13976	1054 5471	829 2048 ± 0	380 415 ± 1	251 427 ± 1	208 419 ± 4	186 427 ± 3	147 436 ± 2	816 2438 ± 14	818 2436 ± 19
1226	tongyi-004	947322	138901	-	1179 2167 ± 0	87 153 ± 0	-	-	-	-	1308 18916 ± 120	1312 18909 ± 126
1227	tongyi-005	1140701	138919	803 2121	1168 2089 ± 0	95 165 ± 1	-	-	-	-	1309 18924 ± 65	1319 20158 ± 103
1228	tongyitrans-001	625339	75069	-	1135 2058 ± 0	250 310 ± 20	-	-	-	-	1297 17769 ± 74	1300 17750 ± 63
1229	tongyitrans-002	625336	75081	-	1134 2058 ± 0	305 356 ± 35	-	-	-	-	1368 29816 ± 281	1301 17799 ± 127
1230	tongyitrans-003	1769974	138883	-	1419 4386 ± 0	444 468 ± 0	-	-	-	-	1445 61793 ± 154	1399 37122 ± 155
1231	toppanidgate-000	671181	711850	737 1798	1321 4096 ± 0	1005 915 ± 1	710 916 ± 1	659 916 ± 1	602 917 ± 1	514 917 ± 1	1353 25262 ± 84	1351 25264 ± 97
1232	toppansecurity-001	1480690	130296	678 1595	1383 4113 ± 0	1283 1238 ± 3	964 1217 ± 0	990 1292 ± 0	862 1223 ± 0	747 1247 ± 1	921 3098 ± 28	942 3207 ± 38

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 50: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1233	toshiba-000	3893310	235400	-	333 1812 ± 0	502 528 ± 3	-	-	-	-	775 2255 ± 60	773 2251 ± 53
1234	toshiba-001	3893310	235400	-	1199 2580 ± 0	633 615 ± 3	-	-	-	-	897 2900 ± 80	895 2881 ± 60
1235	toshiba-002	813606	114260	-	325 1560 ± 0	525 541 ± 0	-	-	-	-	974 3521 ± 369	822 2449 ± 124
1236	toshiba-003	984125	114264	-	324 1560 ± 0	523 540 ± 0	-	-	-	-	808 2390 ± 41	811 2407 ± 81
1237	toshiba-004	599297	27880	680 1596	1062 2056 ± 0	1415 1447 ± 3	1128 1453 ± 2	1113 1457 ± 9	1042 1457 ± 3	919 1479 ± 4	419 1020 ± 25	416 998 ± 32
1238	toshiba-005	599298	61113	-	1064 2056 ± 0	1423 1456 ± 4	1130 1454 ± 2	1116 1461 ± 2	1039 1455 ± 2	904 1459 ± 2	627 1613 ± 34	625 1607 ± 28
1239	toshiba-006	599566	44078	677 1588	1087 2056 ± 0	1436 1481 ± 16	1148 1515 ± 7	1132 1506 ± 6	1072 1521 ± 2	957 1546 ± 30	421 1022 ± 17	427 1022 ± 23
1240	toshiba-007	547545	76675	578 1297	1091 2056 ± 0	793 722 ± 1	605 784 ± 0	552 777 ± 4	491 766 ± 2	457 832 ± 15	298 709 ± 22	313 712 ± 26
1241	toshiba-008	707290	76687	735 1791	1088 2056 ± 0	1079 979 ± 1	770 976 ± 0	727 986 ± 3	679 1005 ± 2	612 1056 ± 14	266 674 ± 68	290 683 ± 71
1242	toshiba-009	651747	76688	689 1639	1061 2056 ± 0	1013 921 ± 1	730 933 ± 1	662 919 ± 2	615 935 ± 3	568 996 ± 21	305 719 ± 48	302 700 ± 31
1243	toshiba-010	1091350	76688	903 2710	1063 2056 ± 0	1381 1393 ± 2	1164 1638 ± 15	1114 1457 ± 1	1020 1426 ± 14	914 1473 ± 19	209 601 ± 25	224 604 ± 23
1244	toshiba-011	1106463	76900	905 2733	1107 2056 ± 0	1365 1379 ± 1	1096 1406 ± 1	1077 1407 ± 1	974 1370 ± 1	900 1448 ± 18	296 708 ± 56	266 641 ± 37
1245	toshiba-012	1151623	85416	829 2253	1095 2056 ± 0	1494 2020 ± 0	1207 2048 ± 0	1195 2049 ± 0	1164 2073 ± 4	1061 2125 ± 5	521 1281 ± 17	537 1315 ± 40
1246	toshiba-013	993674	205625	952 3146	1075 2056 ± 0	1470 1754 ± 0	1175 1752 ± 0	1163 1760 ± 0	1123 1797 ± 0	1031 1947 ± 5	508 1248 ± 23	541 1336 ± 19
1247	toss-000	993640	94527	1014 4182	552 2048 ± 0	1124 1028 ± 0	952 1200 ± 3	953 1246 ± 7	755 1116 ± 2	631 1072 ± 0	386 909 ± 78	392 914 ± 60
1248	toss-001	1006455	93901	1037 4801	693 2048 ± 0	661 631 ± 0	447 629 ± 0	396 625 ± 0	362 636 ± 0	313 661 ± 0	367 855 ± 10	396 918 ± 26
1249	touchlessid-000	92561	64467	110 341	768 2048 ± 0	249 309 ± 5	152 305 ± 2	120 312 ± 5	86 277 ± 4	97 349 ± 17	1380 31935 ± 292	1380 31958 ± 243
1250	touchlessid-001	255274	14355	202 538	933 2048 ± 0	293 344 ± 1	181 347 ± 1	203 414 ± 3	323 595 ± 10	1001 1732 ± 61	681 1806 ± 35	682 1800 ± 35
1251	touchlessid-002	255586	14284	122 367	695 2048 ± 0	327 371 ± 1	209 375 ± 1	225 438 ± 3	367 640 ± 10	959 1548 ± 57	705 1915 ± 41	695 1871 ± 38
1252	touchlessid-003	271397	14226	190 512	498 2048 ± 0	376 410 ± 0	247 419 ± 0	217 429 ± 0	233 479 ± 1	330 685 ± 6	213 608 ± 17	213 596 ± 1
1253	touchlessid-004	386383	14727	258 642	673 2048 ± 0	941 854 ± 0	729 932 ± 2	951 1244 ± 6	1156 2001 ± 2	1158 5013 ± 7	606 1527 ± 28	600 1501 ± 14
1254	transmitsecurity-001	1786905	2031398	604 1363	1445 6144 ± 0	290 342 ± 13	176 344 ± 15	134 332 ± 22	106 312 ± 15	108 370 ± 19	1208 9692 ± 23	1209 9847 ± 34
1255	trueface-000	255123	796861	-	668 2048 ± 0	322 367 ± 8	-	-	-	-	147 482 ± 13	170 528 ± 20
1256	trueface-001	255123	186754	-	94 500 ± 0	351 390 ± 1	-	-	-	-	267 676 ± 26	182 558 ± 50
1257	trueface-002	253947	123116	174 486	341 2000 ± 0	312 360 ± 0	193 361 ± 0	211 423 ± 0	320 590 ± 1	-	36 192 ± 14	38 186 ± 19
1258	trueface-003	346530	24308	1002 3916	871 2048 ± 0	1173 1107 ± 22	502 677 ± 3	517 732 ± 7	595 905 ± 5	-	12 103 ± 11	14 112 ± 29
1259	trueidvng-001	766071	37721	733 1781	1446 6144 ± 0	1073 975 ± 1	779 985 ± 1	731 989 ± 1	690 1016 ± 1	672 1128 ± 2	1396 37129 ± 216	1461 72067 ± 305
1260	truststamp-001	1795742	275073	651 1524	207 768 ± 0	131 194 ± 0	77 191 ± 0	55 187 ± 0	58 210 ± 1	87 338 ± 8	1186 8415 ± 66	1188 8392 ± 71
1261	truststamp-002	1060439	10841	792 2066	562 2048 ± 0	238 300 ± 0	167 330 ± 1	129 325 ± 0	123 335 ± 2	160 452 ± 0	1229 11281 ± 179	1231 11273 ± 203
1262	truststamp-003	1206119	10474	758 1875	904 2048 ± 0	416 445 ± 0	275 445 ± 0	234 449 ± 0	214 462 ± 1	226 539 ± 2	1272 14901 ± 619	1274 14813 ± 472
1263	tsight-000	1218024	107998	881 2579	230 1024 ± 0	1479 1861 ± 0	1191 1855 ± 1	1179 1874 ± 1	1135 1864 ± 1	1023 1877 ± 1	803 2373 ± 4	819 2438 ± 13
1264	ttigroup-000	466606	454674	838 2320	733 2048 ± 0	996 909 ± 2	722 925 ± 8	643 897 ± 1	596 909 ± 3	520 923 ± 2	804 2382 ± 17	808 2387 ± 23
1265	tuputech-000	11476	17185	7 33	851 2048 ± 0	64 122 ± 4	37 120 ± 1	35 142 ± 2	51 196 ± 5	134 411 ± 14	1340 23893 ± 406	1352 25279 ± 406
1266	turingtechvip-001	399874	54535	242 615	479 2048 ± 0	1373 1384 ± 4	1080 1391 ± 1	1064 1393 ± 1	1007 1411 ± 1	918 1476 ± 2	656 1733 ± 19	664 1734 ± 20
1267	turingtechvip-002	167556	140995	385 877	701 2048 ± 0	1441 1493 ± 2	1035 1306 ± 1	1055 1382 ± 1	947 1337 ± 1	883 1426 ± 3	1260 13819 ± 103	1263 13807 ± 137
1268	turkcell-000	271083	133553	256 636	725 2048 ± 0	1178 1110 ± 1	861 1094 ± 0	835 1103 ± 0	769 1126 ± 1	714 1201 ± 1	890 2866 ± 23	894 2873 ± 40
1269	turkcell-001	287616	133870	269 657	981 2048 ± 0	735 682 ± 0	520 695 ± 0	471 688 ± 1	421 692 ± 0	347 702 ± 0	1040 4342 ± 71	1038 4310 ± 72
1270	turkcell-002	542178	139150	253 633	421 2048 ± 0	866 791 ± 1	373 559 ± 1	558 793 ± 3	511 793 ± 2	247 574 ± 0	942 3249 ± 65	945 3231 ± 53
1271	twface-000	661735	11782	919 2796	640 2048 ± 0	954 871 ± 1	677 873 ± 1	628 873 ± 2	572 876 ± 2	503 898 ± 1	598 1504 ± 29	602 1510 ± 34
1272	twface-001	671511	11782	922 2851	504 2048 ± 0	1016 923 ± 1	723 925 ± 2	675 926 ± 1	610 929 ± 2	530 940 ± 2	552 1400 ± 32	558 1402 ± 37
1273	ulsee-001	370519	57261	-	399 2048 ± 0	691 654 ± 2	-	-	-	-	1125 6065 ± 94	1130 6228 ± 77
1274	ultinous-000	90803	503529	-	1364 4100 ± 0	17 53 ± 0	-	-	-	-	1038 4263 ± 50	1037 4262 ± 42
1275	ultinous-001	403572	501651	-	1488 12304 ± 0	141 200 ± 0	-	-	-	-	1395 36357 ± 223	1398 36332 ± 210
1276	uluface-002	0	480761	490 1088	892 2048 ± 0	959 873 ± 42	660 855 ± 9	718 978 ± 24	901 1271 ± 40	1083 2333 ± 68	1312 19207 ± 1114	1308 18501 ± 274

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 51: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1277	uluface-003	97357	529422	562 1264	1208 3072 ± 0	1062 965 ± 11	759 968 ± 10	819 1087 ± 20	984 1387 ± 36	1097 2469 ± 86	1354 26057 ± 195	1360 26865 ± 566
1278	unicc-001	24840	5123261	749 1849	425 2048 ± 0	261 316 ± 4	183 349 ± 4	599 838 ± 14	745 1102 ± 11	804 1312 ± 15	906 2950 ± 29	904 2929 ± 26
1279	unicc-002	277015	1705310	699 1673	841 2048 ± 0	618 609 ± 1	433 620 ± 1	381 612 ± 0	388 660 ± 2	399 763 ± 3	785 2289 ± 13	785 2295 ± 22
1280	unicc-003	361274	1737967	739 1806	765 2048 ± 0	1344 1343 ± 3	1051 1328 ± 1	1033 1345 ± 1	968 1366 ± 1	922 1484 ± 6	796 2339 ± 74	796 2327 ± 29
1281	unicc-004	361274	1737981	736 1793	814 2048 ± 0	1347 1348 ± 2	1057 1334 ± 1	1040 1355 ± 1	993 1397 ± 1	979 1613 ± 4	790 2306 ± 13	791 2304 ± 16
1282	unissey-001	0	1956593	413 934	1282 4096 ± 0	971 880 ± 3	693 892 ± 3	1109 1452 ± 8	1198 3048 ± 12	1189 10017 ± 387	575 1463 ± 35	585 1471 ± 34
1283	unissey-002	0	1443765	322 765	1304 4096 ± 0	808 736 ± 1	580 752 ± 1	737 994 ± 1	1019 1426 ± 1	1135 3331 ± 2	1246 12308 ± 91	1248 12302 ± 137
1284	unissey-003	0	814526	264 650	1229 4096 ± 0	788 718 ± 1	570 744 ± 0	697 956 ± 1	1001 1403 ± 1	1127 3055 ± 2	621 1594 ± 20	620 1570 ± 44
1285	unissey-004	1	1065839	453 1030	1234 4096 ± 0	301 354 ± 2	200 366 ± 0	205 416 ± 0	180 417 ± 0	133 409 ± 0	1037 4261 ± 124	1055 4557 ± 480
1286	unissey-005	0	1125043	464 1050	1233 4096 ± 0	299 352 ± 0	203 369 ± 0	191 402 ± 0	166 397 ± 0	131 408 ± 0	1059 4630 ± 97	1078 4964 ± 161
1287	unissey-006	0	1315567	537 1188	1280 4096 ± 0	1003 914 ± 0	728 930 ± 0	783 1043 ± 0	707 1044 ± 1	605 1048 ± 0	1131 6189 ± 111	1139 6605 ± 186
1288	unissey-007	1	1763338	524 1173	1277 4096 ± 0	1271 1224 ± 22	1014 1283 ± 17	1059 1390 ± 23	997 1401 ± 23	869 1410 ± 23	1346 24488 ± 173	1347 24524 ± 185
1289	uniubi-000	402734	49121	406 915	1093 2056 ± 0	1107 1011 ± 1	802 1009 ± 1	762 1022 ± 1	695 1025 ± 1	606 1051 ± 1	797 2342 ± 8	801 2369 ± 17
1290	universeai-000	341437	1395338	103 320	179 521 ± 0	268 324 ± 0	163 325 ± 0	154 357 ± 0	191 434 ± 0	452 829 ± 0	1262 14062 ± 60	1265 14142 ± 74
1291	upc-001	0	89914	482 1077	286 1052 ± 0	537 551 ± 15	530 703 ± 56	507 724 ± 51	478 751 ± 49	481 863 ± 33	922 3114 ± 44	937 3165 ± 97
1292	urielsoft-000	256261	67932	1034 4700	710 2048 ± 0	137 197 ± 0	101 226 ± 0	99 281 ± 2	93 289 ± 2	217 528 ± 2	238 631 ± 10	248 630 ± 2
1293	useb-000	1002886	493465	634 1437	613 2048 ± 0	630 613 ± 2	412 605 ± 1	389 621 ± 2	379 653 ± 2	471 848 ± 3	754 2153 ± 45	768 2222 ± 115
1294	useb-001	1000960	493816	629 1429	805 2048 ± 0	716 673 ± 4	389 577 ± 4	353 579 ± 8	324 596 ± 9	409 768 ± 2	769 2230 ± 65	787 2299 ± 90
1295	useb-002	1000967	493456	622 1423	503 2048 ± 0	510 532 ± 2	603 778 ± 2	436 657 ± 4	383 657 ± 2	468 843 ± 3	795 2336 ± 47	794 2323 ± 60
1296	useb-003	1001019	493456	643 1484	578 2048 ± 0	490 512 ± 2	337 521 ± 3	311 528 ± 12	289 550 ± 2	365 723 ± 2	758 2172 ± 38	759 2175 ± 75
1297	useb-004	2086479	454237	916 2765	616 2048 ± 0	1358 1369 ± 2	1112 1428 ± 12	1092 1428 ± 5	1079 1557 ± 10	1030 1945 ± 4	789 2303 ± 54	798 2350 ± 94
1298	uxlabs-001	291127	39378	288 700	1230 4096 ± 0	357 395 ± 0	217 387 ± 0	182 388 ± 0	158 390 ± 0	128 396 ± 0	691 1863 ± 31	706 1921 ± 45
1299	uxlabs-003	1051837	37767	688 1638	1244 4096 ± 0	1378 1390 ± 1	1098 1410 ± 5	1060 1391 ± 3	998 1402 ± 6	890 1435 ± 1	635 1645 ± 10	632 1641 ± 37
1300	uzinfocom-000	271124	508385	392 885	1158 2078 ± 0	109 178 ± 0	65 180 ± 0	63 191 ± 0	57 208 ± 12	78 325 ± 0	440 1064 ± 41	447 1055 ± 26
1301	uzinfocom-001	377667	396457	493 1091	1160 2078 ± 0	403 435 ± 0	259 433 ± 0	235 450 ± 0	256 511 ± 0	253 589 ± 1	1440 59486 ± 145	1438 59551 ± 123
1302	vaidio-001	461288	112836	477 1068	967 2048 ± 0	638 616 ± 0	435 621 ± 0	391 622 ± 0	351 623 ± 0	336 690 ± 0	1433 56447 ± 44	1434 56451 ± 63
1303	vcog-002	3229434	118946	988 3666	1501 61504 ± 5	306 357 ± 25	-	-	-	-	1490 296154 ± 3077	1490 296436 ± 4183
1304	vcortex-001	105305	55240	181 502	215 896 ± 0	142 201 ± 2	84 197 ± 0	70 209 ± 3	75 244 ± 6	109 371 ± 19	71 279 ± 4	78 277 ± 4
1305	vd-000	213686	12962	-	271 1028 ± 0	76 134 ± 3	-	-	-	-	578 1467 ± 19	578 1456 ± 16
1306	vd-001	170262	44058	-	1040 2052 ± 0	260 316 ± 6	-	-	-	-	510 1258 ± 38	484 1148 ± 109
1307	vd-002	254498	34389	6 31	170 516 ± 0	742 684 ± 5	504 679 ± 4	458 676 ± 5	422 693 ± 5	390 754 ± 5	81 300 ± 14	92 319 ± 32
1308	vd-003	254505	44051	3 29	1008 2052 ± 0	757 691 ± 5	515 690 ± 5	466 683 ± 4	420 691 ± 5	364 722 ± 5	414 1003 ± 11	421 1001 ± 7
1309	veertec-001	106668	28071	277 674	1138 2060 ± 0	1469 1751 ± 0	1174 1751 ± 0	1160 1749 ± 0	1111 1747 ± 0	1011 1792 ± 15	1049 4453 ± 20	1045 4449 ± 25
1310	veridas-001	196540	190018	-	395 2048 ± 0	713 671 ± 21	-	-	-	-	1111 5748 ± 20	1156 7111 ± 148
1311	veridas-002	193466	190018	-	103 512 ± 0	710 669 ± 20	-	-	-	-	657 1733 ± 81	713 1934 ± 44
1312	veridas-003	293109	141587	-	702 2048 ± 0	983 890 ± 33	-	-	-	-	1096 5484 ± 42	1162 7306 ± 410
1313	veridas-004	196585	160684	-	726 2048 ± 0	729 678 ± 22	574 749 ± 27	1120 1470 ± 52	1201 3228 ± 95	1191 10349 ± 273	1100 5516 ± 42	1167 7425 ± 130
1314	veridas-005	196585	269478	-	730 2048 ± 0	616 606 ± 12	-	-	-	-	1437 56976 ± 200	1448 66298 ± 216
1315	veridas-006	355669	896424	-	926 2048 ± 0	970 880 ± 8	688 885 ± 8	974 1271 ± 18	1177 2242 ± 38	1171 6414 ± 156	1436 56940 ± 149	1447 66077 ± 194
1316	veridas-007	355105	891492	870 2522	914 2048 ± 0	955 872 ± 9	679 875 ± 8	966 1261 ± 18	1176 2238 ± 38	1169 6374 ± 147	261 655 ± 16	278 660 ± 19
1317	veridas-008	1100495	1190915	1088 8633	927 2048 ± 0	1034 944 ± 12	738 945 ± 11	1024 1334 ± 27	1182 2382 ± 48	1173 6959 ± 172	306 723 ± 14	324 731 ± 16
1318	veridas-009	1100503	625120	1092 9163	734 2048 ± 0	591 589 ± 6	378 565 ± 5	350 576 ± 5	310 583 ± 5	319 666 ± 6	248 638 ± 5	265 638 ± 16
1319	veridas-010	1103722	623326	1090 8991	731 2048 ± 0	785 714 ± 4	510 686 ± 4	490 705 ± 6	416 687 ± 4	341 693 ± 5	251 644 ± 2	268 644 ± 4
1320	veridium-000	0	47198	26 98	1496 29399 ± 2045	29 79 ± 0	18 80 ± 0	19 89 ± 0	14 90 ± 0	15 111 ± 0	1448 64880 ± 171	1446 64697 ± 247

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 52: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1321	veridium-001	0	40561	47 142	1187 2489 ± 0	13 44 ± 0	9 45 ± 0	6 48 ± 0	6 50 ± 0	8 72 ± 0	1447 63417 ± 1061	1444 63225 ± 2133
1322	veridium-002	0	52482	53 161	1185 2486 ± 0	4 25 ± 0	2 25 ± 0	3 29 ± 0	2 30 ± 0	3 51 ± 0	1407 43891 ± 177	1408 43918 ± 167
1323	veridium-003	0	25975	41 136	1188 2490 ± 0	6 27 ± 0	3 25 ± 0	2 28 ± 0	3 30 ± 0	4 53 ± 0	1390 34755 ± 140	1392 34828 ± 177
1324	veridium-004	0	30232	64 181	1186 2486 ± 0	14 46 ± 0	10 46 ± 0	7 51 ± 0	7 53 ± 0	10 87 ± 0	1364 28700 ± 167	1368 28723 ± 149
1325	verigram-000	256209	7798	806 2137	819 2048 ± 0	882 807 ± 1	628 821 ± 1	713 972 ± 2	961 1358 ± 3	1113 2848 ± 13	501 1222 ± 17	510 1219 ± 17
1326	verigram-001	282155	11773	887 2638	900 2048 ± 0	702 664 ± 2	500 675 ± 2	596 833 ± 4	841 1202 ± 7	1106 2733 ± 32	638 1664 ± 60	633 1648 ± 56
1327	verigram-002	1588249	20691	789 2057	1228 4096 ± 0	1426 1464 ± 3	1095 1406 ± 0	1125 1486 ± 1	1022 1431 ± 0	971 1586 ± 5	1491 453119 ± 13595	1491 452140 ± 9654
1328	verigram-003	1590775	20692	788 2052	1323 4096 ± 0	754 691 ± 4	503 677 ± 1	460 677 ± 1	424 694 ± 1	388 746 ± 1	1401 38843 ± 1140	1404 38665 ± 1071
1329	verigram-004	1521843	23835	828 2247	1478 8216 ± 0	1454 1575 ± 2	1155 1565 ± 1	1142 1558 ± 1	1080 1558 ± 1	977 1606 ± 1	903 2928 ± 17	902 2915 ± 11
1330	verihubs-inteligensia-000	209562	51877	146 430	465 2048 ± 0	555 567 ± 0	1151 1558 ± 8	1143 1560 ± 8	1086 1568 ± 8	981 1621 ± 8	1331 22351 ± 91	1335 22371 ± 81
1331	verihubs-inteligensia-001	216524	51916	148 437	746 2048 ± 0	549 564 ± 0	375 562 ± 0	338 566 ± 1	298 566 ± 0	264 600 ± 0	1323 21770 ± 84	1327 21735 ± 102
1332	verihubs-inteligensia-002	361105	17045	302 727	898 2048 ± 0	447 471 ± 0	287 458 ± 0	247 459 ± 0	217 463 ± 1	171 471 ± 0	1301 18131 ± 558	1303 17851 ± 197
1333	verijelas-000	254540	10322	719 1736	606 2048 ± 0	264 321 ± 0	164 325 ± 1	133 329 ± 0	122 335 ± 5	103 360 ± 0	1215 10267 ± 143	1215 10218 ± 109
1334	verismart-001	271195	15029	268 657	1153 2068 ± 0	292 343 ± 0	180 346 ± 0	142 346 ± 0	126 345 ± 0	95 346 ± 0	376 885 ± 5	397 926 ± 14
1335	via-000	124422	11151	-	489 2048 ± 0	776 707 ± 8	567 740 ± 5	650 906 ± 41	622 941 ± 40	597 1040 ± 5	402 966 ± 28	426 1021 ± 44
1336	via-001	370255	11151	706 1697	689 2048 ± 0	1060 964 ± 3	804 1011 ± 3	766 1026 ± 4	708 1045 ± 3	678 1137 ± 28	406 983 ± 31	414 989 ± 40
1337	via-004	530941	93634	603 1361	416 2048 ± 0	325 368 ± 0	228 397 ± 1	147 352 ± 1	138 364 ± 2	184 490 ± 1	63 255 ± 7	69 255 ± 8
1338	via-005	510729	93629	830 2257	635 2048 ± 0	521 538 ± 0	344 528 ± 0	309 525 ± 1	273 535 ± 2	265 601 ± 1	53 223 ± 2	53 222 ± 7
1339	via-006	1087232	93594	1009 4074	792 2048 ± 0	956 872 ± 1	696 898 ± 1	682 941 ± 1	618 936 ± 2	566 993 ± 1	54 224 ± 2	55 230 ± 11
1340	via-007	1087263	24041	1051 5345	401 2048 ± 0	1368 1381 ± 2	1073 1383 ± 1	1057 1388 ± 2	1009 1414 ± 1	940 1519 ± 2	105 359 ± 2	122 385 ± 2
1341	viante-000	1244225	19718	654 1532	1083 2056 ± 0	1139 1054 ± 0	835 1059 ± 1	812 1075 ± 1	724 1070 ± 4	636 1087 ± 1	1140 6461 ± 44	1136 6424 ± 32
1342	viante-001	1401078	19722	744 1812	1077 2056 ± 0	1163 1094 ± 1	881 1117 ± 5	848 1119 ± 6	762 1120 ± 3	649 1103 ± 1	930 3158 ± 45	932 3143 ± 53
1343	viante-002	1247402	19718	657 1538	1078 2056 ± 0	1209 1152 ± 1	877 1113 ± 3	840 1109 ± 1	749 1107 ± 2	665 1121 ± 2	936 3213 ± 90	941 3207 ± 90
1344	viante-003	1247402	19705	655 1534	1089 2056 ± 0	1472 1768 ± 2	1179 1787 ± 0	1167 1775 ± 0	1121 1786 ± 8	1010 1791 ± 0	1062 4666 ± 111	1062 4690 ± 83
1345	viante-004	1247402	19705	659 1542	1084 2056 ± 0	1471 1760 ± 0	1186 1840 ± 0	1170 1809 ± 0	1120 1786 ± 0	-	1041 4346 ± 41	1039 4357 ± 44
1346	videmo-000	139643	39470	-	581 2048 ± 0	78 142 ± 5	48 150 ± 4	40 150 ± 6	34 151 ± 4	25 155 ± 8	163 513 ± 16	168 523 ± 38
1347	videmo-001	212051	95063	99 309	522 2048 ± 0	138 199 ± 0	54 164 ± 0	45 164 ± 0	37 164 ± 0	26 165 ± 0	77 296 ± 17	82 288 ± 16
1348	videmo-002	212053	32963	108 333	521 2048 ± 0	139 199 ± 0	57 169 ± 0	48 169 ± 0	38 170 ± 0	28 170 ± 0	48 209 ± 7	48 208 ± 8
1349	videmo-003	173512	43964	153 447	112 512 ± 0	75 133 ± 0	34 111 ± 0	28 115 ± 0	25 114 ± 0	17 115 ± 0	3 75 ± 10	3 75 ± 16
1350	videonetics-001	30875	5963	11 61	136 512 ± 0	192 262 ± 3	127 273 ± 1	227 439 ± 3	532 820 ± 3	1091 2393 ± 43	483 1153 ± 38	481 1142 ± 65
1351	videonetics-002	121981	6289	31 115	1043 2052 ± 0	216 282 ± 5	144 295 ± 1	296 513 ± 4	696 1029 ± 3	1129 3151 ± 46	500 1219 ± 57	521 1262 ± 56
1352	viettelai-000	284612	132031	245 619	662 2048 ± 0	256 315 ± 0	158 317 ± 0	126 321 ± 0	114 322 ± 0	83 334 ± 1	1328 22074 ± 112	1334 22173 ± 163
1353	viettelai-001	528186	132058	538 1189	343 2048 ± 0	786 715 ± 1	-	-	-	-	1333 22740 ± 167	1474 86287 ± 1073
1354	viettelai-002	528186	132049	496 1094	626 2048 ± 0	781 712 ± 1	556 729 ± 0	508 724 ± 0	465 735 ± 0	377 738 ± 5	1356 26231 ± 154	1476 89838 ± 1111
1355	viettelai-003	528138	130904	495 1093	925 2048 ± 0	779 710 ± 1	550 720 ± 0	527 743 ± 3	504 782 ± 0	723 1207 ± 1	658 1743 ± 11	660 1728 ± 7
1356	viettelai-004	1506350	203627	947 3115	1005 2052 ± 0	1501 2910 ± 0	1214 2922 ± 1	1209 2915 ± 0	1193 2917 ± 0	1122 2944 ± 0	517 1274 ± 17	532 1290 ± 32
1357	viettelai-005	1865780	365644	943 3089	1022 2052 ± 0	1484 1908 ± 0	1208 2054 ± 2	1197 2053 ± 2	1161 2051 ± 2	1052 2095 ± 4	559 1413 ± 14	567 1425 ± 20
1358	viettelai-006	1531507	365644	946 3101	1031 2052 ± 0	1485 1910 ± 0	1195 1910 ± 0	1180 1902 ± 0	1149 1925 ± 0	1034 1953 ± 1	549 1398 ± 10	559 1405 ± 13
1359	viettelhightech-000	259471	215557	142 421	442 2048 ± 0	433 461 ± 1	288 461 ± 2	248 461 ± 1	221 467 ± 2	185 494 ± 0	204 599 ± 11	204 591 ± 13
1360	viettelhightech-001	461523	147432	491 1088	817 2048 ± 0	1151 1068 ± 1	848 1084 ± 1	871 1145 ± 4	900 1270 ± 4	1045 2041 ± 2	202 598 ± 14	200 589 ± 5
1361	viettelhightech-002	461606	115515	450 1022	974 2048 ± 0	1463 1700 ± 1	1169 1665 ± 0	1159 1723 ± 1	1142 1893 ± 1	1108 2742 ± 1	334 778 ± 4	341 783 ± 8
1362	vigilantsolutions-000	352218	389974	-	1498 31540 ± 0	975 884 ± 23	-	-	-	-	1302 18201 ± 94	1257 13030 ± 83
1363	vigilantsolutions-001	344685	89145	-	314 1544 ± 0	1011 921 ± 2	-	-	-	-	253 644 ± 13	271 649 ± 16
1364	vigilantsolutions-002	344137	89162	-	1139 2060 ± 0	932 844 ± 2	-	-	-	-	269 679 ± 10	285 680 ± 7

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 53: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE							COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]					TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1365	vigilantsolutions-003	343048	94301	-	[319]1548 ± 0	[910]824 ± 3	-	-	-	-	[244]634 ± 19	[264]638 ± 17
1366	vigilantsolutions-004	343048	125719	-	[320]1548 ± 0	[924]837 ± 6	-	-	-	-	[278]686 ± 14	[295]688 ± 17
1367	vigilantsolutions-005	343048	125719	-	[315]1548 ± 0	[925]837 ± 13	-	-	-	-	[374]874 ± 22	[262]637 ± 16
1368	vigilantsolutions-006	343048	125719	-	[322]1548 ± 0	[929]841 ± 8	-	-	-	-	[392]939 ± 32	[311]711 ± 37
1369	vigilantsolutions-007	255600	125715	-	[321]1548 ± 0	[469]493 ± 6	-	-	-	-	[350]803 ± 35	[351]800 ± 40
1370	vigilantsolutions-008	441835	47824	-	[318]1548 ± 0	[679]647 ± 2	-	-	-	-	[378]889 ± 23	[386]903 ± 44
1371	vigilantsolutions-009	348798	49973	-	[317]1548 ± 0	[636]615 ± 0	[452]631 ± 0	[409]632 ± 1	[361]636 ± 0	[309]659 ± 0	[132]452 ± 11	[139]451 ± 11
1372	vigilantsolutions-010	348798	49973	[361]841	[316]1548 ± 0	[637]615 ± 0	[451]631 ± 0	[411]632 ± 0	[360]636 ± 0	[311]659 ± 0	[152]490 ± 13	[157]488 ± 11
1373	vigilantsolutions-011	255661	49973	[226]591	[323]1548 ± 0	[364]402 ± 0	[244]418 ± 0	[206]418 ± 0	[182]422 ± 0	[158]445 ± 0	[95]339 ± 20	[112]366 ± 37
1374	vinai-000	402391	866522	[457]1034	[511]2048 ± 0	[1169]1099 ± 1	[863]1095 ± 1	[826]1093 ± 1	[744]1099 ± 1	[668]1126 ± 1	[909]2996 ± 20	[916]2993 ± 26
1375	vinbigdata-001	271405	44746	[228]591	[762]2048 ± 0	[1387]1400 ± 5	[1083]1393 ± 2	[1061]1391 ± 2	[989]1393 ± 1	[867]1404 ± 1	[536]1351 ± 50	[536]1310 ± 38
1376	vinbigdata-002	256322	138864	[237]607	[461]2048 ± 0	[558]569 ± 2	[385]572 ± 1	[343]571 ± 1	[301]572 ± 1	[260]596 ± 1	[759]2175 ± 44	[755]2160 ± 53
1377	vinbigdata-003	259968	183557	[112]342	[806]2048 ± 0	[936]849 ± 6	[1038]1312 ± 1	[613]853 ± 2	[940]1321 ± 2	[459]832 ± 2	[446]1074 ± 37	[461]1086 ± 40
1378	vinbigdata-004	1570839	333050	[957]3247	[1249]4096 ± 0	[1217]1162 ± 3	[912]1155 ± 14	[868]1141 ± 2	[812]1171 ± 5	[681]1140 ± 2	[527]1299 ± 17	[519]1252 ± 28
1379	vinbigdata-005	1106052	333050	[833]2294	[611]2048 ± 0	[1256]1210 ± 1	[981]1237 ± 1	[957]1251 ± 1	[872]1234 ± 1	[778]1282 ± 7	[62]251 ± 17	[61]234 ± 16
1380	vinbigdata-006	1106052	333624	[834]2294	[614]2048 ± 0	[1257]1210 ± 2	[957]1205 ± 5	[893]1175 ± 2	[819]1178 ± 3	[715]1202 ± 16	[58]246 ± 16	[59]232 ± 16
1381	vinbigdata-007	1106071	335755	[835]2295	[380]2048 ± 0	[1249]1201 ± 10	[923]1166 ± 3	[958]1252 ± 6	[889]1256 ± 15	[758]1254 ± 20	[61]249 ± 10	[63]238 ± 16
1382	vion-000	228219	7533	[180]498	[1037]2052 ± 0	[279]333 ± 1	-	-	-	-	[1403]39839 ± 3561	[1359]26830 ± 2241
1383	visage-000	49218	70150	[17]72	[111]512 ± 0	[5]27 ± 0	[4]27 ± 0	[4]31 ± 0	[5]38 ± 0	[6]63 ± 0	[767]2220 ± 14	[766]2218 ± 14
1384	visica-001	1	59680	[14]68	[259]1024 ± 0	[79]144 ± 1	[45]143 ± 1	[42]151 ± 1	[47]188 ± 2	[115]383 ± 11	[426]1035 ± 41	[424]1017 ± 56
1385	visica-002	1	63831	[36]126	[842]2048 ± 0	[69]124 ± 1	[40]125 ± 1	[43]156 ± 2	[84]271 ± 4	[500]892 ± 29	[1065]4759 ± 31	[1059]4630 ± 26
1386	visica-003	1	252112	[164]470	[932]2048 ± 0	[234]297 ± 2	[148]297 ± 1	[139]341 ± 2	[224]471 ± 5	[790]1297 ± 32	[1119]5939 ± 47	[1121]5987 ± 81
1387	visionbox-000	176501	190645	-	[823]2048 ± 0	[242]304 ± 7	-	-	-	-	[636]1648 ± 57	[504]1192 ± 42
1388	visionbox-001	256869	190645	-	[494]2048 ± 0	[1082]983 ± 7	[859]1093 ± 46	[1042]1360 ± 68	[1173]2181 ± 105	[1166]5955 ± 281	[485]1161 ± 22	[491]1154 ± 20
1389	visionbox-002	259063	135281	[240]613	[1136]2059 ± 0	[460]482 ± 1	[306]482 ± 0	[267]484 ± 1	[240]492 ± 1	[210]517 ± 3	[719]1969 ± 44	[711]1931 ± 42
1390	visionbox-003	259542	156891	[561]1263	[1152]2068 ± 0	[730]678 ± 2	[506]682 ± 1	[463]682 ± 2	[426]695 ± 2	[382]739 ± 5	[633]1643 ± 56	[634]1649 ± 66
1391	visionbox-004	309730	154797	[621]1422	[1385]4116 ± 0	[334]378 ± 1	[204]369 ± 0	[183]389 ± 5	[171]402 ± 2	[149]437 ± 4	[738]2071 ± 31	[743]2076 ± 41
1392	visionbox-005	309402	197528	[729]1771	[1384]4116 ± 0	[297]350 ± 1	[175]343 ± 1	[146]351 ± 2	[140]367 ± 5	[162]454 ± 19	[740]2074 ± 17	[741]2067 ± 29
1393	visionbox-006	479723	170096	[935]2995	[1379]4107 ± 0	[1022]932 ± 2	[766]973 ± 4	[740]994 ± 6	[677]1003 ± 11	[664]1120 ± 20	[730]2038 ± 74	[728]2022 ± 31
1394	visionbox-007	479723	364416	[861]2456	[1378]4107 ± 0	[641]619 ± 1	[426]616 ± 0	[405]629 ± 2	[396]666 ± 7	[428]794 ± 32	[975]3524 ± 29	[982]3527 ± 40
1395	visionbox-008	479723	449470	[864]2476	[1377]4107 ± 0	[642]619 ± 1	[431]619 ± 1	[403]628 ± 2	[390]663 ± 7	[432]799 ± 36	[994]3704 ± 35	[994]3692 ± 40
1396	visionlabs-001	343661	127922	-	[20]204 ± 0	[1032]943 ± 8	-	-	-	-	[547]1395 ± 45	[486]1148 ± 53
1397	visionlabs-002	591936	46855	-	[65]264 ± 0	[196]265 ± 8	-	-	-	-	[343]796 ± 35	[349]799 ± 31
1398	visionlabs-003	639849	15246	-	[50]260 ± 0	[165]231 ± 5	-	-	-	-	[129]439 ± 23	[141]454 ± 20
1399	visionlabs-004	313016	90858	-	[39]256 ± 0	[244]304 ± 0	-	-	-	-	[359]827 ± 26	[376]865 ± 32
1400	visionlabs-005	369602	16523	-	[131]512 ± 0	[253]313 ± 0	-	-	-	-	[365]848 ± 26	[383]889 ± 37
1401	visionlabs-006	353044	16655	-	[133]512 ± 0	[198]270 ± 0	-	-	-	-	[286]698 ± 19	[327]734 ± 28
1402	visionlabs-007	357204	16643	-	[138]512 ± 0	[201]272 ± 0	-	-	-	-	[401]965 ± 41	[409]972 ± 31
1403	visionlabs-008	706099	19705	-	[104]512 ± 0	[442]467 ± 1	-	-	-	-	[397]955 ± 23	[403]962 ± 25
1404	visionlabs-009	706099	19862	-	[158]513 ± 0	[493]515 ± 41	[296]472 ± 1	[261]474 ± 1	[229]476 ± 1	[215]521 ± 1	[399]957 ± 28	[407]965 ± 32
1405	visionlabs-010	1067280	19357	[391]883	[162]513 ± 0	[802]730 ± 0	[547]717 ± 1	[494]709 ± 0	[445]713 ± 1	[380]739 ± 0	[206]600 ± 41	[247]626 ± 35
1406	visionlabs-011	1067280	19353	[410]923	[160]513 ± 0	[804]731 ± 1	[546]717 ± 1	[495]710 ± 1	[447]714 ± 1	[383]741 ± 1	[174]556 ± 26	[183]559 ± 25
1407	visteam-000	32729	17740	-	[313]1536 ± 0	[42]96 ± 7	[31]102 ± 8	[155]358 ± 19	[693]1022 ± 50	[1143]3987 ± 211	[1137]6361 ± 87	[1141]6668 ± 277
1408	visteam-001	186440	30878	-	[1270]4096 ± 0	[950]869 ± 7	[675]872 ± 6	[851]1121 ± 15	[1107]1719 ± 38	[1150]4375 ± 157	[1157]7054 ± 108	[1154]7025 ± 109

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 54: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480X720	960X1440	1600X2400	3000X4500	GENUINE	IMPOSTOR
1409	visteam-002	186440	30888	-	1265 4096 ± 0	913 829 ± 5	642 832 ± 6	601 839 ± 7	558 853 ± 6	584 1013 ± 14	1153 6952 ± 118	1150 6970 ± 120
1410	visteam-003	215359	33730	177 493	1322 4096 ± 0	1291 1249 ± 4	992 1251 ± 4	970 1266 ± 5	903 1272 ± 5	840 1370 ± 9	1148 6816 ± 111	1146 6816 ± 105
1411	visteam-004	61594	35369	56 168	946 2048 ± 0	241 303 ± 5	153 313 ± 6	98 278 ± 4	91 288 ± 4	111 377 ± 7	1017 3936 ± 72	1024 3938 ± 79
1412	visteam-005	288140	35427	115 347	947 2048 ± 0	1185 1117 ± 6	870 1106 ± 6	802 1060 ± 4	726 1071 ± 4	692 1156 ± 8	1015 3932 ± 97	1020 3932 ± 71
1413	visteam-006	339353	47921	178 493	718 2048 ± 0	1105 1009 ± 1	805 1015 ± 1	776 1035 ± 2	689 1014 ± 1	603 1047 ± 2	861 2711 ± 32	864 2704 ± 41
1414	visteam-007	331673	264832	173 485	686 2048 ± 0	619 611 ± 1	442 625 ± 1	399 626 ± 1	339 615 ± 1	303 647 ± 0	860 2704 ± 38	869 2732 ± 54
1415	visteam-008	582800	264832	304 731	719 2048 ± 0	1231 1179 ± 1	927 1173 ± 1	925 1210 ± 1	833 1190 ± 3	726 1208 ± 1	851 2632 ± 28	852 2647 ± 42
1416	visteam-009	582793	48090	306 731	941 2048 ± 0	1225 1174 ± 1	939 1184 ± 1	928 1218 ± 2	857 1219 ± 2	756 1253 ± 2	853 2668 ± 34	857 2671 ± 38
1417	visteam-010	582793	48090	308 731	948 2048 ± 0	1226 1174 ± 1	955 1203 ± 2	927 1215 ± 1	859 1221 ± 1	754 1252 ± 1	855 2675 ± 51	861 2681 ± 57
1418	vivotek-001	15240	247926	38 128	524 2048 ± 0	48 100 ± 0	30 101 ± 0	27 106 ± 0	31 134 ± 0	51 238 ± 1	1357 26504 ± 57	1358 26520 ± 69
1419	vivotek-002	12769	247949	35 124	568 2048 ± 0	39 93 ± 0	24 93 ± 0	22 98 ± 0	18 95 ± 0	24 150 ± 0	805 2384 ± 19	807 2385 ± 20
1420	vivotek-003	12769	247854	34 123	1052 2052 ± 0	43 97 ± 0	29 100 ± 0	25 101 ± 0	19 95 ± 0	14 100 ± 0	813 2424 ± 80	815 2427 ± 77
1421	vixvizion-000	183453	599571	-	385 2048 ± 0	392 425 ± 24	-	-	-	-	474 1145 ± 25	656 1718 ± 63
1422	vixvizion-001	282680	256875	-	704 2048 ± 0	881 807 ± 29	-	-	-	-	428 1045 ± 22	399 934 ± 45
1423	vixvizion-002	227766	318409	-	942 2048 ± 0	861 786 ± 1	590 766 ± 2	637 885 ± 3	1021 1430 ± 3	1144 4080 ± 10	268 676 ± 16	249 630 ± 20
1424	vixvizion-003	254860	378019	-	615 2048 ± 0	948 865 ± 2	733 938 ± 1	1149 1577 ± 1	1197 3002 ± 3	1187 9521 ± 16	317 738 ± 26	289 683 ± 25
1425	vixvizion-004	254405	380049	-	699 2048 ± 0	648 624 ± 1	399 587 ± 10	400 626 ± 3	322 592 ± 3	358 717 ± 6	325 760 ± 22	304 703 ± 28
1426	vixvizion-005	38886	534579	328 776	838 2048 ± 0	402 433 ± 4	212 381 ± 3	180 383 ± 3	145 373 ± 1	135 411 ± 1	312 731 ± 63	254 632 ± 32
1427	vixvizion-006	594053	396294	601 1359	388 2048 ± 0	966 876 ± 9	635 828 ± 3	577 817 ± 1	536 825 ± 2	487 871 ± 1	208 600 ± 23	230 611 ± 25
1428	vixvizion-007	594053	470119	633 1435	512 2048 ± 0	977 885 ± 35	636 828 ± 1	576 816 ± 1	537 825 ± 1	486 870 ± 1	207 600 ± 28	221 602 ± 34
1429	vnis-000	285395	1404886	207 546	530 2048 ± 0	409 437 ± 1	271 443 ± 1	243 456 ± 2	207 458 ± 2	175 477 ± 2	545 1386 ± 10	556 1382 ± 16
1430	vnpay-000	258919	1404844	182 504	384 2048 ± 0	170 239 ± 1	111 248 ± 0	89 252 ± 0	76 249 ± 0	60 260 ± 0	537 1353 ± 11	545 1353 ± 23
1431	vnpay-001	1044099	3060240	414 937	845 2048 ± 0	1213 1156 ± 1	995 1255 ± 1	959 1252 ± 2	895 1261 ± 2	782 1288 ± 10	560 1421 ± 32	564 1421 ± 26
1432	vnpt-001	266499	535529	-	514 2048 ± 0	477 499 ± 2	319 499 ± 2	278 494 ± 3	247 502 ± 3	202 512 ± 2	731 2049 ± 29	104 337 ± 121
1433	vnpt-002	271649	3203296	-	777 2048 ± 0	811 739 ± 2	557 731 ± 2	525 740 ± 1	470 742 ± 2	401 763 ± 2	329 766 ± 13	333 762 ± 13
1434	vnpt-003	369956	297799	-	1290 4096 ± 0	1334 1315 ± 4	1040 1315 ± 4	1011 1318 ± 2	956 1350 ± 3	884 1428 ± 3	1168 7397 ± 31	1165 7384 ± 29
1435	vnpt-004	370110	240841	435 992	657 2048 ± 0	1282 1238 ± 1	985 1241 ± 1	949 1242 ± 2	925 1307 ± 2	931 1505 ± 2	1029 4047 ± 48	1029 4008 ± 108
1436	vnpt-005	560630	240888	518 1157	655 2048 ± 0	1390 1403 ± 0	1091 1404 ± 6	1073 1403 ± 6	1040 1456 ± 0	983 1630 ± 10	980 3562 ± 23	986 3554 ± 29
1437	vnpt-006	1114807	360198	827 2245	742 2048 ± 0	1308 1277 ± 2	1079 1390 ± 19	1082 1410 ± 14	1059 1481 ± 30	996 1703 ± 69	955 3400 ± 20	961 3393 ± 22
1438	vnpt-007	2113252	360190	984 3622	507 2048 ± 0	1201 1143 ± 1	915 1157 ± 4	890 1170 ± 2	823 1183 ± 3	728 1219 ± 3	967 3486 ± 41	972 3485 ± 35
1439	vnpt-008	1114270	360190	986 3641	538 2048 ± 0	1220 1165 ± 1	940 1186 ± 2	895 1177 ± 9	839 1201 ± 2	744 1244 ± 2	971 3506 ± 16	979 3510 ± 23
1440	vnpt-009	1114270	360190	985 3641	853 2048 ± 0	1221 1166 ± 1	948 1197 ± 6	910 1193 ± 6	848 1209 ± 3	751 1251 ± 4	977 3539 ± 17	985 3551 ± 31
1441	vocalize-001	0	38841	25 96	127 512 ± 0	19 57 ± 0	13 55 ± 0	8 56 ± 0	9 60 ± 0	5 62 ± 0	551 1400 ± 59	550 1365 ± 26
1442	vocord-001	616989	109135	-	1450 6194 ± 0	995 908 ± 16	-	-	-	-	1497 1094730 ± 64282	1497 1107193 ± 66523
1443	vocord-002	918292	648259	-	306 1330 ± 0	853 782 ± 36	-	-	-	-	1474 83063 ± 517	1472 83072 ± 714
1444	vocord-003	900909	1140643	-	211 817 ± 0	232 296 ± 3	-	-	-	-	1415 46683 ± 378	1396 36007 ± 292
1445	vocord-004	1754886	129419	-	213 896 ± 0	643 619 ± 1	-	-	-	-	696 1885 ± 66	700 1889 ± 71
1446	vocord-005	1060445	189974	-	206 768 ± 0	846 771 ± 1	-	-	-	-	664 1768 ± 48	652 1713 ± 77
1447	vocord-006	559457	189913	-	208 768 ± 0	980 886 ± 1	-	-	-	-	725 2020 ± 72	720 1969 ± 62
1448	vocord-007	587489	344995	-	330 1664 ± 0	851 780 ± 2	-	-	-	-	845 2593 ± 83	838 2526 ± 59
1449	vocord-008	603867	345047	-	1201 2688 ± 0	1058 962 ± 2	769 976 ± 2	804 1061 ± 3	873 1236 ± 23	1018 1851 ± 9	912 3015 ± 50	915 2988 ± 62
1450	vocord-009	1380132	201560	1023 4372	336 1920 ± 0	1430 1472 ± 2	1136 1472 ± 1	1140 1549 ± 1	1102 1667 ± 2	1049 2064 ± 2	734 2052 ± 50	738 2056 ± 39
1451	vocord-010	902552	206873	998 3866	294 1088 ± 0	1424 1459 ± 2	1133 1459 ± 1	1117 1463 ± 2	1060 1484 ± 1	951 1535 ± 3	866 2724 ± 31	855 2653 ± 45
1452	vogic-000	237025	23230	168 477	1367 4104 ± 0	473 497 ± 0	318 499 ± 0	283 499 ± 0	245 499 ± 1	189 499 ± 0	593 1490 ± 1	598 1495 ± 2

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 55: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1453	vsoft-001	10202	61965	61 175	952 2048 ± 0	66 123 ± 8	42 131 ± 7	256 468 ± 19	1017 1422 ± 48	1164 5622 ± 229	840 2553 ± 48	844 2559 ± 44
1454	vsoft-002	10202	62518	58 169	386 2048 ± 0	93 160 ± 10	53 164 ± 11	318 537 ± 21	1090 1581 ± 53	1167 6279 ± 228	1028 4030 ± 260	1015 3902 ± 137
1455	vsoft-003	10202	62894	80 258	1131 2057 ± 0	128 191 ± 14	79 191 ± 14	404 628 ± 27	1143 1895 ± 66	1177 7545 ± 376	1291 16609 ± 97	1295 16649 ± 154
1456	vsoft-004	488414	110735	528 1180	1129 2057 ± 0	388 420 ± 13	245 418 ± 15	616 857 ± 28	1170 2135 ± 73	1179 8068 ± 378	1314 19601 ± 136	1317 19776 ± 152
1457	vts-000	256589	169760	705 1697	618 2048 ± 0	464 486 ± 1	305 481 ± 0	268 484 ± 0	235 485 ± 1	209 517 ± 0	1481 124209 ± 352	1481 123652 ± 358
1458	vts-001	293000	475743	246 622	741 2048 ± 0	724 676 ± 1	508 683 ± 6	468 687 ± 3	427 695 ± 2	350 709 ± 2	1207 9620 ± 44	1205 9618 ± 54
1459	vts-002	361977	904603	360 840	770 2048 ± 0	382 416 ± 5	260 434 ± 3	219 432 ± 3	199 449 ± 3	240 556 ± 4	1170 7447 ± 34	1168 7450 ± 54
1460	wicket-000	826392	641802	794 2071	590 2048 ± 0	1400 1419 ± 2	1114 1429 ± 3	1100 1444 ± 4	1046 1460 ± 3	953 1537 ± 6	1443 60976 ± 232	1440 61096 ± 323
1461	winsense-000	270819	32034	-	303 1280 ± 0	217 283 ± 1	-	-	-	-	613 1551 ± 31	612 1532 ± 42
1462	winsense-001	264428	32035	409 922	304 1280 ± 0	839 766 ± 7	834 1058 ± 47	724 983 ± 97	710 1053 ± 119	808 1320 ± 84	631 1631 ± 28	719 1964 ± 171
1463	winsense-002	281379	25780	753 1855	412 2048 ± 0	470 494 ± 2	317 498 ± 1	302 519 ± 1	274 537 ± 1	290 634 ± 1	642 1683 ± 8	642 1685 ± 7
1464	wiseai-001	189467	60781	77 245	499 2048 ± 0	172 240 ± 0	113 251 ± 0	131 328 ± 1	118 327 ± 0	82 332 ± 0	886 2850 ± 29	890 2852 ± 31
1465	wuhantianyu-001	465118	66457	488 1086	621 2048 ± 0	674 642 ± 1	464 642 ± 1	429 644 ± 0	378 652 ± 0	344 697 ± 0	1197 9502 ± 151	1212 9920 ± 253
1466	x-laboratory-000	520020	197310	652 1524	1110 2056 ± 0	883 808 ± 7	695 897 ± 113	653 907 ± 103	584 886 ± 103	323 673 ± 39	307 725 ± 19	329 749 ± 34
1467	x-laboratory-001	625140	398792	748 1844	1118 2056 ± 0	582 586 ± 2	407 596 ± 5	373 603 ± 6	347 620 ± 7	426 793 ± 14	354 813 ± 28	377 872 ± 32
1468	xailient-001	287094	232213	399 901	624 2048 ± 0	472 497 ± 1	341 523 ± 1	312 531 ± 1	253 508 ± 1	273 614 ± 1	97 345 ± 8	130 428 ± 13
1469	xforwardai-000	242457	175556	-	630 2048 ± 0	830 757 ± 6	-	-	-	-	493 1185 ± 44	496 1157 ± 44
1470	xforwardai-001	340100	51163	816 2176	600 2048 ± 0	1236 1180 ± 2	938 1182 ± 1	912 1194 ± 1	831 1186 ± 2	717 1203 ± 1	335 779 ± 17	347 797 ± 13
1471	xforwardai-002	707715	51163	784 2003	1335 4096 ± 0	1033 944 ± 1	737 942 ± 1	686 943 ± 4	616 935 ± 1	546 967 ± 1	556 1406 ± 8	560 1405 ± 13
1472	xm-000	578041	148920	283 684	1006 2052 ± 0	968 878 ± 2	685 882 ± 1	729 988 ± 2	891 1258 ± 3	1095 2434 ± 7	632 1634 ± 17	629 1632 ± 20
1473	yisheng-000	122704	41760	-	796 2048 ± 0	50 103 ± 1	-	-	-	-	337 790 ± 23	343 789 ± 20
1474	yisheng-001	120112	41760	-	369 2048 ± 0	51 103 ± 1	-	-	-	-	383 906 ± 31	387 905 ± 25
1475	yisheng-002	604097	42014	-	307 1348 ± 0	278 332 ± 1	-	-	-	-	222 613 ± 40	199 588 ± 26
1476	yisheng-004	486351	38653	569 1279	1226 3704 ± 0	333 378 ± 12	-	-	-	-	283 693 ± 137	169 526 ± 34
1477	yitu-000	2211068	76146	-	1389 4130 ± 0	714 672 ± 2	-	-	-	-	1392 35352 ± 114	1401 37848 ± 1773
1478	yitu-001	2069440	138901	-	1397 4146 ± 0	822 751 ± 0	-	-	-	-	1398 37810 ± 127	1395 35486 ± 203
1479	yitu-002	1726673	138901	-	1398 4147 ± 0	756 691 ± 0	-	-	-	-	1393 35435 ± 111	1393 35325 ± 131
1480	yitu-003	1525719	138919	991 3737	1166 2082 ± 0	944 860 ± 0	-	-	-	-	1303 18305 ± 71	1306 18286 ± 62
1481	yoonik-000	290414	206059	-	769 2048 ± 0	1028 941 ± 3	758 965 ± 13	709 964 ± 10	641 964 ± 9	551 974 ± 23	465 1116 ± 34	471 1113 ± 54
1482	yoonik-001	346629	265353	-	722 2048 ± 0	1267 1223 ± 3	983 1238 ± 1	945 1238 ± 1	877 1240 ± 1	741 1240 ± 1	294 706 ± 29	296 690 ± 26
1483	yoonik-002	453720	265415	907 2737	724 2048 ± 0	1203 1145 ± 4	884 1123 ± 2	854 1124 ± 2	767 1125 ± 2	669 1126 ± 3	327 761 ± 32	328 736 ± 32
1484	yoonik-003	346691	265415	814 2161	820 2048 ± 0	1089 991 ± 3	775 980 ± 1	725 984 ± 4	651 982 ± 1	558 983 ± 1	276 684 ± 45	284 678 ± 41
1485	yoonik-004	469597	88673	334 792	749 2048 ± 0	1264 1219 ± 1	1015 1284 ± 1	963 1255 ± 1	898 1266 ± 0	752 1251 ± 1	427 1039 ± 42	432 1030 ± 40
1486	yoti-001	267246	470114	507 1125	383 2048 ± 0	1284 1238 ± 4	1042 1316 ± 17	1213 4099 ± 13	1204 4160 ± 45	1156 4774 ± 95	1217 10352 ± 247	1219 10364 ± 246
1487	yoti-002	164043	470114	487 1086	950 2048 ± 0	613 604 ± 1	618 802 ± 3	570 812 ± 4	620 938 ± 15	843 1373 ± 4	1220 10662 ± 805	1218 10320 ± 265
1488	ytu-000	1477360	44032	865 2484	787 2048 ± 0	506 530 ± 0	349 533 ± 0	424 640 ± 0	565 861 ± 2	1032 1949 ± 8	1378 31797 ± 131	1378 31794 ± 133
1489	yuan-000	370472	331437	-	650 2048 ± 0	1181 1112 ± 6	-	-	-	-	875 2777 ± 44	906 2937 ± 87
1490	yuan-001	370473	189558	-	505 2048 ± 0	1370 1383 ± 3	1085 1394 ± 3	1079 1408 ± 3	1047 1461 ± 4	980 1615 ± 4	799 2344 ± 25	795 2325 ± 42
1491	yuan-002	370472	165662	-	800 2048 ± 0	1402 1420 ± 3	1113 1429 ± 4	1134 1511 ± 4	1103 1695 ± 4	1092 2408 ± 5	788 2297 ± 23	792 2310 ± 31
1492	yuan-003	370419	147783	-	707 2048 ± 0	1392 1405 ± 2	1100 1413 ± 3	1103 1446 ± 3	1077 1547 ± 5	1025 1878 ± 5	793 2320 ± 32	784 2287 ± 34
1493	yuan-004	428665	50011	600 1356	1283 4096 ± 0	553 567 ± 0	383 569 ± 0	346 573 ± 0	305 579 ± 0	270 607 ± 0	1113 5816 ± 35	1114 5800 ± 31
1494	yuan-005	258312	145564	363 843	661 2048 ± 0	339 381 ± 0	216 386 ± 0	181 387 ± 2	157 390 ± 4	142 421 ± 3	484 1156 ± 8	505 1196 ± 26
1495	yuan-006	1622733	145572	1022 4371	1298 4096 ± 0	1310 1280 ± 2	1005 1268 ± 1	976 1273 ± 0	902 1272 ± 2	799 1306 ± 3	765 2202 ± 19	762 2190 ± 20
1496	yuan-007	258312	145568	684 1614	516 2048 ± 0	311 359 ± 0	194 361 ± 1	176 379 ± 4	150 380 ± 0	138 417 ± 0	302 715 ± 4	322 726 ± 16

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 56: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

	ALGORITHM	CONFIG	LIBRARY	TEMPLATE						COMPARISON[4]	
	NAME	DATA	DATA	MEMORY	SIZE	GENERATION TIME (ms)[4]				TIME (ns)[5]	
		(KB)[1]	(KB)[2]	(MB)[3]	(B)	MUGSHOT	480x720	960x1440	1600x2400	3000x4500	GENUINE	IMPOSTOR
1497	yuan-008	620369	167833	685 1626	1264 4096 ± 0	1222 1169 ± 0	582 752 ± 0	534 752 ± 3	498 775 ± 4	422 782 ± 0	539 1362 ± 16	554 1377 ± 19
1498	yuan-009	1161457	167833	868 2507	1307 4096 ± 0	1393 1408 ± 2	1106 1418 ± 2	1121 1471 ± 1	985 1387 ± 8	874 1414 ± 12	530 1326 ± 11	540 1336 ± 29
1499	yuan-010	1054332	167833	867 2499	1263 4096 ± 0	1307 1276 ± 1	1097 1406 ± 8	997 1294 ± 1	930 1308 ± 2	847 1375 ± 2	534 1331 ± 22	543 1349 ± 47
1500	yuanli-001	136632	47927	69 190	258 1024 ± 0	77 142 ± 0	46 143 ± 0	38 145 ± 0	33 150 ± 0	42 210 ± 1	741 2080 ± 16	744 2083 ± 22

Notes	
1	The configuration size does not capture static data included in libraries.
2	The library size is the combined total of all files provided in the submission lib folder. These libraries e.g. OpenCV may or may not be installed on any end user's platform natively and would not need to be installed with the algorithm. Some developers put neural network models in their libraries.
3	The memory usage is the peak resident set size reported by the ps system call during template generation.
4	The median template creation times are measured on Intel®Xeon®CPU E5-2630 v4 @ 2.20GHz processors.
5	The comparison durations, in nanoseconds, are estimated using std::chrono::high_resolution_clock which on the machine in (2) counts 1ns clock ticks. Precision is somewhat worse than that however. The ± value is the median absolute deviation times 1.48 for Normal consistency.

Table 57: Summary of algorithms and properties included in this report. The red superscripts give ranking for the quantity in that column.

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1	2060io-001	-		-		0.9149	1369	0.9720	1365	0.9773	1324	0.9768	1170	0.8672	1201	0.9993	815		
2	2060io-002	-		-		0.8904	1367	0.9587	1363	0.9020	1310	1.0000	1321	1.0000	1341	1.0000	915		
3	20face-000	0.1268	936	0.1828	927	0.1748	1263	0.2768	1264	0.1765	1186	0.1864	866	0.0927	949	0.5943	602		
4	20face-001	0.0521	859	0.0732	852	0.1414	1256	0.2549	1259	0.0769	1130	0.1354	837	0.0419	852	0.8953	705		
5	20face-002	-		-		0.0063	765	0.0117	825	0.0076	628	1.0000	1331	1.0000	1343	1.0000	919		
6	3divi-000	-		0.1335	901	-		-		-		-		-		-			
7	3divi-001	0.0777	898	0.1540	914	-		-		-		-		-		-			
8	3divi-002	0.0788	900	0.1539	912	-		-		-		-		-		-			
9	3divi-003	0.0318	821	0.0588	831	0.0389	1163	0.0639	1158	0.0619	1113	0.0928	790	0.0555	893	-			
10	3divi-004	0.0095	551	0.0153	562	0.0097	915	0.0145	903	0.0175	897	0.0330	625	0.0242	752	-			
11	3divi-005	0.0094	549	0.0151	559	0.0078	836	0.0121	835	0.0135	818	0.0231	546	0.0156	626	-			
12	3divi-006	0.0064	431	0.0094	430	0.0047	648	0.0066	652	0.0091	689	0.0191	498	0.0113	517	0.2854	512		
13	3divi-007	0.0024	141	0.0038	149	0.0028	331	0.0034	330	0.0046	451	0.0101	326	0.0082	401	0.2515	491		
14	accurascan-001	0.3670	1002	0.4777	1008	0.5313	1318	0.6155	1310	0.5583	1257	0.6120	1032	0.2569	1054	0.8833	696		
15	accurascan-002	0.4209	1012	0.5213	1018	0.5826	1326	0.6642	1315	0.6295	1261	0.6937	1051	0.3116	1077	0.9162	712		
16	accurascan-003	0.0017	84	0.0028	91	0.0024	160	0.0024	128	0.0029	229	0.0057	136	0.0048	174	0.0629	140		
17	acer-000	0.1393	941	0.9075	1068	0.9981	1384	1.0000	1392	1.0000	1349	1.0000	1292	0.9998	1311	-			
18	acer-001	0.0294	806	0.0504	811	0.0240	1113	0.0463	1120	0.0436	1072	0.0622	746	0.0360	831	-			
19	acer-002	0.0169	708	0.0262	708	0.0103	932	0.0167	953	0.0182	907	0.0281	585	0.0159	629	0.1555	413		
20	acisw-003	0.9682	1067	0.9971	1078	0.7892	1360	0.8738	1354	0.8752	1304	0.8275	1101	0.6698	1159	-			
21	acisw-006	0.2945	990	0.9788	1074	0.6044	1328	0.7588	1326	0.9900	1328	1.0000	1297	0.9999	1318	-			
22	acisw-007	0.4276	1014	0.5493	1025	0.8425	1363	0.9185	1361	0.8424	1295	0.9976	1230	0.9930	1272	0.9999	848		
23	acisw-008	0.0100	565	0.0147	552	0.0094	901	0.0126	845	0.1740	1185	0.6651	1046	0.4545	1116	0.9302	718		
24	adera-001	0.1021	918	0.1757	921	0.1823	1264	0.2967	1267	0.1714	1184	0.6357	1037	0.1127	967	-			
25	adera-002	0.0052	350	0.0071	340	0.0047	641	0.0064	642	0.0087	670	0.0159	444	0.0136	580	-			
26	adera-003	0.0043	291	0.0059	285	0.0036	513	0.0043	463	0.0076	625	0.0151	428	0.0128	558	0.1526	412		
27	adera-004	0.0014	55	0.0022	54	0.0035	500	0.0050	528	0.0023	105	0.0212	519	0.0058	248	0.1469	403		
28	adera-005	0.0017	73	0.0026	77	0.0027	301	0.0030	256	0.0024	117	0.0134	395	0.0049	178	0.1246	361		
29	advance-002	0.0089	527	0.0137	533	0.0073	815	0.0115	822	0.0400	1057	0.0722	769	0.0593	902	-			
30	advance-003	0.0060	411	0.0087	409	0.0052	685	0.0067	655	0.0389	1053	0.4914	995	0.1291	980	0.7998	660		
31	advance-004	0.0083	513	0.0101	464	0.0037	539	0.0054	565	0.0051	490	0.3555	955	0.1088	963	0.8285	668		
32	advance-005	0.0059	400	0.0084	399	0.0043	606	0.0060	609	0.0065	579	0.9998	1278	0.9993	1298	1.0000	937		
33	advance-006	-		-		0.0032	445	0.0042	450	0.0034	319	0.9982	1237	0.5024	1125	1.0000	928		
34	advance-007	-		-		0.0025	199	0.0028	213	0.0025	139	0.0801	778	0.0258	766	0.3283	526		
35	affinsys-ai-000	-		-		0.1154	1245	0.1788	1238	0.1886	1193	0.2159	889	0.1383	989	0.6126	607		
36	afisbiometrics-000	0.0051	349	0.0073	349	0.0030	391	0.0050	533	0.0044	422	0.0077	222	0.0057	244	0.0796	229		
37	afrengine-000	0.6244	1045	0.7336	1052	0.8318	1362	0.9083	1358	0.8122	1287	0.9980	1235	0.9895	1267	1.0000	853		
38	afrengine-001	0.0071	462	0.0129	521	0.0094	906	0.0201	993	0.0097	721	0.9737	1168	0.8179	1192	0.9999	832		
39	afrengine-002	0.0052	355	0.0071	339	0.0032	434	0.0043	468	0.0194	930	0.2125	887	0.0450	863	0.2330	483		
40	afrengine-003	-		-		0.0028	345	0.0031	289	0.0116	778	0.0224	536	0.0208	718	0.0860	251		
41	afrengine-004	-		-		0.0027	319	0.0028	226	0.1475	1174	0.2477	909	0.2470	1048	0.2062	467		
42	afrengine-005	-		-		0.0028	349	0.0031	274	0.0035	328	0.0078	227	0.0059	263	0.0768	216		
43	aifirst-001	0.0119	618	0.0170	595	0.0084	867	0.0127	855	0.0131	808	0.0212	518	0.0138	585	-			
44	aigen-001	0.0124	631	0.0219	661	0.0143	1030	0.0217	1005	0.0236	973	0.8960	1122	0.3255	1083	-			

Table 58: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
45	aigen-002	0.0192	736	0.0343	752	0.0256	1117	0.0402	1097	0.0389	1052	0.9196	1131	0.3876	1100	0.9994	820
46	aigroup-000	-		-		0.0683	1214	0.0774	1178	0.0612	1109	0.9816	1177	0.9069	1218	1.0000	1301
47	aigroup-001	-		-		0.0114	975	0.0192	984	0.0179	900	0.9998	1280	0.9986	1294	1.0000	1258
48	aigroup-002	-		-		0.0034	479	0.0049	525	0.0041	391	0.9104	1127	0.7156	1173	0.9949	785
49	aigroup-003	-		-		0.0026	290	0.0036	373	0.0026	172	0.0510	713	0.0194	698	0.1784	442
50	ailabs-001	0.0158	694	0.0276	716	0.0192	1072	0.0317	1070	0.0352	1037	0.0608	742	0.0434	857	-	
51	aimall-001	0.0176	719	0.0324	741	0.0225	1102	0.0407	1101	-		-		-		-	
52	aimall-002	0.0119	620	0.0167	589	0.0224	1099	0.0411	1102	0.0233	965	0.0373	666	0.0235	745	-	
53	aimall-003	0.0033	215	0.0041	174	0.0033	469	0.0035	361	0.0056	527	0.0109	351	0.0087	422	-	
54	aipractix-000	-		-		0.0649	1208	0.0749	1174	0.0065	580	0.9940	1210	0.9055	1217	0.9999	844
55	aiseemu-001	0.0021	113	0.0029	97	0.0027	298	0.0033	315	0.0038	360	0.0339	638	0.0057	246	0.4819	569
56	aiseemu-002	0.0023	131	0.0032	113	0.0026	252	0.0027	190	0.0036	335	0.0439	684	0.0057	240	0.4259	556
57	aiunionface-000	0.0104	577	0.0154	565	0.0082	859	0.0122	838	0.0141	829	0.0243	553	0.0169	648	-	
58	aize-001	0.0223	762	0.0344	753	0.0199	1079	0.0313	1066	0.0367	1044	0.0522	718	0.0359	829	0.2506	490
59	aize-002	0.0210	752	0.0327	742	0.0280	1127	0.0489	1131	0.0504	1089	0.0692	764	0.0434	856	0.2657	503
60	aize-003	0.0086	520	0.0125	517	0.0072	806	0.0126	849	0.0516	1093	0.6137	1033	0.1038	959	0.9911	774
61	aize-006	-		-		0.0081	853	0.0160	935	0.0133	813	0.9925	1204	0.9473	1232	1.0000	859
62	ajou-001	0.0093	544	0.0147	550	0.0071	805	0.0126	846	0.0173	894	0.0274	579	0.0186	682	-	
63	akiyama-000	0.2564	982	0.3776	990	0.7240	1347	0.8699	1351	0.8899	1307	0.9872	1186	0.9483	1233	-	
64	akiyama-001	0.1311	937	0.2306	948	0.5113	1316	0.6797	1317	0.8748	1303	0.9908	1196	0.9649	1246	-	
65	akiyama-002	-		-		0.0119	985	0.0157	925	0.0104	744	0.9974	1228	0.9791	1256	1.0000	885
66	akiyama-003	-		-		0.0431	1172	0.0516	1137	0.0400	1058	0.2055	883	0.0835	936	0.5380	584
67	akor-000	-		-		0.0022	68	0.0024	111	0.0017	29	0.0035	34	0.0028	20	0.0506	64
68	alchera-000	0.0165	704	0.0243	687	0.0125	998	0.0186	976	0.0204	939	0.0349	645	0.0243	754	-	
69	alchera-001	0.0183	728	0.0299	727	0.0142	1025	0.0234	1022	0.0239	980	0.0388	673	0.0267	777	-	
70	alchera-002	0.0107	588	0.0157	574	0.0104	937	0.0229	1017	0.0144	837	0.0246	557	0.0198	705	-	
71	alchera-003	0.0044	298	0.0055	269	0.0031	409	0.0039	412	0.0042	400	0.0077	224	0.0065	285	0.0777	220
72	alchera-004	0.0035	241	0.0052	256	0.0028	353	0.0039	414	0.0029	234	0.0075	212	0.0044	151	0.0742	196
73	alchera-005	0.0027	166	0.0040	165	0.0026	243	0.0030	250	0.0025	152	0.0055	131	0.0040	120	0.0642	153
74	alchera-006	0.0028	175	0.0039	161	0.0026	253	0.0031	280	0.0025	146	0.0090	283	0.0042	129	0.0867	252
75	alchera-007	0.0034	231	0.0049	239	0.0026	251	0.0031	282	0.0026	177	0.0096	309	0.0044	146	0.0834	247
76	alchera-008	-		-		0.0026	239	0.0030	258	0.0024	137	0.0099	317	0.0043	138	0.0951	297
77	alchera-009	-		-		0.0026	274	0.0029	240	0.0022	86	0.5197	1006	0.2232	1044	0.8470	673
78	alfabeta-001	0.4867	1030	0.5831	1033	0.6855	1339	0.8156	1339	0.8253	1293	0.7765	1077	0.6416	1157	0.9627	733
79	alice-000	0.0119	621	0.0192	631	0.0106	945	0.0170	956	0.0167	880	0.0265	572	0.0150	616	0.1824	444
80	alice-001	0.0034	230	0.0046	217	0.0032	432	0.0037	391	0.0030	259	0.3164	940	0.1088	964	0.8336	669
81	alleyes-000	0.0058	389	0.0090	419	0.0055	708	0.0087	752	0.0068	588	0.0105	339	0.0076	368	-	
82	allgovision-000	0.0346	830	0.0527	818	0.0232	1107	0.0339	1073	0.0372	1048	0.0620	745	0.0443	862	-	
83	alphaface-001	0.0065	436	0.0097	445	0.0039	561	0.0063	638	0.0083	655	-		-		-	
84	alphaface-002	0.0052	354	0.0075	360	0.0030	377	0.0044	477	1.0000	1359	0.0115	369	0.0084	412	-	
85	alttekglobal-000	-		-		0.0031	398	0.0046	488	0.0051	489	0.0097	315	0.0075	356	0.0927	289
86	amplifiedgroup-001	0.5034	1032	0.5848	1034	0.6973	1343	0.8316	1341	0.7807	1282	0.7724	1074	0.6354	1152	-	
87	androvideo-000	0.0243	774	0.0438	789	0.0239	1111	0.0365	1088	0.0483	1083	0.1870	867	0.0635	907	-	
88	anke-000	0.0194	739	0.0260	706	-		-		-		-		-		-	

Table 59: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)													
		CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
89	anke-001	0.0186	732	0.0238	684	-		-		-		-		-		-	
90	anke-002	0.0402	840	0.0759	858	0.0271	1125	0.0484	1129	-		-		-		-	
91	anke-003	0.0131	649	0.0213	655	0.0094	903	0.0175	959	0.0134	815	0.9951	1215	0.0150	615	-	
92	anke-004	0.0080	502	0.0154	564	0.0073	814	0.0112	816	0.0102	739	0.0178	479	0.0118	532	-	
93	anke-005	0.0070	455	0.0109	482	0.0059	739	0.0094	767	0.0105	746	0.0142	406	0.0102	480	-	
94	anyvision-001	0.0716	891	0.0859	869	-		-		-		-		-		-	
95	anyvision-002	0.0660	886	0.0898	875	0.0928	1232	0.1512	1227	0.0899	1145	0.1191	823	0.0801	929	-	
96	anyvision-004	0.0267	792	0.0385	772	0.0258	1119	0.0487	1130	0.0234	970	0.0301	598	0.0191	689	-	
97	anyvision-005	0.0023	128	0.0037	147	0.0027	326	0.0035	351	0.0049	476	0.0084	261	0.0069	312	-	
98	aratek-001	0.0033	224	0.0045	211	0.0028	342	0.0031	291	0.0043	407	0.0092	292	0.0069	315	0.0897	268
99	armatura-001	0.0033	219	0.0042	182	0.0031	407	0.0037	384	0.0056	526	0.0110	352	0.0092	444	0.4625	565
100	armatura-002	0.0041	279	0.0052	253	0.0034	480	0.0044	470	0.0040	377	0.8502	1108	0.0275	782	0.9970	799
101	armatura-003	0.0020	107	0.0029	94	0.0026	267	0.0028	202	0.0025	157	0.0049	102	0.0043	139	0.0608	124
102	arvis-000	-		-		0.3652	1297	0.4777	1292	1.0000	1387	1.0000	1349	1.0000	1367	1.0000	1326
103	asr-000	-		-		0.5523	1320	0.7662	1327	0.8807	1305	1.0000	1326	1.0000	1326	1.0000	921
104	asr-001	-		-		0.0030	375	0.0039	410	0.0027	197	0.3390	946	0.1327	986	0.7167	635
105	asusaics-000	0.0125	637	0.0209	650	0.0085	869	0.0134	871	0.0143	835	0.7189	1057	0.0285	789	-	
106	asusaics-001	0.0125	638	0.0210	651	0.0085	871	0.0134	872	0.0143	836	0.7437	1063	0.0289	794	-	
107	autentika-000	0.1415	943	0.1916	932	0.4130	1305	0.5521	1301	0.4217	1245	0.9998	1276	0.9954	1279	1.0000	936
108	autentika-001	0.0444	850	0.0688	847	0.0987	1234	0.1531	1228	0.1547	1177	1.0000	1293	0.9997	1310	1.0000	904
109	autentika-002	0.0536	863	0.0866	871	0.0657	1211	0.1084	1205	0.1548	1178	0.9993	1254	0.9939	1275	1.0000	912
110	autentika-003	-		-		0.0203	1082	0.0314	1067	0.0404	1061	0.9791	1173	0.9294	1226	0.9994	819
111	authenmetric-002	0.0092	539	0.0134	528	0.0095	907	0.0177	961	0.0192	923	0.0463	697	0.0236	747	-	
112	authenmetric-003	0.0036	252	0.0053	262	0.0039	569	0.0051	539	0.0095	712	0.9930	1205	0.5932	1142	1.0000	878
113	authenmetric-004	0.0027	168	0.0042	185	0.0033	459	0.0036	380	0.0083	658	0.9879	1189	0.4058	1107	1.0000	865
114	authme-001	0.0036	244	0.0049	234	0.0037	541	0.0040	422	0.0030	256	0.3075	933	0.1310	983	0.7046	630
115	authme-002	-		-		0.0032	440	0.0035	355	0.0026	186	0.4284	981	0.2115	1038	0.7861	655
116	aware-000	0.1965	965	0.3090	972	-		-		-		-		-		-	
117	aware-001	0.1965	964	0.3092	973	-		-		-		-		-		-	
118	aware-002	0.1951	963	0.3202	976	-		-		-		-		-		-	
119	aware-003	0.0793	901	0.1161	895	0.1028	1236	0.1708	1234	0.1698	1183	0.2333	900	0.1174	973	-	
120	aware-004	0.0690	889	0.0949	881	0.0837	1227	0.1436	1225	0.1171	1160	0.8137	1091	0.1056	960	-	
121	aware-005	0.0457	853	0.0643	837	0.0603	1203	0.1094	1209	0.0613	1110	0.1075	811	0.0491	875	-	
122	aware-006	0.0487	856	0.0819	865	0.0529	1193	0.1090	1207	0.1011	1155	0.1058	806	0.0502	879	0.5309	581
123	aware-007	0.0101	570	0.0198	642	0.0060	751	0.0133	868	0.0224	956	0.5813	1023	0.1673	1018	0.8871	700
124	aware-008	-		-		0.0062	760	0.0143	900	0.0135	817	0.0332	628	0.0133	568	0.3622	533
125	awiros-001	0.4044	1008	0.4622	1005	0.5530	1321	0.6518	1313	0.2008	1196	0.1994	876	0.1386	991	-	
126	awiros-002	0.1990	968	0.2561	955	0.3319	1293	0.4411	1288	0.3821	1238	0.9938	1208	0.2634	1060	-	
127	awiros-003	-		-		0.0128	1005	0.0158	930	0.9978	1332	1.0000	1318	1.0000	1333	1.0000	930
128	awiros-004	-		-		0.0158	1055	0.0255	1033	0.9981	1335	1.0000	1320	1.0000	1338	1.0000	882
129	aximetria-001	0.0111	598	0.0186	622	0.0110	955	0.0148	911	0.0170	887	0.3928	971	0.2090	1036	0.7924	657
130	ayftech-001	0.0946	913	0.1941	934	0.2438	1279	0.3625	1277	0.1558	1179	0.1589	852	0.0936	951	-	
131	ayftech-003	0.0023	134	0.0037	144	0.0026	275	0.0030	271	0.0072	610	0.5411	1013	0.2515	1050	0.8902	701
132	ayonix-000	0.4351	1017	0.4872	1011	0.6150	1330	0.7510	1325	0.6557	1268	0.6361	1038	0.4981	1123	-	

Table 60: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
133	beethedata-000	0.0127	642	0.0195	635	0.0092	891	0.0157	926	0.0171	890	0.0306	603	0.0204	711	0.1617	425
134	betaface-000	0.8804	1060	0.8998	1064	-		-		-		-		-		-	
135	betaface-002	0.8813	1061	0.8998	1065	-		-		-		-		-		-	
136	beyneai-000	0.0071	467	0.0107	479	0.0104	938	0.0131	864	0.0170	888	0.9837	1180	0.6171	1148	1.0000	851
137	biocube-001	0.5596	1040	0.6834	1046	0.7700	1358	0.8712	1352	0.8446	1296	0.9661	1163	0.7922	1188	0.9846	763
138	biocube-002	0.0612	875	0.0856	868	0.2330	1277	0.2972	1268	0.2365	1213	0.9327	1145	0.6947	1163	0.9989	810
139	bioid-001	-		-		0.0121	990	0.0197	991	0.0148	843	0.9883	1190	0.7807	1185	0.9997	826
140	bioid-002	-		-		0.0127	1003	0.0205	996	0.0158	860	0.7684	1072	0.0224	734	0.9859	765
141	bioid-003	-		-		0.0083	863	0.0137	880	0.0112	769	0.1141	817	0.0135	573	0.3783	542
142	bioid-004	-		-		0.0083	866	0.0136	877	0.0109	760	0.1377	841	0.0139	589	0.5446	587
143	bioidtechswiss-000	0.0066	442	0.0082	393	0.0113	971	0.0225	1012	0.0078	635	0.0139	402	0.0092	439	-	
144	bioidtechswiss-001	0.0054	367	0.0072	344	0.0069	795	0.0124	841	0.0060	549	0.0094	302	0.0065	294	-	
145	bioidtechswiss-002	0.0049	328	0.0067	322	0.0064	769	0.0116	824	0.0067	585	0.0117	375	0.0086	418	-	
146	bioidtechswiss-003	-		-		0.0045	625	0.0050	531	0.0193	927	0.0360	653	0.0356	825	0.1320	371
147	biometric-vision-000	0.0023	129	0.0036	136	0.0028	346	0.0034	344	0.0028	224	0.3109	936	0.1504	1000	0.6849	625
148	biometrik-001	-		-		0.7415	1351	0.8509	1344	0.9980	1334	1.0000	1418	0.9877	1264	1.0000	1412
149	biometrik-002	-		-		0.0641	1207	0.0984	1198	0.0419	1066	0.3401	947	0.0292	798	0.9219	717
150	bm-001	0.7431	1052	0.9494	1072	0.9586	1374	0.9843	1370	0.9049	1311	0.9021	1126	0.8395	1196	-	
151	boetech-001	0.0662	887	0.0802	863	0.0493	1186	0.0791	1180	0.0682	1122	0.1074	810	0.0758	923	0.9951	786
152	boetech-002	0.0535	862	0.0565	827	0.0114	973	0.0136	876	0.0403	1059	0.0650	749	0.0606	903	0.3470	531
153	bresee-000	0.8467	1058	0.9472	1070	0.9819	1375	0.9903	1371	0.9940	1330	0.9919	1199	0.9760	1253	-	
154	bresee-001	0.0085	517	0.0143	542	0.0086	877	0.0153	919	0.0108	755	0.0168	464	0.0115	524	-	
155	bresee-002	0.0079	500	0.0101	459	0.0065	778	0.0079	720	0.0129	800	0.0263	570	0.0224	735	0.2137	474
156	btsdigital-001	-		-		0.0046	635	0.0072	685	0.0077	629	0.0141	404	0.0092	442	0.1256	364
157	bynn-000	-		-		0.0193	1075	0.0234	1021	0.1302	1165	0.9999	1284	0.9996	1306	1.0000	941
158	cakevn-001	-		-		0.0311	1144	0.0677	1163	0.0212	946	0.1427	843	0.0185	679	0.9170	714
159	cakevn-002	-		-		0.0027	305	0.0039	413	0.0026	189	0.0049	103	0.0037	80	0.0569	99
160	camvi-001	0.3822	1006	0.5375	1022	-		-		-		-		-		-	
161	camvi-002	0.0125	636	0.0221	665	0.0089	884	0.0145	905	0.0142	832	0.2650	917	0.0166	646	-	
162	camvi-003	0.0184	729	0.0320	739	0.0134	1014	0.0255	1034	-		-		-		-	
163	camvi-004	0.0171	713	0.0316	738	0.0042	598	0.0049	521	0.0097	723	0.6636	1045	0.0141	593	-	
164	candour-001	0.1048	921	0.3189	975	0.0130	1008	0.0182	968	0.3879	1240	0.9216	1133	0.7071	1168	0.9836	757
165	candour-002	-		-		0.0036	519	0.0061	625	0.1489	1176	0.4137	978	0.2751	1064	0.8795	692
166	candour-003	-		-		0.0040	580	0.0068	662	0.0041	392	0.1321	833	0.0225	737	0.6090	606
167	candour-004	-		-		0.0027	315	0.0039	419	0.0030	249	0.1049	804	0.0121	543	0.5691	594
168	canon-002	0.0034	232	0.0050	244	0.0026	276	0.0033	308	0.0043	416	0.0182	483	0.0065	292	-	
169	canon-003	0.0041	283	0.0059	287	0.0030	376	0.0040	424	0.0040	372	0.0073	205	0.0059	256	0.0735	193
170	canon-004	0.0052	356	0.0091	422	0.0033	467	0.0058	593	0.0037	344	0.0770	776	0.0494	876	0.3192	524
171	canon-005	0.0024	139	0.0035	132	0.0025	218	0.0033	327	0.0052	500	0.0096	312	0.0090	431	0.0542	83
172	cchonolulu-000	0.7910	1056	0.8611	1060	0.9824	1376	0.9926	1373	0.9940	1329	0.9861	1181	0.9534	1235	1.0000	854
173	cchonolulu-001	0.4202	1011	0.4853	1010	0.6298	1331	0.7694	1328	0.7402	1275	0.9720	1167	0.8870	1211	0.9951	787
174	cdachyd-001	-		-		0.7397	1350	0.8335	1342	0.7630	1277	0.7410	1061	0.5756	1140	0.9588	729
175	ceiaufgbr-000	-		-		0.9990	1389	1.0000	1391	0.9998	1343	1.0000	1294	0.9999	1321	1.0000	914
176	ceiec-001	0.0328	825	0.0475	802	0.0295	1141	0.0478	1127	0.0621	1115	0.1059	807	0.0932	950	-	

Table 61: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
177	ceiec-002	0.0161	700	0.0193	632	0.0122	993	0.0164	949	0.0270	997	0.0555	725	0.0472	872	-	
178	ceiec-003	0.0071	468	0.0107	475	0.0061	756	0.0079	725	0.0160	865	0.0316	611	0.0260	770	-	
179	ceiec-004	0.0038	264	0.0051	247	0.0045	632	0.0053	553	0.0062	567	0.3939	973	0.0104	490	-	
180	celloscope-001	-		-		0.1245	1249	0.1840	1239	0.1886	1192	0.2636	916	0.1485	999	0.7055	632
181	chosun-000	0.8481	1059	1.0000	1308	1.0000	1498	0.9999	1389	1.0000	1489	1.0000	1468	1.0000	1499	-	
182	chosun-001	0.0525	860	0.0936	879	0.0742	1218	0.1263	1218	0.0978	1152	1.0000	1322	0.9354	1227	-	
183	chosun-002	0.0390	839	0.0646	838	0.0339	1151	0.0576	1149	0.0455	1079	0.6904	1049	0.1746	1019	-	
184	chtface-001	0.9993	1071	0.9994	1079	0.9999	1392	0.9996	1381	1.0000	1347	1.0000	1298	0.9999	1320	-	
185	chtface-002	0.0150	683	0.0268	710	0.0096	914	0.0140	891	0.0186	912	0.0320	617	0.0194	697	-	
186	chtface-003	0.0091	534	0.0146	547	0.0083	864	0.0128	857	0.0132	809	0.0220	531	0.0149	613	-	
187	chtface-004	0.0046	317	0.0062	304	0.0052	684	0.0080	727	0.0088	681	0.0152	429	0.0106	495	0.1152	341
188	chtface-005	0.0033	217	0.0049	231	0.0029	360	0.0041	438	0.0044	421	0.0317	614	0.0066	297	0.1952	459
189	chtface-006	0.0029	181	0.0043	192	0.0026	283	0.0034	345	0.0040	378	0.2701	921	0.0065	290	0.4464	560
190	chtface-007	-		-		0.0025	216	0.0032	298	0.0040	374	0.2018	878	0.0064	284	0.4751	568
191	chtface-008	-		-		0.0022	74	0.0026	159	0.0025	153	0.1246	827	0.0400	844	0.4384	559
192	chtface-009	-		-		0.0023	102	0.0027	180	0.0024	130	0.3659	961	0.1226	978	0.7714	647
193	canon-000	0.0084	514	0.0156	571	0.0049	663	0.0051	541	0.0275	1005	0.1371	839	0.0138	587	-	
194	canon-001	0.0041	280	0.0061	297	0.0030	396	0.0041	436	0.0048	468	0.0578	731	0.0069	314	-	
195	cist-001	0.0046	312	0.0065	315	0.0042	602	0.0063	636	0.9675	1323	0.9997	1270	0.9994	1300	1.0000	869
196	cist-002	0.0073	474	0.0096	444	0.0085	872	0.0090	759	0.9990	1337	0.9997	1269	0.9996	1308	0.9998	828
197	cist-003	0.0065	437	0.0087	410	0.0046	639	0.0068	661	0.9993	1340	0.9994	1258	0.9993	1299	0.9995	822
198	cist-004	0.0043	297	0.0062	300	0.0031	405	0.0041	443	0.0031	265	0.0363	659	0.0069	317	0.2732	505
199	clearviewai-000	0.0010	29	0.0019	43	0.0024	144	0.0028	209	0.0030	244	0.0058	139	0.0050	183	0.0604	121
200	clearviewai-001	0.0015	59	0.0021	51	0.0024	149	0.0030	260	0.0022	96	0.0745	771	0.0194	696	0.2740	506
201	clearviewai-002	-		-		0.0024	145	0.0031	288	0.0022	94	0.0042	64	0.0037	83	0.0531	72
202	clearviewai-003	-		-		0.0023	84	0.0025	136	0.0022	89	0.0042	66	0.0038	86	0.0533	74
203	closeli-001	0.0136	655	0.0163	581	0.0039	566	0.0054	563	0.0072	608	1.0000	1299	0.0094	452	1.0000	926
204	cloudmatrix-000	0.0192	737	0.0340	749	0.0133	1013	0.0220	1008	0.9837	1326	1.0000	1305	0.0281	783	-	
205	cloudmatrix-001	0.0668	888	0.1141	894	0.0539	1194	0.0905	1194	0.3509	1234	0.9819	1178	0.9010	1215	0.9999	833
206	cloudmatrix-002	0.0075	481	0.0113	495	0.0084	868	0.0120	829	0.9248	1316	0.9997	1268	0.9985	1293	1.0000	935
207	cloudwalk-hr-000	0.0265	791	0.0434	787	0.0340	1152	0.0646	1160	-		-		-		-	
208	cloudwalk-hr-001	0.0044	299	0.0072	347	0.0073	817	0.0108	810	0.0125	793	0.0228	543	0.0145	604	-	
209	cloudwalk-hr-002	0.0043	292	0.0059	284	0.0054	700	0.0076	703	0.0076	626	0.5932	1028	0.0093	447	-	
210	cloudwalk-hr-003	0.0026	164	0.0041	171	0.0040	576	0.0058	592	0.0060	559	0.9992	1251	0.0094	448	-	
211	cloudwalk-hr-004	0.0009	18	0.0018	37	0.0034	473	0.0028	222	0.0052	495	0.9992	1252	0.0093	446	-	
212	cloudwalk-mt-000	0.0100	564	0.0170	594	0.0074	822	0.0118	827	0.0127	796	0.0197	505	0.0129	560	-	
213	cloudwalk-mt-002	0.0064	434	0.0085	403	0.0054	703	0.0098	786	0.0070	600	0.0108	348	0.0076	367	-	
214	cloudwalk-mt-003	0.0013	48	0.0022	52	0.0026	241	0.0027	189	0.0039	364	0.0076	213	0.0067	300	-	
215	cloudwalk-mt-004	0.0009	24	0.0013	14	0.0024	159	0.0021	34	0.0028	212	0.0054	127	0.0050	185	0.0572	101
216	cloudwalk-mt-005	0.0006	11	0.0009	9	0.0025	212	0.0022	61	0.0017	35	0.9286	1139	0.5956	1143	0.9948	784
217	cloudwalk-mt-006	0.0006	10	0.0006	2	0.0023	98	0.0019	1	0.0016	20	0.0032	19	0.0030	37	0.0427	21
218	cloudwalk-mt-007	0.0006	9	0.0007	3	0.0023	105	0.0019	2	0.0016	18	0.0032	16	0.0030	35	0.0419	18
219	clova-000	0.0099	560	0.0150	554	0.0094	904	0.0147	909	0.0136	820	0.0213	522	0.0152	620	-	
220	cmcuni-001	0.0025	153	0.0038	152	0.0025	205	0.0025	153	0.0022	91	0.0052	117	0.0034	57	0.0660	167

Table 62: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
221	codeline-000	0.0057	382	0.0079	378	0.0037	532	0.0053	561	0.2721	1219	1.0000	1302	0.9763	1254	1.0000	938
222	cogent-000	0.0435	848	0.0835	867	-		-		-		-		-		-	
223	cogent-001	0.0176	720	0.0385	773	-		-		-		-		-		-	
224	cogent-002	0.0148	678	0.0321	740	0.0095	909	0.0127	852	-		-		-		-	
225	cogent-003	0.0091	532	0.0188	624	0.0098	917	0.0132	865	0.0187	913	0.0319	615	0.0224	736	-	
226	cogent-004	0.0064	435	0.0116	497	0.0096	912	0.0134	873	0.0157	857	0.0325	621	0.0204	710	-	
227	cogent-005	0.0060	406	0.0112	491	0.0064	773	0.0070	673	0.0095	710	0.0184	487	0.0135	574	-	
228	cogent-006	0.0046	318	0.0059	288	0.0036	503	0.0047	495	0.0058	545	0.0113	364	0.0091	434	0.1522	411
229	cogent-007	0.0022	124	0.0038	150	0.0028	350	0.0031	272	0.0040	380	0.0082	247	0.0067	299	0.1395	385
230	cogent-008	0.0015	60	0.0027	82	0.0023	107	0.0025	144	0.0033	303	0.0063	168	0.0055	227	0.0884	261
231	cognitec-000	0.0116	610	0.0177	608	0.0118	984	0.0167	952	0.0285	1009	0.9924	1203	0.0435	858	-	
232	cognitec-001	0.0126	639	0.0185	620	0.0120	989	0.0168	954	0.0270	995	0.0554	723	0.0357	826	-	
233	cognitec-002	0.0066	439	0.0101	461	0.0079	839	0.0108	808	0.0181	904	0.0317	613	0.0237	748	-	
234	cognitec-003	0.0038	260	0.0052	252	0.0054	705	0.0057	587	0.0225	957	0.0416	682	0.0388	838	0.1397	386
235	cognitec-004	0.0036	246	0.0053	258	0.0053	691	0.0056	578	0.0098	725	0.0202	512	0.0154	621	0.3436	530
236	cognitec-005	0.0022	119	0.0030	104	0.0035	496	0.0041	430	0.0079	637	0.0480	703	0.0233	743	0.2134	472
237	cognitec-006	-		-		0.0034	484	0.0039	418	0.0069	592	0.0160	445	0.0108	501	0.1197	352
238	cognitec-007	-		-		0.0035	502	0.0042	455	0.0069	596	0.0114	367	0.0091	433	0.1112	334
239	cognitec-008	-		-		0.0030	395	0.0036	376	0.0058	538	0.0106	340	0.0091	435	0.0884	260
240	compal-000	-		-		0.0028	357	0.0033	328	0.0039	368	0.0064	173	0.0051	197	0.0736	194
241	compal-001	-		-		0.0073	813	0.0053	556	0.0092	696	0.1965	874	0.0165	643	0.2610	501
242	compal-002	-		-		0.0073	812	0.0053	555	0.0092	695	0.1923	872	0.0165	642	0.2592	497
243	cor-001	0.0075	483	0.0113	492	0.0055	713	0.0084	740	0.0091	691	0.0148	420	0.0092	443	-	
244	coretech-000	0.7699	1055	1.0000	1180	1.0000	1420	0.9999	1390	1.0000	1400	1.0000	1419	1.0000	1382	1.0000	1416
245	coretech-001	0.0052	352	0.0067	325	0.0083	865	0.0092	763	0.0346	1034	0.1363	838	0.0252	761	0.6210	612
246	coretech-002	0.0020	105	0.0036	135	0.0064	775	0.0064	641	0.0118	780	0.0157	442	0.0110	506	0.3091	519
247	coretech-003	-		-		0.0045	630	0.0051	538	0.1155	1158	0.1539	848	0.1515	1001	0.1180	346
248	coretech-004	-		-		0.0044	618	0.0048	506	0.0051	493	0.0093	297	0.0070	322	0.0893	265
249	corsight-001	0.0040	276	0.0057	278	0.0033	468	0.0047	494	0.0045	441	0.0095	305	0.0063	276	-	
250	corsight-002	0.0053	362	0.0068	327	0.0030	385	0.0041	442	0.0039	369	0.0079	231	0.0054	220	0.0747	201
251	corsight-003	0.0026	158	0.0040	168	0.0028	333	0.0045	484	0.0035	332	0.0059	151	0.0046	159	0.0637	149
252	csc-001	0.9158	1063	0.9346	1069	0.9899	1377	0.9993	1379	0.9999	1344	0.9959	1218	0.9906	1269	-	
253	csc-002	0.0099	562	0.0132	525	0.0077	830	0.0142	899	0.0126	794	0.0195	504	0.0146	607	-	
254	csc-003	0.0053	360	0.0065	313	0.0037	534	0.0047	499	0.0074	619	0.0124	385	0.0112	515	0.0928	290
255	ctbcbank-000	0.0168	706	0.0250	697	0.0146	1035	0.0224	1010	0.0211	945	0.8964	1124	0.3779	1097	-	
256	ctbcbank-001	0.0155	688	0.0235	682	0.0148	1042	0.0243	1027	0.0207	940	0.9279	1138	0.3469	1088	-	
257	ctbcbank-002	-		-		0.0031	397	0.0049	517	0.0037	347	0.7785	1078	0.4878	1121	0.9727	742
258	cu-face-002	0.0105	583	0.0116	498	0.0650	1209	0.0568	1146	0.0271	999	0.0139	401	0.0076	363	0.1122	335
259	cu-face-003	0.0036	243	0.0049	238	0.0031	419	0.0037	390	0.0030	252	0.3075	932	0.1310	984	0.7046	629
260	cu-face-004	-		-		0.0032	439	0.0035	354	0.0026	185	0.4284	982	0.2115	1037	0.7861	656
261	cubox-001	0.0064	430	0.0080	381	0.0037	527	0.0055	571	0.0060	550	0.0111	357	0.0077	374	-	
262	cubox-002	0.0034	236	0.0041	170	0.0025	196	0.0025	155	0.0033	302	0.0064	171	0.0058	250	0.0519	66
263	cubox-003	0.0021	110	0.0028	92	0.0024	182	0.0026	170	0.0025	155	0.0046	82	0.0037	84	0.0495	53
264	cubox-004	-		-		0.0025	222	0.0027	194	0.0023	112	0.0044	78	0.0038	87	0.0490	49

Table 63: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
265	cubox-005	-		-		0.0026	258	0.0030	267	0.0025	147	0.0047	92	0.0039	102	0.0515	65
266	cudocommunication-001	0.4777	1026	1.0000	1327	0.4373	1307	0.5360	1298	1.0000	1487	1.0000	1470	1.0000	1493	1.0000	1199
267	cuhkee-001	0.0036	250	0.0045	209	0.0031	421	0.0046	489	0.0051	494	0.0095	306	0.0079	381	-	
268	cvartel-001	-		-		0.0023	77	0.0024	104	0.0020	61	0.0038	48	0.0032	42	0.0466	38
269	cybercore-000	0.0728	893	0.1110	890	0.1521	1259	0.2375	1253	0.1874	1189	0.1907	870	0.1178	975	-	
270	cybercore-001	0.3759	1004	0.5677	1029	0.6928	1342	0.7926	1332	0.8118	1286	0.9291	1142	0.7080	1169	0.9782	749
271	cybercore-002	0.0092	540	0.0119	504	0.0049	661	0.0072	682	0.9105	1314	1.0000	1319	1.0000	1328	1.0000	934
272	cybercore-003	0.0155	687	0.0164	583	0.0032	444	0.0033	324	0.0024	121	0.9719	1166	0.8213	1194	0.9993	816
273	cyberextruder-001	0.1972	966	0.2547	954	0.4686	1314	0.6387	1312	0.3807	1237	0.3806	968	0.2582	1055	-	
274	cyberextruder-002	0.0811	906	0.1336	902	0.1465	1257	0.2266	1252	0.2086	1202	1.0000	1466	1.0000	1501	-	
275	cyberextruder-003	0.0109	593	0.0169	593	0.0071	803	0.0112	818	0.0165	876	0.0410	680	0.0272	781	0.2308	480
276	cyberextruder-004	0.0118	616	0.0181	613	0.0081	850	0.0133	867	0.0191	922	0.0329	624	0.0268	779	0.1904	451
277	cyberextruder-005	-		-		0.0049	660	0.0063	634	0.0055	513	0.0105	337	0.0079	386	0.0806	234
278	cyberextruder-006	-		-		0.0083	860	0.0138	883	0.0100	733	0.0180	481	0.0114	522	0.1276	366
279	cyberlink-000	0.0181	726	0.0274	713	0.0125	1000	0.0180	966	-		-		-		-	
280	cyberlink-001	0.0131	647	0.0210	652	0.0439	1177	0.0517	1138	0.0173	895	0.0333	632	0.0209	719	-	
281	cyberlink-002	0.0114	604	0.0195	633	0.0101	927	0.0163	944	0.0160	866	0.0302	599	0.0192	692	-	
282	cyberlink-003	0.0118	614	0.0192	630	0.0098	920	0.0161	938	0.0153	850	0.0288	592	0.0159	631	-	
283	cyberlink-004	0.0074	478	0.0105	472	0.0068	791	0.0089	755	0.0094	706	0.0176	477	0.0117	529	-	
284	cyberlink-005	0.0060	409	0.0092	427	0.0058	734	0.0067	658	0.0074	615	0.0146	416	0.0105	494	-	
285	cyberlink-006	0.0042	289	0.0054	268	0.0043	605	0.0049	514	0.0052	499	0.0097	314	0.0077	375	-	
286	cyberlink-007	0.0032	207	0.0053	260	0.0041	586	0.0043	461	0.0052	498	0.0243	552	0.0084	413	0.3366	528
287	cyberlink-008	0.0042	288	0.0056	277	0.0038	558	0.0048	507	0.0053	504	0.0099	320	0.0074	350	0.1354	382
288	cyberlink-009	0.0018	87	0.0027	81	0.0047	640	0.0046	486	0.0040	384	0.0086	275	0.0062	273	0.1104	328
289	cyberlink-010	0.0011	36	0.0019	44	0.0041	583	0.0041	431	0.0039	363	0.1829	864	0.0054	221	0.8629	682
290	cyberlink-011	0.0010	28	0.0017	31	0.0035	497	0.0034	333	0.0045	432	0.0080	234	0.0070	325	0.0877	257
291	cyberlink-012	0.0010	27	0.0016	29	0.0036	508	0.0035	359	0.0047	464	0.0099	319	0.0074	351	0.1350	380
292	cyberlink-013	0.0011	38	0.0017	33	0.0037	529	0.0036	371	0.0045	434	0.0090	284	0.0080	390	0.0924	287
293	cyberlink-014	-		-		0.0026	280	0.0027	178	0.0030	242	0.0065	177	0.0052	204	0.0869	254
294	cyberlink-015	-		-		0.0025	207	0.0026	164	0.0028	215	0.0063	167	0.0050	188	0.0835	249
295	cyberlink-016	-		-		0.0025	193	0.0025	141	0.0028	208	0.0063	166	0.0050	184	0.0827	244
296	dahua-000	0.0249	777	0.0477	804	-		-		-		-		-		-	
297	dahua-001	0.0250	782	0.0466	799	0.0228	1105	0.0379	1091	-		-		-		-	
298	dahua-002	0.0129	643	0.0157	573	0.0116	979	0.0153	920	0.0134	814	0.0222	534	0.0170	651	-	
299	dahua-003	0.0052	357	0.0068	329	0.0056	717	0.0062	629	0.0113	772	0.0207	516	0.0185	681	-	
300	dahua-004	0.0045	306	0.0058	281	0.0036	514	0.0048	505	0.0051	491	0.0086	271	0.0070	326	-	
301	dahua-005	0.0031	200	0.0046	214	0.0035	499	0.0049	522	0.0046	445	0.0076	215	0.0062	272	-	
302	dahua-006	0.0027	165	0.0039	160	0.0031	413	0.0039	416	0.0039	367	0.0067	188	0.0058	247	-	
303	dahua-007	0.0017	79	0.0023	57	0.0026	264	0.0032	295	0.0033	295	0.0060	155	0.0054	219	0.0678	173
304	daon-000	0.0095	552	0.0117	500	0.0068	787	0.0077	712	0.0092	700	0.0174	473	0.0137	583	0.1430	395
305	daon-001	-		-		0.0036	509	0.0058	591	0.0050	482	0.0084	256	0.0057	241	0.0805	232
306	daon-002	-		-		0.0030	388	0.0042	449	0.0039	362	0.0146	418	0.0055	229	0.7444	641
307	daon-003	-		-		0.0024	180	0.0029	234	0.0030	250	0.0059	153	0.0048	175	0.0639	151
308	datech-000	0.0020	106	0.0031	111	0.0031	410	0.0041	439	0.0033	301	0.1646	854	0.0667	913	0.4898	572

Table 64: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
309	datech-001	-		-		0.0028	336	0.0034	339	0.0026	192	0.2308	898	0.0818	934	0.5505	588
310	datech-002	-		-		0.0028	335	0.0034	341	0.0026	169	0.3779	966	0.1654	1017	0.7940	658
311	decatur-000	0.0714	890	0.1115	892	0.0608	1204	0.1106	1210	0.0866	1141	1.0000	1313	0.0714	920	-	
312	decatur-001	0.0424	844	0.0711	849	0.0237	1108	0.0458	1116	0.0447	1077	1.0000	1289	0.9969	1285	1.0000	951
313	decloakface-001	0.0795	902	0.1131	893	0.3127	1289	0.4259	1284	0.2118	1205	0.7926	1082	0.6088	1147	0.9589	730
314	decloakface-002	-		-		0.0048	650	0.0077	711	1.0000	1357	0.9998	1277	0.9997	1309	1.0000	929
315	deepglint-001	0.0040	278	0.0062	299	0.0047	646	0.0067	656	0.0069	593	1.0000	1316	1.0000	1327	-	
316	deepglint-002	0.0016	67	0.0027	84	0.0032	430	0.0033	326	0.0043	415	0.0084	262	0.0077	371	-	
317	deepglint-003	0.0027	170	0.0038	151	0.0030	384	0.0032	293	0.0043	405	0.0082	249	0.0076	364	-	
318	deepglint-004	0.0025	150	0.0034	123	0.0039	567	0.0061	624	0.0050	484	0.0091	288	0.0082	400	0.1003	304
319	deepglint-005	0.0052	353	0.0059	290	0.0030	378	0.0031	273	0.0033	307	0.7620	1070	0.1535	1002	0.9912	775
320	deepsea-001	0.0136	657	0.0215	657	0.0142	1027	0.0214	1002	0.0163	870	0.0250	559	0.0192	693	-	
321	deepsense-000	0.0145	673	0.0265	709	0.0113	968	0.0196	989	0.0151	847	0.0215	525	0.0129	561	-	
322	deepsense-001	0.0013	47	0.0019	41	0.0024	170	0.0025	146	0.0027	203	0.0115	370	0.0053	210	0.1394	384
323	deepsense-002	0.0010	26	0.0016	30	0.0024	132	0.0023	93	0.0026	175	0.0052	121	0.0043	141	0.0781	221
324	deepsense-003	0.0009	23	0.0015	24	0.0025	219	0.0024	105	0.0023	110	0.0457	694	0.0097	462	0.2547	494
325	deepsense-004	-		-		0.0023	81	0.0021	40	0.0020	65	0.0111	356	0.0040	123	0.1195	350
326	deepsense-005	-		-		0.0023	79	0.0021	46	0.0019	51	0.0081	242	0.0035	69	0.0922	286
327	deepsense-006	-		-		0.0022	57	0.0022	68	0.0018	42	0.0336	635	0.0059	255	0.2813	511
328	deepsense-007	-		-		0.0022	59	0.0022	67	0.0018	41	0.0035	35	0.0030	38	0.0440	28
329	deepsense-008	-		-		0.0022	62	0.0022	70	0.0018	39	0.0034	33	0.0030	36	0.0434	23
330	dermalog-001	-		0.3049	971	-		-		-		-		-		-	
331	dermalog-002	-		0.3154	974	-		-		-		-		-		-	
332	dermalog-003	0.2355	976	0.2795	964	-		-		-		-		-		-	
333	dermalog-004	0.1810	959	0.2399	951	-		-		-		-		-		-	
334	dermalog-005	0.1526	947	0.1823	926	0.2580	1280	0.4018	1281	-		0.2651	918	0.1585	1008	-	
335	dermalog-006	0.0253	784	0.0369	768	0.0171	1063	0.0283	1053	0.0217	952	0.0358	650	0.0230	739	-	
336	dermalog-007	0.0564	872	0.0798	862	0.1351	1253	0.2471	1255	0.2343	1212	0.2223	891	0.1397	992	-	
337	dermalog-008	0.0096	556	0.0166	588	0.0086	874	0.0133	866	0.0165	875	0.0586	736	0.0226	738	-	
338	dermalog-009	0.0067	445	0.0094	429	0.0051	679	0.0069	666	0.0116	779	0.0312	605	0.0177	667	0.3747	541
339	dermalog-010	0.0030	193	0.0041	172	0.0034	483	0.0037	394	0.0075	620	0.5181	1003	0.2530	1051	0.8853	698
340	dermalog-011	0.0045	305	0.0062	308	0.0035	498	0.0059	602	0.0057	537	0.2242	893	0.0407	847	0.6329	613
341	dermalog-012	0.0080	504	0.0101	463	0.0037	538	0.0057	588	0.0055	519	0.0095	307	0.0068	308	0.0834	248
342	dermalog-013	-		-		0.0036	516	0.0057	585	0.0058	541	0.0103	329	0.0073	340	0.0894	266
343	dermalog-014	-		-		0.0029	367	0.0039	417	0.0037	349	0.0069	195	0.0057	239	0.0646	156
344	dermalog-015	-		-		0.0025	211	0.0025	132	0.0033	293	0.0064	174	0.0054	222	0.0587	111
345	dermalog-016	-		-		0.0023	109	0.0023	88	0.0025	165	0.0050	112	0.0040	114	0.0550	87
346	dicio-001	0.5486	1038	0.6442	1039	0.7516	1353	0.8607	1347	0.8678	1302	0.8268	1099	0.7034	1166	0.9752	747
347	dicio-002	-		-		0.0057	727	0.0095	774	0.0097	724	0.9998	1273	0.9989	1296	1.0000	887
348	didiglobalface-001	0.0055	371	0.0092	425	0.0030	382	0.0045	483	0.0088	674	0.0119	379	0.0085	416	-	
349	didiglobalface-002	0.0033	221	0.0051	249	0.0026	265	0.0034	347	0.0033	300	0.0085	264	0.0047	167	0.1135	338
350	didiglobalface-003	-		-		0.0023	89	0.0025	154	0.0027	199	0.1115	815	0.0052	207	0.5374	583
351	digidata-000	0.0967	914	0.1410	909	0.2596	1281	0.3462	1275	0.0293	1015	0.0363	658	0.0212	721	0.1849	449
352	digidata-001	0.0224	763	0.0352	757	0.0330	1147	0.0570	1148	0.0109	761	0.0481	704	0.0123	548	0.3055	518

Table 65: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
353	digitalbarriers-000	0.3683	1003	0.4630	1006	-		-		-		-		-		-	
354	digitalbarriers-001	0.4683	1024	0.5024	1015	-		-		-		-		-		-	
355	digitalbarriers-002	0.3360	998	0.3690	988	0.0877	1229	0.1557	1229	0.0971	1151	0.0951	796	0.0497	877	-	
356	dps-000	0.0115	607	0.0176	606	0.0149	1045	0.0185	975	0.0173	893	0.0275	581	0.0180	671	0.1757	438
357	dralmey-001	-		-		0.0036	512	0.0052	551	0.0057	532	0.9985	1243	0.9872	1263	1.0000	886
358	dsk-000	0.1526	946	0.2169	942	0.3787	1299	0.5426	1300	0.3115	1226	0.3089	934	0.1994	1032	-	
359	einetworks-000	0.0099	561	0.0180	611	0.0088	882	0.0140	893	0.0130	803	0.0225	539	0.0147	609	-	
360	einetworksindia-000	0.0058	397	0.0074	355	0.0051	678	0.0061	626	0.0074	616	0.0193	502	0.0102	481	0.1819	443
361	einetworksindia-001	0.0042	290	0.0062	301	0.0048	649	0.0064	639	0.0057	535	0.0473	701	0.0092	441	0.3984	552
362	einetworksindia-002	-		-		0.0024	158	0.0027	179	0.0023	102	0.1534	847	0.0513	881	0.4614	563
363	ekin-002	0.1168	929	0.2042	938	0.1530	1260	0.2524	1258	0.1777	1188	0.2773	923	0.1347	987	0.8954	706
364	element-000	0.0023	135	0.0035	133	0.0024	169	0.0026	161	0.0029	233	0.0060	158	0.0048	171	0.0631	144
365	enface-000	0.0028	176	0.0049	236	0.0043	611	0.0072	680	0.0058	546	0.0150	425	0.0090	432	0.7851	654
366	enface-001	0.0072	470	0.0107	477	0.0071	799	0.0138	884	0.0068	589	0.0515	715	0.0094	453	0.9942	782
367	enface-002	0.0033	226	0.0052	255	0.0038	553	0.0073	688	0.0026	191	0.9998	1279	0.9995	1304	1.0000	931
368	enqura-001	-		-		0.0155	1051	0.0144	902	0.0312	1025	0.0597	738	0.0485	874	0.2970	516
369	enqura-002	-		-		0.0082	856	0.0150	914	0.0105	745	0.0193	499	0.0111	510	0.1302	368
370	enqura-003	-		-		0.0023	124	0.0028	199	0.0026	176	0.0049	106	0.0039	105	0.0611	126
371	eocortex-000	0.3485	1000	0.6943	1048	0.1122	1242	0.1574	1230	0.2155	1206	0.2257	897	0.1606	1013	-	
372	ercacat-001	0.0036	253	0.0044	202	0.0033	458	0.0047	500	0.0106	749	0.0202	511	0.0184	677	-	
373	essi-000	-		-		0.0060	747	0.0093	766	0.0110	763	0.0273	578	0.0129	562	0.1965	462
374	ethiopianlmis-000	-		-		0.2967	1285	0.4098	1282	0.9992	1339	1.0000	1295	0.9999	1316	1.0000	943
375	euronovate-001	0.2786	985	0.3608	987	0.4489	1310	0.6105	1309	0.5010	1251	0.5392	1012	0.3769	1095	0.8802	694
376	euronovate-002	0.0256	787	0.0436	788	0.0226	1103	0.0425	1108	0.0301	1020	0.0691	763	0.0319	807	0.3687	537
377	euronovate-003	0.0061	415	0.0088	414	0.0040	581	0.0060	611	0.0065	581	0.7815	1080	0.5034	1126	0.9823	753
378	euronovate-004	0.0082	508	0.0110	483	0.0050	673	0.0077	710	0.0074	617	0.6949	1053	0.3354	1086	0.9791	751
379	everai-000	0.0122	625	0.0225	670	-		-		-		-		-		-	
380	everai-001	0.0085	518	0.0156	570	0.0072	809	0.0114	821	-		-		-		-	
381	everai-002	0.0104	579	0.0159	575	0.0063	767	0.0112	817	0.0182	906	0.0353	647	0.0264	776	-	
382	paravision-003	0.0034	235	0.0050	243	0.0036	521	0.0052	550	0.0092	699	0.0193	501	0.0156	625	-	
383	expasoft-000	0.0427	845	0.0655	842	0.0239	1110	0.0393	1094	0.0673	1121	0.8963	1123	0.3832	1099	-	
384	expasoft-001	0.0328	826	0.0488	808	0.0211	1091	0.0342	1075	0.0629	1116	0.6483	1041	0.2816	1067	-	
385	expasoft-002	0.0170	709	0.0274	714	0.0787	1223	0.0768	1177	0.1629	1180	0.9996	1265	0.9631	1245	1.0000	889
386	f8-001	0.0249	781	0.0336	747	0.0178	1066	0.0232	1019	0.0303	1021	0.0615	744	0.0408	848	-	
387	f8-002	0.0340	828	0.0591	833	0.0213	1093	0.0374	1089	0.0452	1078	0.0760	772	0.0502	880	0.2708	504
388	facehawk-000	1.0000	1476	1.0000	1171	1.0000	1431	1.0000	1422	1.0000	1412	1.0000	1404	1.0000	1384	1.0000	1404
389	facehawk-001	-		-		0.0545	1197	0.0748	1173	0.0751	1128	0.9919	1200	0.9566	1239	1.0000	868
390	faceint-001	-		-		0.0105	942	0.0124	842	0.0313	1026	0.3318	943	0.0557	895	0.9851	764
391	facelocate-001	-		-		0.0664	1212	0.1171	1213	0.0950	1148	0.9219	1134	0.7041	1167	0.9985	805
392	facelocate-002	-		-		0.0335	1148	0.0634	1155	0.0393	1054	0.9988	1249	0.8977	1214	1.0000	870
393	faceonlive-001	0.0269	795	0.0359	763	0.0387	1162	0.0721	1170	0.0246	985	0.0349	644	0.0220	728	0.2408	485
394	faceonlive-002	0.0121	622	0.0135	529	0.0033	463	0.0041	434	0.0037	351	0.9427	1148	0.7927	1189	0.9993	814
395	faceonlive-003	-		-		0.0045	626	0.0050	532	0.0193	928	0.0360	654	0.0356	824	0.1320	370
396	facephi-000	0.0044	301	0.0059	283	0.0047	642	0.0057	589	0.0088	680	1.0000	1333	1.0000	1334	1.0000	947

Table 66: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
397	facephi-001	-		-		0.0031	416	0.0037	389	0.0058	544	0.0111	354	0.0098	467	0.1048	316
398	facephi-002	-		-		0.0024	133	0.0022	63	0.0036	338	0.0093	296	0.0070	319	0.1136	339
399	facephi-003	-		-		0.0023	111	0.0023	85	0.0031	271	0.0073	206	0.0057	245	0.1060	318
400	faceplugin-000	-		-		0.0031	414	0.0042	453	0.0044	427	0.6268	1035	0.0956	952	0.9673	737
401	facesoft-000	0.0085	516	0.0112	489	0.0064	771	0.0107	805	0.0091	690	0.0171	469	0.0107	497	-	
402	facetag-000	0.2836	986	0.4081	997	0.2933	1284	0.4303	1285	0.3448	1232	0.6312	1036	0.3530	1089	0.9753	748
403	facetag-001	0.2920	988	0.4538	1003	0.3220	1291	0.4371	1286	1.0000	1460	1.0000	1424	0.3254	1082	1.0000	1069
404	facetag-002	0.0098	558	0.0147	551	0.0064	776	0.0110	813	0.0116	777	0.0190	497	0.0119	538	0.1334	376
405	facex-001	1.0000	1273	1.0000	1243	1.0000	1393	1.0000	1394	1.0000	1381	1.0000	1356	1.0000	1346	-	
406	facex-002	0.0803	904	0.1404	907	0.1283	1250	0.1979	1246	0.1440	1173	0.1952	873	0.1299	982	0.7427	639
407	facia-001	0.0083	512	0.0124	511	0.0064	774	0.0097	779	0.0098	726	0.9983	1239	0.9849	1260	1.0000	913
408	facia-002	-		-		0.0072	808	0.0109	811	0.0122	787	0.0460	695	0.0135	575	0.4464	561
409	faicetech-000	-		-		0.0026	281	0.0031	278	0.0023	113	0.0116	373	0.0037	82	0.0904	273
410	farfaces-001	0.4890	1031	0.5860	1035	0.5650	1322	0.7268	1322	0.8015	1285	0.7511	1066	0.5892	1141	0.9536	725
411	fastenterprises-000	0.0093	543	0.0151	558	0.0098	918	0.0140	894	0.0209	942	0.2991	928	0.1608	1014	0.6727	622
412	fci-000	-		-		0.0027	296	0.0027	187	0.0028	216	0.1269	830	0.0563	896	0.3886	547
413	fdu-000	0.0386	838	0.0576	829	-		-		-		-		-		-	
414	fdu-001	0.0347	831	0.0439	791	-		-		-		-		-		-	
415	fedu-001	-		-		0.0033	452	0.0047	493	0.0128	797	0.9816	1176	0.9216	1223	0.9999	838
416	fiberhome-nanjing-002	0.0217	758	0.0381	771	0.0874	1228	0.1770	1237	0.0271	998	0.0351	646	0.0188	685	-	
417	fiberhome-nanjing-003	0.0090	528	0.0139	537	0.0082	857	0.0144	901	0.0110	764	0.0174	471	0.0107	499	-	
418	fiberhome-nanjing-004	0.0037	258	0.0056	276	0.0031	408	0.0043	462	0.0043	417	0.0083	255	0.0061	269	-	
419	fincore-000	0.0309	815	0.0502	810	0.0281	1128	0.0510	1135	0.0521	1095	0.0815	780	0.0522	883	0.3100	520
420	firstcreditkz-001	0.0024	146	0.0034	121	0.0024	177	0.0024	129	0.0034	308	1.0000	1386	1.0000	1405	1.0000	1497
421	firstcreditkz-002	0.0018	85	0.0026	76	0.0024	162	0.0024	116	0.0029	240	0.0056	134	0.0049	179	0.0624	137
422	firstcreditkz-003	-		-		0.0023	116	0.0024	121	0.0026	181	0.0104	335	0.0047	164	0.0910	278
423	firstcreditkz-004	-		-		0.0023	97	0.0024	107	0.0024	123	0.0360	655	0.0061	268	0.2315	481
424	foomobi-001	0.4827	1029	0.5795	1030	0.6823	1337	0.8132	1335	0.8217	1290	1.0000	1400	1.0000	1398	1.0000	1376
425	foomobi-002	0.4827	1028	0.5795	1031	0.6823	1338	0.8132	1336	0.8217	1291	1.0000	1441	1.0000	1449	1.0000	964
426	fourthlinecom-001	-		-		0.0442	1178	0.0696	1166	0.0609	1108	0.8783	1115	0.7523	1180	0.9917	776
427	fpt-000	-		-		0.0213	1094	0.0375	1090	0.0364	1043	0.9528	1152	0.8729	1204	0.9993	817
428	fpt-001	-		-		0.0099	921	0.0148	910	0.8158	1288	0.9986	1244	0.9934	1273	1.0000	946
429	fpttelecom-000	-		-		0.0586	1201	0.0761	1176	0.2308	1211	0.9988	1248	0.9937	1274	1.0000	898
430	fpttelecom-001	-		-		0.0036	505	0.0047	492	0.0051	485	0.1693	857	0.0651	909	0.4913	573
431	fraudcom-000	0.0110	596	0.0173	597	0.0083	861	0.0139	889	0.0166	877	0.0258	569	0.0177	664	0.1488	407
432	frpkauai-001	0.0023	132	0.0035	131	0.0026	238	0.0030	266	0.0040	382	0.0080	238	0.0072	335	0.0706	182
433	frpkauai-002	0.0024	142	0.0035	129	0.0024	174	0.0024	124	0.0033	305	0.0065	180	0.0058	251	0.0599	116
434	frpkauai-003	-		-		0.0023	106	0.0023	98	0.0024	116	0.0075	211	0.0039	103	0.0762	211
435	frpkauai-004	-		-		0.0037	526	0.0036	381	0.0069	595	0.0226	540	0.0138	584	0.1354	381
436	frpkauai-005	-		-		0.0023	76	0.0023	80	0.0021	81	0.0082	245	0.0042	130	0.0760	210
437	fujitsulab-000	0.0123	627	0.0212	654	0.0091	889	0.0133	869	0.0251	987	0.4200	980	0.0360	830	-	
438	fujitsulab-001	0.0059	402	0.0082	392	0.0132	1011	0.0258	1036	0.0140	828	0.0246	556	0.0142	596	-	
439	fujitsulab-002	0.0091	536	0.0124	512	0.0105	939	0.0156	923	0.0169	886	0.0345	642	0.0146	608	-	
440	fujitsulab-003	0.0045	309	0.0065	314	0.0057	729	0.0083	736	0.0080	645	0.0154	435	0.0101	473	0.5564	590

Table 67: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
441	g42-intellibrain-001	0.0006	8	0.0009	10	0.0037	531	0.0044	471	0.0030	260	0.0059	149	0.0053	216	0.0750	207
442	g42-intellibrain-002	0.0006	7	0.0011	12	0.0029	358	0.0037	392	0.0028	211	0.0212	520	0.0091	436	0.1398	388
443	geo-000	0.0543	866	0.0814	864	0.1042	1238	0.1592	1231	0.1637	1181	0.8533	1109	0.1800	1024	-	
444	geo-001	0.0180	724	0.0198	641	0.0037	530	0.0055	569	0.0070	598	0.0129	392	0.0094	449	-	
445	geo-002	0.0171	712	0.0187	623	0.0035	491	0.0051	543	0.0064	571	0.0117	376	0.0083	409	0.0938	293
446	geo-003	0.0180	725	0.0313	735	0.0239	1112	0.0552	1140	0.0319	1029	0.0487	707	0.0222	731	0.1962	461
447	geo-004	0.0030	188	0.0041	173	0.0025	220	0.0030	247	0.0035	331	0.0065	178	0.0053	215	0.0648	158
448	gistouch-000	0.0296	808	0.0518	815	0.0342	1153	0.0609	1152	0.0493	1087	1.0000	1334	1.0000	1345	1.0000	952
449	gistouch-001	0.0176	721	0.0312	734	0.8931	1368	0.9735	1366	0.8625	1298	1.0000	1375	1.0000	1415	1.0000	1455
450	glory-000	0.1094	925	0.1286	898	0.2179	1274	0.2656	1262	-		-		-		-	
451	glory-001	0.0902	910	0.1082	888	0.1642	1262	0.2065	1251	0.2186	1207	0.2669	919	0.2089	1035	-	
452	glory-002	0.0241	769	0.0311	733	0.0116	980	0.0151	918	0.0157	859	0.0264	571	0.0188	687	-	
453	glory-003	0.0076	486	0.0125	515	0.0077	833	0.0103	799	0.0130	802	0.0205	513	0.0143	599	-	
454	glory-004	0.0077	489	0.0123	508	0.0074	823	0.0098	787	0.0122	790	0.0193	500	0.0134	572	0.1520	410
455	glory-005	0.0056	376	0.0076	363	0.0054	706	0.0072	683	0.0075	621	0.9237	1136	0.7853	1186	0.9974	801
456	glory-006	0.0050	334	0.0068	326	0.0051	681	0.0070	672	0.0069	597	0.7869	1081	0.5076	1127	0.9719	741
457	glory-007	0.0050	339	0.0068	328	0.0050	671	0.0069	668	0.0068	586	0.0126	390	0.0083	408	0.1066	319
458	gorilla-000	0.0802	903	0.1409	908	-		-		-		-		-		-	
459	gorilla-001	0.0488	858	0.0756	855	0.1218	1248	0.2034	1248	-		-		-		-	
460	gorilla-002	0.0256	788	0.0413	779	0.0478	1183	0.0912	1195	-		-		-		-	
461	gorilla-003	0.0165	705	0.0291	719	0.0205	1088	0.0437	1109	0.0297	1018	0.9658	1162	0.0262	771	-	
462	gorilla-004	0.0138	661	0.0239	686	0.0243	1114	0.0444	1111	0.0275	1004	0.0369	661	0.0204	713	-	
463	gorilla-005	-		-		0.0142	1028	0.0267	1046	0.0228	961	0.0358	649	0.0195	699	-	
464	gorilla-006	0.0105	587	0.0152	561	0.0106	943	0.0203	995	0.0155	852	0.0218	529	0.0136	577	-	
465	gorilla-007	0.0074	477	0.0111	487	0.0065	779	0.0126	847	0.0100	736	0.0151	427	0.0102	477	0.1450	399
466	gorilla-008	0.0058	390	0.0091	421	0.0049	659	0.0079	724	0.0079	642	0.0126	389	0.0091	437	0.1217	357
467	gorilla-009	0.0049	329	0.0072	341	0.0038	547	0.0056	579	0.0065	576	0.0104	331	0.0070	320	0.1103	327
468	gpstechvn-000	0.0202	747	0.0355	760	0.0142	1026	0.0248	1029	0.0208	941	0.0495	709	0.0207	716	0.5098	578
469	graymatics-001	0.1039	919	0.1620	918	0.1344	1252	0.1917	1242	0.1648	1182	0.5160	1001	0.2689	1061	0.9194	715
470	griaule-000	0.0071	464	0.0099	453	0.0050	666	0.0072	678	0.0160	863	0.0304	600	0.0267	778	0.1480	405
471	griaule-001	0.0057	381	0.0078	373	0.0045	631	0.0065	646	0.0070	601	0.7515	1067	0.5106	1129	0.9747	745
472	griaule-002	0.0021	116	0.0032	112	0.0025	233	0.0027	195	0.0034	320	0.0996	802	0.0201	707	0.3978	550
473	gtelots-000	-		-		0.0059	736	0.0098	783	0.0090	686	0.9995	1262	0.9975	1286	1.0000	900
474	gtelots-001	-		-		0.0053	693	0.0102	797	0.0071	602	0.0111	358	0.0070	318	0.1173	344
475	hertasecurity-000	0.0630	880	0.0780	861	0.0503	1189	0.0898	1190	0.0738	1125	0.0693	766	0.0420	853	-	
476	hertasecurity-001	0.0249	779	0.0309	732	0.0105	940	0.0161	937	0.0245	983	0.0447	689	0.0359	828	0.1850	450
477	hertasecurity-002	0.0206	750	0.0315	736	0.0060	743	0.0078	718	0.0253	990	0.0696	767	0.0457	864	0.4565	562
478	hertasecurity-003	0.0079	501	0.0104	469	0.0060	744	0.0078	714	0.0255	991	0.0732	770	0.0460	865	0.4731	566
479	hik-001	0.0096	555	0.0125	514	0.0093	897	0.0164	948	0.0108	756	0.0937	792	0.0127	555	-	
480	hisign-001	0.0036	249	0.0050	241	0.0034	471	0.0046	487	0.0079	641	0.0153	434	0.0133	569	0.1080	321
481	hisign-002	0.0029	185	0.0044	198	0.0027	323	0.0032	307	0.0028	222	0.0409	679	0.0132	567	0.2016	466
482	hisign-003	-		-		0.0030	380	0.0036	369	0.0036	341	0.0582	734	0.0201	708	0.2584	496
483	hyperverge-001	1.0000	1193	1.0000	1335	1.0000	1496	1.0000	1458	1.0000	1499	1.0000	1475	1.0000	1490	-	
484	hyperverge-002	0.0050	335	0.0066	320	0.0035	490	0.0051	537	0.0062	565	0.0107	344	0.0074	352	0.0921	285

Table 68: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
485	hyperverge-003	0.0019	99	0.0030	100	0.0025	201	0.0029	236	0.0027	194	0.0049	110	0.0042	134	0.0576	103
486	hyperverge-004	0.0072	472	0.0116	499	0.0040	574	0.0071	677	0.0058	543	0.0080	236	0.0057	242	0.0621	134
487	hyperverge-005	0.0026	162	0.0039	162	0.0027	292	0.0031	287	0.0036	343	0.0066	182	0.0055	226	0.0658	165
488	hyperverge-006	-		-		0.0025	229	0.0030	252	0.0033	299	0.0060	157	0.0051	199	0.0601	118
489	hyperverge-007	-		-		0.0026	247	0.0030	262	0.0034	317	0.0061	160	0.0052	202	0.0605	122
490	hyundai-roboticslab-000	-		-		0.0037	528	0.0052	547	0.0063	569	1.0000	1416	0.0872	943	1.0000	1418
491	hyundai-roboticslab-001	-		-		0.0048	657	0.0066	653	0.0193	926	0.8620	1110	0.6861	1161	0.9861	766
492	hyundai-roboticslab-002	-		-		0.0039	572	0.0054	564	0.0086	669	0.9589	1155	0.8817	1209	0.9986	807
493	hyundai-roboticslab-003	-		-		0.0024	136	0.0028	205	0.0027	204	0.6384	1039	0.2934	1071	0.8692	685
494	hyundai-roboticslab-004	-		-		0.0023	126	0.0028	203	0.0028	225	0.2635	915	0.0802	930	0.5392	585
495	hzailu-001	0.0122	623	0.0164	584	0.0095	908	0.0196	987	0.0079	639	0.0118	377	0.0090	430	0.1173	345
496	hzailu-002	0.0051	344	0.0072	345	0.0038	560	0.0055	574	0.0040	379	0.0077	216	0.0062	274	0.0928	292
497	hzailu-003	0.0178	723	0.0291	720	0.0031	426	0.0042	458	0.0035	322	0.0061	163	0.0052	201	0.0799	230
498	hzailu-004	0.0175	718	0.0197	639	0.0031	428	0.0042	452	0.0034	316	0.0060	159	0.0051	191	0.0775	219
499	hzailu-005	0.0172	715	0.0197	638	0.0028	347	0.0036	375	0.0032	289	0.0058	140	0.0048	172	0.0751	208
500	hzailu-006	0.0161	701	0.0174	599	0.0024	166	0.0027	174	0.0026	182	0.0049	109	0.0044	152	0.0654	162
501	hzailu-007	-		-		0.0025	194	0.0027	191	0.0026	170	0.0048	98	0.0044	149	0.0653	161
502	hzailu-008	-		-		0.0024	185	0.0026	173	0.0025	162	0.0048	96	0.0044	147	0.0652	160
503	i2v-001	-		-		0.0946	1233	0.1369	1220	0.3989	1242	0.9998	1275	0.9966	1282	1.0000	895
504	i2v-002	-		-		0.0035	494	0.0044	475	0.0037	346	0.9355	1146	0.4823	1119	0.9960	790
505	icehouse-000	-		-		0.0027	293	0.0036	379	0.0047	457	0.0088	281	0.0072	331	0.0726	188
506	icm-002	0.0143	670	0.0249	695	0.0144	1033	0.0256	1035	0.0236	975	0.0386	672	0.0263	775	-	
507	icm-003	0.0138	662	0.0222	668	0.0149	1043	0.0282	1052	0.0227	960	0.0384	670	0.0257	765	0.1938	456
508	icm-004	0.0079	497	0.0120	505	0.0074	820	0.0107	804	0.0091	692	0.0281	586	0.0128	559	0.2778	508
509	icm-005	0.0100	567	0.0140	539	0.0091	885	0.0108	806	0.0169	885	0.0319	616	0.0290	795	0.1775	441
510	icomm-000	0.0019	97	0.0033	118	0.0033	462	0.0044	472	0.0031	278	0.1350	836	0.0552	892	0.4196	555
511	icthtc-000	0.0260	790	0.0396	777	0.0207	1089	0.0339	1074	0.0291	1014	0.0474	702	0.0346	819	-	
512	id3-001	0.1698	953	0.2499	953	-		-		-		-		-		-	
513	id3-002	0.1615	952	0.2393	950	-		-		-		-		-		-	
514	id3-003	0.0361	835	0.0757	857	0.0292	1135	0.0476	1124	0.0447	1076	0.0674	758	0.0396	841	-	
515	id3-004	0.0198	744	0.0344	754	0.0238	1109	0.0423	1107	0.0289	1013	0.0416	681	0.0257	764	-	
516	id3-005	0.0104	580	0.0169	592	0.0080	848	0.0133	870	0.0194	929	0.0334	633	0.0262	772	-	
517	id3-006	0.0072	471	0.0103	466	0.0049	662	0.0074	693	0.0095	708	0.0165	458	0.0119	537	-	
518	id3-007	0.0056	377	0.0078	371	0.0060	749	0.0072	687	0.0275	1002	0.1374	840	0.0519	882	-	
519	id3-008	0.0039	265	0.0055	270	0.0032	442	0.0042	448	0.0081	650	0.0155	437	0.0134	570	0.1186	348
520	id3-009	-		-		0.0035	495	0.0049	523	0.0046	444	0.4689	990	0.0214	725	0.9377	720
521	id3-010	-		-		0.0032	449	0.0045	479	0.0042	403	0.8447	1107	0.2140	1039	0.9932	780
522	id3-011	-		-		0.0032	448	0.0045	480	0.0045	430	0.0162	453	0.0059	260	0.3641	534
523	idbio-000	-		-		0.0044	615	0.0067	657	0.0068	590	0.0557	726	0.0222	732	0.2871	515
524	idemia-003	0.0222	760	0.0316	737	0.0188	1069	0.0325	1071	0.0548	1100	-		-		-	
525	idemia-004	0.0160	698	0.0244	690	0.0199	1080	0.0354	1080	0.0268	993	0.1489	846	0.0263	774	-	
526	idemia-005	0.0132	650	0.0216	660	0.0121	991	0.0218	1007	0.0215	948	0.0323	620	0.0188	686	-	
527	idemia-006	0.0046	320	0.0062	306	0.0047	644	0.0066	648	0.0073	612	0.2882	925	0.0094	454	-	
528	idemia-007	0.0024	140	0.0039	163	0.0032	450	0.0038	404	0.0046	448	0.0092	293	0.0070	327	-	

Table 69: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
529	idemia-008	0.0023	130	0.0032	115	0.0023	119	0.0028	198	0.0034	321	0.0067	187	0.0056	237	0.0765	213		
530	idemia-009	0.0022	126	0.0030	102	0.0022	66	0.0023	100	0.0023	101	0.0046	83	0.0039	98	0.0533	75		
531	idemia-010	0.0015	58	0.0026	71	0.0022	72	0.0020	8	0.0020	63	0.0040	55	0.0033	50	0.0465	37		
532	idemia-011	-		-		0.0021	32	0.0021	41	0.0018	45	0.0077	221	0.0038	91	0.0782	224		
533	identy-000	0.0073	475	0.0095	438	0.0050	667	0.0067	654	0.0071	605	0.8257	1098	0.4310	1111	0.9839	759		
534	identy-001	-		-		0.0029	366	0.0034	329	0.0037	352	0.0066	183	0.0055	228	0.0748	206		
535	identy-002	-		-		0.0027	302	0.0029	246	0.0030	243	0.0058	141	0.0051	194	0.0691	177		
536	identy-003	-		-		0.0025	226	0.0030	254	0.0026	167	0.0050	115	0.0043	135	0.0547	84		
537	idrnd-000	-		-		0.0041	594	0.0053	560	0.0050	479	0.0094	299	0.0075	354	0.1026	310		
538	idrnd-001	-		-		0.0025	232	0.0033	319	0.0032	285	0.0528	720	0.0173	659	0.2243	476		
539	igearx-face-000	0.0091	533	0.0146	548	0.0163	1058	0.0362	1085	0.0399	1056	0.6436	1040	0.3305	1084	0.7832	653		
540	iit-000	0.1516	945	0.1981	935	0.0828	1226	0.1442	1226	0.2214	1208	0.2943	926	0.1877	1027	-			
541	iit-001	0.0104	576	0.0179	610	0.0099	924	0.0142	898	0.1222	1162	0.2677	920	0.2171	1041	-			
542	iit-002	0.0111	601	0.0177	609	0.0085	870	0.0140	892	0.0193	925	0.0332	630	0.0260	768	-			
543	iit-003	0.0082	511	0.0151	560	0.0053	694	0.0084	741	0.0122	789	0.0199	509	0.0137	582	-			
544	imds-software-001	0.0126	640	0.0228	672	0.0130	1007	0.0221	1009	0.0231	963	0.0469	700	0.0199	706	0.4260	557		
545	imds-software-002	0.0048	324	0.0072	346	0.0036	511	0.0052	552	0.0047	462	0.9981	1236	0.0078	378	1.0000	902		
546	imds-software-003	0.0032	204	0.0047	222	0.0028	329	0.0036	378	0.0036	342	0.9987	1245	0.0063	277	1.0000	891		
547	imperial-000	0.0067	446	0.0108	480	0.0080	846	0.0134	874	0.0087	671	0.0581	733	0.0102	482	-			
548	imperial-001	0.0094	546	0.0154	566	0.0072	807	0.0120	831	0.0103	740	0.0190	495	0.0120	541	-			
549	imperial-002	0.0058	391	0.0081	386	0.0055	711	0.0085	744	0.0083	656	0.0157	439	0.0103	483	-			
550	incode-000	0.0766	896	0.1029	887	-		-		-		-		-		-			
551	incode-001	0.0550	869	0.0773	859	-		-		-		-		-		-			
552	incode-002	0.0293	805	0.0548	825	0.0436	1176	0.0785	1179	-		-		-		-			
553	incode-003	0.0142	669	0.0249	696	0.0448	1179	0.0869	1186	0.0354	1038	0.0661	753	0.0320	808	-			
554	incode-004	0.0077	491	0.0132	523	0.0096	913	0.0160	933	0.0171	889	0.0281	584	0.0179	669	-			
555	incode-005	0.0111	600	0.0189	625	0.0156	1053	0.0304	1062	0.0195	932	0.0307	604	0.0172	655	-			
556	incode-006	0.0156	689	0.0273	712	0.0117	981	0.0238	1025	0.0202	937	0.0339	639	0.0164	640	-			
557	incode-007	0.0109	592	0.0155	568	0.0056	721	0.0099	788	0.0107	751	0.0168	463	0.0102	478	-			
558	incode-008	0.0063	426	0.0101	458	0.0046	636	0.0086	750	0.0057	533	0.0104	332	0.0074	345	-			
559	incode-009	0.0044	300	0.0067	324	0.0034	478	0.0051	536	0.0049	477	0.0091	287	0.0067	302	0.0892	264		
560	incode-010	0.0041	281	0.0063	309	0.0028	351	0.0043	460	0.0047	466	0.0077	223	0.0061	267	0.0770	217		
561	incode-011	0.0032	206	0.0044	199	0.0026	272	0.0034	342	0.0032	280	0.0359	651	0.0140	591	0.1840	448		
562	incode-012	0.0027	167	0.0043	186	0.0025	204	0.0035	356	0.0034	313	0.0966	800	0.0385	836	0.3387	529		
563	incode-013	0.0035	242	0.0049	230	0.0032	433	0.0039	415	0.0051	487	0.0094	300	0.0081	398	0.0768	215		
564	incode-014	-		-		0.0022	69	0.0023	94	0.0034	312	0.0066	185	0.0062	275	0.0569	98		
565	incode-015	-		-		0.0021	17	0.0022	69	0.0020	67	0.0040	57	0.0037	78	0.0537	79		
566	incode-016	-		-		0.0021	15	0.0022	65	0.0019	50	0.0038	45	0.0035	68	0.0504	63		
567	infocert-001	0.0105	586	0.0172	596	0.0078	835	0.0125	844	0.0159	861	0.1573	851	0.0565	897	0.5716	595		
568	infocert-002	-		-		1.0000	1469	0.9993	1378	1.0000	1440	1.0000	1442	1.0000	1450	1.0000	967		
569	infocert-003	-		-		0.0202	1081	0.0284	1054	0.0620	1114	0.2624	914	0.1316	985	0.6450	616		
570	infodif-001	-		-		0.6808	1336	0.8174	1340	0.6367	1263	0.7694	1073	0.5999	1145	0.9590	731		
571	innefulabs-000	0.0122	626	0.0199	643	0.0112	965	0.0197	990	0.0222	955	0.0372	665	0.0271	780	-			
572	innominds-001	0.5569	1039	0.6541	1042	0.7642	1357	0.8665	1350	0.7629	1276	0.8200	1093	0.6359	1153	0.9841	761		

Table 70: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
573	innov8tif-000	-		-		0.1842	1265	0.3028	1272	0.2056	1200	0.2348	902	0.1449	995	0.6142	608
574	innov8tif-001	-		-		0.1865	1267	0.2951	1266	0.1976	1195	0.2564	912	0.1593	1009	0.6665	619
575	innov8tif-002	-		-		0.2074	1272	0.2575	1261	0.1162	1159	0.1822	863	0.1782	1021	0.5224	580
576	innovativetechnologyltd-001	0.0578	873	0.0938	880	0.0501	1188	0.0981	1197	0.0592	1107	0.0779	777	0.0422	854	-	
577	innovativetechnologyltd-002	0.0451	851	0.0716	851	0.0541	1195	0.1009	1203	0.0506	1090	0.0682	760	0.0371	833	-	
578	innovatrics-000	0.1105	926	0.1905	931	-		-		-		-		-		-	
579	innovatrics-001	0.1077	924	0.1828	928	-		-		-		-		-		-	
580	innovatrics-002	0.4049	1009	0.6142	1037	-		-		-		-		-		-	
581	innovatrics-003	0.0349	832	0.0483	806	-		-		-		-		-		-	
582	innovatrics-004	0.0194	740	0.0292	721	0.0344	1154	0.0617	1153	0.0562	1102	0.0762	773	0.0412	849	-	
583	innovatrics-005	0.0230	766	0.0353	758	0.0398	1166	0.0721	1171	-		-		-		-	
584	innovatrics-006	0.0058	392	0.0089	418	0.0061	757	0.0096	777	0.0096	718	0.0165	459	0.0103	484	-	
585	innovatrics-007	0.0040	272	0.0054	266	0.0057	724	0.0078	713	0.0079	640	0.0123	382	0.0088	425	-	
586	innovatrics-008	0.0047	321	0.0064	312	0.0038	556	0.0052	549	0.0053	505	0.0088	278	0.0069	316	0.0899	270
587	innovatrics-009	0.0022	118	0.0031	110	0.0028	328	0.0032	304	0.0034	311	0.1165	820	0.0326	809	0.3866	546
588	innovatrics-010	0.0017	83	0.0031	109	0.0027	318	0.0028	204	0.0041	387	0.0078	225	0.0066	298	0.0744	199
589	innovatrics-011	-		-		0.0026	273	0.0027	177	0.0032	287	0.0060	154	0.0044	148	0.0615	130
590	innovatrics-012	-		-		0.0025	208	0.0026	158	0.0034	315	0.9309	1143	0.4977	1122	0.9960	789
591	innovatrics-013	-		-		0.0022	47	0.0021	43	0.0026	171	0.9984	1240	0.3780	1098	1.0000	893
592	innovatrics-014	-		-		0.0021	27	0.0020	6	0.0017	32	0.0033	25	0.0029	22	0.0417	17
593	innovatrics-015	-		-		0.0021	16	0.0020	24	0.0019	54	0.0036	39	0.0031	41	0.0468	39
594	innovatrics-016	-		-		0.0021	20	0.0020	25	0.0018	38	0.0034	31	0.0029	29	0.0451	32
595	innovatrics-017	-		-		0.0020	9	0.0021	38	0.0017	30	0.0033	21	0.0029	24	0.0437	25
596	insightface-000	0.0018	90	0.0027	86	0.0029	359	0.0030	269	0.0038	358	0.0077	220	0.0068	306	0.0765	212
597	insightface-001	0.0009	17	0.0014	19	0.0027	297	0.0024	127	0.0035	330	0.0070	197	0.0065	287	0.0692	178
598	insightface-002	0.0011	39	0.0019	40	0.0027	316	0.0026	163	0.0036	340	0.0069	193	0.0065	286	0.0694	179
599	insightface-003	0.0015	62	0.0021	48	0.0045	629	0.0034	346	0.1298	1164	1.0000	1365	0.9407	1229	1.0000	1268
600	insightface-004	0.0015	64	0.0023	56	0.0052	687	0.0036	377	0.1403	1171	0.9923	1202	0.9128	1219	0.9989	809
601	inspur-000	0.0060	412	0.0078	369	0.7415	1352	0.9093	1359	0.2838	1222	0.9996	1263	0.9976	1287	1.0000	940
602	inspur-001	0.0035	238	0.0049	232	0.0025	206	0.0030	257	0.0029	238	0.9951	1214	0.9592	1242	1.0000	874
603	inspur-002	0.0025	149	0.0039	157	0.0024	135	0.0027	176	0.0031	264	0.0062	165	0.0047	165	0.1112	331
604	inspur-003	-		-		0.0023	101	0.0026	167	0.0031	266	0.0060	156	0.0046	160	0.0812	239
605	intellicloudai-001	0.0142	668	0.0234	680	0.0092	893	0.0145	904	0.0162	868	0.0371	664	0.0171	654	-	
606	intellicloudai-002	0.0059	401	0.0085	401	0.0060	746	0.0069	669	0.0108	754	0.2477	910	0.0171	653	-	
607	intellifusion-001	0.0072	469	0.0094	436	0.0056	723	0.0085	745	0.0111	768	0.0212	521	0.0143	598	-	
608	intellifusion-002	0.0059	399	0.0077	366	0.0040	575	0.0074	692	0.0085	665	0.5352	1010	0.0104	491	-	
609	intellivision-001	0.1335	938	0.2205	945	0.1090	1240	0.1670	1233	0.1385	1168	0.1676	856	0.1170	972	-	
610	intellivision-002	0.1000	915	0.1775	922	0.0610	1205	0.1009	1202	0.0805	1135	0.1074	809	0.0682	914	-	
611	intellivision-003	0.1177	931	0.2006	937	0.0760	1219	0.1244	1217	0.1069	1156	0.1431	844	0.0839	937	0.4940	575
612	intellivision-004	0.0271	796	0.0559	826	0.0294	1139	0.0503	1134	0.0327	1030	0.0461	696	0.0293	800	0.2860	513
613	intellivision-005	0.0105	585	0.0182	616	0.0093	899	0.0141	895	0.0146	840	0.0252	561	0.0172	657	0.1763	440
614	intellivision-006	0.0111	599	0.0174	601	0.0110	956	0.0162	939	0.0181	902	0.0313	606	0.0191	690	0.2416	487
615	intellivision-007	0.0086	519	0.0128	520	0.0098	919	0.0159	931	0.0133	810	0.0218	530	0.0154	622	0.1751	437
616	intellivix-001	0.0064	433	0.0087	413	0.0046	633	0.0063	637	0.0072	606	0.9233	1135	0.7856	1187	0.9977	803

Table 71: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
617	intellivix-002	0.0062	418	0.0085	400	0.0039	564	0.0056	577	0.0060	556	0.3464	950	0.0857	941	0.8759	688
618	intellivix-003	0.0075	482	0.0125	513	0.0052	686	0.0091	762	0.0066	583	0.0297	595	0.0096	461	0.1907	453
619	intellivix-004	0.0032	213	0.0045	210	0.0025	225	0.0031	286	0.0040	376	0.0074	208	0.0063	278	0.0662	168
620	intellivix-005	0.0034	237	0.0048	226	0.0024	143	0.0027	193	0.0035	329	0.0067	186	0.0059	265	0.0609	125
621	intellivix-006	-		-		0.0024	184	0.0030	264	0.0026	173	0.0085	263	0.0040	113	0.0918	282
622	intelresearch-000	0.0307	814	0.0578	830	0.0385	1160	0.0751	1175	0.0409	1063	0.9870	1185	0.5380	1134	-	
623	intelresearch-001	0.0242	771	0.0595	834	0.0129	1006	0.0292	1058	0.0351	1036	0.9993	1253	0.1854	1026	-	
624	intelresearch-002	0.0058	393	0.0082	390	0.0050	675	0.0086	747	0.0136	819	0.0434	683	0.0216	726	-	
625	intelresearch-003	0.0046	316	0.0062	298	0.0038	552	0.0060	612	0.0088	676	0.0168	465	0.0136	578	-	
626	intelresearch-004	0.0025	147	0.0035	127	0.0032	436	0.0038	399	0.0049	478	0.0094	298	0.0072	333	-	
627	intelresearch-005	0.0016	65	0.0023	59	0.0028	327	0.0034	336	0.0042	402	0.0084	258	0.0066	296	0.1433	396
628	intelresearch-006	0.0010	33	0.0015	25	0.0026	289	0.0028	216	0.0032	286	0.8123	1090	0.4742	1117	0.9907	772
629	intelresearch-007	0.0009	20	0.0013	17	0.0025	203	0.0025	133	0.0035	326	0.0069	194	0.0057	238	0.0746	200
630	intelresearch-008	0.0009	19	0.0013	16	0.0025	202	0.0025	134	0.0028	219	0.0304	601	0.0048	173	0.2375	484
631	intelresearch-009	-		-		0.0024	186	0.0025	139	0.0028	221	0.0058	144	0.0046	161	0.0665	169
632	intelresearch-010	-		-		0.0023	112	0.0024	109	0.0029	239	0.0058	145	0.0050	186	0.0683	176
633	intelresearch-011	-		-		0.0023	82	0.0023	72	0.0025	151	0.0049	107	0.0040	121	0.0549	86
634	intelresearch-012	-		-		0.0023	96	0.0023	78	0.0025	143	0.0048	99	0.0040	119	0.0549	85
635	intema-000	0.0012	43	0.0017	35	0.0023	75	0.0022	64	0.0022	93	0.0172	470	0.0061	266	0.0996	303
636	intema-001	0.0010	32	0.0014	22	0.0021	25	0.0020	26	0.0019	57	0.0037	41	0.0030	40	0.0457	35
637	interlinklabs-001	-		-		0.0102	930	0.0205	997	0.0195	931	0.0237	548	0.0115	526	0.1579	415
638	interlinklabs-002	-		-		0.0033	464	0.0049	518	0.0040	373	0.9968	1225	0.9807	1257	1.0000	857
639	interlinklabs-003	-		-		0.0029	370	0.0034	349	0.0046	449	0.0083	253	0.0074	344	0.0620	133
640	intozi-001	0.9982	1070	0.9998	1081	0.9998	1390	1.0000	1395	1.0000	1353	1.0000	1324	1.0000	1337	1.0000	1267
641	intozi-002	-		-		0.0038	550	0.0050	534	0.0054	507	0.9998	1274	0.9989	1297	1.0000	911
642	intsysmsu-000	0.0135	653	0.0204	644	0.0112	964	0.0161	936	0.0148	844	0.0244	554	0.0172	656	-	
643	intsysmsu-001	0.9543	1066	0.9888	1075	0.9923	1378	0.9948	1374	0.9977	1331	0.9955	1216	0.9892	1266	-	
644	intsysmsu-002	0.0130	644	0.0254	699	0.0137	1019	0.0267	1047	0.0160	864	0.0267	574	0.0145	605	-	
645	ionetworks-000	0.0060	410	0.0087	407	0.0044	614	0.0058	597	0.0080	649	0.0144	412	0.0112	513	0.1149	340
646	ionetworks-001	0.0254	785	0.0388	774	0.0435	1173	0.0662	1161	0.0426	1069	0.8936	1121	0.7391	1177	0.9964	793
647	ionetworks-002	0.0045	302	0.0066	319	0.0033	457	0.0047	501	0.0544	1098	0.9648	1160	0.8765	1206	0.9994	818
648	ionetworks-003	-		-		0.0034	485	0.0049	526	0.0043	412	0.0079	230	0.0056	234	0.0812	238
649	ionetworks-004	-		-		0.0034	488	0.0050	530	0.0043	408	0.0079	229	0.0056	232	0.0808	236
650	iqface-000	0.0091	537	0.0143	541	0.0075	824	0.0110	814	0.0171	891	0.2234	892	0.0359	827	-	
651	iqface-001	0.0088	525	0.0117	501	0.0177	1064	0.0230	1018	0.0165	874	0.5514	1015	0.0286	790	-	
652	iqface-003	0.0058	395	0.0079	377	0.0051	680	0.0058	598	0.0104	743	0.0200	510	0.0193	694	-	
653	iqsec-001	-		-		0.7050	1345	0.8569	1346	0.7728	1278	0.9962	1223	0.9840	1258	1.0000	876
654	iqsec-002	-		-		0.7050	1344	0.8569	1345	0.7728	1279	0.9962	1222	0.9841	1259	1.0000	875
655	iqsec-003	-		-		0.9924	1380	0.9984	1377	0.9998	1342	0.9996	1266	0.9981	1289	1.0000	905
656	irex-000	0.0052	351	0.0099	454	0.0056	719	0.0083	738	0.0137	824	0.0163	455	0.0078	377	-	
657	irisid-001	-		-		0.0034	481	0.0048	504	0.0058	542	0.0108	349	0.0079	382	0.1024	308
658	irisid-002	-		-		0.0031	424	0.0042	454	0.0055	517	0.0102	328	0.0077	373	0.1024	309
659	isap-001	0.5092	1033	0.6588	1043	0.6899	1341	0.7978	1333	0.7200	1272	0.7253	1058	0.5373	1133	-	
660	isap-002	0.0114	606	0.0186	621	0.0087	879	0.0151	917	0.0156	856	0.5134	1000	0.0333	813	-	

Table 72: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
661	isityou-000	0.5682	1041	0.7033	1050	1.0000	1419	0.9999	1388	1.0000	1379	1.0000	1336	1.0000	1376	-	
662	istl-001	-		-		0.0041	593	0.0059	608	0.0053	503	0.5416	1014	0.2534	1052	0.8765	689
663	isystems-000	0.1267	935	0.1795	924	-		-		-		-		-		-	
664	isystems-001	0.0149	682	0.0245	692	0.0138	1022	0.0210	1000	0.0209	944	0.0332	629	0.0223	733	-	
665	isystems-002	0.0118	615	0.0182	615	0.0111	960	0.0162	942	0.0166	879	0.0284	588	0.0195	700	-	
666	itmo-001	-		0.4408	1001	-		-		-		-		-		-	
667	itmo-002	0.1607	950	0.2866	967	-		-		-		-		-		-	
668	itmo-003	0.0749	895	0.1398	906	-		-		-		-		-		-	
669	itmo-004	0.0255	786	0.0543	823	-		-		-		-		-		-	
670	itmo-005	0.0182	727	0.0345	755	0.0181	1067	0.0348	1079	0.0225	958	0.0378	668	0.0230	740	-	
671	itmo-006	0.0125	634	0.0220	664	0.0149	1044	0.0266	1043	0.0233	966	0.0383	669	0.0285	788	-	
672	itmo-007	0.0080	503	0.0125	516	0.0107	946	0.0185	973	0.0167	881	0.0222	535	0.0144	602	-	
673	itmo-008	0.0090	531	0.0150	555	0.0058	732	0.0059	605	0.0187	914	0.0355	648	0.0339	814	0.1758	439
674	ivacognitive-001	0.0189	734	0.0351	756	0.0123	994	0.0235	1023	0.0198	934	0.0274	580	0.0155	623	-	
675	iws-000	0.4824	1027	0.5801	1032	0.6859	1340	0.8155	1338	0.8251	1292	0.7756	1076	0.6400	1156	-	
676	jaakit-001	0.5830	1042	0.7146	1051	0.8173	1361	0.8893	1356	0.8950	1308	0.8387	1105	0.7091	1170	0.9733	744
677	jaakit-002	-		-		0.5729	1324	0.6951	1319	0.6495	1266	0.9894	1194	0.9721	1250	0.9999	840
678	jgumainz-000	-		-		0.0104	935	0.0163	945	0.0150	846	0.1982	875	0.0661	910	0.6389	614
679	kaizenvoiz-001	-		-		0.4082	1303	0.5607	1302	0.4678	1248	0.4894	994	0.3770	1096	0.8558	678
680	kakao-001	0.4553	1022	0.5532	1027	0.6580	1334	0.8150	1337	0.4831	1250	1.0000	1485	1.0000	1480	-	
681	kakao-002	0.0625	879	0.1779	923	0.0791	1224	0.1381	1223	0.0636	1117	1.0000	1376	1.0000	1416	-	
682	kakao-003	0.0130	646	0.0185	619	0.0261	1121	0.0464	1121	0.0252	989	0.2380	904	0.0284	787	-	
683	kakao-004	0.0078	492	0.0103	467	0.0059	741	0.0102	794	0.0155	853	0.1182	822	0.0230	741	-	
684	kakao-005	0.0040	270	0.0059	282	0.0036	520	0.0057	586	0.0085	663	0.0239	550	0.0125	551	-	
685	kakao-007	0.0019	98	0.0028	88	0.0024	142	0.0026	165	0.0033	298	0.0061	161	0.0053	211	0.0580	104
686	kakao-008	0.0011	37	0.0018	39	0.0023	87	0.0023	84	0.0021	80	0.0041	62	0.0035	65	0.0475	44
687	kakao-009	0.0009	21	0.0013	15	0.0023	94	0.0023	89	0.0020	68	0.0039	52	0.0034	55	0.0468	40
688	kakaobank-000	0.0041	286	0.0058	279	0.0041	585	0.0058	600	0.0060	558	0.9988	1247	0.3243	1081	1.0000	888
689	kakaobank-001	0.0030	191	0.0046	213	0.0032	446	0.0039	405	0.0049	475	0.9998	1281	0.9994	1301	1.0000	950
690	kakaobank-002	-		-		0.0026	261	0.0036	366	0.0038	355	0.3184	941	0.1174	974	0.6931	627
691	kakaobank-003	-		-		0.0023	117	0.0025	152	0.0029	235	0.0065	175	0.0052	206	0.0733	191
692	kakaopay-001	0.0152	686	0.0252	698	0.0145	1034	0.0270	1048	0.0232	964	0.0344	641	0.0194	695	0.2005	465
693	kasikornlabs-000	0.0112	602	0.0184	618	0.0086	875	0.0137	882	0.0130	805	0.0225	538	0.0148	612	0.1660	430
694	kasikornlabs-001	0.0138	660	0.0206	646	0.0087	878	0.0139	890	0.0142	830	0.0236	547	0.0171	652	0.1684	433
695	kasikornlabs-002	0.0069	453	0.0091	420	0.0048	652	0.0063	632	0.0076	624	0.0144	411	0.0110	508	0.1249	362
696	kasikornlabs-003	0.0057	379	0.0076	362	0.0042	601	0.0056	580	0.0059	547	0.0123	381	0.0098	463	0.1212	356
697	kasikornlabs-004	-		-		0.0041	591	0.0054	566	0.0057	529	0.0112	361	0.0095	455	0.1101	325
698	kasikornlabs-005	-		-		0.0037	523	0.0043	465	0.0054	508	0.0107	345	0.0093	445	0.1069	320
699	kbyai-000	-		-		0.0027	311	0.0032	302	0.0031	275	0.0079	232	0.0049	180	0.0655	163
700	kedacom-000	0.0055	370	0.0081	385	0.0111	963	0.0120	830	0.0415	1064	0.0966	799	0.0686	915	-	
701	keyless-000	-		-		0.0025	221	0.0034	334	0.0031	273	0.1874	868	0.0568	898	0.5543	589
702	kiwitech-000	0.0076	485	0.0105	471	0.0081	851	0.0128	859	0.0096	713	0.0163	454	0.0101	475	-	
703	kloudspot-001	-		-		0.0140	1023	0.0259	1037	0.9999	1346	1.0000	1330	1.0000	1342	1.0000	901
704	kloudspot-002	-		-		0.0255	1116	0.0423	1106	0.0192	924	0.8748	1113	0.7143	1172	0.9929	777

Table 73: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
705	kneron-003	0.0542	865	0.0902	876	0.0346	1155	0.0562	1144	0.0919	1146	0.1251	828	0.0973	954	-	
706	kneron-005	0.0157	691	0.0259	703	0.0126	1002	0.0212	1001	0.0406	1062	0.0693	765	0.0542	889	-	
707	kneron-006	-		-		0.0094	902	0.0125	843	0.1305	1166	0.9751	1169	0.9033	1216	0.9987	808
708	knowutech-000	0.0039	268	0.0055	271	0.0028	355	0.0042	446	0.0042	395	0.0077	218	0.0059	258	0.0820	241
709	kogniza-001	-		-		0.0028	356	0.0031	292	0.0025	150	0.0048	95	0.0042	127	0.0557	89
710	kookmin-001	0.0462	855	0.0750	853	0.0489	1185	0.0842	1183	0.0659	1119	0.8380	1104	0.3212	1080	-	
711	kookmin-002	0.0054	369	0.0077	365	0.0043	608	0.0065	645	0.0123	791	0.7591	1069	0.0198	704	-	
712	koreaid-001	0.0031	198	0.0045	208	0.0026	263	0.0032	299	0.0043	411	0.0083	250	0.0068	310	0.0743	197
713	krungthai-002	0.0105	584	0.0161	578	0.0091	890	0.0141	896	0.7350	1274	0.9889	1192	0.9605	1243	0.9999	837
714	kuke3d-001	0.0058	388	0.0104	468	0.0083	862	0.0093	765	0.0270	996	0.9901	1195	0.8341	1195	1.0000	877
715	kuke3d-002	0.0077	488	0.0135	530	0.0069	794	0.0098	784	0.0111	767	1.0000	1329	1.0000	1336	1.0000	906
716	lebentech-000	0.5940	1043	0.7032	1049	0.8854	1366	0.9511	1362	0.9089	1313	0.9970	1226	0.9861	1262	0.9999	845
717	lebentech-001	0.0304	812	0.9032	1066	0.9998	1391	1.0000	1393	1.0000	1355	1.0000	1310	0.9999	1325	1.0000	942
718	lemalabs-001	0.0111	597	0.0175	603	0.0088	880	0.0142	897	0.0143	833	0.0228	542	0.0140	590	0.1646	427
719	lightvision-000	-		-		0.0146	1038	0.0232	1020	0.0237	976	0.9638	1159	0.7768	1184	0.9986	806
720	lightvision-001	-		-		0.0068	789	0.0096	776	0.0084	661	0.9948	1213	0.9288	1225	1.0000	861
721	lineclova-000	0.0172	714	0.0236	683	0.0109	952	0.0194	986	0.0183	908	0.0291	594	0.0204	712	0.1936	455
722	lineclova-001	0.0025	148	0.0040	166	0.0026	288	0.0034	348	0.0045	443	0.4127	977	0.0080	388	0.9824	754
723	lineclova-002	0.0021	109	0.0035	126	0.0025	191	0.0027	184	0.0041	385	0.0086	269	0.0072	334	0.1595	419
724	lineclova-003	0.0018	92	0.0030	101	0.0028	352	0.0031	275	0.0041	386	0.0083	254	0.0075	361	0.1014	307
725	lookman-002	0.0297	809	0.0547	824	0.0339	1150	0.0562	1143	0.0614	1111	0.0960	798	0.0790	926	-	
726	lookman-004	0.0074	480	0.0099	451	0.0124	997	0.0149	912	0.0430	1071	0.0866	786	0.0694	917	-	
727	lssytech-000	-		-		0.0023	88	0.0023	95	0.0021	82	0.0040	59	0.0036	75	0.0574	102
728	lssytech-001	-		-		0.0021	41	0.0020	19	0.0020	66	0.0039	51	0.0034	56	0.0496	54
729	luxand-000	0.2056	970	0.2814	965	0.4053	1301	0.5365	1299	0.3497	1233	0.3743	964	0.2605	1058	-	
730	luxand-001	0.0116	608	0.0925	877	0.0144	1032	0.0267	1044	1.0000	1393	1.0000	1402	1.0000	1396	1.0000	1372
731	luxand-003	-		-		0.9938	1381	0.9999	1385	0.2905	1223	0.2256	896	0.0849	940	0.4876	570
732	luxand-004	-		-		0.0031	422	0.0042	447	0.0110	762	0.9994	1259	0.9969	1284	1.0000	1017
733	mantra-000	0.0037	255	0.0052	257	0.0054	702	0.0056	582	0.0097	722	0.0181	482	0.0151	617	-	
734	matrixcomsec-000	-		-		0.1032	1237	0.0895	1189	1.0000	1471	1.0000	1489	1.0000	1475	1.0000	1130
735	matrixcomsec-001	-		-		0.0070	796	0.0136	875	0.8826	1306	1.0000	1467	1.0000	1498	1.0000	1175
736	matrixcomsec-002	-		-		0.0091	887	0.0172	957	0.0168	883	0.3770	965	0.1584	1007	0.8585	679
737	maverickresearch-000	-		-		0.0143	1029	0.0192	982	0.0216	949	0.0362	657	0.0250	759	0.2293	477
738	maxvision-000	0.0078	495	0.0106	474	0.0110	957	0.0147	907	0.0368	1046	1.0000	1486	0.1545	1003	1.0000	1105
739	maxvision-001	0.0305	813	0.0528	819	0.1028	1235	0.1921	1243	0.0650	1118	0.3001	929	0.1553	1005	0.6188	611
740	maxvision-002	0.0070	457	0.0107	476	0.0061	755	0.0093	764	0.0080	646	0.5726	1020	0.2943	1072	0.9160	711
741	maxvision-003	0.0056	374	0.0083	396	0.0038	557	0.0060	615	0.0061	561	0.2614	913	0.0650	908	0.5812	597
742	maxvision-004	0.0024	144	0.0038	155	0.0025	237	0.0029	239	0.0029	230	0.4732	991	0.1965	1029	0.8141	664
743	maxvision-005	0.0014	54	0.0026	75	0.0024	151	0.0024	108	0.0025	140	0.1216	824	0.0332	811	0.3922	549
744	maxvision-006	0.0016	70	0.0026	78	0.0024	157	0.0024	113	0.0024	135	0.4012	975	0.1799	1023	0.7634	642
745	mbbankvn-000	-		-		0.0030	379	0.0035	350	0.0026	168	0.3822	969	0.1130	968	0.7052	631
746	megvii-000	0.0159	697	0.0246	694	-		-		-		-		-		-	
747	megvii-001	0.0157	692	0.0244	688	0.0392	1165	0.0671	1162	0.0168	884	0.0455	693	0.0164	639	-	
748	megvii-002	0.0104	581	0.0145	544	0.0225	1100	0.0345	1077	0.0099	732	0.0286	589	0.0240	751	-	

Table 74: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
749	megvii-003	0.0064	429	0.0094	431	0.0136	1018	0.0260	1039	0.0050	480	0.0080	237	0.0059	261	-			
750	megvii-004	0.0020	101	0.0033	119	0.0028	344	0.0035	362	0.0037	354	0.0074	209	0.0068	311	0.0736	195		
751	megvii-005	0.0010	31	0.0015	23	0.0026	254	0.0031	281	0.0019	52	0.0500	710	0.0057	243	0.1991	463		
752	megvii-006	0.0011	34	0.0016	26	0.0026	282	0.0033	320	0.0025	148	0.0050	111	0.0048	169	0.0588	113		
753	megvii-007	0.0006	12	0.0016	28	0.0025	223	0.0028	206	0.0024	131	0.0048	100	0.0047	166	0.0586	107		
754	megvii-008	0.0003	4	0.0008	6	0.0023	120	0.0021	48	0.0024	136	0.0049	104	0.0047	168	0.0541	81		
755	megvii-009	0.0003	3	0.0008	7	0.0023	122	0.0021	49	0.0018	36	0.0149	423	0.0036	71	0.1096	324		
756	meituan-000	0.0197	741	0.0424	785	0.0078	837	0.0074	691	0.0103	741	0.0193	503	0.0164	638	0.1133	337		
757	meituan-001	0.0164	703	0.1886	930	0.0025	210	0.0026	160	0.0030	248	0.0074	207	0.0051	195	0.0791	226		
758	meituan-002	0.0017	77	0.0025	64	0.0024	156	0.0023	79	0.0024	133	0.0067	189	0.0044	154	0.0773	218		
759	meituan-003	0.0017	75	0.0021	49	0.0024	148	0.0023	101	0.0024	128	0.0084	259	0.0046	157	0.0883	259		
760	meituan-004	0.0014	51	0.0021	46	0.0024	150	0.0024	122	0.0023	114	0.0070	196	0.0044	150	0.0813	240		
761	meiya-001	0.0171	711	0.0275	715	0.0159	1057	0.0261	1042	0.0311	1023	0.2250	895	0.0245	756	-			
762	mendaxiatech-000	0.0027	169	0.0036	137	0.0029	361	0.0036	382	0.0031	269	0.0057	138	0.0051	196	-			
763	metsakuurcompany-001	0.0068	451	0.0087	406	0.0068	792	0.0078	716	0.0095	709	0.8972	1125	0.5635	1137	0.9970	798		
764	metsakuurcompany-002	0.0048	323	0.0071	338	0.0030	389	0.0043	467	0.0032	288	0.2059	884	0.0665	912	0.5374	582		
765	metsakuurcompany-003	0.0034	227	0.0051	248	0.0026	259	0.0036	364	0.0028	213	0.2418	905	0.1001	956	0.5866	601		
766	metsakuurcompany-004	-		-		0.0026	286	0.0032	294	0.0032	281	0.4488	987	0.2015	1033	0.8129	663		
767	metsakuurcompany-005	-		-		0.0030	390	0.0041	435	0.0039	365	0.1725	859	0.0054	224	0.8274	667		
768	metsakuurcompany-006	-		-		0.0034	475	0.0061	621	0.0042	393	0.0896	789	0.0399	843	0.2791	509		
769	miaxis-001	0.0068	449	0.0099	452	0.0059	740	0.0097	781	0.0096	717	0.0231	545	0.0109	505	0.1592	416		
770	miaxis-002	0.0745	894	0.0774	860	0.3215	1290	0.4000	1280	0.1485	1175	0.2087	885	0.2058	1034	0.3846	545		
771	miaxis-003	0.0045	304	0.0067	323	0.0037	540	0.0050	529	0.0061	563	0.0139	400	0.0088	423	0.1322	372		
772	miaxis-004	-		-		0.0037	522	0.0046	490	0.0057	534	0.0107	346	0.0084	414	0.1261	365		
773	miaxis-005	-		-		0.0033	460	0.0041	445	0.0053	502	0.0106	342	0.0079	384	0.1236	360		
774	microfocus-000	0.4656	1023	0.5624	1028	-		-		-		-		-		-			
775	microfocus-001	0.4482	1020	0.5524	1026	0.7256	1348	0.8416	1343	0.7301	1273	0.6926	1050	0.5180	1130	-			
776	microfocus-002	0.3605	1001	0.5057	1016	0.5783	1325	0.7223	1321	0.5909	1258	0.5963	1030	0.4160	1108	-			
777	microfocus-003	0.0416	842	0.0607	836	0.0652	1210	0.1025	1204	0.0793	1132	0.2248	894	0.0923	948	0.6153	609		
778	minivision-000	0.0033	214	0.0048	228	0.0038	554	0.0049	513	0.0055	515	0.0094	304	0.0079	385	-			
779	mitek-000	0.6882	1050	0.8192	1058	0.9568	1373	0.9788	1369	0.9545	1321	0.9437	1149	0.8494	1198	0.9931	779		
780	mitek-001	-		-		0.0050	670	0.0068	660	0.0053	501	0.0096	311	0.0077	372	0.1130	336		
781	mitek-002	-		-		0.0039	570	0.0054	562	0.0044	424	0.0085	266	0.0065	291	0.1112	332		
782	mobai-000	0.0360	834	0.0439	792	0.0372	1158	0.0700	1167	0.0367	1045	0.0939	793	0.0795	928	-			
783	mobai-001	0.0199	745	0.0219	662	0.0047	643	0.0061	620	0.0093	705	0.0174	472	0.0138	586	-			
784	mobai-002	-		-		0.0048	653	0.0070	671	0.0088	682	0.5904	1026	0.3167	1079	0.8952	704		
785	mobai-003	-		-		0.0053	697	0.0090	758	0.0689	1123	0.9994	1257	0.9982	1290	1.0000	910		
786	mobbl-000	0.2938	989	0.3861	993	0.5391	1319	0.6888	1318	0.6545	1267	0.8027	1085	0.6207	1149	-			
787	mobbl-001	0.3208	993	0.4375	1000	0.5680	1323	0.7193	1320	0.6282	1260	0.5783	1021	0.3984	1103	0.8864	699		
788	mobbl-002	0.9914	1069	0.9970	1077	0.9355	1370	0.9923	1372	1.0000	1352	1.0000	1300	0.9999	1317	1.0000	918		
789	mobbl-003	0.0087	522	0.0134	527	0.0062	759	0.0087	751	0.0099	728	0.0197	506	0.0122	547	0.1618	426		
790	mobbl-004	-		-		0.0031	420	0.0043	464	0.0088	677	1.0000	1307	1.0000	1329	1.0000	920		
791	mobbl-005	-		-		0.0031	411	0.0044	478	0.0045	429	0.4420	985	0.1465	996	0.7973	659		
792	mobipintech-000	0.0090	530	0.0149	553	0.0039	573	0.0057	583	0.0115	776	0.0465	698	0.0182	674	0.6443	615		

Table 75: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
793	momovn-001	0.1044	920	0.1694	920	0.3253	1292	0.5244	1296	0.3861	1239	0.9995	1261	0.9985	1291	1.0000	924
794	momovn-002	-		-		0.0042	599	0.0060	617	0.0062	566	0.7672	1071	0.5456	1135	0.9731	743
795	momovn-003	-		-		0.0029	363	0.0033	313	0.0056	524	0.0093	295	0.0076	365	0.0612	128
796	moredian-000	0.3874	1007	0.4912	1012	0.9988	1387	0.9999	1387	0.9990	1338	0.9999	1286	0.9998	1315	-	
797	morpho-000	0.1020	917	0.1343	903	-		-		-		-		-		-	
798	morpho-001	-		0.2772	962	-		-		-		-		-		-	
799	morpho-002	0.0377	836	0.0680	846	0.0368	1157	0.0716	1169	-		-		-		-	
800	mukh-001	0.0170	710	0.0285	717	0.0225	1101	0.0405	1100	0.0272	1000	0.0950	795	0.0291	797	0.5846	600
801	mukh-002	0.0269	794	0.0357	761	0.0435	1175	0.0799	1181	0.0143	834	0.0213	523	0.0122	546	0.1908	454
802	mukh-003	0.0077	490	0.0113	494	0.0111	961	0.0224	1011	0.0061	562	0.7813	1079	0.4504	1114	0.9899	771
803	mukh-004	0.0050	336	0.0069	334	0.0051	677	0.0099	790	0.0045	433	0.0083	252	0.0059	262	0.1030	312
804	mukh-005	-		-		0.0023	85	0.0024	130	0.0056	523	0.0161	448	0.0067	301	0.1398	387
805	multimodality-000	0.0034	228	0.0047	221	0.0036	517	0.0044	476	0.0077	631	0.9976	1231	0.4456	1113	1.0000	899
806	multimodality-001	0.0029	183	0.0042	180	0.0031	403	0.0035	353	0.0038	356	0.0071	200	0.0059	257	0.0732	189
807	mvision-001	0.0191	735	0.0233	678	0.0204	1085	0.0356	1081	0.0198	935	0.0337	636	0.0242	753	-	
808	nazhiai-000	0.0040	275	0.0059	289	0.0036	504	0.0048	510	0.0057	536	0.0125	386	0.0083	407	-	
809	ncsgroup-000	-		-		0.0126	1001	0.0226	1013	0.0179	899	0.1715	858	0.0287	792	0.6160	610
810	ncssg-001	0.0207	751	0.1540	913	0.0045	623	0.0067	659	0.0056	528	0.1180	821	0.0301	801	0.3908	548
811	ncssg-002	-		-		0.0032	441	0.0038	401	0.0033	294	0.3144	938	0.1061	961	0.7441	640
812	ncssg-003	-		-		0.0033	461	0.0040	426	0.0033	306	0.2511	911	0.0959	953	0.6553	618
813	ncssg-004	-		-		0.0033	455	0.0041	444	0.0029	241	0.2454	907	0.0860	942	0.6010	603
814	neosystems-001	1.0000	1328	1.0000	1203	0.2987	1287	0.4382	1287	0.5173	1253	0.6570	1043	0.4043	1106	-	
815	neosystems-002	0.2905	987	0.4077	996	0.2028	1271	0.3252	1273	0.4088	1244	0.5519	1016	0.3331	1085	0.9561	727
816	neosystems-003	0.2429	978	0.3349	981	0.1844	1266	0.2999	1270	0.5942	1259	0.3936	972	0.2292	1045	0.8482	674
817	neosystems-004	0.0279	799	0.0495	809	0.0289	1133	0.0585	1151	0.0439	1074	0.9621	1157	0.1296	981	0.9996	824
818	netbridgetech-001	0.4749	1025	0.6599	1044	0.4438	1309	0.5676	1303	0.4491	1247	1.0000	1291	0.9541	1236	-	
819	netbridgetech-002	0.0101	568	0.0166	587	0.0077	832	0.0127	854	0.0133	812	0.8215	1096	0.0523	884	-	
820	neurotechnology-000	-		0.2371	949	-		-		-		-		-		-	
821	neurotechnology-001	0.1393	940	0.2218	946	-		-		-		-		-		-	
822	neurotechnology-002	0.1115	927	0.1663	919	-		-		-		-		-		-	
823	neurotechnology-003	0.0617	876	0.0880	872	-		-		-		-		-		-	
824	neurotechnology-004	0.0146	674	0.0297	725	0.0107	948	0.0164	947	-		-		-		-	
825	neurotechnology-005	0.0141	667	0.0300	728	0.0108	951	0.0163	946	0.0204	938	0.0875	787	0.0246	758	-	
826	neurotechnology-006	0.0098	559	0.0136	532	0.0105	941	0.0182	967	0.0164	872	0.0843	783	0.0214	722	-	
827	neurotechnology-008	0.0091	535	0.0184	617	0.0076	828	0.0116	823	0.0157	858	0.9973	1227	0.1803	1025	-	
828	neurotechnology-009	0.0049	331	0.0087	408	0.0057	728	0.0094	768	0.0079	644	1.0000	1296	0.0101	474	-	
829	neurotechnology-010	0.0048	325	0.0085	402	0.0056	720	0.0090	757	0.0077	634	1.0000	1301	0.0113	516	-	
830	neurotechnology-011	0.0050	340	0.0087	405	0.0061	758	0.0097	780	0.0077	633	0.0404	678	0.0092	440	-	
831	neurotechnology-012	0.0051	347	0.0070	337	0.0038	544	0.0056	581	0.0066	582	0.0112	362	0.0075	358	0.1009	306
832	neurotechnology-013	0.0032	209	0.0045	207	0.0026	285	0.0036	365	0.0037	345	0.0068	190	0.0052	209	0.0747	202
833	neurotechnology-015	0.0022	123	0.0036	140	0.0024	140	0.0028	212	0.0030	245	0.0052	122	0.0041	126	0.0646	157
834	neurotechnology-016	0.0022	127	0.0032	114	0.0023	104	0.0027	185	0.0027	193	0.0049	105	0.0039	111	0.0613	129
835	neurotechnology-017	0.0017	80	0.0031	108	0.0023	83	0.0025	147	0.0025	156	0.0047	93	0.0038	95	0.0612	127
836	neurotechnology-018	0.0020	104	0.0029	93	0.0022	61	0.0024	117	0.0024	132	0.0046	84	0.0037	81	0.0586	108

Table 76: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)													
	Algorithm	CONSTRAINED, COOPERATIVE												LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
837	neurotechnology-019	-		-		0.0022	58	0.0024	102	0.0022	88	0.0042	65	0.0034	52	0.0526	69
838	neurotechnology-020	-		-		0.0023	121	0.0028	232	0.0024	118	0.0043	72	0.0035	63	0.0562	93
839	neurotechnology-021	-		-		0.0022	45	0.0023	90	0.0019	56	0.0035	37	0.0029	27	0.0441	30
840	nextmetalabs-001	-		-		0.1126	1243	0.1962	1245	0.1137	1157	1.0000	1306	0.9998	1312	1.0000	927
841	nextmetalabs-002	-		-		0.0066	783	0.0081	734	0.0097	720	0.9946	1212	0.9565	1238	1.0000	871
842	nextmetalabs-003	-		-		0.0023	131	0.0028	225	0.0028	206	0.9681	1164	0.8769	1207	0.9995	823
843	nextmetalabs-004	-		-		0.0023	130	0.0028	224	0.0029	228	0.0055	130	0.0048	170	0.0716	185
844	nhn-001	0.0066	443	0.0098	446	0.0053	695	0.0079	726	0.0093	701	0.0156	438	0.0109	503	-	
845	nhn-002	0.0068	452	0.0096	439	0.0057	730	0.0087	753	0.0136	823	0.0253	563	0.0186	684	0.1464	401
846	nhn-003	0.0033	218	0.0048	225	0.0027	310	0.0038	398	0.0036	337	0.0198	507	0.0071	329	0.1280	367
847	nhn-004	0.0030	187	0.0044	205	0.0025	224	0.0029	243	0.0030	261	0.0331	626	0.0113	519	0.1598	421
848	nhn-005	0.0028	173	0.0043	188	0.0023	110	0.0027	175	0.0025	141	0.0047	87	0.0038	94	0.0554	88
849	nmetics-001	-		-		0.0026	287	0.0033	323	0.0027	202	0.0052	119	0.0043	136	0.0594	114
850	noblis-000	0.4424	1019	0.5415	1024	-		-		-		-		-		-	
851	nodeflux-000	1.0000	1400	1.0000	1125	1.0000	1440	-		0.9079	1312	-		-		-	
852	nodeflux-001	1.0000	1159	1.0000	1480	1.0000	1458	0.9998	1384	0.5169	1252	1.0000	1452	0.5654	1138	-	
853	nodeflux-002	0.0186	731	0.0340	748	0.0261	1120	0.0451	1113	0.0548	1099	1.0000	1317	1.0000	1335	-	
854	nominder-000	0.0025	157	0.0036	139	0.0029	369	0.0033	318	0.0026	184	0.9919	1201	0.8877	1212	1.0000	881
855	nominder-001	0.0029	182	0.0042	183	0.0031	425	0.0038	400	0.0031	268	0.9975	1229	0.9180	1221	0.9999	846
856	nominder-002	0.0025	156	0.0037	143	0.0033	454	0.0037	388	0.0026	190	0.5785	1022	0.1435	994	0.8381	670
857	nominder-003	-		-		0.0031	401	0.0034	338	0.0026	174	0.7010	1054	0.2159	1040	0.8909	702
858	nominder-004	-		-		0.0028	348	0.0033	314	0.0024	119	0.0316	612	0.0045	155	0.1348	378
859	nominder-005	-		-		0.0025	209	0.0029	238	0.0024	127	0.0046	85	0.0037	85	0.0537	80
860	nominder-006	-		-		0.0027	312	0.0032	297	0.0030	258	0.0059	147	0.0052	203	0.0644	155
861	nominder-007	-		-		0.0027	295	0.0031	277	0.0023	111	0.0072	203	0.0036	70	0.0946	295
862	nominder-008	-		-		0.0026	266	0.0028	230	0.0022	92	0.0069	192	0.0035	66	0.0928	291
863	notiontag-000	0.6669	1048	0.7885	1056	0.3715	1298	0.4978	1295	0.8571	1297	0.8102	1089	0.6460	1158	-	
864	notiontag-001	0.6846	1049	0.8006	1057	0.3955	1300	0.5247	1297	0.8669	1300	0.8313	1102	0.6362	1155	-	
865	notiontag-002	0.0066	440	0.0089	416	0.0045	622	0.0061	622	0.0077	632	0.0137	399	0.0104	488	0.1106	329
866	npt-000	-		-		0.0035	501	0.0050	527	0.0042	404	0.8079	1087	0.5102	1128	0.9695	739
867	nsensecorp-001	0.9909	1068	0.9994	1080	0.9987	1386	0.9999	1386	1.0000	1356	1.0000	1328	1.0000	1339	-	
868	nsensecorp-002	0.4277	1015	0.5375	1021	0.6734	1335	0.7924	1331	0.7194	1271	0.6937	1052	0.5617	1136	0.9392	721
869	nsensecorp-003	0.0251	783	0.0295	724	0.0212	1092	0.0305	1064	0.0131	807	0.2139	888	0.0141	594	0.9968	796
870	nsensecorp-004	0.1370	939	0.1397	905	0.0066	782	0.0094	769	1.0000	1377	1.0000	1338	1.0000	1377	1.0000	1283
871	nsensecorp-005	0.0055	372	0.0080	383	0.0039	565	0.0058	599	0.2723	1220	0.9999	1285	0.9949	1276	1.0000	939
872	ntechlab-000	-		0.0862	870	-		-		-		-		-		-	
873	ntechlab-001	-		0.0825	866	-		-		-		-		-		-	
874	ntechlab-002	0.0558	871	0.0649	839	-		-		-		-		-		-	
875	ntechlab-003	0.0286	803	0.0390	775	-		-		-		-		-		-	
876	ntechlab-004	0.0090	529	0.0132	524	-		-		-		-		-		-	
877	ntechlab-005	0.0093	542	0.0145	546	0.0131	1010	0.0275	1050	-		-		-		-	
878	ntechlab-006	0.0078	494	0.0111	486	0.0112	967	0.0227	1016	0.0108	753	0.0149	424	0.0098	465	-	
879	ntechlab-007	0.0056	375	0.0076	361	0.0073	818	0.0128	858	0.0079	643	0.0144	410	0.0102	476	-	
880	ntechlab-008	0.0041	285	0.0061	293	0.0056	715	0.0108	807	0.0042	401	0.0080	235	0.0063	281	-	

Table 77: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
881	ntechlab-009	0.0039	266	0.0054	267	0.0042	603	0.0063	633	0.0275	1003	0.0674	757	0.0532	886	-			
882	ntechlab-010	0.0013	50	0.0017	32	0.0024	176	0.0029	241	0.0031	270	0.0058	146	0.0050	187	0.0656	164		
883	ntechlab-011	0.0012	45	0.0019	42	0.0024	167	0.0028	228	0.0029	237	0.0055	128	0.0047	163	0.0630	142		
884	ntechlab-012	0.0011	35	0.0016	27	0.0023	128	0.0030	255	0.0026	178	0.0050	114	0.0043	137	0.0587	110		
885	ntnu-000	-		-		0.0040	579	0.0066	650	0.0069	594	0.9942	1211	0.9744	1252	0.9999	849		
886	ocrstudio-001	-		-		0.4396	1308	0.6256	1311	0.3190	1227	0.3128	937	0.1754	1020	0.7785	650		
887	omface-000	0.2573	983	0.3835	992	0.3590	1296	0.4903	1293	0.3956	1241	0.5003	996	0.2595	1057	0.9137	710		
888	omface-001	0.0137	659	0.0212	653	0.0114	977	0.0187	977	0.0174	896	1.0000	1383	0.0214	724	1.0000	1476		
889	omnigarde-000	0.0633	881	0.1002	885	0.1109	1241	0.2042	1249	0.1288	1163	0.5113	999	0.1227	979	0.9604	732		
890	omnigarde-001	0.0168	707	0.0260	704	0.0203	1083	0.0402	1096	0.0243	981	0.0327	622	0.0177	665	0.2164	475		
891	omnigarde-002	0.0033	220	0.0046	219	0.0027	324	0.0039	407	0.0041	388	0.0076	214	0.0059	264	0.0806	235		
892	omnigarde-003	0.0024	138	0.0038	153	0.0024	183	0.0029	233	0.0032	290	0.0059	150	0.0049	182	0.0706	183		
893	omnigarde-004	-		-		0.0022	55	0.0024	106	0.0023	104	0.0044	77	0.0038	96	0.0559	91		
894	omnigarde-005	-		-		0.0022	67	0.0024	115	0.0024	124	0.0045	81	0.0039	109	0.0582	105		
895	onfido-000	0.1472	944	0.2881	968	0.0335	1149	0.0731	1172	0.0515	1092	0.9915	1198	0.9579	1240	1.0000	852		
896	onfido-001	-		-		0.0146	1036	0.0294	1059	0.0190	919	0.0702	768	0.0352	822	0.3117	521		
897	onfido-002	-		-		0.0067	784	0.0098	785	0.0062	568	0.0167	462	0.0091	438	0.1228	359		
898	onfido-003	-		-		0.0045	627	0.0070	670	0.0043	409	0.0164	457	0.0076	366	0.1153	342		
899	onfido-004	-		-		0.0033	470	0.0047	498	0.0032	292	0.0305	602	0.0086	421	0.1450	400		
900	oorudigital-000	-		-		0.0106	944	0.0190	980	0.0146	841	0.9930	1206	0.9585	1241	1.0000	866		
901	ooto-001	-		-		0.0193	1074	0.0290	1057	0.0529	1097	0.0272	576	0.0255	762	0.1956	460		
902	openedge-000	0.0045	307	0.0062	307	0.0049	658	0.0069	667	0.0049	474	0.9889	1193	0.7991	1191	0.9999	842		
903	openface-001	0.1804	958	0.2921	970	0.2878	1283	0.3906	1279	0.2054	1199	0.2338	901	0.1549	1004	0.6531	617		
904	ovisionllc-001	-		-		0.0058	735	0.0081	728	0.8171	1289	1.0000	1327	1.0000	1331	1.0000	903		
905	ovisionllc-002	-		-		0.1963	1268	0.2522	1257	1.0000	1358	1.0000	1332	1.0000	1344	1.0000	909		
906	oz-001	0.0133	651	0.0215	659	0.0109	953	0.0160	934	0.0235	972	1.0000	1491	1.0000	1477	-			
907	oz-002	0.0071	465	0.0099	455	0.0099	925	0.0100	791	0.0139	826	0.0502	712	0.0202	709	-			
908	oz-003	0.0095	553	0.0143	540	0.0054	704	0.0077	709	0.0096	716	0.0175	476	0.0118	533	0.1404	389		
909	oz-004	0.0033	225	0.0049	233	0.0038	559	0.0055	570	0.0081	652	0.0163	456	0.0142	597	-			
910	pa-002	0.1847	960	0.2859	966	-		-		-		-		-		-			
911	pa-003	0.0453	852	0.0679	845	-		-		-		-		-		-			
912	palit-000	0.0062	419	0.0084	398	0.0039	563	0.0055	567	0.0055	518	0.4610	988	0.2468	1047	0.8695	686		
913	palit-001	0.0050	332	0.0068	331	0.0032	447	0.0047	502	0.0045	438	0.0110	353	0.0058	254	0.1446	397		
914	pangiam-000	0.0031	201	0.0043	196	0.0026	244	0.0030	270	0.0038	357	0.0071	202	0.0061	271	0.0667	171		
915	pangiam-001	0.0031	202	0.0044	201	0.0029	362	0.0040	429	0.0028	220	0.0362	656	0.0056	236	0.1711	435		
916	pangiam-002	0.0030	194	0.0042	181	0.0026	278	0.0031	283	0.0030	255	0.1772	860	0.0385	835	0.6076	605		
917	papago-001	0.0067	448	0.0096	442	0.0051	683	0.0077	707	0.0071	603	0.0126	387	0.0086	419	0.1033	314		
918	papil11-000	0.0021	115	0.0030	103	0.0024	134	0.0024	125	0.0025	159	0.0047	90	0.0040	117	0.0536	77		
919	papil11-001	-		-		0.0023	127	0.0024	123	0.0023	100	0.0185	489	0.0040	124	0.1449	398		
920	papil11-002	-		-		0.0023	123	0.0024	118	0.0023	99	0.0130	393	0.0039	106	0.1112	333		
921	papil11-003	-		-		0.0023	113	0.0024	119	0.0023	109	0.0042	67	0.0035	64	0.0529	71		
922	papil11-004	-		-		0.0027	306	0.0029	244	0.0025	161	0.0153	432	0.0058	253	0.0972	299		
923	papsav1923-001	0.0078	496	0.0130	522	0.0068	790	0.0105	801	0.0119	782	0.0221	533	0.0136	579	-			
924	papsav1923-002	0.0021	117	0.0034	120	0.0026	245	0.0030	268	0.0048	470	0.0093	294	0.0086	417	0.0630	141		

Table 78: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
925	papsav1923-003	0.0025	154	0.0035	125	0.0024	178	0.0025	135	0.0034	309	0.0066	181	0.0058	252	0.0603	119
926	paravision-004	0.0030	192	0.0046	215	0.0030	383	0.0036	370	0.0091	694	0.0188	493	0.0173	658	-	
927	paravision-006	0.0016	69	0.0025	69	0.0026	256	0.0026	172	0.0075	622	0.9933	1207	0.0142	595	-	
928	paravision-008	0.0018	88	0.0025	67	0.0024	155	0.0025	145	0.0036	336	0.0070	198	0.0063	280	0.0747	203
929	paravision-010	0.0012	46	0.0021	47	0.0022	53	0.0021	42	0.0027	200	0.0055	129	0.0050	189	0.0534	76
930	paravision-011	0.0008	14	0.0020	45	0.0021	35	0.0020	23	0.0026	179	0.0053	123	0.0049	181	0.0498	55
931	paravision-013	0.0010	30	0.0018	38	0.0021	26	0.0019	4	0.0019	59	0.0041	63	0.0036	74	0.0442	31
932	paravision-015	-		-		0.0021	33	0.0021	47	0.0021	71	0.0043	69	0.0039	100	0.0528	70
933	paravision-017	-		-		0.0021	22	0.0021	31	0.0019	47	0.0038	46	0.0035	59	0.0458	36
934	paravision-018	-		-		0.0020	7	0.0020	21	0.0016	19	0.0034	29	0.0029	30	0.0414	16
935	paravision-020	-		-		0.0020	2	0.0020	14	0.0015	15	0.0033	22	0.0028	19	0.0410	14
936	paravision-022	-		-		0.0020	4	0.0019	3	0.0015	10	0.0032	18	0.0028	17	0.0414	15
937	pensees-001	0.0087	523	0.0133	526	0.0071	801	0.0122	840	0.0145	838	0.0252	562	0.0195	701	-	
938	pentacomp-000	-		-		0.0041	592	0.0061	619	0.0047	459	0.3501	952	0.1211	977	0.7162	634
939	pixelall-002	0.0193	738	0.0340	750	0.0127	1004	0.0209	998	0.0234	971	0.9182	1130	0.2728	1063	-	
940	pixelall-003	0.0074	479	0.0118	502	0.0057	725	0.0079	721	0.0121	786	0.0390	674	0.0170	650	-	
941	pixelall-004	0.0040	271	0.0061	296	0.0055	710	0.0069	665	0.0100	734	0.5663	1019	0.0371	832	-	
942	pixelall-005	0.0038	259	0.0052	254	0.0043	609	0.0051	540	0.0077	630	0.0839	782	0.0136	581	-	
943	pixelall-006	0.0032	212	0.0042	179	0.0032	435	0.0039	409	0.0063	570	0.9960	1220	0.0723	921	1.0000	884
944	pixelall-007	0.0036	248	0.0049	237	0.0039	562	0.0044	469	0.0068	587	0.9873	1187	0.0217	727	1.0000	860
945	pixelall-008	0.0015	56	0.0023	61	0.0034	486	0.0049	512	0.0031	267	0.0057	137	0.0052	200	0.0682	175
946	pixelall-009	0.0018	86	0.0025	68	0.0024	171	0.0026	166	0.0031	277	0.3475	951	0.0053	214	0.9006	707
947	pixelall-010	0.0020	103	0.0030	105	0.0025	215	0.0028	231	0.0031	272	0.3512	953	0.0054	218	0.8813	695
948	pnpsecure-000	-		-		0.0190	1071	0.0274	1049	0.9980	1333	0.9997	1271	0.9996	1307	0.9997	827
949	pnpsecure-001	-		-		0.0050	672	0.0040	425	0.0037	353	0.0166	461	0.0075	357	0.1059	317
950	pnpsecure-002	-		-		0.0025	217	0.0028	220	0.0031	274	0.5367	1011	0.2784	1065	0.8498	676
951	privid-001	0.3350	997	0.5013	1014	0.4327	1306	0.5880	1304	0.9790	1325	1.0000	1308	0.9998	1313	1.0000	948
952	privid-002	0.0460	854	0.0652	841	0.0524	1192	0.0882	1188	0.0671	1120	0.9769	1171	0.8960	1213	0.9999	836
953	privid-003	-		-		0.0030	386	0.0039	406	0.0045	440	0.9996	1267	0.9985	1292	1.0000	894
954	privid-004	-		-		0.0026	277	0.0028	210	0.0034	310	0.9939	1209	0.9657	1247	1.0000	873
955	psl-000	0.0216	754	0.0333	745	-		-		-		-		-		-	
956	psl-001	0.0549	868	0.0927	878	0.0096	911	0.0149	913	-		-		-		-	
957	psl-002	0.0107	589	0.0180	612	0.0089	883	0.0120	833	0.0790	1131	0.0270	575	0.0192	691	-	
958	psl-003	0.0065	438	0.0099	450	0.0055	707	0.0075	699	1.0000	1354	0.0188	494	0.0129	563	-	
959	psl-004	0.0064	428	0.0094	437	0.0035	492	0.0050	535	0.0084	659	0.0145	415	0.0106	496	-	
960	psl-005	0.0060	405	0.0094	432	0.0034	487	0.0048	508	0.0081	651	0.0140	403	0.0103	485	-	
961	psl-006	0.0025	151	0.0039	164	0.0028	343	0.0030	263	0.0160	862	0.0482	705	0.0310	803	-	
962	psl-007	0.0026	163	0.0040	167	0.0027	299	0.0030	259	0.0054	506	0.0101	327	0.0081	396	-	
963	psl-008	0.0026	159	0.0040	169	0.0024	165	0.0028	223	0.0041	390	0.0077	217	0.0055	225	0.0719	186
964	psl-009	0.0161	699	0.0294	723	0.0023	99	0.0025	137	0.0036	339	0.0065	179	0.0048	176	0.0659	166
965	psl-010	0.0017	82	0.0035	128	0.0023	91	0.0025	131	0.0035	327	0.0104	333	0.0052	208	0.0994	302
966	psl-011	0.0013	49	0.0026	74	0.0021	10	0.0021	37	0.0024	122	0.0047	89	0.0035	67	0.0565	96
967	psl-012	0.0017	76	0.0027	85	0.0021	39	0.0023	74	0.0021	72	0.0039	50	0.0032	46	0.0494	51
968	psl-013	-		-		0.0022	46	0.0023	76	0.0021	79	0.0039	54	0.0033	48	0.0499	59

Table 79: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)																	
	Algorithm	CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
969	psl-014	-		-		0.0021	38	0.0023	73	0.0021	77	0.0039	53	0.0033	47	0.0498	56
970	ptakuratsatu-000	0.0060	407	0.0089	417	0.0070	798	0.0104	800	0.0096	719	0.0152	431	0.0100	471	-	
971	pxl-001	0.0488	857	0.0752	854	0.0586	1202	0.1087	1206	0.0946	1147	0.1065	808	0.0625	906	-	
972	pyramid-000	0.0136	654	0.0233	679	0.0117	982	0.0192	983	0.0185	911	0.0322	619	0.0206	715	-	
973	pyramid-001	-		-		0.0025	192	0.0030	265	0.0025	149	0.0099	318	0.0044	145	0.1031	313
974	qazbs-000	0.0058	385	0.0083	394	0.0046	638	0.0072	681	0.0130	804	0.0244	555	0.0196	702	0.7784	649
975	qazsmartvisionai-000	0.0008	16	0.0011	11	0.0021	14	0.0022	52	0.0018	43	0.0035	36	0.0030	34	0.0441	29
976	qazsmartvisionai-001	0.0003	2	0.0008	8	0.0020	8	0.0020	28	0.0015	14	0.0031	13	0.0027	13	0.0394	11
977	qazsmartvisionai-002	-		-		0.0020	3	0.0020	12	0.0014	3	0.0028	3	0.0025	4	0.0380	5
978	qazsmartvisionai-003	-		-		0.0020	5	0.0020	13	0.0015	5	0.0029	7	0.0025	6	0.0381	7
979	qazsmartvisionai-004	-		-		0.0020	6	0.0020	18	0.0014	2	0.0028	2	0.0025	2	0.0377	4
980	qluevision-001	0.0223	761	0.0419	781	0.0205	1087	0.0343	1076	0.0327	1031	0.8762	1114	0.7413	1178	0.9908	773
981	qnap-000	0.0149	680	0.0228	673	0.0155	1050	0.0267	1045	0.0238	979	0.8329	1103	0.0396	840	0.9992	813
982	qnap-001	0.0148	677	0.0215	658	0.0103	933	0.0162	941	0.0183	910	0.0301	597	0.0186	683	0.1995	464
983	qnap-002	0.0122	624	0.0191	627	0.0075	827	0.0095	773	0.0146	839	0.0281	587	0.0184	676	0.1829	446
984	qnap-003	0.0637	882	0.0657	844	0.0058	733	0.0078	715	0.0082	654	0.9985	1242	0.9658	1248	1.0000	897
985	qnap-004	0.0046	315	0.0061	295	0.0038	548	0.0059	606	0.0042	399	0.9960	1219	0.9442	1230	1.0000	867
986	qnap-005	0.0033	222	0.0047	223	0.0068	788	0.0071	676	0.0096	715	0.0184	488	0.0163	636	0.0948	296
987	qnap-006	-		-		0.0030	392	0.0041	440	0.0044	423	0.0088	280	0.0065	293	0.0799	231
988	quantasoft-003	0.0081	507	0.0113	493	0.0056	718	0.0076	702	0.0091	693	0.0161	450	0.0107	500	0.1193	349
989	rankone-000	-		0.2760	960	-		-		-		-		-		-	
990	rankone-001	-		0.4198	998	-		-		-		-		-		-	
991	rankone-002	0.1547	948	0.2170	943	-		-		-		-		-		-	
992	rankone-003	0.1175	930	0.1838	929	-		-		-		-		-		-	
993	rankone-004	0.0548	867	0.1015	886	-		-		-		-		-		-	
994	rankone-005	0.0405	841	0.0757	856	0.0170	1062	0.0299	1061	-		-		-		-	
995	rankone-006	0.0242	770	0.0460	797	0.0119	986	0.0188	979	0.0315	1027	0.0539	722	0.0385	837	-	
996	rankone-007	0.0197	743	0.0366	766	0.0113	970	0.0177	962	0.0286	1010	0.0454	692	0.0333	812	-	
997	rankone-008	0.0124	630	0.0232	676	0.0082	855	0.0107	803	0.0188	916	0.0320	618	0.0244	755	-	
998	rankone-009	0.0087	521	0.0119	503	0.0065	777	0.0086	749	0.0088	678	0.0161	447	0.0121	544	-	
999	rankone-010	0.0079	499	0.0112	490	0.0061	752	0.0081	732	0.0088	675	0.0149	421	0.0117	531	-	
1000	rankone-011	0.0049	326	0.0075	359	0.0038	543	0.0048	509	0.0060	555	0.0143	409	0.0080	391	0.1839	447
1001	rankone-012	0.0043	294	0.0058	280	0.0031	429	0.0038	396	0.0047	458	0.0081	241	0.0065	288	0.0842	250
1002	rankone-013	0.0028	171	0.0041	175	0.0026	248	0.0033	316	0.0028	226	0.0055	132	0.0040	115	0.0725	187
1003	rankone-014	0.0016	72	0.0021	50	0.0024	138	0.0027	183	0.0022	95	0.0047	91	0.0035	61	0.0666	170
1004	rankone-015	0.0009	25	0.0014	20	0.0024	146	0.0026	162	0.0024	134	0.0044	79	0.0037	76	0.0635	148
1005	realnetworks-000	0.0325	822	0.0463	798	-		-		-		-		-		-	
1006	realnetworks-001	0.0315	820	0.0455	795	0.0920	1231	0.1938	1244	-		-		-		-	
1007	realnetworks-002	0.0248	776	0.0358	762	0.0513	1190	0.1127	1212	0.0371	1047	0.0614	743	0.0316	806	-	
1008	realnetworks-003	0.0259	789	0.0372	769	0.0541	1196	0.1208	1216	0.0378	1050	0.0578	732	0.0306	802	-	
1009	realnetworks-004	0.0075	484	0.0101	460	0.0066	780	0.0097	782	0.0108	759	0.0187	491	0.0131	565	0.1430	394
1010	realnetworks-005	0.0070	456	0.0093	428	0.0063	764	0.0089	756	0.0092	698	0.0161	451	0.0104	489	0.1249	363
1011	realnetworks-006	0.0040	273	0.0056	274	0.8657	1365	0.9705	1364	0.0059	548	0.0112	359	0.0085	415	0.0911	279
1012	realnetworks-007	0.0031	199	0.0051	250	0.0028	340	0.0035	357	0.0048	471	0.0091	286	0.0074	347	0.0945	294

Table 80: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1013	realnetworks-008	0.0022	125	0.0039	158	0.0038	545	0.0045	481	0.0055	512	0.0100	325	0.0080	395	0.0889	262
1014	realsense-001	-		-		0.0023	90	0.0025	149	0.0023	115	0.0046	86	0.0039	99	0.0537	78
1015	realsense-002	-		-		0.0022	60	0.0023	91	0.0023	103	0.0045	80	0.0038	90	0.0531	73
1016	realsense-003	-		-		0.0022	54	0.0022	59	0.0022	98	0.0044	75	0.0037	79	0.0522	67
1017	rebs-000	0.0030	190	0.0043	195	0.0024	154	0.0032	296	0.0024	120	0.3069	931	0.1170	971	0.7667	644
1018	rebs-001	0.0037	254	0.0049	235	0.0024	179	0.0033	311	0.0025	142	0.0257	568	0.0096	460	0.1487	406
1019	recognito-000	0.0004	5	0.0006	1	0.0021	34	0.0022	66	0.0016	22	0.0555	724	0.0207	717	0.1943	457
1020	recognito-001	0.0003	1	0.0007	5	0.0021	24	0.0022	57	0.0016	16	0.0663	754	0.0257	763	0.1825	445
1021	regula-000	0.0184	730	0.0376	770	0.0103	934	0.0185	972	0.0120	784	0.9983	1238	0.0231	742	1.0000	908
1022	regula-001	0.0072	473	0.0107	478	0.0102	931	0.0179	964	0.0123	792	0.0333	631	0.0174	660	0.3711	540
1023	regula-002	-		-		0.0053	690	0.0058	596	0.0044	428	0.9999	1283	0.9995	1302	1.0000	907
1024	regula-003	-		-		0.0056	714	0.0059	607	0.0060	557	0.5284	1008	0.4298	1110	0.8213	666
1025	remarkai-000	0.0147	676	0.0257	702	0.0102	928	0.0158	929	0.0163	871	0.0577	730	0.0185	678	-	
1026	remarkai-001	0.0144	671	0.0256	700	0.0102	929	0.0159	932	0.0162	869	0.0582	735	0.0185	680	-	
1027	remarkai-002	0.0151	684	0.0197	640	0.0075	825	0.0108	809	0.0119	783	0.0187	492	0.0127	554	-	
1028	remarkai-003	0.0047	322	0.0063	310	0.0033	466	0.0049	516	0.0054	509	0.0100	324	0.0072	336	0.0908	276
1029	rendip-000	0.0055	373	0.0077	367	0.0048	655	0.0060	616	0.0080	647	0.0142	408	0.0110	509	0.1027	311
1030	revealmedia-005	0.0050	342	0.0074	354	0.0050	668	0.0068	663	0.0075	623	0.0124	383	0.0104	492	0.0895	267
1031	revealmedia-006	0.0040	277	0.0067	321	0.0041	589	0.0056	576	0.0056	521	0.0085	268	0.0068	305	0.0872	256
1032	roc-016	0.0009	22	0.0014	21	0.0023	95	0.0024	110	0.0024	125	0.0042	68	0.0035	60	0.0558	90
1033	roc-017	-		-		0.0023	86	0.0025	151	0.0021	78	0.0039	49	0.0032	43	0.0542	82
1034	roc-018	0.0007	13	0.0012	13	0.0022	56	0.0022	60	0.0018	44	0.0034	32	0.0029	25	0.0493	50
1035	roc-019	-		-		0.0021	42	0.0021	36	0.0016	23	0.0032	15	0.0027	15	0.0439	27
1036	roc-020	-		-		0.0021	18	0.0021	29	0.0016	21	0.0032	17	0.0027	14	0.0452	33
1037	roc-021	-		-		0.0021	11	0.0020	20	0.0015	11	0.0030	12	0.0026	9	0.0409	13
1038	rokid-000	0.0093	545	0.0145	545	0.0073	816	0.0102	796	0.0164	873	0.0280	583	0.0214	723	-	
1039	rokid-001	0.0105	582	0.0162	579	0.0094	905	0.0163	943	0.0181	903	0.0276	582	0.0165	644	-	
1040	s1-001	0.0314	819	0.0651	840	0.0252	1115	0.0357	1082	0.0444	1075	0.0653	750	0.0429	855	-	
1041	s1-002	0.0095	554	0.0144	543	0.0112	966	0.0196	988	0.0234	969	0.0371	663	0.0282	786	-	
1042	s1-003	0.0051	346	0.0073	350	0.0044	617	0.0063	635	0.0052	497	0.0096	310	0.0070	321	0.0870	255
1043	s1-004	0.0053	359	0.0080	384	0.0038	546	0.0059	604	0.0057	531	0.0103	330	0.0073	342	0.0899	271
1044	s1-005	0.0024	145	0.0036	138	0.0025	231	0.0029	242	0.0026	187	0.0048	101	0.0038	92	0.0638	150
1045	s1-006	0.0029	184	0.0044	200	0.0028	330	0.0033	309	0.0035	324	0.0073	204	0.0044	153	0.7160	633
1046	s1-007	0.0023	136	0.0056	273	0.0024	139	0.0028	200	0.0025	154	0.0048	97	0.0039	101	0.0608	123
1047	s1-008	-		-		0.0093	898	0.0137	879	0.0179	901	0.9118	1128	0.0286	791	0.9999	839
1048	s1-009	-		-		0.0023	129	0.0023	97	0.0022	87	0.0043	70	0.0037	77	0.0504	62
1049	saffe-001	0.4339	1016	0.5261	1019	0.7539	1355	0.8736	1353	0.7977	1284	0.9810	1175	0.7435	1179	-	
1050	saffe-002	0.0119	619	0.0206	645	0.0107	950	0.0177	960	0.0244	982	0.9998	1272	0.2785	1066	-	
1051	samsungsds-000	0.0046	319	0.0069	332	0.0132	1012	0.0081	729	0.0099	729	0.0179	480	0.0162	635	0.1101	326
1052	samsungsds-001	0.0015	63	0.0026	73	0.0023	100	0.0023	96	0.0024	126	0.1660	855	0.0536	887	0.4995	577
1053	samsungsds-002	0.0017	81	0.0027	79	0.0023	93	0.0022	53	0.0021	76	0.0043	71	0.0036	73	0.0586	109
1054	samtech-000	0.3418	999	0.4425	1002	-		-		-		-		-		-	
1055	samtech-001	0.0197	742	0.0365	764	0.0146	1040	0.0241	1026	0.0238	978	0.0394	675	0.0251	760	-	
1056	samtech-002	0.0084	515	0.0127	519	0.0093	895	0.0158	927	0.0113	773	0.0442	686	0.0132	566	0.4110	554

Table 81: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1057	scanalitix-001	-		-		0.9968	1383	0.9997	1382	1.0000	1348	1.0000	1314	1.0000	1330	1.0000	922
1058	scanovate-001	0.0175	716	0.0331	744	0.0163	1060	0.0248	1028	0.2476	1214	0.3801	967	0.3740	1094	-	
1059	scanovate-002	0.0175	717	0.0355	759	0.0146	1037	0.0286	1055	0.0269	994	0.0301	596	0.0178	668	-	
1060	scanovate-003	0.0054	365	0.0080	382	0.0054	698	0.0072	686	0.0312	1024	0.0599	739	0.0568	899	-	
1061	sdc-000	0.0303	811	0.0526	817	0.0572	1200	0.1094	1208	0.0867	1142	0.6230	1034	0.3682	1093	0.8797	693
1062	seamfix-001	-		-		0.1973	1269	0.2878	1265	0.2613	1217	0.3156	939	0.1976	1031	0.8176	665
1063	securifai-001	0.4538	1021	0.6142	1036	0.5844	1327	0.7428	1323	0.7051	1270	0.9961	1221	0.9558	1237	-	
1064	securifai-002	0.7557	1054	0.8574	1059	0.4550	1311	0.5953	1305	0.9860	1327	0.9796	1174	0.9158	1220	-	
1065	securifai-003	0.4086	1010	0.7577	1054	0.7233	1346	0.8070	1334	0.7787	1281	1.0000	1323	0.9988	1295	1.0000	925
1066	securifai-004	0.0136	656	0.0192	629	0.0064	770	0.0099	789	0.0115	775	0.0272	577	0.0127	556	0.4881	571
1067	securifai-005	0.0125	633	0.0190	626	0.0080	847	0.0126	848	0.0134	816	0.9861	1182	0.9205	1222	1.0000	858
1068	securifai-006	0.0140	666	0.0196	636	0.0067	786	0.0102	793	0.0113	771	0.9888	1191	0.9239	1224	1.0000	863
1069	securifai-007	0.0124	629	0.0175	604	0.0064	772	0.0105	802	0.0108	752	0.9999	1282	0.9966	1283	1.0000	923
1070	securifai-008	-		-		0.0060	750	0.0102	795	0.2096	1203	0.9649	1161	0.8757	1205	0.9990	811
1071	securitypitch-000	-		-		0.0286	1131	0.0382	1092	0.0393	1055	0.9979	1234	0.9881	1265	1.0000	896
1072	sensetime-001	0.0063	424	0.0092	424	0.0130	1009	-		0.0252	988	0.0886	788	0.0886	946	-	
1073	sensetime-002	0.0068	450	0.0098	447	0.0143	1031	1.0000	1397	0.0278	1007	0.0502	711	0.0502	878	-	
1074	sensetime-003	0.0021	114	0.0027	83	0.0027	291	0.0027	192	0.0051	492	0.0100	323	0.0089	427	-	
1075	sensetime-004	0.0026	161	0.0038	154	0.0022	63	0.0023	81	0.0042	396	0.0082	246	0.0078	376	-	
1076	sensetime-005	0.0019	94	0.0029	96	0.0022	51	0.0021	39	0.0023	107	0.0044	74	0.0039	97	0.0564	94
1077	sensetime-006	0.0014	53	0.0024	62	0.0021	37	0.0020	16	0.0021	69	0.0040	58	0.0036	72	0.0525	68
1078	sensetime-007	0.0012	44	0.0022	53	0.0021	40	0.0020	22	0.0018	40	0.0034	27	0.0029	32	0.0479	47
1079	sensetime-008	0.0008	15	0.0014	18	0.0021	19	0.0020	7	0.0018	46	0.0036	38	0.0033	49	0.0504	61
1080	sensetime-010	-		-		0.0022	44	0.0021	35	0.0018	37	0.0034	30	0.0029	31	0.0437	24
1081	sensetime-011	-		-		0.0022	43	0.0020	27	0.0017	28	0.0033	24	0.0029	23	0.0438	26
1082	sensetime-012	-		-		0.0022	52	0.0023	83	0.0015	9	0.0029	9	0.0026	8	0.0372	1
1083	serendipity-000	0.0046	310	0.0069	335	0.0042	604	0.0056	575	0.0073	613	0.9129	1129	0.7189	1175	0.9952	788
1084	sertis-000	0.0118	617	0.0208	648	0.0080	843	0.0127	851	0.0110	766	0.0176	478	0.0114	520	-	
1085	sertis-001	0.0113	603	0.0182	614	0.0204	1086	0.0403	1099	0.0216	950	0.0763	775	0.0282	785	-	
1086	sertis-002	0.0049	327	0.0061	292	0.0039	571	0.0061	627	0.0055	514	0.0099	321	0.0070	323	0.0921	284
1087	sertis-003	0.0049	330	0.0059	291	0.0037	533	0.0058	595	0.0055	520	0.0100	322	0.0072	337	0.0881	258
1088	seventhsense-000	0.0067	447	0.0099	456	0.0045	620	0.0065	647	0.0093	702	0.0169	468	0.0124	550	0.1325	374
1089	seventhsense-001	0.0034	233	0.0047	224	0.0025	230	0.0031	279	0.0029	236	0.0338	637	0.0109	504	0.2135	473
1090	seventhsense-002	0.0036	251	0.0050	242	0.0028	339	0.0036	368	0.0035	325	0.0811	779	0.0183	675	0.5444	586
1091	seventhsense-003	0.0035	240	0.0046	216	0.0026	271	0.0033	310	0.0062	564	0.0448	690	0.0175	662	0.2594	498
1092	seventhsense-005	0.0031	203	0.0045	206	0.0028	332	0.0030	248	0.0028	218	0.0466	699	0.0165	645	0.1949	458
1093	seventhsense-006	-		-		0.0029	365	0.0034	343	0.0035	333	0.2052	882	0.0802	931	0.4742	567
1094	shaman-000	0.9297	1065	0.9774	1073	0.9990	1388	0.9997	1383	0.9999	1345	1.0000	1303	0.9999	1323	-	
1095	shaman-001	0.3346	996	0.4616	1004	0.2368	1278	0.3723	1278	0.3574	1235	0.3527	954	0.2304	1046	-	
1096	shu-001	0.0103	574	0.0140	538	0.0293	1138	0.0688	1165	0.0172	892	0.0182	484	0.0120	540	-	
1097	shu-002	-		0.0079	379	0.0146	1039	0.0308	1065	1.0000	1351	0.0183	485	0.0115	523	-	
1098	shu-003	0.0028	172	0.0041	177	0.0050	665	0.0088	754	0.0081	653	0.0133	394	0.0094	450	-	
1099	shufti-001	-		-		0.0053	689	0.0083	737	0.0085	664	1.0000	1461	0.0098	466	1.0000	1016
1100	siat-002	0.0091	538	0.0126	518	0.0109	954	0.0190	981	0.0276	1006	0.0516	716	0.0464	870	-	

Table 82: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1101	siat-004	0.0067	444	0.0099	449	0.0152	1048	1.0000	1396	0.0275	1001	0.4823	993	0.4823	1118	-	
1102	siat-005	0.0021	112	0.0038	148	0.0059	738	0.0049	519	0.0742	1126	0.9623	1158	0.6801	1160	0.9833	756
1103	sightropic-000	-		-		0.0028	354	0.0038	403	0.0029	232	0.2207	890	0.0810	933	0.5137	579
1104	sightropic-001	-		-		0.0052	688	0.0049	520	0.0050	483	0.0094	301	0.0081	399	0.0672	172
1105	sightropic-002	-		-		0.0027	317	0.0037	387	0.0028	207	0.0051	116	0.0038	88	0.0599	117
1106	sjtu-001	0.0051	345	0.0080	380	0.0211	1090	0.0446	1112	0.0131	806	0.0175	474	0.0110	507	-	
1107	sjtu-002	0.0053	364	0.0078	374	0.0138	1020	0.0296	1060	0.0084	660	0.0136	397	0.0095	456	-	
1108	sjtu-003	0.0017	78	0.0033	116	0.0030	387	0.0037	385	0.0058	539	0.0104	334	0.0081	397	-	
1109	sjtu-004	0.0014	52	0.0025	65	0.0027	300	0.0028	229	0.0046	447	0.0086	272	0.0073	341	0.0906	274
1110	sjtu-005	0.0029	180	0.0036	142	0.0025	235	0.0027	186	0.0044	419	0.0120	380	0.0072	332	0.1325	373
1111	sjtu-006	-		-		0.0027	308	0.0028	207	0.0044	426	0.0081	244	0.0071	330	0.0715	184
1112	sjtu-007	-		-		0.0023	125	0.0023	71	0.0040	375	0.0251	560	0.0126	553	0.1694	434
1113	sjtu-008	-		-		0.0022	49	0.0023	75	0.0029	227	0.0078	228	0.0055	230	0.0748	204
1114	sjtu-009	-		-		0.0022	50	0.0021	50	0.0028	210	0.5621	1017	0.0397	842	0.6795	624
1115	sjtu-010	-		-		0.0021	23	0.0020	9	0.0027	201	0.1050	805	0.0528	885	0.3290	527
1116	sktelecom-000	0.0038	263	0.0054	265	0.0031	399	0.0051	542	0.0042	394	0.3418	948	0.0061	270	0.9499	724
1117	smargtech-001	-		-		0.4604	1313	0.6005	1307	0.5247	1254	0.5858	1024	0.3098	1076	0.9826	755
1118	smartbiometrik-001	0.5485	1037	0.6442	1040	0.7550	1356	0.8611	1349	0.8677	1301	0.8270	1100	0.7030	1165	0.9752	746
1119	smartengines-000	0.6240	1044	0.7562	1053	0.9552	1372	0.9784	1368	0.9515	1320	0.9288	1141	0.8200	1193	0.9930	778
1120	smartengines-001	0.6434	1046	0.7666	1055	0.9446	1371	0.9750	1367	0.9387	1318	0.9556	1153	0.8647	1200	0.9972	800
1121	smartvist-000	0.0912	911	0.1587	915	0.1163	1246	0.1841	1240	0.1397	1170	0.9372	1147	0.7107	1171	0.9990	812
1122	smartvist-001	0.0137	658	0.0244	689	0.0099	923	0.0158	928	0.0151	848	0.1349	835	0.0419	851	0.5589	592
1123	smilart-001	0.6476	1047	0.8779	1061	-		-		-		-		-		-	
1124	smilart-002	0.2440	979	0.3532	986	-		-		0.3785	1236	0.4145	979	0.2611	1059	-	
1125	smilart-003	0.6944	1051	0.8836	1062	0.0695	1215	0.1193	1214	0.0894	1144	0.1221	825	0.0737	922	-	
1126	sodec-000	0.0033	223	0.0044	204	0.0040	577	0.0053	558	0.0054	510	0.0096	308	0.0080	387	0.0915	280
1127	softtech-000	-		-		0.0024	173	0.0023	92	0.0022	90	0.0043	73	0.0038	89	0.0641	152
1128	sparktek-000	-		-		0.0025	188	0.0025	143	0.0031	276	0.0058	143	0.0053	212	0.0643	154
1129	sparktek-001	-		-		0.0021	29	0.0020	15	0.0017	31	0.0034	26	0.0029	26	0.0470	41
1130	sparsh-001	0.0018	93	0.0029	98	0.0026	257	0.0028	219	0.0027	195	0.8806	1116	0.6313	1151	0.9963	791
1131	sqisoft-001	0.1220	933	0.2088	939	0.1978	1270	0.3386	1274	0.2111	1204	0.2798	924	0.1474	998	0.7820	652
1132	sqisoft-002	0.0082	510	0.0124	509	0.0051	676	0.0086	748	0.0102	738	0.0183	486	0.0122	545	0.1498	408
1133	sqisoft-003	0.0041	282	0.0055	272	0.0026	242	0.0032	306	0.0039	370	1.0000	1422	1.0000	1462	1.0000	1050
1134	staqu-000	0.0139	663	0.0208	649	0.0104	936	0.0145	906	0.0156	854	0.8063	1086	0.1408	993	-	
1135	starhybrid-001	0.0108	590	0.0138	534	0.0081	849	0.0113	819	0.0152	849	0.0265	573	0.0189	688	-	
1136	stcon-000	0.0040	274	0.0056	275	0.0031	415	0.0047	496	0.0048	472	0.9863	1183	0.3562	1091	0.9999	841
1137	stcon-001	0.0040	269	0.0053	264	0.0030	393	0.0045	482	0.0047	456	0.8632	1112	0.1788	1022	0.9892	769
1138	stcon-002	0.0031	196	0.0043	190	0.0025	198	0.0034	337	0.0030	254	0.9988	1246	0.9855	1261	1.0000	879
1139	stcon-003	0.0038	261	0.0050	245	0.0024	141	0.0032	303	0.0023	106	0.1082	813	0.0350	821	0.3982	551
1140	stcon-004	-		-		0.0022	65	0.0025	150	0.0020	64	0.0519	717	0.0145	603	0.2564	495
1141	stcon-005	-		-		0.0021	30	0.0021	32	0.0017	26	0.0336	634	0.0115	525	0.1725	436
1142	stcon-006	-		-		0.0021	31	0.0021	33	0.0017	25	0.0034	28	0.0028	18	0.0425	20
1143	stcon-007	-		-		0.0021	28	0.0021	30	0.0017	24	0.0033	23	0.0028	16	0.0424	19
1144	stcon-008	-		-		0.0021	21	0.0020	5	0.0015	7	0.0029	11	0.0026	7	0.0390	10

Table 83: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1145	stcon-009	-		-		0.0021	13	0.0020	10	0.0015	4	0.0029	5	0.0025	5	0.0380	6
1146	stcon-010	-		-		0.0021	12	0.0020	11	0.0015	6	0.0028	4	0.0025	3	0.0376	3
1147	stellartecs-001	-		-		0.0046	637	0.0060	614	0.0046	446	0.0658	752	0.0181	673	0.3818	544
1148	stellartecs-003	-		-		0.0028	338	0.0036	367	0.0031	279	0.0146	417	0.0065	289	0.1313	369
1149	stengg-000	0.0059	403	0.0082	389	0.0037	537	0.0060	613	0.0050	481	0.5065	998	0.0792	927	0.9348	719
1150	sukshi-000	0.5409	1035	0.6612	1045	0.4556	1312	0.6567	1314	0.9296	1317	0.8898	1120	0.7384	1176	0.9899	770
1151	sumsub-000	-		-		0.0041	588	0.0043	466	0.0038	359	0.5301	1009	0.2536	1053	0.9068	708
1152	sumsub-001	-		-		0.0023	92	0.0025	156	0.0028	223	0.0052	118	0.0044	142	0.0633	145
1153	suprema-000	0.0064	432	0.0092	423	0.0081	852	0.0096	778	0.0139	827	0.0254	566	0.0220	729	-	
1154	suprema-001	0.0041	287	0.0053	259	0.0038	555	0.0047	497	0.0060	552	0.0111	355	0.0095	459	0.0907	275
1155	suprema-002	0.0030	189	0.0041	178	0.0034	474	0.0040	421	0.0045	431	0.0085	265	0.0072	338	0.0781	223
1156	suprema-003	0.0028	178	0.0041	176	0.0034	472	0.0039	408	0.0030	257	0.3095	935	0.0580	901	0.6672	620
1157	suprema-004	0.0024	137	0.0035	130	0.0032	438	0.0036	372	0.0028	205	0.0053	124	0.0045	156	0.0625	138
1158	suprema-005	0.0022	120	0.0036	134	0.0034	489	0.0031	285	0.0026	188	0.0050	113	0.0044	144	0.0588	112
1159	supremaid-001	0.0053	363	0.0073	352	0.0045	621	0.0066	651	0.0099	731	0.0186	490	0.0148	611	0.1040	315
1160	supremaid-002	0.0063	427	0.0094	433	0.0044	612	0.0062	630	0.0072	611	0.0229	544	0.0095	458	0.2986	517
1161	surrey-cvssp-000	0.9084	1062	0.9909	1076	0.9923	1379	0.9950	1375	0.9981	1336	0.9994	1256	0.9979	1288	0.9998	831
1162	surrey-cvssp-001	1.0000	1394	1.0000	1121	0.0077	834	0.0079	723	0.0266	992	0.3822	970	0.0551	891	0.7285	636
1163	surrey-cvssp-002	0.0019	95	0.0028	87	0.0027	321	0.0028	208	0.0024	129	0.5978	1031	0.1206	976	0.8488	675
1164	surrey-cvssp-003	0.0016	71	0.0028	90	0.0022	64	0.0024	112	0.0022	97	0.3048	930	0.0539	888	0.5830	599
1165	surrey-cvssp-004	-		-		0.0029	372	0.0025	157	0.0025	160	0.0690	762	0.0076	369	0.2741	507
1166	swsam-001	0.0268	793	0.0476	803	0.0271	1124	0.0460	1117	0.0584	1106	0.7745	1075	0.5013	1124	0.9642	734
1167	synesis-000	0.1752	955	0.3241	979	-		-		-		-		-		-	
1168	synesis-001	0.1777	956	1.0000	1420	1.0000	1479	-		-		-		-		-	
1169	synesis-003	0.8375	1057	0.8857	1063	0.8548	1364	0.9178	1360	-		-		-		-	
1170	synesis-004	0.0310	816	0.0480	805	0.0476	1182	0.0443	1110	0.0764	1129	-		-		-	
1171	synesis-005	0.0147	675	0.0226	671	0.0153	1049	0.0226	1014	0.0213	947	0.0375	667	0.0262	773	-	
1172	synesis-006	0.0070	458	0.0096	441	0.0107	947	0.0166	951	-		0.0128	391	0.0089	426	-	
1173	synesis-007	0.0050	337	0.0073	353	0.0062	763	0.0076	701	-		0.0105	336	0.0080	392	-	
1174	synology-000	0.0149	681	0.0238	685	0.0148	1041	0.0261	1040	0.0221	954	0.0331	627	0.0209	720	-	
1175	synology-001	0.0214	753	0.0307	731	0.0266	1123	0.0513	1136	0.0283	1008	0.8884	1118	0.2589	1056	-	
1176	synology-002	0.0104	578	0.0153	563	0.0107	949	0.0184	969	0.0189	918	0.2032	880	0.0180	670	-	
1177	syscake-000	-		-		0.0091	886	0.0155	922	0.0142	831	0.0287	590	0.0145	606	0.1682	432
1178	szpu-000	-		-		0.0034	476	0.0037	386	0.0039	366	0.8444	1106	0.3886	1101	0.8849	697
1179	sztu-000	0.0092	541	0.0139	536	0.0091	888	0.0201	994	0.0136	821	0.0685	761	0.0118	535	-	
1180	sztu-001	0.0031	195	0.0043	197	0.0025	213	0.0028	221	0.0051	486	0.0113	365	0.0089	428	0.9963	792
1181	t4isb-000	0.0058	386	0.0087	411	0.0041	597	0.0064	644	0.0083	657	0.0157	440	0.0103	486	0.1196	351
1182	tahakom-001	-		-		0.0223	1098	0.0364	1086	0.0361	1042	1.0000	1304	0.9998	1314	1.0000	916
1183	tahakom-002	-		-		0.0380	1159	0.0456	1114	0.0403	1060	0.4790	992	0.2494	1049	0.8529	677
1184	tahakom-003	-		-		0.0024	147	0.0028	214	0.0045	436	0.0088	277	0.0078	380	0.0604	120
1185	taiwanmobile-001	-		-		0.2136	1273	0.3014	1271	0.3408	1231	0.9439	1150	0.7559	1181	0.9966	795
1186	taiwanmobile-002	-		-		0.0088	881	0.0114	820	0.0108	757	0.9776	1172	0.8683	1202	0.9998	830
1187	tcc-000	-		-		0.1374	1254	0.2052	1250	0.6334	1262	0.9985	1241	0.9897	1268	0.9999	835
1188	tcc-001	-		-		0.0120	988	0.0217	1006	0.2620	1218	0.9958	1217	0.9791	1255	1.0000	862

Table 84: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.			
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1189	tcv-001	-		-		0.0301	1142	0.0556	1141	0.0570	1104	0.1222	826	0.0786	924	0.3172	523		
1190	tech5-000	0.0114	605	0.0160	576	-		-		-		-		-		-			
1191	tech5-001	0.0130	645	0.0176	607	0.0218	1096	0.0419	1104	0.0156	855	-		-		-			
1192	tech5-002	0.0046	313	0.0063	311	0.0113	972	0.0216	1004	0.0233	967	0.9609	1156	0.7704	1183	-			
1193	tech5-003	0.0053	361	0.0070	336	0.0099	922	0.0185	971	0.0095	711	0.6670	1047	0.1024	958	-			
1194	tech5-004	0.0123	628	0.0234	681	0.0086	876	0.0162	940	0.0065	578	0.0112	360	0.0082	403	-			
1195	tech5-005	0.0054	368	0.0072	342	0.0069	793	0.0122	839	0.0060	551	0.0094	303	0.0066	295	-			
1196	tech5-007	0.0020	100	0.0029	95	0.0024	153	0.0028	201	0.0034	318	0.8622	1111	0.5335	1132	0.9867	767		
1197	tech5-008	0.0020	102	0.0031	106	0.0026	255	0.0028	215	0.0026	180	0.9238	1137	0.7170	1174	0.9984	804		
1198	tech5-009	-		-		0.0041	590	0.0040	420	0.0045	435	0.0087	276	0.0080	389	0.0821	242		
1199	techainer-001	0.0061	413	0.0092	426	0.0046	634	0.0073	689	0.0060	553	0.9575	1154	0.8567	1199	0.9995	821		
1200	techainer-002	-		-		0.0058	731	0.0121	836	0.0136	822	0.0853	785	0.0313	804	0.1610	423		
1201	techainer-003	-		-		0.0029	364	0.0033	325	0.0046	455	0.0084	257	0.0074	348	0.0623	136		
1202	techbiometricid-001	-		-		0.0877	1230	0.1663	1232	0.1000	1154	0.5946	1029	0.3065	1074	0.9792	752		
1203	techbiometricid-002	-		-		0.0060	745	0.0091	760	0.8984	1309	1.0000	1325	1.0000	1340	1.0000	945		
1204	techsign-000	0.0325	823	0.0511	813	0.0435	1174	0.0710	1168	0.0746	1127	0.1104	814	0.0841	938	0.4288	558		
1205	techsign-001	0.0110	594	0.0196	637	0.0067	785	0.0120	832	0.0087	673	0.2475	908	0.0883	945	0.6767	623		
1206	telexper-000	-		-		0.0056	716	0.0077	706	0.0104	742	0.9976	1232	0.8691	1203	0.9999	843		
1207	tevian-000	0.0871	908	0.1289	900	-		-		-		-		-		-			
1208	tevian-001	0.0813	907	0.1266	897	-		-		-		-		-		-			
1209	tevian-002	0.0279	800	0.0392	776	-		-		-		-		-		-			
1210	tevian-003	0.0217	756	0.0298	726	0.0230	1106	0.0484	1128	-		-		-		-			
1211	tevian-004	0.0228	764	0.0304	729	0.0226	1104	0.0478	1125	0.0128	799	0.0228	541	0.0138	588	-			
1212	tevian-005	0.0043	296	0.0062	303	0.0057	726	0.0085	746	0.0070	599	0.0135	396	0.0119	536	-			
1213	tevian-006	0.0045	308	0.0061	294	0.0045	628	0.0066	649	0.0046	453	0.0091	285	0.0075	360	-			
1214	tevian-007	0.0019	96	0.0027	80	0.0032	443	0.0041	433	0.0045	437	0.0086	270	0.0078	379	0.0925	288		
1215	tevian-008	0.0012	41	0.0017	34	0.0033	451	0.0042	457	0.0042	397	0.0081	240	0.0068	307	0.0909	277		
1216	tiger-001	0.3120	992	0.3974	994	-		-		-		-		-		-			
1217	tiger-002	0.0658	884	0.0889	874	0.1083	1239	0.1766	1236	0.0952	1149	0.1568	850	0.0661	911	-			
1218	tiger-003	0.0313	818	0.0602	835	0.0188	1070	0.0359	1084	0.0344	1033	-		-		-			
1219	tiger-004	0.0779	899	0.1393	904	0.0488	1184	0.0905	1193	0.0800	1133	0.1640	853	0.0617	904	-			
1220	tiger-005	0.0624	878	0.2450	952	0.0292	1136	0.0556	1142	0.0430	1070	1.0000	1288	0.9964	1280	1.0000	880		
1221	tiger-006	0.0066	441	0.0101	462	0.0050	674	0.0075	700	0.0089	684	0.0158	443	0.0117	530	0.1466	402		
1222	tiger-007	-		-		0.0041	596	0.0063	631	0.0088	679	0.7340	1060	0.3935	1102	0.9464	723		
1223	tinkoff-001	0.0145	672	0.0244	691	0.0318	1145	0.0636	1156	0.0236	974	1.0000	1458	0.0339	815	1.0000	1032		
1224	tngd-001	-		-		0.0827	1225	0.1404	1224	0.0981	1153	0.1409	842	0.0808	932	0.7653	643		
1225	tngd-002	-		-		0.1471	1258	0.2504	1256	0.1879	1191	0.7463	1064	0.6359	1154	0.9555	726		
1226	tnitech-000	0.0025	152	0.0037	145	0.0033	456	0.0035	358	0.0025	145	0.8893	1119	0.6283	1150	0.9883	768		
1227	tongyi-004	0.0428	846	0.0540	820	-		-		-		-		-		-			
1228	tongyi-005	0.0073	476	0.0146	549	0.0187	1068	0.0421	1105	0.0161	867	0.0215	524	0.0149	614	-			
1229	tongyitrans-001	0.0617	877	0.0715	850	-		-		-		-		-		-			
1230	tongyitrans-002	0.0437	849	0.0657	843	-		-		-		-		-		-			
1231	tongyitrans-003	0.0249	780	0.0474	801	-		-		-		-		-		-			
1232	toppanidgate-000	0.0021	108	0.0033	117	0.0026	250	0.0028	211	0.0039	371	0.0075	210	0.0068	303	0.0631	143		

Table 85: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1233	toppansecurity-001	-		-		0.0025	195	0.0027	197	0.0032	282	0.0059	152	0.0051	193	0.0499	58
1234	toshiba-000	0.1906	962	0.2896	969	-		-		-		-		-		-	
1235	toshiba-001	0.1063	923	0.1926	933	-		-		-		-		-		-	
1236	toshiba-002	0.0134	652	0.0222	666	0.0097	916	0.0154	921	-		0.0327	623	0.0158	628	-	
1237	toshiba-003	0.0125	635	0.0214	656	0.0085	873	0.0131	863	-		0.0241	551	0.0151	619	-	
1238	toshiba-004	0.0030	186	0.0042	184	0.0025	214	0.0027	196	0.0034	314	0.0063	170	0.0053	217	0.0700	181
1239	toshiba-005	0.0023	133	0.0037	146	0.0024	161	0.0026	168	0.0072	609	0.0141	405	0.0130	564	0.0918	283
1240	toshiba-006	0.0022	122	0.0035	124	0.0024	172	0.0025	148	0.0027	196	0.7425	1062	0.3070	1075	0.9691	738
1241	toshiba-007	0.0016	68	0.0023	60	0.0024	181	0.0023	77	0.0021	73	0.0629	747	0.0056	233	0.2301	478
1242	toshiba-008	0.0012	42	0.0017	36	0.0024	164	0.0022	62	0.0019	55	0.0108	347	0.0039	110	0.0899	269
1243	toshiba-009	-		-		0.0024	152	0.0021	51	0.0019	53	0.0256	567	0.0055	231	0.1500	409
1244	toshiba-010	-		-		0.0024	137	0.0021	44	0.0017	33	0.0316	609	0.0053	213	0.1331	375
1245	toshiba-011	-		-		0.0024	163	0.0022	56	0.0019	49	0.0037	44	0.0033	51	0.0453	34
1246	toshiba-012	-		-		0.0022	71	0.0022	55	0.0019	48	0.0040	56	0.0034	53	0.0503	60
1247	toshiba-013	-		-		0.0021	36	0.0022	54	0.0017	34	0.0288	591	0.0046	158	0.0917	281
1248	toss-000	-		-		0.0023	78	0.0022	58	0.0021	75	0.0041	61	0.0035	58	0.0474	42
1249	toss-001	-		-		0.0023	108	0.0024	103	0.0021	83	0.0040	60	0.0035	62	0.0494	52
1250	touchlessid-000	0.3296	995	0.4804	1009	0.4111	1304	0.6026	1308	0.5324	1256	0.9996	1264	0.9964	1281	1.0000	890
1251	touchlessid-001	0.0076	487	0.0104	470	0.0680	1213	0.0842	1184	0.1358	1167	1.0000	1290	0.9995	1303	1.0000	949
1252	touchlessid-002	0.0070	459	0.0096	443	0.0407	1168	0.0469	1122	0.4234	1246	0.9991	1250	0.9953	1278	1.0000	883
1253	touchlessid-003	0.0061	414	0.0081	387	0.0040	582	0.0053	559	0.0056	525	0.0155	436	0.0082	402	0.1370	383
1254	touchlessid-004	-		-		0.0035	493	0.0049	511	0.0043	413	0.0077	219	0.0058	249	0.0793	228
1255	transmitsecurity-001	-		-		0.0051	682	0.0078	717	0.0089	685	0.0151	426	0.0111	512	0.1181	347
1256	trueface-000	0.0249	778	0.4321	999	0.0119	987	0.0180	965	0.0297	1019	0.7467	1065	0.1602	1012	-	
1257	trueface-001	0.0204	749	0.0438	790	0.0095	910	0.0138	887	0.0154	851	0.0253	564	0.0169	649	-	
1258	trueface-002	0.0060	408	0.0096	440	0.0048	654	0.0061	623	0.0112	770	0.0198	508	0.0155	624	-	
1259	trueface-003	0.0070	461	0.0094	434	0.0053	692	0.0081	733	0.0122	788	0.0217	528	0.0159	630	0.1204	354
1260	trueidvng-001	0.0063	422	0.0077	368	0.0033	465	0.0044	473	0.0046	450	0.0086	273	0.0069	313	0.0809	237
1261	truststamp-001	0.0242	772	0.0484	807	0.0194	1076	0.0359	1083	0.0381	1051	0.8808	1117	0.6898	1162	0.9948	783
1262	truststamp-002	-		-		0.0080	842	0.0138	888	0.0128	798	0.9718	1165	0.8779	1208	0.9998	829
1263	truststamp-003	-		-		0.0047	647	0.0070	674	0.0073	614	0.7132	1056	0.4538	1115	0.9578	728
1264	tsight-000	-		-		0.0020	1	0.0020	17	0.0014	1	0.0028	1	0.0025	1	0.0373	2
1265	ttigroup-000	-		-		0.0028	341	0.0034	332	0.0043	418	0.9208	1132	0.7608	1182	0.9975	802
1266	tuputech-000	0.3218	994	0.3696	989	-		-		0.3237	1228	0.4304	983	0.2973	1073	-	
1267	turingtechvip-001	0.0330	827	0.0540	821	0.0458	1181	0.1007	1200	0.4715	1249	0.9286	1140	0.8448	1197	0.9964	794
1268	turingtechvip-002	0.0126	641	0.0163	580	0.0092	894	0.0118	826	0.2264	1210	1.0000	1388	0.9925	1271	0.9999	847
1269	turkcell-000	0.1134	928	0.1288	899	0.0770	1222	0.1112	1211	0.2570	1215	1.0000	1287	0.9999	1322	1.0000	917
1270	turkcell-001	0.0043	295	0.0066	317	0.0037	542	0.0055	573	0.0043	406	0.0440	685	0.0082	404	0.3148	522
1271	turkcell-002	-		-		0.0036	510	0.0053	557	0.0044	420	0.0371	662	0.0075	355	0.2871	514
1272	twface-000	0.0051	343	0.0072	348	0.0041	587	0.0058	590	0.0071	604	0.0153	433	0.0100	468	0.1595	418
1273	twface-001	0.0036	247	0.0051	246	0.0031	417	0.0038	395	0.0049	473	0.0091	290	0.0075	359	0.0831	246
1274	ulsee-001	0.0151	685	0.0246	693	0.0113	969	0.0185	974	0.0187	915	0.6766	1048	0.0181	672	-	
1275	ultinous-000	0.2343	975	0.3484	984	-		-		-		-		-		-	
1276	ultinous-001	0.2485	980	0.4003	995	-		-		-		-		-		-	

Table 86: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1277	uluface-002	0.0081	506	0.0123	507	0.0071	800	0.0095	775	0.0107	750	1.0000	1379	0.0140	592	-			
1278	uluface-003	0.0100	566	0.0150	556	0.0079	838	0.0128	856	-		-		-		-			
1279	unicc-001	0.5101	1034	0.6167	1038	0.6577	1333	0.7894	1330	0.9208	1315	0.9866	1184	0.9386	1228	0.9999	850		
1280	unicc-002	0.2153	972	0.2190	944	0.0417	1170	0.0526	1139	0.0891	1143	0.1321	834	0.1125	966	0.3813	543		
1281	unicc-003	0.0217	757	0.0260	705	0.0093	900	0.0136	878	0.0129	801	0.0216	526	0.0143	600	0.2091	468		
1282	unicc-004	-		-		0.0055	709	0.0077	708	0.0076	627	0.0147	419	0.0102	479	0.1421	393		
1283	unissey-001	0.0095	550	0.0160	577	0.0134	1015	0.0150	915	0.0147	842	0.0253	565	0.0163	637	0.1651	429		
1284	unissey-002	0.0094	547	0.0151	557	0.0079	840	0.0110	812	0.0114	774	0.4424	986	0.1914	1028	0.8051	661		
1285	unissey-003	0.0057	380	0.0082	388	0.0047	645	0.0082	735	0.0067	584	0.5179	1002	0.2863	1069	0.8684	684		
1286	unissey-004	0.0043	293	0.0066	318	0.0037	524	0.0052	546	0.0048	469	0.5272	1007	0.2903	1070	0.8777	691		
1287	unissey-005	-		-		0.0037	525	0.0052	545	0.0086	668	0.0161	449	0.0134	571	0.1006	305		
1288	unissey-006	-		-		0.0073	810	0.0121	837	0.0288	1011	0.0529	721	0.0464	869	0.2808	510		
1289	unissey-007	-		-		0.3124	1288	0.4571	1289	0.6912	1269	0.9823	1179	0.8827	1210	1.0000	864		
1290	uniubi-000	-		-		0.0037	536	0.0064	643	0.0047	460	0.9321	1144	0.0147	610	0.9969	797		
1291	universeai-000	-		-		0.0023	118	0.0025	142	0.0043	414	0.0082	248	0.0075	353	0.0616	131		
1292	upc-001	0.0234	767	0.0519	816	0.0291	1134	0.0490	1132	0.0294	1016	0.2316	899	0.0389	839	-			
1293	urielsoft-000	-		-		0.0026	262	0.0032	300	0.0043	410	0.0083	251	0.0068	309	0.0744	198		
1294	useb-000	0.0158	693	0.0222	669	0.0123	996	0.0168	955	0.0182	905	0.0289	593	0.0167	647	0.1614	424		
1295	useb-001	0.0035	239	0.0053	261	0.0027	309	0.0033	322	0.0027	198	0.0085	267	0.0044	143	0.0806	233		
1296	useb-002	0.0028	177	0.0039	159	0.0027	307	0.0031	284	0.0025	144	0.0090	282	0.0042	128	0.0867	253		
1297	useb-003	-		-		0.0026	240	0.0030	249	0.0025	138	0.0091	289	0.0042	131	0.0901	272		
1298	useb-004	-		-		0.0025	228	0.0028	227	0.0022	85	0.5197	1005	0.2232	1043	0.8470	672		
1299	uxlabs-001	0.0534	861	0.0570	828	0.0118	983	0.0131	862	0.0237	977	0.0399	677	0.0288	793	0.2097	470		
1300	uxlabs-003	0.2133	971	0.2143	940	0.0386	1161	0.0457	1115	0.0374	1049	0.3343	944	0.1554	1006	0.6941	628		
1301	uzinfocom-000	-		-		0.0070	797	0.0120	834	0.0086	667	0.9874	1188	0.9469	1231	0.9999	834		
1302	uzinfocom-001	-		-		0.0029	371	0.0038	397	0.0041	389	0.9963	1224	0.9725	1251	1.0000	872		
1303	vaidio-001	-		-		0.0135	1016	0.0166	950	0.0079	638	0.6587	1044	0.4279	1109	0.9123	709		
1304	vcog-002	0.7522	1053	0.9033	1067	-		-		-		-		-		-			
1305	vcortex-001	0.0131	648	0.0207	647	0.0192	1073	0.0315	1068	0.2238	1209	0.3586	957	0.3536	1090	0.3569	532		
1306	vd-000	0.3790	1005	0.4973	1013	-		-		-		-		-		-			
1307	vd-001	0.0243	773	0.0452	794	0.0271	1126	0.0402	1098	0.0424	1068	-		-		-			
1308	vd-002	0.0429	847	0.0704	848	0.0569	1199	0.0844	1185	0.0801	1134	0.0937	791	0.0577	900	-			
1309	vd-003	0.0199	746	0.0222	667	0.0115	978	0.0130	861	0.0138	825	0.0239	549	0.0177	666	0.1647	428		
1310	veertec-001	-		-		0.0564	1198	0.0984	1199	0.0845	1139	0.1076	812	0.0691	916	0.4622	564		
1311	veridas-001	0.1998	969	0.2724	959	0.2987	1286	0.4587	1290	0.2599	1216	0.2454	906	0.1357	988	-			
1312	veridas-002	0.1733	954	0.2257	947	0.2617	1282	0.4147	1283	0.2073	1201	0.2028	879	0.1112	965	-			
1313	veridas-003	0.0557	870	0.0983	883	0.0734	1217	0.1267	1219	0.0694	1124	0.0951	797	0.0480	873	-			
1314	veridas-004	0.0281	801	0.0467	800	0.0353	1156	0.0643	1159	0.0424	1067	0.0644	748	0.0342	816	-			
1315	veridas-005	0.0599	874	0.1110	891	0.0294	1140	0.0639	1157	0.0811	1137	0.1130	816	0.0548	890	-			
1316	veridas-006	0.0098	557	0.0167	590	0.0079	841	0.0127	850	0.0127	795	0.0217	527	0.0151	618	0.1573	414		
1317	veridas-007	0.0063	425	0.0083	395	0.0044	616	0.0058	594	0.0080	648	0.0152	430	0.0120	542	0.1199	353		
1318	veridas-008	0.0032	211	0.0045	212	0.0030	381	0.0033	317	0.0085	666	0.0206	514	0.0143	601	0.1228	358		
1319	veridas-009	0.0031	197	0.0043	187	0.0030	394	0.0035	352	0.0055	511	0.0106	341	0.0100	469	0.0959	298		
1320	veridas-010	-		-		0.0027	320	0.0033	321	0.0042	398	0.0080	239	0.0073	339	0.0649	159		

Table 87: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1321	veridium-000	0.0726	892	0.1248	896	0.5226	1317	0.6652	1316	0.6425	1265	0.8150	1092	0.7989	1190	0.8772	690		
1322	veridium-001	0.0274	797	0.0368	767	0.0292	1137	0.0475	1123	0.0488	1084	0.0673	756	0.0463	867	0.3691	539		
1323	veridium-002	0.0343	829	0.0427	786	0.0283	1130	0.0460	1118	0.0489	1085	0.0673	755	0.0464	868	0.3658	535		
1324	veridium-003	-		-		0.0286	1132	0.0462	1119	0.0492	1086	0.0679	759	0.0470	871	0.3672	536		
1325	veridium-004	-		-		0.0141	1024	0.0254	1031	0.0217	951	0.0315	607	0.0197	703	0.2126	471		
1326	verigram-000	0.0032	205	0.0043	191	0.0031	402	0.0034	335	0.0093	704	0.0175	475	0.0164	641	0.1158	343		
1327	verigram-001	0.0032	210	0.0044	203	0.0027	303	0.0032	301	0.0030	246	0.9995	1260	0.9953	1277	1.0000	944		
1328	verigram-002	0.0079	498	0.0089	415	0.0038	549	0.0040	423	0.0060	554	0.0113	366	0.0107	498	0.0634	146		
1329	verigram-003	0.0101	569	0.0109	481	0.0034	477	0.0039	411	0.0037	348	0.0070	199	0.0063	279	0.0559	92		
1330	verigram-004	-		-		0.0023	115	0.0023	82	0.0021	70	1.0000	1312	0.9999	1319	1.0000	932		
1331	verihubs-inteligensia-000	0.0070	460	0.0098	448	0.0048	656	0.0076	705	0.0092	697	0.0160	446	0.0117	528	0.1348	379		
1332	verihubs-inteligensia-001	0.0071	463	0.0114	496	0.0050	669	0.0076	704	0.0096	714	0.0165	460	0.0114	521	0.1409	391		
1333	verihubs-inteligensia-002	0.0028	174	0.0043	189	0.0025	236	0.0030	251	0.0047	461	0.0092	291	0.0083	410	0.0983	301		
1334	verijelas-000	0.2488	981	0.3431	982	0.4861	1315	0.6004	1306	0.0811	1136	0.1148	818	0.0440	859	0.4077	553		
1335	verismart-001	-		-		0.0044	613	0.0068	664	0.0064	574	0.1877	869	0.0080	393	0.8127	662		
1336	via-000	0.0216	755	0.0365	765	0.0177	1065	0.0287	1056	0.0296	1017	0.0572	728	0.0290	796	-			
1337	via-001	0.0149	679	0.0229	675	0.0114	976	0.0177	963	0.0183	909	0.4056	976	0.0176	663	-			
1338	via-004	0.0099	563	0.0135	531	0.0031	423	0.0044	474	0.0055	516	0.4009	974	0.0080	394	0.9840	760		
1339	via-005	0.0063	421	0.0082	391	0.0031	404	0.0037	393	0.0032	284	0.3659	960	0.0068	304	0.8709	687		
1340	via-006	-		-		0.0025	187	0.0027	181	0.0025	163	0.5904	1027	0.0315	805	0.9455	722		
1341	via-007	-		-		0.0027	313	0.0033	312	0.0028	214	0.0052	120	0.0043	140	0.0617	132		
1342	viante-000	0.0004	6	0.0007	4	0.0026	269	0.0026	171	0.0017	27	0.0033	20	0.0029	28	0.0475	45		
1343	viante-001	-		-		0.0025	234	0.0025	138	0.0016	17	0.0031	14	0.0028	21	0.0432	22		
1344	viante-002	-		-		0.0025	197	0.0024	114	0.0015	12	0.0029	10	0.0027	12	0.0399	12		
1345	viante-003	-		-		0.0025	190	0.0023	86	0.0015	8	0.0029	6	0.0026	10	0.0388	8		
1346	viante-004	-		-		0.0025	200	0.0023	87	0.0015	13	0.0029	8	0.0027	11	0.0390	9		
1347	videmo-000	0.0298	810	0.0423	783	0.0155	1052	0.0260	1038	0.0246	984	0.0397	676	0.0239	750	-			
1348	videmo-001	0.0295	807	0.0417	780	0.0164	1061	0.0261	1041	0.0355	1039	0.0603	741	0.0442	861	0.2428	488		
1349	videmo-002	0.0158	695	0.0288	718	0.0149	1046	0.0249	1030	0.0230	962	0.3429	949	0.1468	997	0.8393	671		
1350	videmo-003	-		-		0.0082	858	0.0138	886	0.0133	811	0.0220	532	0.0119	539	0.1596	420		
1351	videonetics-001	0.5483	1036	0.6446	1041	0.7517	1354	0.8607	1348	0.8664	1299	0.8255	1097	0.6956	1164	-			
1352	videonetics-002	0.4274	1013	0.5329	1020	0.6081	1329	0.7438	1324	0.7775	1280	0.7297	1059	0.5756	1139	-			
1353	viettelai-000	0.0051	348	0.0068	330	0.0054	699	0.0064	640	0.9451	1319	1.0000	1482	1.0000	1488	1.0000	1222		
1354	viettelai-001	0.0054	366	0.0076	364	0.0049	664	0.0079	722	0.0064	573	0.0225	537	0.0083	406	1.0000	1021		
1355	viettelai-002	-		-		0.0029	368	0.0041	437	0.0069	591	0.0126	388	0.0074	349	0.1095	323		
1356	viettelai-003	-		-		0.0034	482	0.0041	441	0.2038	1197	0.3193	942	0.1067	962	0.8681	683		
1357	viettelai-004	-		-		0.0023	114	0.0028	218	0.0023	108	0.0044	76	0.0034	54	0.0566	97		
1358	viettelai-005	-		-		0.0023	103	0.0027	188	0.0021	74	0.0038	47	0.0030	39	0.0499	57		
1359	viettelai-006	-		-		0.0022	73	0.0024	126	0.0020	62	0.0037	42	0.0029	33	0.0481	48		
1360	viettelhightech-000	0.0117	613	0.0166	586	0.0110	958	0.0198	992	0.0167	882	0.0249	558	0.0158	627	0.1604	422		
1361	viettelhightech-001	-		-		0.0031	418	0.0040	428	0.0032	291	0.0063	169	0.0051	192	0.0781	222		
1362	viettelhightech-002	-		-		0.0031	400	0.0042	459	0.0033	296	0.0061	164	0.0051	190	0.0758	209		
1363	vigilantsolutions-000	-		0.6881	1047	-		-		-		-		-		-			
1364	vigilantsolutions-001	-		0.3475	983	-		-		-		-		-		-			

Table 88: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)															
		CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1365	vigilantsolutions-002	0.2290	974	0.3214	977	-		-		-		-		-		-	
1366	vigilantsolutions-003	0.1878	961	0.2673	958	-		-		-		-		-		-	
1367	vigilantsolutions-004	0.1780	957	0.2598	956	-		-		-		-		-		-	
1368	vigilantsolutions-005	0.1613	951	0.4702	1007	0.0163	1059	0.0279	1051	-		-		-		-	
1369	vigilantsolutions-006	0.1264	934	0.3221	978	0.0150	1047	0.0254	1032	0.0493	1088	0.2991	927	0.0403	846	-	
1370	vigilantsolutions-007	0.0202	748	0.0307	730	0.0136	1017	0.0227	1015	0.0356	1041	1.0000	1350	1.0000	1368	-	
1371	vigilantsolutions-008	0.0156	690	0.0233	677	0.0101	926	0.0147	908	0.0355	1040	0.0492	708	0.0293	799	-	
1372	vigilantsolutions-009	0.0117	612	0.0165	585	0.0075	826	0.0101	792	0.0219	953	0.0385	671	0.0238	749	-	
1373	vigilantsolutions-010	0.0109	591	0.0164	582	0.0074	821	0.0095	772	0.0209	943	0.0365	660	0.0233	744	0.2465	489
1374	vigilantsolutions-011	0.0124	632	0.0176	605	0.0073	811	0.0095	771	0.0196	933	0.0360	652	0.0221	730	0.2633	502
1375	vinai-000	0.0081	505	0.0124	510	0.0045	619	0.0072	684	0.0089	683	0.1814	862	0.0112	514	-	
1376	vinbigdata-001	0.2576	984	0.2763	961	0.1404	1255	0.1988	1247	0.1407	1172	0.1150	819	0.0703	919	0.3688	538
1377	vinbigdata-002	0.0102	573	0.0175	602	0.0071	804	0.0084	742	0.0090	688	0.8017	1084	0.3134	1078	0.9936	781
1378	vinbigdata-003	0.0771	897	0.0992	884	0.1202	1247	0.1756	1235	0.2762	1221	0.2369	903	0.0100	470	0.8914	703
1379	vinbigdata-004	-		-		0.0031	406	0.0046	485	0.0037	350	0.1476	845	0.0063	282	0.4917	574
1380	vinbigdata-005	-		-		0.0023	80	0.0024	120	0.0022	84	0.0068	191	0.0038	93	0.0699	180
1381	vinbigdata-006	-		-		0.0022	70	0.0023	99	0.0019	60	0.0037	43	0.0032	44	0.0479	46
1382	vinbigdata-007	-		-		0.0022	48	0.0021	45	0.0019	58	0.0036	40	0.0032	45	0.0475	43
1383	vion-000	0.0419	843	0.0590	832	0.0422	1171	0.0478	1126	0.0581	1105	0.0968	801	0.0847	939	-	
1384	visage-000	0.0933	912	0.1441	911	0.1316	1251	0.2416	1254	0.1395	1169	0.1920	871	0.1001	955	-	
1385	visica-001	-		-		0.2293	1276	0.3588	1276	0.2936	1224	0.3564	956	0.1967	1030	0.7795	651
1386	visica-002	-		-		0.0520	1191	0.0898	1191	0.2960	1225	0.8202	1094	0.5956	1144	0.9838	758
1387	visica-003	-		-		0.3326	1294	0.4772	1291	0.3237	1229	0.3591	958	0.2203	1042	0.7317	637
1388	visionbox-000	0.0293	804	0.0541	822	0.0197	1078	0.0339	1072	0.0349	1035	0.0593	737	0.0329	810	-	
1389	visionbox-001	0.0159	696	0.0270	711	0.0111	962	0.0173	958	0.0190	920	0.0315	608	0.0205	714	-	
1390	visionbox-002	0.0058	387	0.0079	375	0.0060	742	0.0074	690	0.0084	662	0.0149	422	0.0113	518	0.0979	300
1391	visionbox-003	0.0057	384	0.0075	357	0.0062	761	0.0083	739	0.0100	735	0.9915	1197	0.9625	1244	1.0000	856
1392	visionbox-004	0.0046	314	0.0075	358	0.0060	748	0.0075	698	0.0056	522	0.0316	610	0.0070	324	0.2528	492
1393	visionbox-005	-		-		0.0041	584	0.0057	584	0.0046	454	0.0109	350	0.0088	424	0.0767	214
1394	visionbox-006	-		-		0.0063	766	0.0075	695	0.0048	467	0.0116	372	0.0094	451	0.0789	225
1395	visionbox-007	-		-		0.0063	768	0.0075	694	0.0047	465	0.0116	374	0.0095	457	0.0793	227
1396	visionbox-008	-		-		0.0056	722	0.0072	679	0.0044	425	0.0115	371	0.0089	429	0.0828	245
1397	visionlabs-001	0.1049	922	0.1803	925	-		-		-		-		-		-	
1398	visionlabs-002	0.0326	824	0.0507	812	-		-		-		-		-		-	
1399	visionlabs-003	0.0178	722	0.0293	722	-		-		-		-		-		-	
1400	visionlabs-004	0.0219	759	0.0457	796	-		-		-		-		-		-	
1401	visionlabs-005	0.0088	524	0.0156	572	0.0048	651	0.0070	675	-		-		-		-	
1402	visionlabs-006	0.0037	256	0.0066	316	0.0041	595	0.0060	610	0.0061	560	0.0106	343	0.0076	370	-	
1403	visionlabs-007	0.0038	262	0.0048	229	0.0036	515	0.0048	503	0.0057	530	0.0097	313	0.0071	328	-	
1404	visionlabs-008	0.0026	160	0.0036	141	0.0031	427	0.0040	427	0.0045	439	0.0079	233	0.0064	283	-	
1405	visionlabs-009	0.0018	89	0.0025	66	0.0026	249	0.0029	245	0.0035	323	0.0064	172	0.0054	223	-	
1406	visionlabs-010	0.0017	74	0.0024	63	0.0026	268	0.0030	261	0.0033	304	0.0061	162	0.0052	205	-	
1407	visionlabs-011	0.0012	40	0.0022	55	0.0024	175	0.0026	169	0.0028	209	0.0053	125	0.0046	162	0.0635	147
1408	visteam-000	0.9200	1064	0.9489	1071	0.9959	1382	0.9984	1376	0.9994	1341	0.9978	1233	0.9914	1270	-	

Table 89: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1409	visteam-001	0.4417	1018	0.5385	1023	0.6410	1332	0.7788	1329	0.6386	1264	0.5904	1025	0.4023	1105	-	
1410	visteam-002	0.1564	949	0.2789	963	0.1581	1261	0.2567	1260	0.1776	1187	0.2090	886	0.1021	957	0.5820	598
1411	visteam-003	0.0804	905	0.2166	941	0.0613	1206	0.1204	1215	0.0963	1150	0.1269	831	0.0441	860	0.6049	604
1412	visteam-004	0.0541	864	0.5202	1017	0.0406	1167	0.0827	1182	0.1879	1190	0.1795	861	0.0347	820	0.9673	736
1413	visteam-005	0.0235	768	0.0333	746	0.0265	1122	0.0583	1150	0.0341	1032	0.0524	719	0.0259	767	0.2413	486
1414	visteam-006	0.0188	733	0.0256	701	0.0158	1056	0.0365	1087	0.0198	936	0.0340	640	0.0174	661	0.1904	452
1415	visteam-007	0.0312	817	0.0448	793	0.0416	1169	0.1008	1201	0.0466	1080	0.0602	740	0.0281	784	0.2601	499
1416	visteam-008	0.0378	837	0.0515	814	0.0391	1164	0.0937	1196	0.0437	1073	0.0567	727	0.0260	769	0.2533	493
1417	visteam-009	-		-		0.0138	1021	0.0304	1063	0.0166	878	0.9993	1255	0.9719	1249	0.9997	825
1418	visteam-010	-		-		0.0156	1054	0.0347	1078	0.0189	917	1.0000	1309	0.9996	1305	1.0000	855
1419	vivotek-001	-		-		0.9982	1385	0.9994	1380	1.0000	1350	1.0000	1311	0.9999	1324	1.0000	892
1420	vivotek-002	-		-		0.0761	1221	0.1380	1222	0.9587	1322	1.0000	1315	1.0000	1332	1.0000	933
1421	vivotek-003	-		-		0.0761	1220	0.1380	1221	0.1178	1161	0.1552	849	0.0823	935	0.4943	576
1422	vixvizion-000	0.0642	883	0.0882	873	0.0497	1187	0.0905	1192	0.0848	1140	0.1026	803	0.0618	905	-	
1423	vixvizion-001	0.0245	775	0.0407	778	0.0257	1118	0.0497	1133	0.0514	1091	0.0656	751	0.0400	845	-	
1424	vixvizion-002	0.0062	417	0.0086	404	0.0053	696	0.0075	696	0.0121	785	0.0207	515	0.0161	633	-	
1425	vixvizion-003	0.0059	404	0.0084	397	0.0059	737	0.0081	730	0.0119	781	0.0209	517	0.0162	634	-	
1426	vixvizion-004	0.0063	423	0.0094	435	0.0055	712	0.0081	731	0.0098	727	0.0157	441	0.0111	511	0.1417	392
1427	vixvizion-005	0.0276	798	0.0420	782	0.0302	1143	0.0629	1154	0.0288	1012	0.0447	688	0.0235	746	0.2304	479
1428	vixvizion-006	0.0082	509	0.0122	506	0.0093	896	0.0194	985	0.0099	730	0.0169	467	0.0108	502	0.1338	377
1429	vixvizion-007	0.0110	595	0.0191	628	0.0080	845	0.0157	924	0.0101	737	0.0190	496	0.0118	534	0.1407	390
1430	vnis-000	0.1397	942	0.1608	917	0.2184	1275	0.2976	1269	0.5277	1255	0.3603	959	0.2714	1062	0.6929	626
1431	vnpay-000	0.2380	977	0.2614	957	0.4054	1302	0.4968	1294	0.3993	1243	0.4389	984	0.4006	1104	0.7693	646
1432	vnpay-001	-		-		0.0026	270	0.0030	253	0.0065	577	0.0124	384	0.0116	527	0.0826	243
1433	vnpt-001	0.3117	991	0.3523	985	0.3474	1295	0.2747	1263	0.3405	1230	0.5015	997	0.4827	1120	-	
1434	vnpt-002	0.0351	833	0.0424	784	0.0220	1097	0.0316	1069	0.0471	1081	0.0817	781	0.0698	918	0.2610	500
1435	vnpt-003	0.0117	611	0.0138	535	0.0040	578	0.0058	601	0.0087	672	0.0161	452	0.0126	552	0.1110	330
1436	vnpt-004	0.0058	394	0.0078	372	0.0037	535	0.0053	554	0.0051	488	0.4640	989	0.1384	990	0.7753	648
1437	vnpt-005	0.0036	245	0.0052	251	0.0027	304	0.0031	276	0.0036	334	0.0066	184	0.0056	235	0.0621	135
1438	vnpt-006	0.0034	234	0.0050	240	0.0027	294	0.0035	363	0.0026	183	0.0049	108	0.0040	122	0.0585	106
1439	vnpt-007	-		-		0.0025	227	0.0029	237	0.0025	158	0.0047	88	0.0039	108	0.0564	95
1440	vnpt-008	-		-		0.0026	246	0.0029	235	0.0025	164	0.0047	94	0.0040	112	0.0570	100
1441	vnpt-009	-		-		0.0030	374	0.0042	451	0.0030	253	0.0053	126	0.0042	132	0.0628	139
1442	vocalize-001	-		-		0.7356	1349	0.8750	1355	0.7830	1283	0.7067	1055	0.5268	1131	0.9164	713
1443	vocord-001	-		0.1415	910	-		-		-		-		-		-	
1444	vocord-002	0.0228	765	0.0341	751	-		-		-		-		-		-	
1445	vocord-003	0.1003	916	0.1092	889	-		-		-		-		-		-	
1446	vocord-004	0.0101	571	0.0174	600	0.0196	1077	0.0384	1093	-		-		-		-	
1447	vocord-005	0.0089	526	0.0155	569	0.0216	1095	0.0414	1103	-		-		-		-	
1448	vocord-006	0.0062	420	0.0102	465	0.0082	854	0.0151	916	0.0475	1082	0.0852	784	0.0788	925	-	
1449	vocord-007	0.0039	267	0.0053	263	0.0061	754	0.0094	770	0.0520	1094	0.0939	794	0.0910	947	-	
1450	vocord-008	0.0029	179	0.0038	156	0.0042	600	0.0055	568	0.0045	442	0.0086	274	0.0073	343	-	
1451	vocord-009	0.0022	121	0.0029	99	0.0036	506	0.0046	491	0.0052	496	0.0098	316	0.0086	420	-	
1452	vocord-010	0.0024	143	0.0031	107	0.0036	507	0.0049	524	0.0025	166	0.0065	176	0.0040	116	0.0679	174

Table 90: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	FALSE NON-MATCH RATE (FNMR)																	
		CONSTRAINED, COOPERATIVE																LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER			
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06			
1453	vogic-000	-		-		0.0043	607	0.0061	628	0.0078	636	0.8206	1095	0.6006	1146	0.9843	762		
1454	vsoft-001	-		-		0.0707	1216	0.0877	1187	0.8390	1294	0.9519	1151	0.9506	1234	0.9655	735		
1455	vsoft-002	-		-		0.0451	1180	0.0679	1164	0.0550	1101	0.3671	962	0.1633	1015	0.8586	680		
1456	vsoft-003	-		-		0.0326	1146	0.0563	1145	0.0526	1096	0.3671	963	0.1633	1016	0.8586	681		
1457	vsoft-004	-		-		0.0061	753	0.0103	798	0.0090	687	0.1830	865	0.0344	818	0.5669	593		
1458	vts-000	0.0103	575	0.0174	598	0.0080	844	0.0129	860	0.0250	986	0.0450	691	0.0372	834	-			
1459	vts-001	0.0033	216	0.0048	227	0.0027	314	0.0036	374	0.0032	283	0.6519	1042	0.3563	1092	0.9218	716		
1460	vts-002	-		-		0.0025	189	0.0032	305	0.0030	247	0.0113	363	0.0042	133	0.1593	417		
1461	wicket-000	0.0018	91	0.0028	89	0.0024	168	0.0027	182	0.0031	263	0.7968	1083	0.4340	1112	0.9788	750		
1462	winsense-000	0.0140	665	0.0228	674	0.0125	999	0.0215	1003	0.0226	959	0.8091	1088	0.0461	866	-			
1463	winsense-001	0.0062	416	0.0099	457	0.0092	892	0.0210	999	0.0093	703	0.0144	414	0.0098	464	-			
1464	winsense-002	0.0050	341	0.0073	351	0.0038	551	0.0059	603	0.0064	572	0.0118	378	0.0084	411	-			
1465	wiseai-001	0.0658	885	0.0964	882	0.7743	1359	0.8956	1357	0.1967	1194	0.7526	1068	0.3419	1087	0.9695	740		
1466	wuhantianyu-001	0.0163	702	0.0262	707	0.0281	1129	0.0569	1147	0.0316	1028	0.0486	706	0.0344	817	0.2097	469		
1467	x-laboratory-000	0.0071	466	0.0106	473	0.0123	995	0.0138	885	0.0419	1065	0.5629	1018	0.2852	1068	-			
1468	x-laboratory-001	0.0059	398	0.0110	484	0.0054	701	0.0078	719	0.0094	707	0.0142	407	0.0100	472	-			
1469	xailient-001	-		-		0.0026	279	0.0031	290	0.0029	231	0.0056	133	0.0039	104	0.0599	115		
1470	xforwardai-000	0.0050	338	0.0072	343	0.0036	518	0.0051	544	0.0074	618	0.0136	398	0.0105	493	-			
1471	xforwardai-001	0.0021	111	0.0034	122	0.0027	322	0.0028	217	0.0046	452	0.0088	279	0.0079	383	-			
1472	xforwardai-002	0.0016	66	0.0023	58	0.0026	284	0.0025	140	0.0040	381	0.0081	243	0.0074	346	-			
1473	xm-000	0.0015	57	0.0026	72	0.0031	412	0.0038	402	0.0058	540	0.0105	338	0.0082	405	-			
1474	yisheng-000	0.1215	932	0.1990	936	-		-		-		-		-		-			
1475	yisheng-001	0.0892	909	0.1600	916	-		-		-		-		-		-			
1476	yisheng-002	0.2241	973	0.3795	991	-		-		-		-		-		-			
1477	yisheng-004	0.1988	967	0.3329	980	0.1147	1244	0.1849	1241	0.2044	1198	-		-		-			
1478	yitu-000	0.0284	802	0.0330	743	-		-		-		-		-		-			
1479	yitu-001	0.0050	333	0.0074	356	-		-		-		-		-		-			
1480	yitu-002	0.0025	155	0.0043	194	-		-		-		-		-		-			
1481	yitu-003	0.0015	61	0.0026	70	0.0066	781	0.0085	743	0.0064	575	0.0114	368	0.0103	487	-			
1482	yoonik-000	0.0070	454	0.0112	488	0.0074	819	0.0118	828	0.0564	1103	0.2013	877	0.1160	969	-			
1483	yoonik-001	0.0057	378	0.0079	376	0.0043	610	0.0061	618	0.0307	1022	0.0762	774	0.0556	894	-			
1484	yoonik-002	0.0052	358	0.0062	305	0.0029	373	0.0034	340	0.0615	1112	0.1279	832	0.1166	970	0.2329	482		
1485	yoonik-003	0.0034	229	0.0047	220	0.0032	437	0.0037	383	0.0816	1138	0.2033	881	0.1601	1011	0.3210	525		
1486	yoonik-004	0.0041	284	0.0059	286	0.0033	453	0.0052	548	0.0040	383	0.0071	201	0.0051	198	0.0732	190		
1487	yoti-001	0.0139	664	0.0195	634	0.0110	959	0.0187	978	0.0191	921	0.1255	829	0.0415	850	0.5578	591		
1488	yoti-002	-		-		0.0076	829	0.0137	881	0.0106	747	0.3375	945	0.1599	1010	0.7672	645		
1489	ytu-000	0.0057	383	0.0087	412	0.0121	992	0.0238	1024	0.0047	463	0.0078	226	0.0059	259	-			
1490	yuan-000	0.0101	572	0.0168	591	0.0077	831	0.0127	853	0.0178	898	0.5189	1004	0.0245	757	-			
1491	yuan-001	0.0116	609	0.0220	663	0.0114	974	0.0184	970	0.0149	845	0.0574	729	0.0160	632	-			
1492	yuan-002	0.0094	548	0.0154	567	0.0071	802	0.0110	815	0.0108	758	0.0348	643	0.0127	557	0.5738	596		
1493	yuan-003	0.0078	493	0.0111	485	0.0062	762	0.0091	761	0.0106	748	0.0511	714	0.0123	549	0.7337	638		
1494	yuan-004	0.0058	396	0.0078	370	0.0039	568	0.0055	572	0.0234	968	0.0442	687	0.0353	823	0.1673	431		
1495	yuan-005	0.0037	257	0.0046	218	0.0027	325	0.0035	360	0.0033	297	0.2706	922	0.0876	944	0.6695	621		
1496	yuan-006	0.0045	303	0.0062	302	0.0032	431	0.0049	515	0.0038	361	0.0084	260	0.0049	177	0.1211	355		

Table 91: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

		FALSE NON-MATCH RATE (FNMR)															
	Algorithm	CONSTRAINED, COOPERATIVE														LESS CONSTRAINED, NON-COOP.	
	Name	VISAMC		VISA		MUGSHOT		MUGSHOT12+YRS		VISABORDER		BORDER		BORDER		KIOSKBORDER	
	FMR	0.0001		1E-06		1E-05		1E-05		1E-06		1E-06		1E-05		1E-06	
1497	yuan-007	0.0046	311	0.0069	333	0.0045	624	0.0075	697	0.0072	607	0.0144	413	0.0076	362	0.1081	322
1498	yuan-008	0.0032	208	0.0043	193	0.0026	260	0.0034	331	0.0028	217	0.0059	148	0.0039	107	0.0891	263
1499	yuan-009	-		-		0.0028	337	0.0042	456	0.0030	251	0.0057	135	0.0040	118	0.0733	192
1500	yuan-010	-		-		0.0028	334	0.0041	432	0.0030	262	0.0058	142	0.0041	125	0.0748	205
1501	yuanli-001	-		-		0.0203	1084	0.0399	1095	0.0110	765	0.0168	466	0.0135	576	0.1476	404

Table 92: FNMR is the proportion of mated comparisons below a threshold set to achieve the FMR given in the header on the fourth row. FMR is the proportion of impostor comparisons at or above that threshold. The light grey values give rank over all algorithms in that column. The pink columns use only same-sex impostors; others are selected regardless of demographics. The exception, in the green column, uses "matched-covariates" i.e. impostors of the same sex, age group, and country of birth. The second pink column includes effects of extended ageing. Missing entries for border, visa, mugshot and wild images generally mean the algorithm did not run to completion. The VISA *columns compare images described in section 2.1. The* MUGSHOT *columns compare images described in section 2.5. The* VISA-BORDER *column compare images described in section 2.2 with those of section 2.4. The* BORDER *column compares images described in section 2.4. The* KIOSK-BORDER *columns compare images described in section 2.6 with those of section 2.4.*

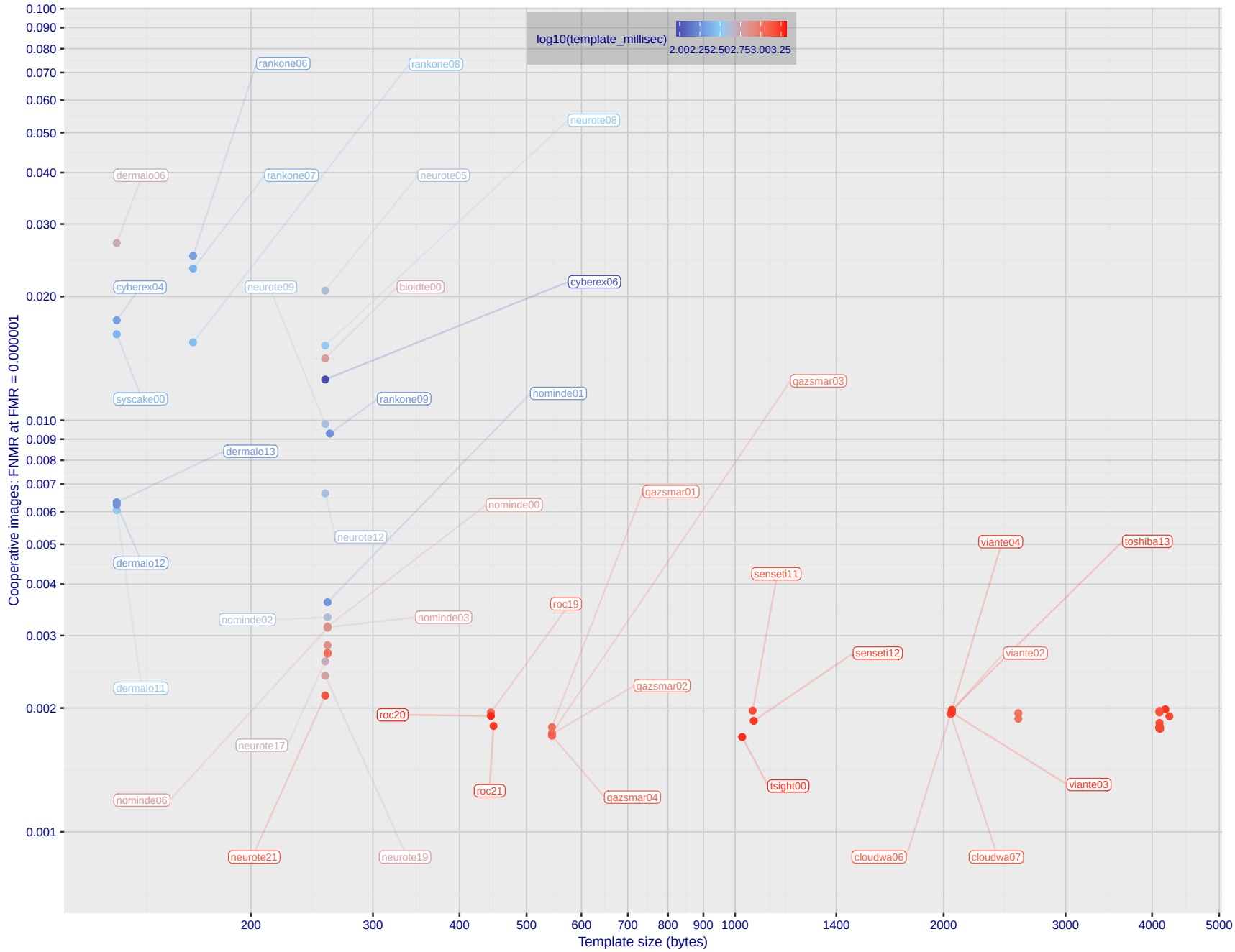


Figure 1: The points show false non-match rates (FNMR) versus the size of the encoded template. FNMR is the geometric mean of FNMR values for visa-border and mugshot images (from Figs. 225 and 296) at the false match rate (FMR) given in the y-axis label. The color of the points encodes template generation time - which spans at least one order of magnitude. Durations are measured on a single core of a c. 2016 Intel Xeon CPU E5-2630 v4 running at 2.20GHz. Algorithms with poor FNMR are omitted.

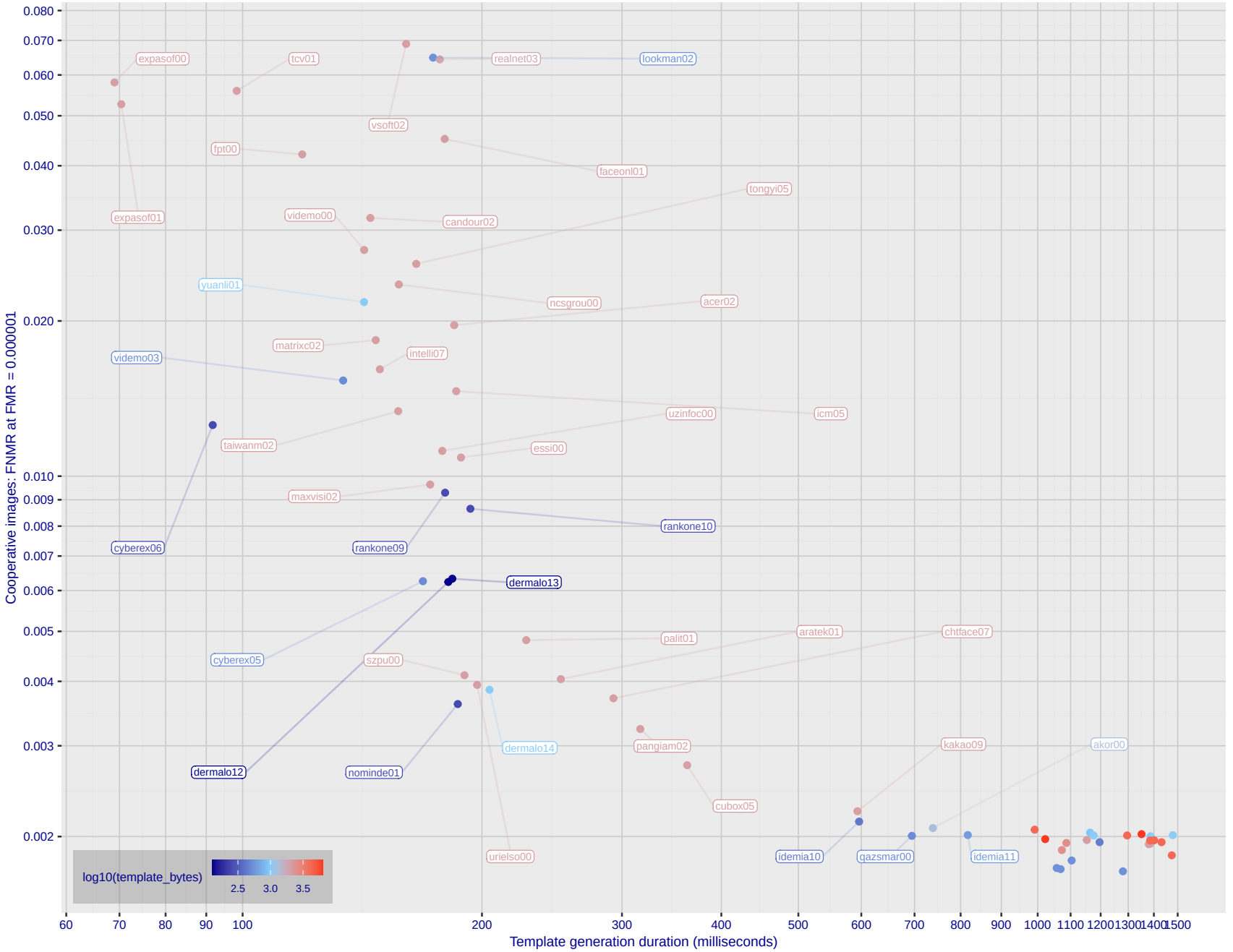


Figure 2: The points show false non-match rates (FNMR) versus the duration of the template generation operation. FNMR is the geometric mean of FNMR values for visa-border and mugshot images (from Figs. 225 and 296) at a false match rate (FMR) given in the y-axis label. Template generation time is a median estimated over 640 x 480 pixel portraits. It is measured on a single core of a c. 2016 Intel Xeon CPU E5-2630 v4 running at 2.20GHz. The color of the points encodes template size - which spans two orders of magnitude. Algorithms with poor FNMR are omitted.

1 Metrics

1.1 Core accuracy

Given a vector of N genuine scores, u, the false non-match rate (FNMR) is computed as the proportion below some threshold, T:

$$\text{FNMR}(T) = 1 - \frac{1}{N}\sum_{i=1}^{N} H(u_i - T) \tag{1}$$

where $H(x)$ is the unit step function, and $H(0)$ taken to be 1.

Similarly, given a vector of N impostor scores, v, the false match rate (FMR) is computed as the proportion above T:

$$\text{FMR}(T) = \frac{1}{N}\sum_{i=1}^{N} H(v_i - T) \tag{2}$$

The threshold, T, can take on any value. We typically generate a set of thresholds from quantiles of the observed impostor scores, v, as follows. Given some interesting false match rate range, $[\text{FMR}_L, \text{FMR}_U]$, we form a vector of K thresholds corresponding to FMR measurements evenly spaced on a logarithmic scale

$$T_k = Q_v(1 - \text{FMR}_k) \tag{3}$$

where Q is the quantile function, and FMR_k comes from

$$\log_{10}\text{FMR}_k = \log_{10}\text{FMR}_L + \frac{k}{K}\ [\log_{10}\text{FMR}_U - \log_{10}\text{FMR}_L] \tag{4}$$

Error tradeoff characteristics are plots of FNMR(T) vs. FMR(T). These are plotted with $\text{FMR}_U \to 1$ and FMR_L as low as is sustained by the number of impostor comparisons, N. This is somewhat higher than the "rule of three" limit $3/N$ because samples are not independent, due to re-use of images.

1.2 Multi-template scoring methodology

There are some scenarios when one or more people exist and are detected in an image, and some of the proposed test images include K $>$ 1 persons for some images and situations where the subject of interest may or may not be the foreground face (largest face in the image). The NIST FRTE 1:1 API supports this by allowing generation of multiple templates representing each person detected in an image. When this occurs, NIST will match all templates generated from the enrollment image with all templates generated from the verification image and use the **maximum** similarity score across all template comparisons. This scoring approach will be used in our calculation of FMR and FNMR (this applies to both genuine and imposter comparisons).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

2 Datasets

2.1 Visa images

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^5.
- The number of subjects with two images is on the order of 10^4.
- The images have geometry in reasonable conformance with the ISO/IEC 19794-5 Full Frontal image type. Pose is generally excellent.
- The images are of size 252x300 pixels. The mean interocular distance (IOD) is 69 pixels.
- The images are of subjects from greater than 100 countries, with significant imbalance due to visa issuance patterns.
- The images are of subjects of all ages, including children, again with imbalance due to visa issuance demand.
- Many of the images are live capture. A substantial number of the images are photographs of paper photographs.
- When these images are input to the algorithm, they are labelled as being of type "ISO" - see Table 4 of the FRTE API.

2.2 Application images

- The number of images is on the order of 10^6.
- The number of subjects is on the order of 10^6.
- The number of subjects with two images is on the order of 10^6.
- The images have geometry in good conformance with the ISO/IEC 19794-5 Full Frontal image type. Pose is generally excellent.
- The images are of size 300x300 pixels. The mean interocular distance (IOD) is 61 pixels.
- The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.
- The images are of subjects of adults.
- All of the images are live capture.
- When these images are input to the algorithm, they are labelled as being of type "ISO" - see Table 4 of the FRTE API.

2.3 Application images with head yaw

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^5.
- The number of subjects with two images is on the order of 10^5.
- The images have geometry in good conformance with the ISO/IEC 19794-5 Full Frontal image type *except* the yaw angle is between 25 and 85 degrees. Our pose estimates are approximate, with an angular error that increases with yaw. The angular estimates will be improved over time.
- The images are of size 300x300 pixels. The mean interocular distance (IOD), if frontal, would be about pixels, but reduces with cosine of yaw.
- The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The images are of subjects of adults.
- ▷ All of the images are live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

2.4 Border crossing images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^6.
- ▷ The number of subjects with two images is on the order of 10^6.
- ▷ The images are taken with at camera oriented by an attendant toward a cooperating subject. This is done under time constraints so there are role, pitch and yaw angle variations. Also background illumination is sometimes strong, so the face is under-exposed. There is some perspective distortion due to close range images. Some faces are partially cropped.
- ▷ The images have mean IOD of 38 pixels.
- ▷ The images are of subjects of adults and childen aged 12 or above.
- ▷ The images are of subjects from greater than 100 countries, with significant imbalance due to population and immigration patterns.
- ▷ The images are all live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

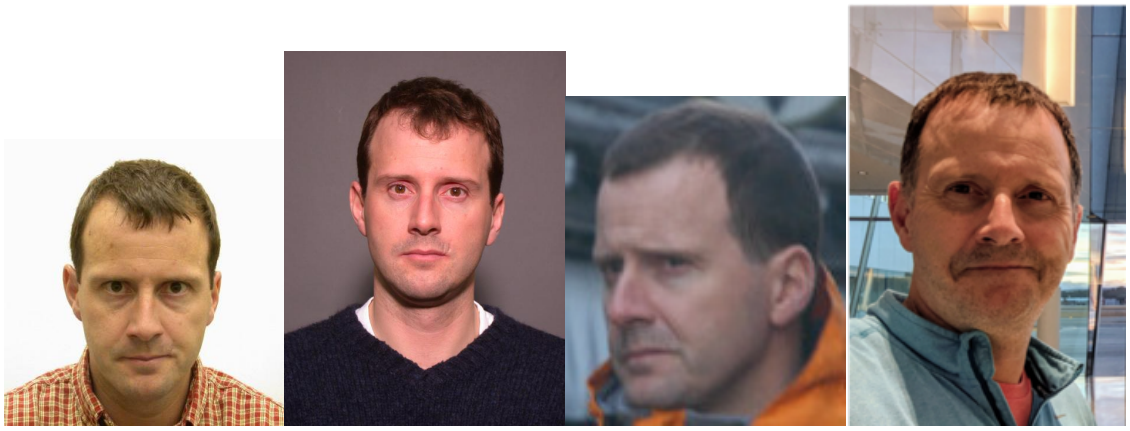
2.5 Mugshot images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^6.
- ▷ The number of subjects with two images is on the order of 10^6.
- ▷ The images have geometry in reasonable conformance with the ISO/IEC 19794-5 Full Frontal image type.
- ▷ The images are of variable sizes. The median IOD is 105 pixels. The mean IOD is 113 pixels. The 1-st, 5-th, 10-th, 25-th, 75-th, 90-th and 99-th percentiles are 34, 58, 70, 87, 121, 161 and 297 pixels.
- ▷ The images are of subjects from the United States.
- ▷ The images are of adults.
- ▷ The images are all live capture.
- ▷ When these images are input to the algorithm, they are labelled as being of type ”mugshot” - see Table 4 of the FRTE API.

2.6 Kiosk images

- ▷ The number of images is on the order of 10^6.
- ▷ The number of subjects is on the order of 10^5.
- ▷ The number of subjects with multiple images is the order of 10^5.

FNMR(T) “False non-match rate”
FMR(T) “False match rate”



(a) Visa (b) Mugshot (c) Wild (d) Border

Figure 3: The figure gives simulated samples of image types used in this report.

- The images are taken at kiosk equipped with a camera intended to capture a centered face. However the images have specific quality defects arising from the camera triggering before the subject looks at it. These are downward pitch of the face relative to the optical axis; cropping of the forehead; and cropping of left or right part of the face. Partial cropping affects perhaps 10% of the images. Resolution does not vary widely.
- The images are of adults.
- The images have mean IOD of 44 pixels, with maximum below 75, and minimum when both eyes are present above 25 pixels.
- All of the images are live capture, none are scanned.
- When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

2.7 Wild images

- The number of images is on the order of 10^5.
- The number of subjects is on the order of 10^4.
- The number of subjects with two images on the order of 10^4.
- The images include many photojournalism-style images. Images are given to the algorithm using a variable but generally tight crop of the head. Resolution varies very widely. The images are very unconstrained, with wide yaw and pitch pose variation. Faces can be occluded, including hair and hands.
- The images are of adults.
- All of the images are live capture, none are scanned.
- When these images are input to the algorithm, they are labelled as being of type ”WILD” - see Table 4 of the FRTE API.

3 Results

3.1 Test goals

- To state absolute accuracy for different kinds of images, including those with and without subject cooperation.

FNMR(T) “False non-match rate”
FMR(T) “False match rate”

▷ To state comparative accuracy, across algorithms.

3.2 Test design

Method: For visa images:

▷ The comparisons are of visa photos against visa photos.
▷ The number of genuine comparisons is on the order of 10^4.
▷ The number of impostor comparisons is on the order of 10^{10}.
▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates. However, later analysis is conducted on subsets.
▷ The number of persons is on the order of 10^5.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For mugshot images:

▷ The comparisons are of mugshot photos against mugshot photos.
▷ The number of genuine comparisons is on the order of 10^6.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.
▷ The number of persons is on the order of 10^6.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For visa-border comparisons:

▷ The comparisons are of visa-like frontals against border crossing webcam photos.
▷ The number of genuine comparisons is on the order of 10^6.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.
▷ The number of persons is on the order of 10^6.
▷ The number of images used to make a template is one.
▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For visa-border non-frontal yaw comparisons:

▷ The comparisons are of visa-like images with yaw 25 to 85 degrees against border crossing webcam photos.
▷ The number of genuine comparisons is on the order of 10^5.
▷ The number of impostor comparisons is on the order of 10^8.
▷ The impostors are paired by sex, but not by age or other covariates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The number of persons is on the order of 10^5.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For kiosk-border comparisons:

- ▷ The comparisons are of visa-like frontals against kiosk-style photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The impostors are paired by sex, but not by age or other covariates.
- ▷ The number of persons is on the order of 10^5.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two, corresponding to simple one-to-one verification.

Method: For border-border comparisons:

- ▷ The comparisons are of border crossing webcam photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The impostors are paired by sex, but not by age or other covariates.
- ▷ The number of persons is on the order of 10^6.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For wild images:

- ▷ The comparisons are of wild photos against wild photos.
- ▷ The number of genuine comparisons is on the order of 10^6.
- ▷ The number of impostor comparisons is on the order of 10^8.
- ▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates.
- ▷ The number of persons is on the order of 10^4.
- ▷ The number of images used to make a template is one.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.

Method: For child exploitation images:

- ▷ The comparisons are of unconstrained child exploitation photos against others of the same type.
- ▷ The number of genuine comparisons is on the order of 10^4.
- ▷ The number of impostor comparisons is on the order of 10^7.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

- ▷ The comparisons are fully zero-effort, meaning impostors are paired without attention to sex, age or other covariates.
- ▷ The number of persons is on the order of 10^3.
- ▷ The number of images used to make 1 template is 1.
- ▷ The number of templates used to make each comparison score is two corresponding to simple one-to-one verification.
- ▷ We produce two performance statements. First, is a DET as used for visa and mugshot images. The second is a cumulative match characteristic (CMC) summarizing a simulated one-to-many search process. This is done as follows.
  - We regard M enrollment templates as items in a gallery.
  - These M templates come from $M > N$ individuals, because multiple images of a subject are present in the gallery under separate identifiers.
  - We regard the verification templates as search templates.
  - For each search we compute the rank of the highest scoring mate.
  - This process should properly be conducted with a 1:N algorithm, such as those tested in NIST IR 8009. We use the 1:1 algorithms in a simulated 1:N mode here to a) better reflect what a child exploitation analyst does, and b) to show algorithm efficacy is better than that revealed in the verification DETs.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.3 Failure to enroll

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1	2060io-001	0.0160	1333	0.0998	1338	0.4698	981	0.0193	1393	-	1138
2	2060io-002	0.0000	1054	0.0015	961	0.0238	656	0.0028	1346	-	1286
3	20face-000	0.0000	811	0.0008	706	0.0217	605	0.0000	355	0.0004	613
4	20face-001	0.0000	793	0.0008	704	0.0000	10	0.0000	360	0.0004	607
5	20face-002	0.0000	827	0.0008	705	0.0217	604	0.0000	353	-	1263
6	3divi-000	-	1468	-	1428	-	1232	-	1426	0.0008	933
7	3divi-001	-	1371	-	1436	-	1257	-	1475	0.0007	923
8	3divi-002	-	1350	-	1457	-	1334	-	1495	0.0010	952
9	3divi-003	0.0001	1243	0.0059	1232	-	1161	0.0007	1226	0.0006	892
10	3divi-004	0.0001	1242	0.0059	1233	-	1404	0.0008	1241	0.0006	899
11	3divi-005	0.0000	971	0.0008	674	-	1031	0.0000	506	0.0002	271
12	3divi-006	0.0000	698	0.0007	622	0.0214	600	0.0001	728	0.0002	280
13	3divi-007	0.0000	596	0.0007	621	0.0214	599	0.0001	730	0.0002	278
14	accurascan-001	0.0001	1232	0.0060	1237	0.0838	940	0.0006	1214	0.0022	1030
15	accurascan-002	0.0001	1231	0.0060	1238	0.0838	941	0.0006	1215	0.0022	1031
16	accurascan-003	0.0000	858	0.0009	793	0.0226	623	0.0000	504	0.0003	395
17	acer-000	0.0000	1080	0.0024	1079	-	1347	0.0002	926	0.0004	737
18	acer-001	0.0000	606	0.0011	869	-	1305	0.0001	647	0.0004	646
19	acer-002	0.0000	1139	0.0008	687	0.0191	536	0.0003	1017	0.0004	643
20	acisw-003	0.0000	141	0.0000	232	-	1467	0.0000	213	0.0000	212
21	acisw-006	0.0000	416	0.0000	96	-	1187	0.0000	52	0.0000	87
22	acisw-007	0.0000	310	0.0000	44	0.0000	29	0.0000	137	0.0000	45
23	acisw-008	0.0000	743	0.0009	771	0.0173	475	0.0004	1121	0.0004	435
24	adera-001	0.0000	1081	0.0034	1159	-	1454	0.0003	1040	0.0005	862
25	adera-002	0.0000	1083	0.0034	1158	-	1388	0.0003	1039	0.0005	861
26	adera-003	0.0000	1085	0.0034	1157	0.0403	839	0.0003	1038	0.0005	857
27	adera-004	0.0000	550	0.0008	684	0.0202	557	0.0003	1074	0.0004	566
28	adera-005	0.0000	831	0.0008	683	0.0202	556	0.0003	1072	0.0004	563
29	advance-002	0.0000	711	0.0013	936	-	1080	0.0000	600	0.0004	637
30	advance-003	0.0000	1048	0.0012	895	0.0247	671	0.0001	794	0.0004	739
31	advance-004	0.0001	1239	0.0010	838	0.0157	439	0.0008	1251	0.0006	901
32	advance-005	0.0000	822	0.0003	323	0.0074	241	0.0002	913	0.0003	290
33	advance-006	0.0001	1234	0.0012	909	0.0202	555	0.0008	1248	-	1351
34	advance-007	0.0000	869	0.0007	624	0.0212	587	0.0002	861	-	1254
35	affinsys-ai-000	0.0000	1149	0.0068	1241	0.0641	900	0.0016	1310	-	1139
36	afisbiometrics-000	0.0000	552	0.0008	662	0.0213	597	0.0000	376	0.0004	648
37	afrengine-000	0.0000	703	0.0015	966	0.0254	694	0.0008	1249	0.0004	505
38	afrengine-001	0.0000	846	0.0007	639	0.0183	500	0.0005	1156	0.0004	742
39	afrengine-002	0.0000	667	0.0007	641	0.0183	501	0.0005	1158	0.0004	744
40	afrengine-003	0.0000	635	0.0099	1270	0.0384	823	0.0005	1157	-	1452
41	afrengine-004	0.0000	821	0.1602	1342	0.0528	871	0.0005	1155	-	1272
42	afrengine-005	0.0000	623	0.0012	906	0.0232	647	0.0006	1199	-	1336
43	aifirst-001	0.0000	448	0.0000	66	-	1132	0.0000	21	0.0000	71
44	aigen-001	0.0000	131	0.0000	231	-	1461	0.0000	204	0.0000	224
45	aigen-002	0.0000	209	0.0000	172	0.0000	105	0.0000	147	0.0000	176
46	aigroup-000	0.0000	327	0.0005	434	0.0103	305	0.0000	85	-	1118
47	aigroup-001	0.0000	898	0.0005	452	0.0104	308	0.0001	857	-	1359
48	aigroup-002	0.0000	81	0.0003	354	0.0083	273	0.0001	736	-	1312
49	aigroup-003	0.0000	65	0.0000	126	0.0000	82	0.0000	235	-	1325
50	ailabs-001	0.0000	843	0.0090	1266	-	1217	0.0007	1239	0.0005	797
51	aimall-001	0.0000	254	0.0000	197	-	1422	0.0000	171	0.0000	189
52	aimall-002	0.0000	1086	0.0043	1195	-	1325	0.0012	1286	0.0005	850
53	aimall-003	0.0000	1006	0.0012	910	-	1310	0.0004	1096	0.0005	784
54	aipractix-000	0.0021	1305	0.0021	1053	0.0175	481	0.0390	1400	-	1406
55	aiseemu-001	0.0000	216	0.0000	179	0.0000	108	0.0000	150	0.0000	175
56	aiseemu-002	0.0000	159	0.0000	220	0.0000	136	0.0000	193	0.0000	207
57	aiunionface-000	0.0000	36	0.0000	168	-	1378	0.0000	283	0.0000	170
58	aize-001	0.0001	1241	0.0040	1177	0.0652	902	0.0026	1342	0.0022	1029

Table 93: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
59	aize-002	0.0000	509	0.0014	954	0.0230	641	0.0005	1195	0.0004	612
60	aize-003	0.0000	116	0.0000	258	0.0001	160	0.0000	249	0.0000	241
61	aize-006	0.0000	750	0.0004	373	0.0090	290	0.0001	854	-	1166
62	ajou-001	0.0000	714	0.0020	1040	-	1060	0.0001	733	0.0004	748
63	akiyama-000	0.0000	238	0.0000	209	-	1434	0.0000	187	0.0000	194
64	akiyama-001	0.0000	443	0.0000	110	-	1228	0.0000	66	0.0000	103
65	akiyama-002	0.0000	1183	0.0015	964	0.0213	591	0.0044	1364	-	1230
66	akiyama-003	0.0325	1345	0.0079	1254	0.0216	602	0.0270	1397	-	1239
67	akor-000	0.0000	848	0.0004	427	0.0112	339	0.0000	531	-	1218
68	alchera-000	0.0000	1196	0.0041	1190	-	1475	0.0004	1118	0.0014	1007
69	alchera-001	0.0000	1197	0.0041	1189	-	1165	0.0004	1117	0.0014	1006
70	alchera-002	0.0000	833	0.0008	716	-	1235	0.0001	816	0.0004	492
71	alchera-003	0.0001	1262	0.0013	932	0.0317	753	0.0002	952	0.0004	666
72	alchera-004	0.0000	610	0.0009	767	0.0228	633	0.0001	824	0.0004	518
73	alchera-005	0.0000	753	0.0009	764	0.0228	629	0.0001	819	0.0004	506
74	alchera-006	0.0000	656	0.0009	770	0.0228	635	0.0001	821	0.0004	529
75	alchera-007	0.0000	765	0.0009	763	0.0228	628	0.0001	818	0.0004	500
76	alchera-008	0.0000	781	0.0009	762	0.0228	627	0.0001	817	-	1101
77	alchera-009	0.0000	671	0.0004	423	0.0112	338	0.0002	933	-	1416
78	alfabeta-001	0.0005	1286	0.0650	1327	0.2142	971	0.0024	1335	0.0018	1020
79	alice-000	0.0000	163	0.0006	509	0.0133	378	0.0000	448	0.0004	499
80	alice-001	0.0000	340	0.0006	506	0.0133	377	0.0000	439	0.0004	478
81	alleyes-000	0.0000	589	0.0010	817	-	1275	0.0002	859	0.0004	696
82	allgovision-000	0.0007	1295	0.0062	1239	-	1119	0.0026	1340	0.0052	1069
83	alphaface-001	0.0000	638	0.0012	886	-	1442	0.0000	612	0.0004	703
84	alphaface-002	0.0000	818	0.0012	884	-	1178	0.0000	605	0.0004	694
85	alttekglobal-000	0.0000	598	0.0020	1034	0.0471	855	0.0000	619	-	1311
86	amplifiedgroup-001	0.0114	1328	0.1023	1339	-	1009	0.0189	1392	0.0279	1084
87	androvideo-000	0.0000	386	0.0000	98	-	1193	0.0000	55	0.0000	98
88	anke-000	-	1411	-	1479	-	1431	-	1467	0.0021	1025
89	anke-001	-	1383	-	1484	-	1457	-	1470	0.0047	1068
90	anke-002	-	1427	-	1378	-	1069	0.0004	1103	0.0009	939
91	anke-003	-	1390	-	1489	-	1470	0.0001	758	0.0004	581
92	anke-004	0.0000	852	0.0011	864	-	1136	0.0001	761	0.0004	709
93	anke-005	0.0000	555	0.0012	892	-	1332	0.0001	804	0.0004	736
94	anyvision-001	-	1474	-	1433	-	1249	-	1427	0.0091	1077
95	anyvision-002	0.0060	1317	0.0230	1298	-	1497	0.0070	1377	0.0090	1076
96	anyvision-004	0.0000	1055	0.0017	1001	-	1194	0.0001	802	0.0004	623
97	anyvision-005	0.0000	609	0.0013	919	-	1313	0.0000	524	0.0004	494
98	aratek-001	0.0000	19	0.0000	147	0.0000	93	0.0000	262	0.0000	149
99	armatura-001	0.0000	1106	0.0021	1054	0.0257	700	0.0005	1178	0.0005	809
100	armatura-002	0.0000	1041	0.0018	1008	0.0206	567	0.0003	1075	0.0004	745
101	armatura-003	0.0000	557	0.0012	897	0.0333	786	0.0004	1111	0.0004	595
102	arvis-000	0.0008	1299	0.0014	945	0.0195	544	0.0083	1379	-	1197
103	asr-000	0.0000	972	0.0007	623	0.0152	420	0.0002	976	-	1131
104	asr-001	0.0000	969	0.0006	555	0.0138	385	0.0002	997	-	1141
105	asusaics-000	0.0000	298	0.0000	40	-	1084	0.0000	131	0.0000	49
106	asusaics-001	0.0000	11	0.0000	157	-	1343	0.0000	274	0.0000	153
107	autentika-000	0.0000	87	0.0000	125	0.0000	81	0.0000	228	0.0000	117
108	autentika-001	0.0000	230	0.0000	199	0.0000	127	0.0000	180	0.0000	197
109	autentika-002	0.0000	499	0.0000	86	0.0000	59	0.0000	36	0.0000	85
110	autentika-003	0.0000	458	0.0000	73	0.0000	48	0.0000	25	-	1212
111	authenmetric-002	0.0000	232	0.0000	206	-	1425	0.0000	178	0.0000	201
112	authenmetric-003	0.0000	502	0.0000	89	0.0000	60	0.0000	45	0.0000	81
113	authenmetric-004	0.0000	323	0.0000	10	0.0000	7	0.0000	86	0.0000	16
114	authme-001	0.0000	1147	0.0009	784	0.0245	669	0.0003	1063	0.0005	786
115	authme-002	0.0000	899	0.0007	648	0.0179	488	0.0001	799	-	1380
116	aware-000	-	1398	-	1493	-	1473	-	1473	0.0005	805

Table 94: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
117	aware-001	-	1440	-	1355	-	1008	-	1438	0.0002	261
118	aware-002	-	1423	-	1372	-	1058	-	1449	0.0006	872
119	aware-003	-	1413	-	1475	-	1409	0.0016	1308	0.0013	996
120	aware-004	0.0000	993	0.0023	1068	-	1131	0.0002	890	0.0005	780
121	aware-005	0.0000	915	0.0020	1038	-	1309	0.0001	856	0.0004	716
122	aware-006	0.0000	874	0.0009	751	0.0249	677	0.0000	532	0.0004	622
123	aware-007	0.0000	755	0.0012	898	0.0325	772	0.0000	543	0.0004	658
124	aware-008	0.0000	769	0.0012	904	0.0327	774	0.0000	544	-	1110
125	awiros-001	0.0039	1309	0.0369	1315	-	1004	0.0386	1399	0.0872	1089
126	awiros-002	0.0000	1145	0.0038	1173	-	1208	0.0007	1235	0.0012	995
127	awiros-003	0.0000	454	0.0000	72	0.0000	45	0.0000	18	-	1222
128	awiros-004	0.0000	193	0.0000	182	0.0000	112	0.0000	154	-	1415
129	aximetria-001	0.0000	987	0.0010	845	0.0217	606	0.0001	852	0.0004	609
130	ayftech-001	0.0002	1270	0.0046	1202	-	1360	0.0043	1363	0.0011	960
131	ayftech-003	0.0000	89	0.0006	568	0.0123	358	0.0000	255	0.0000	143
132	ayonix-000	0.0053	1314	0.0341	1311	-	994	0.0113	1386	0.0137	1079
133	beethedata-000	0.0005	1283	0.0042	1192	0.0366	808	0.0002	897	0.0010	948
134	betaface-000	-	1355	-	1454	-	1324	-	1492	0.0011	965
135	betaface-002	-	1465	-	1413	-	1176	-	1415	0.0010	950
136	beyneai-000	0.0000	396	0.0000	106	0.0000	69	0.0000	59	0.0000	97
137	biocube-001	0.0006	1290	0.0391	1316	0.1207	952	0.0015	1302	0.0020	1024
138	biocube-002	0.0000	883	0.0046	1205	0.0721	928	0.0008	1247	0.0012	985
139	bioid-001	0.0005	1285	0.0019	1029	0.0324	771	0.0005	1168	-	1097
140	bioid-002	0.0000	265	0.0000	38	0.0000	25	0.0000	126	-	1160
141	bioid-003	0.0000	235	0.0000	210	0.0000	130	0.0000	188	-	1433
142	bioid-004	0.0000	451	0.0000	65	0.0000	44	0.0000	19	-	1217
143	bioidtechswiss-000	0.0000	588	0.0008	668	-	1274	0.0003	1032	0.0004	752
144	bioidtechswiss-001	0.0000	537	0.0007	612	-	1348	0.0000	484	0.0004	681
145	bioidtechswiss-002	0.0000	761	0.0007	618	-	1012	0.0000	491	0.0004	660
146	bioidtechswiss-003	0.0000	1012	0.0202	1294	0.0980	947	0.0005	1171	-	1454
147	biometric-vision-000	0.0000	764	0.0005	448	0.0107	325	0.0002	981	0.0004	532
148	biometrik-001	0.0000	231	0.0000	205	0.0000	129	0.0000	177	-	1444
149	biometrik-002	0.0000	491	0.0000	85	0.0000	56	0.0000	40	-	1251
150	bm-001	0.0000	292	0.0000	50	-	1112	0.0000	308	0.0000	53
151	boetech-001	0.0087	1324	0.0272	1302	0.2117	968	0.0032	1350	0.0160	1080
152	boetech-002	0.0087	1323	0.0272	1303	0.2117	969	0.0032	1351	0.0160	1081
153	bresee-000	0.0000	812	0.0010	831	-	1205	0.0002	898	0.0003	359
154	bresee-001	0.0000	792	0.0010	830	-	1027	0.0002	899	0.0003	353
155	bresee-002	0.0000	1076	0.0020	1046	0.0219	613	0.0008	1240	0.0004	576
156	btsdigital-001	0.0000	405	0.0009	748	0.0265	709	0.0000	419	-	1265
157	bynn-000	0.0130	1330	0.0048	1209	0.0089	285	0.0112	1385	-	1403
158	cakevn-001	0.0000	501	0.0007	629	0.0130	372	0.0000	47	-	1238
159	cakevn-002	0.0000	263	0.0008	660	0.0163	457	0.0000	122	-	1163
160	camvi-001	-	1372	-	1439	-	1262	-	1479	0.0010	956
161	camvi-002	0.0000	14	0.0000	156	-	1346	0.0000	273	0.0000	152
162	camvi-003	-	1496	-	1409	-	1170	0.0000	43	0.0000	80
163	camvi-004	0.0000	195	0.0000	255	-	1395	0.0000	153	0.0000	184
164	candour-001	0.0000	164	0.0000	238	0.0000	149	0.0000	220	0.0000	230
165	candour-002	0.0000	228	0.0000	201	0.0000	128	0.0000	179	-	1436
166	candour-003	0.0000	242	0.0000	190	0.0000	116	0.0000	166	-	1430
167	candour-004	0.0000	296	0.0000	48	0.0000	34	0.0000	141	-	1189
168	canon-002	0.0000	204	0.0000	184	-	1403	0.0000	159	0.0000	178
169	canon-003	0.0000	734	0.0008	661	0.0234	649	0.0000	590	0.0004	638
170	canon-004	0.0000	534	0.0008	663	0.0234	650	0.0000	591	0.0004	651
171	canon-005	0.0000	444	0.0044	1197	0.0269	716	0.0000	468	0.0004	540
172	cchonolulu-000	0.0054	1315	0.0395	1318	0.2802	976	0.0036	1354	0.0012	983
173	cchonolulu-001	0.0056	1316	0.0414	1319	0.2894	978	0.0038	1360	0.0012	987
174	cdachyd-001	0.0005	1284	0.0639	1323	0.2112	967	0.0023	1330	-	1478

Table 95: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
175	ceiaufgbr-000	0.0000	375	0.0000	22	0.0000	11	0.0000	103	-	1123
176	ceiec-001	-	1494	-	1411	-	1167	0.0029	1348	0.0023	1035
177	ceiec-002	0.0002	1269	0.0056	1229	-	1230	0.0036	1355	0.0031	1051
178	ceiec-003	0.0000	88	0.0013	938	-	1264	0.0001	673	0.0004	678
179	ceiec-004	0.0000	369	0.0008	701	-	1028	0.0000	507	0.0004	502
180	celloscope-001	0.0000	1124	0.0056	1228	0.0540	876	0.0022	1327	-	1446
181	chosun-000	0.0000	149	0.0000	214	-	1441	0.0000	190	0.0000	210
182	chosun-001	0.0000	93	0.0000	144	-	1314	0.0000	260	0.0000	137
183	chosun-002	0.0000	134	0.0000	229	-	1462	0.0000	203	0.0000	223
184	chtface-001	0.0000	18	0.0000	149	-	1321	0.0000	264	0.0000	150
185	chtface-002	0.0000	1136	0.0021	1056	-	1253	0.0002	1009	0.0007	915
186	chtface-003	0.0000	1061	0.0018	1012	-	1270	0.0001	689	0.0006	873
187	chtface-004	0.0000	413	0.0017	993	0.0320	758	0.0000	555	0.0004	693
188	chtface-005	0.0000	201	0.0017	994	0.0320	759	0.0000	560	0.0004	701
189	chtface-006	0.0000	328	0.0017	992	0.0320	757	0.0000	558	0.0004	687
190	chtface-007	0.0000	186	0.0017	995	0.0320	760	0.0000	562	-	1476
191	chtface-008	0.0000	652	0.0009	733	0.0221	616	0.0000	502	-	1493
192	chtface-009	0.0000	741	0.0009	732	0.0221	615	0.0000	496	-	1180
193	cib-000	0.0000	395	0.0000	102	-	1202	0.0001	827	0.0000	94
194	cib-001	0.0000	402	0.0000	94	-	1175	0.0000	51	0.0000	91
195	cist-001	0.0000	33	0.0005	489	0.0087	283	0.0000	287	0.0000	169
196	cist-002	0.0000	445	0.0048	1207	0.0128	366	0.0000	65	0.0000	102
197	cist-003	0.0000	40	0.0011	859	0.0343	797	0.0000	291	0.0000	166
198	cist-004	0.0000	357	0.0005	453	0.0108	328	0.0000	105	0.0000	33
199	clearviewai-000	0.0000	582	0.0003	332	0.0081	262	0.0000	576	0.0003	316
200	clearviewai-001	0.0000	561	0.0006	529	0.0174	477	0.0001	628	0.0004	468
201	clearviewai-002	0.0000	976	0.0009	790	0.0244	663	0.0001	651	-	1280
202	clearviewai-003	0.0000	982	0.0009	791	0.0244	664	0.0001	653	-	1297
203	closeli-001	0.0000	479	0.0000	63	0.0000	39	0.0000	12	0.0000	60
204	cloudmatrix-000	0.0000	984	0.0012	899	-	1250	0.0001	654	0.0004	459
205	cloudmatrix-001	0.0000	988	0.0028	1100	0.0225	619	0.0001	648	0.0004	450
206	cloudmatrix-002	0.0000	914	0.0028	1101	0.0225	620	0.0001	663	0.0004	465
207	cloudwalk-hr-000	-	1377	-	1447	-	1286	0.0003	1055	0.0008	929
208	cloudwalk-hr-001	0.0000	569	0.0007	638	-	1357	0.0001	635	0.0004	570
209	cloudwalk-hr-002	0.0000	807	0.0007	634	-	1197	0.0002	936	0.0004	711
210	cloudwalk-hr-003	0.0000	631	0.0008	708	-	1466	0.0001	687	0.0004	498
211	cloudwalk-hr-004	0.0000	556	0.0011	880	-	1371	0.0004	1105	0.0003	414
212	cloudwalk-mt-000	0.0000	809	0.0008	676	-	1203	0.0002	942	0.0004	710
213	cloudwalk-mt-002	0.0000	622	0.0003	324	-	1297	0.0001	634	0.0004	463
214	cloudwalk-mt-003	0.0000	790	0.0007	597	-	1023	0.0002	927	0.0004	707
215	cloudwalk-mt-004	0.0000	866	0.0009	722	0.0191	534	0.0002	955	0.0004	740
216	cloudwalk-mt-005	0.0000	587	0.0005	468	0.0130	373	0.0003	1011	0.0004	714
217	cloudwalk-mt-006	0.0000	751	0.0006	513	0.0158	440	0.0002	928	0.0004	708
218	cloudwalk-mt-007	0.0000	545	0.0006	516	0.0158	441	0.0002	931	0.0004	717
219	clova-000	0.0000	1115	0.0022	1064	-	1054	0.0006	1212	0.0005	787
220	cmcuni-001	0.0000	251	0.0005	476	0.0119	353	0.0000	174	0.0000	185
221	codeline-000	0.0000	72	0.0000	131	0.0000	83	0.0000	238	0.0000	121
222	cogent-000	-	1391	-	1492	-	1471	-	1471	0.0005	836
223	cogent-001	-	1379	-	1445	-	1289	-	1484	0.0004	762
224	cogent-002	-	1393	-	1483	-	1448	0.0002	869	0.0004	765
225	cogent-003	0.0000	1069	0.0018	1013	-	1426	0.0001	688	0.0004	618
226	cogent-004	0.0000	500	0.0000	90	-	1166	0.0000	46	0.0000	82
227	cogent-005	0.0000	339	0.0000	1	-	986	0.0000	77	0.0000	7
228	cogent-006	0.0000	48	0.0000	161	0.0000	100	0.0000	278	0.0000	162
229	cogent-007	0.0000	1066	0.0000	253	0.0000	155	0.0000	522	0.0000	247
230	cogent-008	0.0000	436	0.0010	846	0.0304	744	0.0000	579	0.0004	438
231	cognitec-000	0.0005	1280	0.0112	1277	-	1184	0.0007	1231	0.0007	922
232	cognitec-001	0.0003	1276	0.0110	1276	-	1492	0.0008	1250	0.0010	944

Table 96: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
233	cognitec-002	0.0001	1216	0.0069	1243	-	1333	0.0003	1070	0.0005	802
234	cognitec-003	0.0001	1217	0.0194	1292	0.0820	937	0.0003	1059	0.0005	817
235	cognitec-004	0.0001	1218	0.0037	1171	0.0580	883	0.0003	1060	0.0005	810
236	cognitec-005	0.0000	1126	0.0026	1084	0.0550	878	0.0002	881	0.0005	796
237	cognitec-006	0.0000	1120	0.0016	985	0.0511	861	0.0002	858	-	1309
238	cognitec-007	0.0000	1209	0.0025	1083	0.0736	930	0.0002	902	-	1302
239	cognitec-008	0.0000	1072	0.0030	1129	0.0523	869	0.0001	828	-	1137
240	compal-000	0.0000	128	0.0000	221	0.0000	140	0.0000	205	-	1466
241	compal-001	0.0000	1010	0.0060	1235	0.0320	761	0.0012	1284	-	1337
242	compal-002	0.0000	1013	0.0060	1236	0.0320	762	0.0012	1283	-	1450
243	cor-001	0.0000	662	0.0006	531	-	1482	0.0002	985	0.0004	635
244	coretech-000	0.0000	403	0.0000	93	0.0000	62	0.0000	50	0.0000	90
245	coretech-001	0.0000	1189	0.0033	1153	0.0677	911	0.0005	1192	0.0011	975
246	coretech-002	0.0000	21	0.0013	942	0.0301	738	0.0000	375	0.0004	650
247	coretech-003	0.0000	129	0.0012	911	0.0282	721	0.0000	501	-	1467
248	coretech-004	0.0000	3	0.0013	943	0.0301	739	0.0000	377	-	1370
249	corsight-001	0.0000	619	0.0006	556	-	1296	0.0001	835	0.0004	579
250	corsight-002	0.0000	669	0.0005	497	0.0152	422	0.0001	769	0.0004	598
251	corsight-003	0.0000	594	0.0006	558	0.0175	480	0.0001	741	0.0004	614
252	csc-001	0.0000	1165	0.0030	1127	-	1459	0.0002	885	0.0005	789
253	csc-002	0.0015	1303	0.0033	1144	-	1359	0.0006	1220	0.0006	904
254	csc-003	0.0015	1304	0.0033	1145	0.0445	853	0.0006	1219	0.0006	905
255	ctbcbank-000	0.0001	1223	0.0051	1216	-	1078	0.0011	1273	0.0019	1021
256	ctbcbank-001	0.0000	1141	0.0036	1169	-	1358	0.0005	1187	0.0010	945
257	ctbcbank-002	0.0000	137	0.0006	500	0.0150	415	0.0000	414	-	1462
258	cu-face-002	0.0000	363	0.0000	27	0.0000	19	0.0000	113	0.0000	24
259	cu-face-003	0.0000	1074	0.0009	782	0.0245	668	0.0003	1020	0.0004	706
260	cu-face-004	0.0000	928	0.0007	650	0.0179	489	0.0001	796	-	1498
261	cubox-001	0.0000	455	0.0000	71	-	1134	0.0000	16	0.0000	74
262	cubox-002	0.0000	978	0.0006	544	0.0159	444	0.0002	984	0.0005	854
263	cubox-003	0.0000	102	0.0010	837	0.0210	583	0.0002	882	0.0004	679
264	cubox-004	0.0000	1051	0.0009	781	0.0193	539	0.0003	1082	-	1098
265	cubox-005	0.0000	1044	0.0009	785	0.0193	540	0.0003	1084	-	1443
266	cudocommunication-001	0.0000	82	0.0000	121	0.0000	80	0.0000	226	0.0000	120
267	cuhkee-001	0.0000	649	0.0011	875	-	1495	0.0000	370	0.0004	603
268	cvartel-001	0.0000	13	0.0004	375	0.0096	301	0.0000	378	-	1361
269	cybercore-000	0.0000	643	0.0073	1247	-	1447	0.0001	717	0.0005	804
270	cybercore-001	0.0000	1088	0.0001	275	0.0014	170	0.0002	877	0.0002	270
271	cybercore-002	0.0000	1097	0.0001	274	0.0014	169	0.0002	874	0.0002	266
272	cybercore-003	0.0000	710	0.0003	334	0.0060	209	0.0005	1194	0.0003	323
273	cyberextruder-001	0.0029	1308	0.0293	1304	-	1450	0.0024	1333	0.0029	1048
274	cyberextruder-002	0.0013	1300	0.0840	1335	-	1391	0.0027	1343	0.0028	1044
275	cyberextruder-003	0.0000	1090	0.0077	1251	0.0887	943	0.0001	838	0.0006	894
276	cyberextruder-004	0.0000	1084	0.0097	1268	0.1025	949	0.0001	811	0.0007	907
277	cyberextruder-005	0.0000	380	0.0000	23	0.0000	14	0.0000	100	-	1126
278	cyberextruder-006	0.0000	91	0.0000	141	0.0000	90	0.0000	254	-	1348
279	cyberlink-000	-	1352	-	1459	-	1344	0.0006	1218	0.0008	927
280	cyberlink-001	-	1401	-	1494	-	1477	0.0073	1378	0.0005	792
281	cyberlink-002	-	1477	-	1419	-	1212	0.0004	1090	0.0004	769
282	cyberlink-003	0.0000	414	0.0007	628	-	1188	0.0001	643	0.0004	510
283	cyberlink-004	0.0000	639	0.0010	854	-	1443	0.0004	1095	0.0003	416
284	cyberlink-005	0.0000	791	0.0009	750	-	1029	0.0003	1076	0.0003	406
285	cyberlink-006	0.0000	111	0.0005	437	-	1293	0.0000	346	0.0003	363
286	cyberlink-007	0.0000	432	0.0003	301	0.0067	228	0.0000	324	0.0003	361
287	cyberlink-008	0.0000	153	0.0004	413	0.0106	319	0.0000	336	0.0003	367
288	cyberlink-009	0.0000	44	0.0004	411	0.0106	318	0.0000	350	0.0003	364
289	cyberlink-010	0.0000	300	0.0004	406	0.0106	314	0.0000	333	0.0003	355
290	cyberlink-011	0.0000	399	0.0004	407	0.0106	315	0.0000	323	0.0003	358

Table 97: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
291	cyberlink-012	0.0000	140	0.0004	414	0.0106	320	0.0000	338	0.0003	368
292	cyberlink-013	0.0000	169	0.0004	415	0.0106	321	0.0000	341	0.0003	369
293	cyberlink-014	0.0000	417	0.0004	408	0.0106	316	0.0000	325	-	1300
294	cyberlink-015	0.0000	75	0.0004	409	0.0106	317	0.0000	345	-	1315
295	cyberlink-016	0.0000	333	0.0004	405	0.0106	313	0.0000	326	-	1109
296	dahua-000	-	1453	-	1370	-	1047	-	1447	0.0000	27
297	dahua-001	-	1395	-	1482	-	1451	0.0000	194	0.0000	206
298	dahua-002	-	1405	-	1472	-	1400	0.0024	1336	0.0022	1032
299	dahua-003	0.0000	289	0.0019	1019	-	1107	0.0002	943	0.0003	309
300	dahua-004	0.0000	6	0.0009	789	-	1339	0.0001	798	0.0003	288
301	dahua-005	0.0000	442	0.0000	246	-	1226	0.0000	517	0.0003	360
302	dahua-006	0.0000	60	0.0000	249	-	1366	0.0000	586	0.0003	413
303	dahua-007	0.0000	156	0.0000	250	0.0000	157	0.0000	582	0.0003	415
304	daon-000	0.0000	1153	0.0028	1109	0.0577	882	0.0014	1296	0.0015	1010
305	daon-001	0.0000	335	0.0009	728	0.0206	566	0.0000	490	-	1105
306	daon-002	0.0000	205	0.0004	426	0.0094	295	0.0000	158	-	1409
307	daon-003	0.0000	342	0.0008	679	0.0174	478	0.0000	391	-	1096
308	datech-000	0.0000	885	0.0004	359	0.0089	288	0.0000	515	0.0003	295
309	datech-001	0.0000	712	0.0004	358	0.0089	287	0.0000	520	-	1161
310	datech-002	0.0000	872	0.0004	360	0.0089	289	0.0000	516	-	1247
311	decatur-000	0.0000	960	0.0020	1035	-	990	0.0004	1136	0.0005	777
312	decatur-001	0.0000	535	0.0009	777	0.0194	542	0.0001	698	0.0004	568
313	decloakface-001	0.0000	788	0.0016	983	0.0392	829	0.0001	746	0.0005	864
314	decloakface-002	0.0000	189	0.0011	865	0.0218	610	0.0001	632	-	1412
315	deepglint-001	0.0000	290	0.0000	47	-	1110	0.0000	144	0.0000	51
316	deepglint-002	0.0000	566	0.0004	412	-	1381	0.0002	957	0.0004	549
317	deepglint-003	0.0000	528	0.0004	410	-	1335	0.0002	956	0.0004	546
318	deepglint-004	0.0000	707	0.0005	439	0.0130	368	0.0002	977	0.0004	504
319	deepglint-005	0.0000	1017	0.0019	1022	0.0438	850	0.0006	1210	0.0006	902
320	deepsea-001	0.0000	2	0.0000	154	-	1337	0.0000	270	0.0000	157
321	deepsense-000	0.0000	59	0.0006	560	-	1363	0.0000	431	0.0004	427
322	deepsense-001	0.0000	132	0.0006	569	0.0191	529	0.0000	445	0.0004	445
323	deepsense-002	0.0000	382	0.0006	567	0.0191	528	0.0000	437	0.0004	436
324	deepsense-003	0.0000	426	0.0004	394	0.0110	330	0.0000	358	0.0003	326
325	deepsense-004	0.0000	654	0.0007	614	0.0179	490	0.0001	706	-	1490
326	deepsense-005	0.0000	322	0.0005	494	0.0144	401	0.0001	669	-	1115
327	deepsense-006	0.0000	69	0.0005	442	0.0113	341	0.0000	451	-	1329
328	deepsense-007	0.0000	385	0.0005	441	0.0113	340	0.0000	436	-	1282
329	deepsense-008	0.0000	171	0.0005	443	0.0113	342	0.0000	449	-	1491
330	dermalog-001	-	1469	-	1427	-	1233	-	1425	0.0013	998
331	dermalog-002	-	1426	-	1375	-	1065	-	1452	0.0013	997
332	dermalog-003	-	1464	-	1418	-	1199	-	1418	0.0025	1038
333	dermalog-004	-	1421	-	1382	-	1081	-	1456	0.0090	1075
334	dermalog-005	-	1478	0.0031	1133	-	1214	0.0013	1292	0.0041	1062
335	dermalog-006	0.0005	1282	0.0031	1137	-	987	0.0013	1291	0.0041	1063
336	dermalog-007	0.0000	645	0.0006	512	-	1491	0.0000	563	0.0002	275
337	dermalog-008	0.0000	1121	0.0031	1136	-	1311	0.0006	1203	0.0003	300
338	dermalog-009	0.0000	1130	0.0031	1135	0.0148	409	0.0006	1201	0.0003	298
339	dermalog-010	0.0000	1132	0.0031	1134	0.0148	408	0.0006	1200	0.0003	296
340	dermalog-011	0.0000	1179	0.0005	445	0.0116	347	0.0001	646	0.0003	317
341	dermalog-012	0.0000	590	0.0003	344	0.0068	232	0.0003	1027	0.0005	869
342	dermalog-013	0.0000	680	0.0002	297	0.0051	197	0.0002	868	-	1393
343	dermalog-014	0.0000	600	0.0003	343	0.0068	231	0.0003	1026	-	1306
344	dermalog-015	0.0000	687	0.0003	346	0.0068	234	0.0003	1025	-	1439
345	dermalog-016	0.0000	8	0.0003	345	0.0068	233	0.0001	627	-	1368
346	dicio-001	0.0005	1288	0.0649	1324	0.2136	970	0.0024	1332	0.0012	984
347	dicio-002	0.0000	225	0.0007	605	0.0185	510	0.0000	426	-	1442
348	didiglobalface-001	0.0000	613	0.0012	885	-	1290	0.0000	614	0.0004	697

Table 98: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
349	didiglobalface-002	0.0000	747	0.0012	882	0.0247	674	0.0000	608	0.0004	690
350	didiglobalface-003	0.0000	693	0.0012	888	0.0247	675	0.0000	611	-	1432
351	digidata-000	0.0000	575	0.0023	1071	0.0375	819	0.0004	1148	0.0006	889
352	digidata-001	0.0000	706	0.0023	1069	0.0375	817	0.0004	1146	0.0006	887
353	digitalbarriers-000	-	1381	-	1449	-	1299	-	1488	0.0019	1022
354	digitalbarriers-001	-	1451	-	1368	-	1045	-	1445	0.0018	1019
355	digitalbarriers-002	0.0001	1251	0.0045	1199	-	1083	0.0028	1345	0.0027	1041
356	dps-000	0.0000	221	0.0000	178	0.0000	109	0.0000	149	0.0000	174
357	dralmey-001	0.0000	727	0.0007	599	0.0185	509	0.0000	422	-	1147
358	dsk-000	0.0000	463	0.0000	75	-	1142	0.0000	24	0.0000	69
359	einetworks-000	0.0000	1142	0.0017	999	-	1306	0.0002	939	0.0005	843
360	einetworksindia-000	0.0000	1111	0.0018	1009	0.0516	863	0.0002	1003	0.0005	845
361	einetworksindia-001	0.0000	1107	0.0018	1010	0.0516	864	0.0002	1005	0.0005	846
362	einetworksindia-002	0.0000	571	0.0006	518	0.0159	446	0.0000	621	-	1384
363	ekin-002	0.0000	187	0.0000	256	0.0004	164	0.0000	340	0.0000	240
364	element-000	0.0000	632	0.0008	714	0.0192	537	0.0000	512	0.0003	375
365	enface-000	0.0000	338	0.0012	907	0.0305	746	0.0000	542	0.0004	605
366	enface-001	0.0000	428	0.0012	905	0.0304	745	0.0000	469	0.0004	588
367	enface-002	0.0000	758	0.0004	374	0.0084	276	0.0000	408	0.0004	432
368	enqura-001	0.0003	1271	0.0203	1295	0.1737	963	0.0023	1329	-	1229
369	enqura-002	0.0000	266	0.0000	33	0.0000	22	0.0000	119	-	1152
370	enqura-003	0.0000	787	0.0005	436	0.0123	359	0.0000	443	-	1144
371	eocortex-000	0.0095	1326	0.0602	1322	-	1318	0.0094	1382	0.0059	1070
372	ercacat-001	0.0000	199	0.0005	481	-	1399	0.0000	551	0.0003	374
373	essi-000	0.0000	744	0.0011	860	0.0240	658	0.0002	919	-	1184
374	ethiopianlmis-000	0.0001	1220	0.0019	1028	0.0159	448	0.0009	1257	-	1479
375	euronovate-001	0.0255	1341	0.0102	1273	0.0517	866	0.0021	1323	0.0004	756
376	euronovate-002	0.0000	775	0.0010	802	0.0265	708	0.0000	440	0.0004	686
377	euronovate-003	0.0000	615	0.0006	523	0.0159	443	0.0002	935	0.0004	698
378	euronovate-004	0.0000	1116	0.0016	976	0.0236	651	0.0015	1303	0.0006	890
379	everai-000	-	1407	-	1473	-	1405	-	1464	0.0004	472
380	everai-001	-	1362	-	1461	-	1356	0.0004	1122	0.0004	763
381	everai-002	-	1409	-	1469	-	1394	0.0002	946	0.0004	470
382	everai-paravision-003	0.0000	975	0.0008	715	-	1201	0.0002	879	0.0004	512
383	expasoft-000	0.0000	237	0.0000	207	-	1432	0.0000	185	0.0000	195
384	expasoft-001	0.0000	477	0.0000	53	-	1121	0.0000	2	0.0000	64
385	expasoft-002	0.0000	365	0.0000	31	0.0000	21	0.0000	112	0.0000	28
386	f8-001	0.0003	1274	0.0059	1231	-	1385	0.0035	1353	0.0030	1050
387	f8-002	0.0000	1200	0.0150	1286	0.0685	915	0.0005	1150	0.0013	1002
388	facehawk-000	0.0000	1073	0.0357	1312	0.0720	927	0.0021	1324	0.0009	938
389	facehawk-001	0.0299	1343	0.0098	1269	0.0236	652	0.0253	1396	-	1326
390	faceint-001	0.0001	1240	0.0190	1290	0.0898	944	0.0014	1301	-	1418
391	facelocate-001	0.0000	77	0.0019	1025	0.0329	778	0.0000	575	-	1310
392	facelocate-002	0.0000	585	0.0025	1081	0.0387	827	0.0001	633	-	1327
393	faceonlive-001	0.0000	1171	0.0029	1116	0.0481	858	0.0013	1294	0.0011	961
394	faceonlive-002	0.0002	1267	0.0009	779	0.0075	248	0.0008	1244	0.0008	935
395	faceonlive-003	0.0000	1033	0.0202	1293	0.0980	946	0.0005	1170	-	1249
396	facephi-000	0.0000	512	0.0004	362	0.0090	291	0.0001	767	0.0004	451
397	facephi-001	0.0000	925	0.0034	1156	0.0499	860	0.0003	1042	-	1497
398	facephi-002	0.0000	688	0.0016	981	0.0286	728	0.0002	953	-	1435
399	facephi-003	0.0000	244	0.0006	546	0.0153	423	0.0001	740	-	1429
400	faceplugin-000	0.0000	949	0.0026	1095	0.0636	898	0.0001	787	-	1162
401	facesoft-000	0.0000	84	0.0000	124	-	1260	0.0000	230	0.0000	115
402	facetag-000	0.0000	211	0.0000	175	0.0000	107	0.0000	145	0.0000	177
403	facetag-001	0.0000	389	0.0000	101	0.0000	67	0.0000	54	0.0000	100
404	facetag-002	0.0000	456	0.0000	69	0.0000	46	0.0000	15	0.0000	72
405	facex-001	0.0001	1260	0.0360	1314	-	1498	0.0047	1367	0.0027	1043
406	facex-002	0.0001	1261	0.0360	1313	0.2663	974	0.0047	1366	0.0027	1042

Table 99: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
407	facia-001	0.0000	118	0.0005	438	0.0122	357	0.0001	625	0.0004	493
408	facia-002	0.0000	1001	0.0017	997	0.0185	508	0.0002	996	-	1385
409	faicetech-000	0.0000	877	0.0007	615	0.0130	371	0.0002	992	-	1237
410	farfaces-001	0.0000	1134	0.0007	617	0.0061	211	0.0003	1049	0.0003	341
411	fastenterprises-000	0.0000	565	0.0082	1260	0.0169	470	0.0001	837	0.0003	365
412	fci-000	0.0000	383	0.0000	99	0.0000	66	0.0000	56	-	1283
413	fdu-000	-	1376	-	1452	-	1316	-	1489	0.0011	978
414	fdu-001	-	1378	-	1446	-	1284	-	1486	0.0009	943
415	fedu-001	0.0000	32	0.0000	167	0.0000	101	0.0000	286	-	1389
416	fiberhome-nanjing-002	0.0000	903	0.0006	563	-	1369	0.0001	712	0.0004	522
417	fiberhome-nanjing-003	0.0000	378	0.0004	384	-	1037	0.0000	101	0.0003	303
418	fiberhome-nanjing-004	0.0000	507	0.0004	385	-	1173	0.0000	41	0.0003	305
419	fincore-000	0.0000	683	0.0008	710	0.0185	513	0.0001	631	0.0004	683
420	firstcreditkz-001	0.0000	1011	0.0019	1033	0.0321	765	0.0000	603	0.0004	601
421	firstcreditkz-002	0.0000	668	0.0010	848	0.0232	648	0.0000	550	0.0004	525
422	firstcreditkz-003	0.0000	1146	0.0010	839	0.0219	611	0.0000	529	-	1471
423	firstcreditkz-004	0.0000	713	0.0004	401	0.0124	360	0.0000	329	-	1157
424	foomobi-001	0.0007	1293	0.0000	25	0.0000	16	0.0020	1320	0.0011	962
425	foomobi-002	0.0007	1292	0.0000	46	0.0000	31	0.0020	1321	0.0011	963
426	fourthlinecom-001	0.0000	943	0.0004	398	0.0078	255	0.0005	1165	-	1425
427	fpt-000	0.0001	1227	0.0007	625	0.0076	250	0.0006	1222	-	1394
428	fpt-001	0.0001	1230	0.0005	485	0.0067	229	0.0006	1206	-	1410
429	fpttelecom-000	0.0000	210	0.0000	176	0.0000	106	0.0000	146	-	1405
430	fpttelecom-001	0.0001	1228	0.0023	1074	0.0198	546	0.0009	1260	-	1253
431	fraudcom-000	0.0000	940	0.0055	1225	0.0308	750	0.0002	950	0.0004	684
432	frpkauai-001	0.0000	1021	0.0024	1080	0.0360	806	0.0001	658	0.0004	720
433	frpkauai-002	0.0000	1027	0.0019	1031	0.0321	763	0.0000	597	0.0004	587
434	frpkauai-003	0.0000	705	0.0006	532	0.0145	404	0.0000	495	-	1165
435	frpkauai-004	0.0001	1250	0.0051	1214	0.0730	929	0.0011	1274	-	1284
436	frpkauai-005	0.0000	752	0.0006	533	0.0145	405	0.0000	497	-	1176
437	fujitsulab-000	0.0000	1186	0.0033	1149	-	1345	0.0005	1169	0.0002	262
438	fujitsulab-001	0.0000	318	0.0014	952	-	1098	0.0002	962	0.0003	393
439	fujitsulab-002	0.0000	288	0.0009	740	-	1108	0.0001	795	0.0003	304
440	fujitsulab-003	0.0000	7	0.0008	682	0.0166	463	0.0001	760	0.0001	257
441	g42-intellibrain-001	0.0000	359	0.0000	29	0.0000	17	0.0000	115	0.0000	26
442	g42-intellibrain-002	0.0000	25	0.0000	152	0.0000	95	0.0000	267	0.0000	146
443	geo-000	0.0000	320	0.0000	12	-	1007	0.0000	88	0.0000	17
444	geo-001	0.0000	778	0.0011	862	-	995	0.0000	598	0.0004	583
445	geo-002	0.0000	880	0.0015	963	0.0332	784	0.0001	622	0.0004	734
446	geo-003	0.0000	876	0.0010	813	0.0244	665	0.0000	386	0.0004	722
447	geo-004	0.0000	777	0.0005	492	0.0138	387	0.0001	725	0.0004	530
448	gistouch-000	0.0000	1198	0.0188	1289	0.1345	955	0.0018	1314	0.0007	908
449	gistouch-001	0.0000	354	0.0002	279	0.0027	181	0.0001	715	0.0003	314
450	glory-000	-	1386	-	1485	-	1463	0.0053	1372	0.0013	1000
451	glory-001	0.0160	1332	0.0314	1308	-	1025	0.0051	1371	0.0010	949
452	glory-002	0.0003	1272	0.0045	1198	-	1105	0.0015	1304	0.0011	977
453	glory-003	0.0000	1015	0.0027	1099	-	1397	0.0004	1104	0.0005	795
454	glory-004	0.0000	997	0.0020	1044	0.0345	799	0.0001	774	0.0004	723
455	glory-005	0.0000	909	0.0020	1045	0.0345	800	0.0001	781	0.0004	724
456	glory-006	0.0000	991	0.0020	1043	0.0345	798	0.0001	771	0.0004	721
457	glory-007	0.0000	1025	0.0020	1051	0.0403	840	0.0002	958	0.0005	778
458	gorilla-000	-	1403	-	1499	-	1490	-	1474	0.0004	776
459	gorilla-001	-	1497	-	1412	-	1174	0.0003	1079	0.0004	767
460	gorilla-002	-	1473	-	1434	-	1248	0.0003	1081	0.0004	770
461	gorilla-003	-	1481	-	1422	-	1222	0.0003	1080	0.0004	768
462	gorilla-004	0.0000	90	0.0008	694	-	1303	0.0000	479	0.0003	338
463	gorilla-005	0.0000	305	0.0008	702	-	1090	0.0000	472	0.0003	335
464	gorilla-006	0.0000	184	0.0006	581	-	1484	0.0000	428	0.0003	343

Table 100: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
465	gorilla-007	0.0000	754	0.0009	795	0.0252	689	0.0001	694	0.0004	659
466	gorilla-008	0.0000	772	0.0009	794	0.0259	701	0.0001	693	0.0004	673
467	gorilla-009	0.0000	673	0.0010	825	0.0276	719	0.0001	659	0.0004	654
468	gpstechvn-000	0.0000	257	0.0000	194	0.0000	123	0.0000	169	0.0000	188
469	graymatics-001	0.0000	262	0.0010	799	0.0210	579	0.0001	832	0.0004	557
470	griaule-000	0.0000	1161	0.0026	1096	0.0531	874	0.0004	1147	0.0010	946
471	griaule-001	0.0000	471	0.0012	908	0.0366	811	0.0000	416	0.0004	665
472	griaule-002	0.0000	404	0.0007	627	0.0209	570	0.0000	593	0.0004	578
473	gtelots-000	0.0000	1018	0.0003	351	0.0081	269	0.0004	1101	-	1150
474	gtelots-001	0.0000	1036	0.0009	754	0.0177	484	0.0004	1115	-	1321
475	hertasecurity-000	0.0133	1331	0.0077	1252	-	1115	0.0025	1339	0.0243	1082
476	hertasecurity-001	0.0000	167	0.0000	257	0.0000	158	0.0000	398	0.0001	248
477	hertasecurity-002	0.0000	223	0.0000	203	0.0000	124	0.0000	396	0.0000	235
478	hertasecurity-003	0.0000	162	0.0000	239	0.0000	148	0.0000	343	0.0000	237
479	hik-001	0.0000	261	0.0000	260	-	1074	0.0000	123	0.0000	43
480	hisign-001	0.0000	45	0.0000	170	0.0000	103	0.0000	288	0.0000	167
481	hisign-002	0.0000	934	0.0006	548	0.0150	416	0.0001	788	0.0003	401
482	hisign-003	0.0000	930	0.0008	711	0.0237	654	0.0001	789	-	1489
483	hyperverge-001	0.0000	1195	0.0072	1244	-	1374	0.0015	1306	0.0014	1005
484	hyperverge-002	0.0000	49	0.0008	678	0.0210	585	0.0002	995	0.0004	569
485	hyperverge-003	0.0000	277	0.0008	671	0.0210	580	0.0002	994	0.0004	555
486	hyperverge-004	0.0000	544	0.0008	707	0.0218	607	0.0002	945	0.0004	591
487	hyperverge-005	0.0000	467	0.0008	672	0.0210	581	0.0002	989	0.0004	560
488	hyperverge-006	0.0000	510	0.0008	673	0.0210	582	0.0002	991	-	1242
489	hyperverge-007	0.0000	439	0.0008	677	0.0210	584	0.0002	993	-	1292
490	hyundai-roboticslab-000	0.0000	326	0.0011	877	0.0246	670	0.0001	757	-	1119
491	hyundai-roboticslab-001	0.0000	67	0.0105	1275	0.1972	964	0.0007	1234	-	1330
492	hyundai-roboticslab-002	0.0000	95	0.0020	1050	0.0299	737	0.0002	974	-	1344
493	hyundai-roboticslab-003	0.0000	291	0.0005	444	0.0136	381	0.0000	364	-	1186
494	hyundai-roboticslab-004	0.0000	495	0.0007	607	0.0200	547	0.0000	384	-	1246
495	hzailu-001	0.0000	1112	0.0016	982	0.0449	854	0.0003	1071	0.0005	849
496	hzailu-002	0.0000	1110	0.0015	970	0.0424	843	0.0003	1073	0.0005	848
497	hzailu-003	0.0000	722	0.0004	365	0.0081	264	0.0002	873	0.0003	379
498	hzailu-004	0.0000	548	0.0004	368	0.0081	267	0.0002	876	0.0003	388
499	hzailu-005	0.0000	646	0.0004	369	0.0081	268	0.0002	875	0.0003	391
500	hzailu-006	0.0000	849	0.0004	366	0.0081	265	0.0002	871	0.0003	381
501	hzailu-007	0.0000	963	0.0004	364	0.0081	263	0.0003	1033	-	1102
502	hzailu-008	0.0000	905	0.0004	367	0.0081	266	0.0003	1035	-	1328
503	i2v-001	0.0000	1109	0.0018	1016	0.0341	796	0.0001	716	-	1183
504	i2v-002	0.0000	1026	0.0012	881	0.0244	666	0.0000	538	-	1287
505	icehouse-000	0.0000	256	0.0000	195	0.0000	121	0.0000	172	-	1422
506	icm-002	0.0000	105	0.0001	266	-	1291	0.0000	243	0.0000	232
507	icm-003	0.0000	104	0.0001	265	0.0023	176	0.0000	245	0.0000	231
508	icm-004	0.0000	1158	0.0033	1155	0.0698	919	0.0006	1216	0.0010	957
509	icm-005	0.0002	1266	0.0117	1280	0.1223	953	0.0018	1317	0.0014	1009
510	icomm-000	0.0000	644	0.0004	392	0.0098	304	0.0000	574	0.0003	344
511	icthtc-000	0.0001	1258	0.0047	1206	-	1148	0.0028	1347	0.0029	1045
512	id3-001	-	1487	-	1403	-	1145	-	1409	0.0043	1066
513	id3-002	-	1459	-	1361	-	1034	-	1442	0.0032	1052
514	id3-003	0.0001	1219	0.0052	1219	-	1038	0.0016	1309	0.0011	973
515	id3-004	0.0000	1211	0.0052	1217	-	1144	0.0015	1307	0.0011	974
516	id3-005	0.0000	770	0.0074	1249	-	1021	0.0002	896	0.0005	793
517	id3-006	0.0000	1028	0.0009	788	-	1239	0.0004	1119	0.0005	835
518	id3-007	0.0000	697	0.0041	1183	-	1411	0.0001	764	0.0004	632
519	id3-008	0.0000	188	0.0006	554	0.0184	505	0.0001	829	0.0004	442
520	id3-009	0.0000	953	0.0003	353	0.0088	284	0.0001	776	-	1187
521	id3-010	0.0000	387	0.0001	277	0.0030	182	0.0000	466	-	1288
522	id3-011	0.0000	415	0.0005	440	0.0119	354	0.0000	465	-	1262

Table 101: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
523	idbio-000	0.0000	409	0.0008	688	0.0189	517	0.0000	606	-	1259
524	idemia-003	-	1397	-	1501	-	1496	0.0000	450	0.0004	476
525	idemia-004	0.0000	834	0.0006	528	-	1238	0.0000	489	0.0004	490
526	idemia-005	0.0000	268	0.0004	416	-	1059	0.0000	395	0.0003	380
527	idemia-006	0.0000	101	0.0004	421	-	1283	0.0000	400	0.0003	386
528	idemia-007	0.0000	234	0.0004	424	-	1429	0.0000	397	0.0003	390
529	idemia-008	0.0000	516	0.0004	418	0.0078	259	0.0000	382	0.0003	383
530	idemia-009	0.0000	57	0.0004	402	0.0077	253	0.0000	402	0.0003	389
531	idemia-010	0.0000	313	0.0004	357	0.0074	242	0.0000	135	0.0003	334
532	idemia-011	0.0000	452	0.0003	303	0.0055	202	0.0000	17	-	1224
533	identy-000	0.0000	1199	0.0020	1047	0.0152	419	0.0012	1280	0.0004	728
534	identy-001	0.0000	1205	0.0013	939	0.0398	835	0.0002	983	-	1365
535	identy-002	0.0000	1208	0.0013	940	0.0401	836	0.0002	979	-	1437
536	identy-003	0.0000	1151	0.0011	870	0.0247	673	0.0001	729	-	1457
537	idrnd-000	0.0000	902	0.0020	1037	0.0566	880	0.0003	1085	-	1376
538	idrnd-001	0.0000	459	0.0006	534	0.0182	494	0.0000	417	-	1211
539	igearx-face-000	0.0000	663	0.0006	508	0.0153	425	0.0004	1113	0.0004	672
540	iit-000	-	1435	-	1387	-	1092	0.0007	1224	0.0011	966
541	iit-001	0.0000	1020	0.0806	1334	-	1099	0.0001	850	0.0004	640
542	iit-002	0.0000	1154	0.0021	1052	-	1180	0.0009	1259	0.0005	859
543	iit-003	0.0000	815	0.0008	709	-	1177	0.0000	505	0.0004	453
544	imds-software-001	0.0000	135	0.0000	228	0.0000	143	0.0000	200	0.0000	220
545	imds-software-002	0.0000	191	0.0000	180	0.0000	110	0.0000	156	0.0000	182
546	imds-software-003	0.0000	126	0.0000	222	0.0000	139	0.0000	206	0.0000	215
547	imperial-000	0.0000	344	0.0000	5	-	1000	0.0000	83	0.0000	1
548	imperial-001	-	1374	-	1450	-	1308	0.0000	253	0.0000	142
549	imperial-002	0.0000	214	0.0000	177	-	1389	0.0000	151	0.0000	173
550	incode-000	-	1472	-	1430	-	1244	-	1429	0.0035	1055
551	incode-001	-	1445	-	1351	-	989	-	1430	0.0035	1054
552	incode-002	-	1482	-	1421	-	1225	0.0004	1140	0.0007	910
553	incode-003	-	1488	-	1402	-	1146	0.0004	1139	0.0007	909
554	incode-004	-	1406	-	1474	-	1406	0.0004	1141	0.0007	911
555	incode-005	0.0000	108	0.0008	670	-	1292	0.0001	708	0.0004	644
556	incode-006	0.0000	717	0.0005	458	-	1063	0.0001	810	0.0004	420
557	incode-007	0.0000	974	0.0009	757	-	1200	0.0002	901	0.0004	538
558	incode-008	0.0000	923	0.0009	760	-	1439	0.0002	907	0.0004	550
559	incode-009	0.0000	964	0.0009	756	0.0255	695	0.0002	903	0.0004	531
560	incode-010	0.0000	912	0.0009	758	0.0255	696	0.0002	910	0.0004	541
561	incode-011	0.0000	926	0.0009	761	0.0255	698	0.0002	909	0.0004	552
562	incode-012	0.0000	908	0.0009	759	0.0255	697	0.0002	911	0.0004	543
563	incode-013	0.0001	1263	0.0012	900	0.0254	693	0.0005	1179	0.0004	754
564	incode-014	0.0000	309	0.0006	564	0.0166	462	0.0001	843	-	1172
565	incode-015	0.0000	384	0.0007	644	0.0195	543	0.0000	435	-	1281
566	incode-016	0.0000	227	0.0010	801	0.0268	714	0.0000	540	-	1438
567	infocert-001	0.0000	1052	0.0059	1234	0.0424	844	0.0001	699	0.0006	875
568	infocert-002	0.0084	1322	0.0041	1191	0.0084	277	0.0085	1380	-	1258
569	infocert-003	0.0089	1325	0.0042	1193	0.0077	252	0.0087	1381	-	1133
570	infodif-001	0.0000	962	0.0010	844	0.0242	662	0.0003	1058	-	1095
571	innefulabs-000	0.0000	704	0.0024	1078	-	1073	0.0003	1047	0.0005	828
572	innominds-001	0.0000	1094	0.0030	1126	0.0637	899	0.0010	1271	0.0008	932
573	innov8tif-000	0.0000	1144	0.0029	1123	0.0657	904	0.0004	1133	-	1413
574	innov8tif-001	0.0000	1143	0.0029	1122	0.0657	903	0.0004	1134	-	1333
575	innov8tif-002	0.0258	1342	0.0862	1337	0.4368	980	0.1312	1401	-	1129
576	innovativetechnologyltd-001	0.0001	1255	0.0050	1213	-	1162	0.0024	1337	0.0025	1037
577	innovativetechnologyltd-002	0.0000	998	0.0046	1200	-	1341	0.0057	1374	0.0005	833
578	innovatrics-000	-	1485	-	1400	-	1135	-	1408	0.0004	733
579	innovatrics-001	-	1470	-	1429	-	1237	-	1424	0.0004	735
580	innovatrics-002	-	1353	-	1460	-	1350	-	1497	0.0004	466

Table 102: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
581	innovatrics-003	-	1357	-	1463	-	1372	-	1501	0.0005	785
582	innovatrics-004	0.0000	955	0.0017	998	-	1087	0.0000	601	0.0004	674
583	innovatrics-005	-	1457	-	1362	-	1032	0.0000	599	0.0004	661
584	innovatrics-006	0.0000	814	0.0009	783	-	1207	0.0000	534	0.0004	454
585	innovatrics-007	0.0000	800	0.0007	636	-	1189	0.0001	623	0.0003	337
586	innovatrics-008	0.0000	682	0.0009	775	0.0204	558	0.0000	539	0.0004	429
587	innovatrics-009	0.0000	503	0.0005	433	0.0142	396	0.0000	44	0.0000	239
588	innovatrics-010	0.0000	695	0.0019	1024	0.0393	830	0.0001	696	0.0004	443
589	innovatrics-011	0.0000	636	0.0001	267	0.0017	174	0.0000	335	-	1451
590	innovatrics-012	0.0000	748	0.0004	430	0.0118	348	0.0000	521	-	1170
591	innovatrics-013	0.0000	716	0.0001	270	0.0027	180	0.0000	299	-	1153
592	innovatrics-014	0.0000	529	0.0007	604	0.0165	458	0.0000	454	-	1366
593	innovatrics-015	0.0000	591	0.0008	698	0.0236	653	0.0000	430	-	1323
594	innovatrics-016	0.0000	664	0.0006	507	0.0177	486	0.0000	312	-	1475
595	innovatrics-017	0.0000	659	0.0006	540	0.0189	520	0.0000	368	-	1487
596	insightface-000	0.0000	249	0.0000	193	0.0000	119	0.0000	176	0.0000	187
597	insightface-001	0.0000	155	0.0000	217	0.0000	135	0.0000	195	0.0000	202
598	insightface-002	0.0000	297	0.0000	41	0.0000	26	0.0000	132	0.0000	50
599	insightface-003	0.0000	276	0.0000	34	0.0000	24	0.0000	120	0.0000	35
600	insightface-004	0.0000	10	0.0000	158	0.0000	98	0.0000	275	0.0000	154
601	inspur-000	0.0000	27	0.0000	153	0.0000	97	0.0000	265	0.0000	147
602	inspur-001	0.0000	229	0.0000	200	0.0000	126	0.0000	181	0.0000	198
603	inspur-002	0.0000	15	0.0000	160	0.0000	99	0.0000	272	0.0000	156
604	inspur-003	0.0000	114	0.0000	139	0.0000	89	0.0000	251	-	1335
605	intellicloudai-001	0.0000	472	0.0000	57	-	1120	0.0000	3	0.0000	65
606	intellicloudai-002	0.0000	158	0.0008	692	-	1449	0.0000	511	0.0004	444
607	intellifusion-001	0.0000	884	0.0005	486	-	1151	0.0001	679	0.0003	397
608	intellifusion-002	0.0000	437	0.0000	254	-	1218	0.0000	296	0.0000	105
609	intellivision-001	0.0042	1312	0.0296	1305	-	1487	0.0048	1368	0.0042	1065
610	intellivision-002	0.0000	1203	0.0046	1201	-	1413	0.0012	1282	0.0005	870
611	intellivision-003	0.0000	661	0.0012	896	0.0308	748	0.0003	1034	0.0004	775
612	intellivision-004	0.0000	736	0.0011	866	0.0266	712	0.0002	1000	0.0004	758
613	intellivision-005	0.0001	1248	0.0041	1184	0.0528	870	0.0005	1174	0.0007	918
614	intellivision-006	0.0001	1246	0.0041	1186	0.0528	872	0.0005	1176	0.0007	920
615	intellivision-007	0.0001	1247	0.0041	1187	0.0528	873	0.0005	1175	0.0007	921
616	intellivix-001	0.0000	466	0.0000	74	0.0000	49	0.0000	23	0.0000	68
617	intellivix-002	0.0000	483	0.0009	780	0.0184	506	0.0000	11	0.0000	59
618	intellivix-003	0.0000	245	0.0000	191	0.0000	117	0.0000	167	0.0000	192
619	intellivix-004	0.0000	523	0.0022	1059	0.0333	787	0.0000	32	0.0000	78
620	intellivix-005	0.0000	34	0.0022	1060	0.0333	788	0.0000	285	0.0000	172
621	intellivix-006	0.0000	4	0.0007	608	0.0180	491	0.0000	269	-	1369
622	intelresearch-000	0.0000	691	0.0006	499	-	1427	0.0000	573	0.0003	403
623	intelresearch-001	0.0000	1180	0.0082	1259	-	1185	0.0005	1183	0.0010	947
624	intelresearch-002	0.0000	863	0.0022	1057	-	1123	0.0000	567	0.0004	424
625	intelresearch-003	0.0000	825	0.0006	520	-	1183	0.0000	434	0.0004	509
626	intelresearch-004	0.0000	538	0.0006	525	-	1349	0.0000	456	0.0004	521
627	intelresearch-005	0.0000	621	0.0006	524	0.0144	403	0.0000	453	0.0004	516
628	intelresearch-006	0.0000	349	0.0000	251	0.0004	165	0.0000	328	0.0004	503
629	intelresearch-007	0.0000	411	0.0007	589	0.0168	466	0.0000	320	0.0004	511
630	intelresearch-008	0.0000	391	0.0006	578	0.0169	471	0.0000	322	0.0004	539
631	intelresearch-009	0.0000	58	0.0007	649	0.0232	646	0.0000	349	-	1372
632	intelresearch-010	0.0000	653	0.0010	847	0.0335	790	0.0000	523	-	1492
633	intelresearch-011	0.0000	867	0.0006	571	0.0175	479	0.0000	486	-	1203
634	intelresearch-012	0.0000	824	0.0007	616	0.0191	535	0.0000	488	-	1261
635	intema-000	0.0000	465	0.0005	455	0.0126	362	0.0000	577	0.0004	449
636	intema-001	0.0000	742	0.0004	400	0.0106	322	0.0000	332	0.0003	407
637	interlinklabs-001	0.0000	1156	0.0004	420	0.0085	279	0.0005	1163	-	1320
638	interlinklabs-002	0.0000	1155	0.0004	422	0.0085	280	0.0005	1164	-	1363

Table 103: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
639	interlinklabs-003	0.0000	907	0.0031	1140	0.0249	679	0.0003	1052	-	1316
640	intozi-001	0.0000	745	0.0005	493	0.0126	361	0.0001	853	0.0004	423
641	intozi-002	0.0000	485	0.0000	60	0.0000	41	0.0000	9	-	1198
642	intsysmsu-000	-	1436	-	1386	-	1093	0.0004	1112	0.0012	979
643	intsysmsu-001	0.0000	388	0.0010	826	-	1196	0.0001	731	0.0004	624
644	intsysmsu-002	0.0000	122	0.0010	828	-	1453	0.0001	734	0.0004	634
645	ionetworks-000	0.0000	54	0.0016	989	0.0387	826	0.0004	1087	0.0005	798
646	ionetworks-001	0.0000	260	0.0081	1256	0.0666	905	0.0000	124	0.0000	39
647	ionetworks-002	0.0000	272	0.0003	315	0.0048	193	0.0000	116	0.0000	37
648	ionetworks-003	0.0000	56	0.0004	376	0.0079	261	0.0000	282	-	1374
649	ionetworks-004	0.0000	301	0.0004	363	0.0086	282	0.0000	130	-	1179
650	iqface-000	0.0000	146	0.0000	212	-	1436	0.0000	192	0.0000	209
651	iqface-001	0.0000	99	0.0000	145	-	1320	0.0000	314	0.0000	139
652	iqface-003	0.0000	1148	0.0076	1250	-	1410	0.0006	1202	0.0005	867
653	iqsec-001	0.0000	279	0.0763	1333	0.1617	960	0.0000	139	-	1191
654	iqsec-002	0.0000	373	0.0763	1332	0.1617	959	0.0000	96	-	1130
655	iqsec-003	0.0000	150	0.0000	213	0.0000	132	0.0000	189	-	1456
656	irex-000	0.0000	1056	0.0009	778	-	1181	0.0000	569	0.0005	781
657	irisid-001	0.0000	612	0.0016	979	0.0441	851	0.0000	620	-	1347
658	irisid-002	0.0000	628	0.0016	980	0.0441	852	0.0000	618	-	1469
659	isap-001	0.0000	46	0.0000	163	-	1354	0.0000	279	0.0000	163
660	isap-002	0.0000	424	0.0000	118	-	1246	0.0000	72	0.0000	113
661	isityou-000	0.0068	1320	0.0316	1309	-	1493	0.0023	1328	0.0010	954
662	istl-001	0.0000	1102	0.0005	496	0.0162	456	0.0010	1269	-	1094
663	isystems-000	-	1408	-	1468	-	1390	-	1462	0.0010	953
664	isystems-001	0.0000	1170	0.0035	1166	-	1160	0.0010	1266	0.0007	917
665	isystems-002	0.0000	1168	0.0035	1165	-	997	0.0010	1267	0.0007	916
666	itmo-001	-	1479	-	1424	-	1220	-	1423	0.0047	1067
667	itmo-002	-	1358	-	1466	-	1373	-	1500	0.0029	1049
668	itmo-003	-	1476	-	1420	-	1209	-	1419	0.0008	925
669	itmo-004	-	1414	-	1477	-	1416	-	1466	0.0004	764
670	itmo-005	-	1430	-	1391	-	1104	0.0005	1159	0.0002	267
671	itmo-006	0.0000	240	0.0015	969	-	1408	0.0004	1142	0.0004	631
672	itmo-007	0.0000	37	0.0009	735	-	1377	0.0003	1077	0.0000	171
673	itmo-008	0.0000	243	0.0135	1282	0.1239	954	0.0024	1338	0.0000	193
674	ivacognitive-001	0.0000	961	0.0011	868	-	993	0.0001	655	0.0004	731
675	iws-000	0.0005	1287	0.0650	1326	-	1251	0.0024	1334	0.0012	986
676	jaakit-001	0.0008	1297	0.0858	1336	0.2713	975	0.0042	1361	0.0021	1028
677	jaakit-002	0.0000	525	0.0000	81	0.0000	55	0.0000	29	-	1228
678	jgumainz-000	0.0001	1237	0.0027	1097	0.0519	867	0.0006	1197	-	1419
679	kaizenvoiz-001	0.0000	1137	0.0000	20	0.0000	13	0.0037	1356	-	1120
680	kakao-001	-	1388	-	1490	-	1465	0.0002	964	0.0005	816
681	kakao-002	0.0000	1016	0.0057	1230	-	1392	0.0002	963	0.0005	814
682	kakao-003	0.0000	314	0.0000	45	-	1094	0.0000	134	0.0000	47
683	kakao-004	0.0000	20	0.0000	148	-	1323	0.0000	263	0.0000	148
684	kakao-005	0.0000	421	0.0000	247	-	1240	0.0000	75	0.0000	243
685	kakao-007	0.0000	144	0.0007	587	0.0165	459	0.0001	719	0.0004	526
686	kakao-008	0.0000	130	0.0009	747	0.0209	577	0.0001	718	0.0004	527
687	kakao-009	0.0000	41	0.0009	746	0.0209	576	0.0001	722	0.0004	523
688	kakaobank-000	0.0000	302	0.0000	39	0.0000	27	0.0000	129	0.0000	48
689	kakaobank-001	0.0000	397	0.0006	522	0.0176	482	0.0000	388	0.0004	455
690	kakaobank-002	0.0000	630	0.0010	855	0.0303	743	0.0000	474	-	1461
691	kakaobank-003	0.0000	647	0.0013	941	0.0397	832	0.0000	447	-	1500
692	kakaopay-001	0.0000	994	0.0013	934	0.0322	767	0.0001	666	0.0004	741
693	kasikornlabs-000	0.0000	1185	0.0035	1163	0.0713	925	0.0004	1138	0.0012	993
694	kasikornlabs-001	0.0001	1257	0.0050	1212	0.0885	942	0.0006	1221	0.0035	1057
695	kasikornlabs-002	0.0000	1182	0.0033	1147	0.0698	921	0.0004	1127	0.0012	980
696	kasikornlabs-003	0.0000	1178	0.0033	1150	0.0698	923	0.0004	1128	0.0012	981

Table 104: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
697	kasikornlabs-004	0.0000	1177	0.0033	1148	0.0698	922	0.0004	1131	-	1314
698	kasikornlabs-005	0.0000	1181	0.0033	1146	0.0698	920	0.0004	1125	-	1207
699	kbyai-000	0.0000	766	0.0013	918	0.0265	711	0.0001	644	-	1111
700	kedacom-000	0.0000	381	0.0000	100	-	1190	0.0000	57	0.0000	99
701	keyless-000	0.0000	541	0.0010	822	0.0308	749	0.0000	483	-	1364
702	kiwitech-000	0.0000	601	0.0009	731	-	1263	0.0004	1129	0.0005	791
703	kloudspot-001	0.0000	1035	0.0010	820	0.0211	586	0.0001	800	-	1235
704	kloudspot-002	0.0000	241	0.0006	572	0.0159	447	0.0002	949	-	1423
705	kneron-003	0.0239	1339	0.0306	1306	-	1088	0.0044	1365	0.0016	1017
706	kneron-005	0.0000	1174	0.0226	1297	-	1472	0.0006	1198	0.0005	823
707	kneron-006	0.0000	1138	0.1109	1340	0.0686	916	0.0012	1281	-	1243
708	knowutech-000	0.0000	795	0.0008	664	0.0215	601	0.0000	527	0.0004	636
709	kogniza-001	0.0000	684	0.0012	913	0.0337	795	0.0003	1064	-	1392
710	kookmin-001	0.0000	17	0.0000	159	-	1351	0.0000	271	0.0000	155
711	kookmin-002	0.0000	76	0.0000	132	-	1280	0.0000	237	0.0000	123
712	koreaid-001	0.0000	480	0.0023	1072	0.0371	814	0.0000	588	0.0005	779
713	krungthai-002	0.0000	527	0.0005	469	0.0111	333	0.0002	940	0.0003	412
714	kuke3d-001	0.0000	348	0.0000	26	0.0000	15	0.0000	110	0.0000	32
715	kuke3d-002	0.0000	400	0.0000	105	0.0000	70	0.0000	58	0.0000	96
716	lebentech-000	0.0042	1310	0.0029	1124	0.0252	688	0.0051	1370	0.0066	1072
717	lebentech-001	0.0000	1210	0.0012	912	0.0317	754	0.0004	1137	0.0005	842
718	lemalabs-001	0.0000	190	0.0005	490	0.0141	394	0.0002	929	0.0004	473
719	lightvision-000	0.0000	441	0.0000	109	0.0000	72	0.0000	67	-	1291
720	lightvision-001	0.0000	484	0.0000	61	0.0000	40	0.0000	10	-	1200
721	lineclova-000	0.0000	474	0.0000	56	0.0000	37	0.0000	5	0.0000	66
722	lineclova-001	0.0000	362	0.0000	28	0.0000	18	0.0000	114	0.0000	25
723	lineclova-002	0.0000	97	0.0007	588	0.0181	493	0.0000	258	0.0000	141
724	lineclova-003	0.0000	1119	0.0023	1066	0.0700	924	0.0002	1001	0.0005	782
725	lookman-002	0.0000	394	0.0000	103	-	1204	0.0000	60	0.0000	93
726	lookman-004	0.0000	71	0.0000	129	-	1273	0.0000	231	0.0000	126
727	lssytech-000	0.0000	677	0.0006	561	0.0177	483	0.0000	549	-	1395
728	lssytech-001	0.0000	871	0.0004	386	0.0092	293	0.0000	537	-	1250
729	luxand-000	0.0000	350	0.0000	24	-	1042	0.0000	108	0.0000	30
730	luxand-001	0.0000	1057	0.0016	987	0.0366	810	0.0006	1208	0.0005	838
731	luxand-003	0.0000	1047	0.0016	986	0.0366	809	0.0006	1209	-	1164
732	luxand-004	0.0000	536	0.0003	340	0.0061	213	0.0003	1013	-	1367
733	mantra-000	0.0001	1221	0.0041	1188	0.0680	914	0.0003	1045	0.0004	759
734	matrixcomsec-000	0.0002	1264	1.0000	1496	1.0000	1478	0.0336	1398	-	1486
735	matrixcomsec-001	0.0000	246	0.0009	739	0.0252	687	0.0000	165	-	1428
736	matrixcomsec-002	0.0000	757	0.0009	741	0.0252	690	0.0001	726	-	1114
737	maverickresearch-000	0.0000	1022	0.0096	1267	0.0574	881	0.0013	1293	-	1178
738	maxvision-000	0.0000	343	0.0000	244	0.0000	3	0.0000	84	0.0000	3
739	maxvision-001	0.0000	497	0.0000	88	0.0000	57	0.0000	38	0.0000	86
740	maxvision-002	0.0000	686	0.0009	726	0.0229	639	0.0002	865	0.0004	599
741	maxvision-003	0.0000	672	0.0009	725	0.0229	638	0.0002	864	0.0004	597
742	maxvision-004	0.0000	856	0.0009	723	0.0229	636	0.0002	867	0.0004	586
743	maxvision-005	0.0000	558	0.0009	724	0.0229	637	0.0002	870	0.0004	594
744	maxvision-006	0.0000	730	0.0007	654	0.0226	622	0.0002	888	0.0004	585
745	mbbankvn-000	0.0000	774	0.0004	390	0.0098	303	0.0000	570	-	1106
746	megvii-000	-	1492	-	1396	-	1128	-	1404	0.0003	310
747	megvii-001	0.0000	281	0.0006	526	-	1100	0.0007	1232	0.0004	483
748	megvii-002	0.0000	464	0.0006	527	-	1143	0.0054	1373	0.0004	487
749	megvii-003	0.0000	844	0.0010	849	-	1221	0.0002	975	0.0004	712
750	megvii-004	0.0000	660	0.0010	823	-	1480	0.0002	908	0.0004	656
751	megvii-005	0.0000	531	0.0010	809	0.0206	564	0.0002	972	0.0004	700
752	megvii-006	0.0000	785	0.0010	804	0.0206	561	0.0002	968	0.0004	688
753	megvii-007	0.0000	895	0.0010	807	0.0206	563	0.0002	967	0.0004	692
754	megvii-008	0.0000	689	0.0010	810	0.0206	565	0.0002	970	0.0004	702

Table 105: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
755	megvii-009	0.0000	732	0.0010	806	0.0206	562	0.0002	969	0.0004	691
756	meituan-000	0.0000	271	0.0001	271	0.0013	168	0.0000	394	0.0002	274
757	meituan-001	0.0000	586	0.0014	955	0.0295	733	0.0001	742	0.0004	669
758	meituan-002	0.0000	532	0.0013	930	0.0251	685	0.0001	744	0.0004	652
759	meituan-003	0.0000	543	0.0013	929	0.0251	684	0.0001	743	0.0004	649
760	meituan-004	0.0000	854	0.0013	928	0.0251	683	0.0001	738	0.0004	642
761	meiya-001	0.0000	1162	0.0028	1110	-	1236	0.0004	1144	0.0010	955
762	mendaxiatech-000	0.0000	553	0.0010	808	0.0206	560	0.0002	980	0.0004	699
763	metsakuurcompany-001	0.0000	178	0.0011	858	0.0208	569	0.0002	965	0.0004	528
764	metsakuurcompany-002	0.0000	447	0.0000	68	0.0000	43	0.0000	20	0.0000	70
765	metsakuurcompany-003	0.0000	124	0.0000	225	0.0000	138	0.0000	207	0.0000	216
766	metsakuurcompany-004	0.0000	208	0.0000	173	0.0000	104	0.0000	148	-	1401
767	metsakuurcompany-005	0.0000	513	0.0000	76	0.0000	51	0.0000	26	-	1231
768	metsakuurcompany-006	0.0000	177	0.0005	449	0.0128	364	0.0000	564	-	1494
769	miaxis-001	0.0000	607	0.0013	924	0.0262	705	0.0001	836	0.0003	327
770	miaxis-002	0.0448	1348	0.1162	1341	0.2565	973	0.2128	1403	0.0347	1086
771	miaxis-003	0.0000	699	0.0013	925	0.0262	706	0.0001	833	0.0003	328
772	miaxis-004	0.0000	519	0.0000	79	0.0000	52	0.0000	35	-	1227
773	miaxis-005	0.0000	324	0.0000	9	0.0000	6	0.0000	87	-	1112
774	microfocus-000	-	1467	-	1415	-	1186	-	1416	0.0016	1013
775	microfocus-001	0.0001	1252	0.0053	1223	-	1500	0.0008	1246	0.0016	1015
776	microfocus-002	0.0001	1253	0.0053	1222	-	1488	0.0008	1245	0.0016	1014
777	microfocus-003	0.0001	1245	0.0049	1211	0.0819	936	0.0007	1236	0.0015	1012
778	minivision-000	0.0000	47	0.0000	162	-	1353	0.0000	280	0.0000	164
779	mitek-000	0.0000	1042	0.0029	1115	0.0521	868	0.0002	934	0.0003	351
780	mitek-001	0.0000	957	0.0020	1036	0.0566	879	0.0003	1083	-	1174
781	mitek-002	0.0000	360	0.0012	901	0.0373	816	0.0000	559	-	1135
782	mobai-000	0.0000	1059	0.0114	1278	-	1364	0.0003	1056	0.0012	989
783	mobai-001	0.0000	986	0.0040	1176	-	1138	0.0001	766	0.0012	988
784	mobai-002	0.0000	981	0.0008	666	0.0149	411	0.0002	872	-	1264
785	mobai-003	0.0000	117	0.0000	248	0.0001	159	0.0000	248	-	1338
786	mobbl-000	0.0116	1329	0.0720	1331	-	1243	0.0119	1387	0.0063	1071
787	mobbl-001	0.0000	1150	0.0052	1218	0.0678	912	0.0002	886	0.0005	839
788	mobbl-002	0.0000	1173	0.0029	1118	0.0633	897	0.0002	954	0.0009	941
789	mobbl-003	0.0000	1175	0.0029	1117	0.0633	896	0.0002	951	0.0009	940
790	mobbl-004	0.0000	194	0.0012	903	0.0312	752	0.0000	473	-	1414
791	mobbl-005	0.0000	798	0.0009	736	0.0217	603	0.0000	442	-	1127
792	mobipintech-000	0.0000	406	0.0000	92	0.0000	63	0.0000	49	0.0000	89
793	momovn-001	0.0015	1302	0.0010	798	0.0103	307	0.0011	1275	0.0008	928
794	momovn-002	0.0015	1301	0.0010	797	0.0103	306	0.0011	1277	-	1194
795	momovn-003	0.0000	959	0.0031	1139	0.0249	678	0.0003	1050	-	1104
796	moredian-000	0.0000	861	0.0009	730	-	1117	0.0004	1126	0.0005	790
797	morpho-000	-	1480	-	1423	-	1224	-	1422	0.0000	101
798	morpho-001	-	1439	-	1356	-	1010	-	1437	0.0000	20
799	morpho-002	-	1471	-	1431	-	1241	0.0000	556	0.0004	489
800	mukh-001	0.0000	68	0.0010	827	0.0154	434	0.0001	759	0.0003	291
801	mukh-002	0.0000	481	0.0022	1065	0.0513	862	0.0002	878	0.0004	575
802	mukh-003	0.0000	593	0.0003	310	0.0060	210	0.0001	855	0.0003	399
803	mukh-004	0.0000	701	0.0013	933	0.0327	773	0.0000	581	0.0004	422
804	mukh-005	0.0000	676	0.0009	792	0.0245	667	0.0000	510	-	1397
805	multimodality-000	0.0000	407	0.0000	95	0.0000	64	0.0000	48	0.0000	92
806	multimodality-001	0.0000	226	0.0009	721	0.0259	702	0.0000	182	0.0000	199
807	mvision-001	0.0000	64	0.0000	127	-	1267	0.0000	234	0.0000	124
808	nazhiai-000	0.0000	42	0.0000	171	-	1382	0.0000	289	0.0000	168
809	ncsgroup-000	0.0000	192	0.0000	183	0.0000	111	0.0000	155	-	1417
810	ncssg-001	0.0000	173	0.0000	240	0.0000	151	0.0000	223	0.0000	226
811	ncssg-002	0.0000	255	0.0000	196	0.0000	122	0.0000	170	-	1421
812	ncssg-003	0.0000	725	0.0006	549	0.0149	412	0.0003	1022	-	1148

Table 106: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
813	ncssg-004	0.0000	568	0.0006	553	0.0149	413	0.0003	1024	-	1381
814	neosystems-001	0.0000	440	0.0000	245	-	1223	0.0013	1290	0.9994	1093
815	neosystems-002	0.0000	427	0.0000	115	0.0000	76	0.0000	70	0.0000	110
816	neosystems-003	0.0000	123	0.0000	226	0.0000	137	0.0000	210	0.0000	219
817	neosystems-004	0.0000	393	0.0000	104	0.0000	68	0.0000	61	0.0000	95
818	netbridgetech-001	0.0000	264	0.0000	37	-	1077	0.0000	127	0.0000	38
819	netbridgetech-002	0.0000	330	0.0000	16	-	1019	0.0000	93	0.0000	15
820	neurotechnology-000	-	1431	-	1390	-	1106	-	1458	0.0000	55
821	neurotechnology-001	-	1363	-	1462	-	1365	-	1498	0.0000	161
822	neurotechnology-002	-	1375	-	1453	-	1312	-	1490	0.0000	138
823	neurotechnology-003	-	1493	-	1397	-	1130	-	1405	0.0005	794
824	neurotechnology-004	-	1361	-	1467	-	1380	0.0000	432	0.0001	258
825	neurotechnology-005	-	1399	-	1497	-	1474	0.0004	1100	0.0004	572
826	neurotechnology-006	0.0000	1049	0.0006	547	-	1102	0.0004	1099	0.0004	558
827	neurotechnology-008	0.0000	1003	0.0006	545	-	1277	0.0004	1098	0.0004	542
828	neurotechnology-009	0.0000	945	0.0008	669	-	1412	0.0002	894	0.0004	497
829	neurotechnology-010	0.0000	786	0.0001	268	-	1044	0.0000	572	0.0002	264
830	neurotechnology-011	0.0000	918	0.0013	916	-	1294	0.0002	889	0.0003	400
831	neurotechnology-012	0.0000	1117	0.0010	857	0.0242	661	0.0001	801	0.0004	628
832	neurotechnology-013	0.0000	408	0.0008	713	0.0185	512	0.0000	354	0.0001	249
833	neurotechnology-015	0.0000	239	0.0004	371	0.0082	270	0.0000	168	0.0000	234
834	neurotechnology-016	0.0000	356	0.0003	322	0.0069	236	0.0000	104	0.0000	233
835	neurotechnology-017	0.0000	166	0.0003	348	0.0065	225	0.0000	219	0.0000	238
836	neurotechnology-018	0.0000	540	0.0003	355	0.0074	243	0.0000	455	0.0004	440
837	neurotechnology-019	0.0000	658	0.0004	397	0.0115	345	0.0000	339	-	1480
838	neurotechnology-020	0.0000	563	0.0004	395	0.0115	346	0.0000	351	-	1387
839	neurotechnology-021	0.0000	303	0.0004	399	0.0115	344	0.0000	331	-	1177
840	nextmetalabs-001	0.0000	341	0.0000	4	0.0000	2	0.0000	76	-	1107
841	nextmetalabs-002	0.0000	222	0.0000	204	0.0000	125	0.0000	183	-	1441
842	nextmetalabs-003	0.0000	879	0.0006	583	0.0184	507	0.0000	385	-	1245
843	nextmetalabs-004	0.0000	762	0.0008	719	0.0255	699	0.0000	392	-	1116
844	nhn-001	0.0000	837	0.0019	1018	-	1245	0.0001	691	0.0004	750
845	nhn-002	0.0000	275	0.0004	429	0.0091	292	0.0000	493	0.0003	315
846	nhn-003	0.0000	1019	0.0000	35	0.0000	23	0.0001	846	0.0004	663
847	nhn-004	0.0000	1029	0.0000	116	0.0000	75	0.0001	845	0.0004	668
848	nhn-005	0.0000	735	0.0000	51	0.0000	32	0.0000	535	0.0004	482
849	nmetics-001	0.0000	1176	0.0011	863	0.0230	642	0.0002	1004	-	1400
850	noblis-000	-	1448	-	1352	-	996	-	1436	0.0000	2
851	nodeflux-000	-	1356	-	1456	-	1329	0.0001	770	0.0002	285
852	nodeflux-001	0.0000	283	0.0016	978	-	1101	0.0001	768	0.0002	283
853	nodeflux-002	0.0000	823	0.0261	1301	-	1182	0.0008	1243	0.0005	832
854	nominder-000	0.0000	580	0.0003	300	0.0053	200	0.0000	478	0.0003	385
855	nominder-001	0.0000	796	0.0003	299	0.0053	199	0.0000	471	0.0003	377
856	nominder-002	0.0000	851	0.0006	514	0.0119	349	0.0002	948	0.0003	396
857	nominder-003	0.0000	771	0.0005	474	0.0095	299	0.0002	892	-	1103
858	nominder-004	0.0000	860	0.0007	619	0.0130	370	0.0002	988	-	1210
859	nominder-005	0.0000	358	0.0009	737	0.0206	559	0.0000	492	-	1136
860	nominder-006	0.0000	977	0.0017	990	0.0379	822	0.0003	1067	-	1275
861	nominder-007	0.0000	919	0.0006	576	0.0119	351	0.0003	1066	-	1332
862	nominder-008	0.0000	929	0.0006	580	0.0119	350	0.0003	1065	-	1483
863	notiontag-000	0.0000	125	0.0000	223	-	1456	0.0000	208	0.0000	217
864	notiontag-001	0.0000	62	0.0000	166	-	1368	0.0027	1344	0.0000	159
865	notiontag-002	0.0000	311	0.0000	43	0.0000	30	0.0000	136	0.0000	44
866	npt-000	0.0000	317	0.0002	291	0.0042	188	0.0001	755	-	1173
867	nsensecorp-001	0.0000	1194	0.0024	1076	-	1055	0.0014	1300	0.0101	1078
868	nsensecorp-002	0.0000	708	0.0009	734	0.0219	612	0.0003	1018	0.0011	958
869	nsensecorp-003	0.0000	39	0.0000	262	0.0002	161	0.0000	458	0.0007	924
870	nsensecorp-004	0.0406	1347	0.0035	1162	0.0181	492	0.0016	1311	0.0760	1087

Table 107: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
871	nsensecorp-005	0.0000	250	0.0000	192	0.0000	120	0.0000	175	0.0000	186
872	ntechlab-000	-	1365	-	1441	-	1276	-	1483	0.0016	1016
873	ntechlab-001	-	1456	-	1360	-	1024	-	1440	0.0005	800
874	ntechlab-002	-	1486	-	1401	-	1139	-	1410	0.0005	801
875	ntechlab-003	-	1447	-	1353	-	999	-	1435	0.0005	799
876	ntechlab-004	-	1491	-	1399	-	1127	-	1407	0.0004	507
877	ntechlab-005	-	1450	-	1366	-	1039	0.0000	409	0.0004	480
878	ntechlab-006	-	1463	-	1417	-	1198	0.0000	321	0.0004	437
879	ntechlab-007	0.0000	916	0.0019	1020	-	1285	0.0001	661	0.0004	464
880	ntechlab-008	0.0000	932	0.0005	435	-	1483	0.0001	672	0.0004	474
881	ntechlab-009	0.0000	1050	0.0009	752	-	991	0.0001	803	0.0004	417
882	ntechlab-010	0.0000	723	0.0005	484	0.0107	326	0.0001	754	0.0004	447
883	ntechlab-011	0.0000	31	0.0003	304	0.0057	205	0.0000	587	0.0004	428
884	ntechlab-012	0.0000	151	0.0003	305	0.0057	206	0.0000	583	0.0004	431
885	ntnu-000	0.0000	107	0.0004	361	0.0075	249	0.0001	840	-	1342
886	ocrstudio-001	0.0000	1164	0.0034	1160	0.0534	875	0.0001	806	-	1350
887	omface-000	0.0000	133	0.0000	230	0.0000	141	0.0000	202	0.0000	222
888	omface-001	0.0000	115	0.0000	252	0.0000	156	0.0000	250	0.0000	128
889	omnigarde-000	0.0000	875	0.0008	657	0.0213	595	0.0000	464	0.0004	621
890	omnigarde-001	0.0000	773	0.0008	656	0.0213	594	0.0000	470	0.0004	619
891	omnigarde-002	0.0000	595	0.0008	658	0.0213	596	0.0000	477	0.0004	626
892	omnigarde-003	0.0000	121	0.0007	652	0.0212	590	0.0000	347	0.0004	515
893	omnigarde-004	0.0000	374	0.0007	651	0.0213	592	0.0000	361	-	1124
894	omnigarde-005	0.0000	148	0.0007	653	0.0213	593	0.0000	367	-	1453
895	onfido-000	0.0000	1167	0.0040	1175	0.0804	935	0.0004	1088	0.0012	982
896	onfido-001	0.0000	728	0.0006	573	0.0183	496	0.0000	301	-	1146
897	onfido-002	0.0000	882	0.0006	574	0.0183	497	0.0000	292	-	1233
898	onfido-003	0.0000	804	0.0006	575	0.0183	498	0.0000	295	-	1290
899	onfido-004	0.0000	629	0.0006	577	0.0183	499	0.0000	310	-	1474
900	oorudigital-000	0.0000	1034	0.0002	285	0.0016	173	0.0002	990	-	1240
901	ooto-001	0.0000	429	0.0000	114	0.0000	77	0.0000	69	-	1296
902	openedge-000	0.0000	554	0.0014	944	0.0212	588	0.0003	1041	0.0004	680
903	openface-001	0.0000	1091	0.0104	1274	0.0668	906	0.0004	1114	0.0006	903
904	ovisionllc-001	0.0000	449	0.0000	67	0.0000	42	0.0000	22	-	1221
905	ovisionllc-002	0.0000	476	0.0000	54	0.0000	38	0.0000	1	-	1206
906	oz-001	0.0000	1053	0.0011	878	-	1043	0.0006	1213	0.0004	554
907	oz-002	0.0000	176	0.0003	306	-	1501	0.0000	429	0.0003	352
908	oz-003	0.0000	522	0.0002	282	0.0042	187	0.0000	318	0.0003	289
909	oz-004	0.0000	1114	0.0003	318	0.0041	186	0.0000	371	0.0002	263
910	pa-002	-	1475	-	1432	-	1247	-	1428	0.0000	111
911	pa-003	-	1415	-	1476	-	1418	-	1465	0.0000	190
912	palit-000	0.0000	617	0.0005	479	0.0134	380	0.0002	916	0.0004	517
913	palit-001	0.0000	802	0.0007	655	0.0201	552	0.0002	914	0.0004	513
914	pangiam-000	0.0000	207	0.0021	1055	0.0364	807	0.0001	624	0.0005	788
915	pangiam-001	0.0000	412	0.0005	487	0.0128	365	0.0002	1007	0.0003	410
916	pangiam-002	0.0000	490	0.0003	338	0.0072	238	0.0000	433	0.0003	346
917	papago-001	0.0000	980	0.0008	675	0.0159	445	0.0002	1002	0.0004	577
918	papil11-000	0.0000	832	0.0008	712	0.0192	538	0.0000	518	0.0003	411
919	papil11-001	0.0000	694	0.0005	451	0.0133	376	0.0000	425	-	1426
920	papil11-002	0.0000	657	0.0004	372	0.0115	343	0.0000	369	-	1482
921	papil11-003	0.0000	168	0.0003	349	0.0077	254	0.0000	342	-	1501
922	papil11-004	0.0000	797	0.0008	665	0.0227	626	0.0003	1046	-	1128
923	papsav1923-001	0.0000	740	0.0007	620	-	1085	0.0001	727	0.0002	277
924	papsav1923-002	0.0000	731	0.0018	1014	0.0268	715	0.0000	547	0.0004	641
925	papsav1923-003	0.0000	999	0.0019	1032	0.0321	764	0.0000	604	0.0004	592
926	paravision-004	0.0000	1039	0.0007	642	-	1469	0.0002	930	0.0004	600
927	paravision-006	0.0000	366	0.0000	19	-	1026	0.0000	98	0.0000	22
928	paravision-008	0.0000	79	0.0010	816	0.0201	550	0.0001	707	0.0004	439

Table 108: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
929	paravision-010	0.0000	299	0.0010	811	0.0201	548	0.0001	704	0.0004	434
930	paravision-011	0.0000	5	0.0010	818	0.0201	551	0.0001	709	0.0004	441
931	paravision-013	0.0000	316	0.0010	812	0.0201	549	0.0001	705	0.0004	433
932	paravision-015	0.0000	508	0.0009	773	0.0266	713	0.0000	596	-	1241
933	paravision-017	0.0000	826	0.0009	727	0.0220	614	0.0001	685	-	1260
934	paravision-018	0.0000	233	0.0005	495	0.0142	400	0.0000	366	-	1431
935	paravision-020	0.0000	51	0.0004	380	0.0107	324	0.0000	379	-	1383
936	paravision-022	0.0000	273	0.0004	377	0.0107	323	0.0000	362	-	1154
937	pensees-001	0.0000	627	0.0000	224	-	1455	0.0000	209	0.0000	218
938	pentacomp-000	0.0001	1233	0.0010	796	0.0206	568	0.0004	1102	-	1122
939	pixelall-002	0.0000	83	0.0000	120	-	1258	0.0000	225	0.0000	119
940	pixelall-003	0.0000	196	0.0000	181	-	1398	0.0000	152	0.0000	183
941	pixelall-004	0.0000	63	0.0000	128	-	1265	0.0000	236	0.0000	125
942	pixelall-005	0.0000	252	0.0000	198	-	1417	0.0000	173	0.0000	191
943	pixelall-006	0.0000	26	0.0000	151	0.0000	96	0.0000	266	0.0000	145
944	pixelall-007	0.0000	505	0.0000	91	0.0000	61	0.0000	42	0.0000	83
945	pixelall-008	0.0000	286	0.0008	696	0.0247	672	0.0000	138	0.0000	56
946	pixelall-009	0.0000	202	0.0000	185	0.0000	114	0.0000	161	0.0000	181
947	pixelall-010	0.0000	136	0.0000	227	0.0000	142	0.0000	201	0.0000	221
948	pnpsecure-000	0.0000	392	0.0030	1125	0.0390	828	0.0003	1012	-	1274
949	pnpsecure-001	0.0000	307	0.0023	1067	0.0328	775	0.0002	893	-	1181
950	pnpsecure-002	0.0000	570	0.0006	510	0.0148	410	0.0001	682	-	1379
951	privid-001	0.0001	1259	0.0176	1288	0.0598	890	0.0021	1325	0.0021	1027
952	privid-002	0.0198	1336	0.0040	1179	0.0386	825	0.0014	1297	0.0011	967
953	privid-003	0.0000	1005	0.0002	287	0.0030	183	0.0001	822	-	1308
954	privid-004	0.0000	780	0.0001	278	0.0025	178	0.0001	680	-	1100
955	psl-000	-	1444	-	1348	-	985	-	1432	0.0000	8
956	psl-001	-	1441	-	1357	-	1013	0.0000	95	0.0000	12
957	psl-002	-	1501	-	1407	-	1157	0.0000	31	0.0000	76
958	psl-003	0.0000	887	0.0007	591	-	1153	0.0000	568	0.0004	611
959	psl-004	0.0000	763	0.0007	590	-	1014	0.0000	571	0.0004	606
960	psl-005	0.0000	853	0.0007	585	-	1137	0.0000	461	0.0004	559
961	psl-006	0.0000	737	0.0002	283	-	1113	0.0000	143	0.0003	287
962	psl-007	0.0000	614	0.0007	596	-	1288	0.0000	584	0.0003	362
963	psl-008	0.0000	584	0.0003	312	0.0068	230	0.0000	373	0.0003	348
964	psl-009	0.0000	836	0.0004	383	0.0095	298	0.0000	298	0.0004	426
965	psl-010	0.0000	702	0.0004	382	0.0095	297	0.0000	305	0.0004	421
966	psl-011	0.0000	720	0.0003	308	0.0063	215	0.0000	303	0.0003	378
967	psl-012	0.0000	840	0.0003	316	0.0067	227	0.0000	297	0.0003	384
968	psl-013	0.0000	835	0.0003	309	0.0063	216	0.0000	557	-	1293
969	psl-014	0.0000	721	0.0003	307	0.0063	214	0.0000	304	-	1145
970	ptakuratsatu-000	0.0000	549	0.0007	637	-	1328	0.0001	626	0.0003	339
971	pxl-001	0.0000	1215	0.0044	1196	-	1005	0.0005	1177	0.0022	1033
972	pyramid-000	0.0001	1249	0.0041	1185	-	1211	0.0005	1173	0.0007	919
973	pyramid-001	0.0000	685	0.0008	680	0.0231	643	0.0001	805	-	1440
974	qazbs-000	0.0000	430	0.0009	749	0.0265	710	0.0000	420	0.0004	562
975	qazsmartvisionai-000	0.0000	576	0.0003	352	0.0083	274	0.0000	348	0.0003	350
976	qazsmartvisionai-001	0.0000	886	0.0003	328	0.0064	220	0.0000	317	0.0003	324
977	qazsmartvisionai-002	0.0000	626	0.0003	330	0.0064	222	0.0000	337	-	1468
978	qazsmartvisionai-003	0.0000	847	0.0003	327	0.0064	219	0.0000	316	-	1219
979	qazsmartvisionai-004	0.0000	820	0.0003	329	0.0064	221	0.0000	319	-	1266
980	qluevision-001	0.0000	1038	0.0008	686	0.0153	424	0.0008	1242	0.0004	771
981	qnap-000	0.0000	197	0.0007	640	0.0174	476	0.0002	905	0.0002	259
982	qnap-001	0.0000	651	0.0000	259	0.0002	162	0.0000	565	0.0001	252
983	qnap-002	0.0000	1152	0.0033	1143	0.0761	932	0.0004	1097	0.0002	260
984	qnap-003	0.0000	165	0.0016	975	0.0402	838	0.0000	595	0.0001	251
985	qnap-004	0.0000	506	0.0016	974	0.0402	837	0.0000	592	0.0001	250
986	qnap-005	0.0001	1229	0.0078	1253	0.0581	884	0.0010	1268	0.0005	866

Table 109: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
987	qnap-006	0.0000	995	0.0018	1006	0.0427	846	0.0002	941	-	1252
988	quantasoft-003	0.0000	1064	0.0015	972	0.0355	803	0.0005	1167	0.0006	891
989	rankone-000	-	1416	-	1379	-	1072	-	1454	0.0003	354
990	rankone-001	-	1368	-	1435	-	1252	-	1478	0.0000	246
991	rankone-002	-	1354	-	1455	-	1322	-	1493	0.0000	244
992	rankone-003	-	1351	-	1458	-	1342	-	1494	0.0000	245
993	rankone-004	-	1490	-	1398	-	1126	-	1406	0.0000	242
994	rankone-005	-	1389	-	1488	-	1468	0.0000	212	0.0000	211
995	rankone-006	-	1417	-	1381	-	1076	0.0000	121	0.0000	41
996	rankone-007	0.0000	1214	0.0048	1208	-	1419	0.0003	1048	0.0004	727
997	rankone-008	0.0000	1193	0.0049	1210	-	1428	0.0003	1010	0.0004	430
998	rankone-009	0.0000	370	0.0000	17	-	1030	0.0000	97	0.0000	21
999	rankone-010	0.0000	524	0.0000	82	-	1158	0.0000	30	0.0000	77
1000	rankone-011	0.0000	175	0.0000	242	0.0000	153	0.0000	221	0.0000	228
1001	rankone-012	0.0000	236	0.0000	208	0.0000	131	0.0000	186	0.0000	196
1002	rankone-013	0.0000	248	0.0005	456	0.0126	363	0.0000	413	0.0003	302
1003	rankone-014	0.0000	284	0.0005	446	0.0129	367	0.0000	410	0.0002	265
1004	rankone-015	0.0000	106	0.0000	138	0.0000	84	0.0000	244	0.0000	135
1005	realnetworks-000	-	1442	-	1359	-	1018	-	1439	0.0003	318
1006	realnetworks-001	-	1419	-	1384	-	1079	0.0004	1091	0.0003	319
1007	realnetworks-002	0.0000	1014	0.0003	342	-	1485	0.0004	1092	0.0003	322
1008	realnetworks-003	0.0000	1030	0.0003	341	-	1219	0.0004	1089	0.0003	320
1009	realnetworks-004	0.0000	724	0.0003	302	0.0052	198	0.0000	302	0.0002	281
1010	realnetworks-005	0.0000	625	0.0002	293	0.0045	192	0.0000	311	0.0002	286
1011	realnetworks-006	0.0000	893	0.0002	294	0.0045	190	0.0000	293	0.0002	282
1012	realnetworks-007	0.0000	675	0.0013	935	0.0425	845	0.0000	334	0.0004	596
1013	realnetworks-008	0.0000	806	0.0002	295	0.0045	191	0.0000	294	0.0002	284
1014	realsense-001	0.0000	865	0.0006	570	0.0177	485	0.0000	460	-	1199
1015	realsense-002	0.0000	220	0.0006	537	0.0168	469	0.0000	423	-	1398
1016	realsense-003	0.0000	696	0.0006	543	0.0169	472	0.0000	412	-	1424
1017	rebs-000	0.0000	325	0.0005	447	0.0130	369	0.0000	441	0.0004	479
1018	rebs-001	0.0000	294	0.0005	464	0.0144	402	0.0000	509	0.0004	574
1019	recognito-000	0.0000	760	0.0003	313	0.0056	203	0.0000	526	0.0003	376
1020	recognito-001	0.0000	888	0.0003	314	0.0056	204	0.0000	525	0.0003	382
1021	regula-000	0.0000	78	0.0000	119	0.0000	79	0.0000	227	0.0000	118
1022	regula-001	0.0000	174	0.0000	243	0.0000	152	0.0000	222	0.0000	229
1023	regula-002	0.0000	66	0.0009	786	0.0283	722	0.0000	233	-	1324
1024	regula-003	0.0000	55	0.0009	787	0.0283	723	0.0000	276	-	1382
1025	remarkai-000	-	1437	-	1389	-	1097	0.0000	133	0.0000	46
1026	remarkai-001	0.0000	9	0.0000	155	-	1340	0.0000	268	0.0000	158
1027	remarkai-002	0.0000	185	0.0000	237	-	1489	0.0000	217	0.0000	225
1028	remarkai-003	0.0000	564	0.0007	613	0.0187	515	0.0000	566	0.0004	495
1029	rendip-000	0.0000	1023	0.0016	984	0.0293	732	0.0002	904	0.0004	757
1030	revealmedia-005	0.0000	1046	0.0007	630	0.0189	522	0.0009	1258	0.0004	772
1031	revealmedia-006	0.0000	1	0.0009	774	0.0238	655	0.0001	751	0.0004	682
1032	roc-016	0.0000	182	0.0000	261	0.0006	166	0.0000	218	0.0000	236
1033	roc-017	0.0000	496	0.0003	356	0.0073	240	0.0000	404	-	1256
1034	roc-018	0.0000	859	0.0006	562	0.0160	450	0.0000	459	0.0004	448
1035	roc-019	0.0000	574	0.0007	611	0.0187	514	0.0000	514	-	1373
1036	roc-020	0.0000	709	0.0006	541	0.0157	438	0.0000	330	-	1159
1037	roc-021	0.0000	269	0.0000	264	0.0010	167	0.0000	300	-	1151
1038	rokid-000	0.0000	183	0.0072	1245	-	1481	0.0001	720	0.0005	824
1039	rokid-001	0.0000	334	0.0013	927	-	1022	0.0000	91	0.0000	13
1040	s1-001	0.0000	1213	0.0073	1248	-	1067	0.0013	1287	0.0007	912
1041	s1-002	0.0000	939	0.0089	1265	-	1387	0.0001	791	0.0005	831
1042	s1-003	0.0000	390	0.0002	286	0.0040	185	0.0007	1229	0.0003	308
1043	s1-004	0.0000	215	0.0000	263	0.0003	163	0.0000	609	0.0001	256
1044	s1-005	0.0000	120	0.0004	388	0.0120	355	0.0001	662	0.0002	268

Table 110: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1045	s1-006	0.0000	274	0.0003	311	0.0074	245	0.0001	645	0.0002	272
1046	s1-007	0.0000	53	0.0006	517	0.0110	332	0.0001	664	0.0002	269
1047	s1-008	0.0000	719	0.0018	1017	0.0252	691	0.0005	1196	-	1155
1048	s1-009	0.0000	514	0.0006	515	0.0110	331	0.0001	649	-	1232
1049	saffe-001	0.0000	493	0.0000	84	-	1163	0.0000	39	0.0000	84
1050	saffe-002	0.0000	435	0.0000	108	-	1215	0.0000	63	0.0000	106
1051	samsungsds-000	0.0000	1004	0.0055	1226	0.0646	901	0.0038	1358	0.0005	822
1052	samsungsds-001	0.0000	353	0.0005	483	0.0146	406	0.0001	703	0.0003	392
1053	samsungsds-002	0.0000	368	0.0004	393	0.0119	352	0.0001	701	0.0003	370
1054	samtech-000	-	1424	-	1374	-	1062	-	1448	0.0042	1064
1055	samtech-001	0.0001	1244	0.0032	1142	-	1268	0.0004	1130	0.0008	926
1056	samtech-002	0.0000	1105	0.0018	1011	0.0516	865	0.0002	1006	0.0005	847
1057	scanalitix-001	0.0000	829	0.0005	457	0.0095	300	0.0001	844	-	1301
1058	scanovate-001	0.0208	1337	0.2388	1344	-	1338	0.0024	1331	0.0014	1004
1059	scanovate-002	0.0000	948	0.0018	1007	-	1070	0.0000	607	0.0004	749
1060	scanovate-003	0.0000	954	0.0233	1299	0.3371	979	0.0006	1204	0.0004	766
1061	sdc-000	0.0000	1190	0.0035	1161	0.0678	913	0.0005	1191	0.0011	972
1062	seamfix-001	0.0000	198	0.0000	187	0.0000	113	0.0000	163	-	1407
1063	securifai-001	0.0000	85	0.0000	123	-	1259	0.0000	313	0.0000	116
1064	securifai-002	0.0000	345	0.0000	8	-	1001	0.0000	82	0.0000	6
1065	securifai-003	0.0000	367	0.0000	18	0.0000	9	0.0000	99	0.0000	23
1066	securifai-004	0.0000	109	0.0000	136	0.0000	85	0.0000	241	0.0000	134
1067	securifai-005	0.0000	410	0.0000	97	0.0000	65	0.0000	53	0.0000	88
1068	securifai-006	0.0000	152	0.0000	219	0.0000	133	0.0000	199	0.0000	205
1069	securifai-007	0.0000	347	0.0000	6	0.0000	4	0.0000	80	0.0000	4
1070	securifai-008	0.0000	112	0.0000	137	0.0000	87	0.0000	242	-	1339
1071	securitypitch-000	0.0175	1335	0.0055	1227	0.0108	327	0.0137	1389	-	1464
1072	sensetime-001	-	1425	-	1376	-	1066	0.0000	494	0.0004	573
1073	sensetime-002	0.0004	1279	0.0082	1258	-	1331	0.0011	1279	0.0005	860
1074	sensetime-003	0.0000	567	0.0011	874	-	1355	0.0000	315	0.0004	593
1075	sensetime-004	0.0000	746	0.0011	872	-	1091	0.0000	307	0.0004	584
1076	sensetime-005	0.0000	218	0.0004	381	0.0106	312	0.0000	499	0.0003	342
1077	sensetime-006	0.0000	482	0.0004	378	0.0106	310	0.0000	487	0.0003	336
1078	sensetime-007	0.0000	29	0.0004	379	0.0106	311	0.0000	503	0.0003	340
1079	sensetime-008	0.0000	308	0.0007	635	0.0250	681	0.0000	306	0.0003	408
1080	sensetime-010	0.0000	160	0.0005	482	0.0111	336	0.0000	552	-	1449
1081	sensetime-011	0.0000	30	0.0005	480	0.0111	335	0.0000	554	-	1391
1082	sensetime-012	0.0000	285	0.0005	477	0.0111	334	0.0000	548	-	1195
1083	serendipity-000	0.0000	892	0.0008	681	0.0202	554	0.0001	841	0.0004	610
1084	sertis-000	0.0000	315	0.0007	626	-	1095	0.0000	617	0.0004	534
1085	sertis-001	0.0000	450	0.0004	431	-	1133	0.0002	947	0.0005	868
1086	sertis-002	0.0000	293	0.0007	603	0.0152	418	0.0000	602	0.0004	535
1087	sertis-003	0.0000	776	0.0008	695	0.0183	495	0.0000	519	0.0004	604
1088	seventhsense-000	0.0000	559	0.0006	566	0.0184	504	0.0001	638	0.0004	629
1089	seventhsense-001	0.0000	604	0.0006	565	0.0184	503	0.0001	636	0.0004	627
1090	seventhsense-002	0.0000	475	0.0003	350	0.0076	251	0.0000	615	0.0004	425
1091	seventhsense-003	0.0000	679	0.0003	337	0.0083	275	0.0000	411	0.0002	279
1092	seventhsense-005	0.0000	881	0.0002	296	0.0065	223	0.0000	418	0.0003	307
1093	seventhsense-006	0.0000	637	0.0002	298	0.0065	224	0.0000	427	-	1458
1094	shaman-000	0.0000	526	0.0000	80	-	1159	0.0000	28	0.0000	75
1095	shaman-001	0.0000	259	0.0000	36	-	1071	0.0000	125	0.0000	40
1096	shu-001	0.0000	1160	0.0022	1063	-	1278	0.0010	1265	0.0006	886
1097	shu-002	0.0000	946	0.0010	832	-	1414	0.0005	1152	0.0004	725
1098	shu-003	0.0000	253	0.0007	601	-	1415	0.0001	640	0.0003	332
1099	shufti-001	0.0000	738	0.0005	491	0.0131	374	0.0001	851	-	1190
1100	siat-002	0.0000	579	0.0012	902	-	1266	0.0000	541	0.0004	589
1101	siat-004	0.0000	640	0.0011	871	-	1444	0.0000	500	0.0004	580
1102	siat-005	0.0000	319	0.0000	13	0.0000	154	0.0000	90	0.0000	19

Table 111: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]							
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1103	sightropic-000	0.0000	927	0.0006	557	0.0138	386	0.0002	998	-	1499
1104	sightropic-001	0.0000	979	0.0037	1170	0.0261	704	0.0007	1238	-	1270
1105	sightropic-002	0.0000	803	0.0002	280	0.0022	175	0.0001	790	-	1289
1106	sjtu-001	0.0000	966	0.0010	852	-	1040	0.0005	1166	0.0004	747
1107	sjtu-002	0.0000	947	0.0010	833	-	1420	0.0005	1151	0.0004	726
1108	sjtu-003	0.0000	361	0.0005	498	-	1050	0.0000	580	0.0003	313
1109	sjtu-004	0.0000	35	0.0000	169	0.0000	102	0.0000	284	0.0003	312
1110	sjtu-005	0.0000	1201	0.0000	83	0.0000	54	0.0000	33	0.0006	885
1111	sjtu-006	0.0000	562	0.0026	1085	0.0298	734	0.0002	883	-	1386
1112	sjtu-007	0.0000	258	0.0002	290	0.0054	201	0.0000	309	-	1420
1113	sjtu-008	0.0000	43	0.0003	321	0.0074	247	0.0000	290	-	1388
1114	sjtu-009	0.0000	98	0.0003	320	0.0074	244	0.0000	257	-	1346
1115	sjtu-010	0.0000	270	0.0003	319	0.0074	246	0.0000	117	-	1158
1116	sktelecom-000	0.0000	533	0.0008	697	0.0190	526	0.0000	585	0.0004	653
1117	smargtech-001	0.0000	247	0.0000	189	0.0000	118	0.0000	164	-	1427
1118	smartbiometrik-001	0.0005	1281	0.0649	1325	0.2147	972	0.0017	1313	0.0008	934
1119	smartengines-000	0.0066	1319	0.0150	1285	0.1656	961	0.0022	1326	0.0013	1001
1120	smartengines-001	0.0003	1275	0.0073	1246	0.0714	926	0.0007	1230	0.0005	834
1121	smartvist-000	0.0000	282	0.0026	1094	0.0357	805	0.0002	860	0.0011	970
1122	smartvist-001	0.0000	828	0.0005	454	0.0069	235	0.0001	813	0.0003	357
1123	smilart-001	-	1369	-	1437	-	1255	-	1476	0.0273	1083
1124	smilart-002	0.0000	1184	0.0036	1167	-	1327	-	1491	0.0011	968
1125	smilart-003	0.0003	1273	0.0100	1271	-	1336	0.0014	1295	0.0013	1003
1126	sodec-000	0.0000	457	0.0000	70	0.0000	47	0.0000	14	0.0000	73
1127	softtech-000	0.0000	143	0.0006	582	0.0166	461	0.0001	793	-	1463
1128	sparktek-000	0.0000	1103	0.0016	977	0.0356	804	0.0001	692	-	1294
1129	sparktek-001	0.0000	634	0.0004	387	0.0092	294	0.0000	561	-	1455
1130	sparsh-001	0.0000	278	0.0007	584	0.0168	464	0.0000	363	0.0004	484
1131	sqisoft-001	0.0000	418	0.0003	336	0.0078	256	0.0000	389	0.0003	398
1132	sqisoft-002	0.0000	492	0.0003	331	0.0078	257	0.0000	405	0.0003	409
1133	sqisoft-003	0.0000	517	0.0003	339	0.0078	260	0.0000	403	0.0003	347
1134	staqu-000	0.0000	52	0.0000	164	-	1361	0.0000	277	0.0000	165
1135	starhybrid-001	0.0001	1254	0.0033	1154	-	1206	0.0009	1256	0.0023	1036
1136	stcon-000	0.0000	870	0.0017	1003	0.0301	740	0.0000	463	0.0003	371
1137	stcon-001	0.0000	547	0.0017	1004	0.0301	741	0.0000	480	0.0003	373
1138	stcon-002	0.0000	581	0.0011	861	0.0226	624	0.0000	452	0.0003	321
1139	stcon-003	0.0000	419	0.0005	460	0.0142	398	0.0000	438	0.0004	491
1140	stcon-004	0.0000	142	0.0005	463	0.0142	399	0.0000	446	-	1465
1141	stcon-005	0.0000	304	0.0005	459	0.0142	397	0.0000	444	-	1185
1142	stcon-006	0.0000	180	0.0007	595	0.0191	533	0.0000	475	-	1488
1143	stcon-007	0.0000	22	0.0007	592	0.0191	530	0.0000	481	-	1356
1144	stcon-008	0.0000	28	0.0007	593	0.0191	531	0.0000	482	-	1353
1145	stcon-009	0.0000	179	0.0007	594	0.0191	532	0.0000	476	-	1484
1146	stcon-010	0.0000	401	0.0006	579	0.0184	502	0.0000	467	-	1276
1147	stellartecs-001	0.0000	992	0.0012	890	0.0324	769	0.0003	1021	-	1201
1148	stellartecs-003	0.0000	913	0.0012	891	0.0324	770	0.0003	1023	-	1307
1149	stengg-000	0.0000	50	0.0013	931	0.0350	801	0.0000	513	0.0004	615
1150	sukshi-000	0.0000	425	0.0000	117	0.0000	74	0.0000	71	0.0000	112
1151	sumsub-000	0.0000	1067	0.0018	1015	0.0189	521	0.0011	1278	-	1481
1152	sumsub-001	0.0000	592	0.0007	609	0.0197	545	0.0000	344	-	1322
1153	suprema-000	0.0000	951	0.0017	996	-	1061	0.0002	944	0.0004	664
1154	suprema-001	0.0000	1002	0.0027	1098	0.0471	856	0.0003	1029	0.0004	753
1155	suprema-002	0.0000	897	0.0010	842	0.0271	718	0.0002	895	0.0004	547
1156	suprema-003	0.0000	542	0.0008	703	0.0231	644	0.0000	401	0.0004	544
1157	suprema-004	0.0000	855	0.0014	948	0.0299	735	0.0000	381	0.0004	537
1158	suprema-005	0.0000	924	0.0014	949	0.0299	736	0.0005	1161	0.0004	671
1159	supremaid-001	0.0000	648	0.0020	1042	0.0330	780	0.0001	735	0.0004	755
1160	supremaid-002	0.0000	830	0.0020	1041	0.0330	779	0.0001	732	0.0004	751

Table 112: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1161	surrey-cvssp-000	0.0000	331	0.0000	15	0.0000	8	0.0000	92	0.0000	14
1162	surrey-cvssp-001	0.0173	1334	0.0007	606	0.0179	487	0.0011	1276	0.0015	1011
1163	surrey-cvssp-002	0.0000	1127	0.0006	530	0.0156	435	0.0001	697	0.0004	582
1164	surrey-cvssp-003	0.0000	438	0.0009	738	0.0165	460	0.0000	62	0.0001	253
1165	surrey-cvssp-004	0.0000	726	0.0015	968	0.0311	751	0.0004	1094	-	1149
1166	swsam-001	0.0000	355	0.0012	914	0.0263	707	0.0000	106	0.0000	34
1167	synesis-000	-	1495	-	1410	-	1168	-	1413	0.0007	914
1168	synesis-001	-	1402	-	1498	-	1479	0.0035	1352	0.0021	1026
1169	synesis-003	-	1412	-	1478	-	1430	0.0102	1383	0.1184	1091
1170	synesis-004	-	1434	-	1385	-	1089	0.0164	1390	0.0035	1056
1171	synesis-005	0.0000	605	0.0019	1021	-	1304	0.0001	711	0.0005	783
1172	synesis-006	0.0000	504	0.0003	347	-	1172	0.0000	578	0.0003	297
1173	synesis-007	0.0000	546	0.0013	926	-	1326	0.0002	966	0.0004	590
1174	synology-000	0.0000	147	0.0000	211	-	1438	0.0000	191	0.0000	208
1175	synology-001	0.0000	422	0.0000	112	-	1242	0.0000	74	0.0000	108
1176	synology-002	0.0000	267	0.0000	32	-	1057	0.0000	118	0.0000	36
1177	syscake-000	0.0008	1298	0.0017	1002	0.0287	730	0.0007	1225	-	1357
1178	szpu-000	0.0000	287	0.0014	959	0.0330	781	0.0001	657	-	1193
1179	sztu-000	0.0000	489	0.0000	64	-	1129	0.0000	7	0.0000	62
1180	sztu-001	0.0000	433	0.0000	107	0.0000	71	0.0000	64	0.0000	104
1181	t4isb-000	0.0000	473	0.0000	55	0.0000	36	0.0000	4	0.0000	67
1182	tahakom-001	0.0025	1306	0.0020	1049	0.0139	388	0.0020	1319	-	1188
1183	tahakom-002	0.0329	1346	0.0083	1261	0.0185	511	0.0238	1395	-	1140
1184	tahakom-003	0.0000	376	0.0000	21	0.0000	12	0.0000	102	-	1121
1185	taiwanmobile-001	0.0000	1058	0.0023	1075	0.0331	783	0.0006	1205	-	1208
1186	taiwanmobile-002	0.0000	1037	0.0015	962	0.0238	657	0.0004	1135	-	1349
1187	tcc-000	0.0000	808	0.0699	1330	0.0478	857	0.0000	356	-	1273
1188	tcc-001	0.0000	1204	0.0004	391	0.0089	286	0.0005	1172	-	1268
1189	tcv-001	0.0000	817	0.0013	915	0.0152	421	0.0001	830	-	1269
1190	tech5-000	-	1432	-	1393	-	1109	-	1461	0.0003	394
1191	tech5-001	-	1410	-	1470	-	1393	0.0004	1116	0.0003	366
1192	tech5-002	-	1387	-	1491	-	1464	0.0001	684	0.0003	292
1193	tech5-003	0.0000	968	0.0005	465	-	1048	0.0001	683	0.0003	306
1194	tech5-004	0.0000	583	0.0008	667	-	1271	0.0003	1031	0.0004	743
1195	tech5-005	0.0000	896	0.0007	647	-	1156	0.0000	462	0.0004	676
1196	tech5-007	0.0000	681	0.0014	956	0.0305	747	0.0000	424	0.0004	524
1197	tech5-008	0.0000	212	0.0007	586	0.0168	465	0.0000	365	0.0004	496
1198	tech5-009	0.0000	1009	0.0030	1131	0.0542	877	0.0002	938	-	1334
1199	techainer-001	0.0000	1159	0.0004	419	0.0085	278	0.0005	1160	0.0004	667
1200	techainer-002	0.0000	1157	0.0004	425	0.0085	281	0.0005	1162	-	1496
1201	techainer-003	0.0000	921	0.0031	1141	0.0249	680	0.0003	1051	-	1472
1202	techbiometricid-001	0.0000	154	0.0000	218	0.0000	134	0.0000	196	-	1445
1203	techbiometricid-002	0.0000	206	0.0000	188	0.0000	115	0.0000	157	-	1408
1204	techsign-000	0.0007	1291	0.0334	1310	0.2093	966	0.0020	1322	0.0011	964
1205	techsign-001	0.0000	864	0.0008	718	0.0253	692	0.0002	918	0.0004	608
1206	telexper-000	0.0000	911	0.0006	559	0.0147	407	0.0003	1068	-	1304
1207	tevian-000	-	1420	-	1383	-	1082	-	1455	0.0012	994
1208	tevian-001	-	1498	-	1404	-	1147	-	1411	0.0012	991
1209	tevian-002	-	1428	-	1377	-	1068	-	1451	0.0005	865
1210	tevian-003	-	1454	-	1369	-	1051	0.0003	1030	0.0005	863
1211	tevian-004	0.0000	1063	0.0024	1077	-	1315	0.0002	891	0.0005	856
1212	tevian-005	0.0001	1256	0.0041	1182	-	1384	0.0006	1207	0.0006	906
1213	tevian-006	0.0000	434	0.0012	893	-	1213	0.0003	1043	0.0005	829
1214	tevian-007	0.0000	739	0.0015	973	0.0429	847	0.0002	937	0.0004	639
1215	tevian-008	0.0000	573	0.0006	511	0.0109	329	0.0000	457	0.0003	331
1216	tiger-001	-	1366	-	1443	-	1279	-	1482	0.0006	893
1217	tiger-002	0.0000	551	0.0010	821	-	1330	0.0001	783	0.0004	565
1218	tiger-003	0.0000	891	-	1406	-	1154	0.0001	772	0.0004	561

Table 113: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1219	tiger-004	0.0000	1207	0.0022	1061	-	1386	0.0001	777	0.0004	738
1220	tiger-005	0.0000	789	0.0009	776	0.0194	541	0.0001	695	0.0004	553
1221	tiger-006	0.0000	941	0.0011	873	0.0396	831	0.0001	820	0.0004	773
1222	tiger-007	0.0000	906	0.0051	1215	0.0201	553	0.0002	912	-	1318
1223	tinkoff-001	0.0000	904	0.0008	693	0.0171	473	0.0001	797	0.0004	564
1224	tngd-001	0.0000	715	0.0009	720	0.0218	608	0.0000	546	-	1156
1225	tngd-002	0.0000	139	0.0000	233	0.0000	144	0.0000	215	-	1460
1226	tnitech-000	0.0000	810	0.0003	335	0.0060	208	0.0005	1193	0.0003	325
1227	tongyi-004	-	1380	-	1444	-	1287	-	1485	0.0006	898
1228	tongyi-005	0.0000	203	0.0000	186	-	1402	0.0000	162	0.0000	180
1229	tongyitrans-001	-	1392	-	1480	-	1440	-	1468	0.0040	1061
1230	tongyitrans-002	-	1438	-	1388	-	1096	-	1457	0.0040	1060
1231	tongyitrans-003	-	1422	-	1373	-	1056	-	1450	0.0006	896
1232	toppanidgate-000	0.0000	602	0.0008	685	0.0232	645	0.0004	1093	0.0004	625
1233	toppansecurity-001	0.0000	900	0.0020	1039	0.0161	455	0.0001	690	-	1371
1234	toshiba-000	-	1433	-	1394	-	1111	-	1460	0.0000	54
1235	toshiba-001	-	1382	-	1448	-	1301	-	1487	0.0000	129
1236	toshiba-002	0.0000	61	0.0000	165	-	1367	0.0000	281	0.0000	160
1237	toshiba-003	0.0000	38	0.0001	269	-	1379	0.0001	786	0.0001	255
1238	toshiba-004	0.0000	511	0.0000	77	0.0000	50	0.0000	27	0.0000	79
1239	toshiba-005	0.0000	650	0.0004	404	0.0050	196	0.0001	809	0.0003	333
1240	toshiba-006	0.0000	733	0.0004	403	0.0050	195	0.0001	808	0.0003	329
1241	toshiba-007	0.0000	937	0.0005	461	0.0082	271	0.0002	986	0.0004	616
1242	toshiba-008	0.0000	935	0.0005	462	0.0082	272	0.0002	987	0.0004	617
1243	toshiba-009	0.0000	783	0.0004	428	0.0070	237	0.0002	973	-	1099
1244	toshiba-010	0.0000	990	0.0004	389	0.0064	217	0.0002	961	-	1205
1245	toshiba-011	0.0000	624	0.0006	521	0.0137	384	0.0002	960	-	1473
1246	toshiba-012	0.0000	670	0.0004	432	0.0120	356	0.0000	610	-	1411
1247	toshiba-013	0.0000	938	0.0004	396	0.0078	258	0.0000	528	-	1404
1248	toss-000	0.0000	460	0.0009	744	0.0209	574	0.0001	713	-	1216
1249	toss-001	0.0000	420	0.0009	745	0.0209	575	0.0001	714	-	1299
1250	touchlessid-000	0.0042	1311	0.0133	1281	0.2009	965	0.0018	1316	0.0032	1053
1251	touchlessid-001	0.0000	200	0.0036	1168	0.0923	945	0.0000	160	0.0000	179
1252	touchlessid-002	0.0000	351	0.0043	1194	0.1087	950	0.0000	109	0.0000	31
1253	touchlessid-003	0.0000	768	0.0010	824	0.0287	729	0.0002	863	0.0004	732
1254	touchlessid-004	0.0000	1206	0.0013	937	0.0398	834	0.0002	982	-	1303
1255	transmitsecurity-001	0.0000	819	0.0009	742	0.0226	625	0.0000	533	-	1267
1256	trueface-000	0.0000	642	0.0000	216	-	1446	0.0000	198	0.0000	203
1257	trueface-001	0.0000	983	0.0038	1174	-	1234	0.0007	1237	0.0005	830
1258	trueface-002	0.0000	952	0.0046	1203	-	1064	0.0003	1015	0.0005	852
1259	trueface-003	0.0000	973	0.0046	1204	0.0397	833	0.0003	1014	0.0005	855
1260	trueidvng-001	0.0000	985	0.0020	1048	0.0385	824	0.0002	924	0.0005	821
1261	truststamp-001	0.0000	805	0.0010	815	0.0189	524	0.0000	407	0.0003	372
1262	truststamp-002	0.0000	816	0.0010	814	0.0189	523	0.0000	406	-	1271
1263	truststamp-003	0.0000	666	0.0010	819	0.0189	525	0.0000	415	-	1477
1264	tsight-000	0.0000	794	0.0003	326	0.0064	218	0.0000	327	-	1132
1265	ttigroup-000	0.0000	521	0.0000	78	0.0000	53	0.0000	34	-	1225
1266	tuputech-000	0.0003	1277	0.0116	1279	-	1352	-	1496	0.0081	1074
1267	turingtechvip-001	0.0001	1235	0.0007	633	0.0061	212	0.0007	1228	0.0006	882
1268	turingtechvip-002	0.0001	1236	0.0017	1000	0.0097	302	0.0007	1227	0.0006	881
1269	turkcell-000	0.0110	1327	0.0234	1300	0.0350	802	0.0103	1384	0.0306	1085
1270	turkcell-001	0.0000	170	0.0002	289	0.0034	184	0.0001	710	0.0003	293
1271	turkcell-002	0.0000	23	0.0002	281	0.0026	179	0.0001	637	-	1355
1272	twface-000	0.0000	138	0.0000	234	0.0000	145	0.0000	214	0.0000	213
1273	twface-001	0.0000	110	0.0000	135	0.0000	86	0.0000	240	0.0000	133
1274	ulsee-001	0.0000	100	0.0000	133	-	1282	0.0000	246	0.0000	131
1275	ultinous-000	-	1460	-	1364	-	1035	-	1441	0.0003	345
1276	ultinous-001	-	1364	-	1440	-	1269	-	1480	0.0003	349

Table 114: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1277	uluface-002	0.0000	486	0.0000	59	-	1125	0.0000	8	0.0000	57
1278	uluface-003	0.0000	461	0.0001	276	-	1141	0.0002	866	0.0002	273
1279	unicc-001	0.0000	127	0.0012	889	0.0230	640	0.0004	1143	0.0004	571
1280	unicc-002	0.0316	1344	0.0082	1257	0.0188	516	0.0226	1394	0.0936	1090
1281	unicc-003	0.0007	1294	0.0022	1058	0.0094	296	0.0026	1341	0.0029	1046
1282	unicc-004	0.0000	1202	0.0012	887	0.0066	226	0.0012	1285	-	1352
1283	unissey-001	0.0000	295	0.0000	49	0.0000	33	0.0000	142	0.0000	52
1284	unissey-002	0.0000	364	0.0000	30	0.0000	20	0.0000	111	0.0000	29
1285	unissey-003	0.0000	377	0.0008	659	0.0191	527	0.0001	702	0.0004	501
1286	unissey-004	0.0000	161	0.0009	755	0.0240	660	0.0001	756	0.0004	602
1287	unissey-005	0.0000	518	0.0009	753	0.0240	659	0.0001	752	-	1234
1288	unissey-006	0.0000	620	0.0220	1296	0.0583	885	0.0003	1054	-	1331
1289	unissey-007	0.0000	801	0.0101	1272	0.0376	820	0.0004	1086	-	1285
1290	uniubi-000	0.0000	498	0.0000	87	0.0000	58	0.0000	37	-	1255
1291	universeai-000	0.0000	1082	0.0030	1128	0.0321	766	0.0000	613	-	1470
1292	upc-001	0.0000	1108	0.0003	325	-	1302	0.0003	1019	0.0003	387
1293	urielsoft-000	0.0000	462	0.0023	1073	0.0371	815	0.0000	589	-	1215
1294	useb-000	0.0000	539	0.0009	769	0.0228	634	0.0001	826	0.0004	520
1295	useb-001	0.0000	603	0.0009	766	0.0228	631	0.0001	823	0.0004	519
1296	useb-002	0.0000	894	0.0009	765	0.0228	630	0.0001	815	0.0004	508
1297	useb-003	0.0000	608	0.0009	768	0.0228	632	0.0001	825	-	1345
1298	useb-004	0.0000	729	0.0004	417	0.0112	337	0.0002	932	-	1196
1299	uxlabs-001	0.0000	145	0.0000	235	0.0000	146	0.0000	211	0.0000	214
1300	uxlabs-003	0.0000	24	0.0000	150	0.0000	94	0.0000	261	0.0000	151
1301	uzinfocom-000	0.0000	379	0.0005	450	0.0132	375	0.0001	762	-	1125
1302	uzinfocom-001	0.0000	446	0.0002	288	0.0044	189	0.0001	677	-	1220
1303	vaidio-001	0.0000	873	0.0006	551	0.0157	437	0.0043	1362	-	1257
1304	vcog-002	-	1446	-	1350	-	988	-	1431	0.0019	1023
1305	vcortex-001	0.0000	1140	0.2295	1343	0.0802	934	0.0017	1312	0.0006	880
1306	vd-000	-	1418	-	1380	-	1075	-	1453	0.0000	42
1307	vd-001	0.0000	1172	0.0030	1132	-	992	0.0004	1120	0.0009	937
1308	vd-002	0.0000	157	0.0000	215	1.0000	1445	0.0000	197	0.0000	204
1309	vd-003	0.0001	1238	0.0041	1181	0.0676	910	0.0030	1349	0.0029	1047
1310	veertec-001	0.0006	1289	0.0140	1284	0.1098	951	0.0019	1318	-	1167
1311	veridas-001	-	1373	-	1451	-	1307	0.0001	782	0.0005	808
1312	veridas-002	0.0000	1087	0.0026	1093	-	1370	0.0001	785	0.0005	812
1313	veridas-003	0.0000	1089	0.0026	1087	-	1362	0.0001	784	0.0005	811
1314	veridas-004	0.0000	1104	0.0026	1092	-	1171	0.0001	773	0.0005	807
1315	veridas-005	0.0000	1093	0.0026	1090	-	1486	0.0001	780	0.0005	818
1316	veridas-006	0.0000	1099	0.0026	1086	0.0595	886	0.0001	775	0.0005	806
1317	veridas-007	0.0000	1095	0.0026	1089	0.0595	888	0.0001	779	0.0005	815
1318	veridas-008	0.0000	1096	0.0026	1088	0.0595	887	0.0001	778	0.0005	813
1319	veridas-009	0.0000	1101	0.0026	1091	0.0597	889	0.0001	700	0.0005	820
1320	veridas-010	0.0000	989	0.0011	876	0.0212	589	0.0004	1132	-	1213
1321	veridium-000	0.0061	1318	0.5956	1346	0.2889	977	0.0050	1369	0.0009	936
1322	veridium-001	0.0001	1225	0.0087	1264	0.1615	958	0.0014	1299	0.0006	874
1323	veridium-002	0.0000	1212	0.0083	1262	0.1591	956	0.0010	1262	0.0005	851
1324	veridium-003	0.0001	1224	0.0087	1263	0.1615	957	0.0014	1298	-	1447
1325	veridium-004	0.0001	1226	0.0028	1111	0.0832	939	0.0005	1180	-	1354
1326	verigram-000	0.0000	1031	0.0068	1240	0.0822	938	0.0003	1078	0.0005	837
1327	verigram-001	0.0000	944	0.0003	333	0.0060	207	0.0002	959	0.0003	402
1328	verigram-002	0.0000	1135	0.0052	1221	0.0378	821	0.0010	1264	0.0006	900
1329	verigram-003	0.0000	1187	0.0025	1082	0.0288	731	0.0009	1261	0.0008	930
1330	verigram-004	0.0000	910	0.0001	273	0.0016	172	0.0001	814	-	1305
1331	verihubs-inteligensia-000	0.0000	841	0.0029	1113	0.0669	908	0.0001	652	0.0004	677
1332	verihubs-inteligensia-001	0.0000	690	0.0029	1114	0.0669	909	0.0001	660	0.0004	685
1333	verihubs-inteligensia-002	0.0000	889	0.0029	1112	0.0669	907	0.0001	650	0.0004	675
1334	verijelas-000	0.0000	850	0.0023	1070	0.0375	818	0.0004	1145	0.0006	888

Table 115: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1335	verismart-001	0.0000	312	0.0002	292	0.0049	194	0.0001	656	-	1171
1336	via-000	0.0000	329	0.0000	14	-	1016	0.0000	94	0.0000	11
1337	via-001	0.0000	336	0.0000	3	-	984	0.0000	78	0.0000	9
1338	via-004	0.0000	337	0.0000	2	0.0000	1	0.0000	79	0.0000	10
1339	via-005	0.0000	470	0.0000	52	0.0000	35	0.0000	6	0.0000	63
1340	via-006	0.0000	306	0.0000	42	0.0000	28	0.0000	128	-	1182
1341	via-007	0.0000	611	0.0010	856	0.0318	756	0.0003	1028	-	1343
1342	viante-000	0.0000	922	0.0008	700	0.0168	468	0.0001	749	0.0004	774
1343	viante-001	0.0000	917	0.0008	699	0.0168	467	0.0001	750	-	1341
1344	viante-002	0.0000	641	0.0007	646	0.0154	428	0.0001	748	-	1448
1345	viante-003	0.0000	692	0.0007	645	0.0154	427	0.0001	747	-	1434
1346	viante-004	0.0000	878	0.0007	643	0.0154	426	0.0001	745	-	1244
1347	videmo-000	0.0000	996	0.0019	1026	-	1169	0.0003	1053	0.0012	992
1348	videmo-001	0.0000	1070	0.0170	1287	0.0332	785	0.0010	1272	0.0011	976
1349	videmo-002	0.0000	520	0.0006	550	0.0189	518	0.0001	723	0.0004	488
1350	videmo-003	0.0000	398	0.0006	552	0.0189	519	0.0001	724	-	1279
1351	videonetics-001	0.0004	1278	0.0309	1307	-	1300	0.0015	1305	0.0010	951
1352	videonetics-002	0.0000	1032	0.0459	1320	-	1118	0.0006	1211	0.0005	858
1353	viettelai-000	0.0000	423	0.0000	111	0.0000	73	0.0000	73	0.0000	107
1354	viettelai-001	0.0000	950	0.0010	805	0.0161	454	0.0000	594	0.0004	481
1355	viettelai-002	0.0000	1000	0.0019	1023	0.0334	789	0.0002	971	-	1358
1356	viettelai-003	0.0000	901	0.0011	879	0.0218	609	0.0001	642	-	1377
1357	viettelai-004	0.0000	749	0.0005	475	0.0139	389	0.0000	508	-	1168
1358	viettelai-005	0.0000	674	0.0005	467	0.0137	383	0.0000	498	-	1402
1359	viettelai-006	0.0000	862	0.0005	466	0.0137	382	0.0000	485	-	1209
1360	viettelhightech-000	0.0000	1128	0.0019	1030	0.0368	812	0.0007	1233	0.0005	853
1361	viettelhightech-001	0.0000	560	0.0014	947	0.0434	849	0.0001	848	-	1390
1362	viettelhightech-002	0.0000	616	0.0014	946	0.0434	848	0.0001	847	-	1340
1363	vigilantsolutions-000	-	1499	-	1405	-	1149	-	1412	0.0007	913
1364	vigilantsolutions-001	-	1462	-	1416	-	1195	-	1417	0.0004	695
1365	vigilantsolutions-002	-	1452	-	1367	-	1046	-	1444	0.0004	689
1366	vigilantsolutions-003	-	1360	-	1464	-	1375	-	1499	0.0004	469
1367	vigilantsolutions-004	-	1400	-	1495	-	1476	-	1472	0.0004	475
1368	vigilantsolutions-005	-	1461	-	1365	-	1036	0.0001	670	0.0004	446
1369	vigilantsolutions-006	0.0000	1077	0.0028	1104	-	1210	0.0001	668	0.0004	458
1370	vigilantsolutions-007	0.0000	1068	0.0028	1107	-	1401	0.0001	671	0.0004	471
1371	vigilantsolutions-008	0.0000	1079	0.0028	1102	-	1164	0.0001	665	0.0004	452
1372	vigilantsolutions-009	0.0000	1075	0.0028	1103	-	1192	0.0001	667	0.0004	456
1373	vigilantsolutions-010	0.0000	1060	0.0028	1106	0.0609	892	0.0001	675	0.0004	467
1374	vigilantsolutions-011	0.0000	1062	0.0028	1105	0.0609	891	0.0001	674	0.0004	461
1375	vinai-000	0.0000	119	0.0000	140	-	1298	0.0000	247	0.0000	130
1376	vinbigdata-001	0.0000	92	0.0000	142	0.0000	91	0.0000	252	0.0000	144
1377	vinbigdata-002	0.0000	103	0.0015	967	0.0250	682	0.0000	530	0.0004	715
1378	vinbigdata-003	0.0000	782	0.0007	602	0.0151	417	0.0000	616	0.0004	477
1379	vinbigdata-004	0.0000	597	0.0005	473	0.0141	393	0.0000	399	-	1313
1380	vinbigdata-005	0.0000	813	0.0005	471	0.0141	391	0.0000	387	-	1278
1381	vinbigdata-006	0.0000	839	0.0005	472	0.0141	392	0.0000	390	-	1295
1382	vinbigdata-007	0.0000	890	0.0005	470	0.0141	390	0.0000	383	-	1226
1383	vion-000	0.0050	1313	0.0392	1317	-	1433	0.0130	1388	0.0078	1073
1384	visage-000	0.0000	1133	0.0054	1224	-	1140	0.0009	1254	0.0006	883
1385	visica-001	0.0000	1169	0.0068	1242	0.0692	917	0.0010	1263	-	1204
1386	visica-002	0.0000	956	0.0080	1255	0.1006	948	0.0004	1124	-	1175
1387	visica-003	0.0000	958	0.0022	1062	0.0419	842	0.0004	1123	-	1108
1388	visionbox-000	0.0000	1192	0.0033	1152	-	1150	0.0005	1189	0.0011	971
1389	visionbox-001	0.0000	1191	0.0033	1151	-	1020	0.0005	1190	0.0011	969
1390	visionbox-002	0.0000	74	0.0017	991	0.0270	717	0.0000	536	0.0004	761
1391	visionbox-003	0.0000	1007	0.0010	800	0.0209	578	0.0001	737	0.0004	647
1392	visionbox-004	0.0000	965	0.0013	917	0.0248	676	0.0001	807	0.0004	719

Table 116: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1393	visionbox-005	0.0000	1065	0.0037	1172	0.0301	742	0.0005	1149	-	1459
1394	visionbox-006	0.0000	1131	0.0040	1178	0.0331	782	0.0007	1223	-	1277
1395	visionbox-007	0.0000	1113	0.0041	1180	0.0324	768	0.0006	1217	-	1319
1396	visionbox-008	0.0000	1071	0.0035	1164	0.0280	720	0.0001	849	-	1143
1397	visionlabs-001	-	1455	-	1371	-	1052	-	1446	0.0014	1008
1398	visionlabs-002	-	1458	-	1363	-	1033	-	1443	0.0026	1040
1399	visionlabs-003	-	1370	-	1438	-	1256	-	1477	0.0025	1039
1400	visionlabs-004	-	1449	-	1354	-	1003	-	1434	0.0008	931
1401	visionlabs-005	-	1385	-	1486	-	1460	0.0002	1008	0.0023	1034
1402	visionlabs-006	-	1396	-	1500	-	1494	0.0003	1062	0.0005	819
1403	visionlabs-007	0.0000	942	0.0014	953	-	1407	0.0003	1057	0.0005	803
1404	visionlabs-008	0.0000	799	0.0015	965	-	1191	0.0002	999	0.0004	760
1405	visionlabs-009	0.0000	756	0.0010	803	-	1006	0.0001	629	0.0004	620
1406	visionlabs-010	0.0000	1043	0.0009	743	-	1424	0.0001	812	0.0004	633
1407	visionlabs-011	0.0000	469	0.0006	542	0.0156	436	0.0001	676	0.0004	485
1408	visteam-000	0.0000	1188	0.0031	1138	-	1229	0.0005	1182	0.0011	959
1409	visteam-001	0.0000	1045	0.0014	950	-	1435	0.0002	906	0.0004	655
1410	visteam-002	0.0000	1040	0.0014	951	0.0285	727	0.0002	900	0.0004	657
1411	visteam-003	0.0000	665	0.0010	836	0.0225	621	0.0001	641	0.0004	551
1412	visteam-004	0.0000	784	0.0010	835	0.0225	618	0.0001	753	0.0004	533
1413	visteam-005	0.0000	572	0.0010	834	0.0224	617	0.0001	681	0.0004	548
1414	visteam-006	0.0000	845	0.0008	689	0.0209	571	0.0000	357	0.0004	457
1415	visteam-007	0.0000	578	0.0008	691	0.0209	573	0.0000	374	0.0004	462
1416	visteam-008	0.0000	599	0.0008	690	0.0209	572	0.0000	372	0.0004	460
1417	visteam-009	0.0000	857	0.0010	853	0.0328	777	0.0000	352	-	1214
1418	visteam-010	0.0000	759	0.0010	851	0.0328	776	0.0000	359	-	1113
1419	vivotek-001	0.0000	280	0.0006	536	0.0161	452	0.0000	140	-	1192
1420	vivotek-002	0.0000	352	0.0006	535	0.0161	451	0.0000	107	-	1142
1421	vivotek-003	0.0000	372	0.0006	538	0.0161	453	0.0000	545	-	1134
1422	vixvizion-000	0.0002	1265	0.0028	1108	-	1114	0.0010	1270	0.0012	990
1423	vixvizion-001	0.0000	970	0.0030	1130	-	1053	0.0001	763	0.0004	662
1424	vixvizion-002	0.0000	1008	0.0018	1005	-	1317	0.0000	553	0.0004	645
1425	vixvizion-003	0.0000	70	0.0000	130	-	1272	0.0000	232	0.0000	127
1426	vixvizion-004	0.0000	321	0.0000	11	0.0000	5	0.0000	89	0.0000	18
1427	vixvizion-005	0.0000	172	0.0000	241	0.0000	150	0.0000	224	0.0000	227
1428	vixvizion-006	0.0000	431	0.0000	113	0.0000	78	0.0000	68	0.0000	109
1429	vixvizion-007	0.0000	94	0.0000	143	0.0000	92	0.0000	259	0.0000	136
1430	vnis-000	0.0007	1296	0.0695	1329	0.0756	931	0.0038	1359	0.0009	942
1431	vnpay-000	0.0000	1078	0.0657	1328	0.0693	918	0.0037	1357	0.0006	876
1432	vnpay-001	0.0000	577	0.0052	1220	0.0497	859	0.0001	765	-	1375
1433	vnpt-001	0.0652	1349	0.2829	1345	-	1011	0.2116	1402	0.1598	1092
1434	vnpt-002	0.0000	678	0.0002	284	0.0024	177	0.0003	1061	0.0003	301
1435	vnpt-003	0.0000	332	0.0004	370	0.0105	309	0.0002	862	0.0004	418
1436	vnpt-004	0.0000	530	0.0006	519	0.0160	449	0.0002	887	0.0004	567
1437	vnpt-005	0.0000	487	0.0006	501	0.0154	429	0.0002	917	0.0004	536
1438	vnpt-006	0.0000	16	0.0006	504	0.0154	432	0.0002	922	0.0004	545
1439	vnpt-007	0.0000	213	0.0006	505	0.0154	433	0.0002	920	-	1396
1440	vnpt-008	0.0000	12	0.0006	503	0.0154	431	0.0002	923	-	1362
1441	vnpt-009	0.0000	73	0.0006	502	0.0154	430	0.0002	921	-	1317
1442	vocalize-001	0.0216	1338	0.0193	1291	0.1691	962	0.0068	1376	-	1495
1443	vocord-001	-	1484	-	1425	-	1231	-	1420	0.0038	1059
1444	vocord-002	-	1443	-	1349	-	983	-	1433	0.0037	1058
1445	vocord-003	-	1429	-	1392	-	1103	-	1459	0.0776	1088
1446	vocord-004	-	1359	-	1465	-	1376	0.0005	1181	0.0003	311
1447	vocord-005	-	1394	-	1481	-	1452	0.0013	1289	0.0003	404
1448	vocord-006	-	1500	-	1408	-	1155	0.0003	1044	0.0003	356
1449	vocord-007	0.0000	967	0.0535	1321	-	1041	0.0001	831	0.0004	419
1450	vocord-008	0.0000	618	0.0015	971	-	1295	0.0003	1069	0.0001	254

Table 117: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

	Algorithm	Failure to Enrol Rate[1]									
	Name	APPLICATION		BORDER		KIOSK		MUGSHOT		VISA	
	Name	SEC. 2.2		SEC. 2.4		SEC. 2.6		SEC. 2.5		SEC. 2.1	
1451	vocord-009	0.0000	842	0.0006	539	-	1216	0.0001	839	0.0003	299
1452	vocord-010	0.0000	931	0.0005	488	0.0141	395	0.0002	925	0.0003	405
1453	vogic-000	0.0000	868	0.0001	272	0.0015	171	0.0002	884	-	1202
1454	vsoft-001	0.0069	1321	0.8292	1347	0.8200	982	0.0173	1391	-	1298
1455	vsoft-002	0.0026	1307	0.0029	1120	0.0613	894	0.0059	1375	-	1236
1456	vsoft-003	0.0000	1163	0.0029	1121	0.0613	895	0.0005	1154	-	1360
1457	vsoft-004	0.0000	1166	0.0029	1119	0.0613	893	0.0005	1153	-	1169
1458	vts-000	0.0000	1024	0.0011	867	-	1049	0.0001	842	0.0004	746
1459	vts-001	0.0000	371	0.0003	317	0.0073	239	0.0000	393	0.0003	294
1460	vts-002	0.0000	453	0.0008	717	0.0150	414	0.0000	380	-	1223
1461	wicket-000	0.0000	718	0.0009	729	0.0260	703	0.0000	421	0.0004	556
1462	winsense-000	0.0000	478	0.0000	62	-	1122	0.0000	13	0.0000	61
1463	winsense-001	0.0000	224	0.0000	202	-	1423	0.0000	184	0.0000	200
1464	winsense-002	0.0000	86	0.0000	122	-	1261	0.0000	229	0.0000	114
1465	wiseai-001	0.0001	1222	0.0137	1283	0.0768	933	0.0018	1315	0.0018	1018
1466	wuhantianyu-001	0.0000	217	0.0007	610	0.0159	442	0.0001	630	0.0004	630
1467	x-laboratory-000	0.0247	1340	0.0000	174	-	1383	0.0005	1188	0.0002	276
1468	x-laboratory-001	0.0000	633	0.0012	894	-	1437	0.0001	792	0.0004	729
1469	xailient-001	0.0000	181	0.0000	236	0.0000	147	0.0000	216	-	1485
1470	xforwardai-000	0.0000	96	0.0000	146	-	1319	0.0000	256	0.0000	140
1471	xforwardai-001	0.0000	779	0.0007	631	-	998	0.0003	1036	0.0004	704
1472	xforwardai-002	0.0000	767	0.0007	632	-	1017	0.0003	1037	0.0004	705
1473	xm-000	0.0000	515	0.0007	600	-	1152	0.0001	639	0.0003	330
1474	yisheng-000	-	1466	-	1414	-	1179	-	1414	0.0006	877
1475	yisheng-001	-	1404	-	1471	-	1396	-	1463	0.0006	878
1476	yisheng-002	-	1384	-	1487	-	1458	-	1469	0.0006	879
1477	yisheng-004	0.0002	1268	-	1358	-	1015	0.0013	1288	0.0006	895
1478	yitu-000	-	1483	-	1426	-	1227	-	1421	0.0013	999
1479	yitu-001	-	1489	-	1395	-	1116	0.0009	1255	0.0006	897
1480	yitu-002	-	1367	-	1442	-	1281	-	1481	0.0000	122
1481	yitu-003	0.0000	488	0.0000	58	-	1124	0.0009	1253	0.0000	58
1482	yoonik-000	0.0000	655	0.0019	1027	-	1499	0.0001	721	0.0004	730
1483	yoonik-001	0.0000	80	0.0014	957	-	1254	0.0001	834	0.0004	713
1484	yoonik-002	0.0000	933	0.0010	829	0.0284	726	0.0003	1016	0.0006	871
1485	yoonik-003	0.0000	936	0.0009	772	0.0214	598	0.0002	880	0.0004	670
1486	yoonik-004	0.0000	468	0.0012	883	0.0251	686	0.0001	678	0.0004	486
1487	yoti-001	0.0000	920	0.0014	958	0.0418	841	0.0008	1252	0.0006	884
1488	yoti-002	0.0000	494	0.0007	598	0.0171	474	0.0001	739	-	1248
1489	ytu-000	0.0000	700	0.0010	850	-	1421	0.0002	978	0.0004	718
1490	yuan-000	0.0000	346	0.0000	7	-	1002	0.0000	81	0.0000	5
1491	yuan-001	0.0000	1098	0.0010	840	-	1086	0.0005	1184	0.0005	826
1492	yuan-002	0.0000	1092	0.0010	843	0.0284	725	0.0005	1186	0.0005	827
1493	yuan-003	0.0000	1100	0.0010	841	0.0284	724	0.0005	1185	0.0005	825
1494	yuan-004	0.0000	113	0.0000	134	0.0000	88	0.0000	239	0.0000	132
1495	yuan-005	0.0000	838	0.0005	478	0.0134	379	0.0002	915	0.0004	514
1496	yuan-006	0.0000	1122	0.0014	960	0.0369	813	0.0004	1109	0.0005	840
1497	yuan-007	0.0000	1125	0.0013	923	0.0336	794	0.0004	1107	0.0005	844
1498	yuan-008	0.0000	1123	0.0013	921	0.0336	792	0.0004	1108	0.0005	841
1499	yuan-009	0.0000	1118	0.0013	922	0.0336	793	0.0004	1110	-	1378
1500	yuan-010	0.0000	1129	0.0013	920	0.0336	791	0.0004	1106	-	1117
1501	yuanli-001	0.0000	219	0.0016	988	0.0318	755	0.0001	686	-	1399

Table 118: FTE is the proportion of failed template generation attempts. Failures can occur because the software throws an exception, or because the software electively refuses to process the input image. This would typically occur if a face is not detected. FTE is measured as the number of function calls that give EITHER a non-zero error code OR that give a "small" template. This is defined as one whose size is less than 0.3 times the median template size for that algorithm. This second rule is needed because some algorithms incorrectly fail to return a non-zero error code when template generation fails.
A hyphen "-" indicates the dataset was not produced. [1]*The effects of FTE are included in the accuracy results of this report by regarding any template comparison involving a failed template to produce a low similarity score. Thus higher FTE results in higher FNMR and lower FMR.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.4 Recognition accuracy

Core algorithm accuracy is stated via:

- ▷ **Cooperative subjects**
  - The summary table of Figure 92;
  - The visa image DETs of Figure 225;
  - The mugshot DETs of Figure 296;
  - The mugshot ageing profiles of Figure 813;
  - The human-difficult pairs of Figure 115
- ▷ **Non-cooperative subjects**
  - The photojournalism DET of Figure 332

Figure 635 shows dependence of false match rate on algorithm score threshold. This allows a deployer to set a threshold to target a particular false match rate appropriate to the security objectives of the application.

Figure 634 likewise shows FMR(T) but for mugshots, and specially four subsets of the population.

Note that in both the mugshot and visa sets false match rates vary with the ethnicity, age, and sex, of the enrollee and impostor. For example figure 402 summarizes FMR for impostors paired from four groups black females, black males, white females, white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

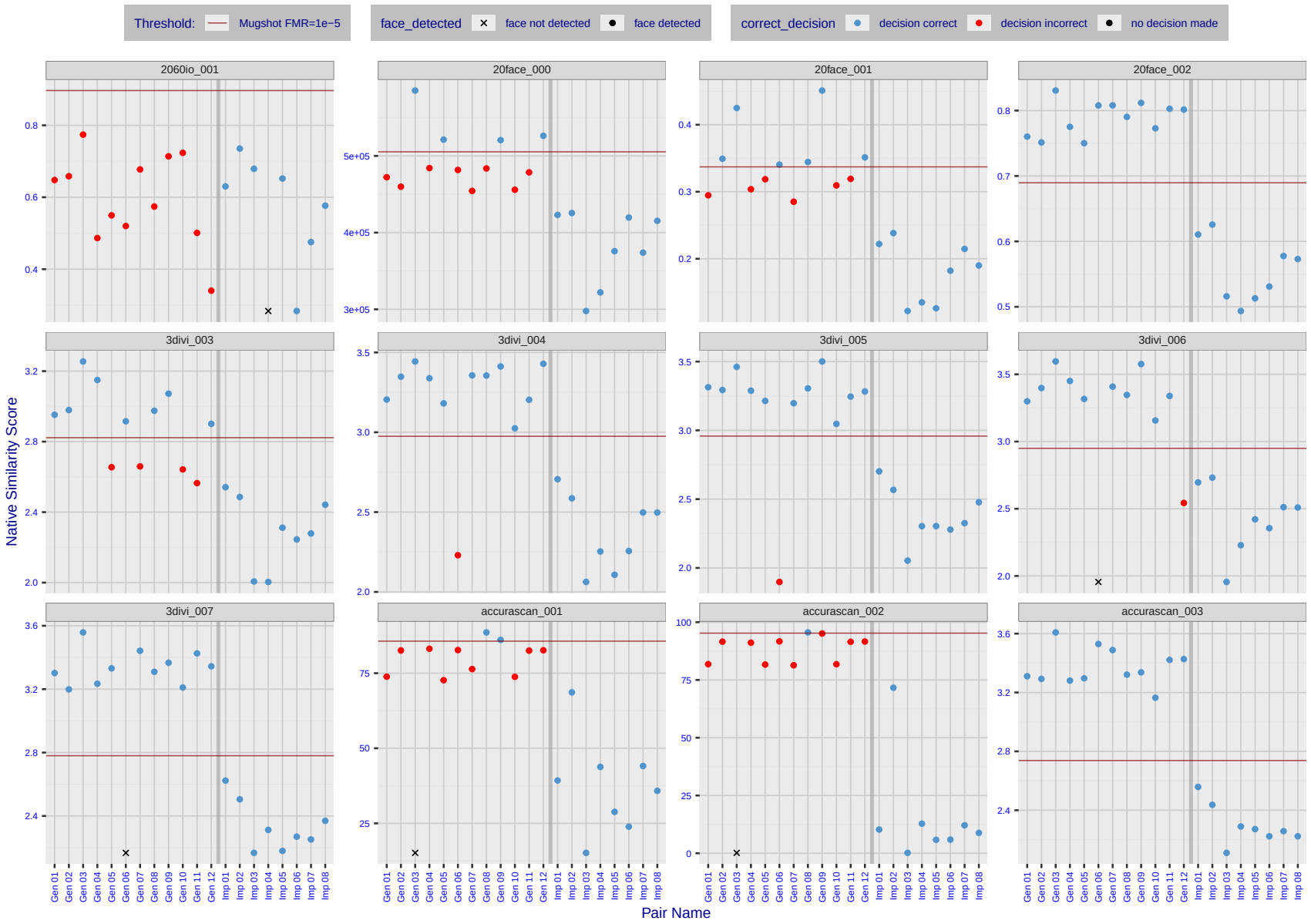


Figure 4: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

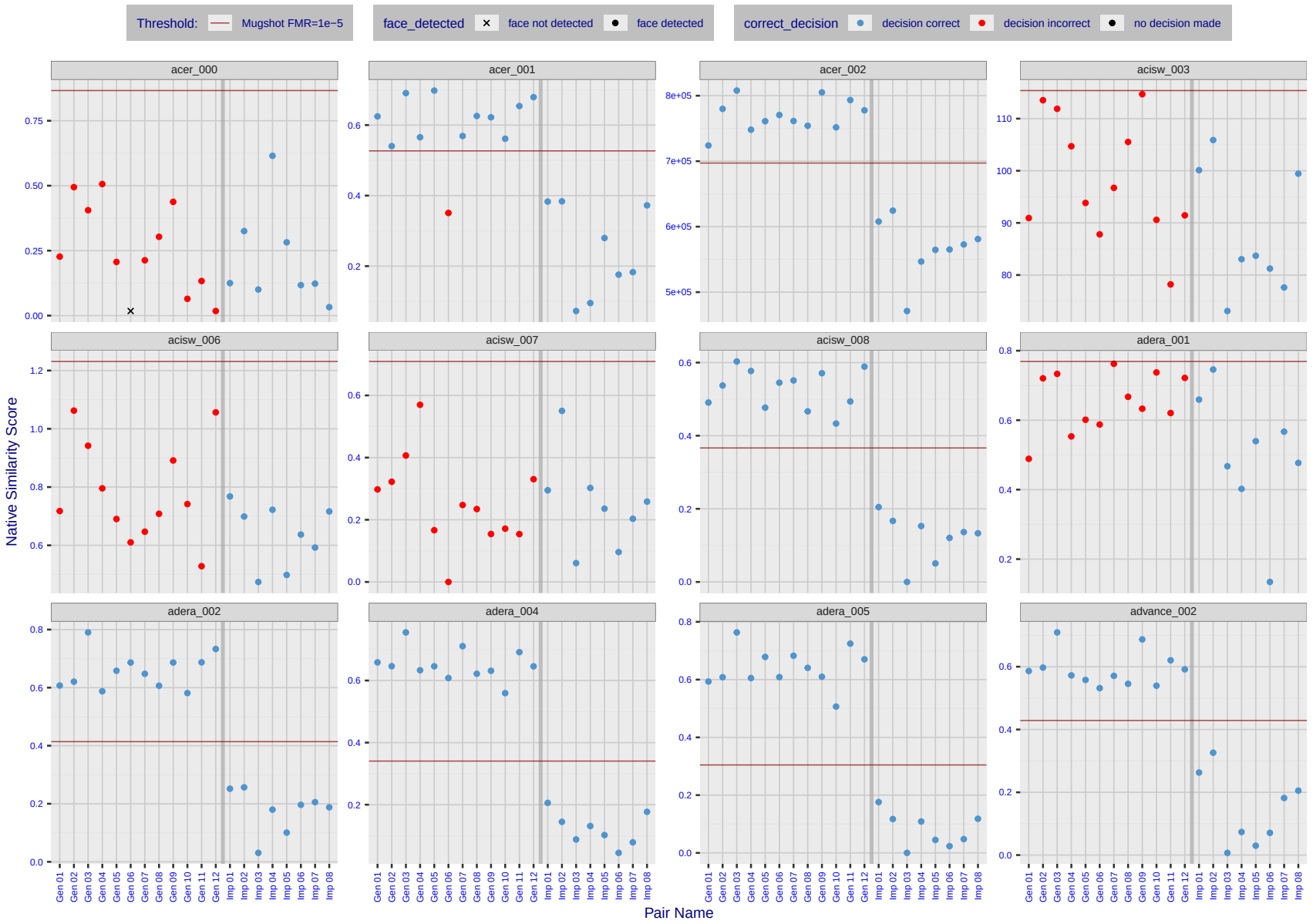


Figure 5: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

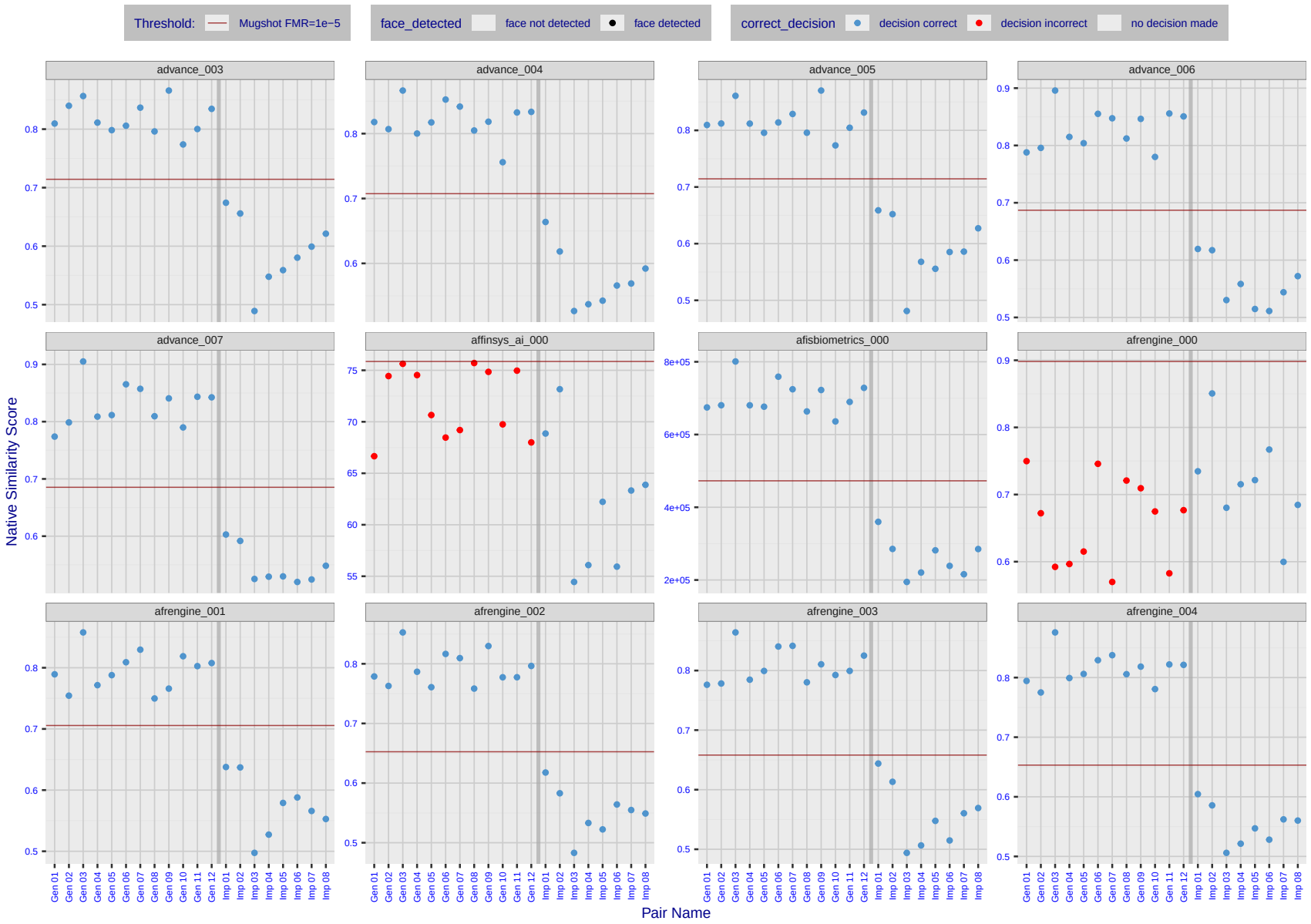


Figure 6: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

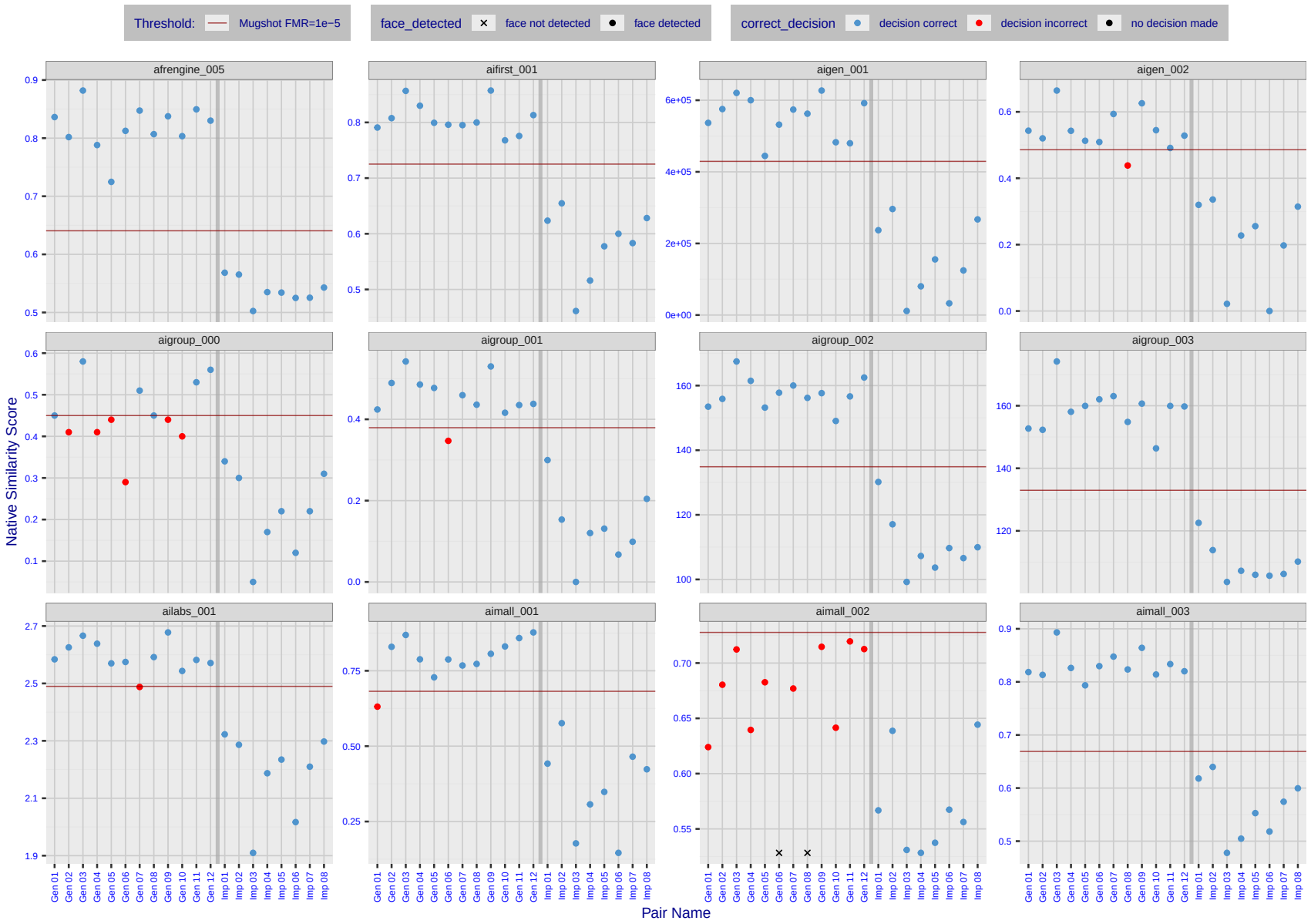


Figure 7: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

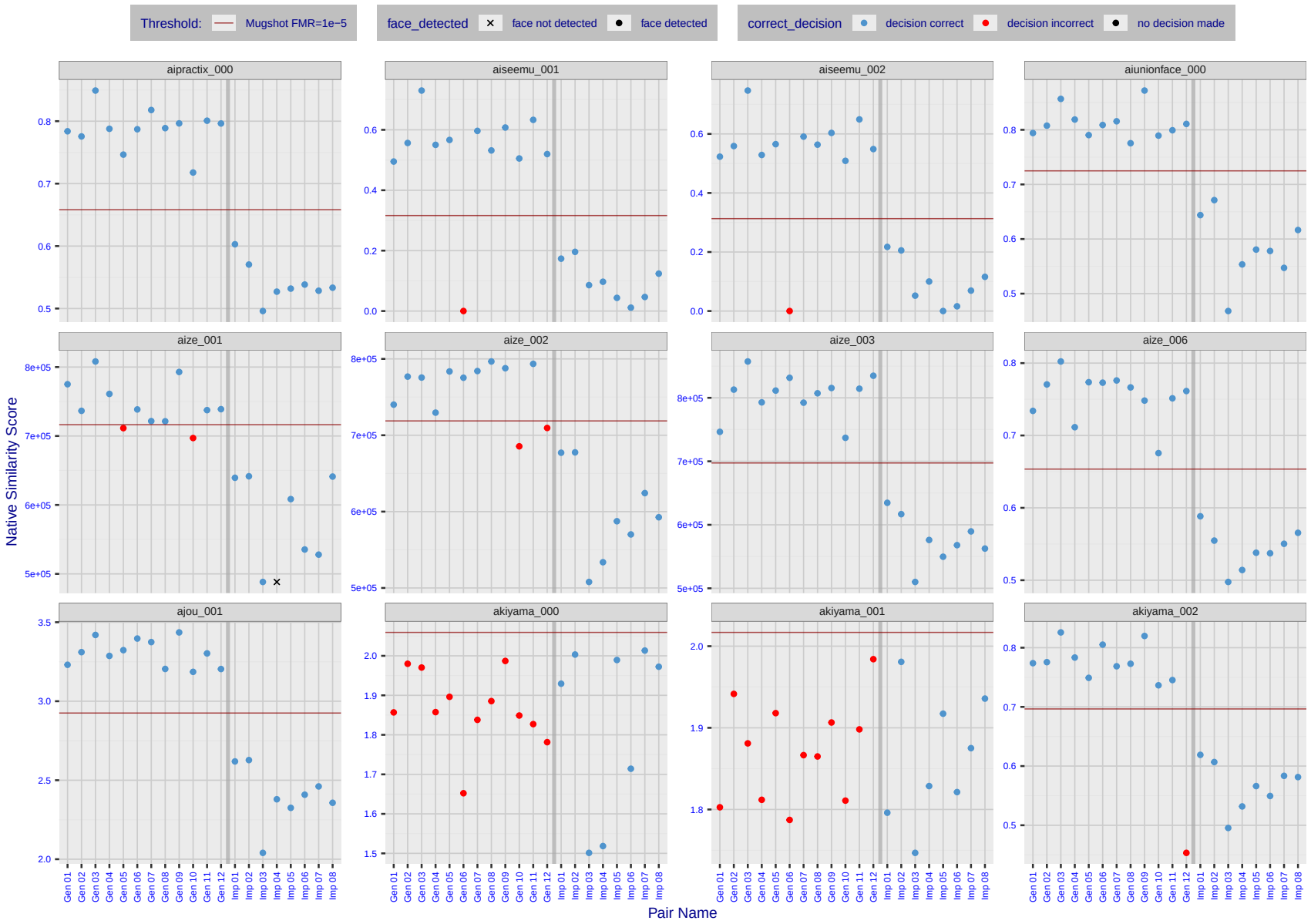


Figure 8: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

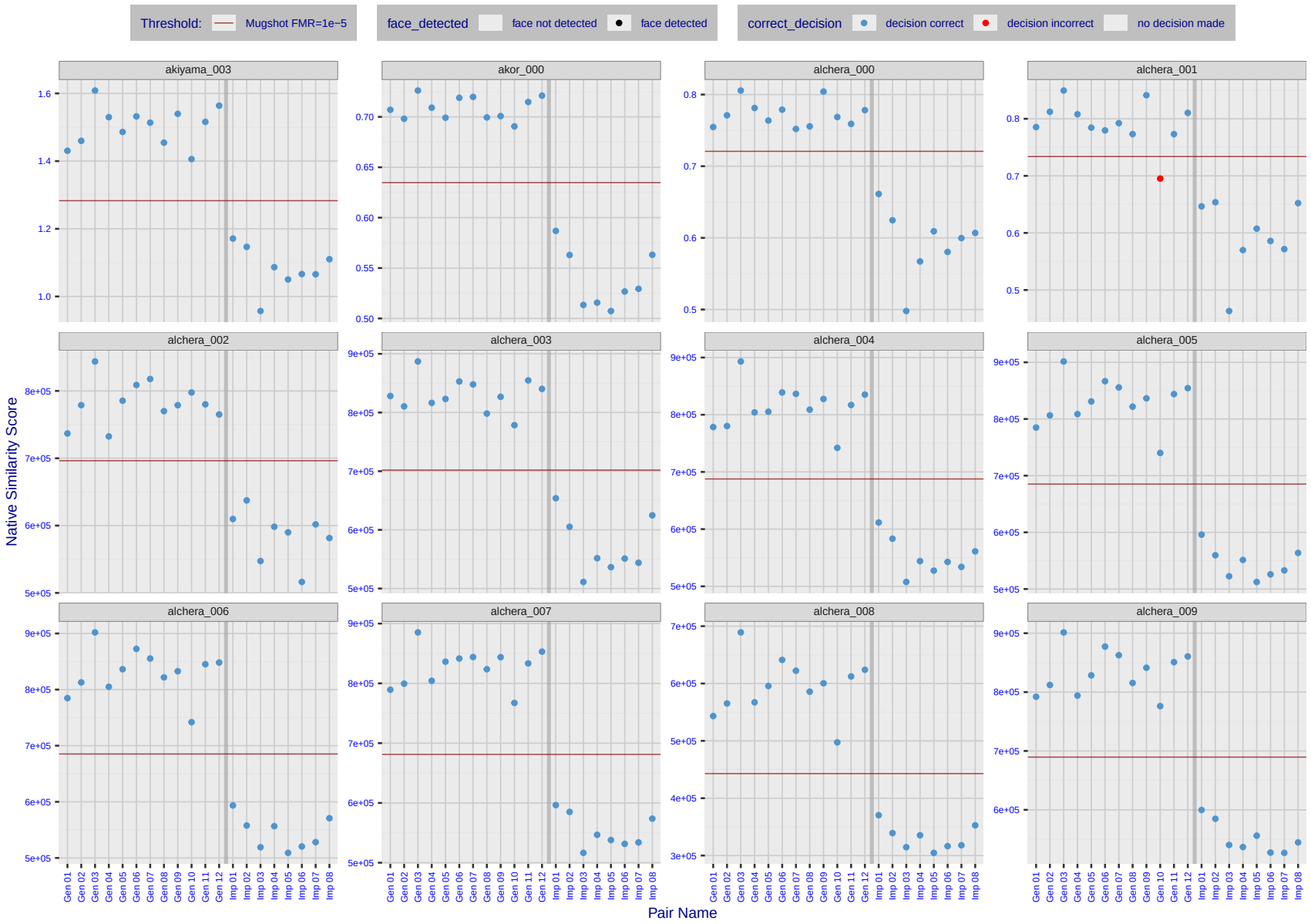


Figure 9: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

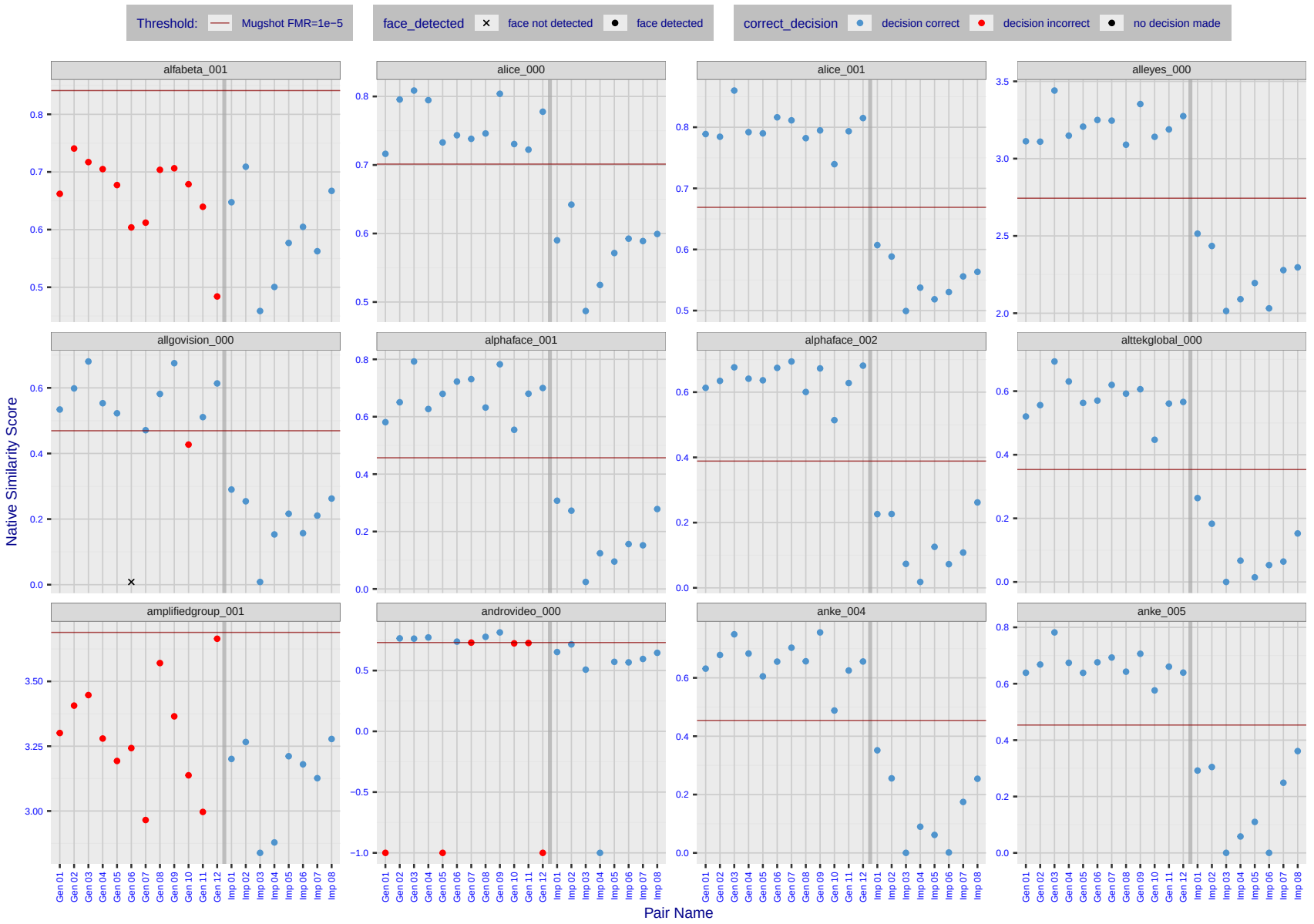


Figure 10: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

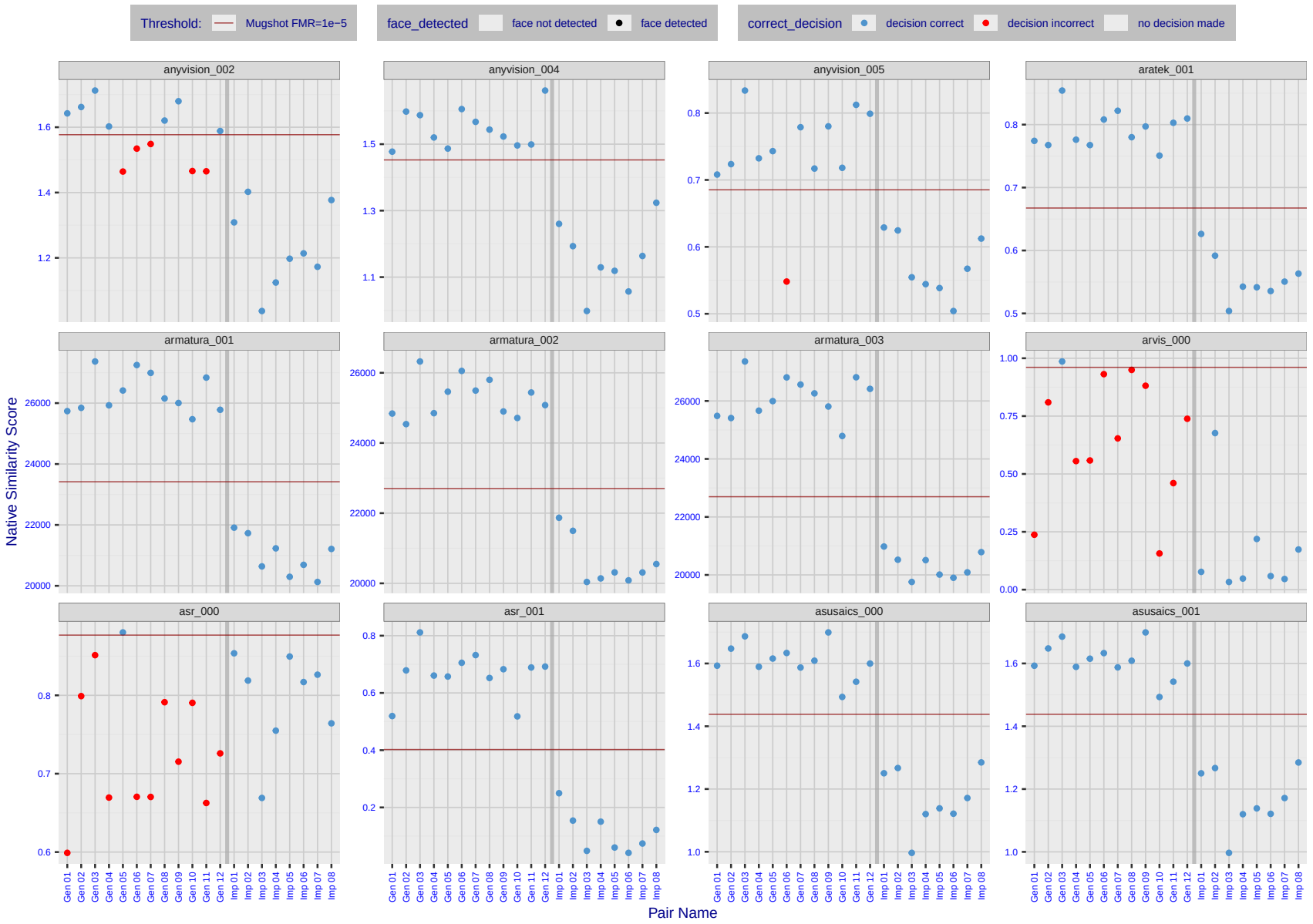


Figure 11: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

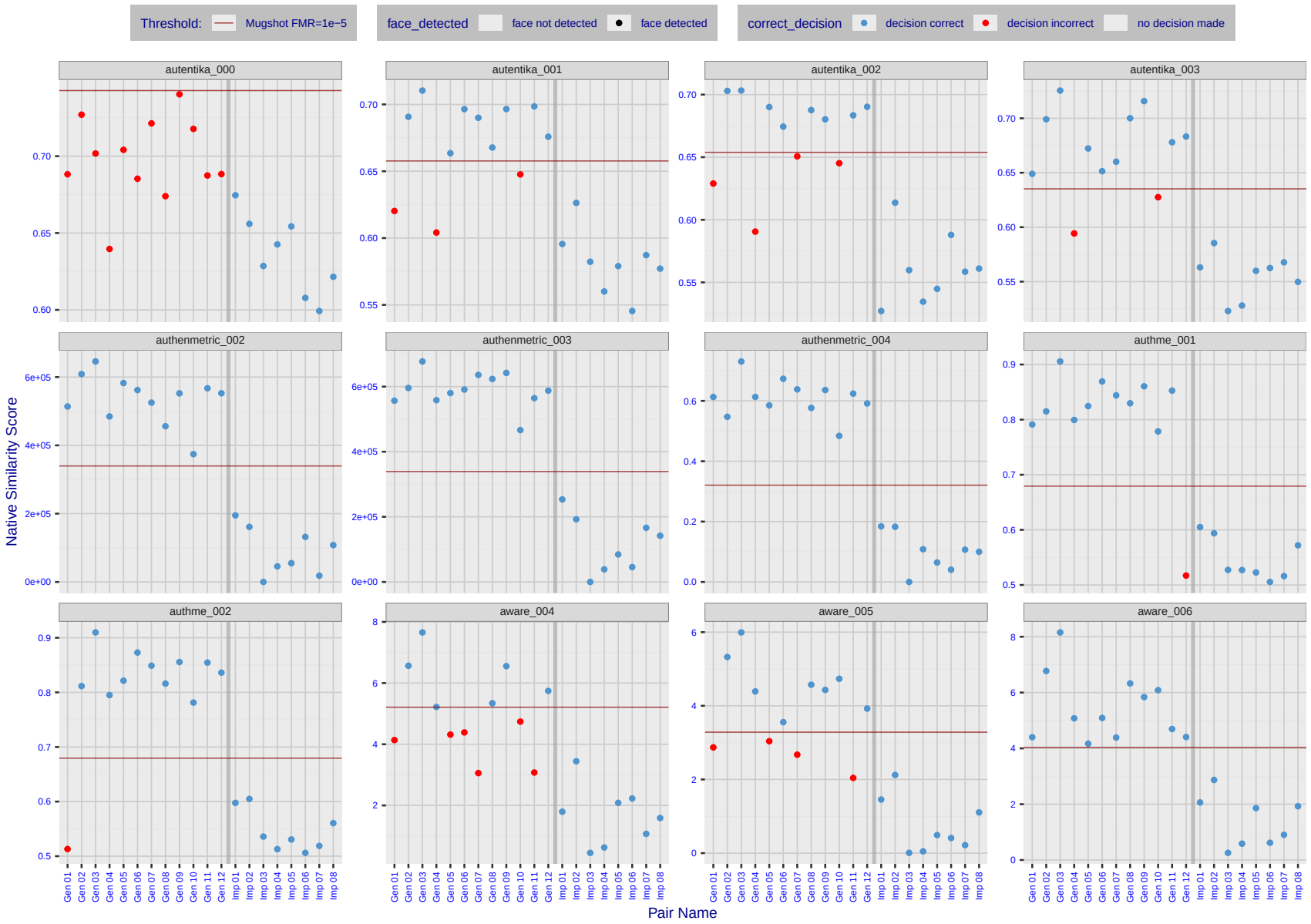


Figure 12: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

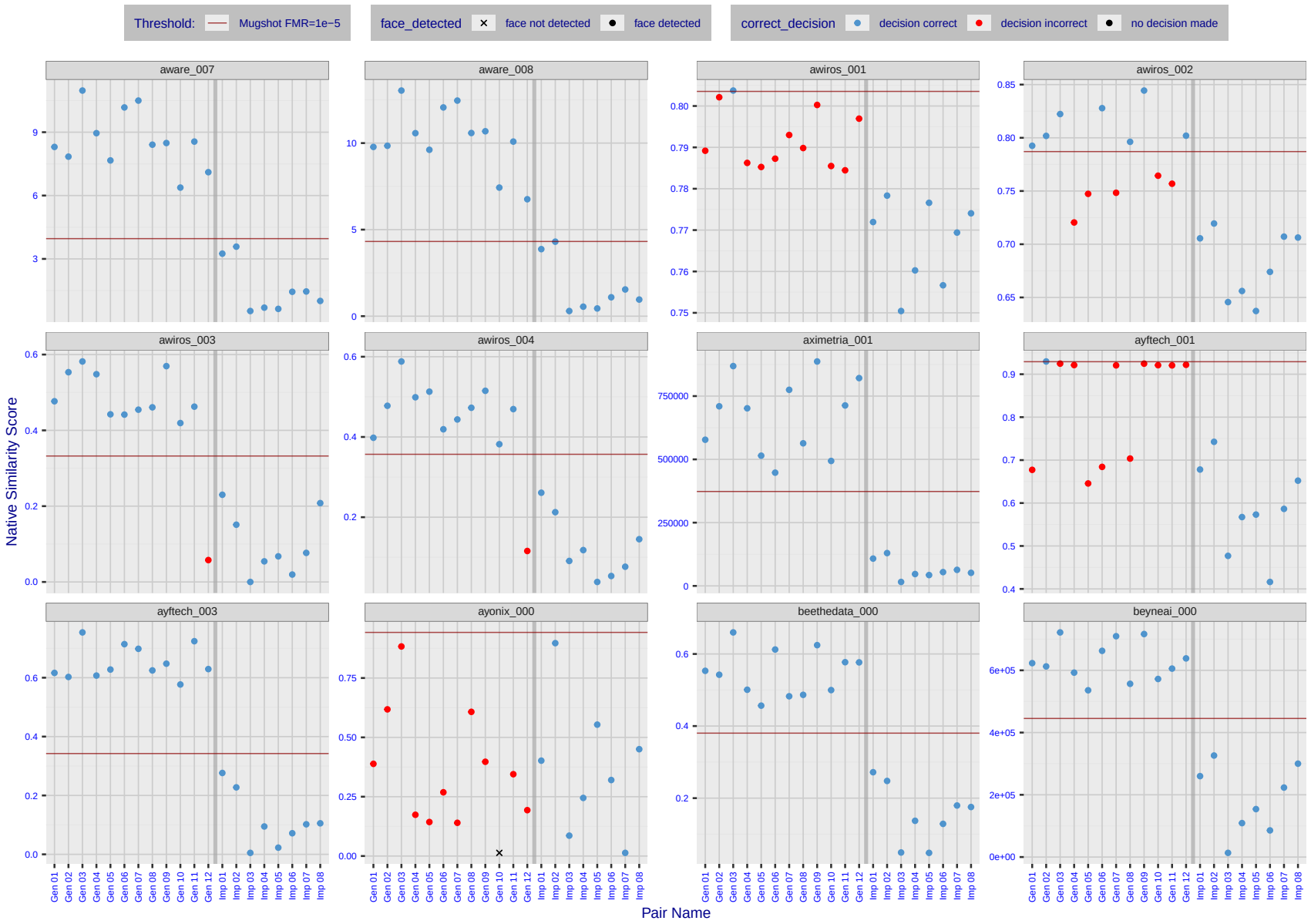


Figure 13: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

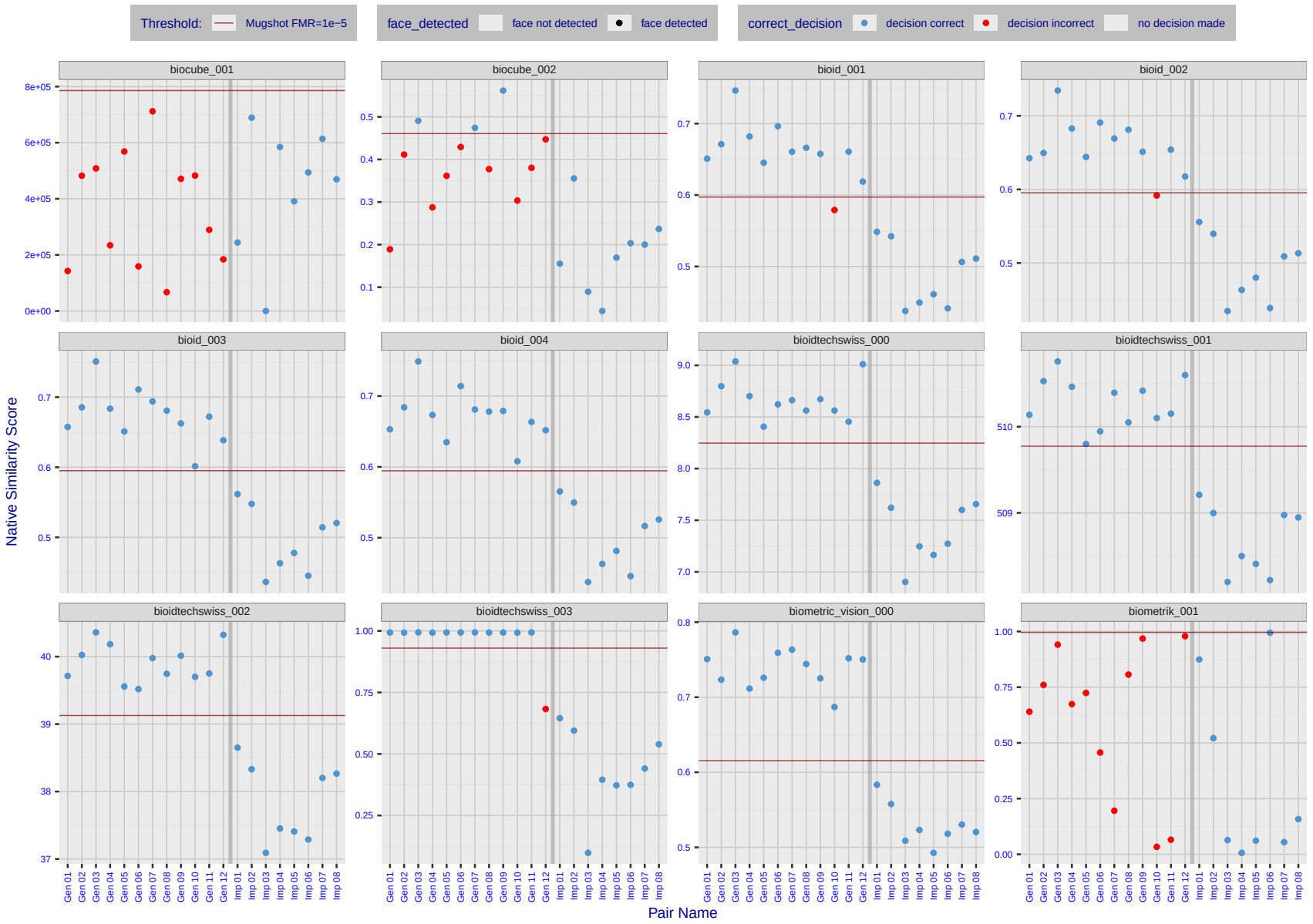


Figure 14: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

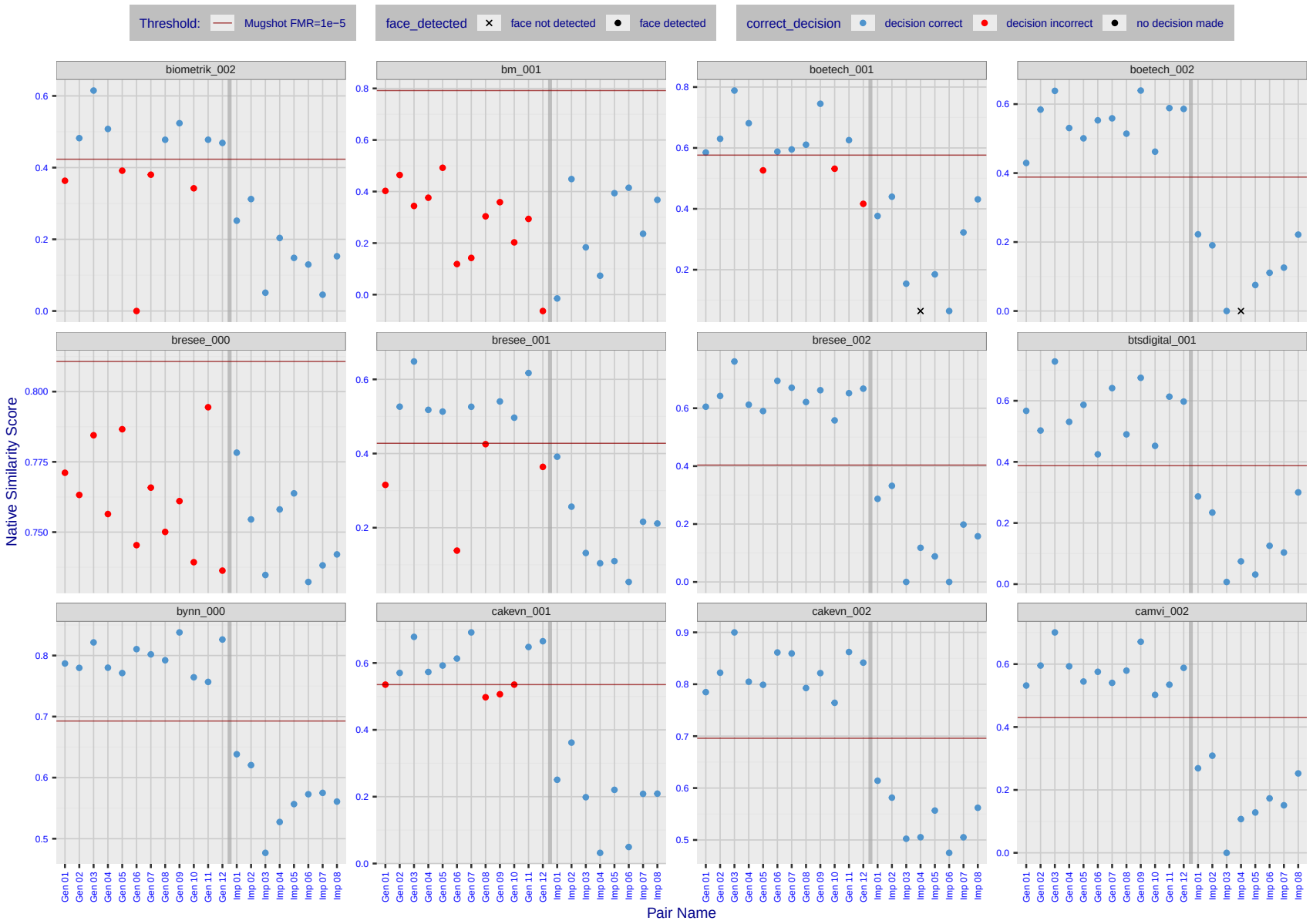


Figure 15: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

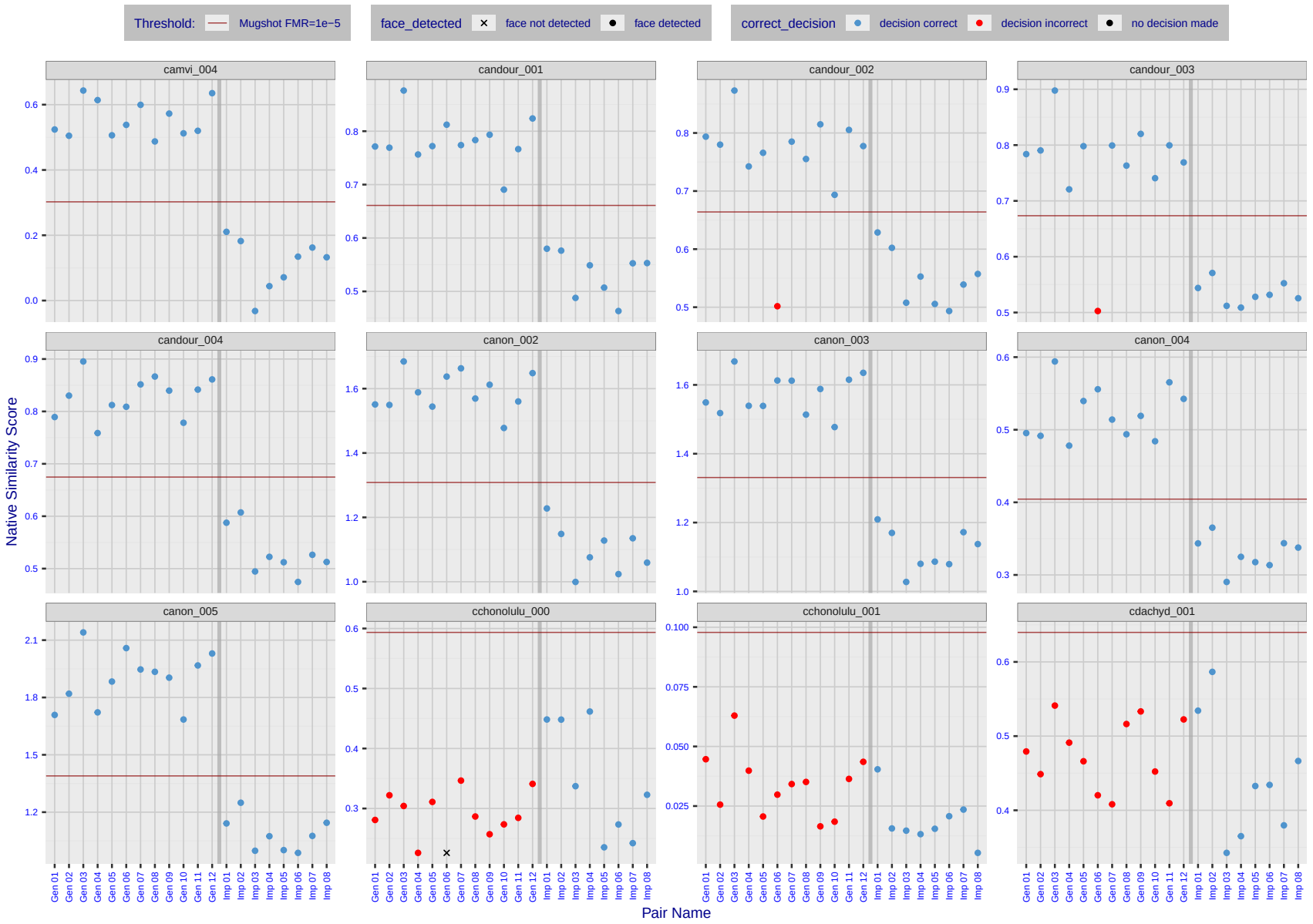


Figure 16: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

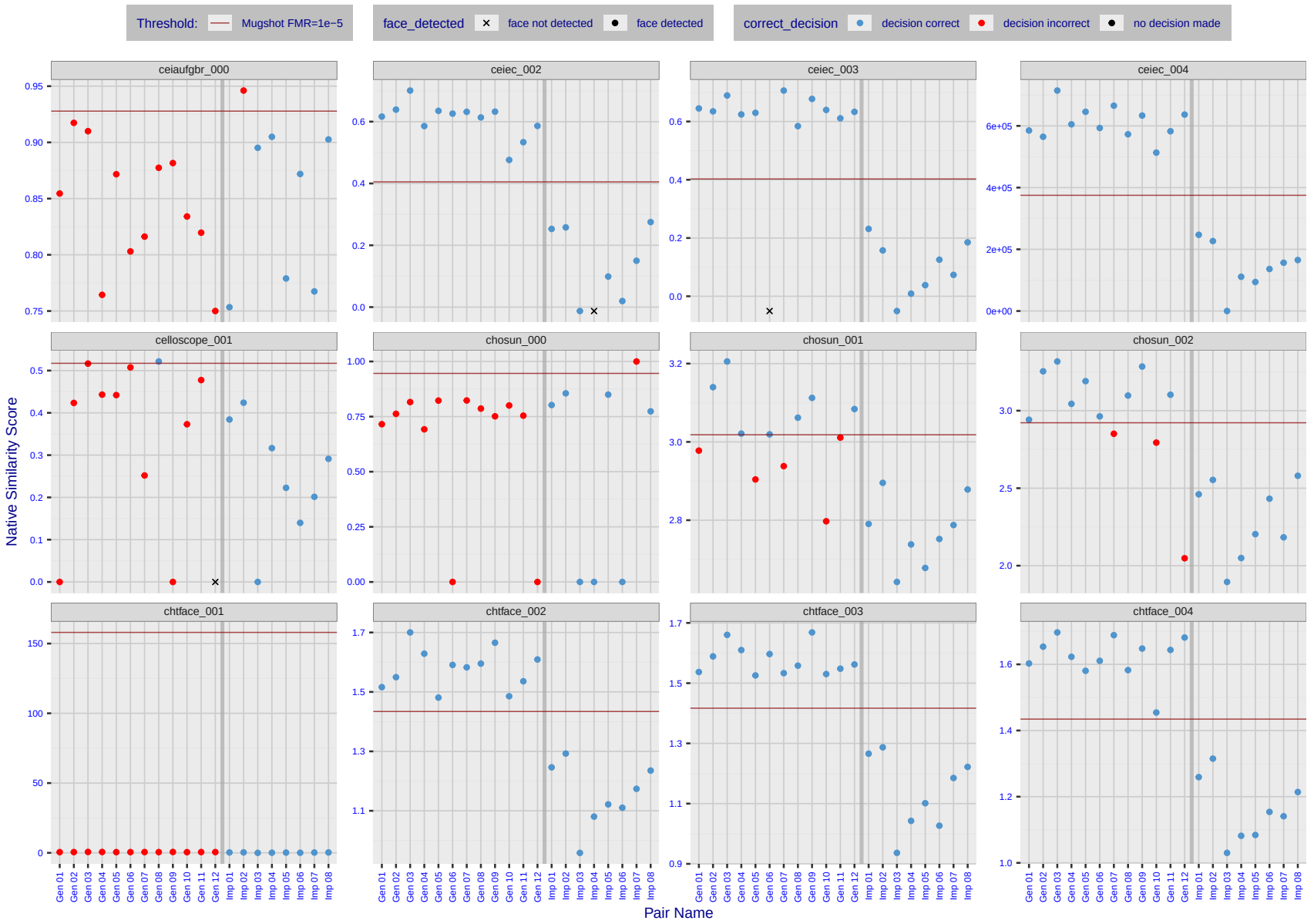


Figure 17: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

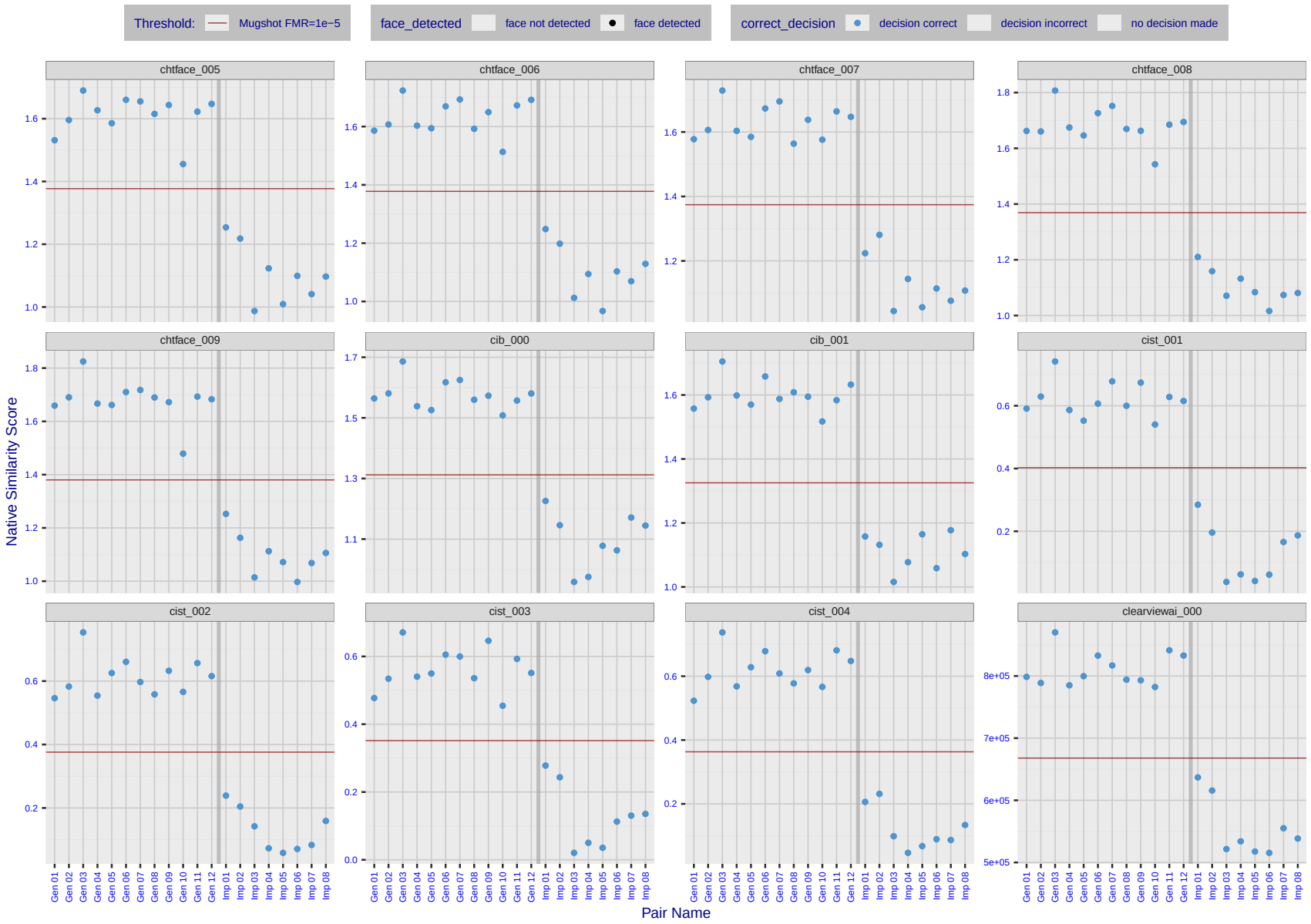


Figure 18: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

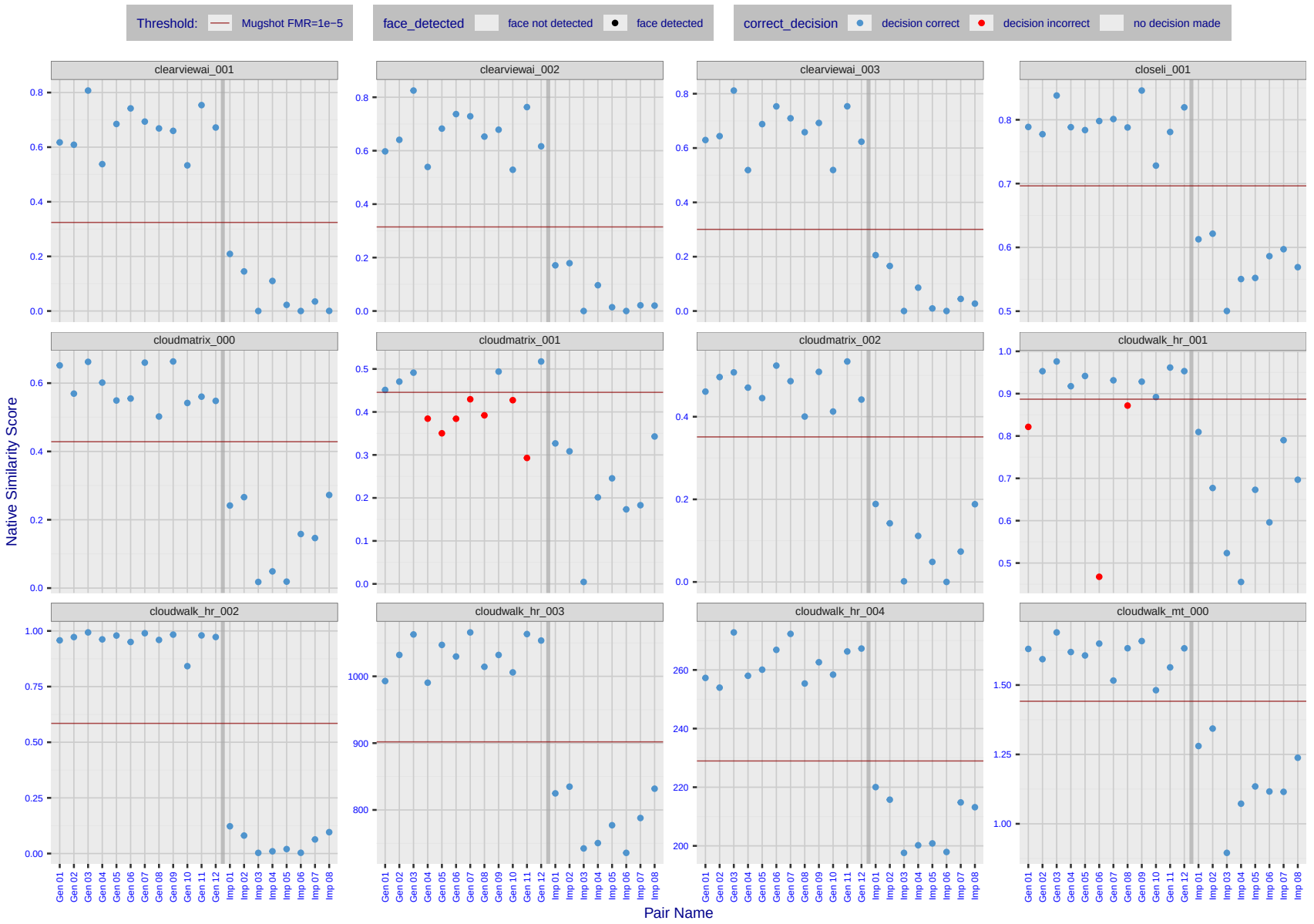


Figure 19: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

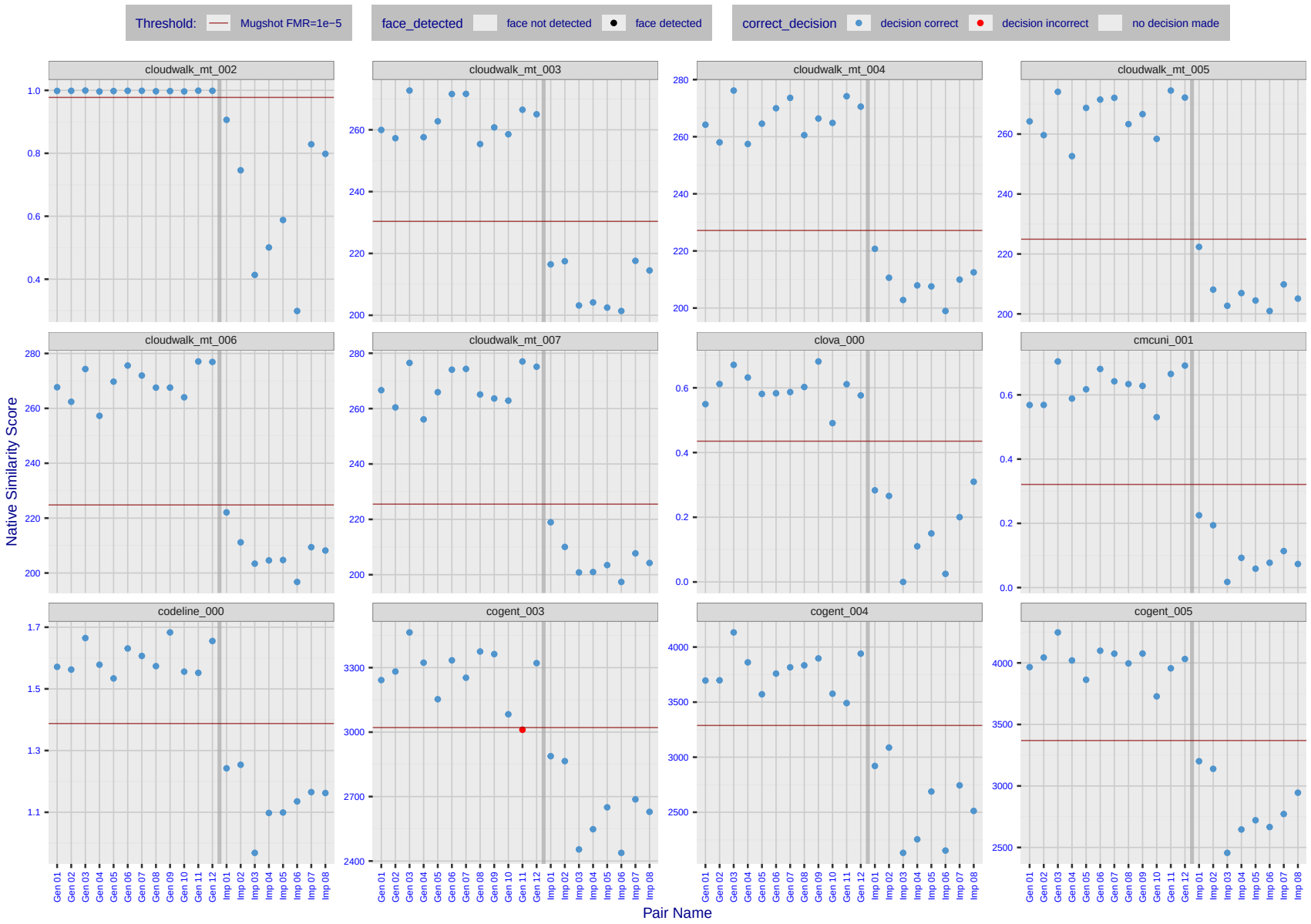


Figure 20: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

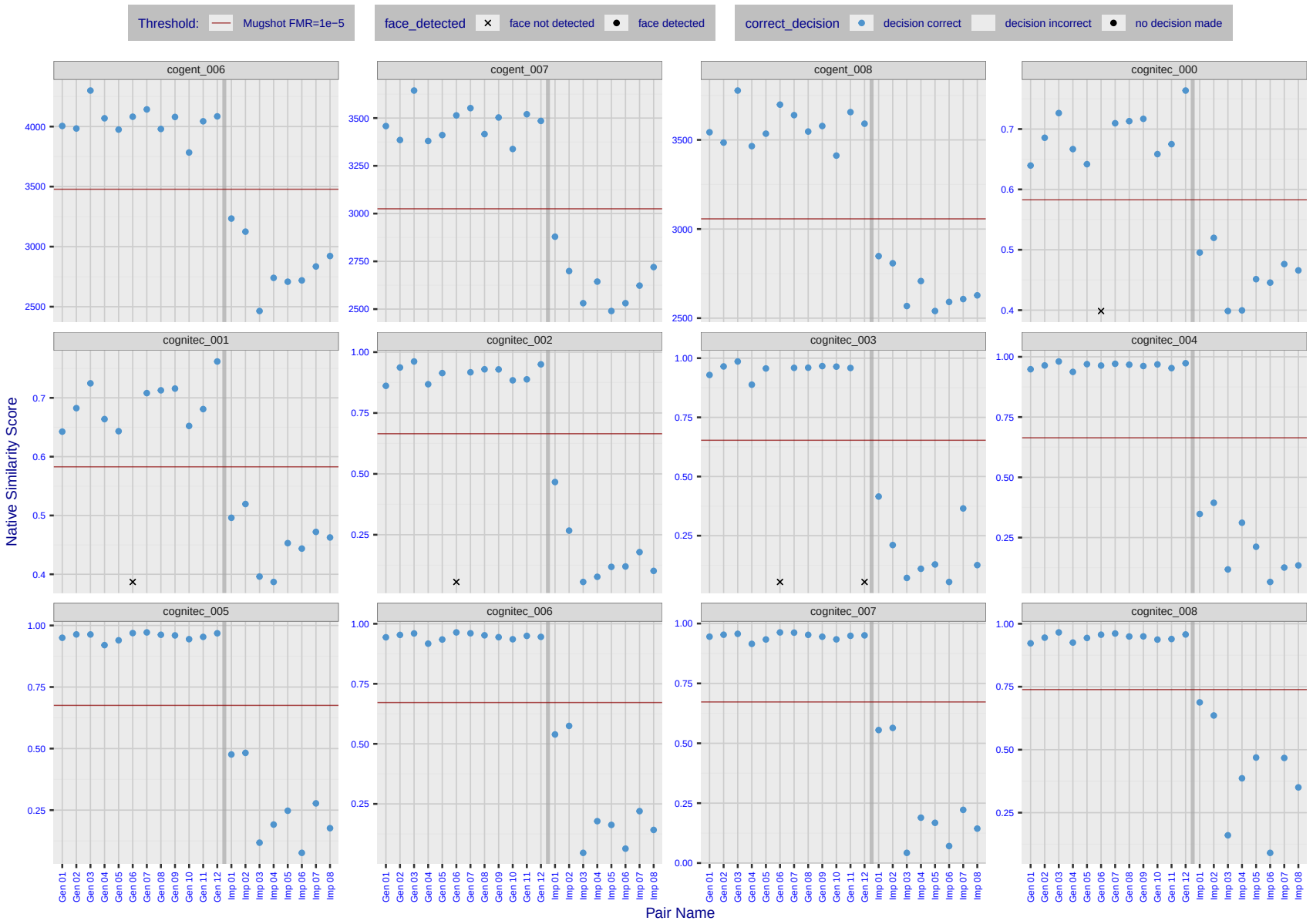


Figure 21: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

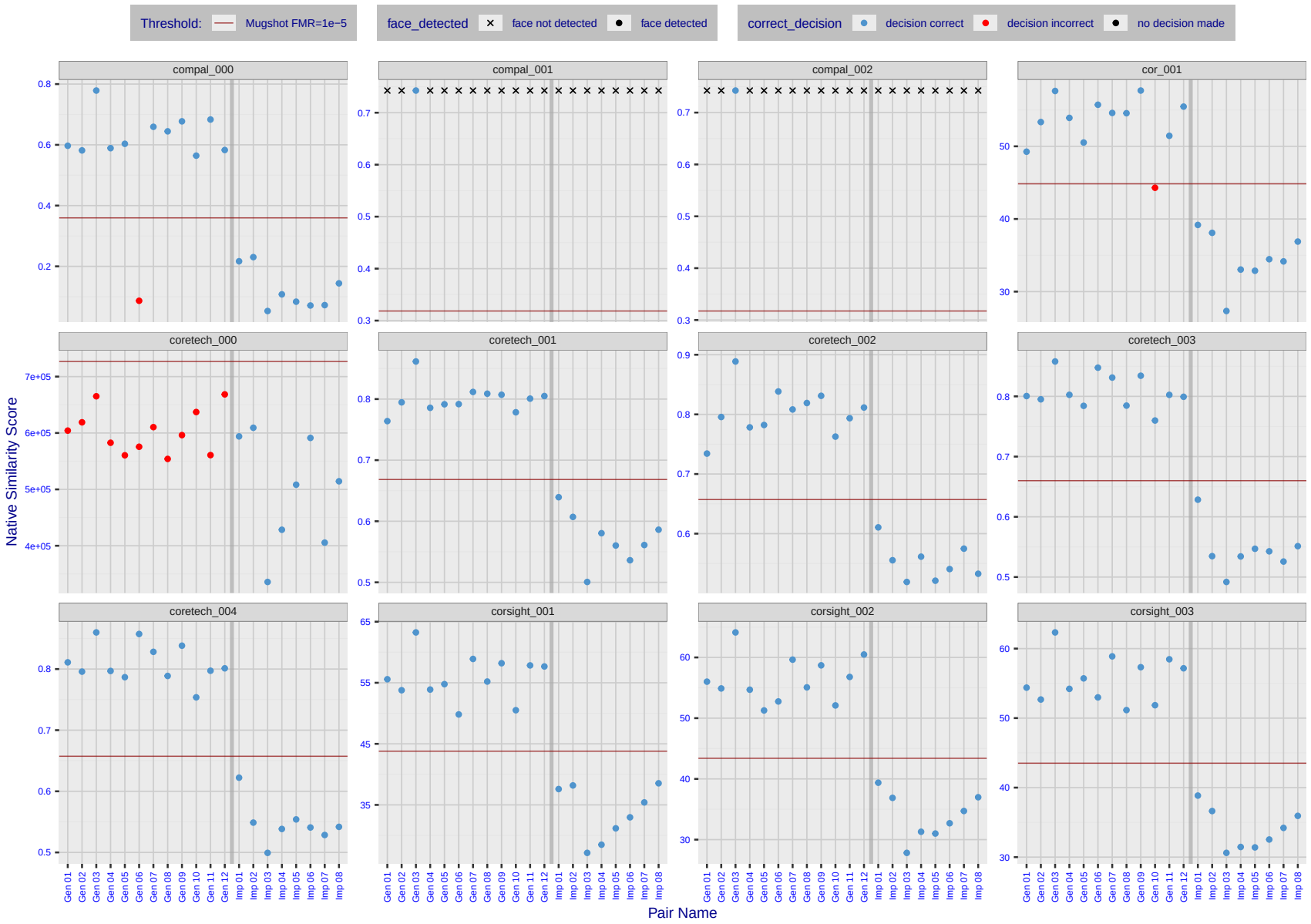


Figure 22: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

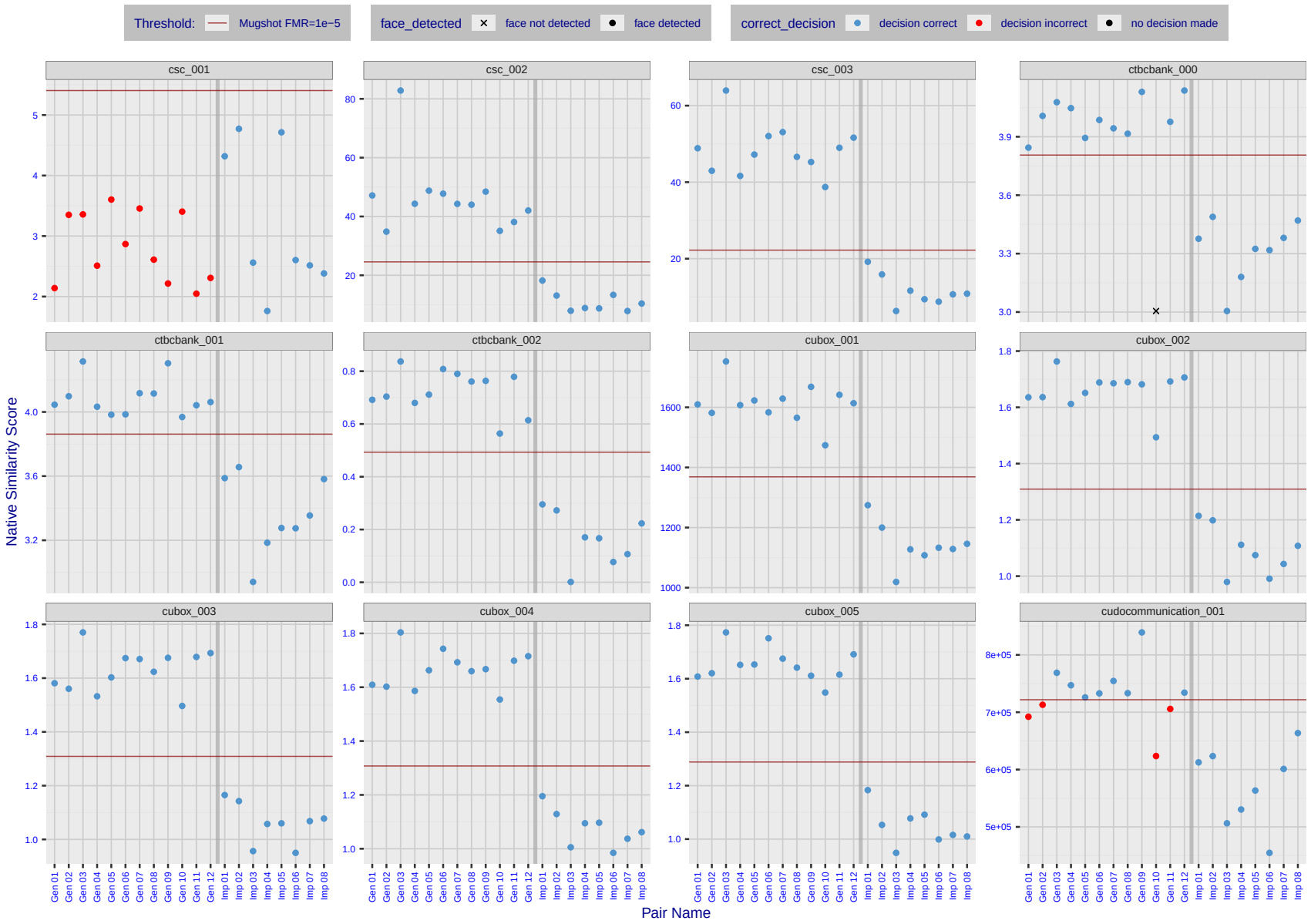


Figure 23: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

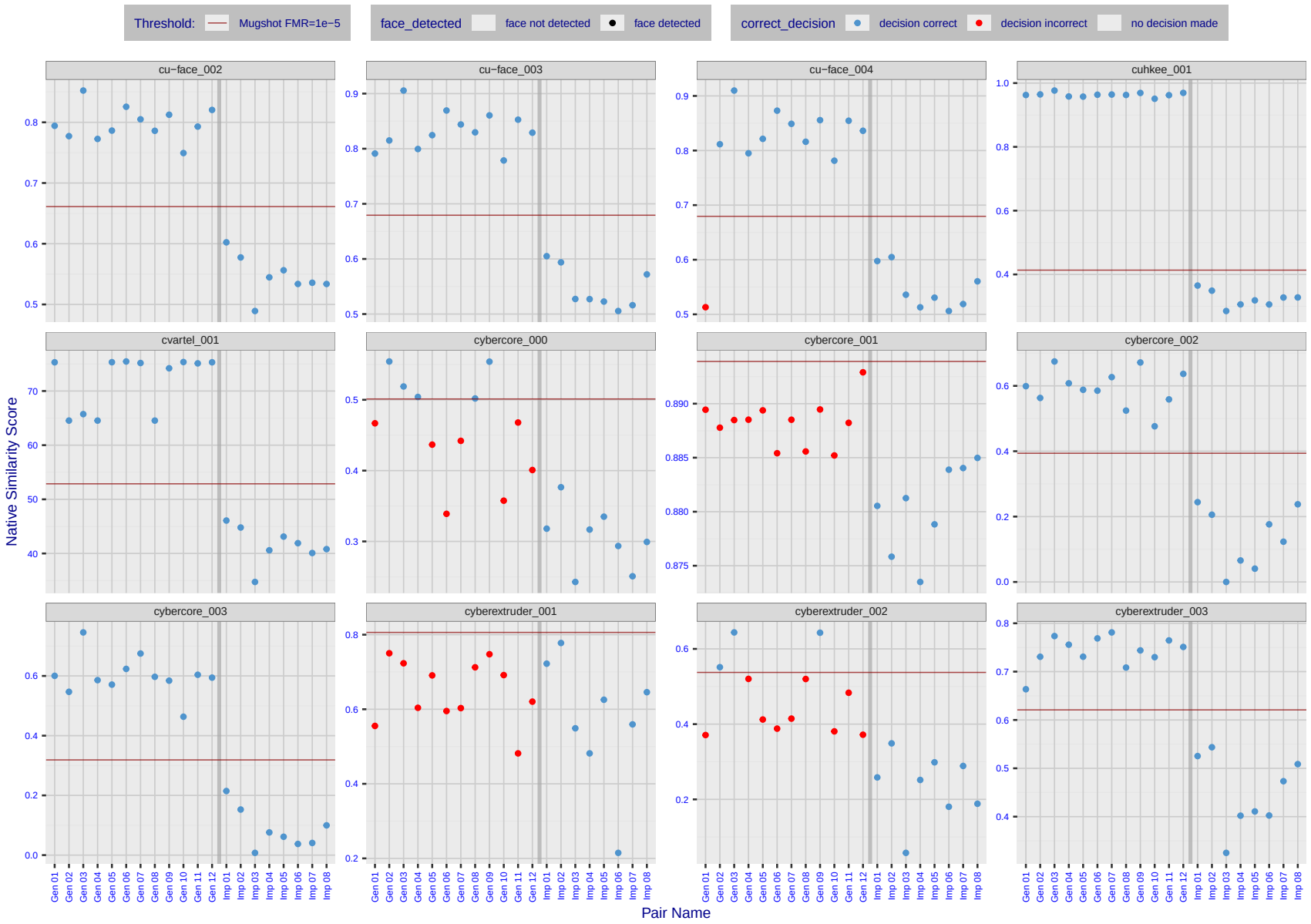


Figure 24: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

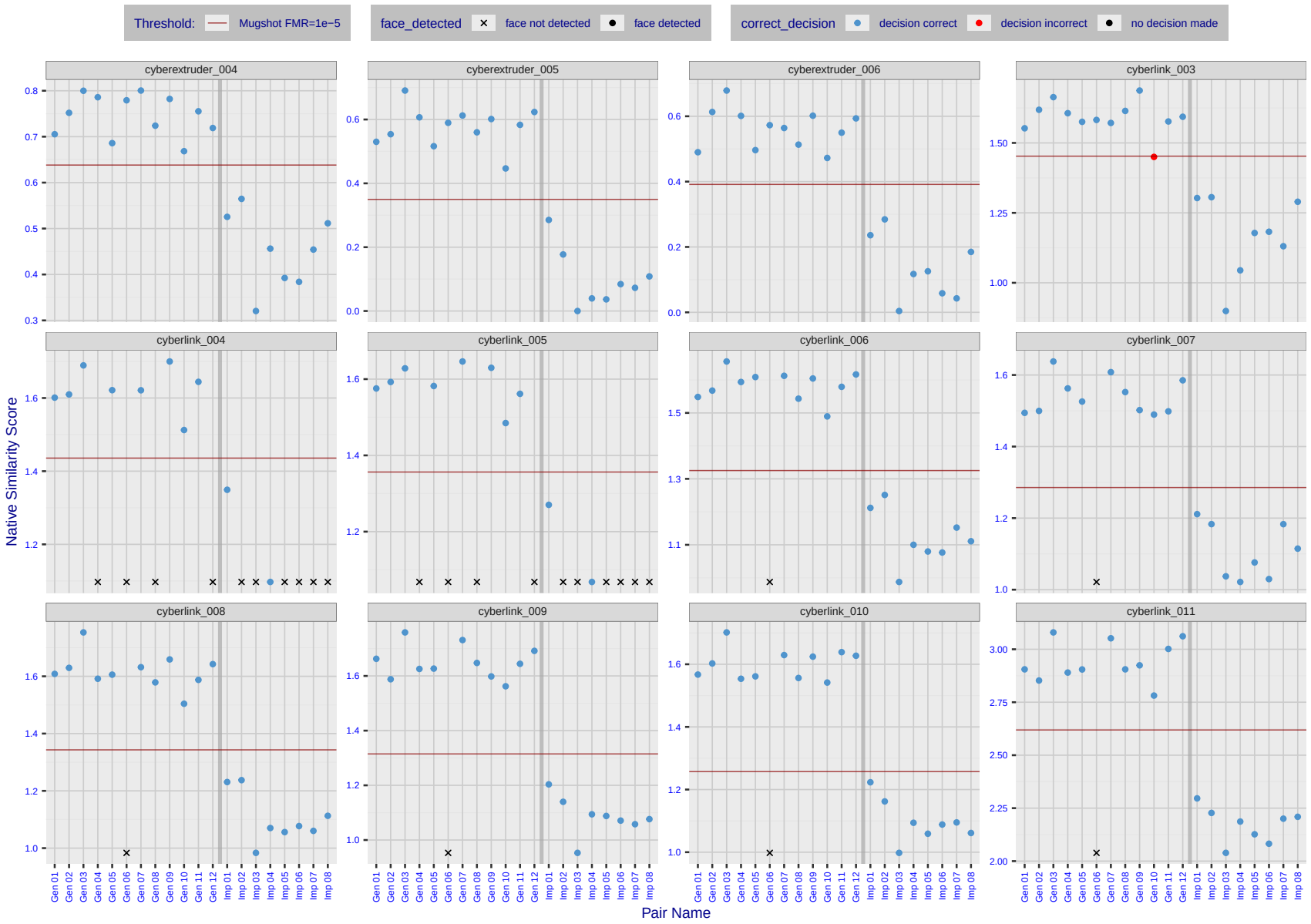


Figure 25: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

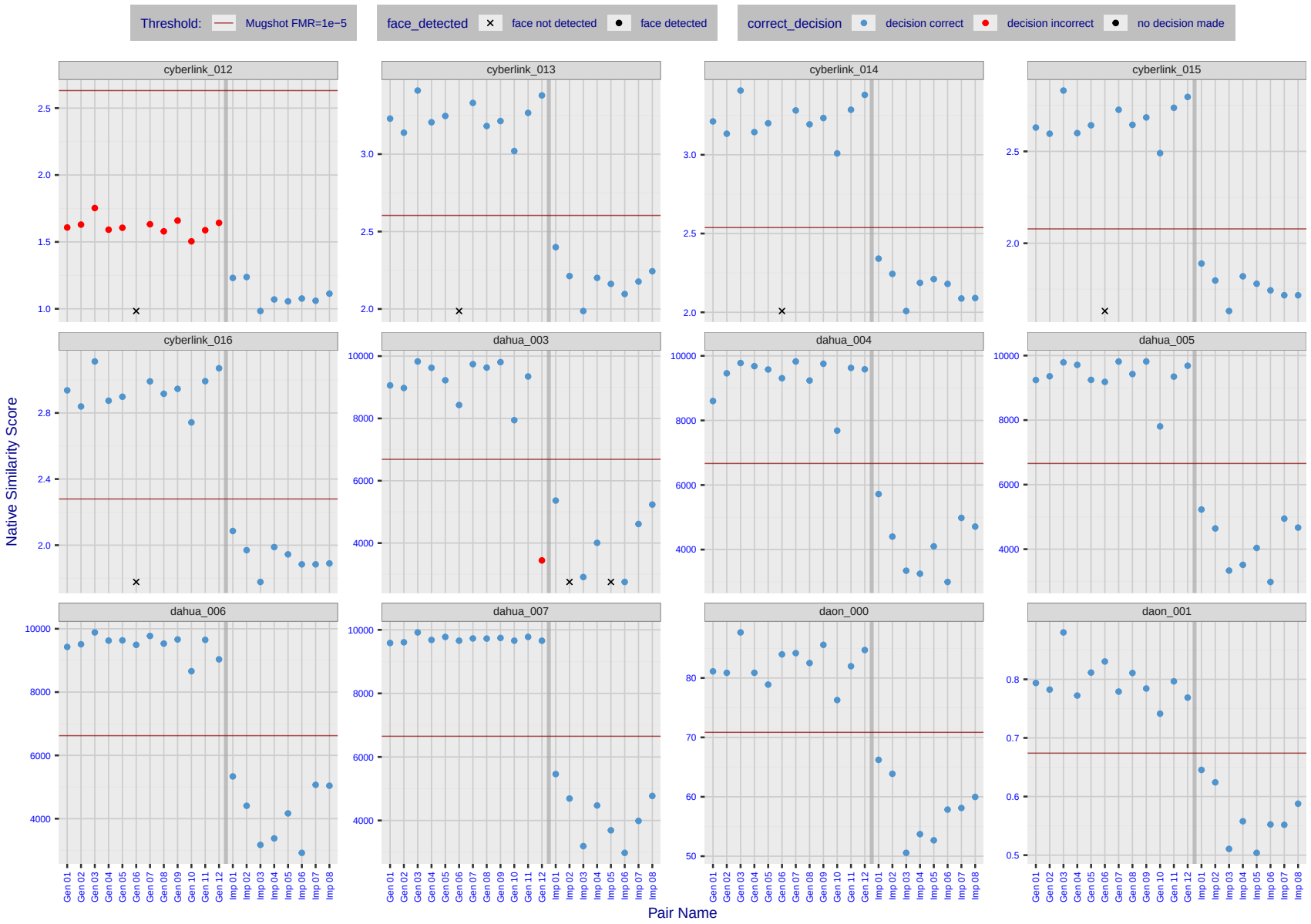


Figure 26: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

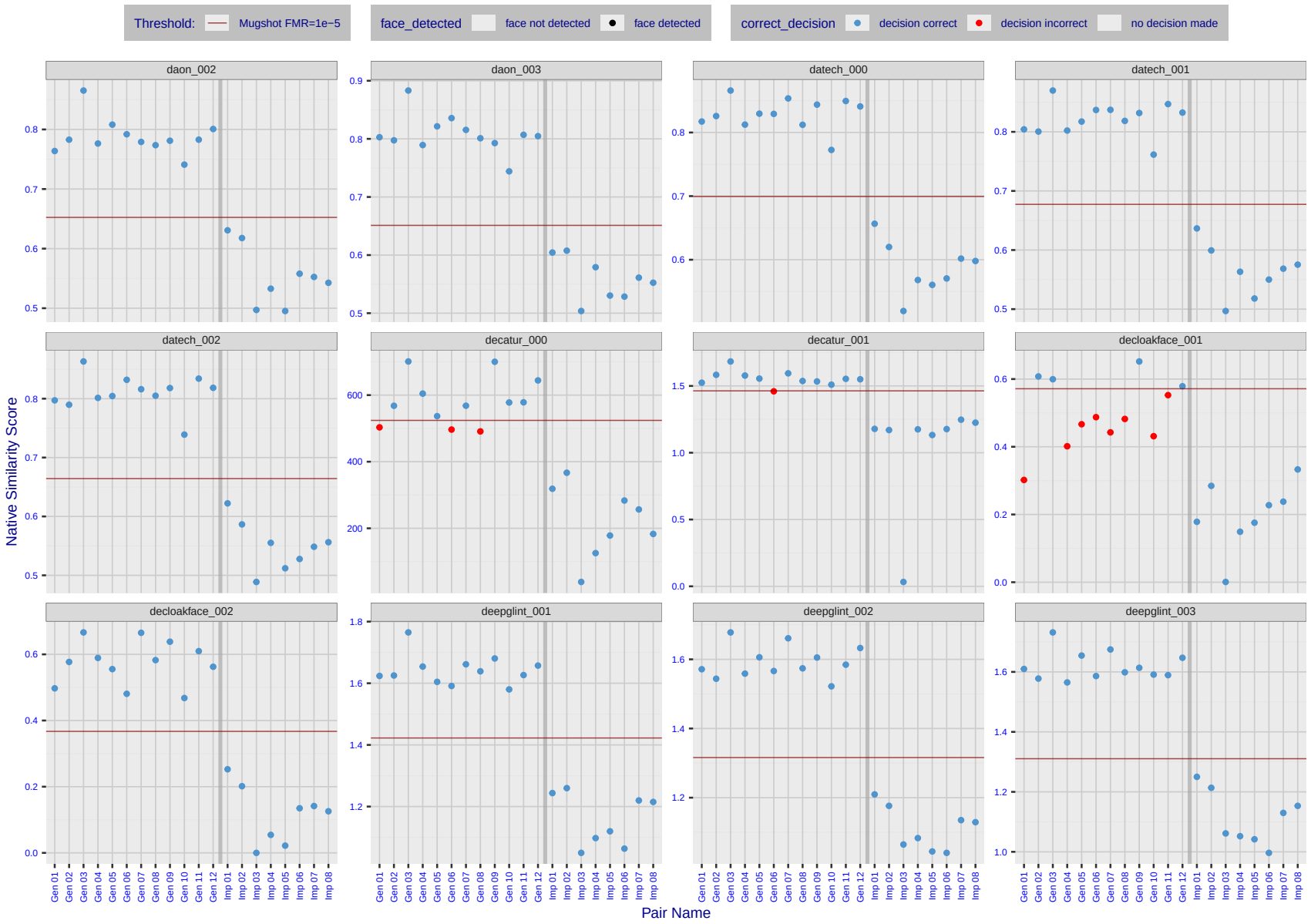


Figure 27: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

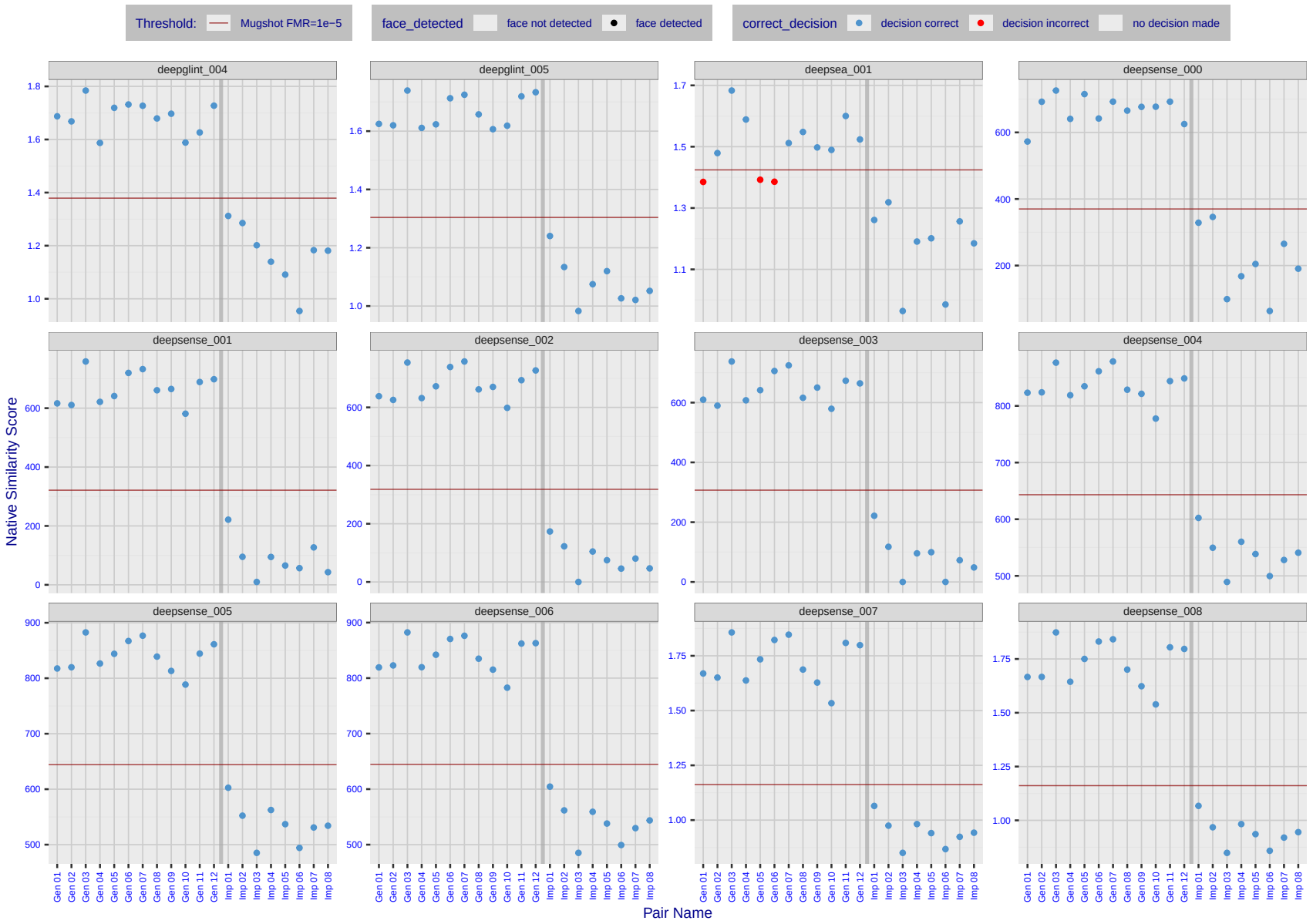


Figure 28: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

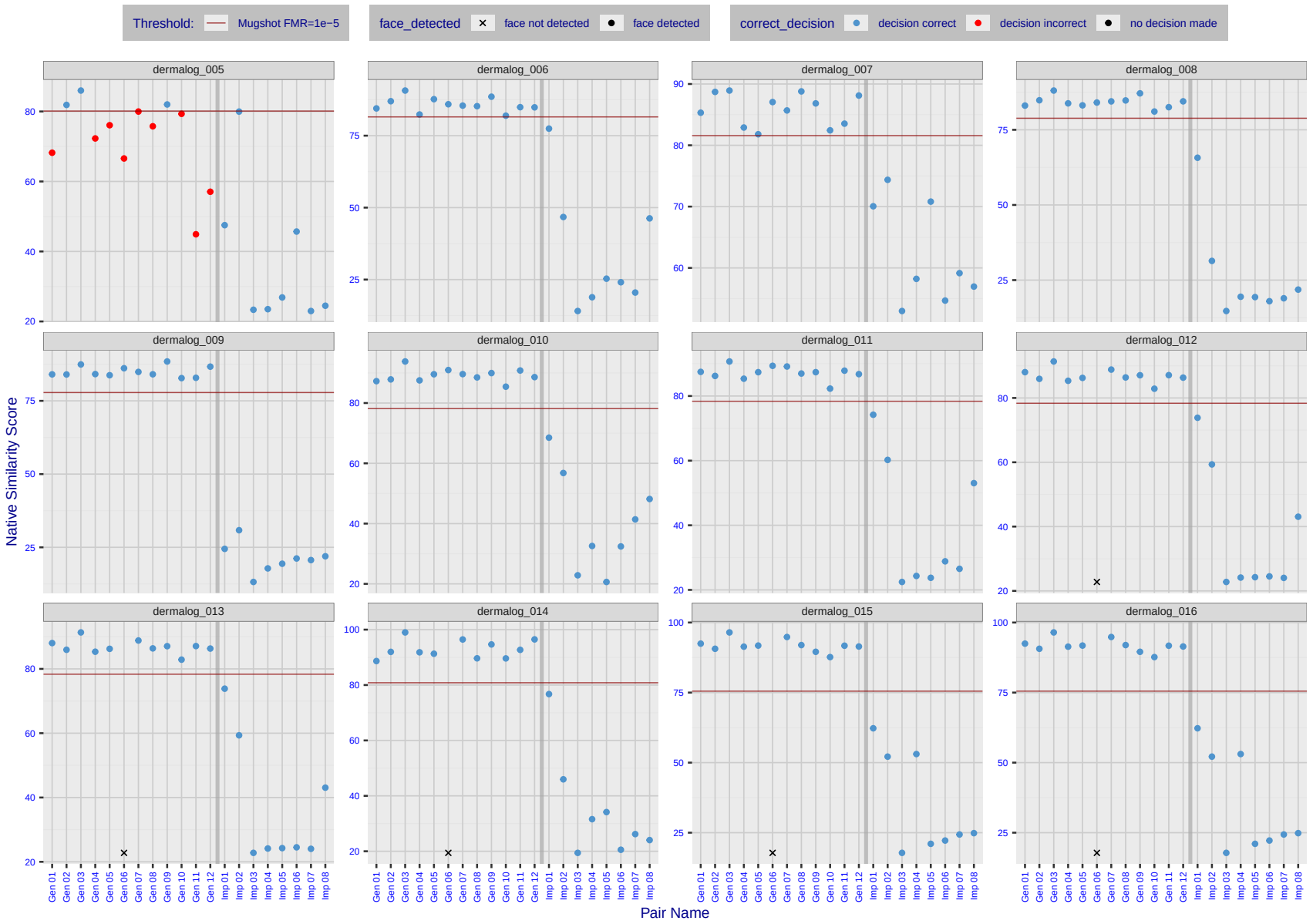


Figure 29: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

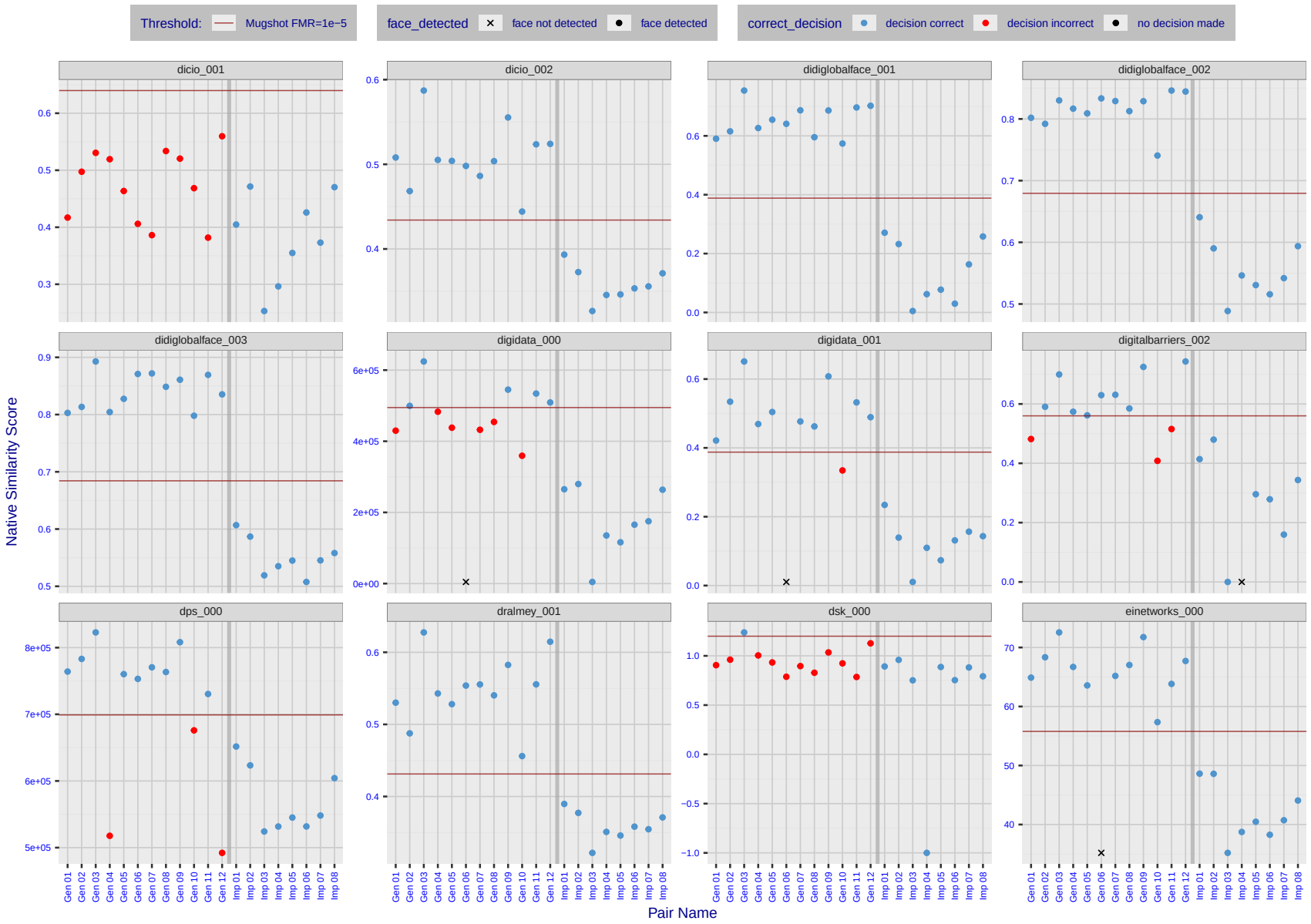


Figure 30: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

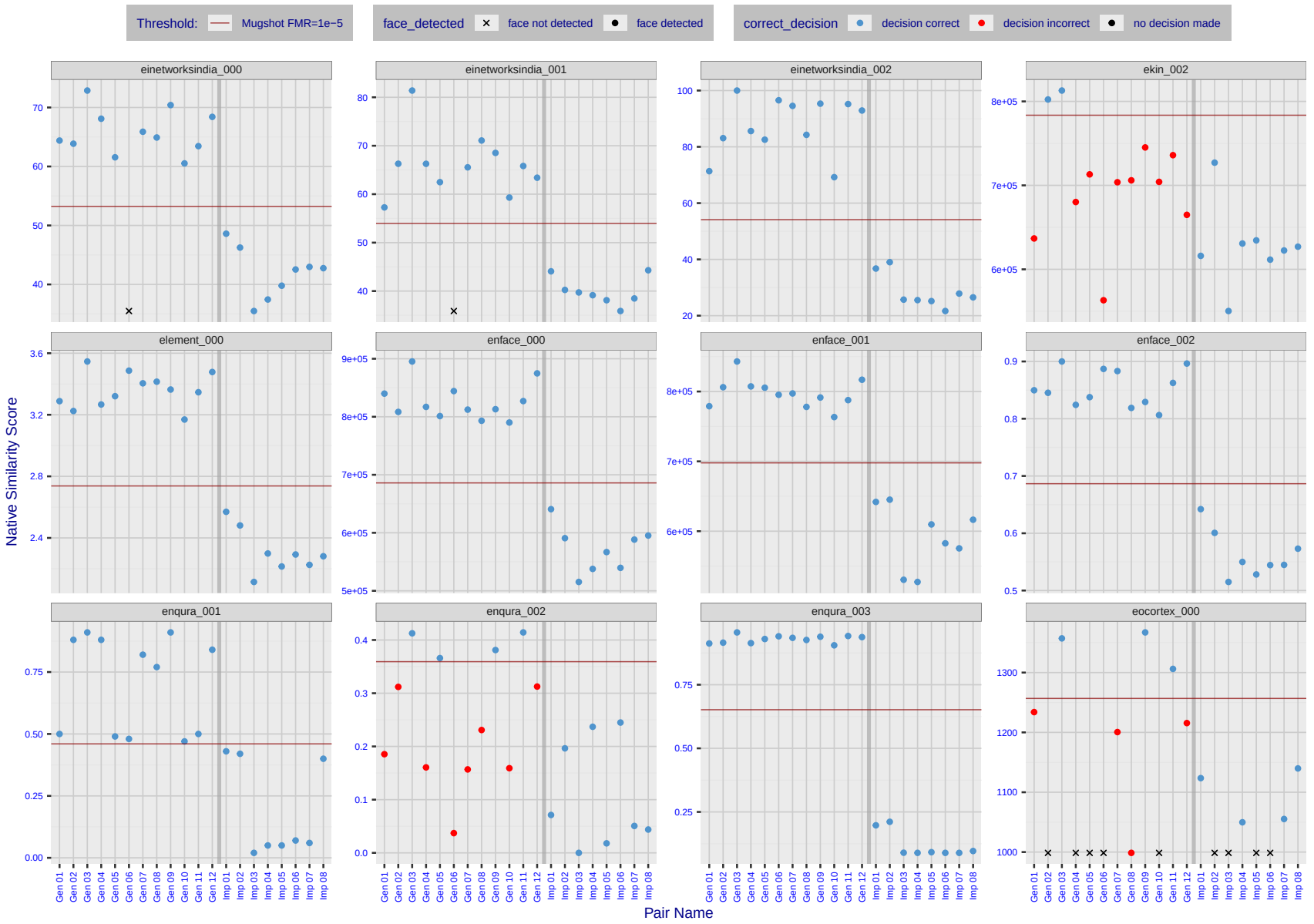


Figure 31: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

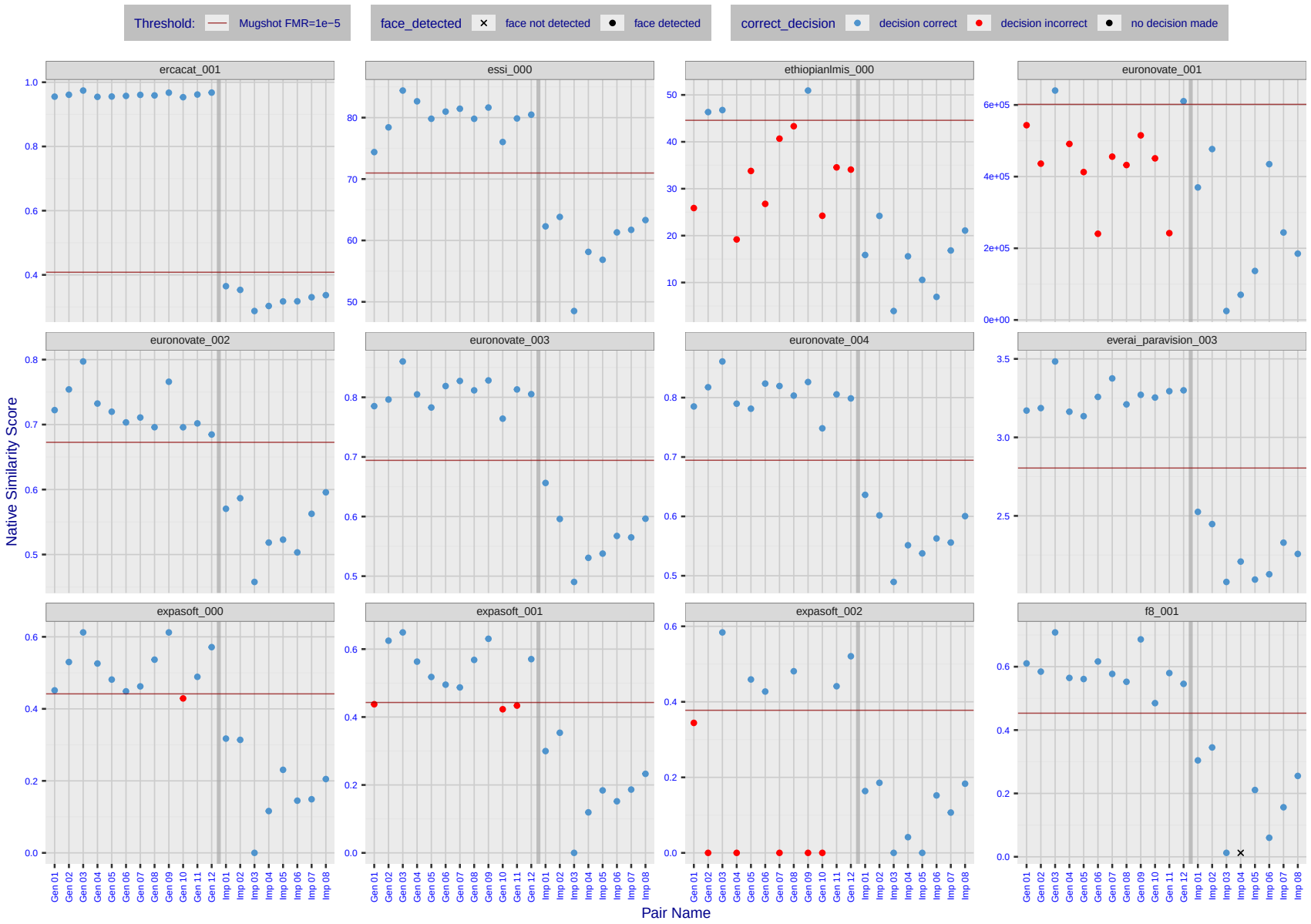


Figure 32: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

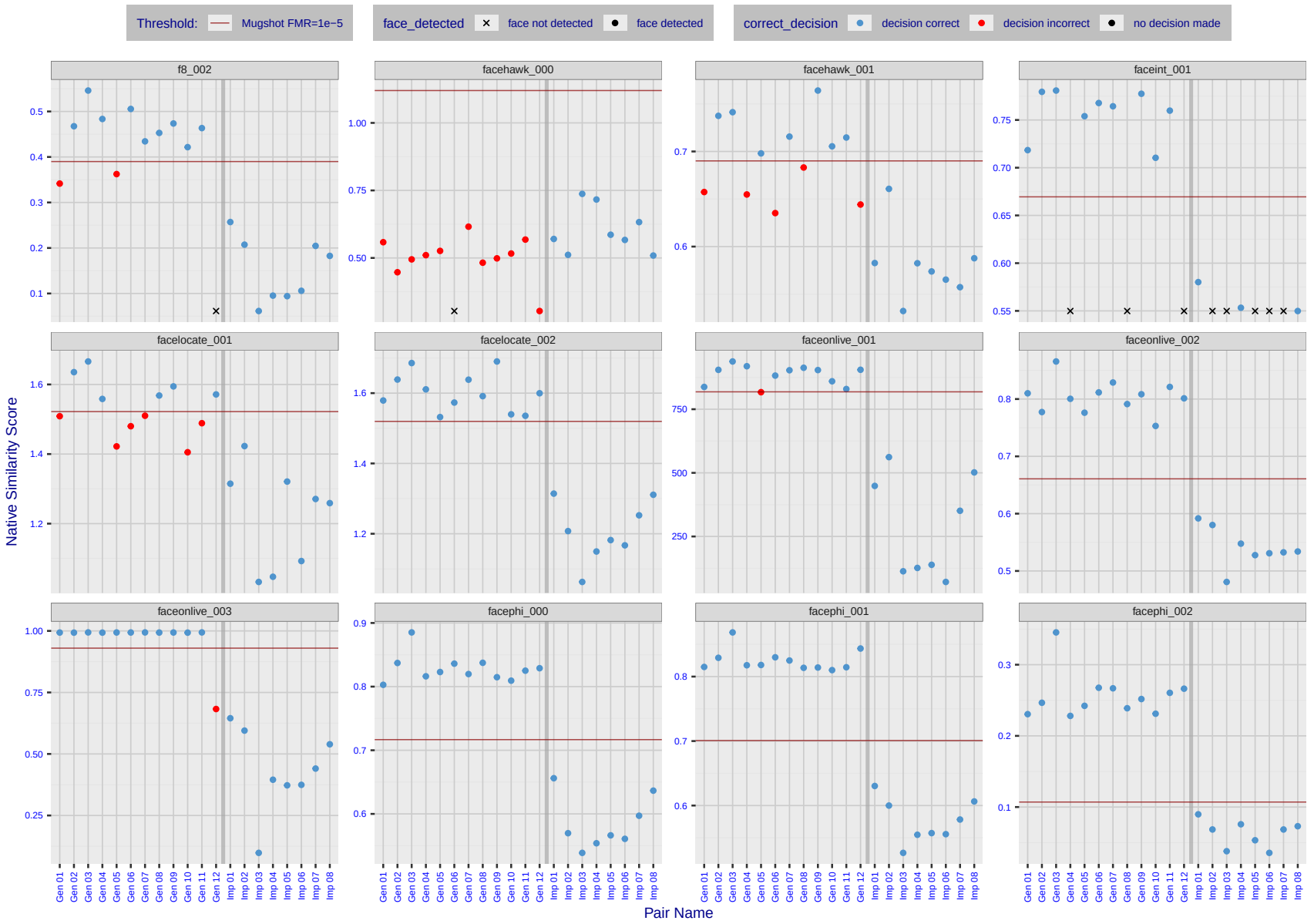


Figure 33: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

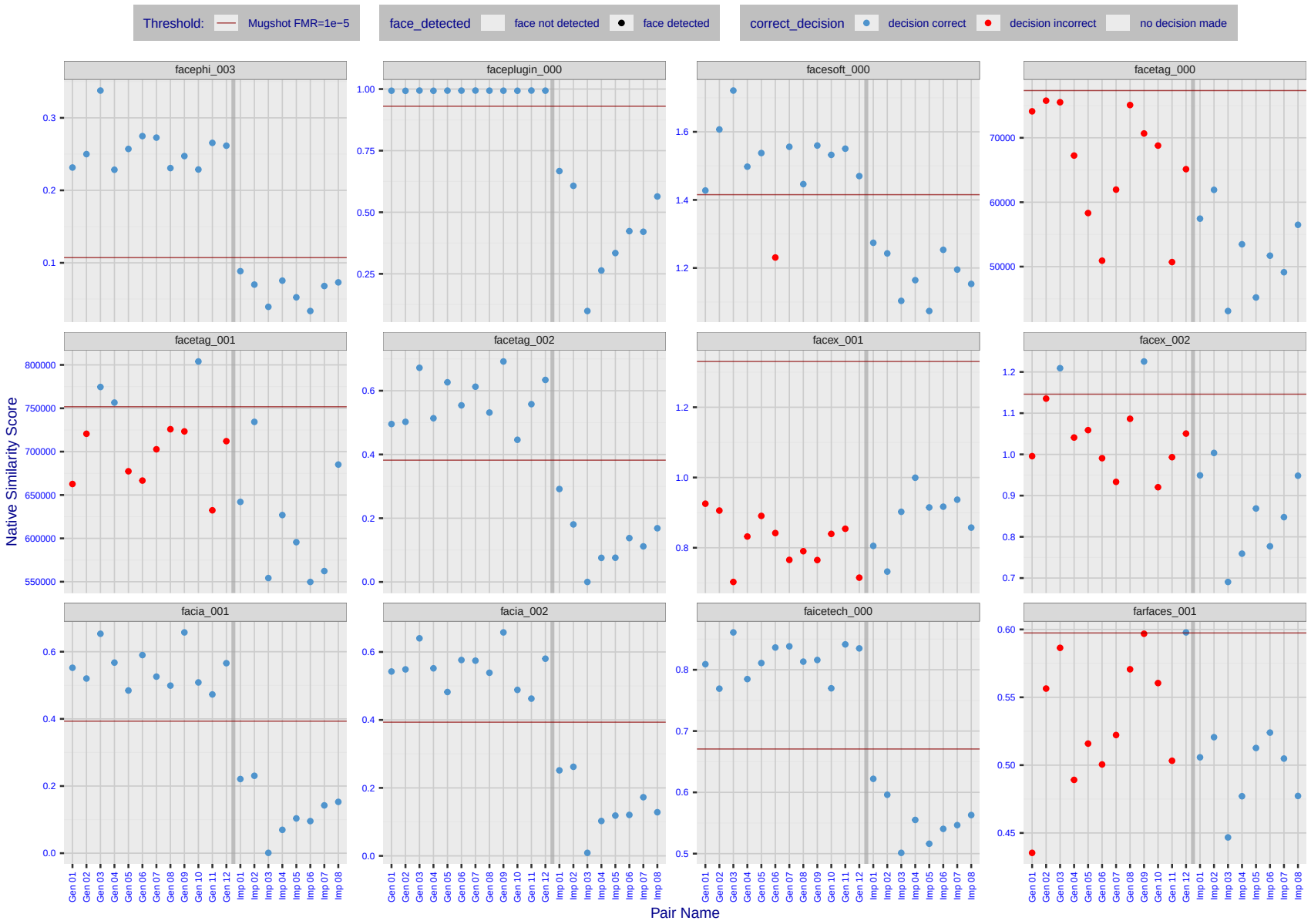


Figure 34: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

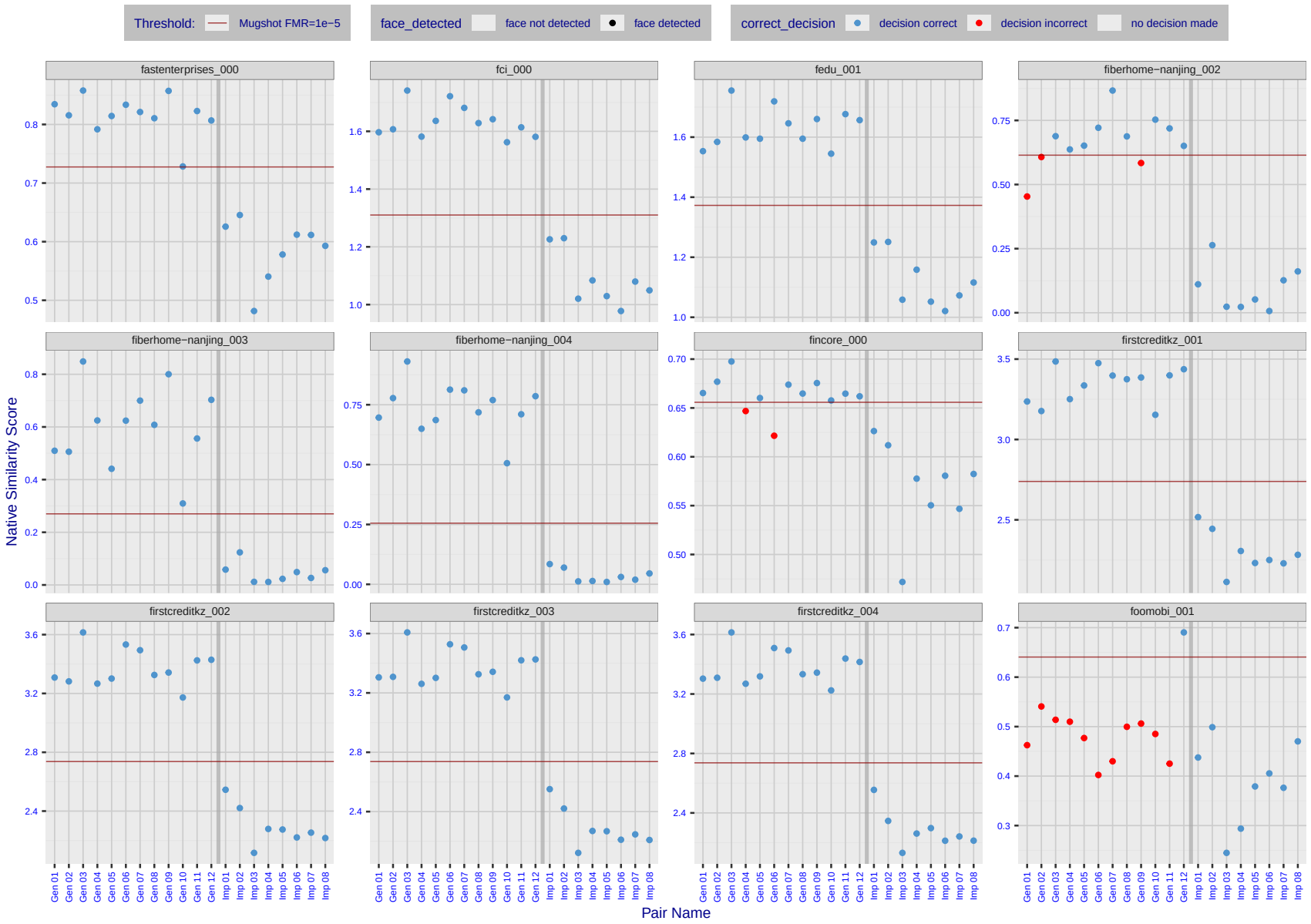


Figure 35: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

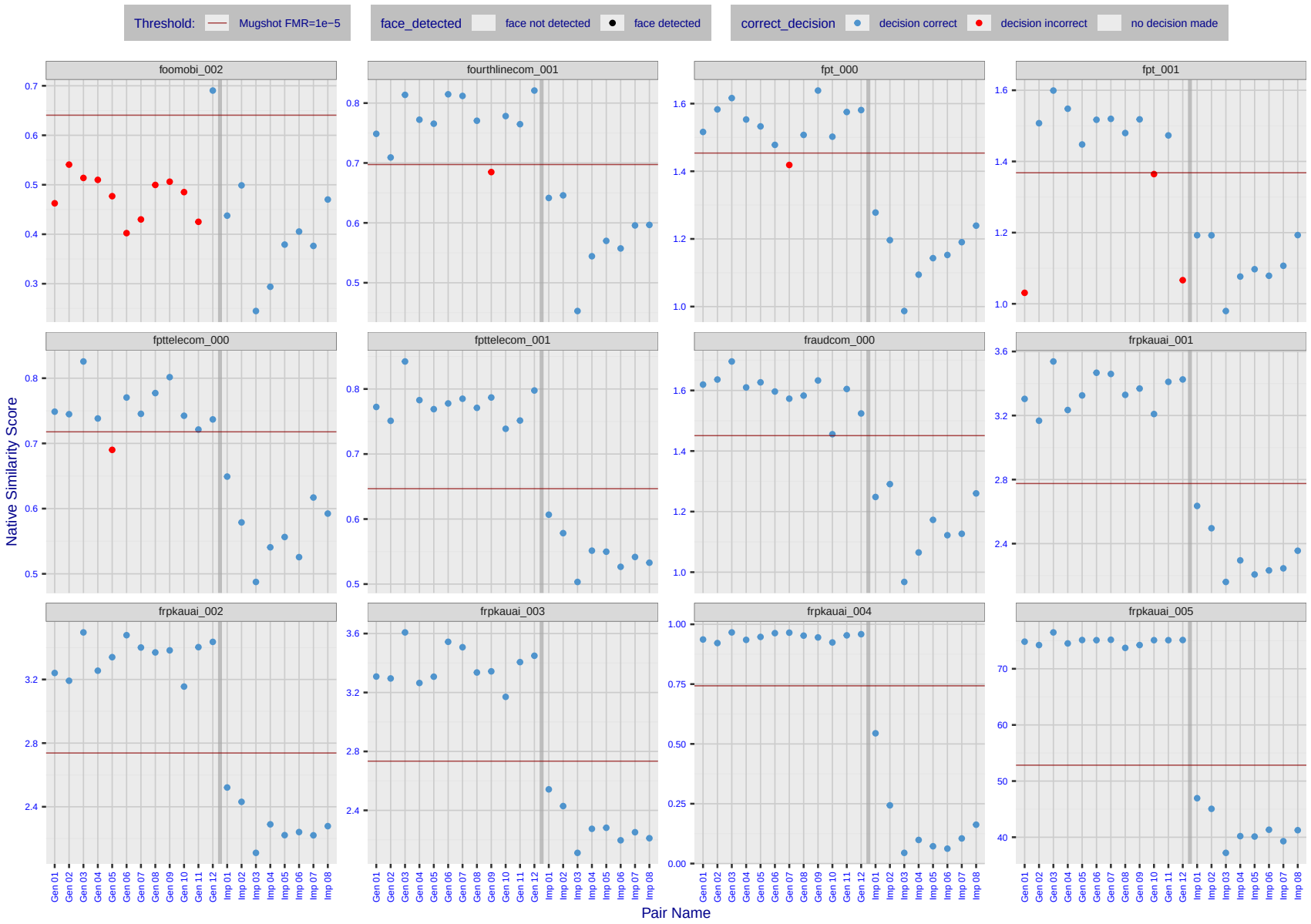


Figure 36: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

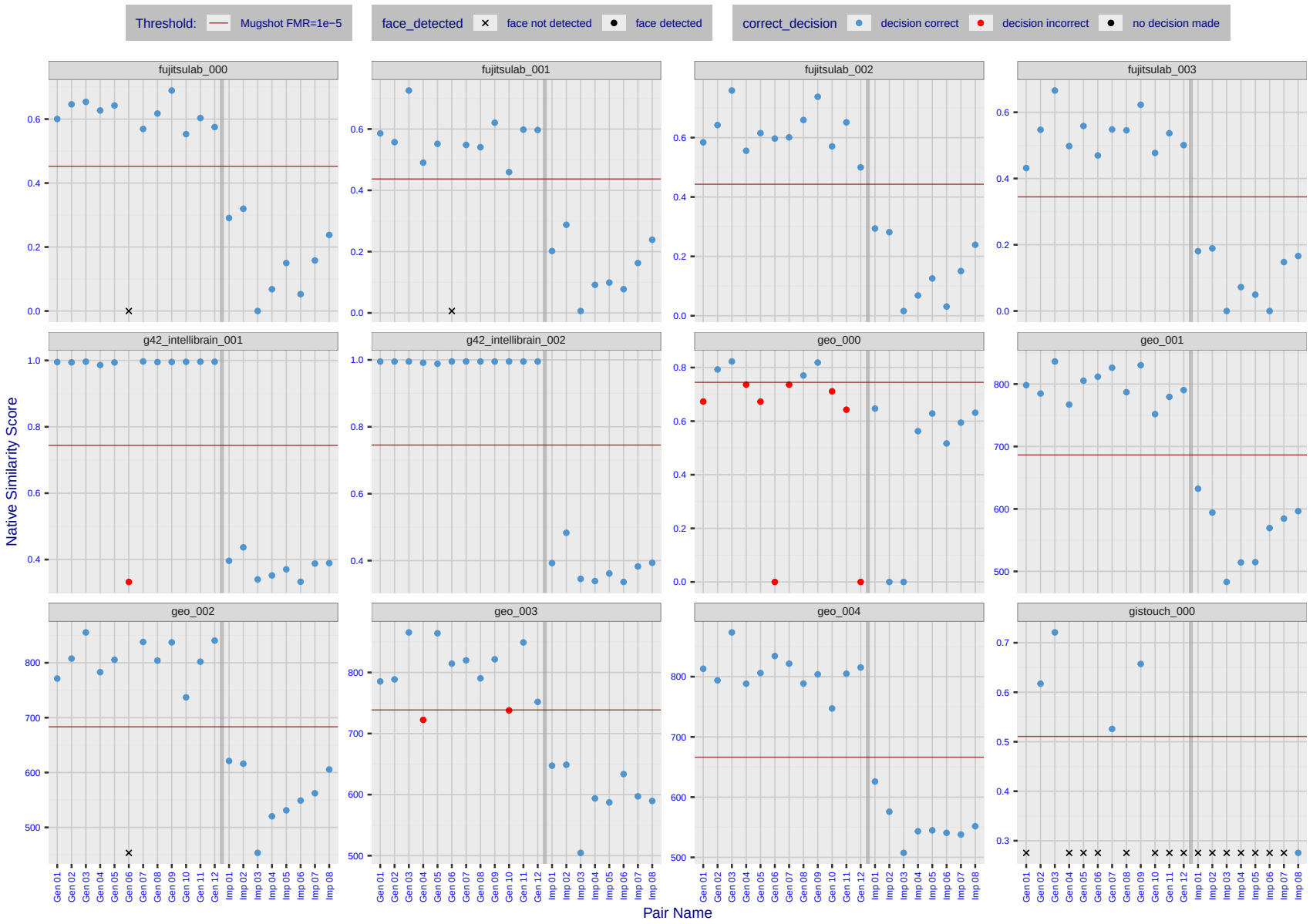


Figure 37: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

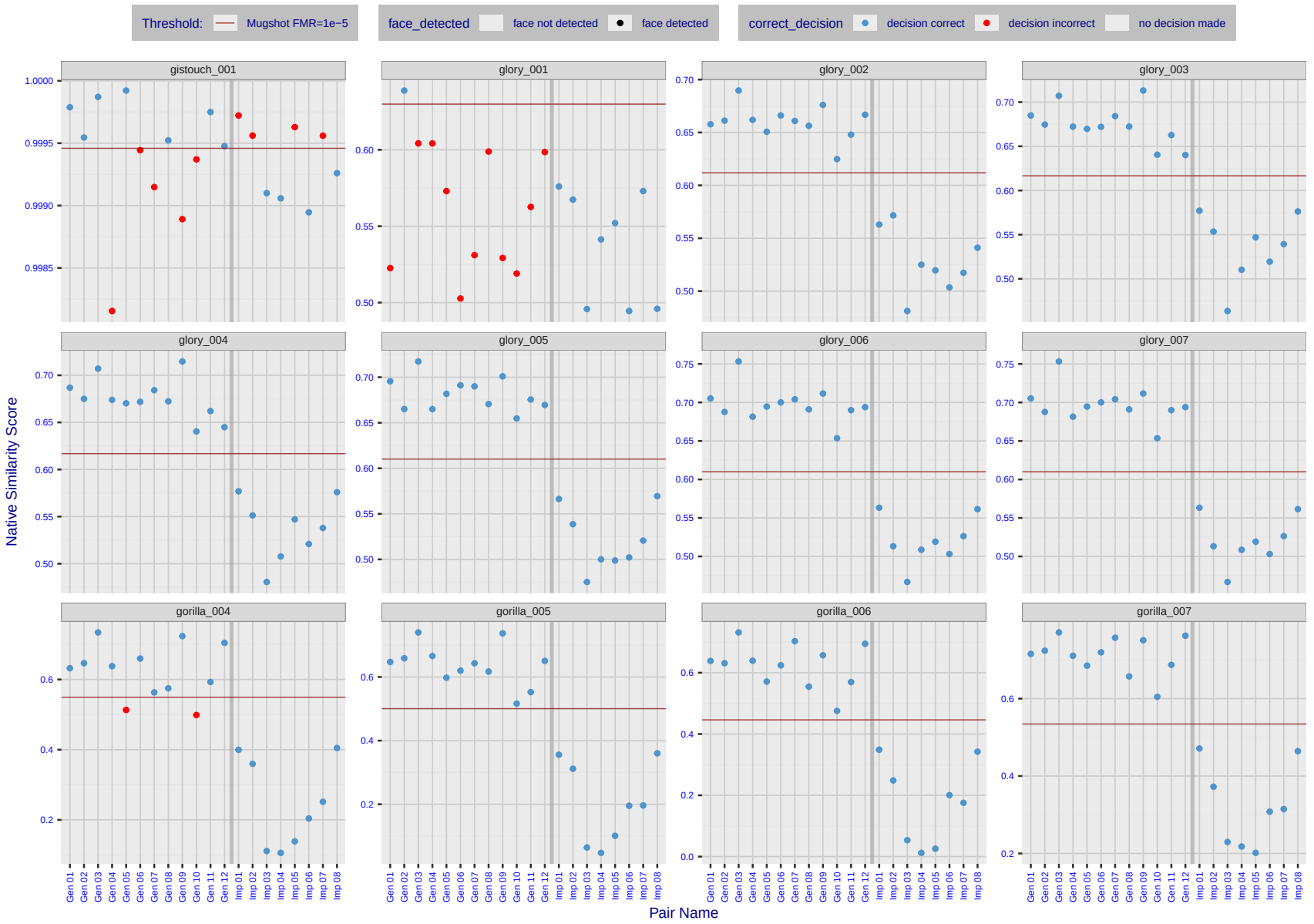


Figure 38: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

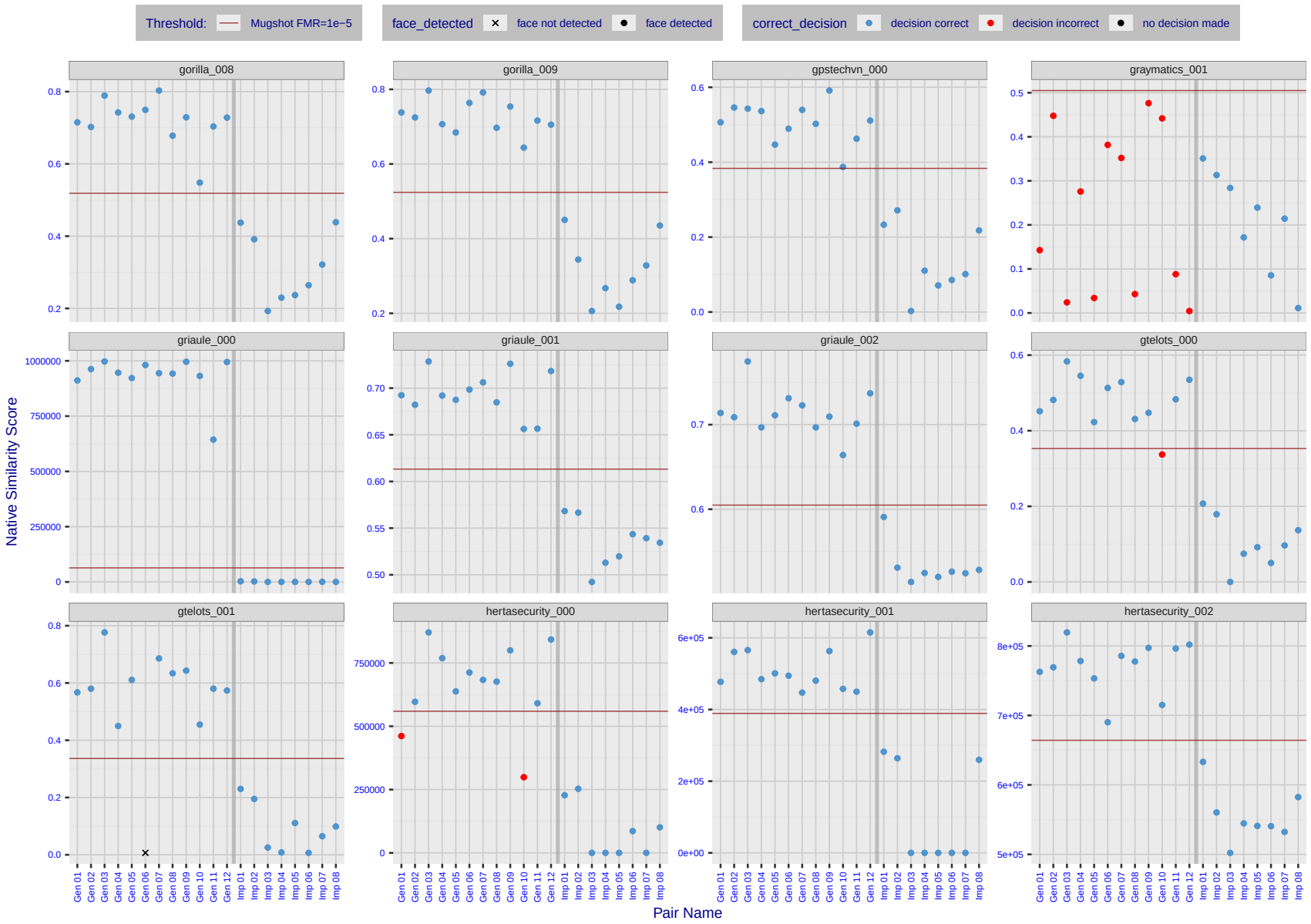


Figure 39: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

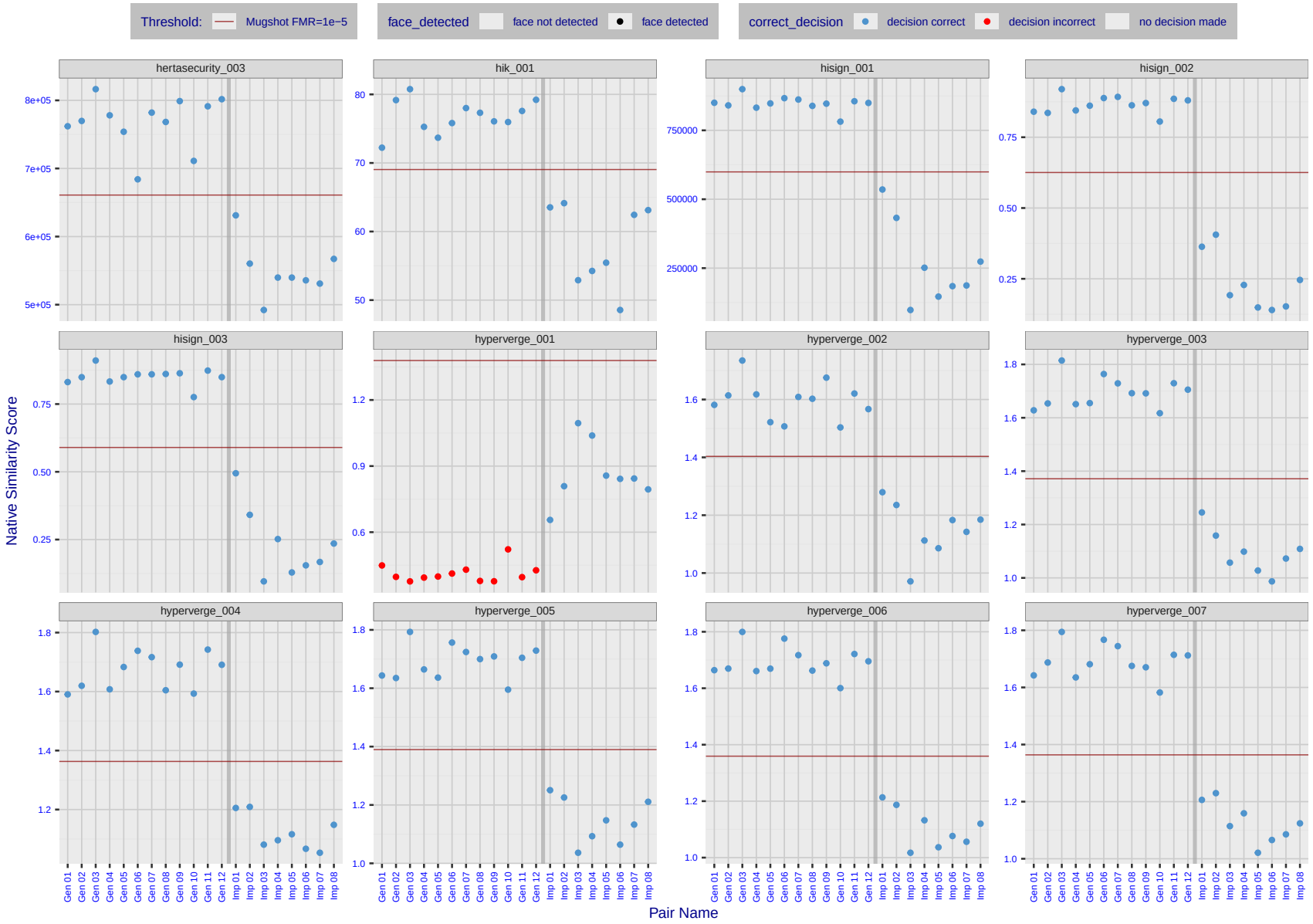


Figure 40: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

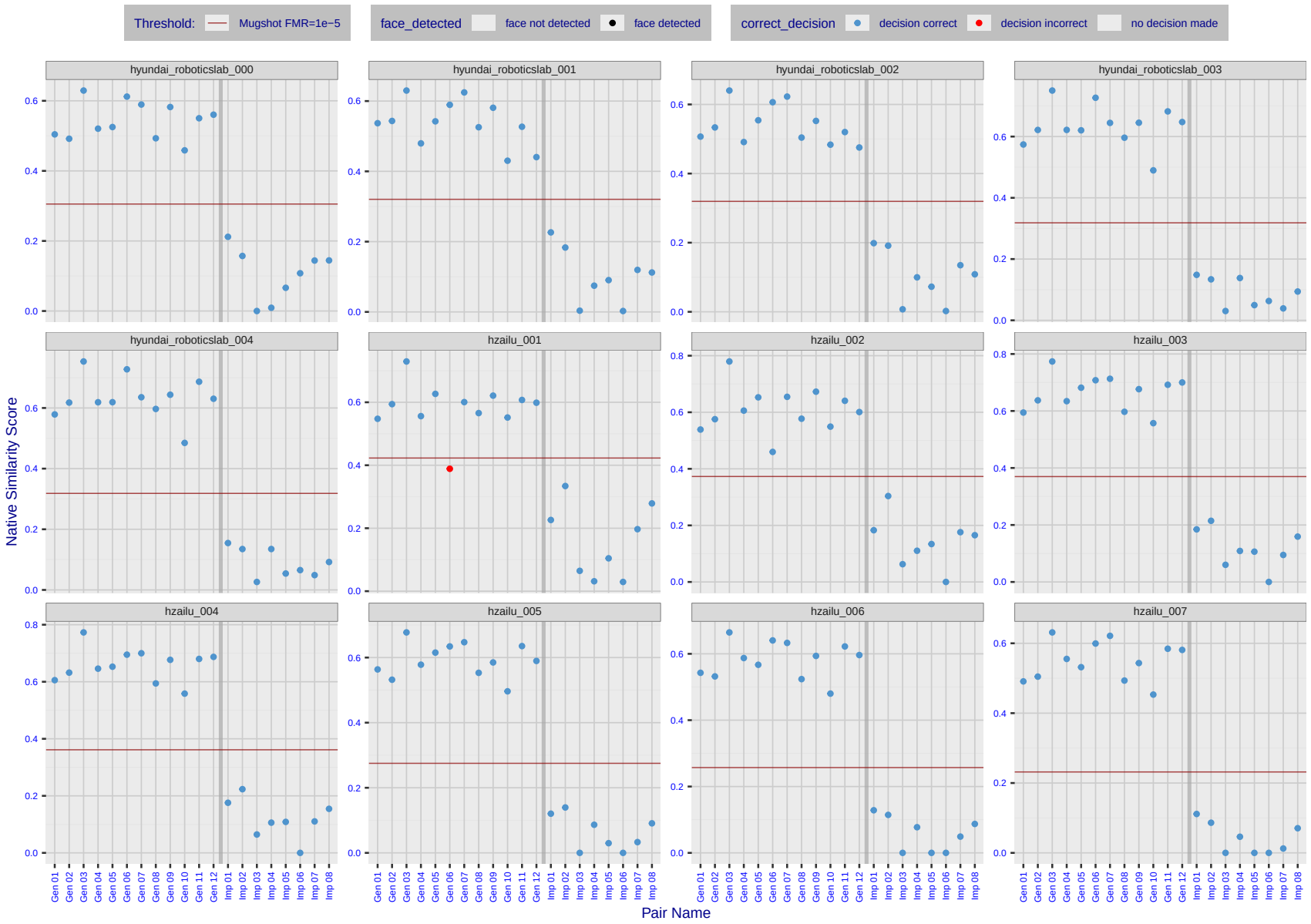


Figure 41: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

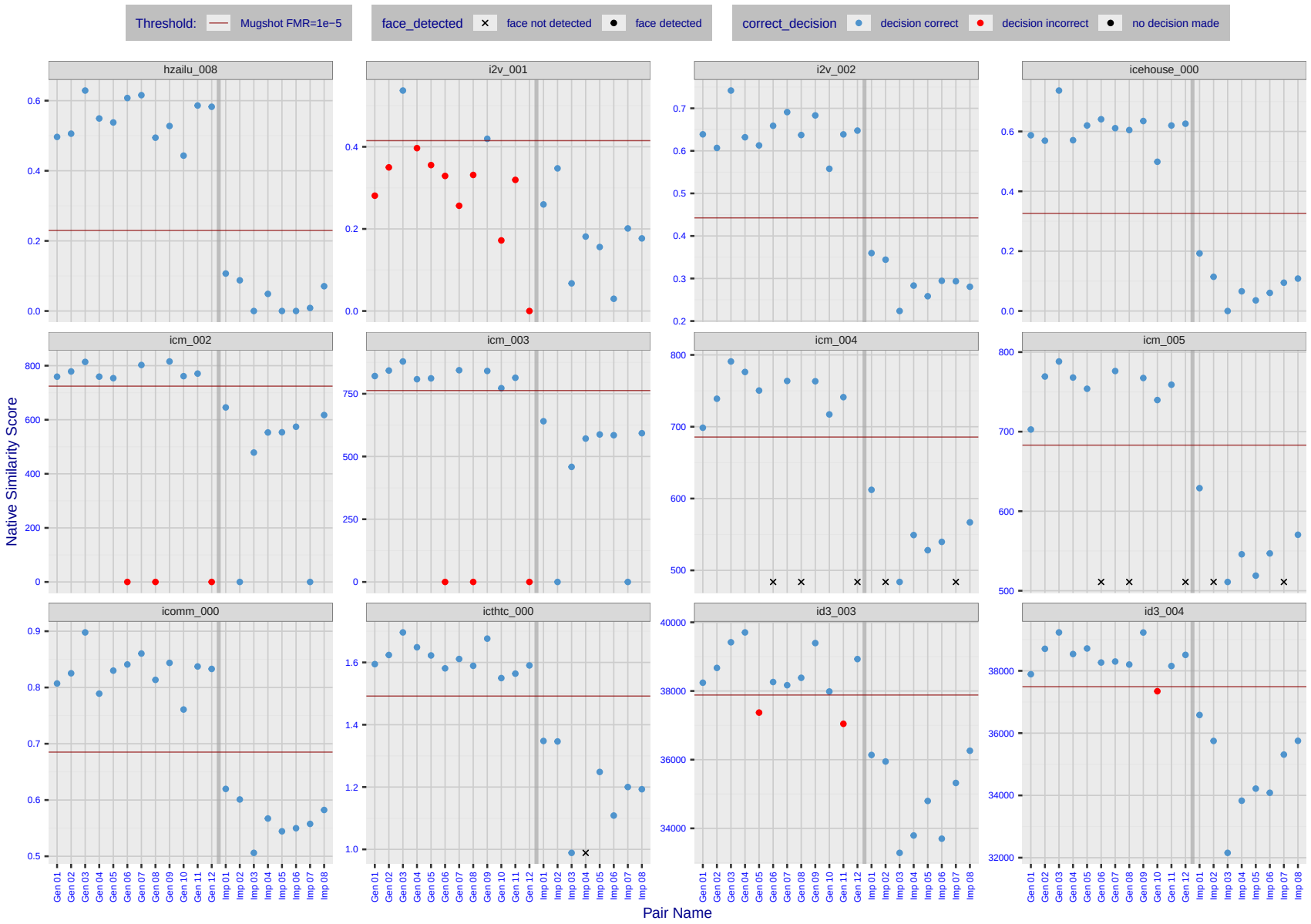


Figure 42: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

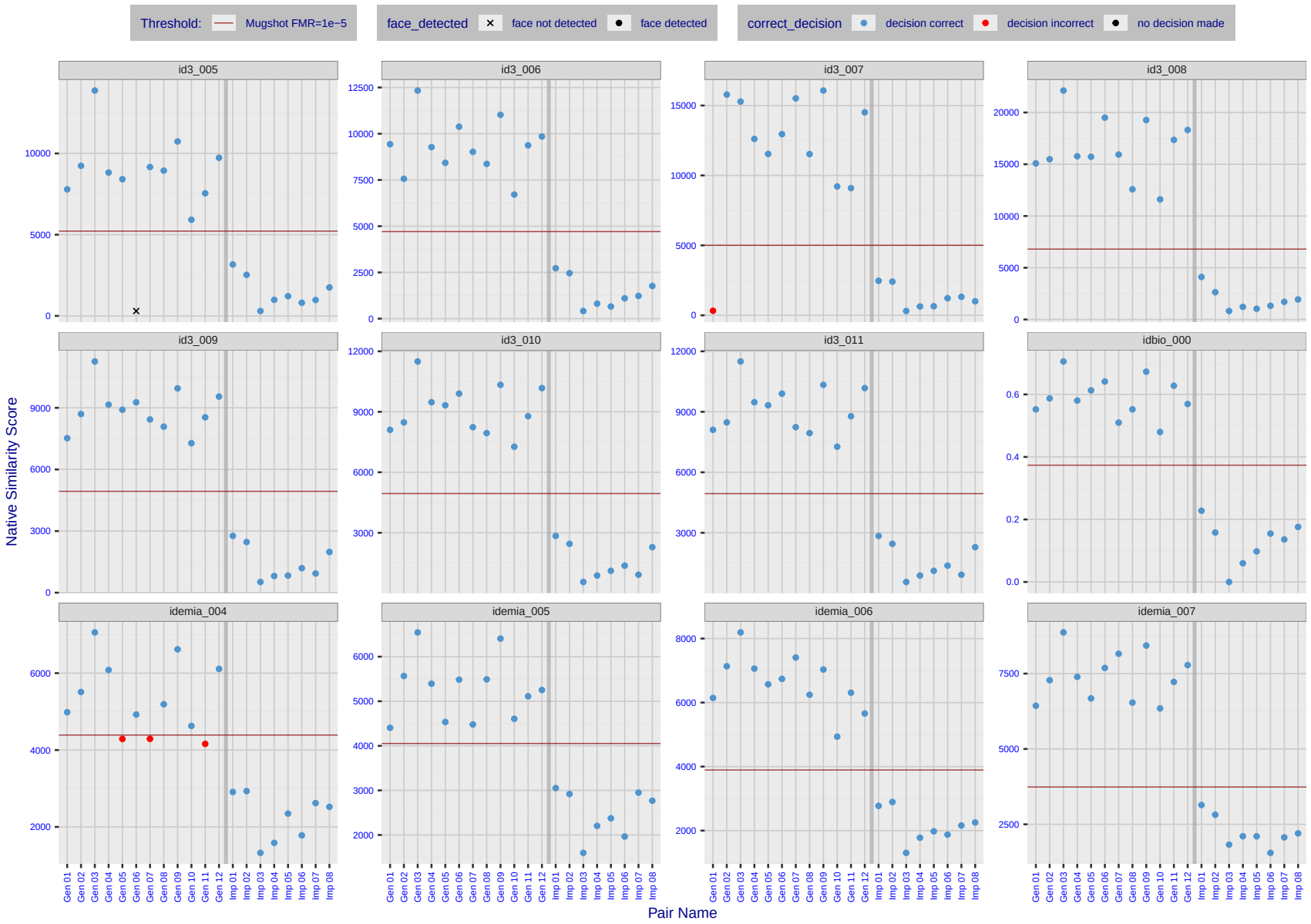


Figure 43: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

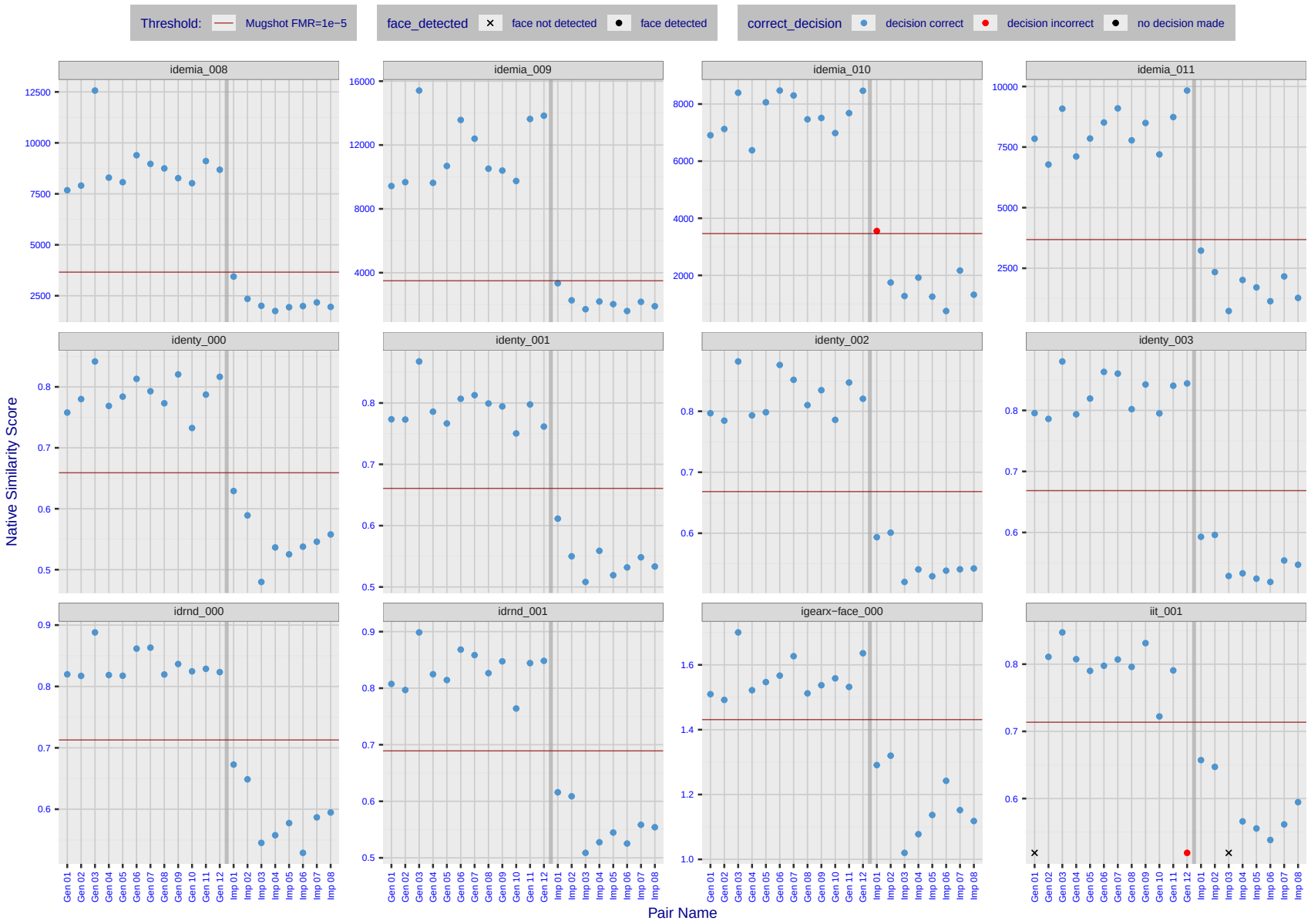


Figure 44: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

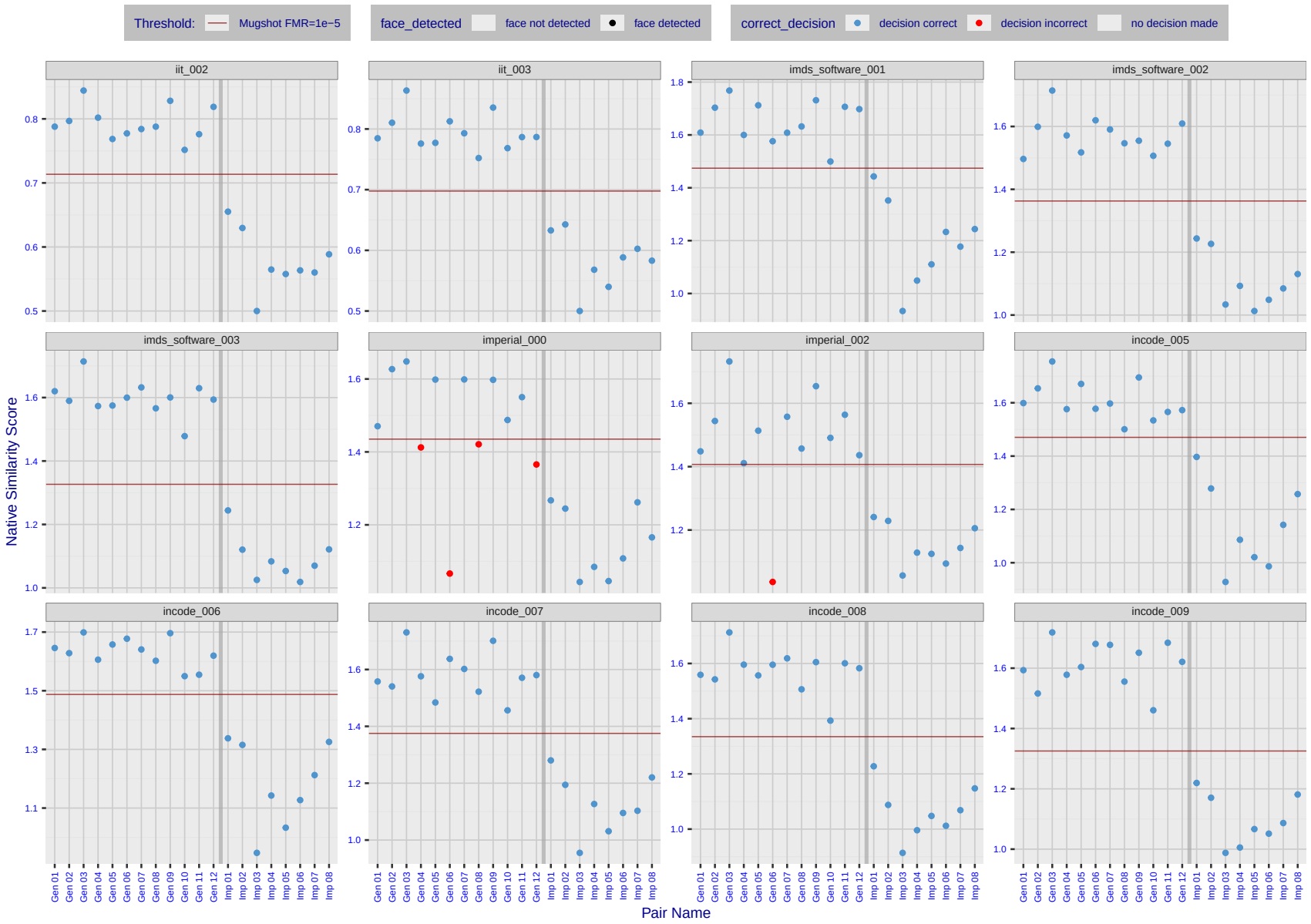


Figure 45: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

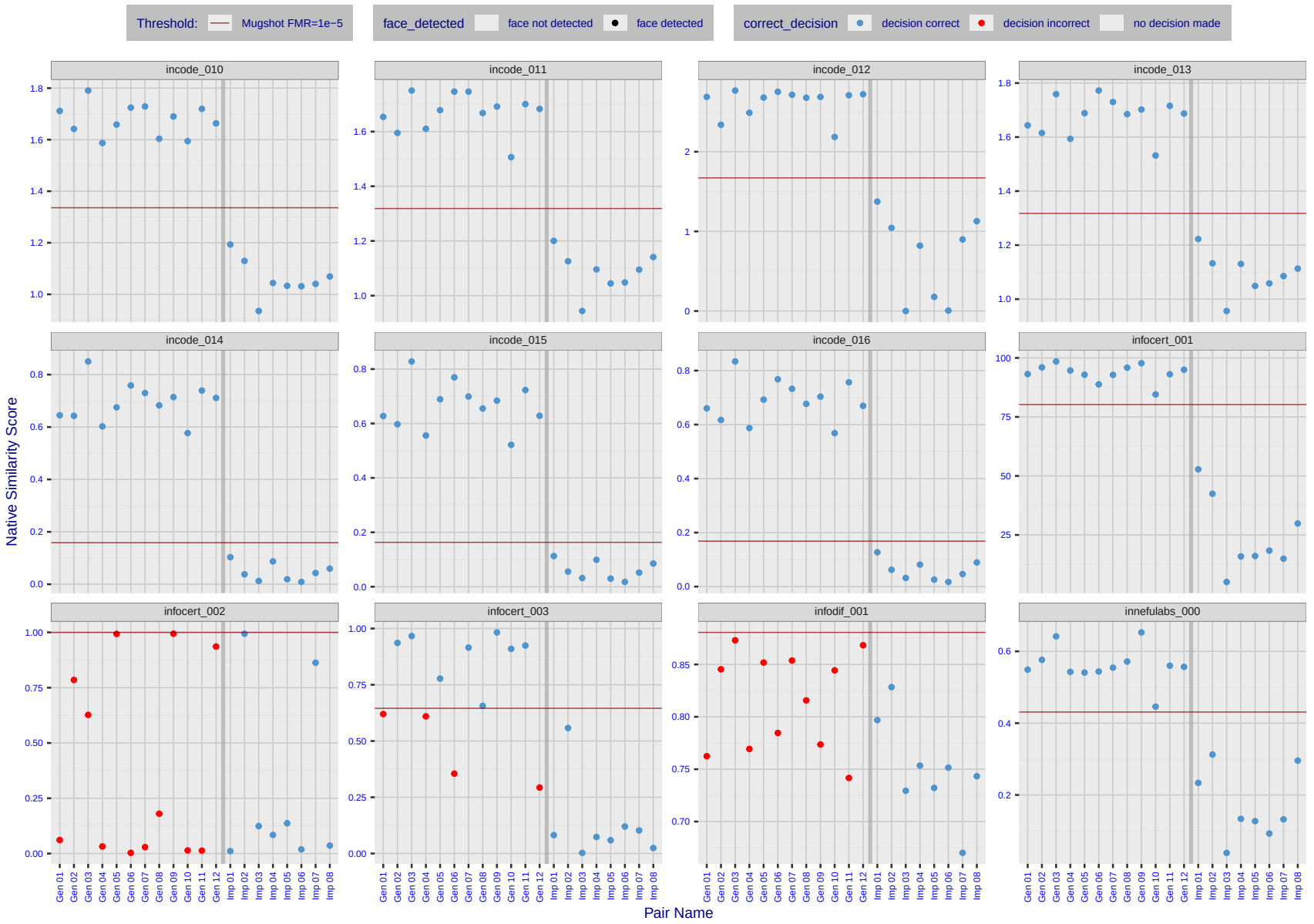


Figure 46: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

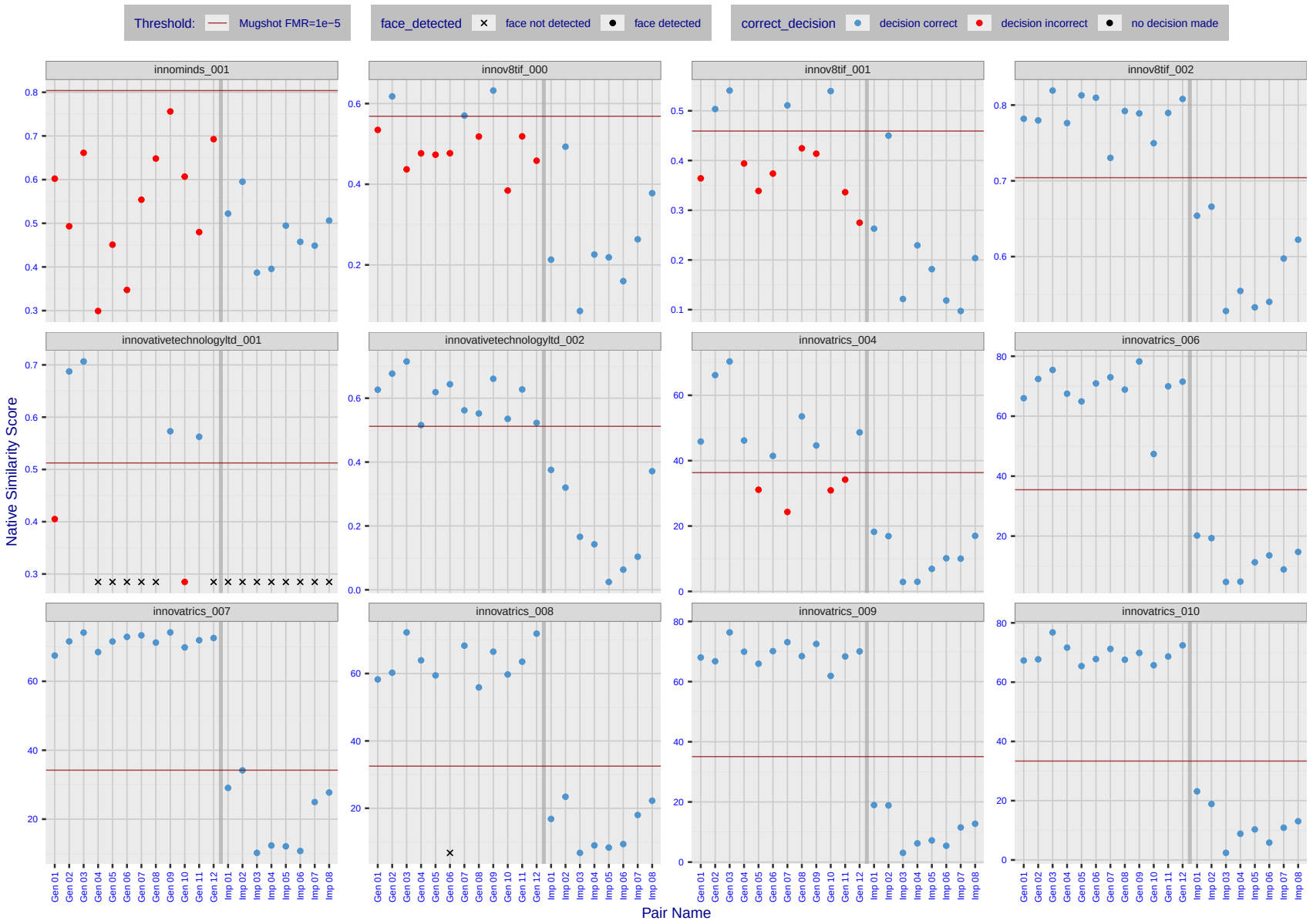


Figure 47: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

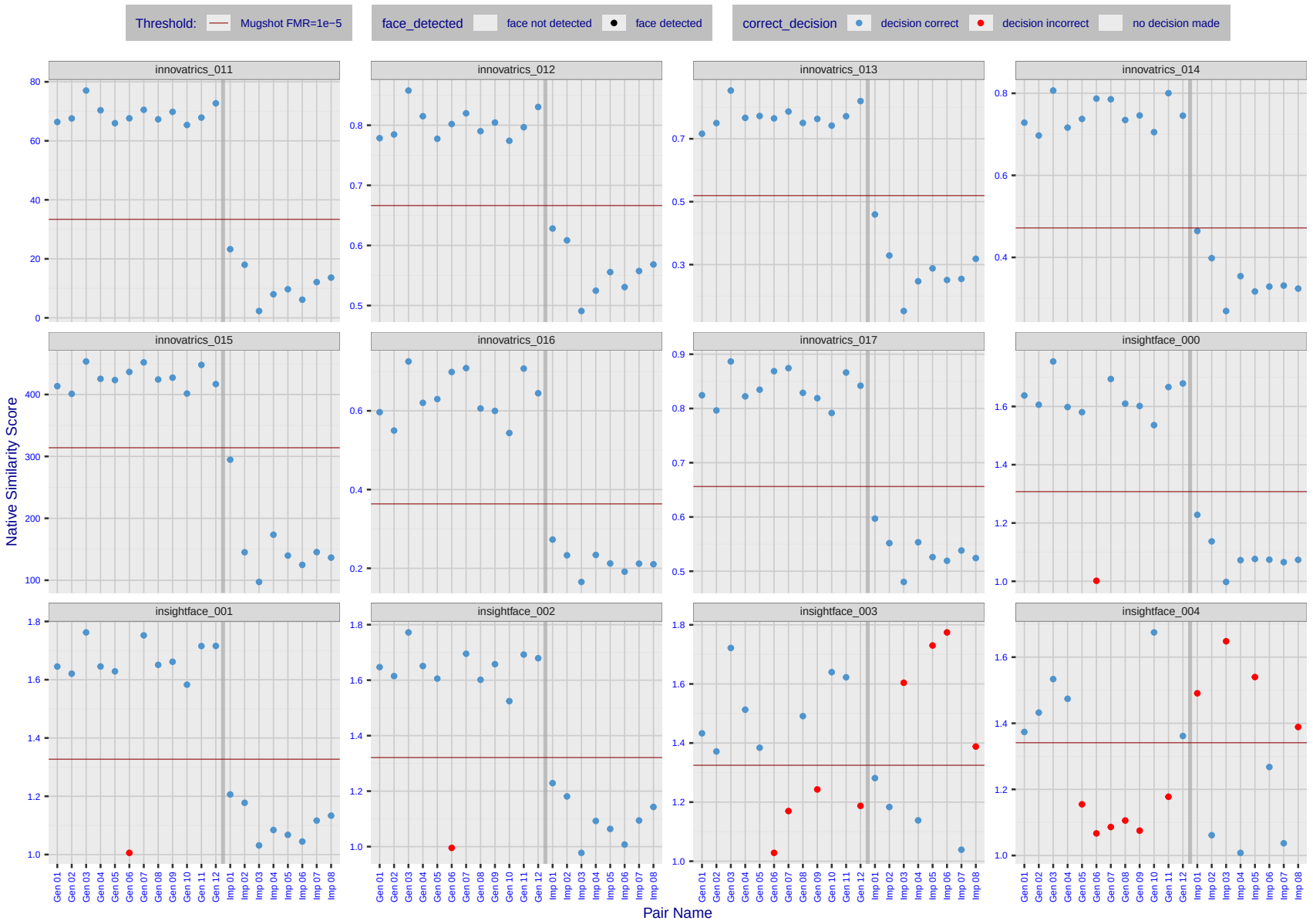


Figure 48: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

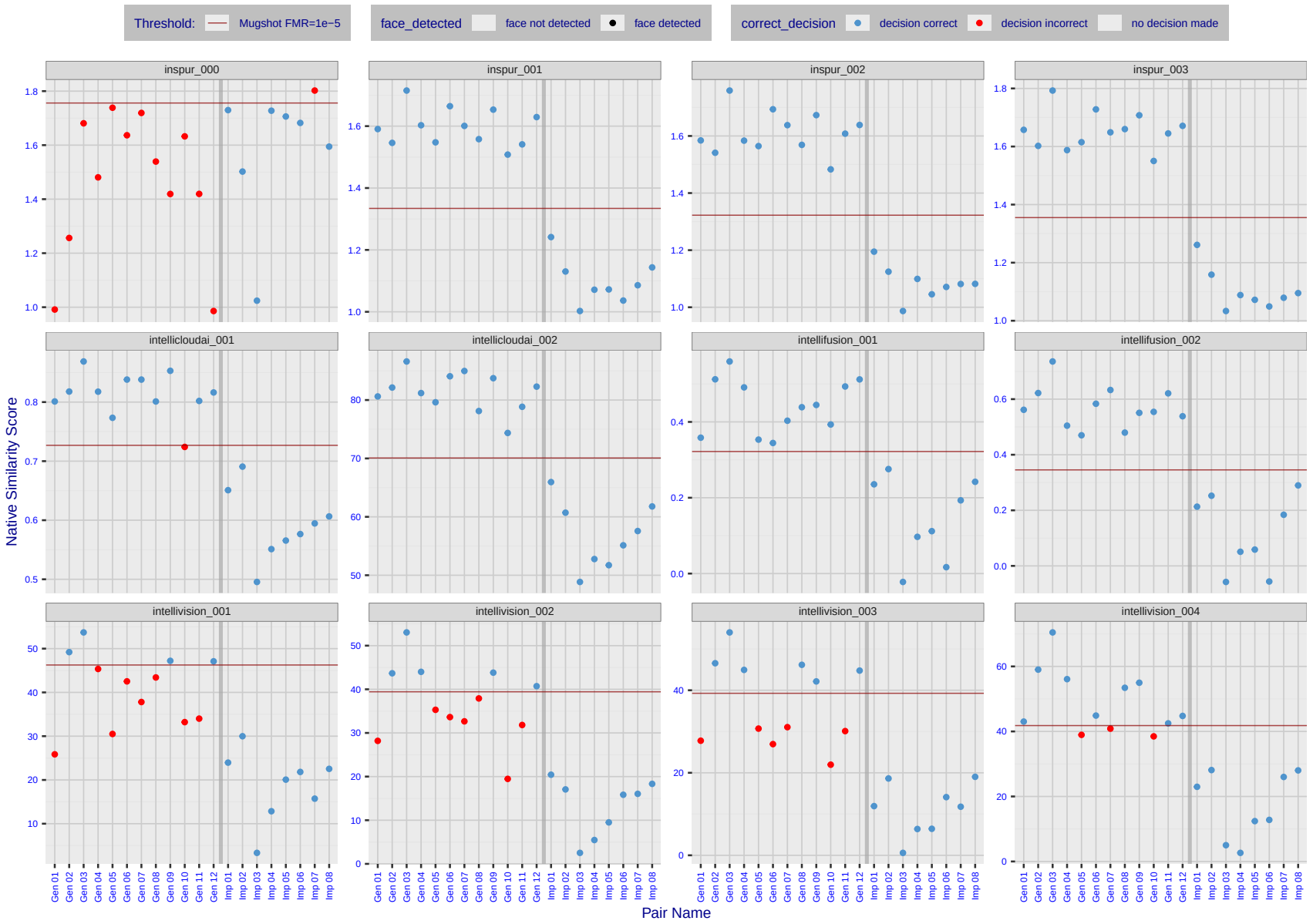


Figure 49: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

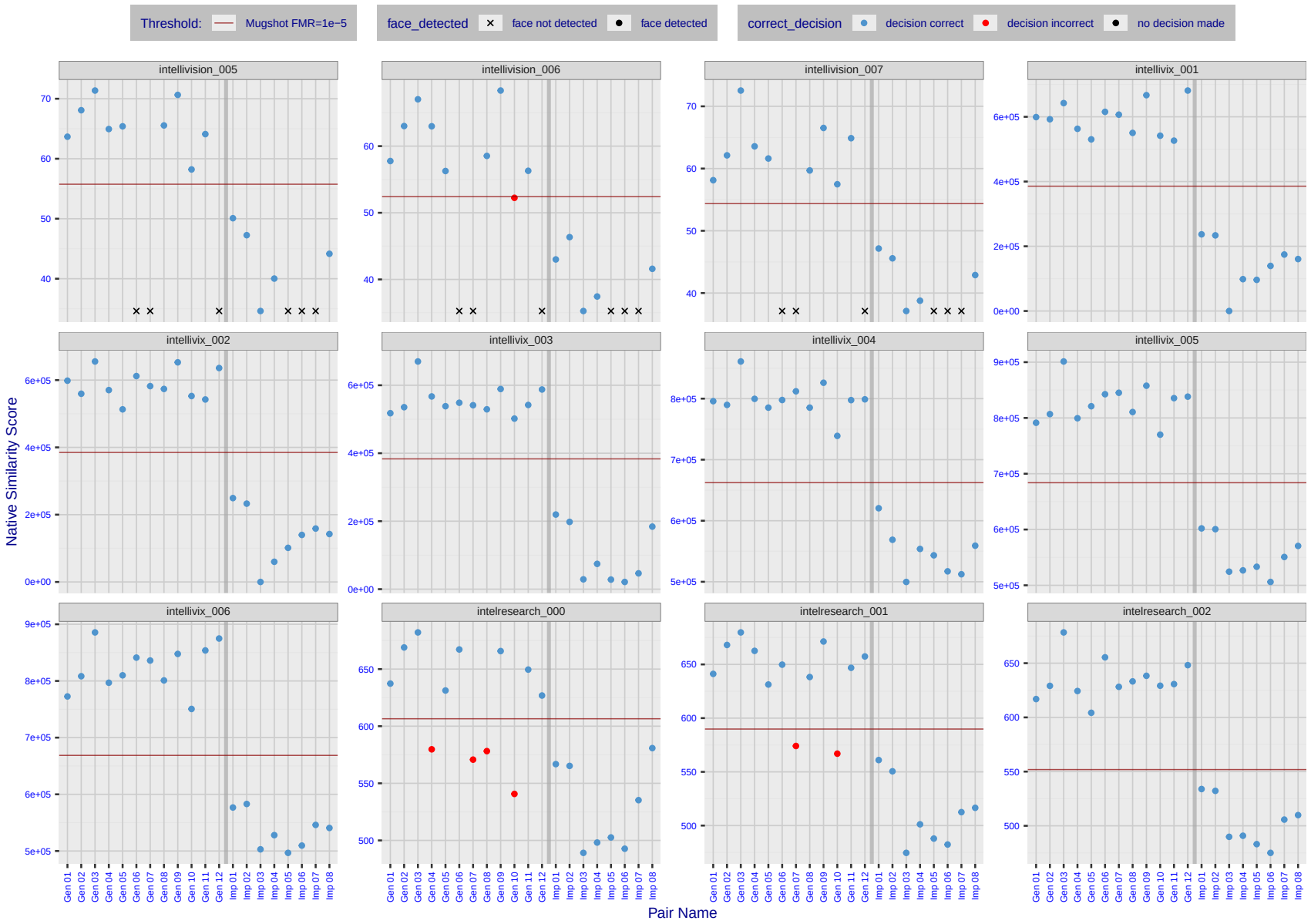


Figure 50: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

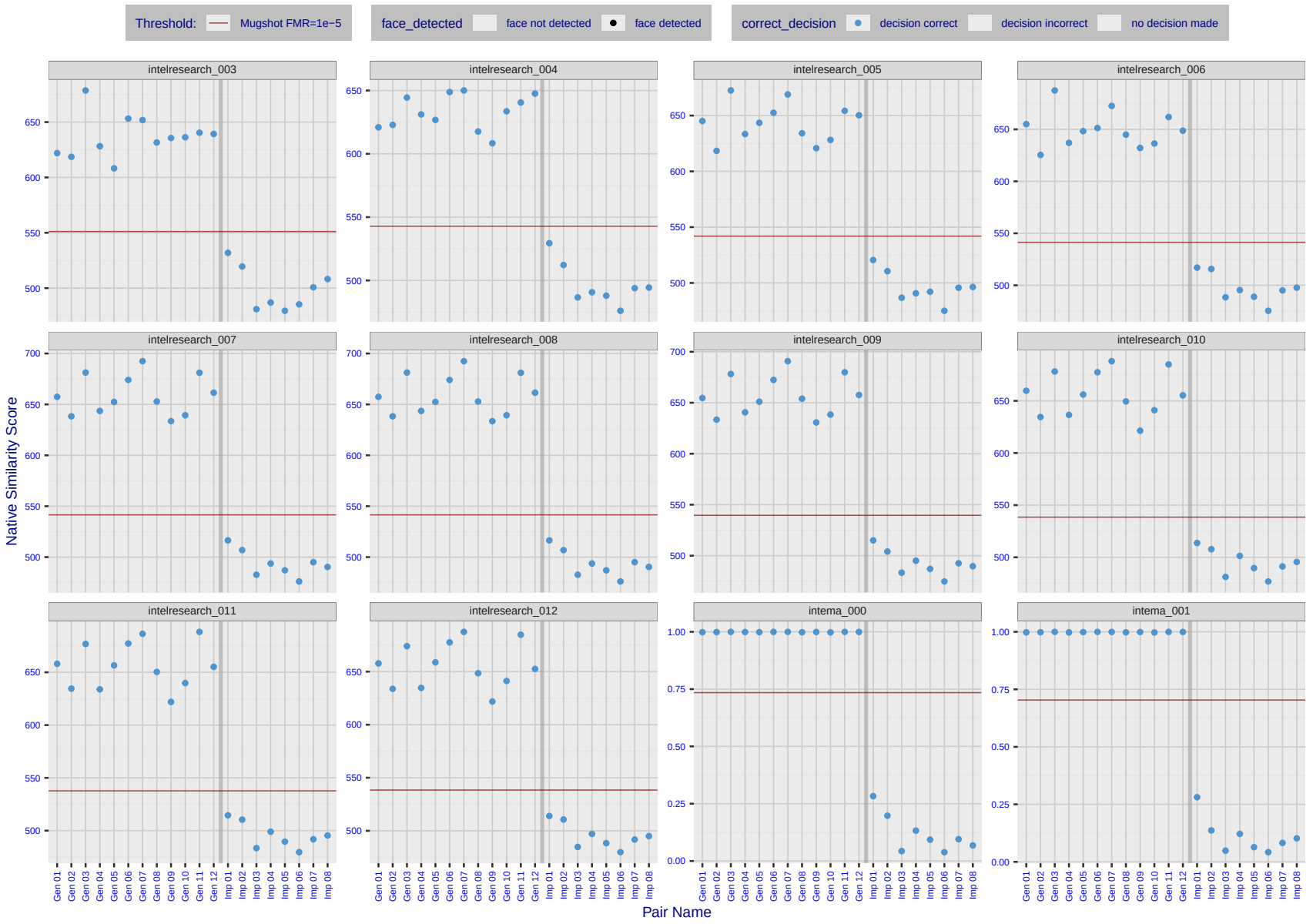


Figure 51: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

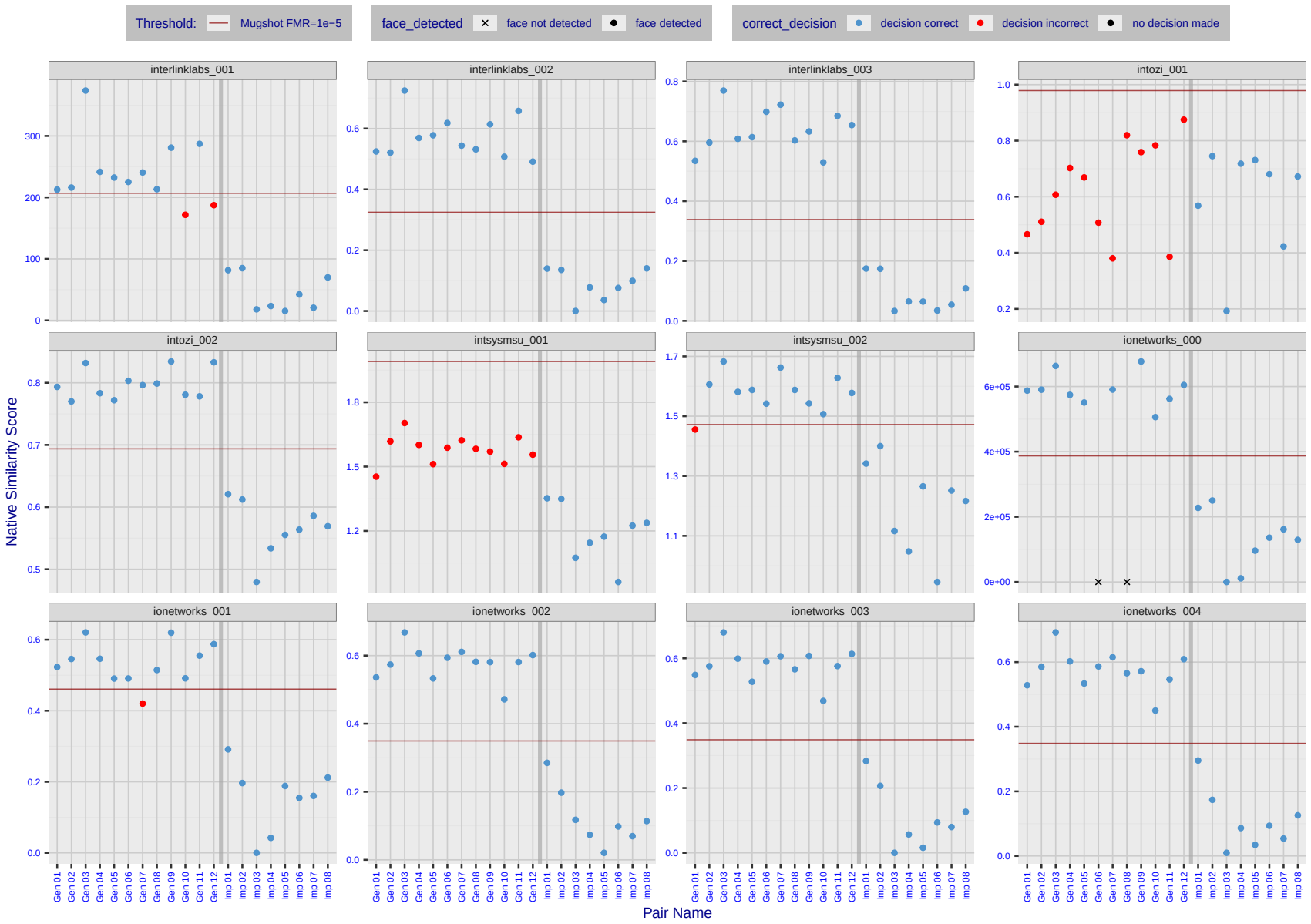


Figure 52: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

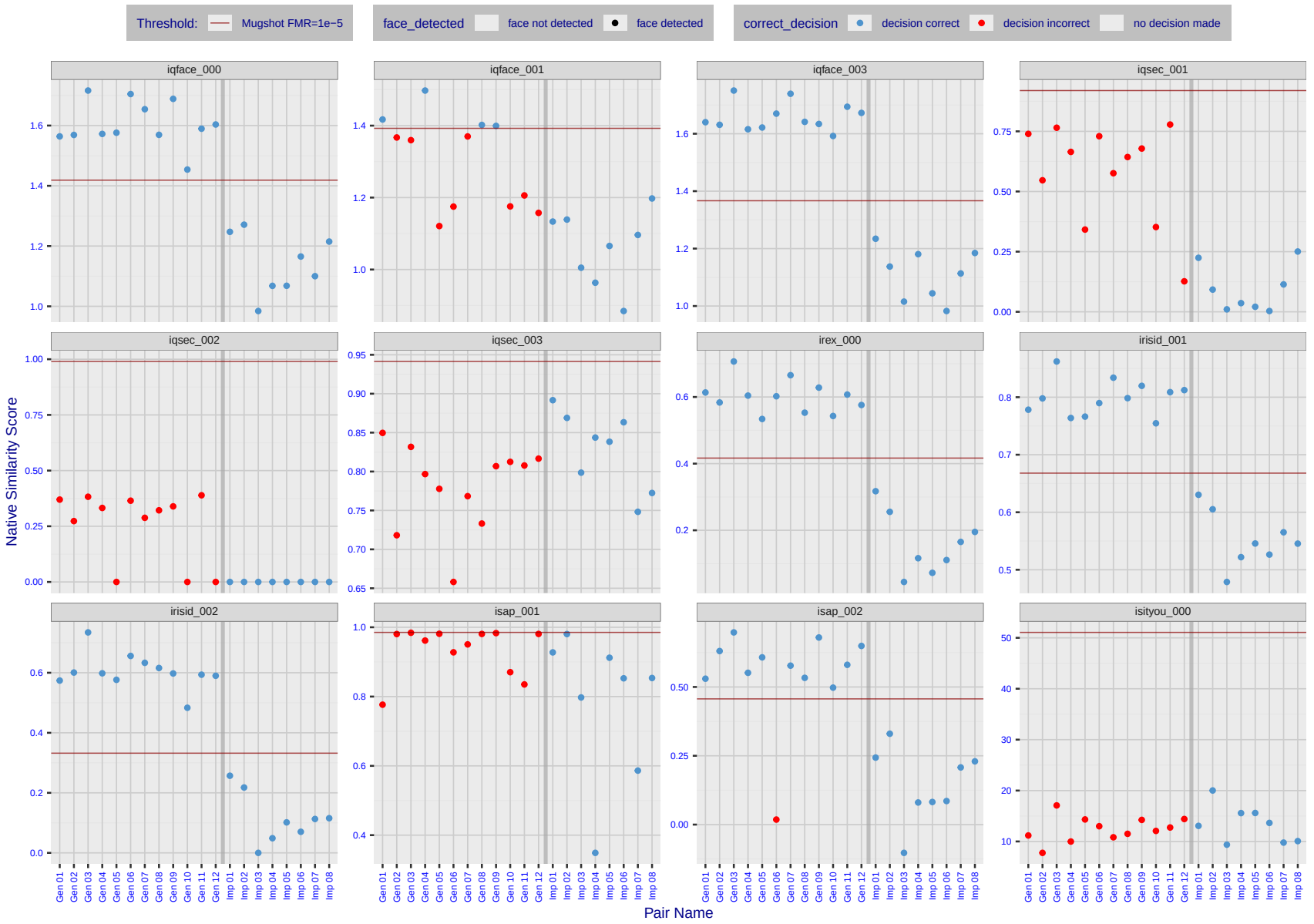


Figure 53: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

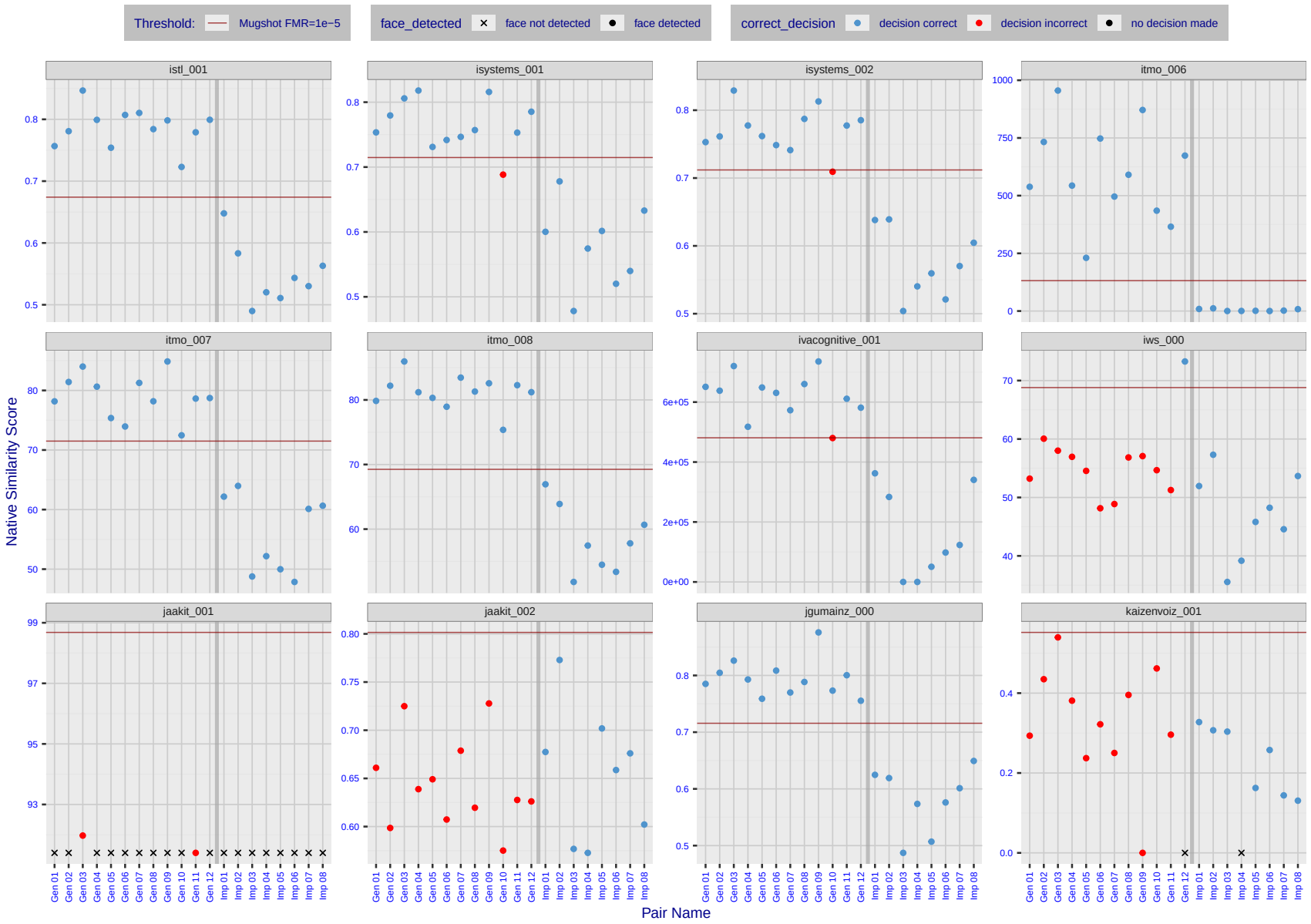


Figure 54: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

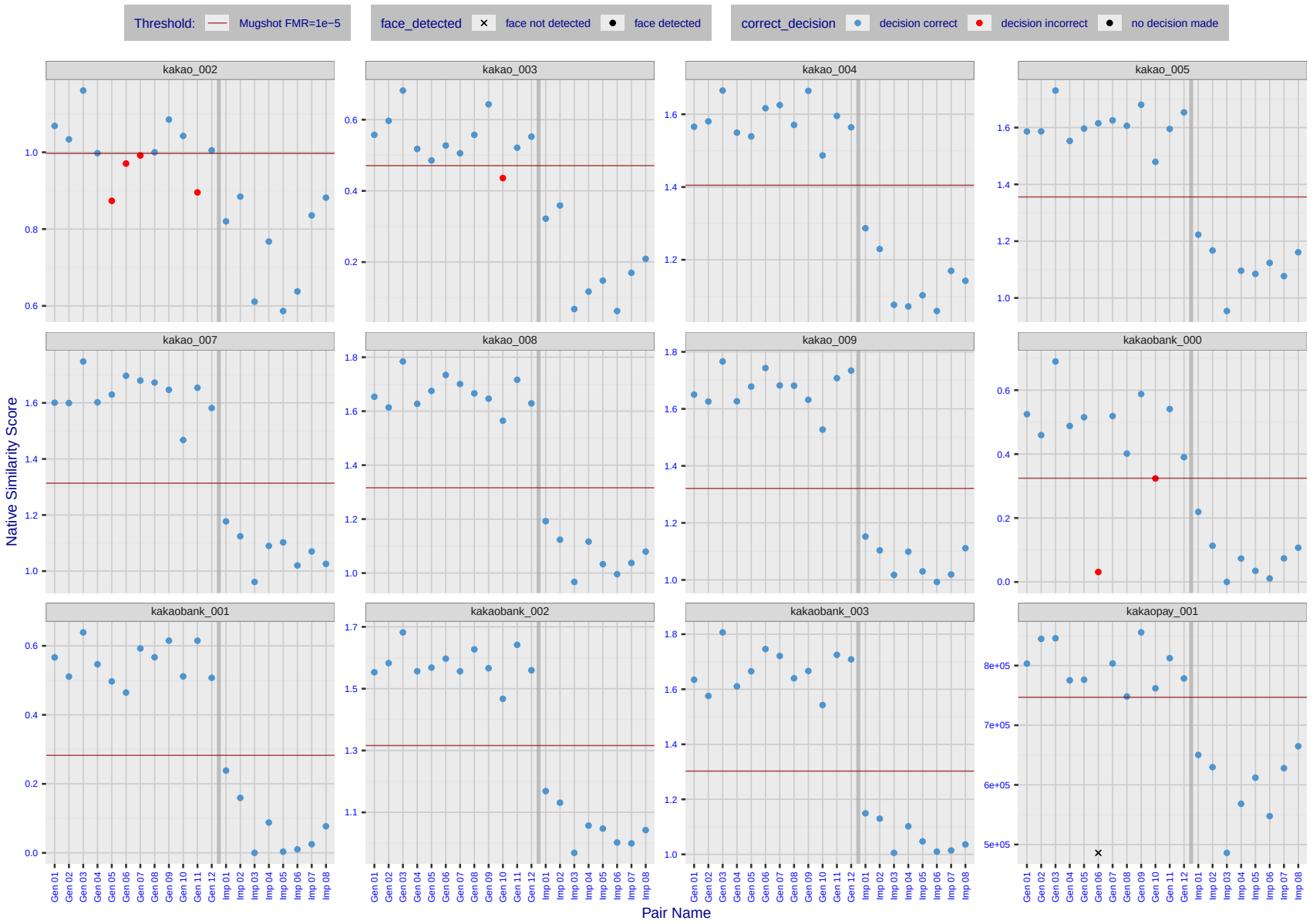


Figure 55: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

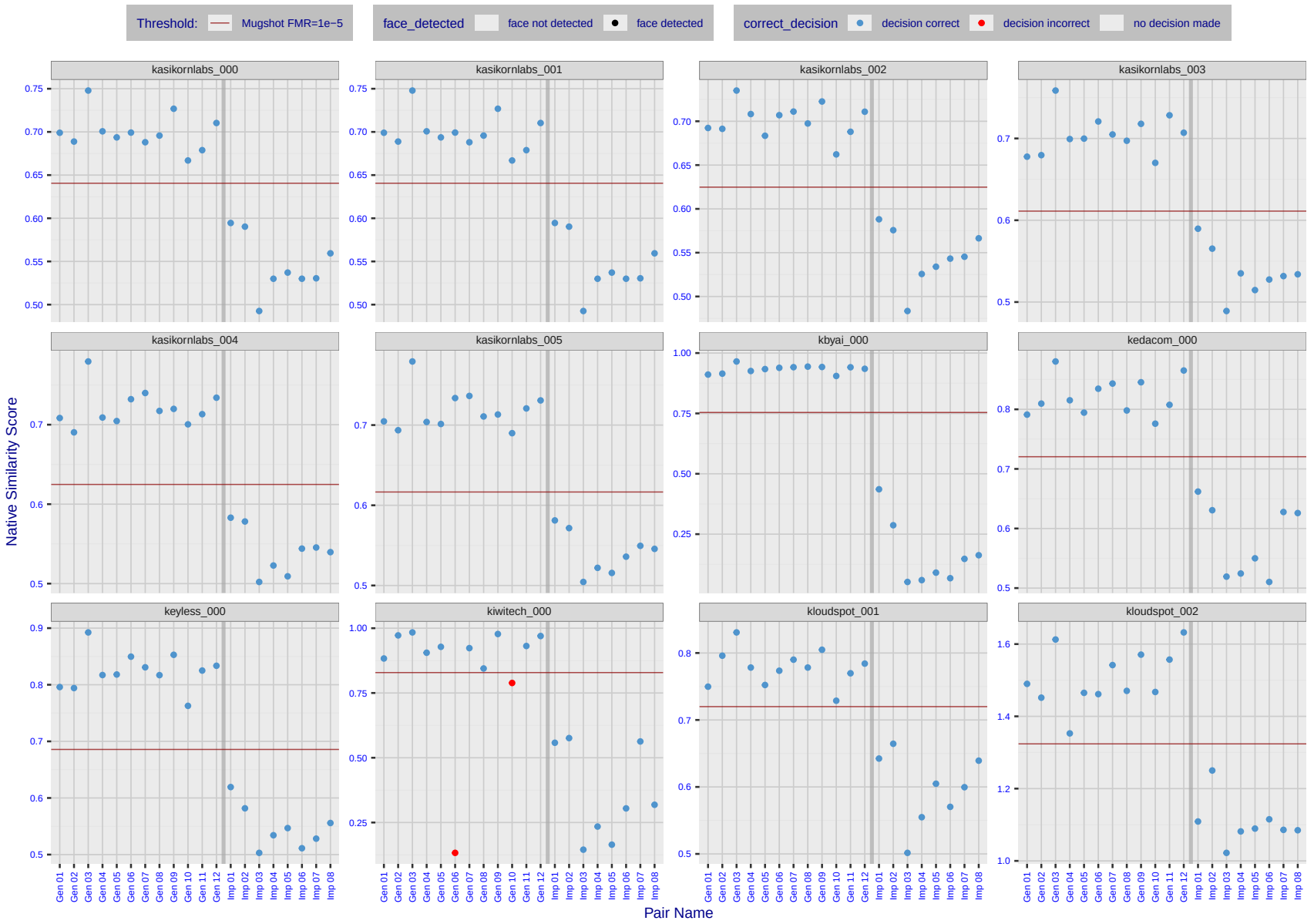


Figure 56: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

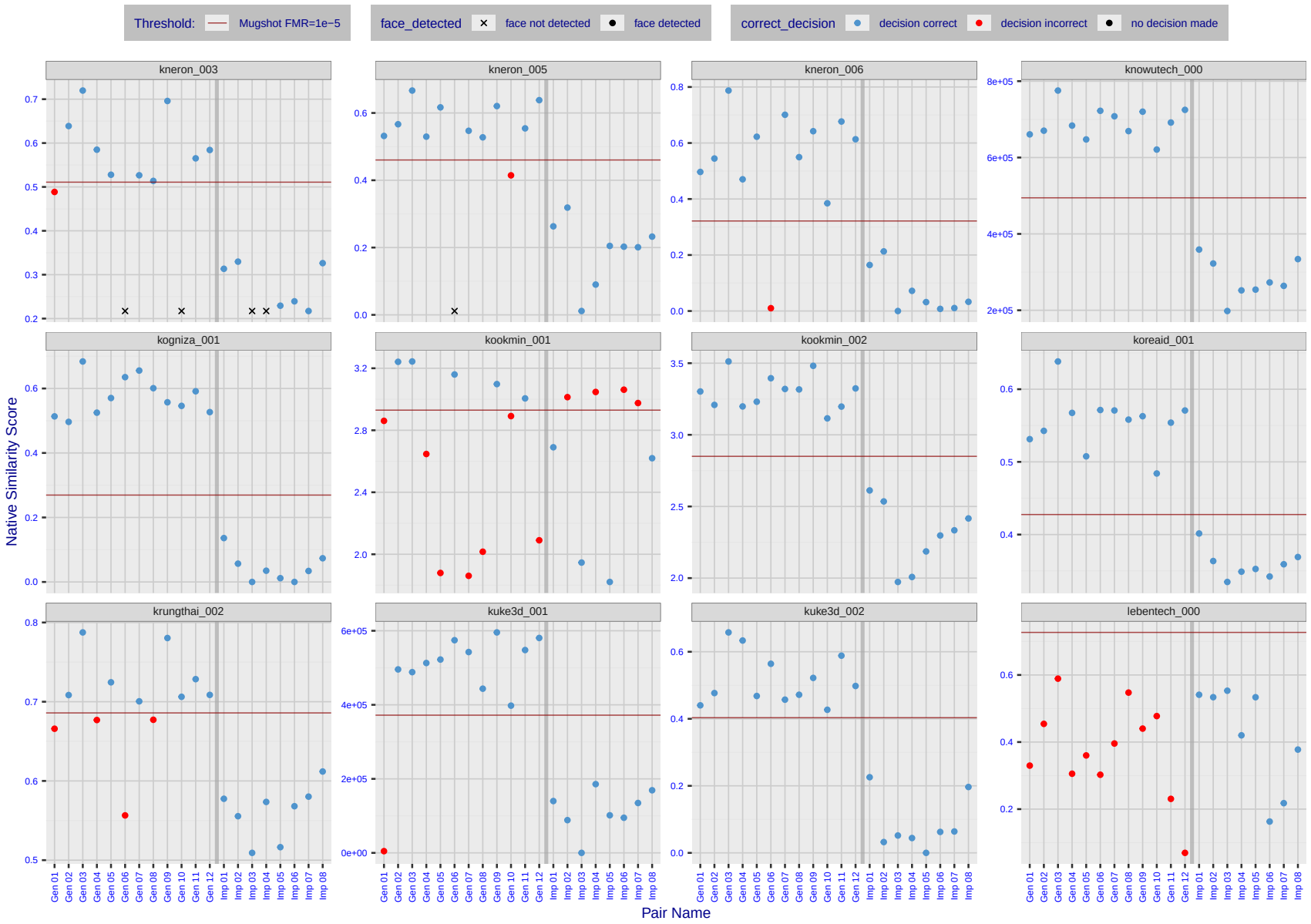


Figure 57: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

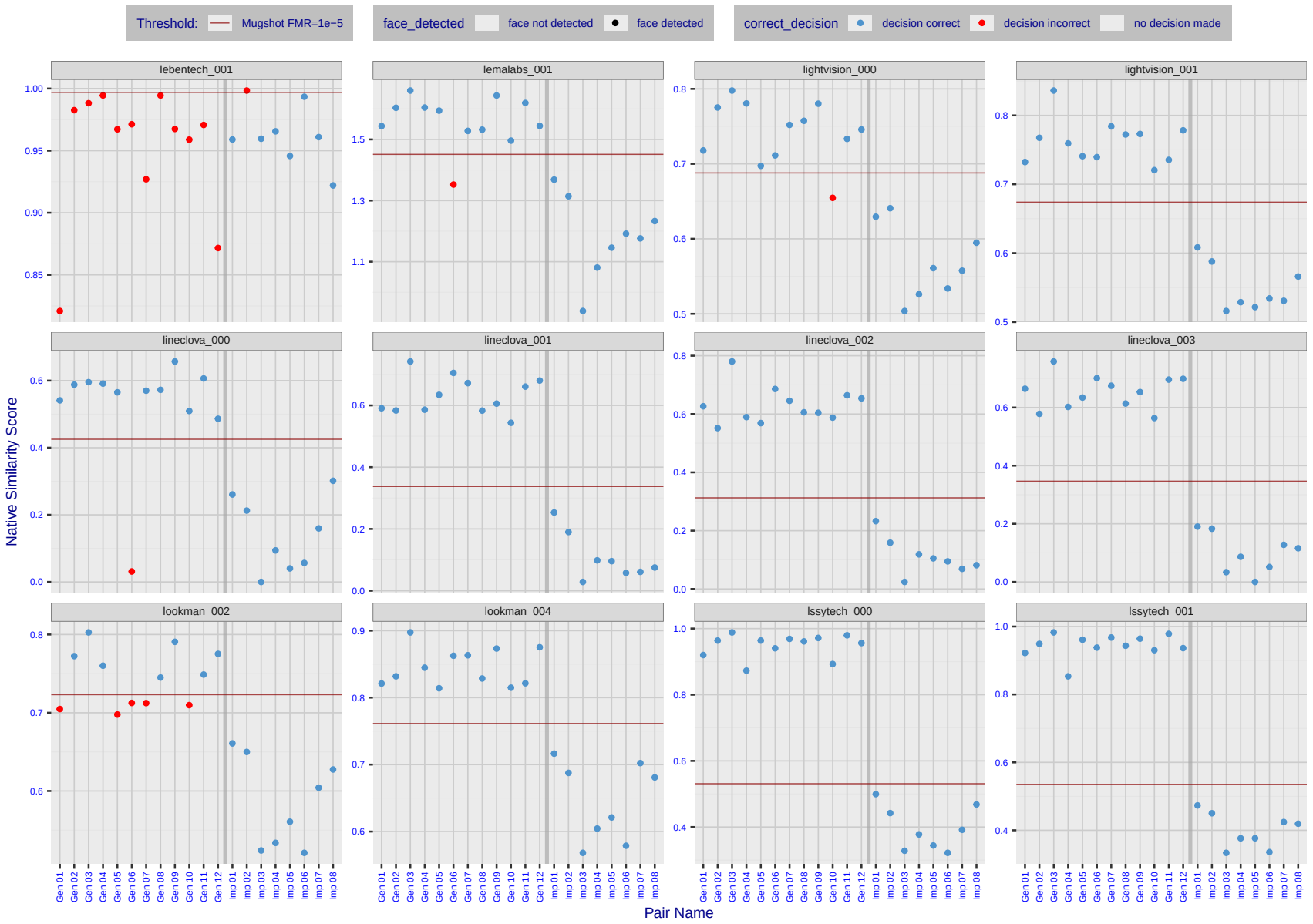


Figure 58: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

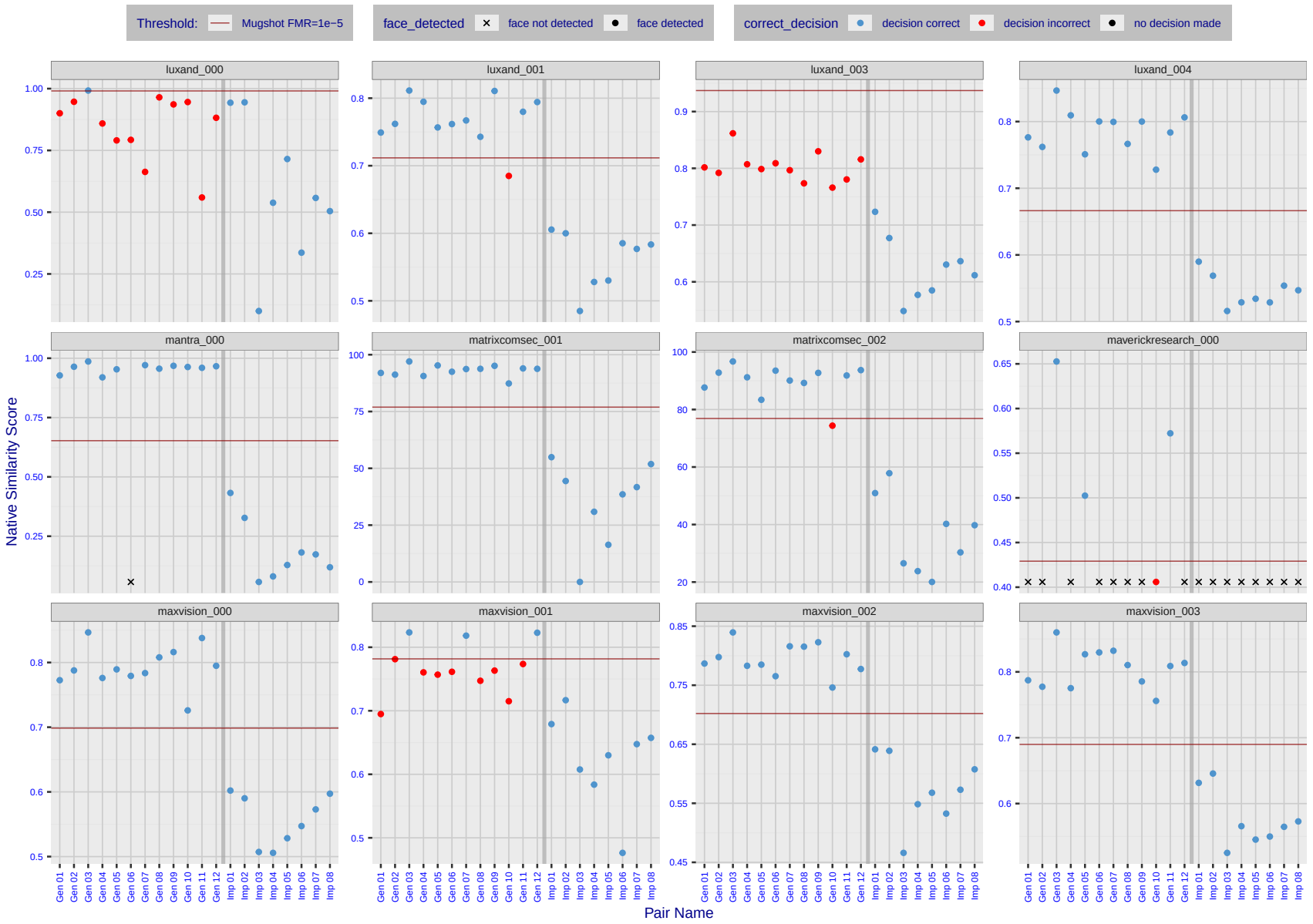


Figure 59: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

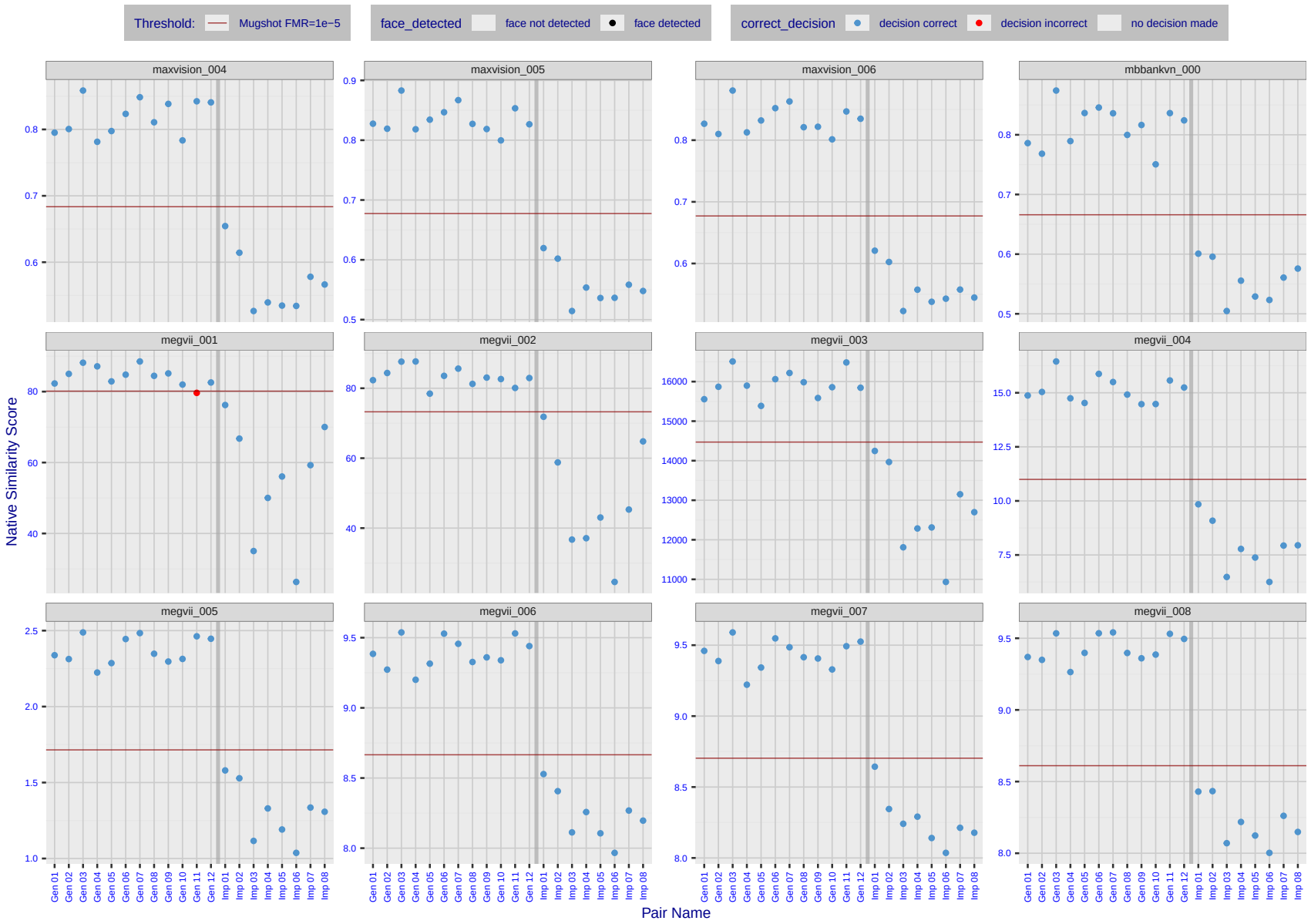


Figure 60: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

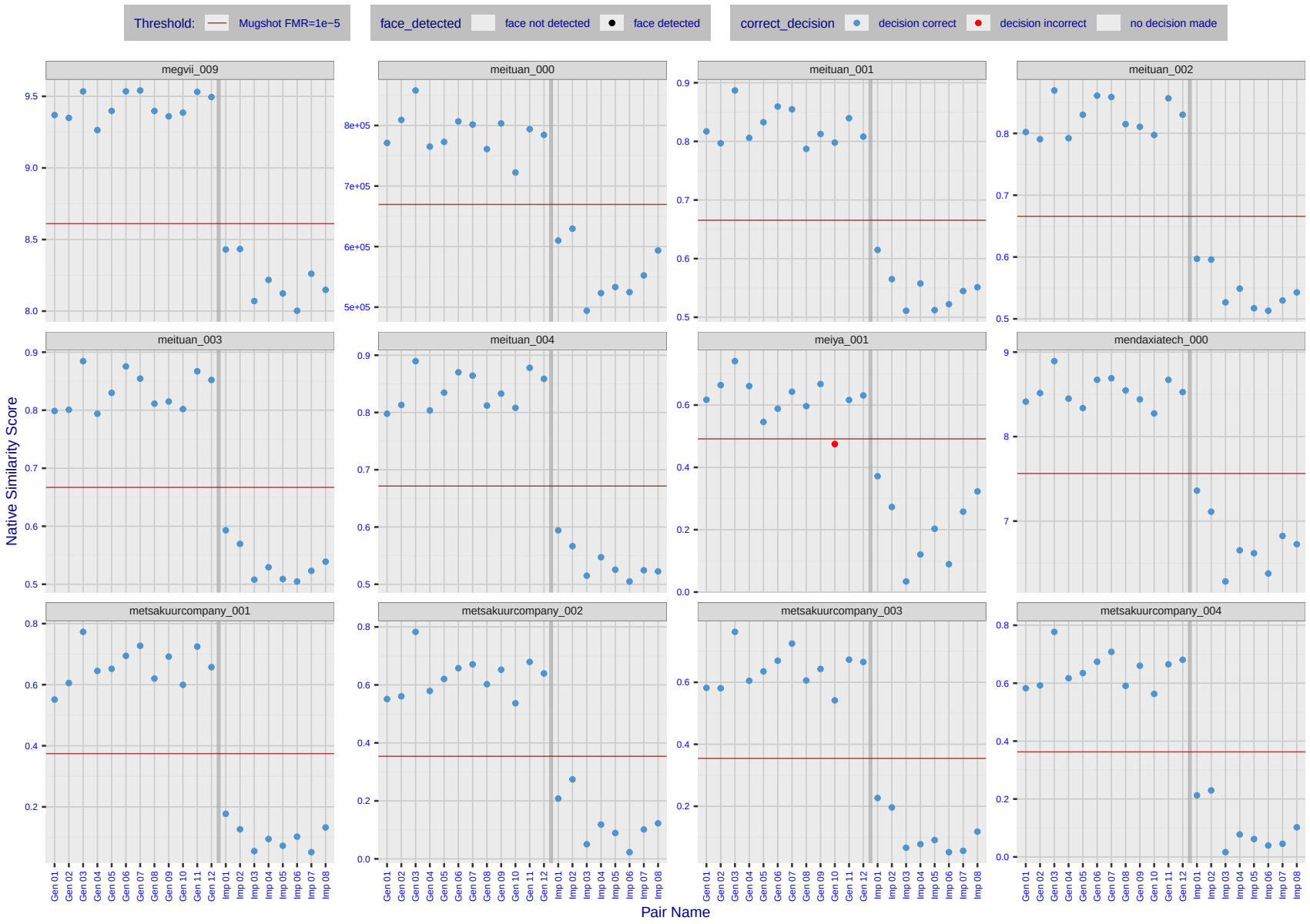


Figure 61: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

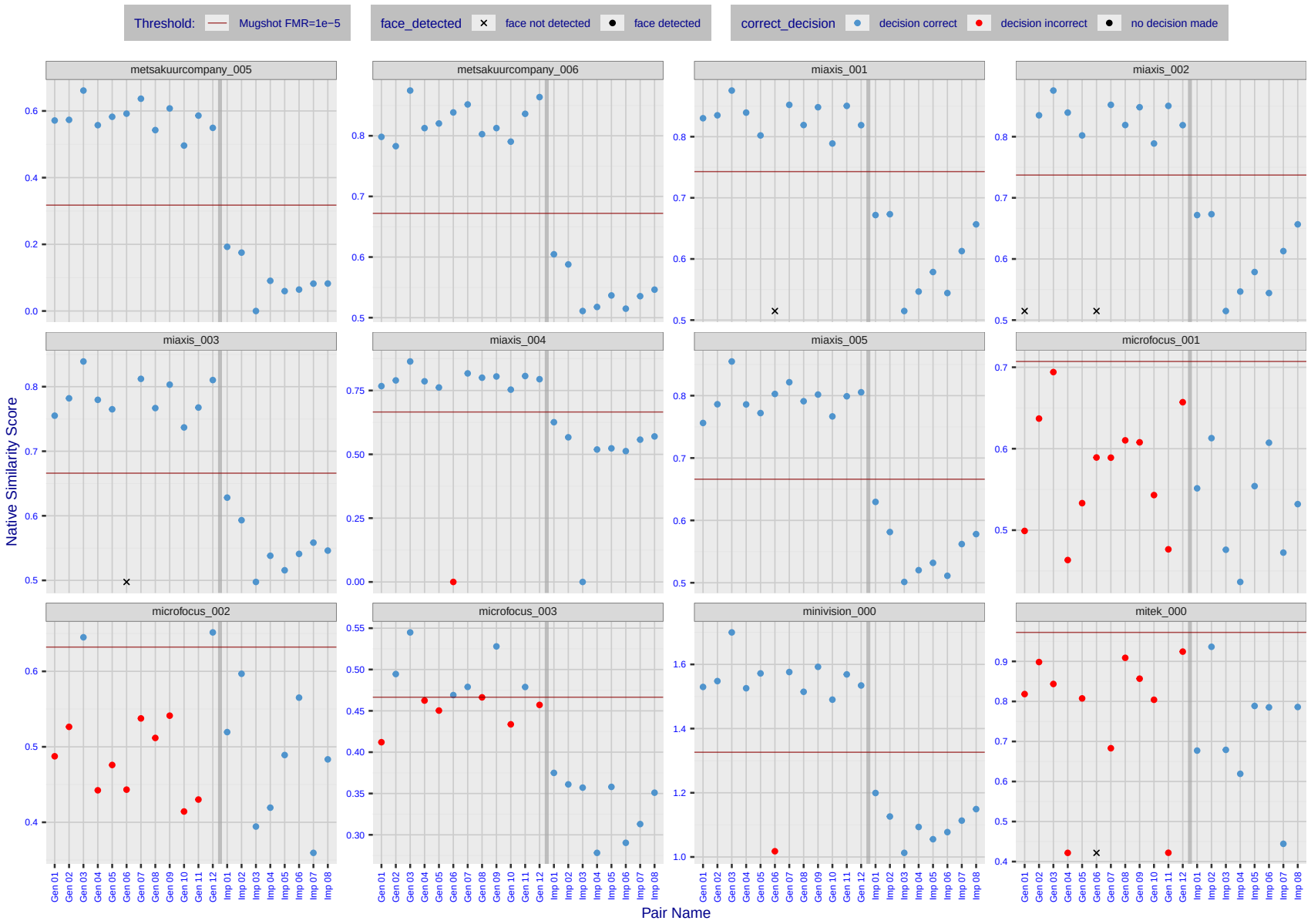


Figure 62: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

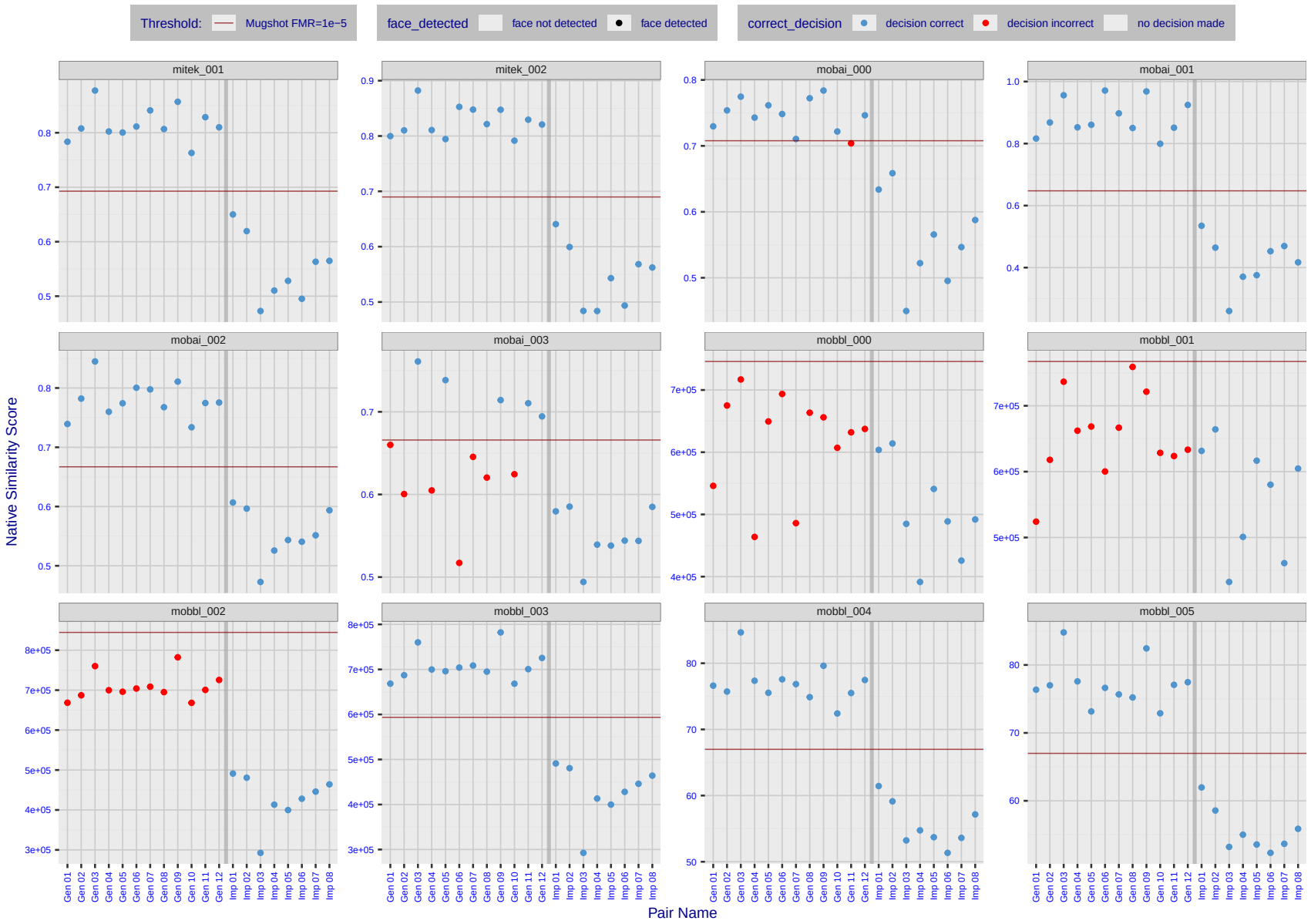


Figure 63: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

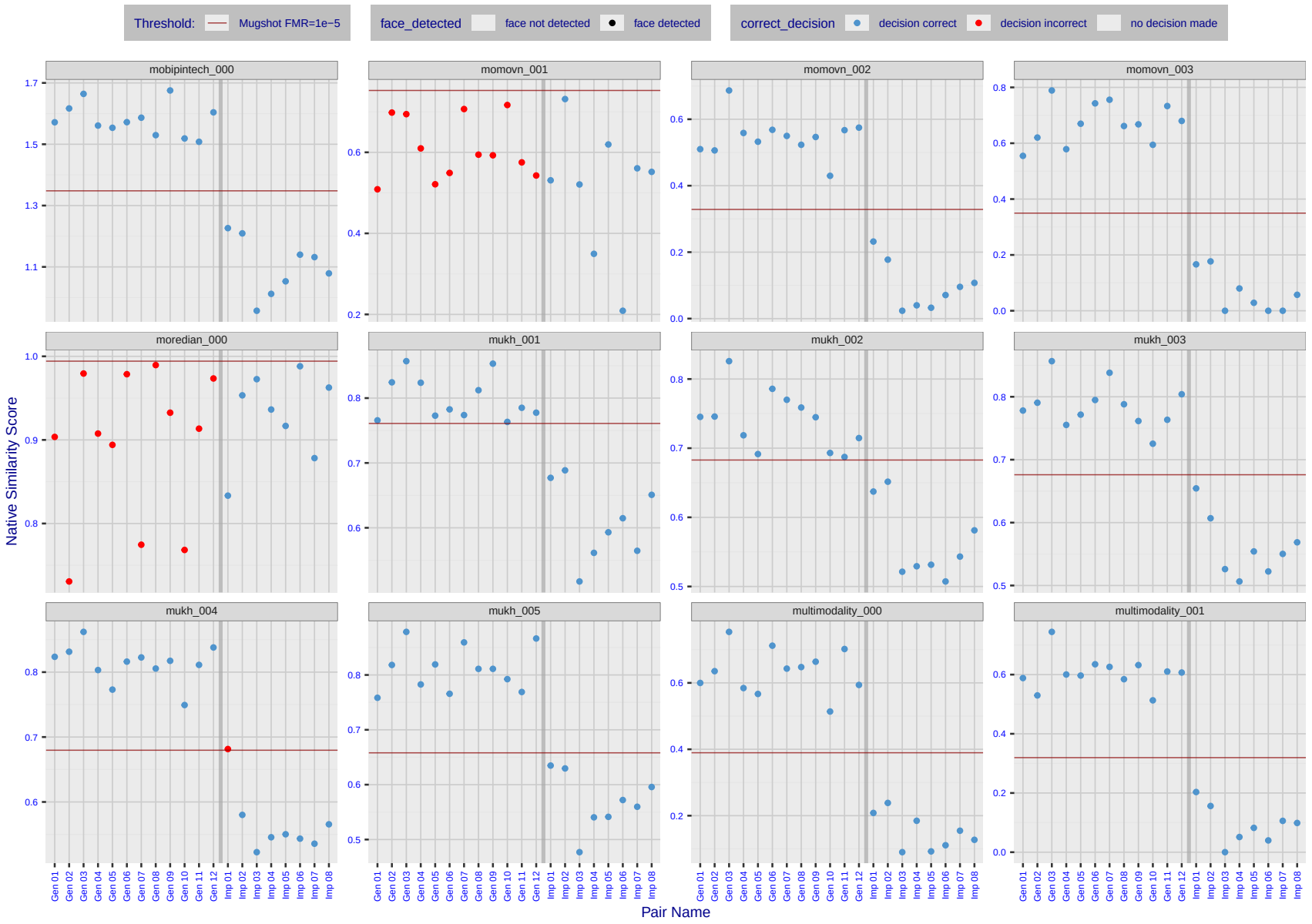


Figure 64: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

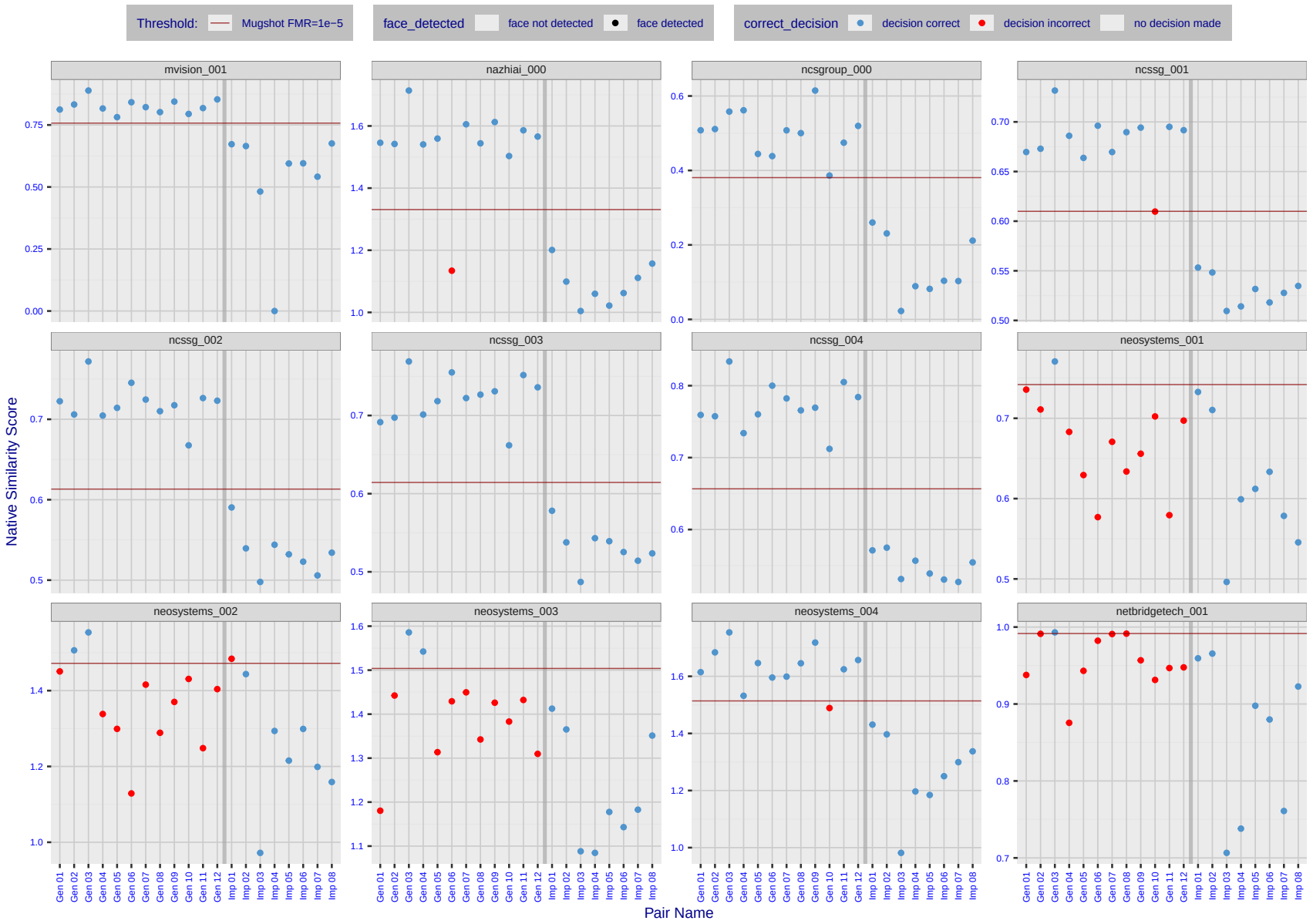


Figure 65: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

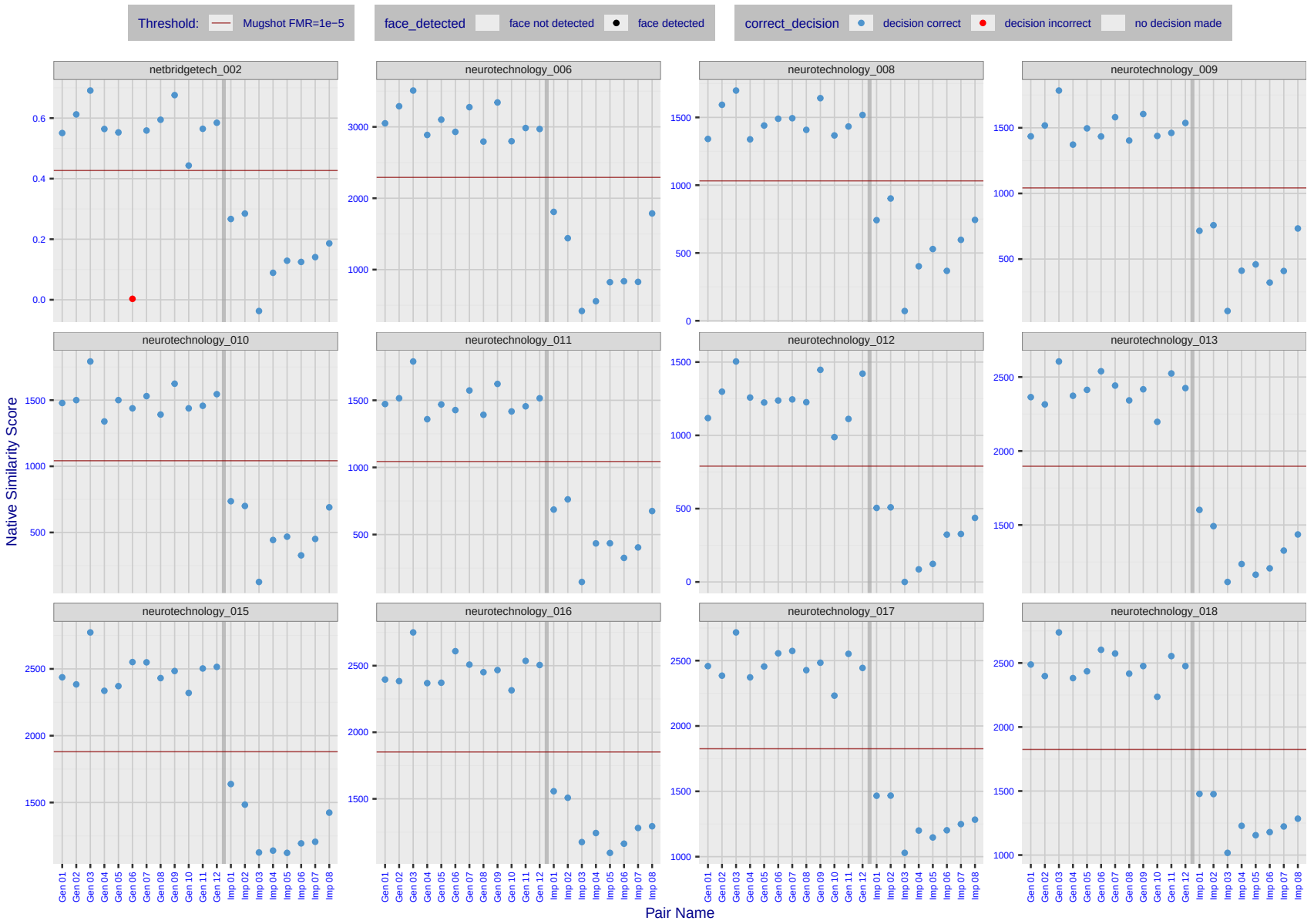


Figure 66: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

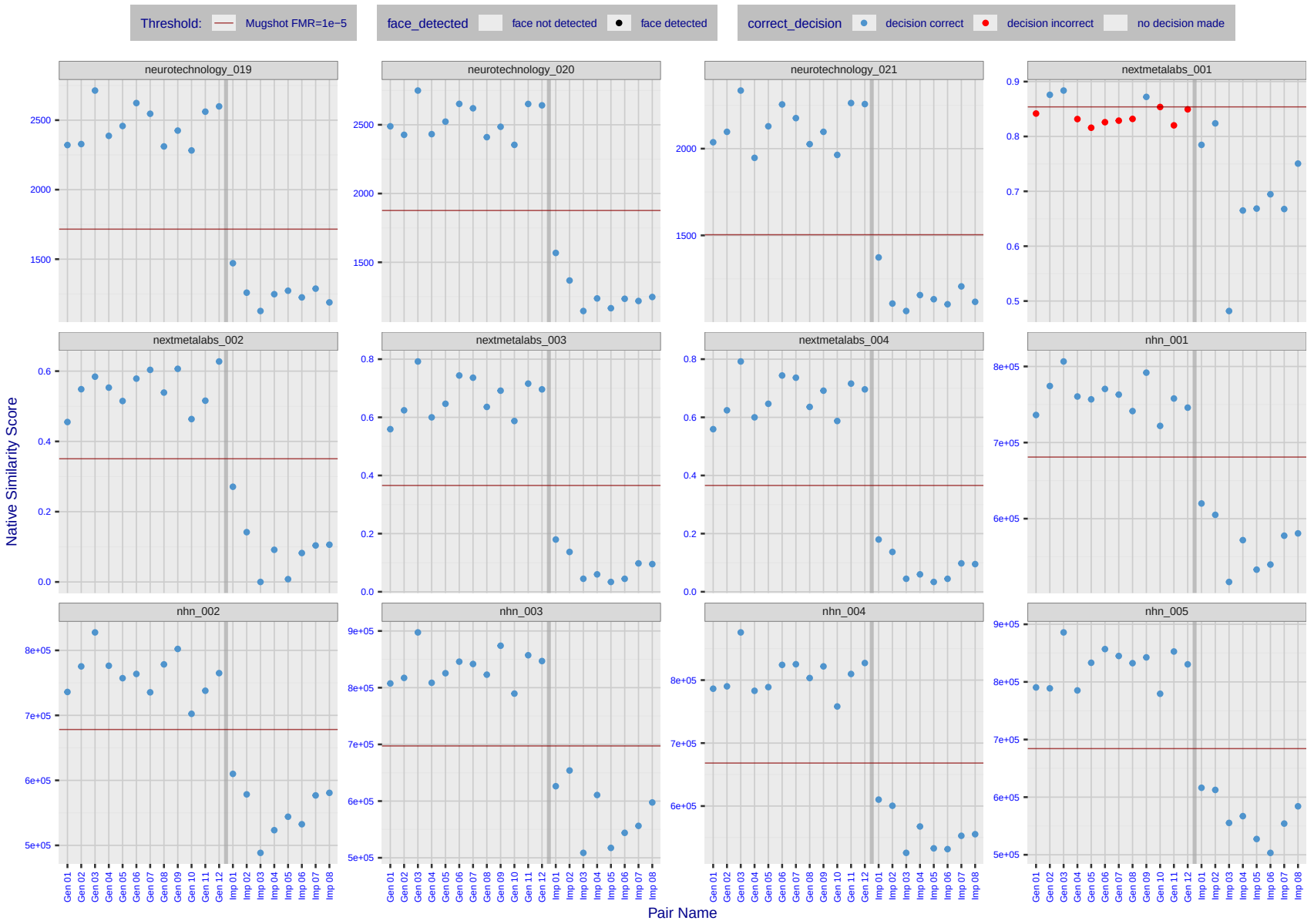


Figure 67: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

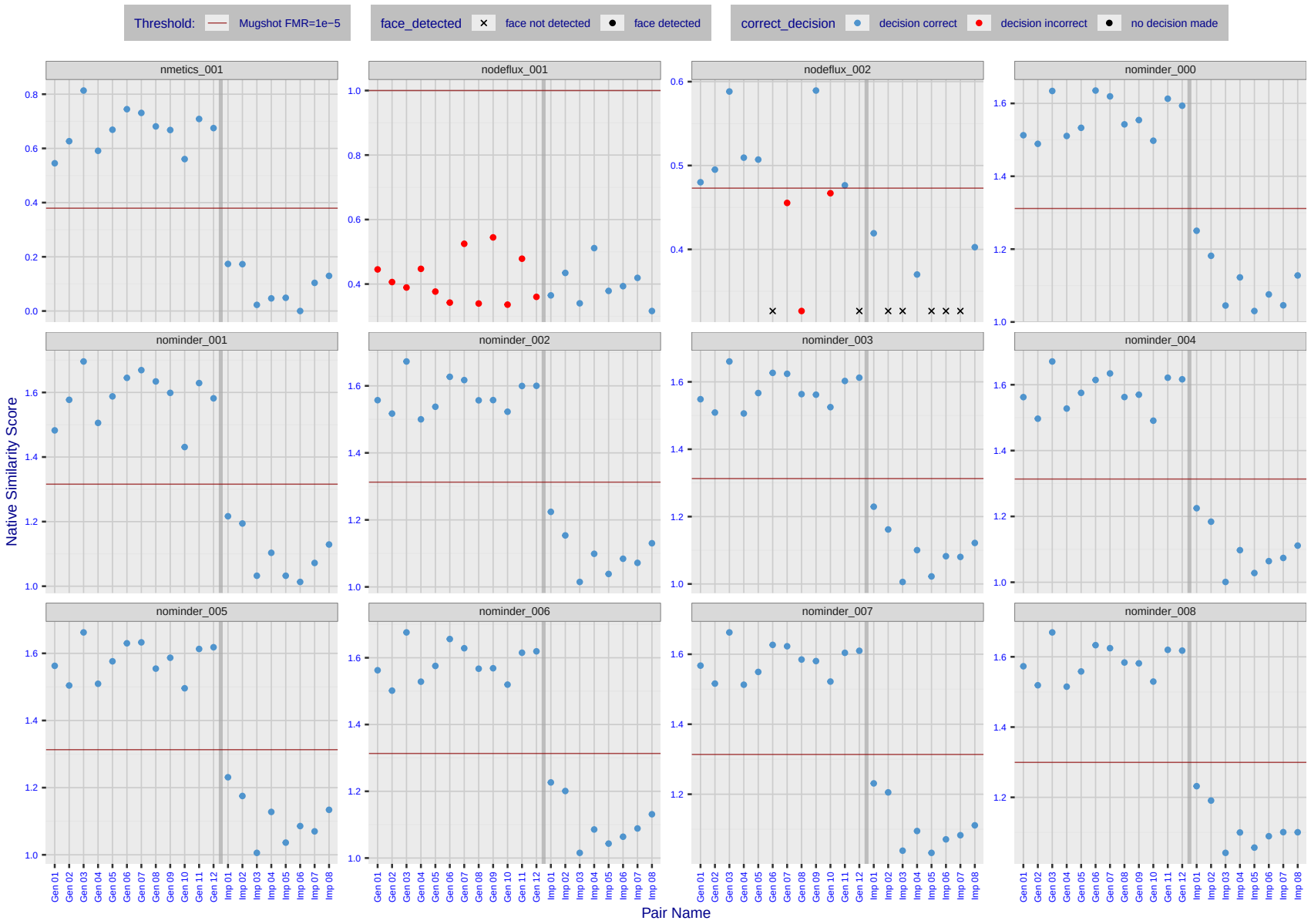


Figure 68: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

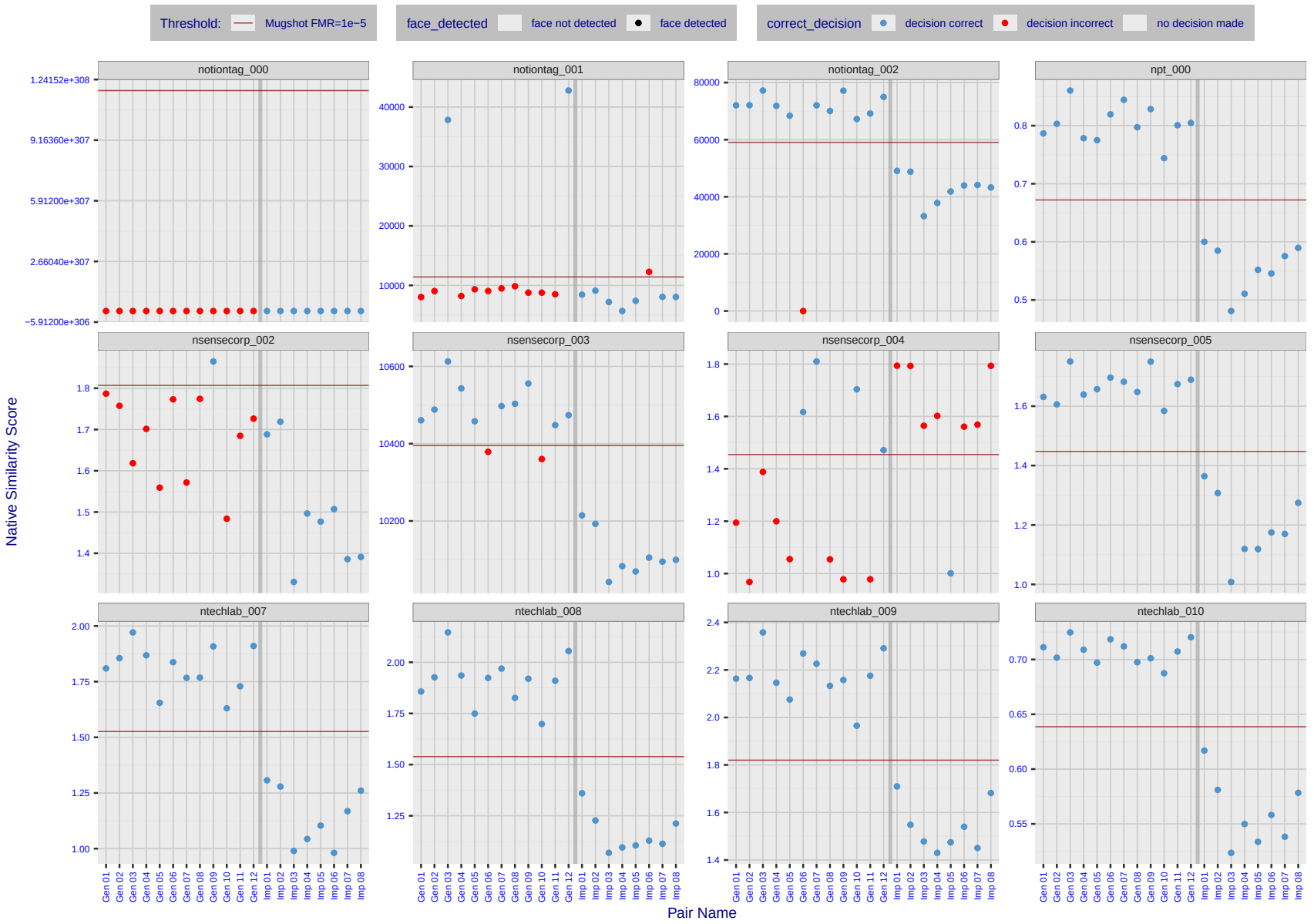


Figure 69: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

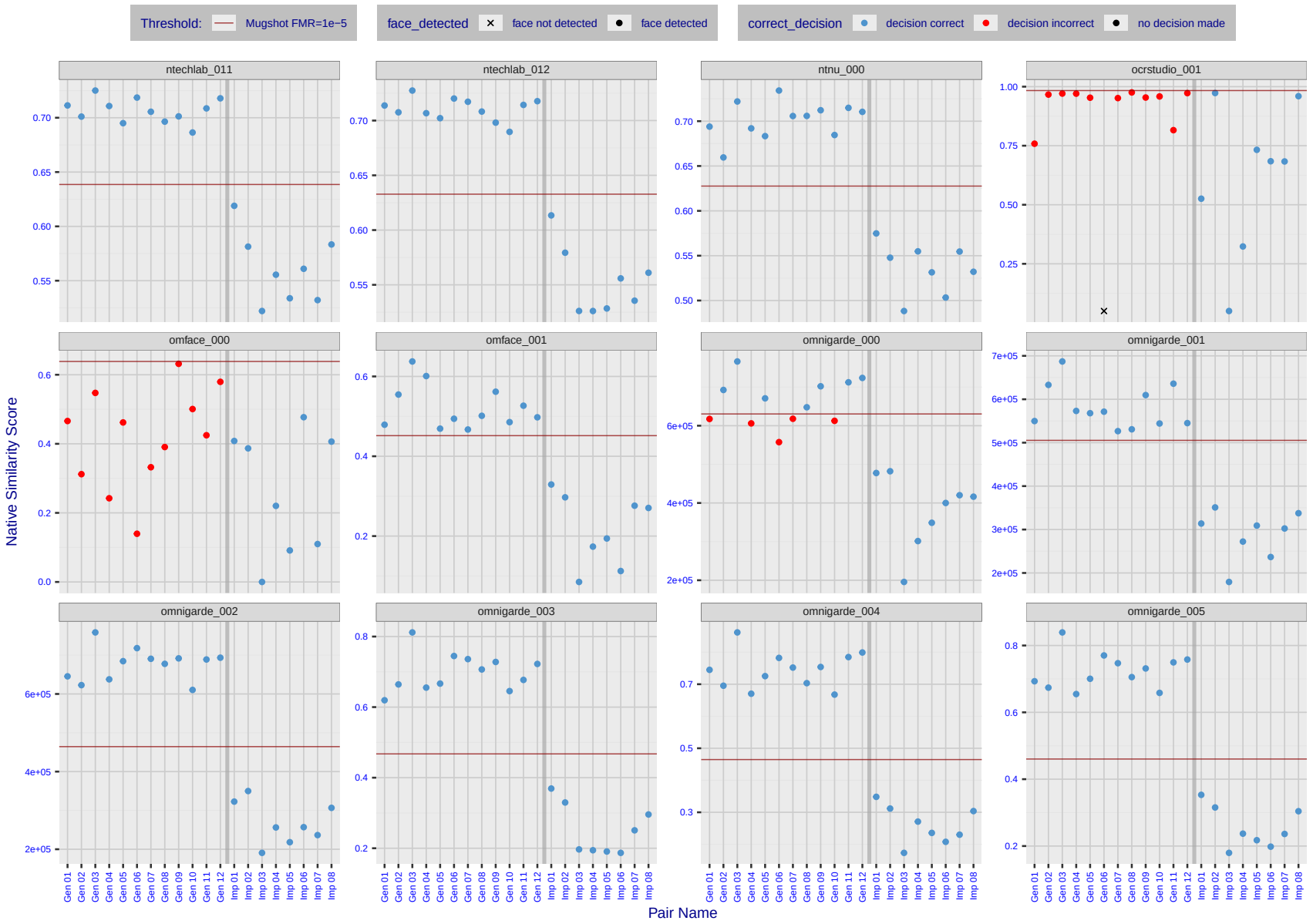


Figure 70: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

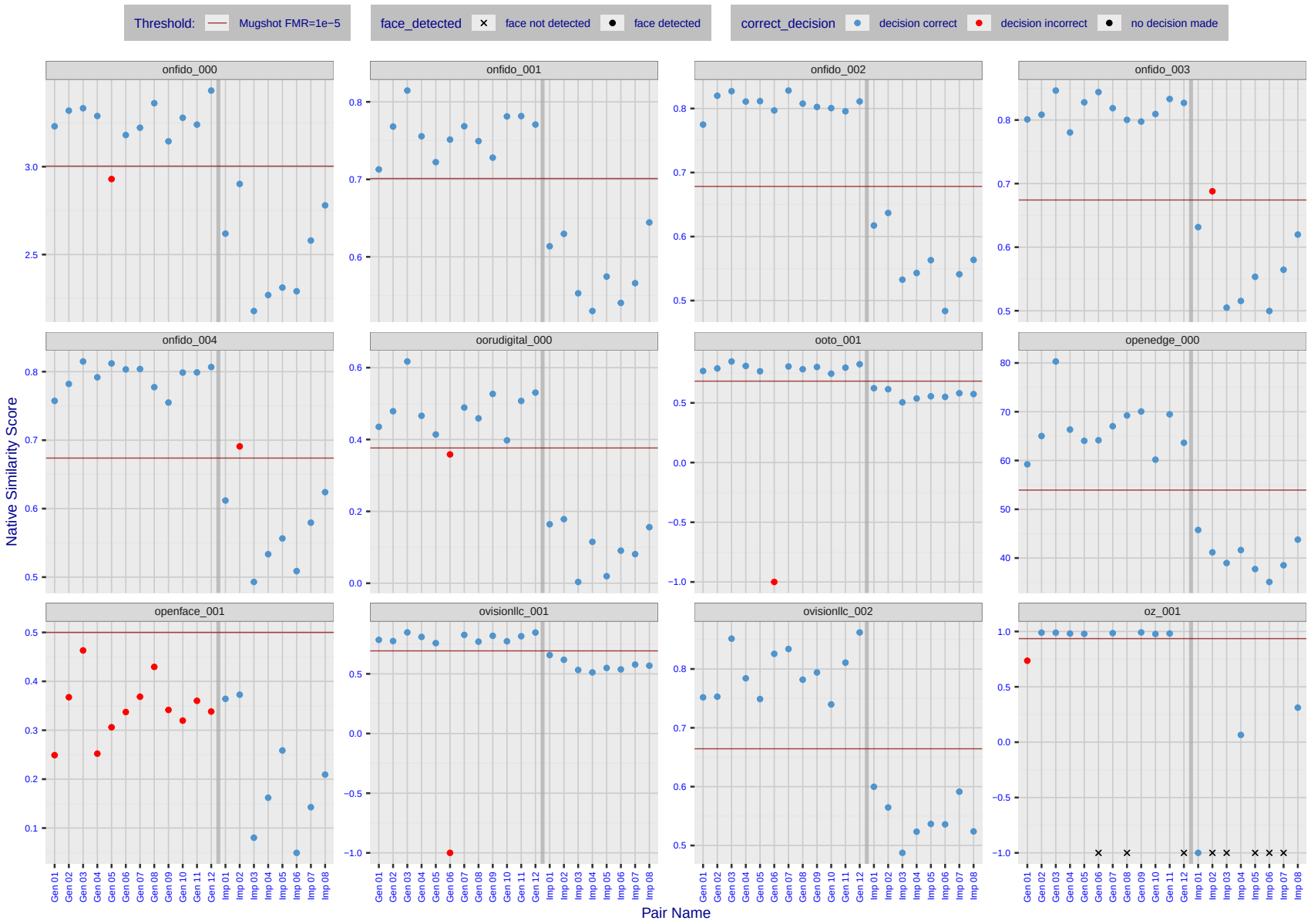


Figure 71: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

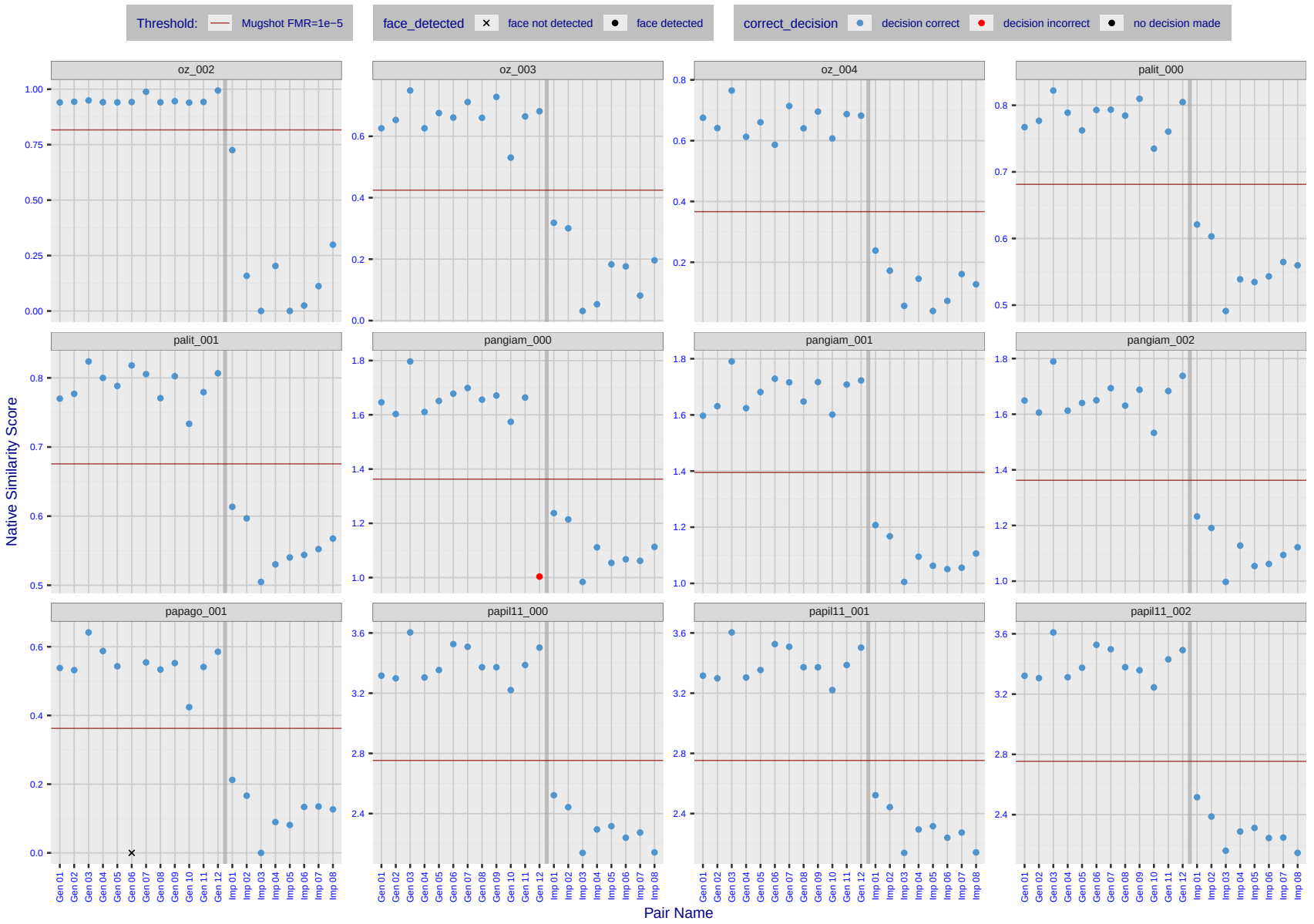


Figure 72: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

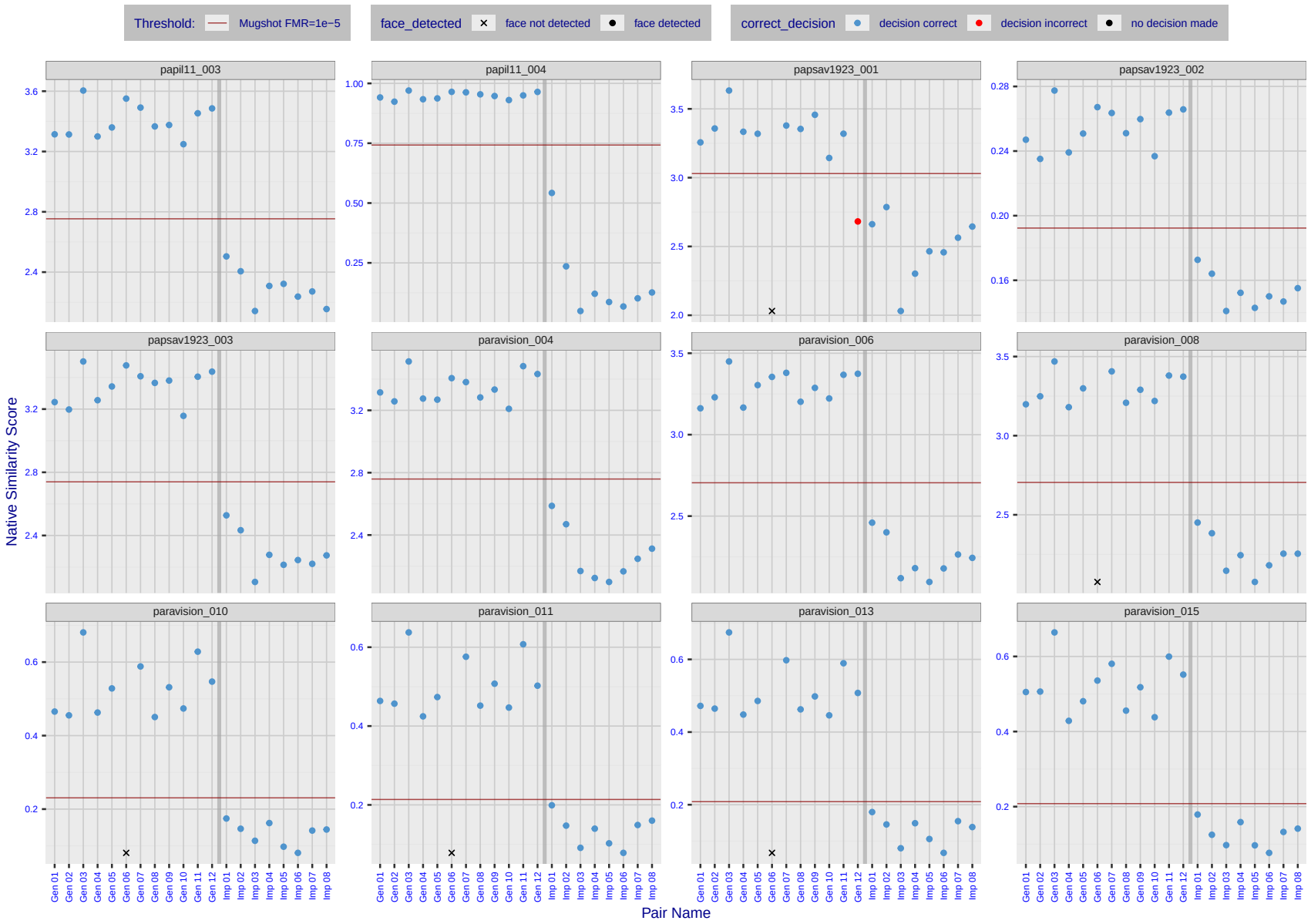


Figure 73: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

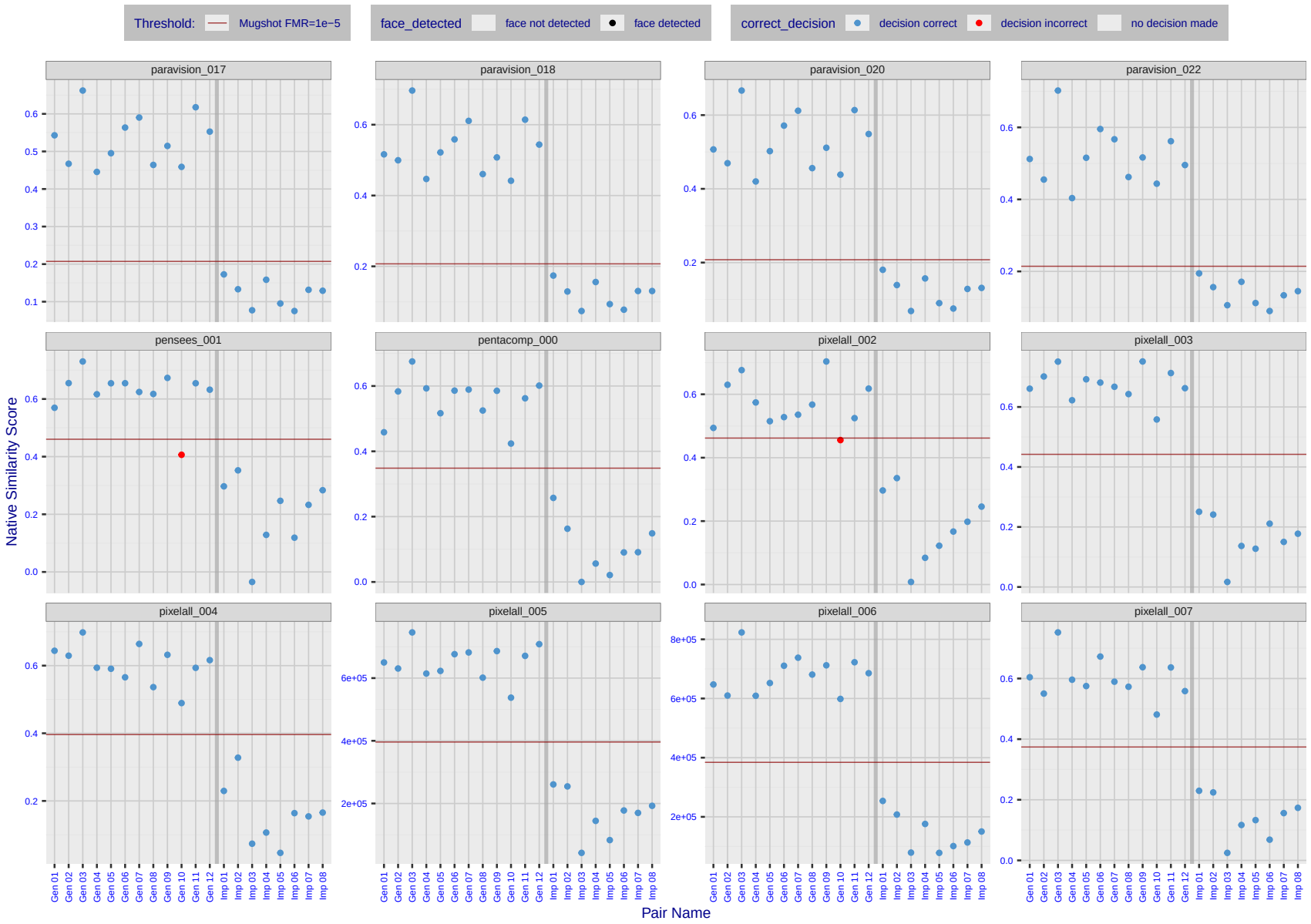


Figure 74: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

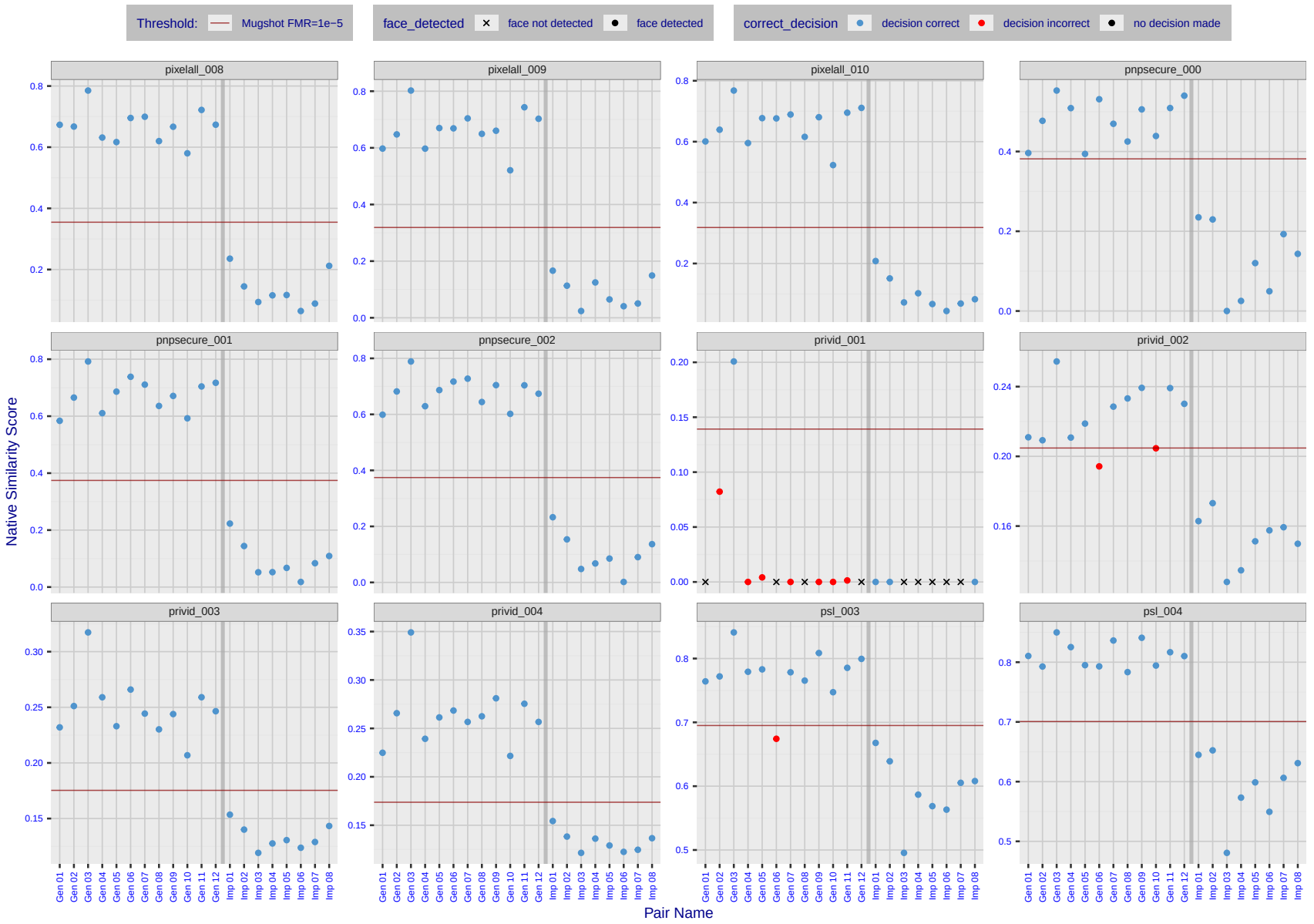


Figure 75: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

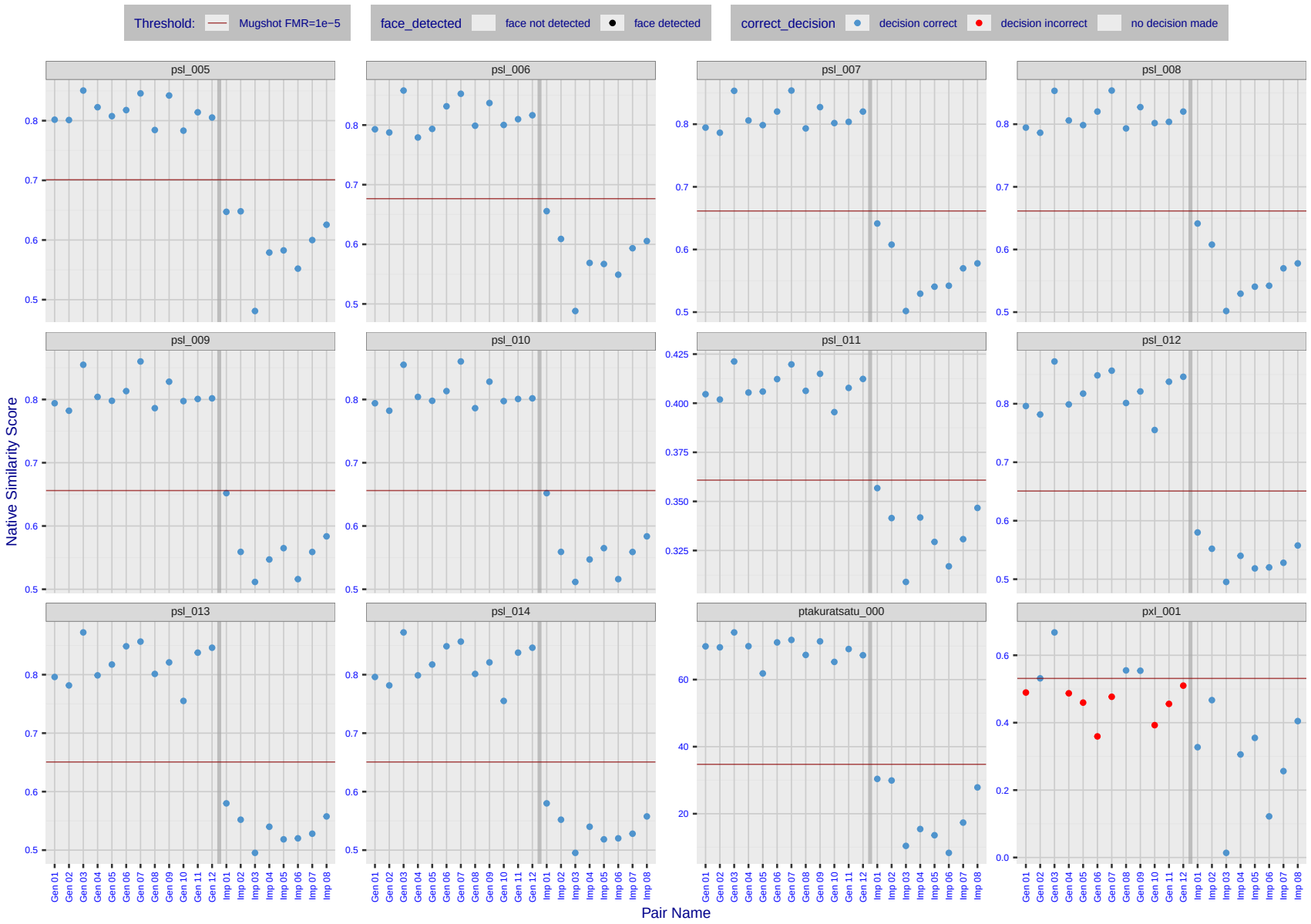


Figure 76: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

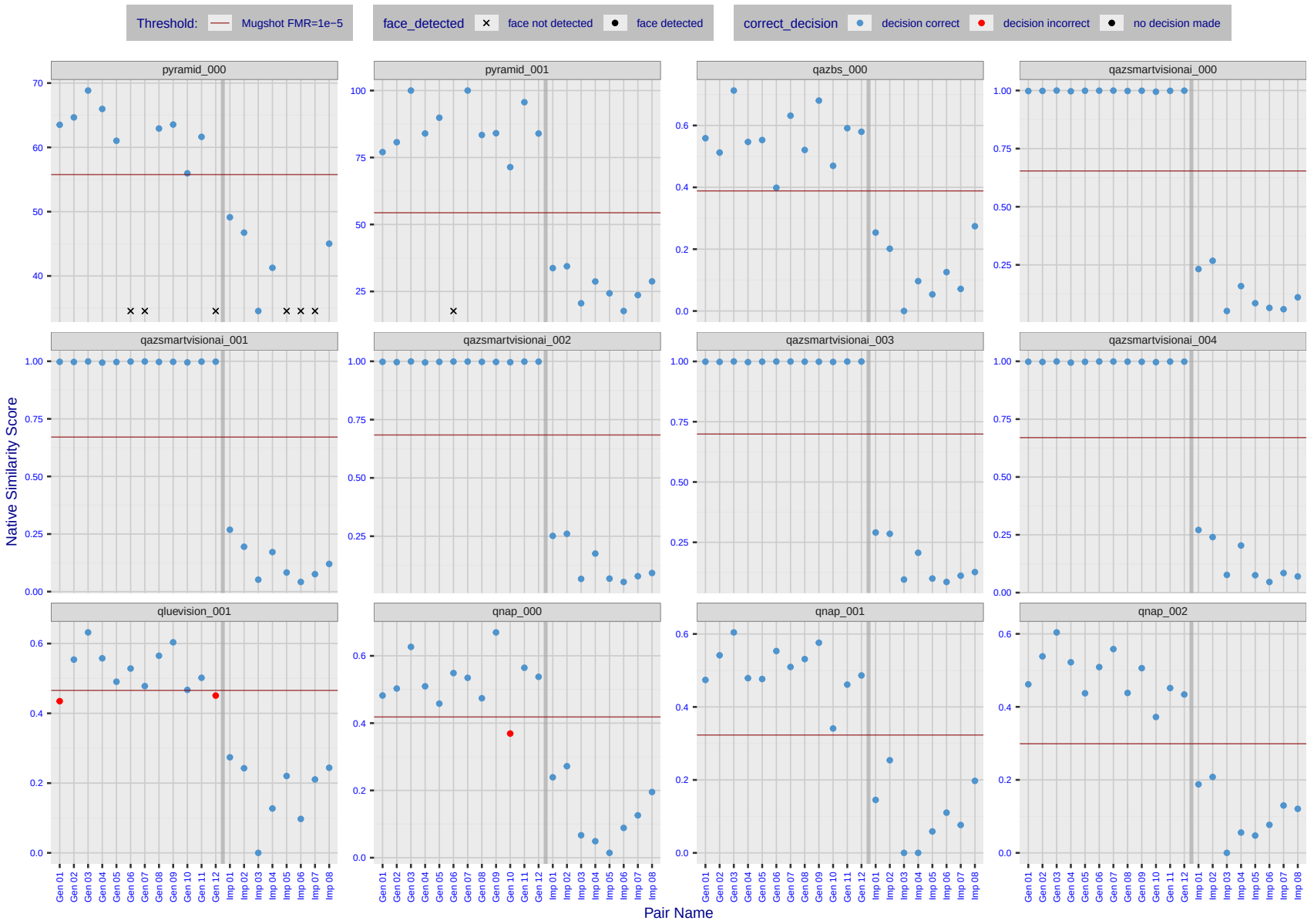


Figure 77: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

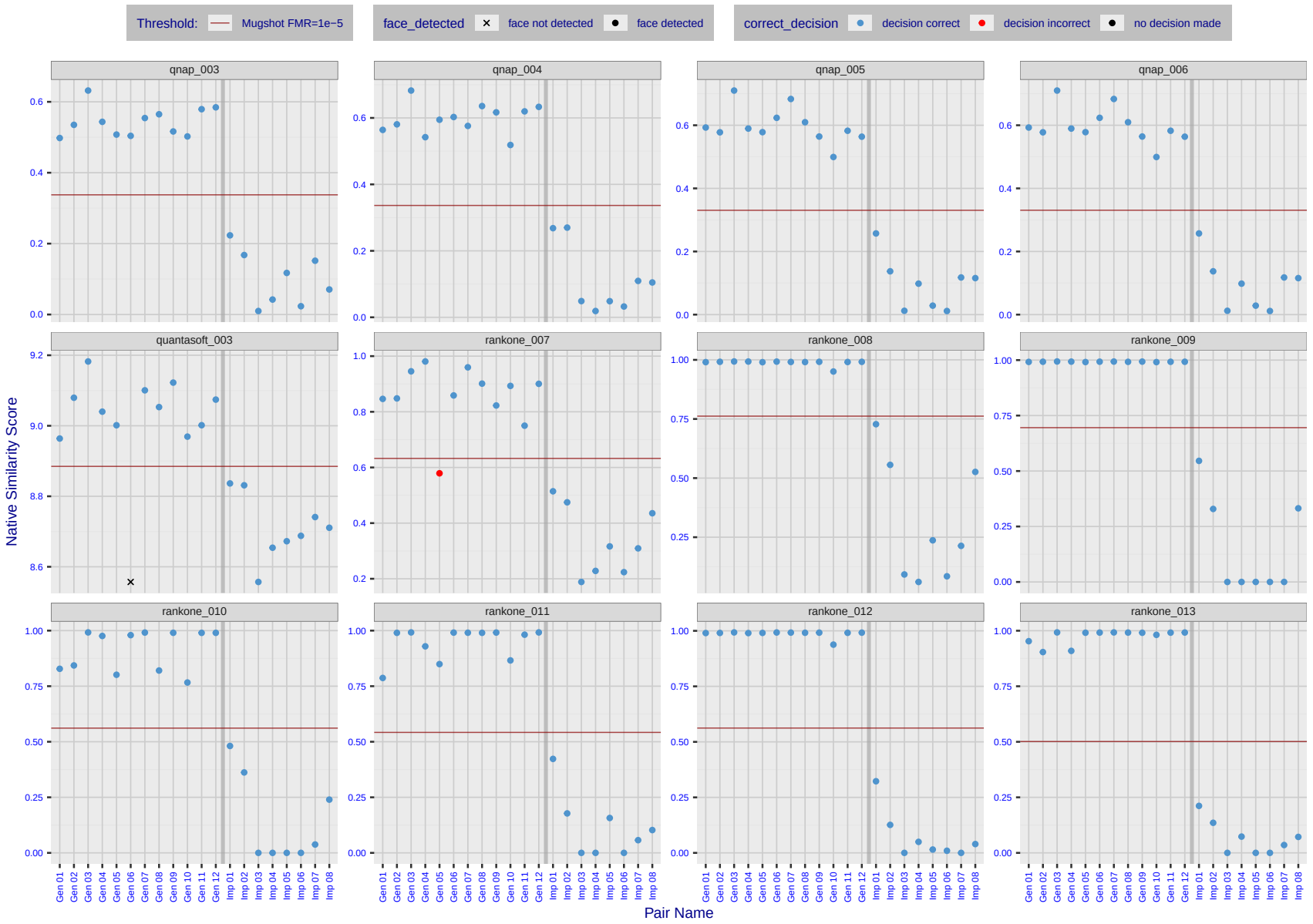


Figure 78: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

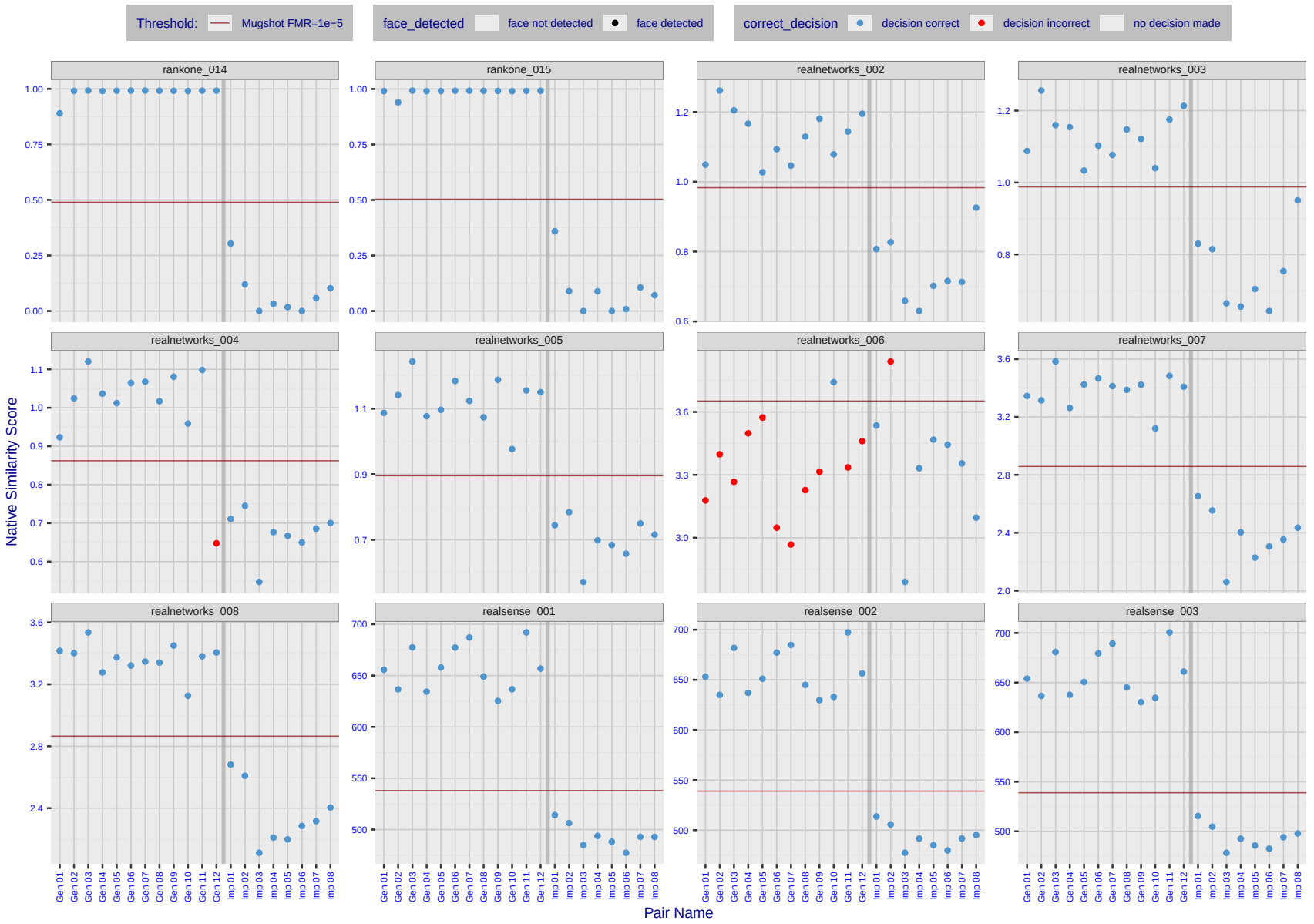


Figure 79: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

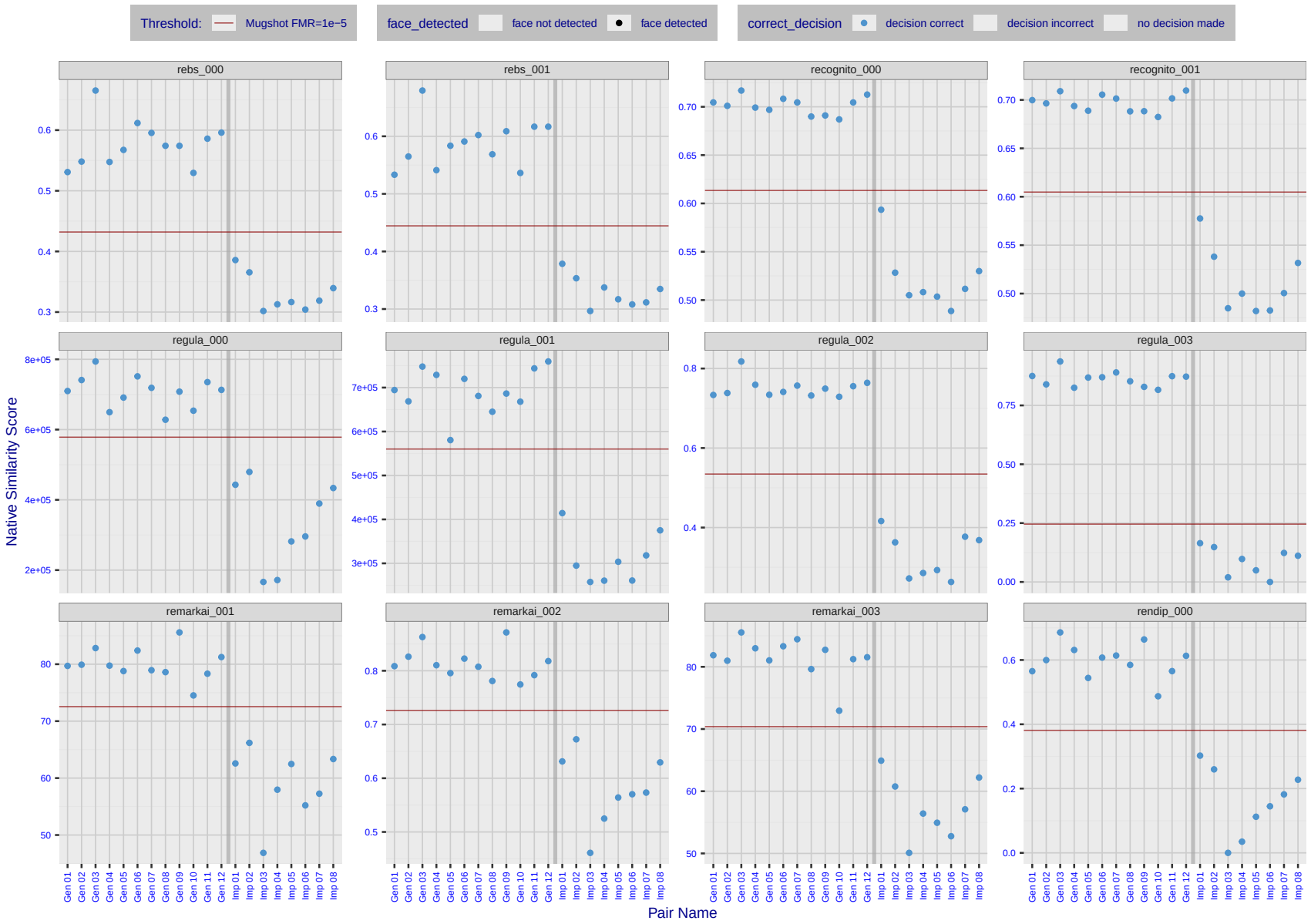


Figure 80: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

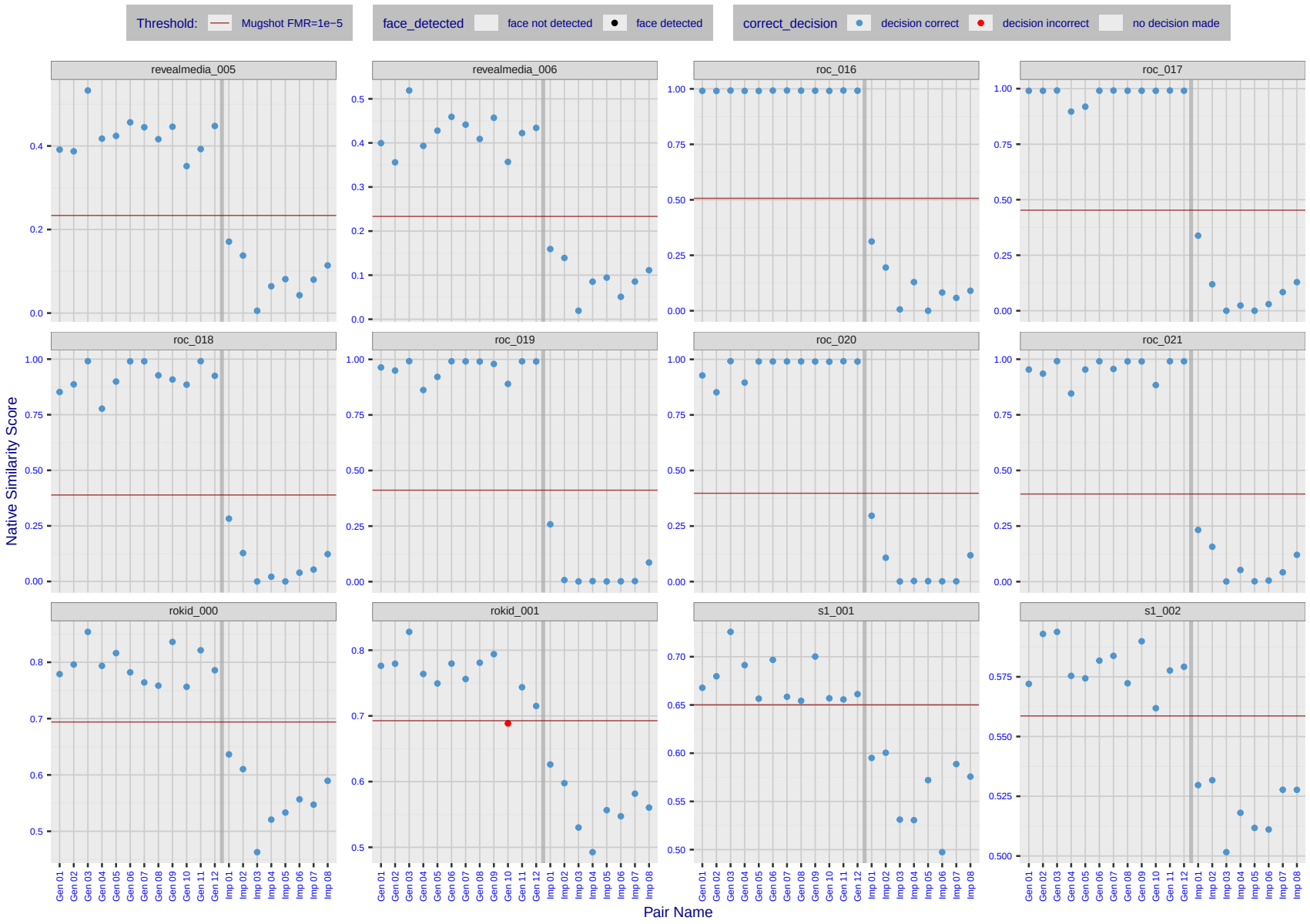


Figure 81: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

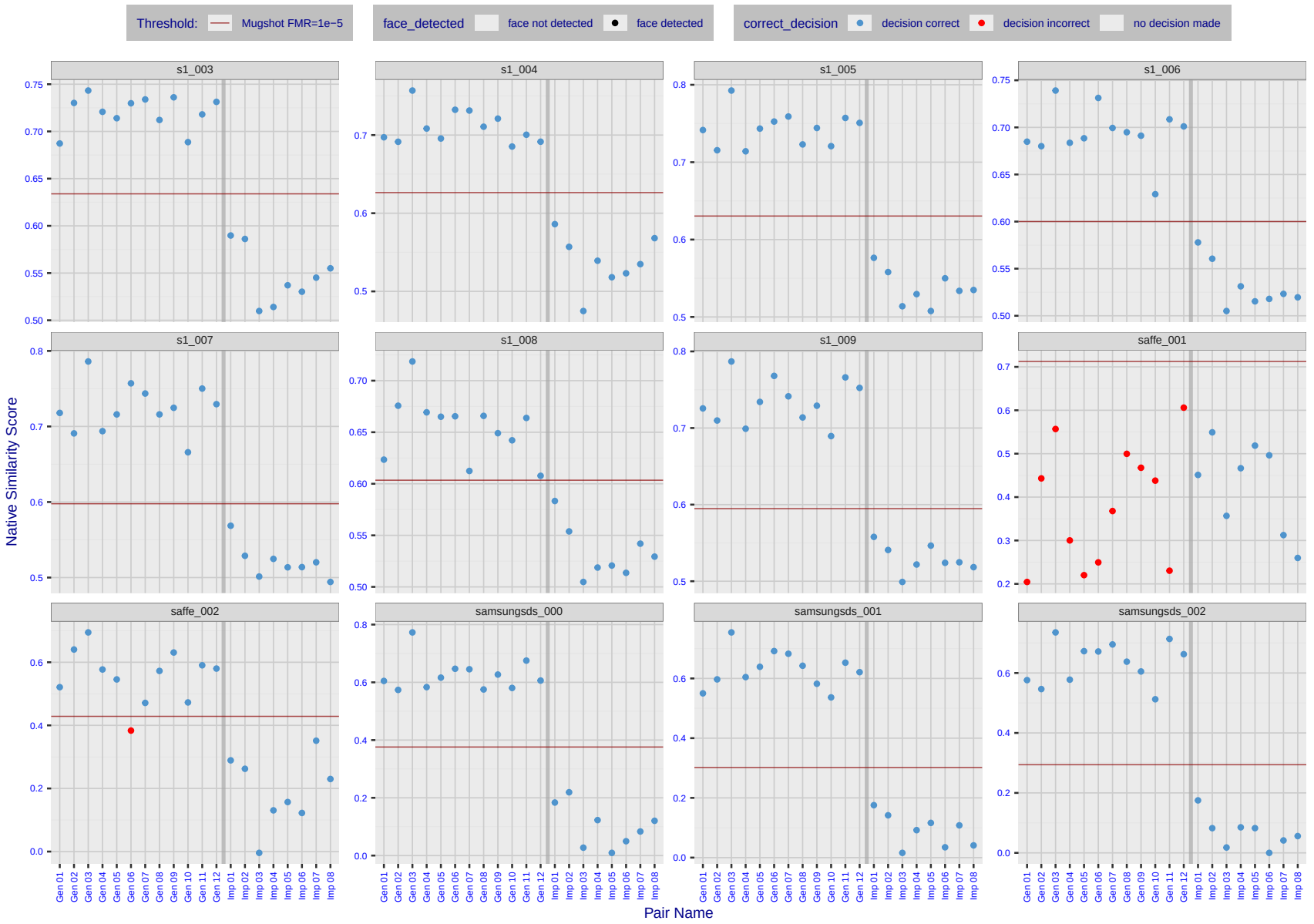


Figure 82: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

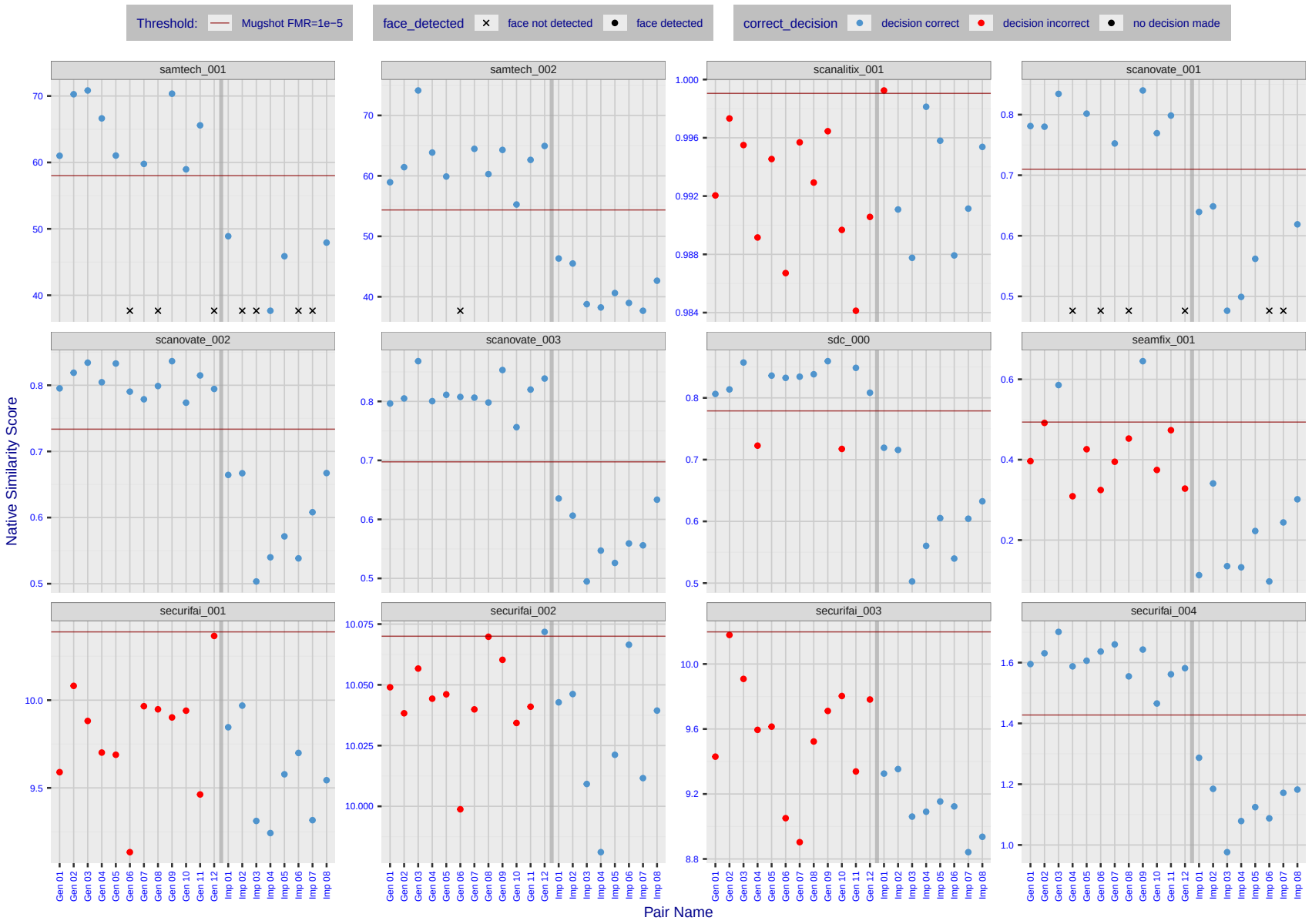


Figure 83: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

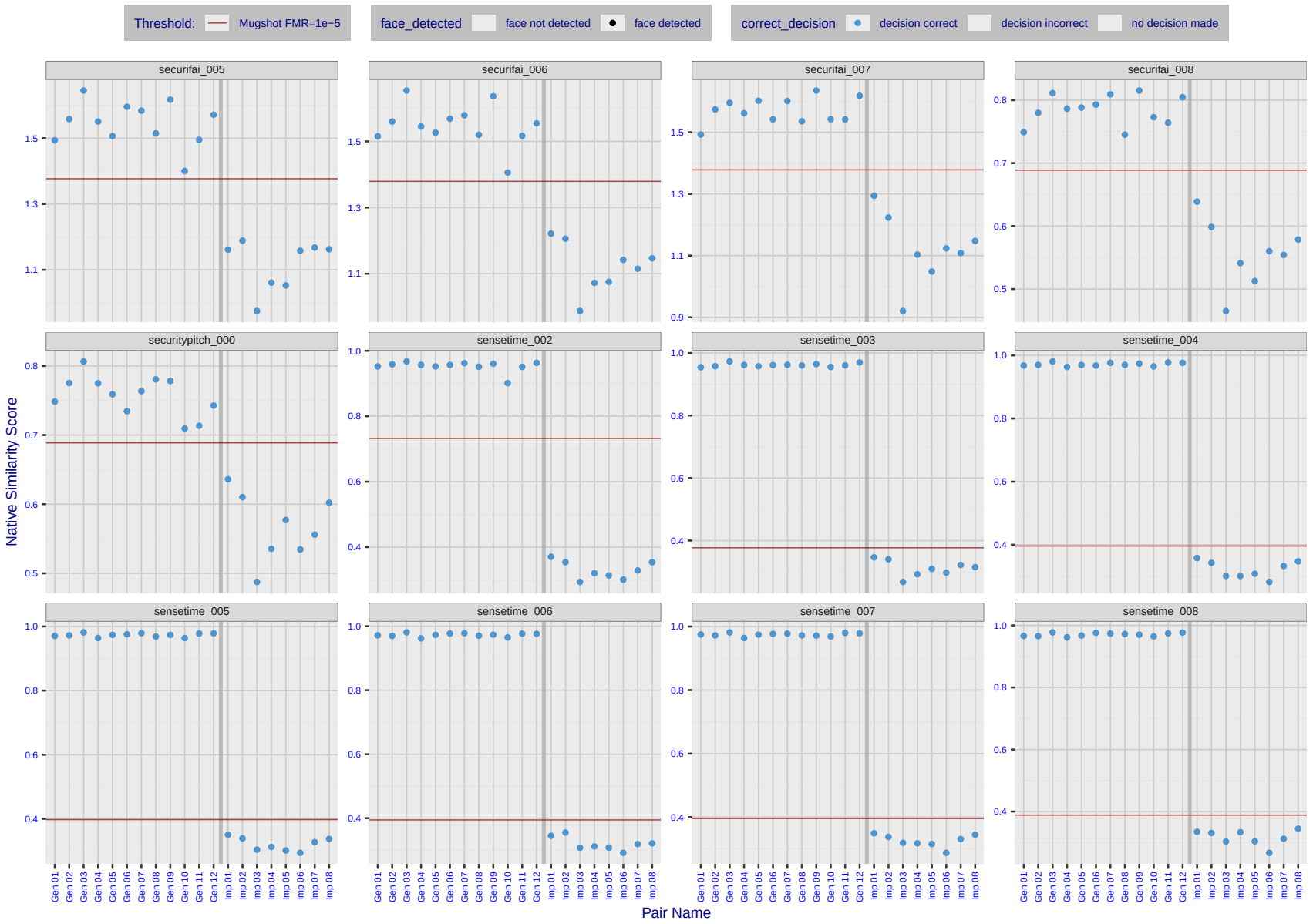


Figure 84: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

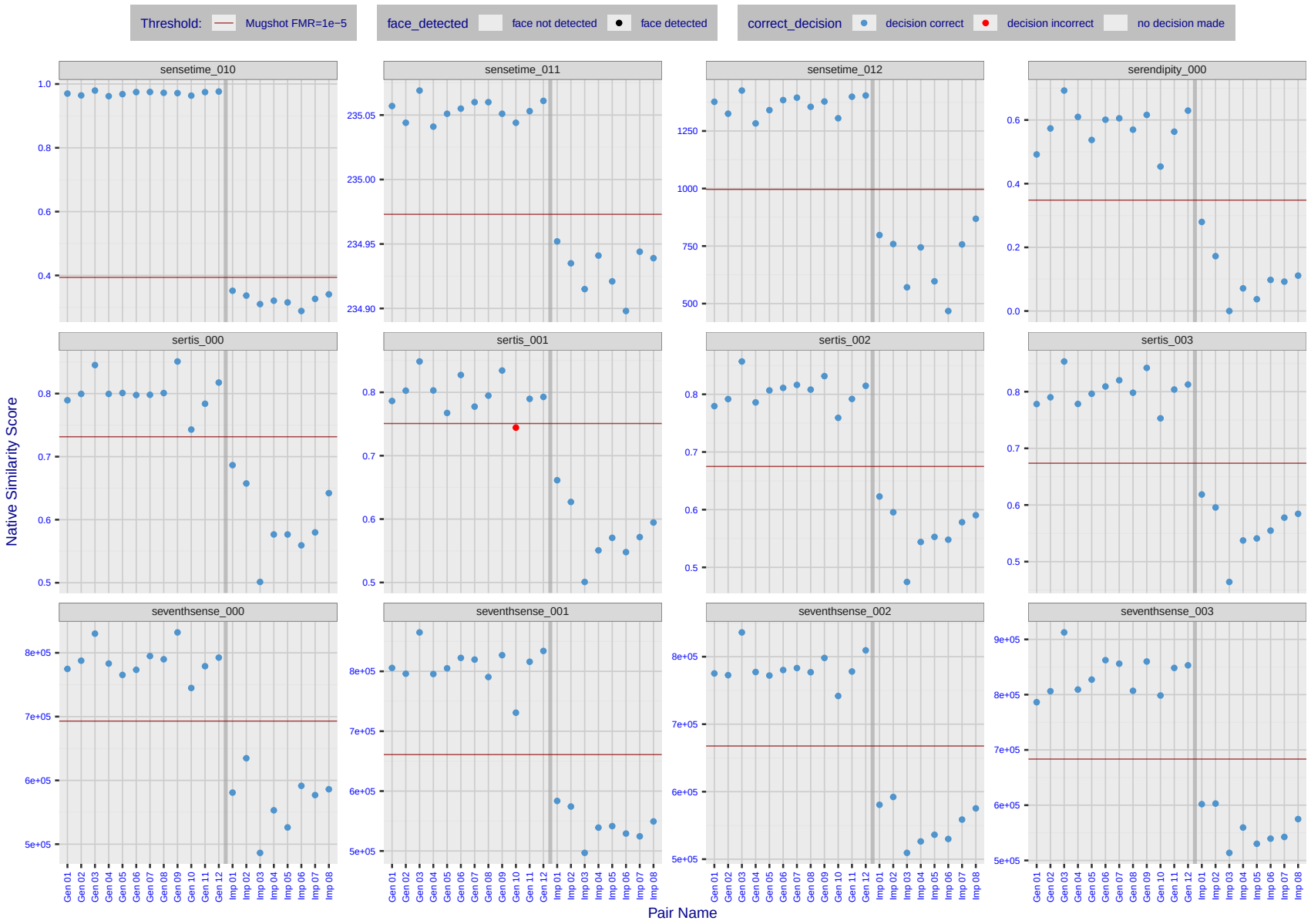


Figure 85: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

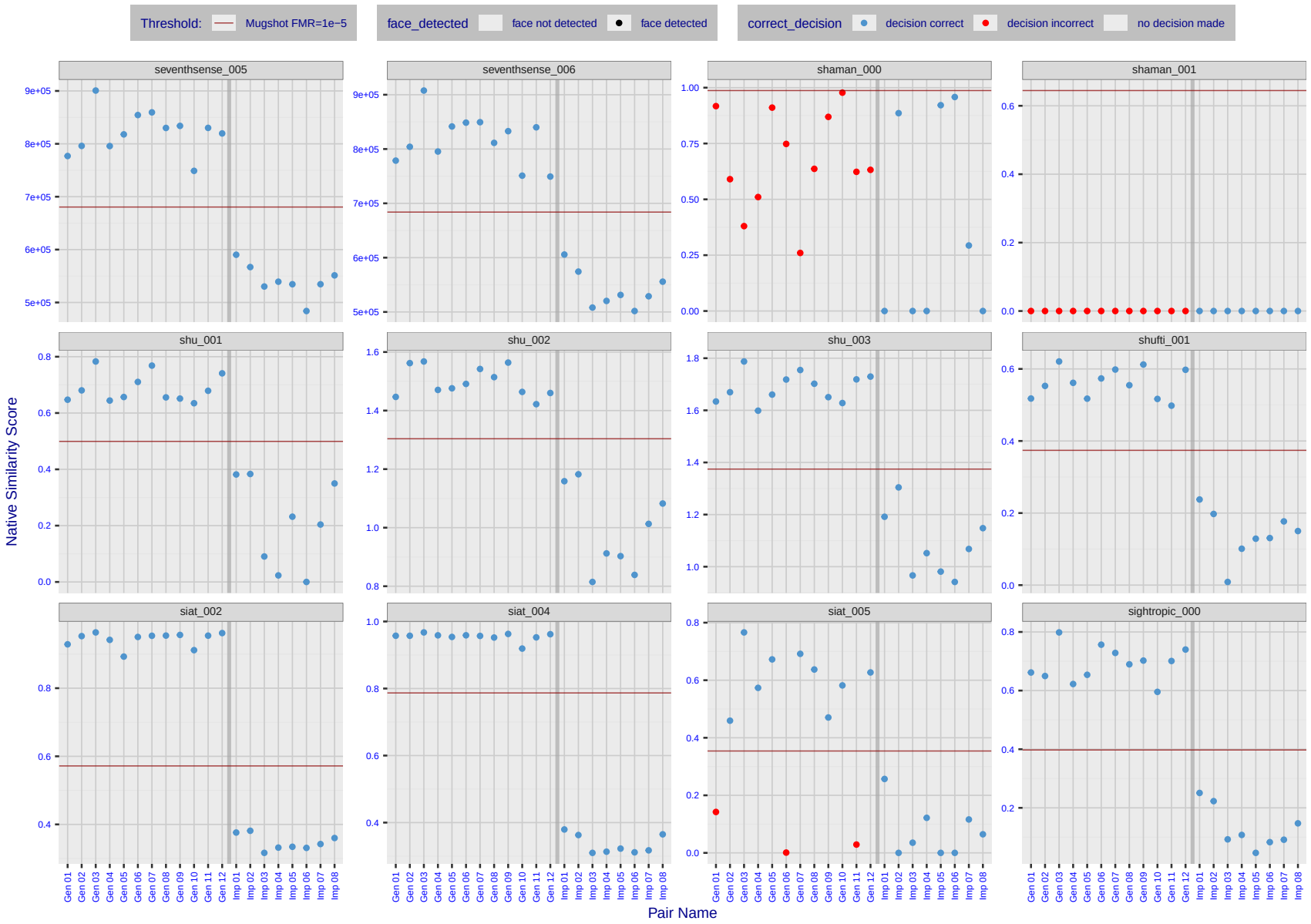


Figure 86: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

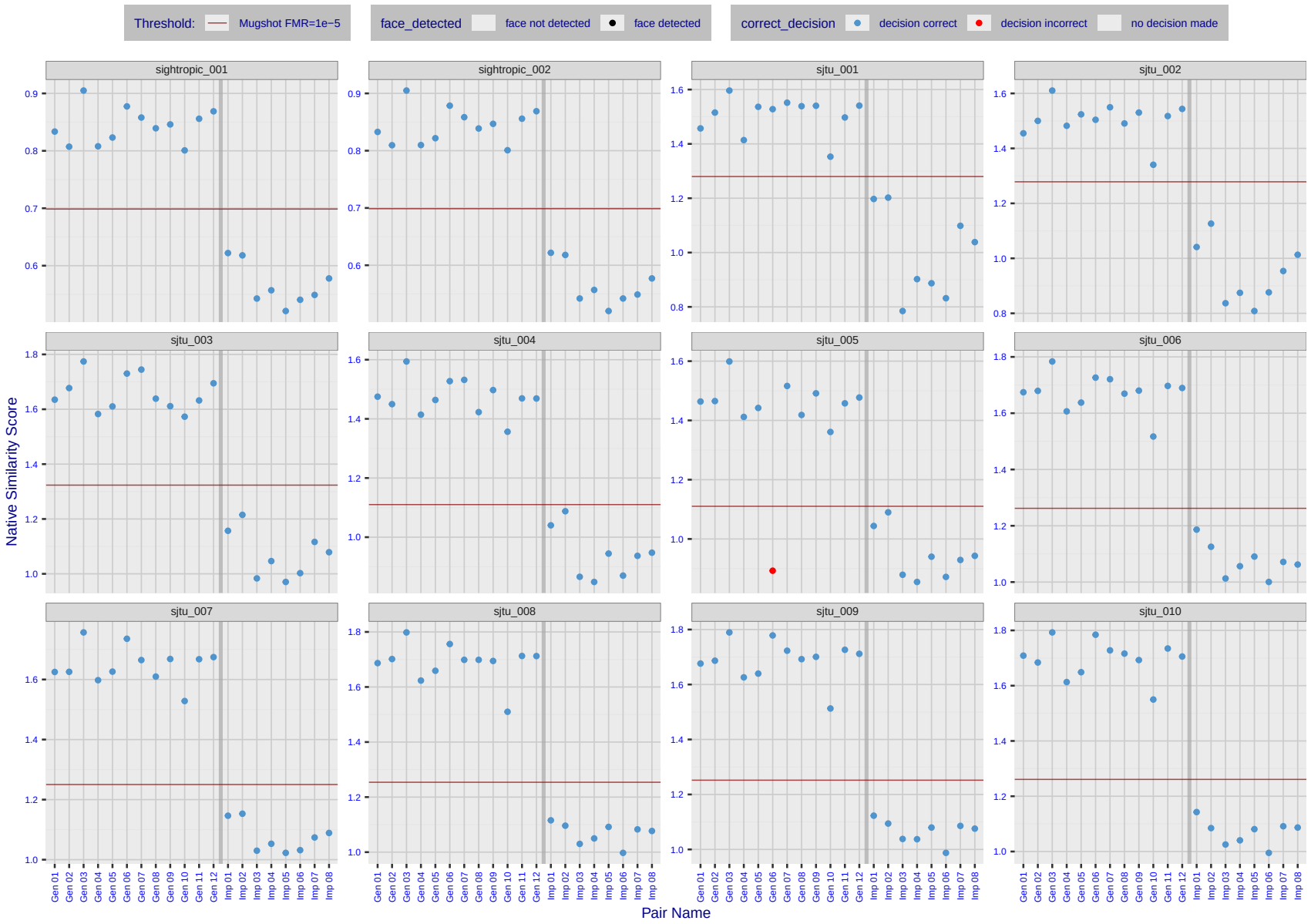


Figure 87: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

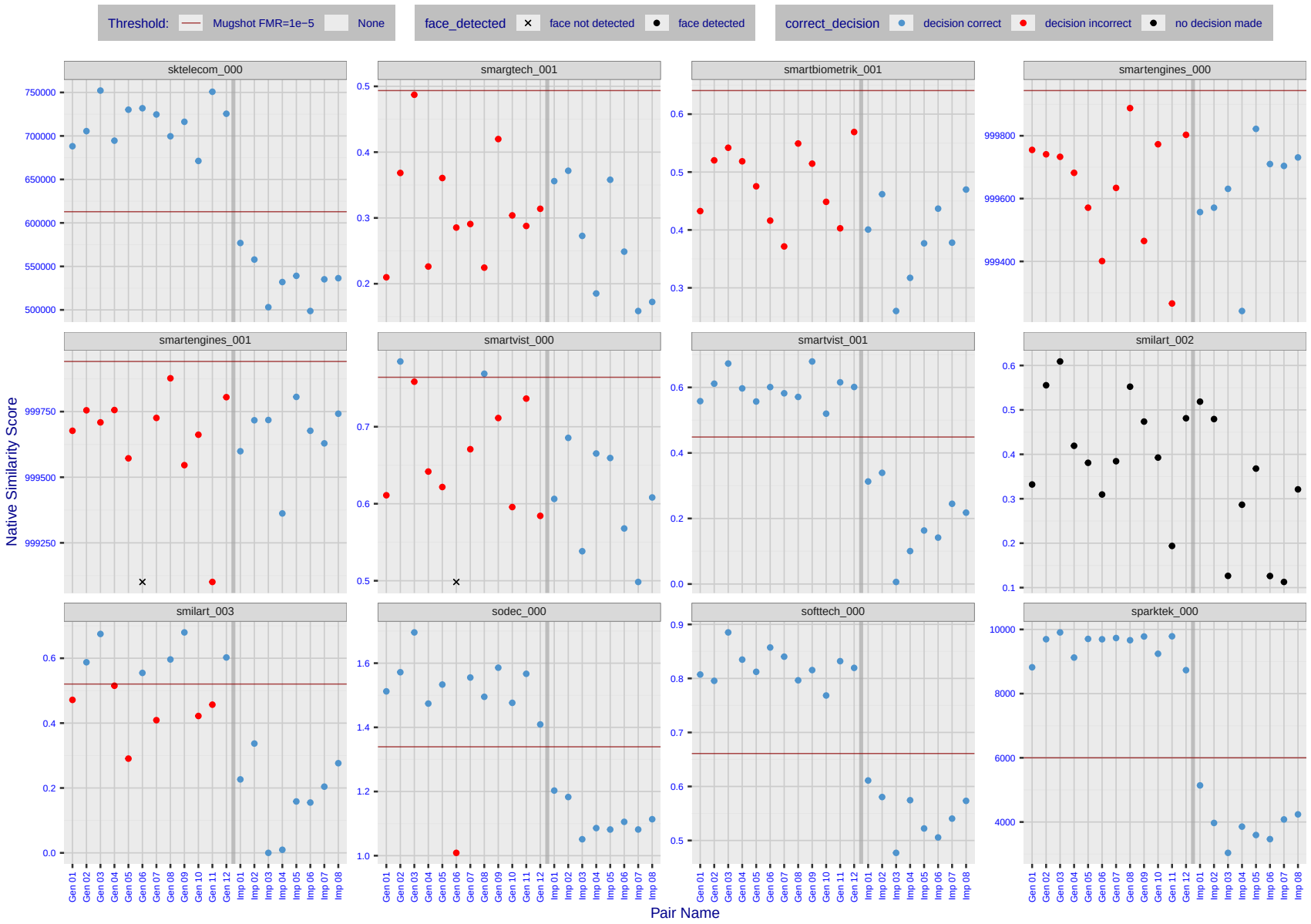


Figure 88: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

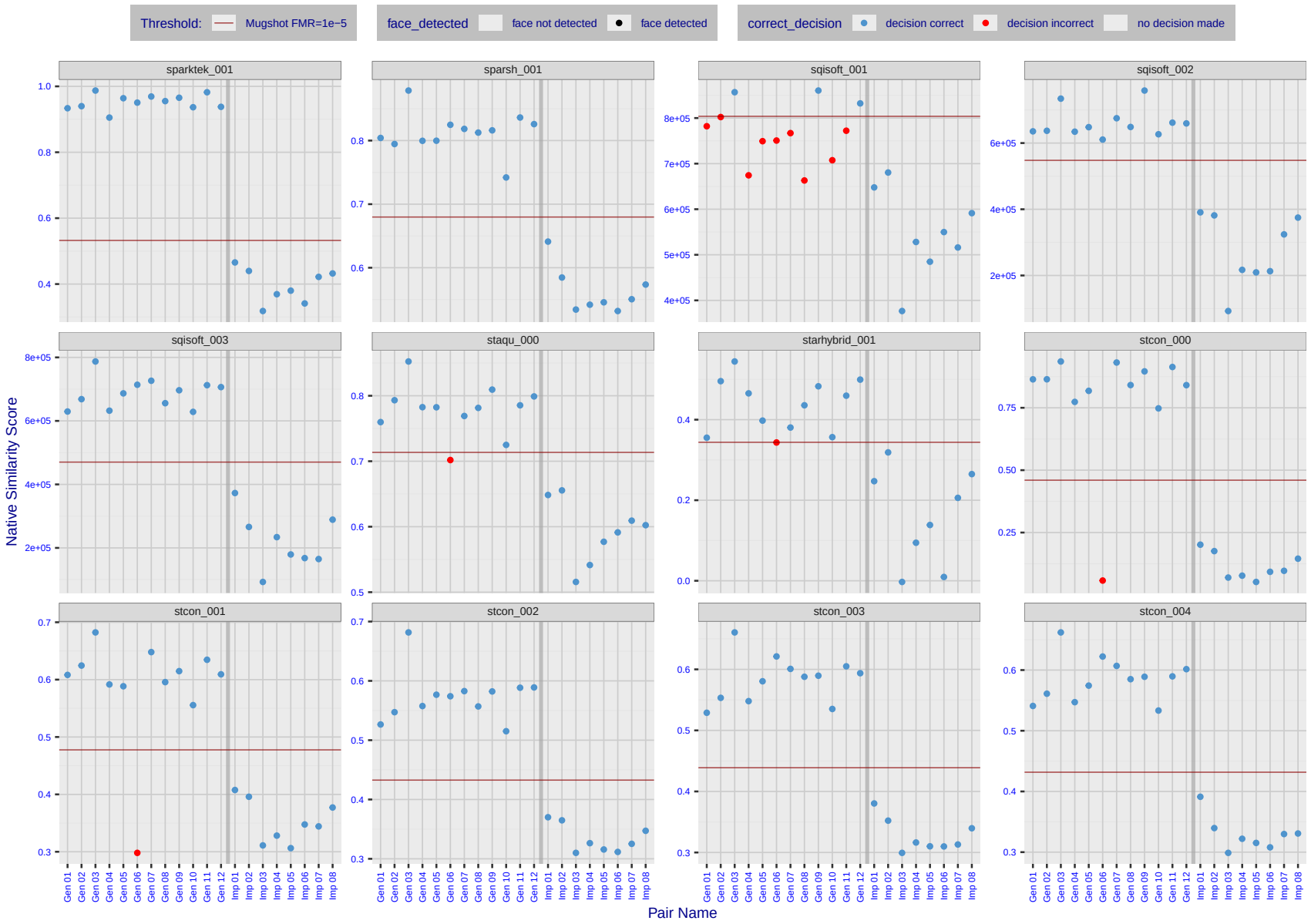


Figure 89: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

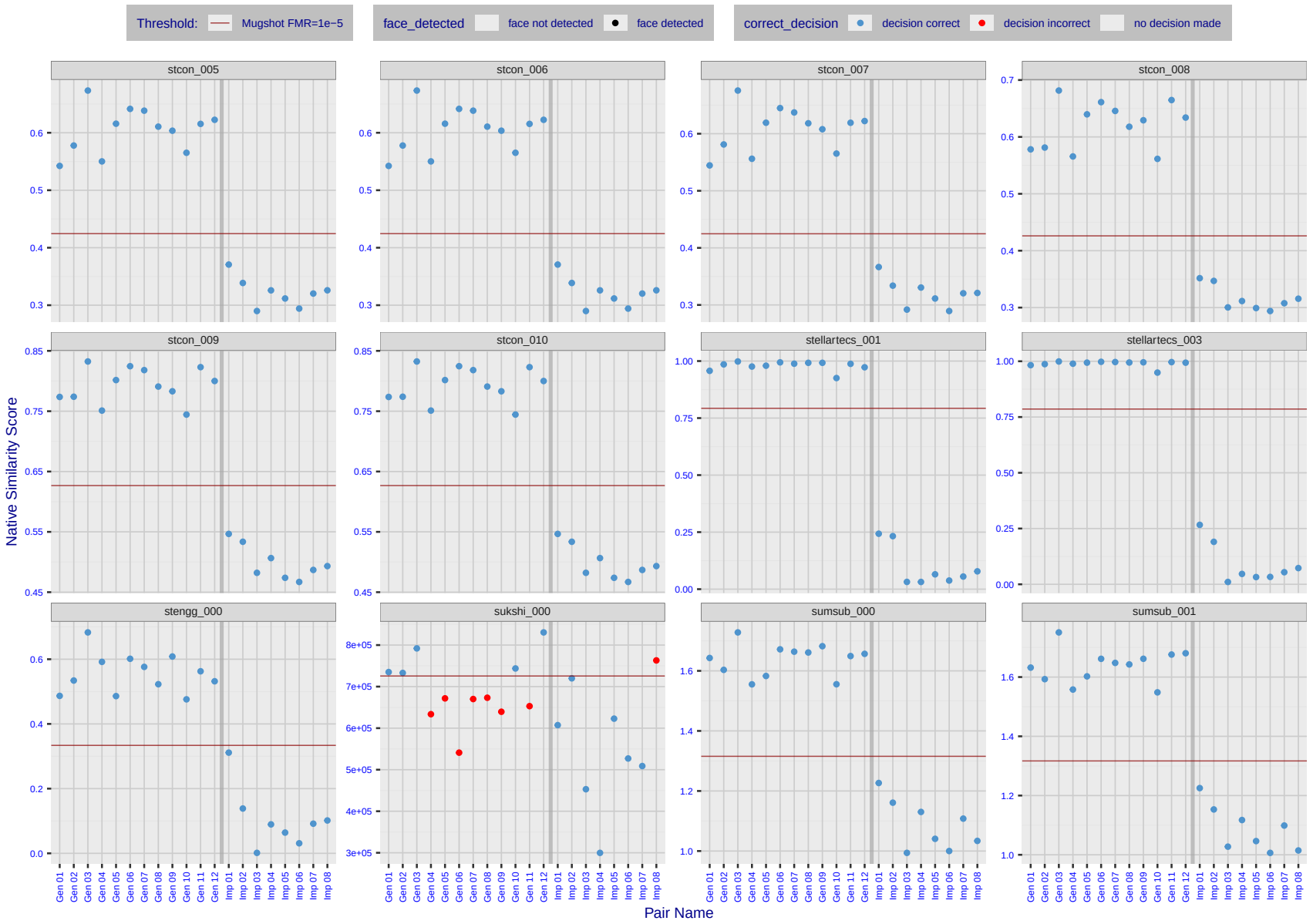


Figure 90: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

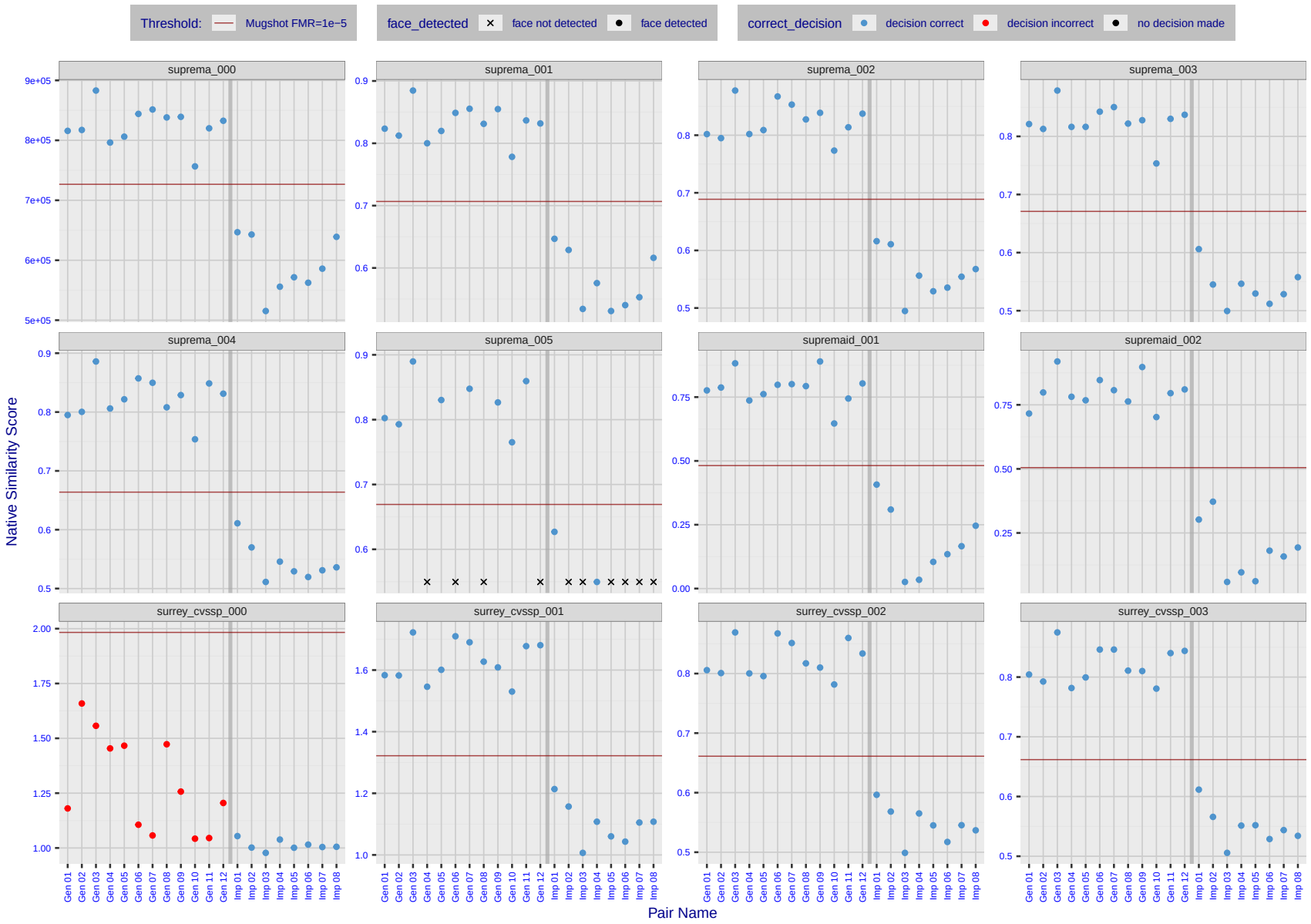


Figure 91: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

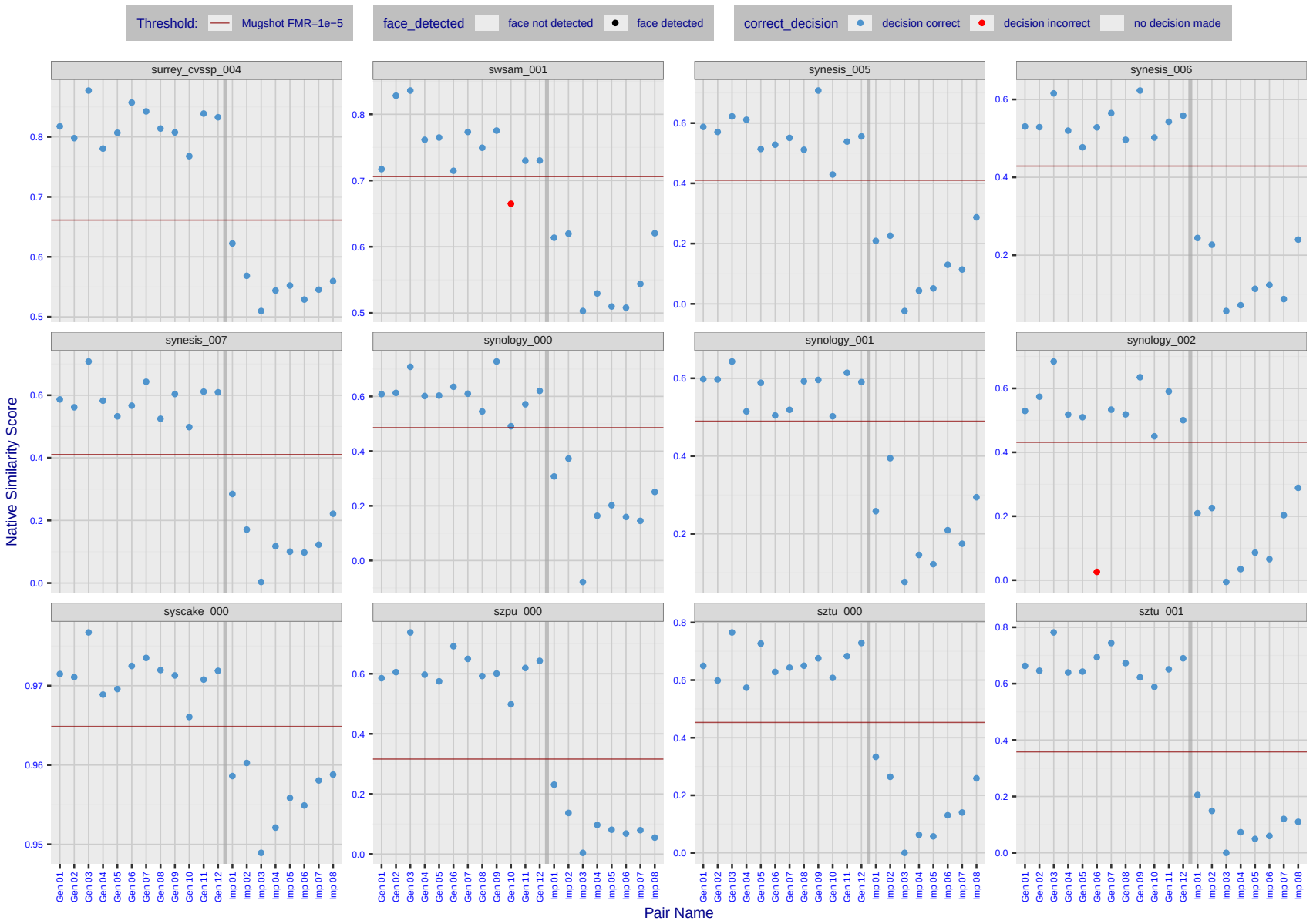


Figure 92: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

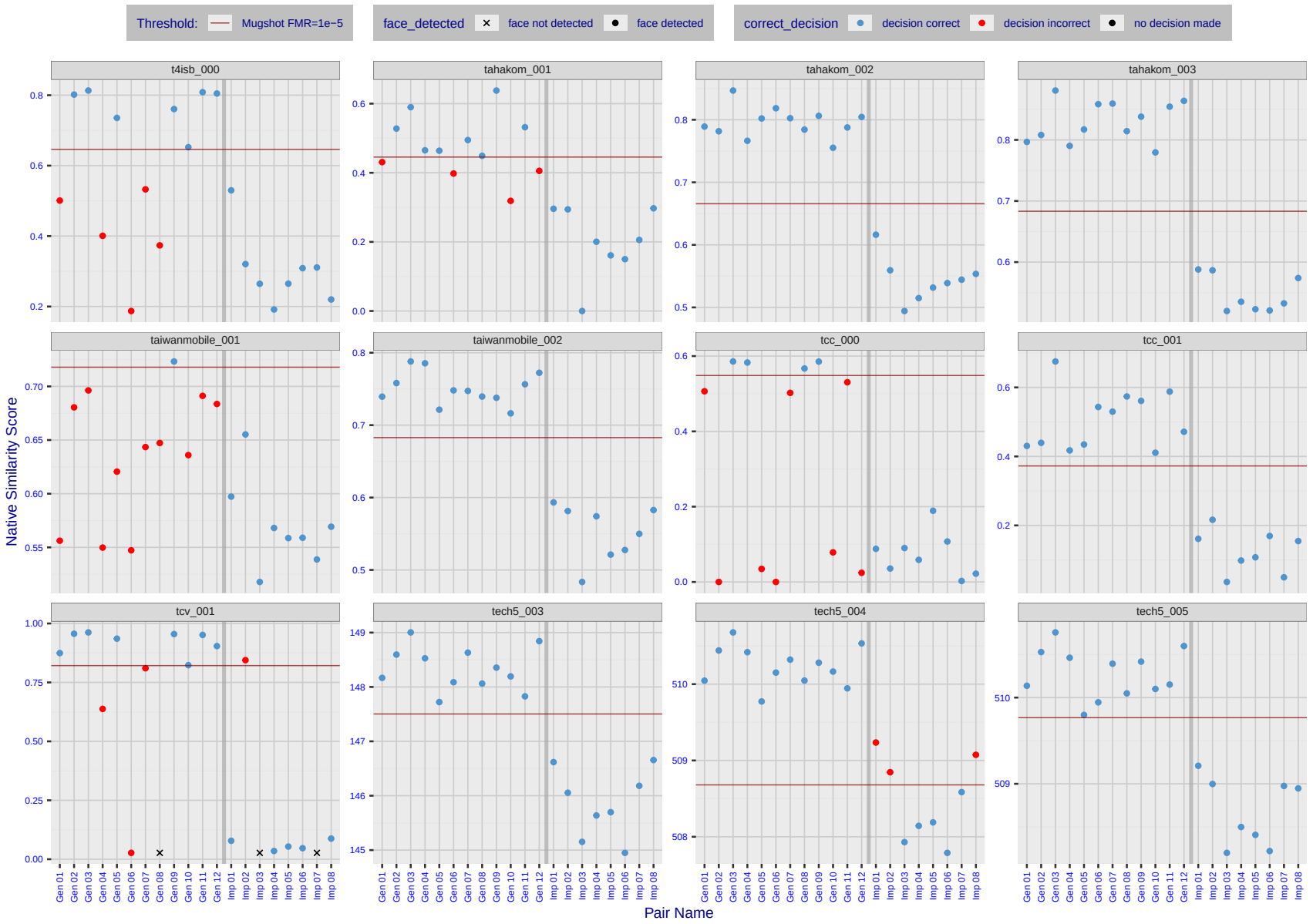


Figure 93: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

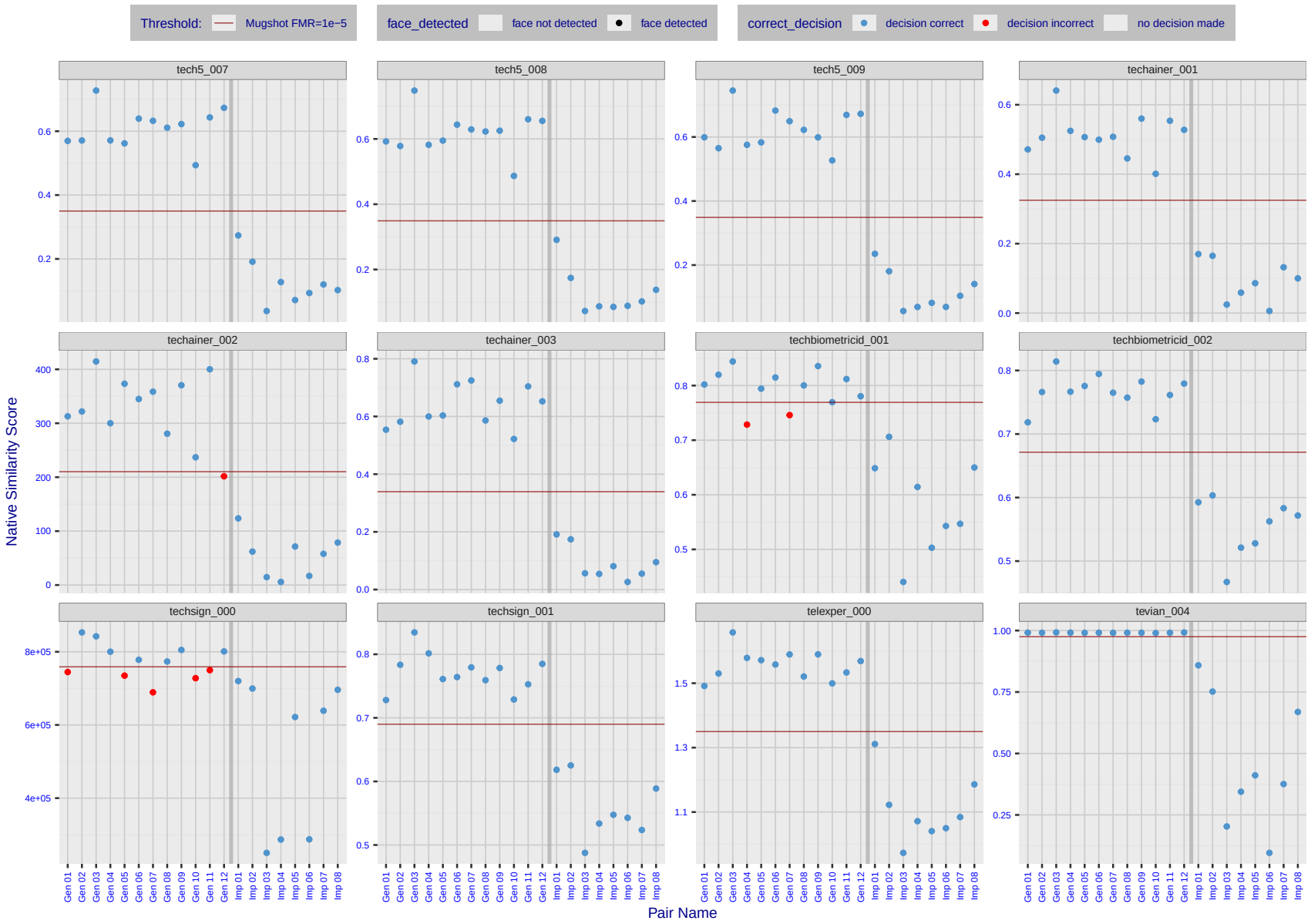


Figure 94: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

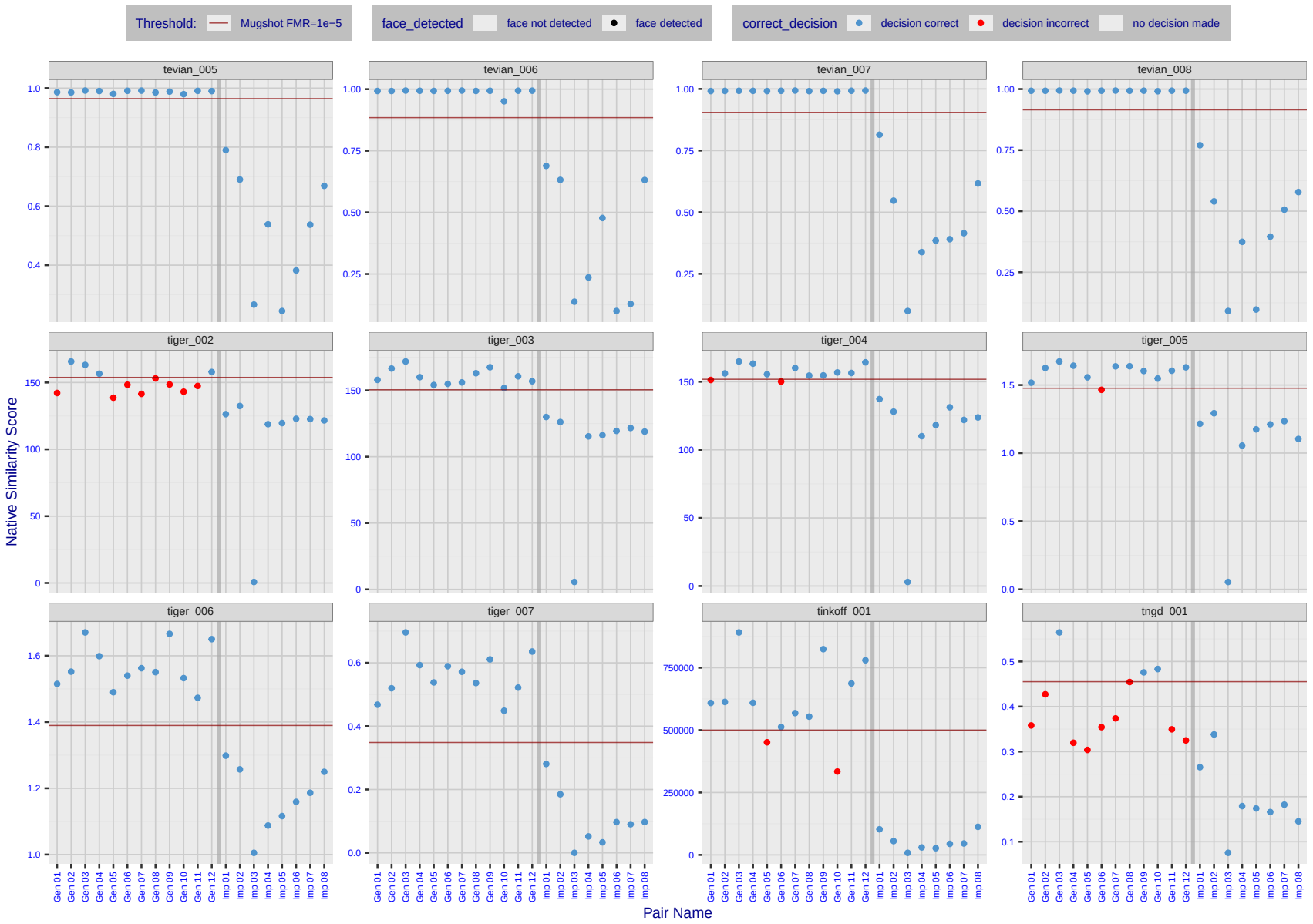


Figure 95: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

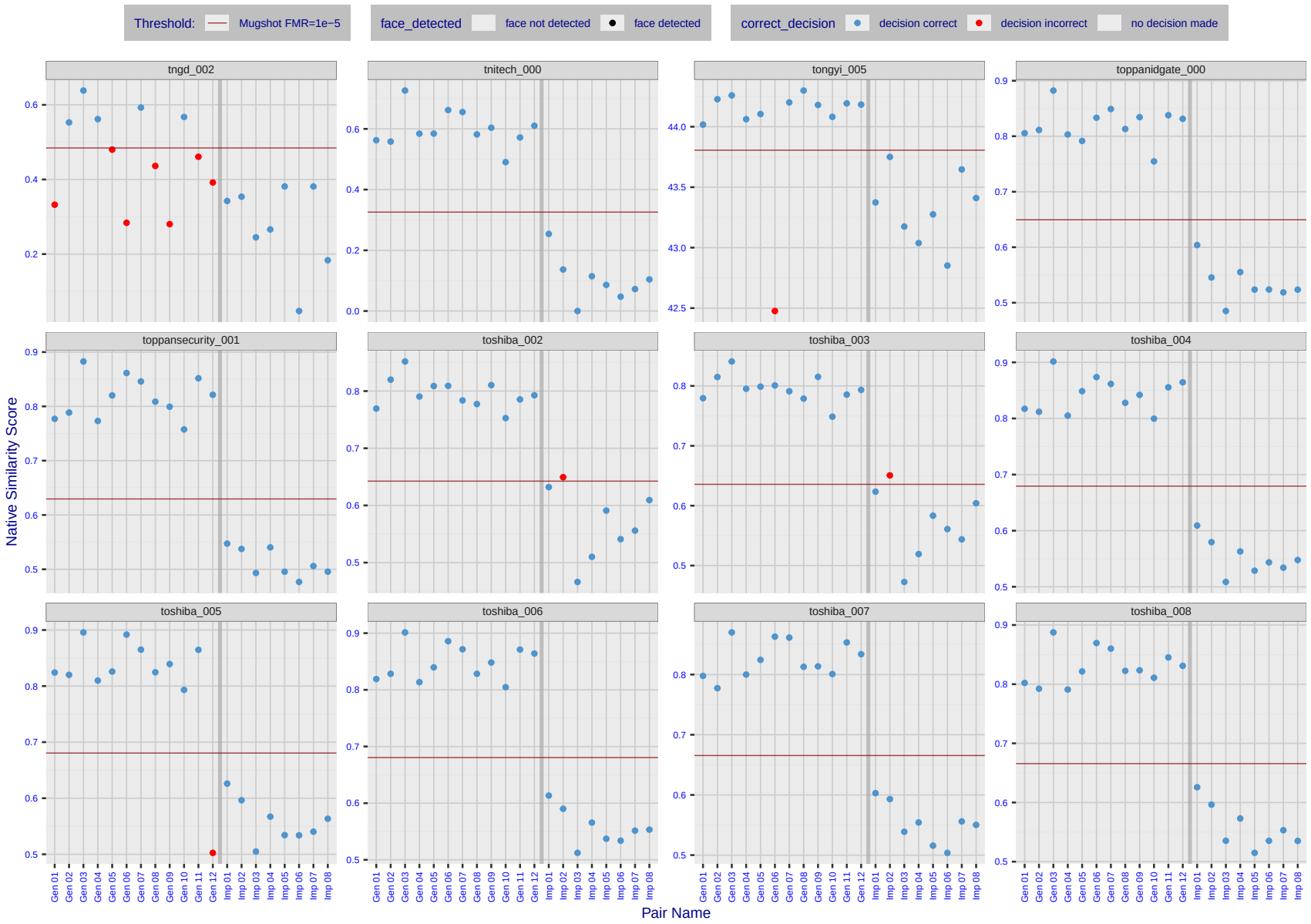


Figure 96: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

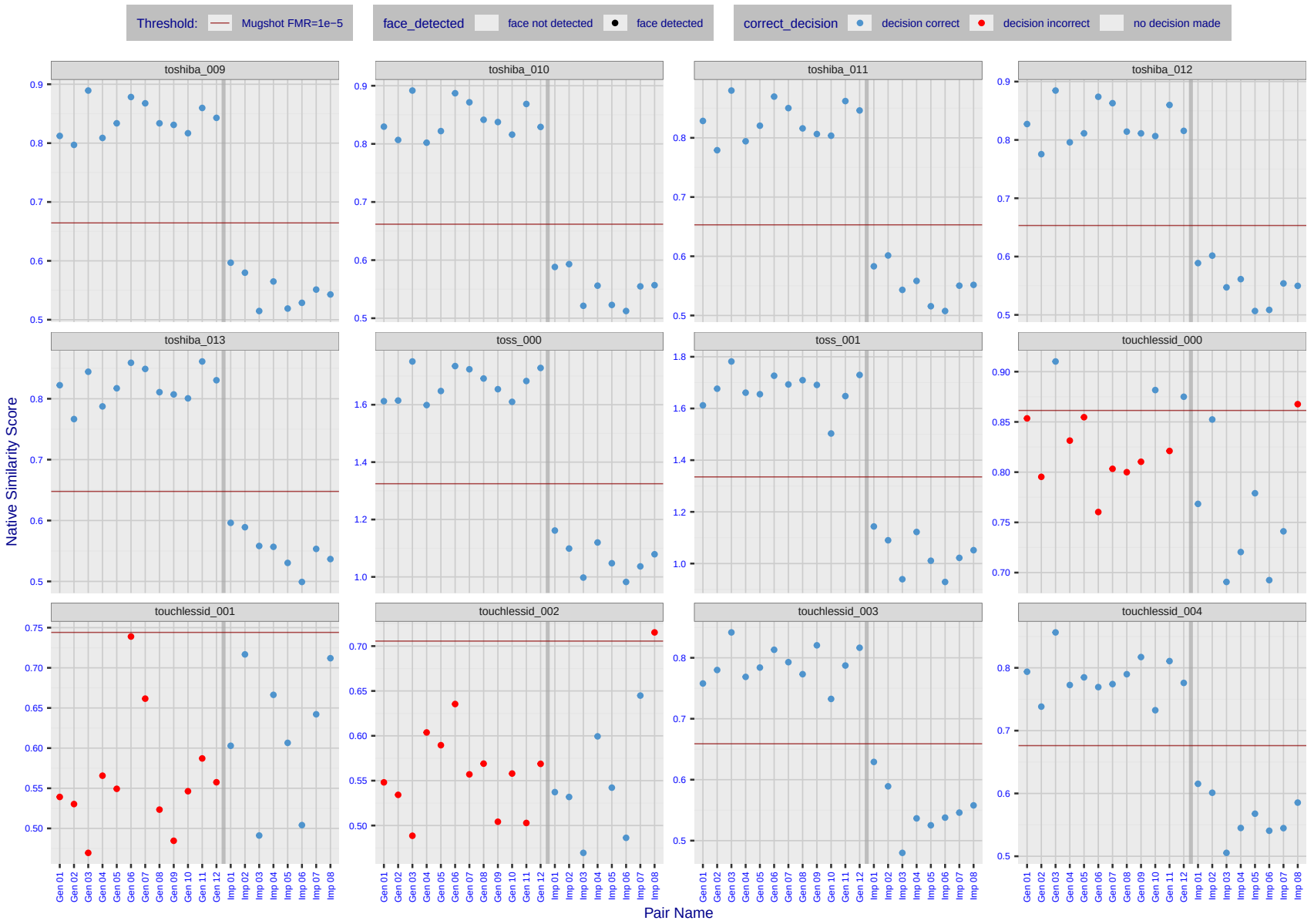


Figure 97: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

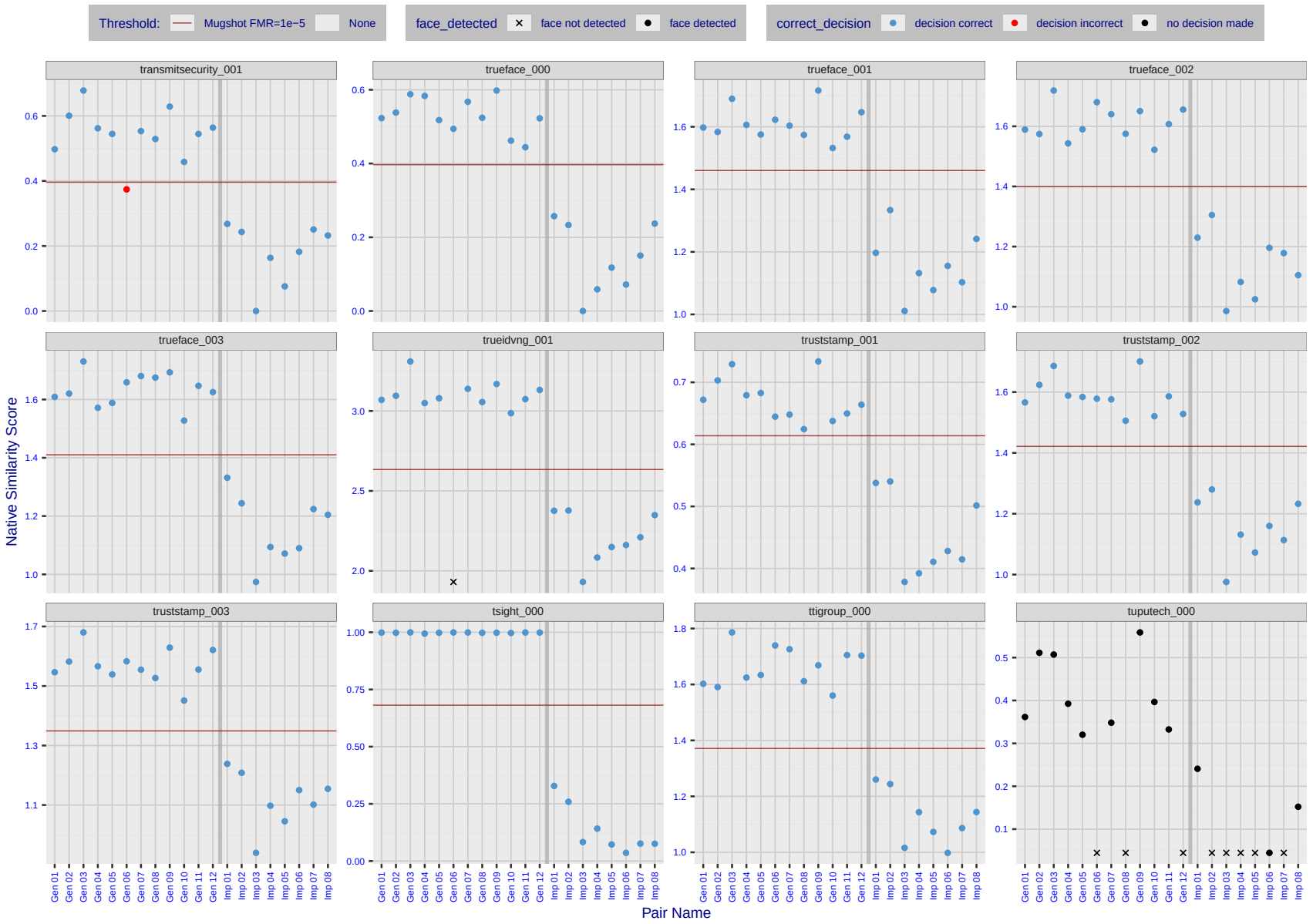


Figure 98: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

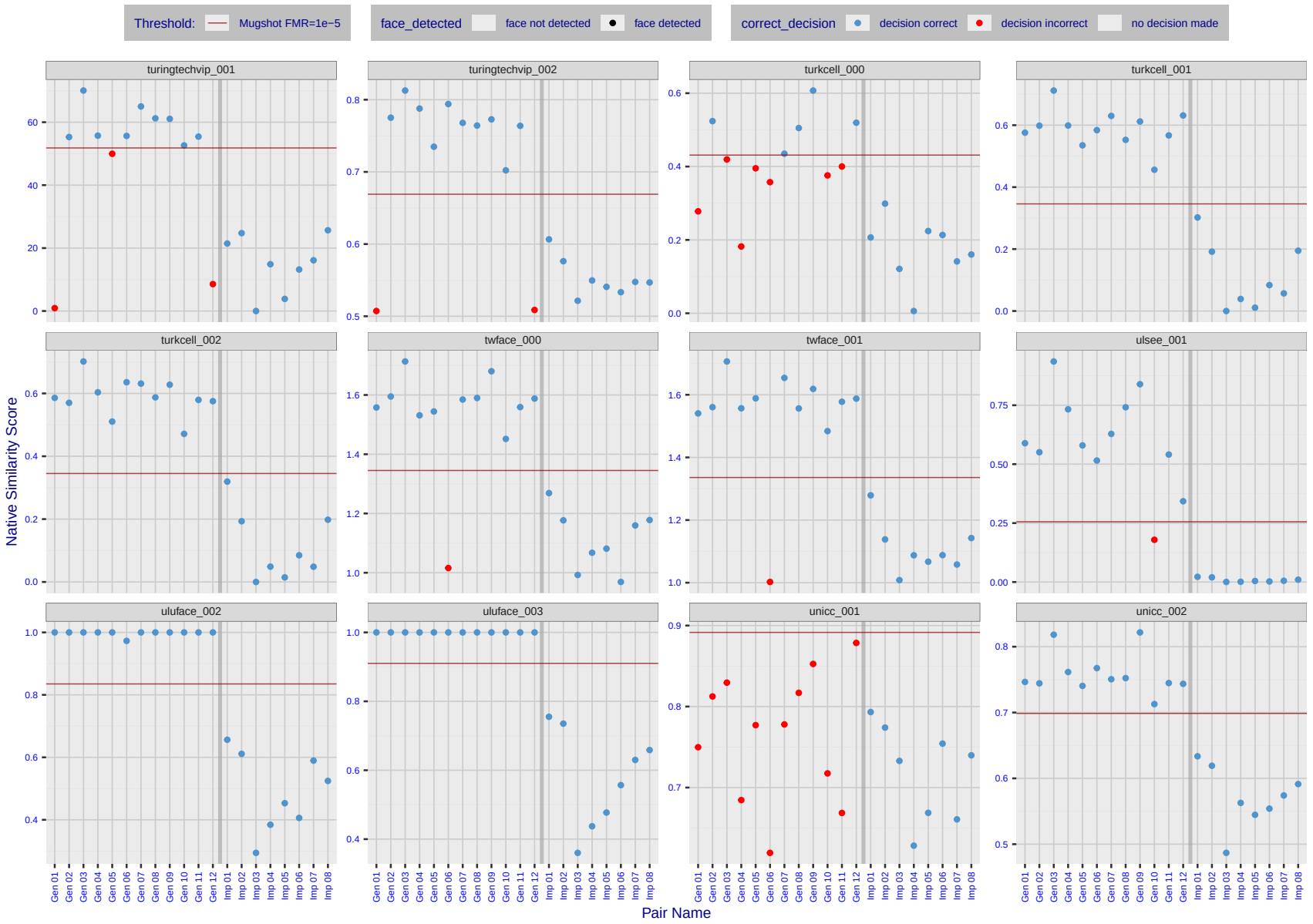


Figure 99: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

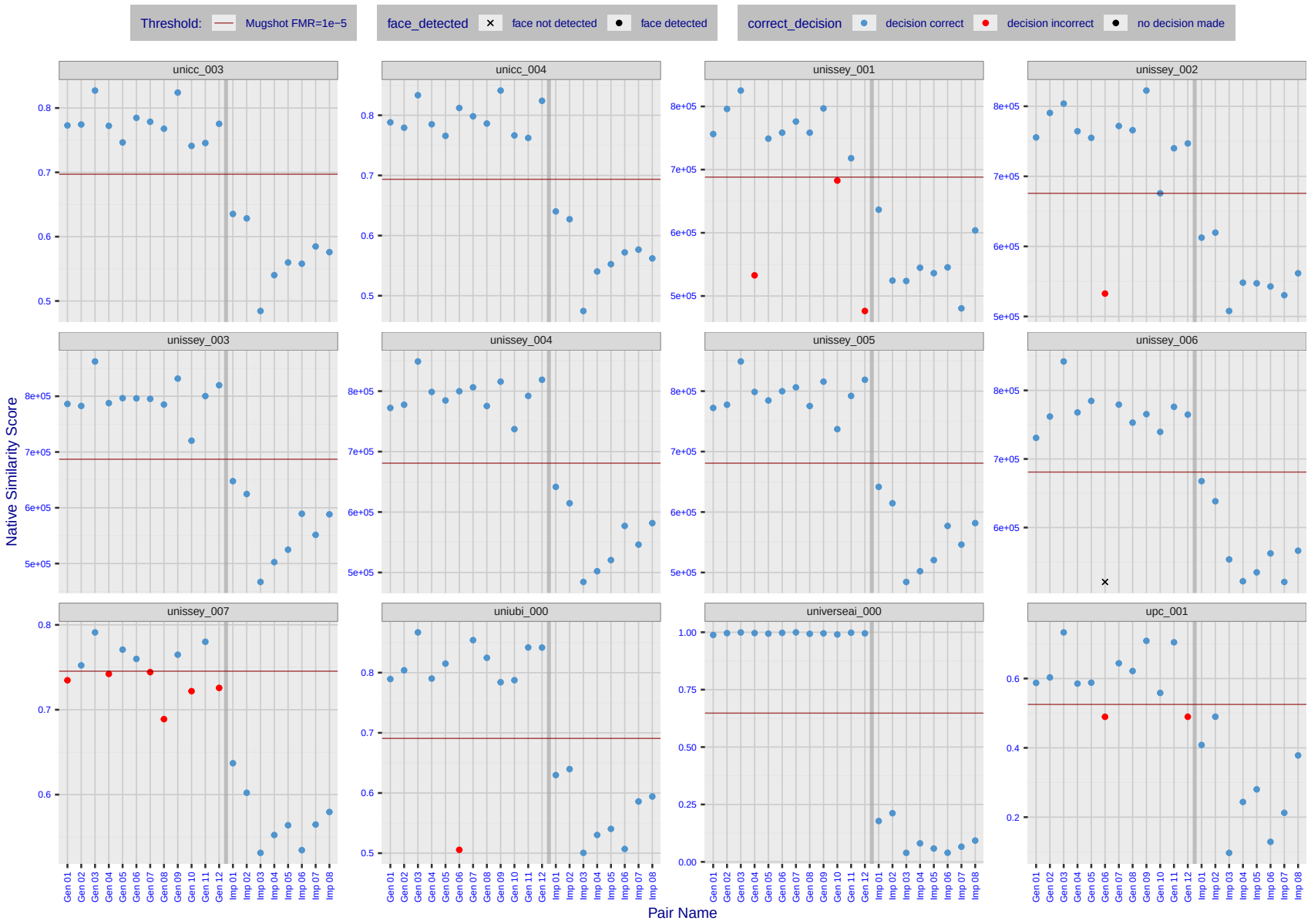


Figure 100: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

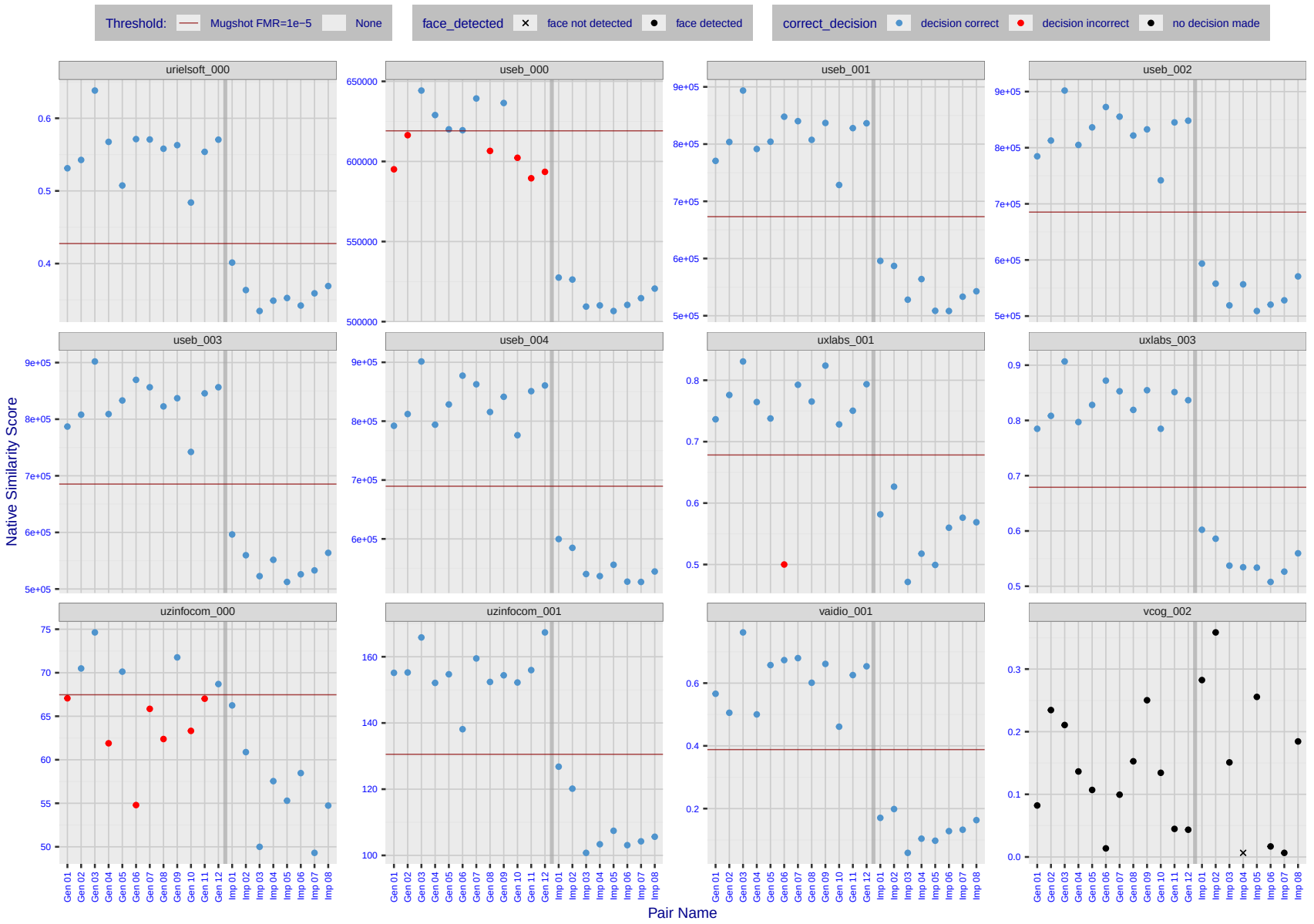


Figure 101: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

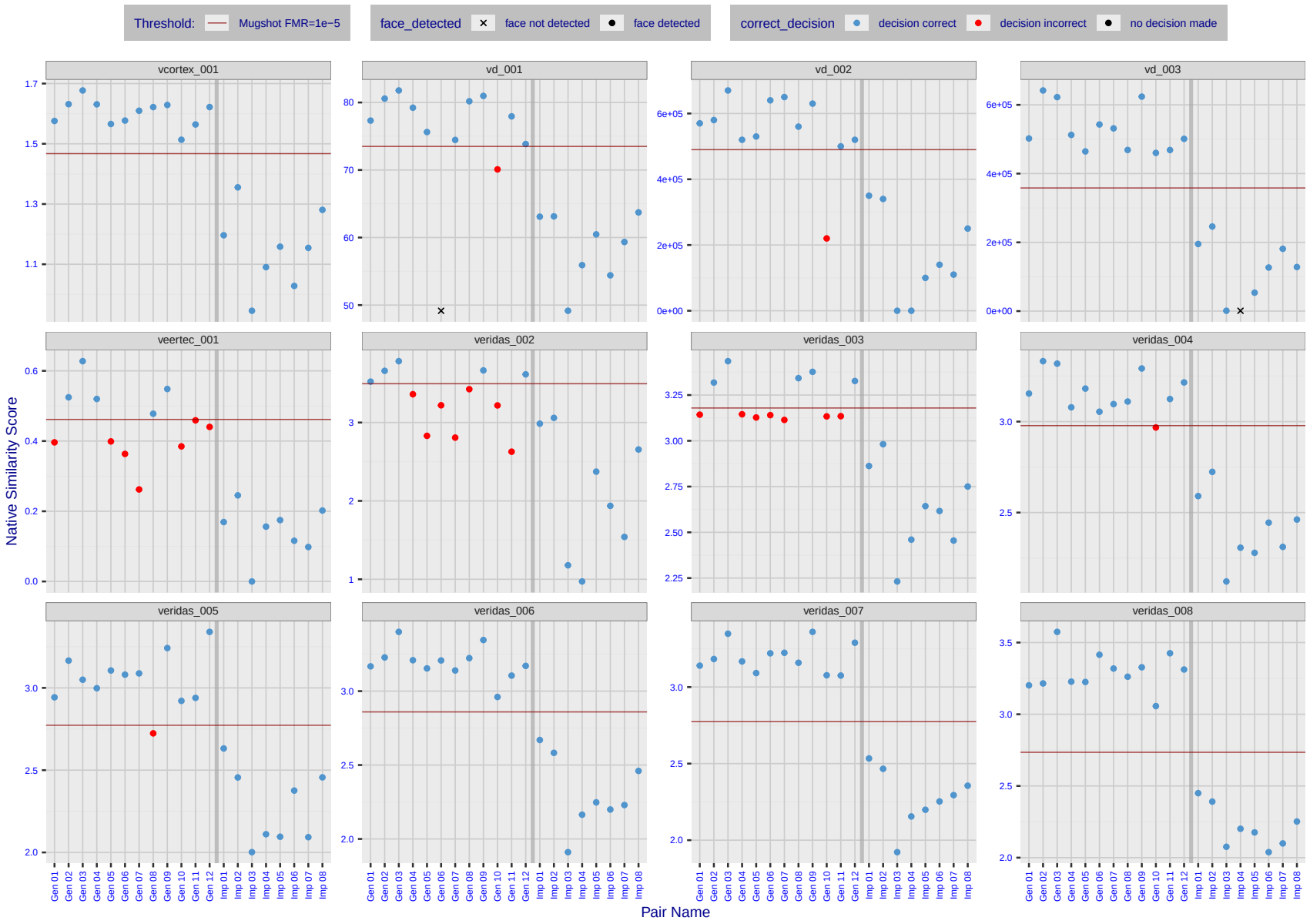


Figure 102: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

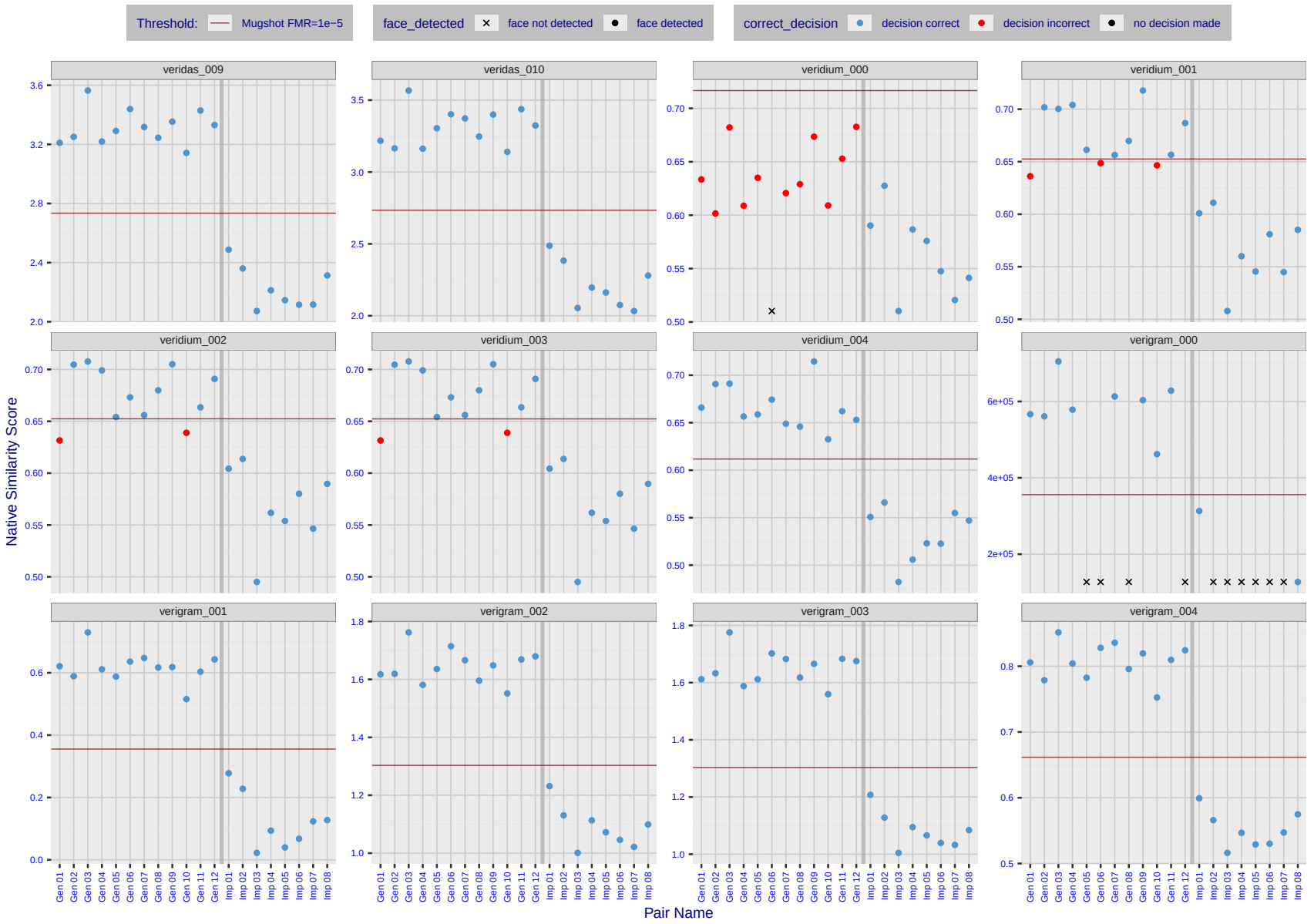


Figure 103: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

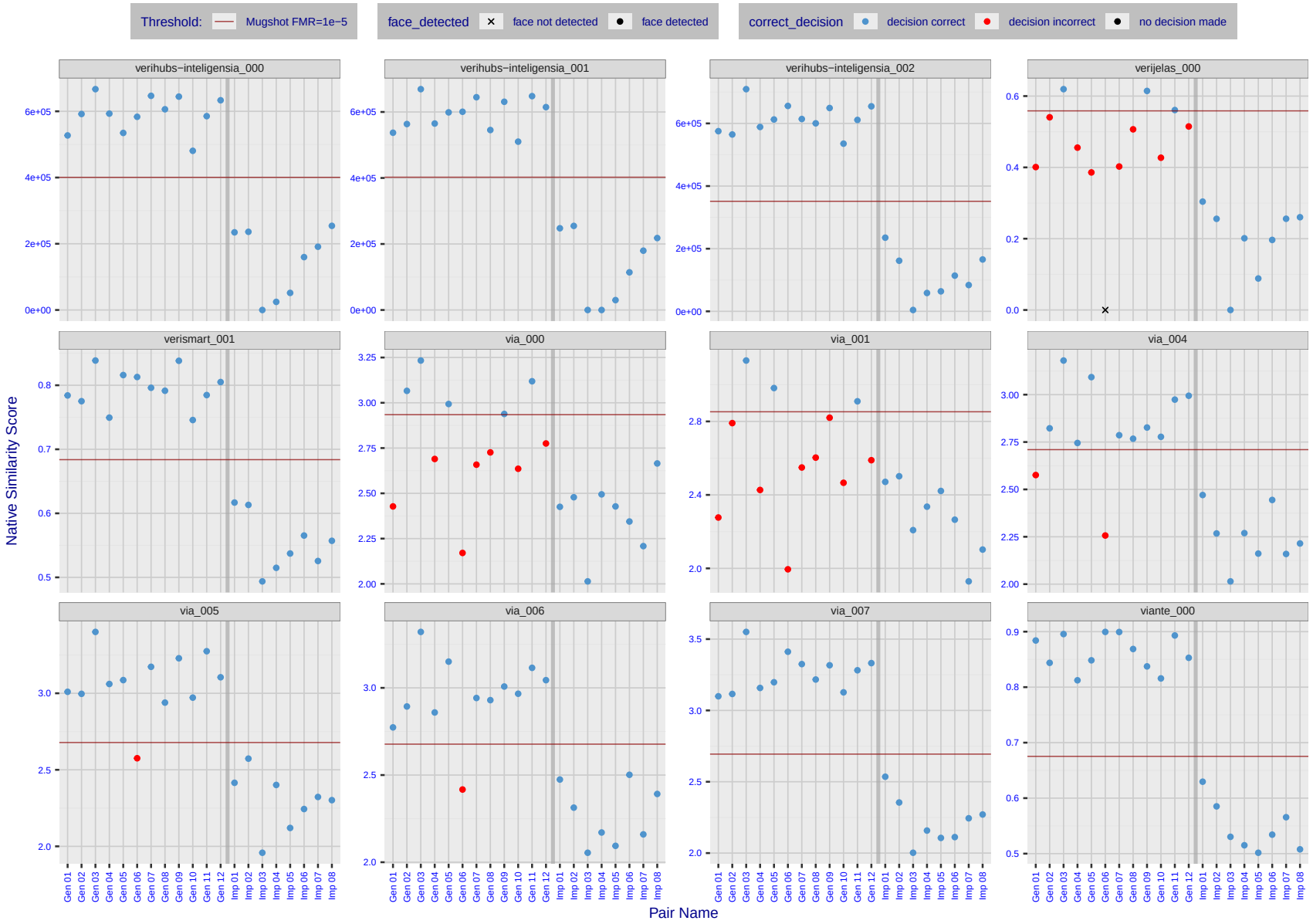


Figure 104: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

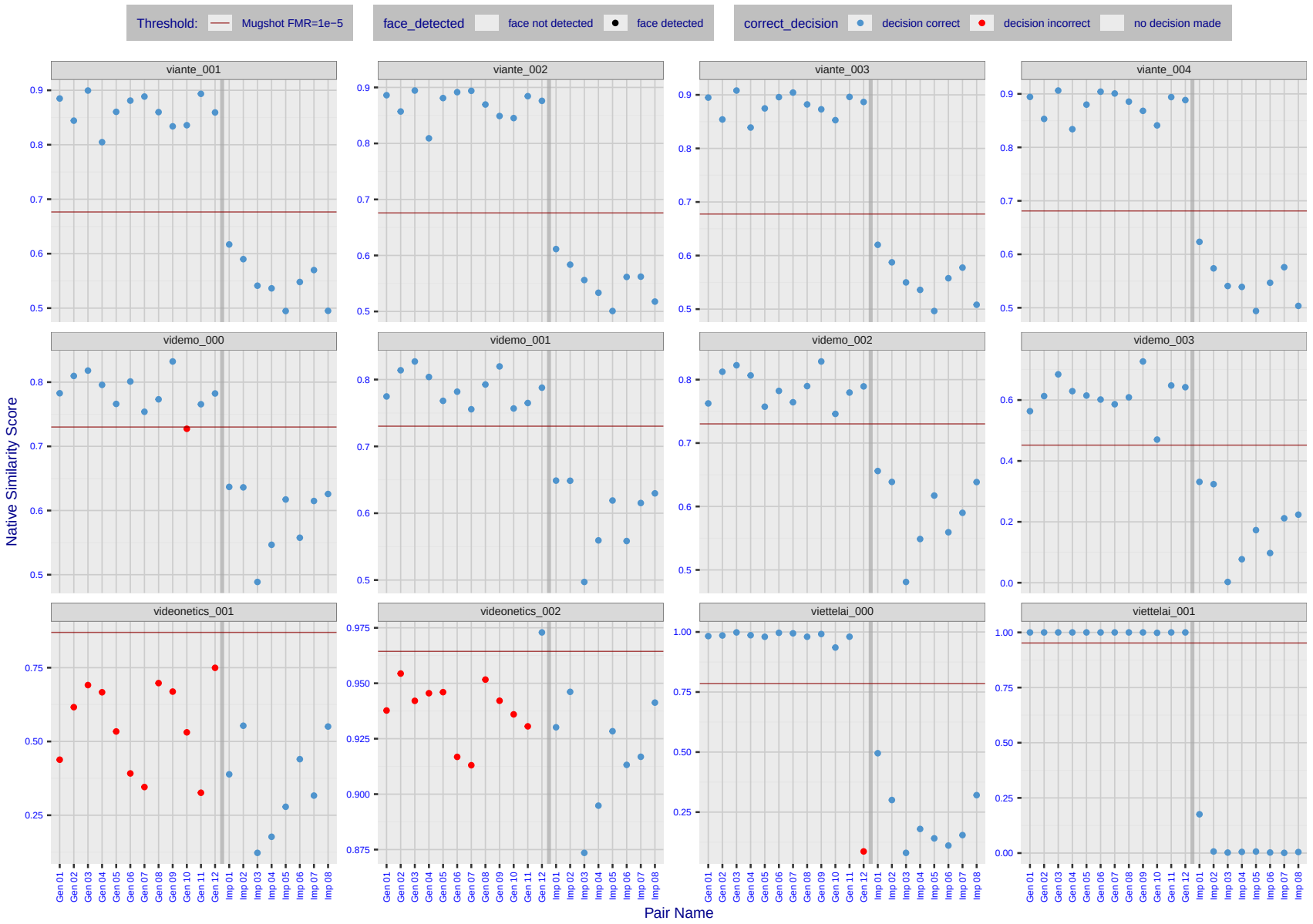


Figure 105: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

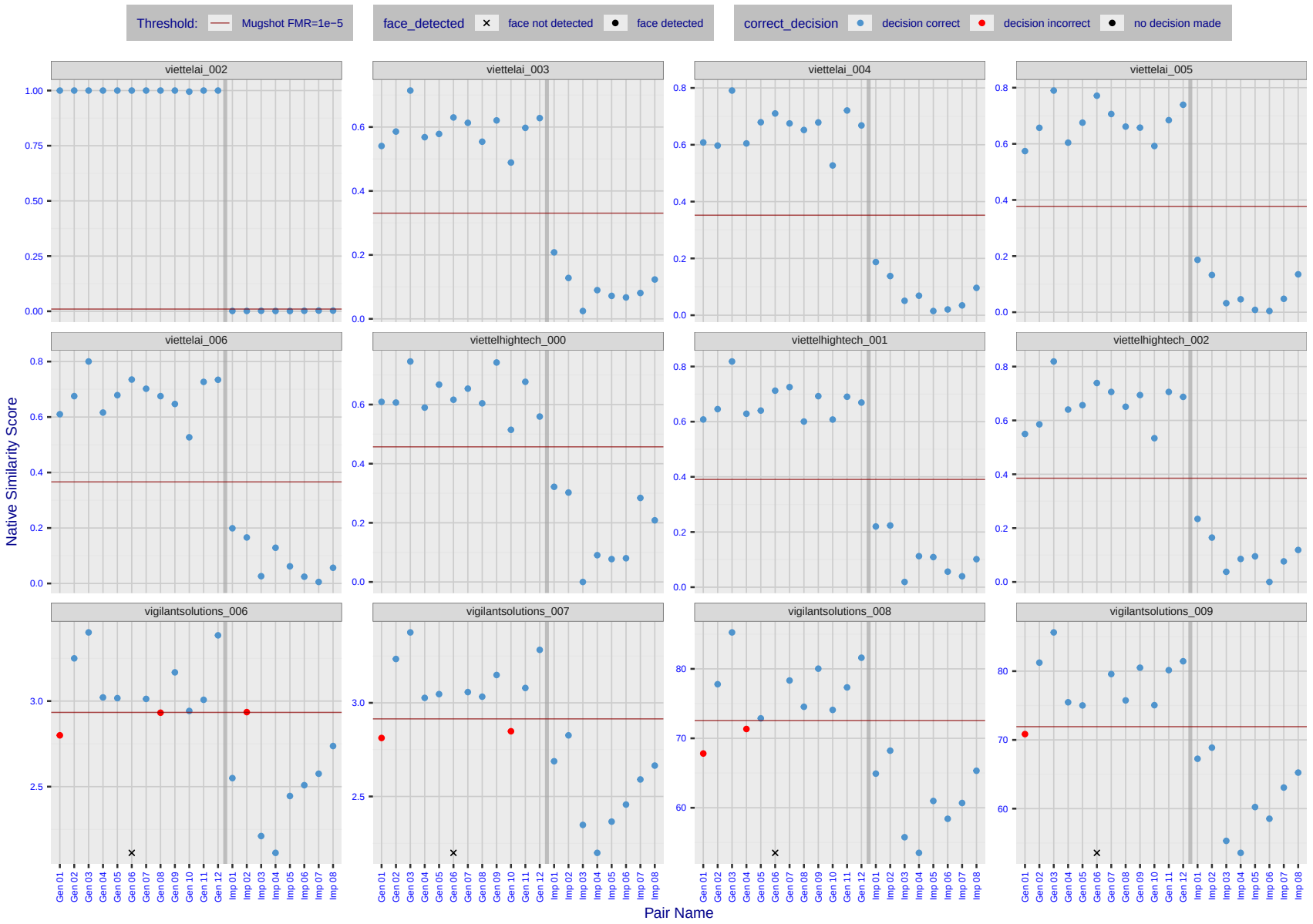


Figure 106: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

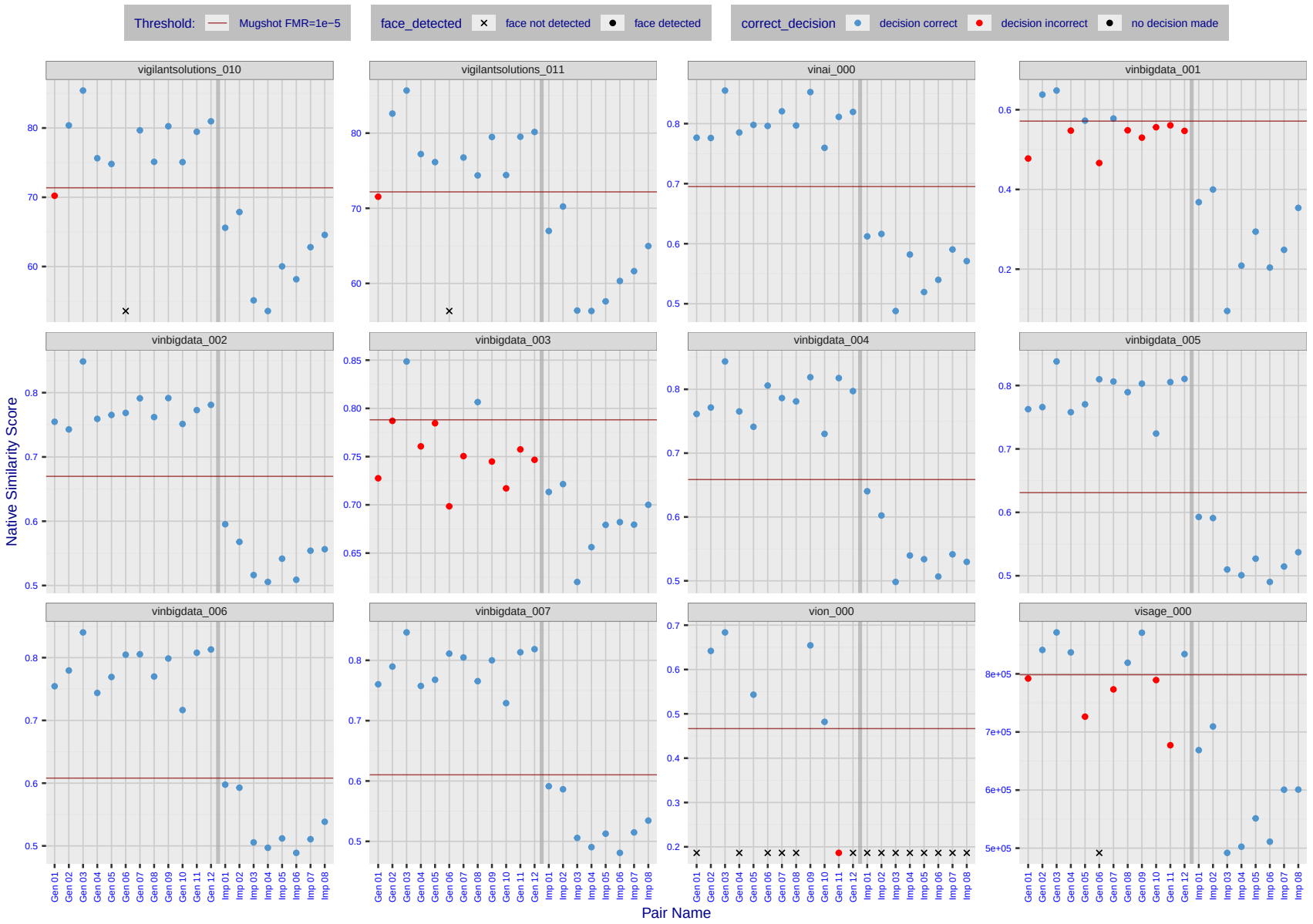


Figure 107: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

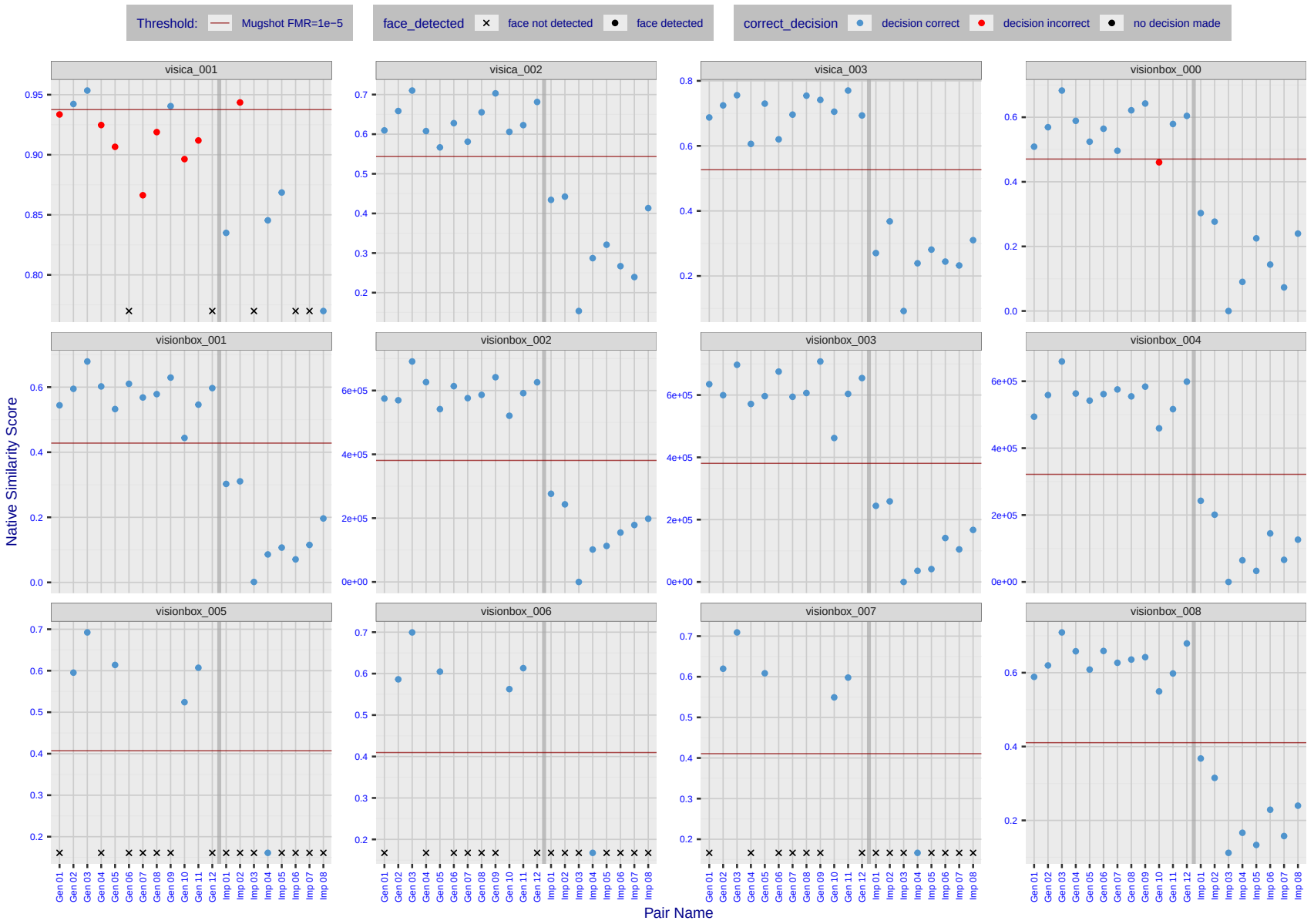


Figure 108: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

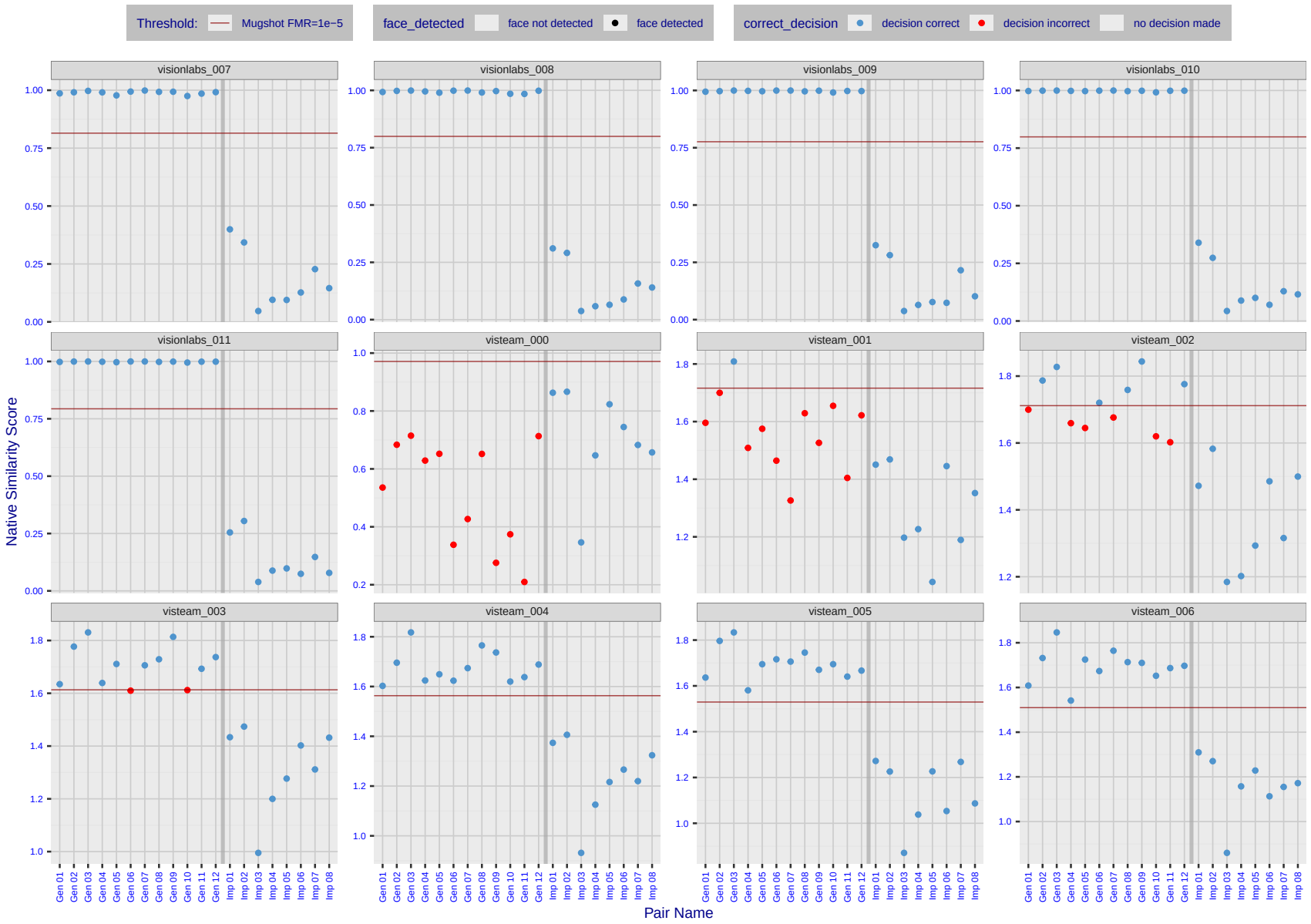


Figure 109: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

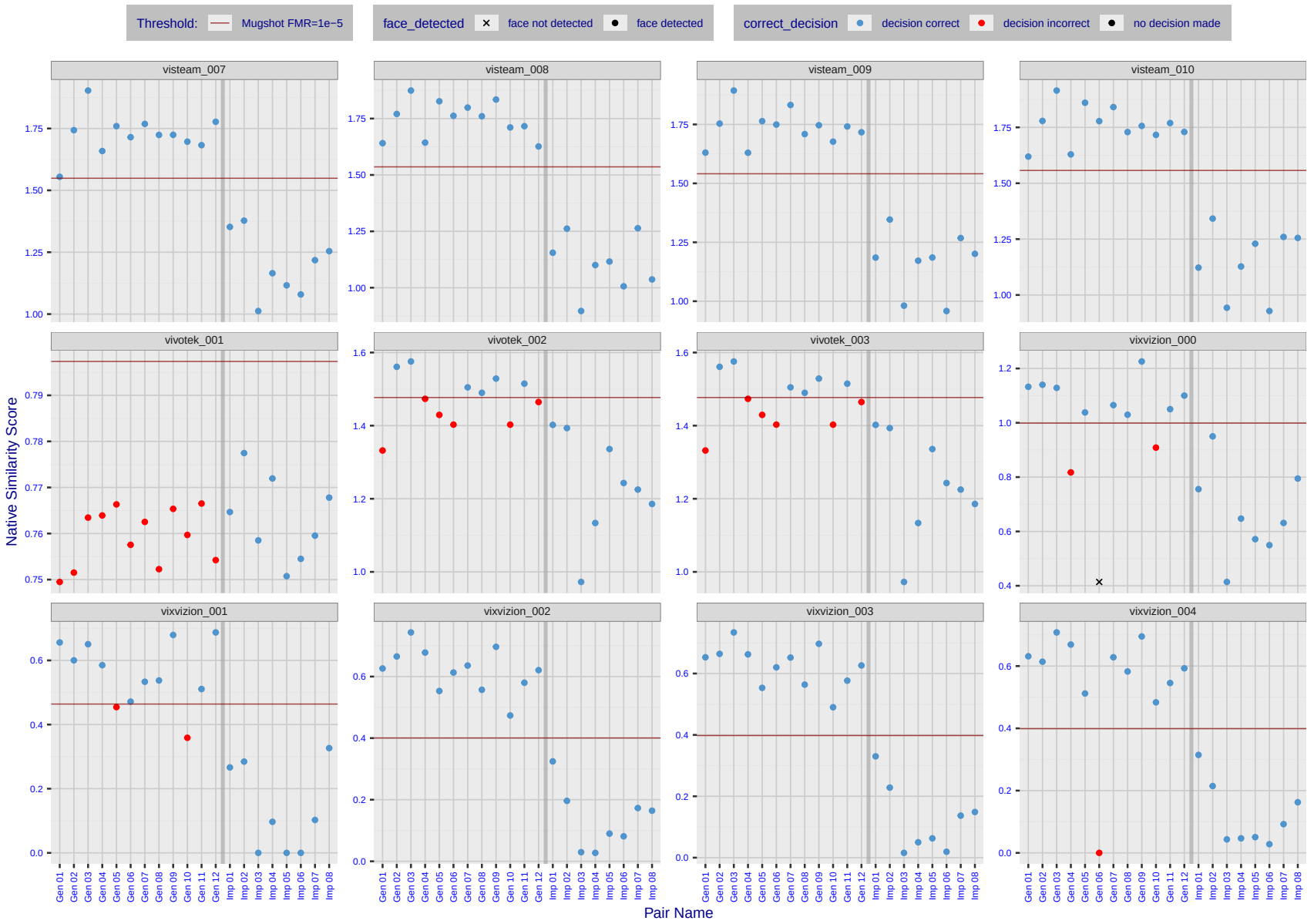


Figure 110: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

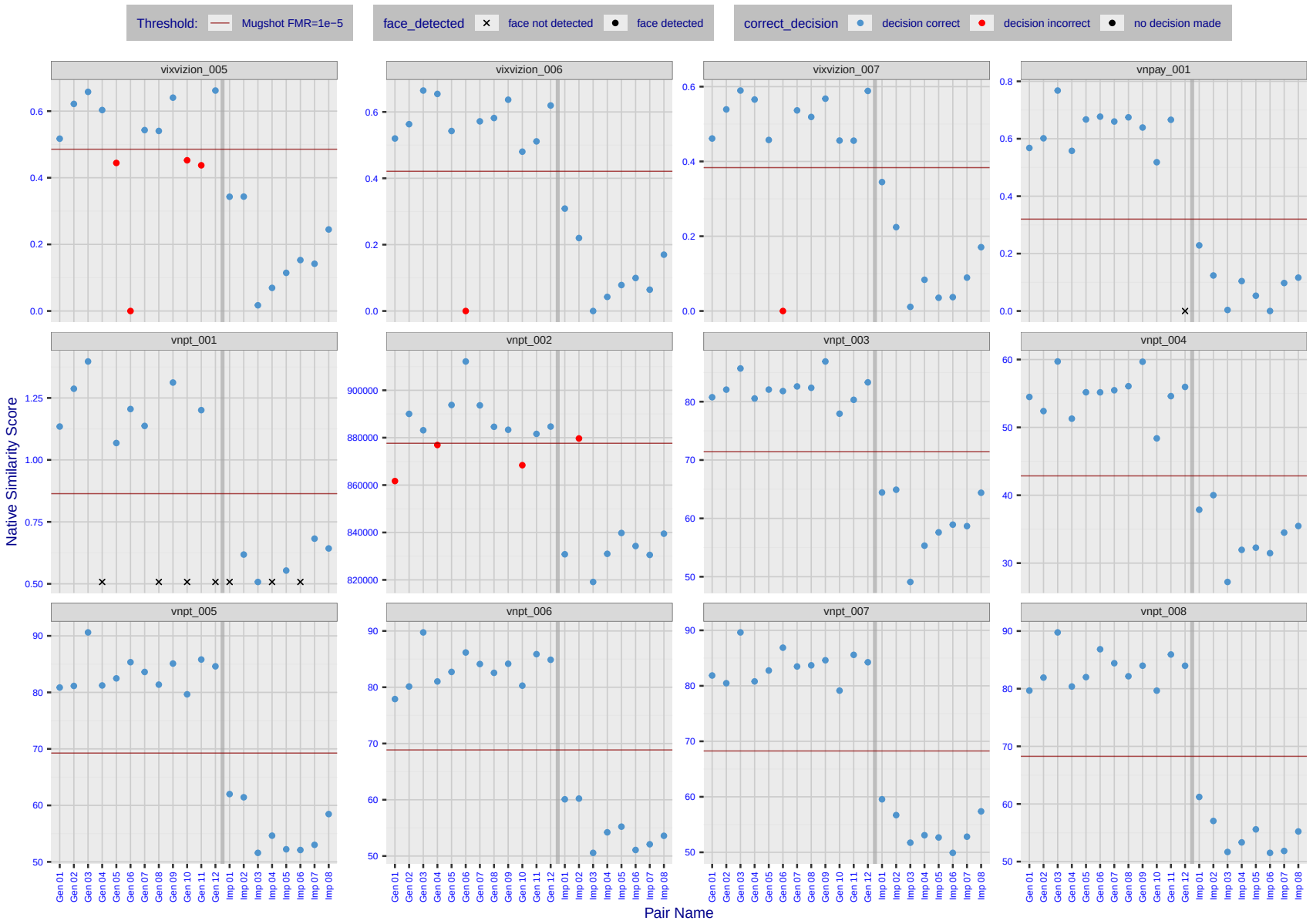


Figure 111: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

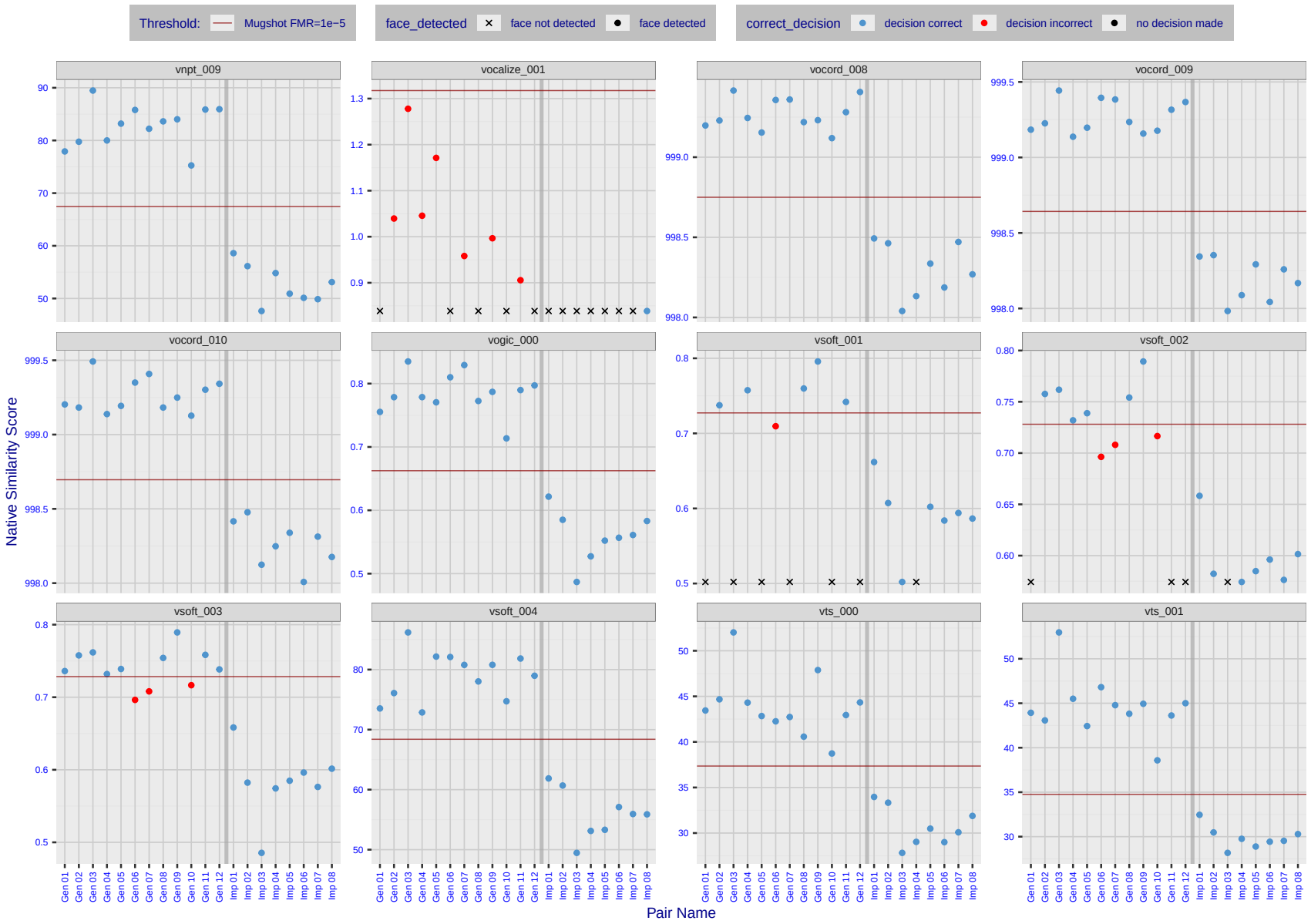


Figure 112: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

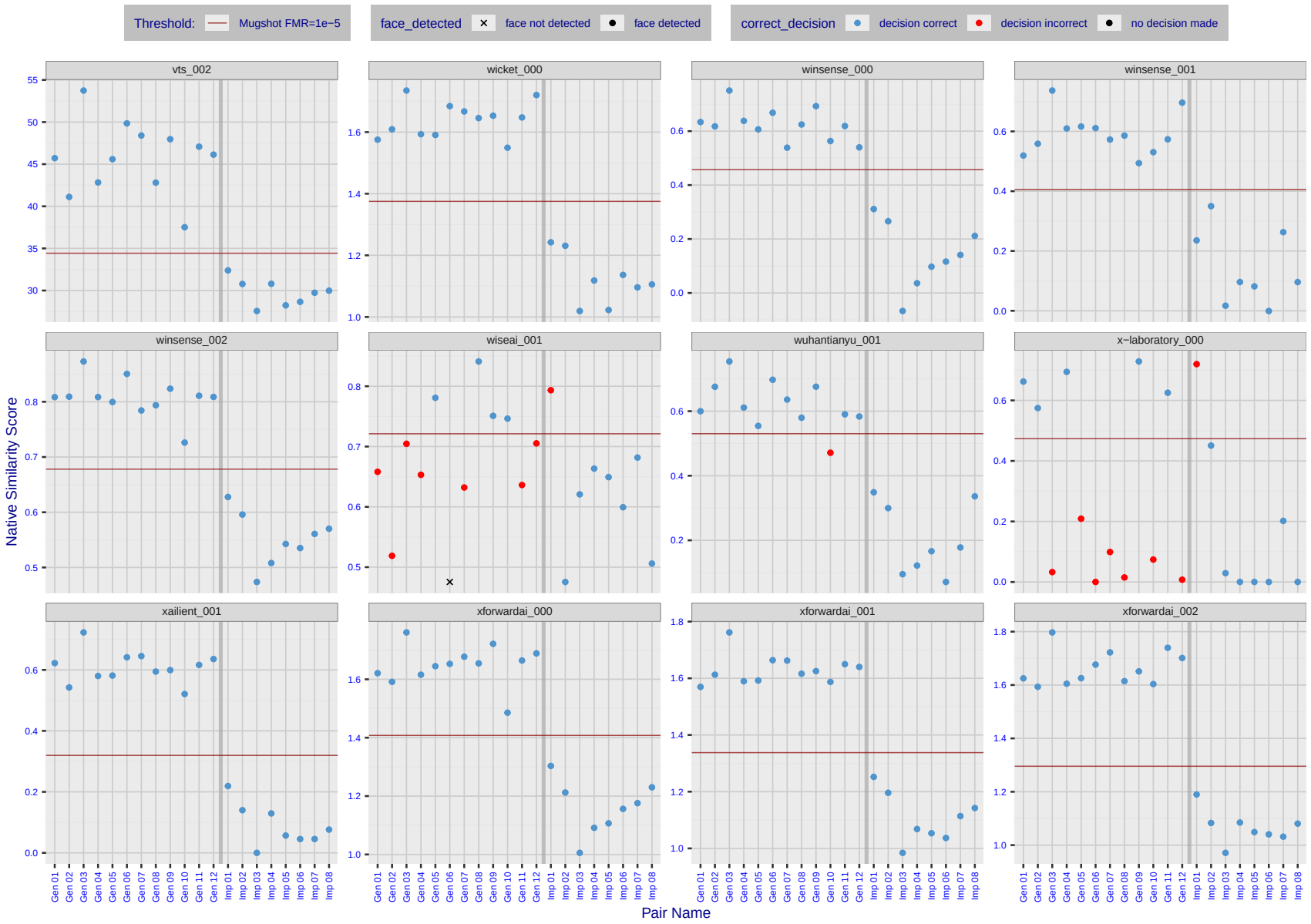


Figure 113: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

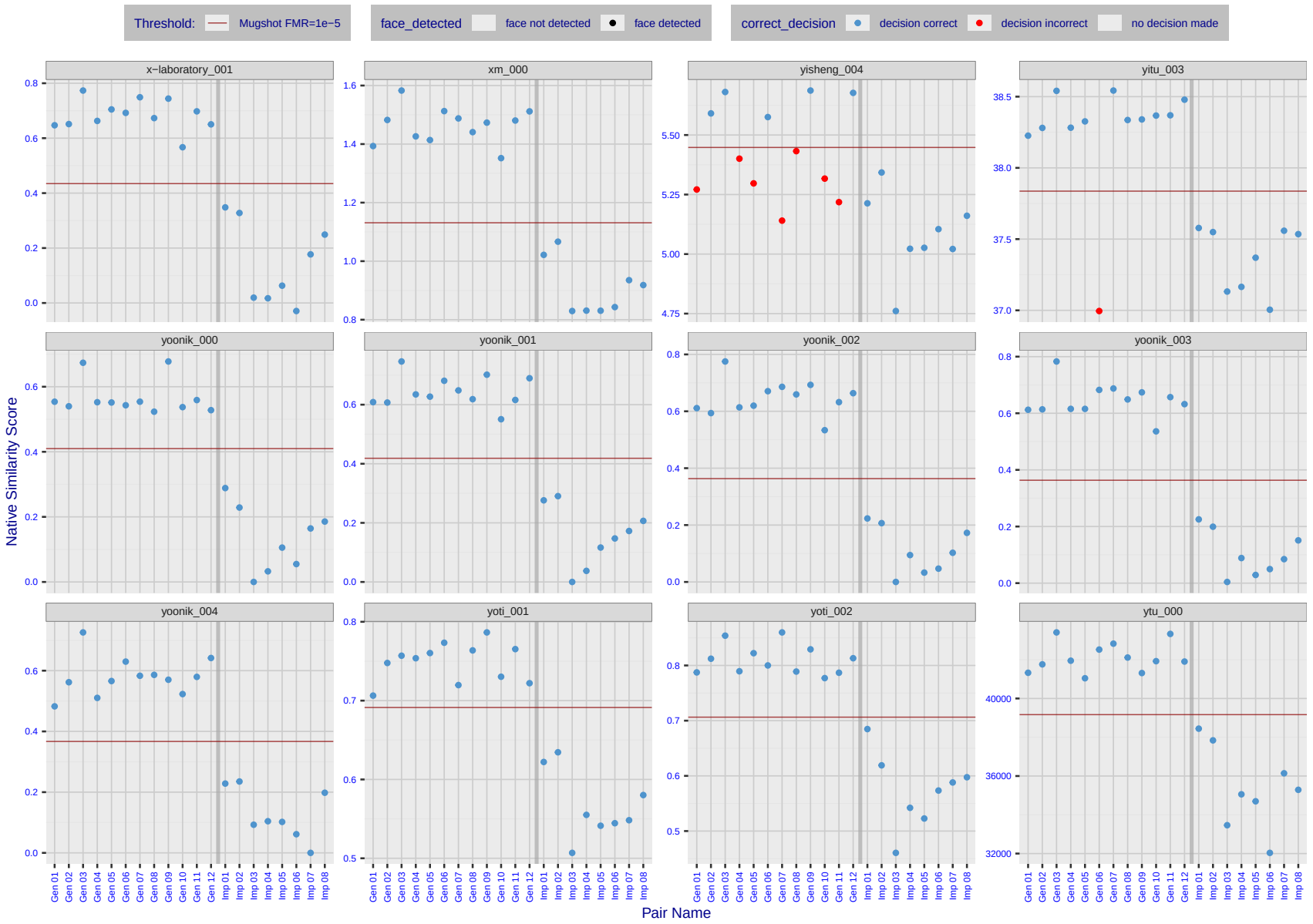


Figure 114: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

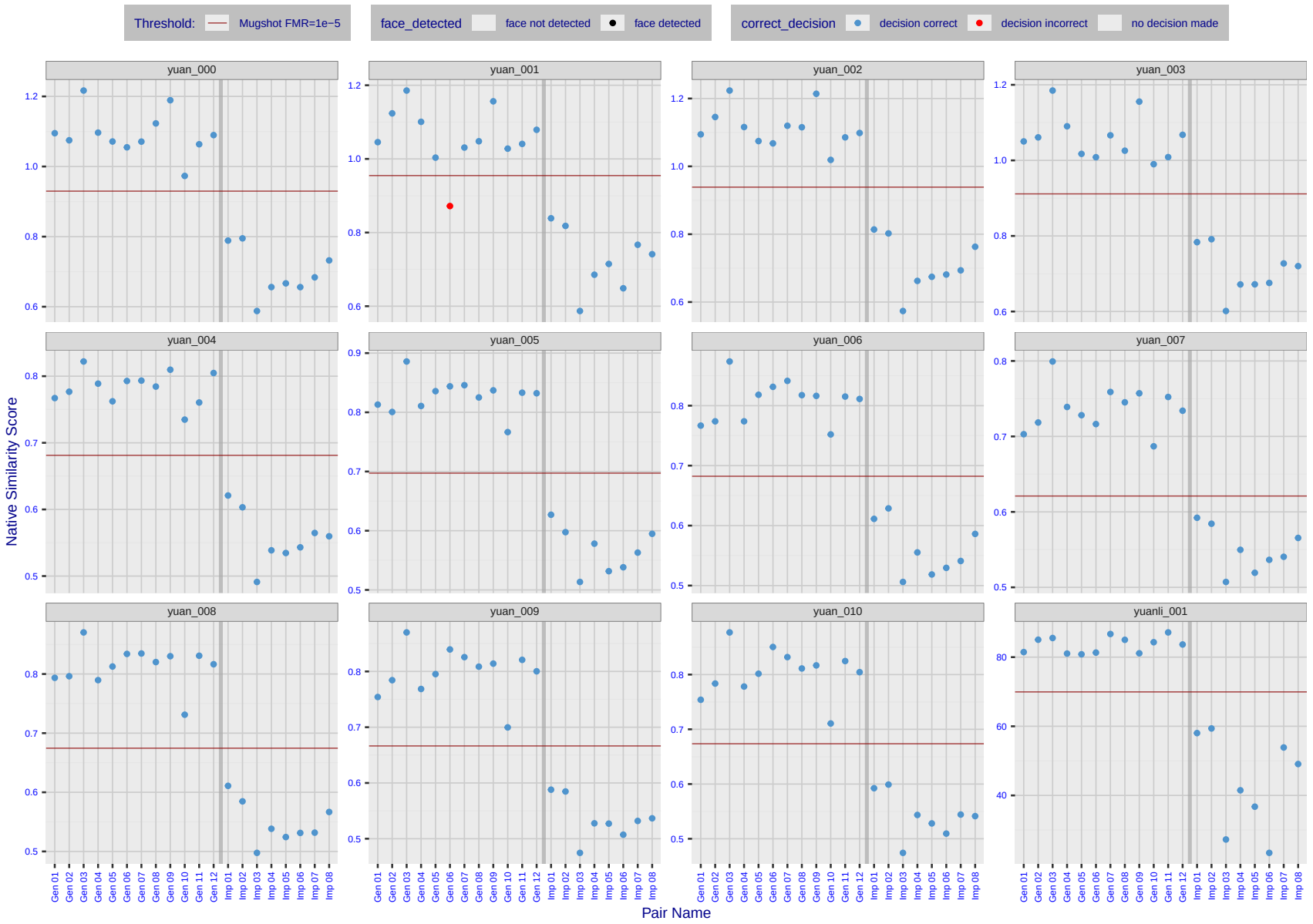


Figure 115: The figure shows algorithms' similarity scores for 12 genuine and 8 impostor image pairs used in a May 2018 paper by Phillips et al. ([1]). The threshold (red horizontal line) is a value calibrated to give FMR = 0.0001 on mugshot images. Points above the threshold correspond to pairs determined to be genuine, and points below the threshold correspond to pairs determined to be impostors. If the determined class (genuine or impostor) matches the real class, points will be blue; if not, red. An X represents face detection failure in either of the images in the pair. Note that the sample size (n=20) is small, and the figure may change substantially if larger or different sets are used. The images can be viewed on p. 13 of the Appendix, where Gen 01 corresponds to Same-Identity Pair 1, Gen 02 corresponds to Same-Identity Pair 2, and so on.

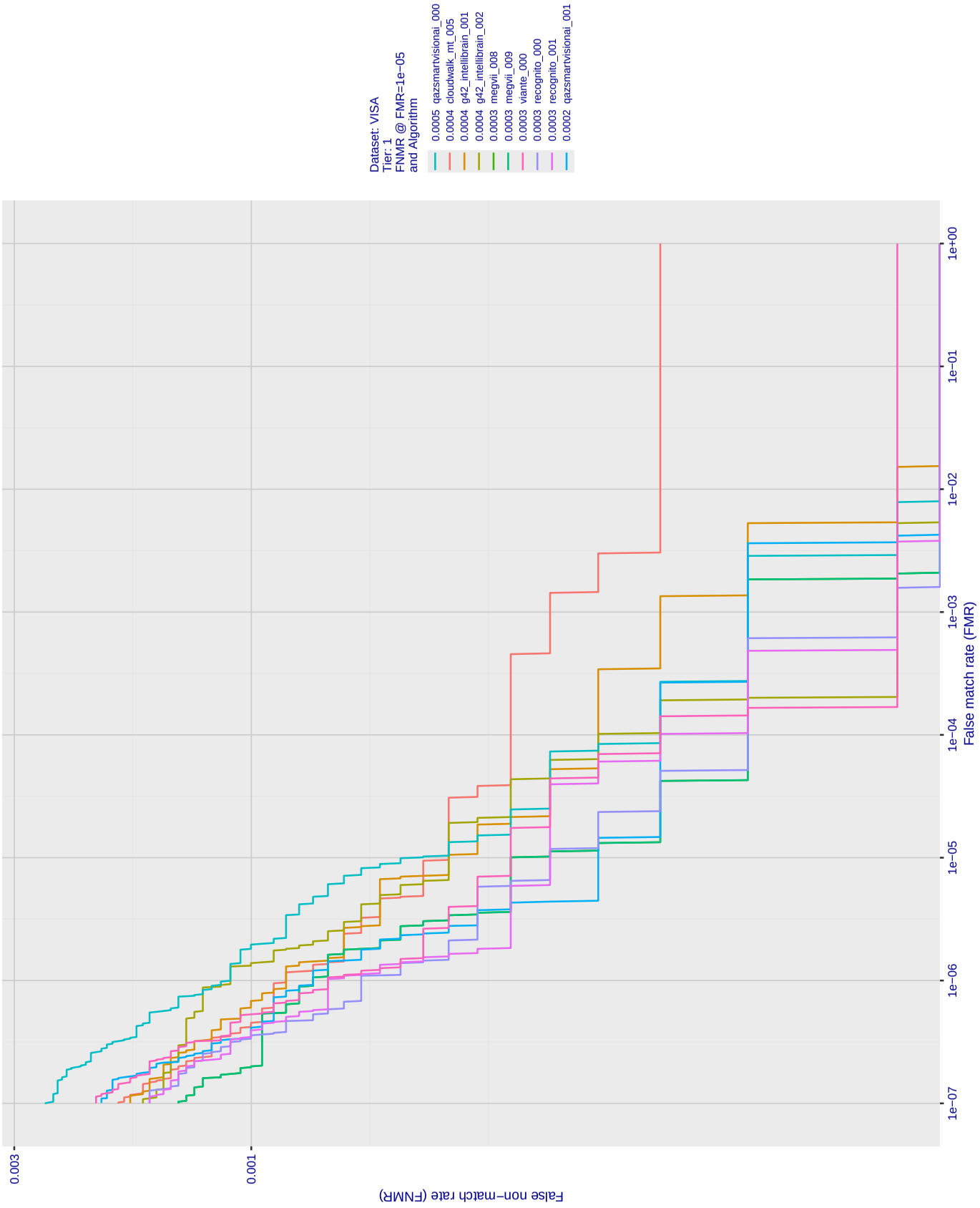


Figure 116: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

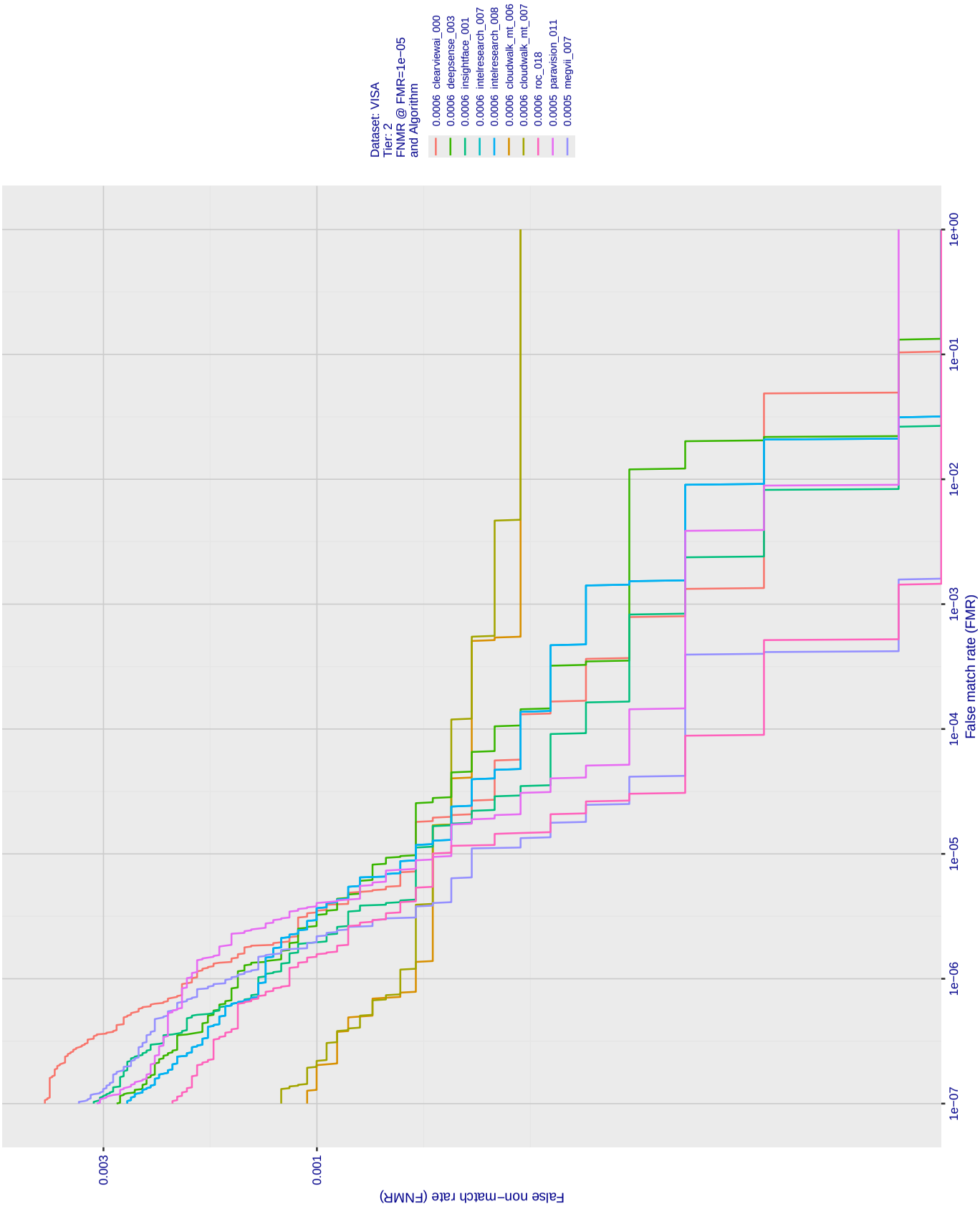


Figure 117: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

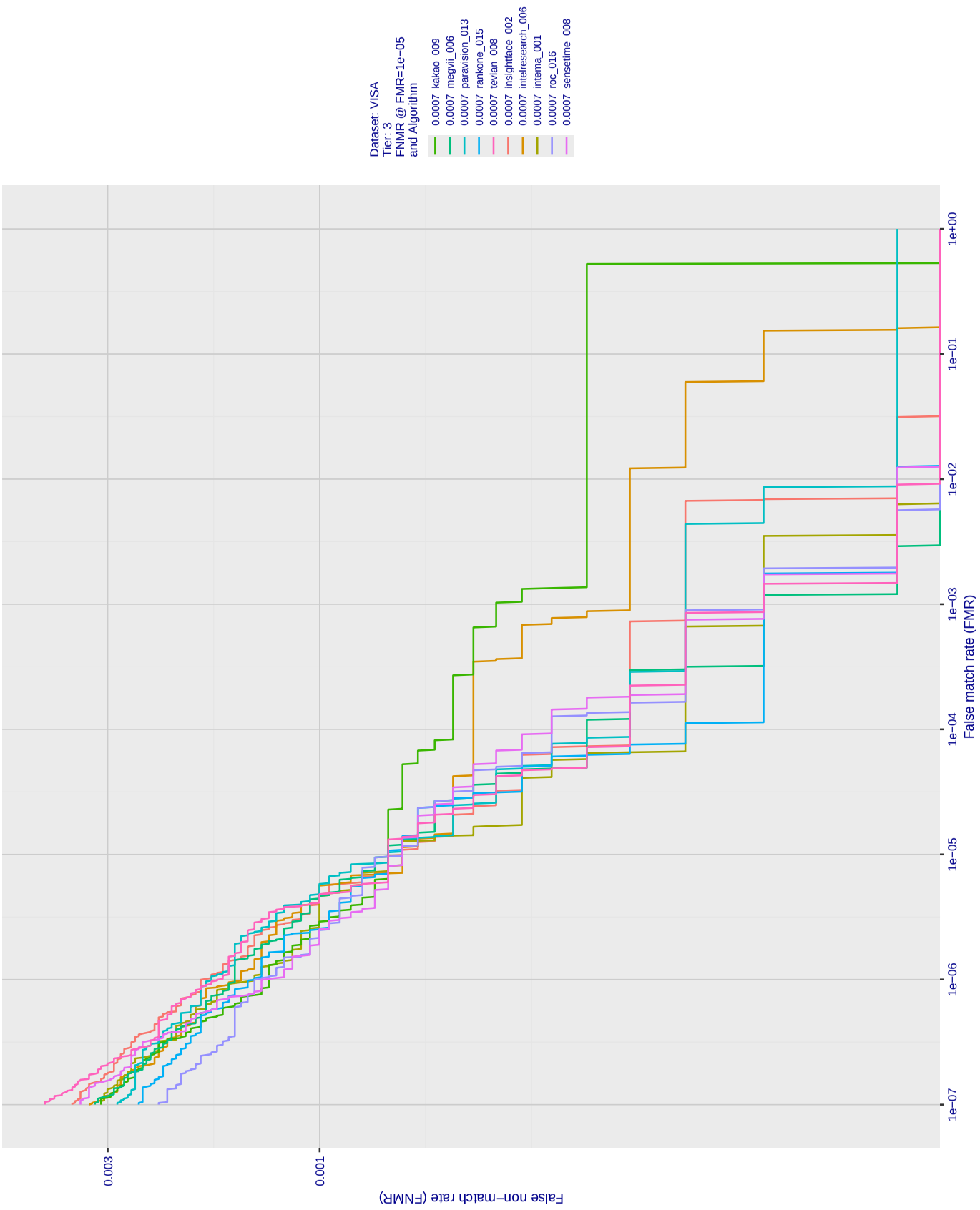


Figure 118: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

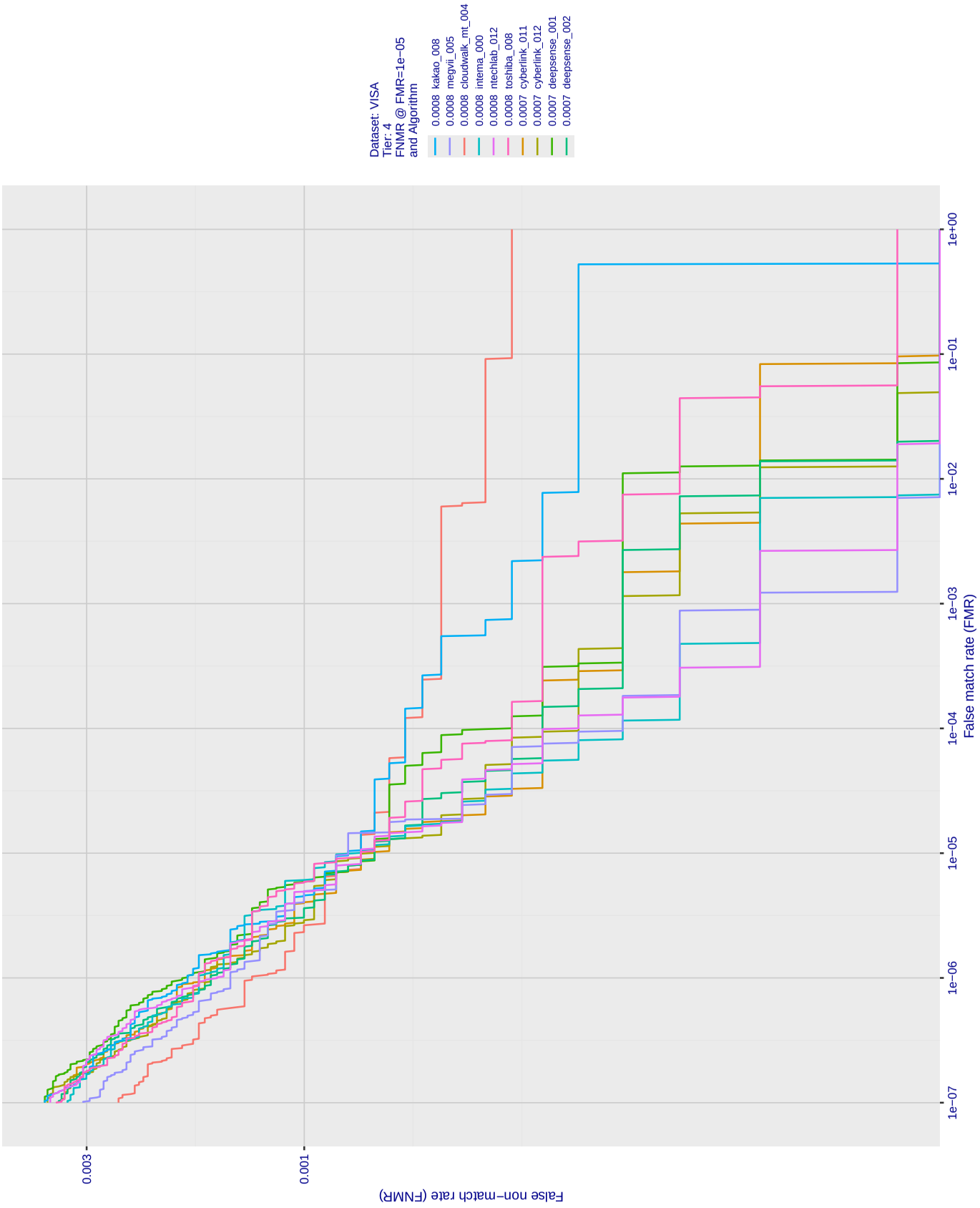


Figure 119: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

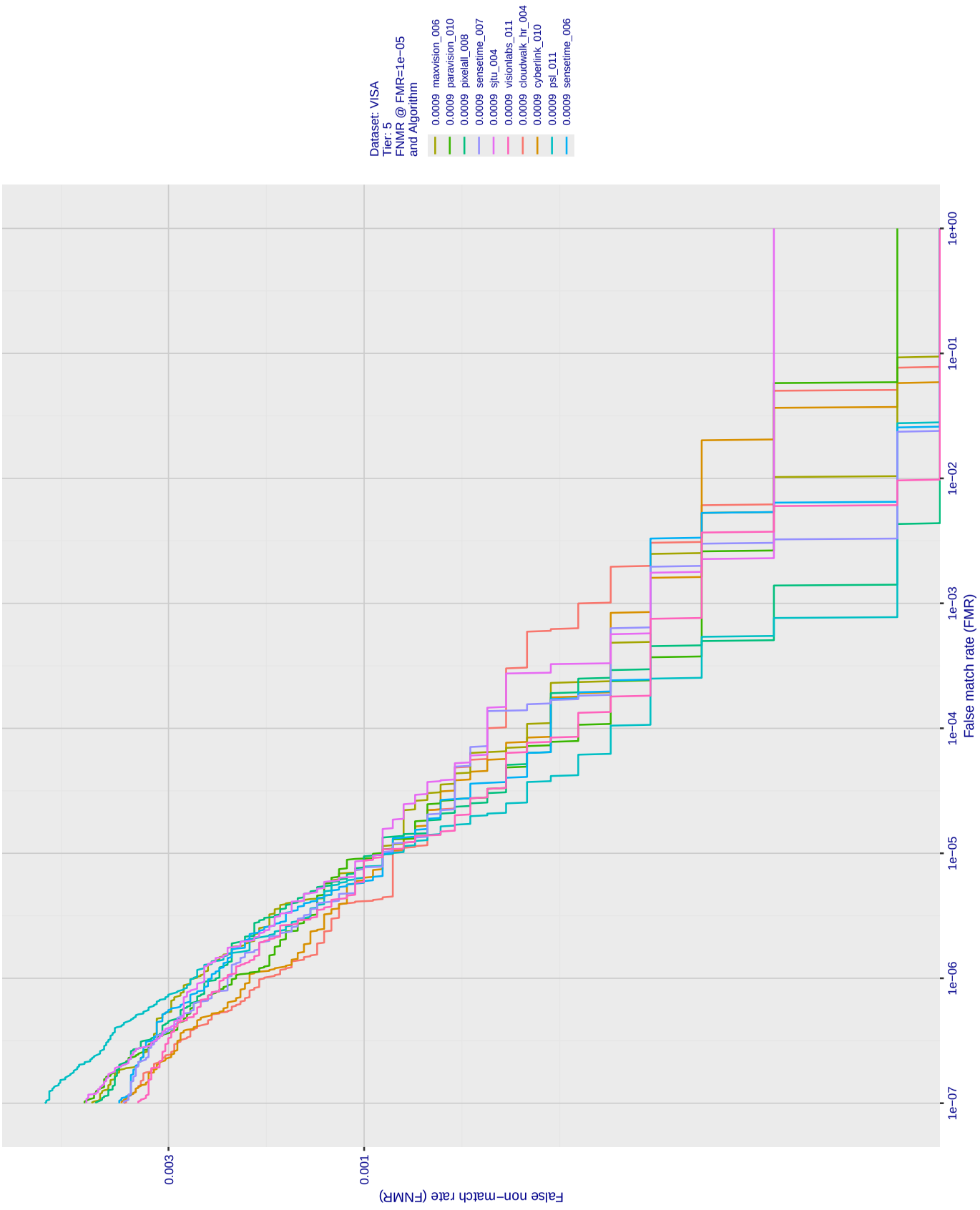


Figure 120: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

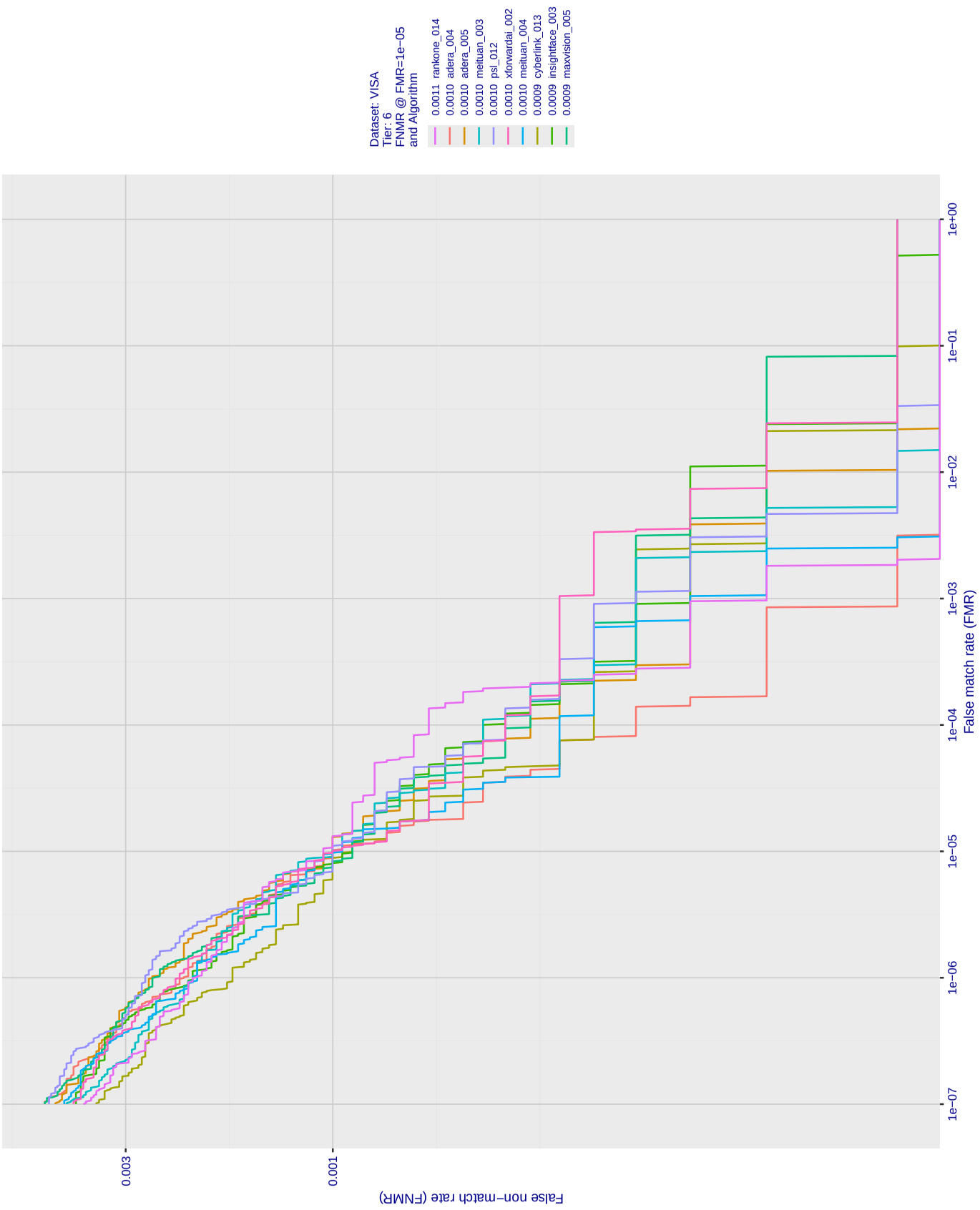


Figure 121: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

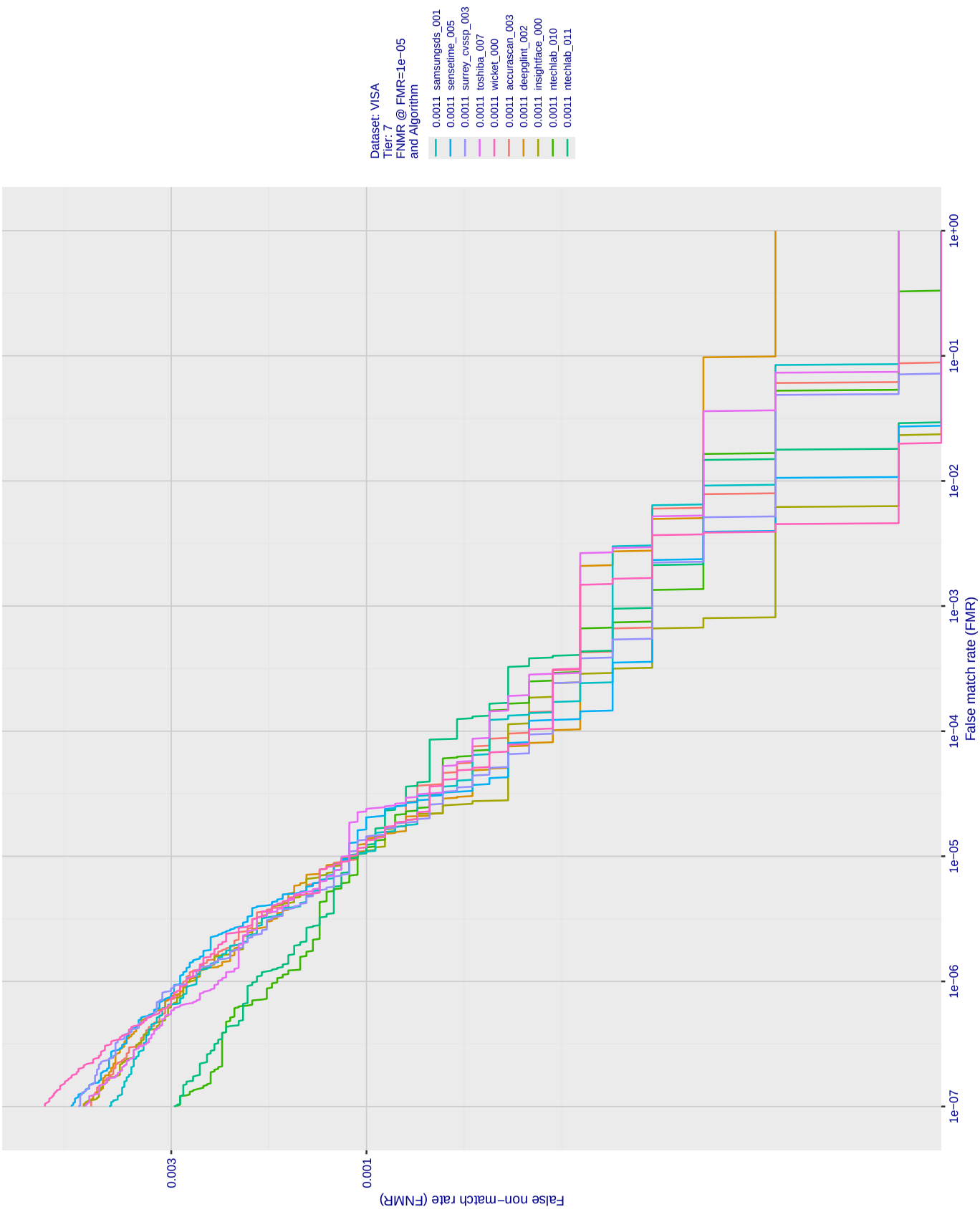


Figure 122: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

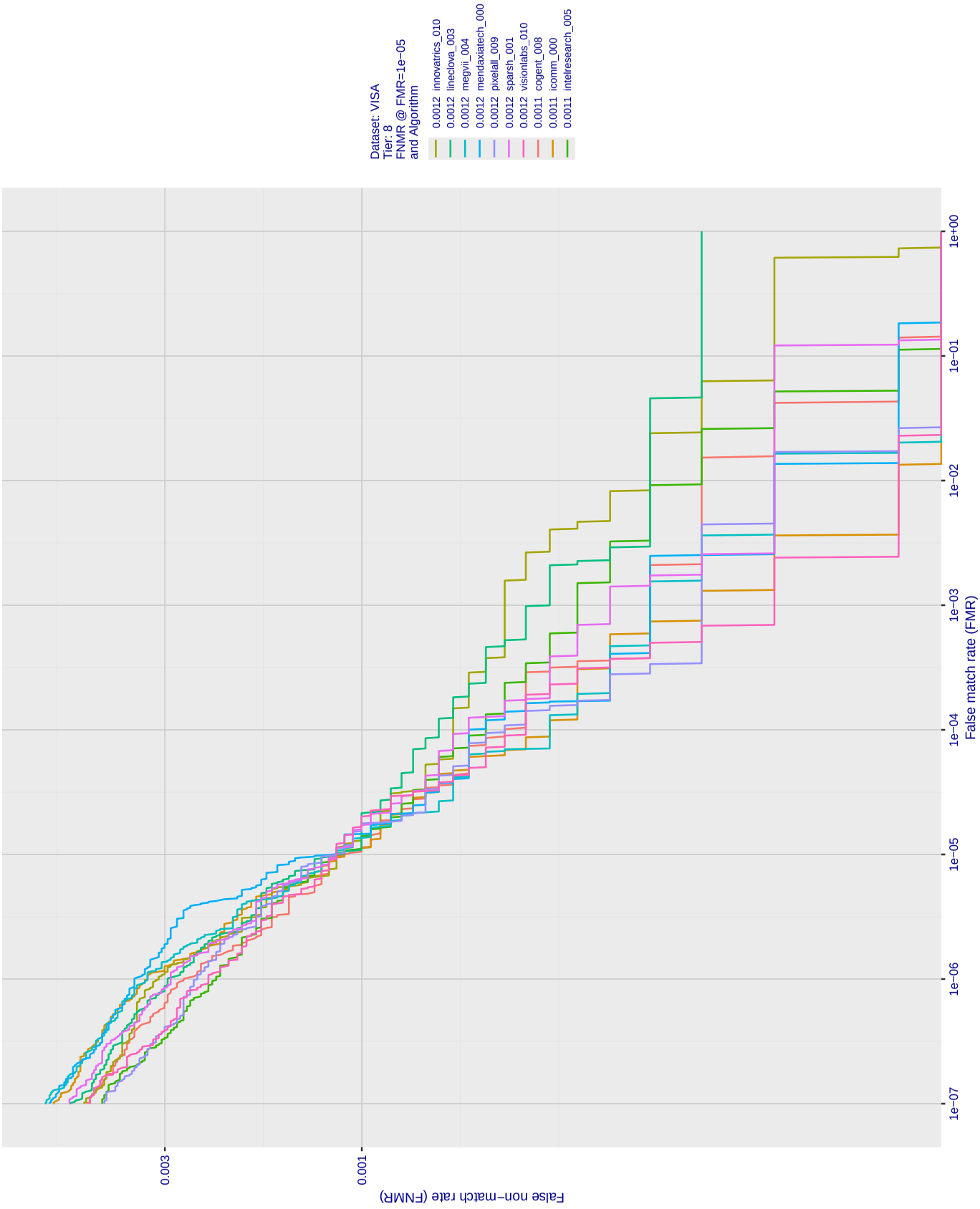


Figure 123: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

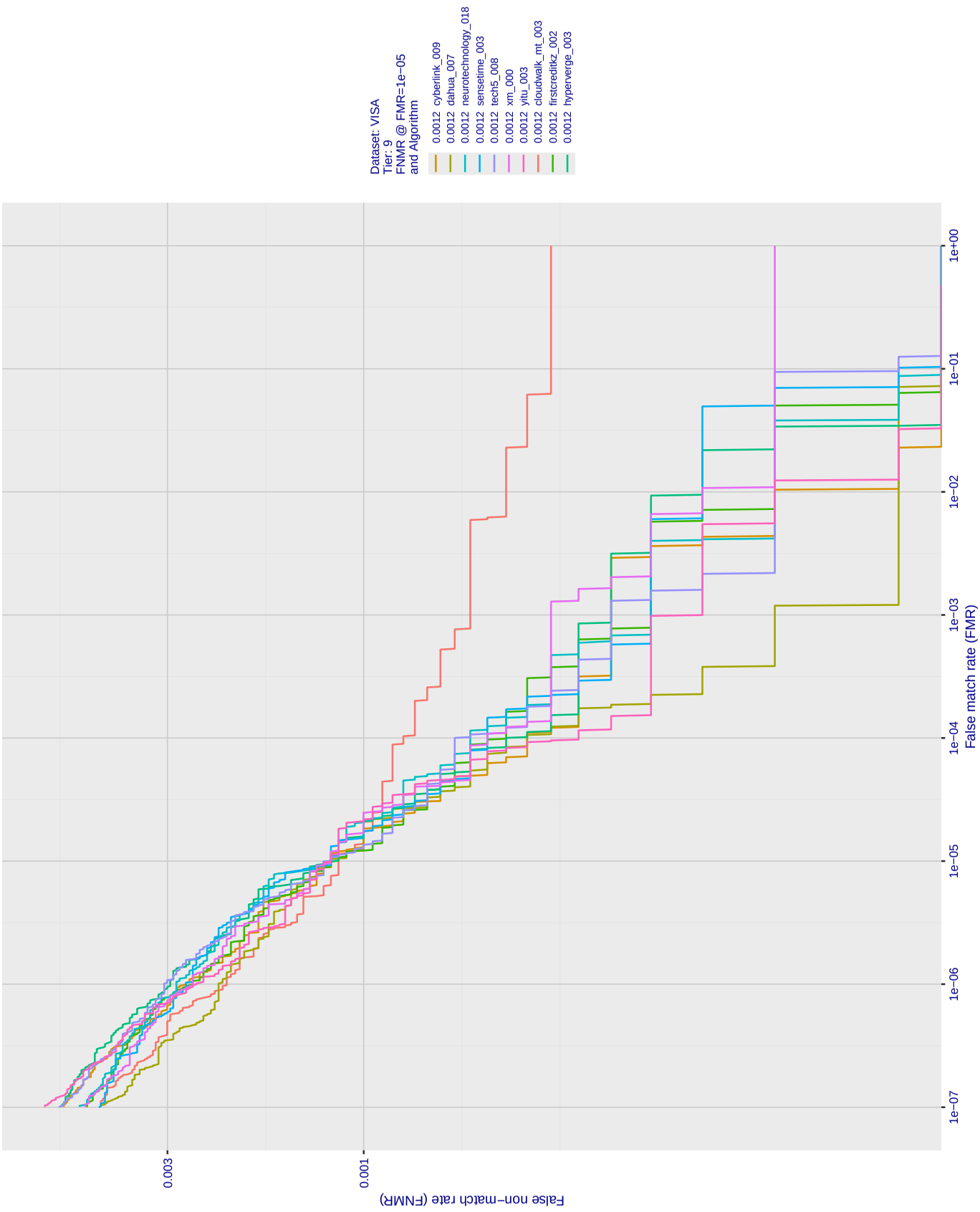


Figure 124: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

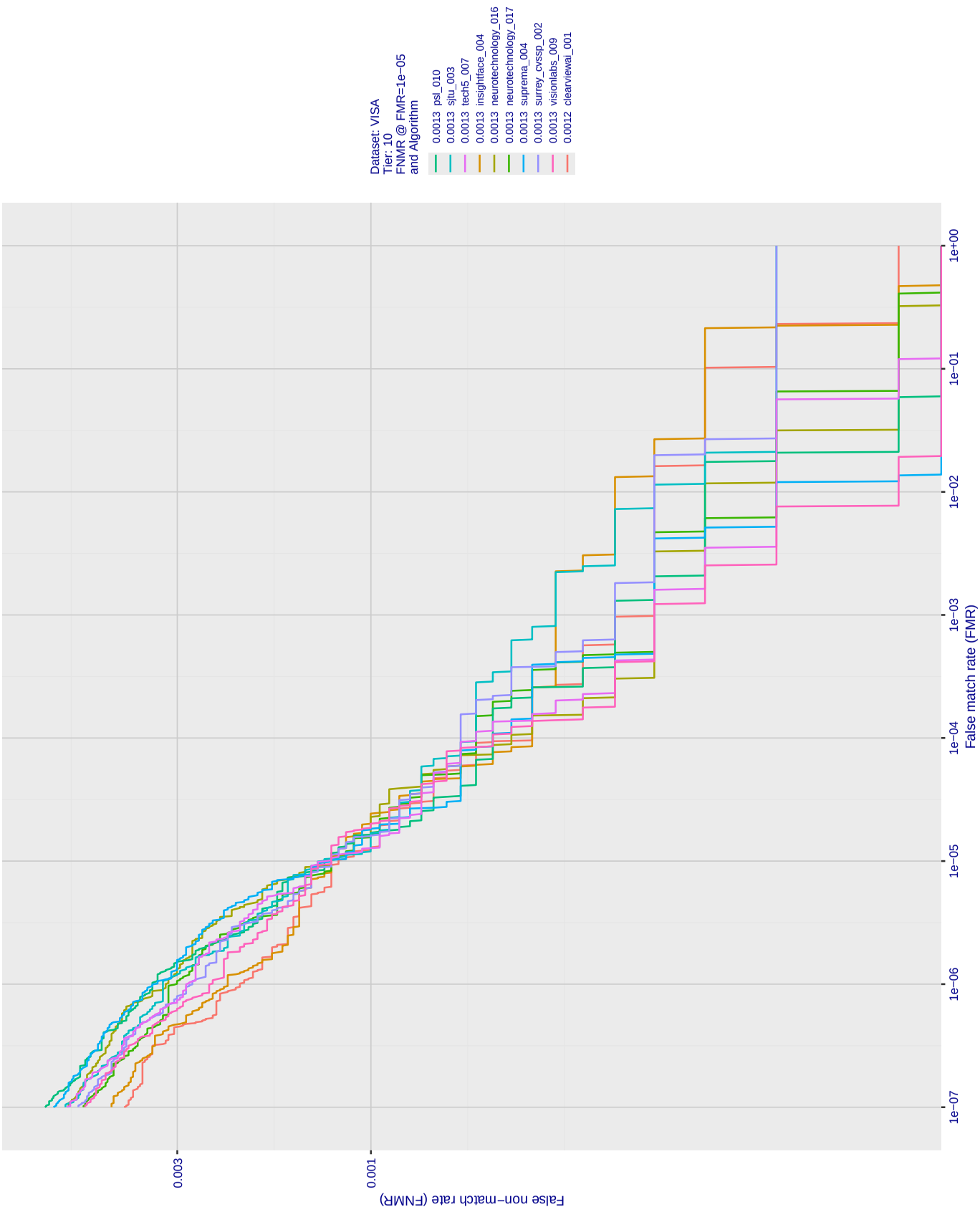


Figure 125: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

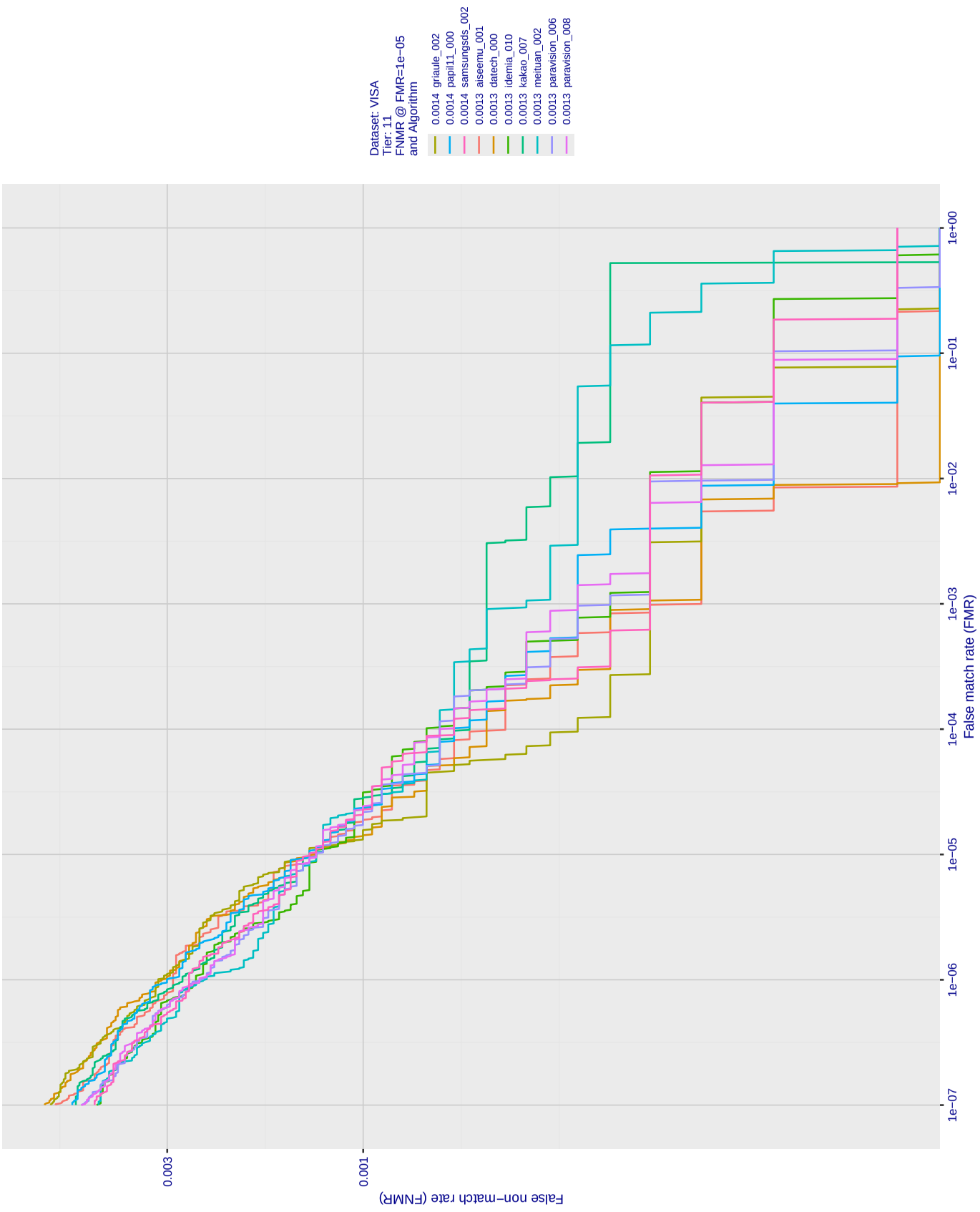


Figure 126: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

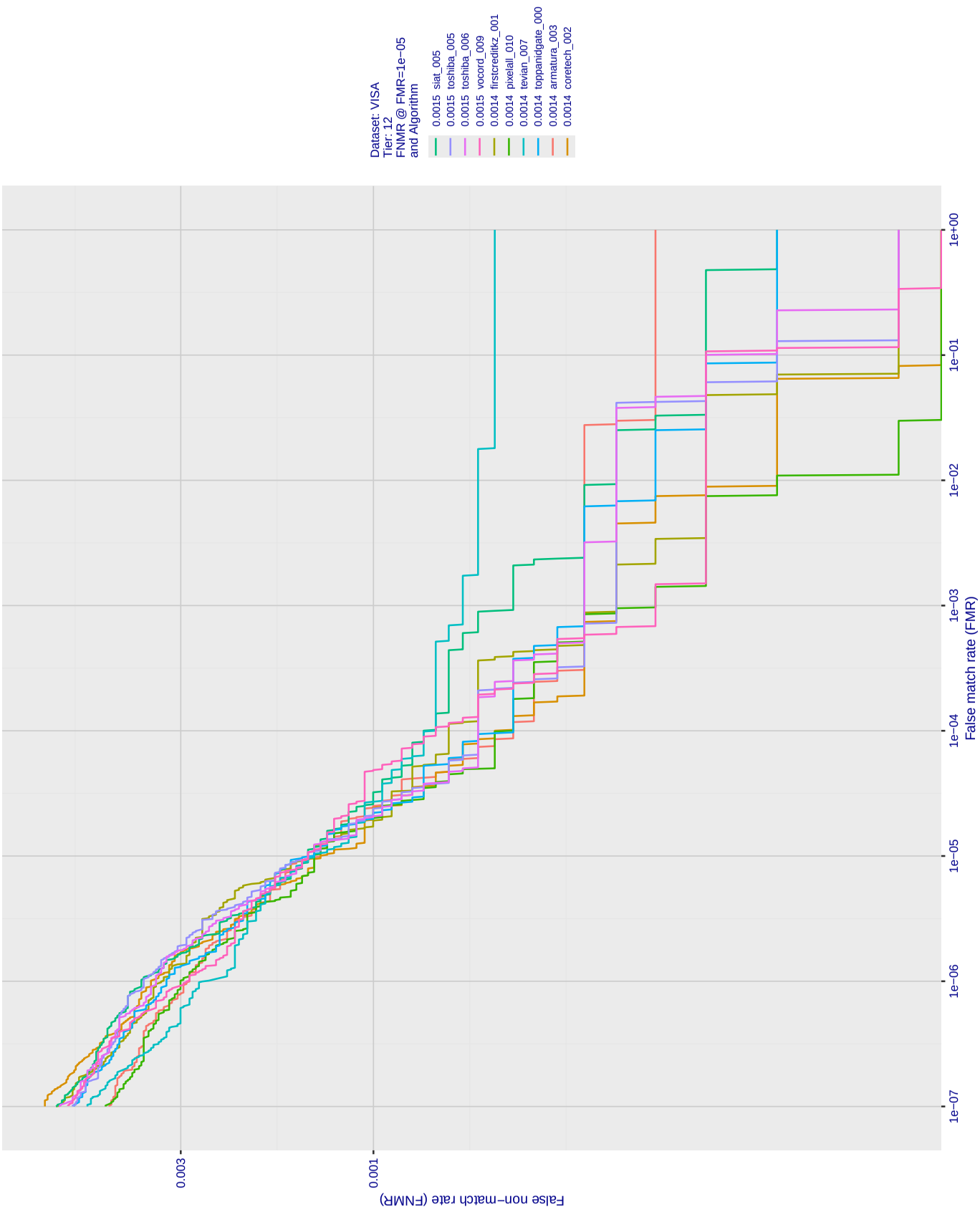


Figure 127: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

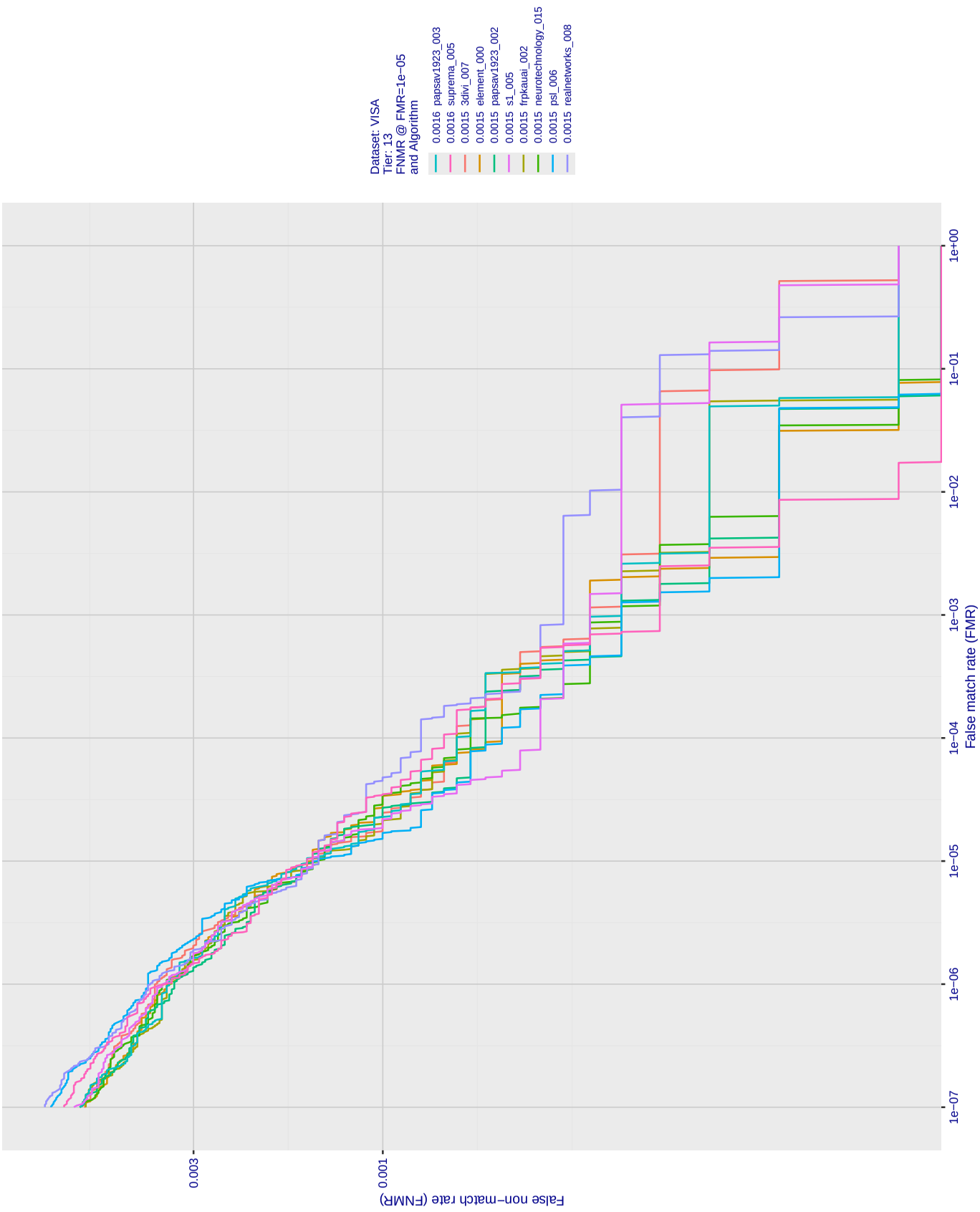


Figure 128: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

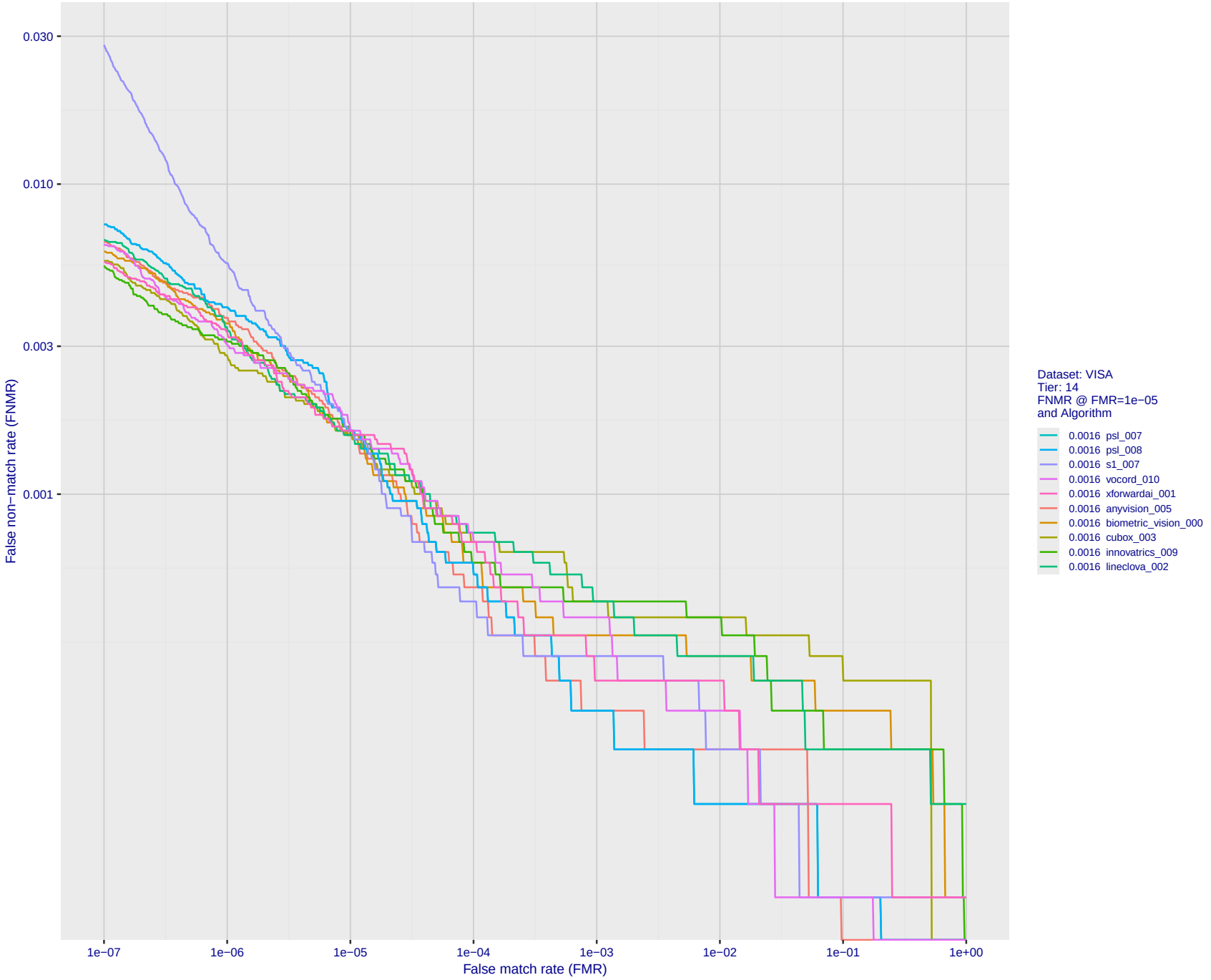


Figure 129: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

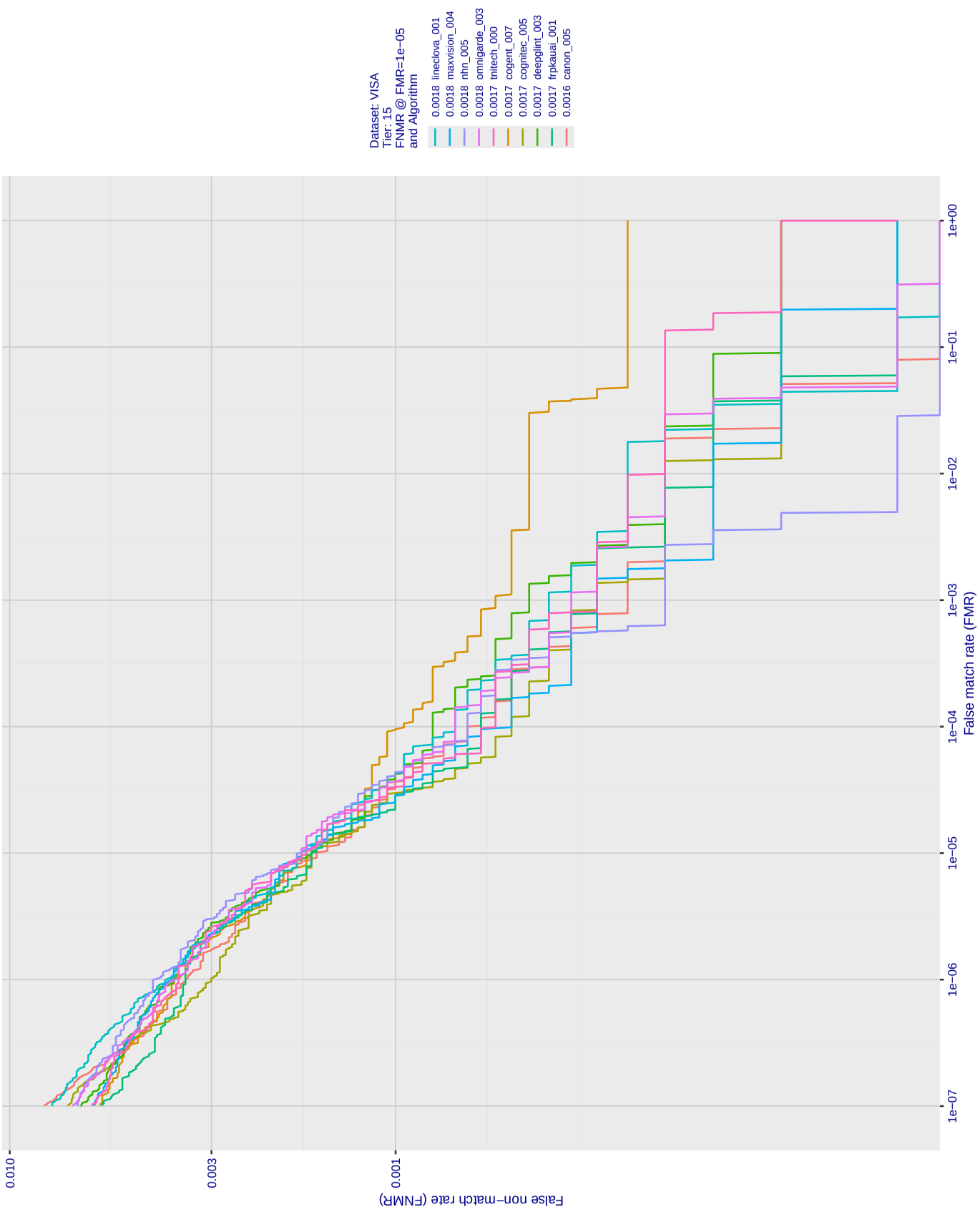


Figure 130: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

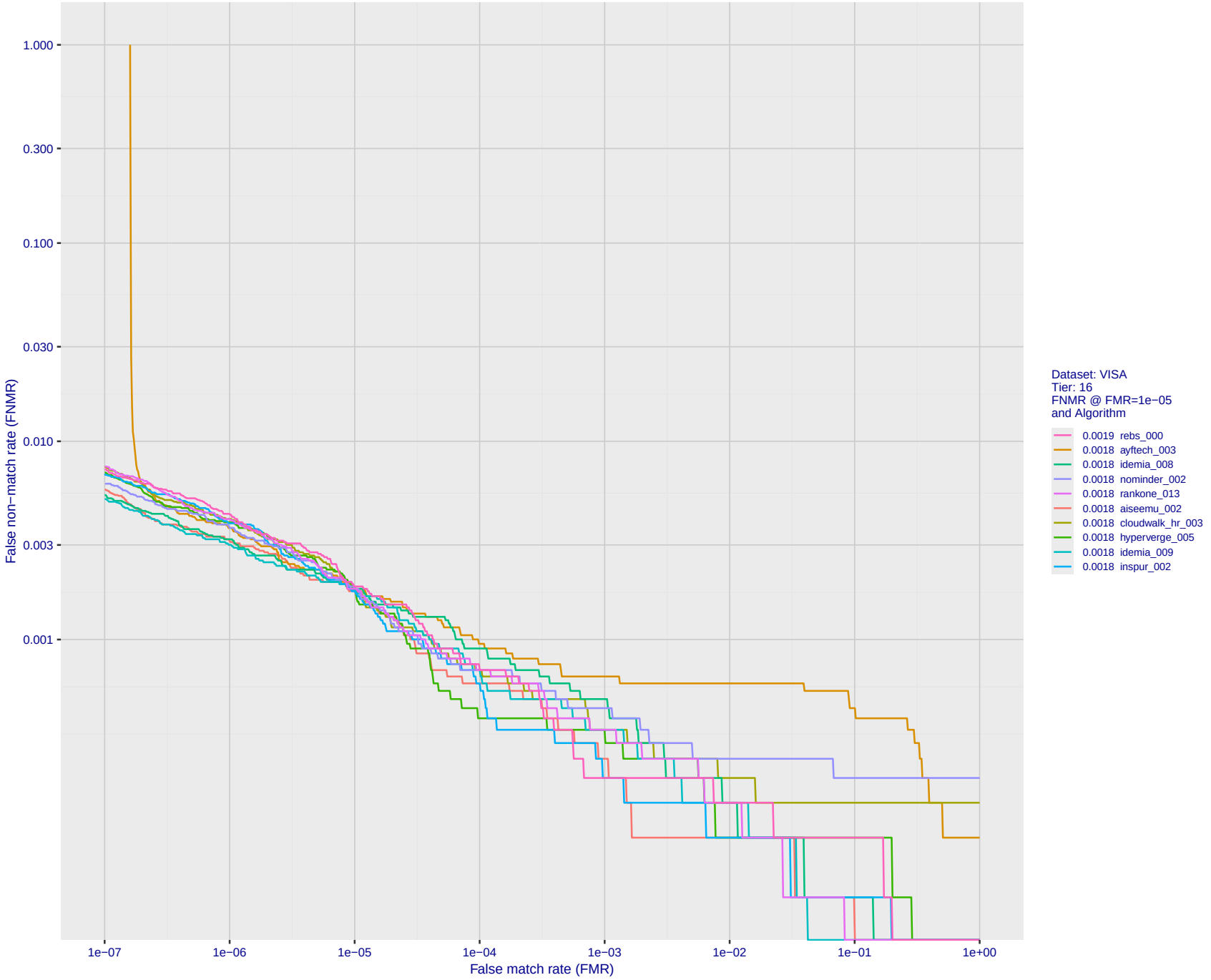


Figure 131: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

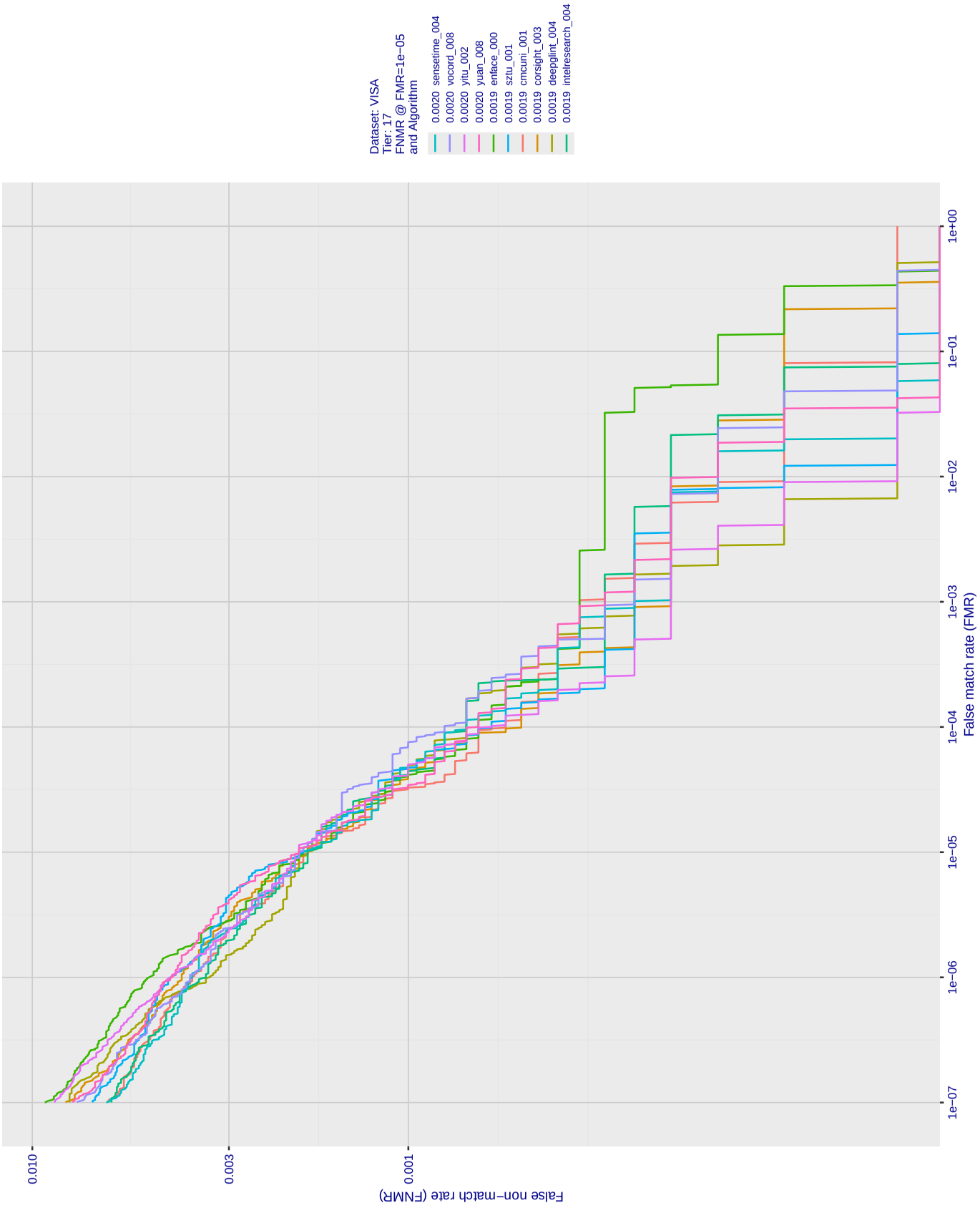


Figure 132: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

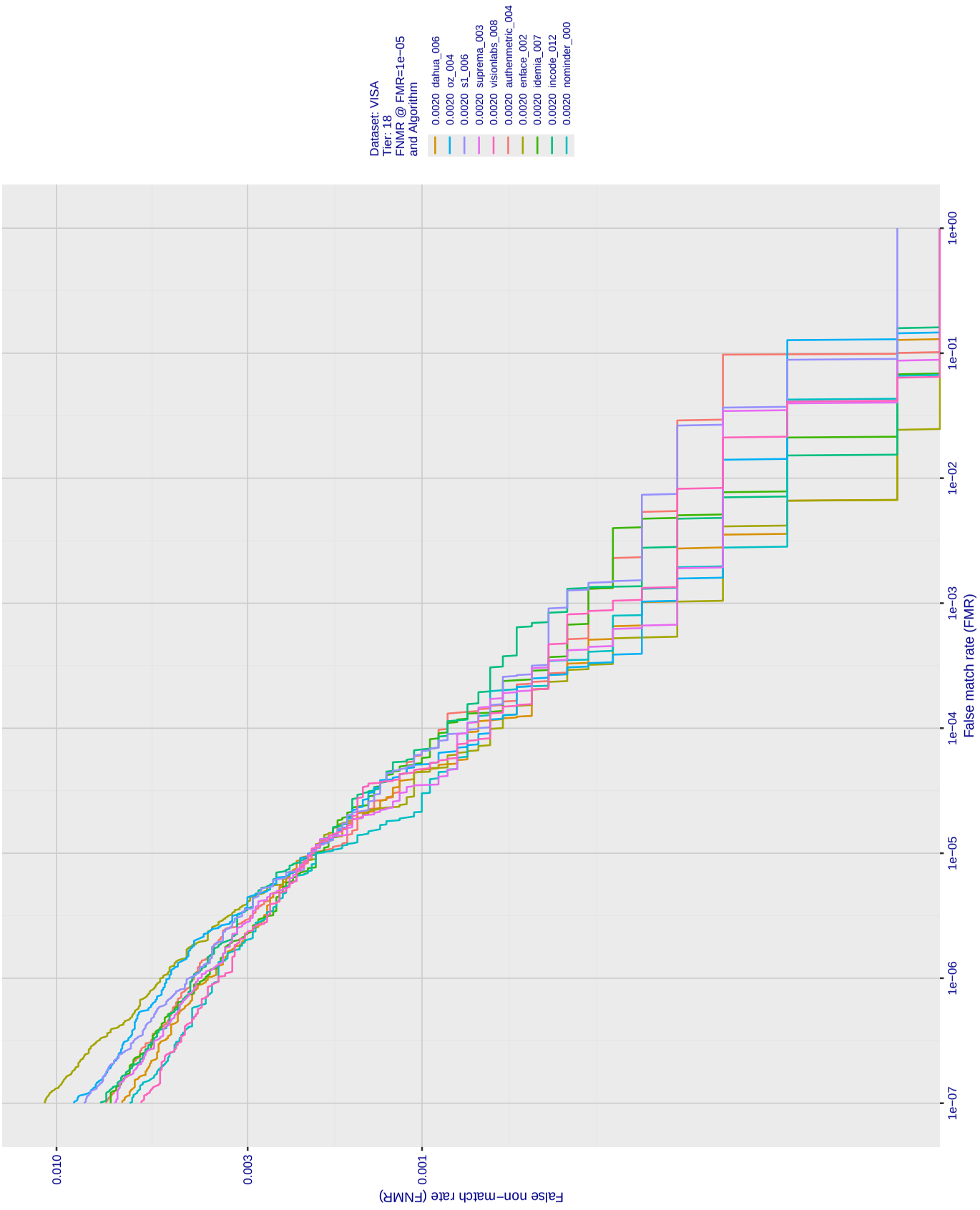


Figure 133: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

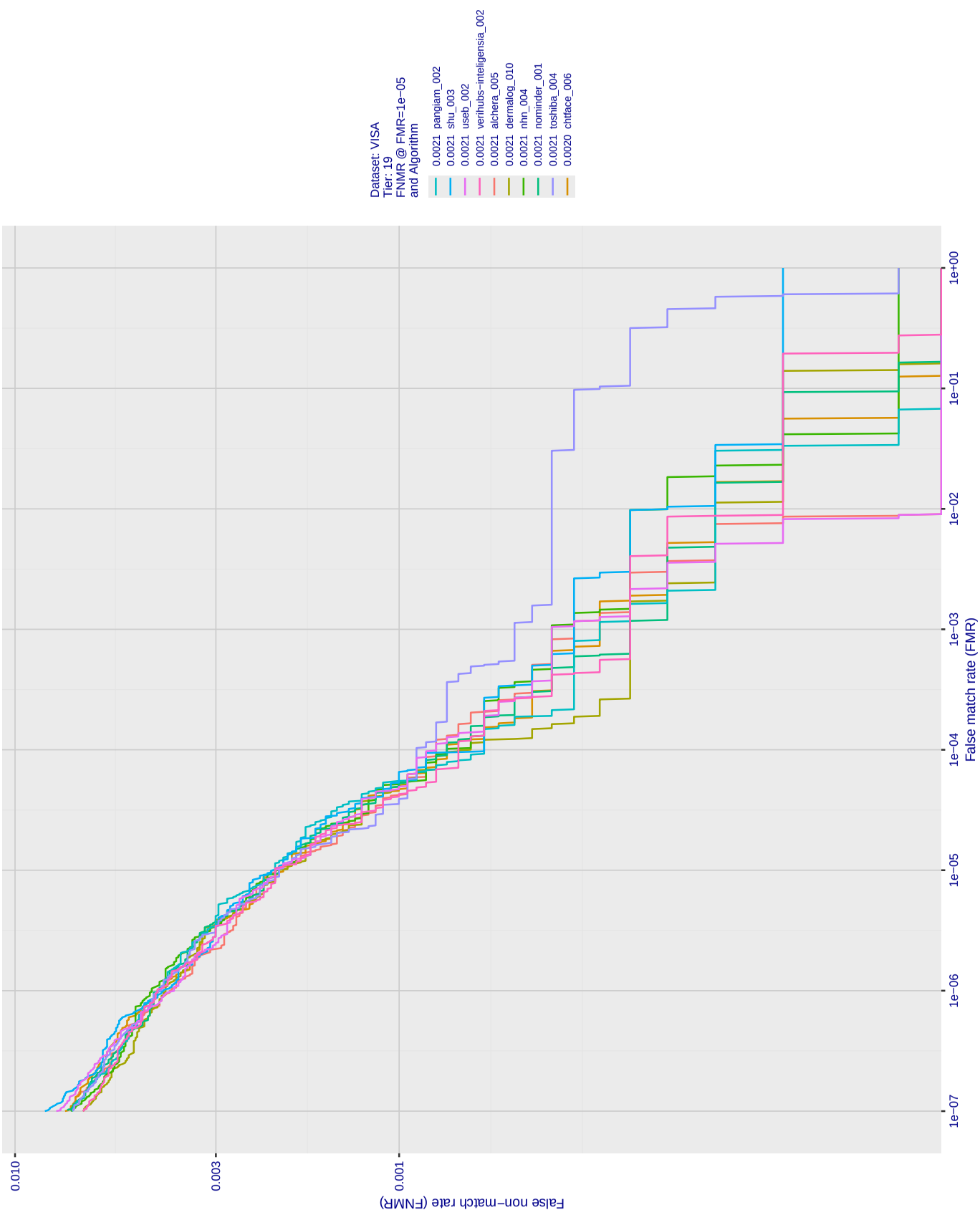


Figure 134: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

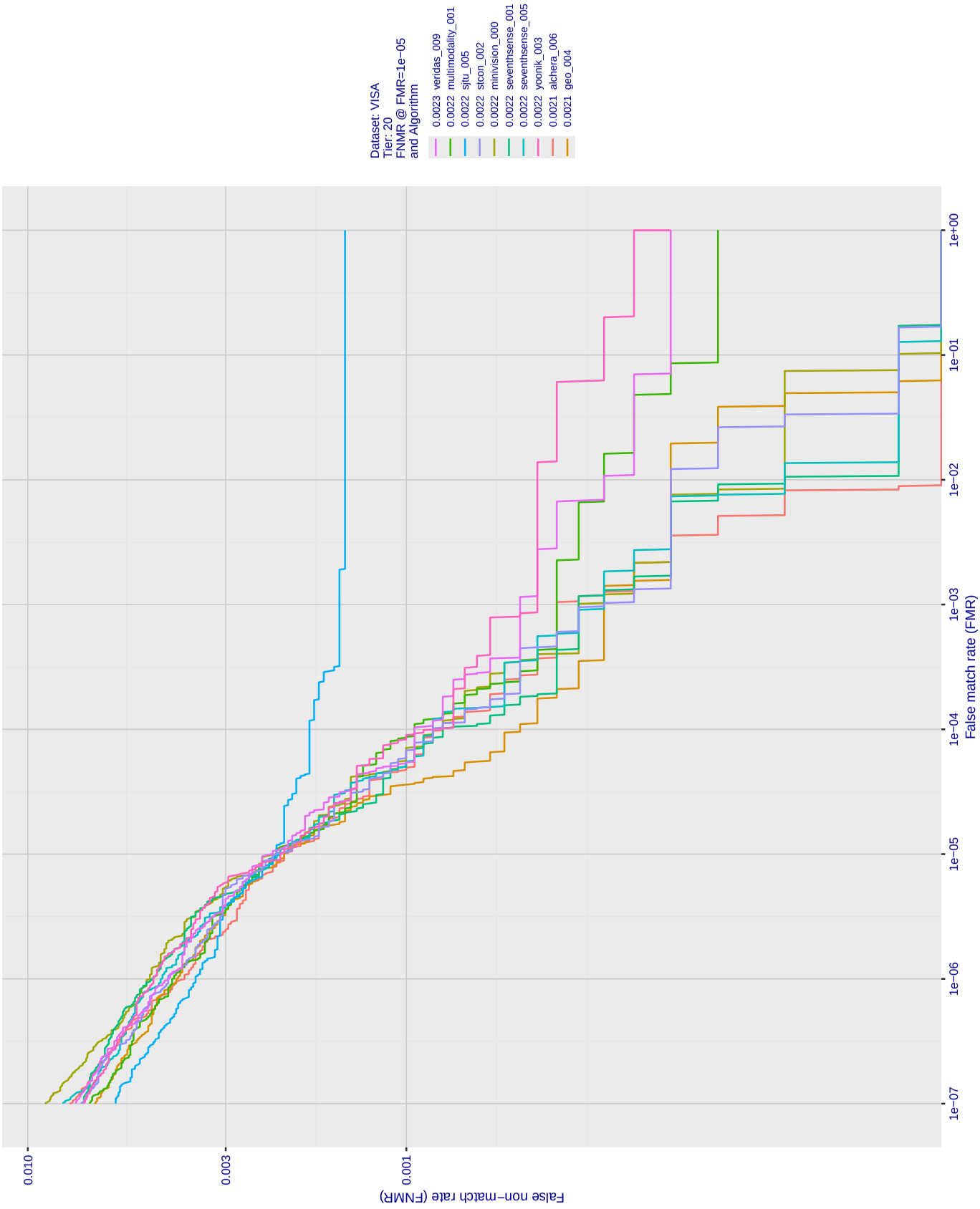


Figure 135: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

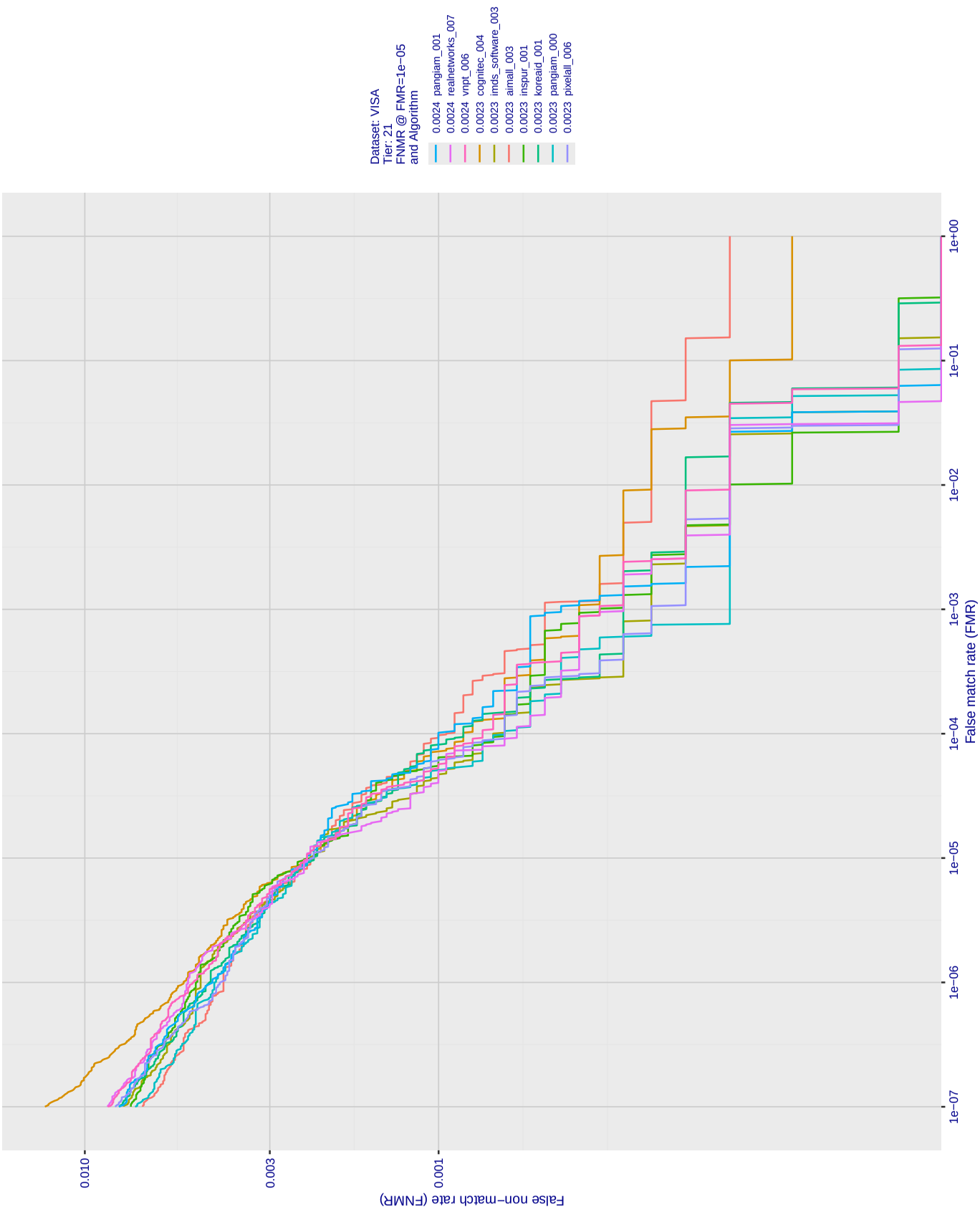


Figure 136: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

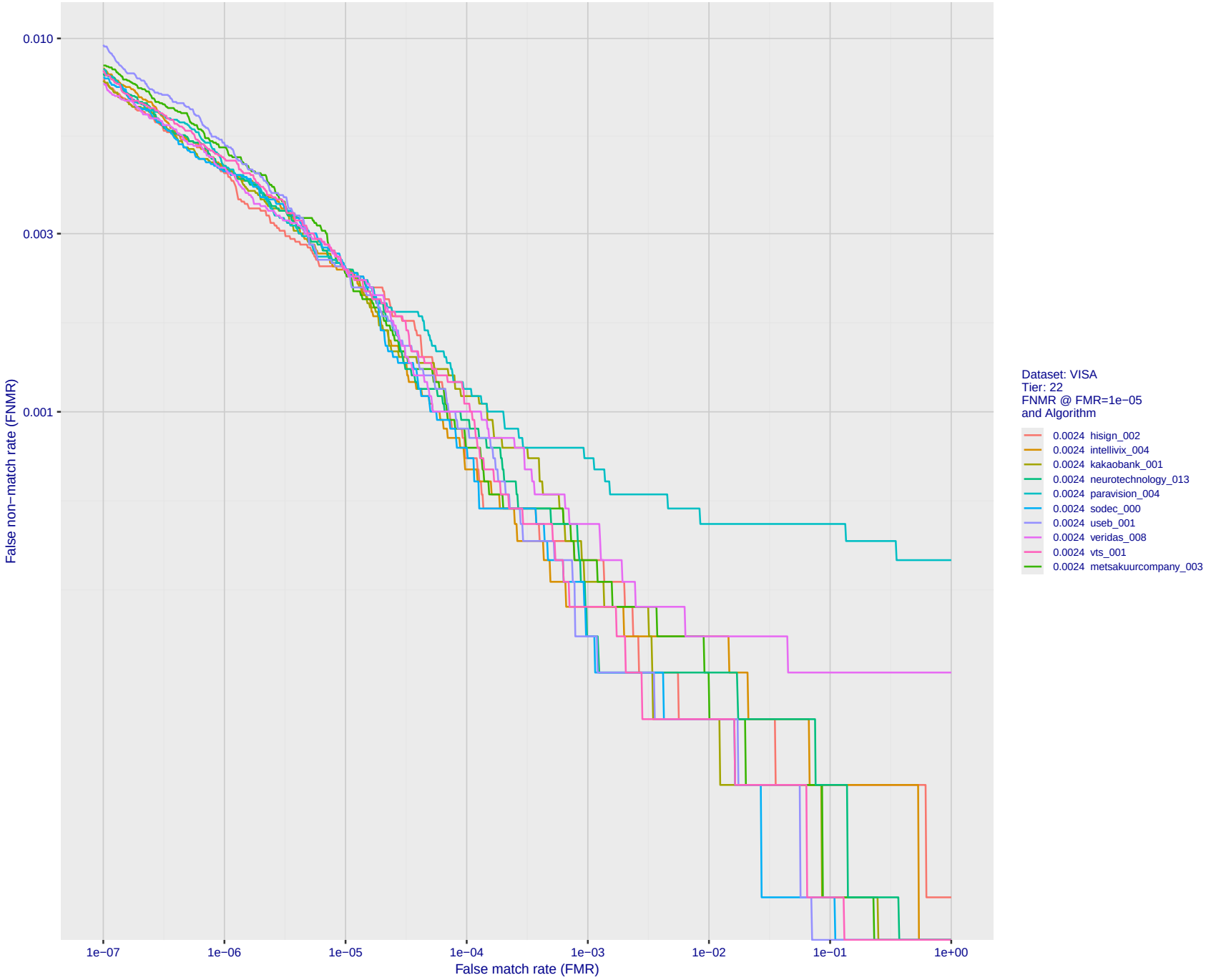


Figure 137: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

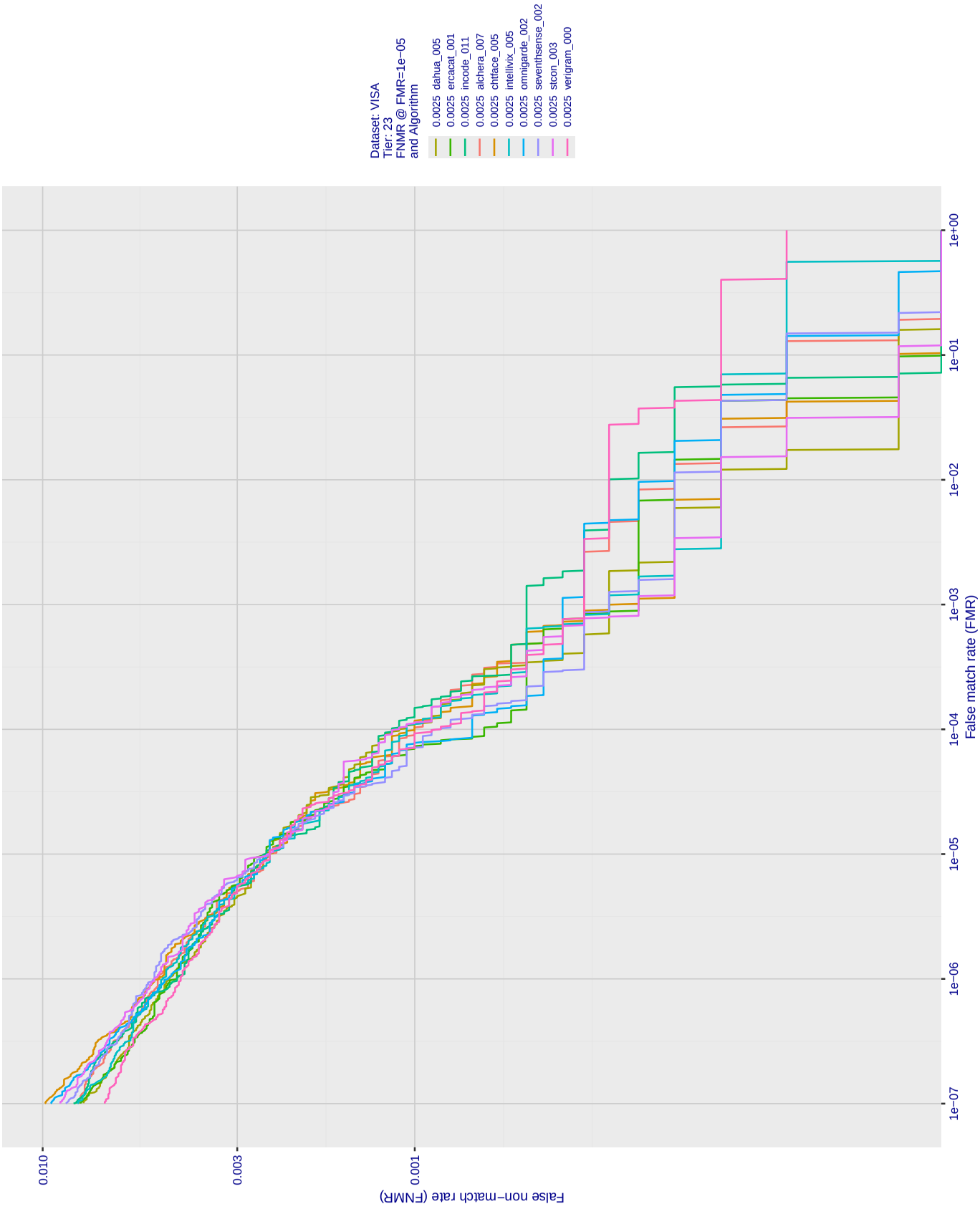


Figure 138: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

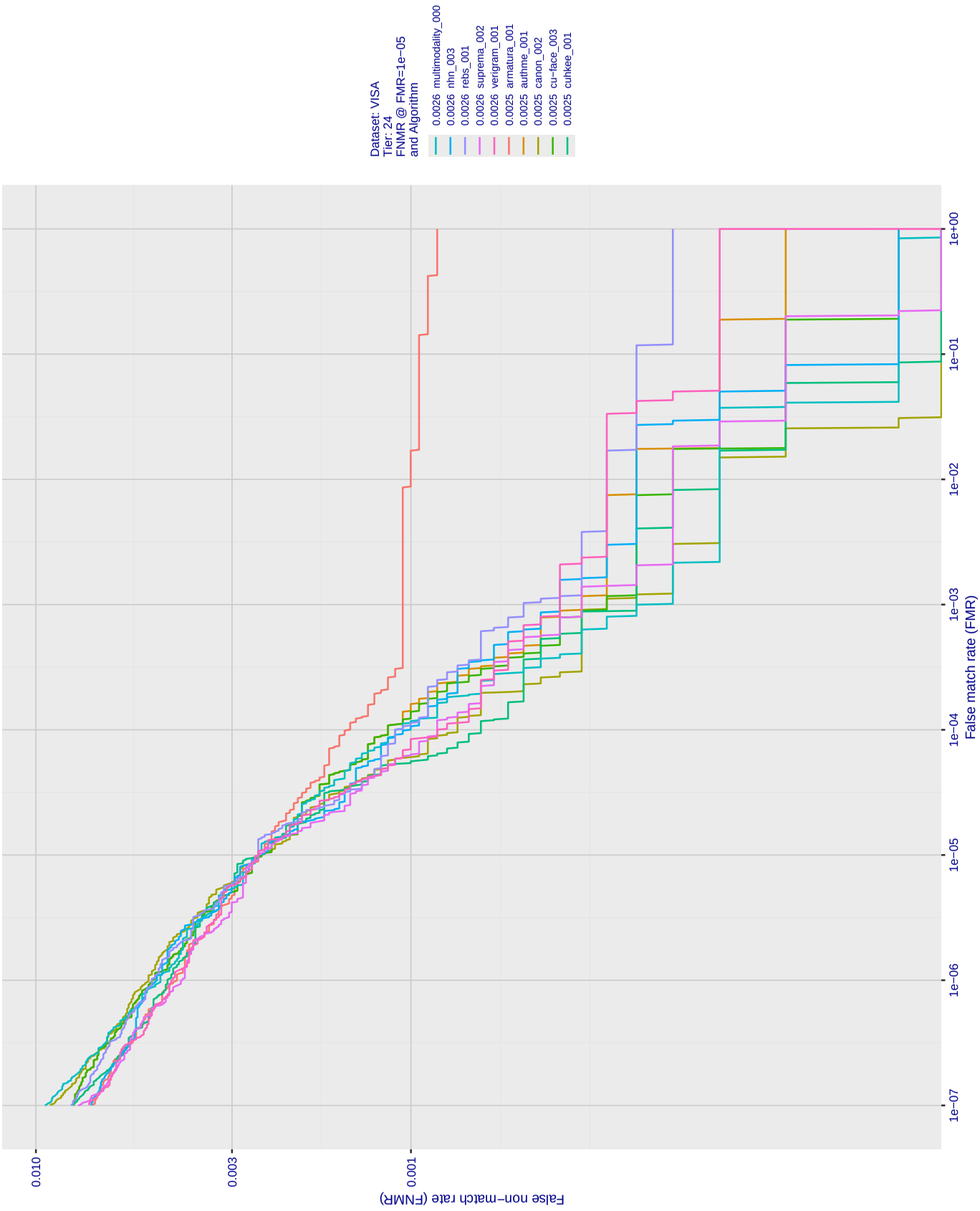


Figure 139: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

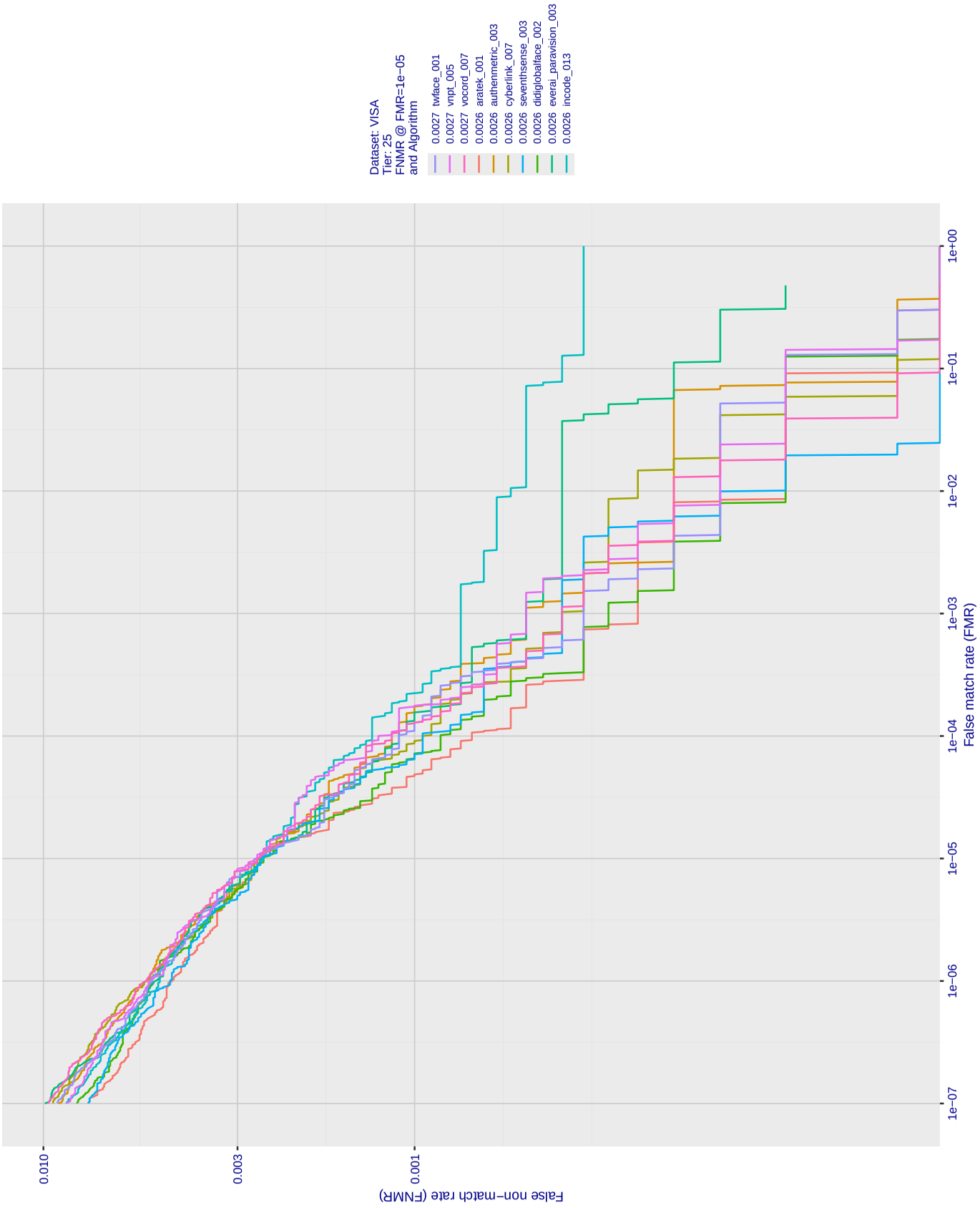


Figure 140: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

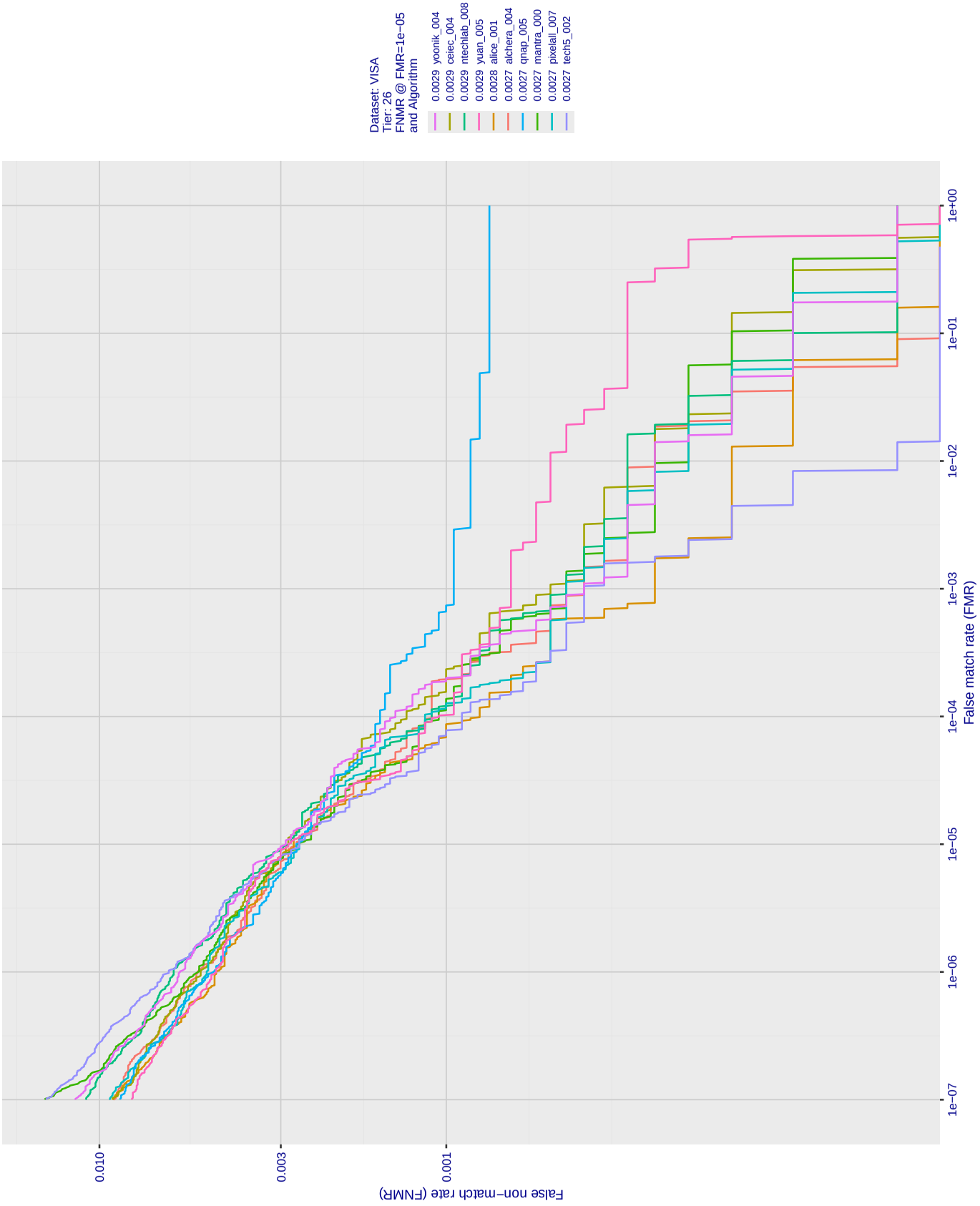


Figure 141: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

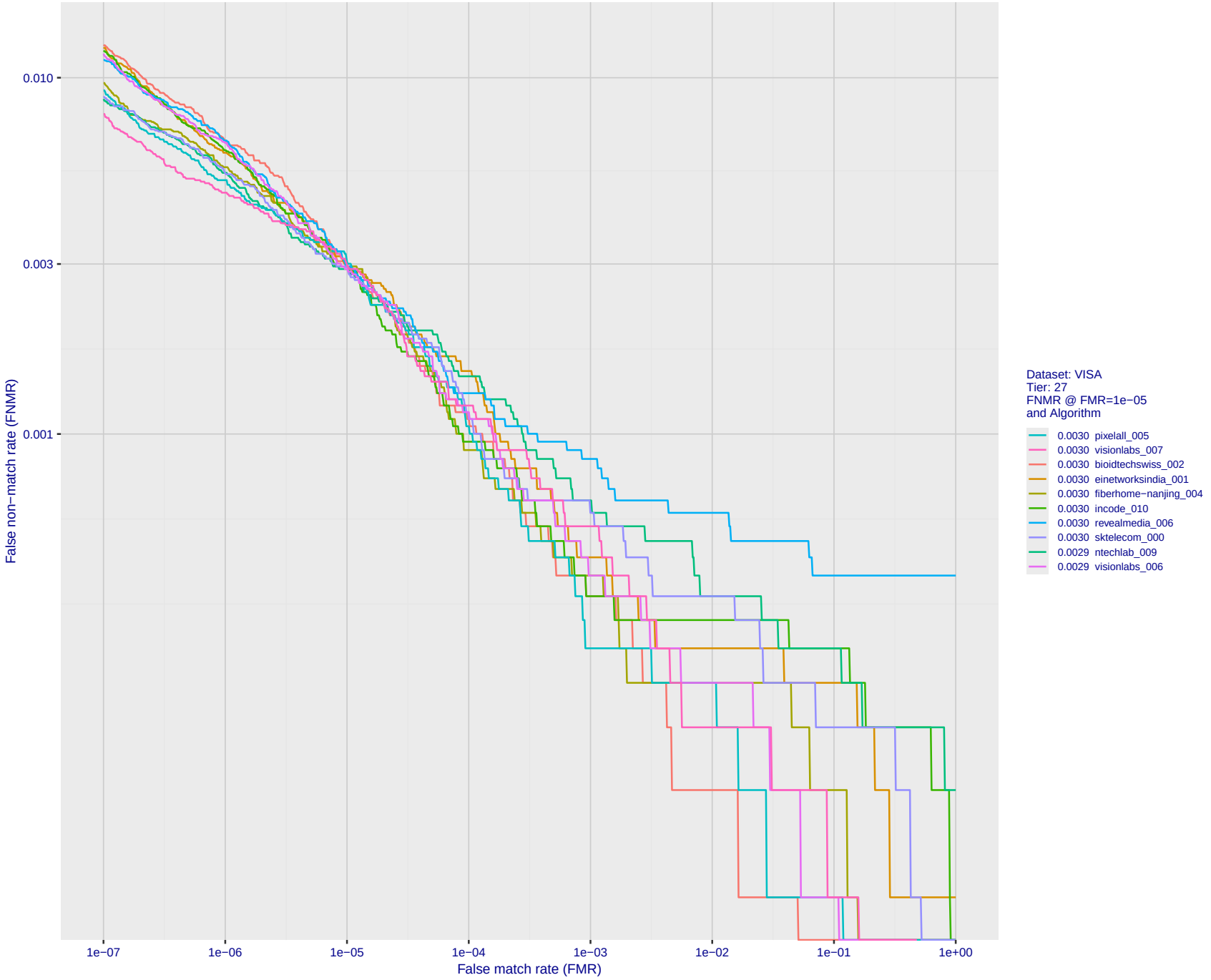


Figure 142: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

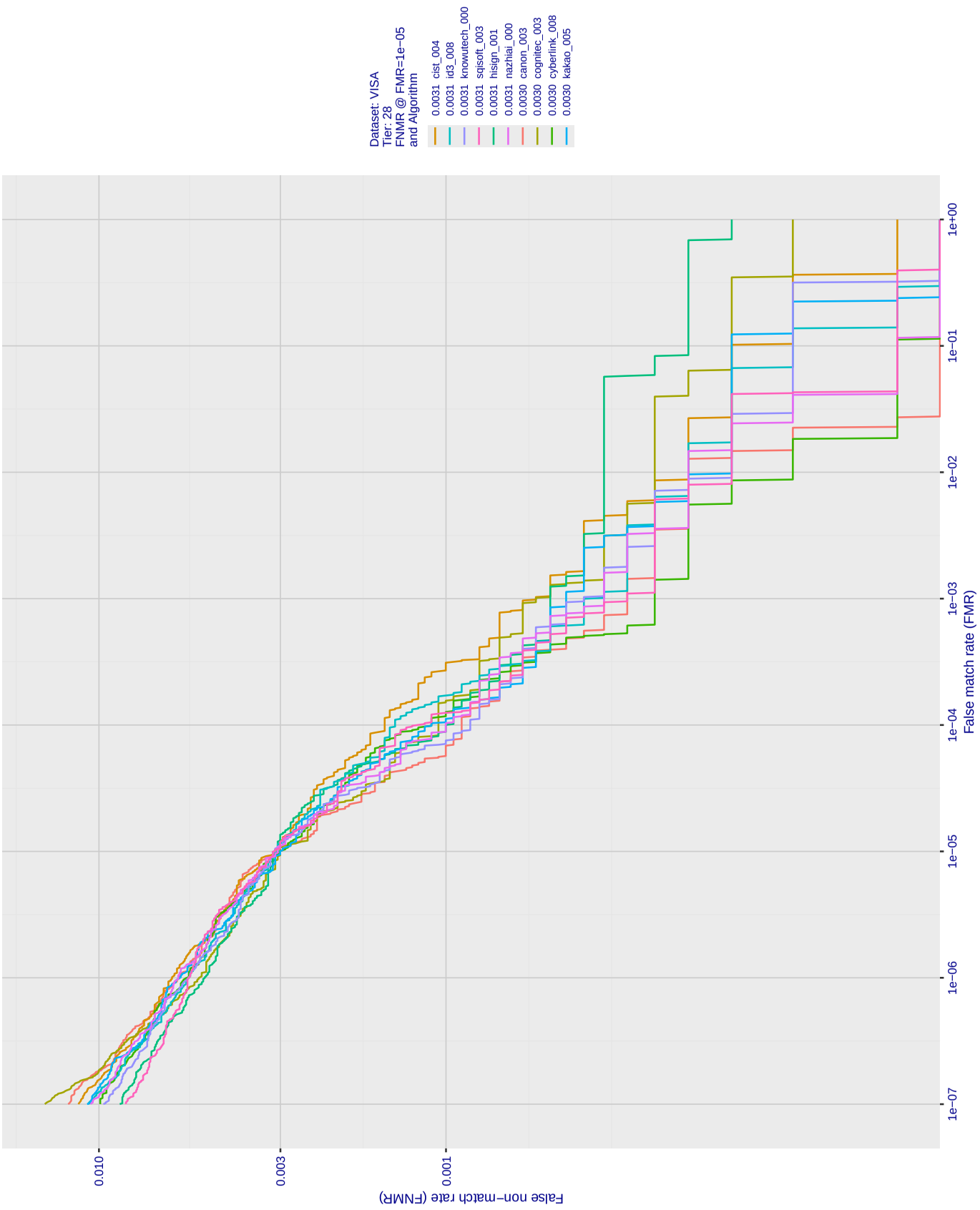


Figure 143: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

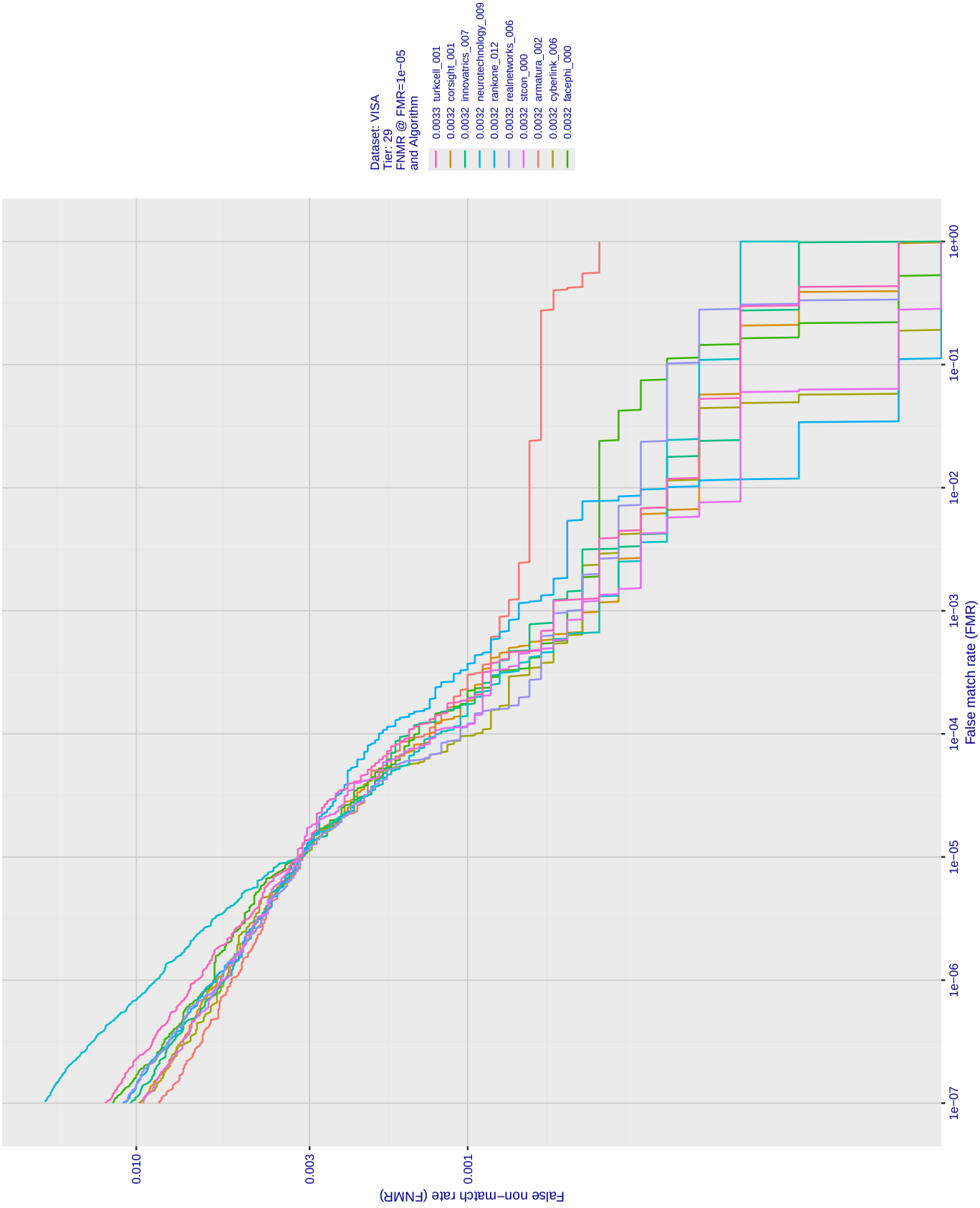


Figure 144: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

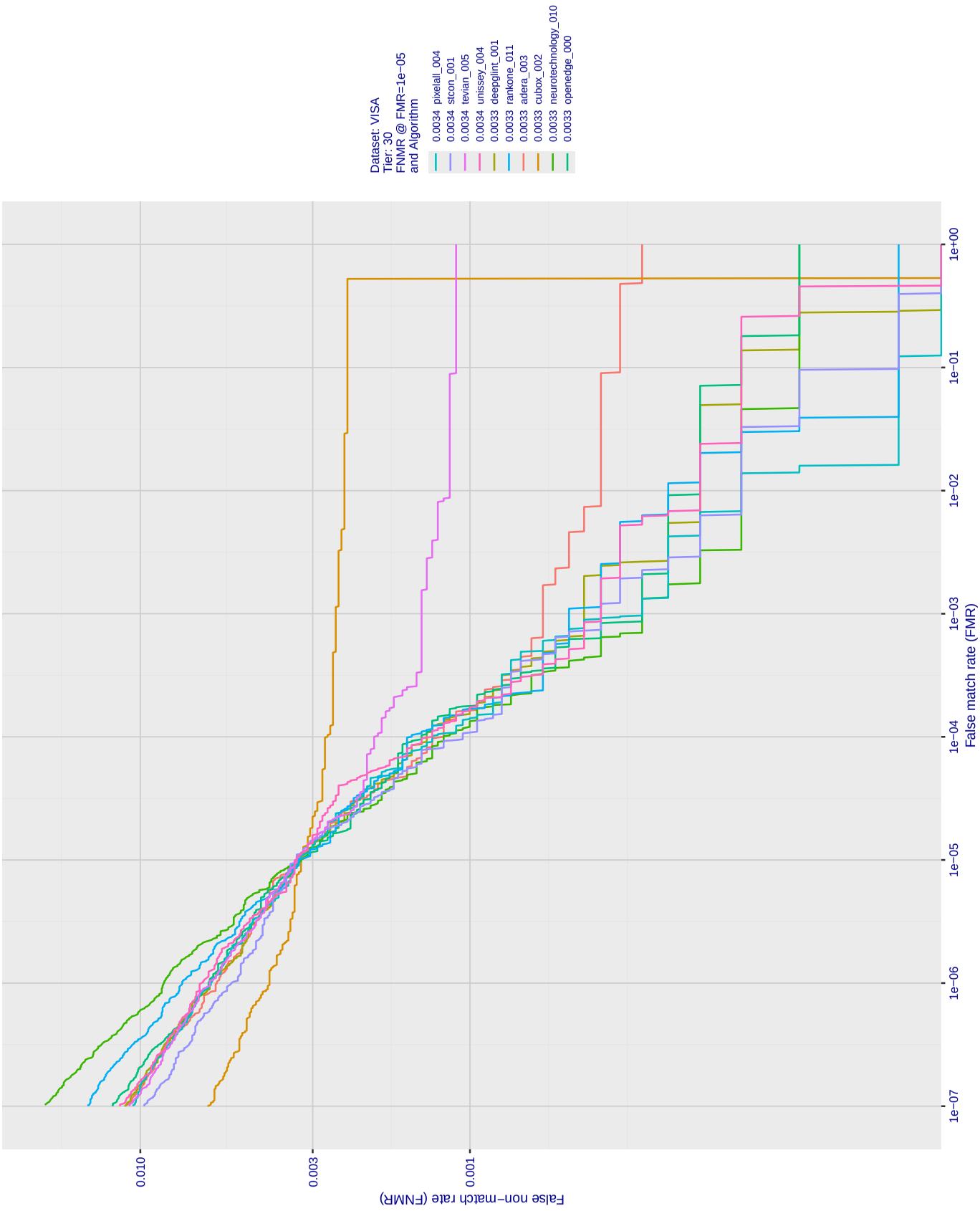


Figure 145: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

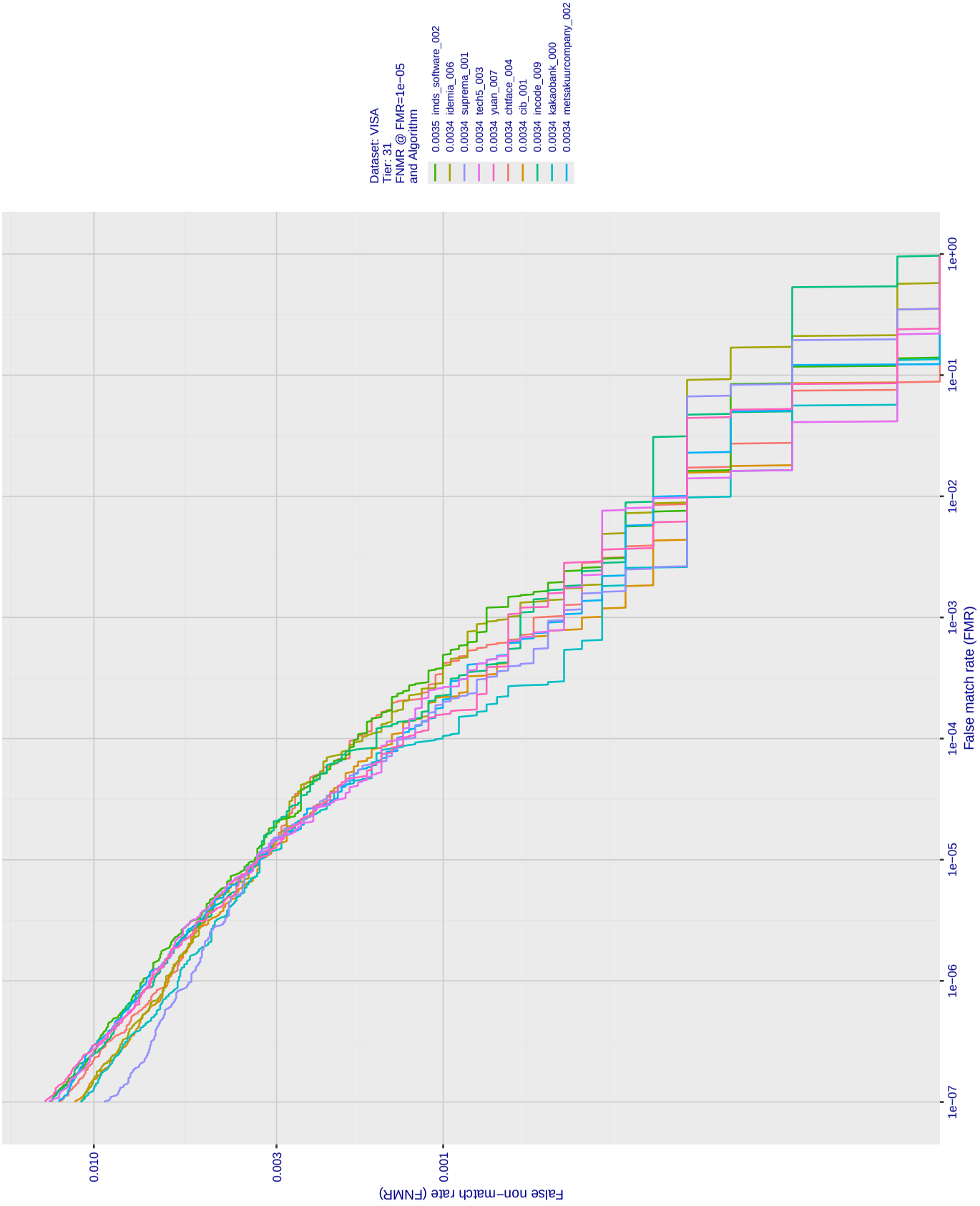


Figure 146: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

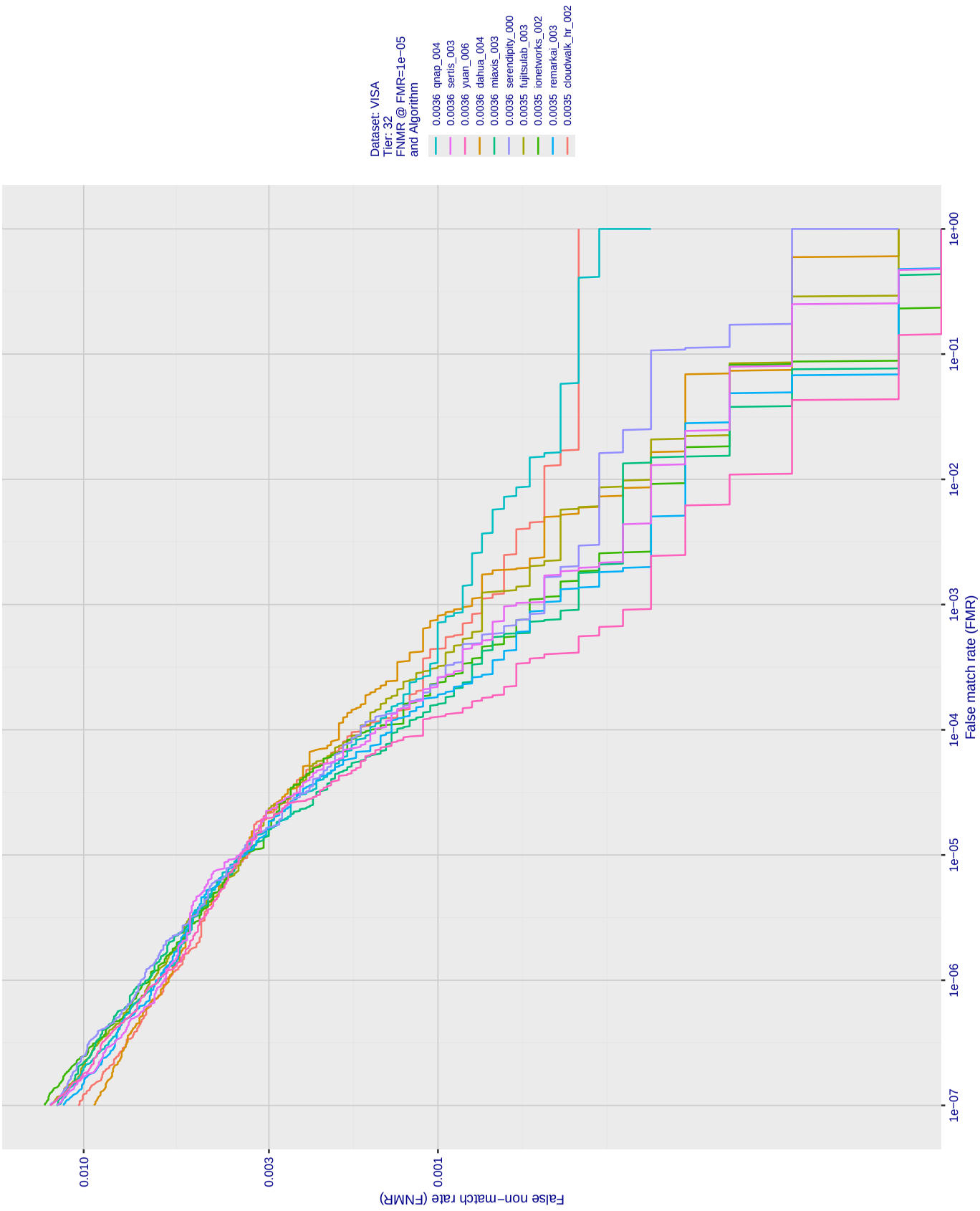


Figure 147: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

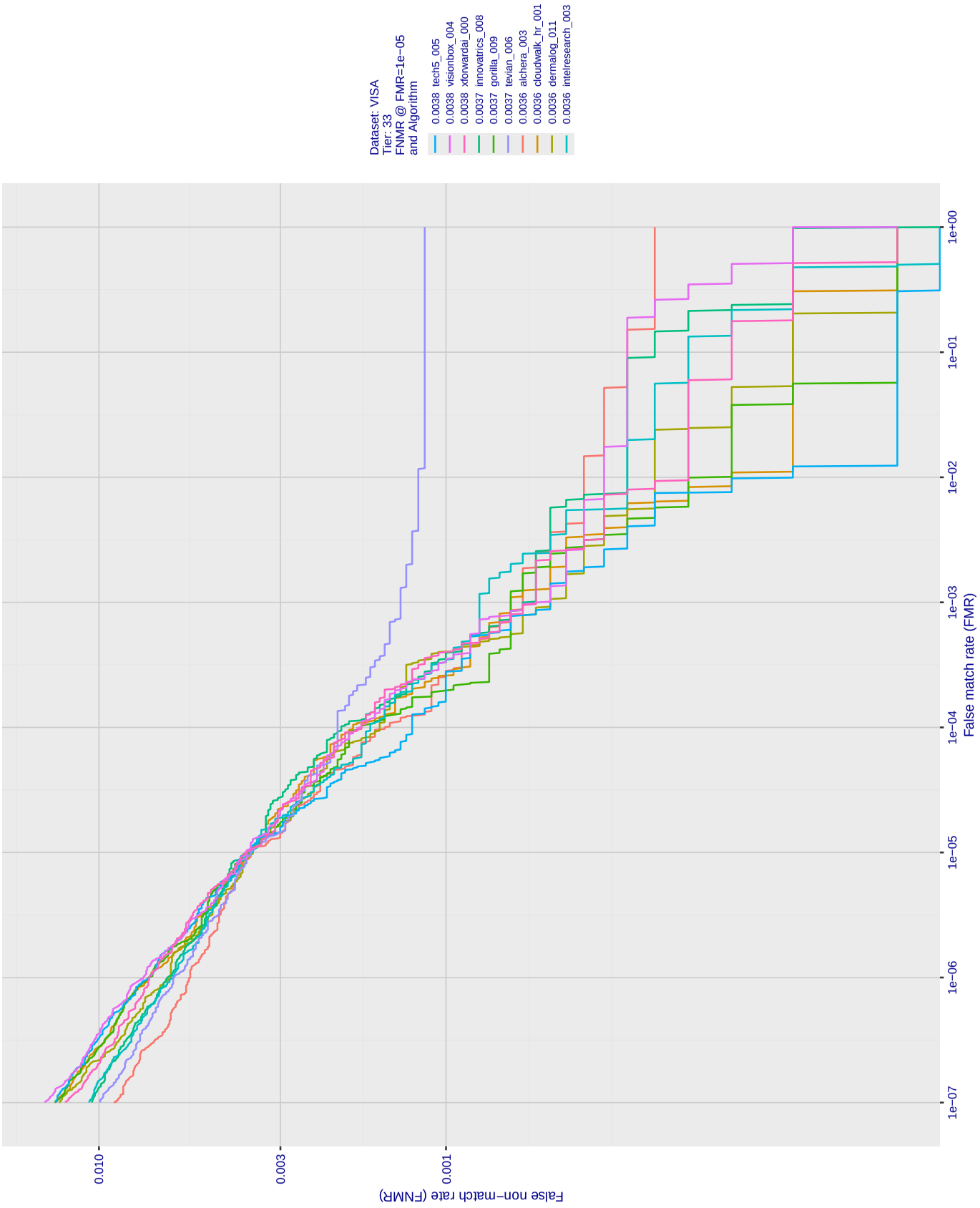


Figure 148: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

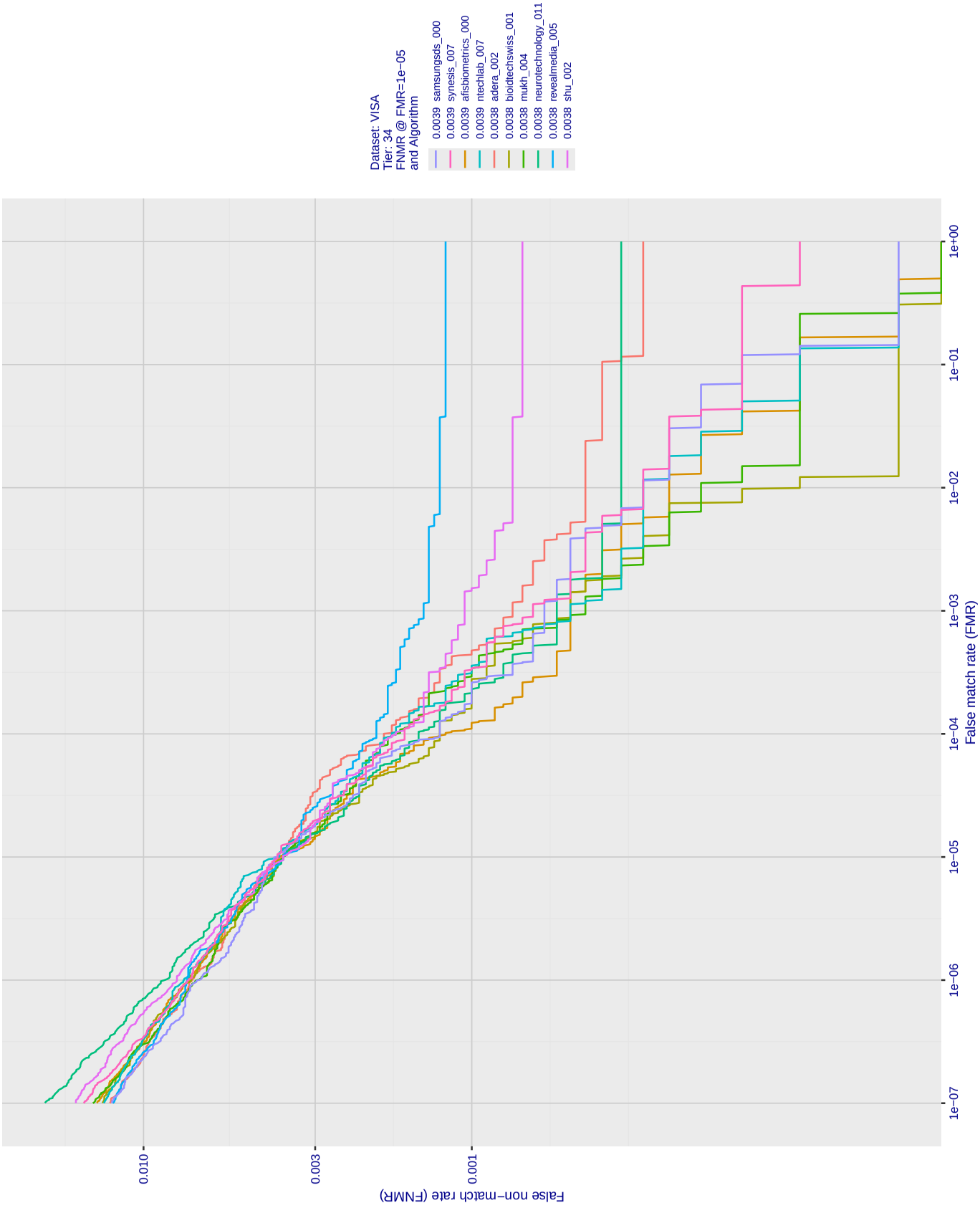


Figure 149: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

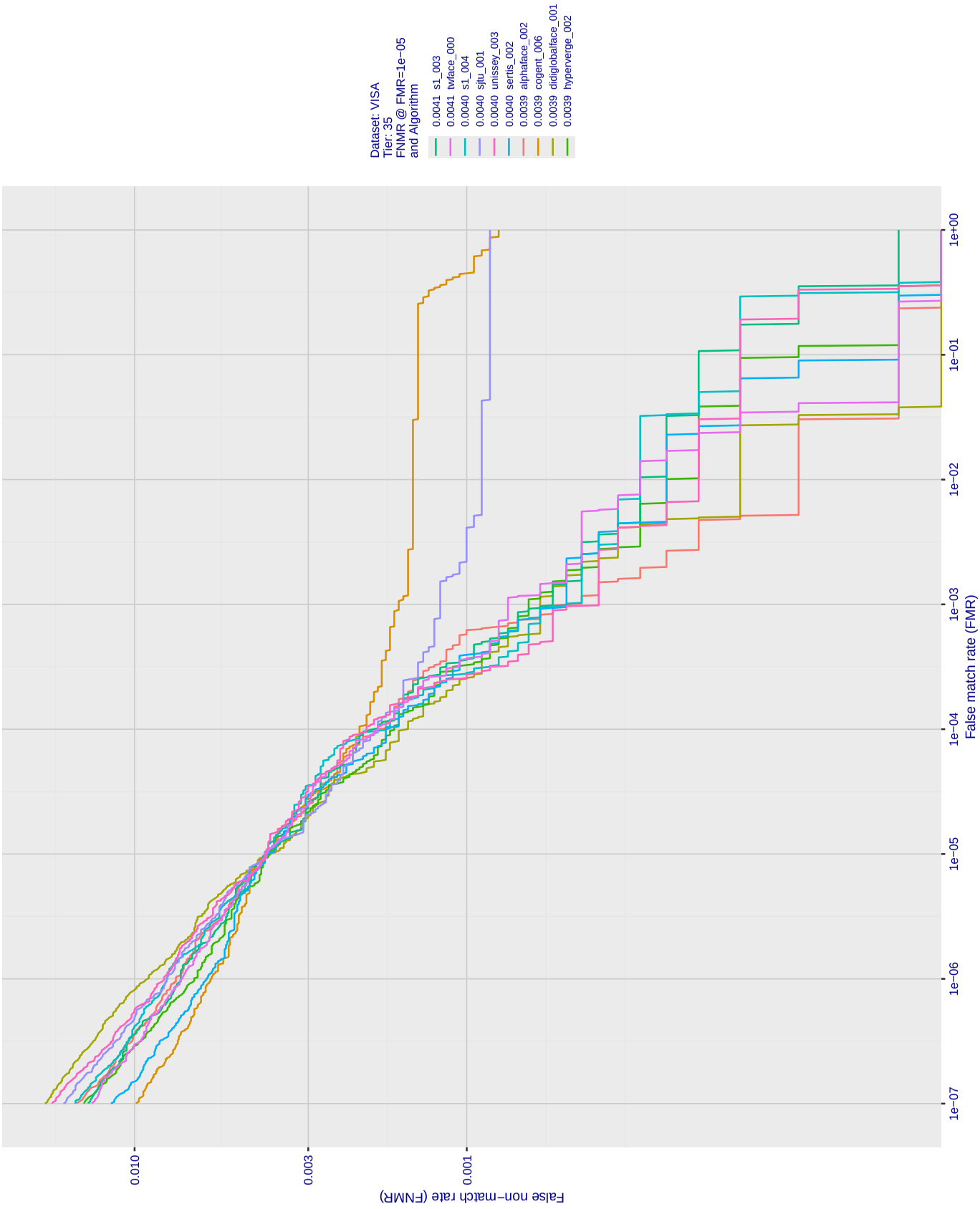


Figure 150: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

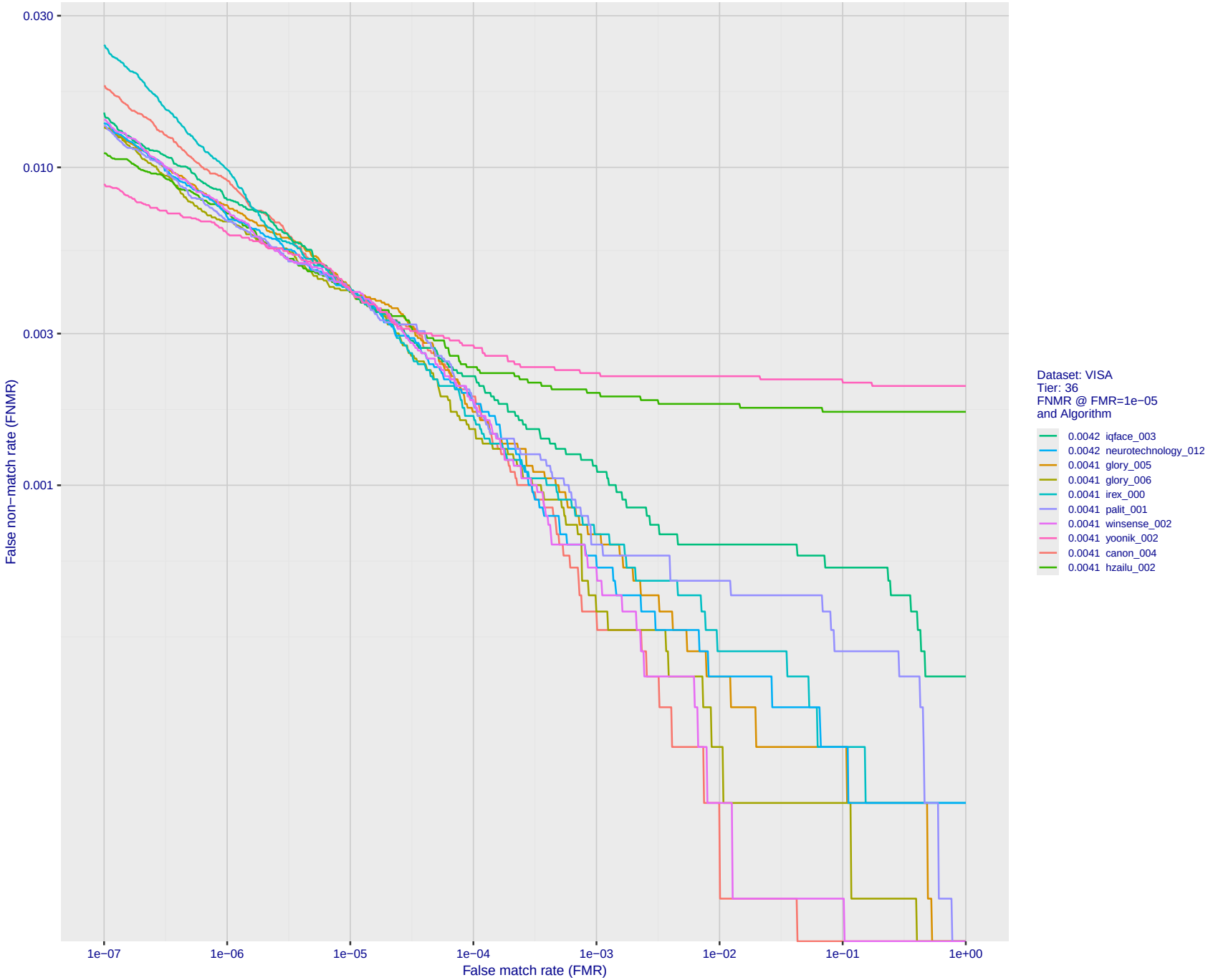


Figure 151: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

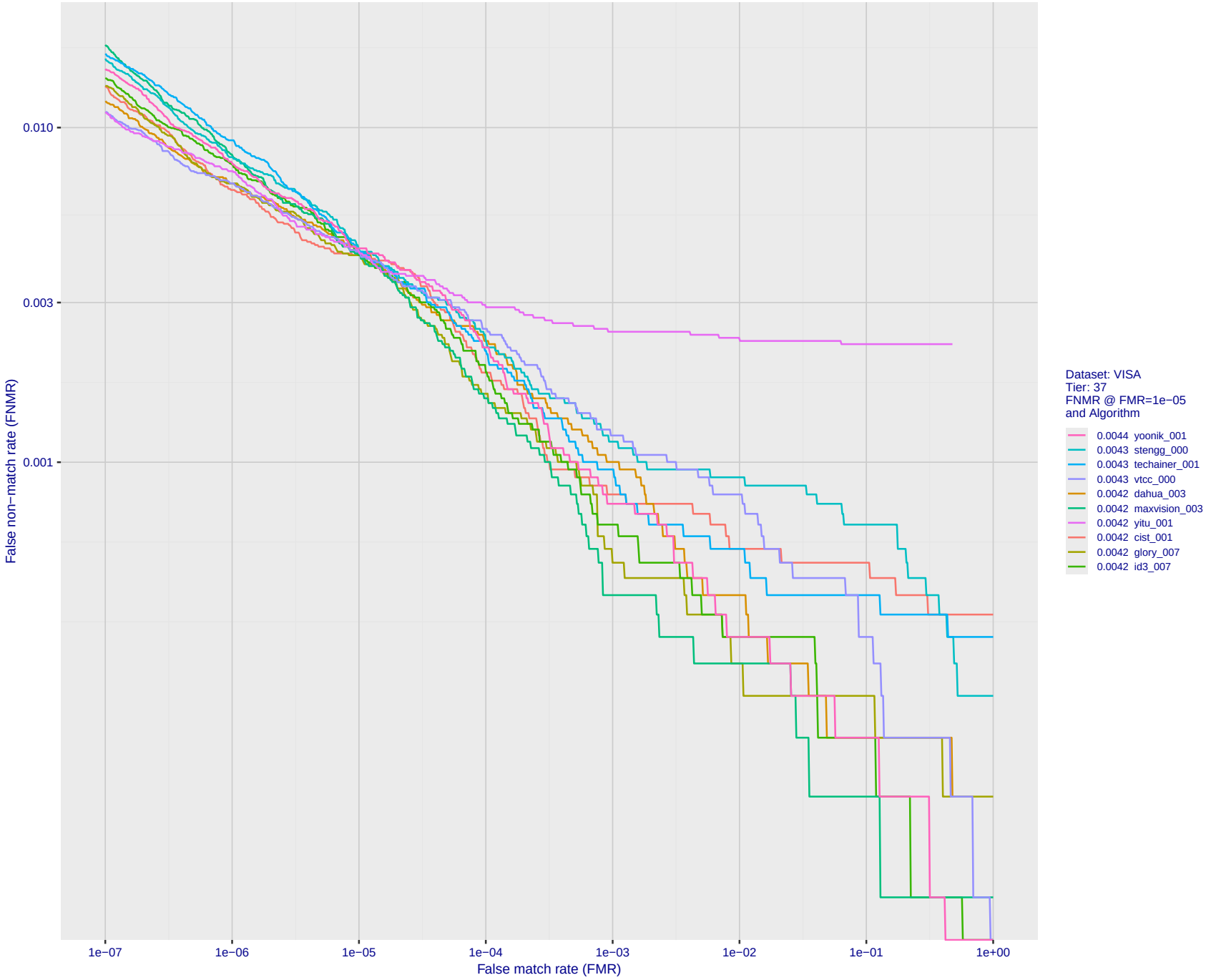


Figure 152: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

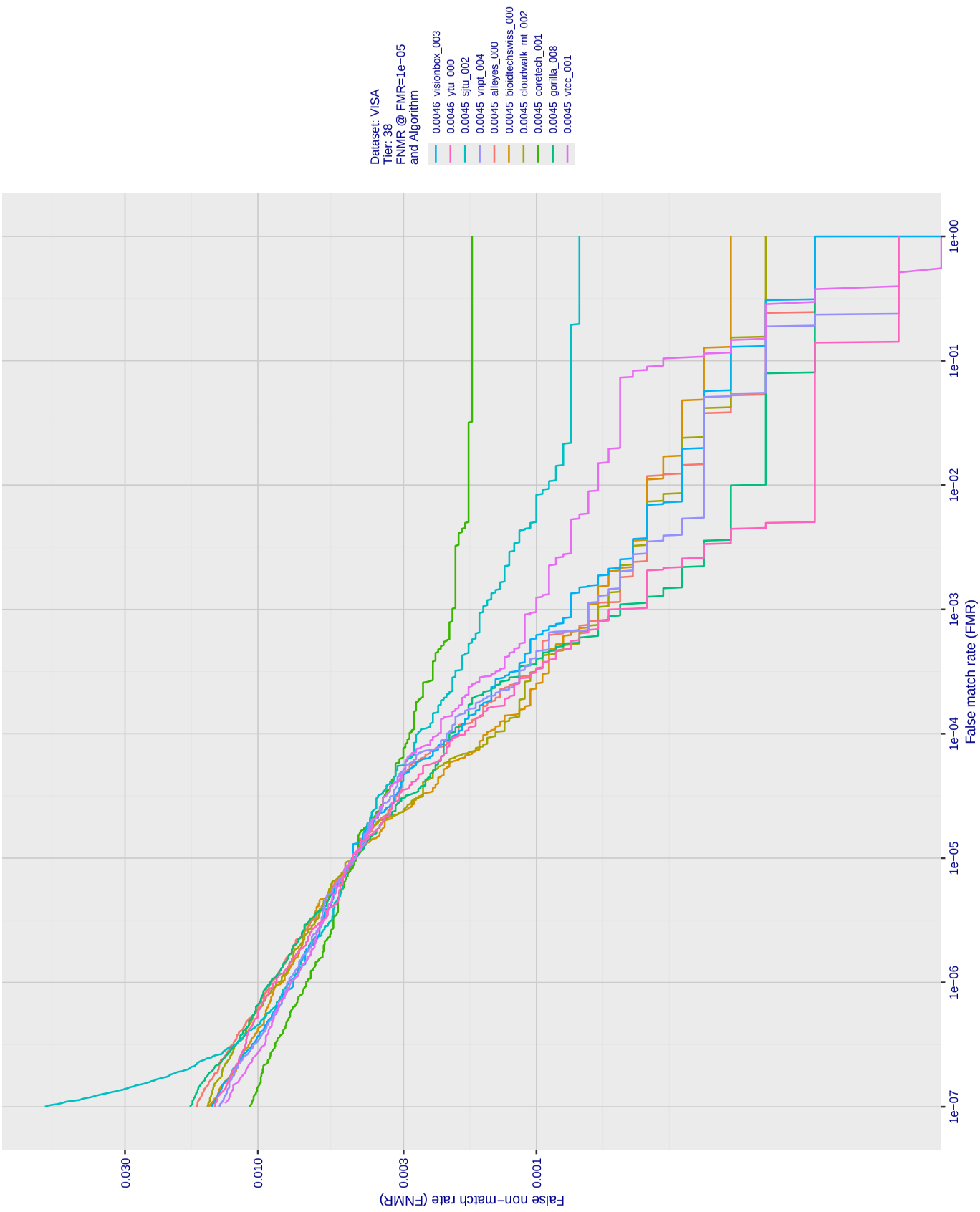


Figure 153: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

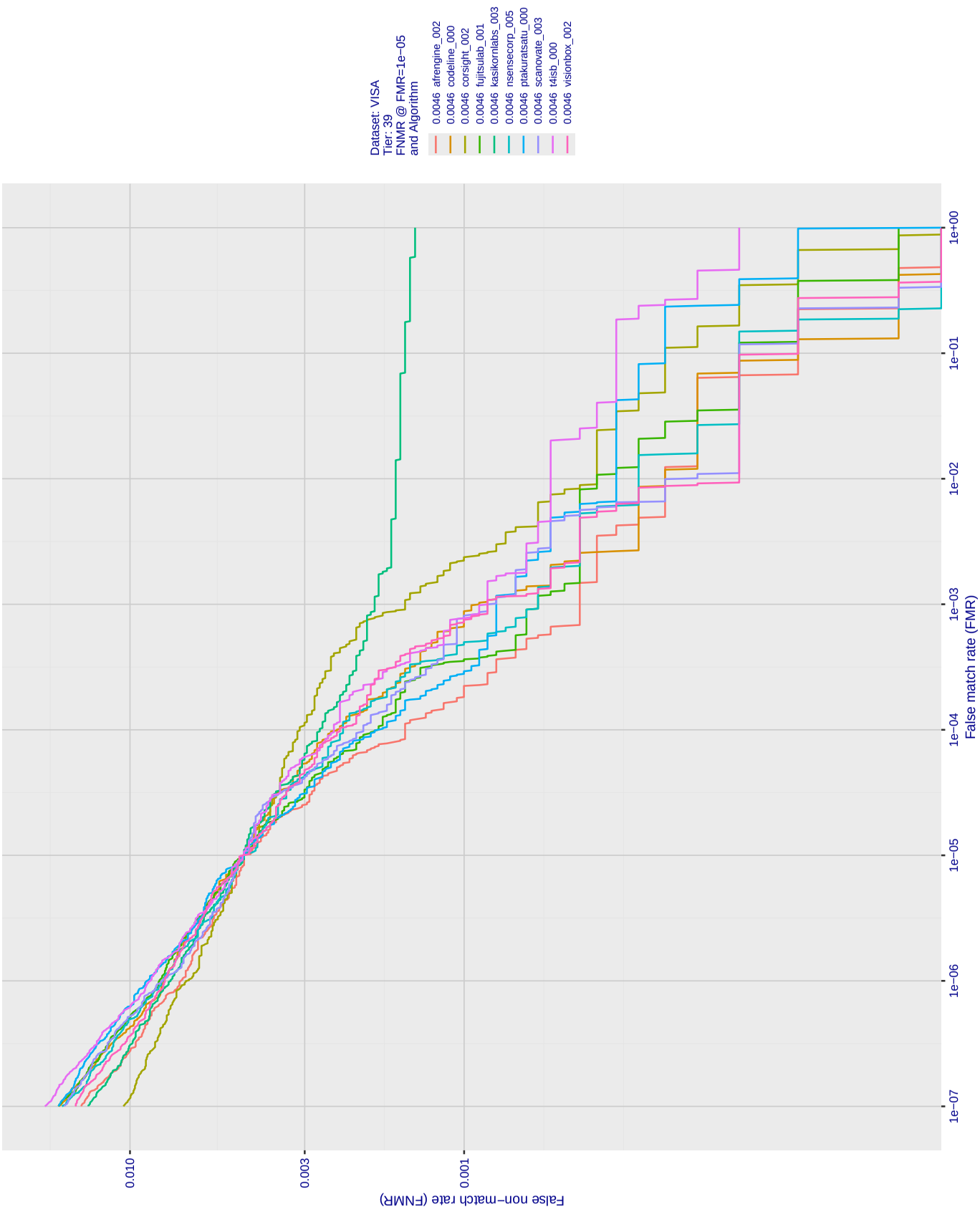


Figure 154: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

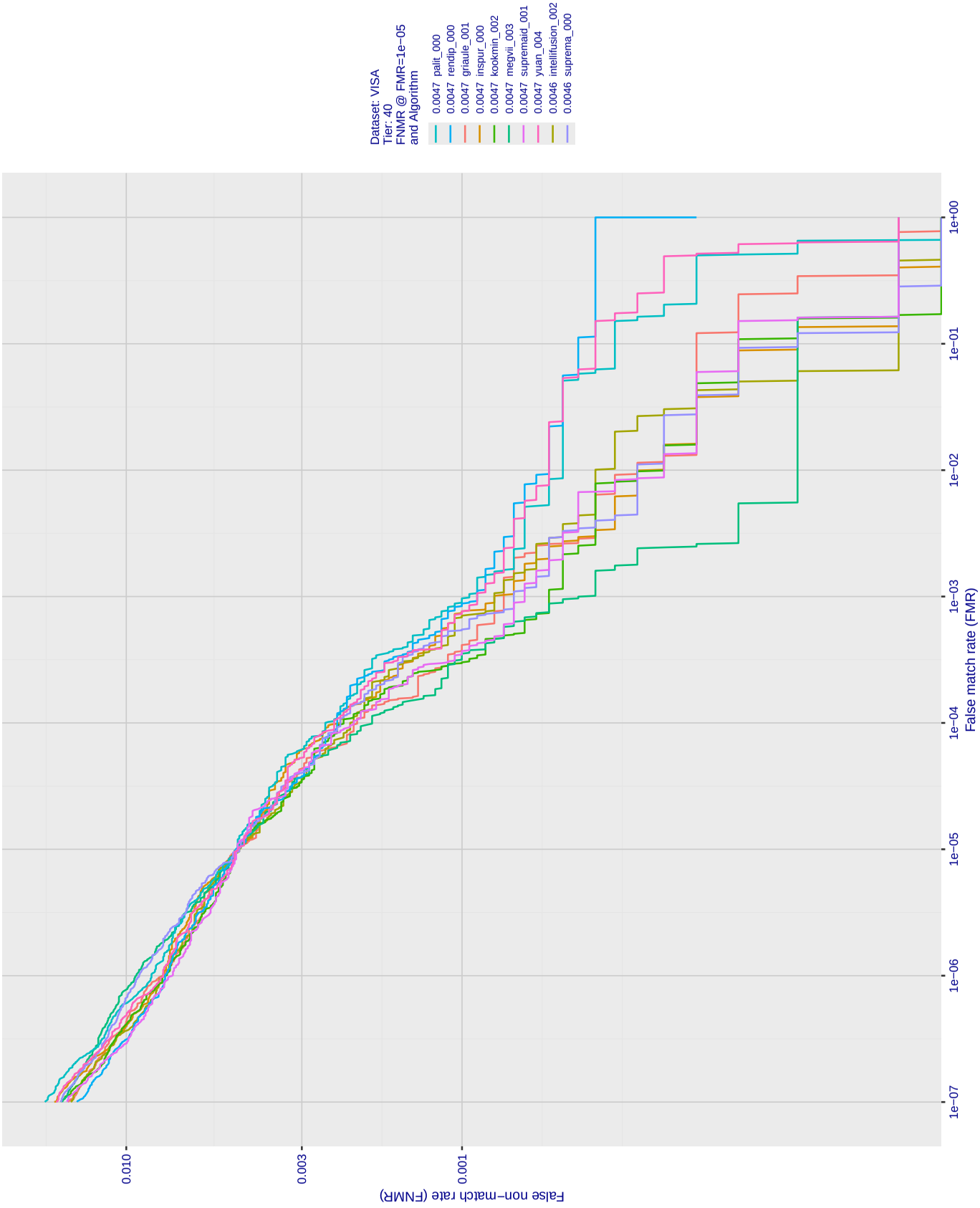


Figure 155: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

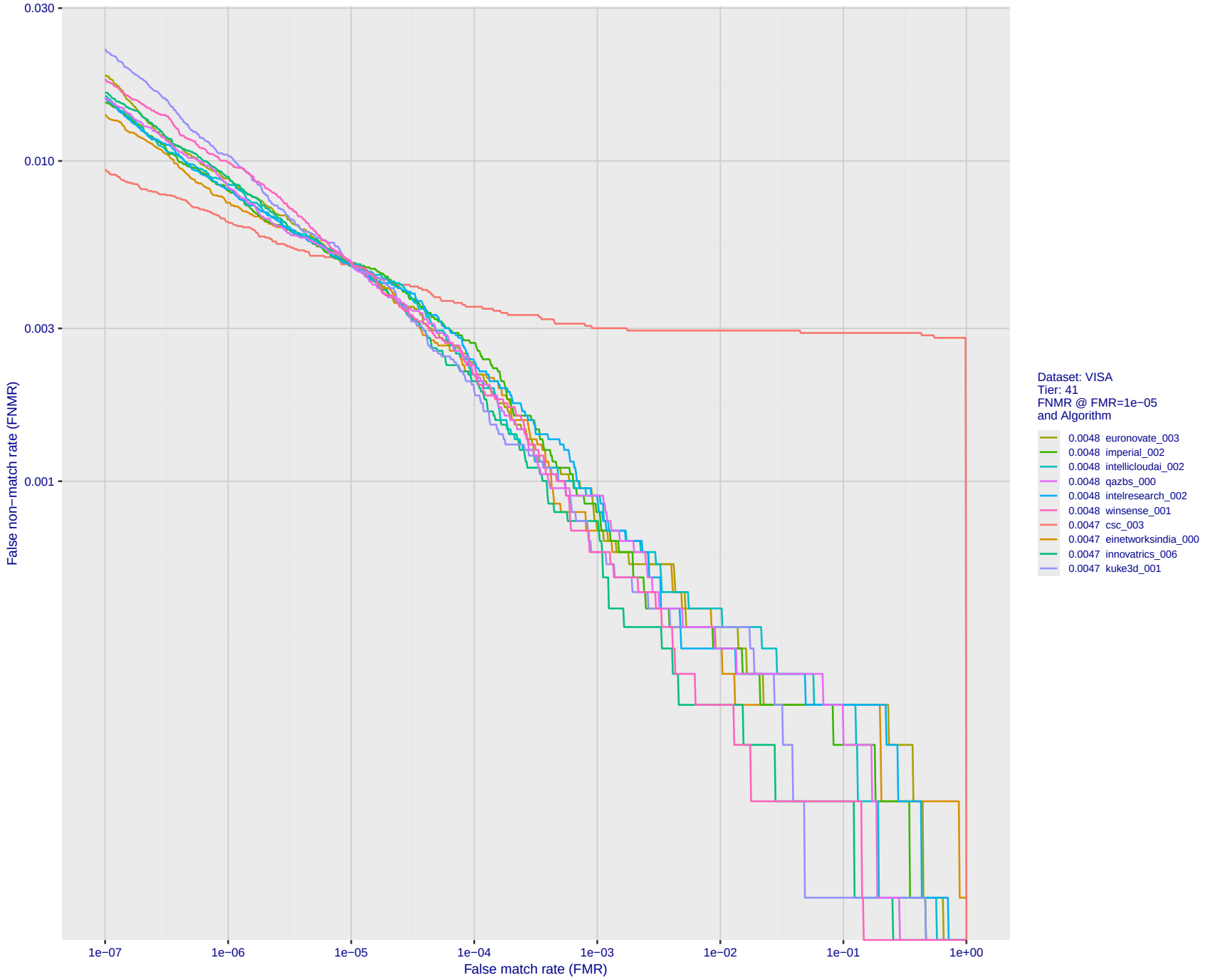


Figure 156: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

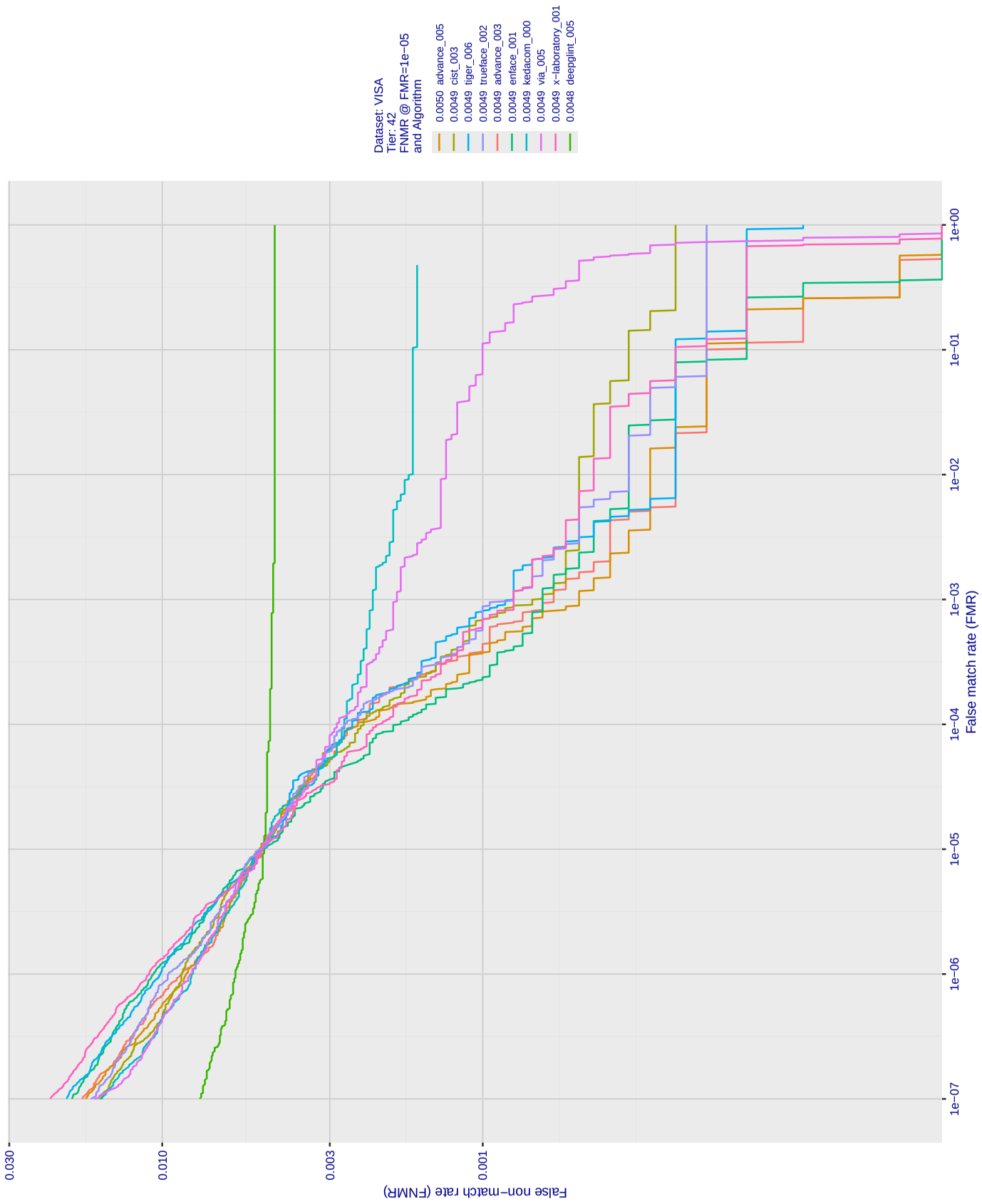


Figure 157: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

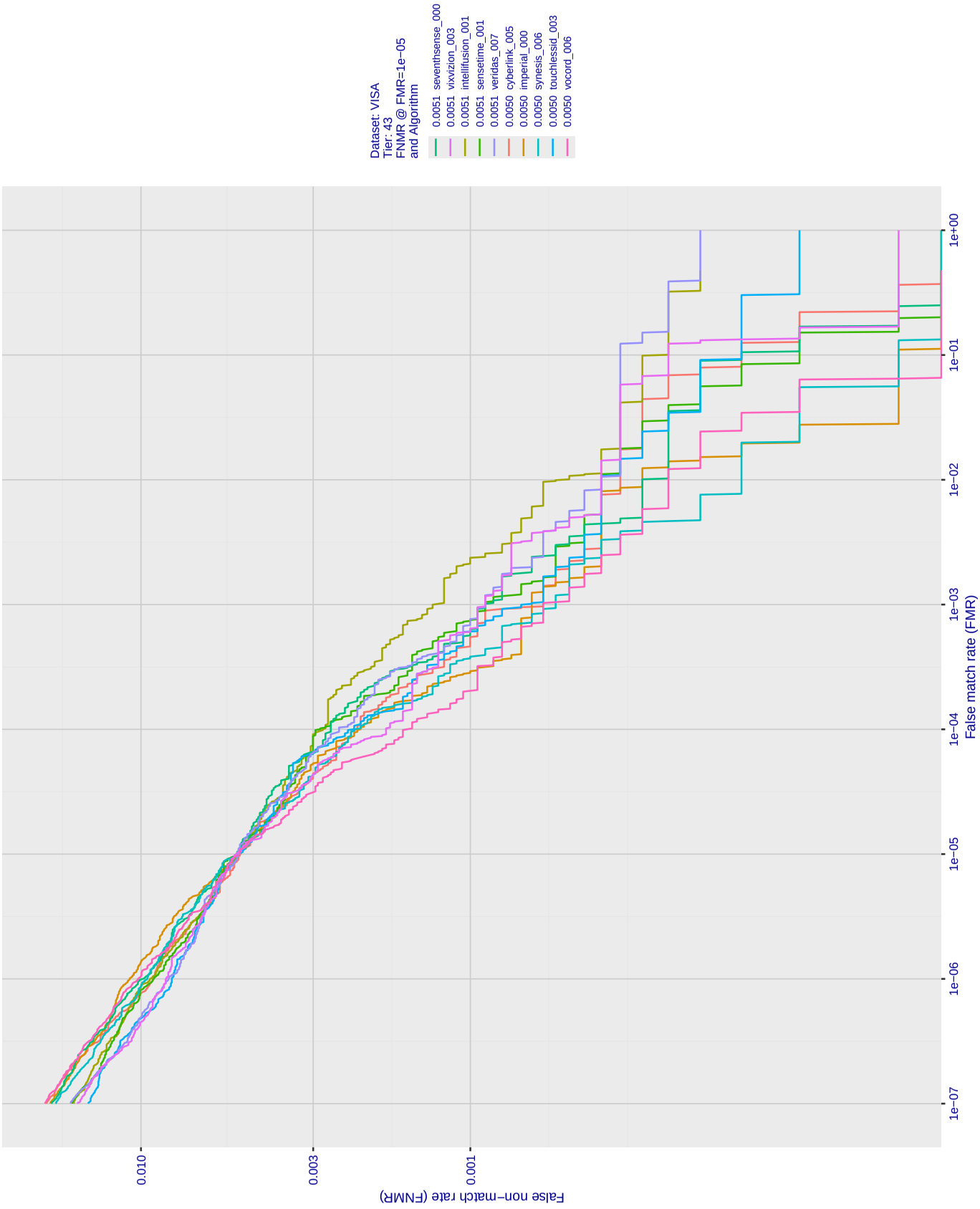


Figure 158: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

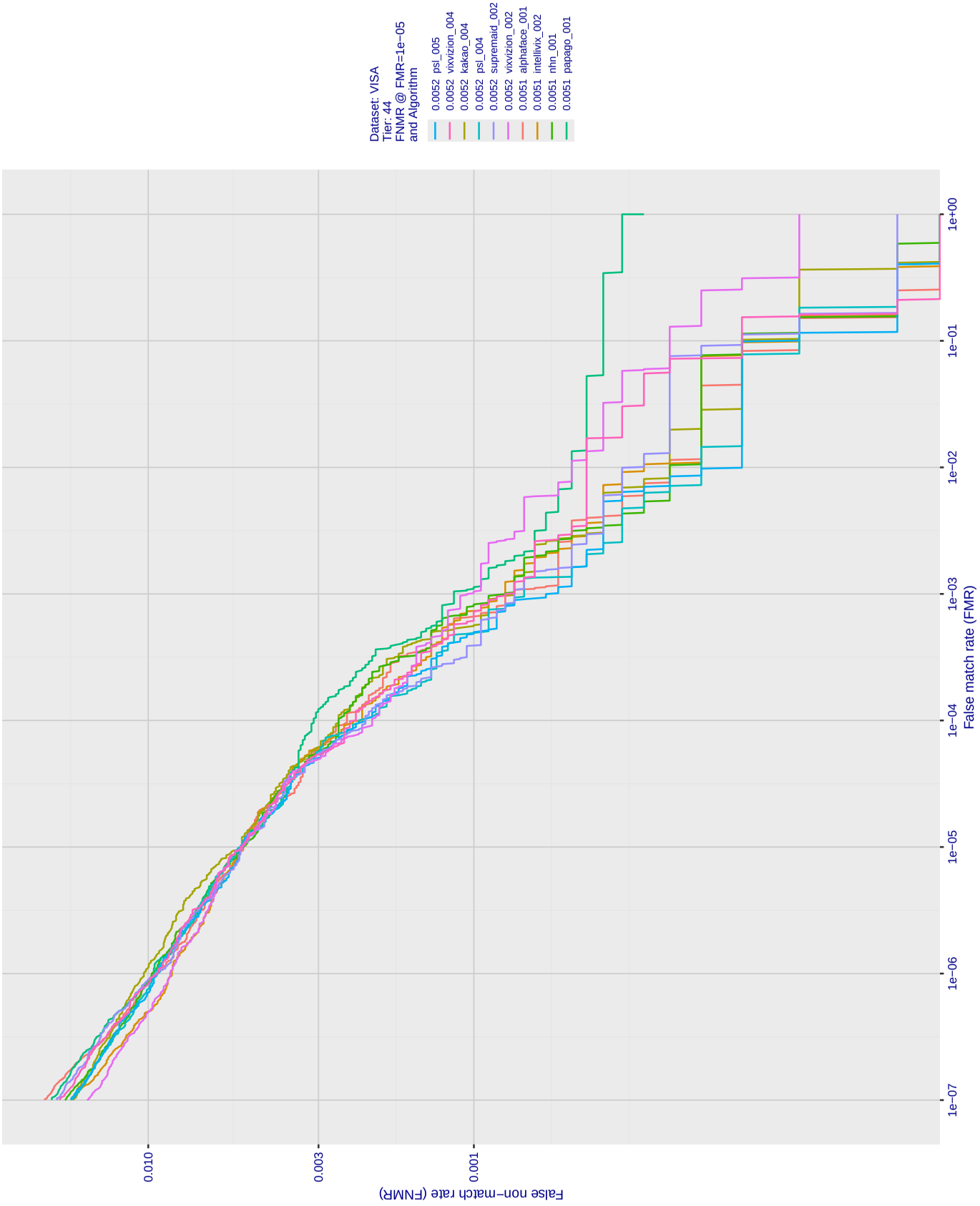


Figure 159: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

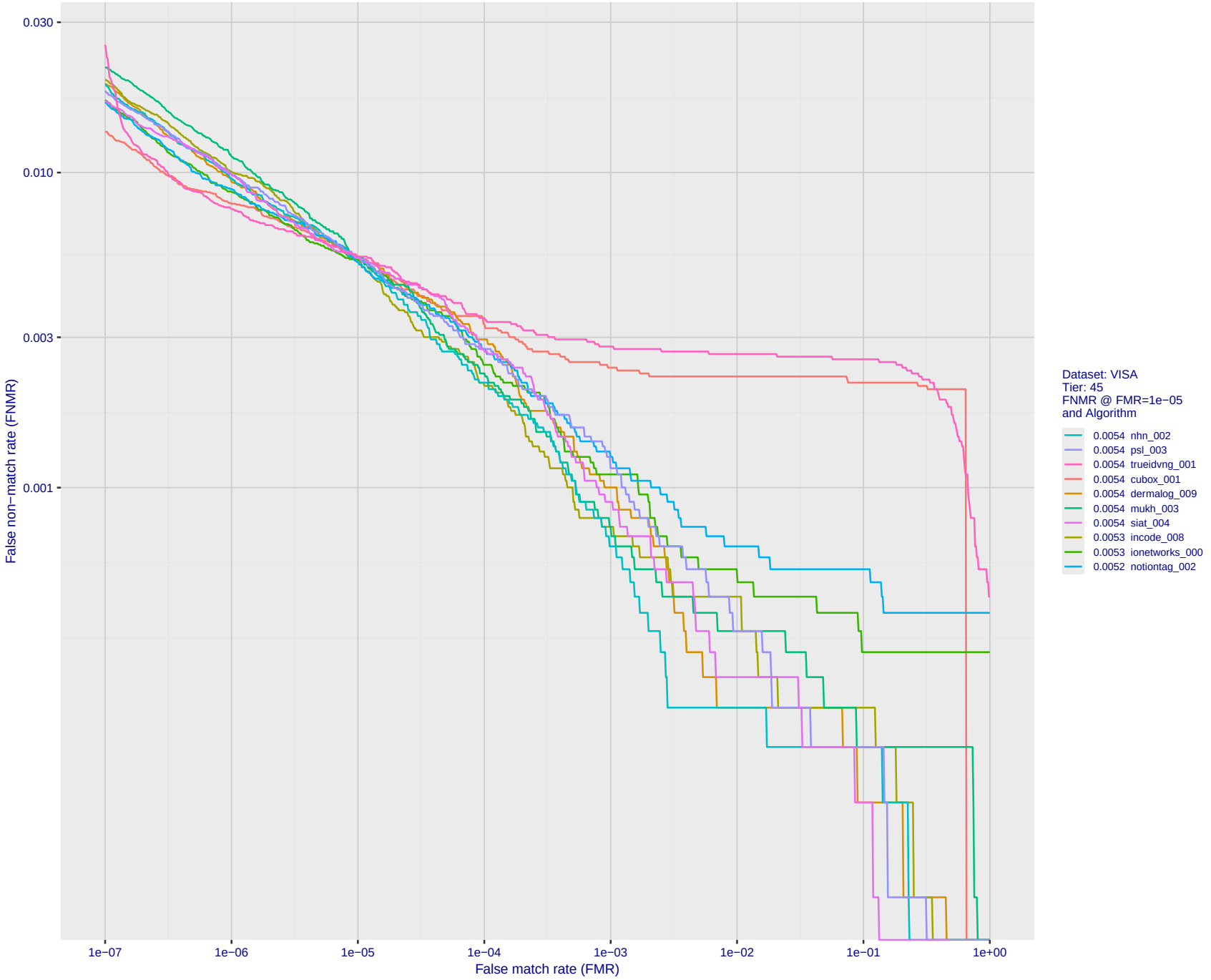


Figure 160: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

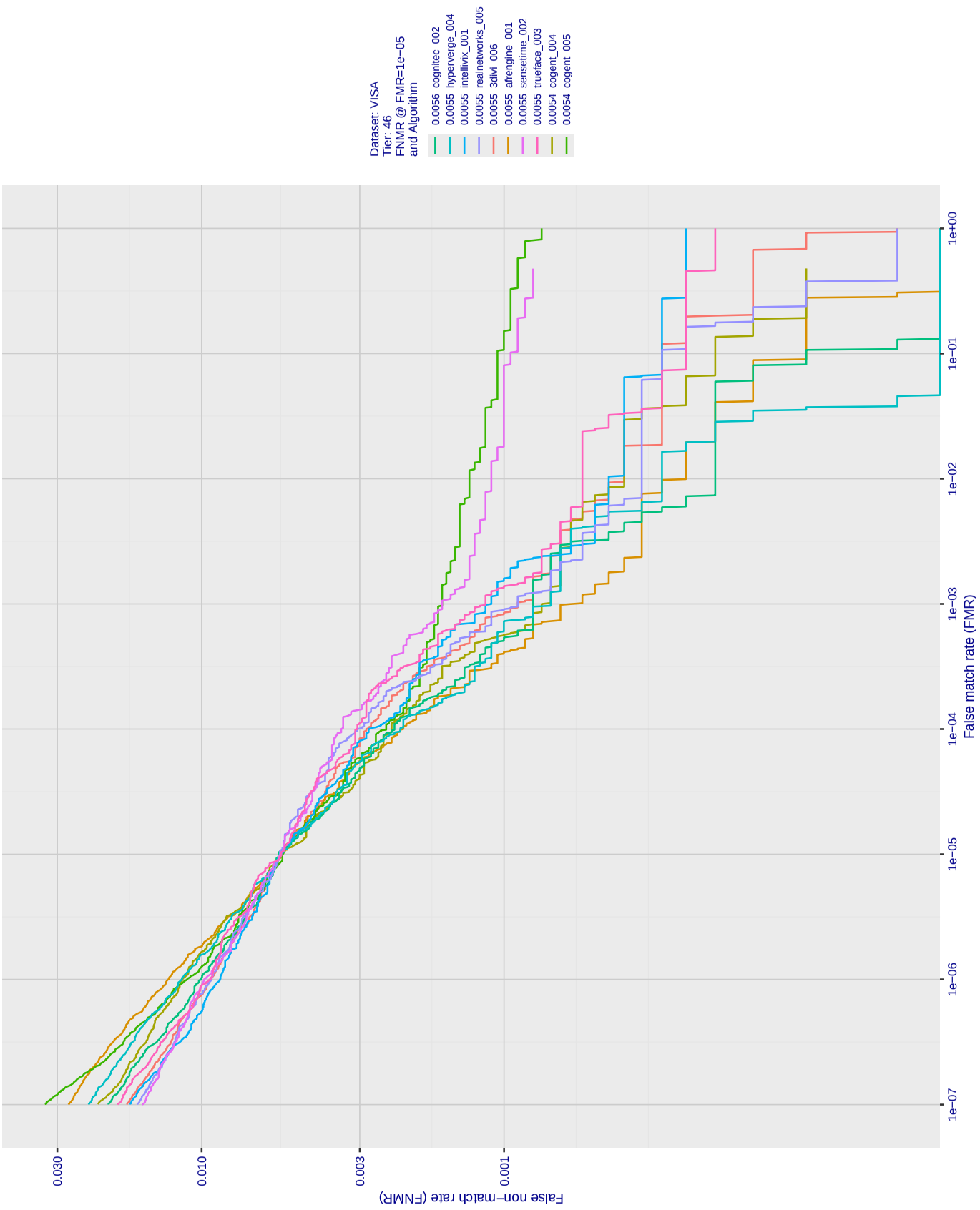


Figure 161: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

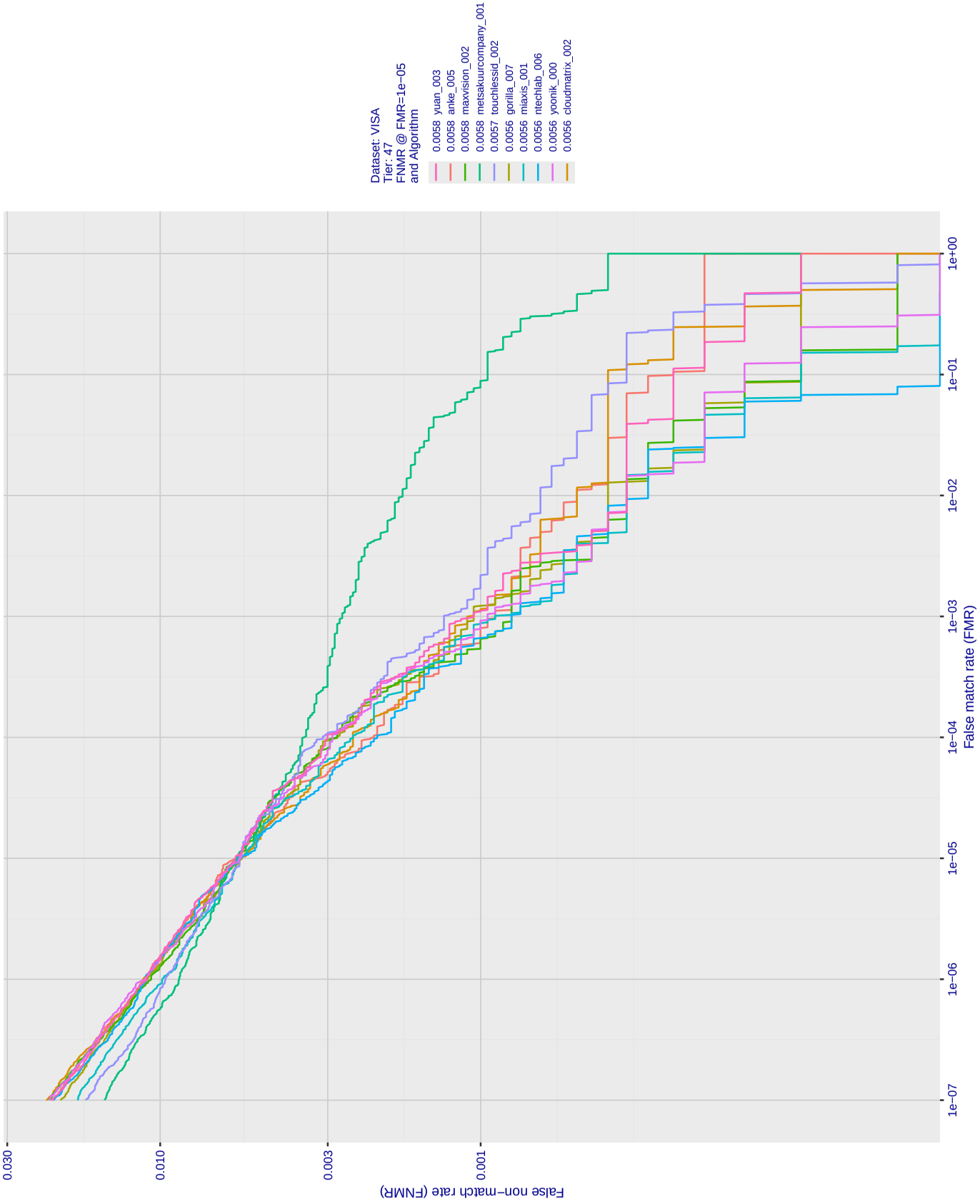


Figure 162: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

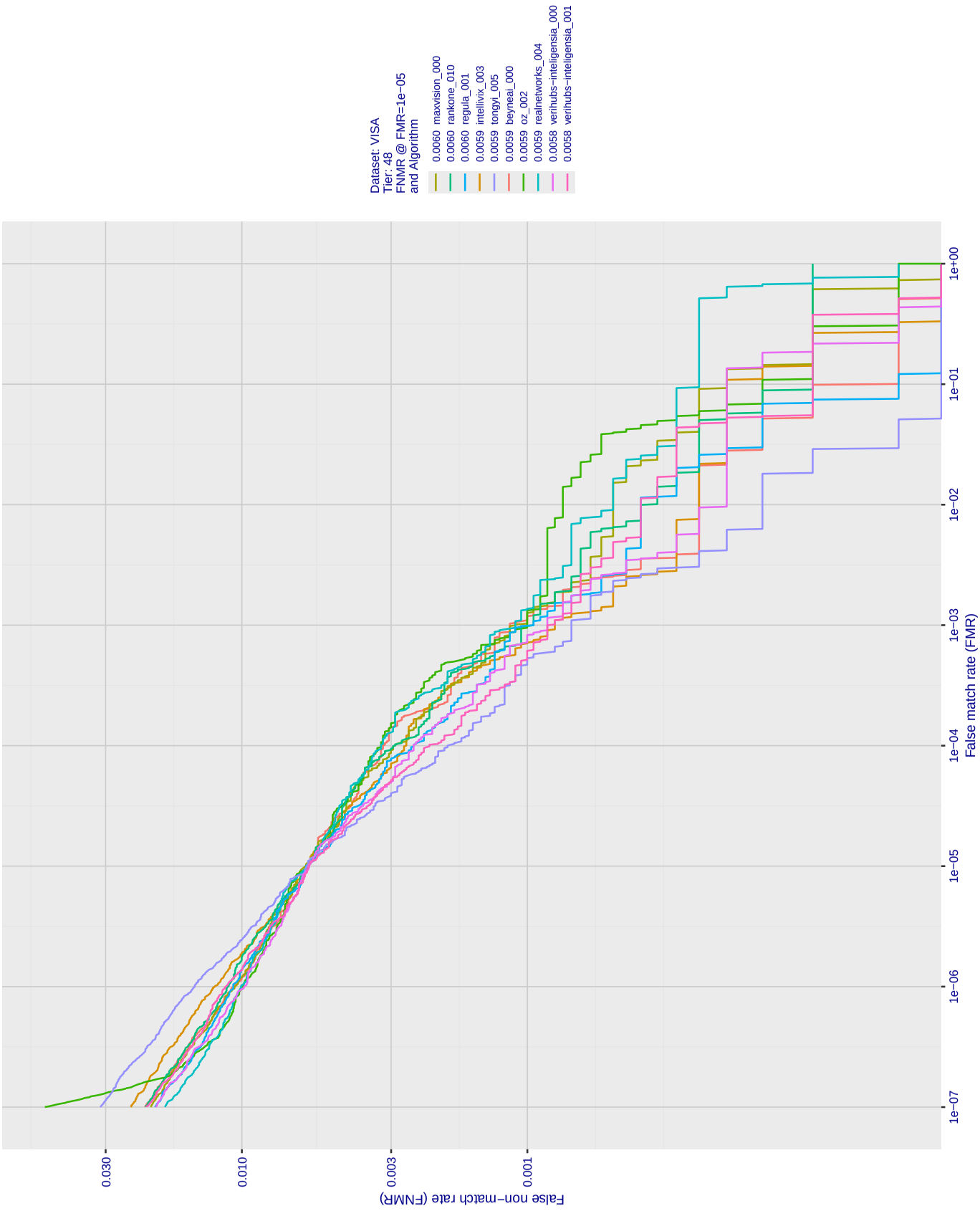


Figure 163: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

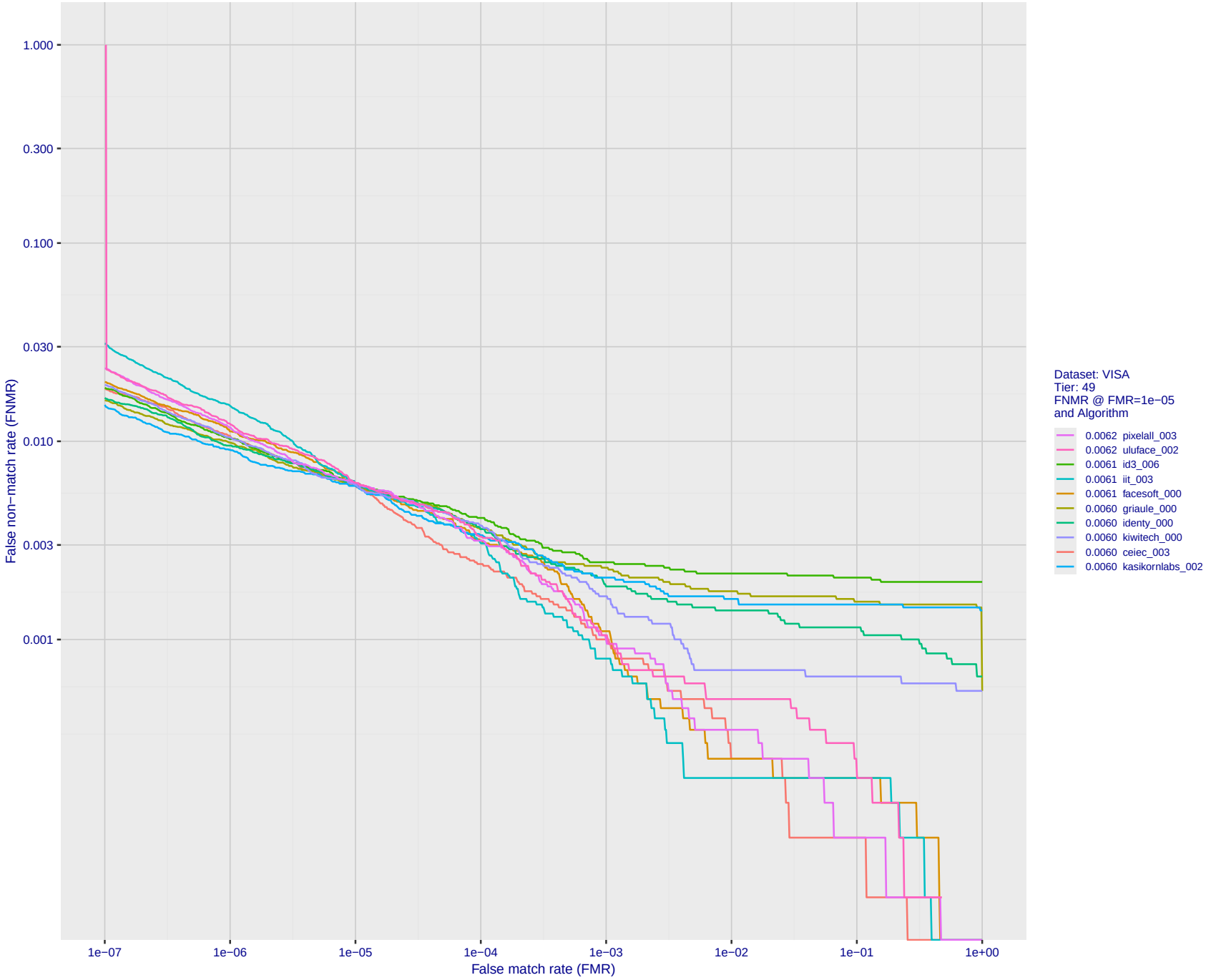


Figure 164: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

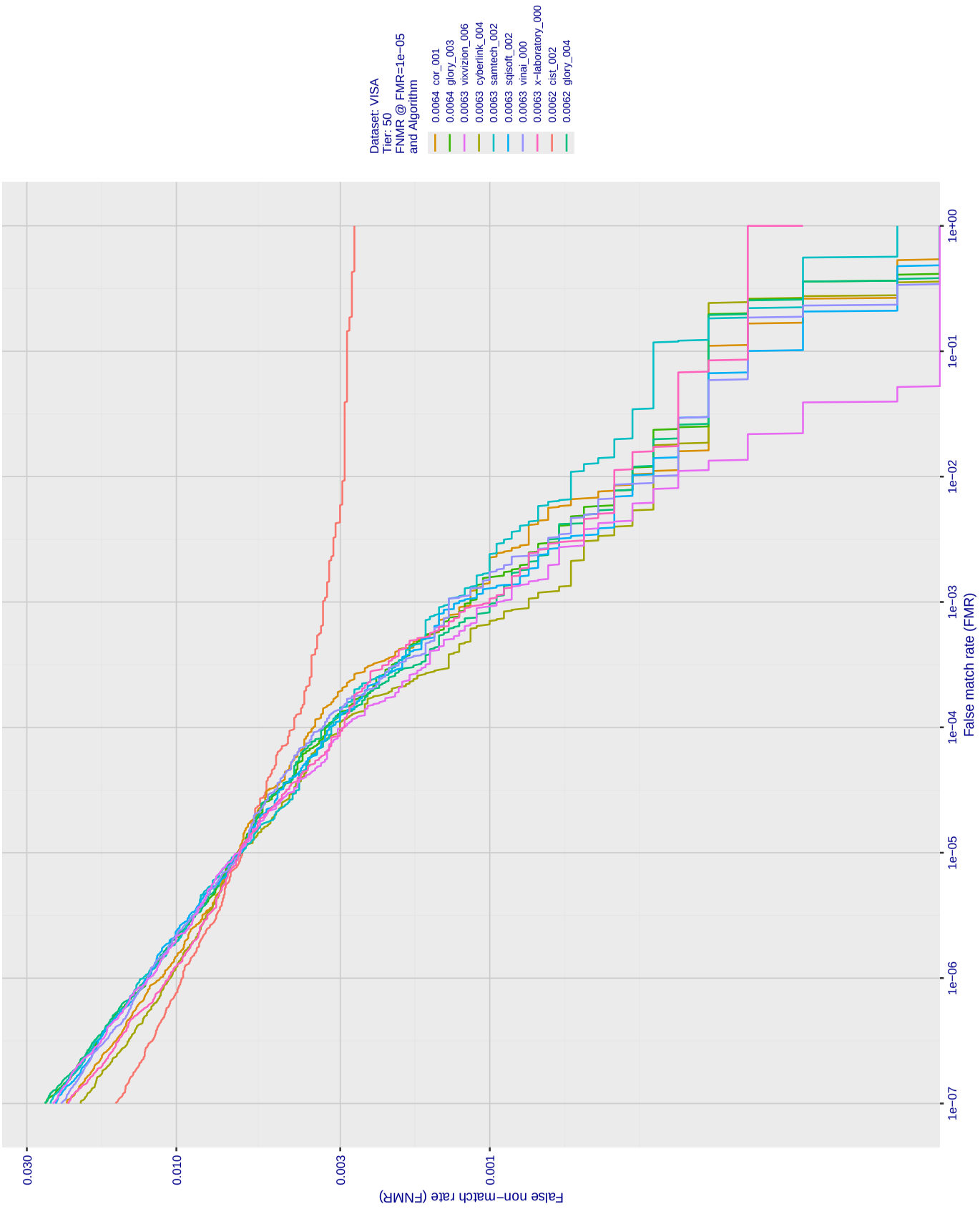


Figure 165: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

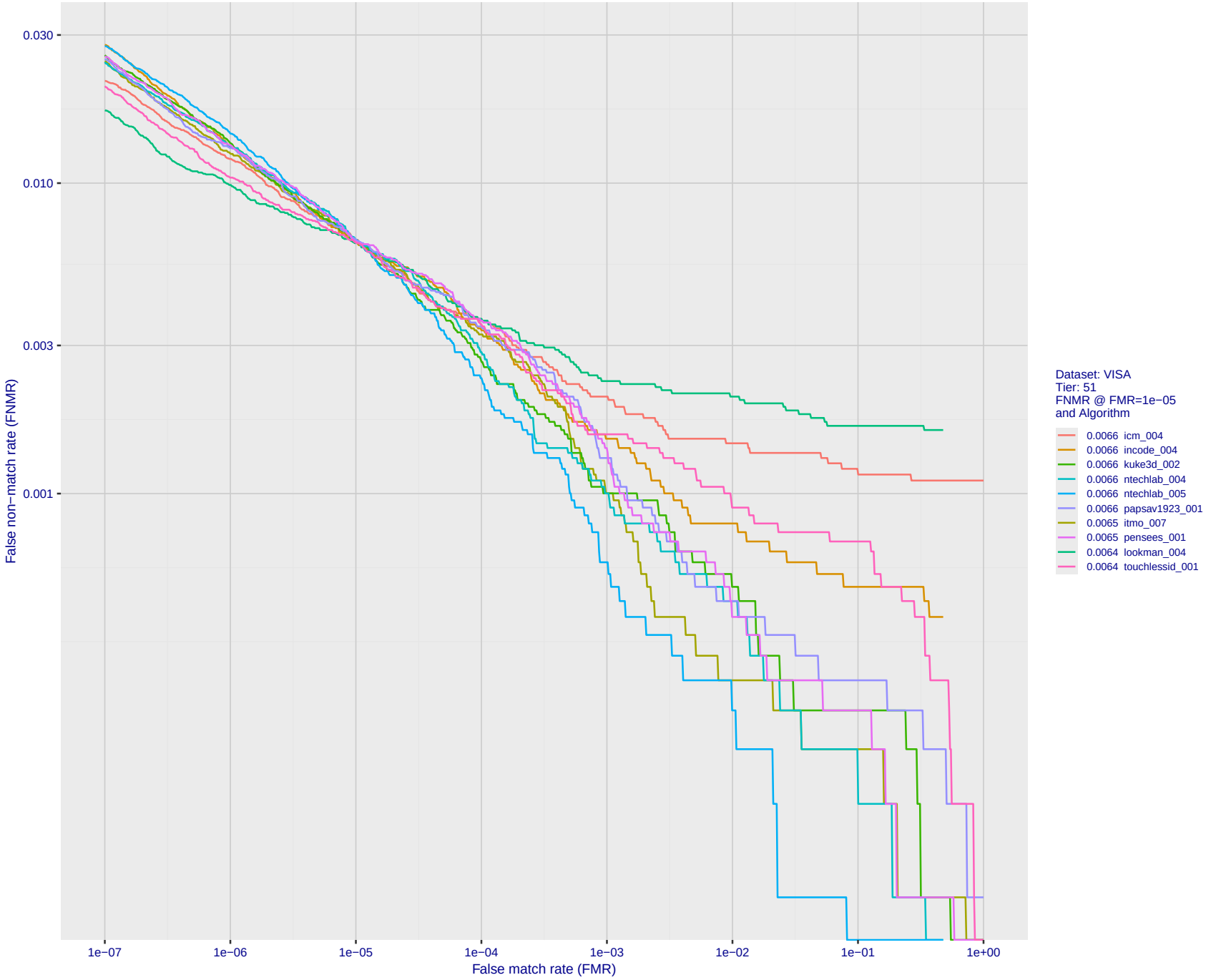


Figure 166: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

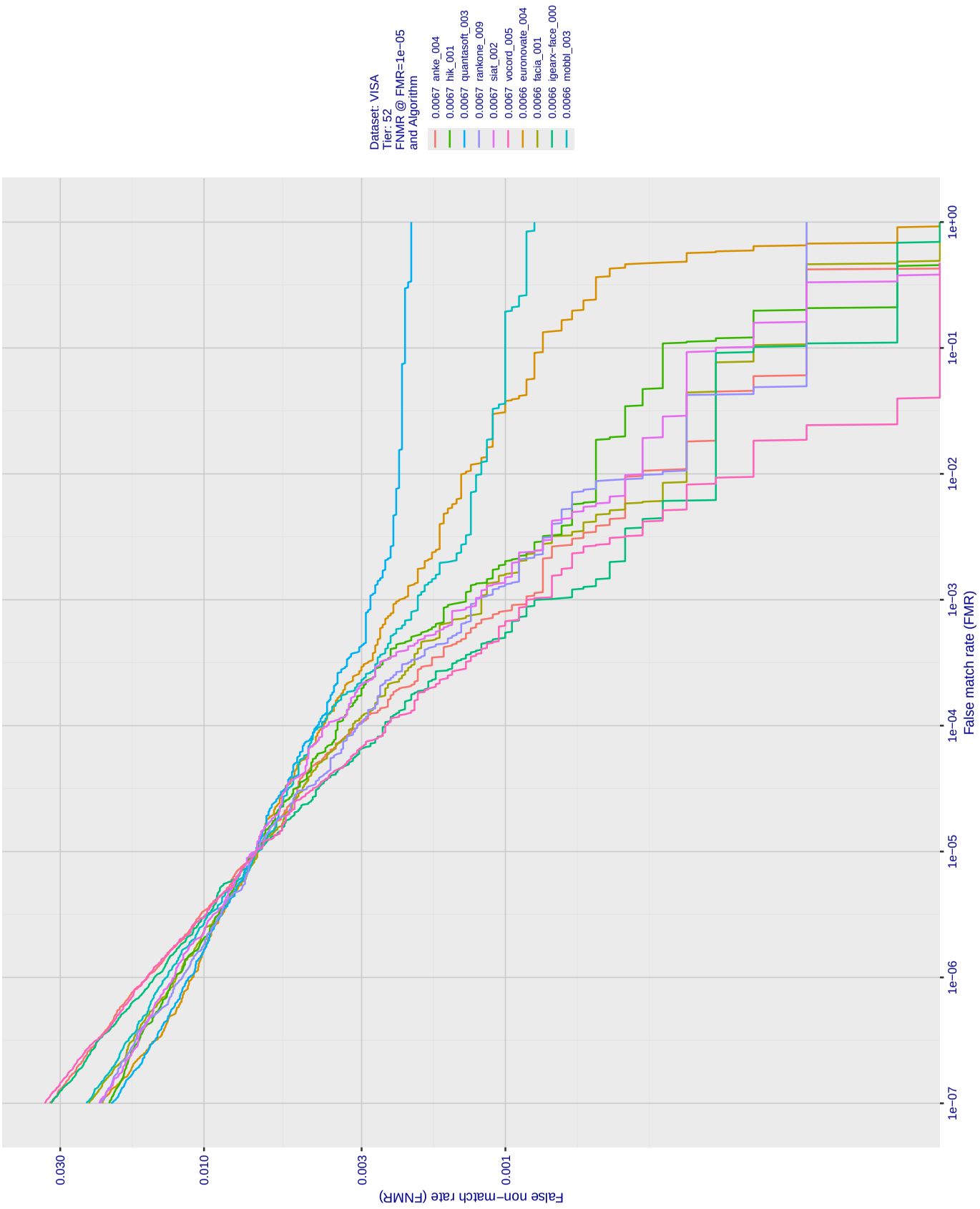


Figure 167: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

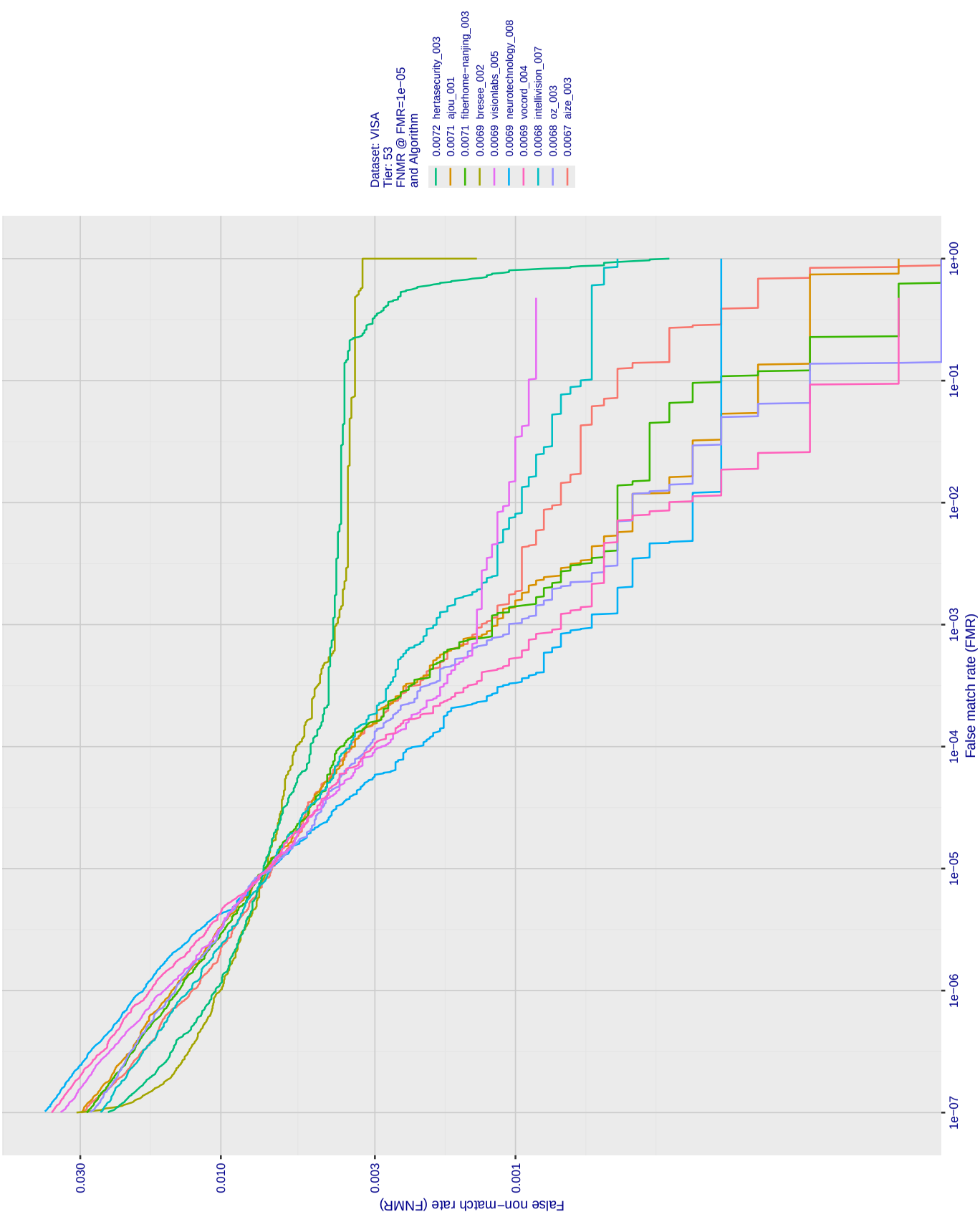


Figure 168: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

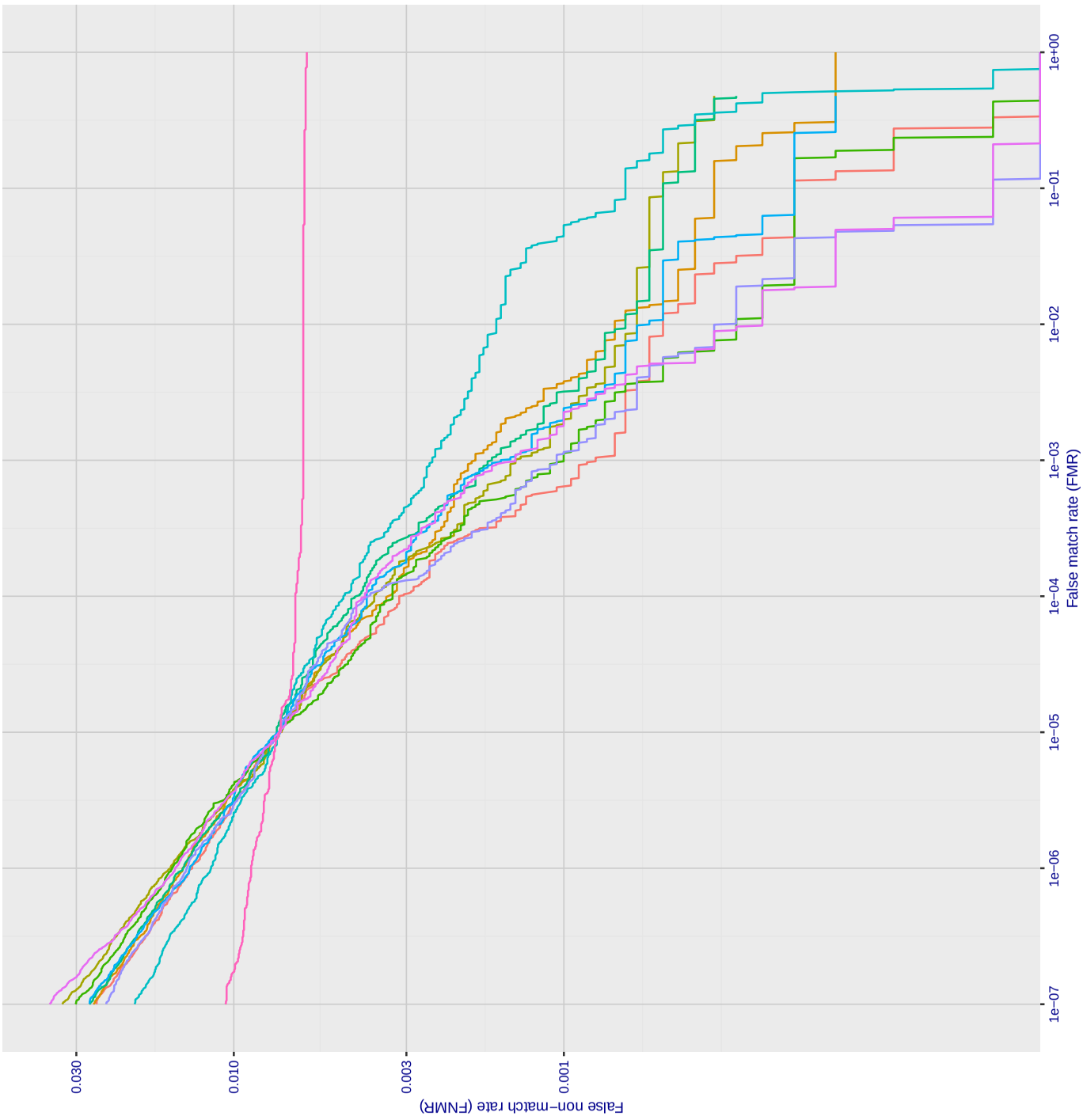


Figure 169: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

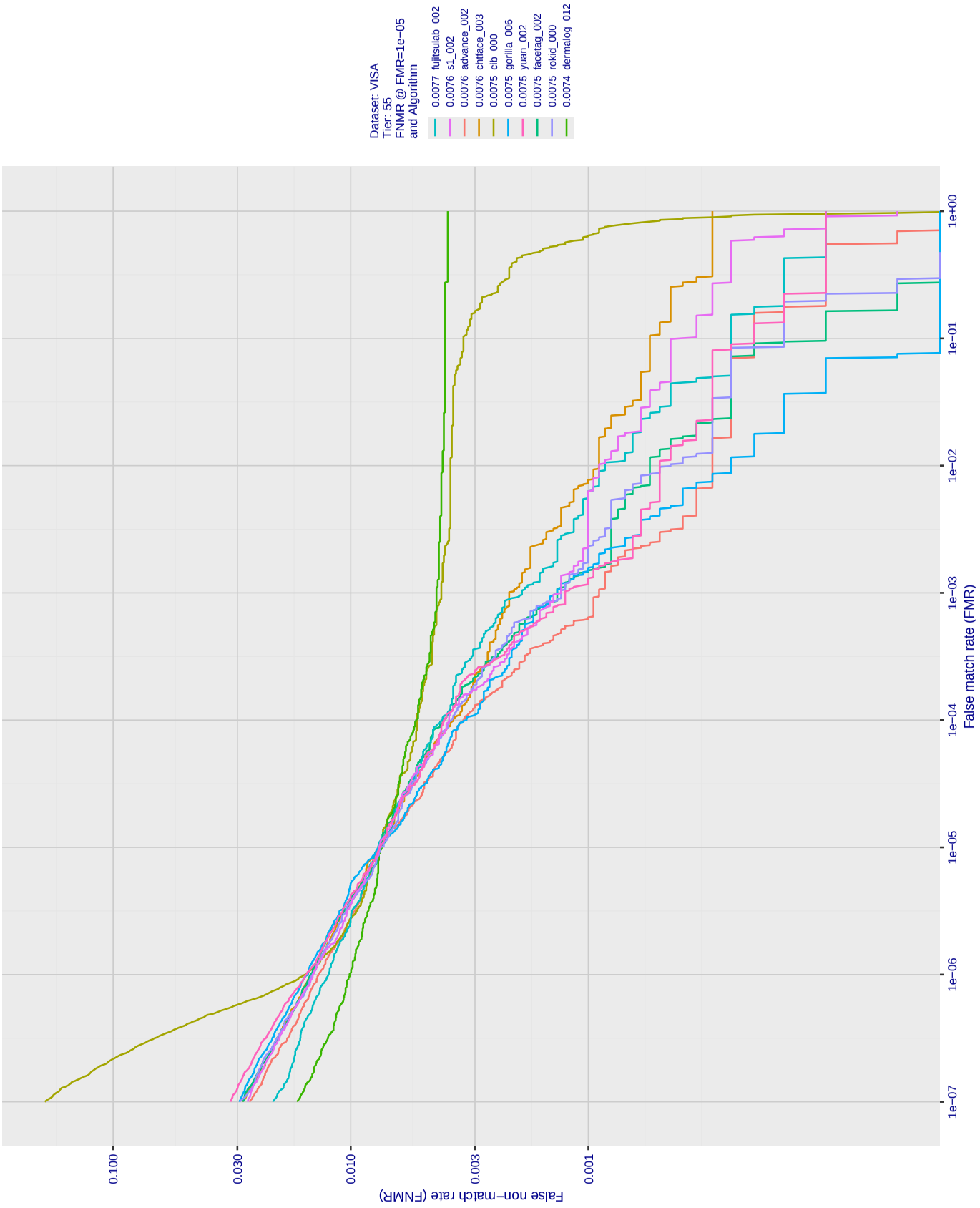


Figure 170: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

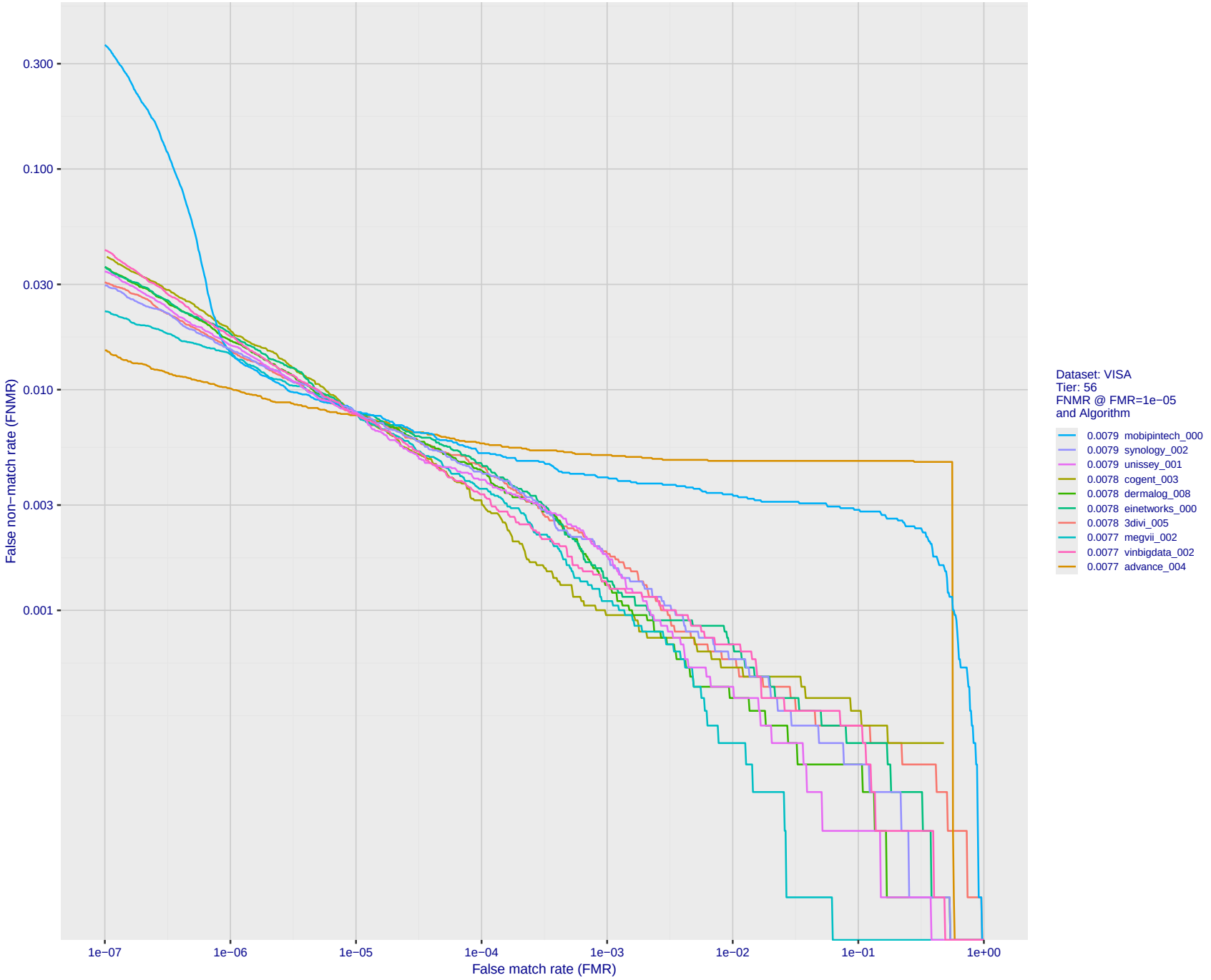


Figure 171: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

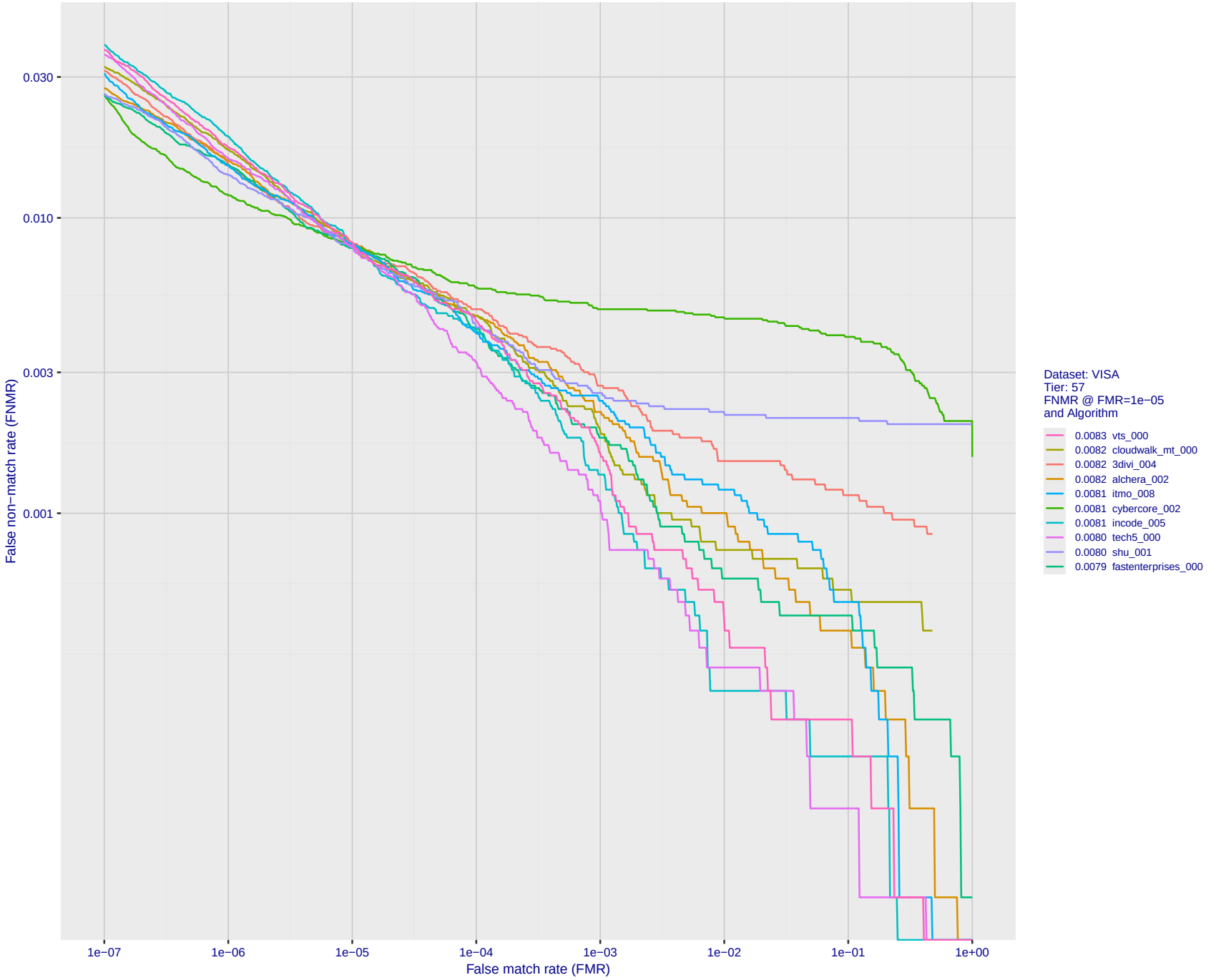


Figure 172: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

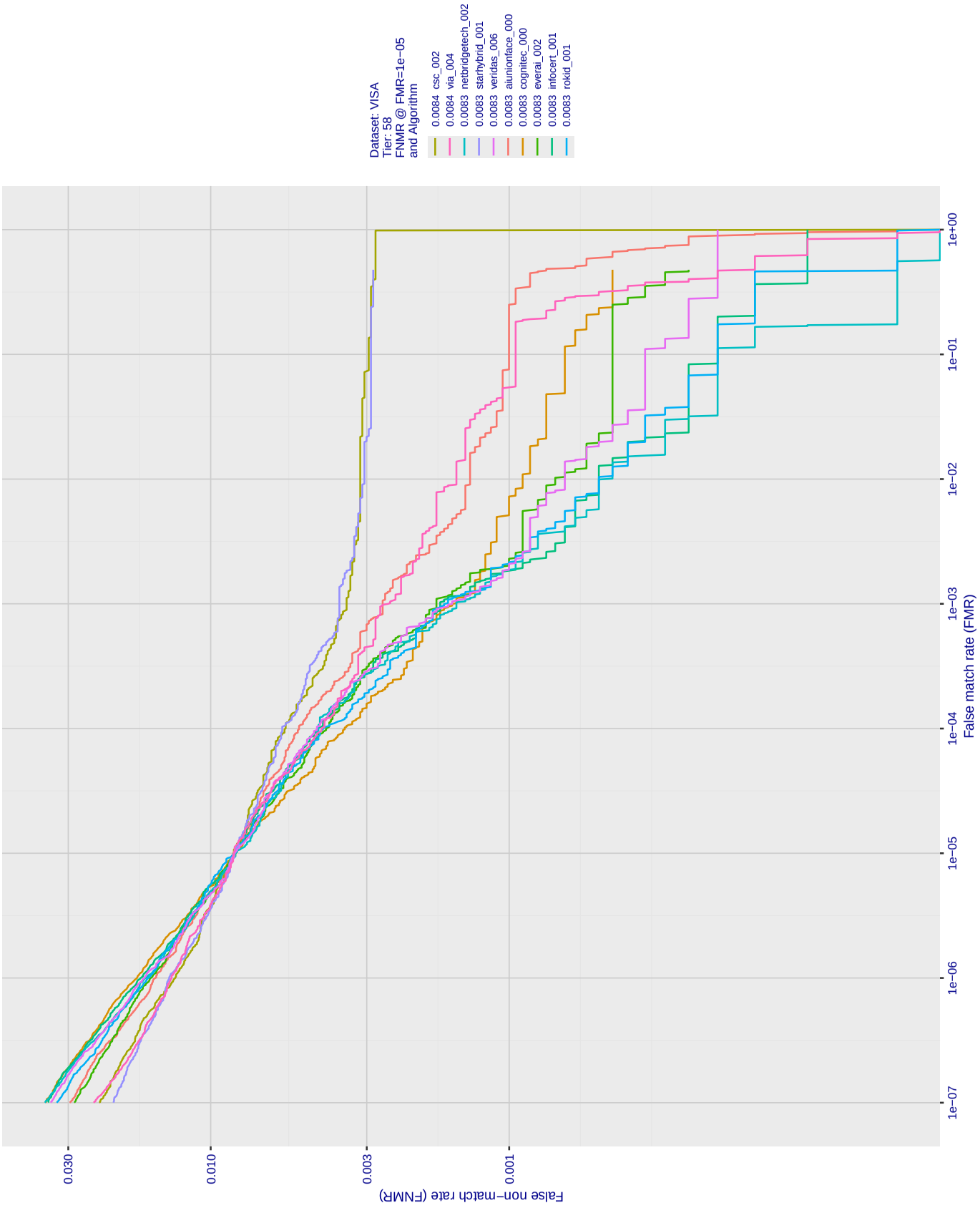


Figure 173: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

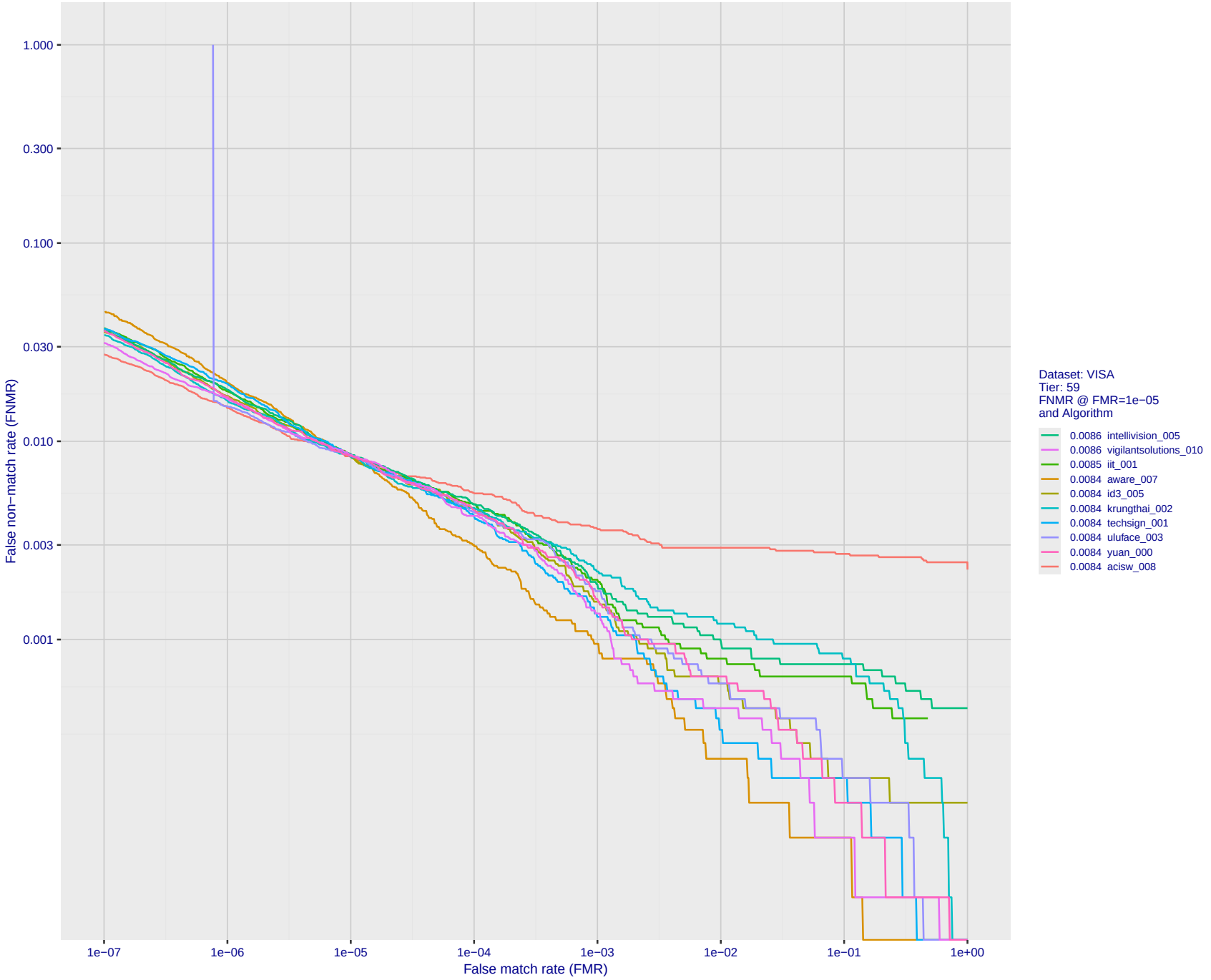


Figure 174: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

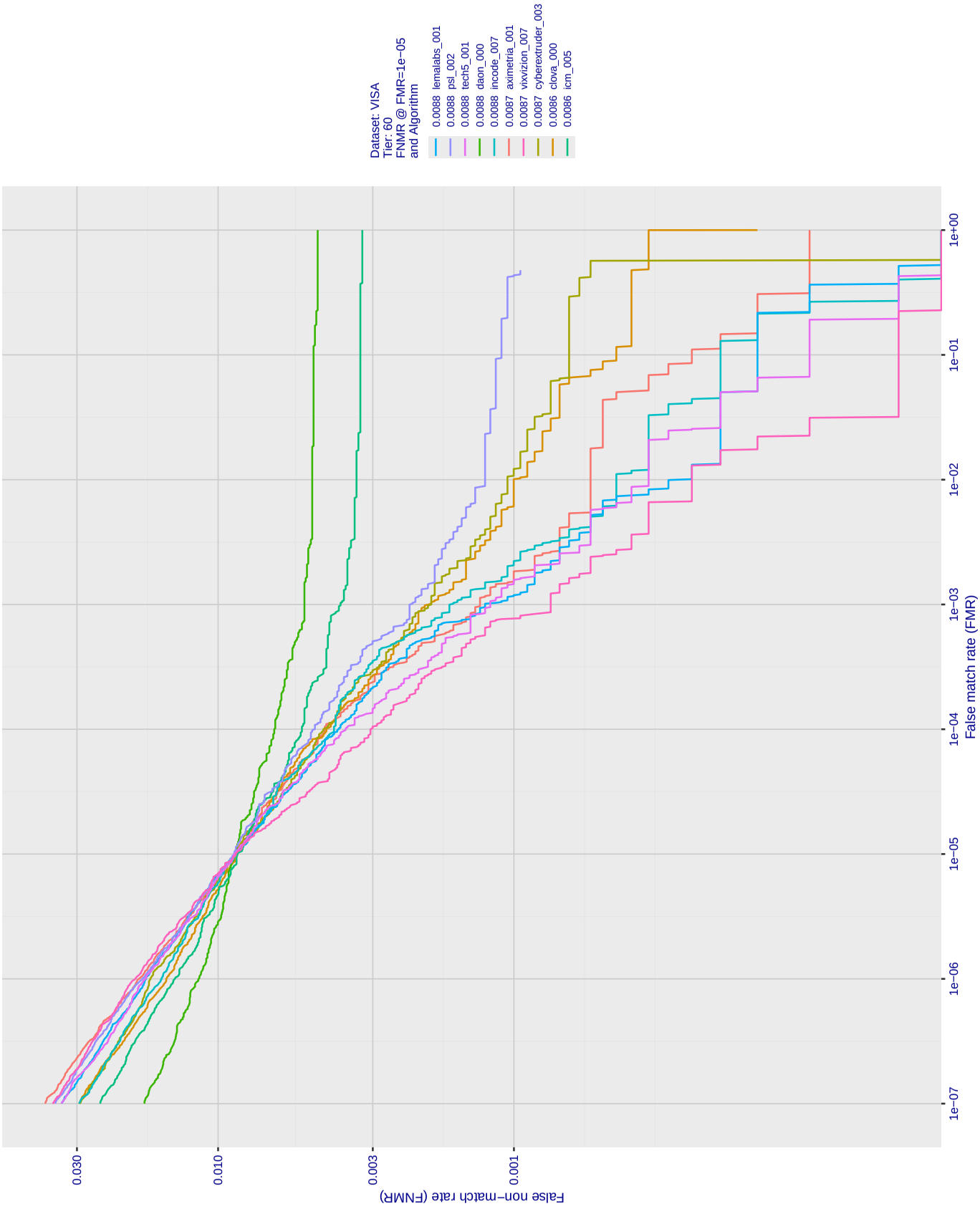


Figure 175: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

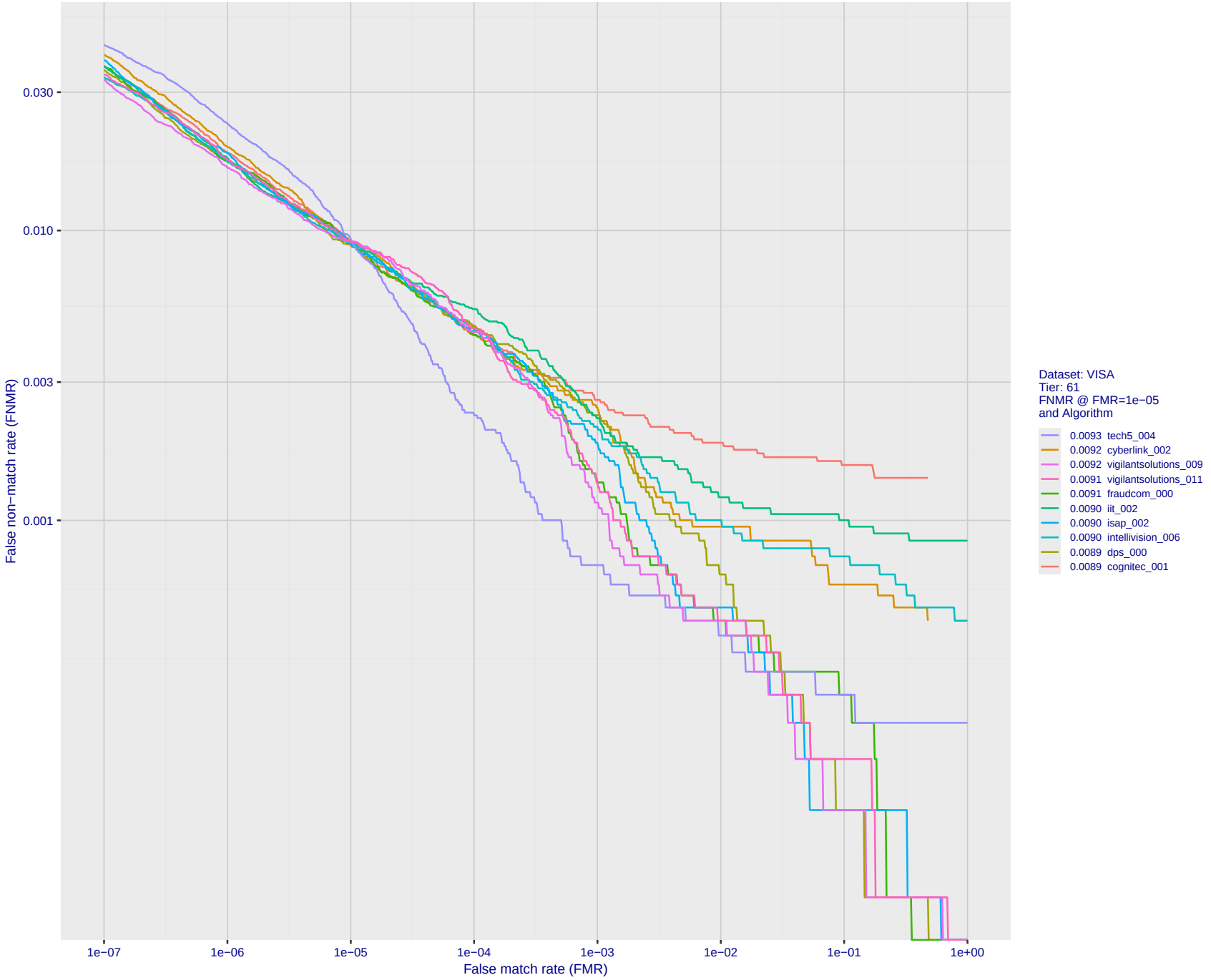


Figure 176: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

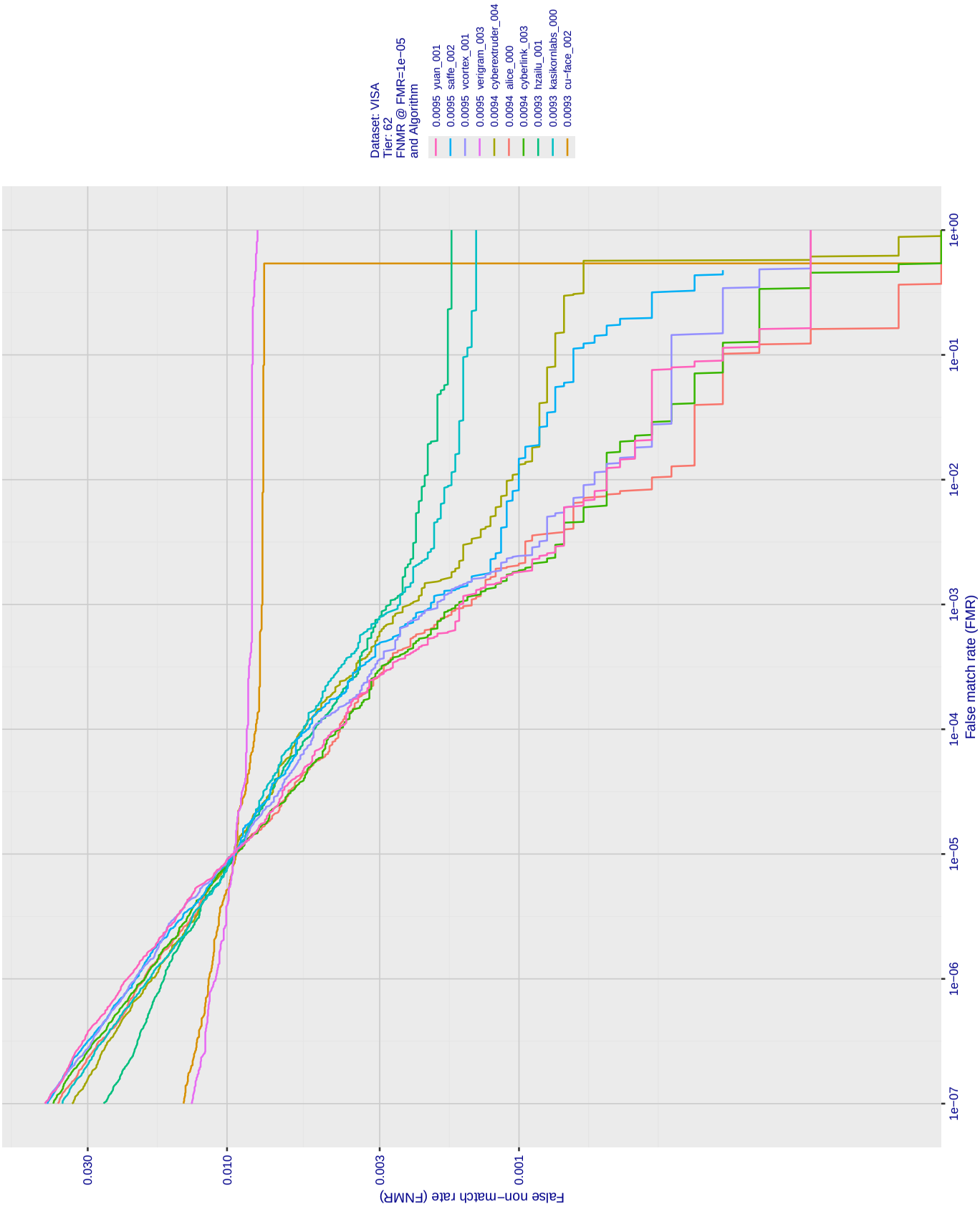


Figure 177: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

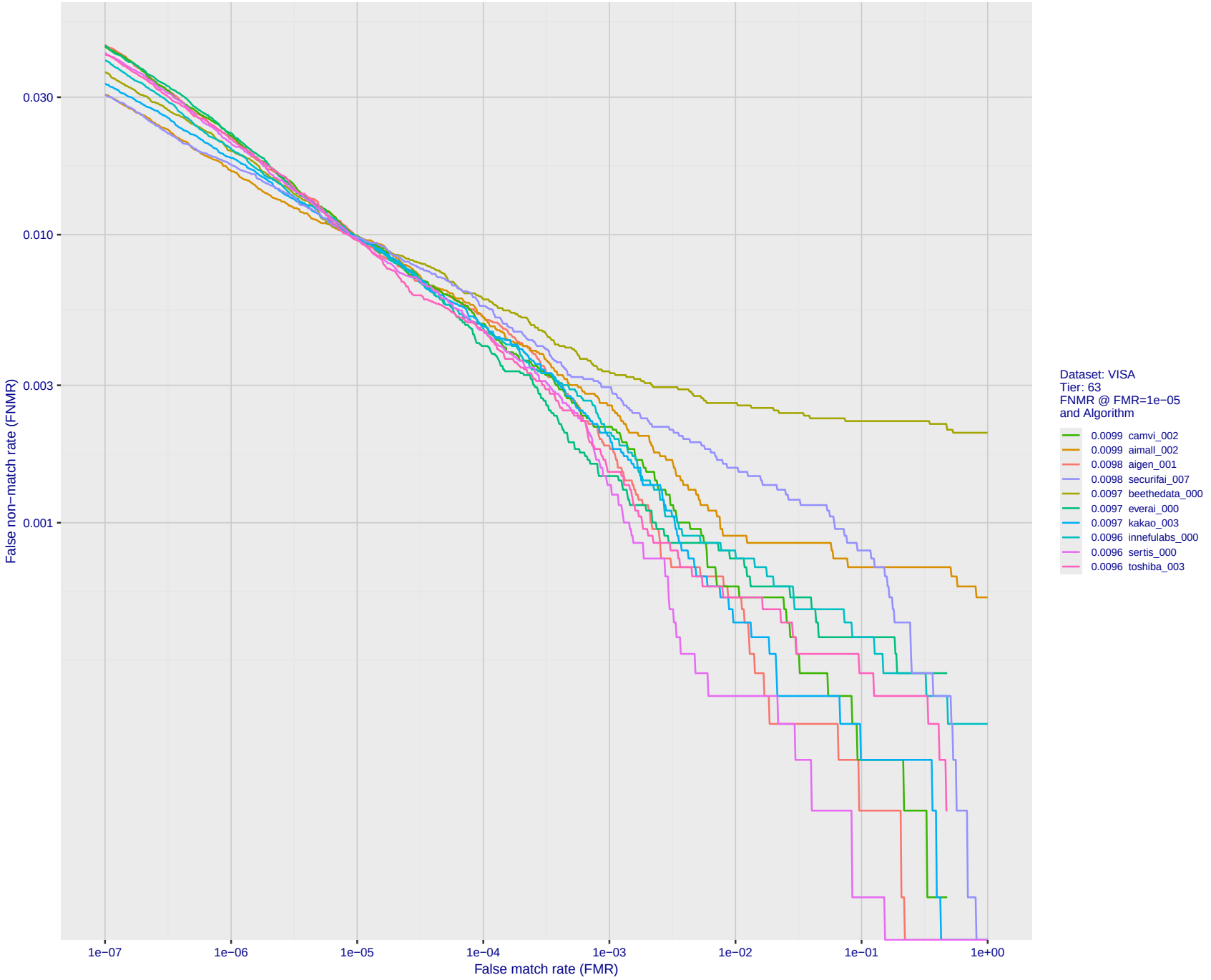


Figure 178: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

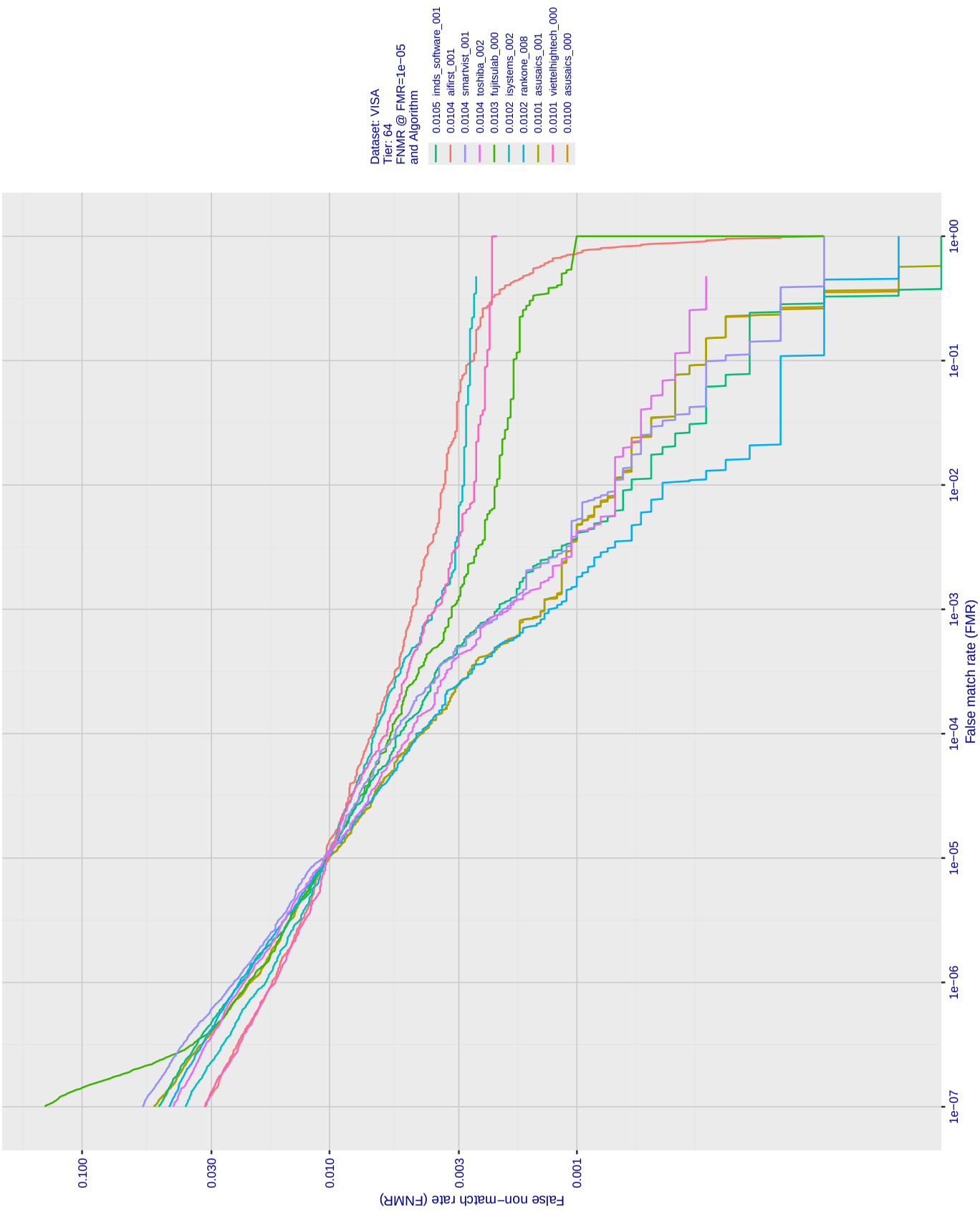


Figure 179: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

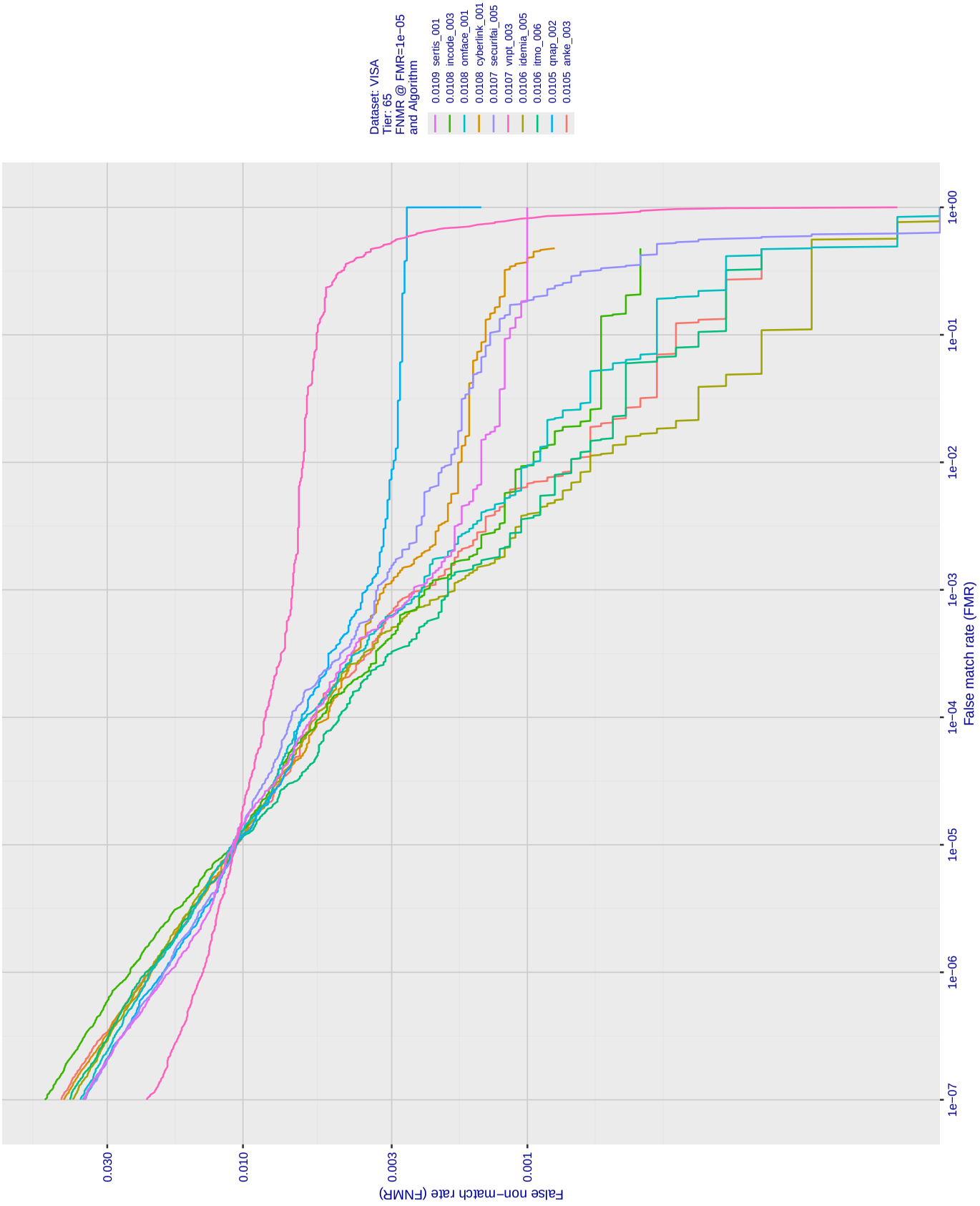


Figure 180: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

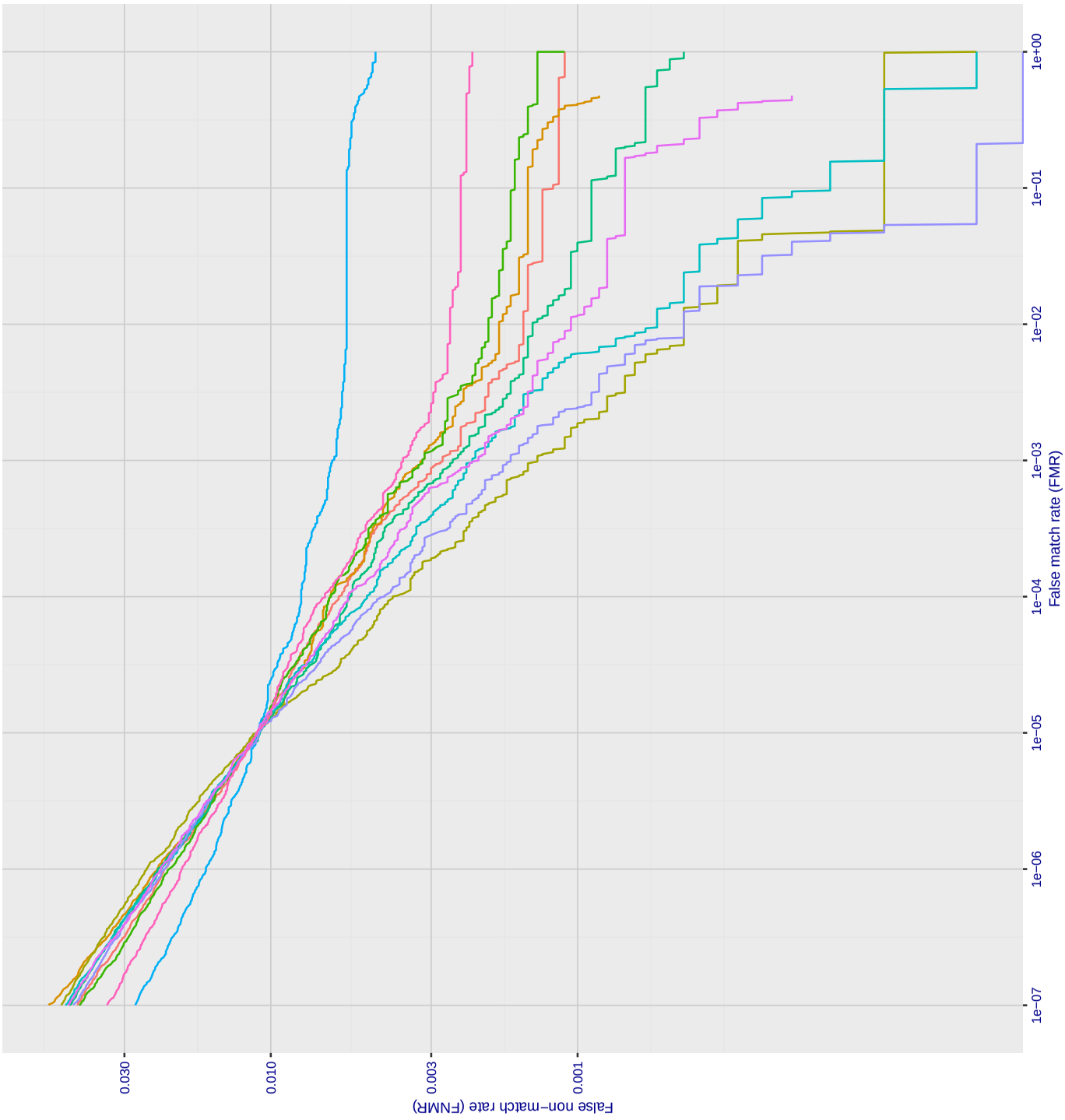


Figure 181: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

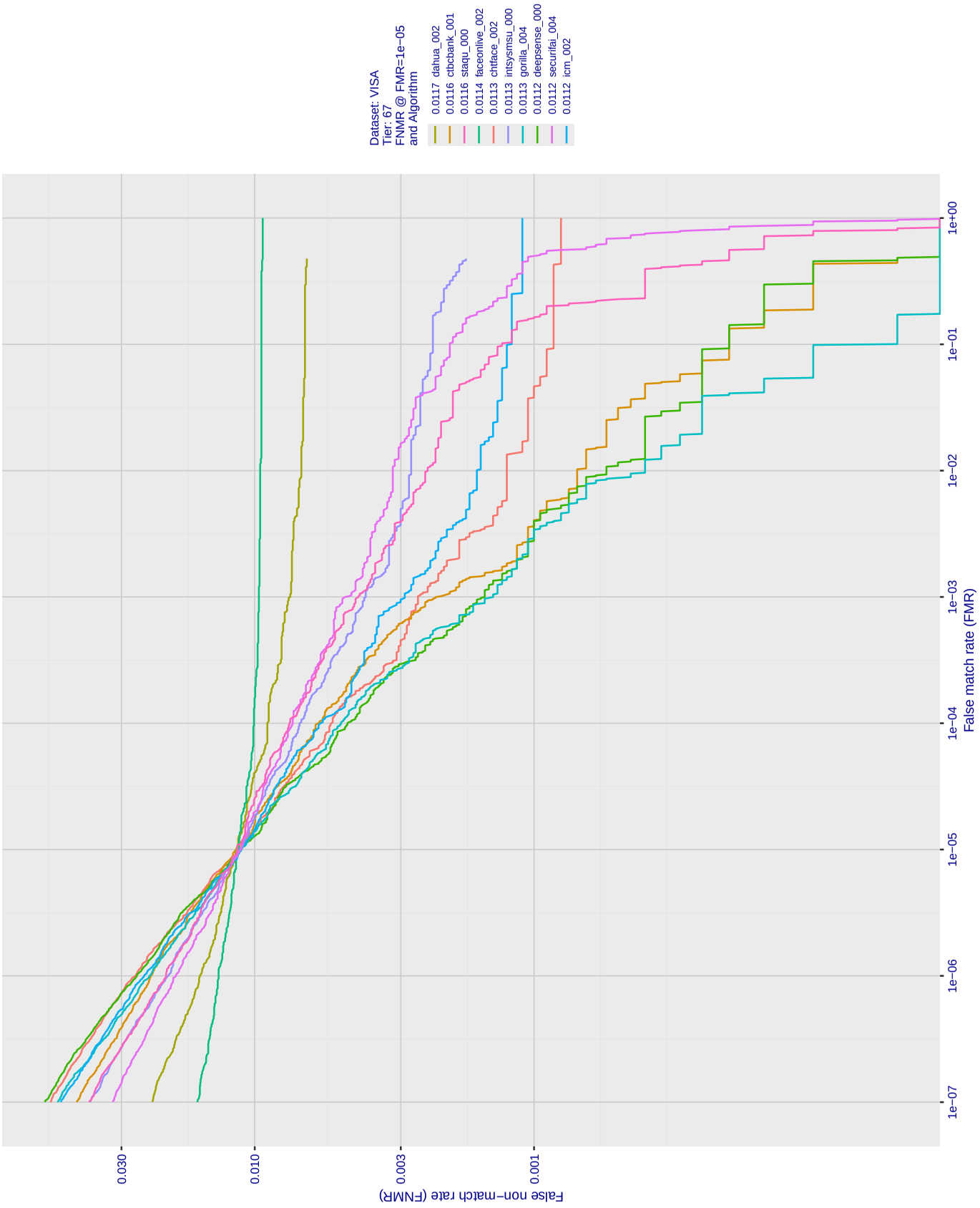


Figure 182: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

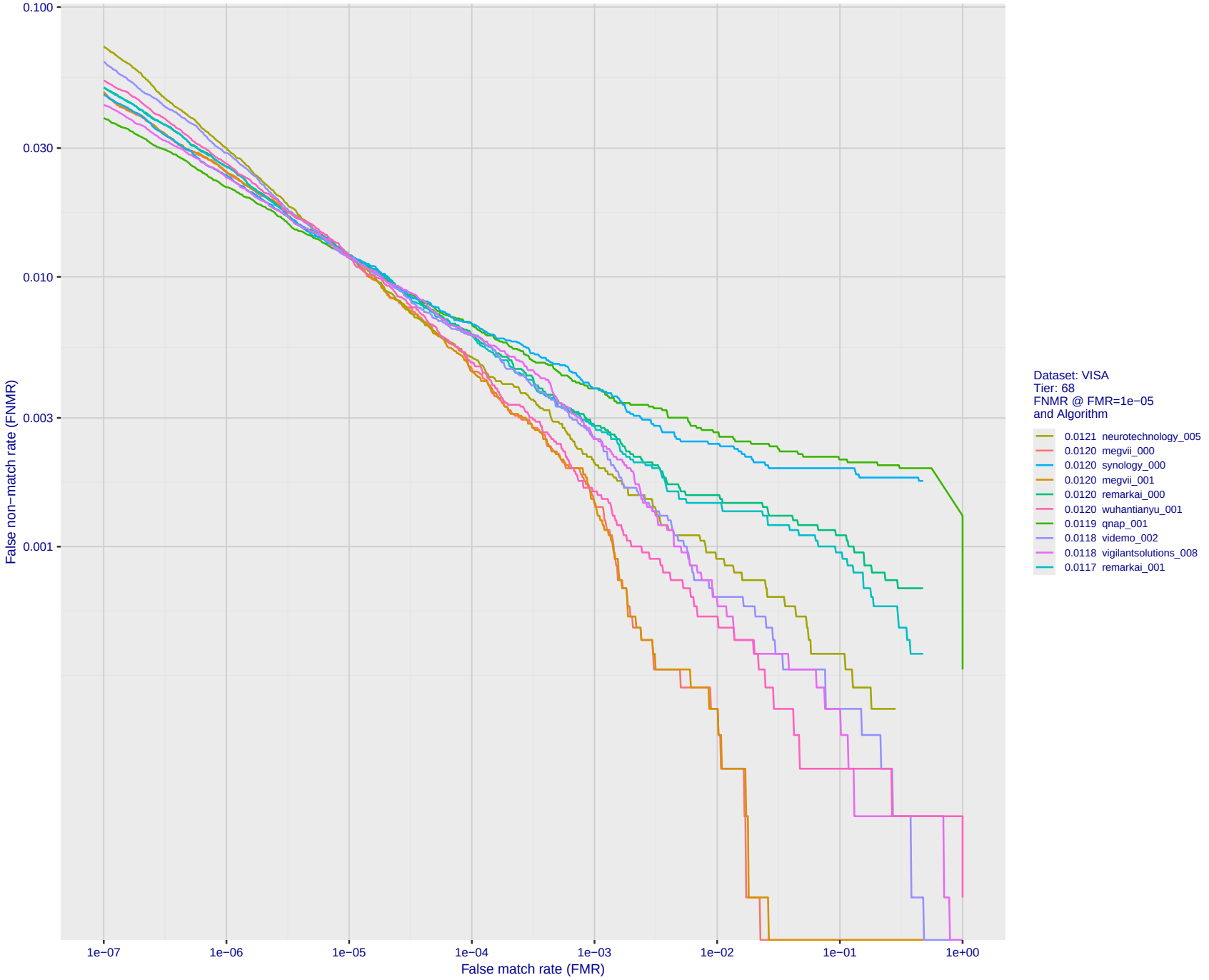


Figure 183: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

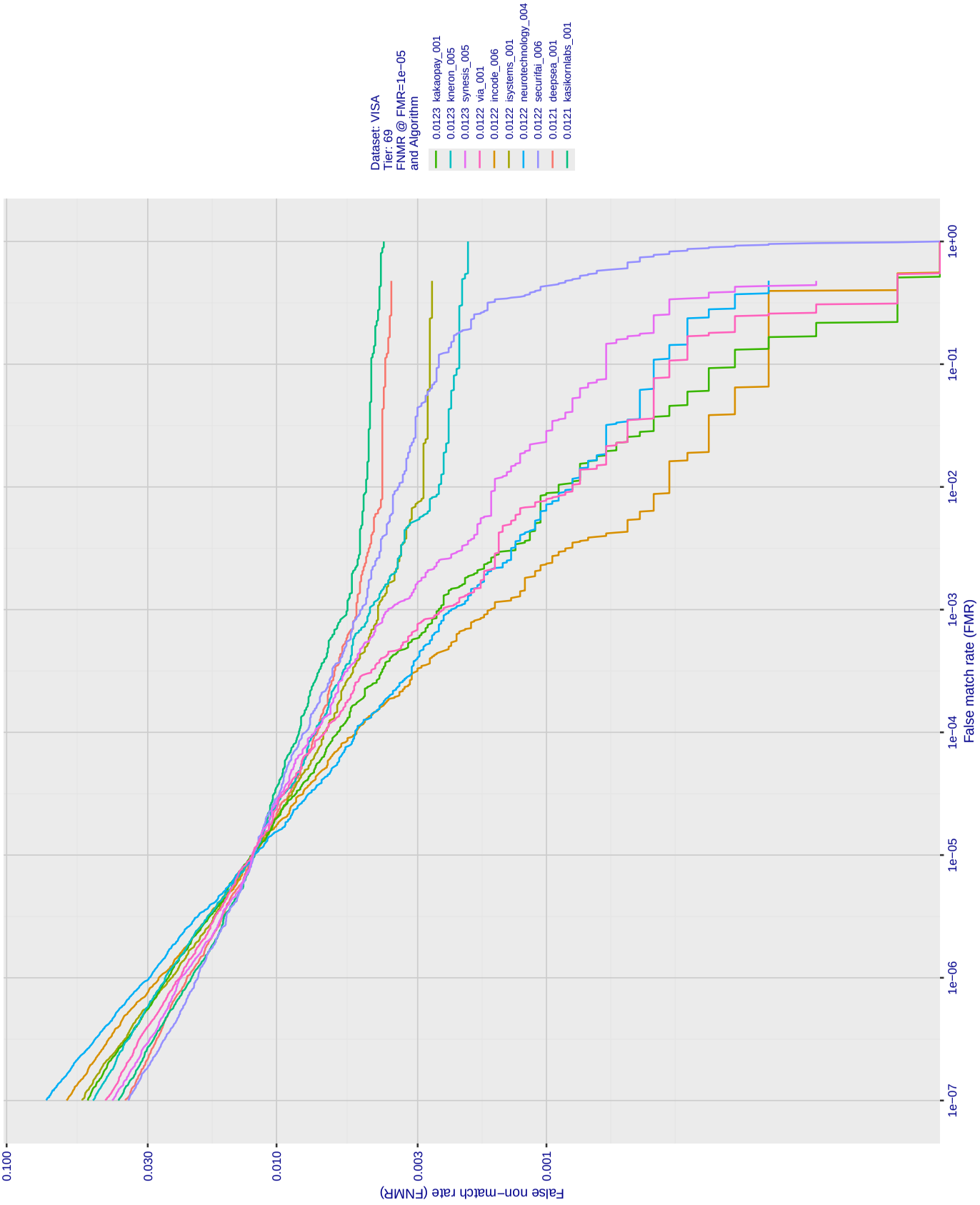


Figure 184: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

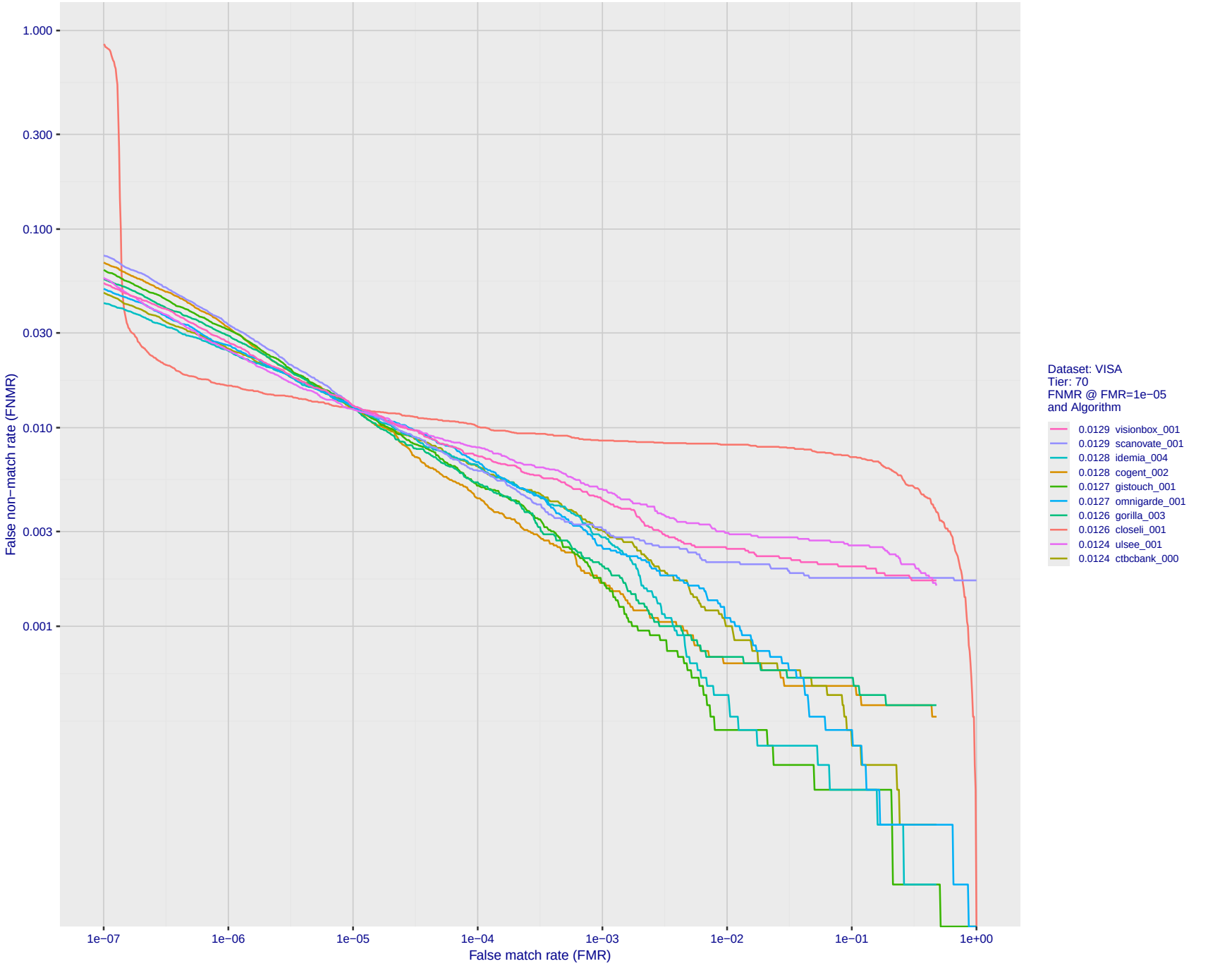


Figure 185: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

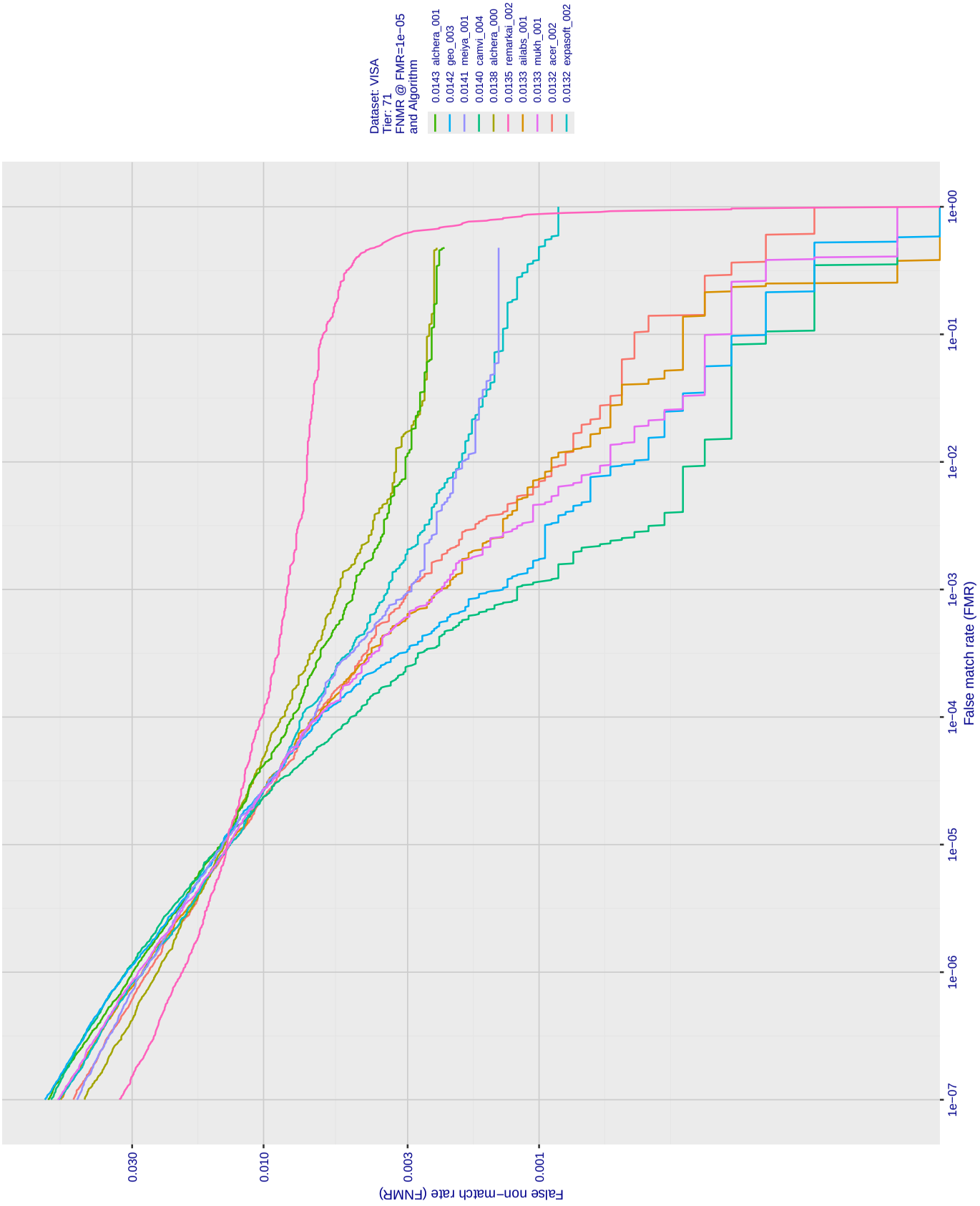


Figure 186: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

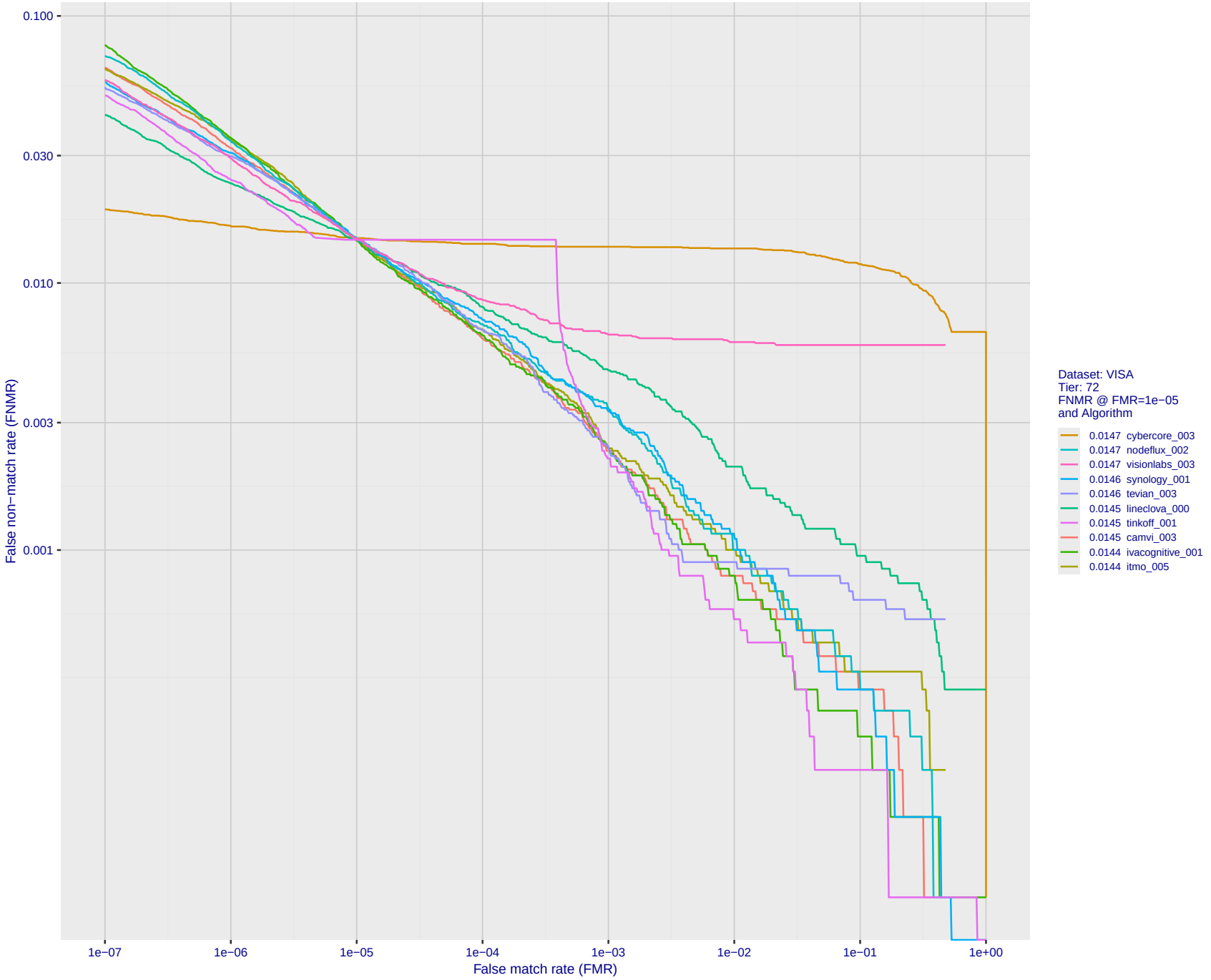


Figure 187: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

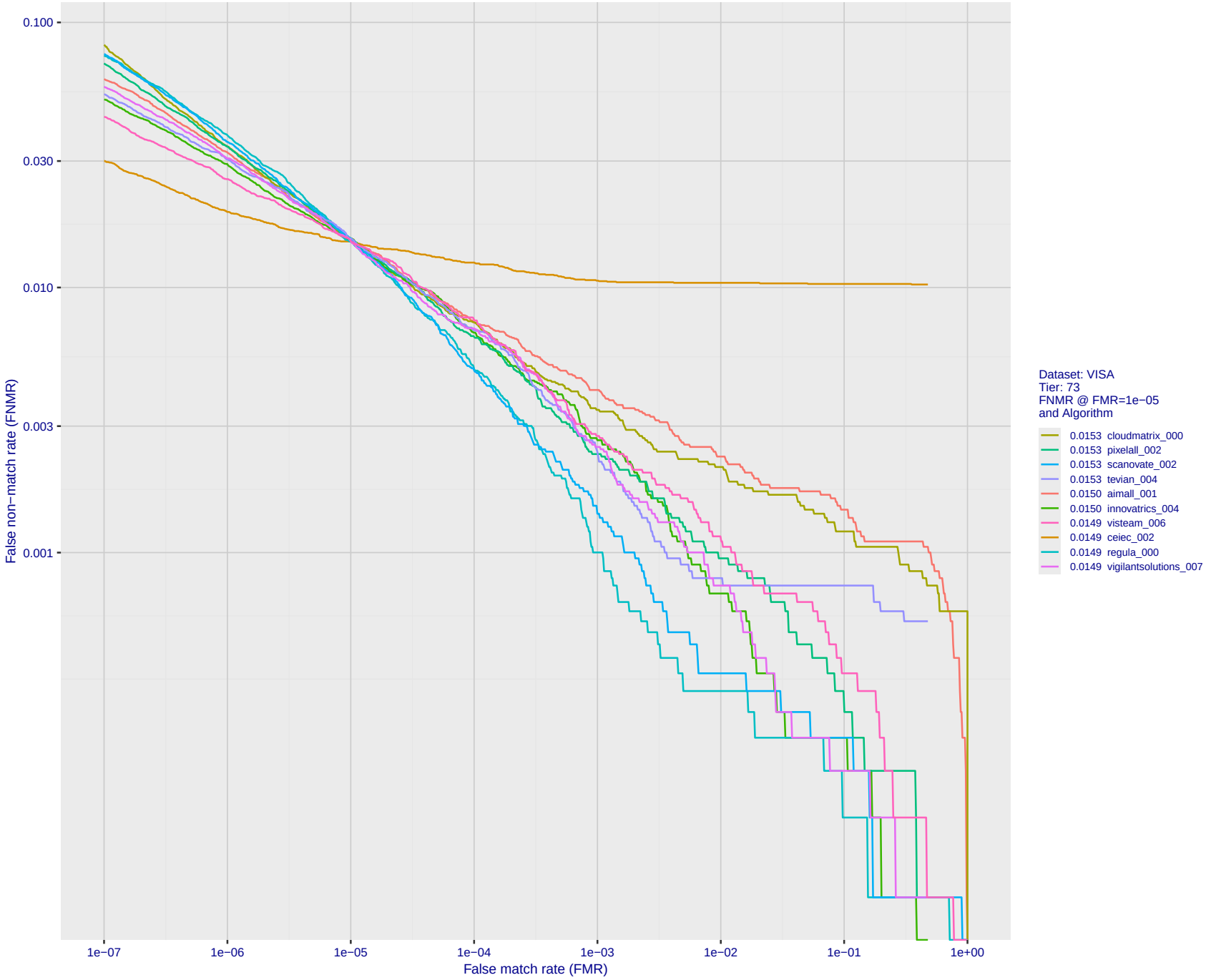


Figure 188: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

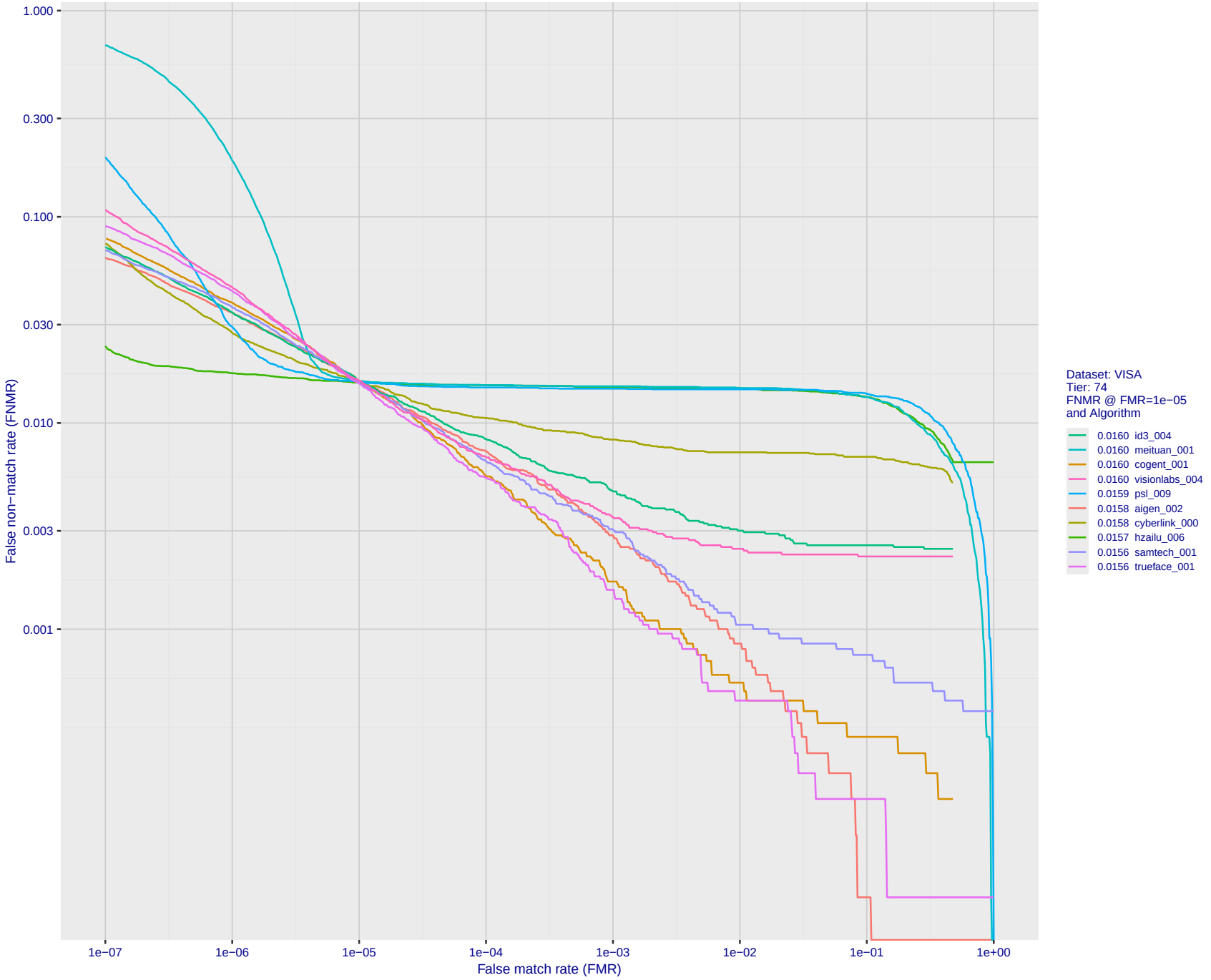


Figure 189: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

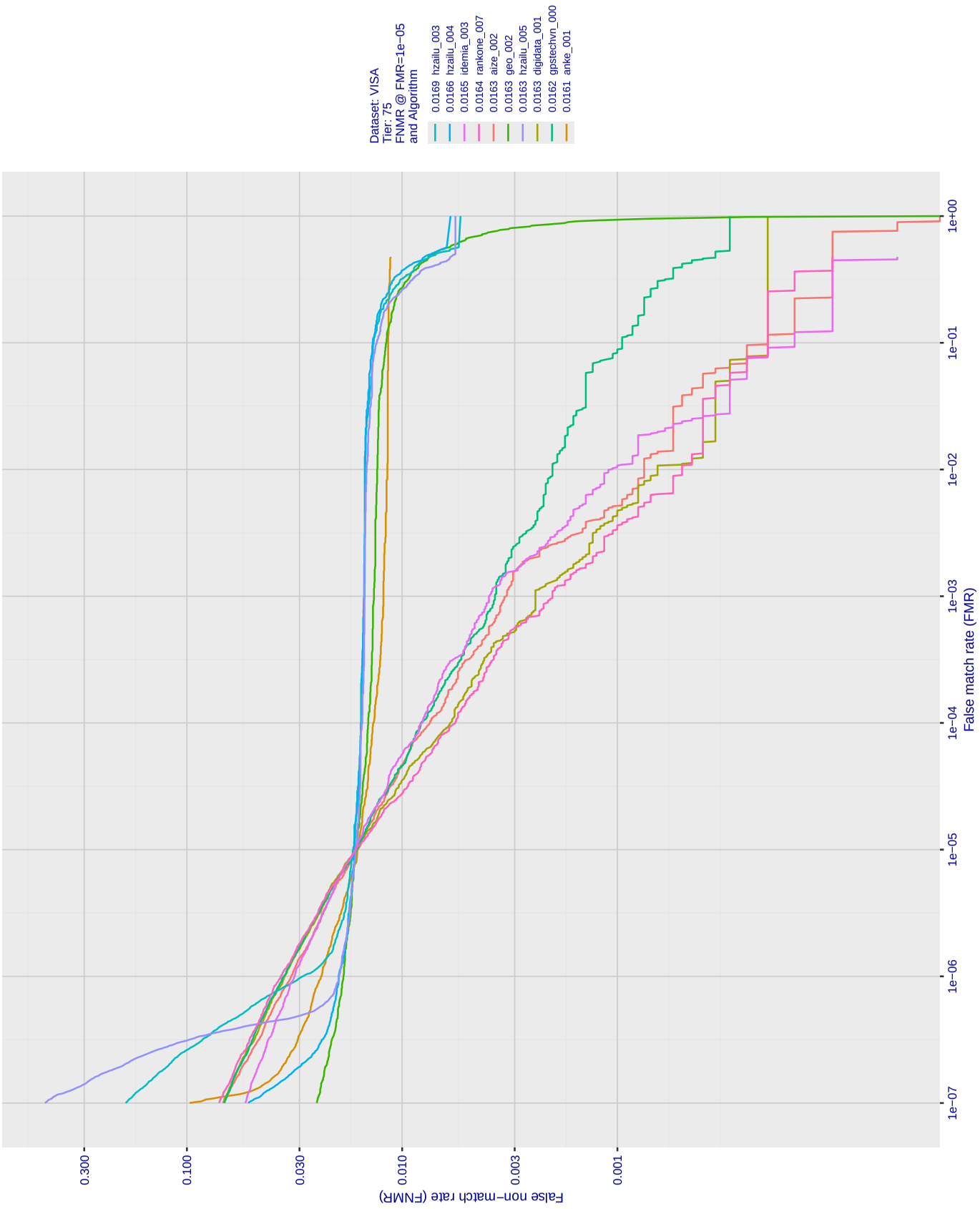


Figure 190: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

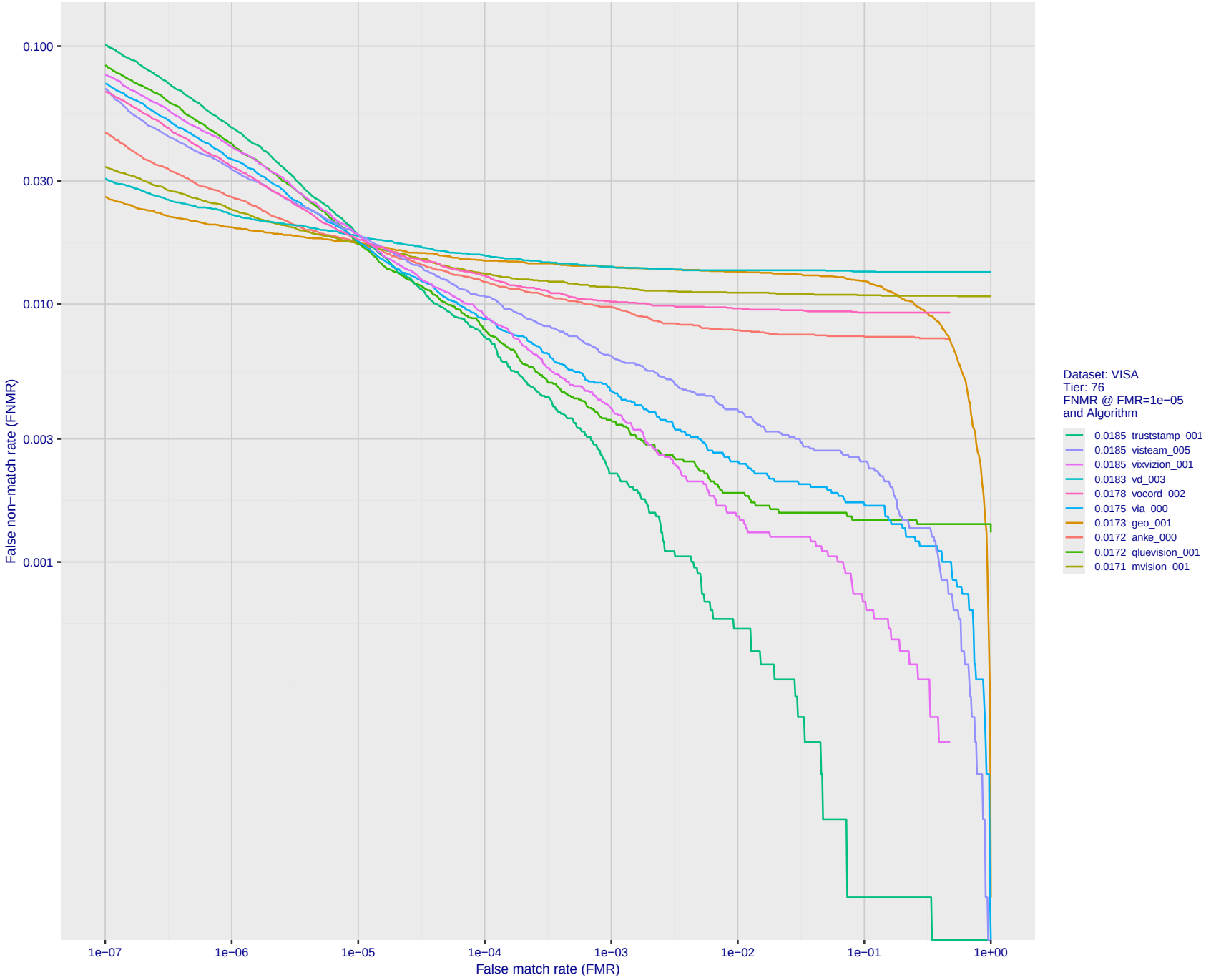


Figure 191: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

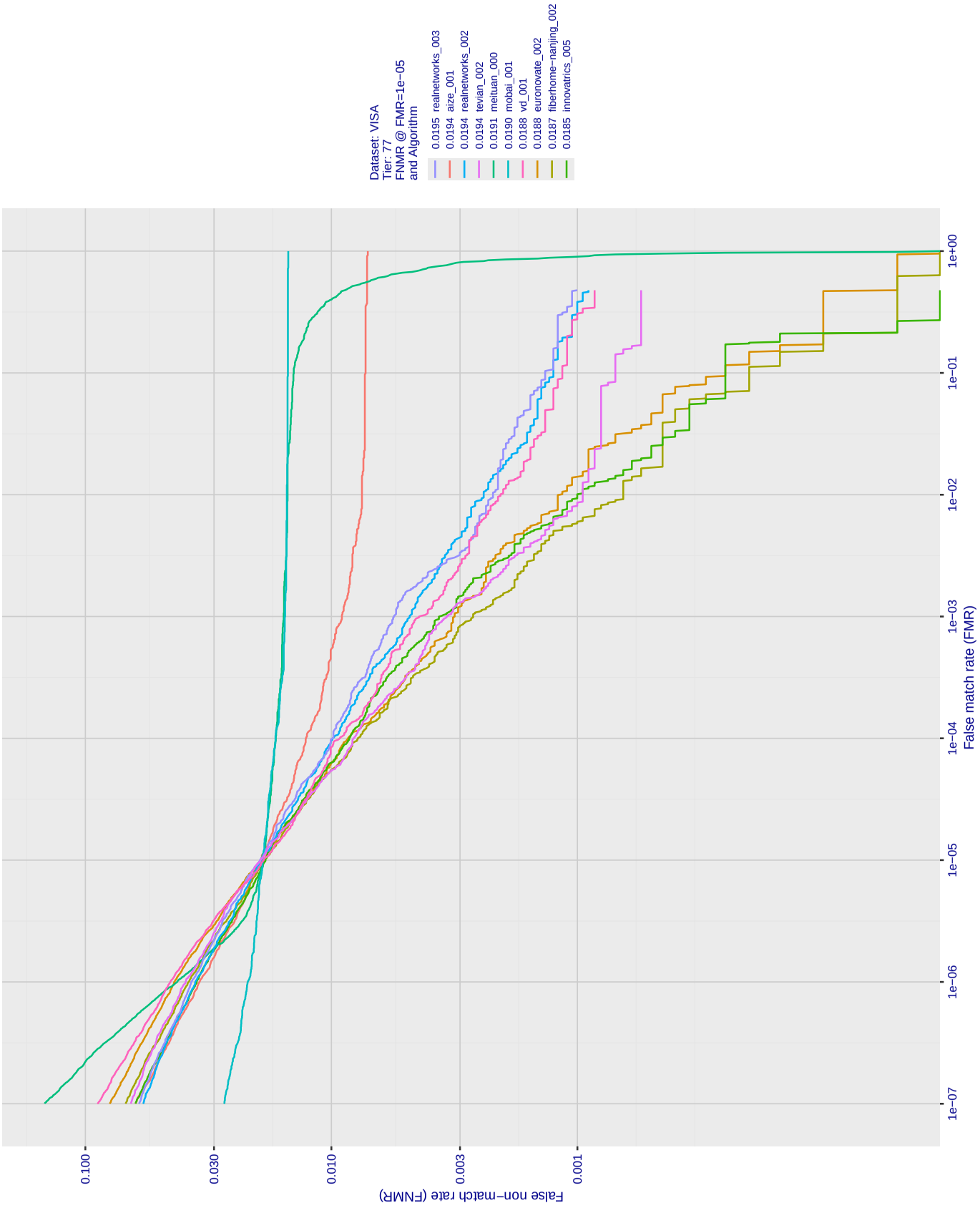


Figure 192: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

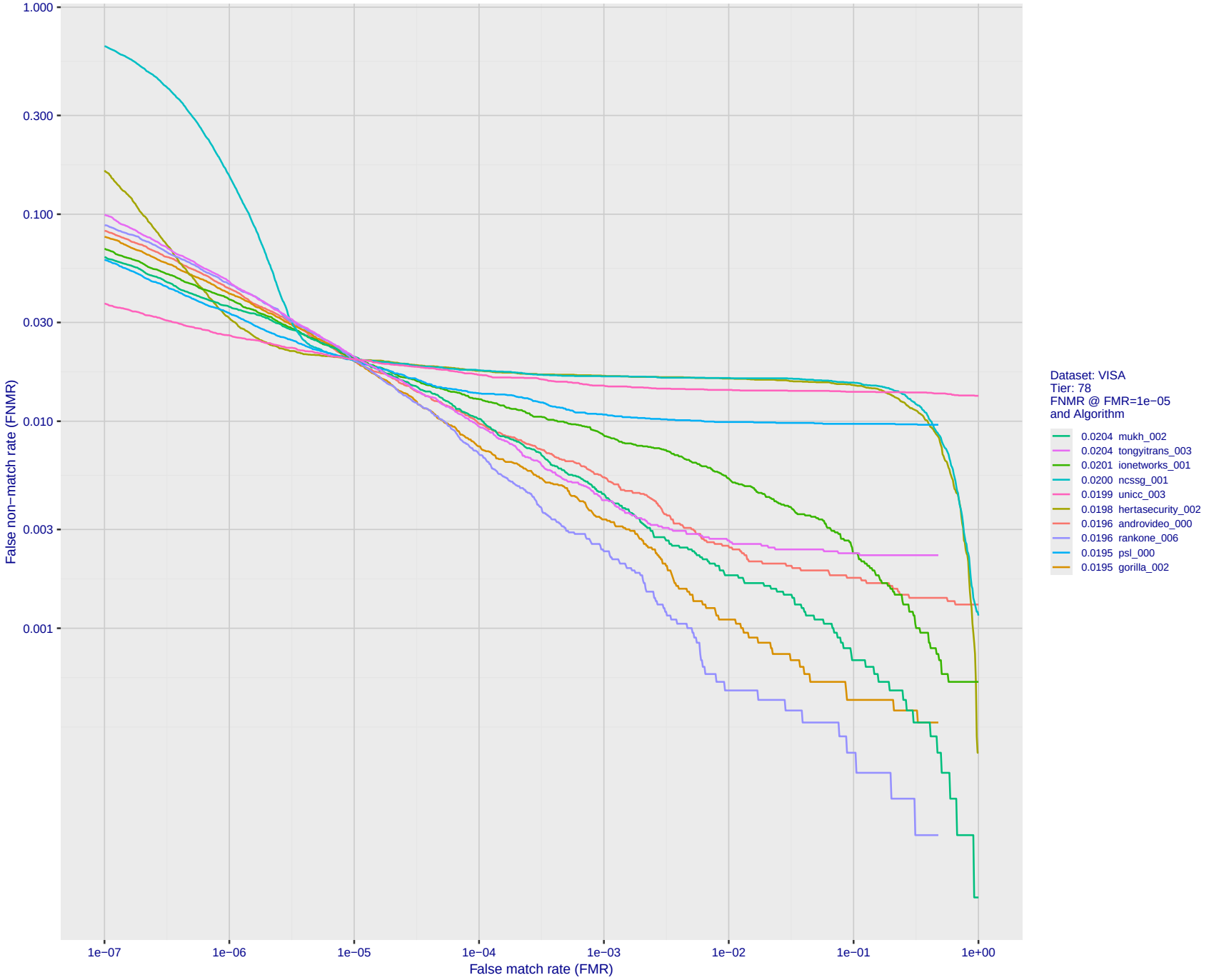


Figure 193: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

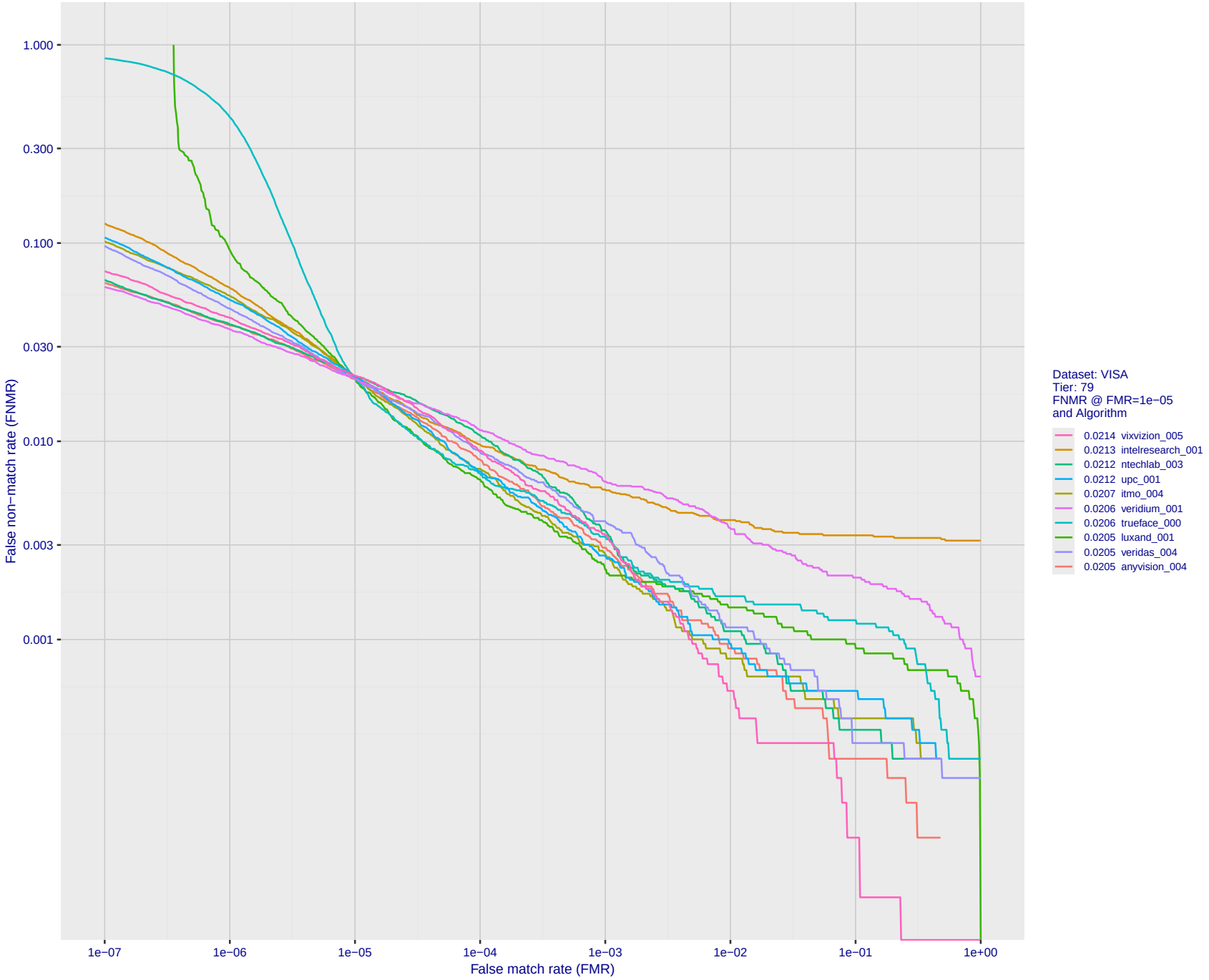


Figure 194: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

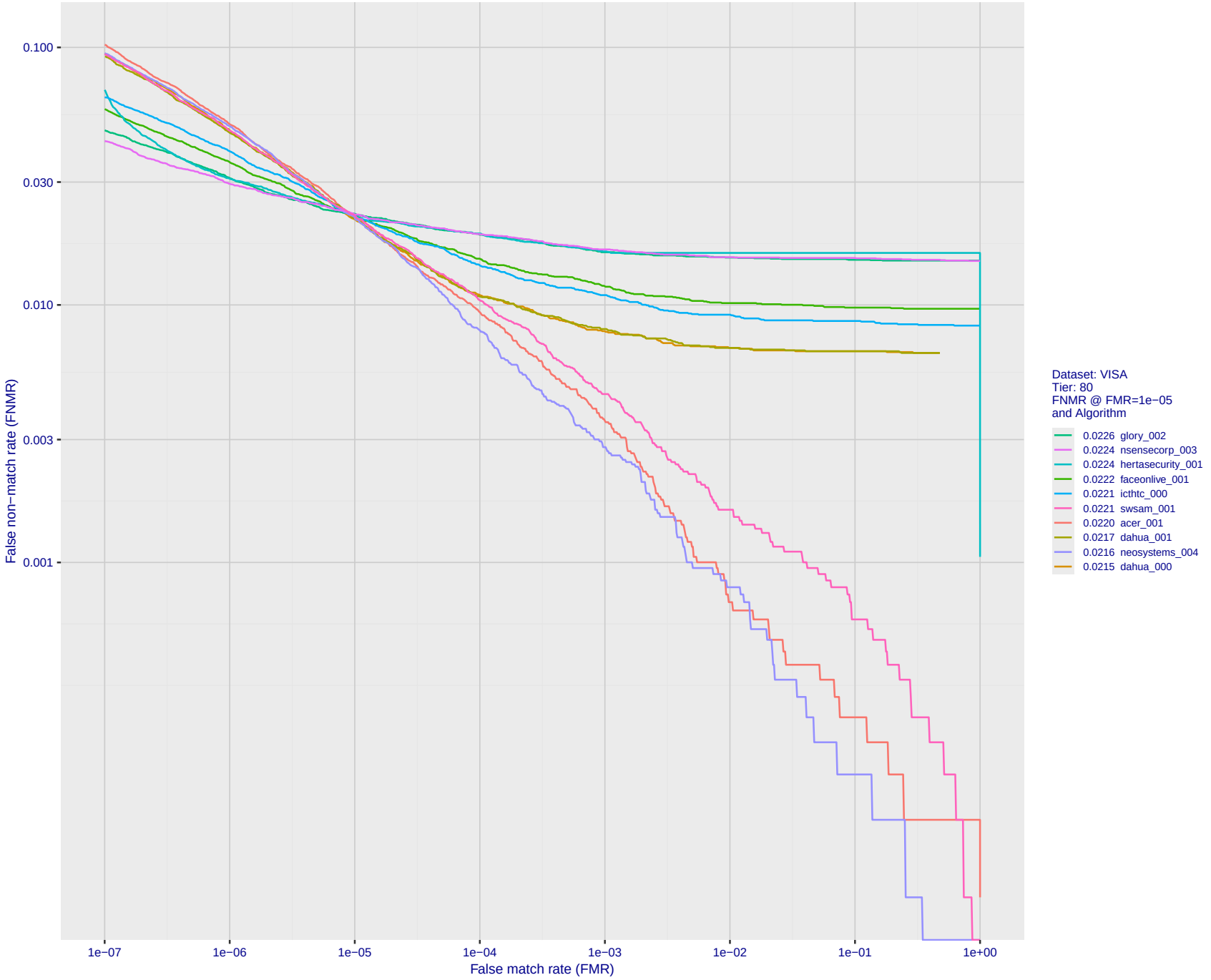


Figure 195: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

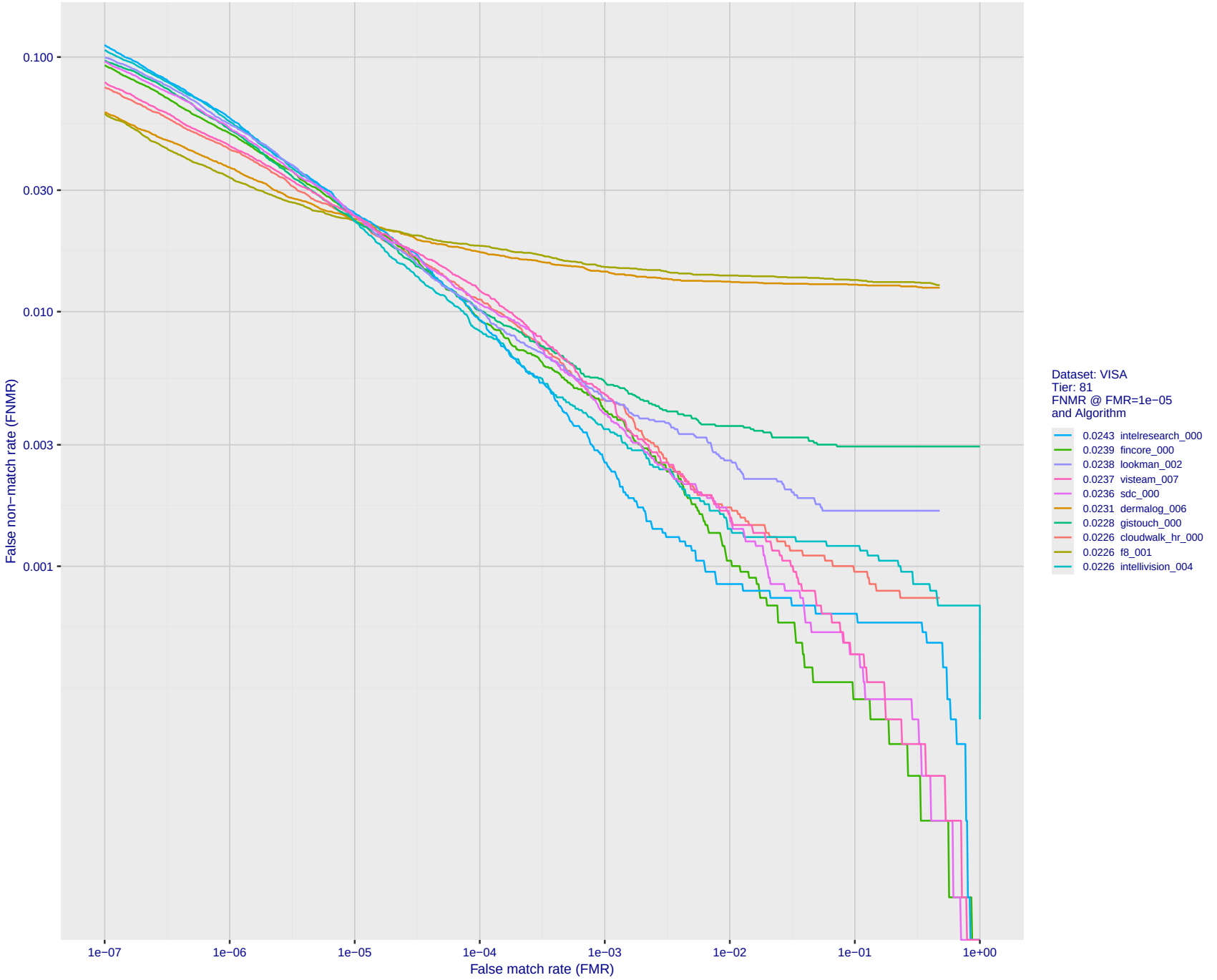


Figure 196: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

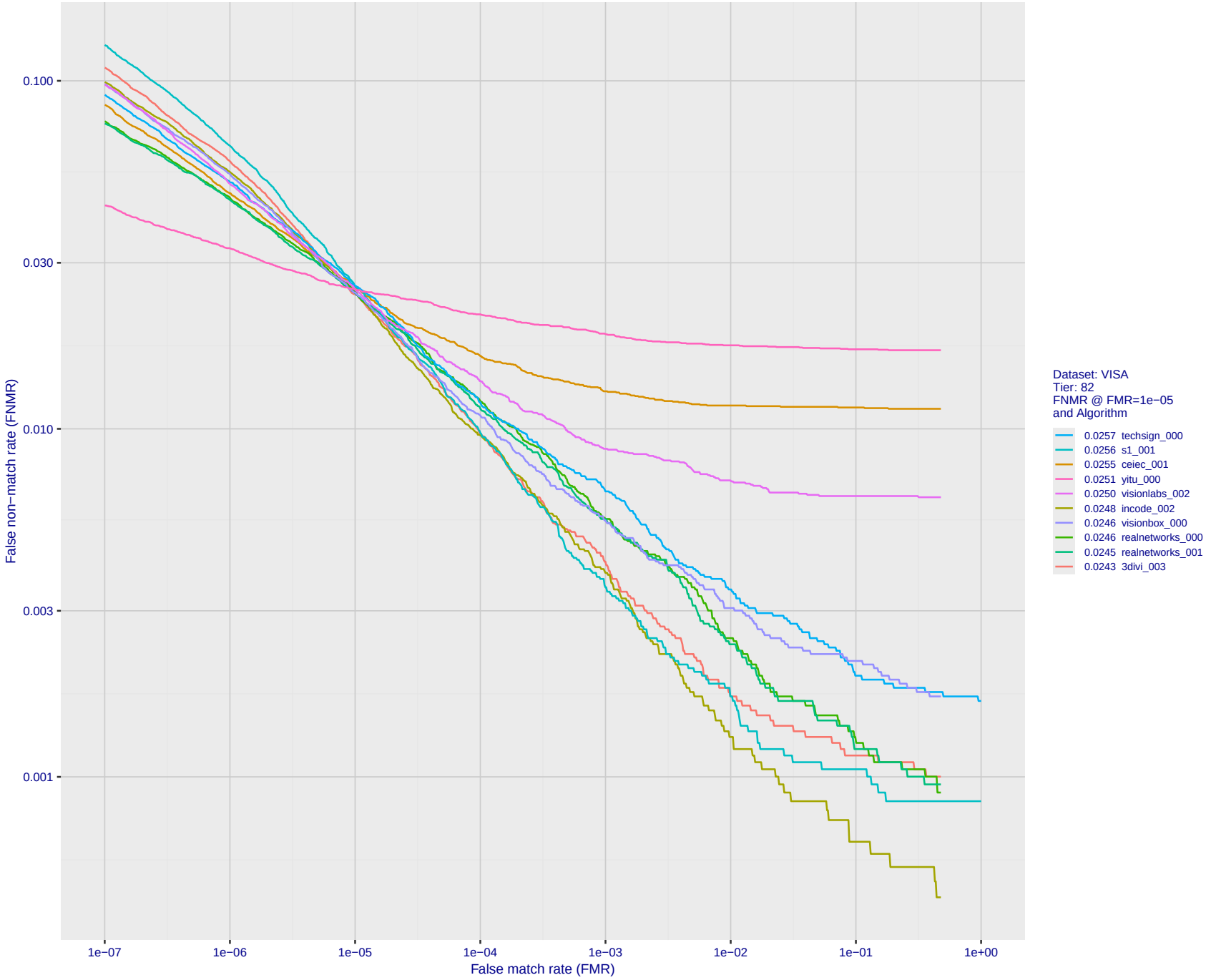


Figure 197: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

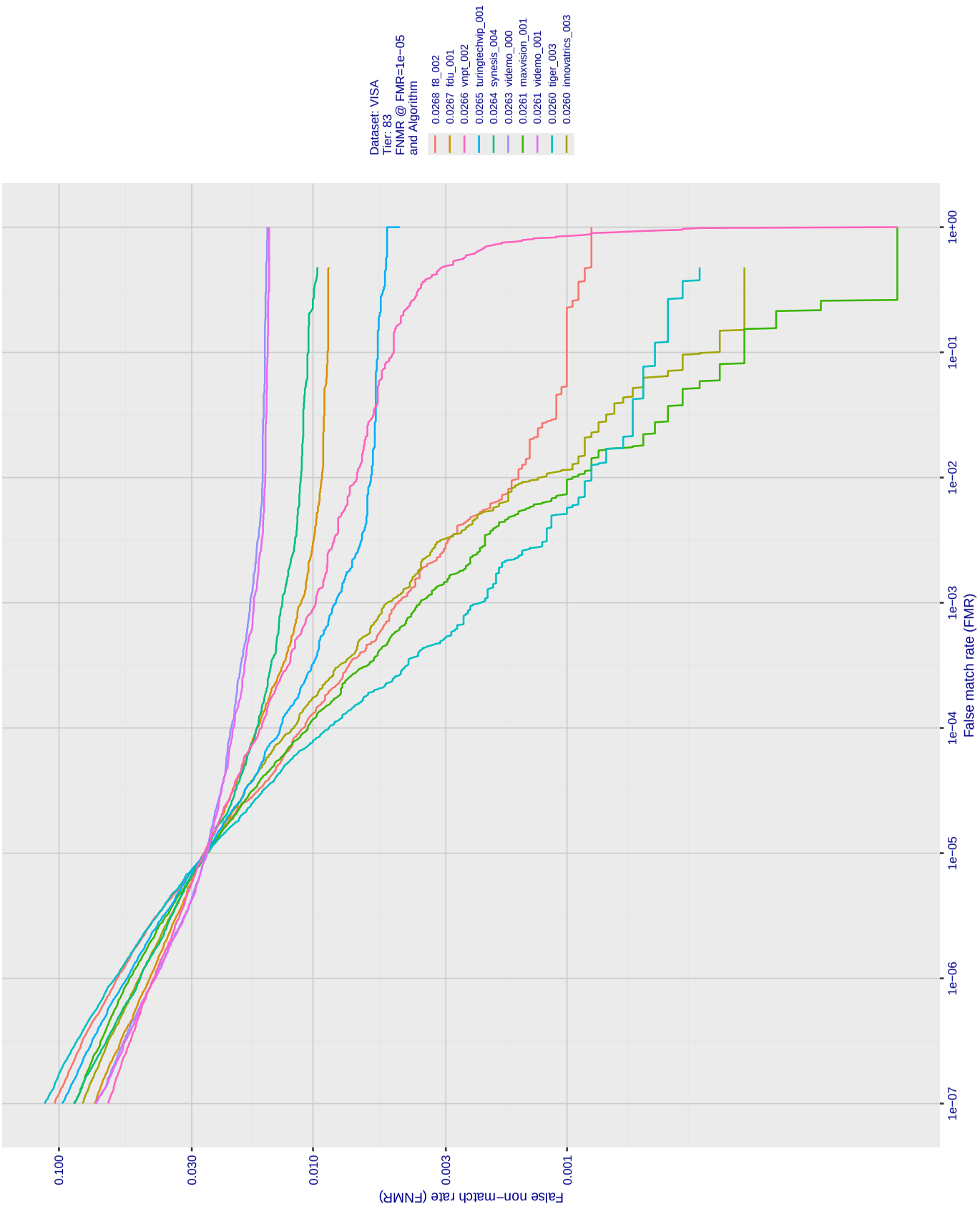


Figure 198: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

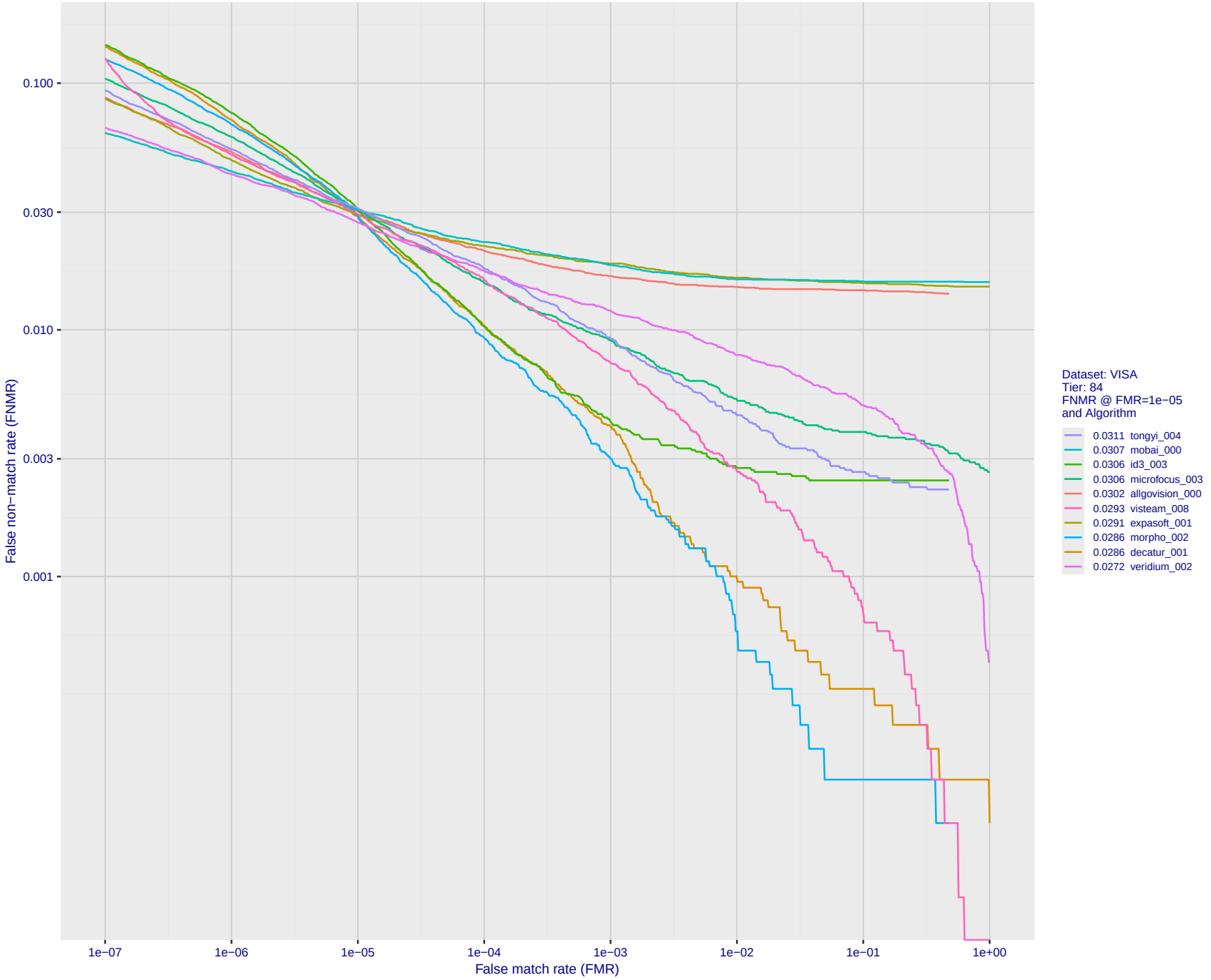


Figure 199: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

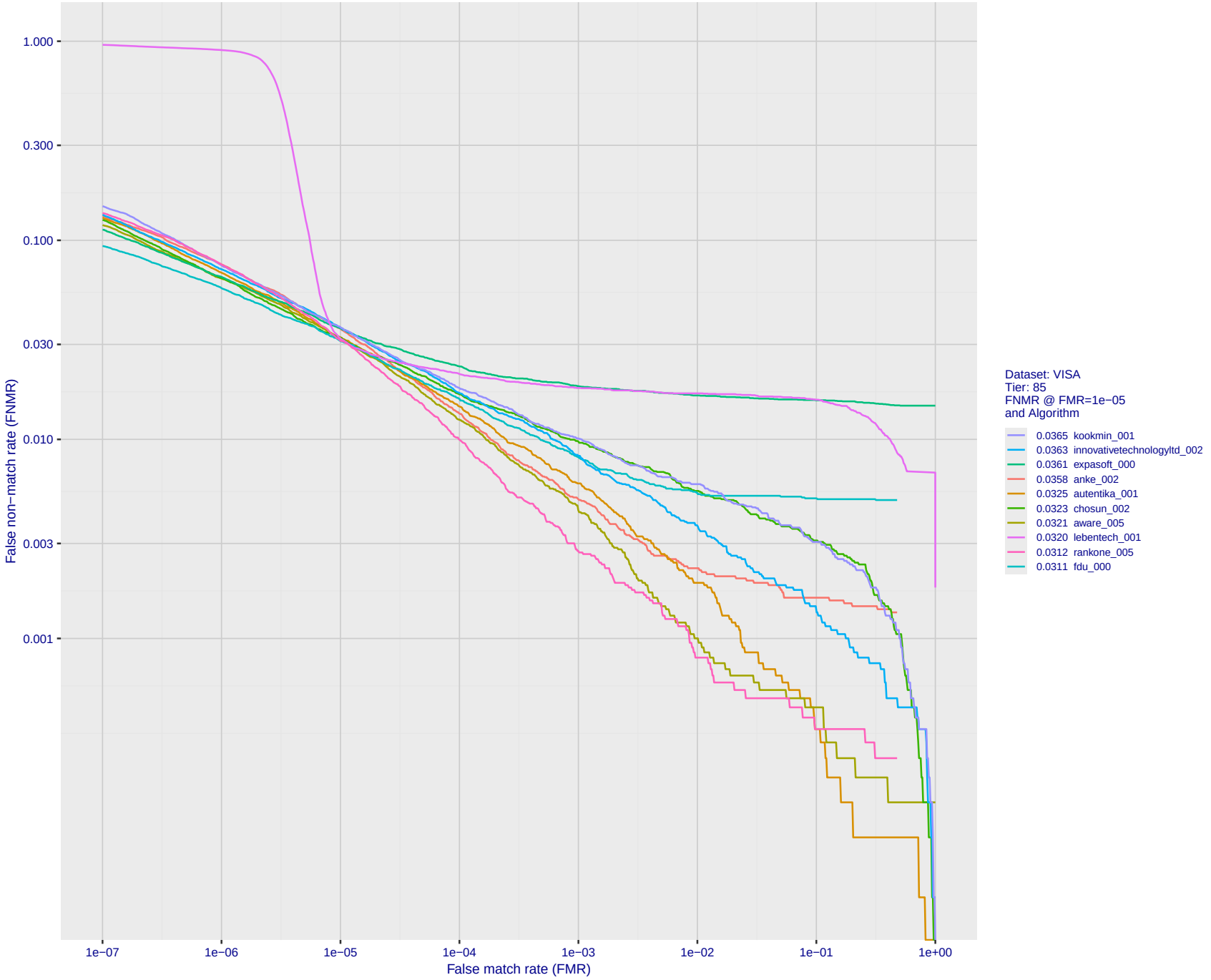


Figure 200: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

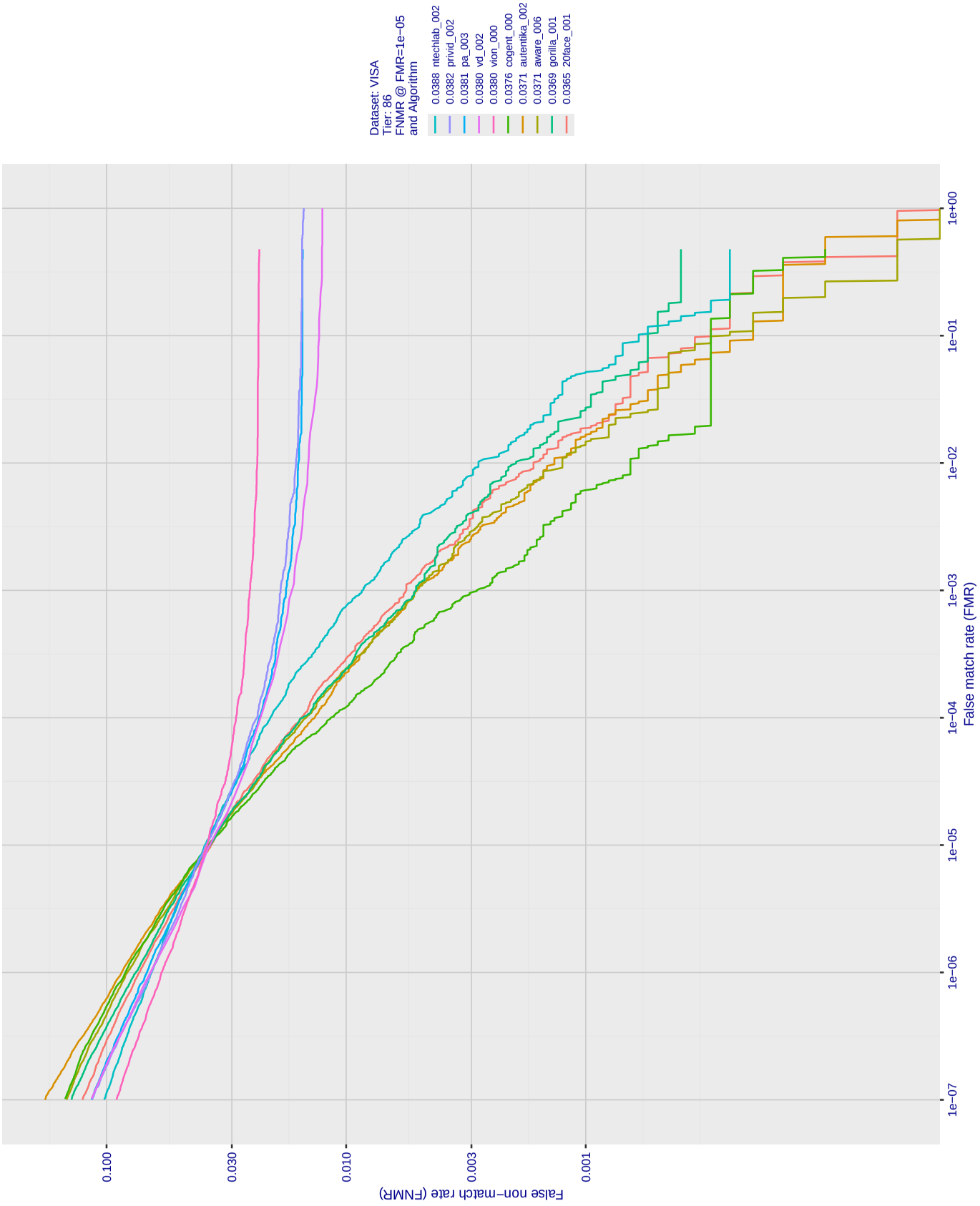


Figure 201: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

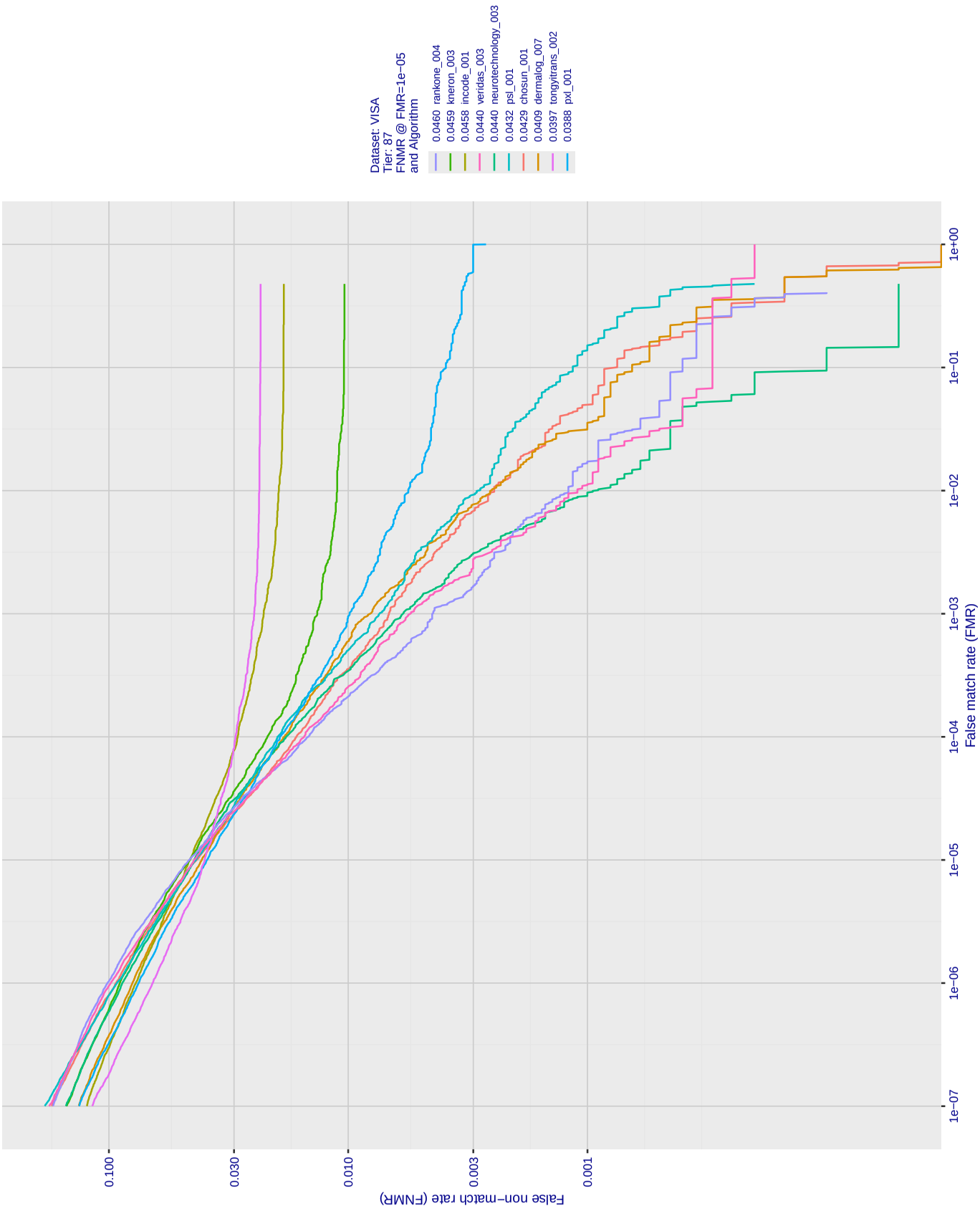


Figure 202: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

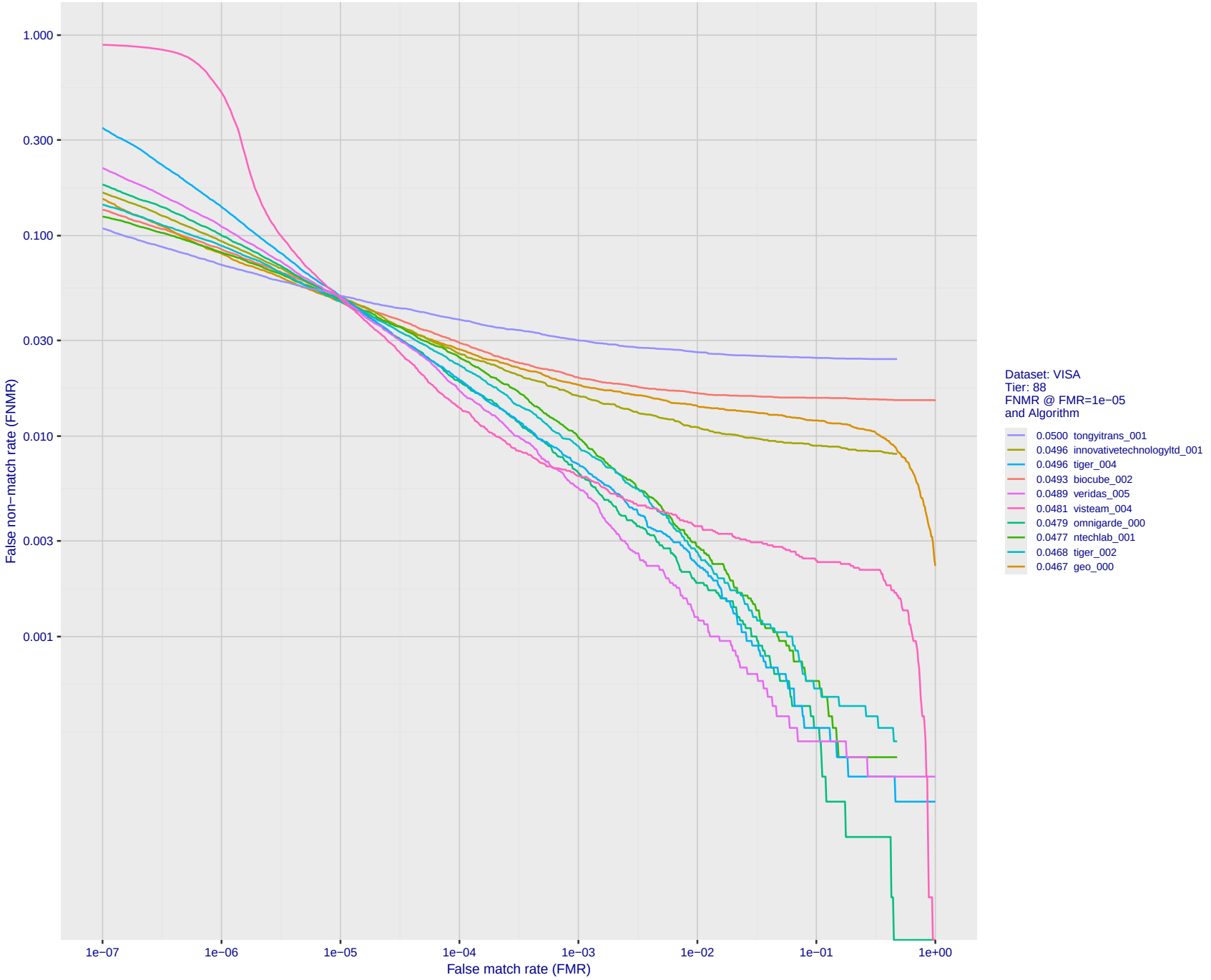


Figure 203: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

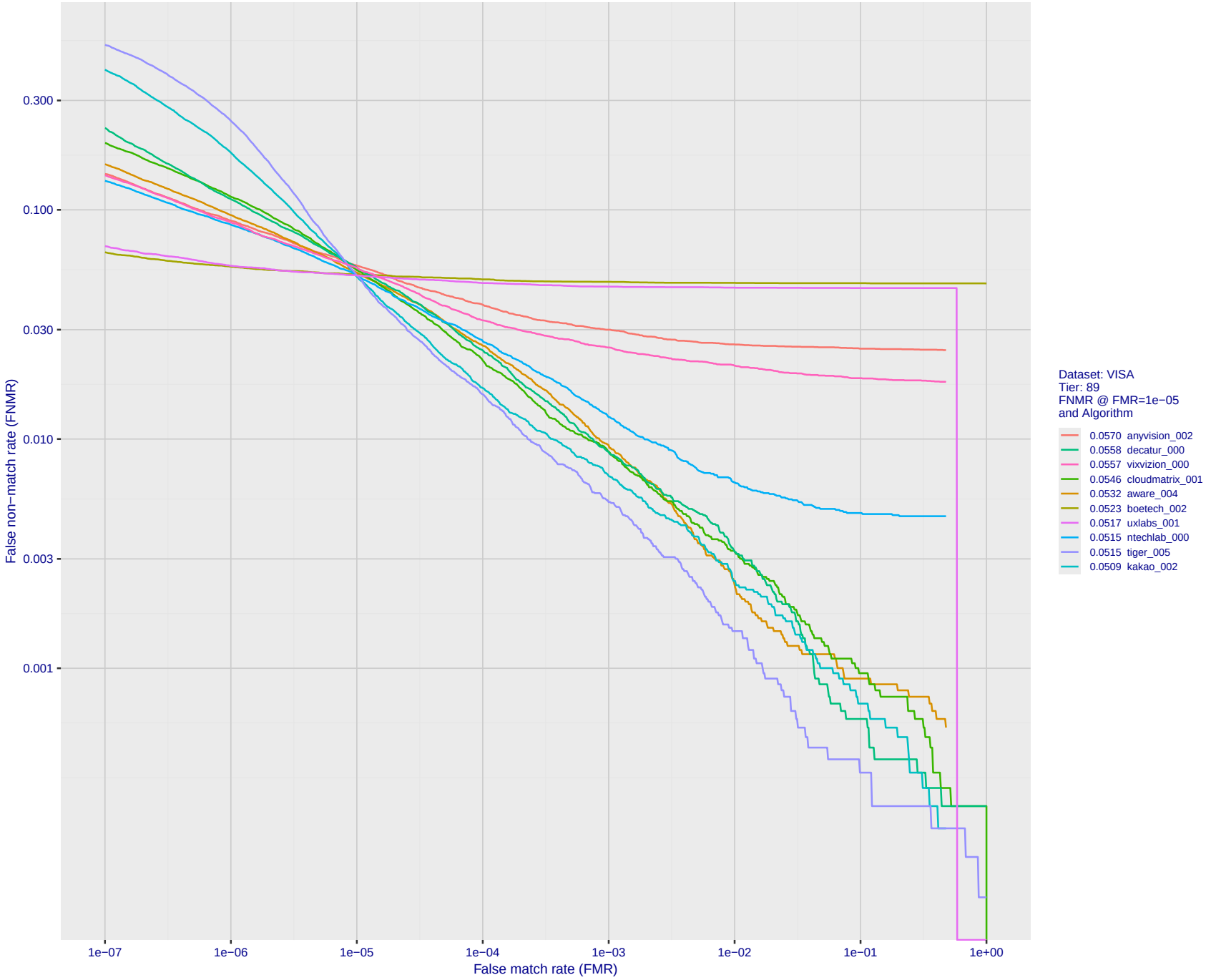


Figure 204: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

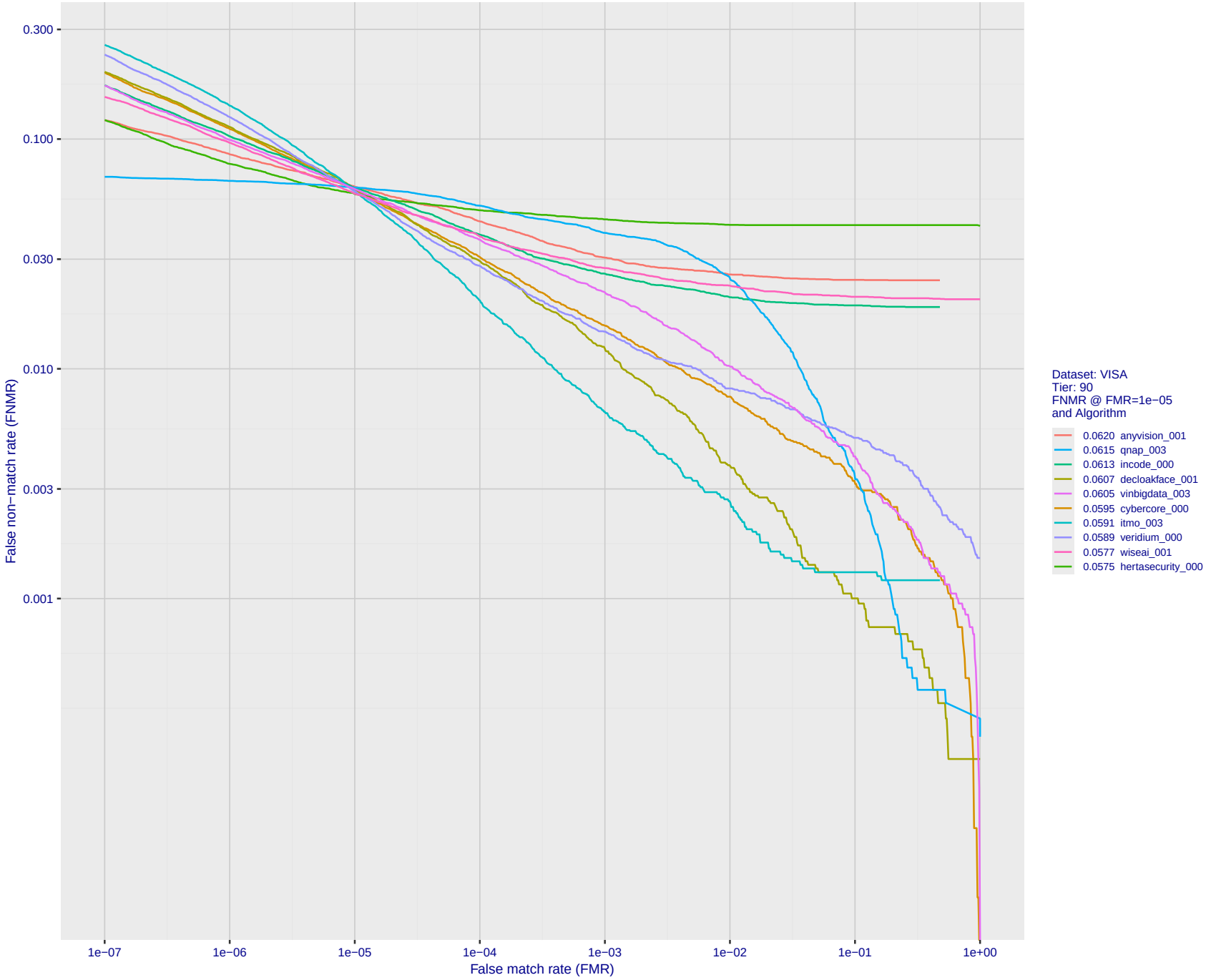


Figure 205: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

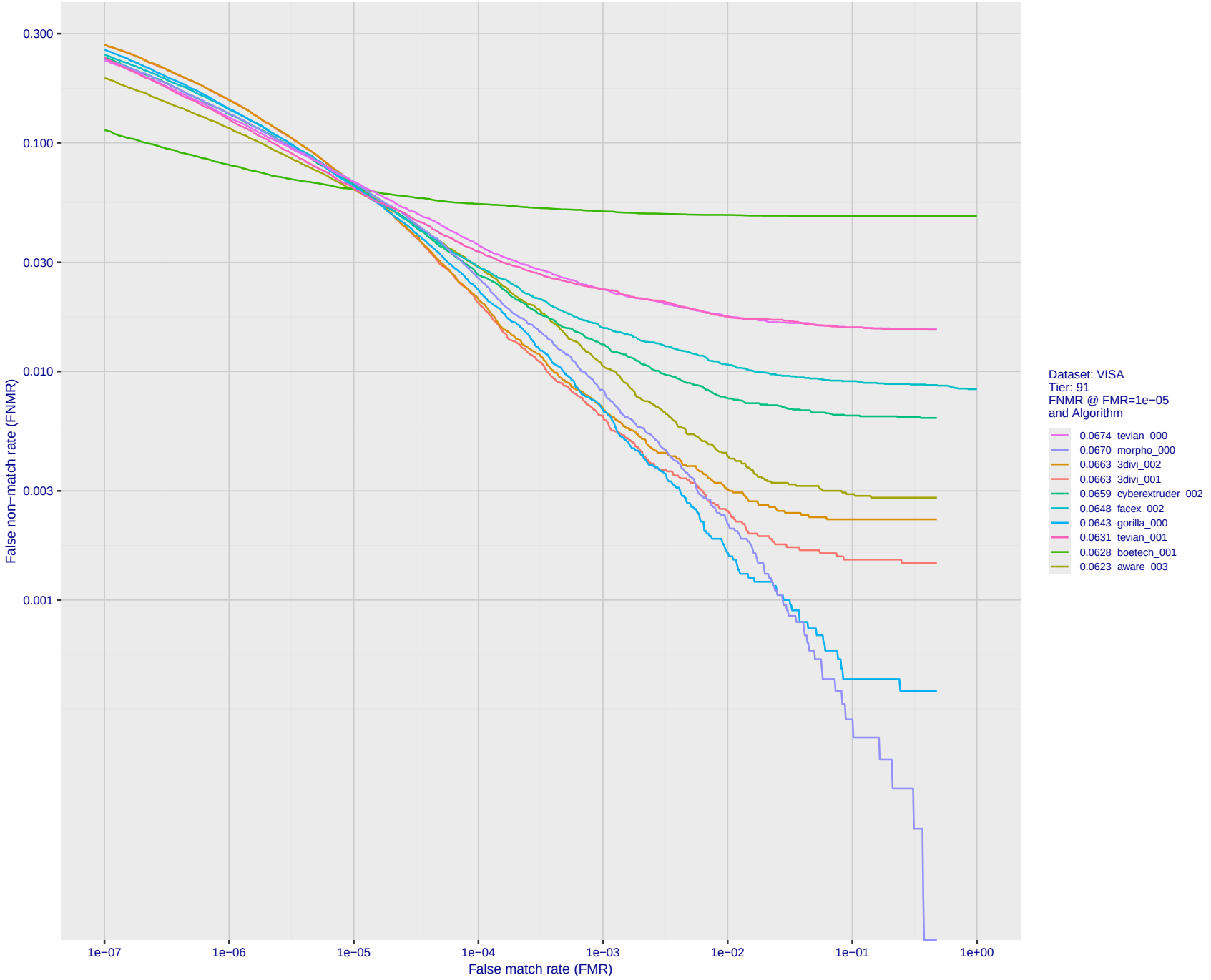


Figure 206: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

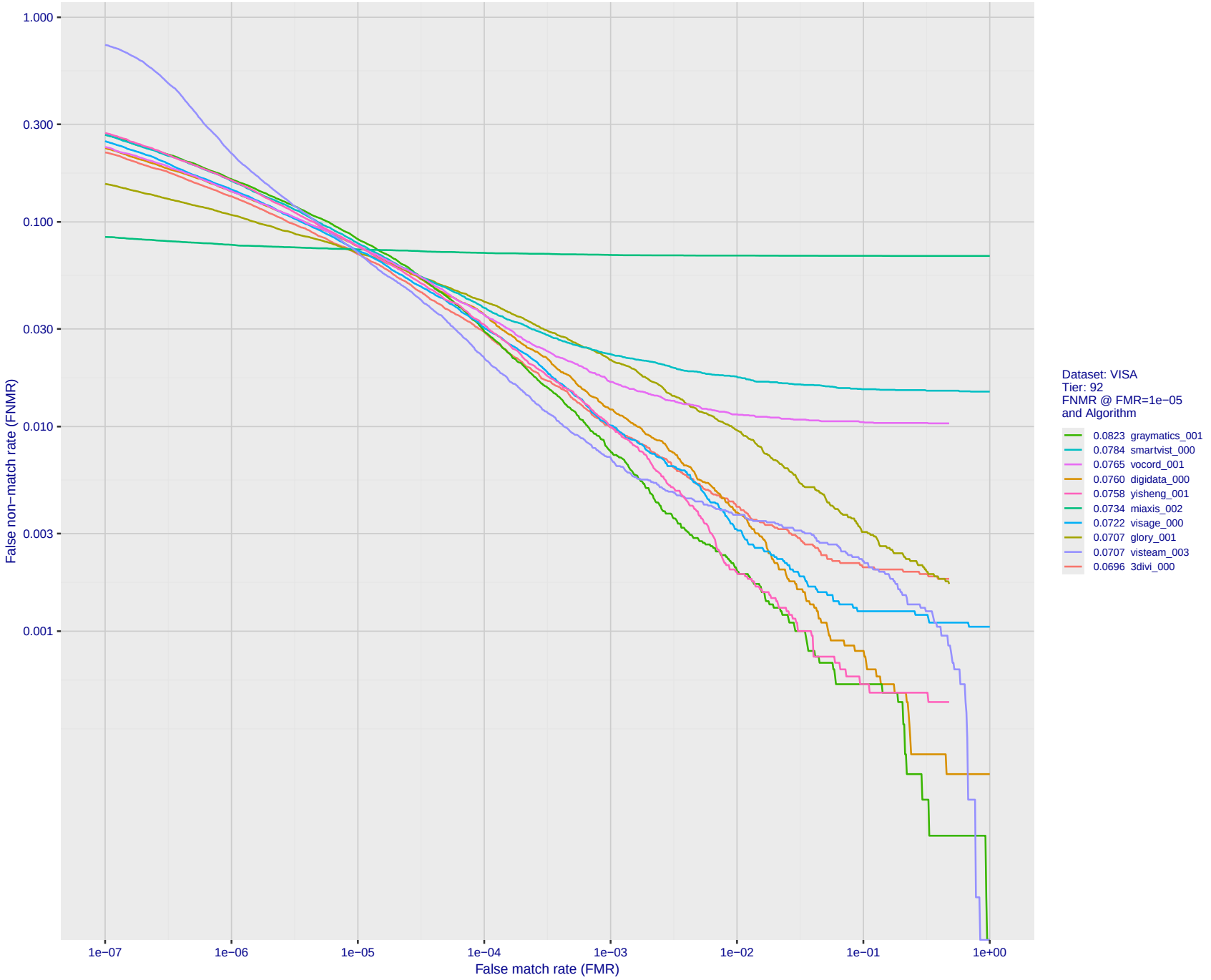


Figure 207: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

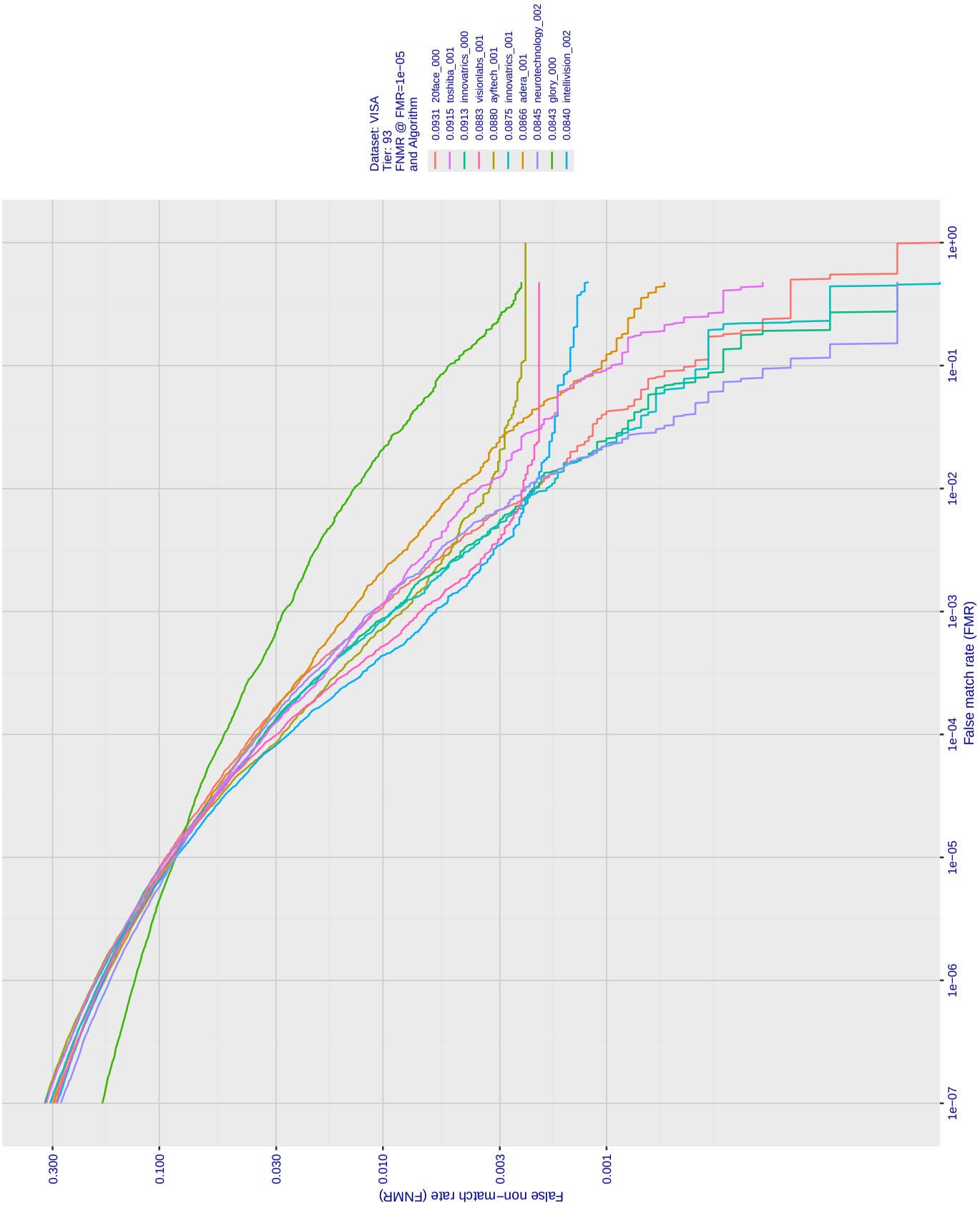


Figure 208: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

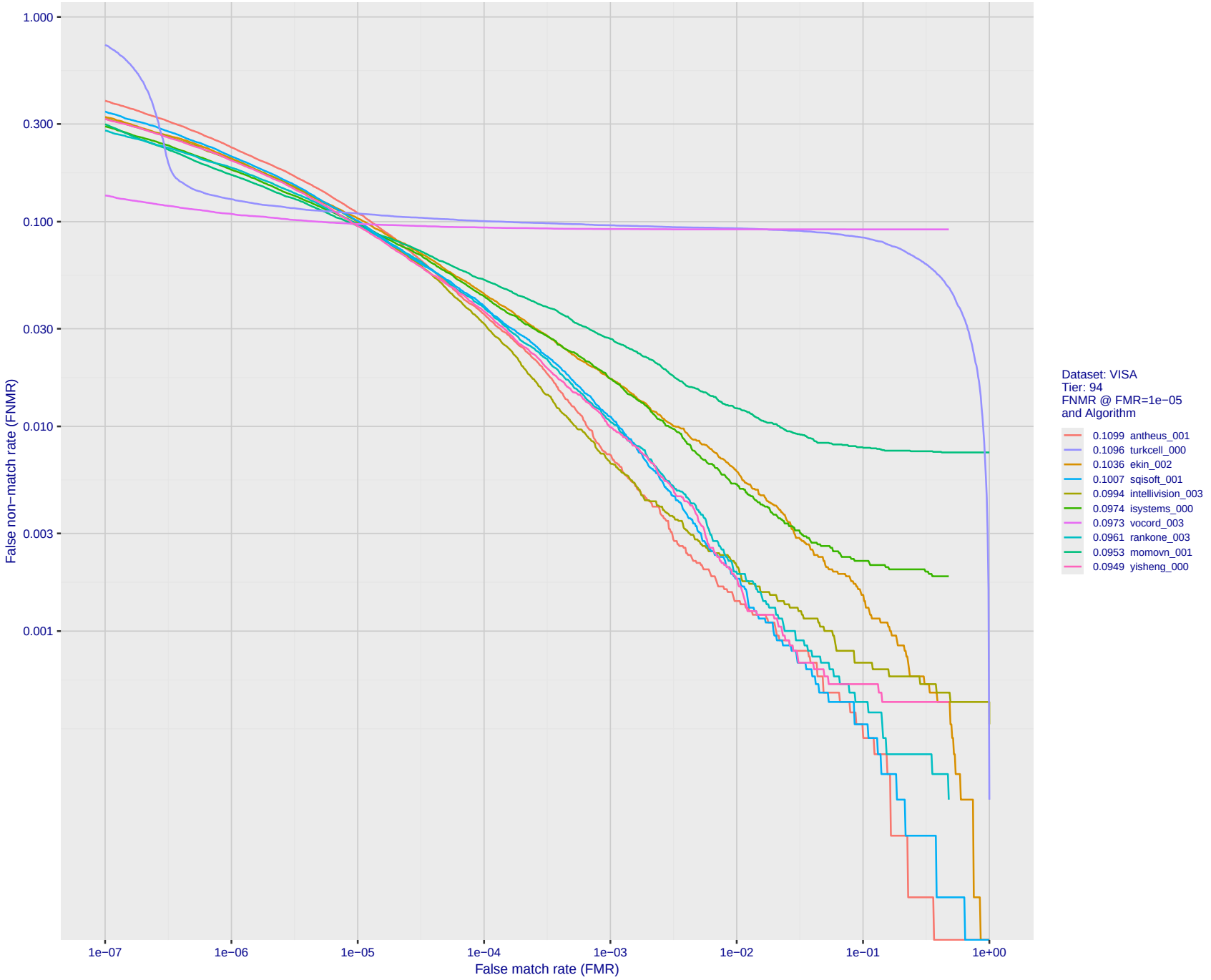


Figure 209: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

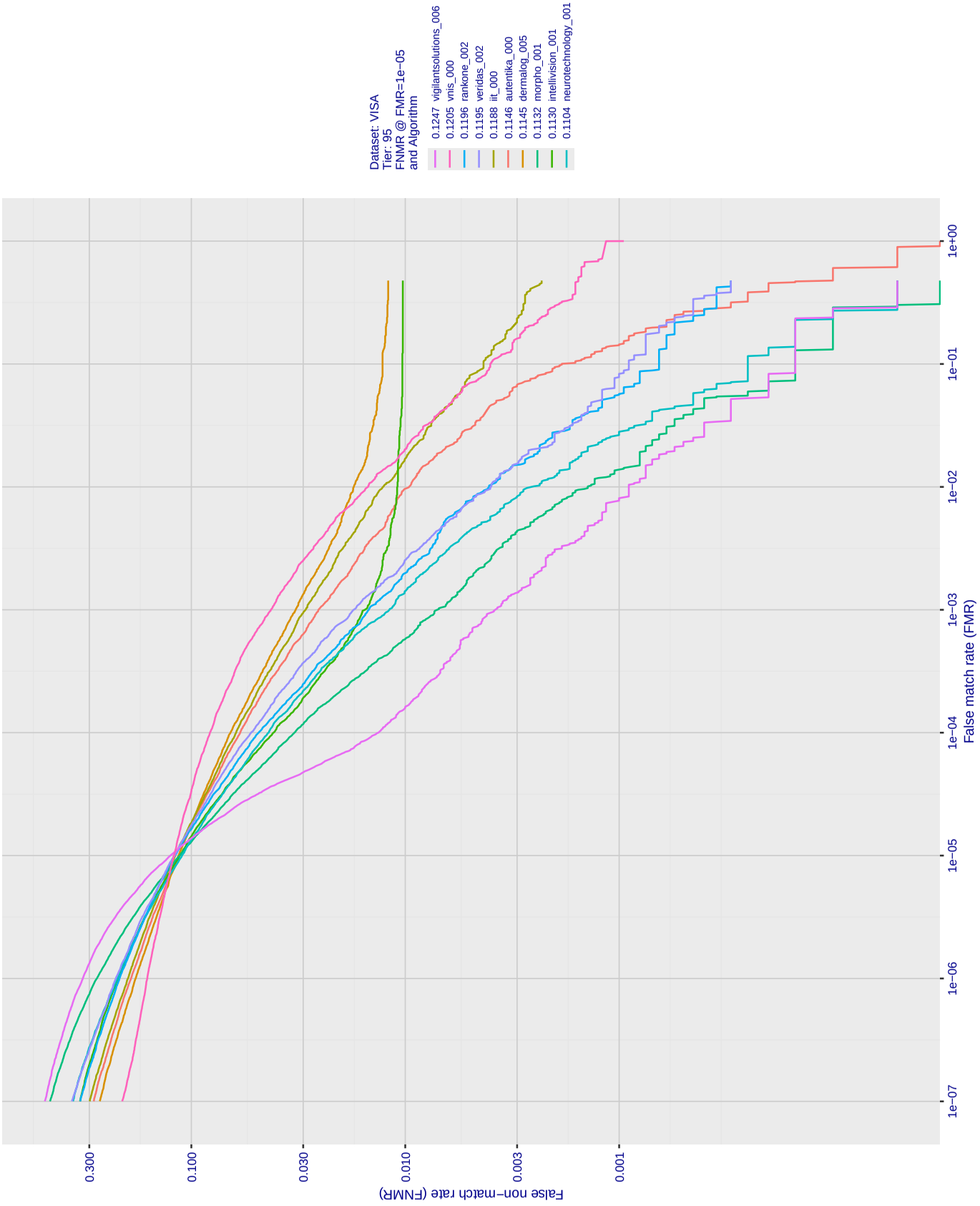


Figure 210: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

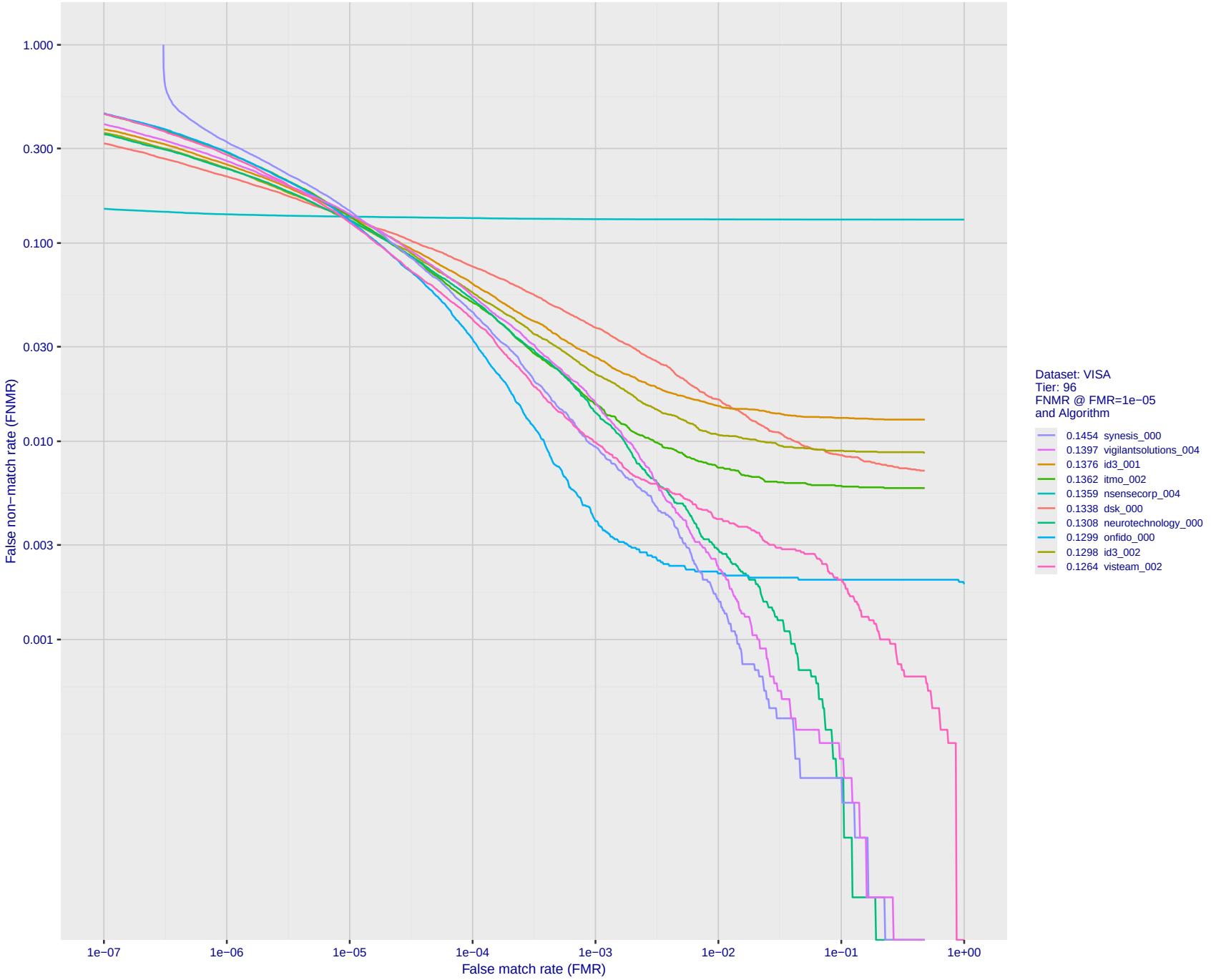


Figure 211: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

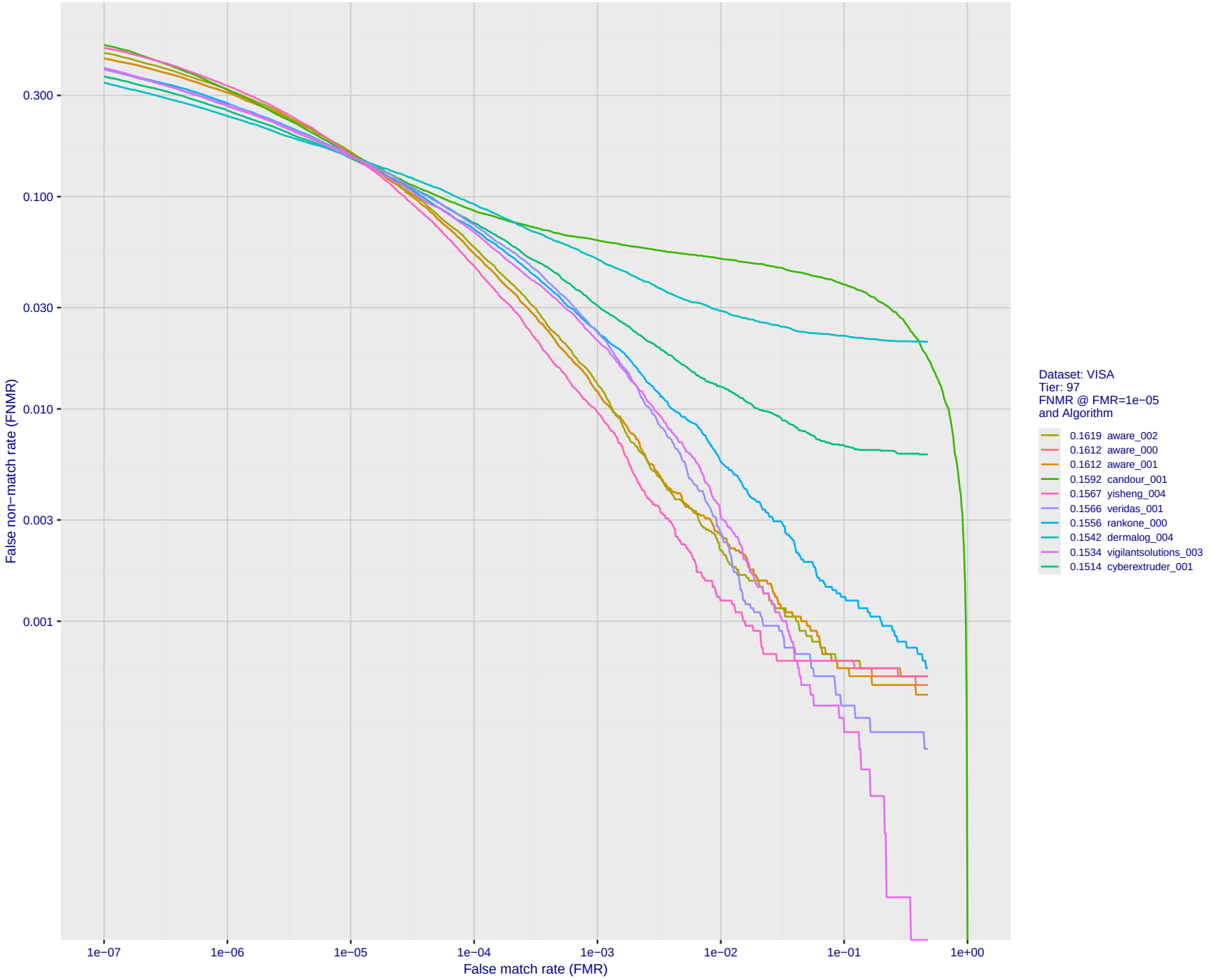


Figure 212: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

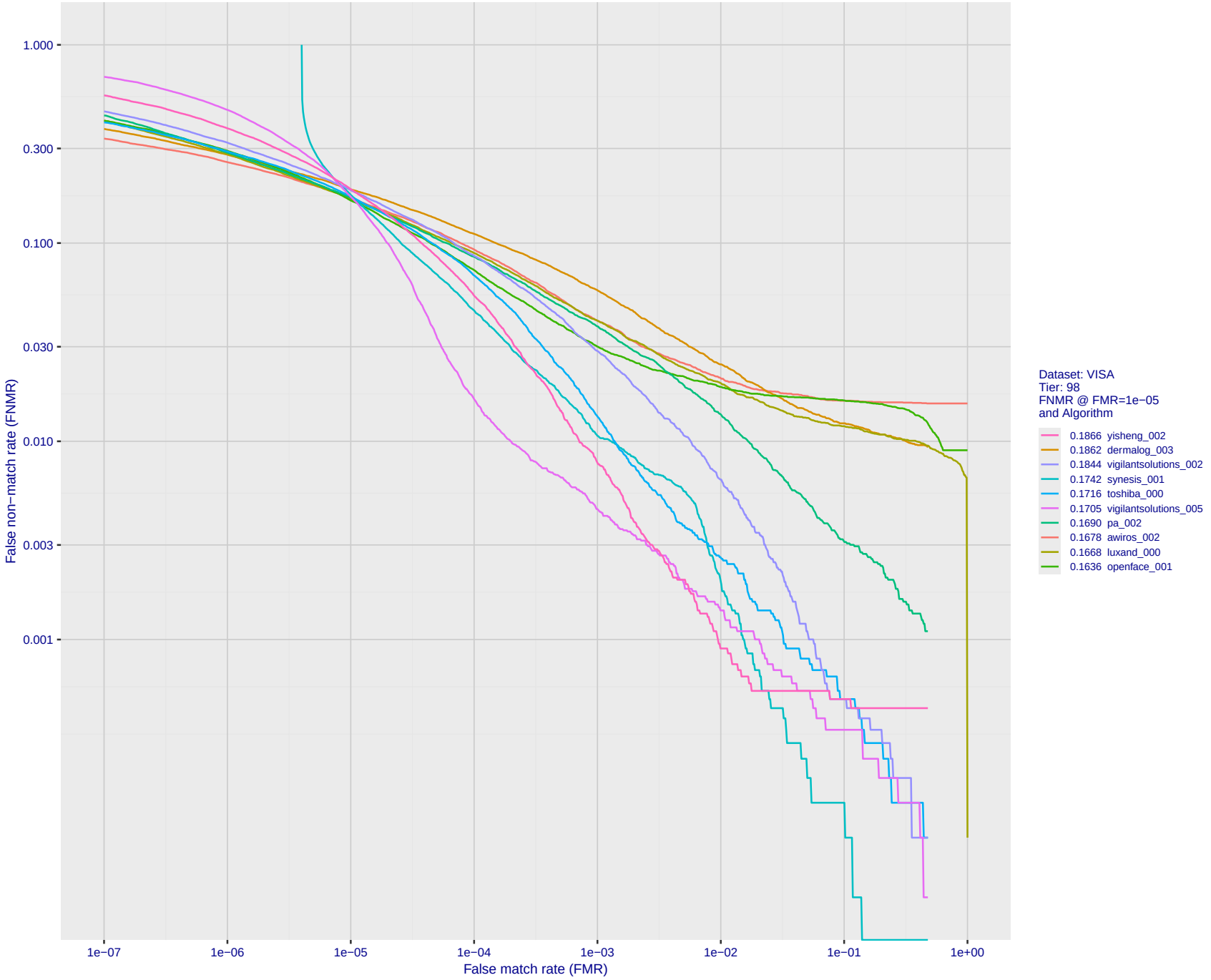


Figure 213: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

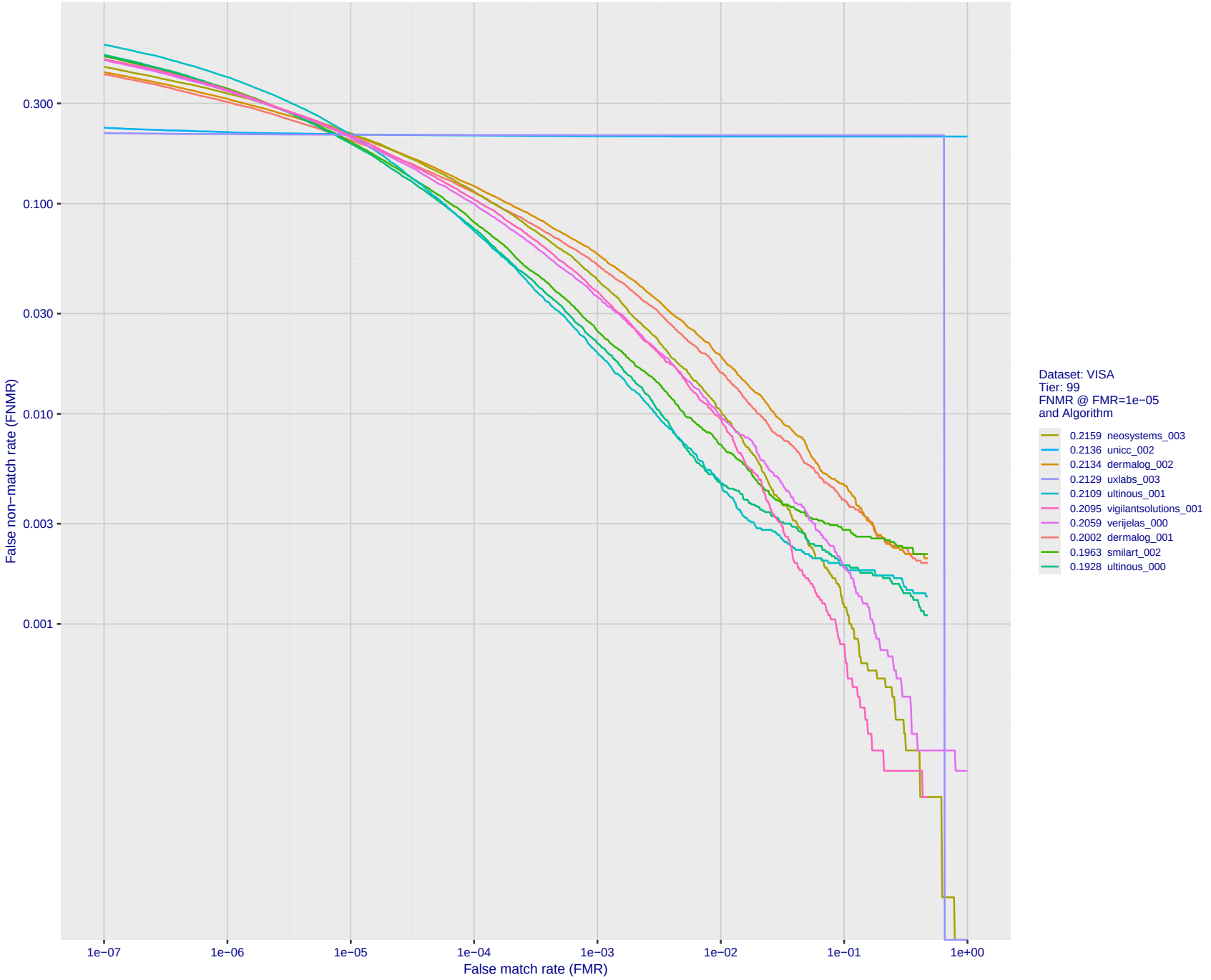


Figure 214: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

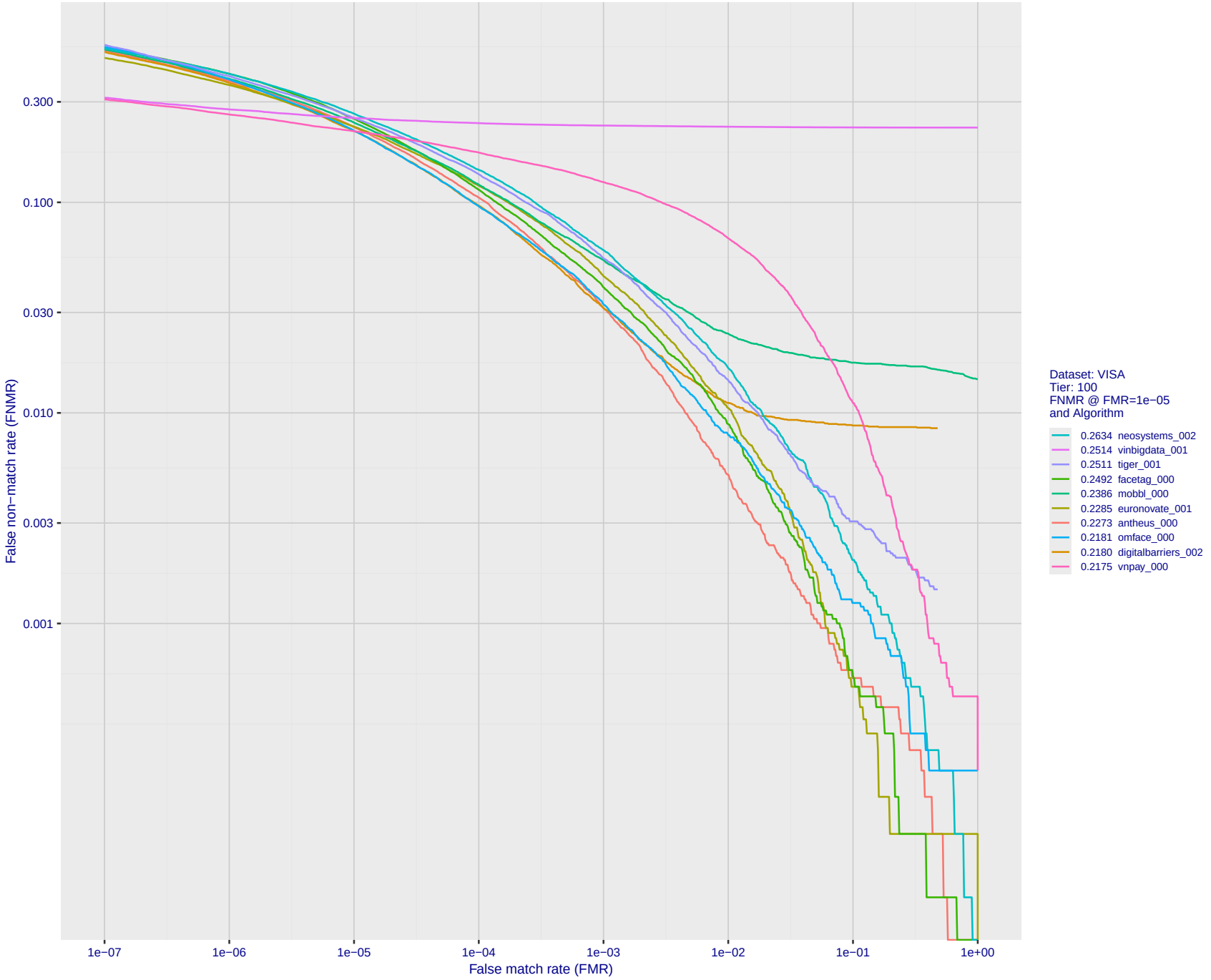


Figure 215: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

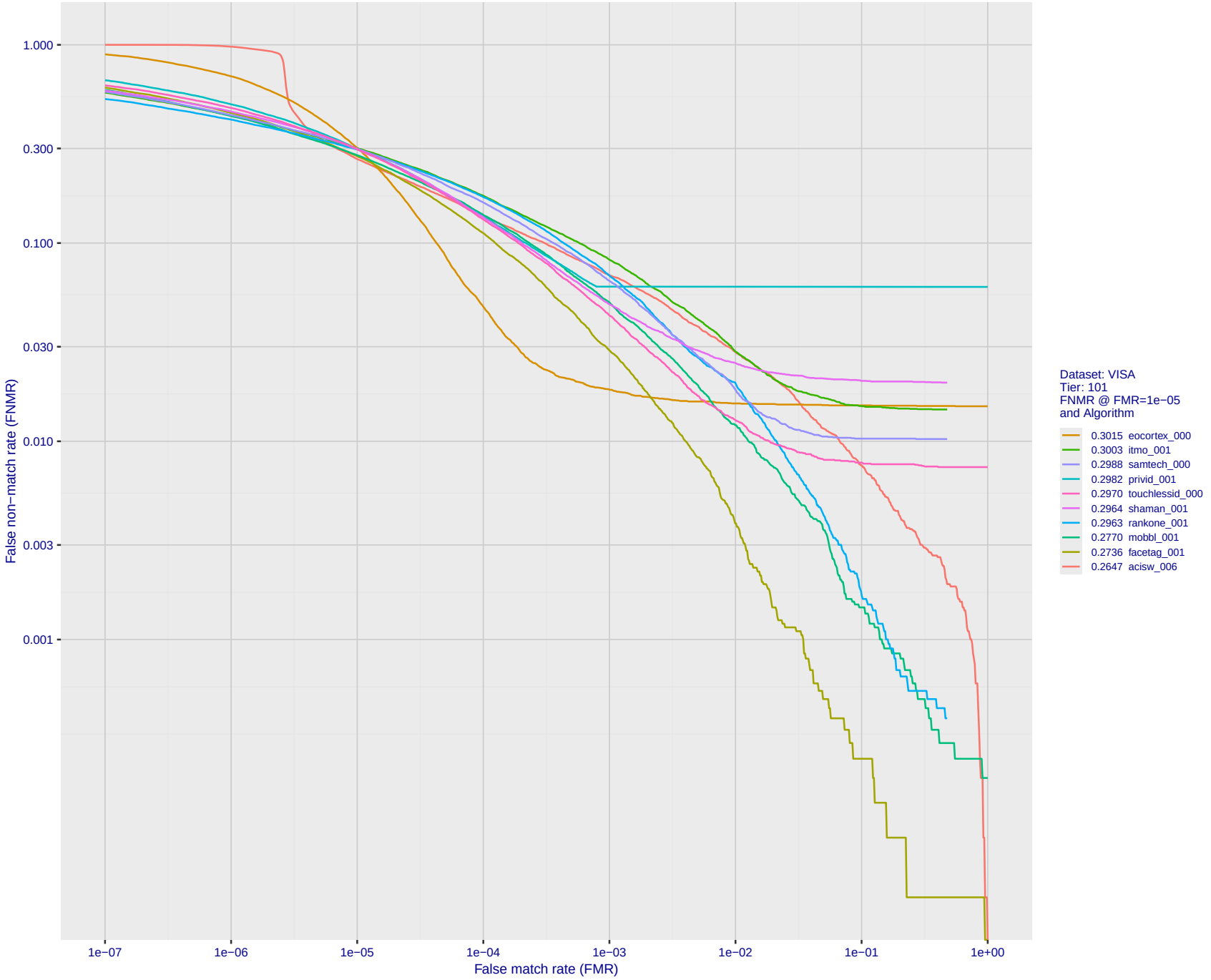


Figure 216: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

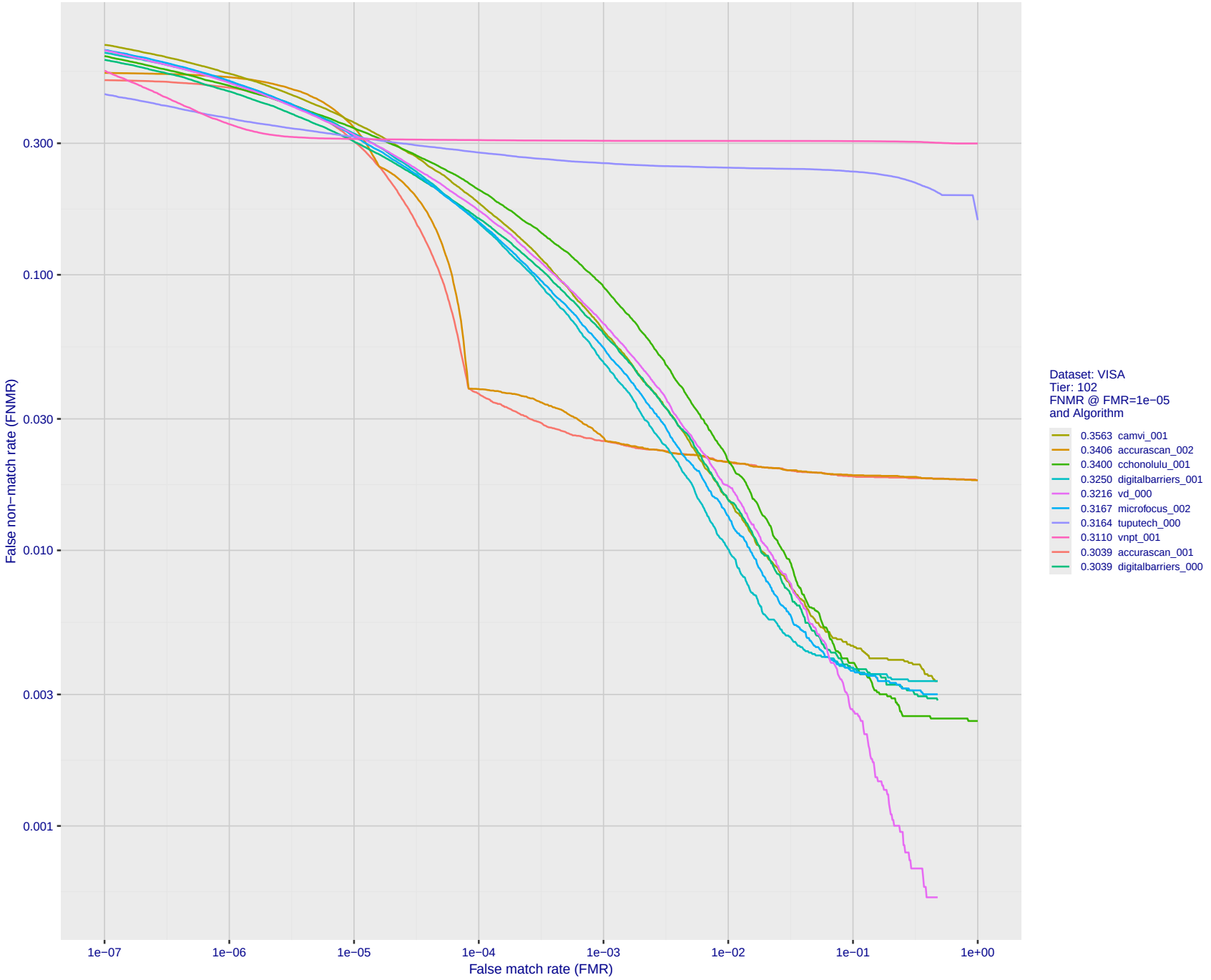


Figure 217: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

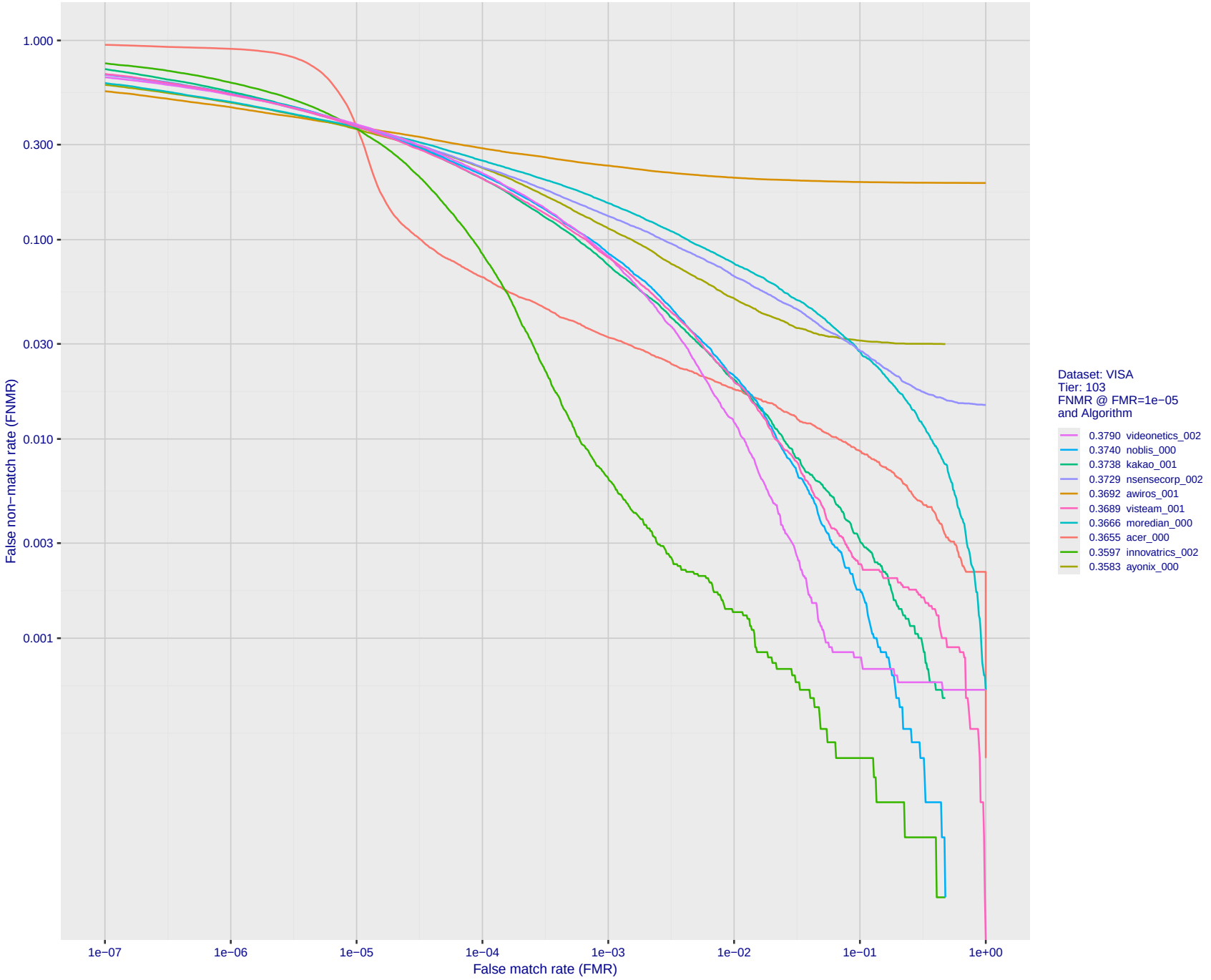


Figure 218: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

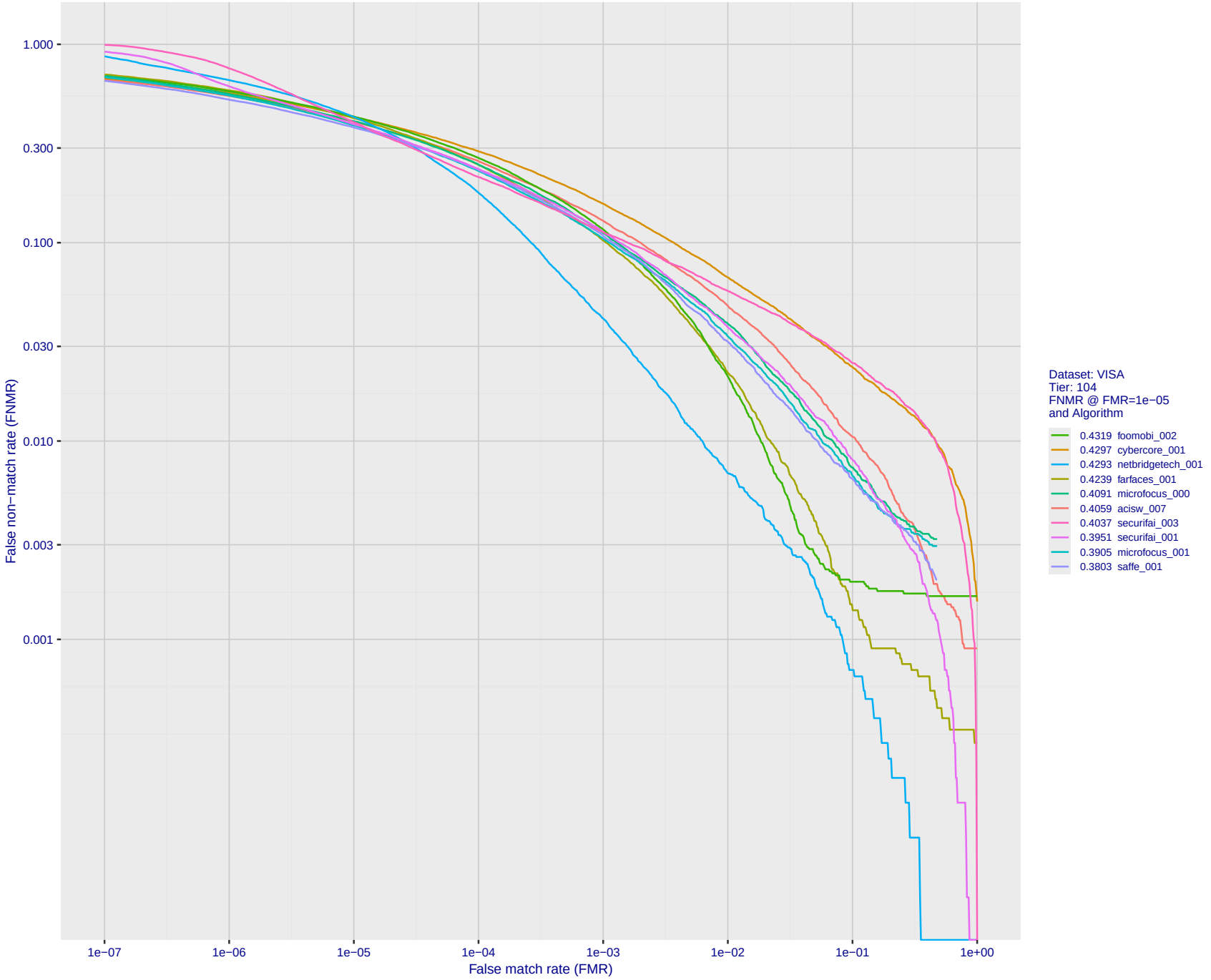


Figure 219: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

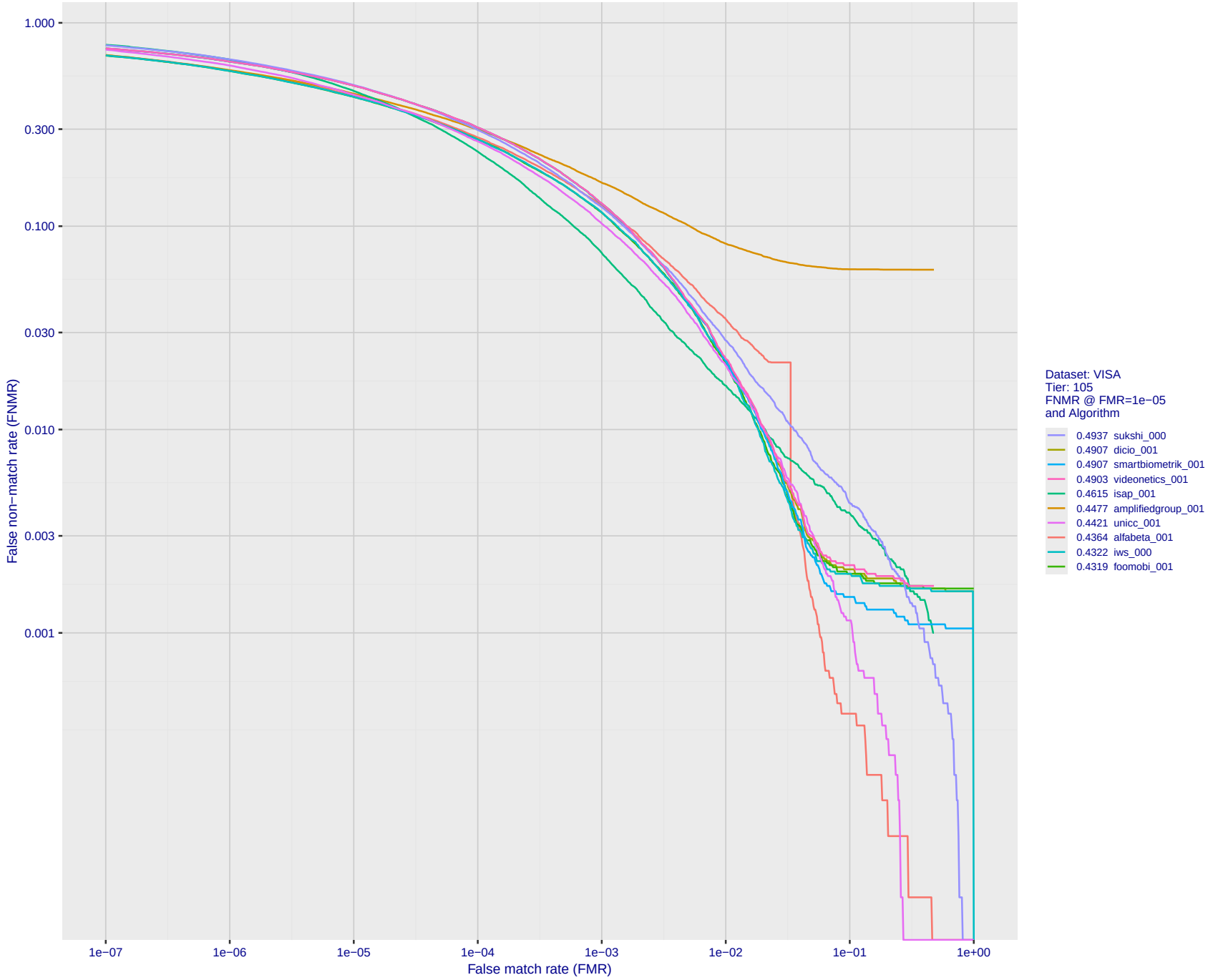


Figure 220: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

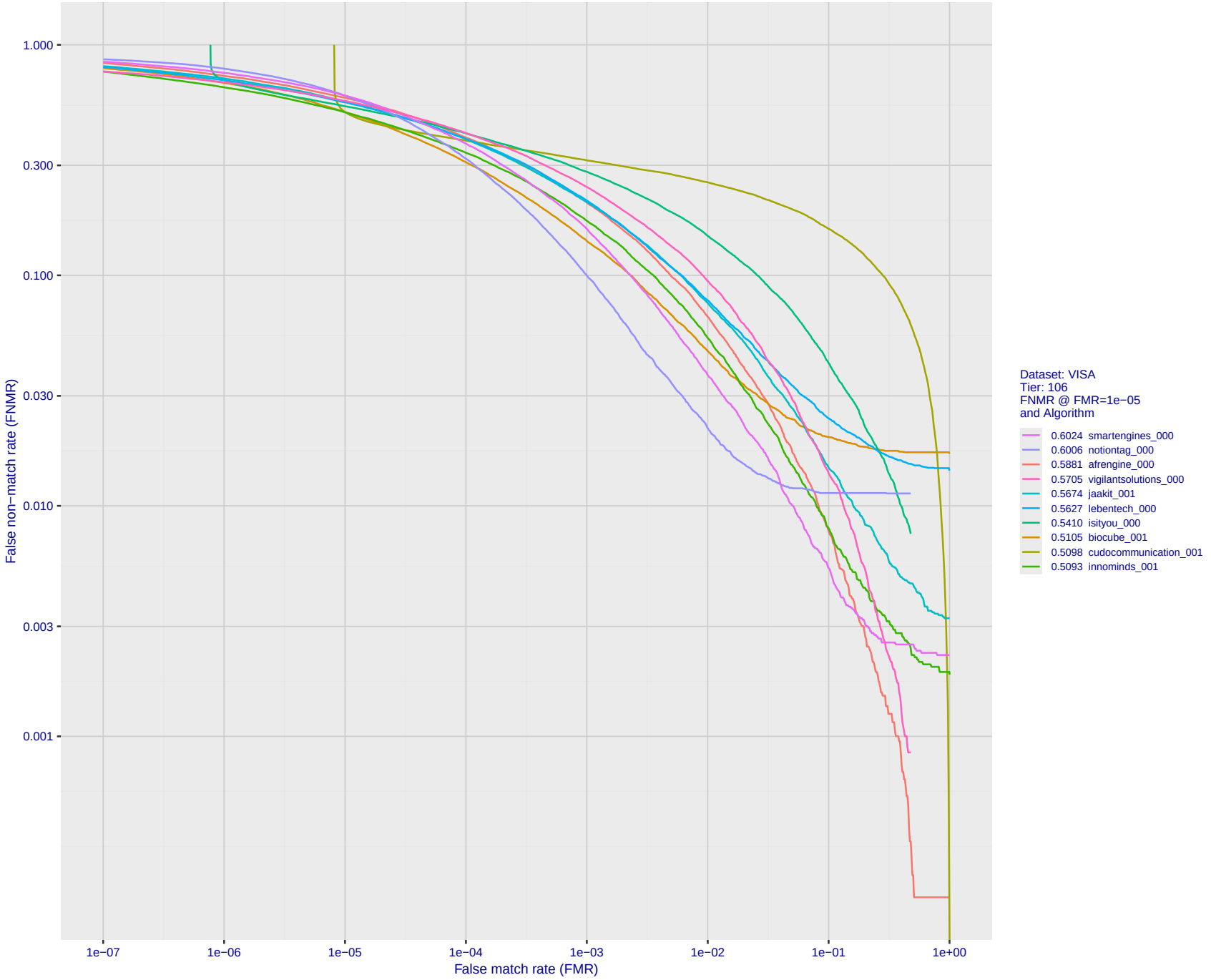


Figure 221: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

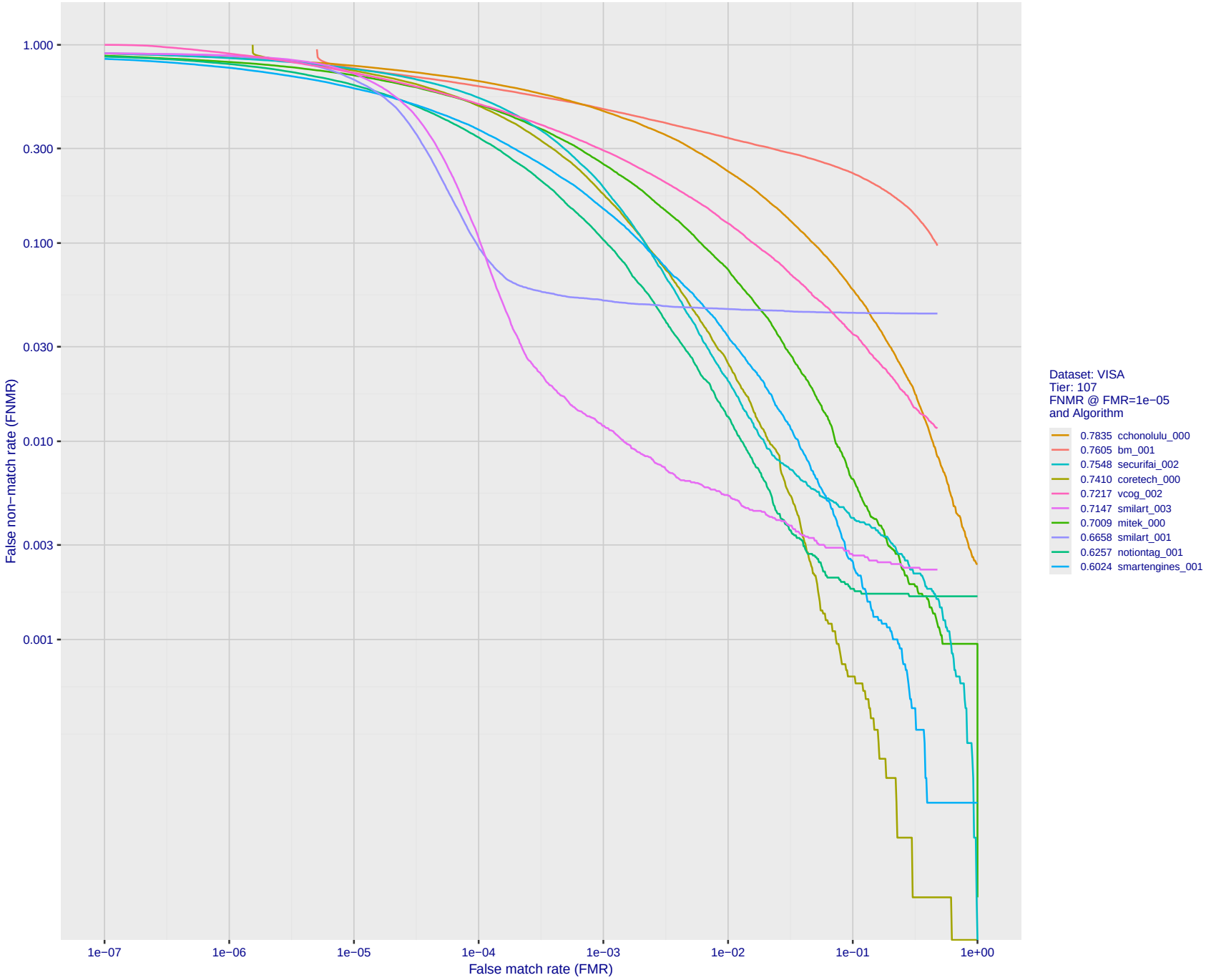


Figure 222: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

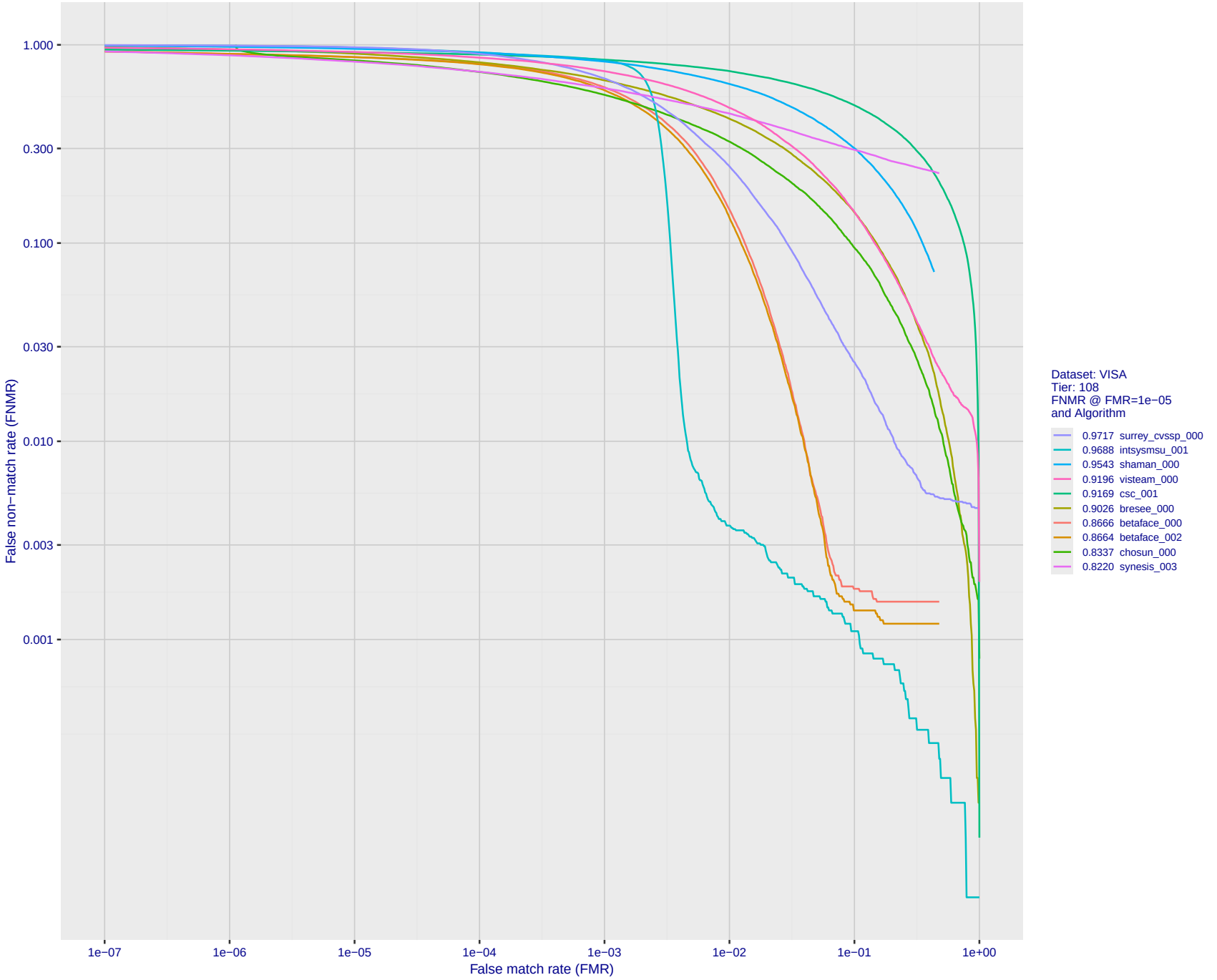


Figure 223: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

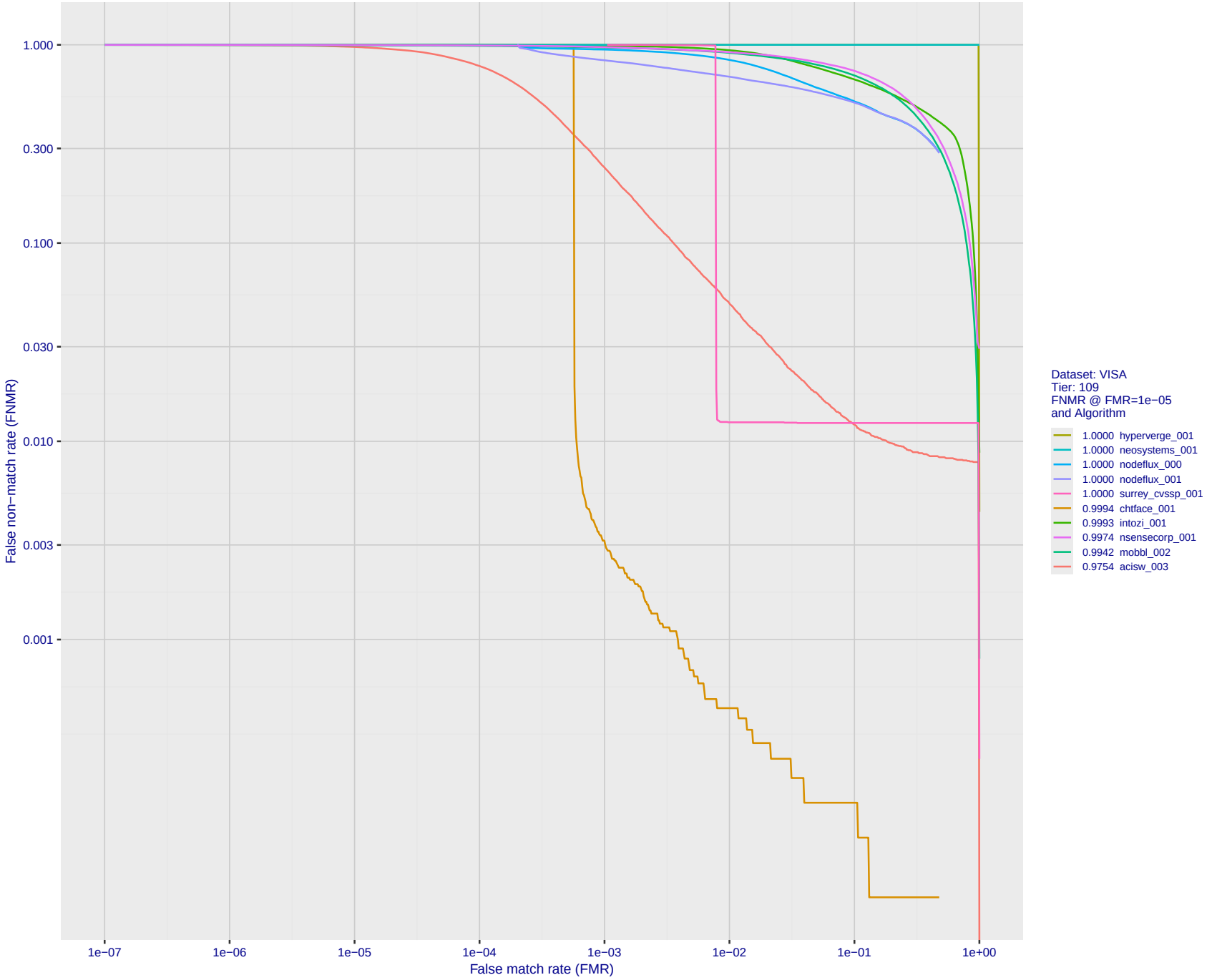


Figure 224: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

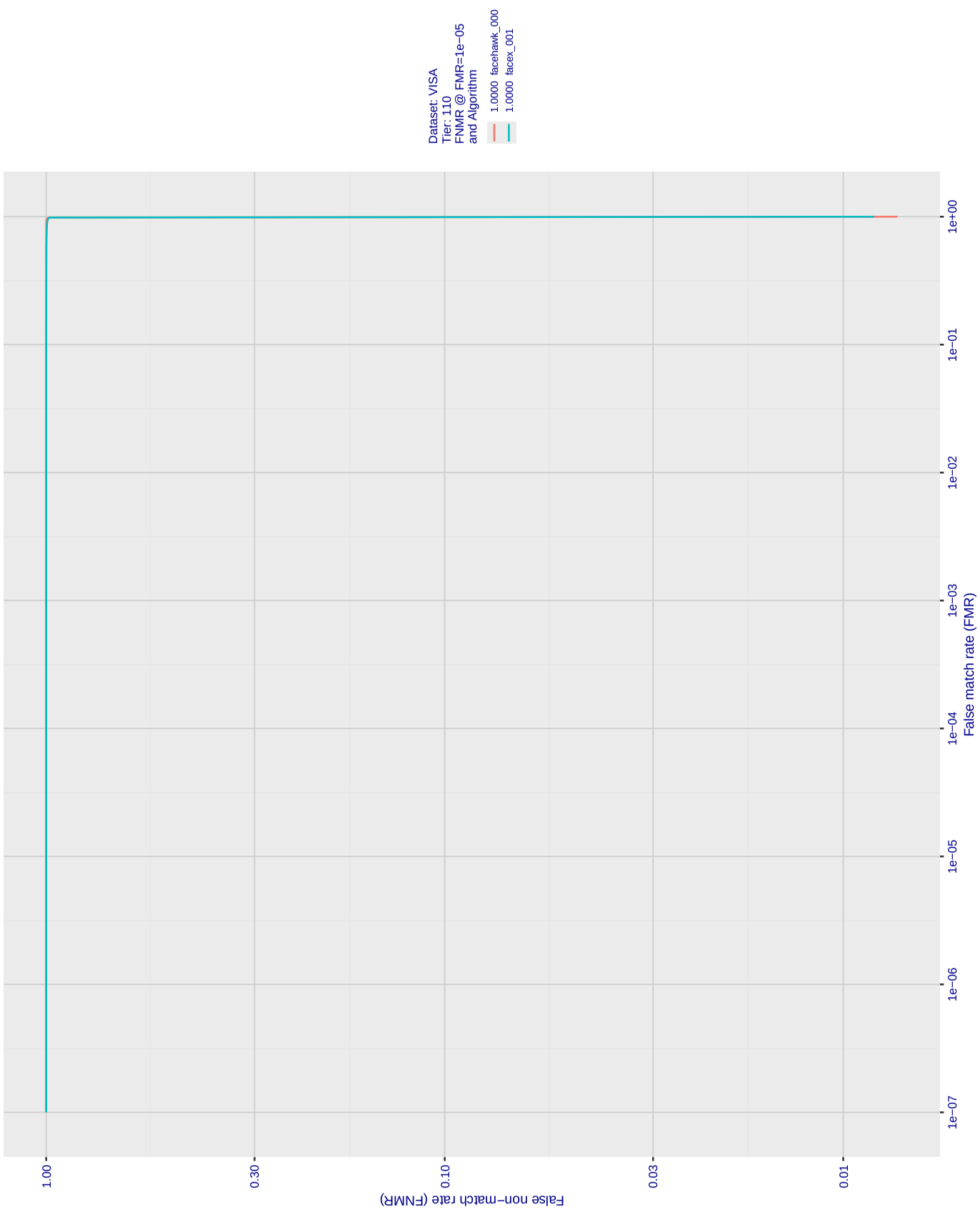


Figure 225: For the visa images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show many decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

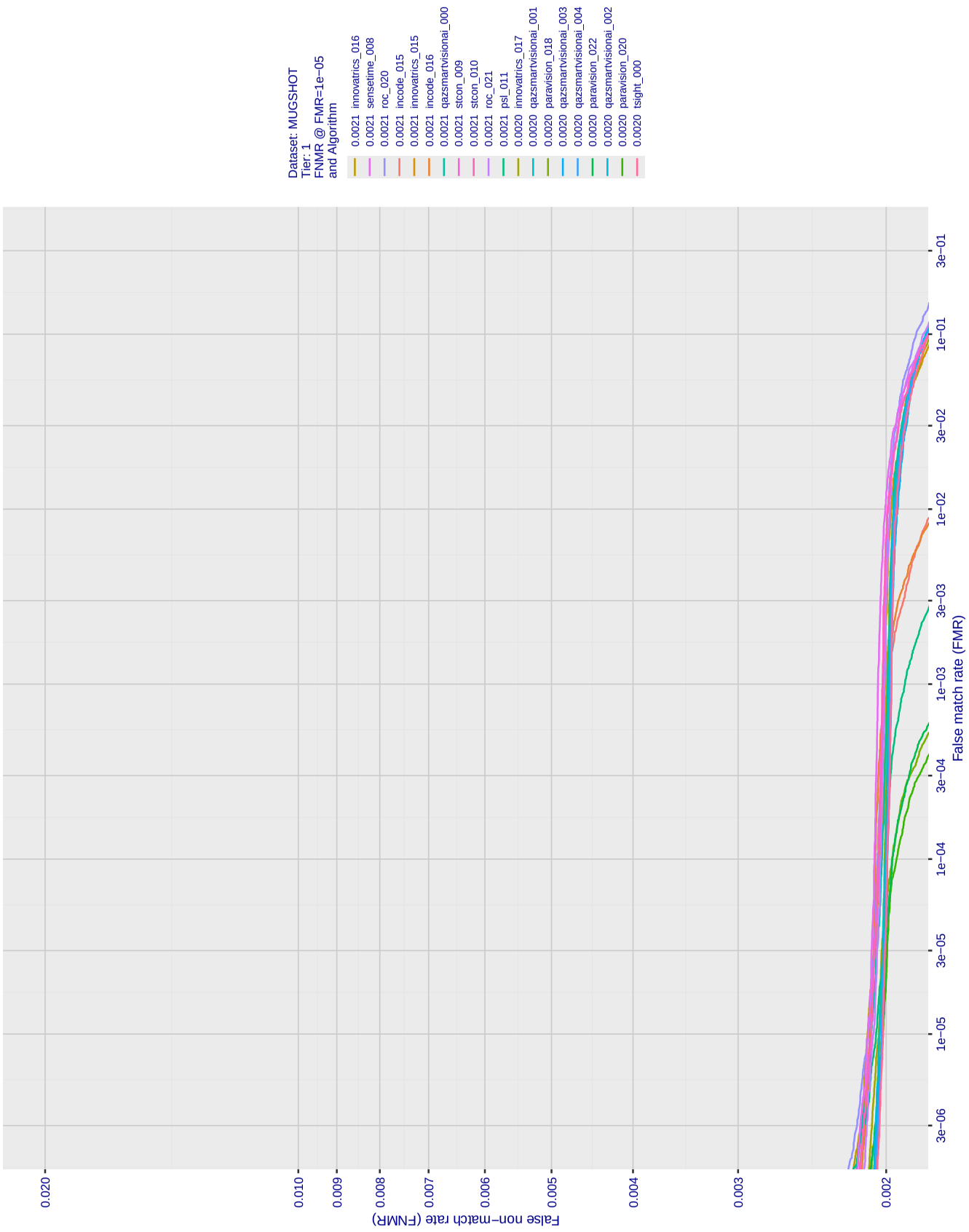


Figure 226: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

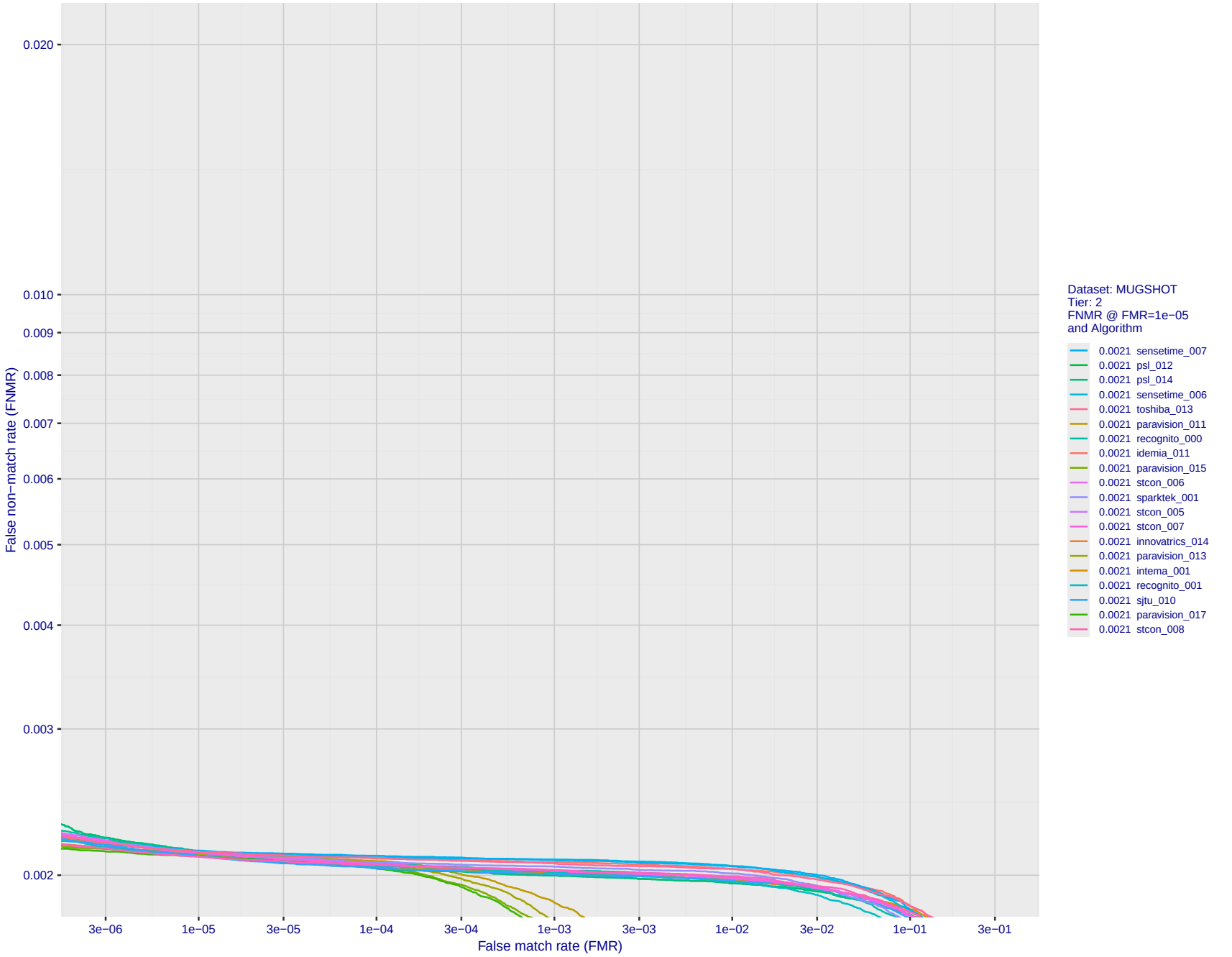


Figure 227: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

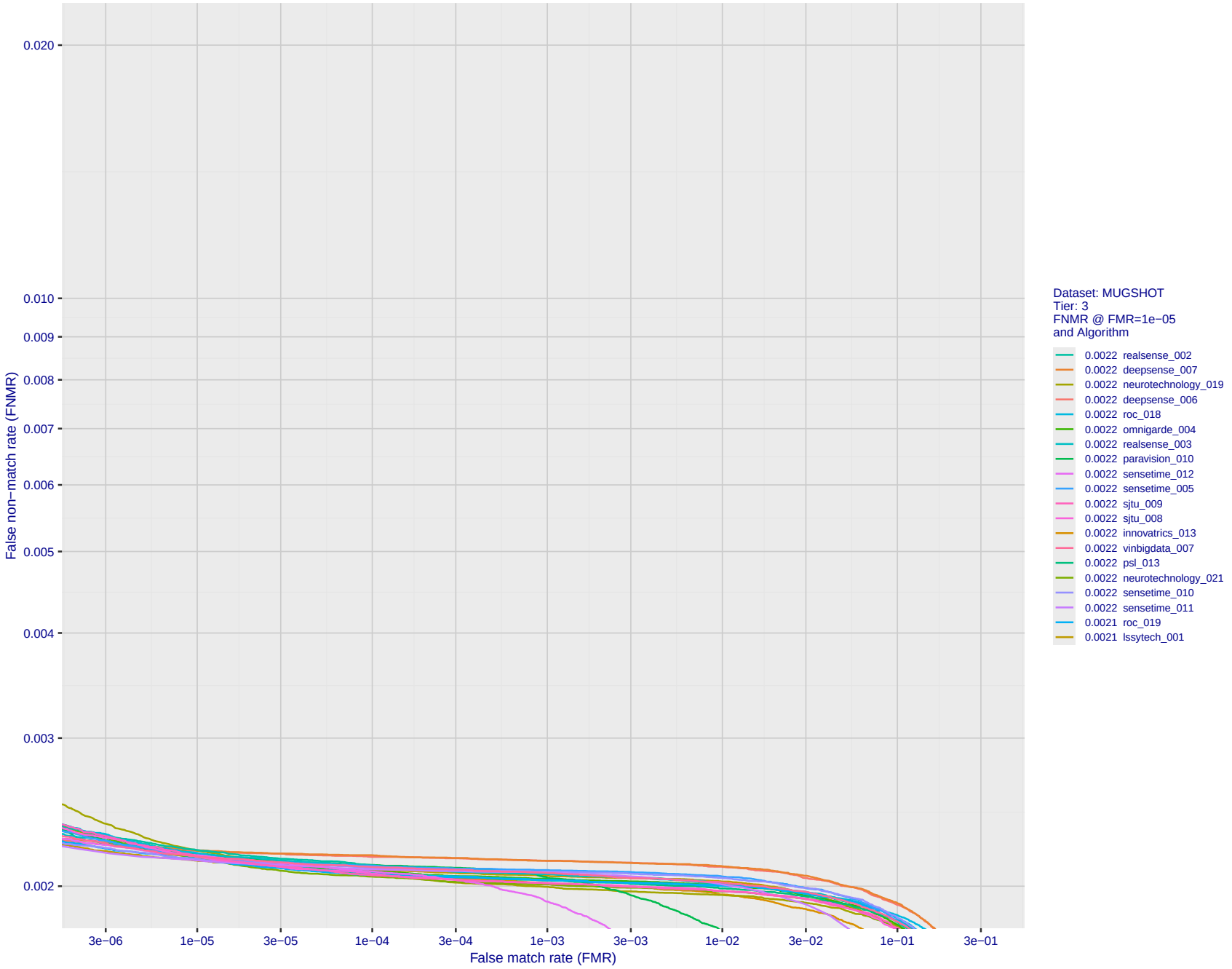


*Figure 228: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

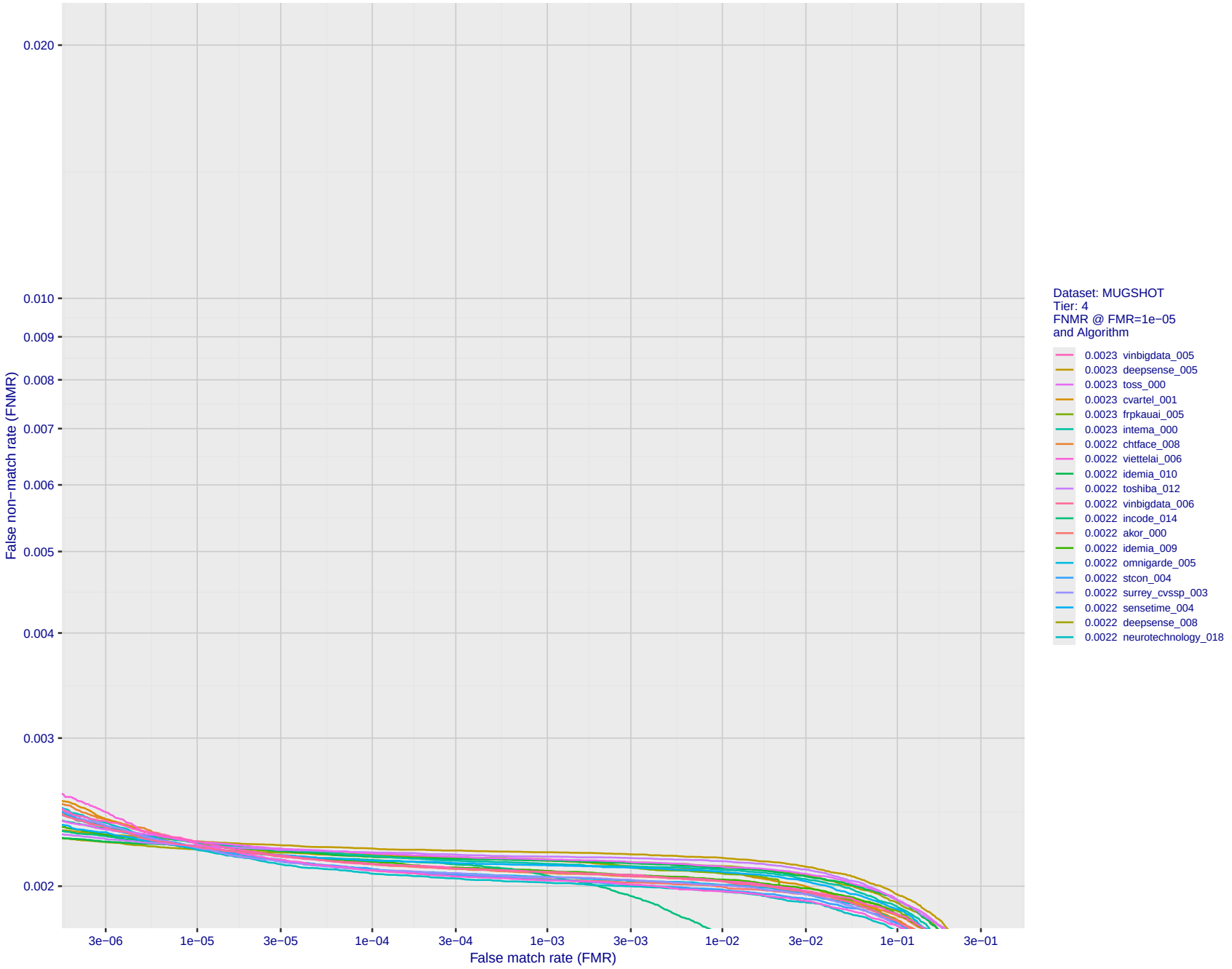


*Figure 229: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

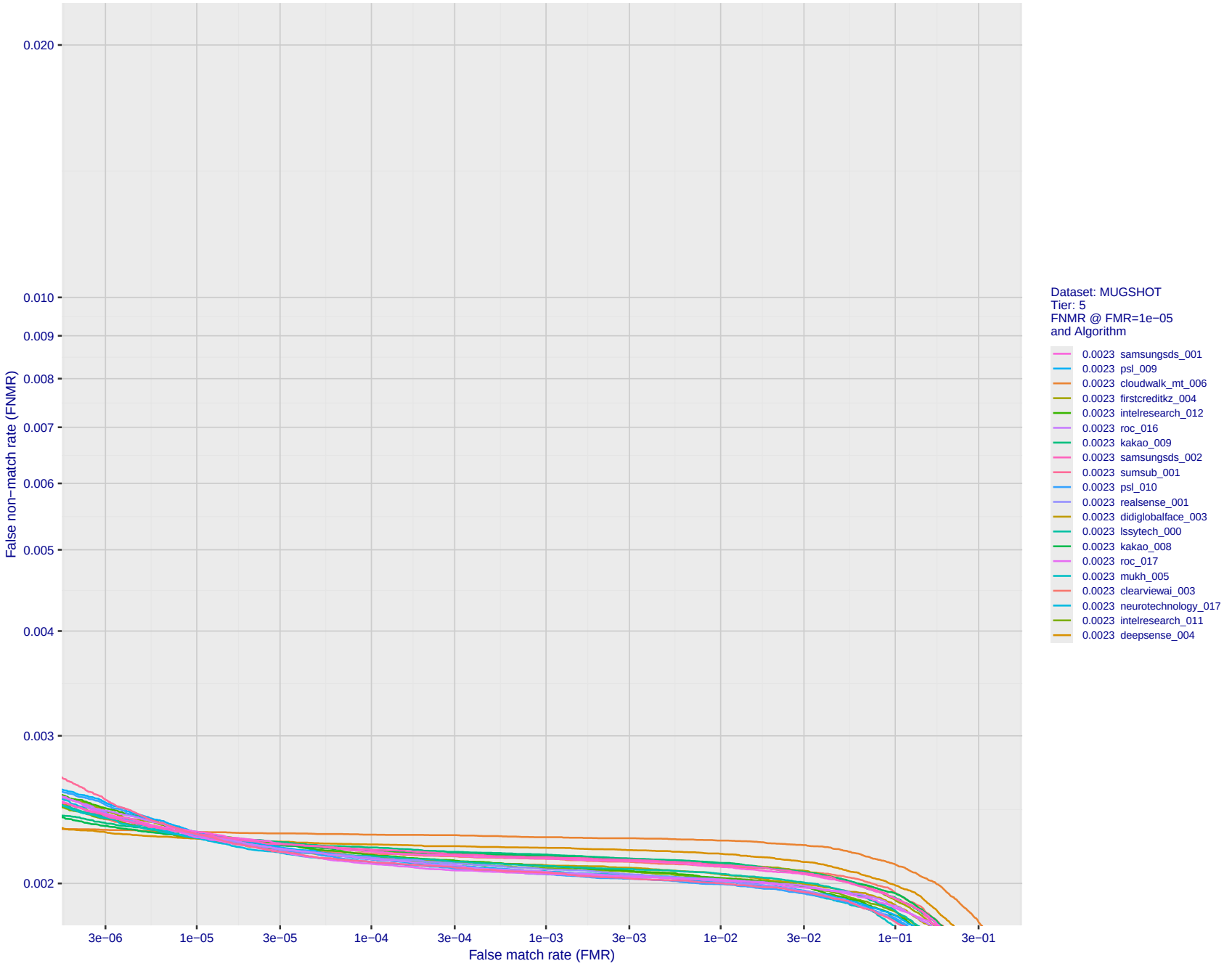


Figure 230: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

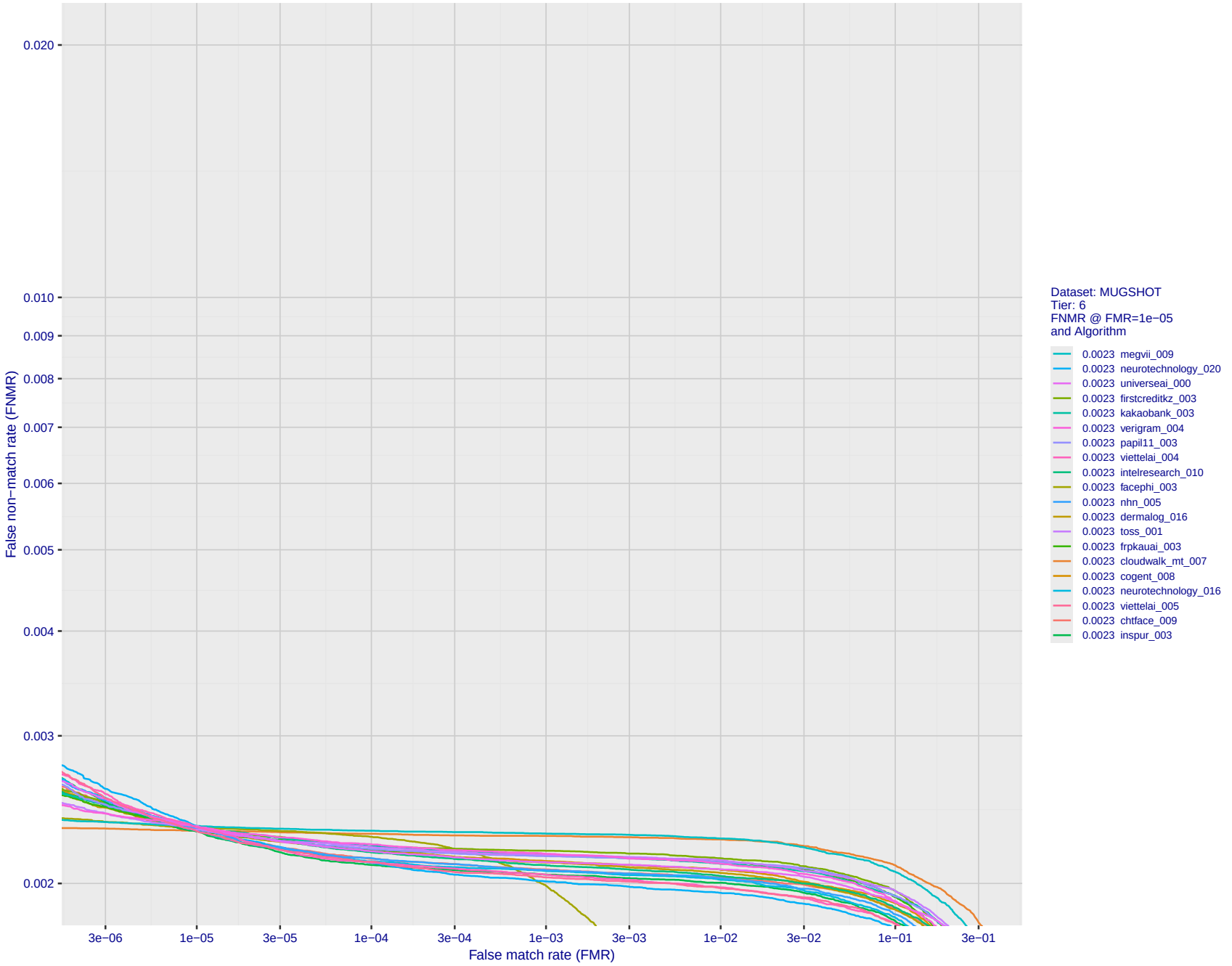


Figure 231: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

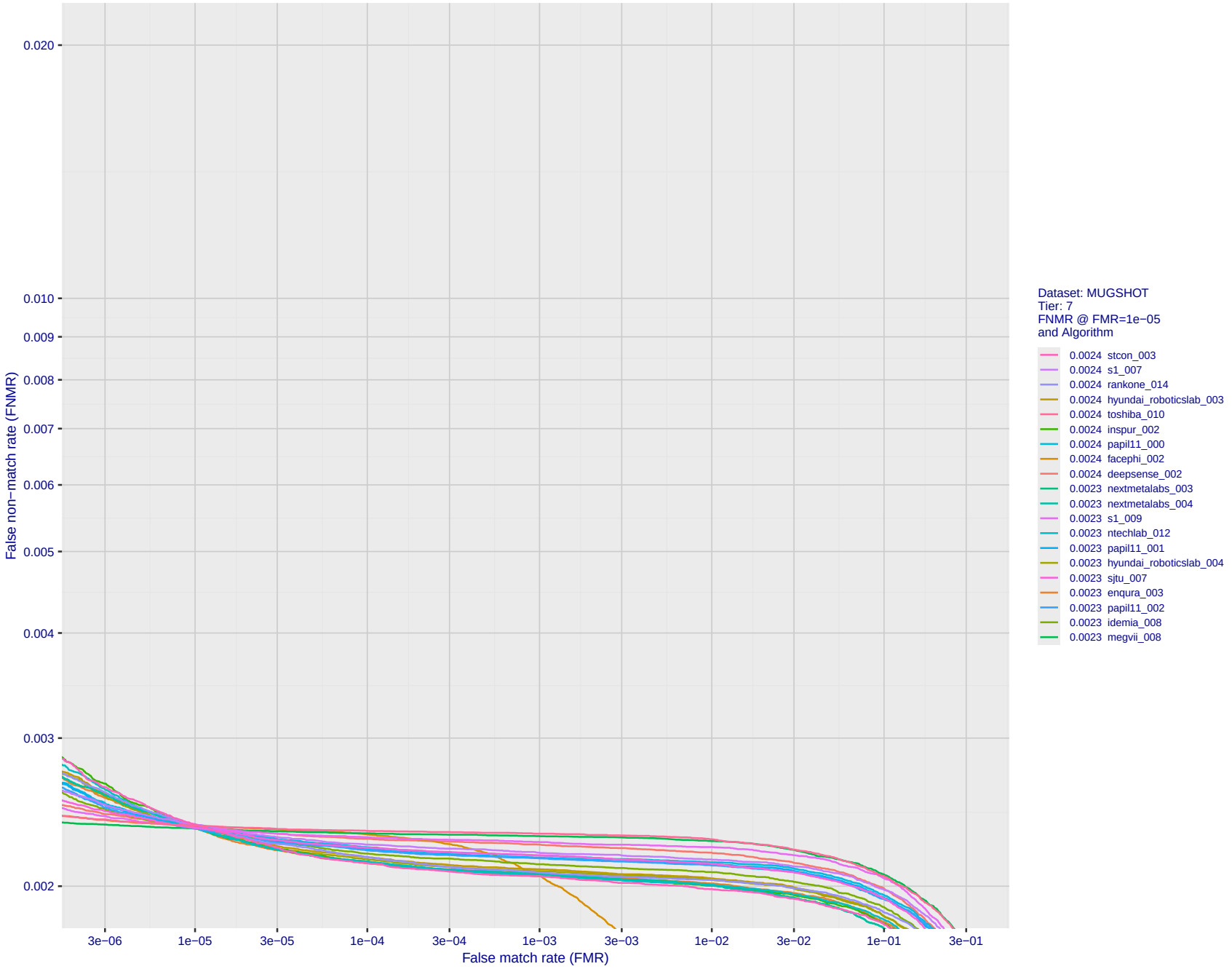


Figure 232: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

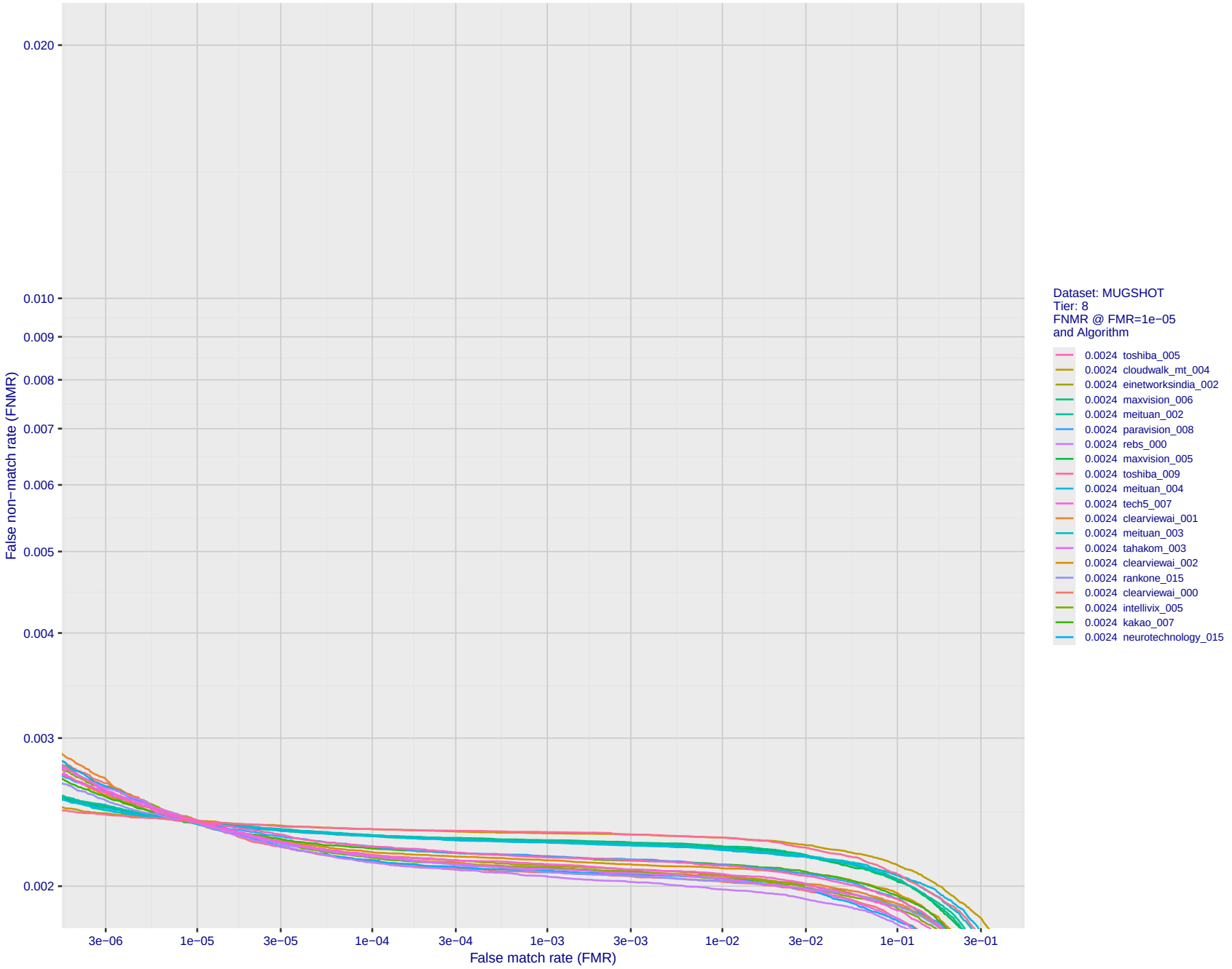


Figure 233: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

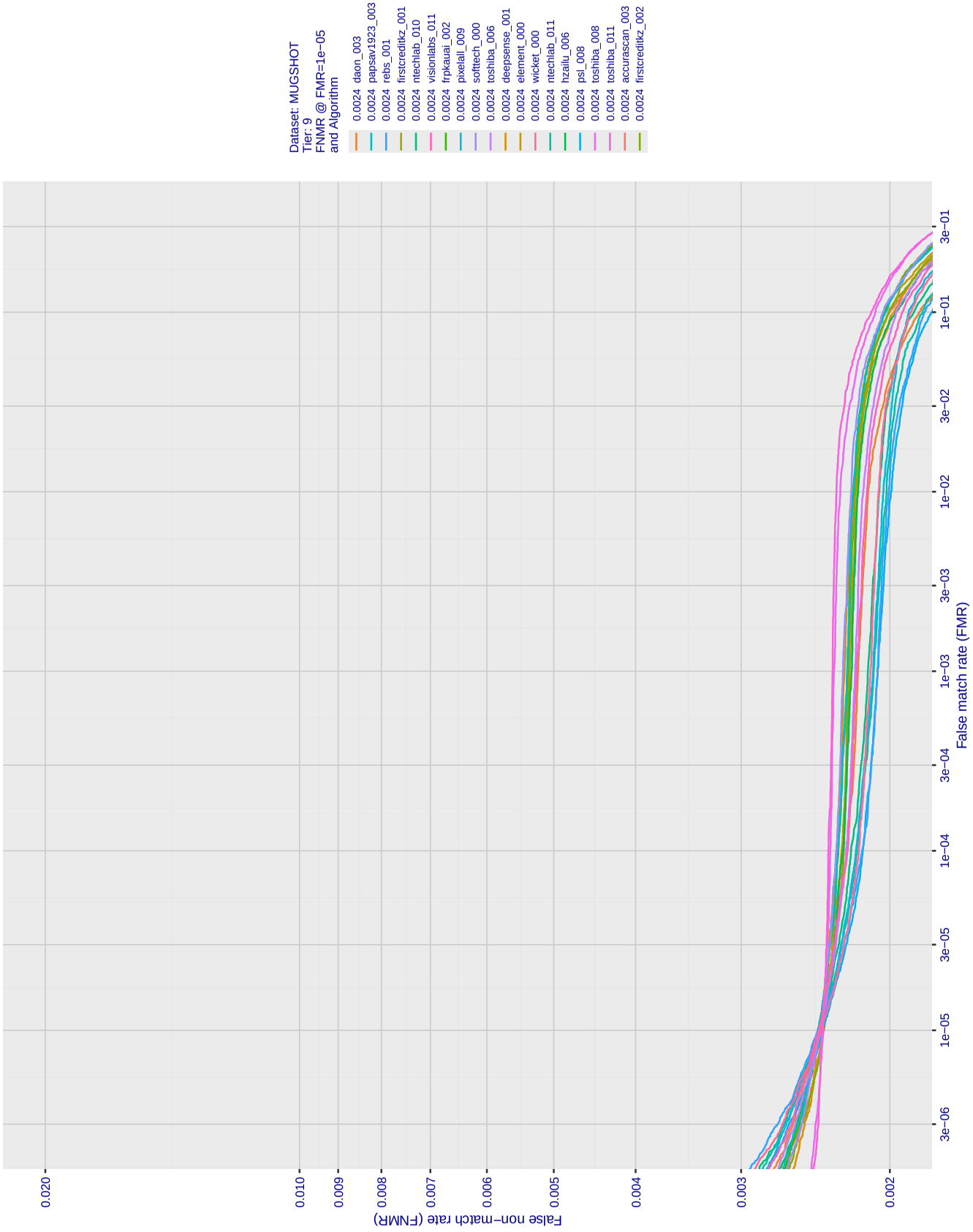


Figure 234: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

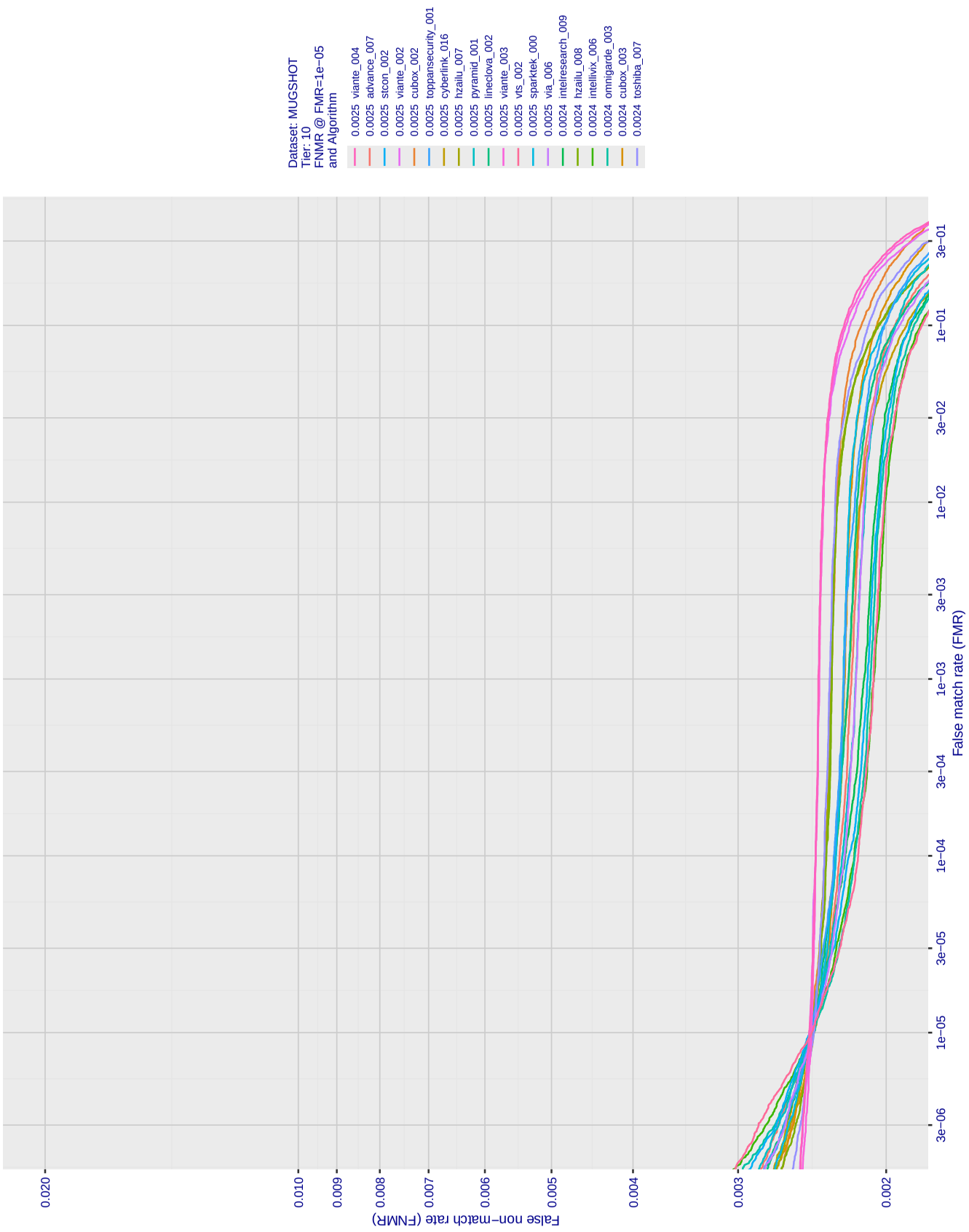


Figure 235: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

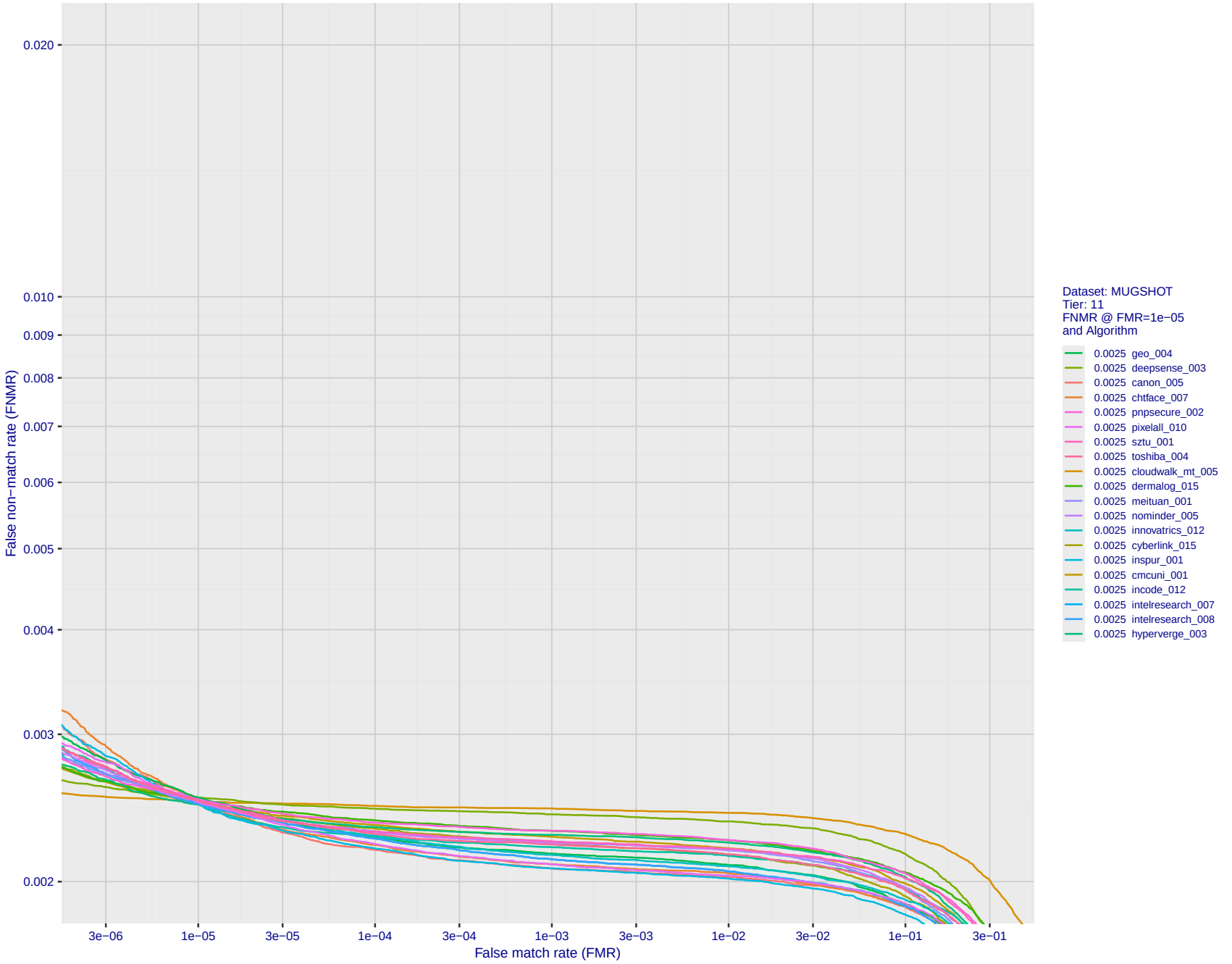


Figure 236: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

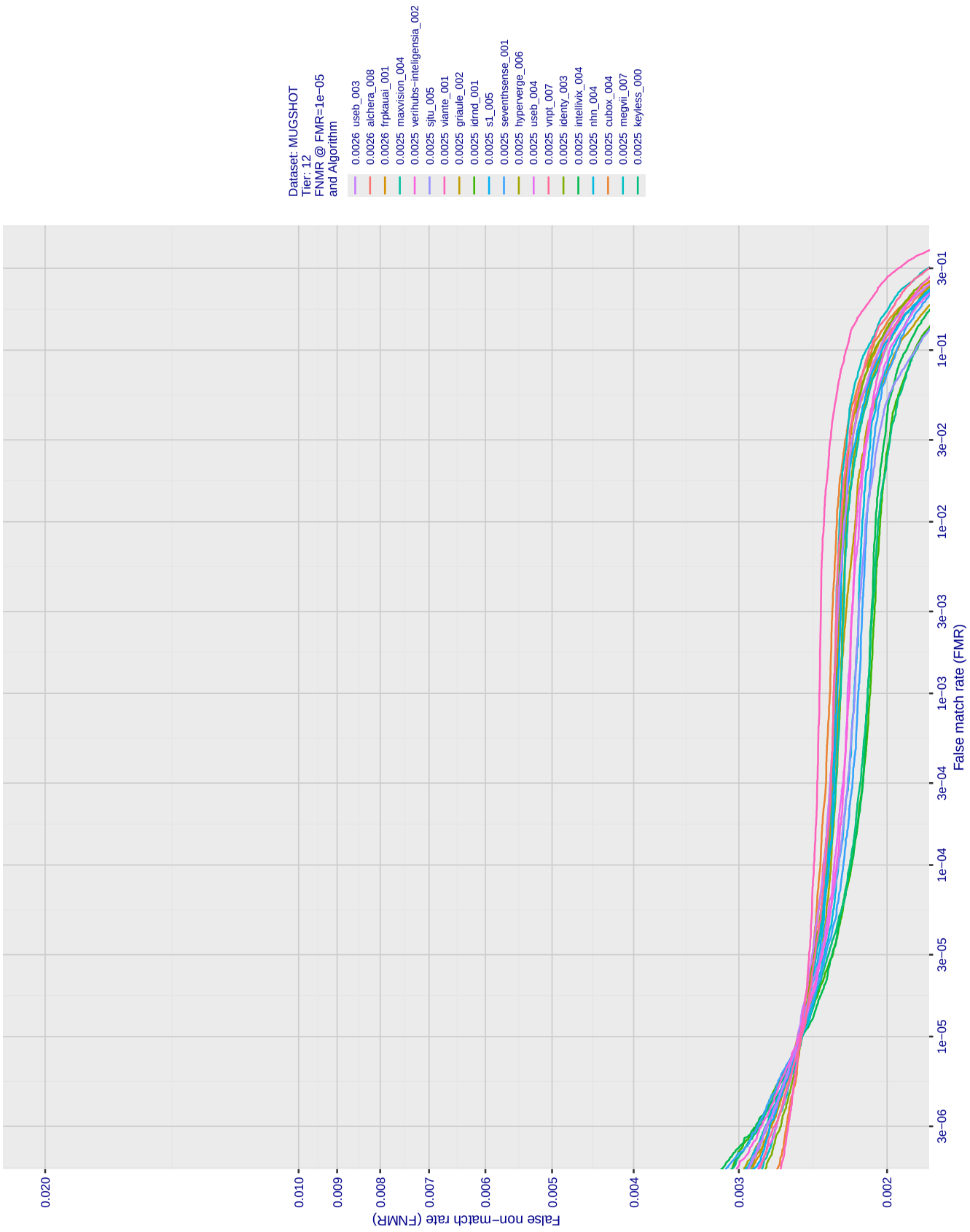


Figure 237: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

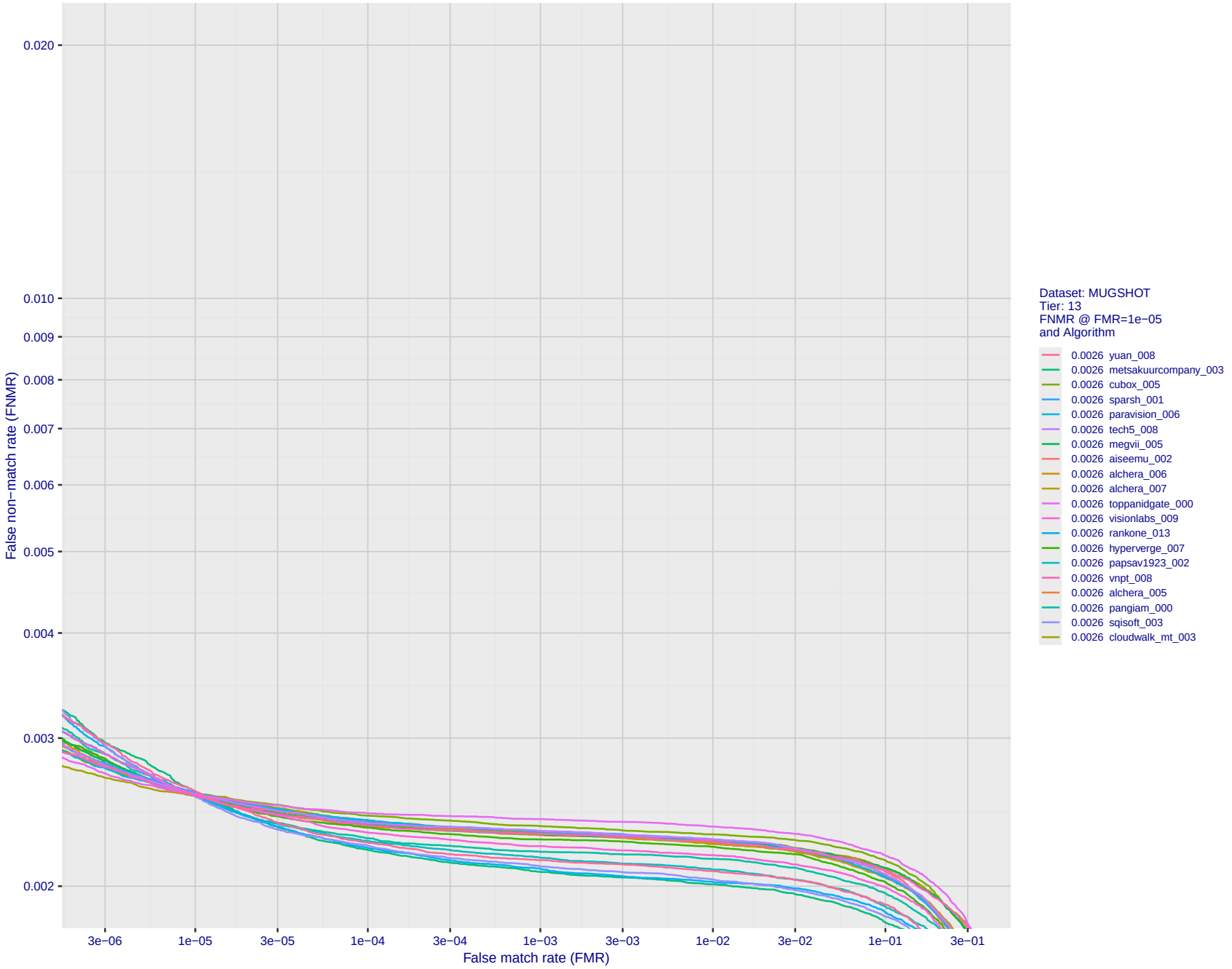


Figure 238: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

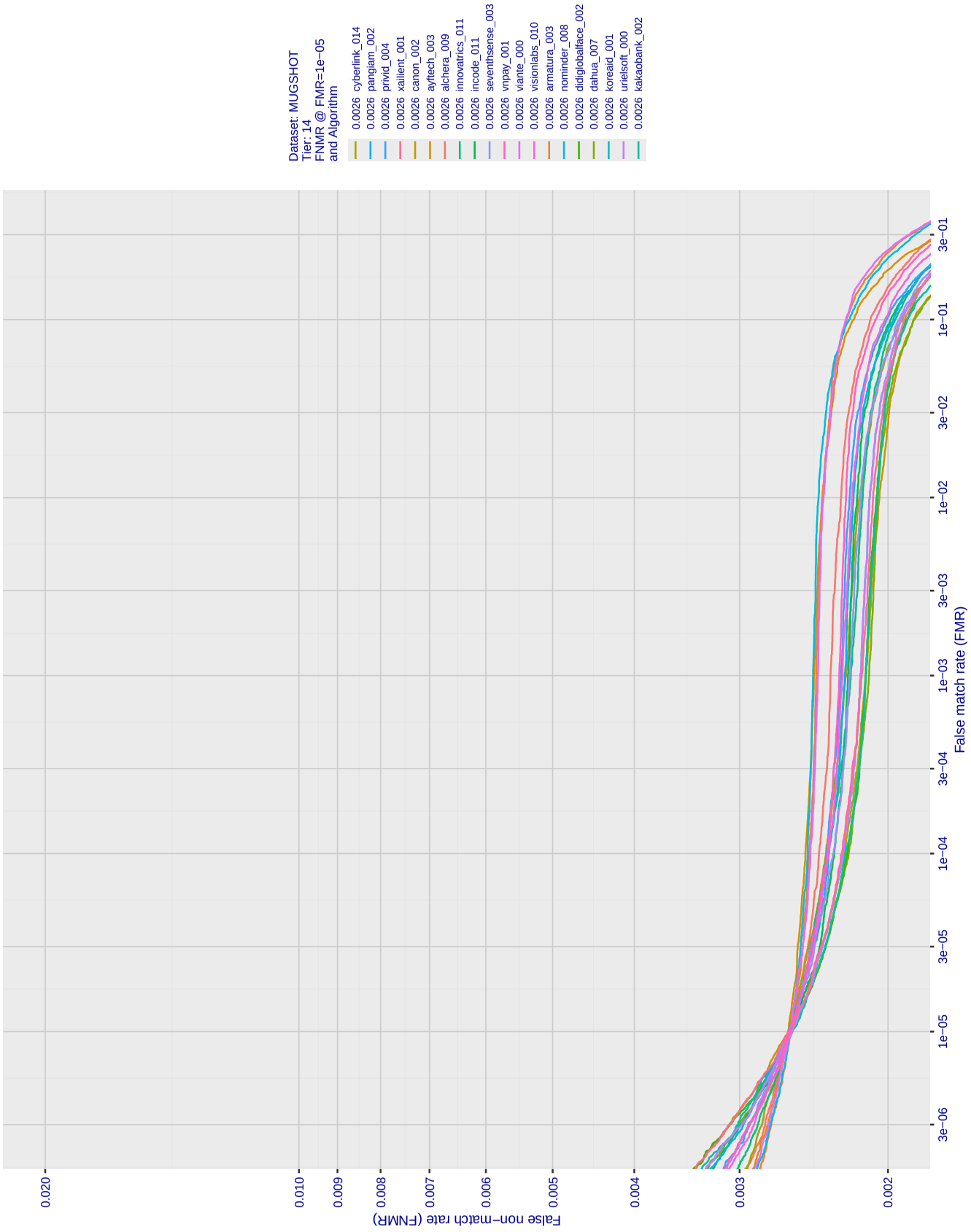


Figure 239: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

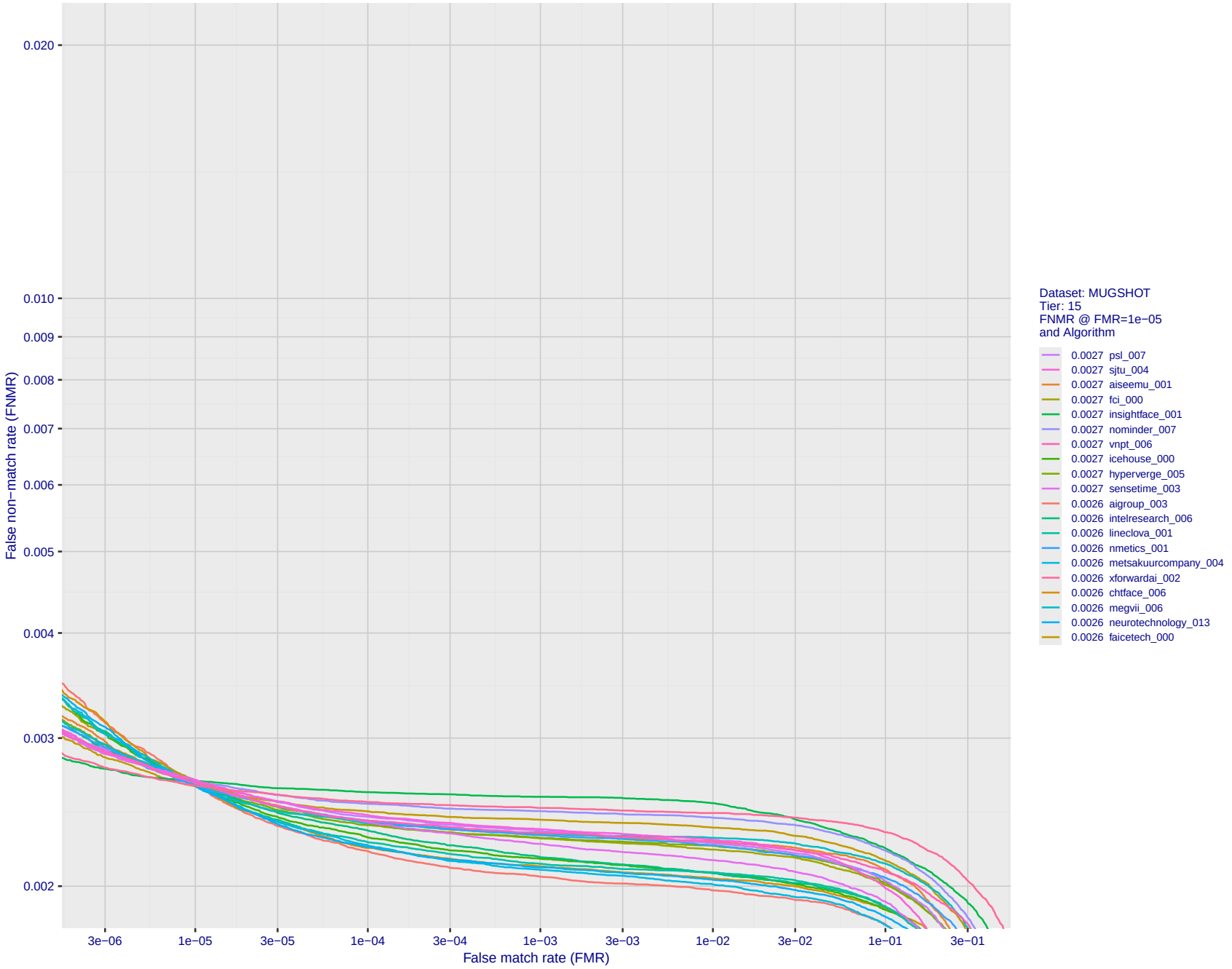


Figure 240: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

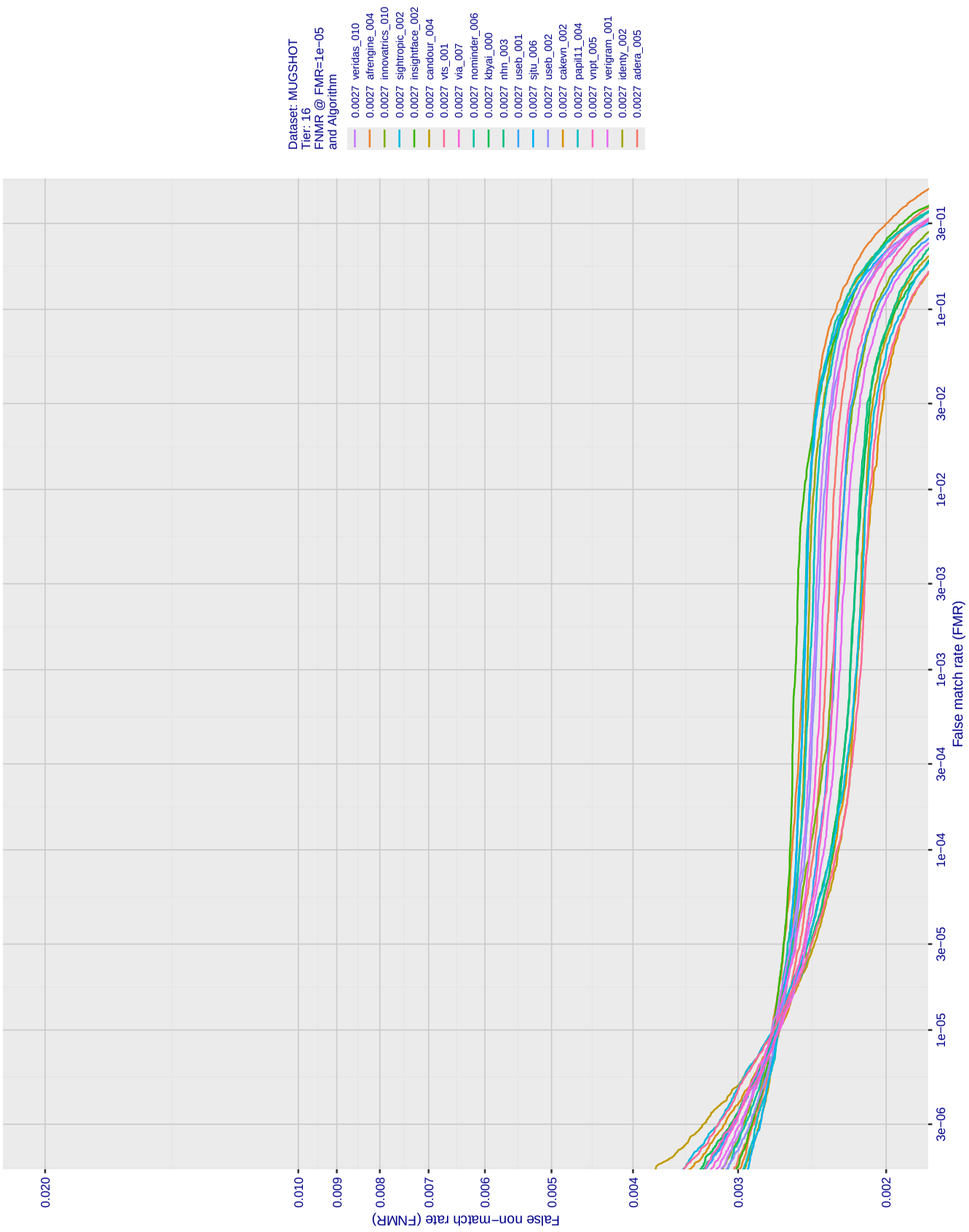


Figure 241: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

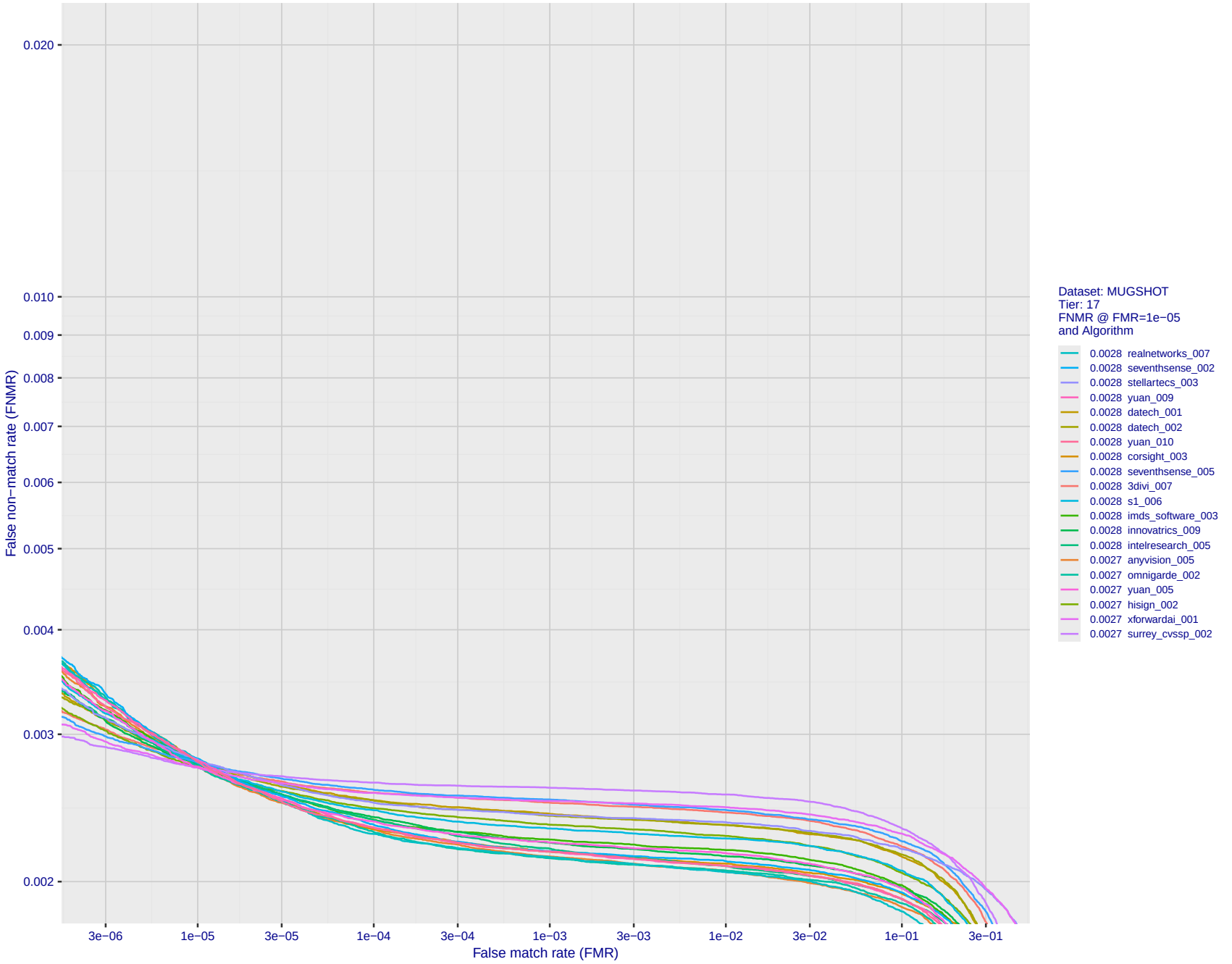


Figure 242: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

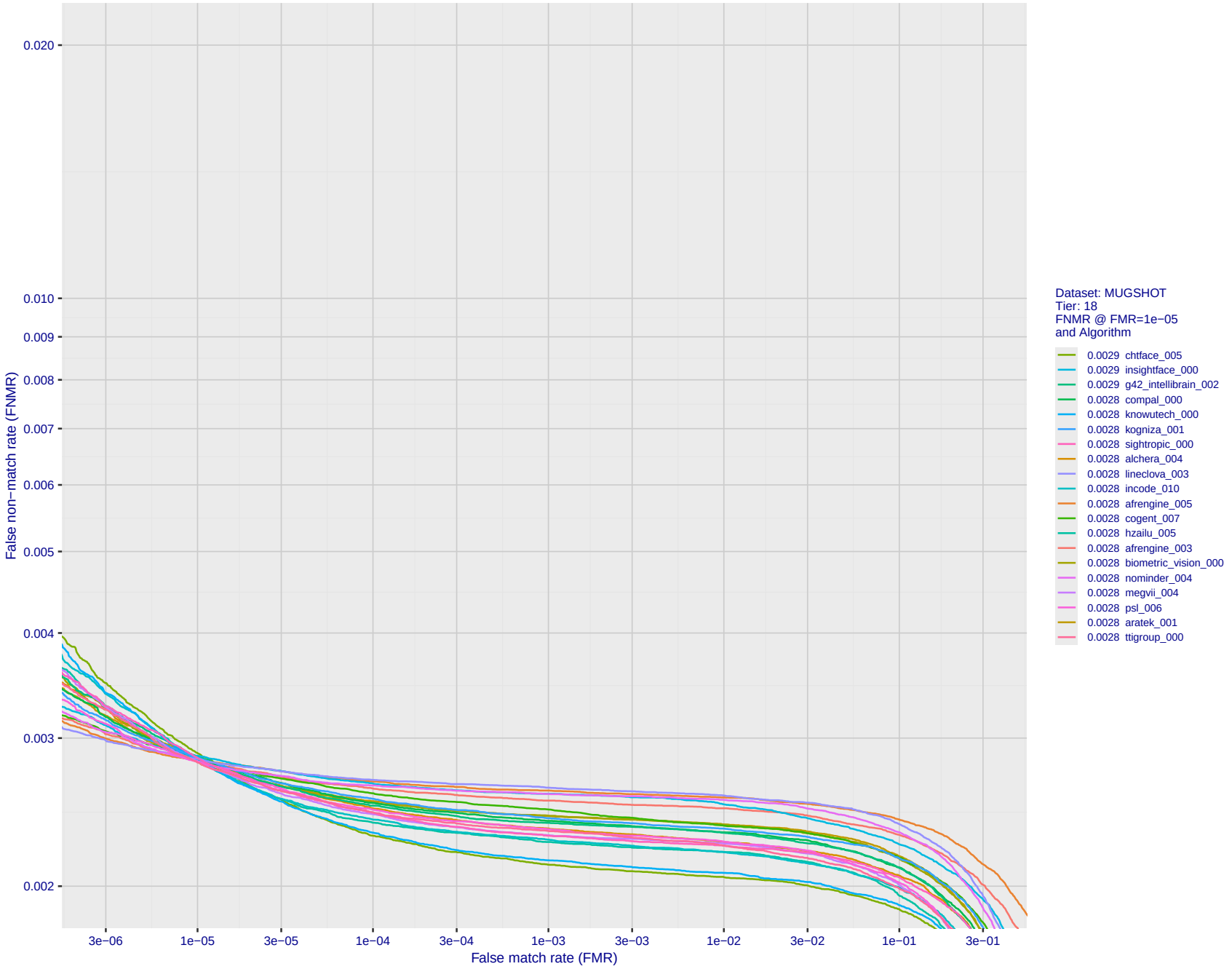


Figure 243: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

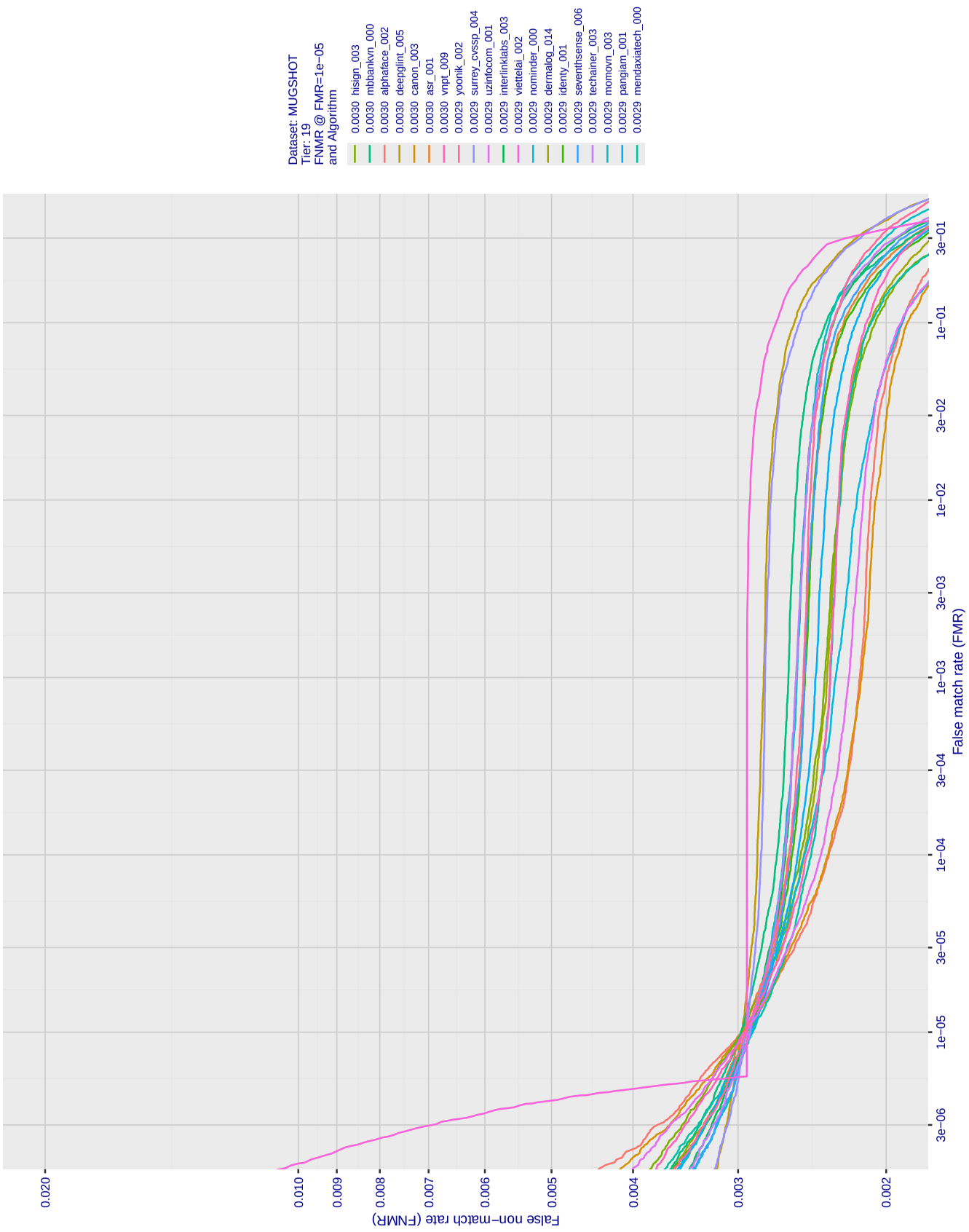


Figure 244: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

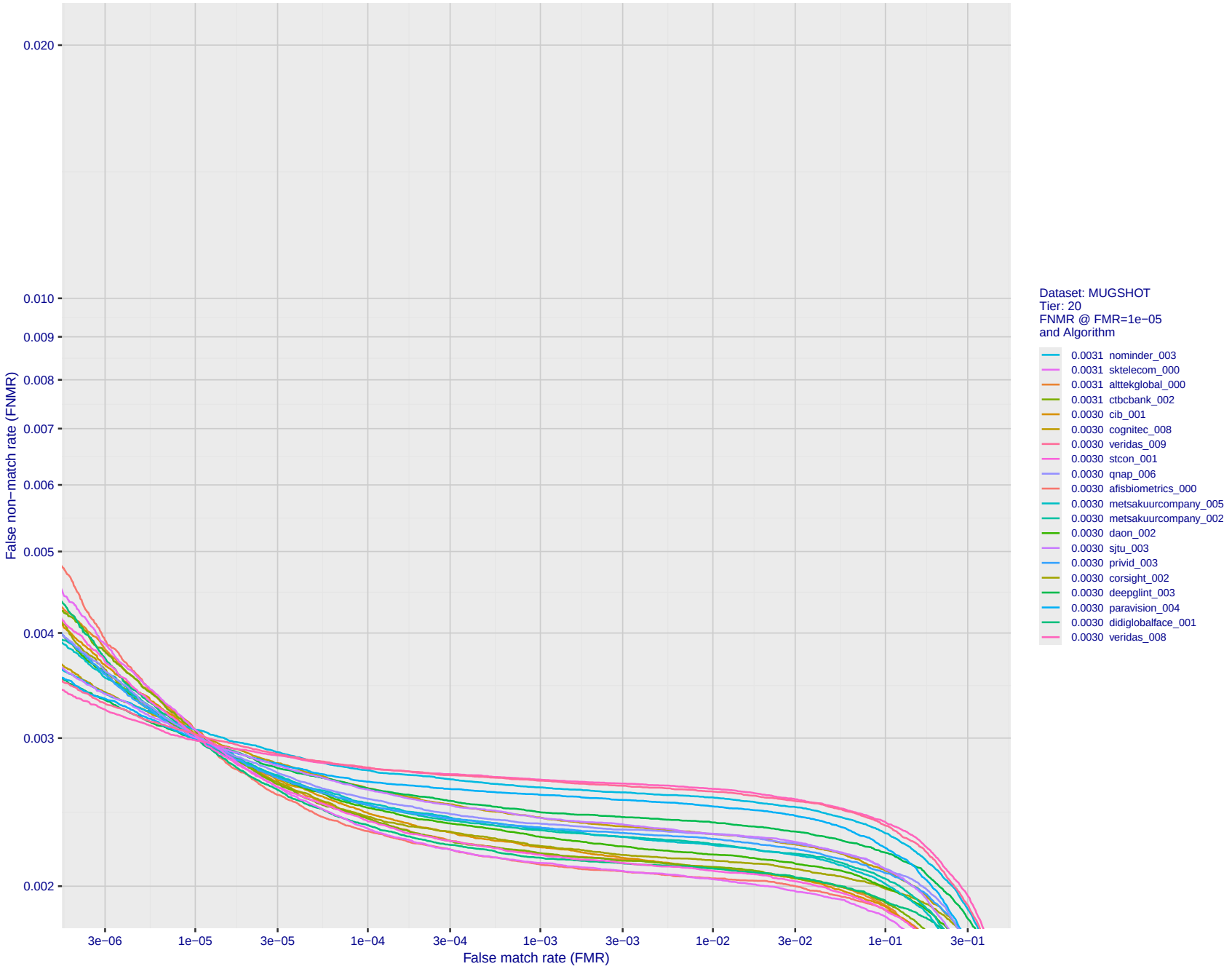


Figure 245: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

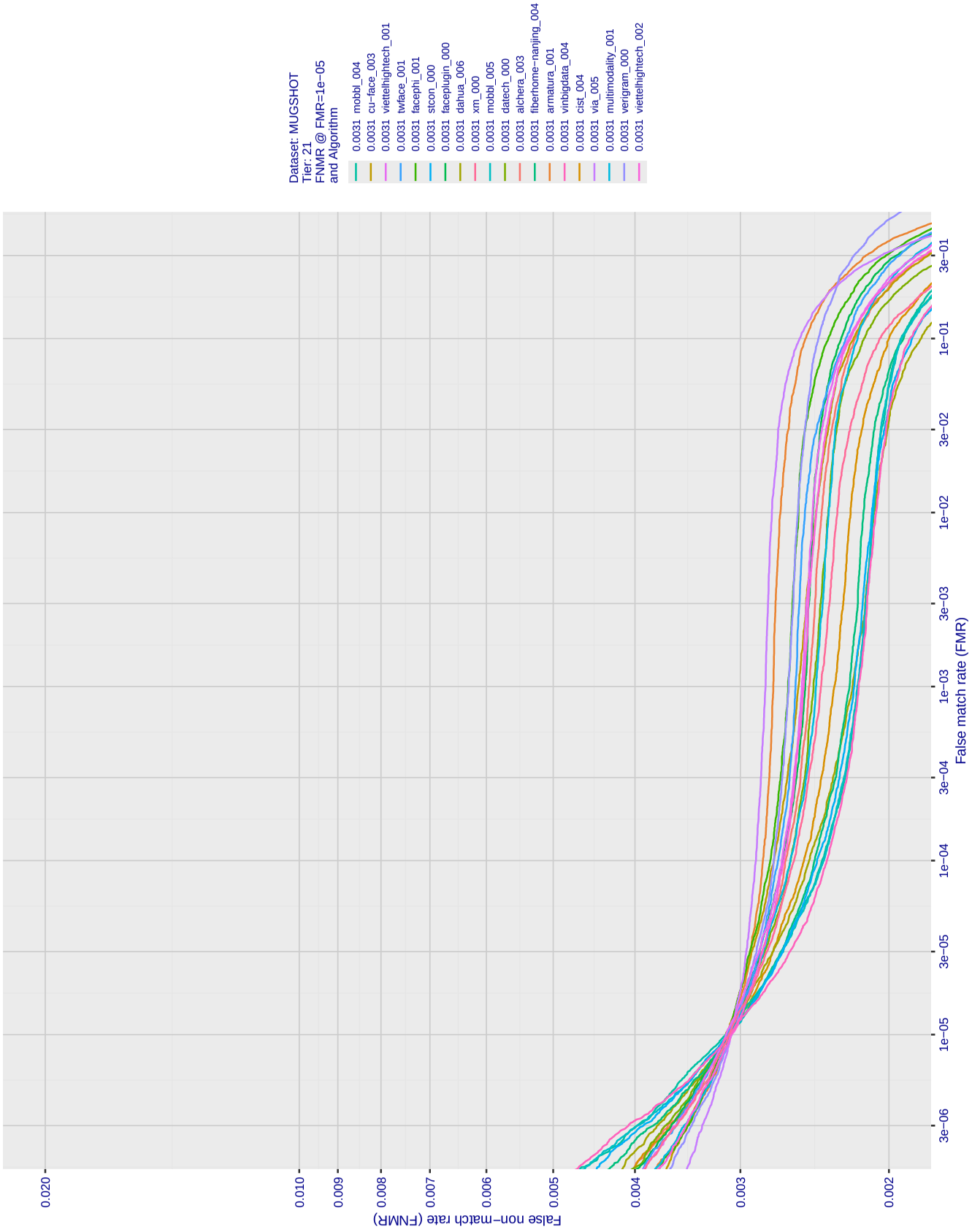


Figure 246: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

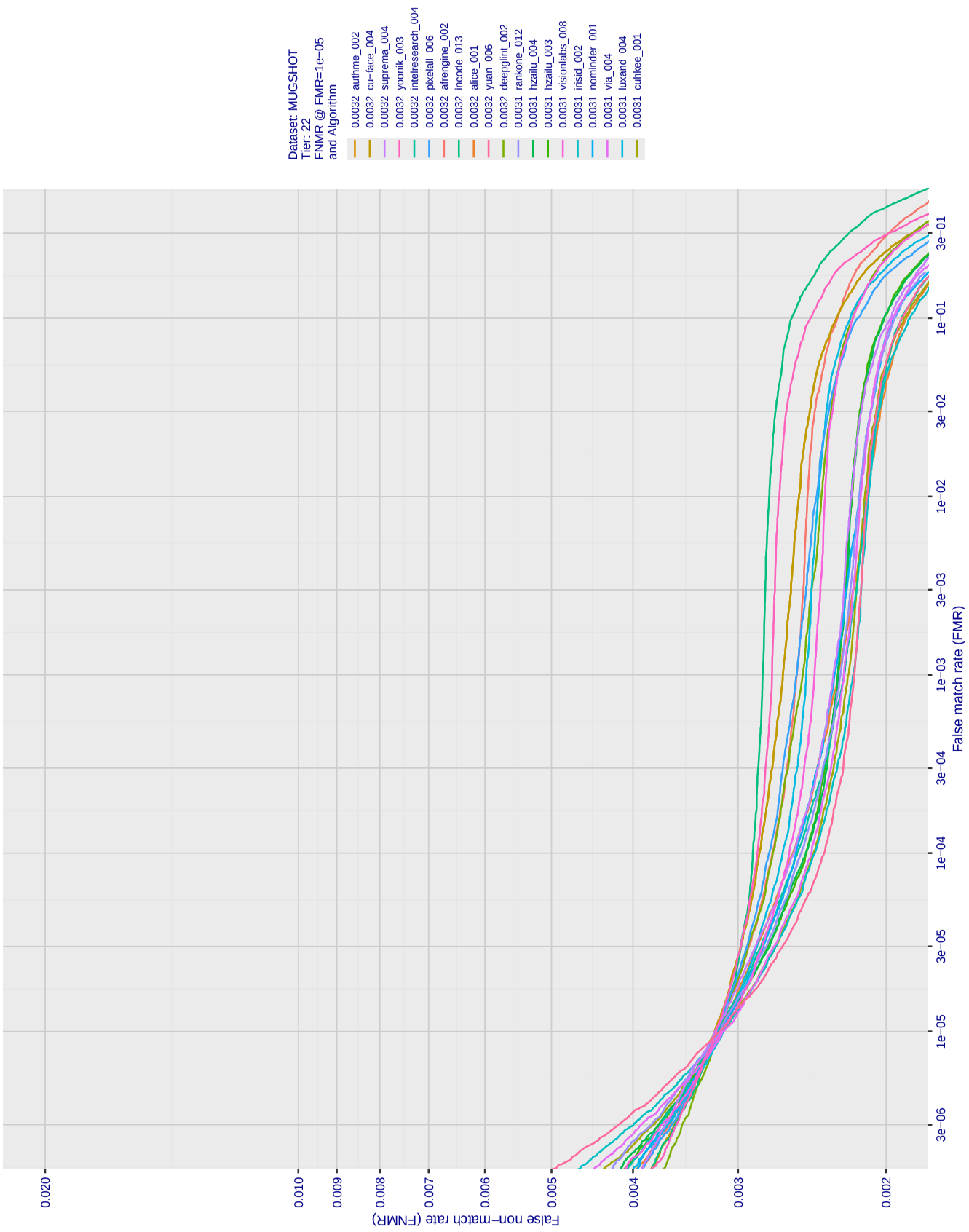


Figure 247: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

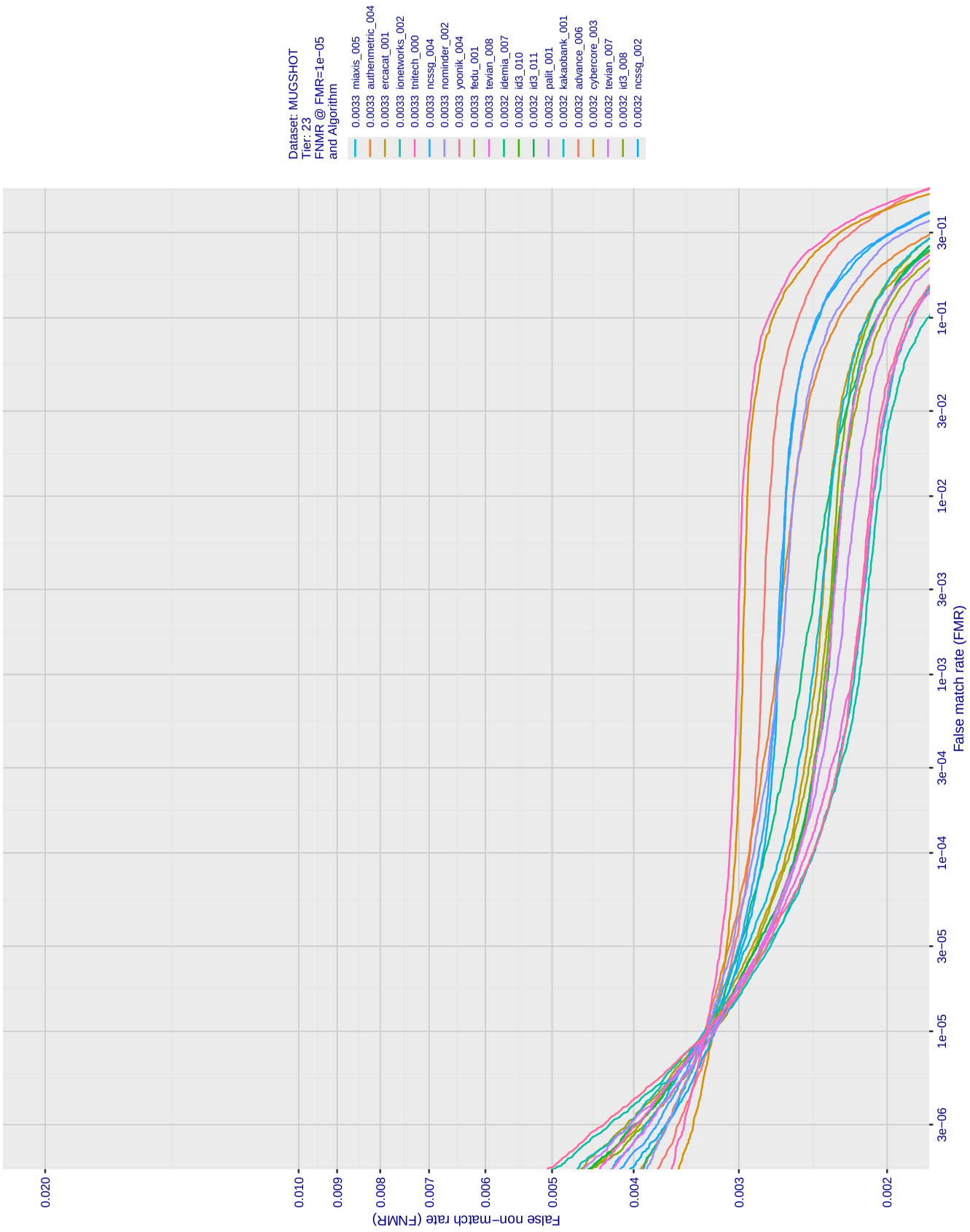


Figure 248: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

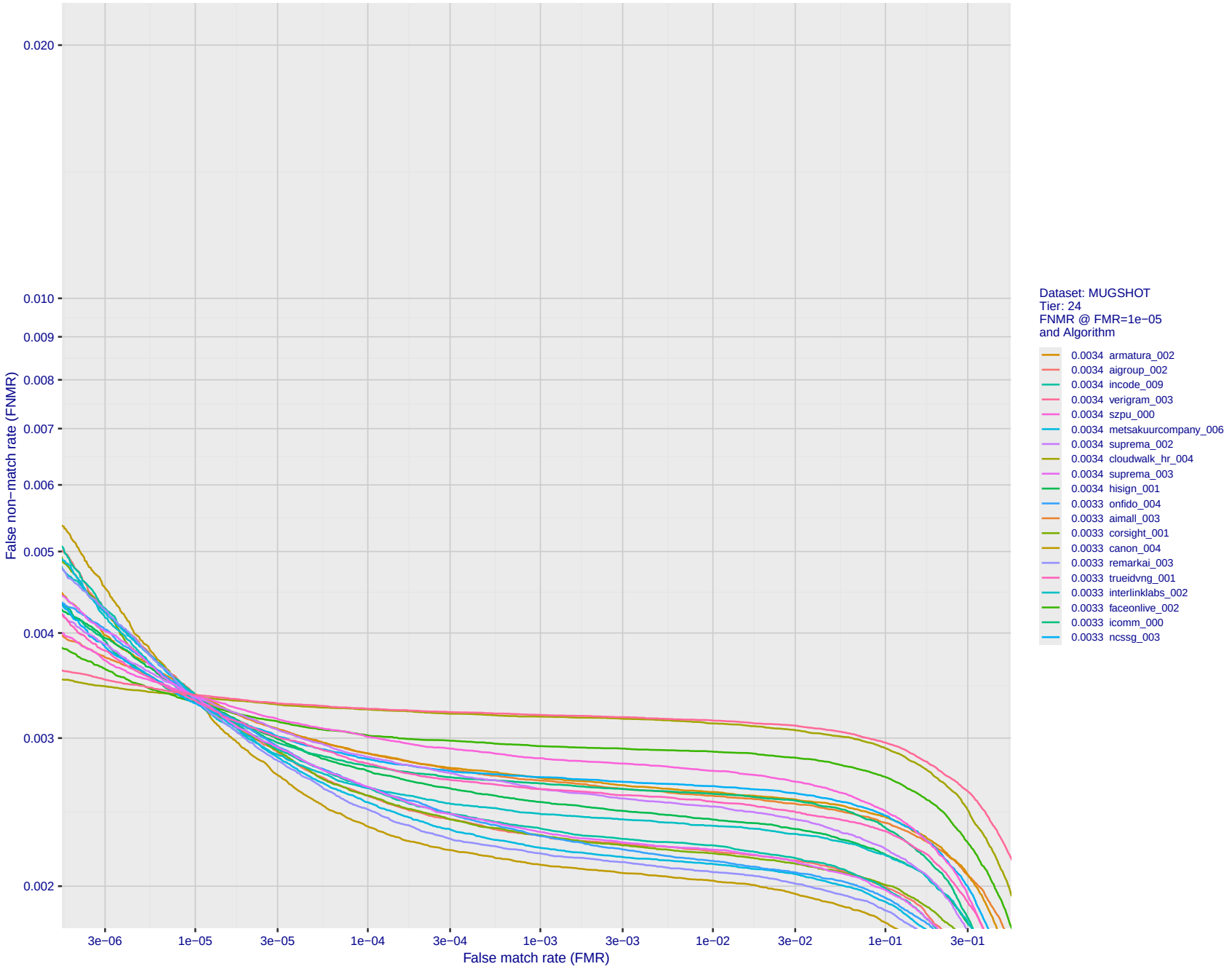


*Figure 249: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

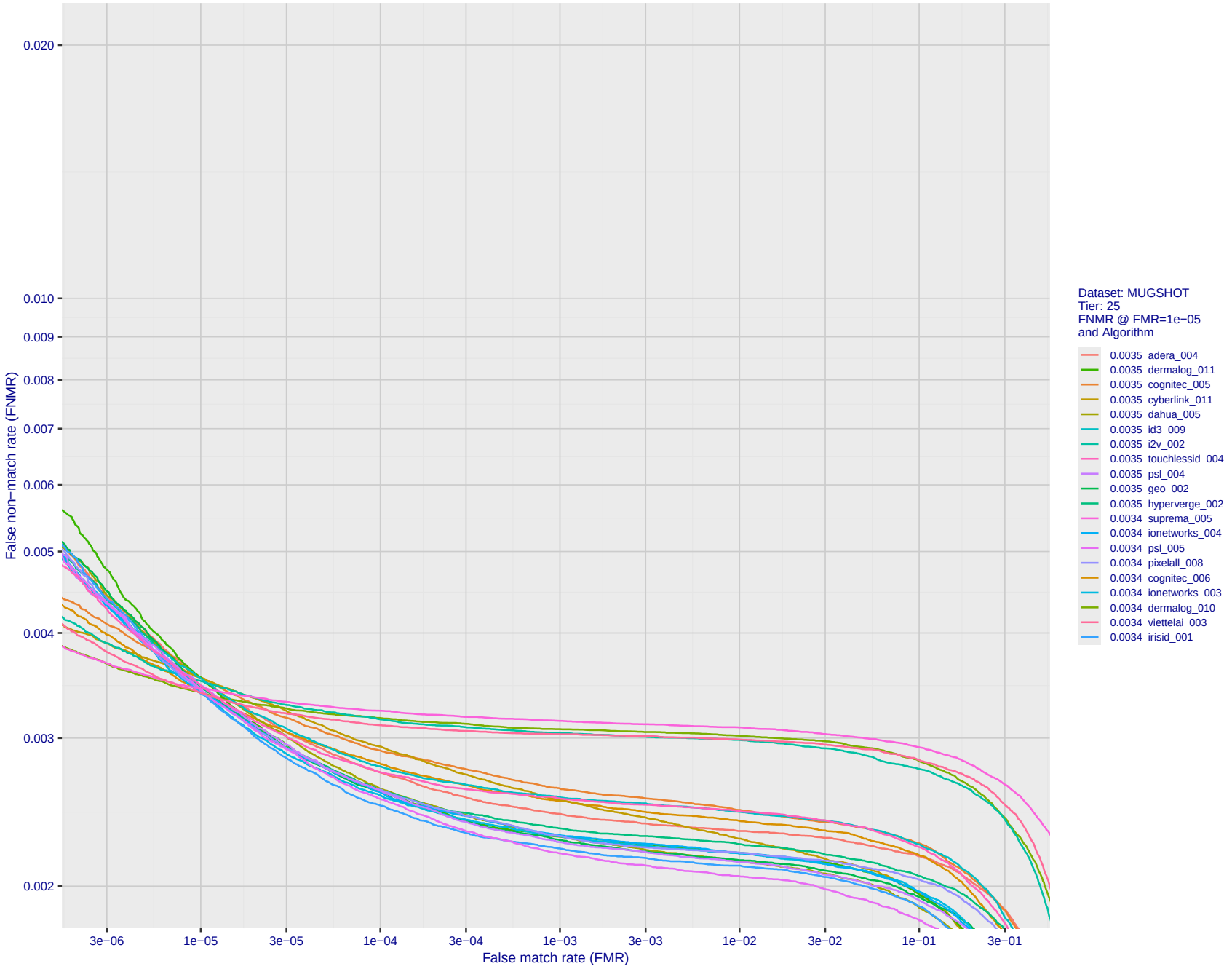


Figure 250: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

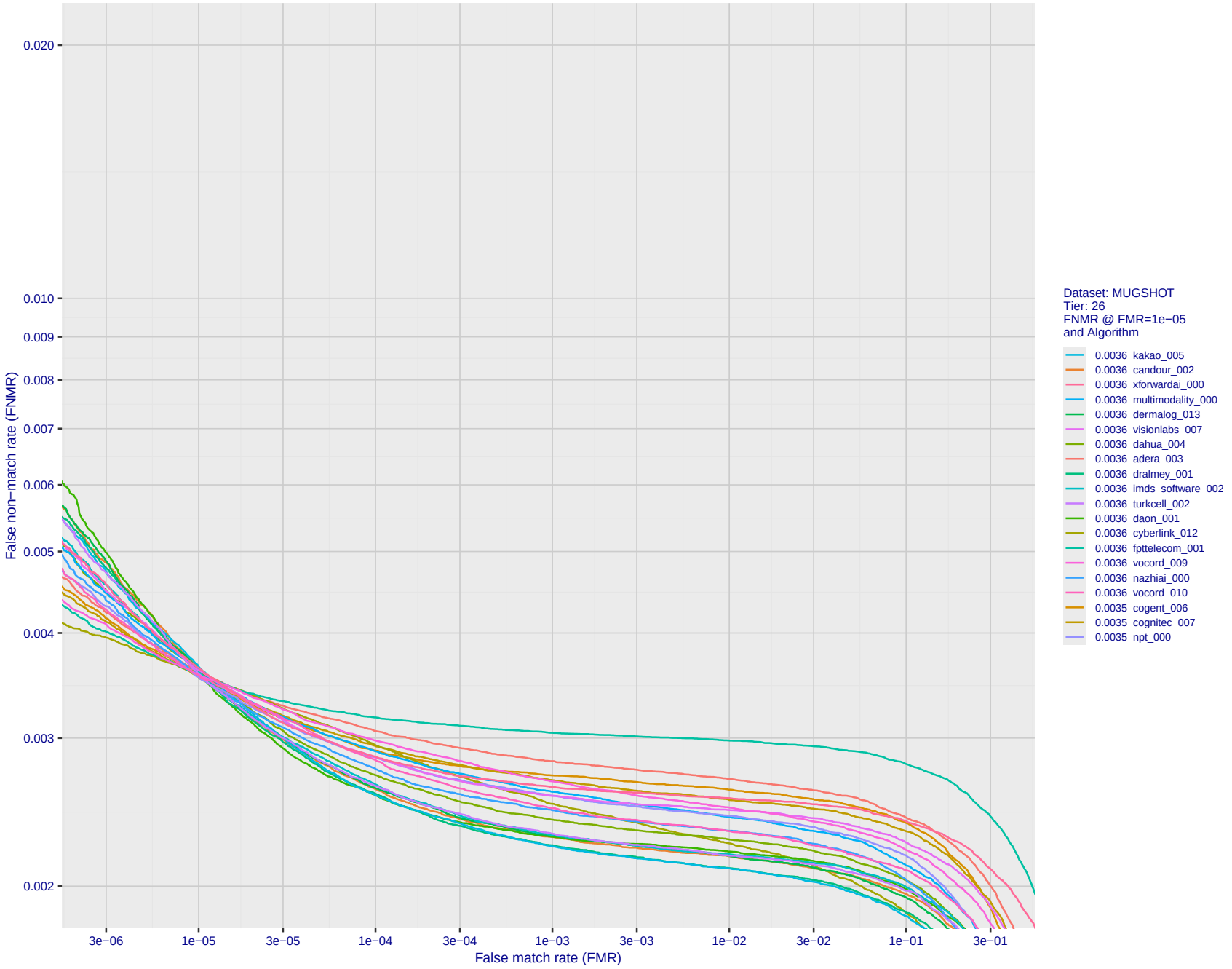


Figure 251: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

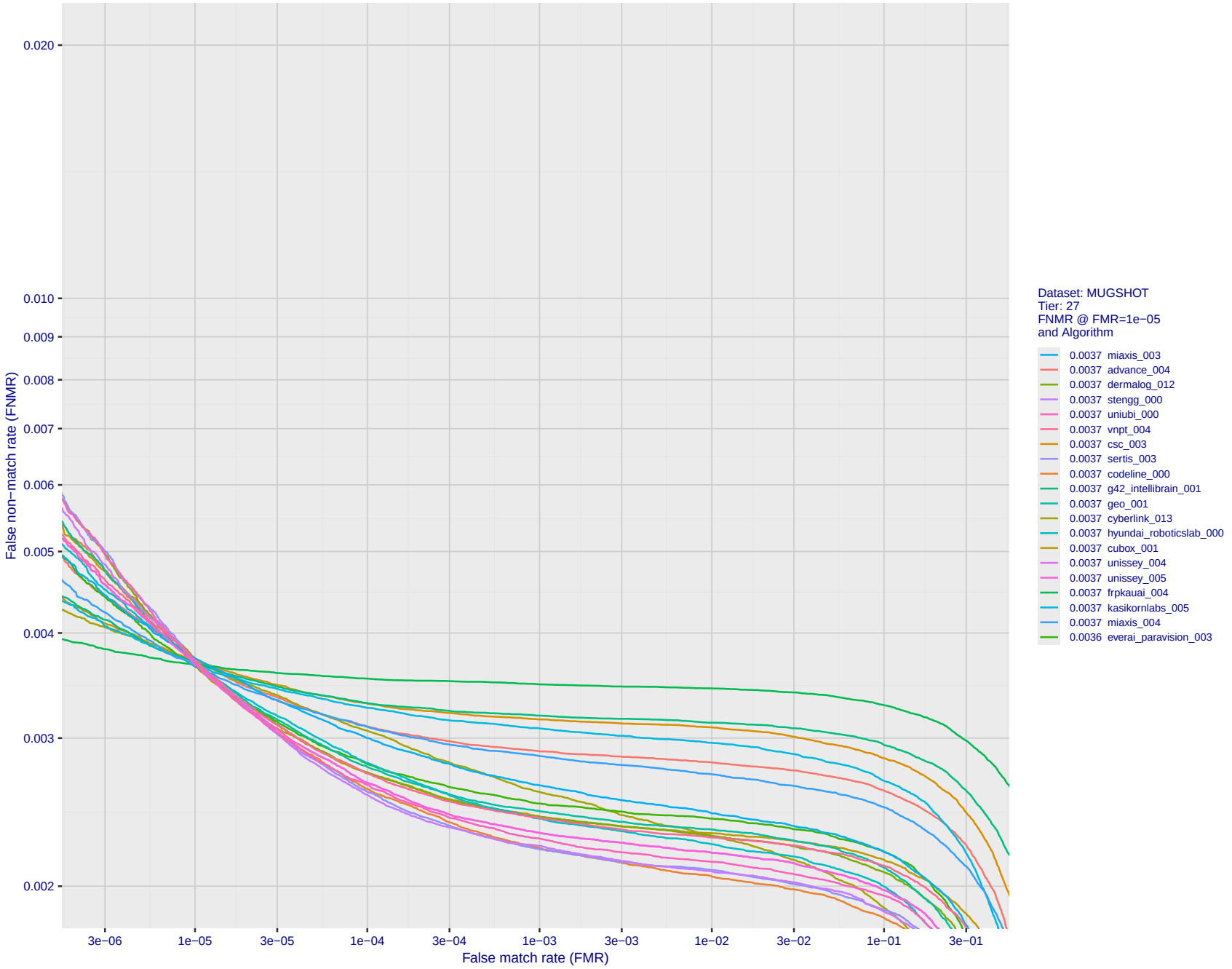


Figure 252: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

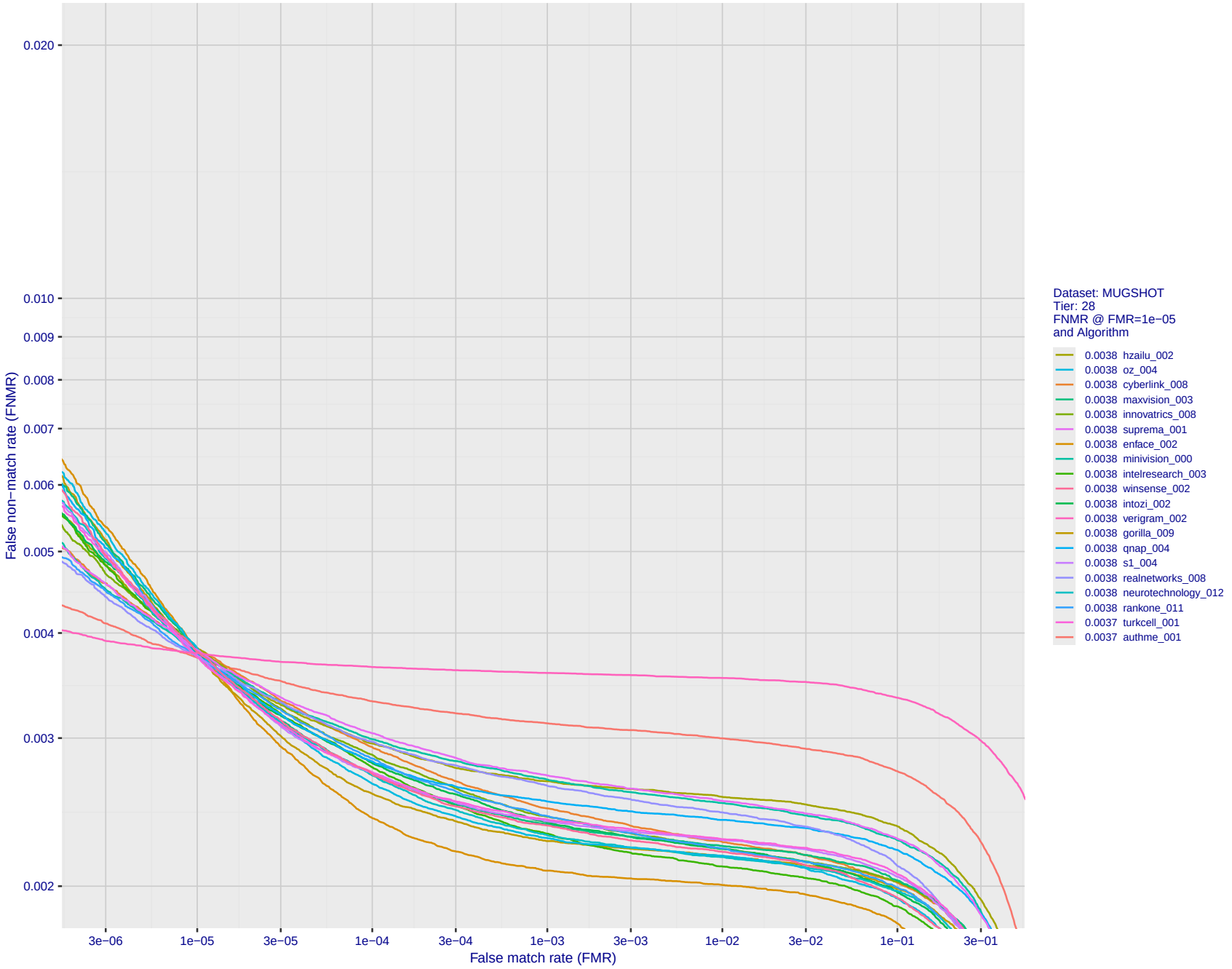


Figure 253: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

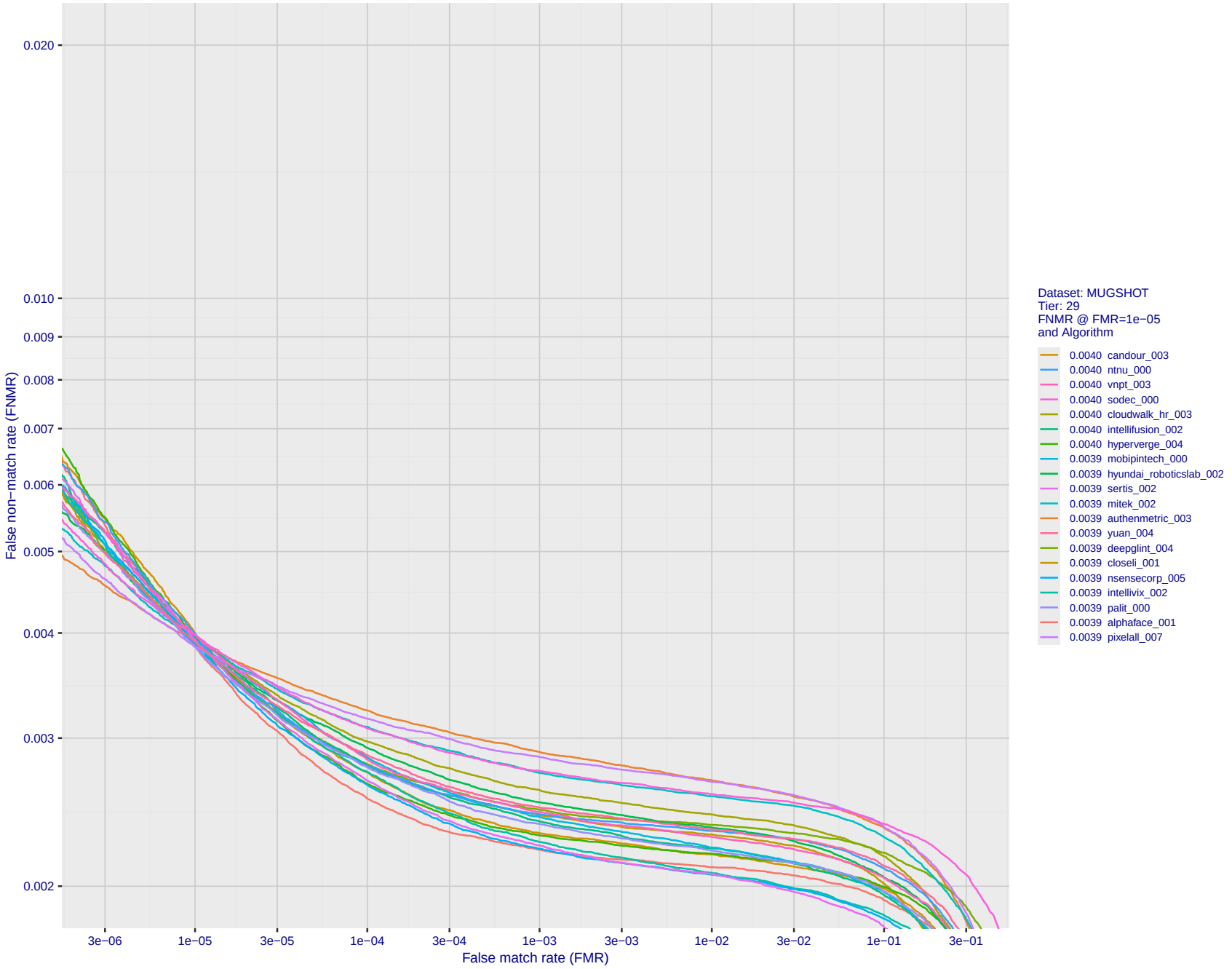


*Figure 254: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

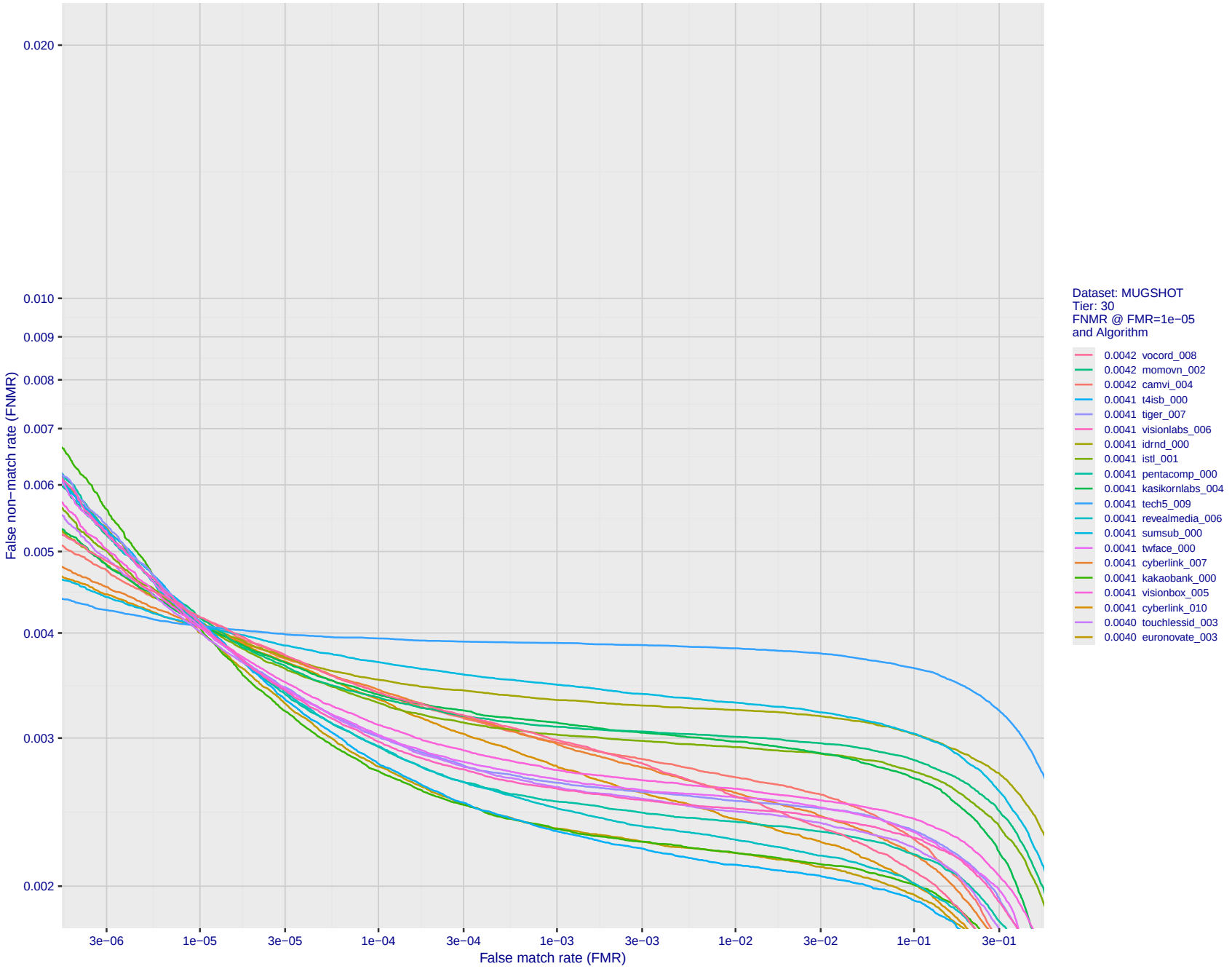


Figure 255: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

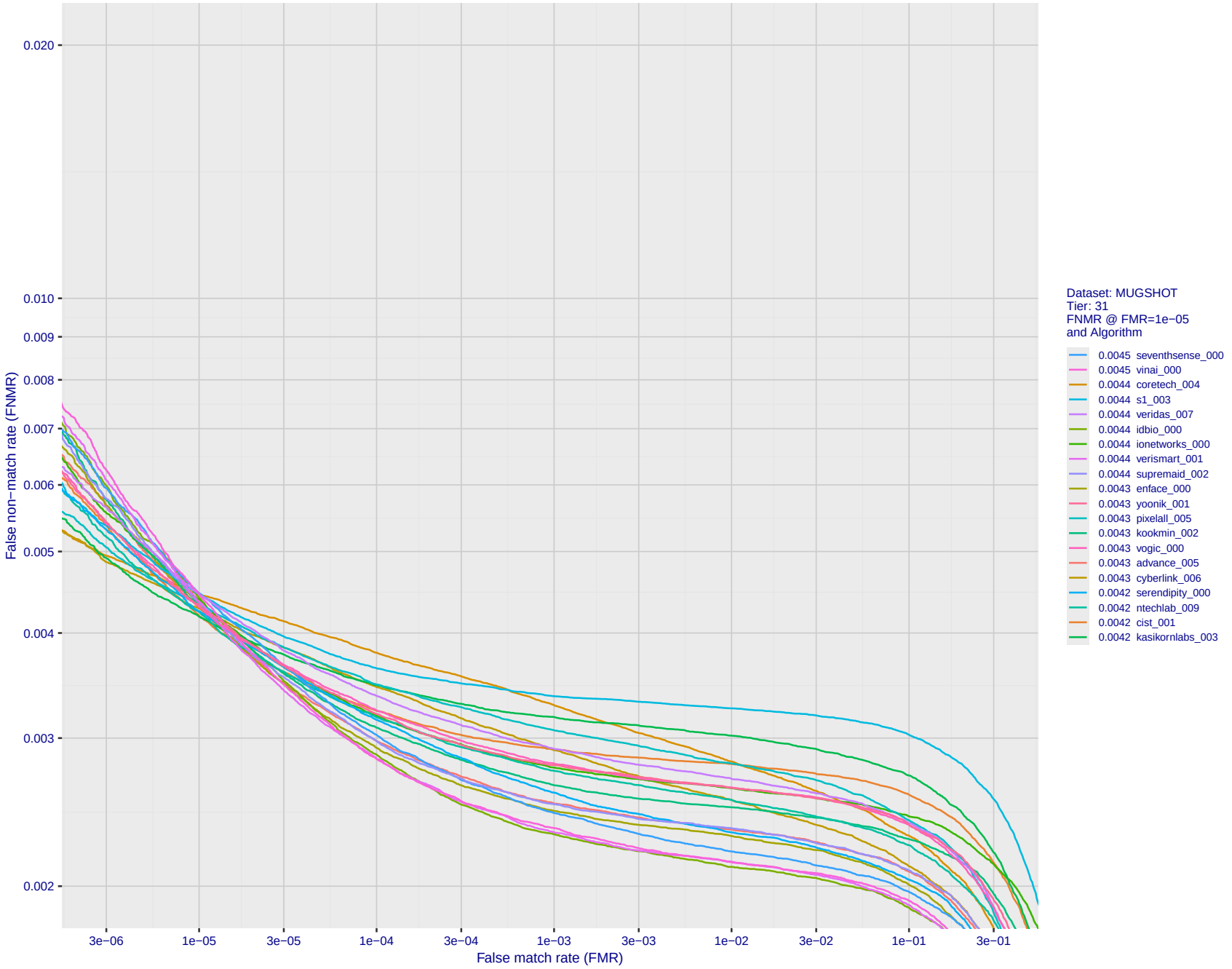


Figure 256: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

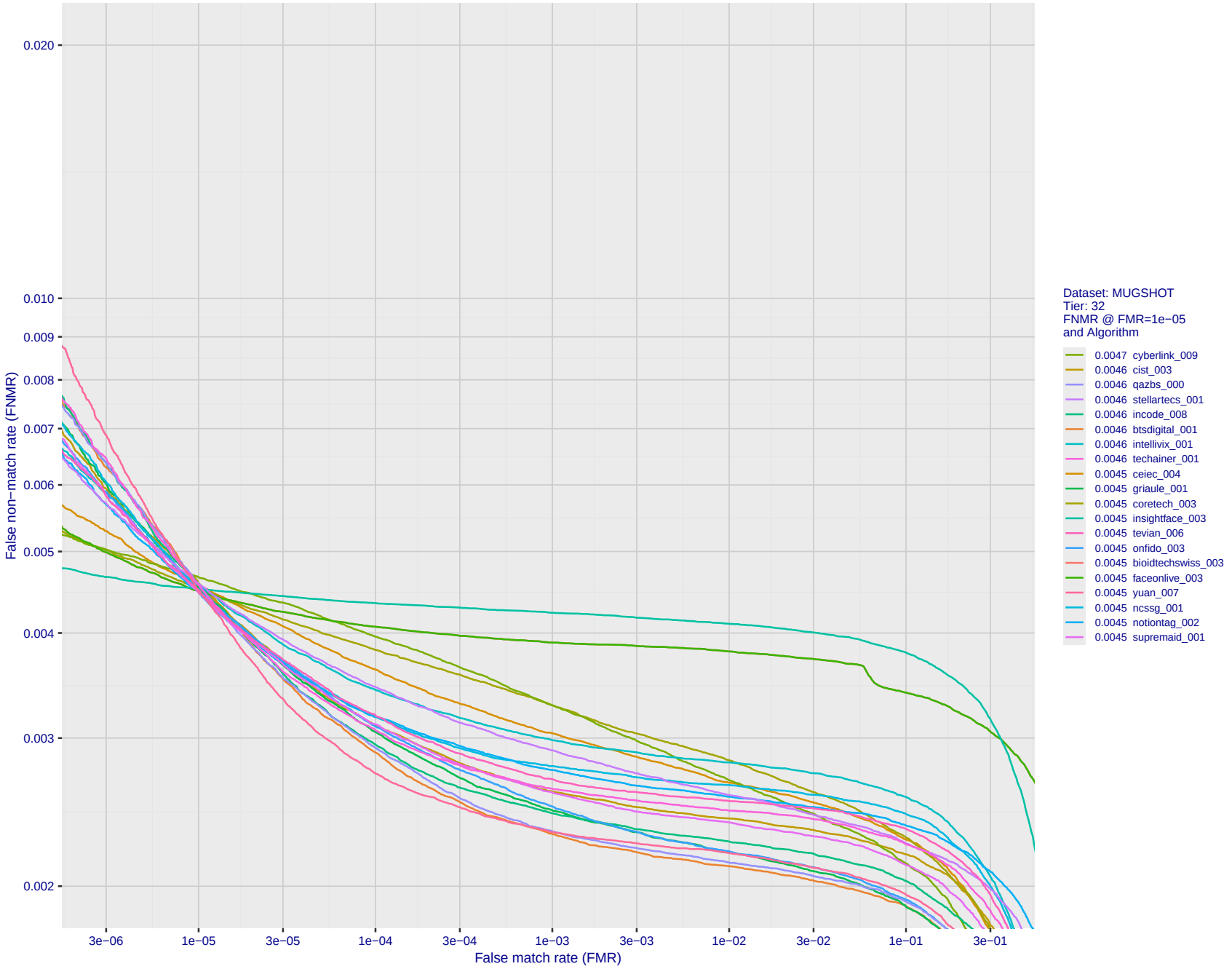


Figure 257: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

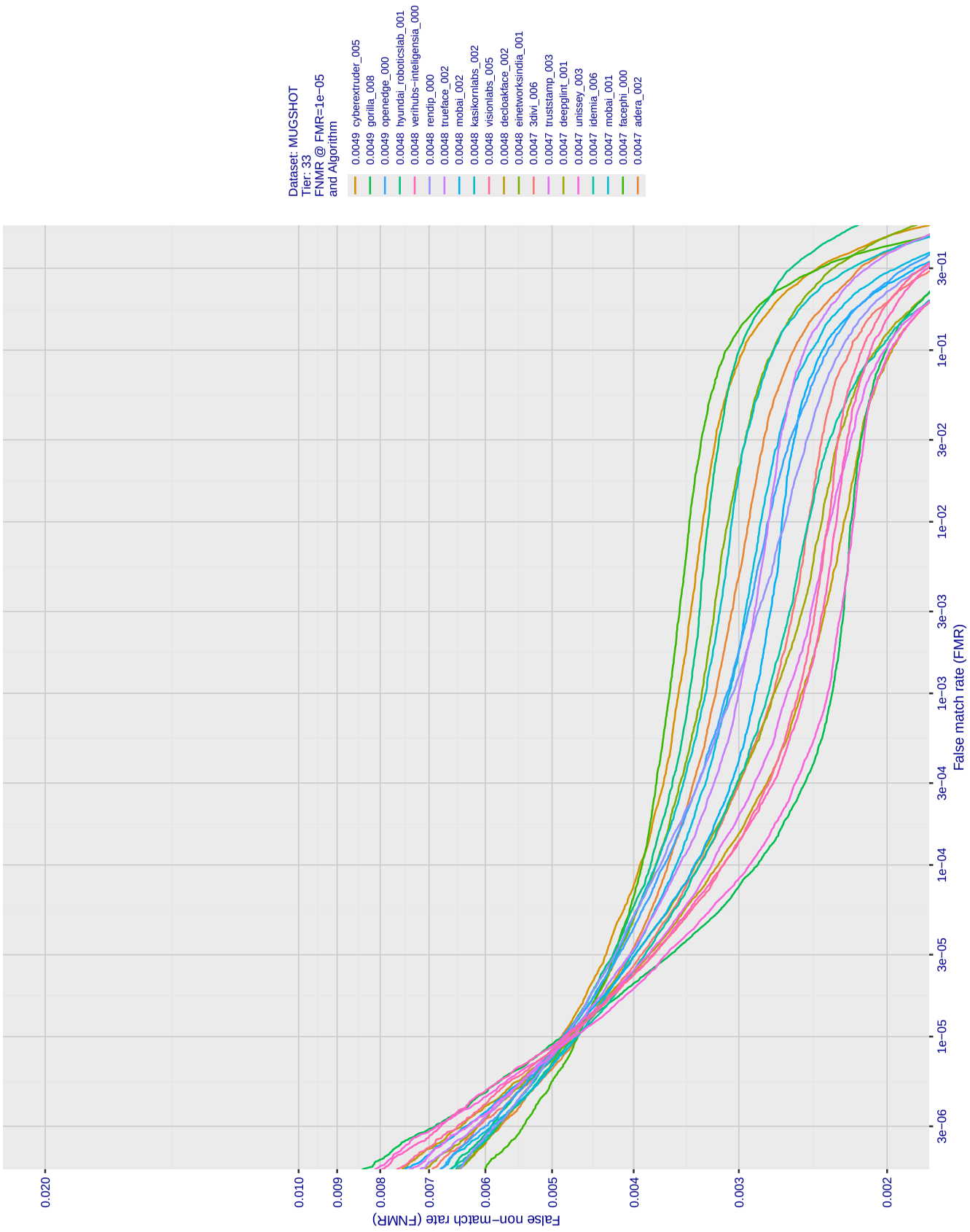


Figure 258: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

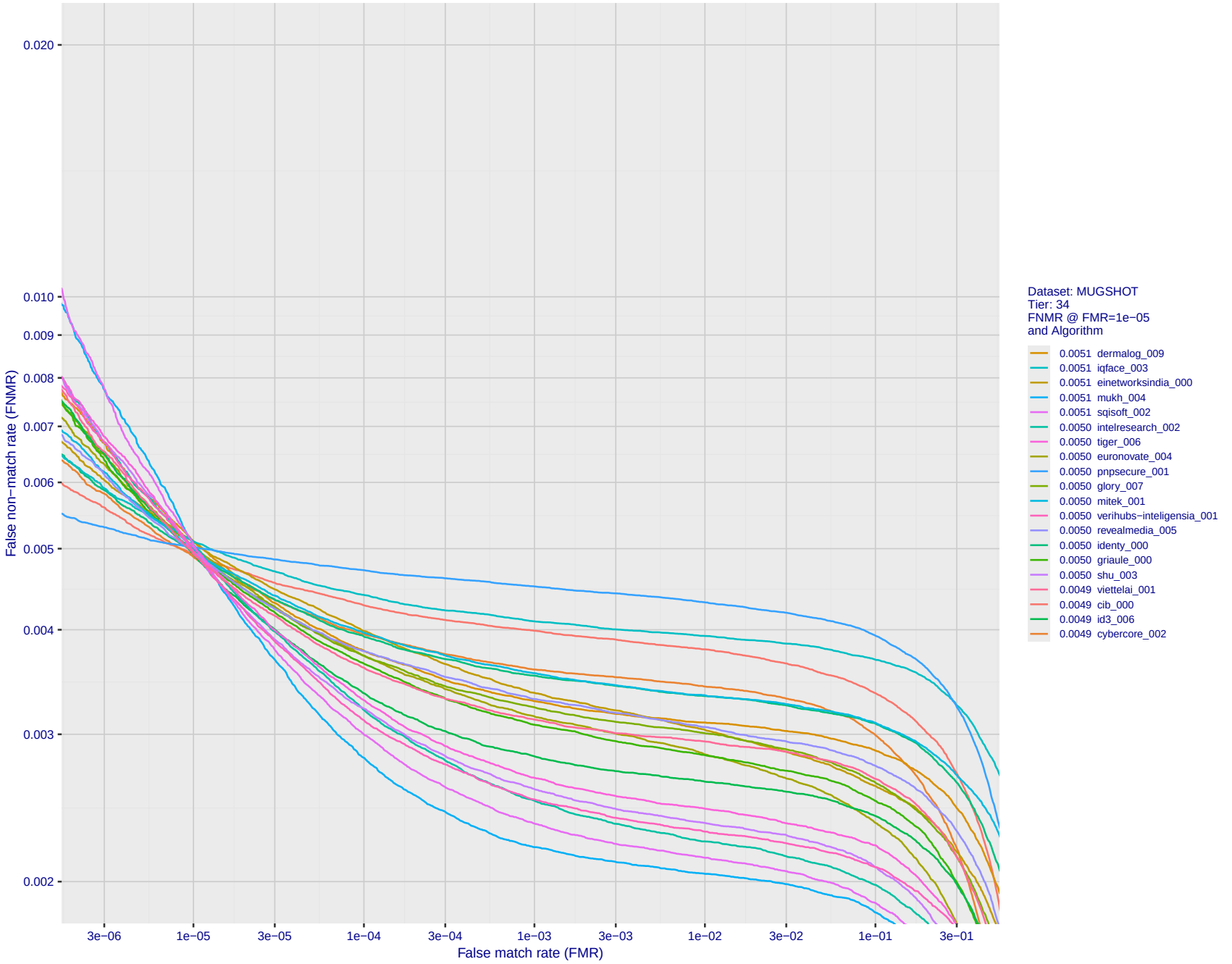


Figure 259: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

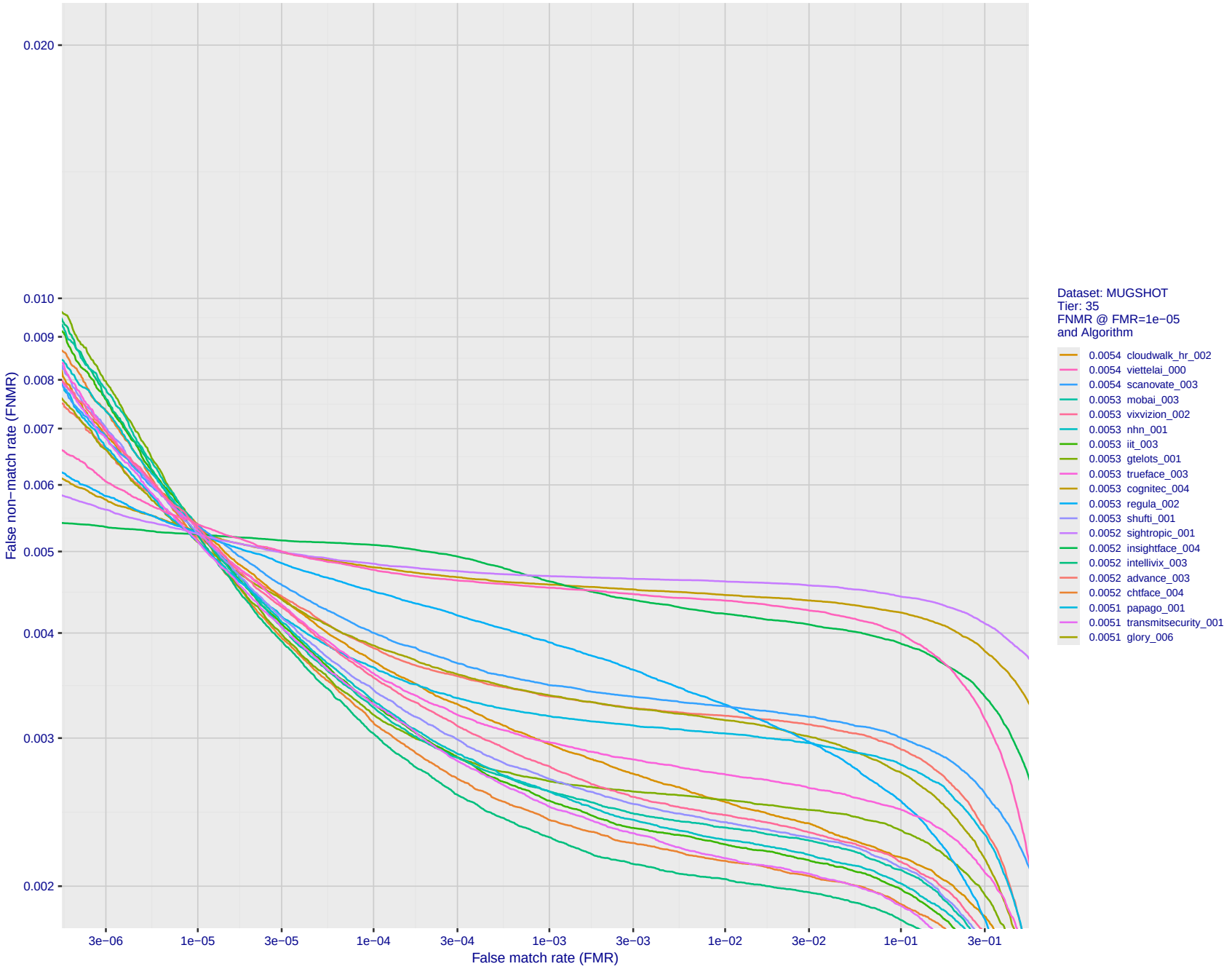


Figure 260: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

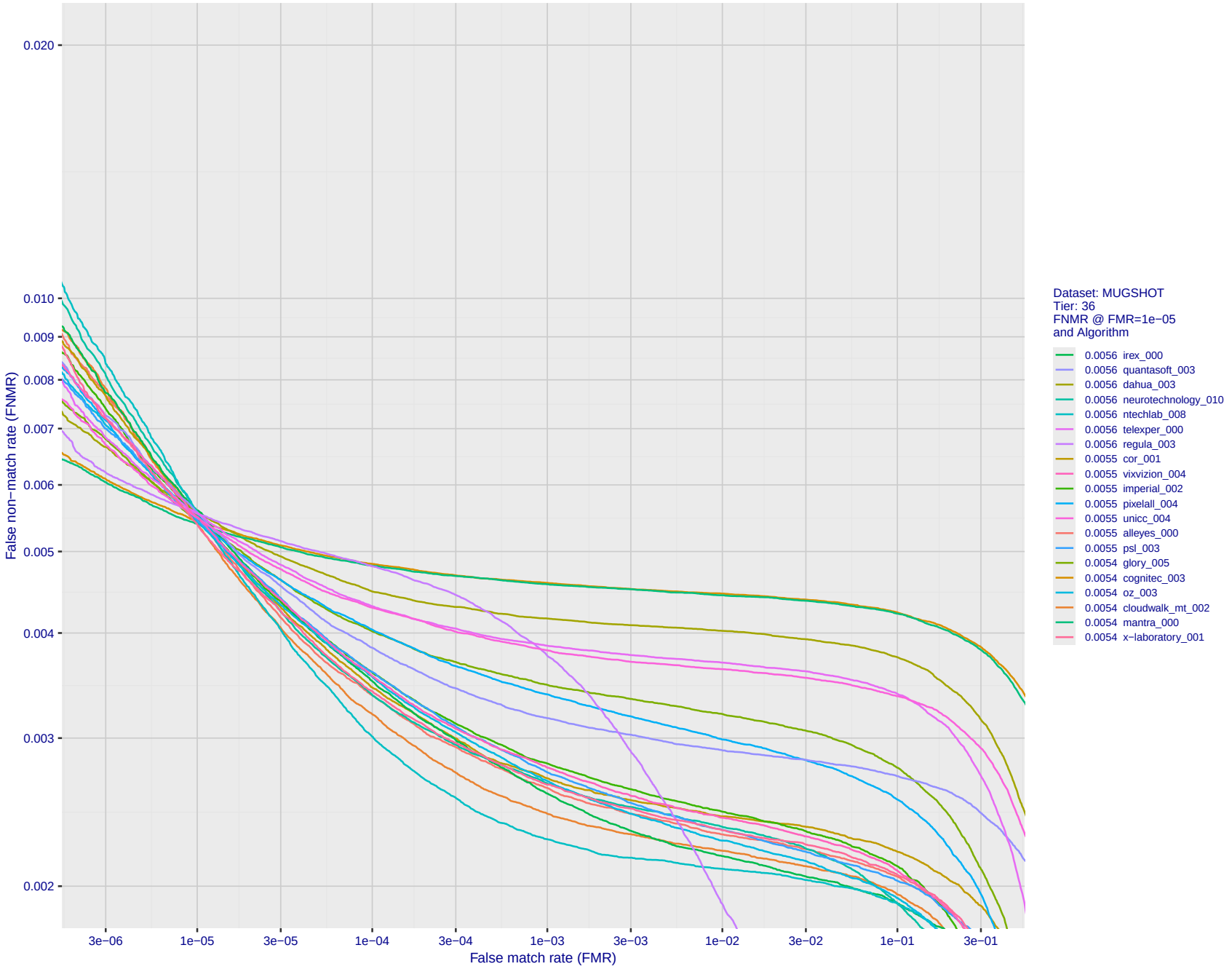


Figure 261: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

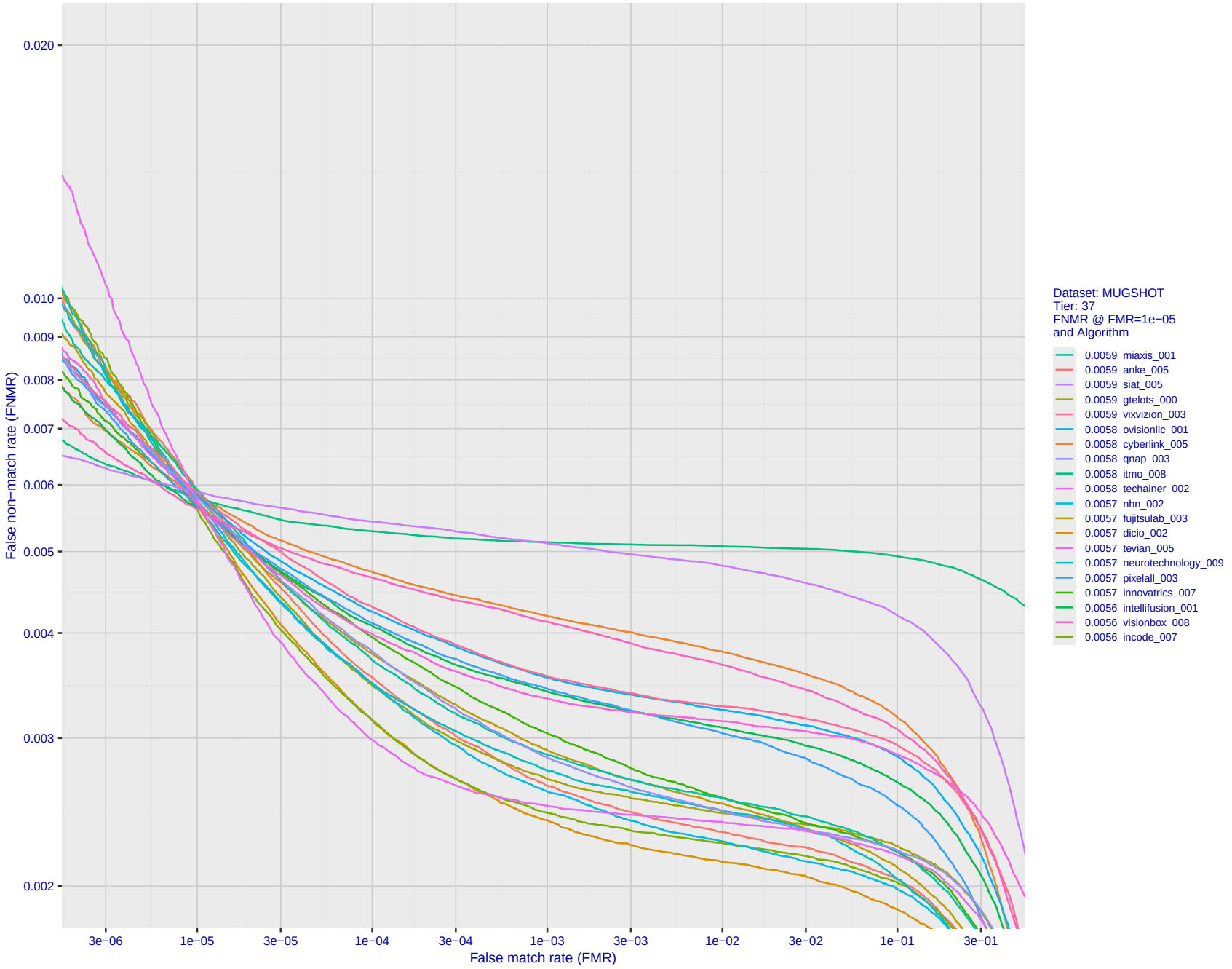


Figure 262: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

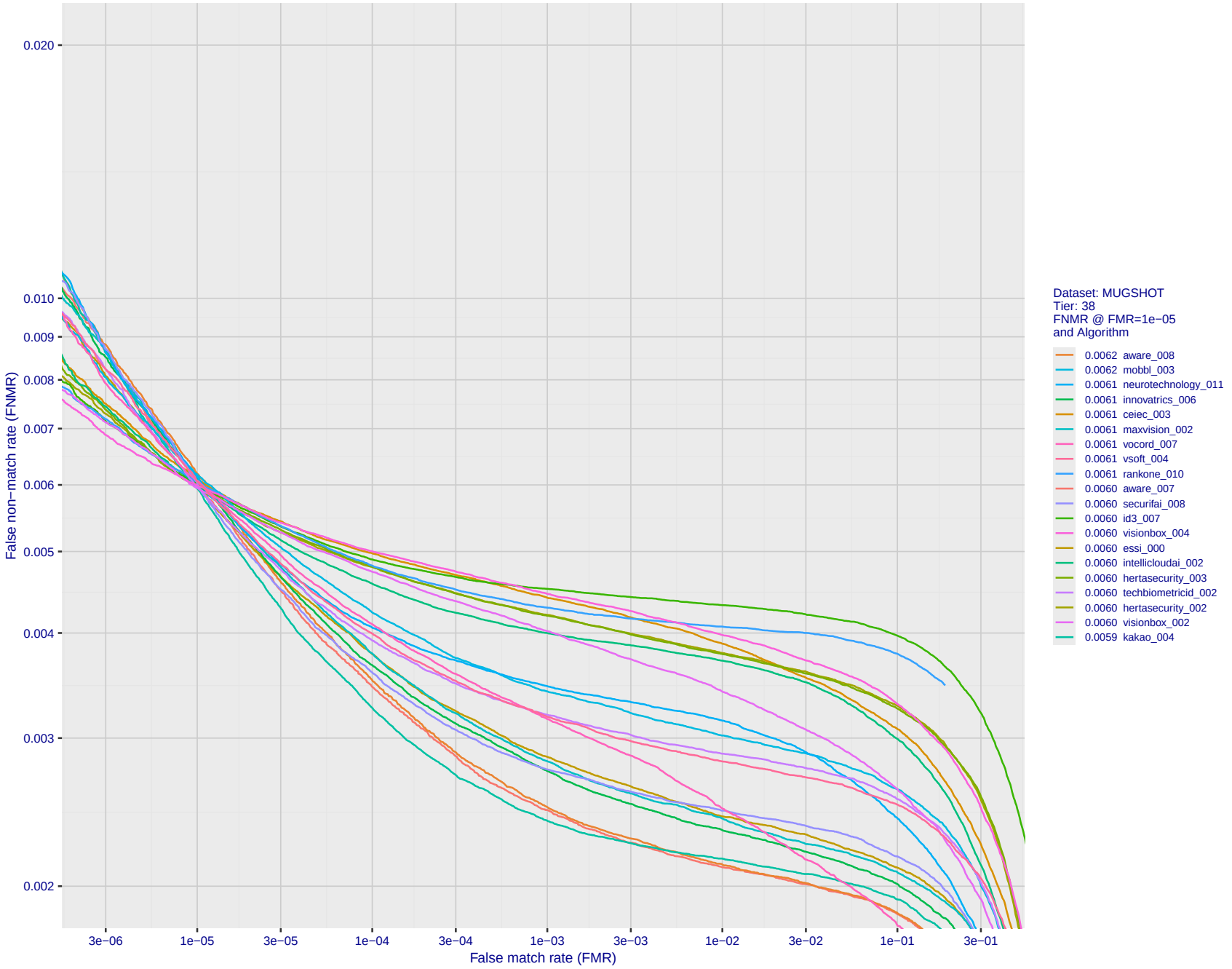


Figure 263: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

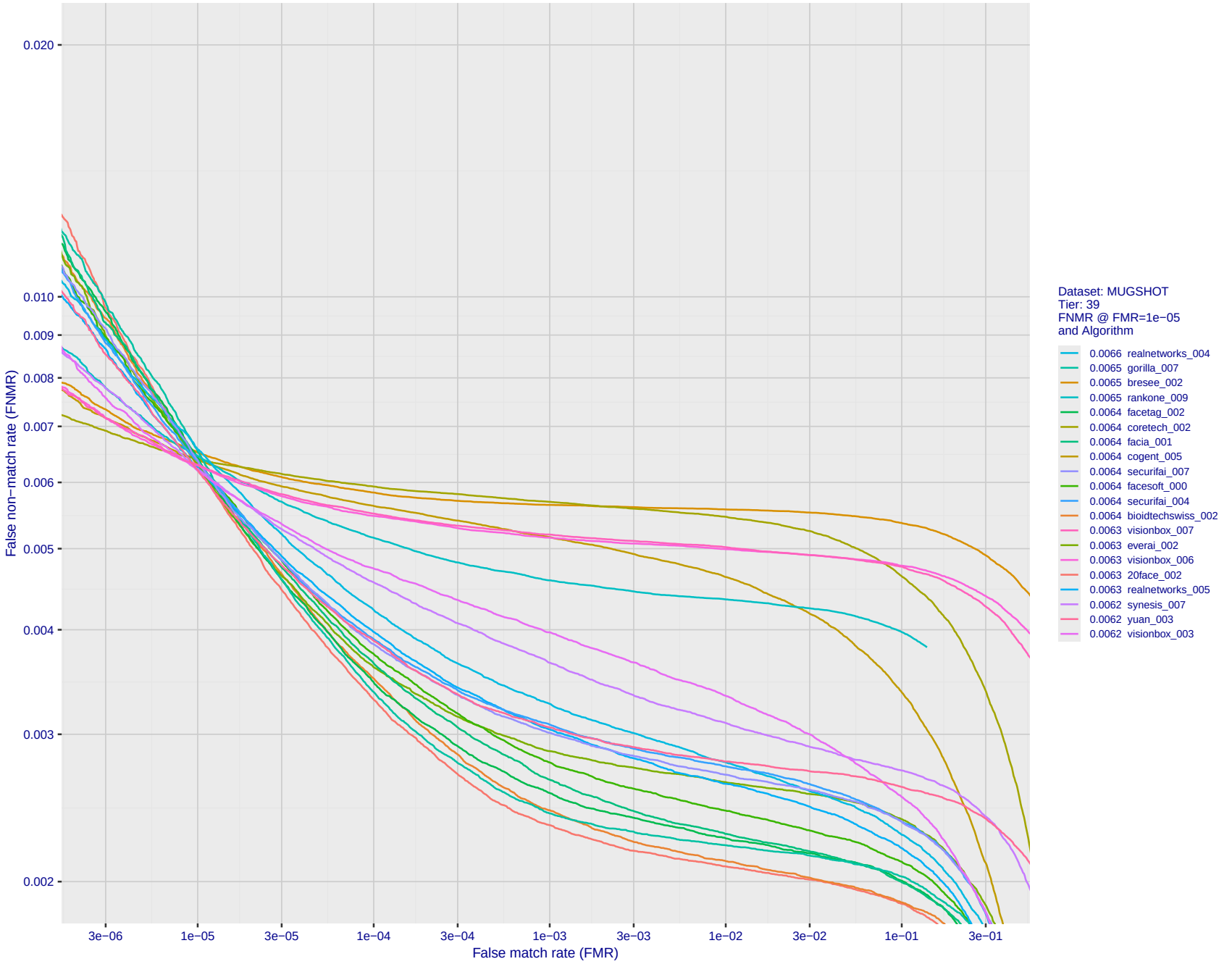


Figure 264: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

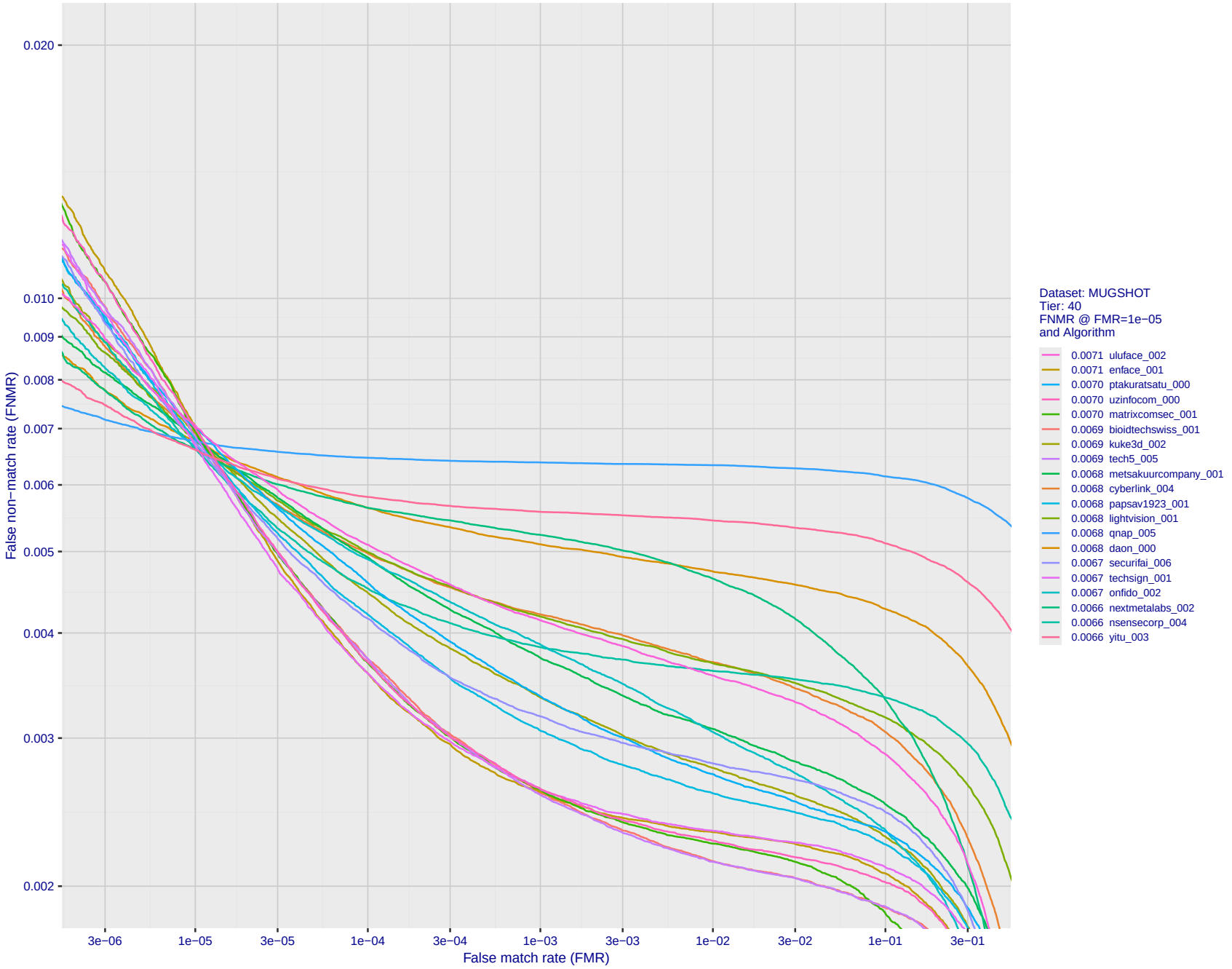


Figure 265: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

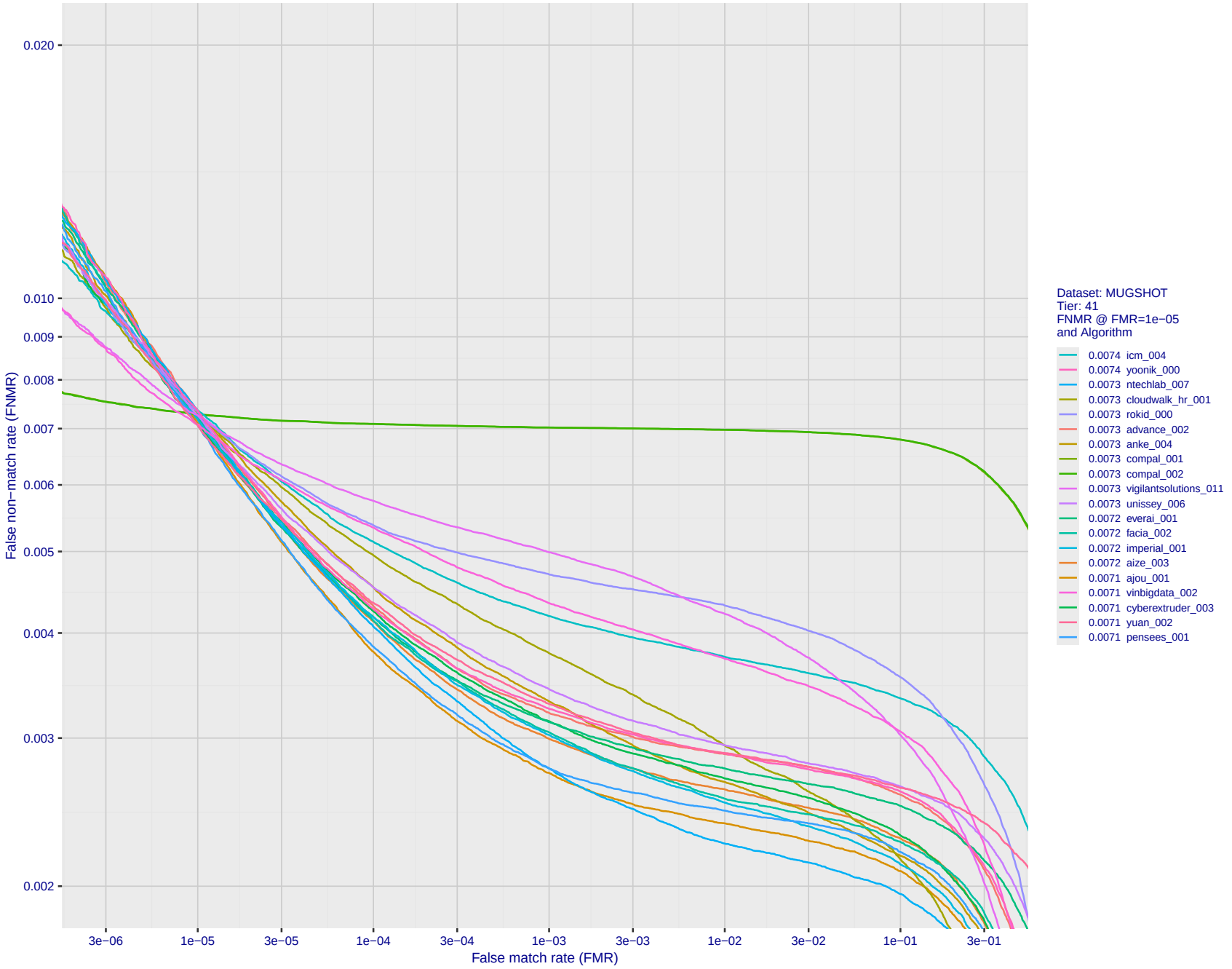


Figure 266: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

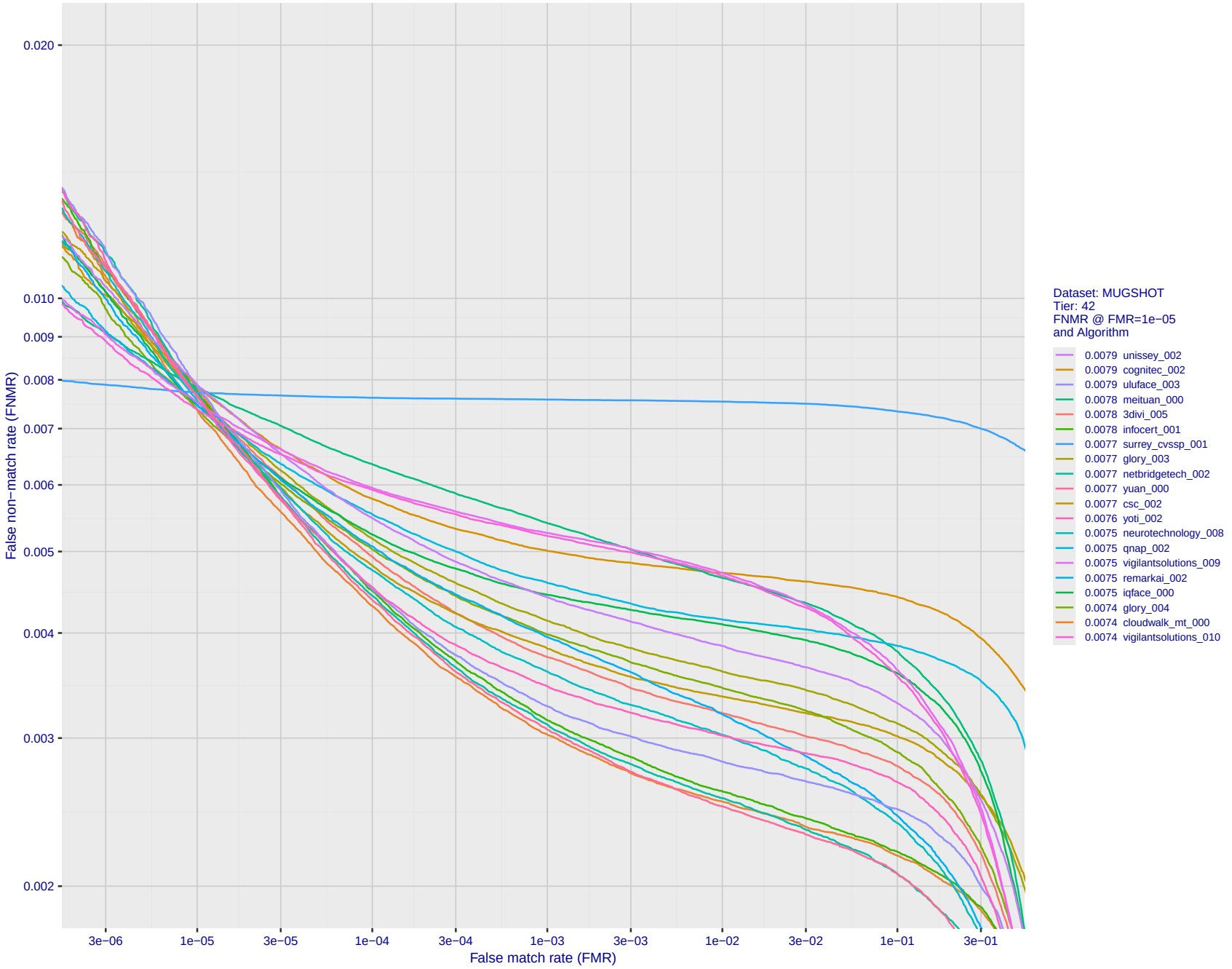


Figure 267: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

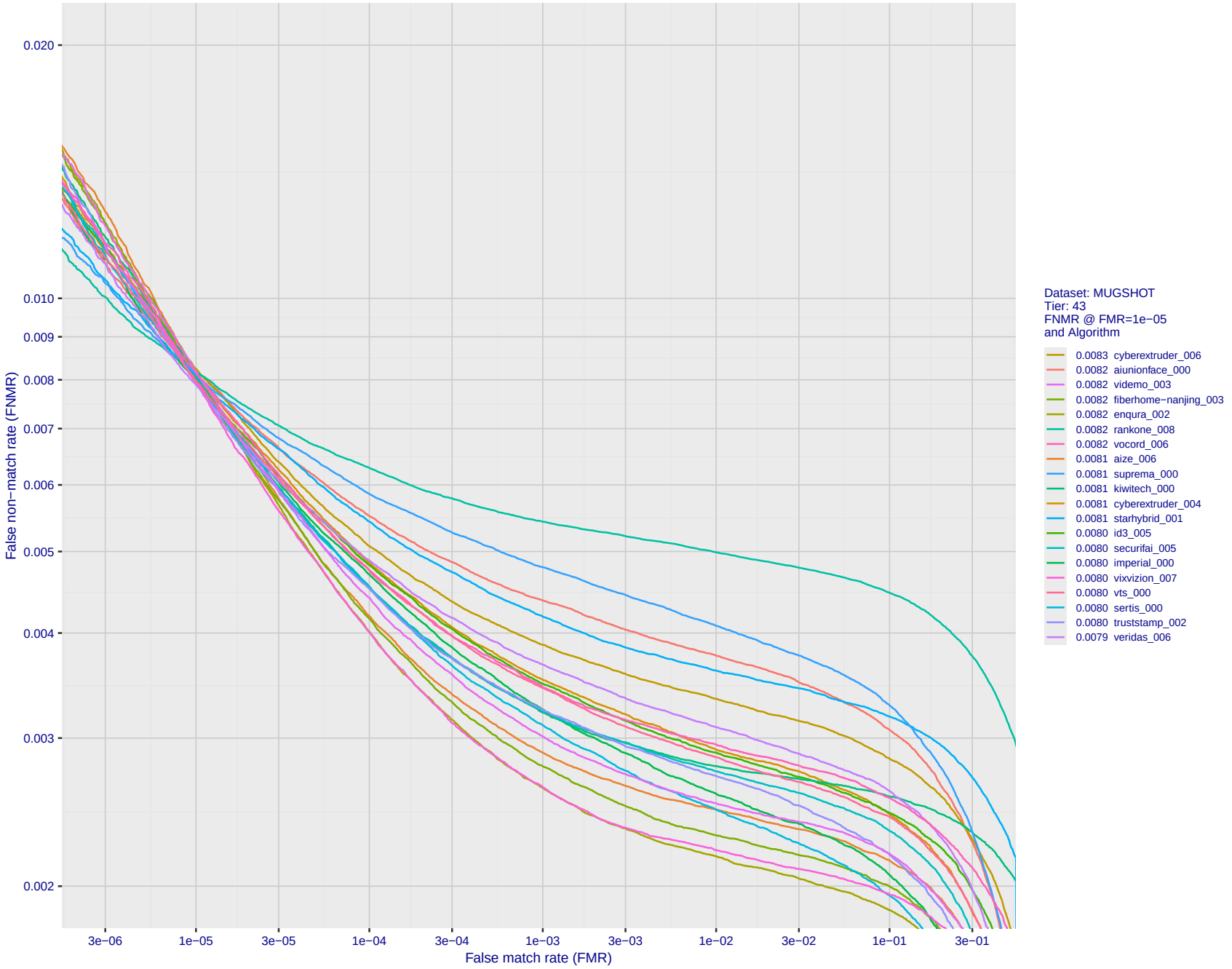


*Figure 268: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR*.*

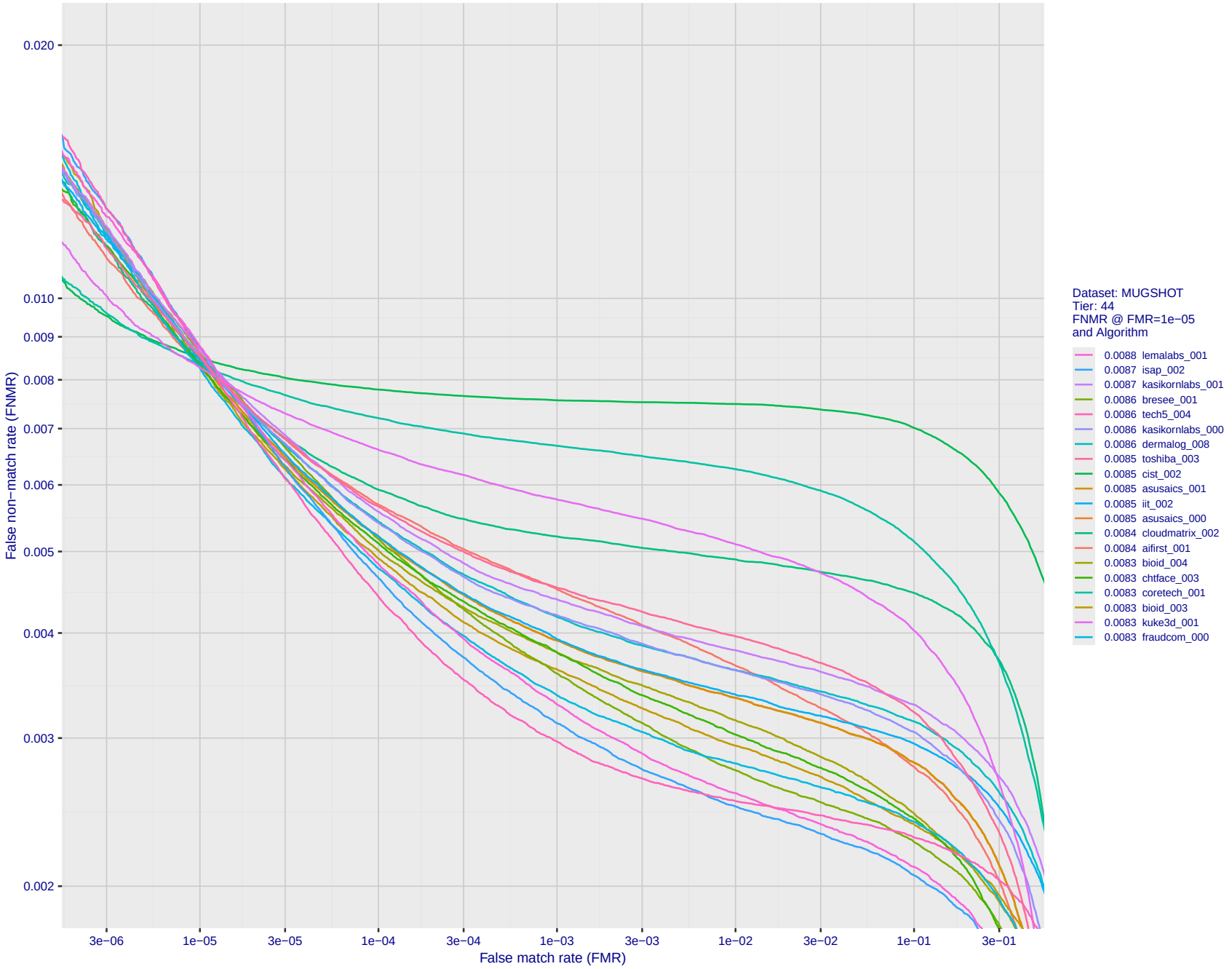


Figure 269: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

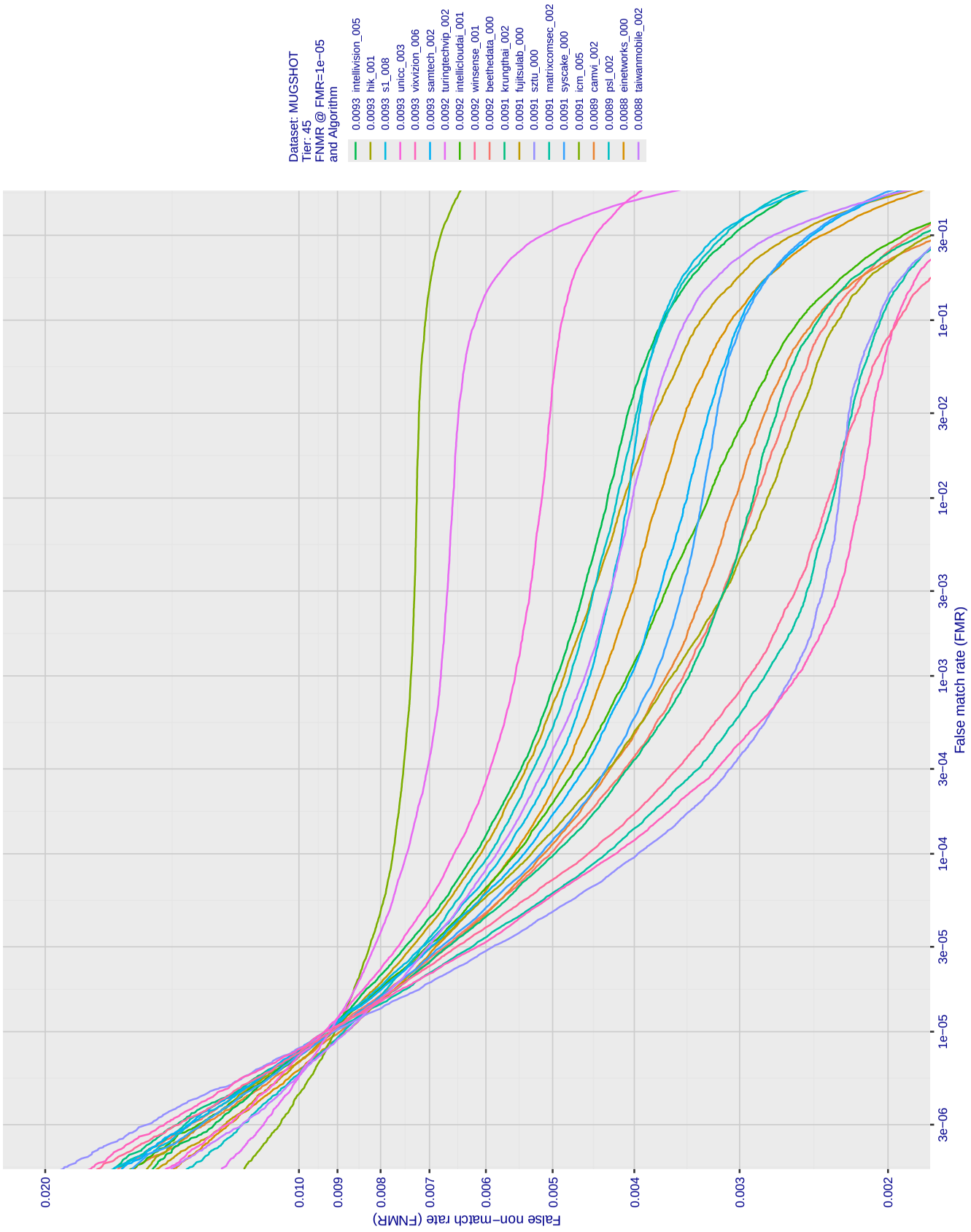


Figure 270: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

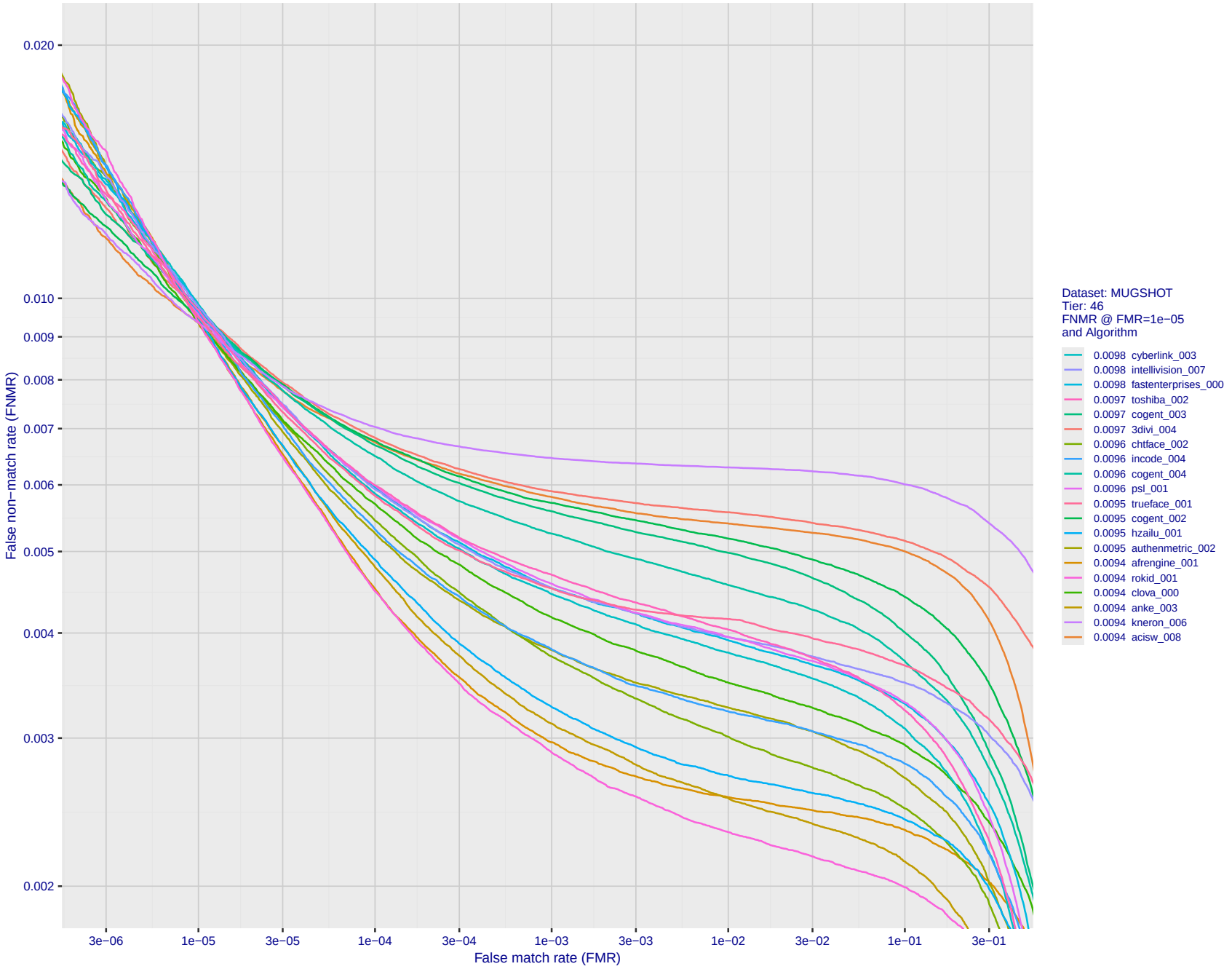


Figure 271: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

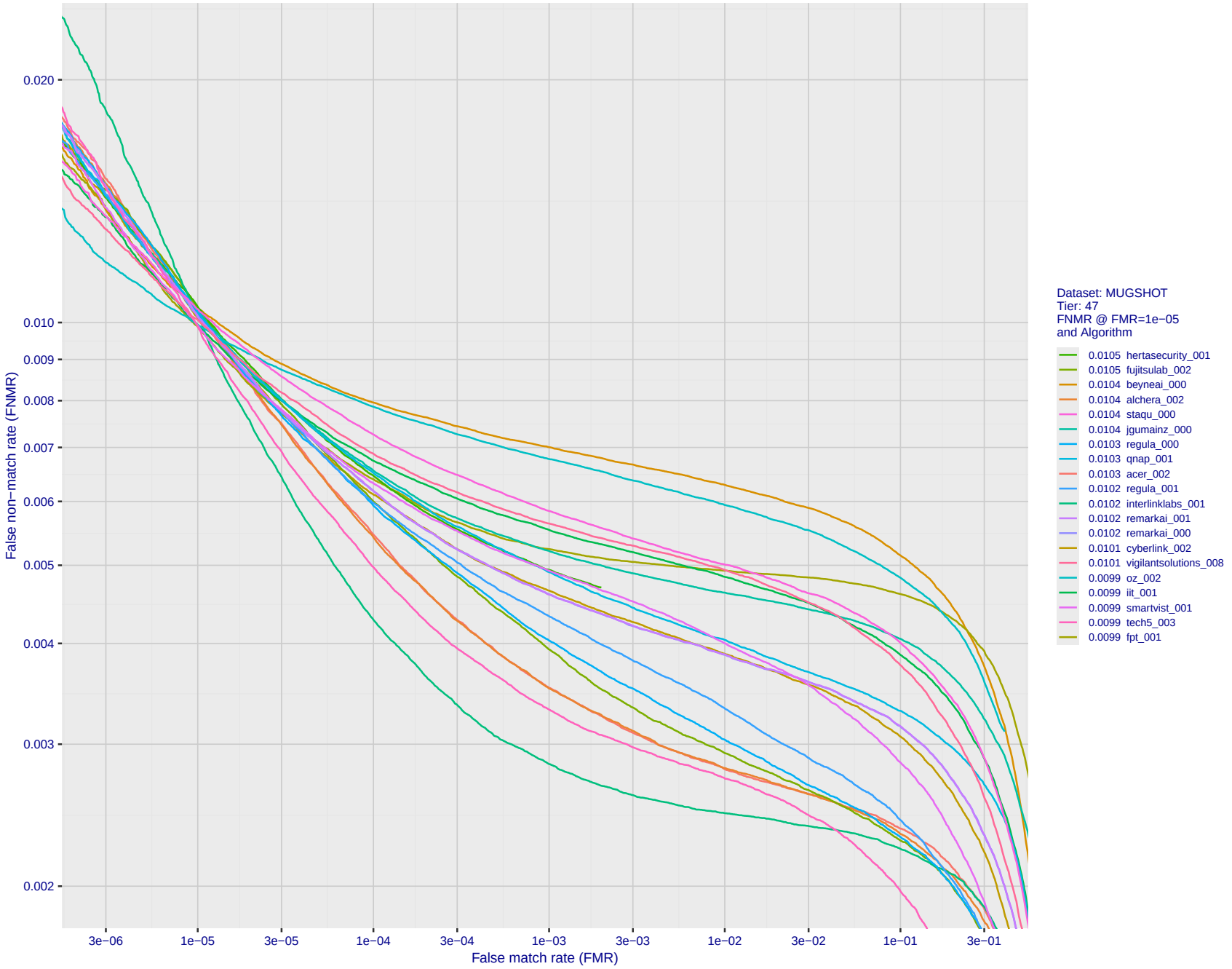


Figure 272: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

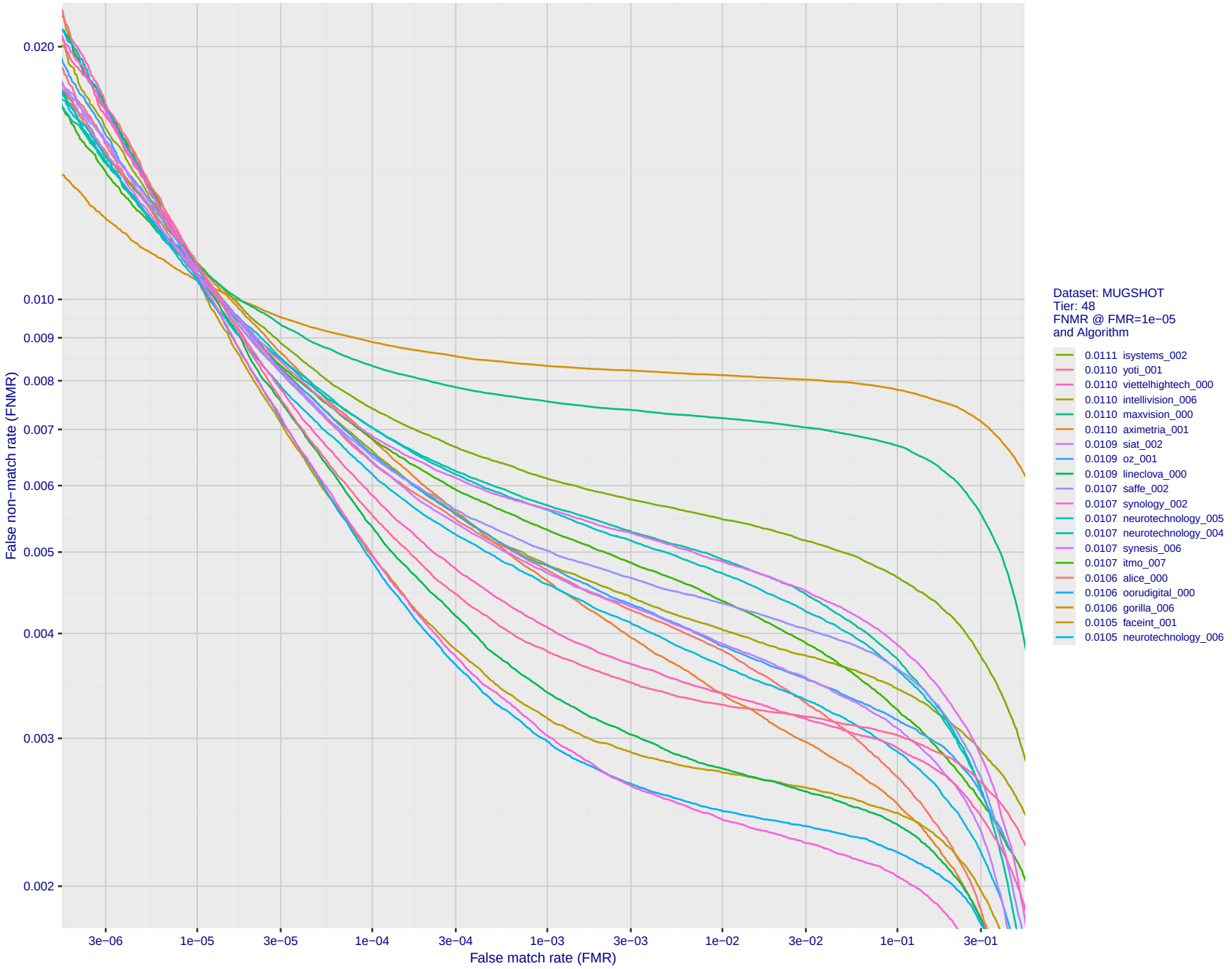


Figure 273: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

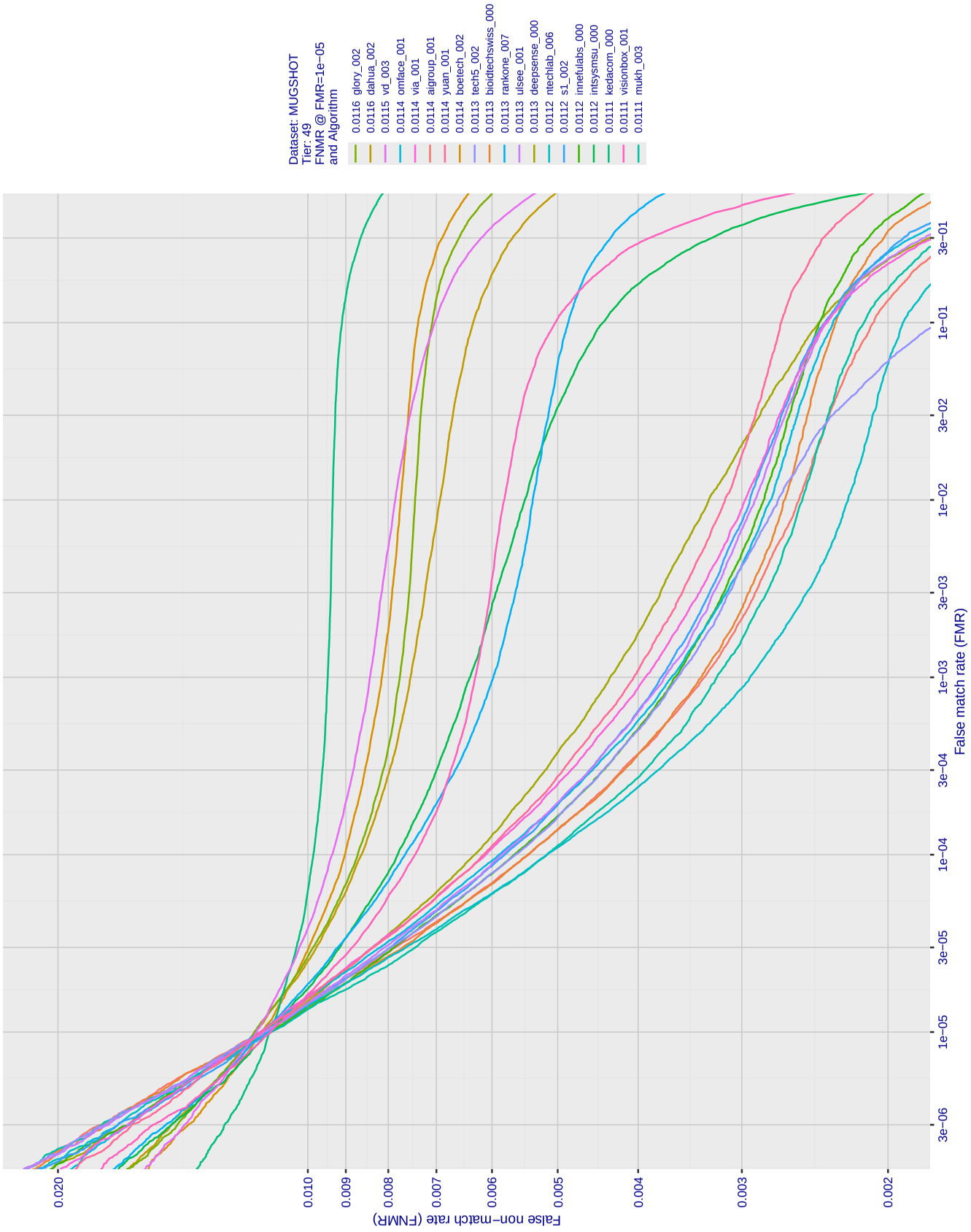


Figure 274: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

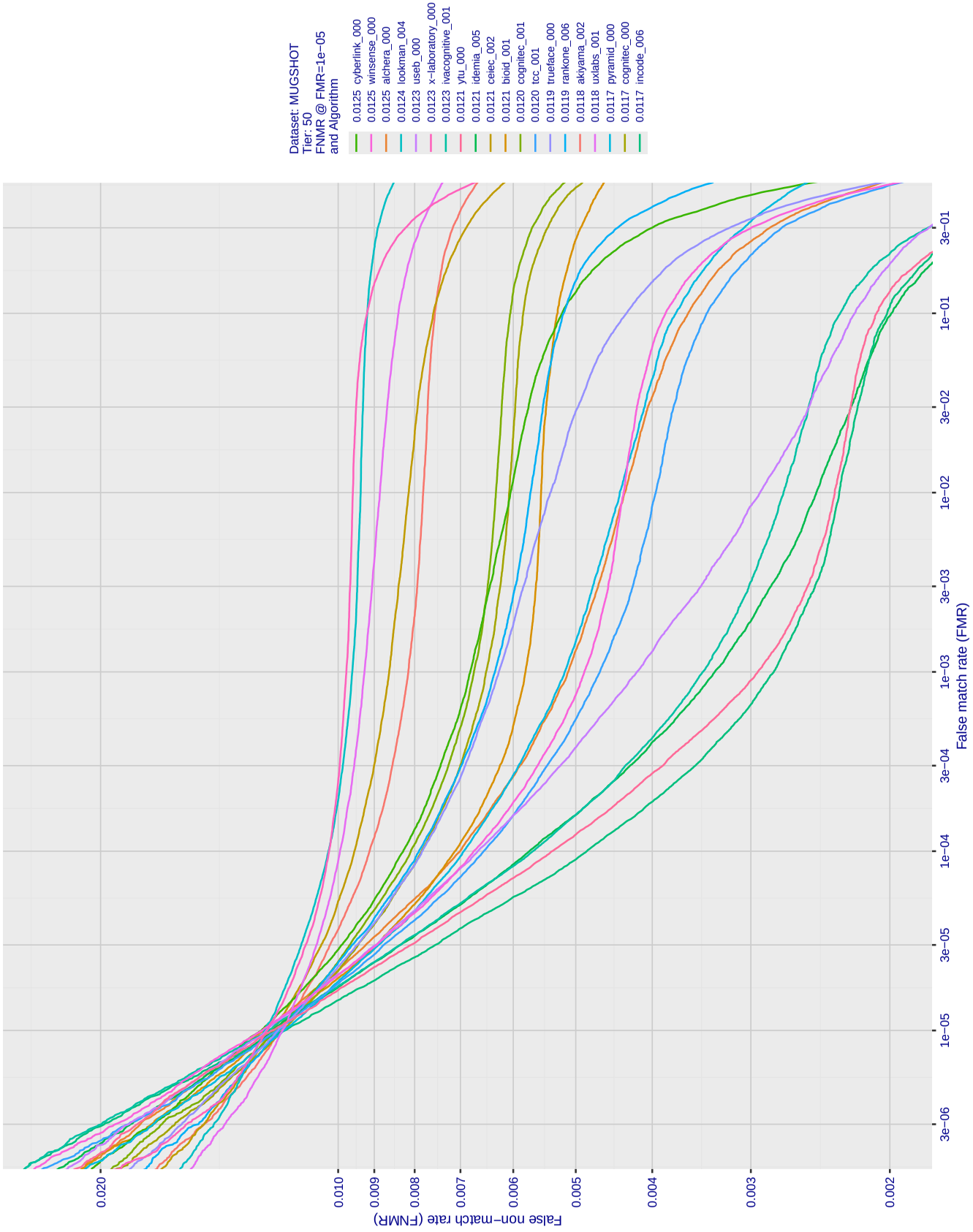


Figure 275: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

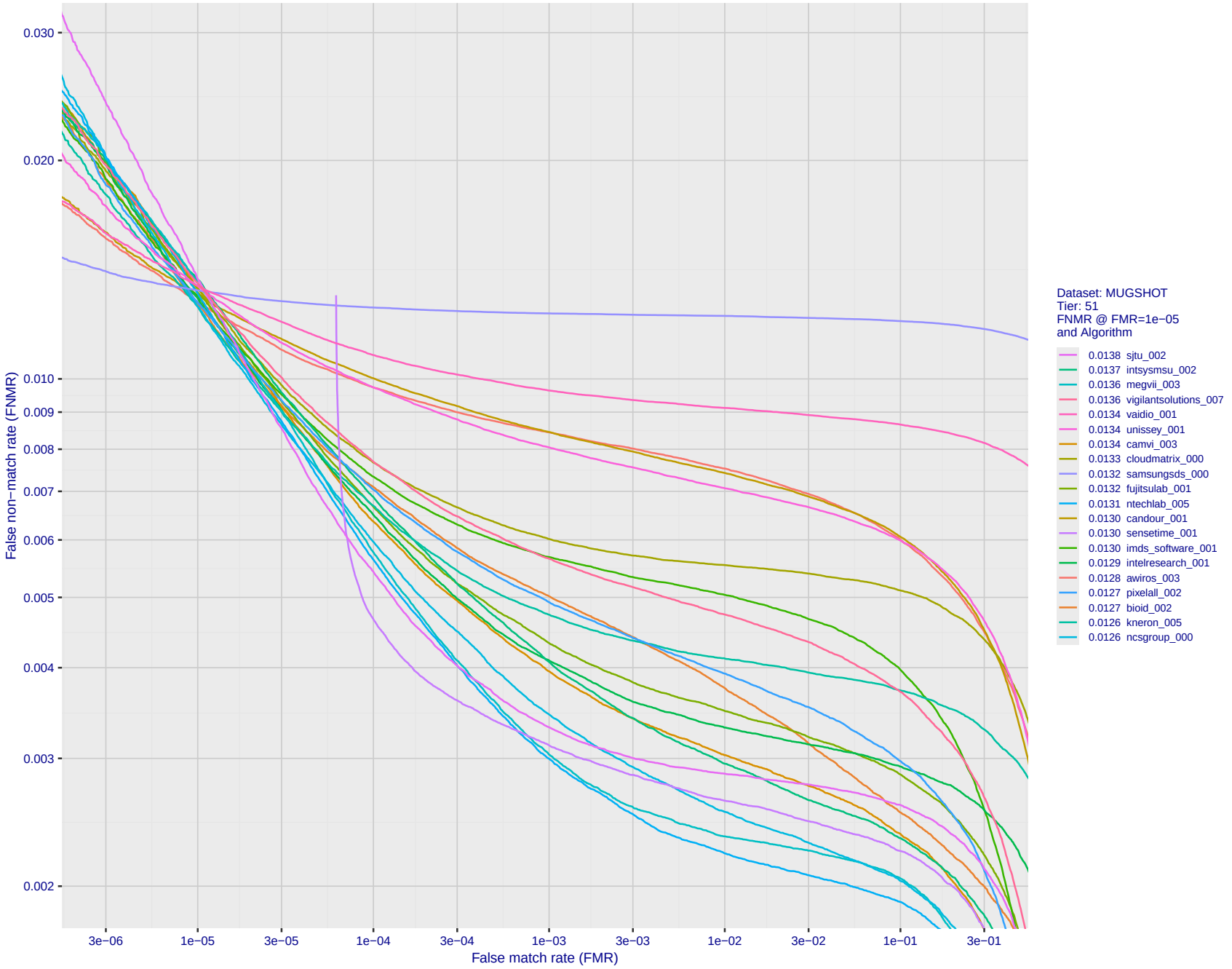


Figure 276: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

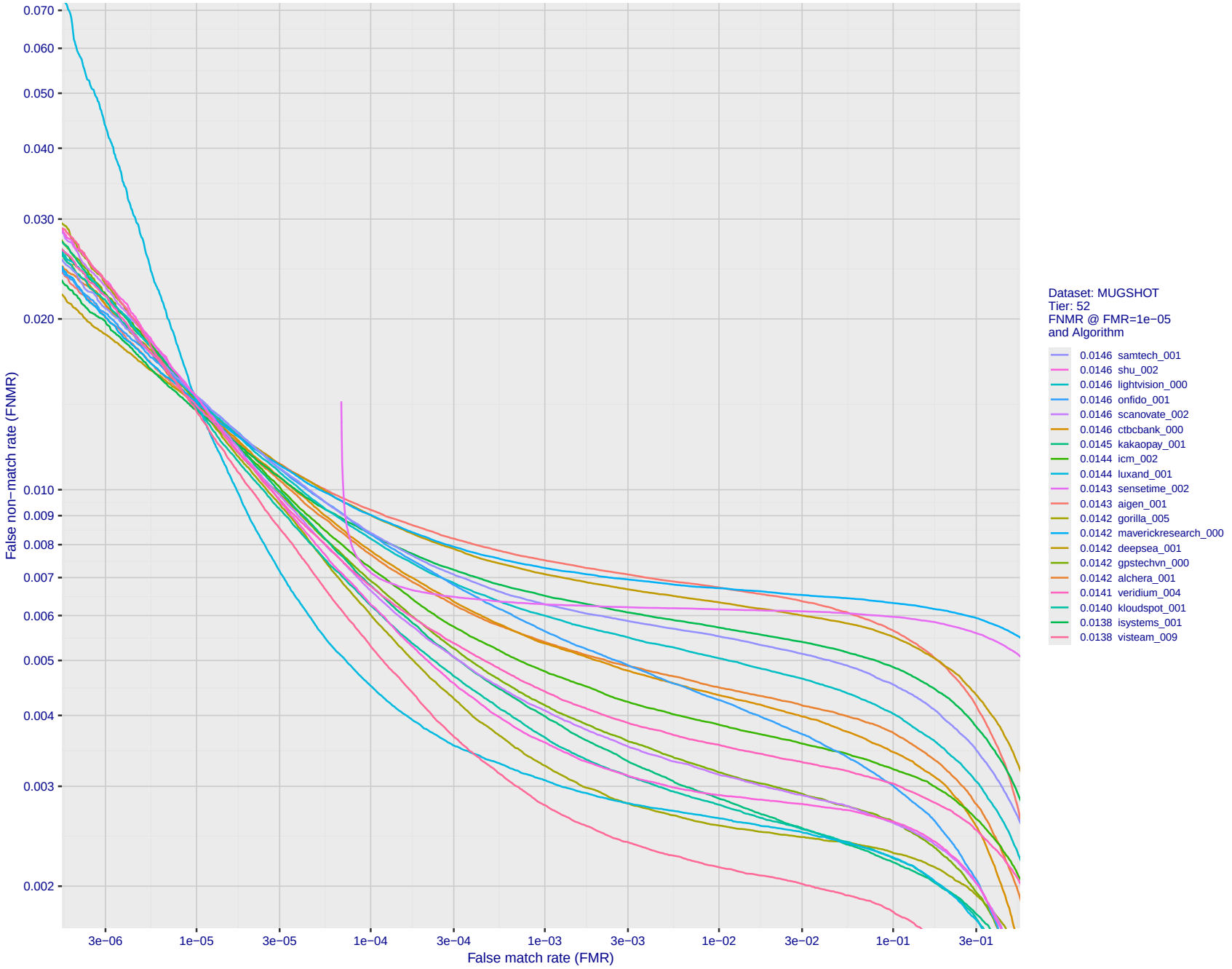


*Figure 277: For the mugshot images, detection error tradeoff (*DET*) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of* FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

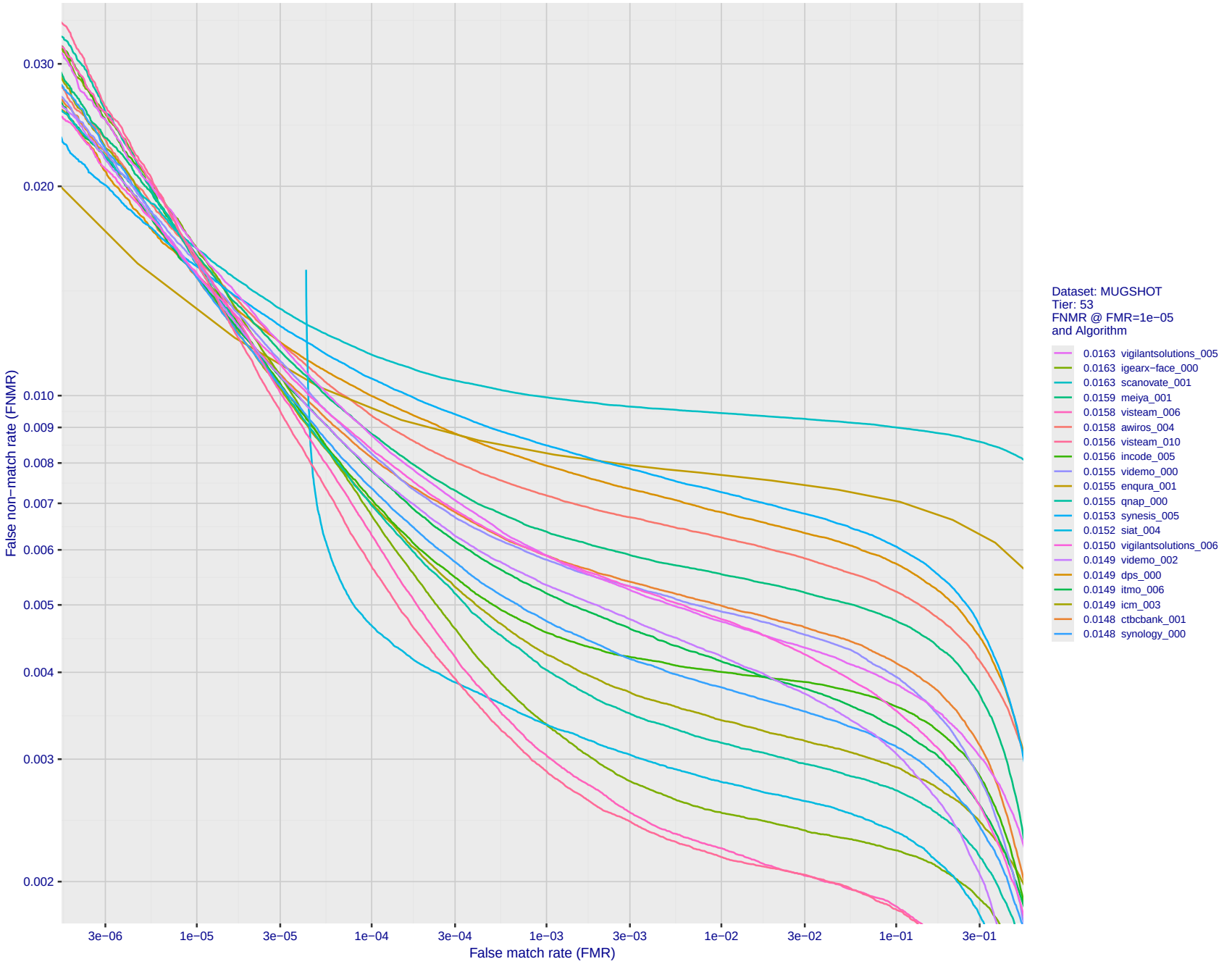


Figure 278: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

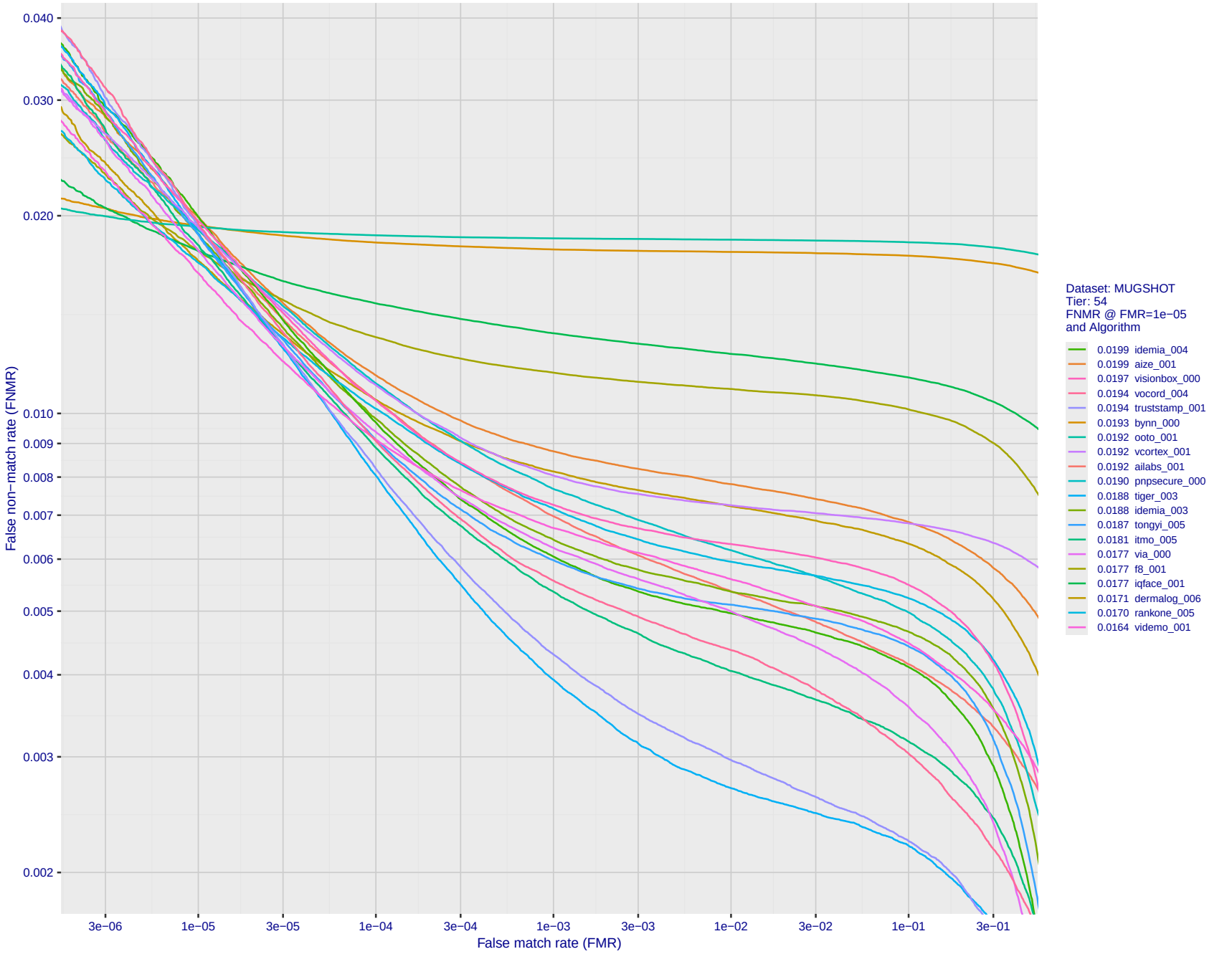


Figure 279: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

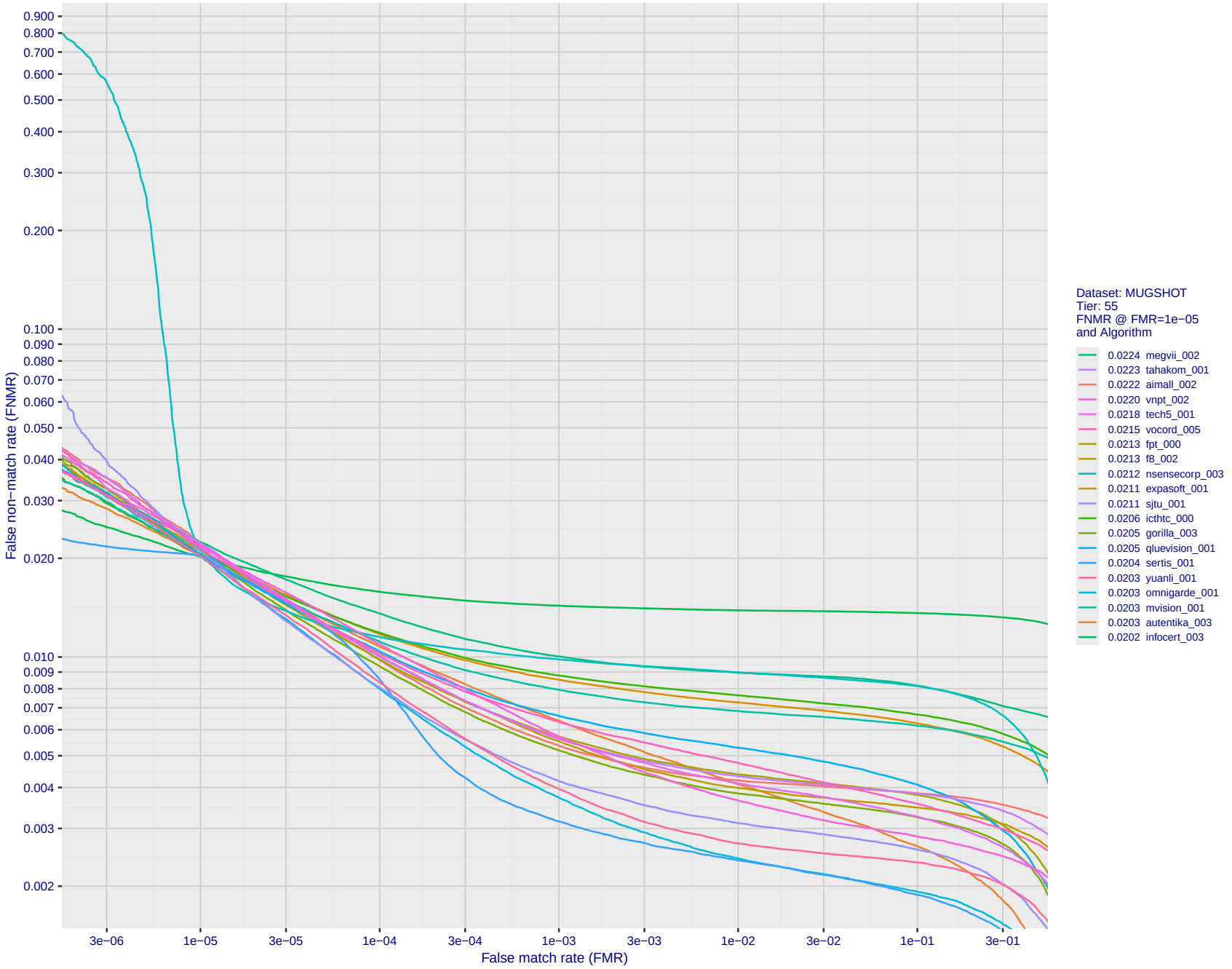


Figure 280: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

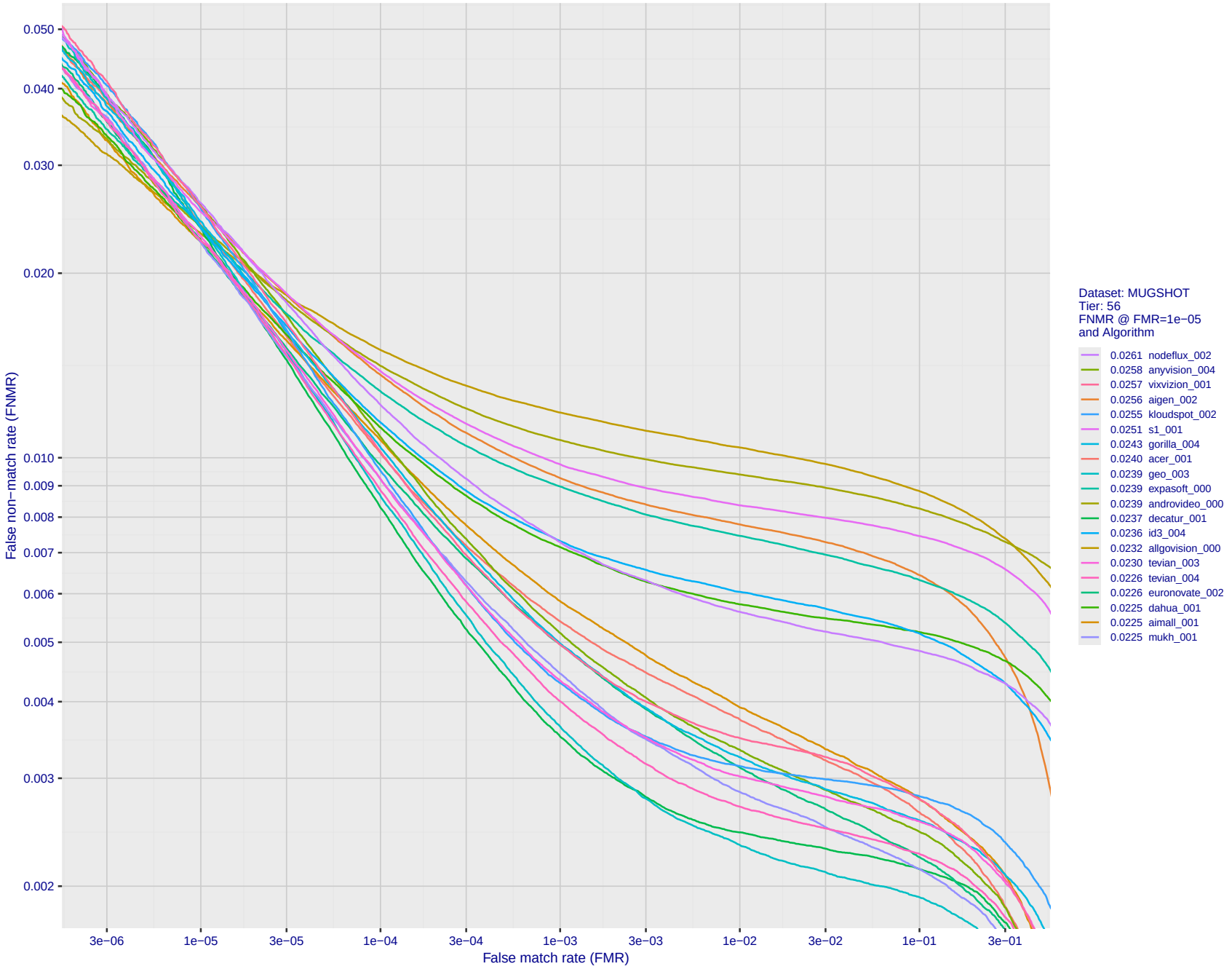


Figure 281: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/11/06 13:03:11

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

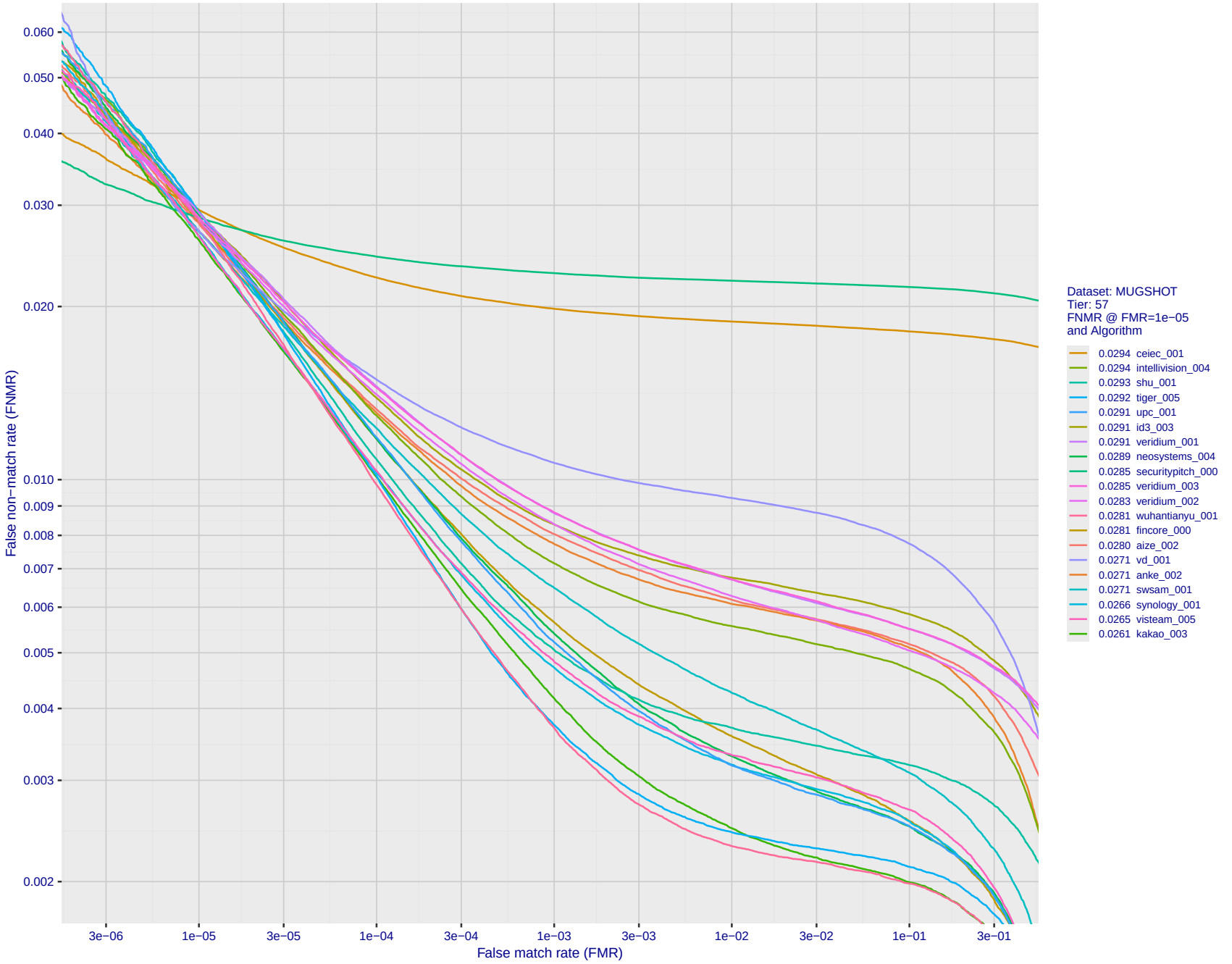


Figure 282: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

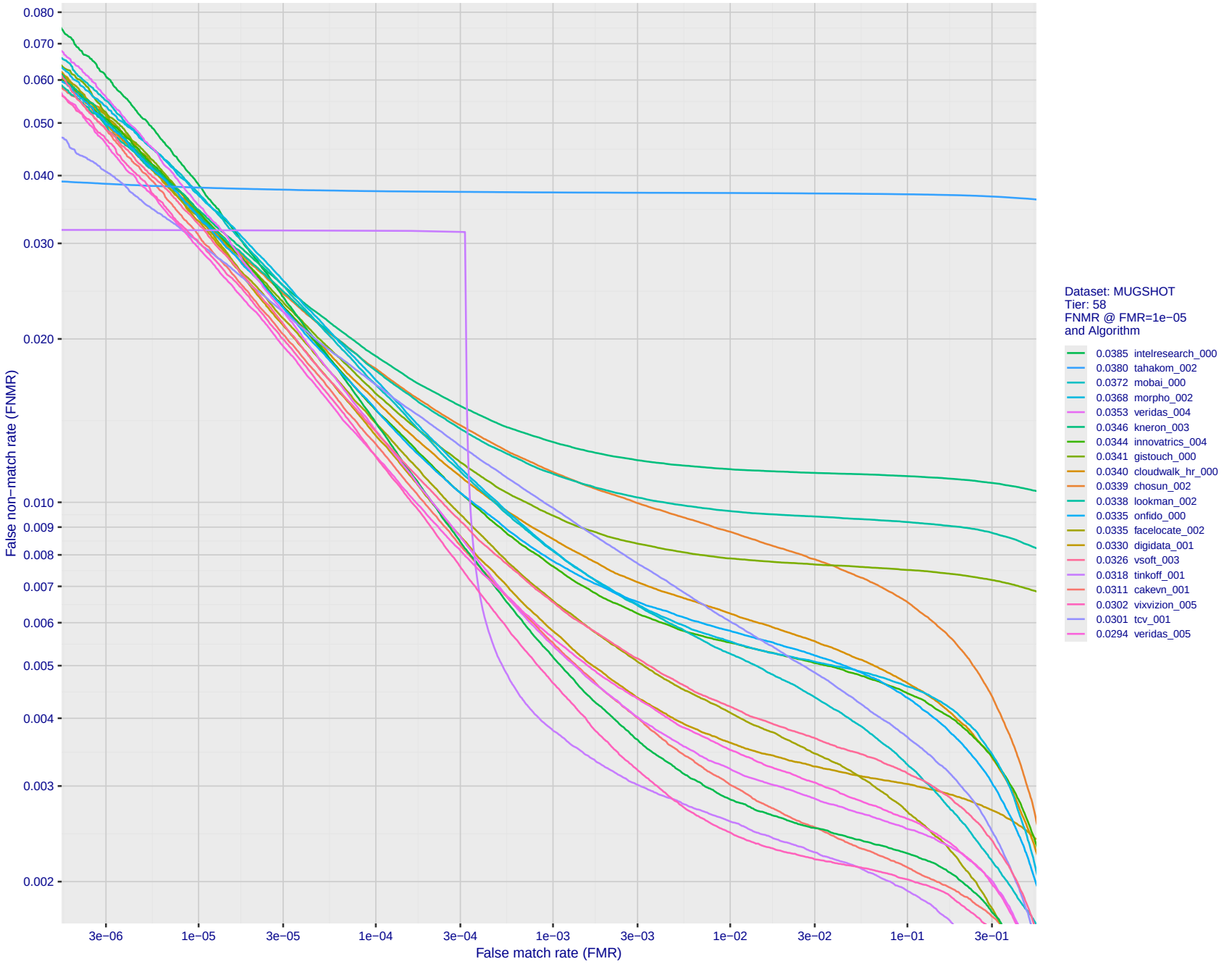


Figure 283: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

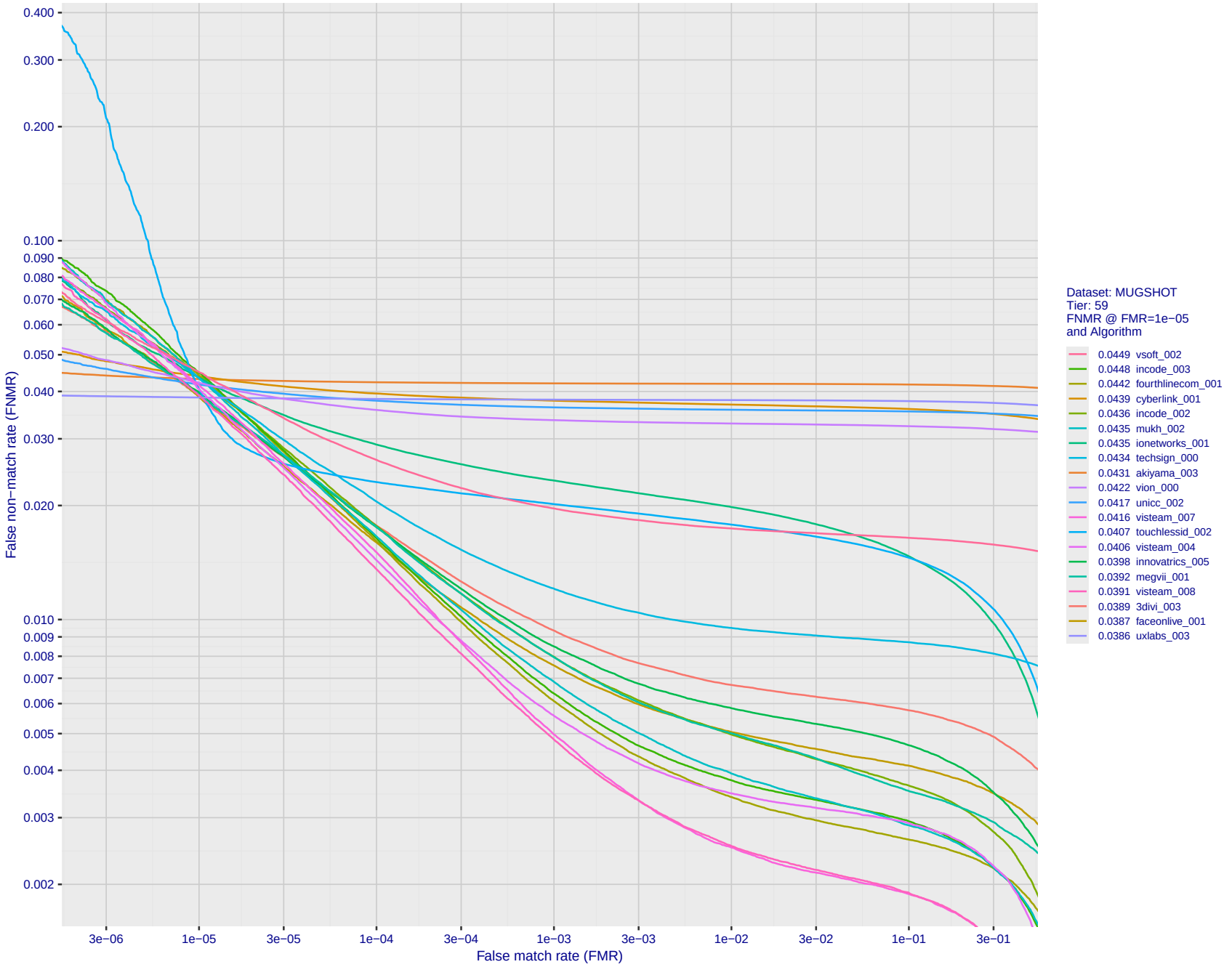


Figure 284: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

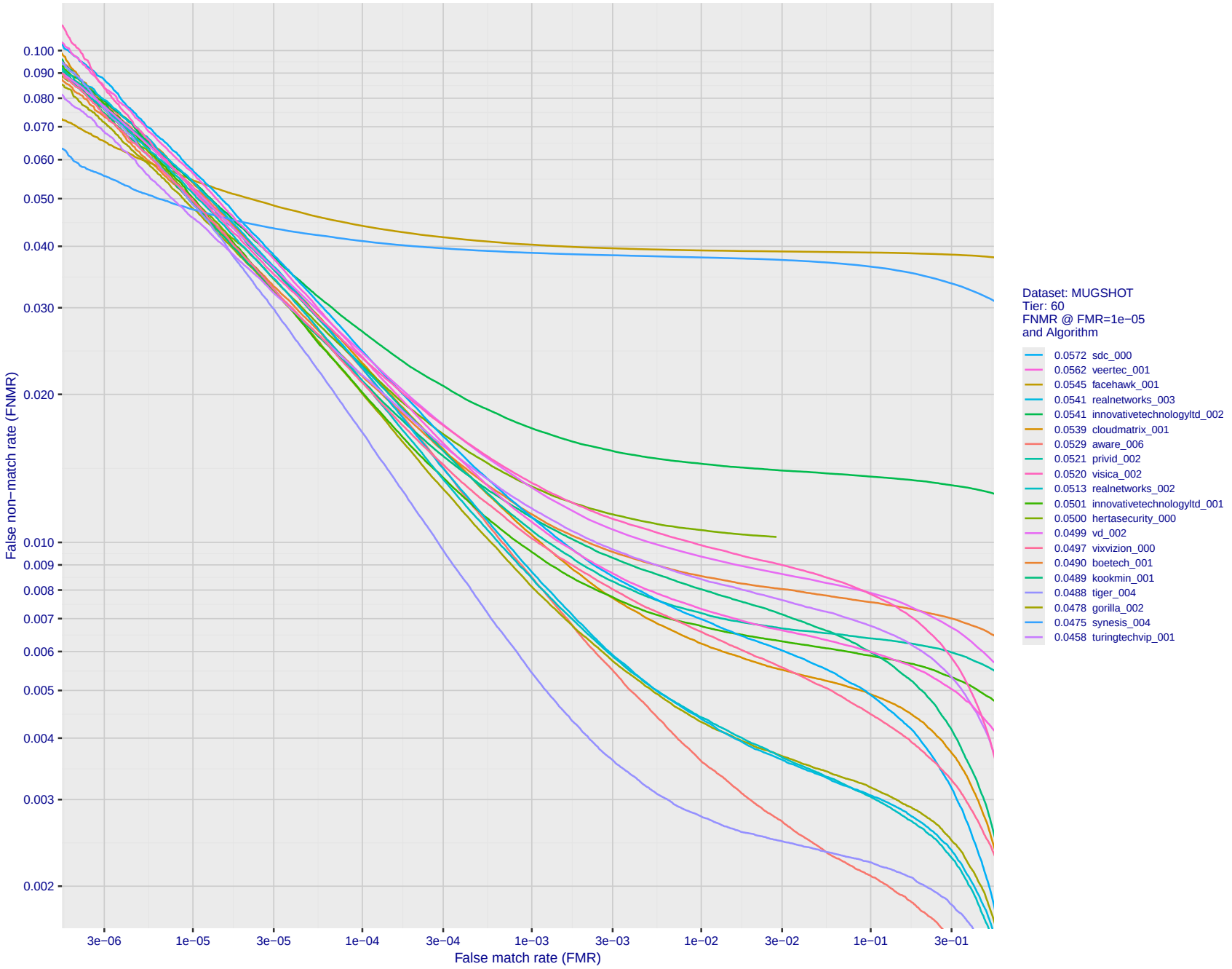


Figure 285: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

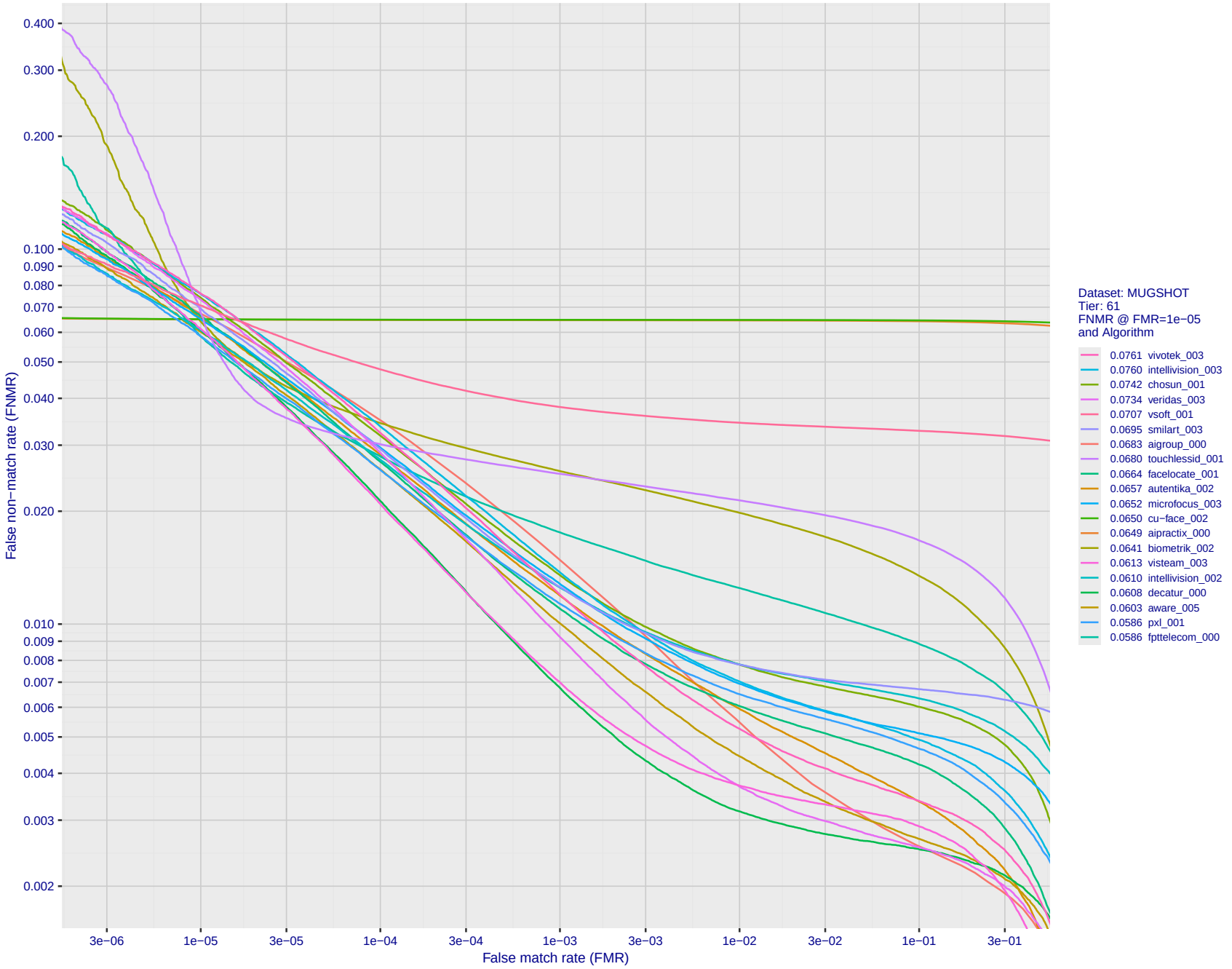


Figure 286: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

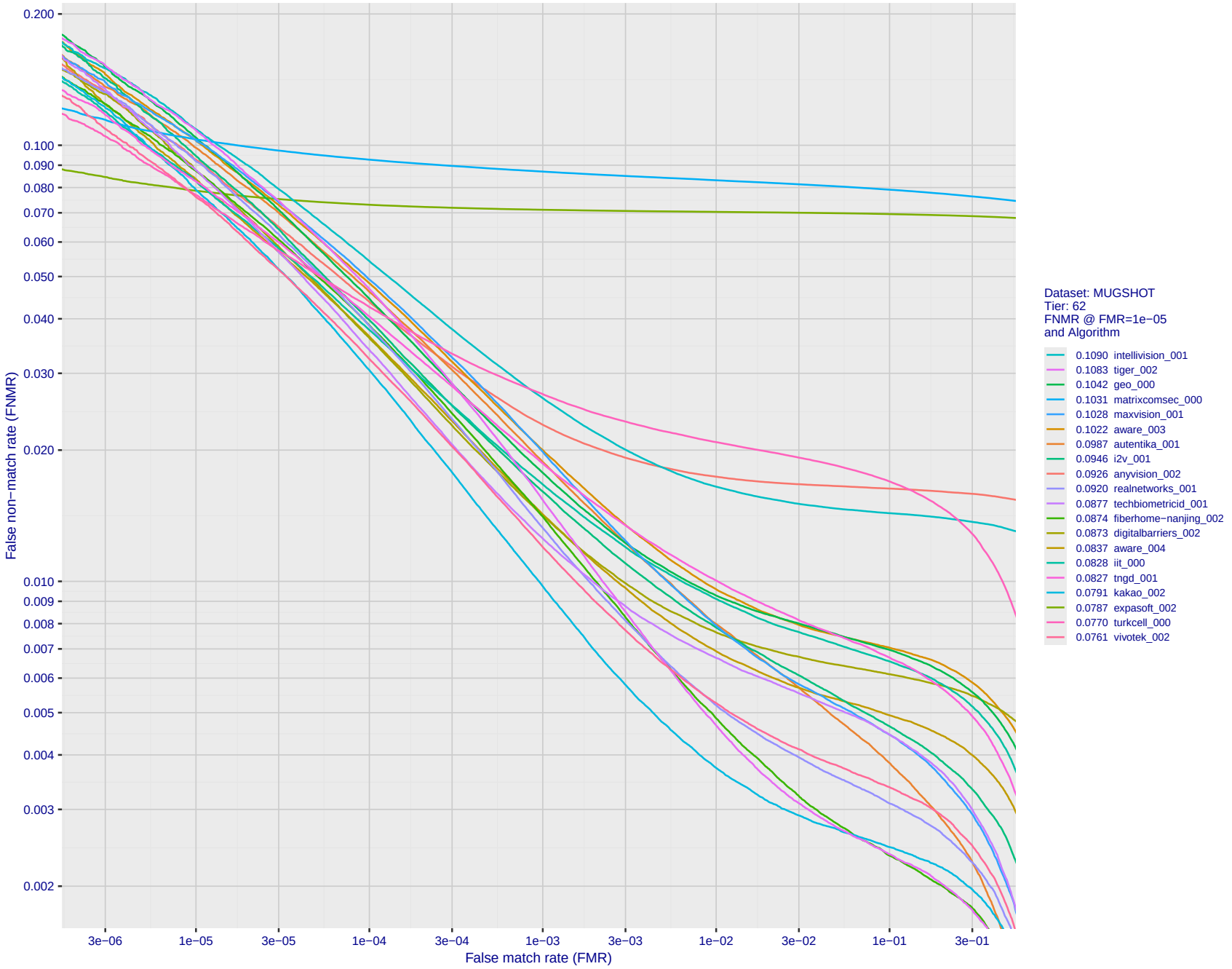


Figure 287: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

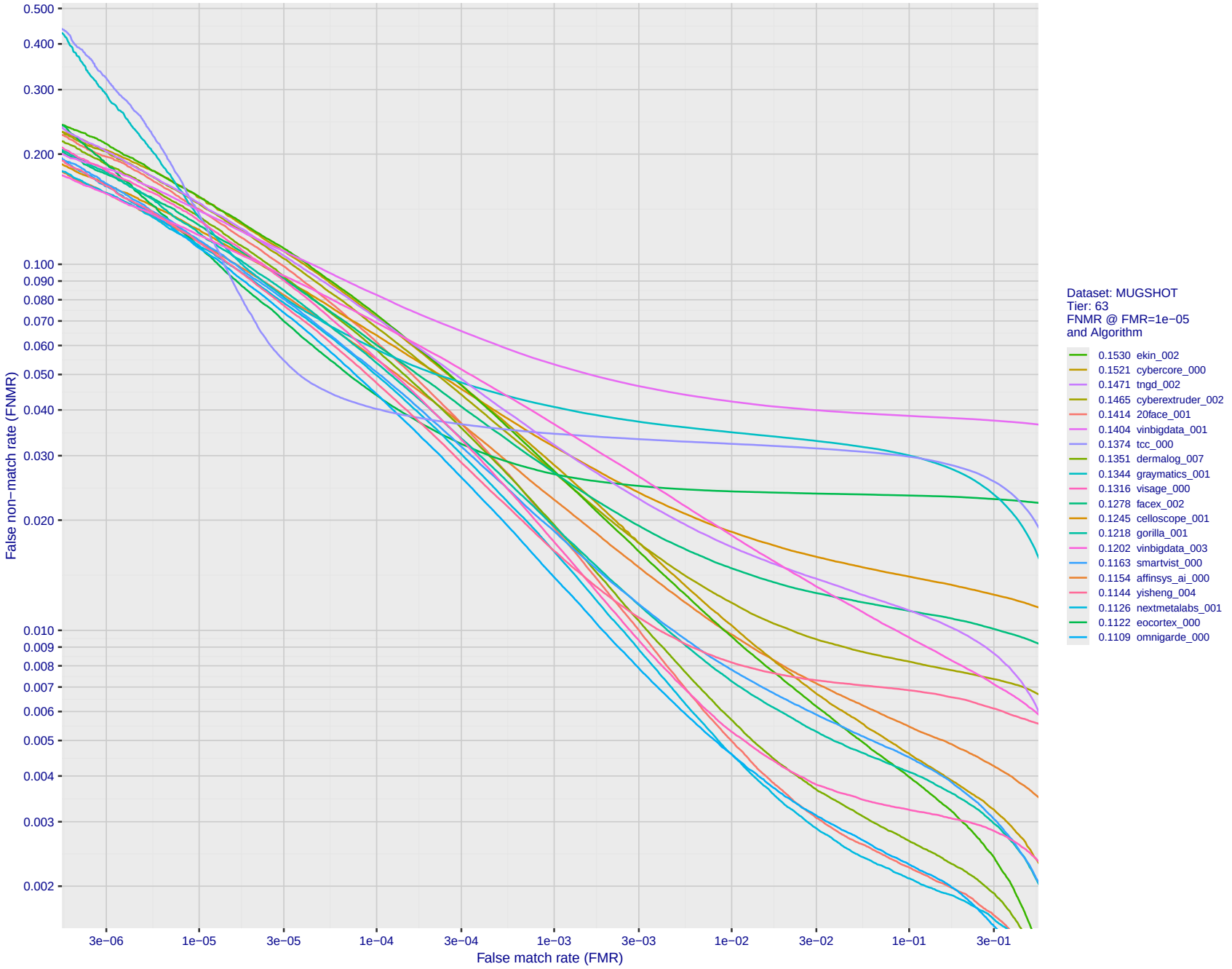


Figure 288: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

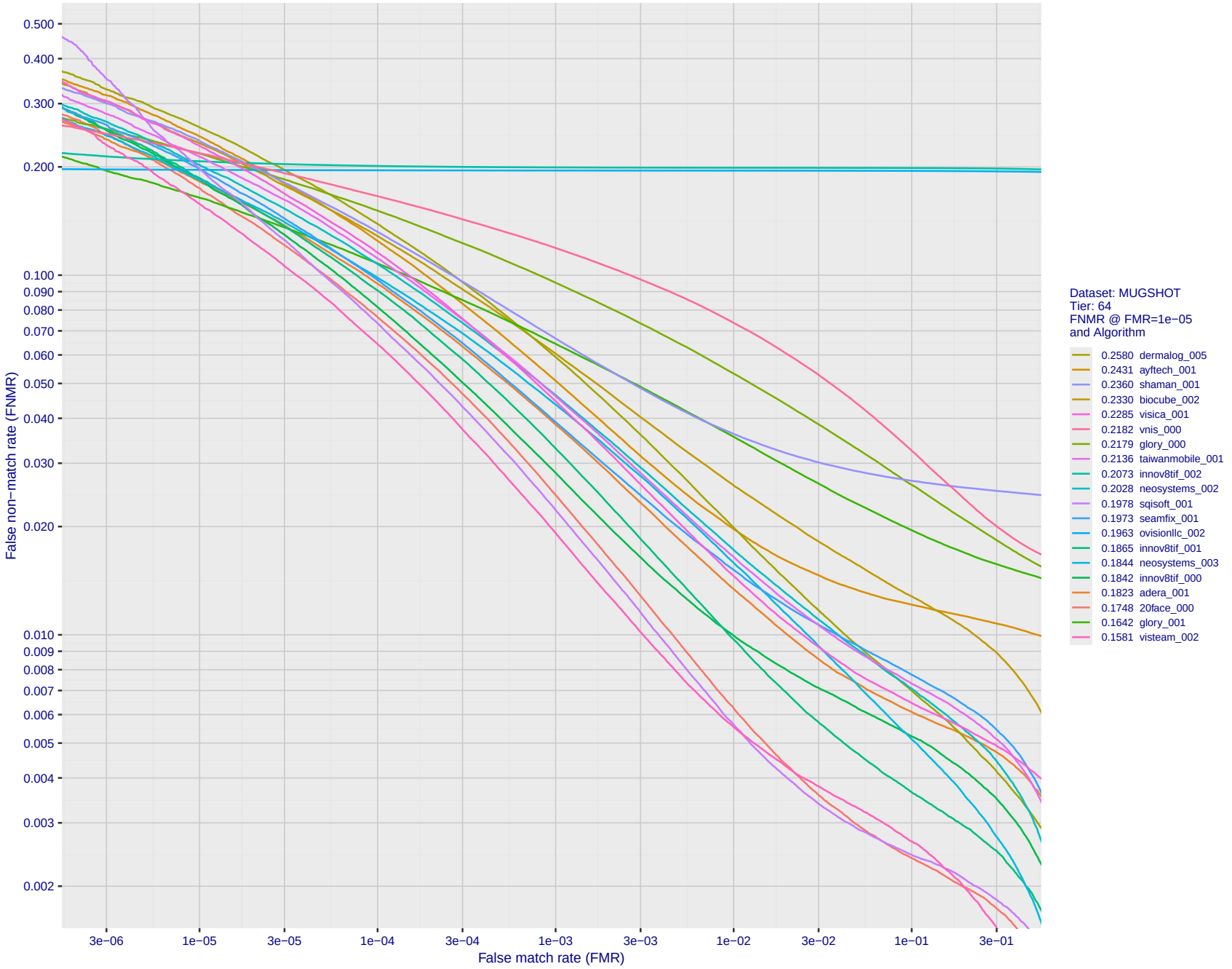


Figure 289: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

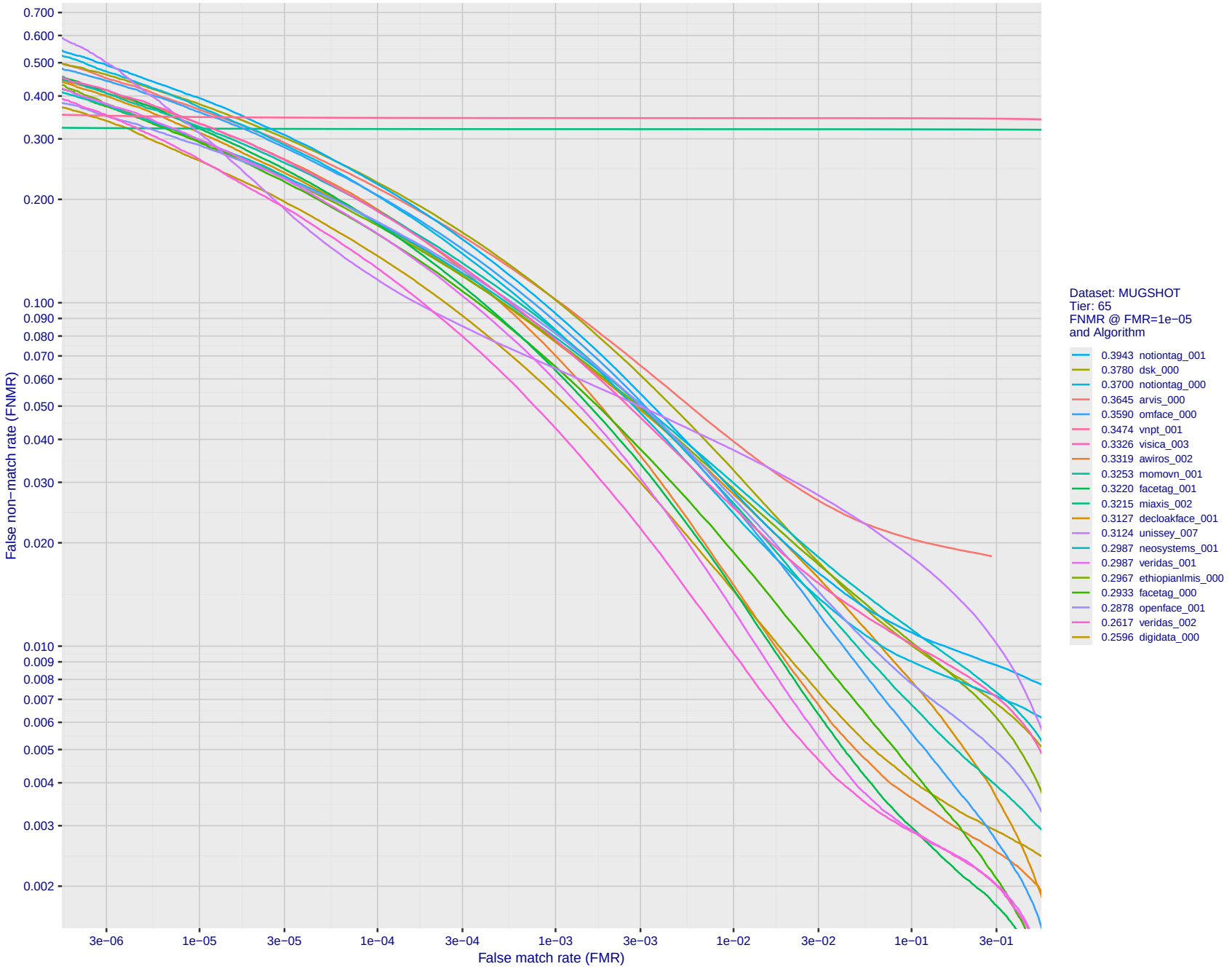


Figure 290: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

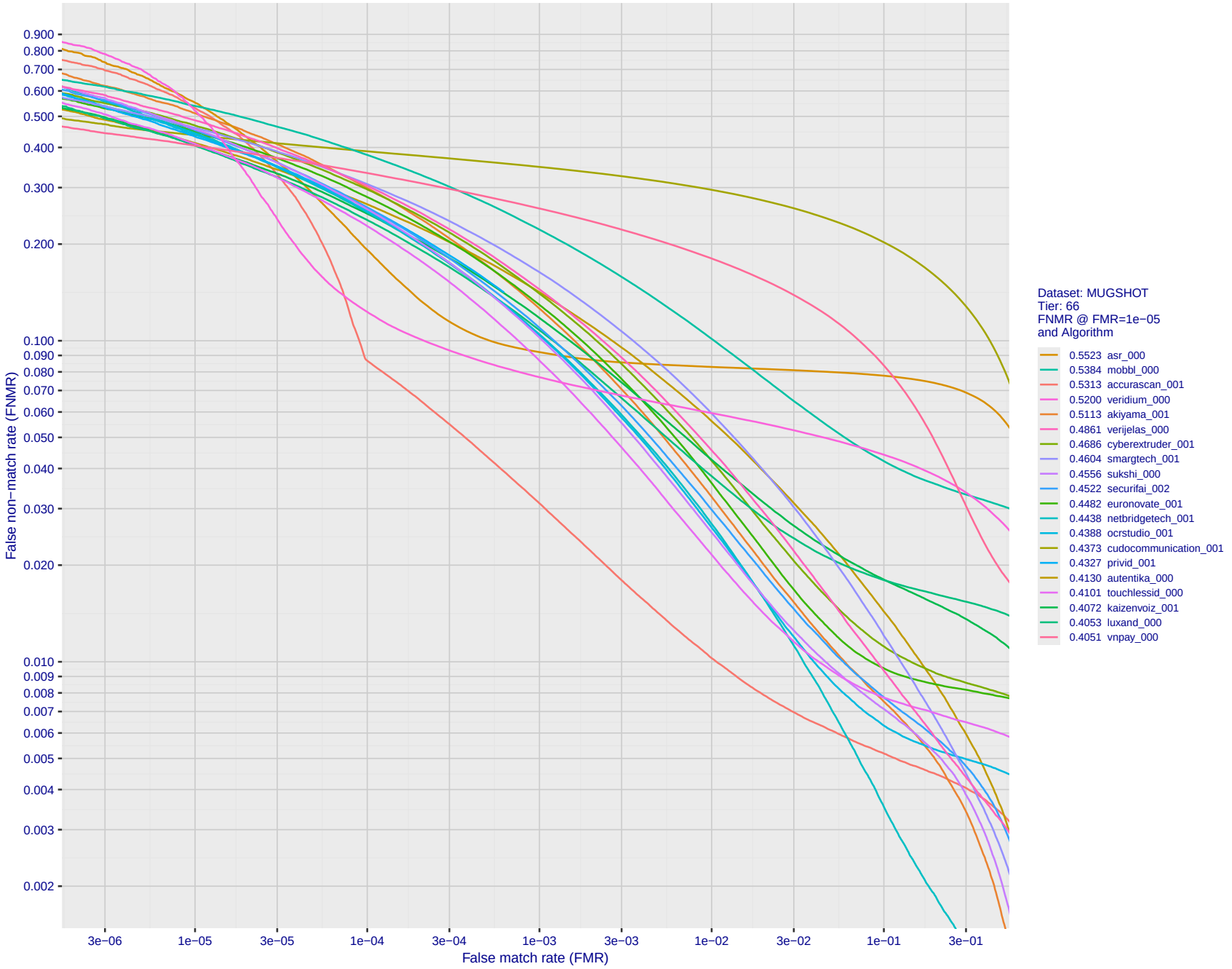


Figure 291: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

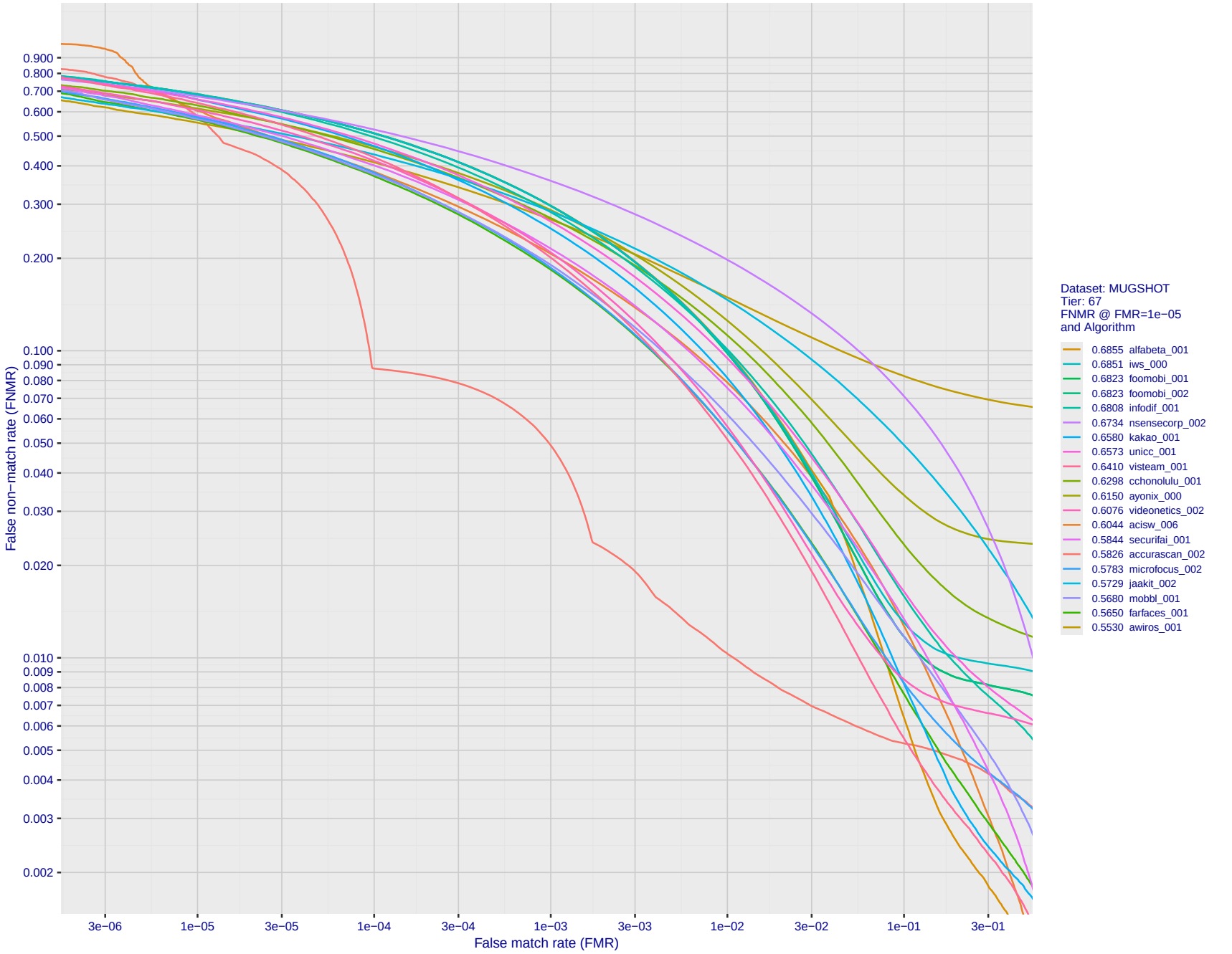


Figure 292: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

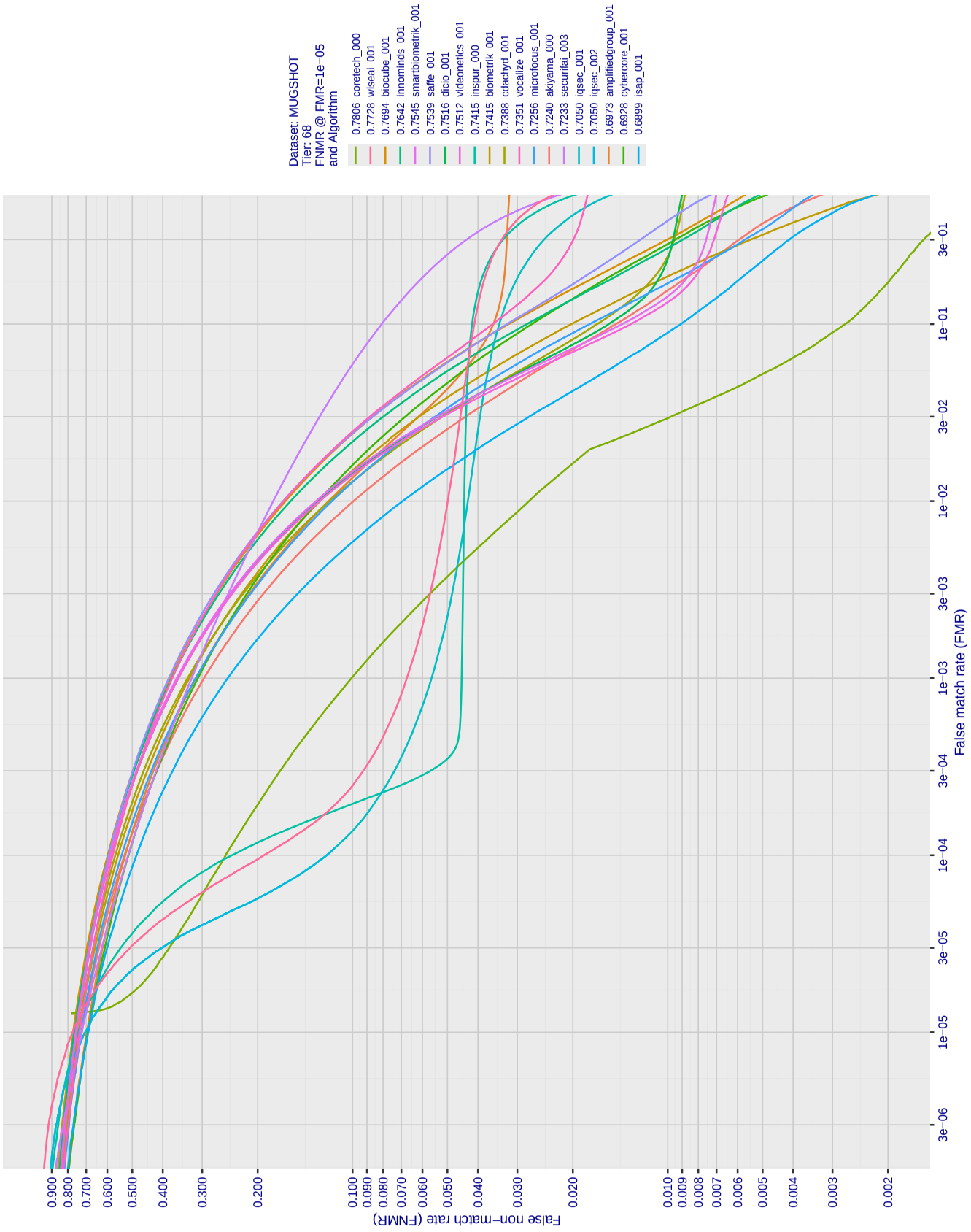


Figure 293: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

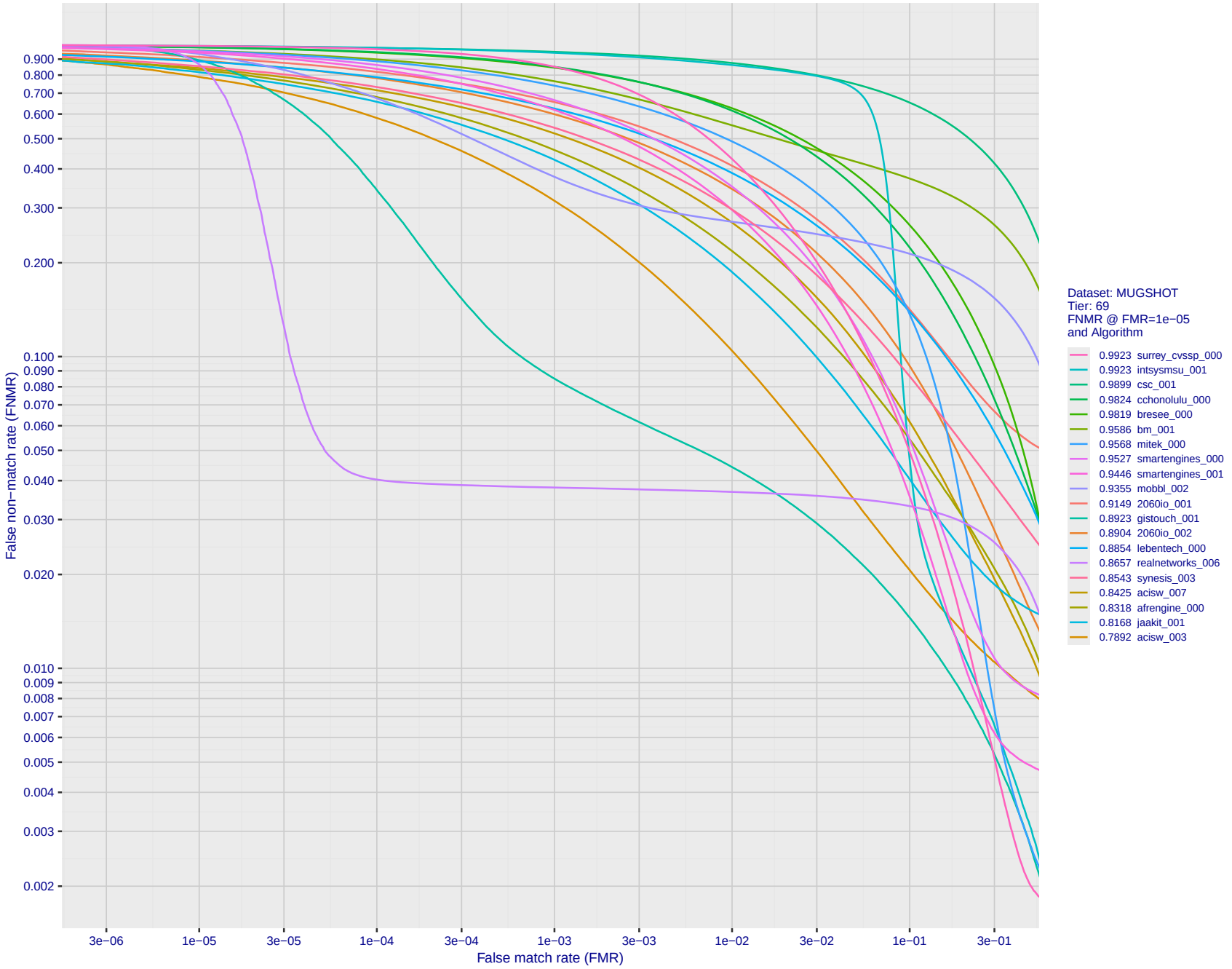


Figure 294: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

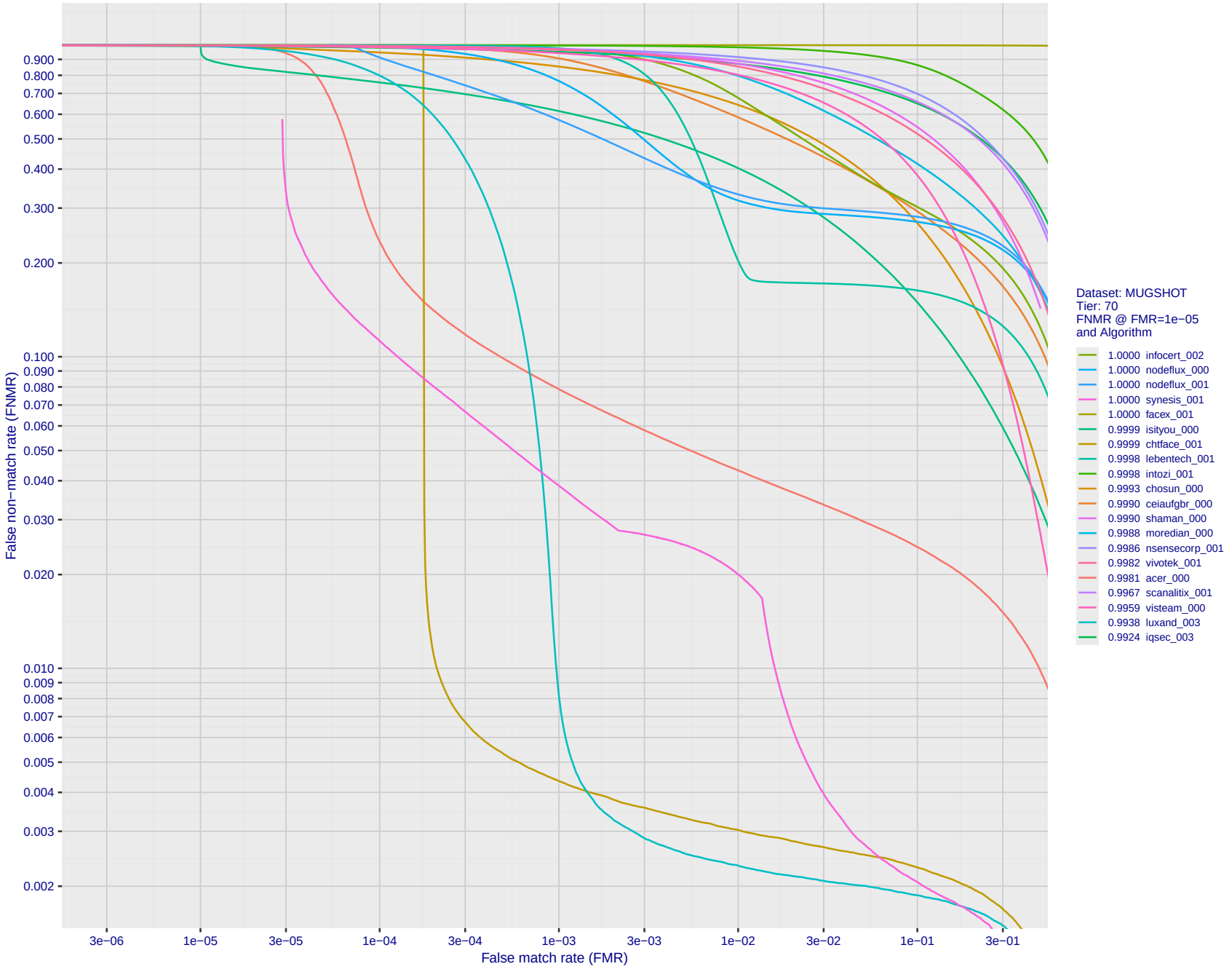


Figure 295: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

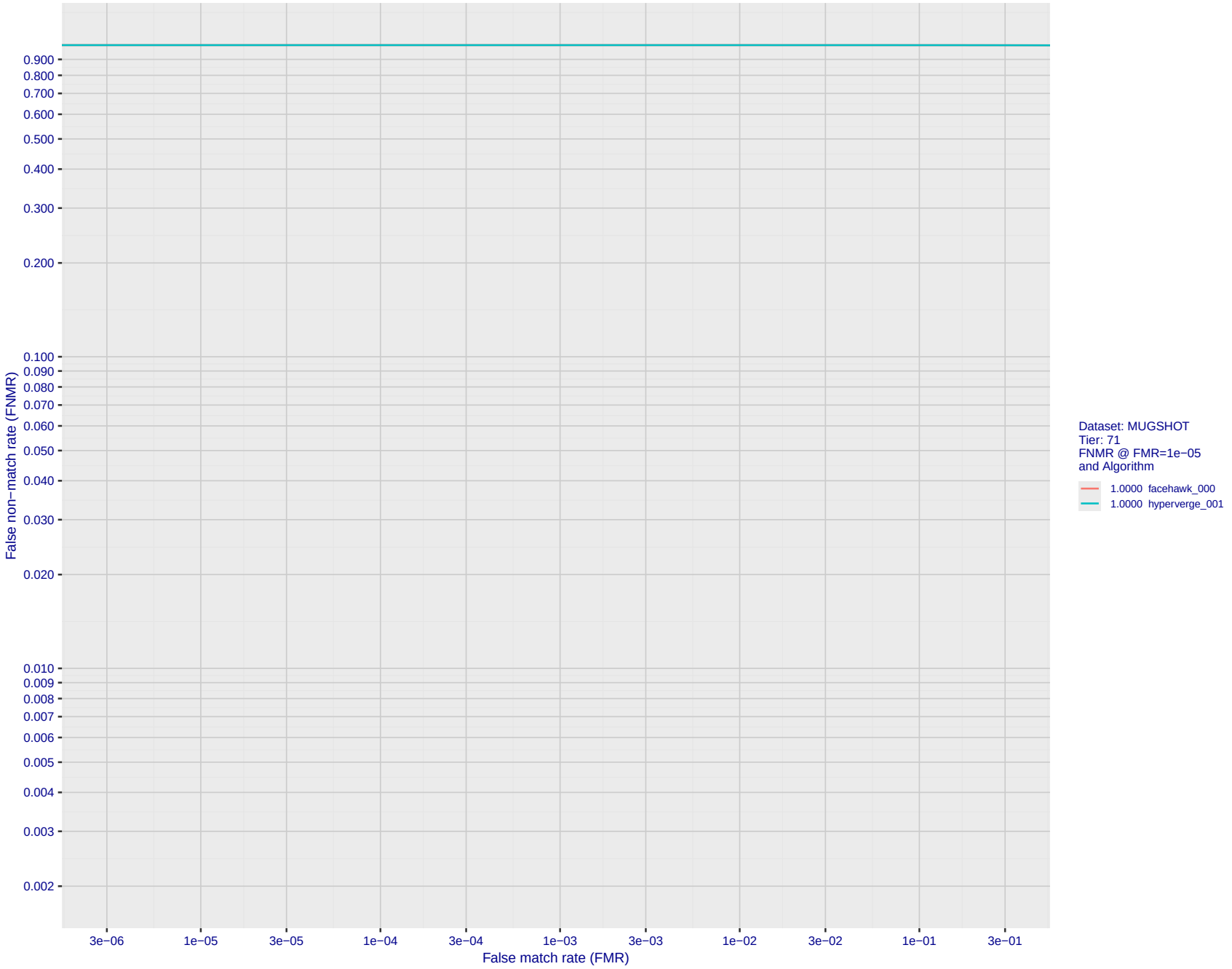


Figure 296: For the mugshot images, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show decades of FMR.

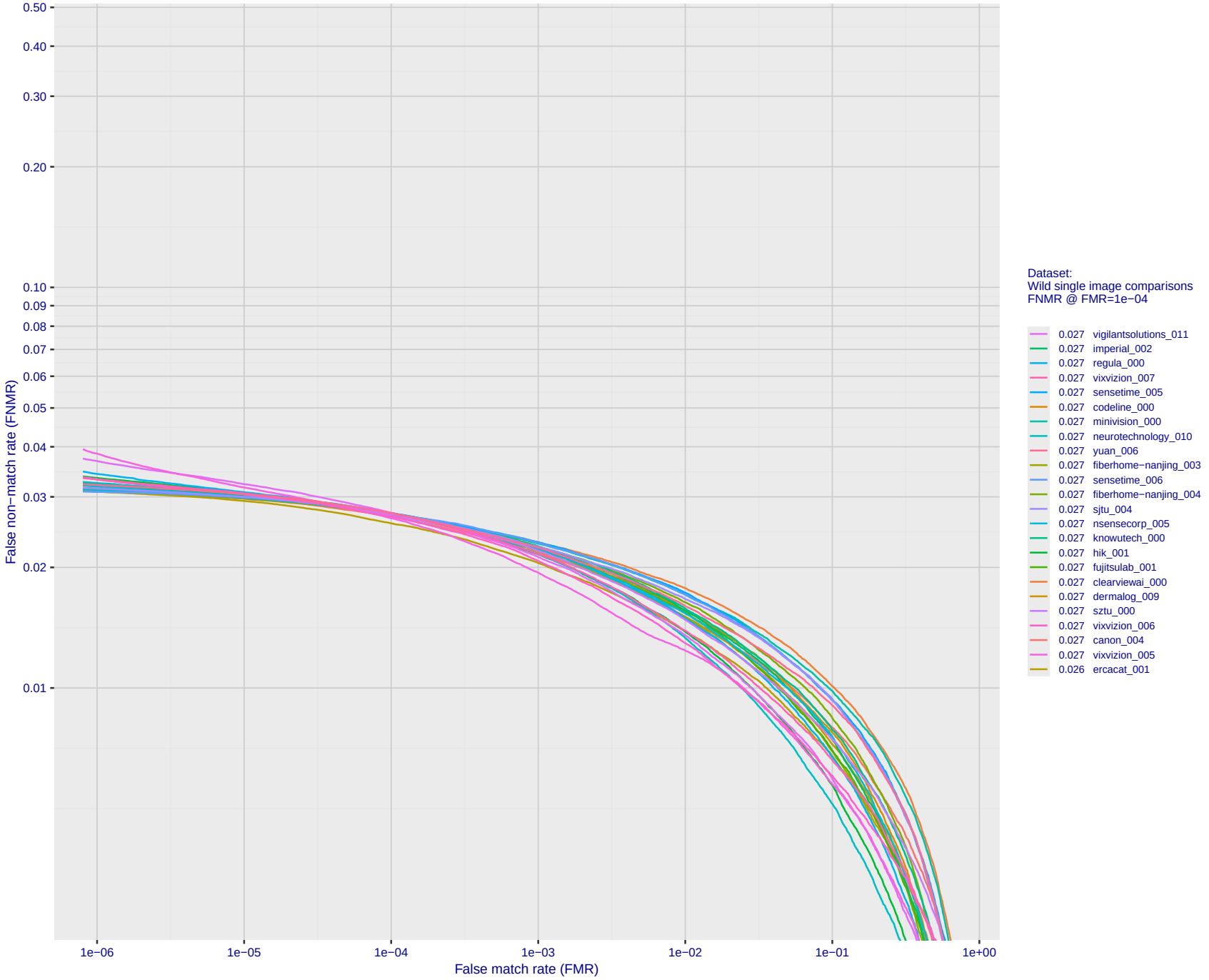


Figure 297: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

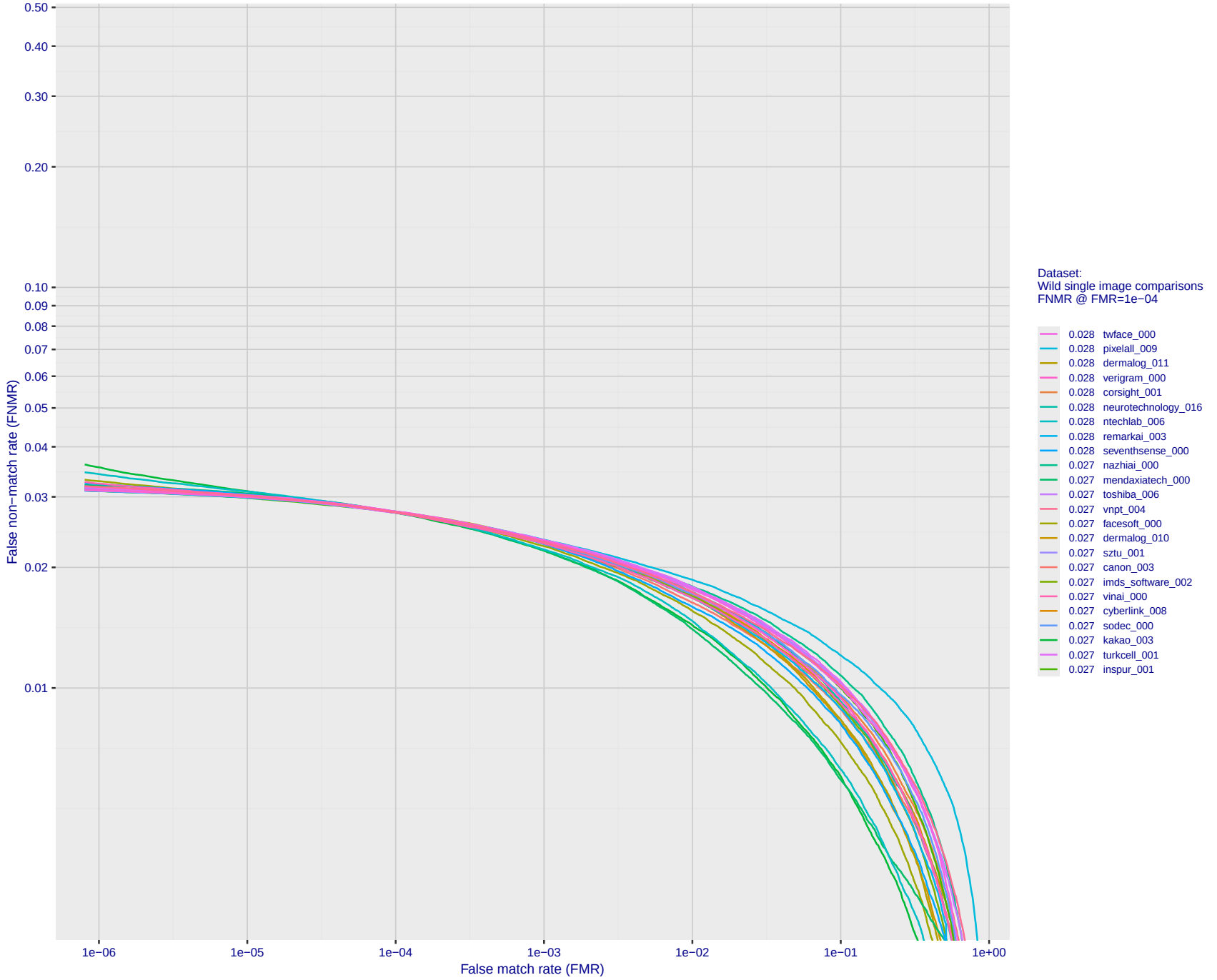


Figure 298: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

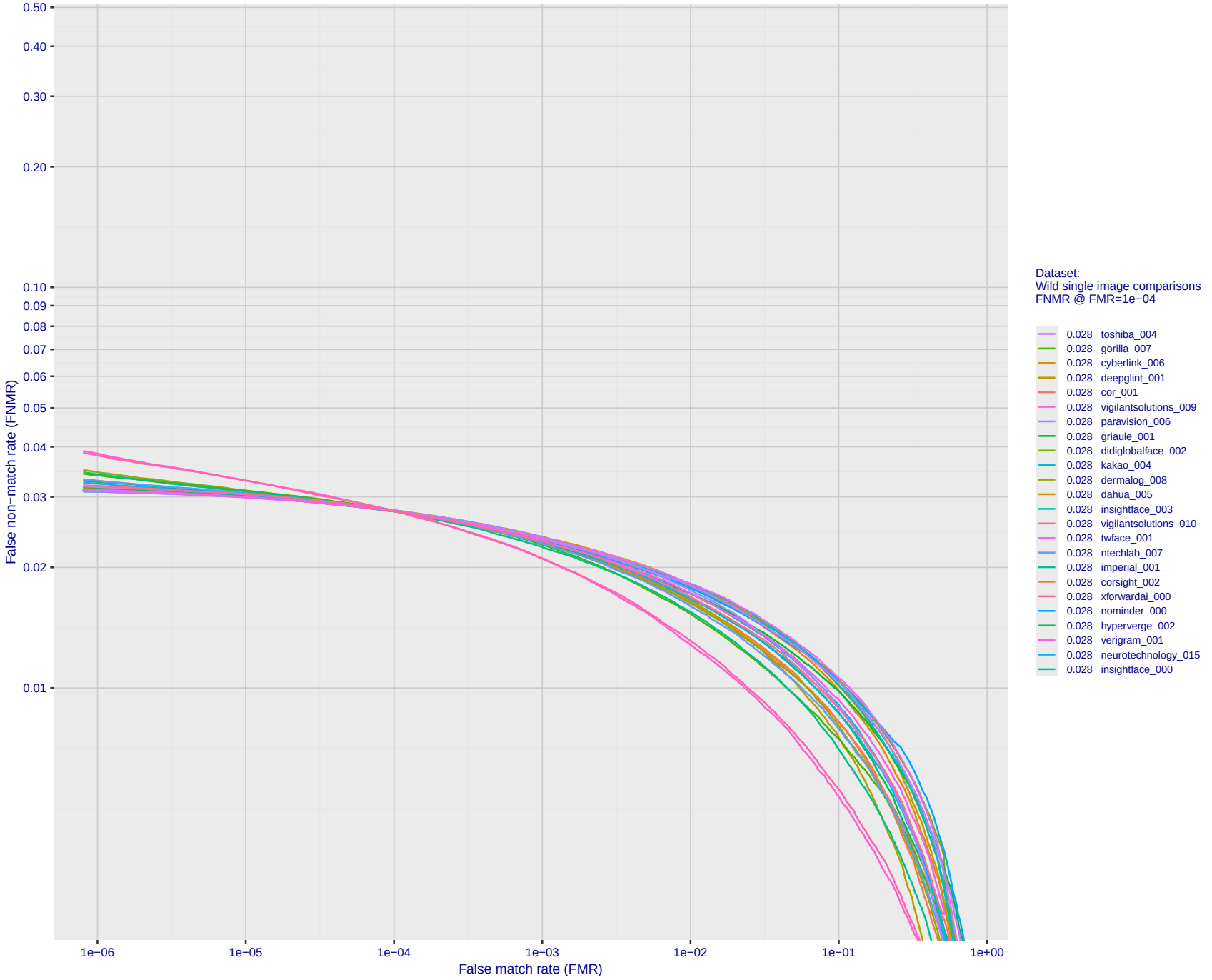


Figure 299: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

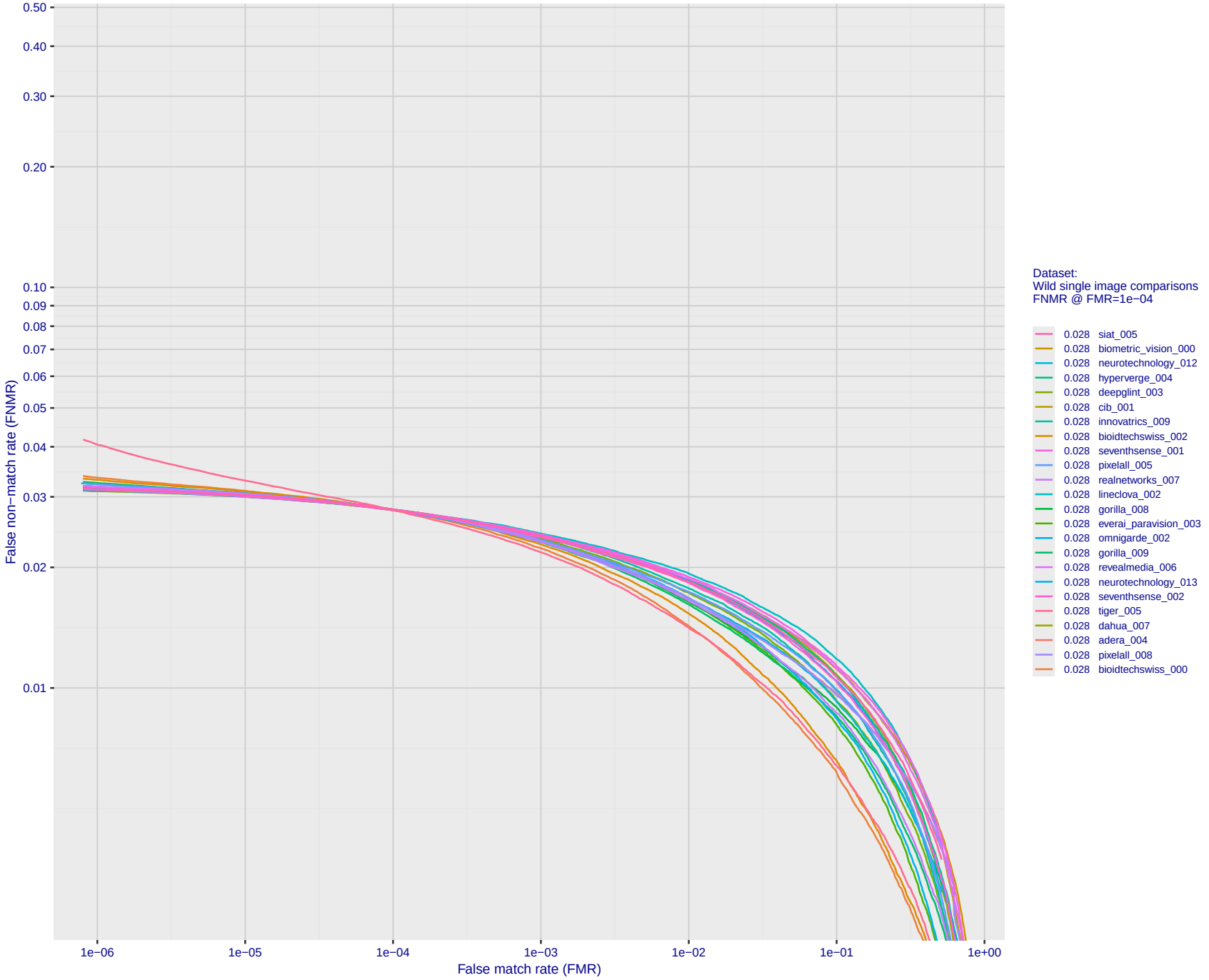


Figure 300: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

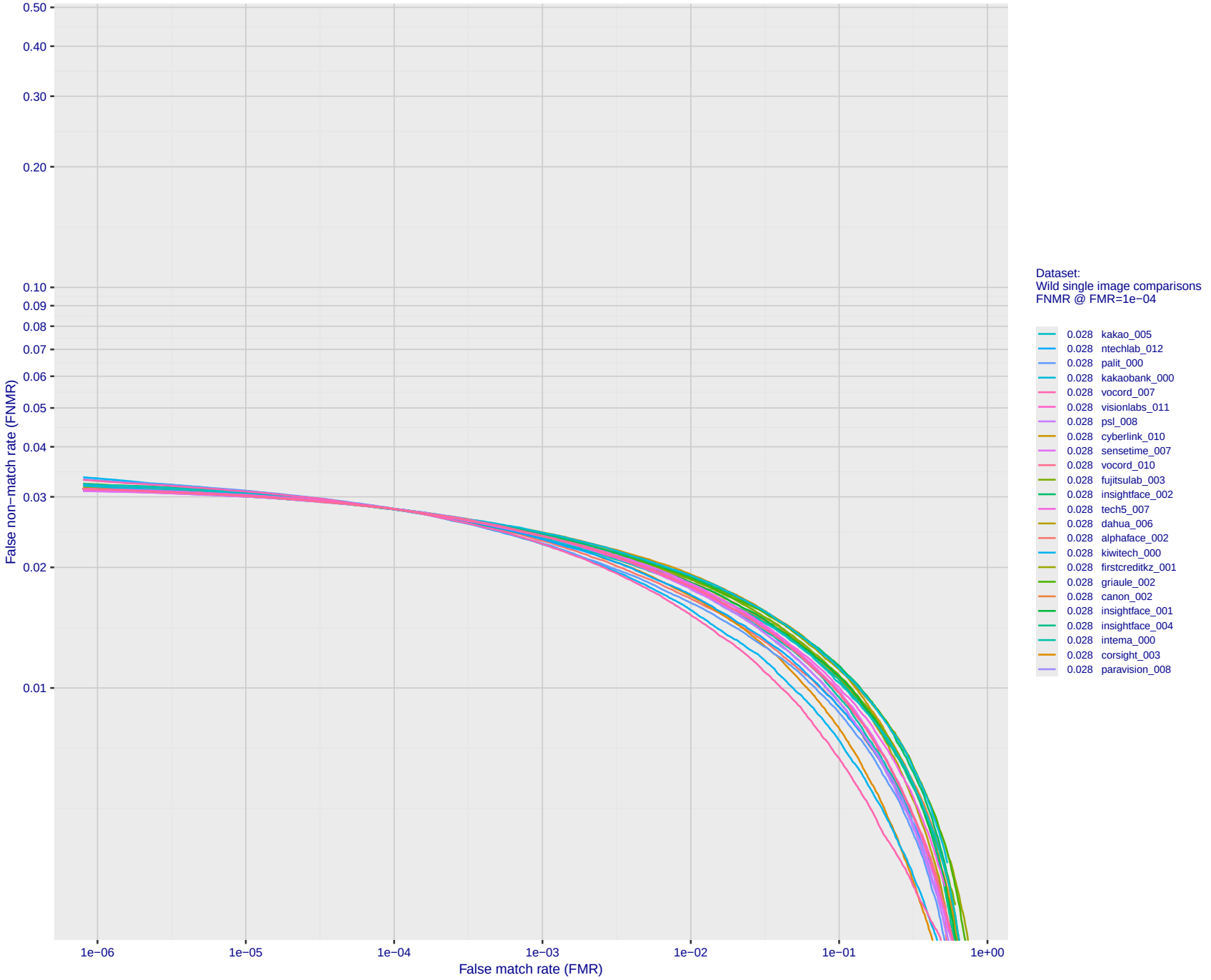


Figure 301: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

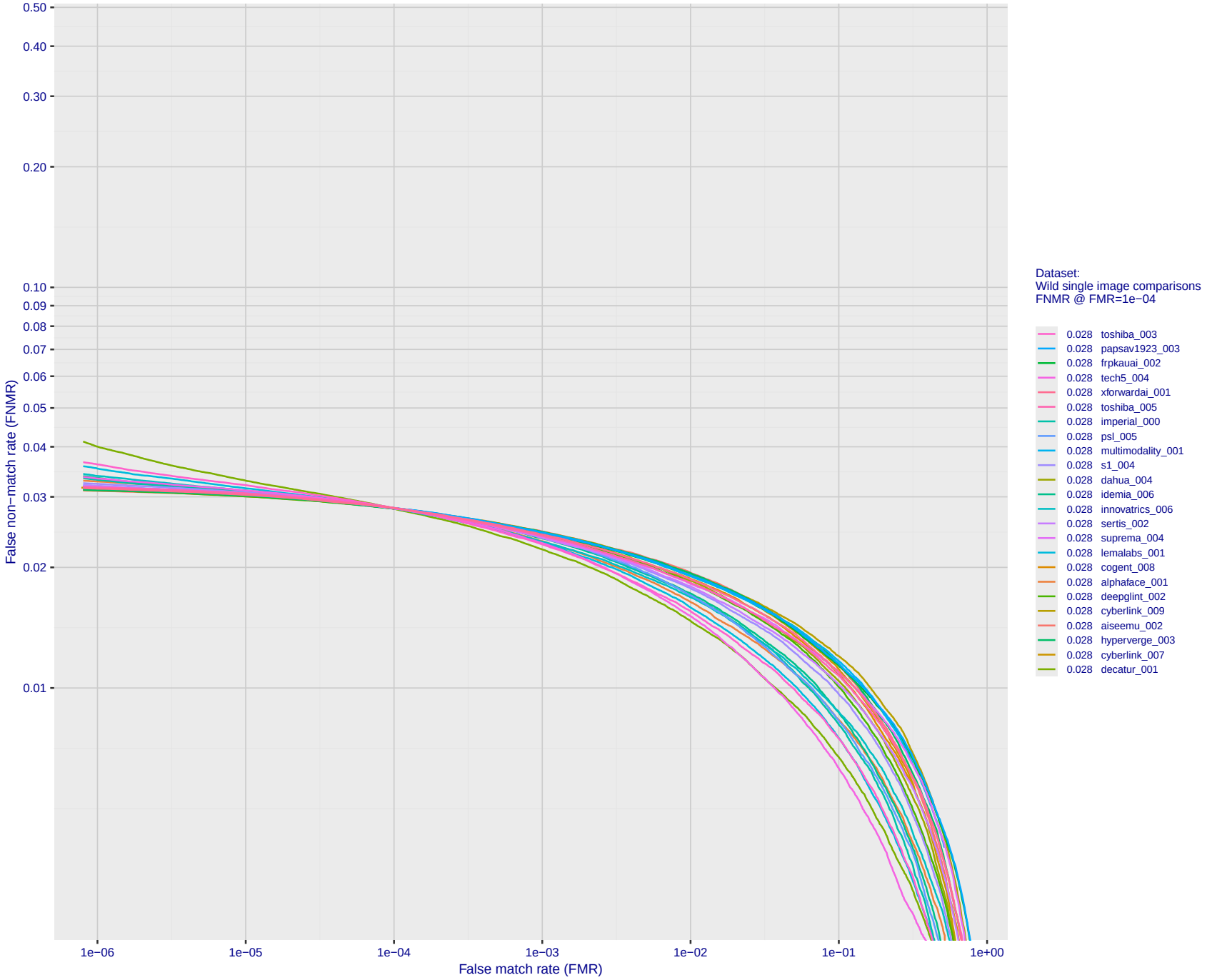


Figure 302: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

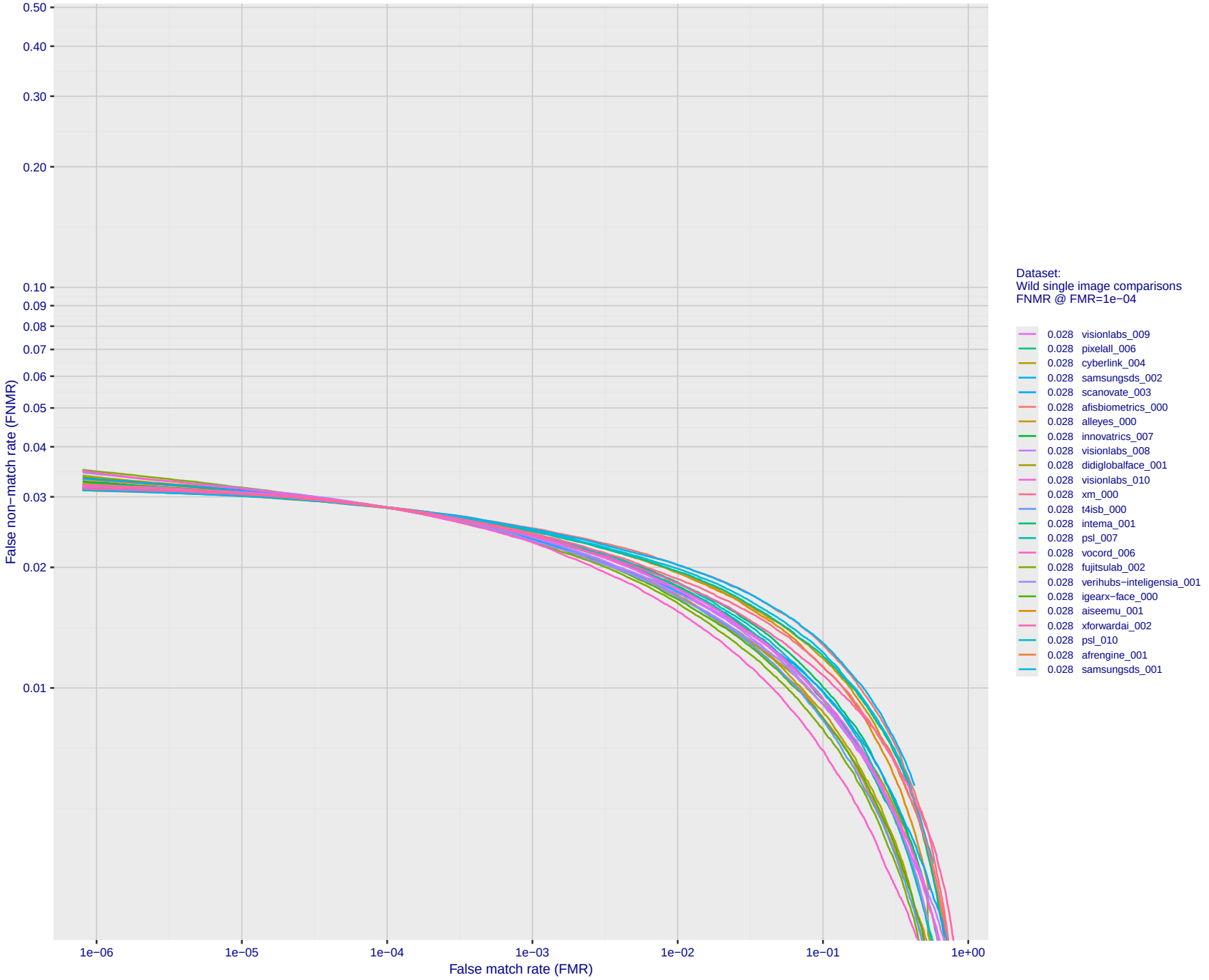


Figure 303: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

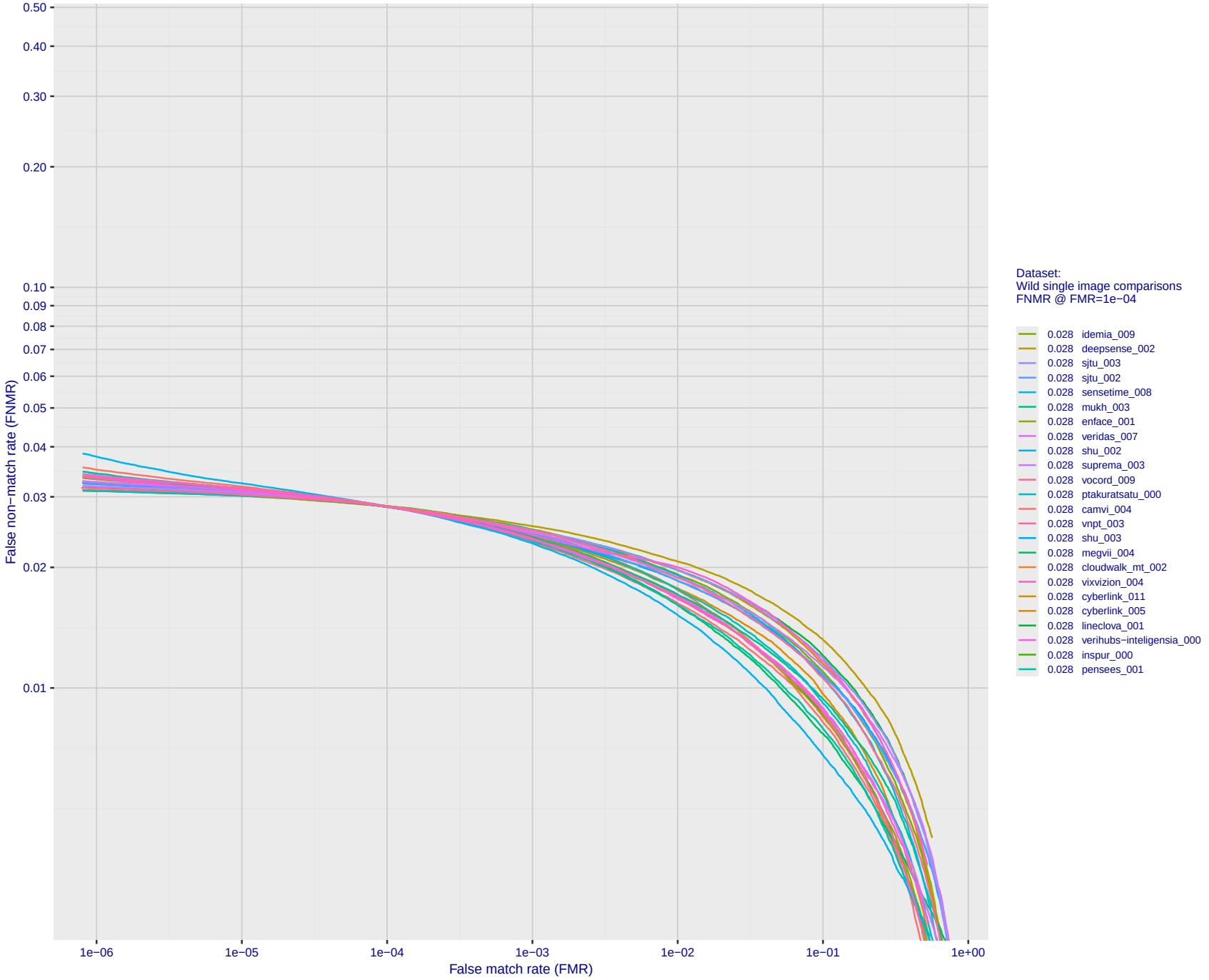


Figure 304: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

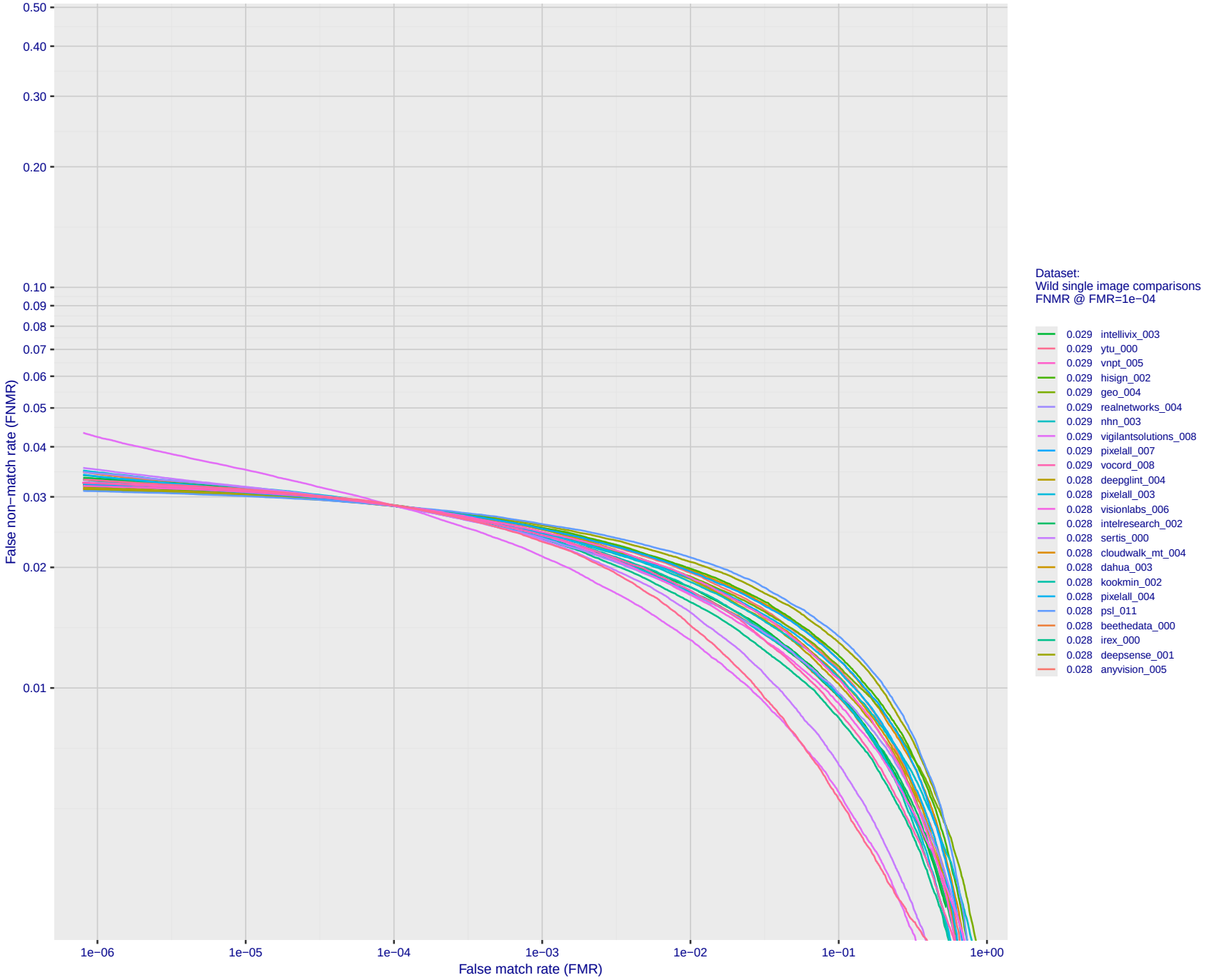


Figure 305: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

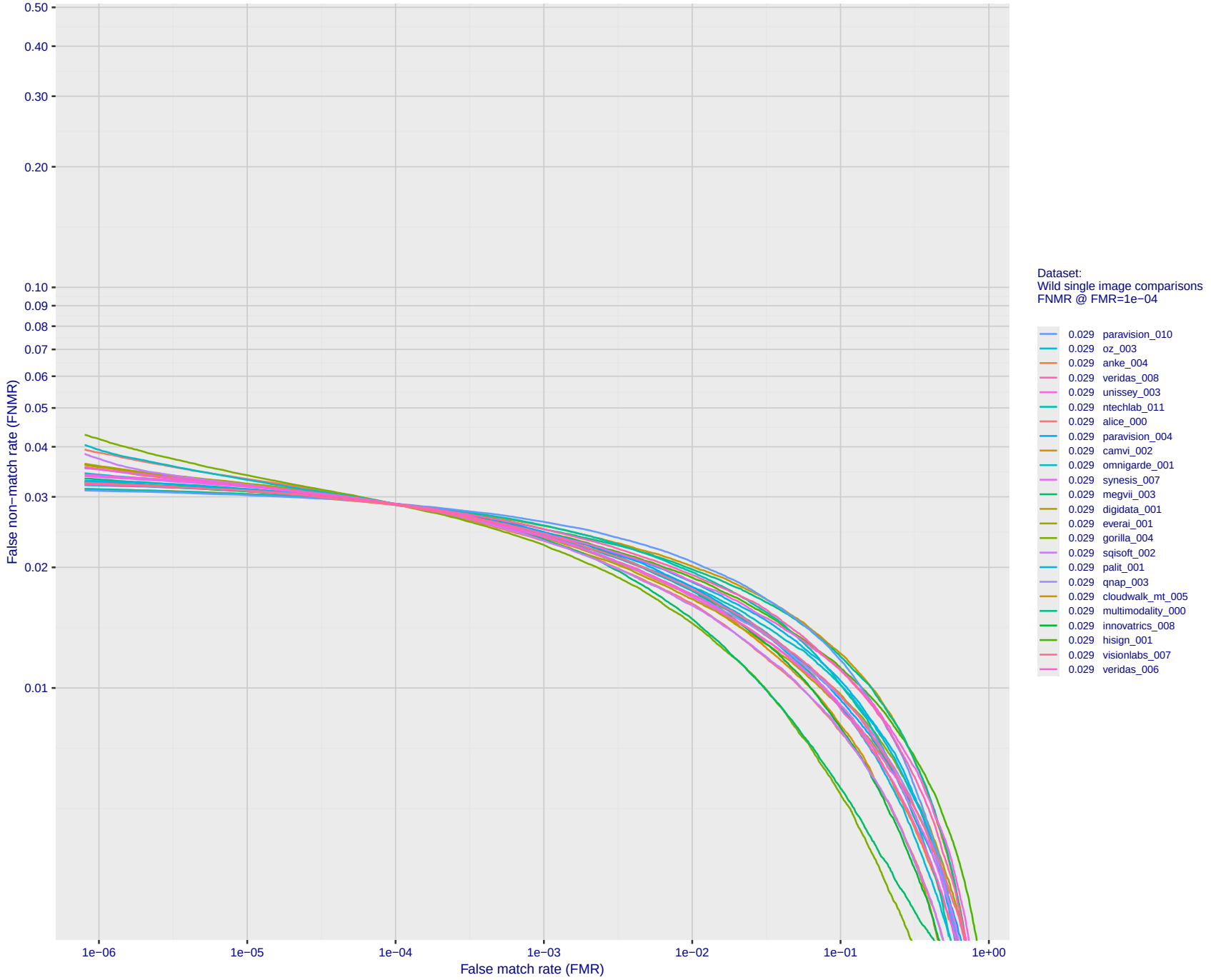


Figure 306: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

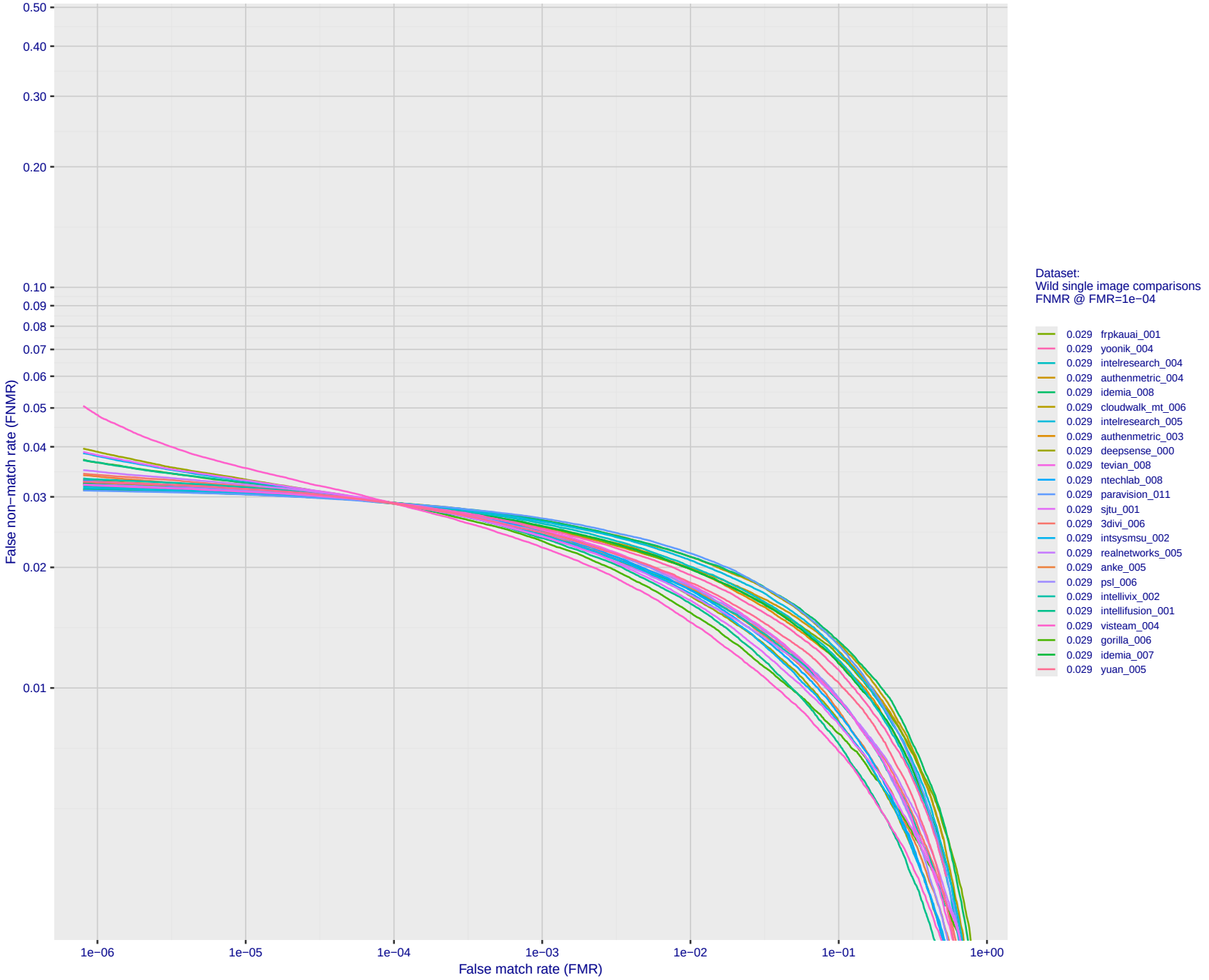


Figure 307: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

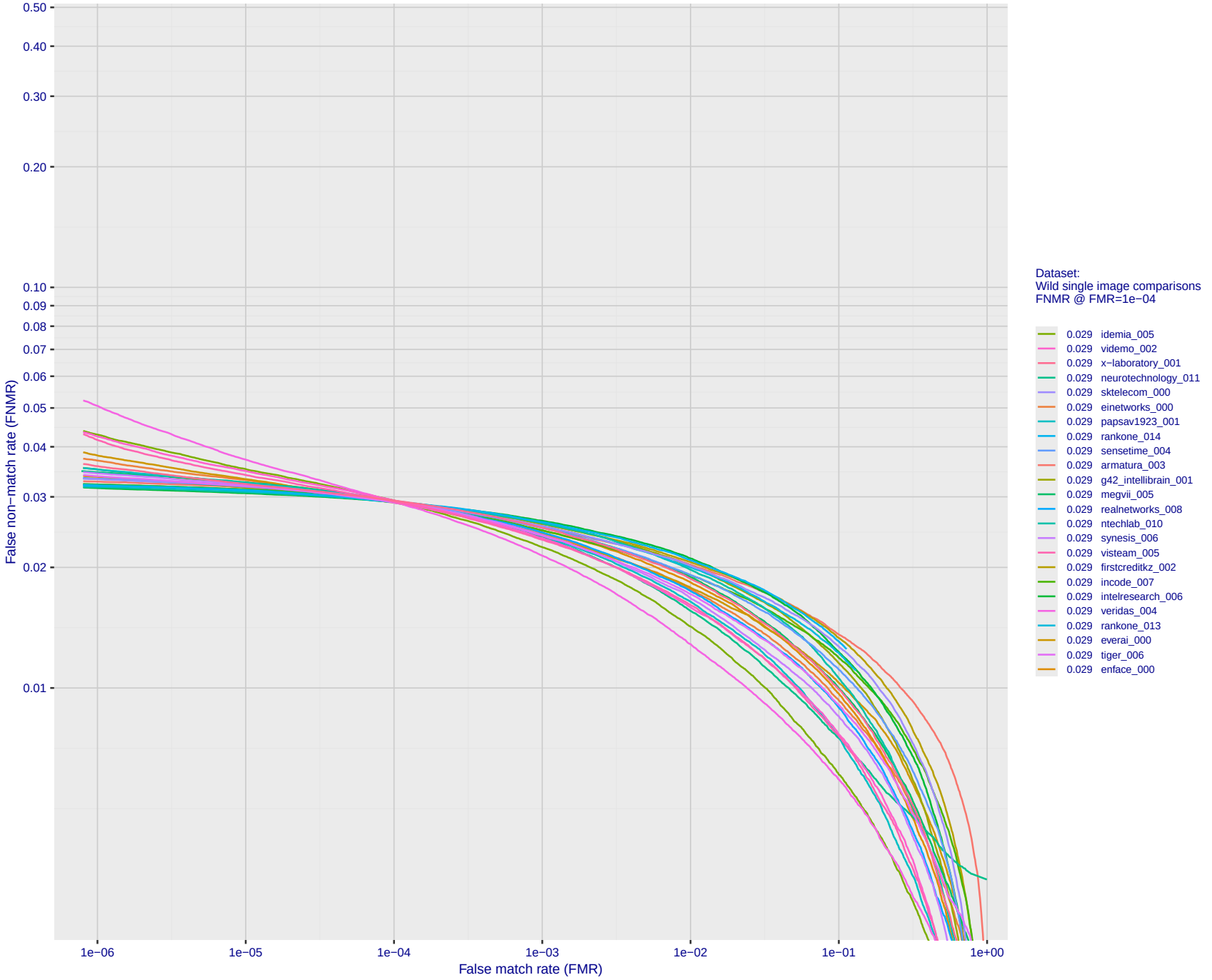


Figure 308: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

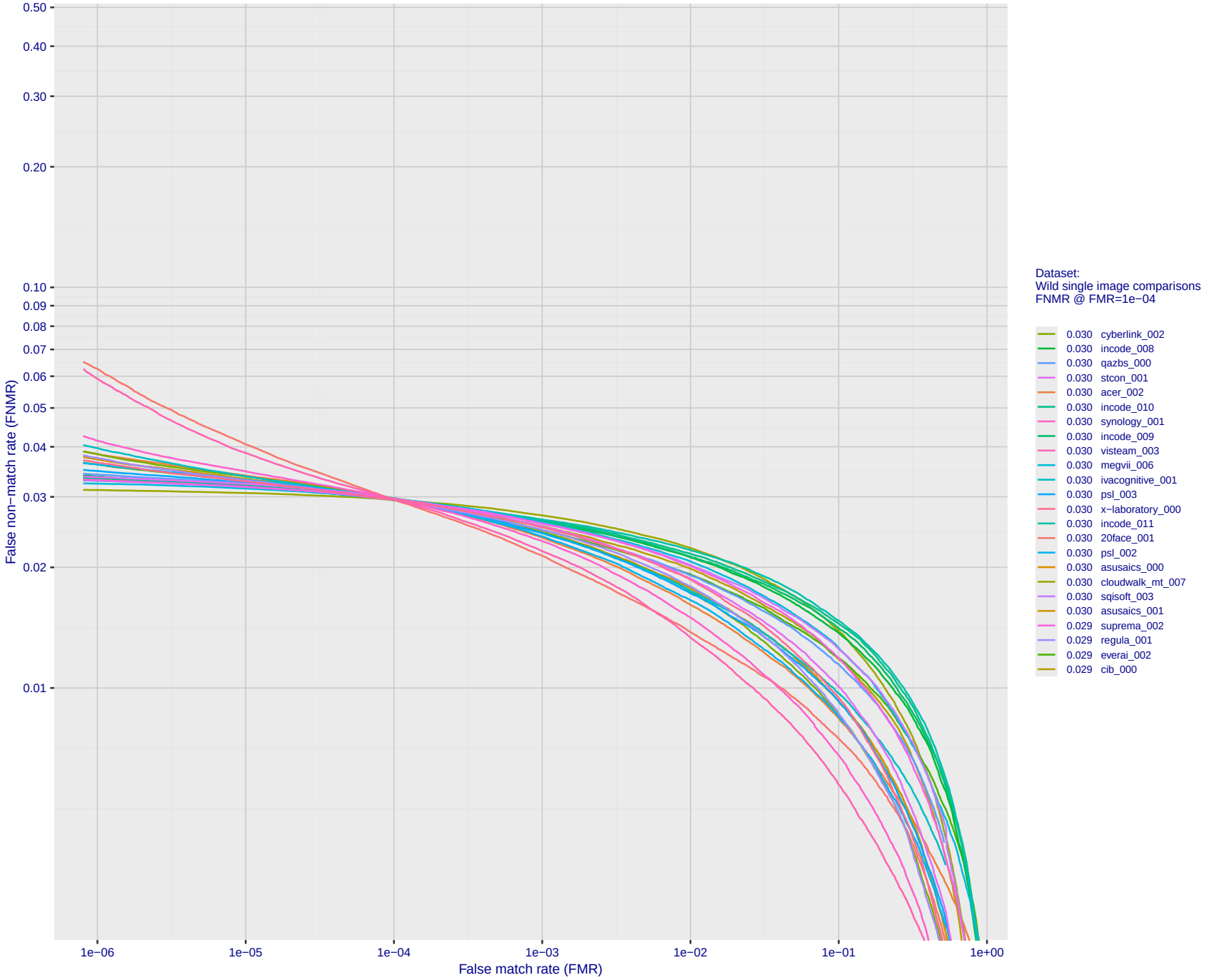


Figure 309: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

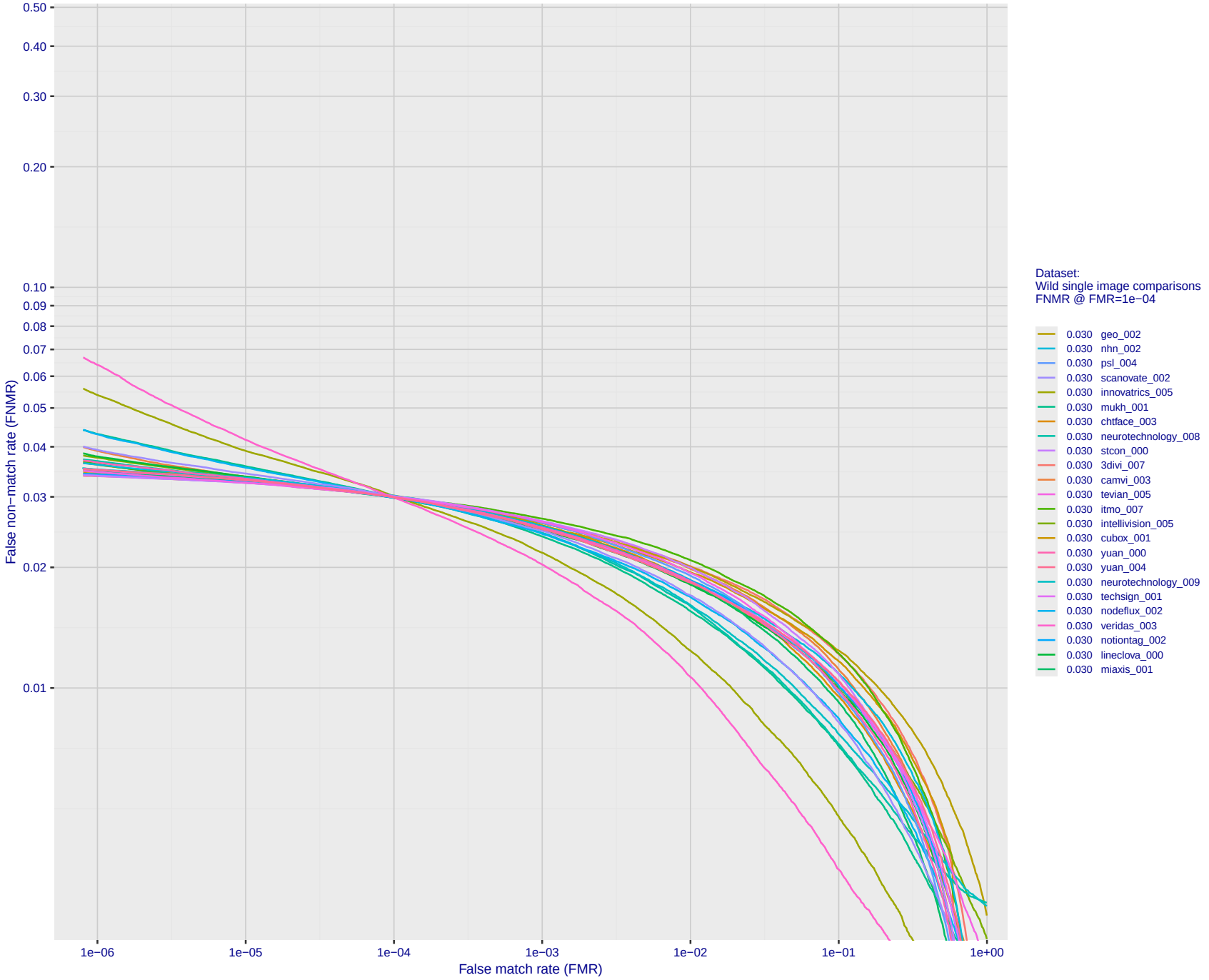


Figure 310: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

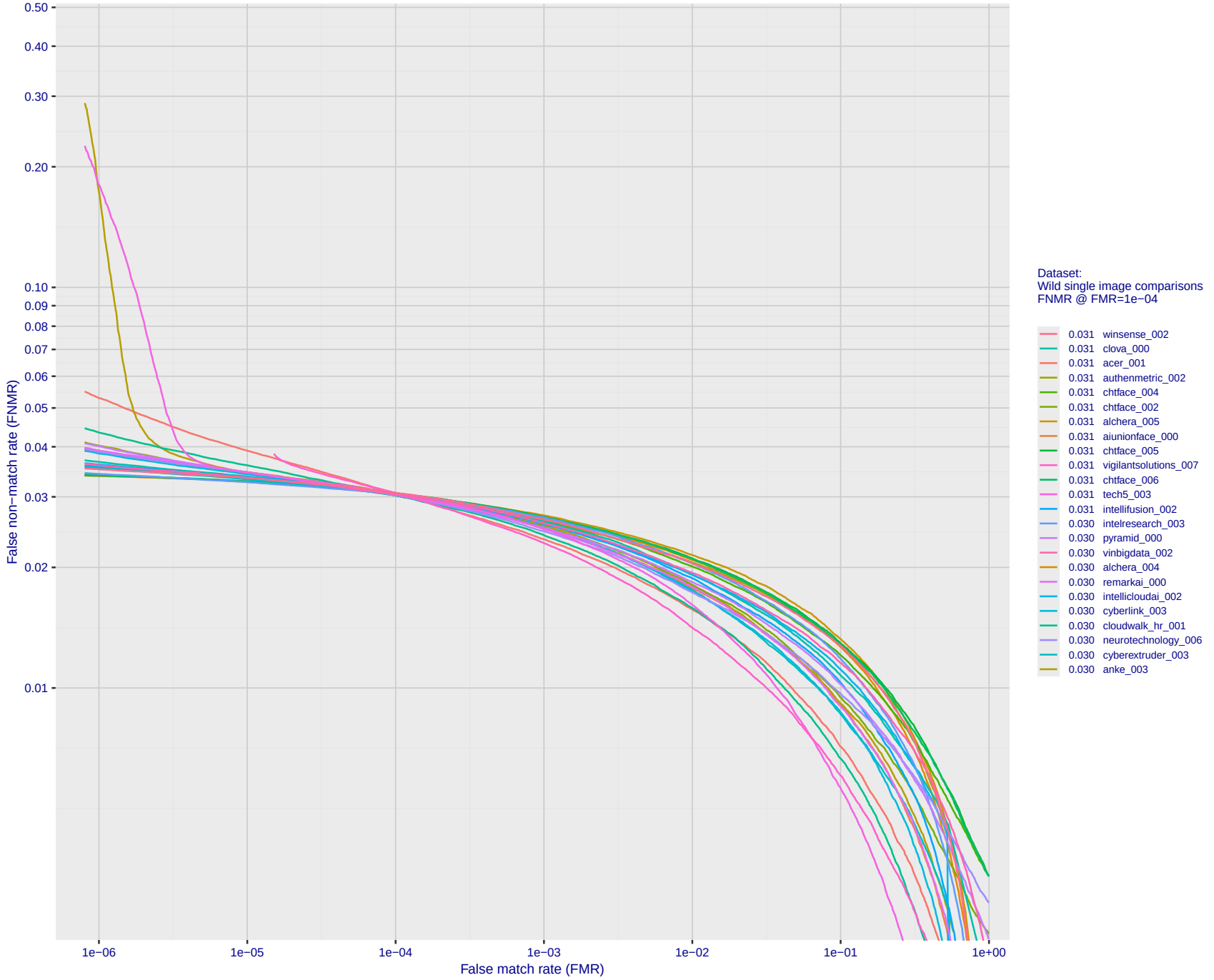


Figure 311: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

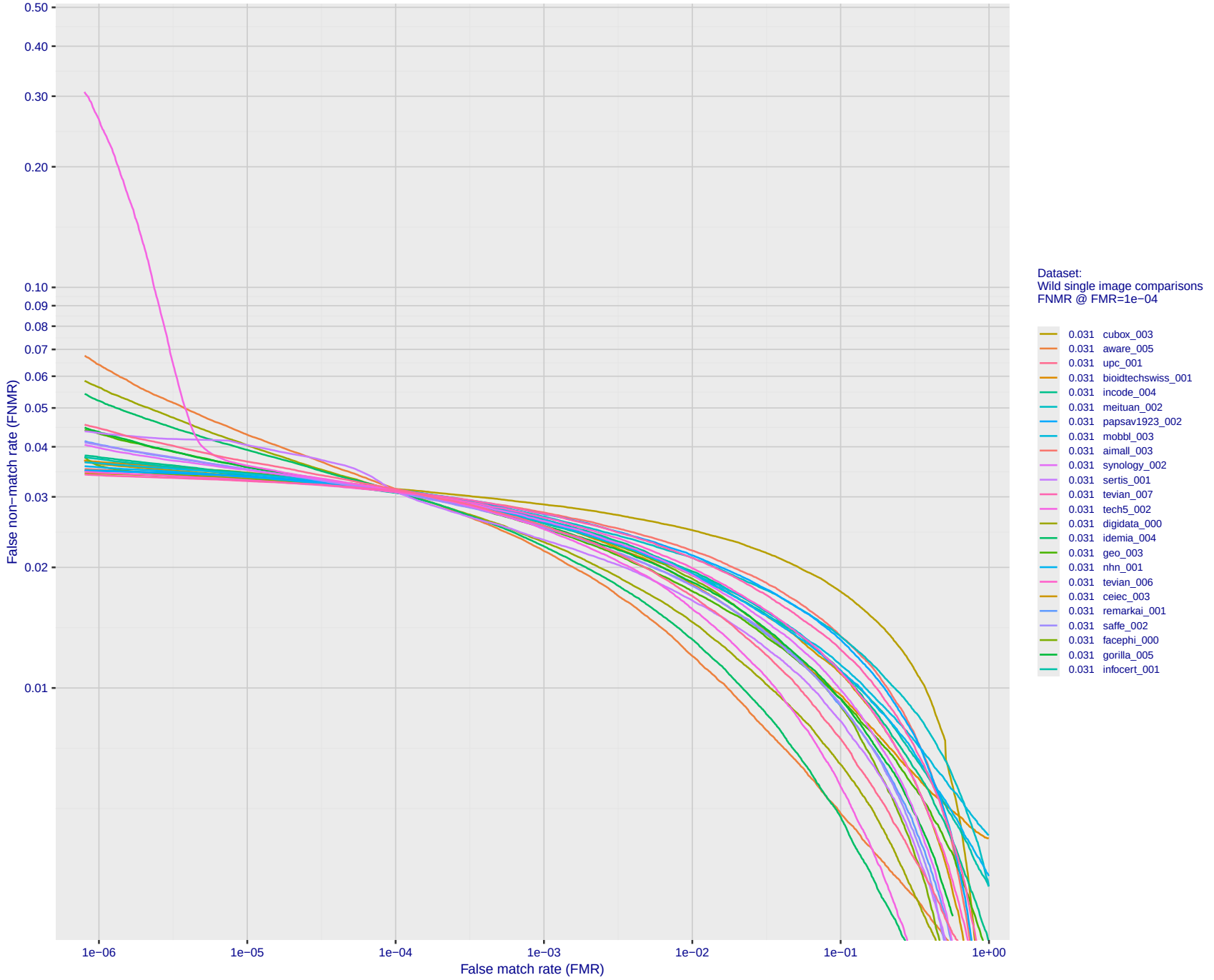


Figure 312: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

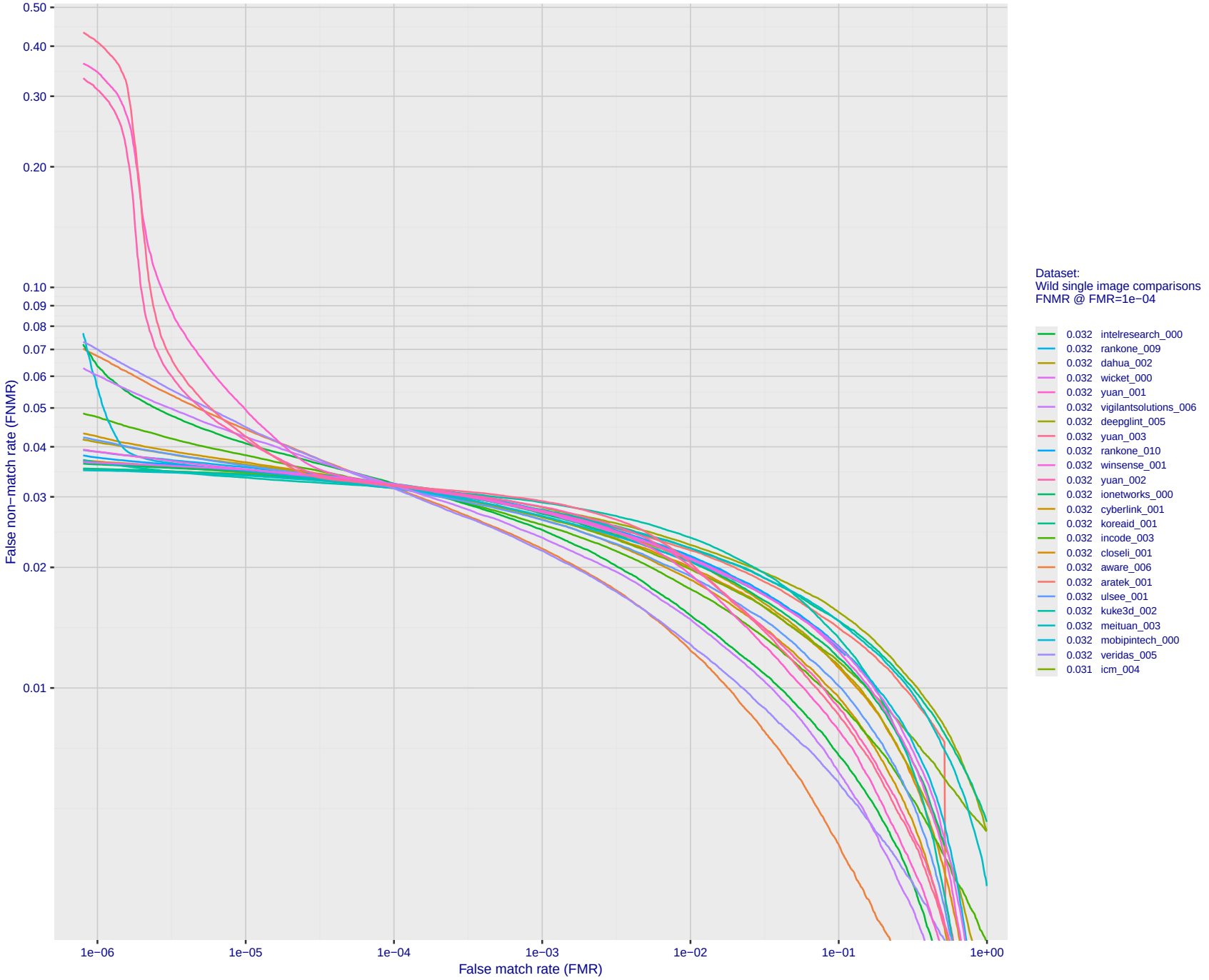


Figure 313: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

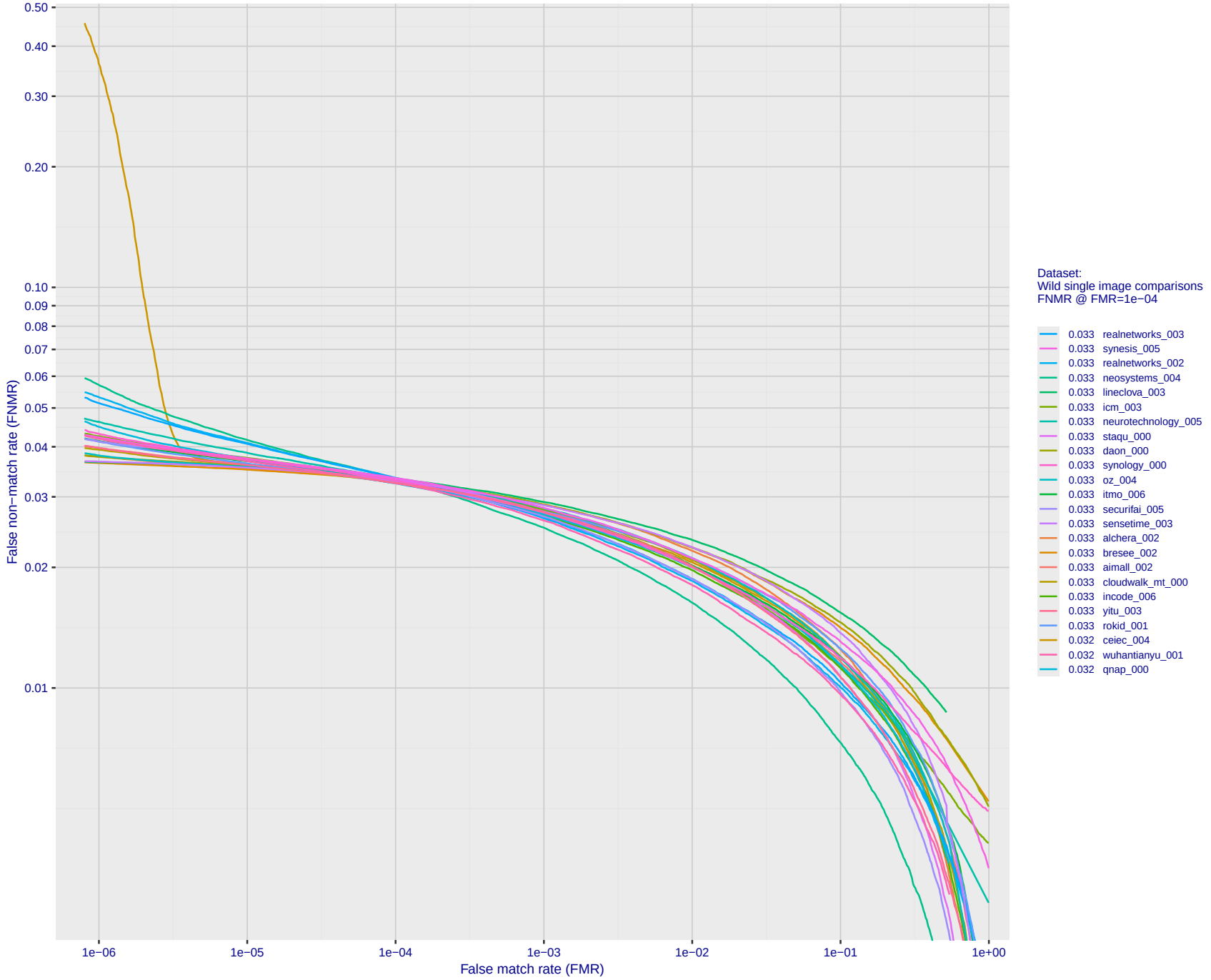


Figure 314: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

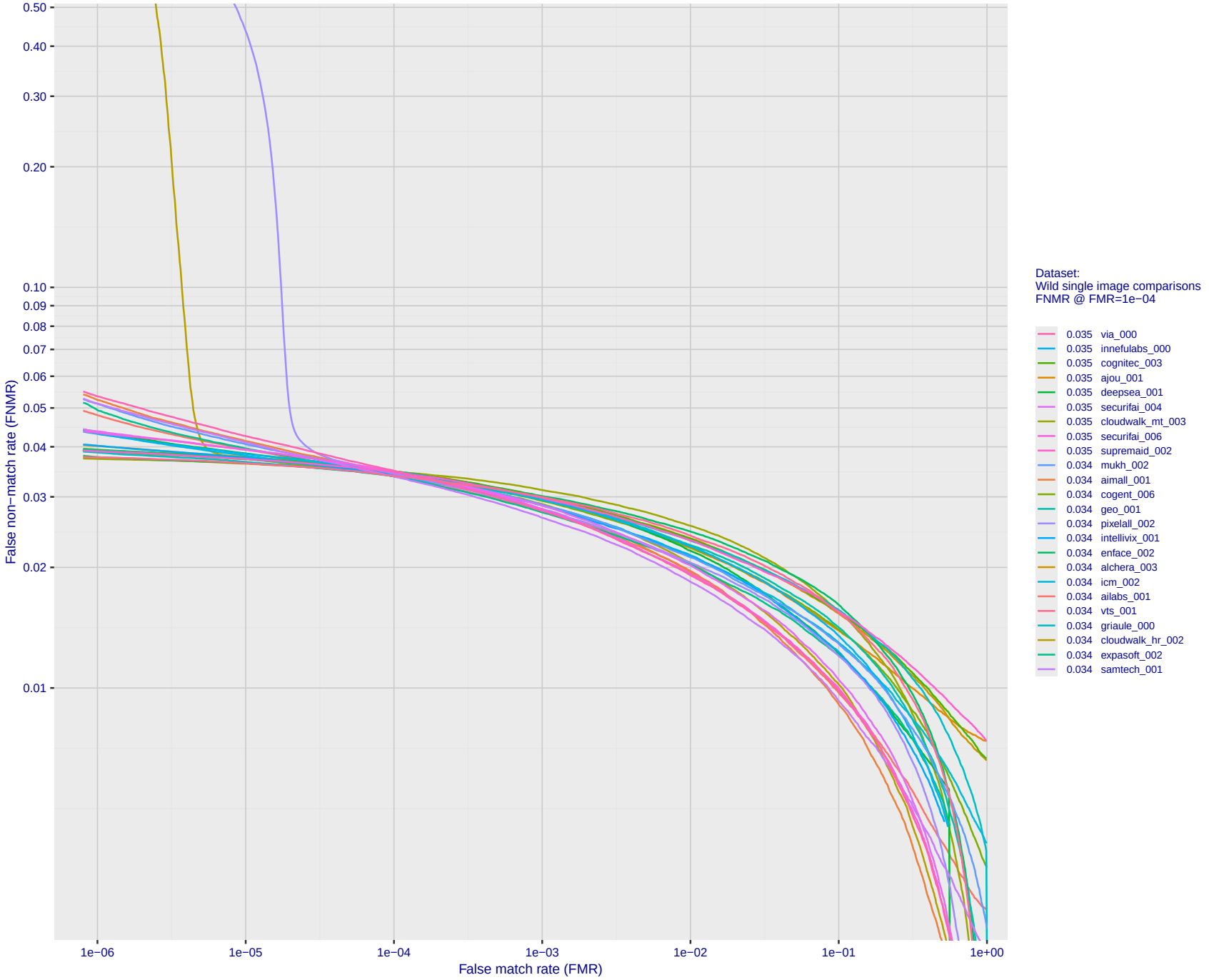


Figure 315: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

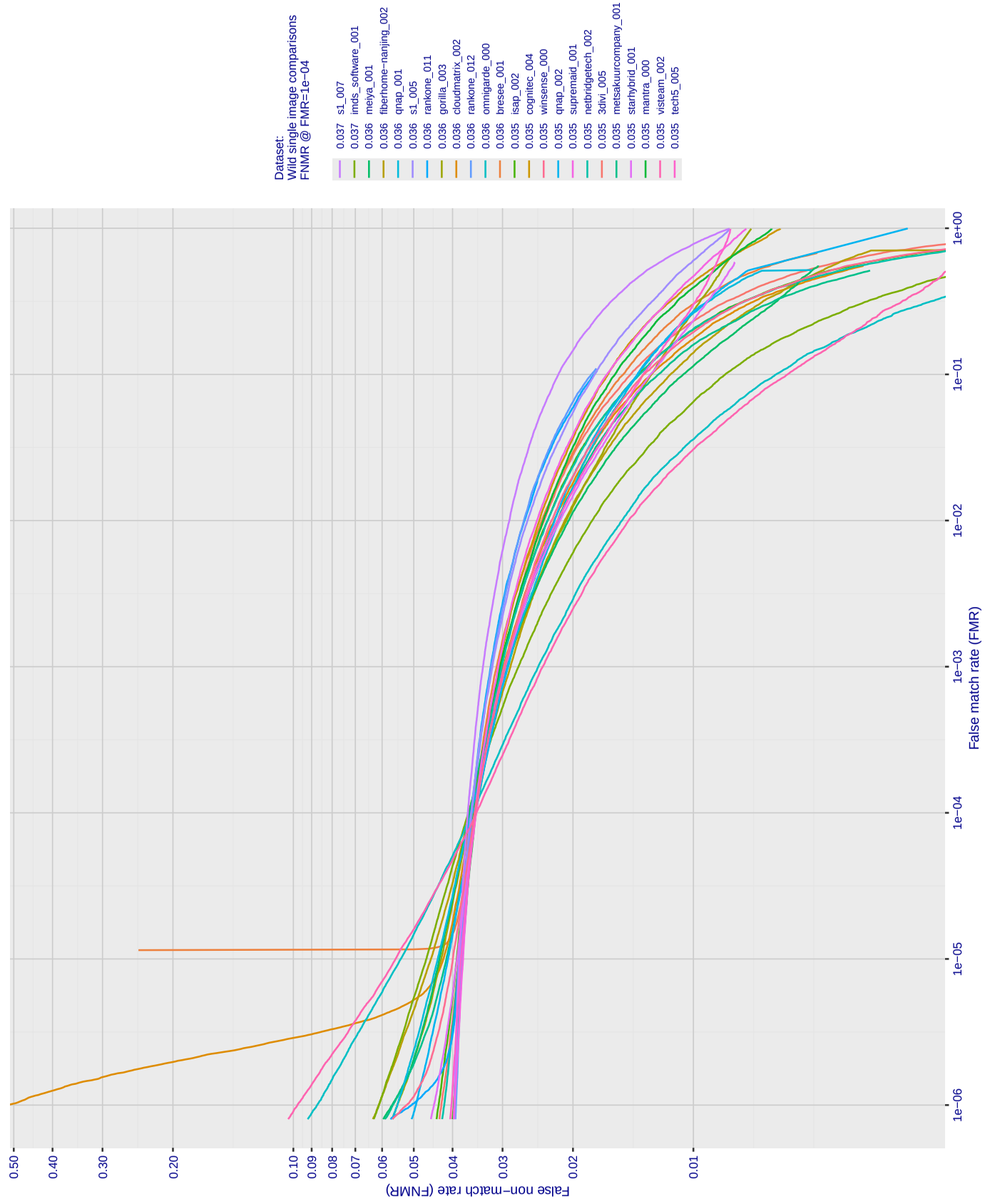


Figure 316: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

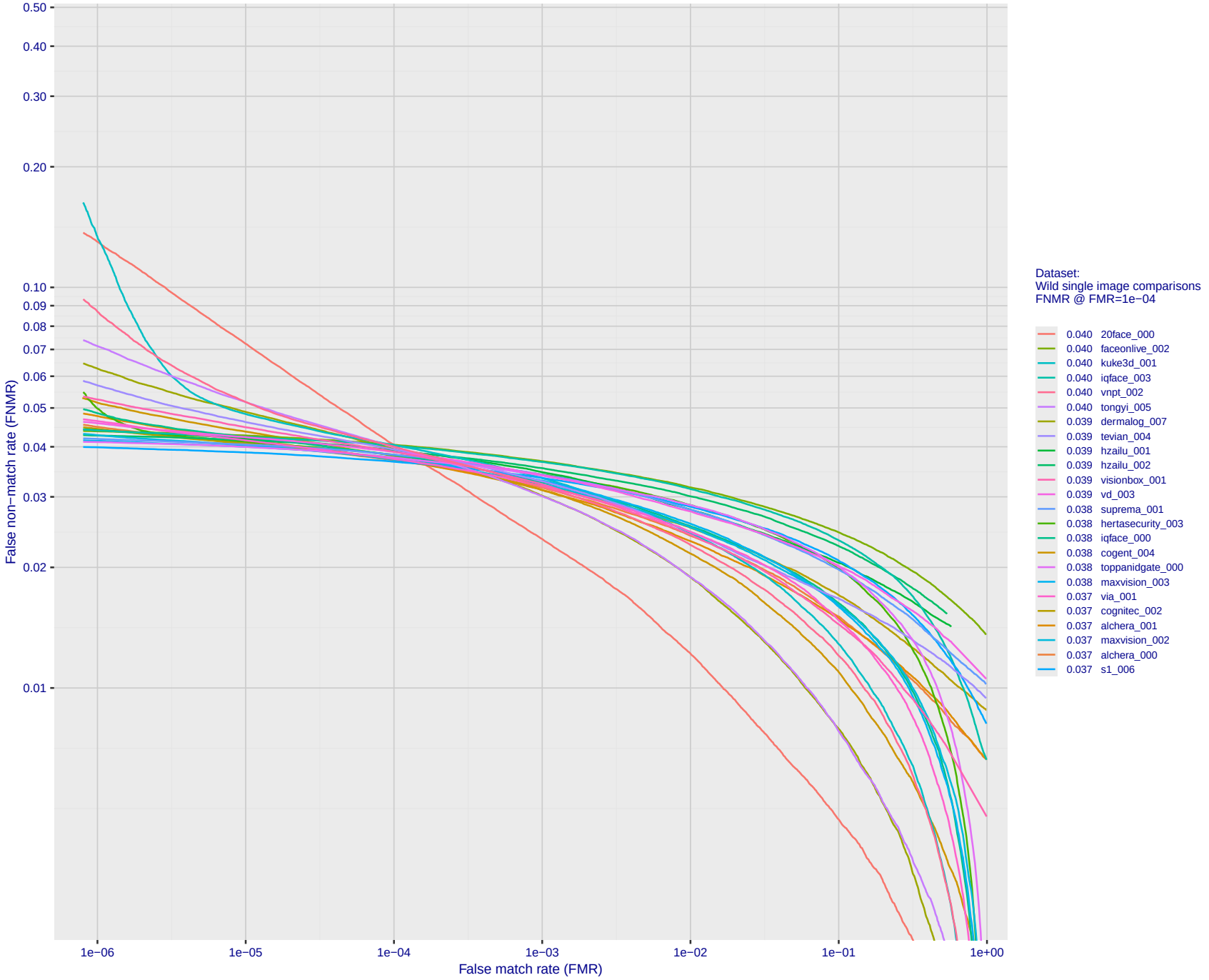


Figure 317: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

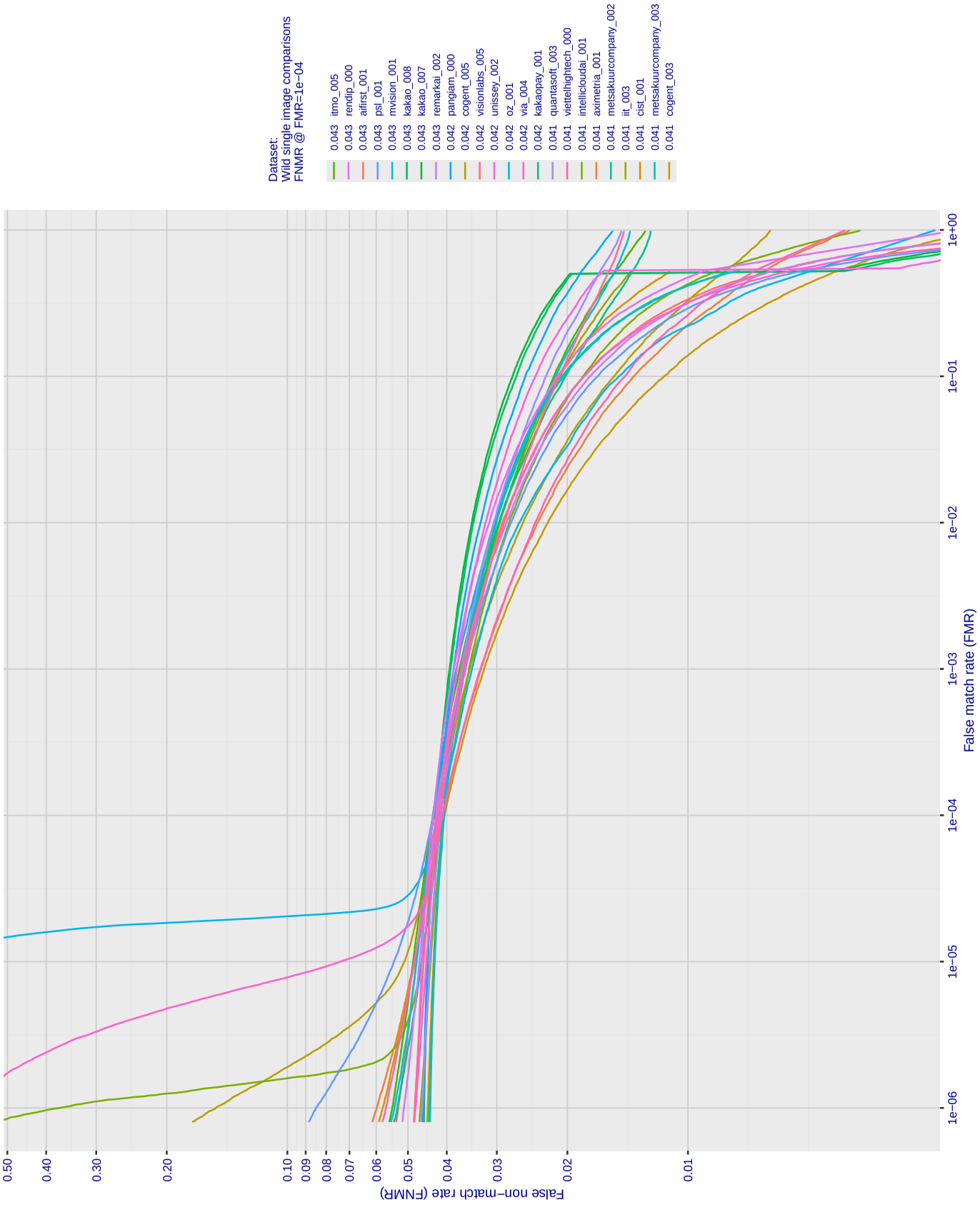


Figure 318: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

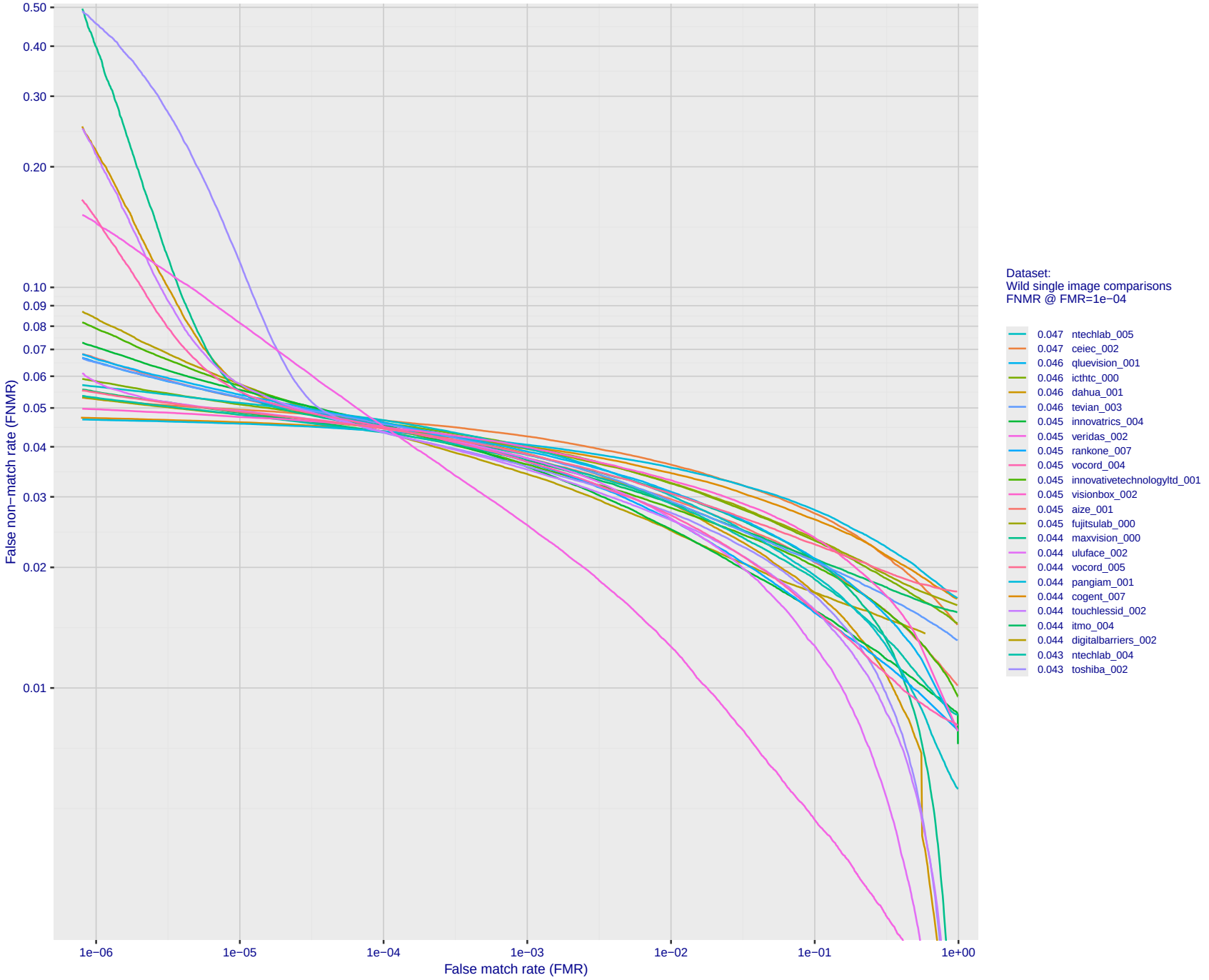


Figure 319: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

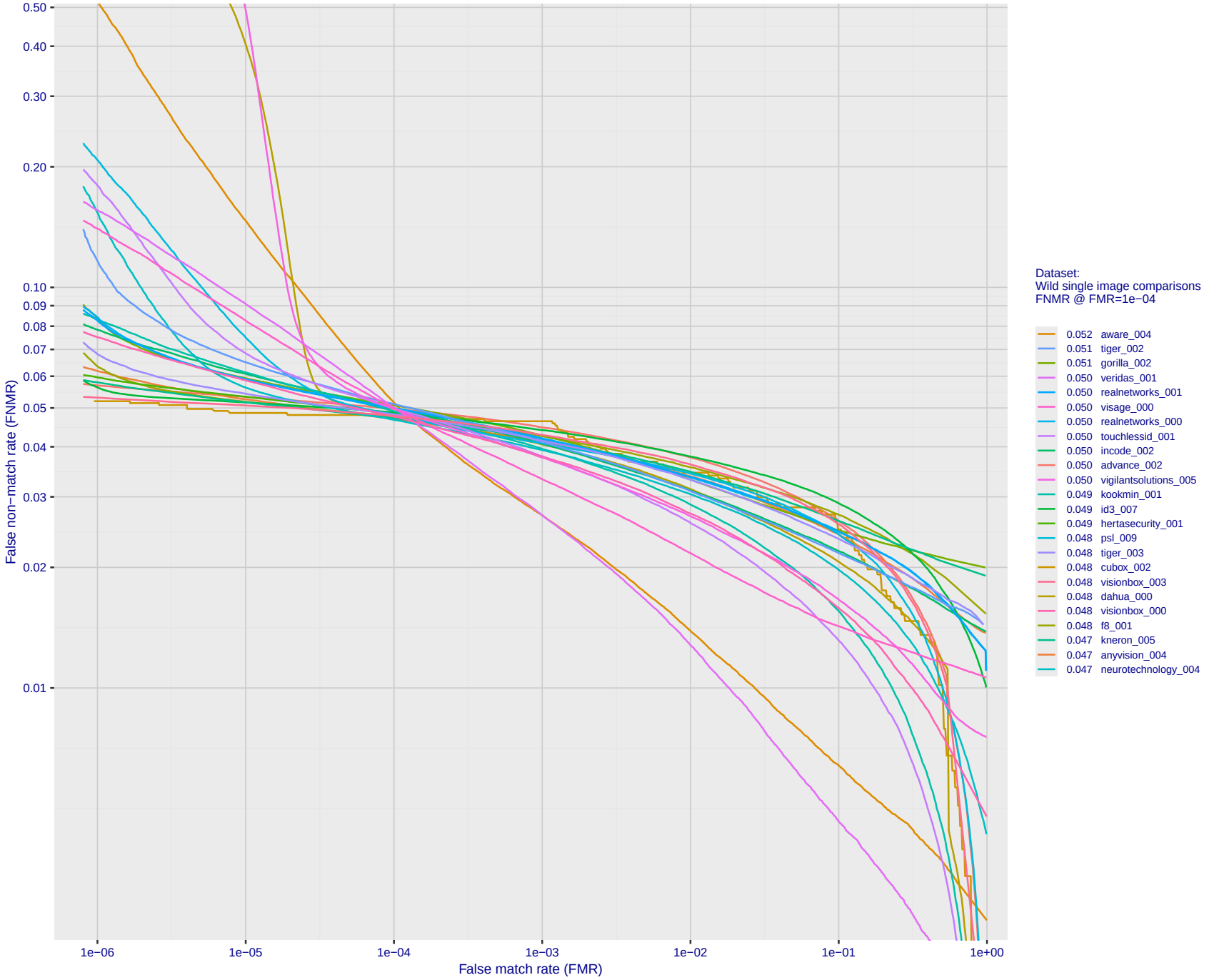


Figure 320: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

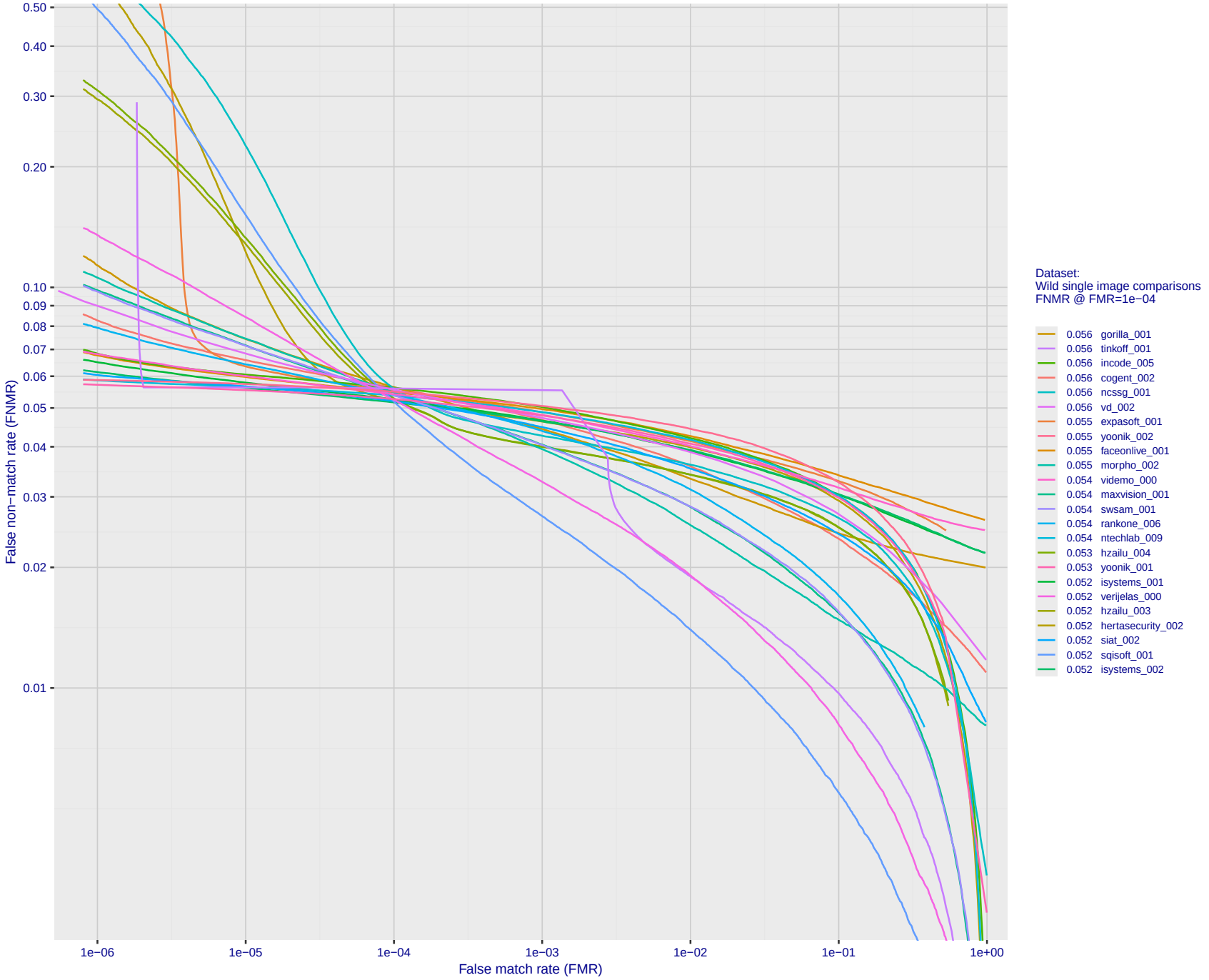


Figure 321: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

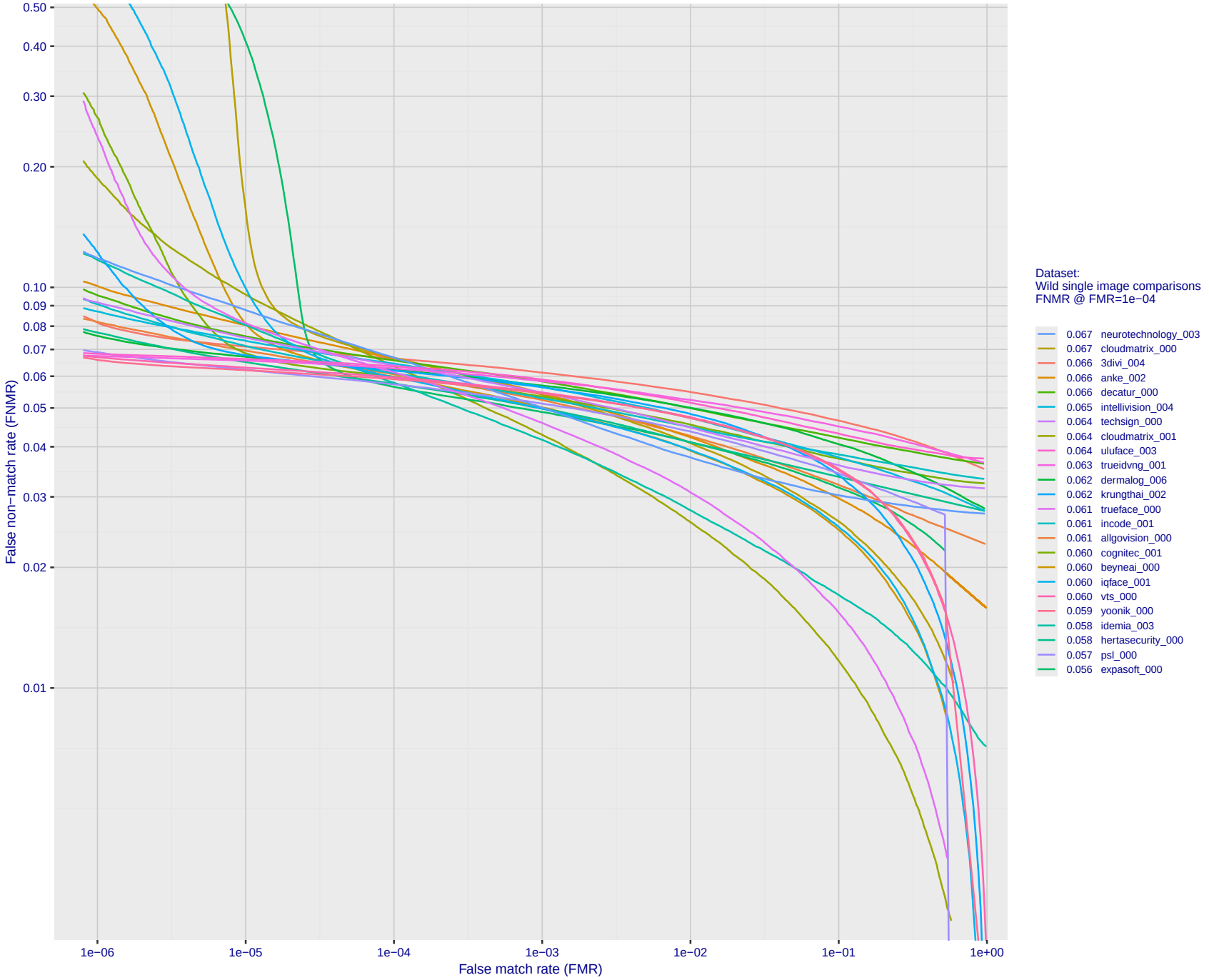


Figure 322: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

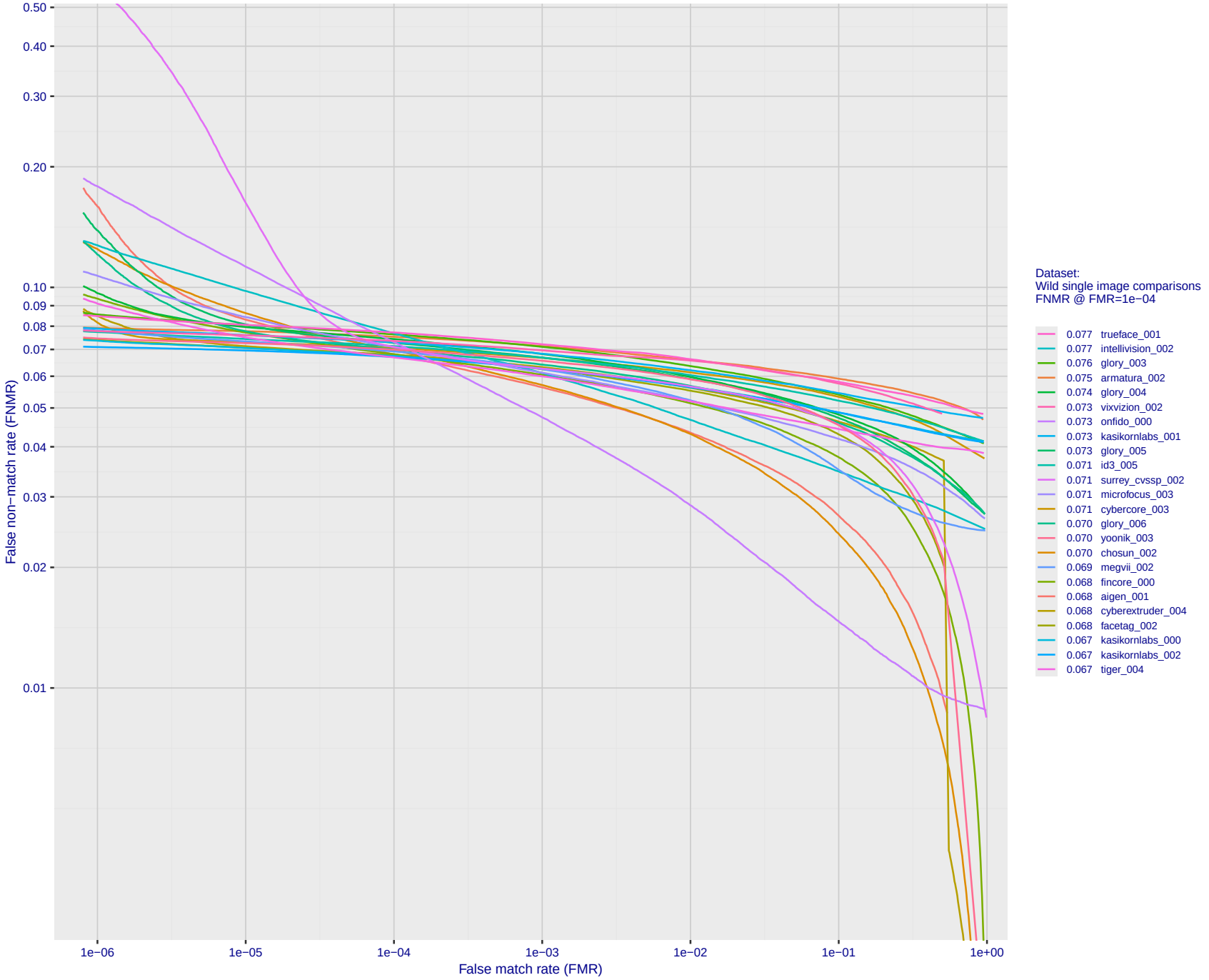


Figure 323: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

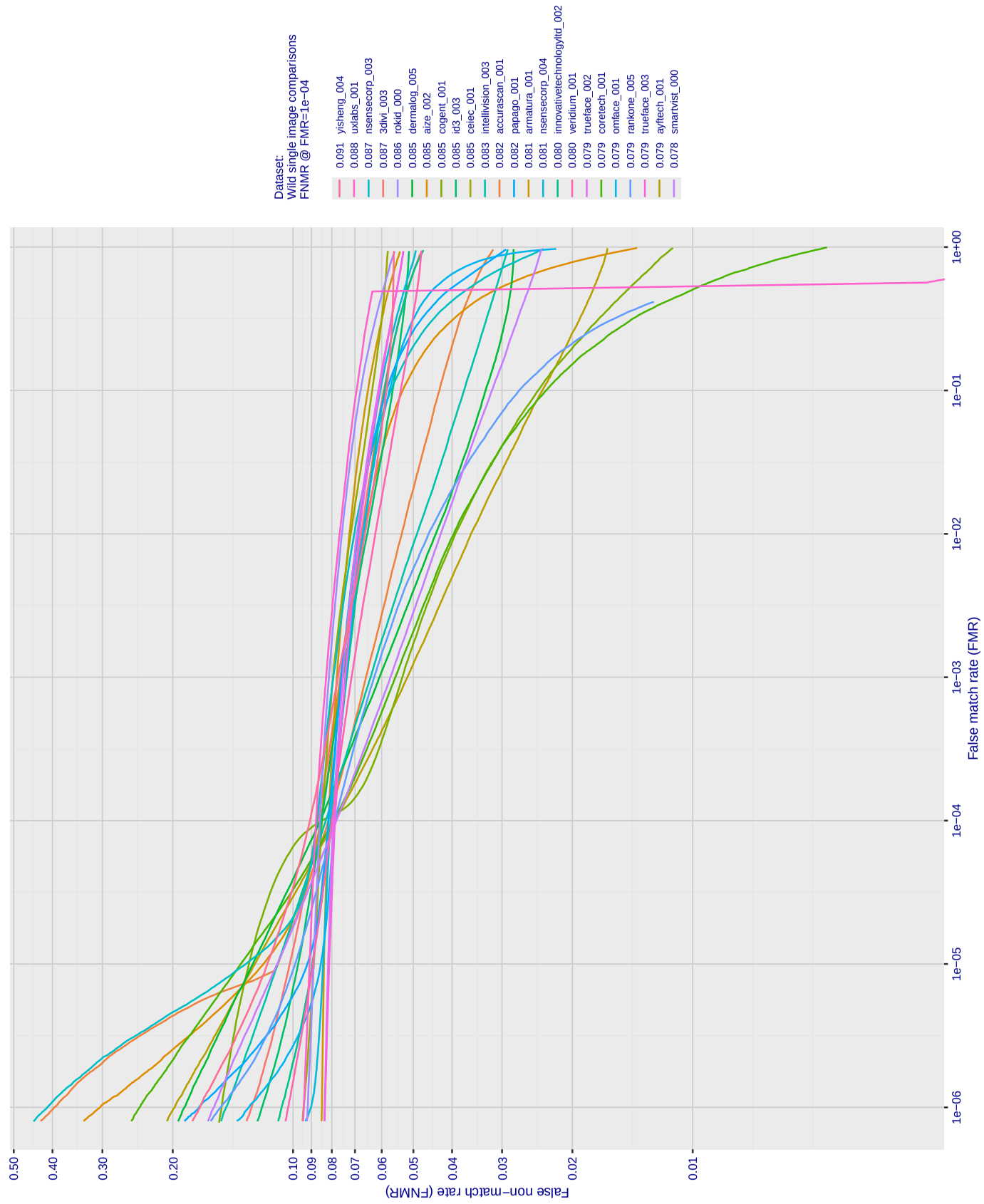


Figure 324: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

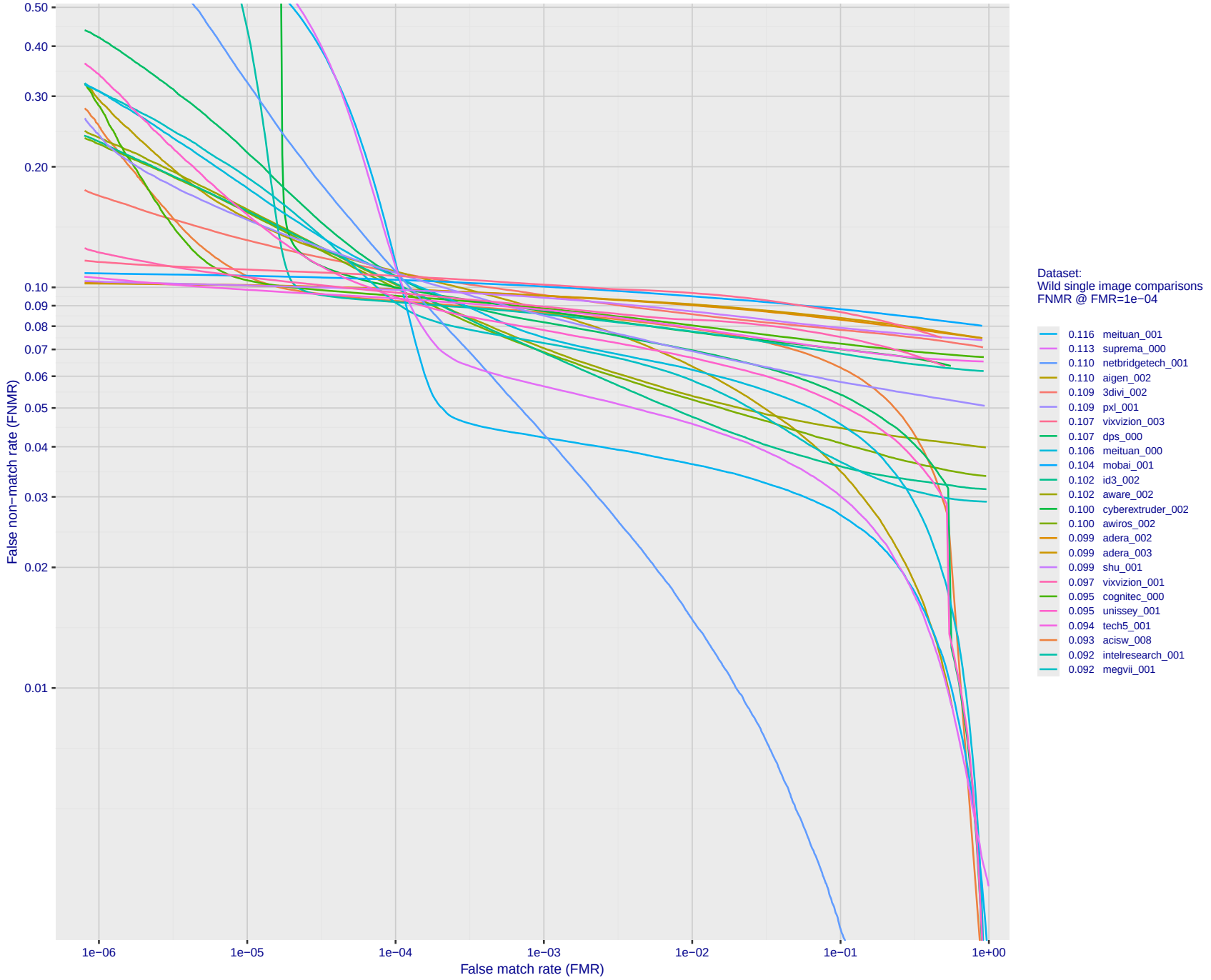


Figure 325: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

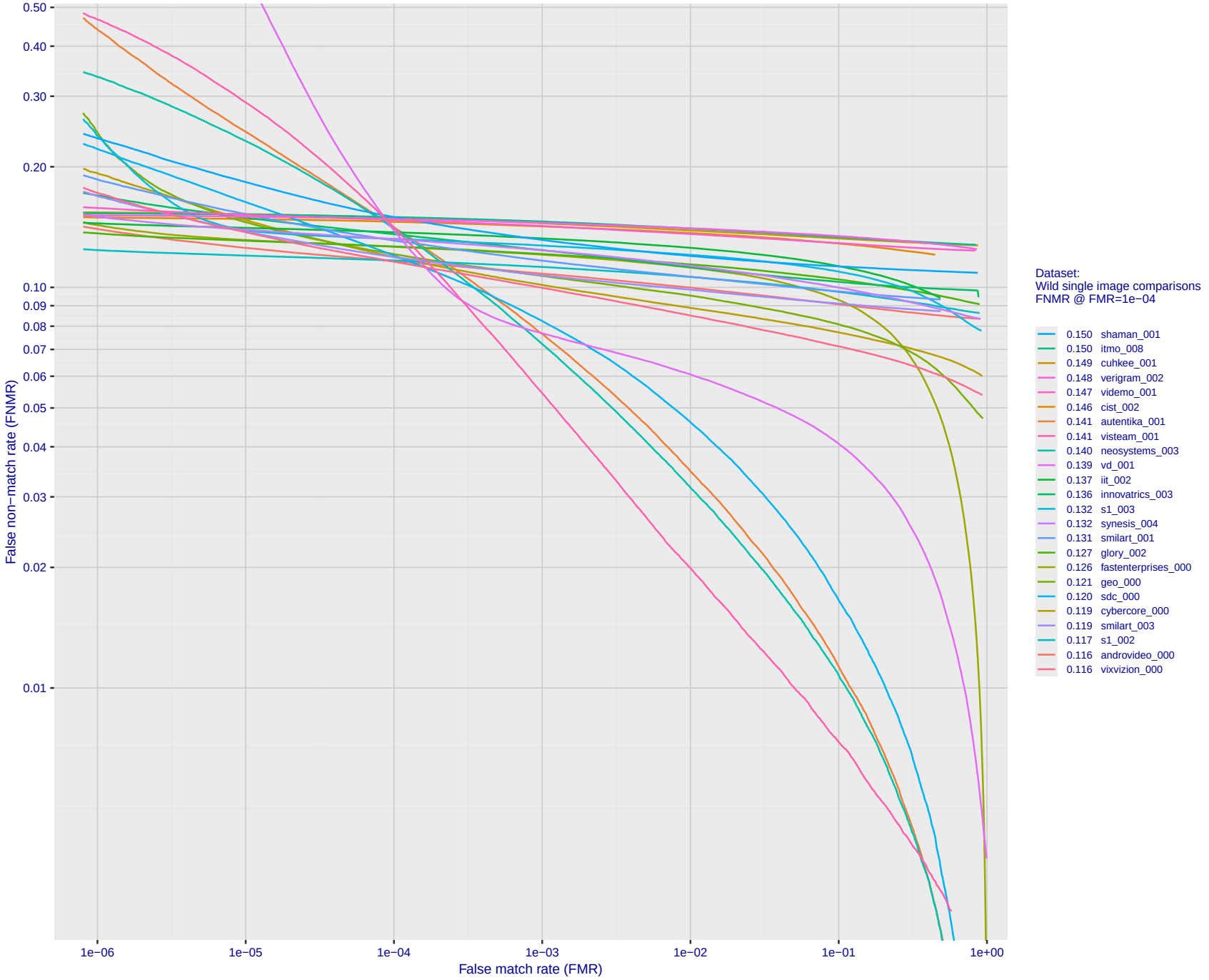


Figure 326: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

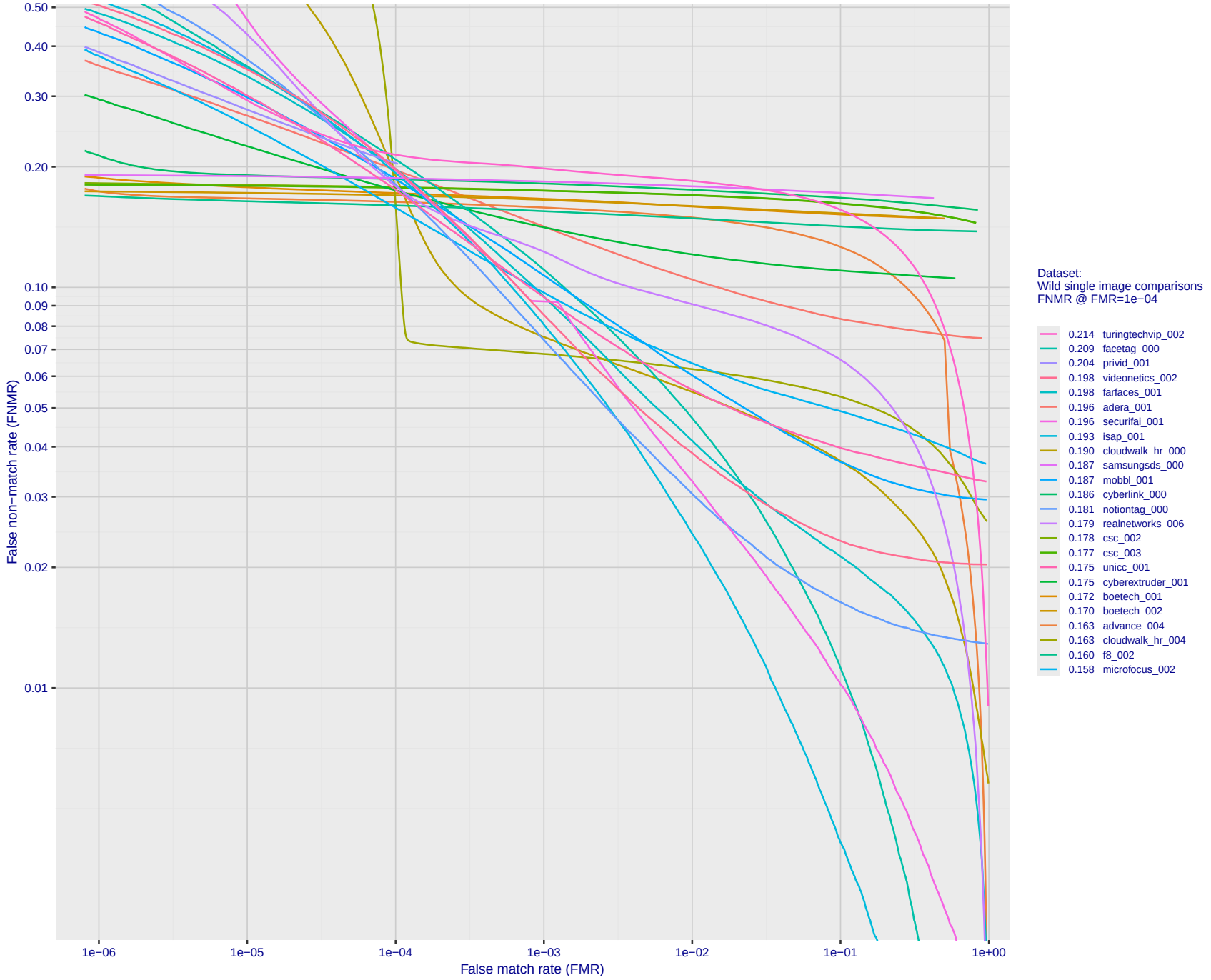


Figure 327: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

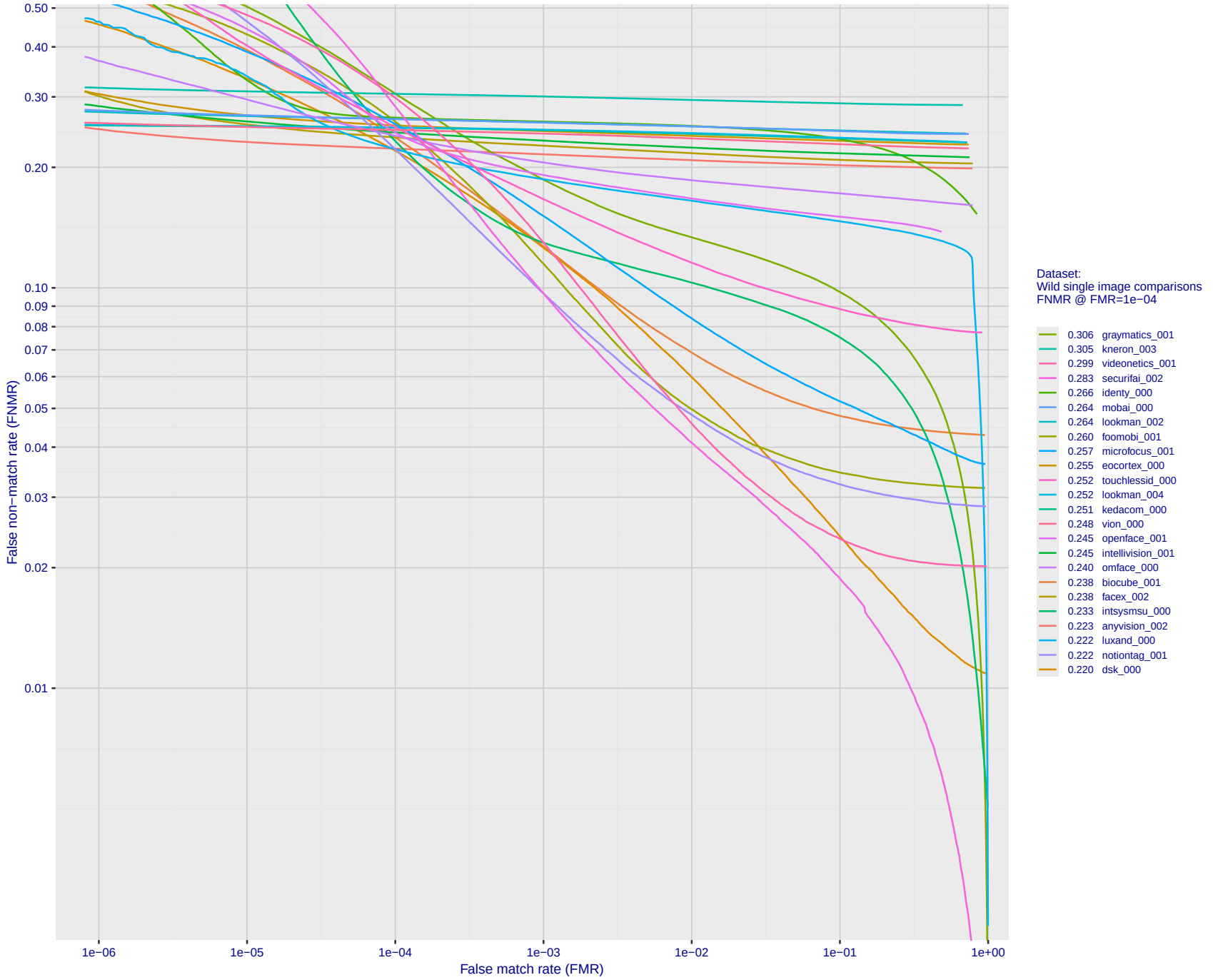


Figure 328: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

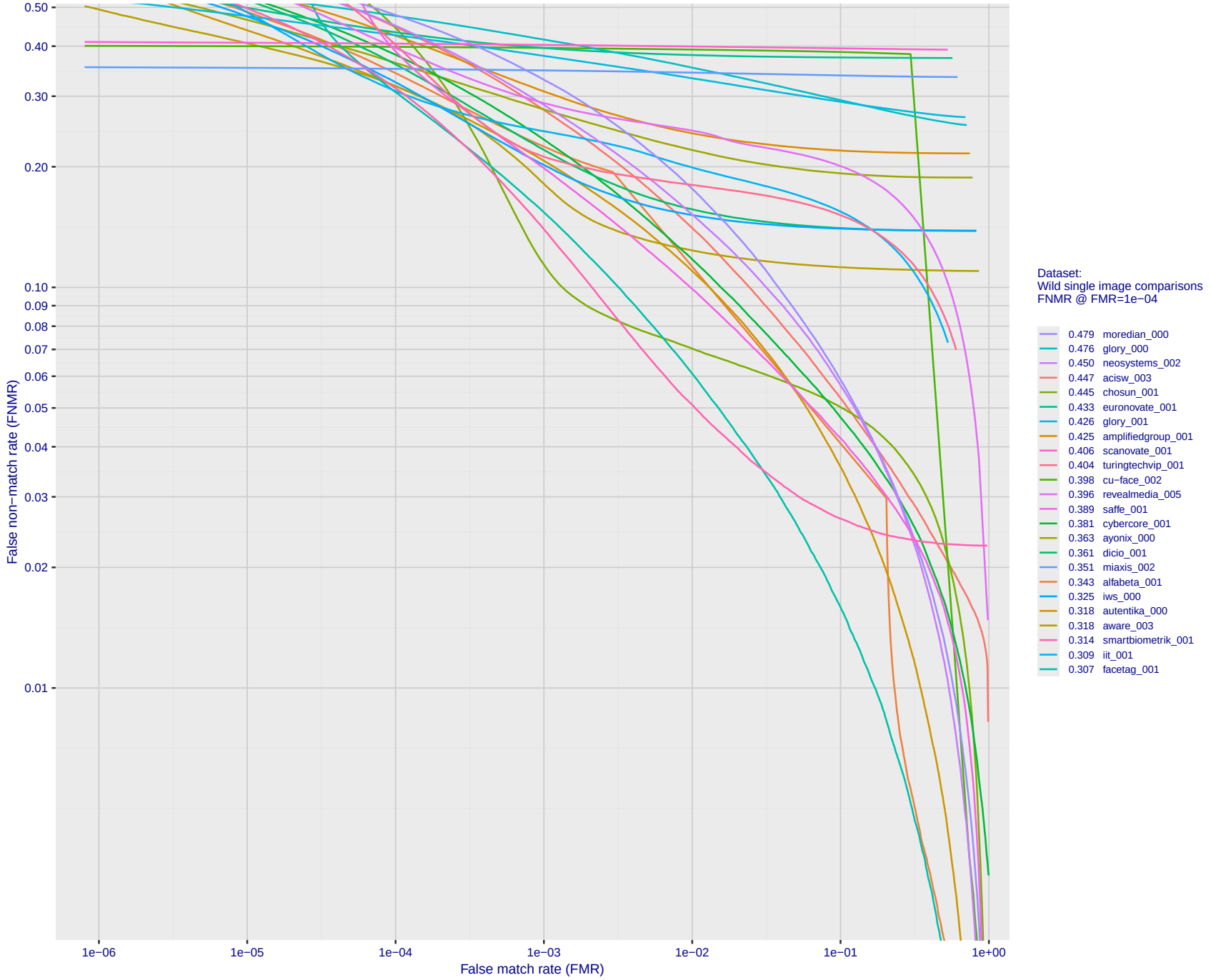


Figure 329: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

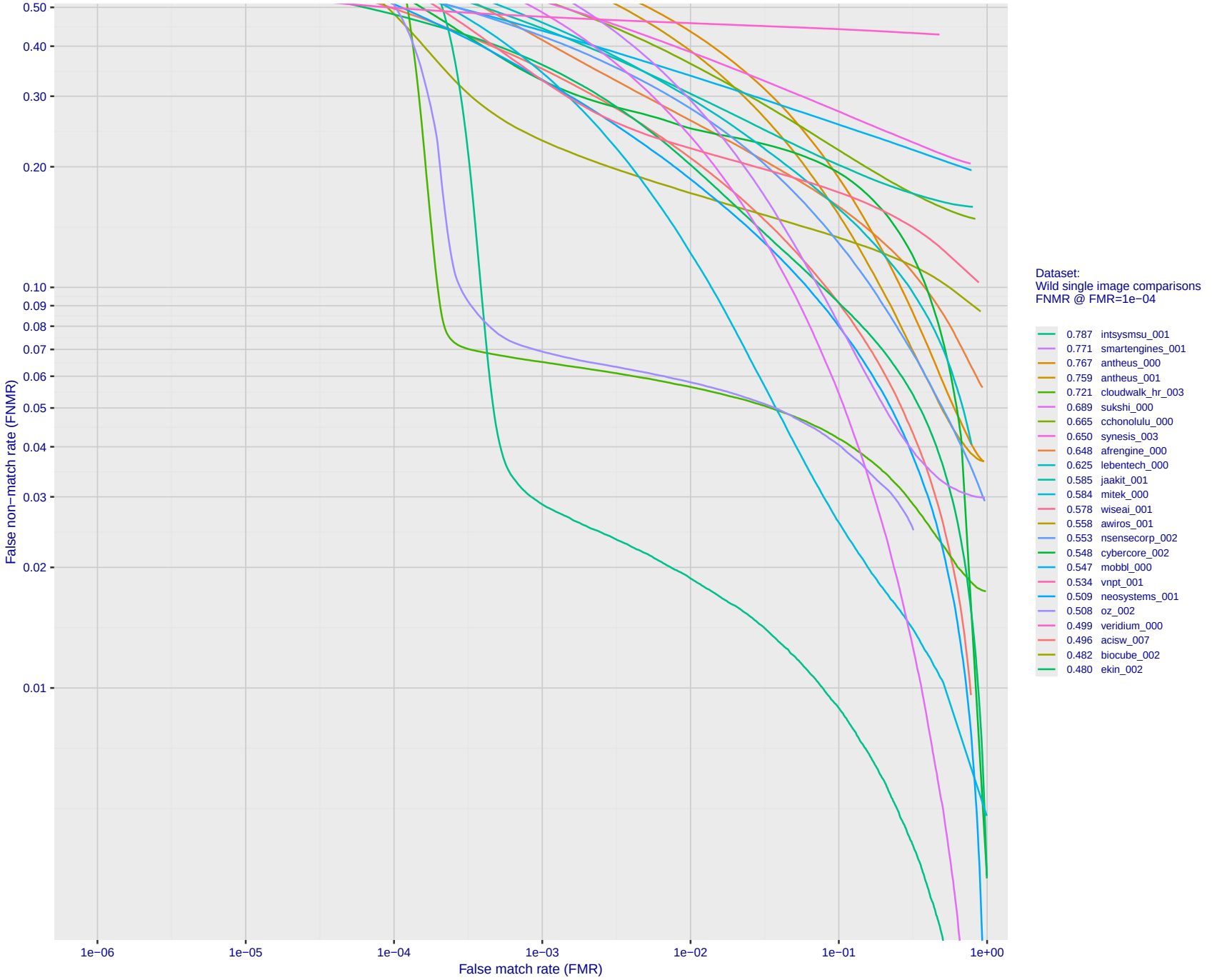


Figure 330: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

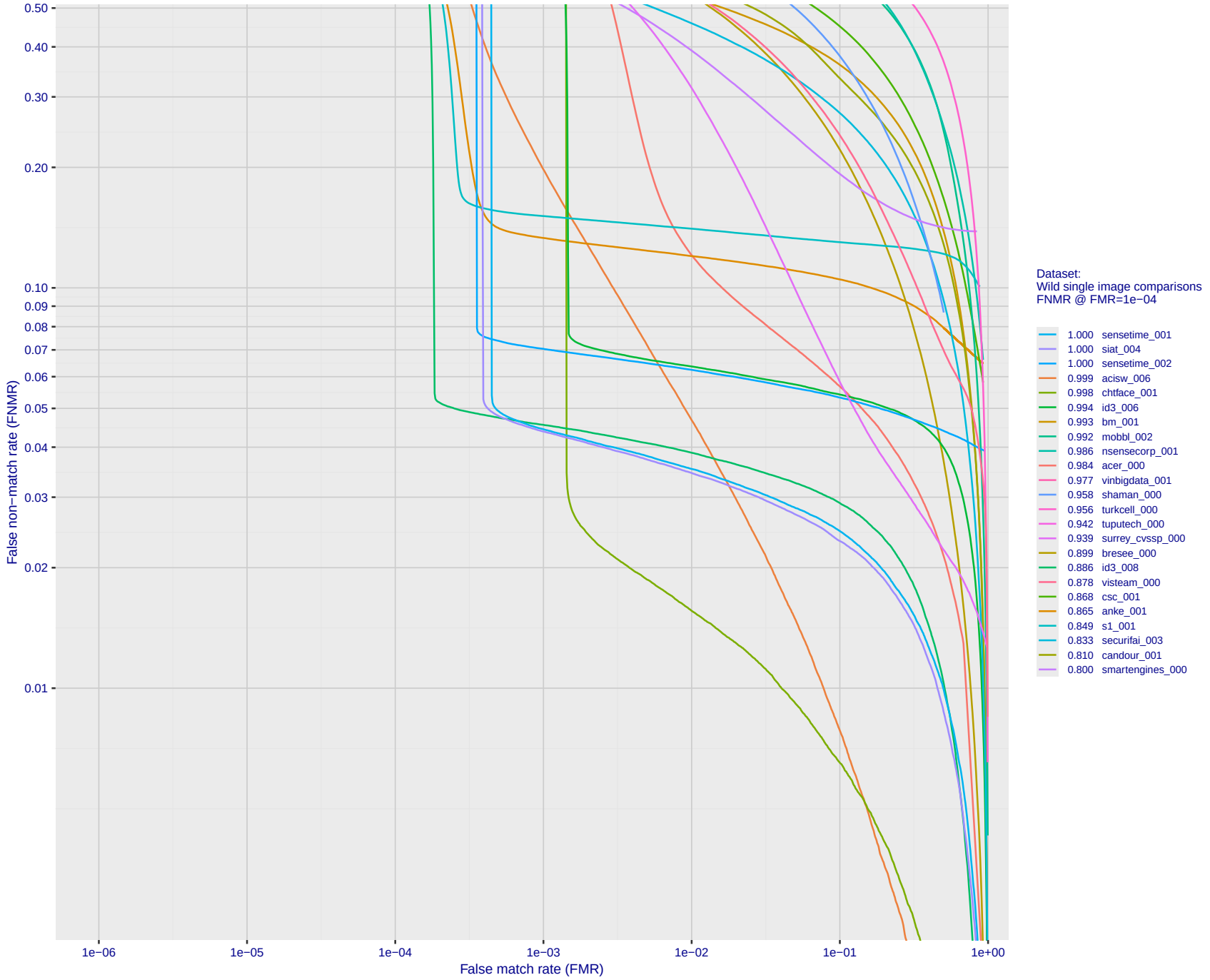


Figure 331: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

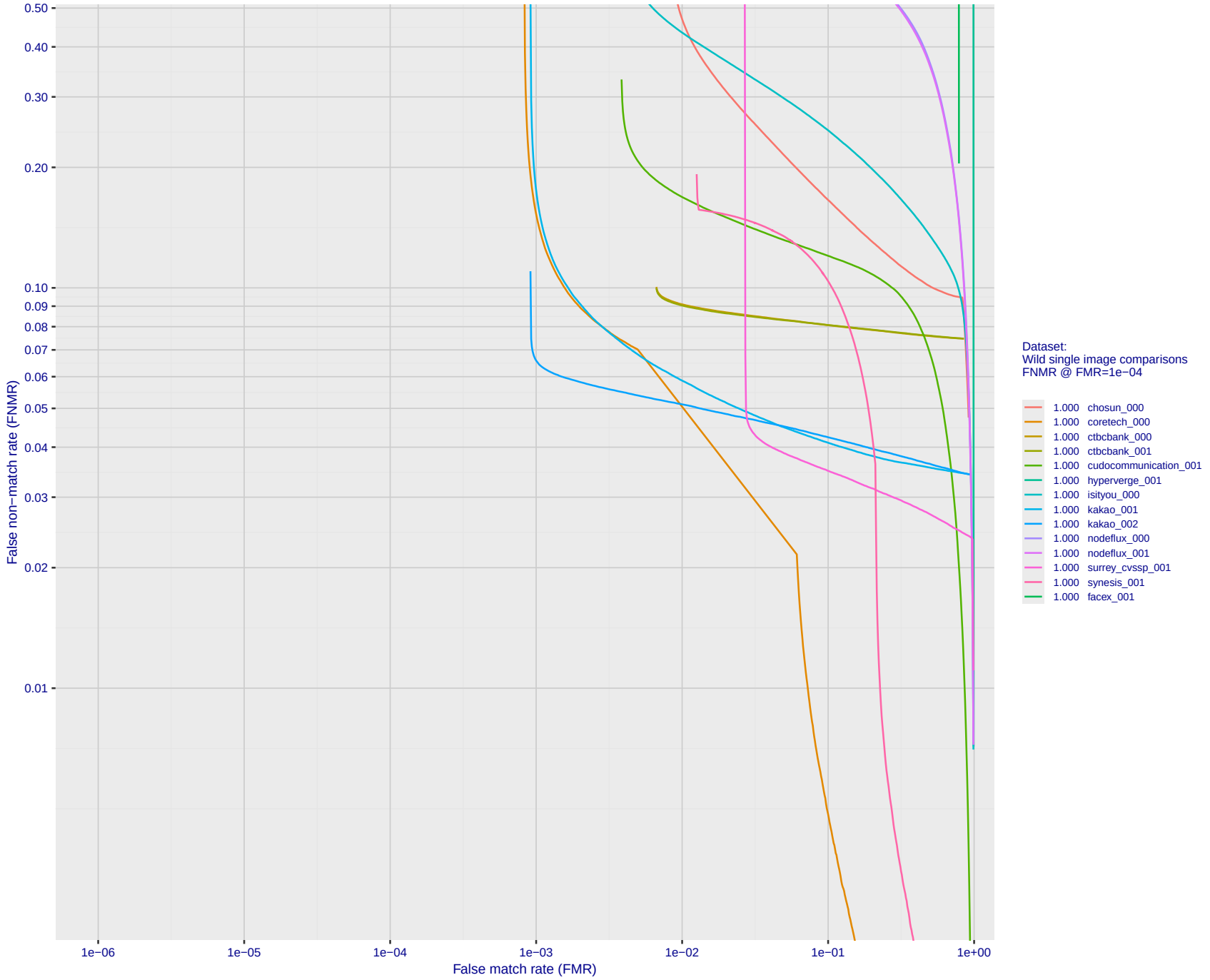


Figure 332: For the 2018 wild image comparisons, detection error tradeoff (DET) characteristics showing false non-match rate vs. false match rate plotted parametrically on threshold, T. The scales are logarithmic in order to show several decades of FMR.

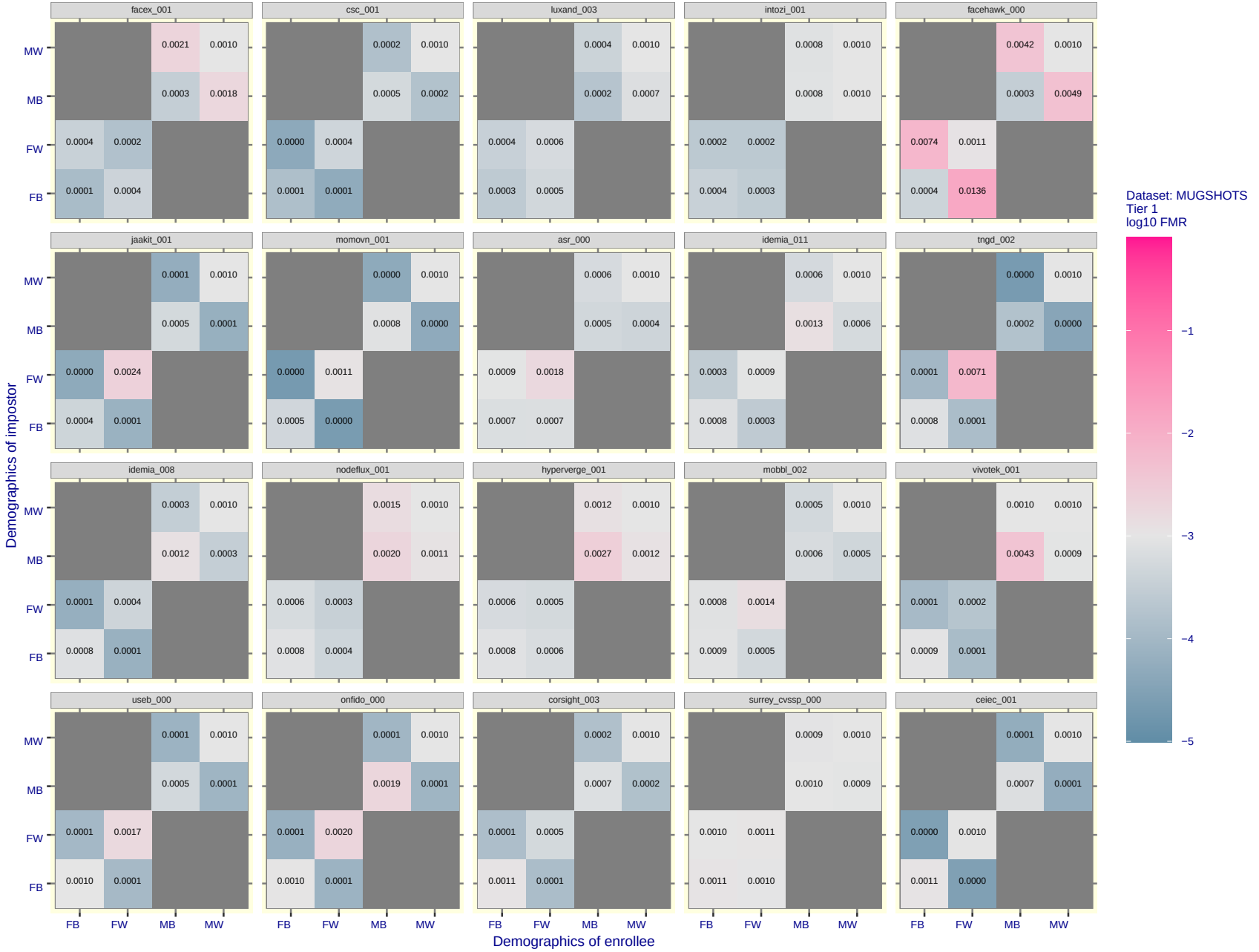


Figure 333: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

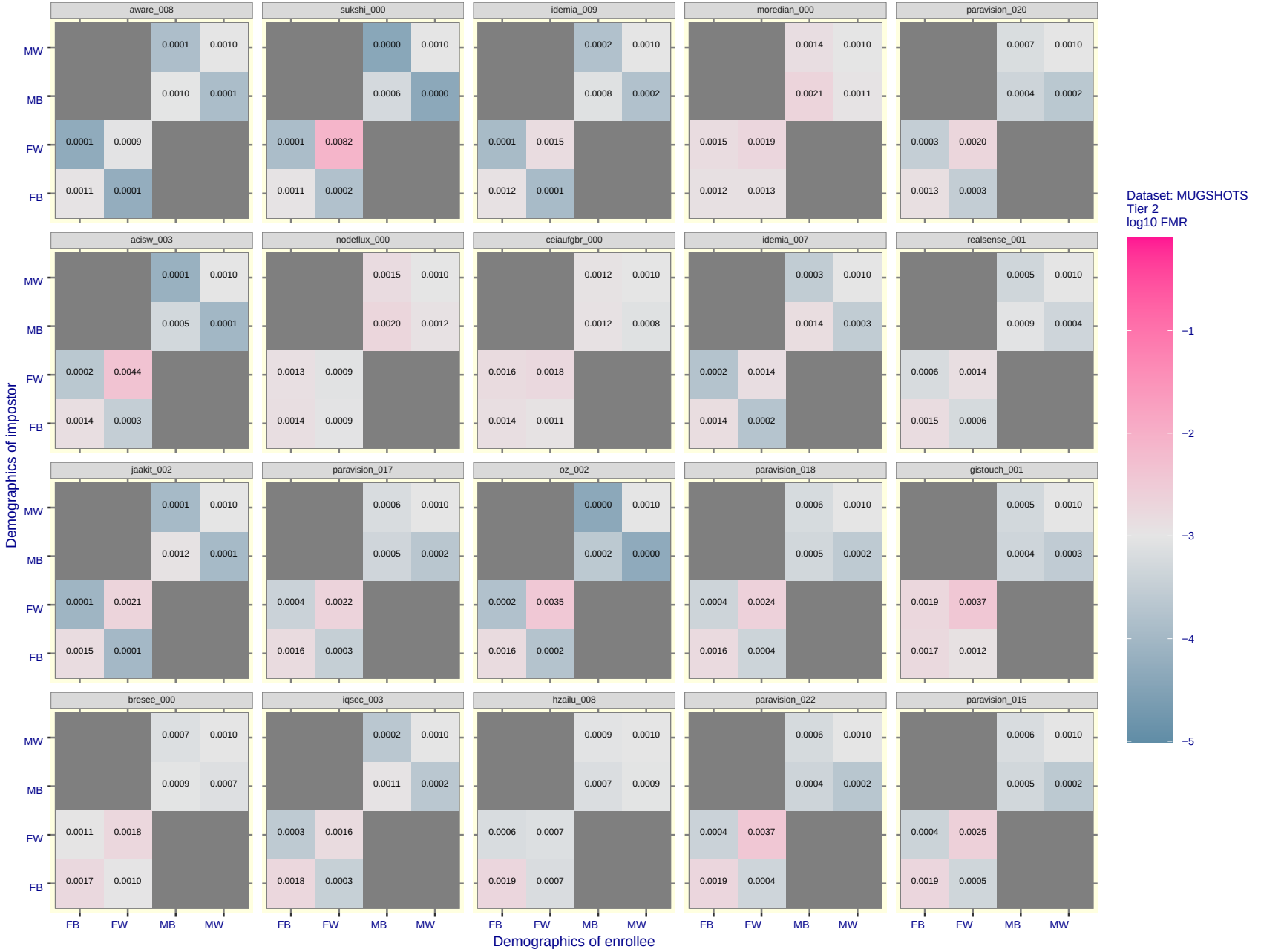


Figure 334: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

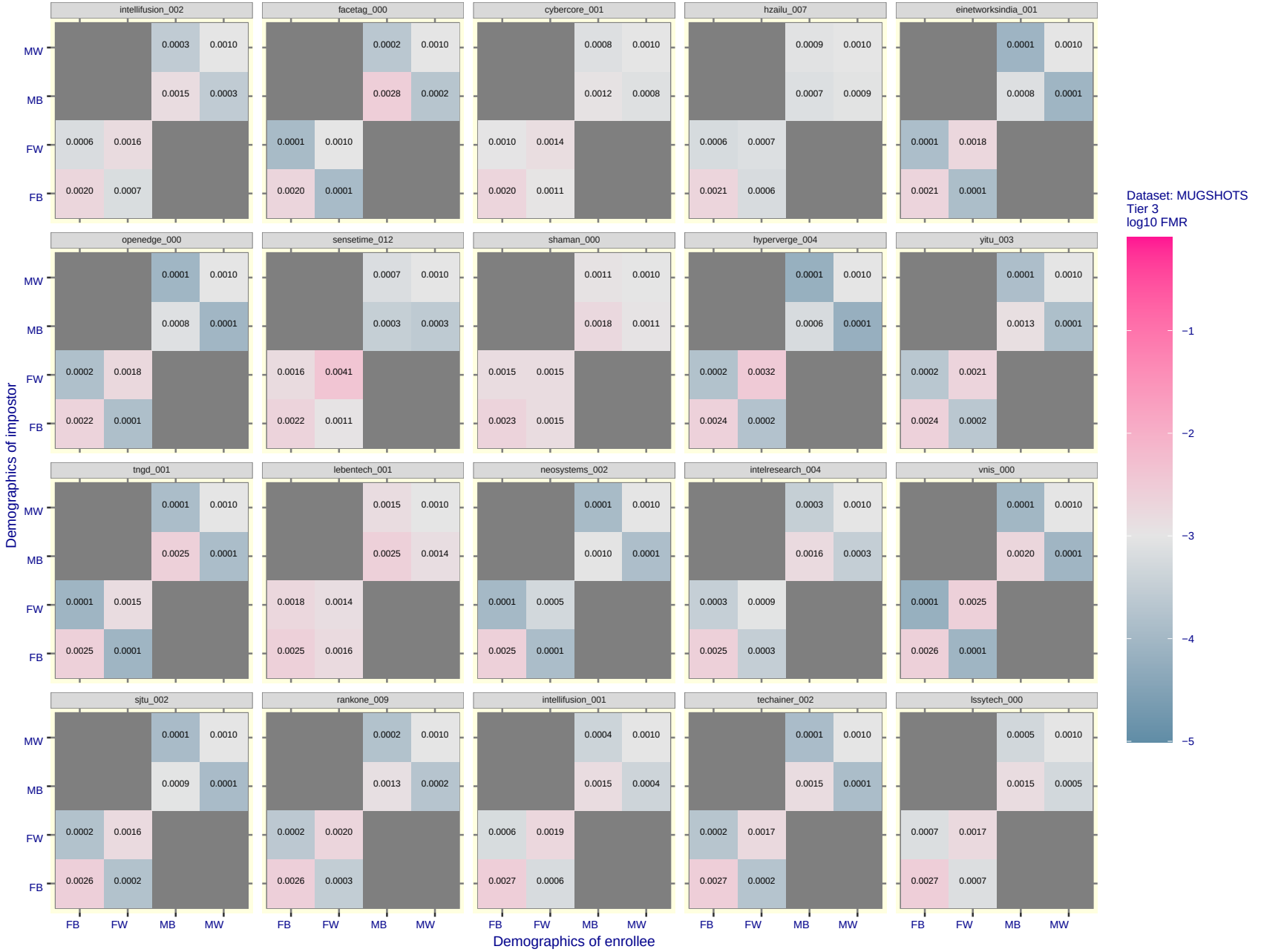


Figure 335: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

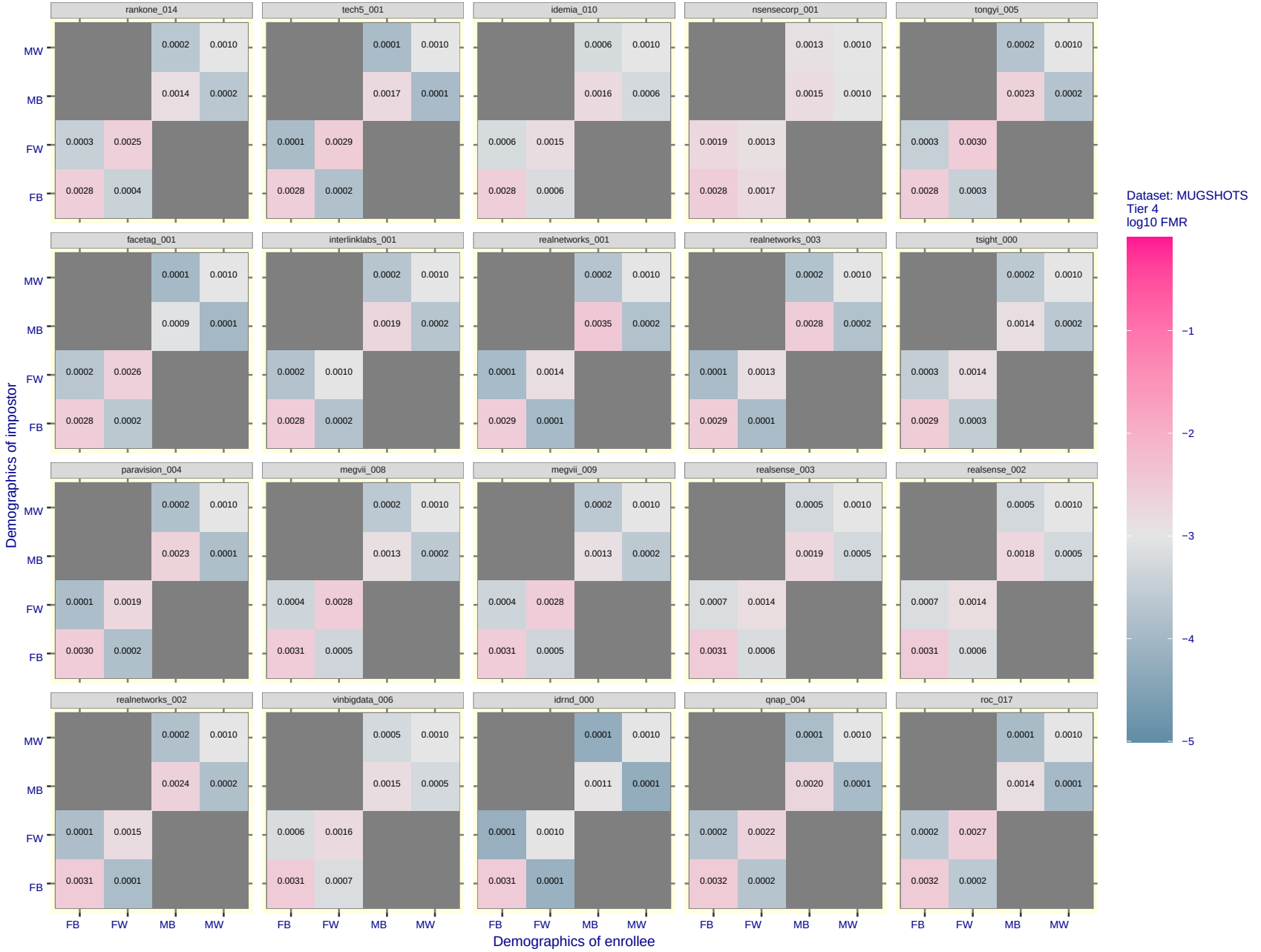


Figure 336: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

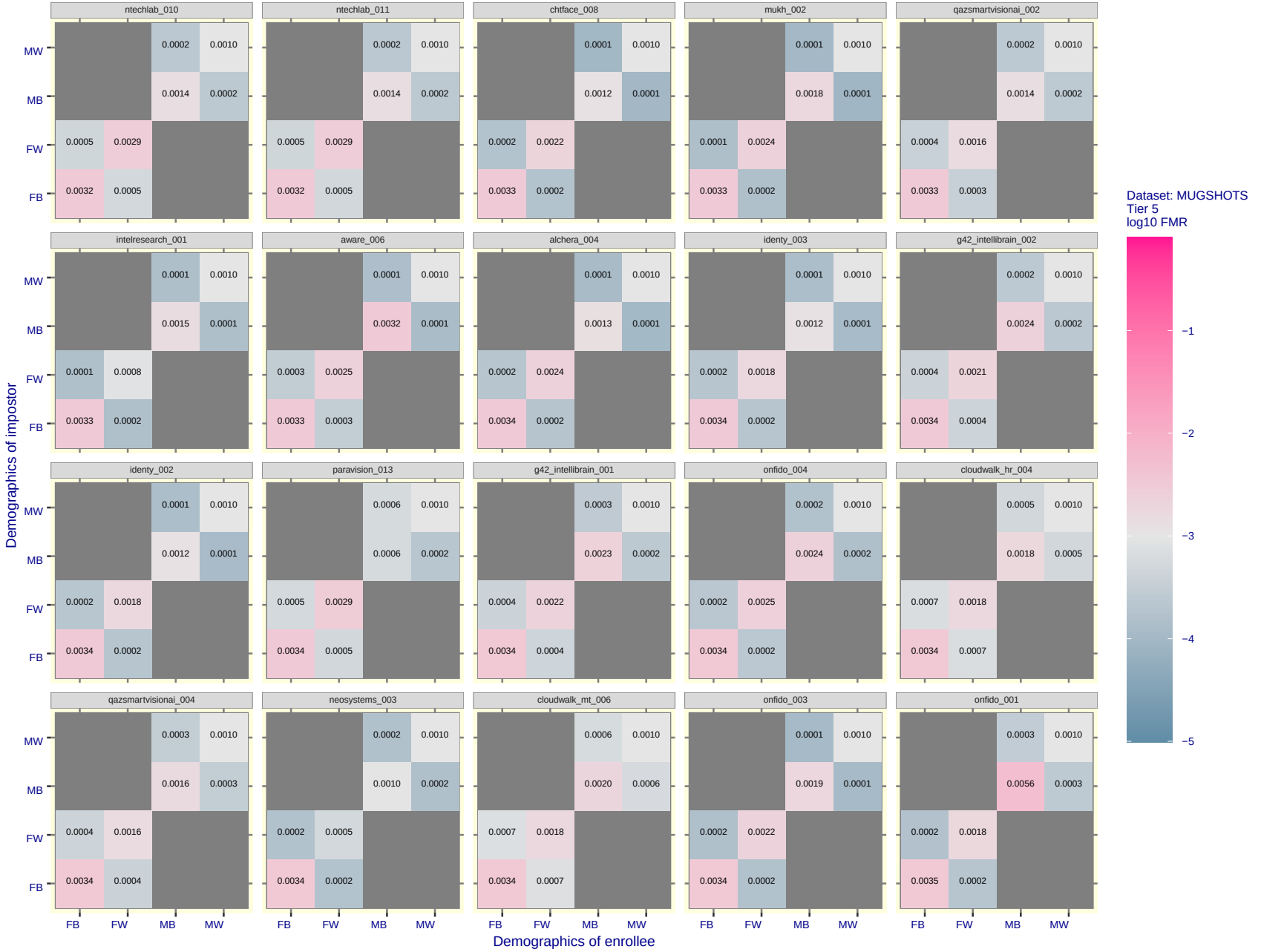


Figure 337: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

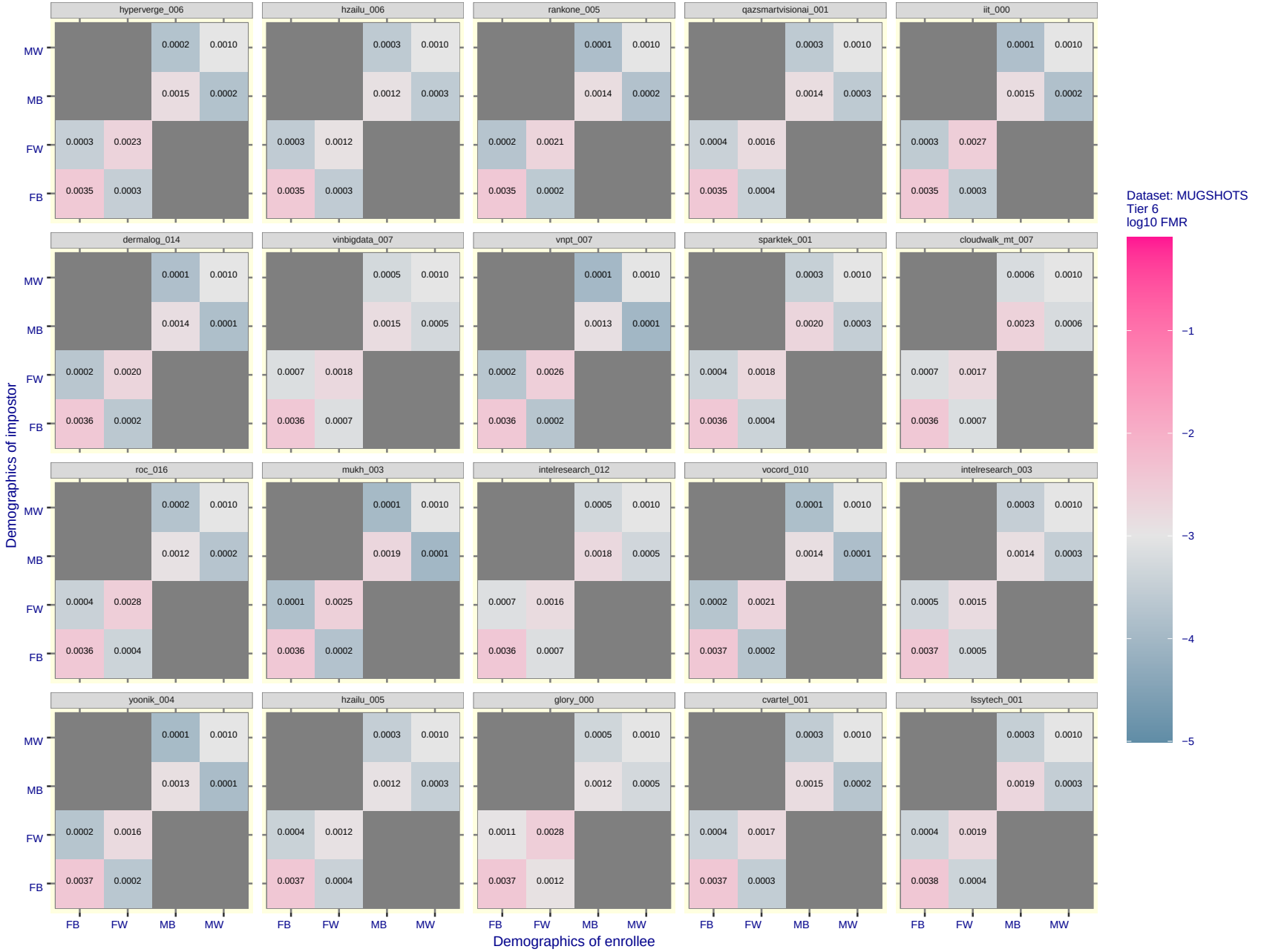


Figure 338: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

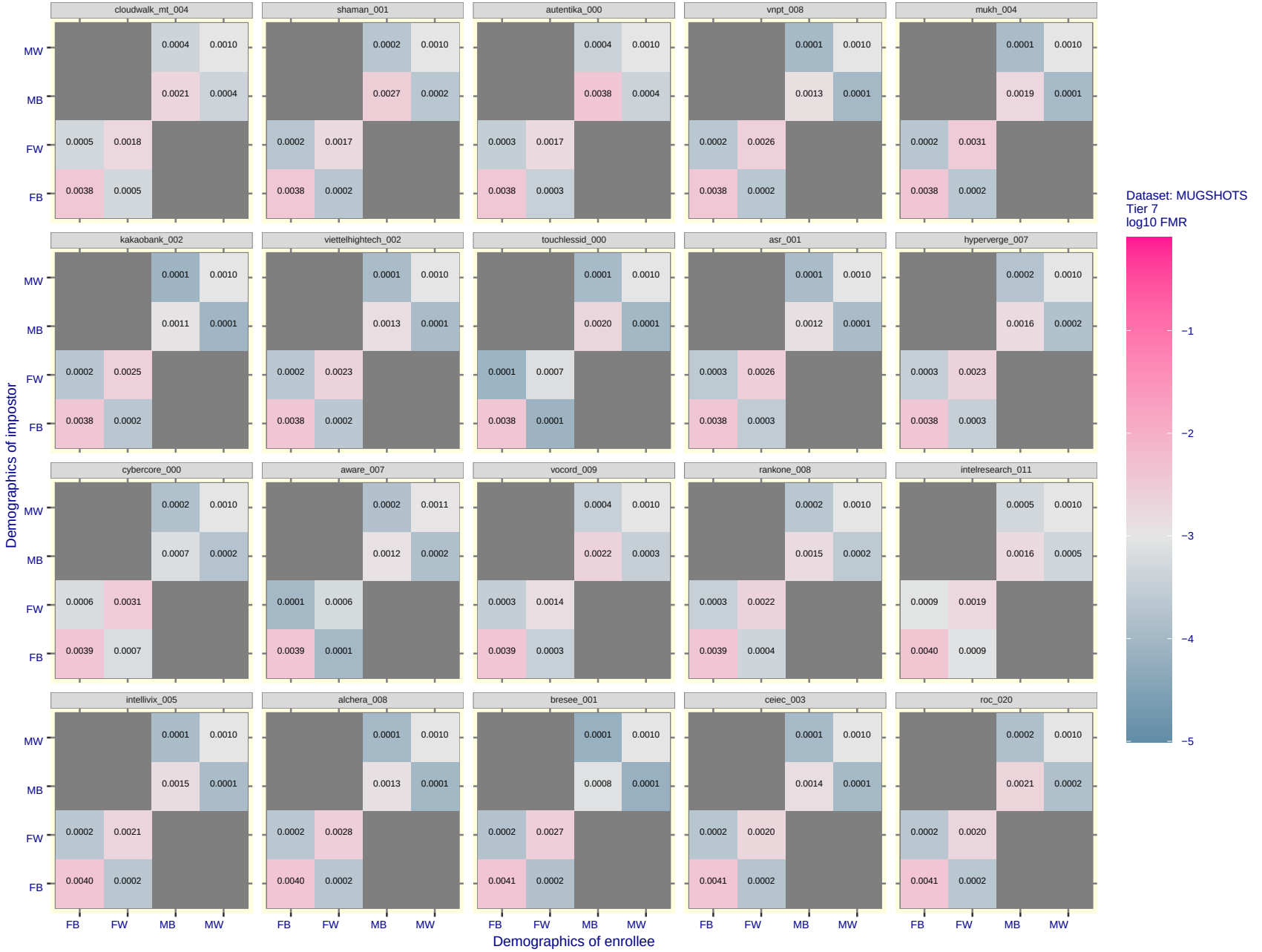


Figure 339: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

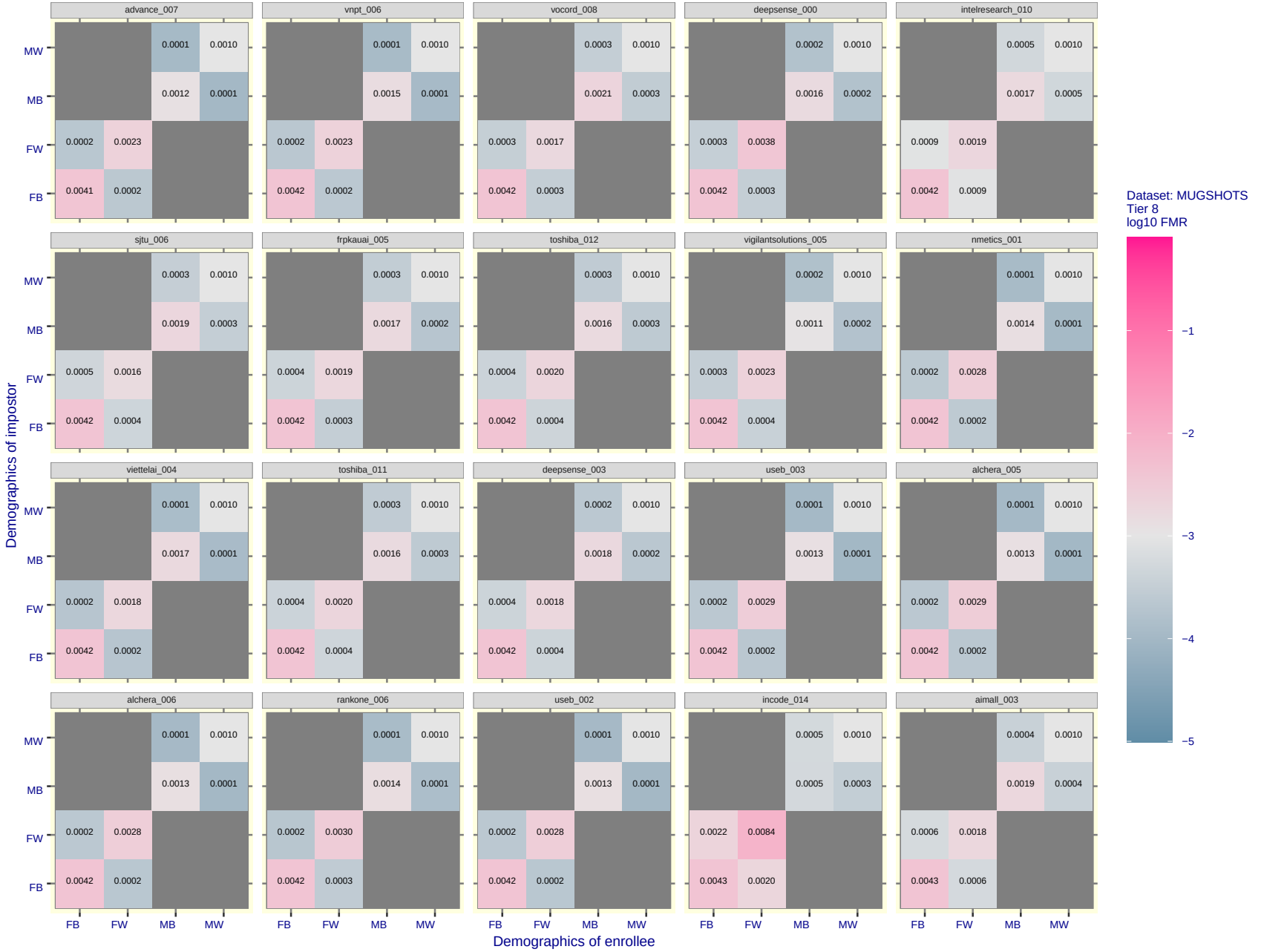


Figure 340: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

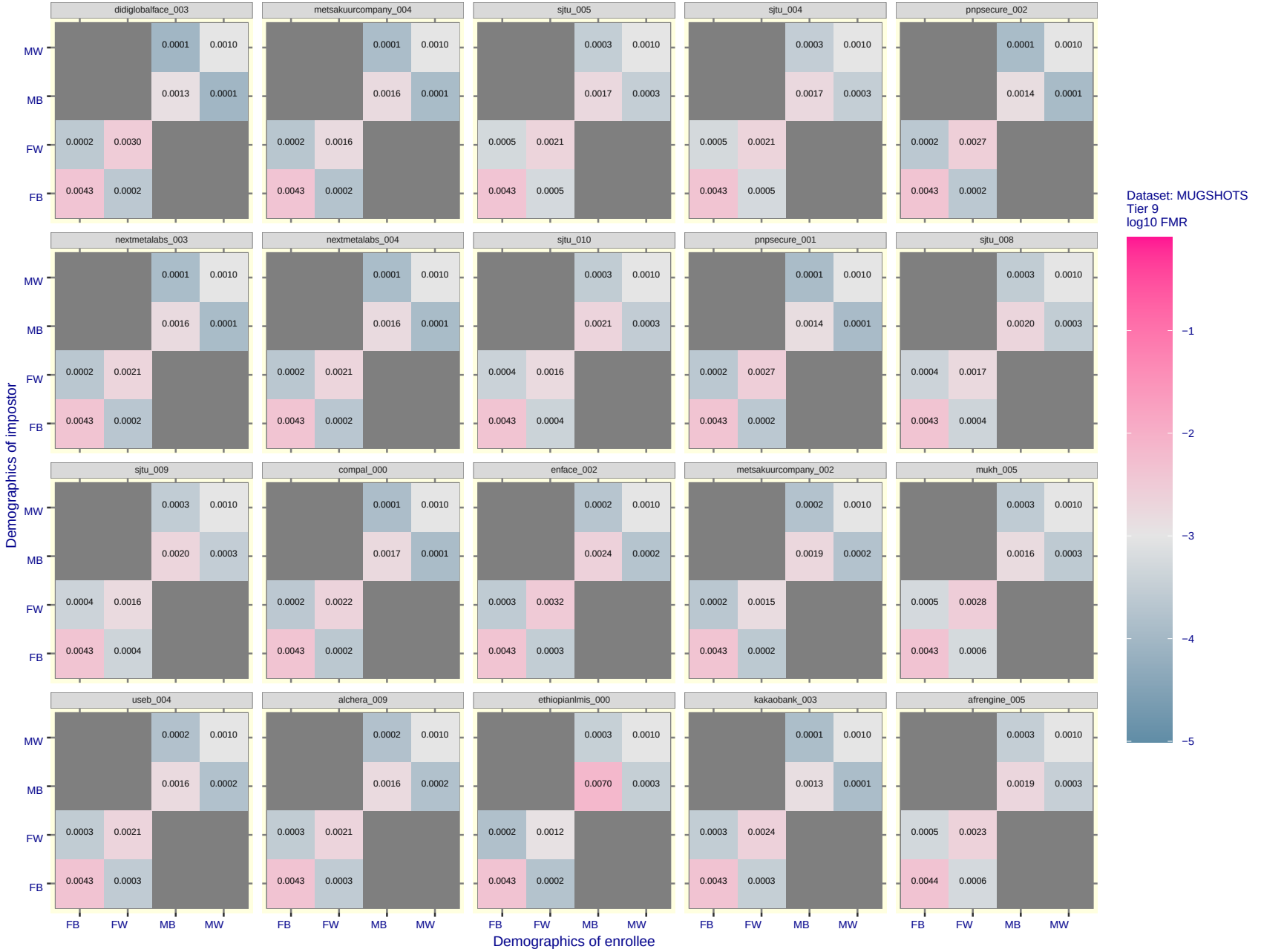


Figure 341: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

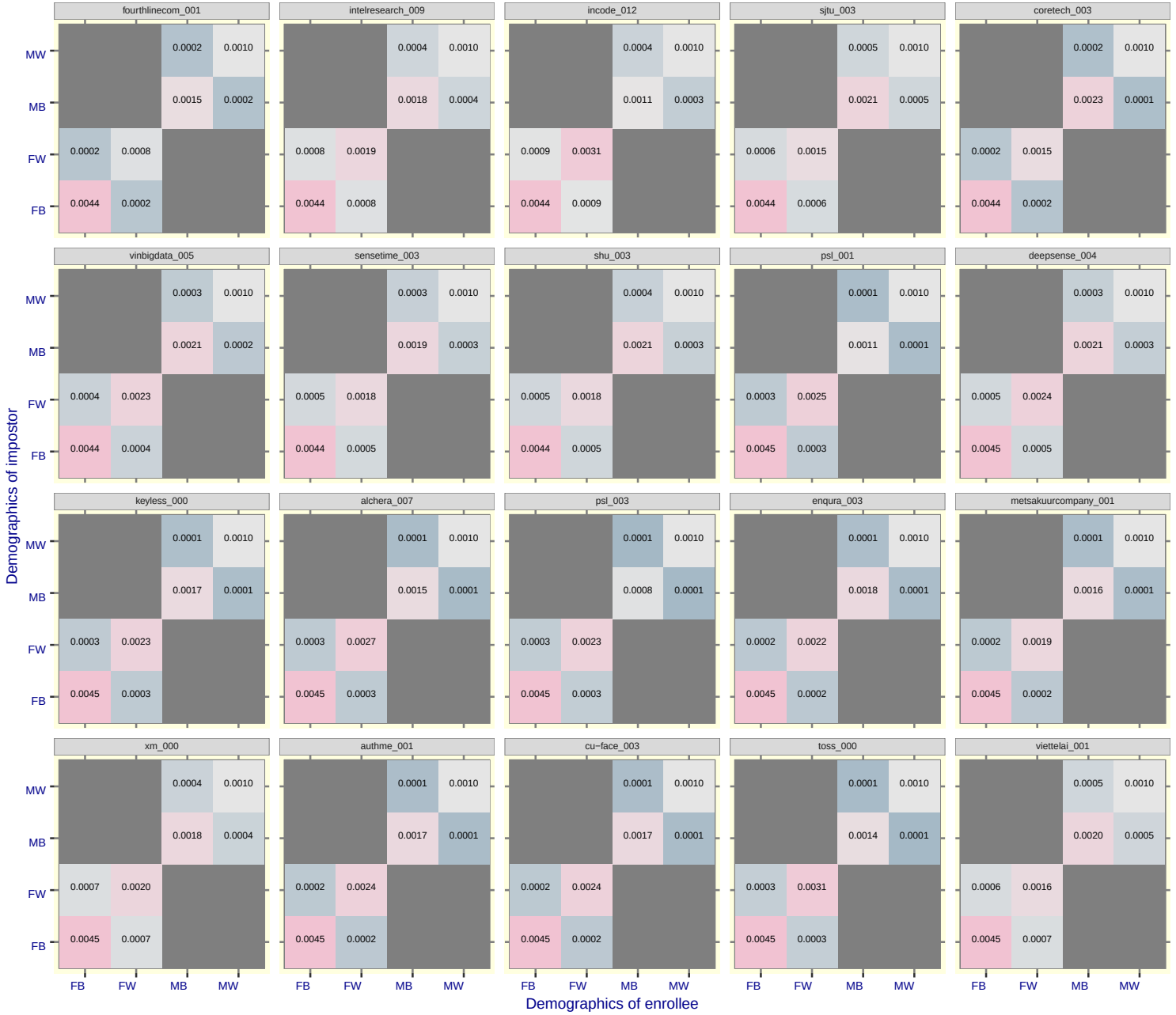


Figure 342: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

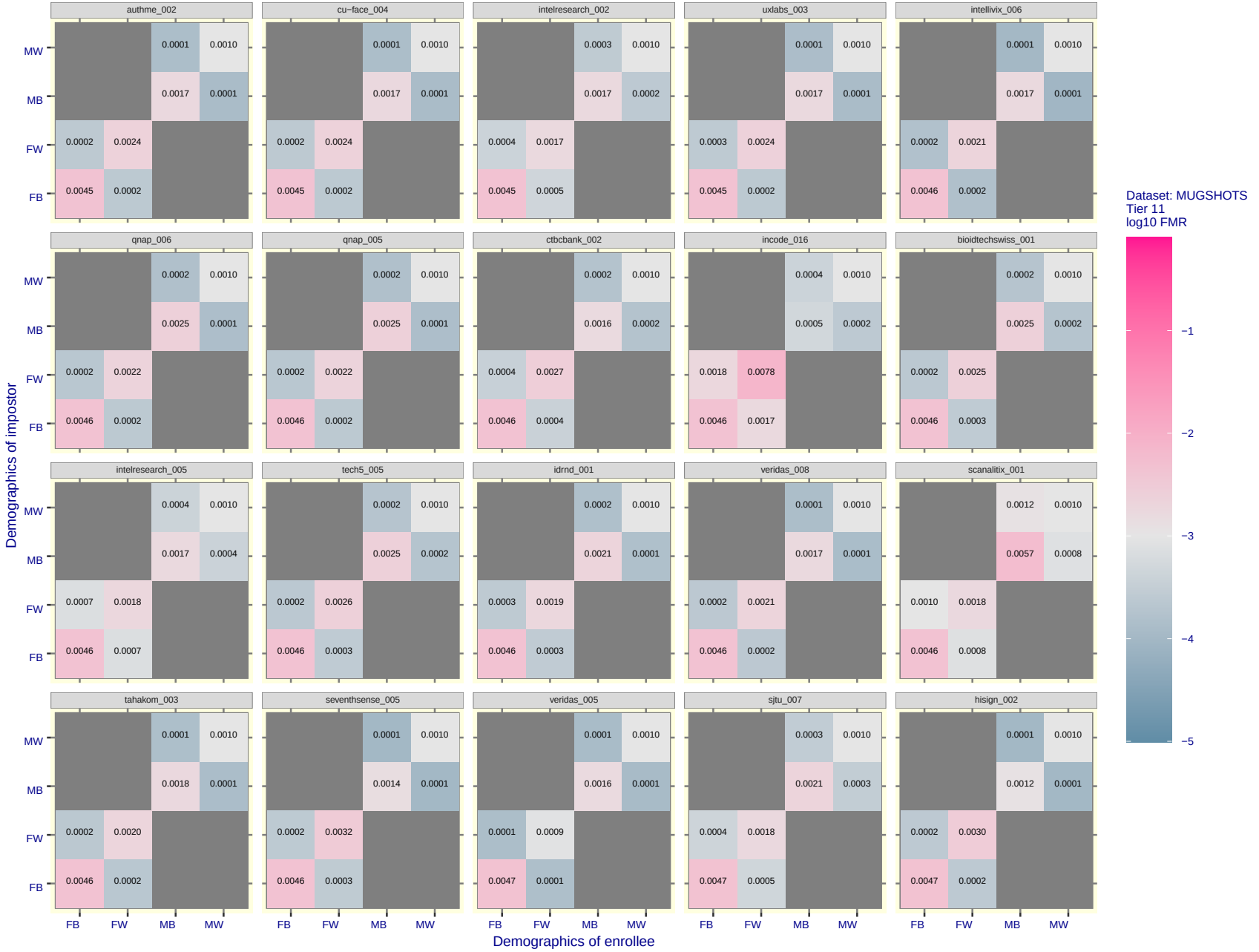


Figure 343: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

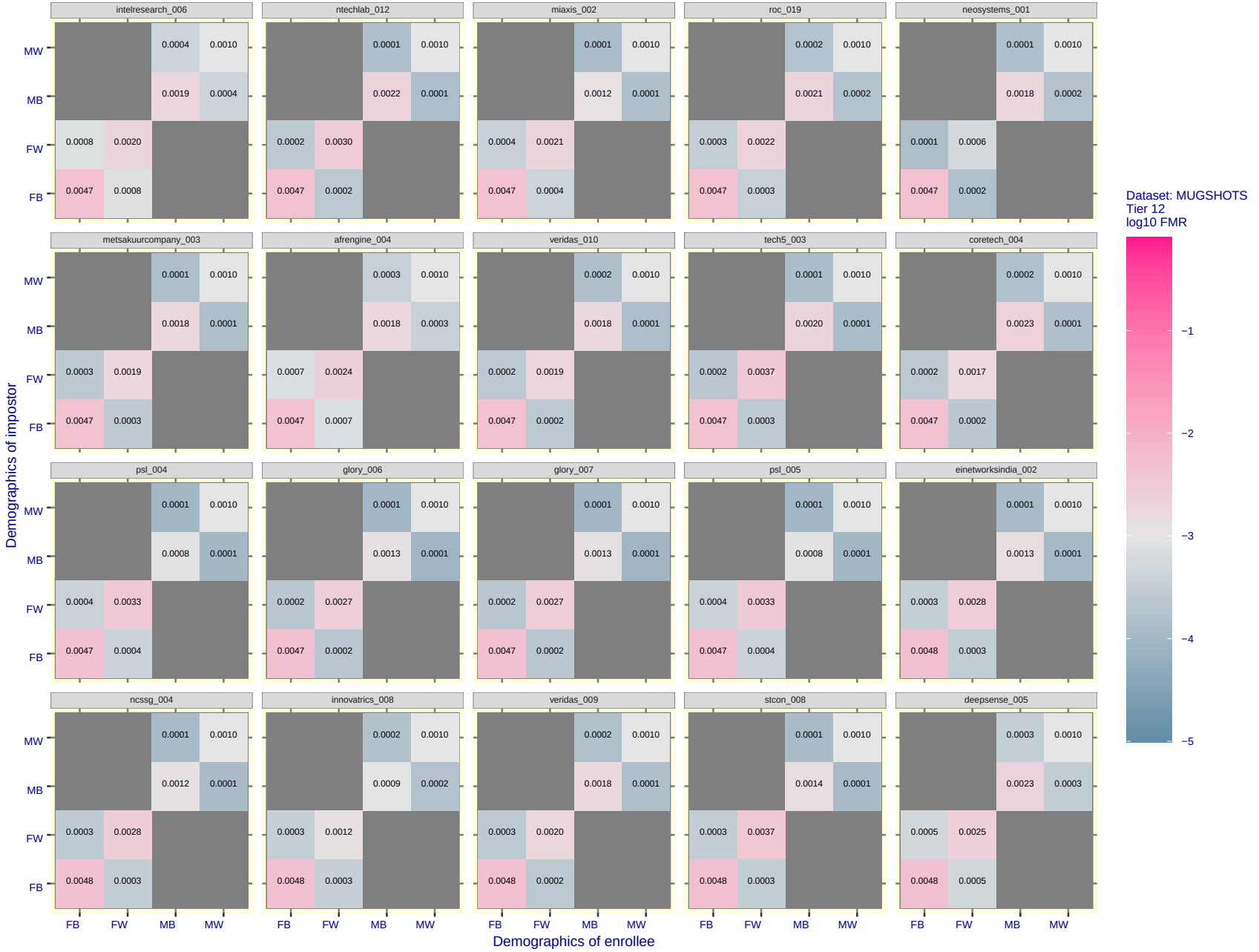


Figure 344: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

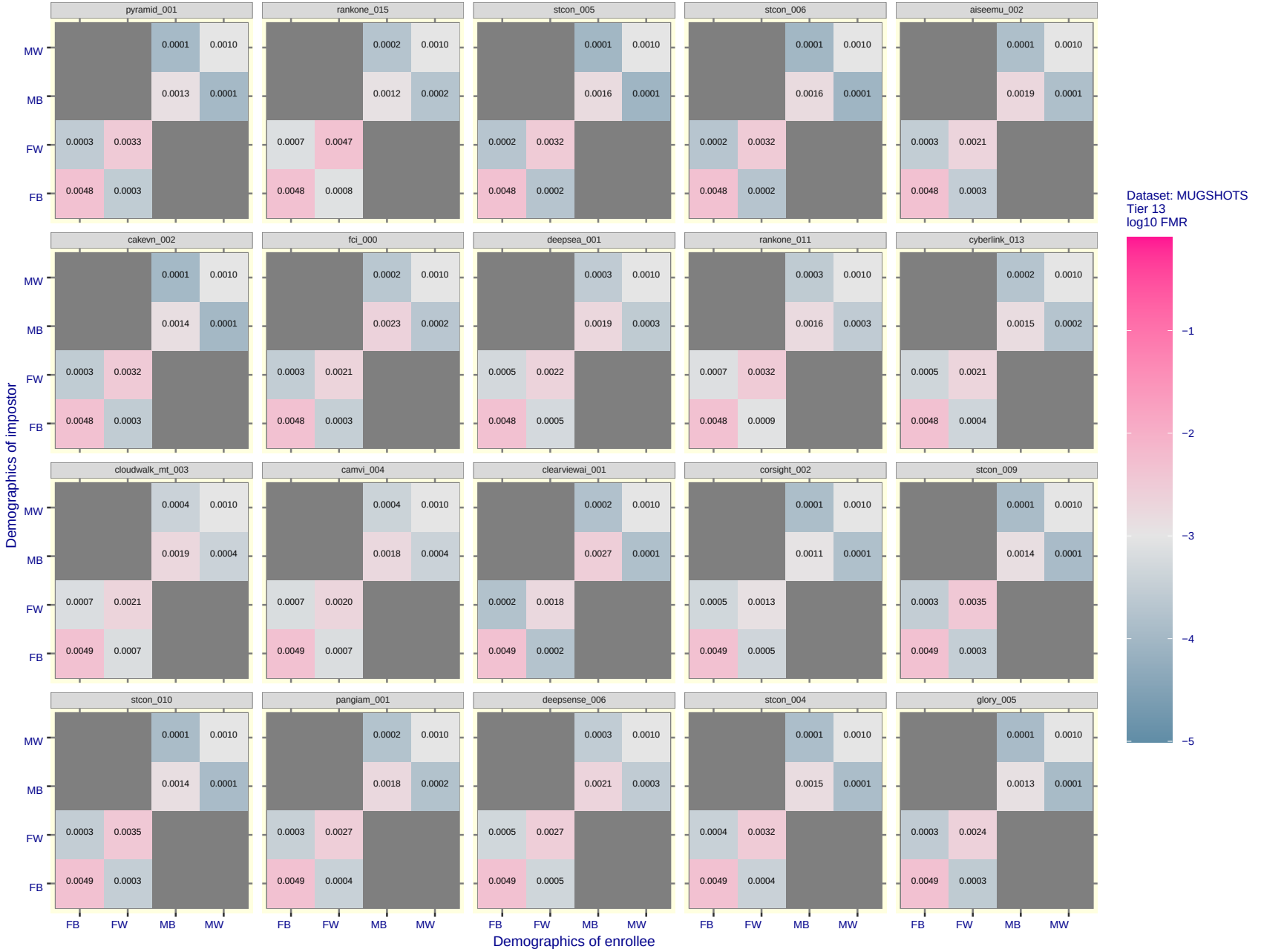


Figure 345: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

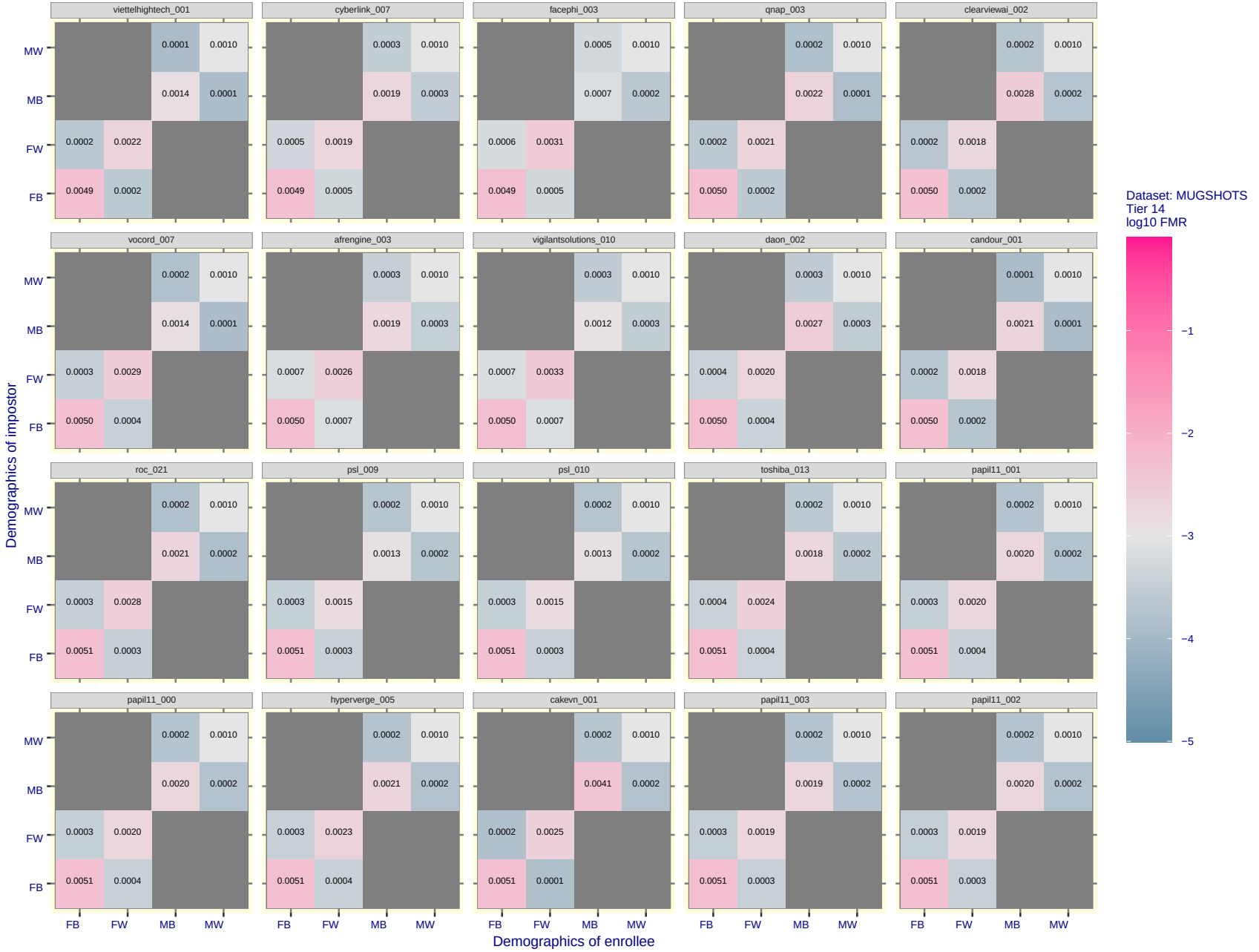


Figure 346: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

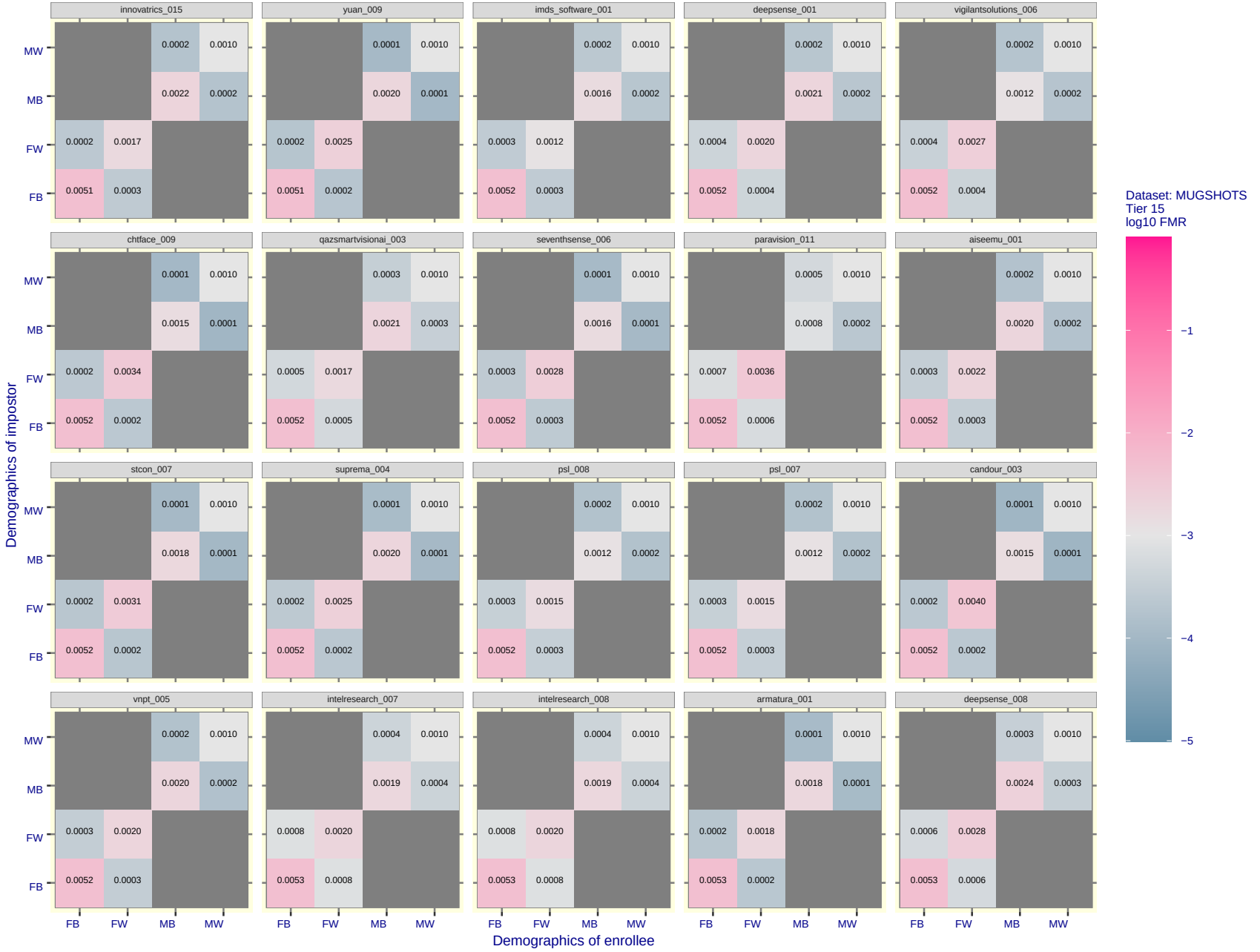


Figure 347: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

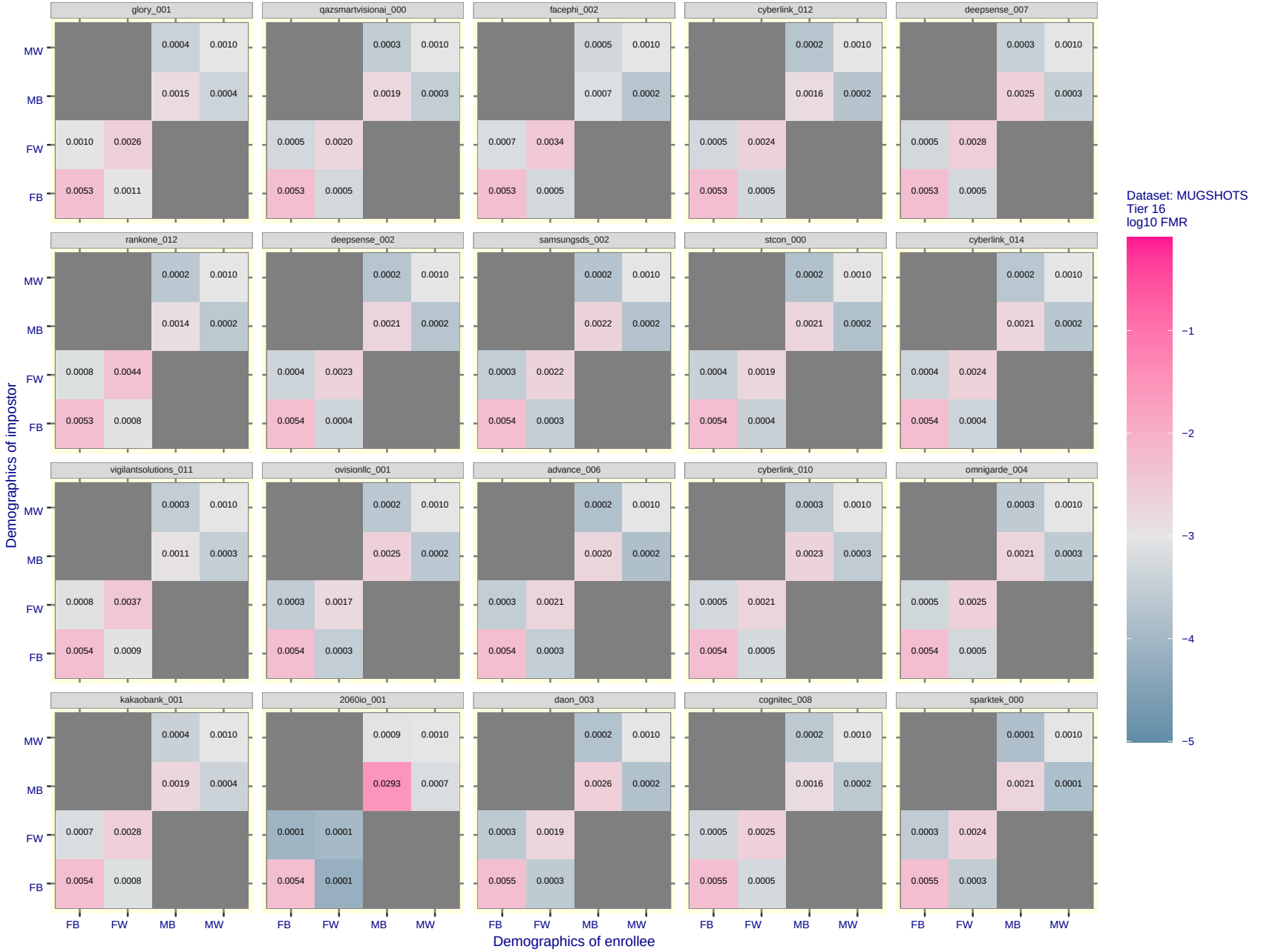


Figure 348: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

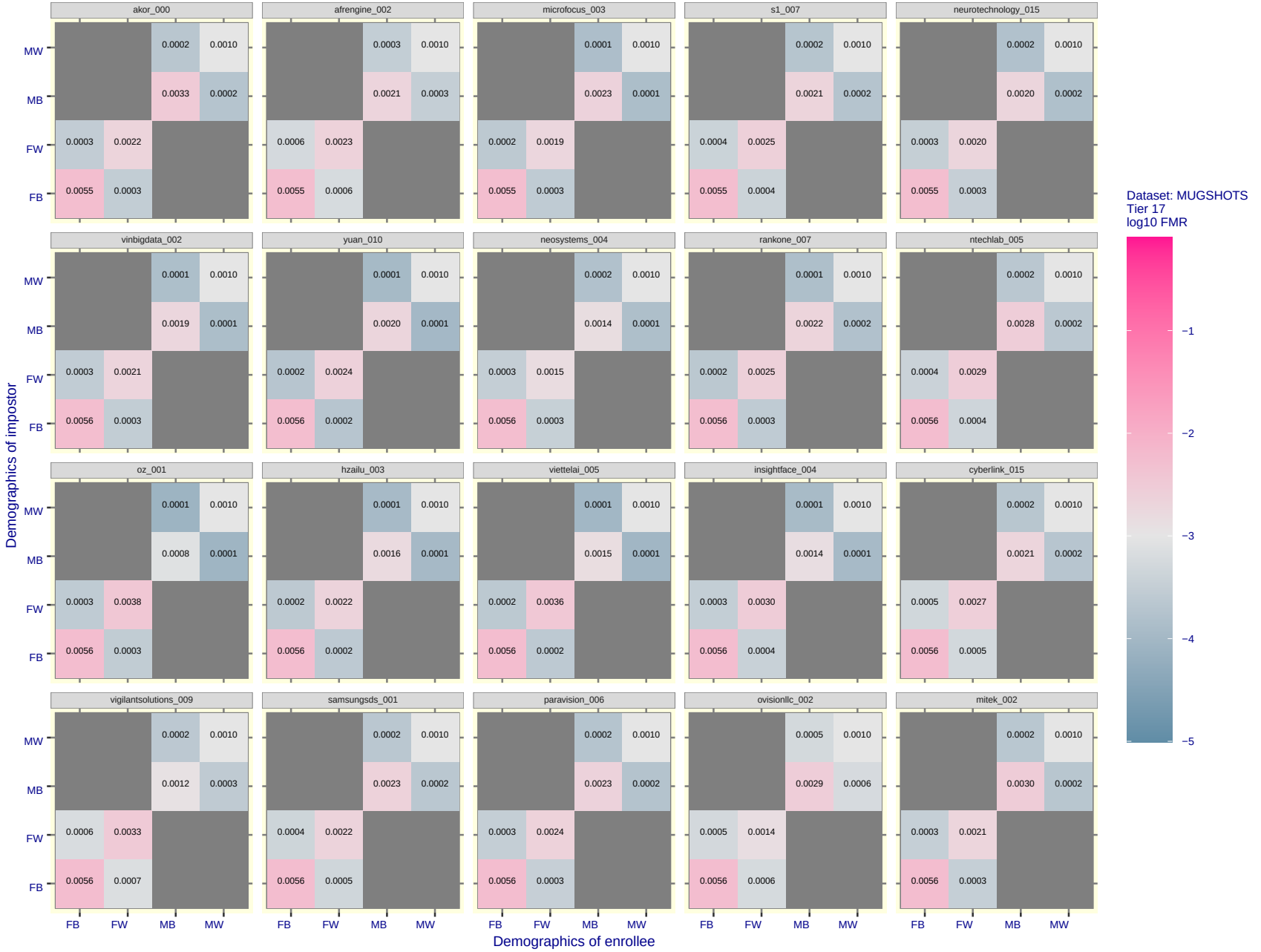


Figure 349: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

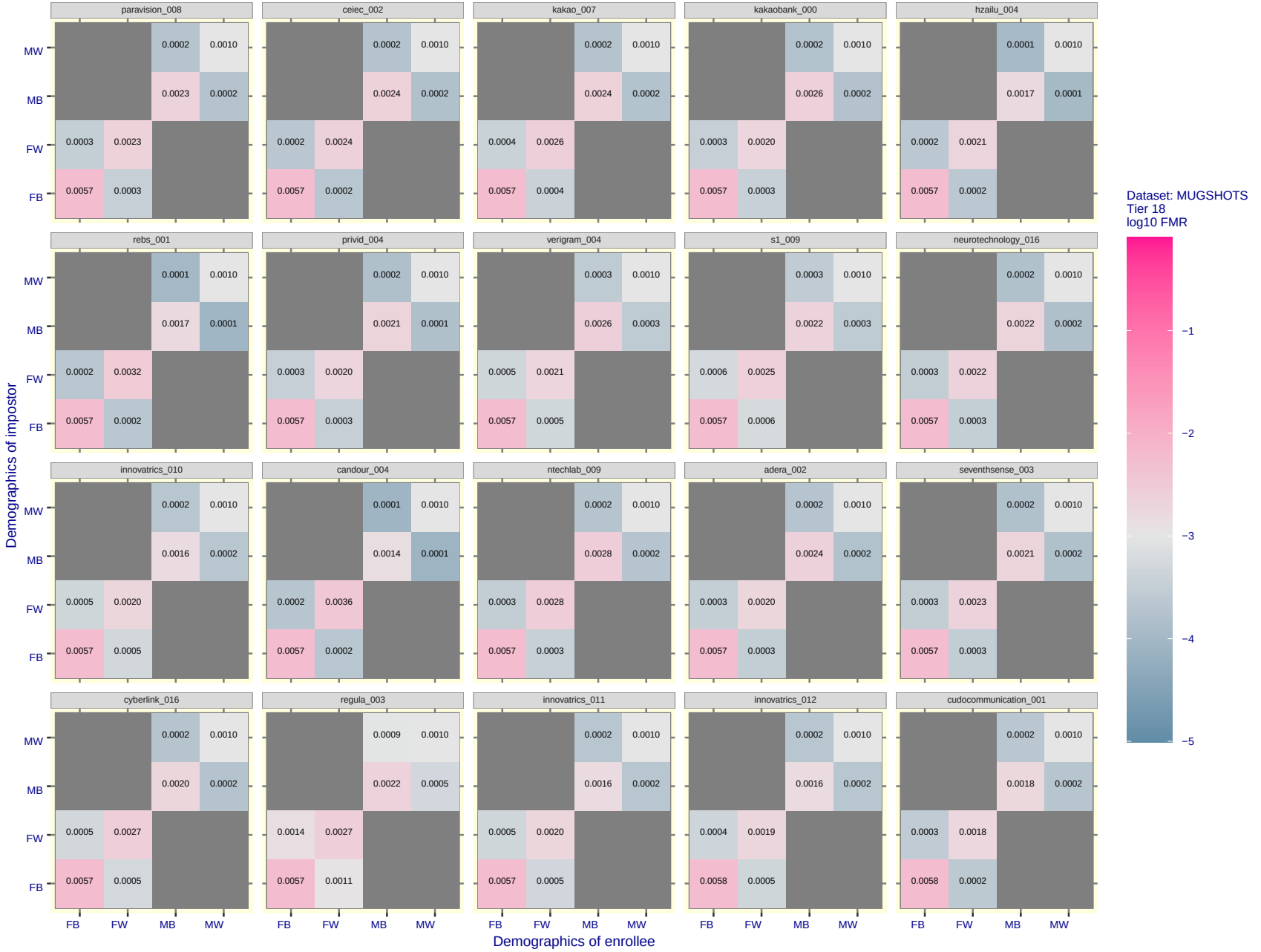


Figure 350: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

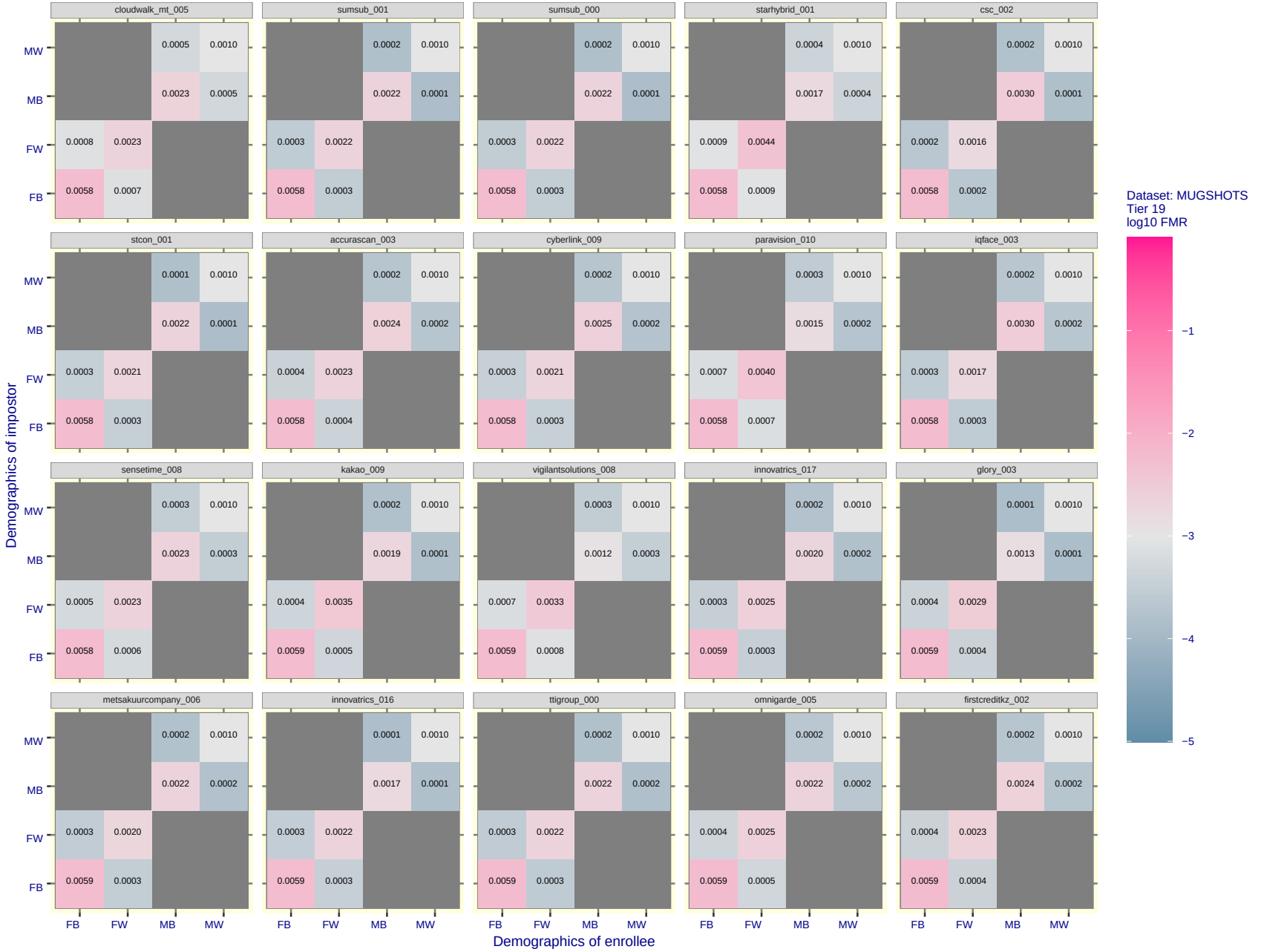


Figure 351: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

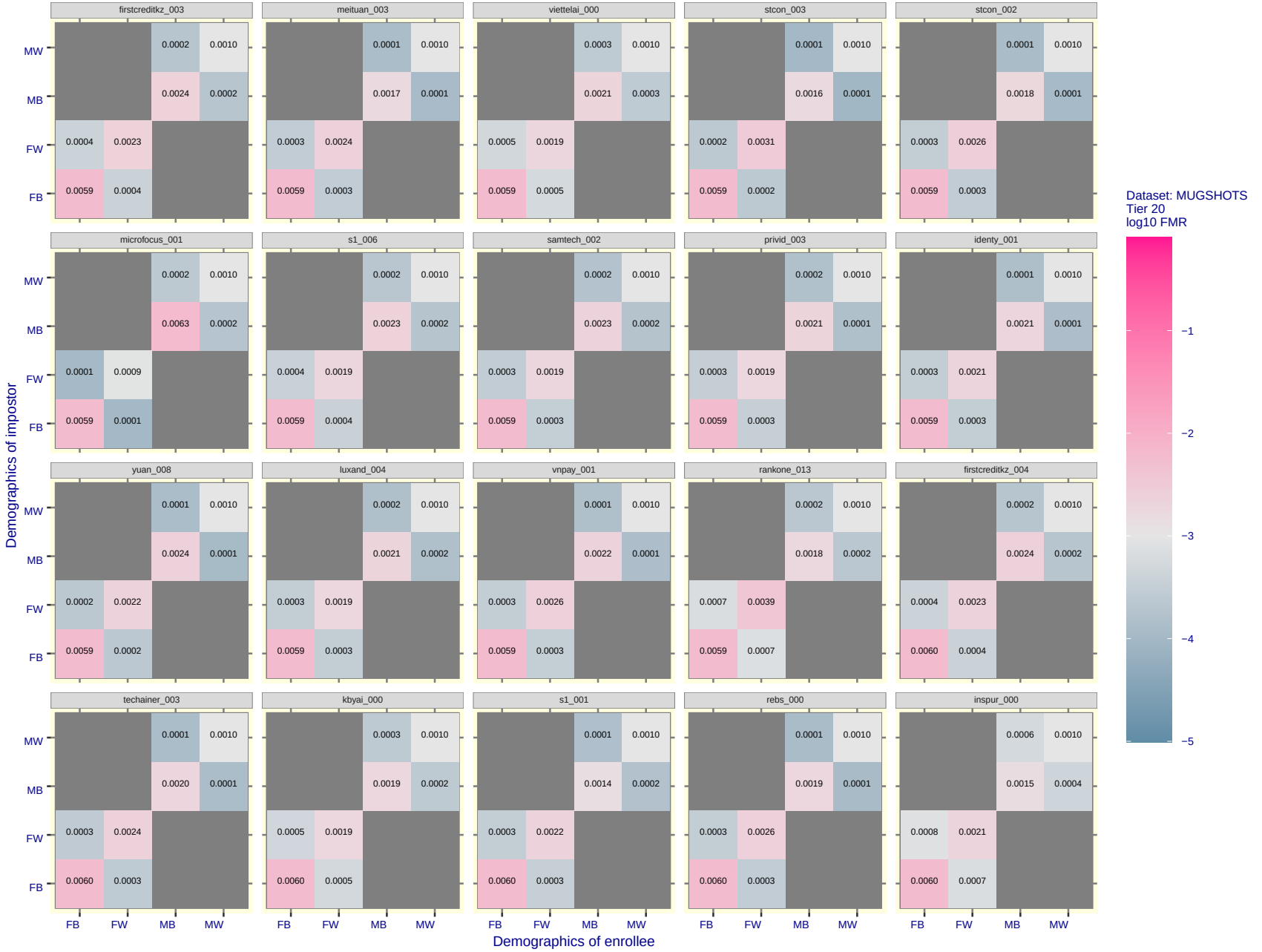


Figure 352: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

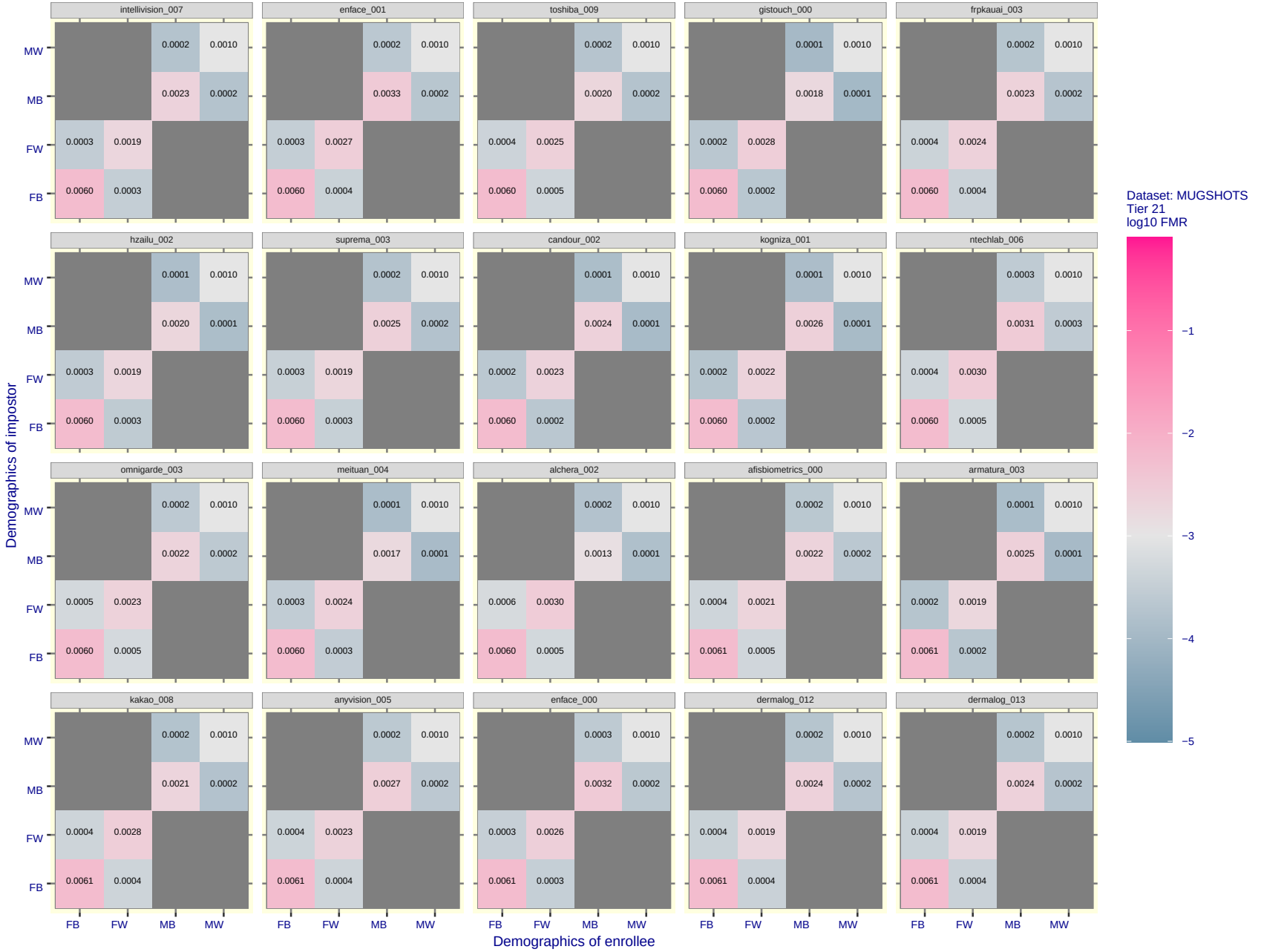


Figure 353: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

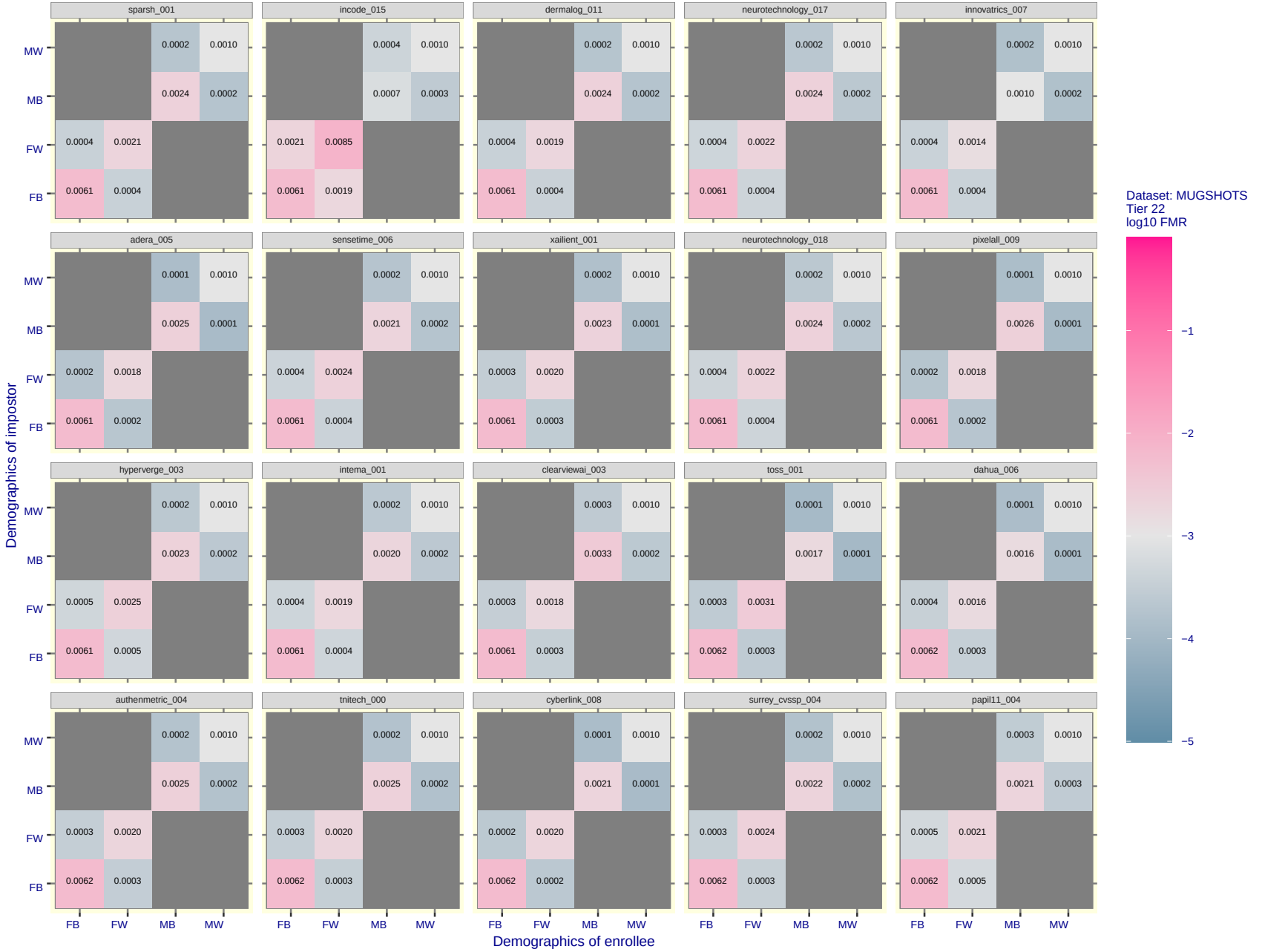


Figure 354: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

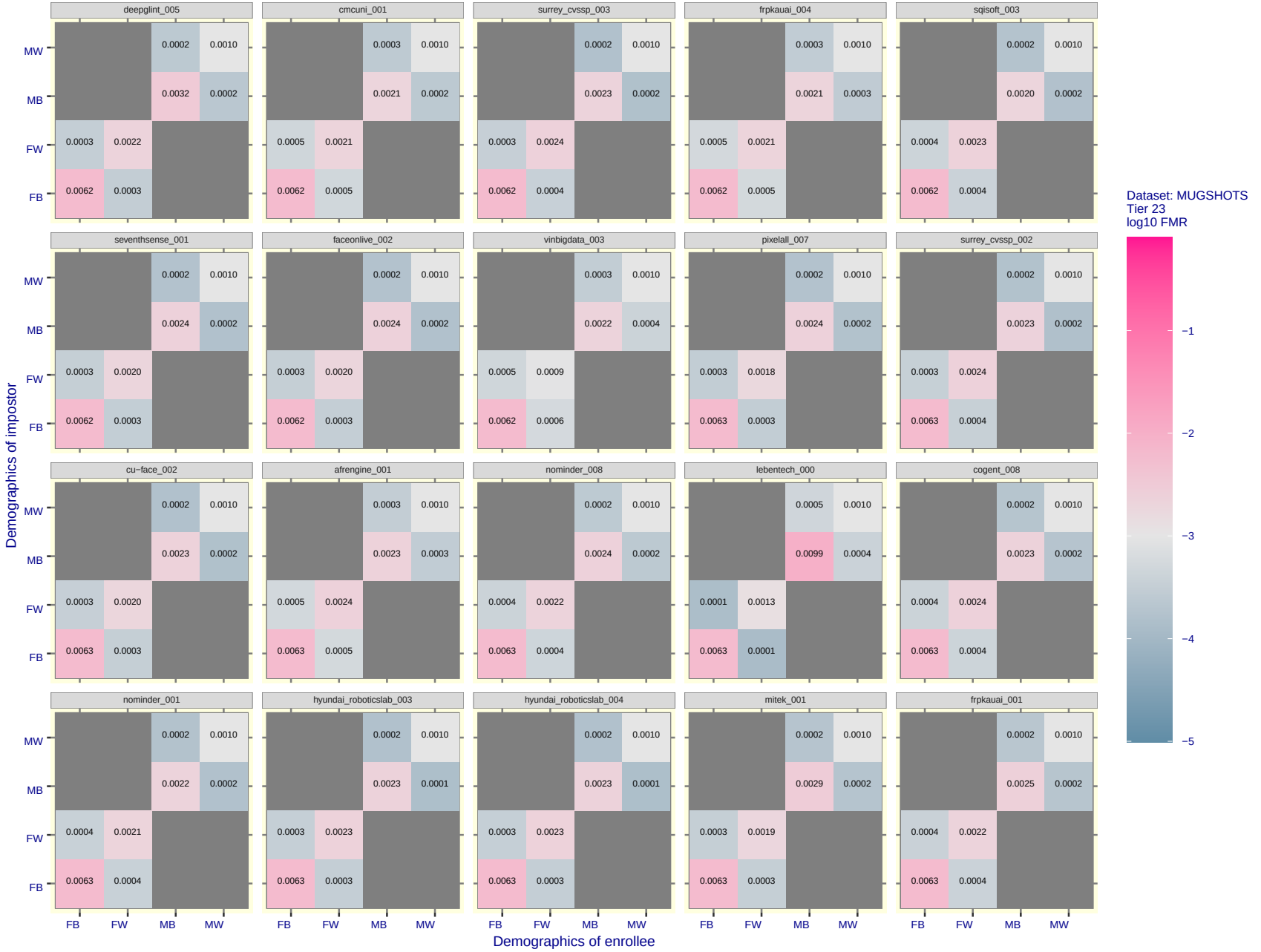


Figure 355: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

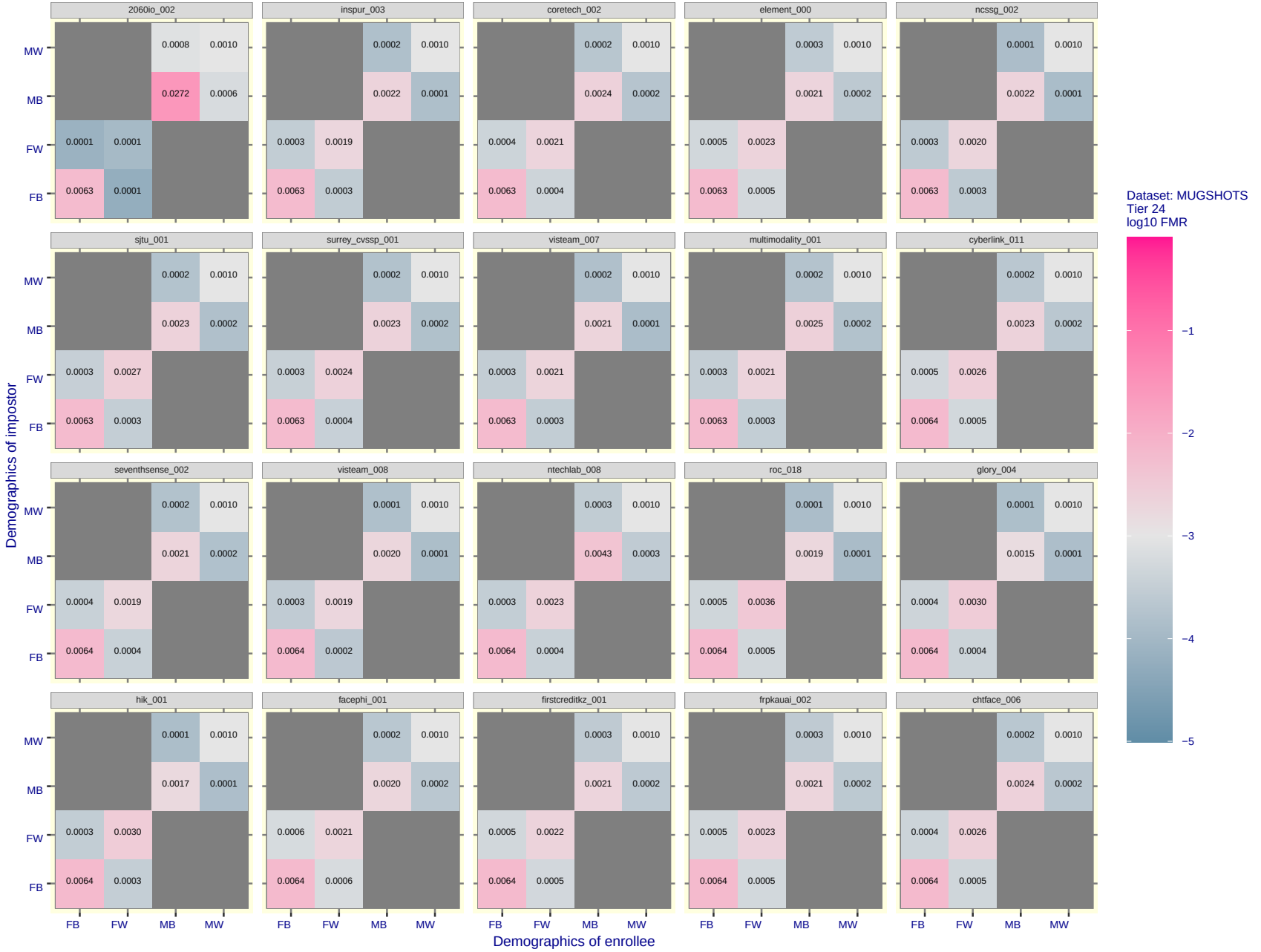


Figure 356: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

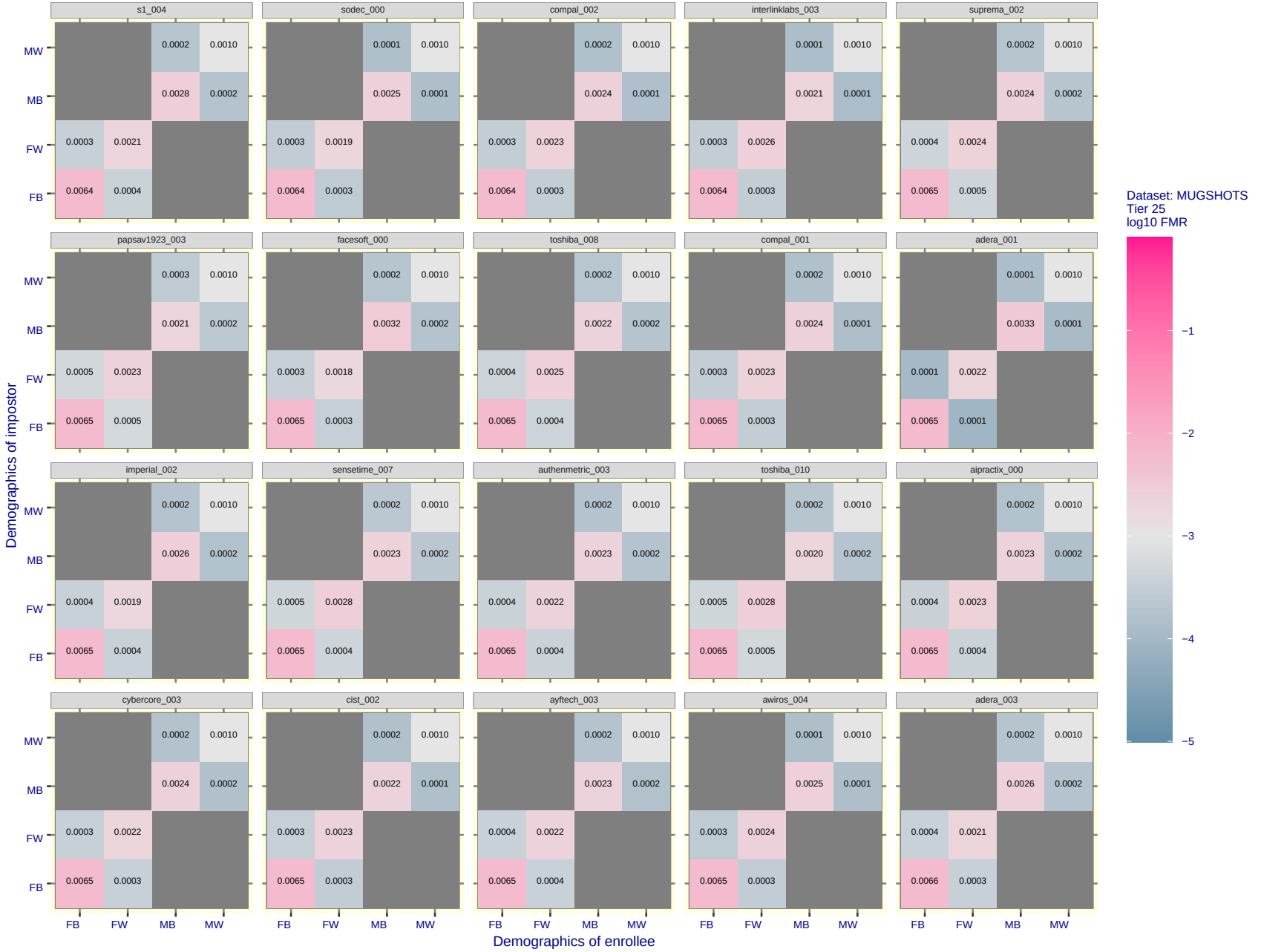


Figure 357: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

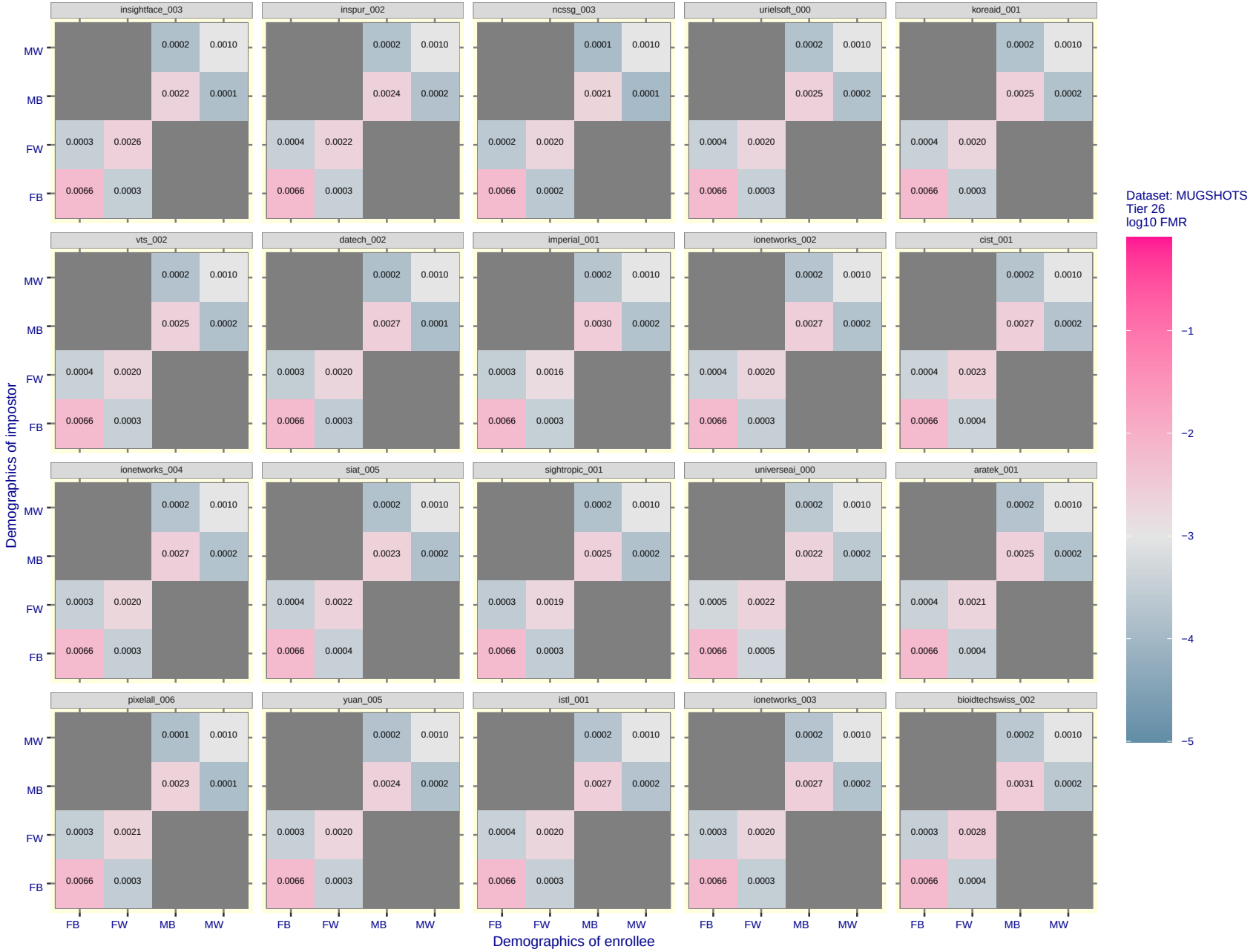


Figure 358: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

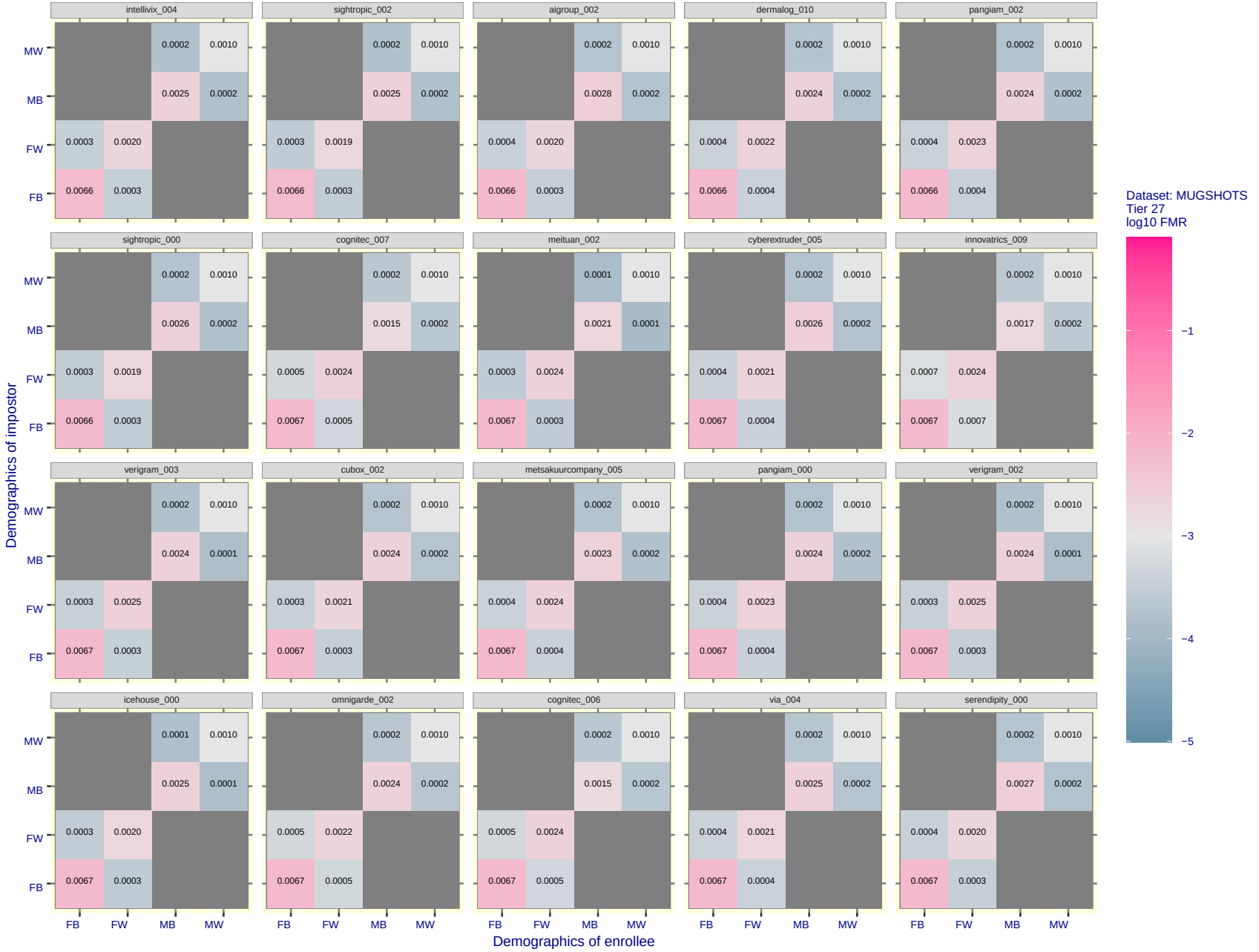


Figure 359: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

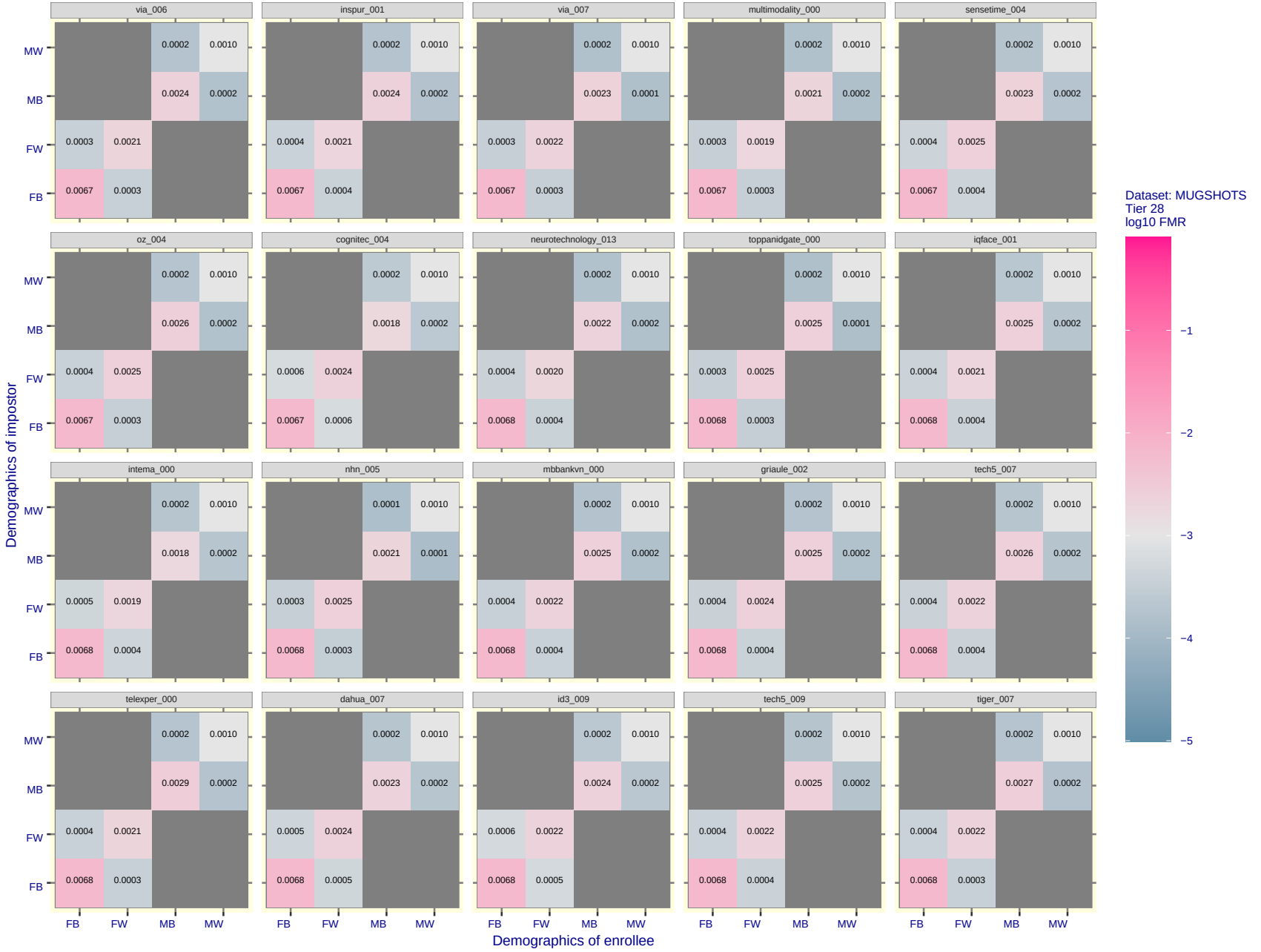


Figure 360: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

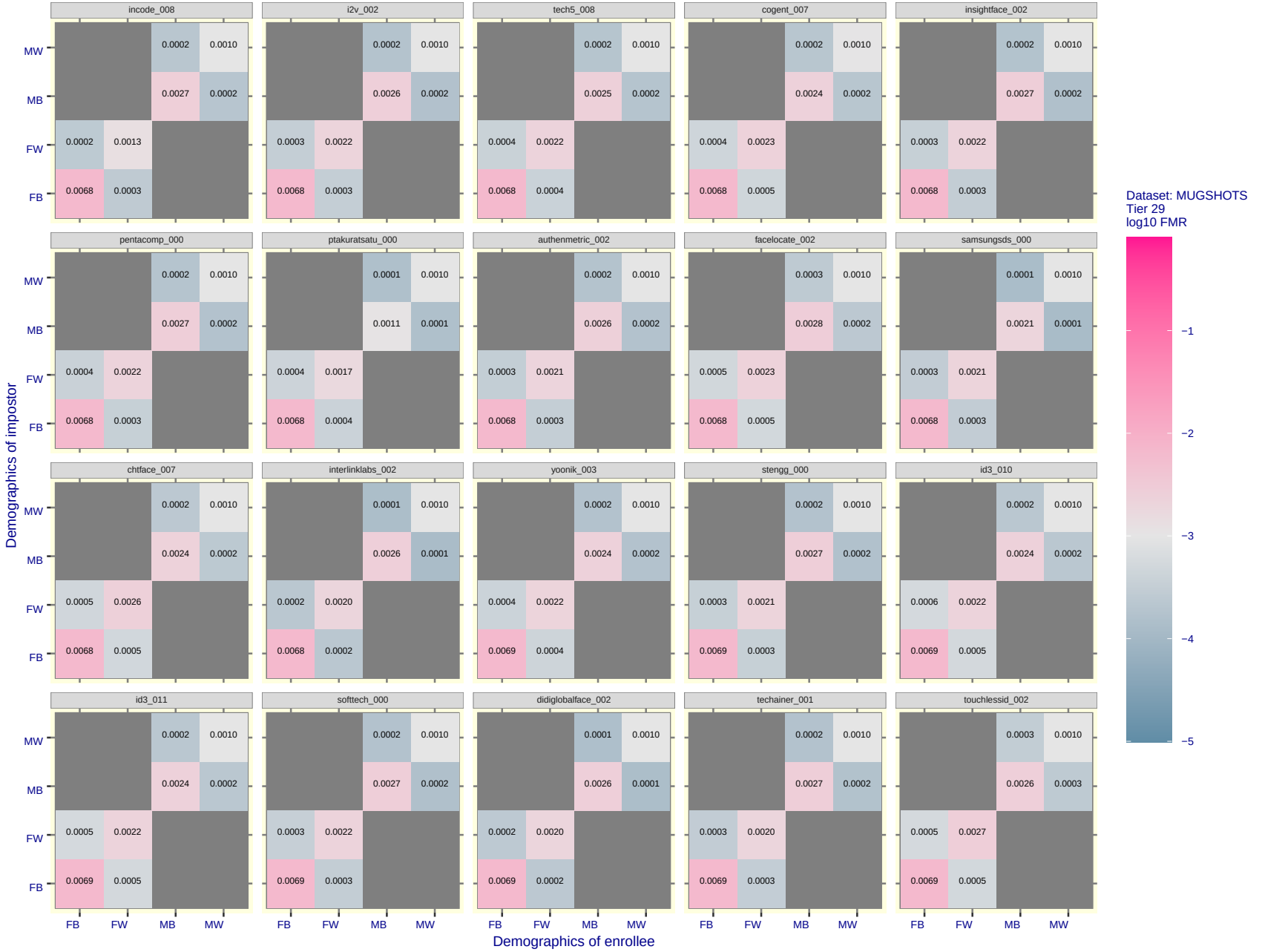


Figure 361: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

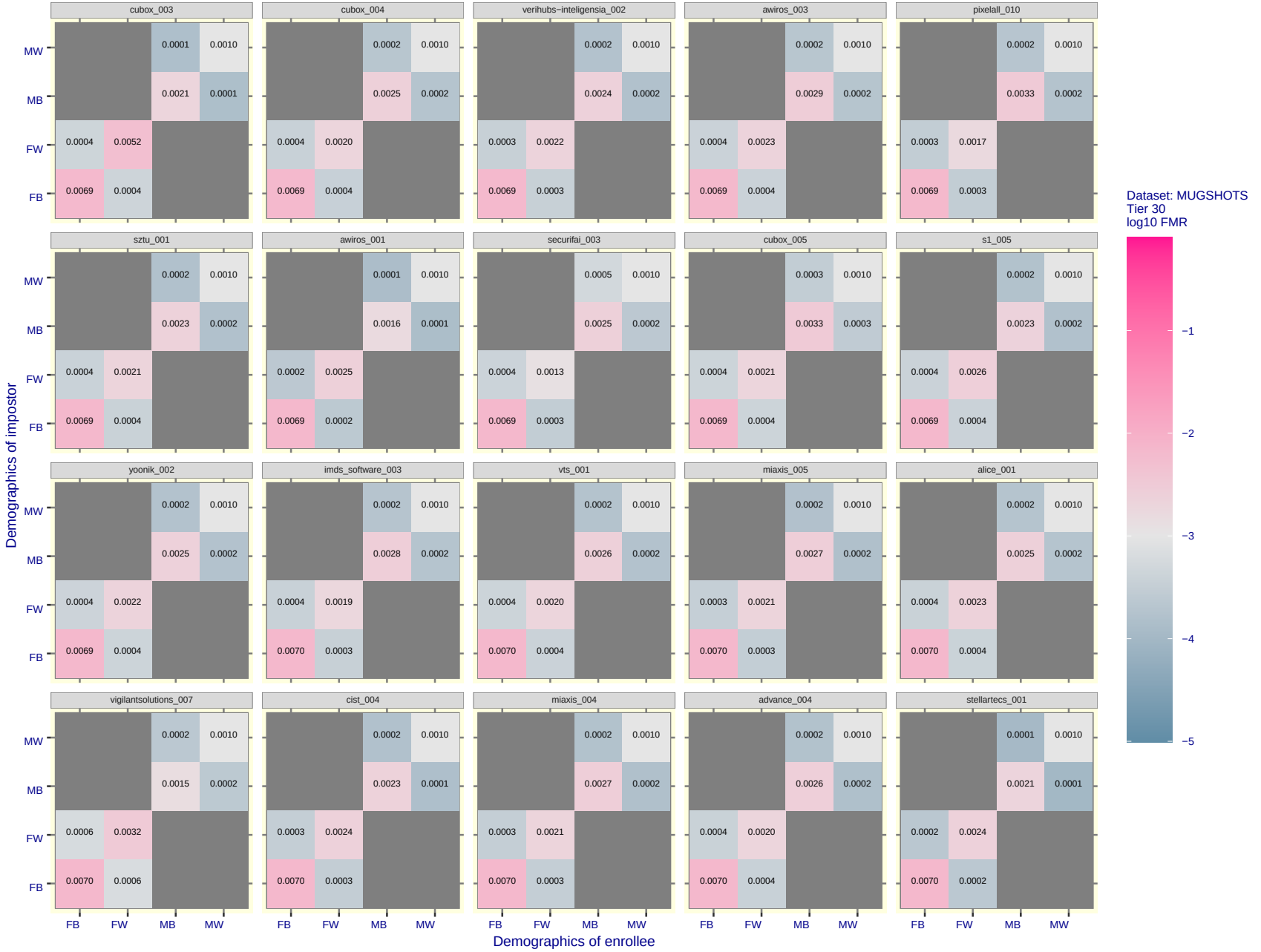


Figure 362: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

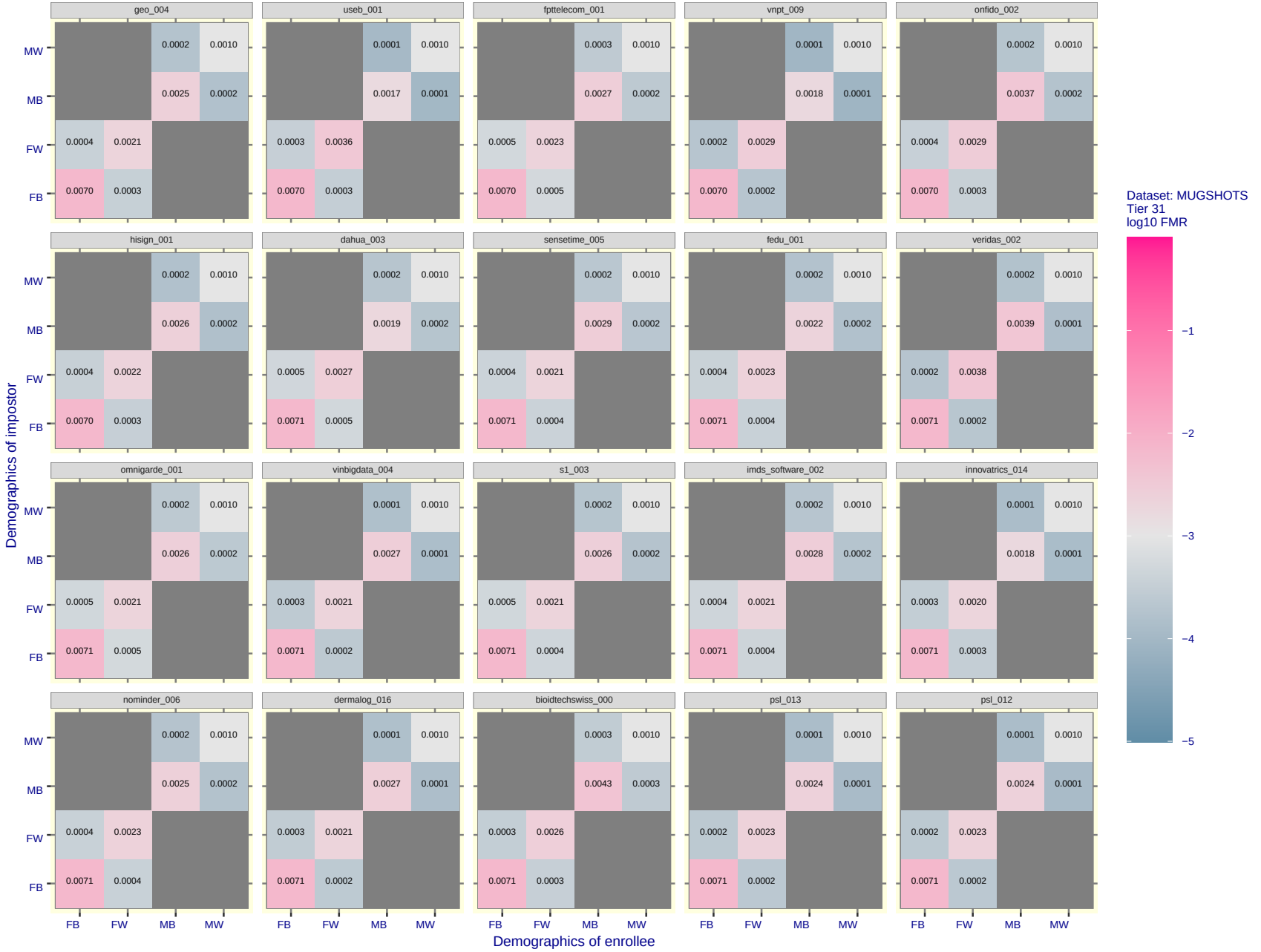


Figure 363: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

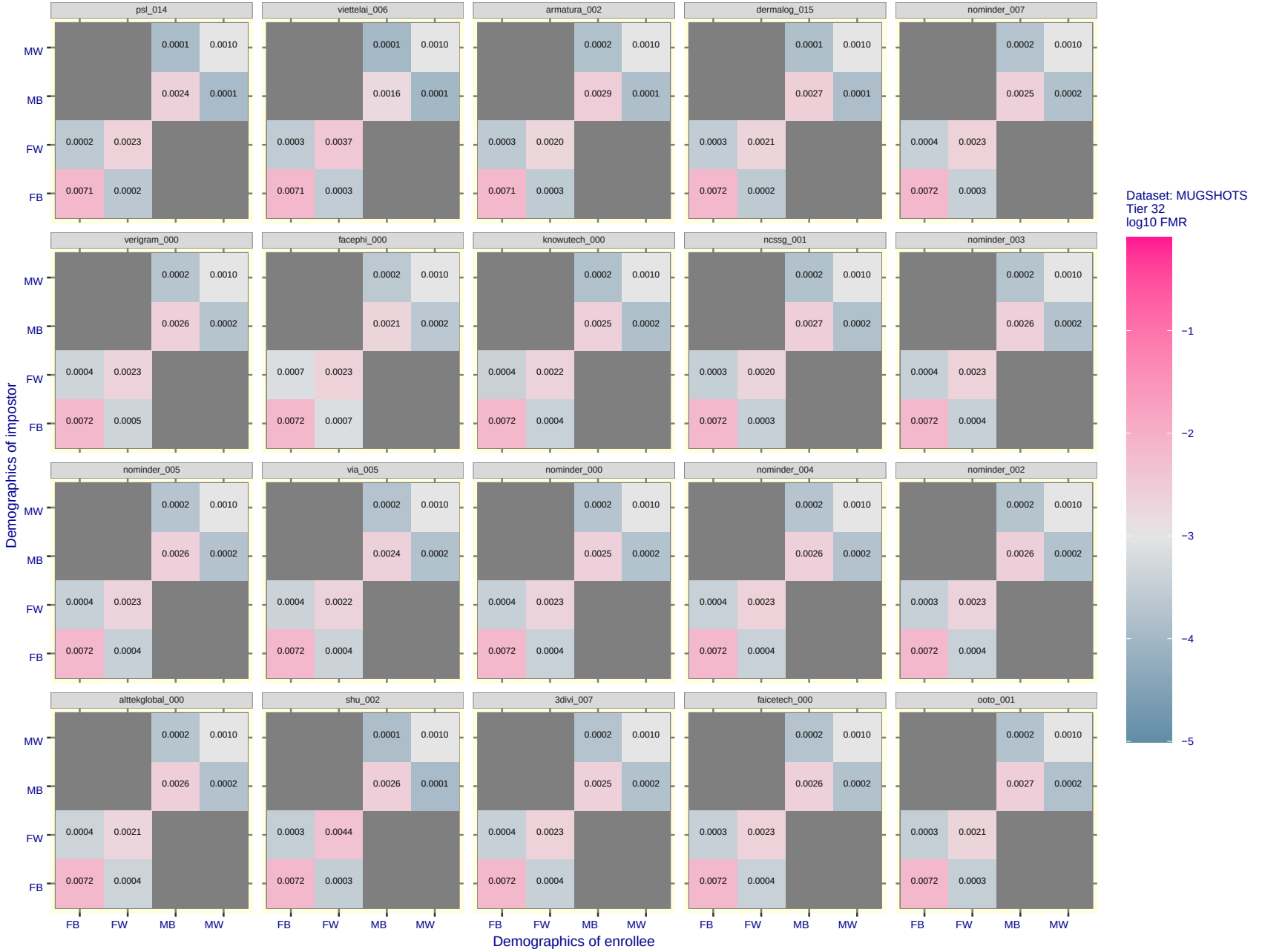


Figure 364: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

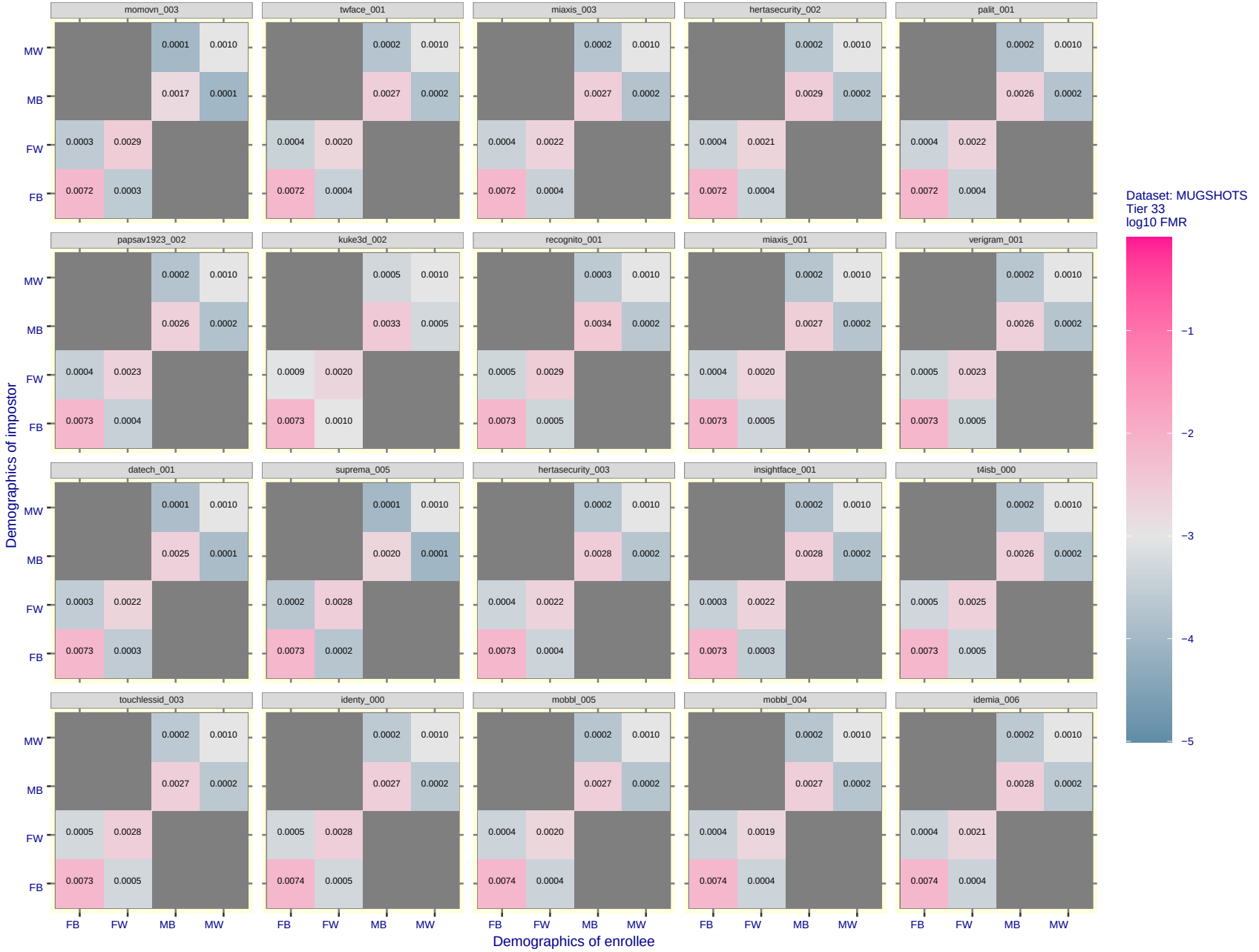


Figure 365: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

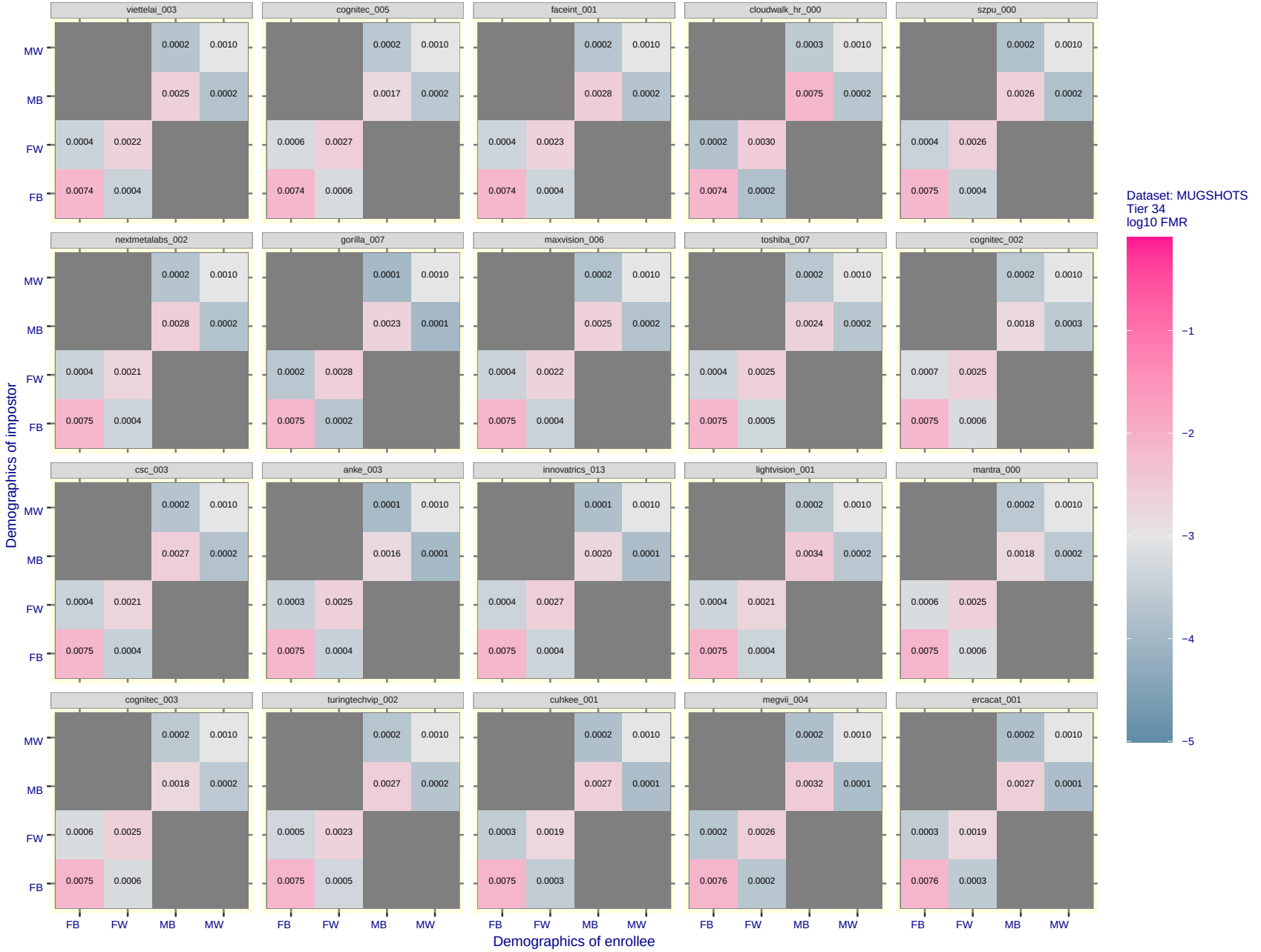


Figure 366: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

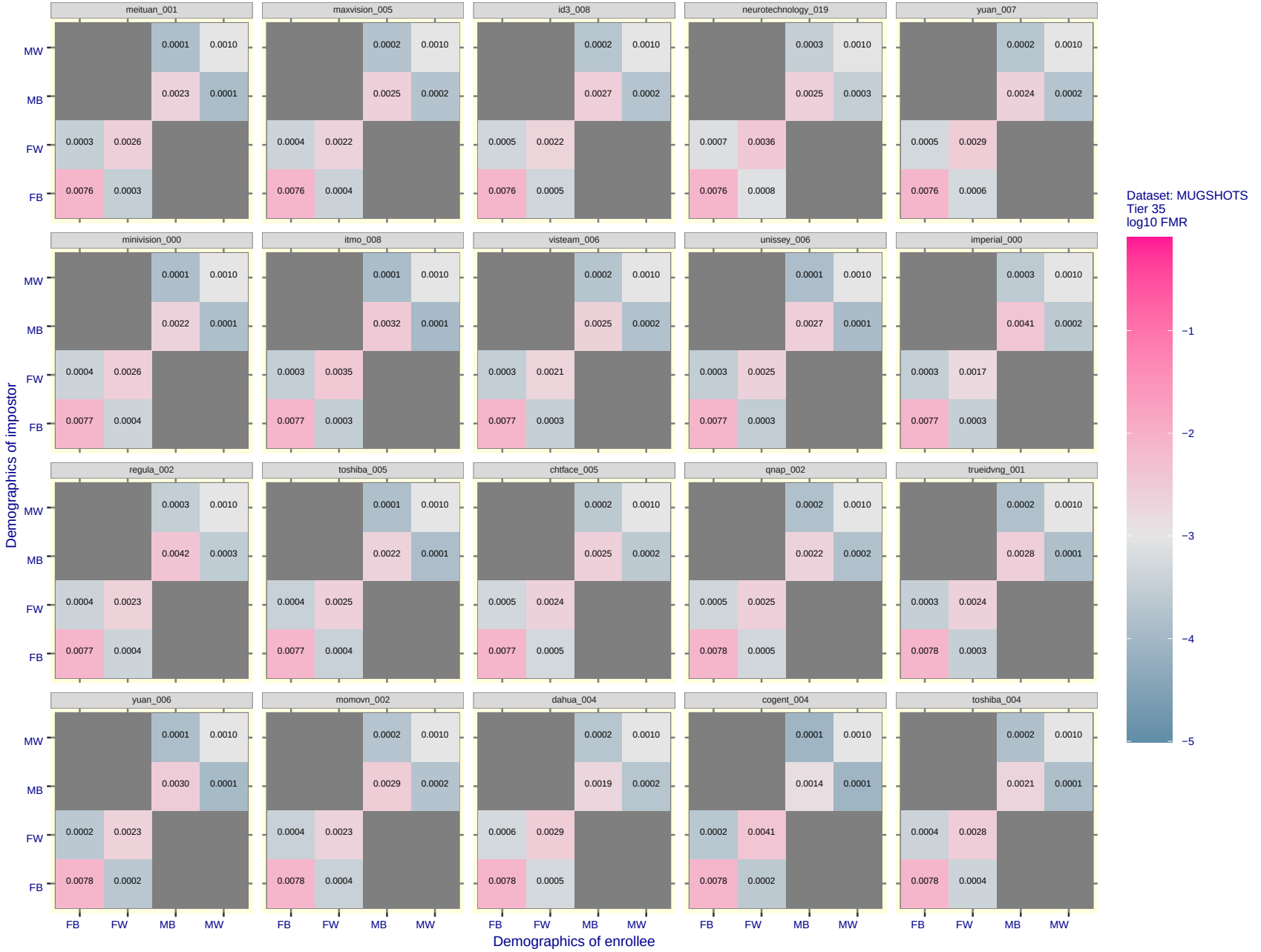


Figure 367: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

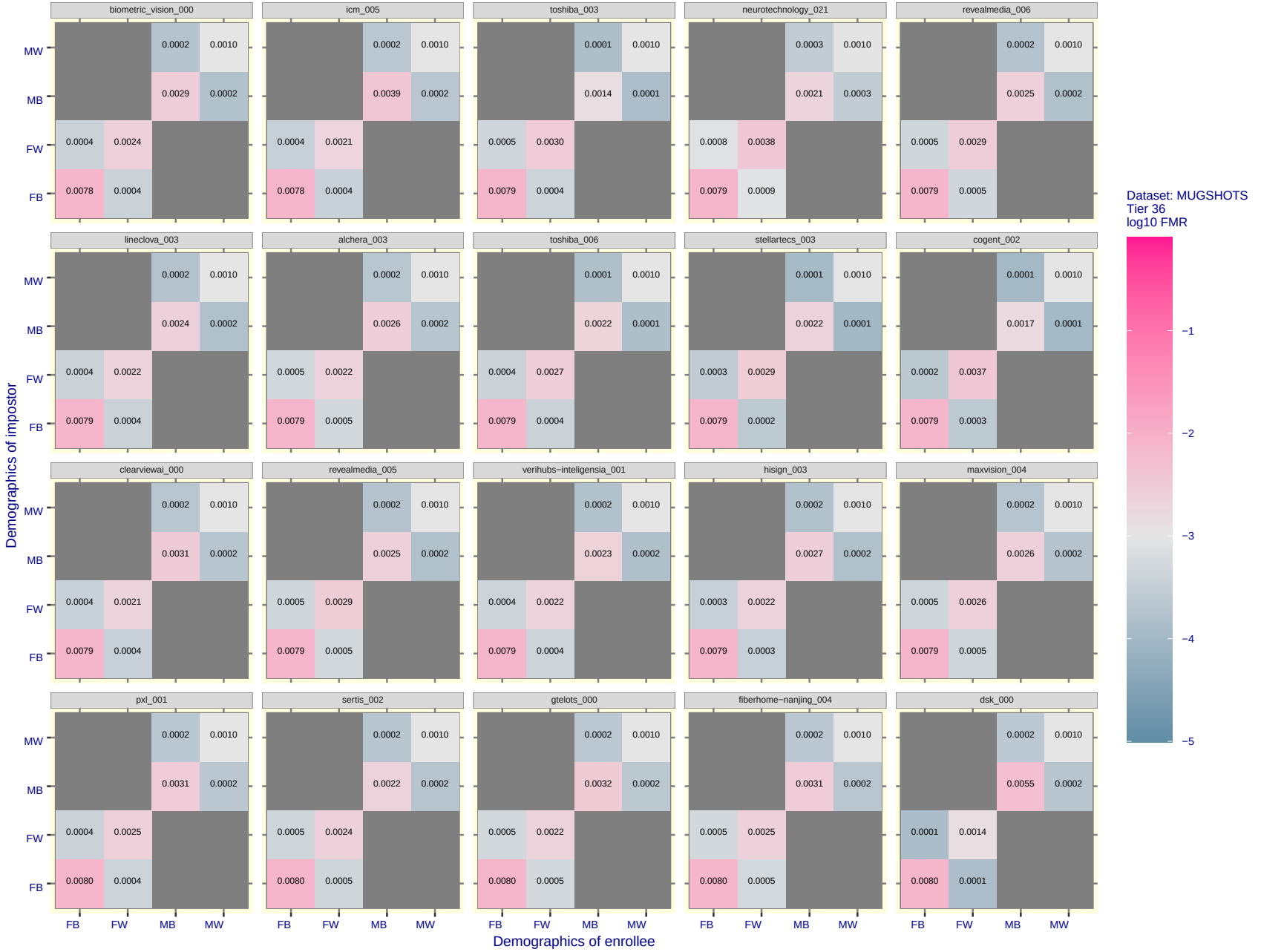


Figure 368: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

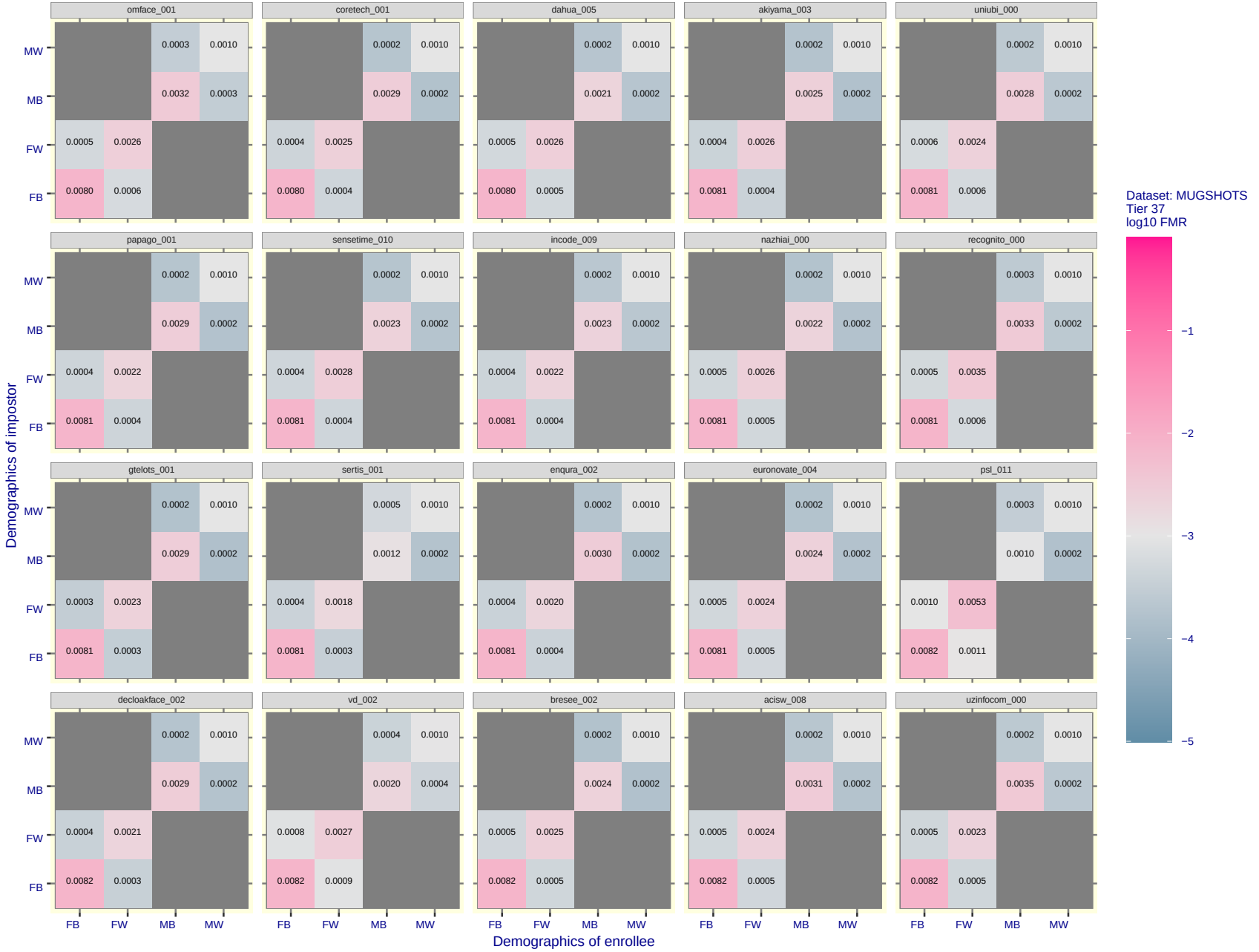


Figure 369: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

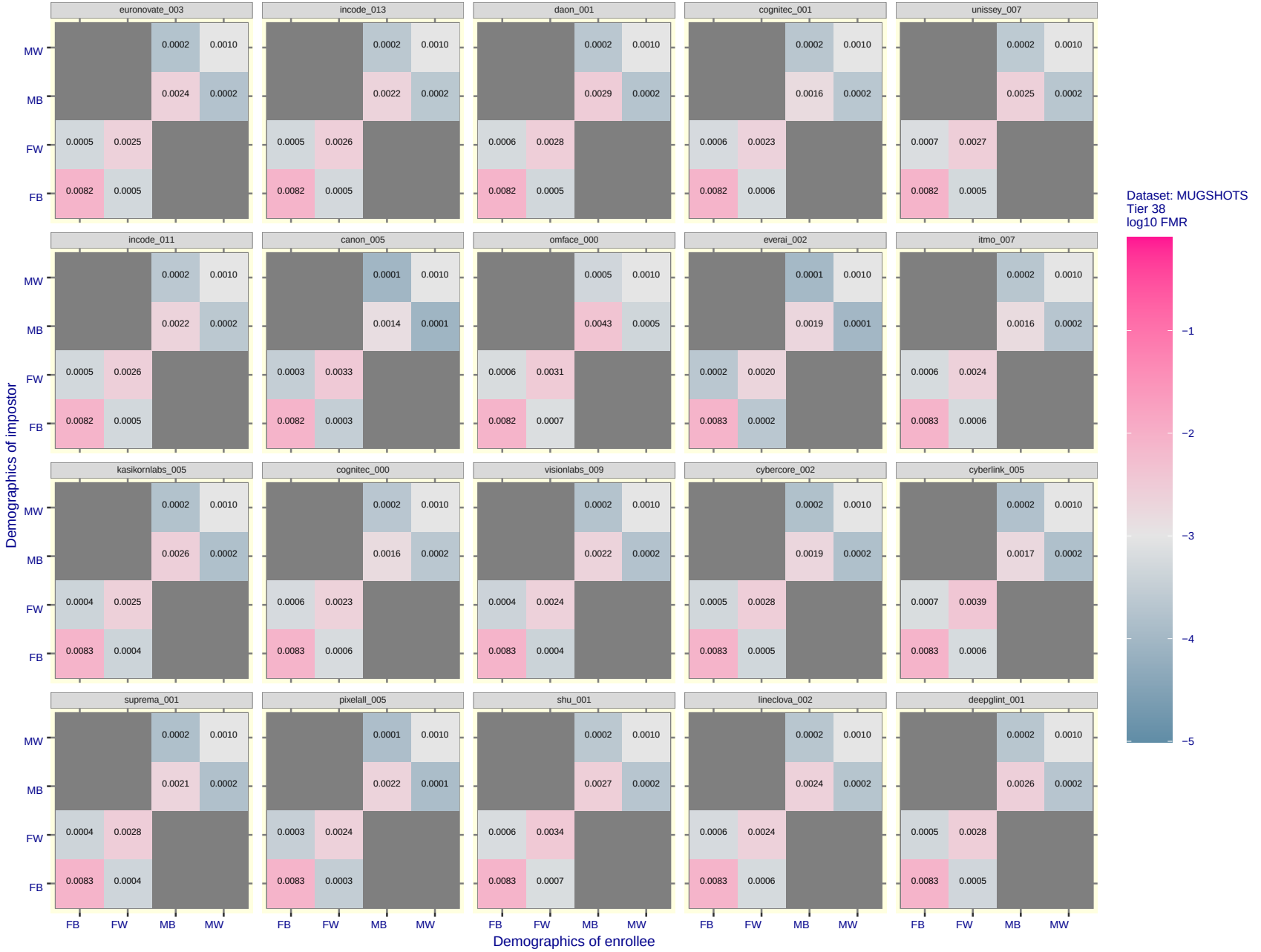


Figure 370: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

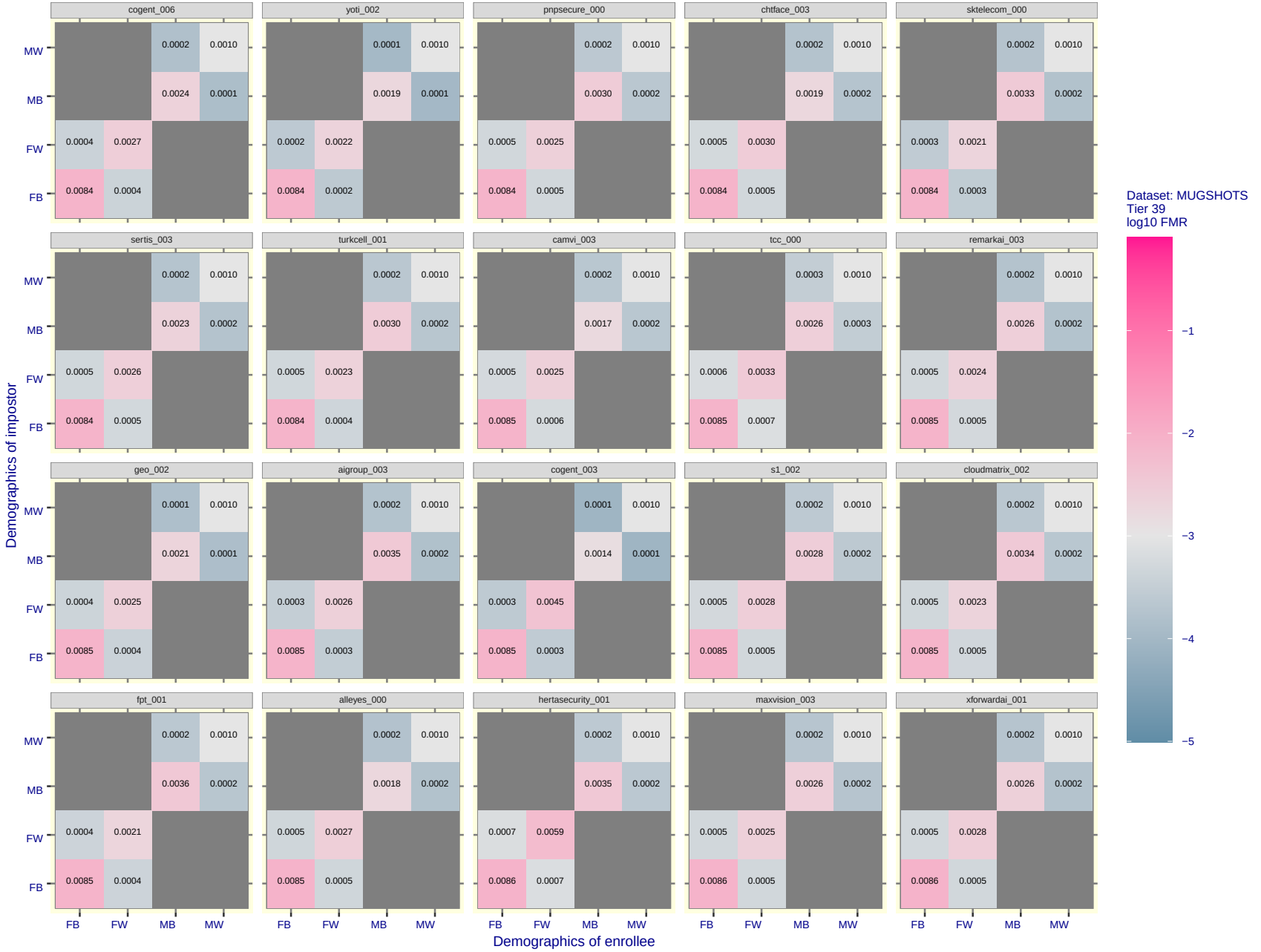


Figure 371: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

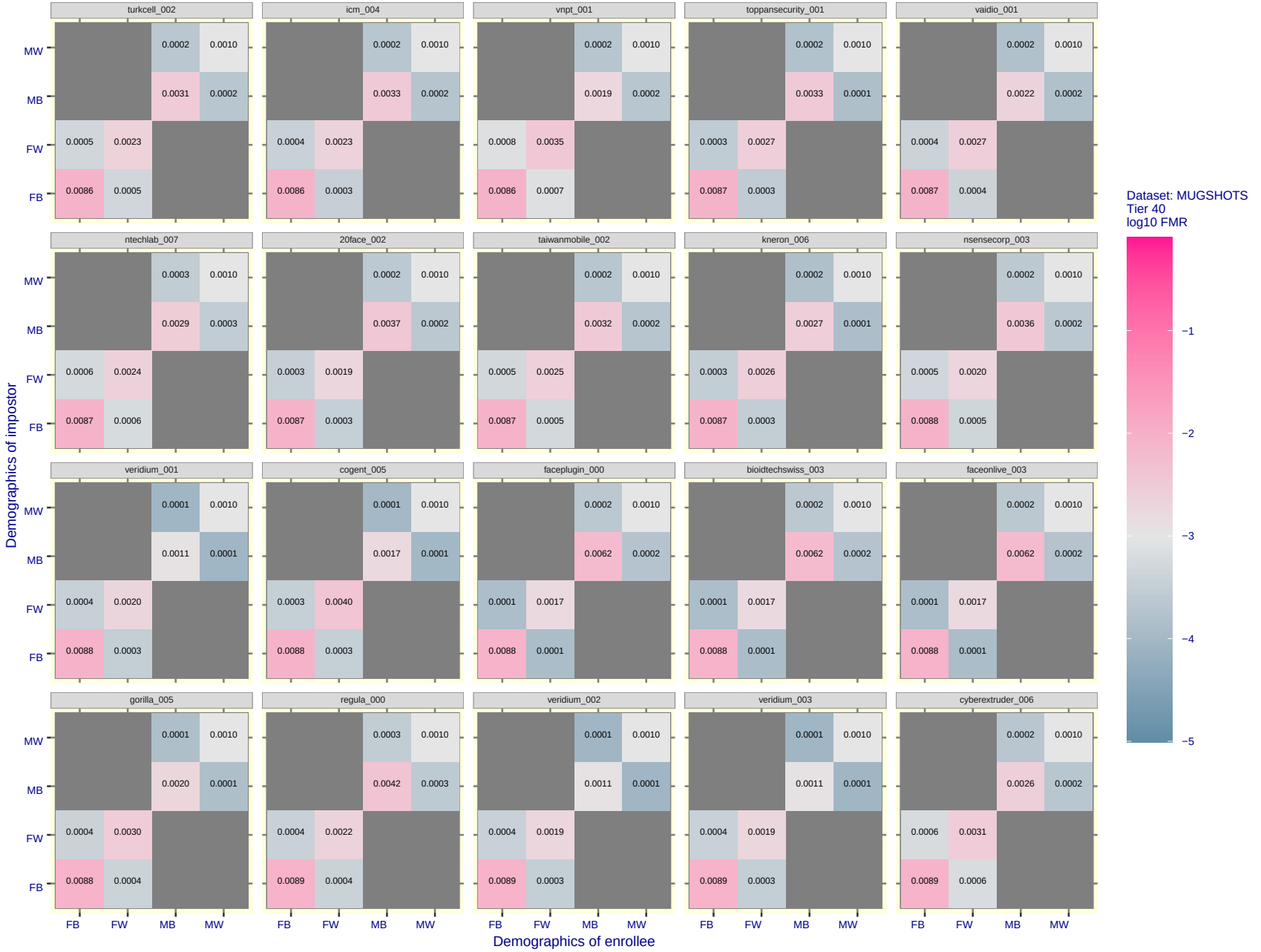


Figure 372: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

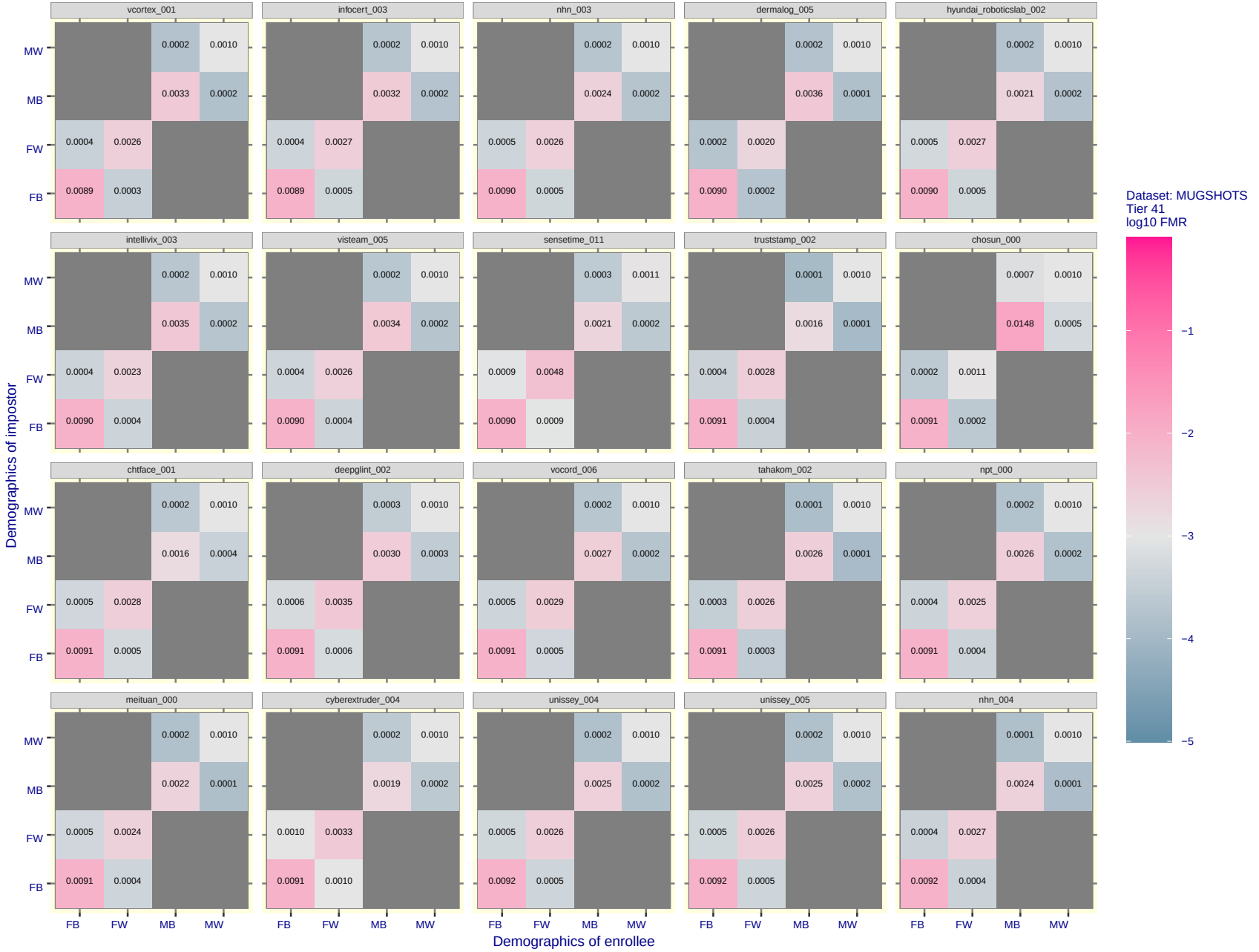


Figure 373: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

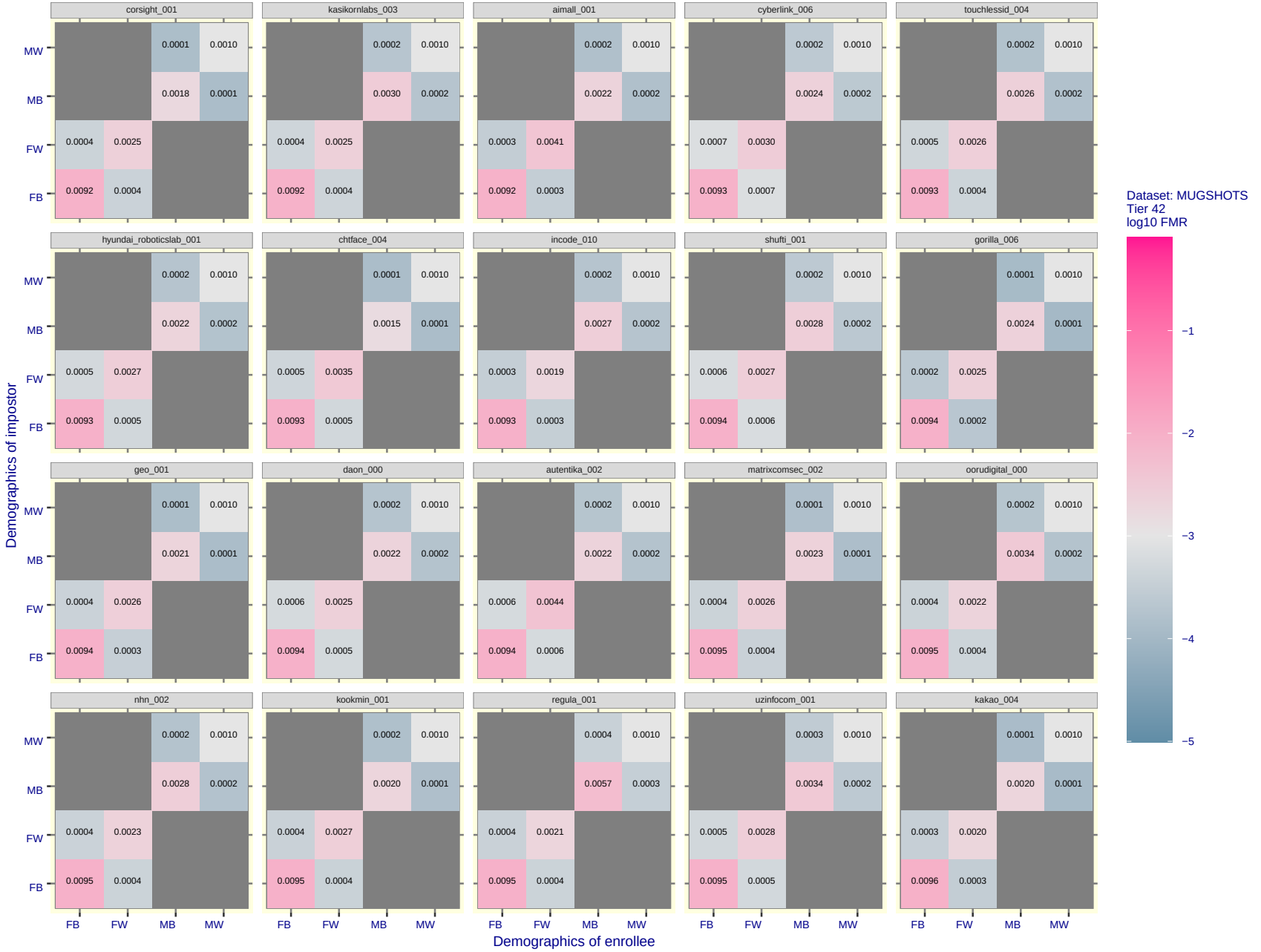


Figure 374: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

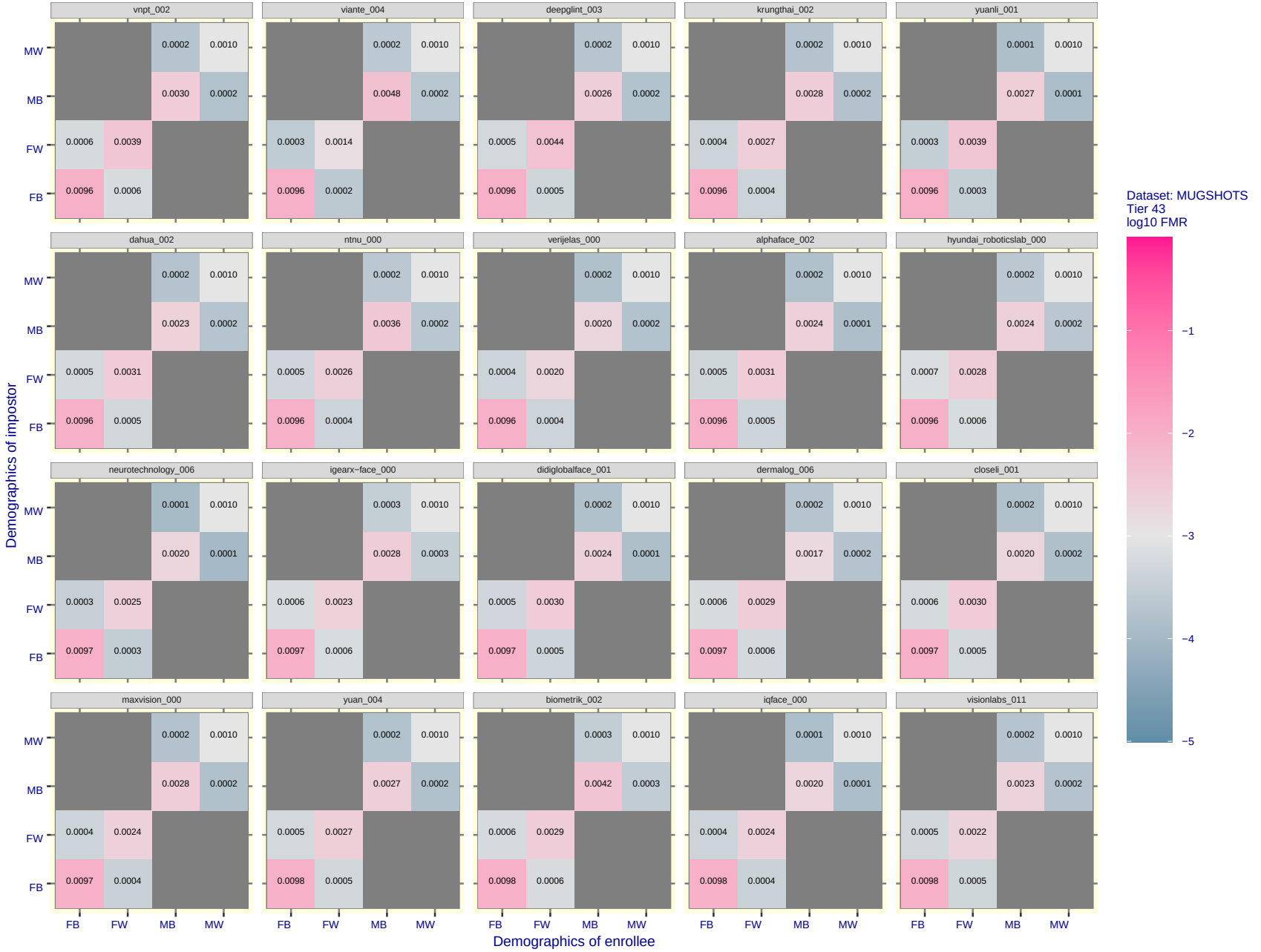


Figure 375: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

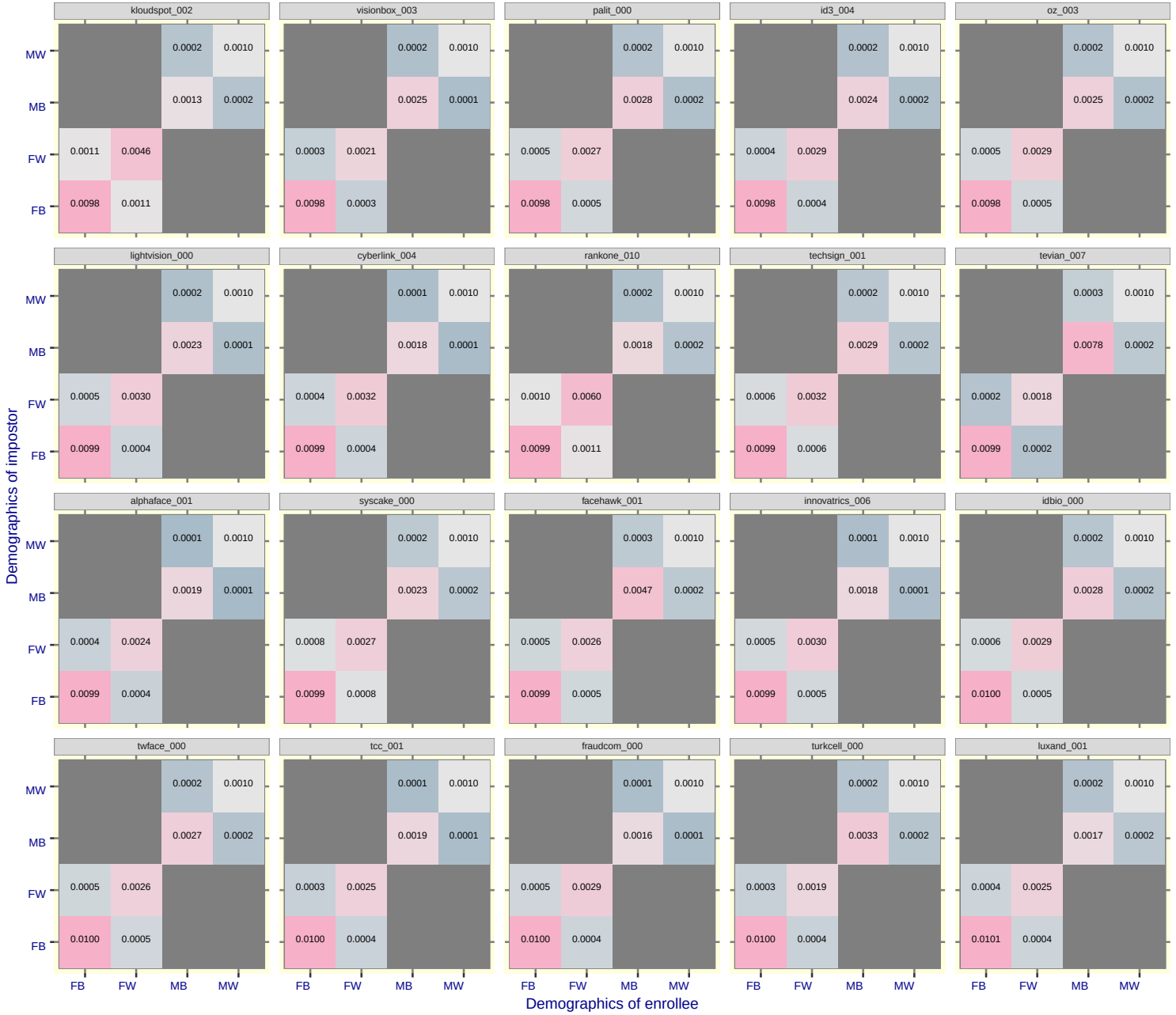


Figure 376: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

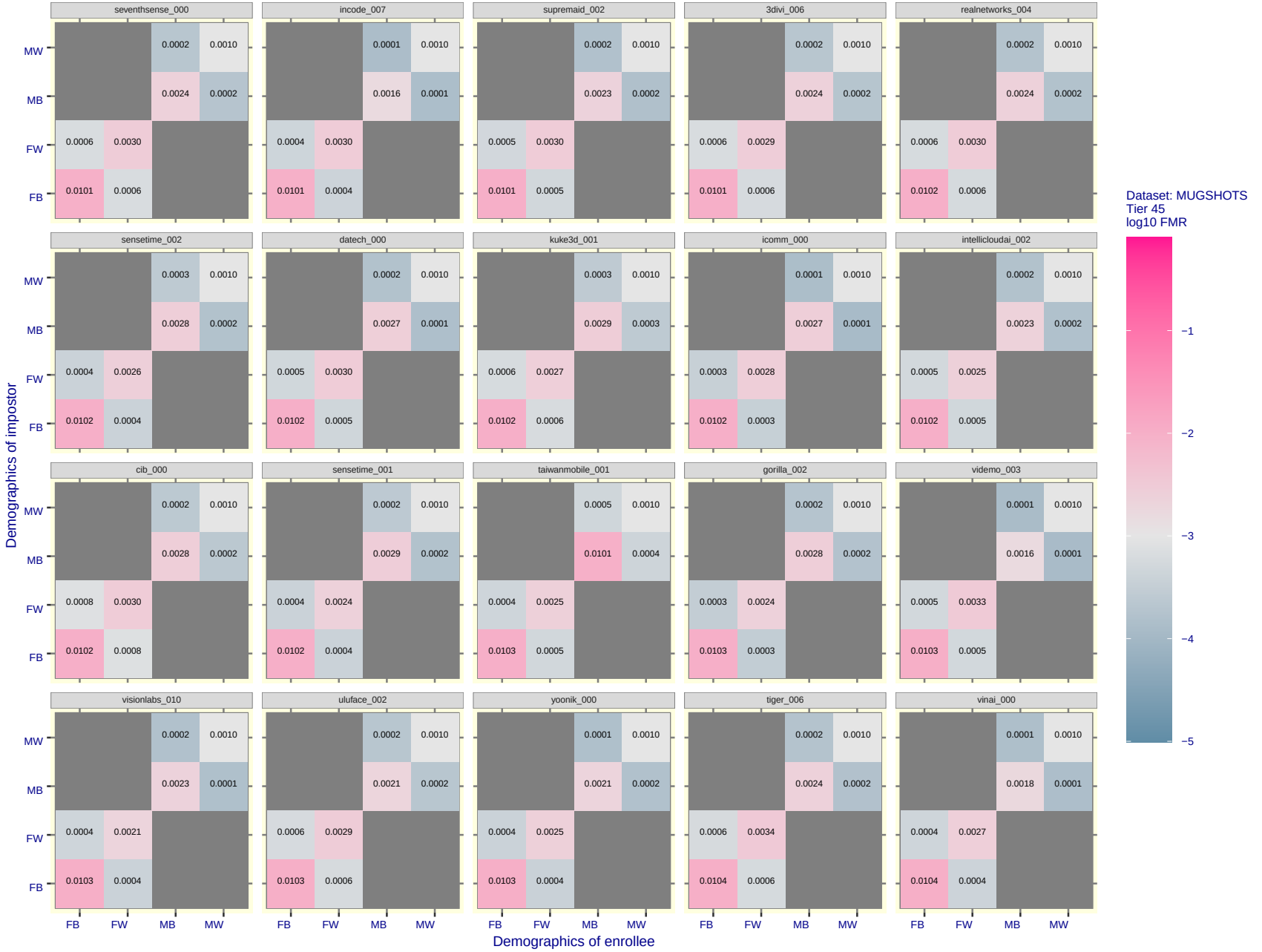


Figure 377: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

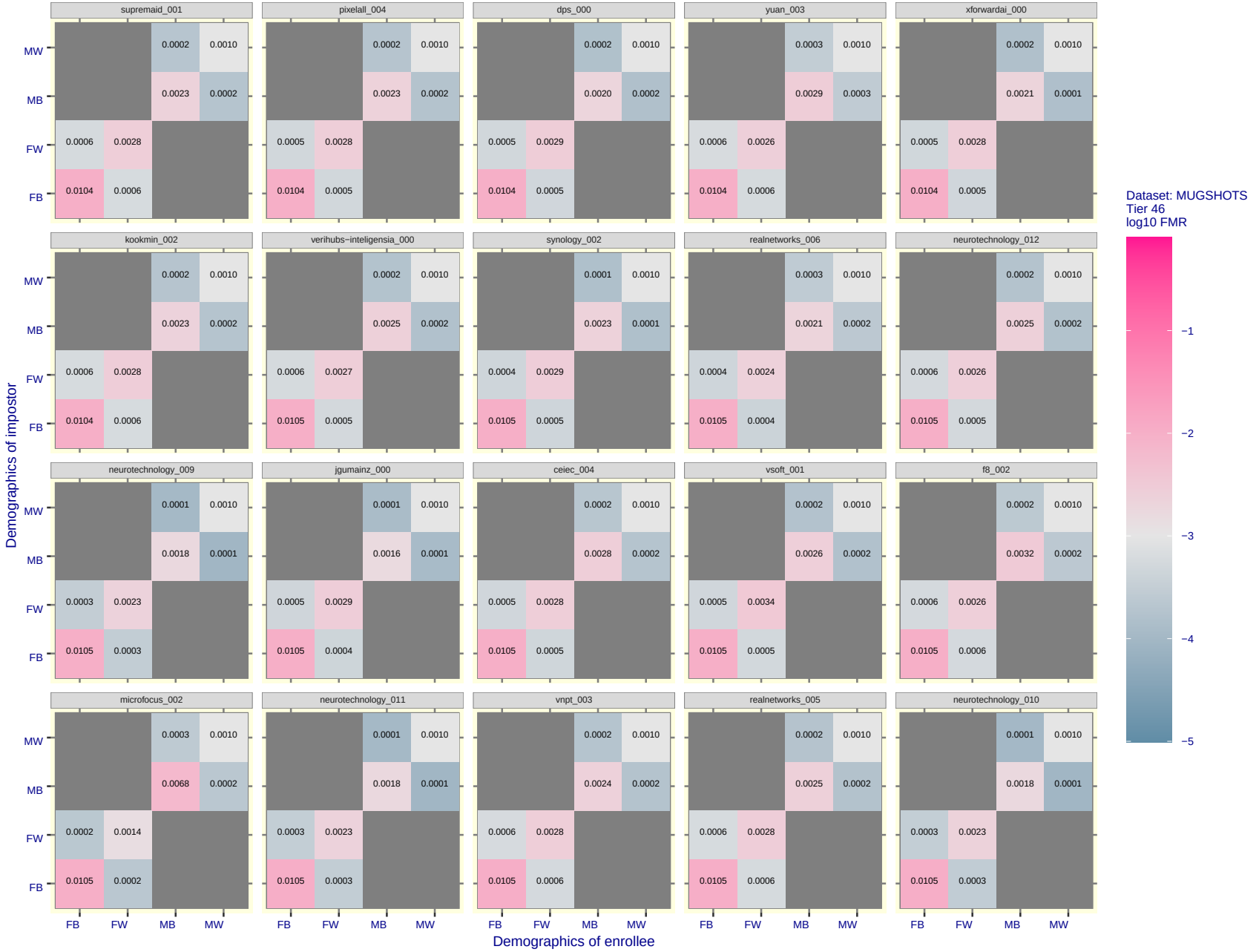


Figure 378: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

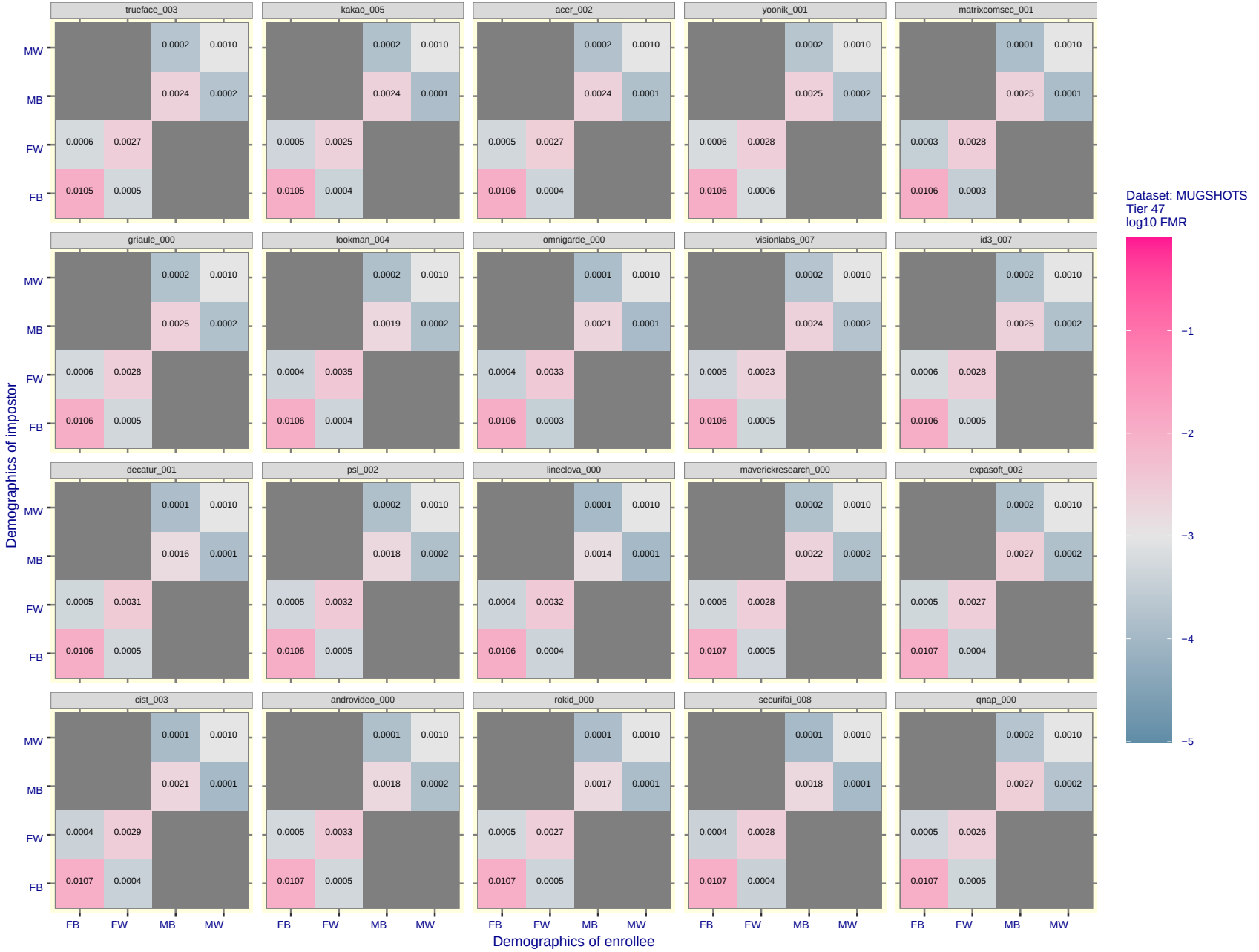


Figure 379: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

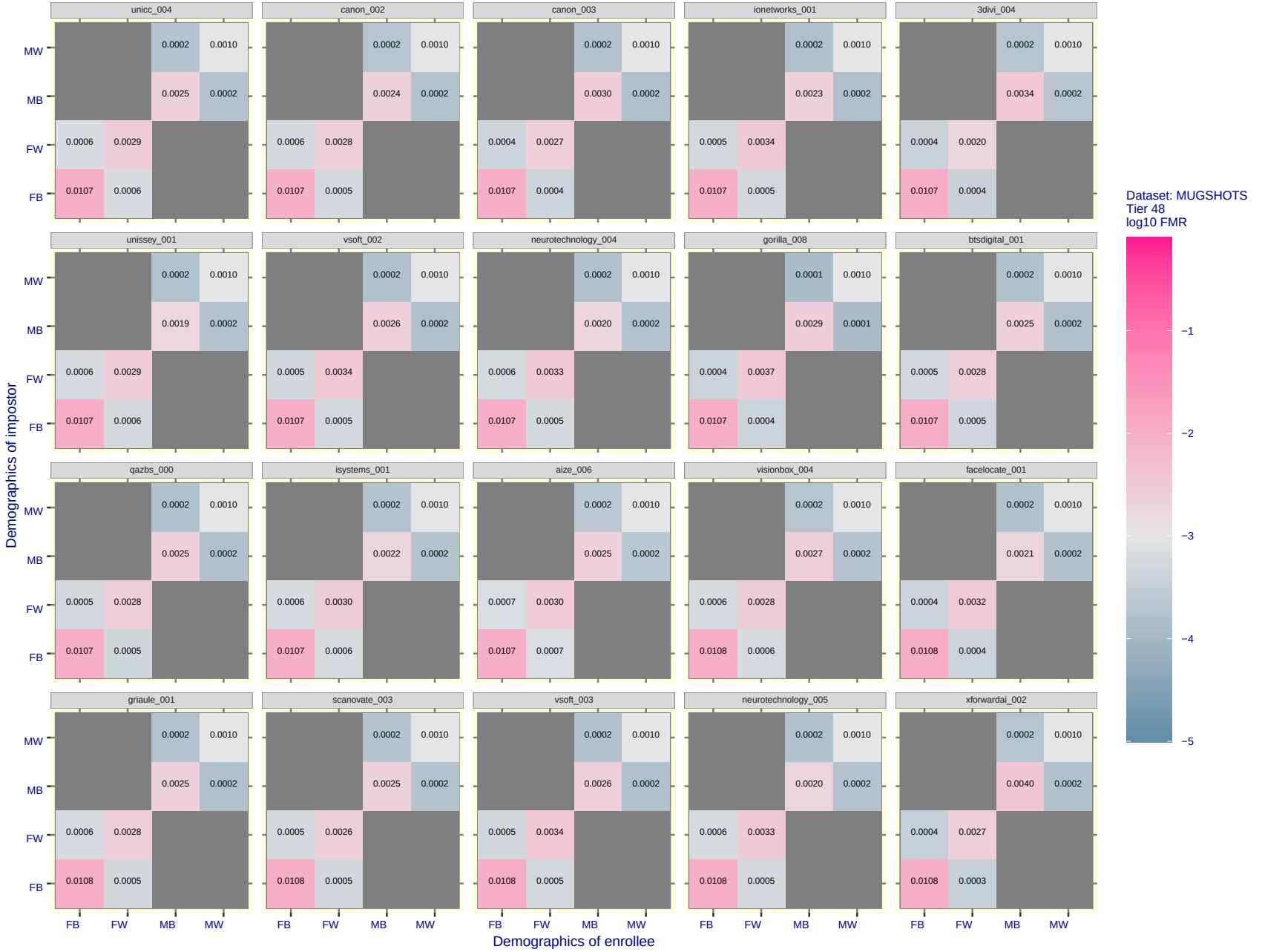


Figure 380: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

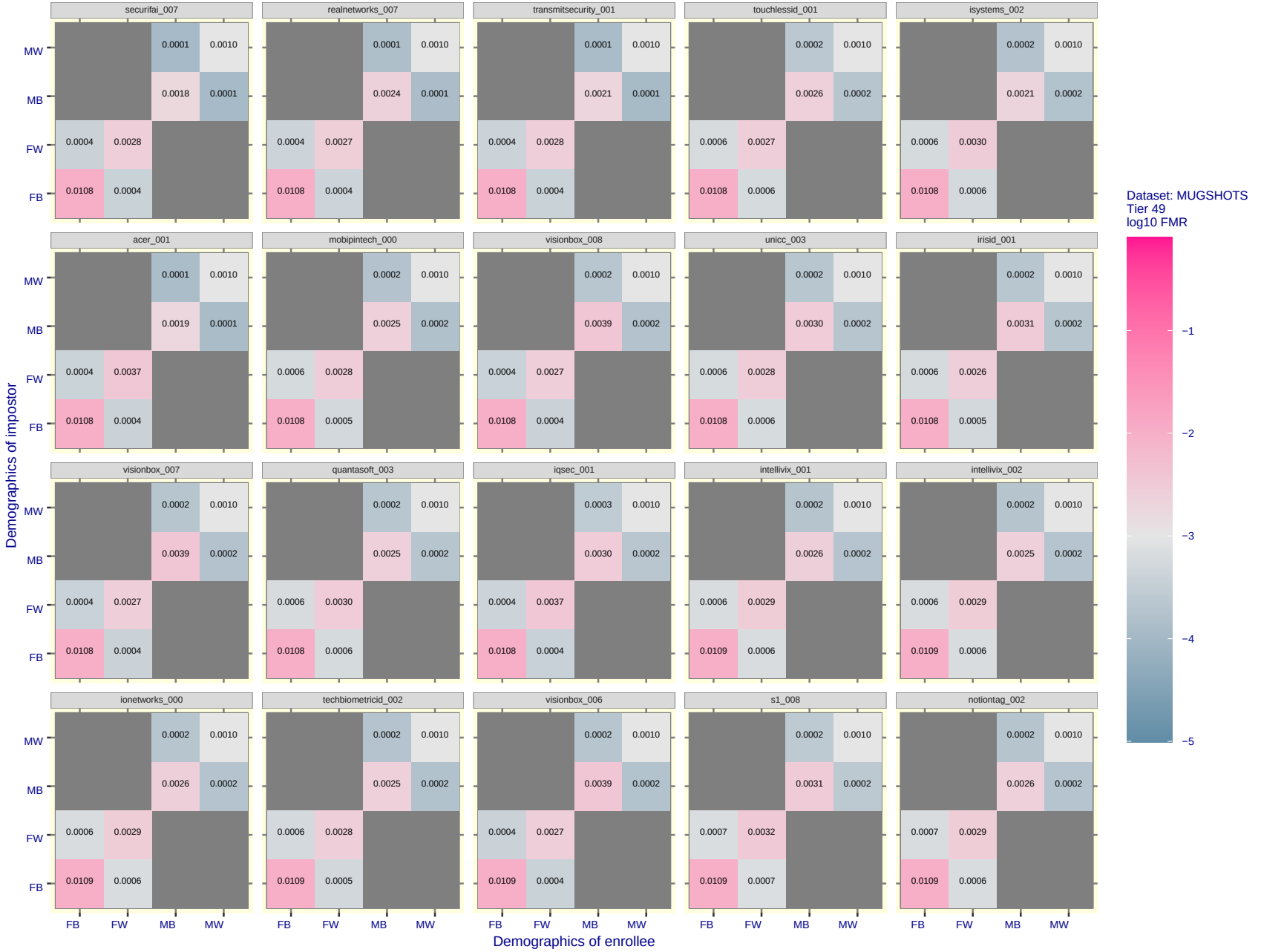


Figure 381: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

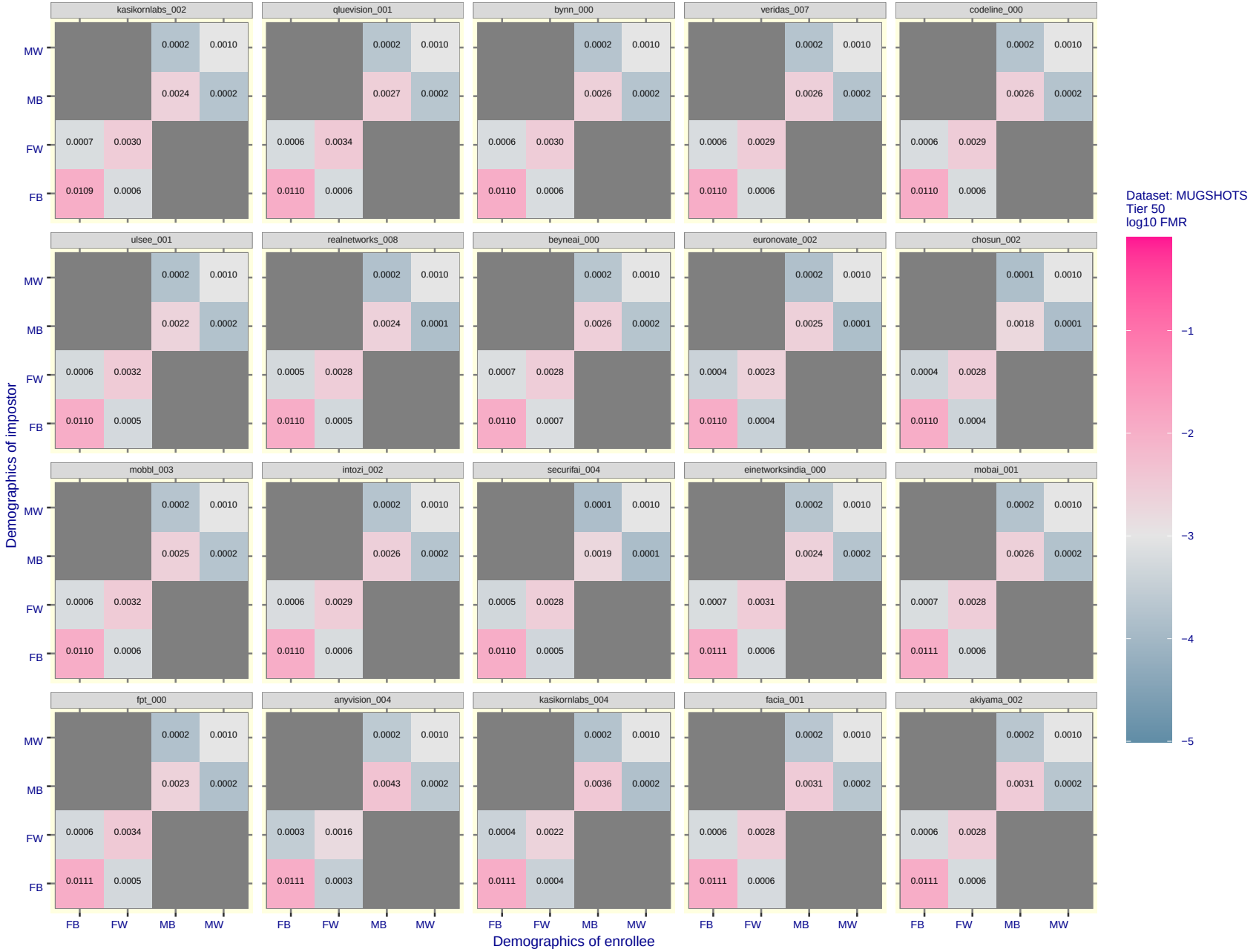


Figure 382: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

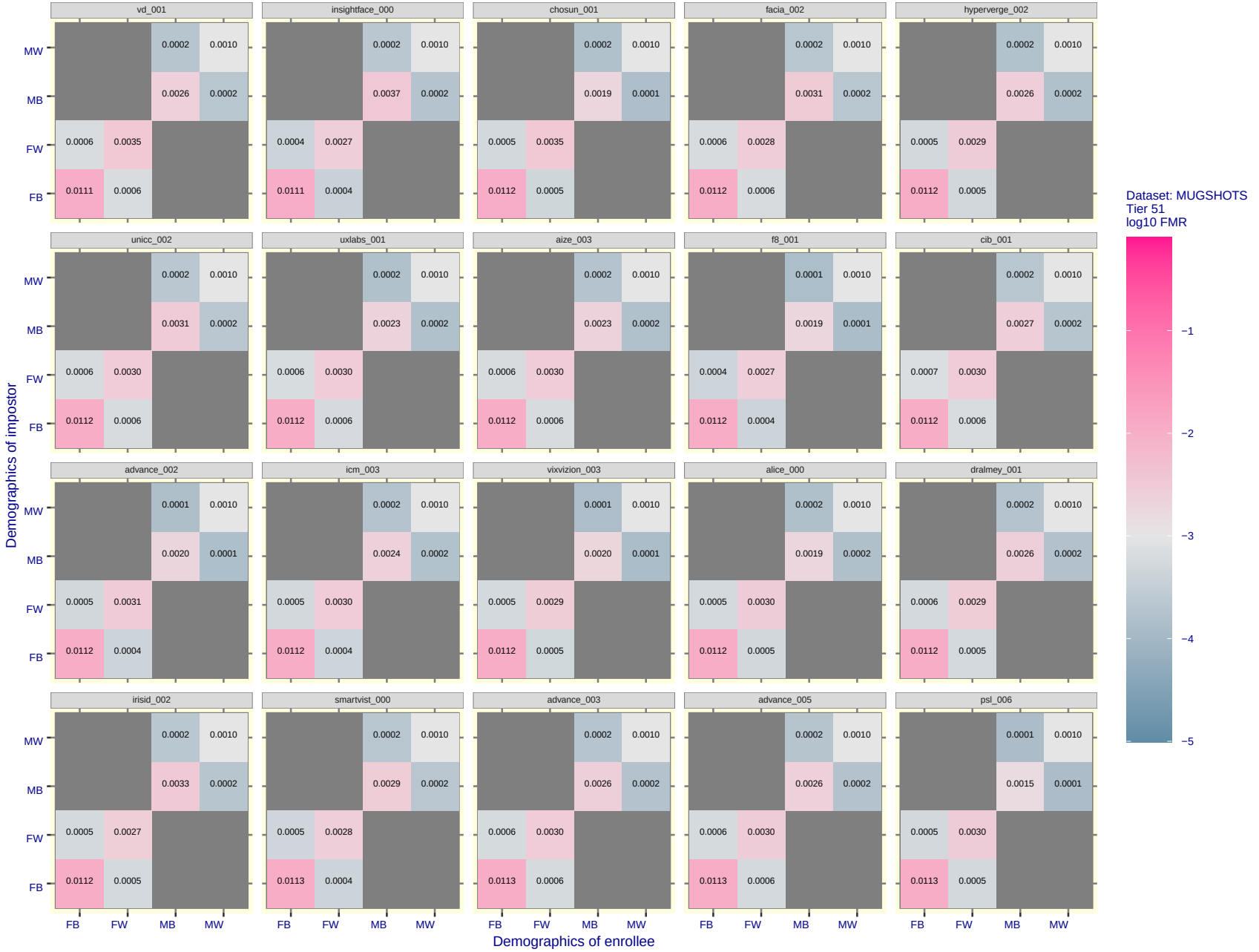


Figure 383: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

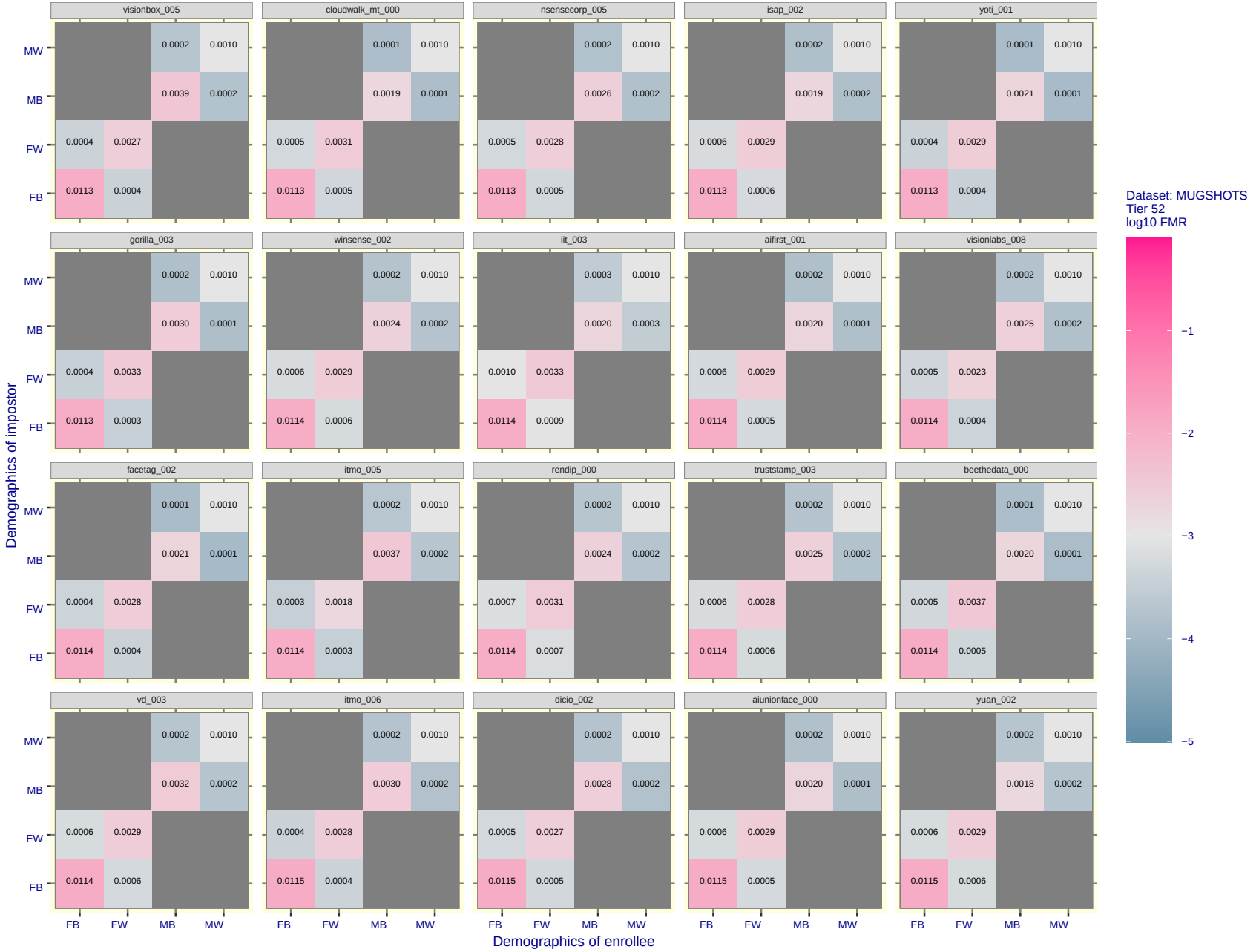


Figure 384: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

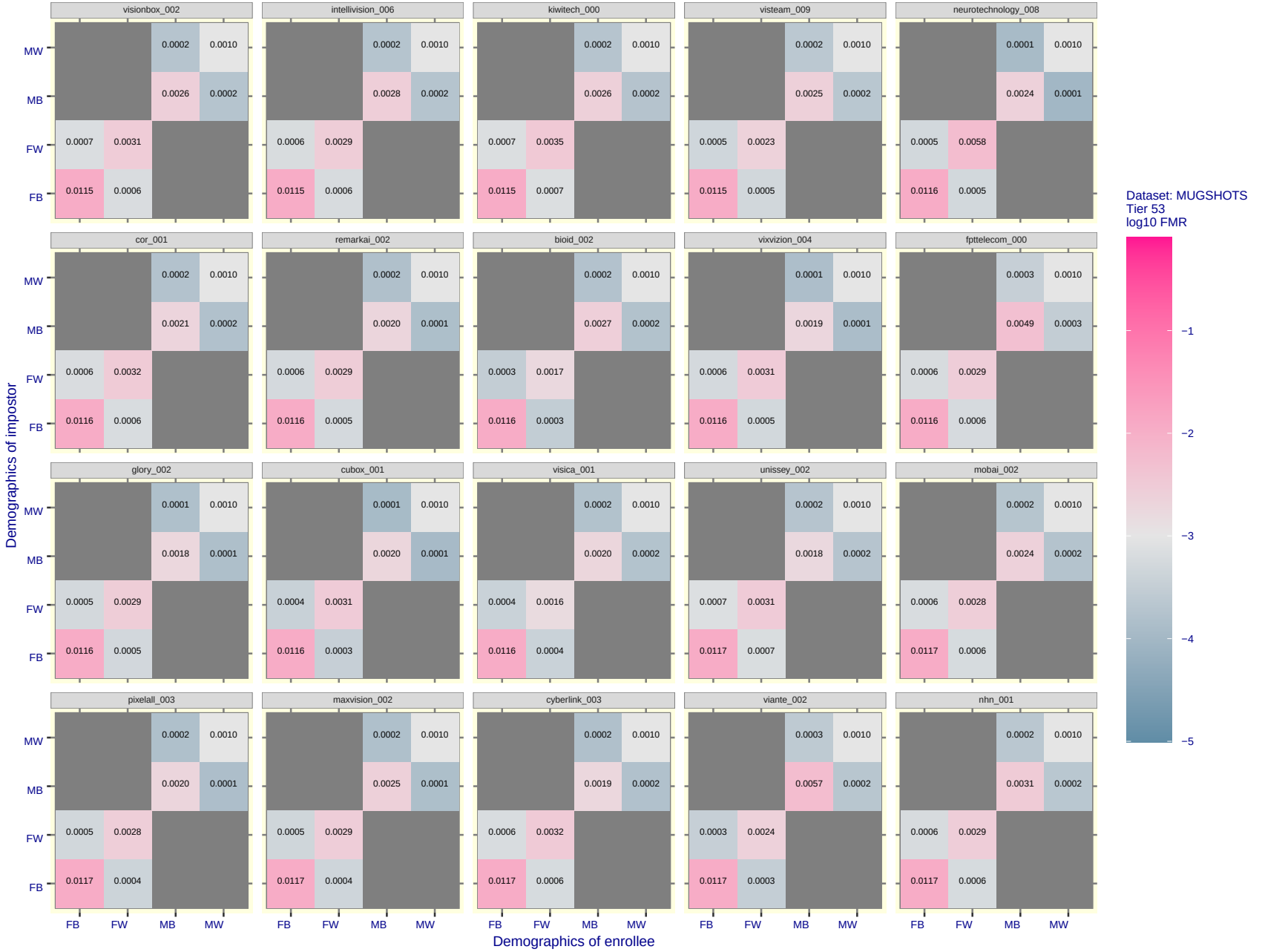


Figure 385: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

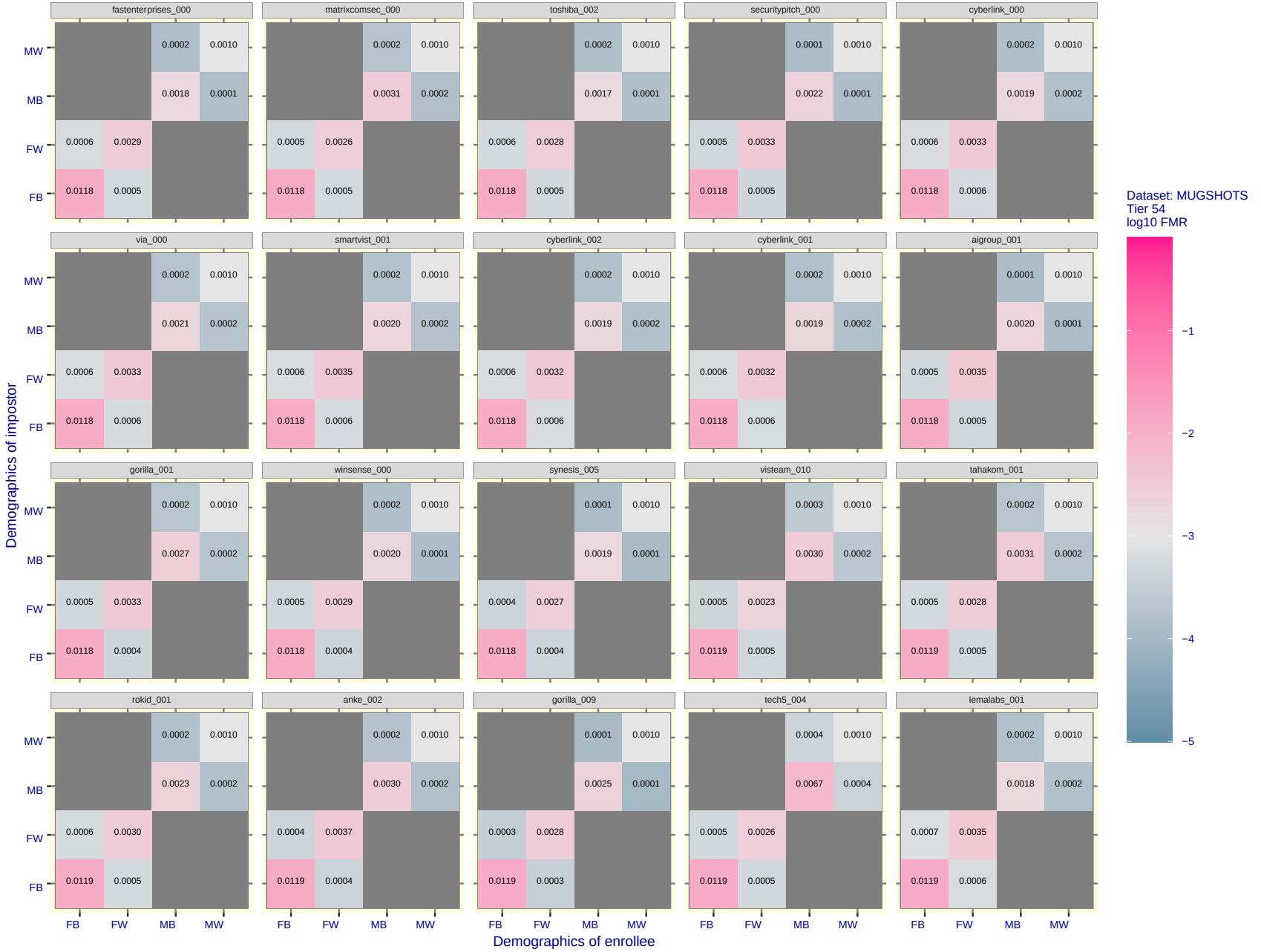


Figure 386: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

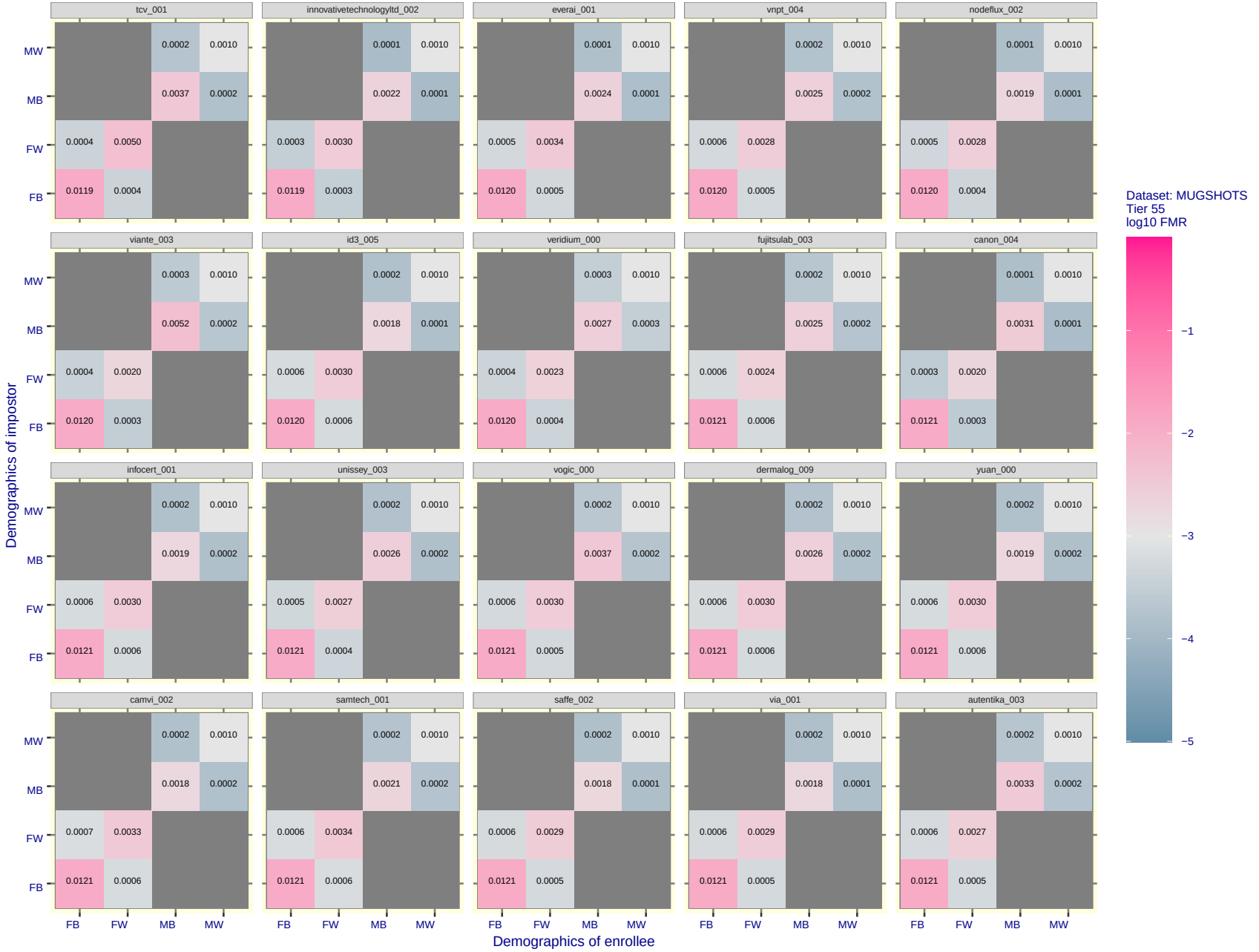


Figure 387: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

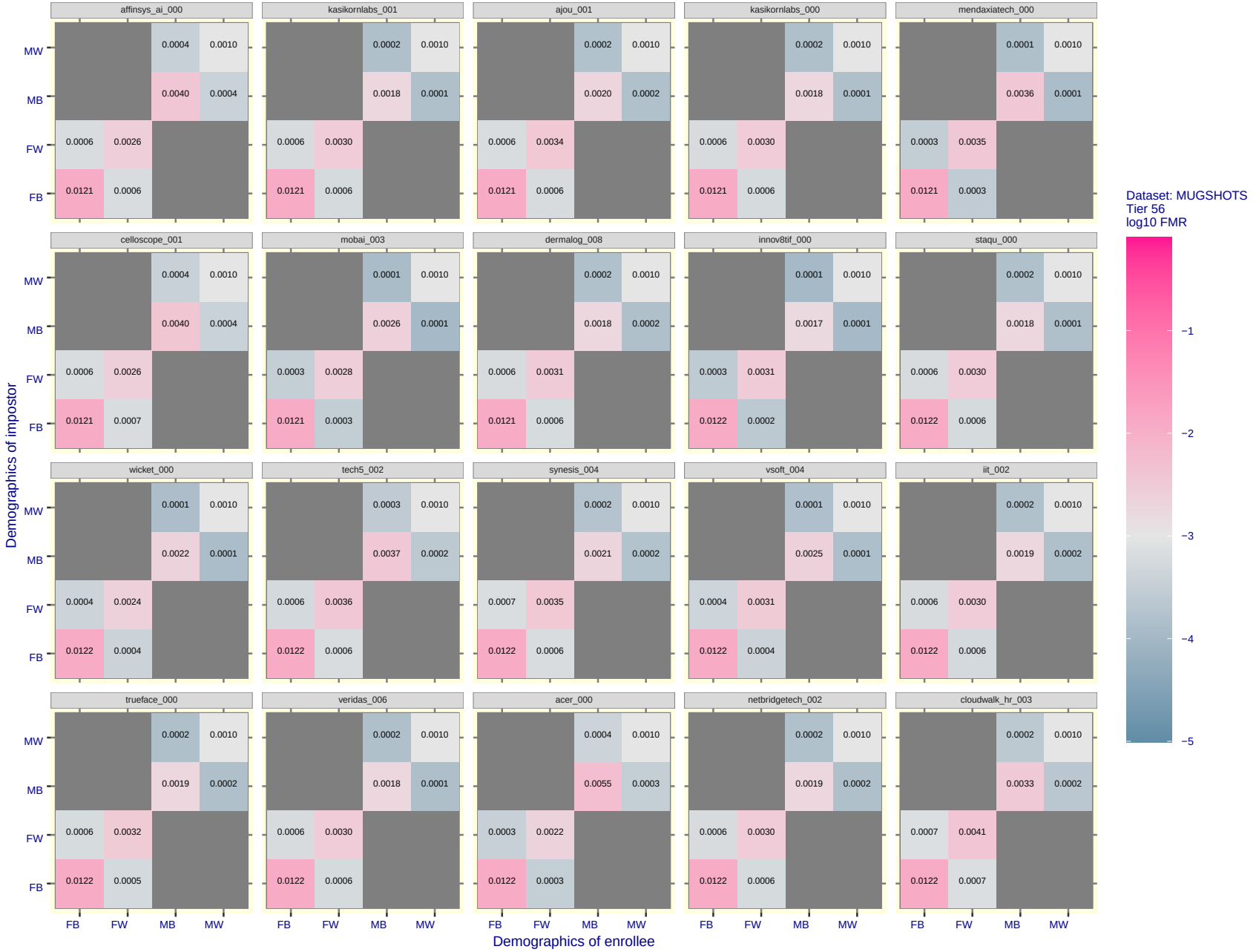


Figure 388: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

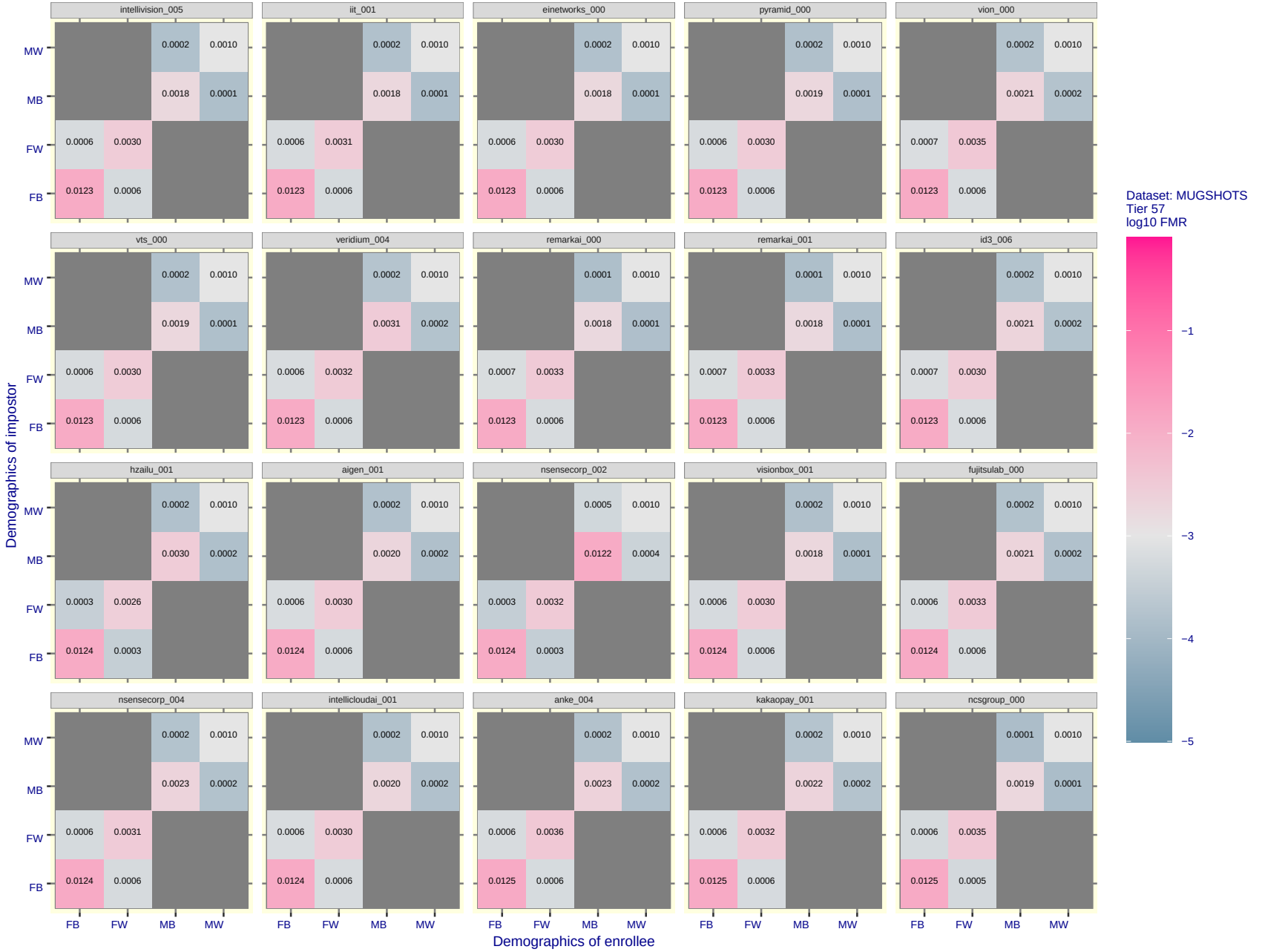


Figure 389: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

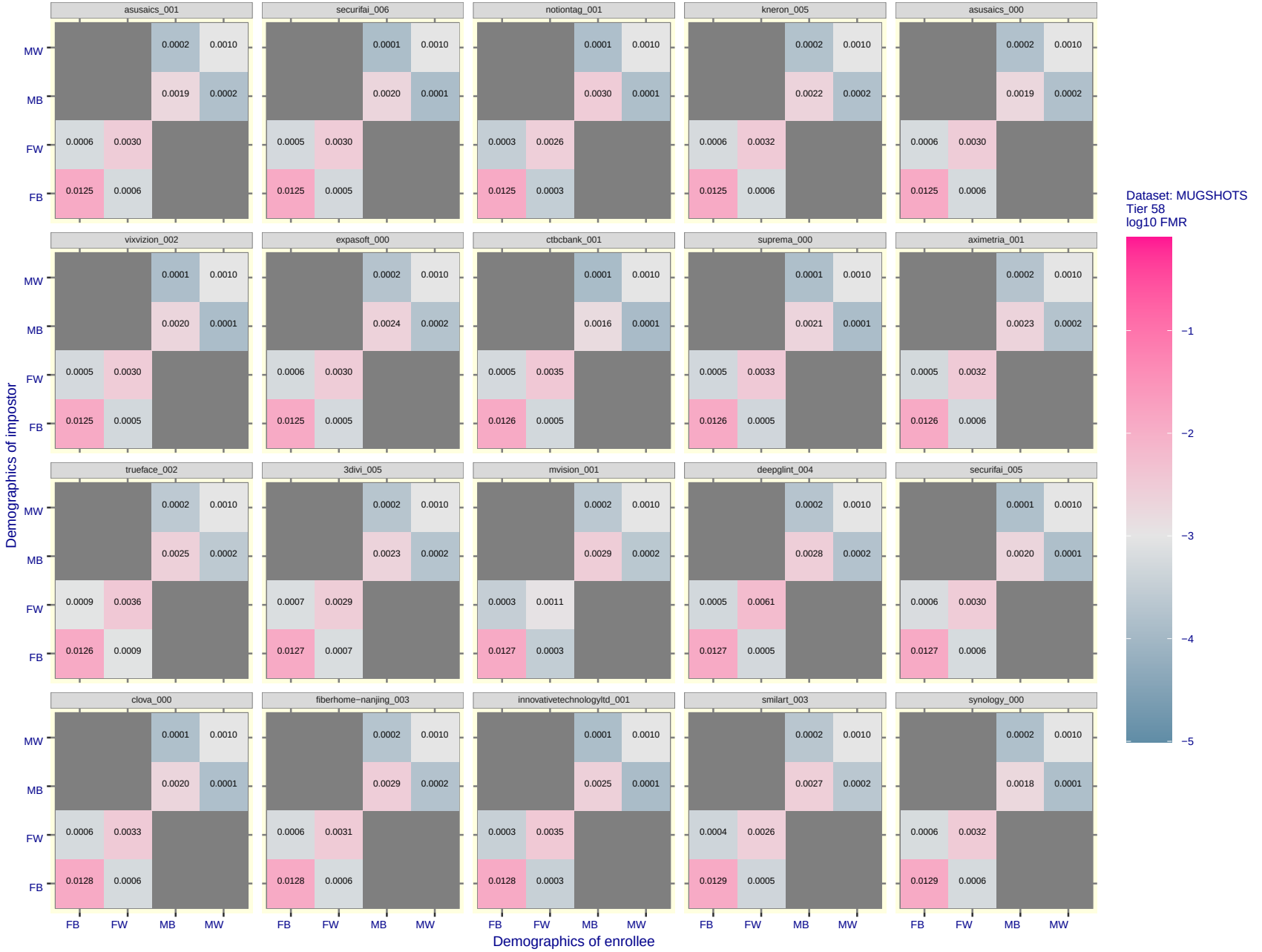


Figure 390: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

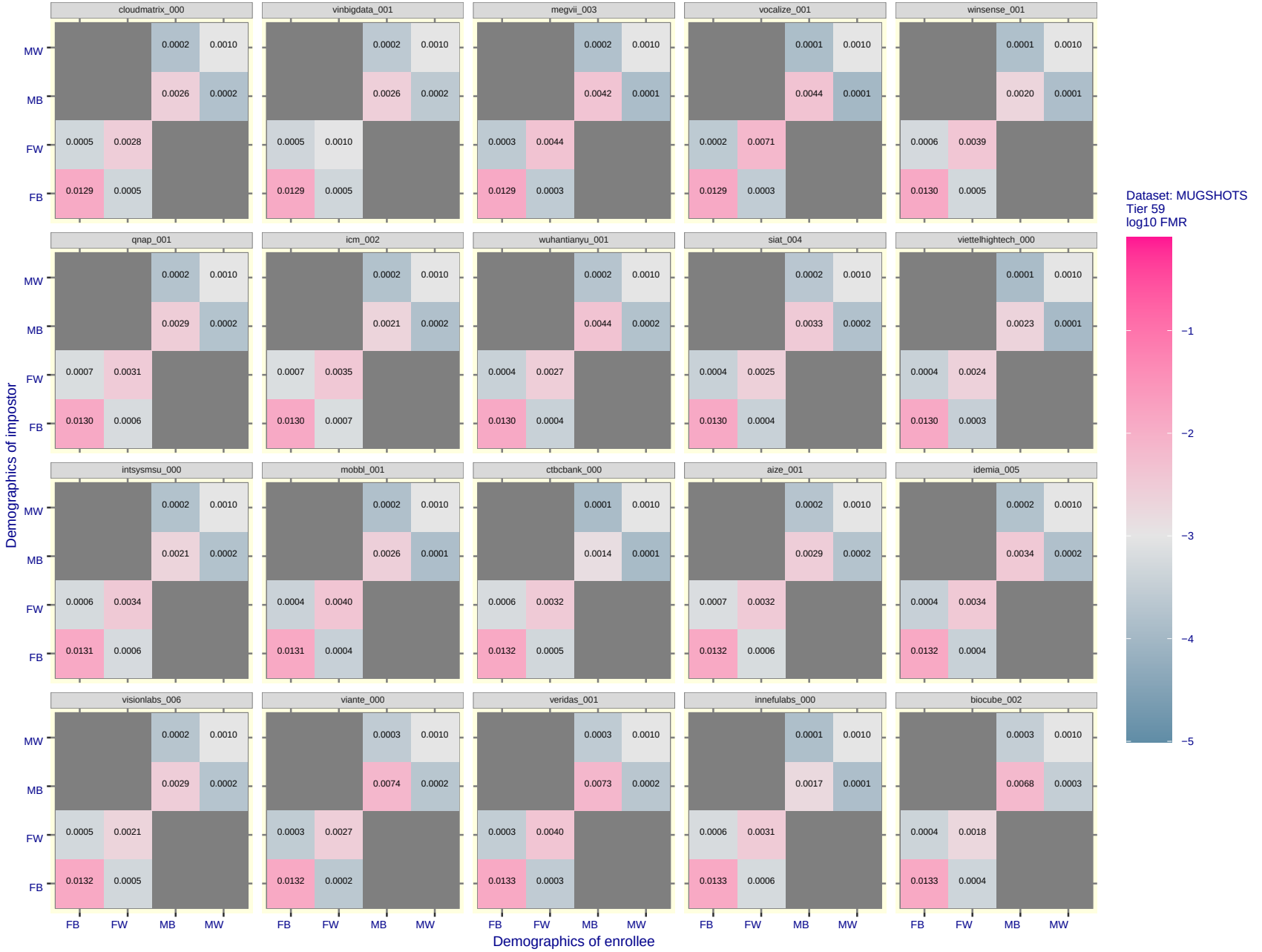


Figure 391: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

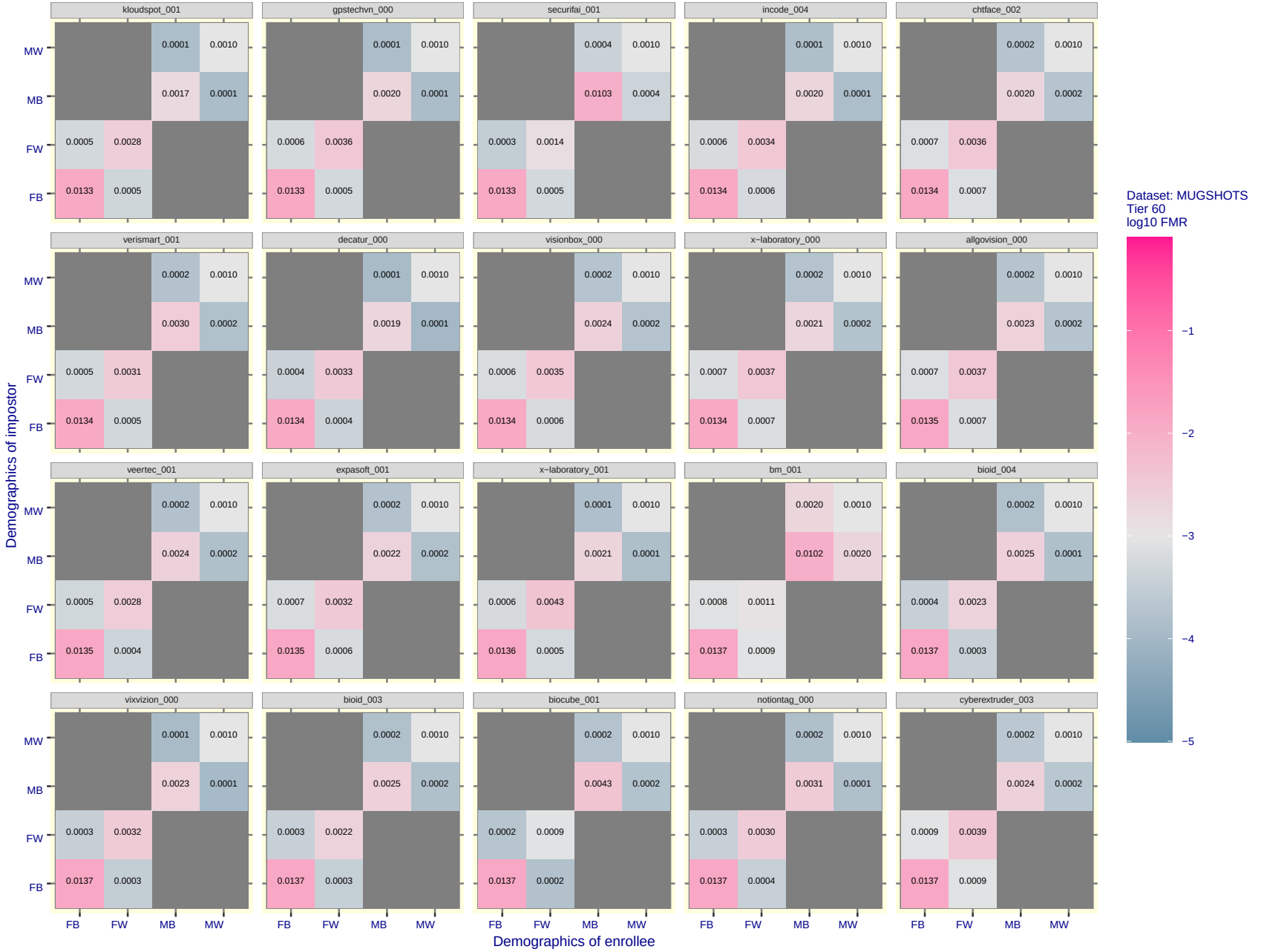


Figure 392: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

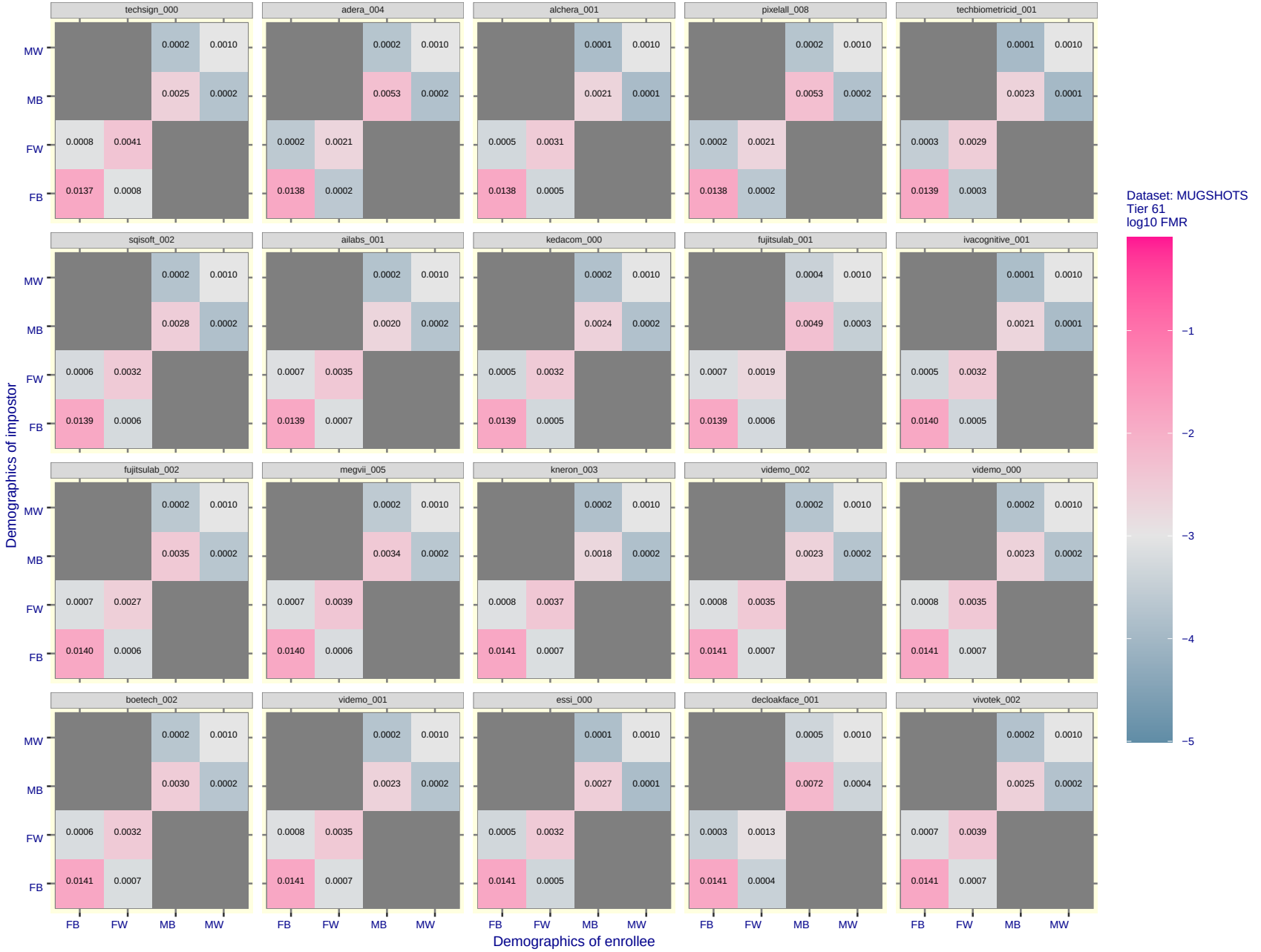


Figure 393: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

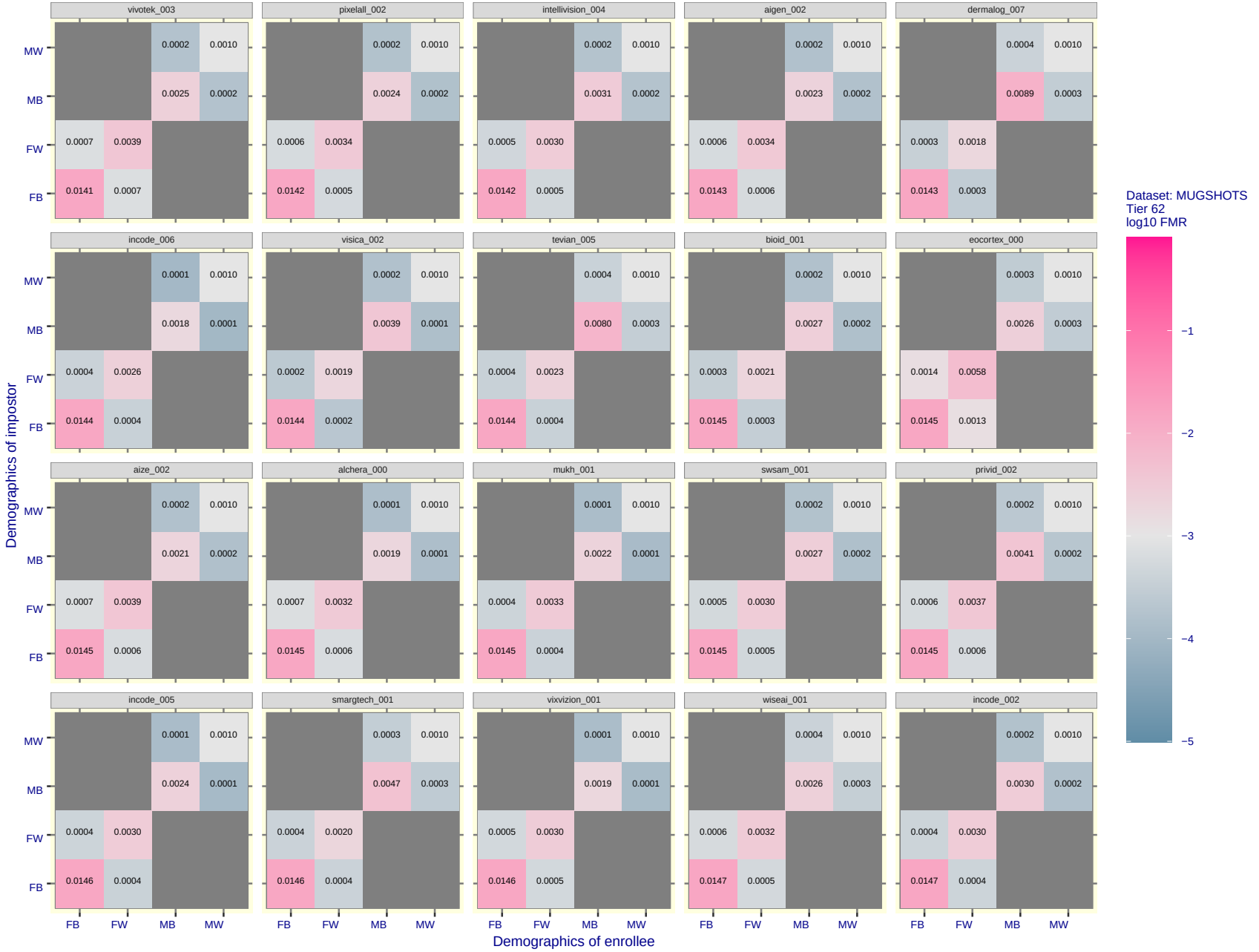


Figure 394: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

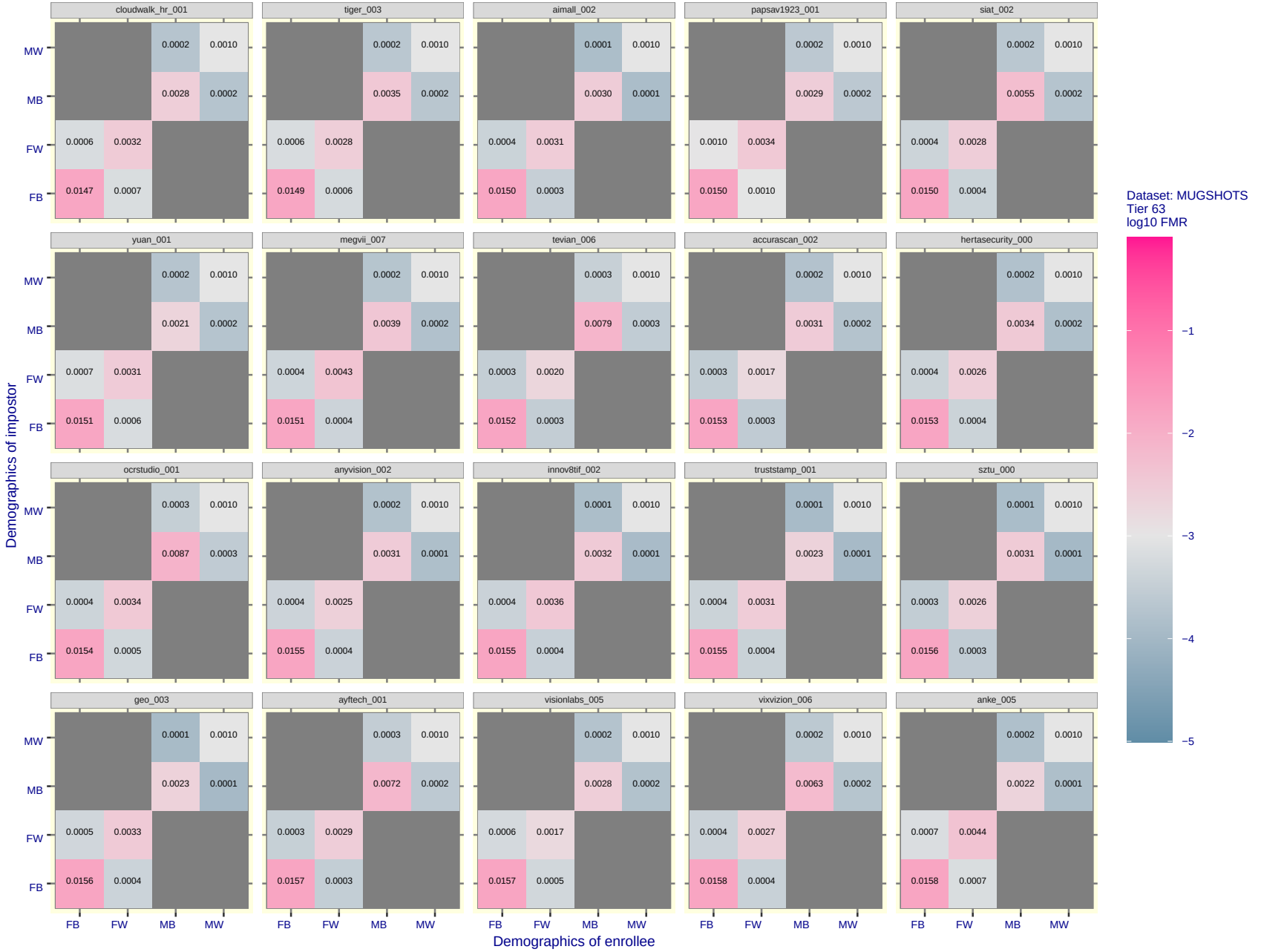


Figure 395: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

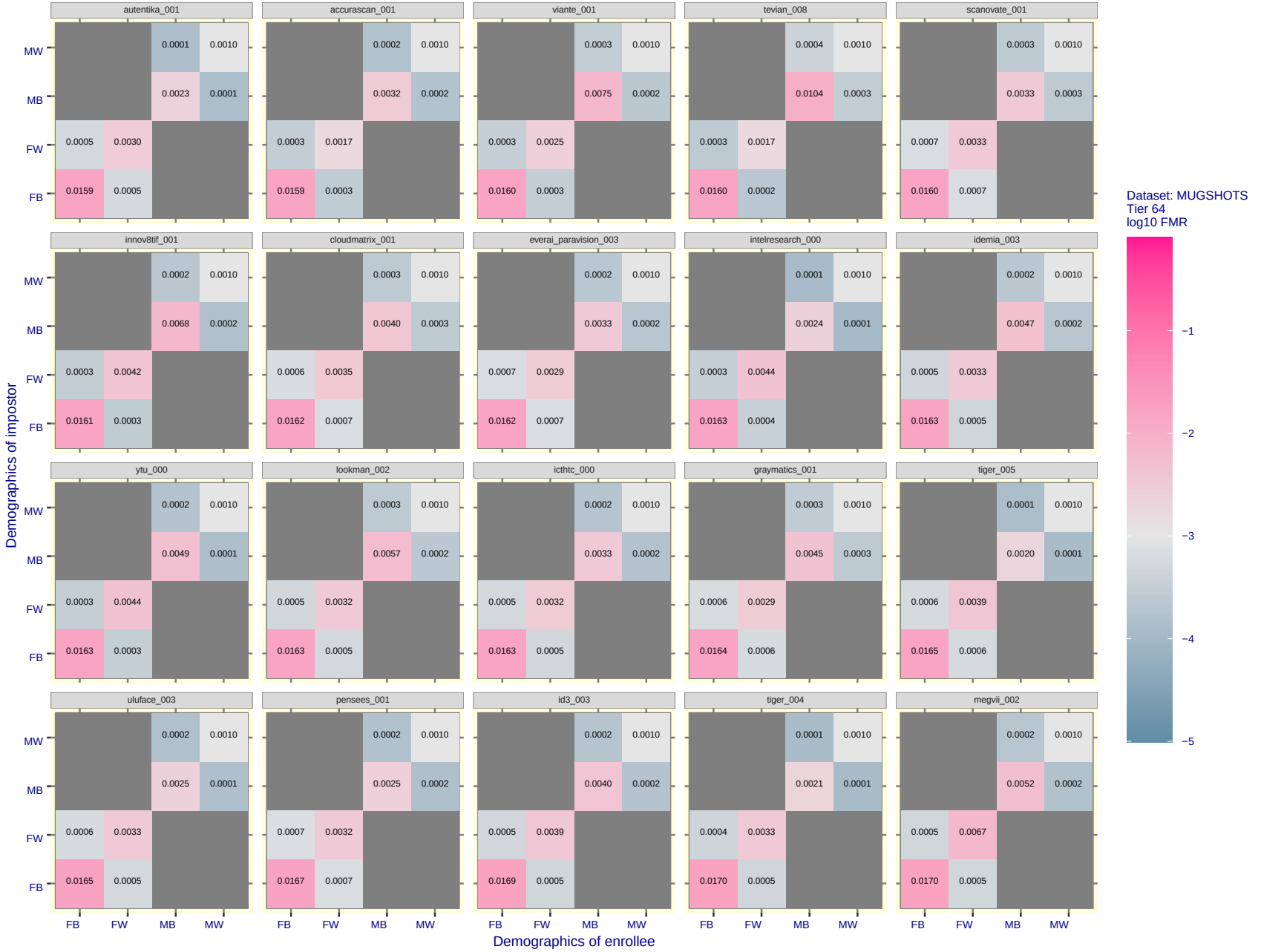


Figure 396: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

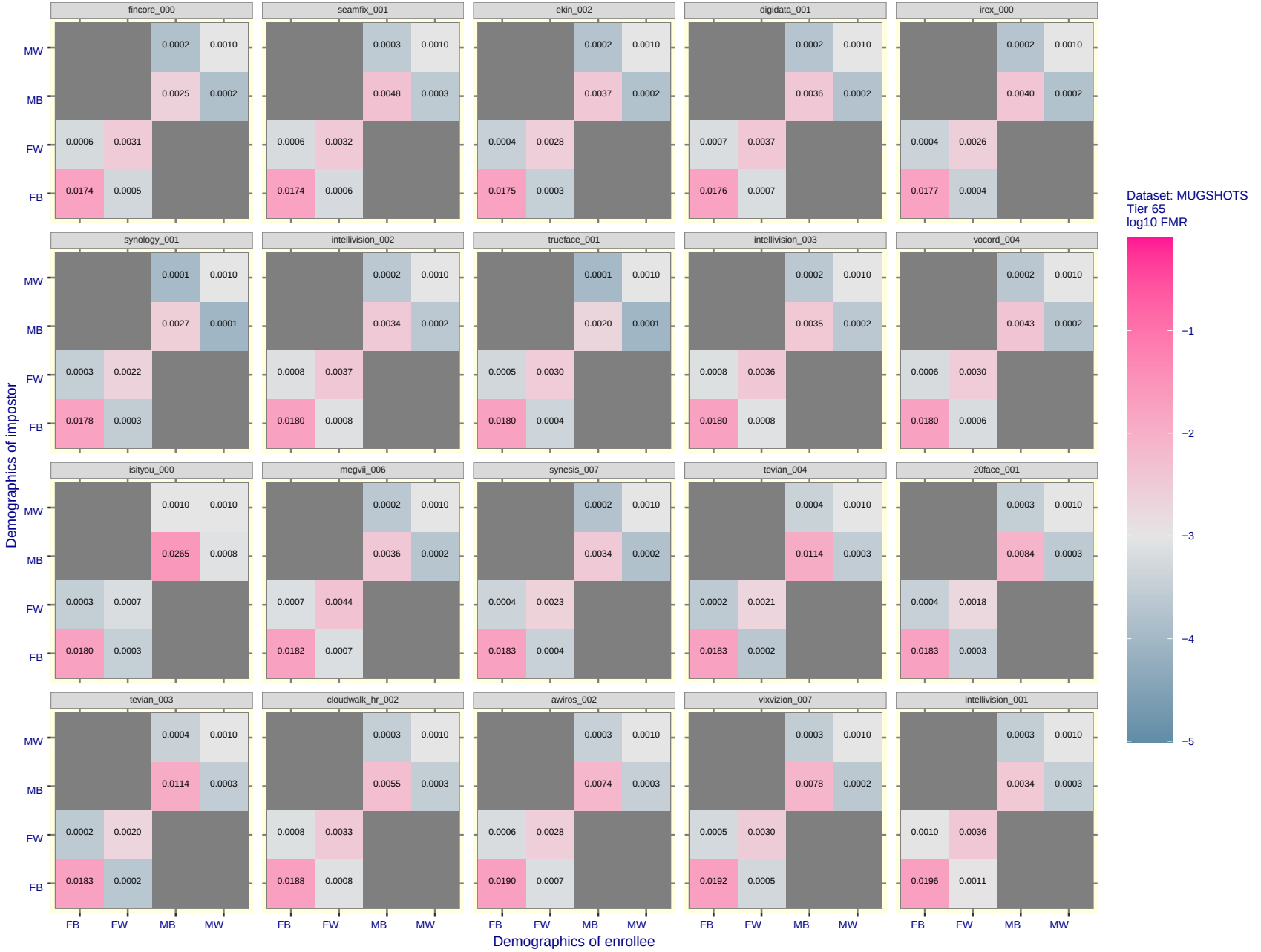


Figure 397: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

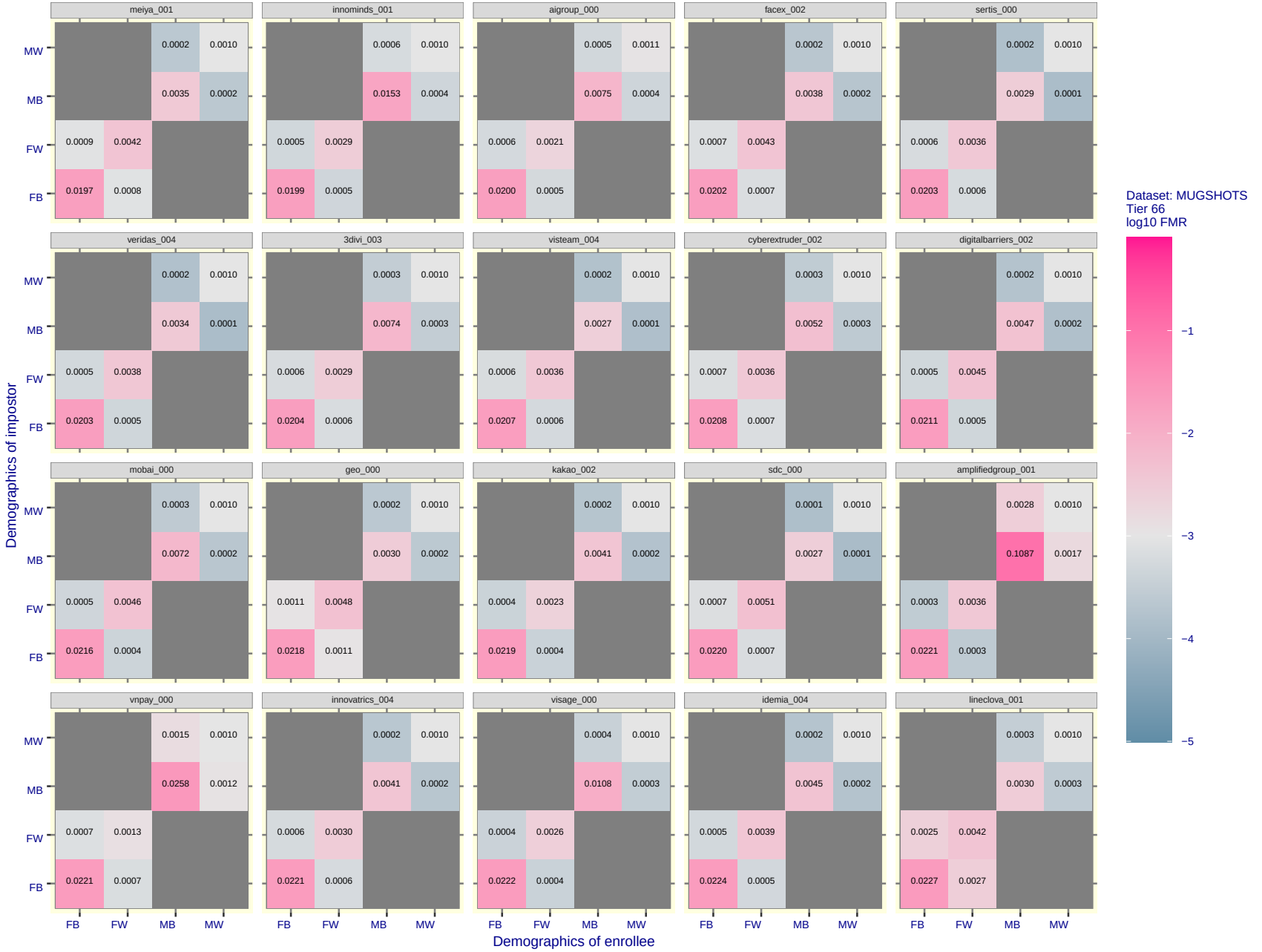


Figure 398: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

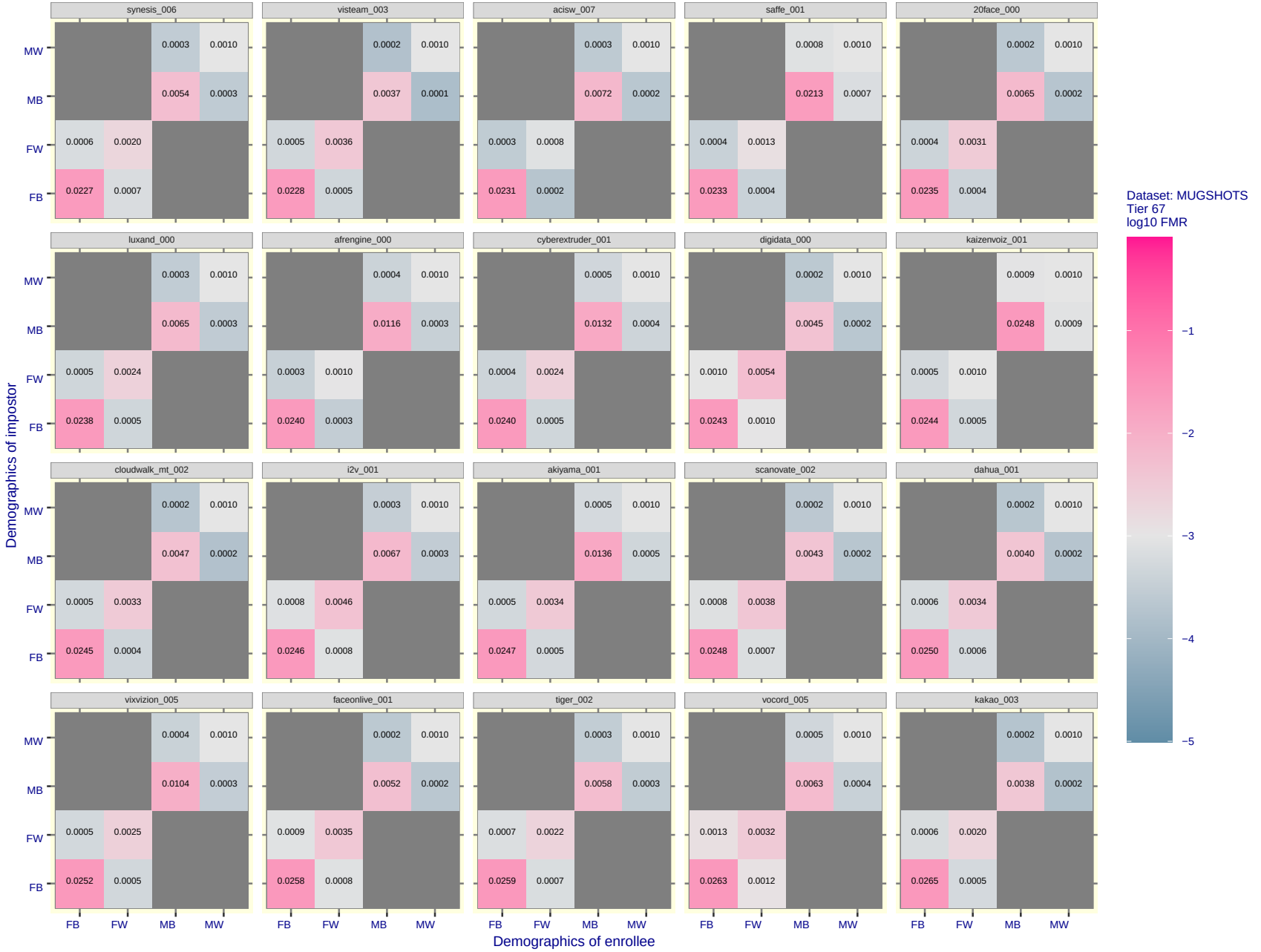


Figure 399: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

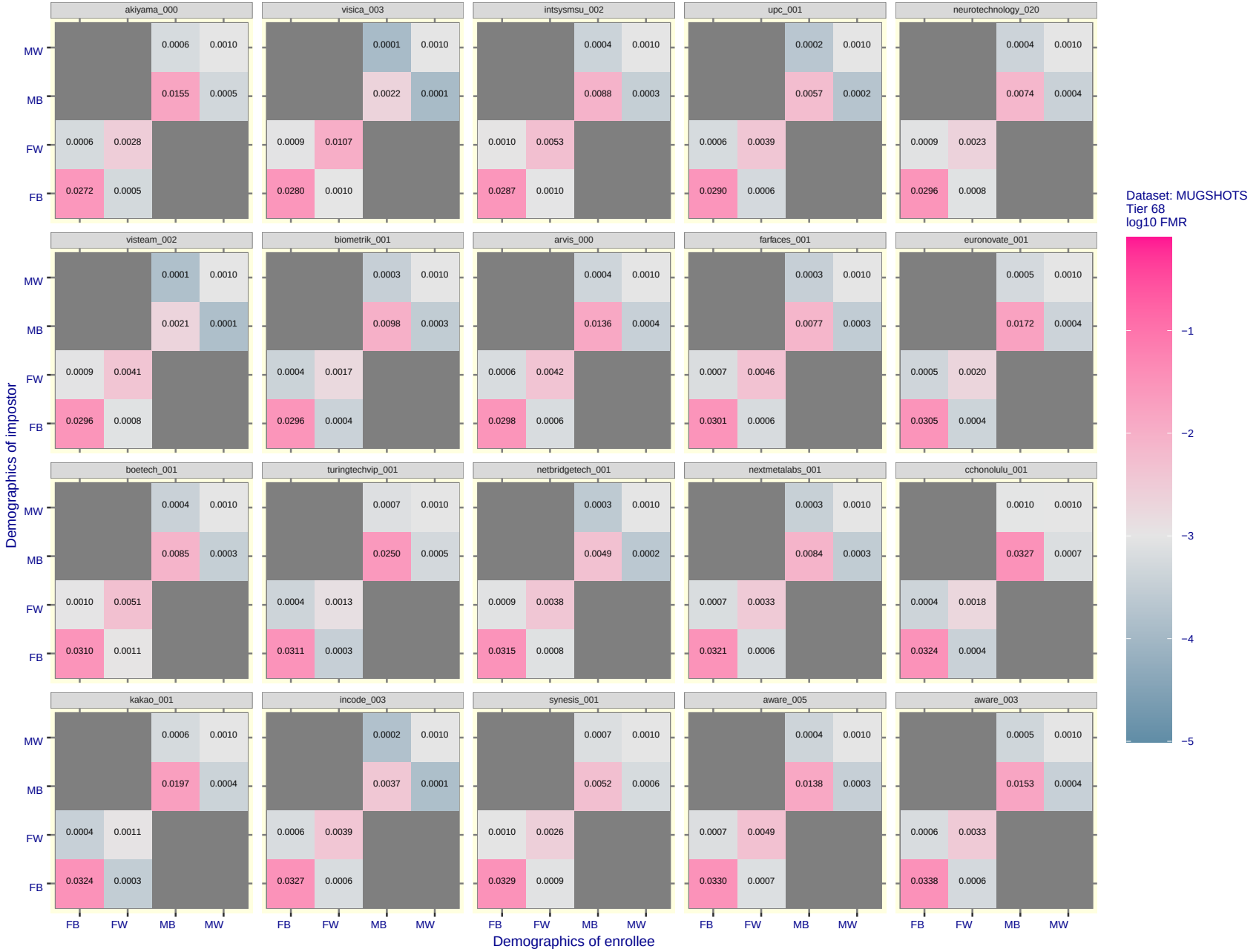


Figure 400: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

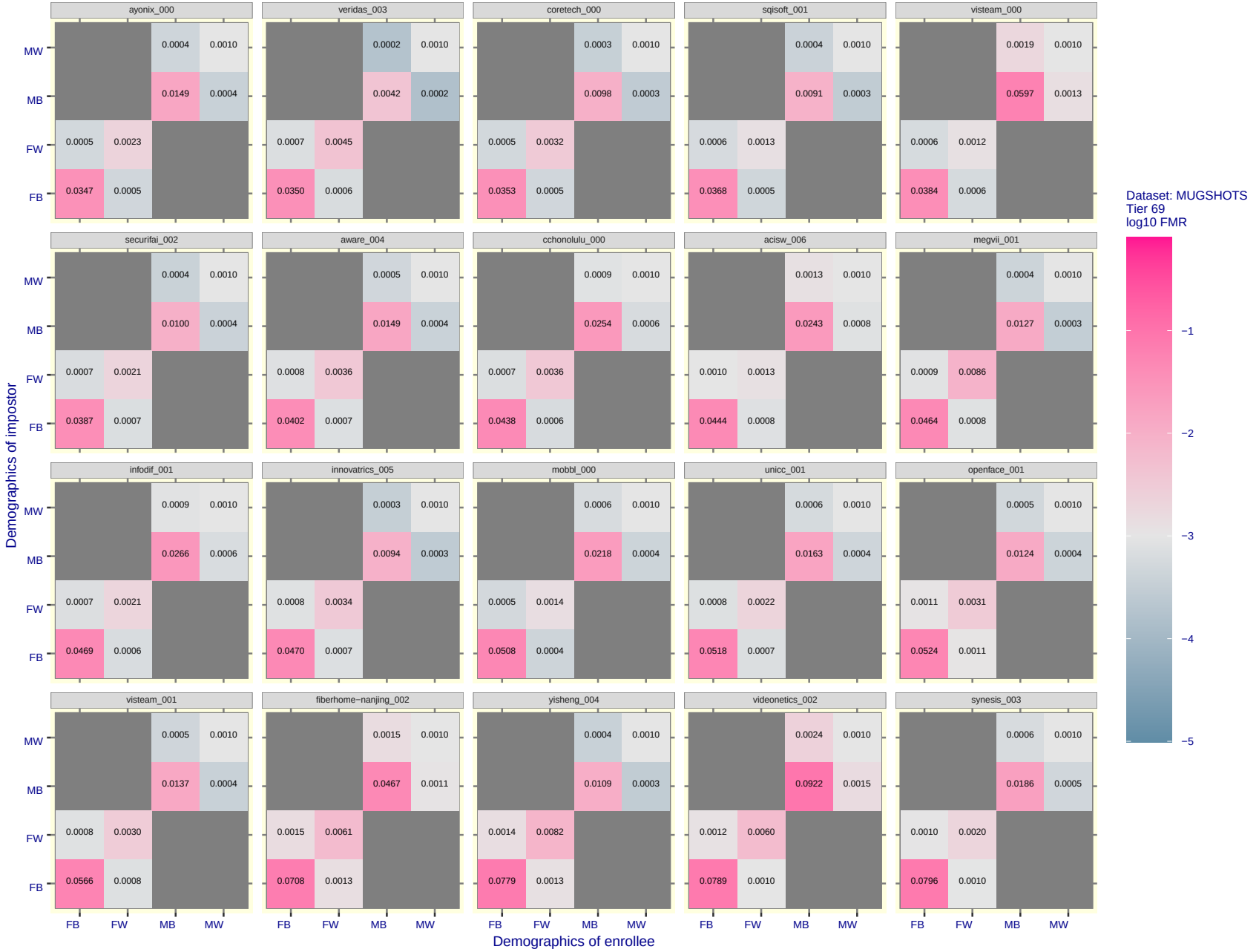


Figure 401: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

Figure 402: For the mugshot images, FMR *for same-sex impostor pairs of images annotated with codes for black female, black male, white female, white male. The threshold is set for each algorithm to give* FMR *= 0.001 for white males which is the demographic that usually gives the lowest* FMR*. This means the top right box is the same color in all panels. The panels are sorted over multiple pages in order of* FMR *on black females, which is the demographc that usually gives the highest* FMR*.*

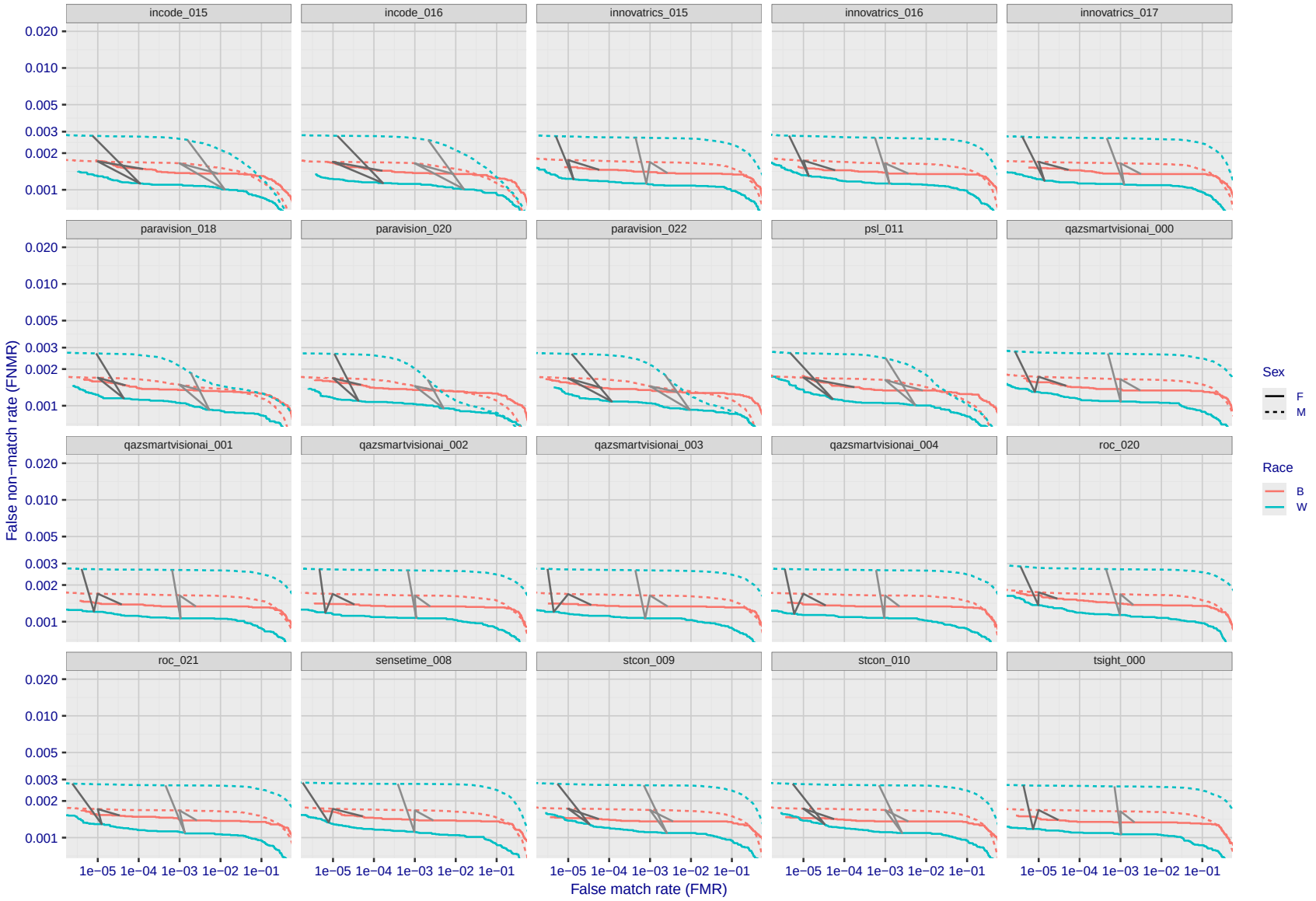


Figure 403: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

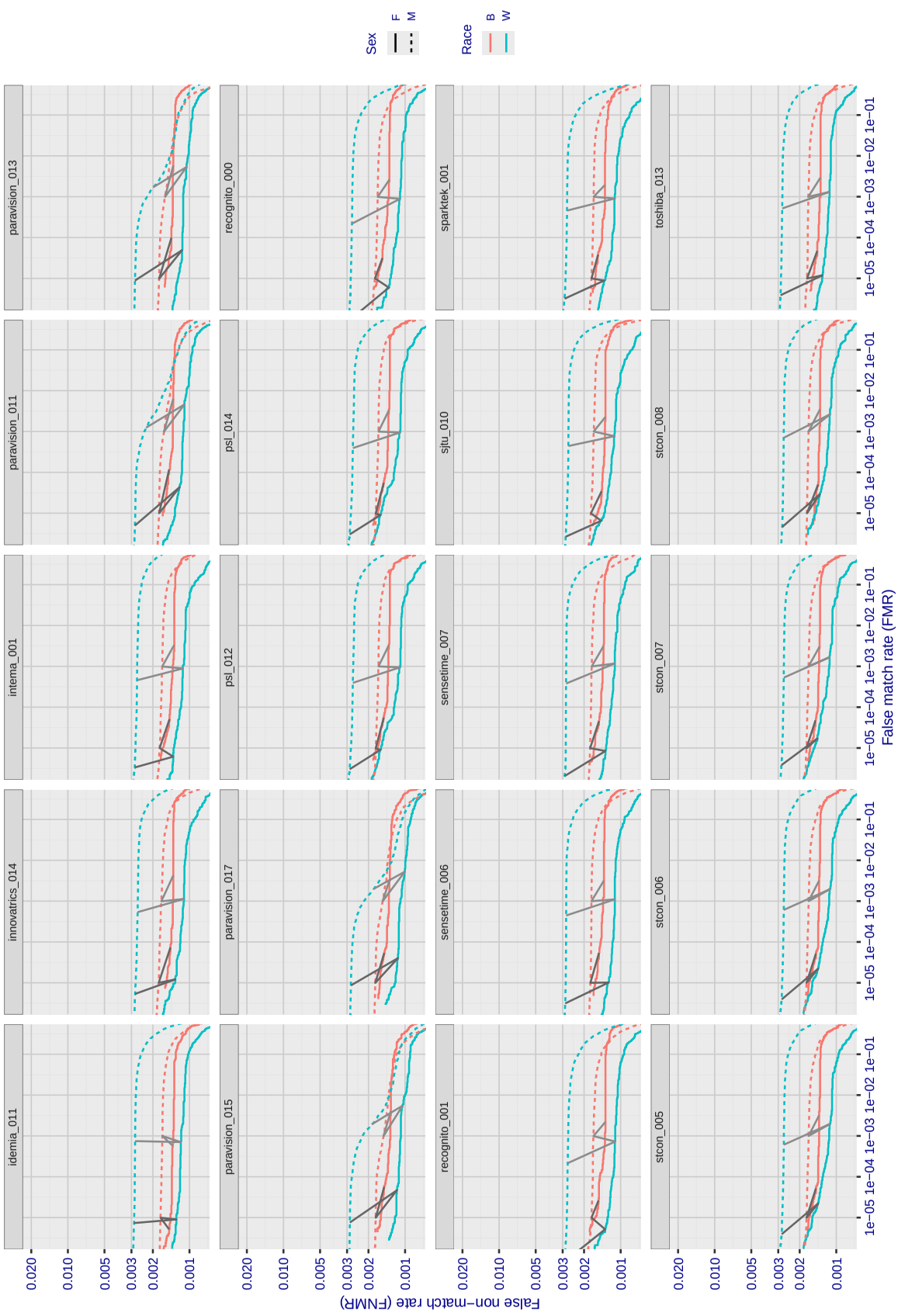


Figure 404: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

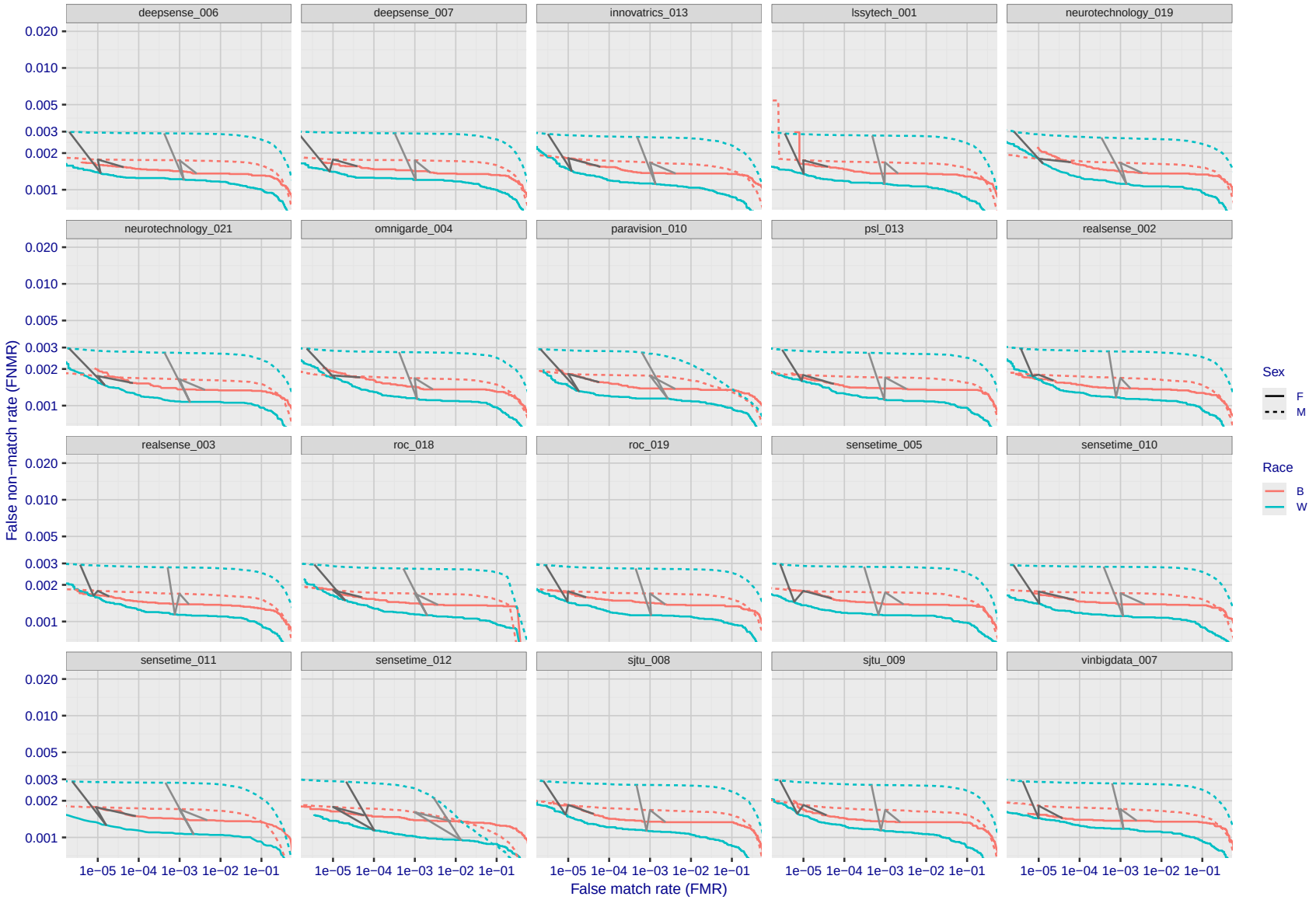


Figure 405: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

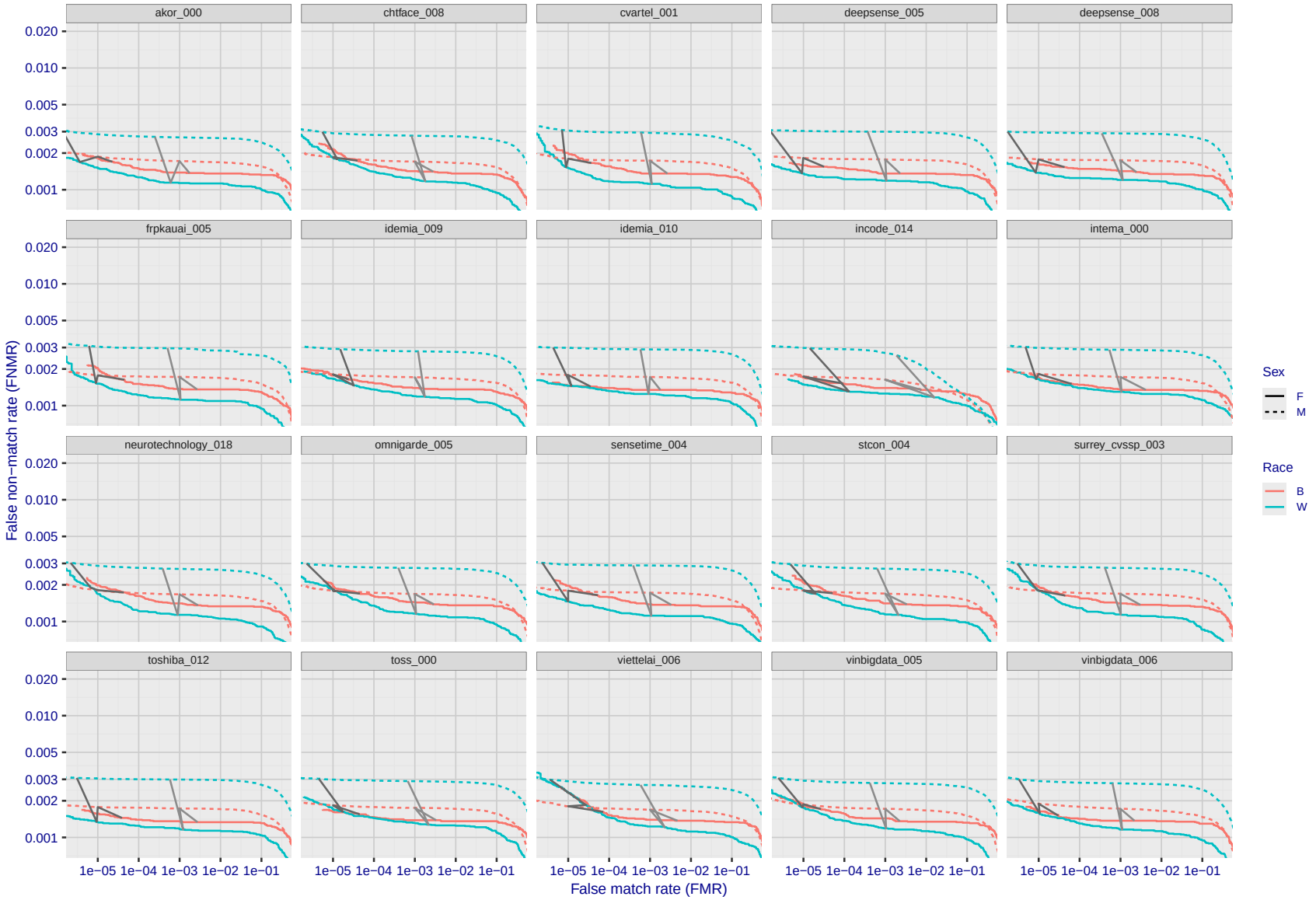


Figure 406: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

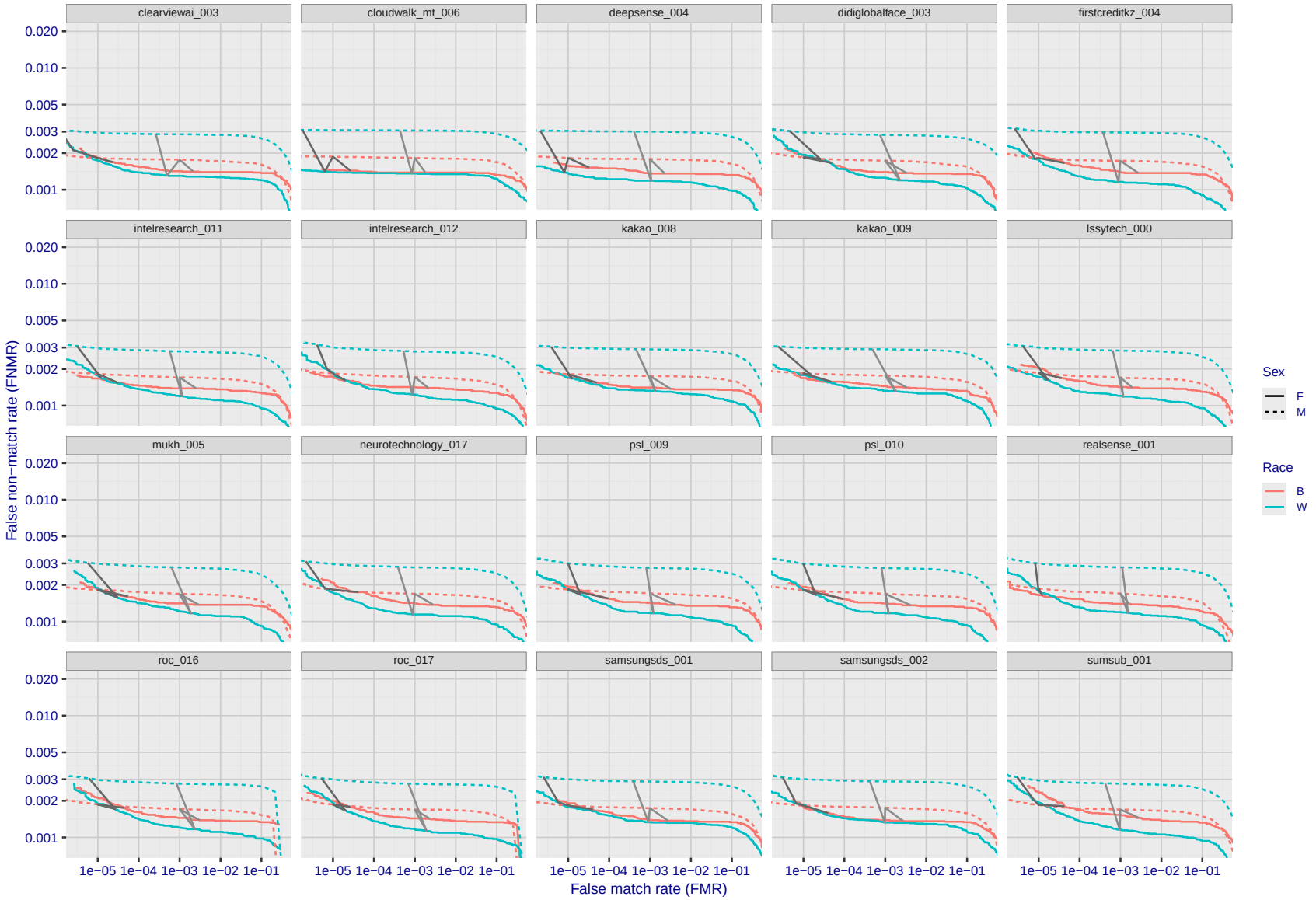


Figure 407: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

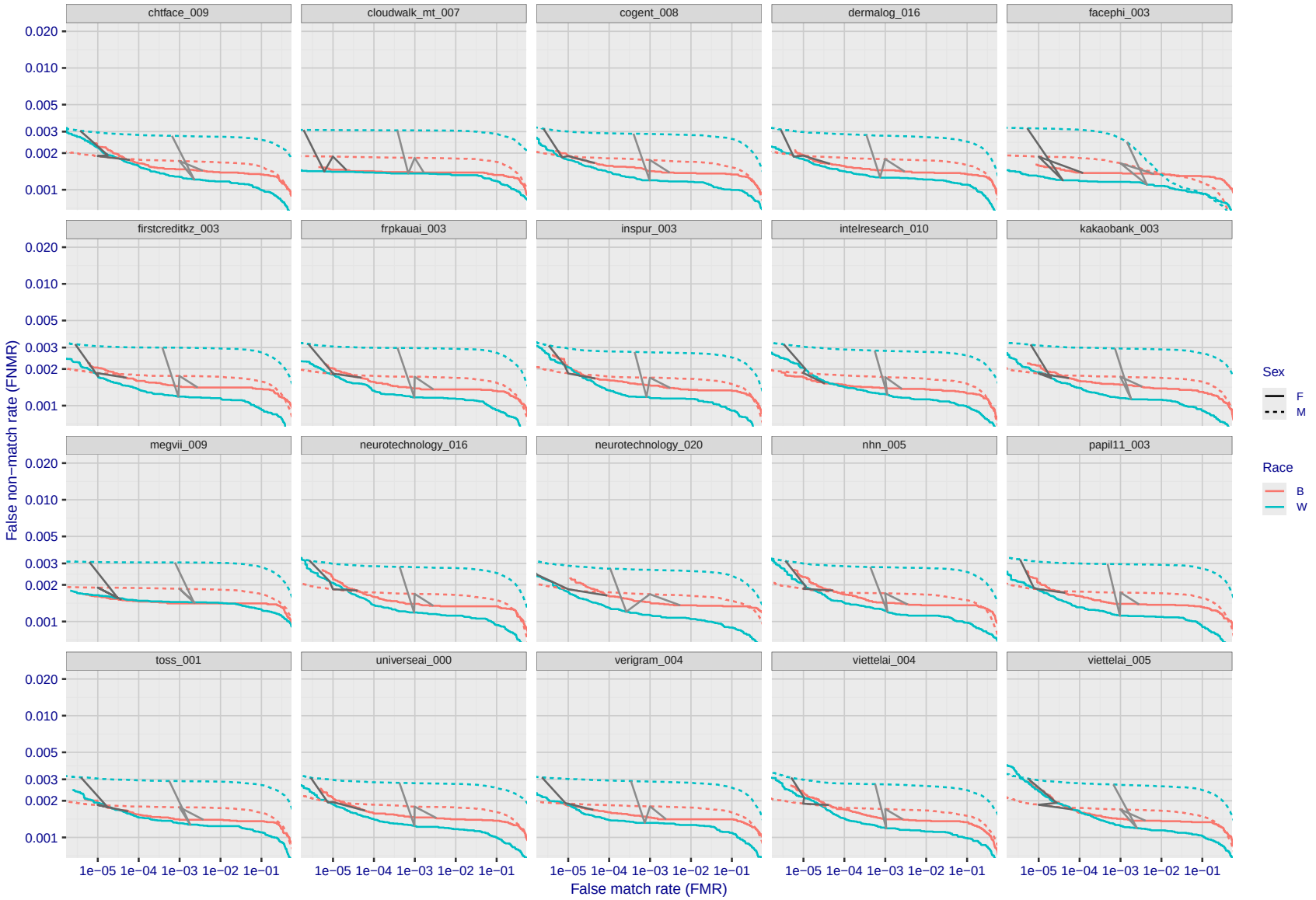


Figure 408: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

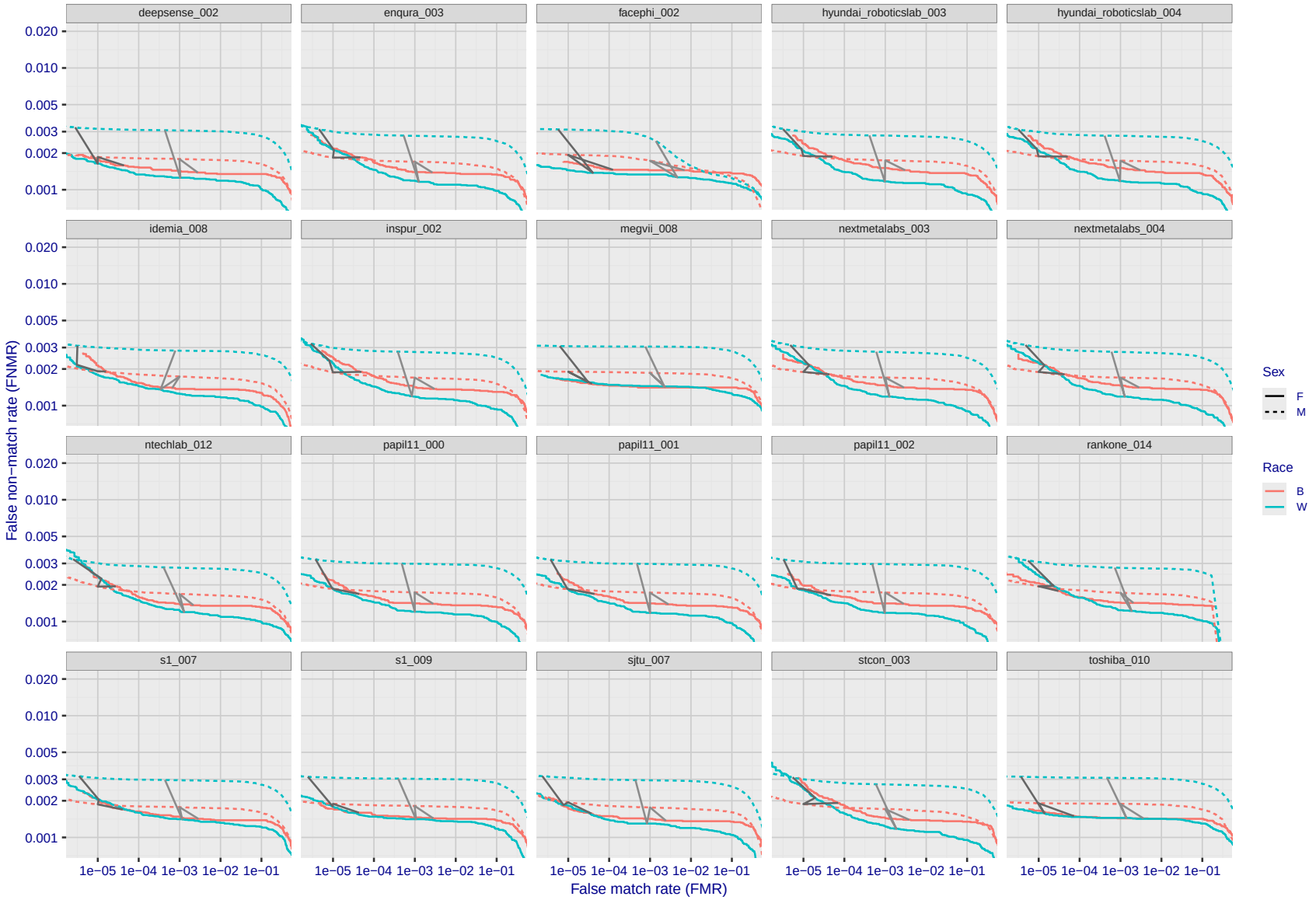


Figure 409: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

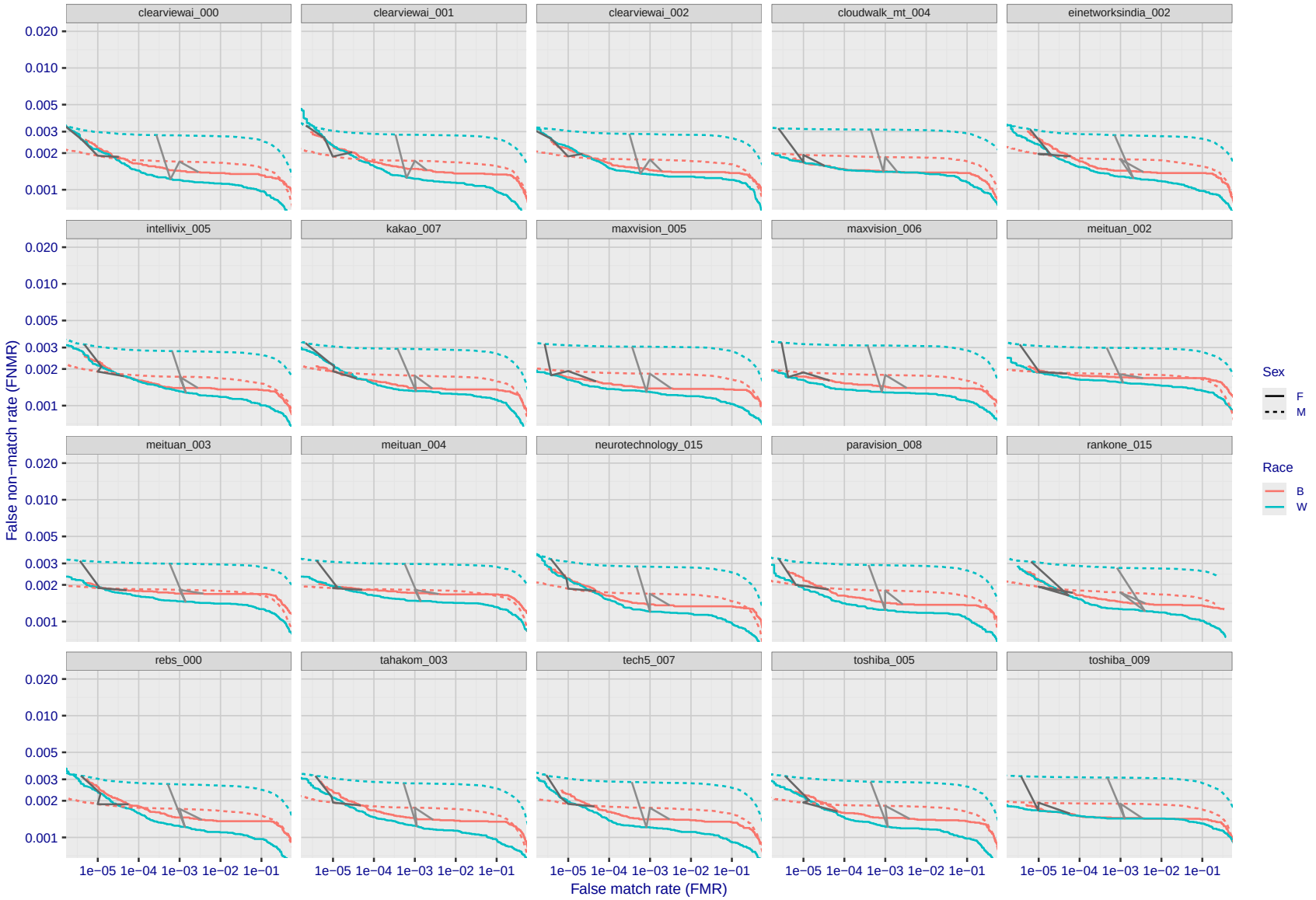


Figure 410: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

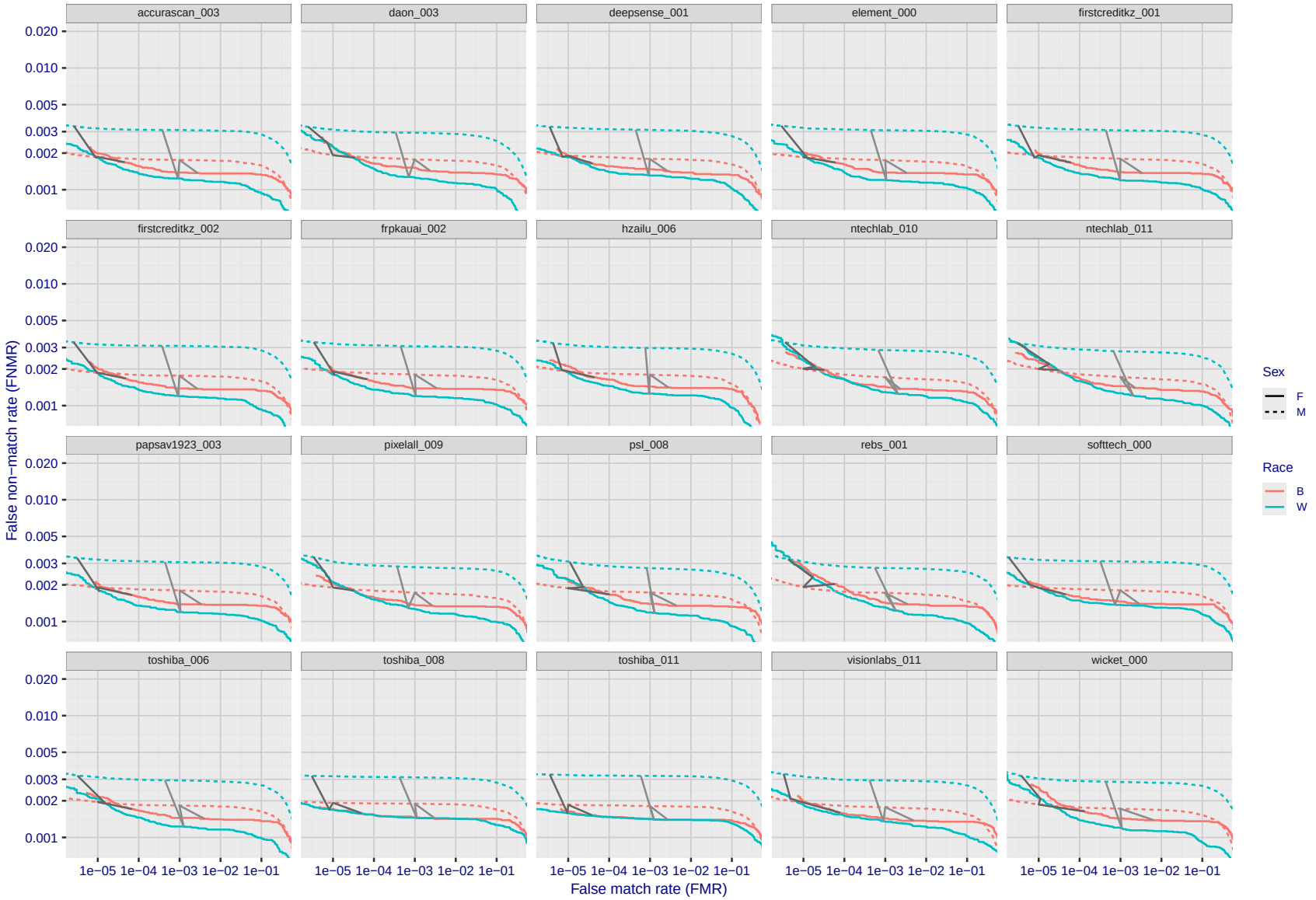


Figure 411: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

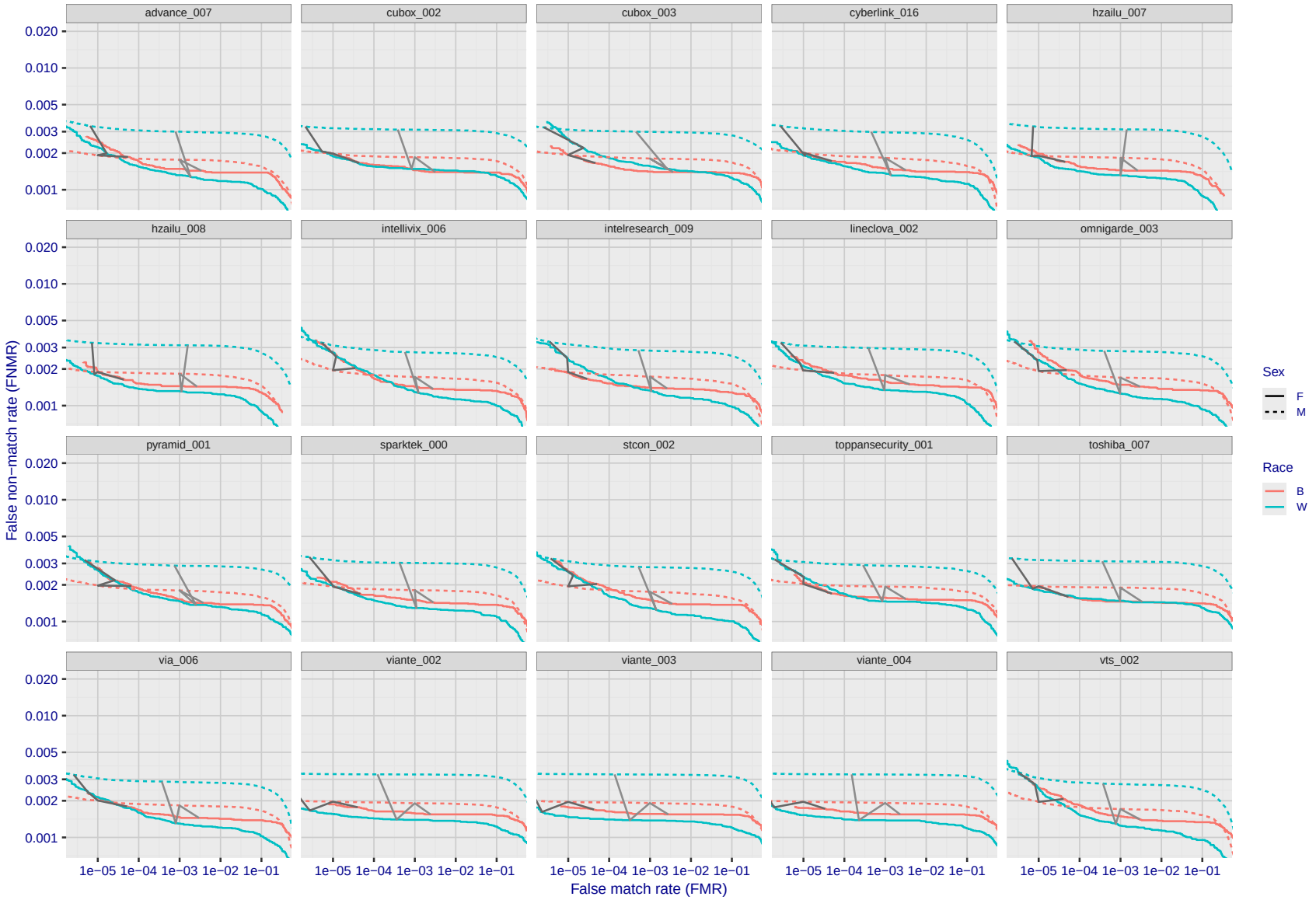


Figure 412: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

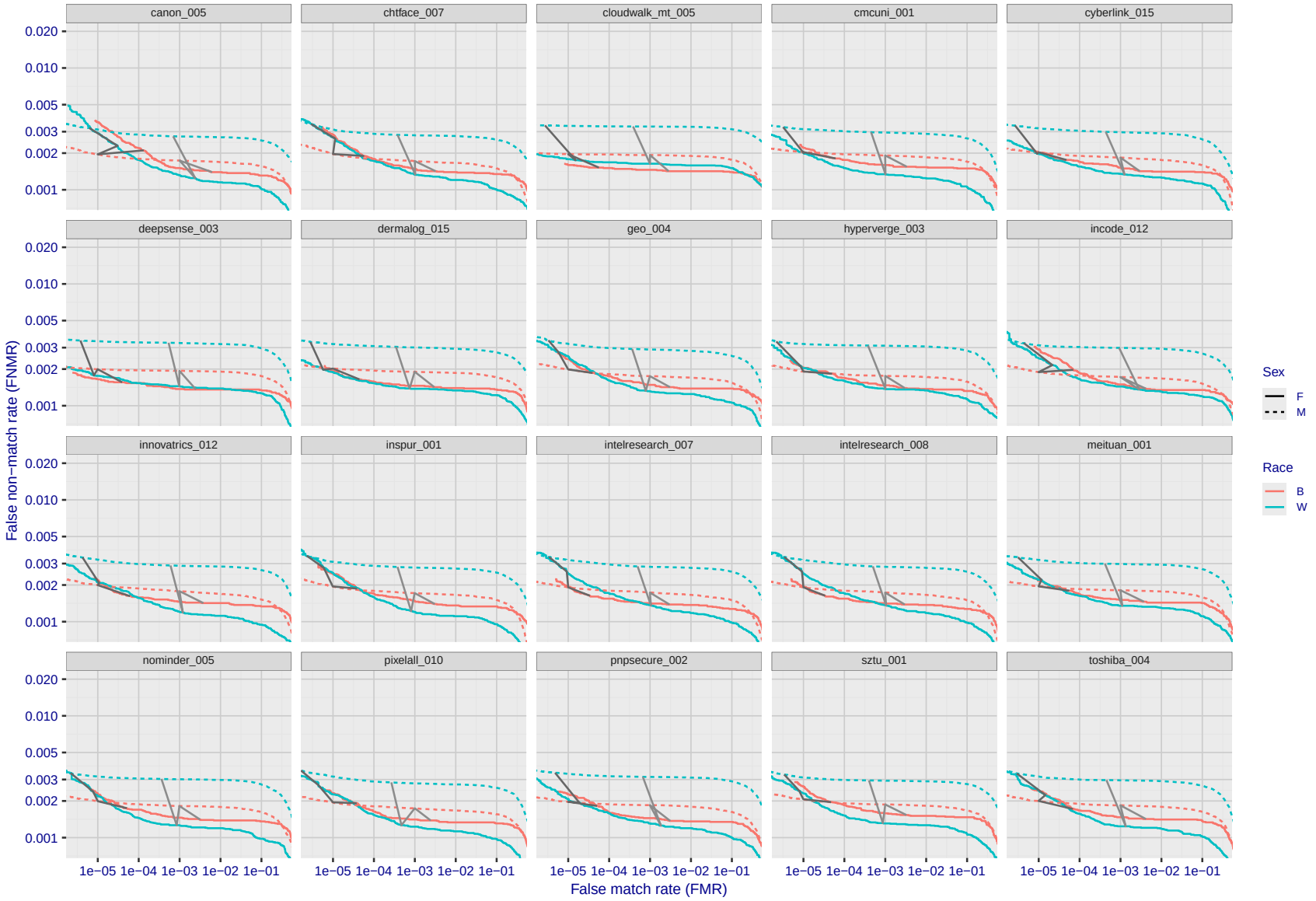


Figure 413: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

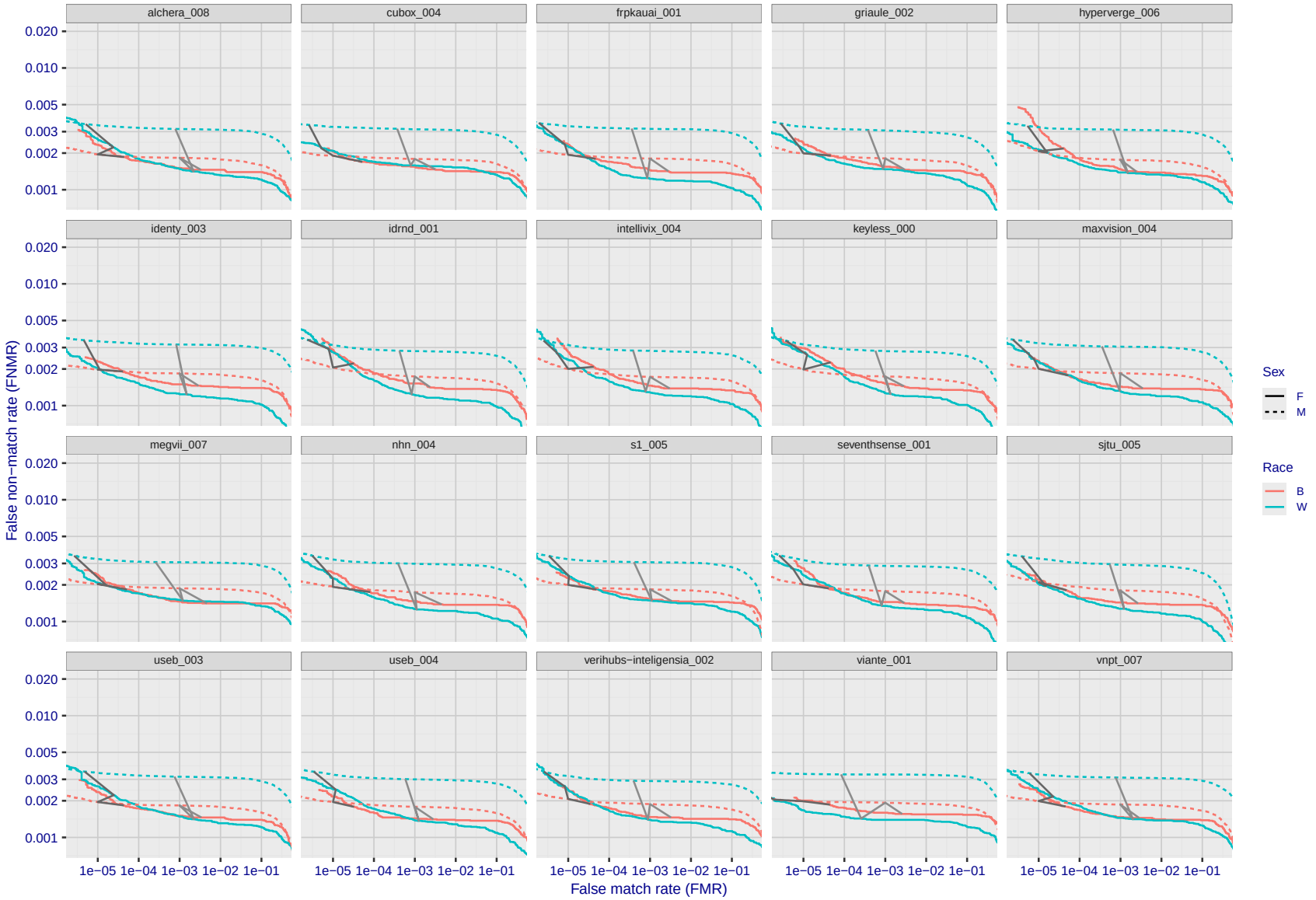


Figure 414: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

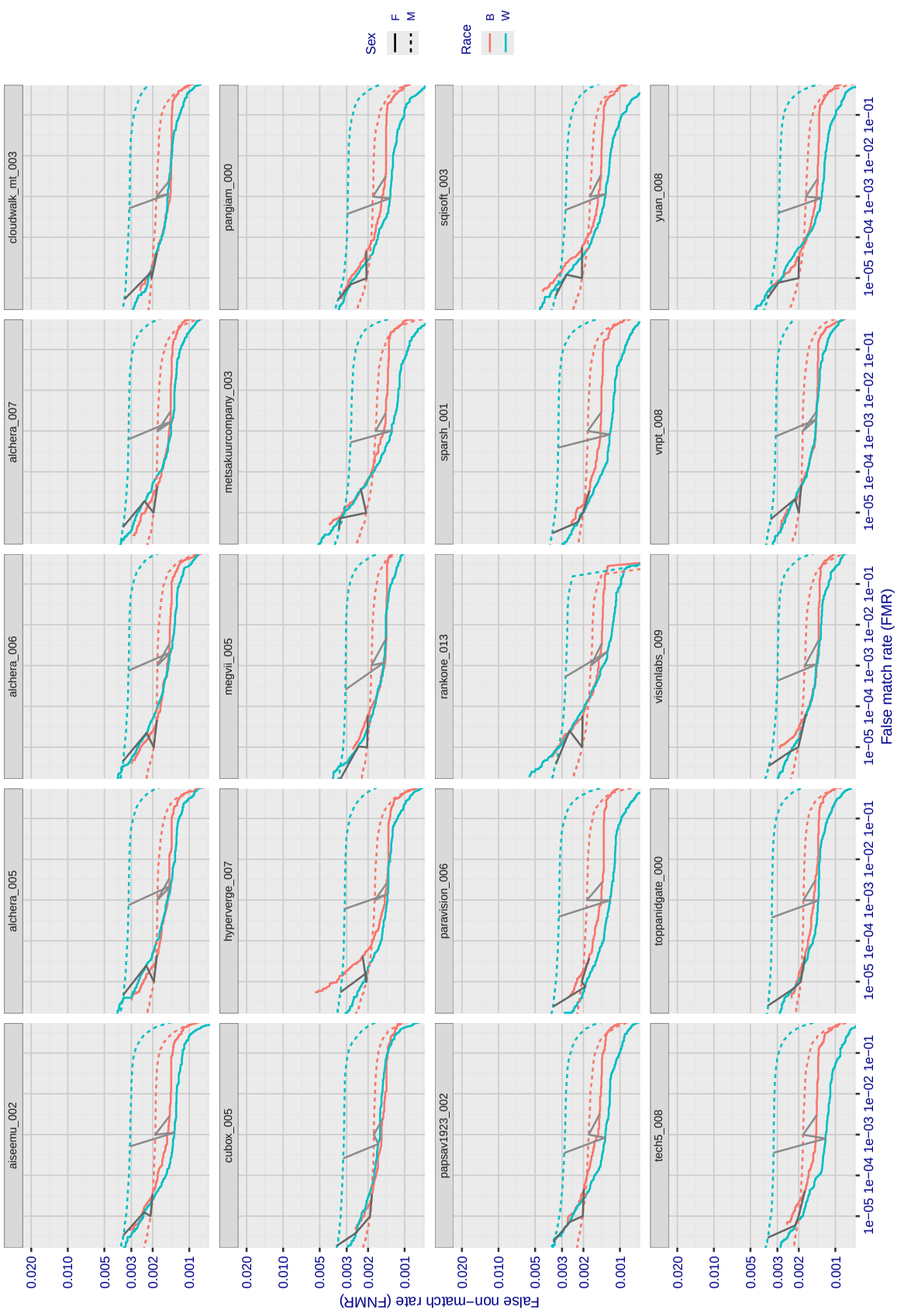


Figure 415: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

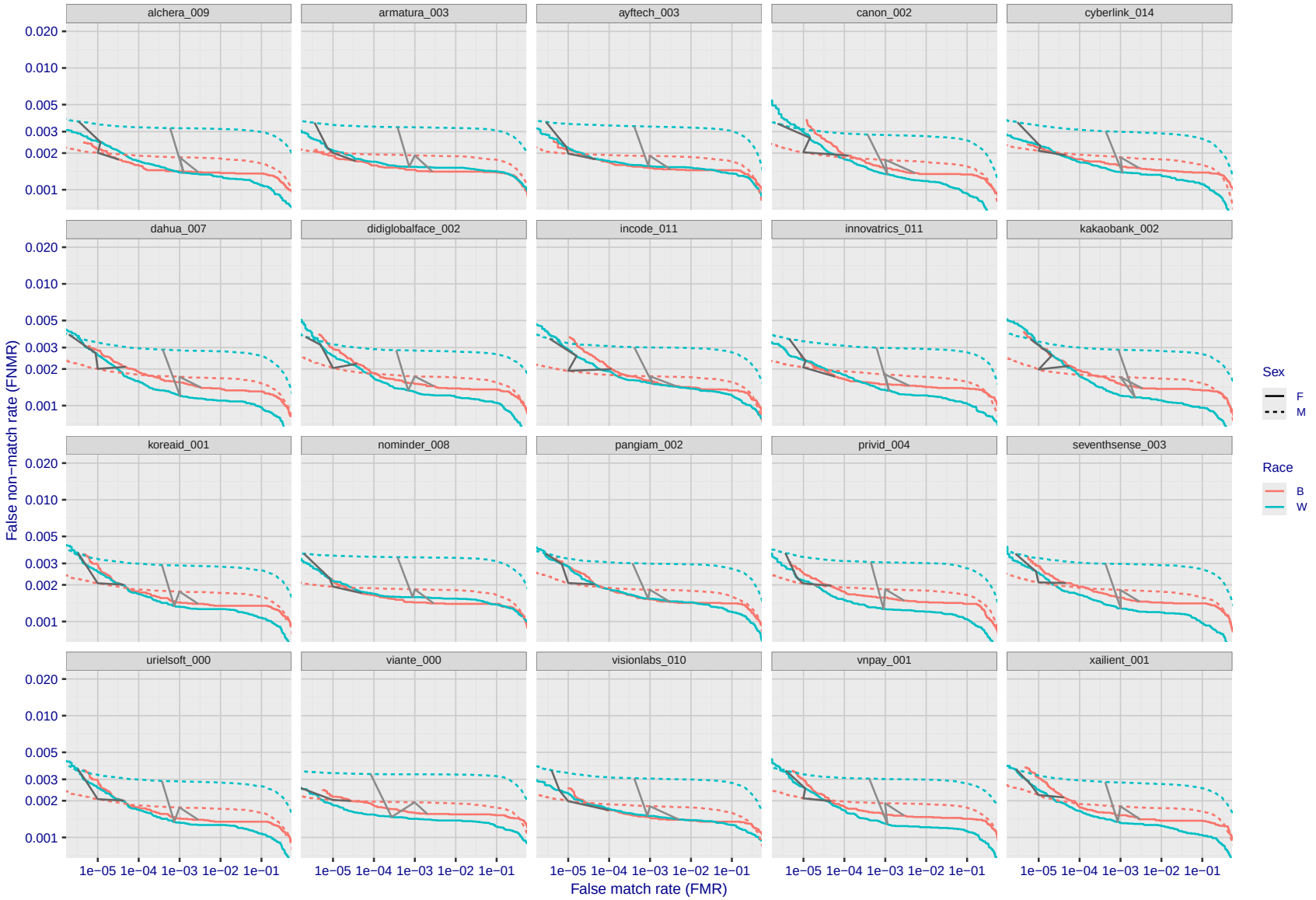


Figure 416: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

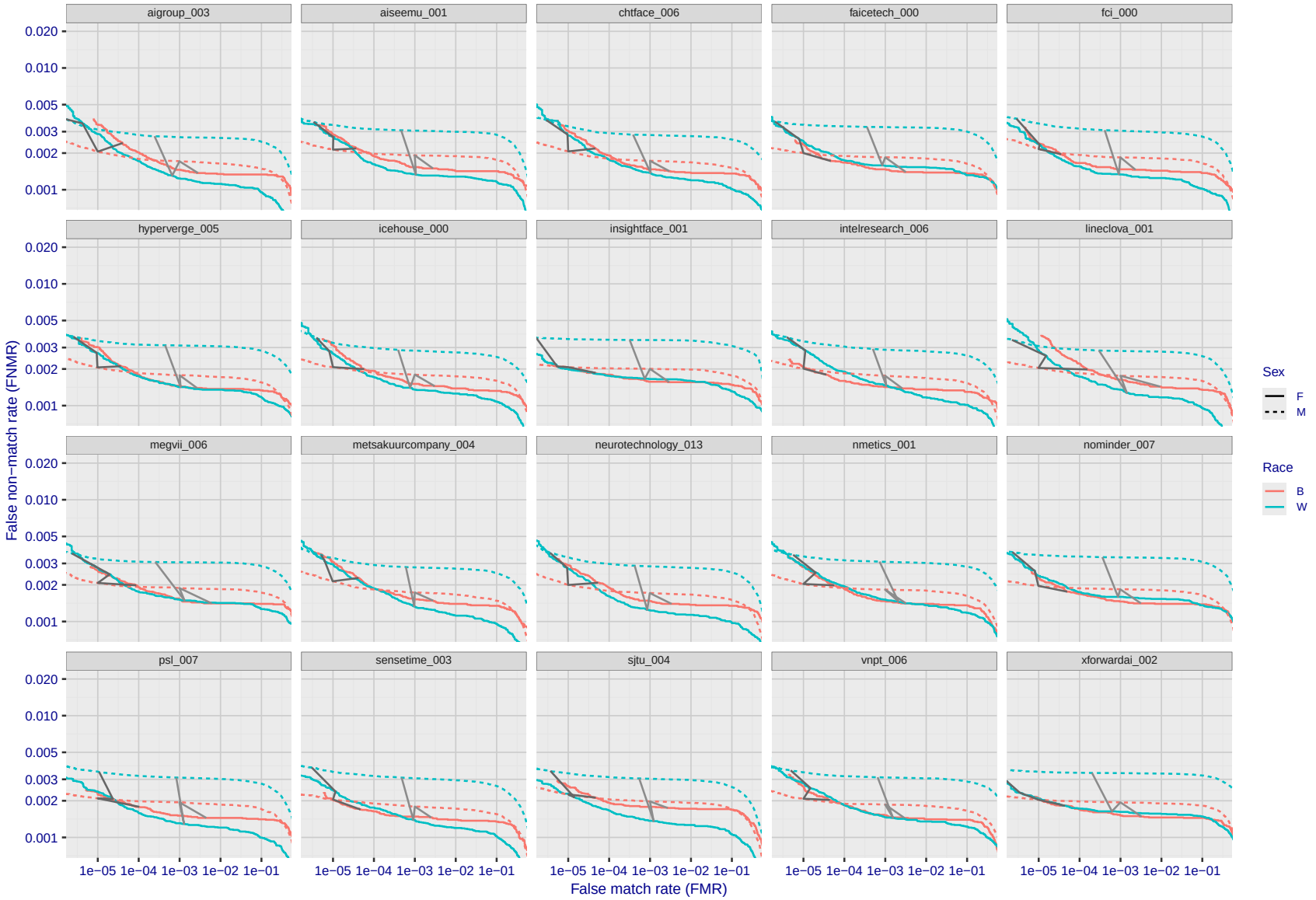


Figure 417: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

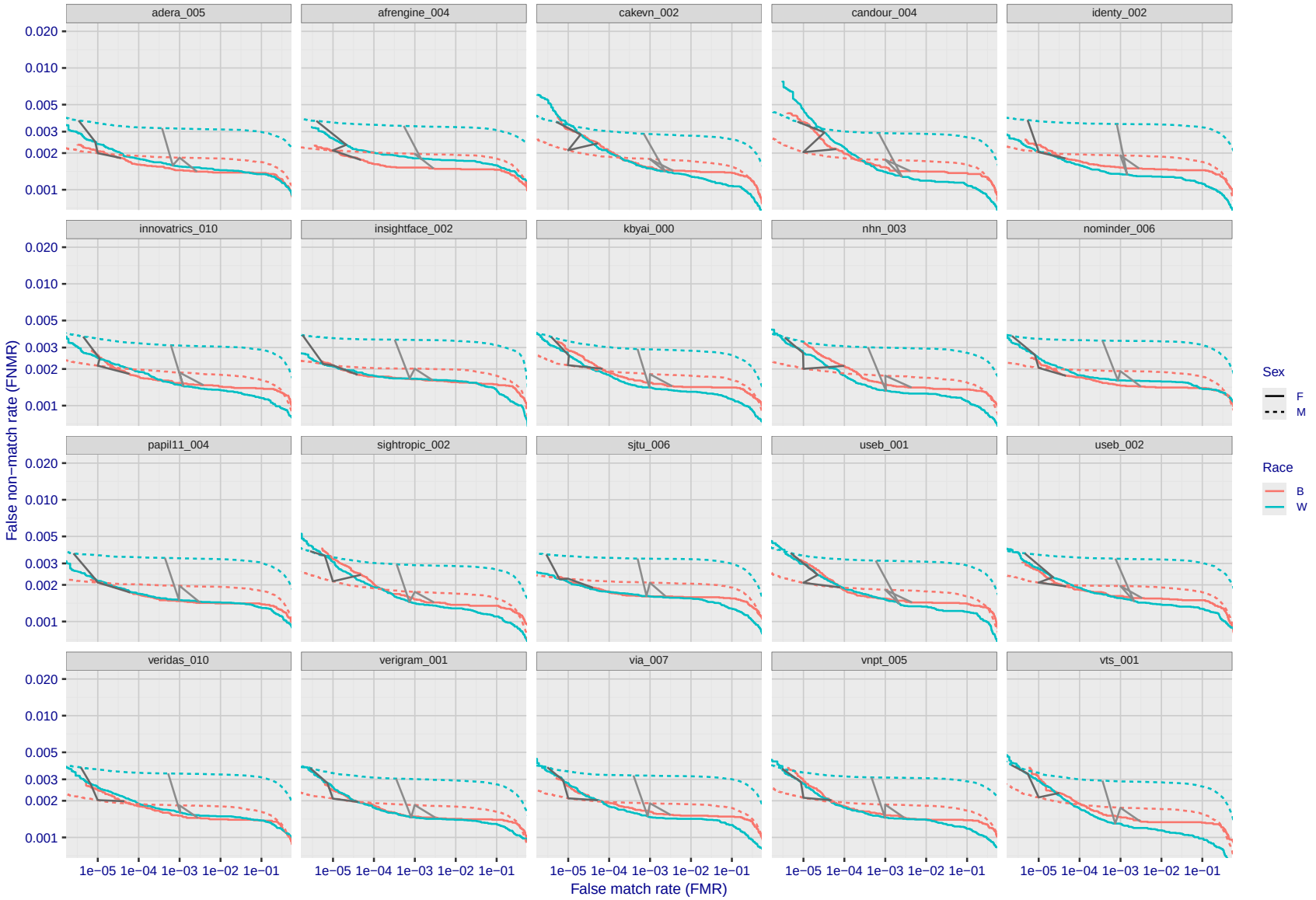


Figure 418: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

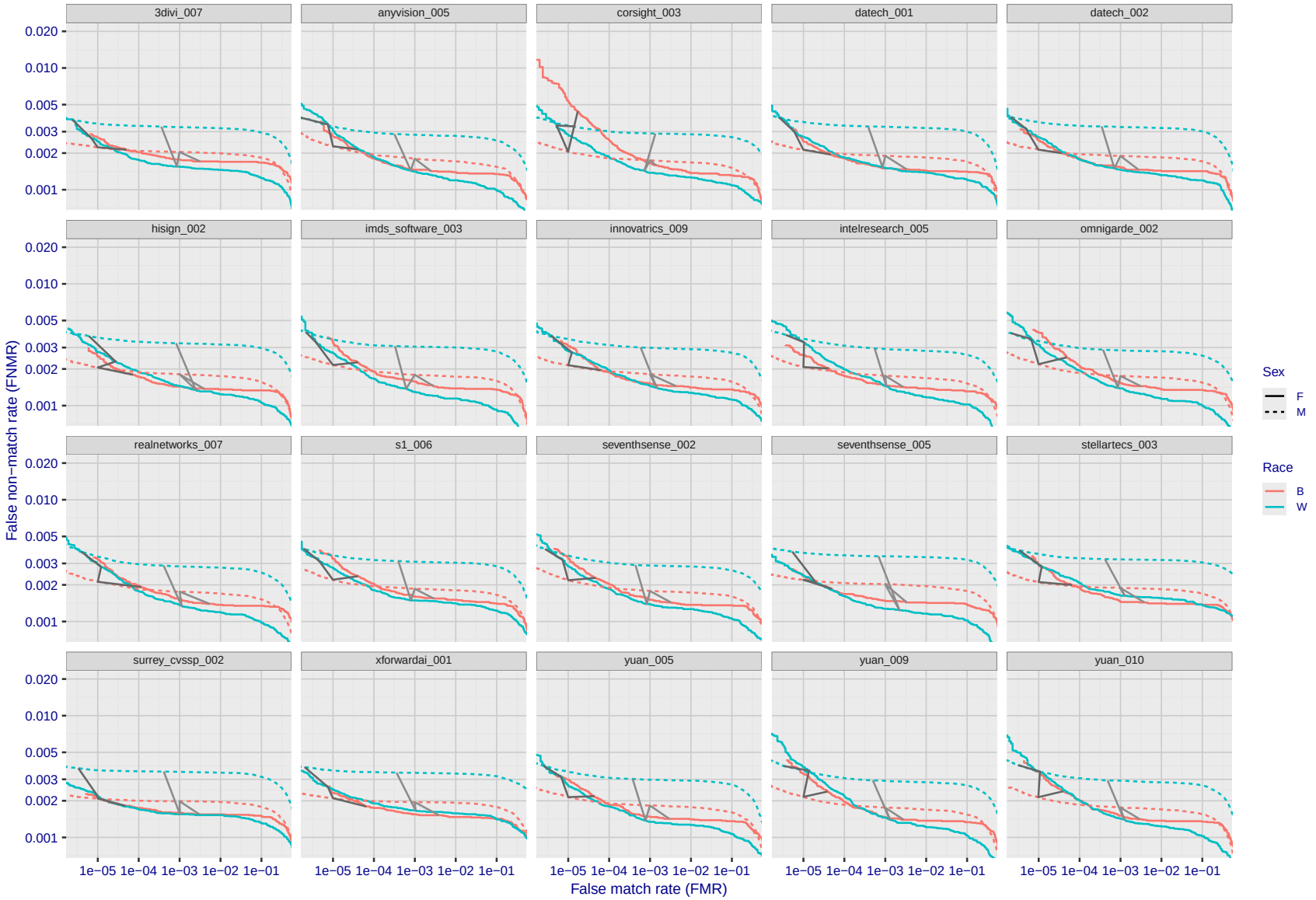


Figure 419: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

2026/11/06 13:03:11

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

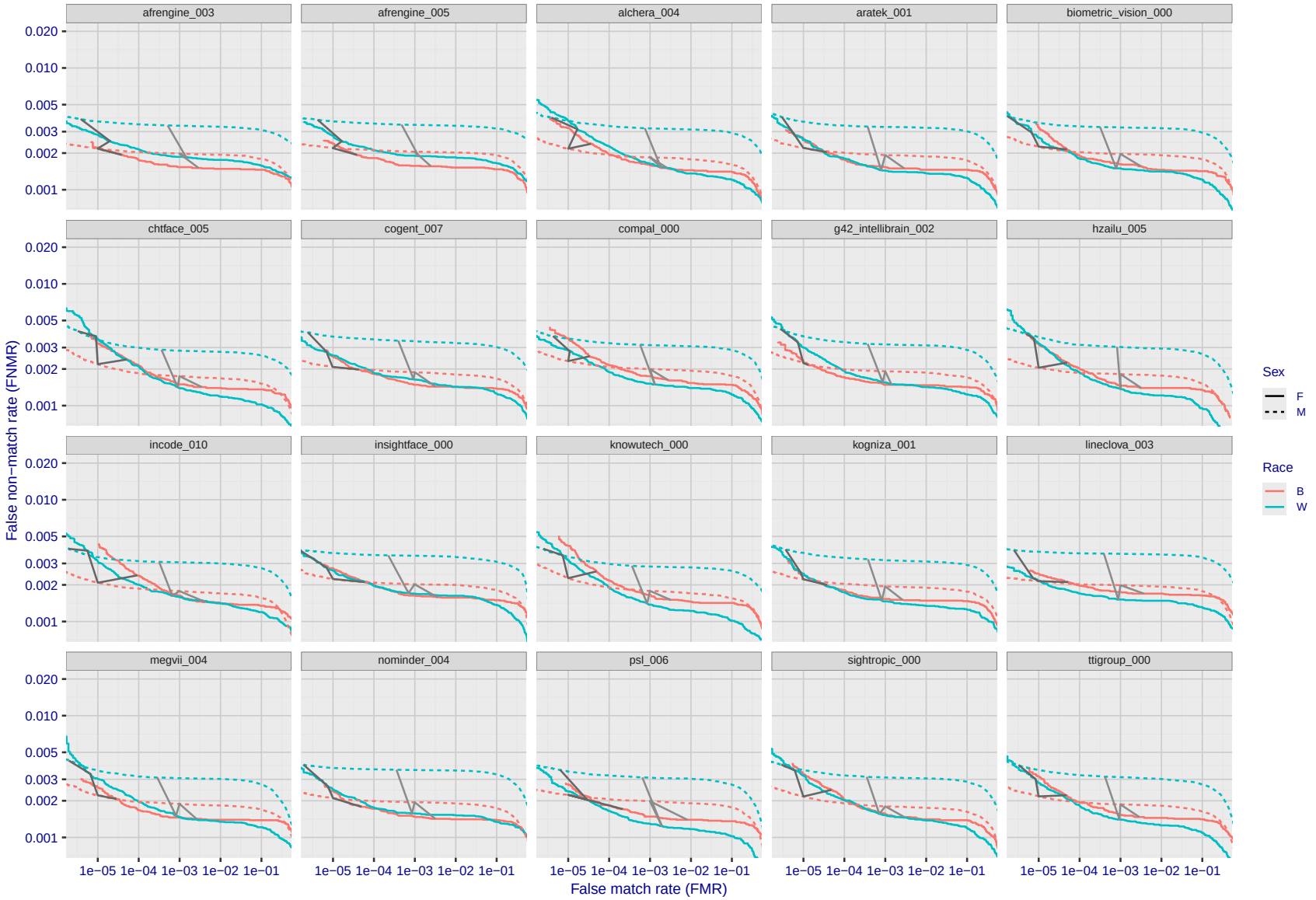


Figure 420: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

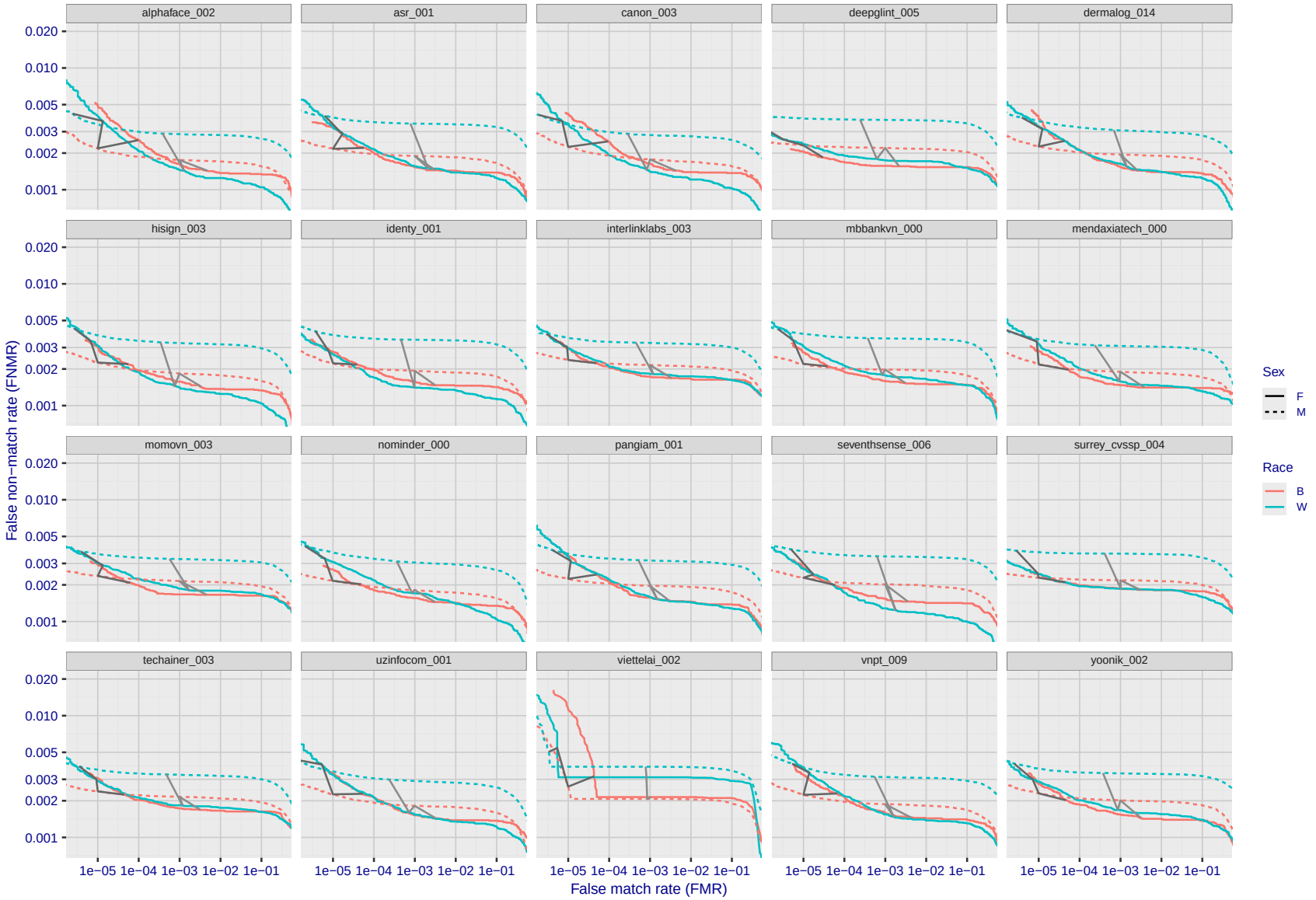


Figure 421: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

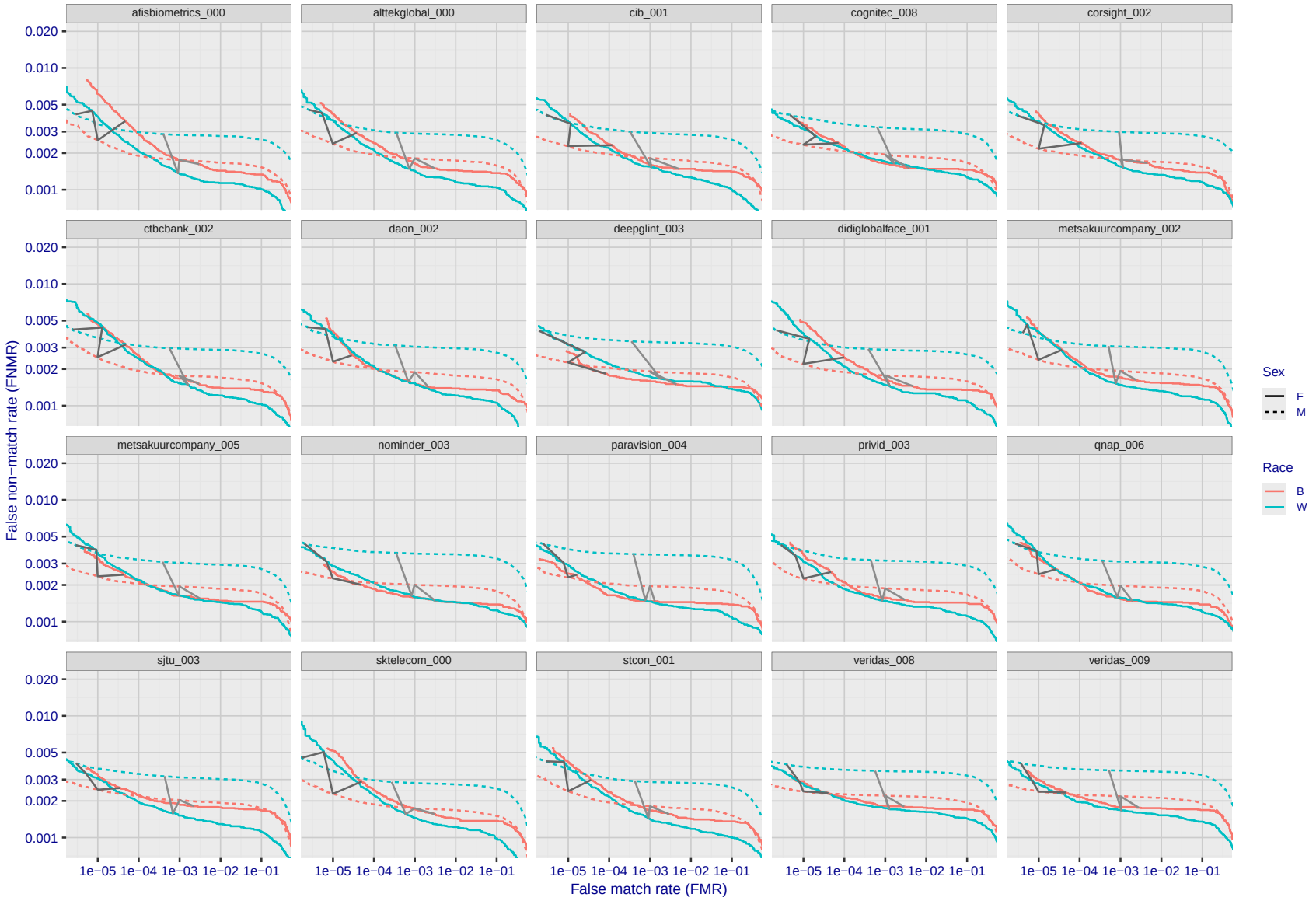


Figure 422: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

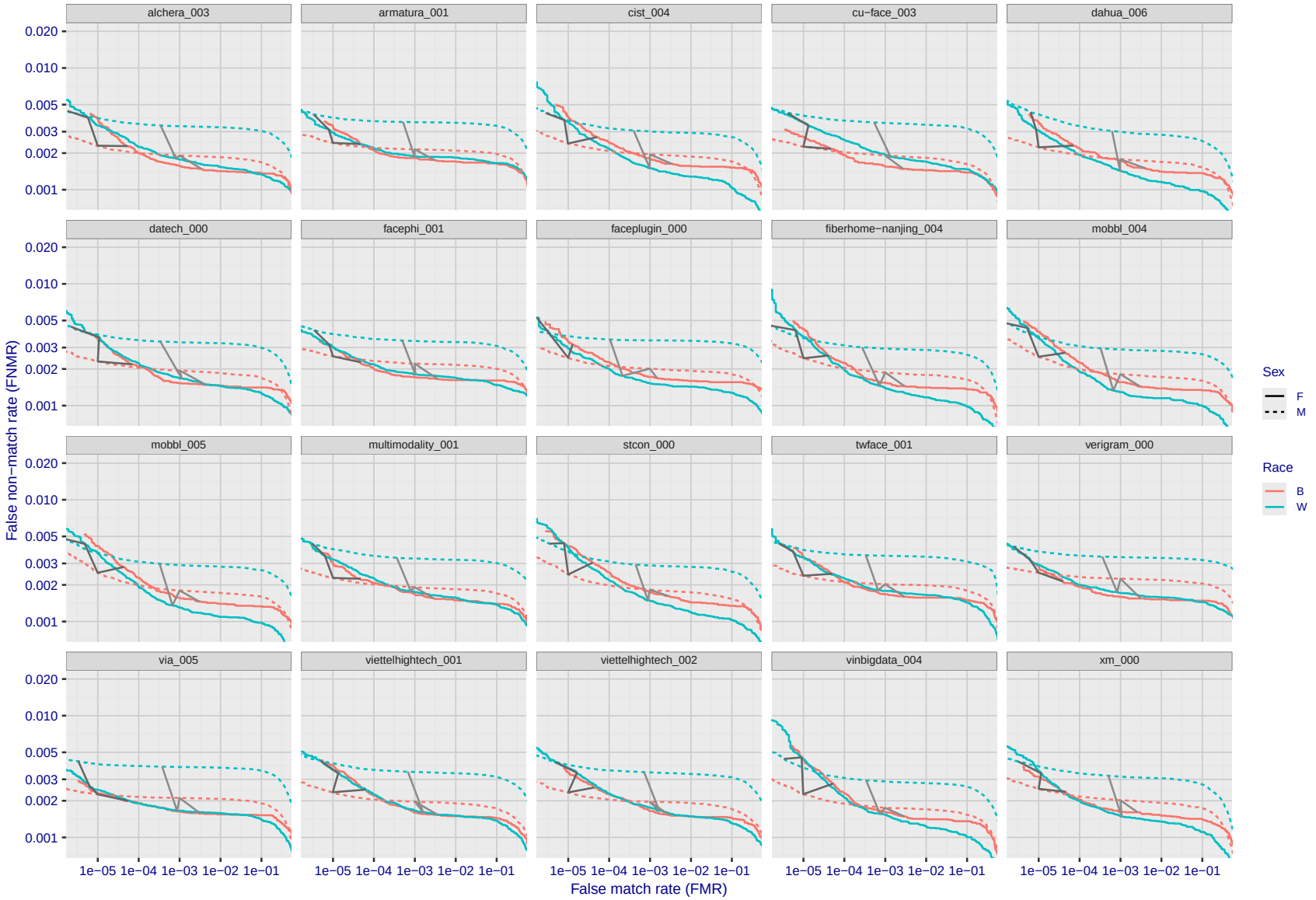


Figure 423: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

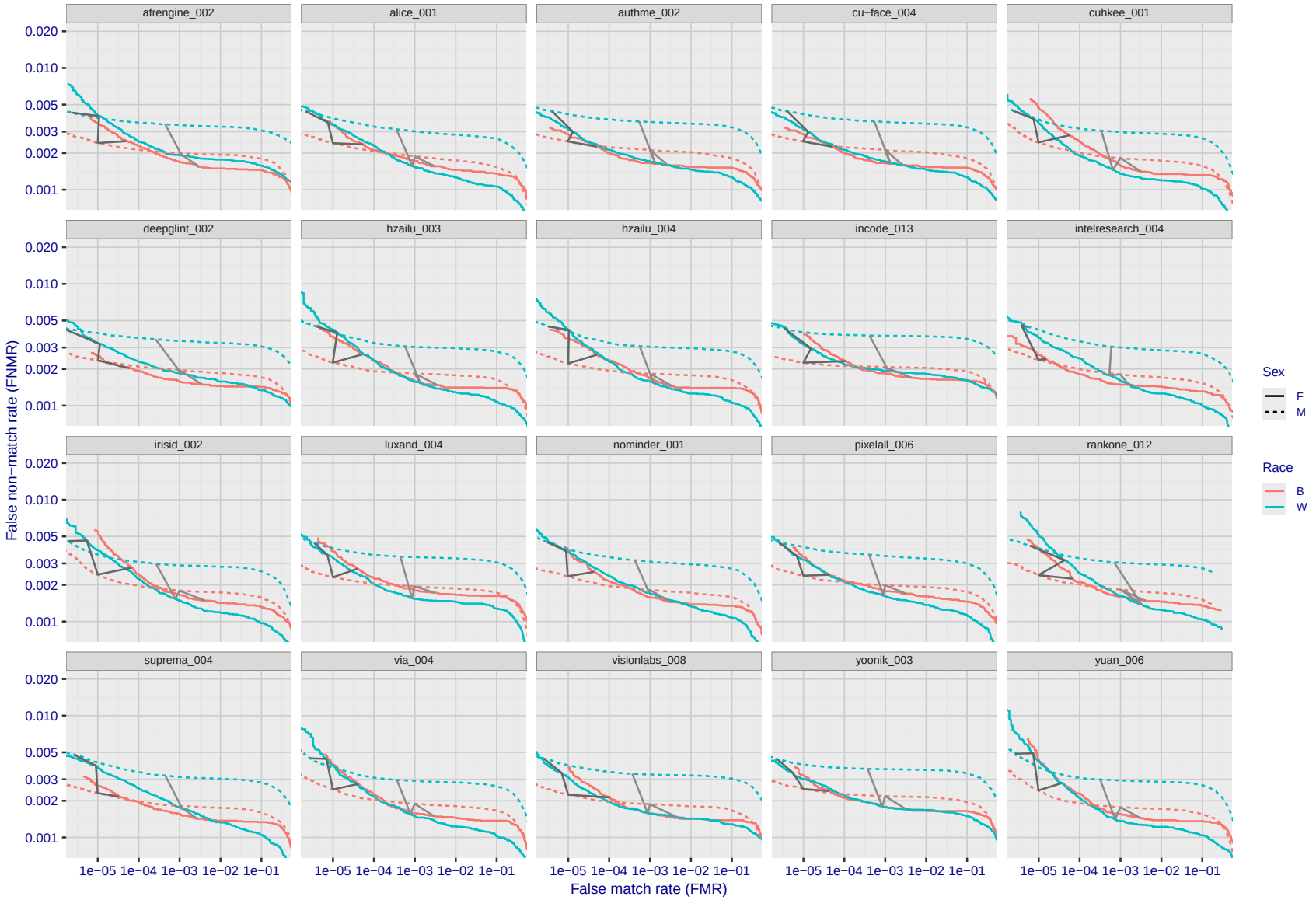


Figure 424: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

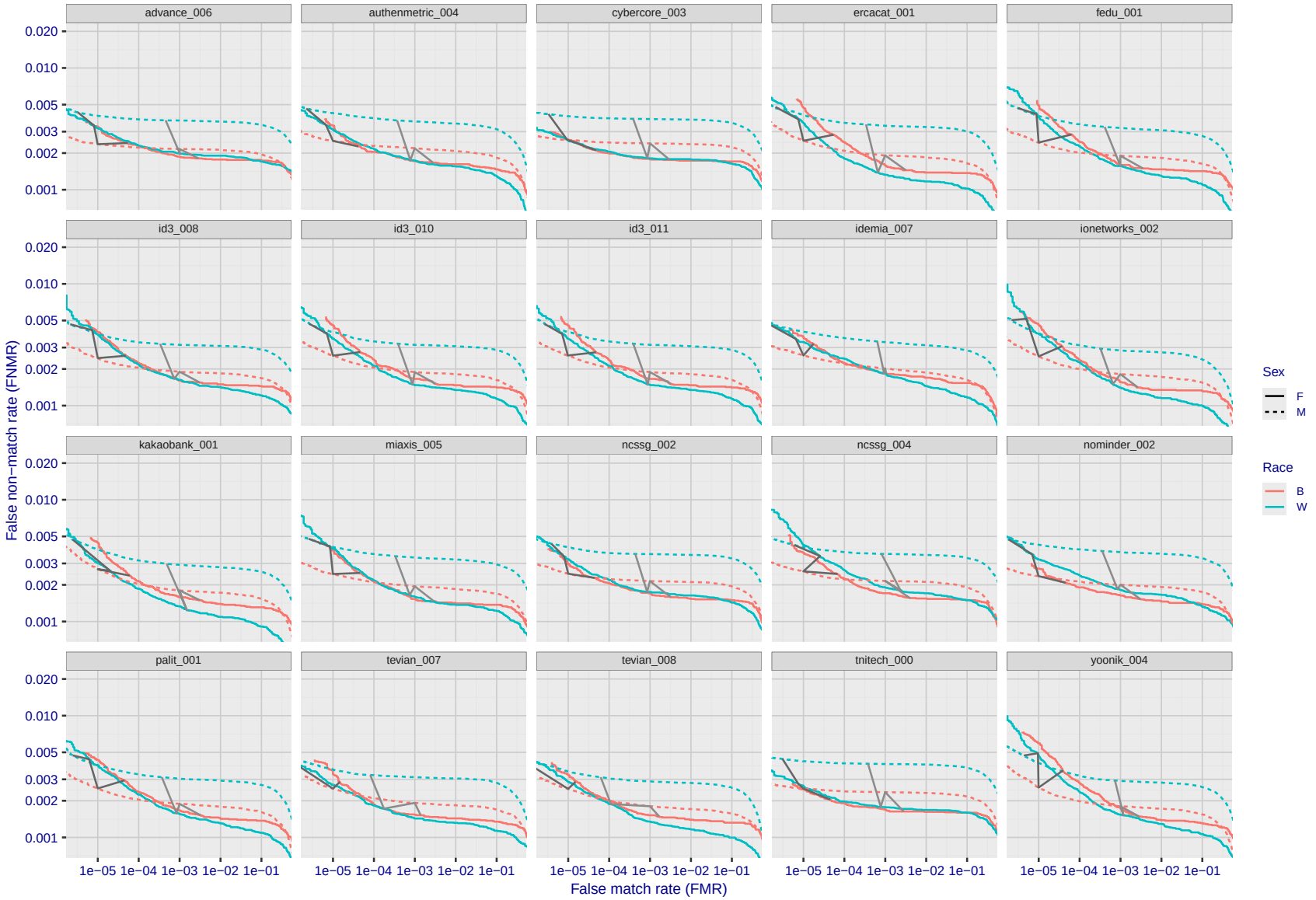


Figure 425: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

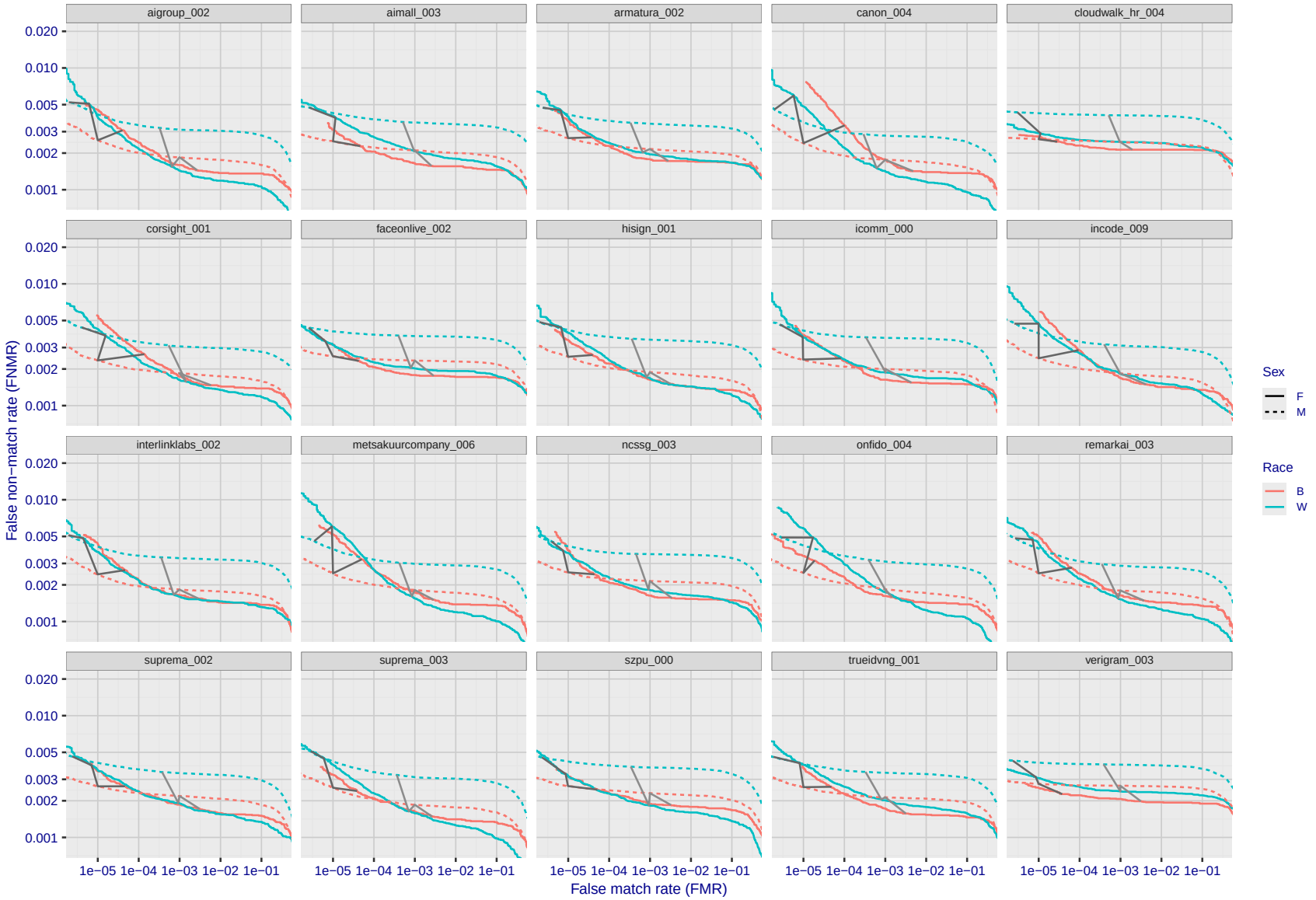


Figure 426: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

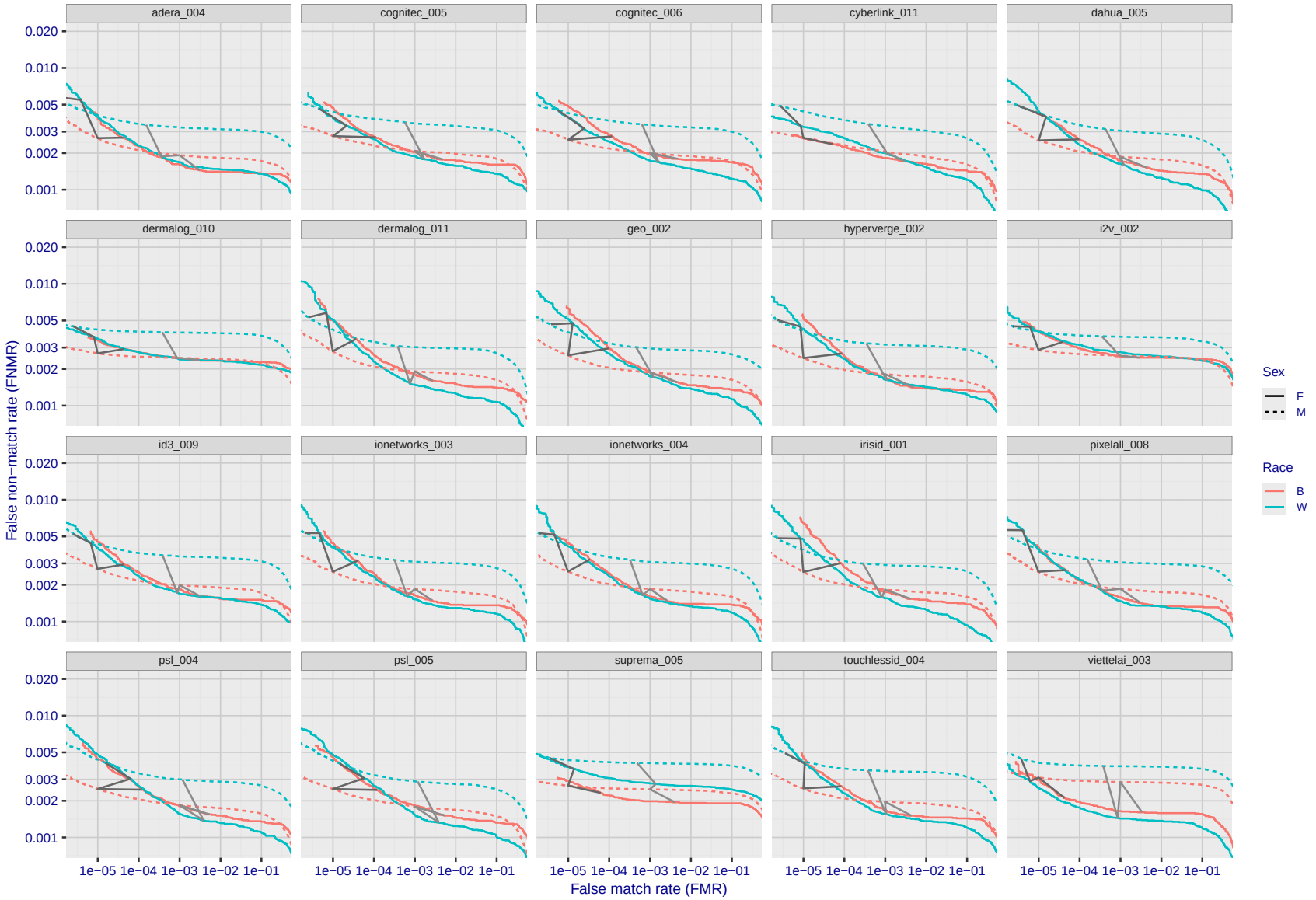


Figure 427: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

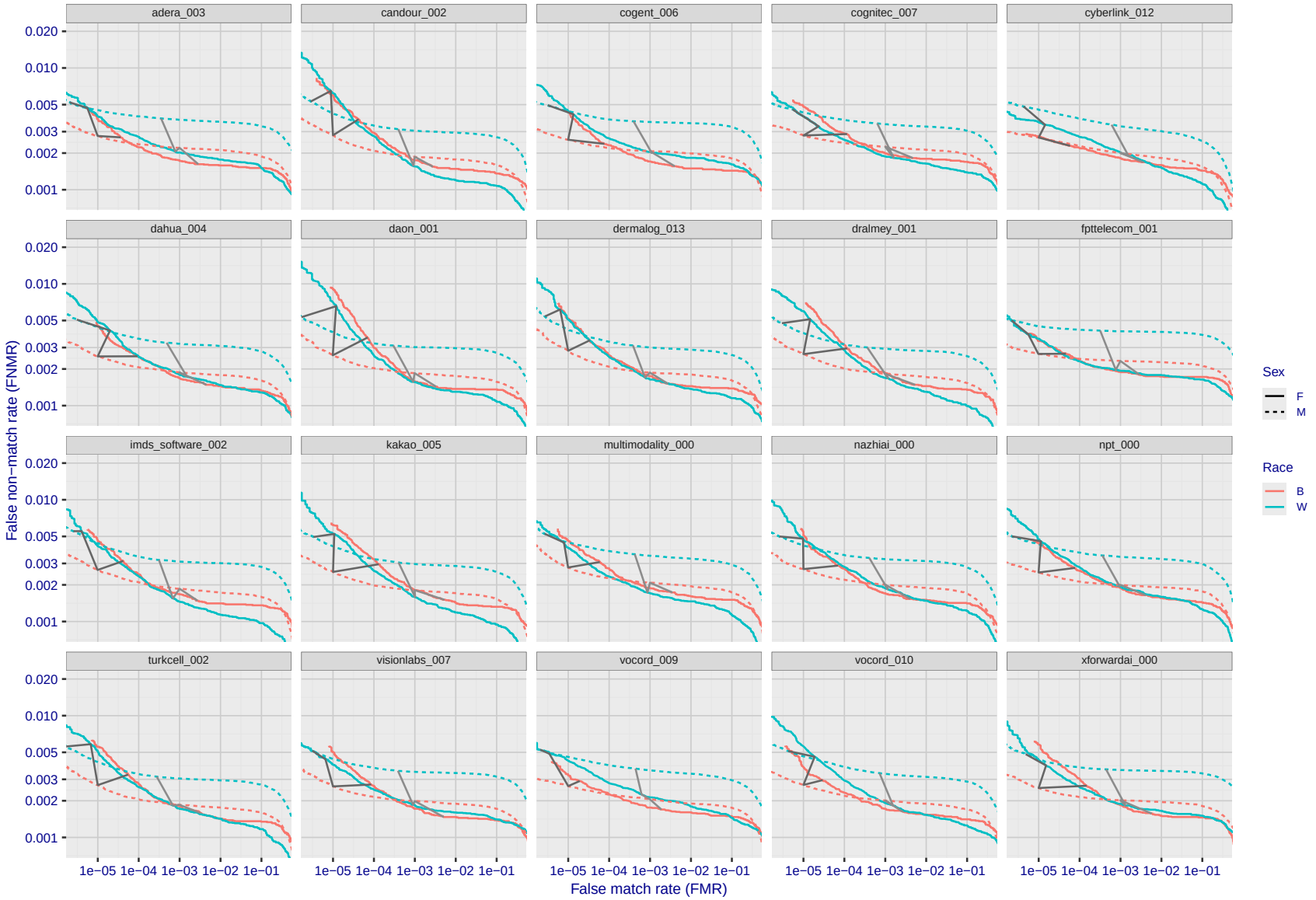


Figure 428: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

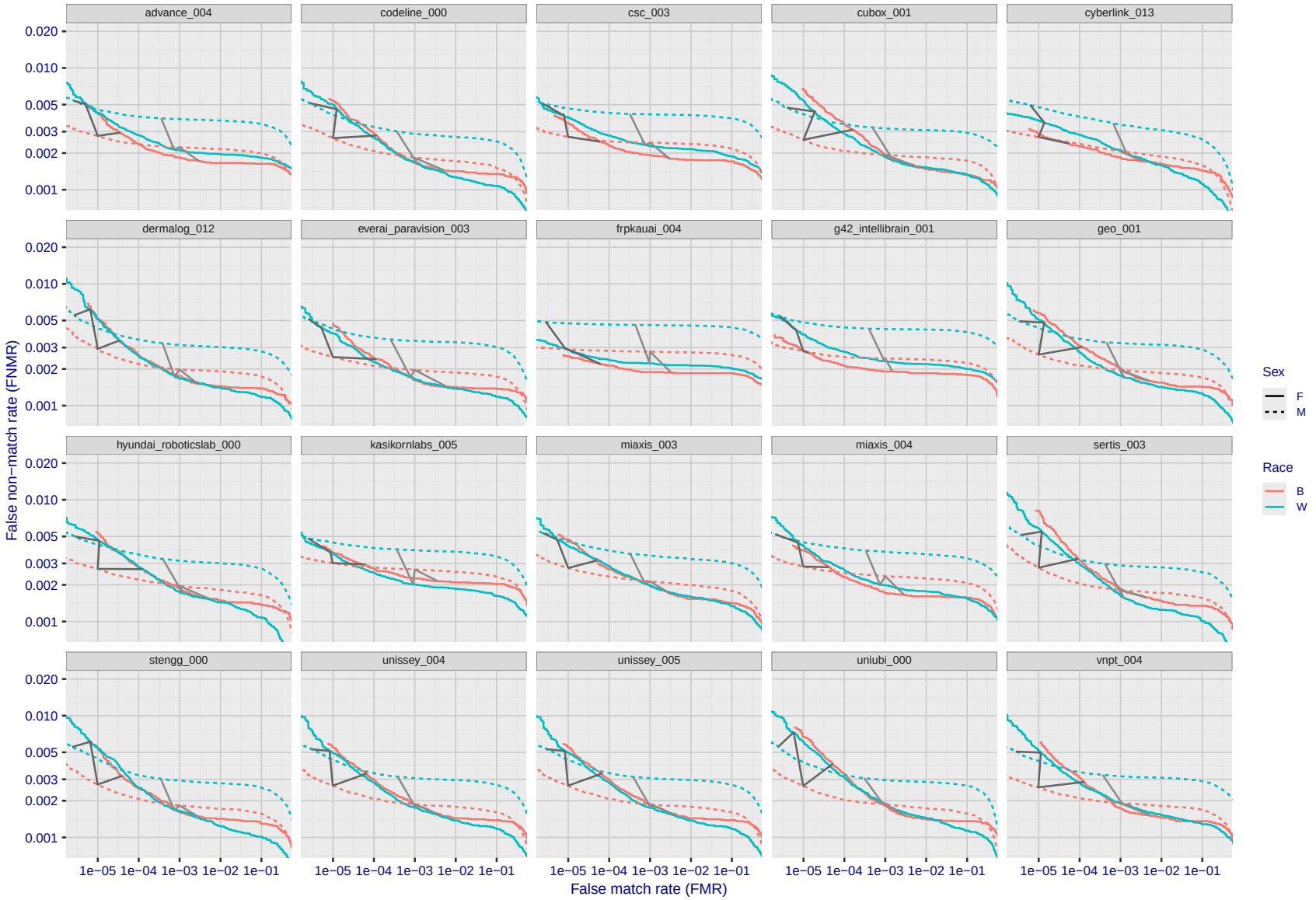


Figure 429: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

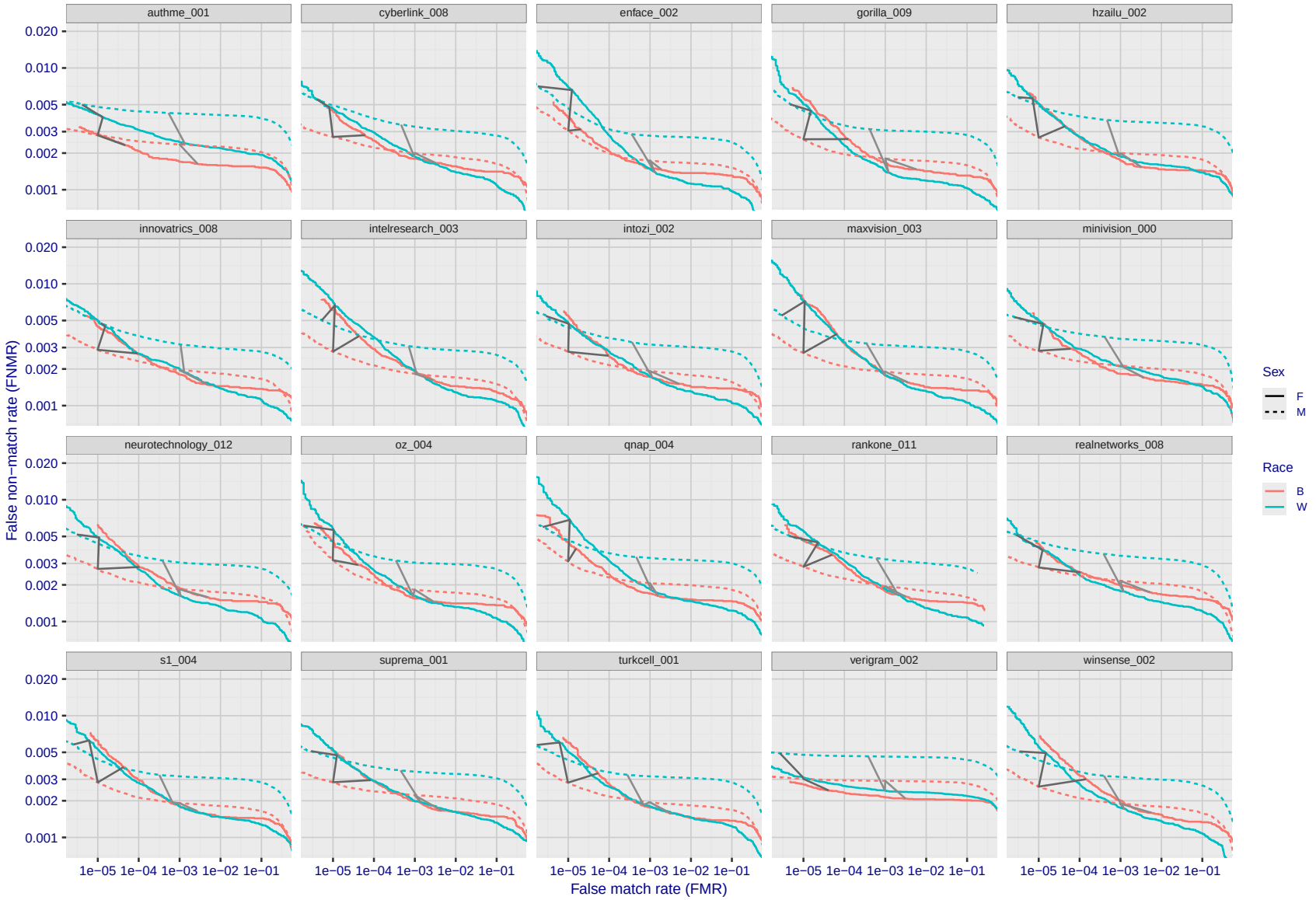


Figure 430: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

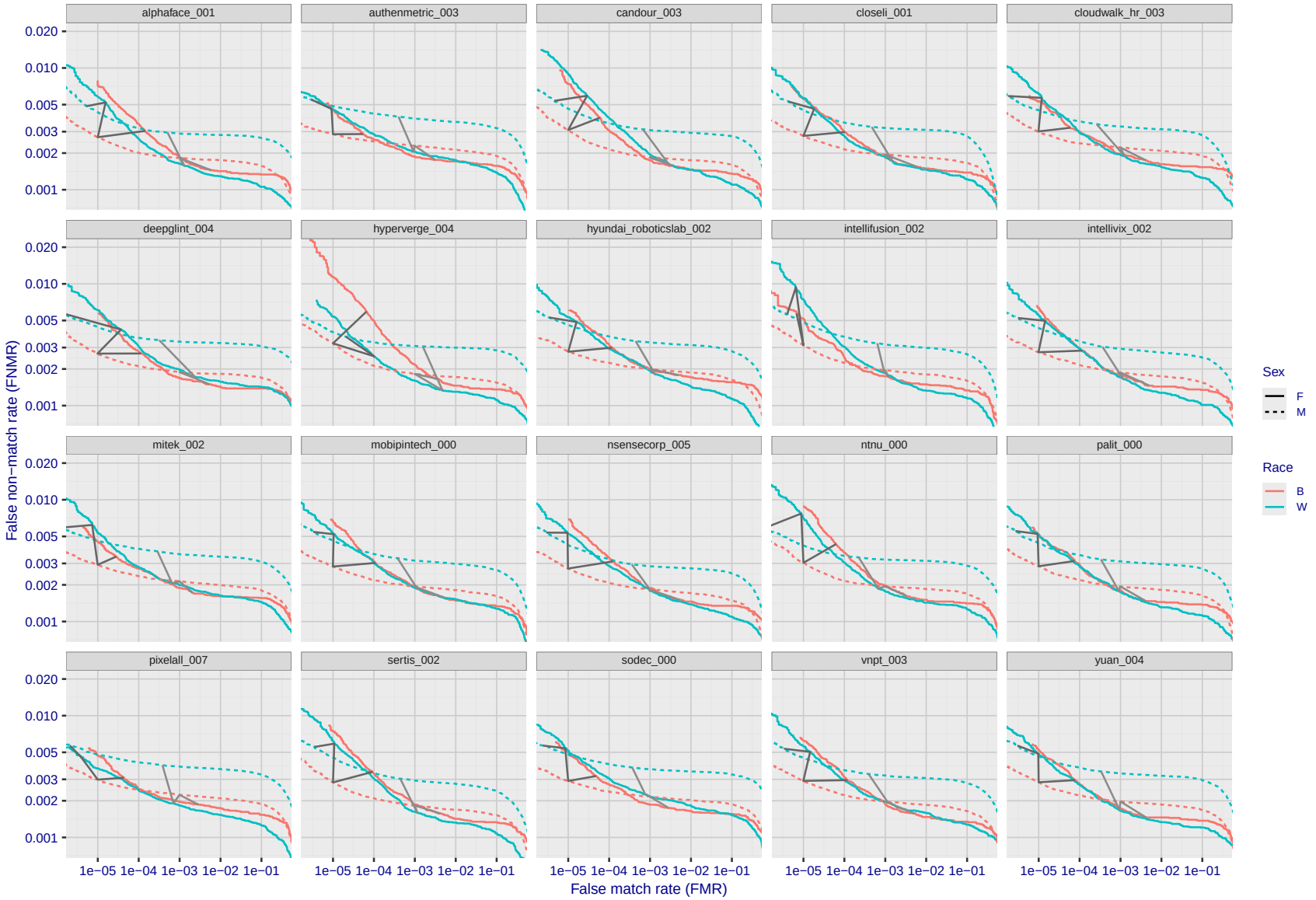


Figure 431: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

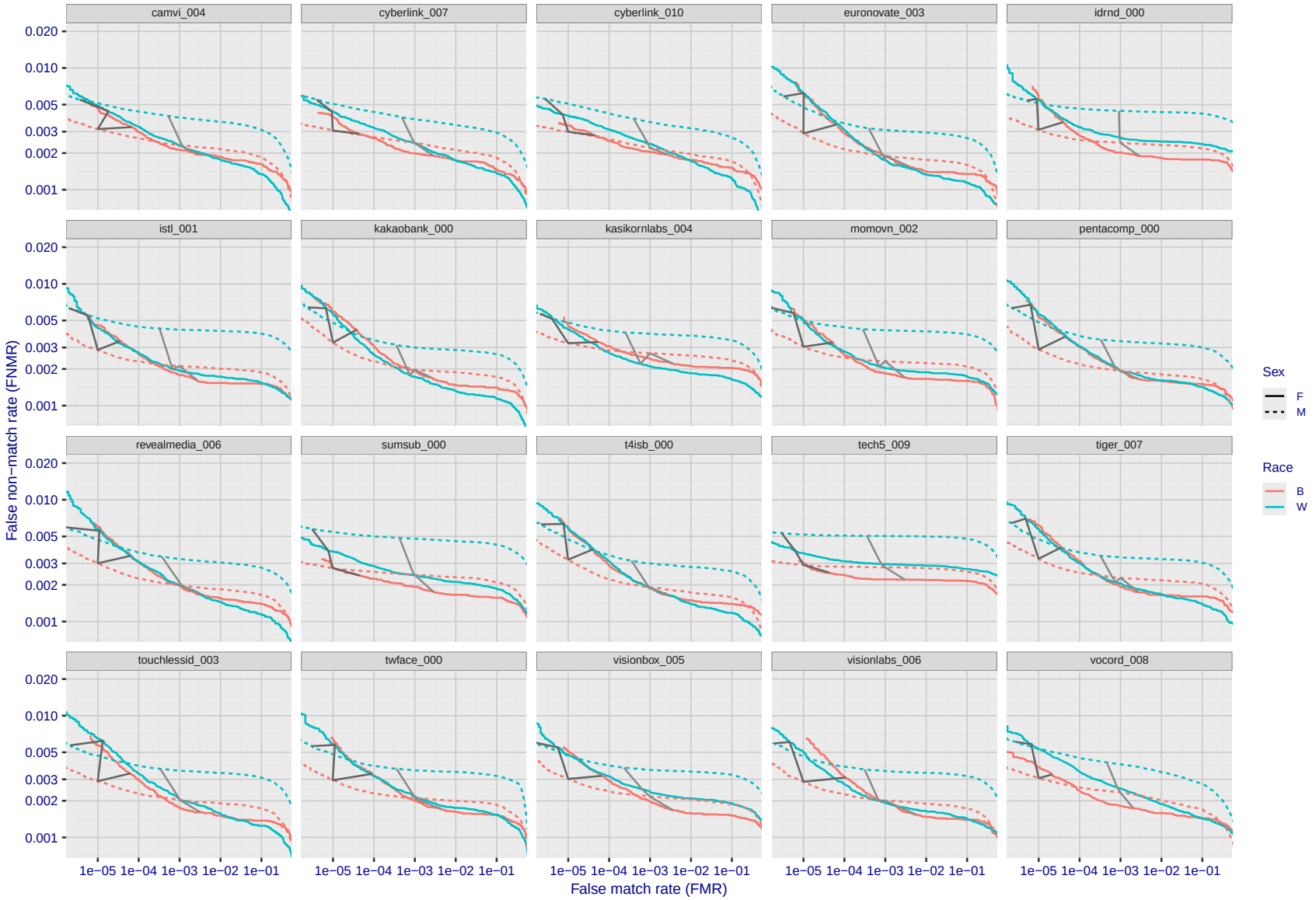


Figure 432: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

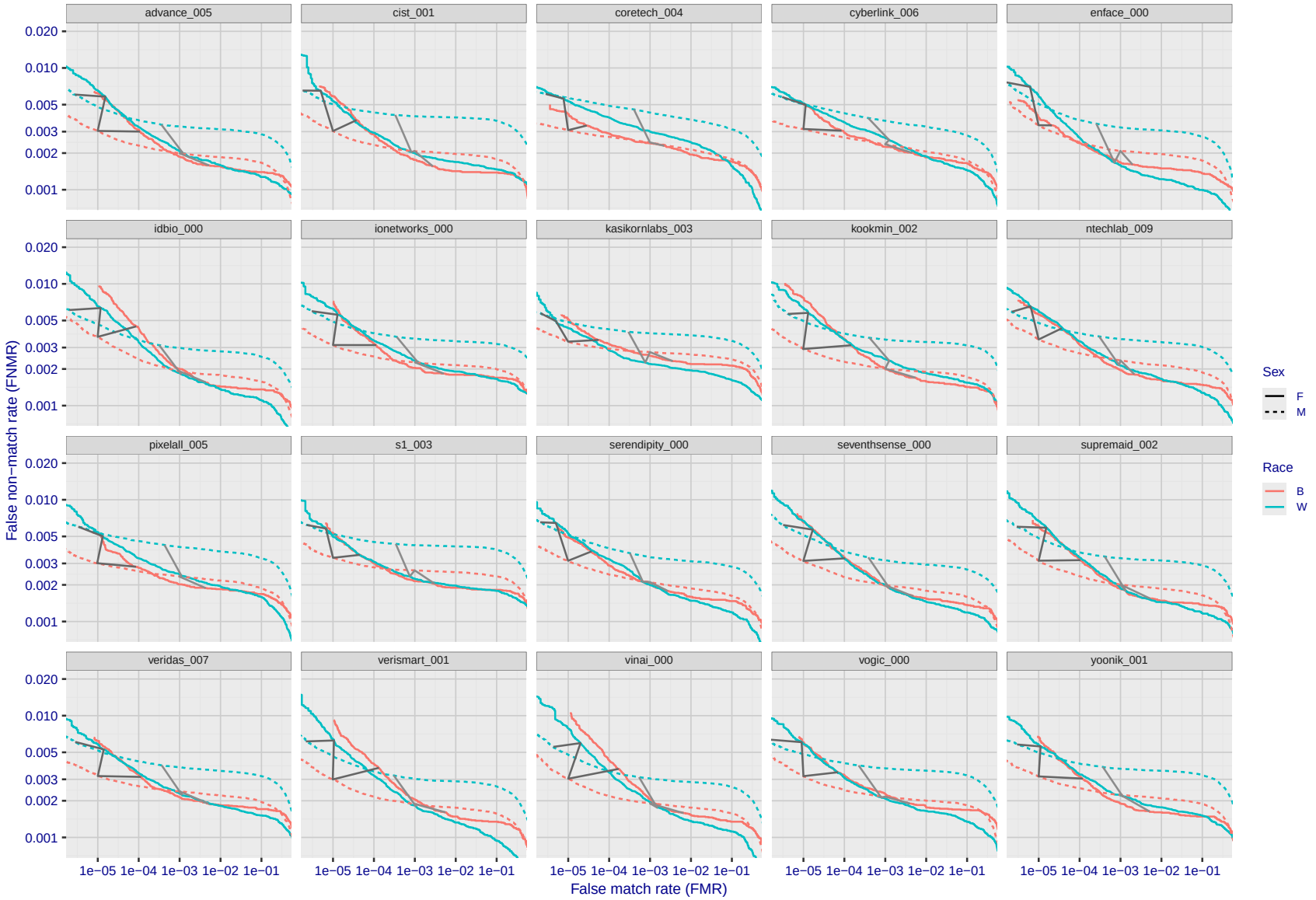


Figure 433: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

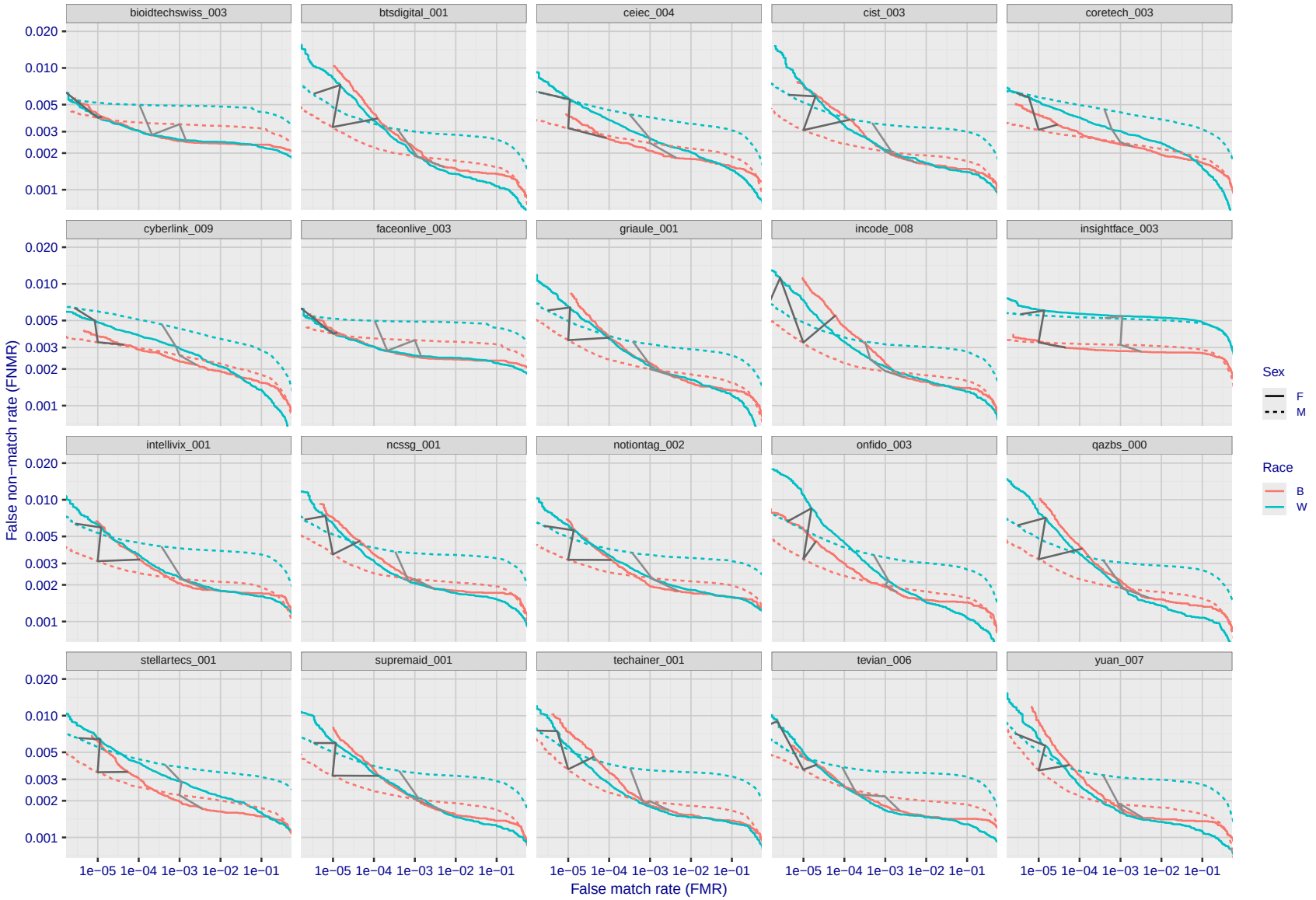


Figure 434: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

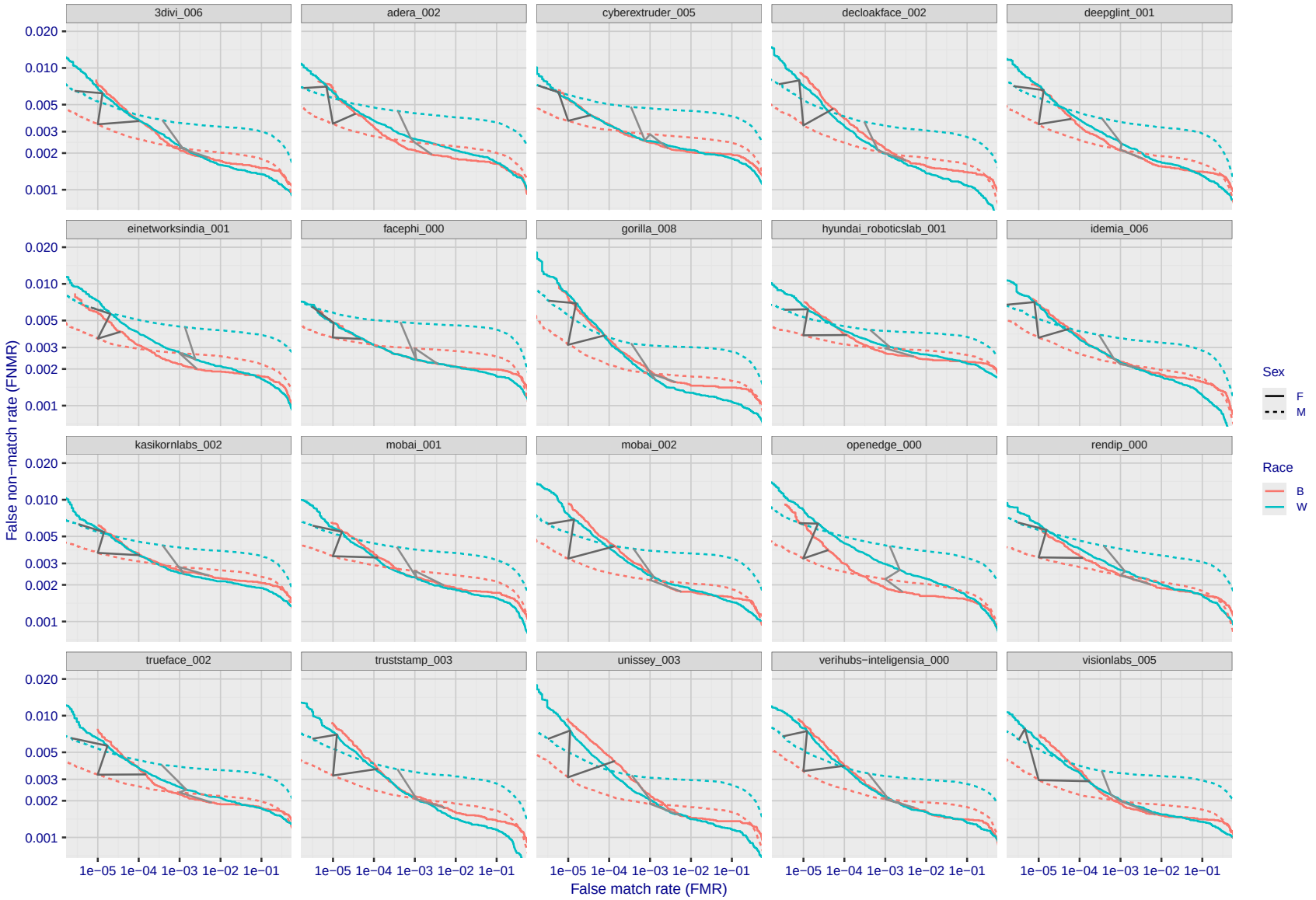


Figure 435: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

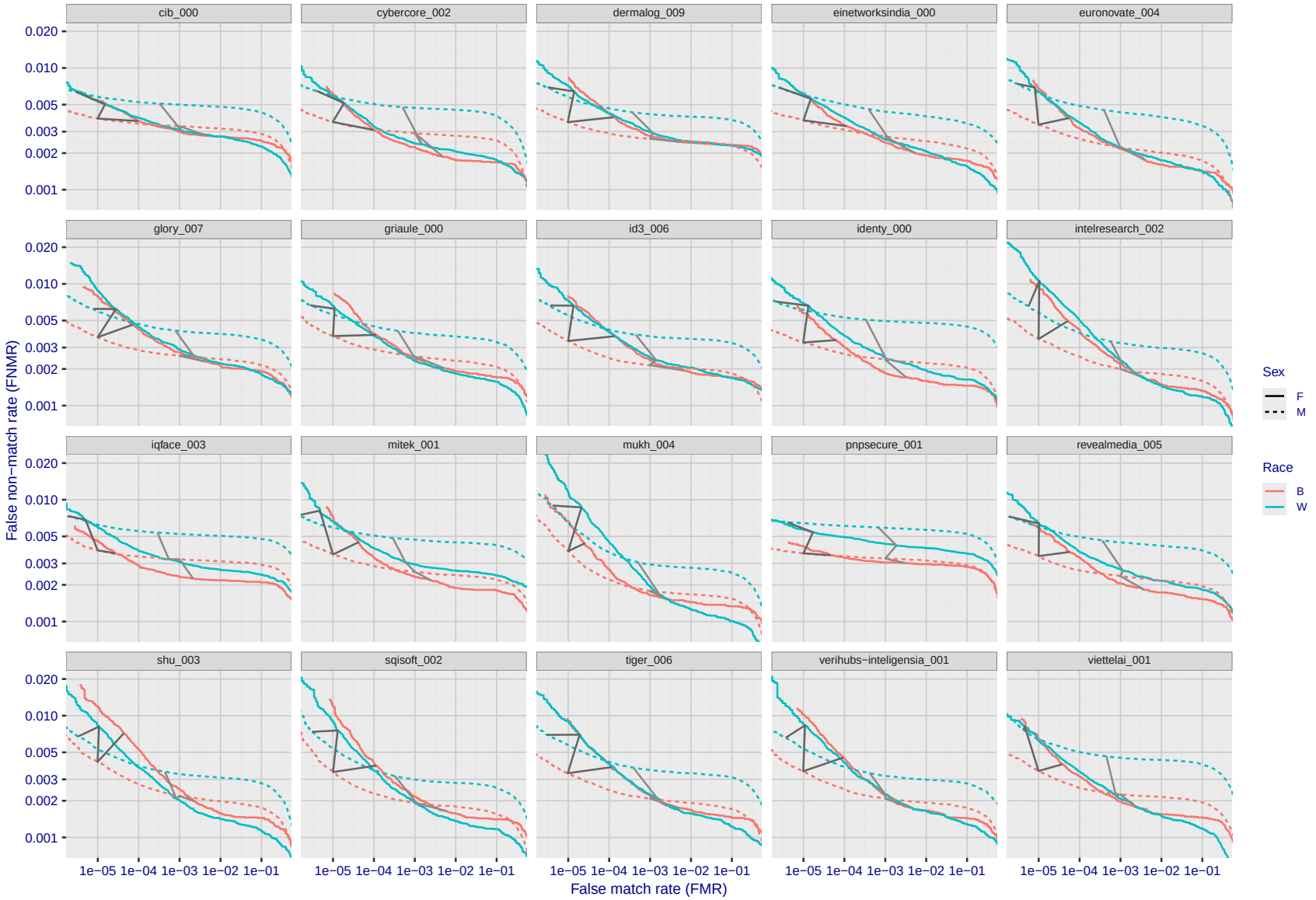


Figure 436: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

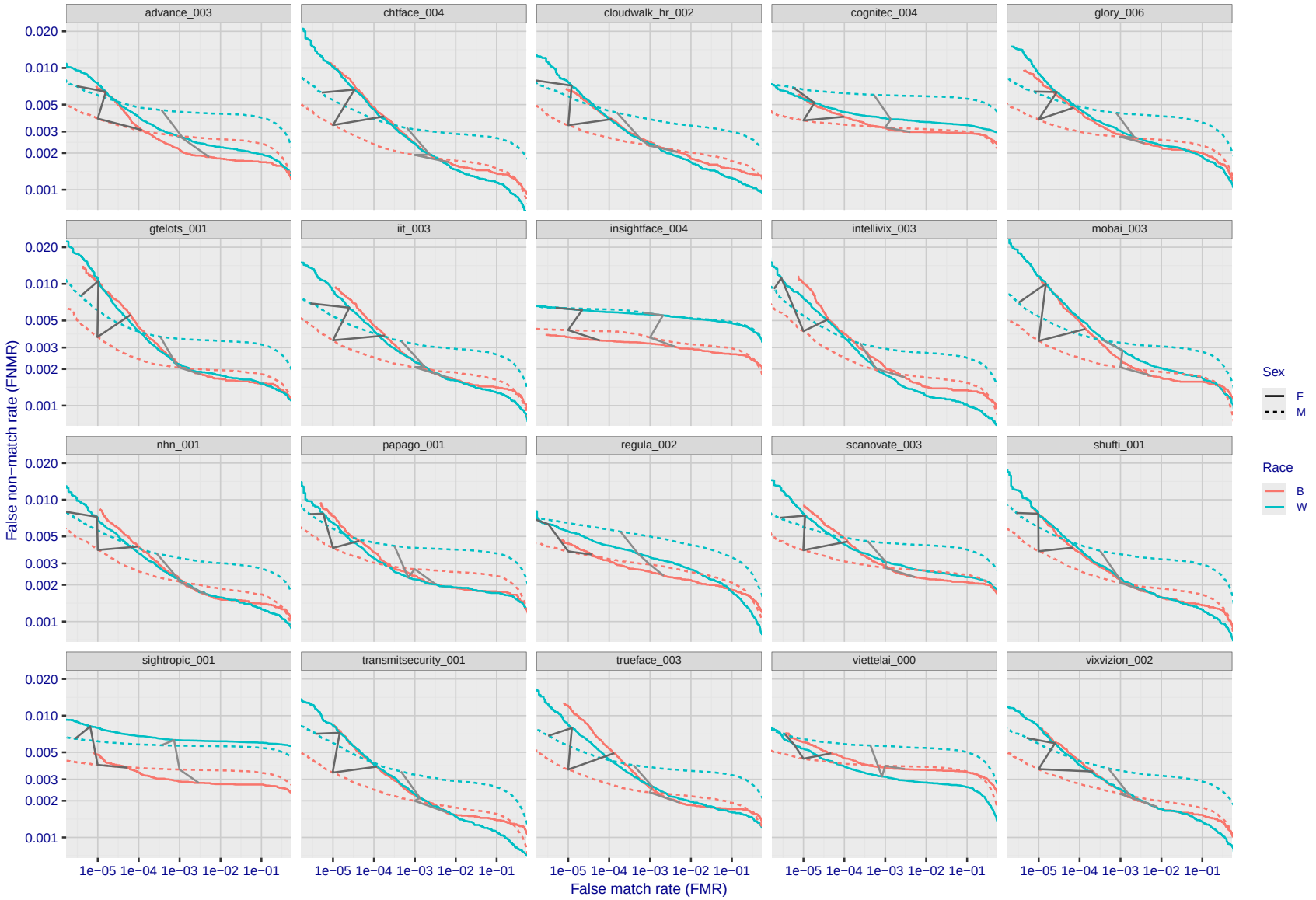


Figure 437: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

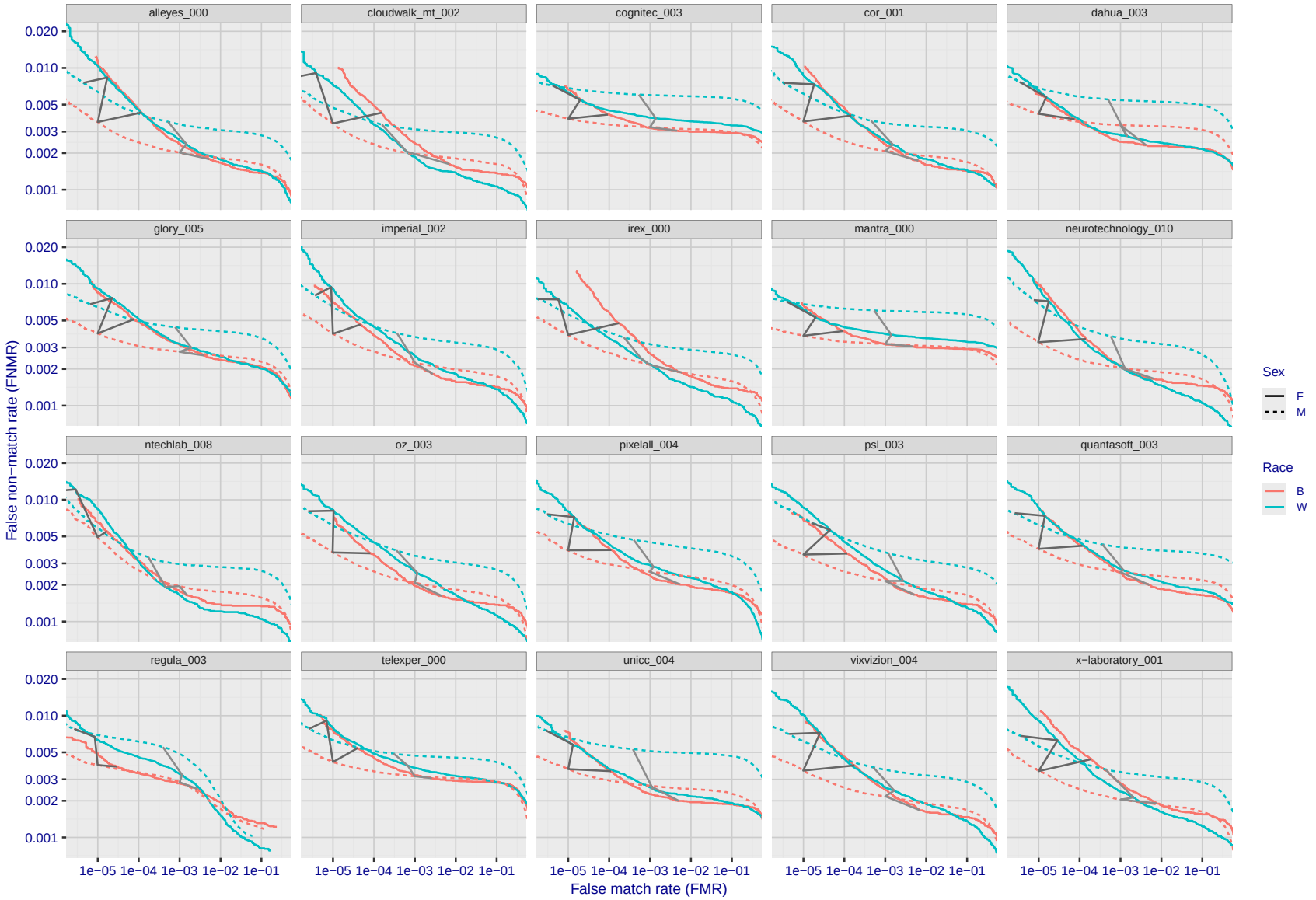


Figure 438: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

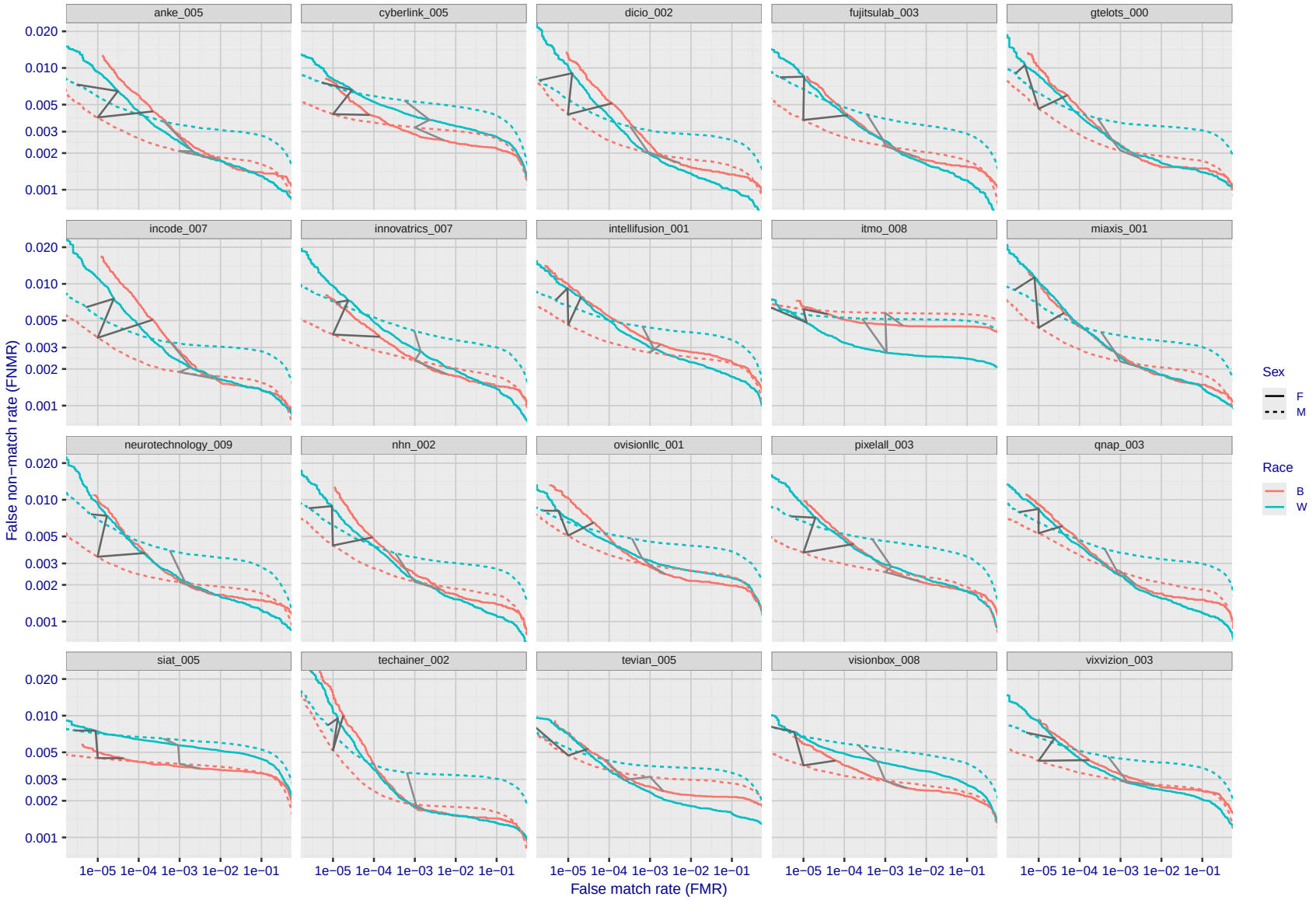


Figure 439: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

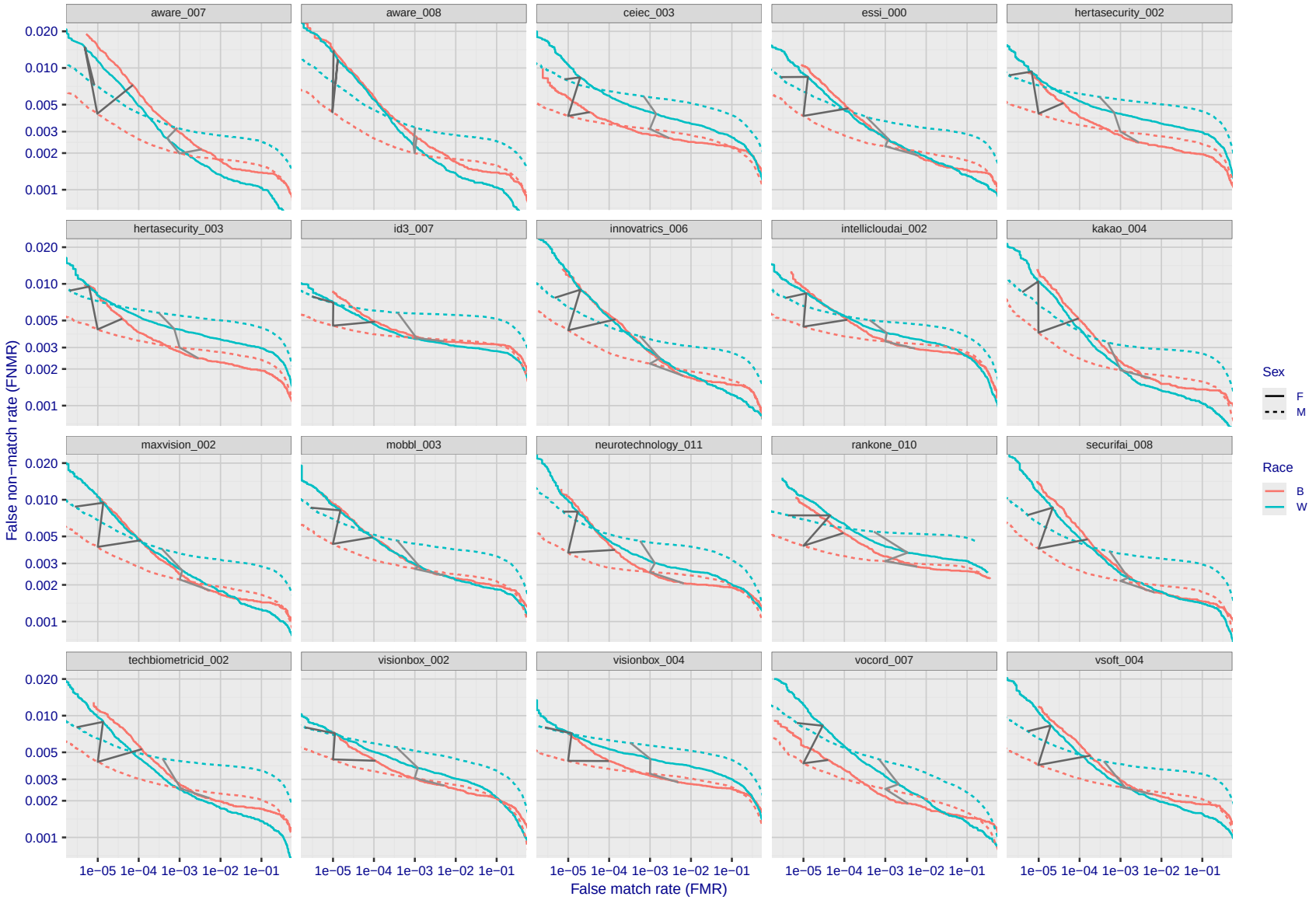


Figure 440: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

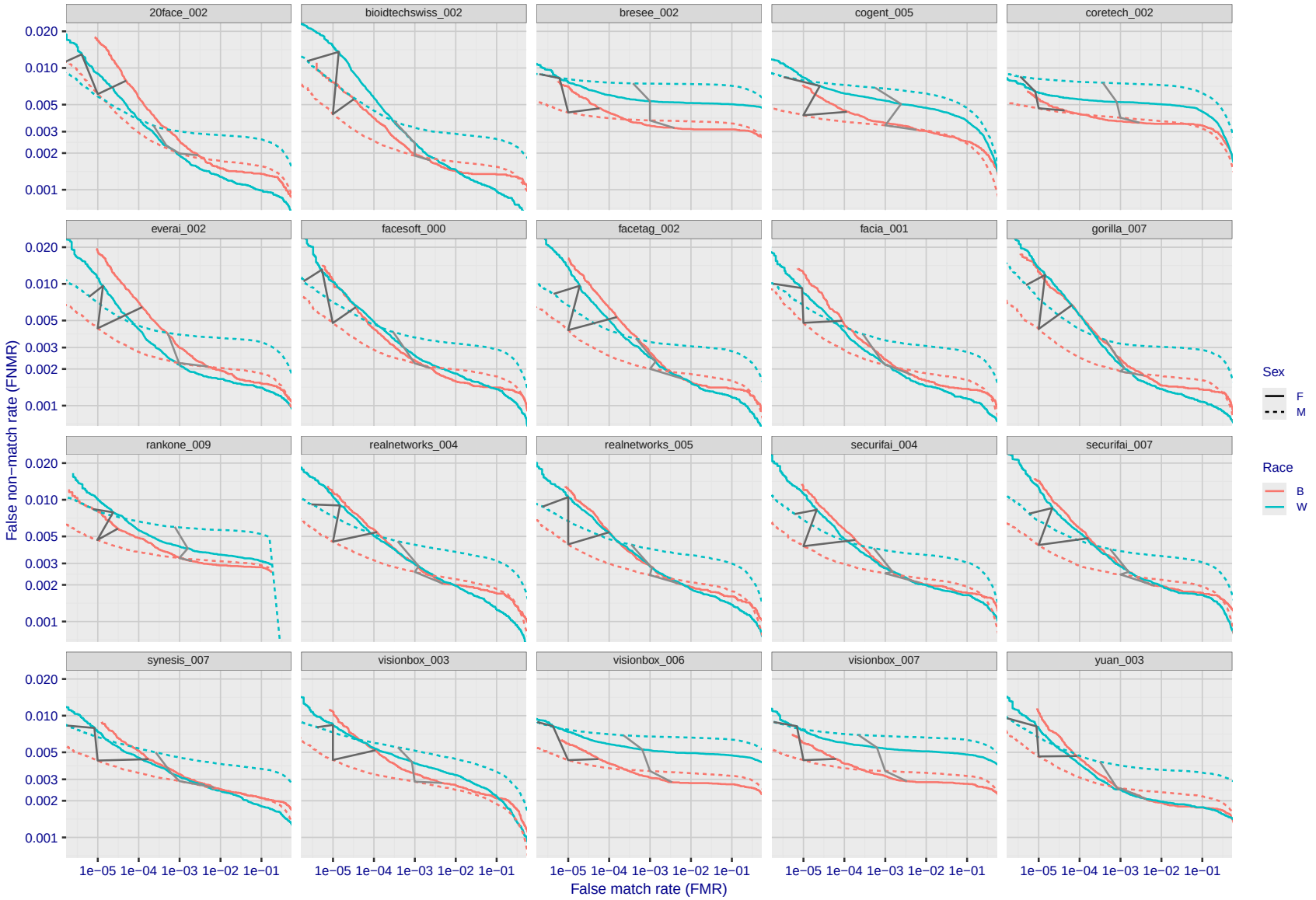


Figure 441: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

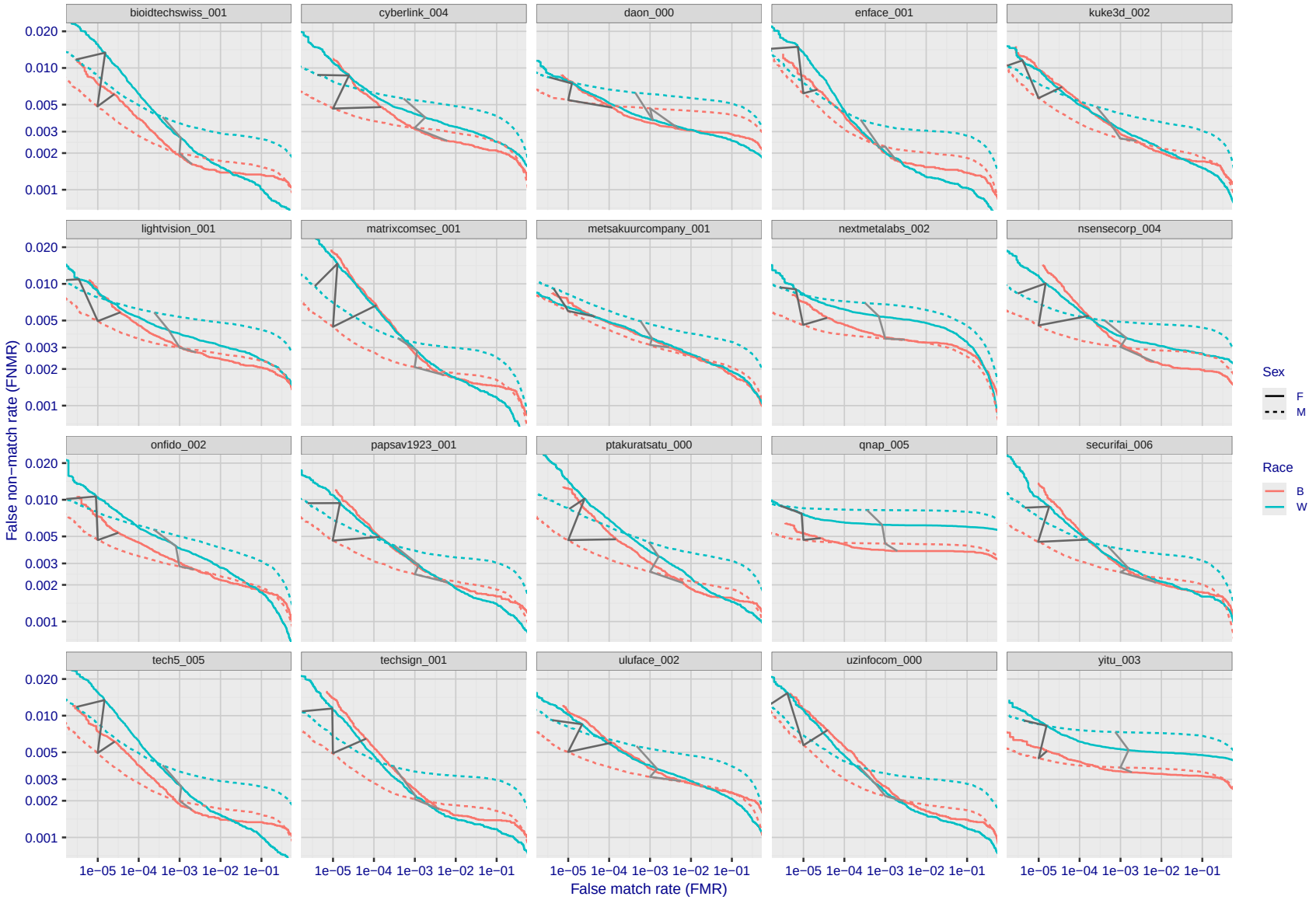


Figure 442: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

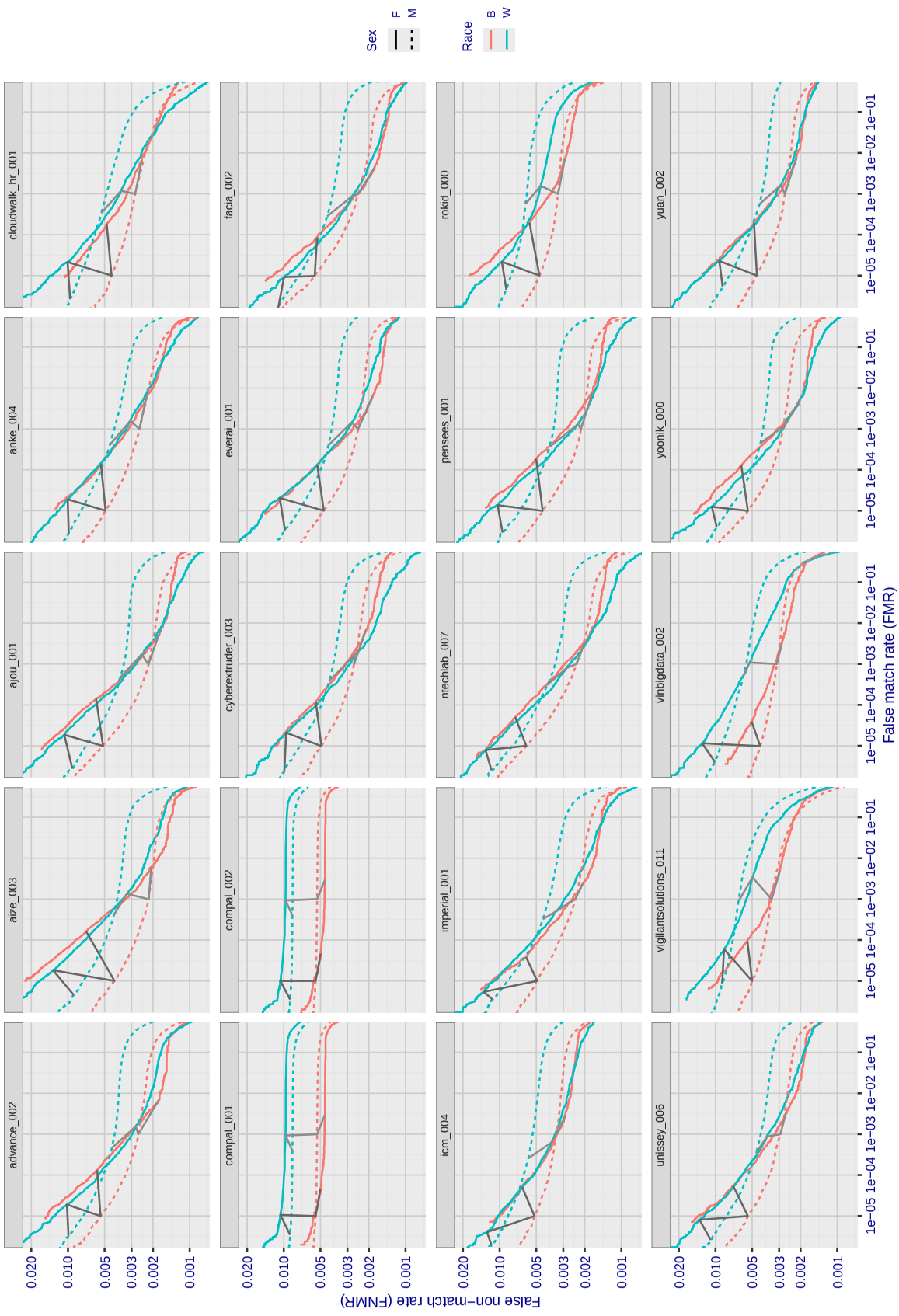


Figure 443: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

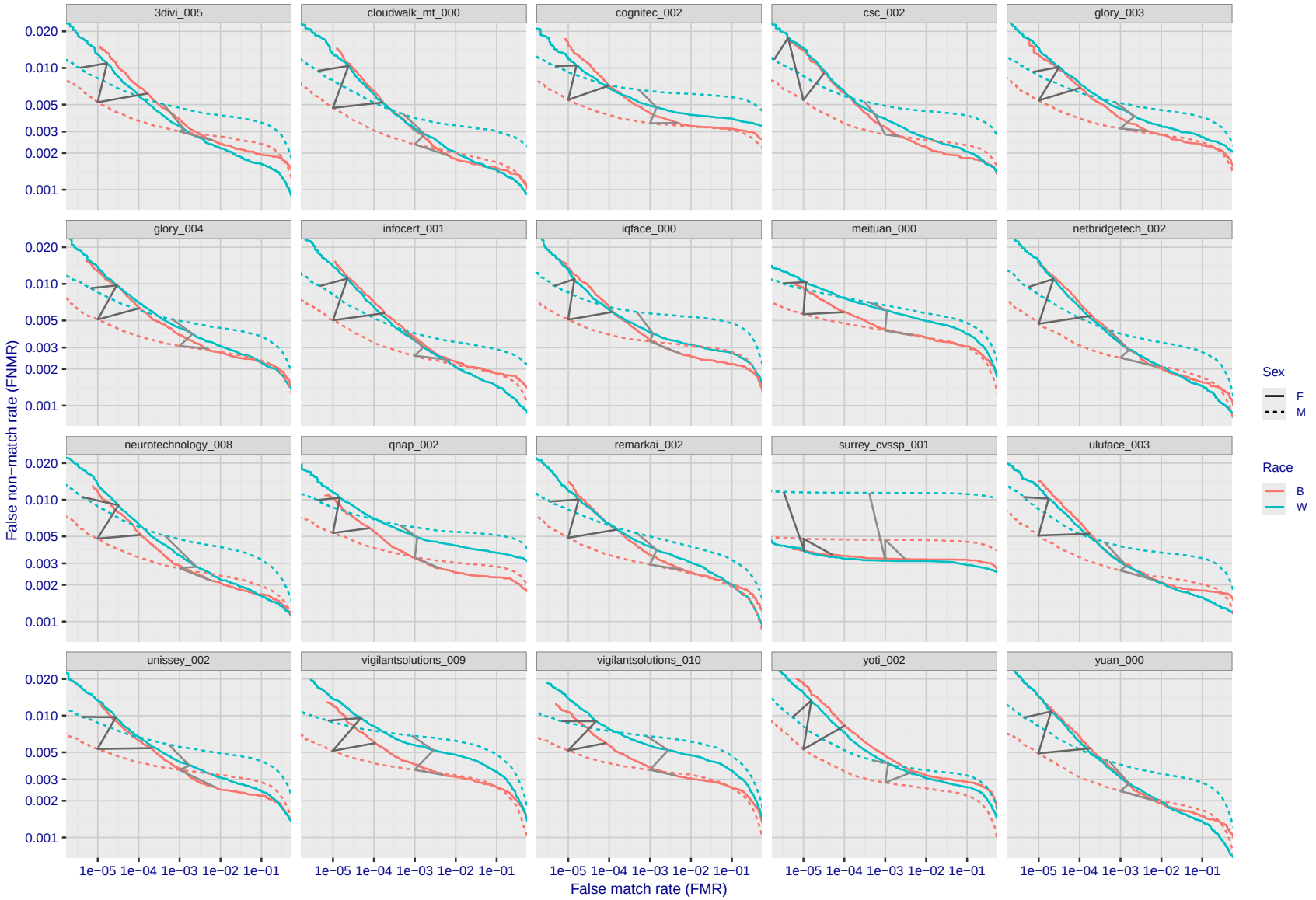


Figure 444: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

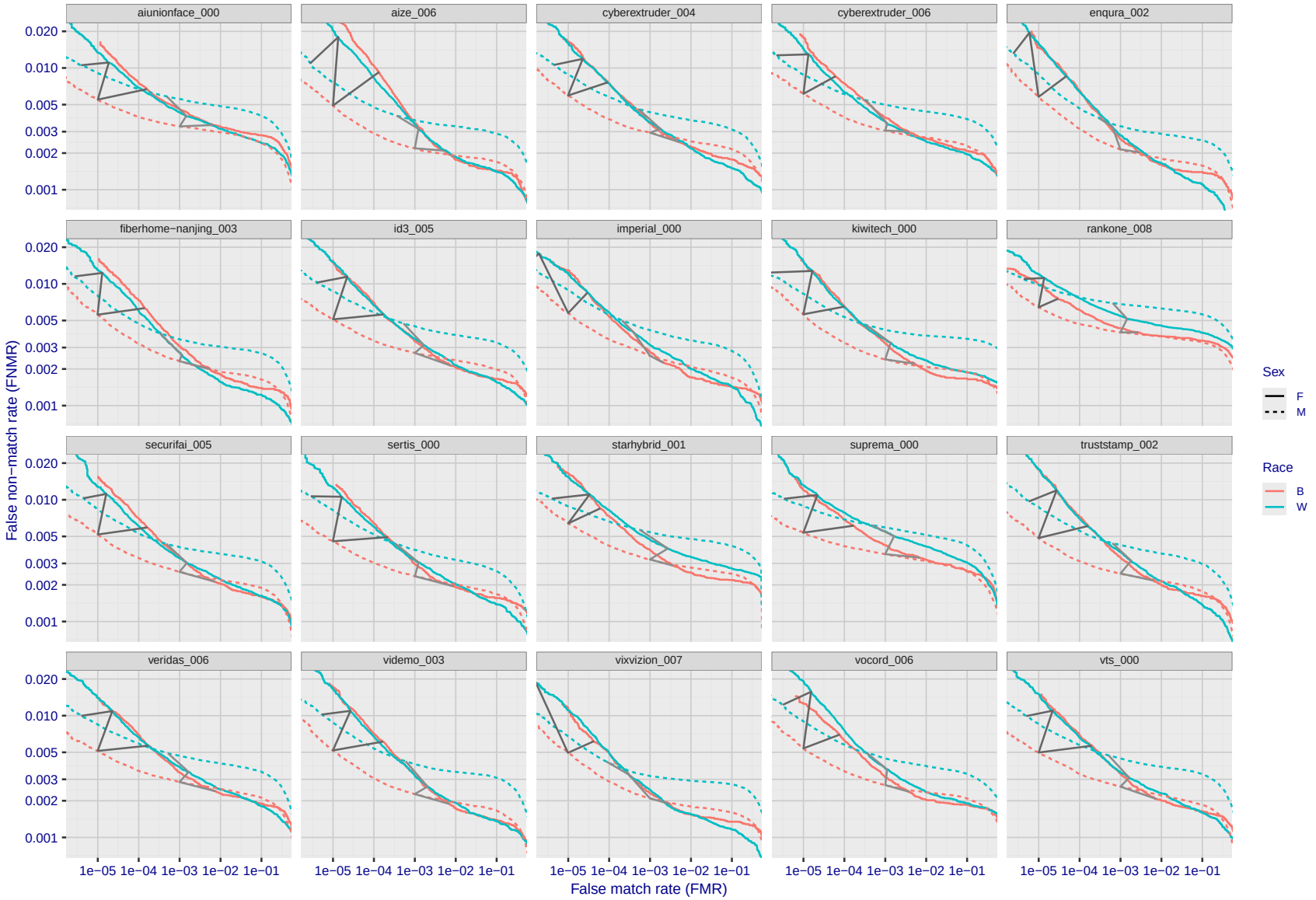


Figure 445: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

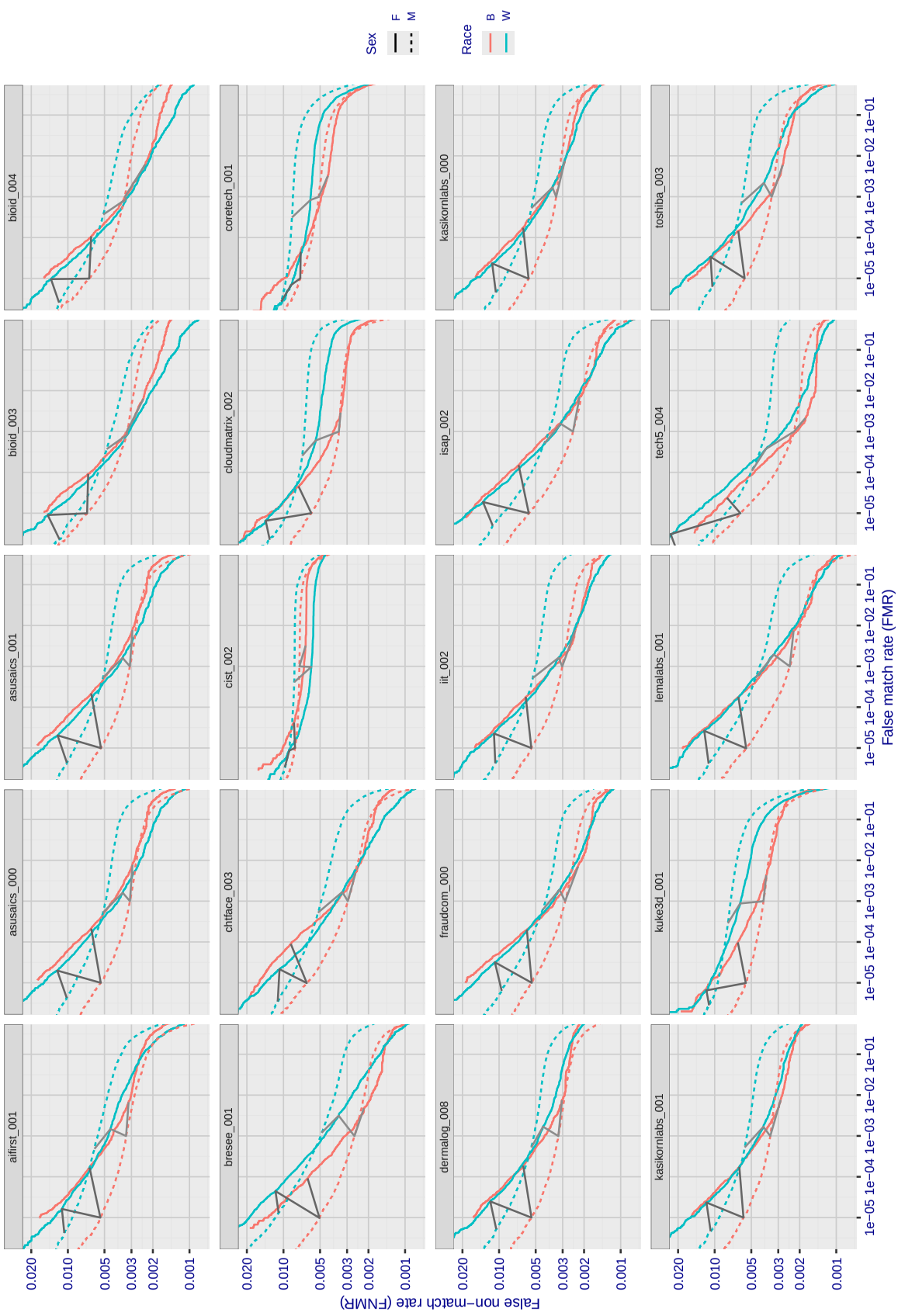


Figure 446: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

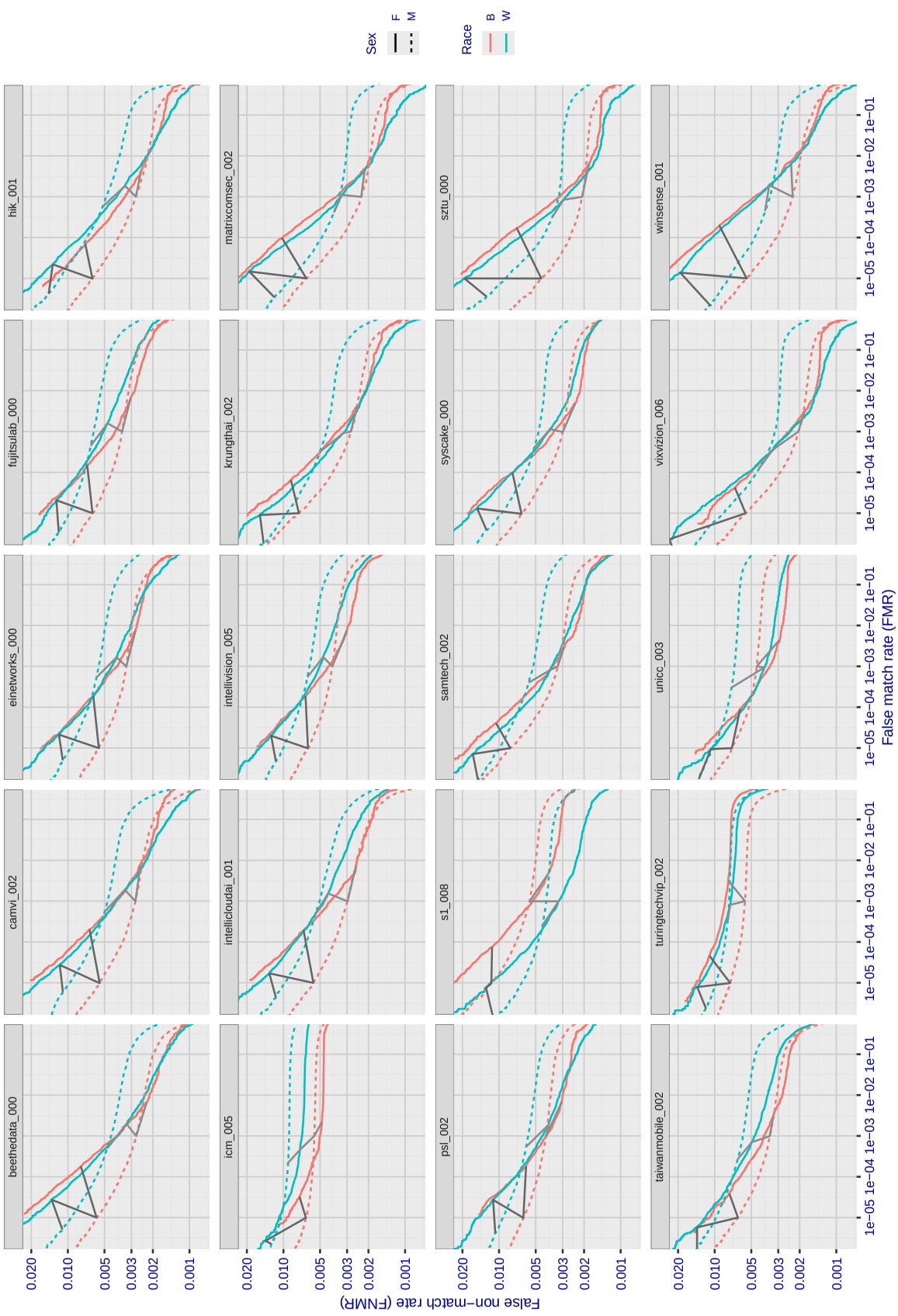


Figure 447: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

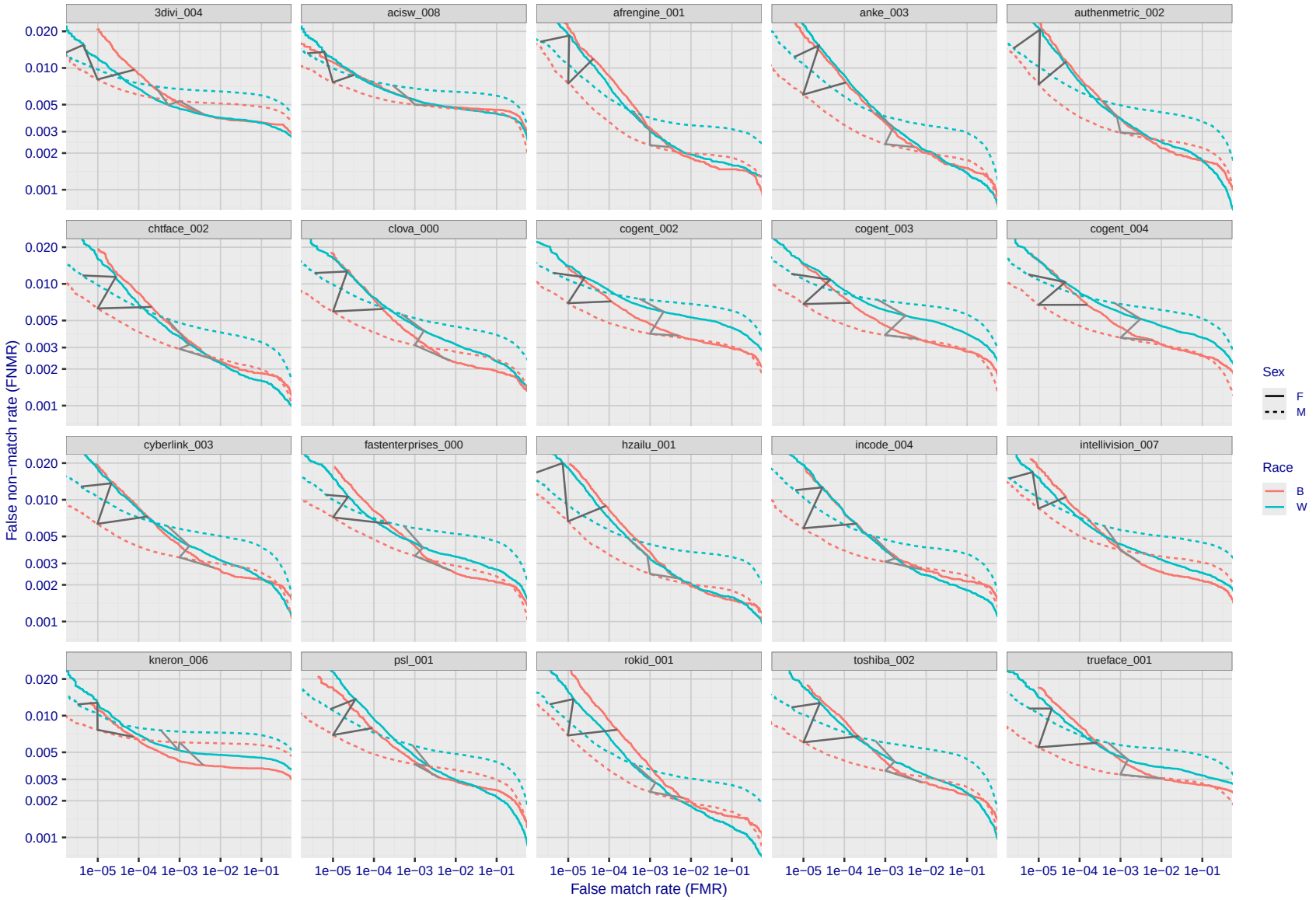


Figure 448: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

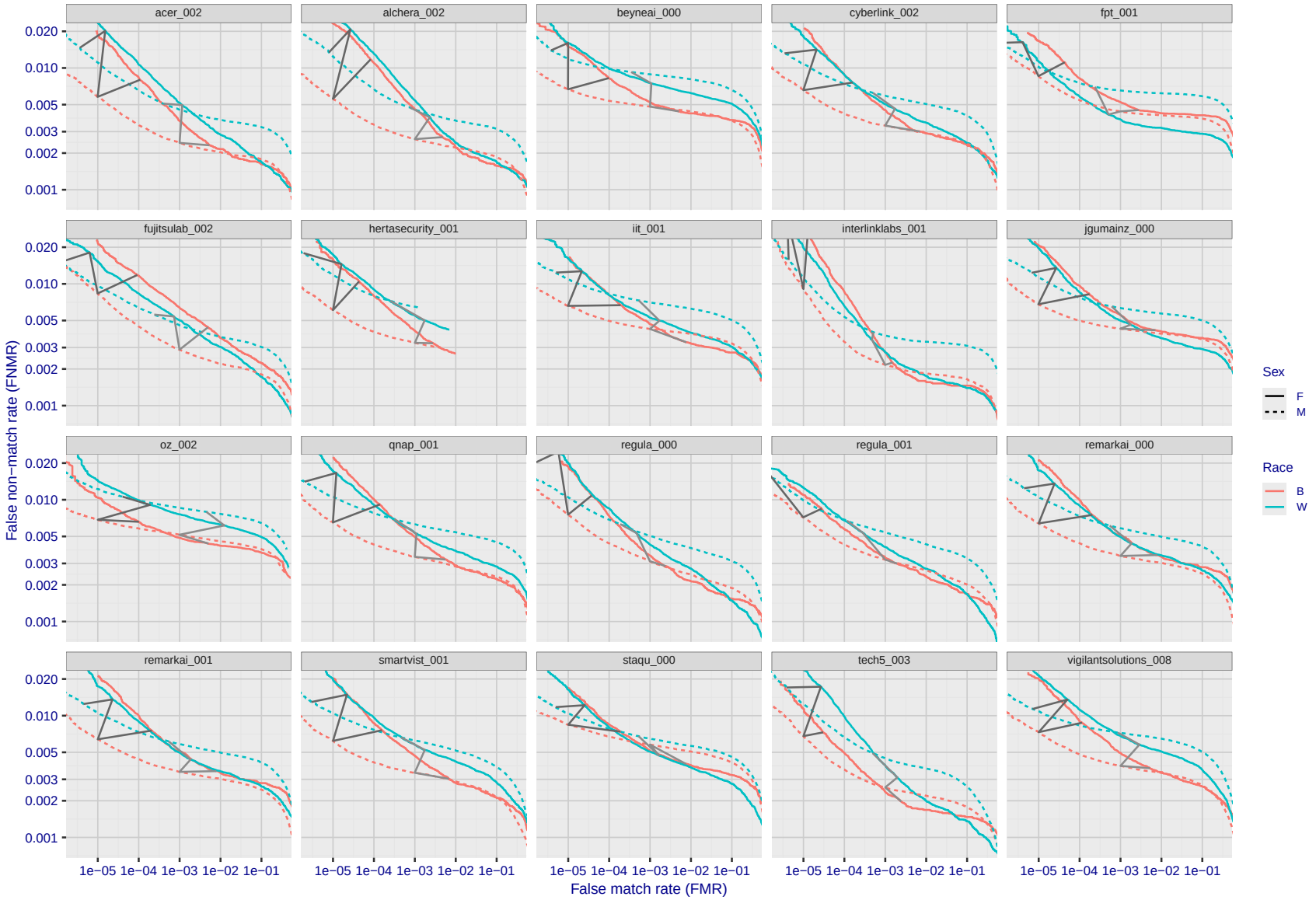


Figure 449: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

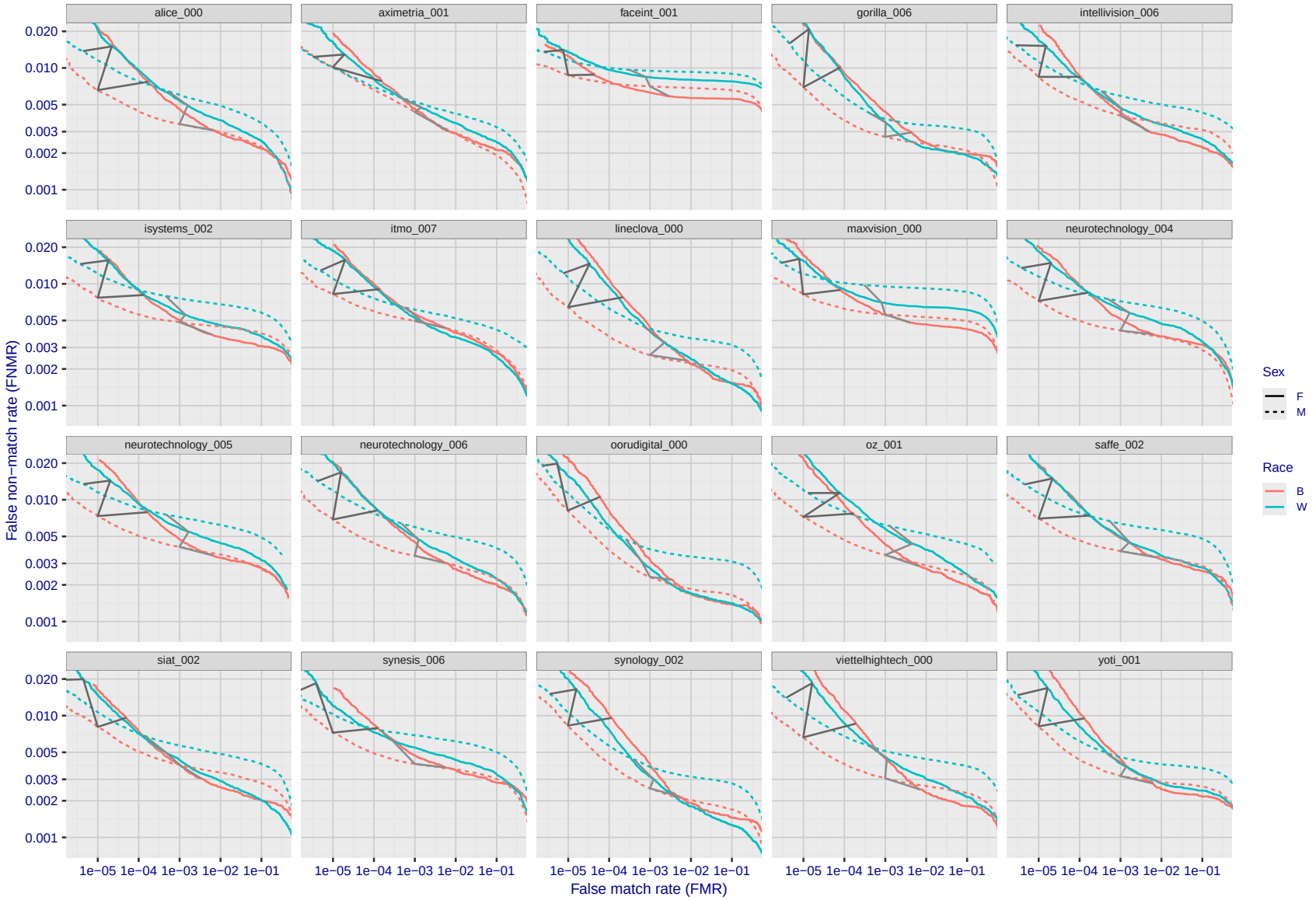


Figure 450: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

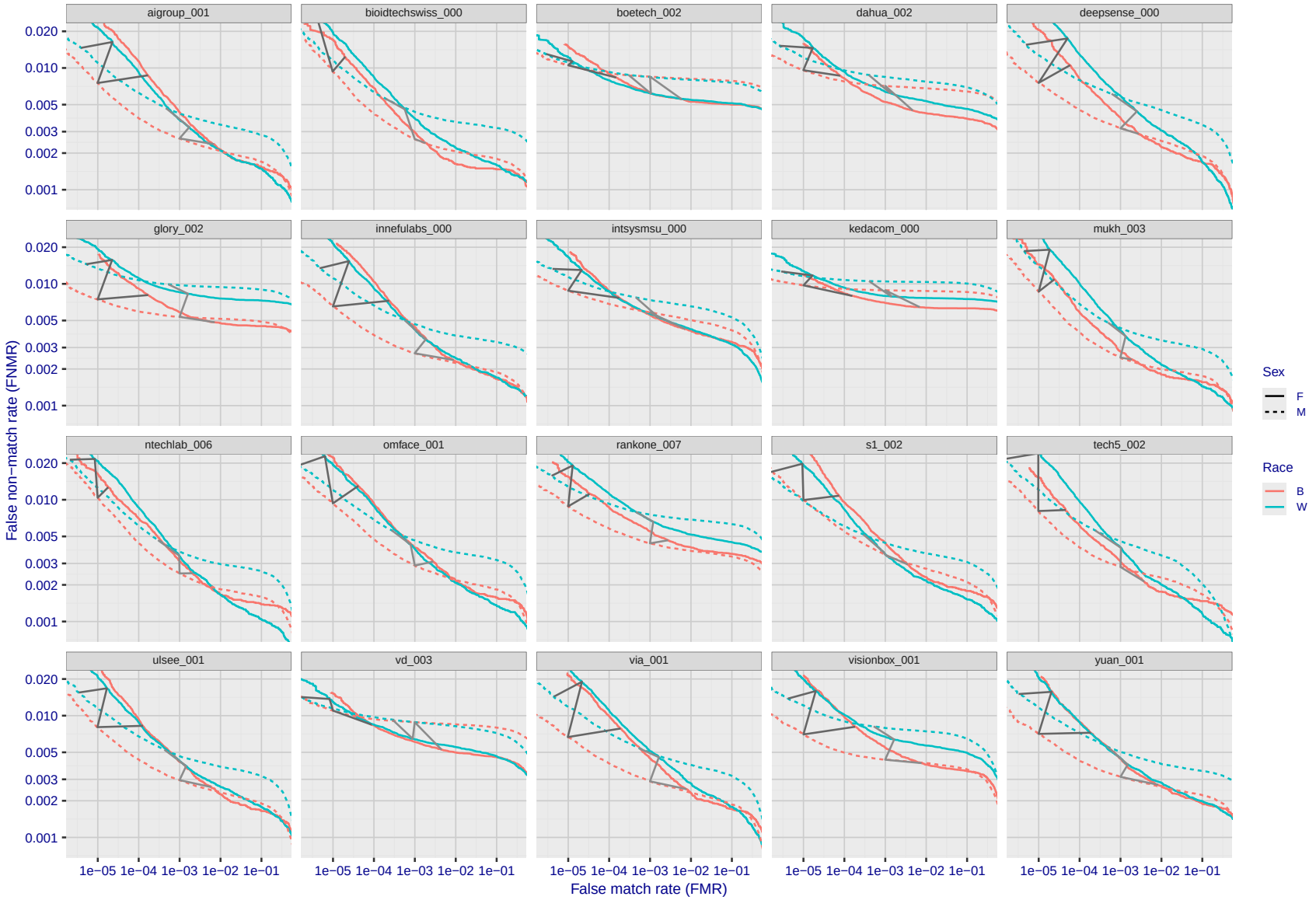


Figure 451: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

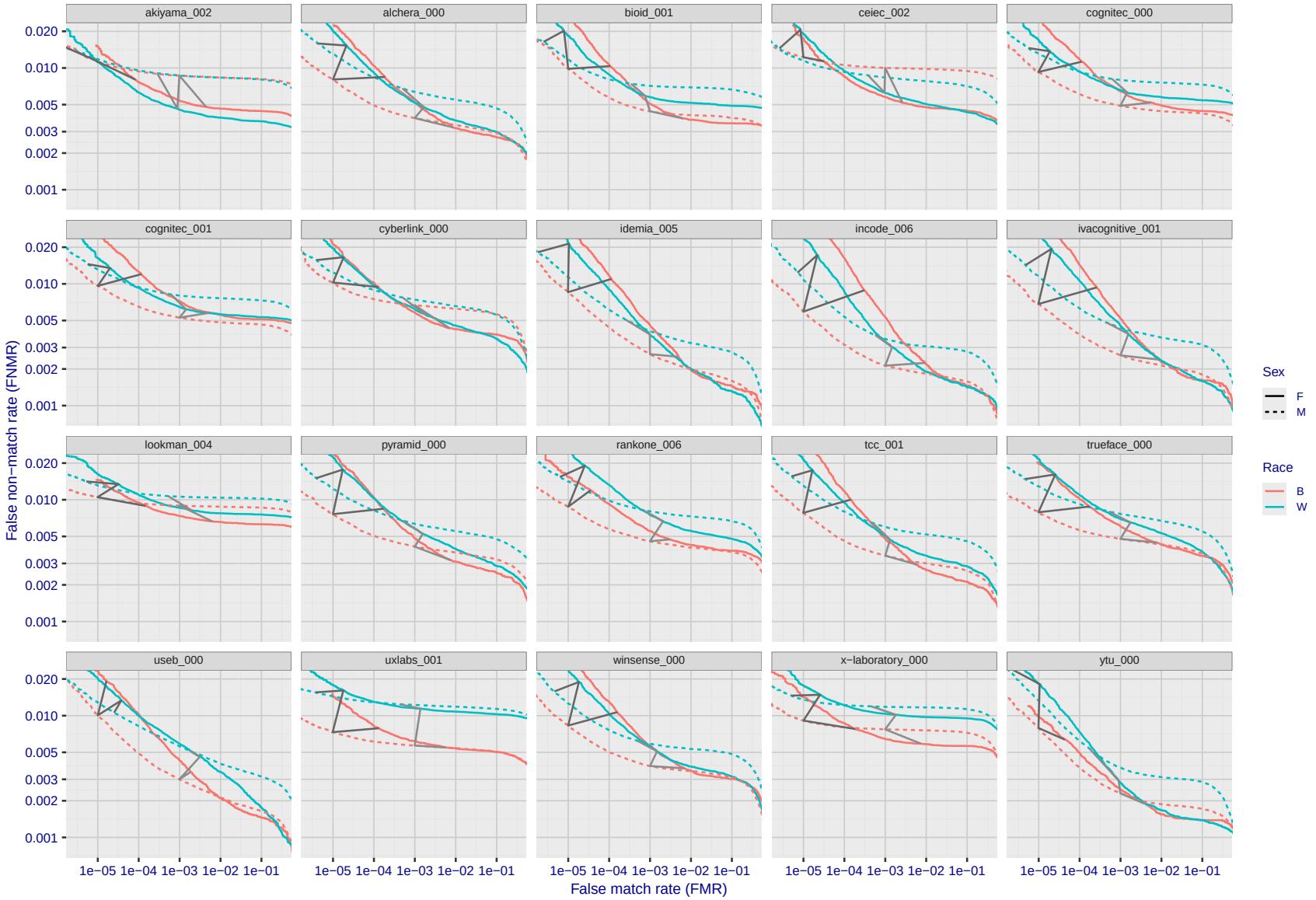


Figure 452: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

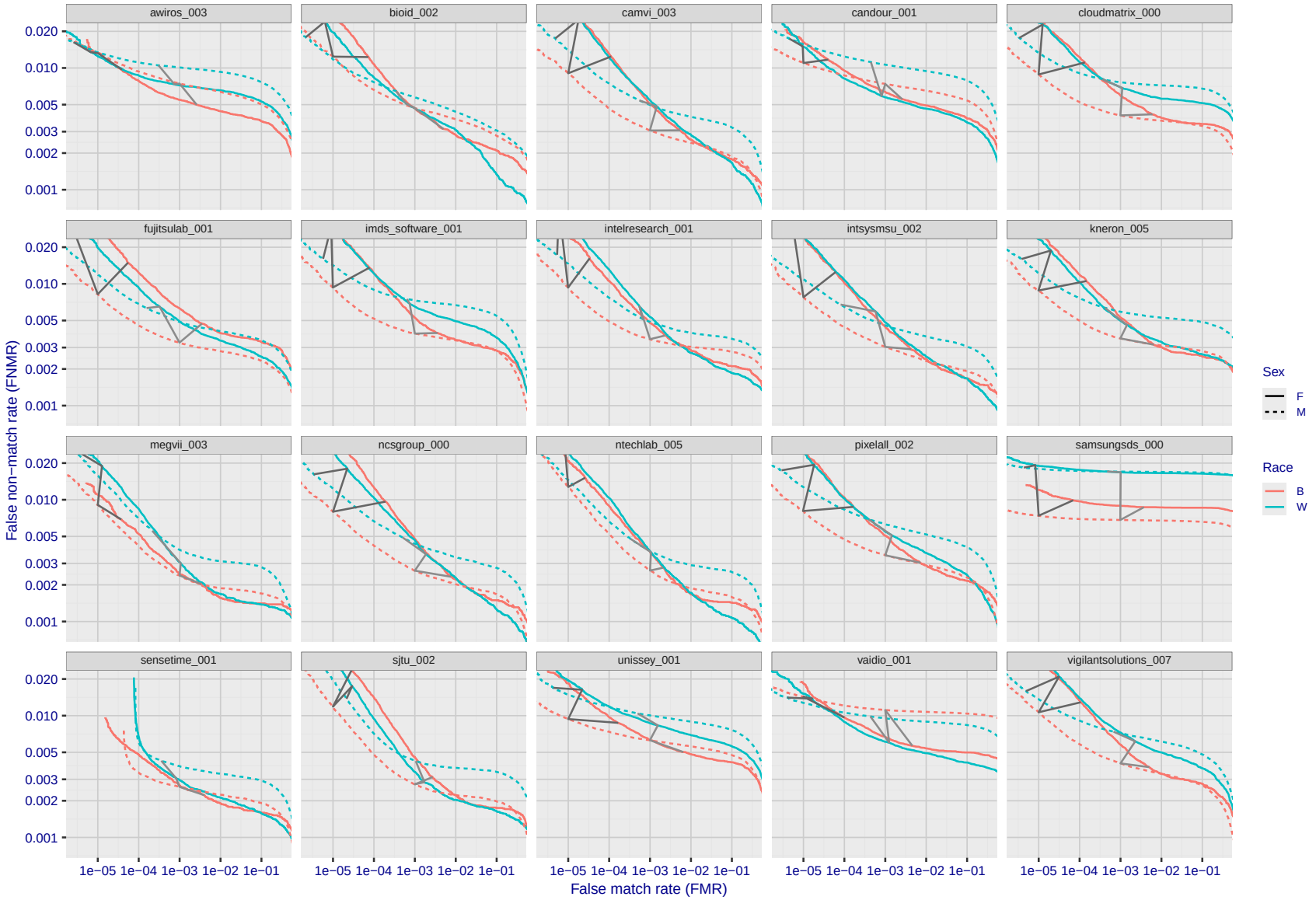


Figure 453: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

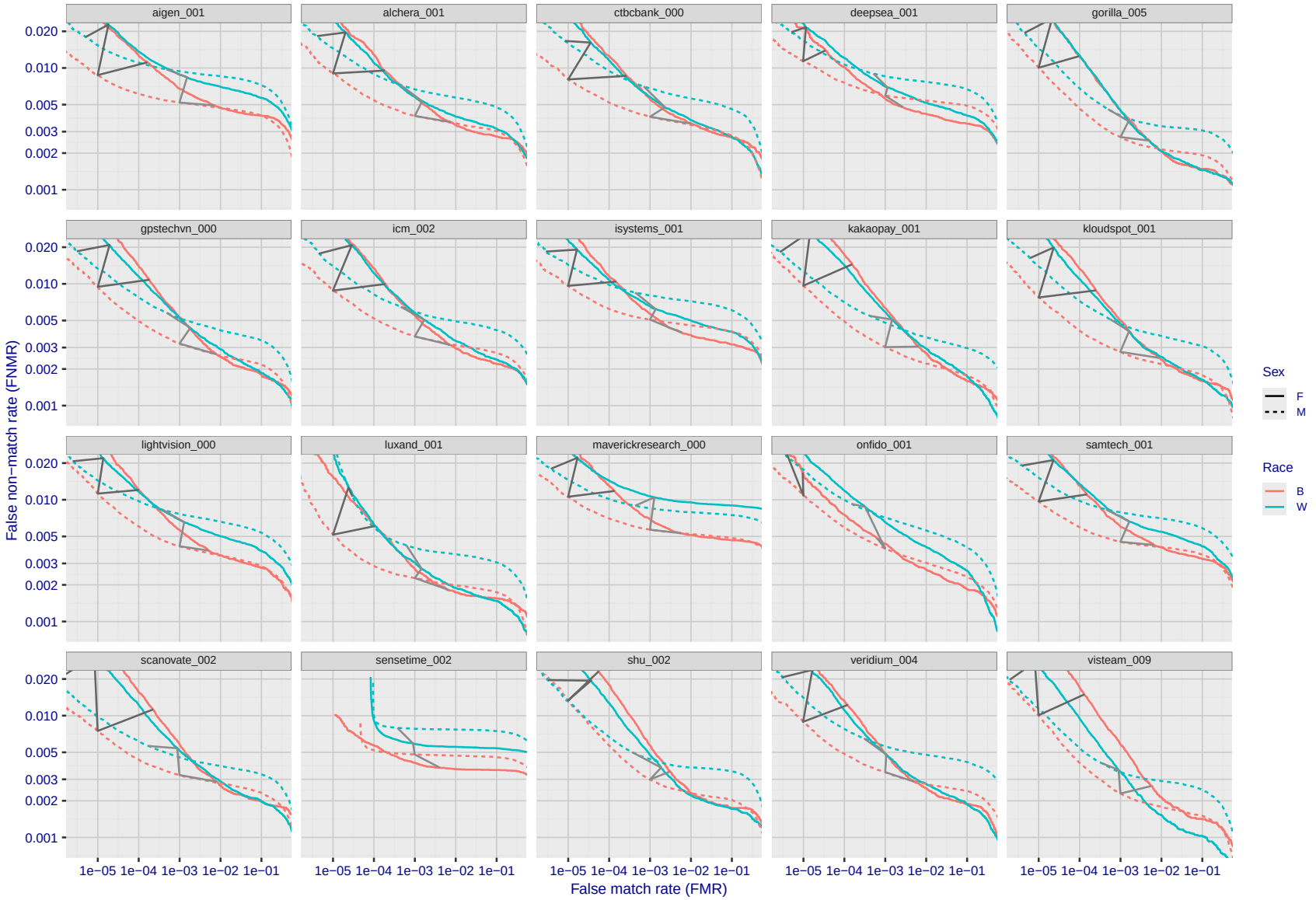


Figure 454: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

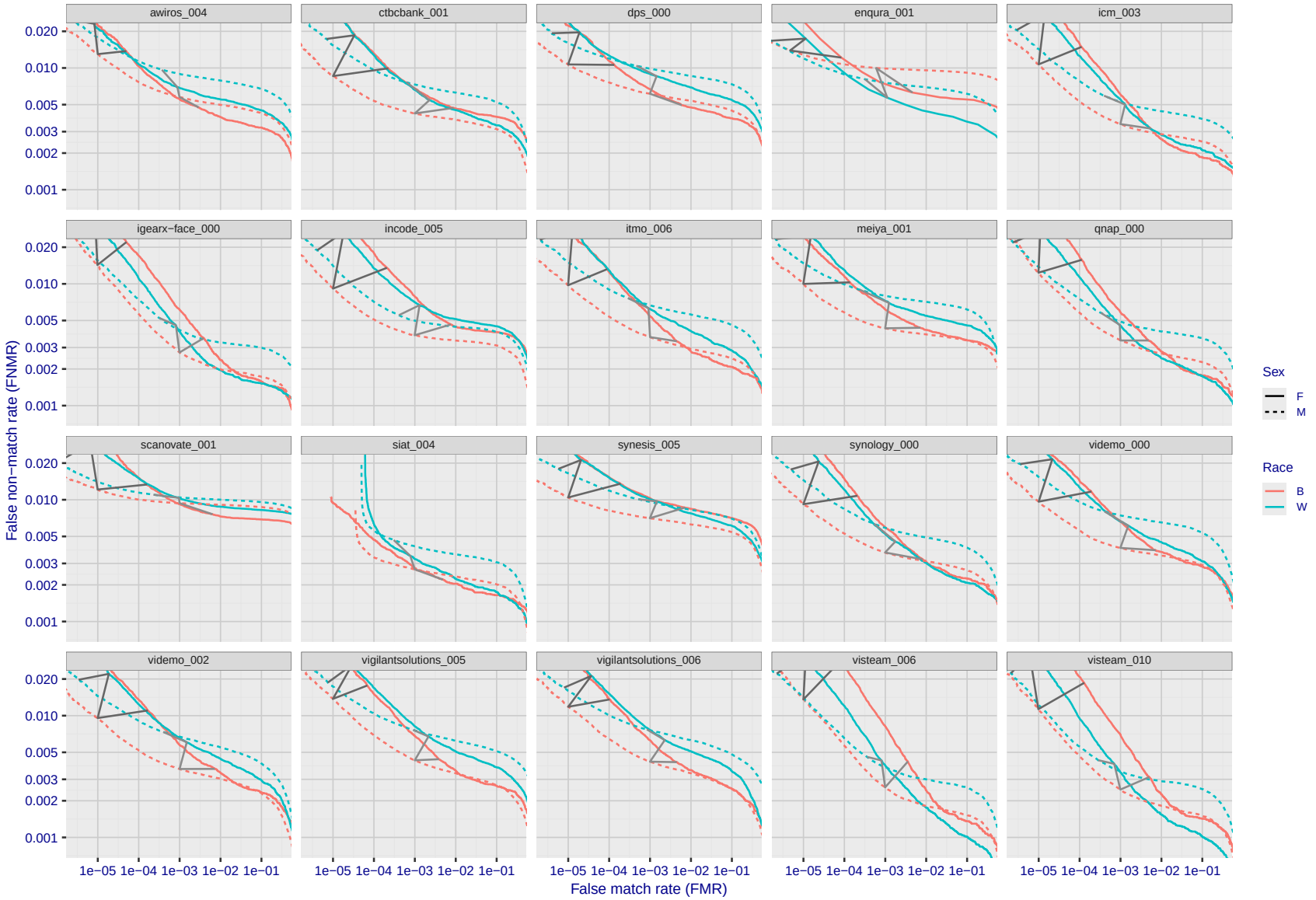


Figure 455: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

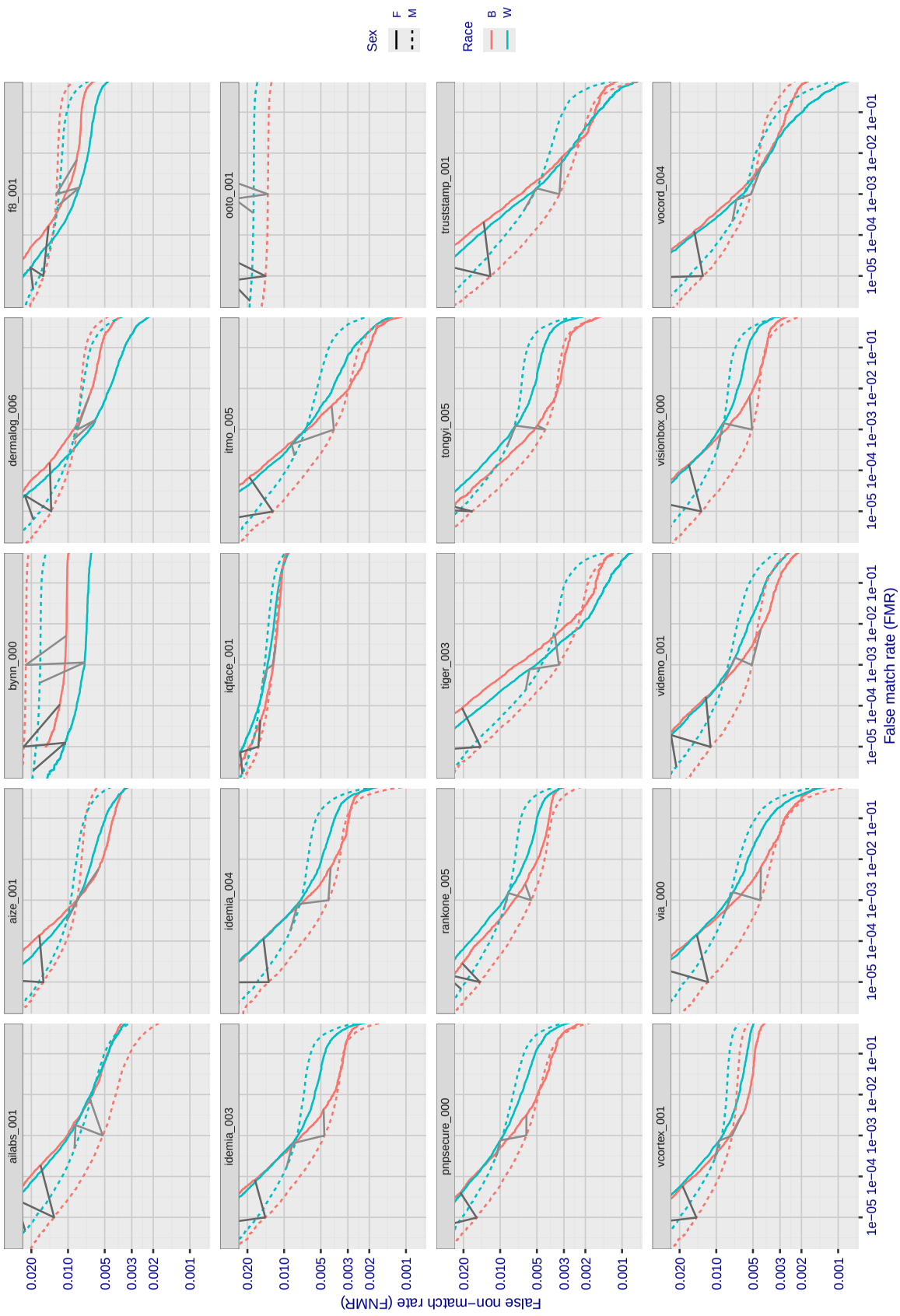


Figure 456: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

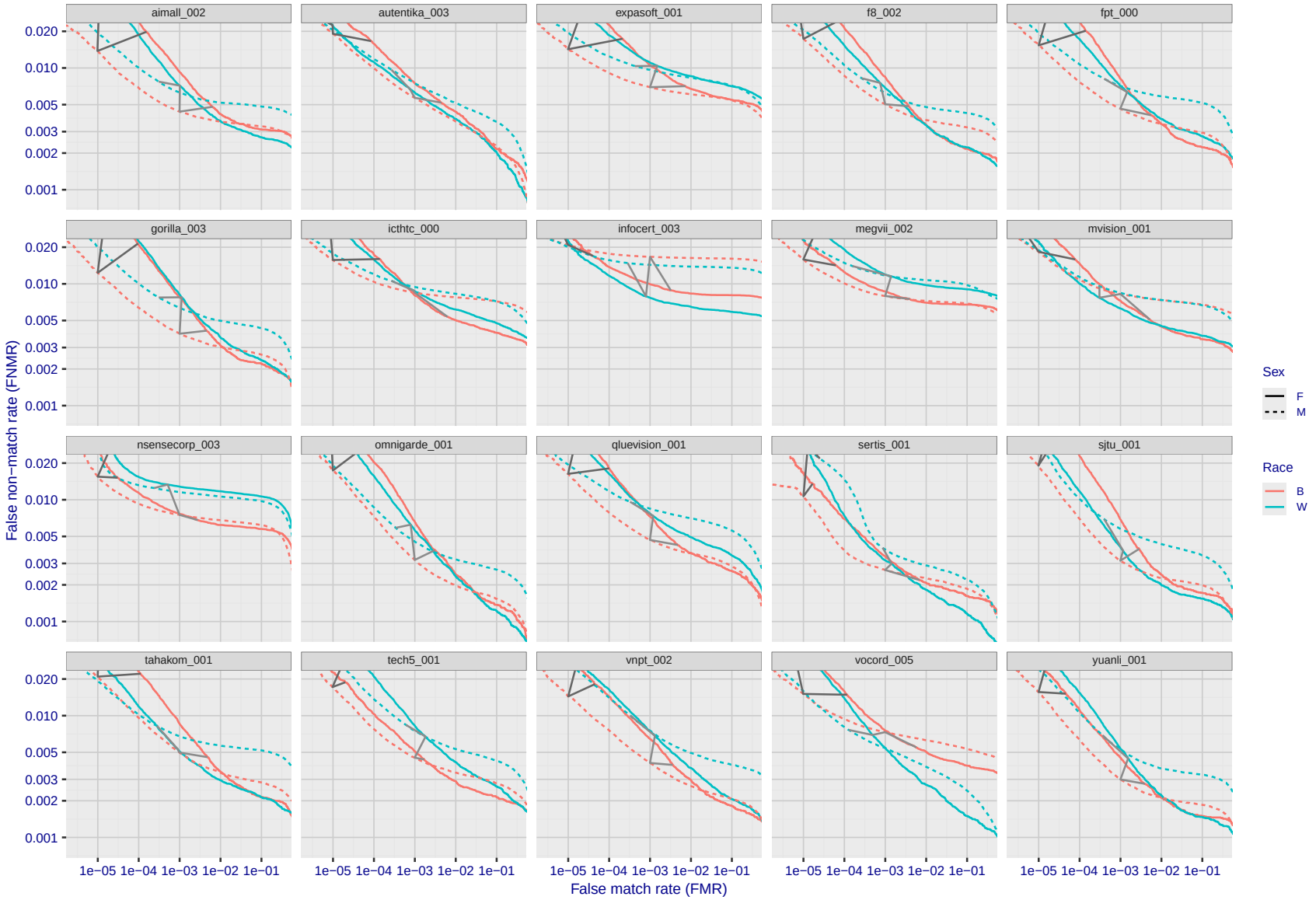


Figure 457: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

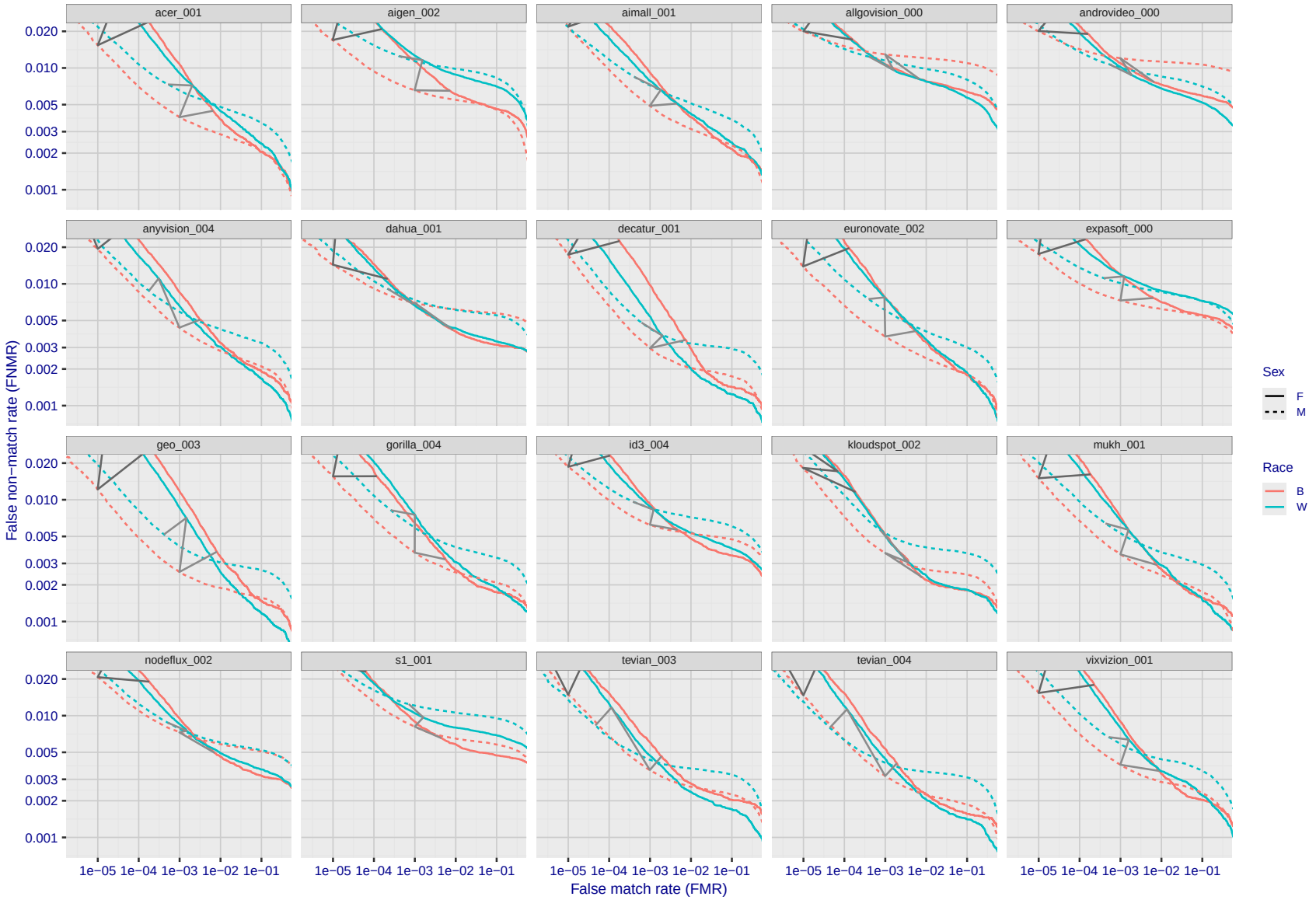


Figure 458: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

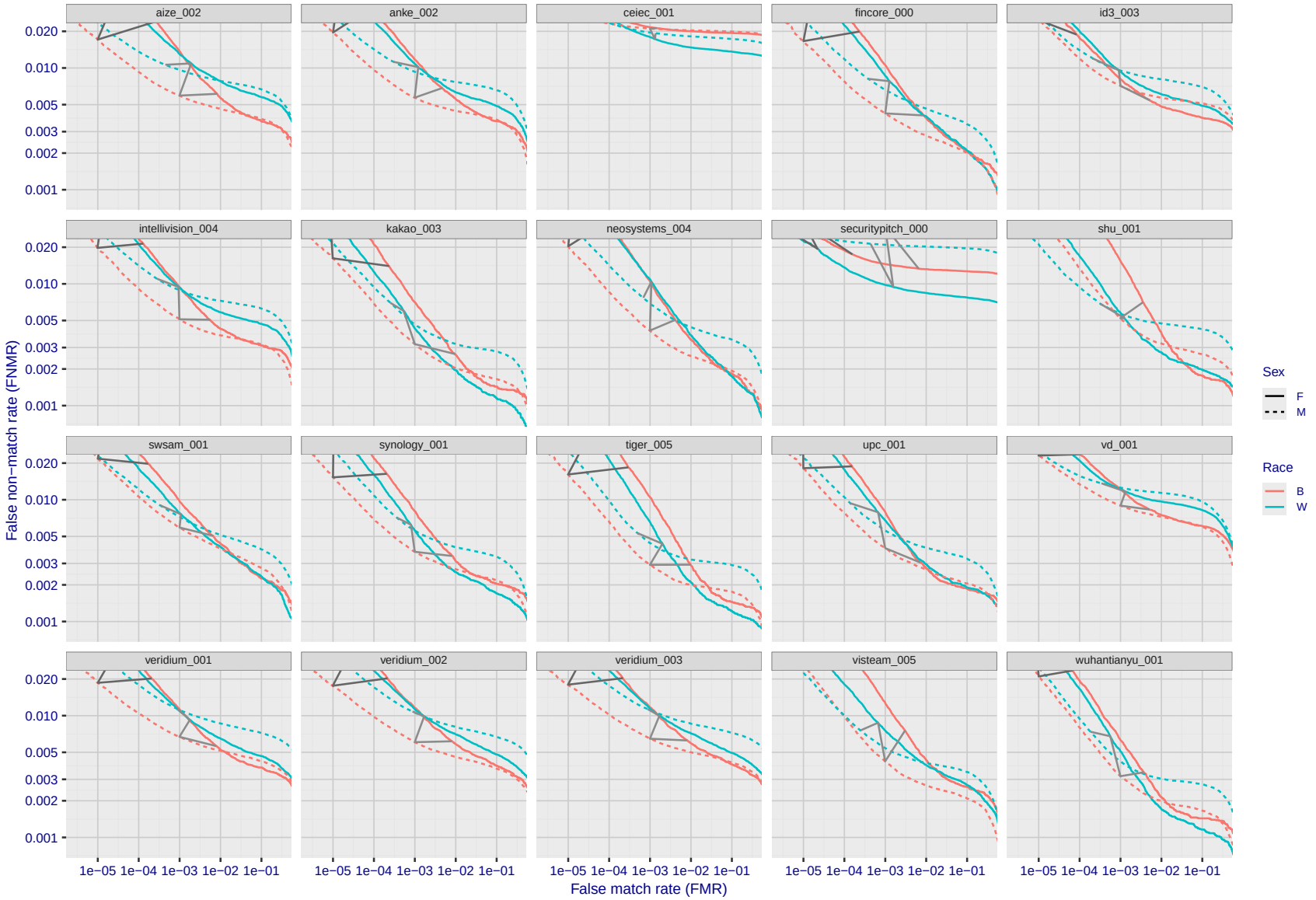


Figure 459: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

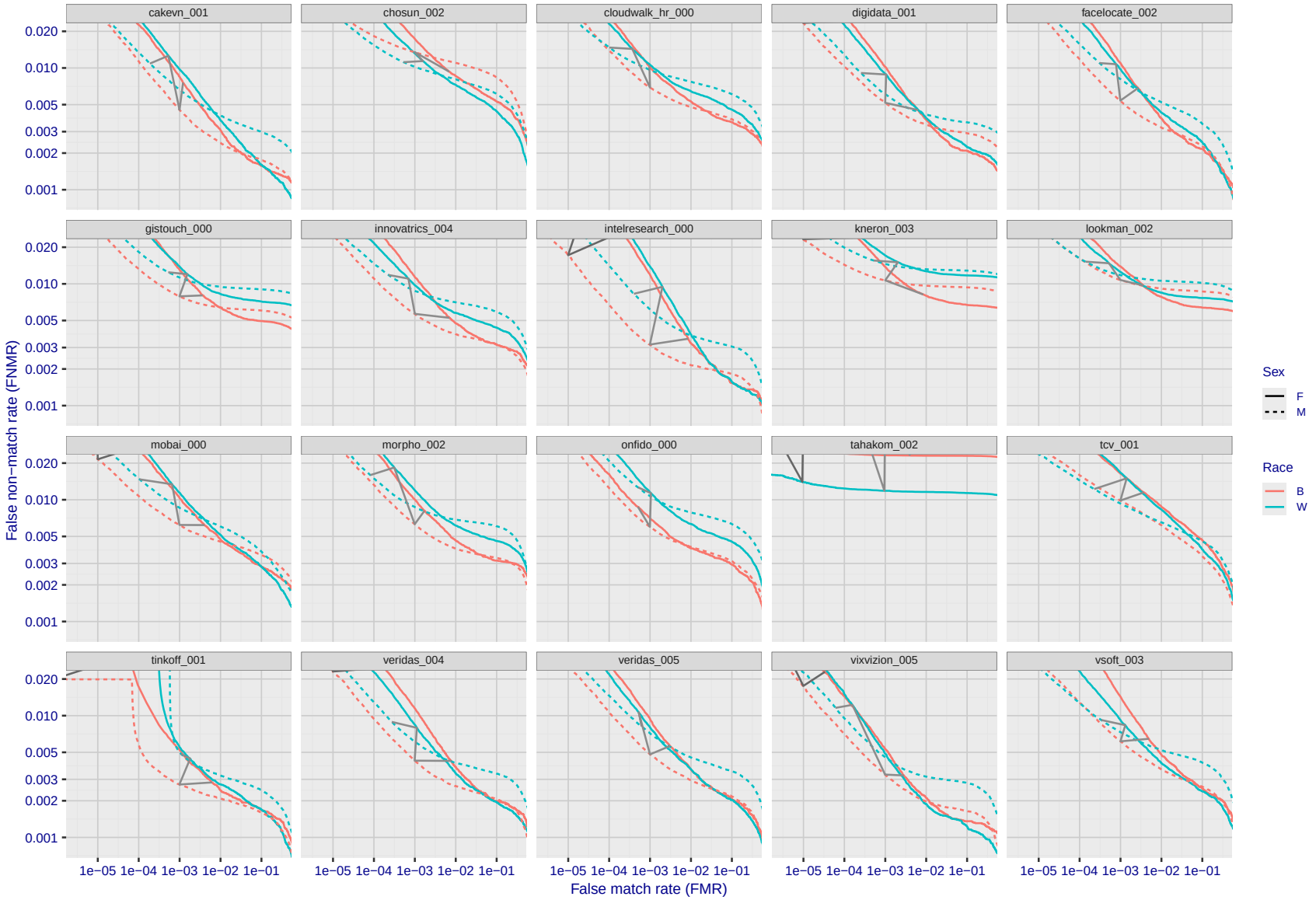


Figure 460: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

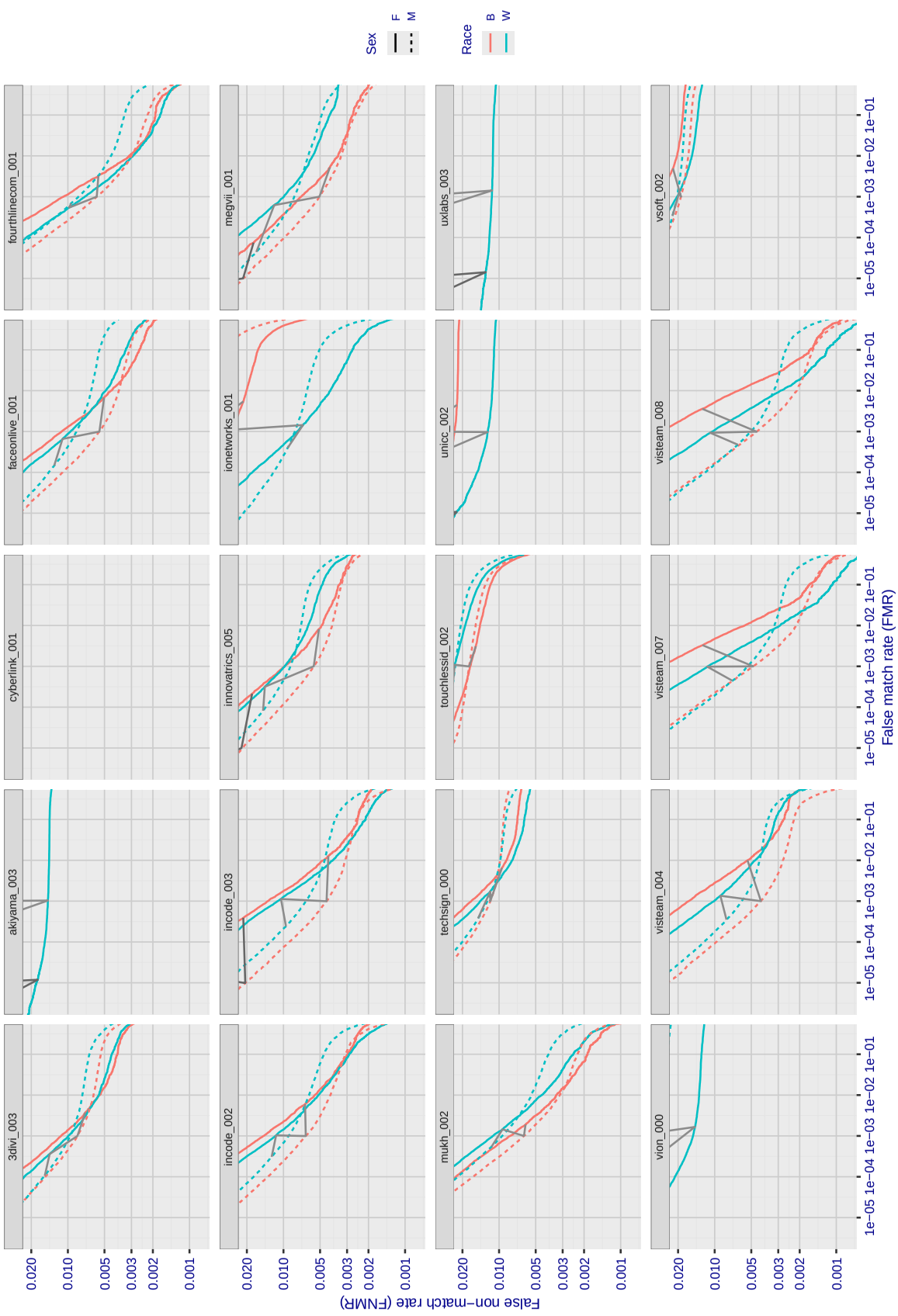


Figure 461: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

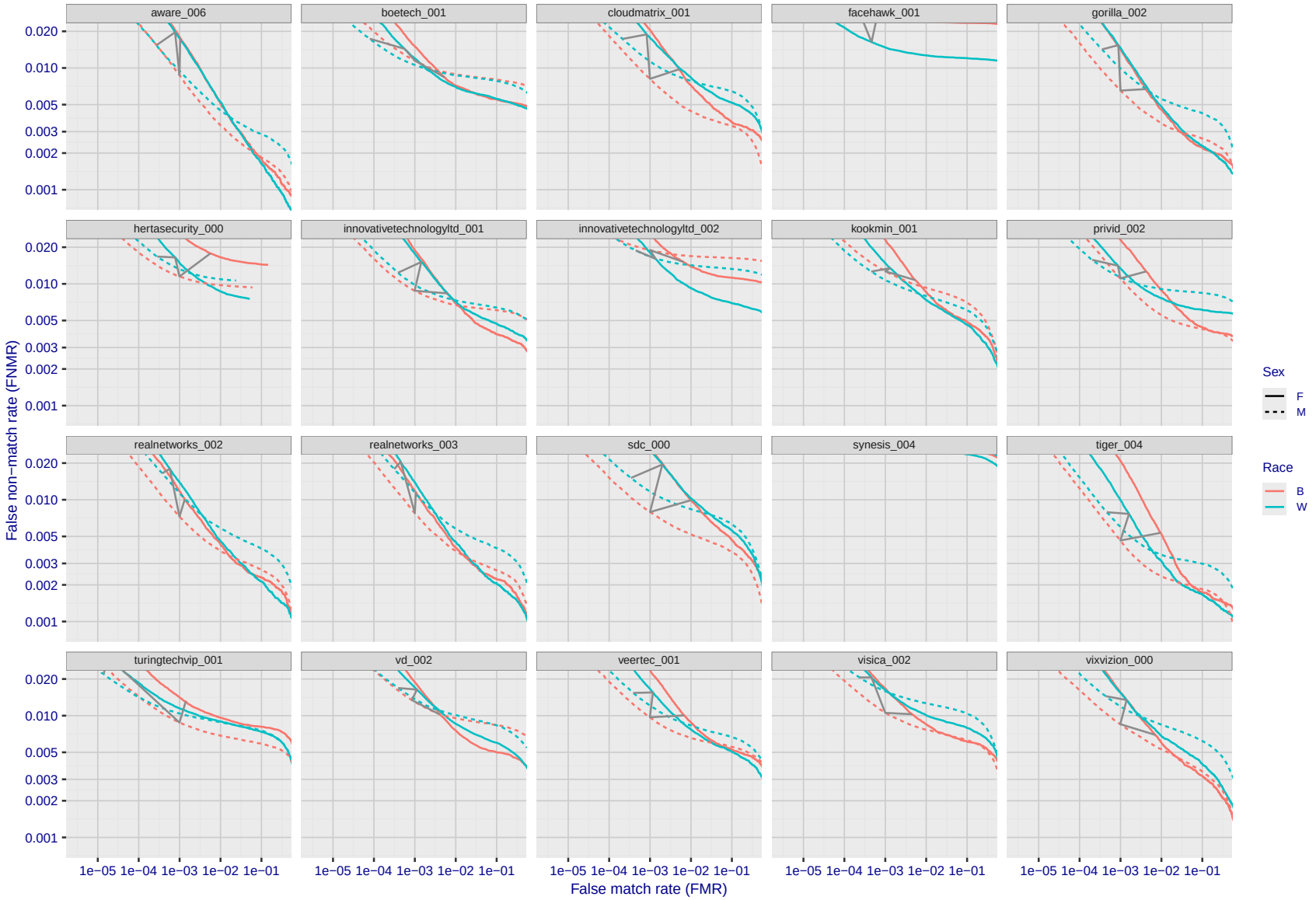


Figure 462: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

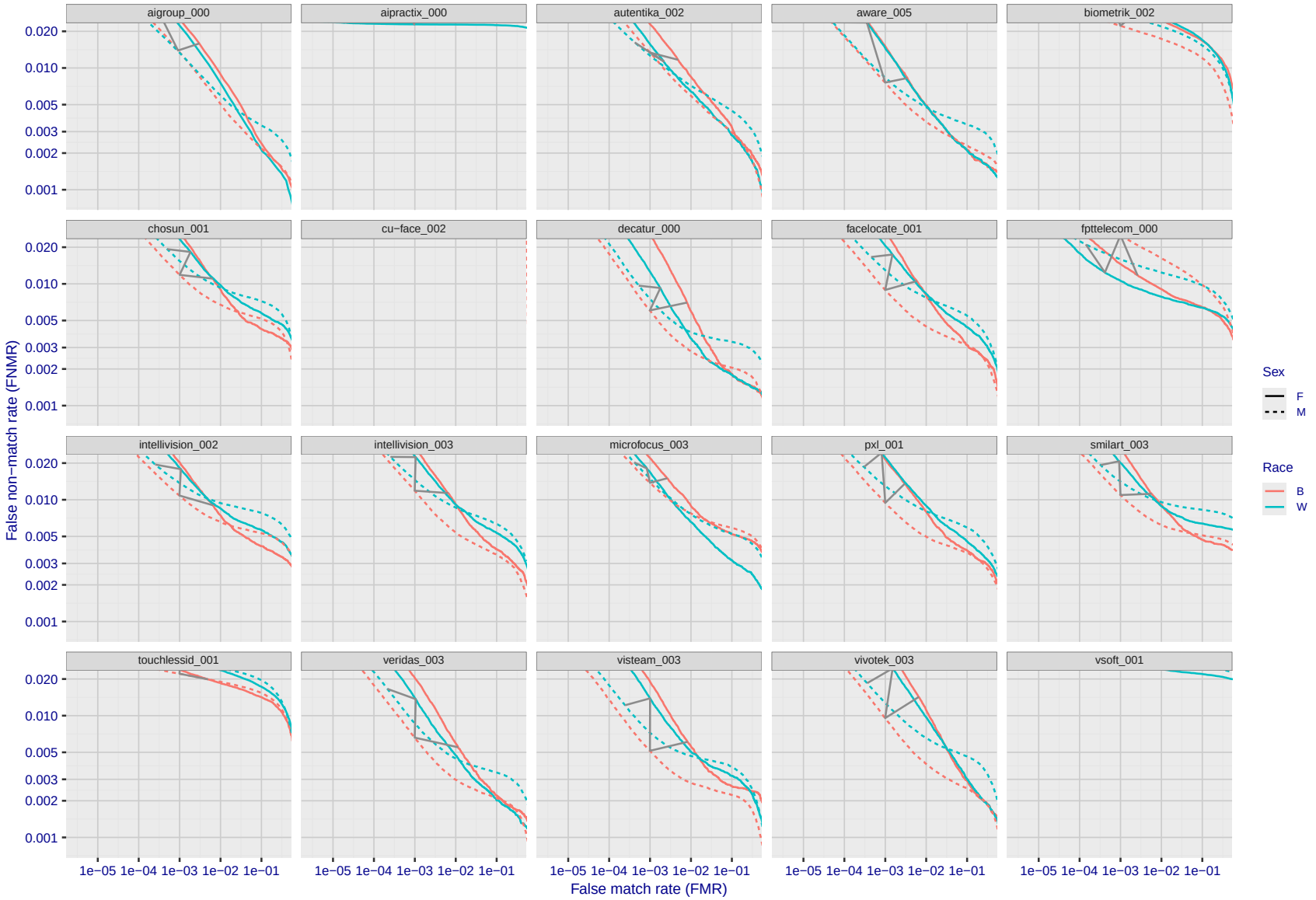


Figure 463: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

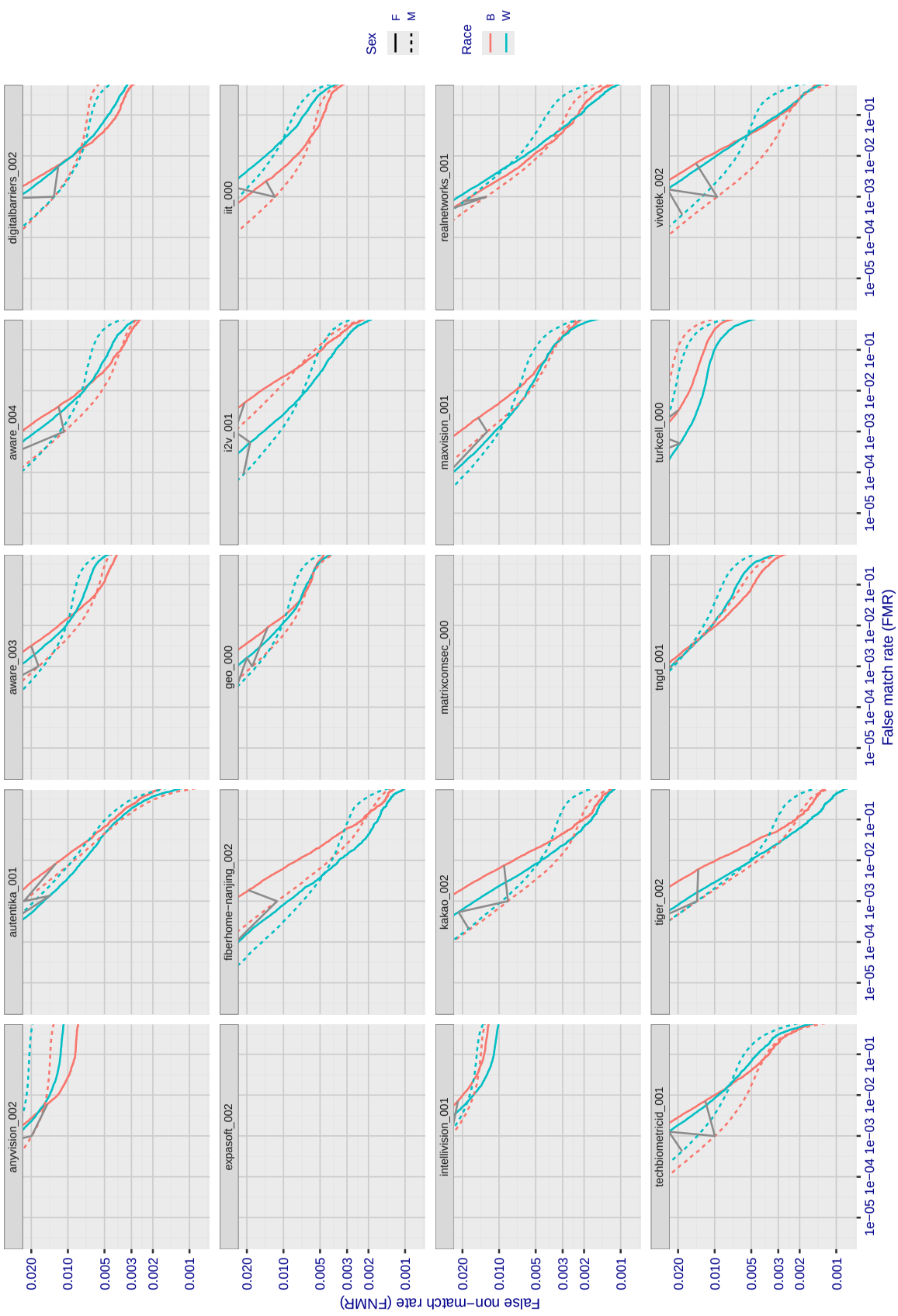


Figure 464: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

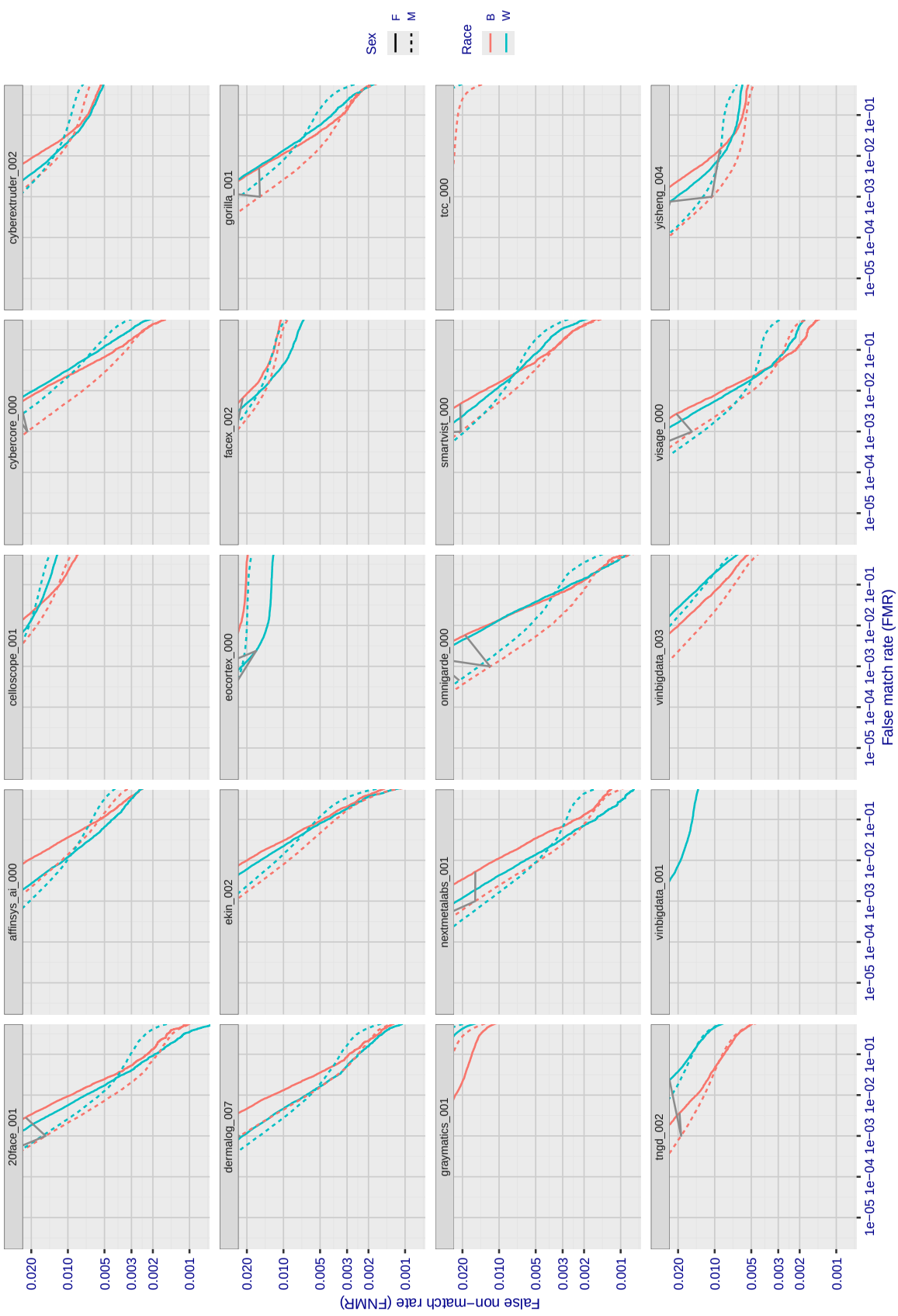


Figure 465: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

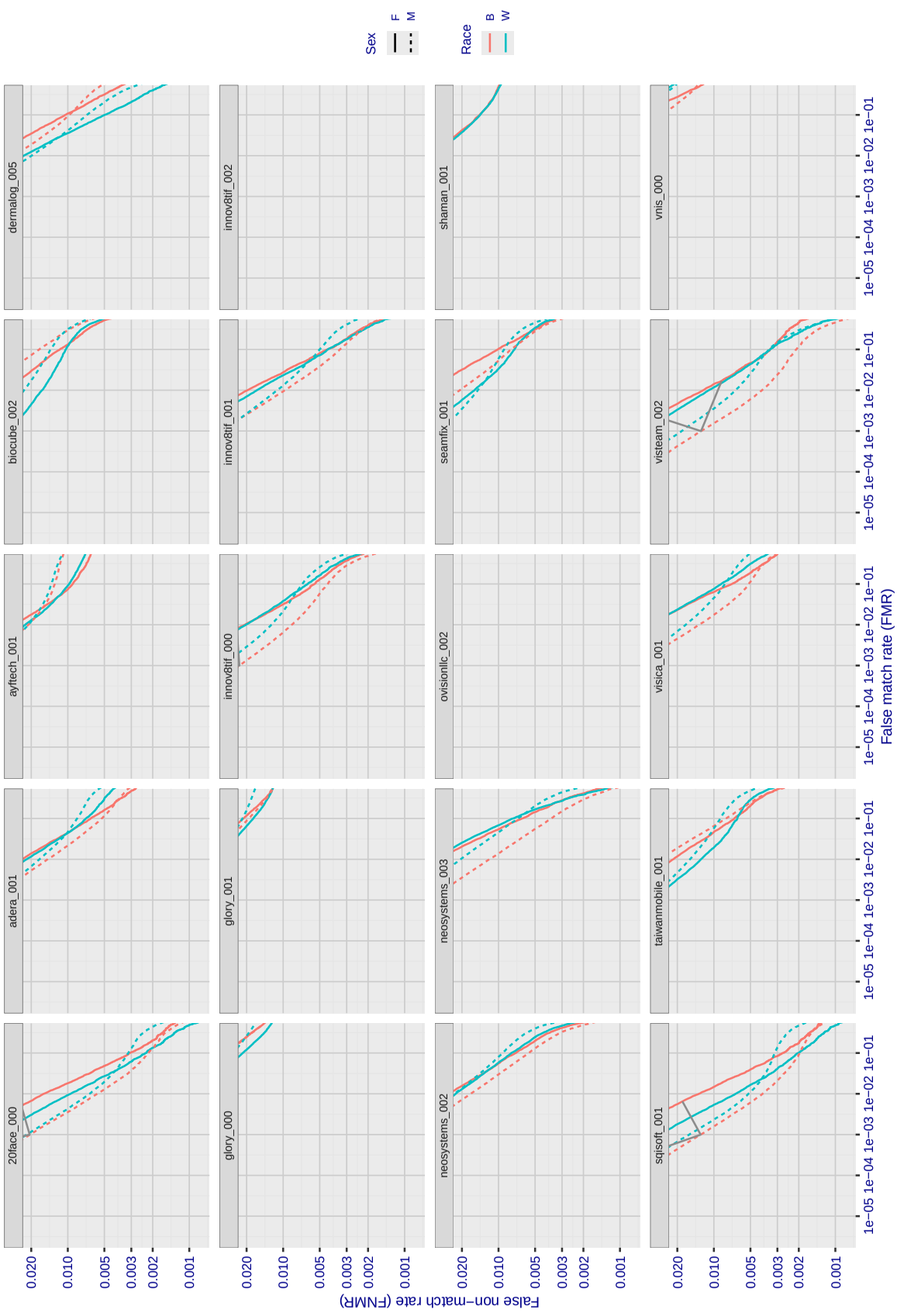


Figure 466: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

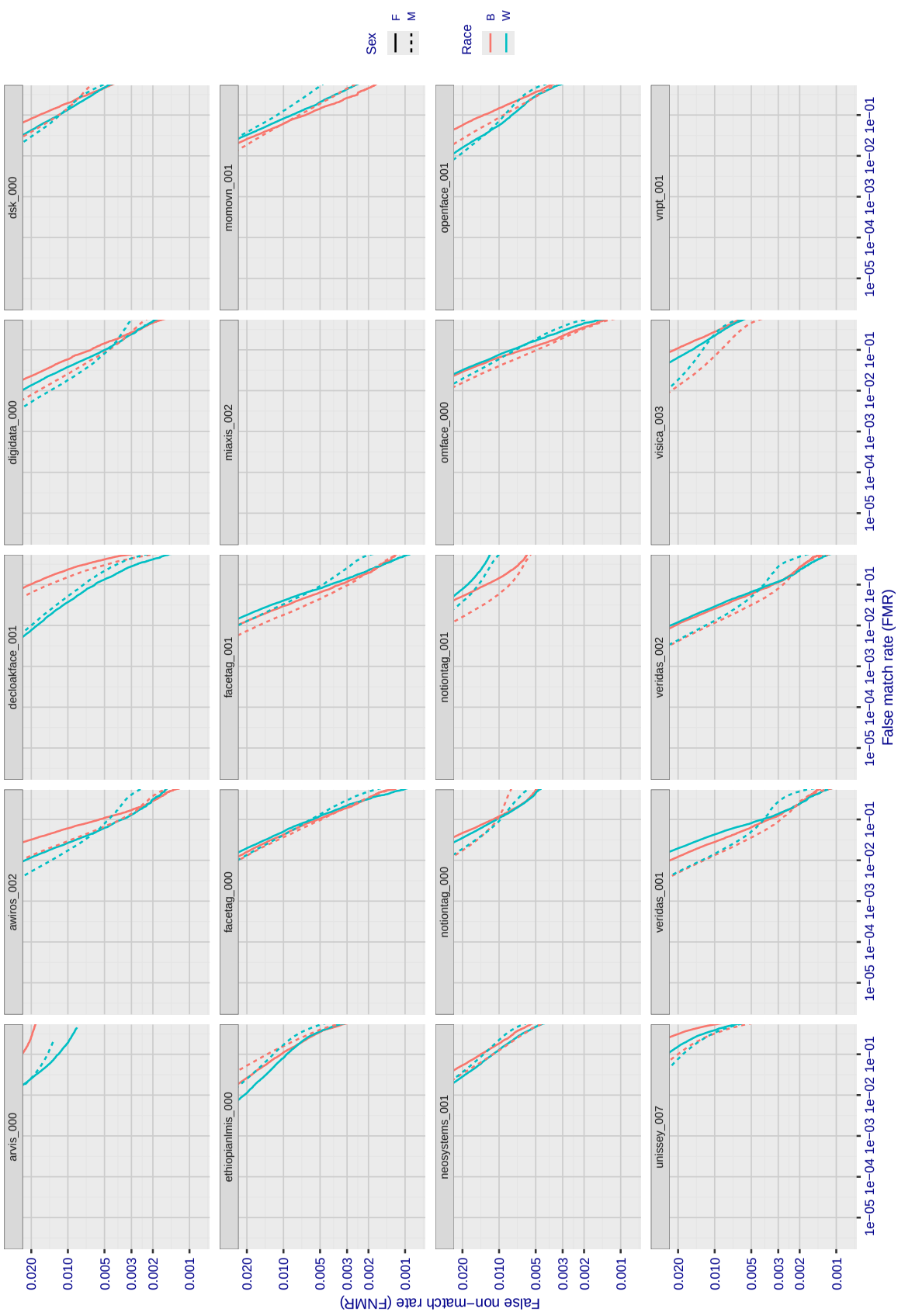


Figure 467: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

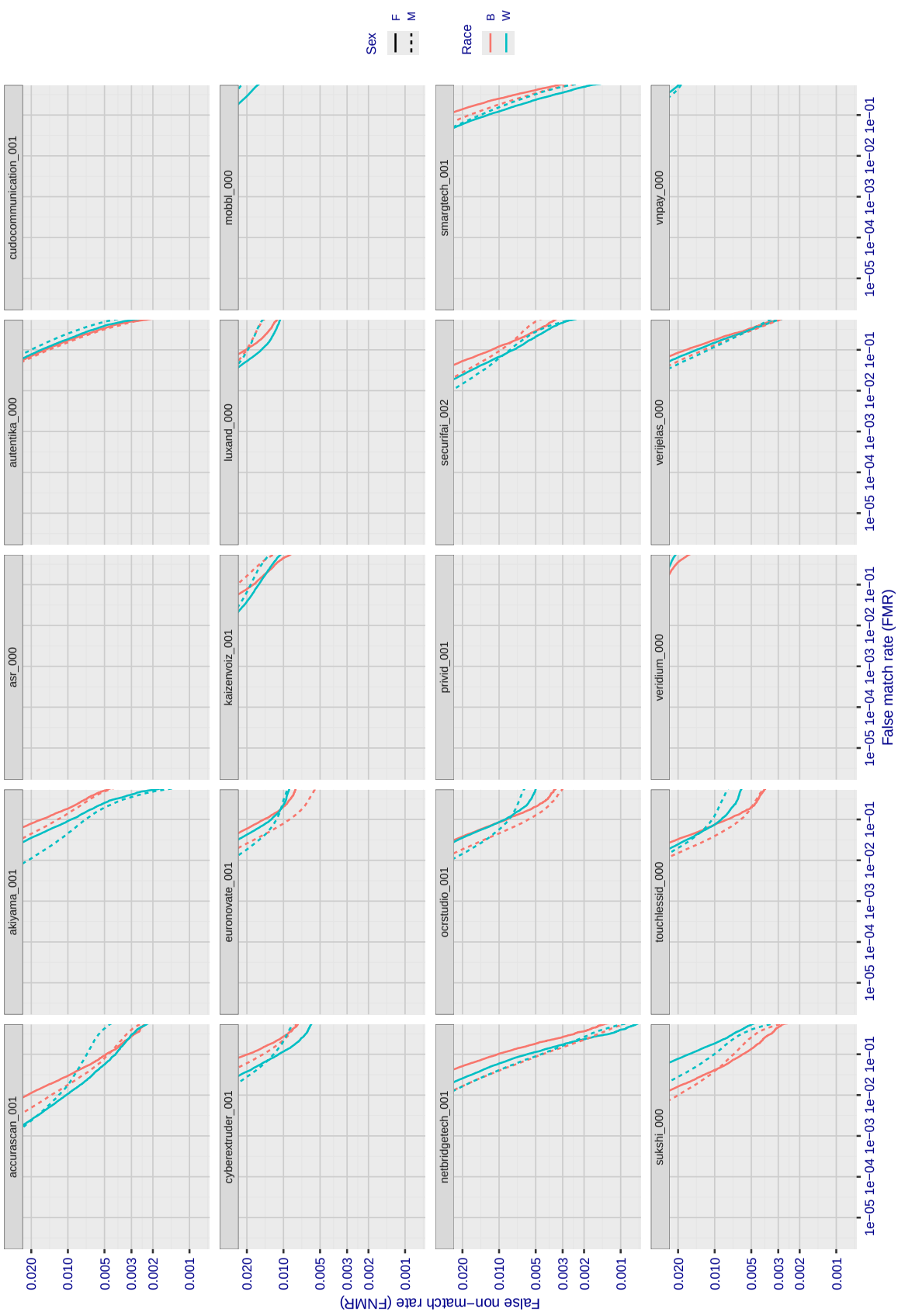


Figure 468: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

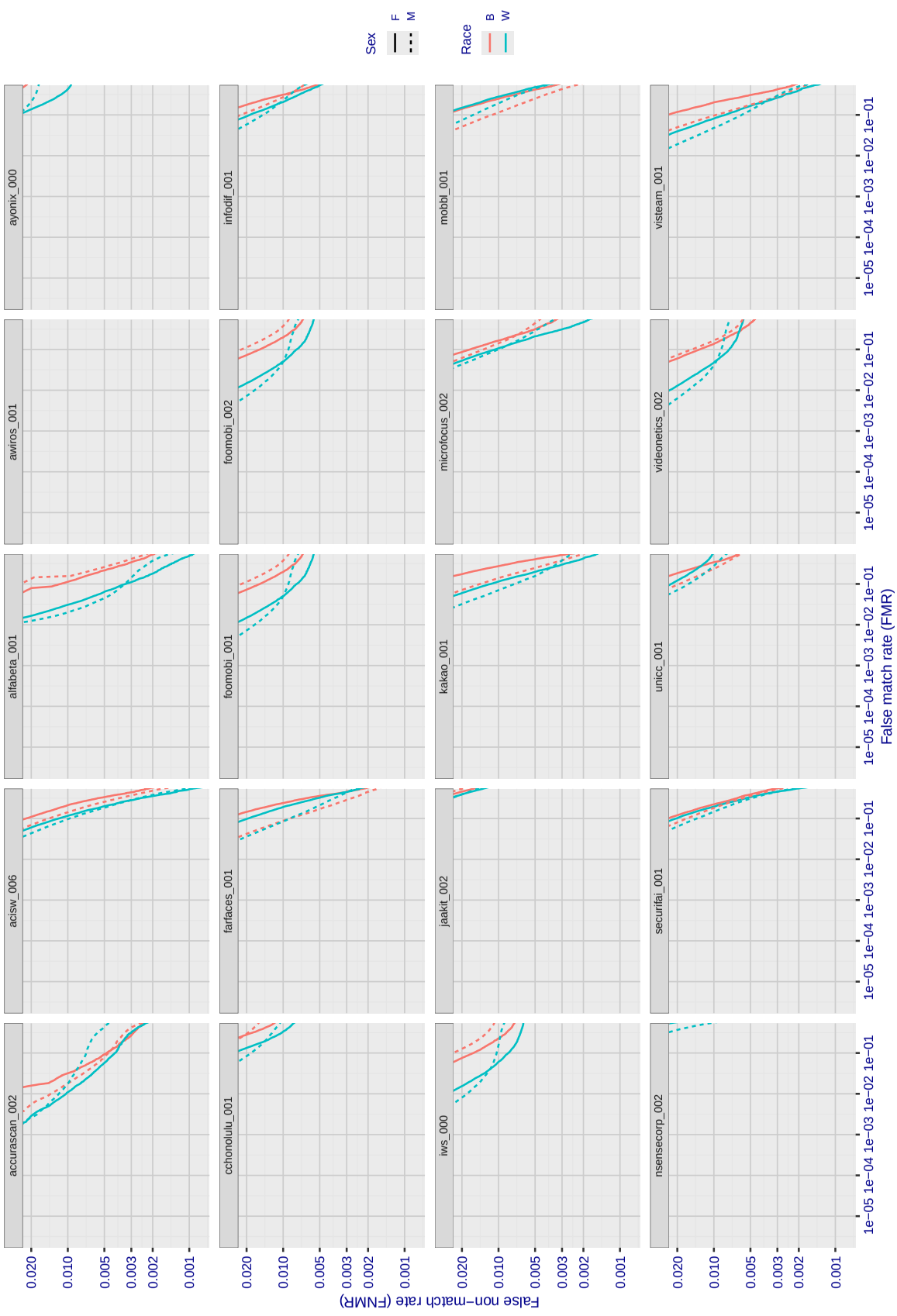


Figure 469: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

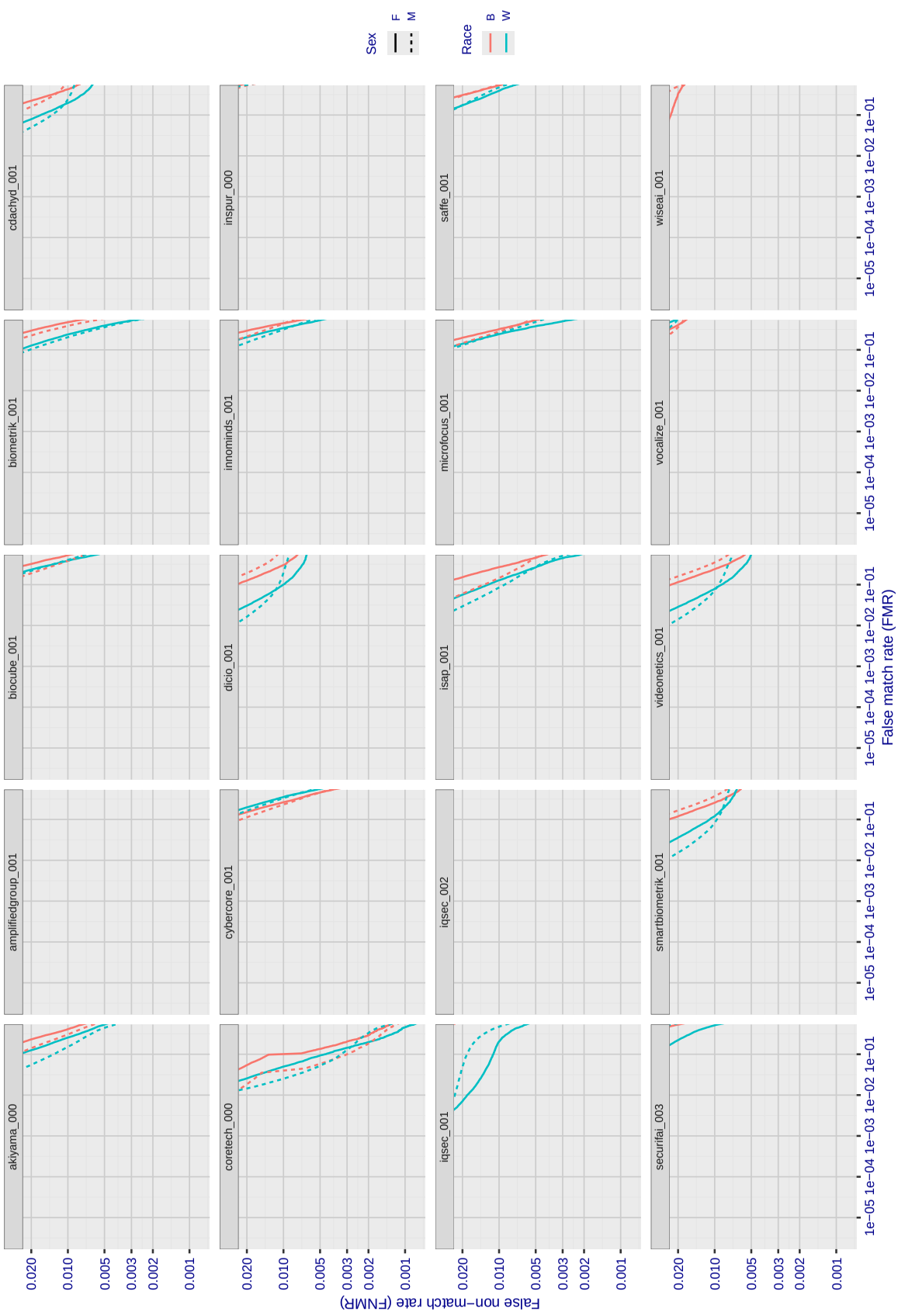


Figure 470: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

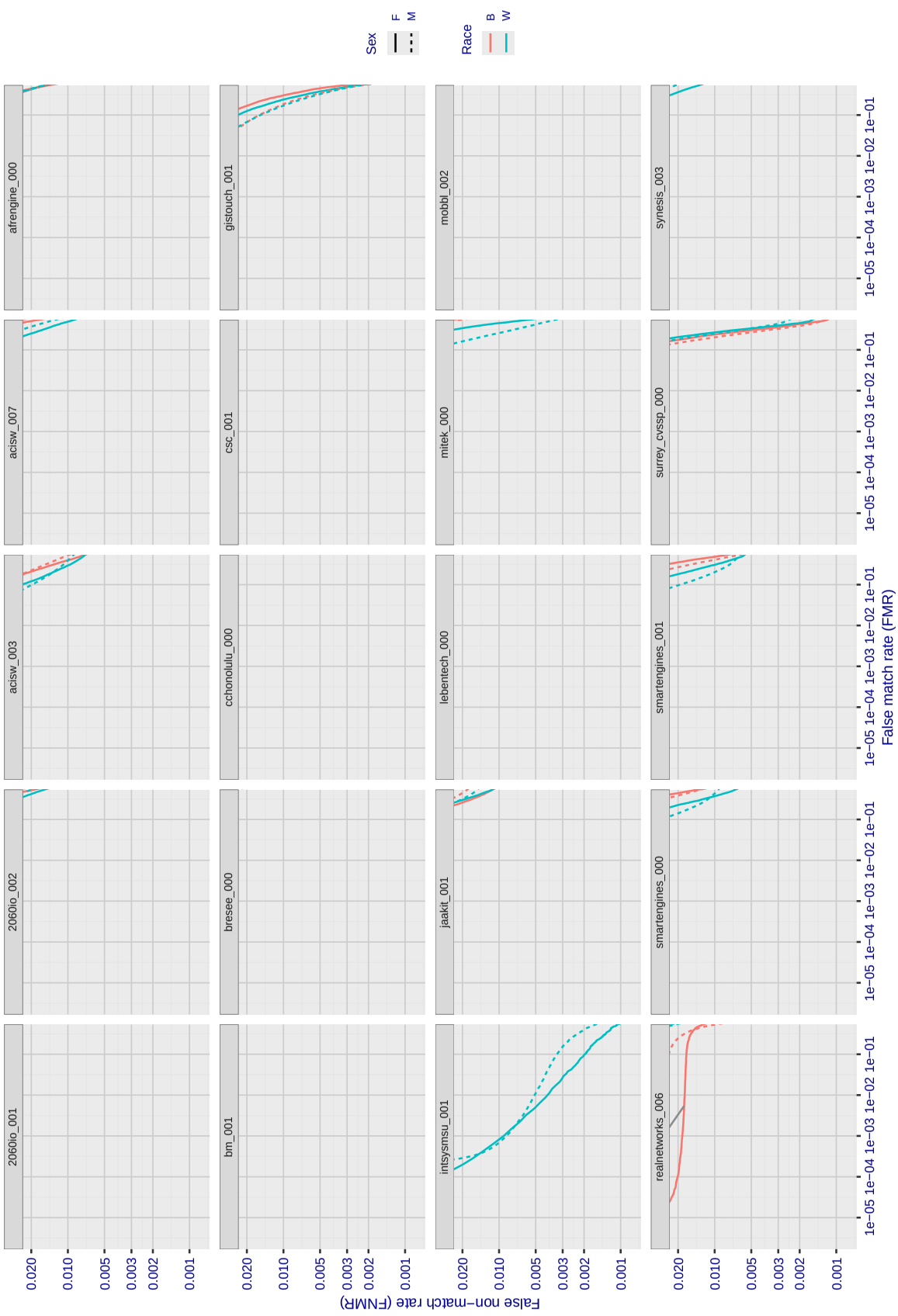


Figure 471: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is higher than for women at a fixed threshold but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

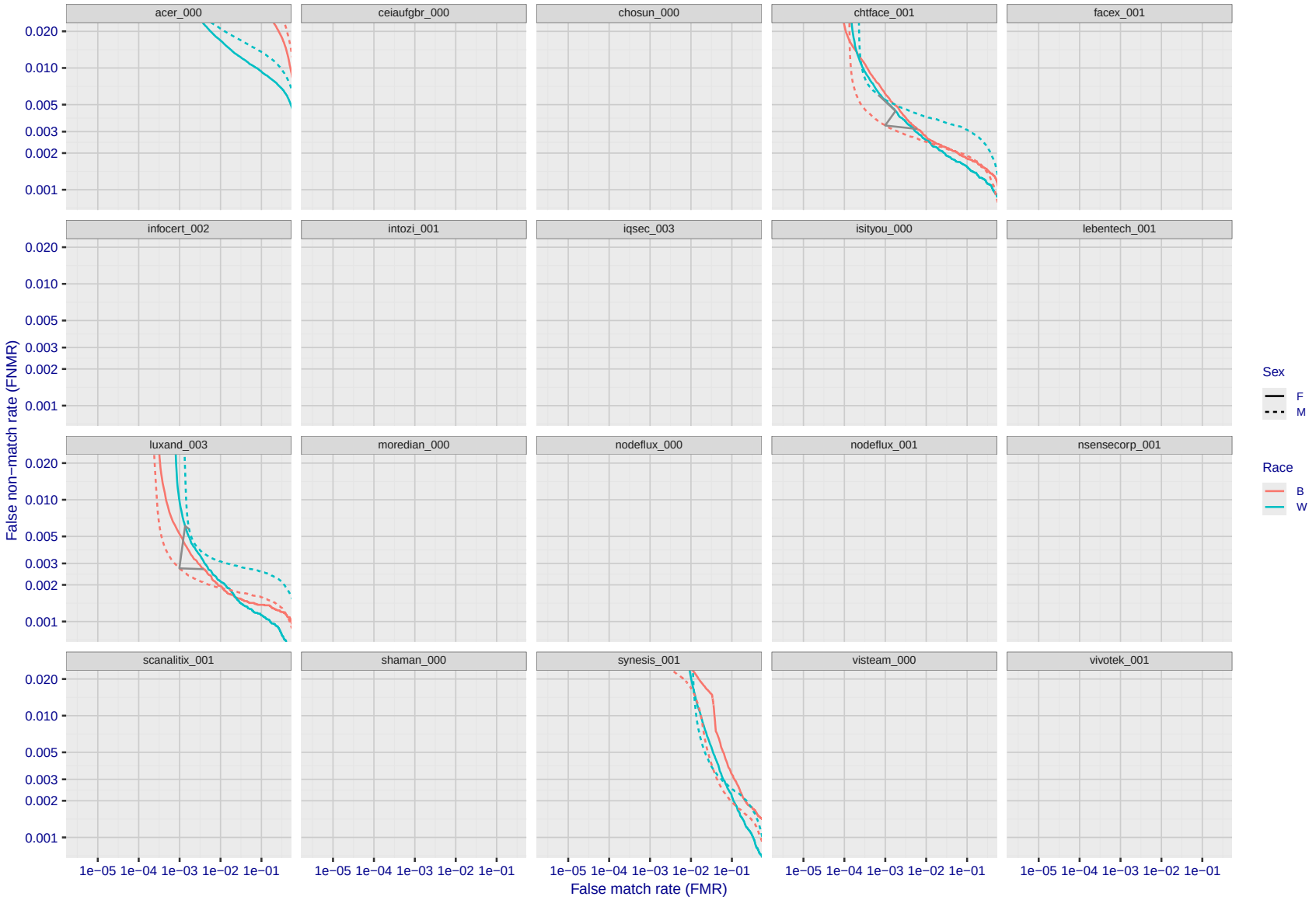


Figure 472: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

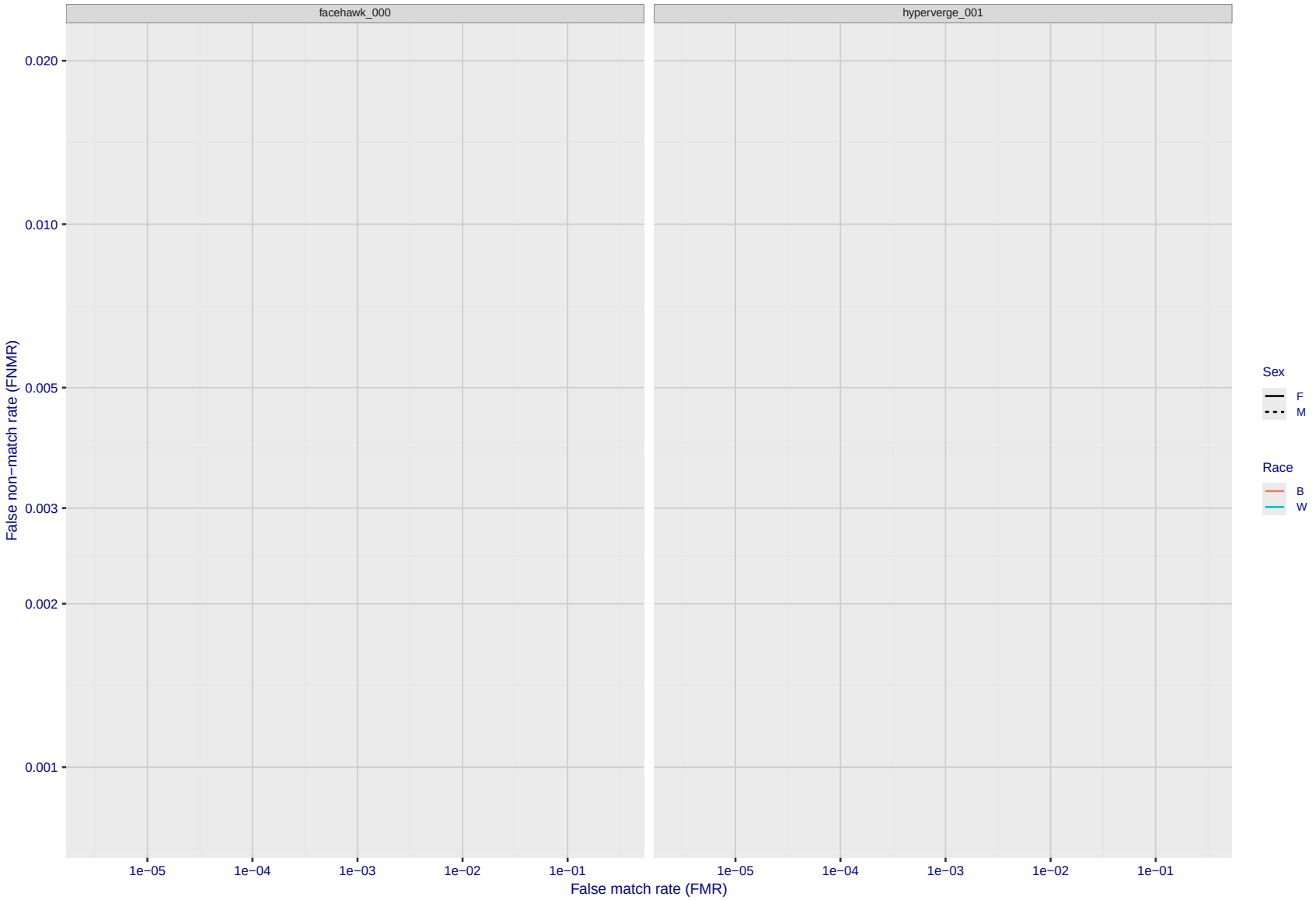


Figure 473: For the mugshot images, error tradeoff characteristics for white females, black females, black males and white males. The Z-shaped grey lines correspond to fixed thresholds, showing both *FNMR and FMR vary at one T value. Note: Many of the plots will naively be read as saying women gives worse error rates than men because the solid traces lie above the dotted ones. However, this is misleading and incomplete: The grey lines show the traces reveal horizontal shifts. Thus for the cogent-003 algorithm FNMR for men is* higher *than for women* at a fixed threshold *but, at the same time, FMR is higher for women - see Figure 634. As access control systems almost always operate at a fixed threshold, the naive interpretation is incorrect.*

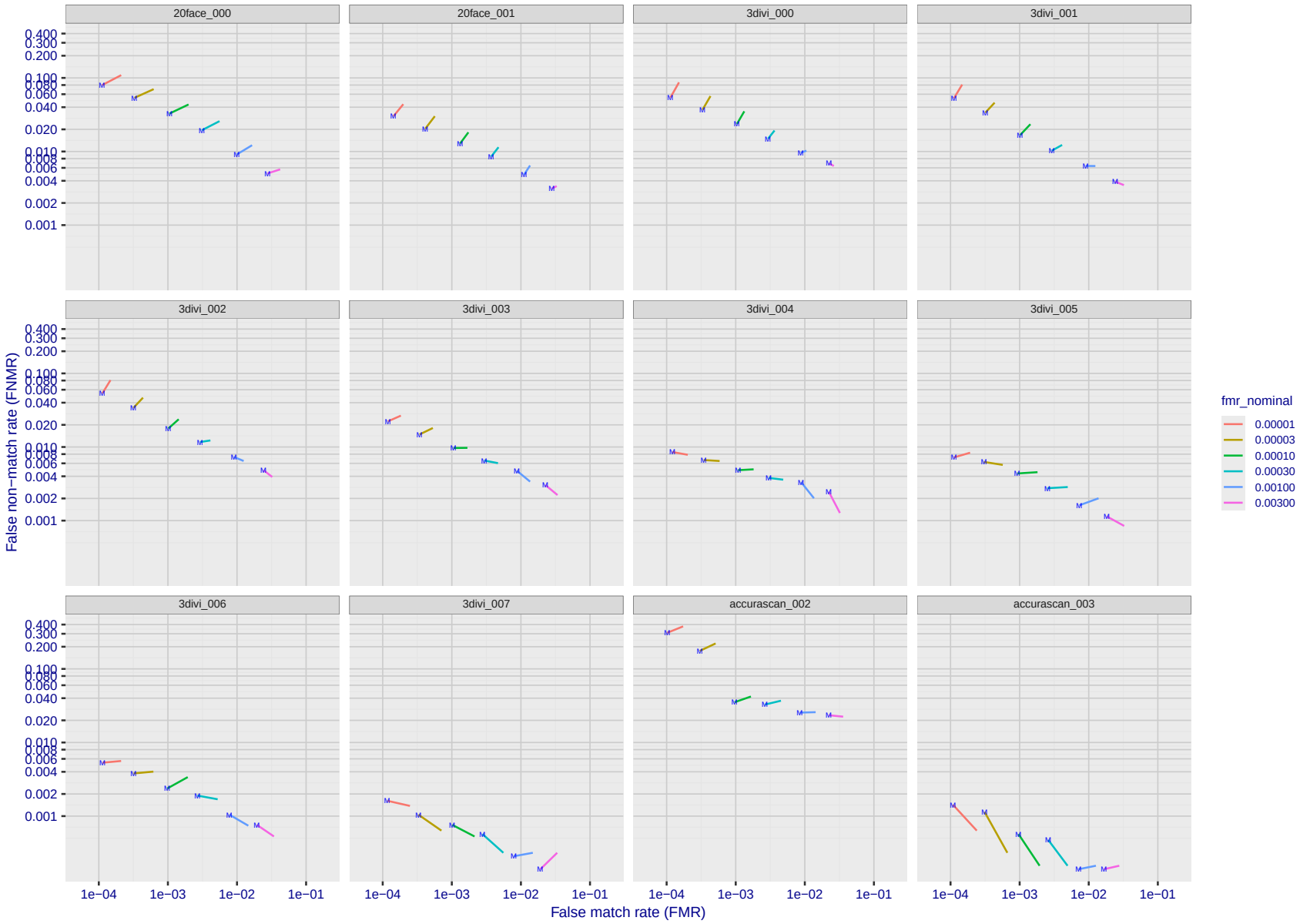


Figure 474: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

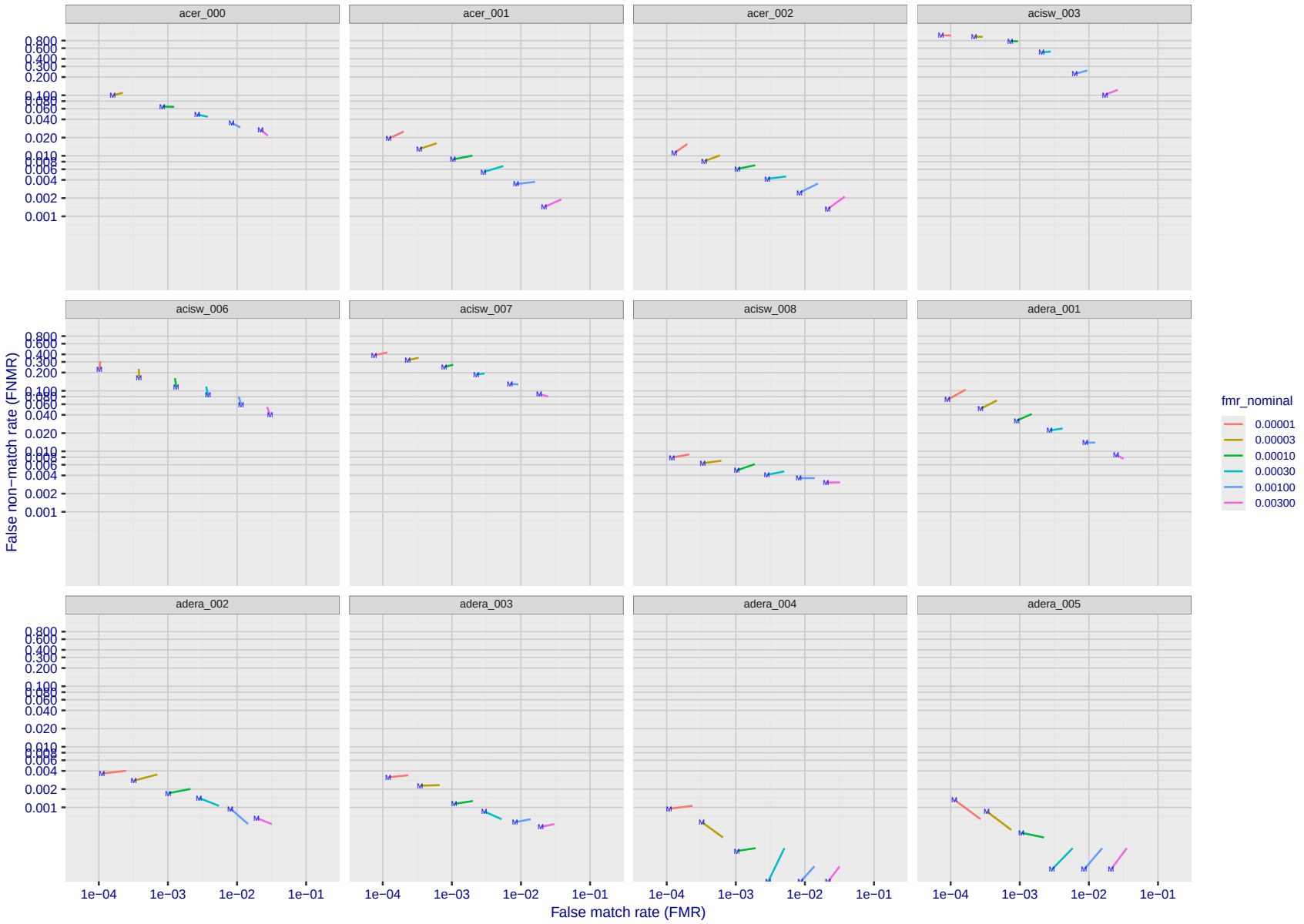


Figure 475: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

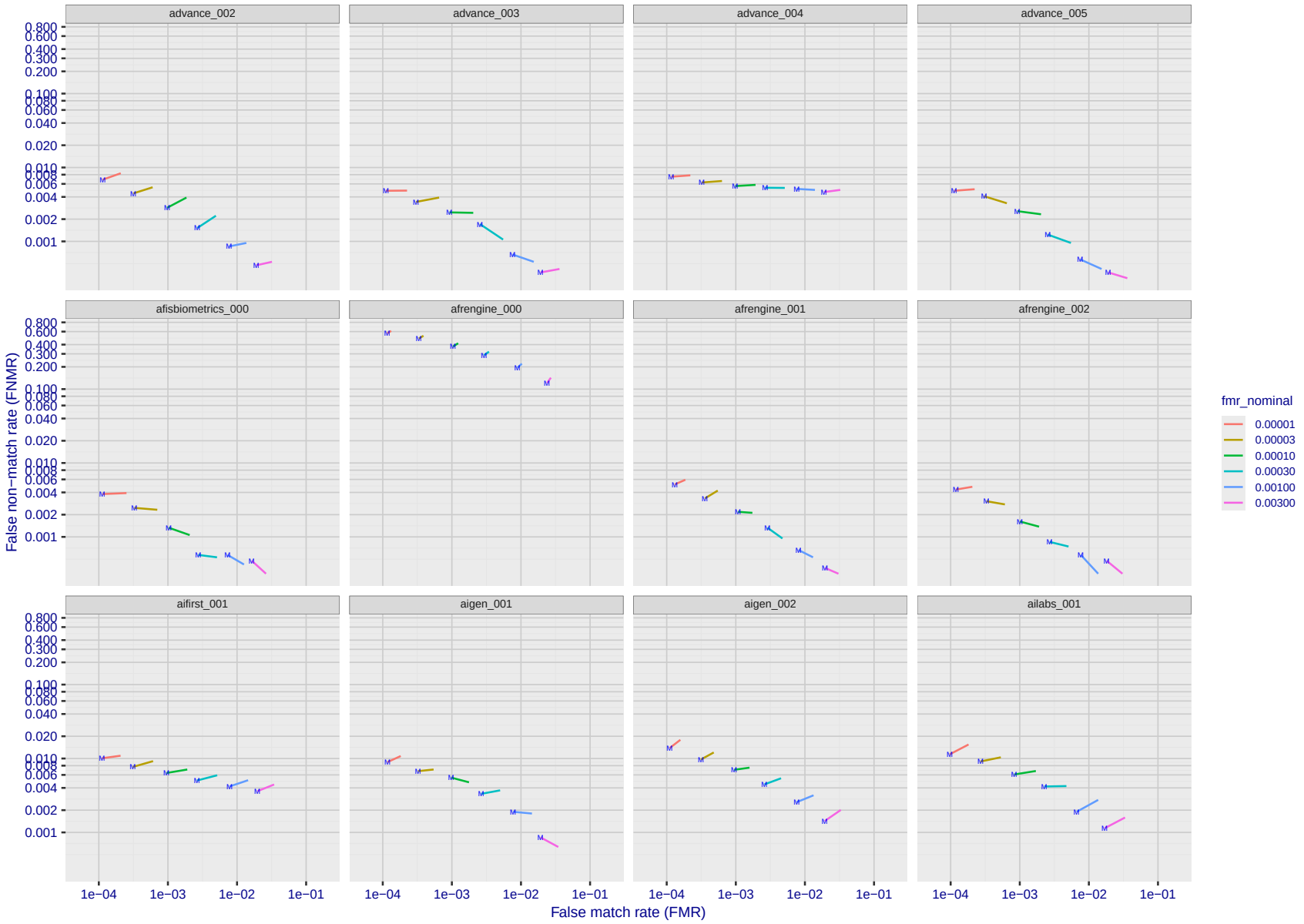


Figure 476: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

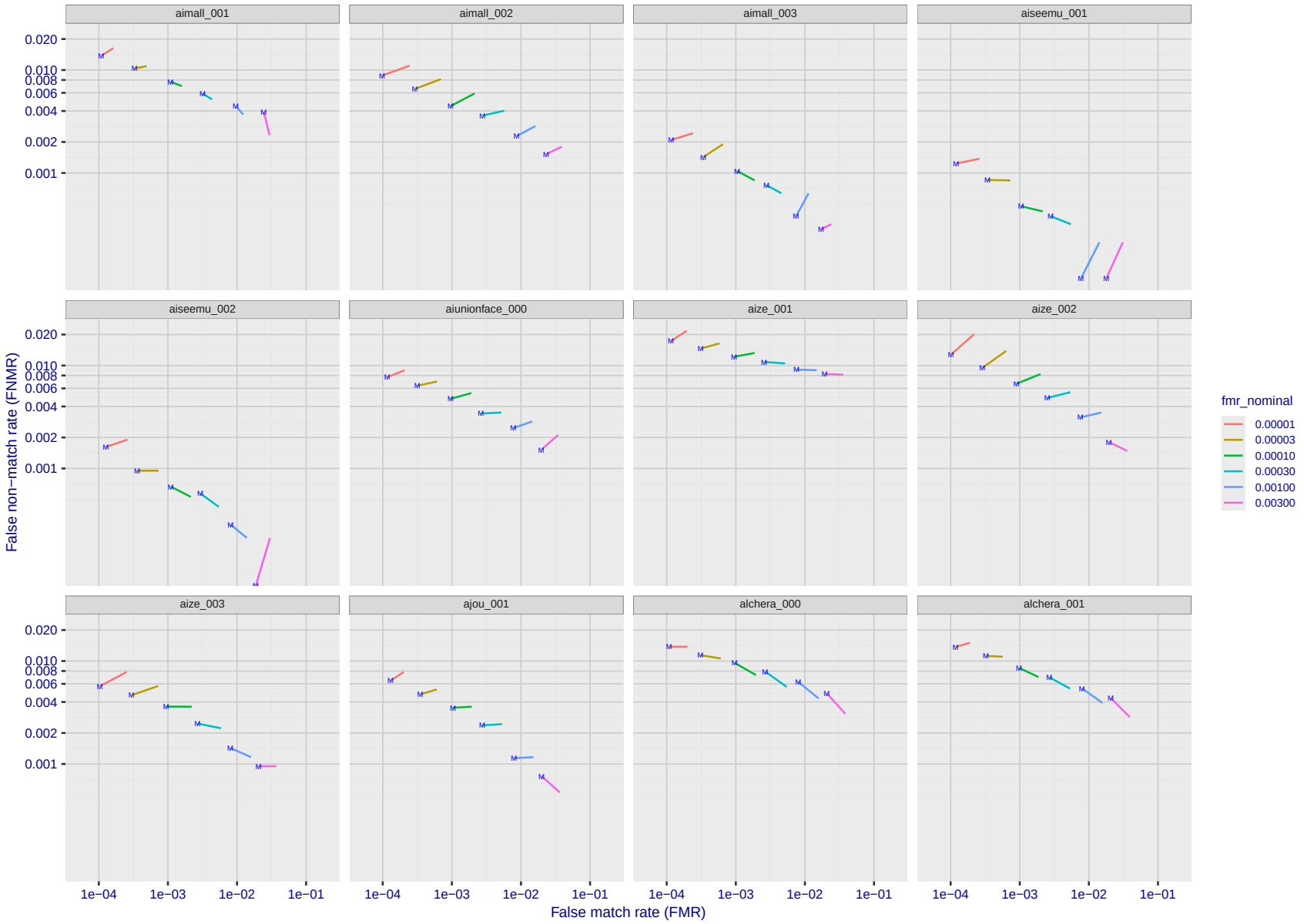


Figure 477: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

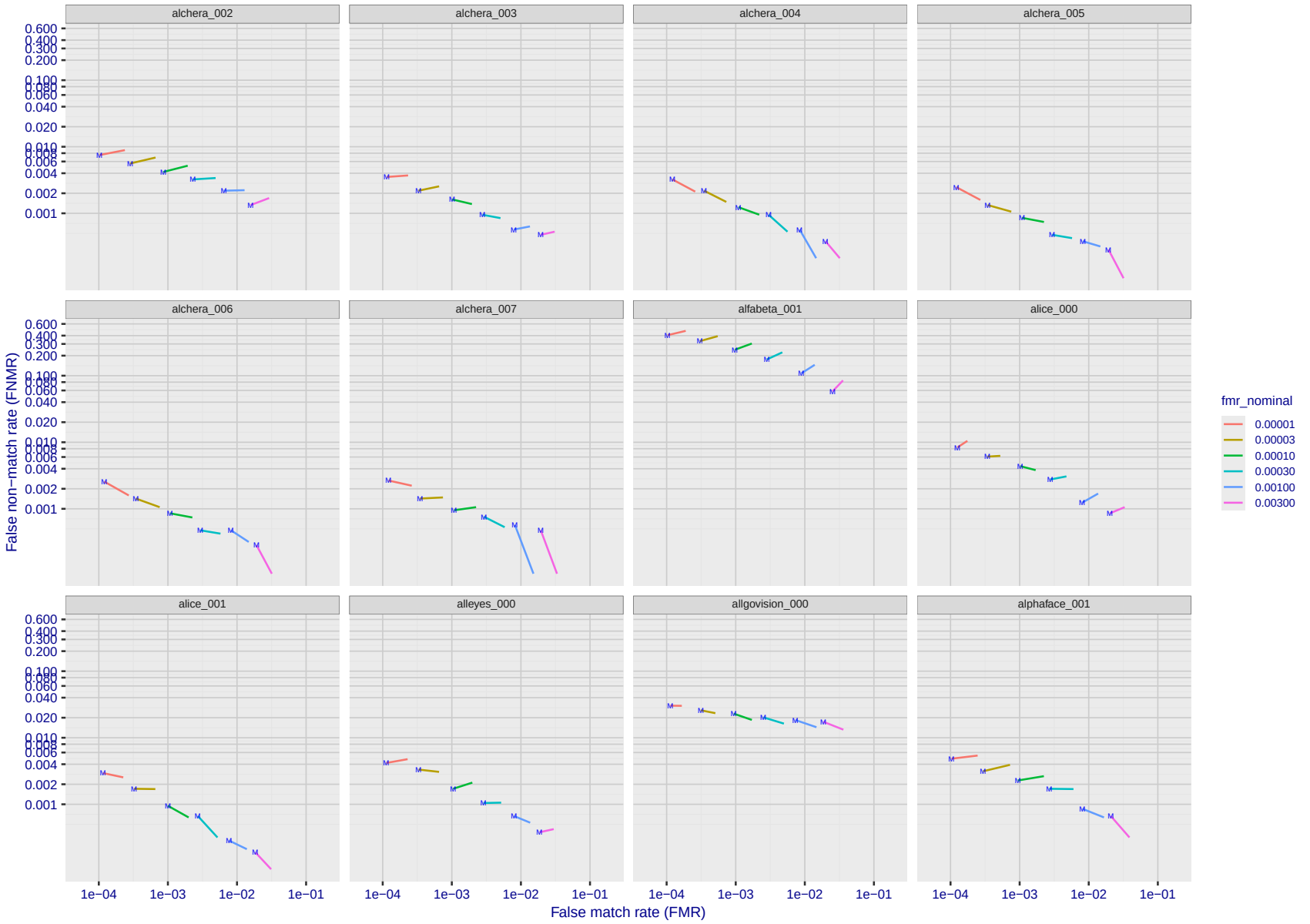


Figure 478: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

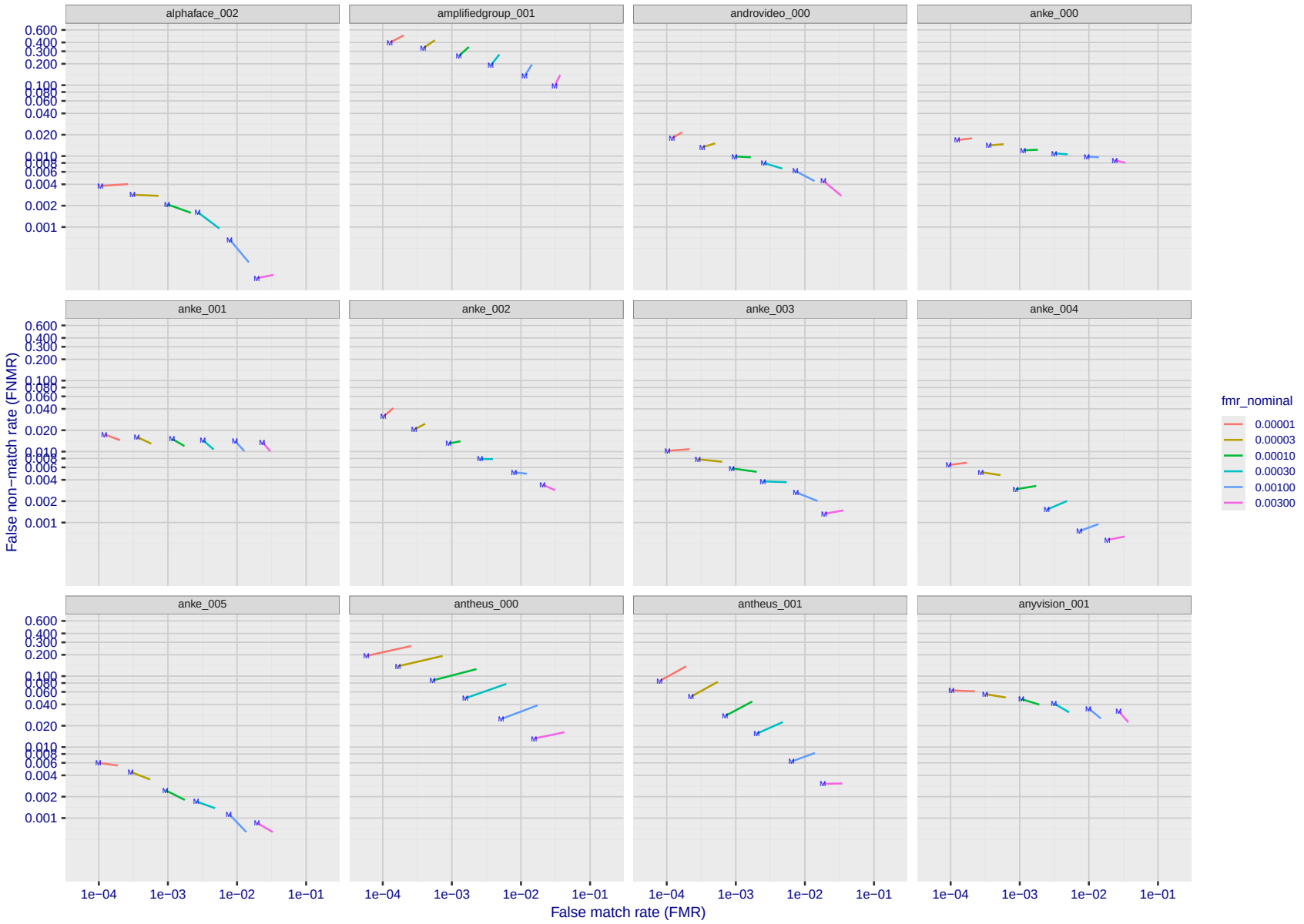


Figure 479: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

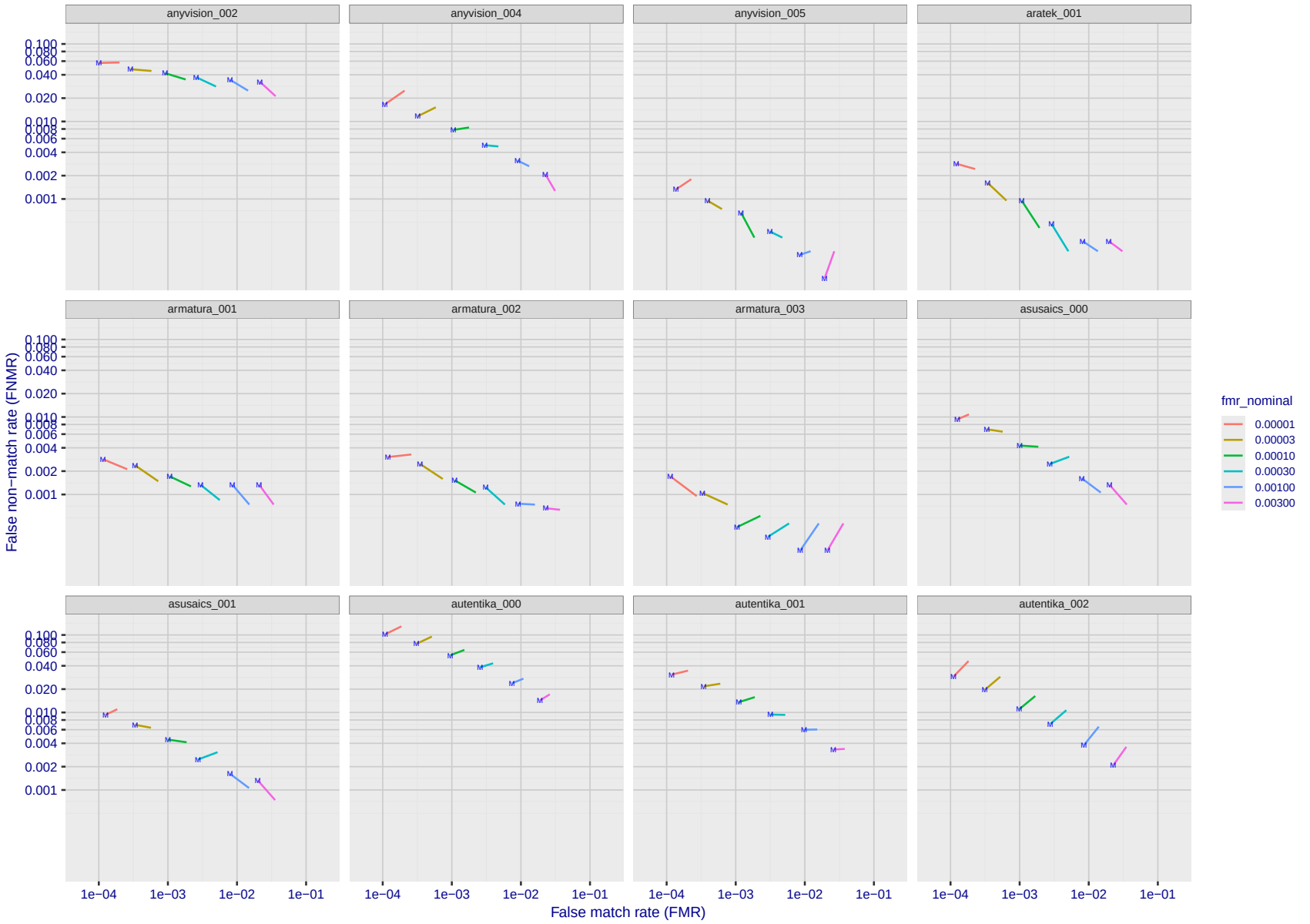


Figure 480: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

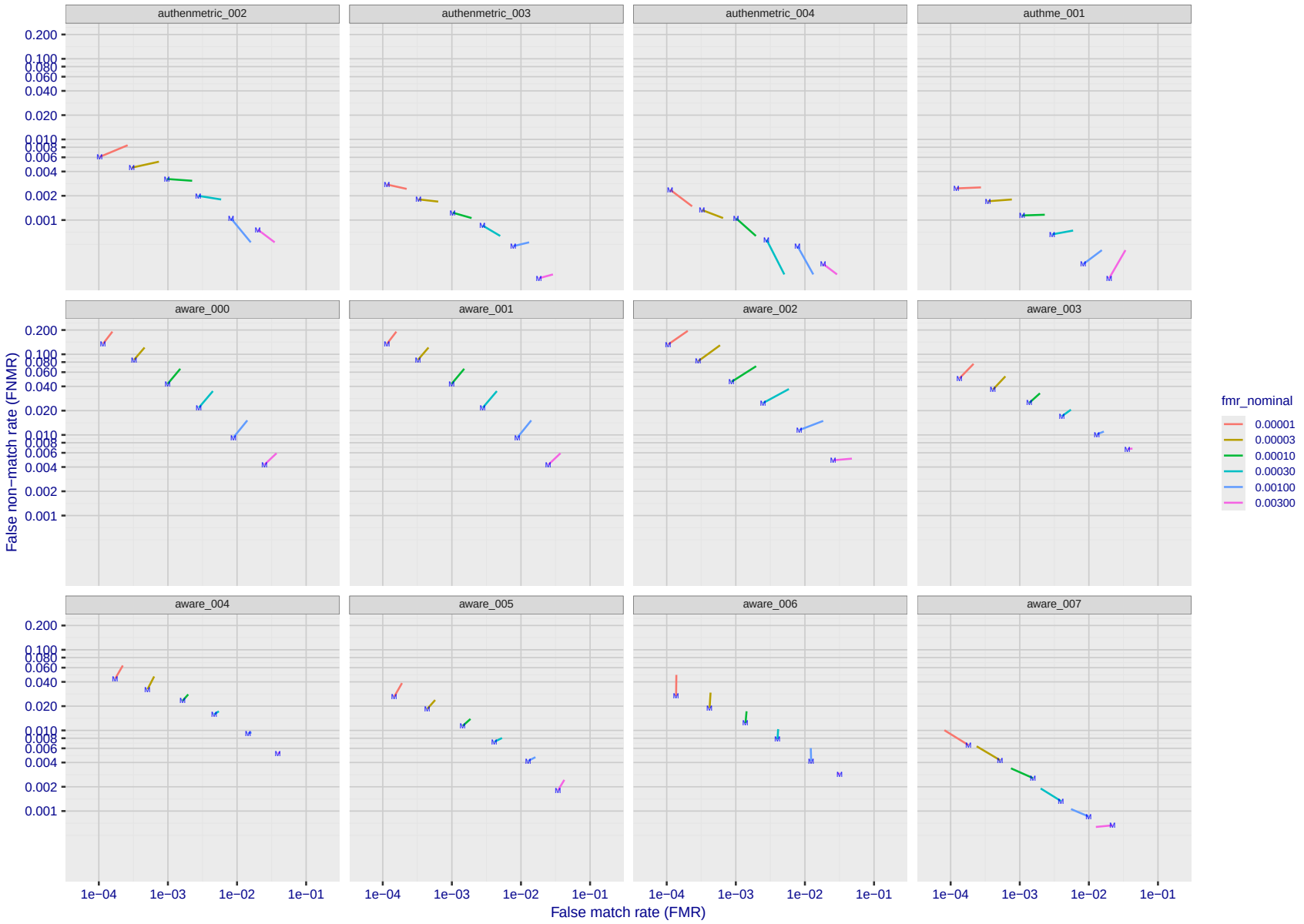


Figure 481: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

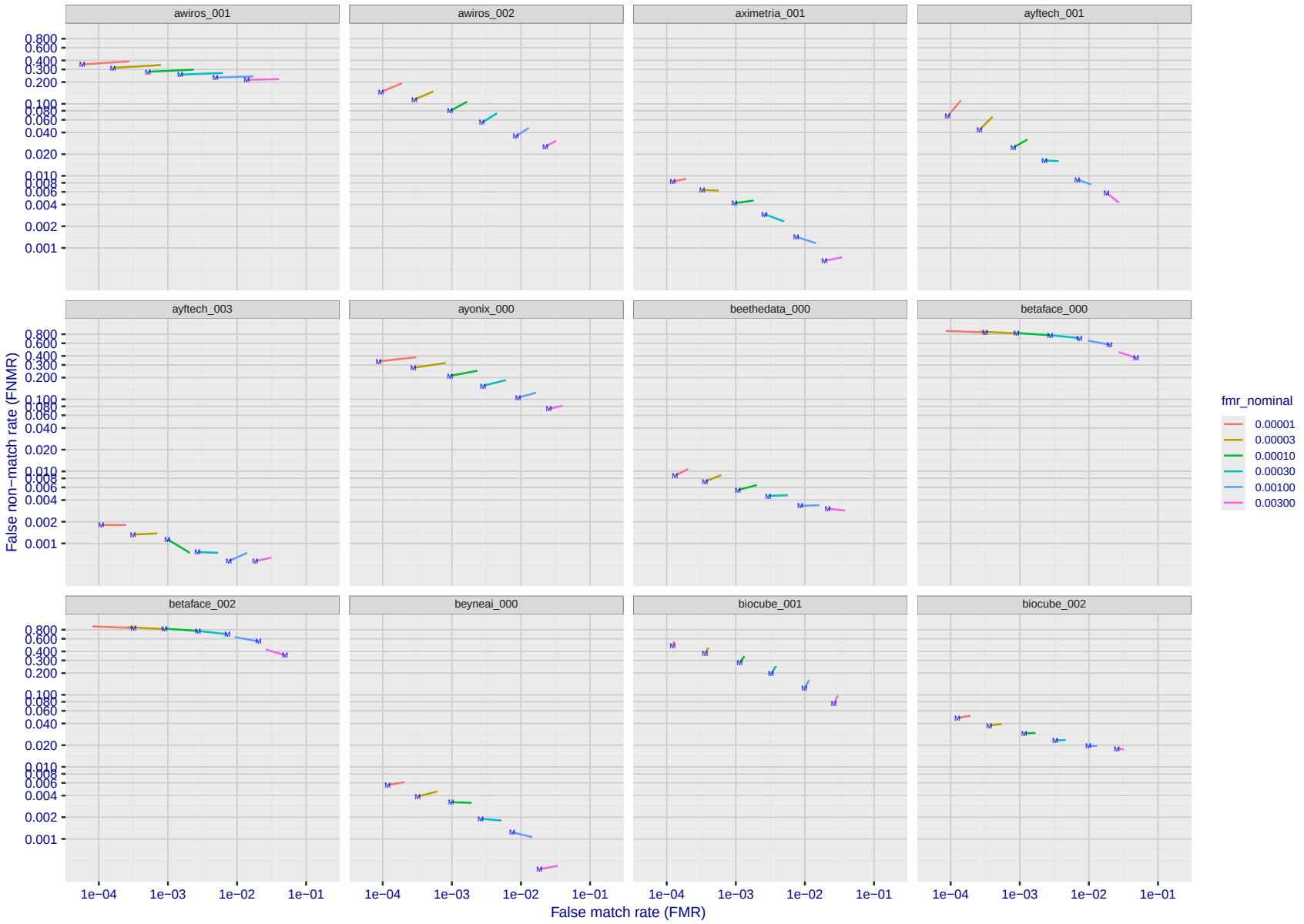


Figure 482: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

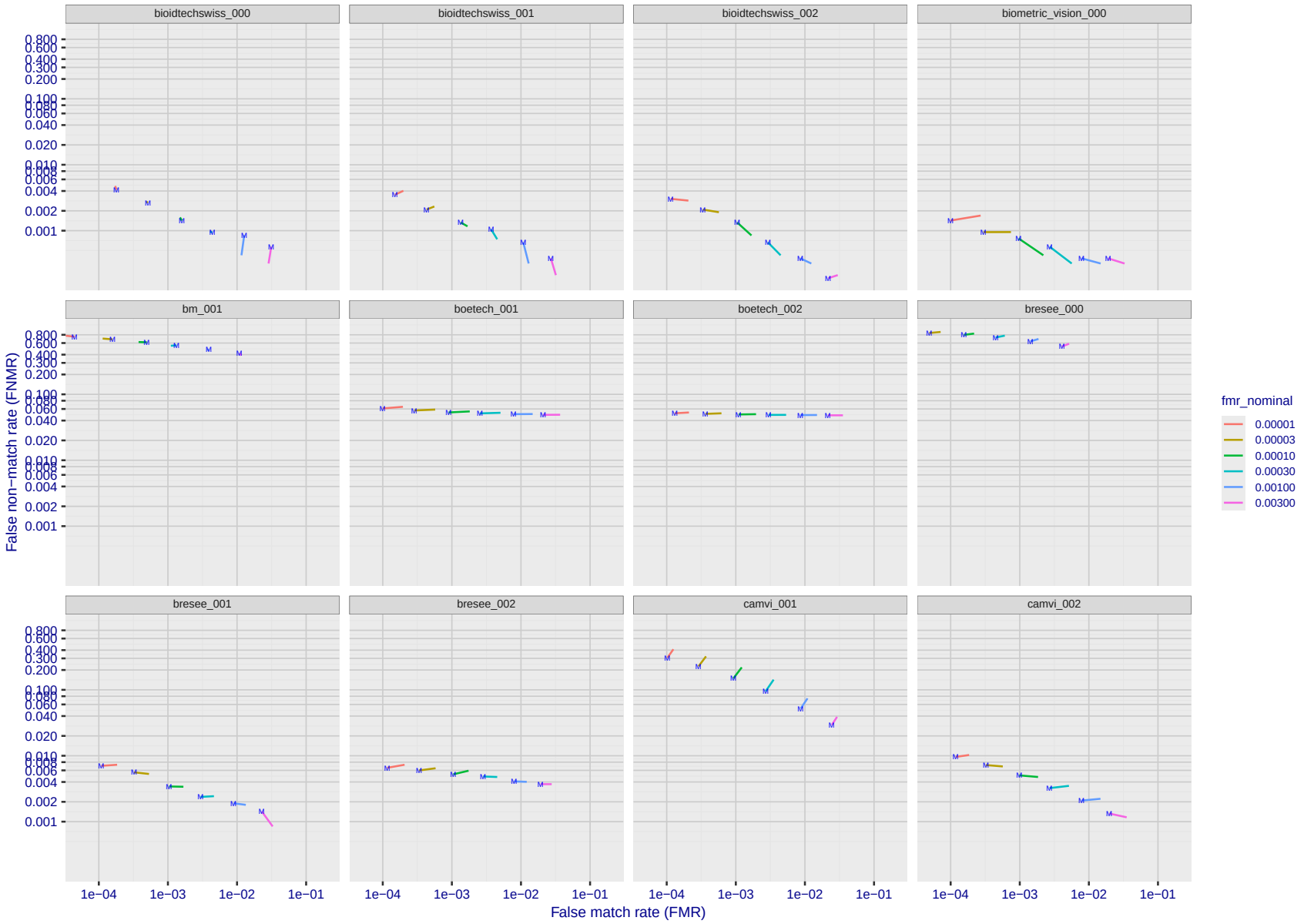


Figure 483: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

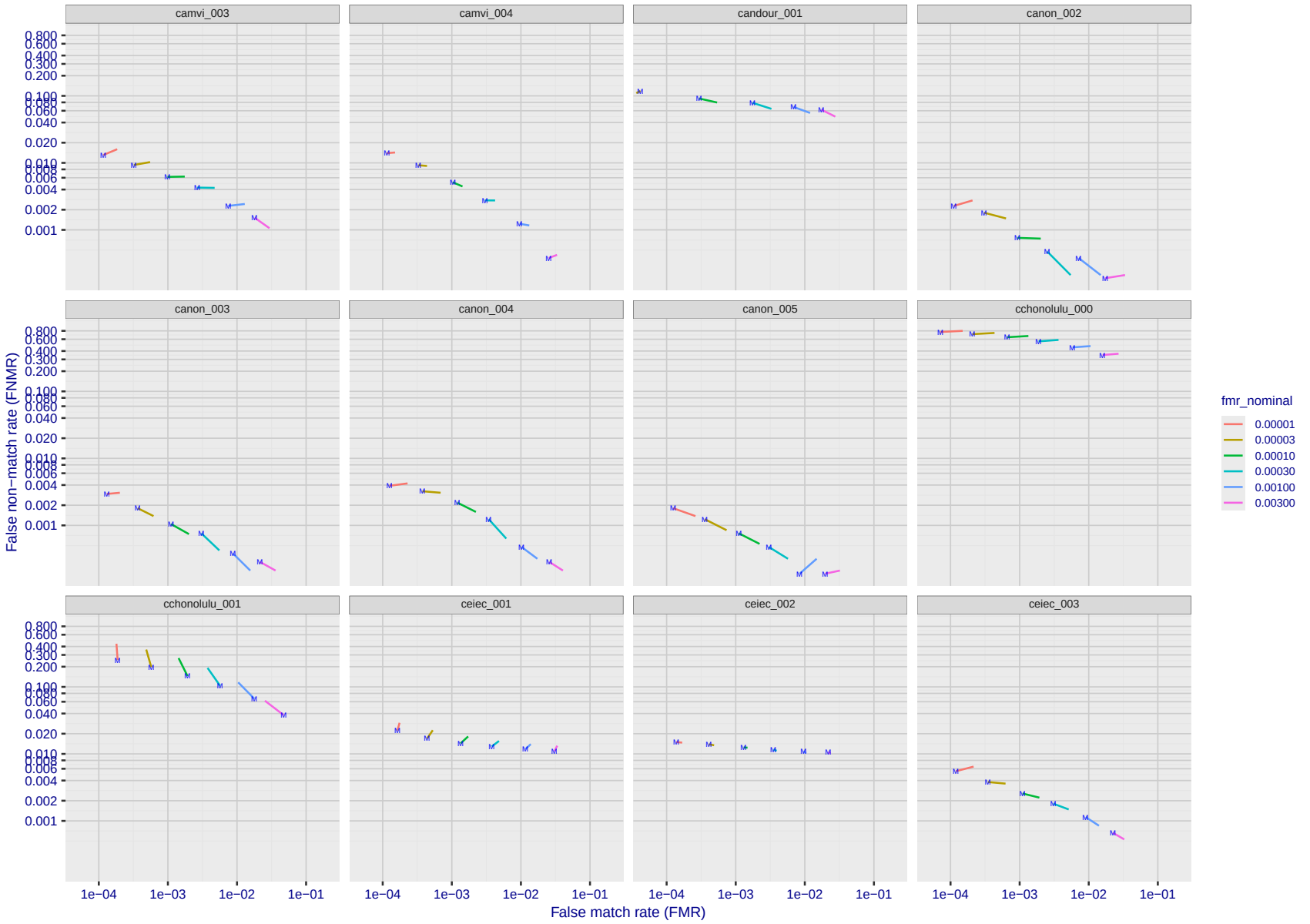


Figure 484: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

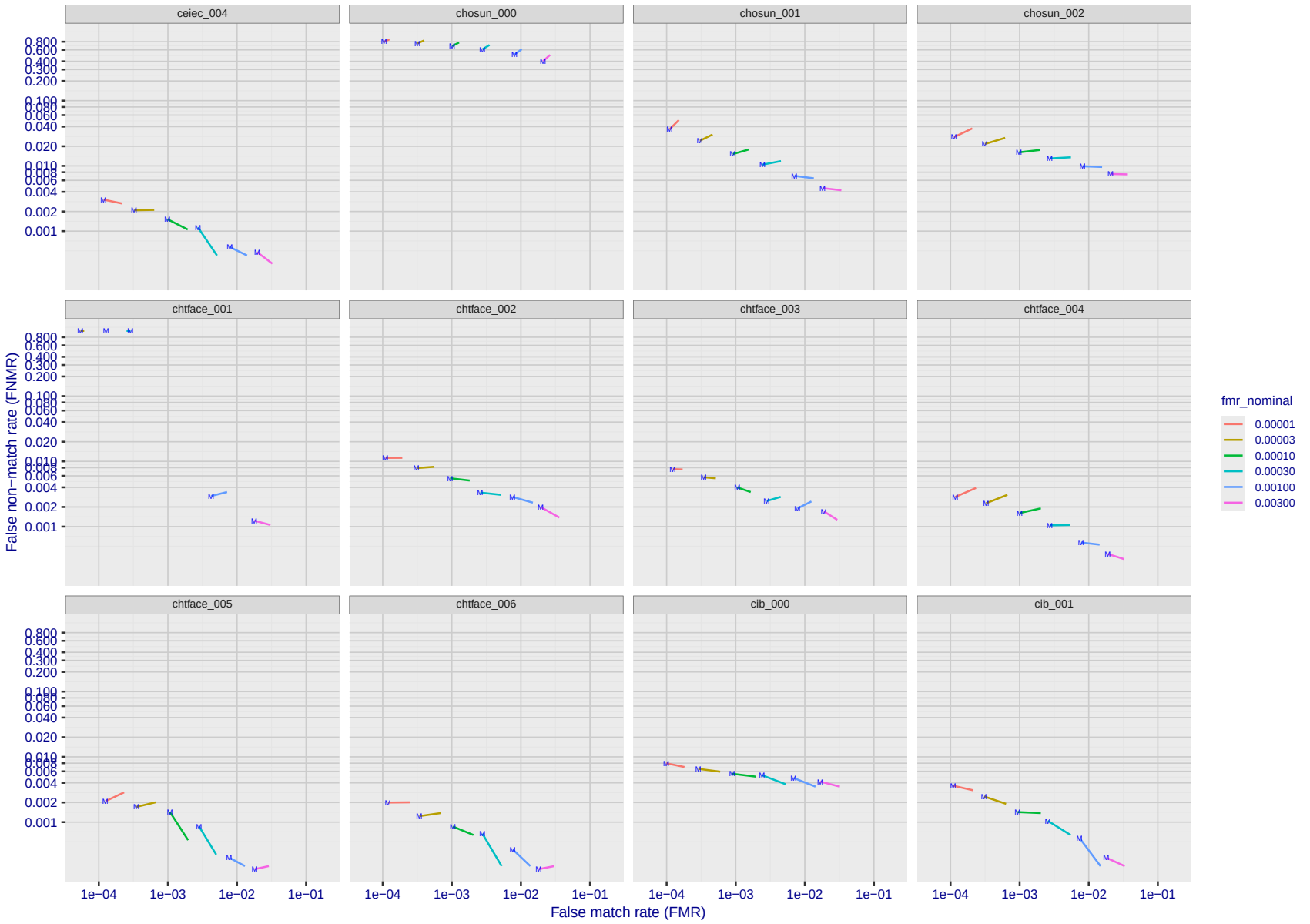


Figure 485: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

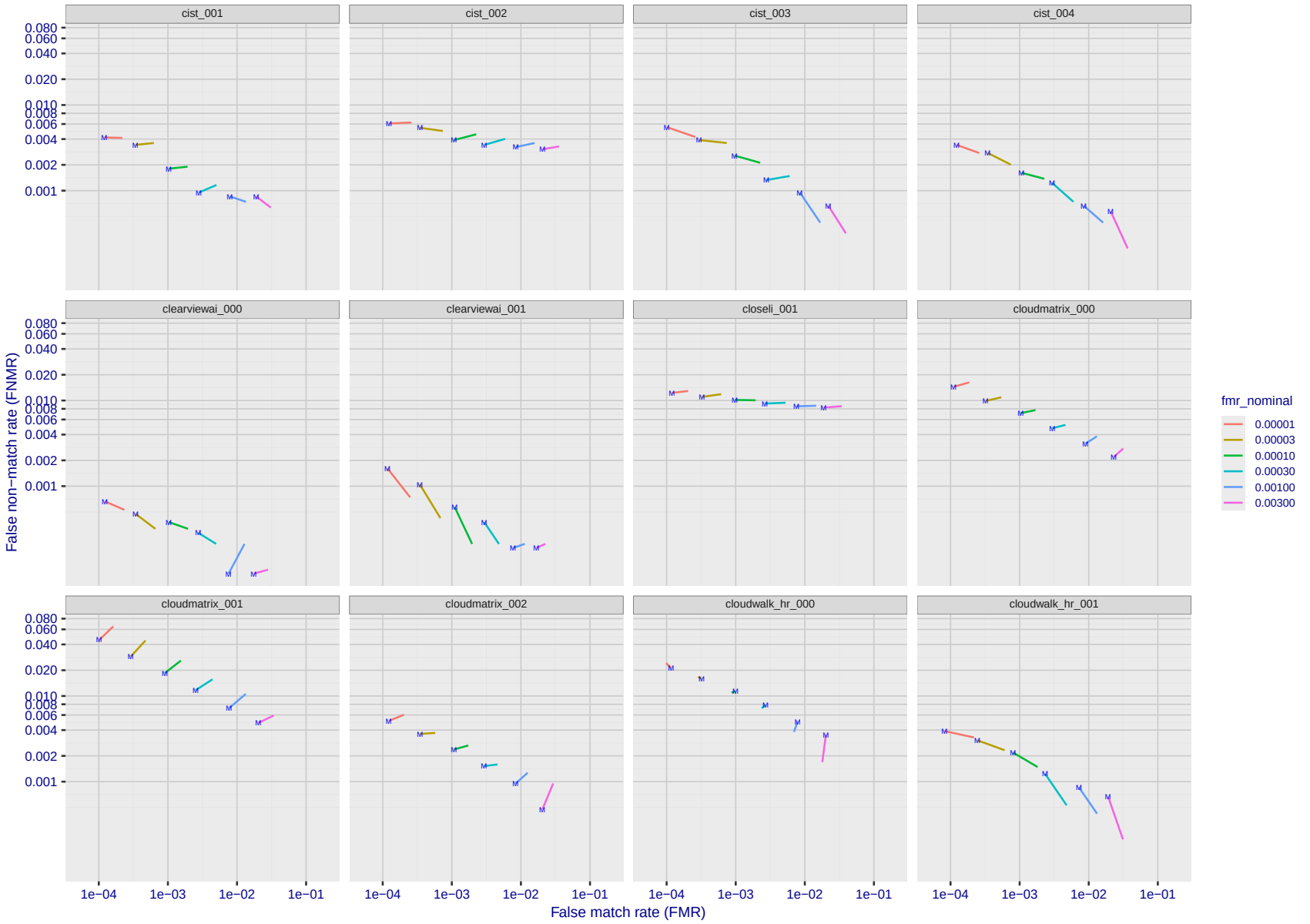


Figure 486: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

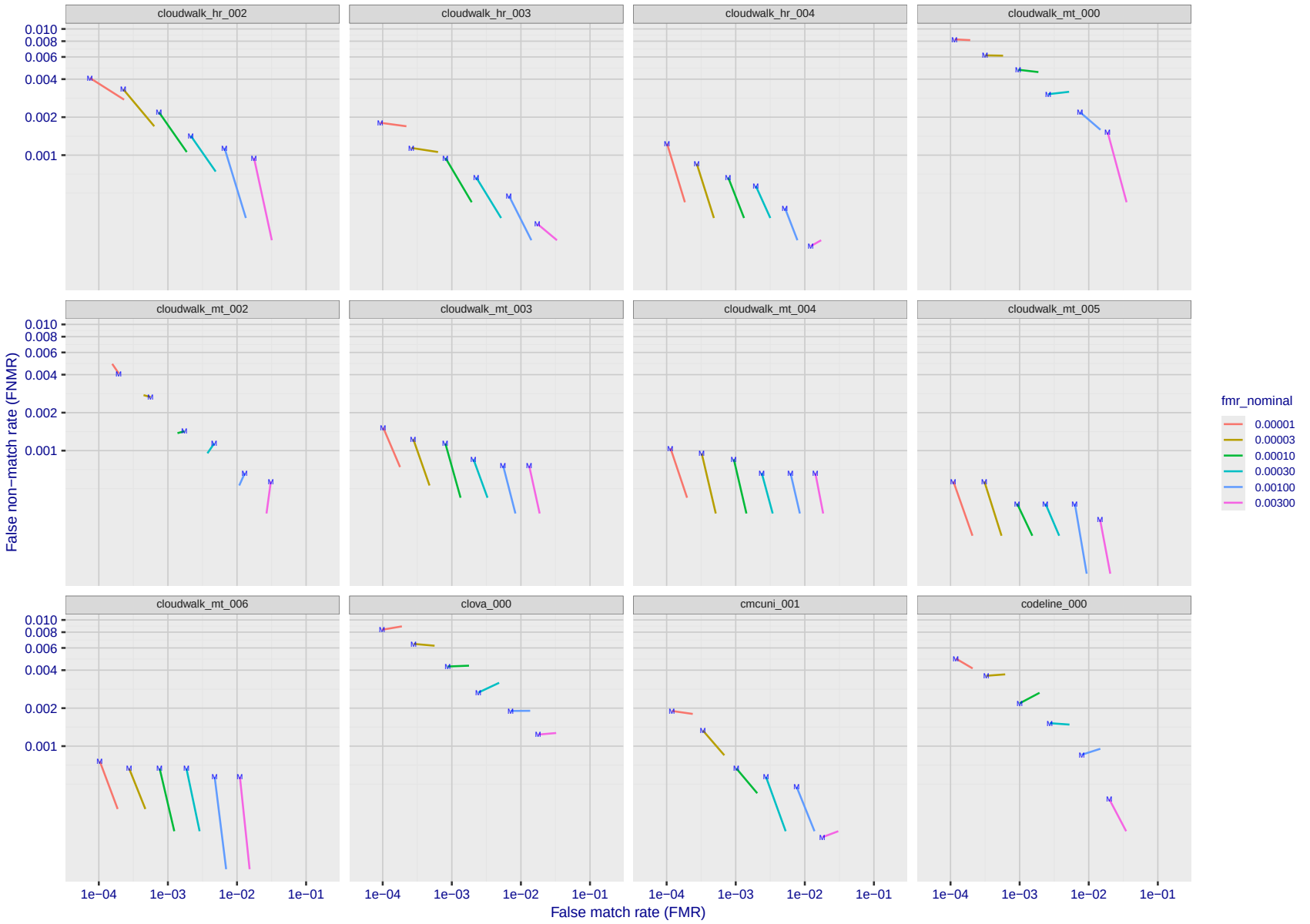


Figure 487: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

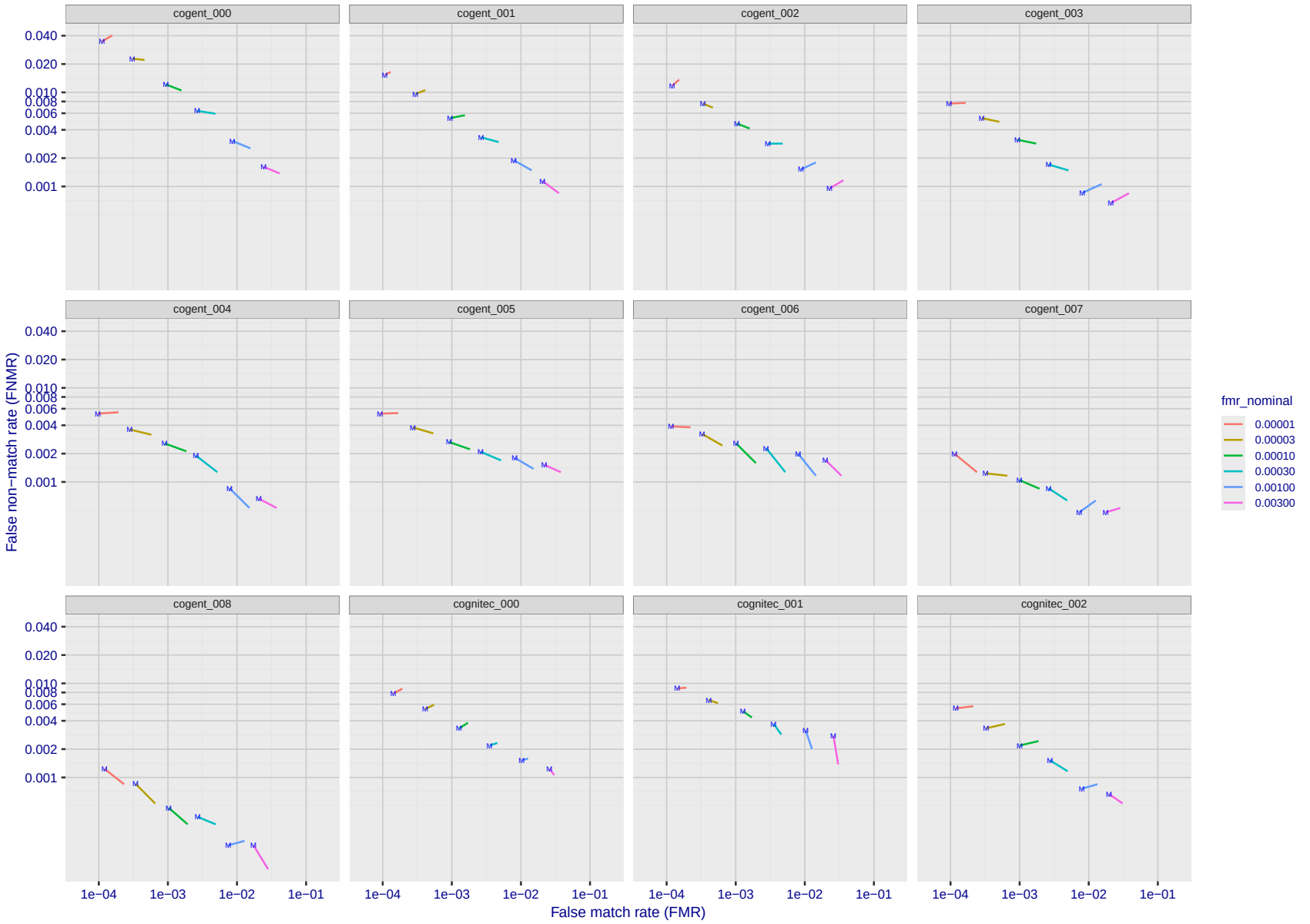


Figure 488: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

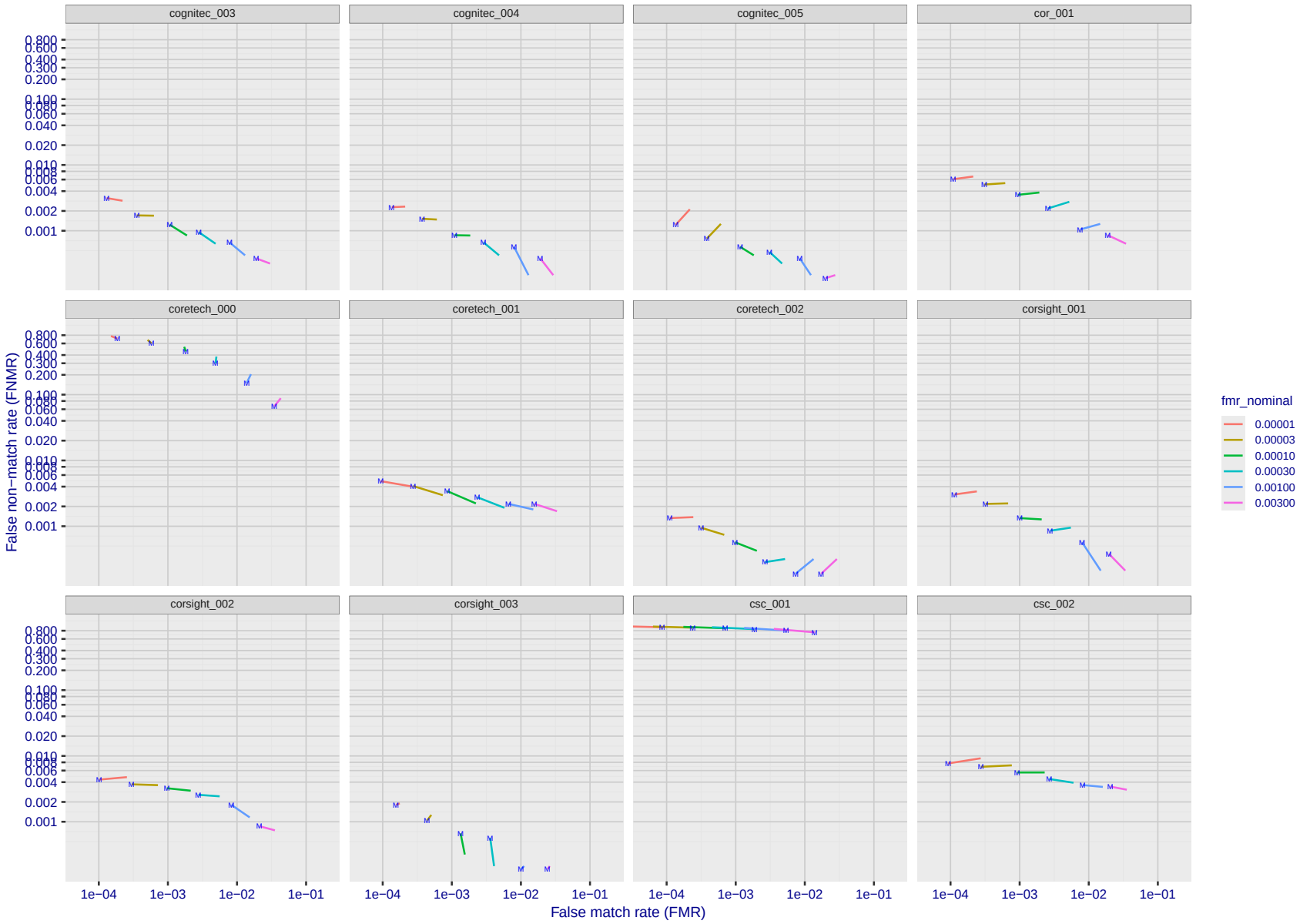


Figure 489: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

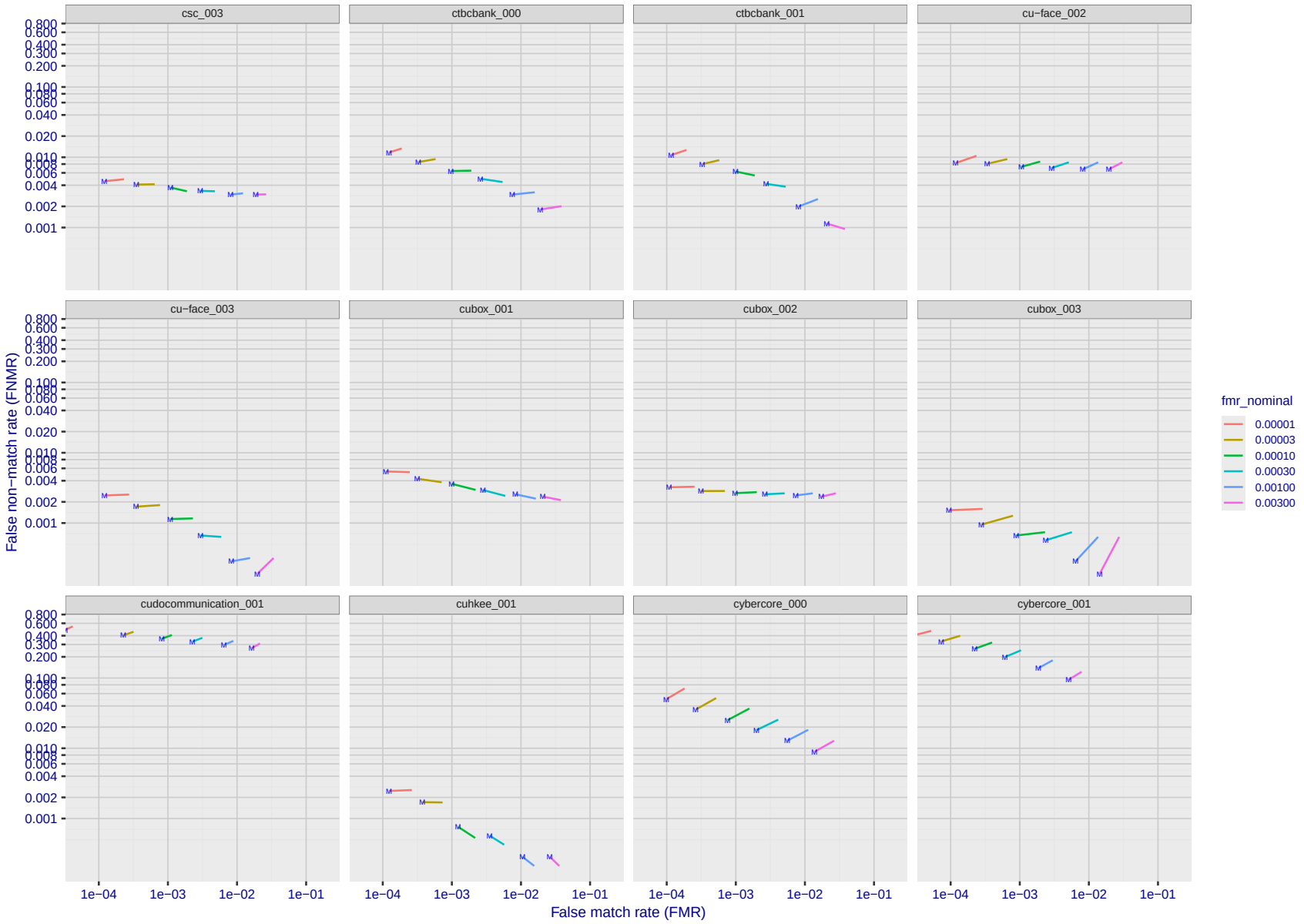


Figure 490: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

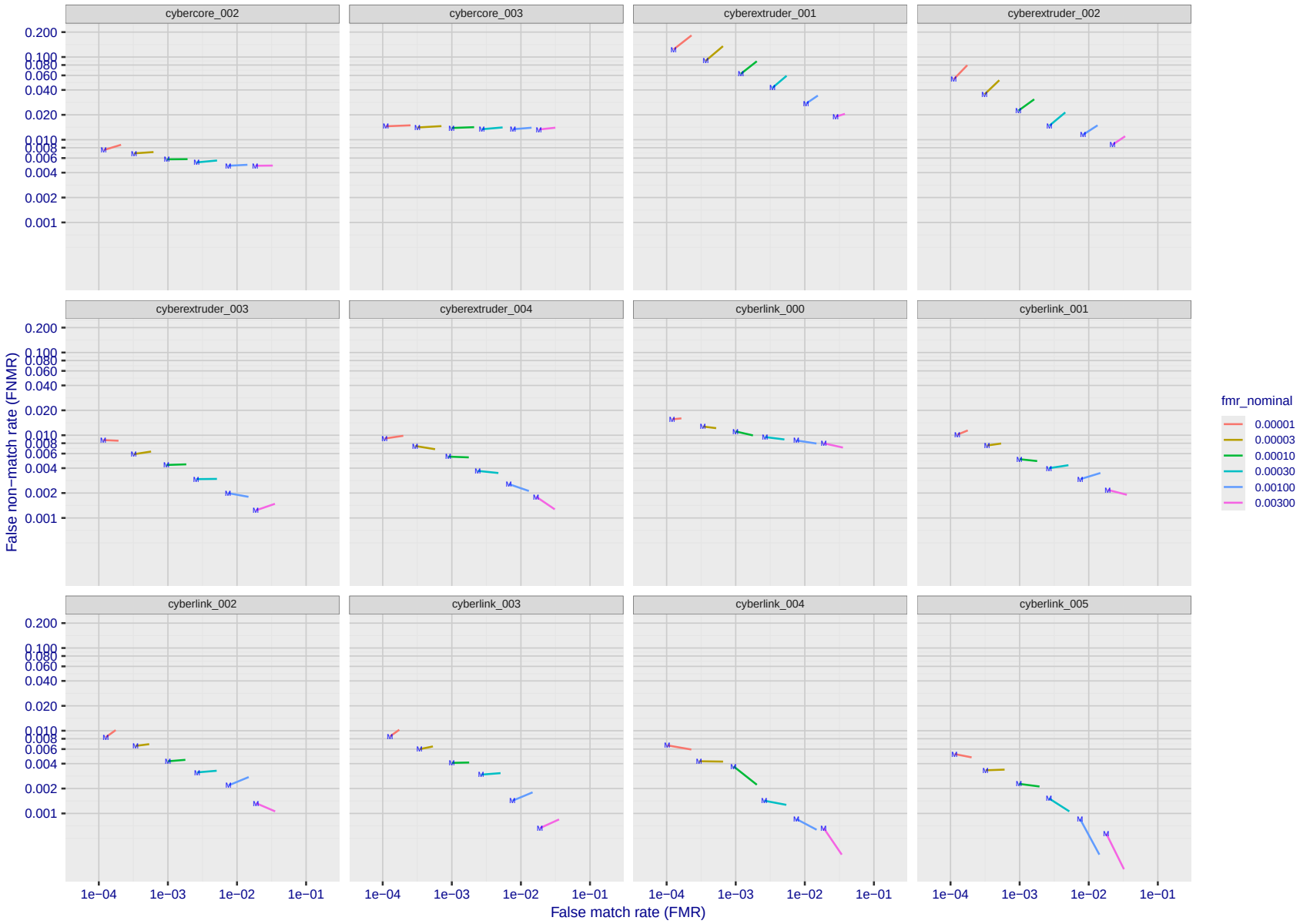


Figure 491: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

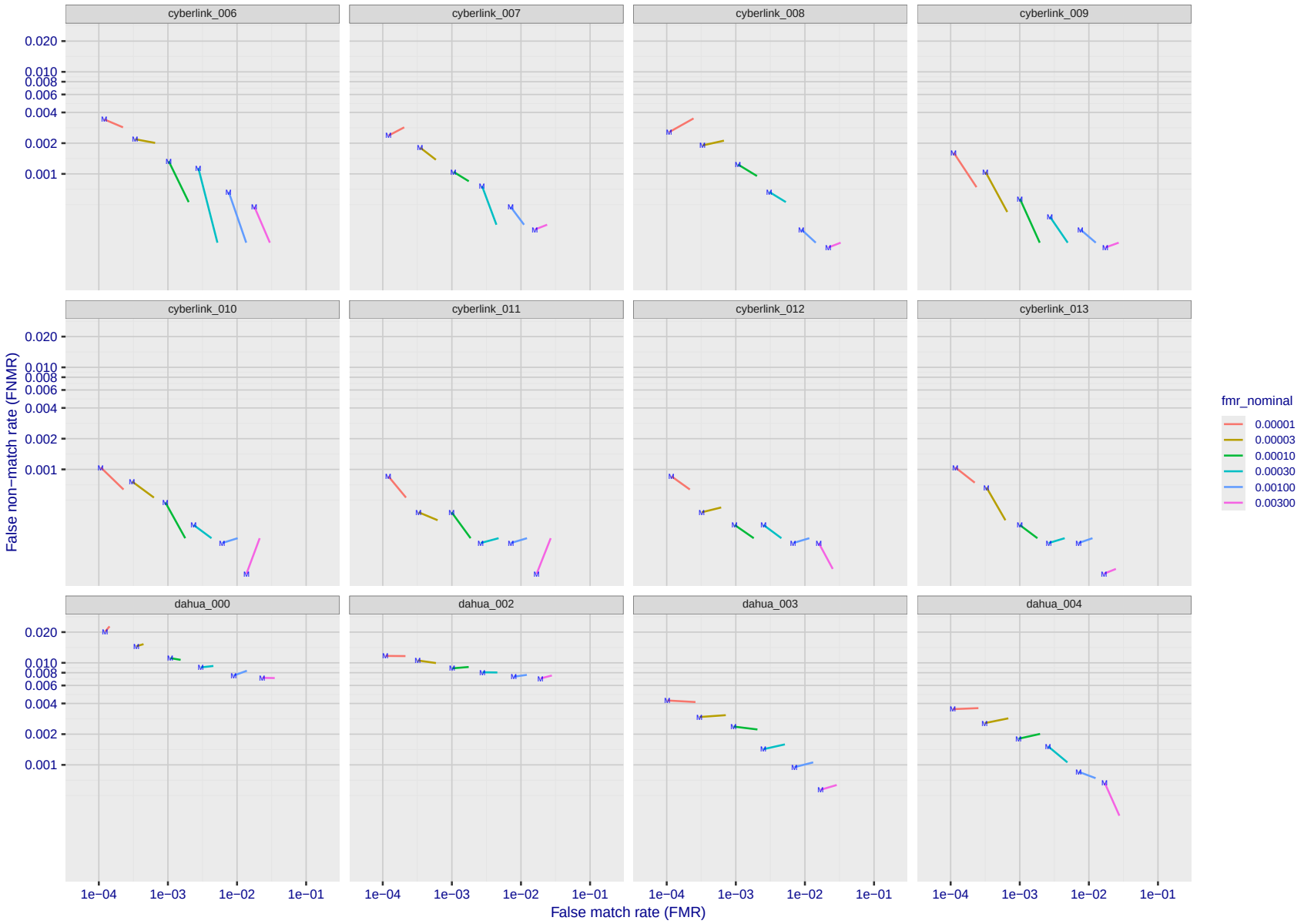


Figure 492: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

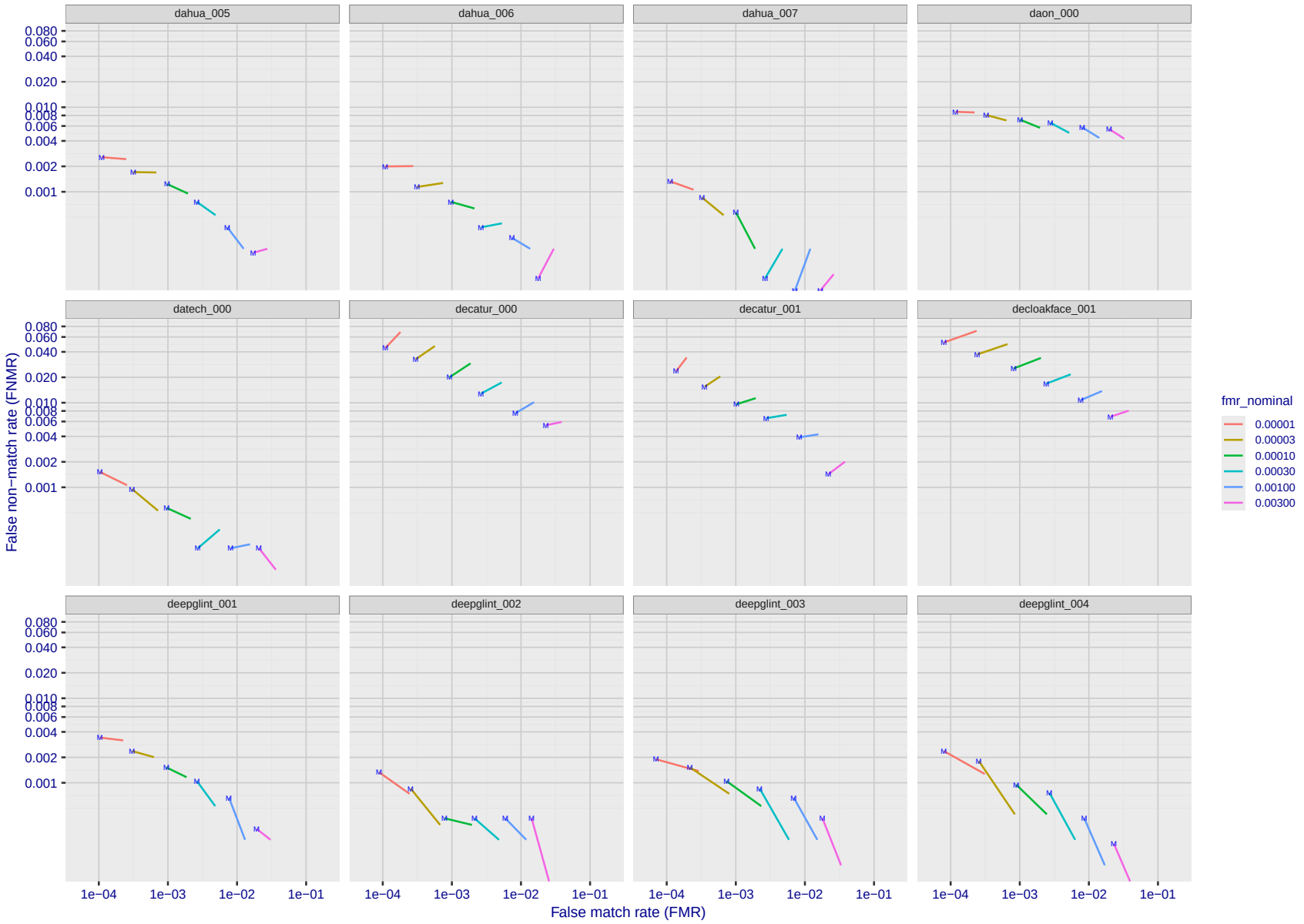


Figure 493: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

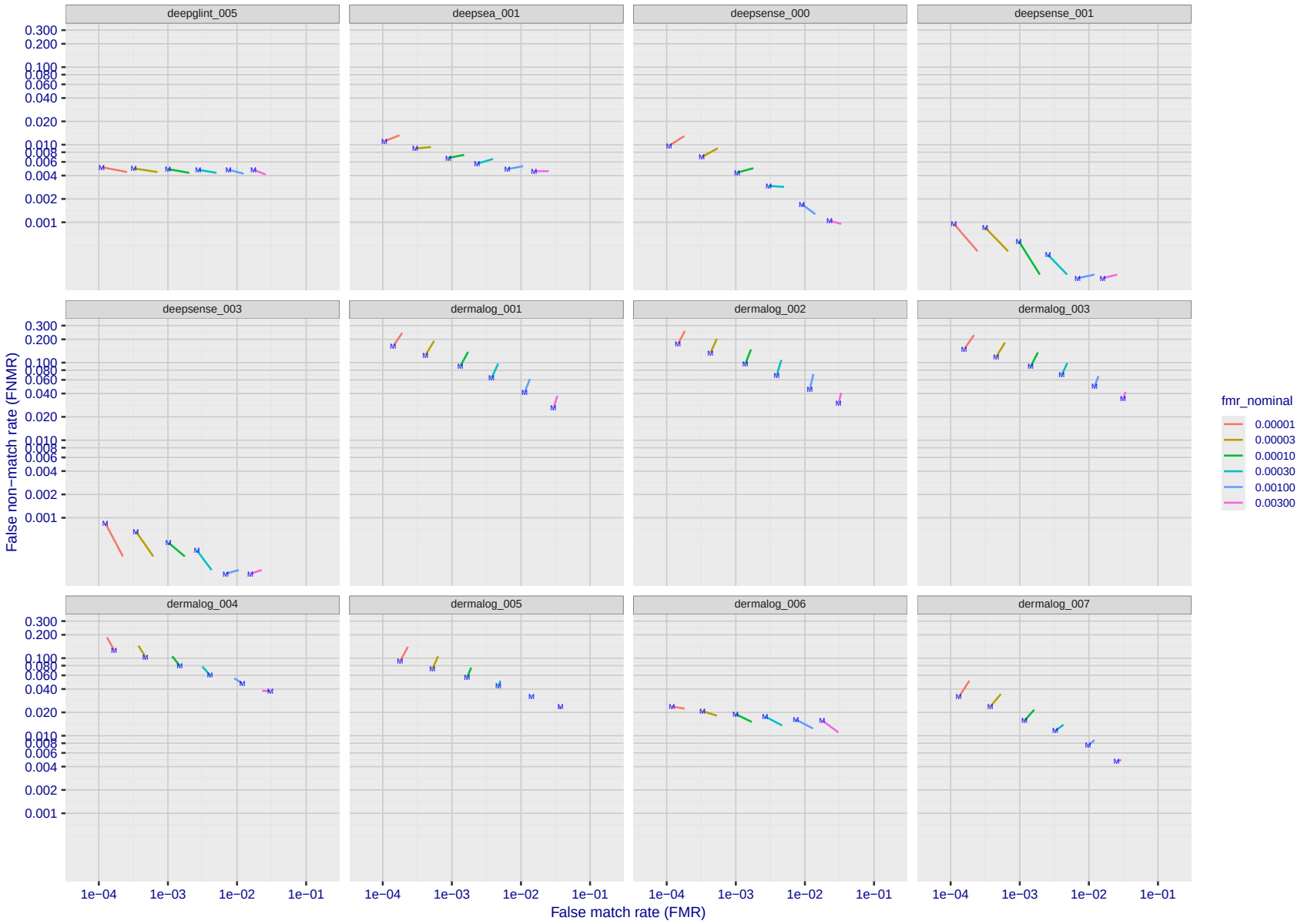


Figure 494: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

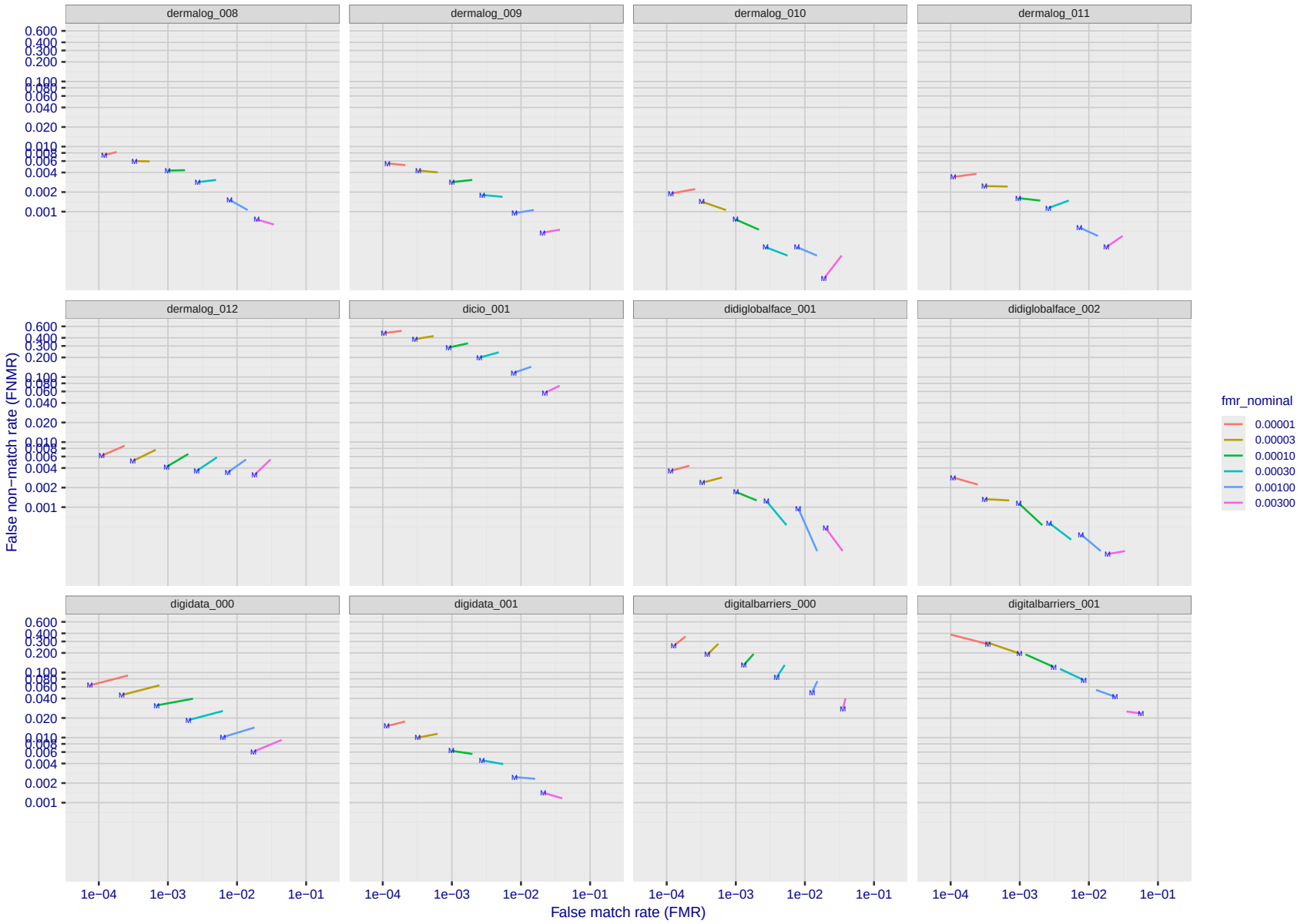


Figure 495: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

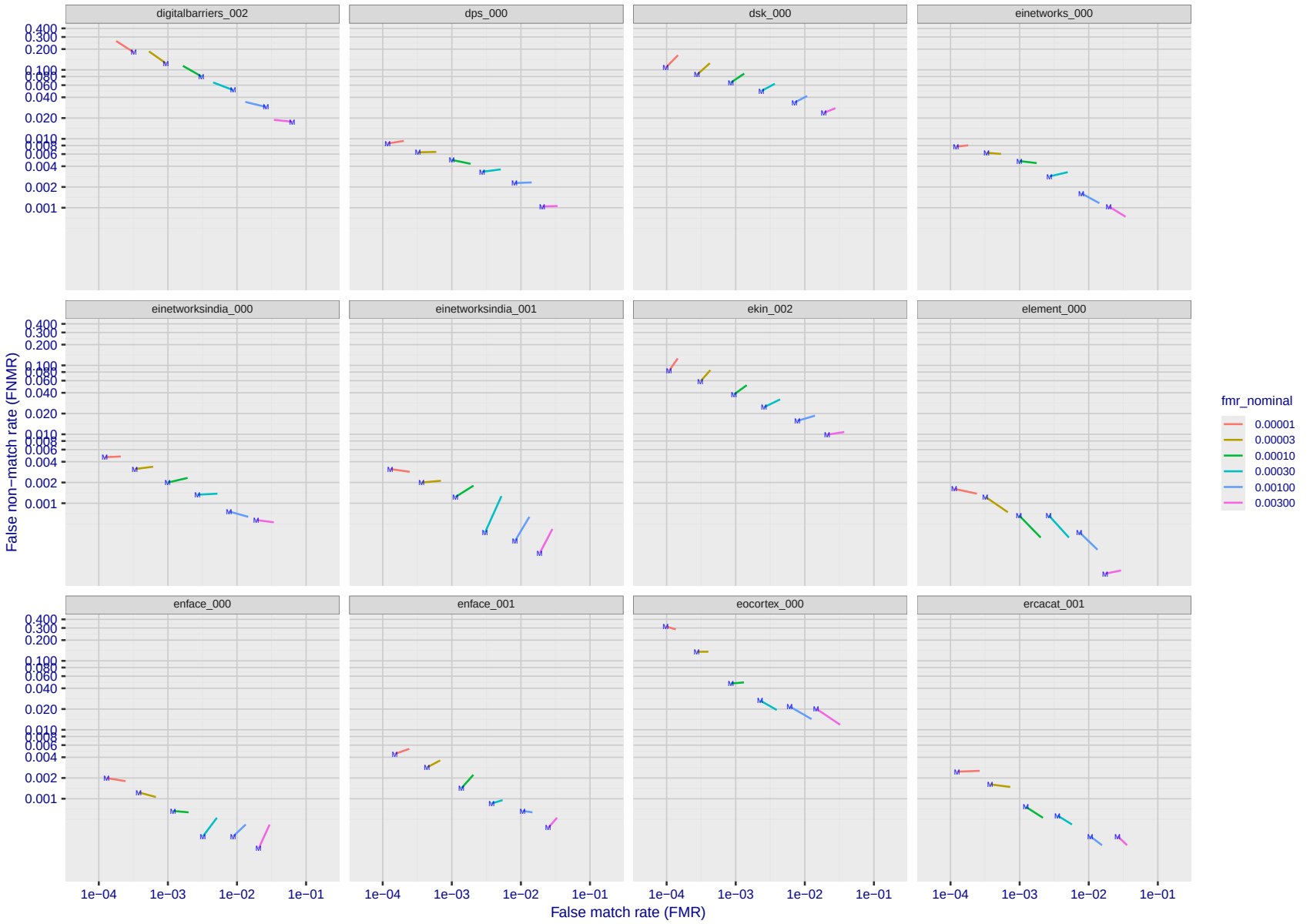


Figure 496: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

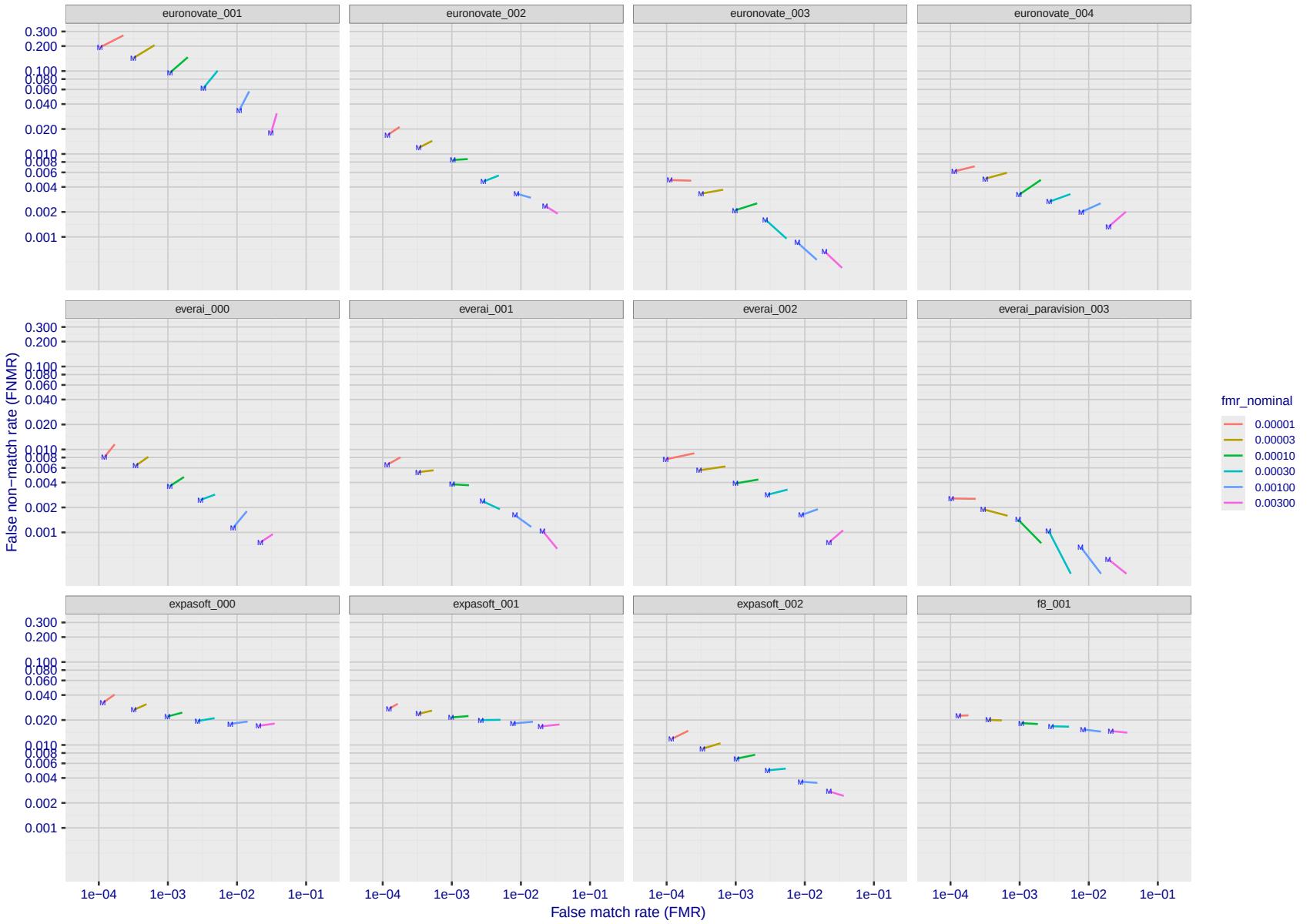


Figure 497: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

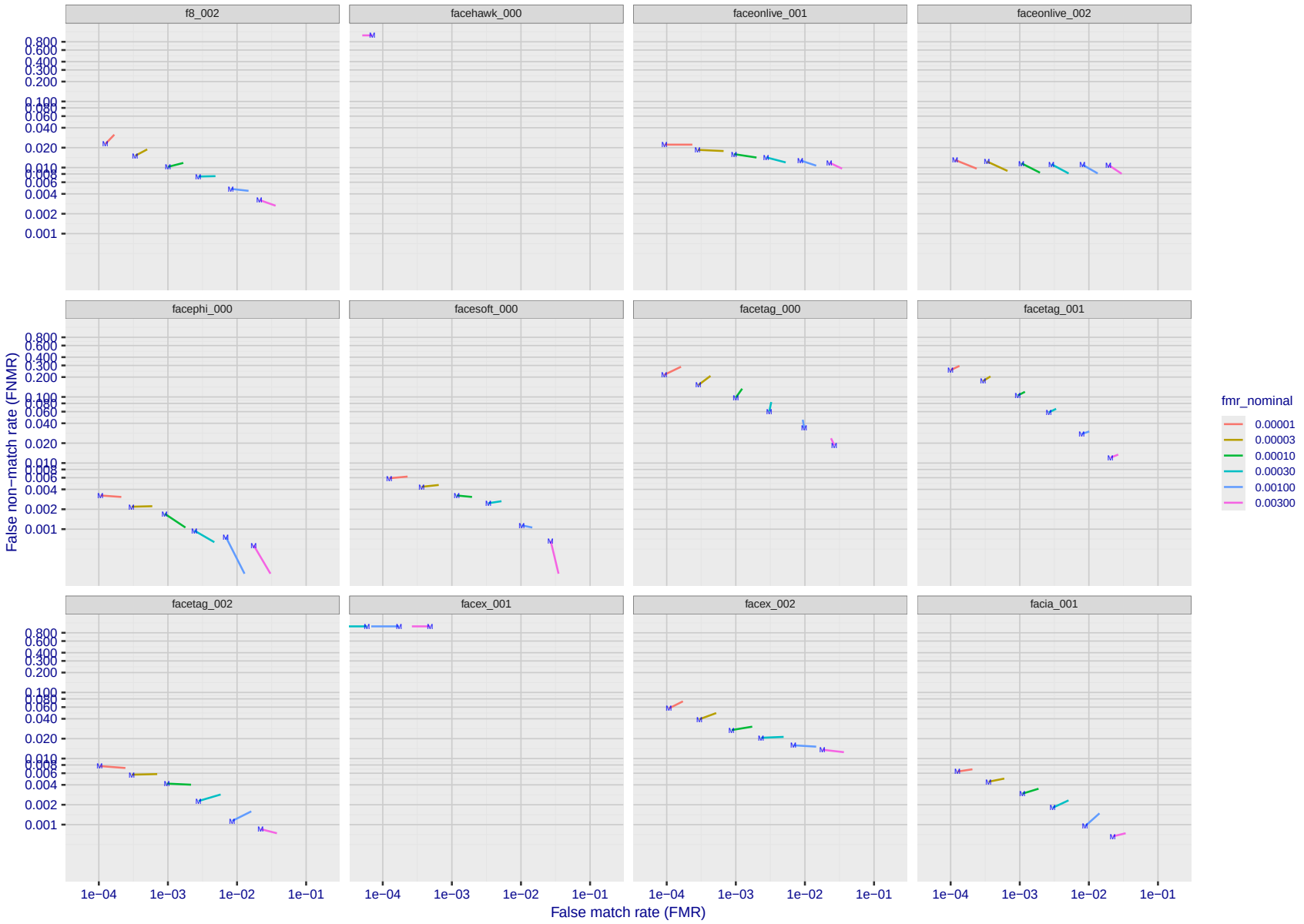


Figure 498: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

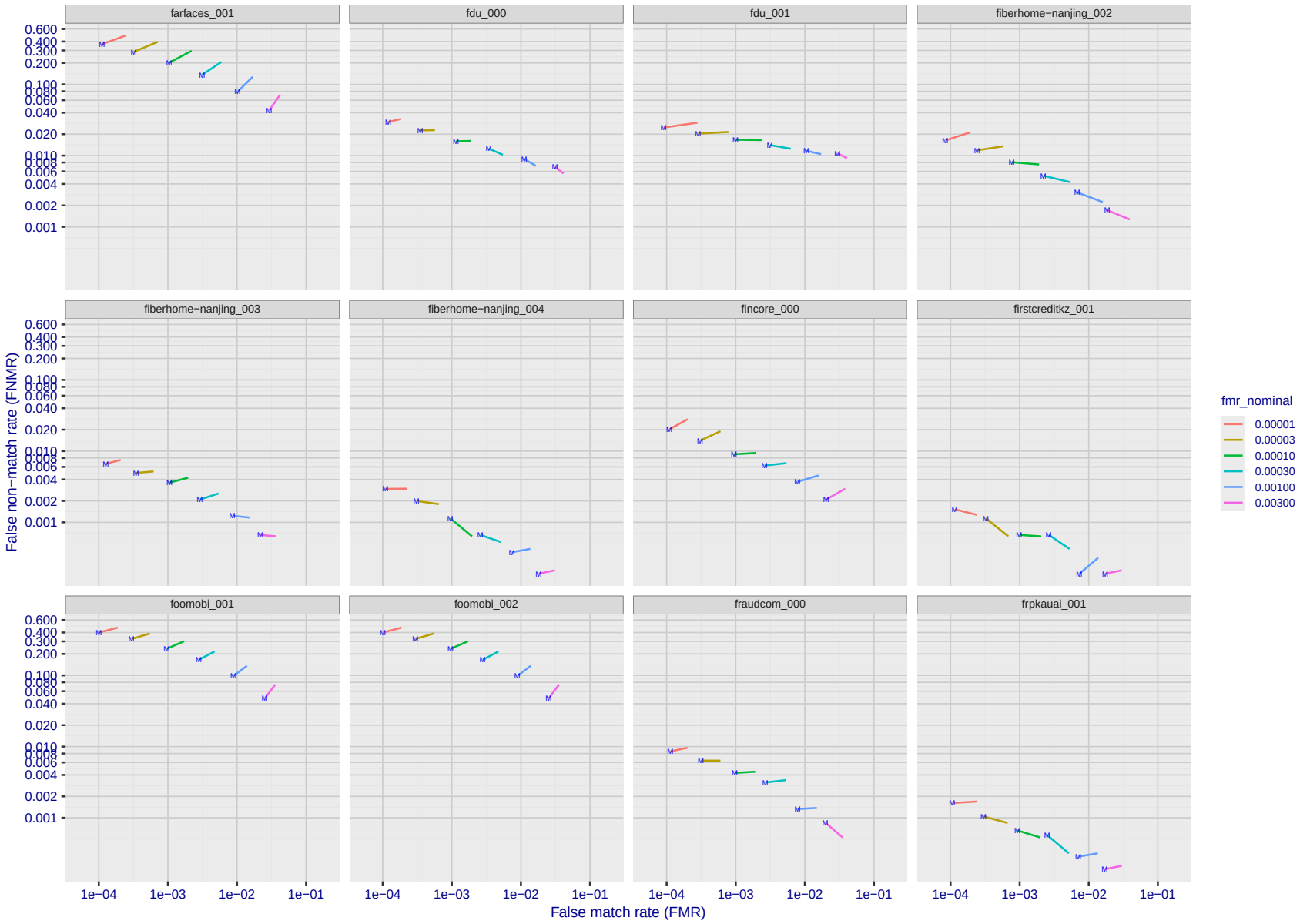


Figure 499: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

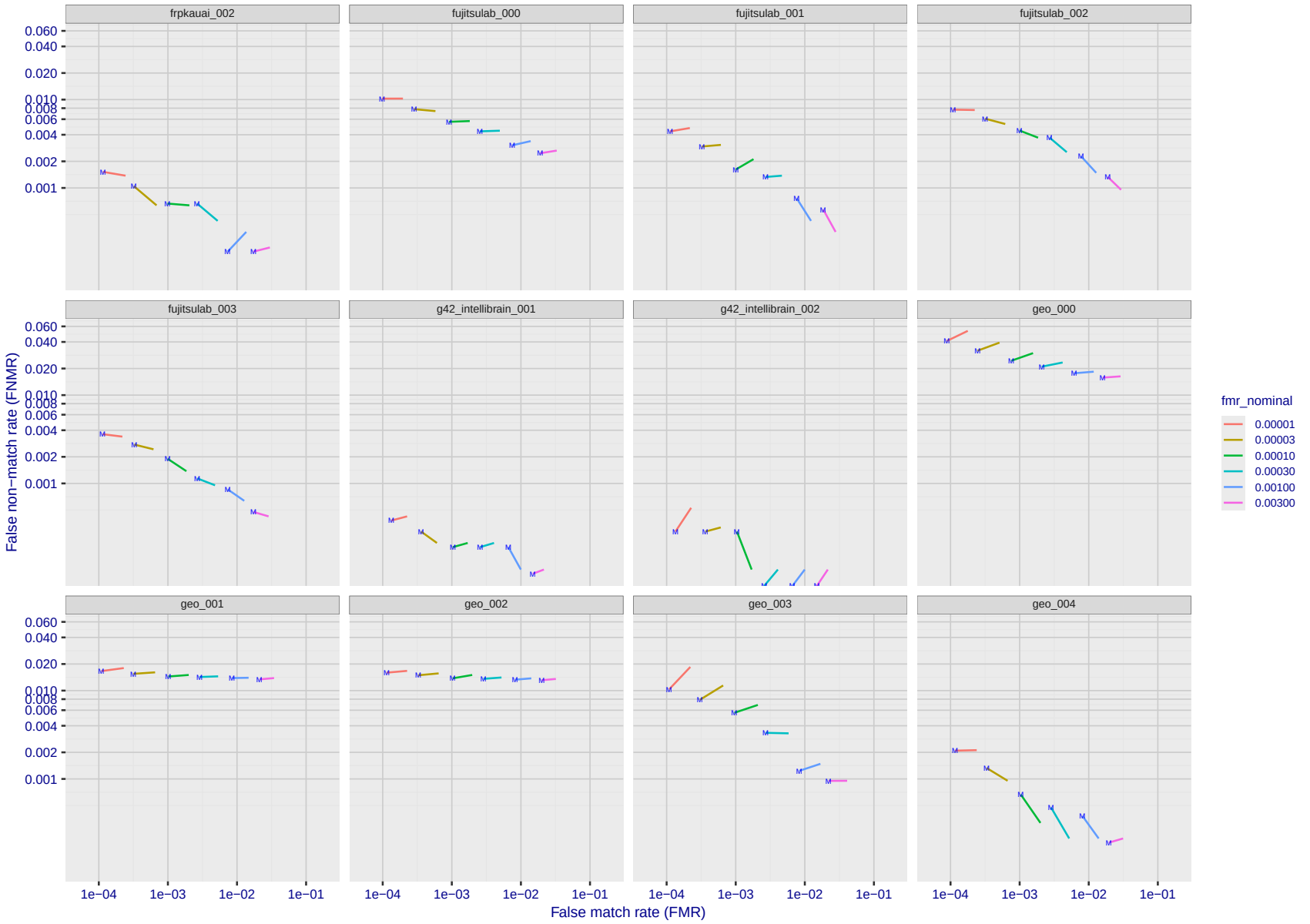


Figure 500: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

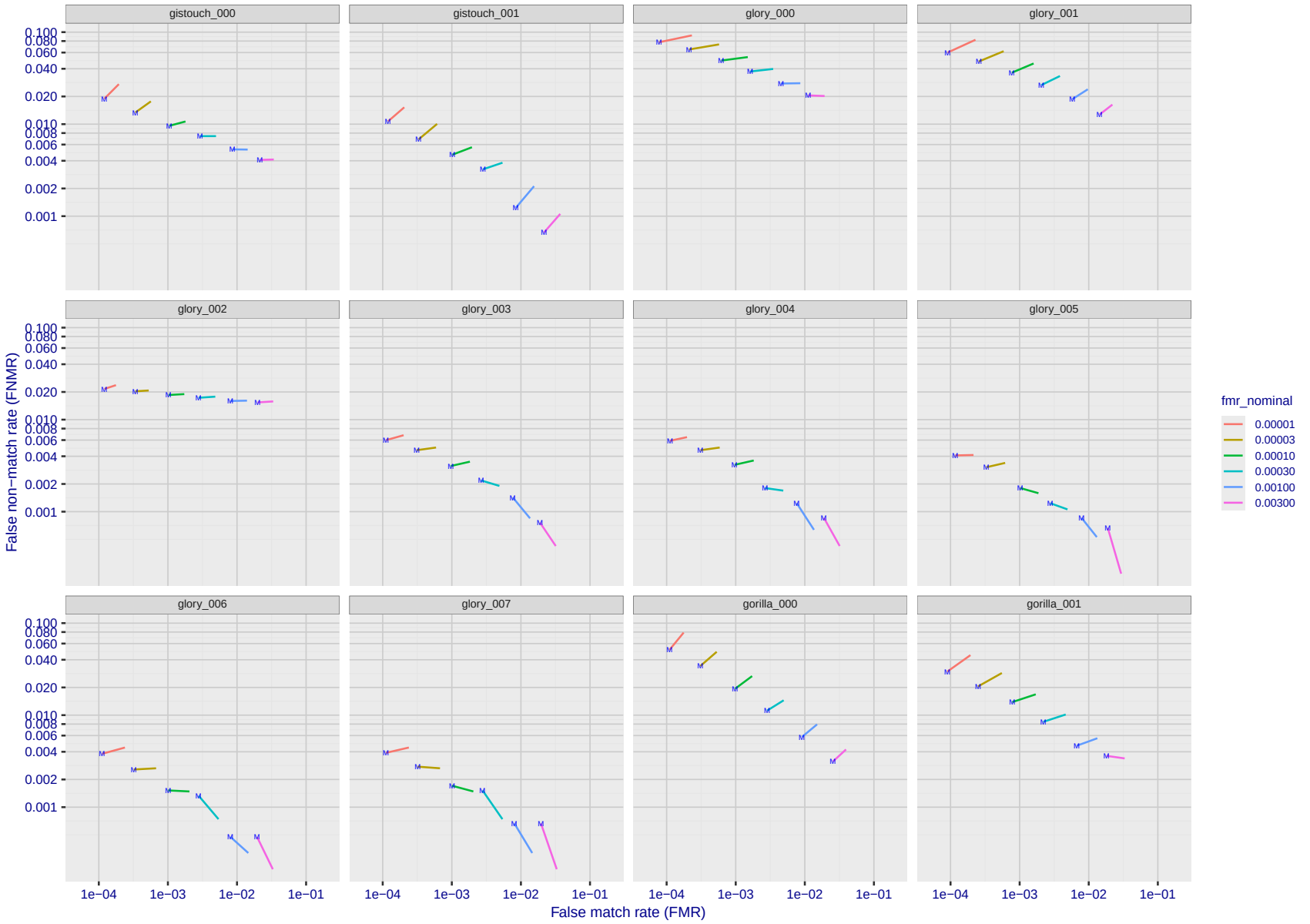


Figure 501: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

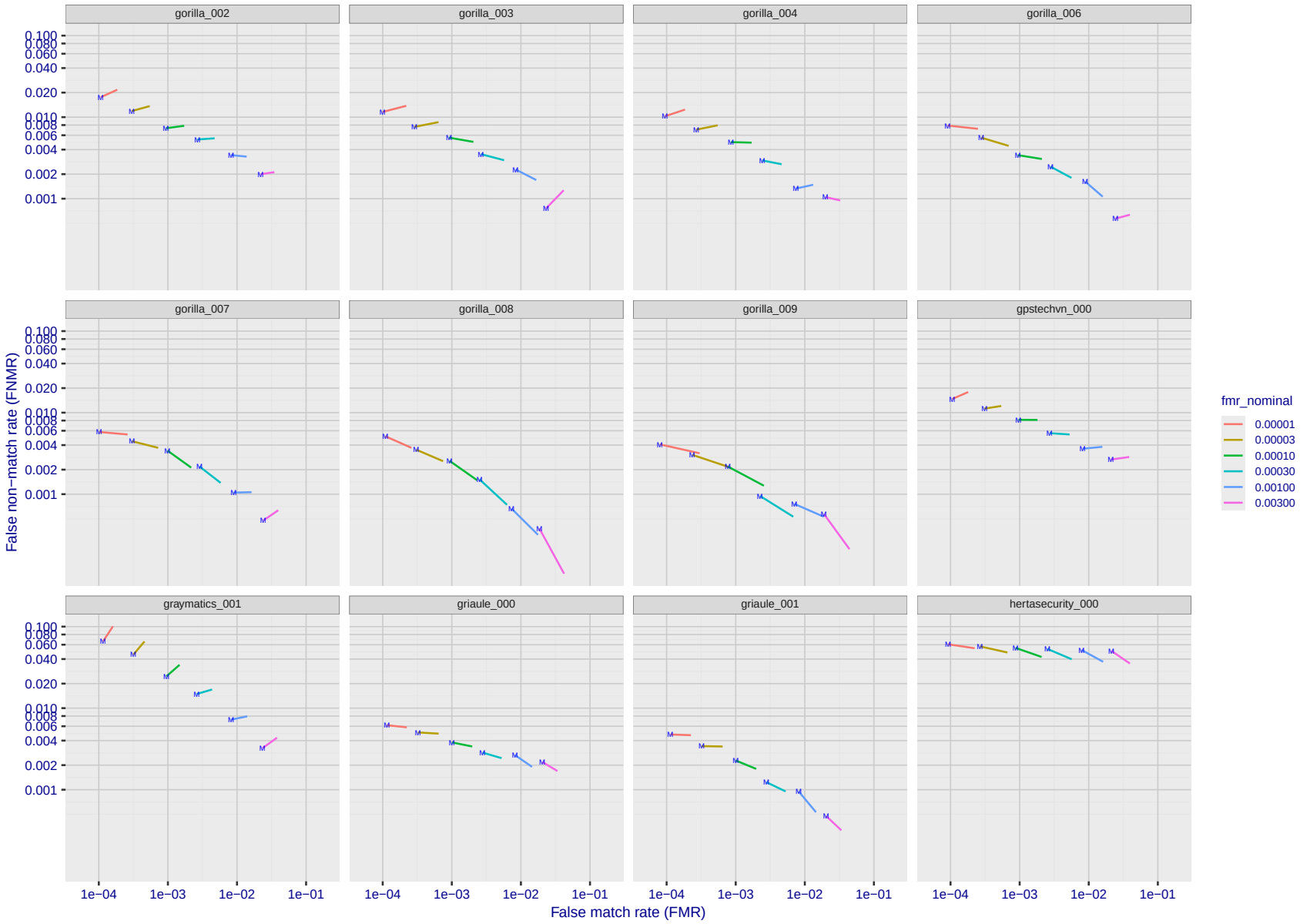


Figure 502: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

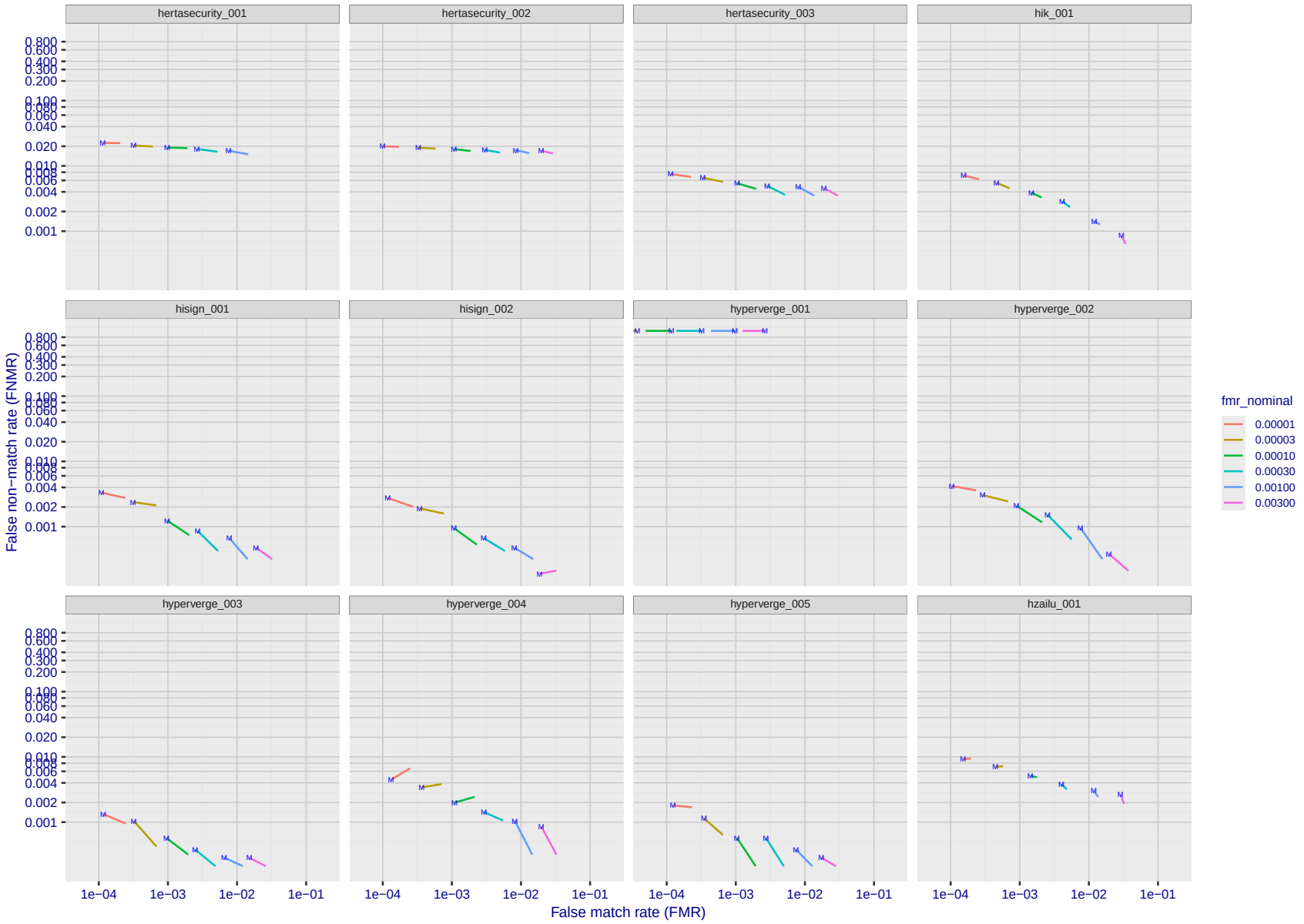


Figure 503: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

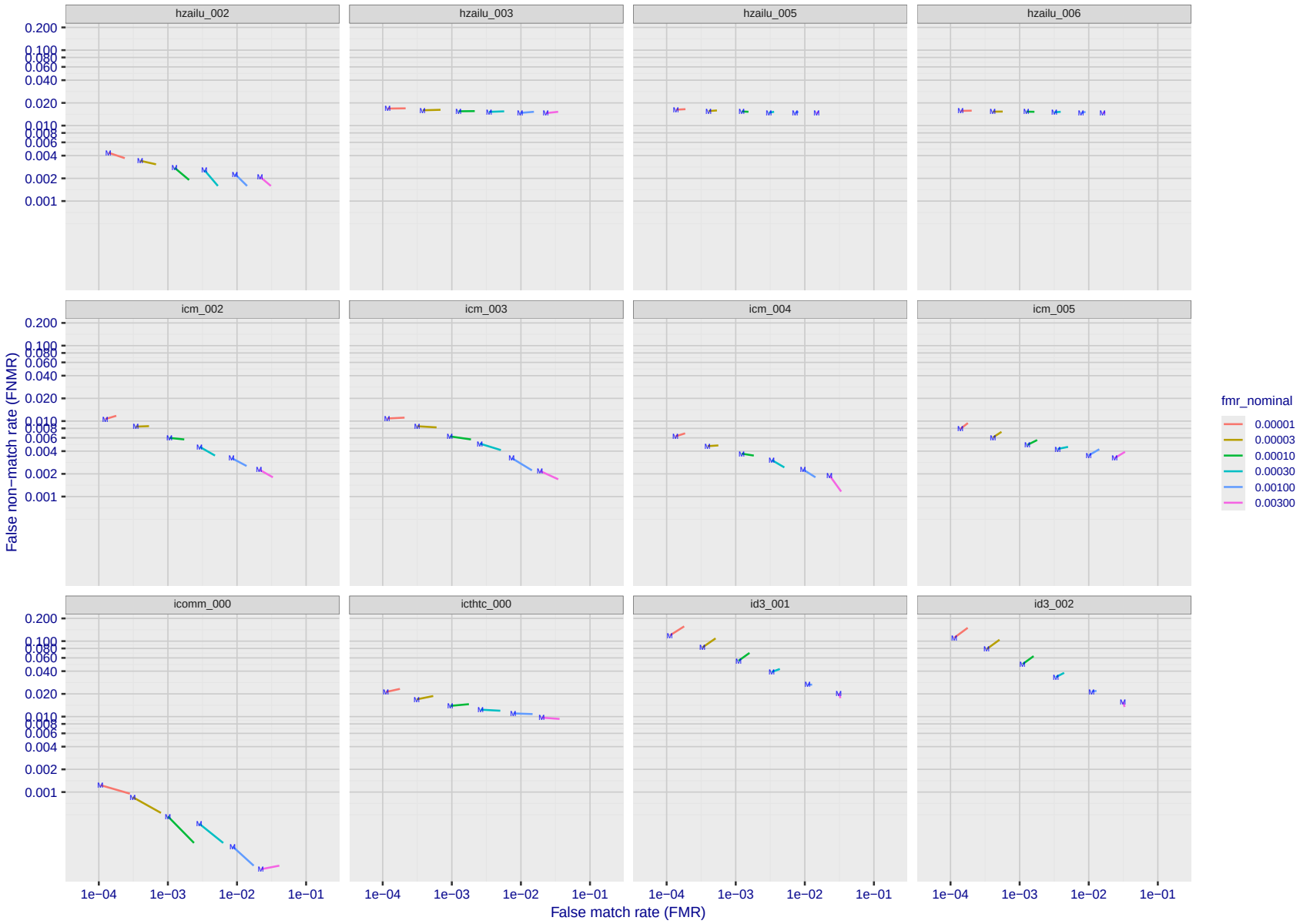


Figure 504: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

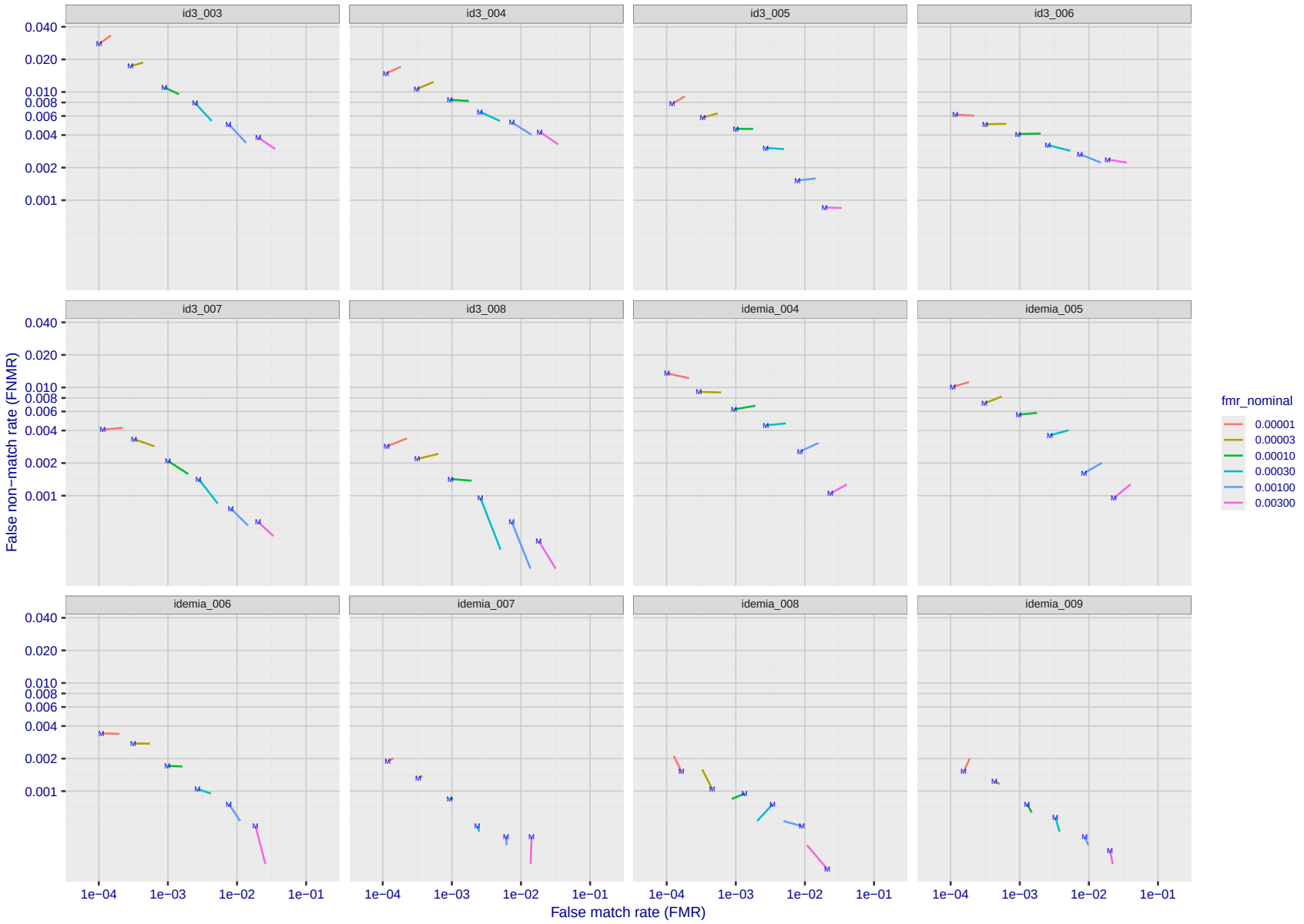


Figure 505: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

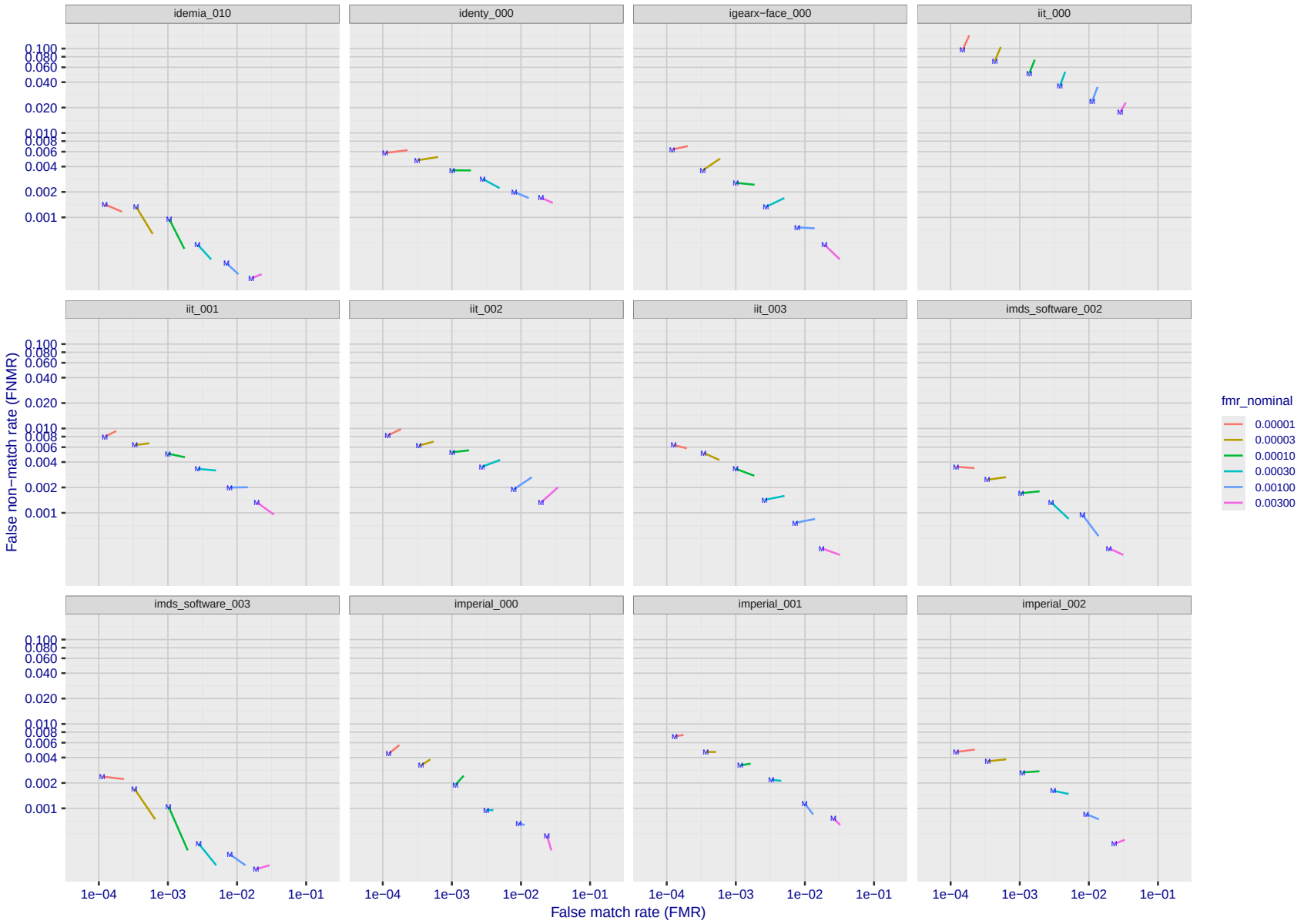


Figure 506: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

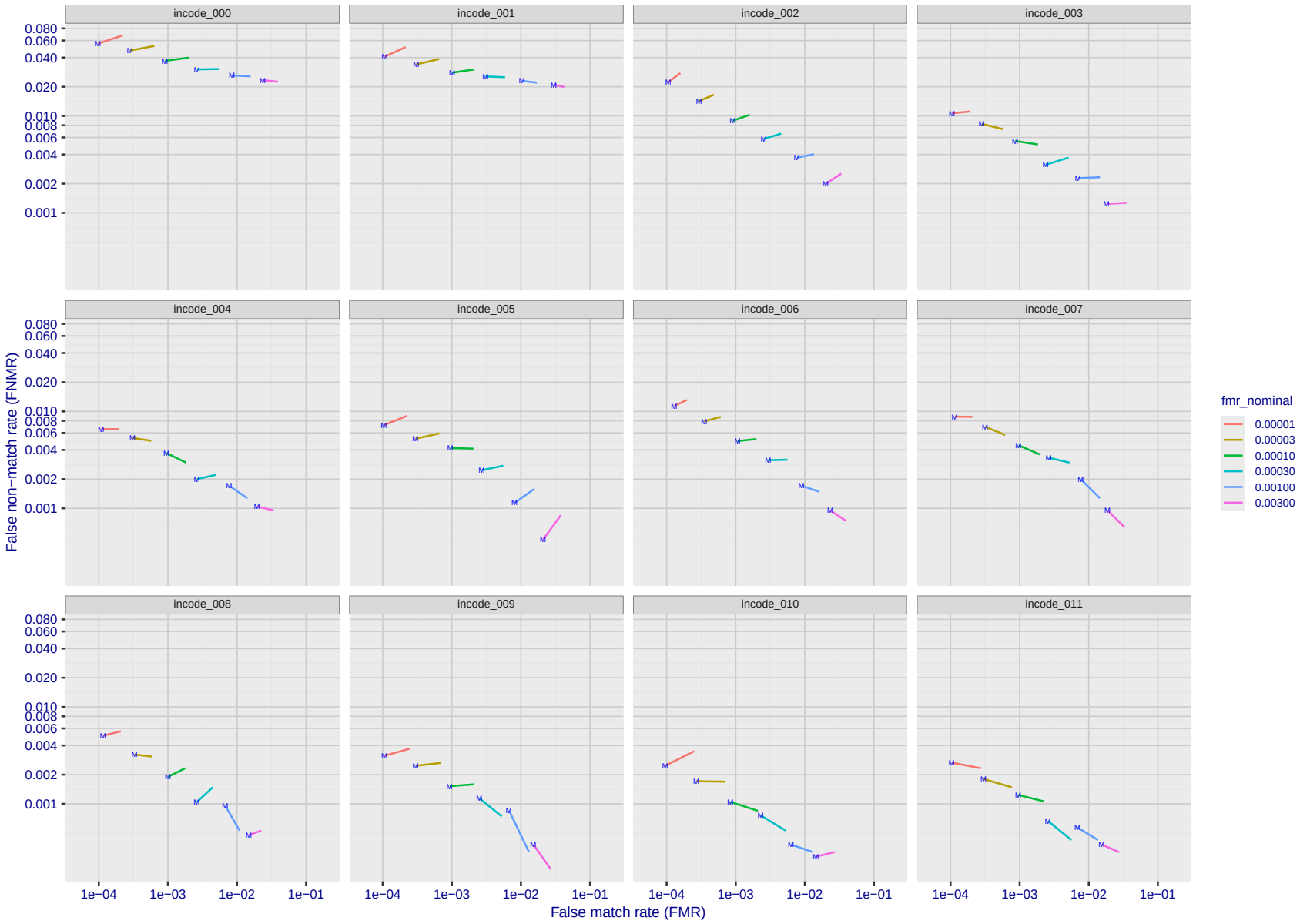


Figure 507: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

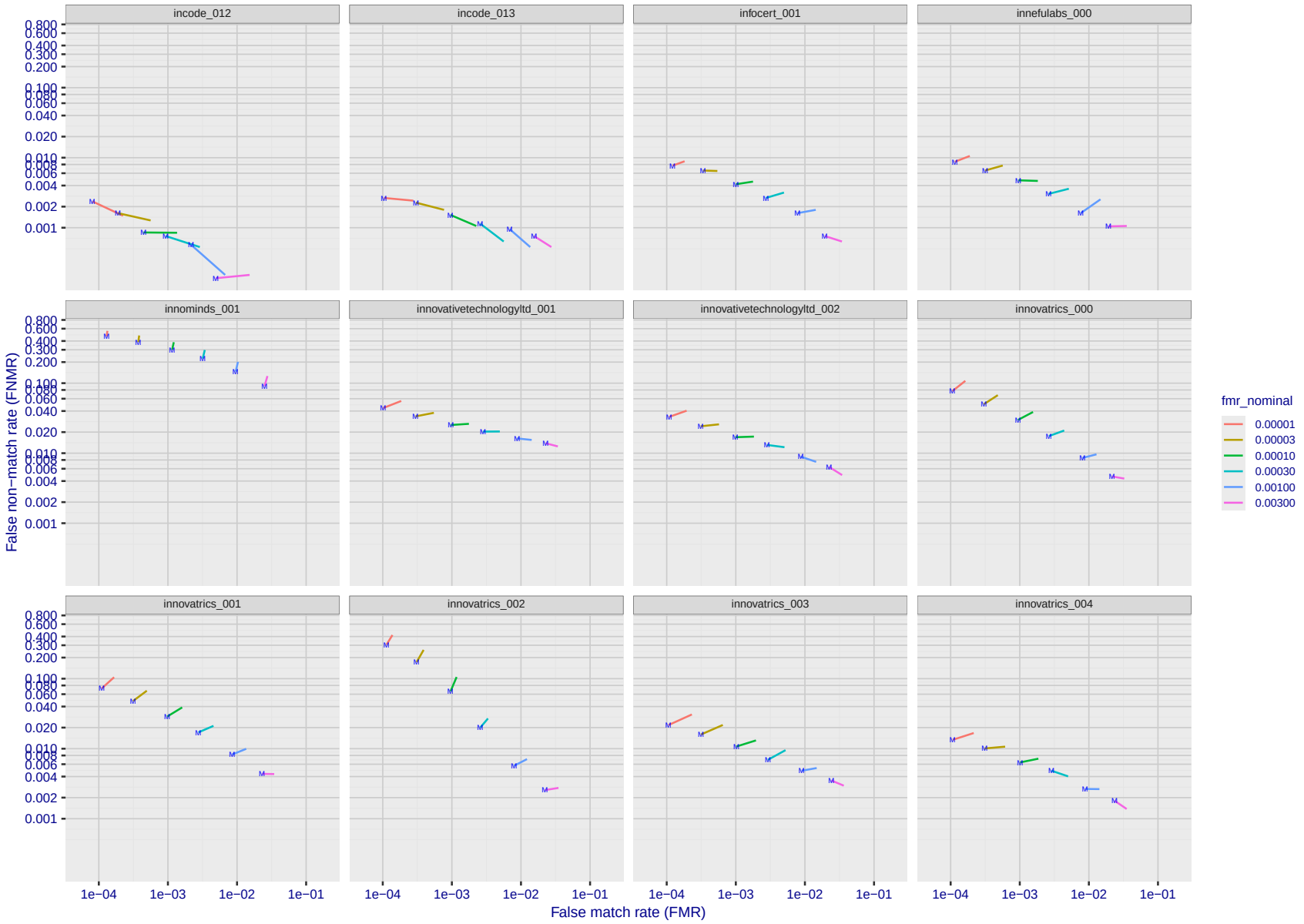


Figure 508: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

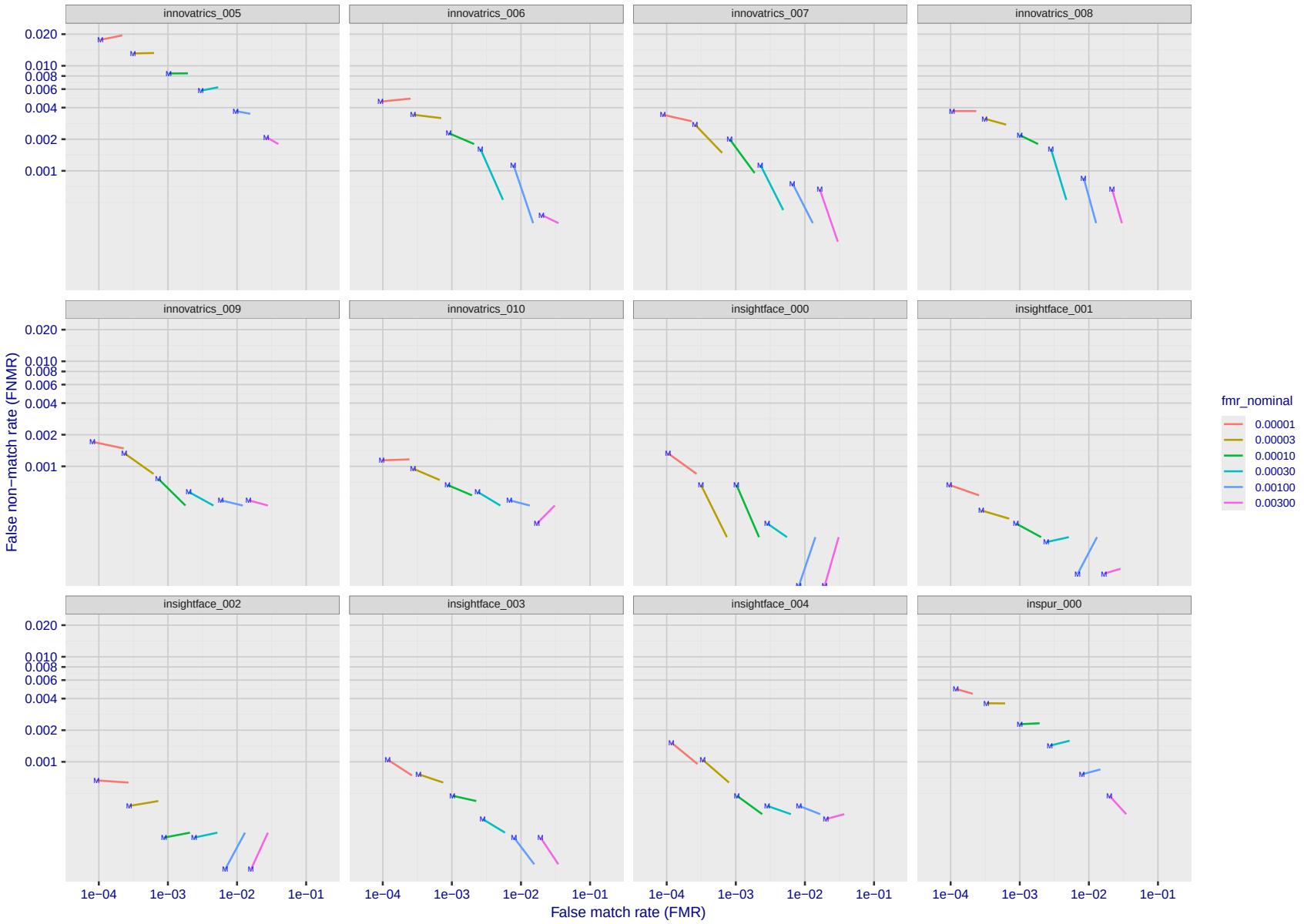


Figure 509: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

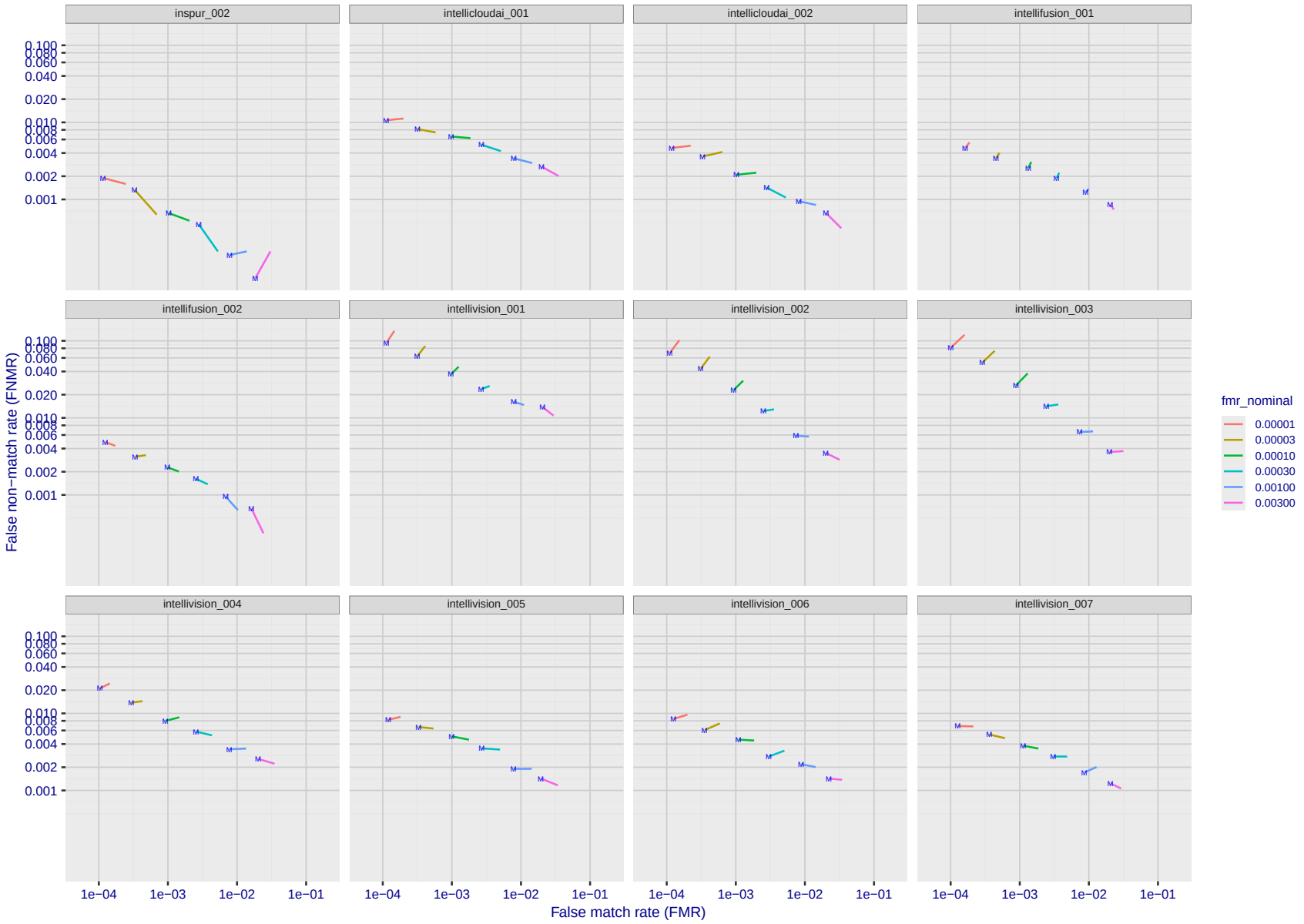


Figure 510: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

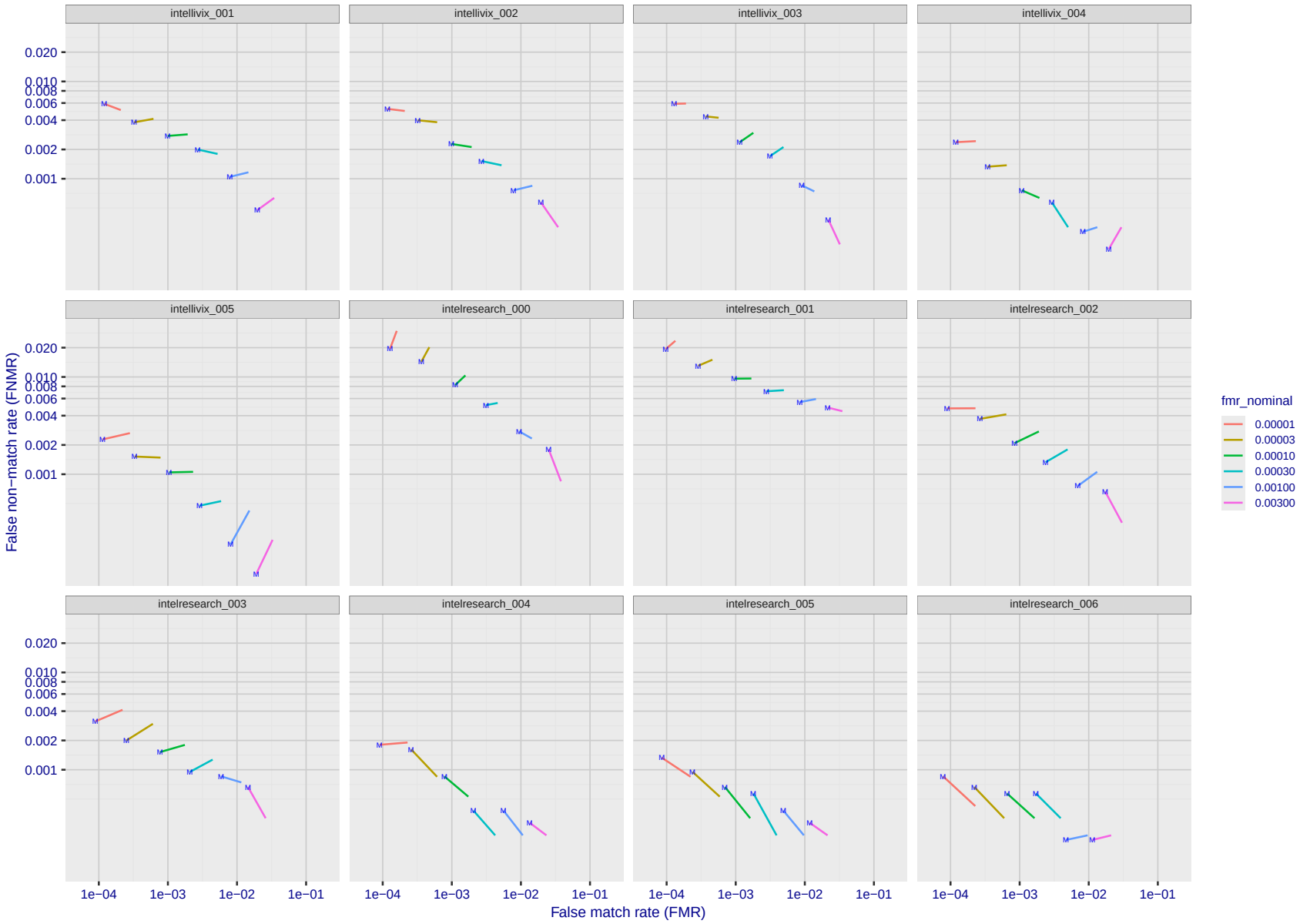


Figure 511: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

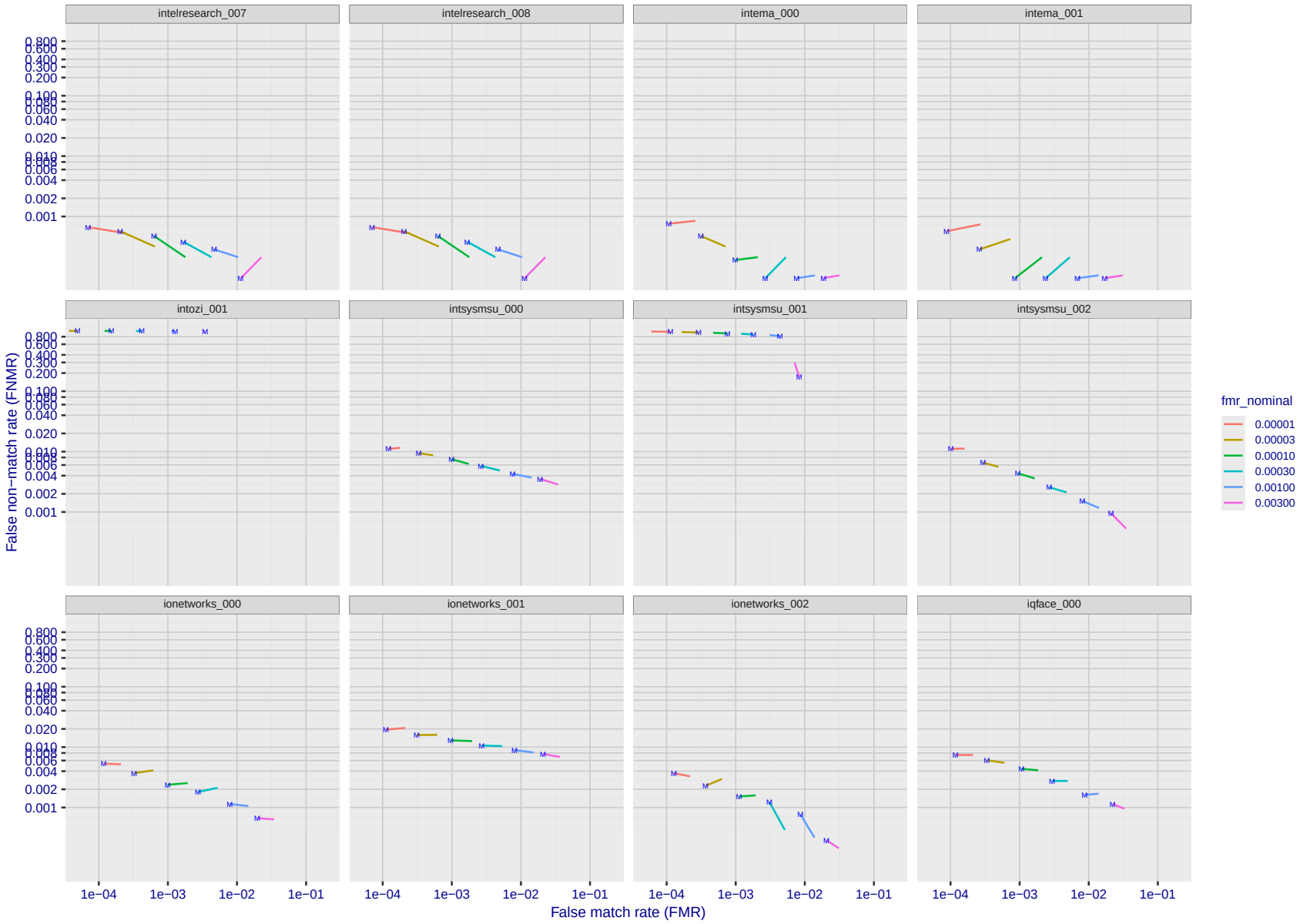


Figure 512: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

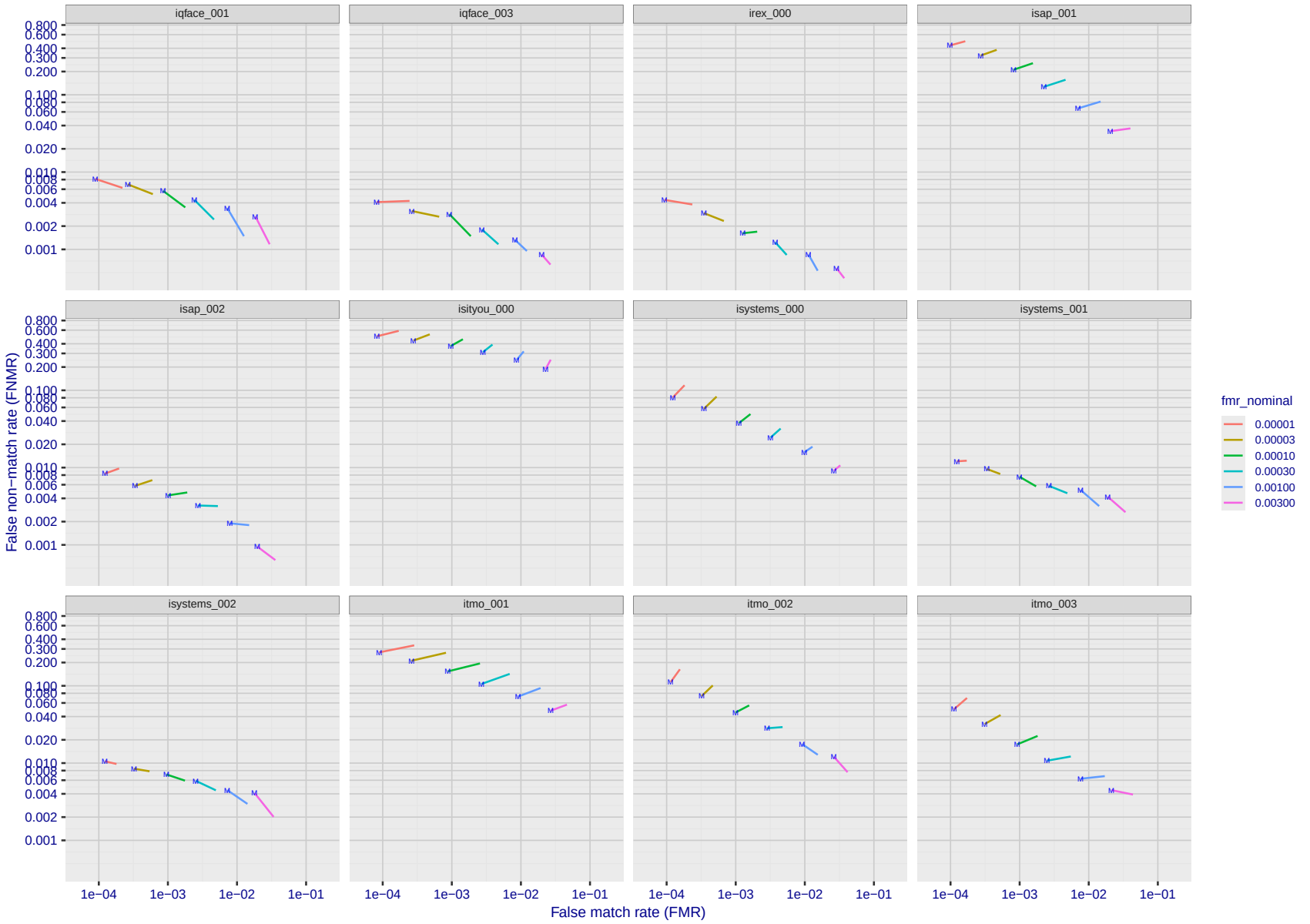


Figure 513: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

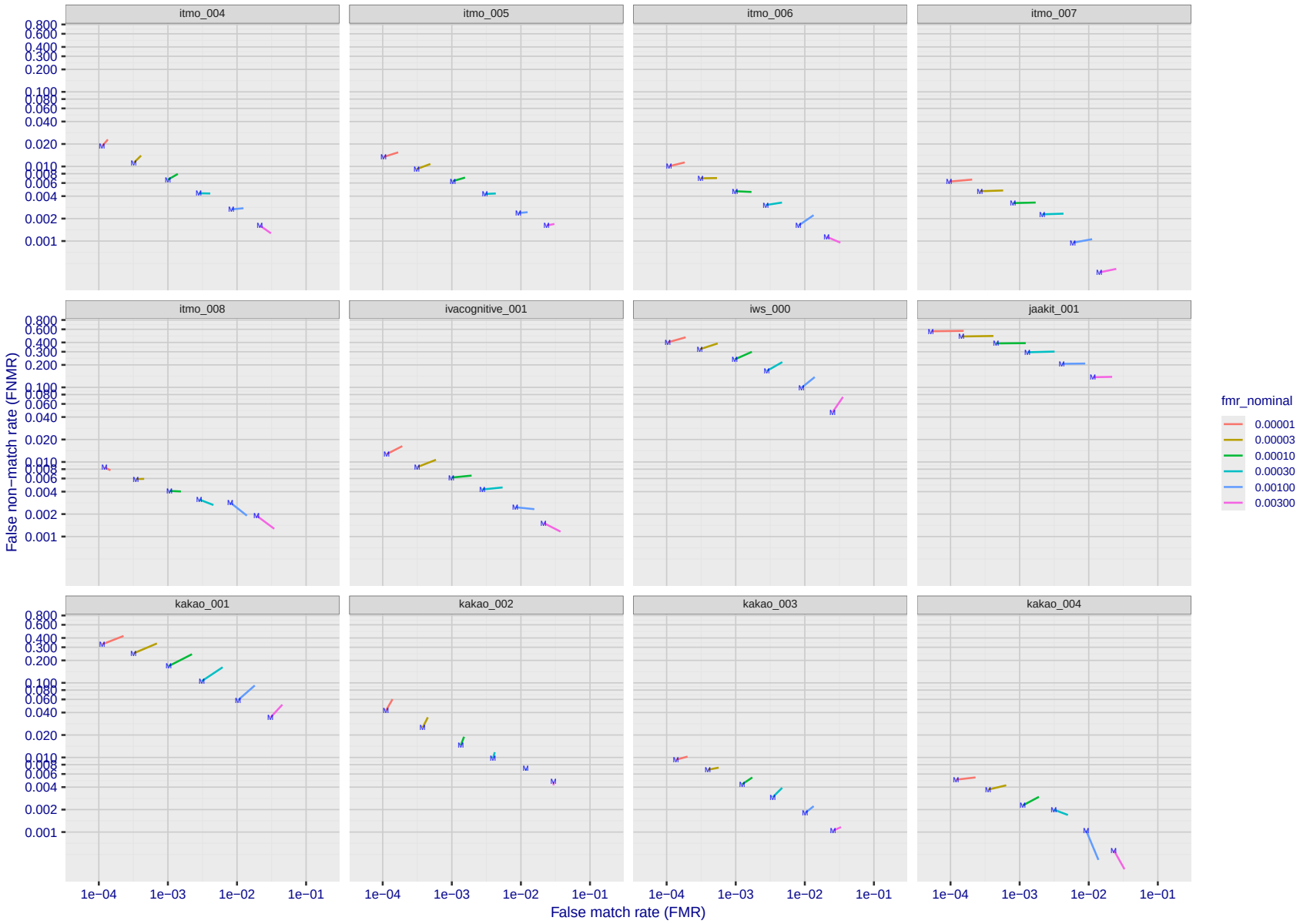


Figure 514: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

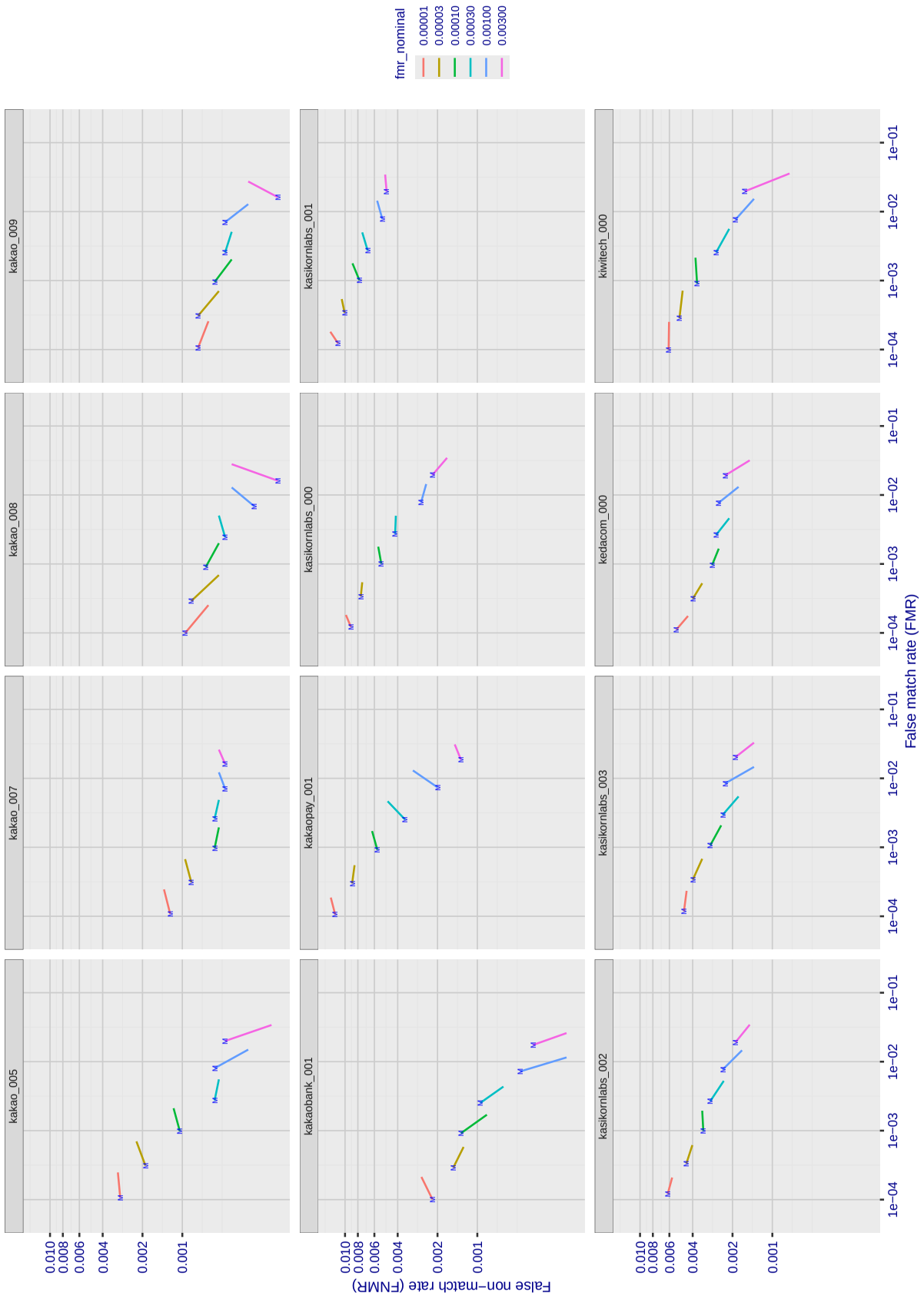


Figure 515: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

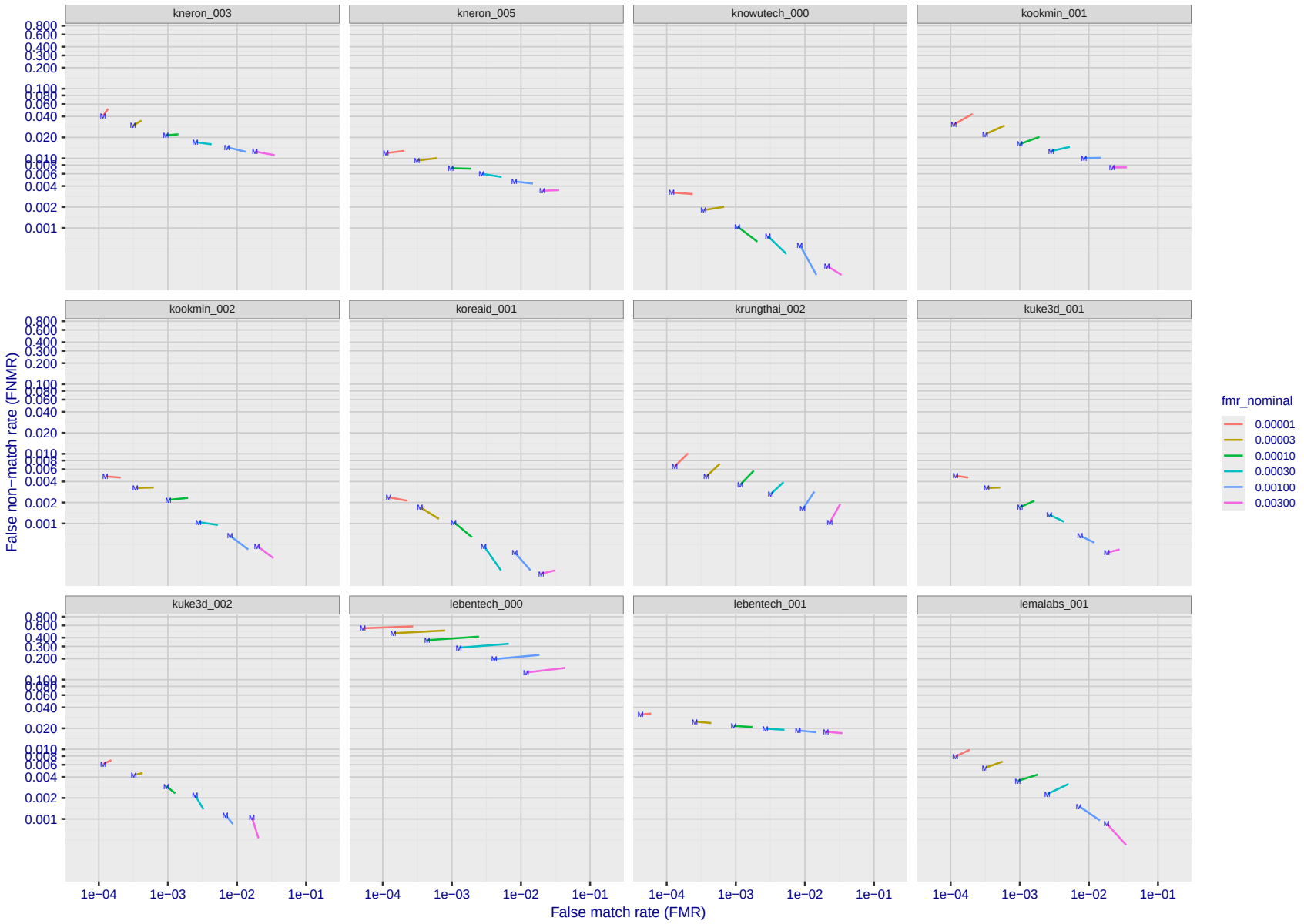


Figure 516: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

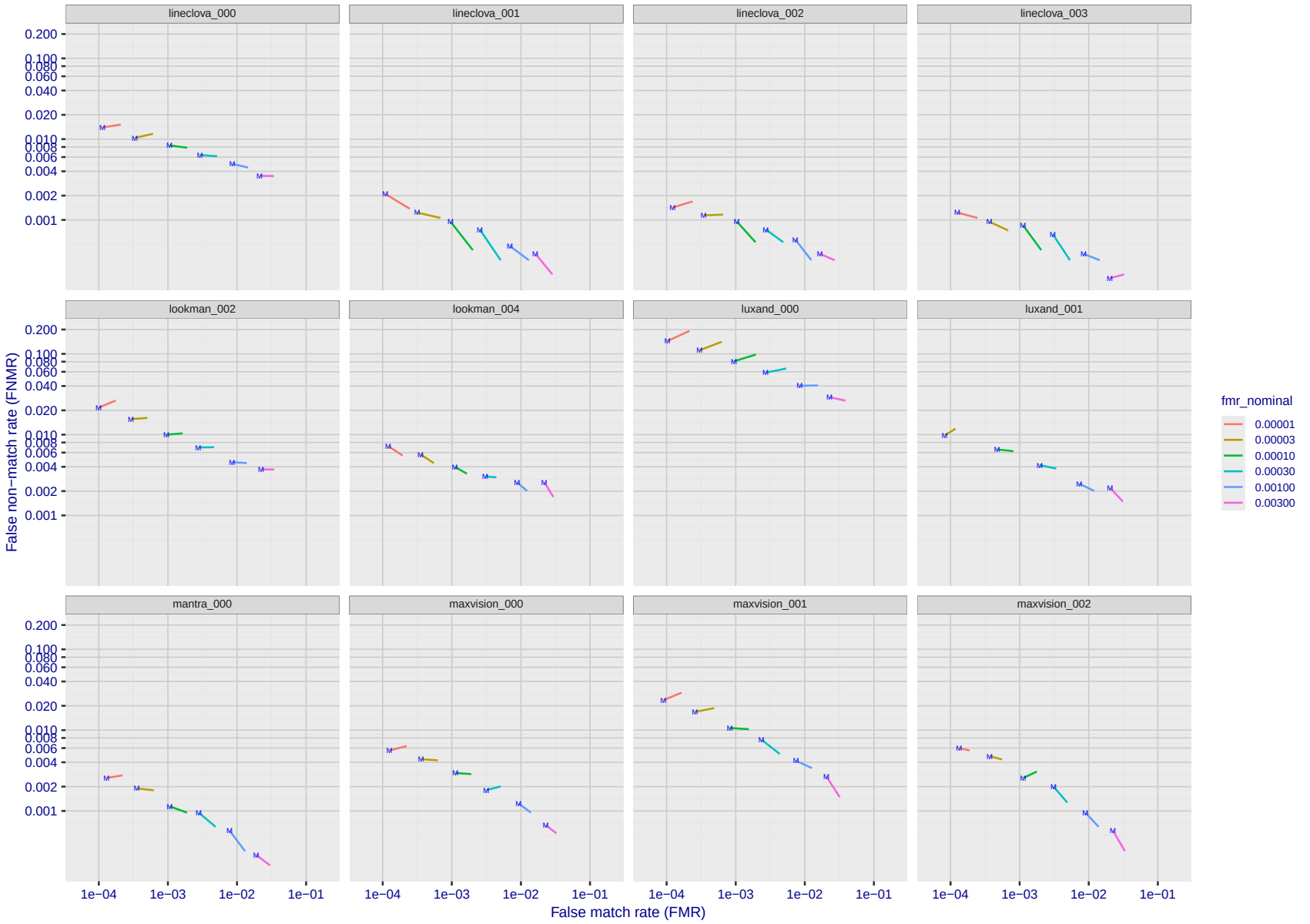


Figure 517: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

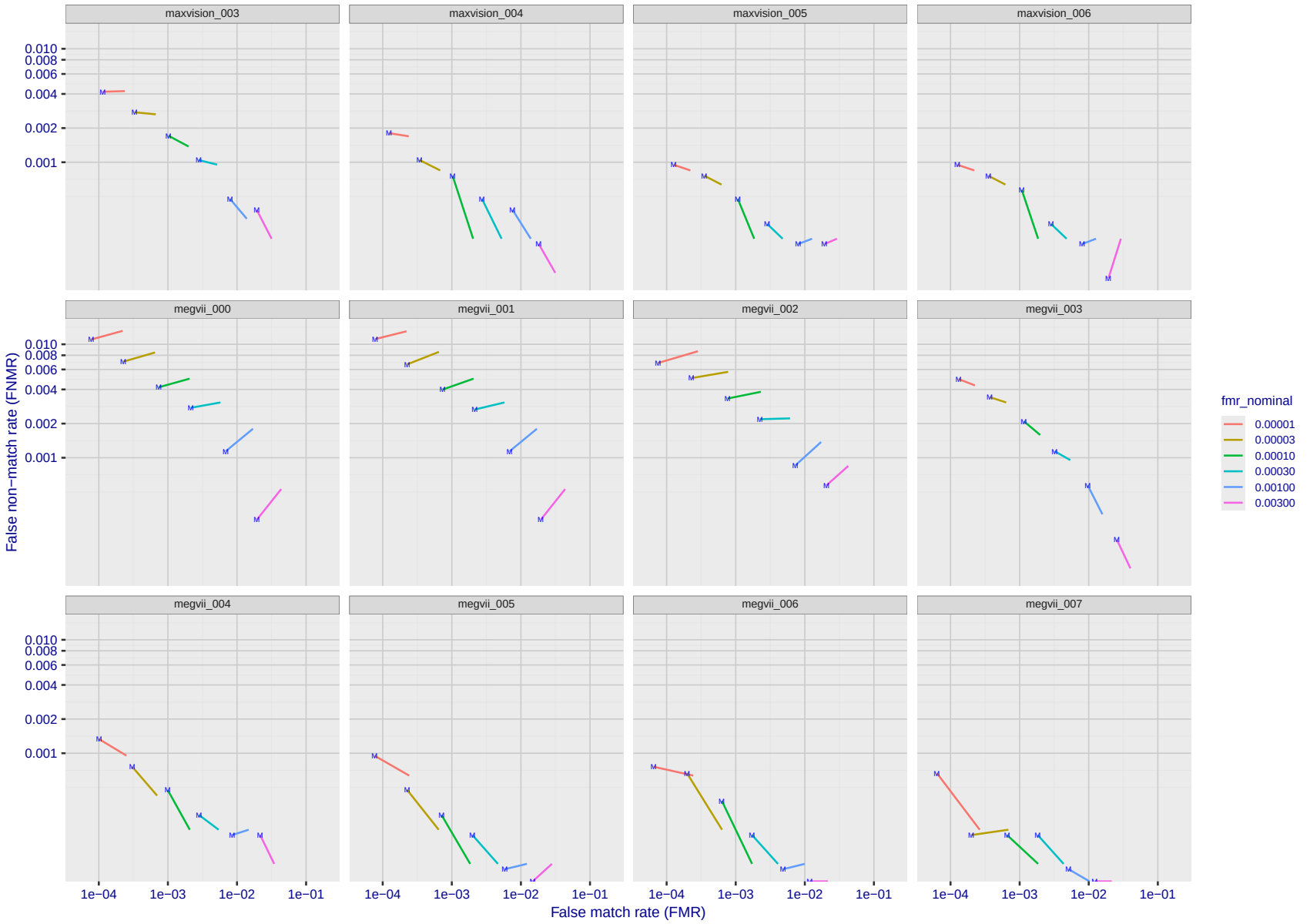


Figure 518: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

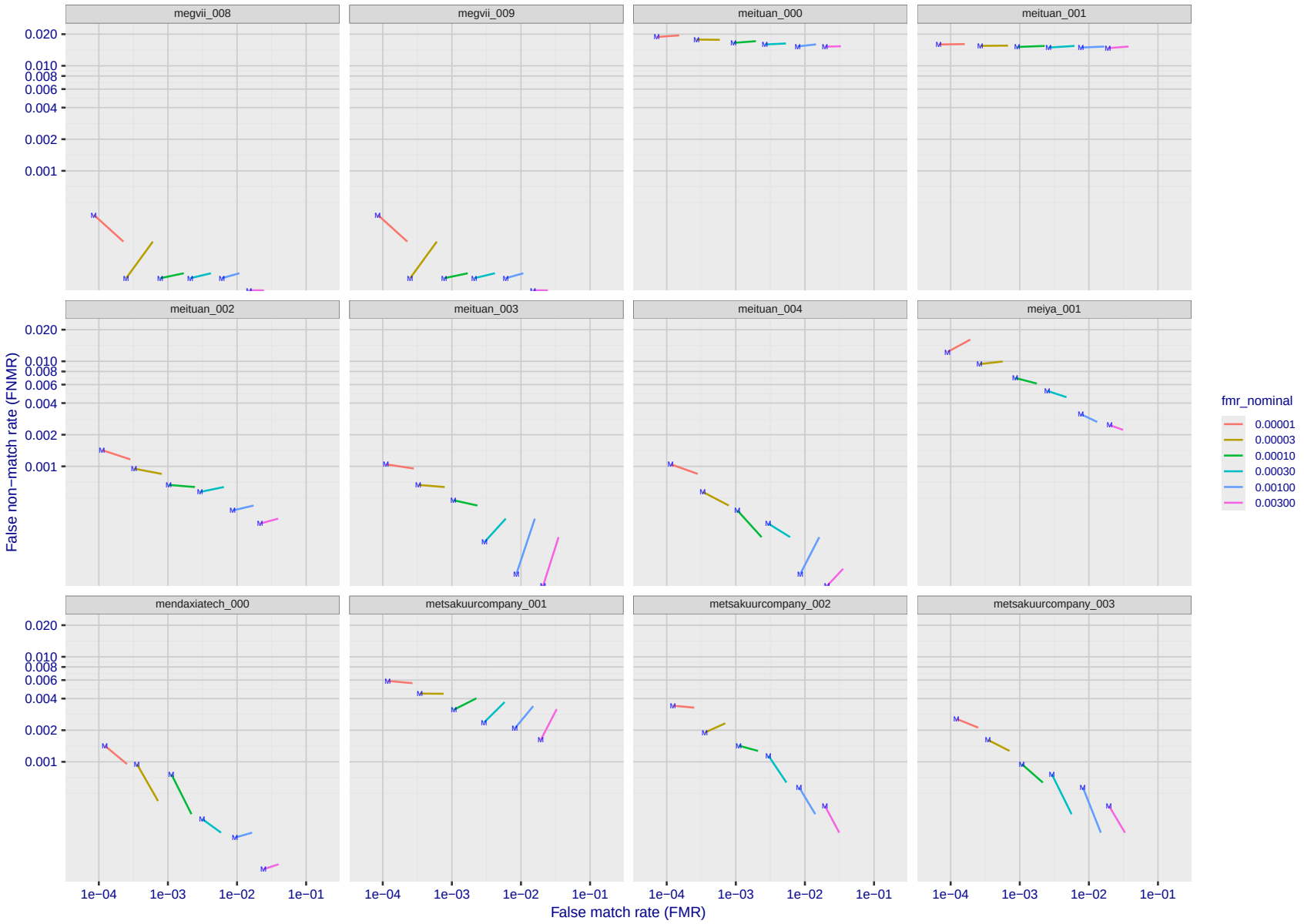


Figure 519: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

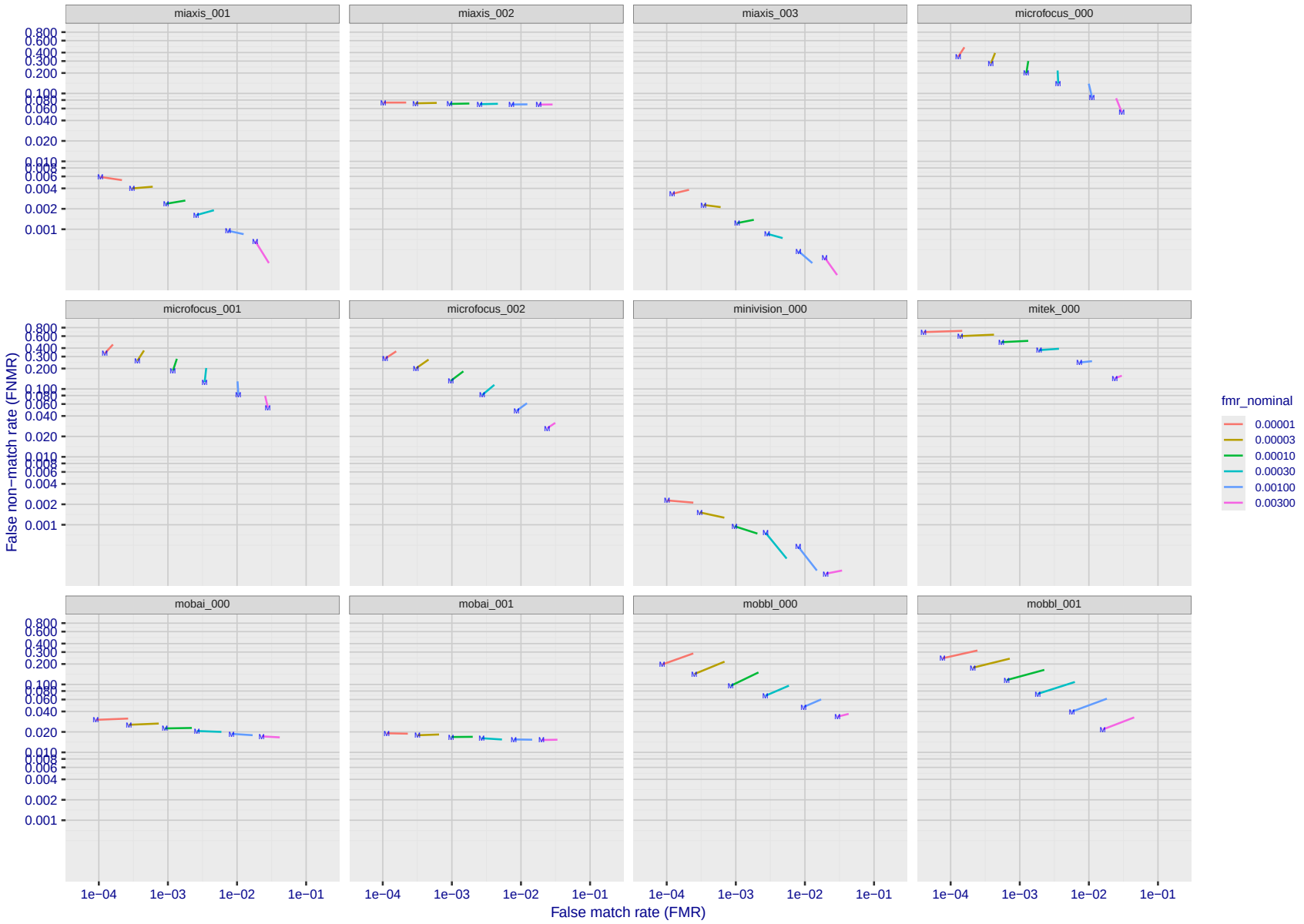


Figure 520: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

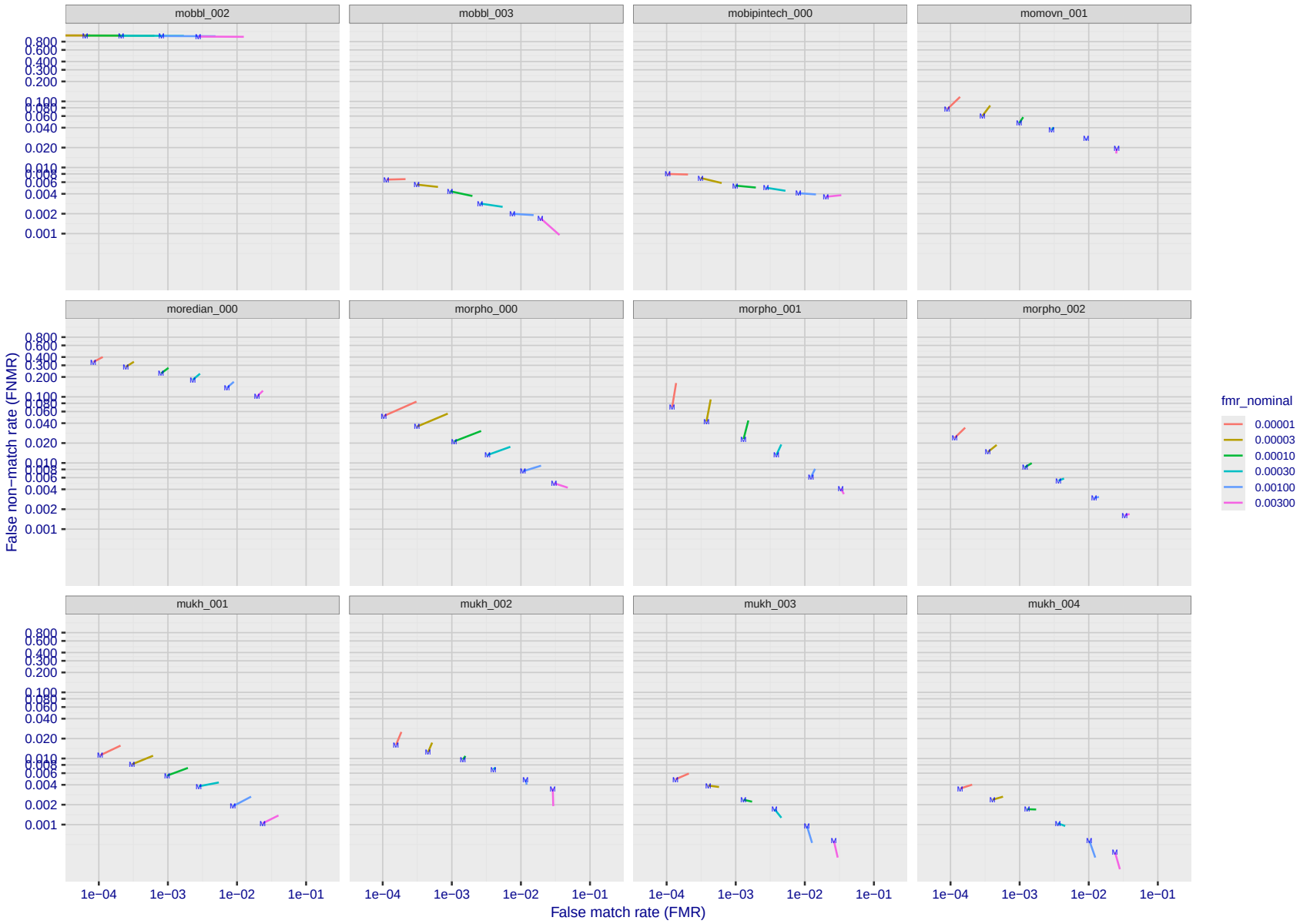


Figure 521: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

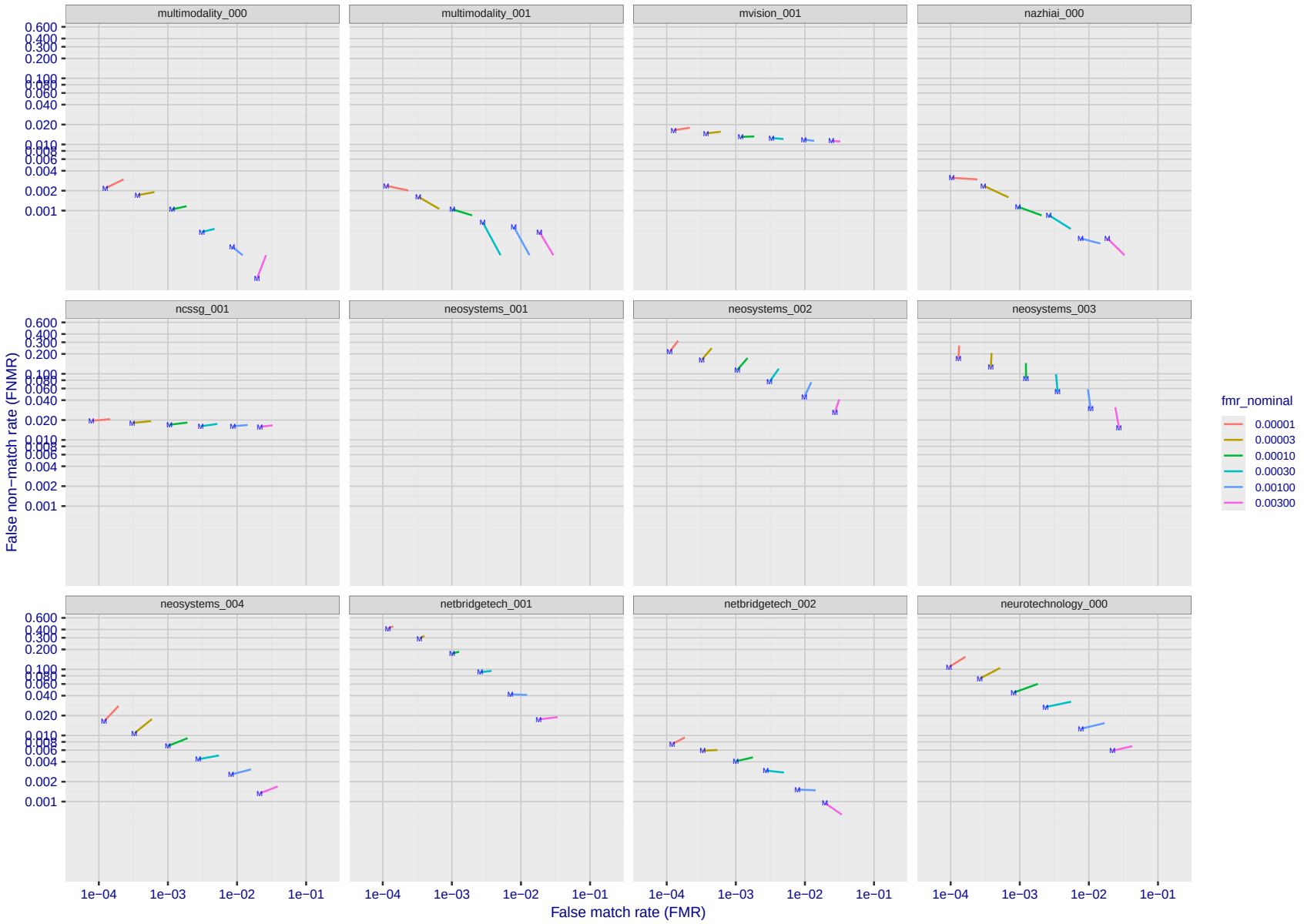


Figure 522: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

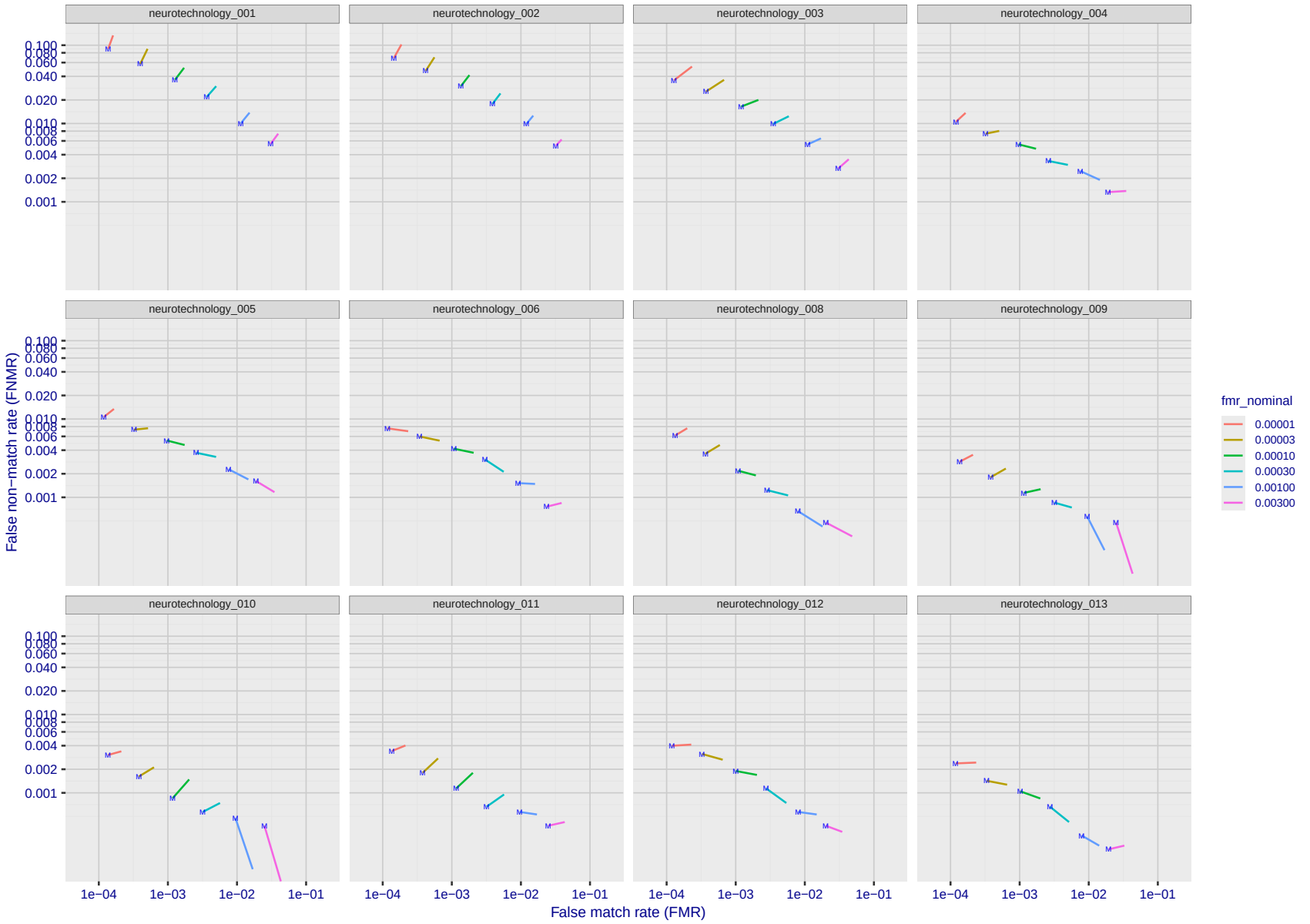


Figure 523: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

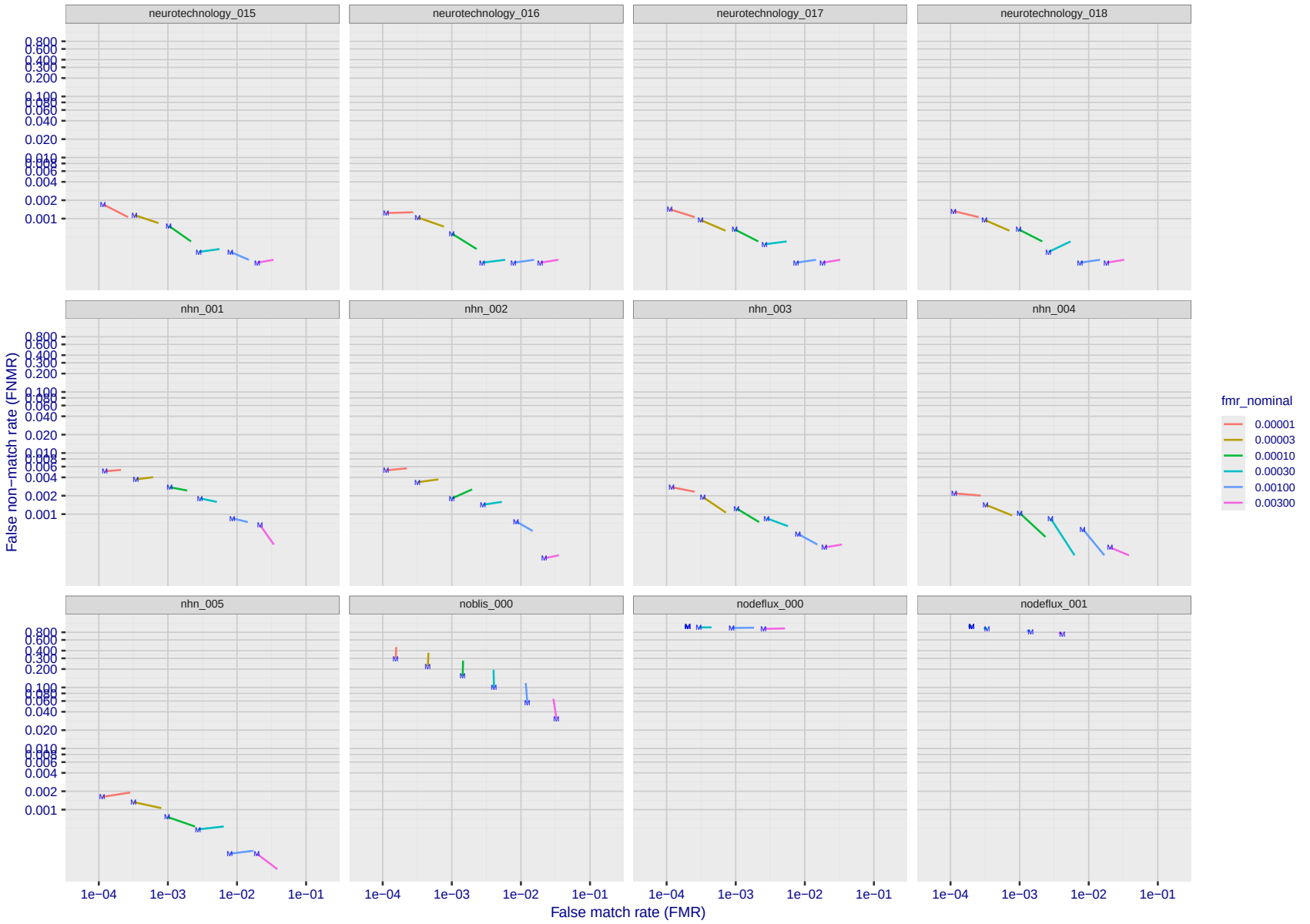


Figure 524: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

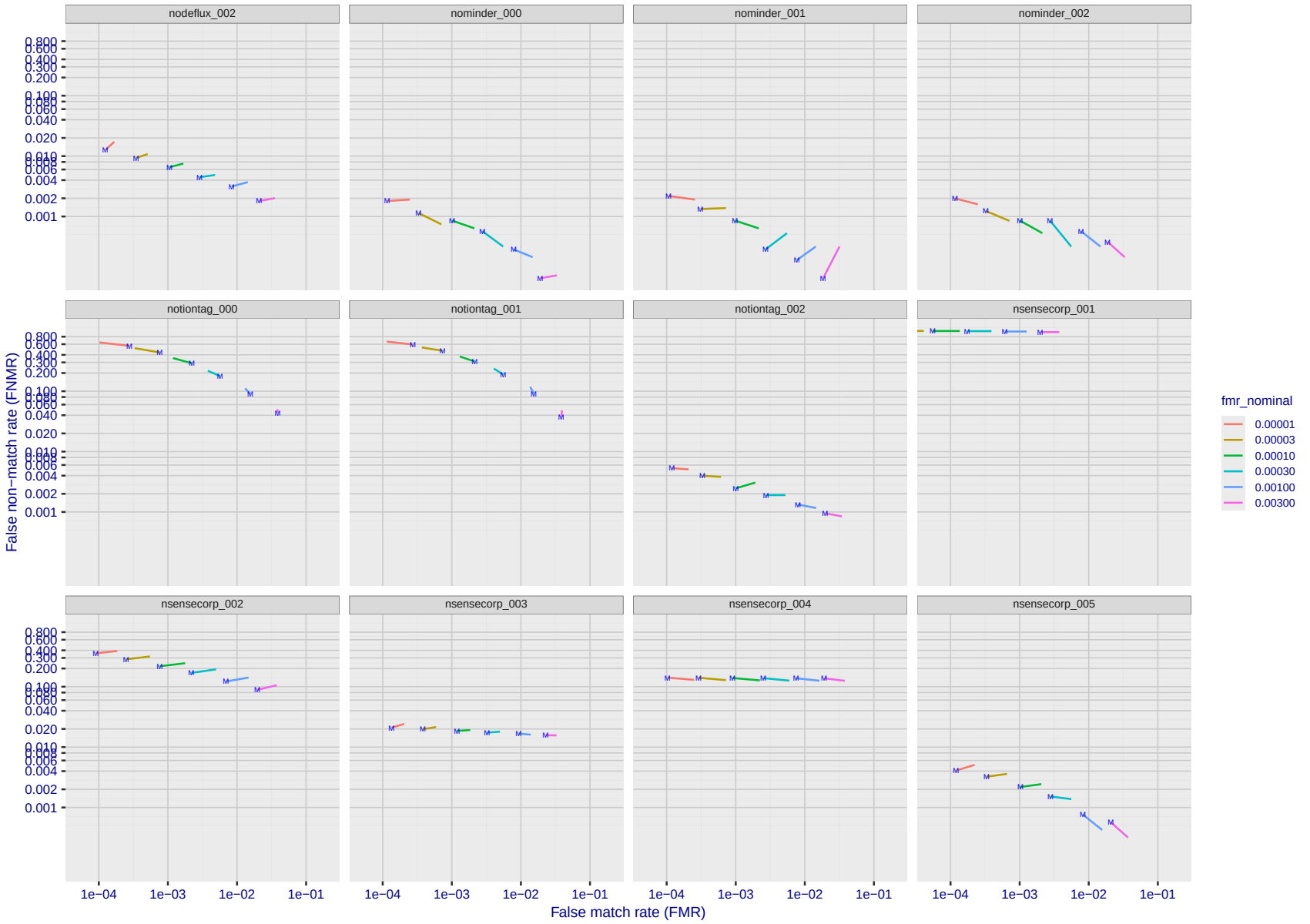


Figure 525: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

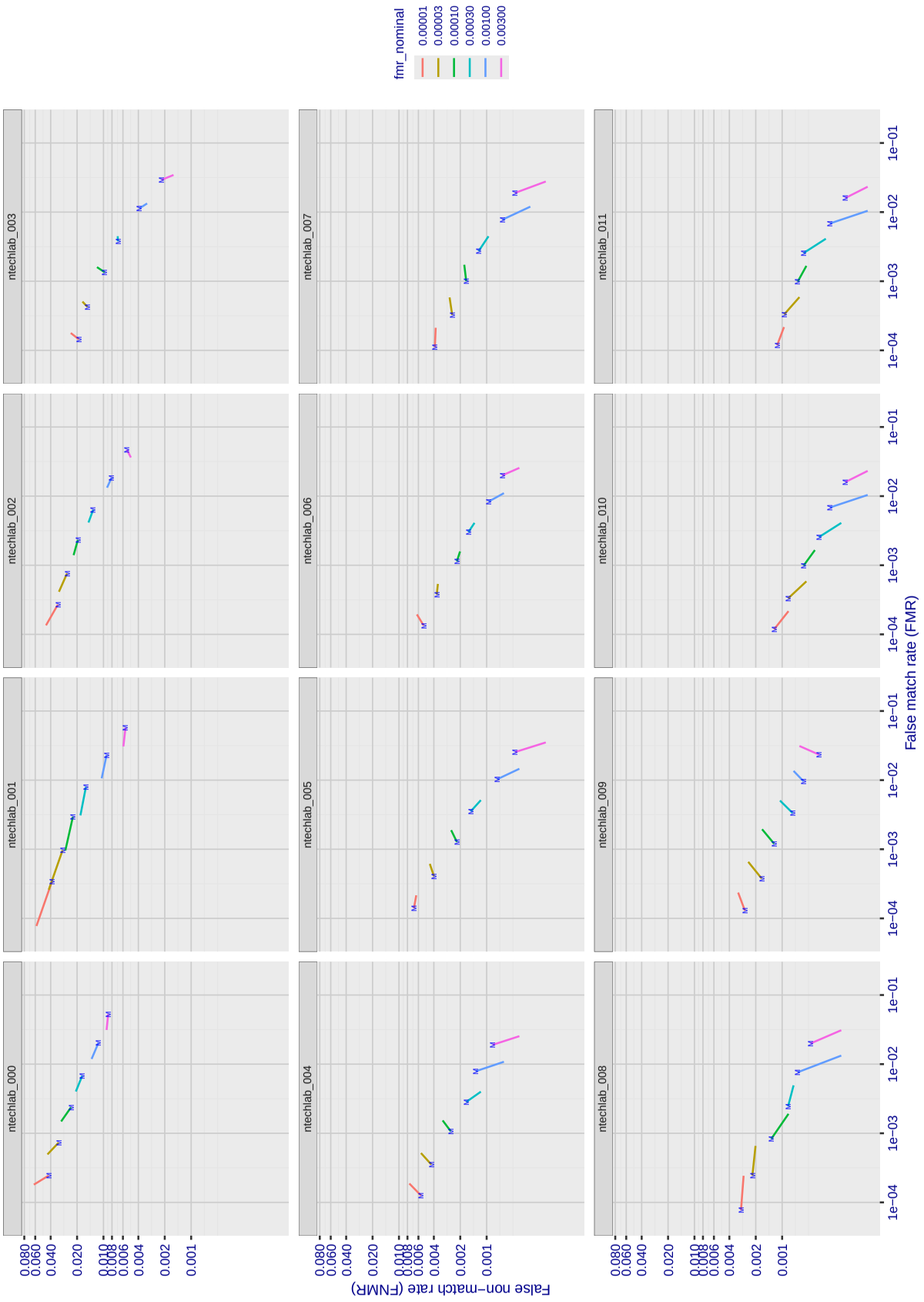


Figure 526: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between $(FMR,FNMR)_{\text{MALE}}$ and $(FMR,FNMR)_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

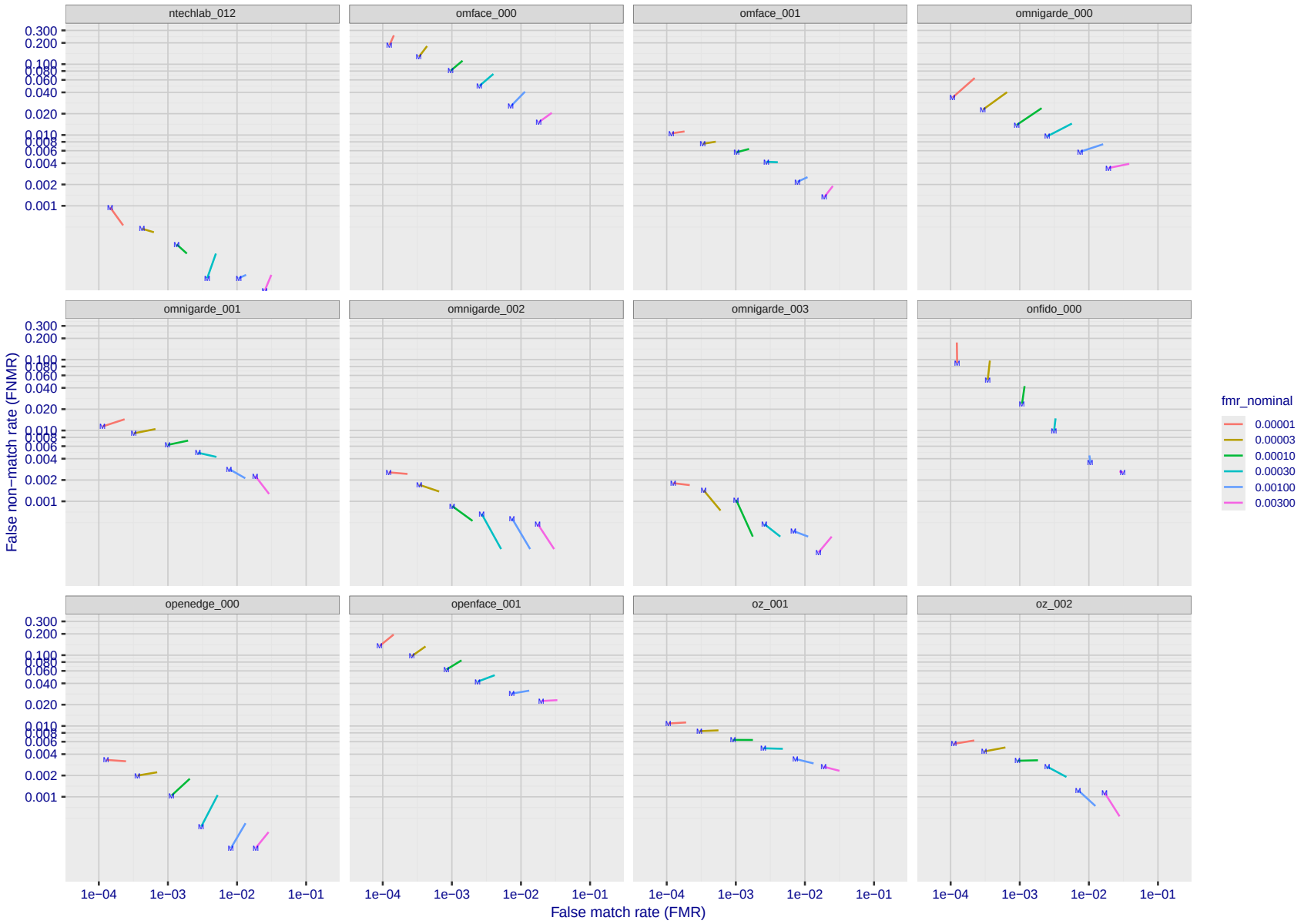


Figure 527: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

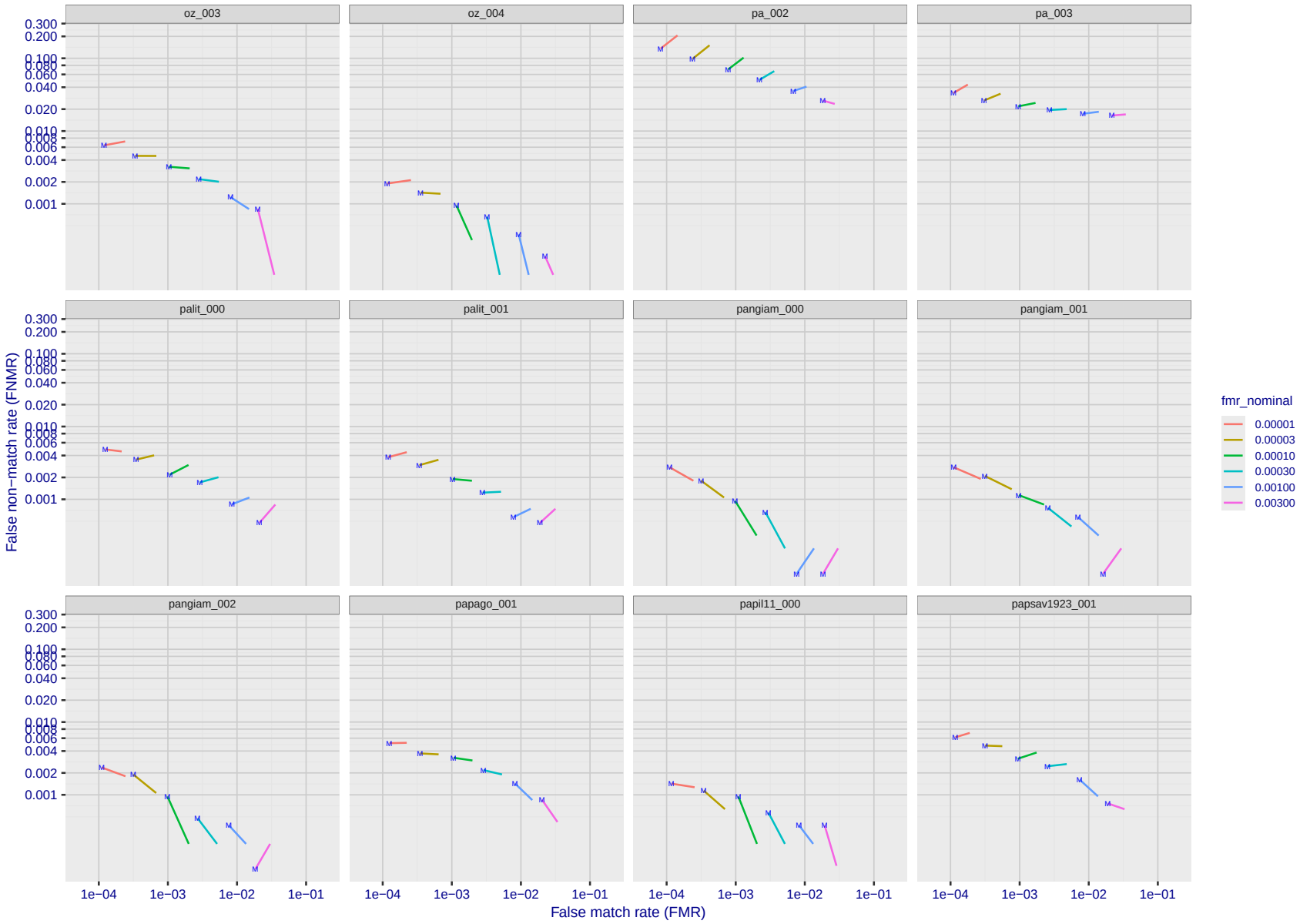


Figure 528: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

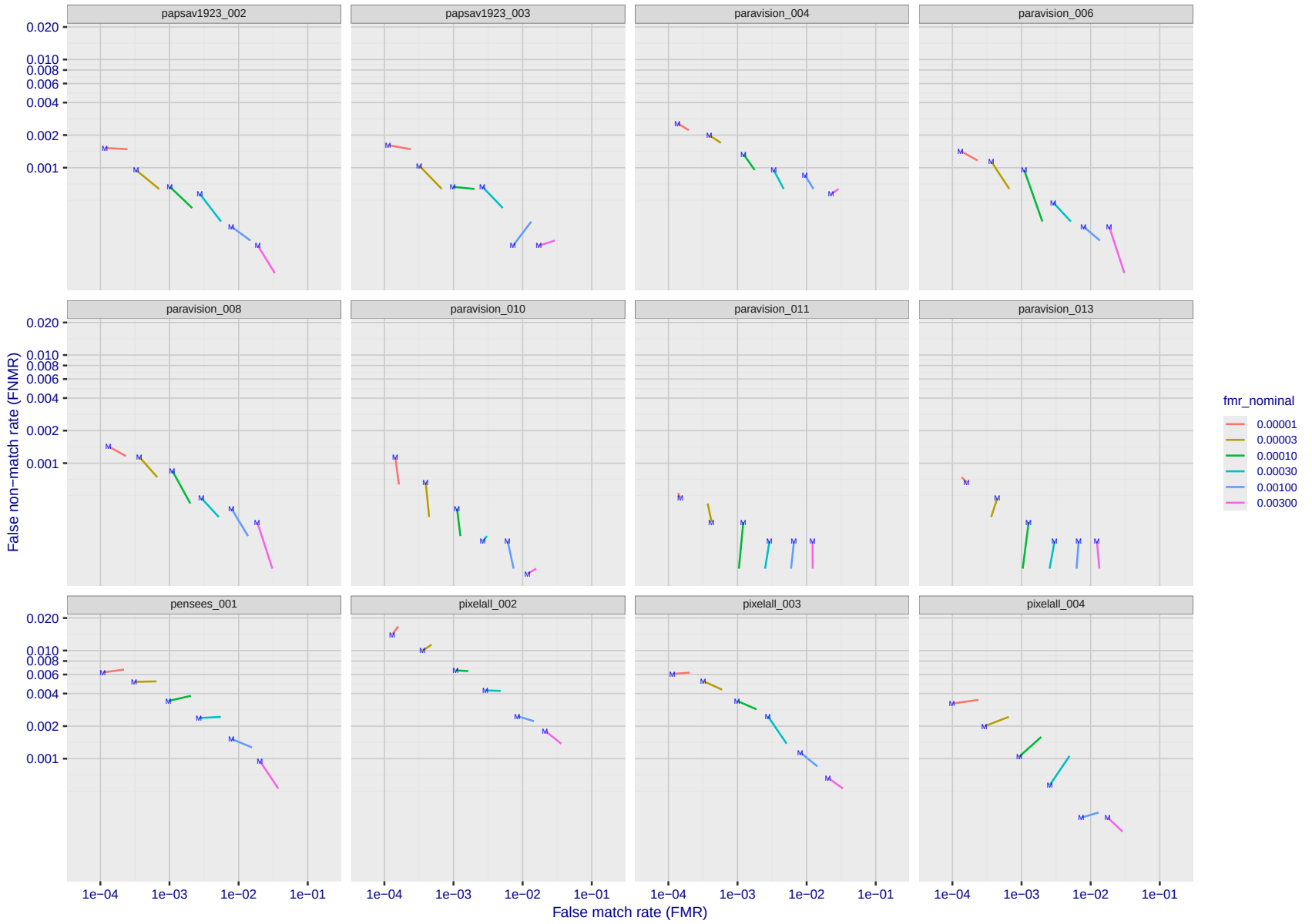


Figure 529: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

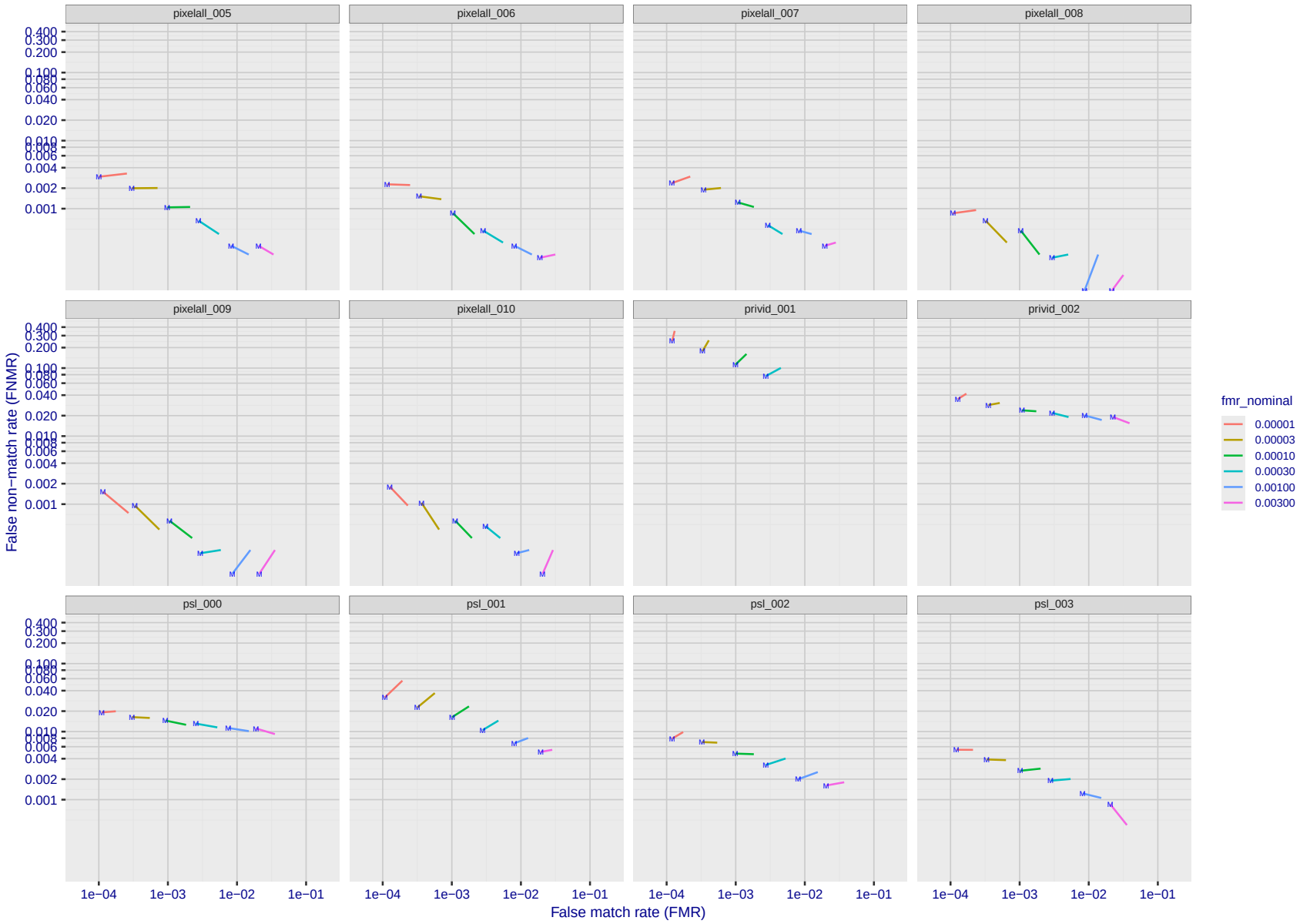


Figure 530: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

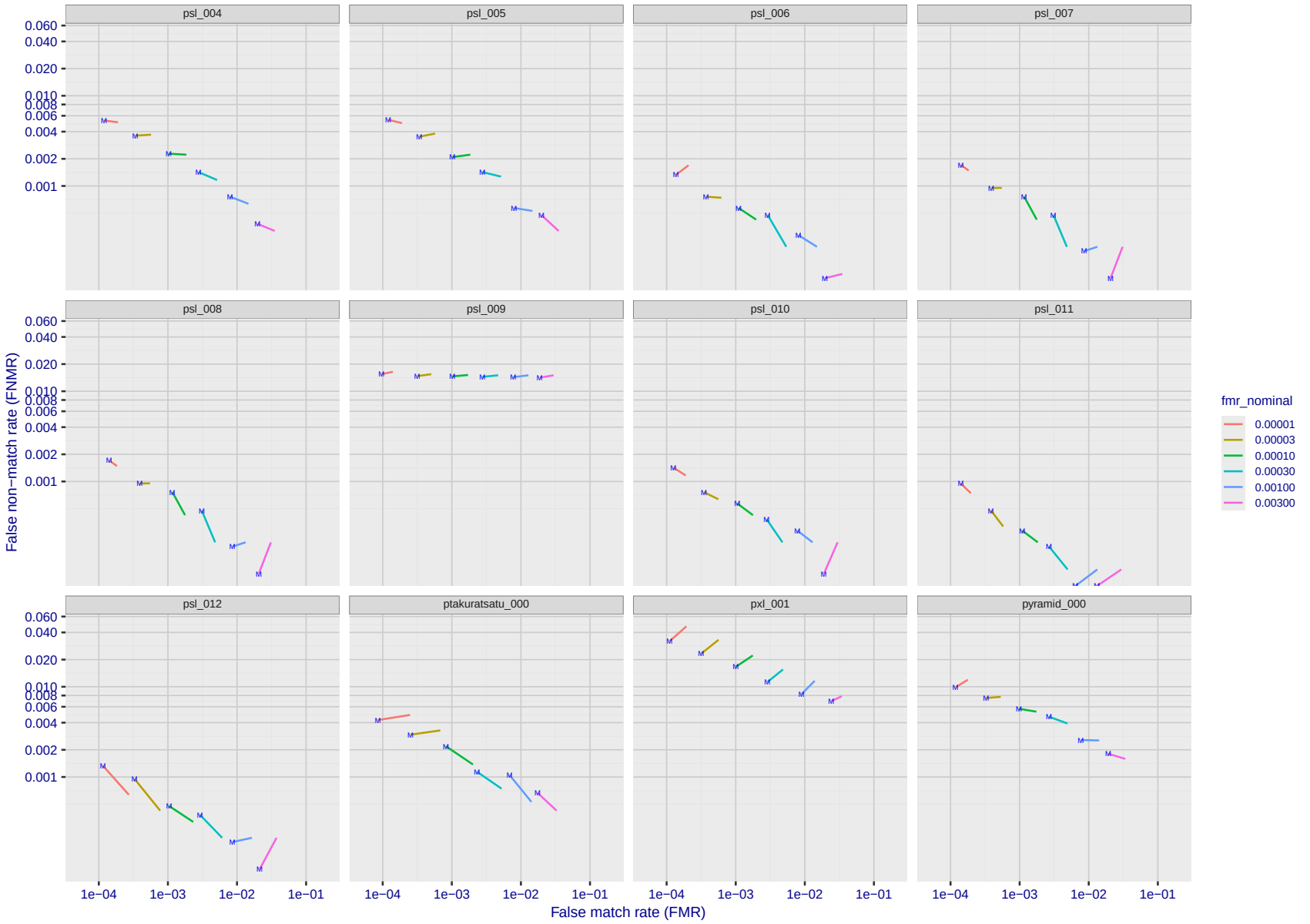


Figure 531: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

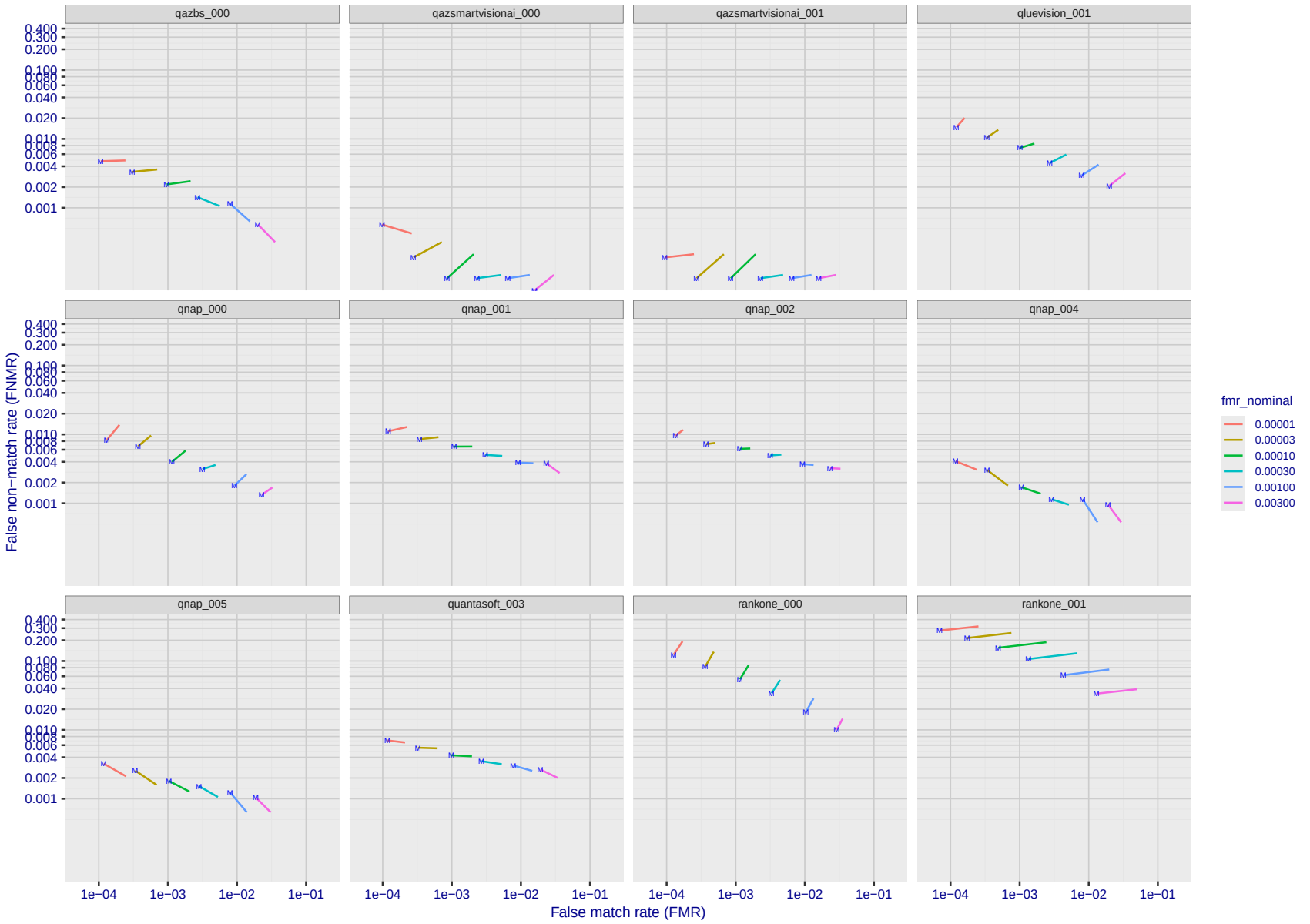


Figure 532: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

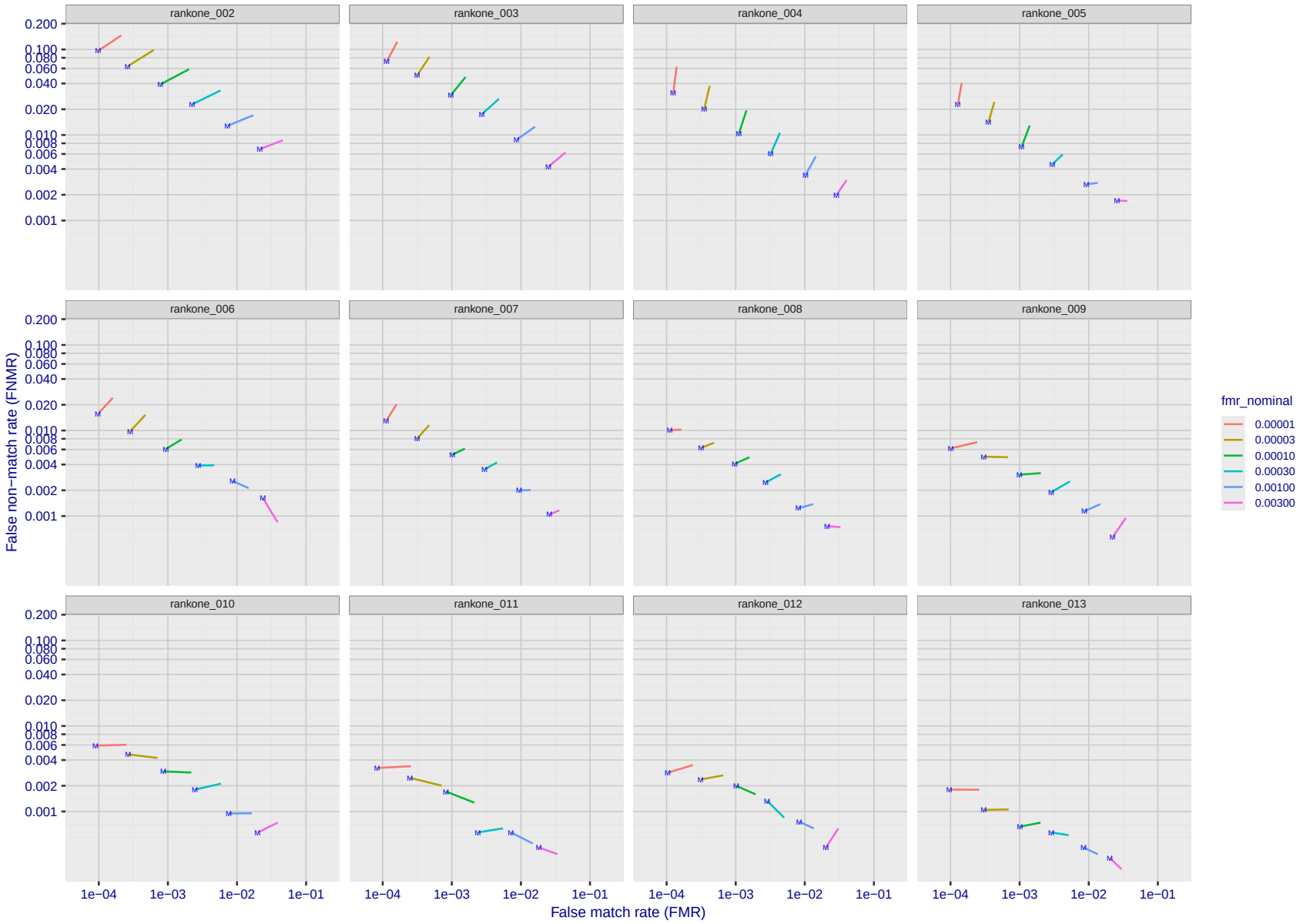


Figure 533: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

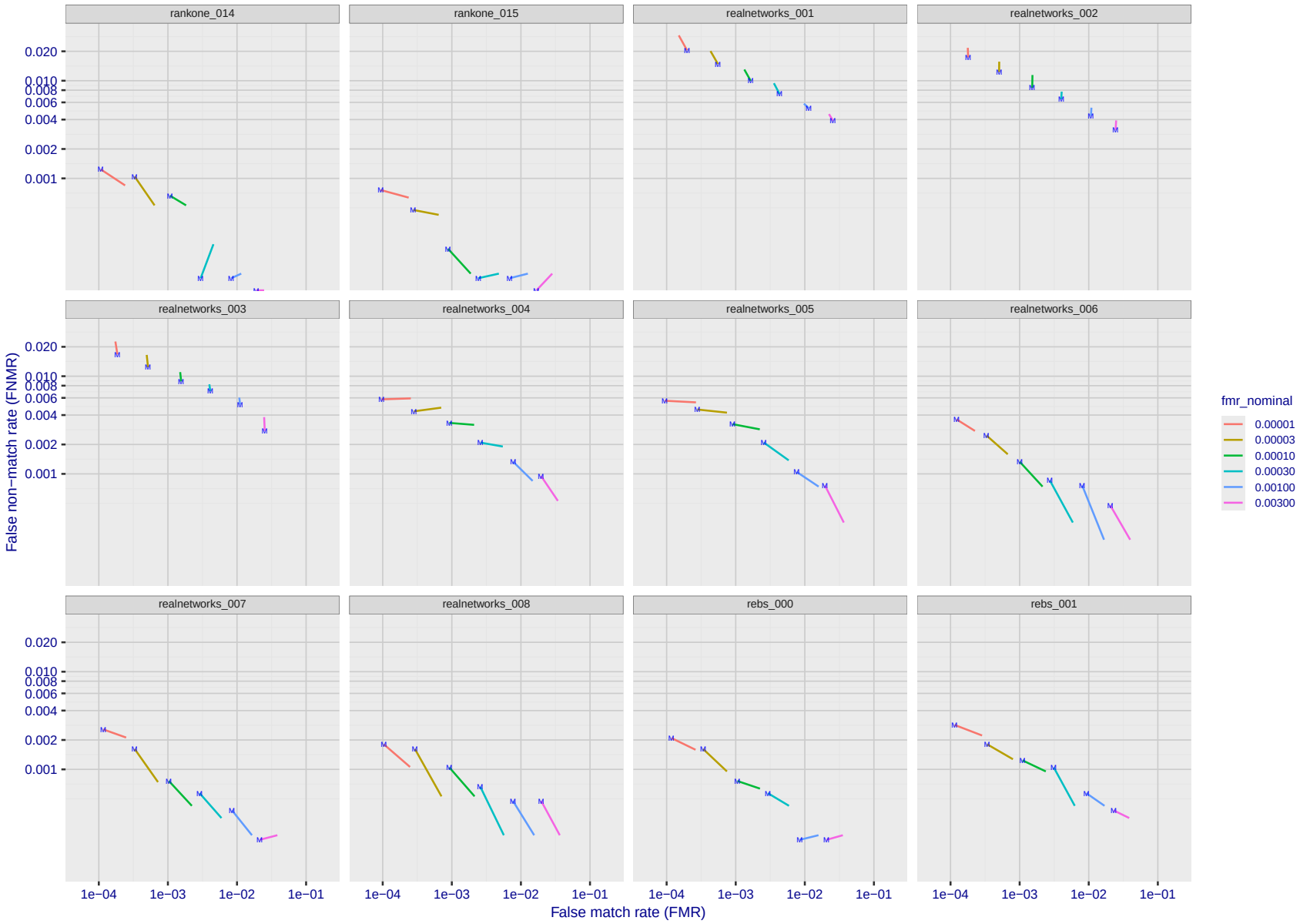


Figure 534: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

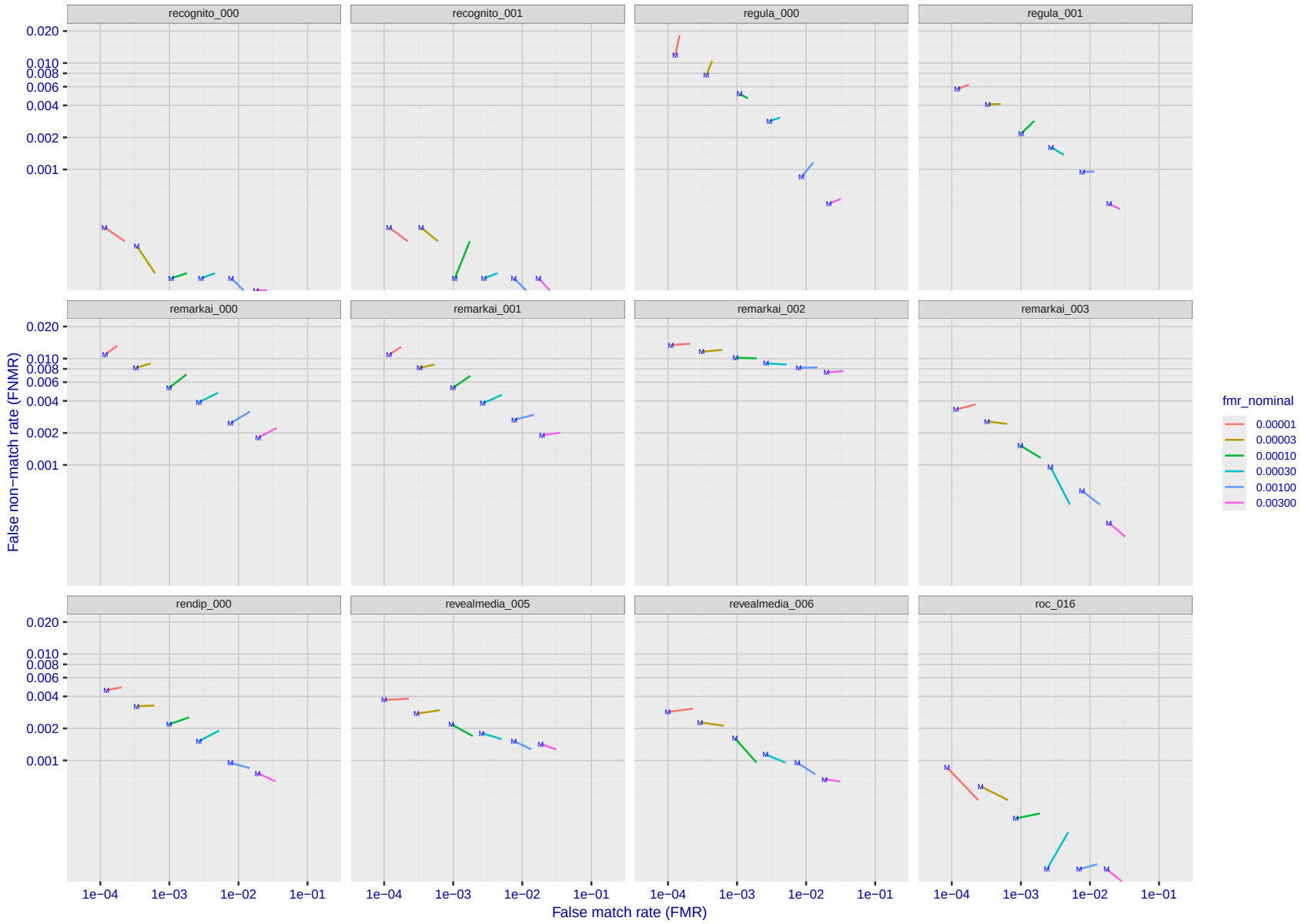


Figure 535: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

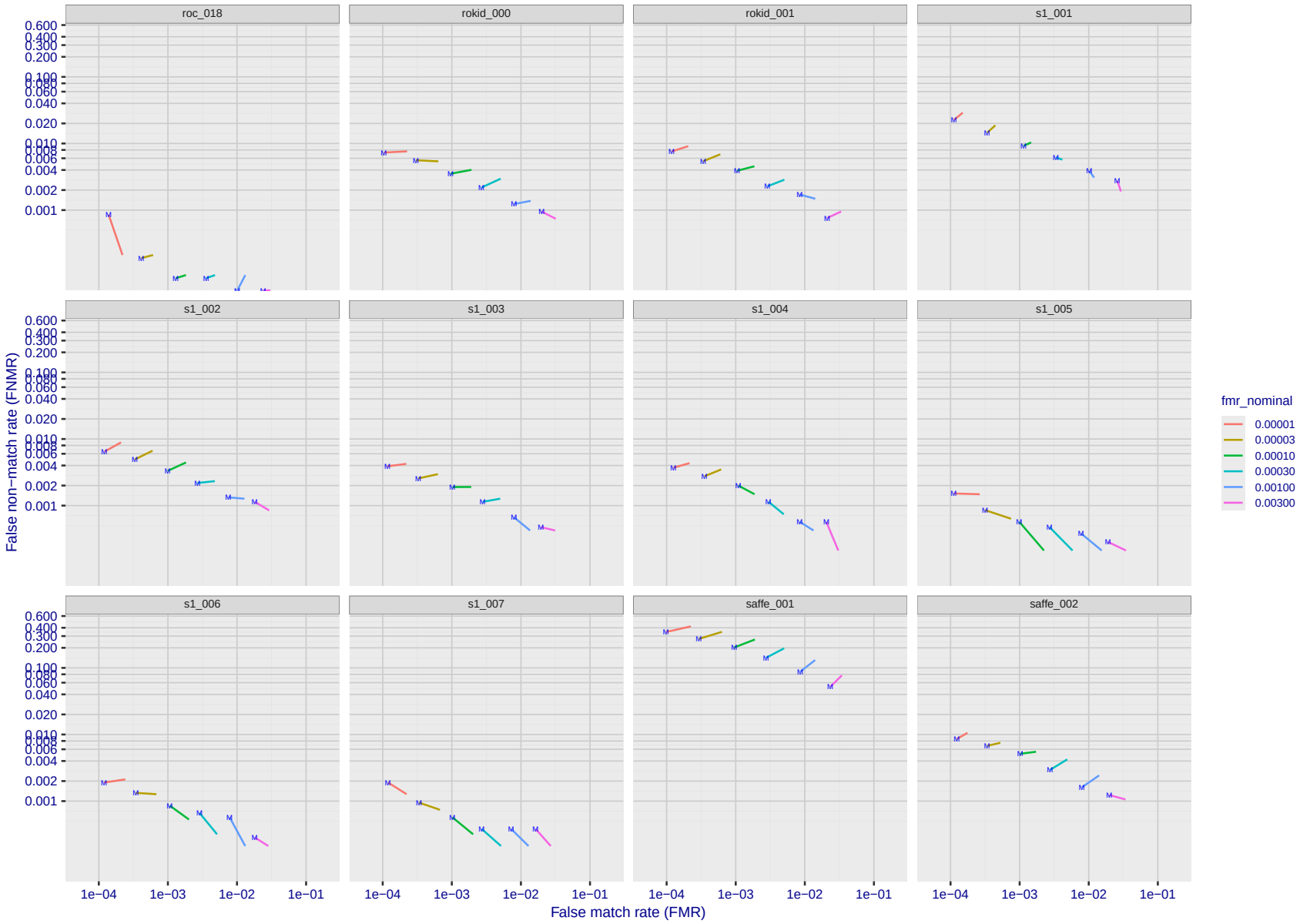


Figure 536: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

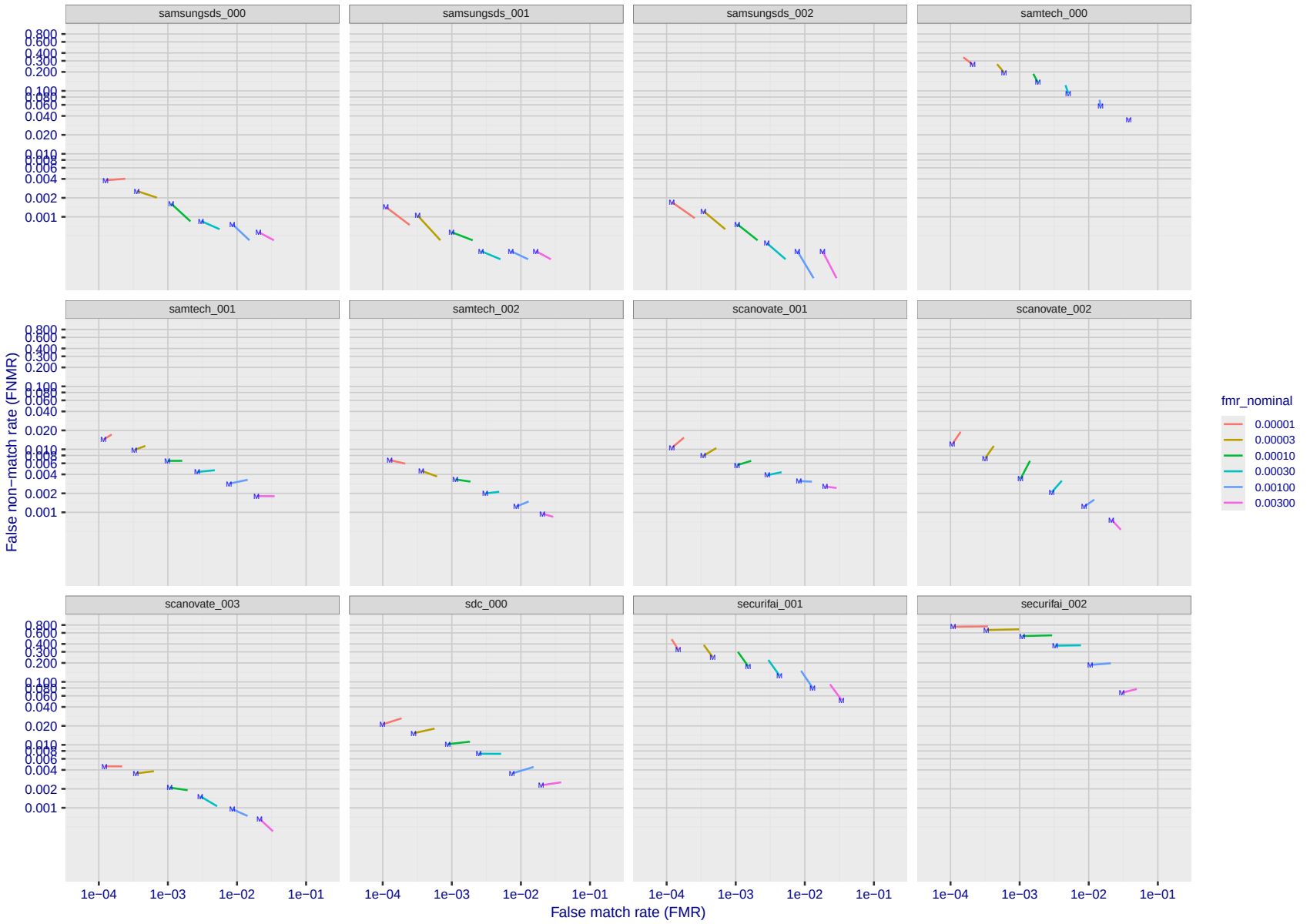


Figure 537: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

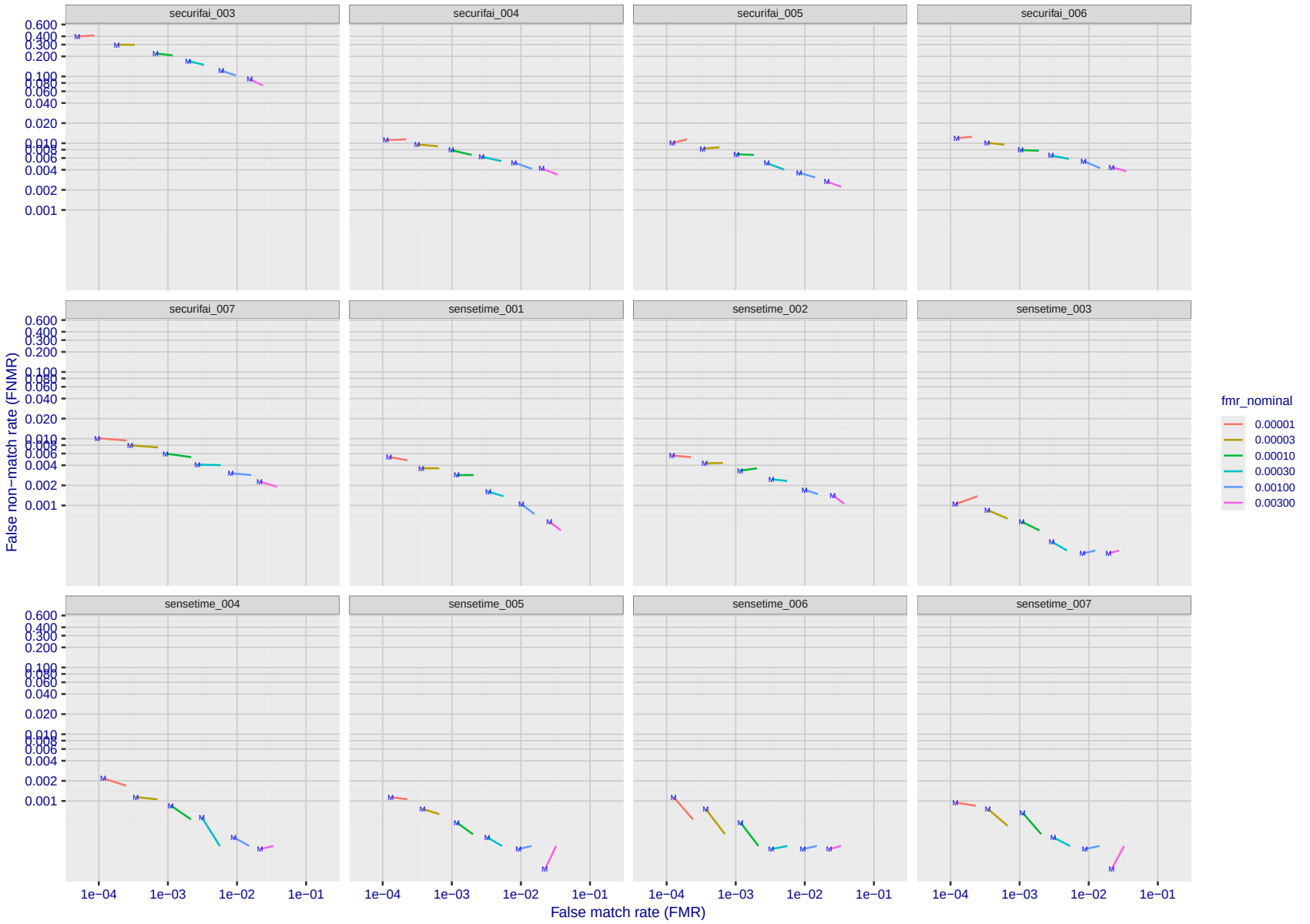


Figure 538: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

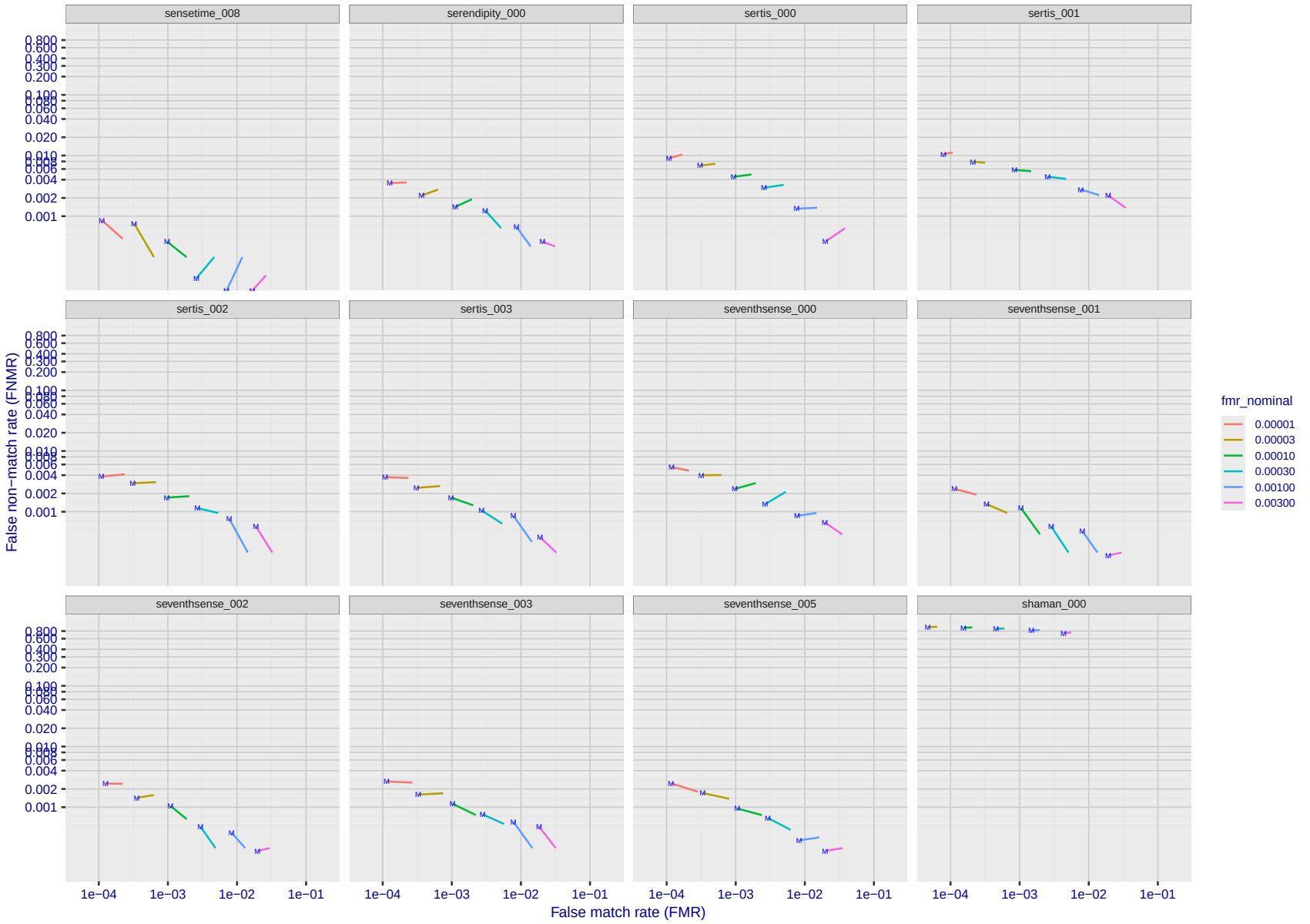


Figure 539: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

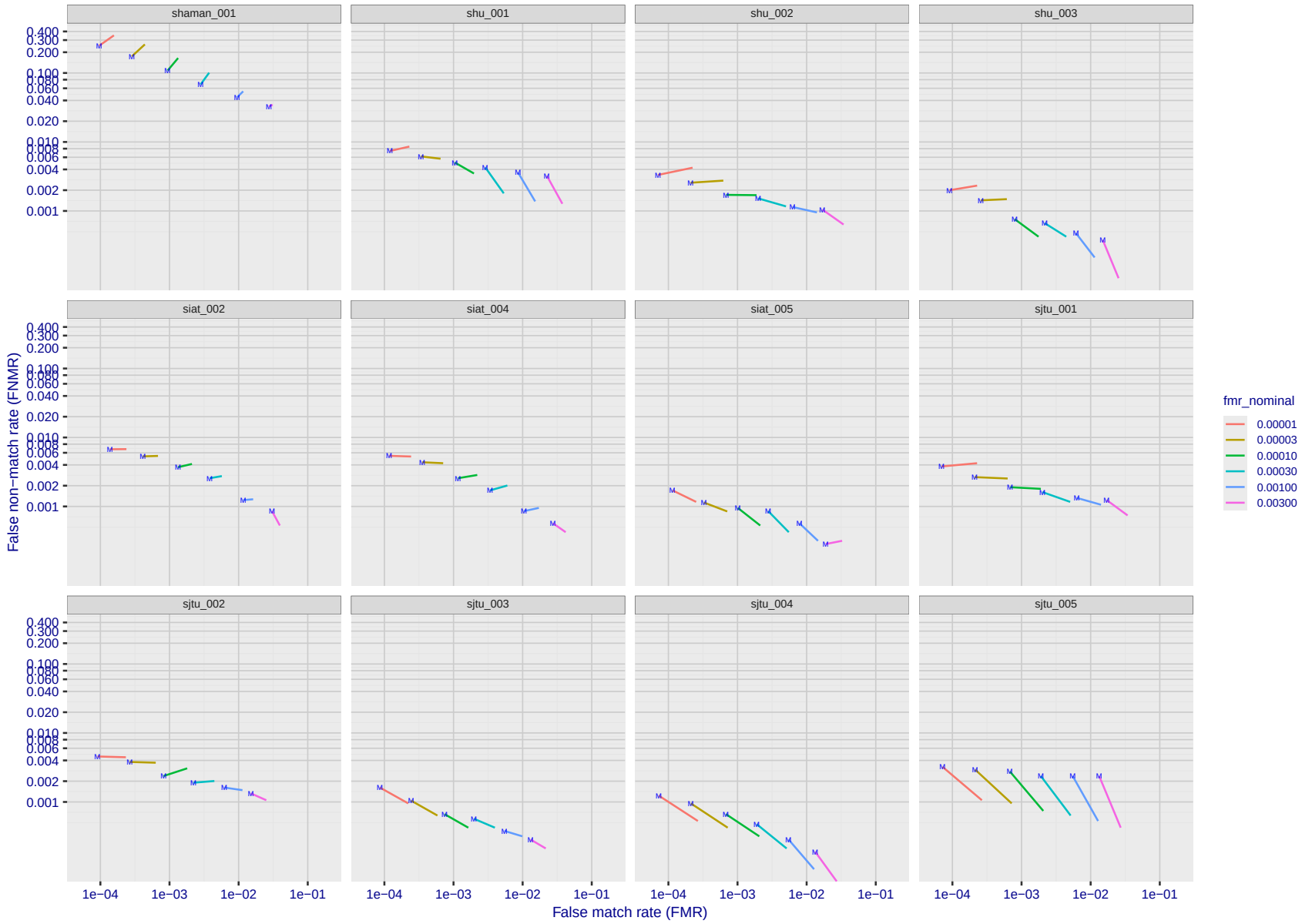


Figure 540: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

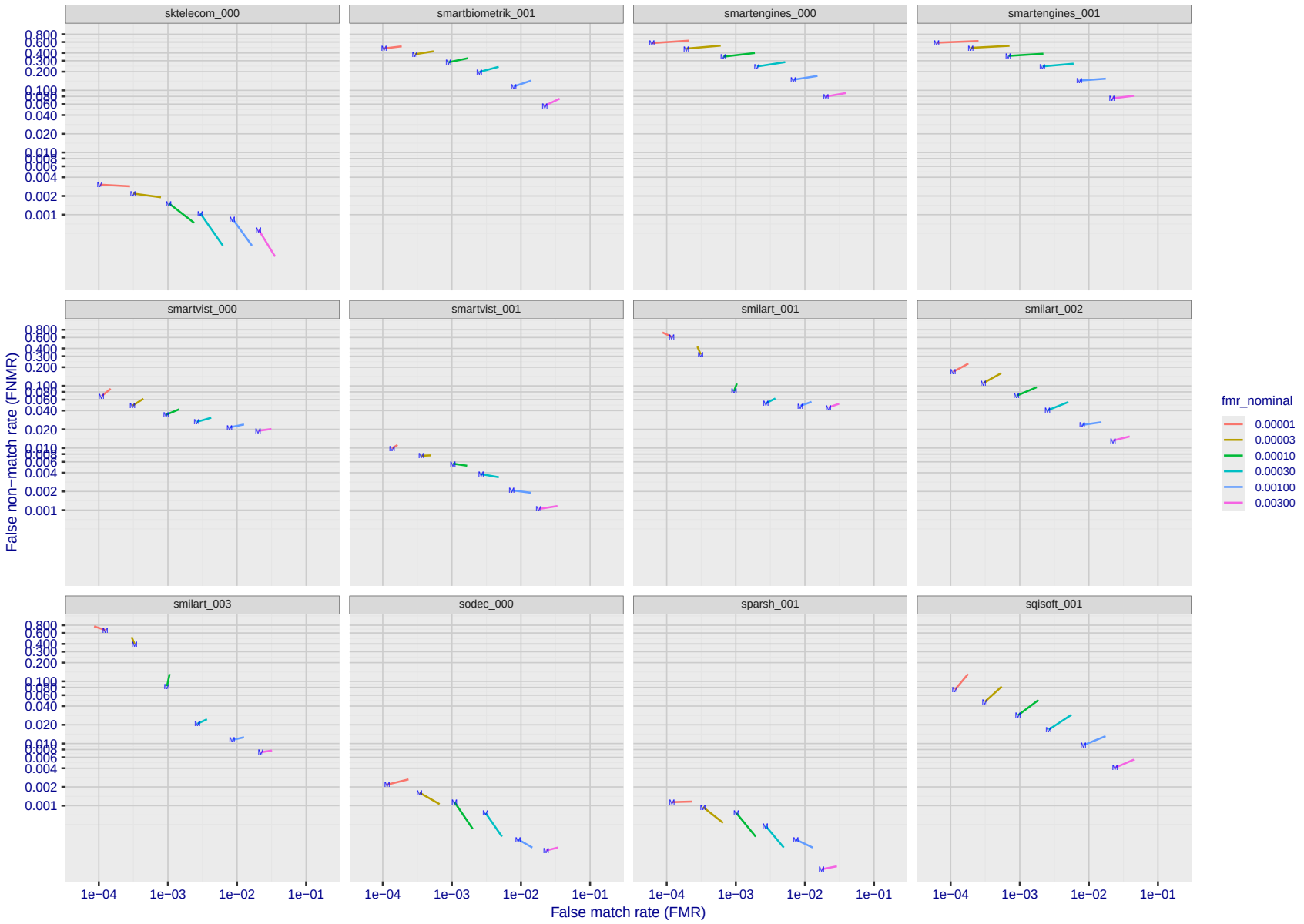


Figure 541: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

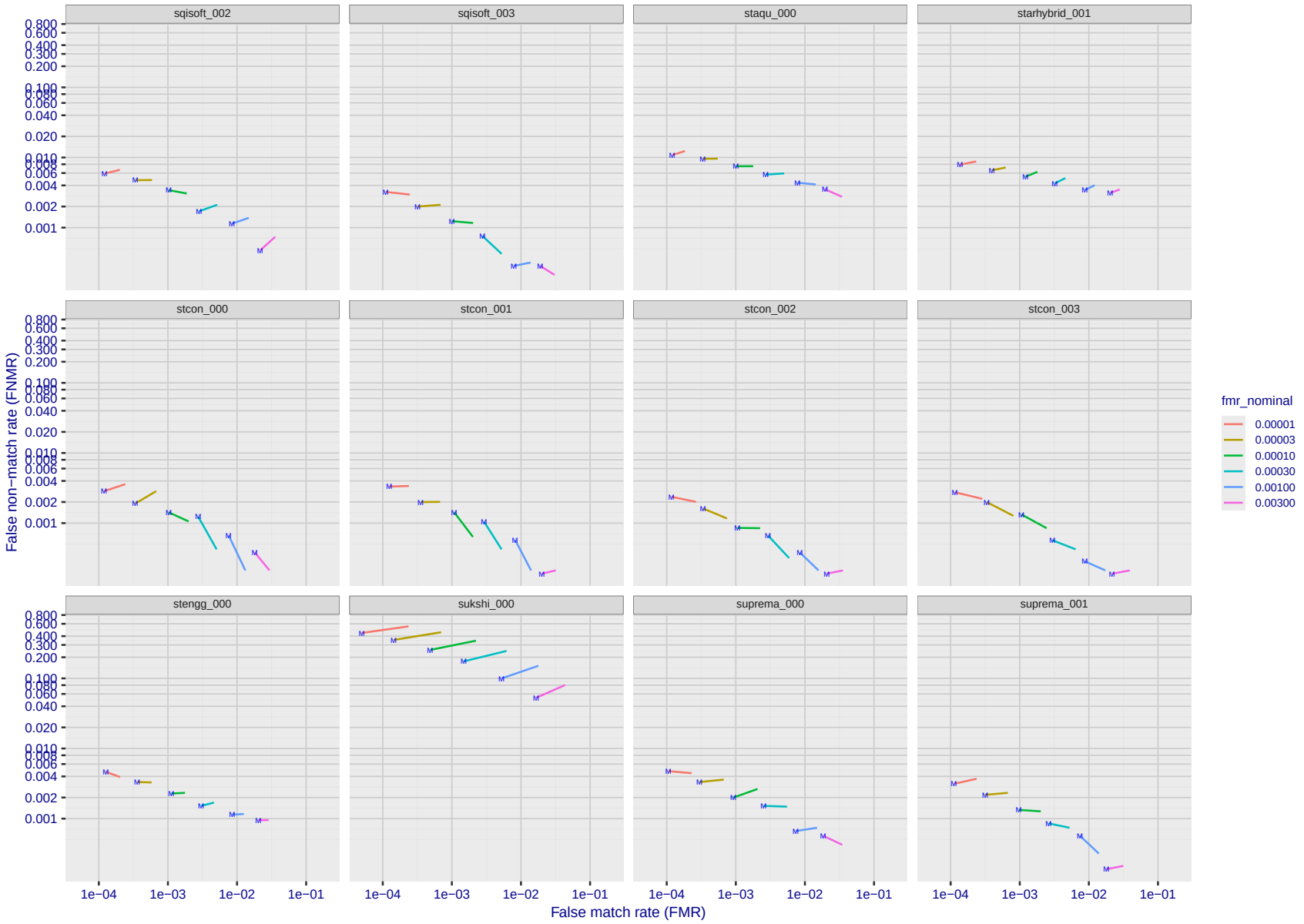


Figure 542: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

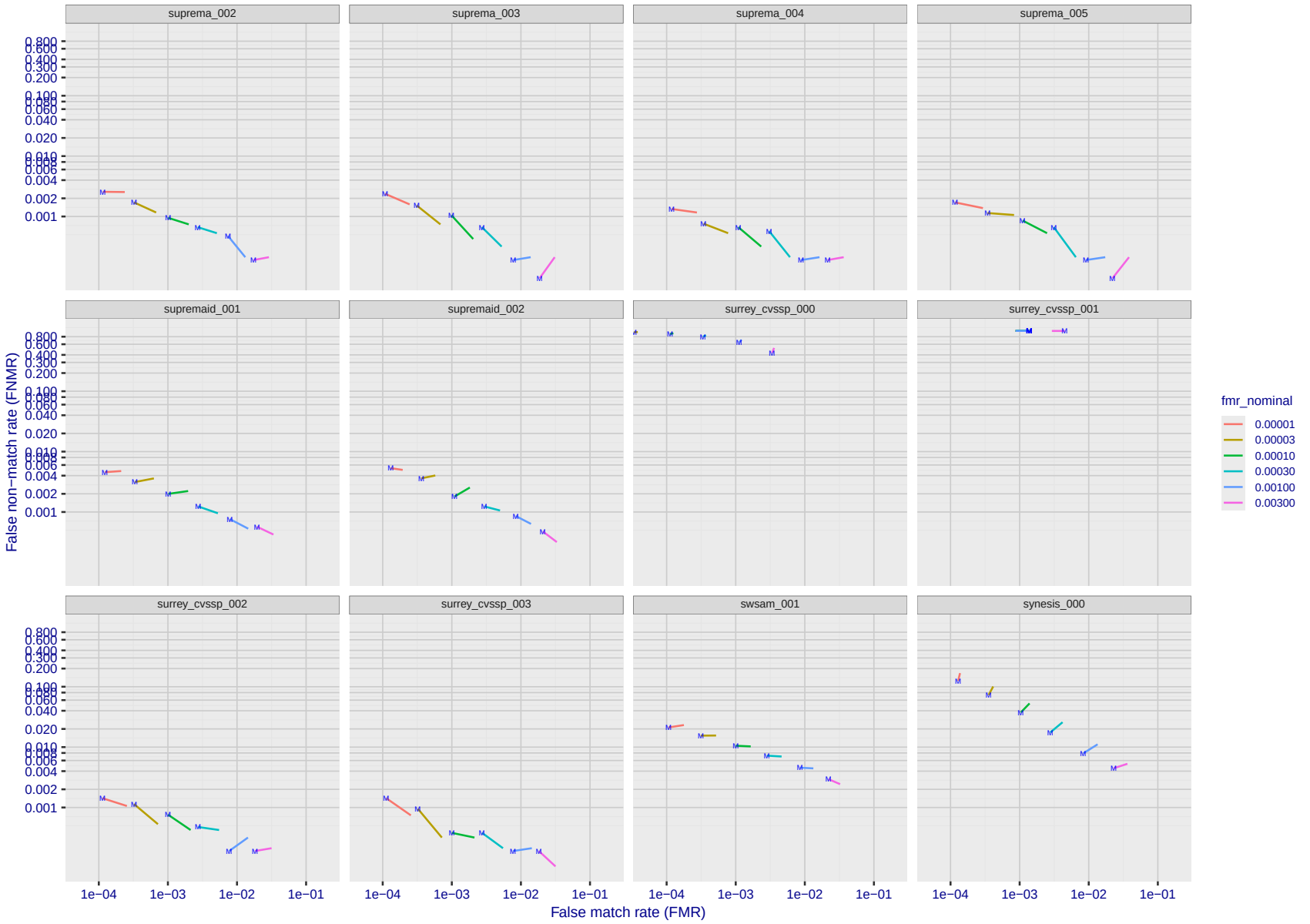


Figure 543: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

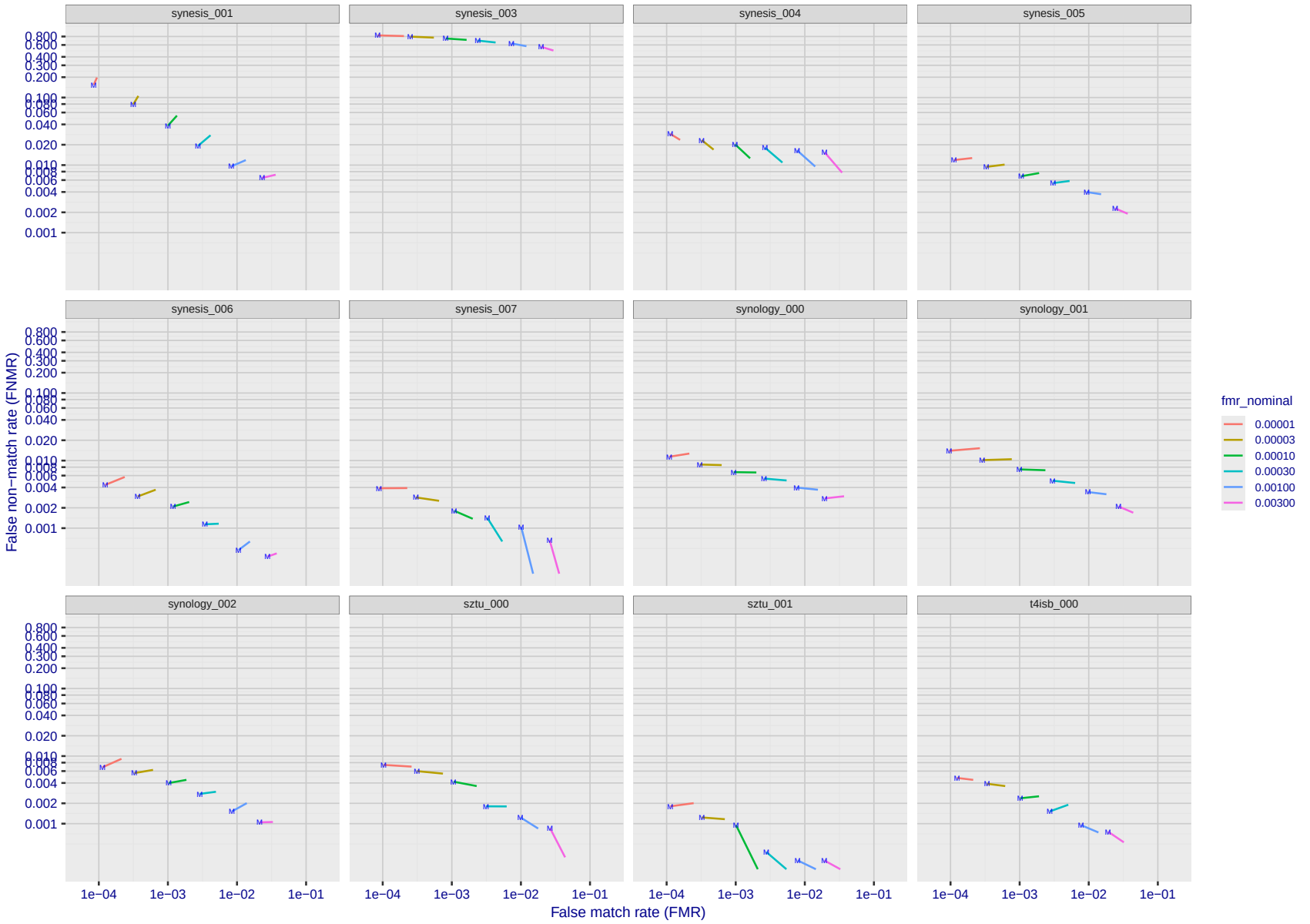


Figure 544: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

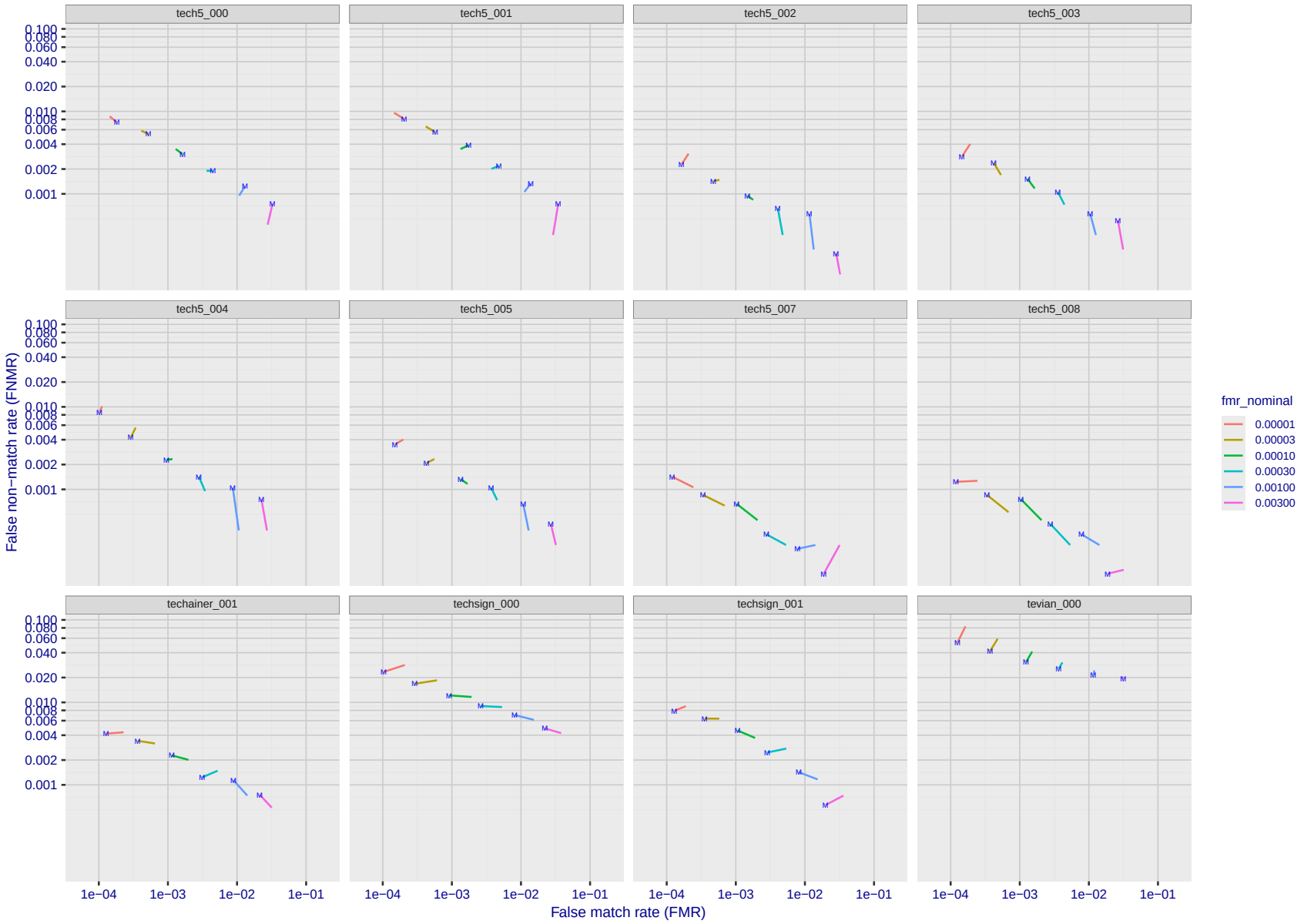


Figure 545: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

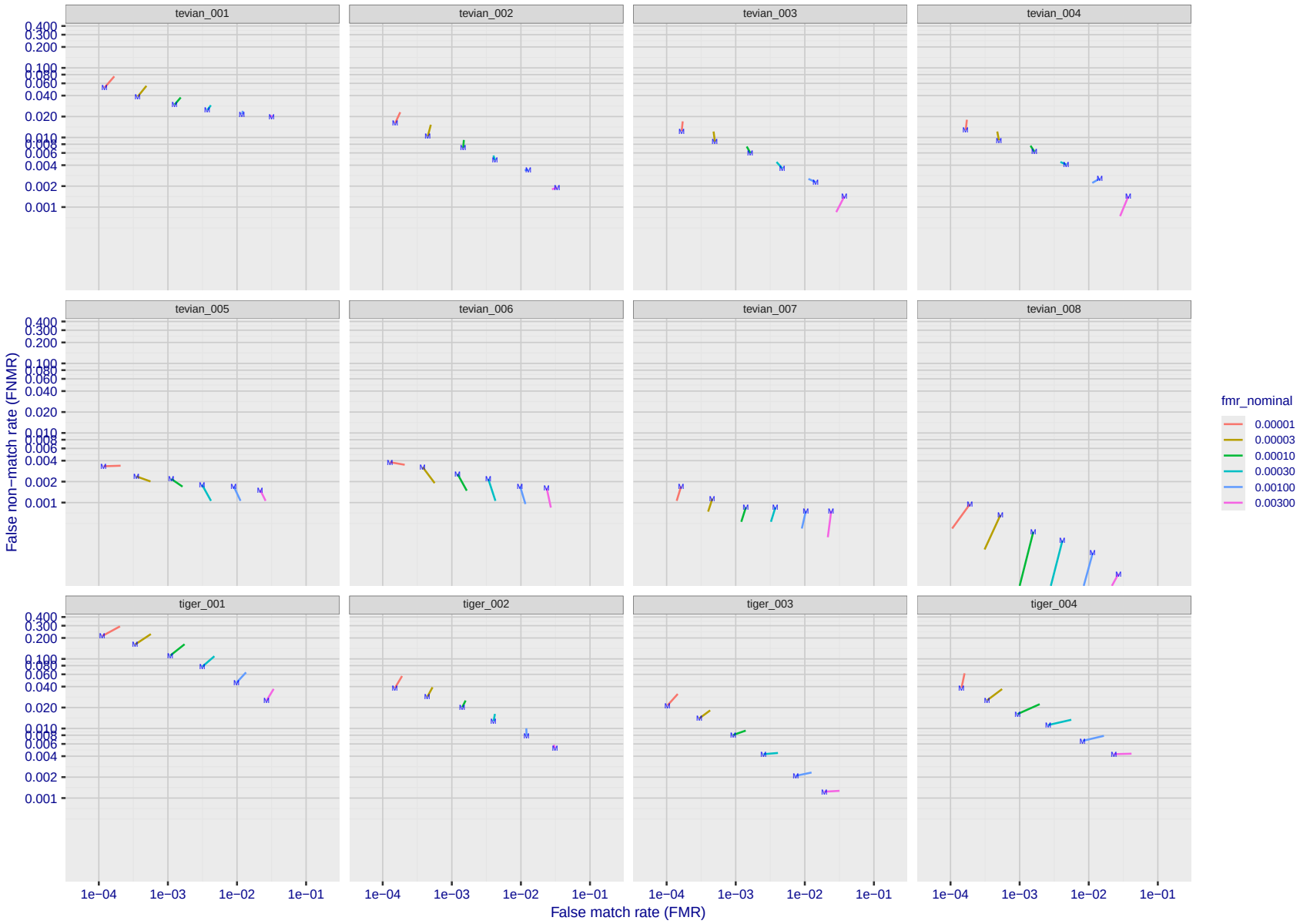


Figure 546: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

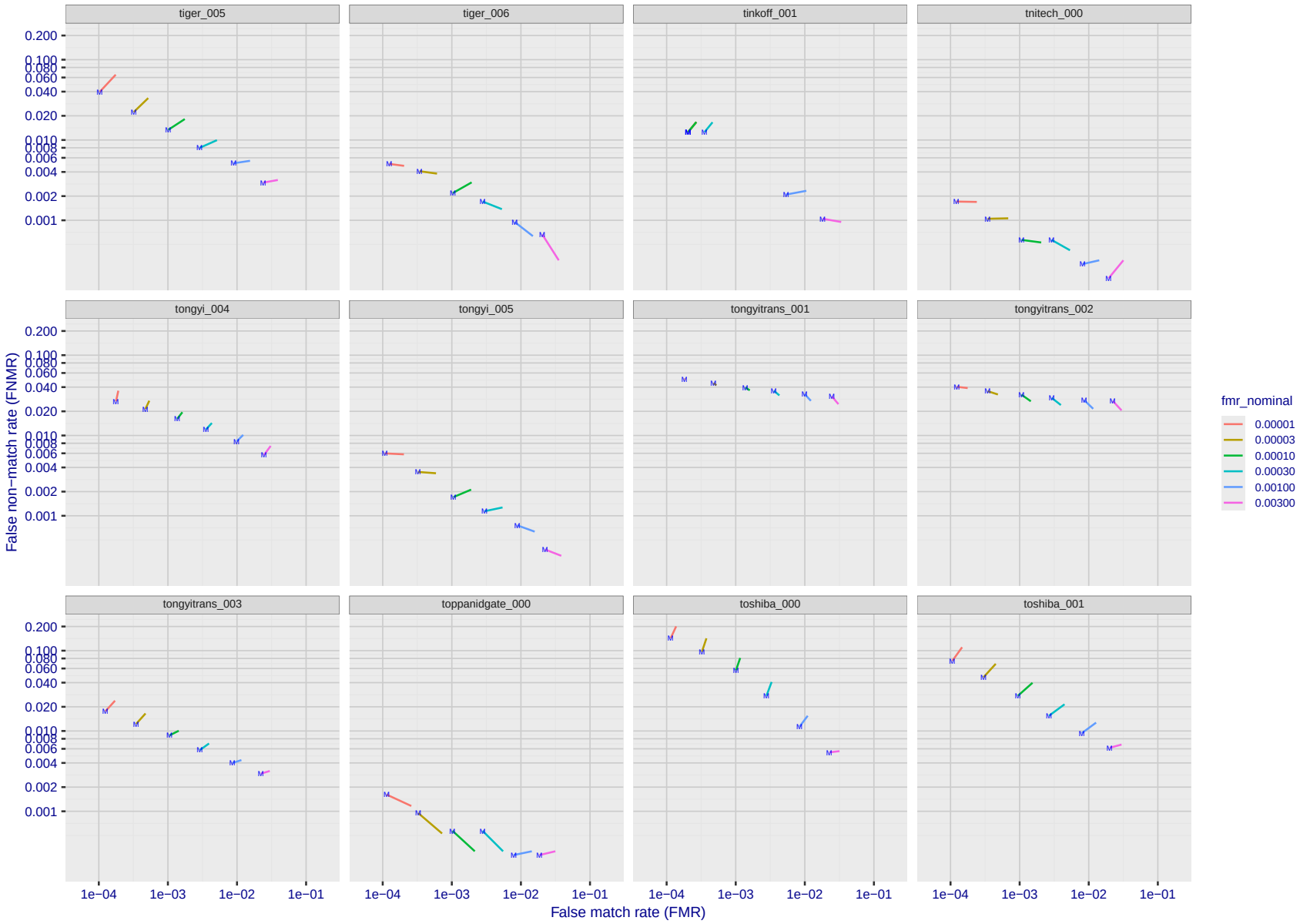


Figure 547: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

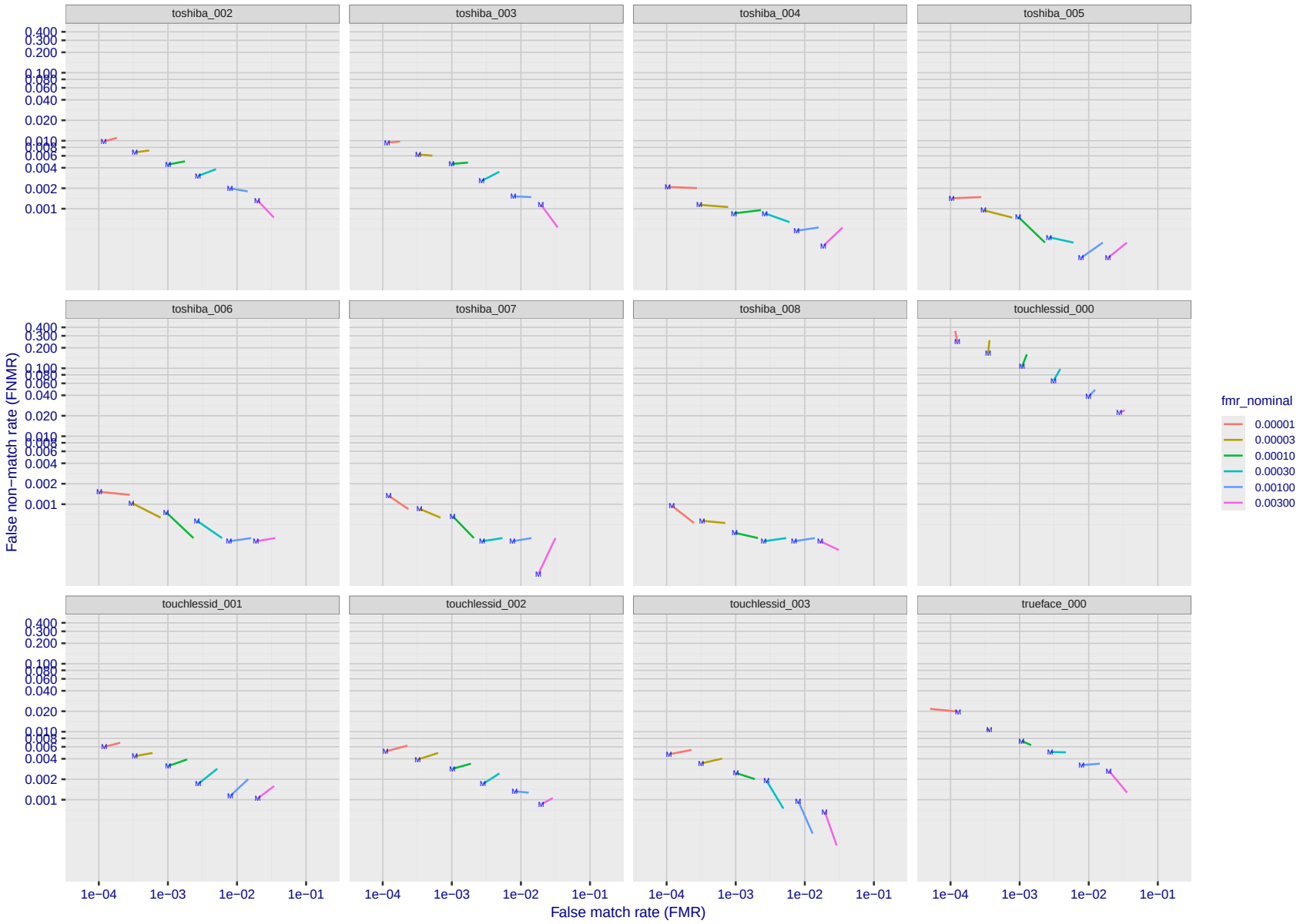


Figure 548: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

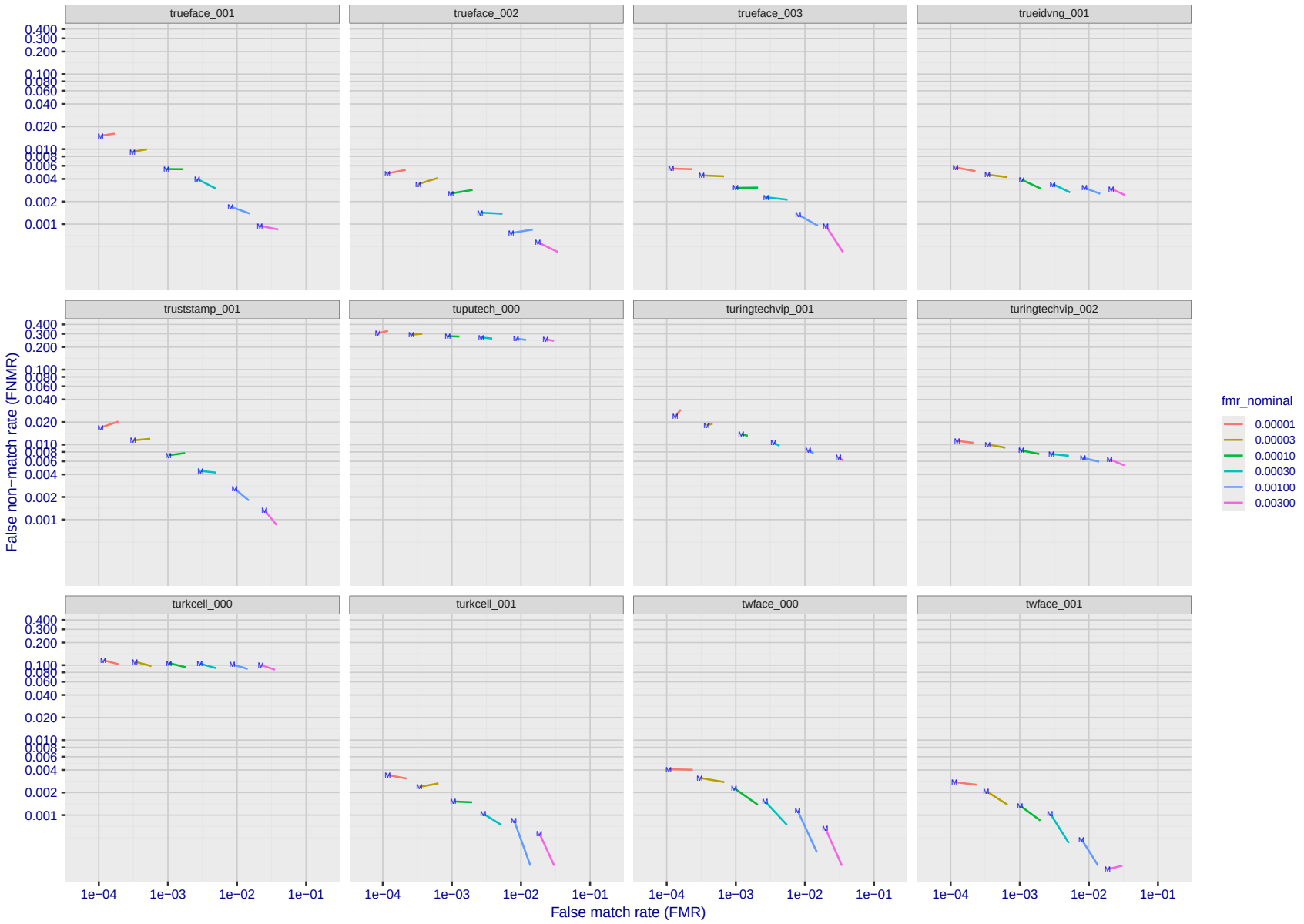


Figure 549: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

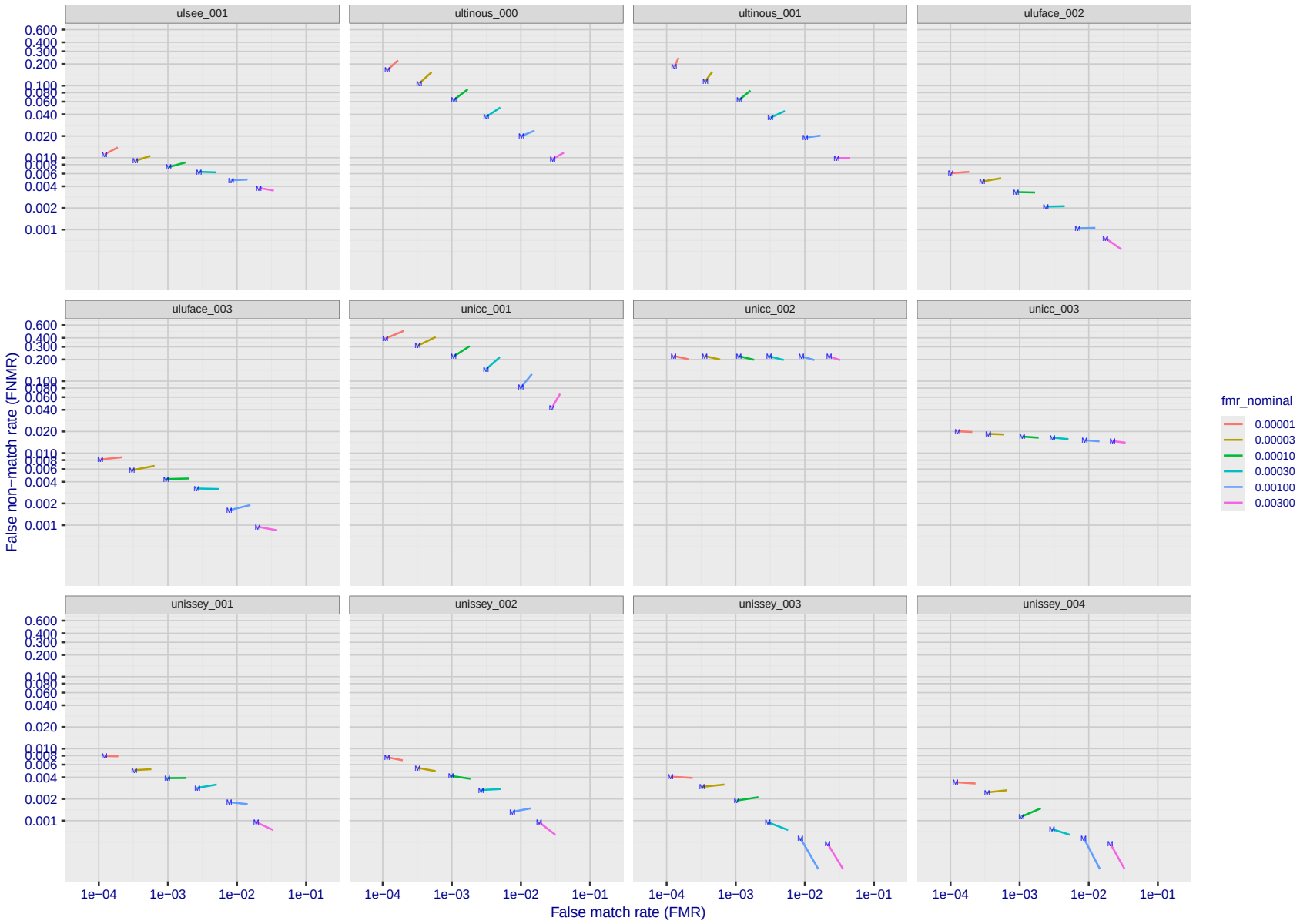


Figure 550: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

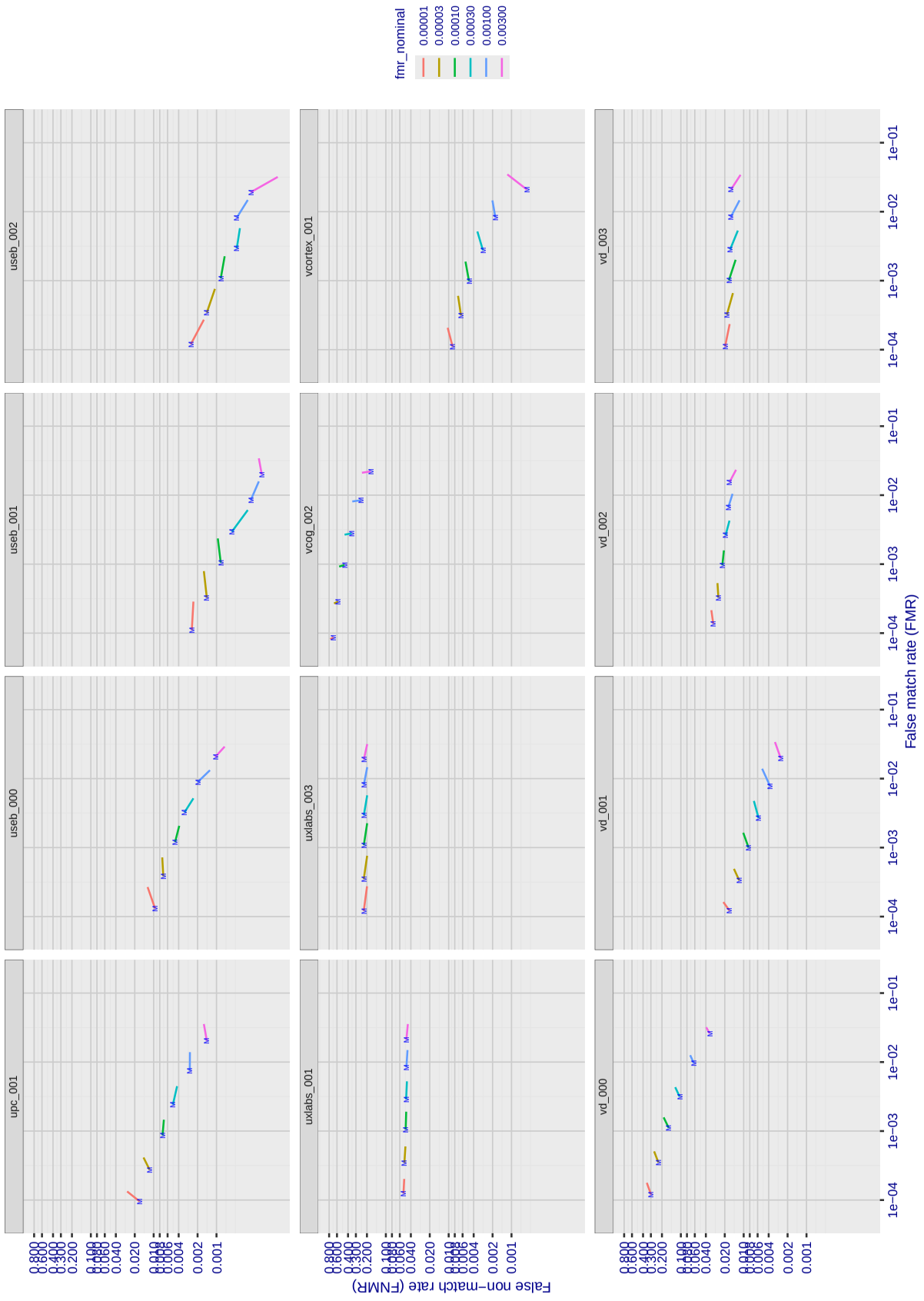


Figure 551: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ *and (FMR,FNMR)*$_{\text{FEMALE}}$ *showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

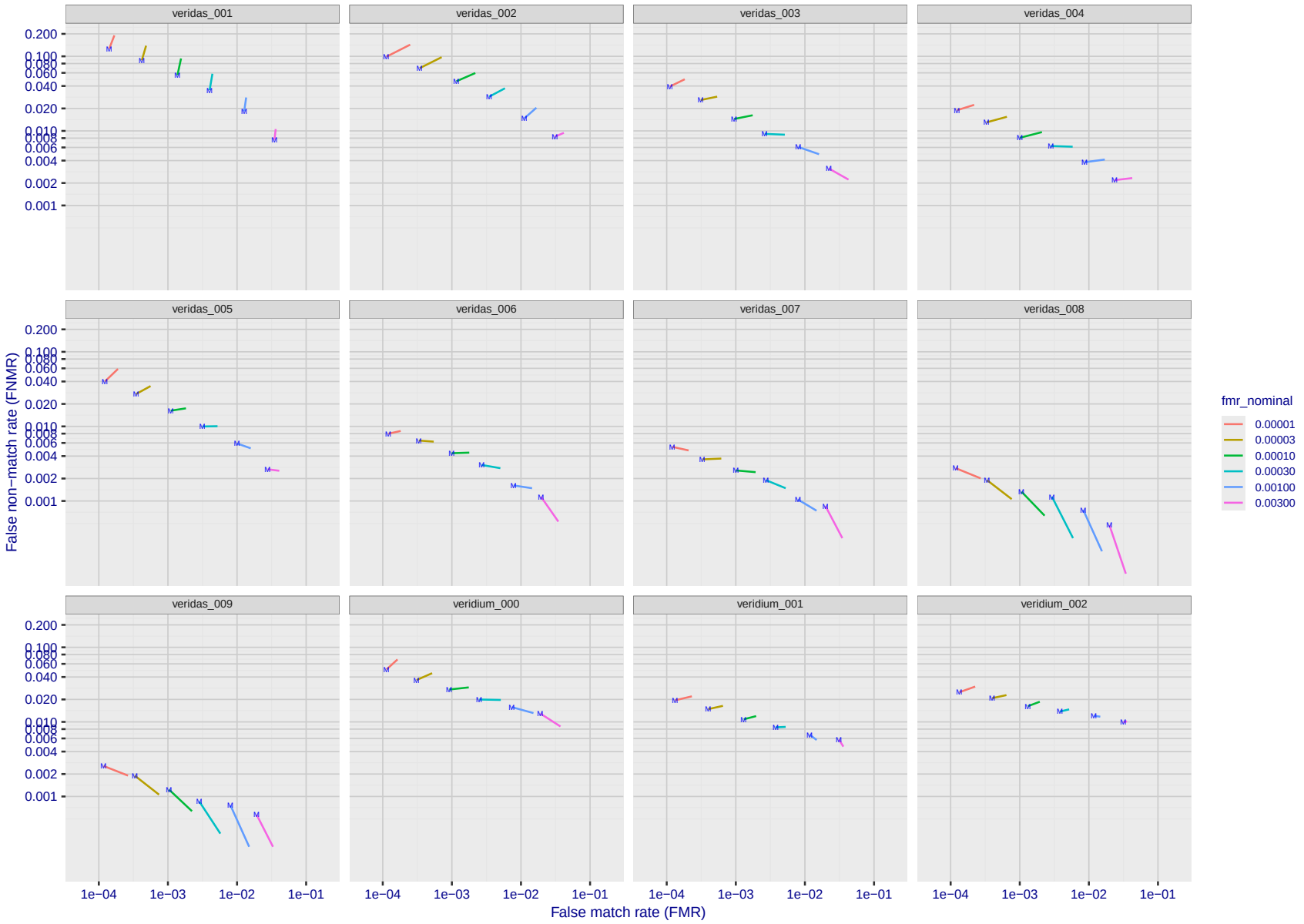


Figure 552: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

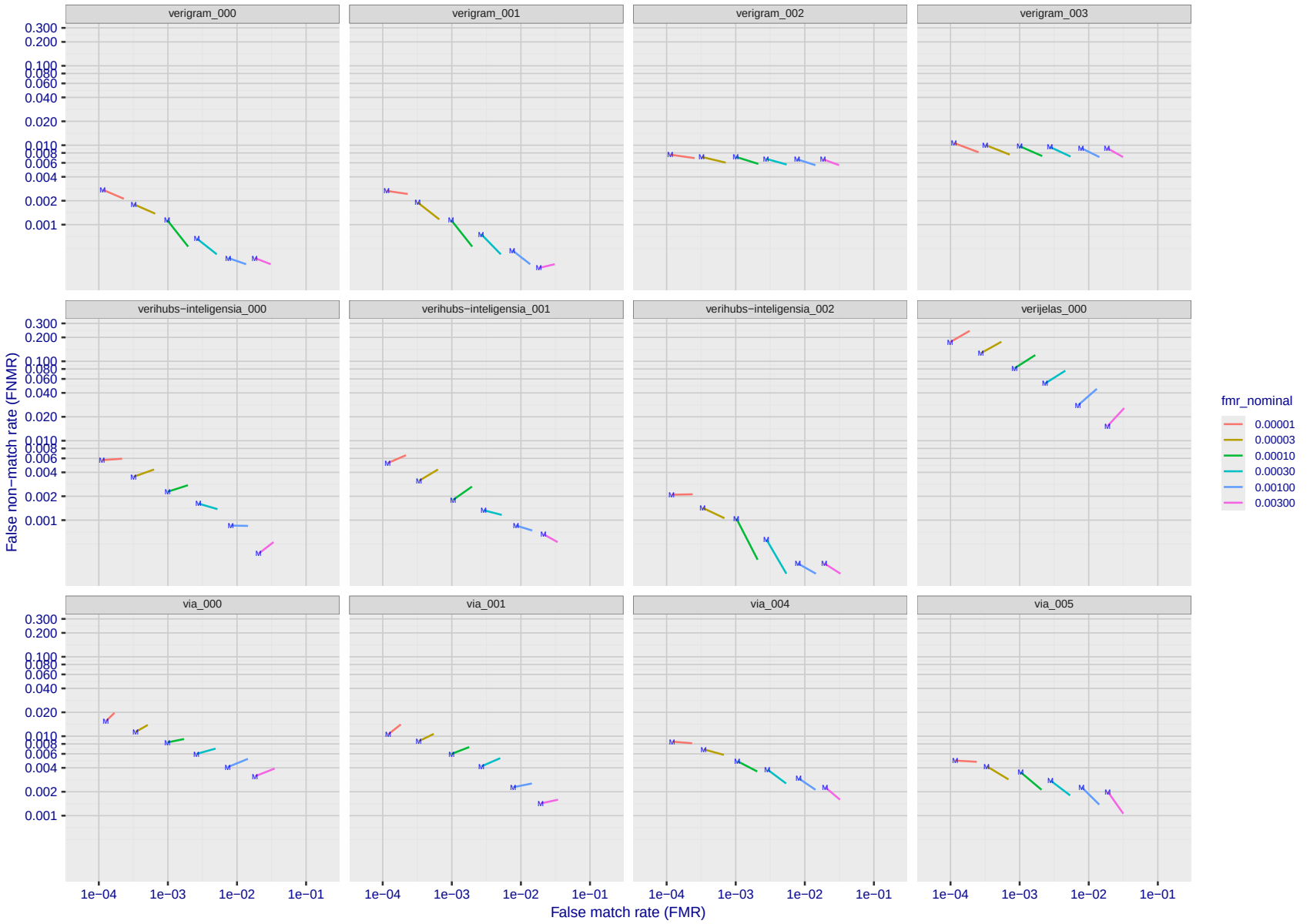


Figure 553: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

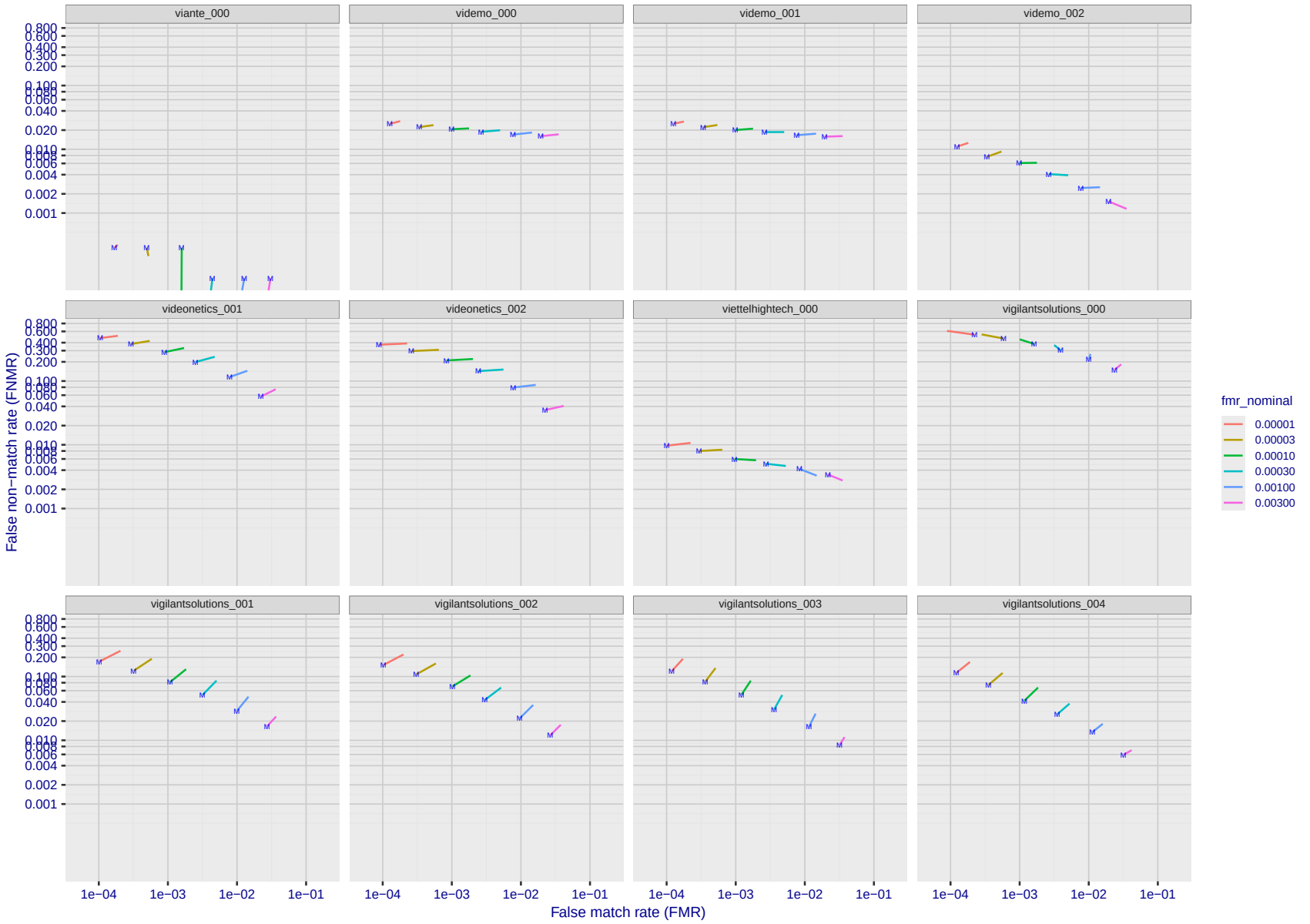


Figure 554: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

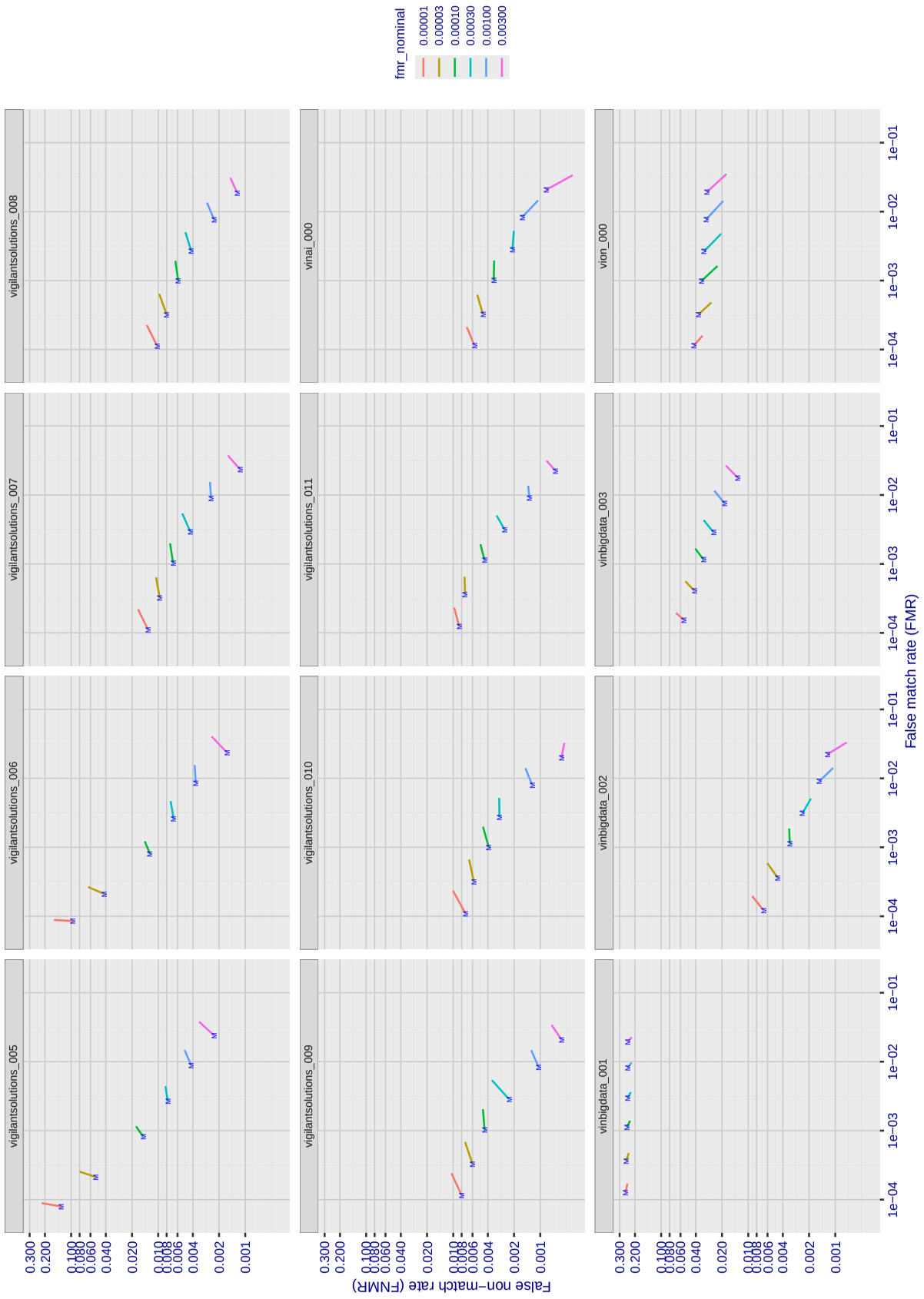


Figure 555: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between $(FMR,FNMR)_{\text{MALE}}$ and $(FMR,FNMR)_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

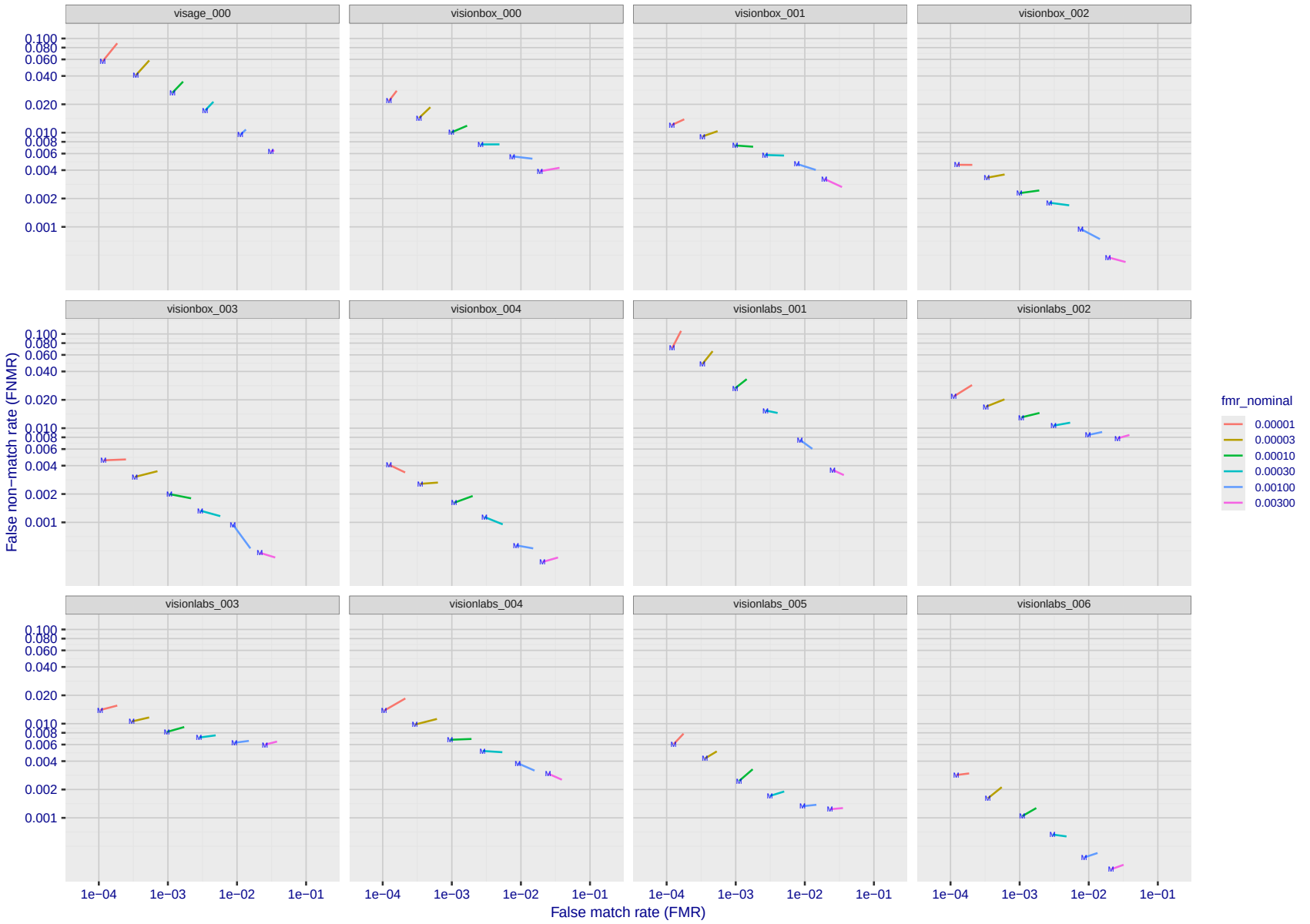


Figure 556: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

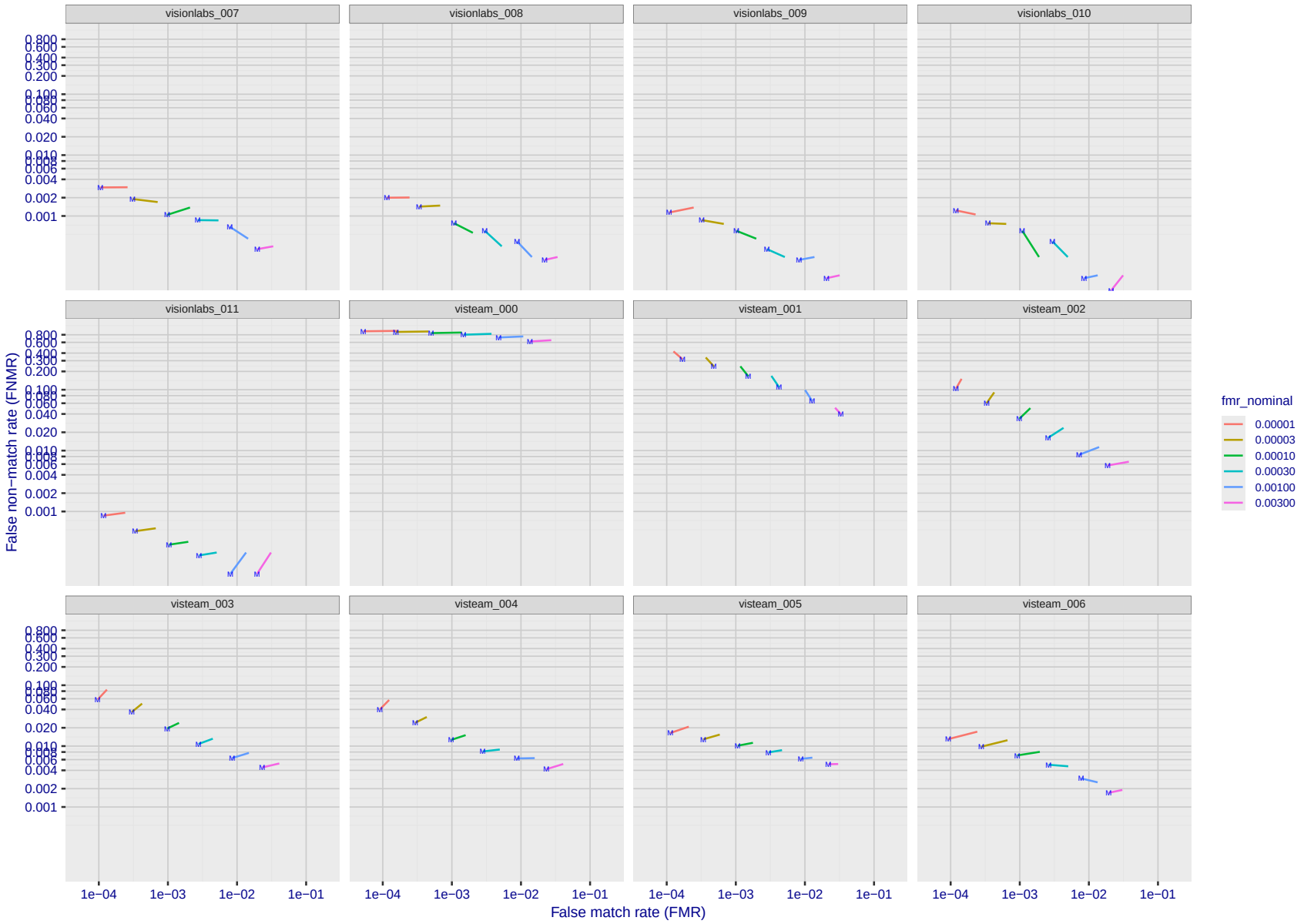


Figure 557: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

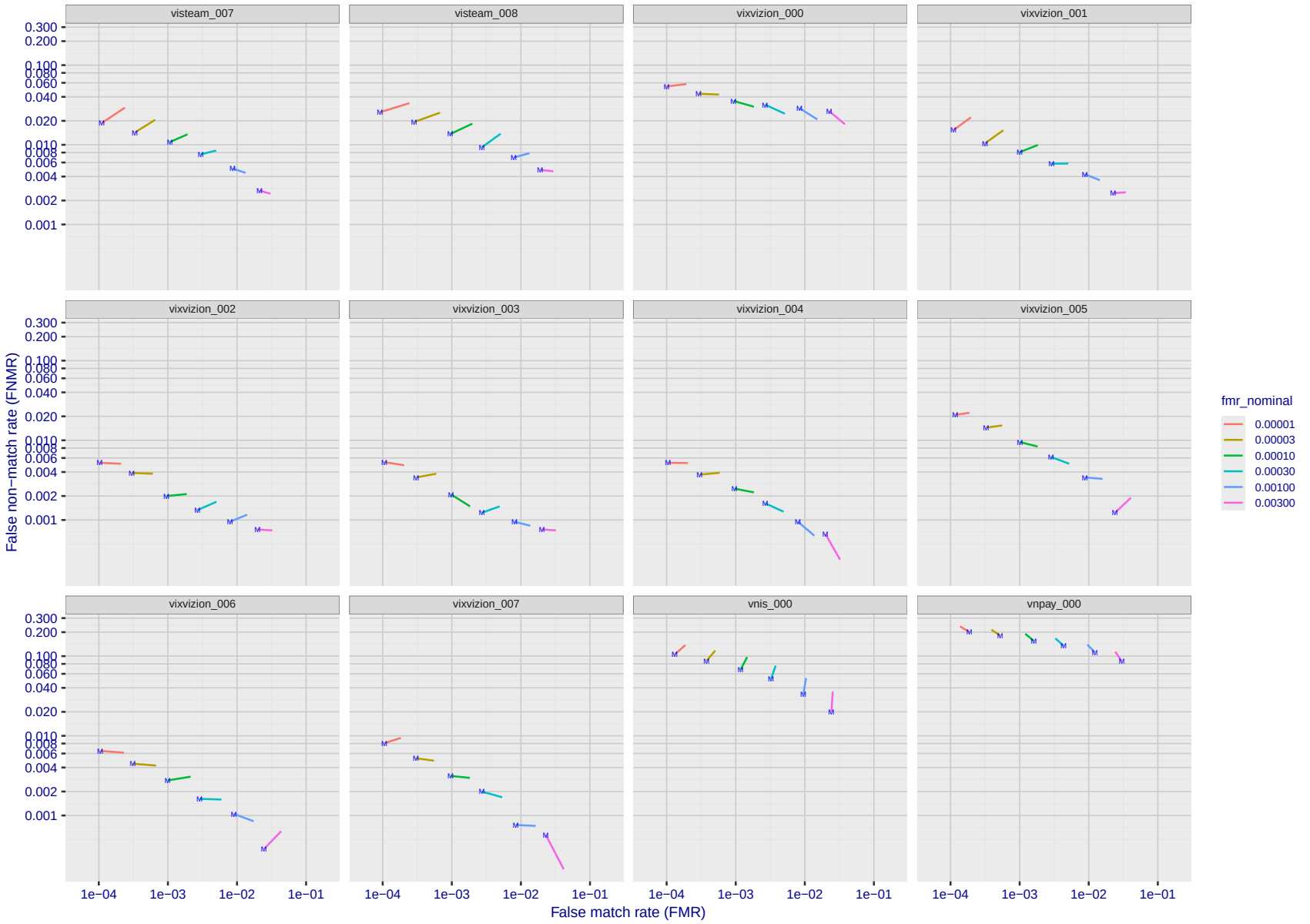


Figure 558: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

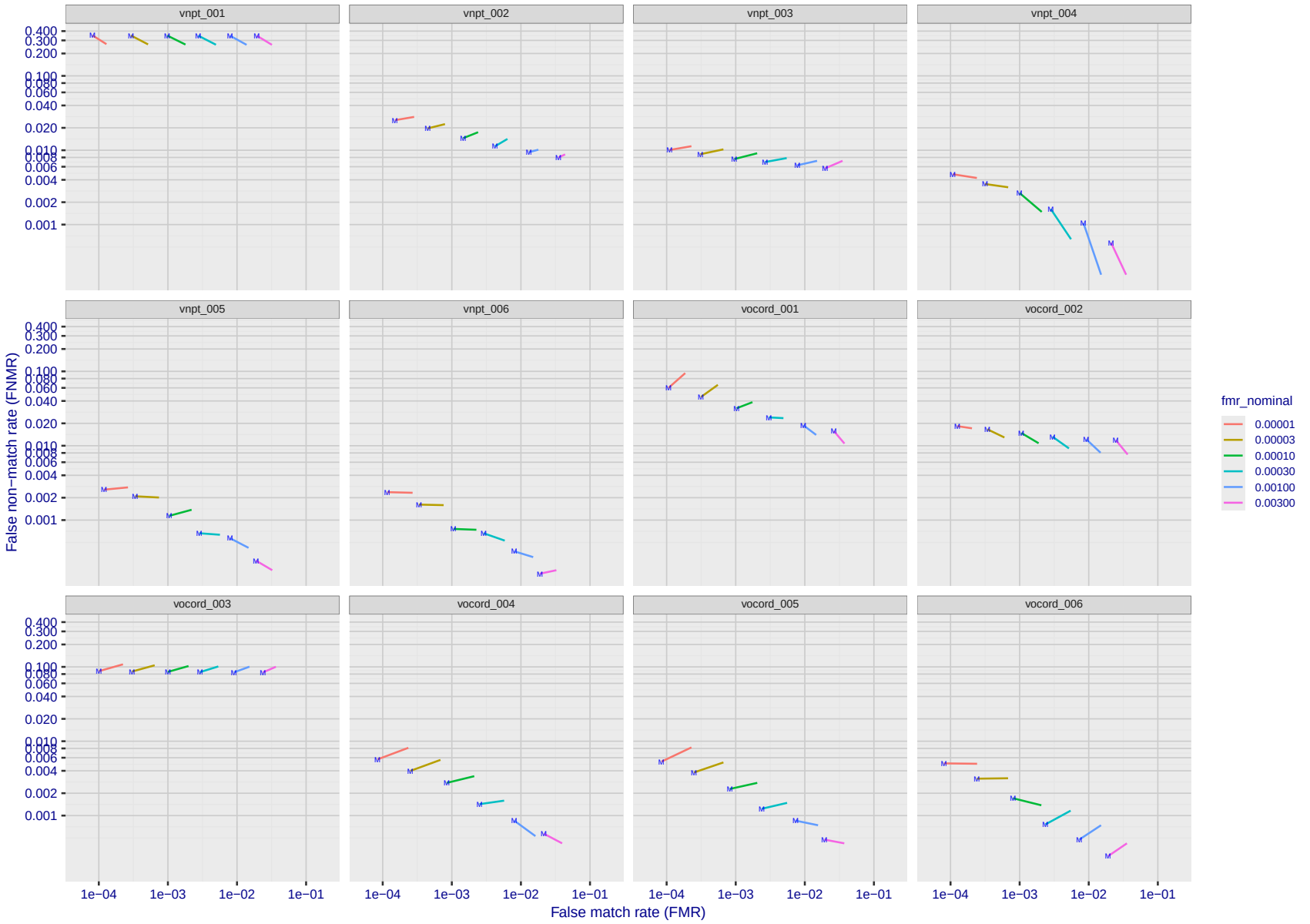


Figure 559: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

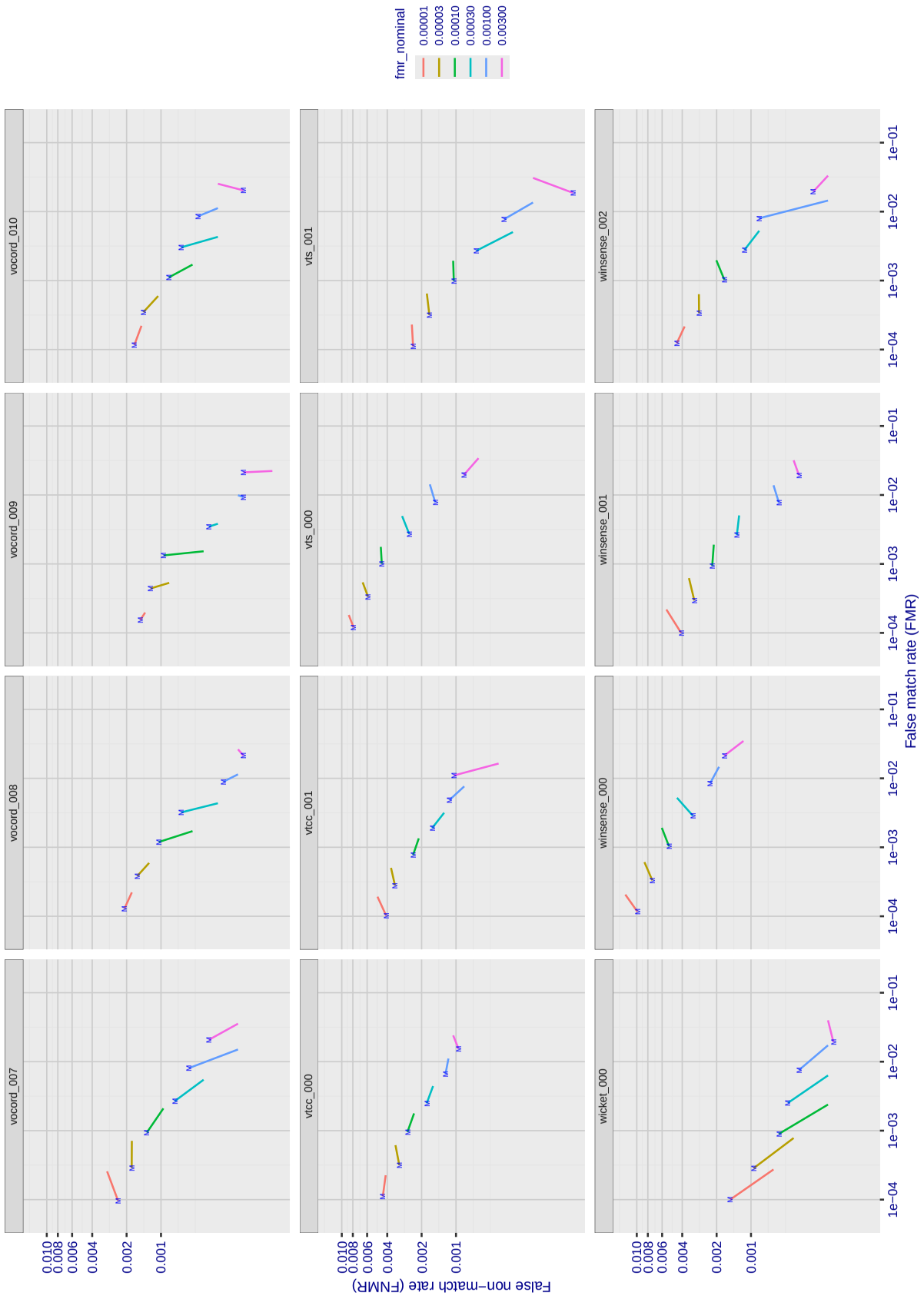


Figure 560: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

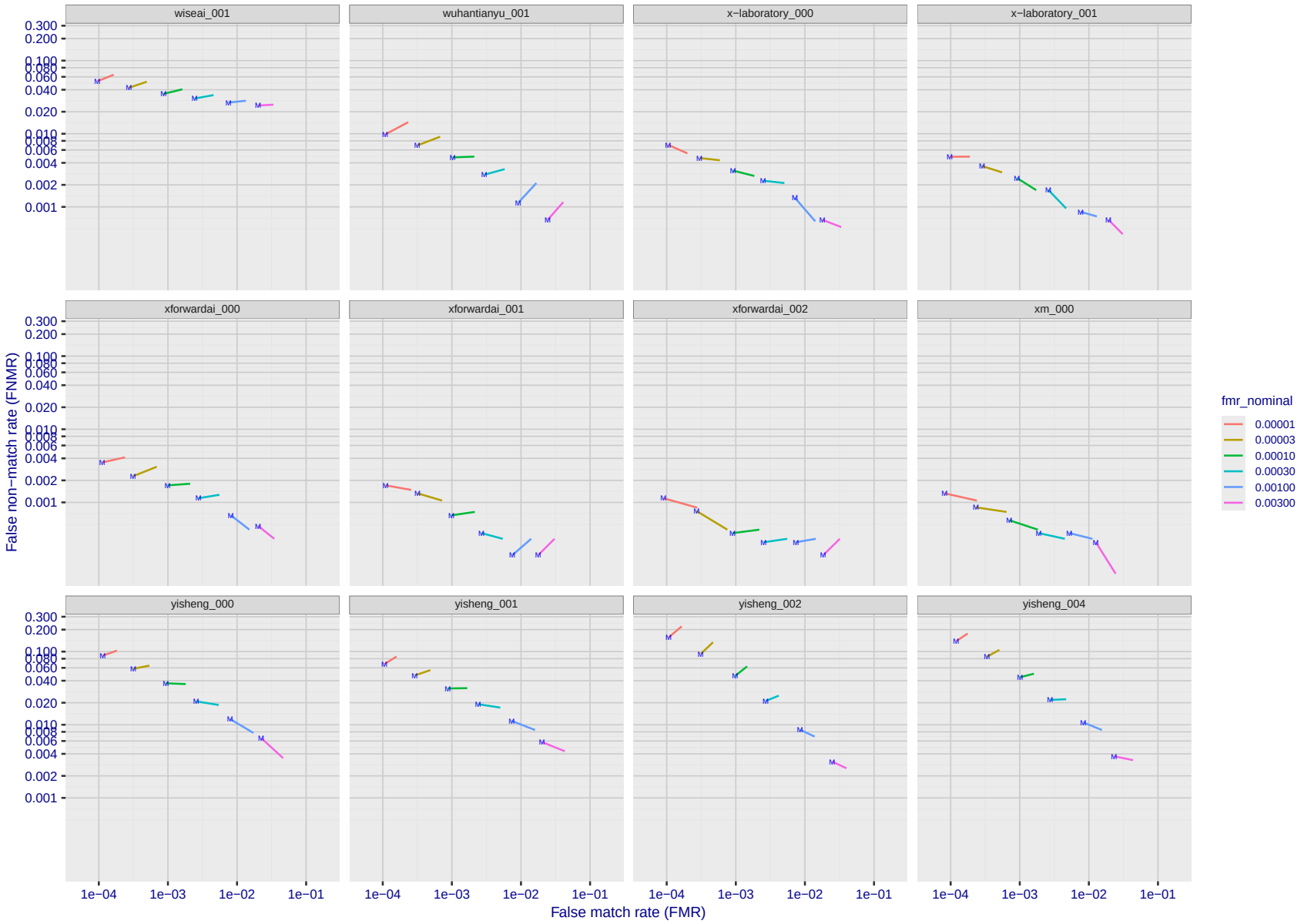


Figure 561: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

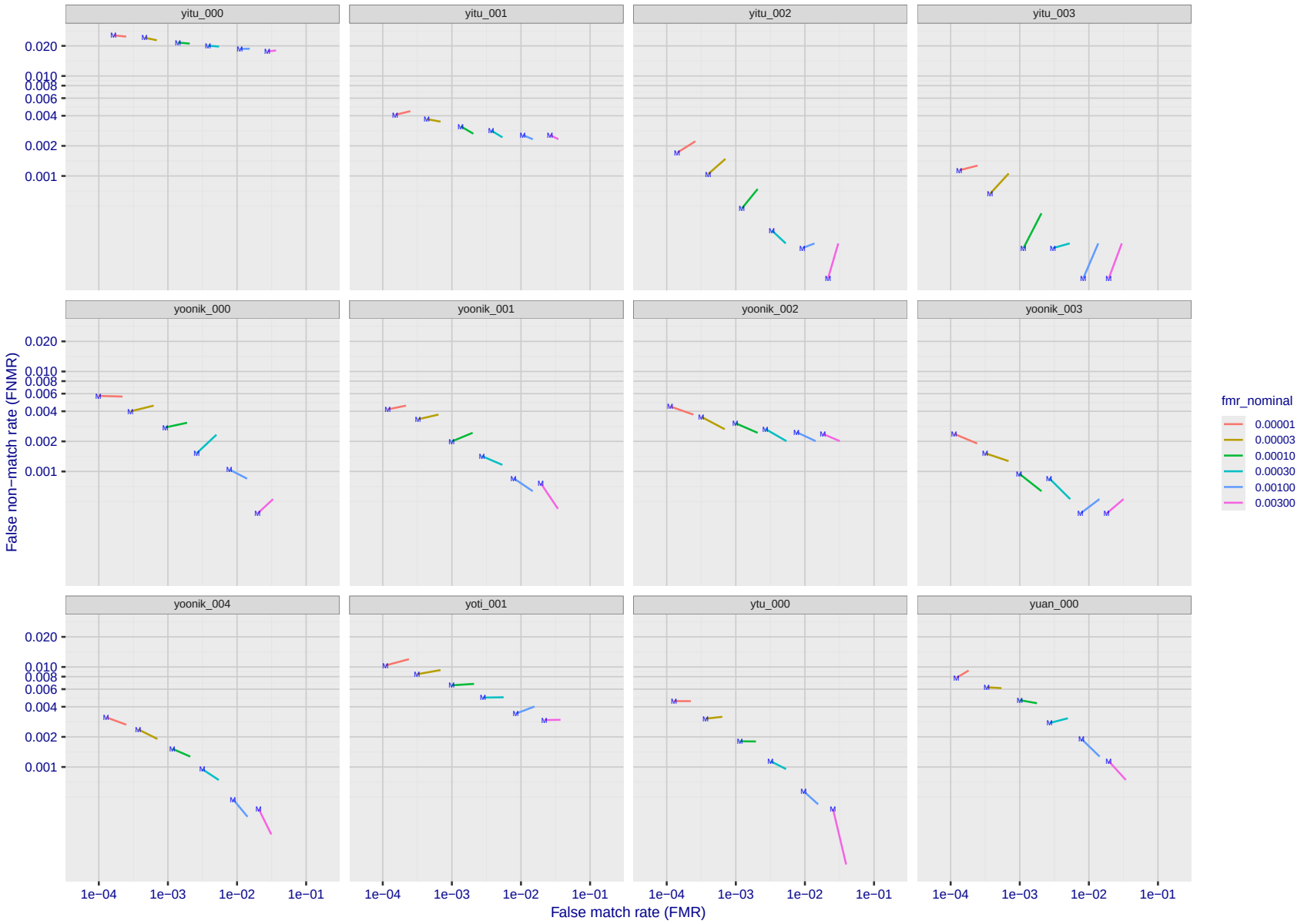


Figure 562: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

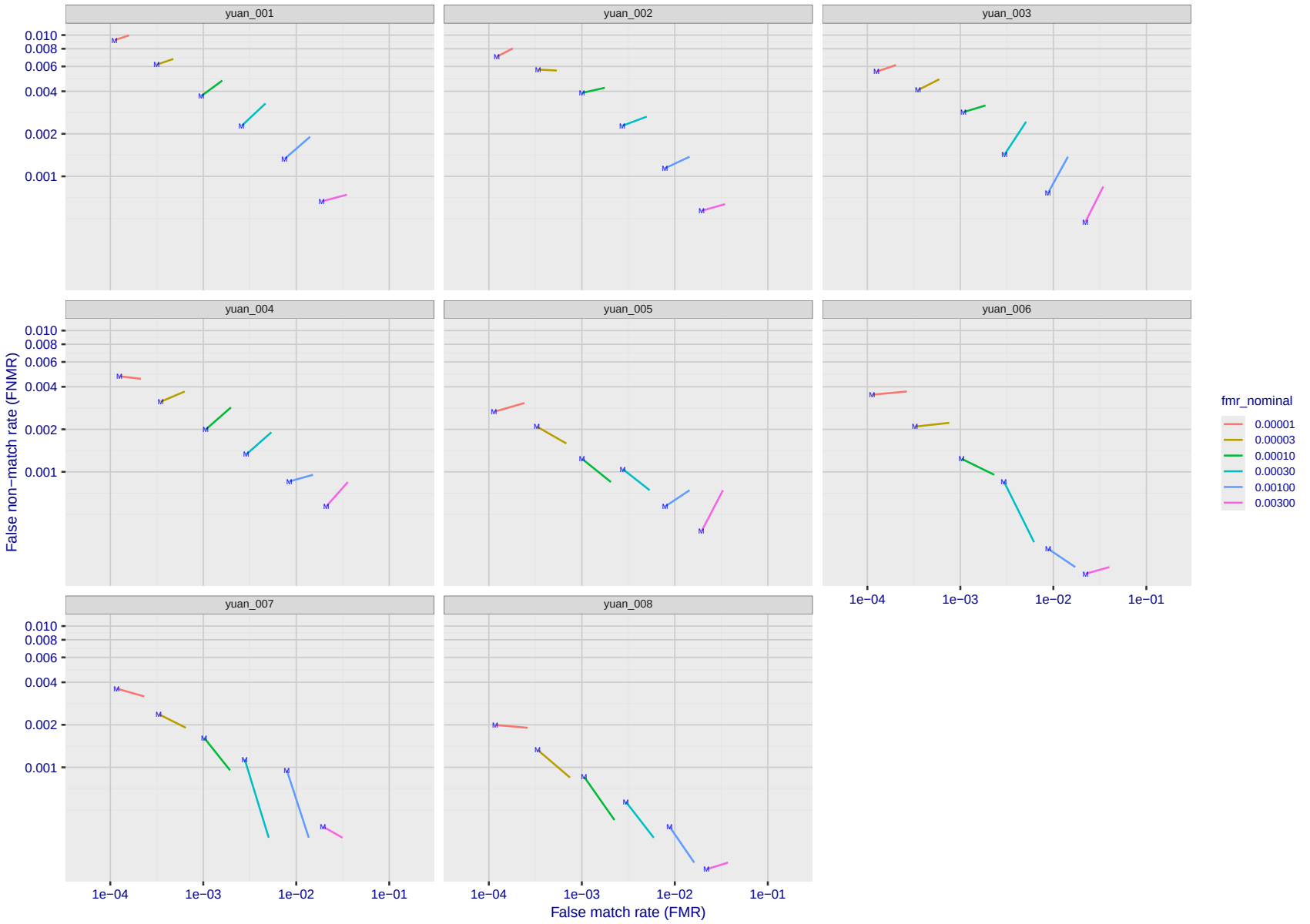


Figure 563: For the visa images, FNMR and FMR at six operating points along the DET characteristic. At each point a line is drawn between (FMR,FNMR)$_{\text{MALE}}$ and (FMR,FNMR)$_{\text{FEMALE}}$ showing how which sex has lower FMR and/or FNMR. The "M" label denotes male, the other end of the line corresponds to female. The six operating thresholds are selected to give the nominal false match rates given in the legend, and are computed over all impostor pairs regardless of age, sex, and place of birth. The plotted FMR values are broadly an order of magnitude larger than the nominal rates because FMR is computed over demographically-matched impostor pairs i.e individuals of the same sex, from the same geographic region (see section 3.6.1), and the same age group (see section 3.6.2).

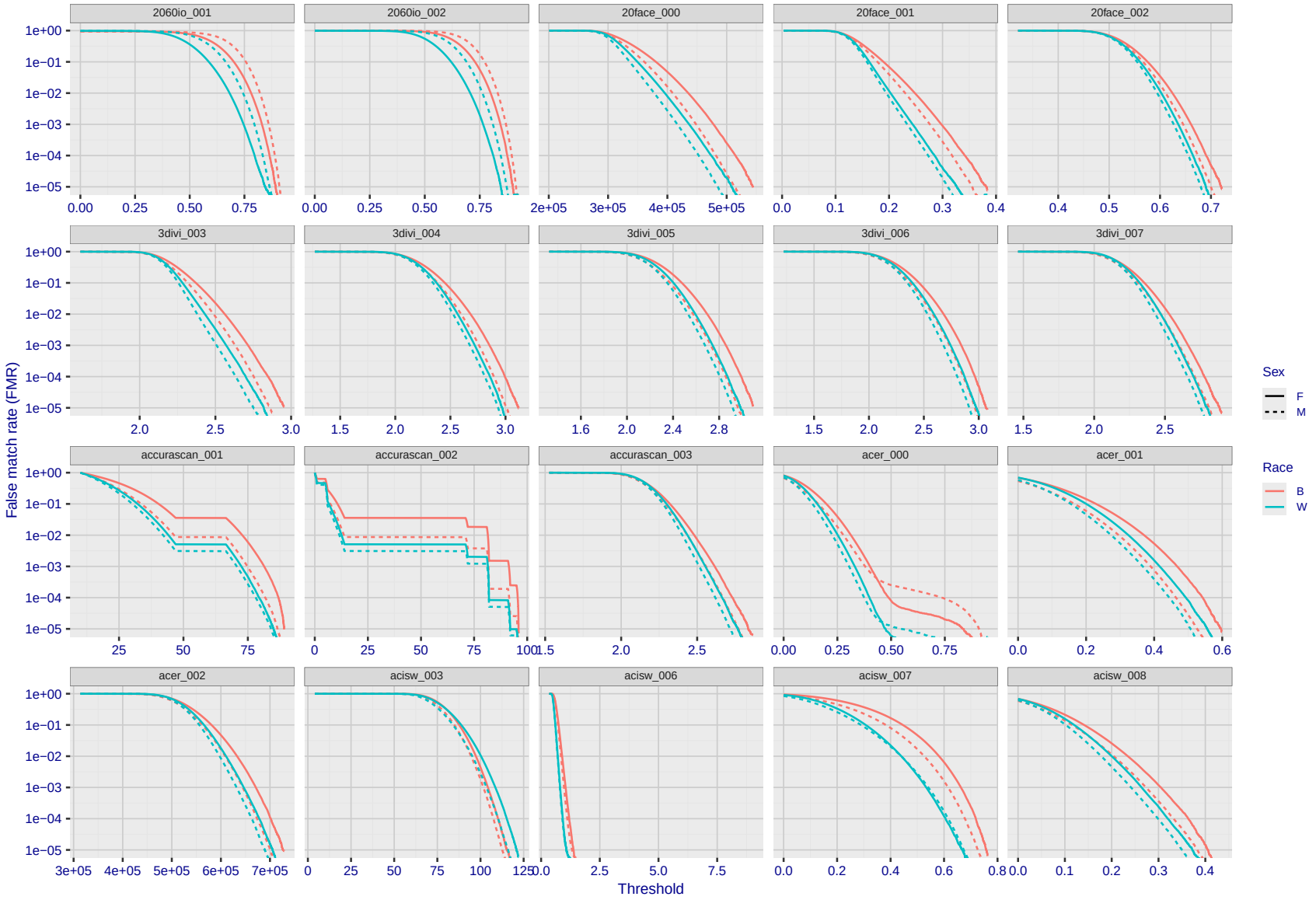


Figure 564: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

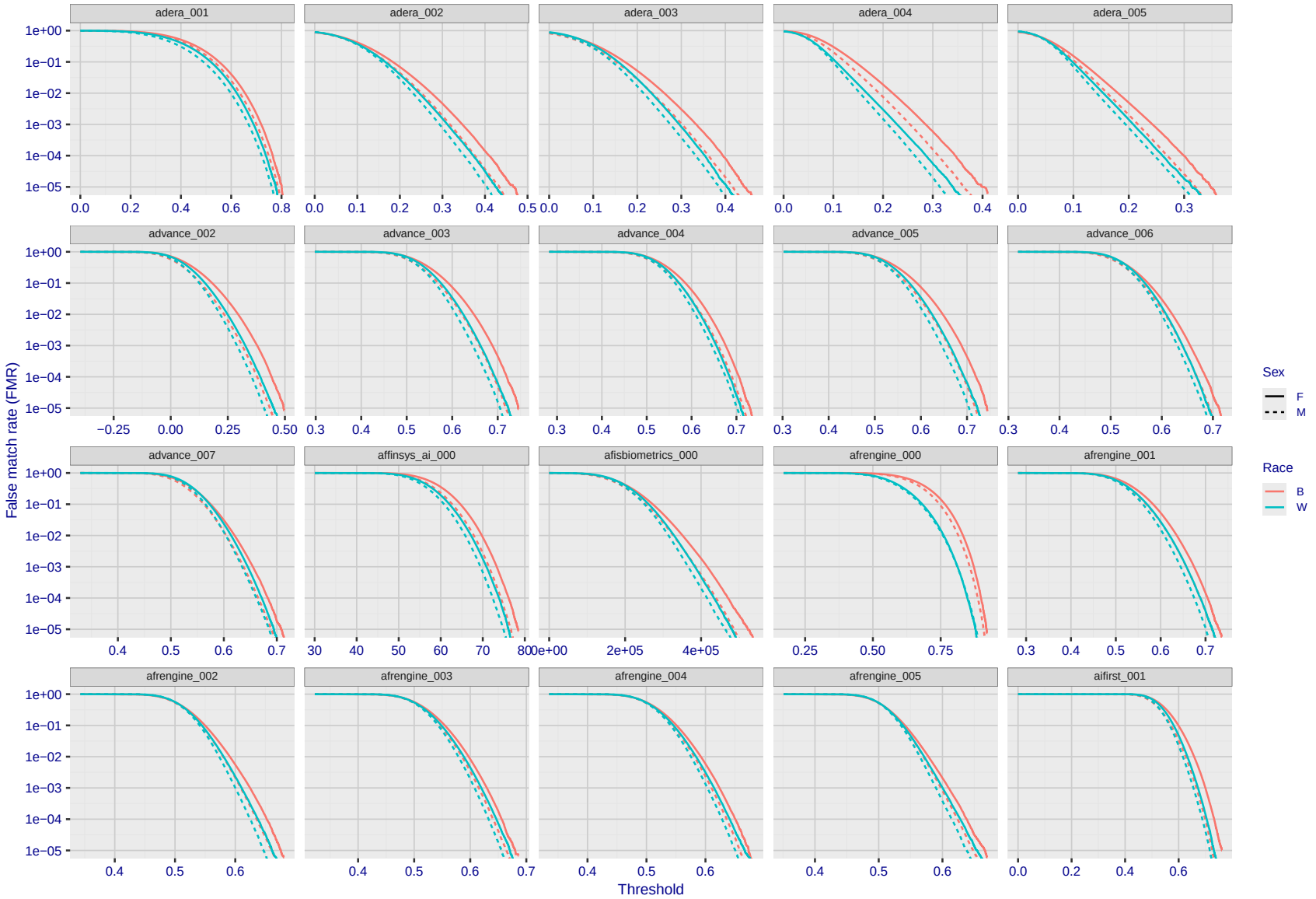


Figure 565: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

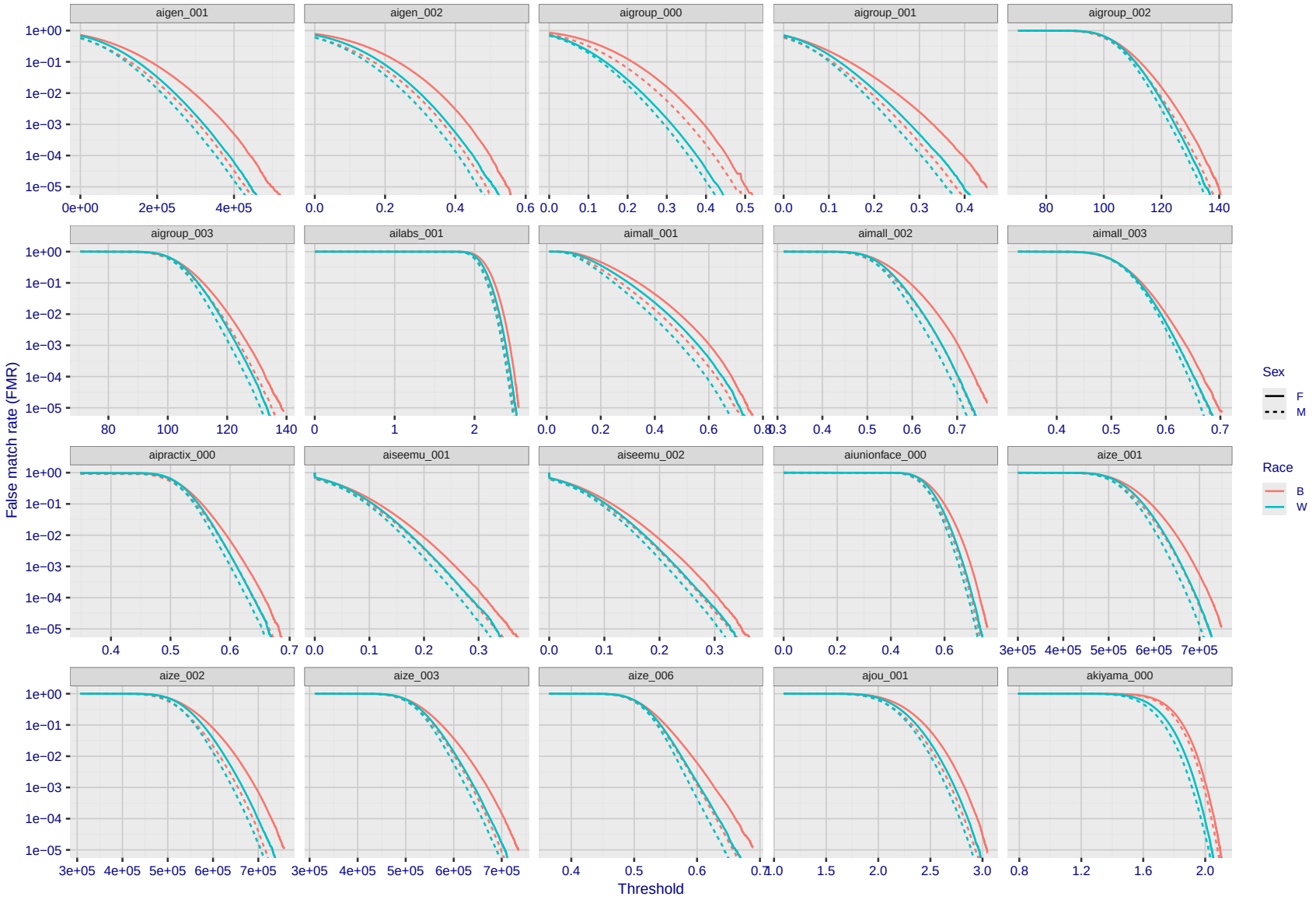


Figure 566: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

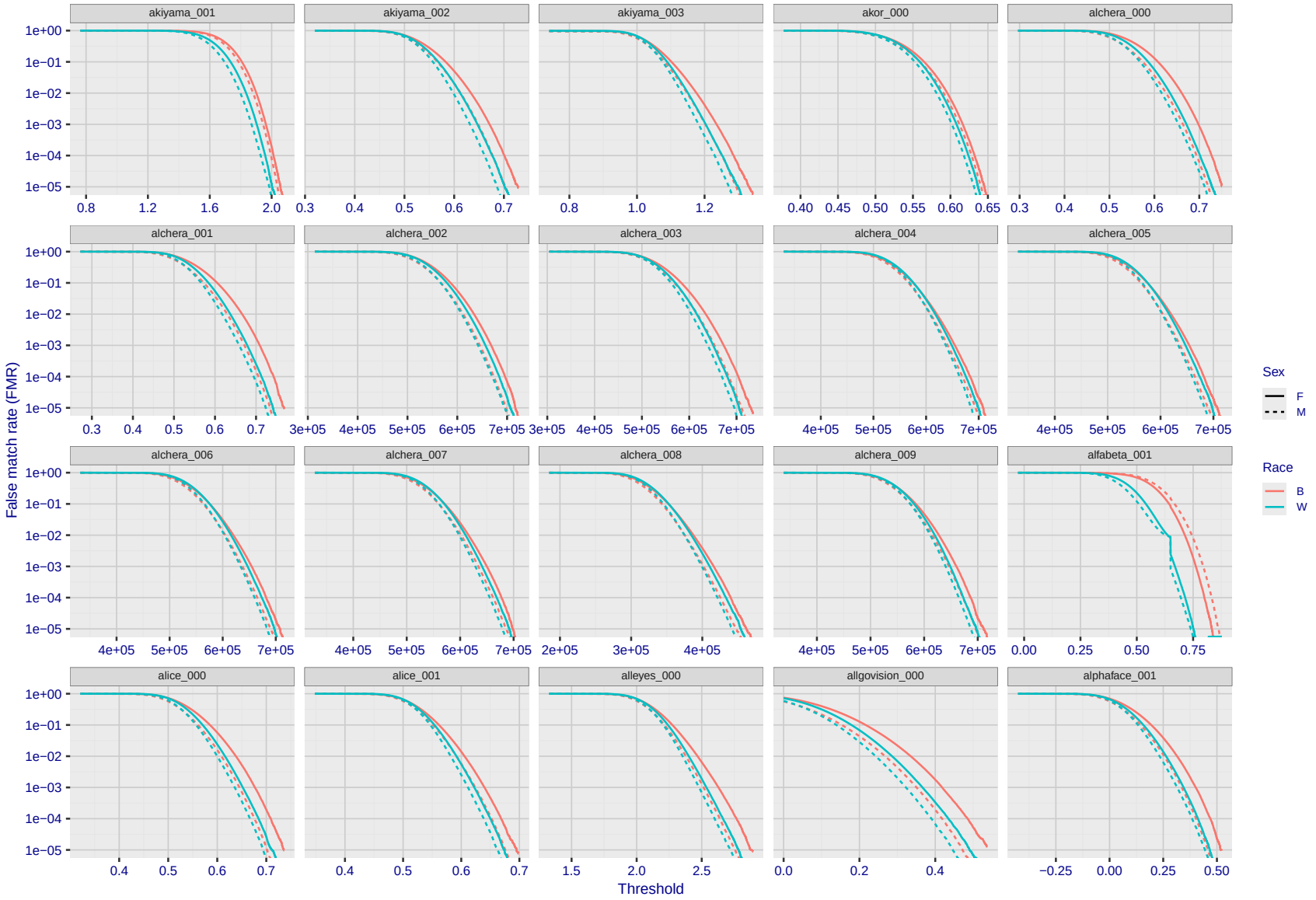


Figure 567: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

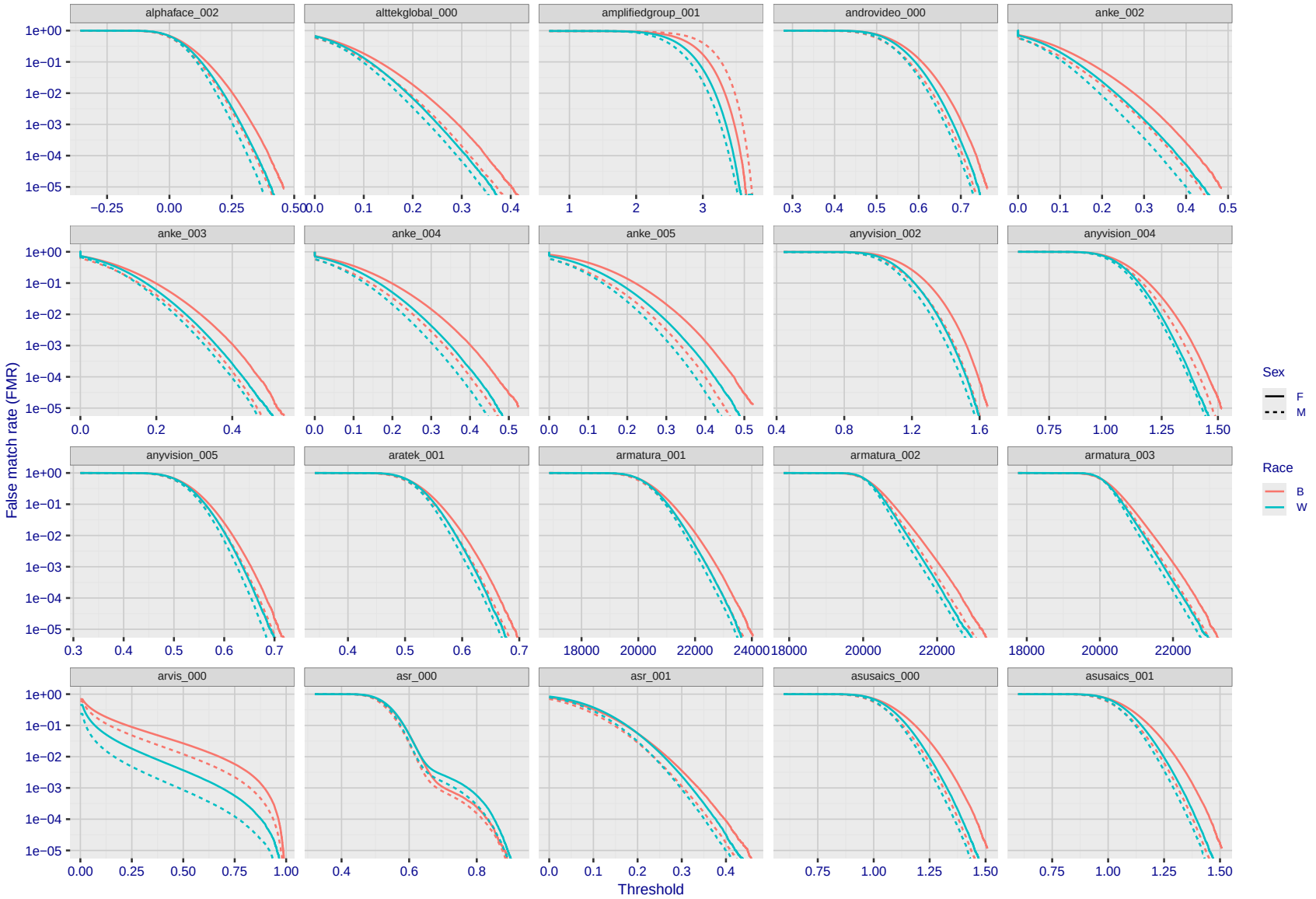


Figure 568: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

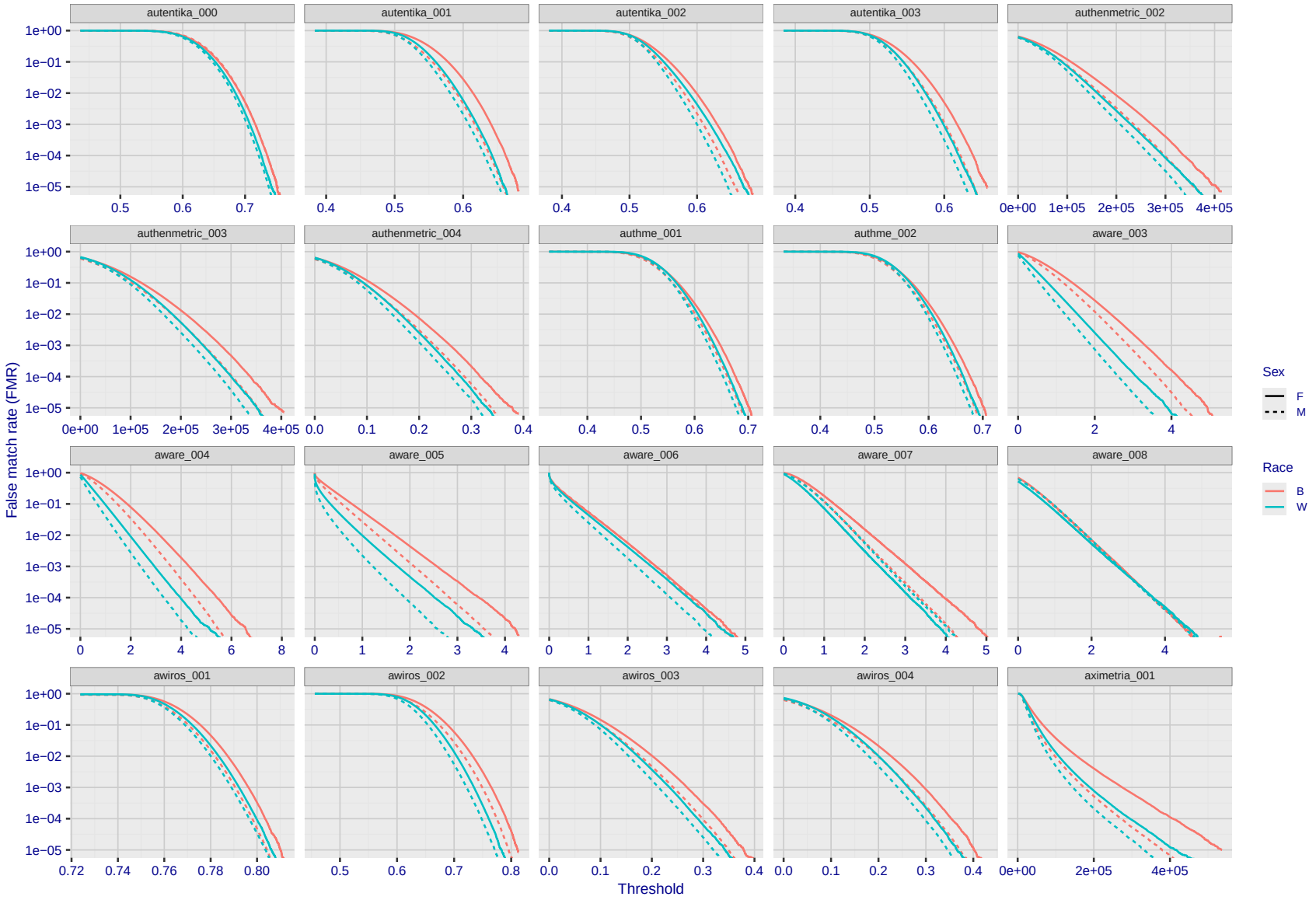


Figure 569: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

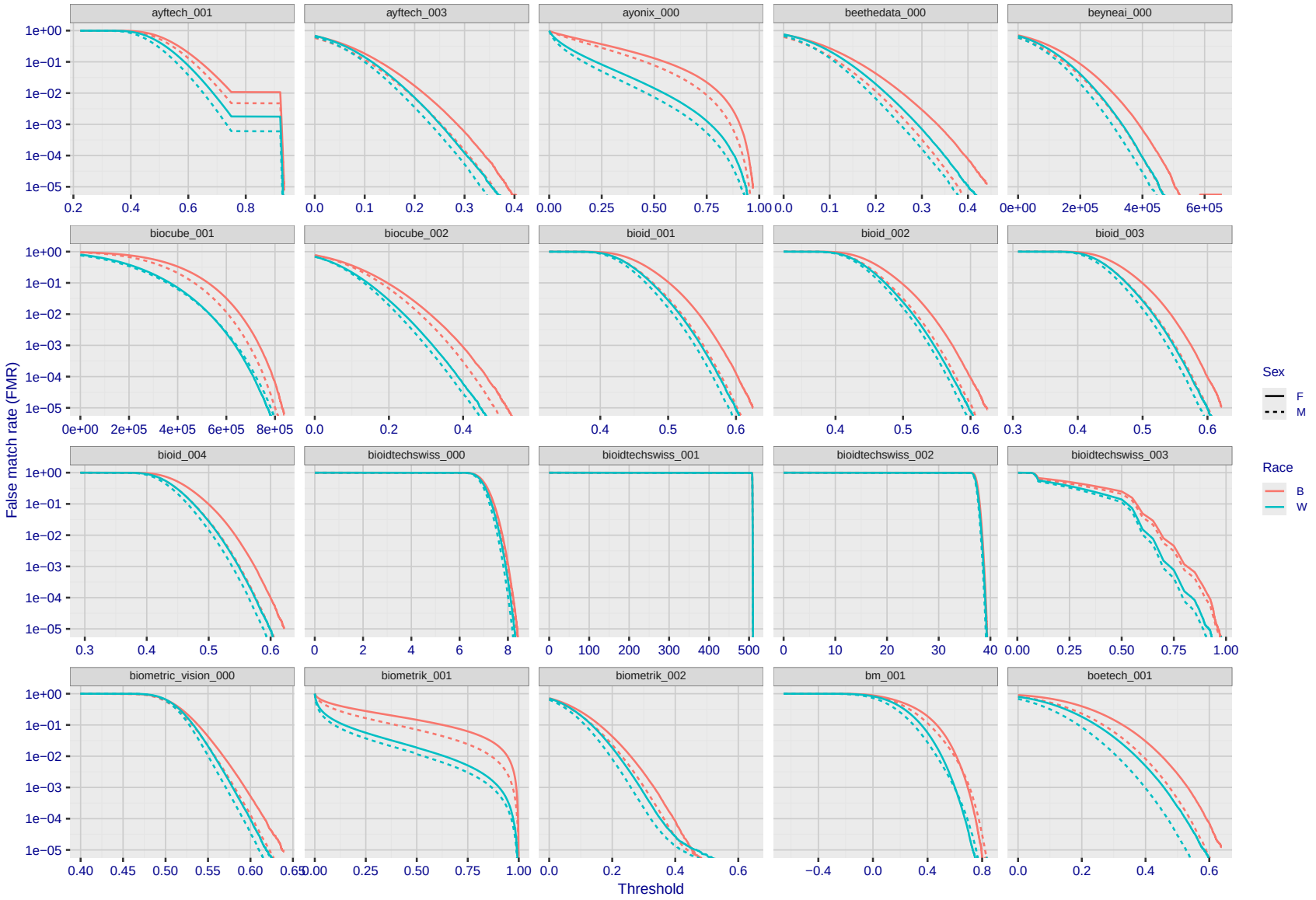


Figure 570: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

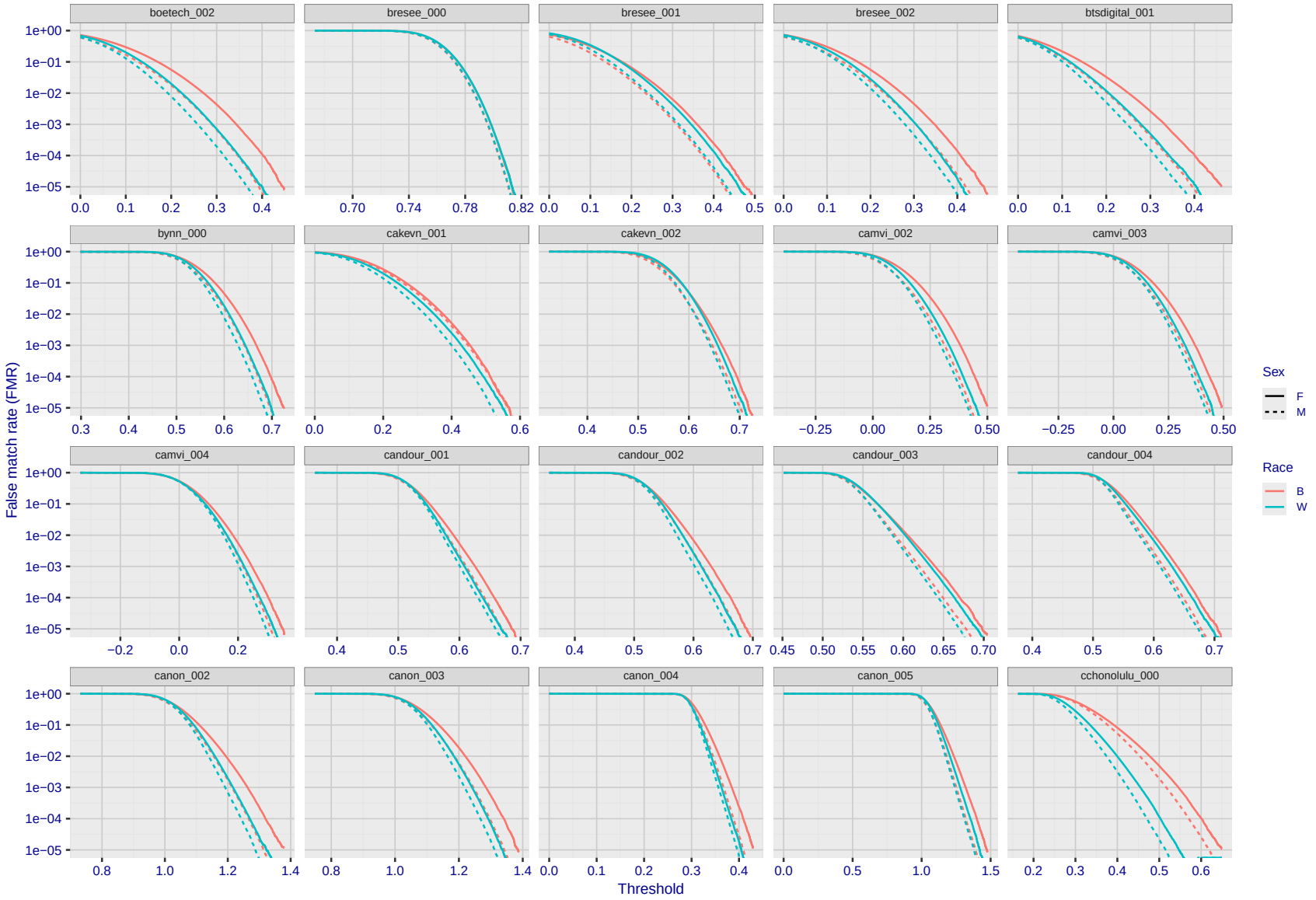


Figure 571: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

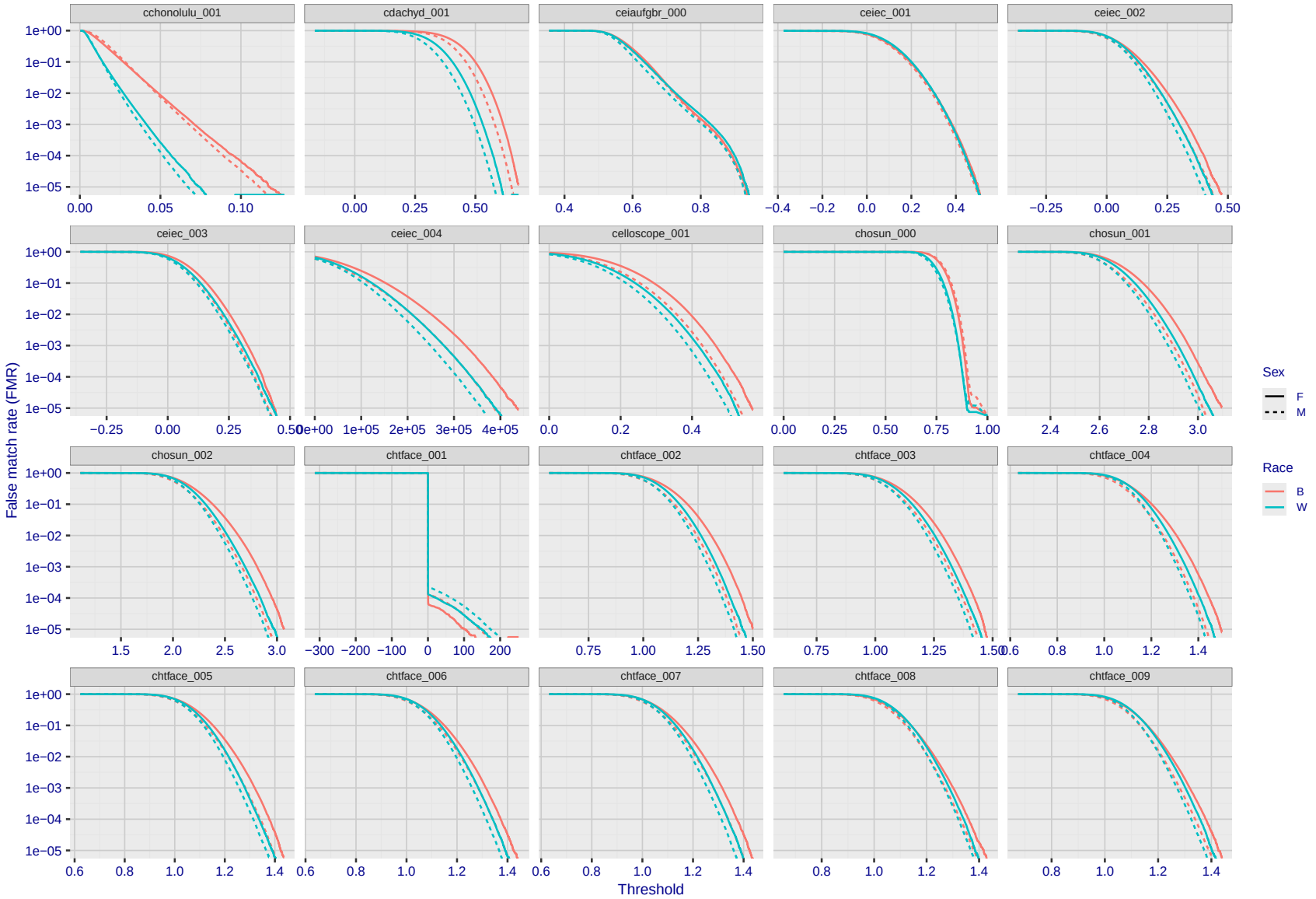


Figure 572: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

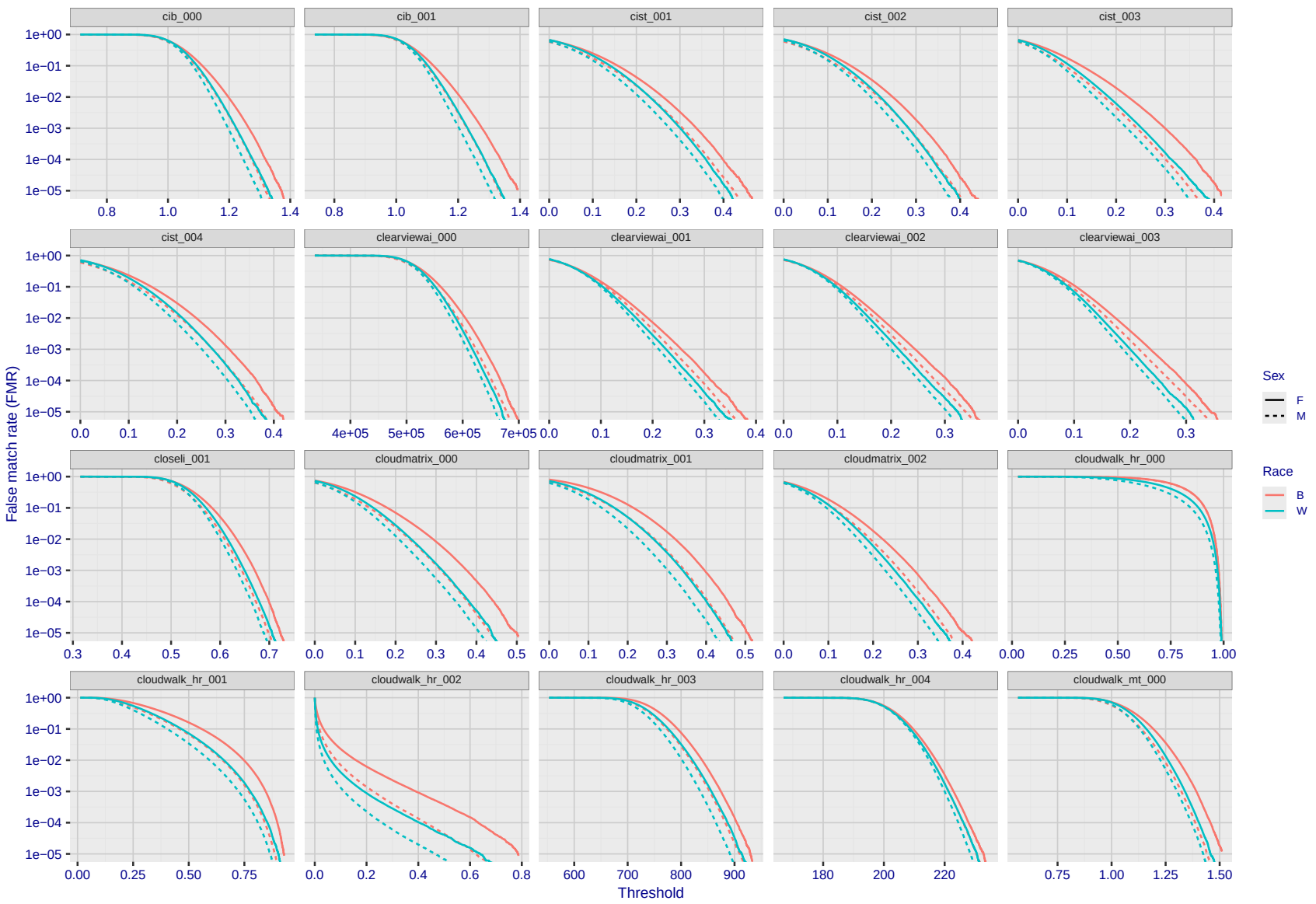


Figure 573: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

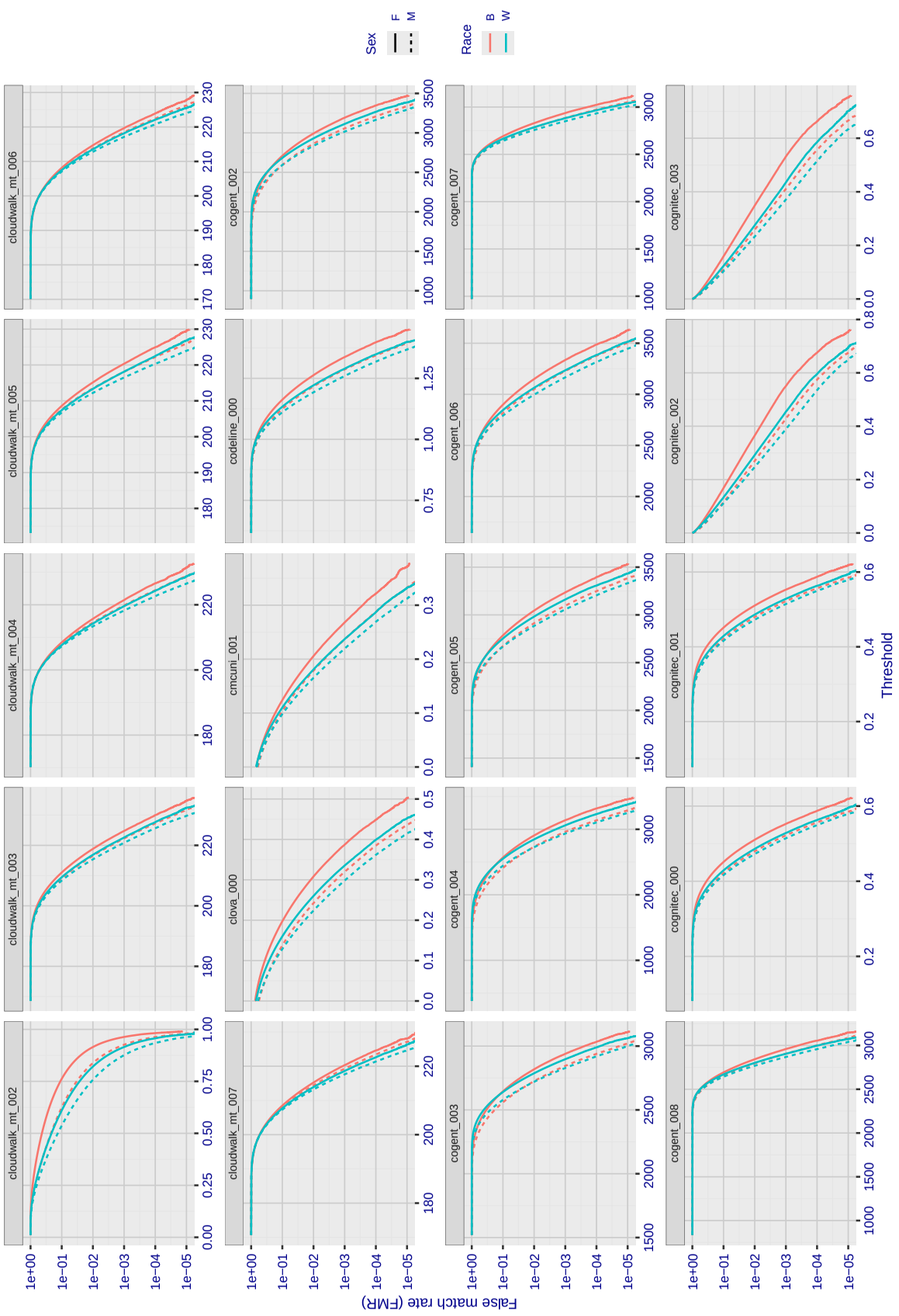


Figure 574: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

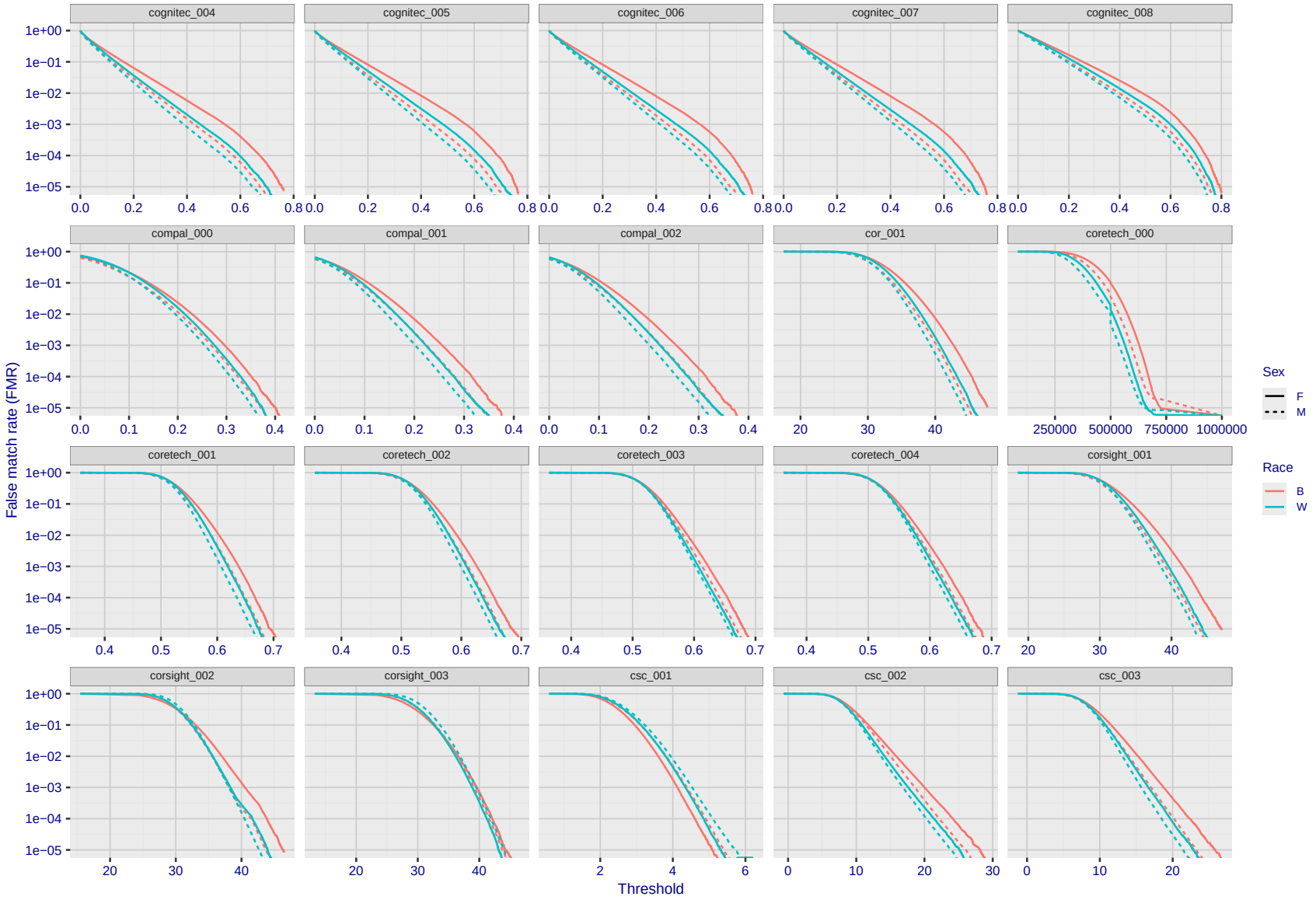


Figure 575: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

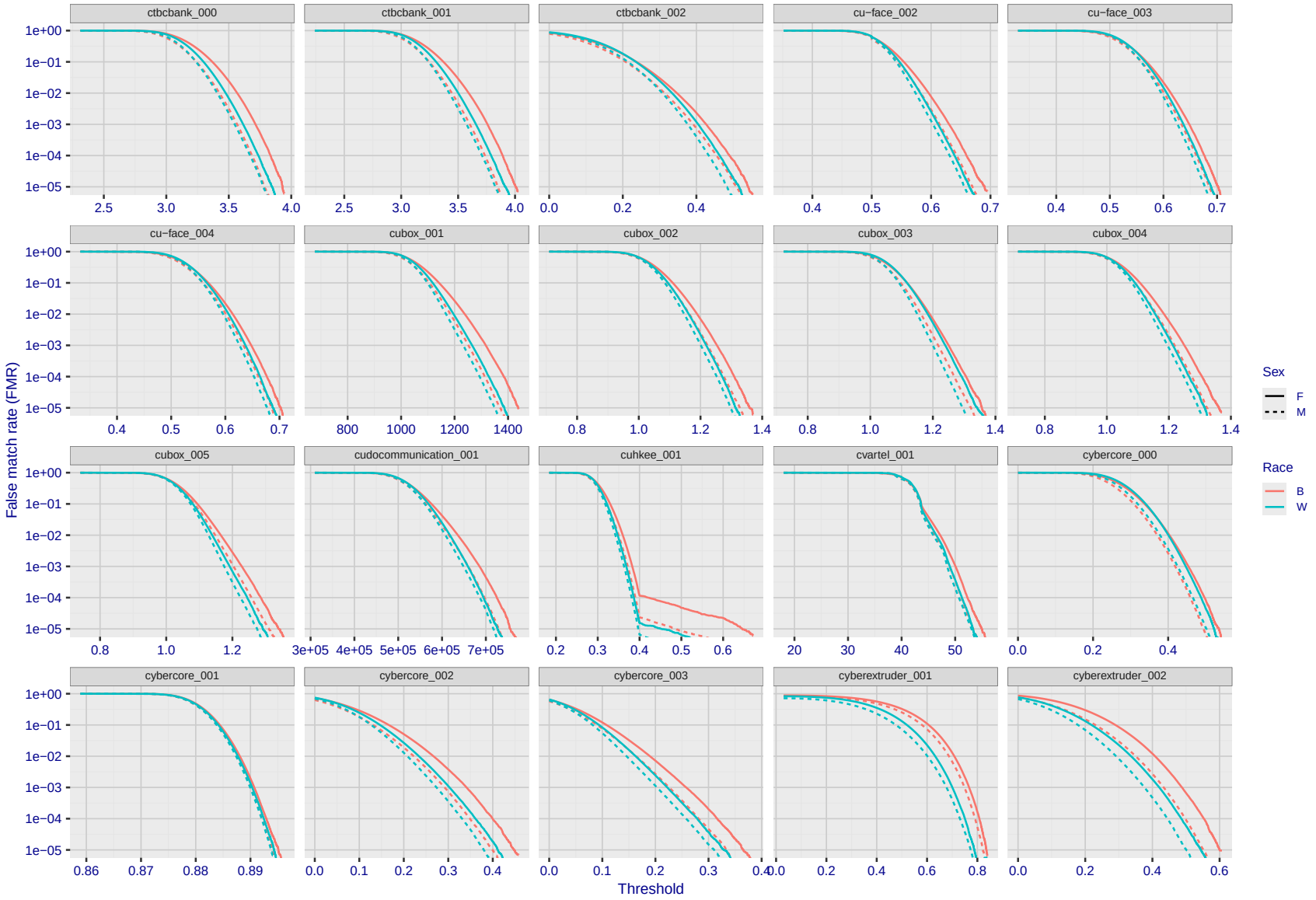


Figure 576: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

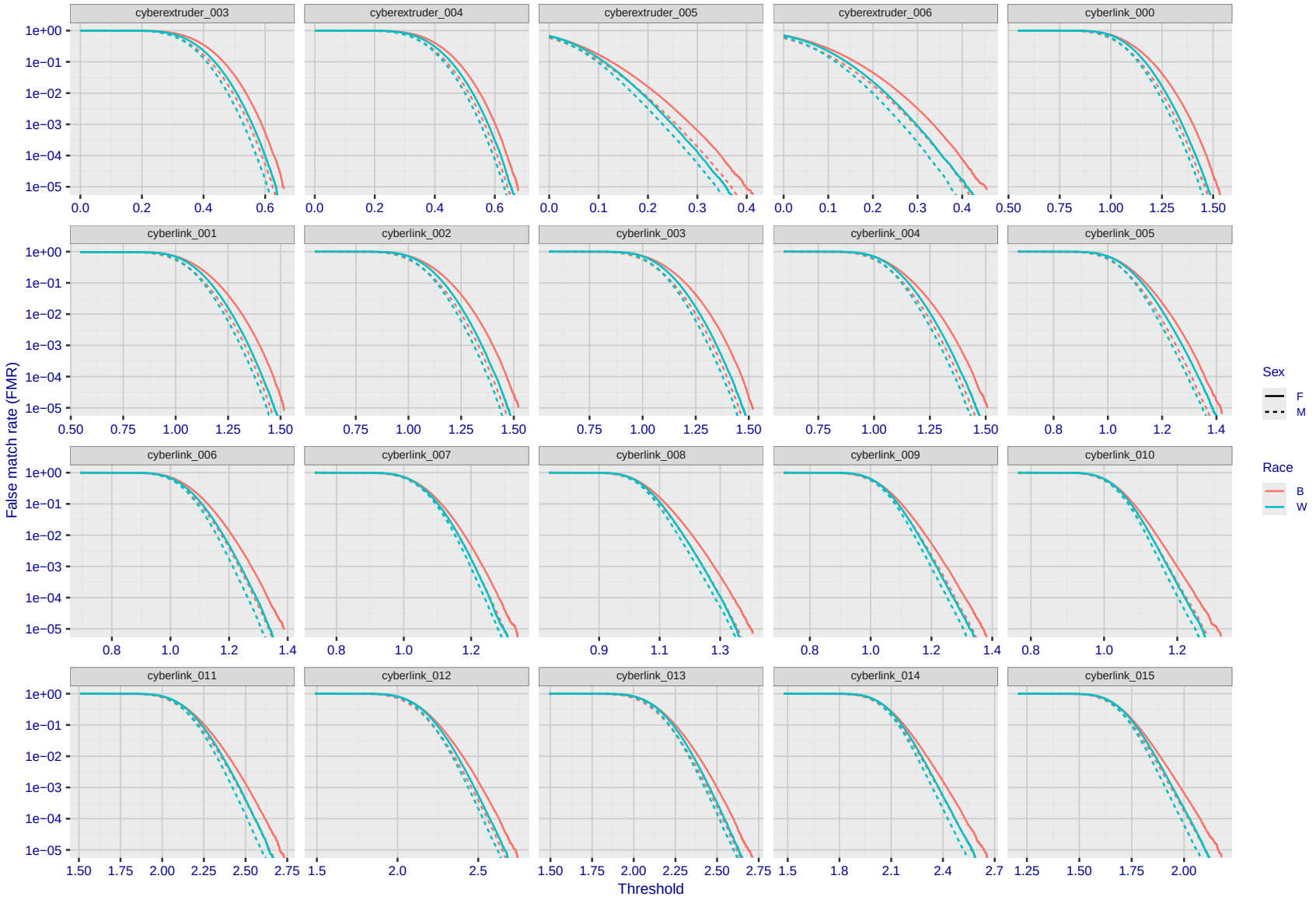


Figure 577: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

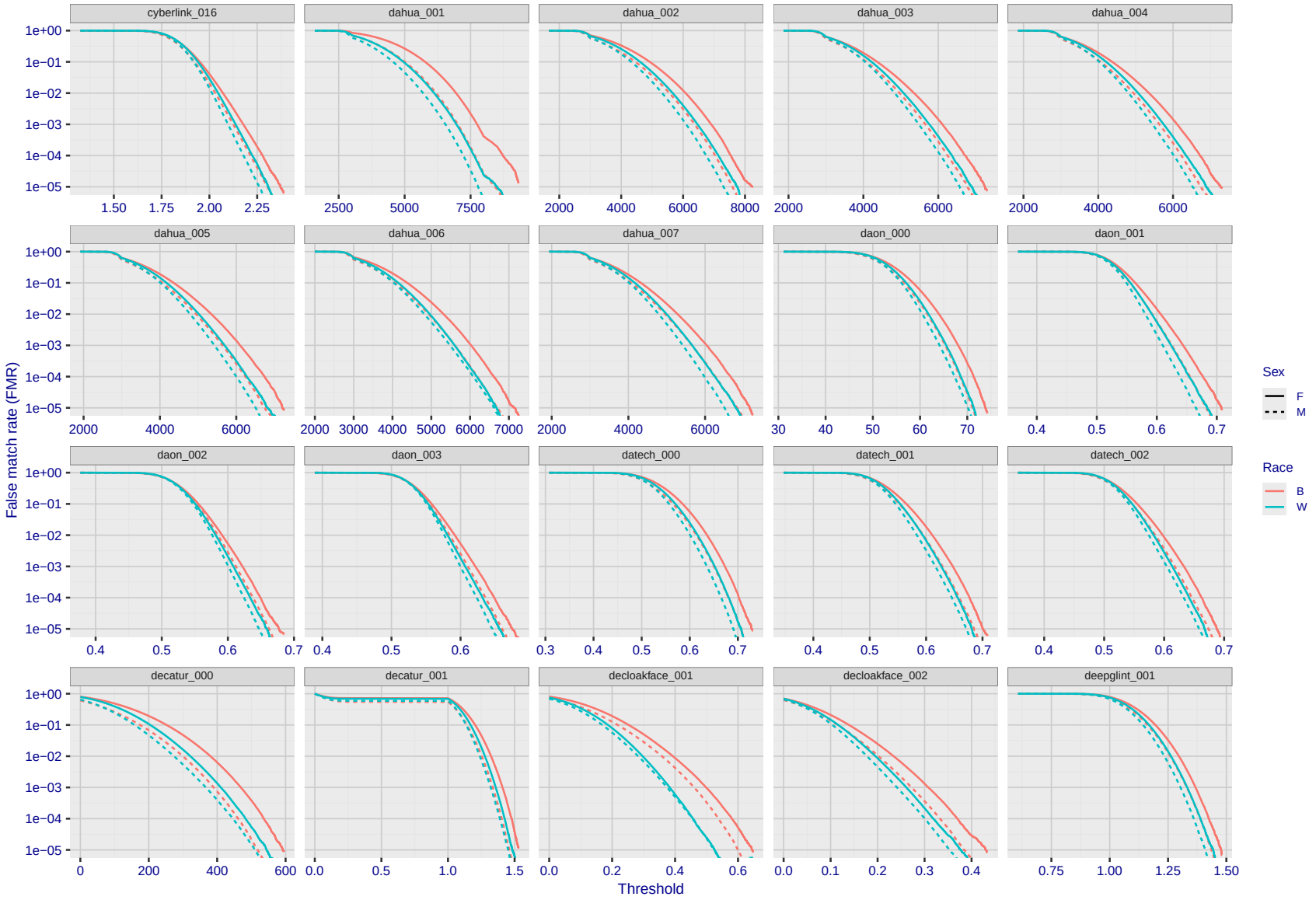


Figure 578: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

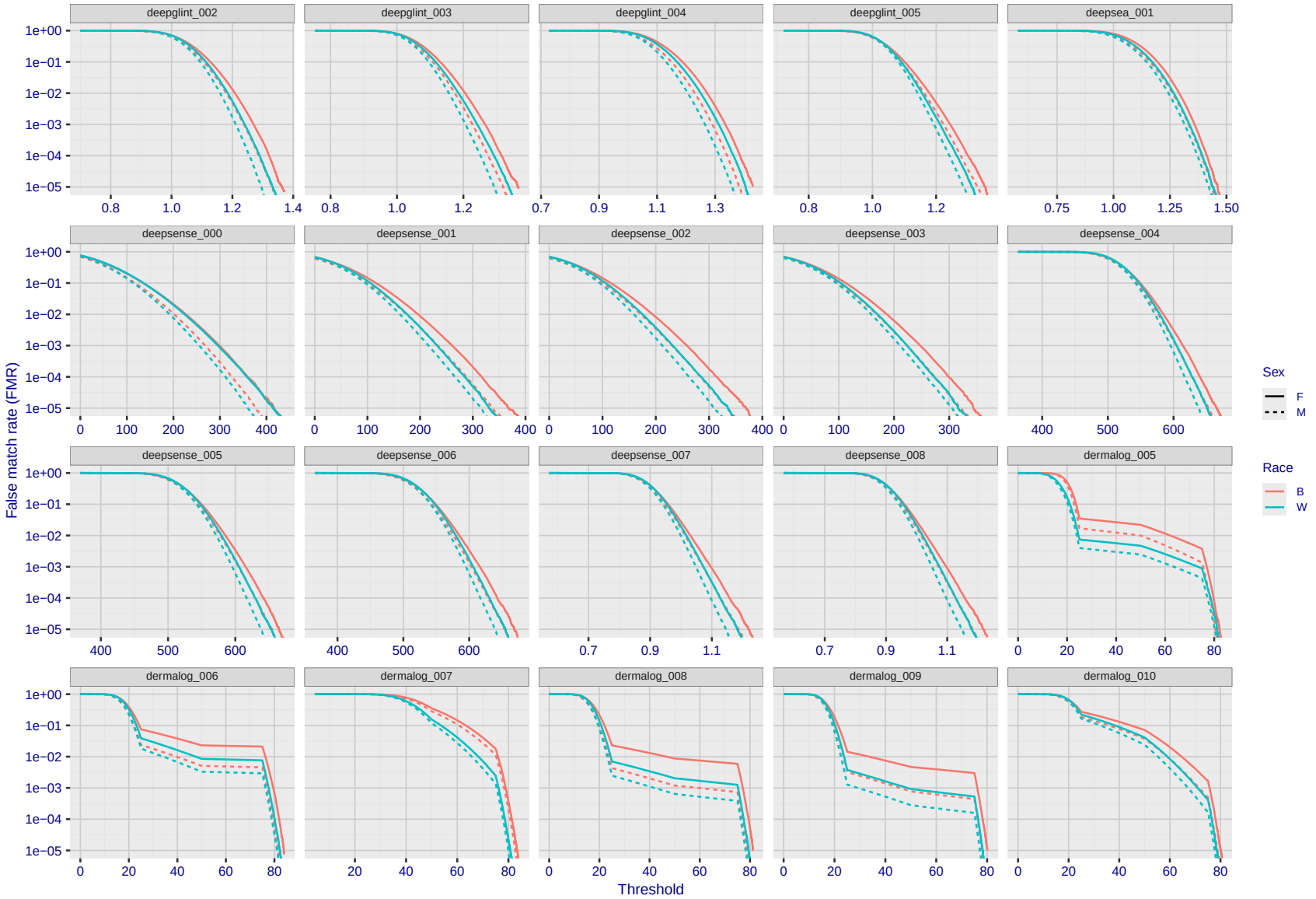


Figure 579: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

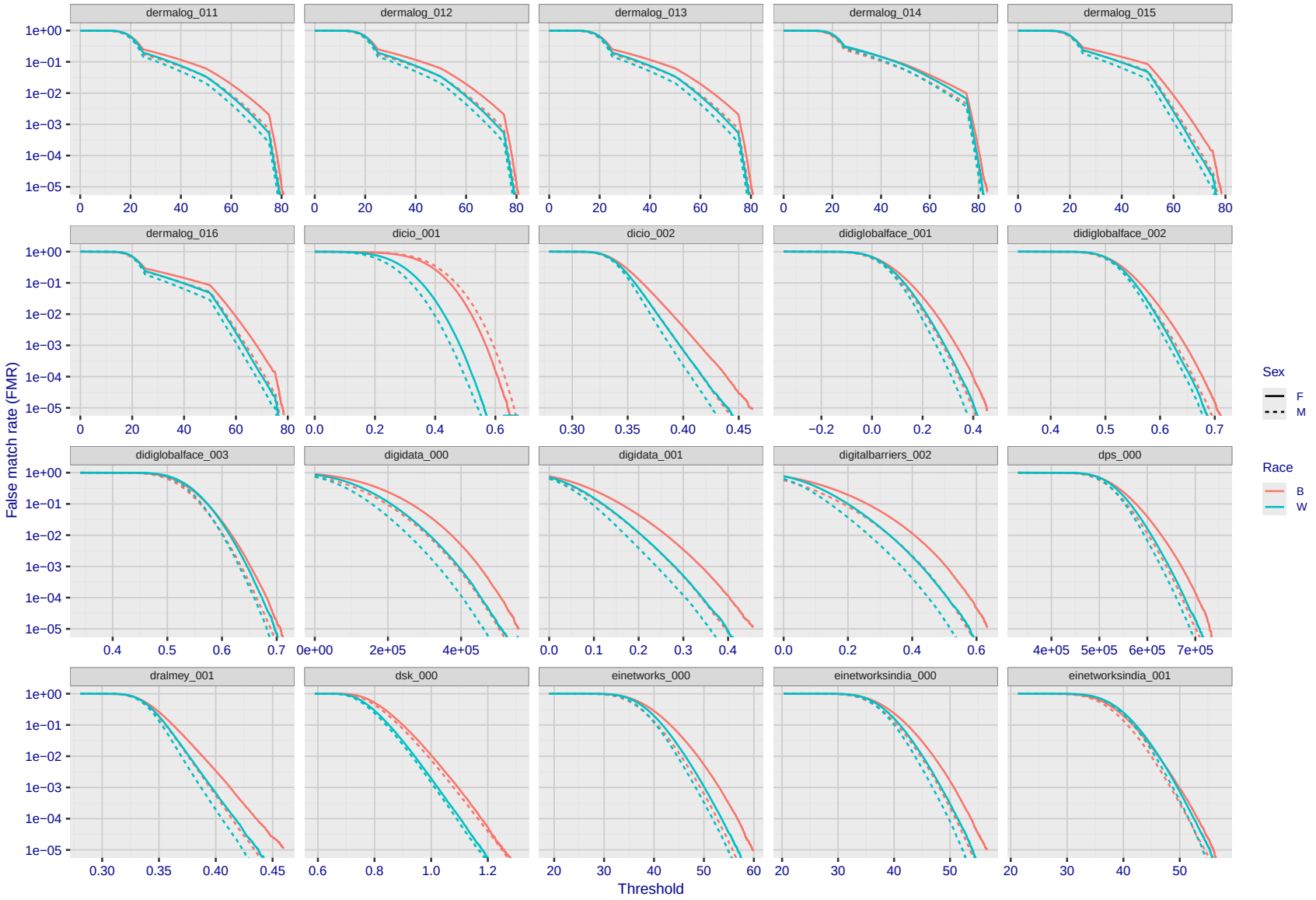


Figure 580: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

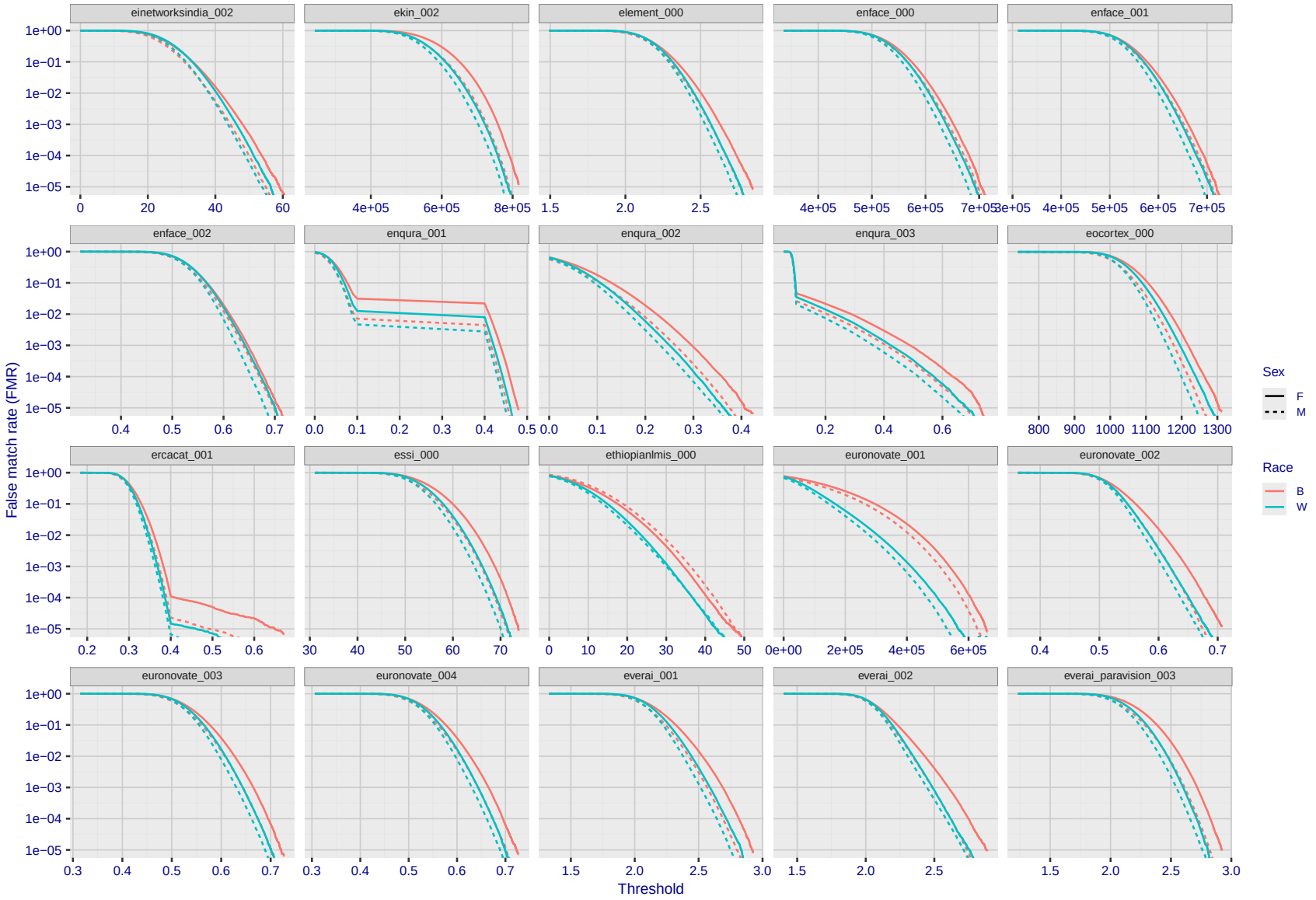


Figure 581: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

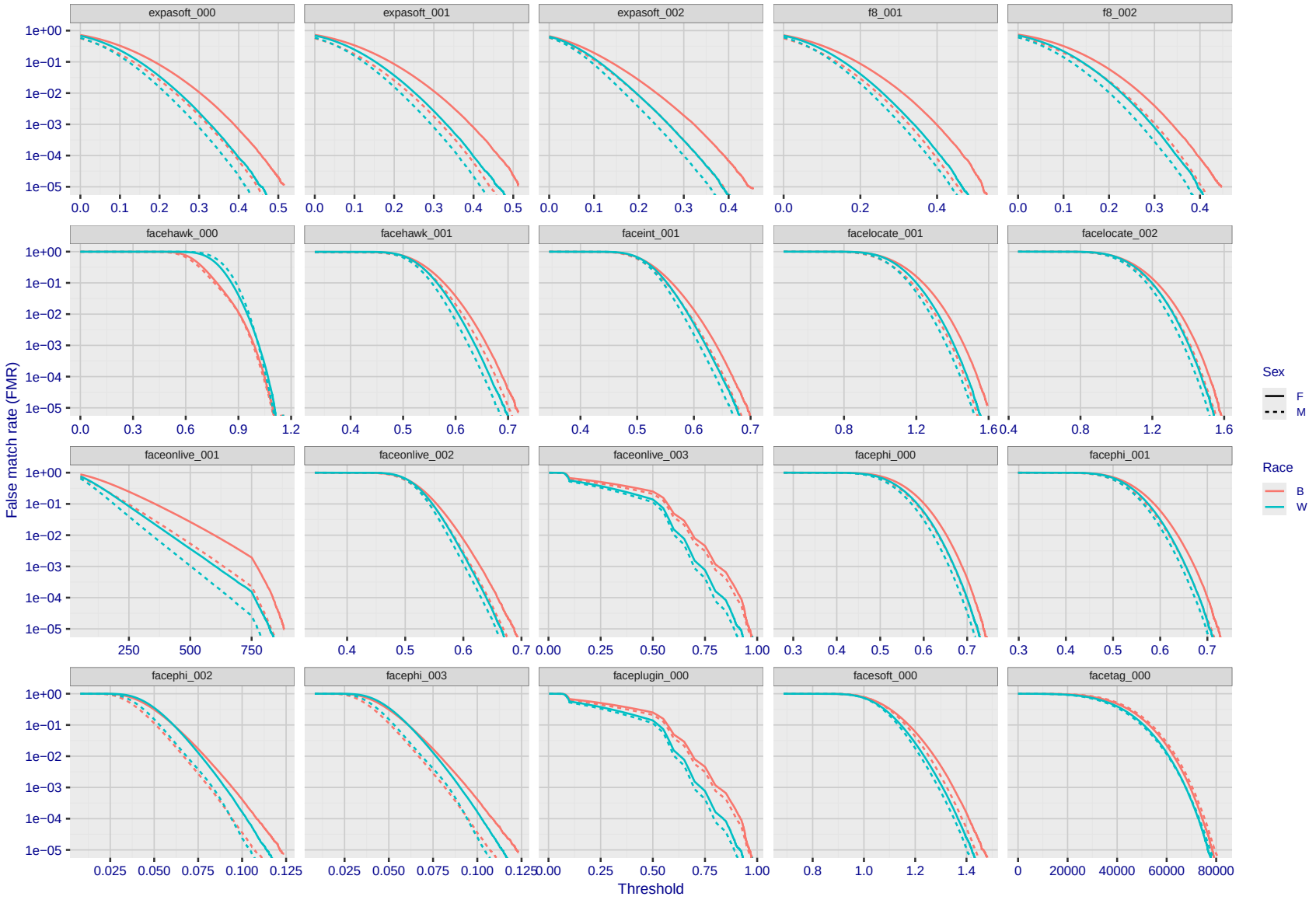


Figure 582: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

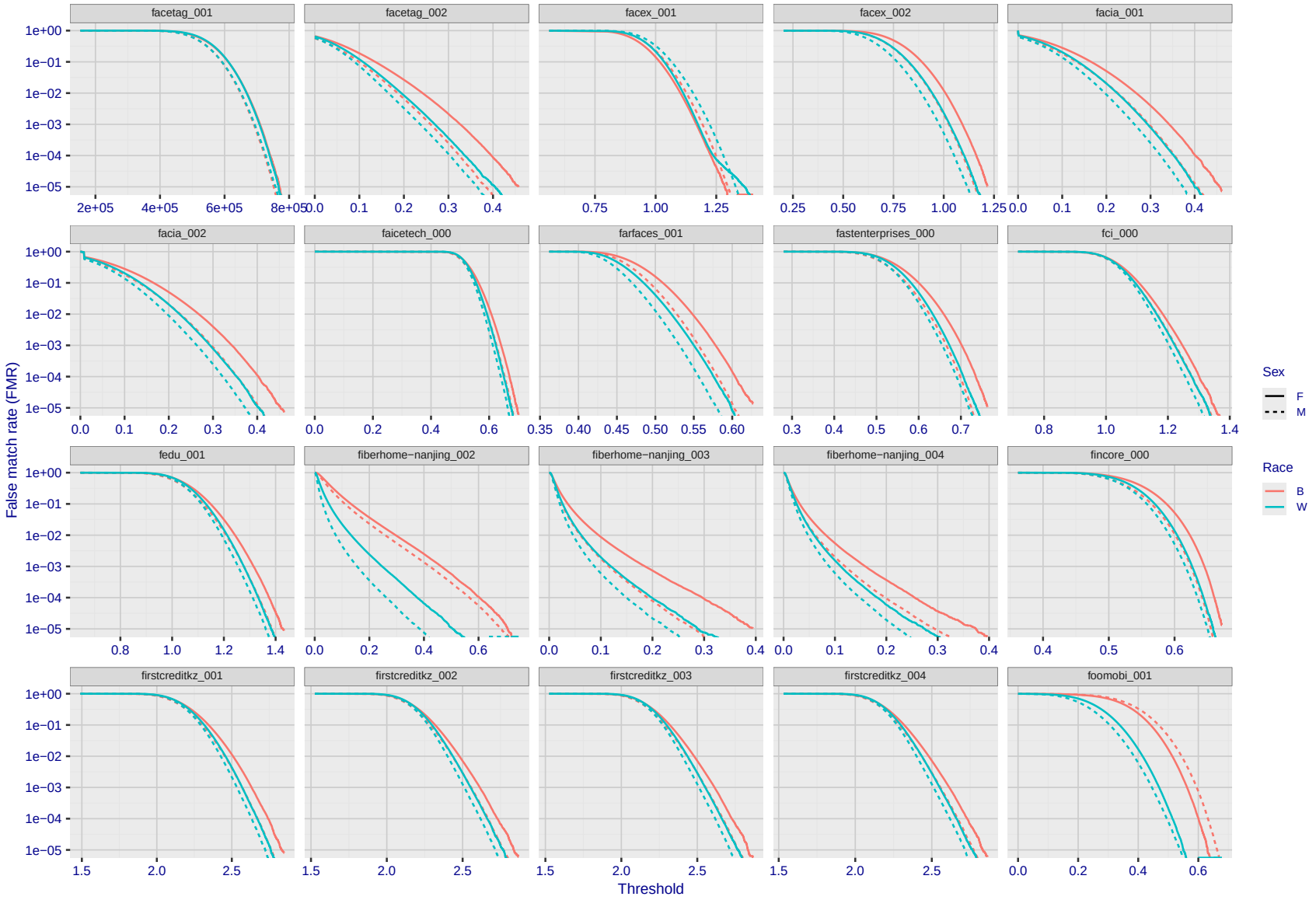


Figure 583: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

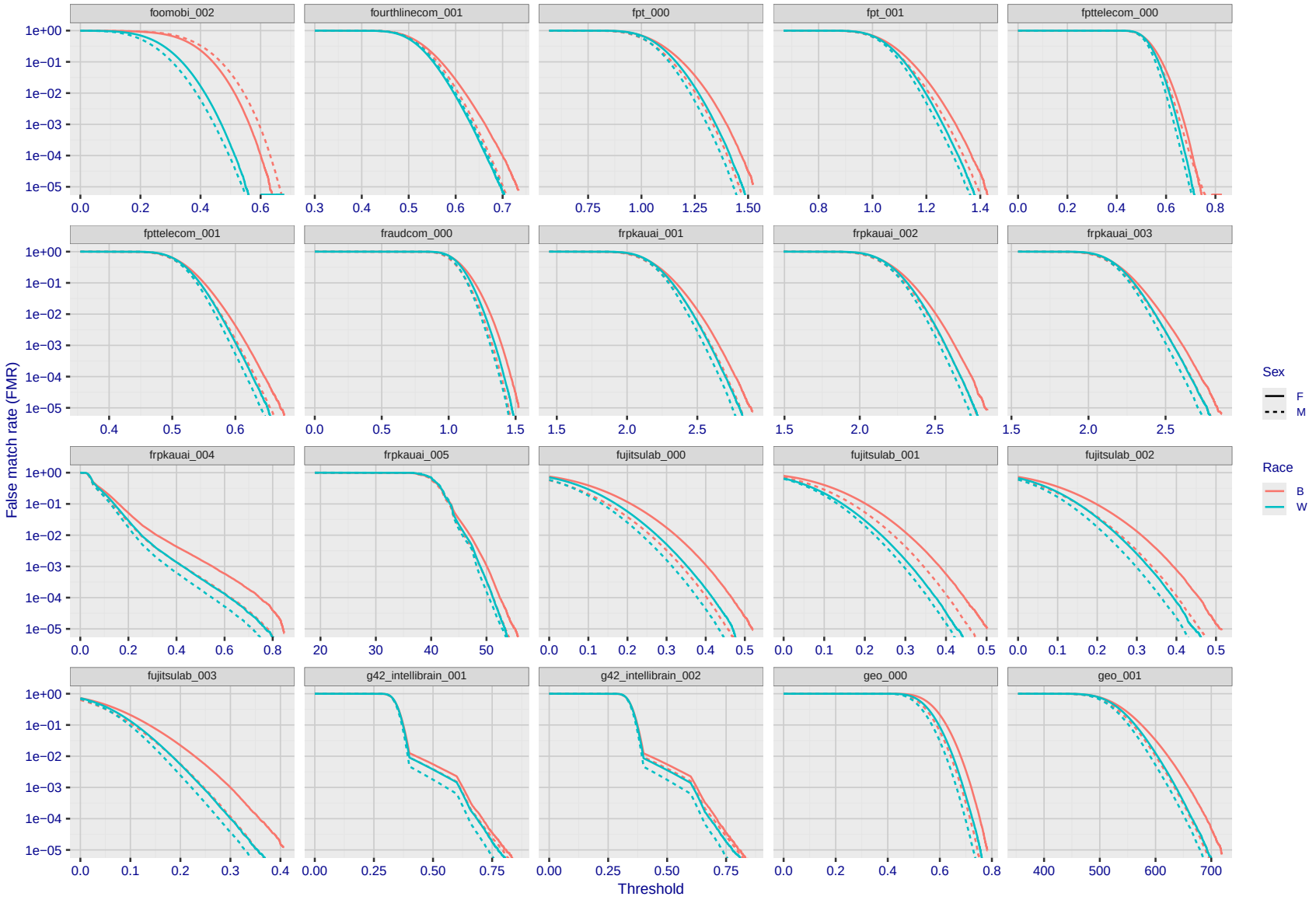


Figure 584: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

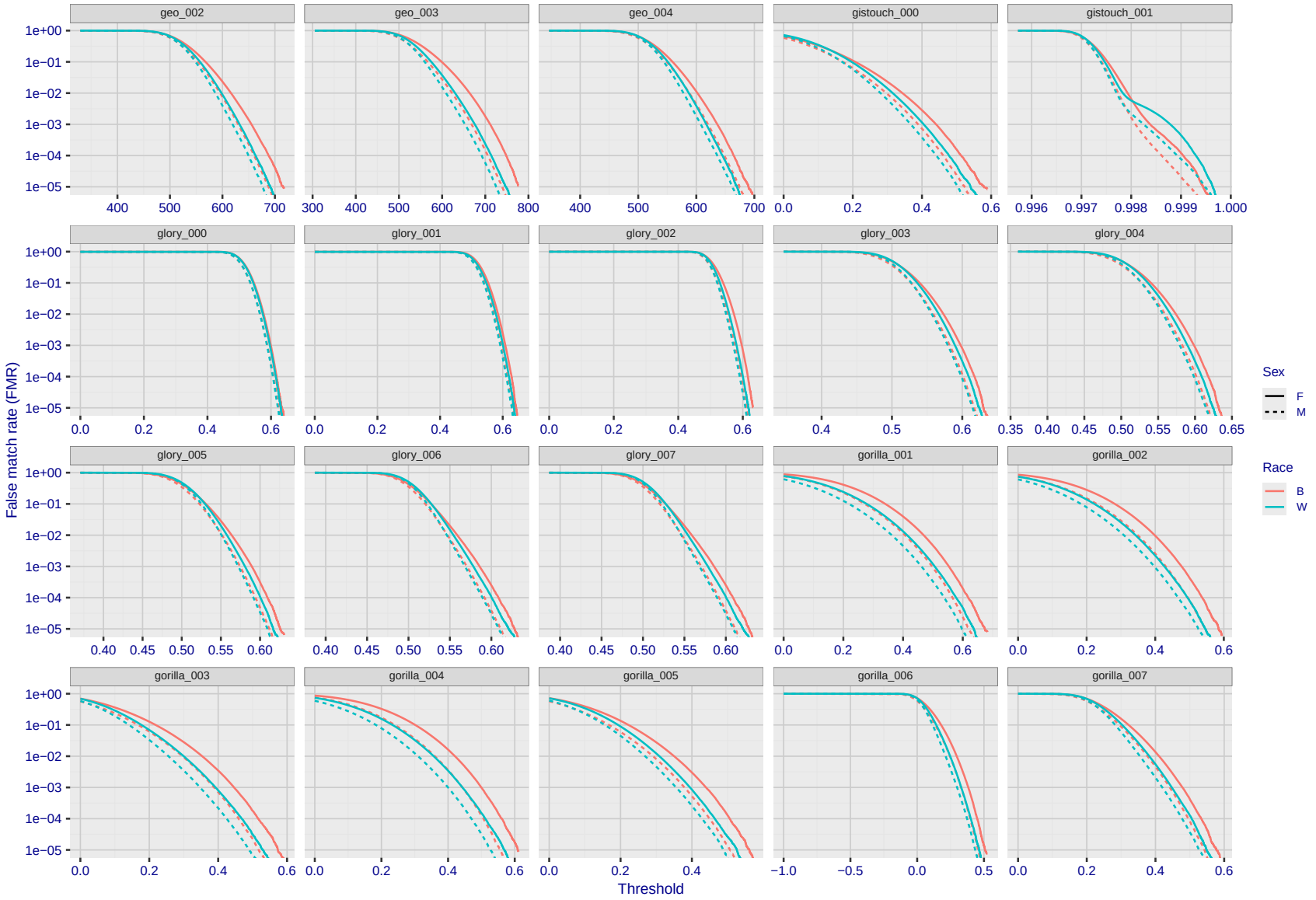


Figure 585: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

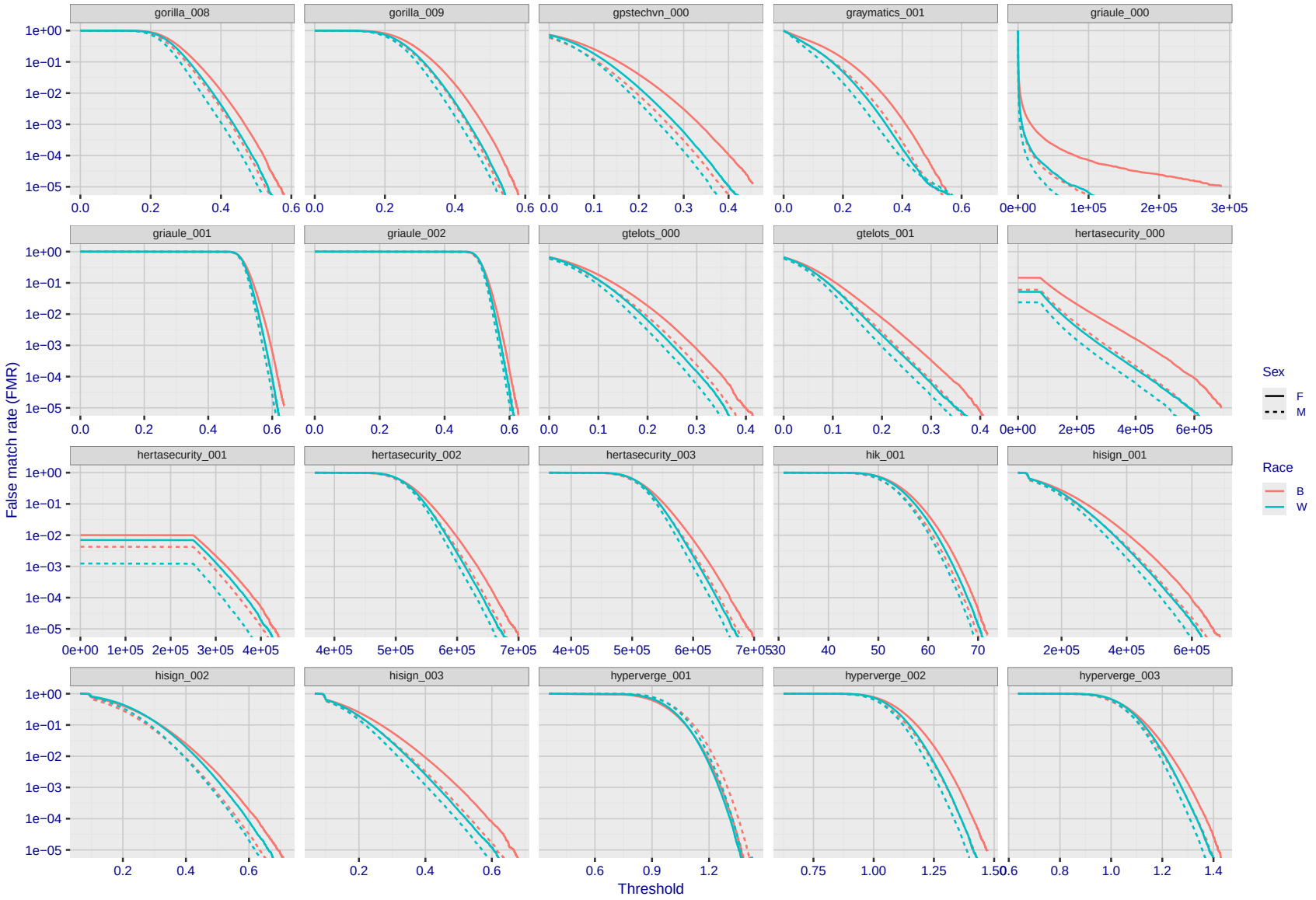


Figure 586: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

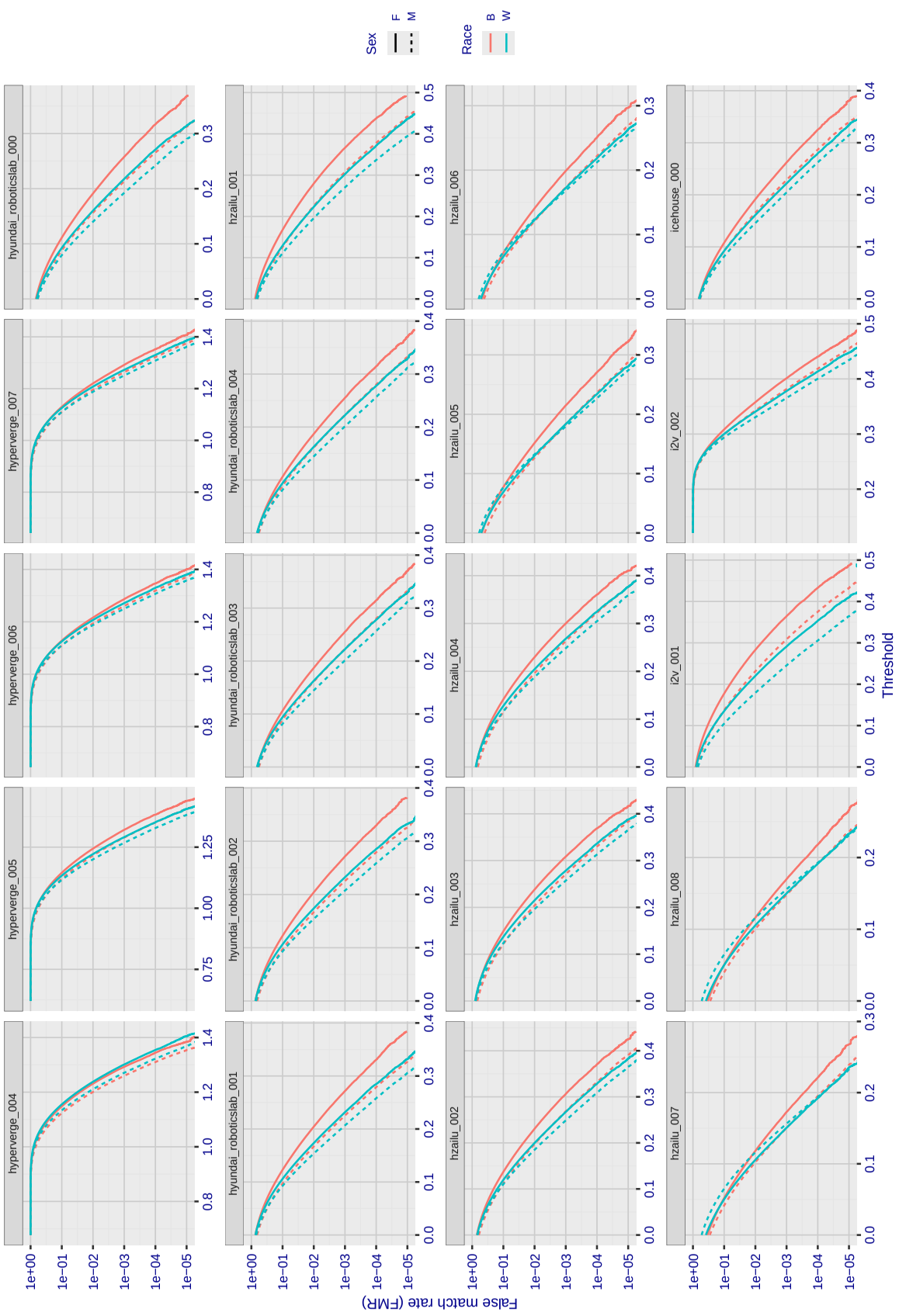


Figure 587: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

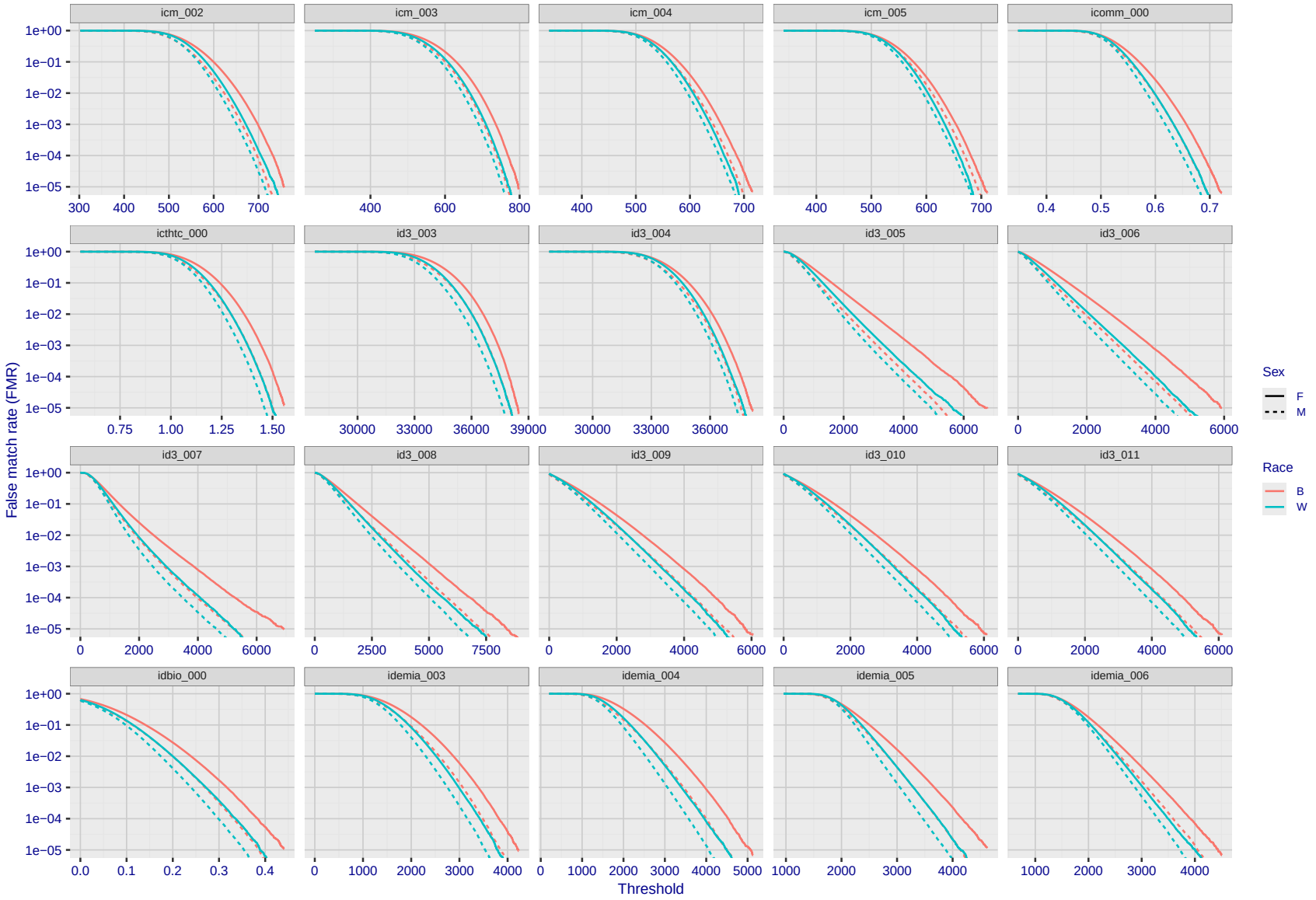


Figure 588: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

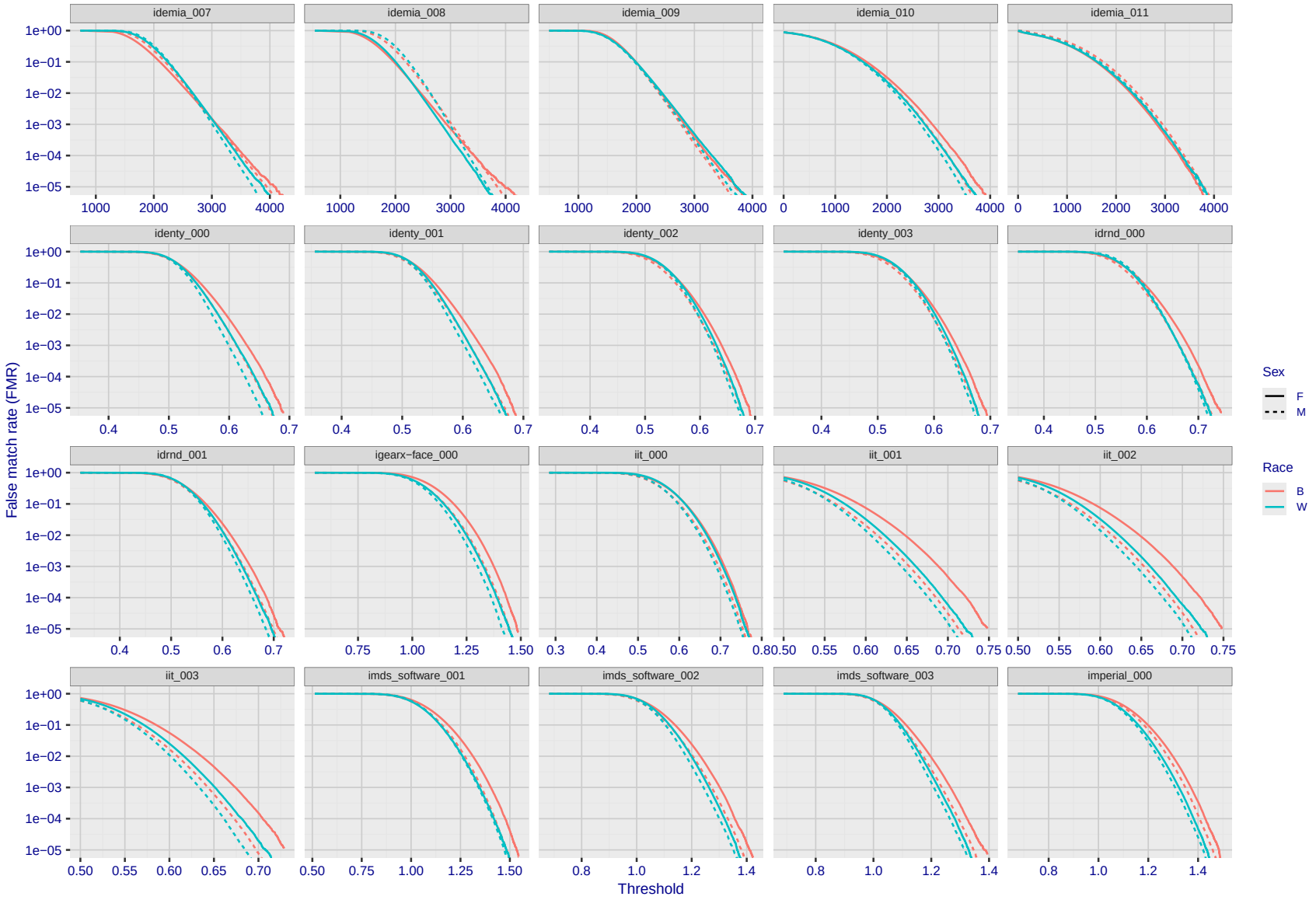


Figure 589: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

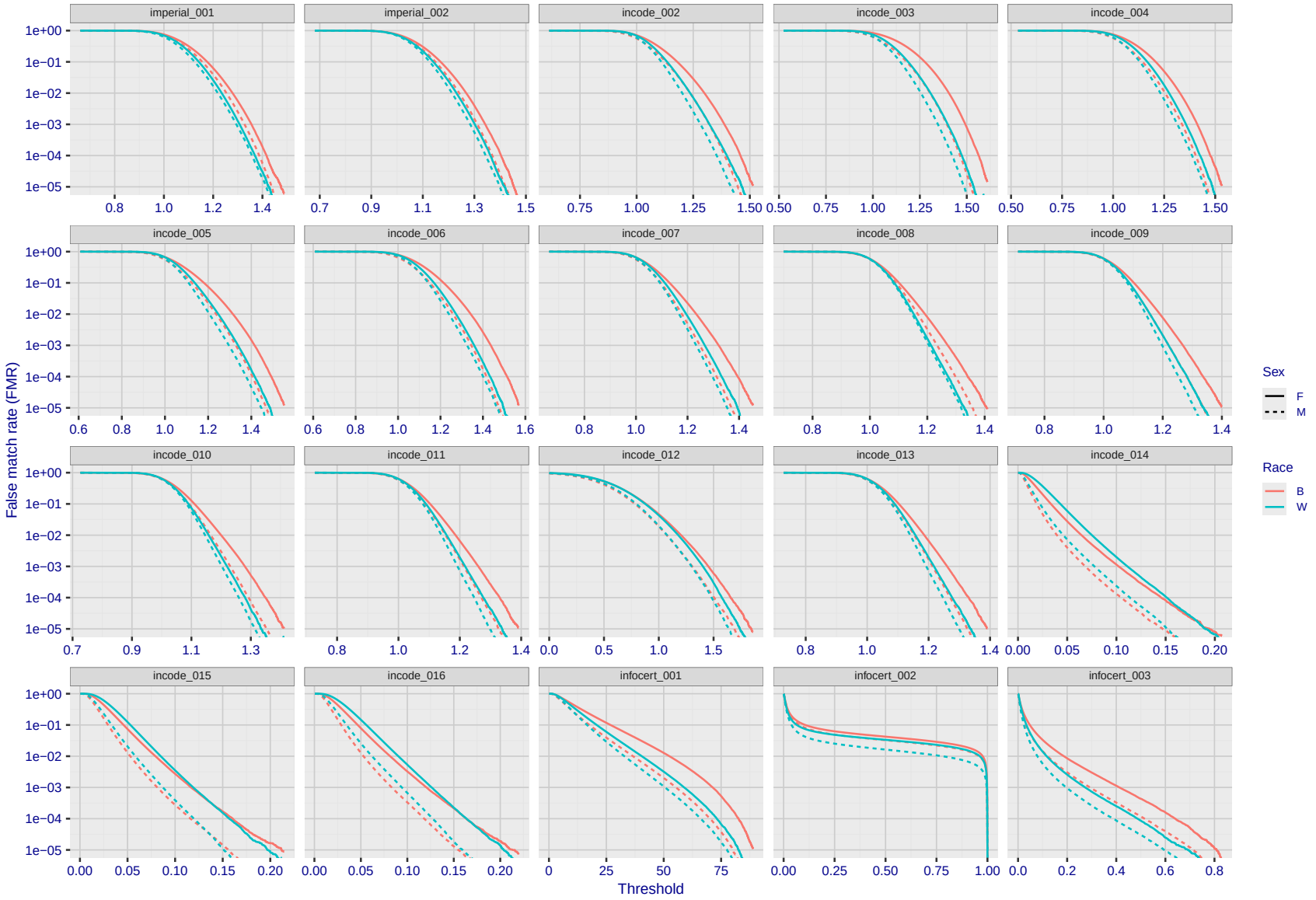


Figure 590: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

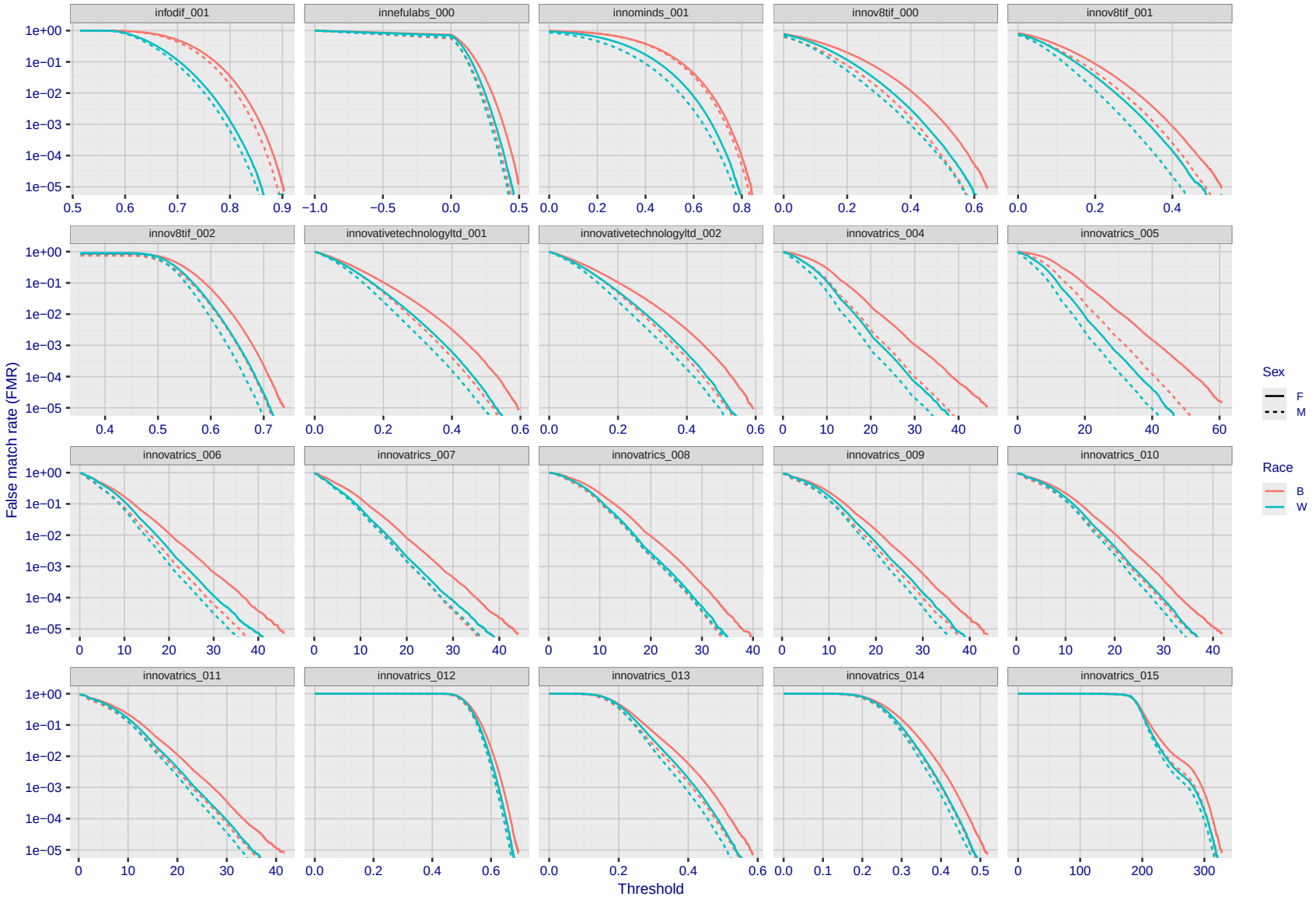


Figure 591: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

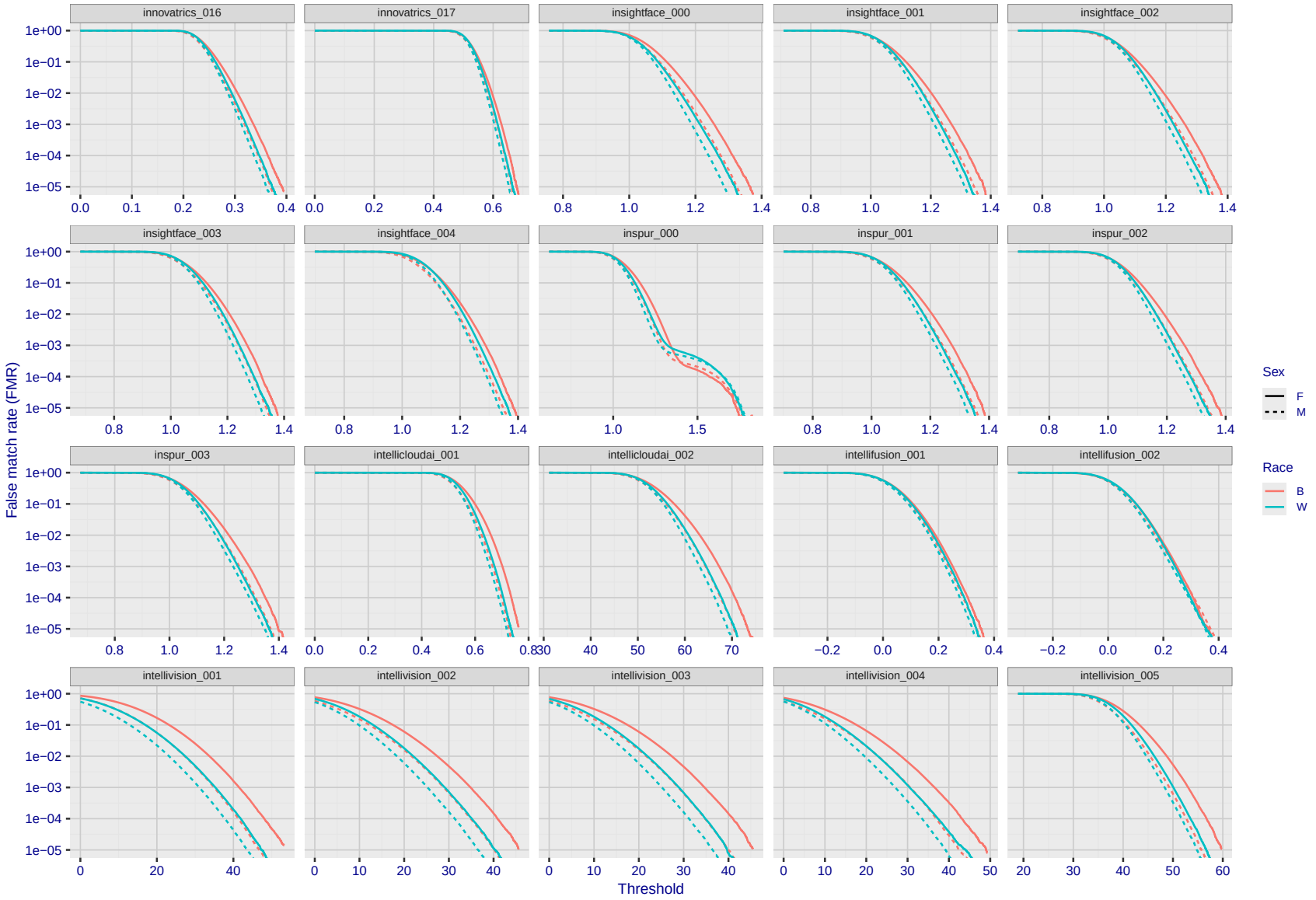


Figure 592: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

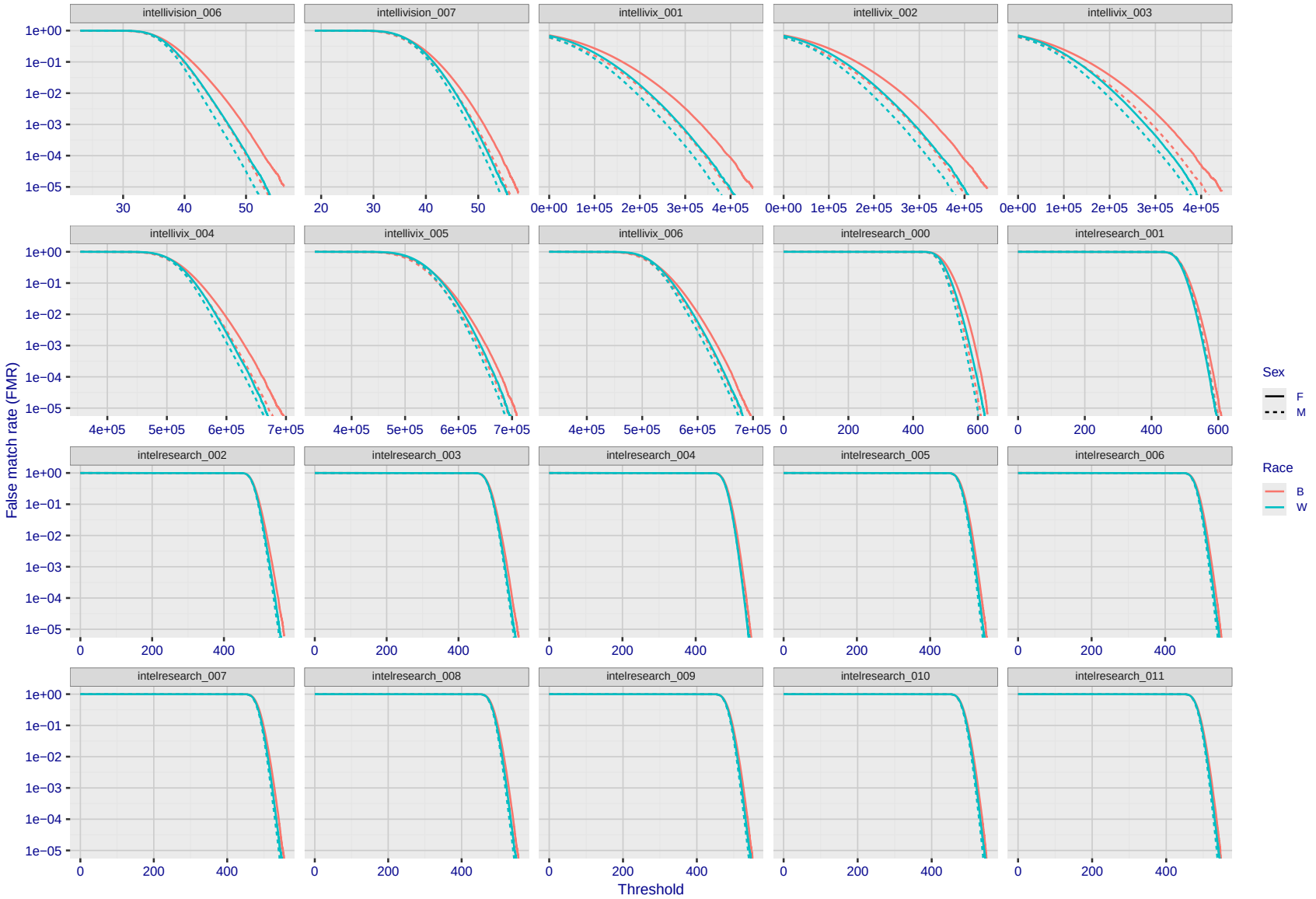


Figure 593: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

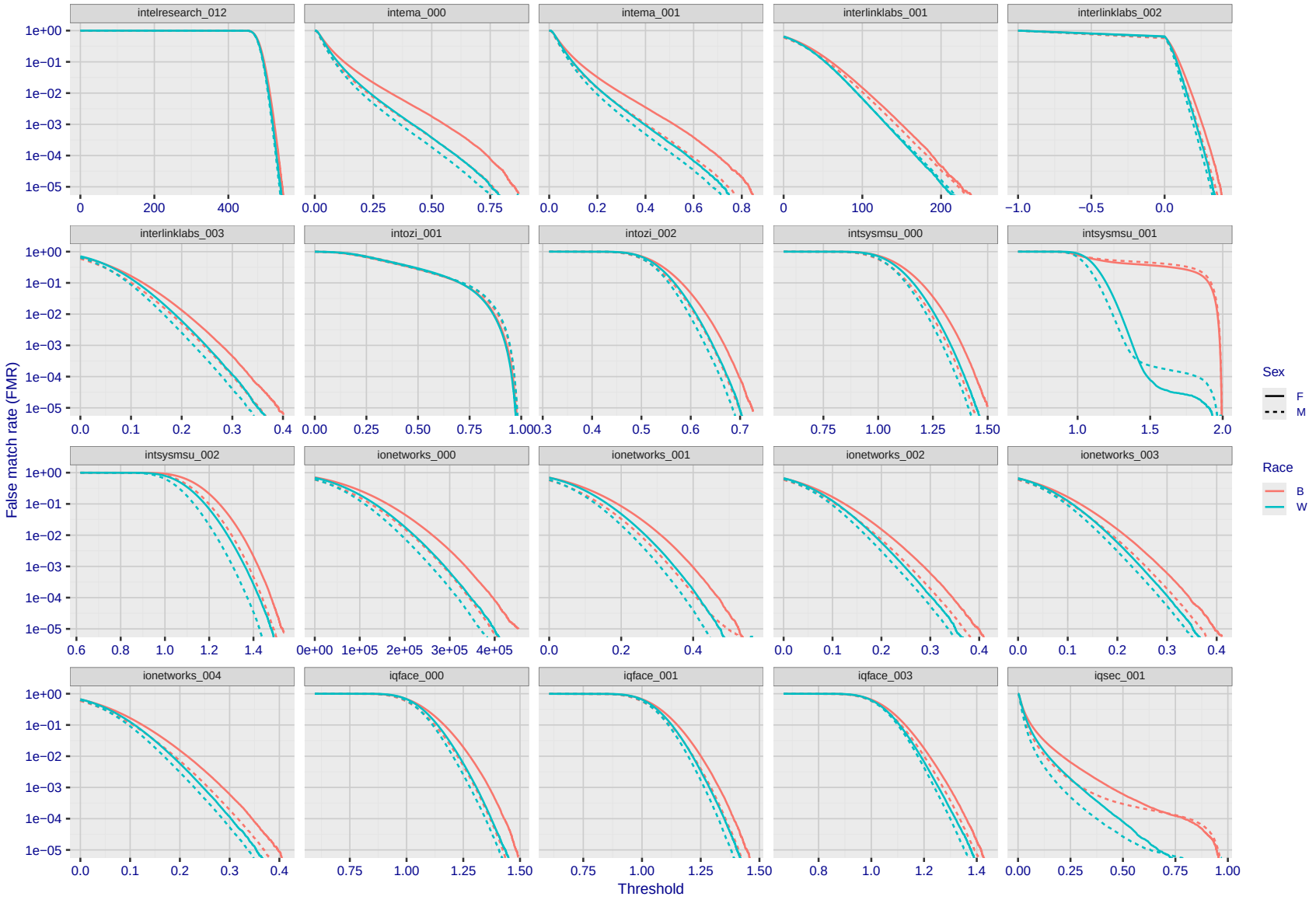


Figure 594: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

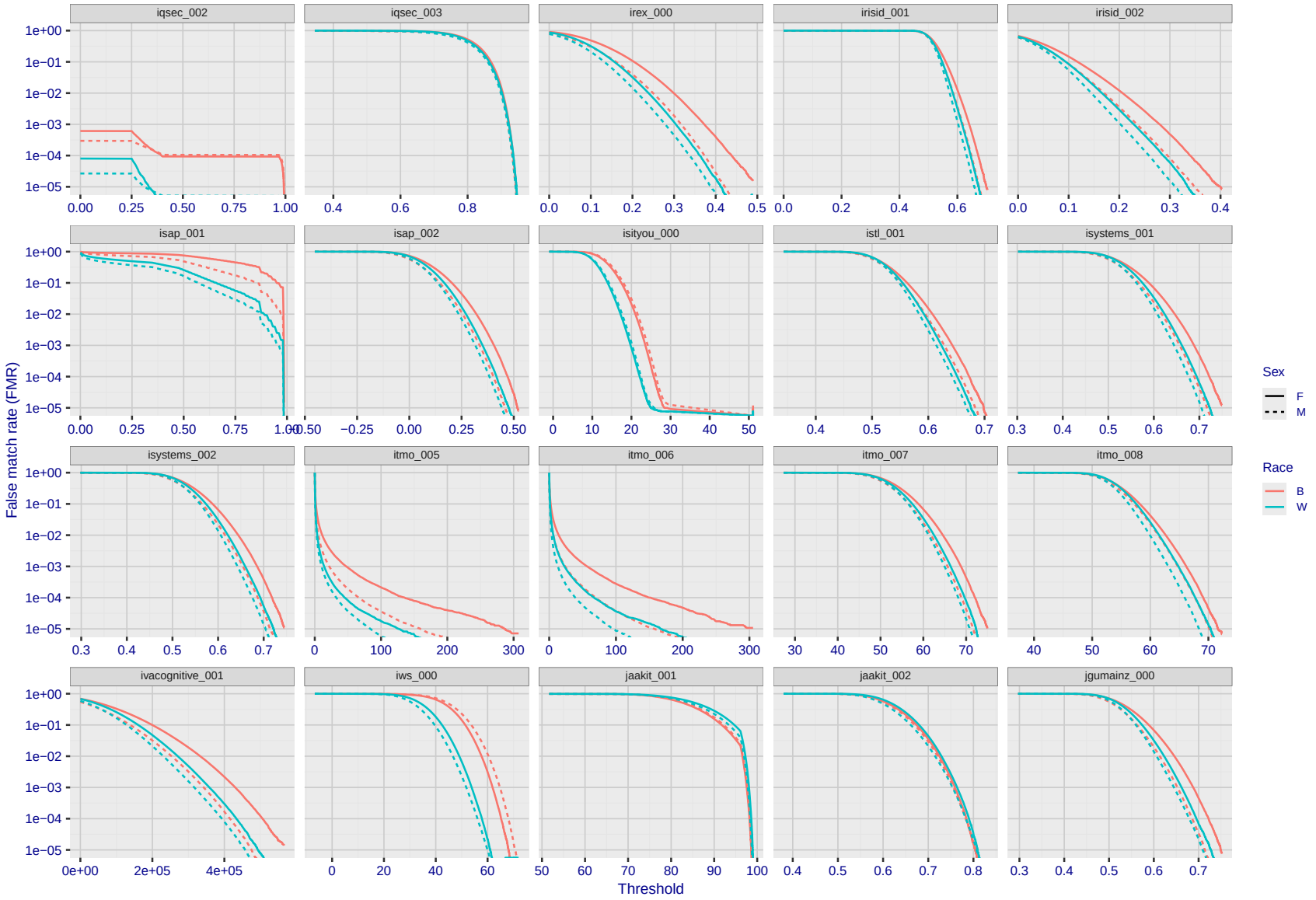


Figure 595: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

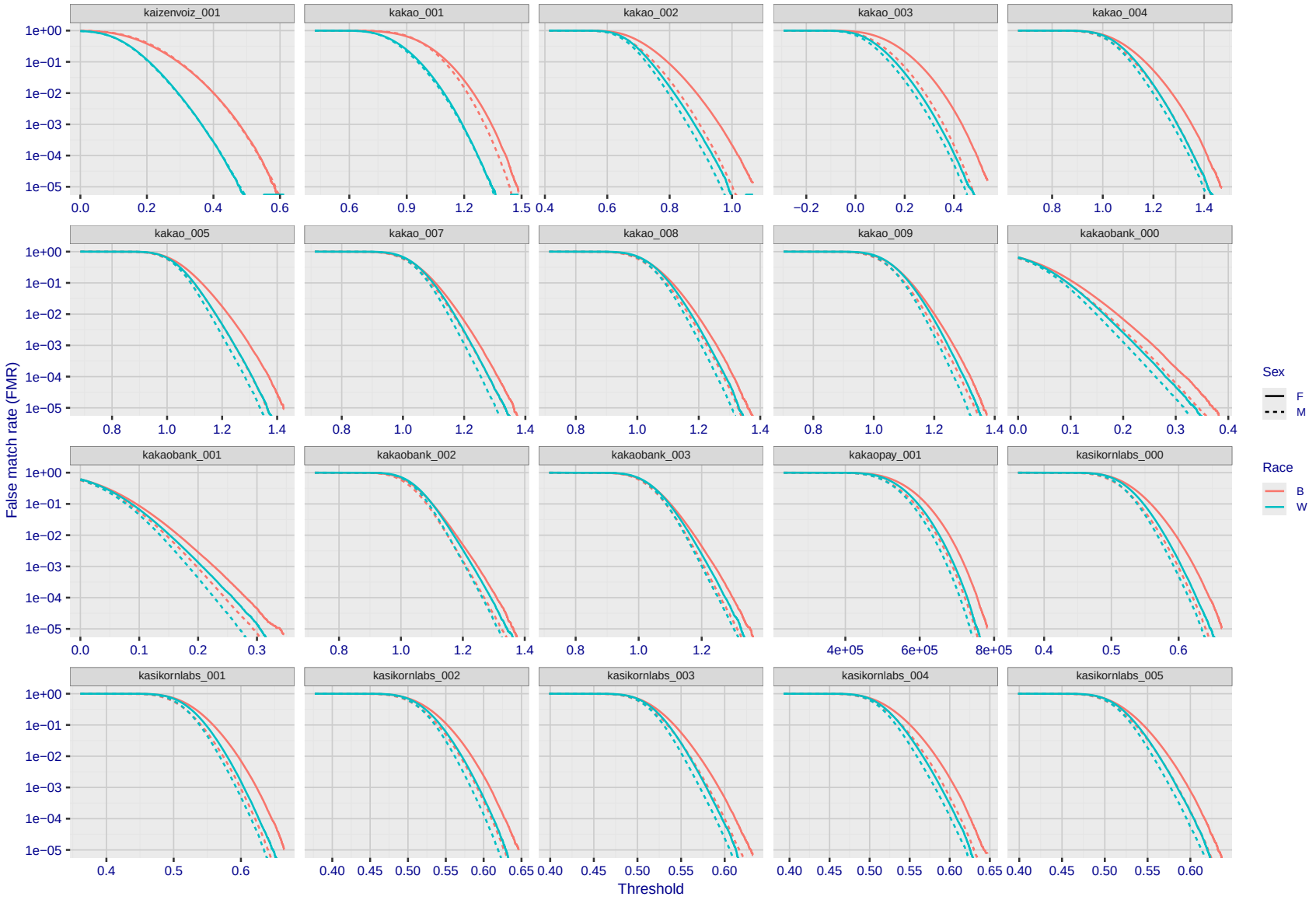


Figure 596: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

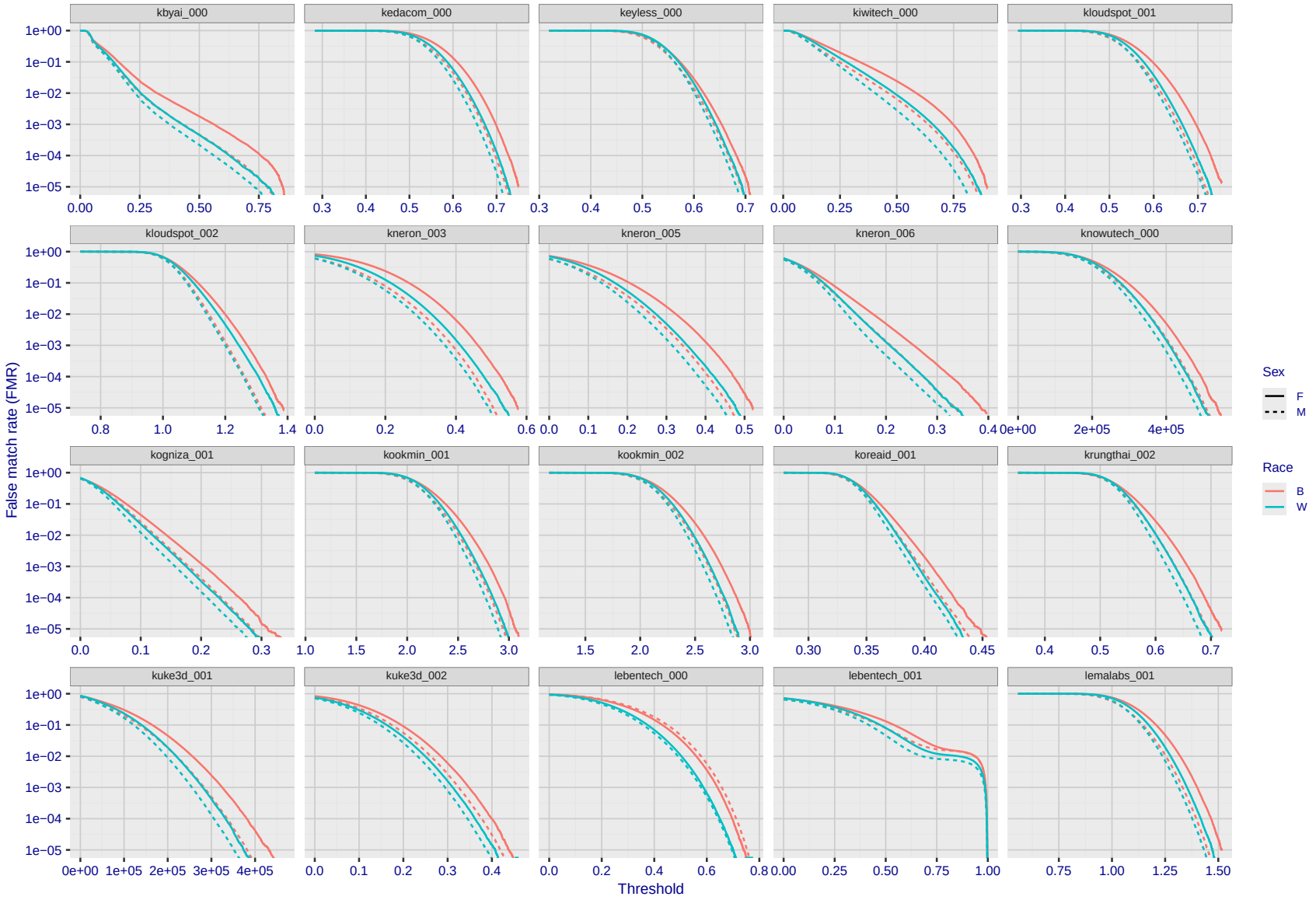


Figure 597: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

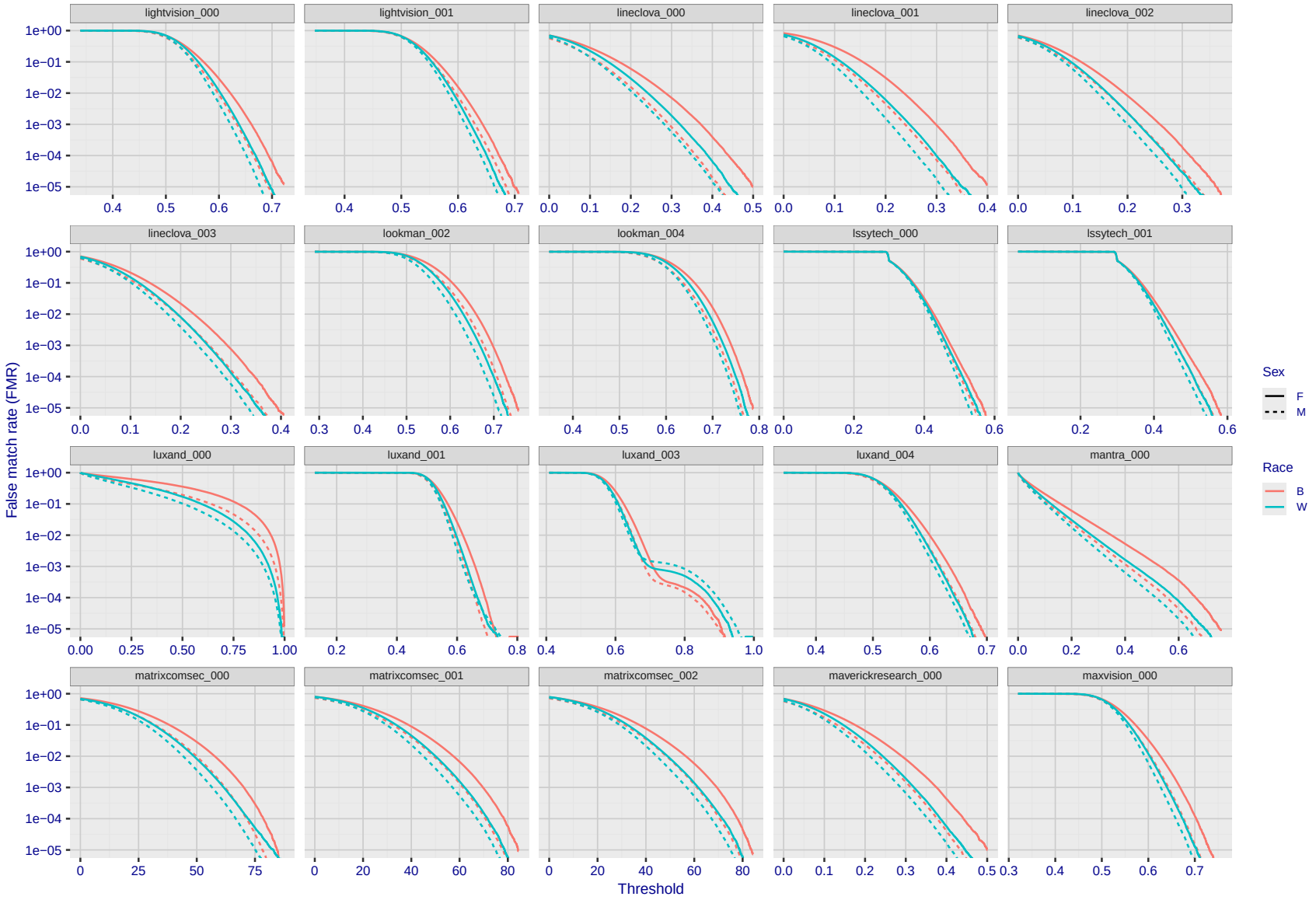


Figure 598: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

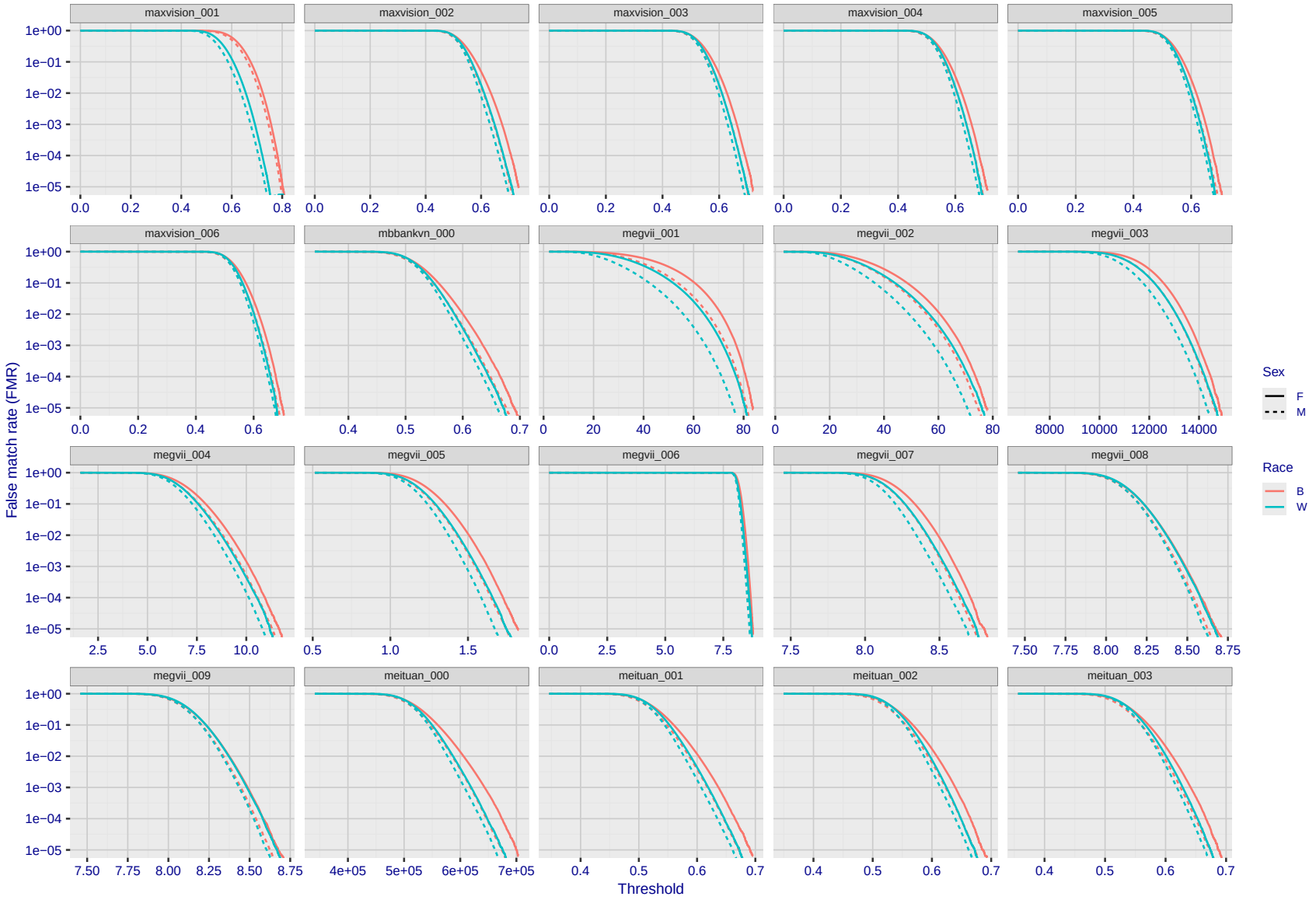


Figure 599: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

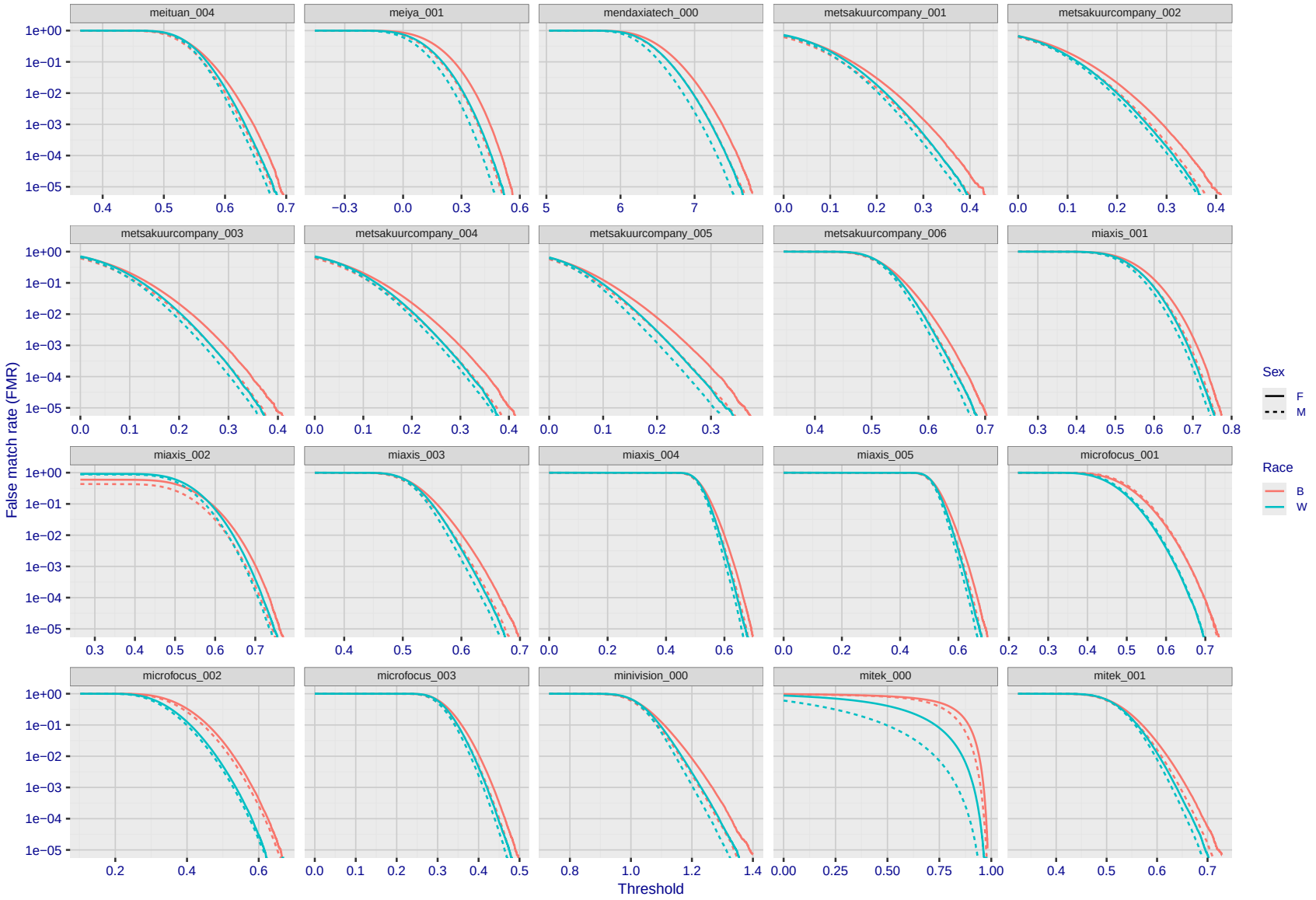


Figure 600: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

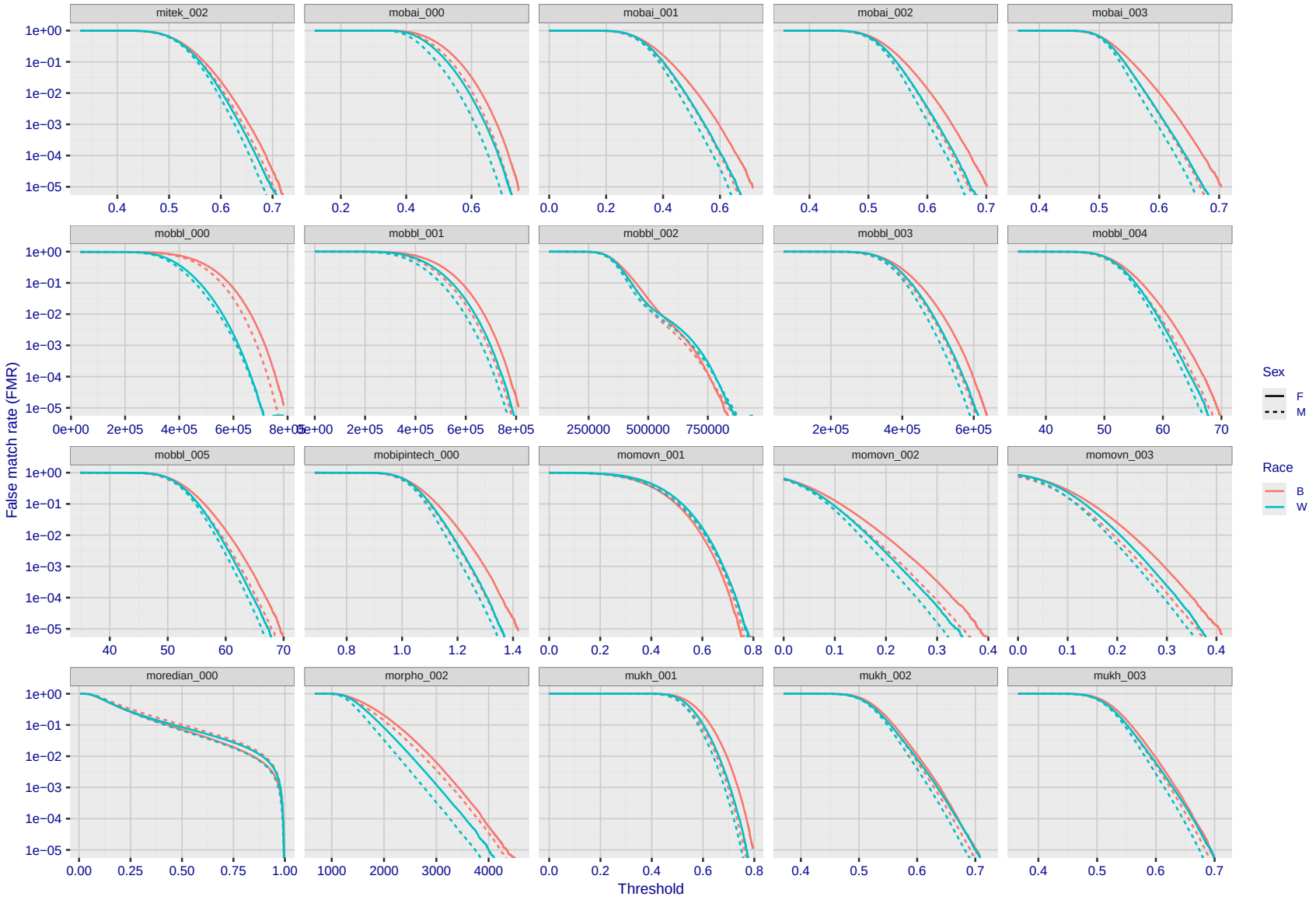


Figure 601: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

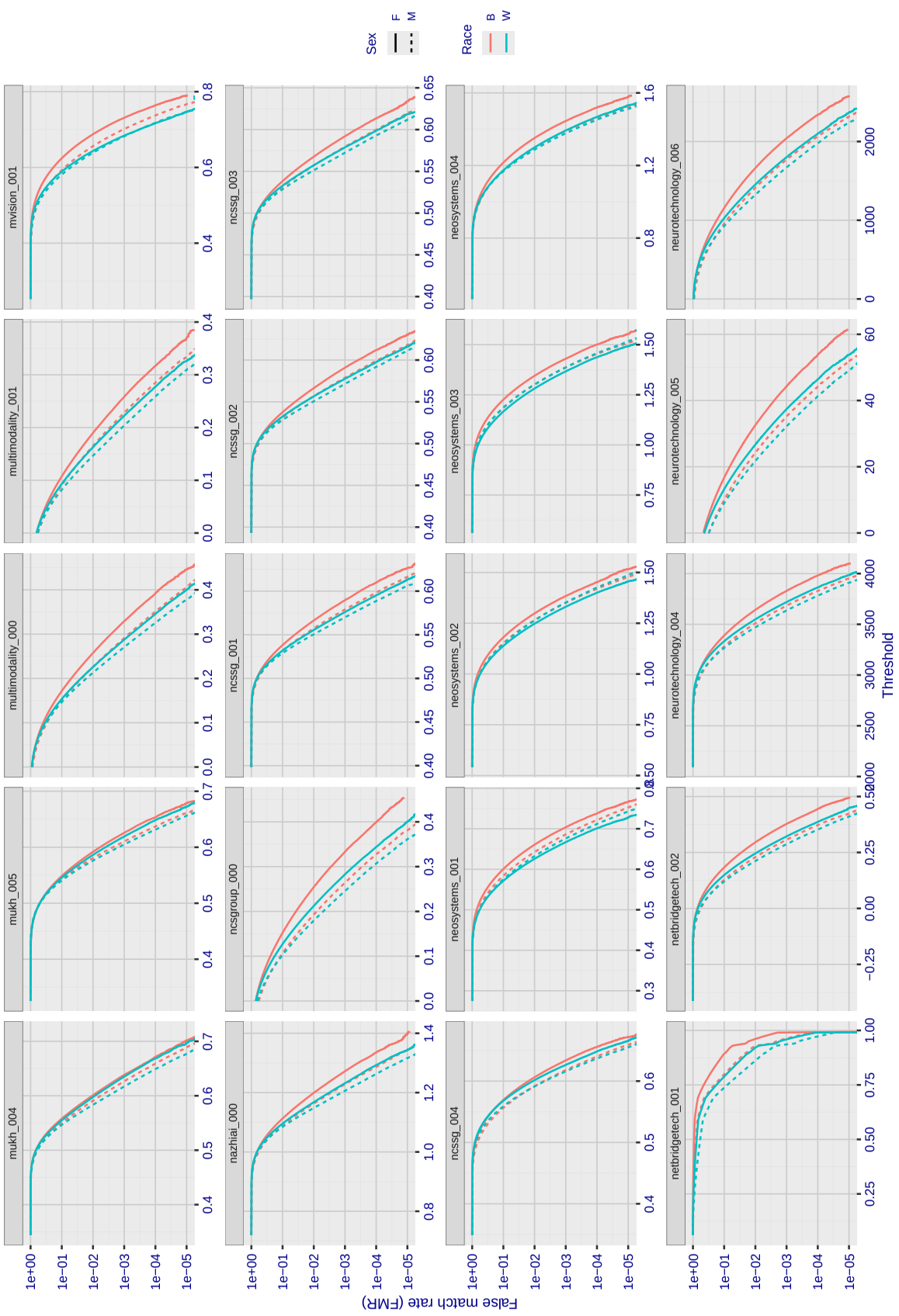


Figure 602: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

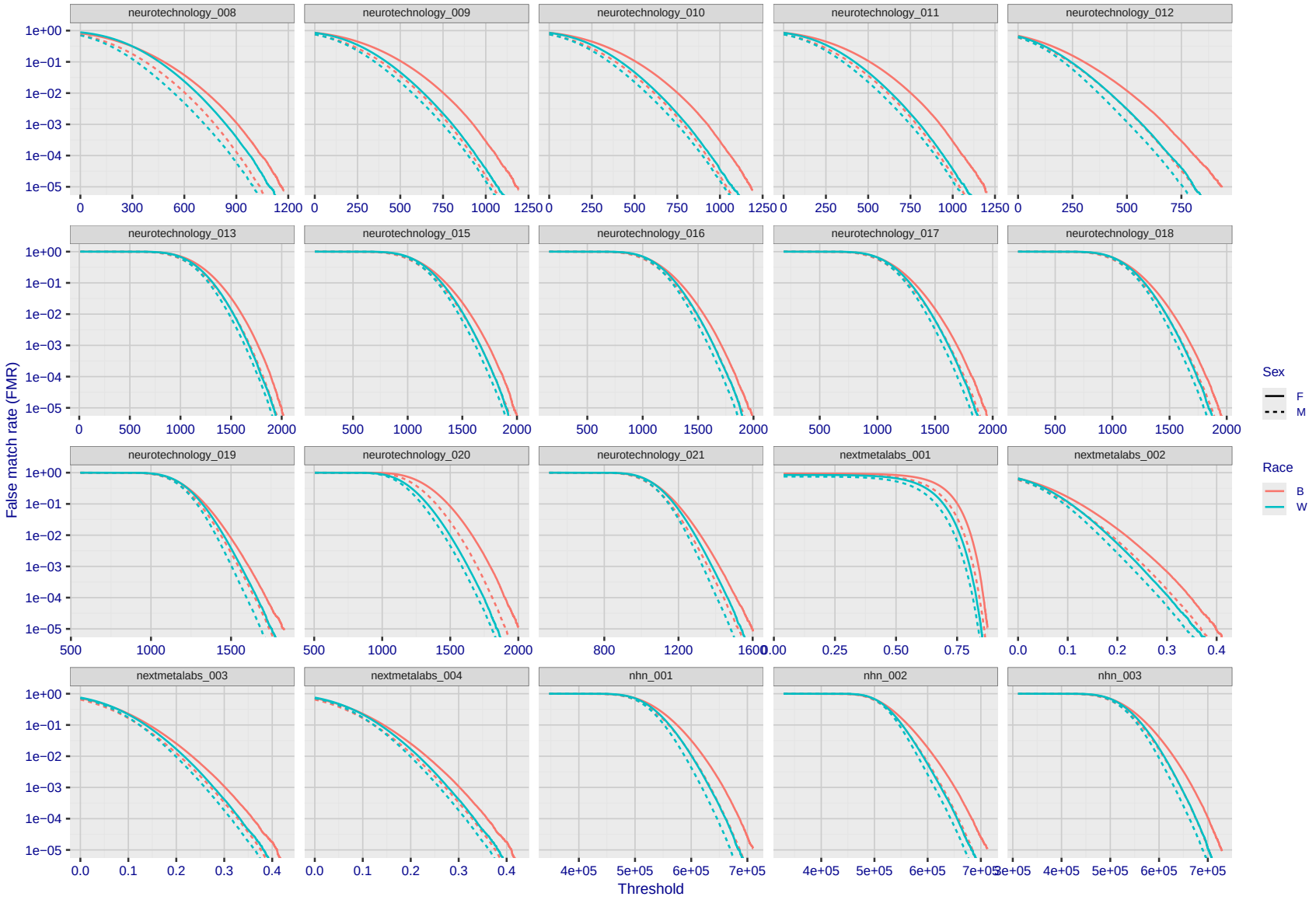


Figure 603: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

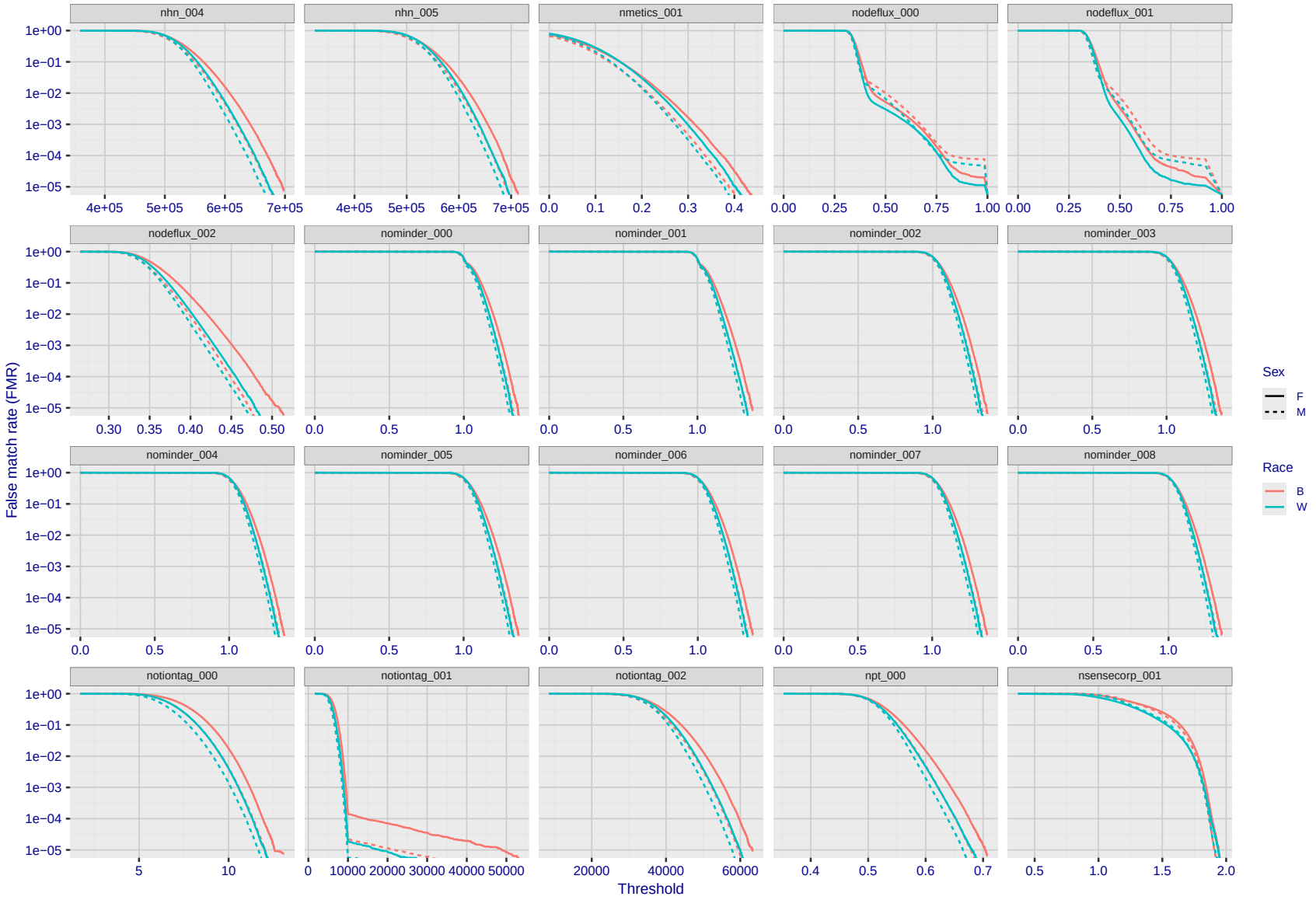


Figure 604: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

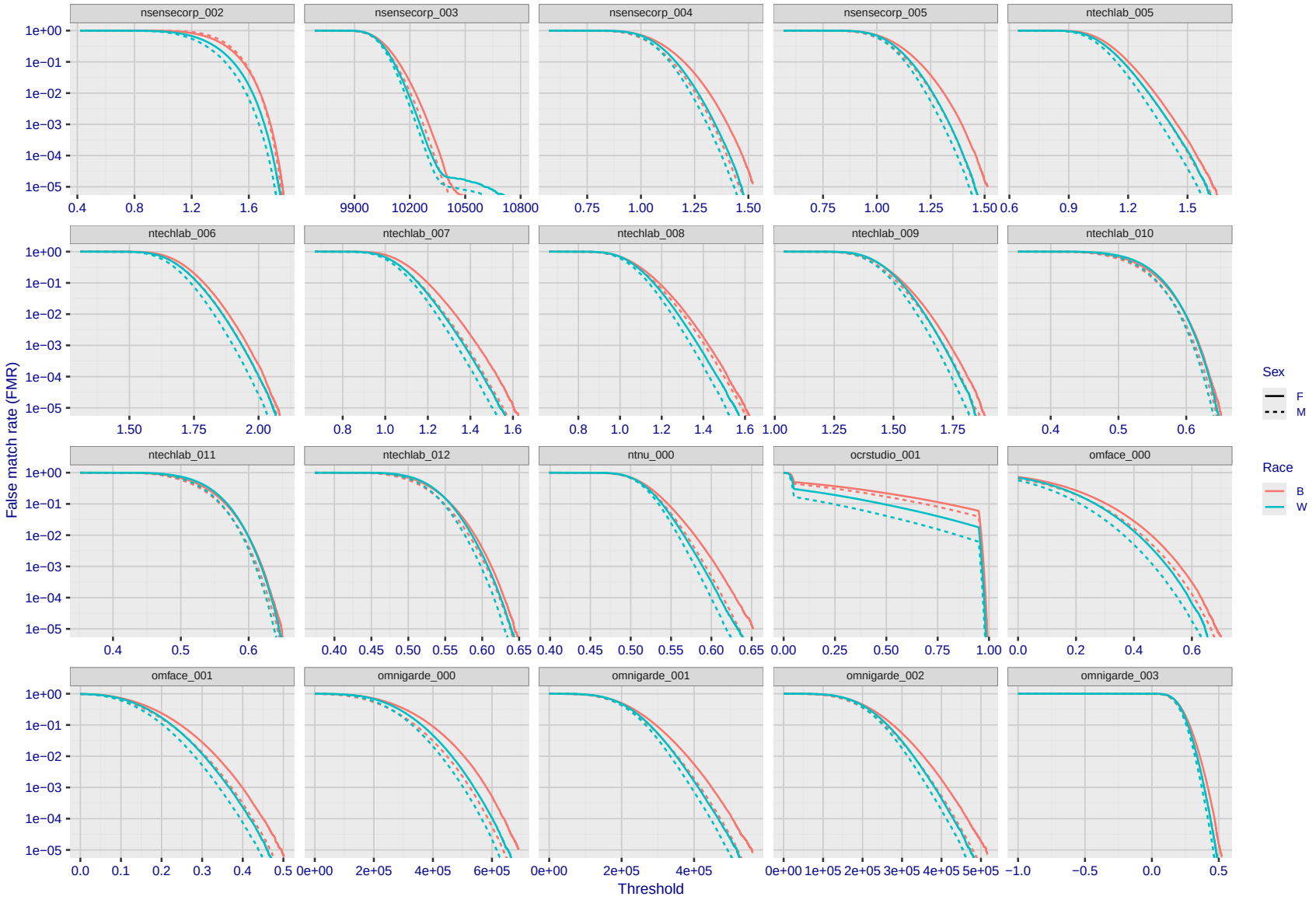


Figure 605: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

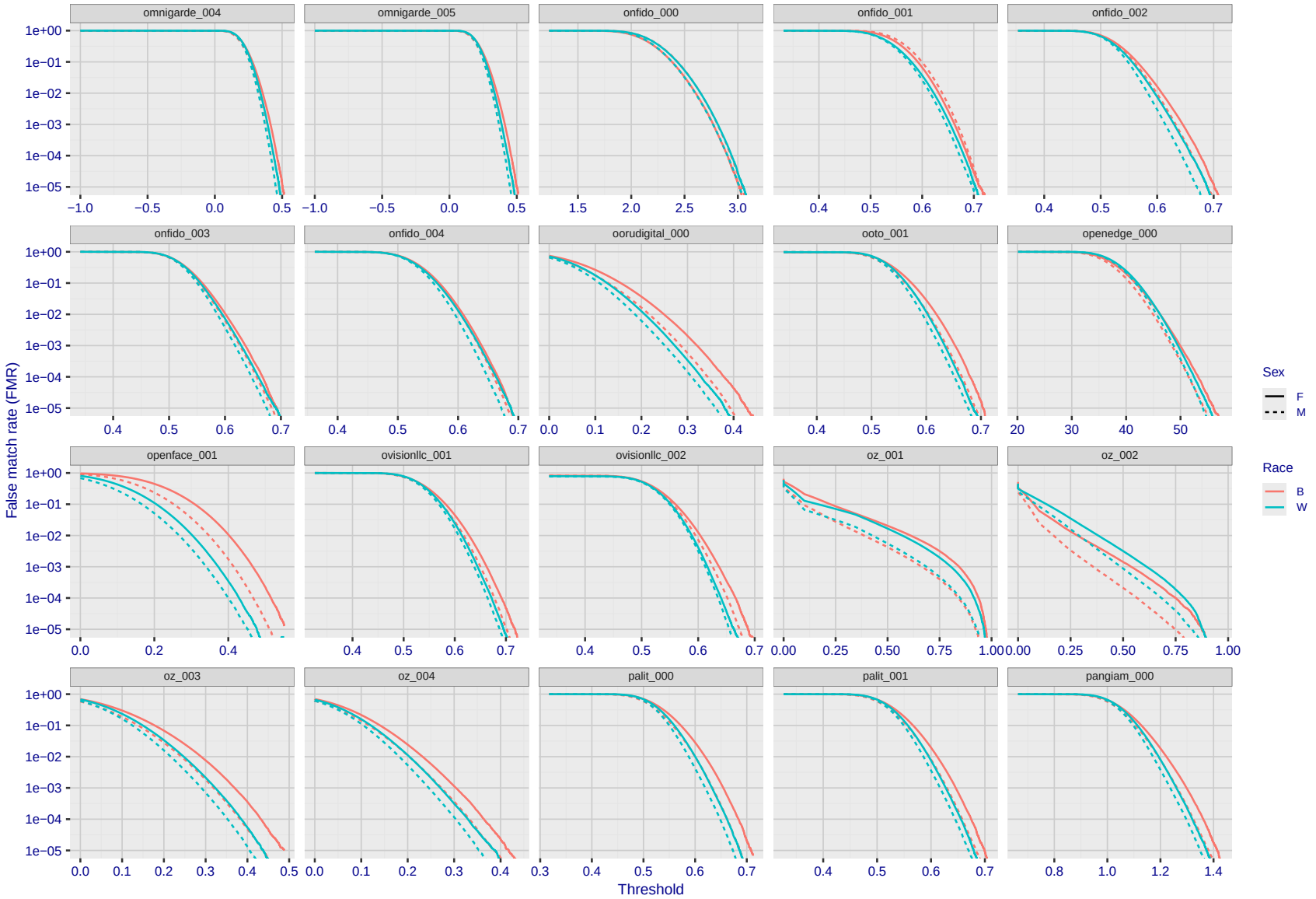


Figure 606: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

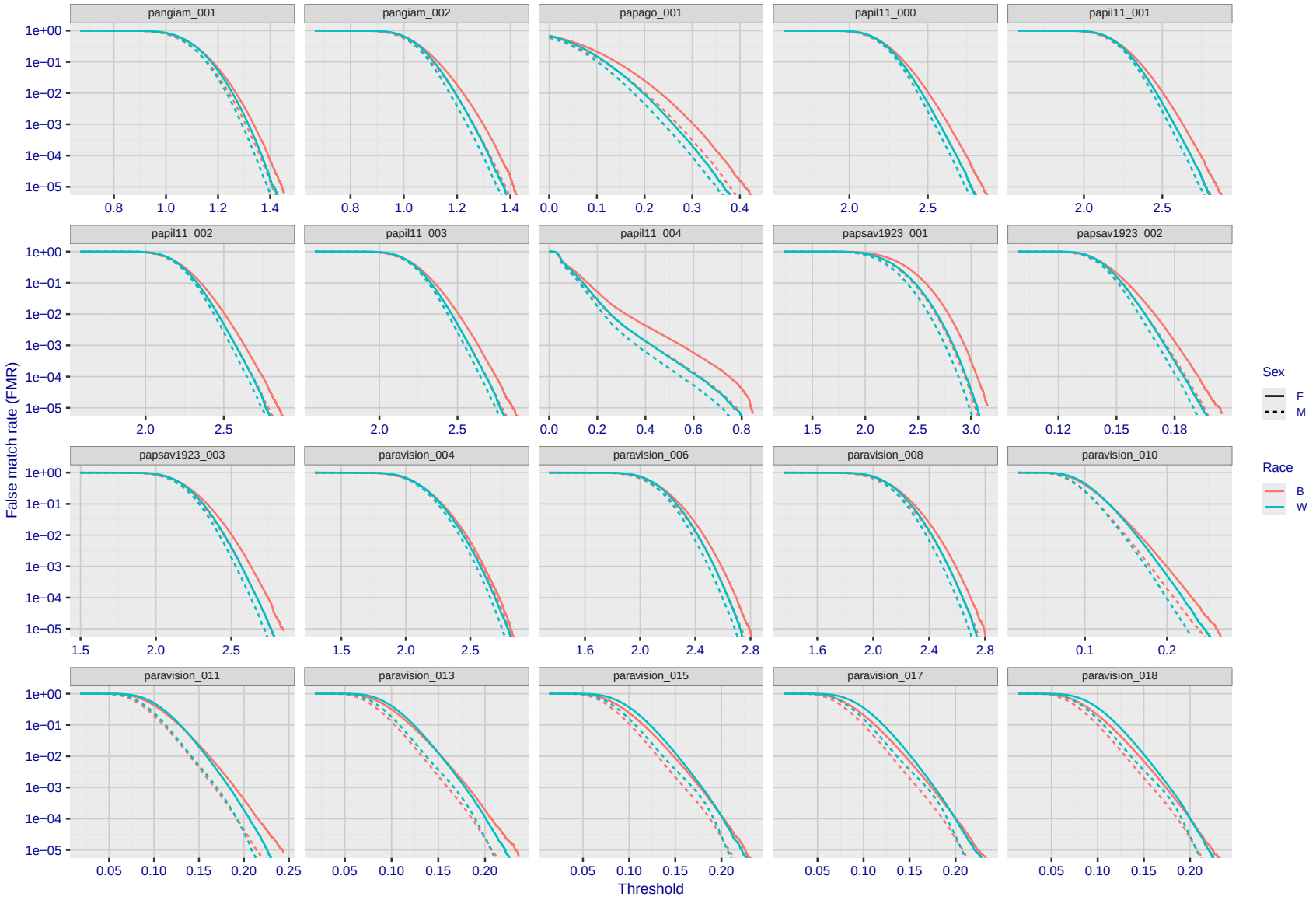


Figure 607: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

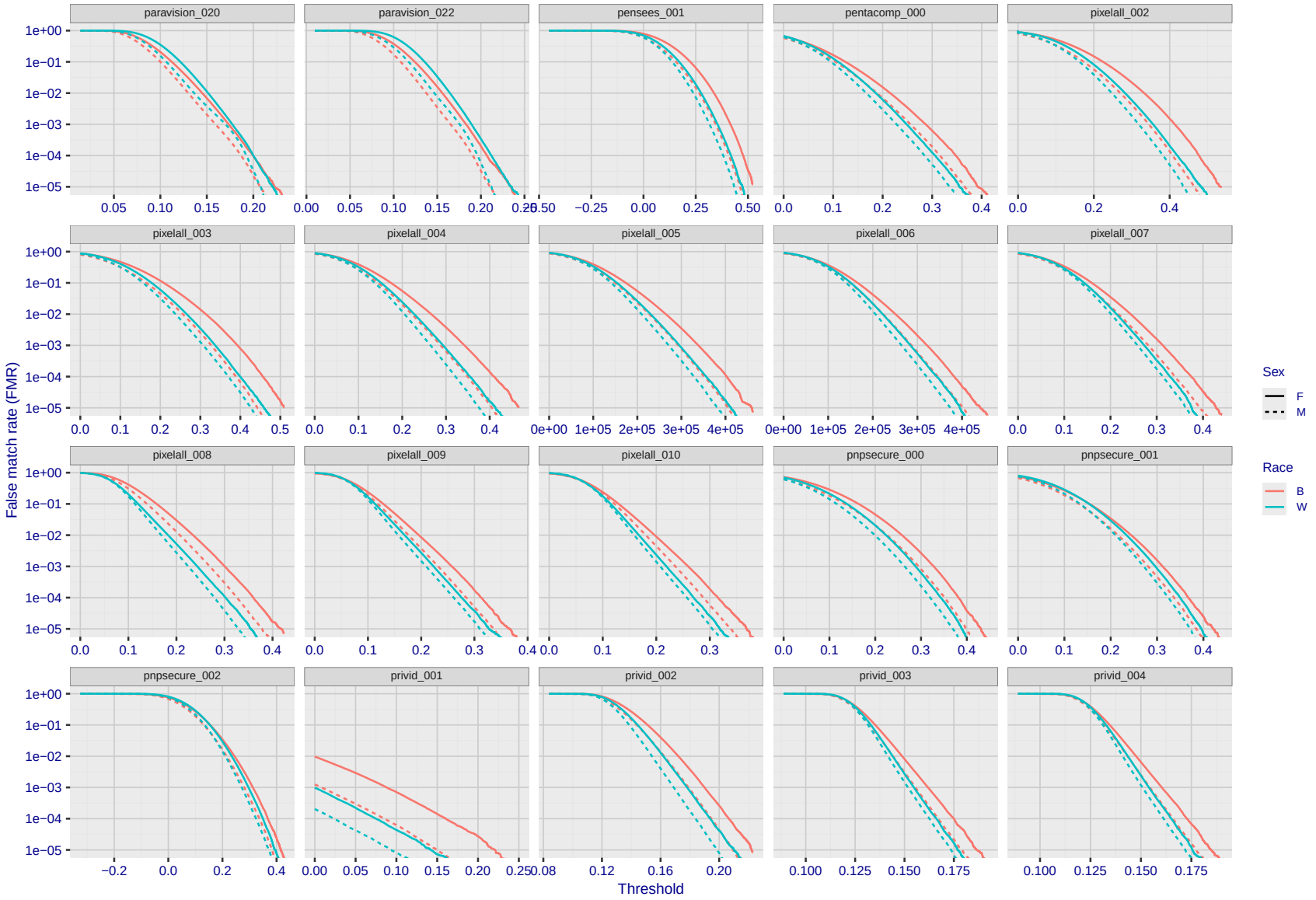


Figure 608: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

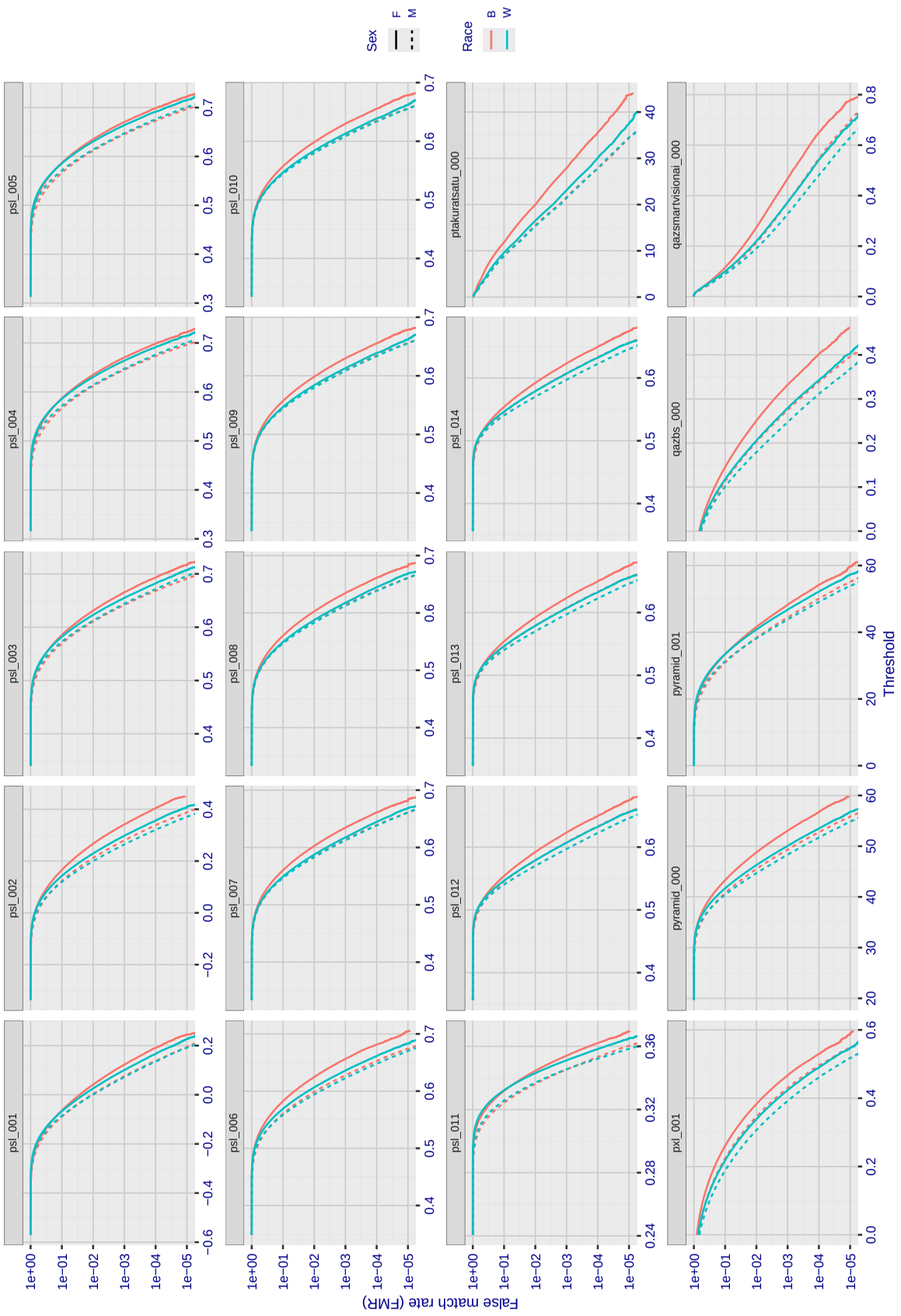


Figure 609: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

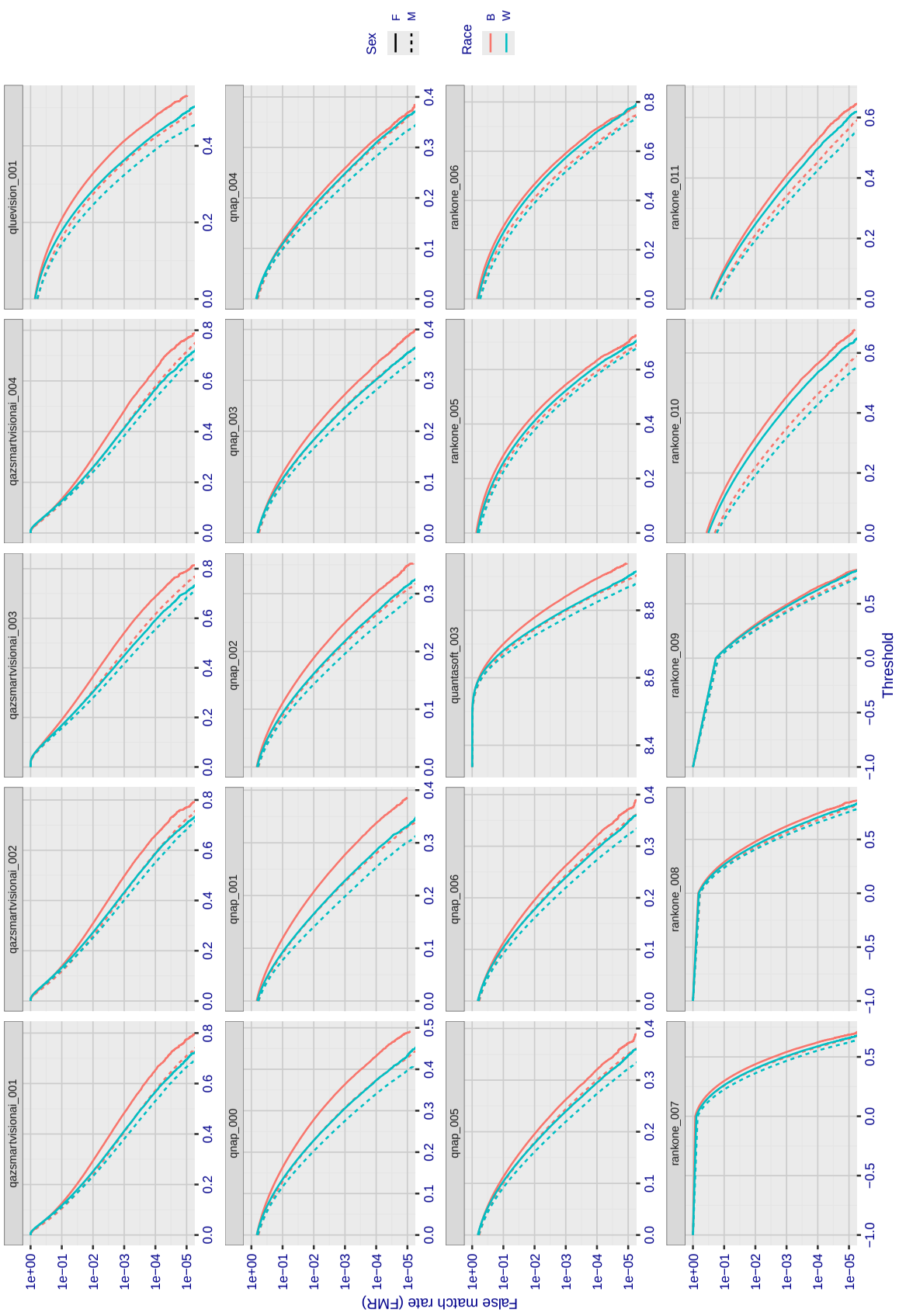


Figure 610: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

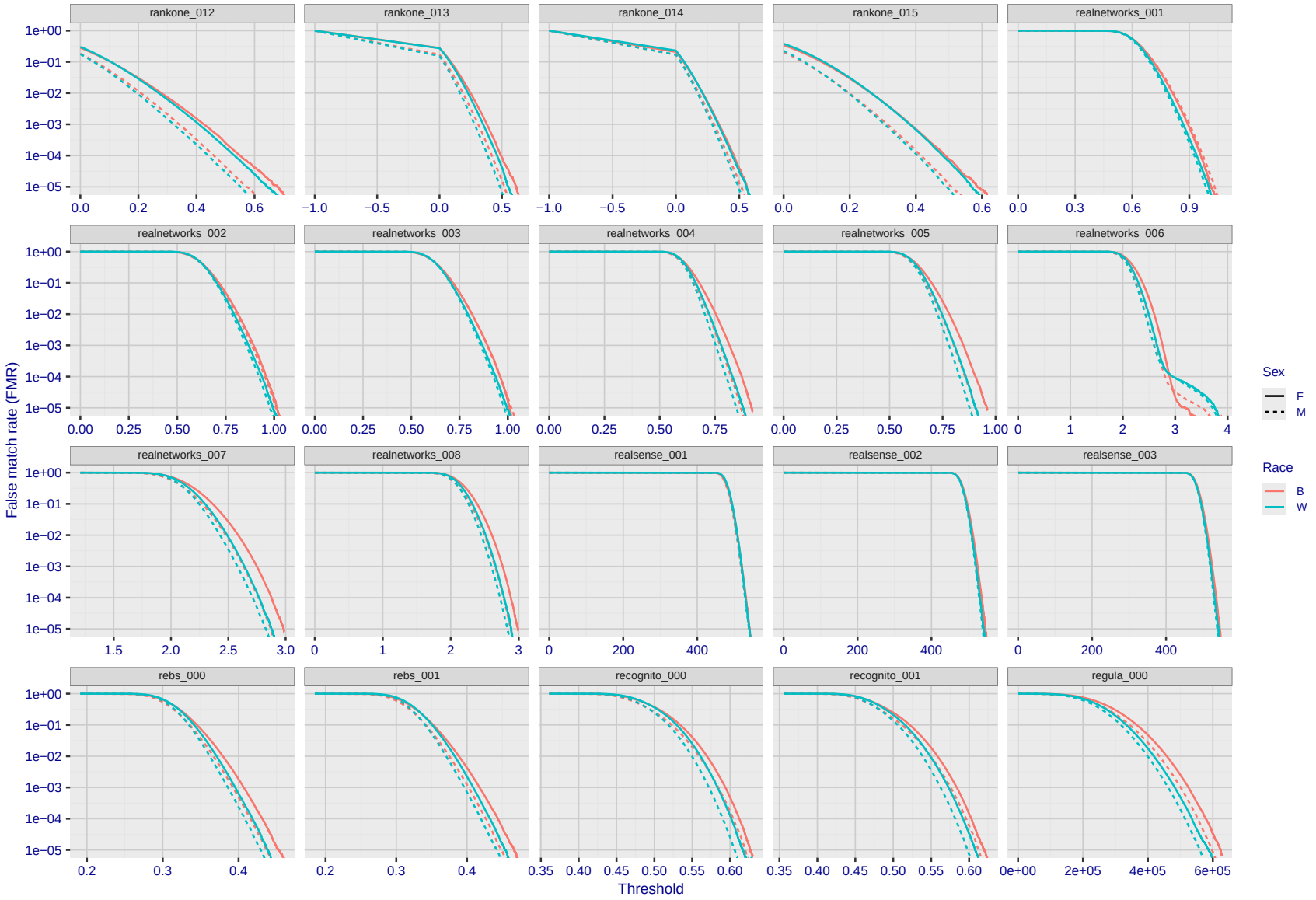


Figure 611: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

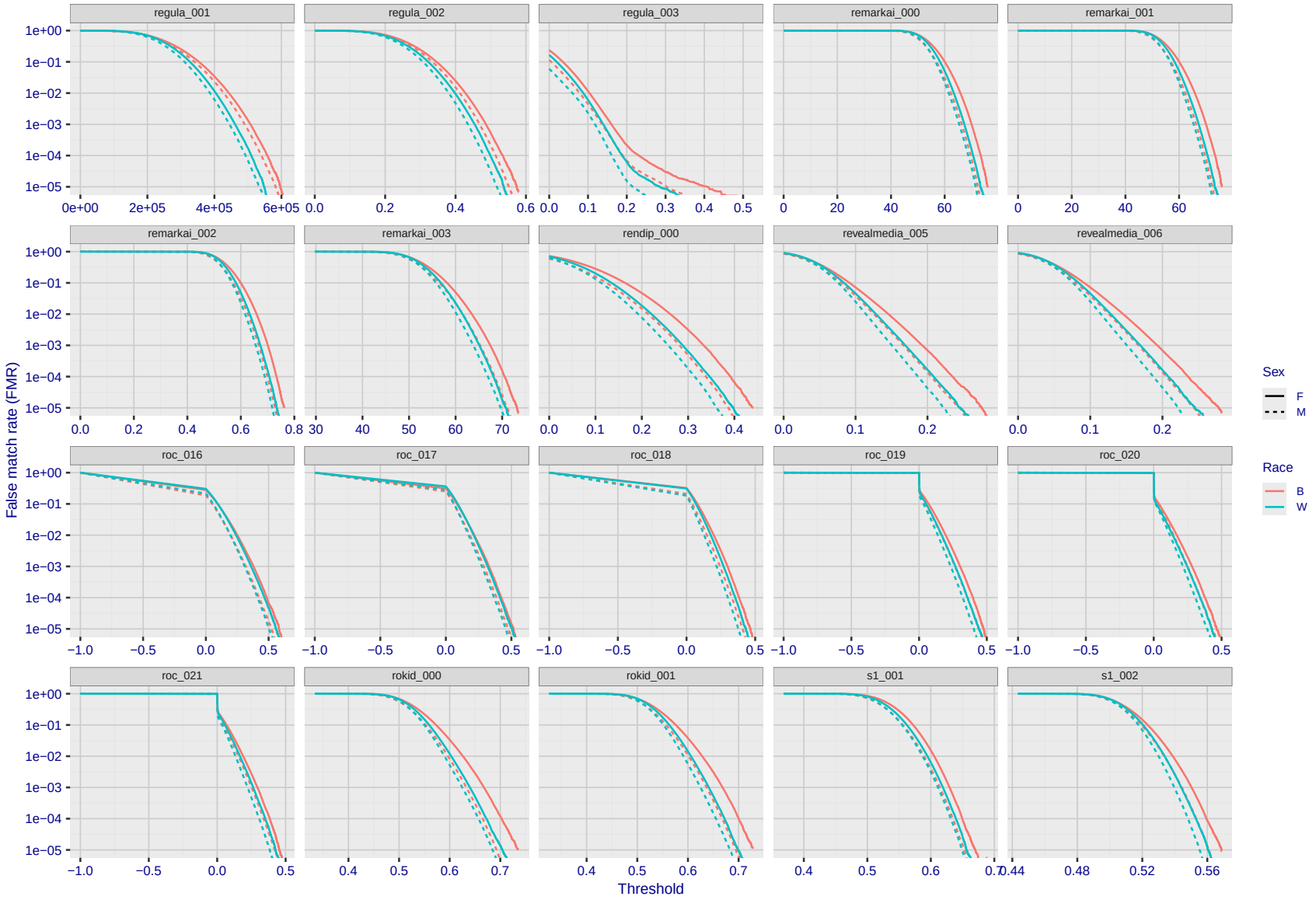


Figure 612: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

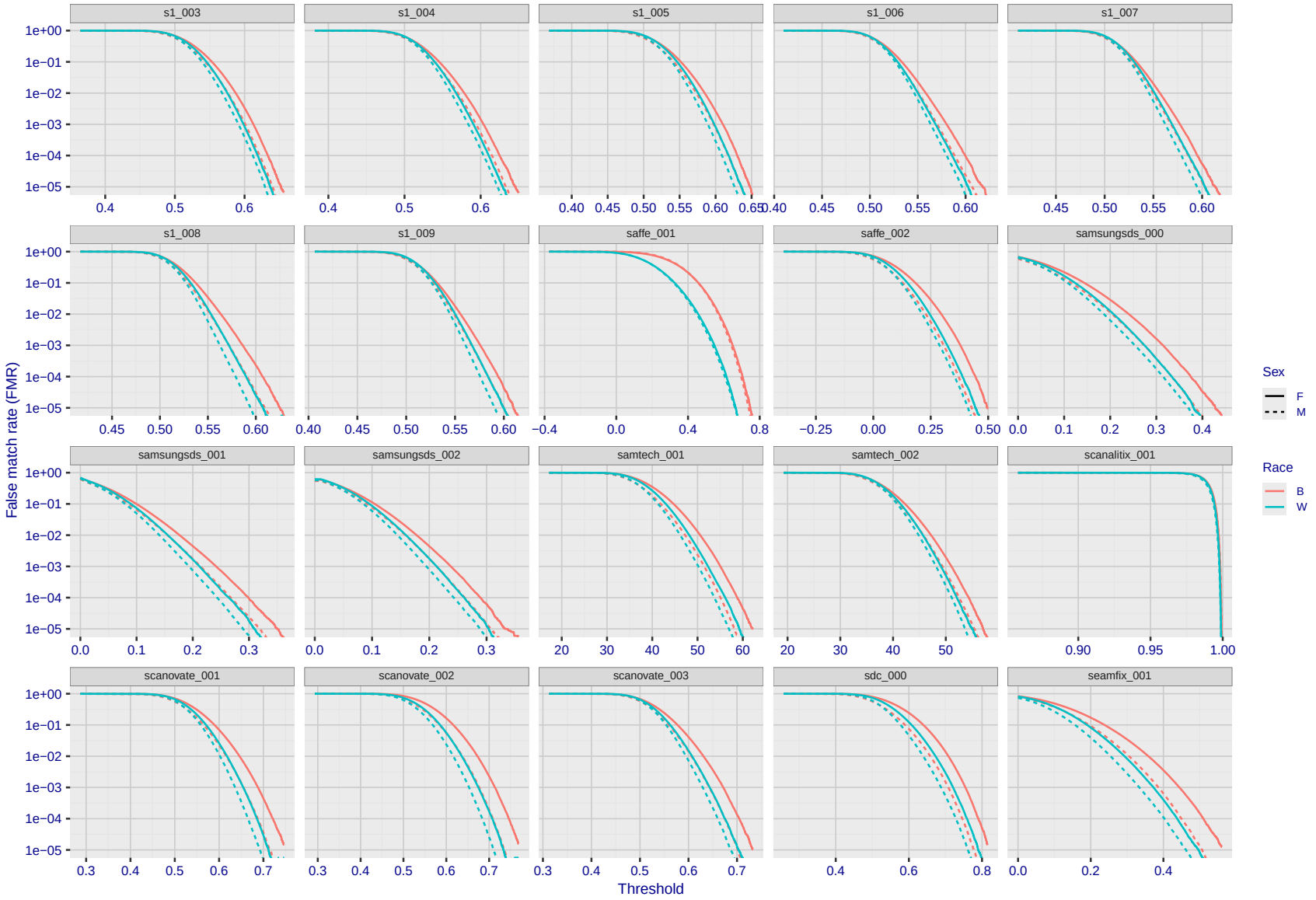


Figure 613: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

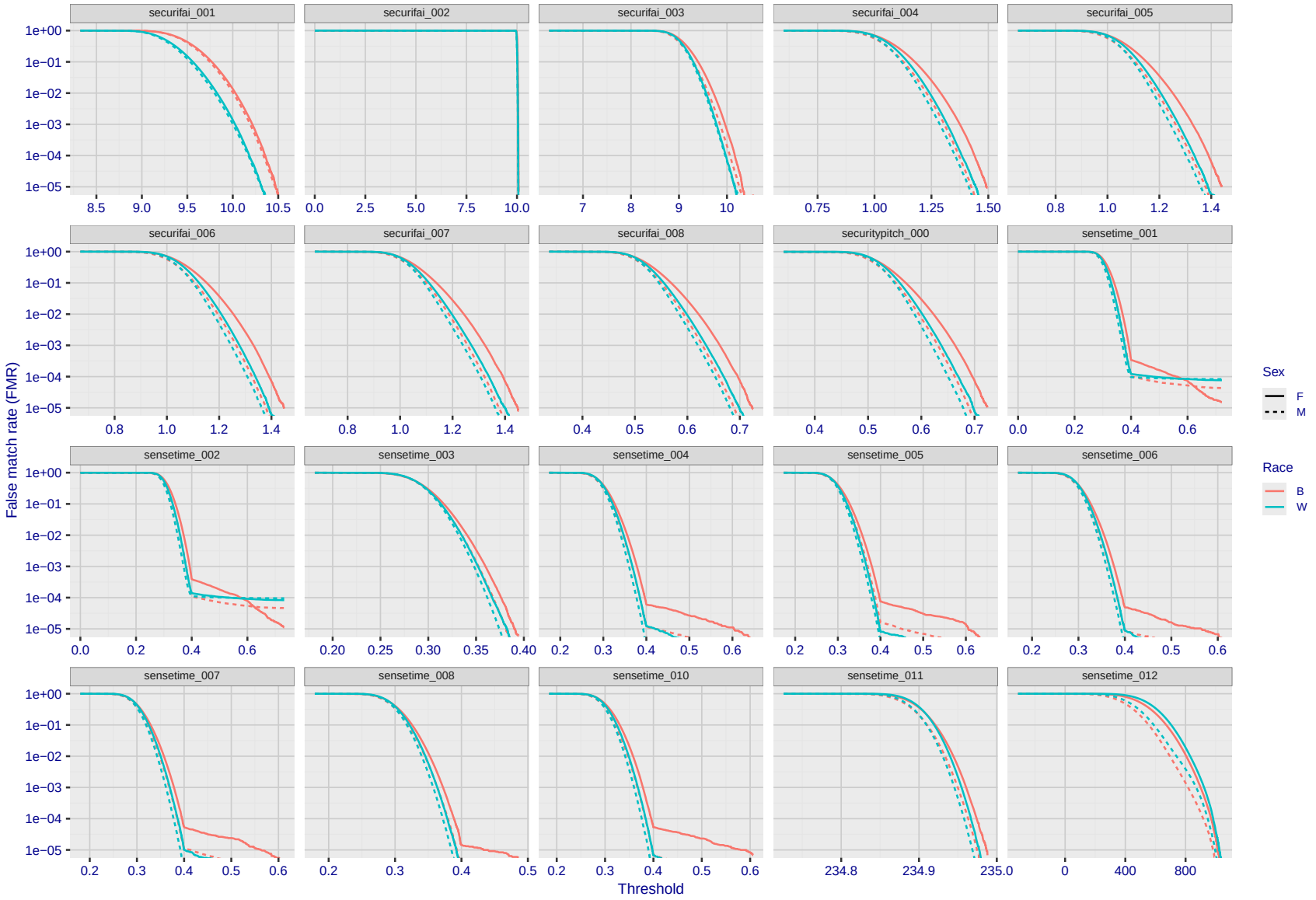


Figure 614: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

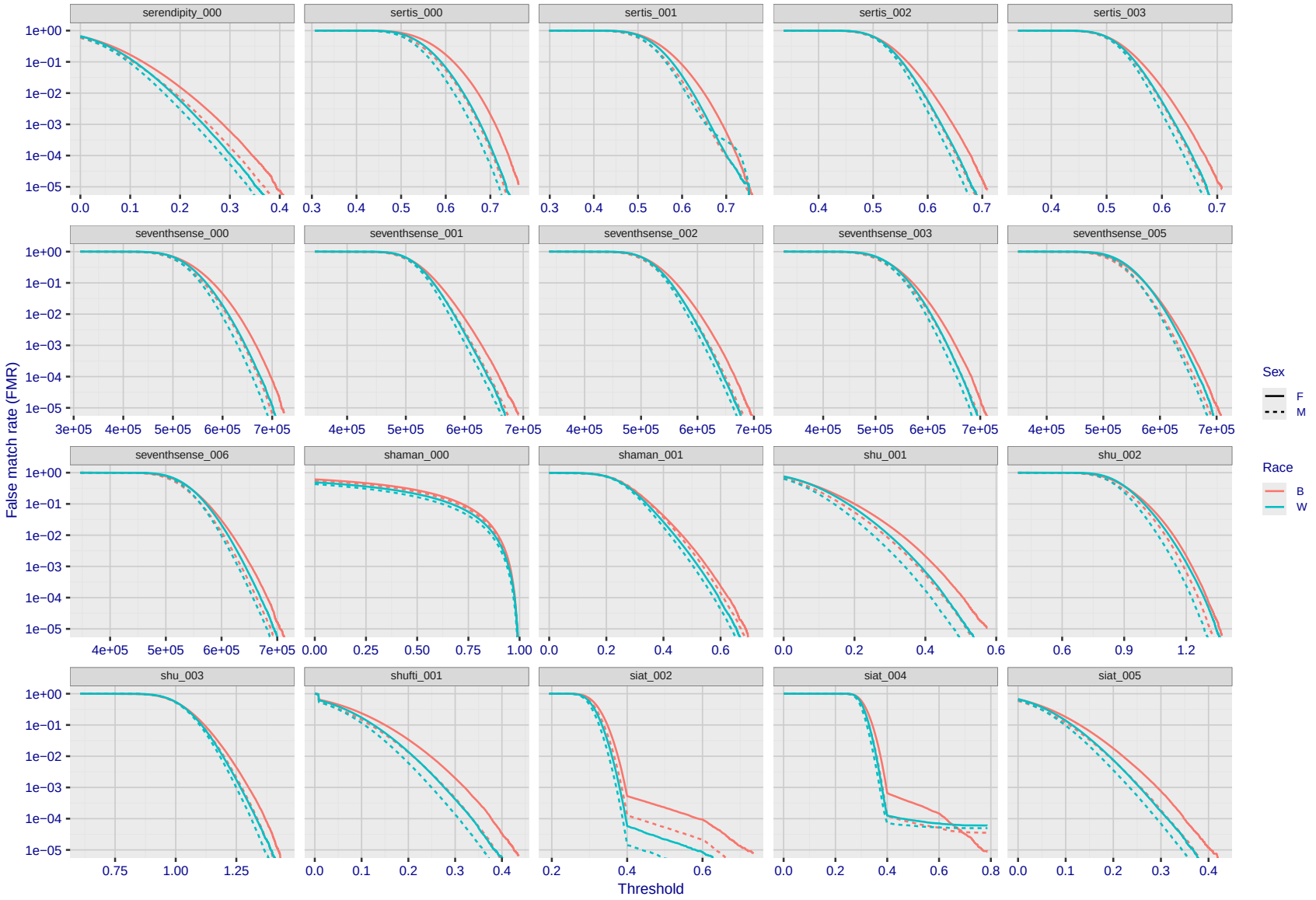


Figure 615: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

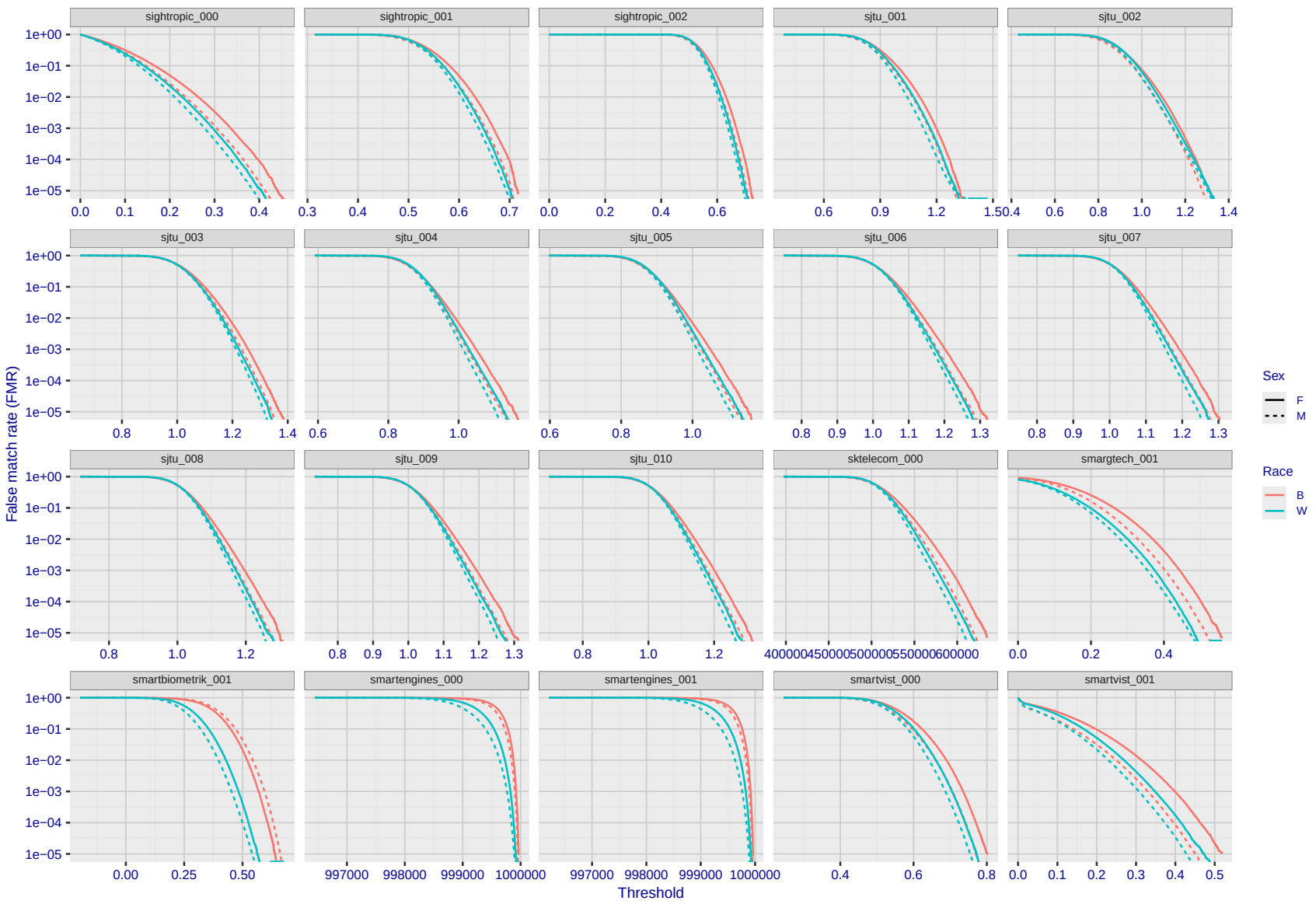


Figure 616: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

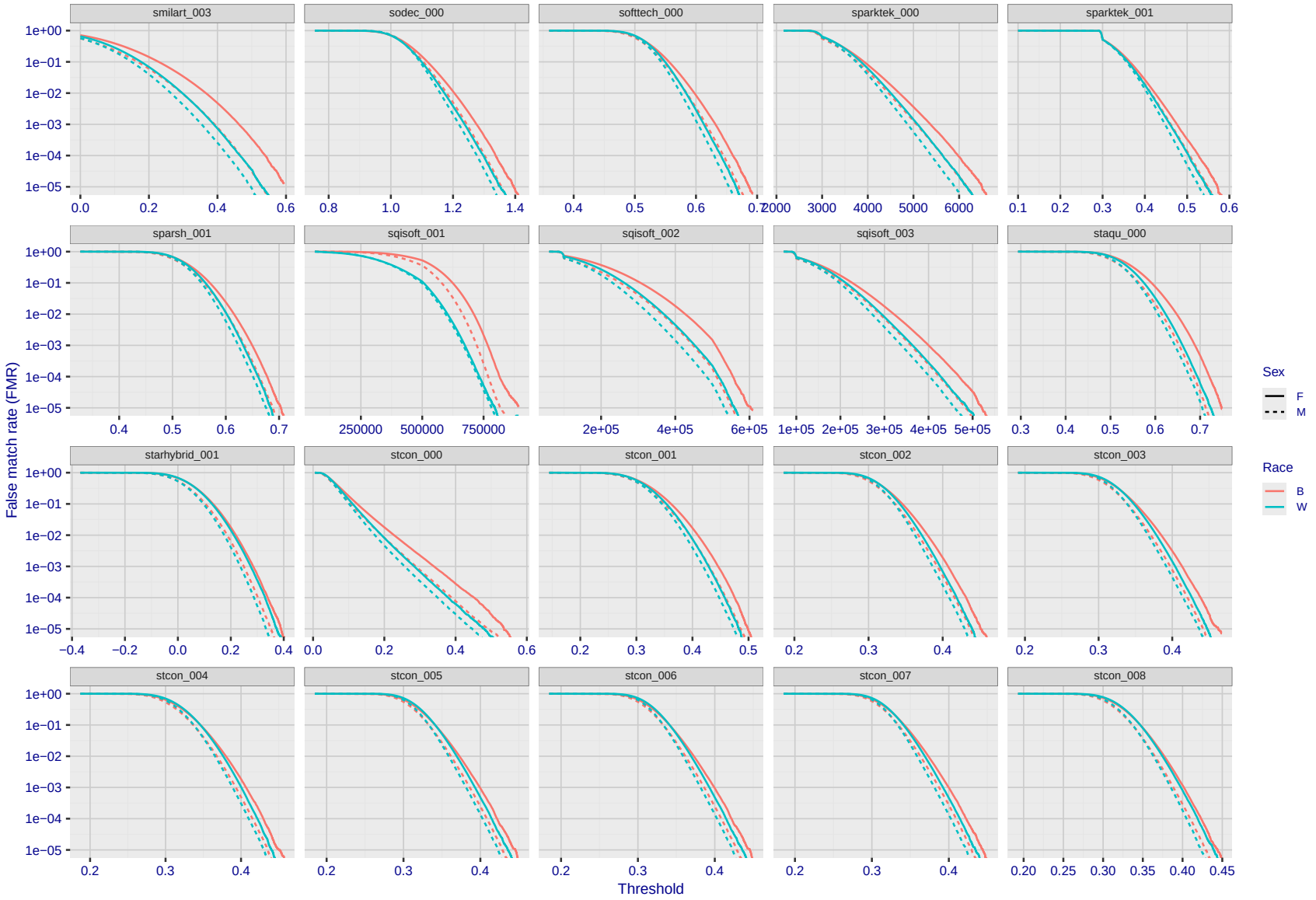


Figure 617: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

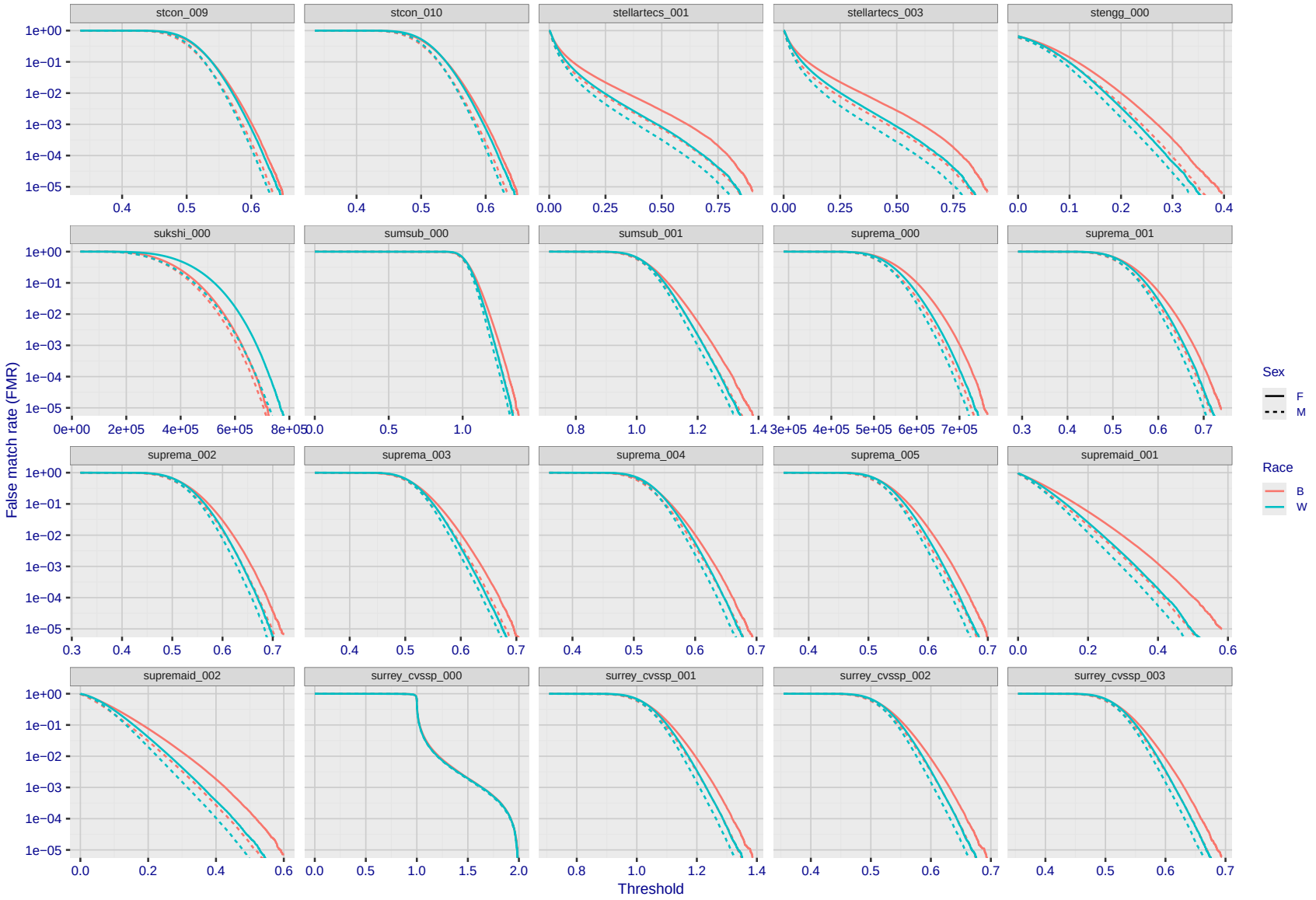


Figure 618: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

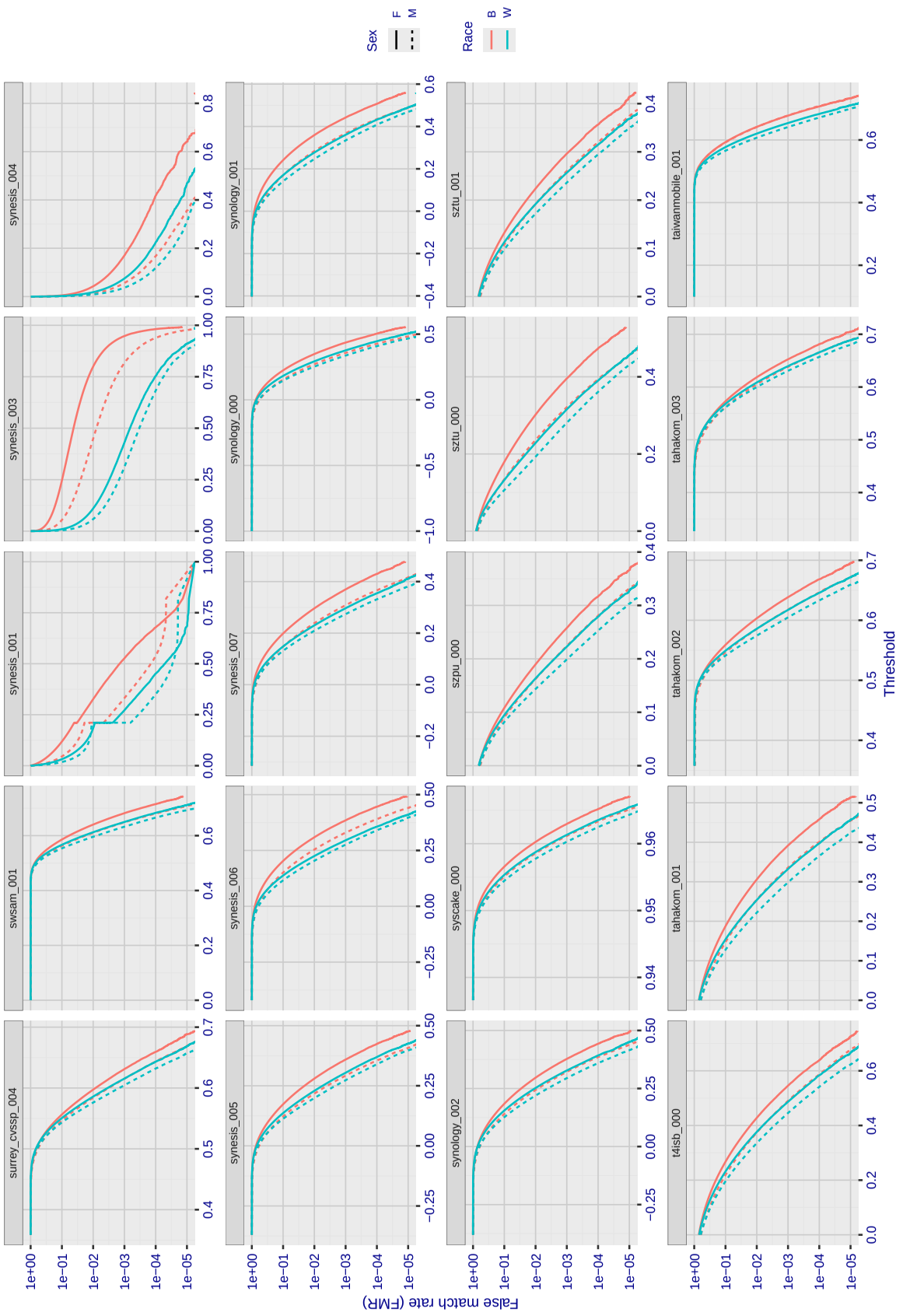


Figure 619: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

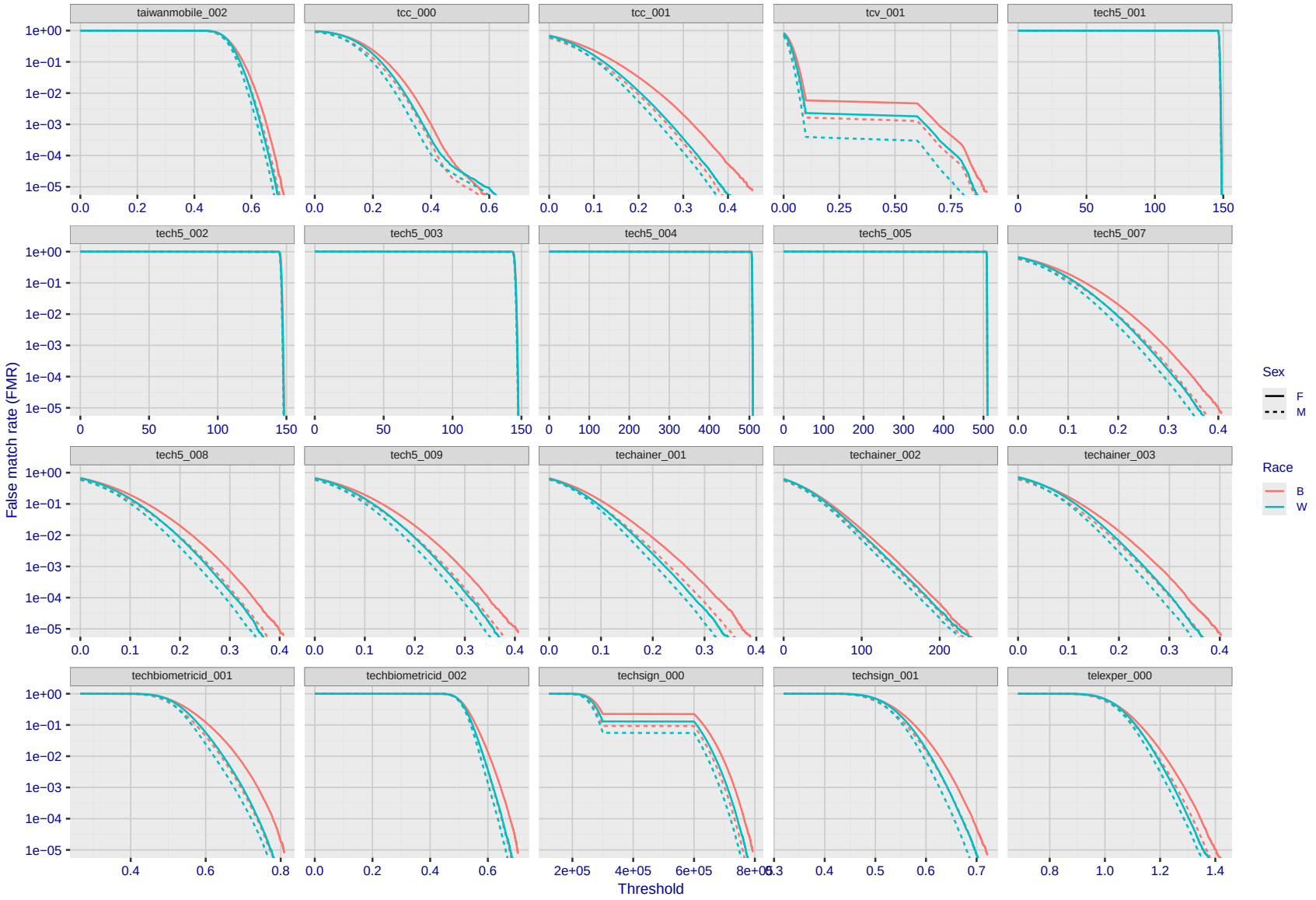


Figure 620: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

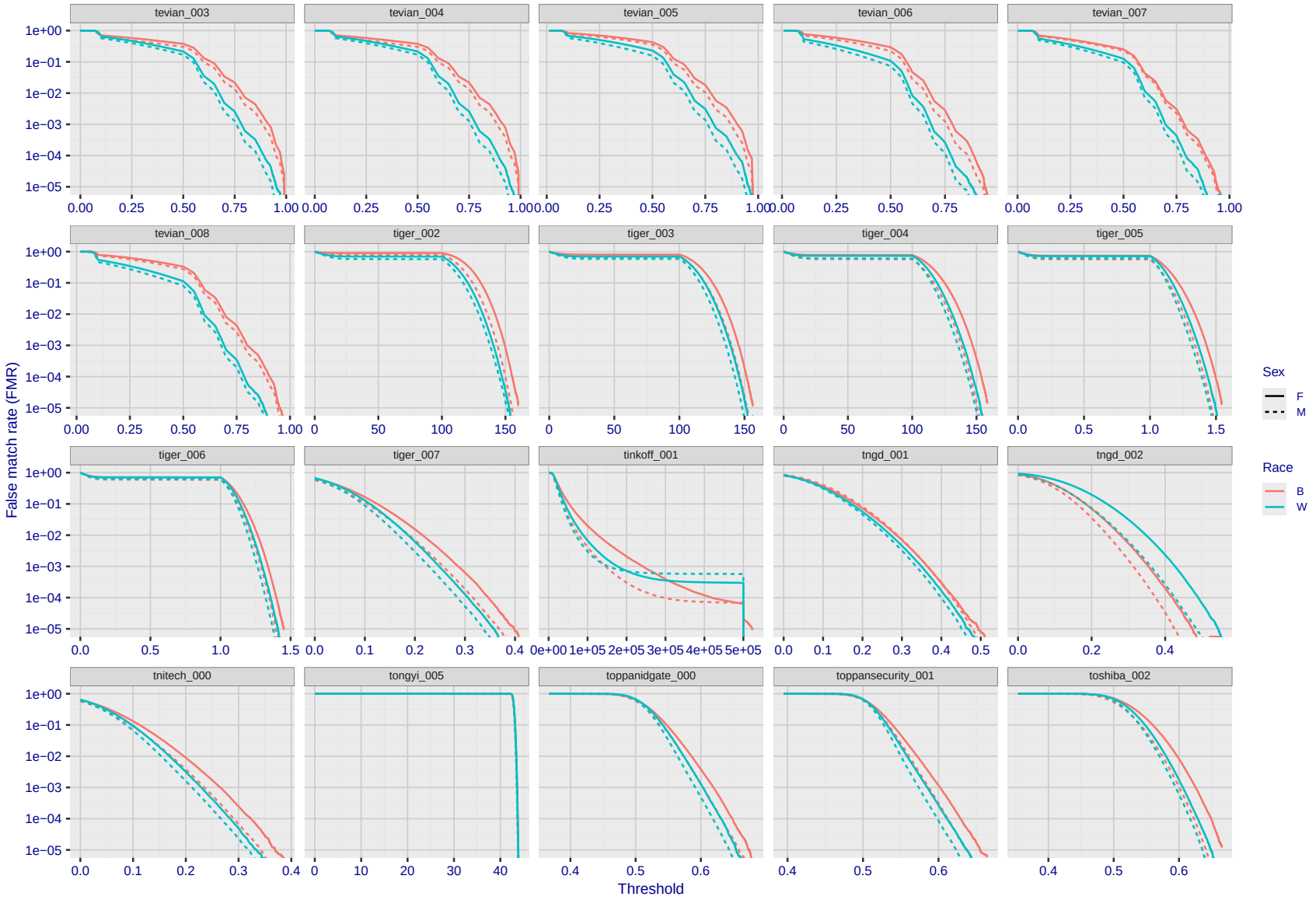


Figure 621: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

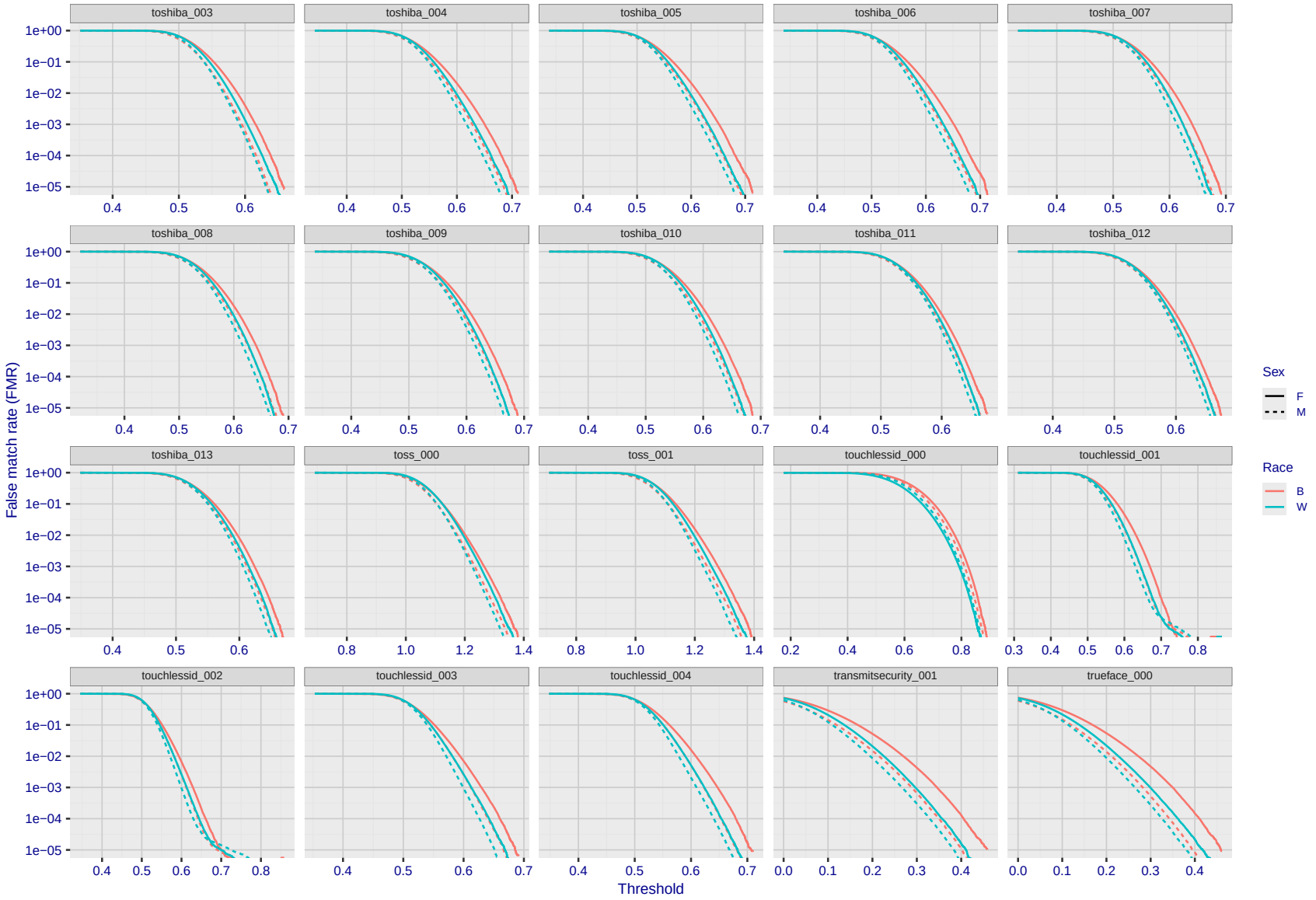


Figure 622: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

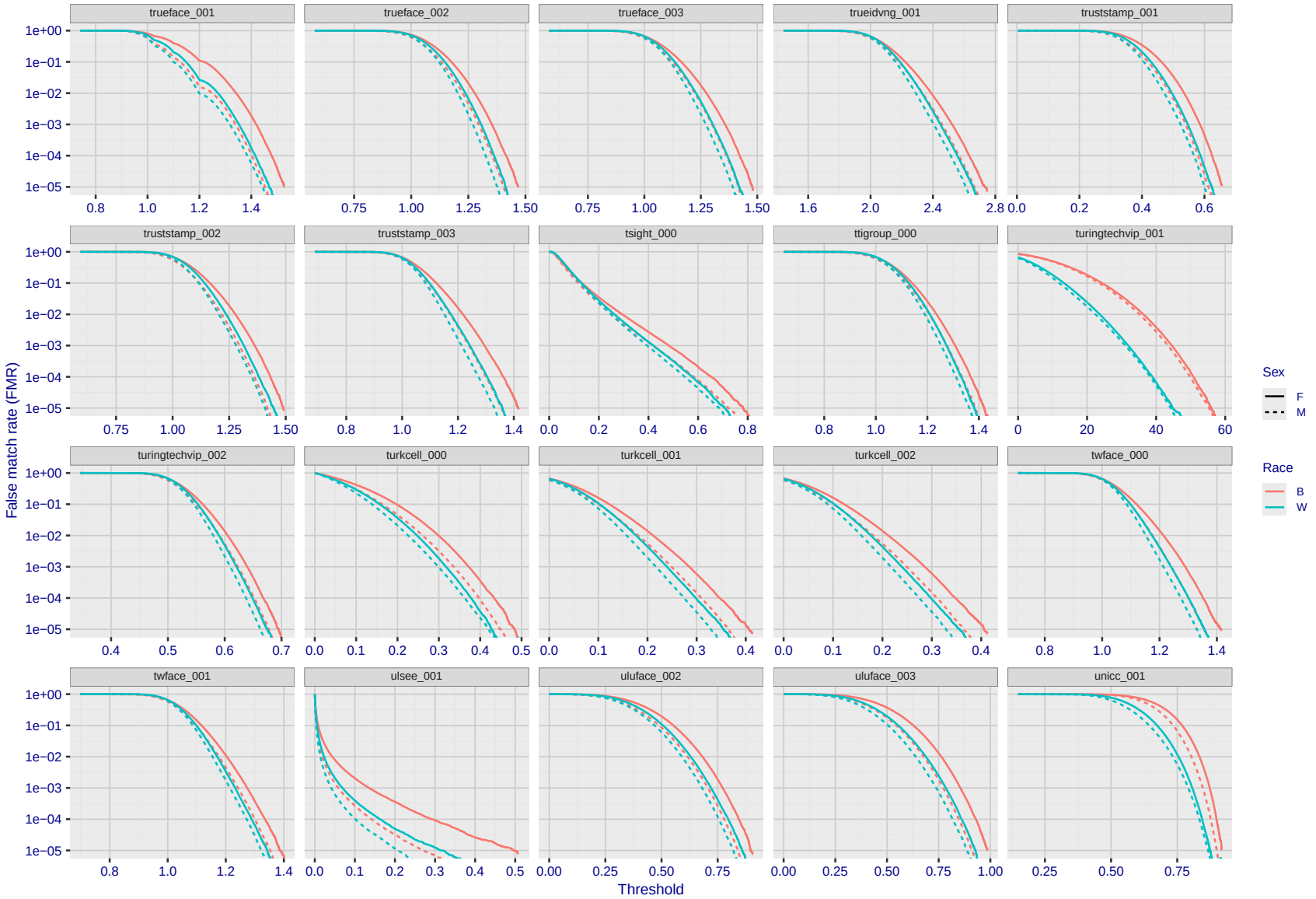


Figure 623: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

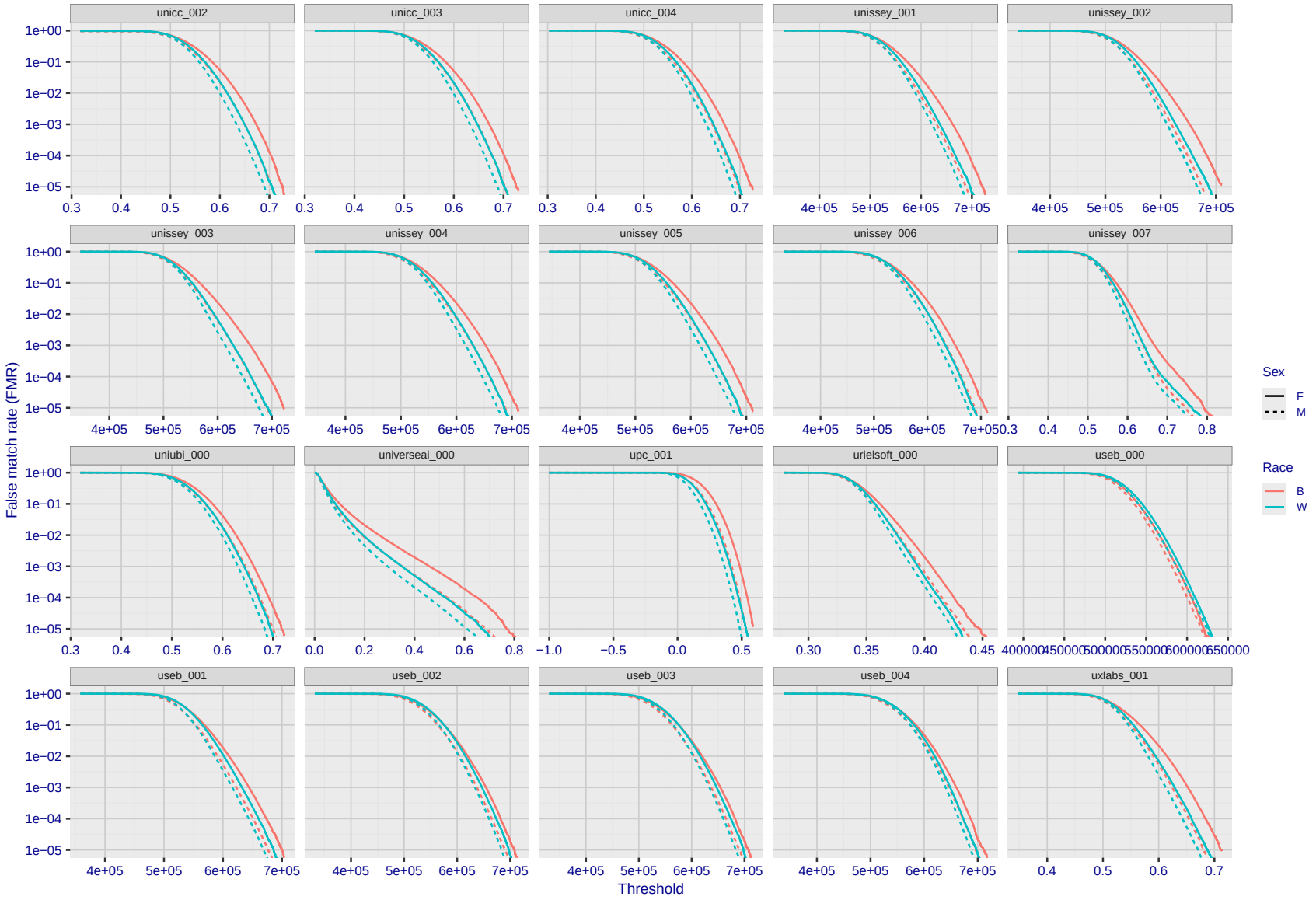


Figure 624: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

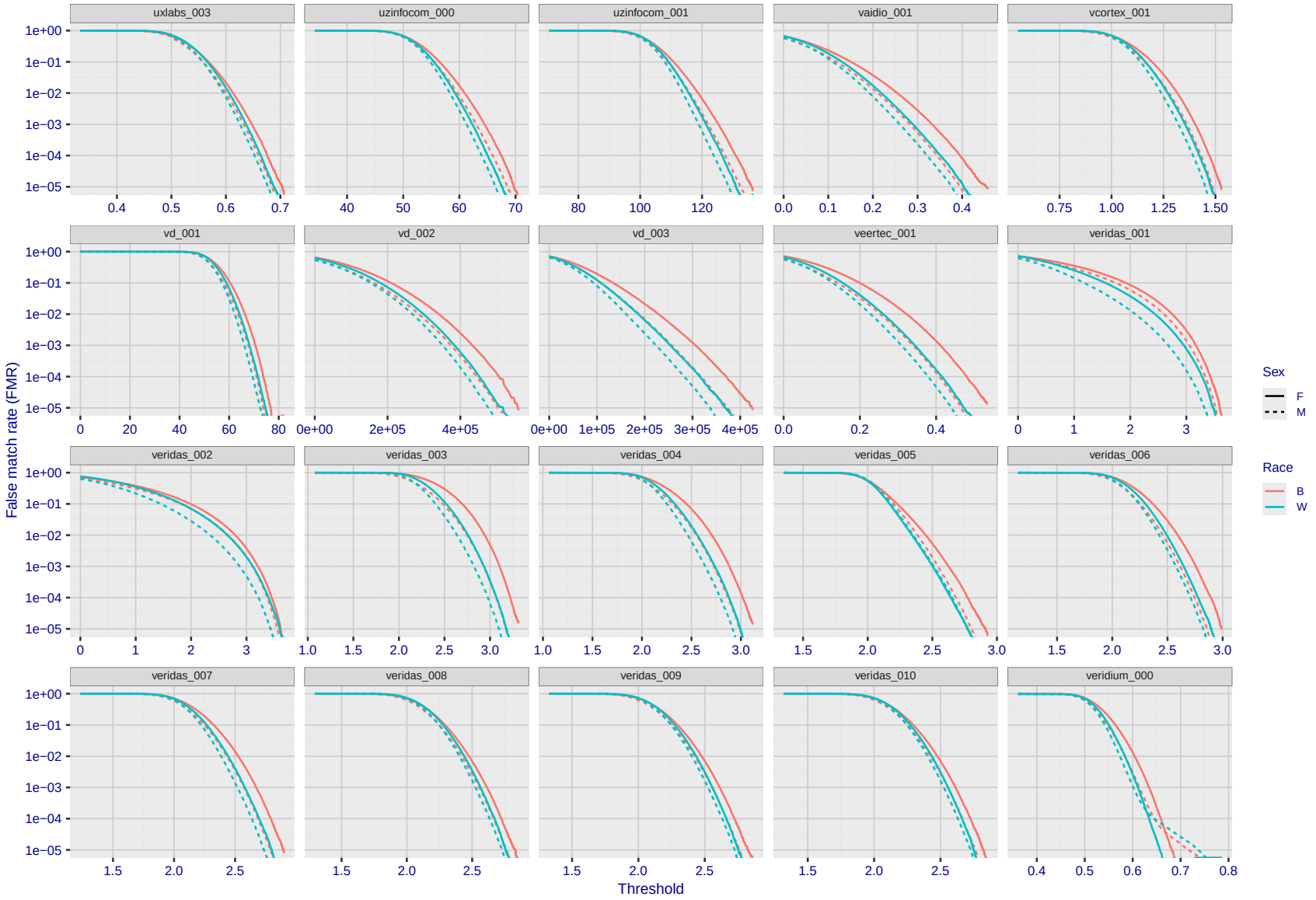


Figure 625: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

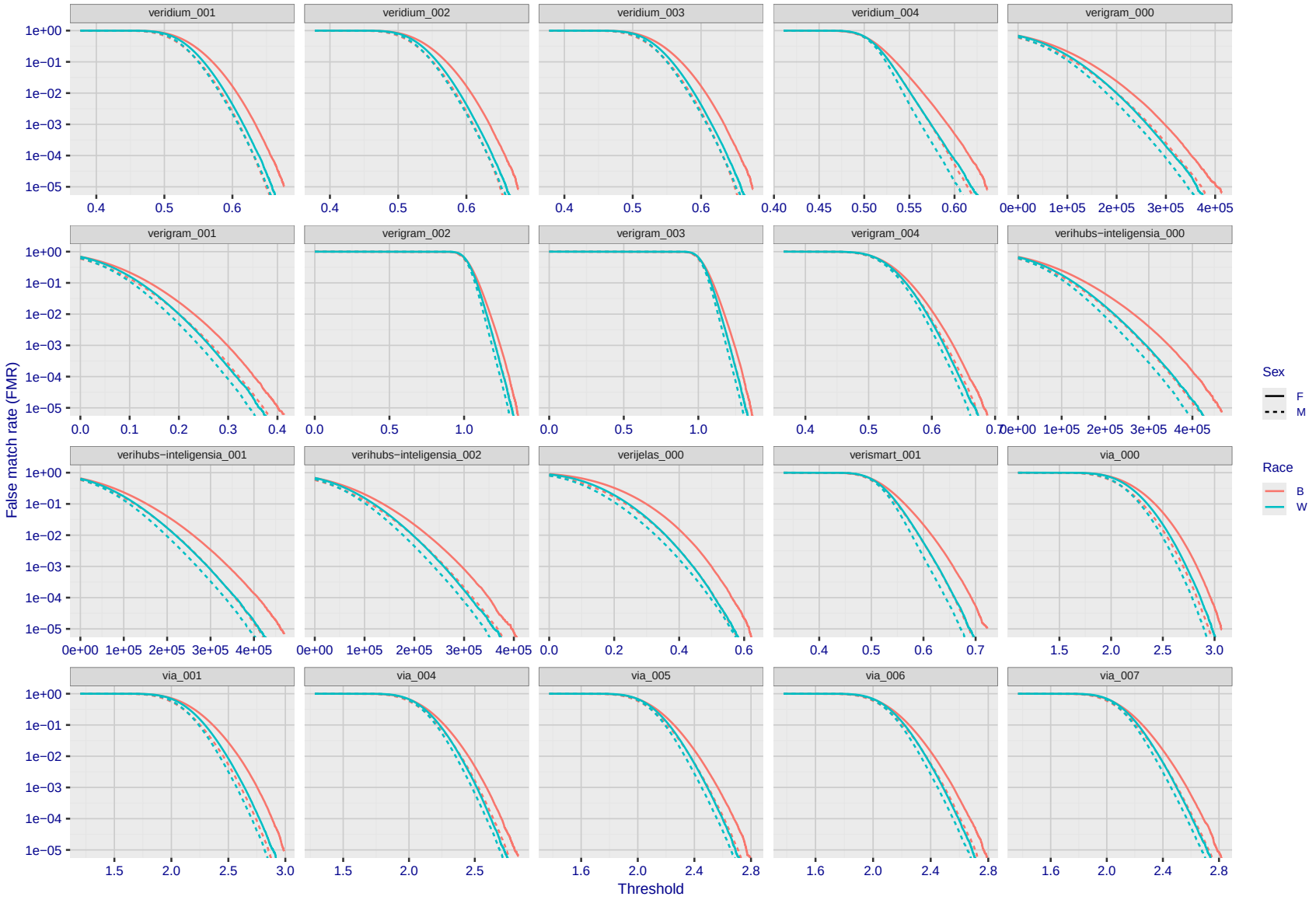


Figure 626: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

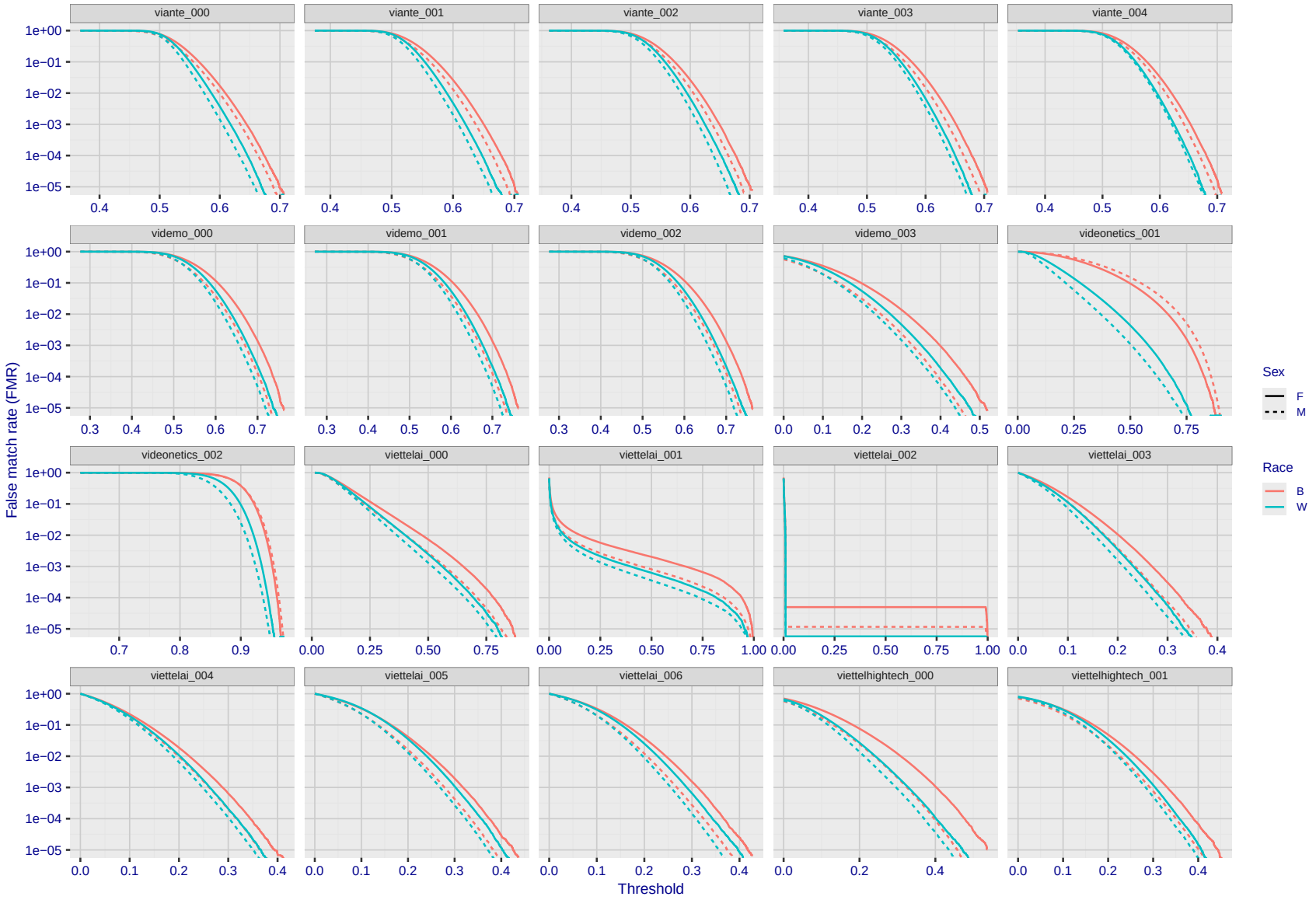


Figure 627: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

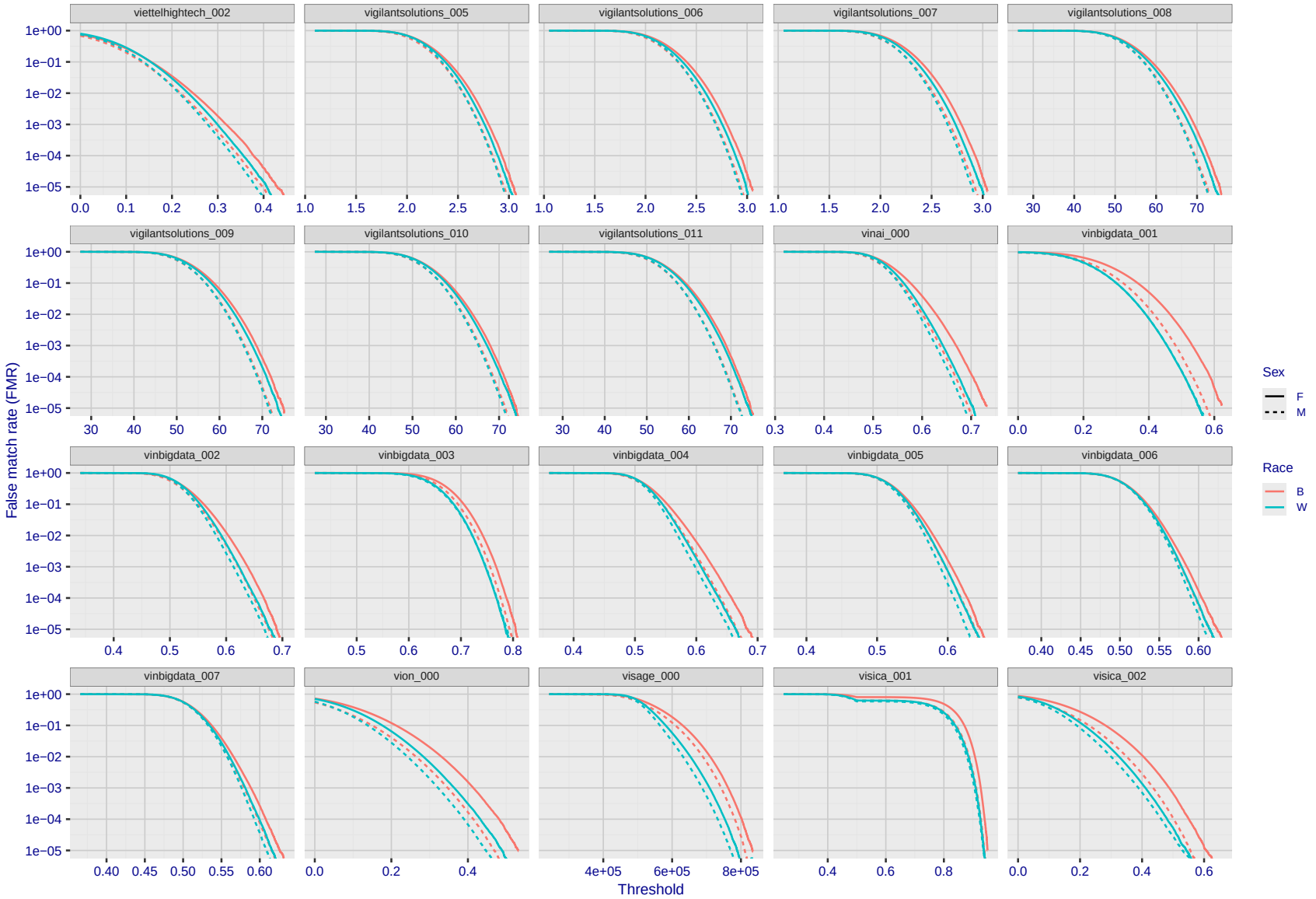


Figure 628: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

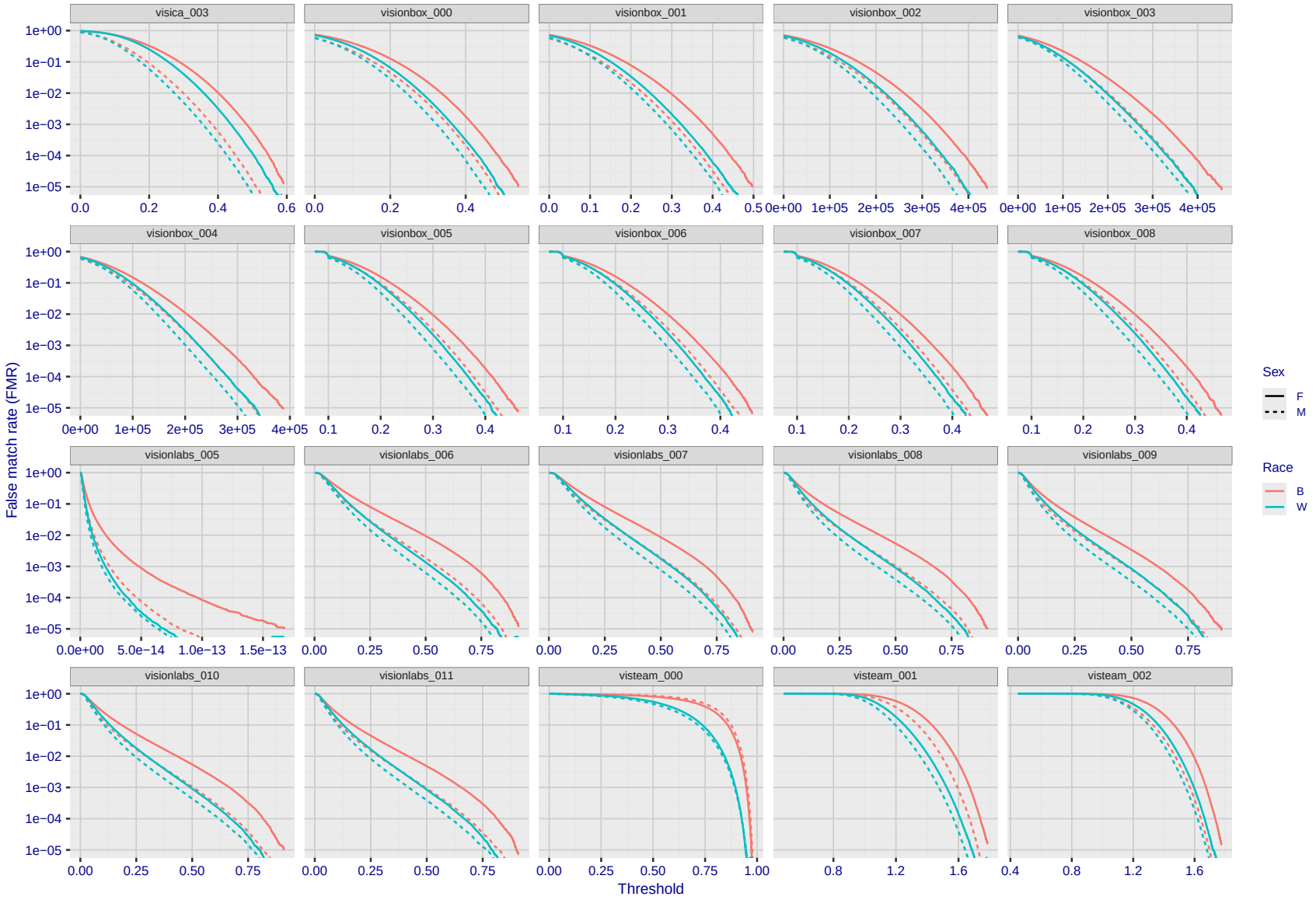


Figure 629: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

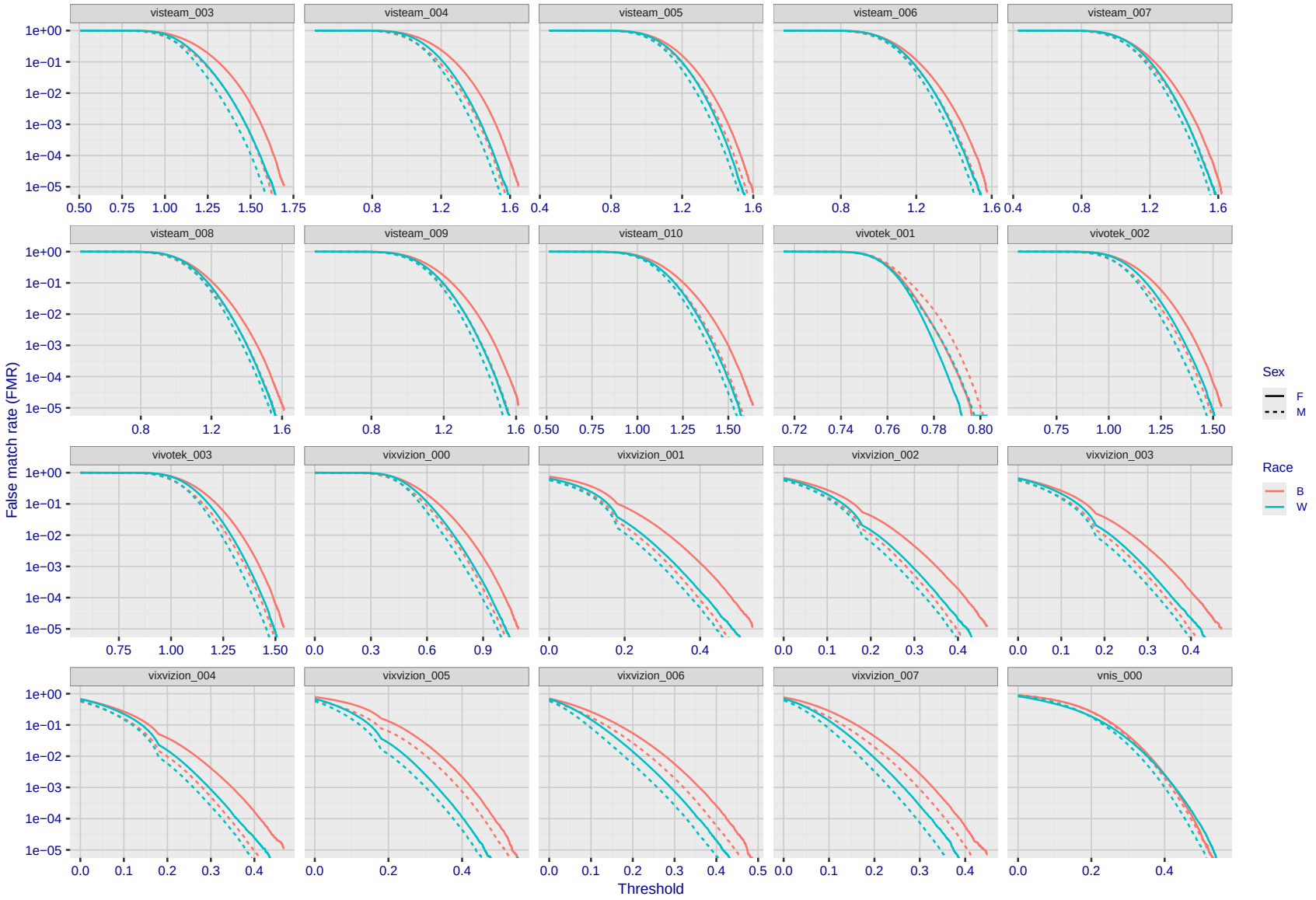


Figure 630: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

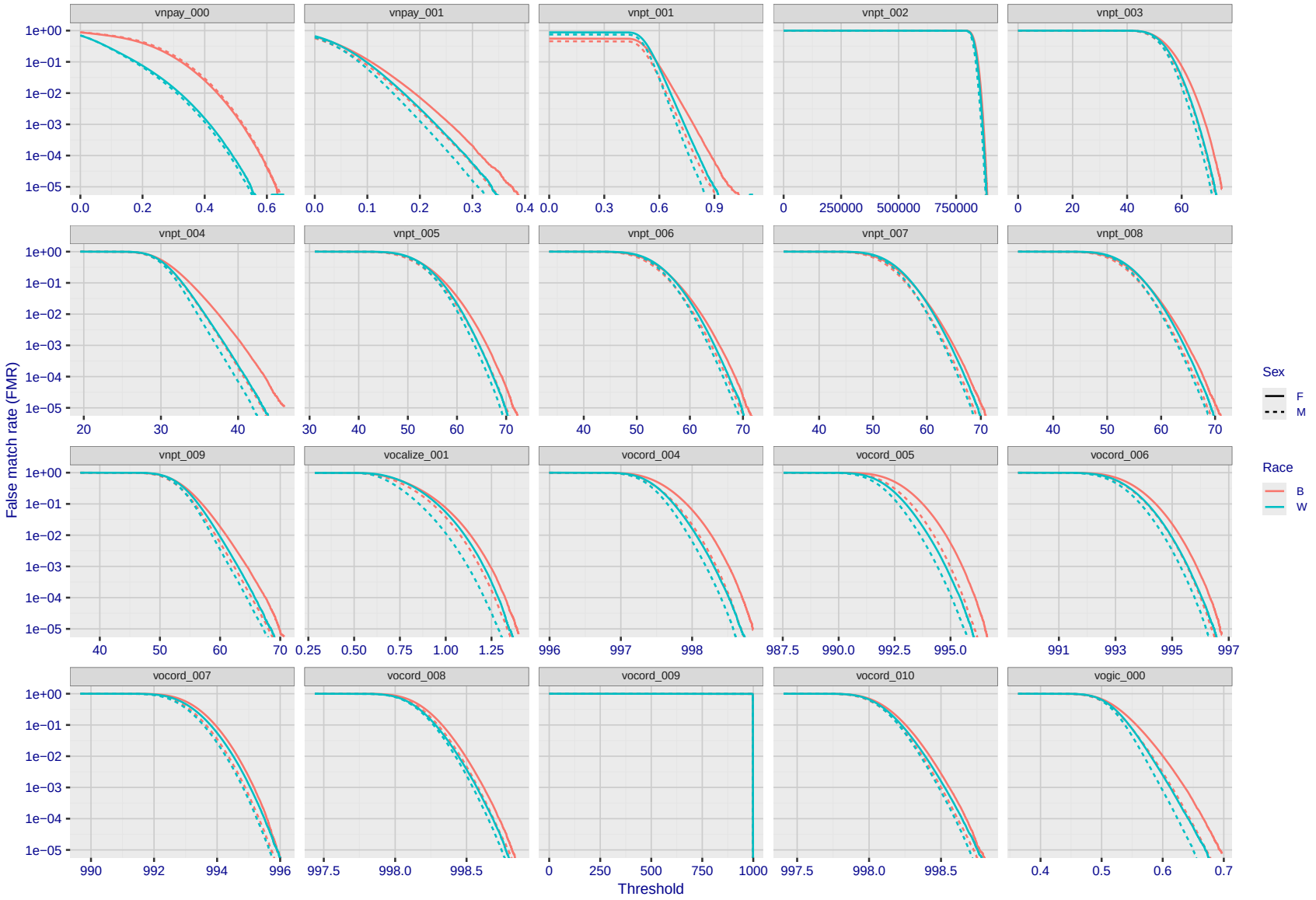


Figure 631: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

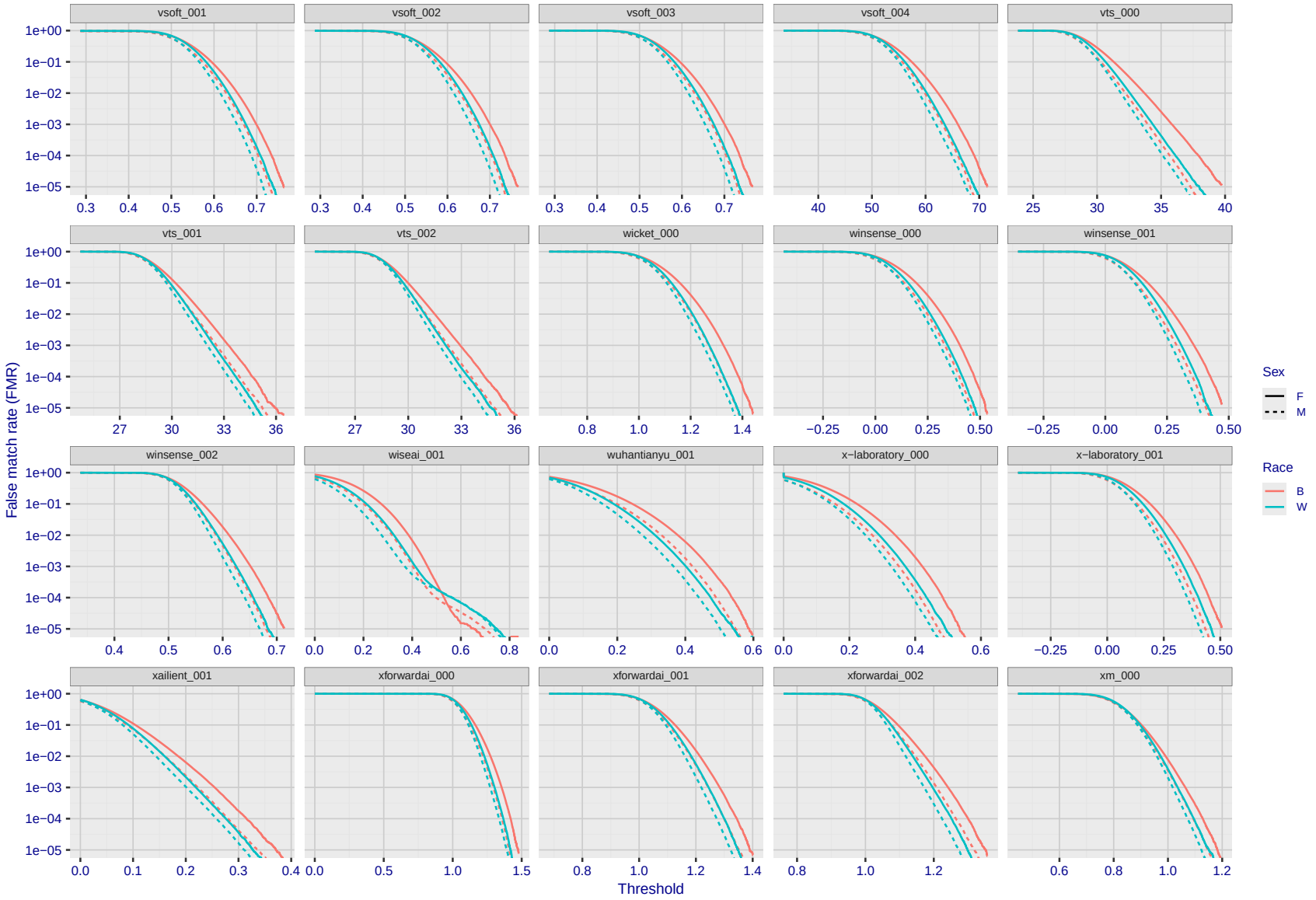


Figure 632: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

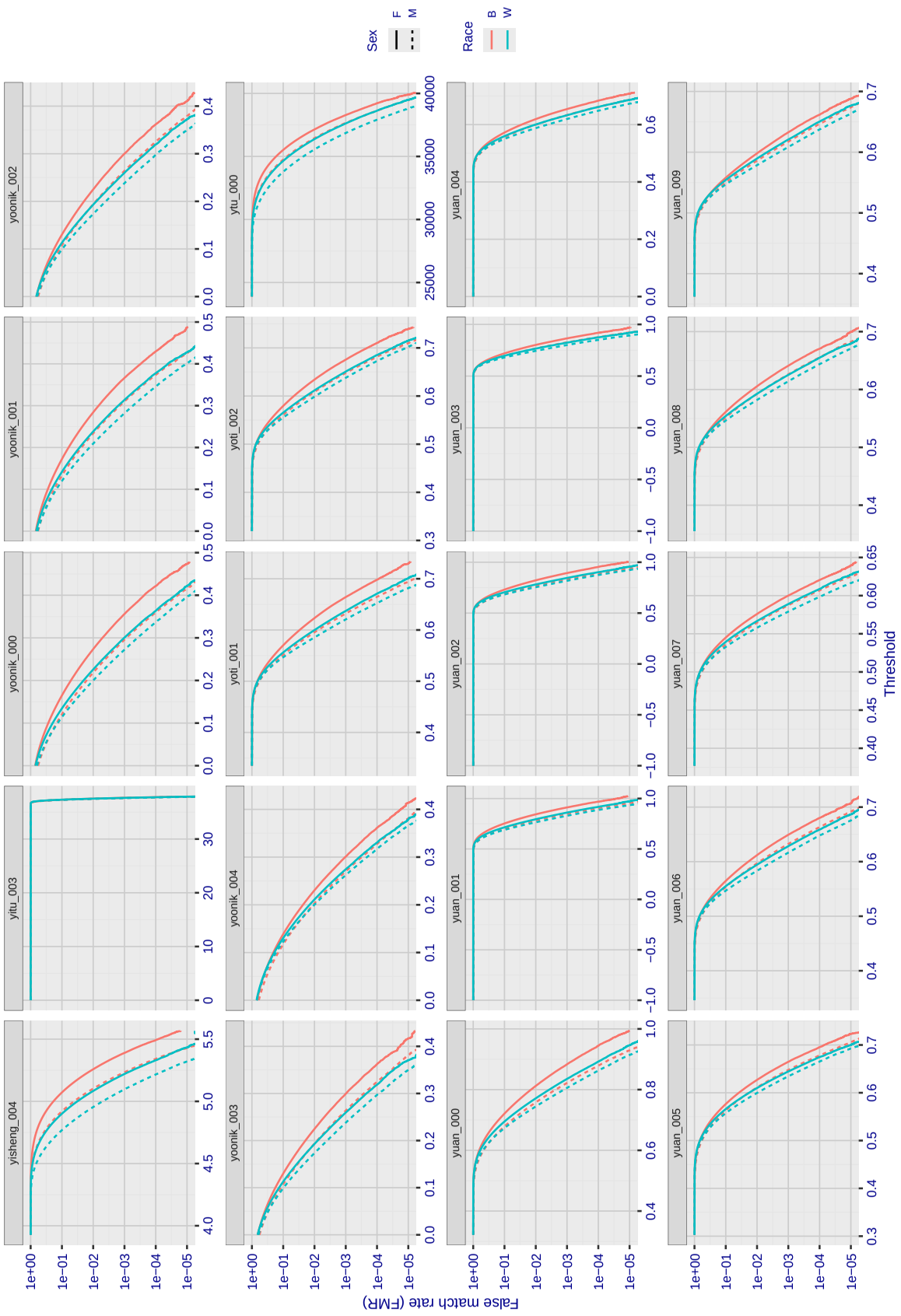


Figure 633: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

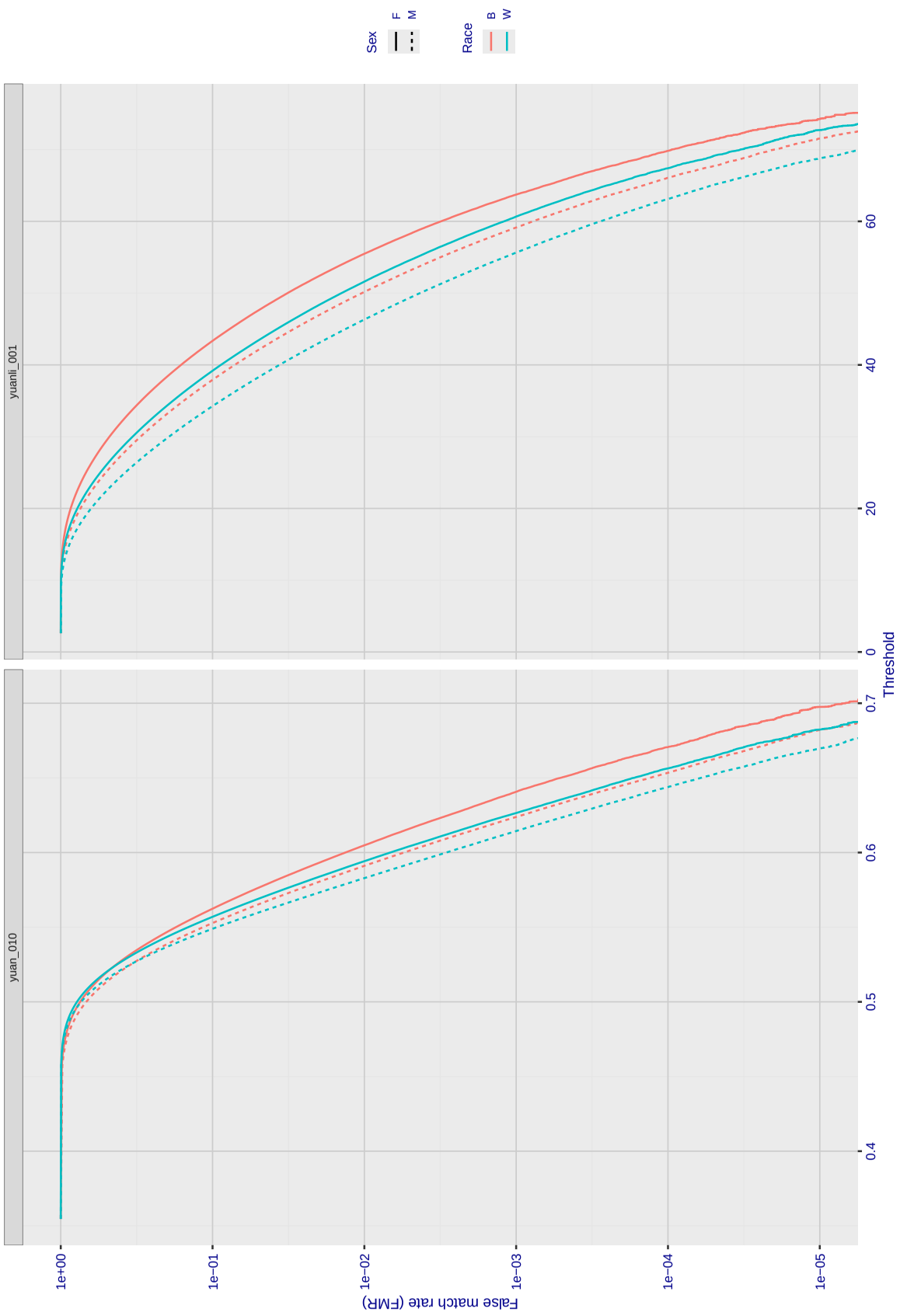


Figure 634: For the mugshot images, the false match calibration curves show false match rate vs. threshold. Separate curves appear for white females, black females, black males and white males.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

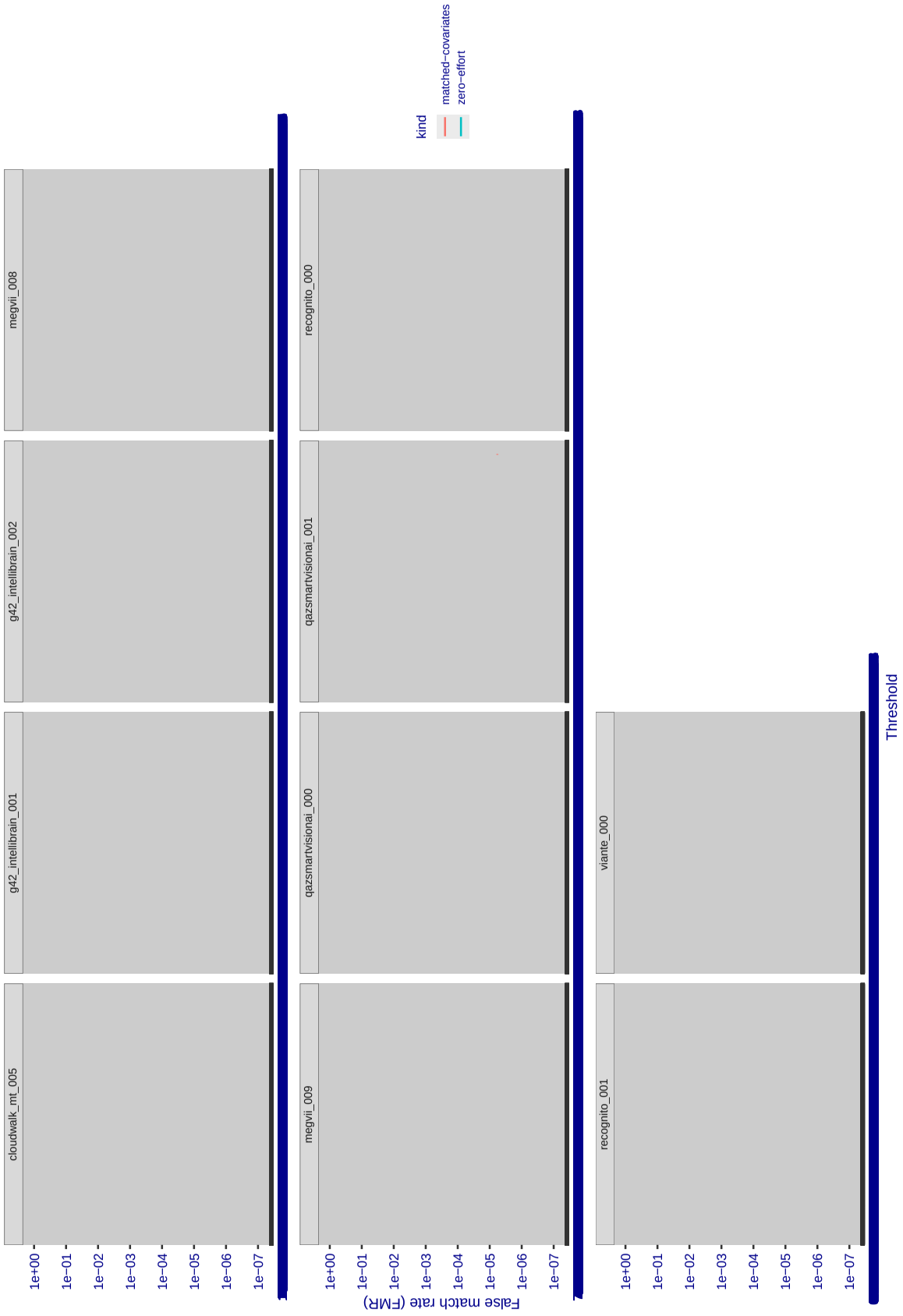


Figure 635: For the visa images, the false match calibration curves show FMR vs. threshold, T. The blue (lower) curves are for zero-effort impostors (i.e. comparing all images against all). The red (upper) curves are for persons of the same-sex, same-age, and same national-origin. This shows that FMR is underestimated (by a factor of 10 or more) by using a zero-effort impostor calculation to calibrate T. As shown later (sec. 3.6), FMR is higher for demographic-matched impostors.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.5 Genuine distribution stability

3.5.1 Effect of birth place on the genuine distribution

Background: Both skin tone and bone structure vary geographically. Prior studies have reported variations in FNMR and FMR.

Goal: To measure false non-match rate (FNMR) variation with country of birth.

Methods: Thresholds are determined that give FMR = $\{0.001, 0.0001\}$ over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores. Only those countries with at least 140 individuals are included in the analysis.

Results: Figure 725 shows FNMR by country of birth for the two thresholds.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

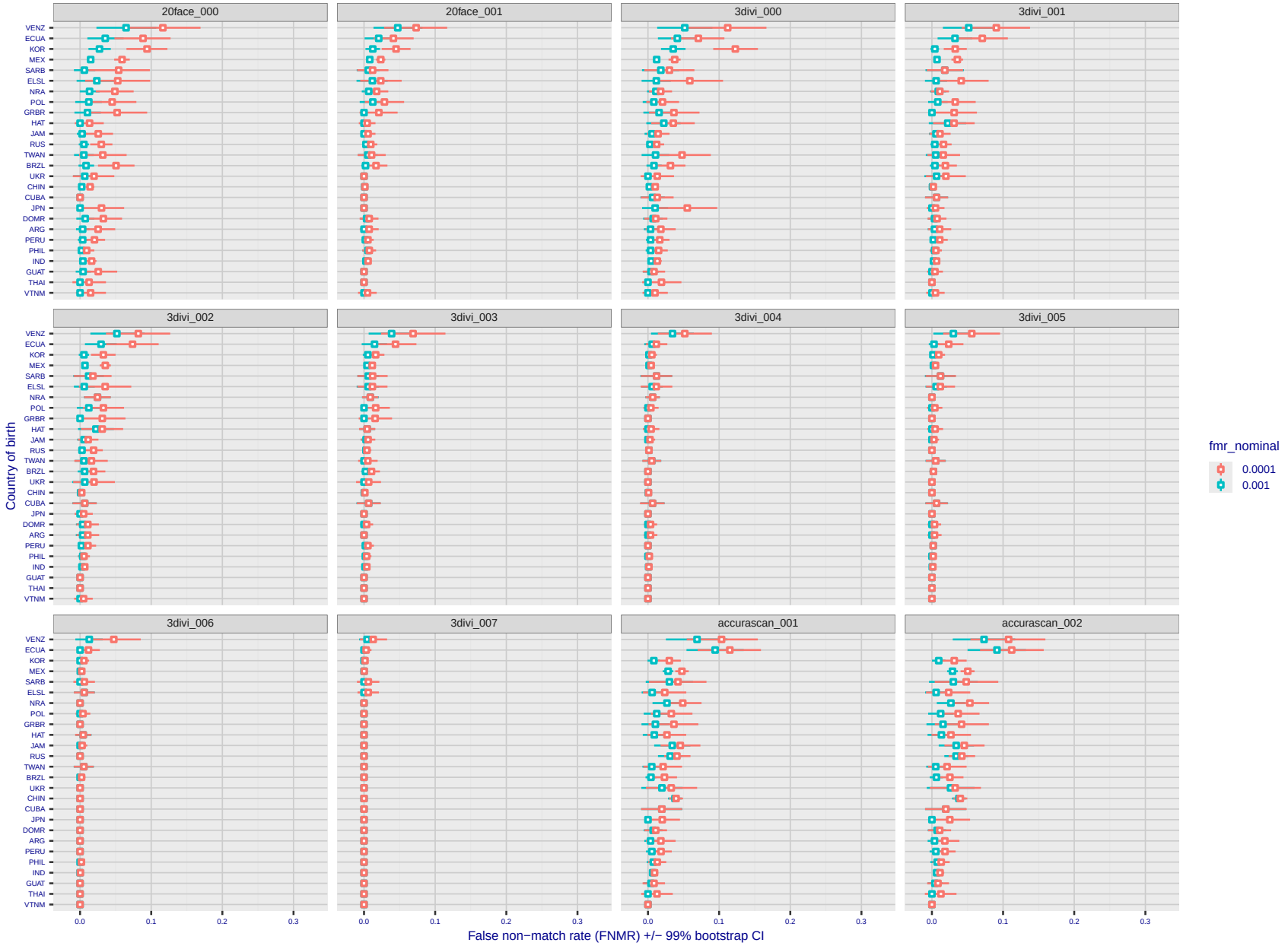


Figure 636: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 637: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

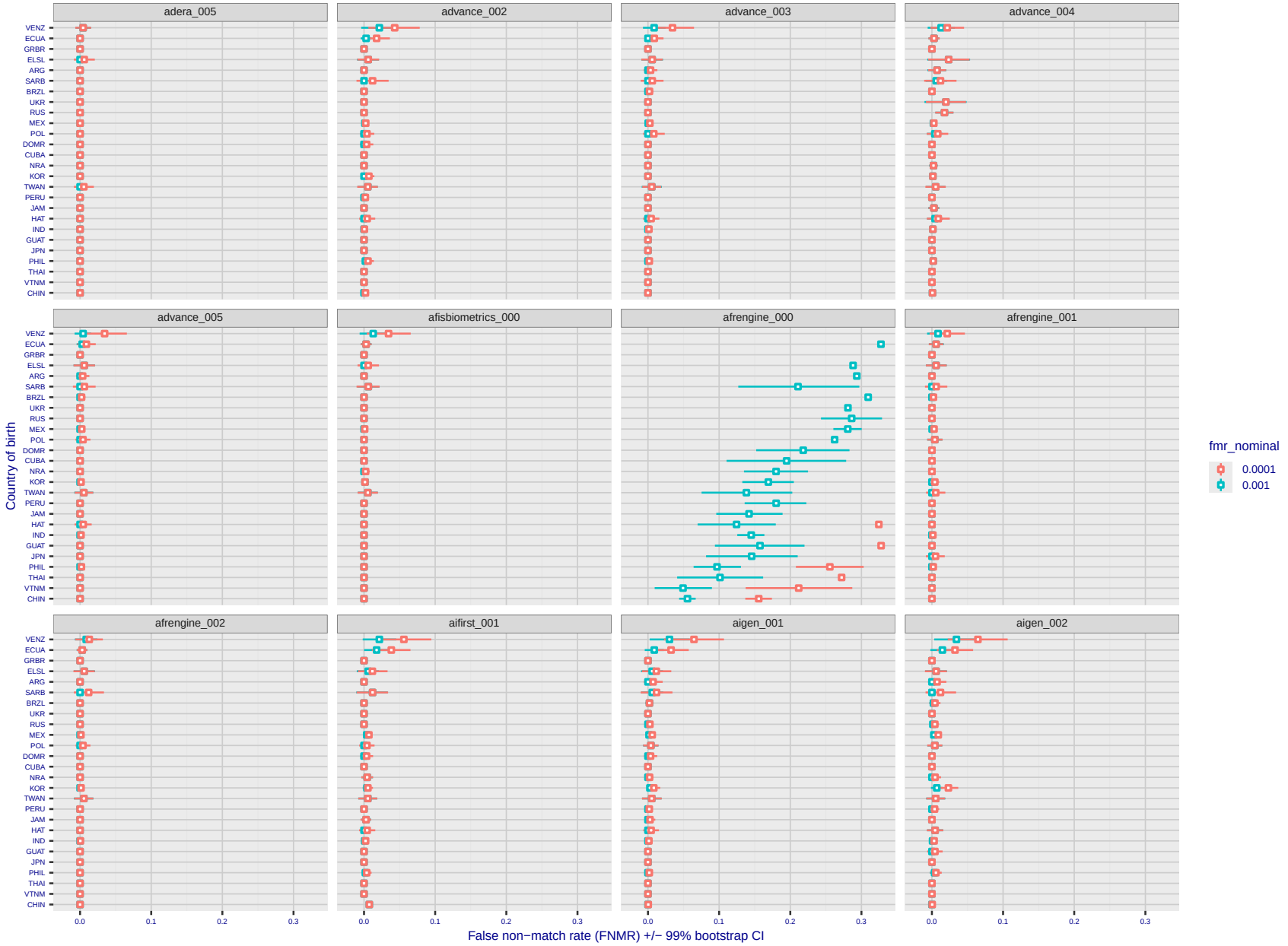


Figure 638: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

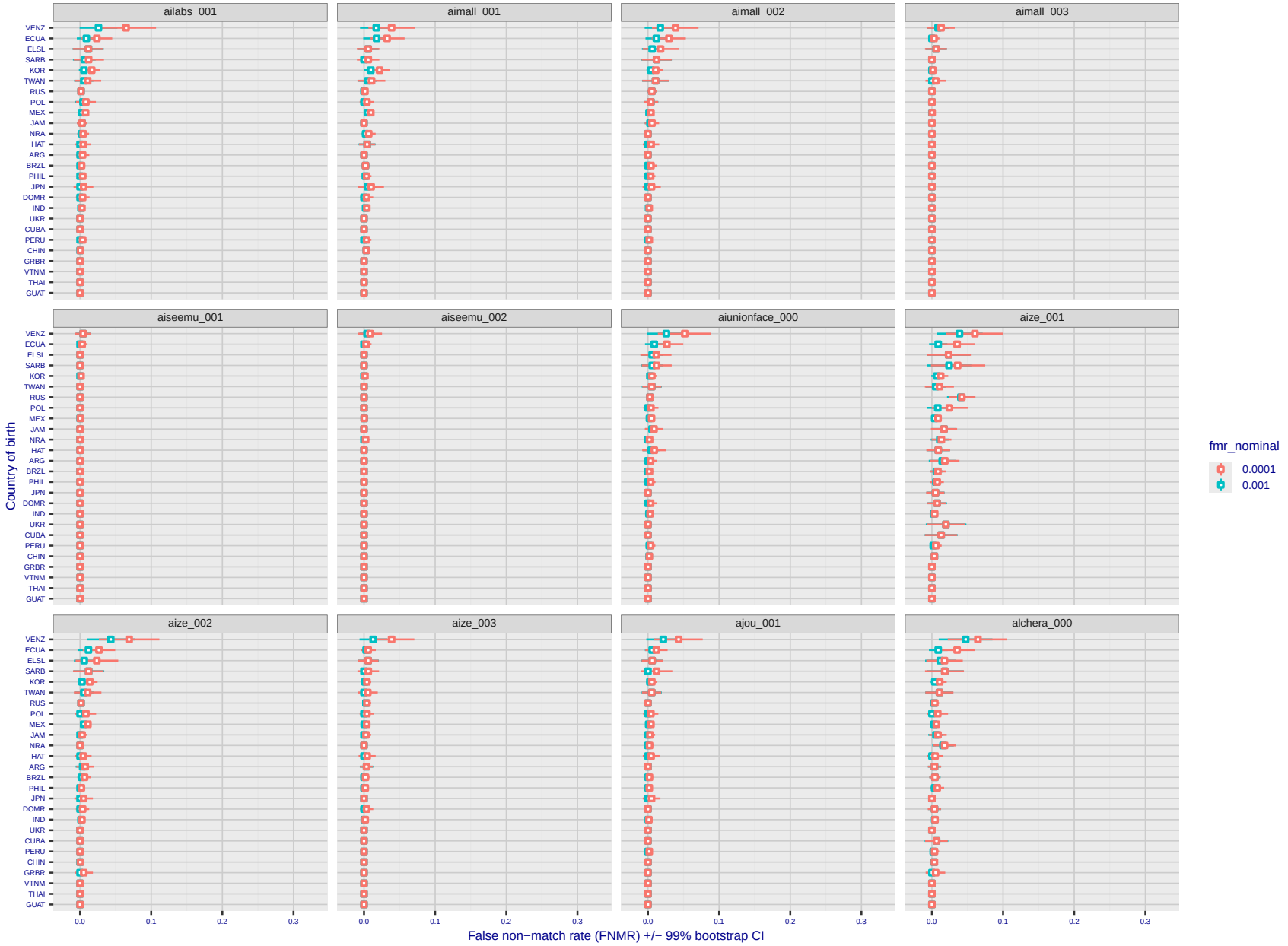


Figure 639: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 640: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

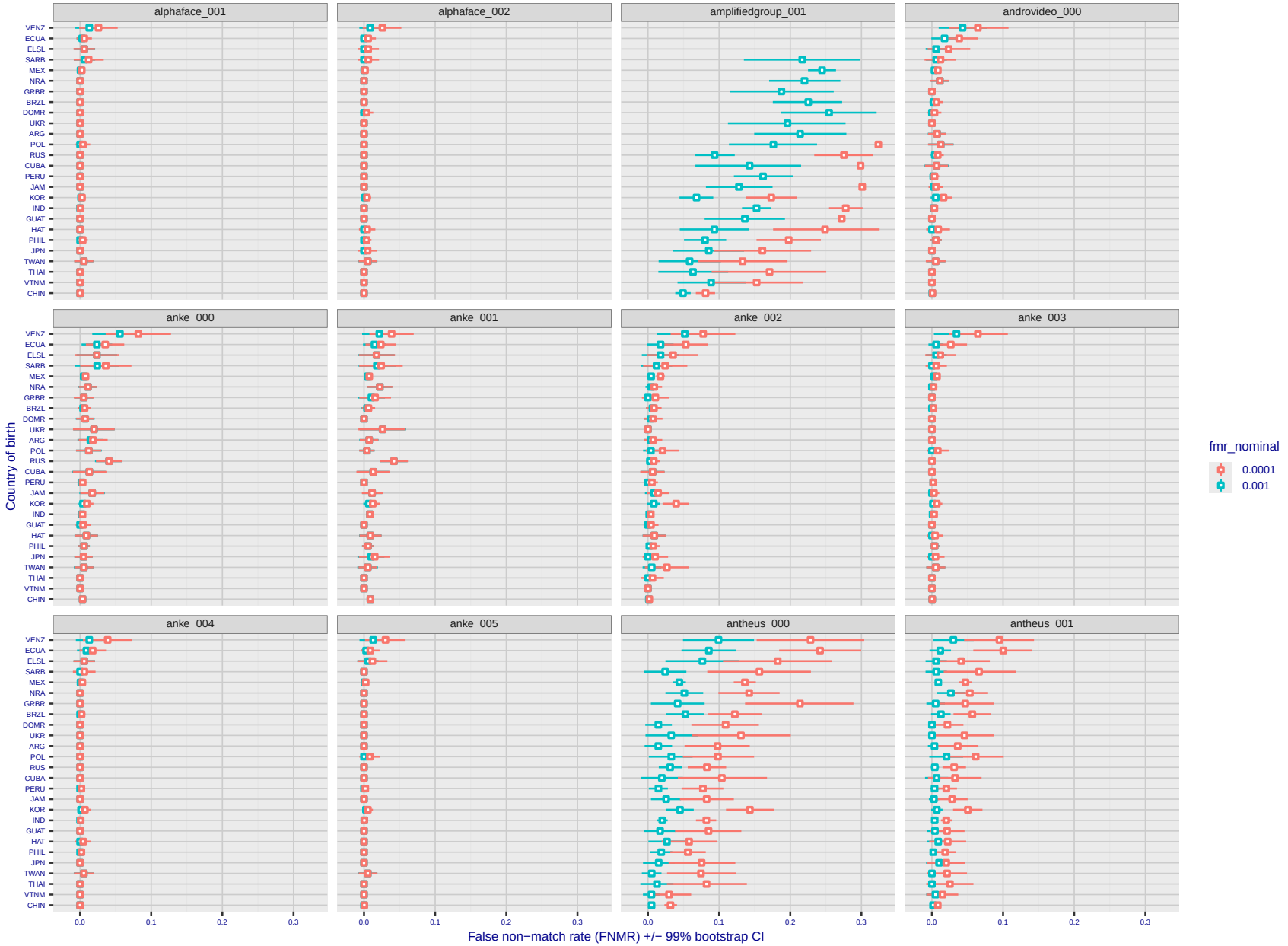


Figure 641: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 642: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

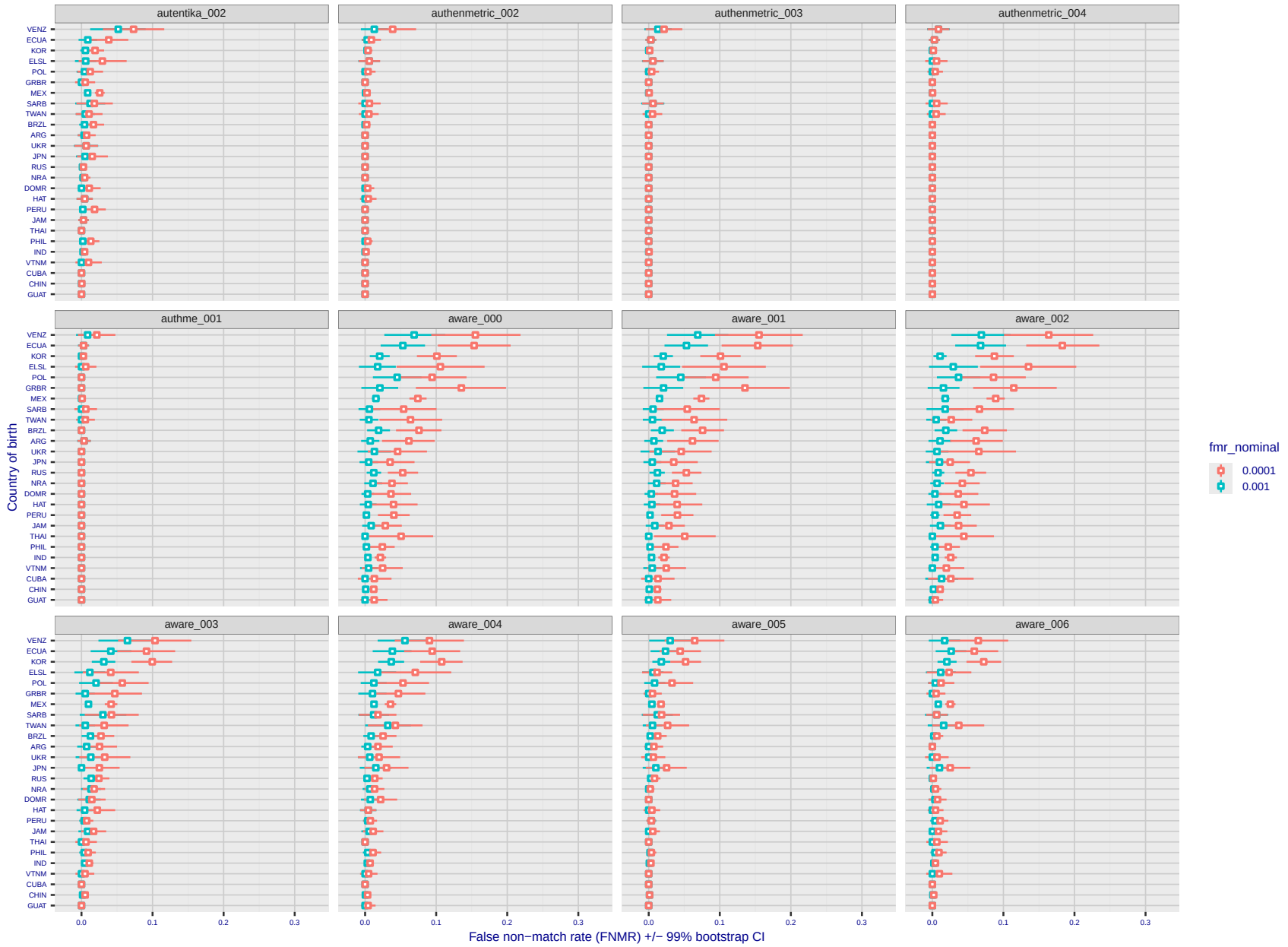


Figure 643: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

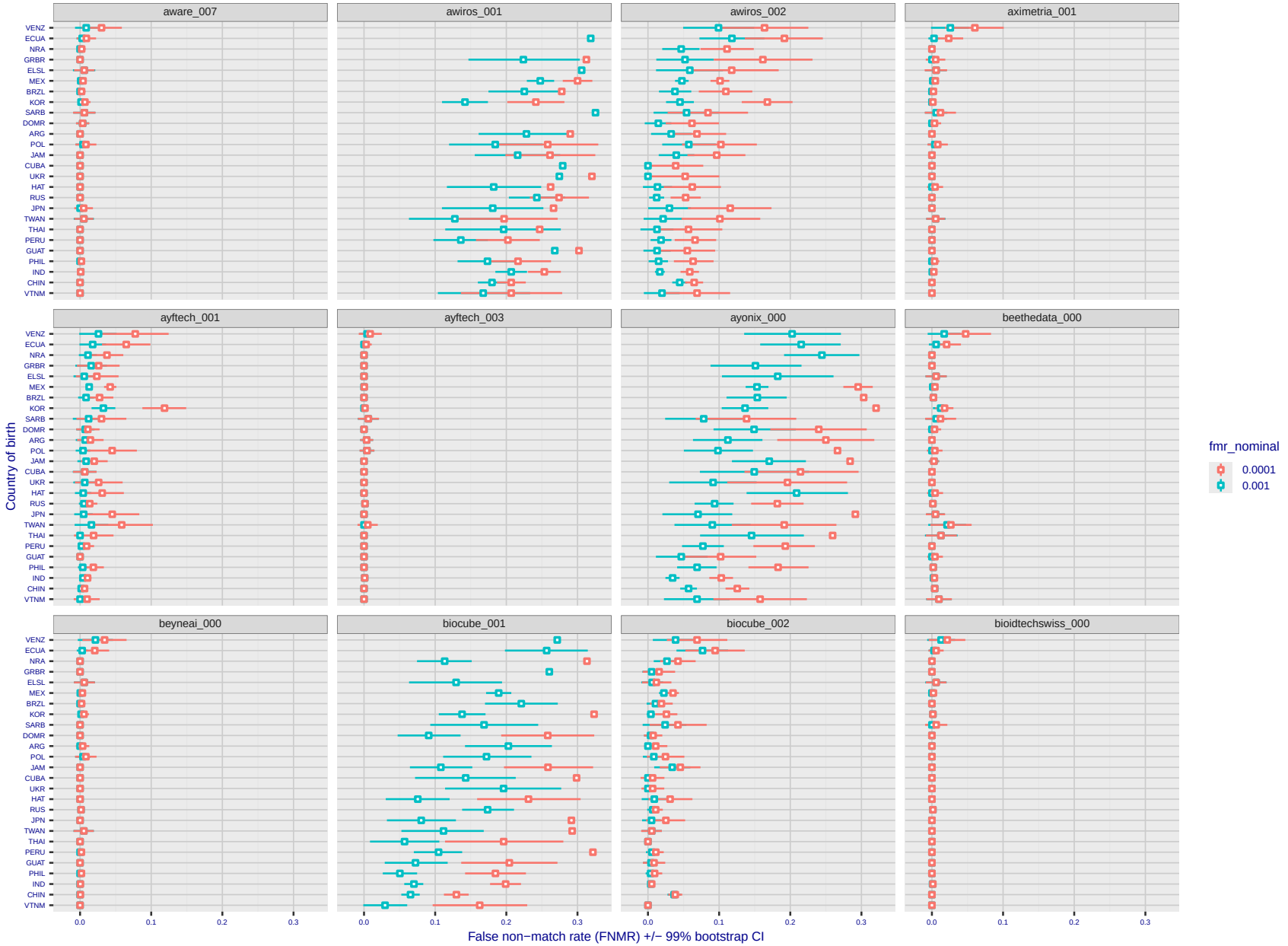


Figure 644: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 645: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 646: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

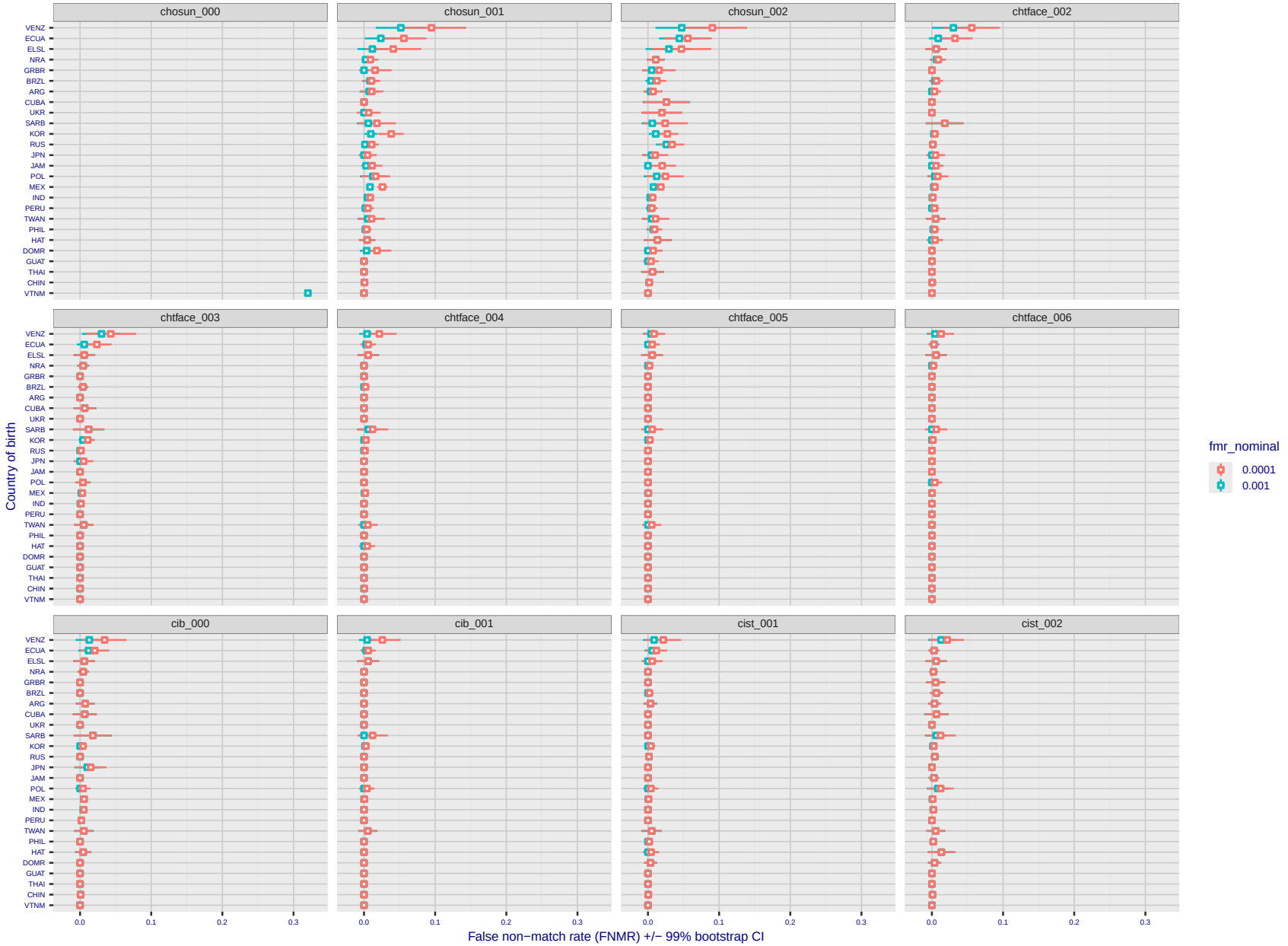


Figure 647: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

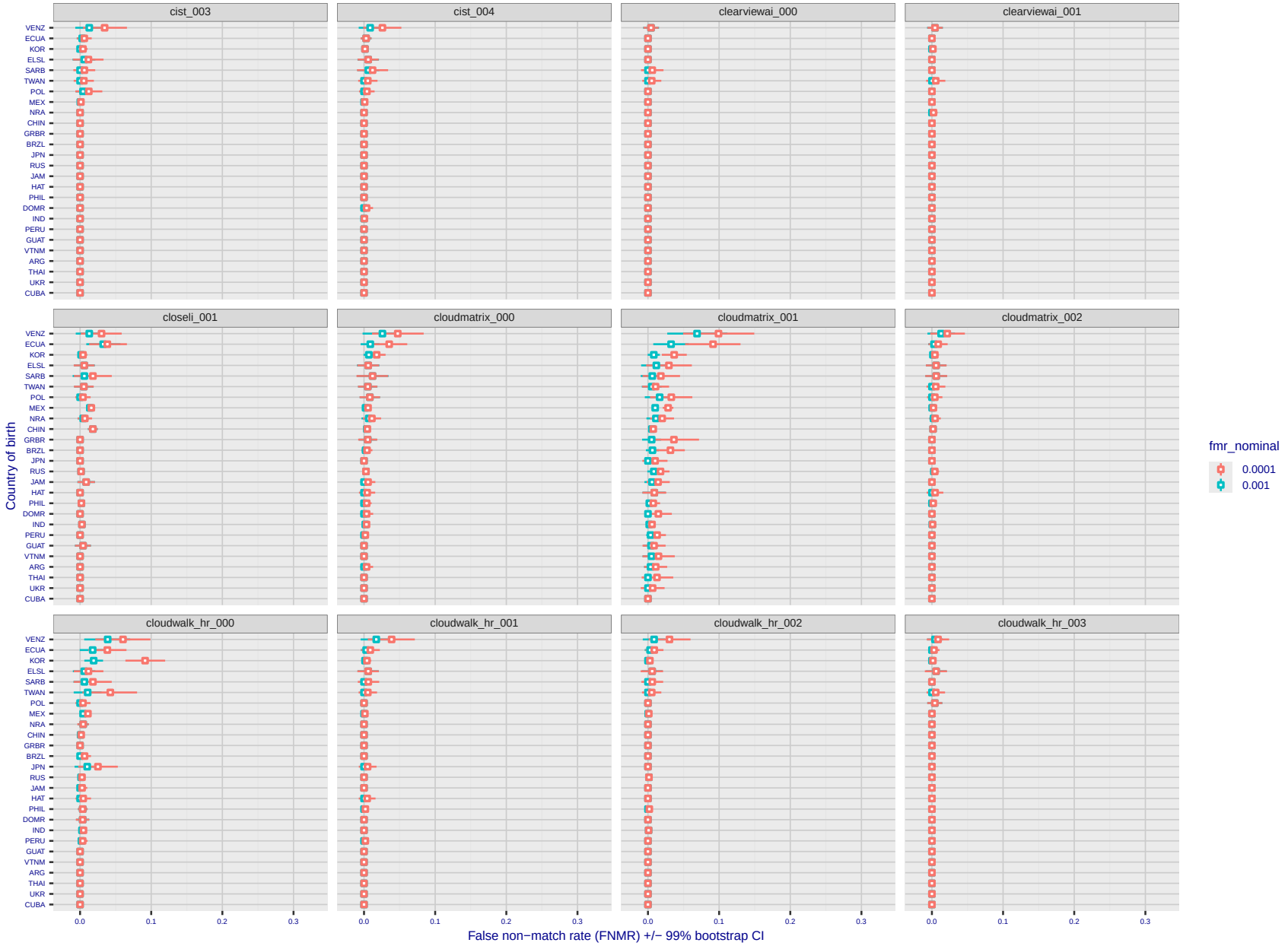


Figure 648: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

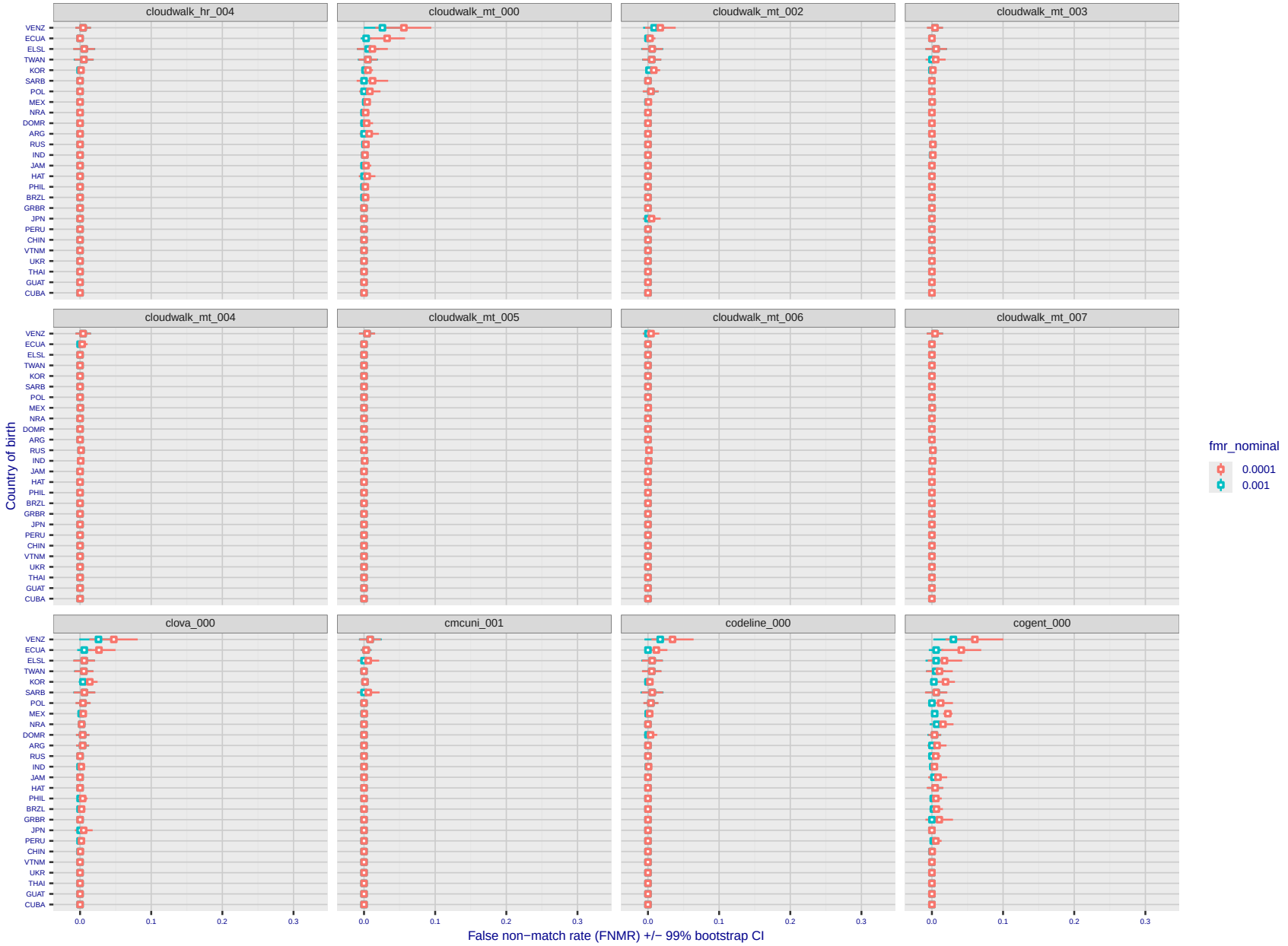


Figure 649: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

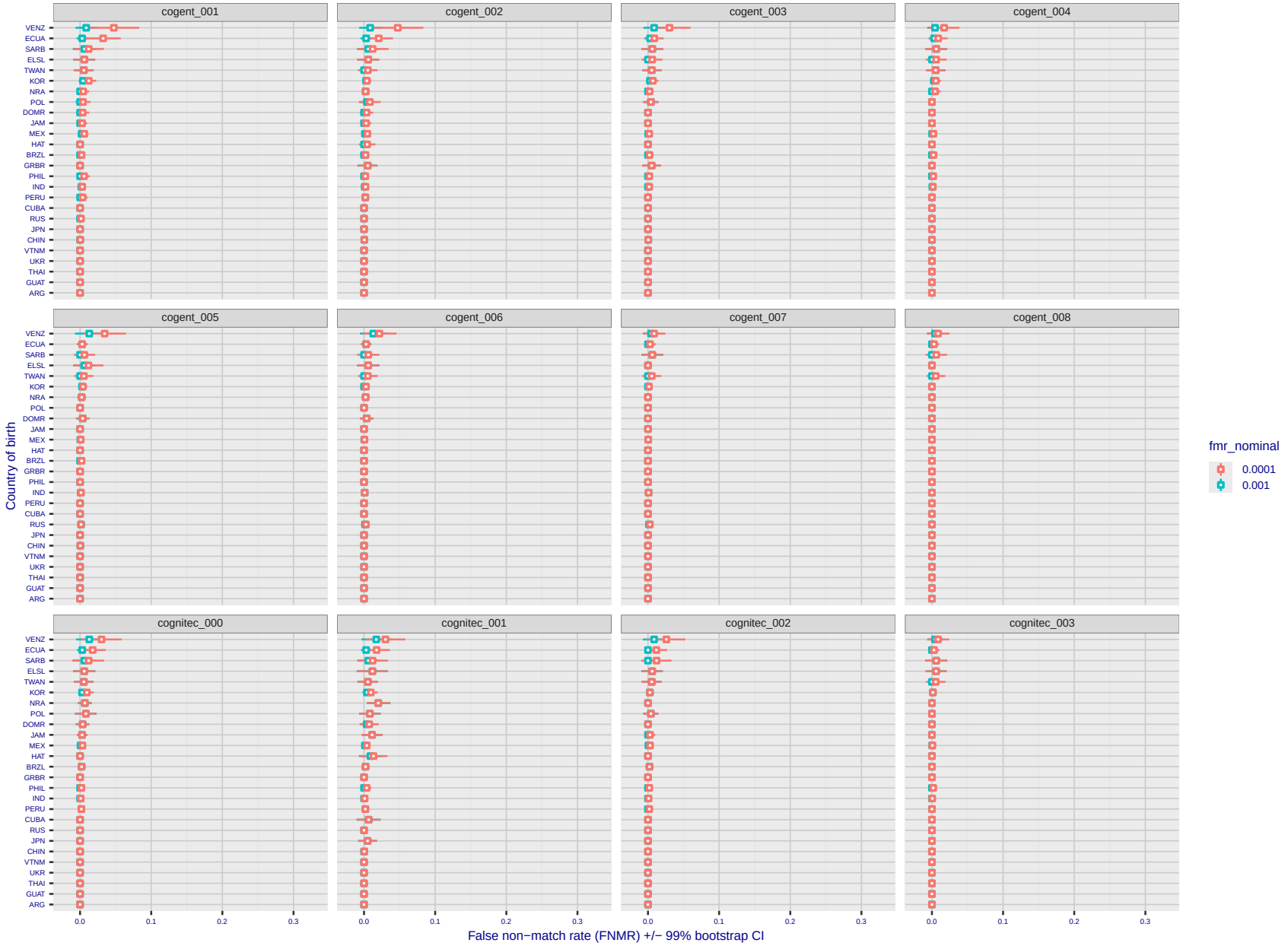


Figure 650: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

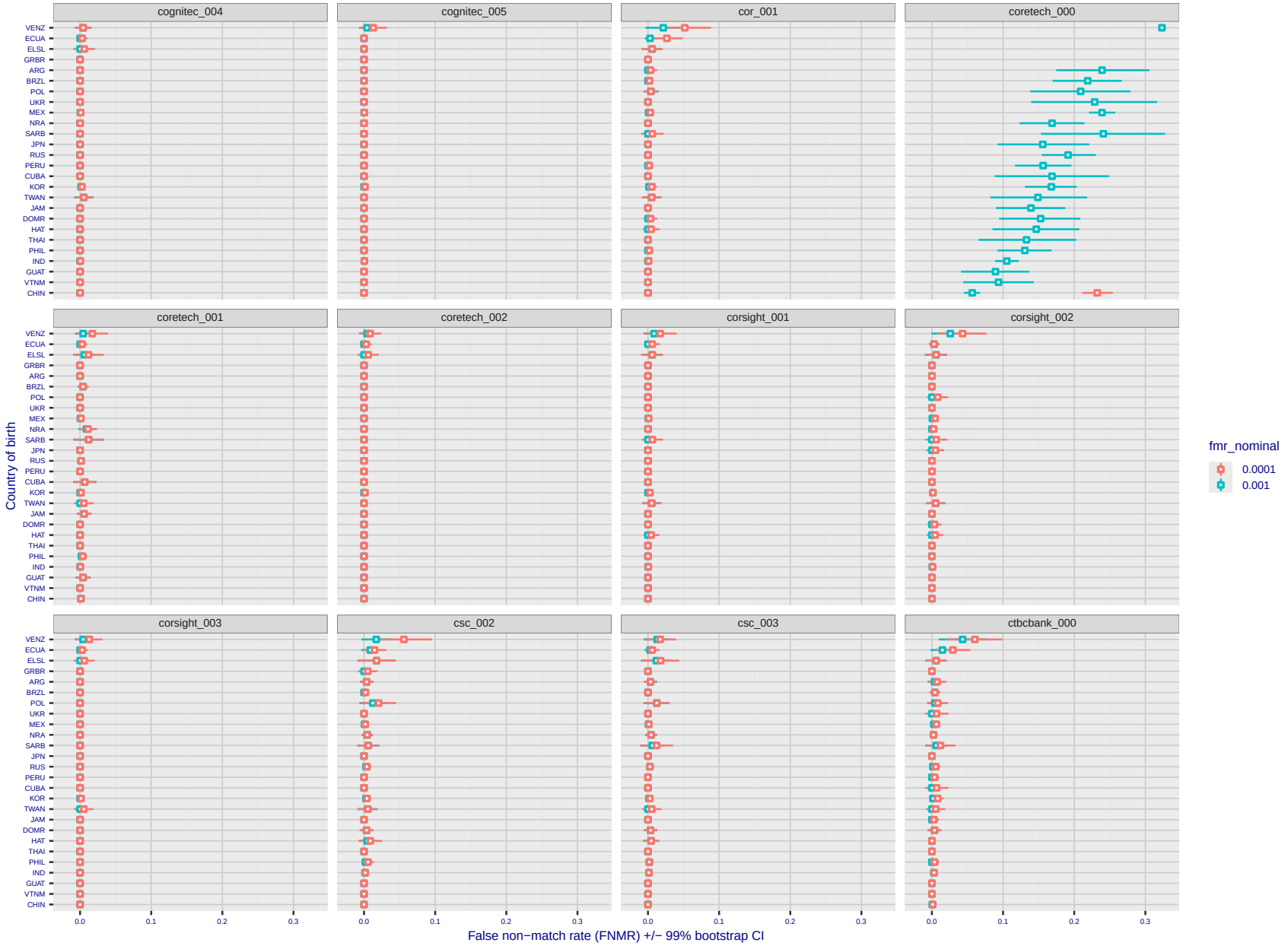


Figure 651: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

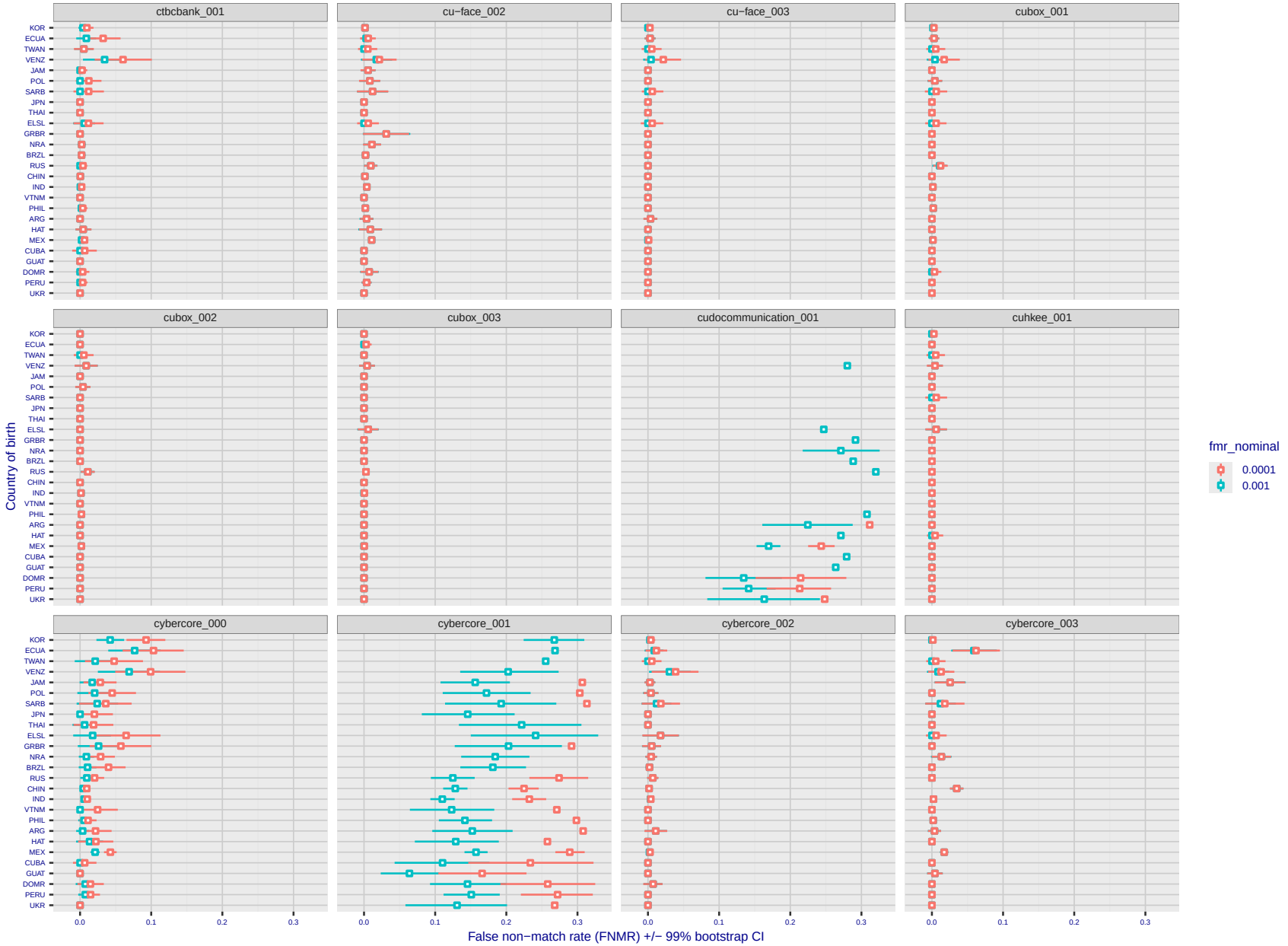


Figure 652: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 653: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

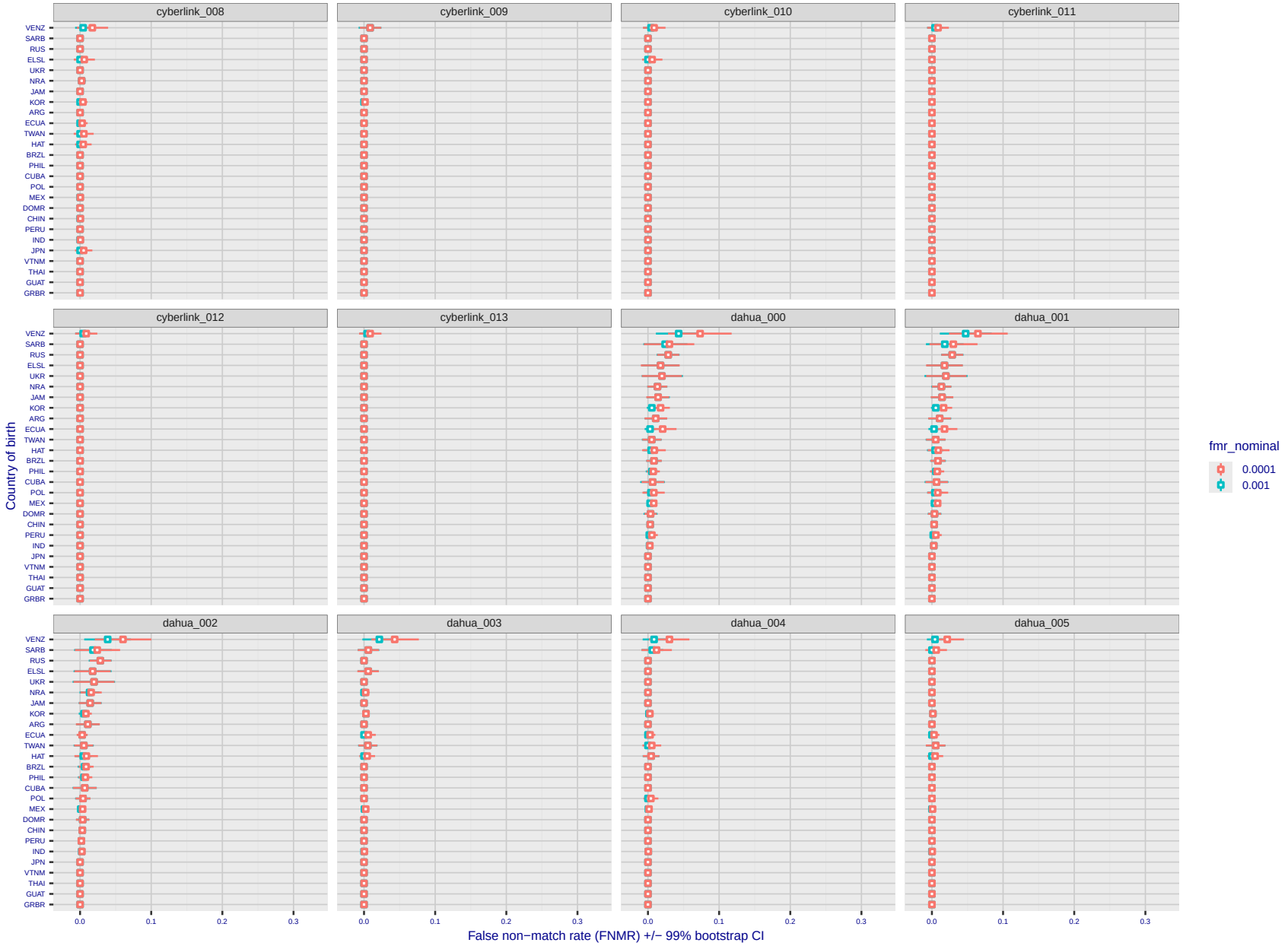


Figure 654: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 655: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

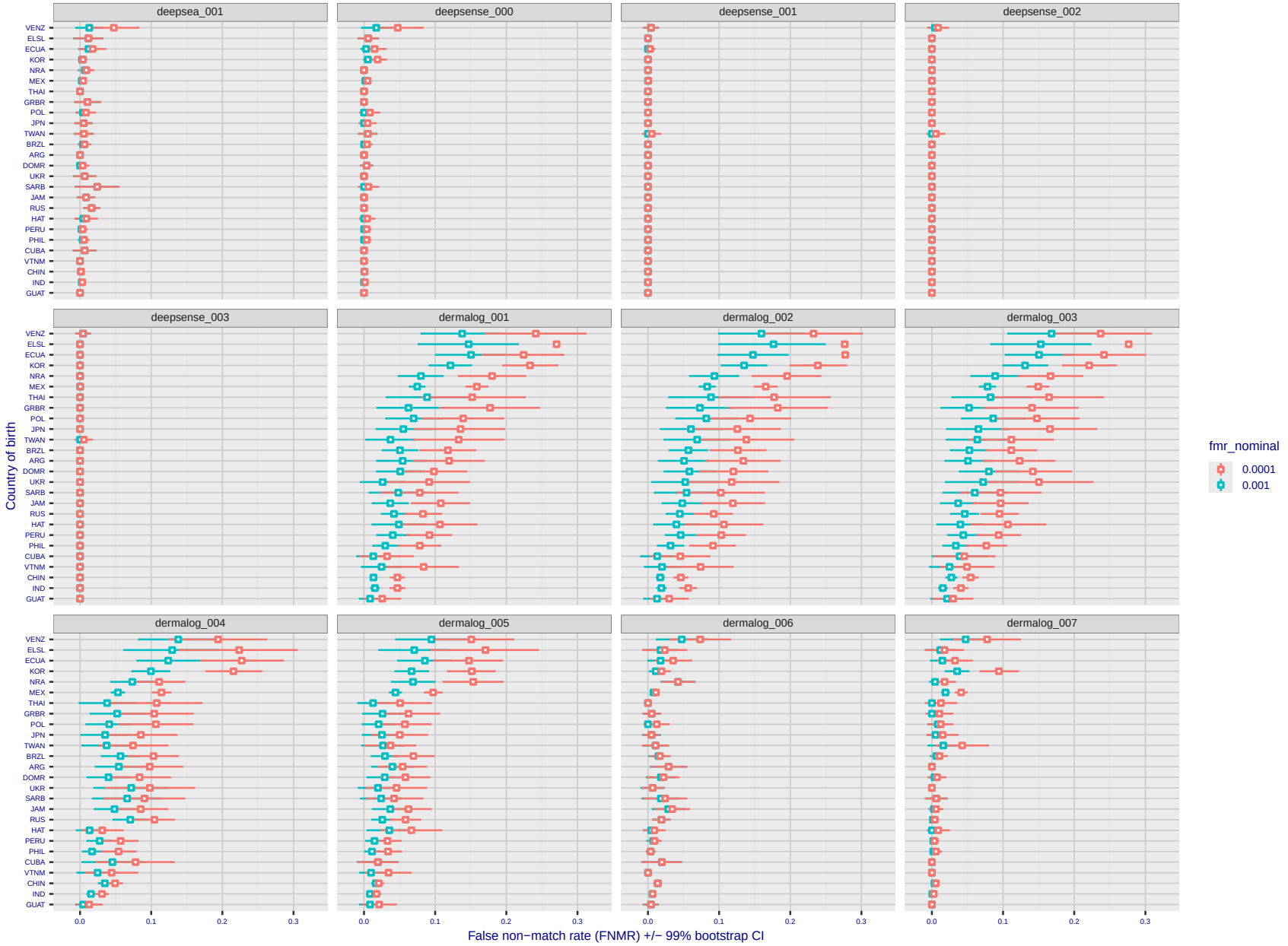


Figure 656: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

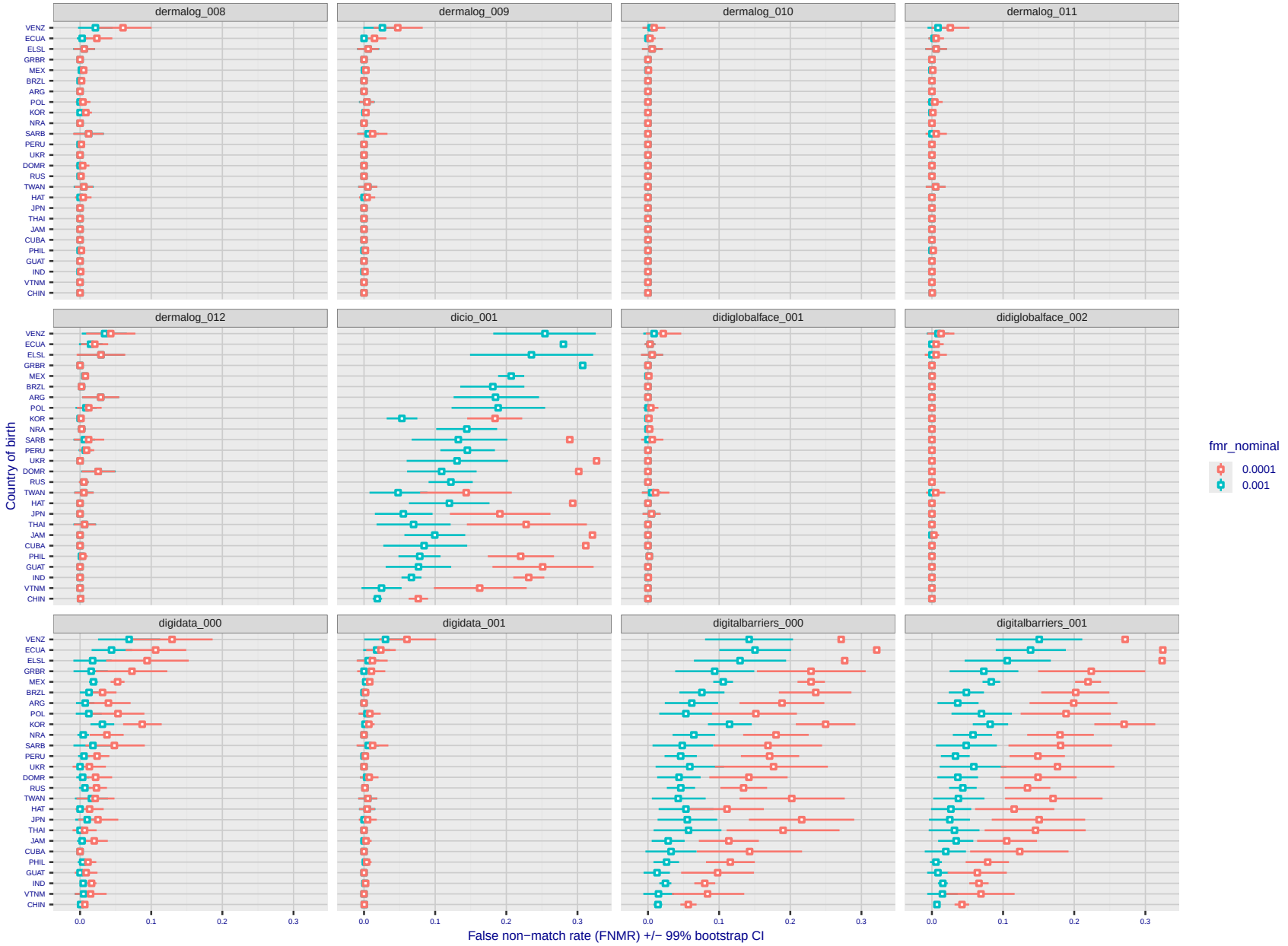


Figure 657: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 658: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 659: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 660: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

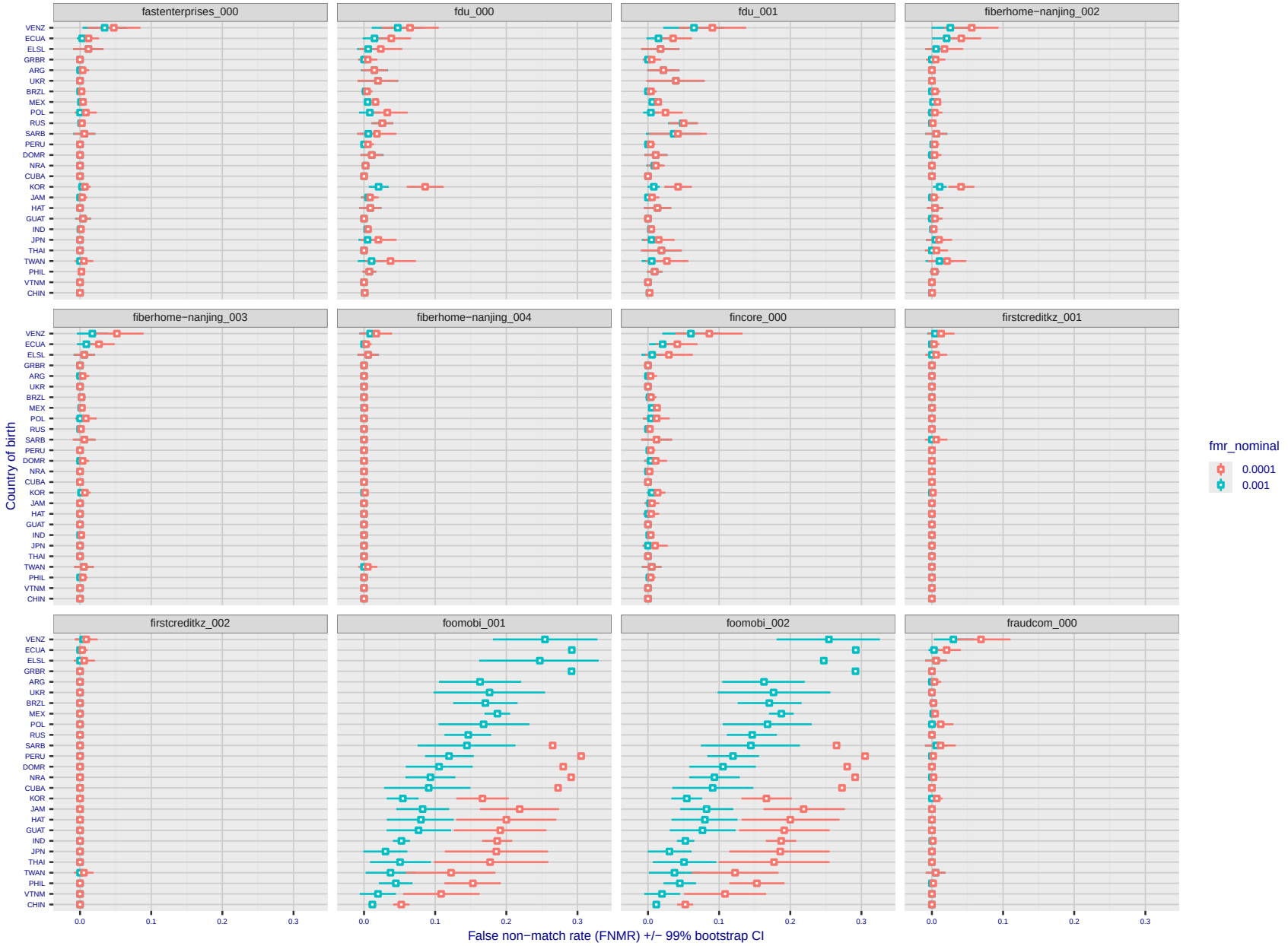


Figure 661: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

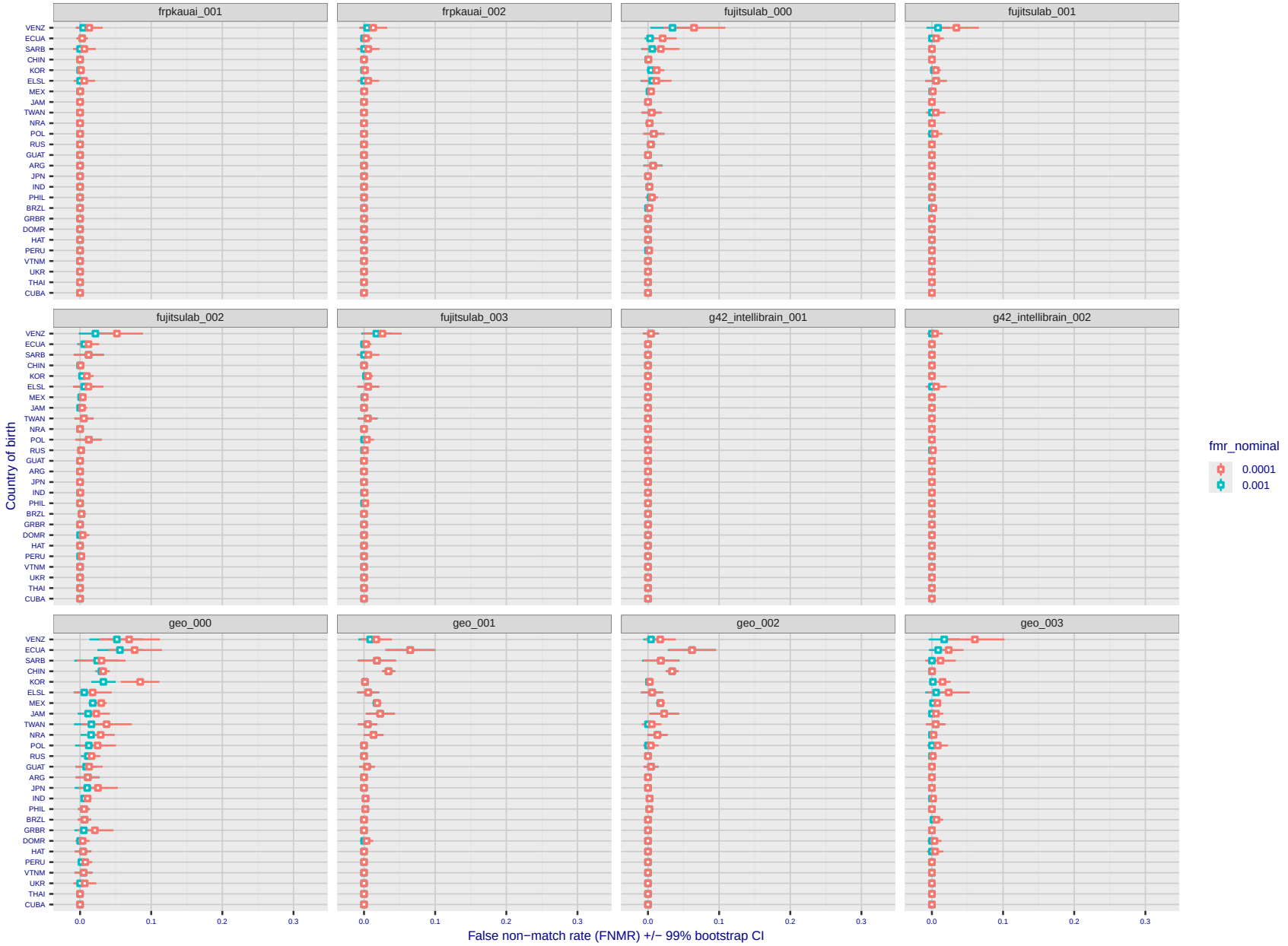


Figure 662: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 663: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

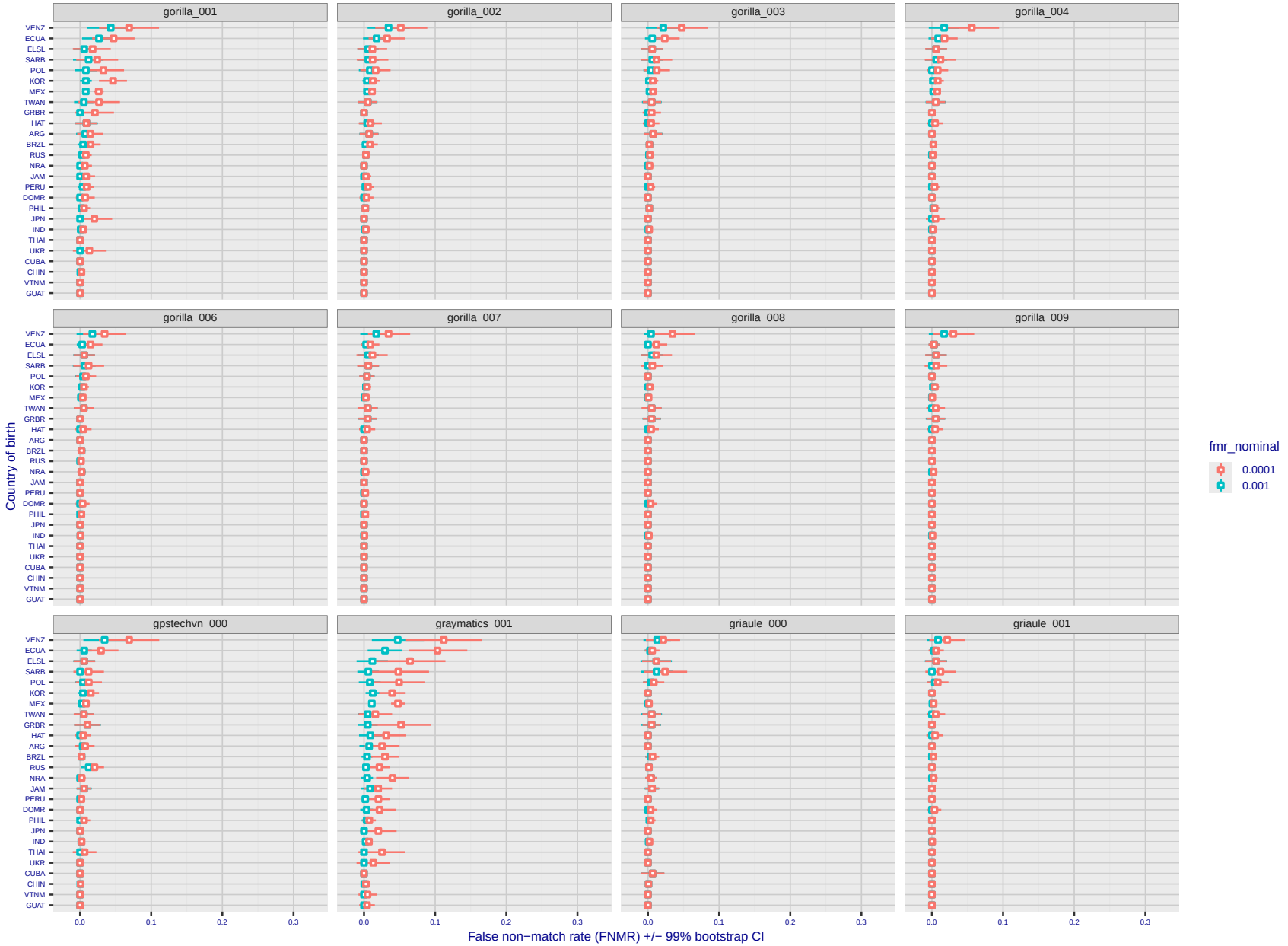


Figure 664: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 665: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 666: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

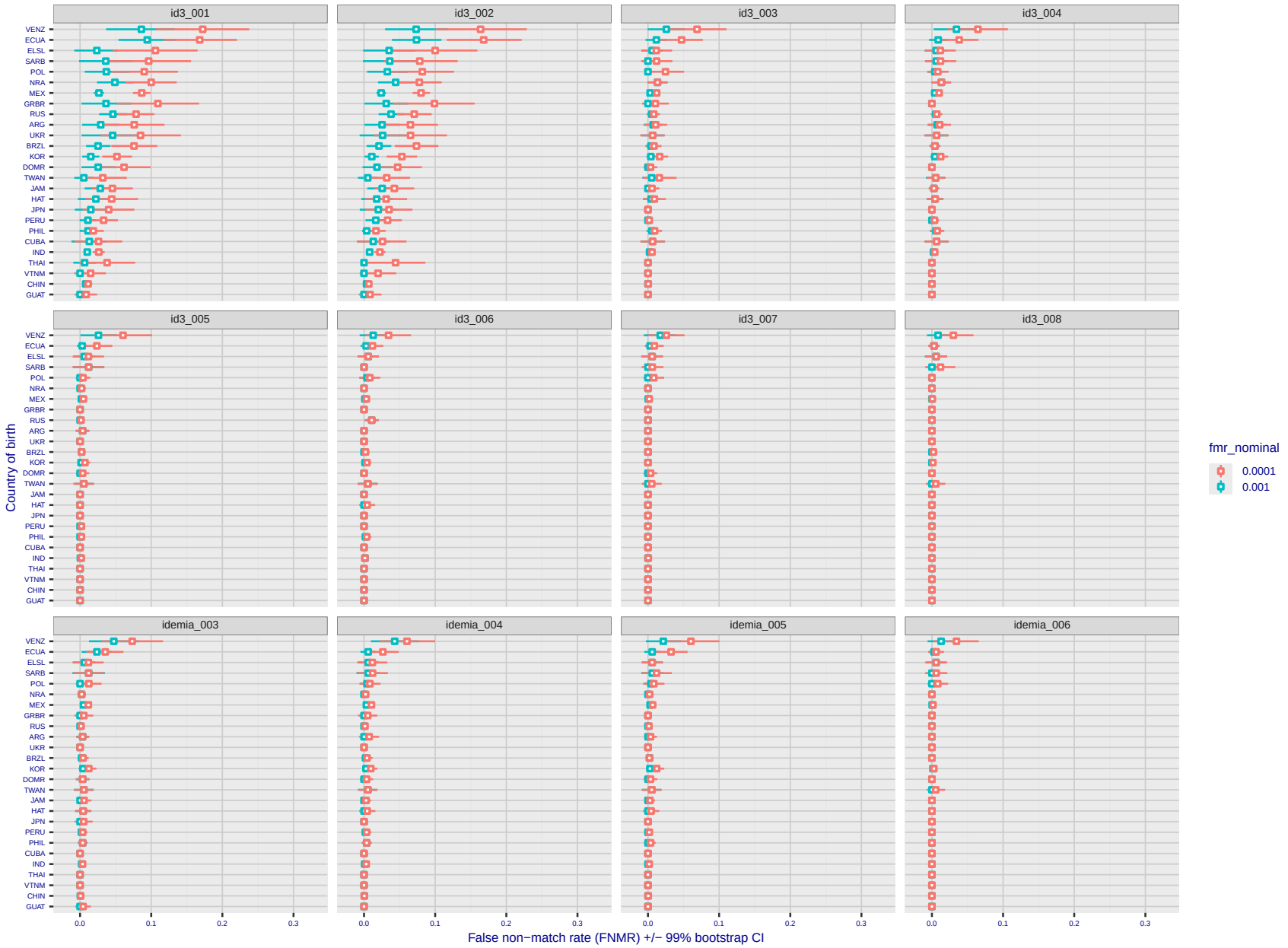


Figure 667: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

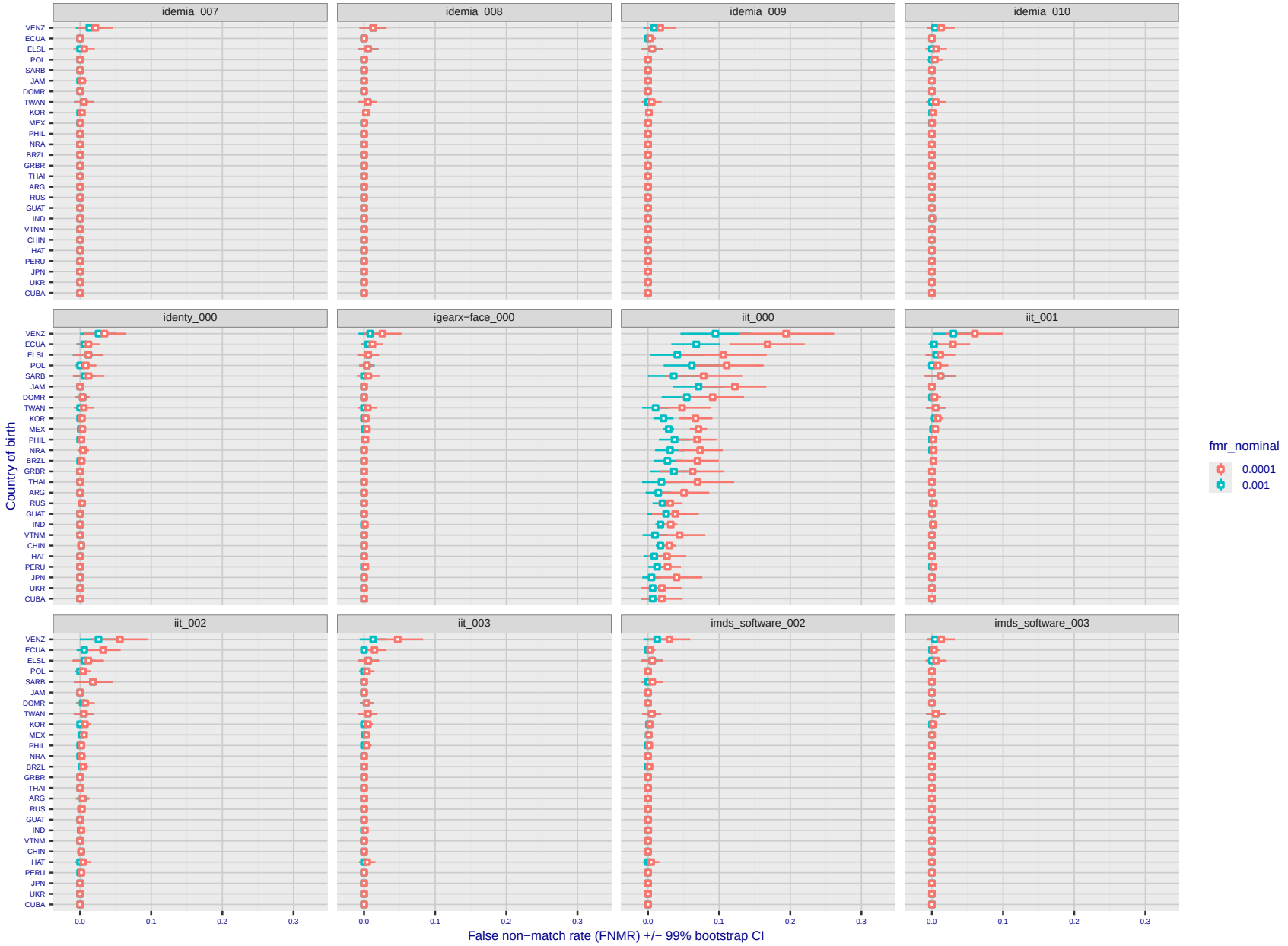


Figure 668: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

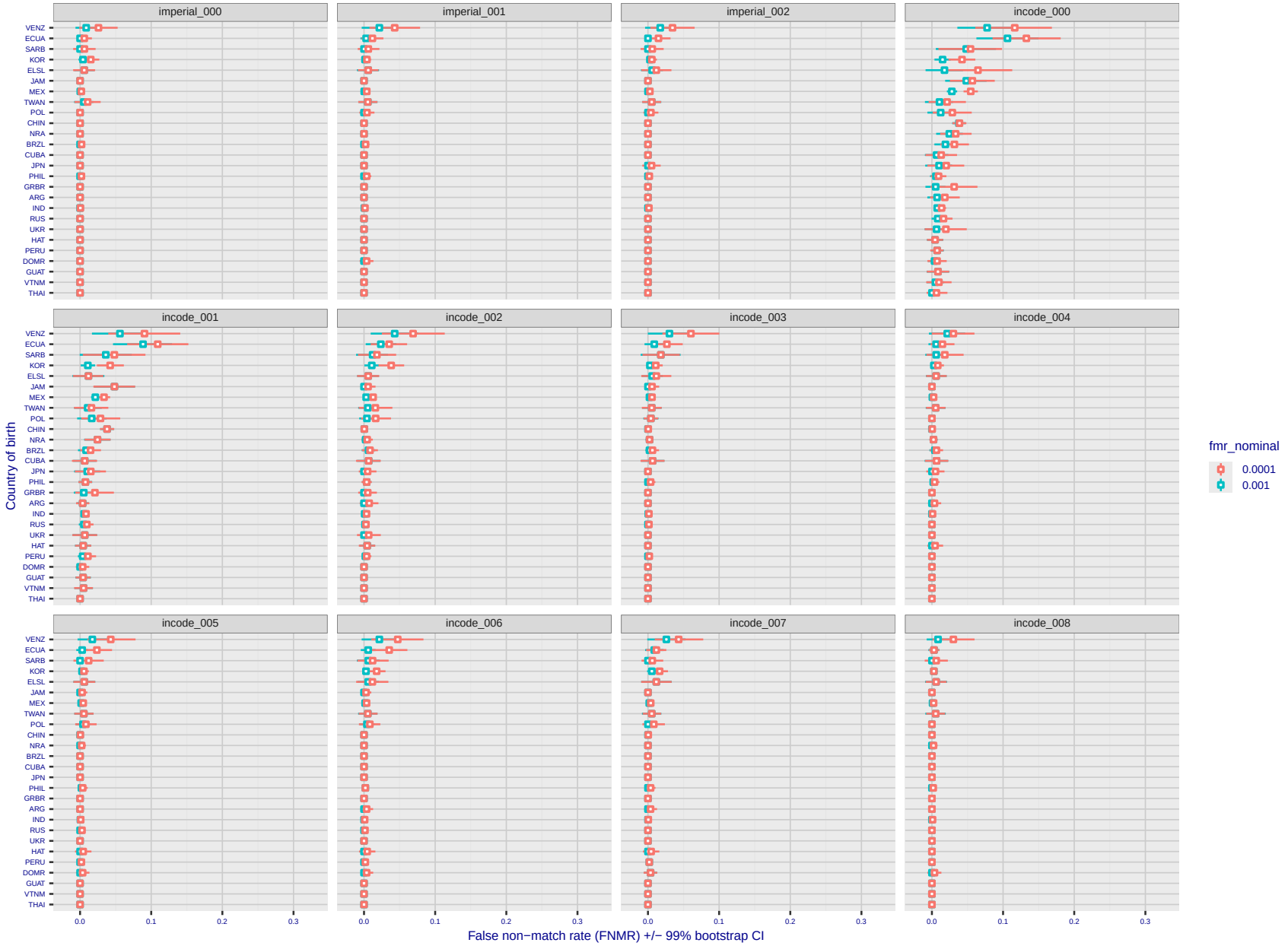


Figure 669: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

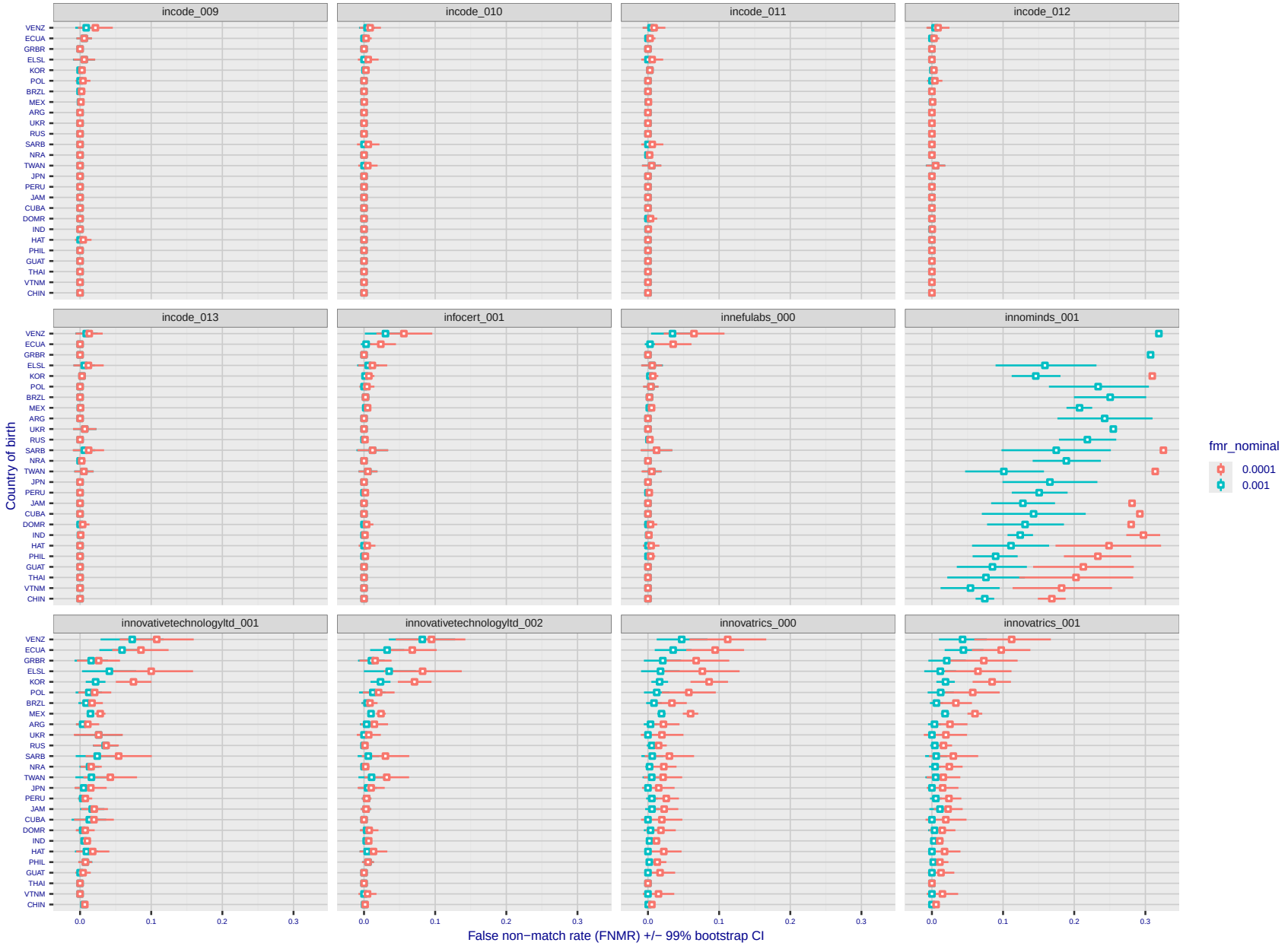


Figure 670: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

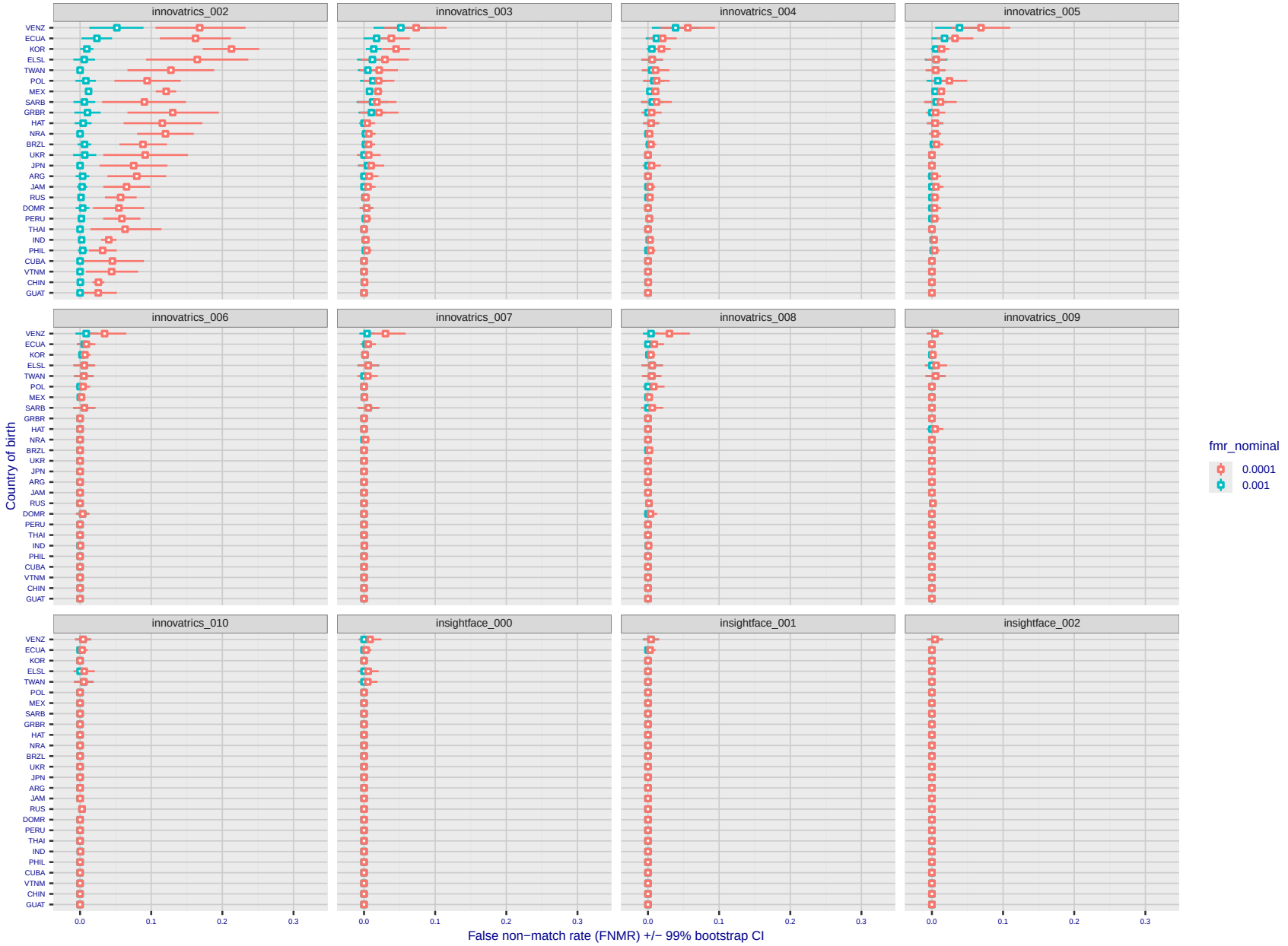


Figure 671: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

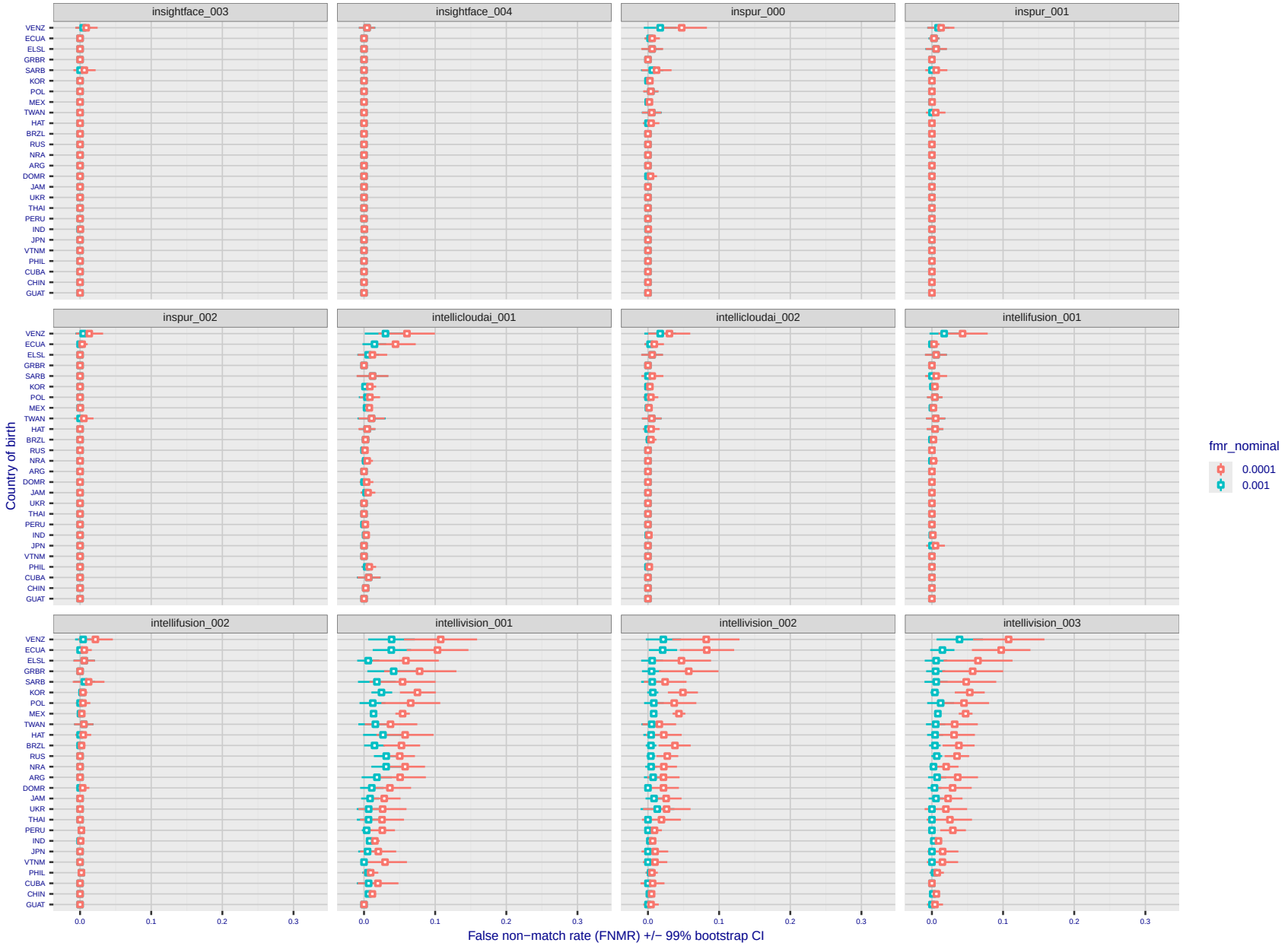


Figure 672: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

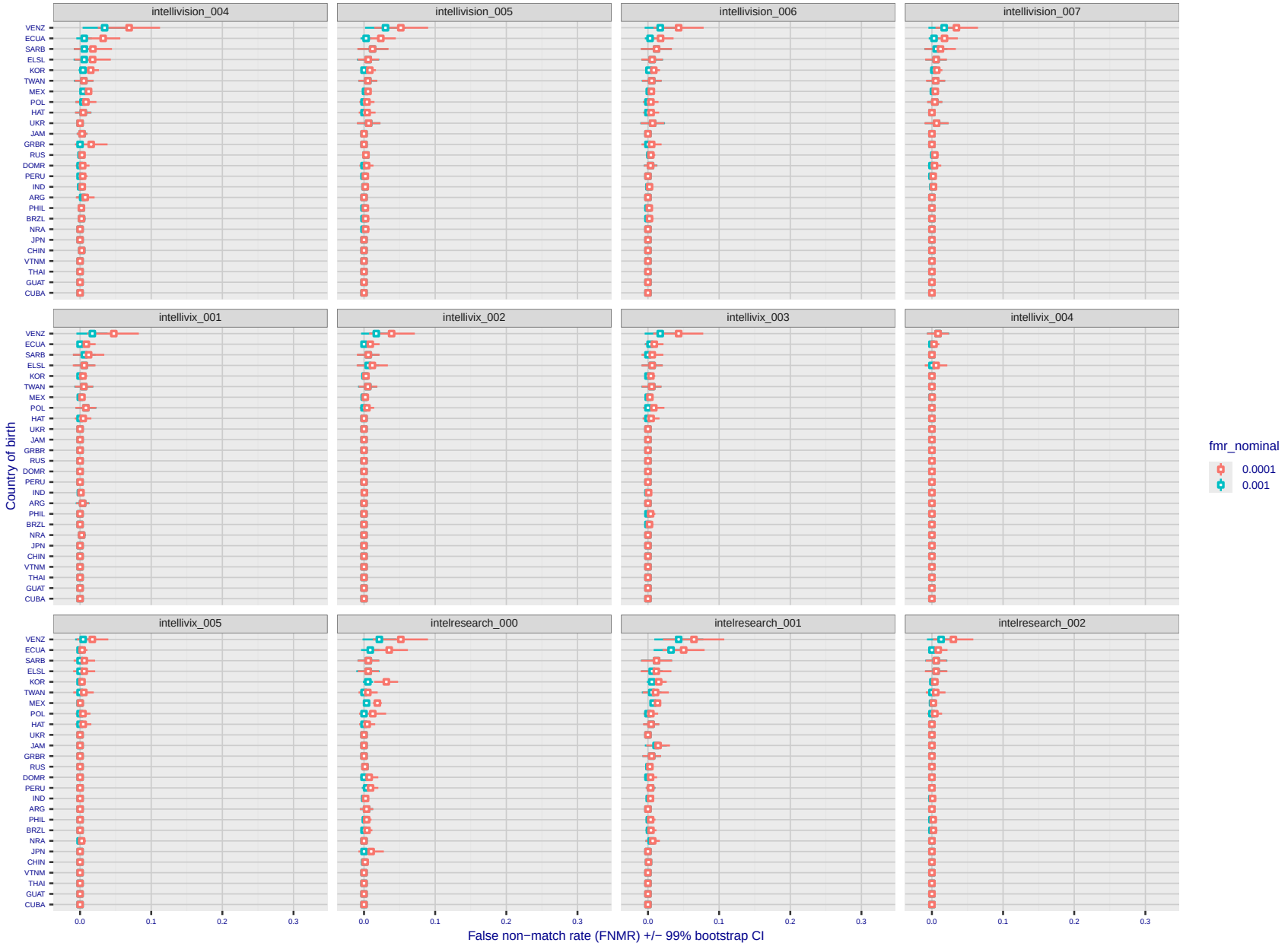


Figure 673: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 674: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

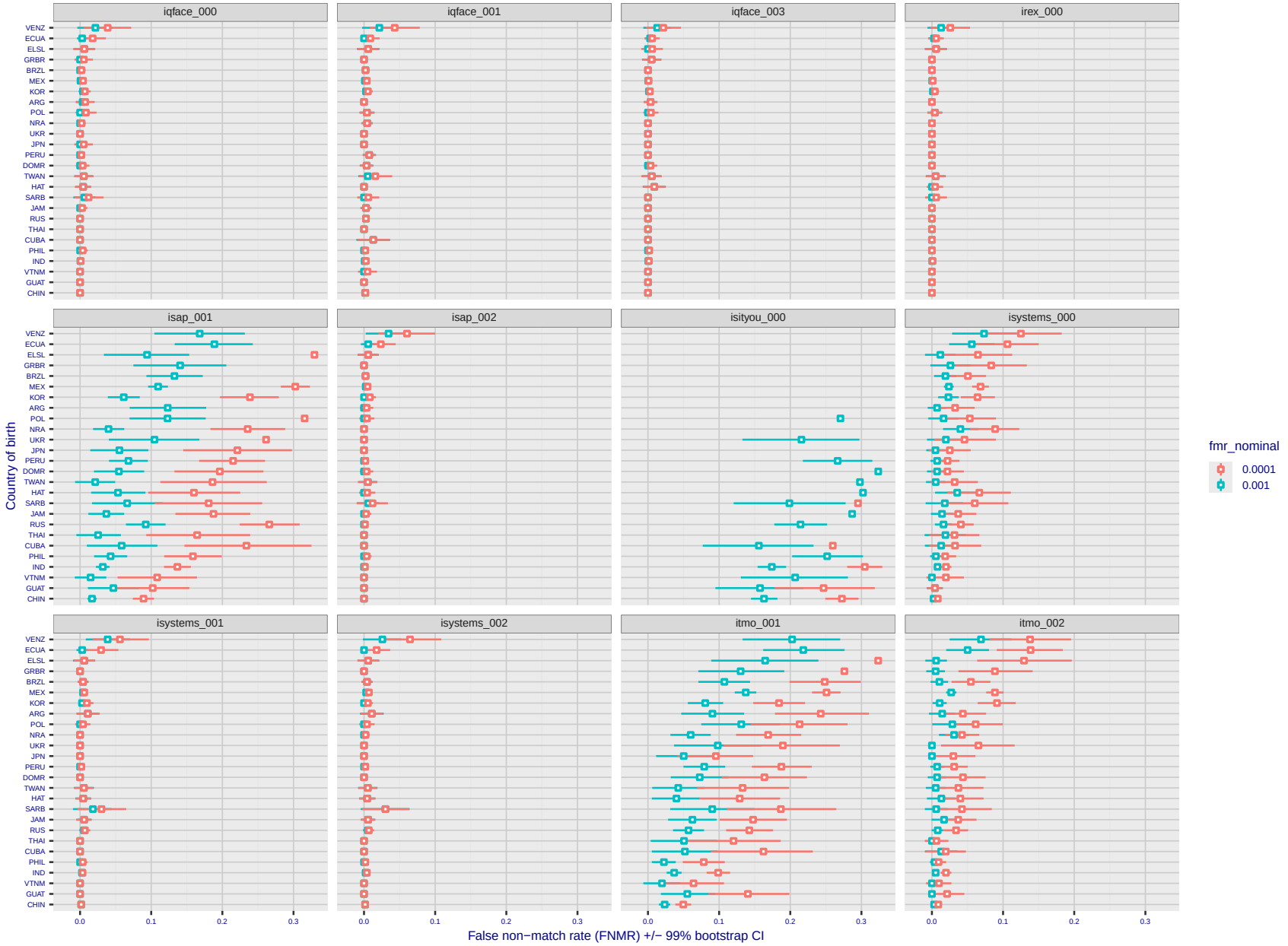


Figure 675: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

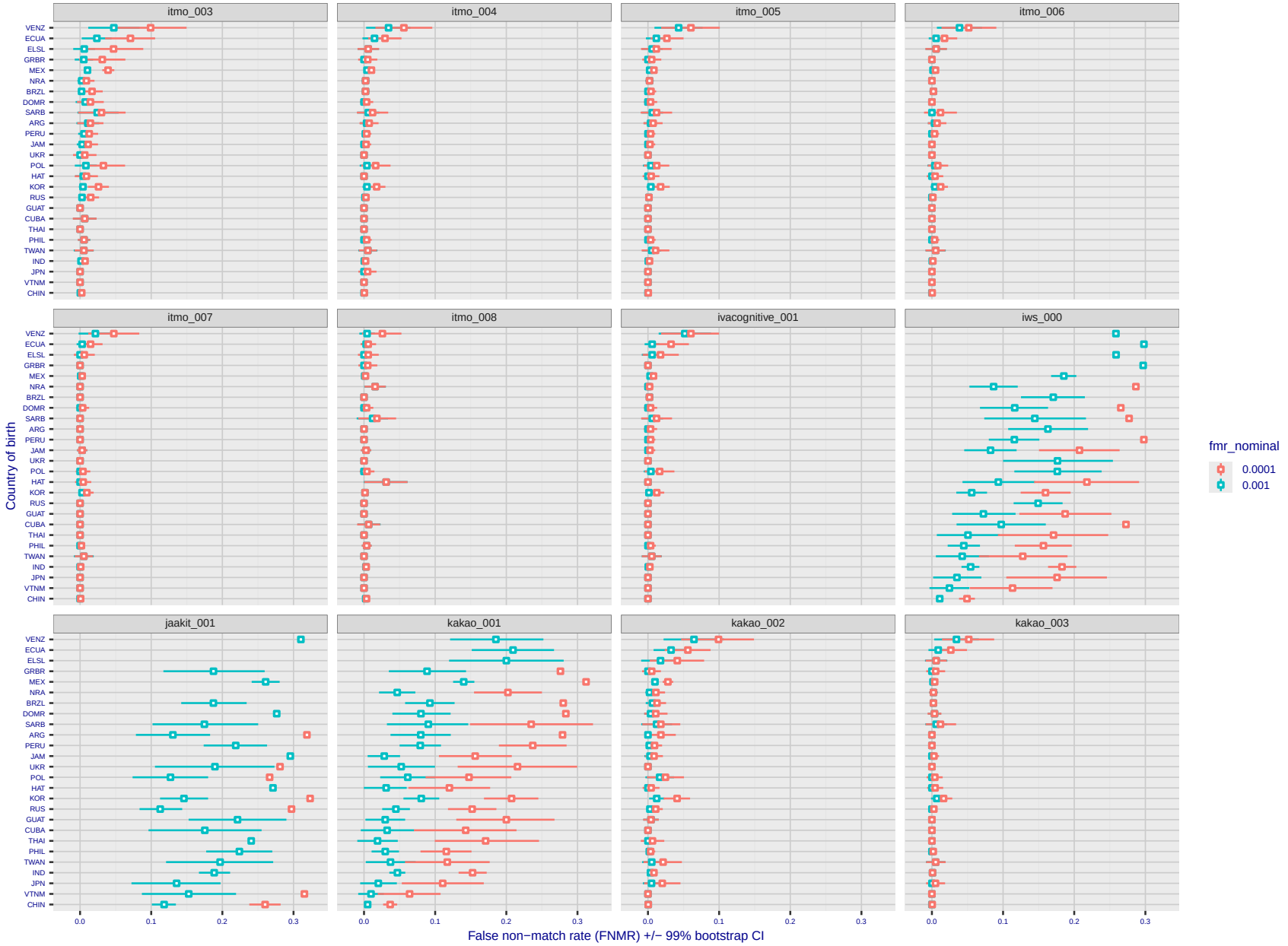


Figure 676: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

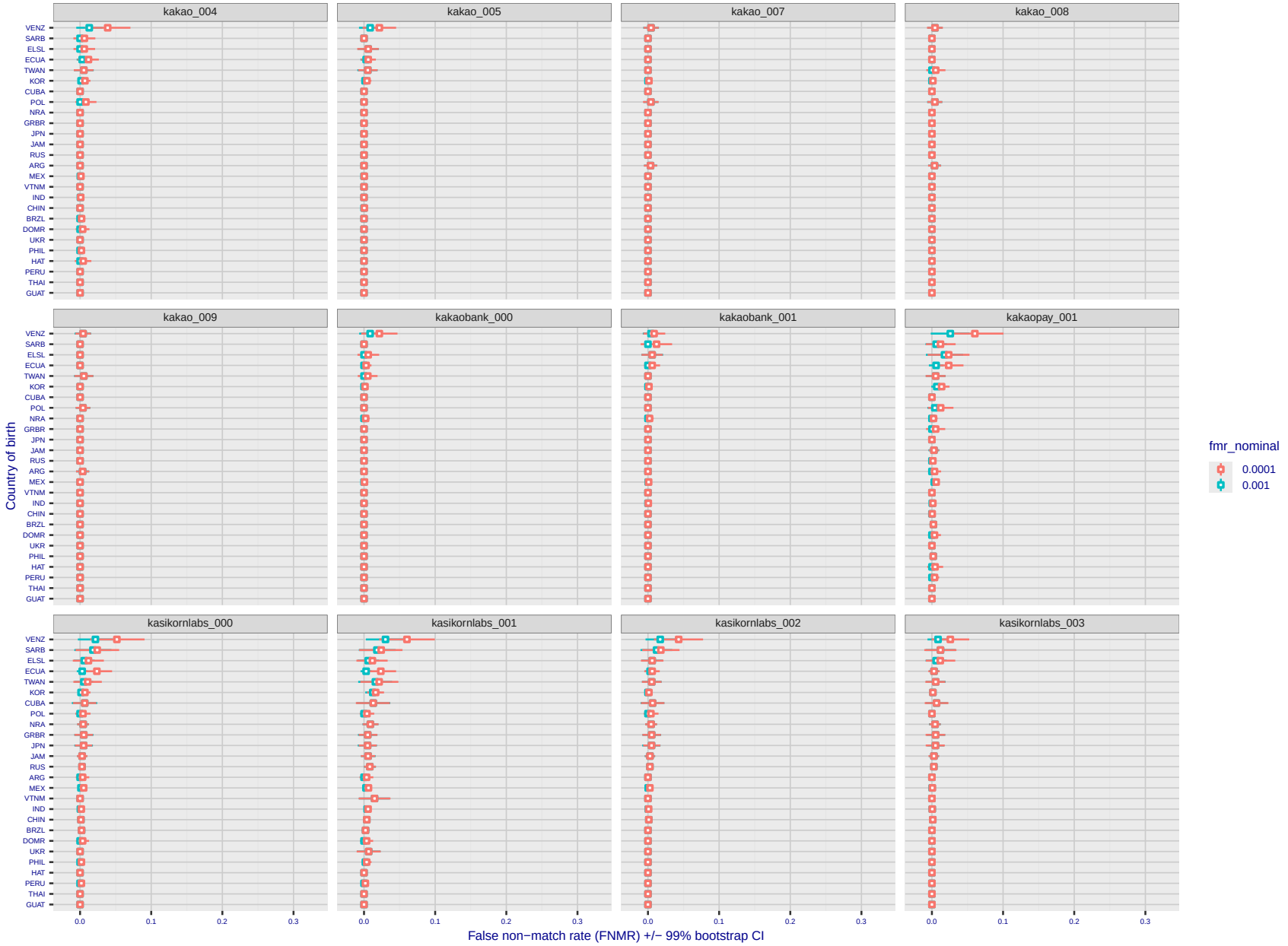


Figure 677: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

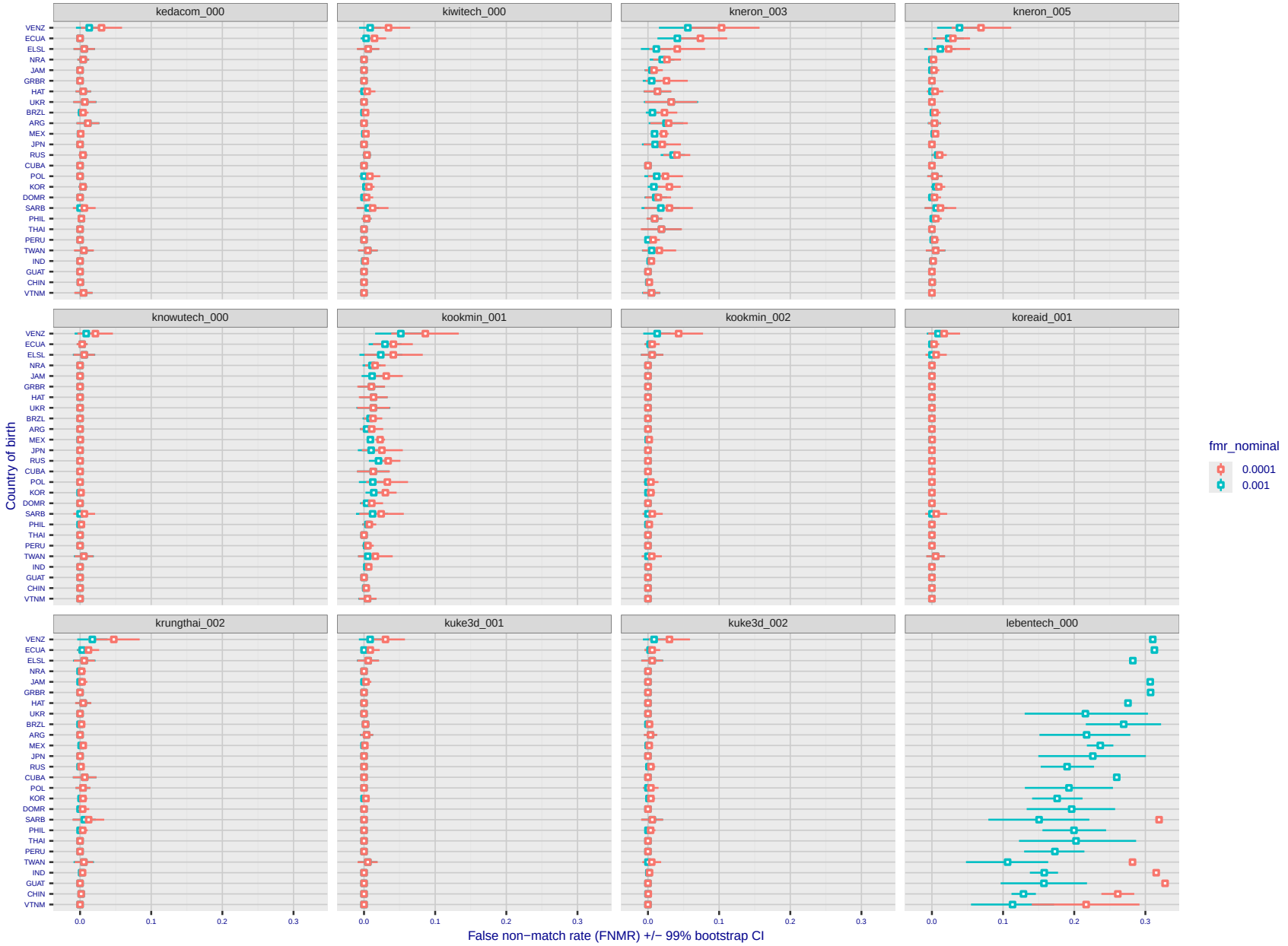


Figure 678: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

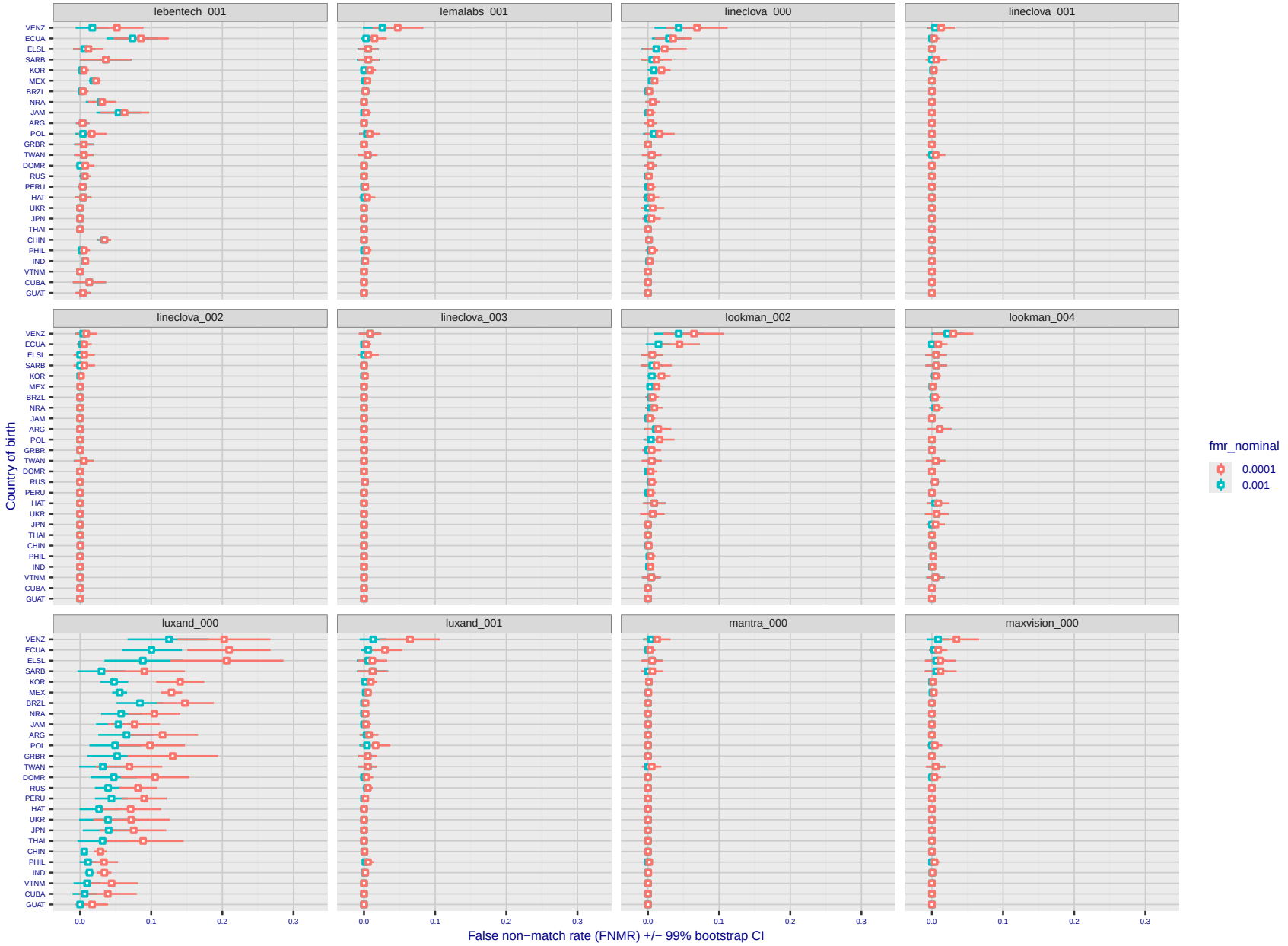


Figure 679: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

2026/11/06 13:03:11
FNMR(T) "False non-match rate"
FMR(T) "False match rate"

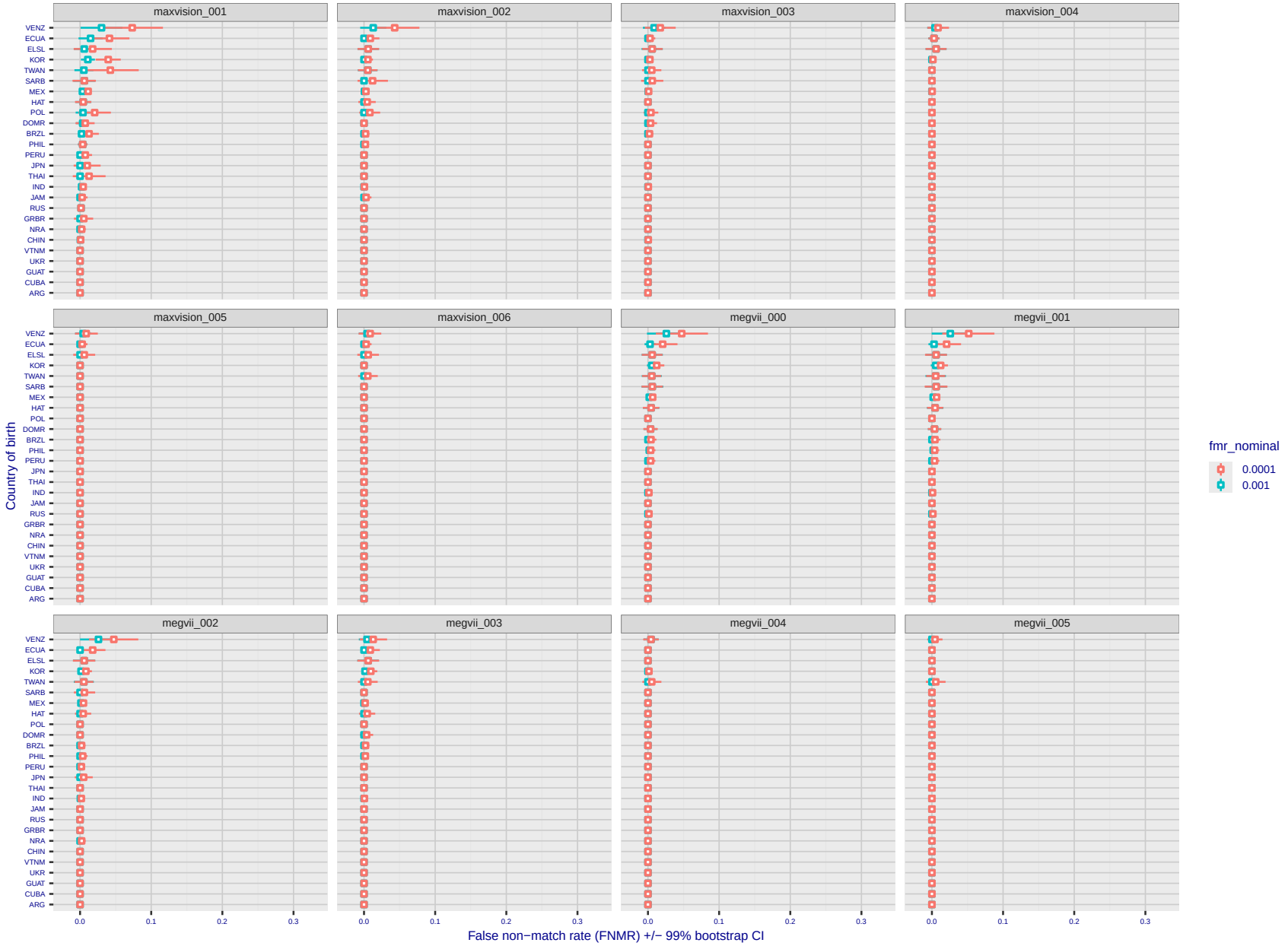


Figure 680: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

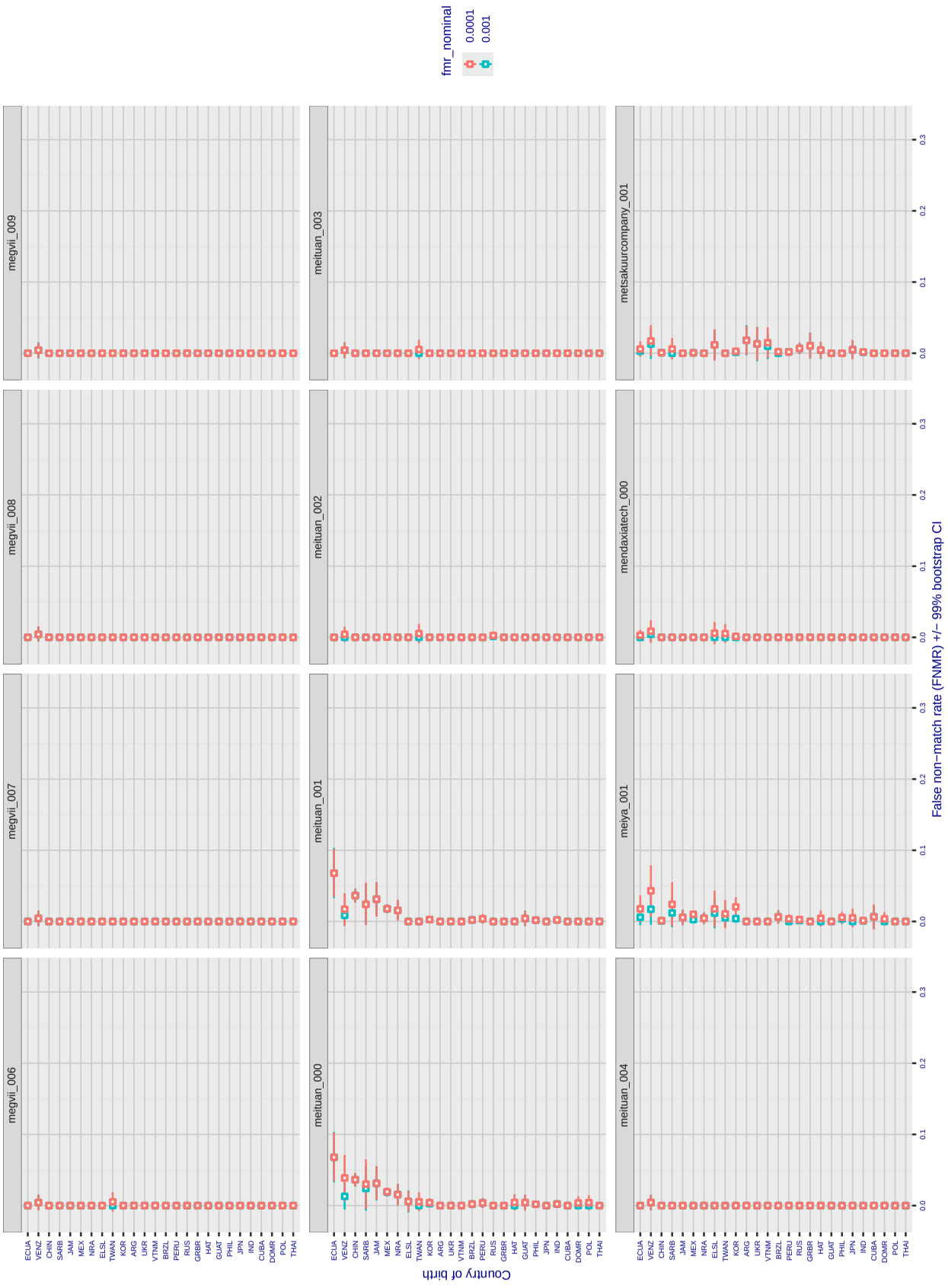


Figure 681: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 682: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 683: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

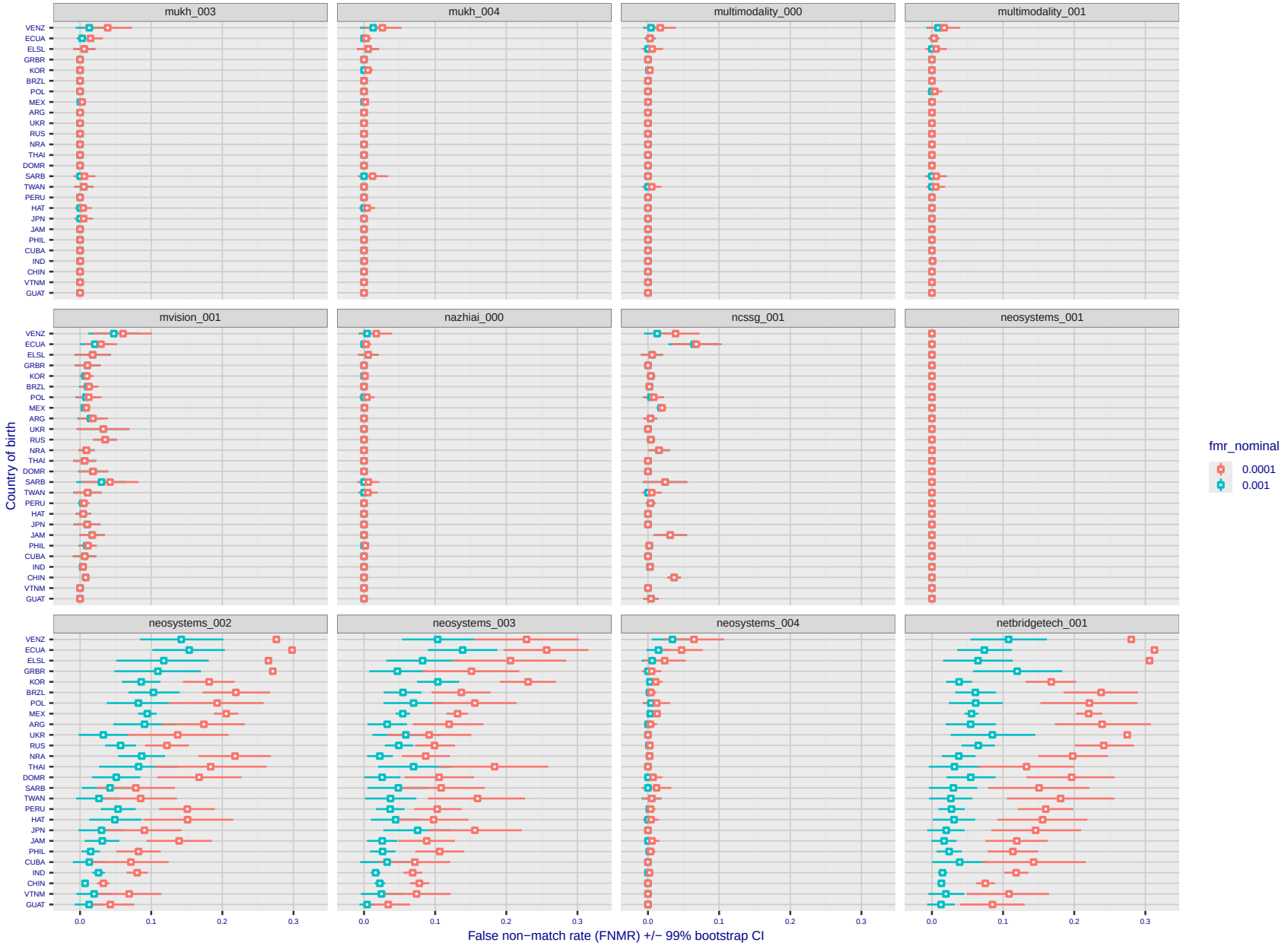


Figure 684: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

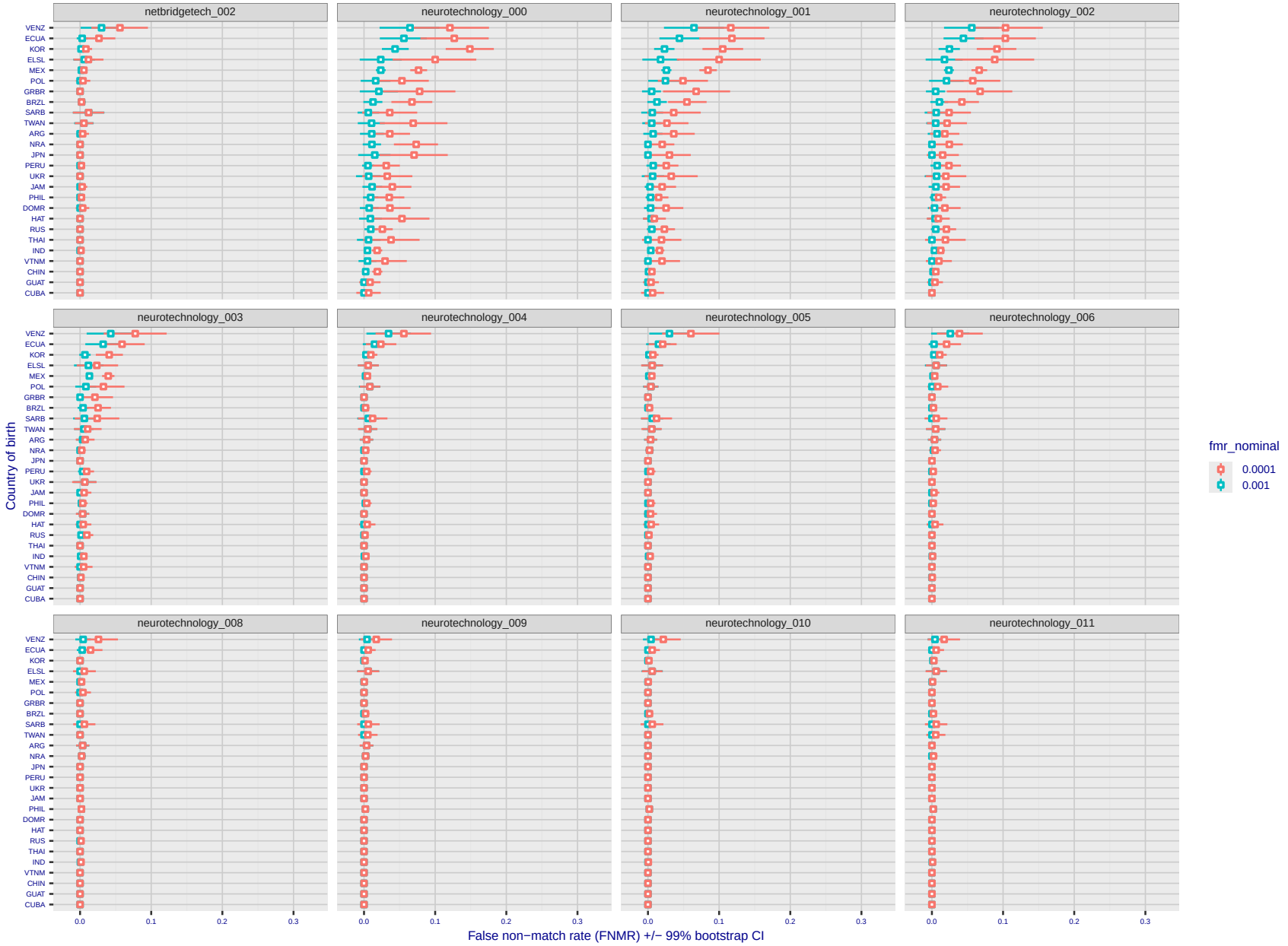


Figure 685: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

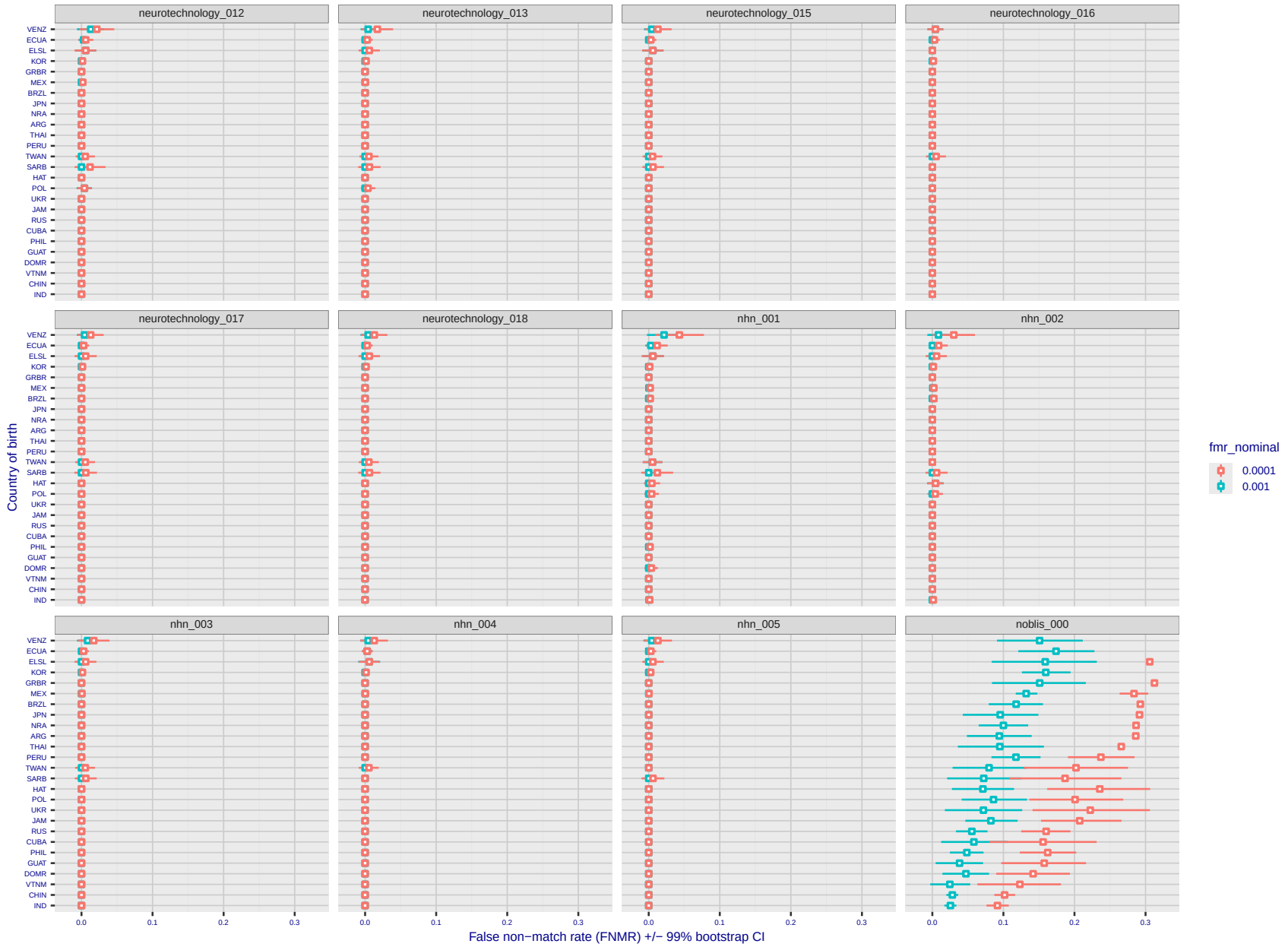


Figure 686: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 687: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

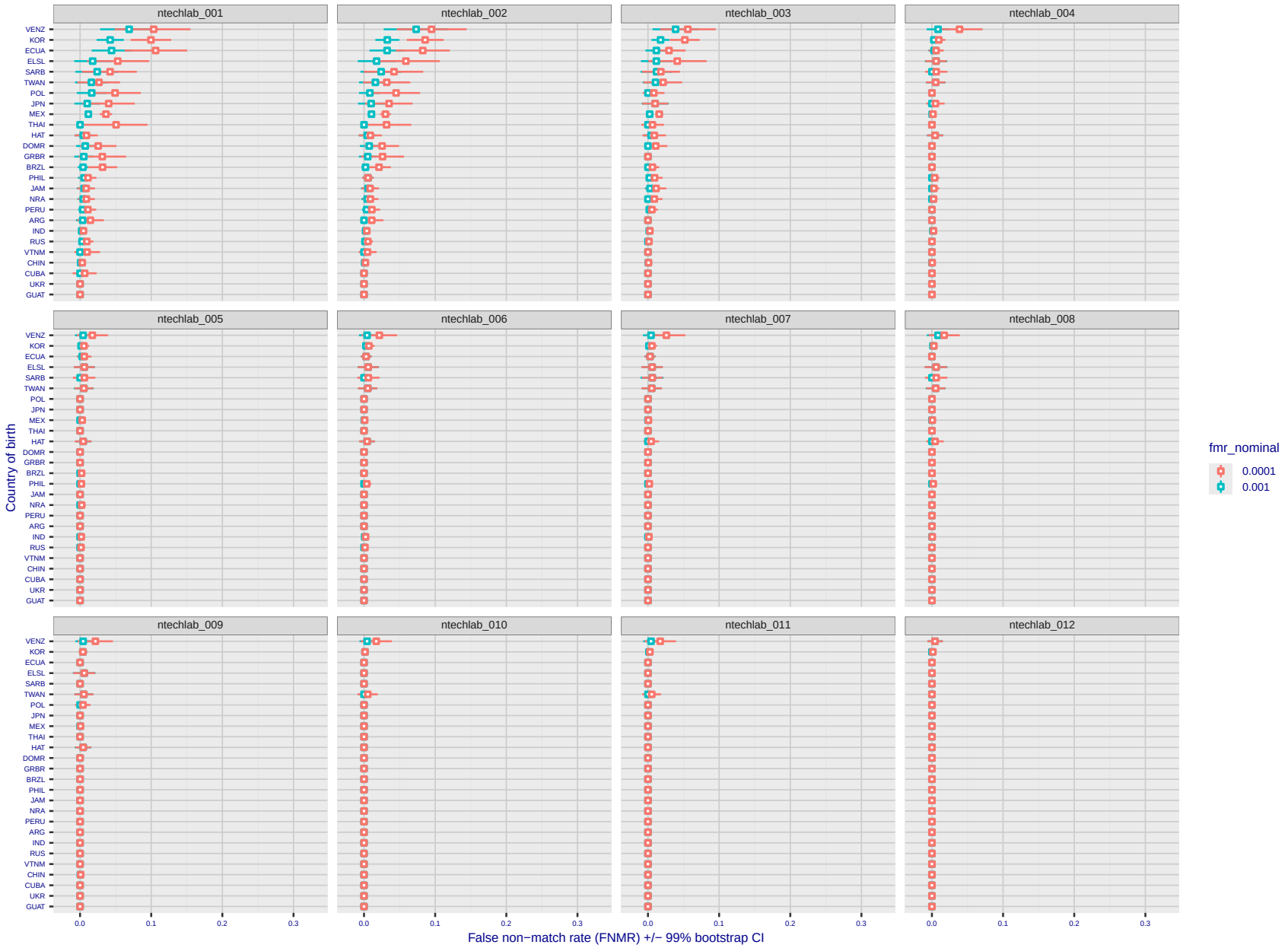


Figure 688: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 689: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 690: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

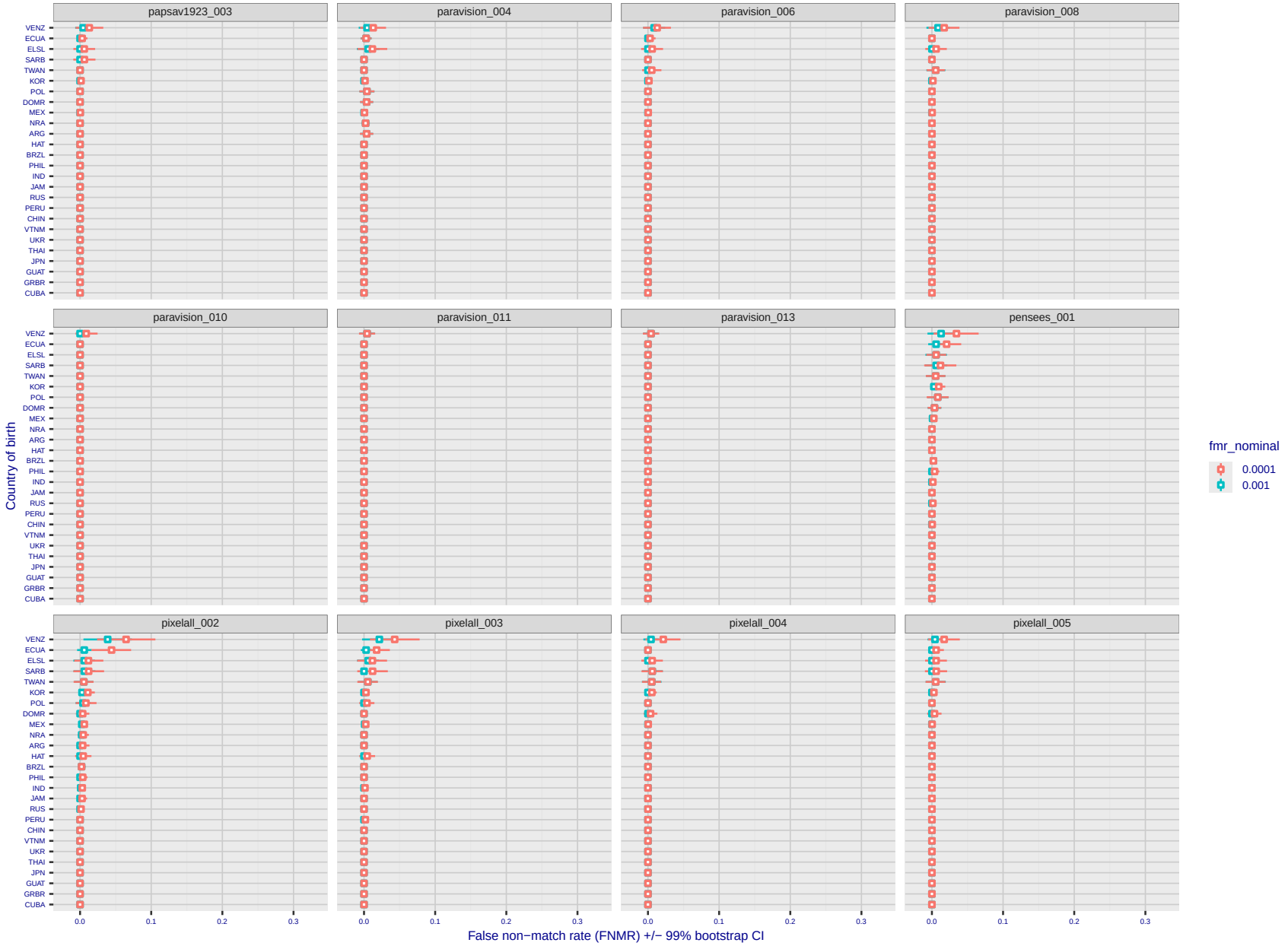


Figure 691: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 692: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

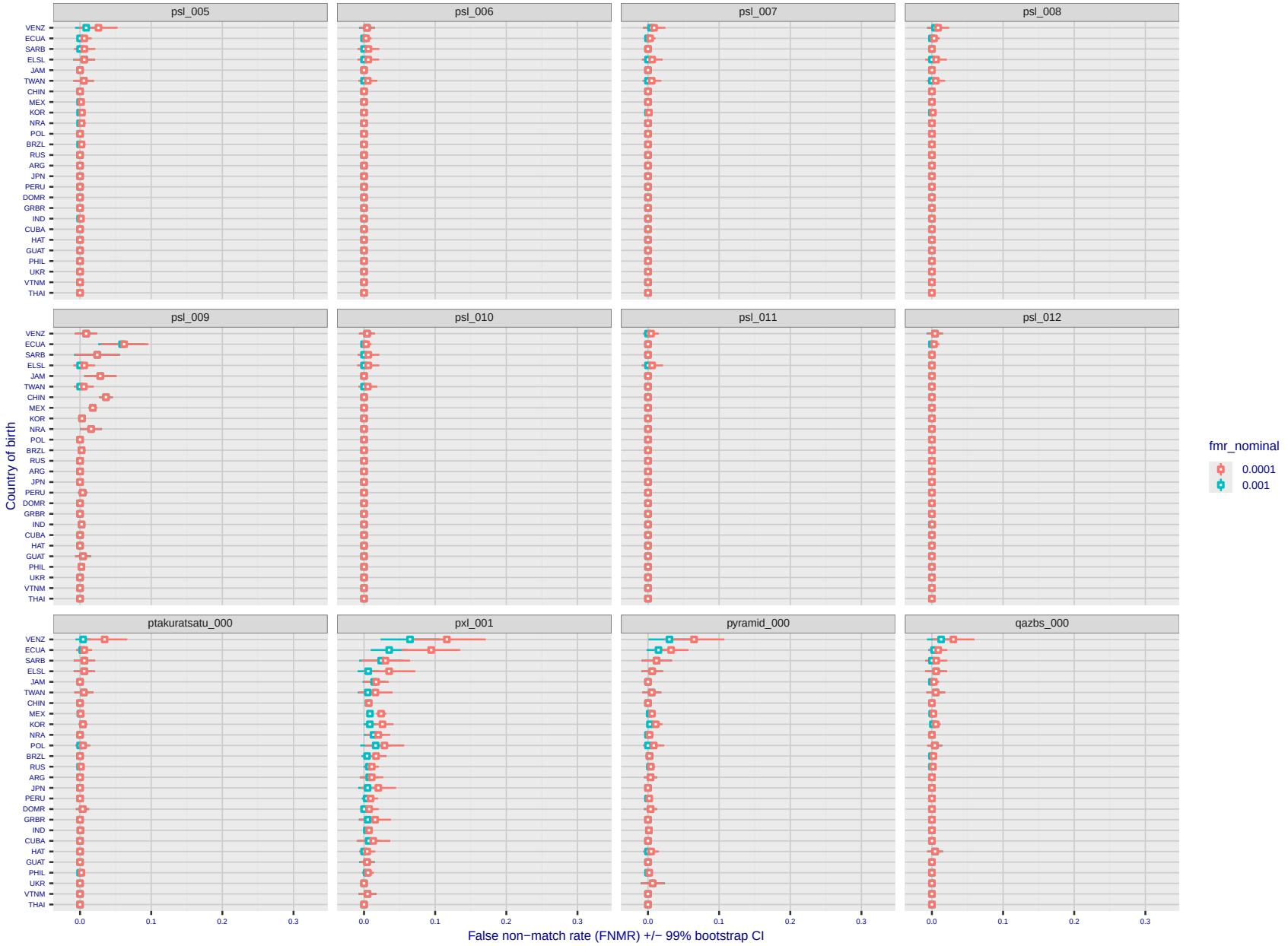


Figure 693: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 694: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

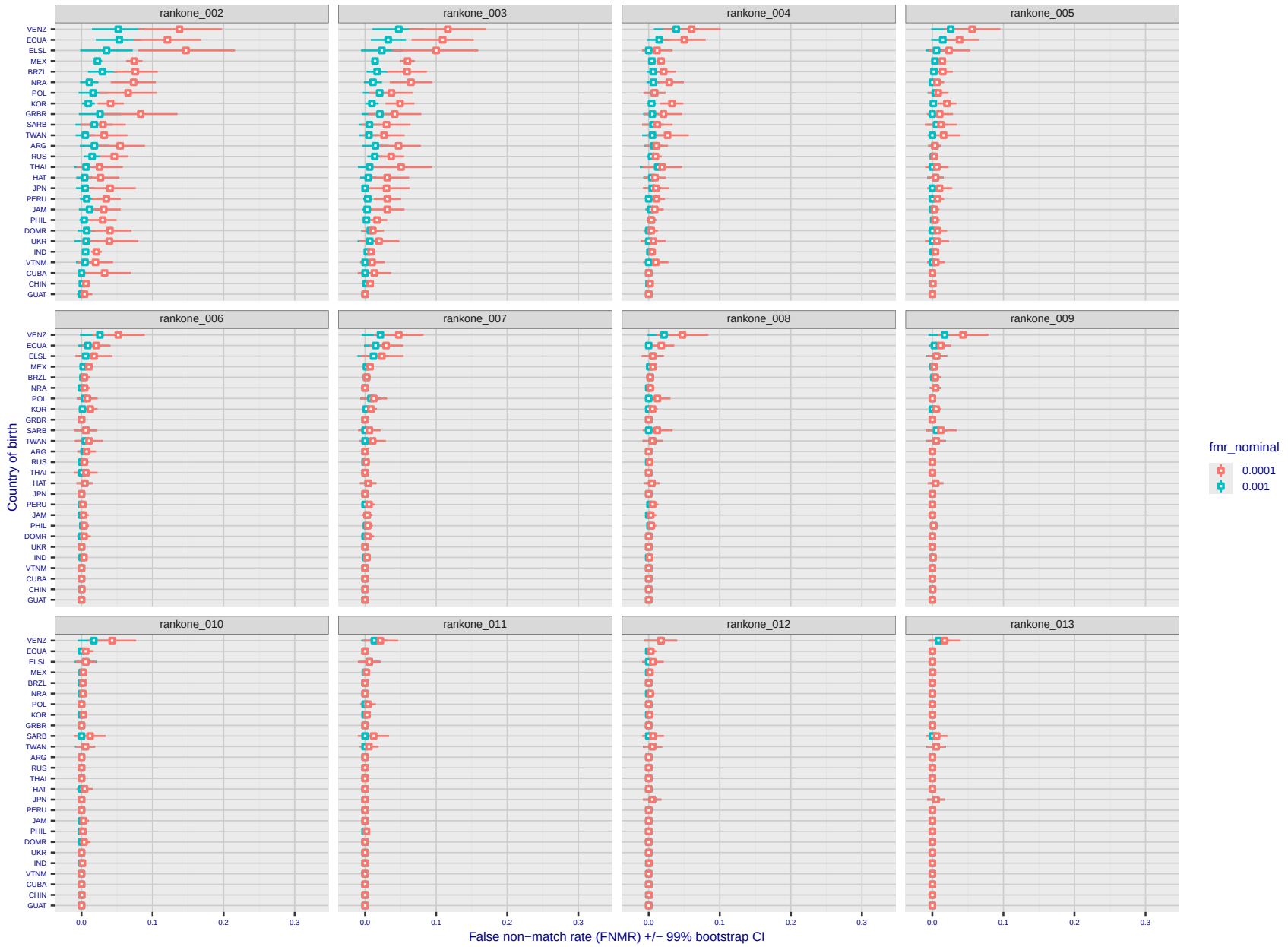


Figure 695: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

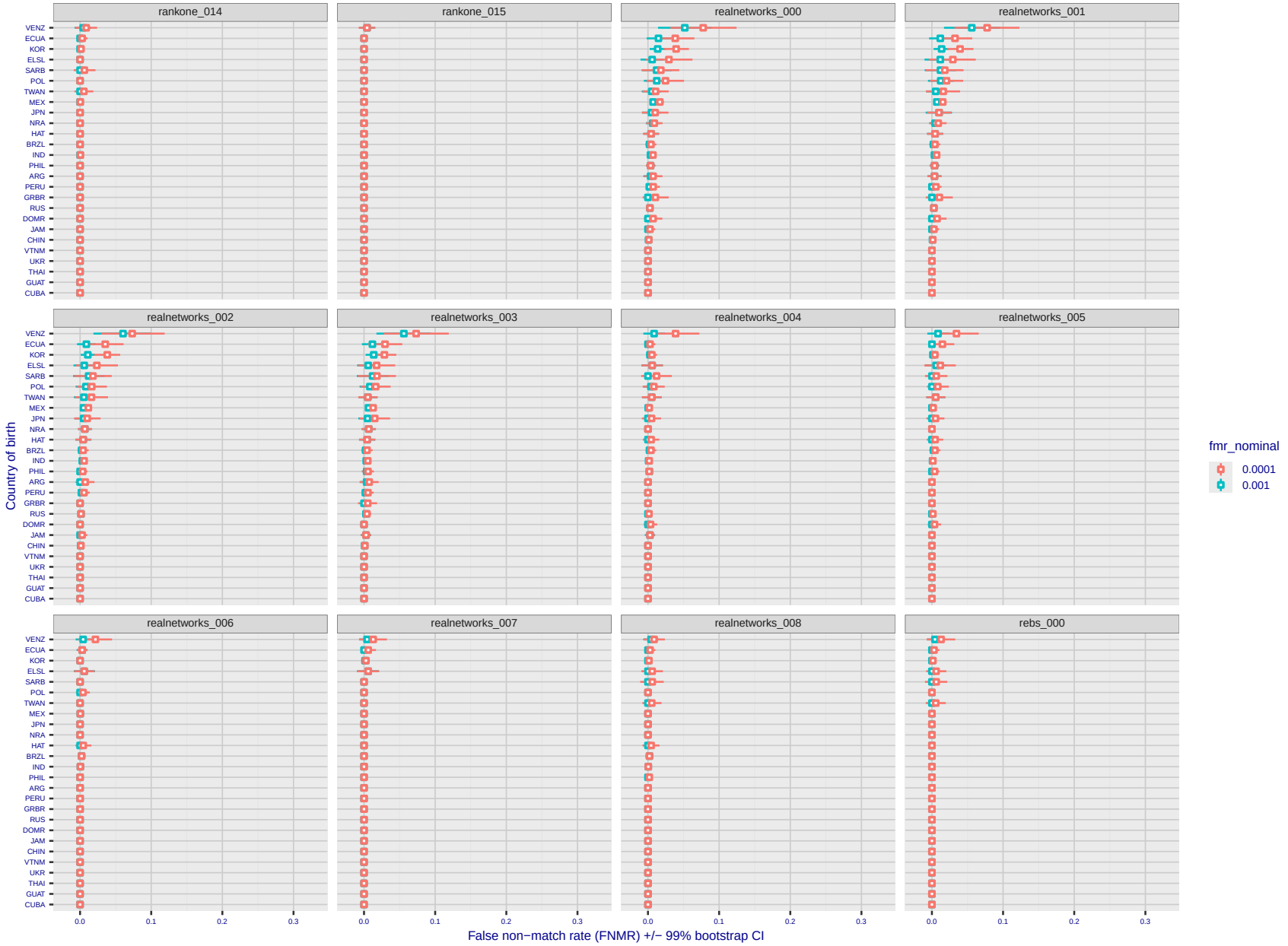


Figure 696: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

2026/11/06 13:03:11

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

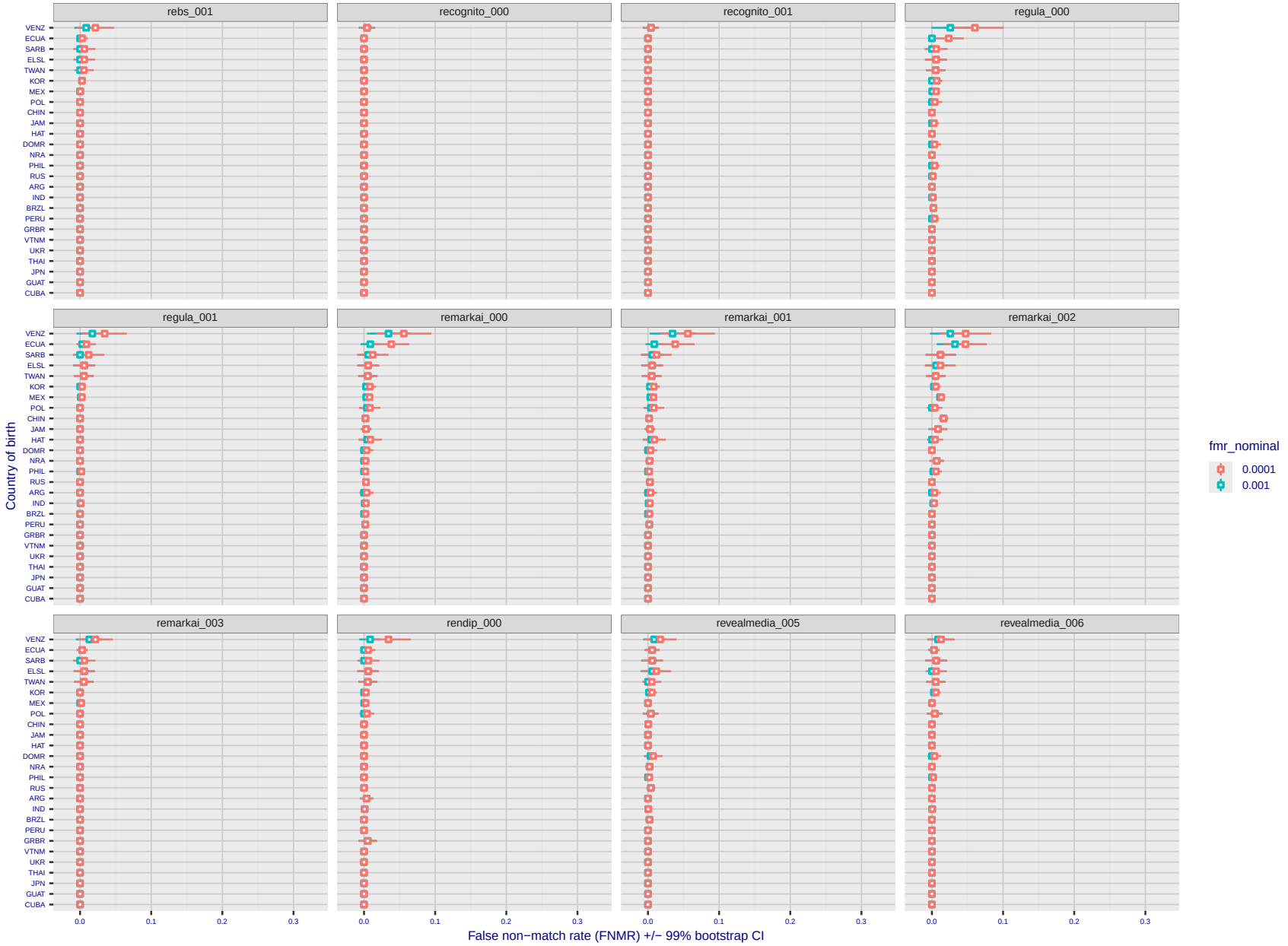


Figure 697: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

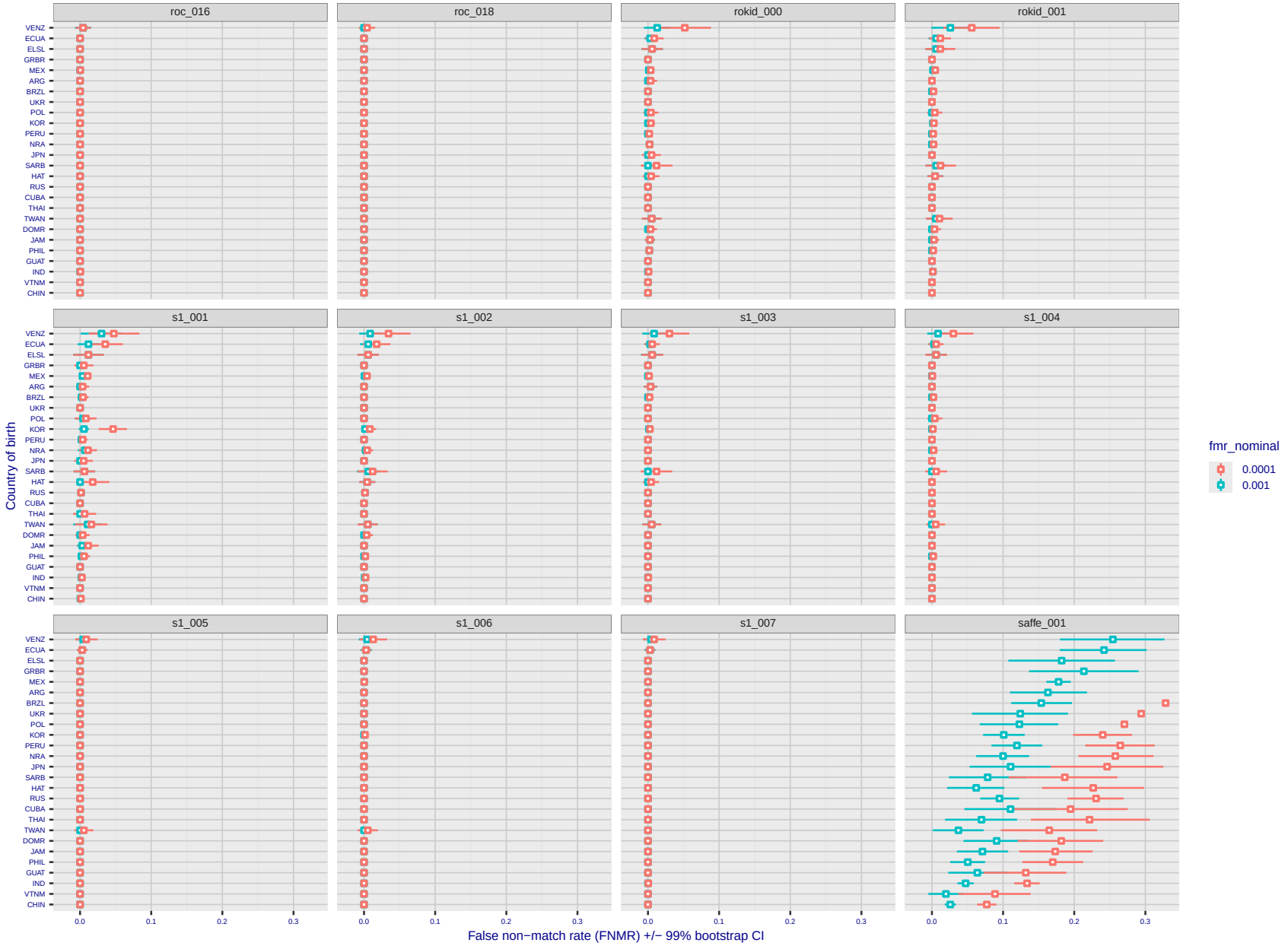


Figure 698: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

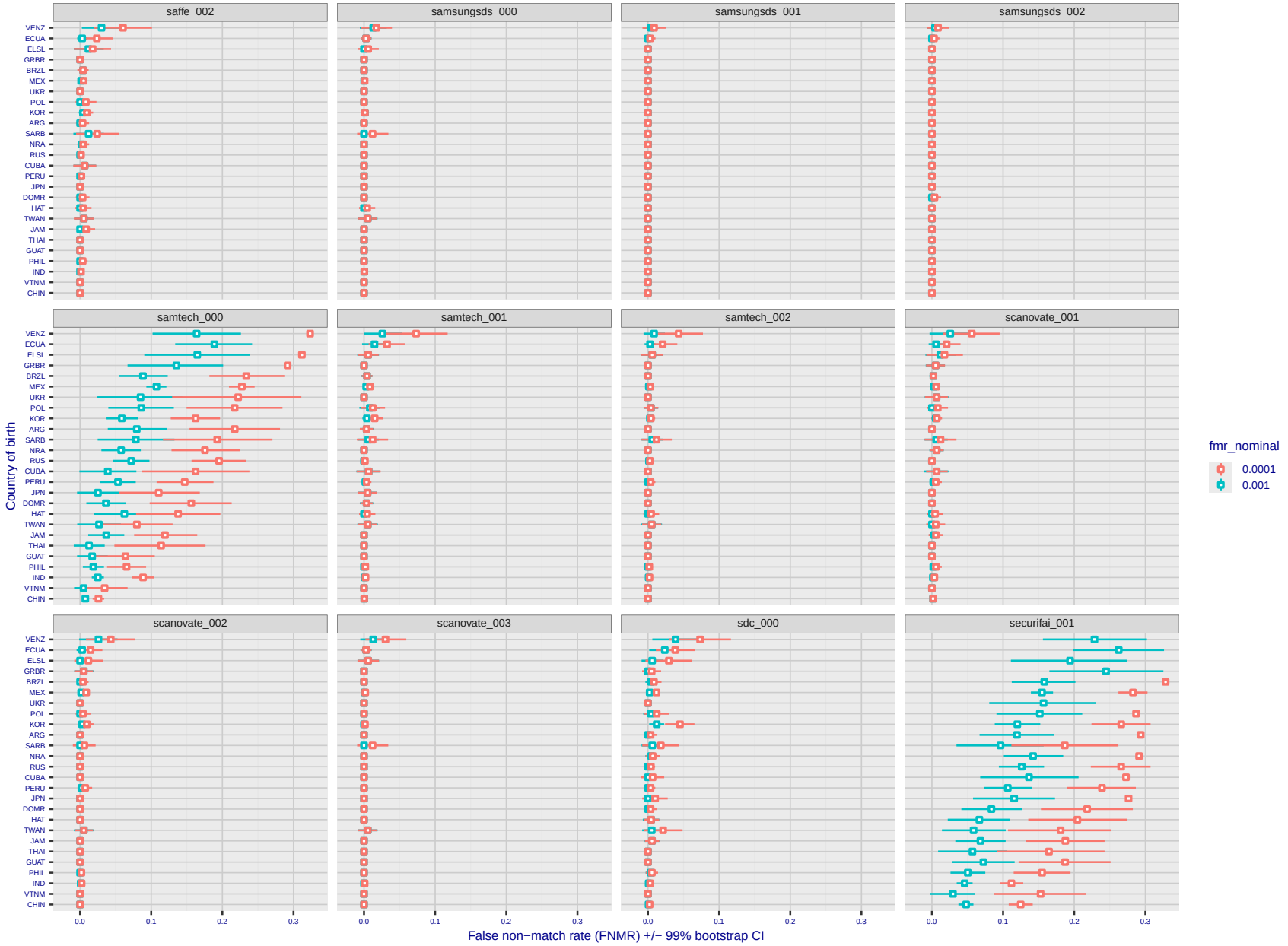


Figure 699: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 700: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

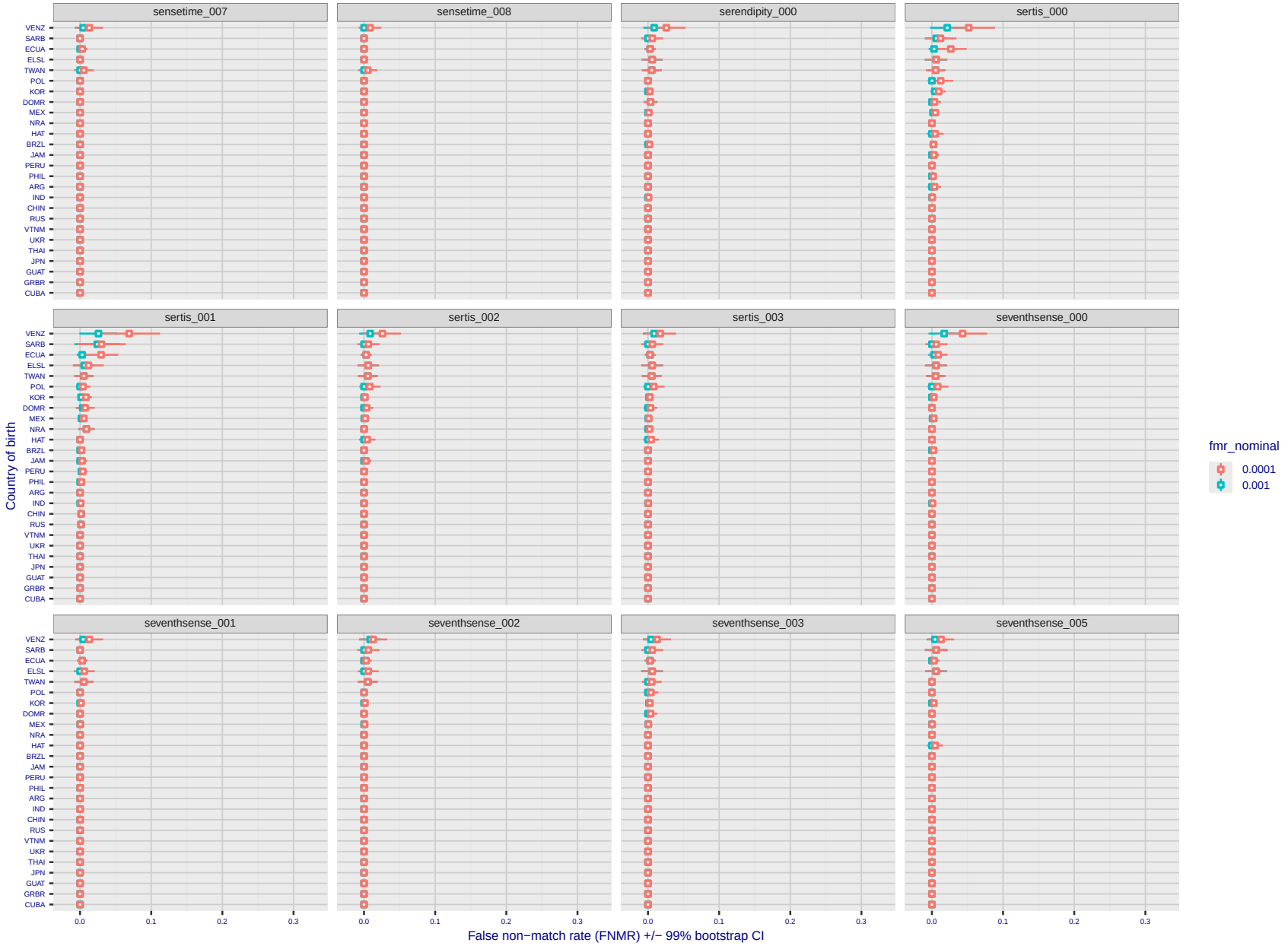


Figure 701: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 702: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 703: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

2026/11/06 13:03:11 FNMR(T) "False non-match rate" FMR(T) "False match rate"

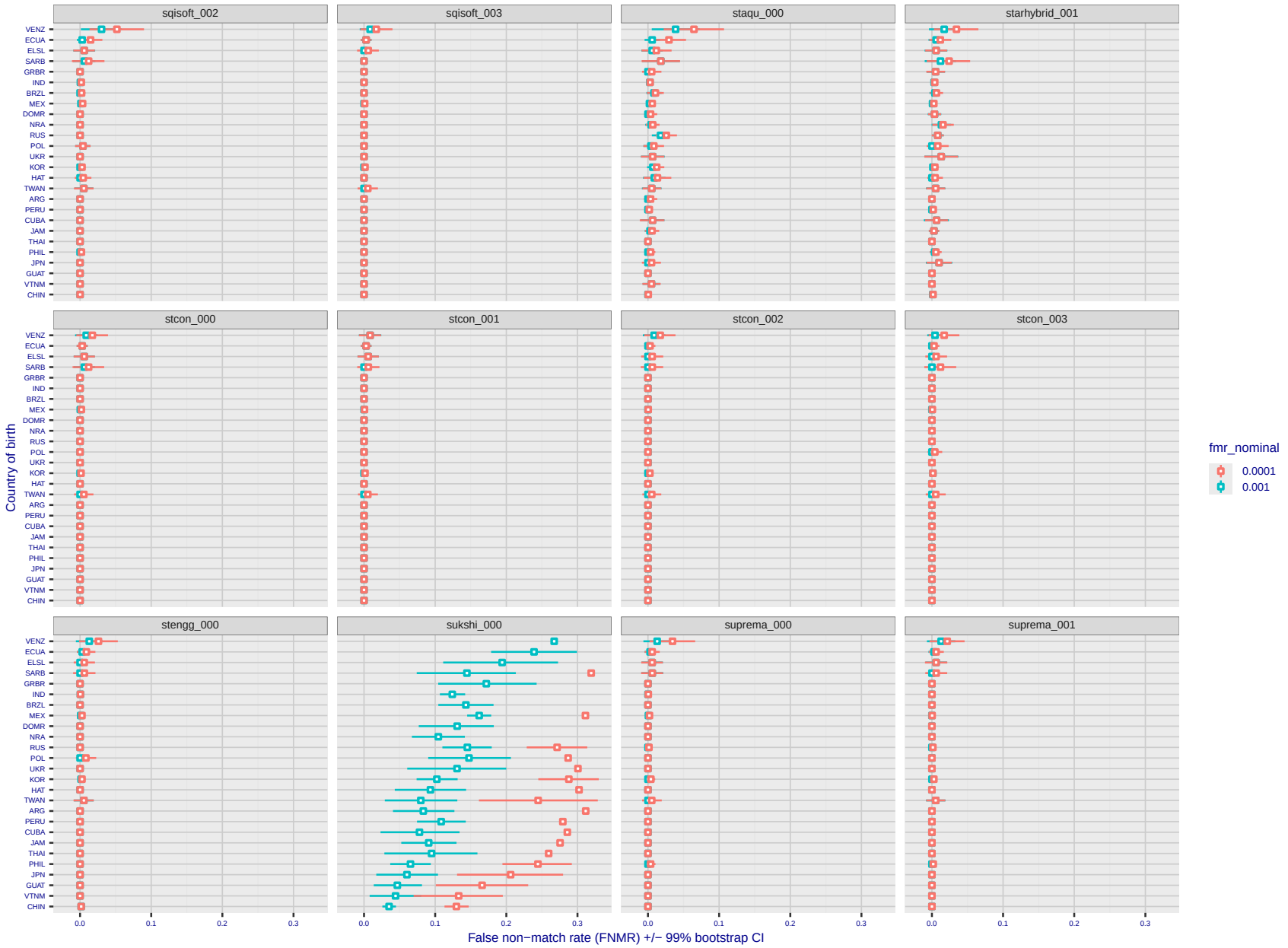


Figure 704: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

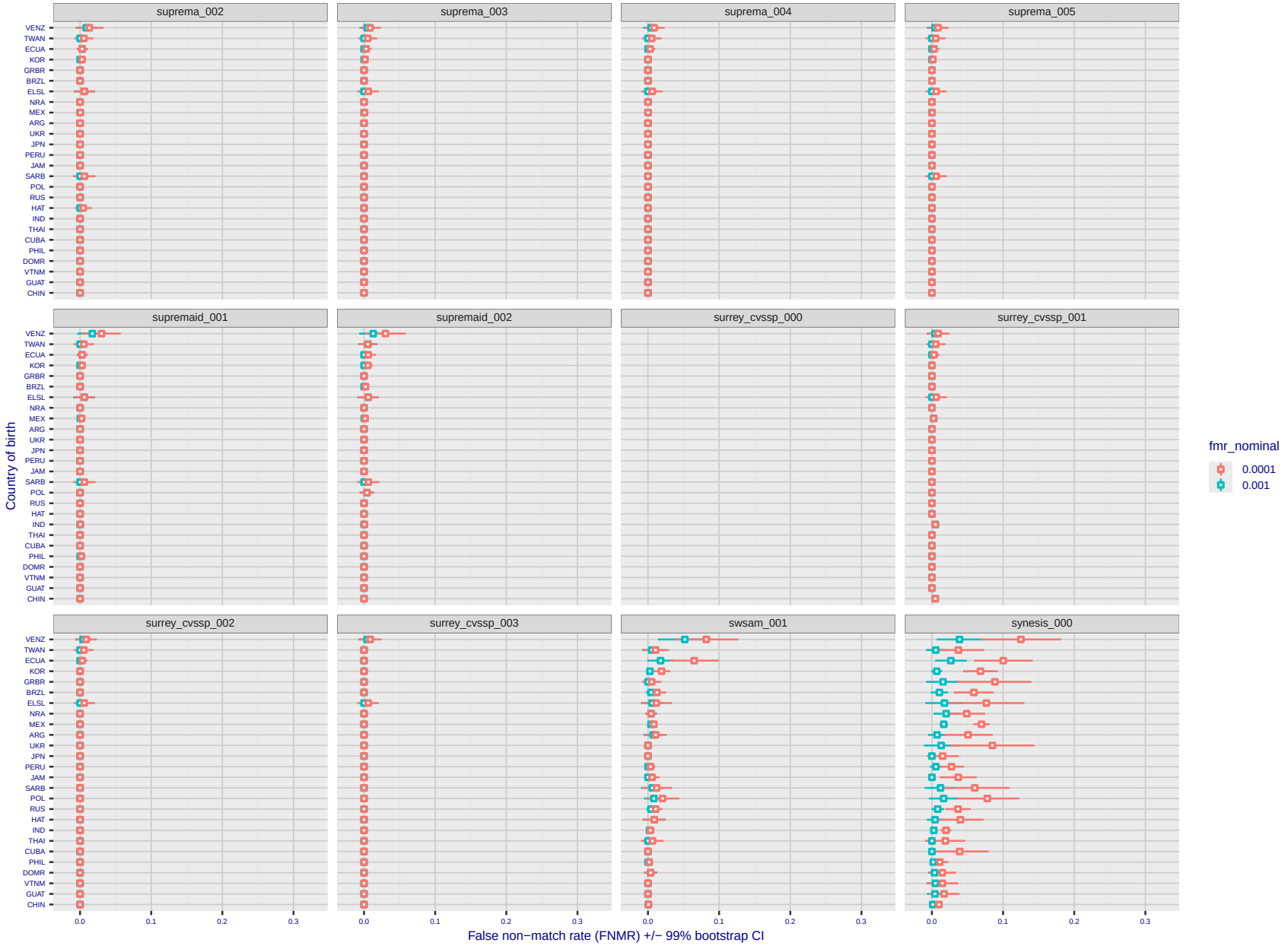


Figure 705: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 706: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

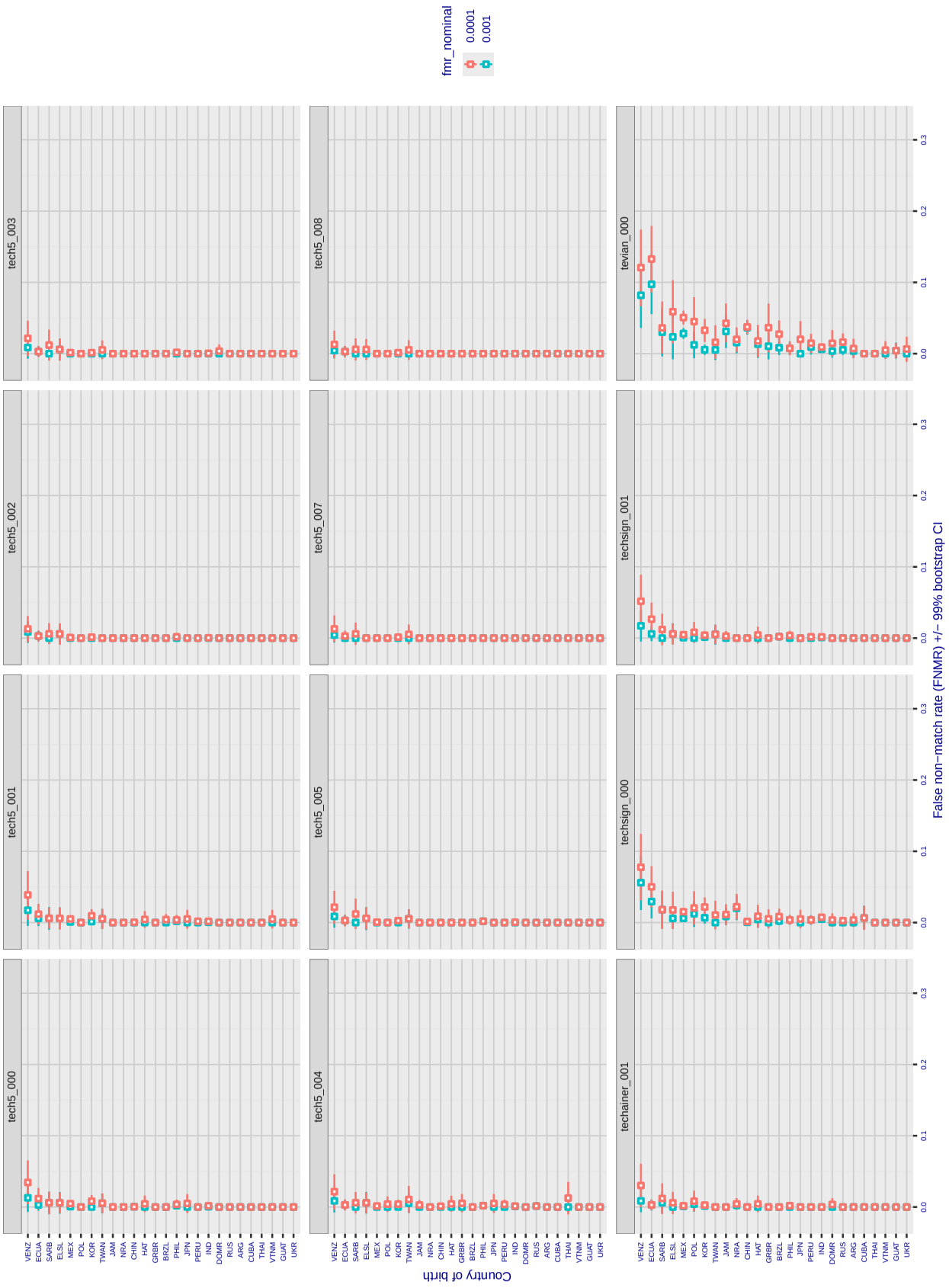


Figure 707: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = {0.001, 0.0001} computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

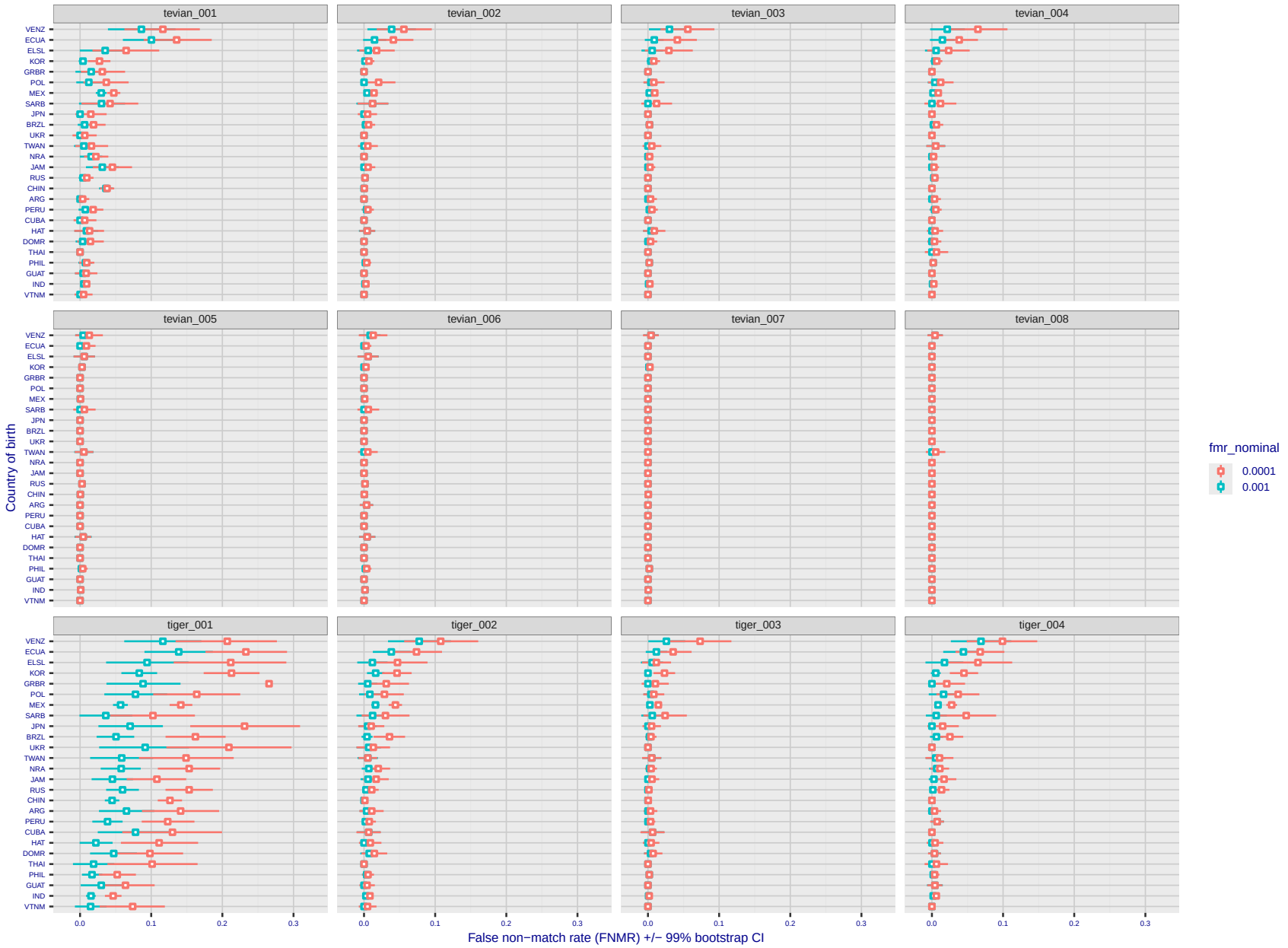


Figure 708: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 709: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

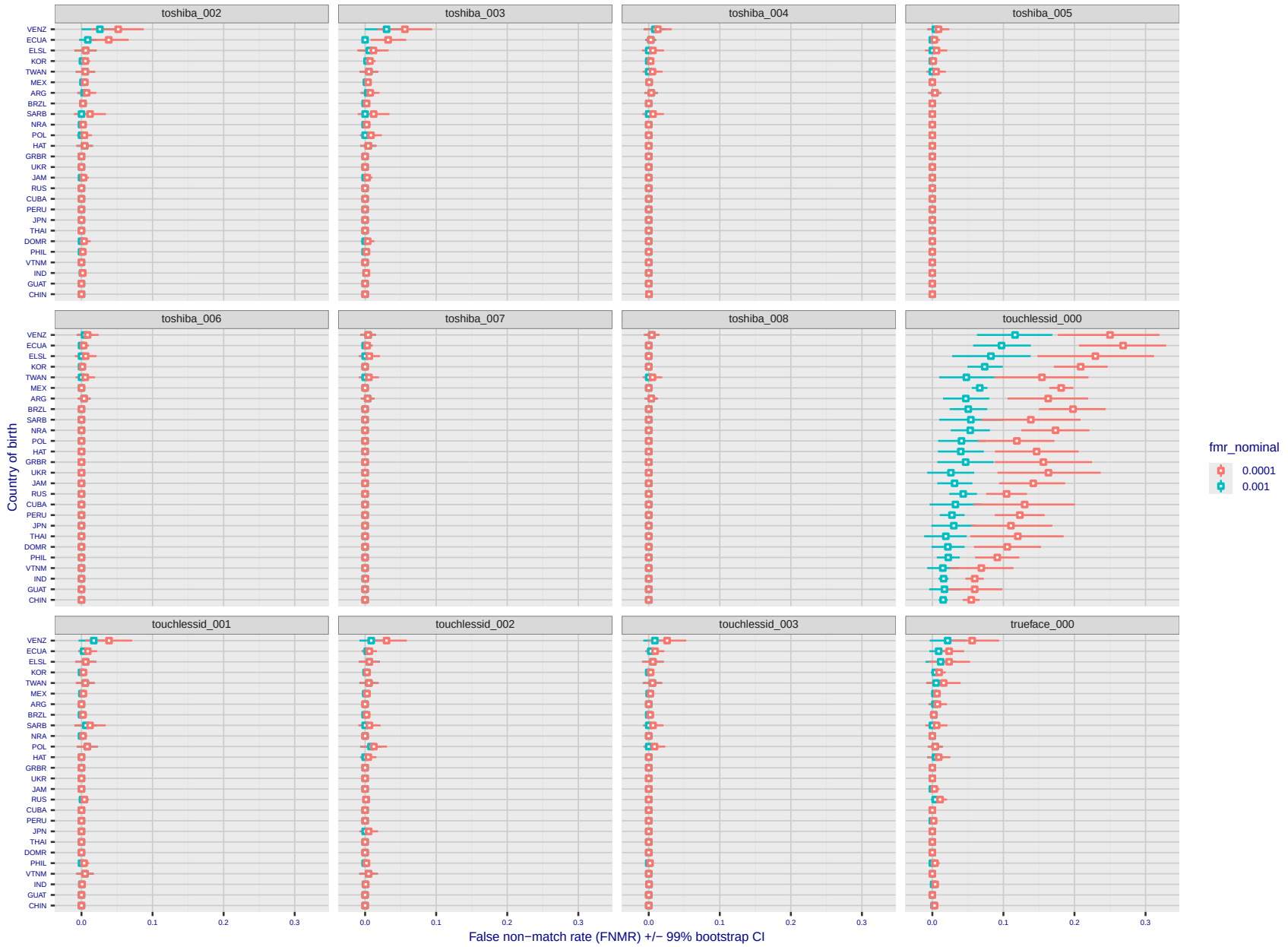


Figure 710: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 711: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

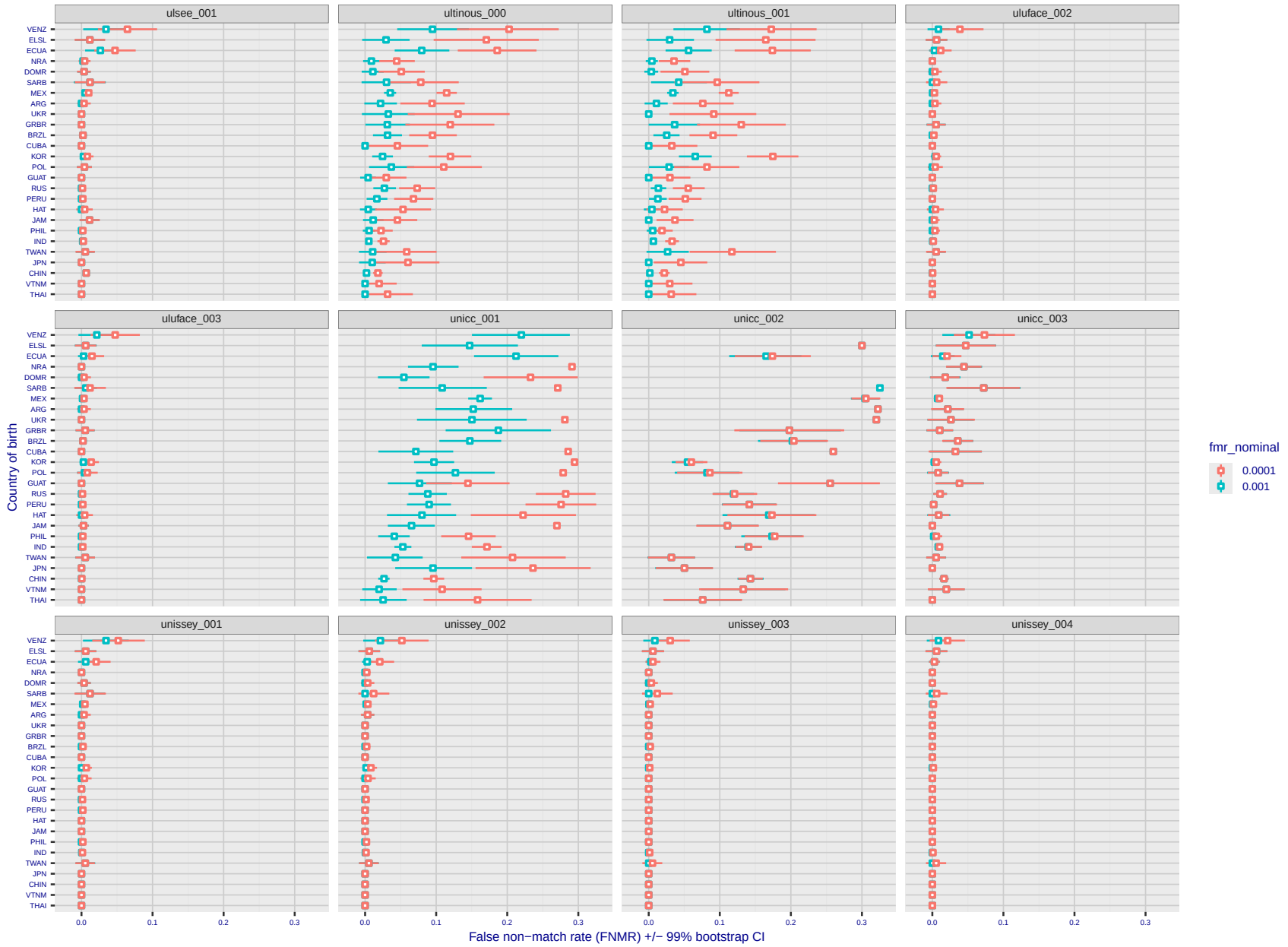


Figure 712: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

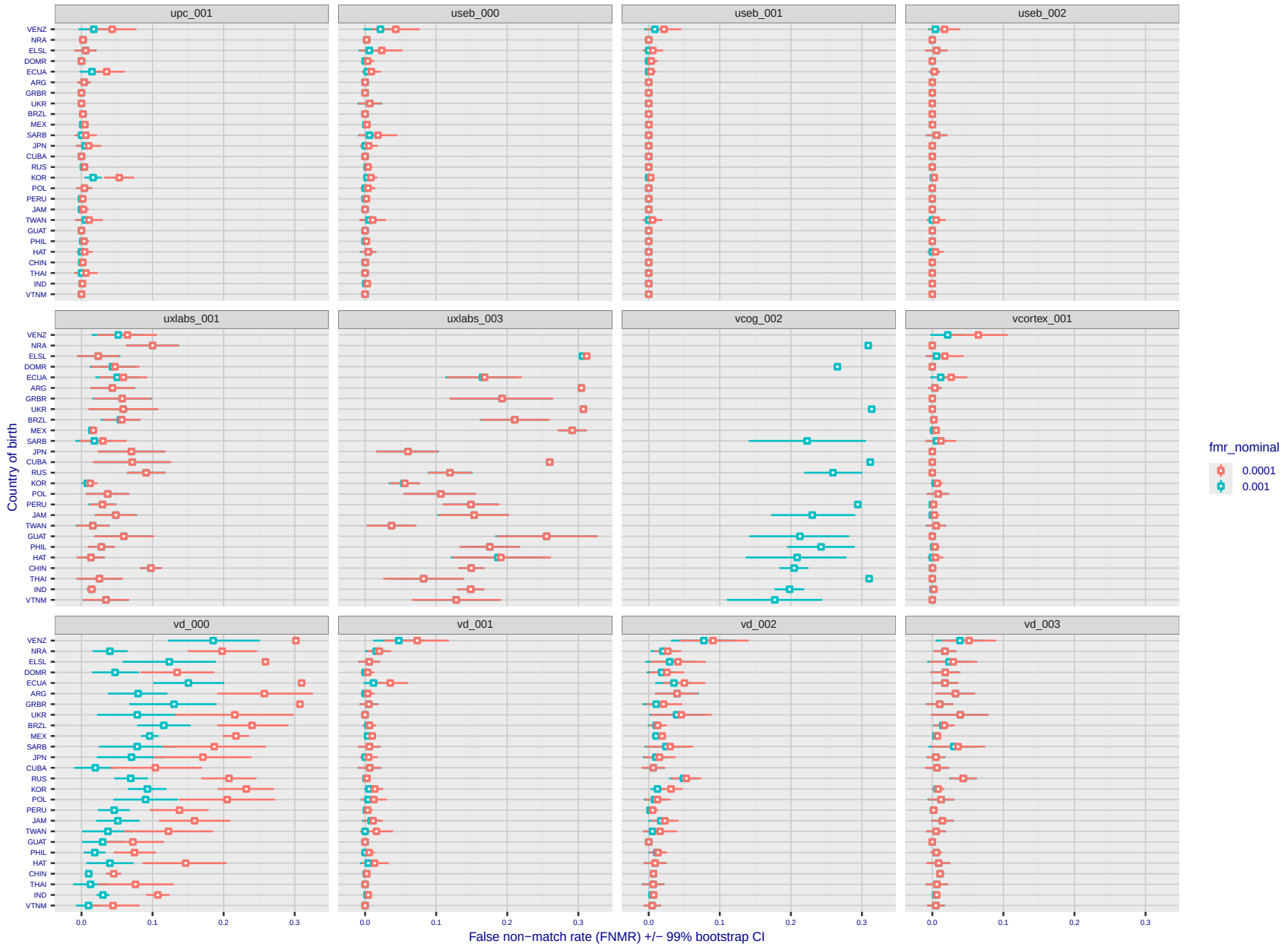


Figure 713: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

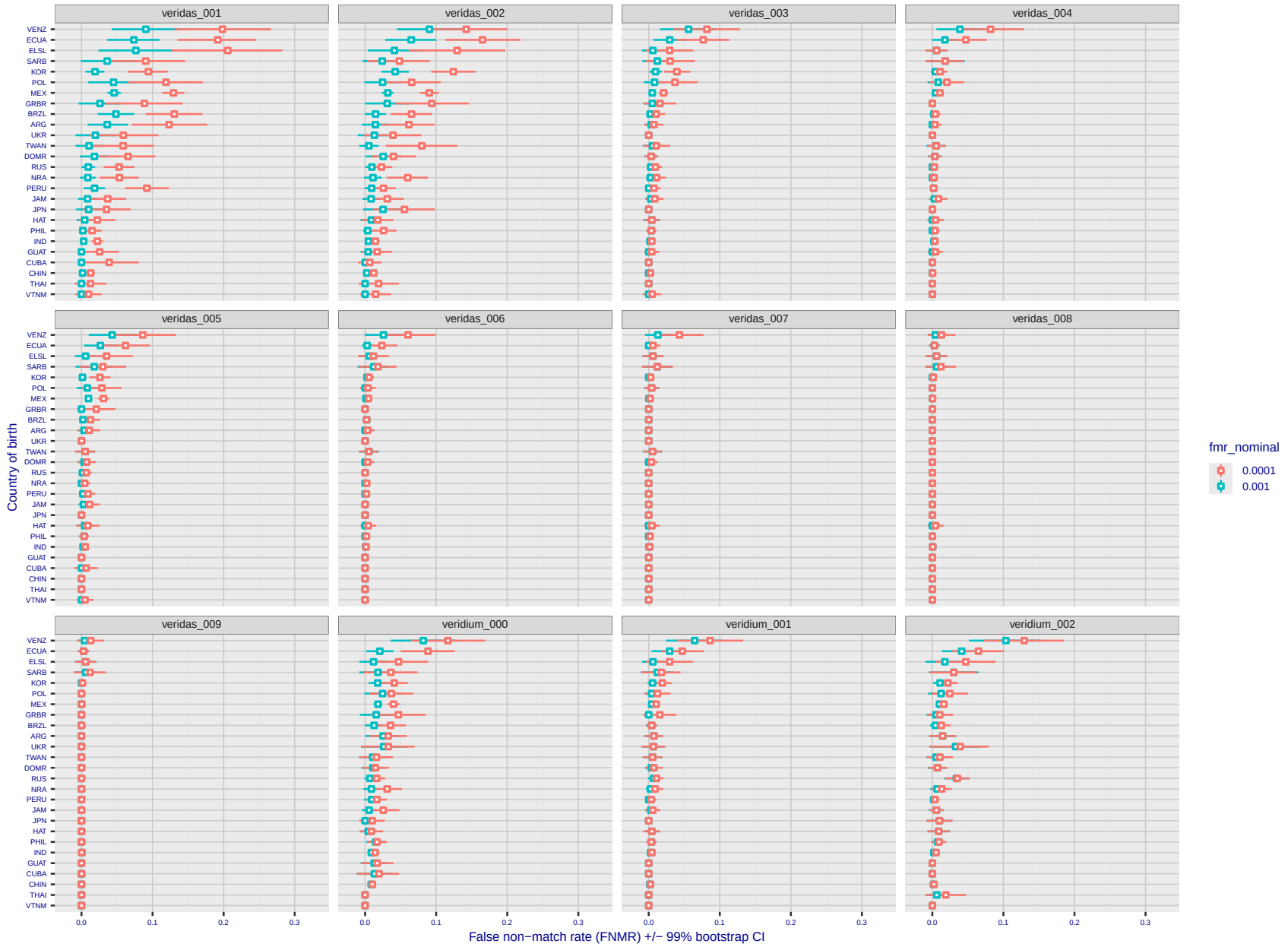


Figure 714: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

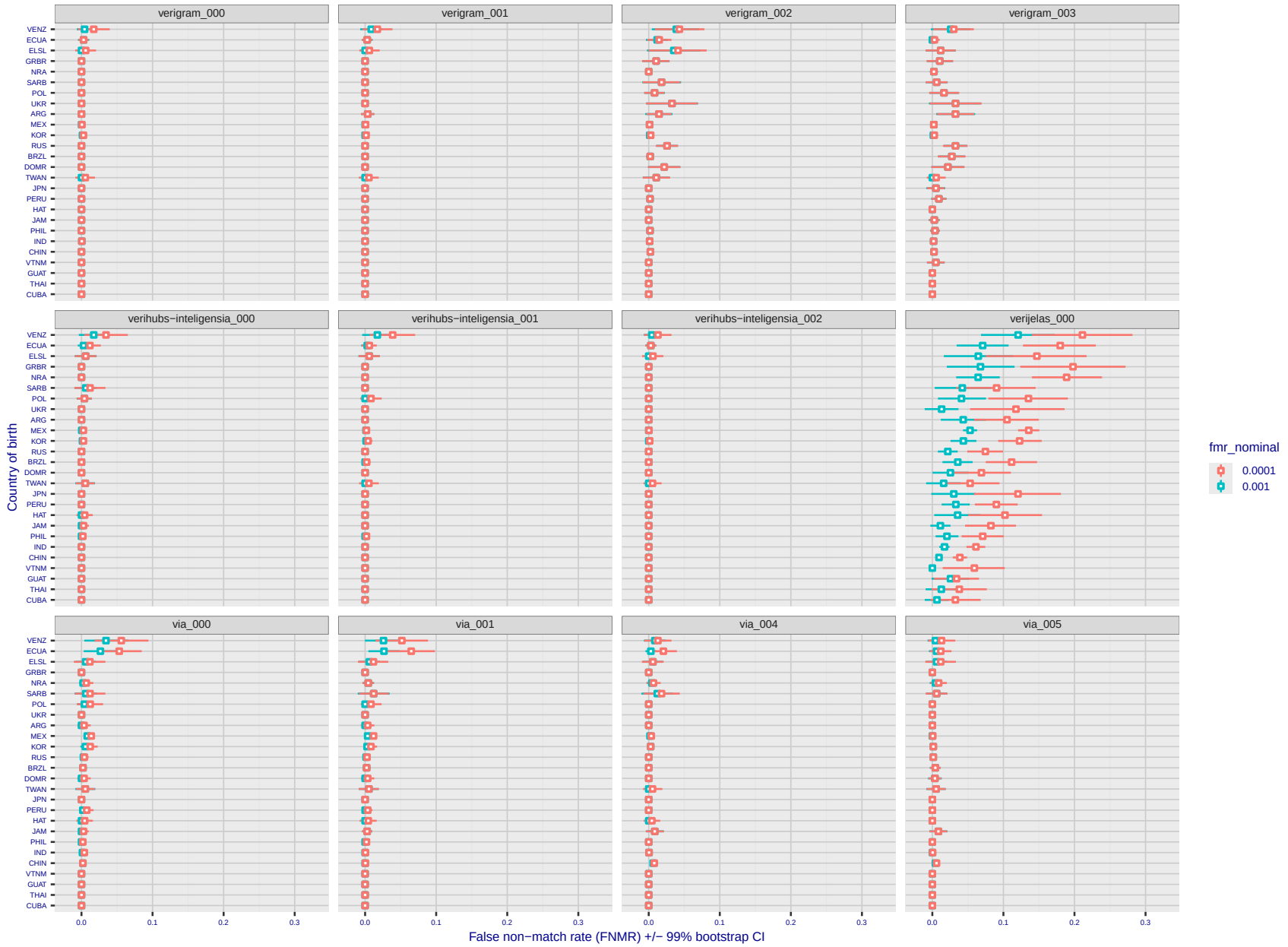


Figure 715: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

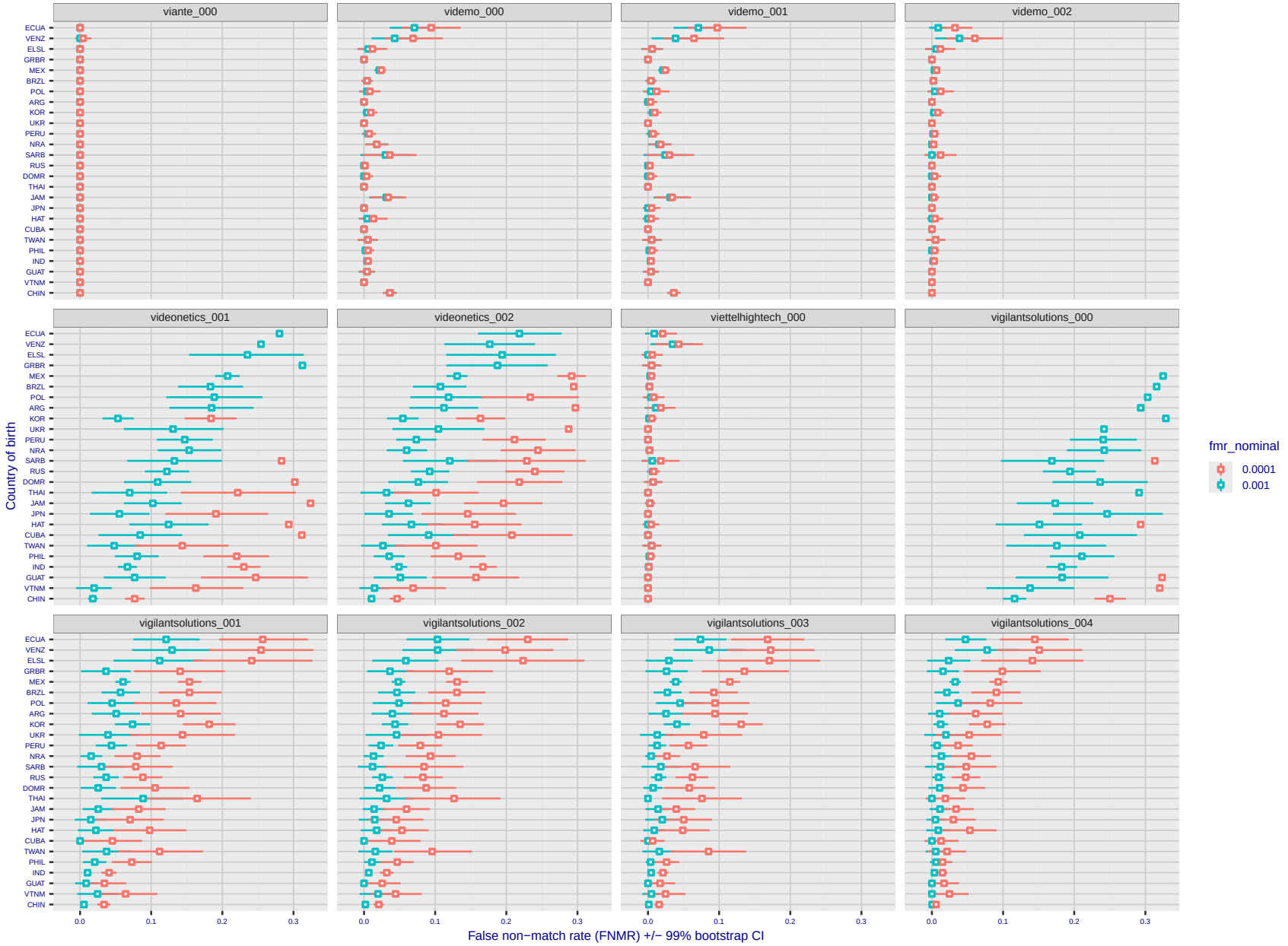


Figure 716: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

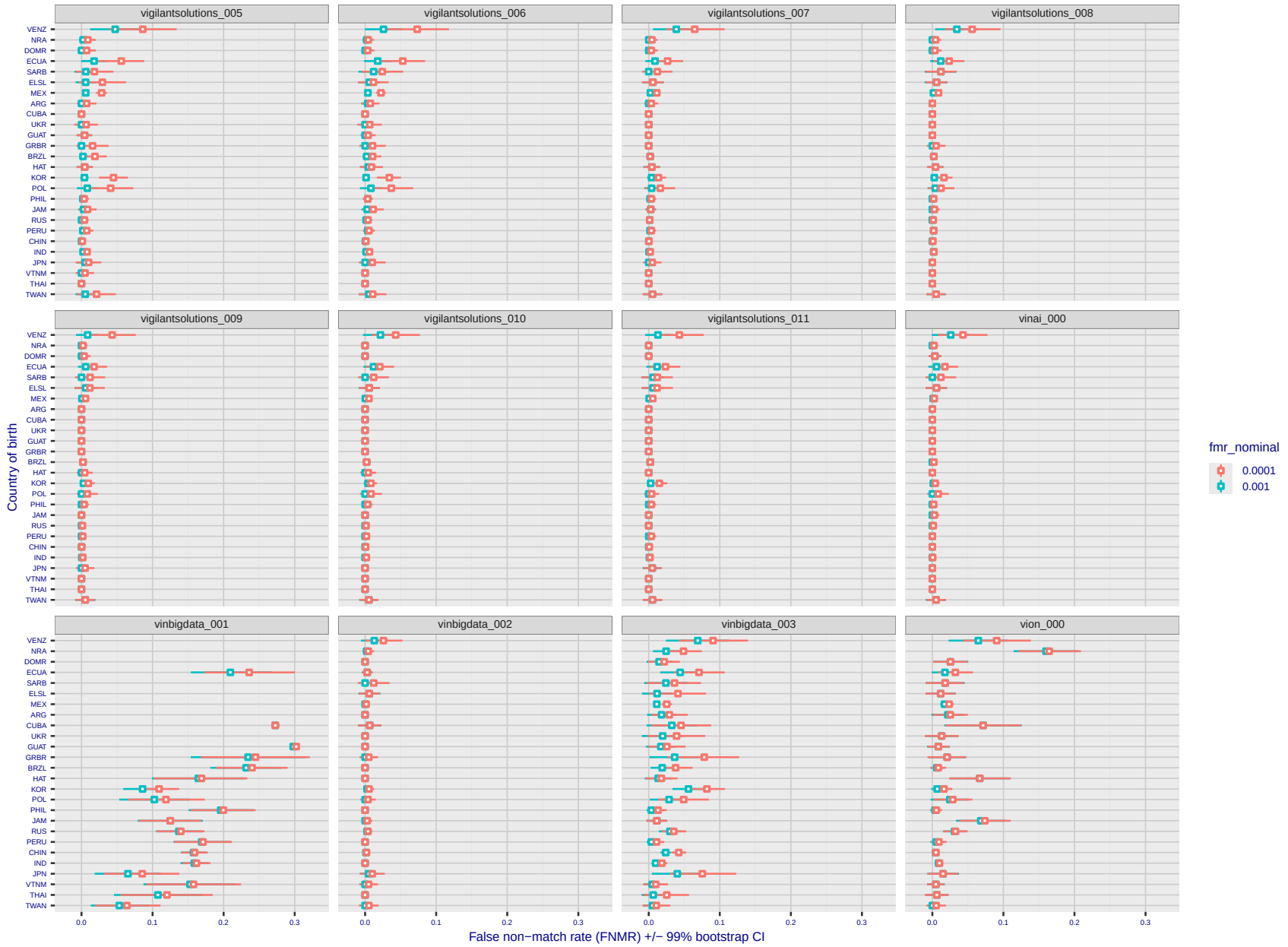


Figure 717: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

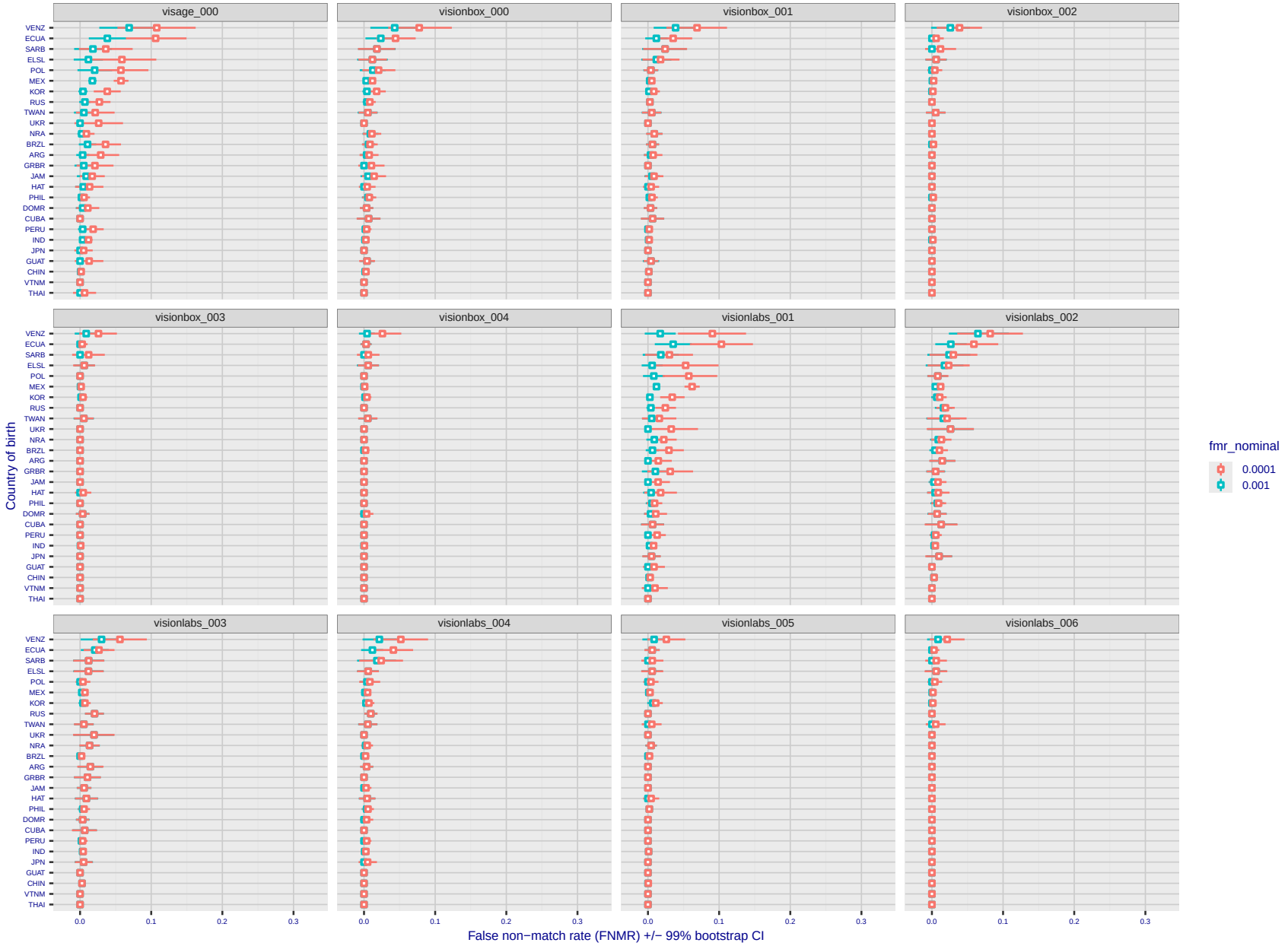


Figure 718: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

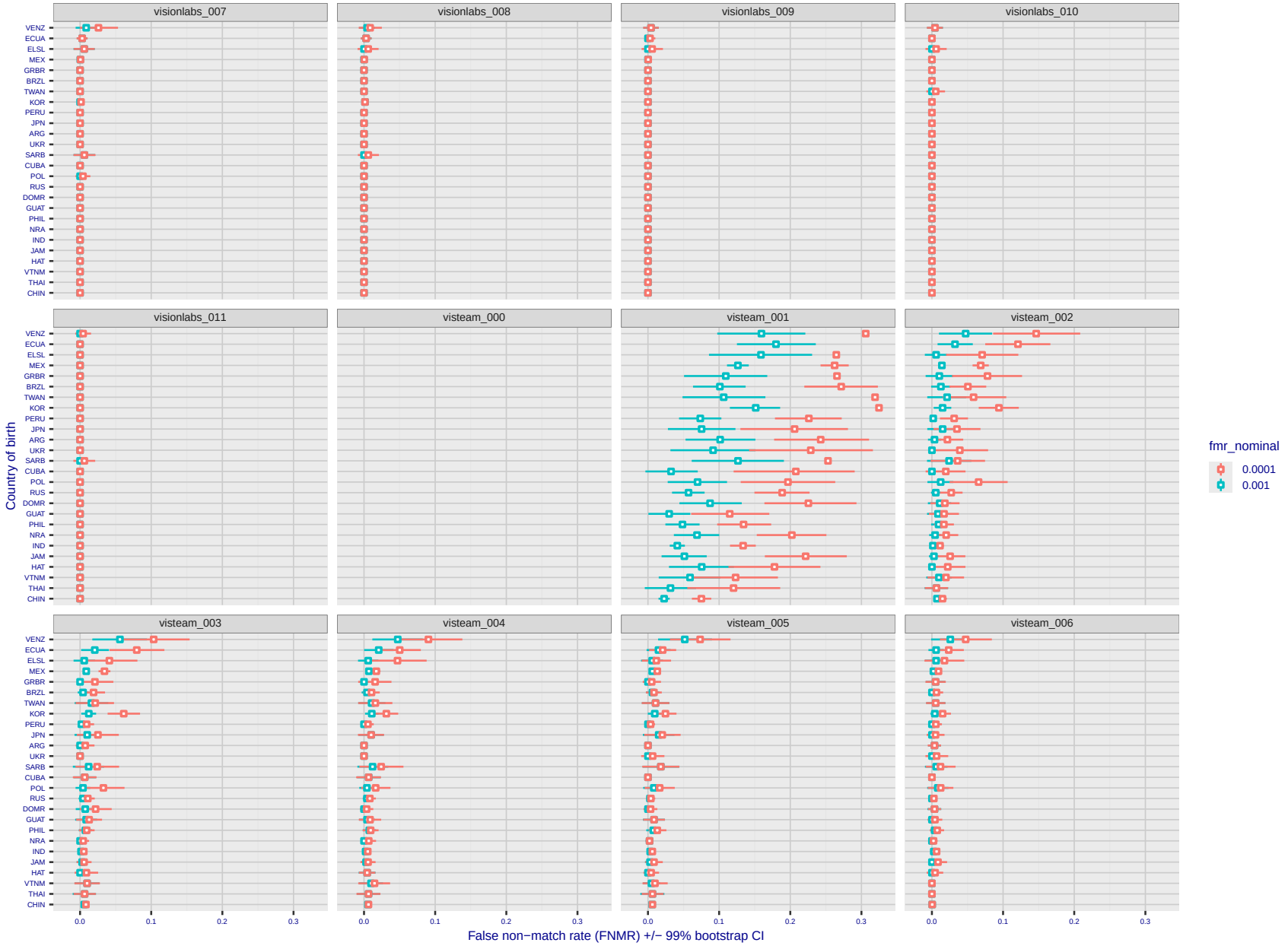


Figure 719: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 720: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

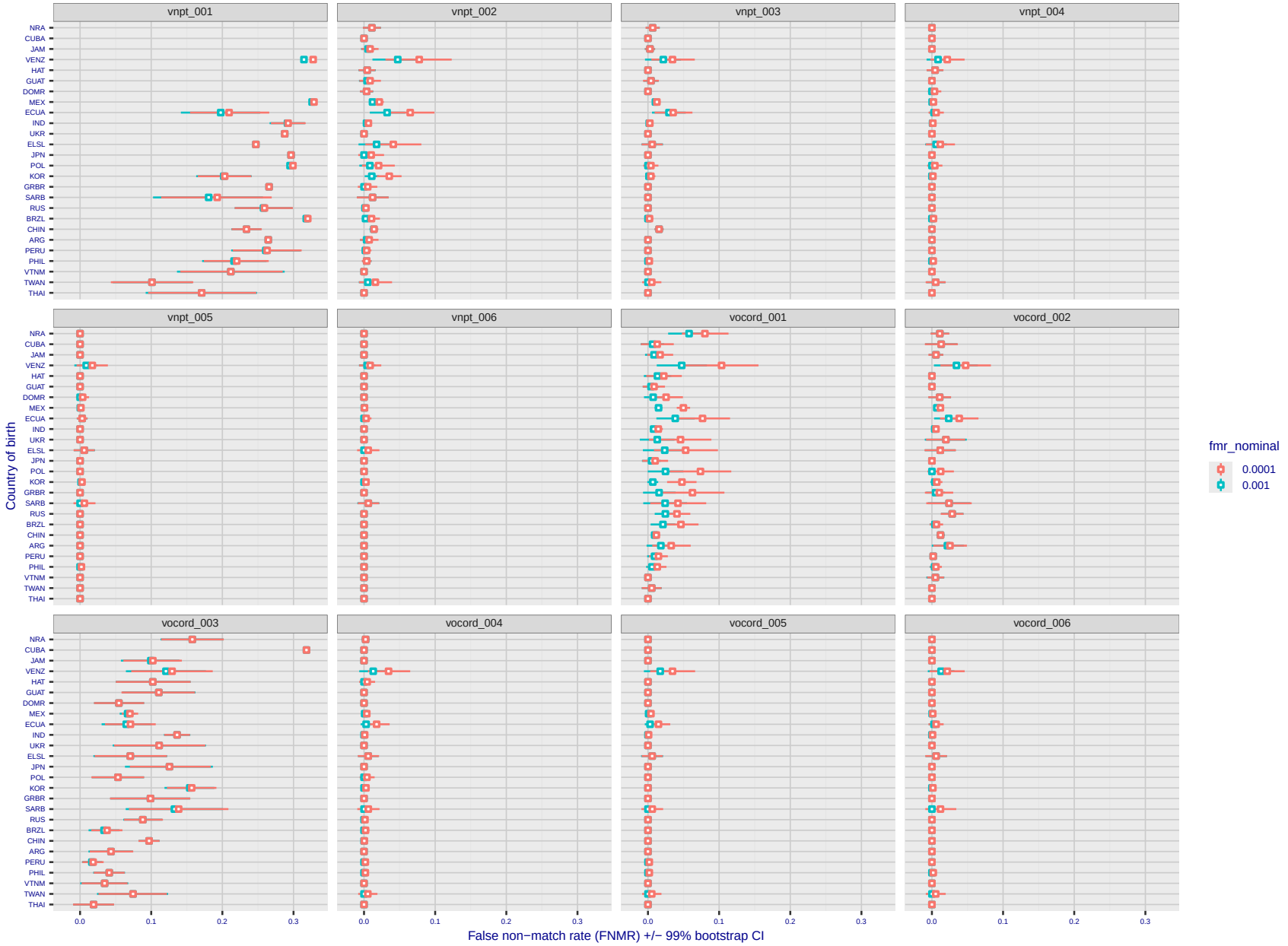


Figure 721: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

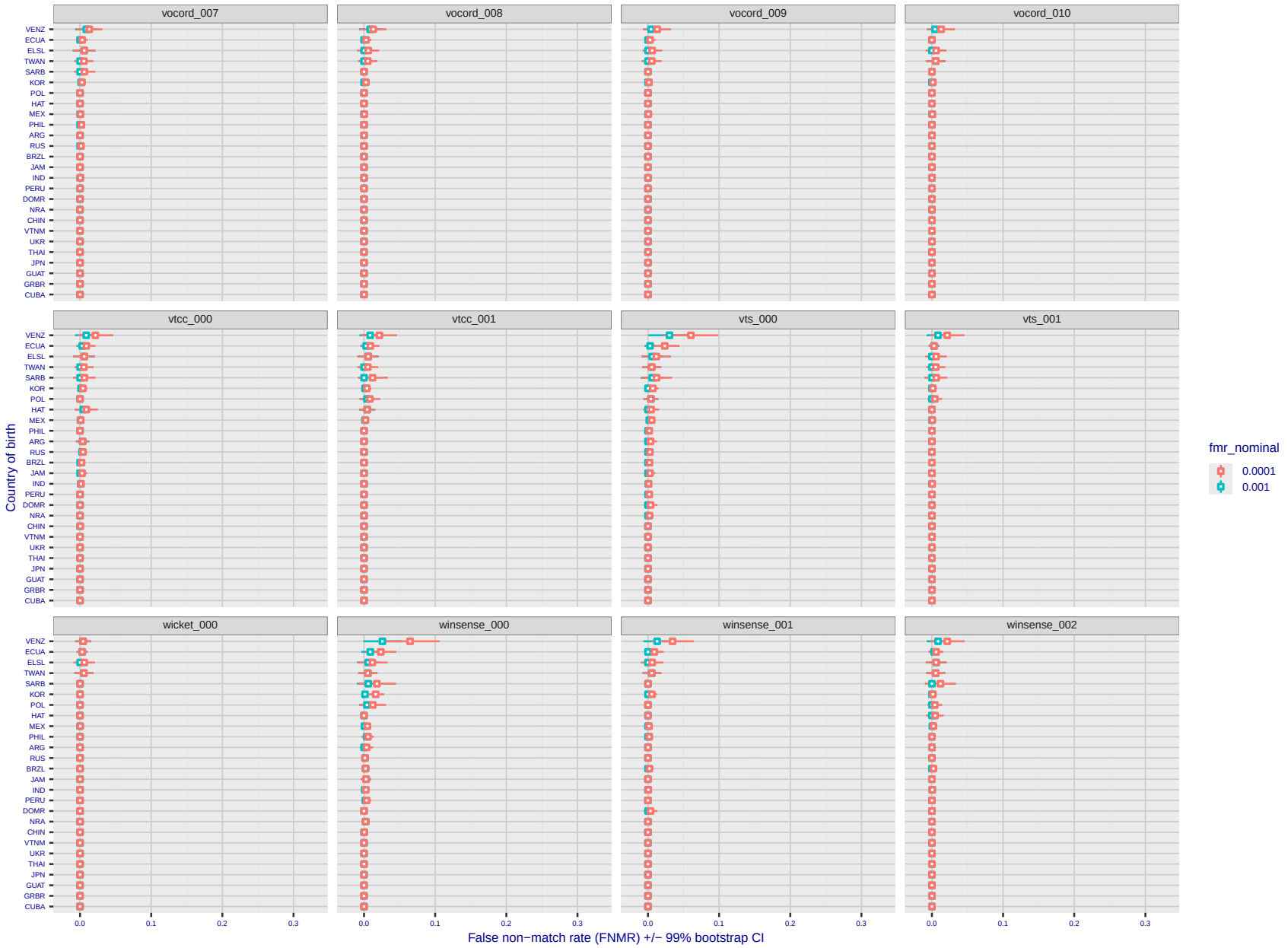


Figure 722: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

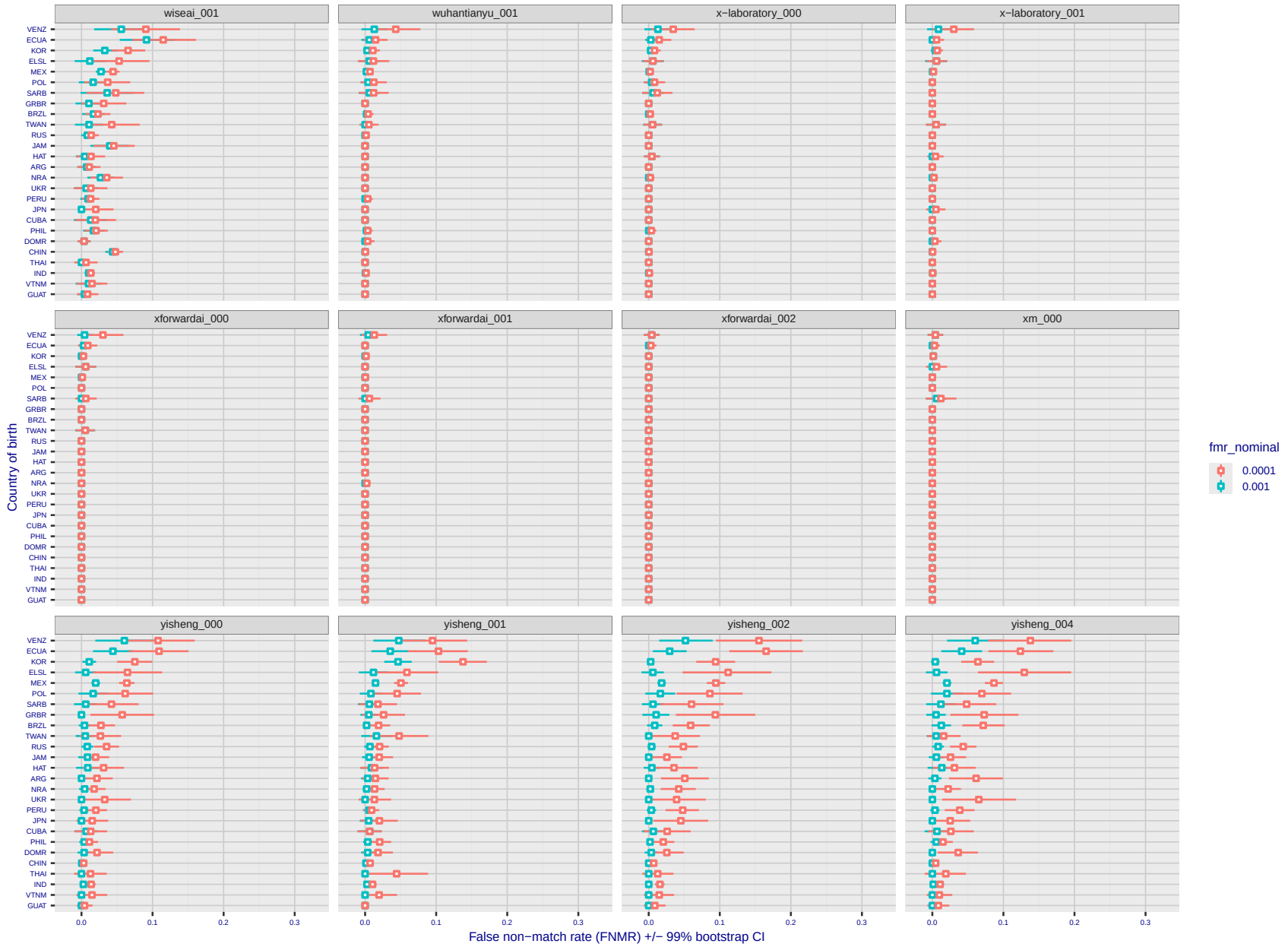


Figure 723: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

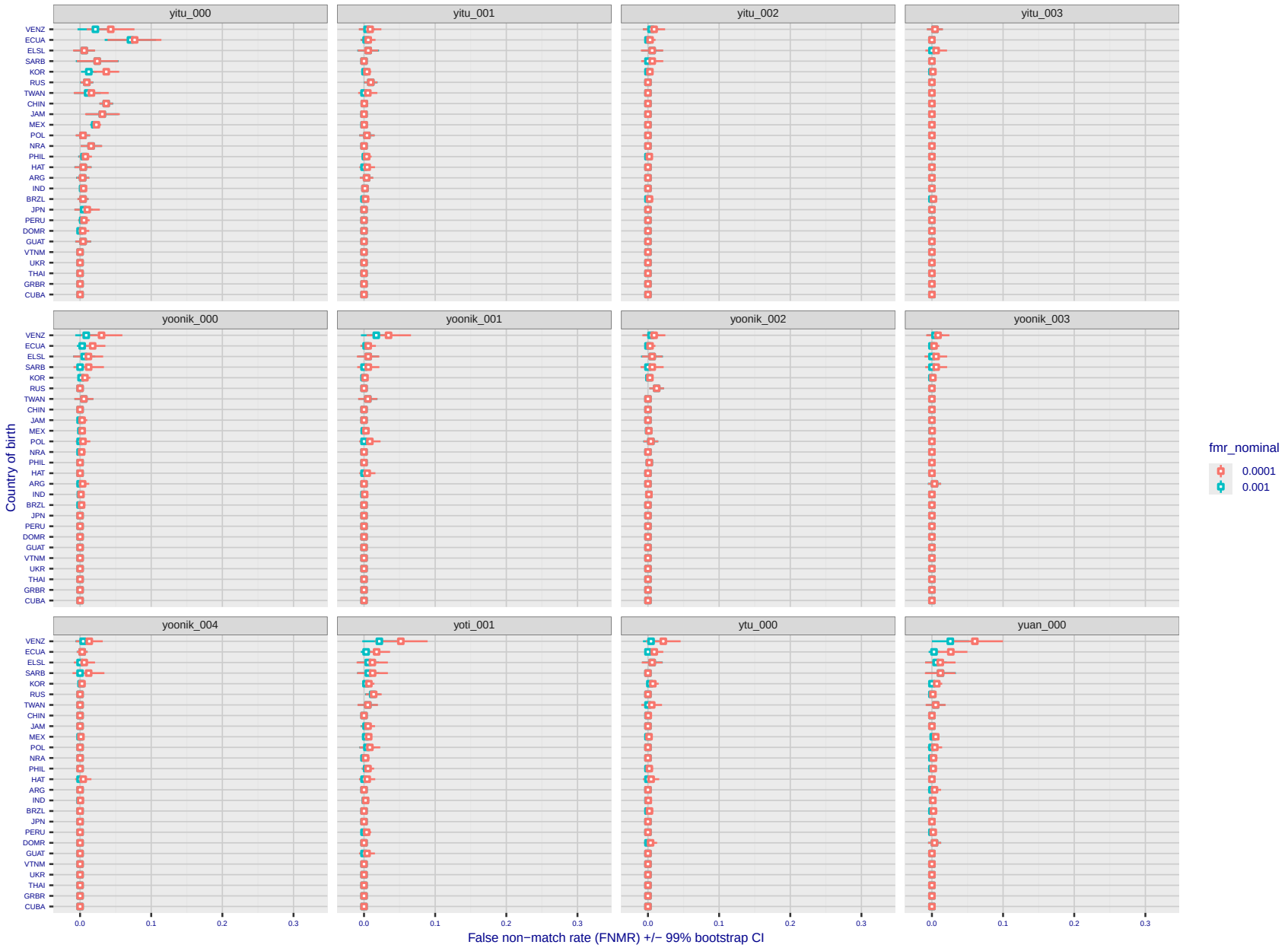


Figure 724: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

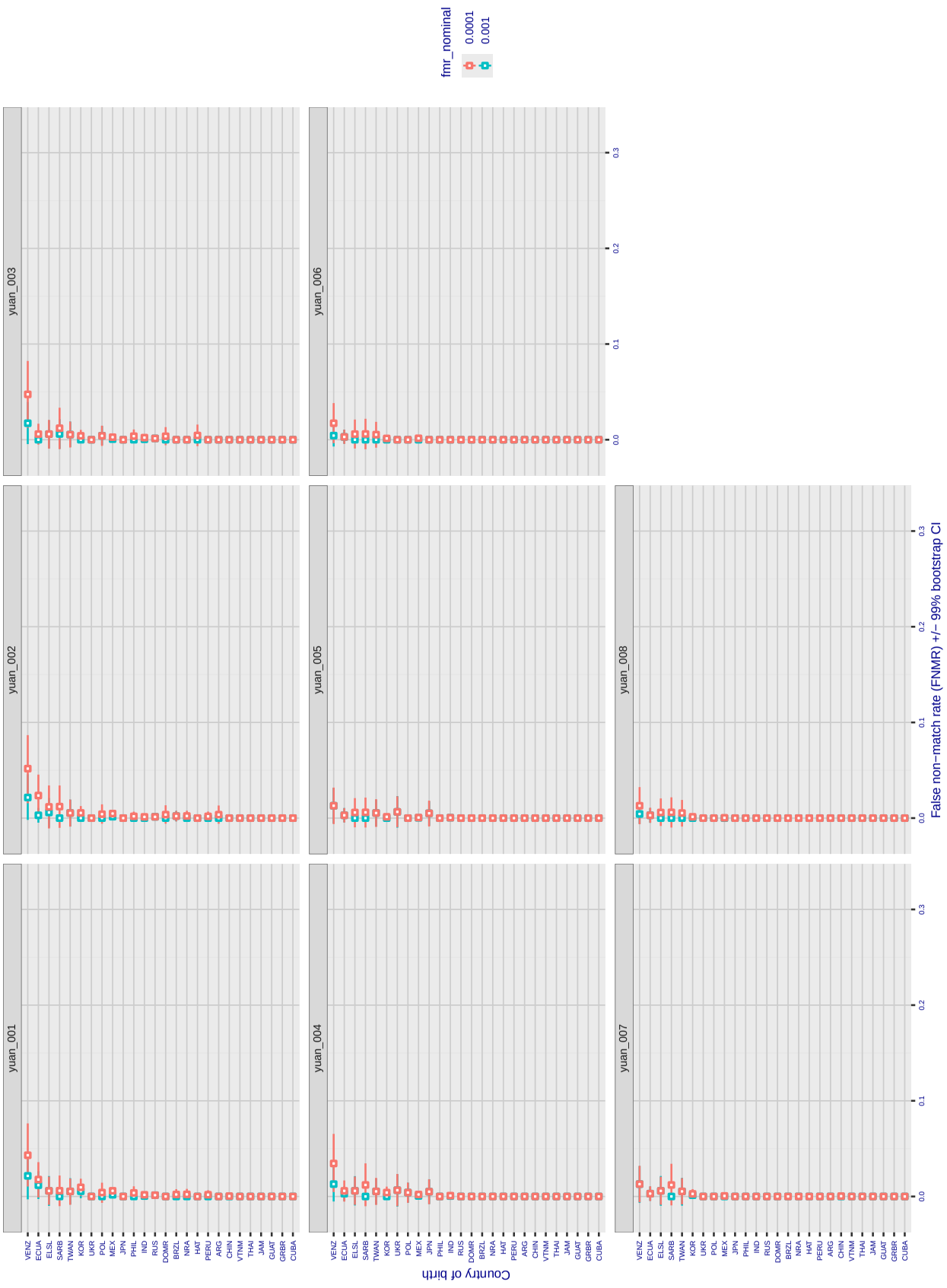


Figure 725: For the visa images, the dots show FNMR by country of birth for two globally set operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.1. The figures shows an order of magnitude variation in FNMR across country of birth; these effects are likely due quality variations, then demographics like age and race. The error rates in some cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Caveats: The results may not relate to subject-specific properties. Instead they could reflect image-specific quality differences, which could occur due to collection protocol or software processing variations.

3.5.2 Effect of ageing

Background: Faces change appearance throughout life. This change gradually reduces similarity of a new image to an earlier image. Face recognition algorithms give reduced similarity scores and more frequent false rejections.

Goal: To quantify false non-match rates (FNMR) as a function of elapsed time in an adult population.

Methods: Using the mugshot images, a threshold is set to give FMR = 0.00001 over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores.

Results: For the visa images, Figure 813 shows how false non-match rates for genuine users, as a function of age group.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

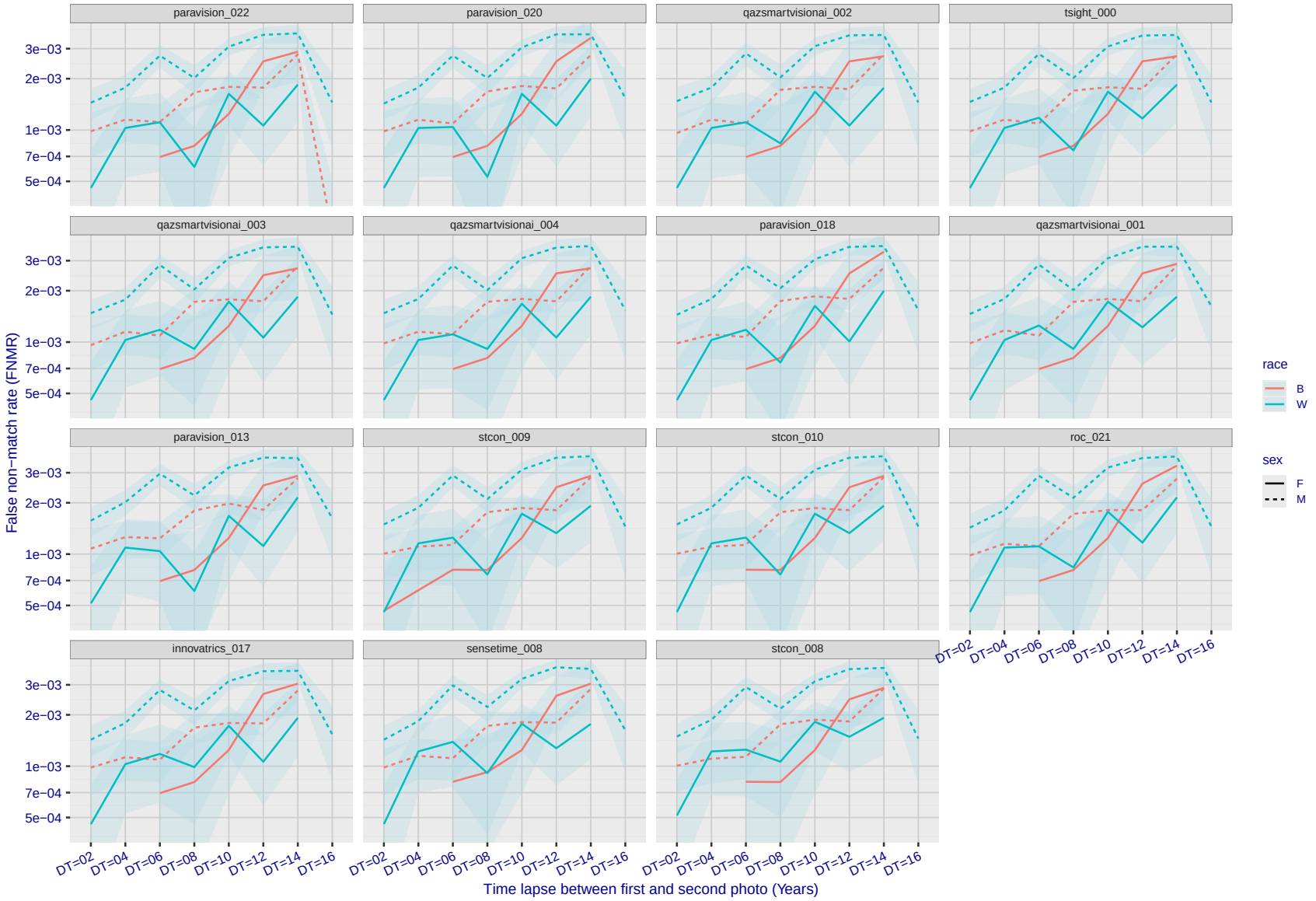


Figure 726: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

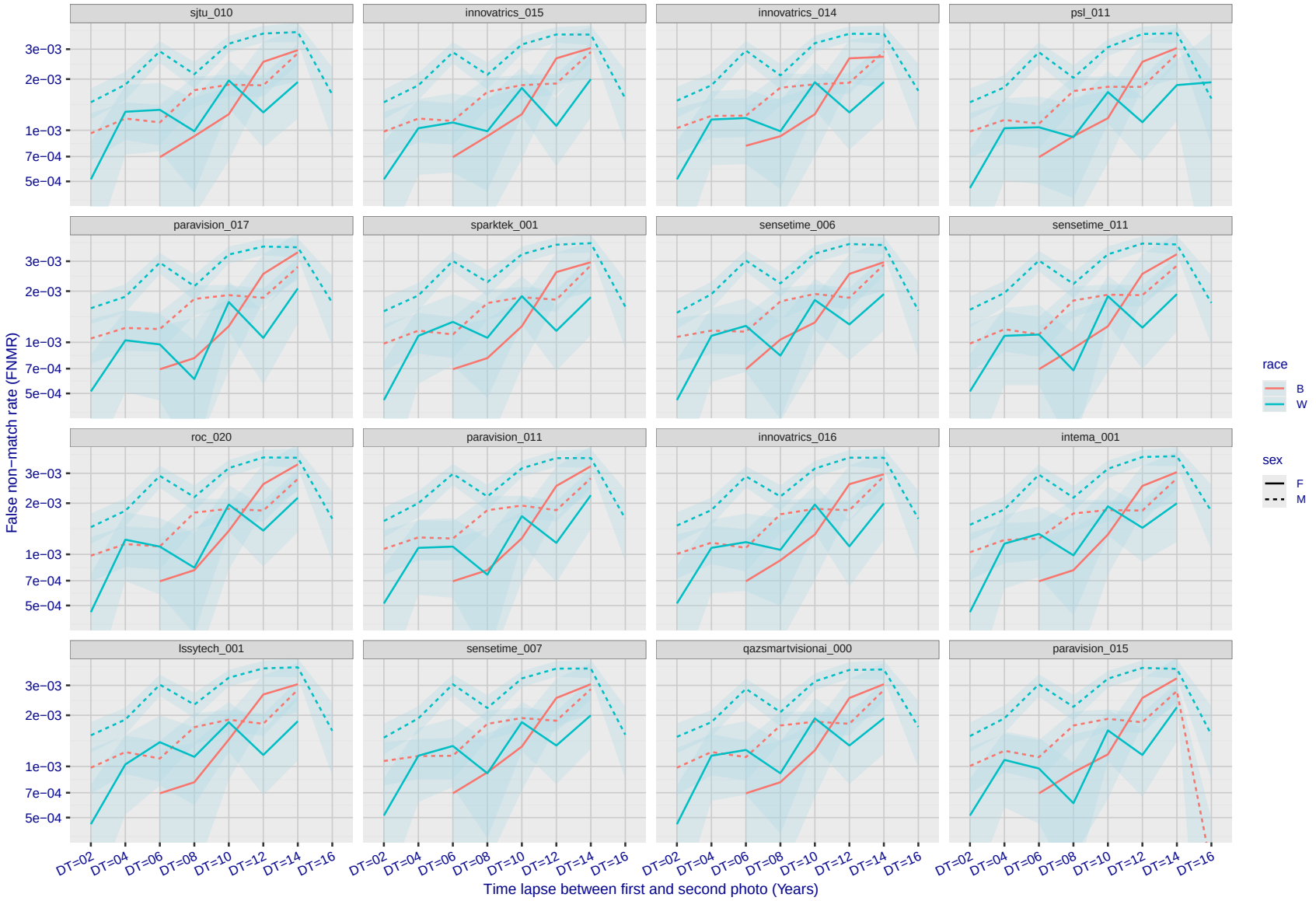


Figure 727: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

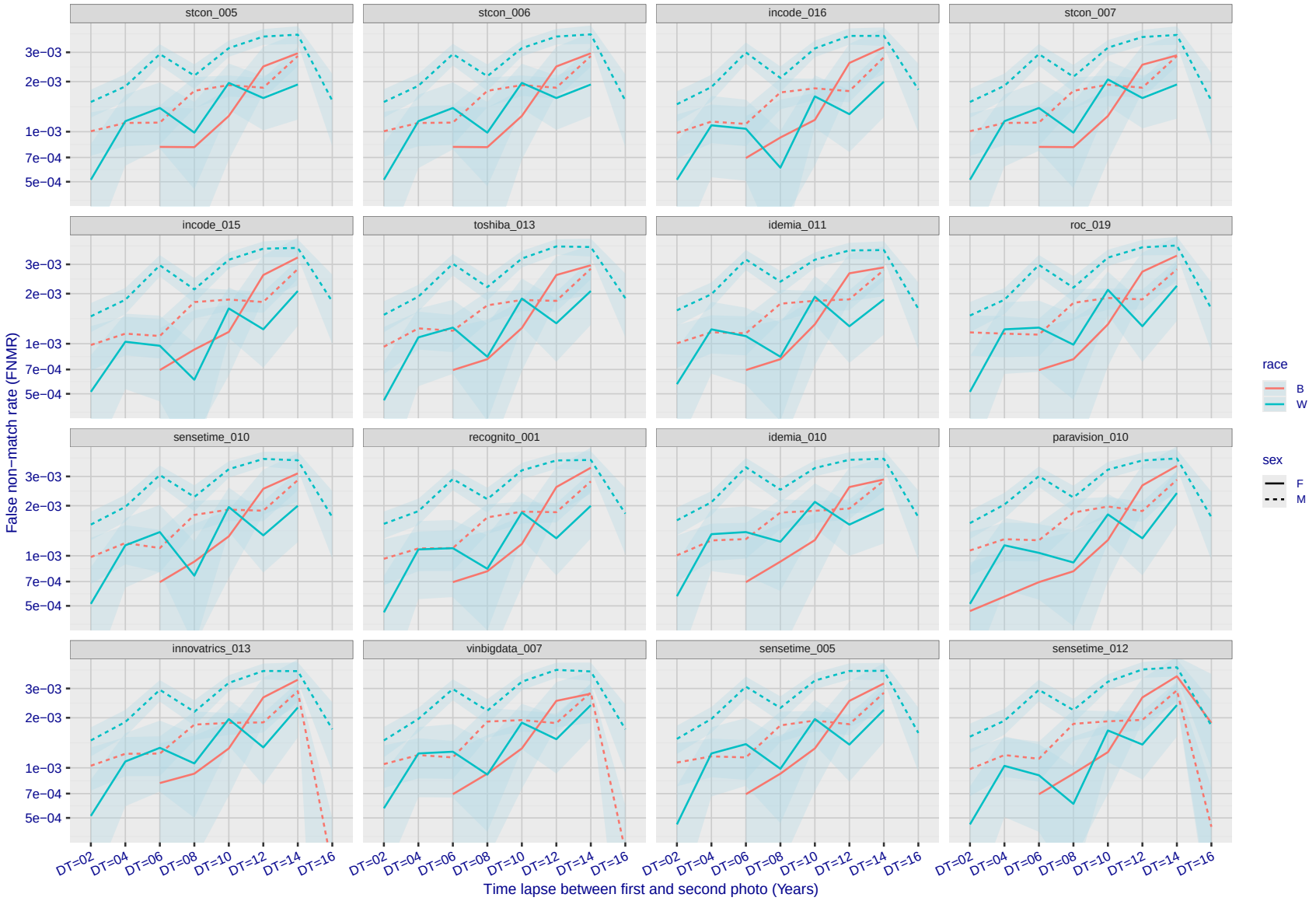


Figure 728: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

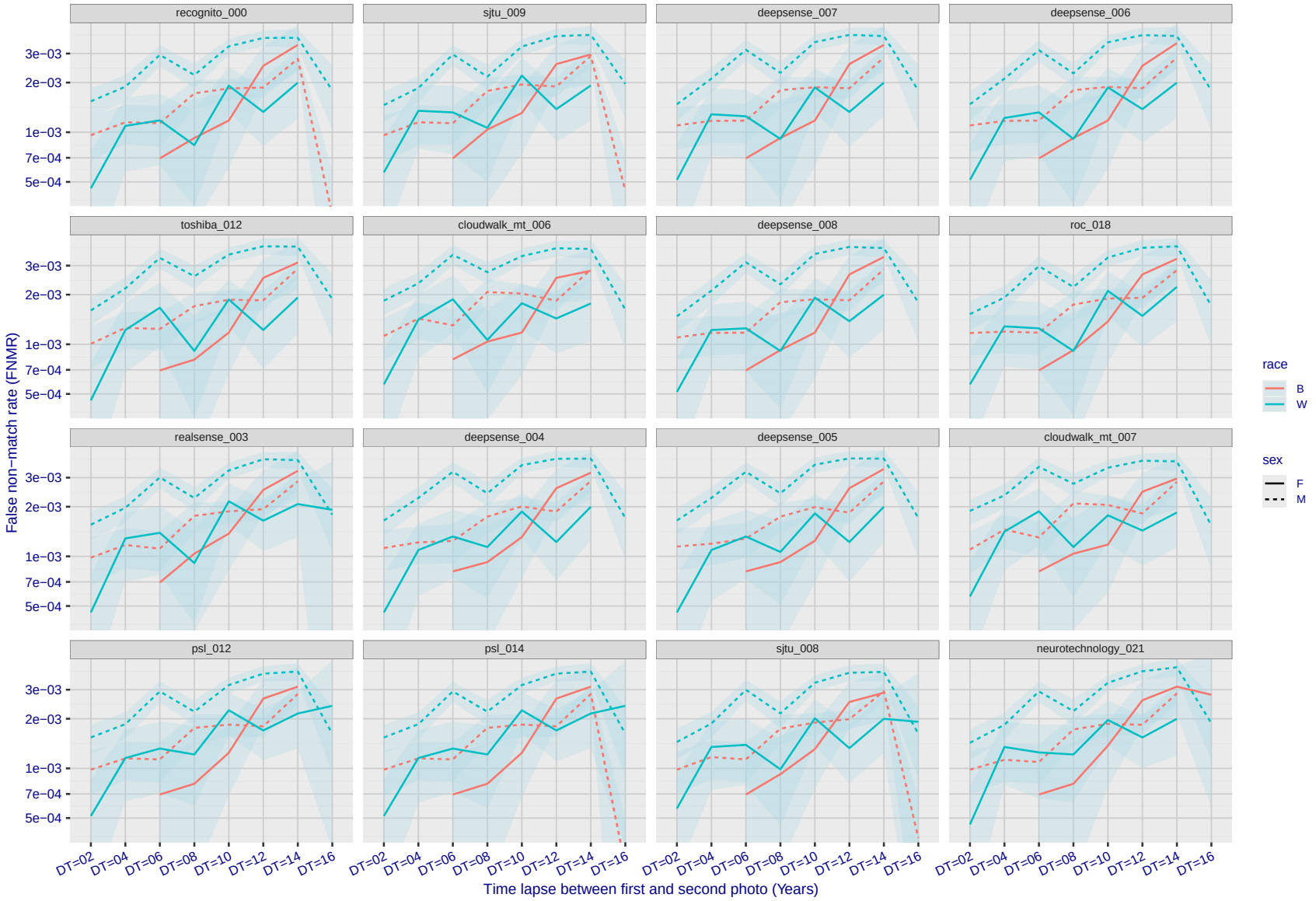


Figure 729: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

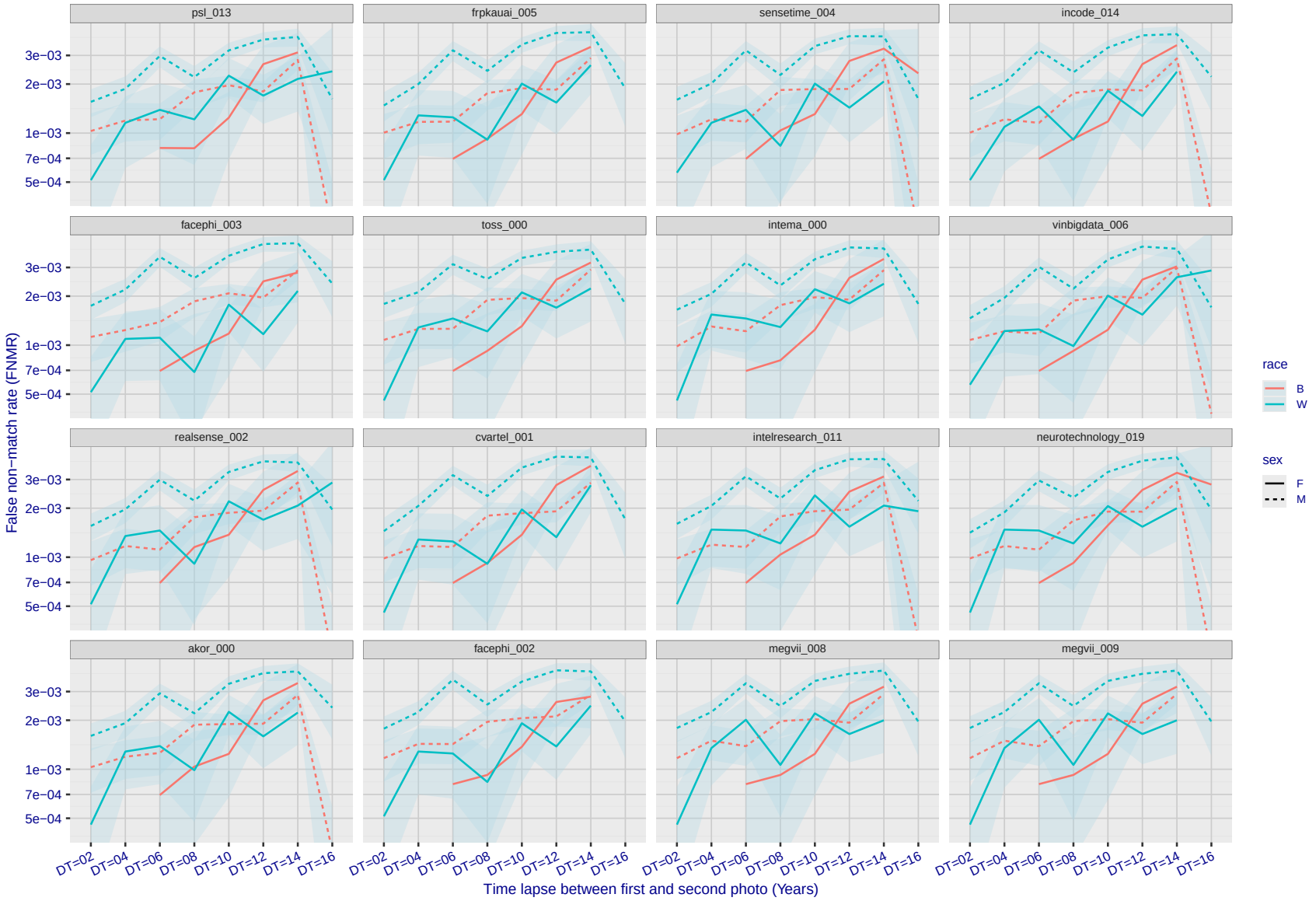


Figure 730: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

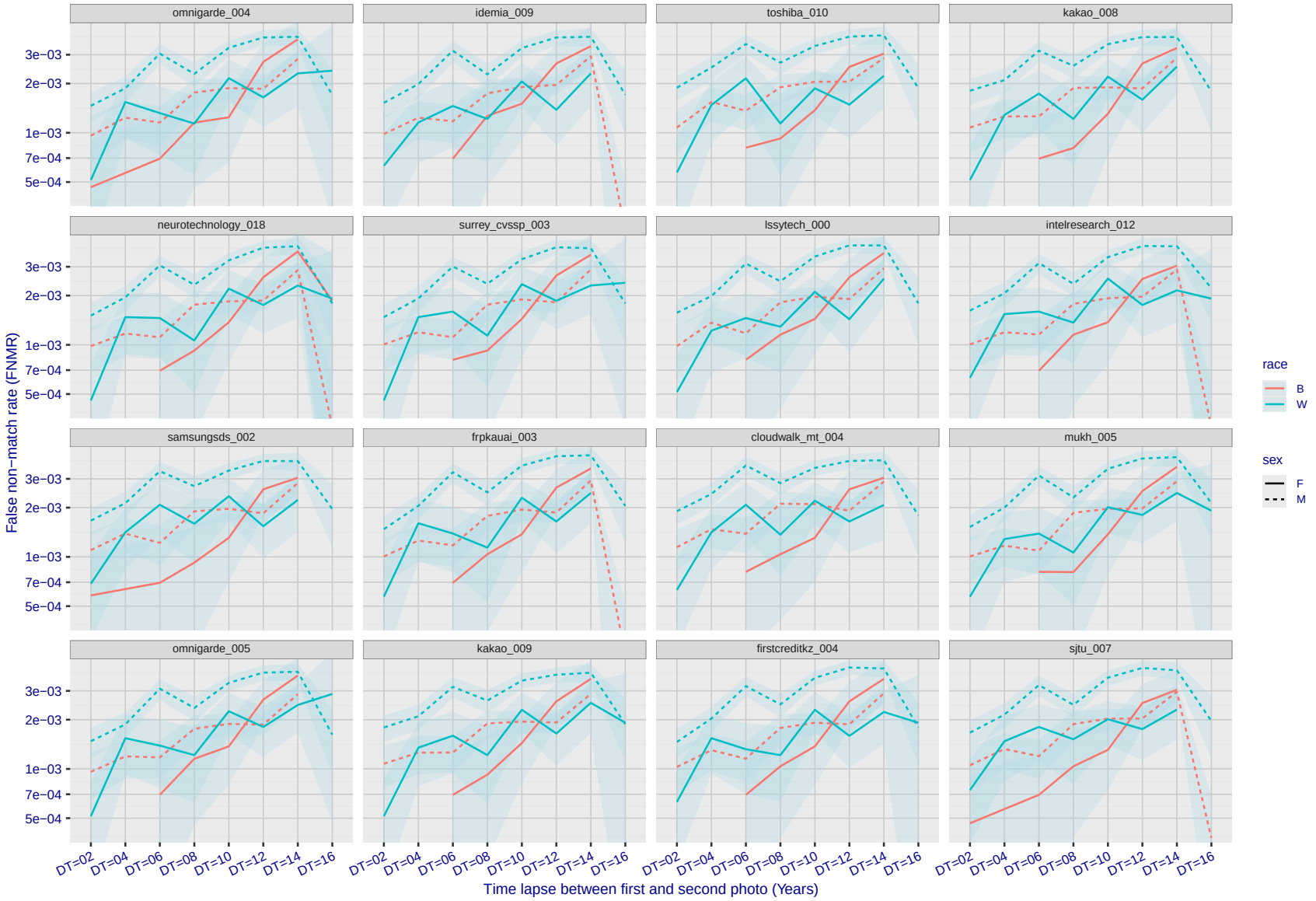


Figure 731: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

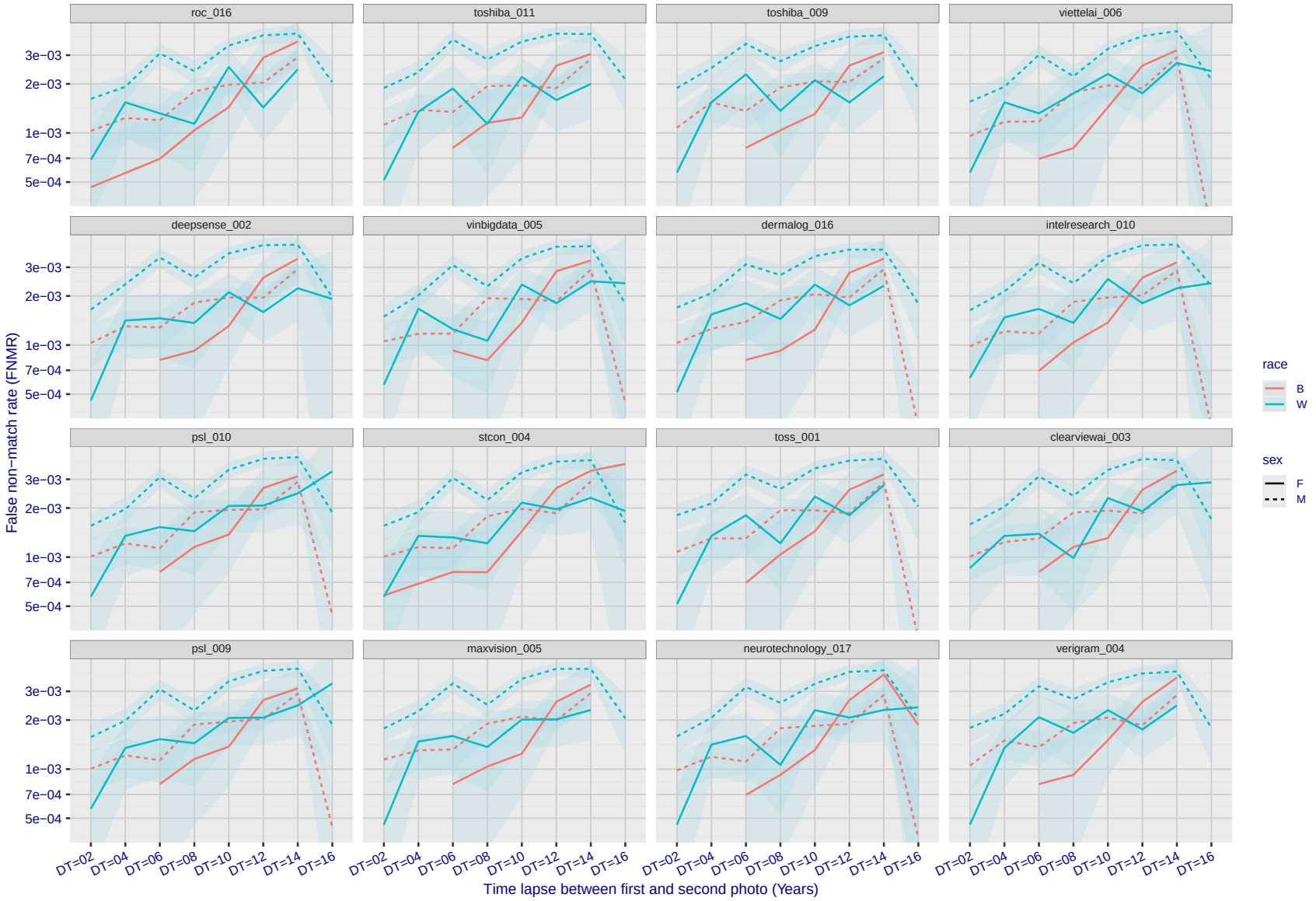


Figure 732: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

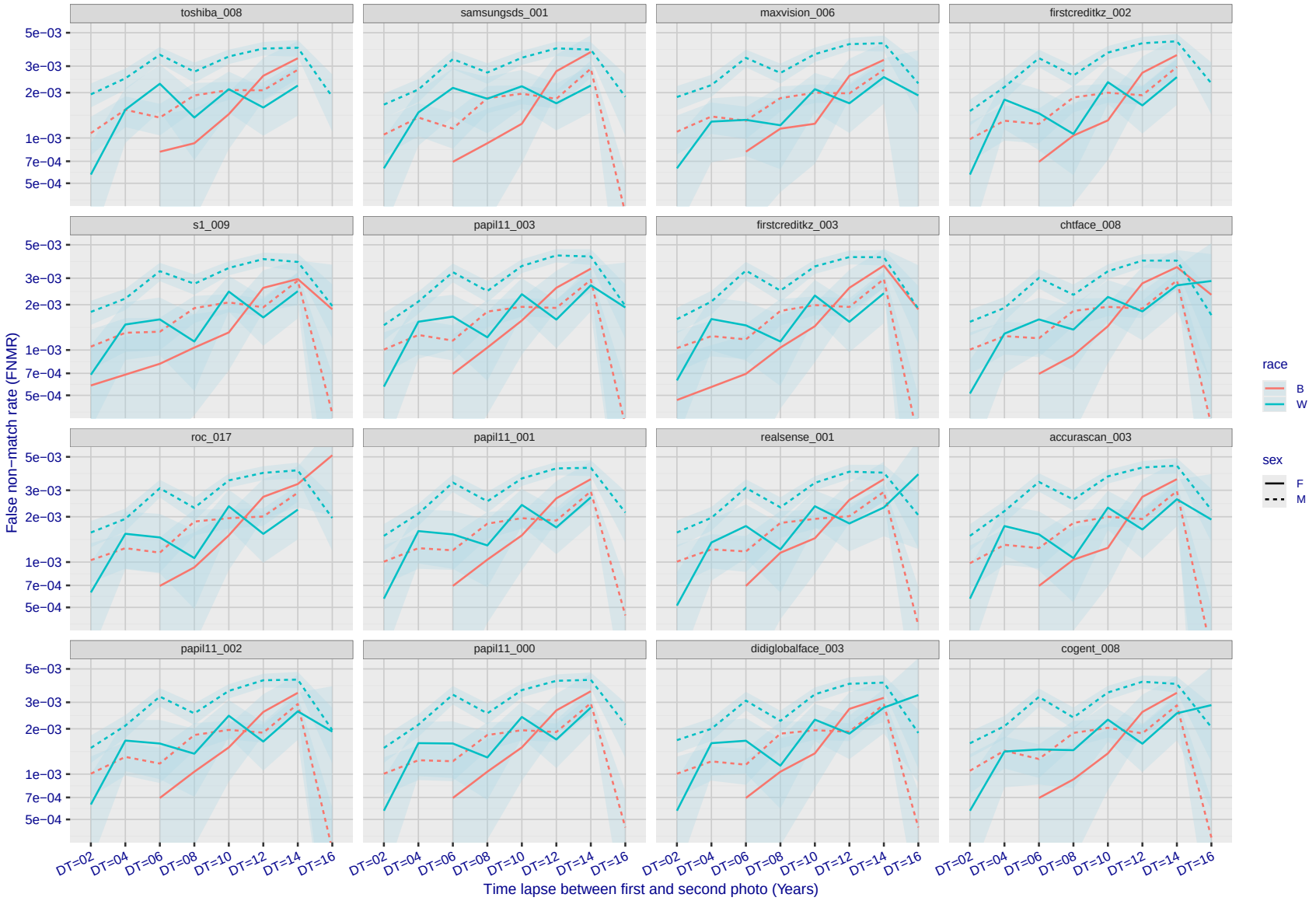


Figure 733: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

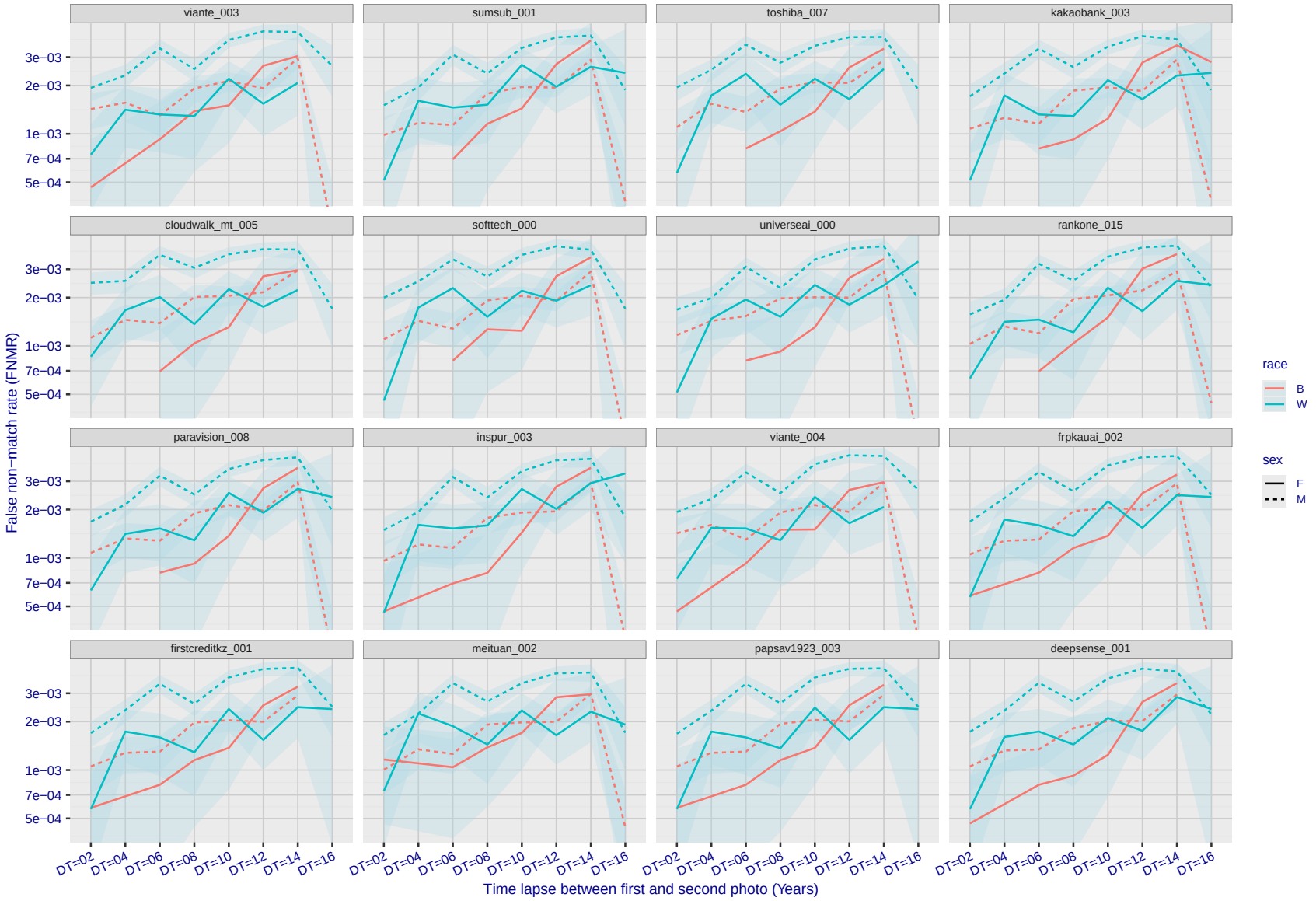


Figure 734: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

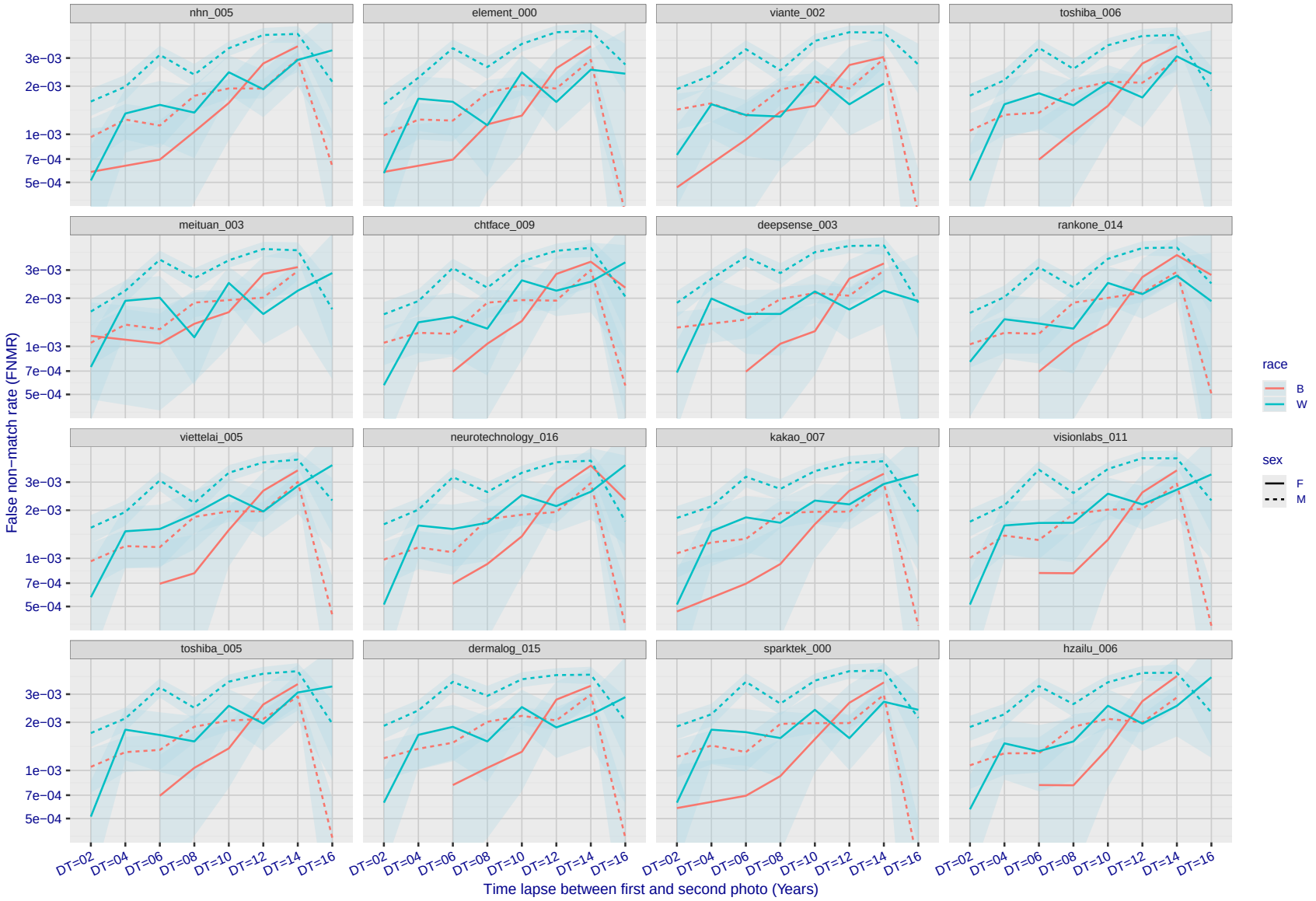


Figure 735: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

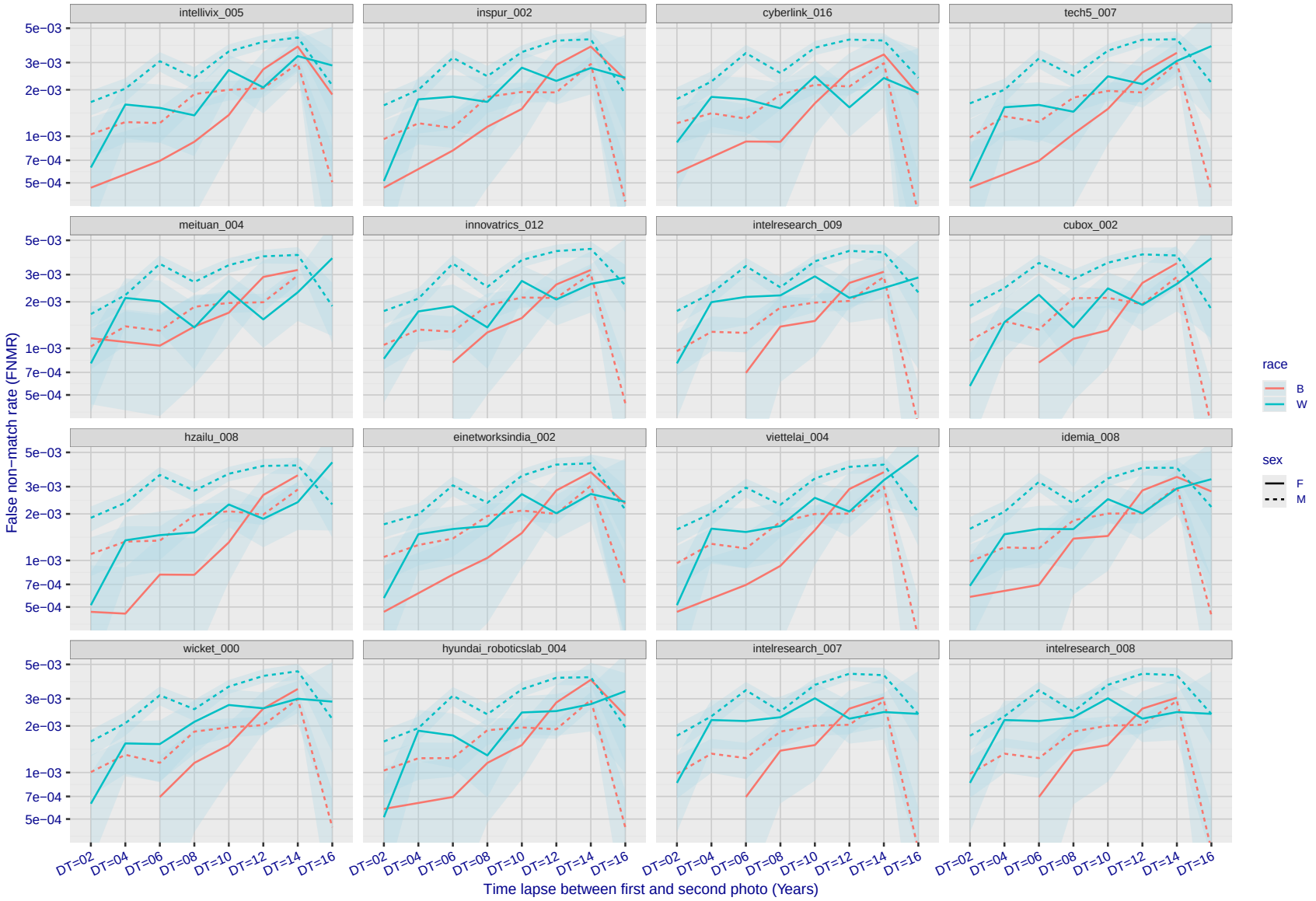


Figure 736: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

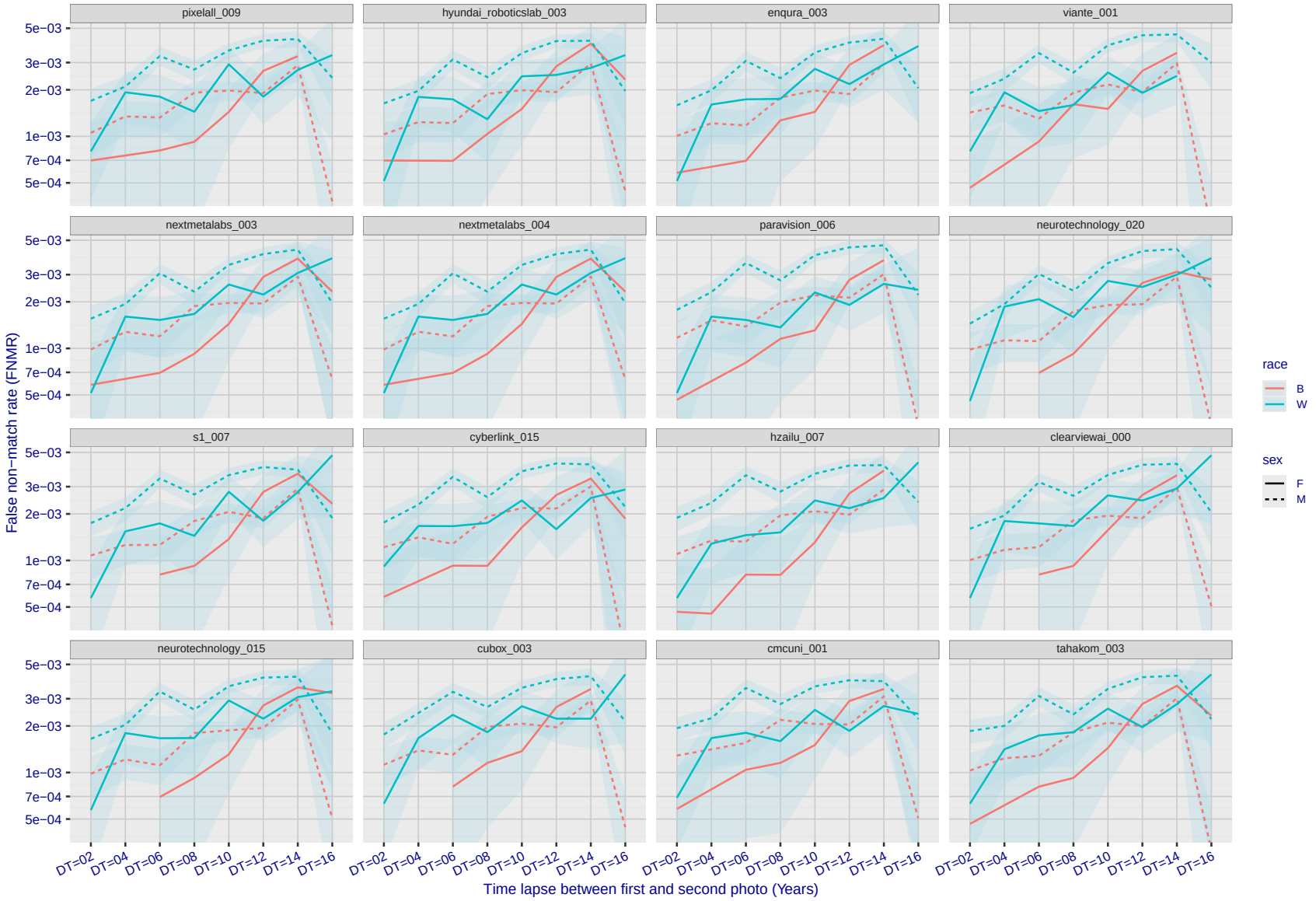


Figure 737: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

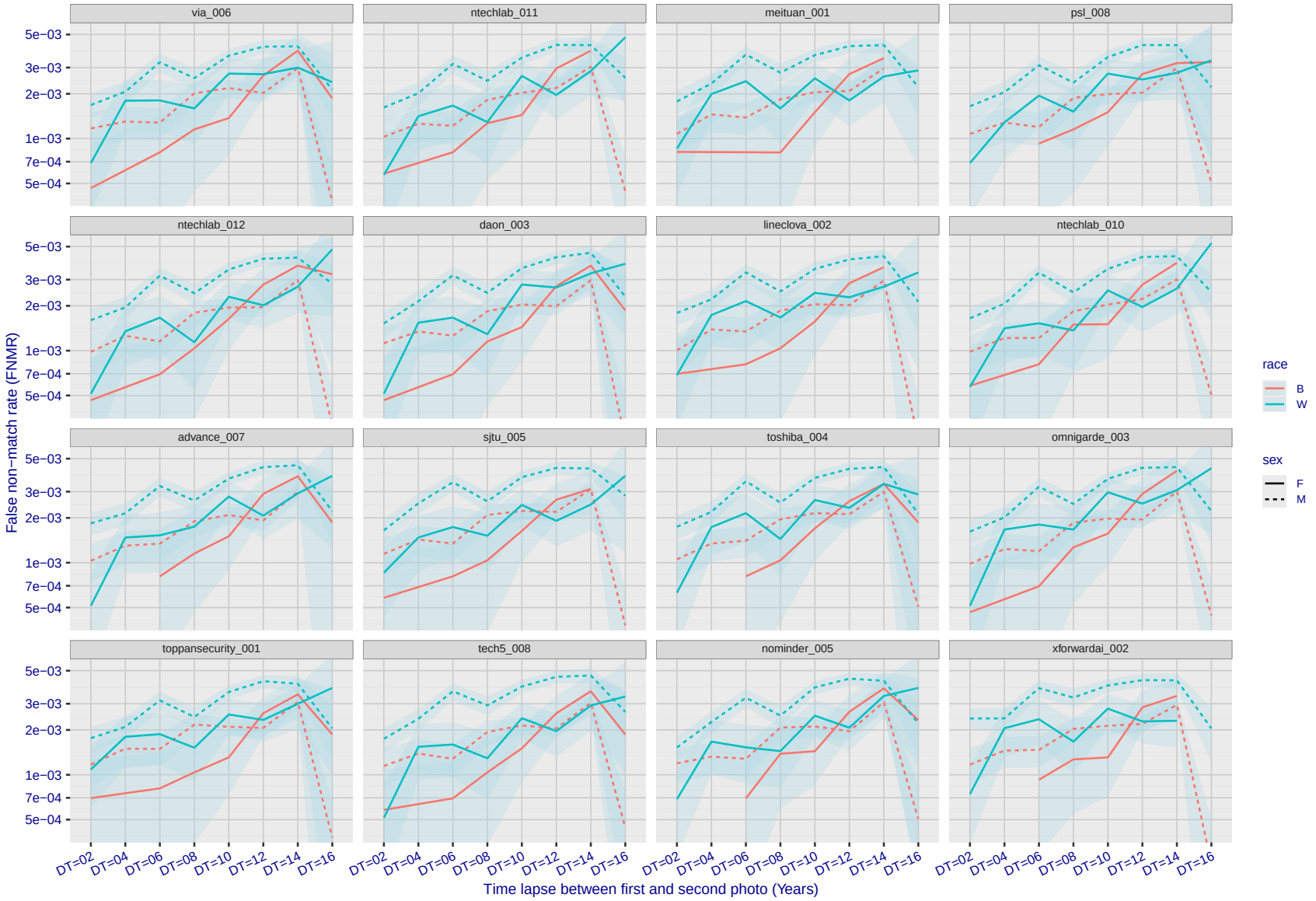


Figure 738: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

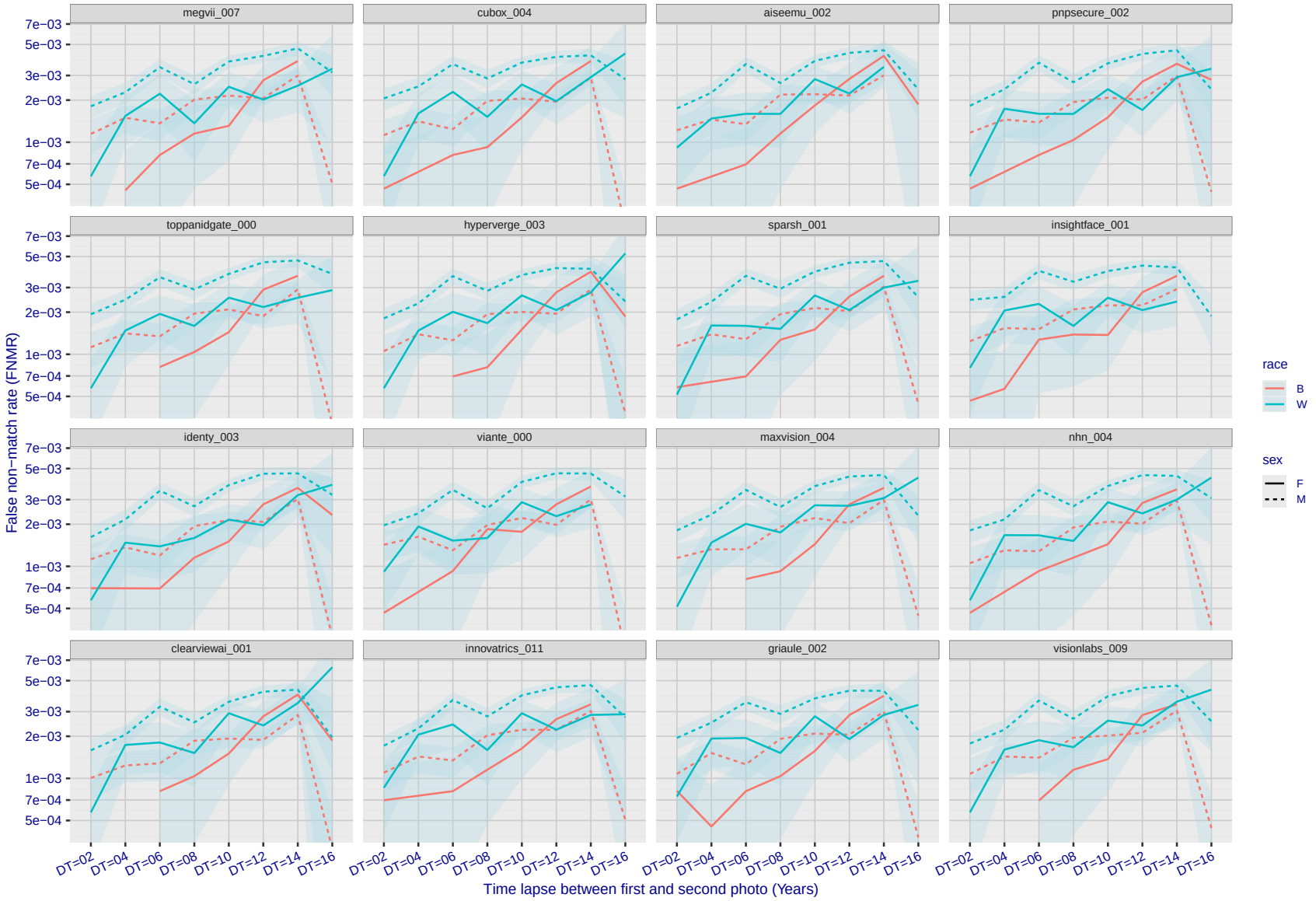


Figure 739: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

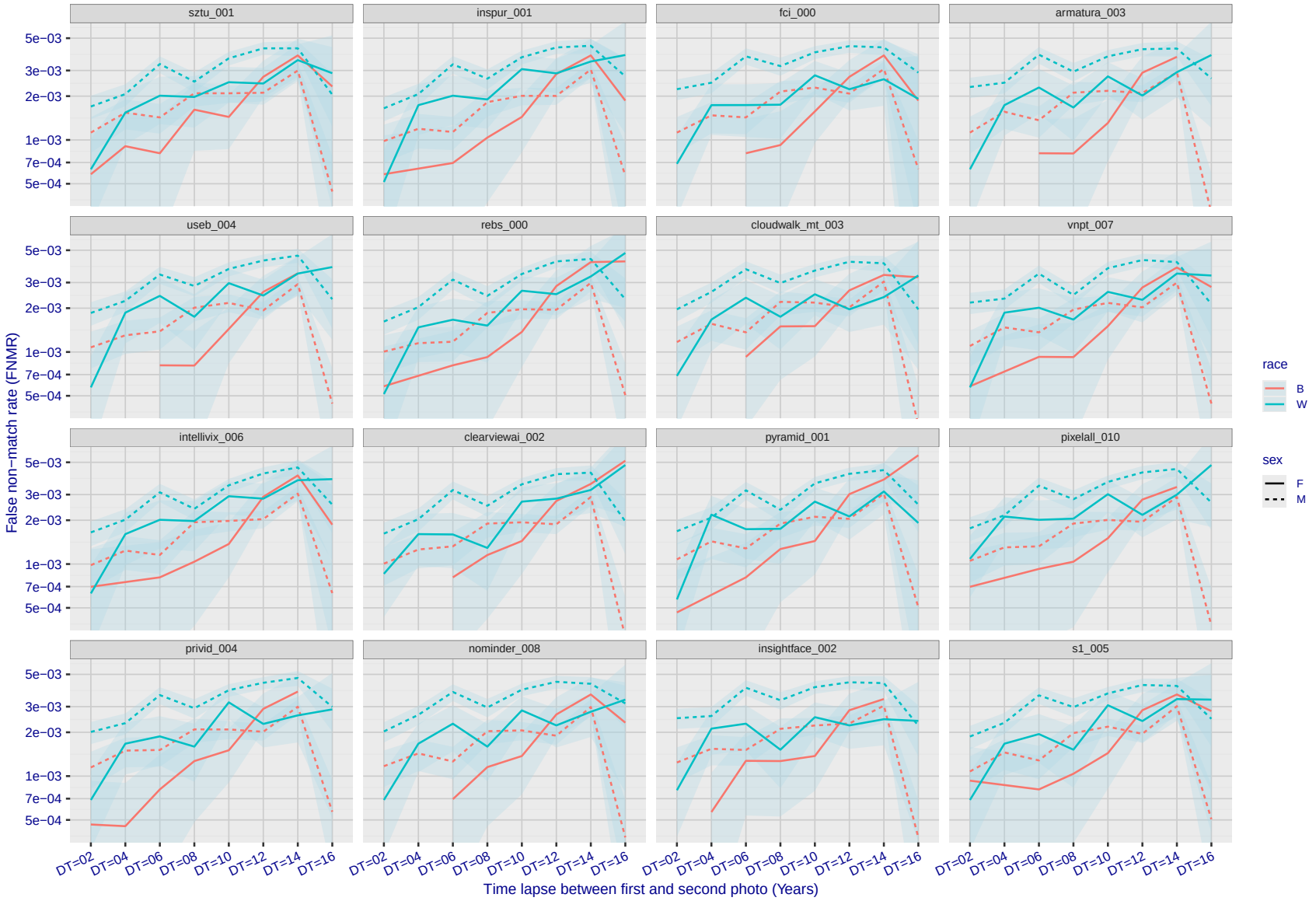


Figure 740: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

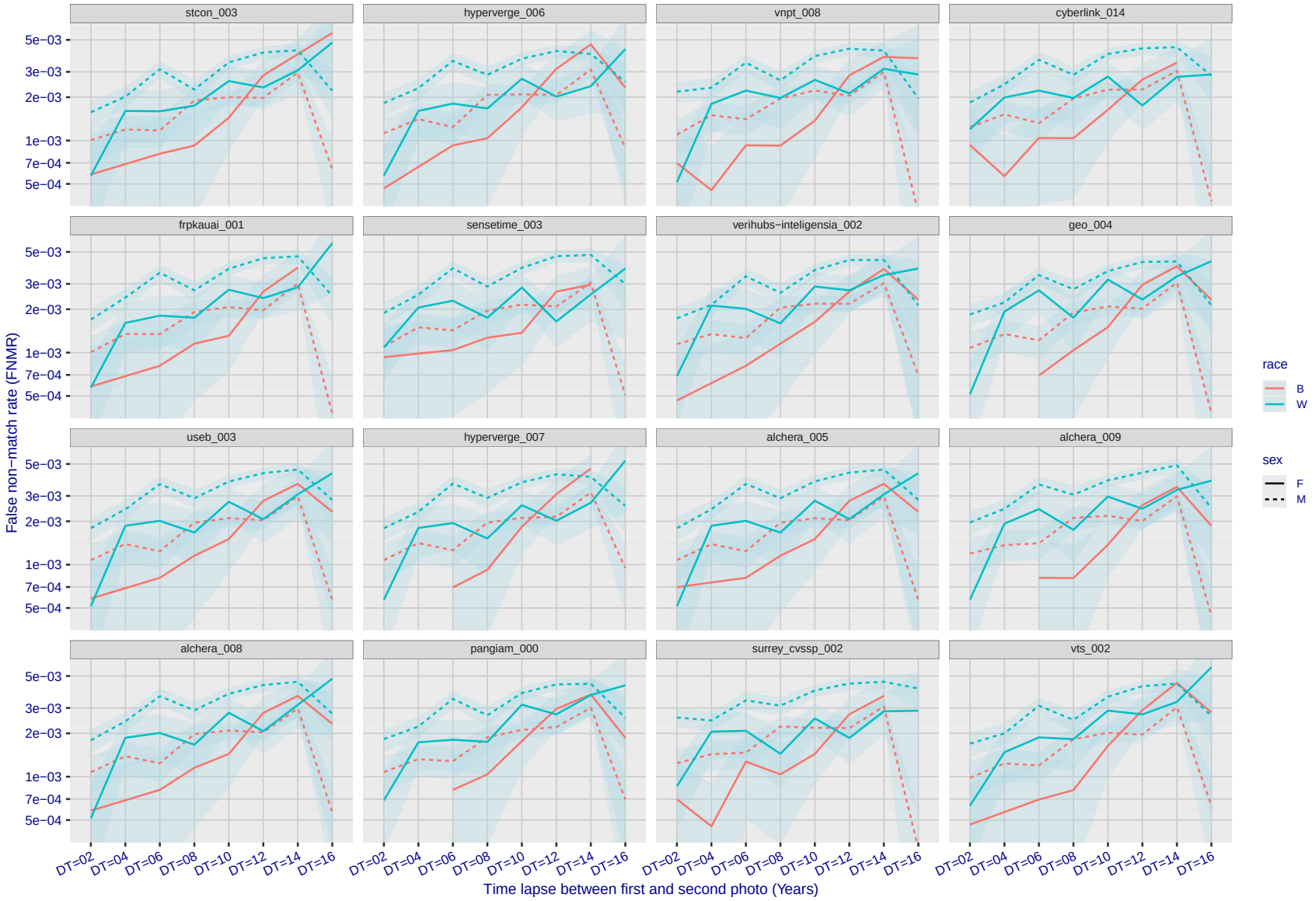


Figure 741: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\bar{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

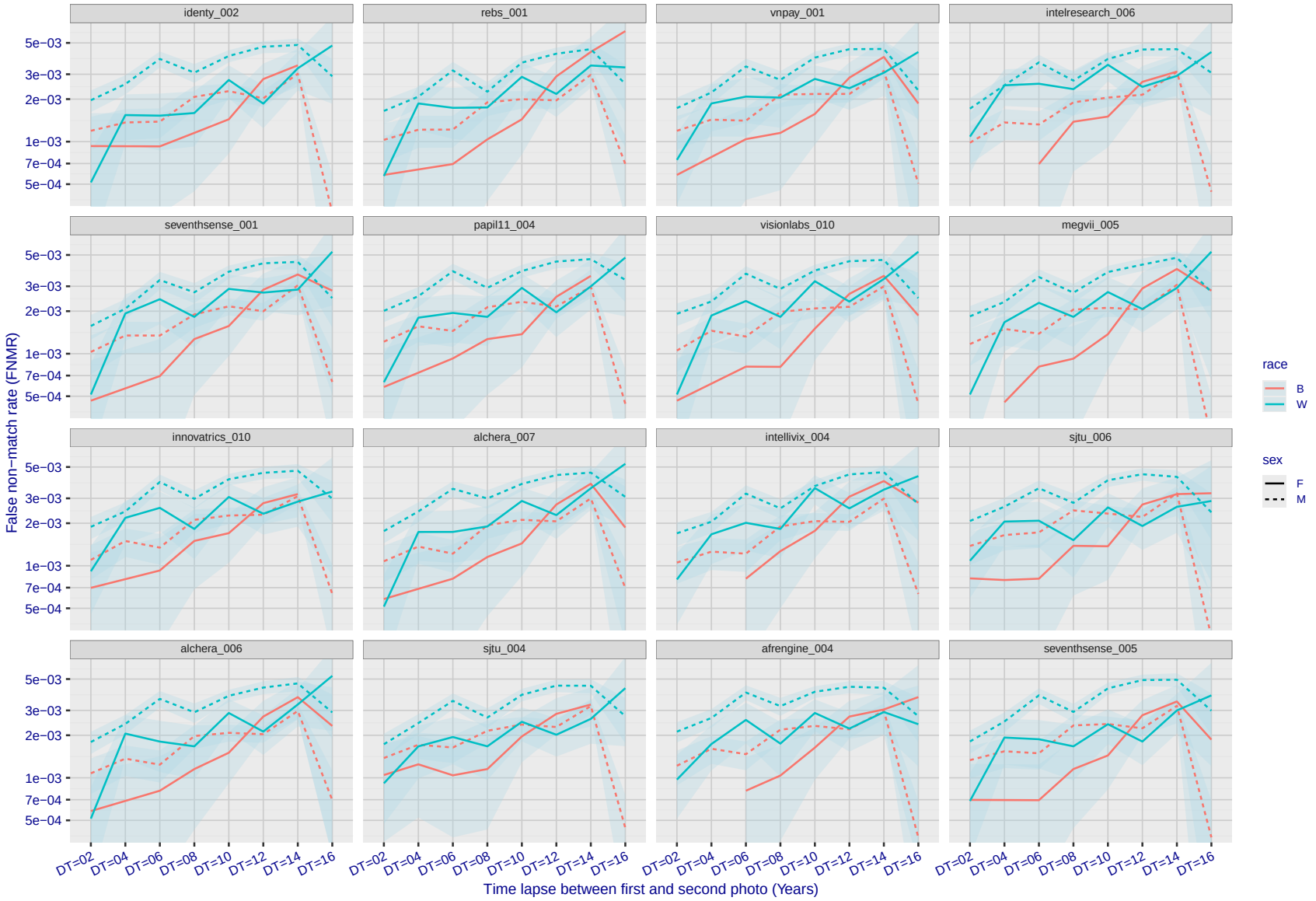


Figure 742: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

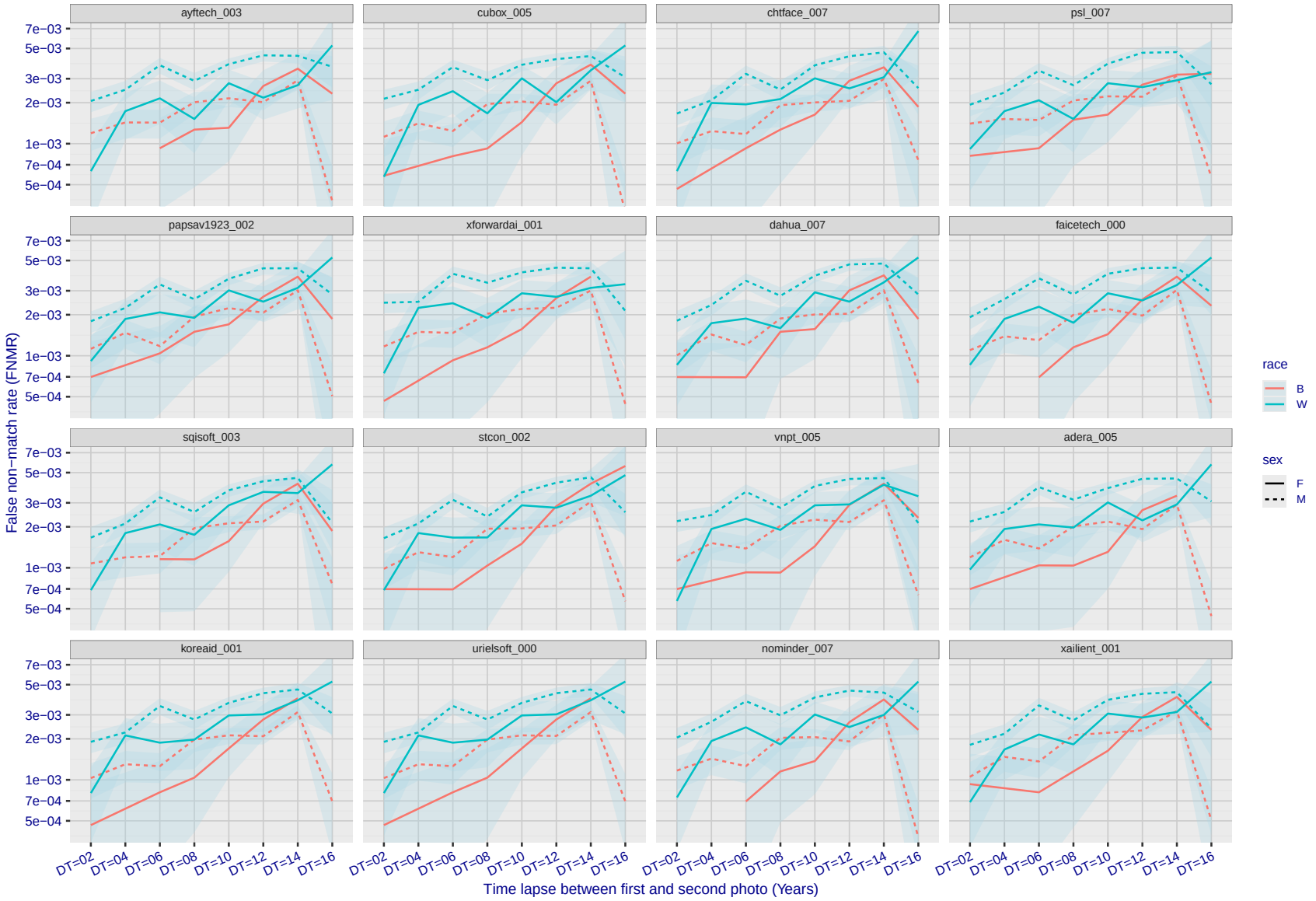


Figure 743: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

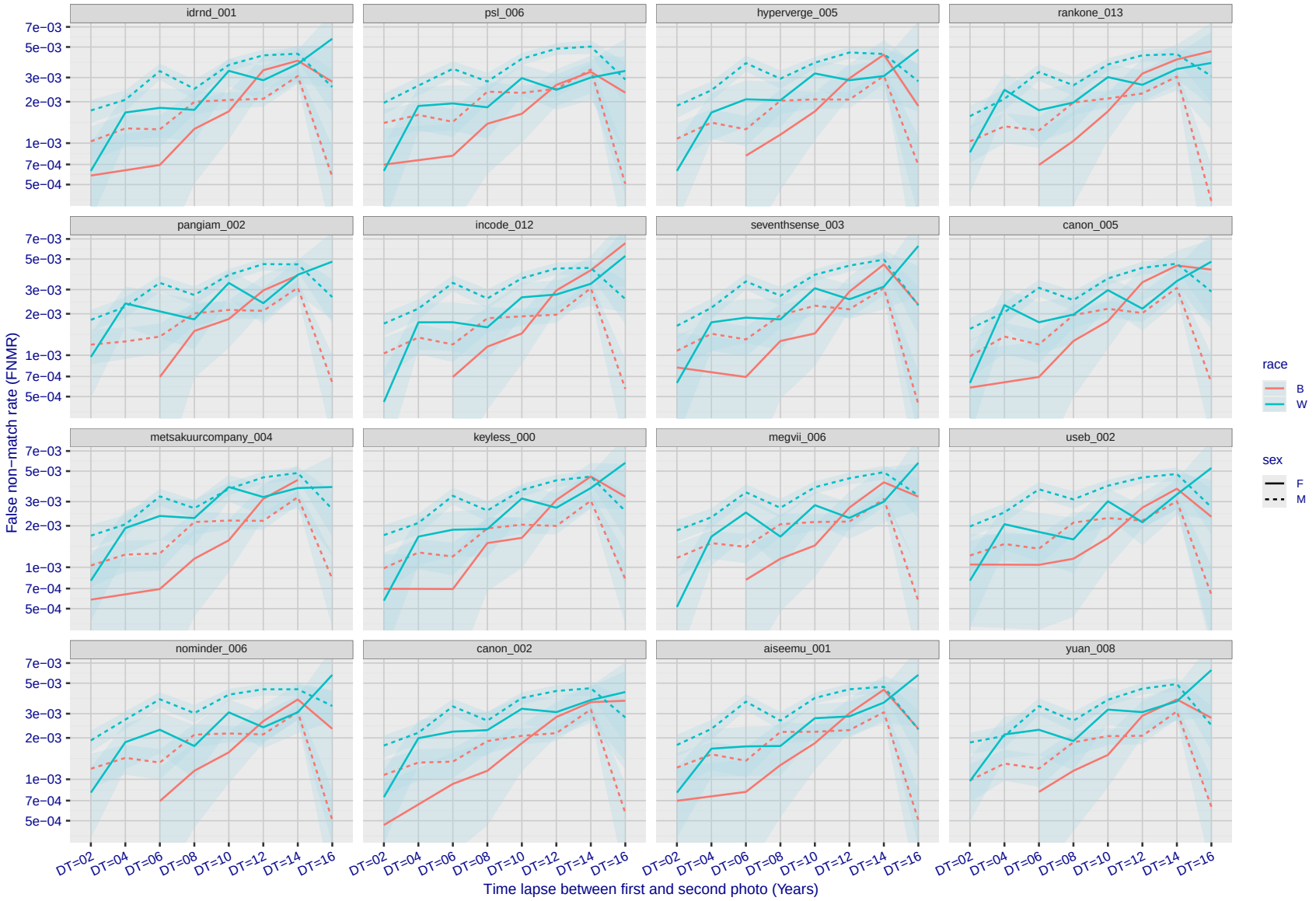


Figure 744: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

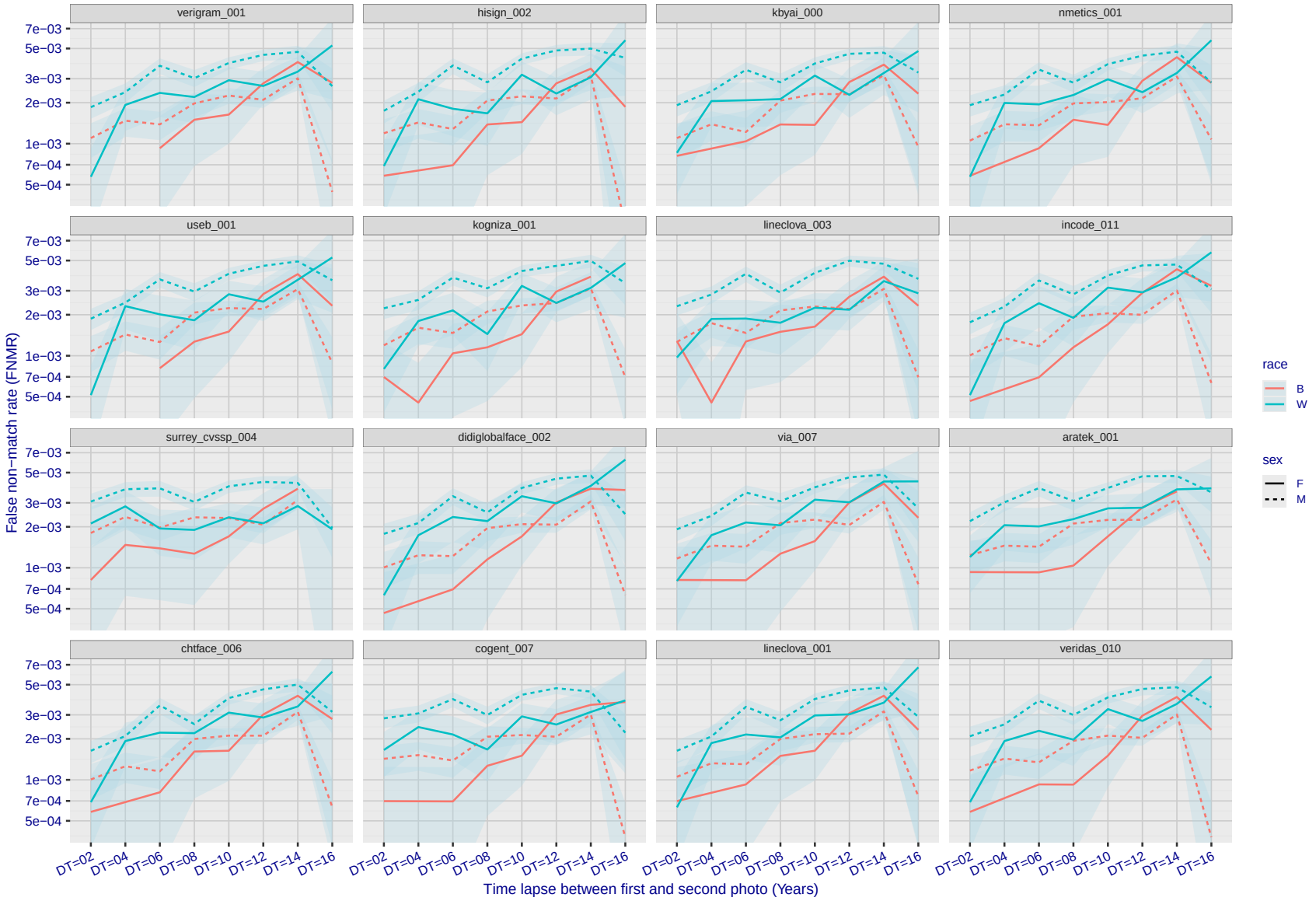


Figure 745: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\bar{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

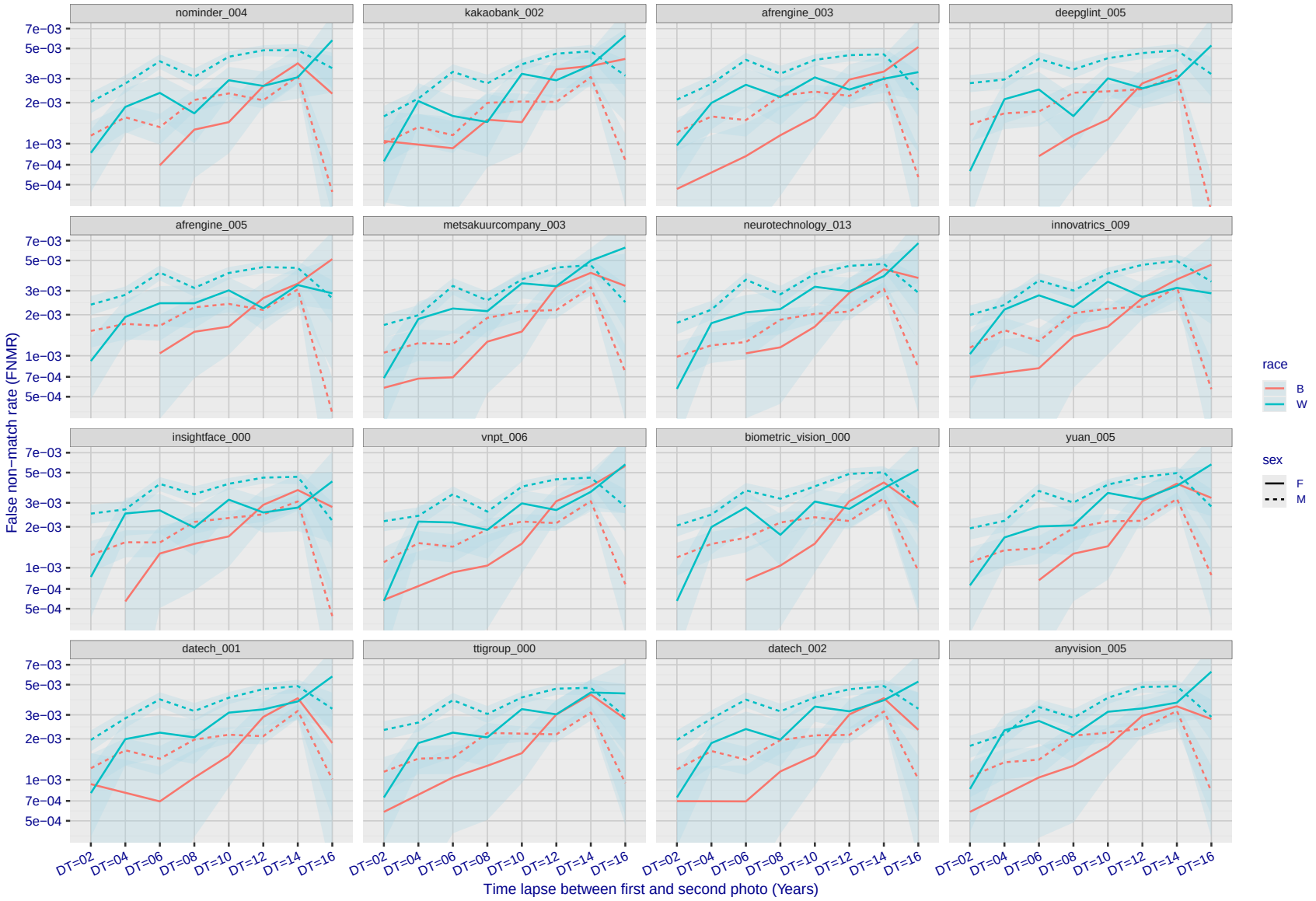


Figure 746: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

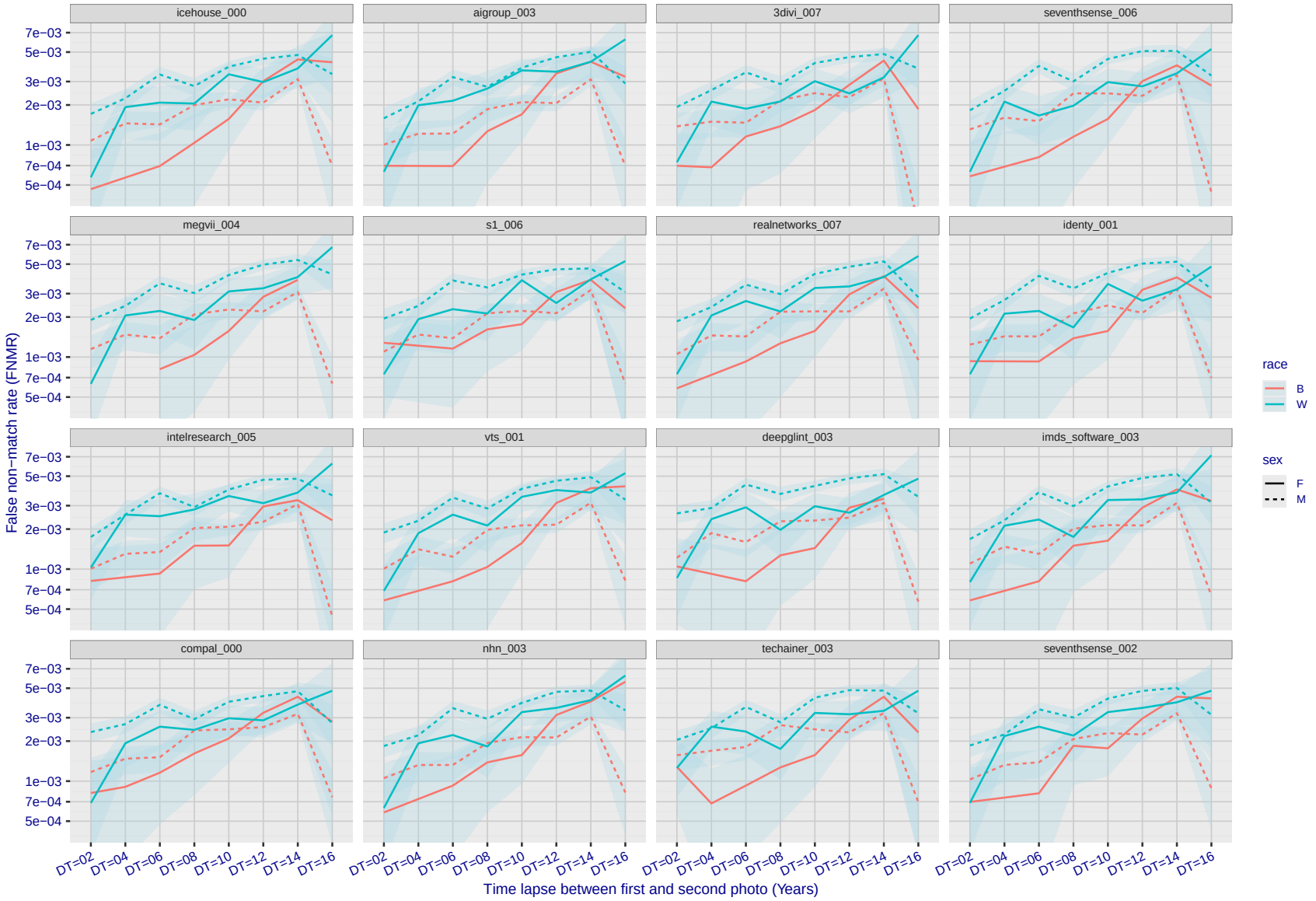


Figure 747: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

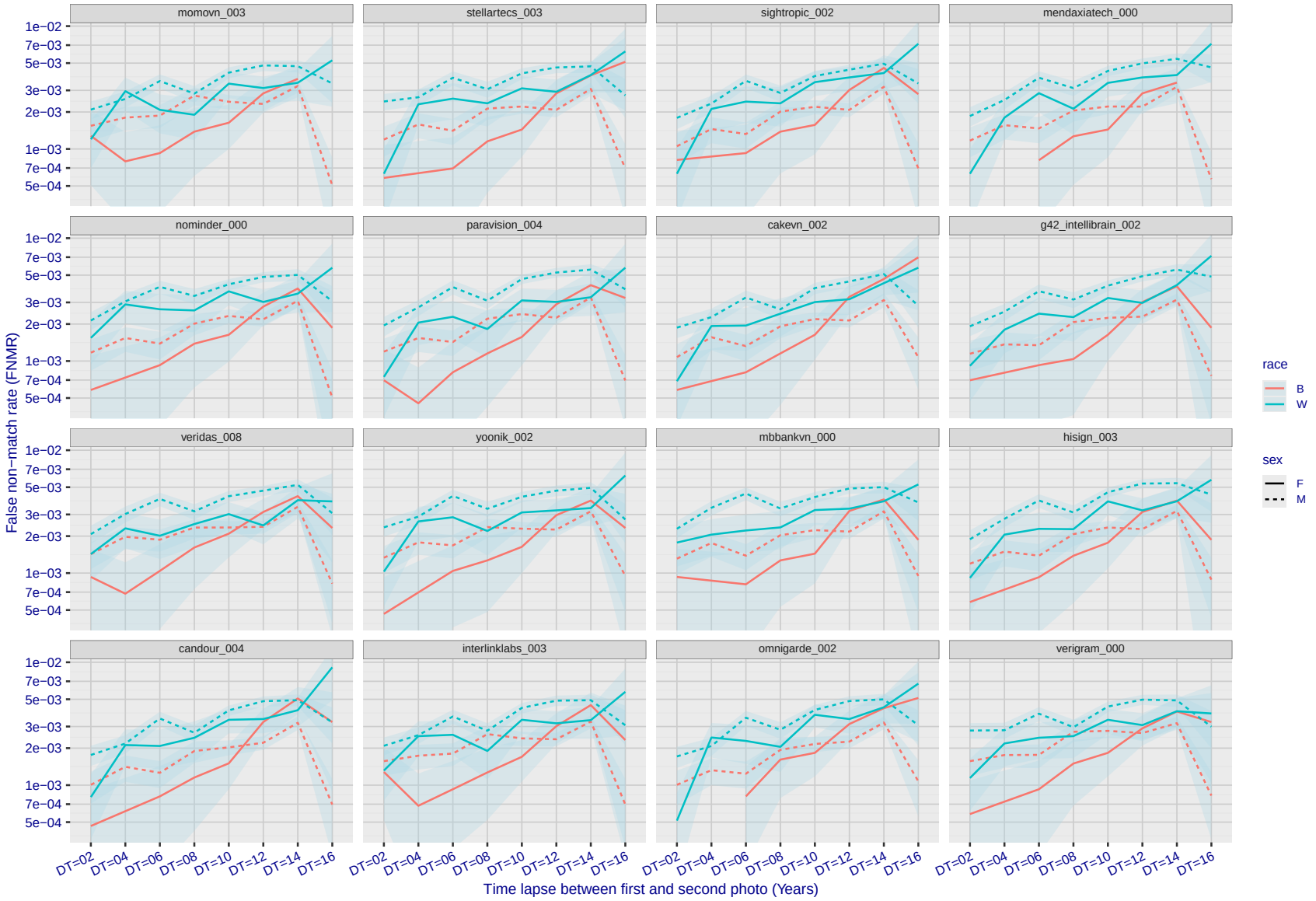


Figure 748: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

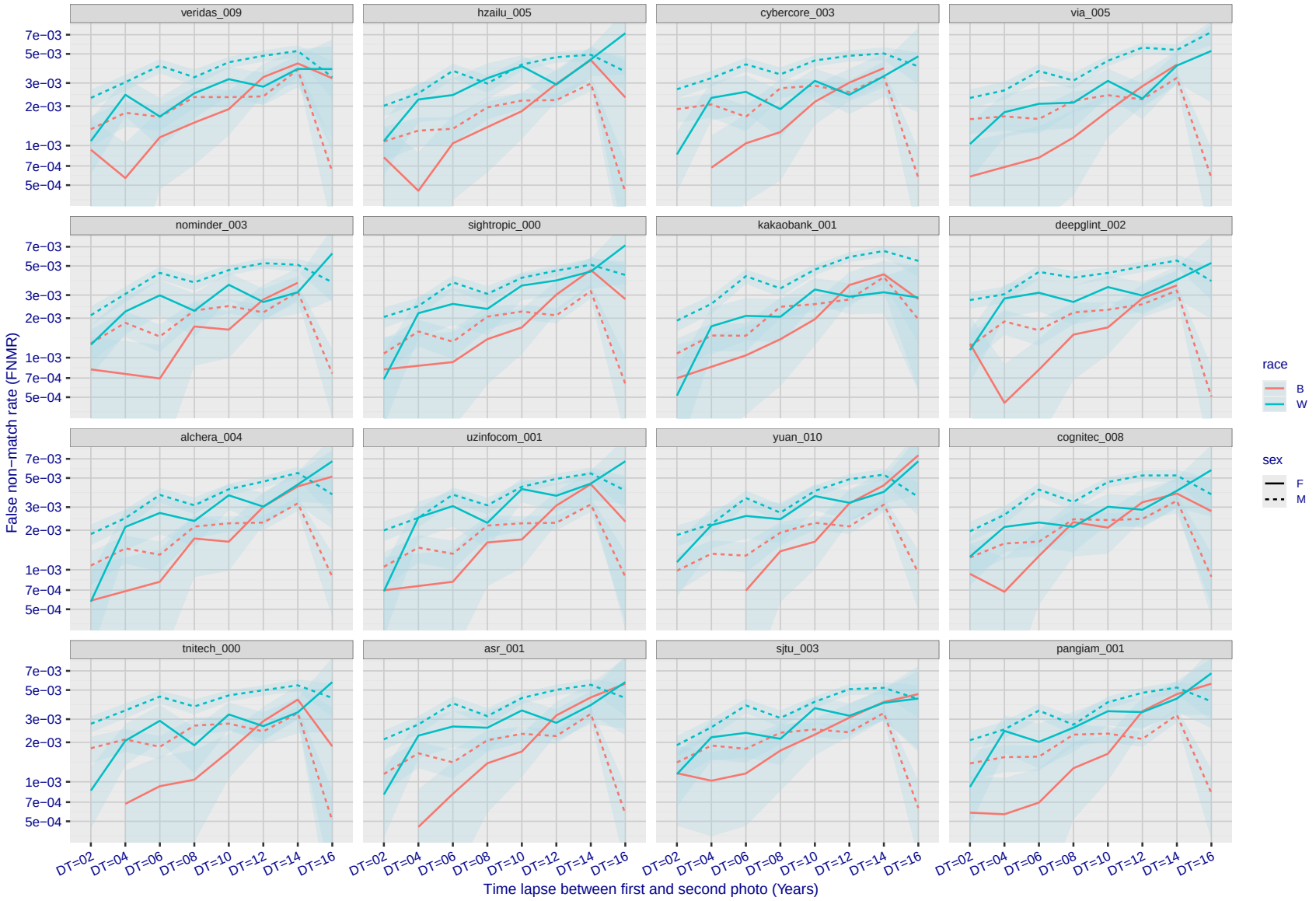


Figure 749: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

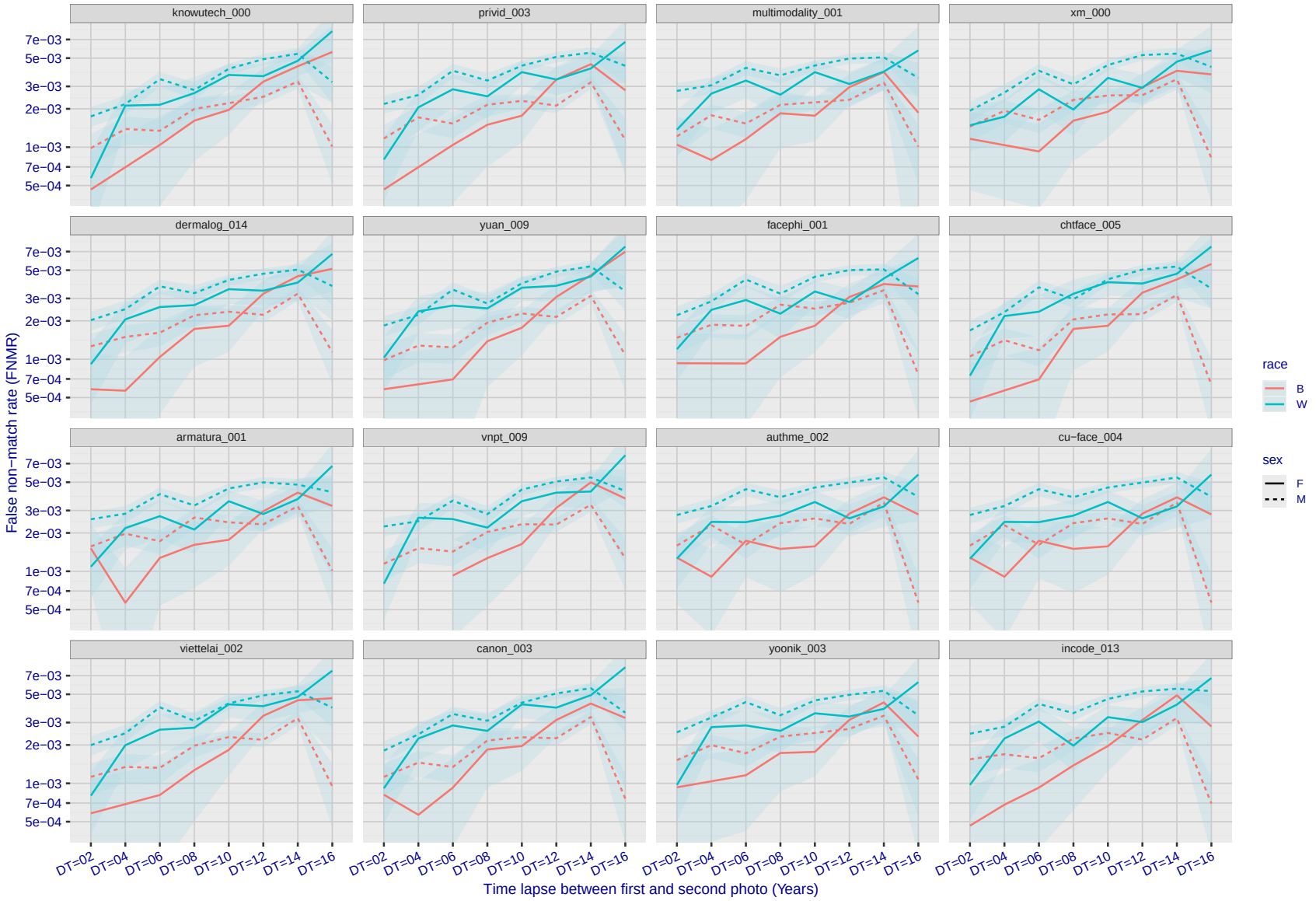


Figure 750: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

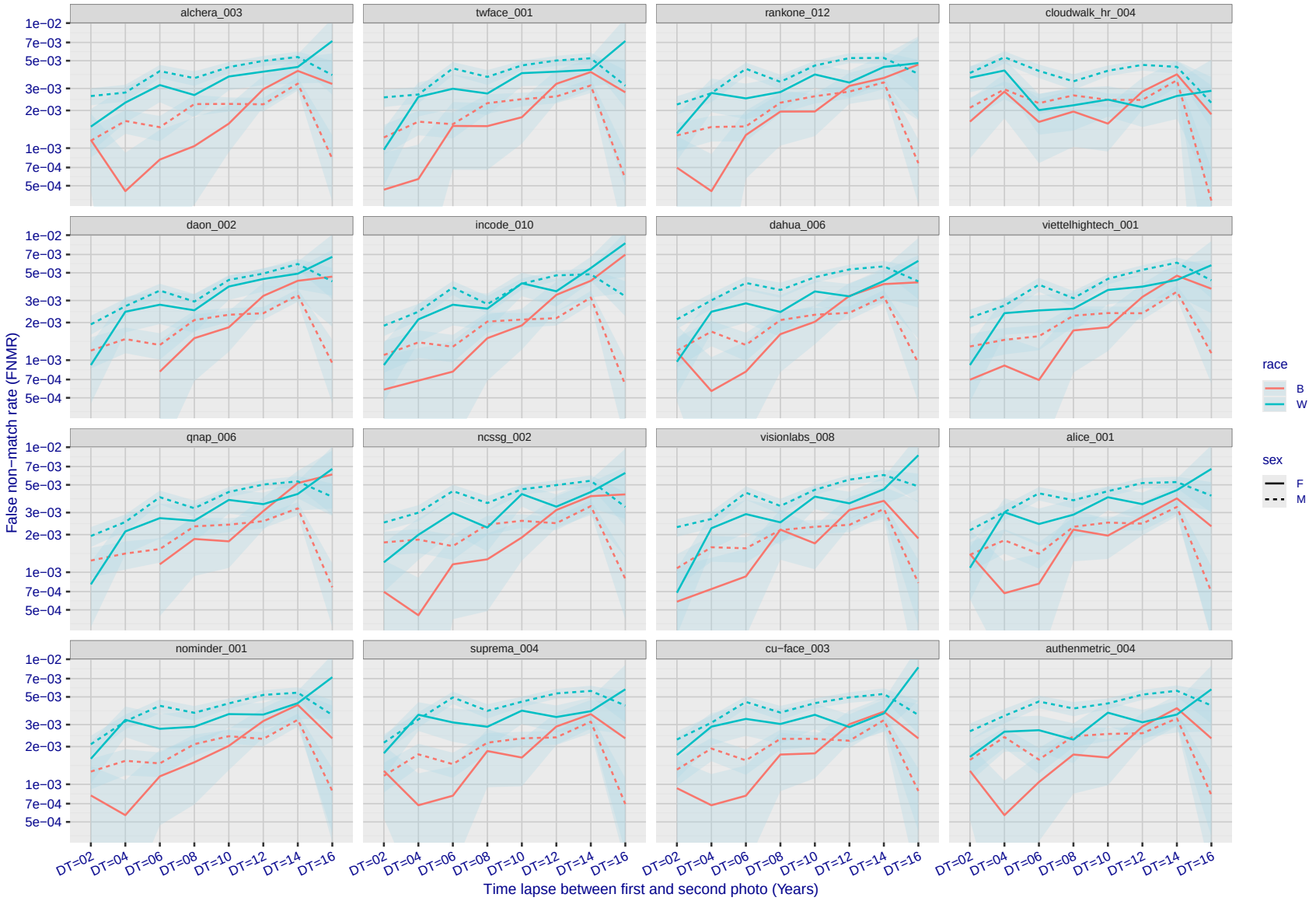


Figure 751: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

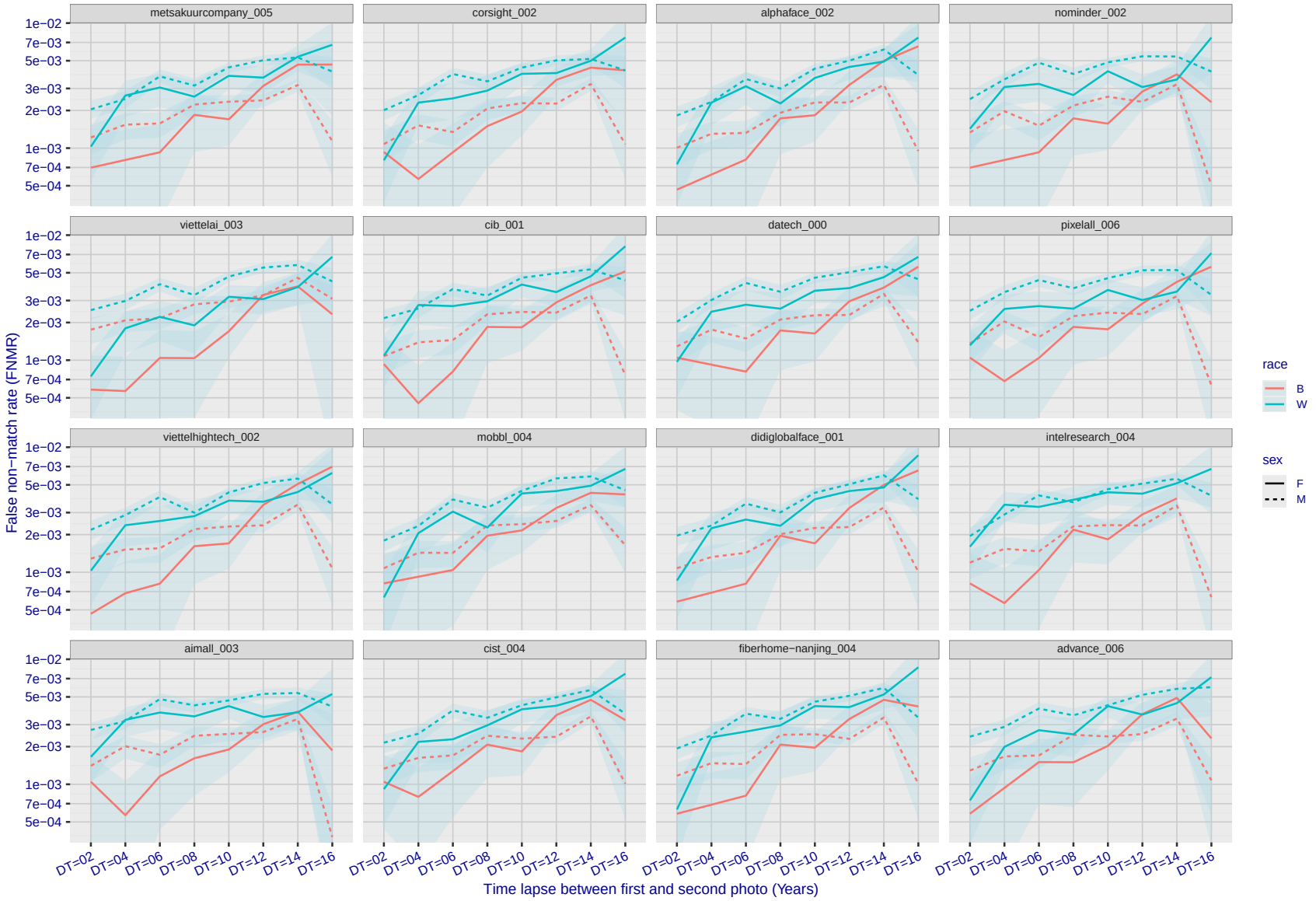


Figure 752: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

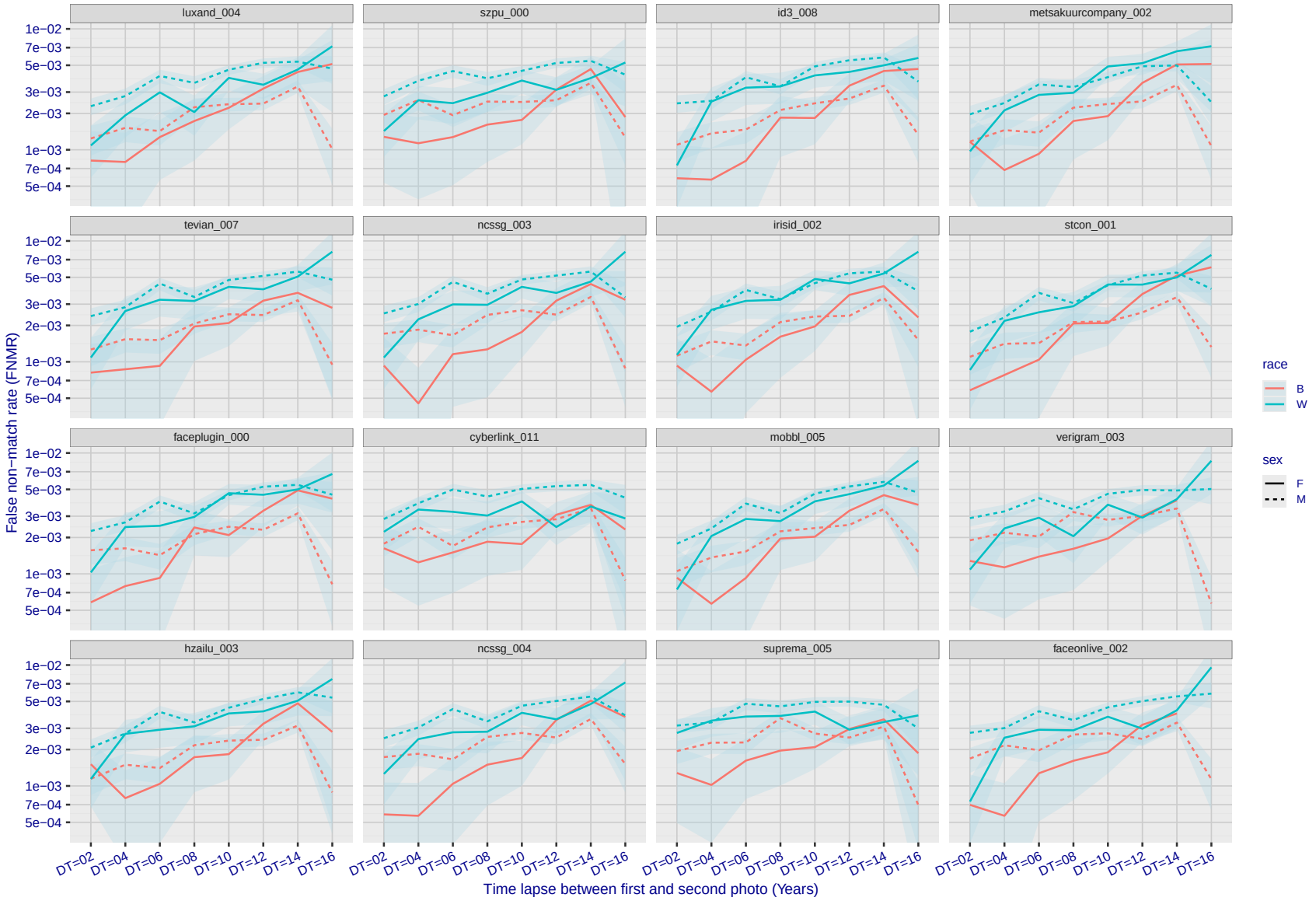


Figure 753: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

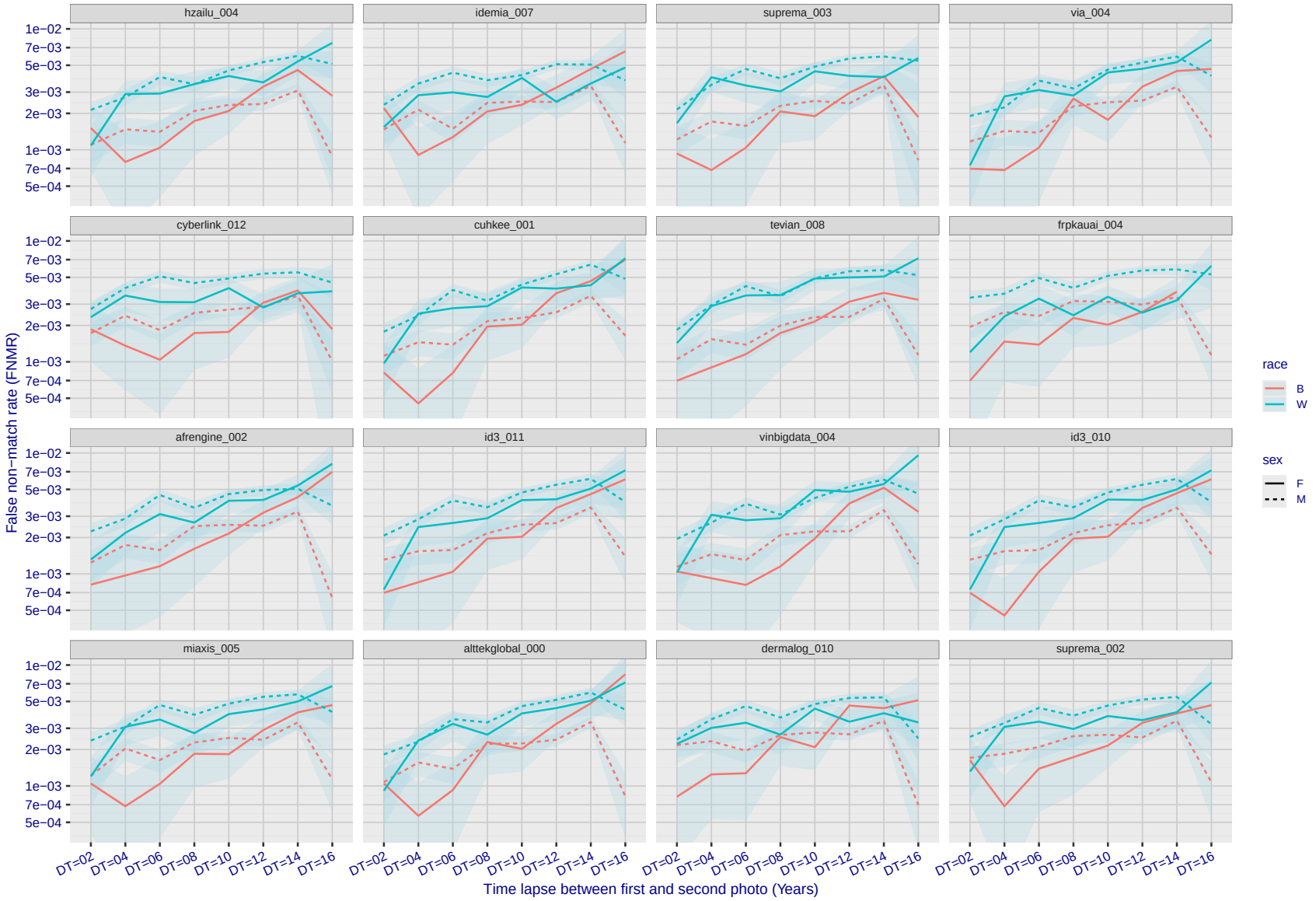


Figure 754: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

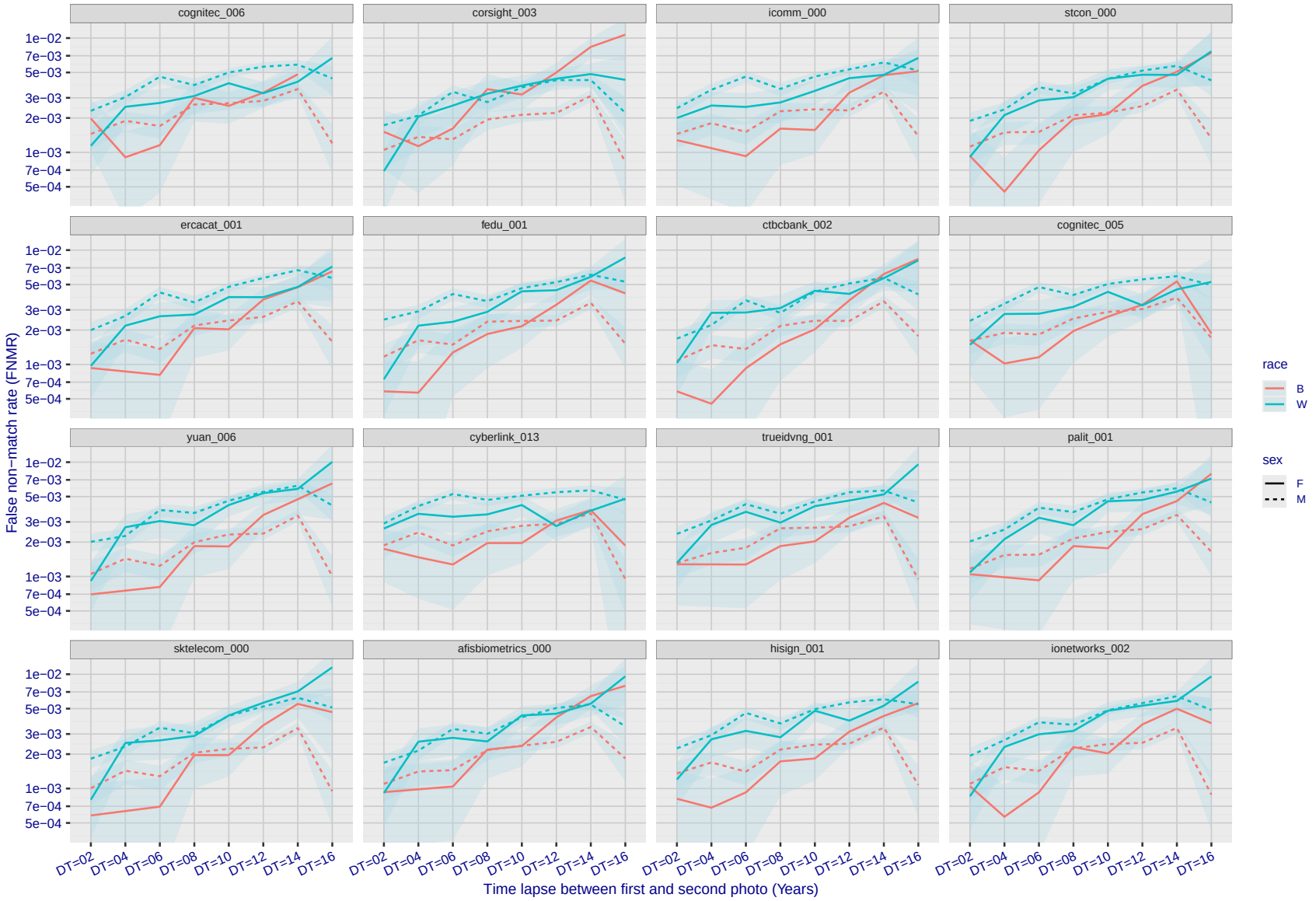


Figure 755: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\tilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

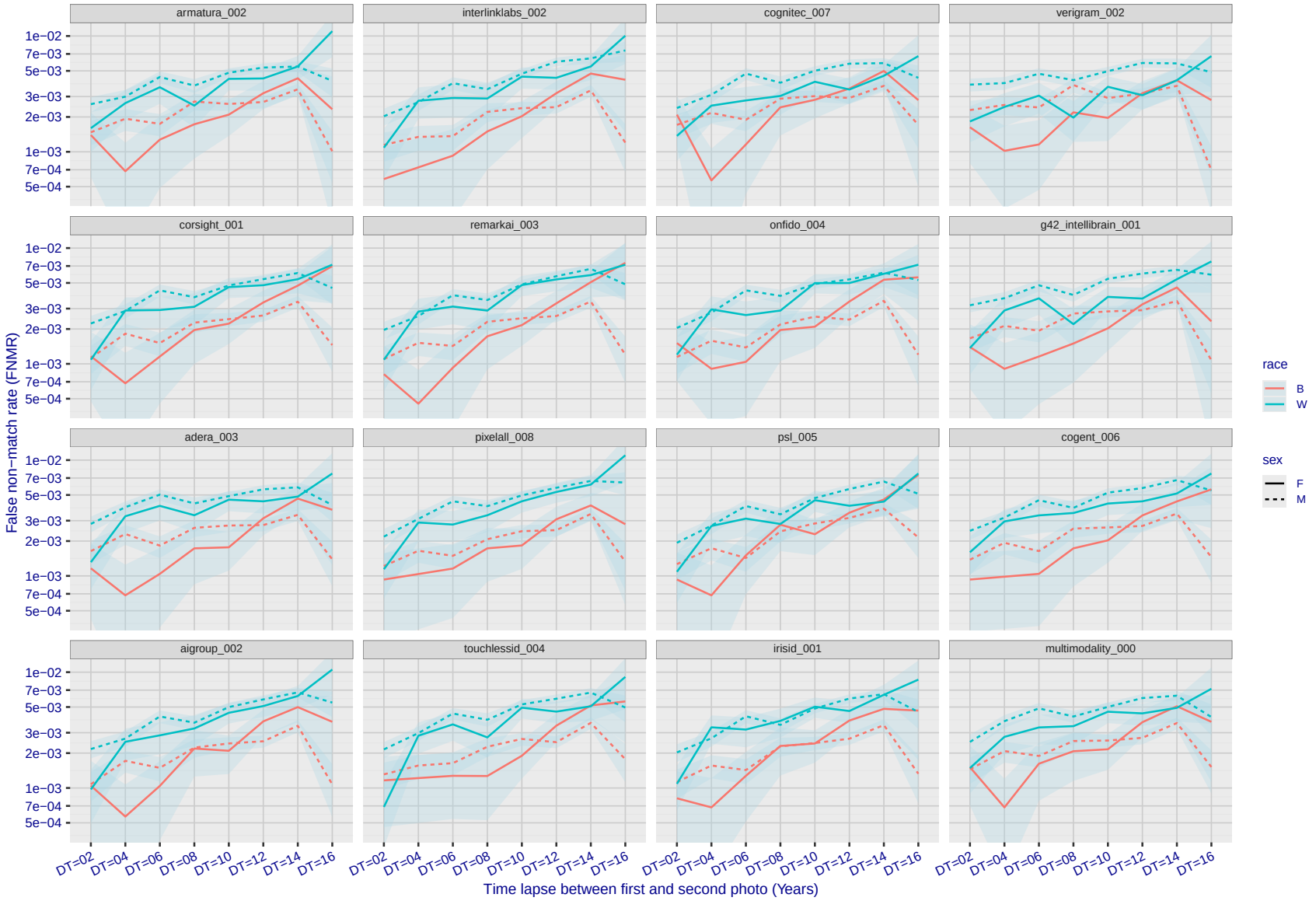


Figure 756: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

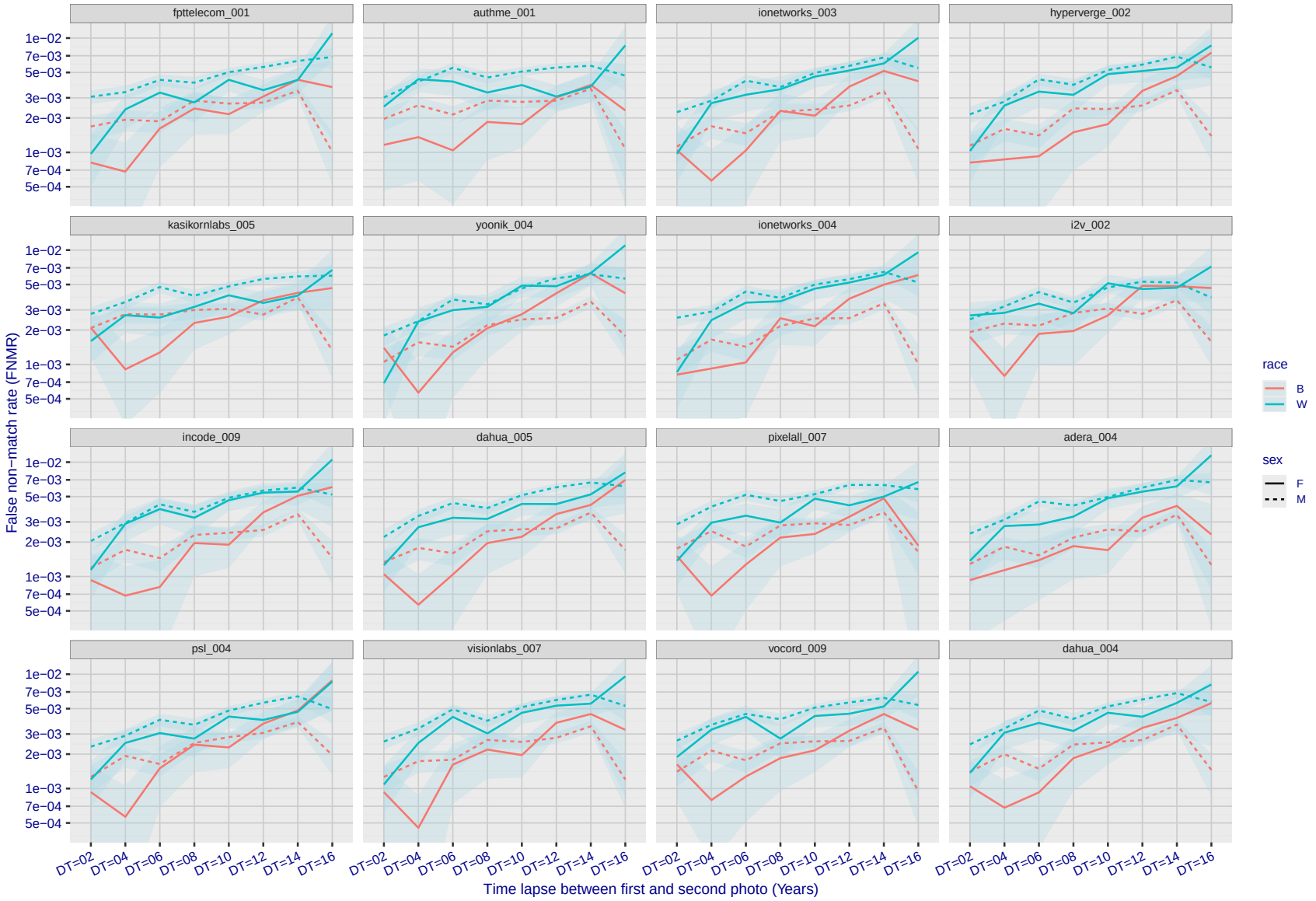


Figure 757: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

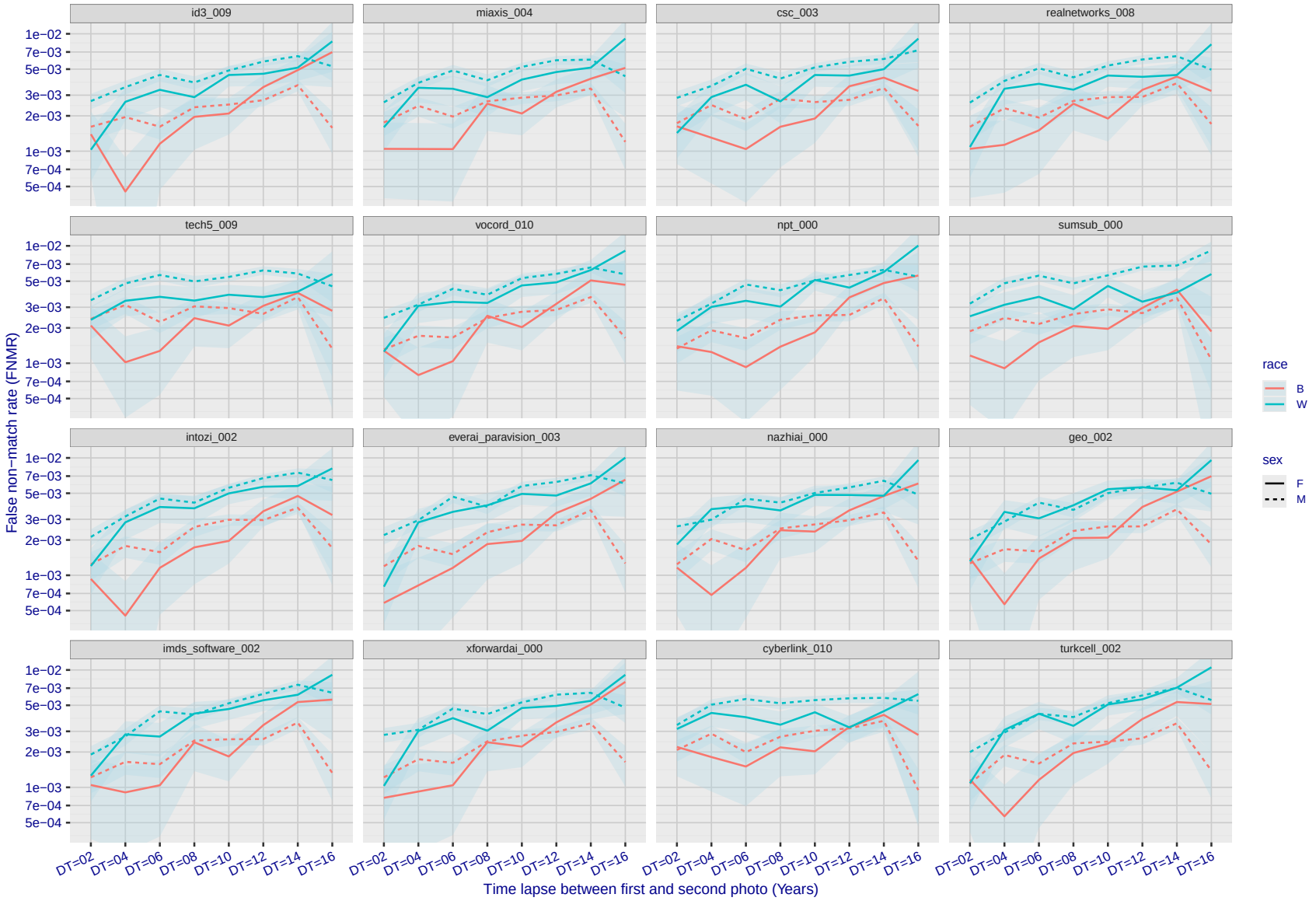


Figure 758: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

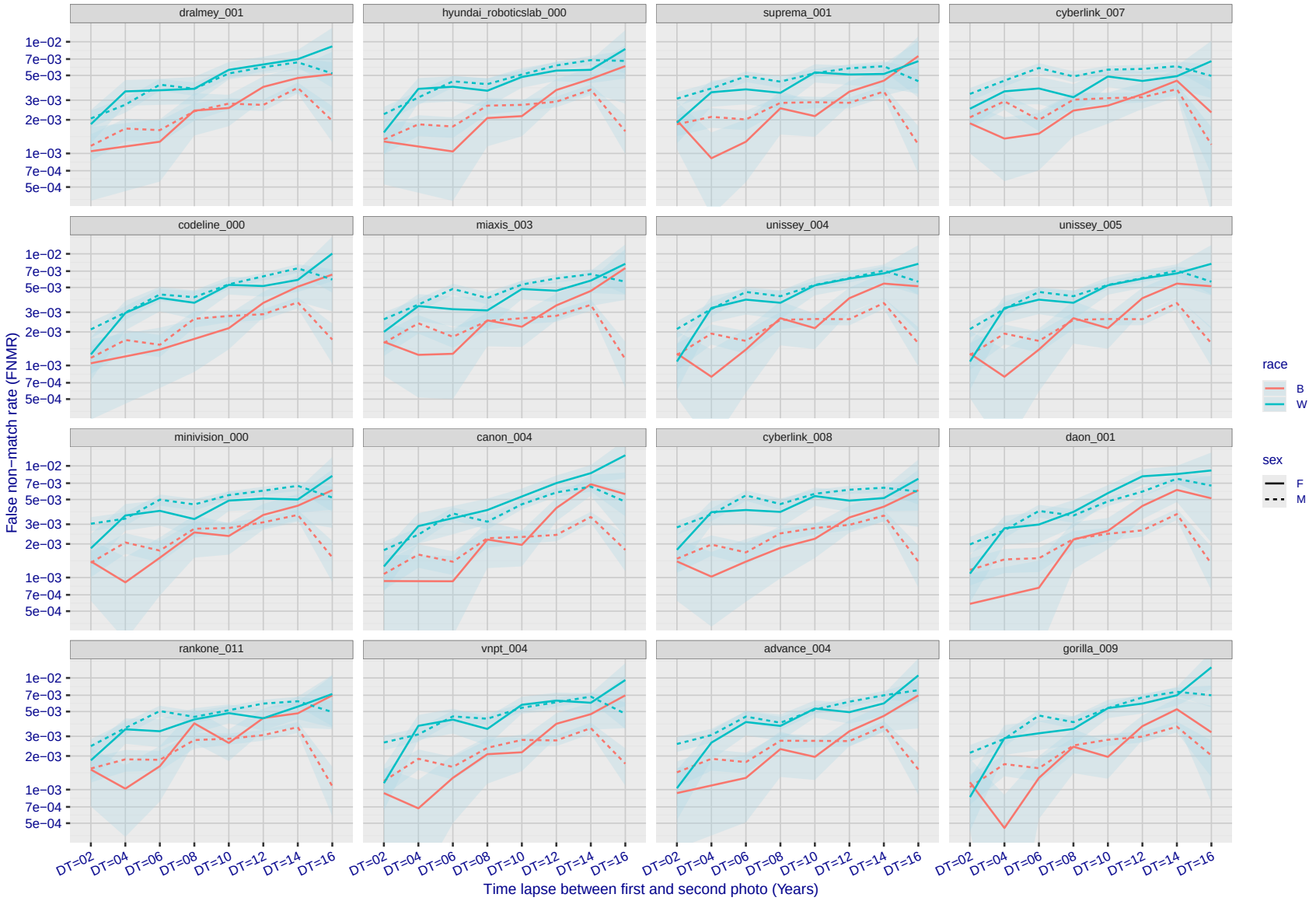


Figure 759: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

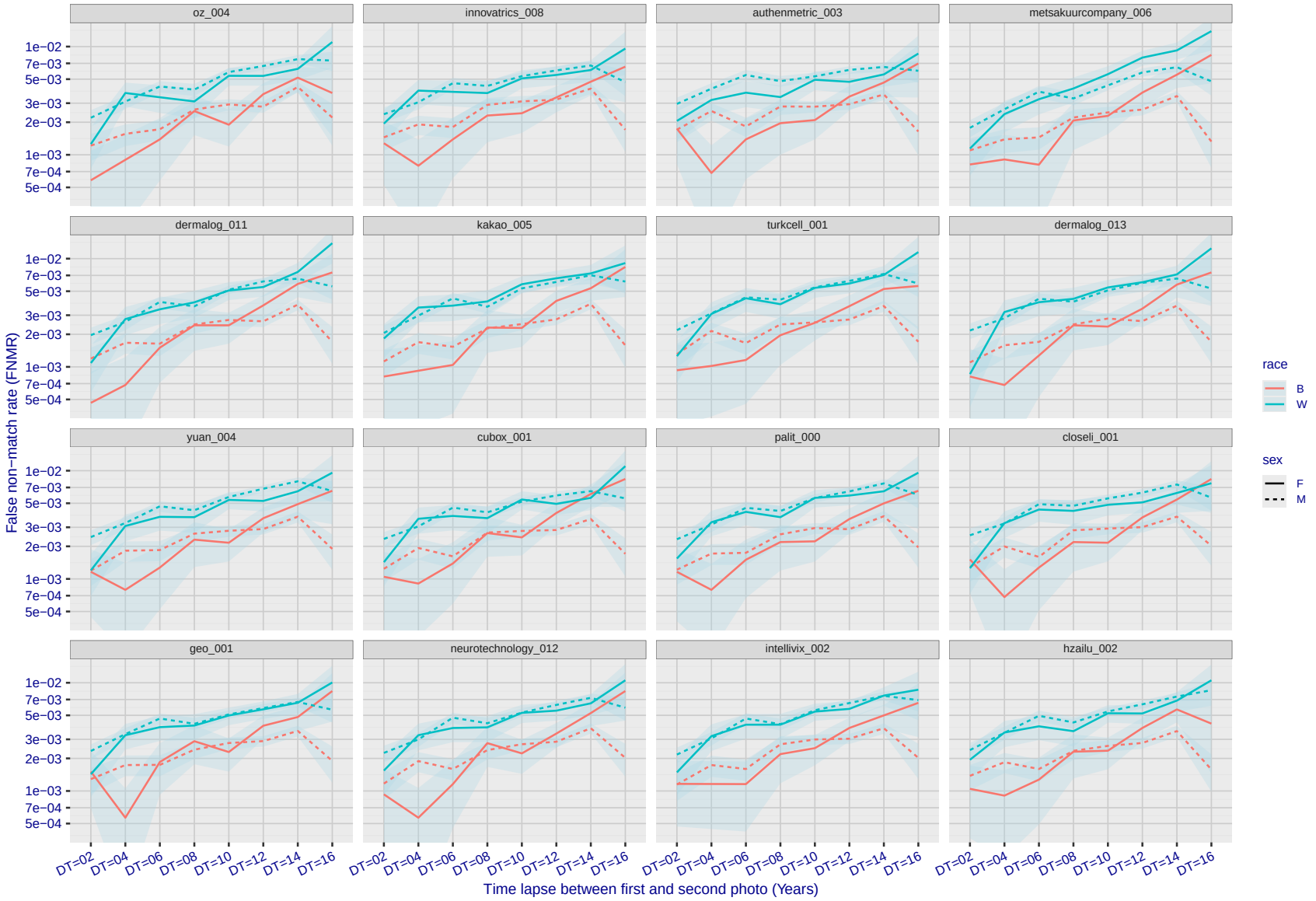


Figure 760: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

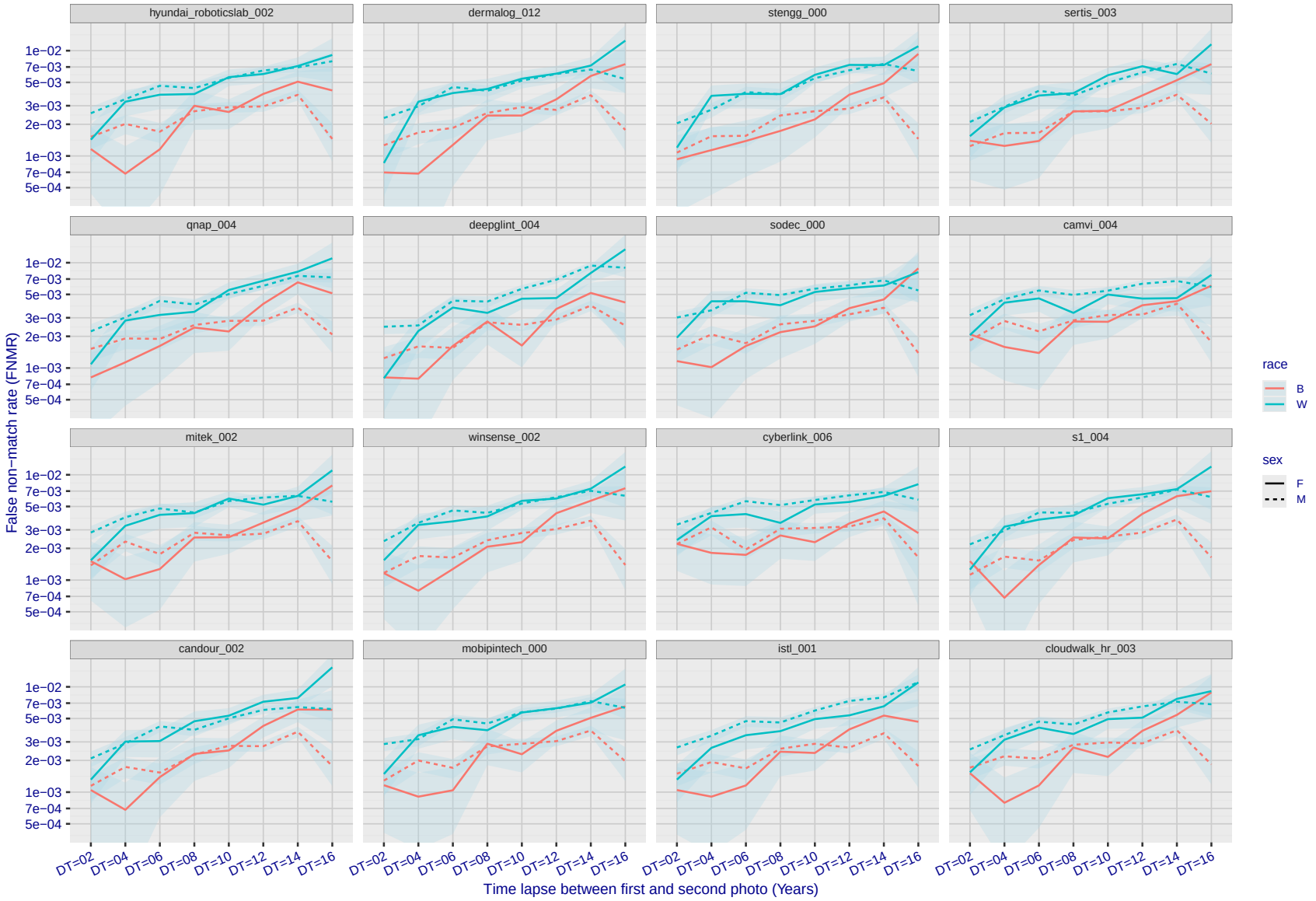


Figure 761: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

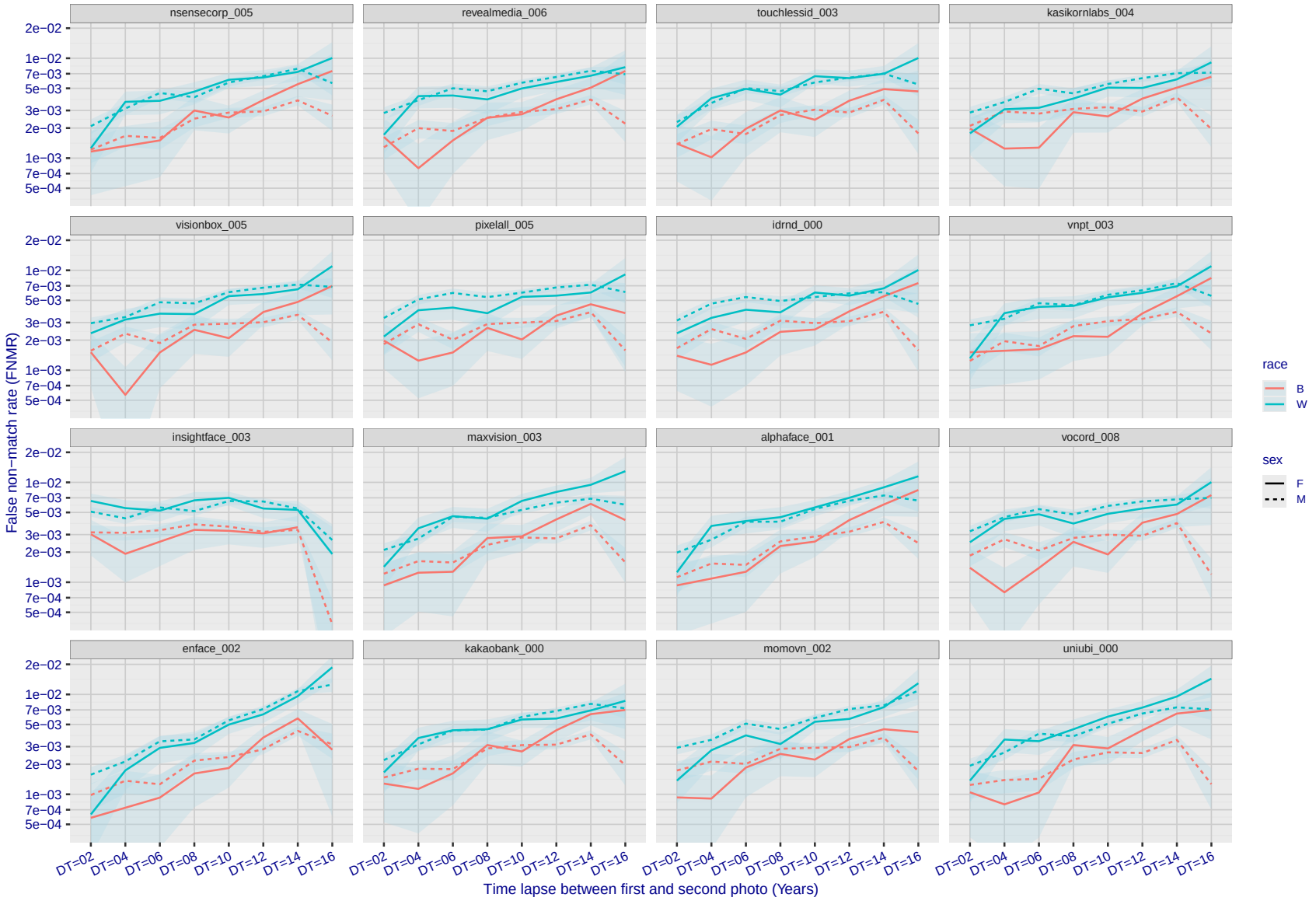


Figure 762: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

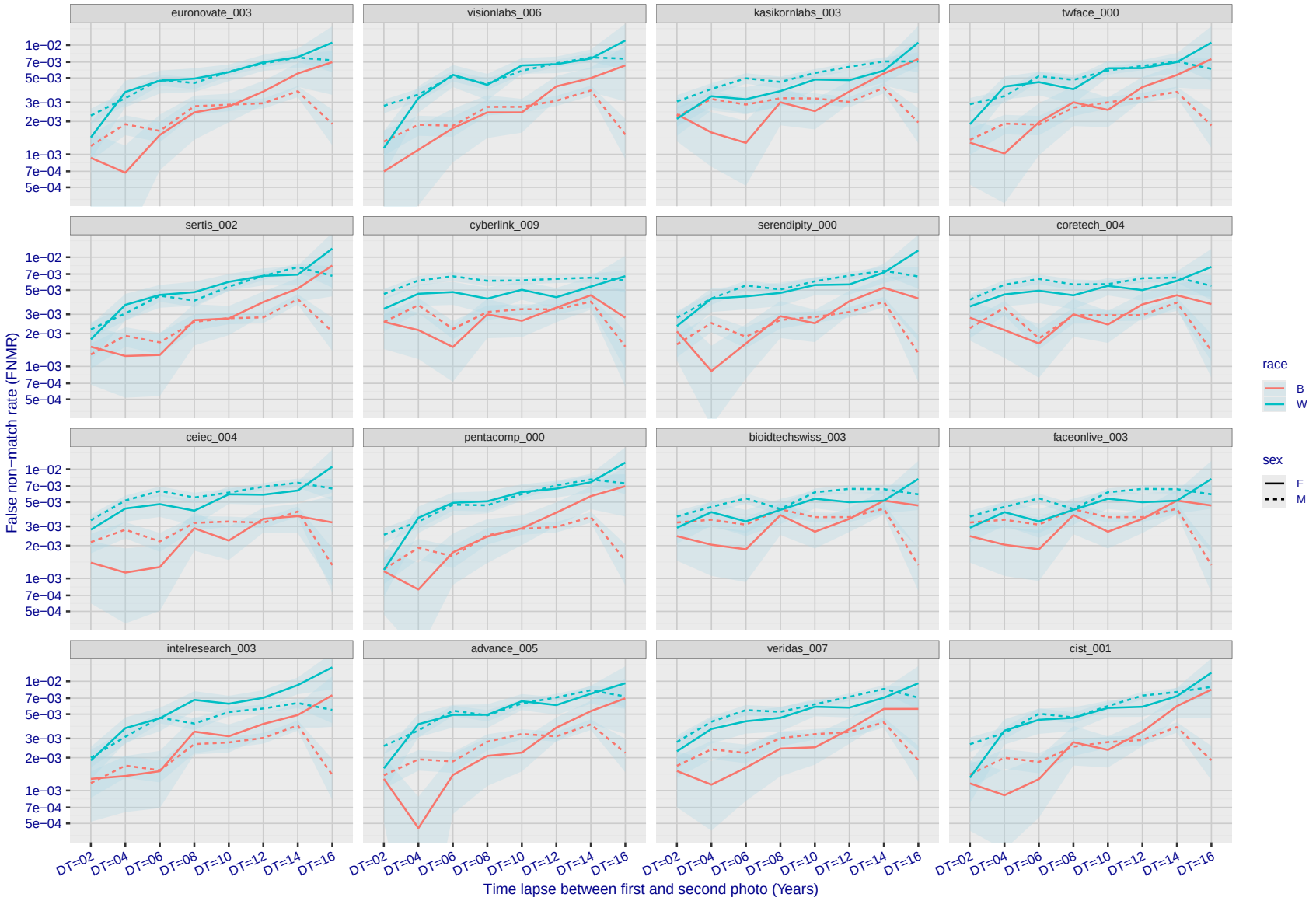


Figure 763: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

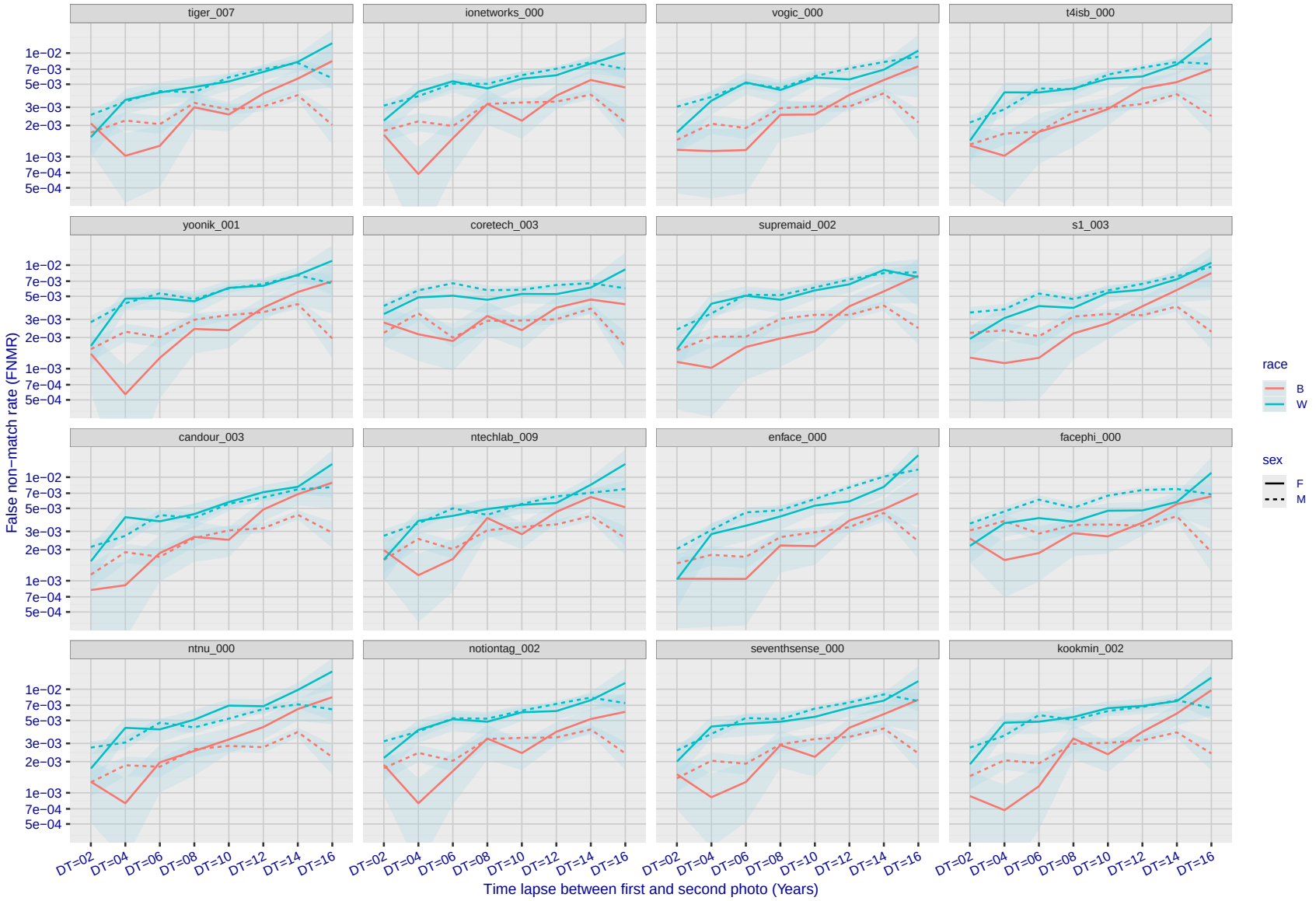


Figure 764: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

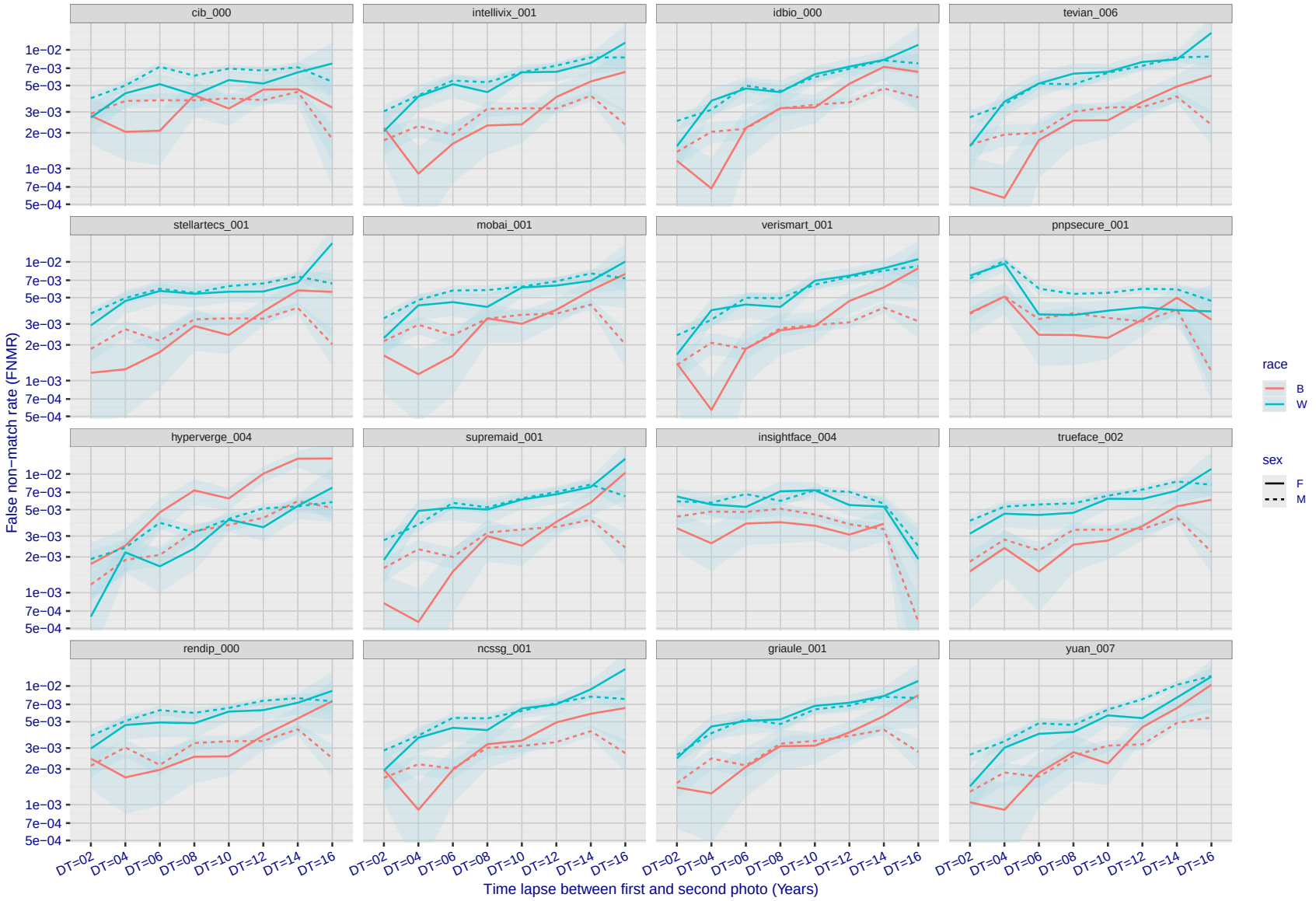


Figure 765: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

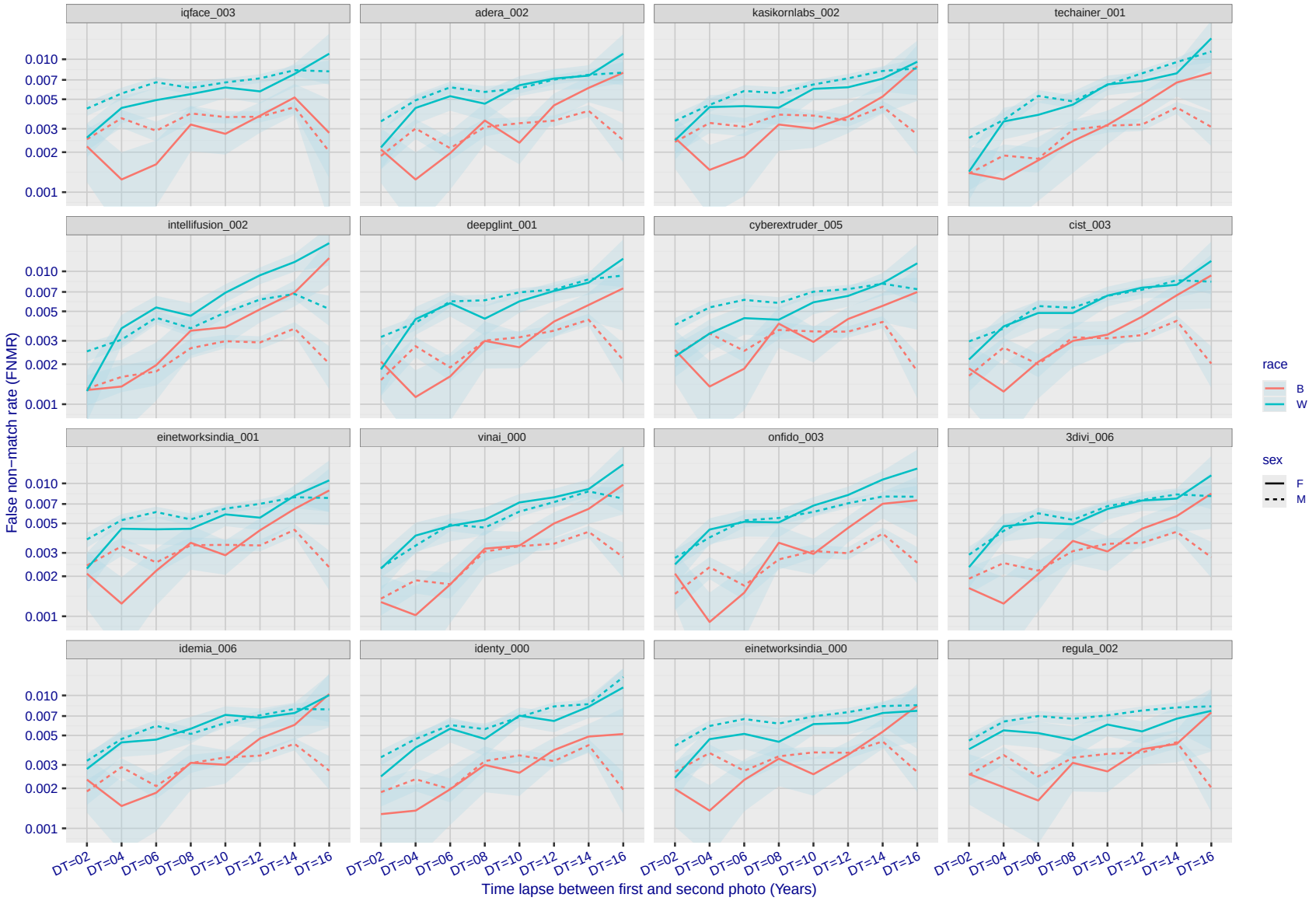


Figure 766: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

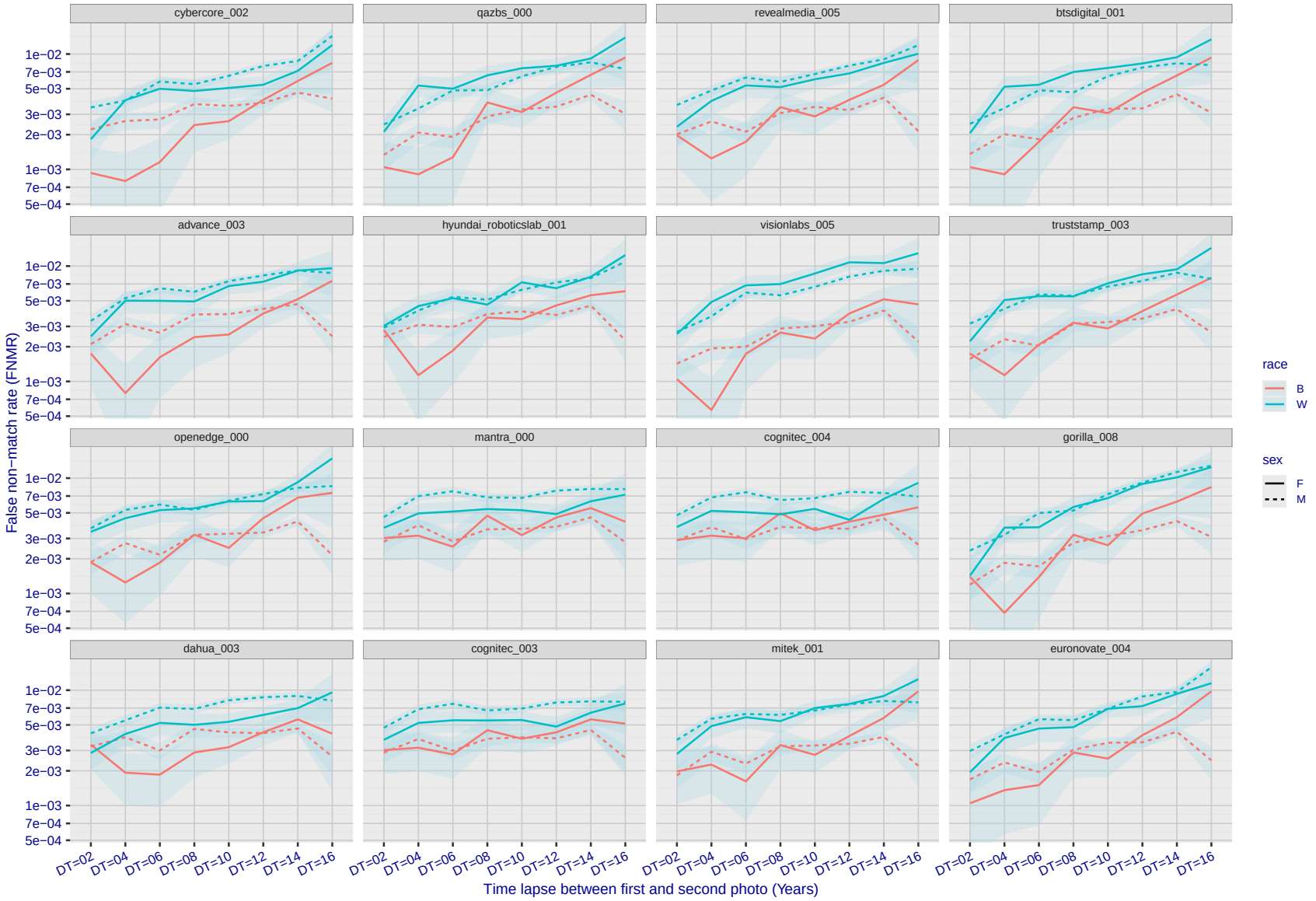


Figure 767: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

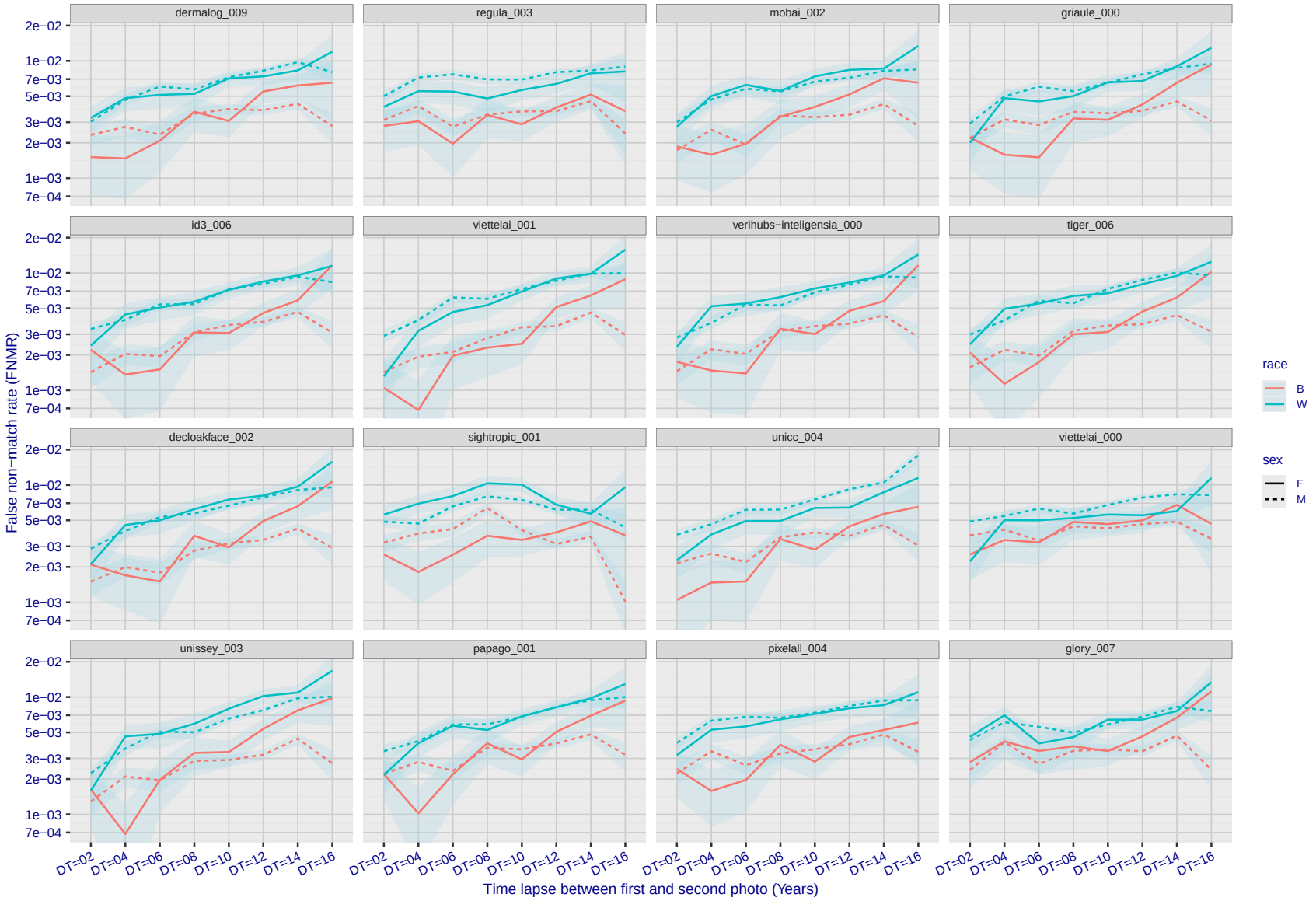


Figure 768: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

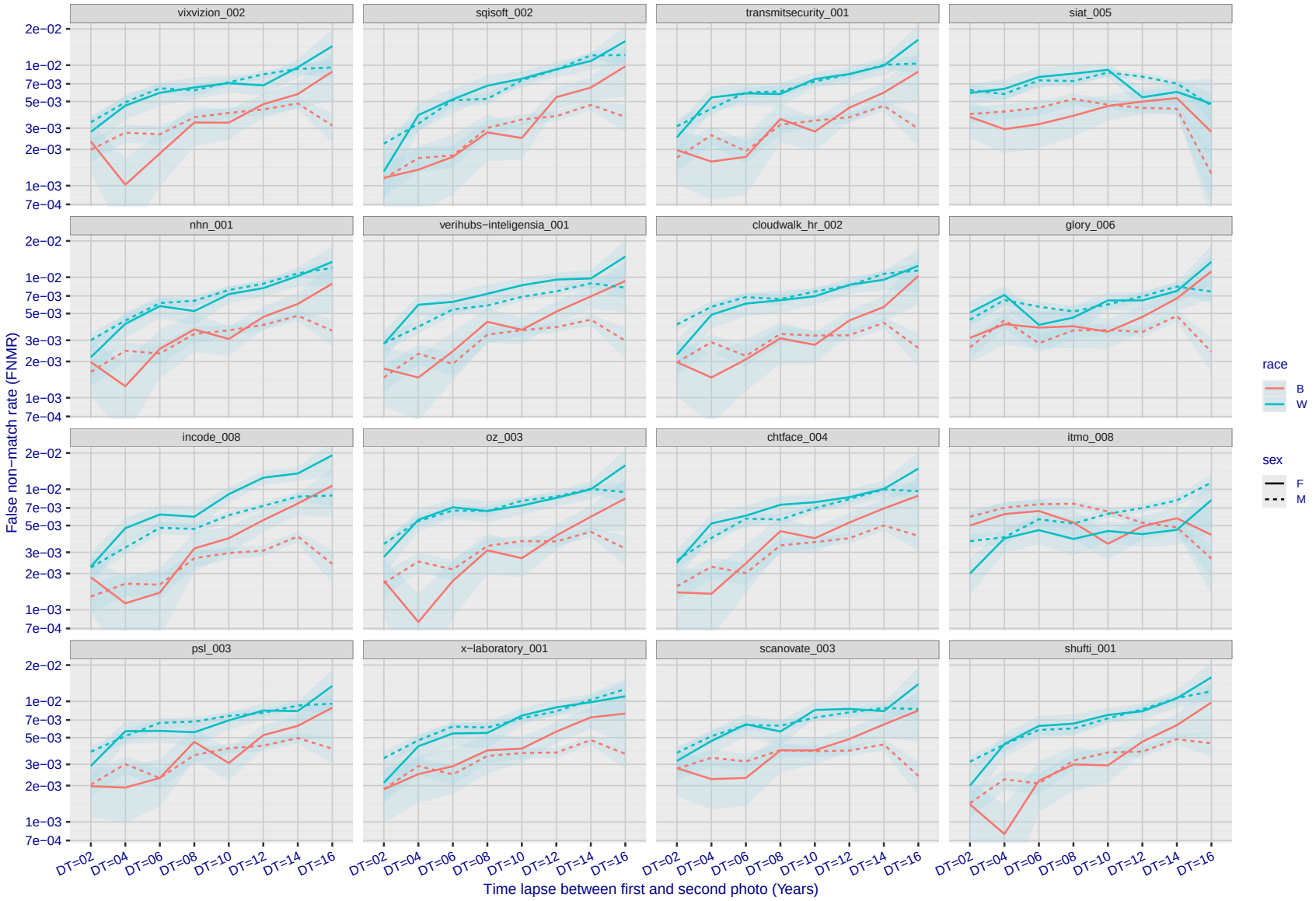


Figure 769: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

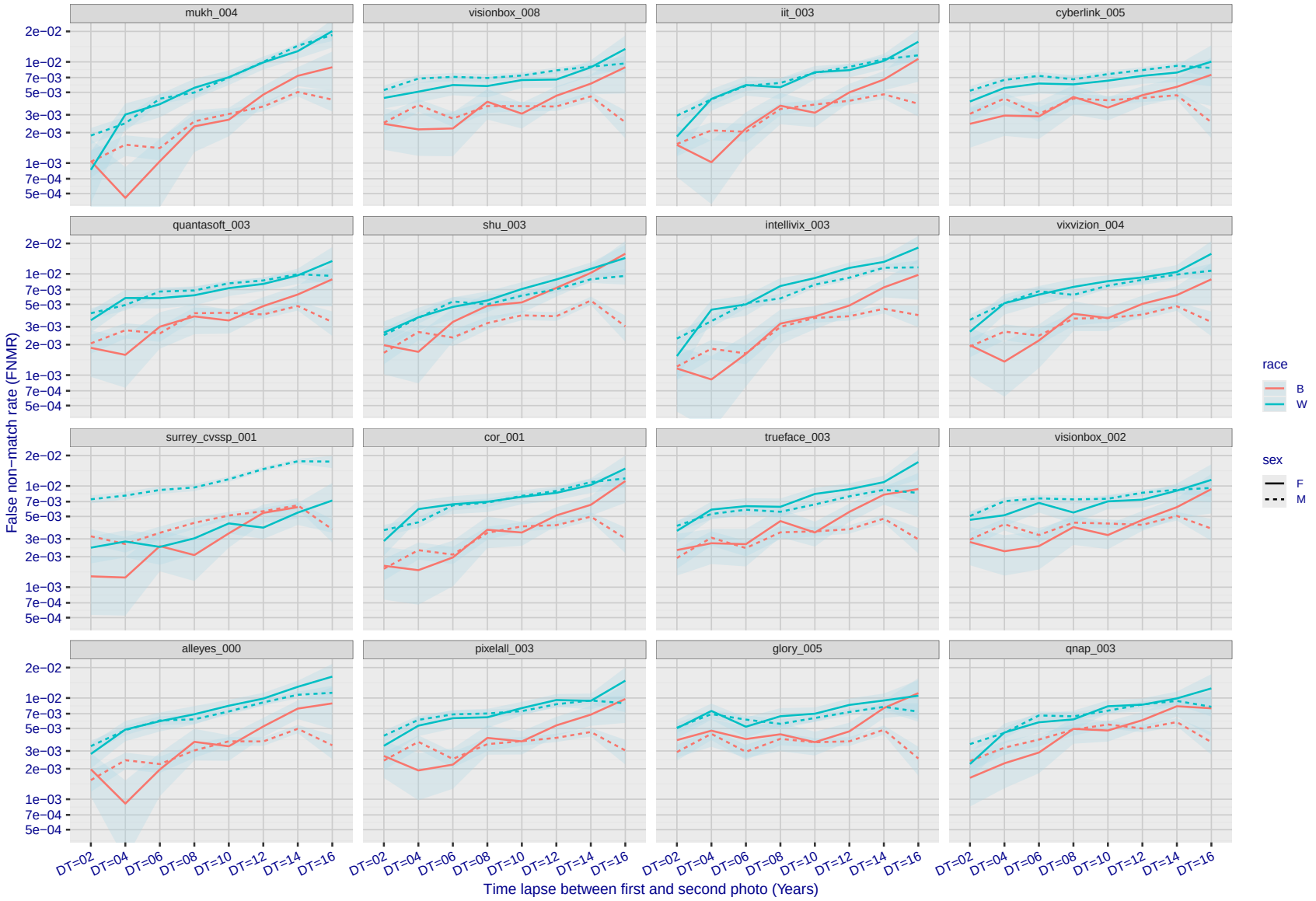


Figure 770: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

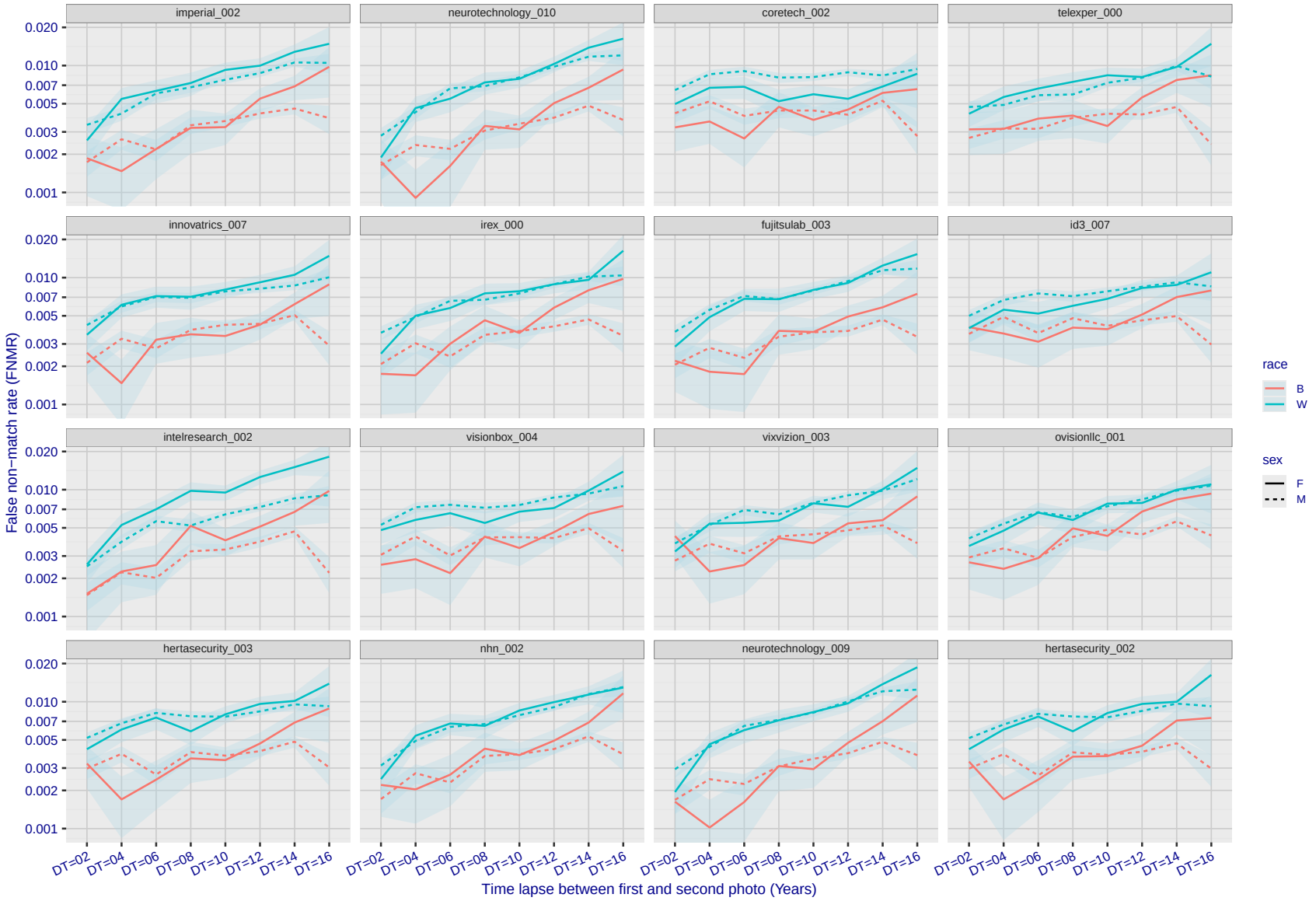


Figure 771: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

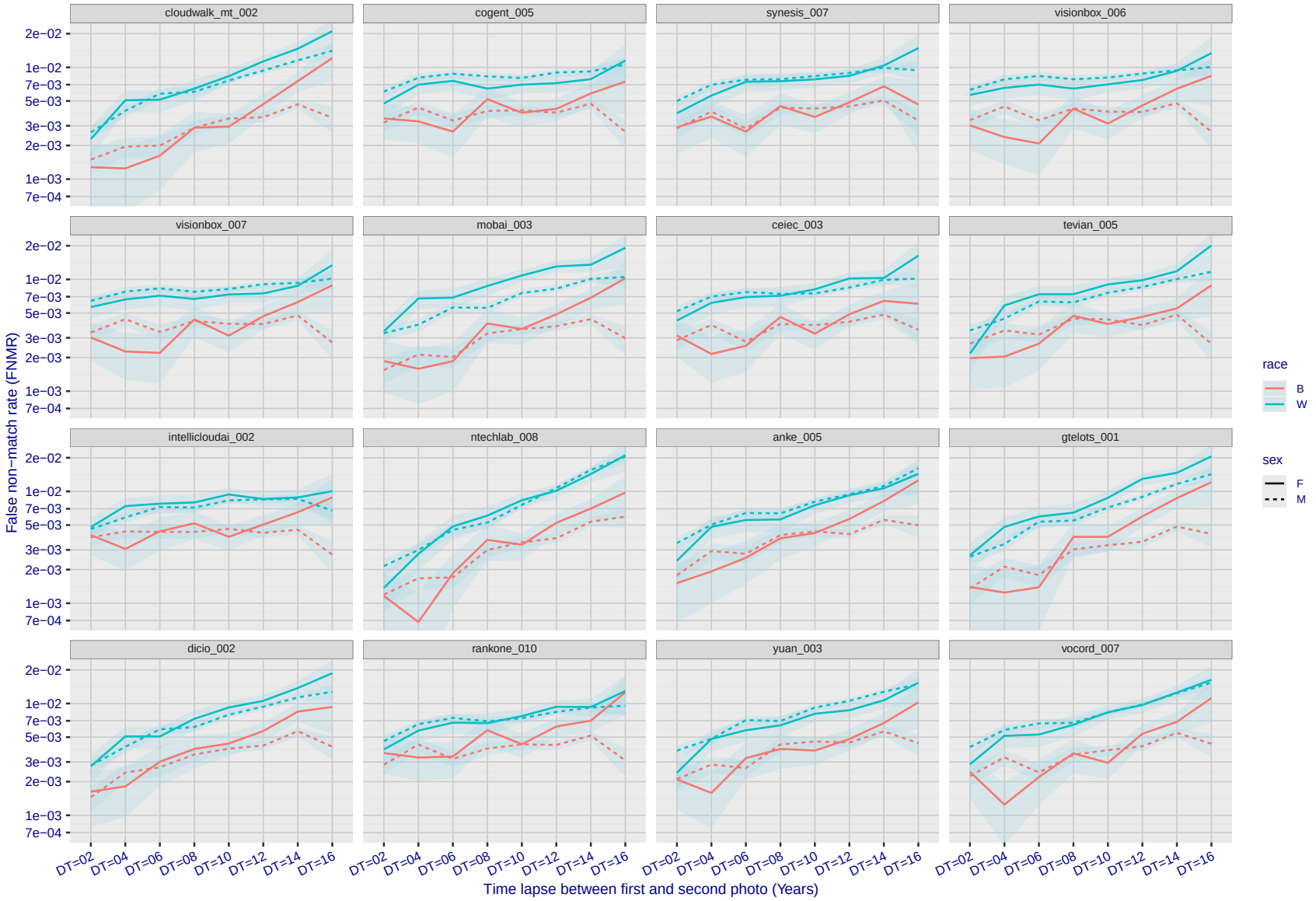


Figure 772: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

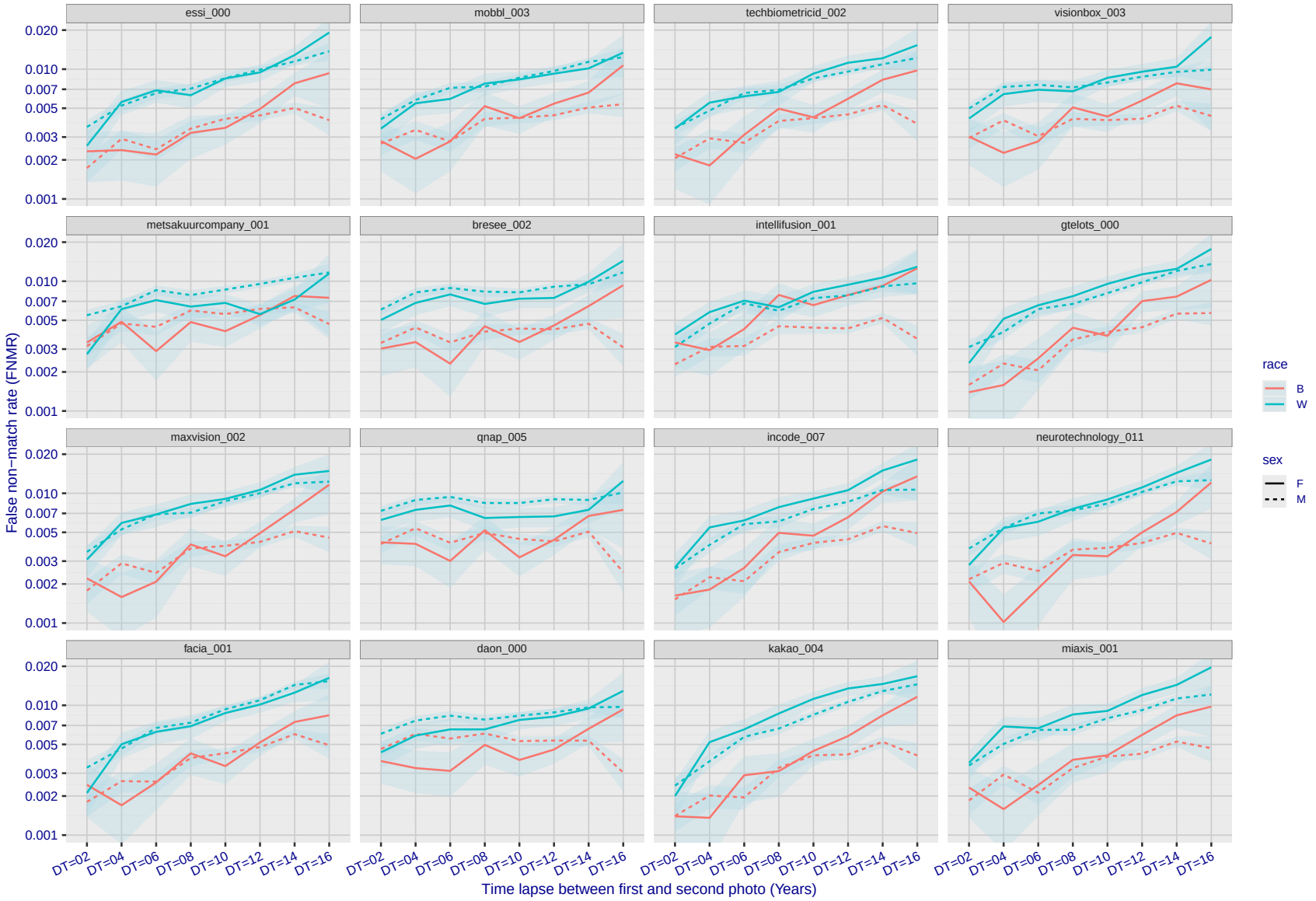


Figure 773: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

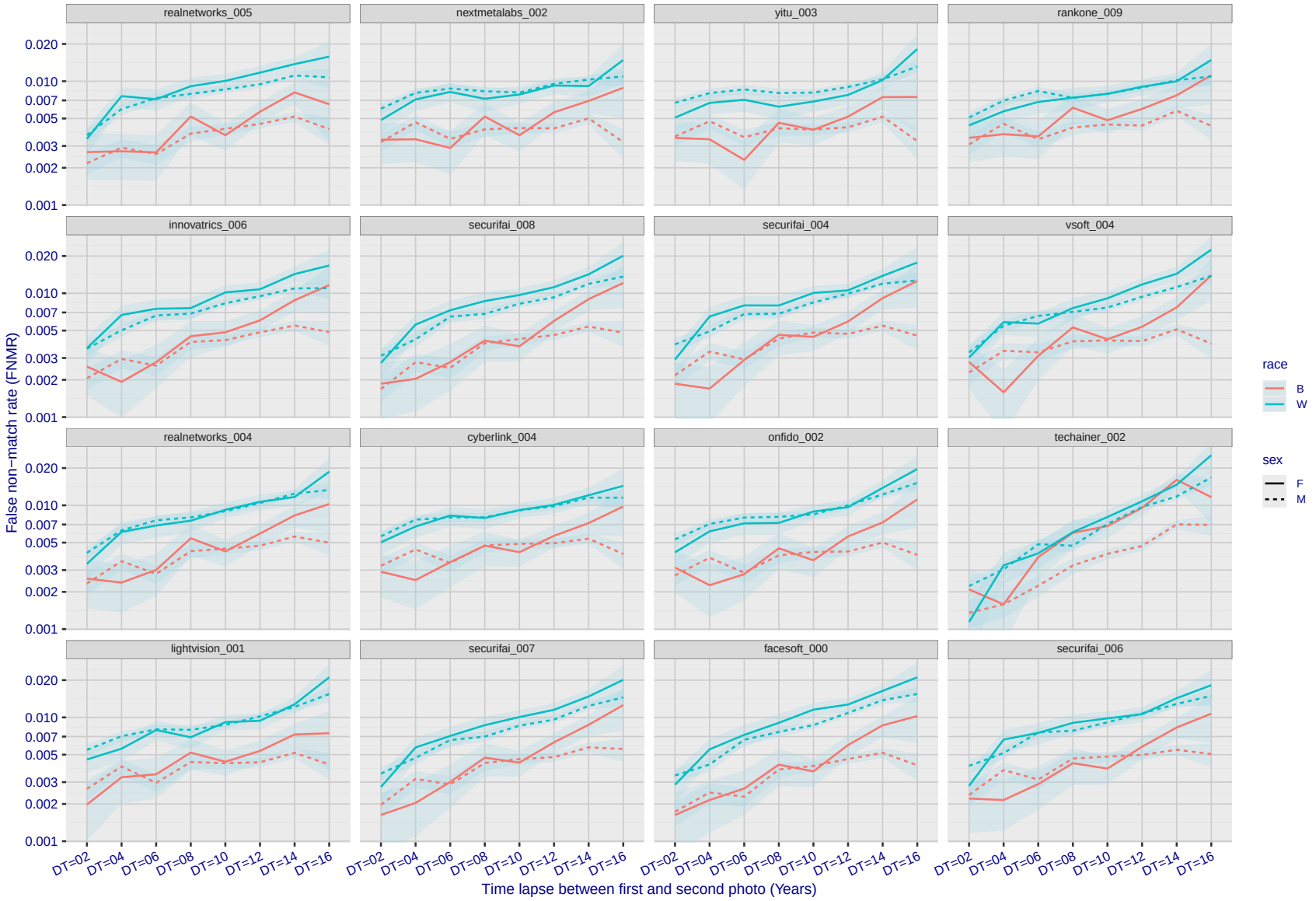


Figure 774: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

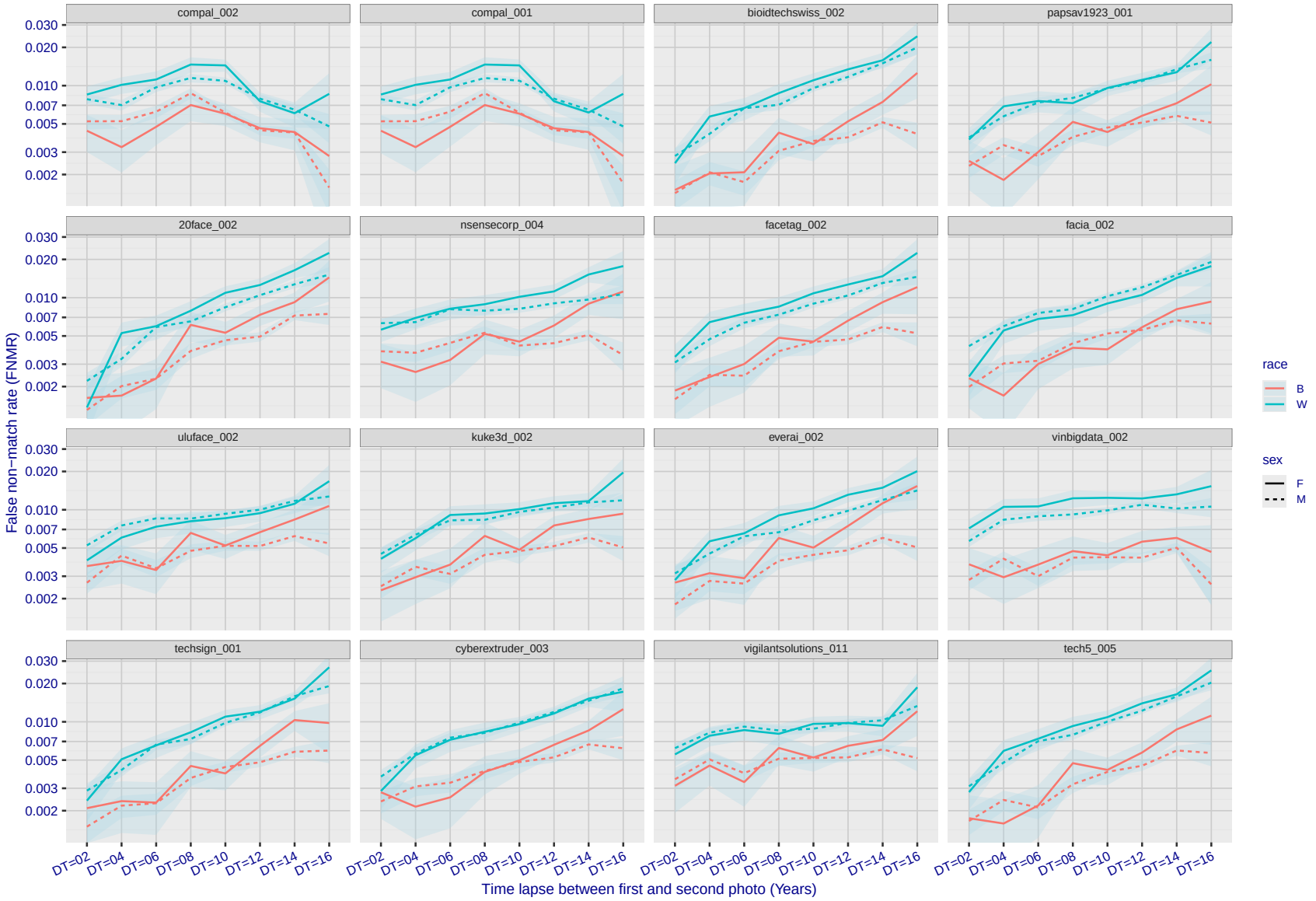


Figure 775: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

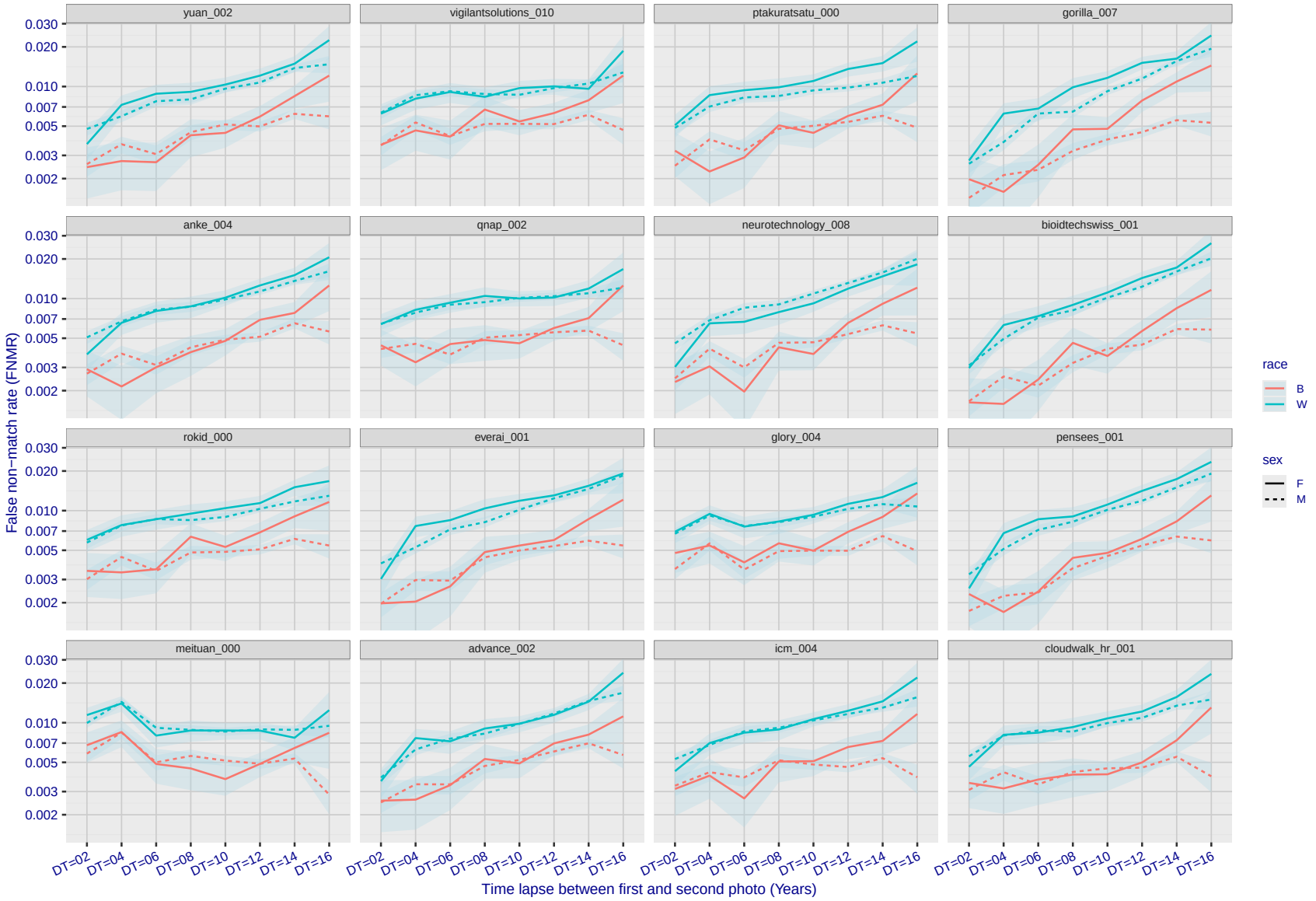


Figure 776: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

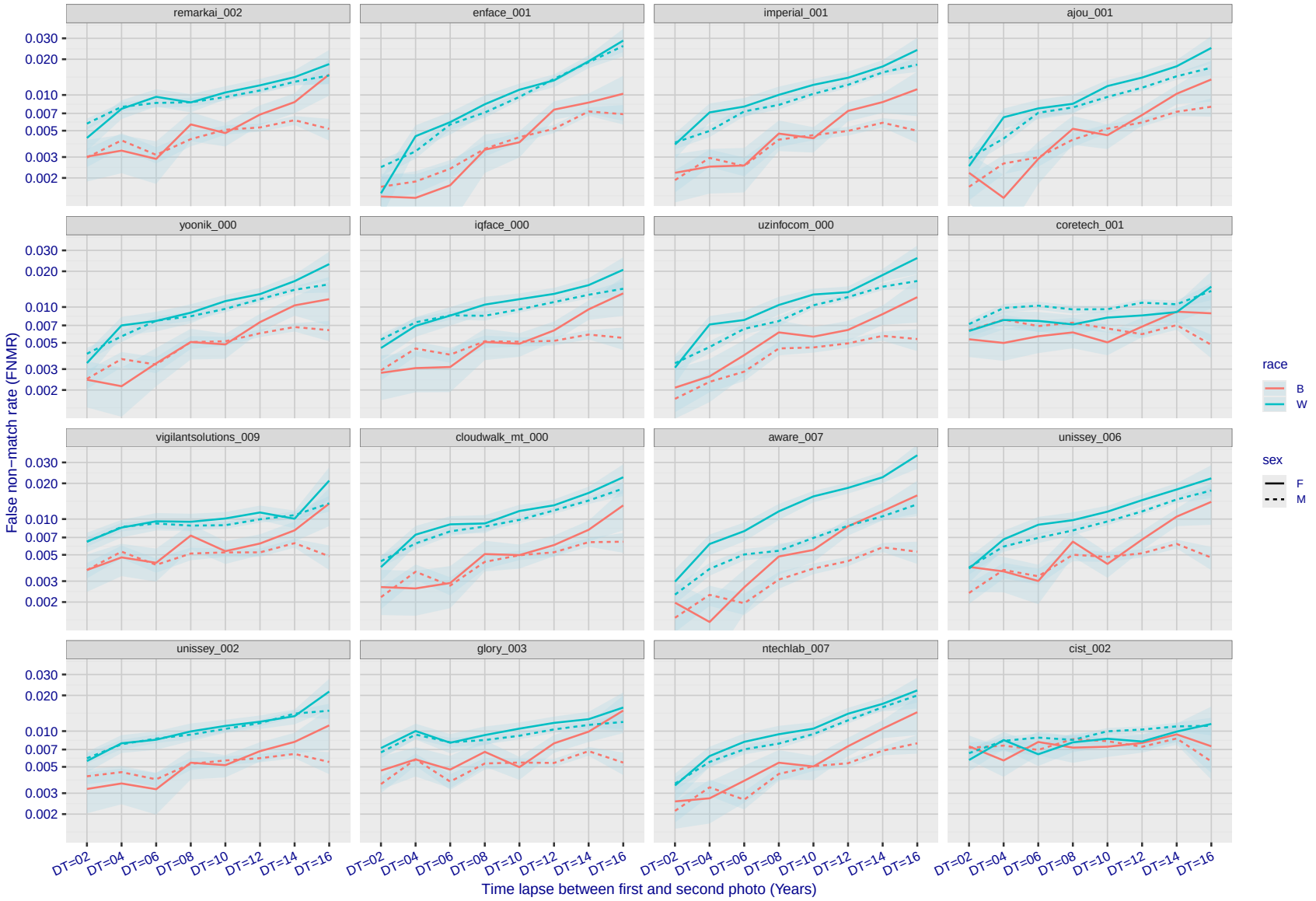


Figure 777: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

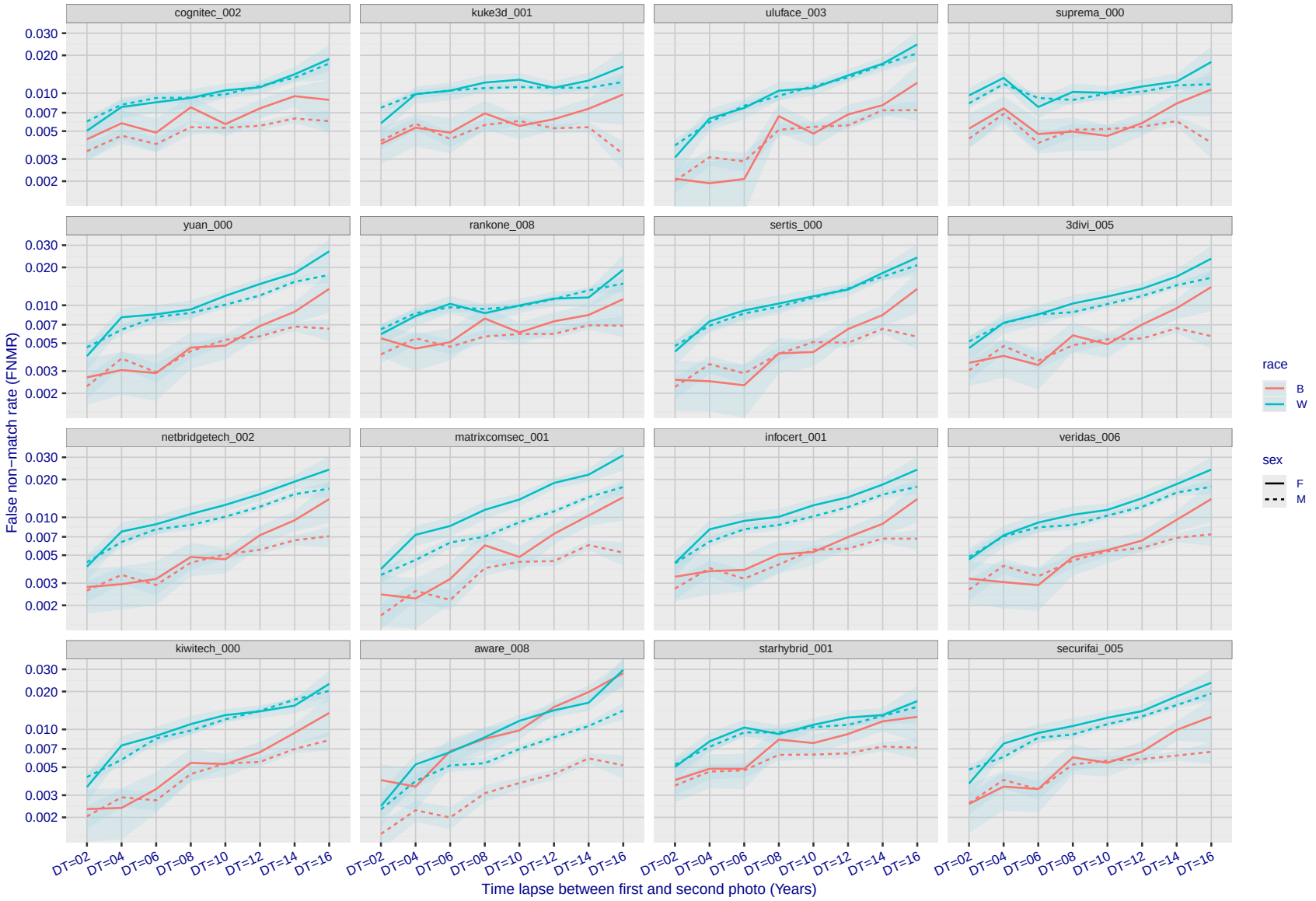


Figure 778: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

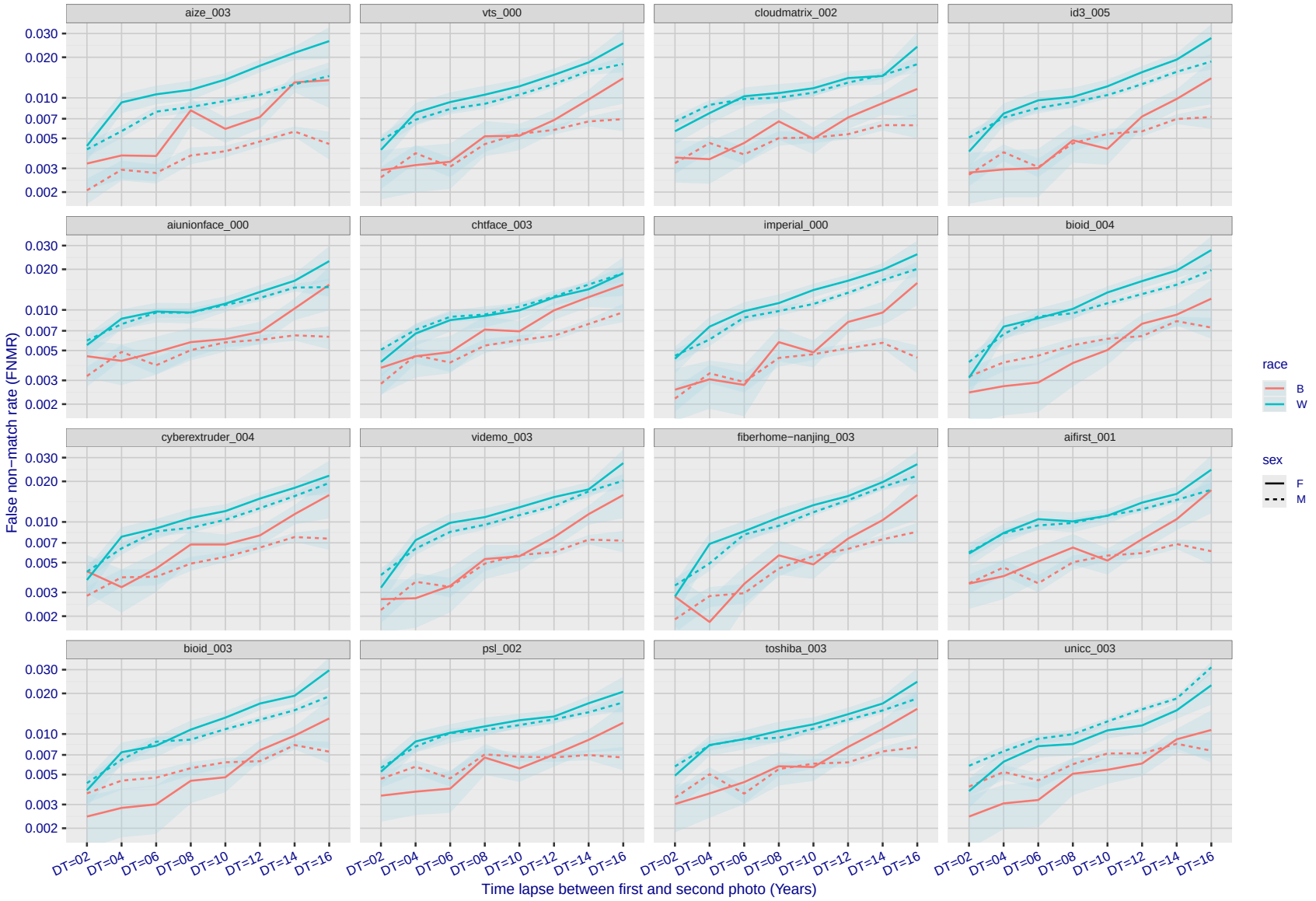


Figure 779: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

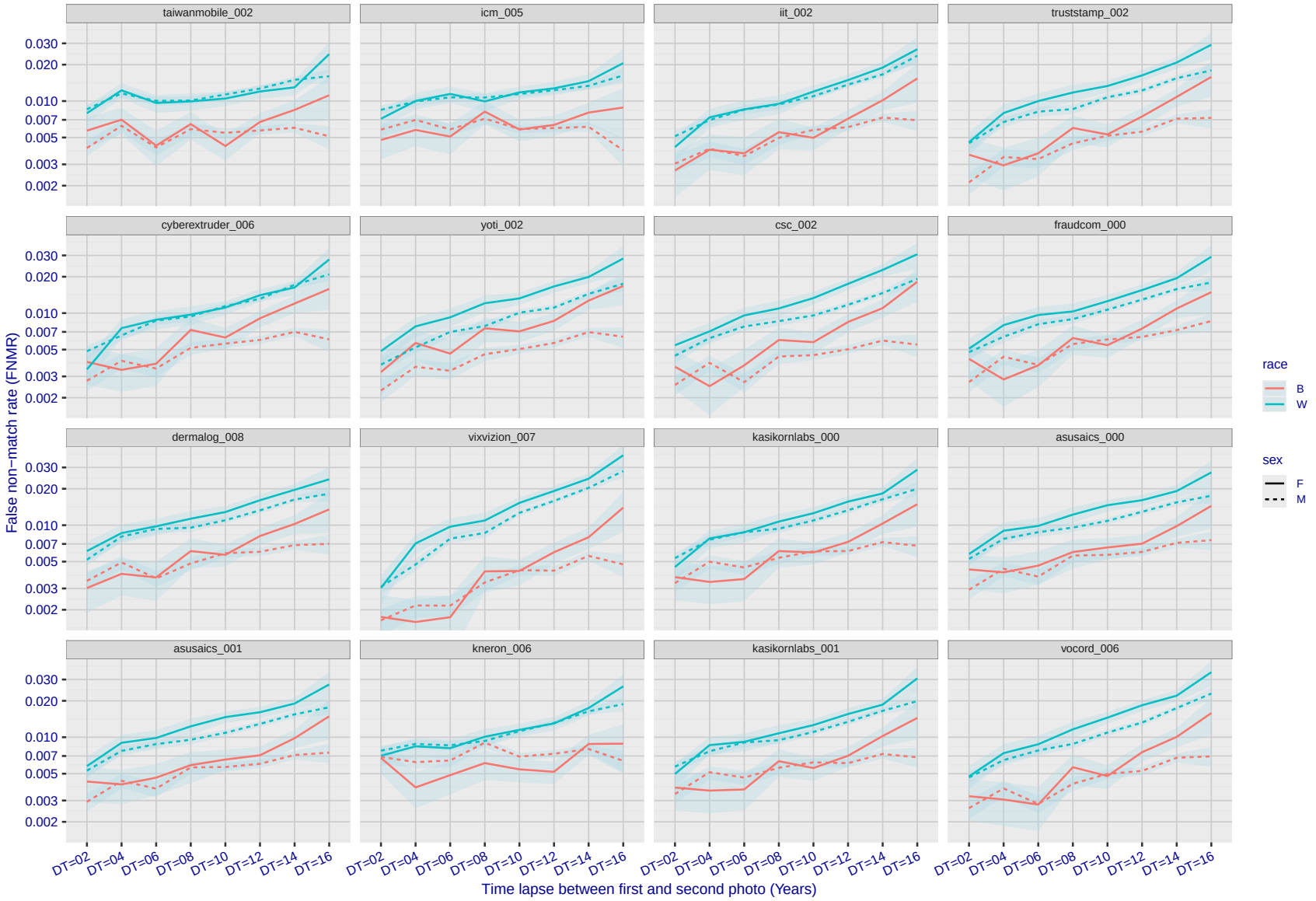


Figure 780: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

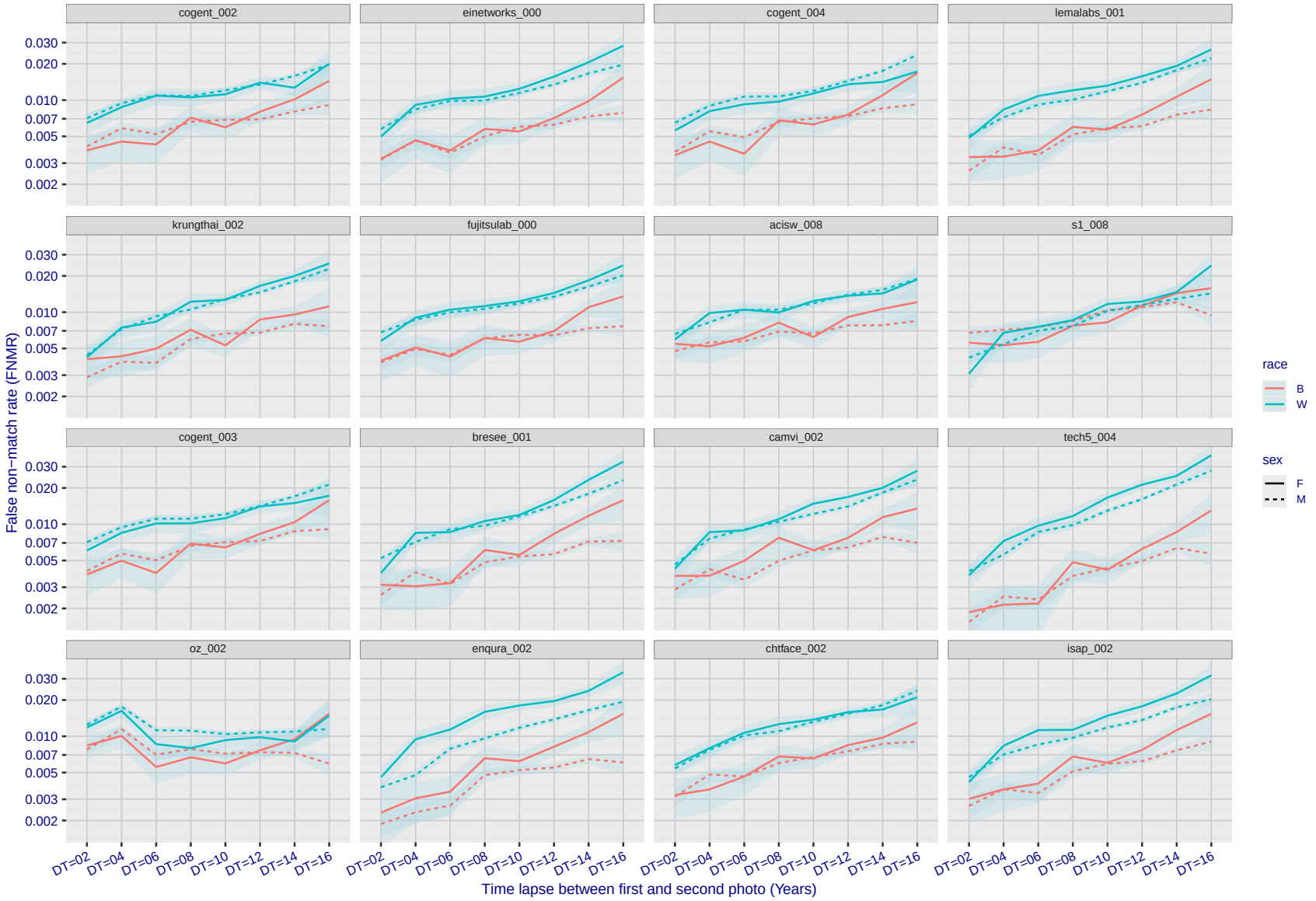


Figure 781: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

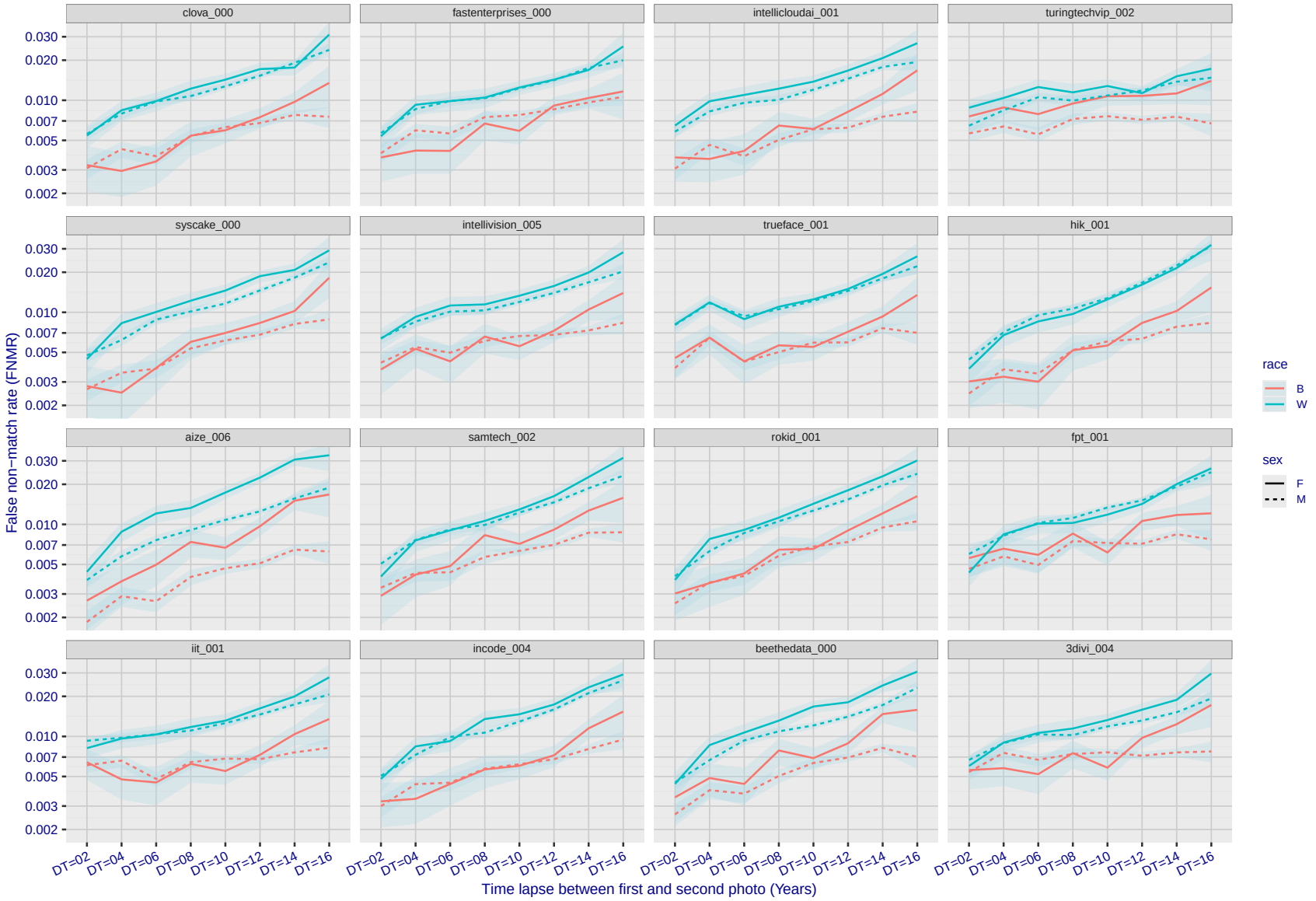


Figure 782: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

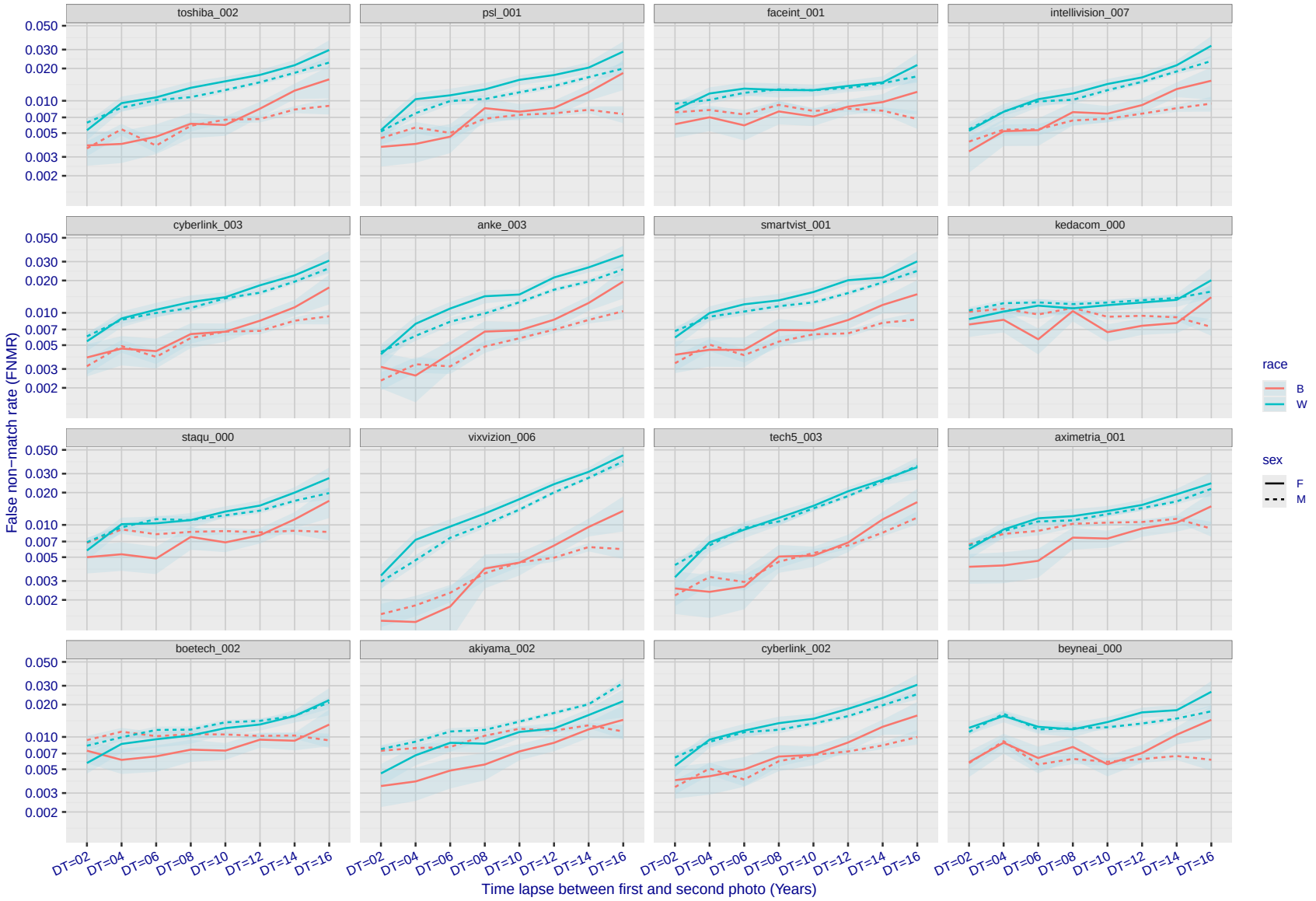


Figure 783: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

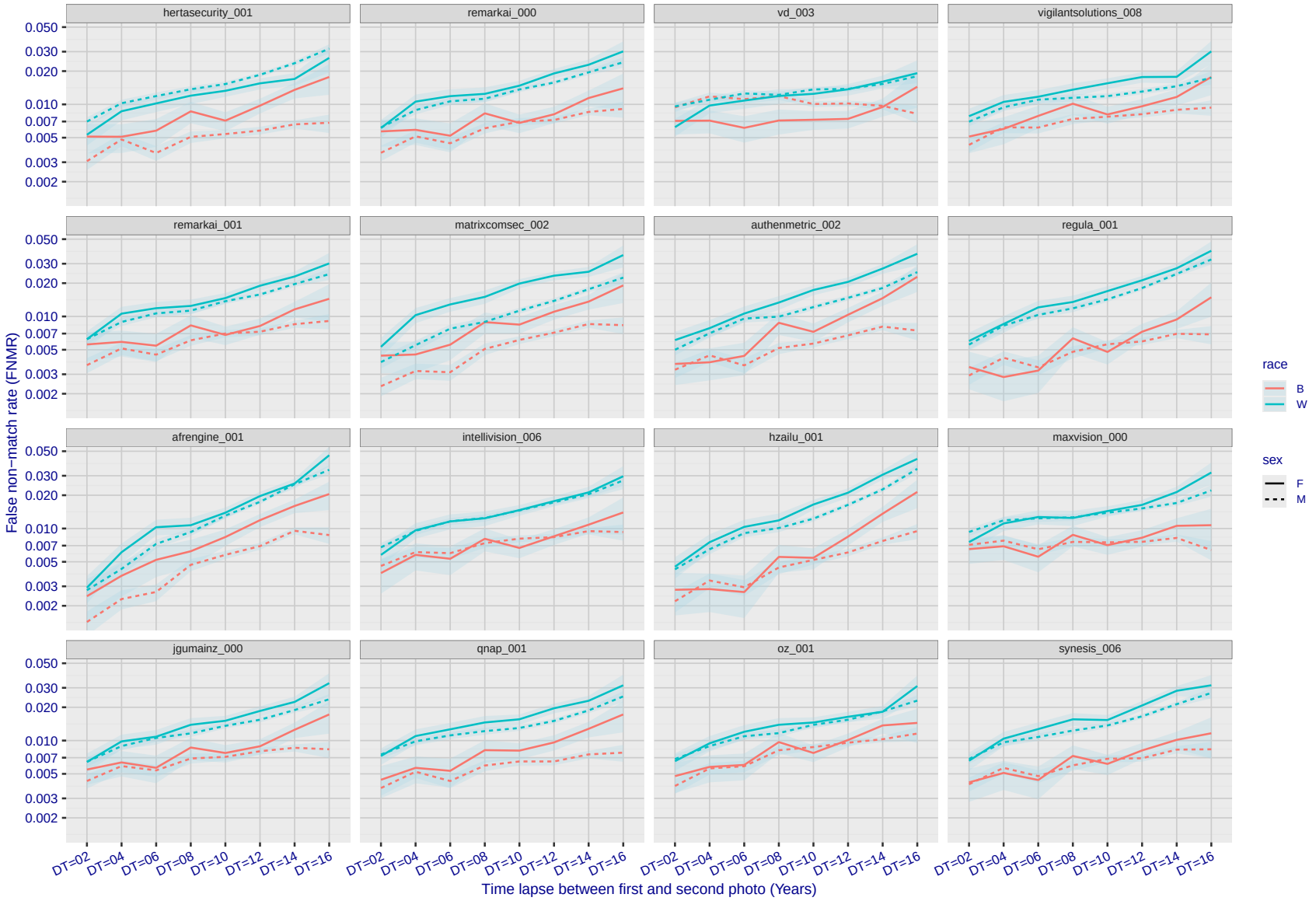


Figure 784: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

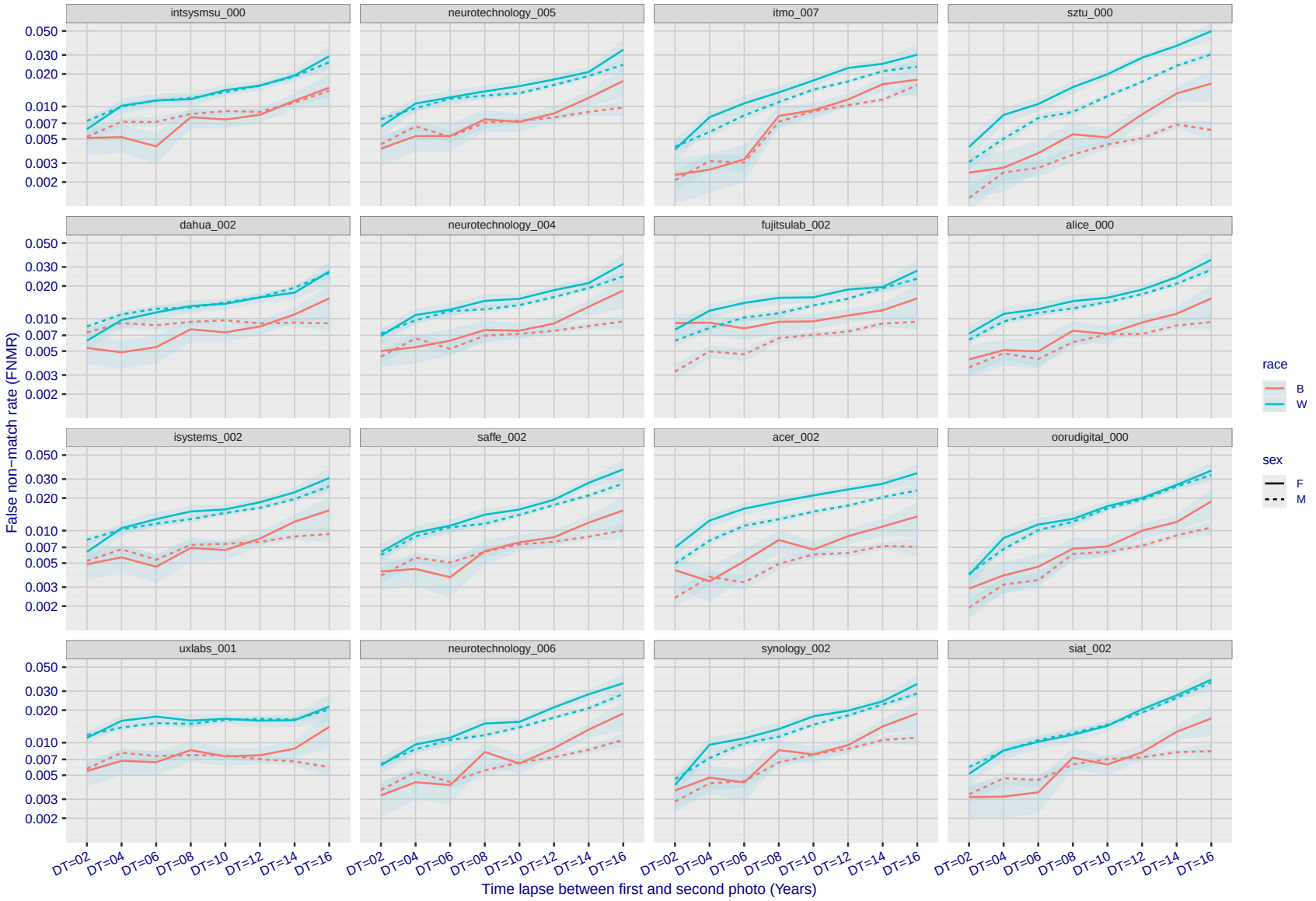


Figure 785: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

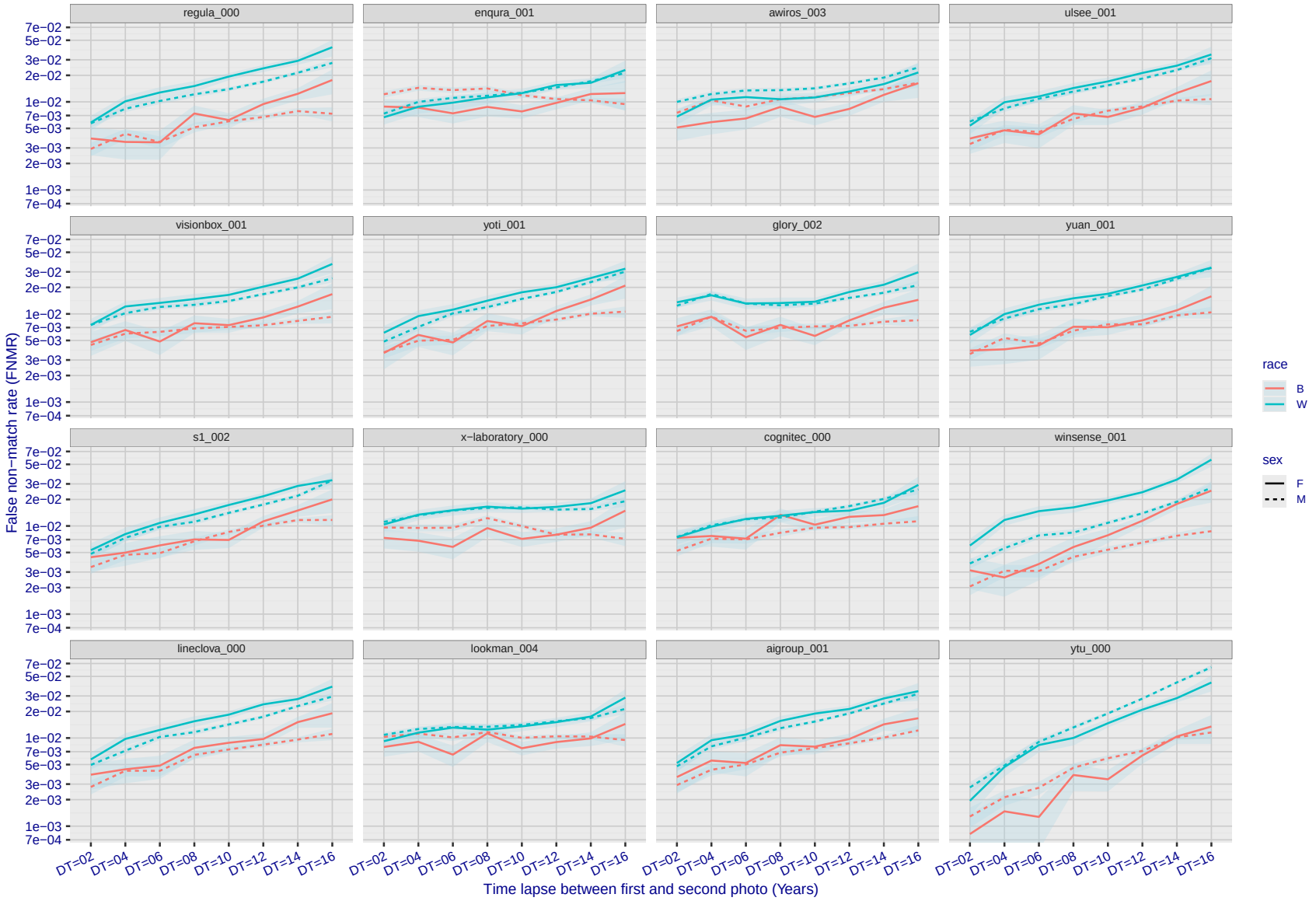


Figure 786: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

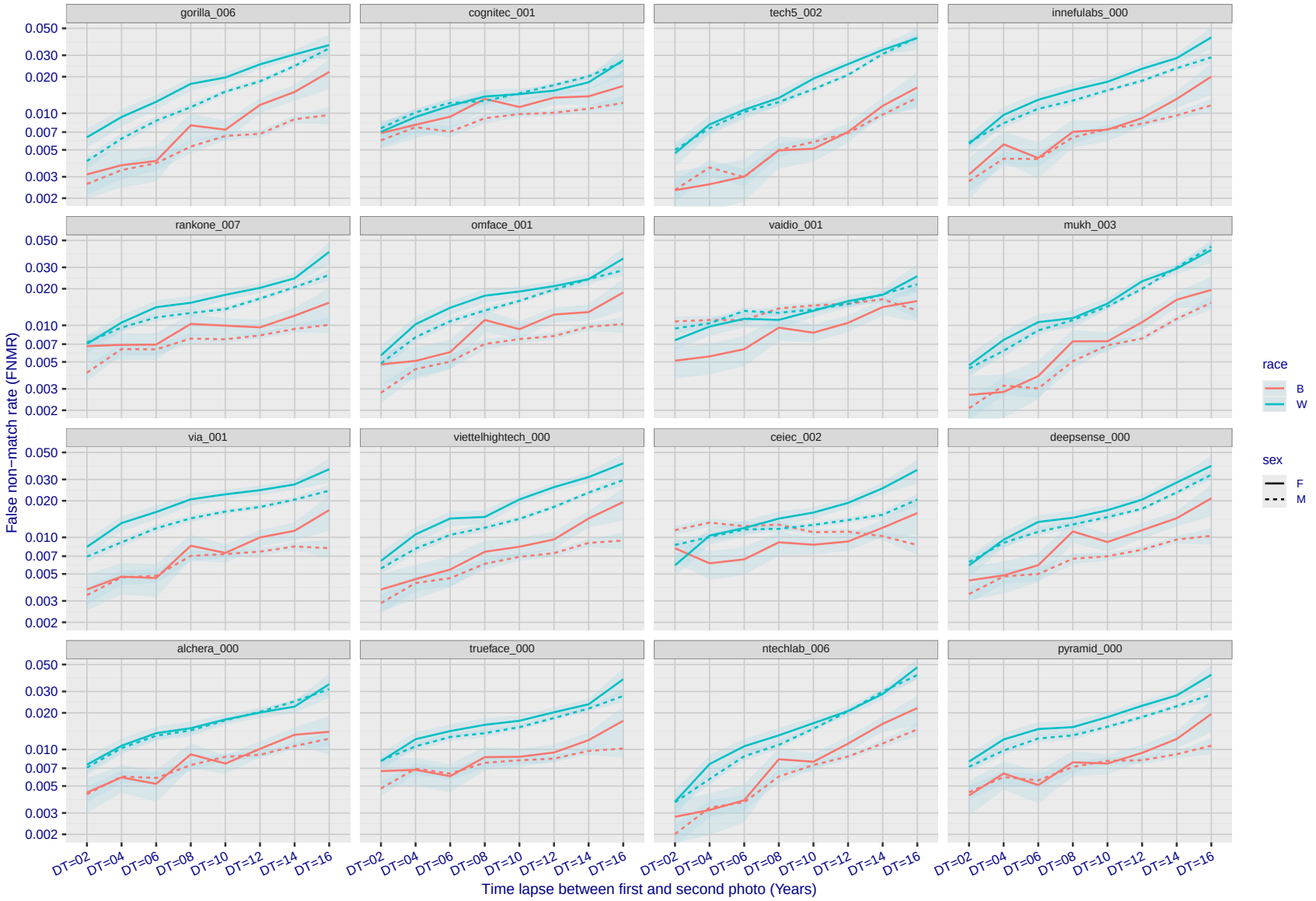


Figure 787: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

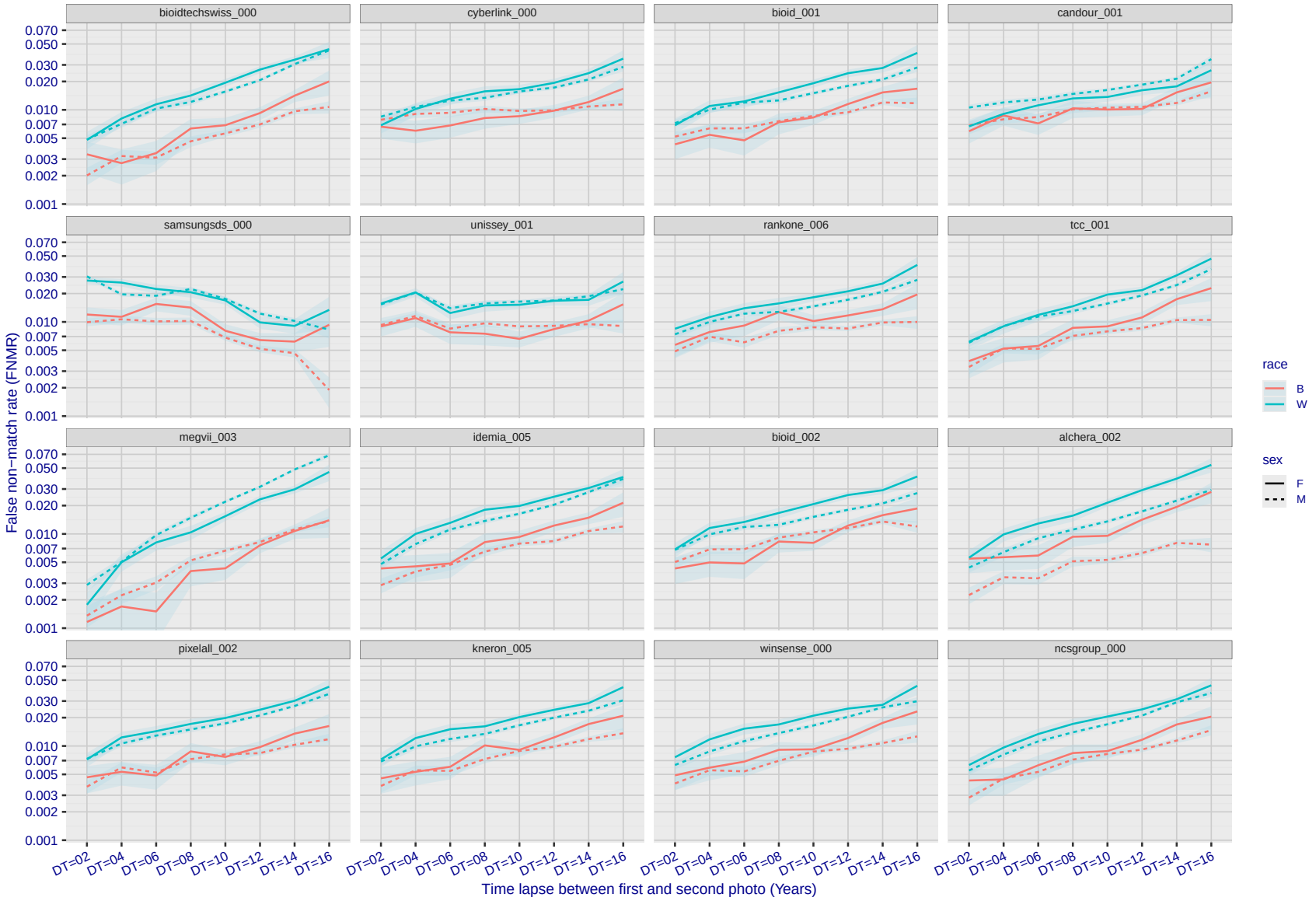


Figure 788: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

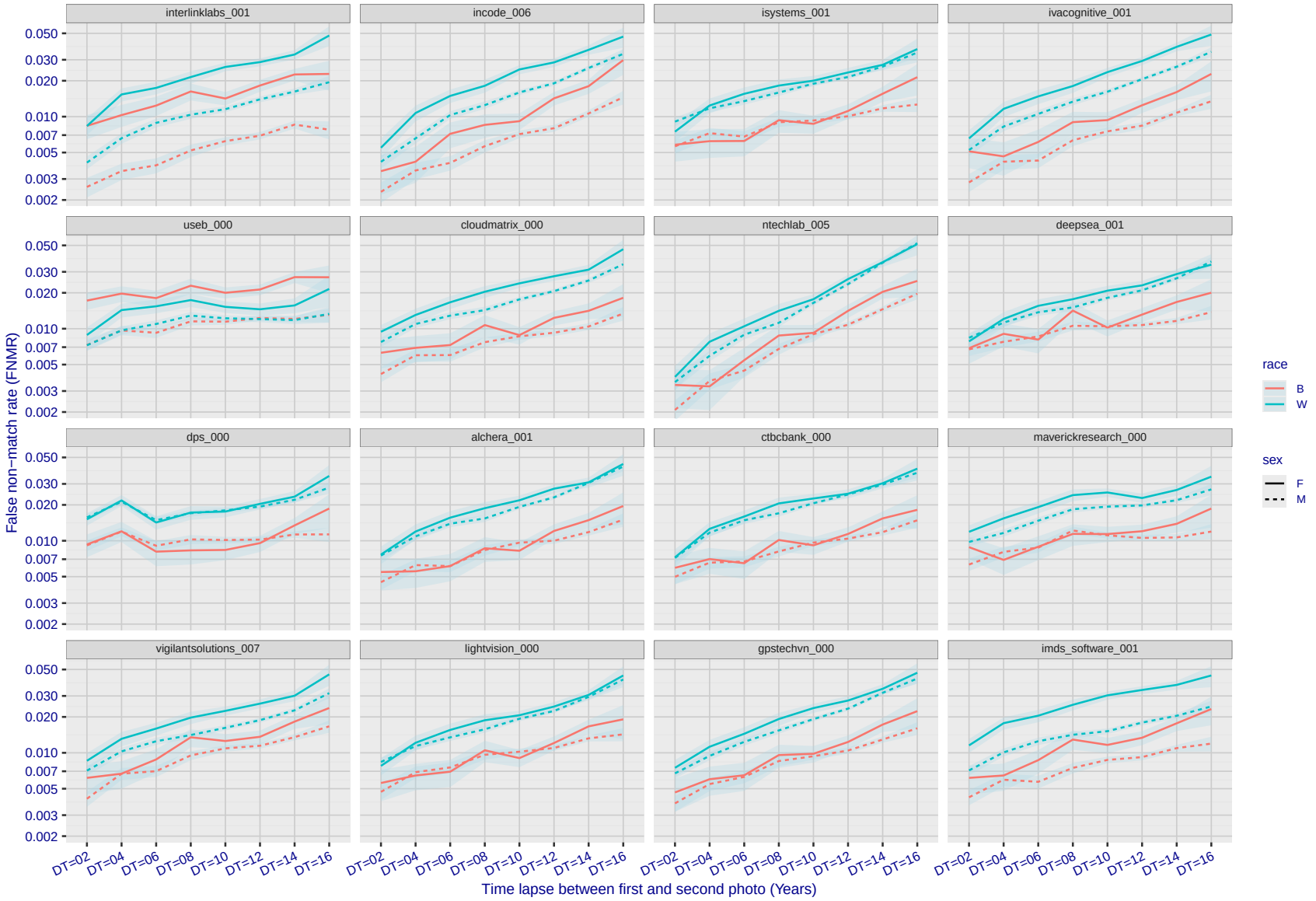


Figure 789: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

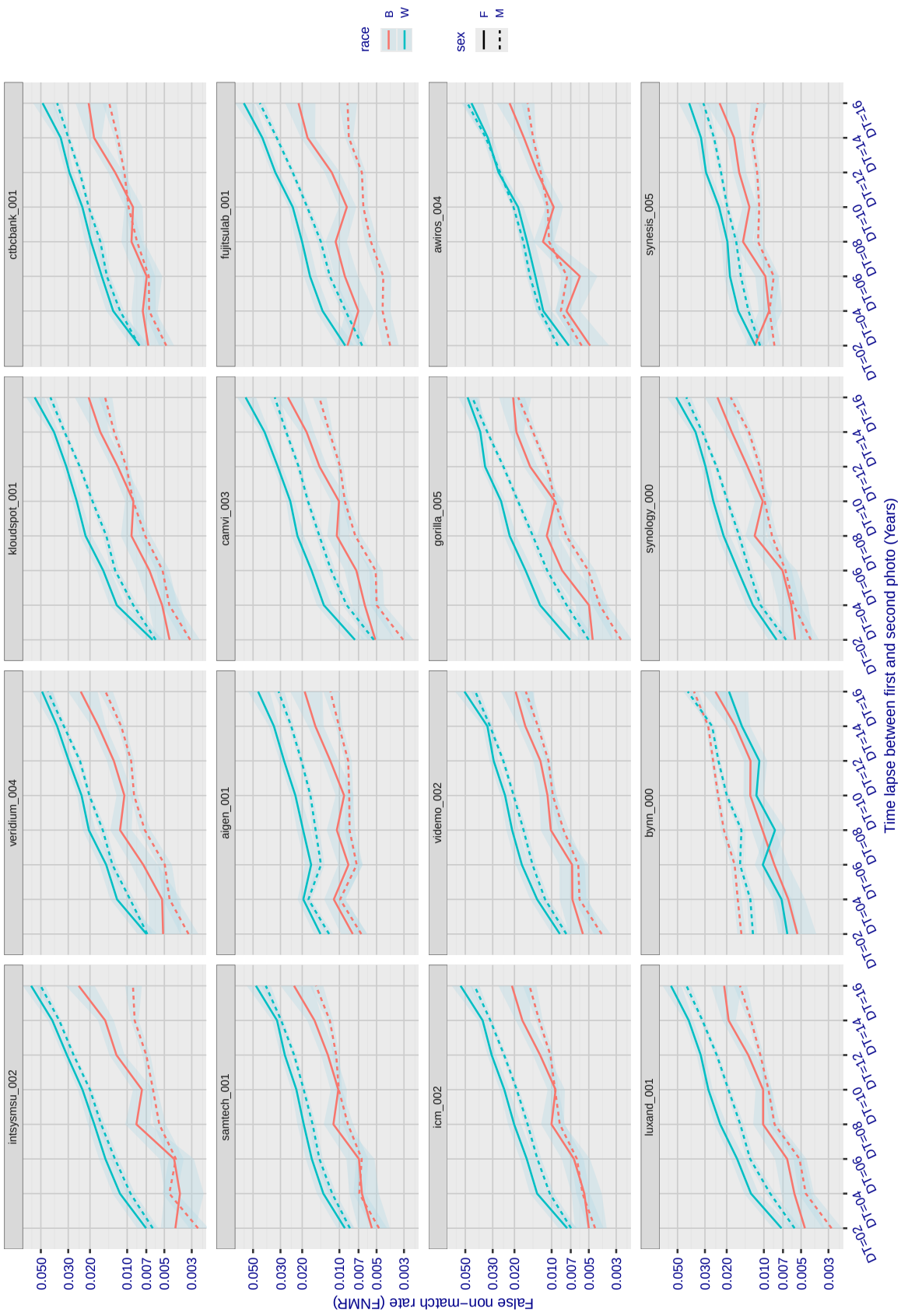


Figure 790: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

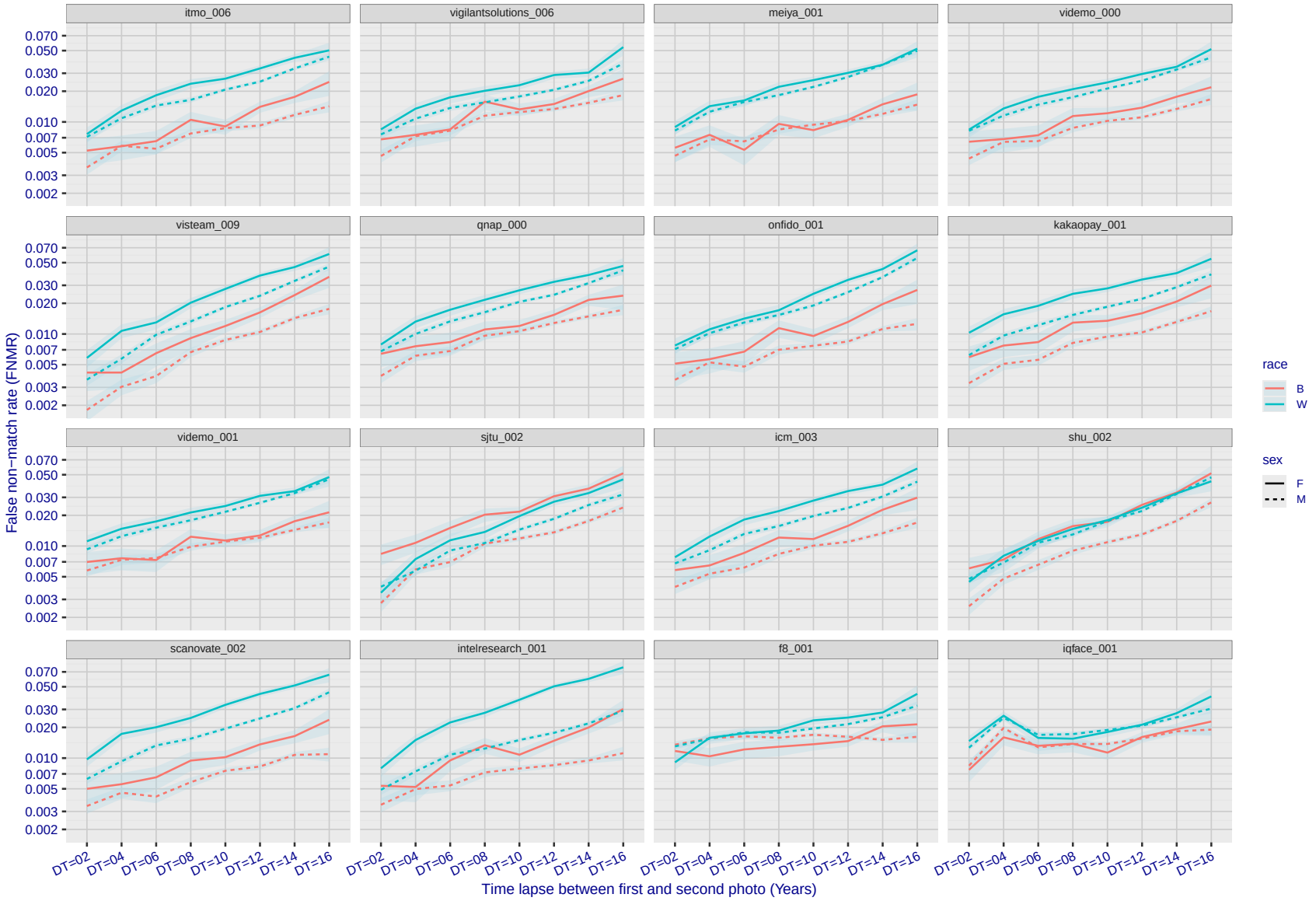


Figure 791: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

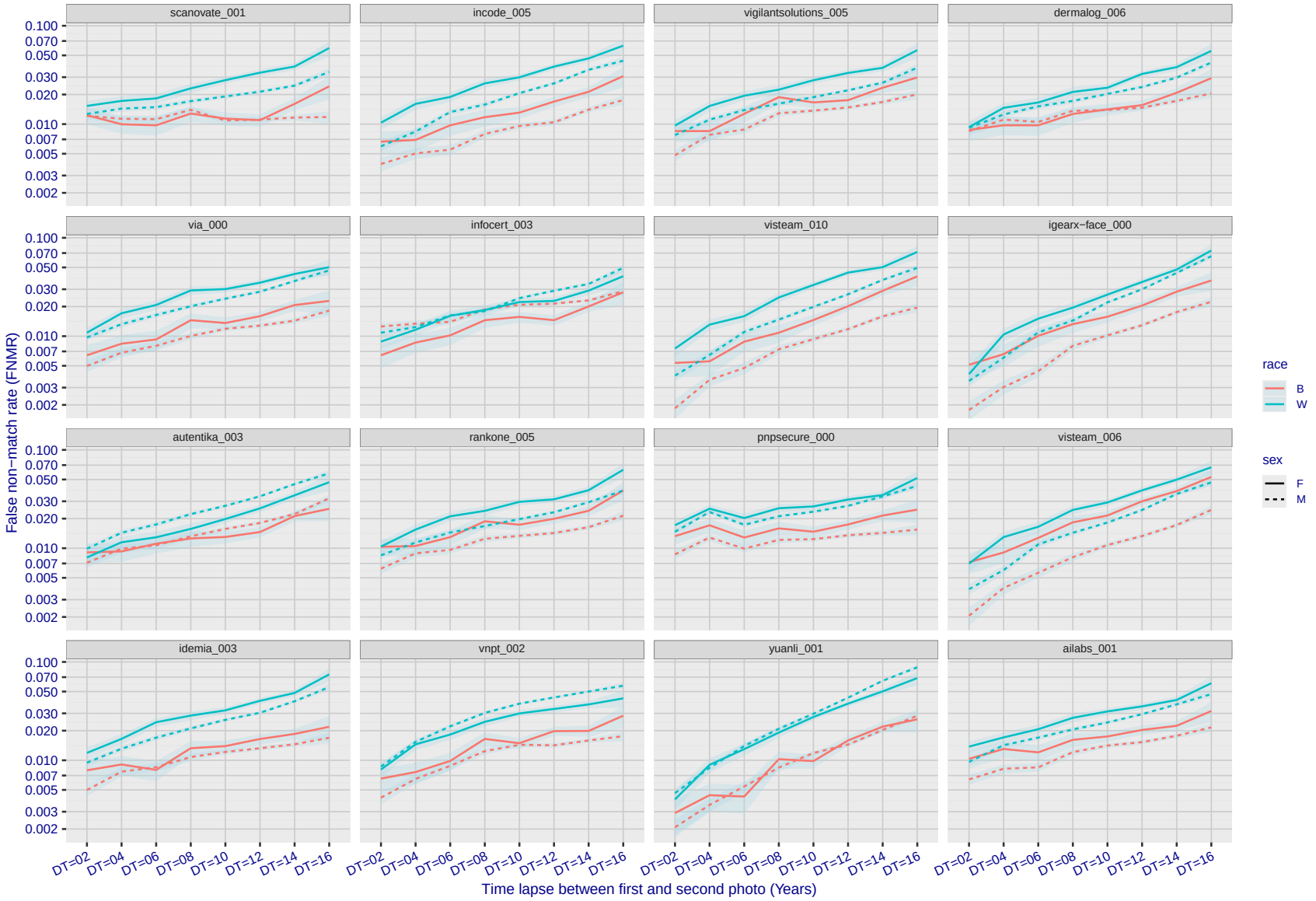


Figure 792: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

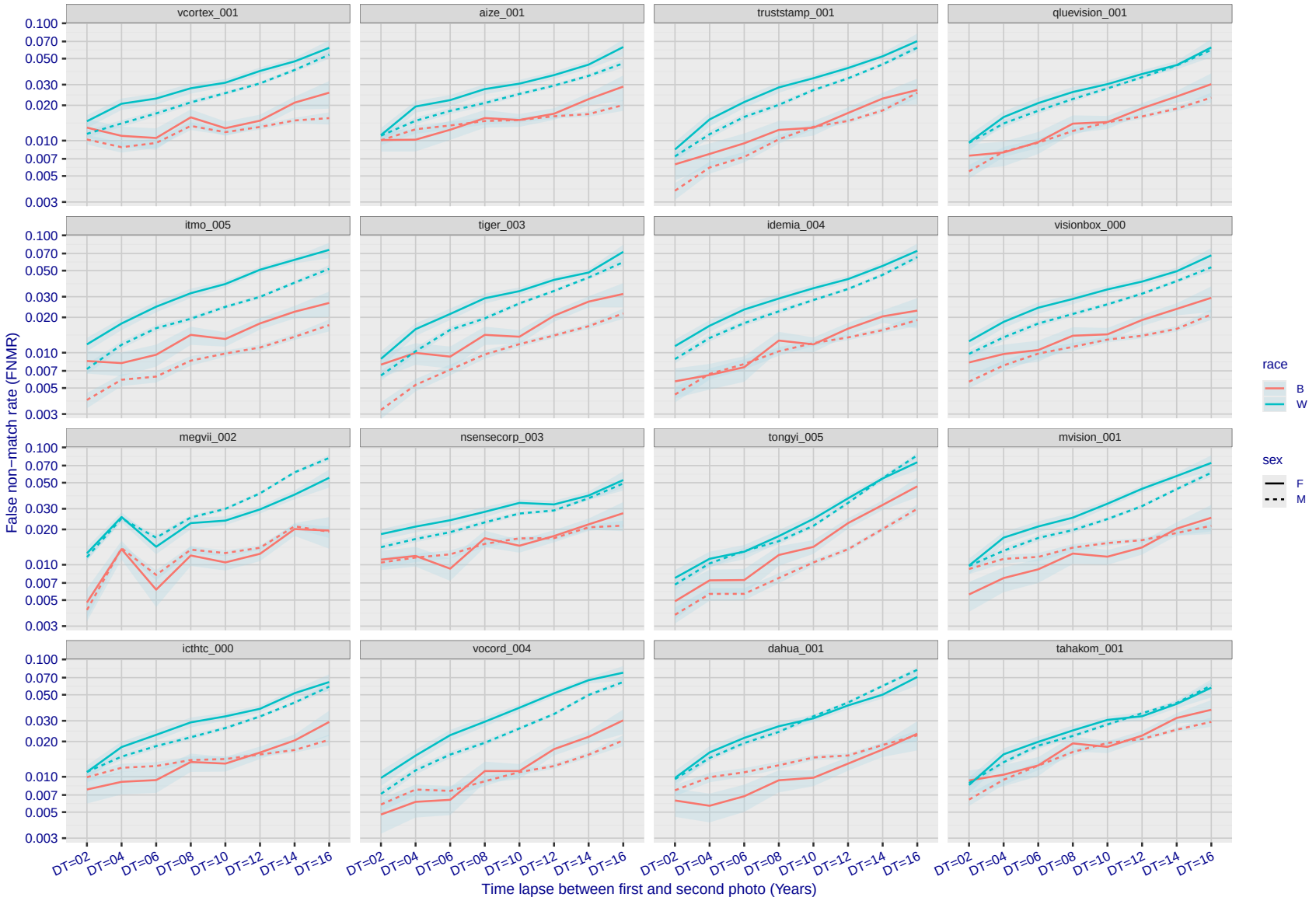


Figure 793: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

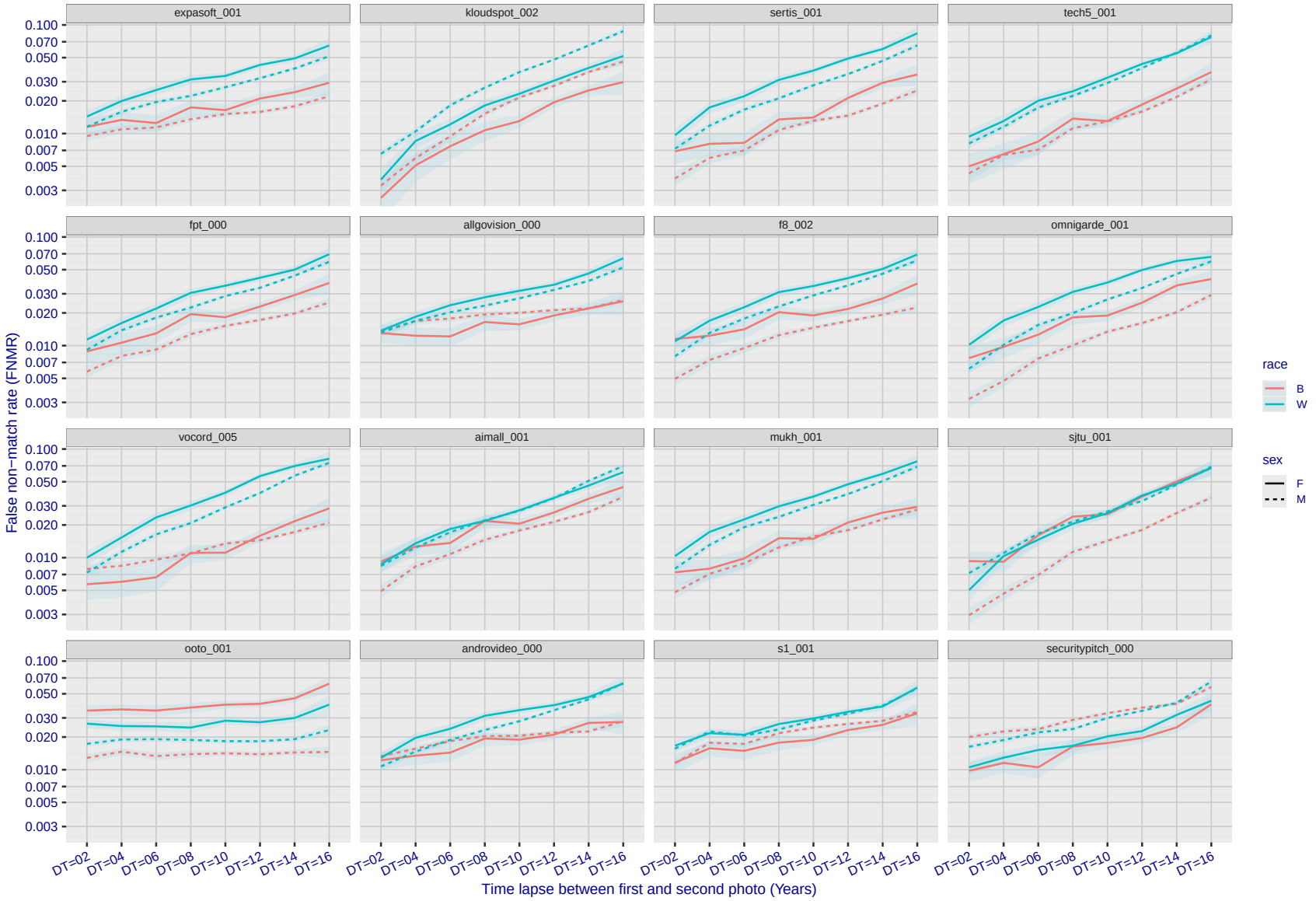


Figure 794: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widehat{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

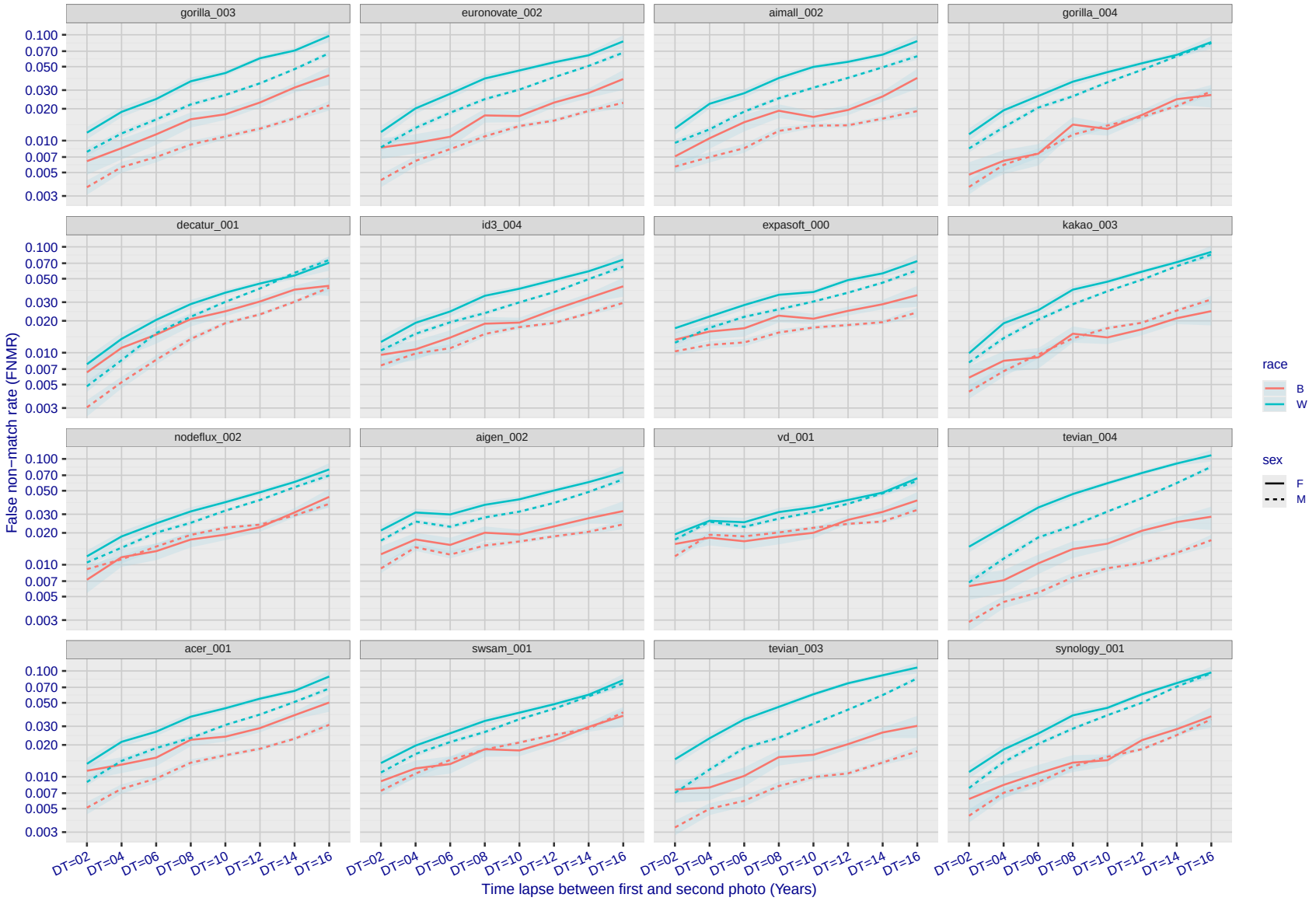


Figure 795: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

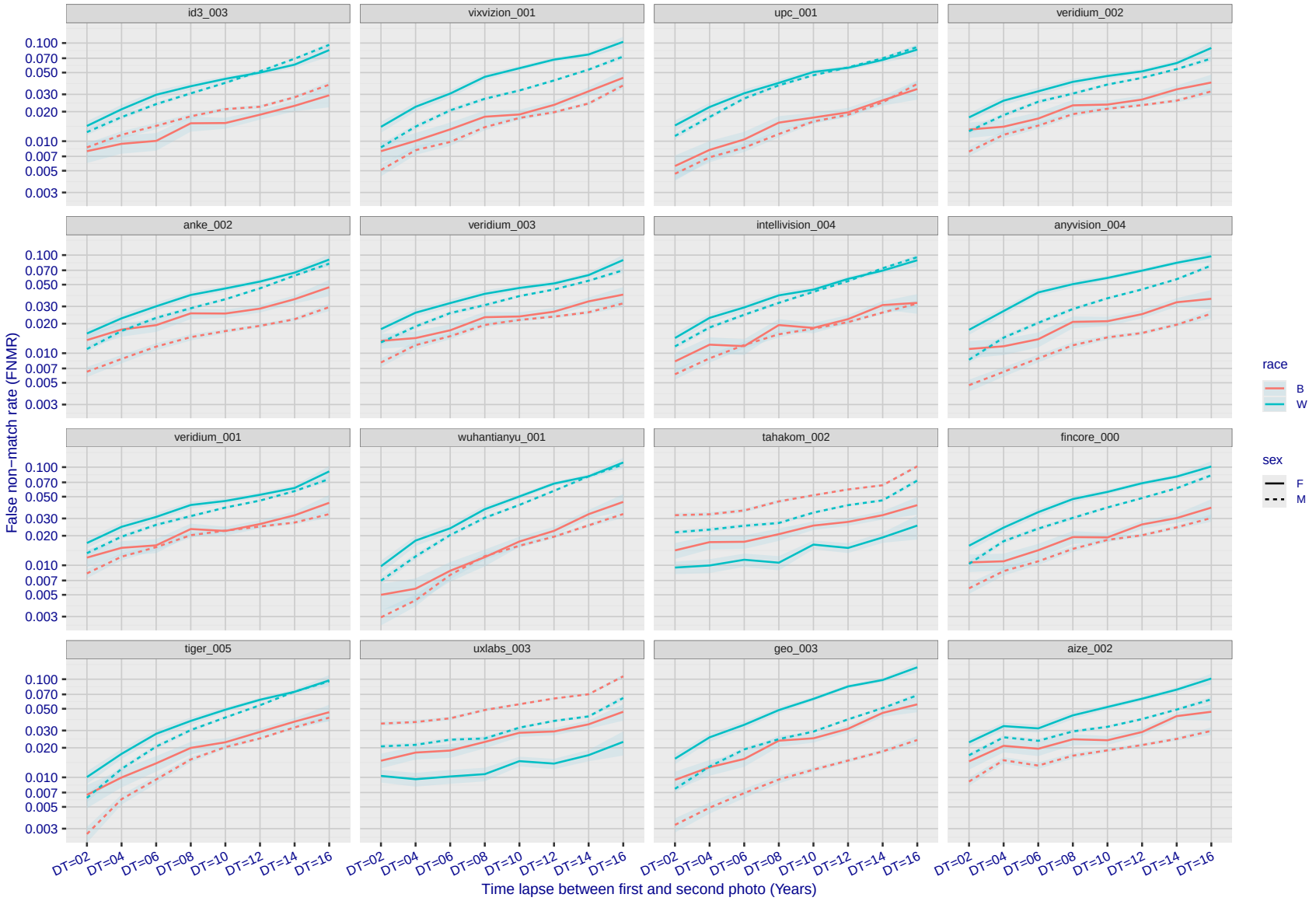


Figure 796: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

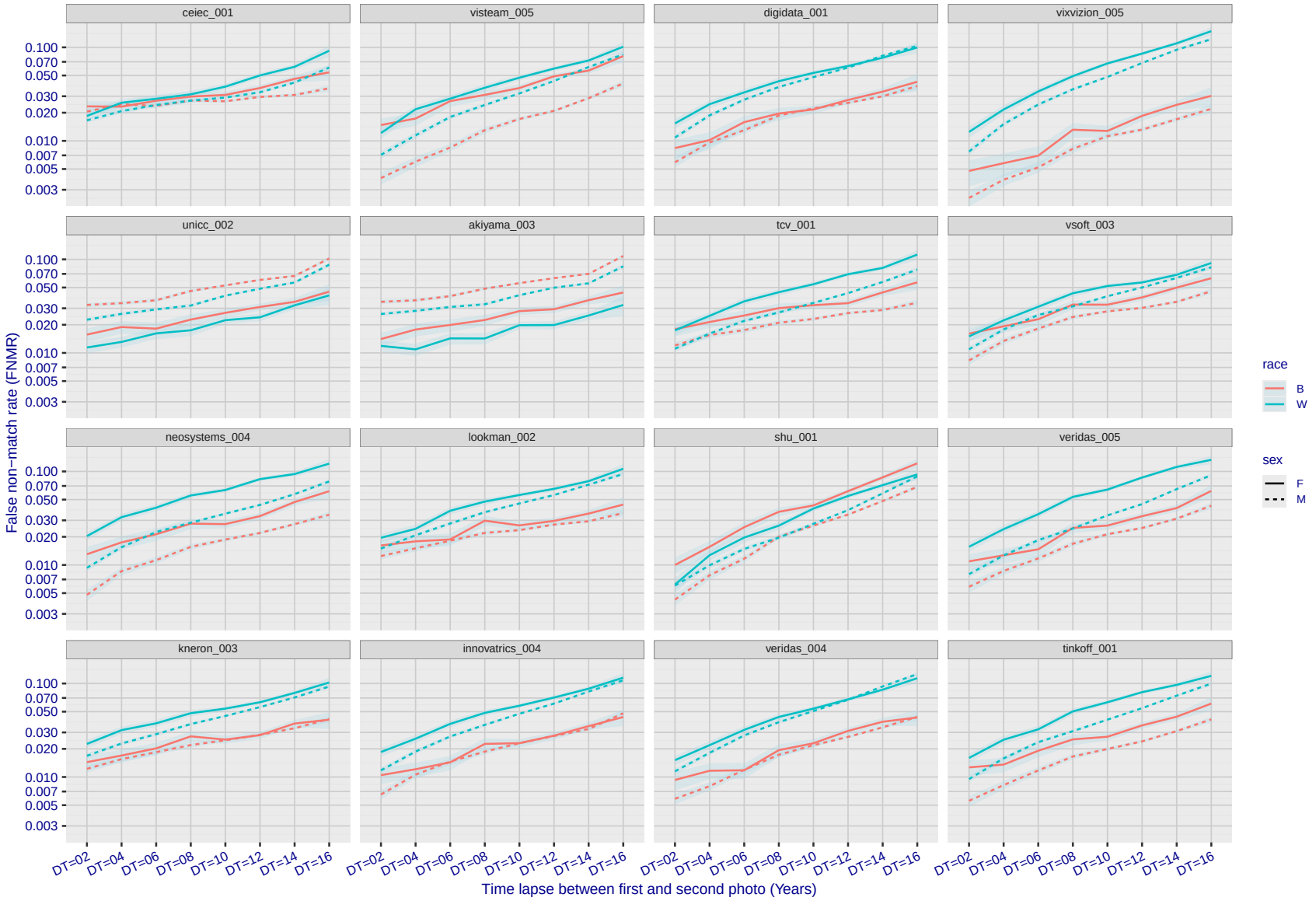


Figure 797: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

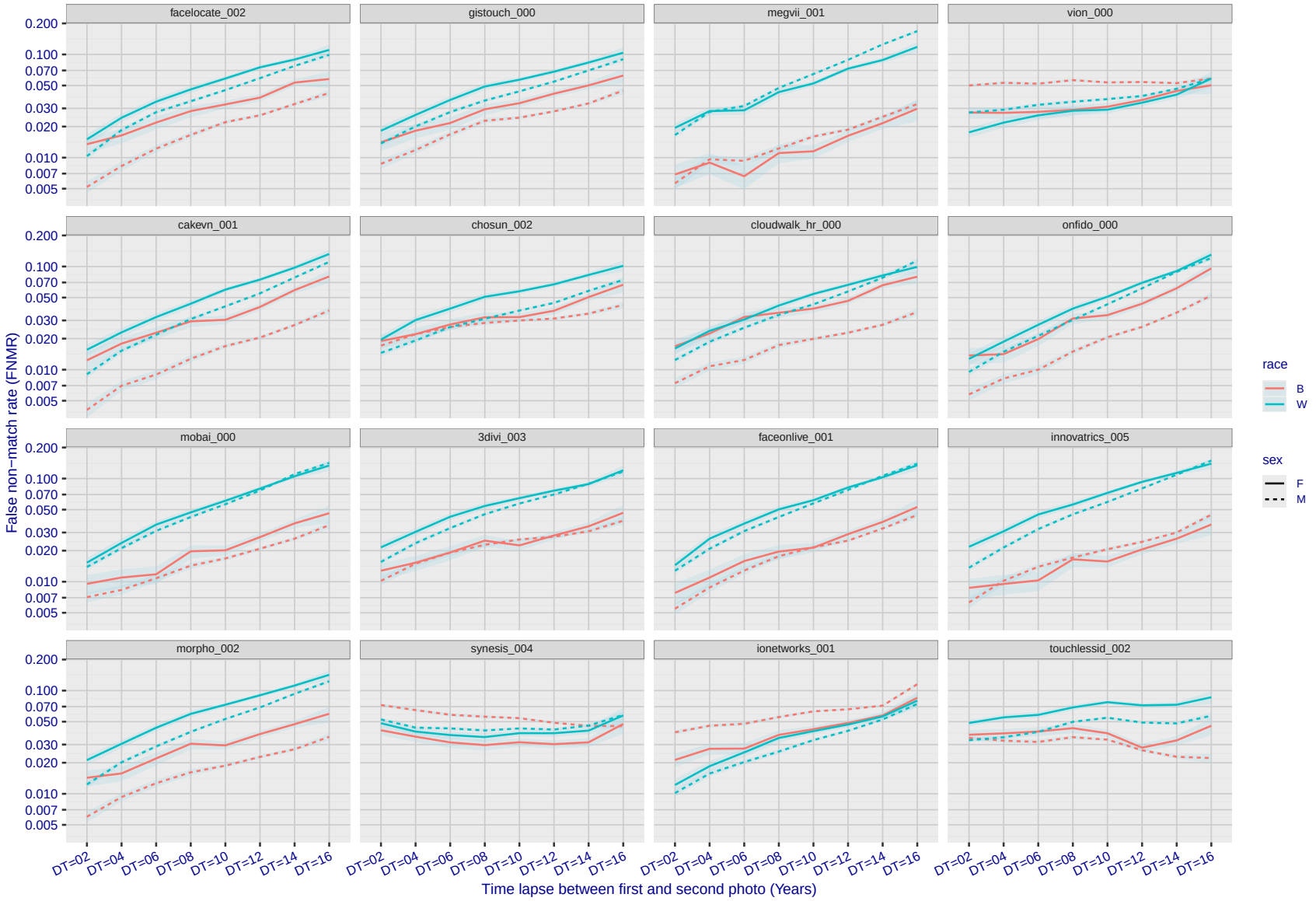


Figure 798: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

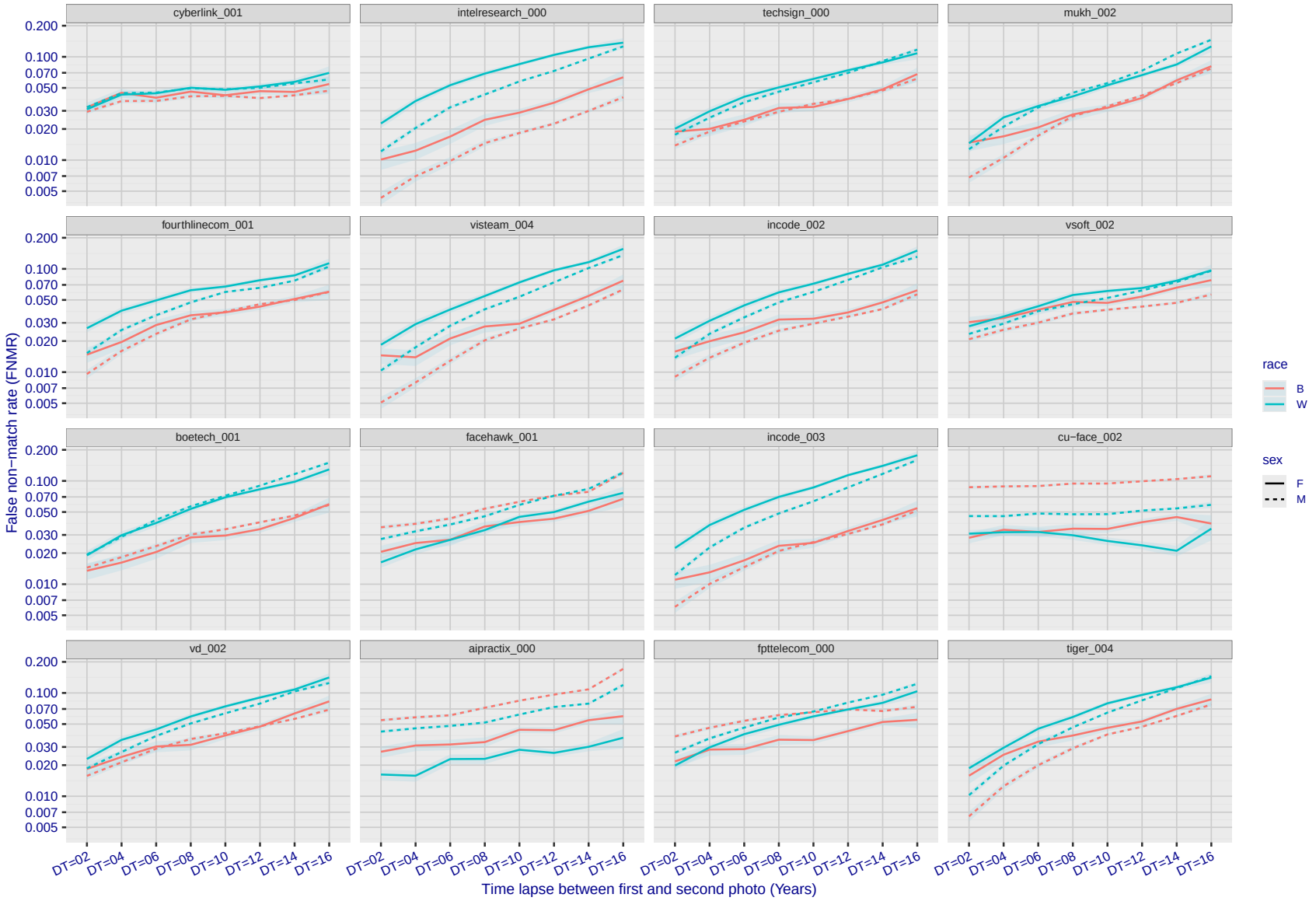


Figure 799: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

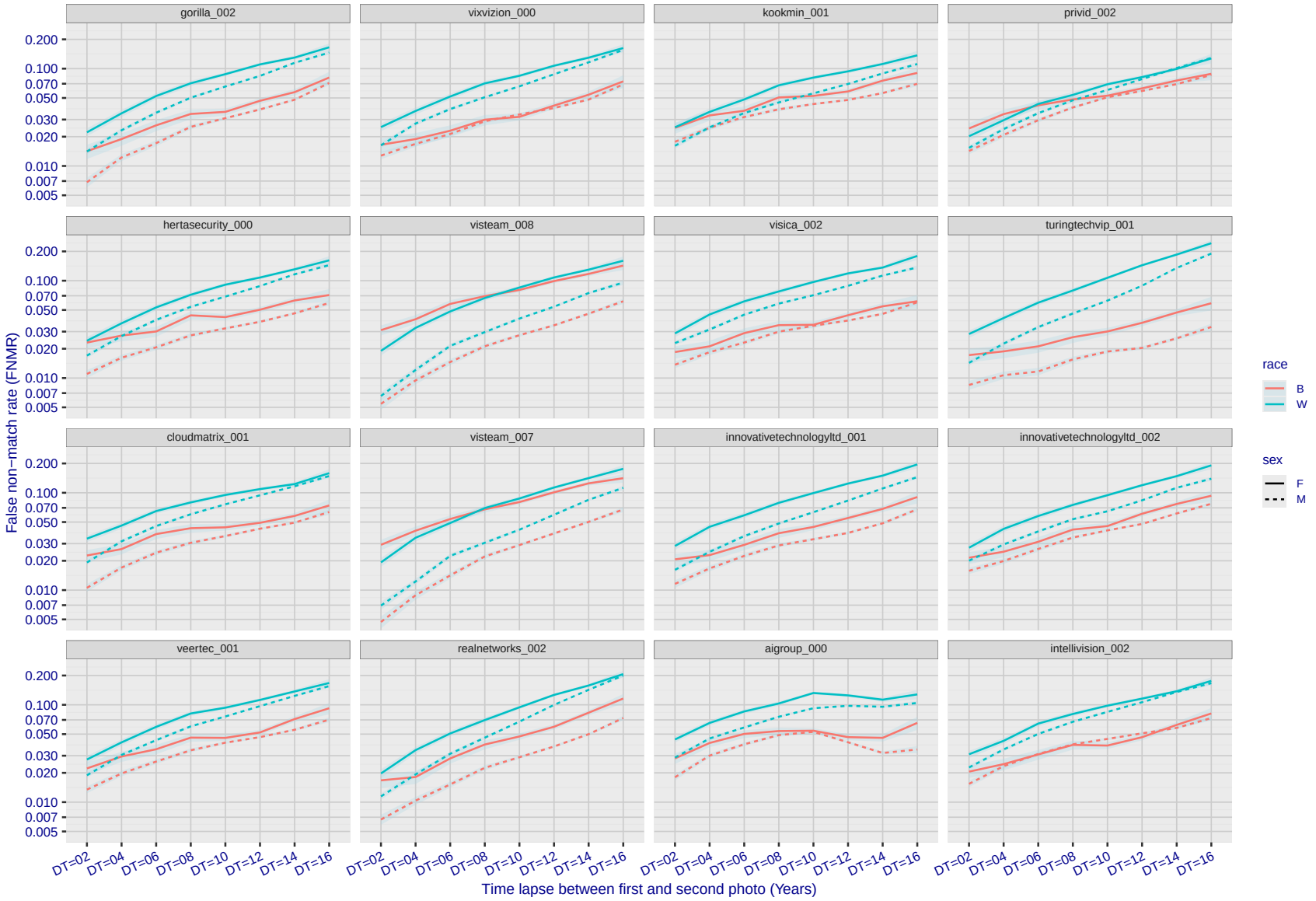


Figure 800: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

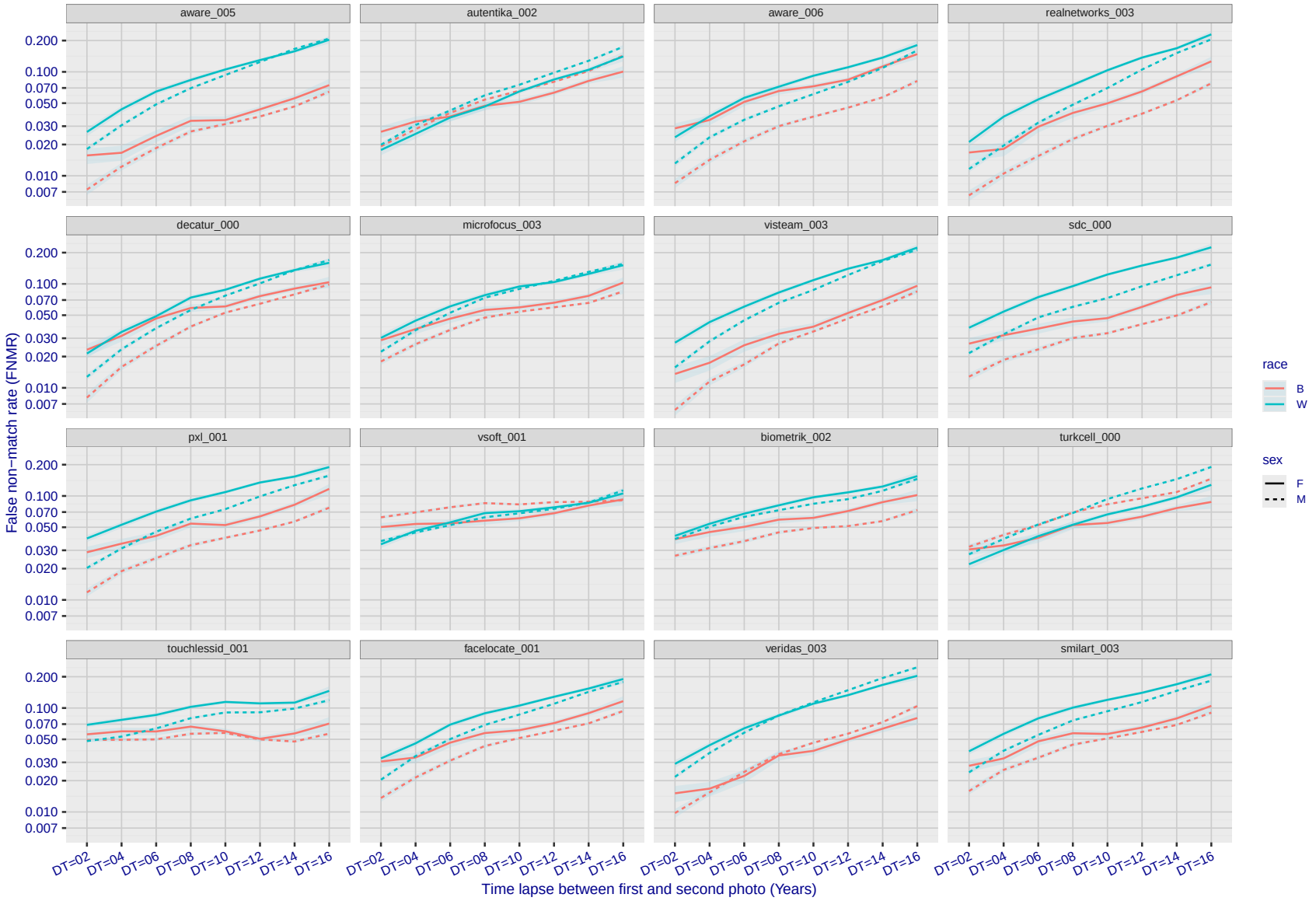


Figure 801: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

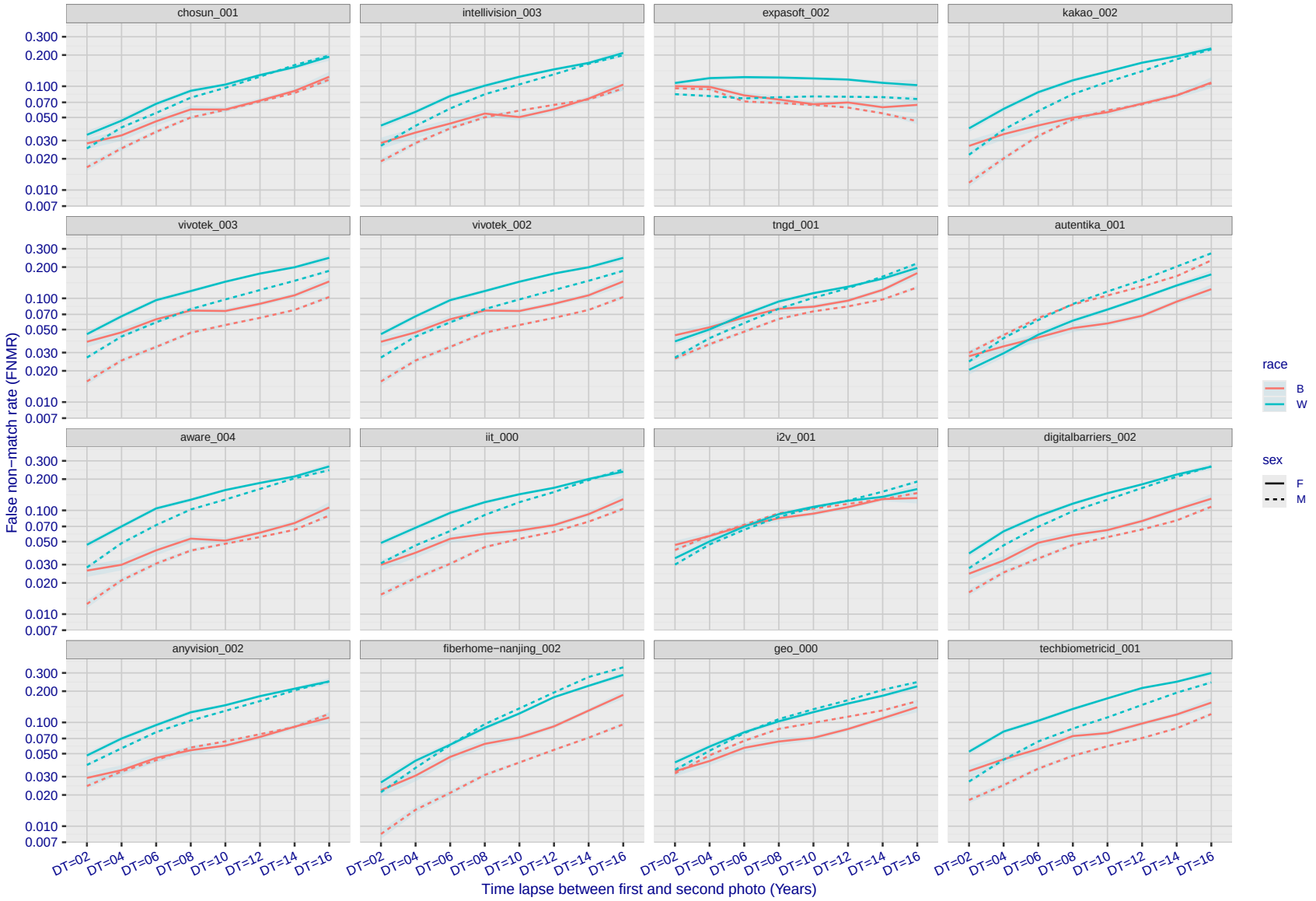


Figure 802: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

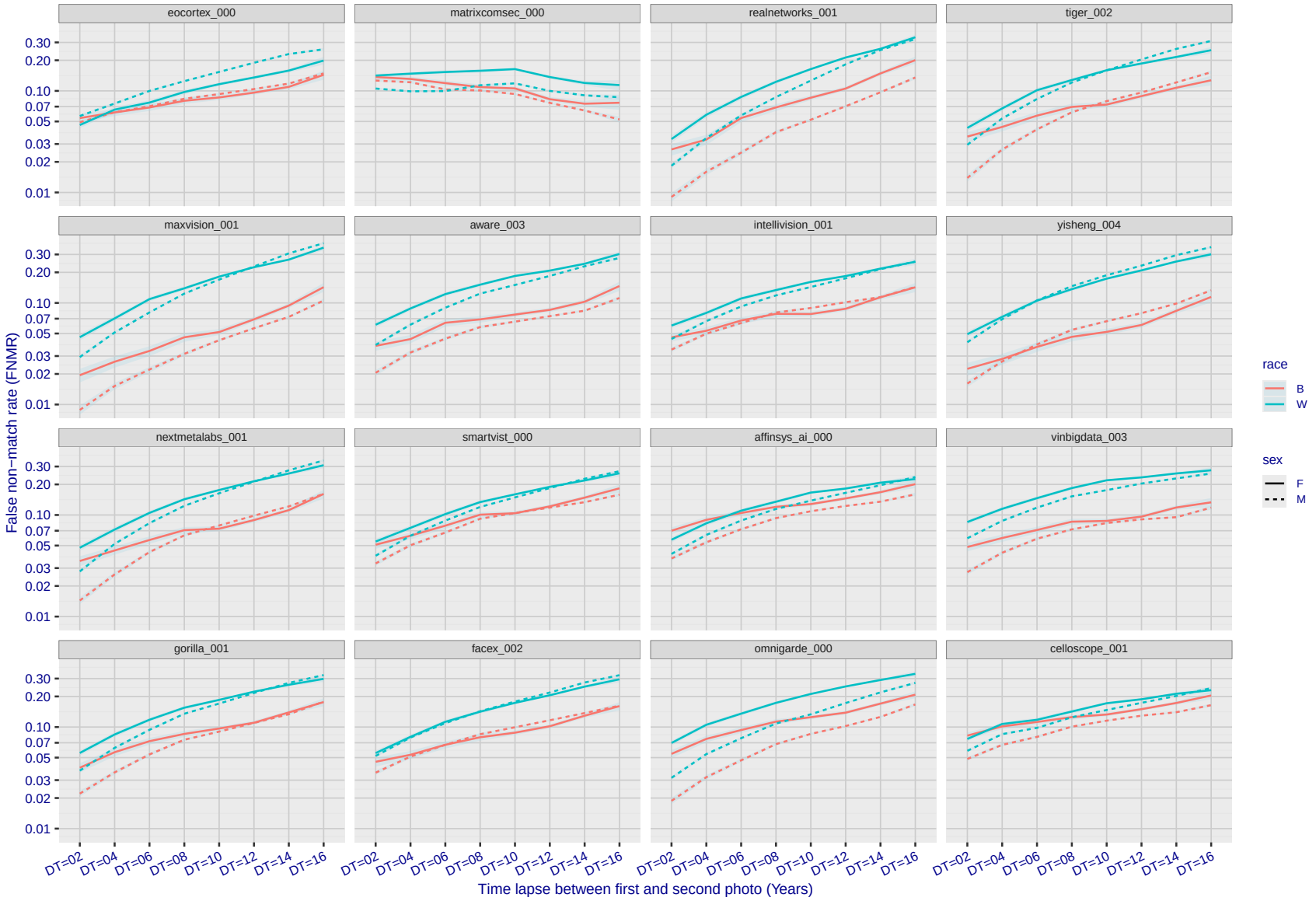


Figure 803: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

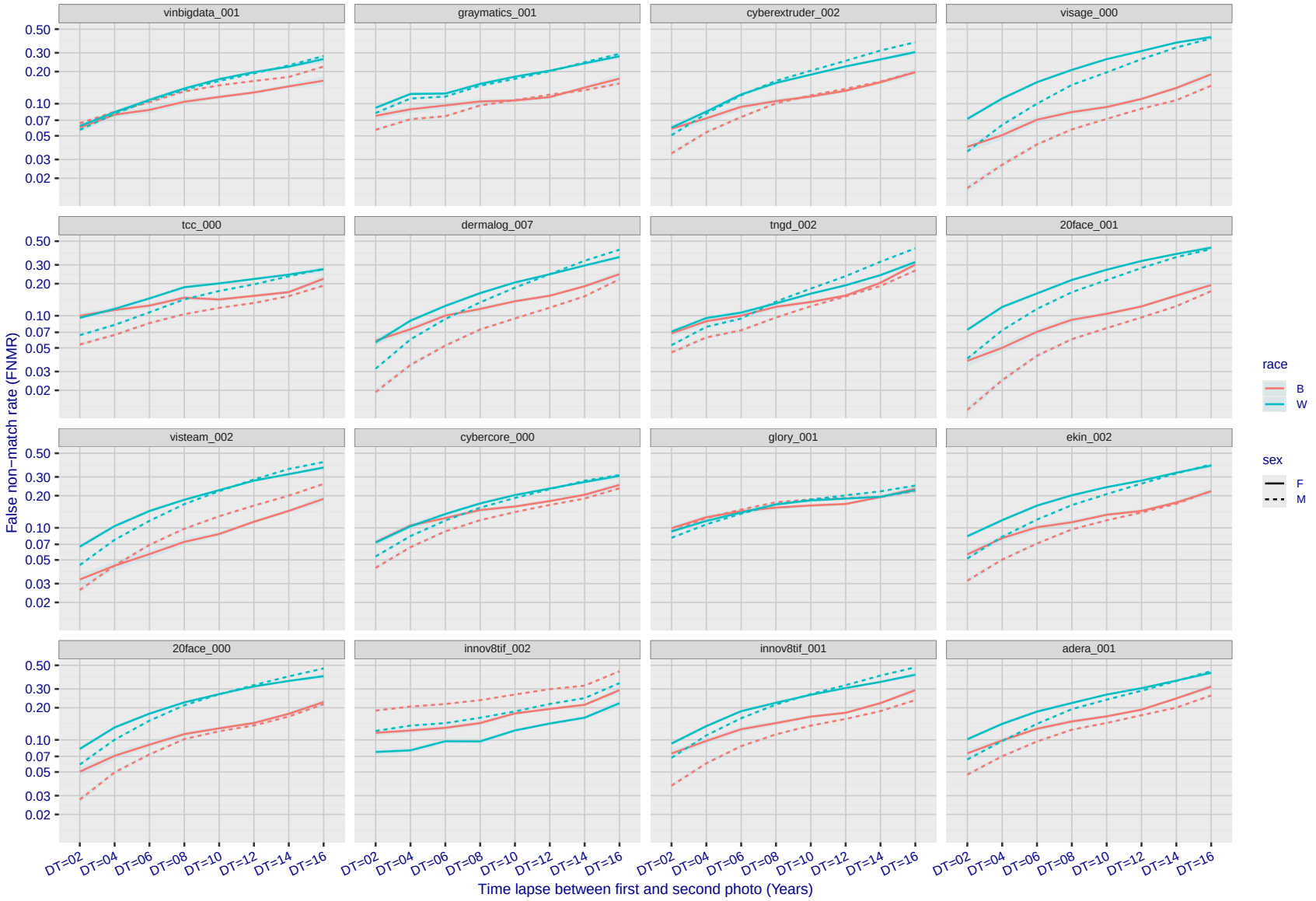


Figure 804: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

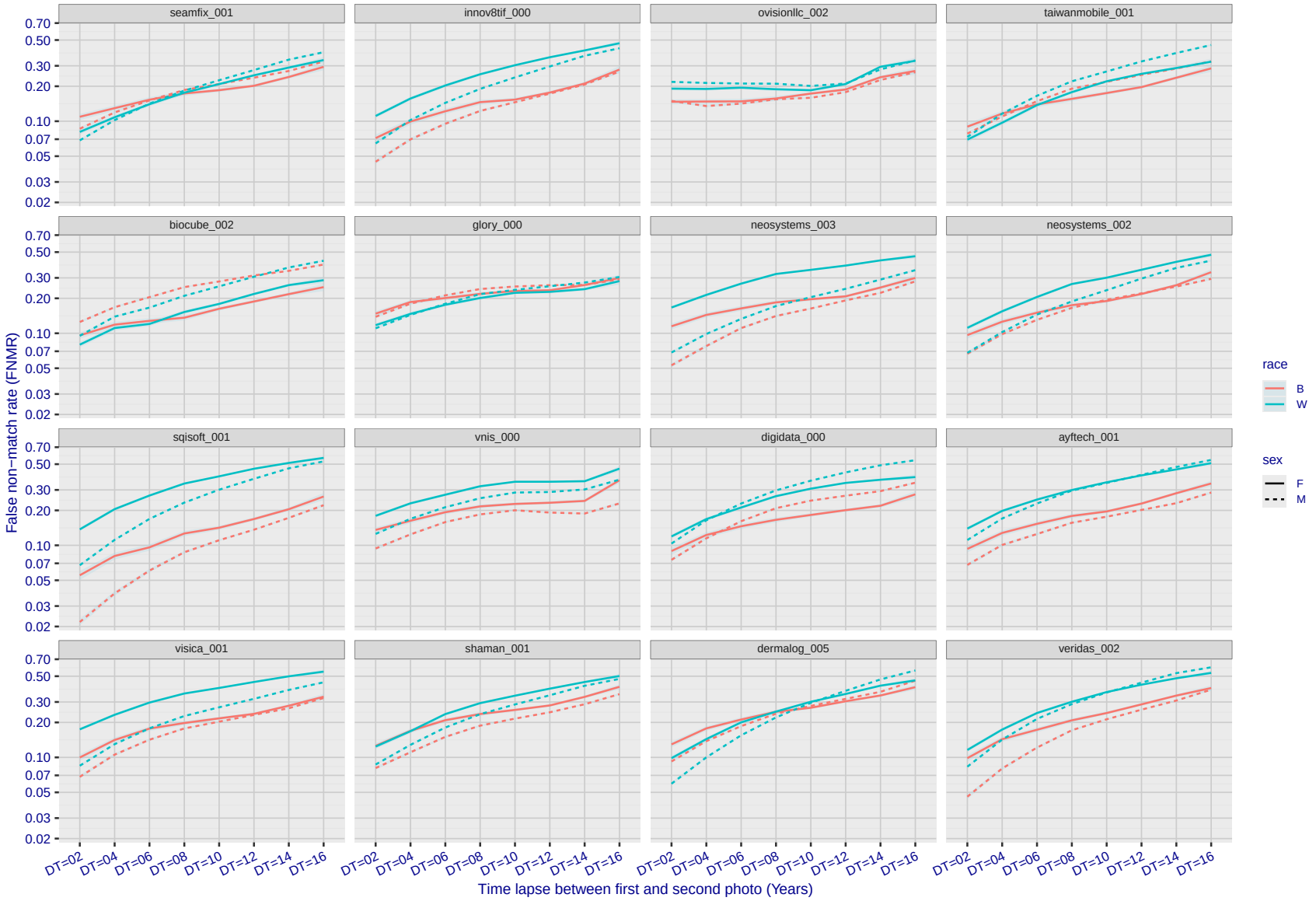


Figure 805: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

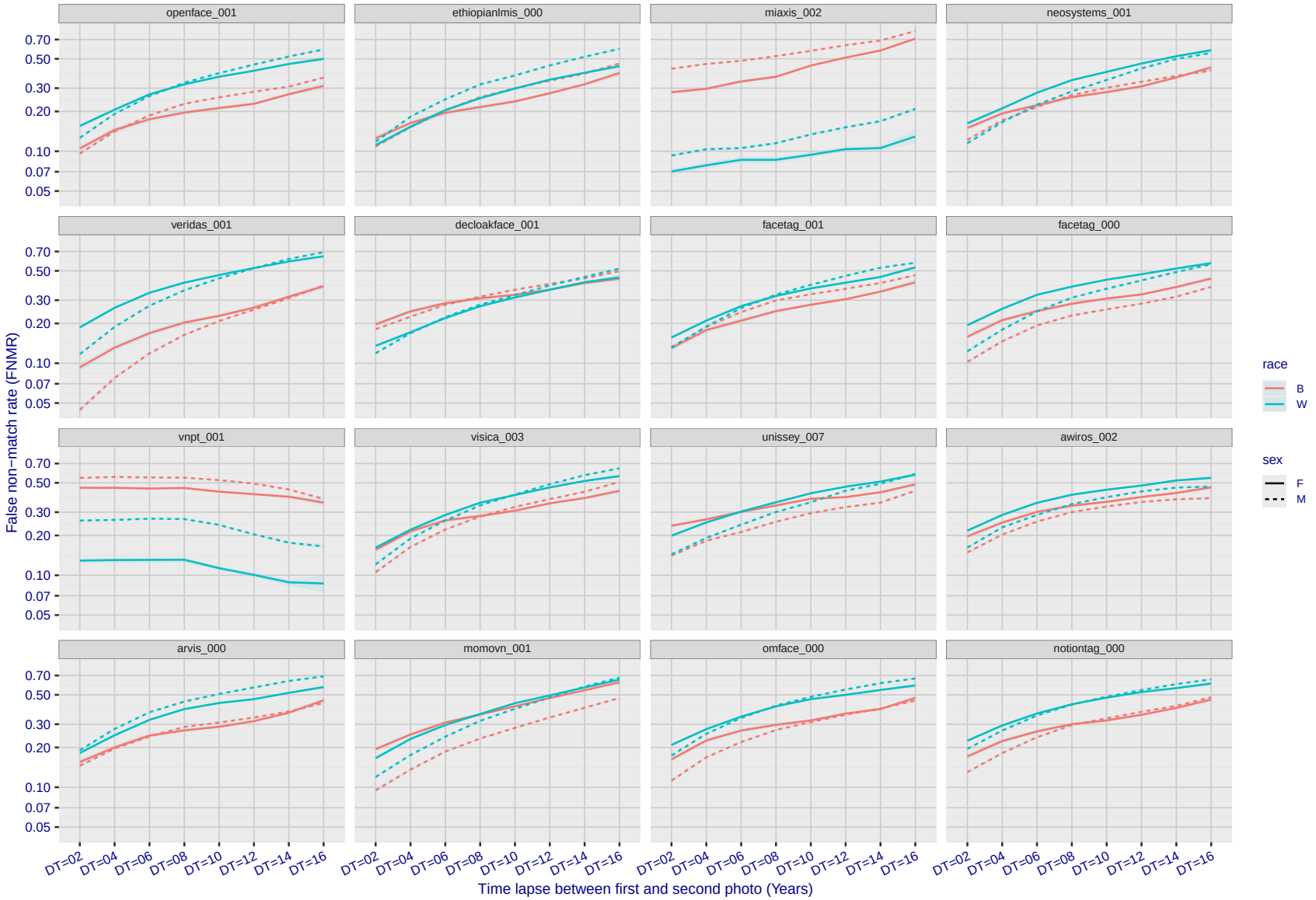


Figure 806: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all ($\widetilde{10}^8$) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

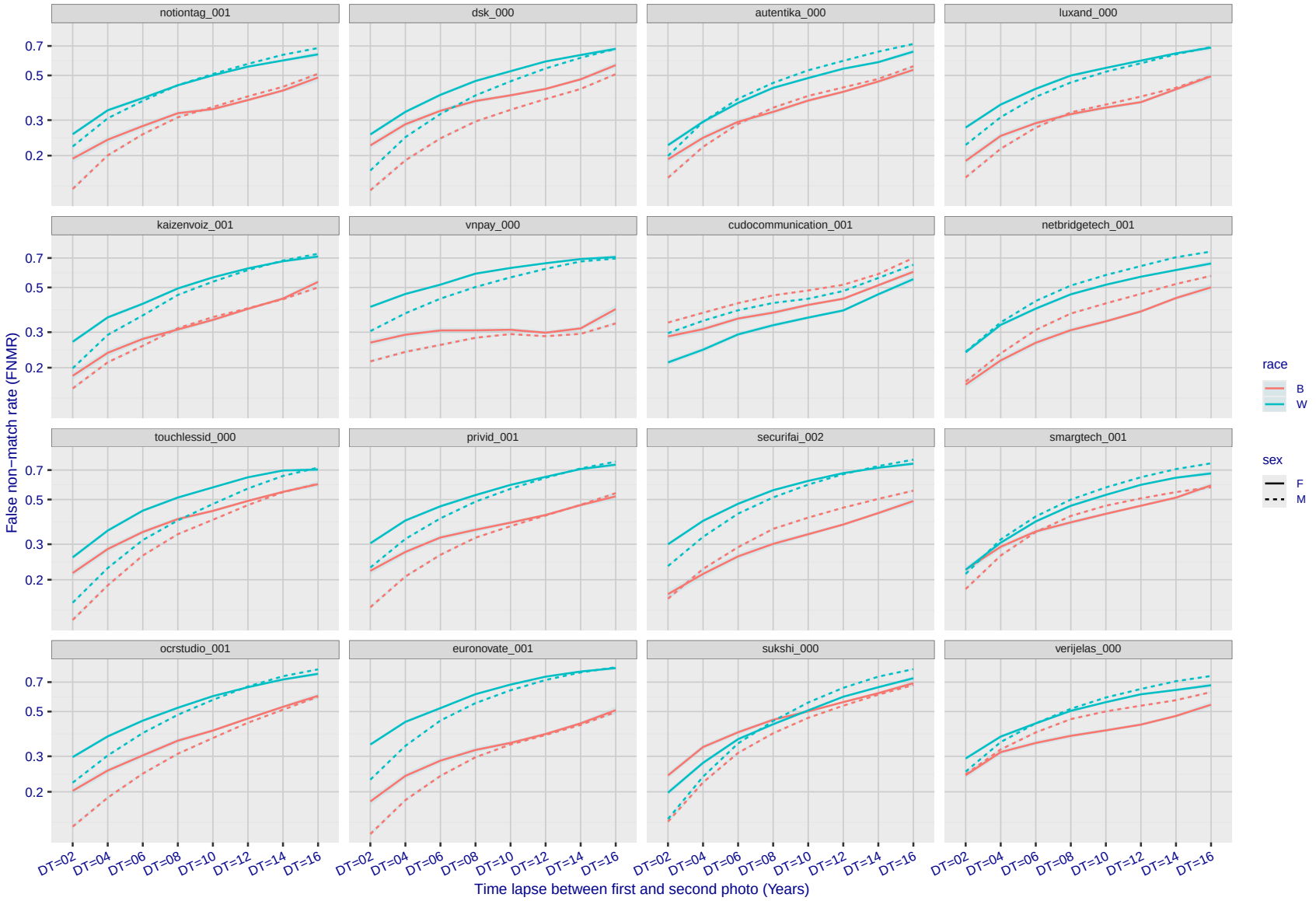


Figure 807: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

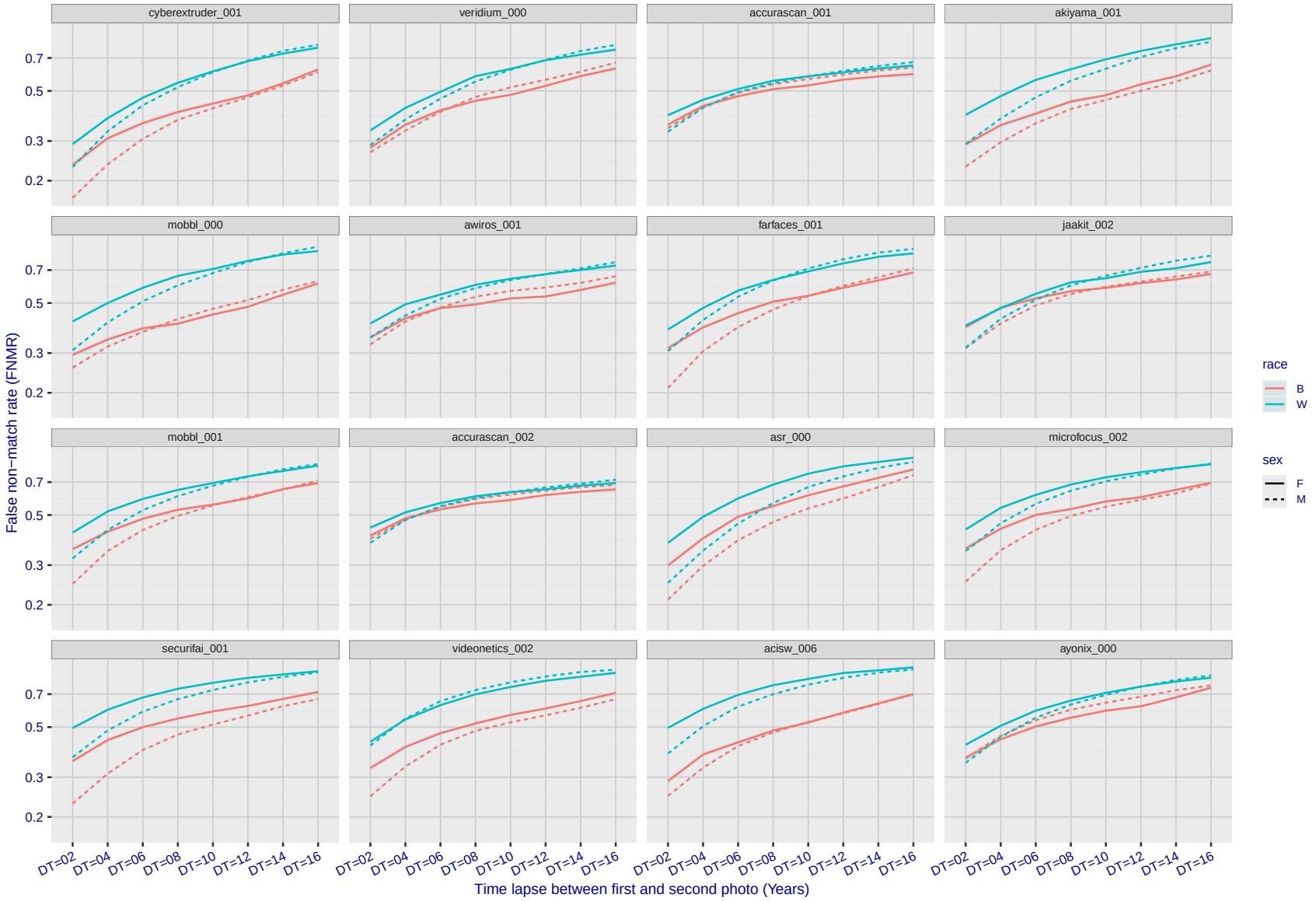


Figure 808: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

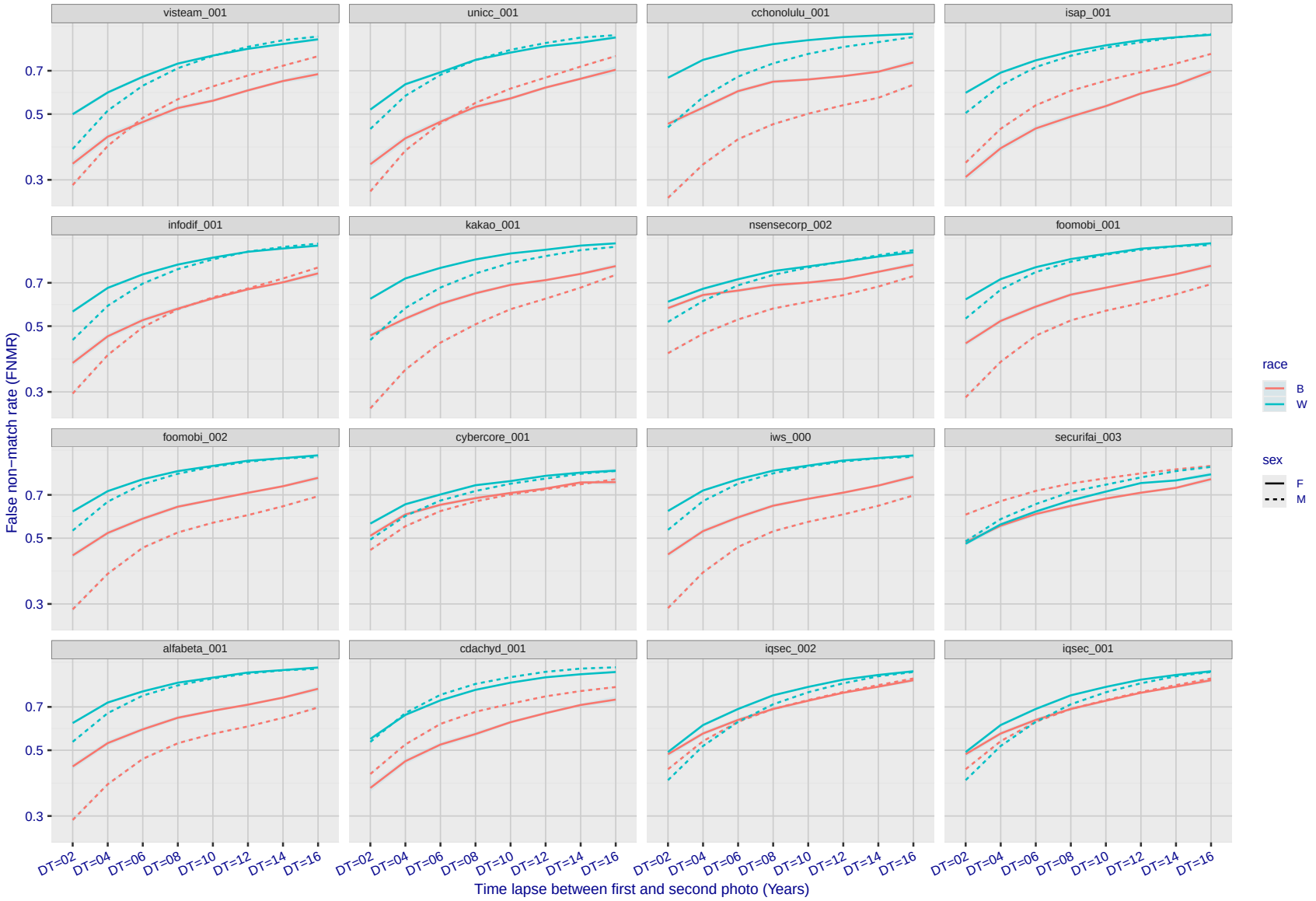


Figure 809: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

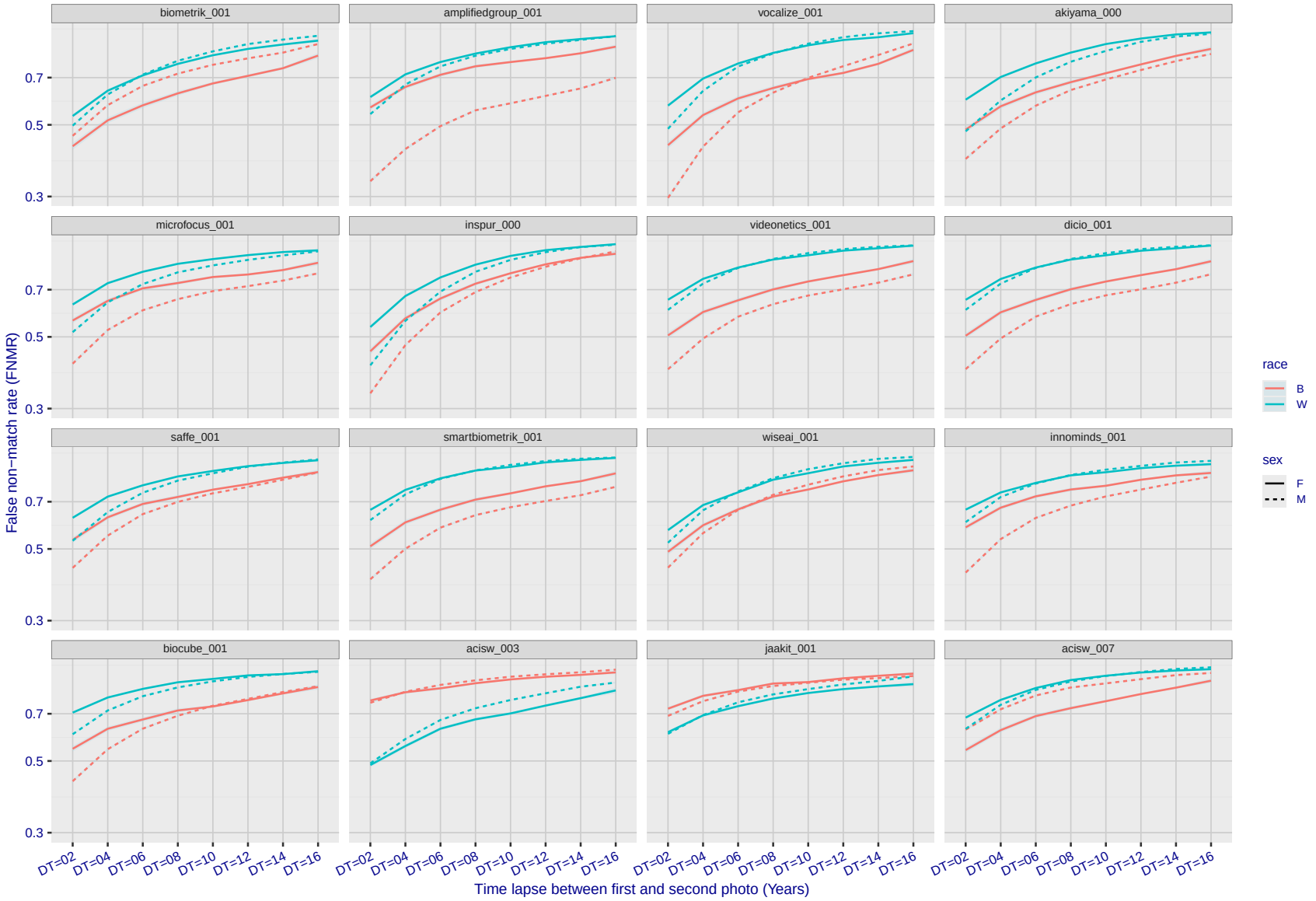


Figure 810: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

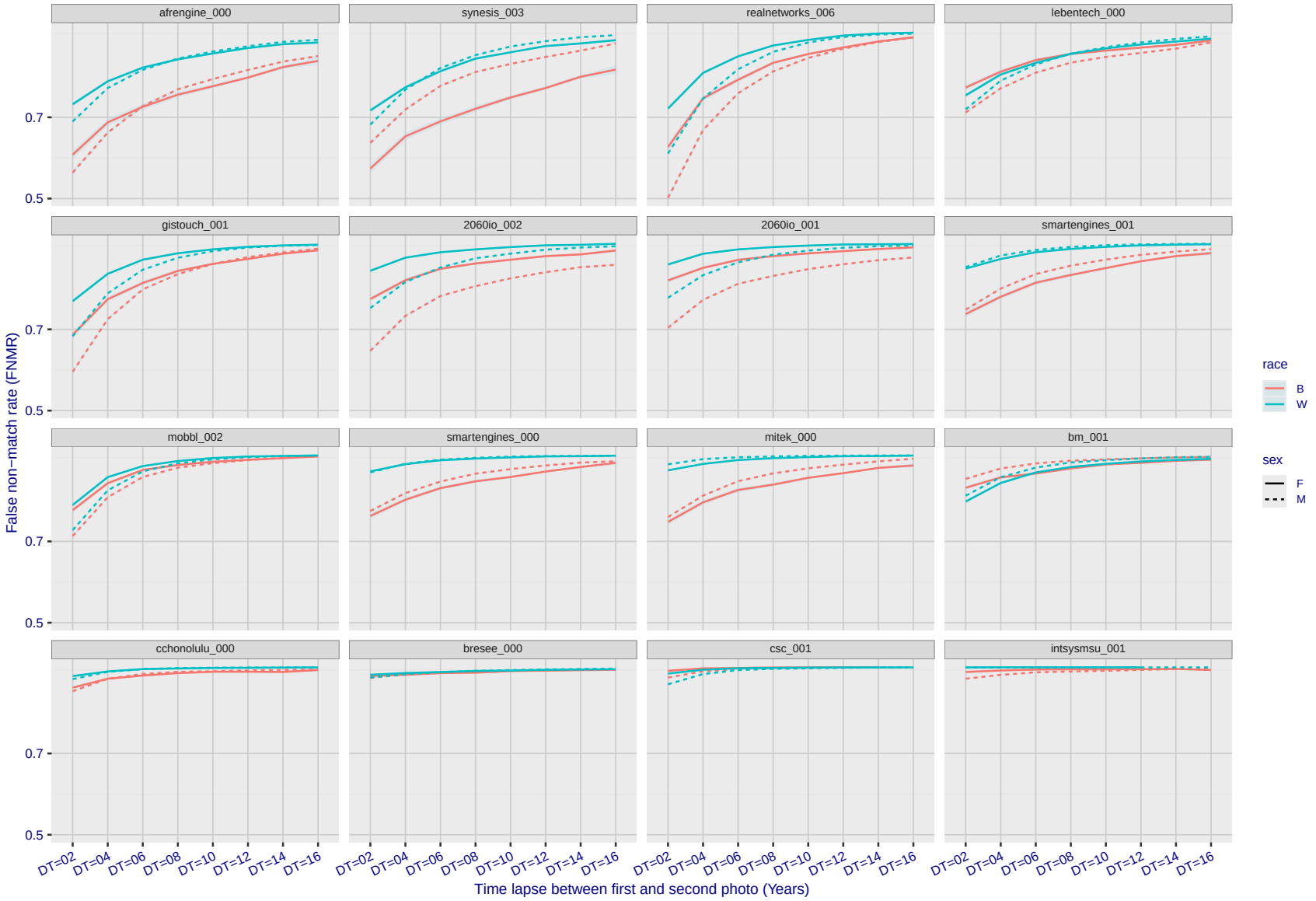


Figure 811: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

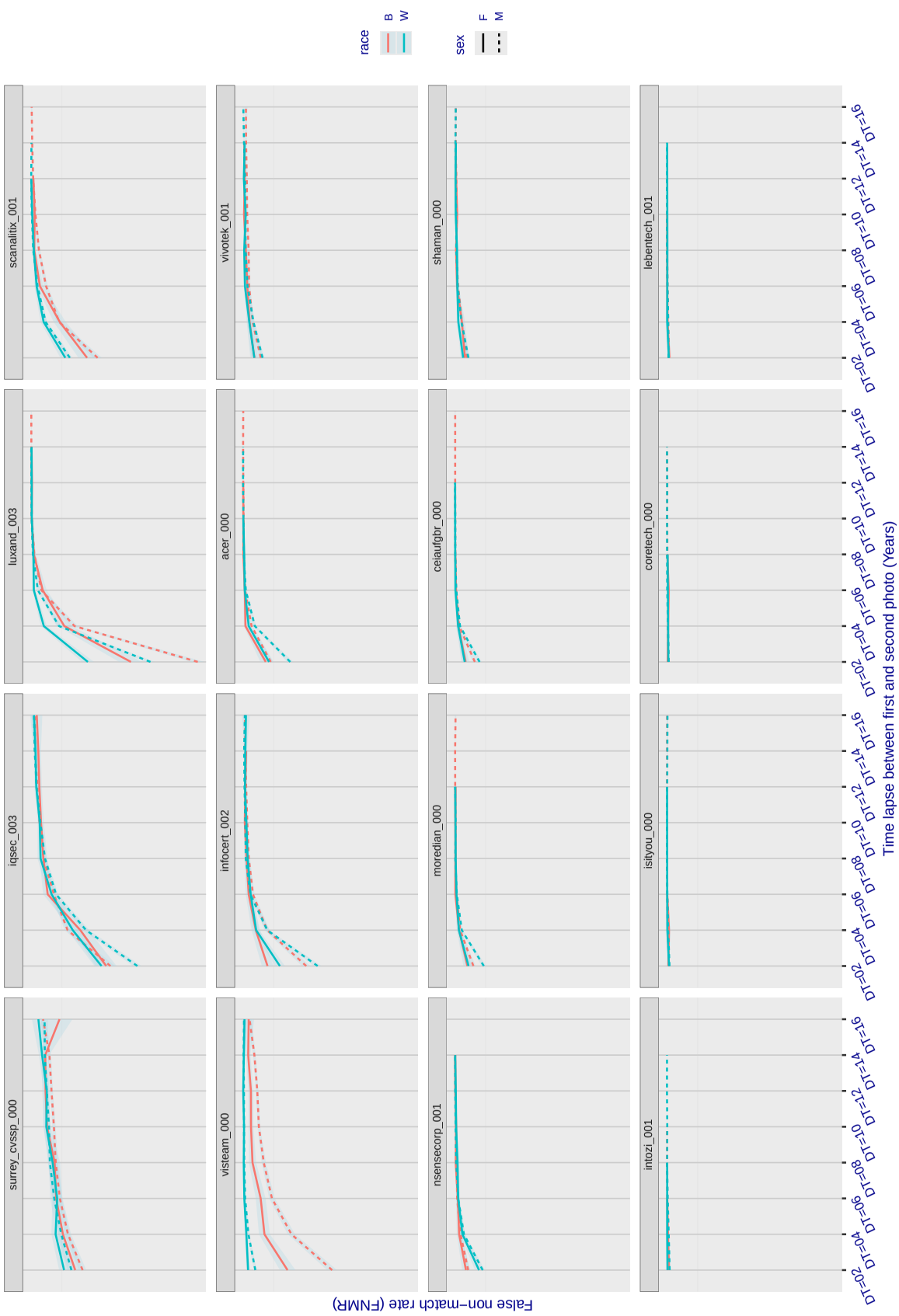


Figure 812: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all (10^8) impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR < 0.001) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

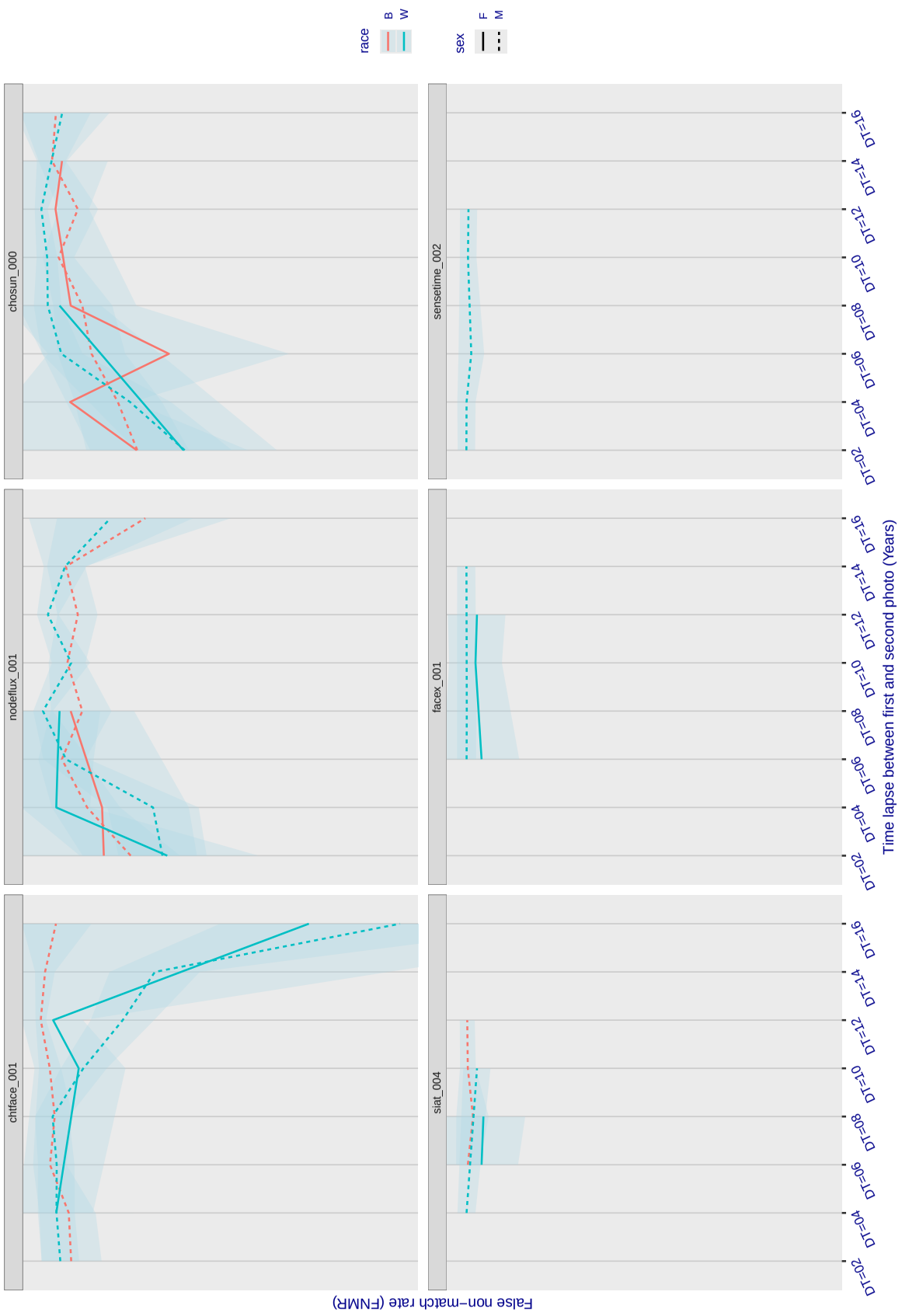


Figure 813: For the mugshot images, FNMR *as a function of elapsed time between initial enrollment and second verification images. The panels appear most accurate first, and vertical scale changes on each page. The four traces correspond to images annotated with codes for black female, black male, white female, white male. The threshold is fixed for each algorithm to give FMR = 0.00001 over all* (10^8) *impostor comparisons. For short time-lapses, the most accurate algorithms give very few errors (FNMR* < 0.001*) so that the uncertainty estimates are high.*

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.5.3 Effect of age on genuine subjects

Background: Faces change appearance throughout life. Face recognition algorithms have previously beeen reported to give better accuracy on older individuals (See NIST IR 8009).

Goal: To quantify false non-match rates (FNMR) as a function of age, without an ageing component.

Methods: Using the visa images, which span fewer than five years, thresholds are determined that give FMR = 0.001 and 0.0001 over the entire impostor set. Then FNMR is measured over 1000 bootstrap replications of the genuine scores.

Results: For the visa images, Figure 904 shows how false non-match rates for genuine users, as a function of age group.

The notable aspects are:

- ▷ Younger subjects give considerably higher FNMR. This is likely due to rapid growth and change in facial appearance.
- ▷ FNMR trends down throughout life. The last bin, AGE $>$ 72, contains fewer than 140 mated pairs, and may be affected by small sample size.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 814: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 815: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 816: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 817: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 818: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

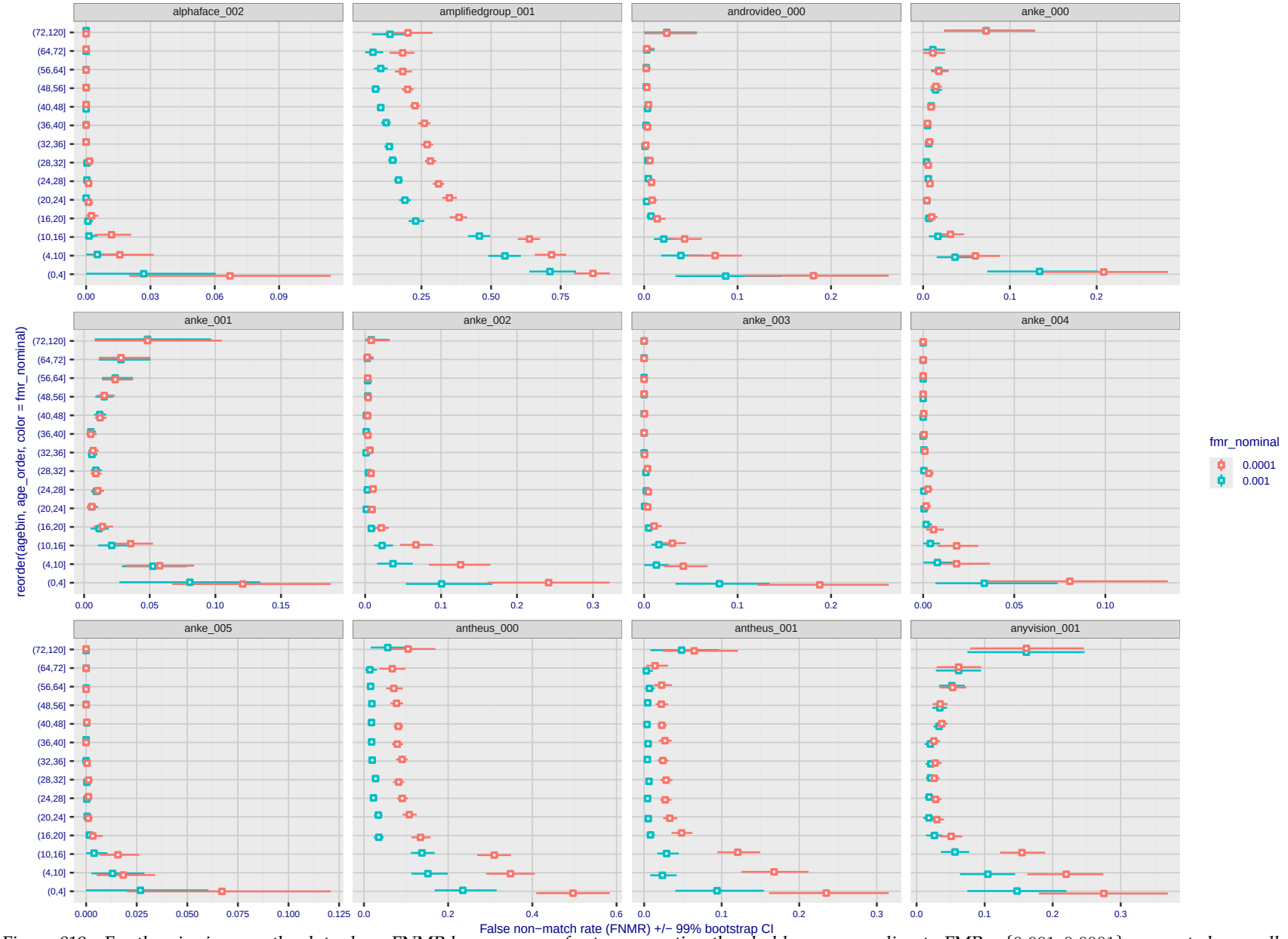


Figure 819: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 820: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 821: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 822: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 823: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 824: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

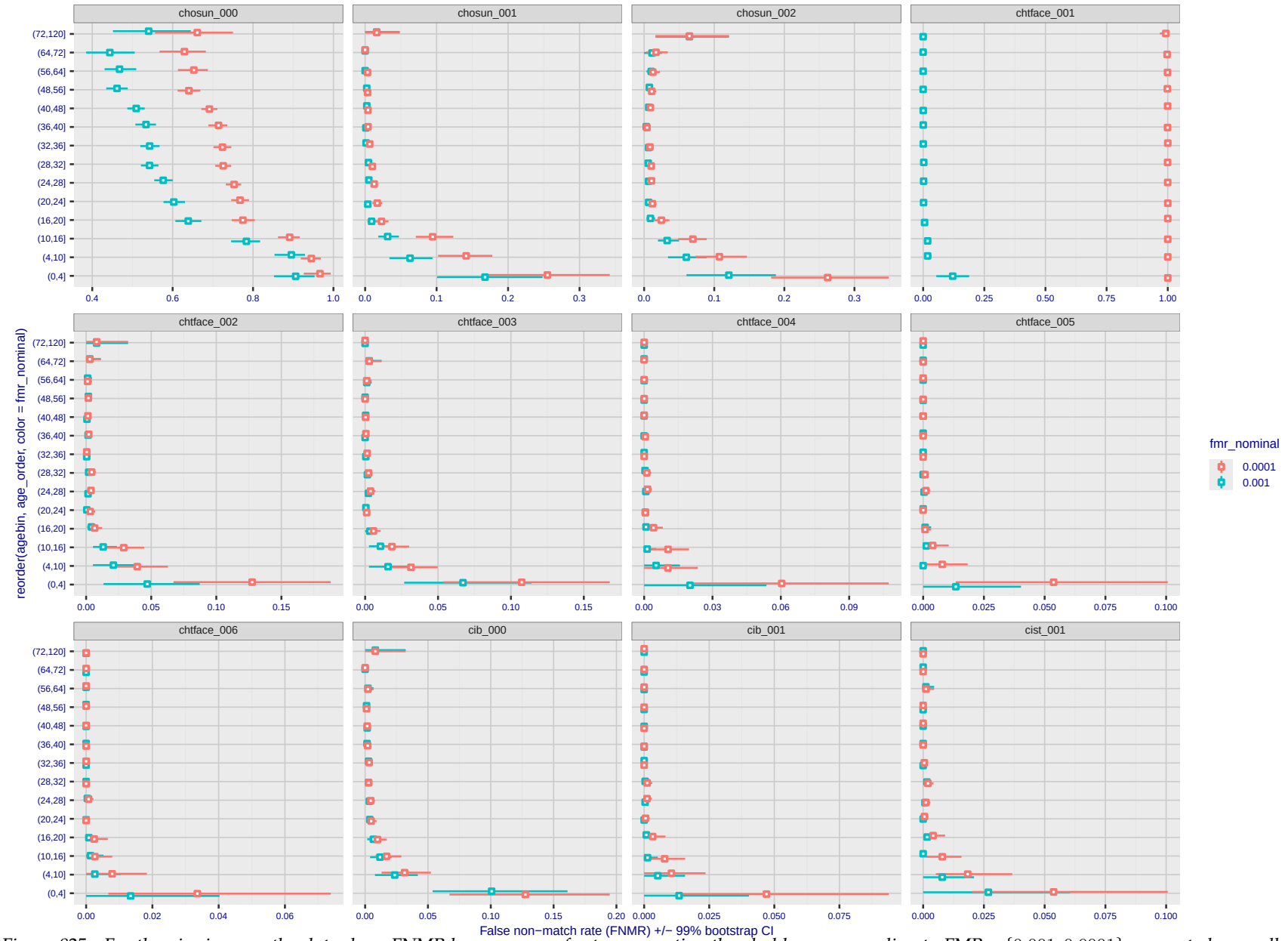


Figure 825: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 826: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 827: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

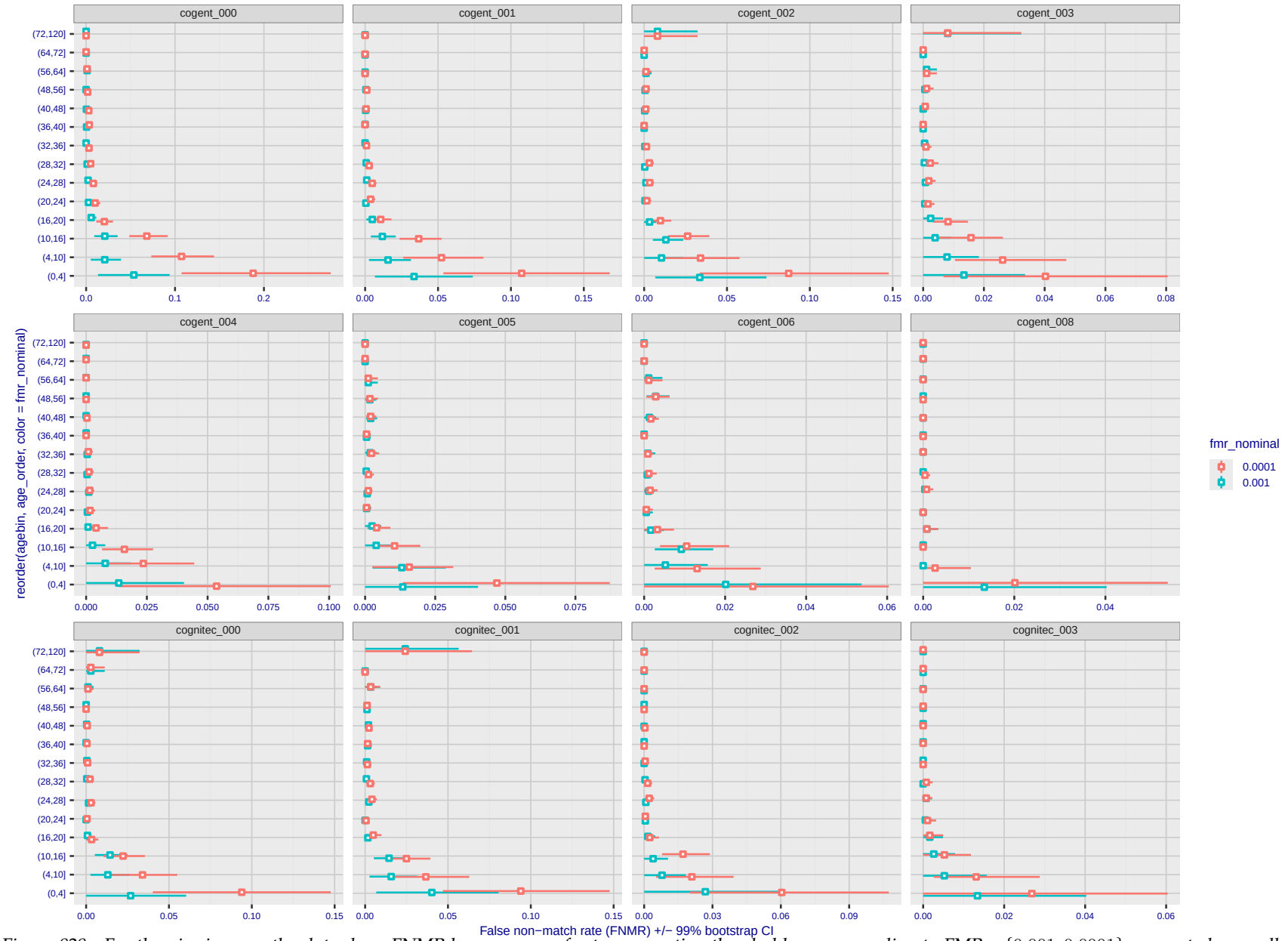


Figure 828: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 829: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 830: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 831: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

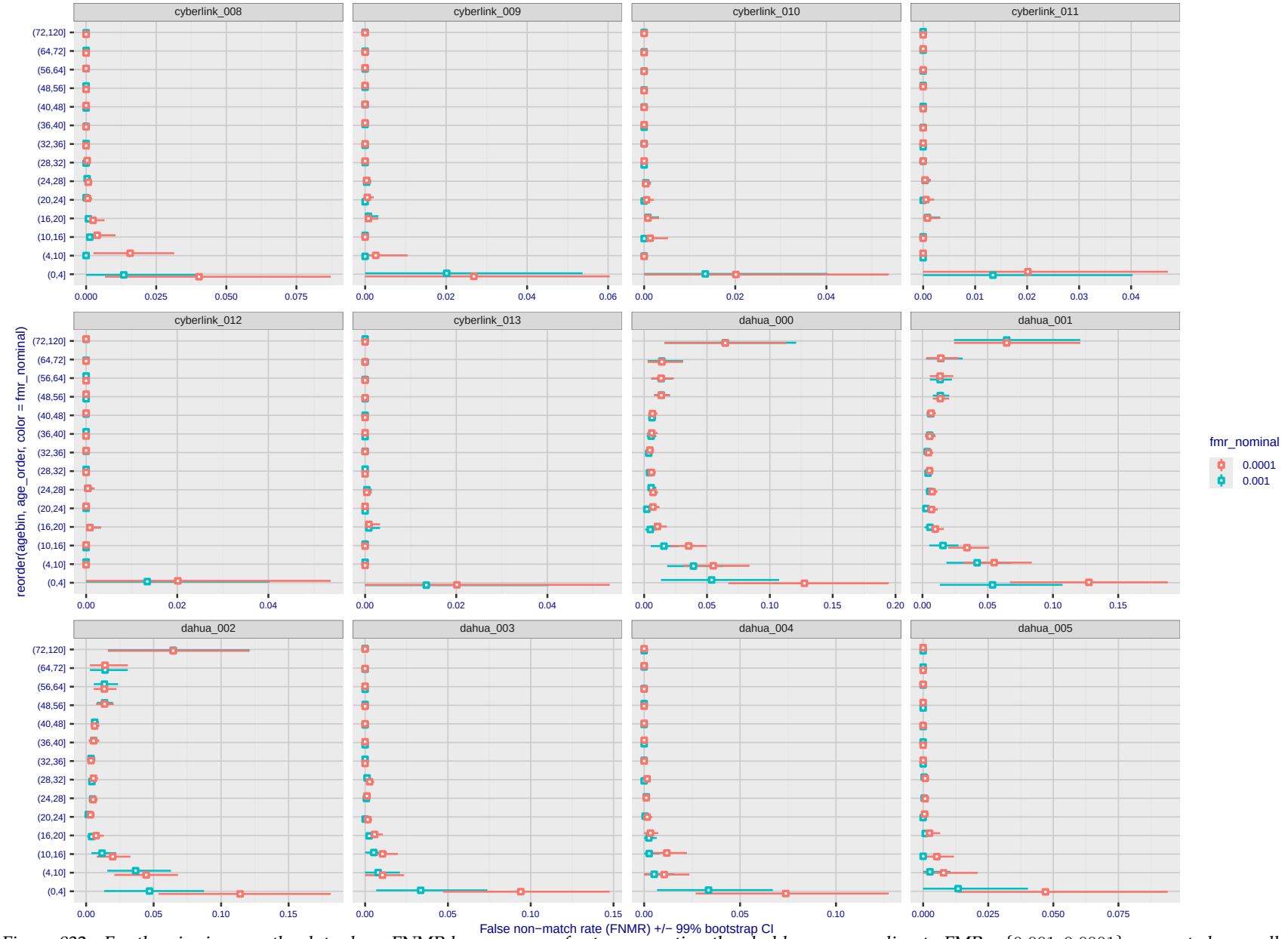


Figure 832: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

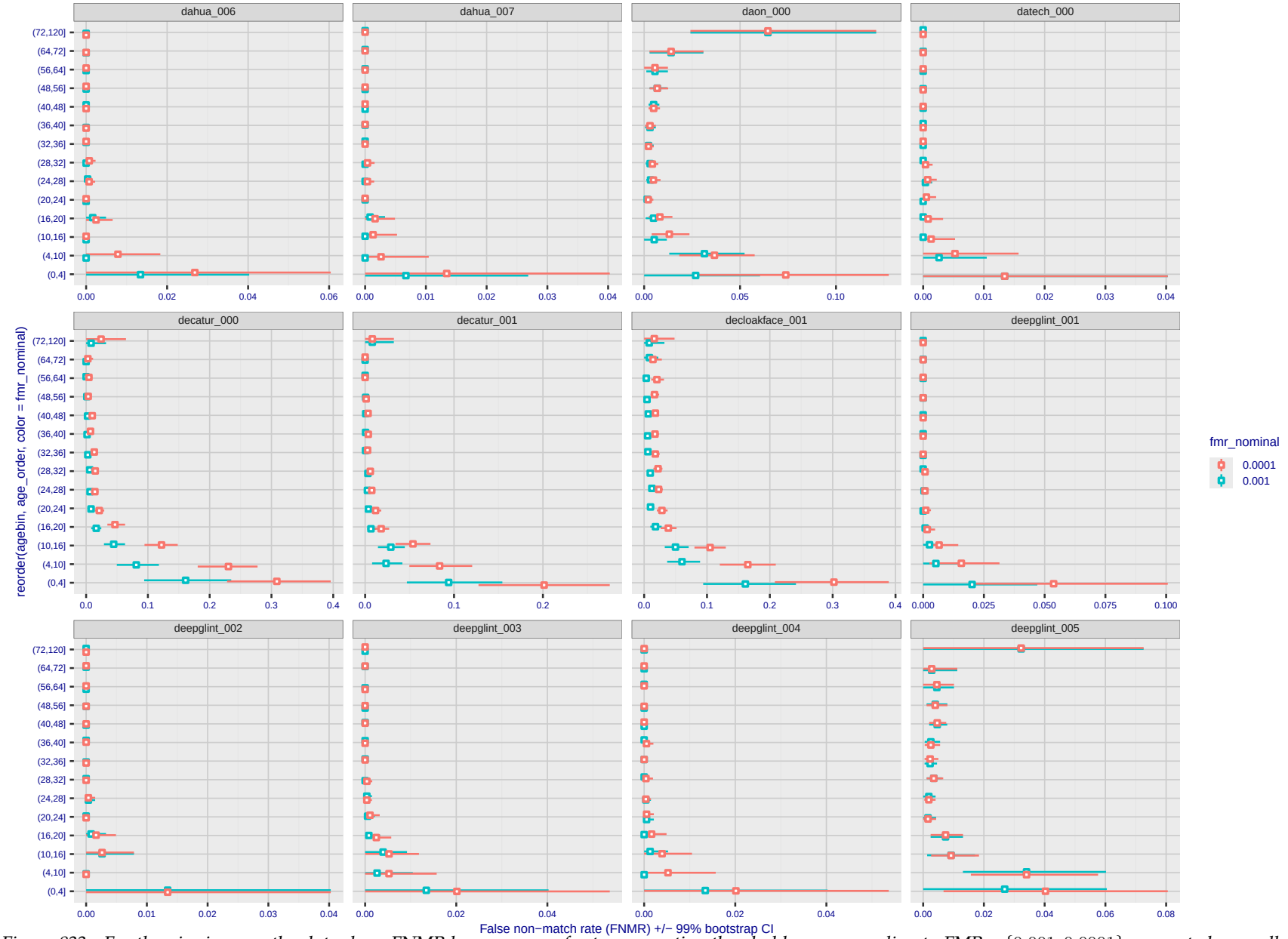


Figure 833: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 834: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

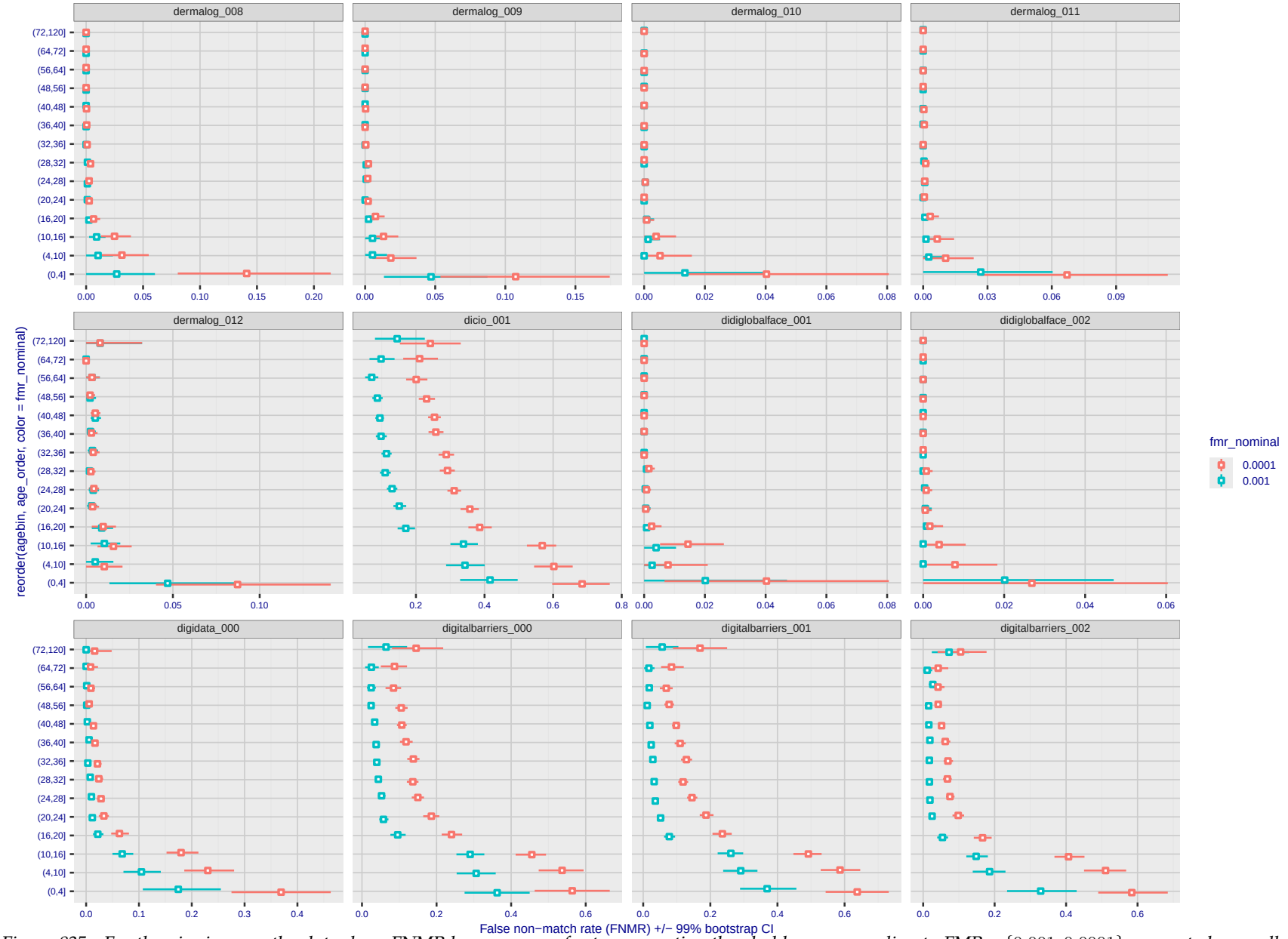


Figure 835: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 836: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

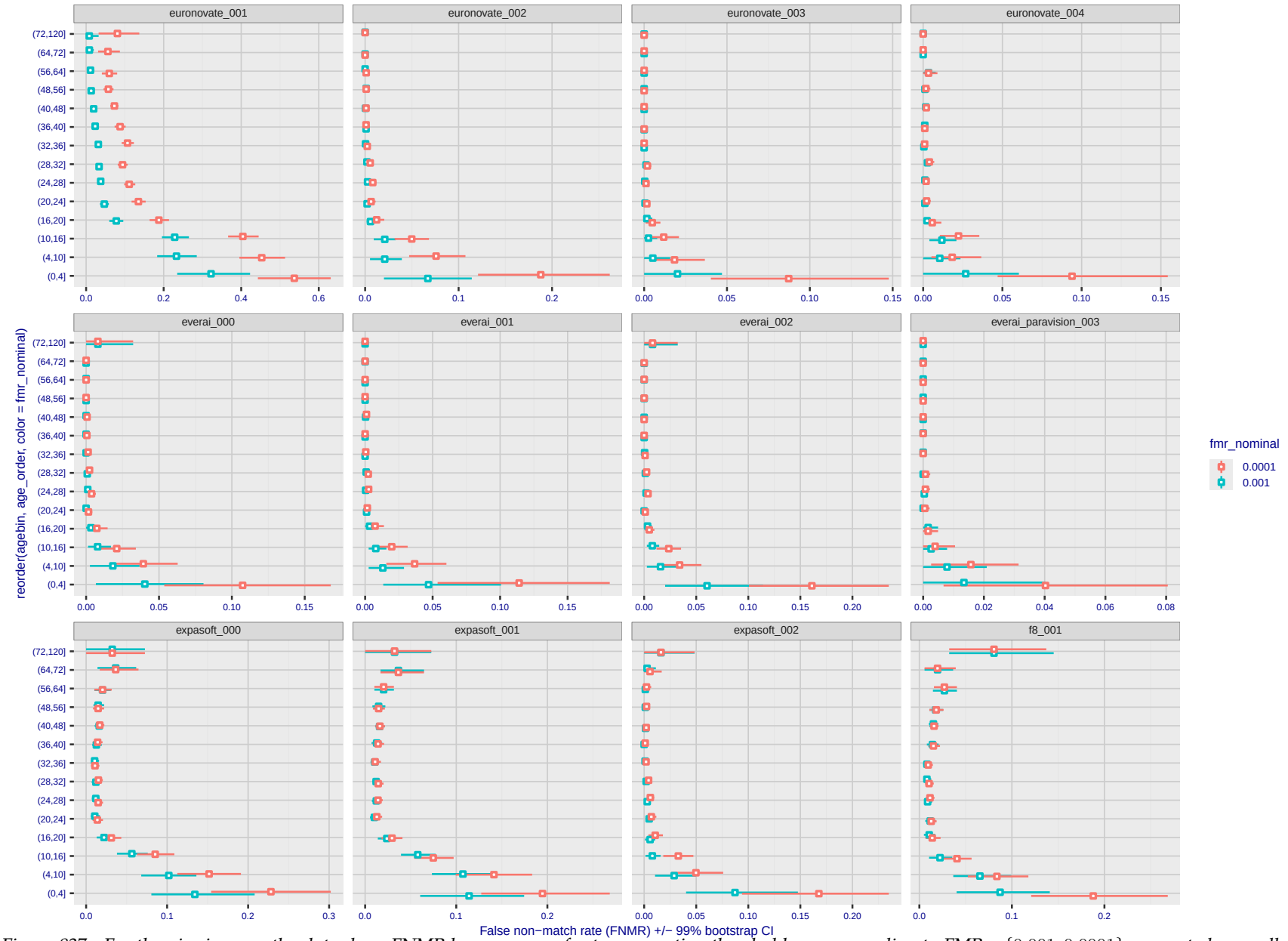


Figure 837: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 838: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

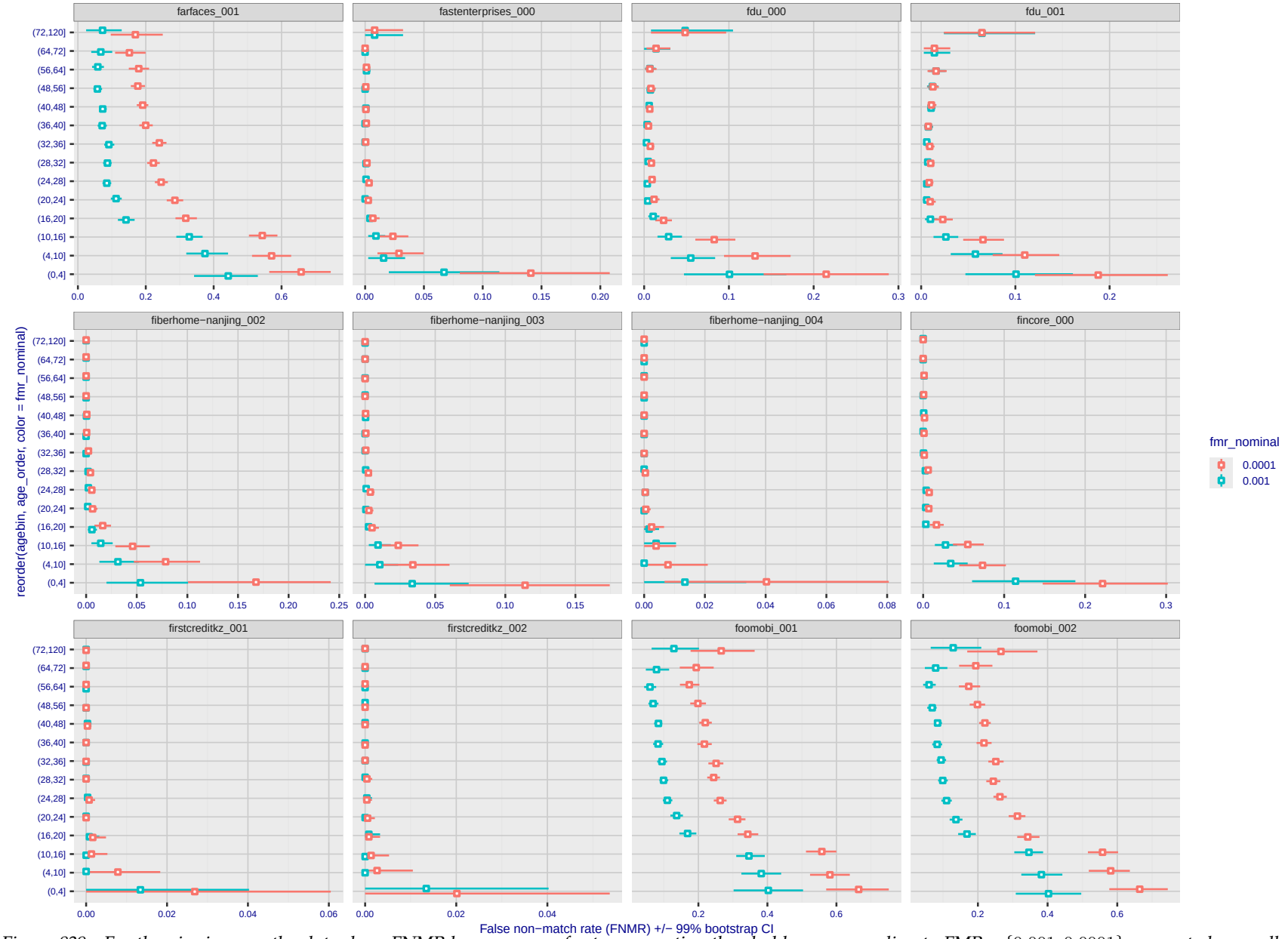


Figure 839: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 840: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 841: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 842: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 843: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 844: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

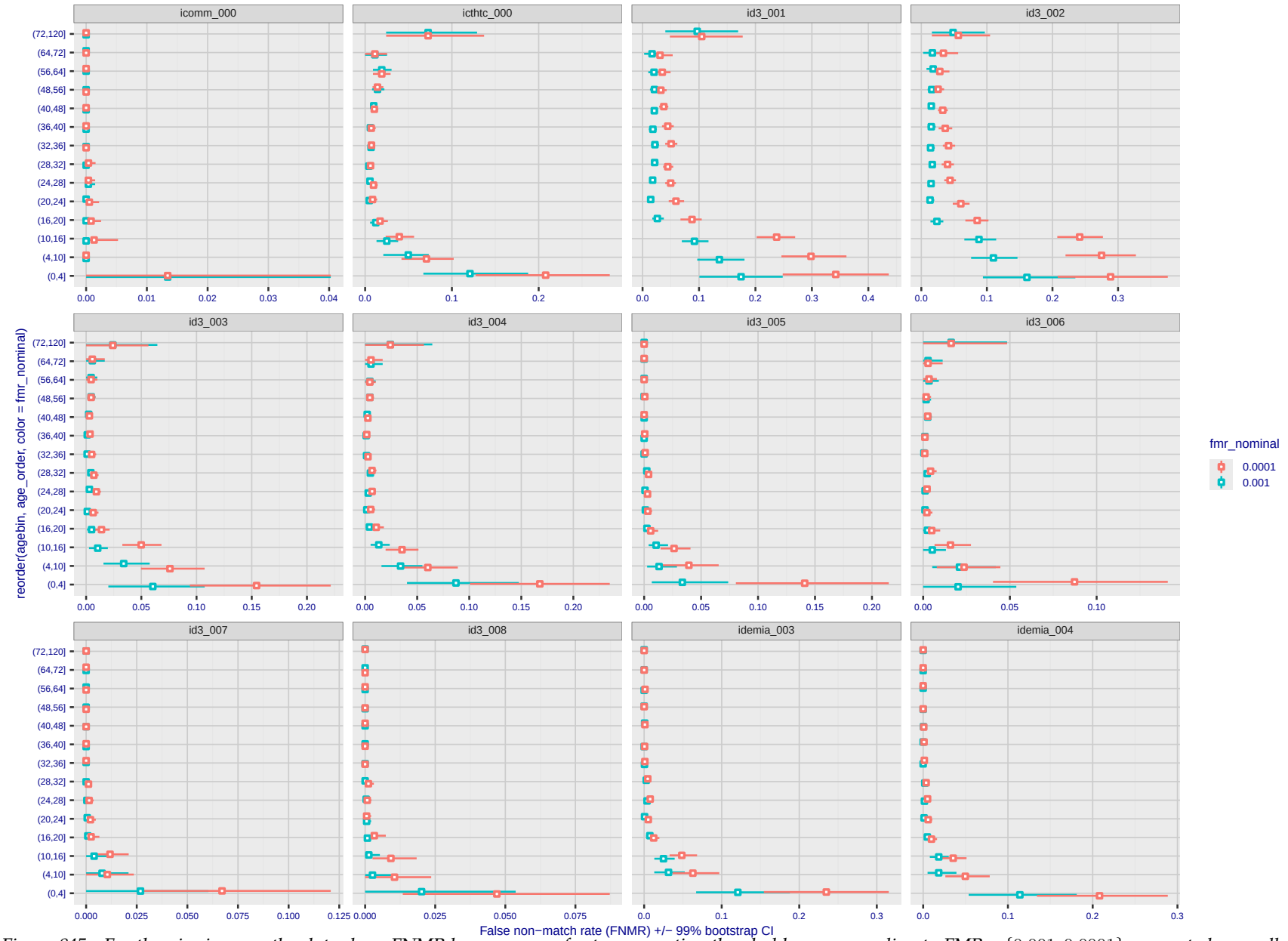


Figure 845: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

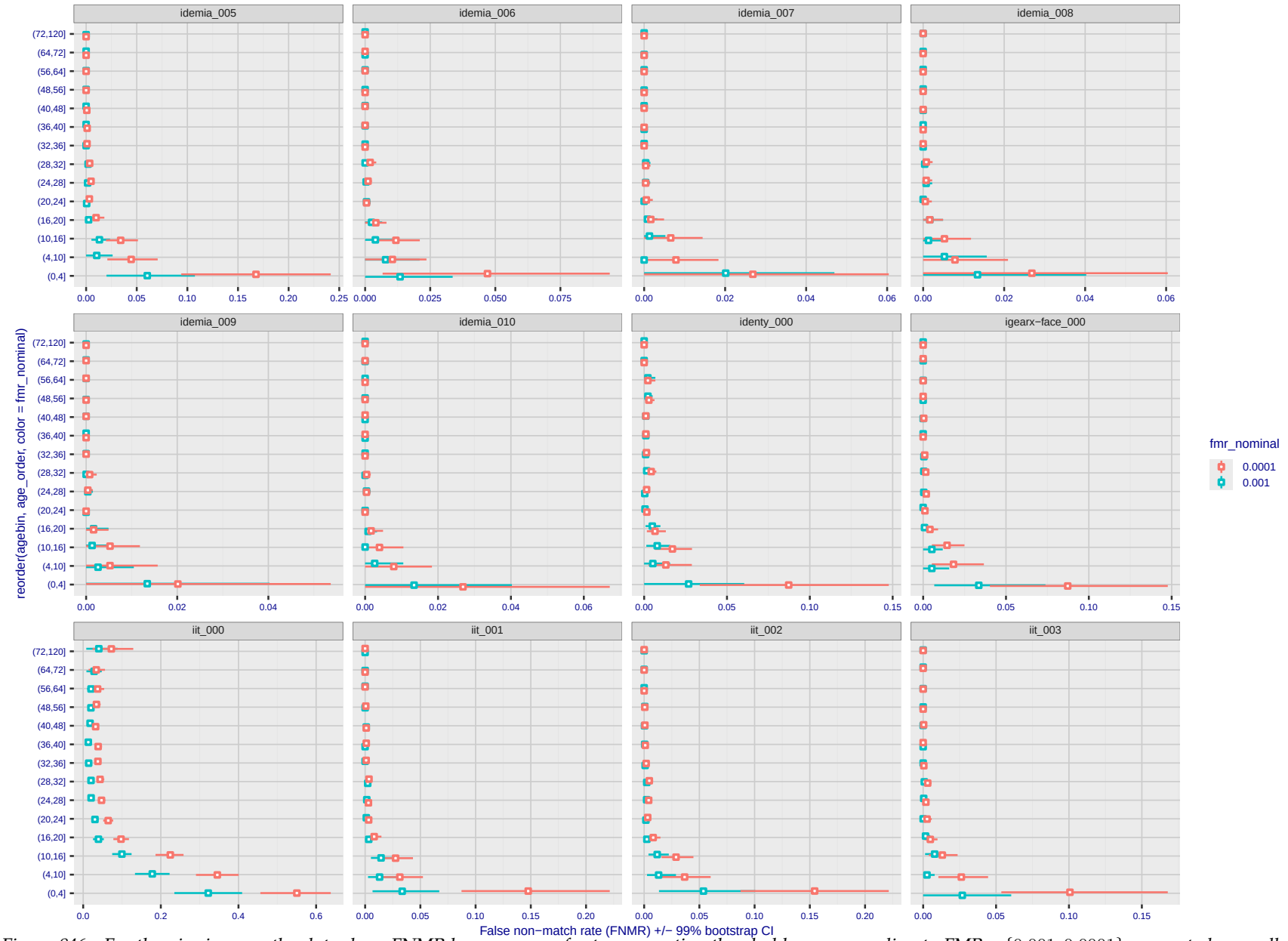


Figure 846: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 847: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

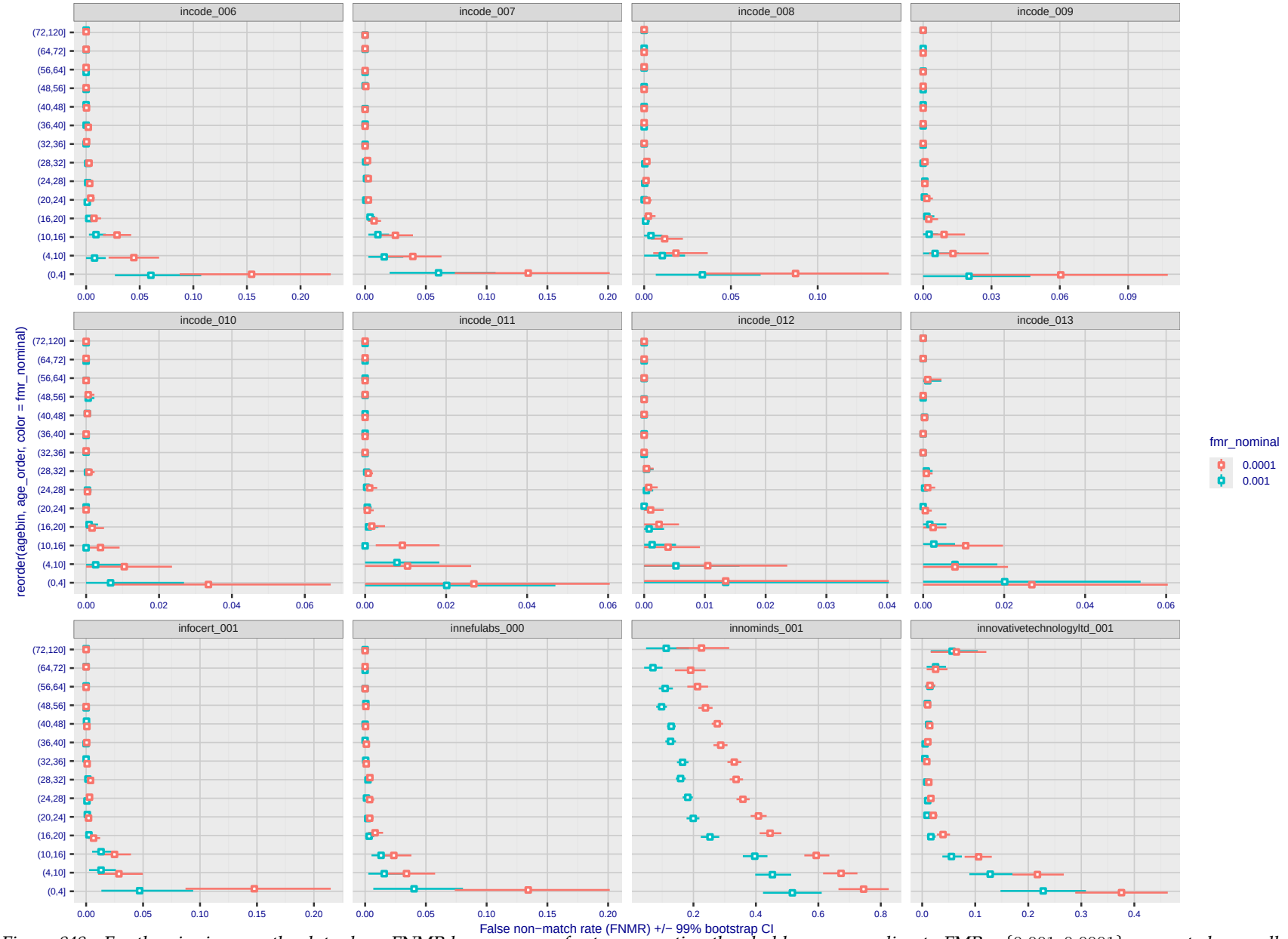


Figure 848: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 849: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

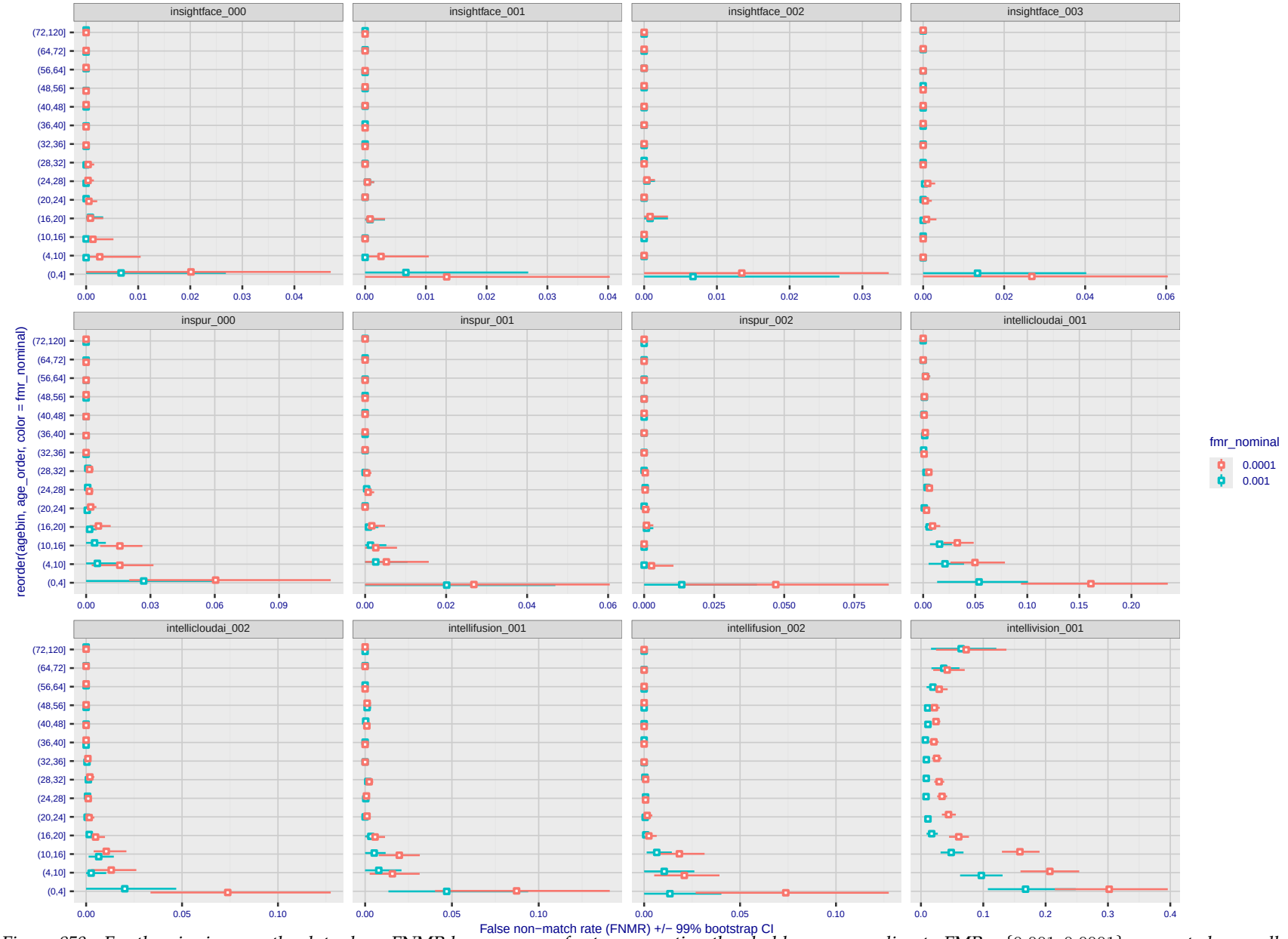


Figure 850: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

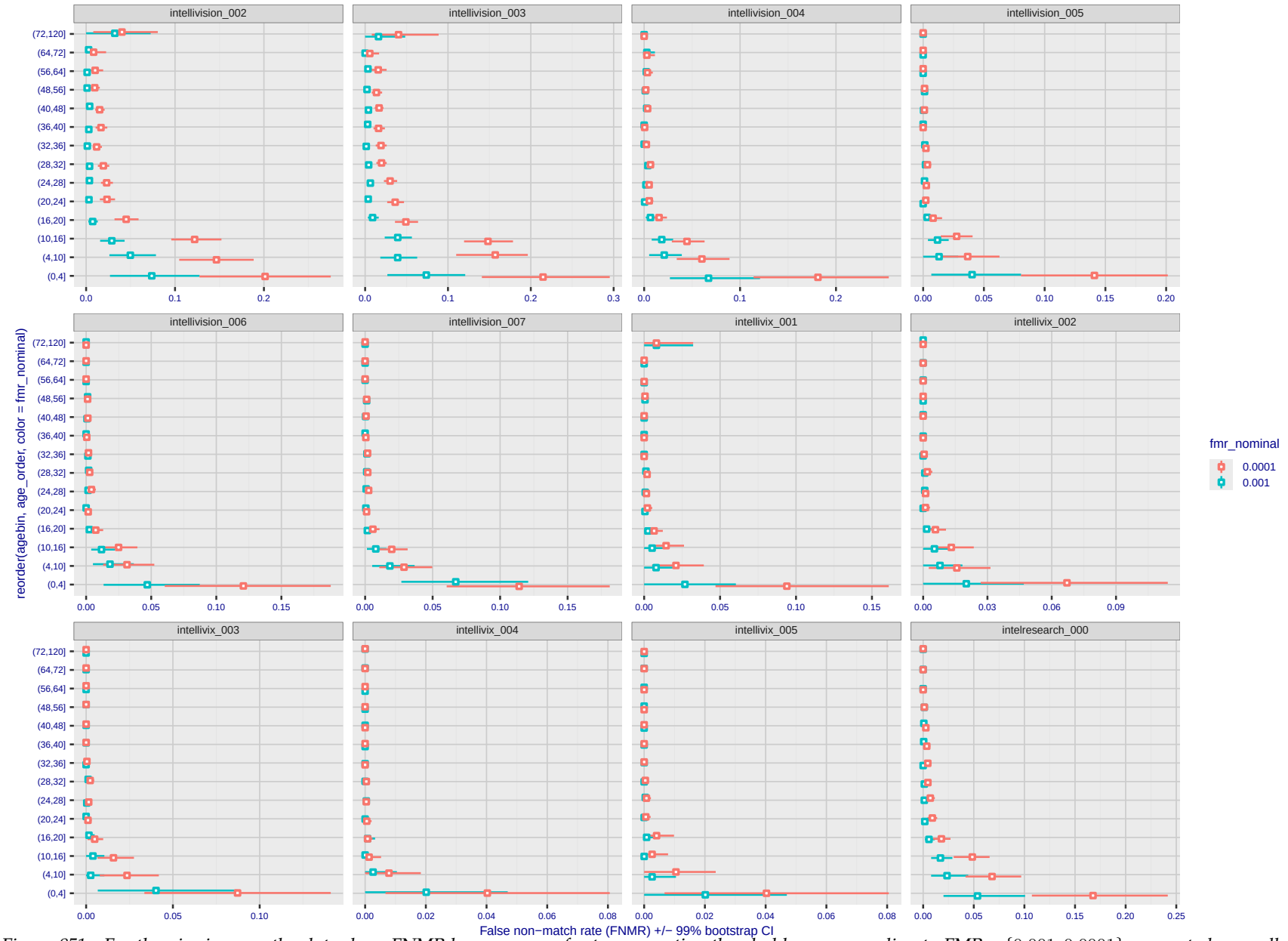


Figure 851: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

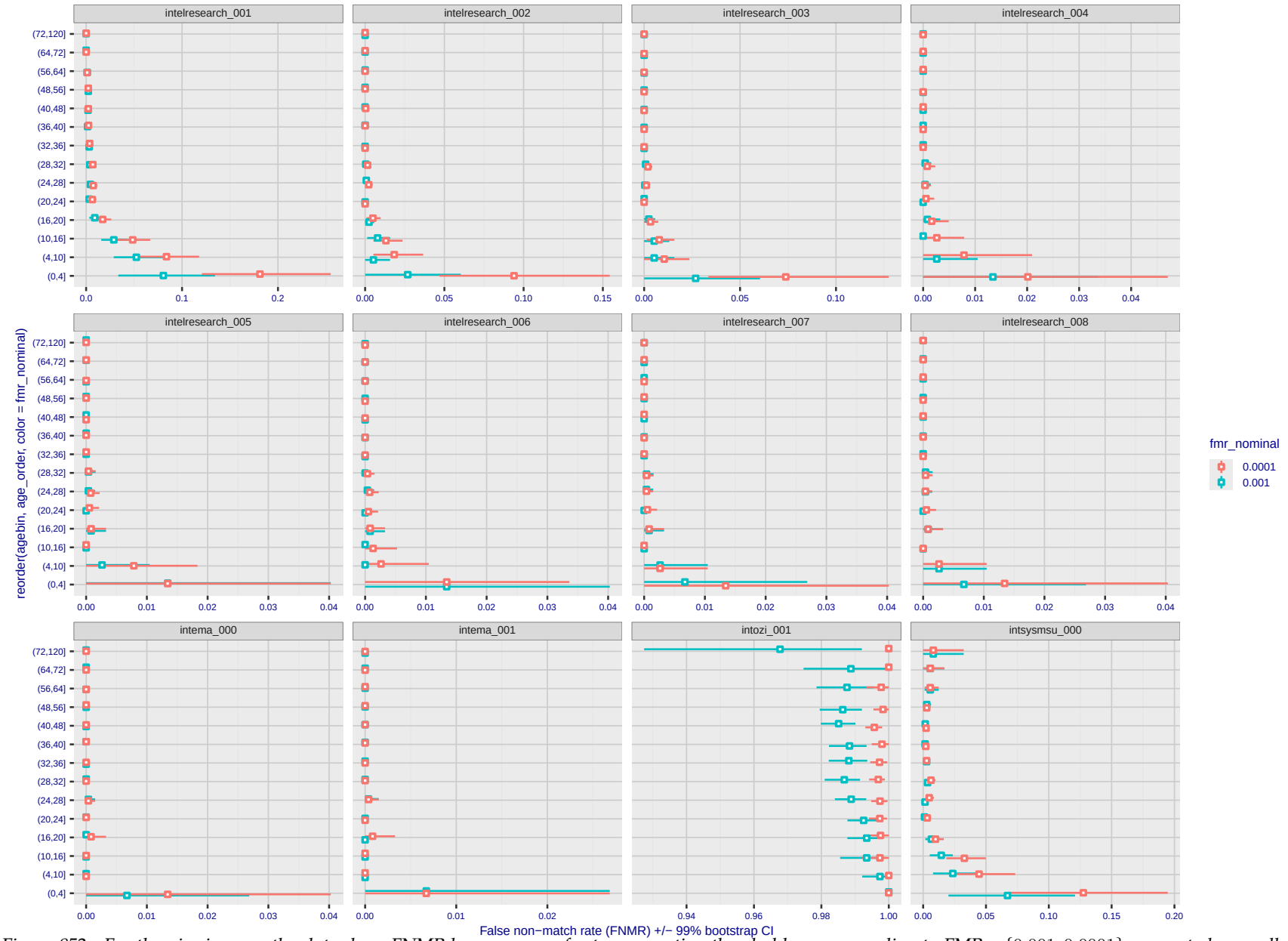


Figure 852: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 853: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 854: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 855: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 856: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 857: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

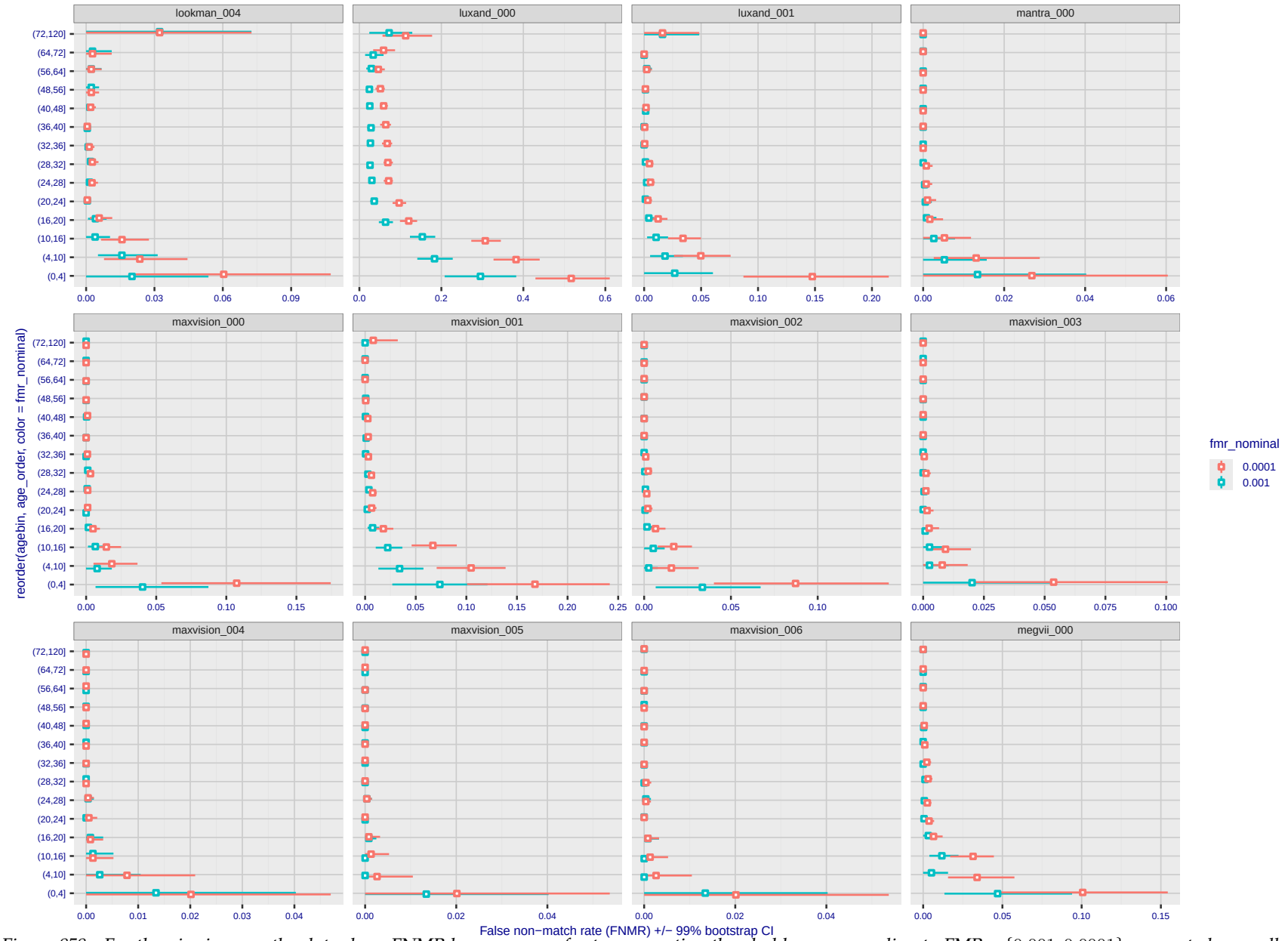


Figure 858: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

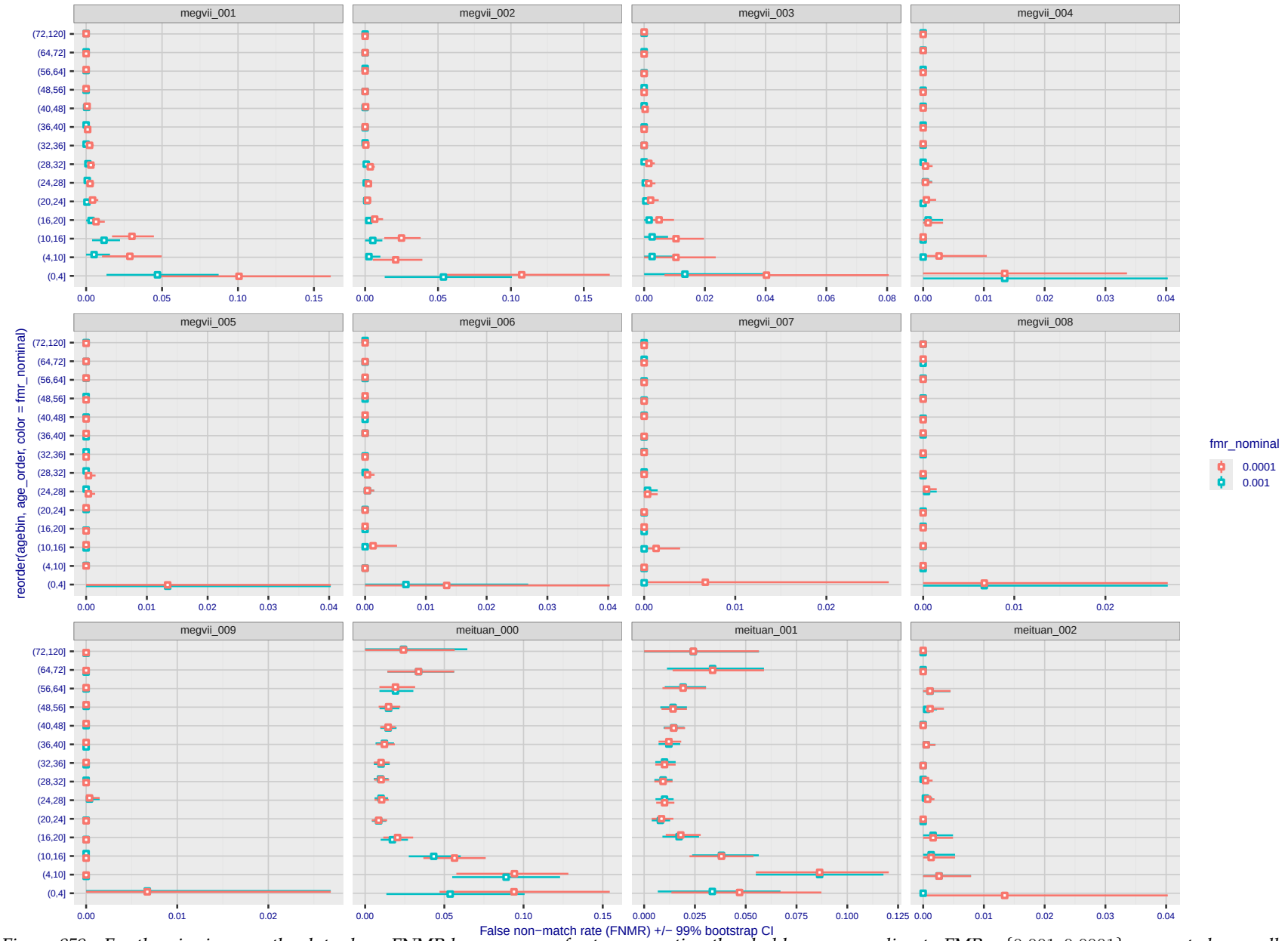


Figure 859: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 860: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

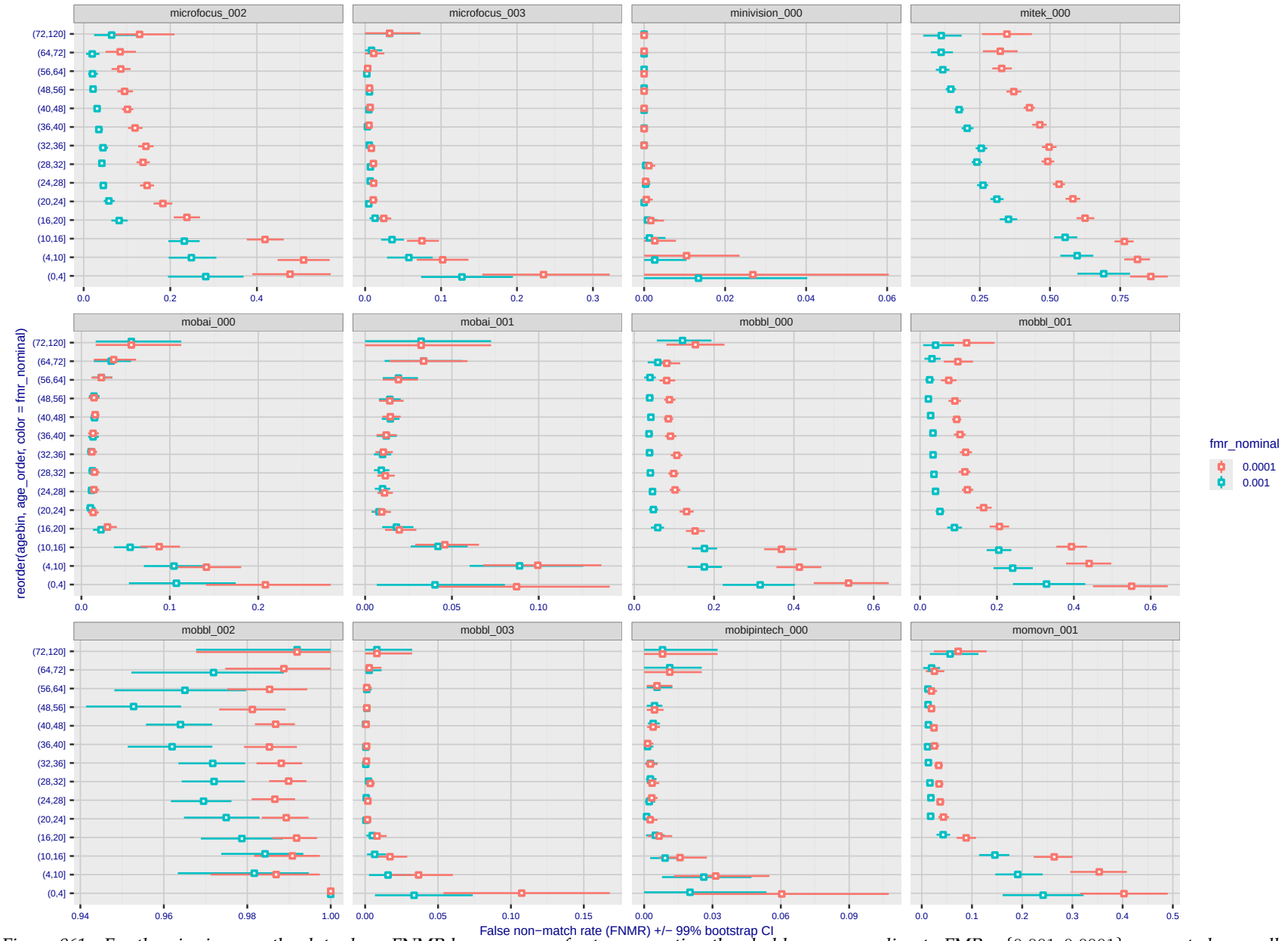


Figure 861: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 862: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

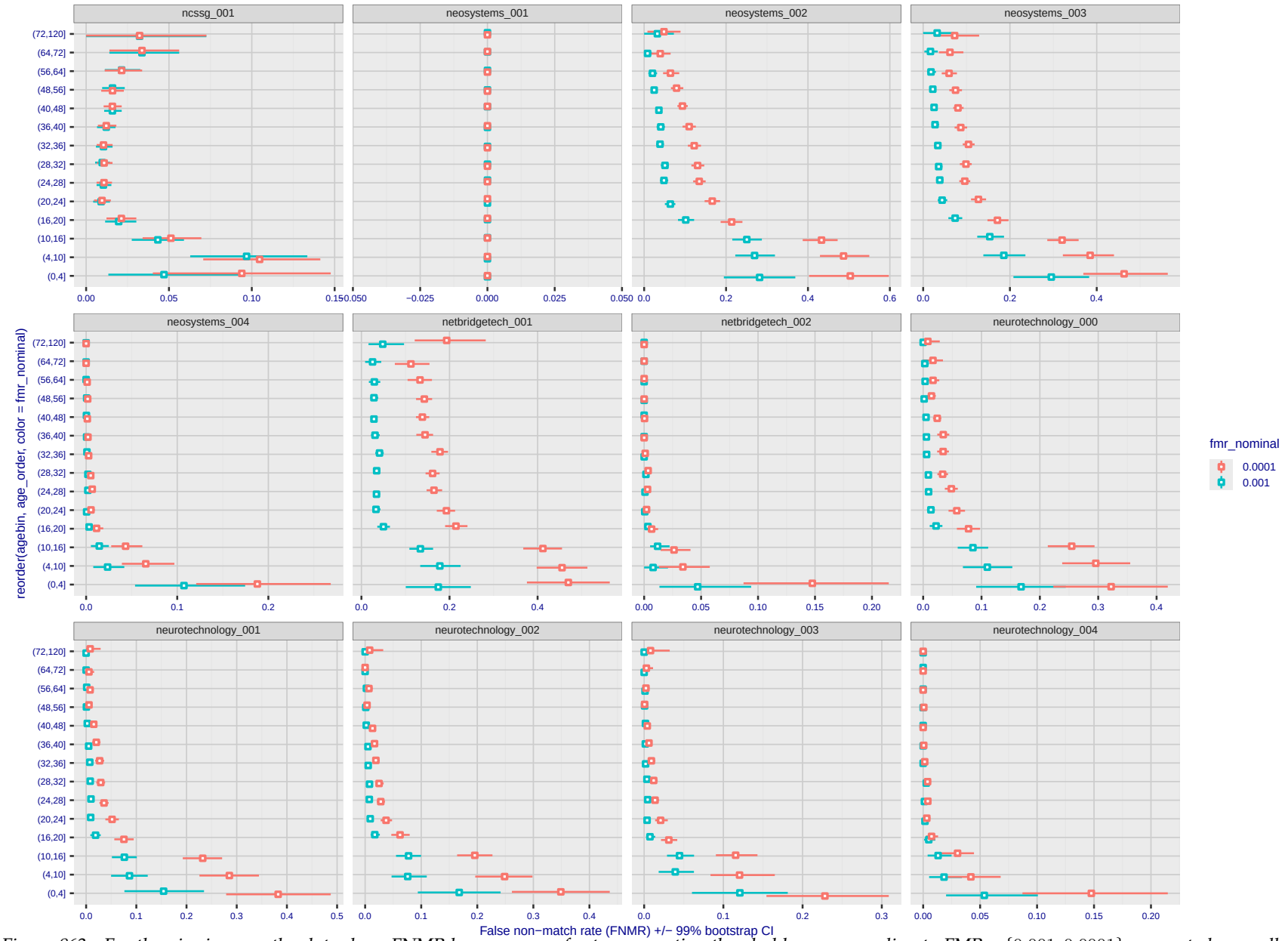


Figure 863: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

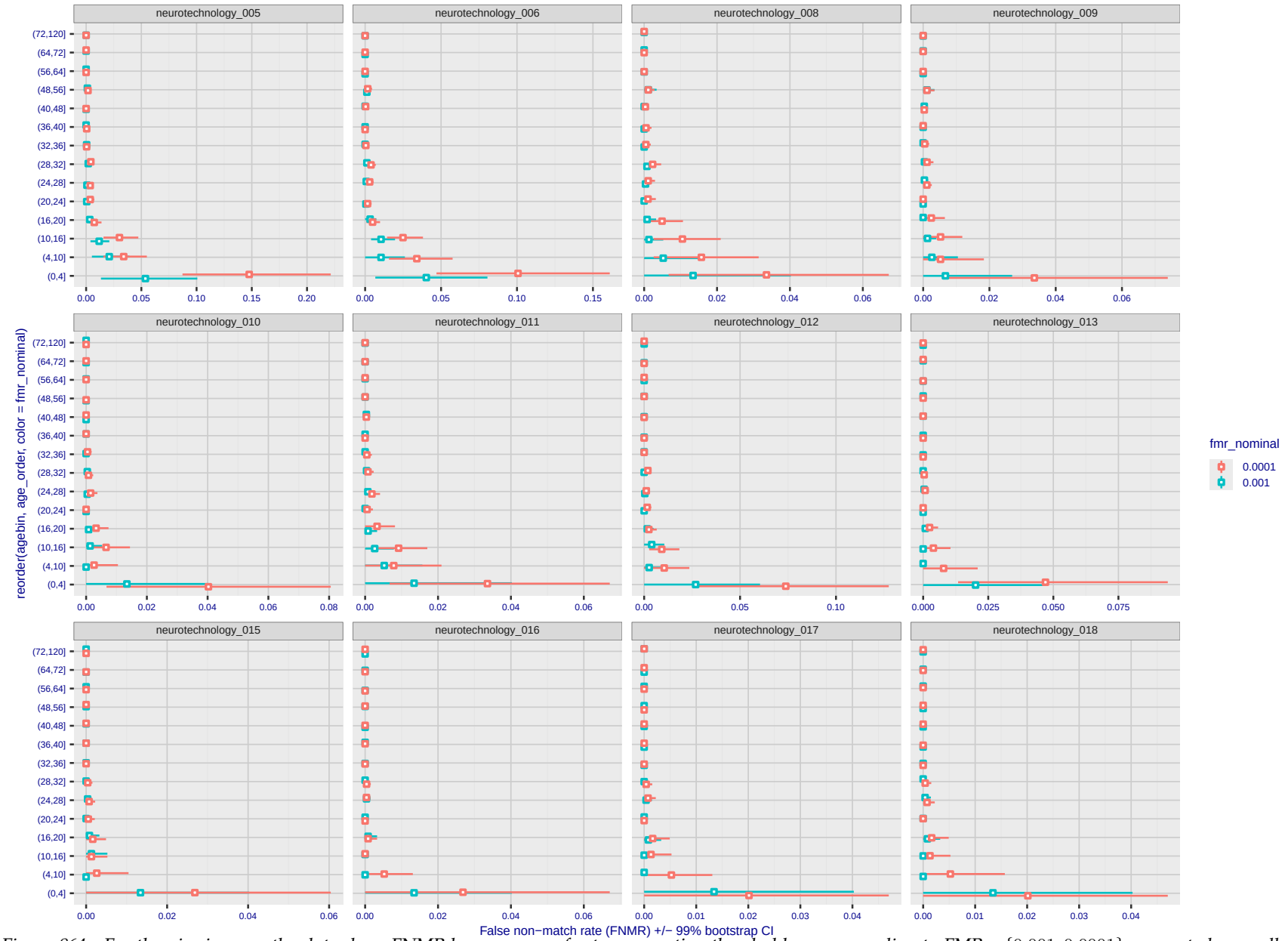


Figure 864: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 865: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 866: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

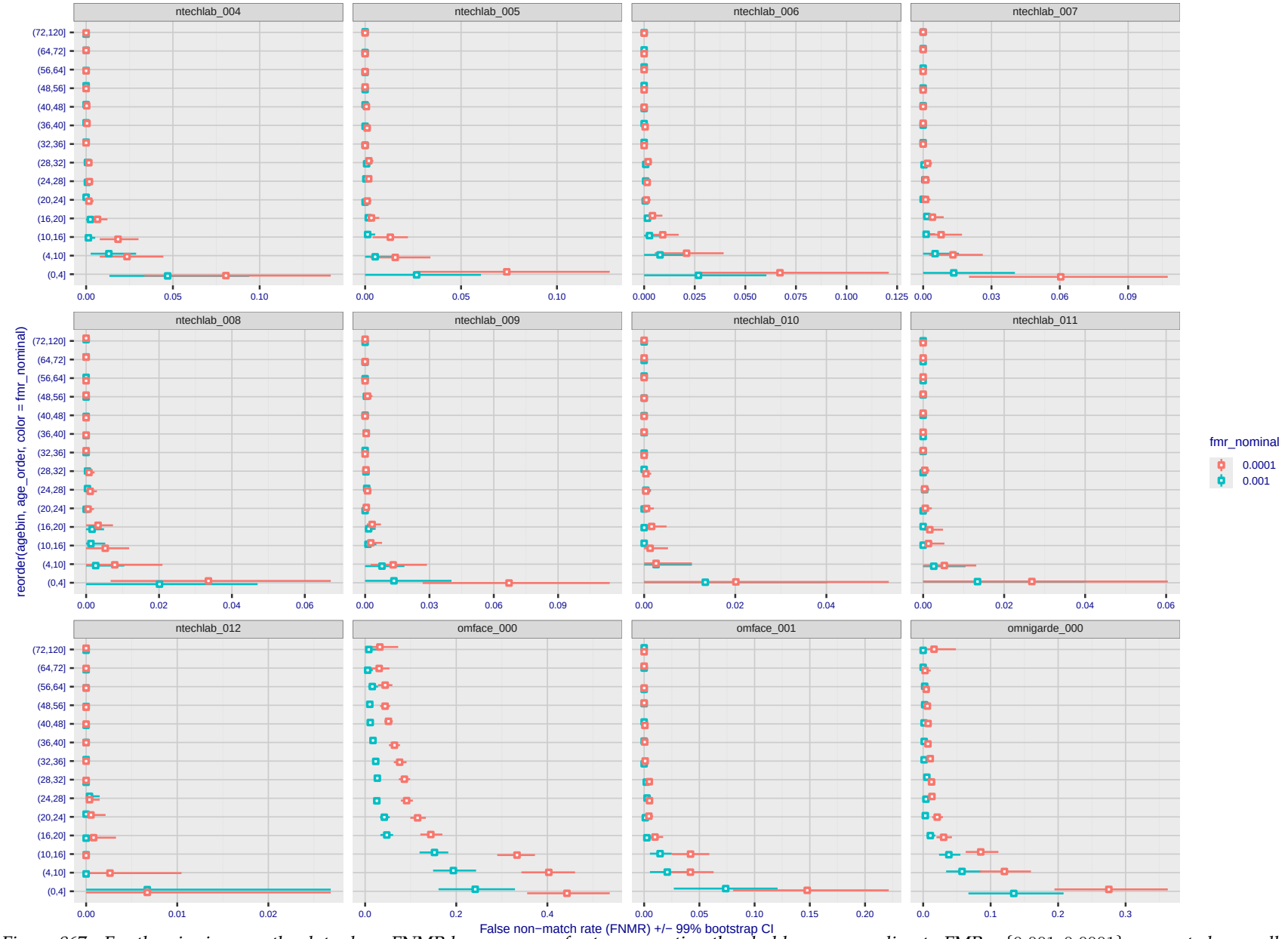


Figure 867: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 868: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

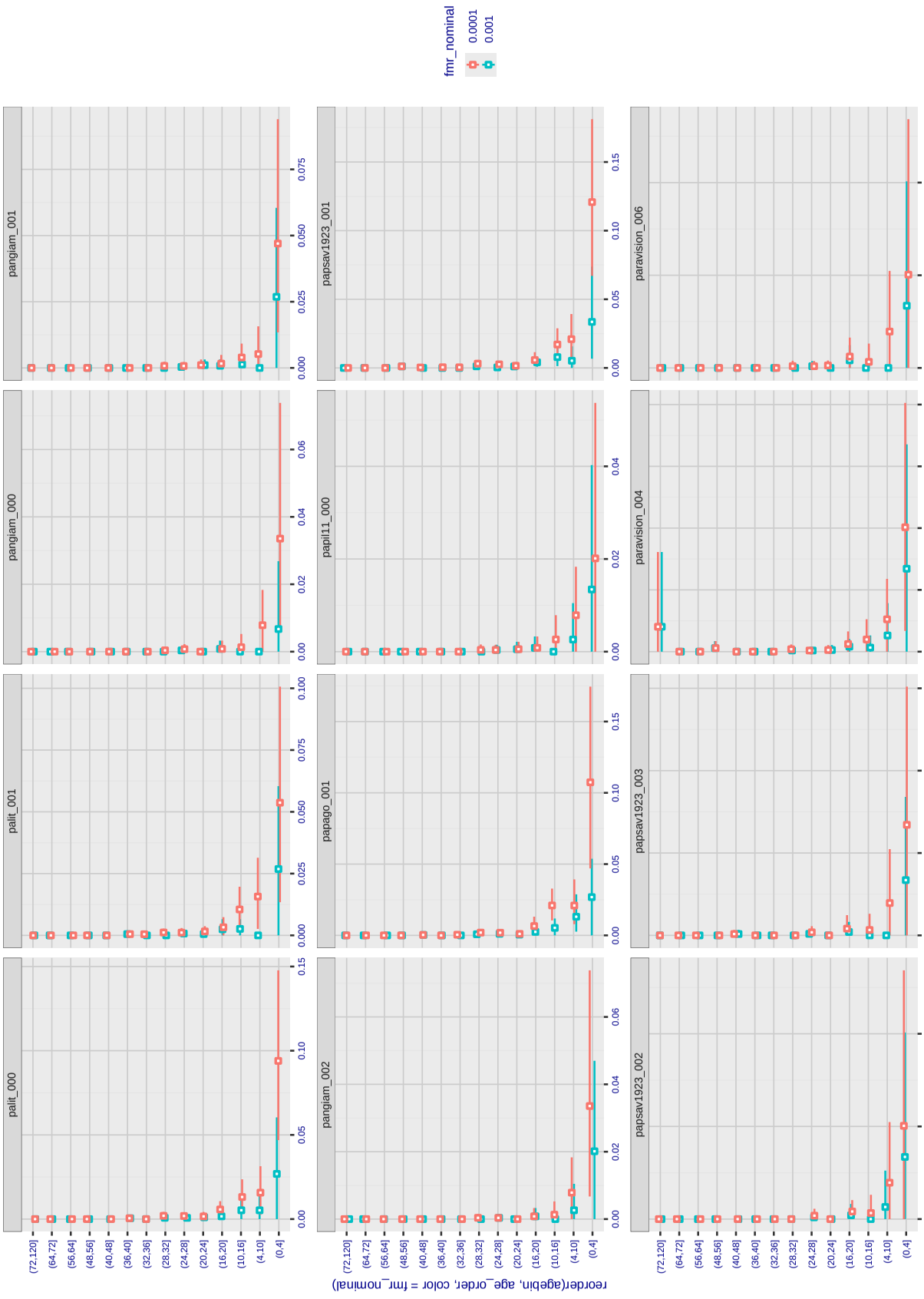


Figure 869: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

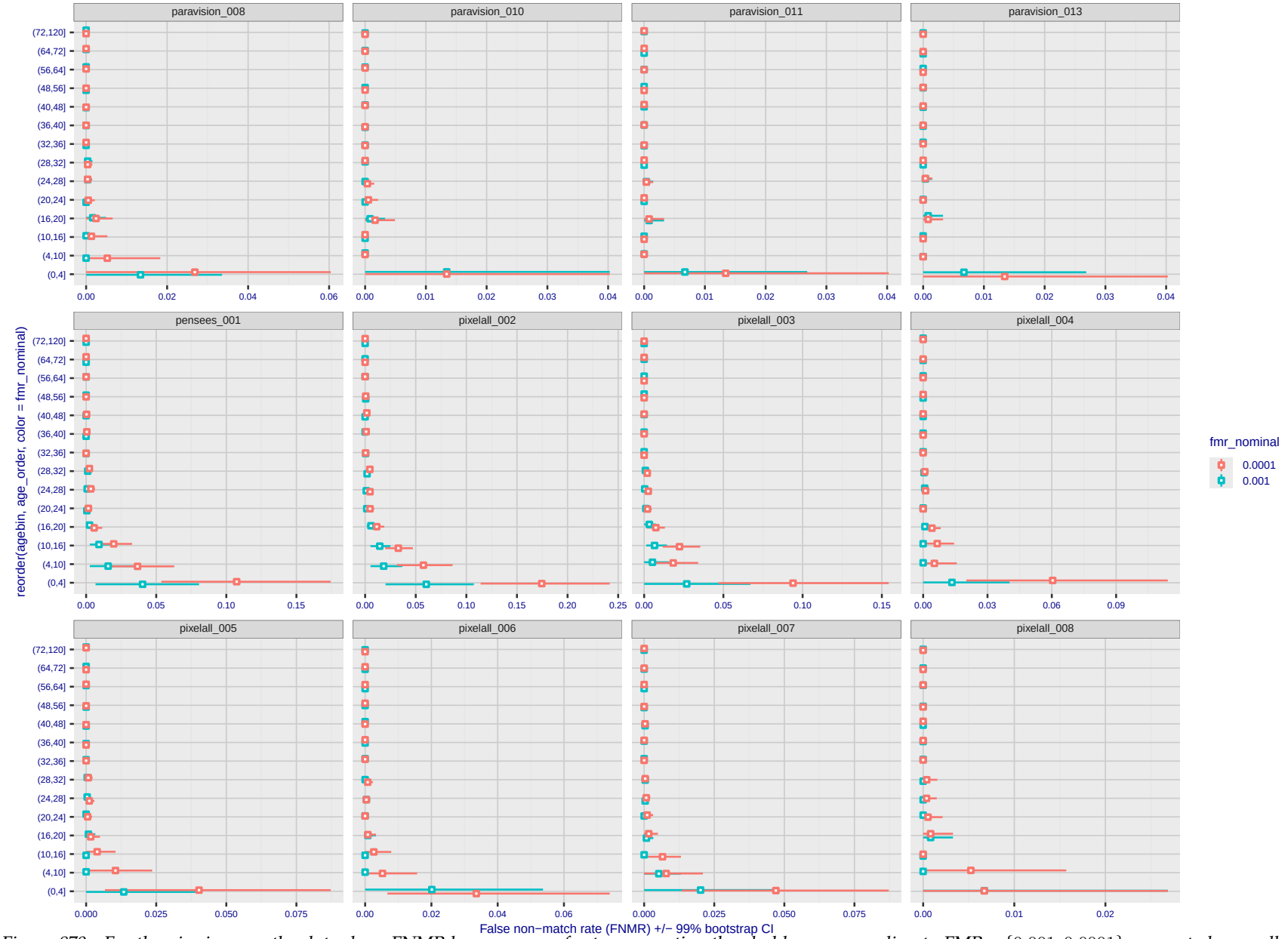


Figure 870: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 871: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

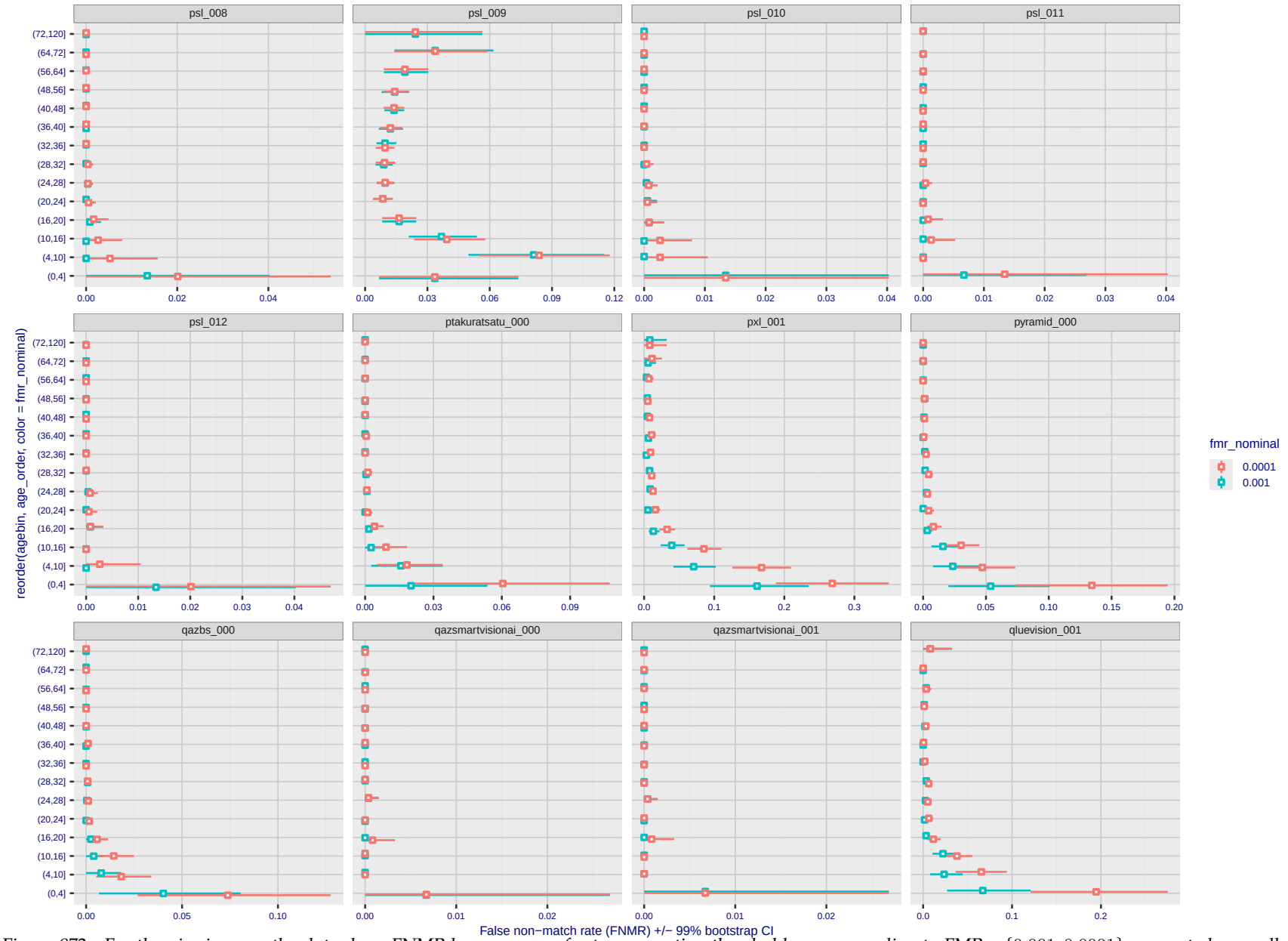


Figure 872: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 873: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

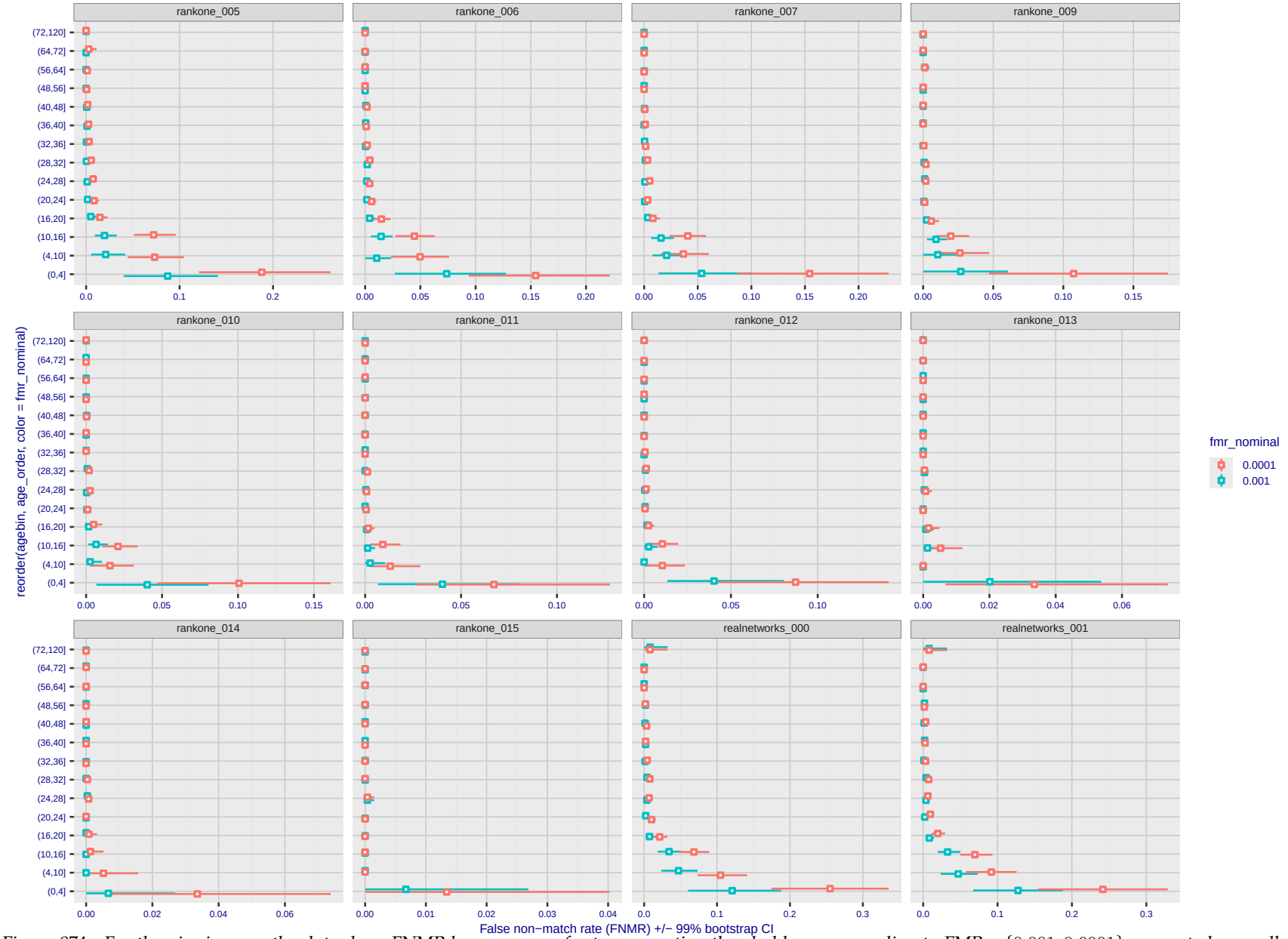


Figure 874: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

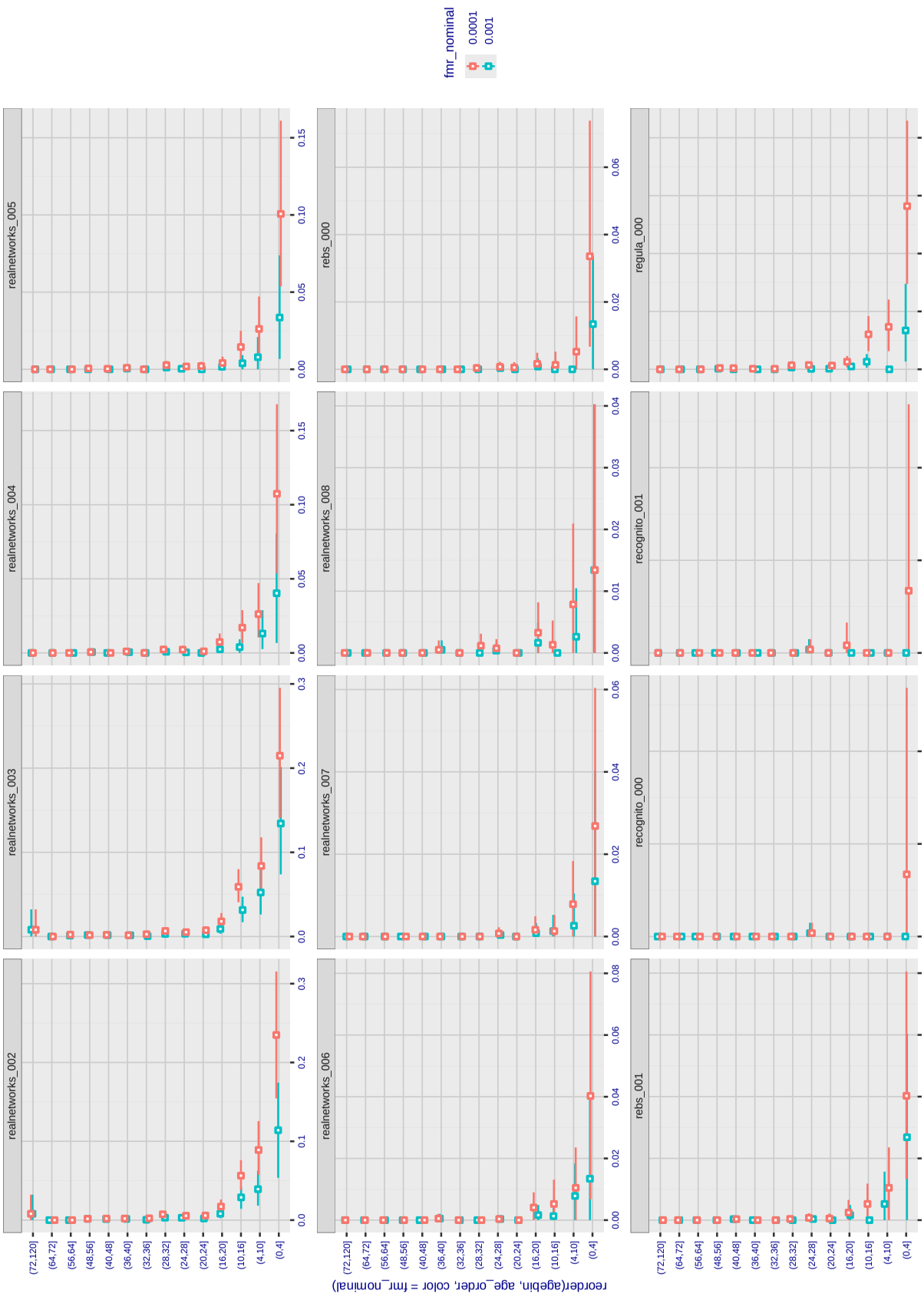


Figure 875: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = {0.001, 0.0001} computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 876: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

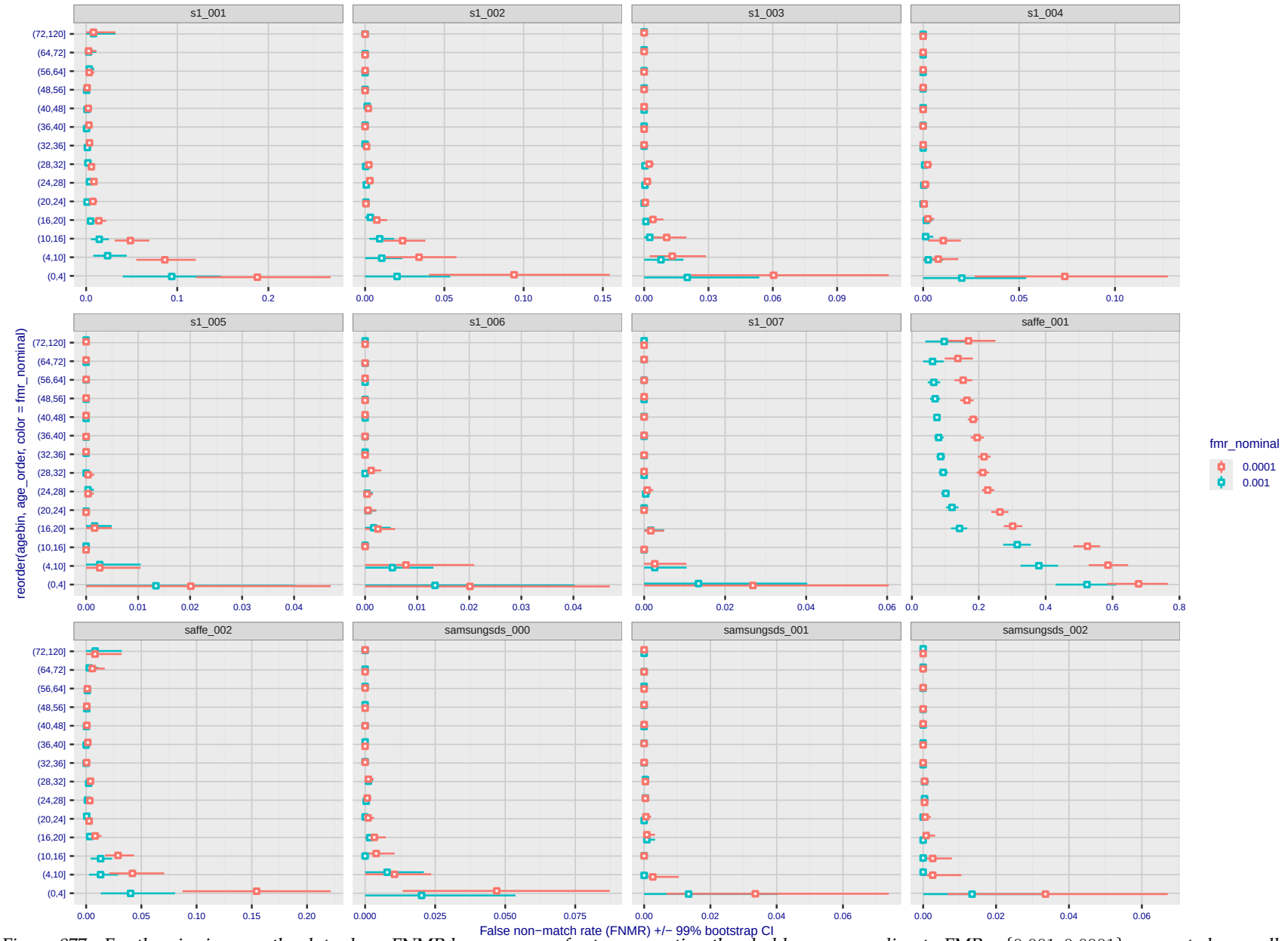


Figure 877: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

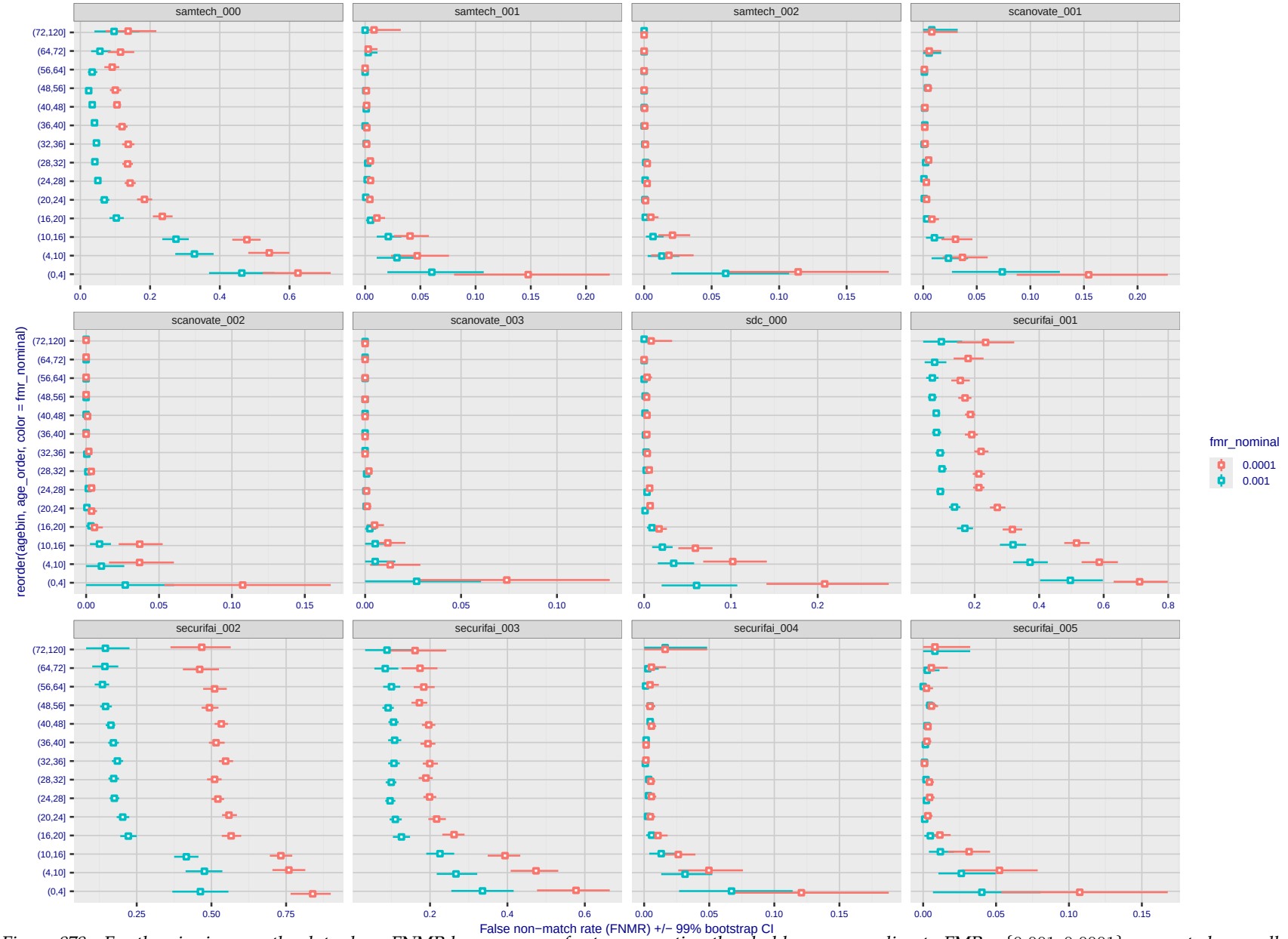


Figure 878: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

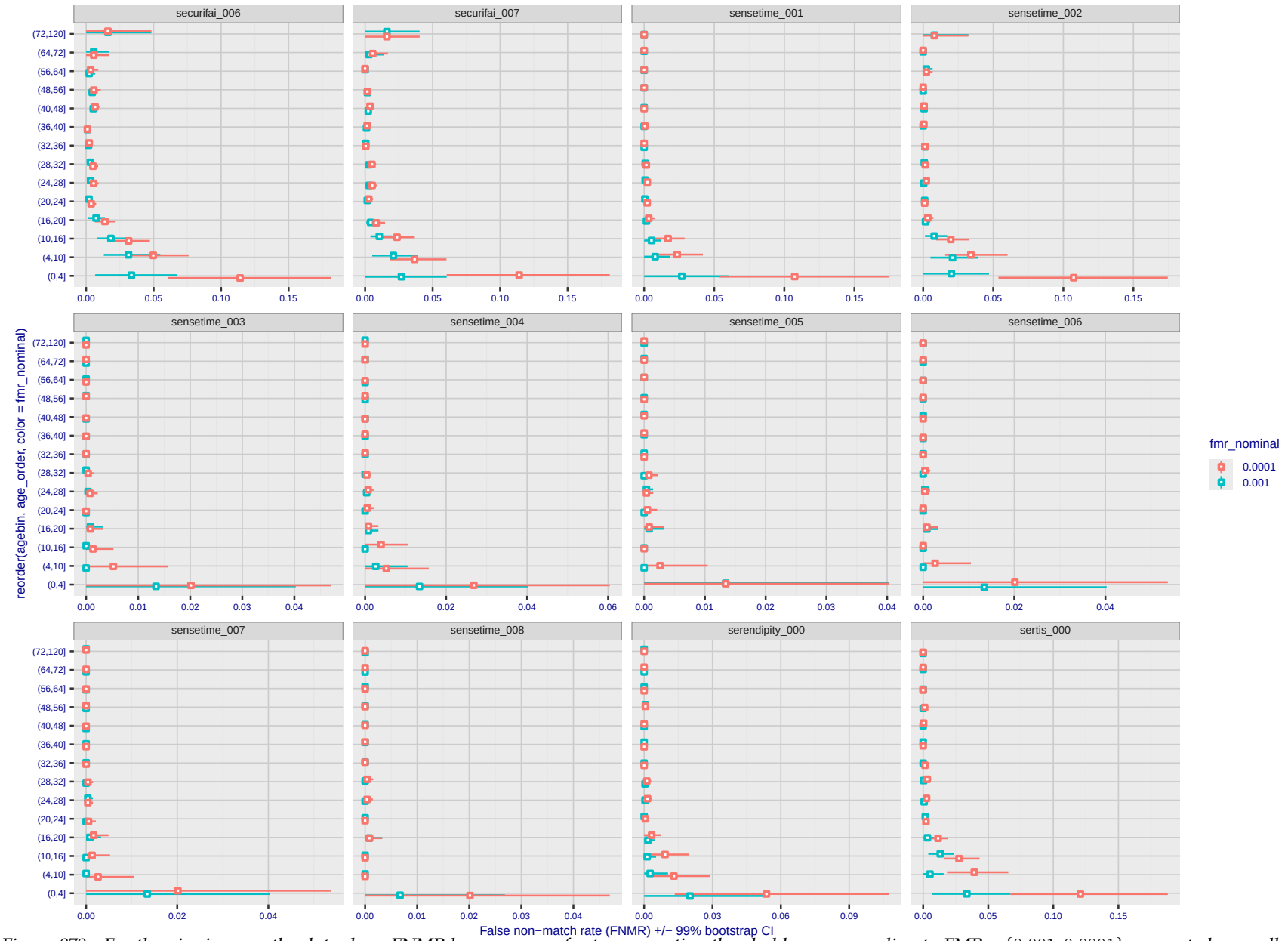


Figure 879: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

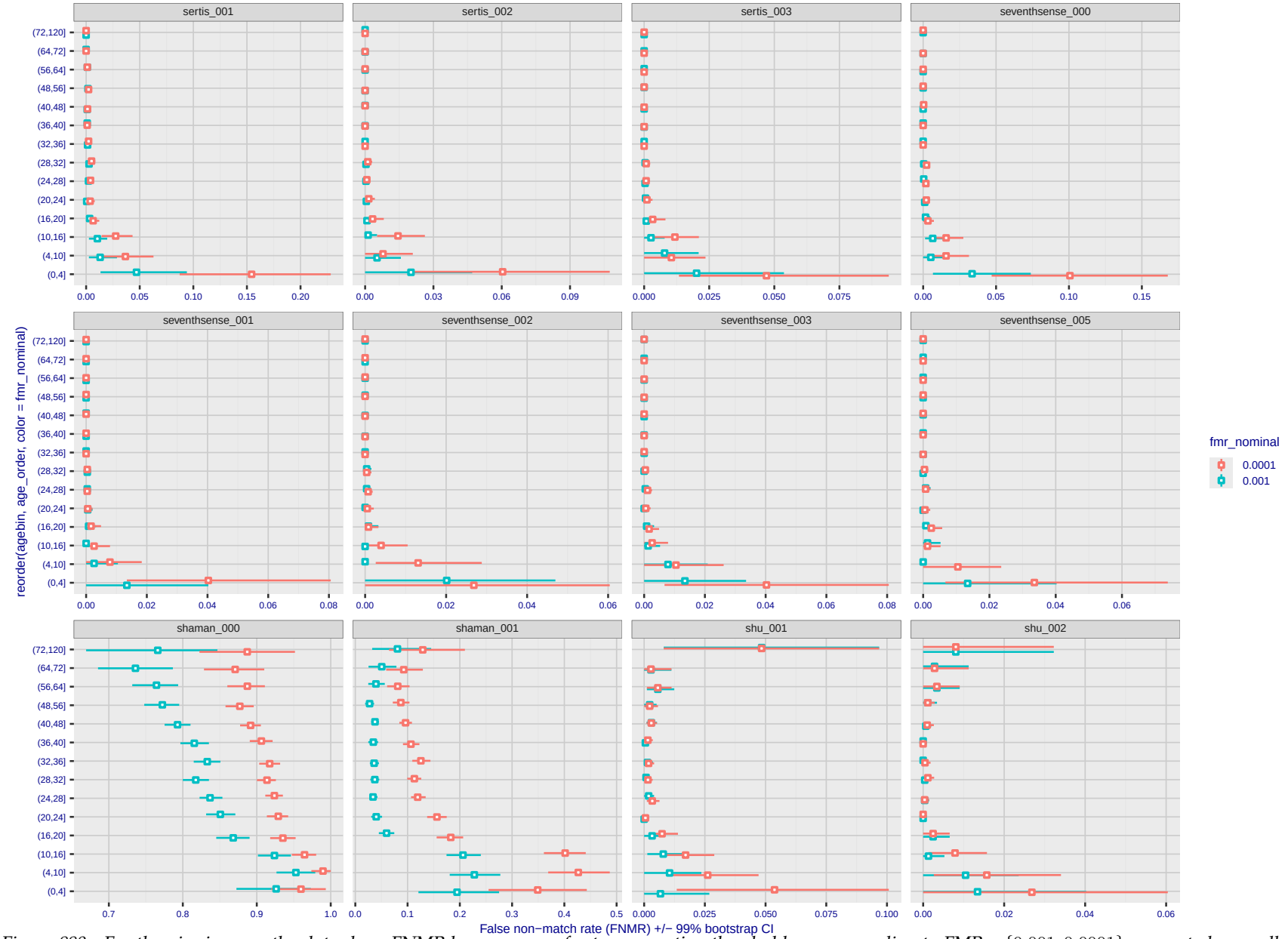


Figure 880: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 881: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 882: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

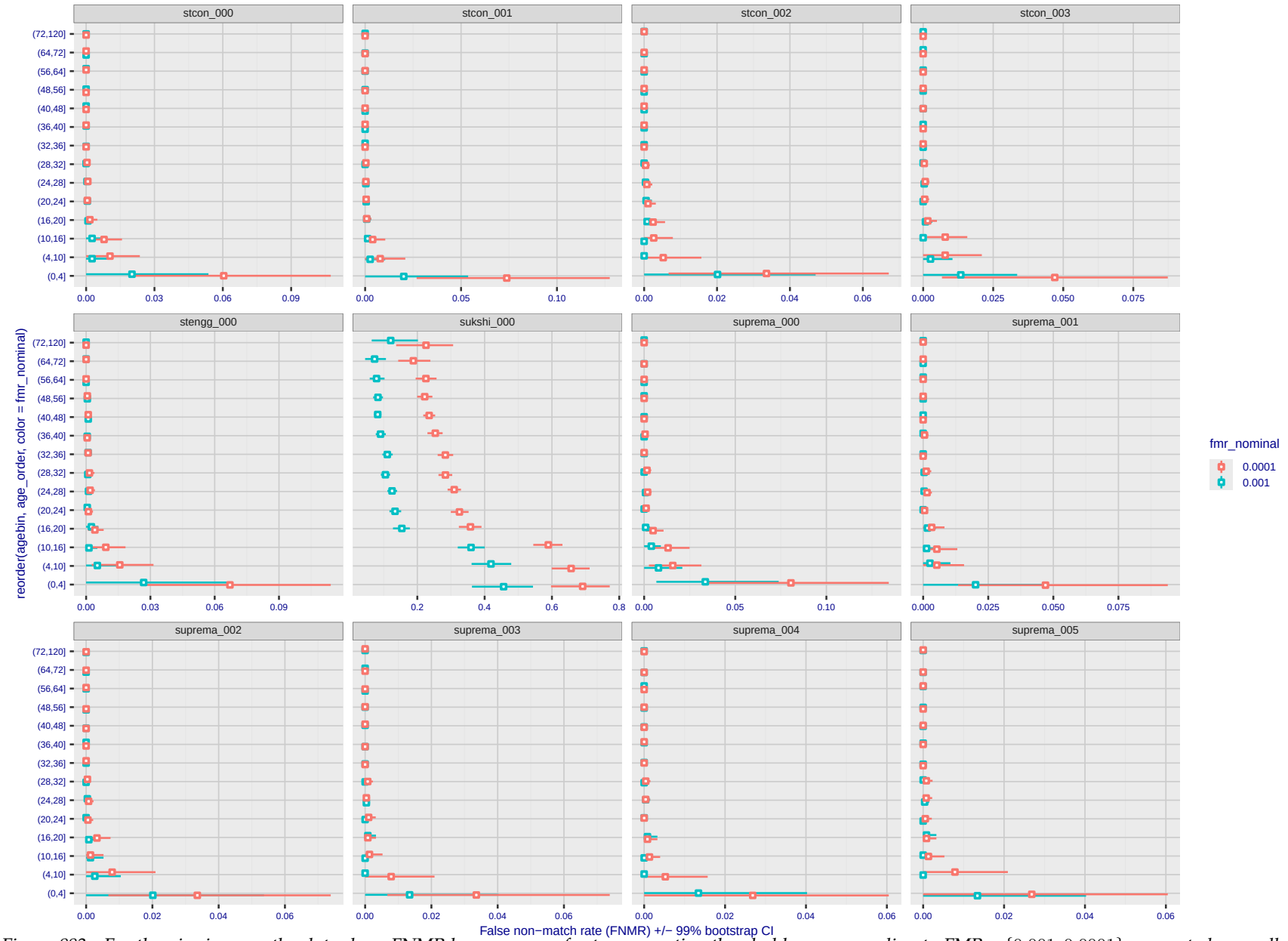


Figure 883: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

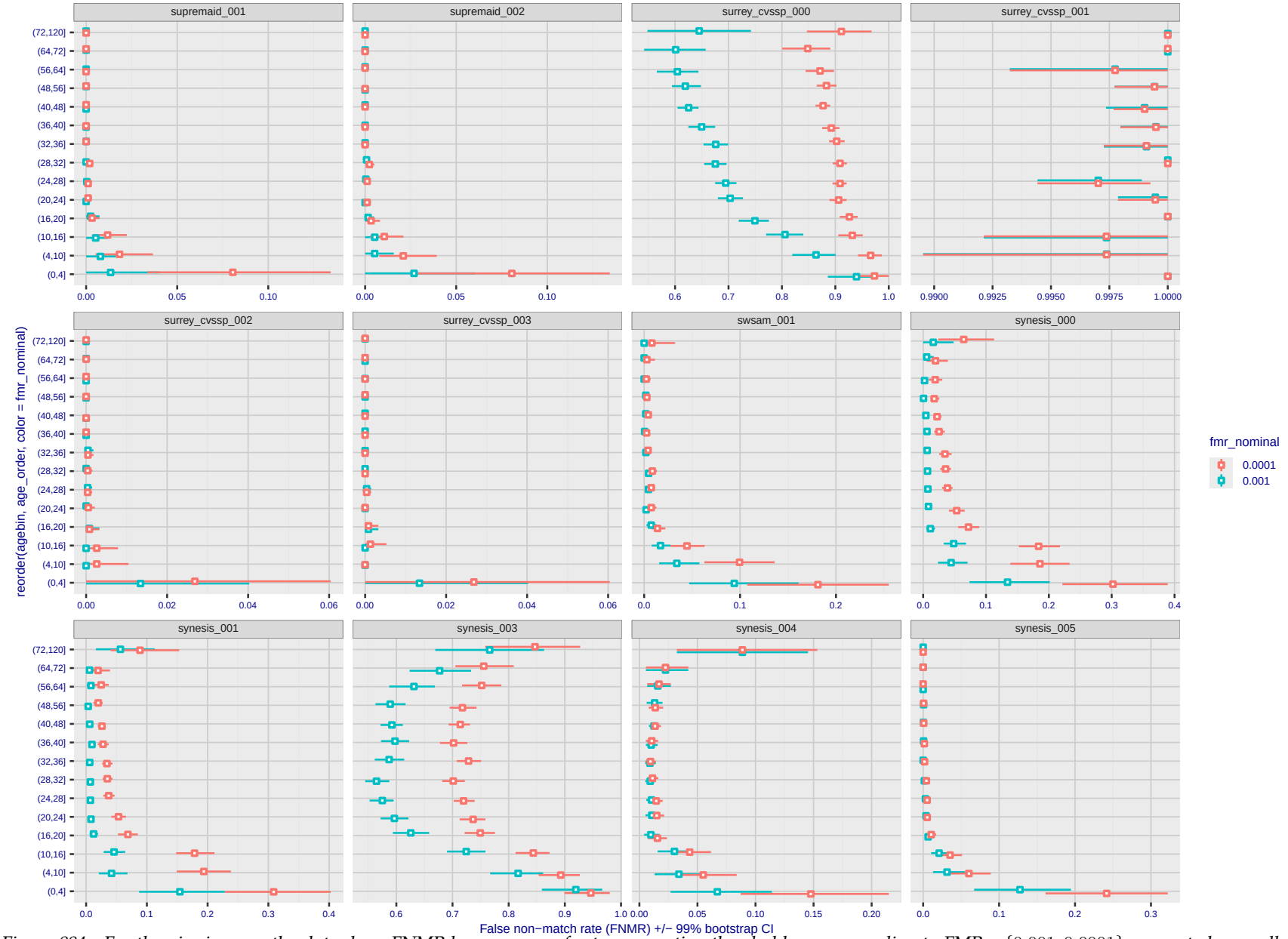


Figure 884: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 885: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 886: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 887: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

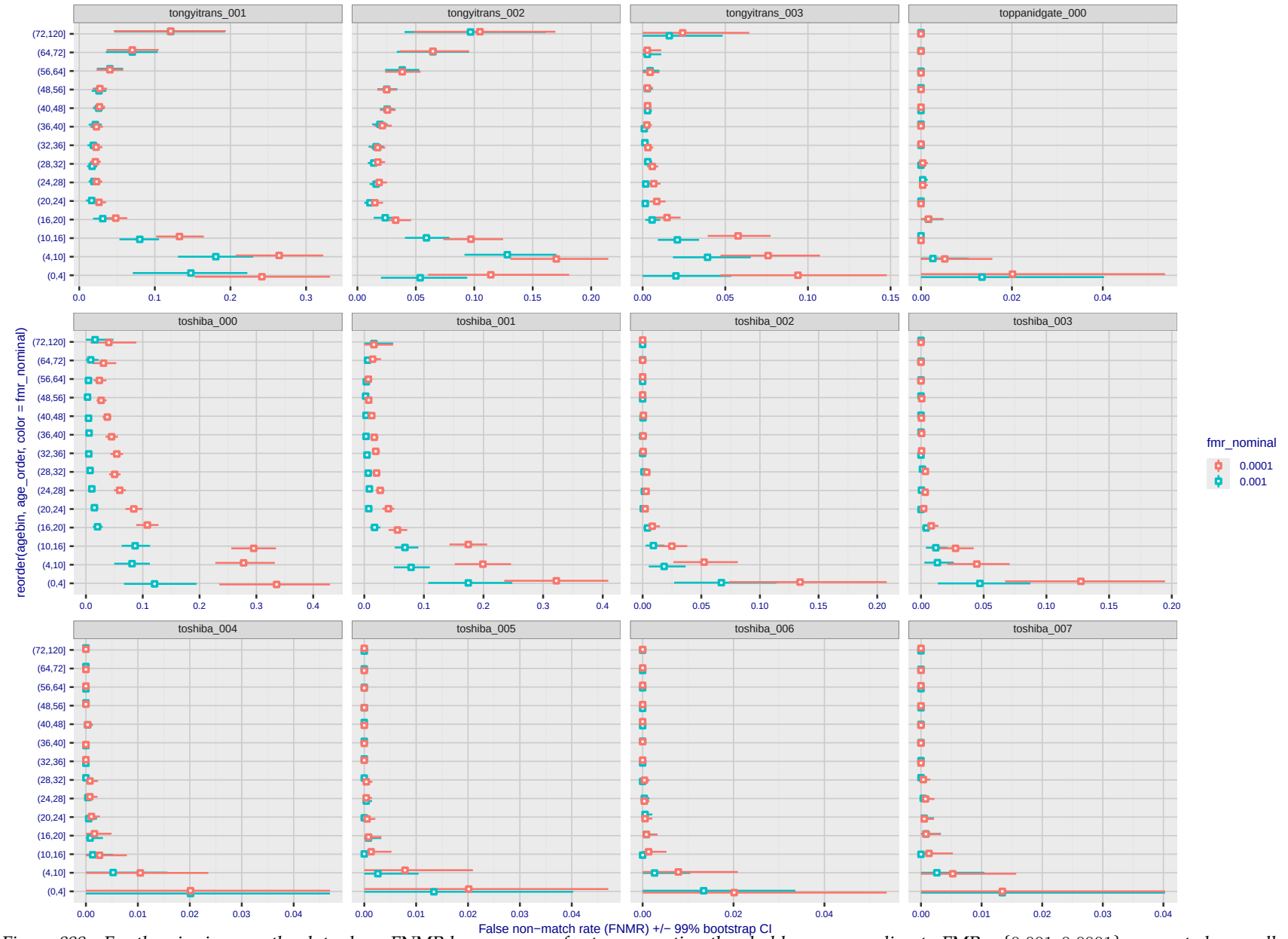


Figure 888: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

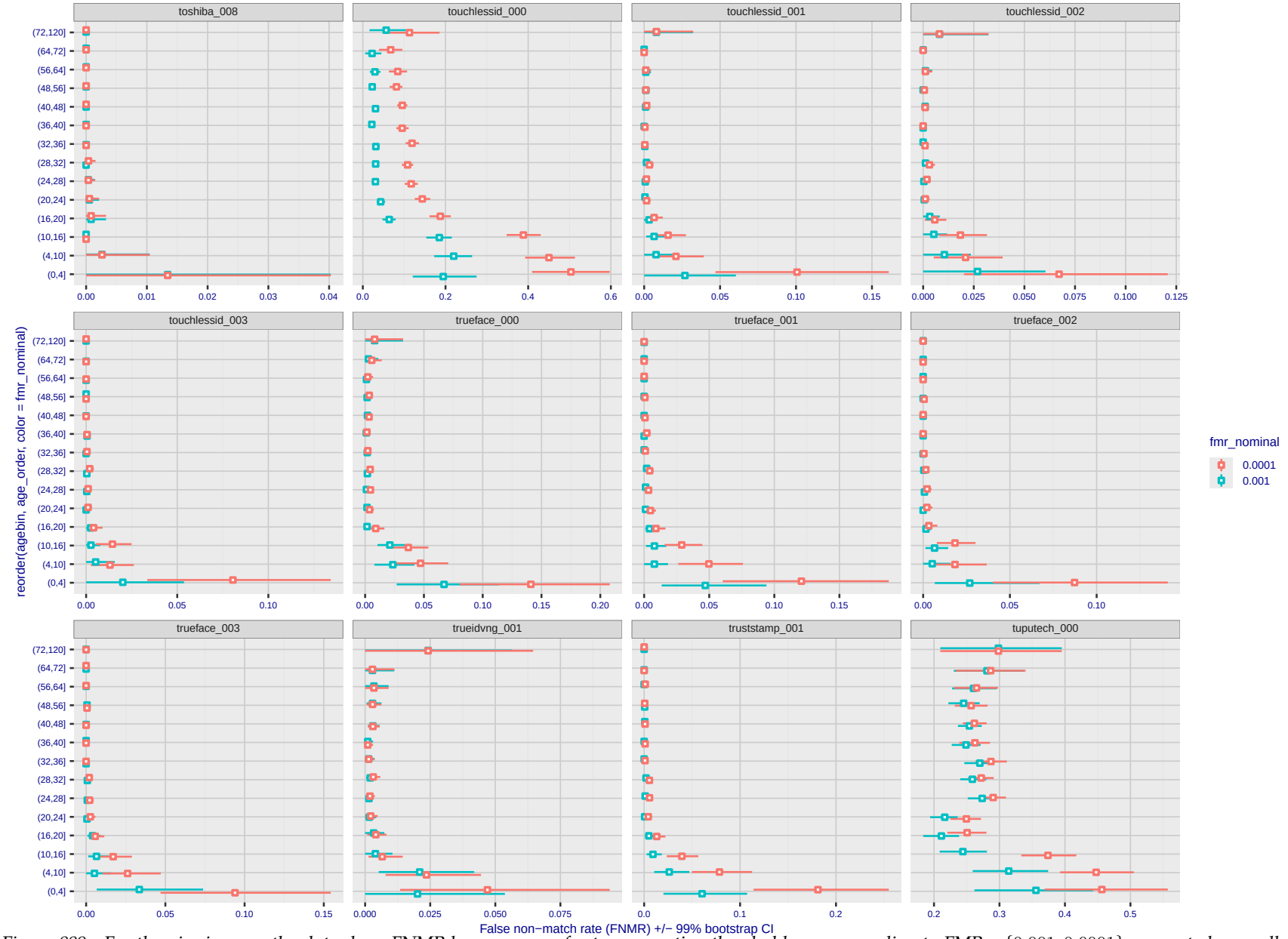


Figure 889: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 890: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 891: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

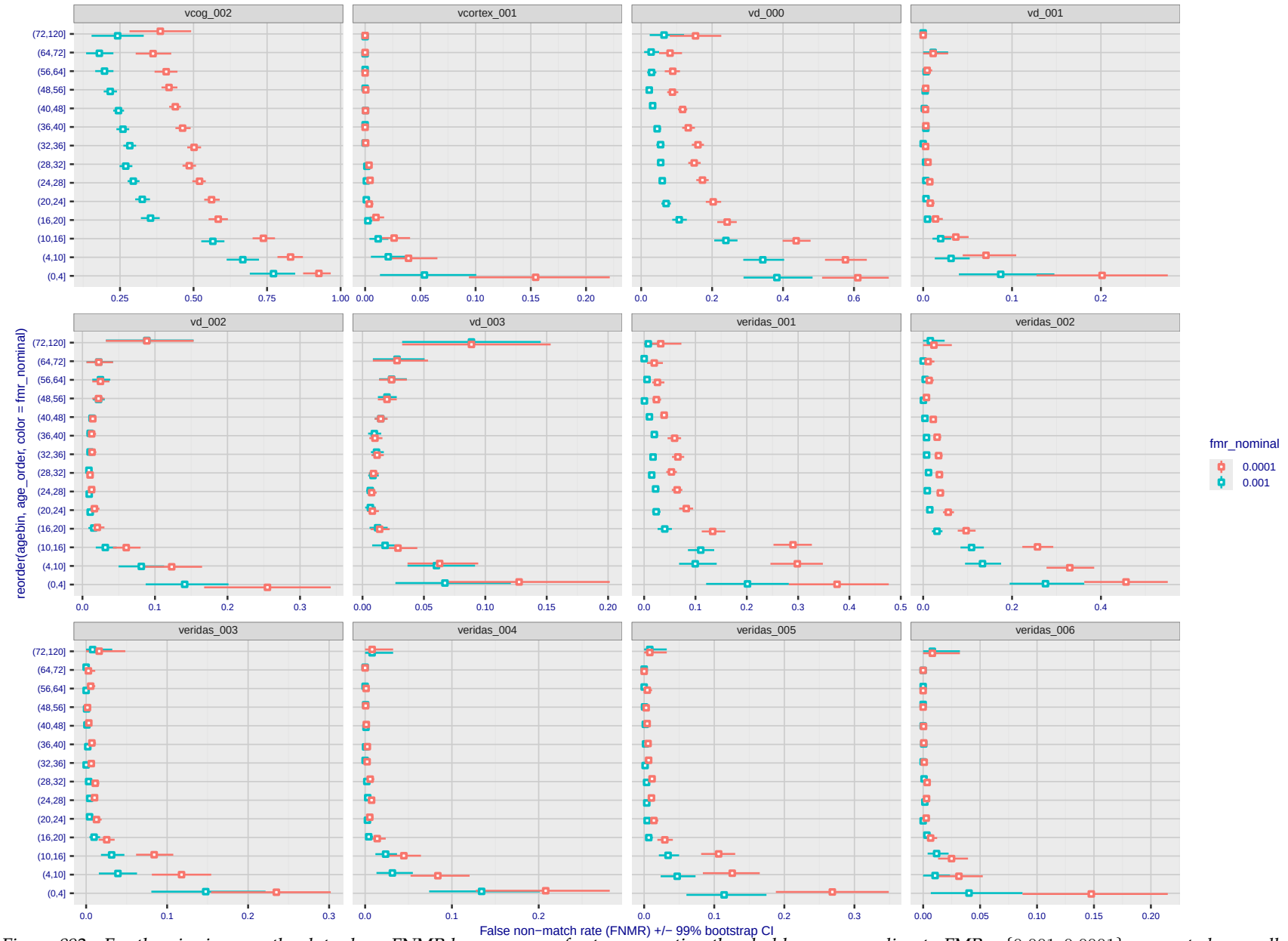


Figure 892: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 893: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 894: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

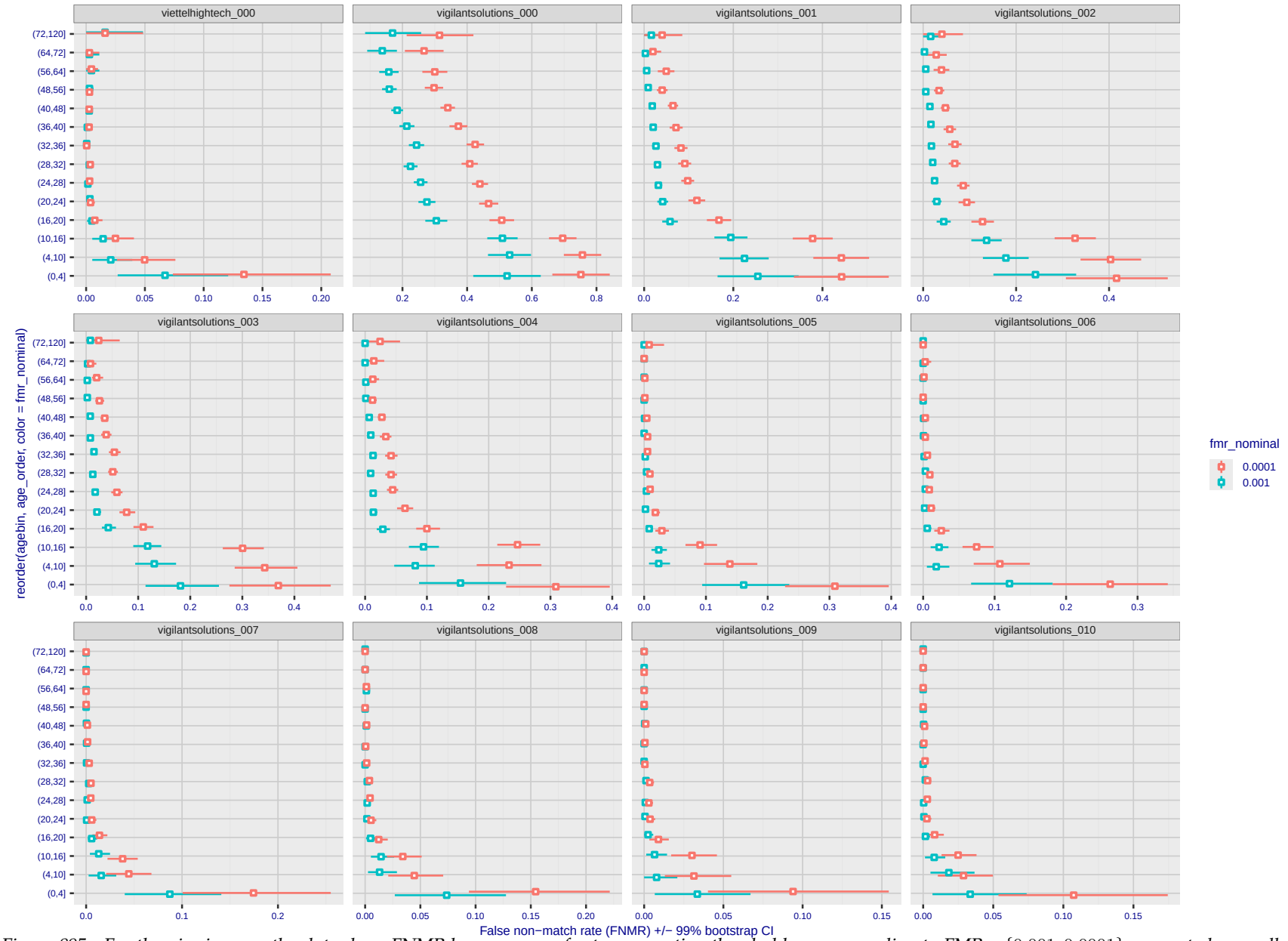


Figure 895: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 896: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

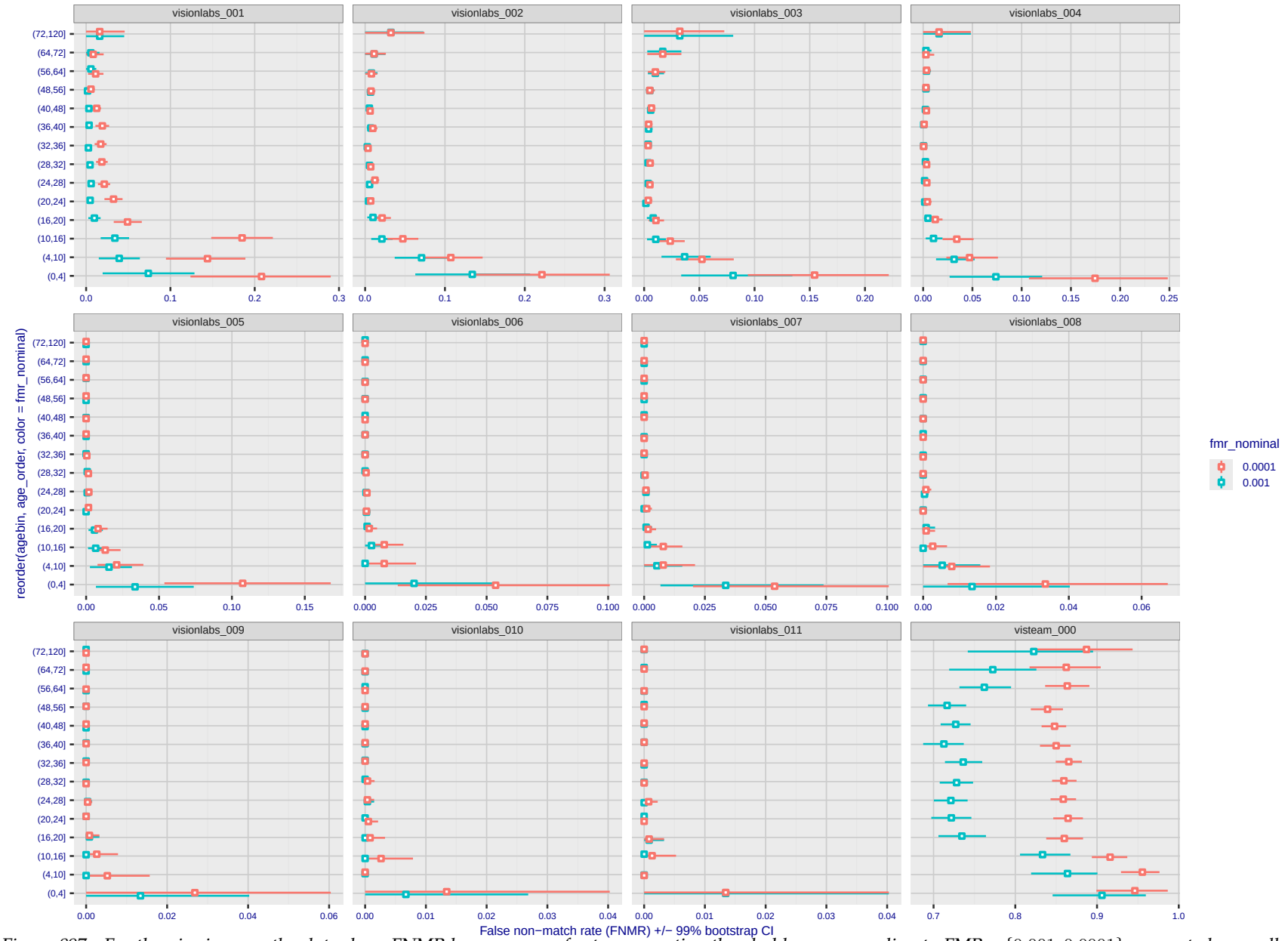


Figure 897: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Figure 898: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 899: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 900: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

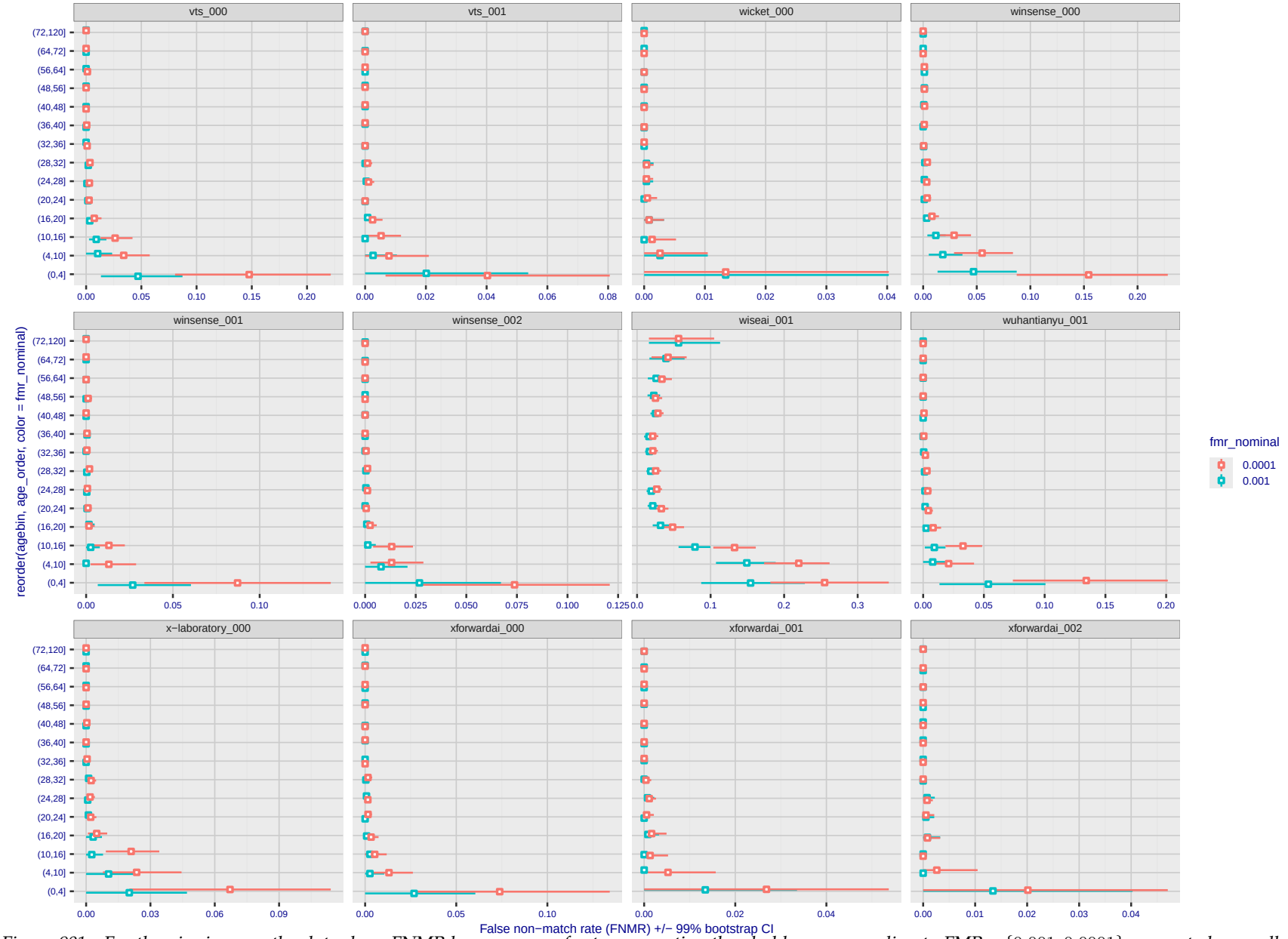


Figure 901: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Figure 902: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

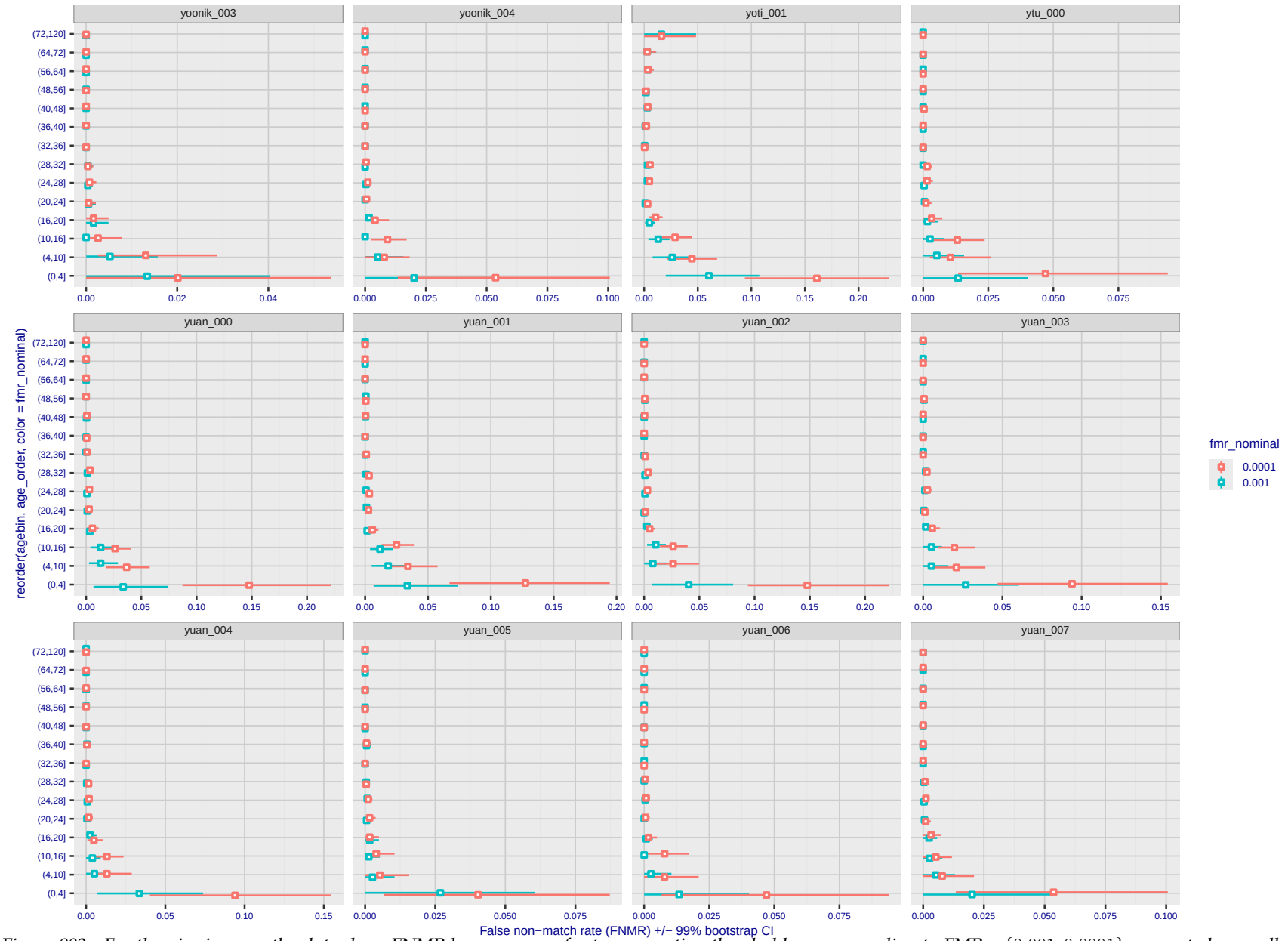


Figure 903: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

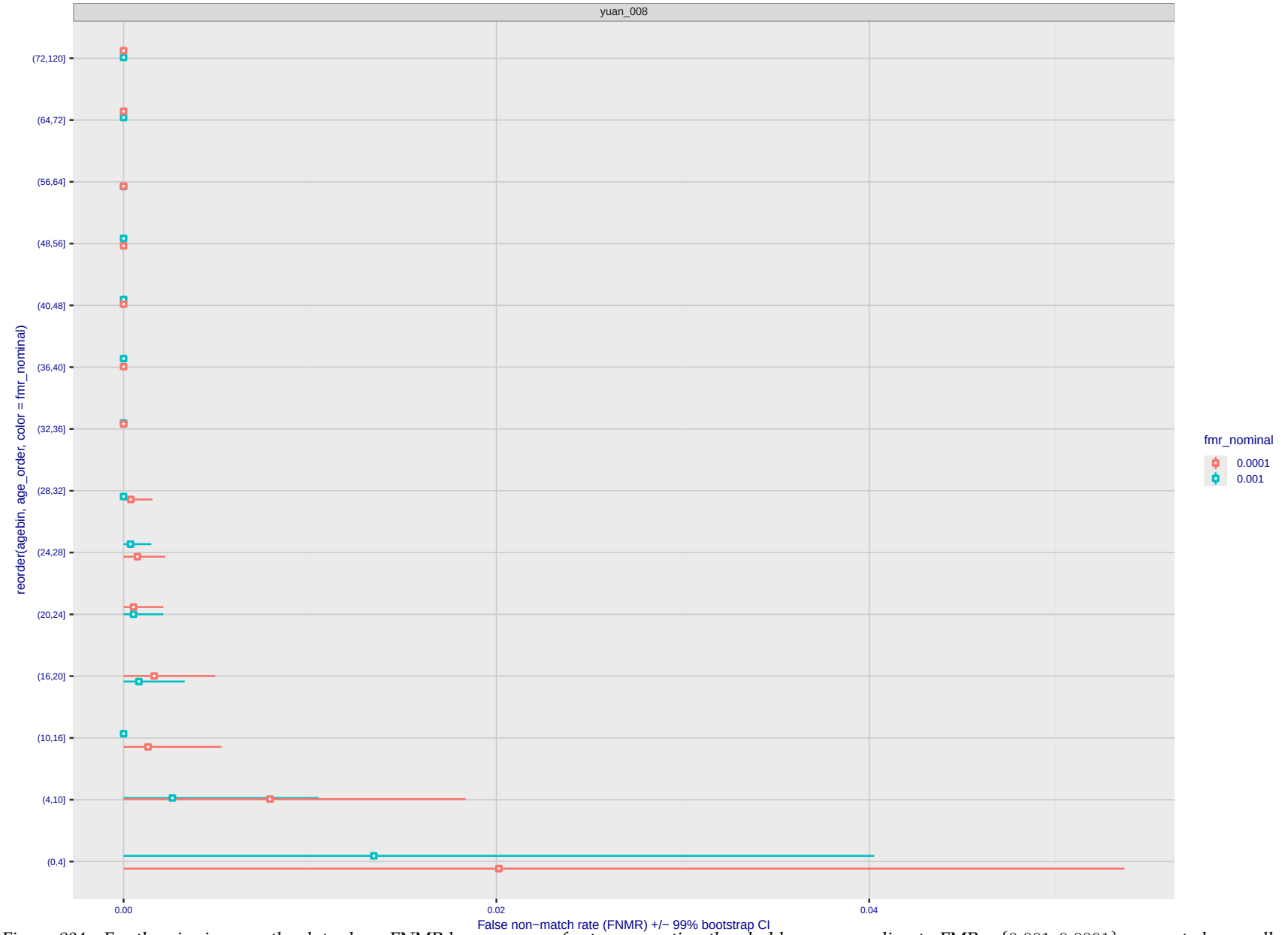


Figure 904: For the visa images, the dots show FNMR by age group for two operating thresholds corresponding to FMR = $\{0.001, 0.0001\}$ computed over all on the order of 10^{10} impostor scores. The FMR in each bin will vary also - see subsequent impostor heatmaps in sec. 3.6.2. Given a pair of face images taken at different times, we assign the comparison to the bin that is the arithmetic average of the subject's ages. This plot shows only the effect of age, not ageing. The number of comparisons in each bin is generally in the thousands, however the first and last bins are computed over 149 and 124 respectively. The error rates in some (adult) cases are zero, and in others the DET is flat so the error rates at the two thresholds are identical. The lines span 1% and 99% of boostrap replicated FNMR estimates.

Caveats: None.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.6 Impostor distribution stability

3.6.1 Effect of birth place on the impostor distribution

Background: Facial appearance varies geographically, both in terms of skin tone, cranio-facial structure and size. This section addresses whether false match rates vary intra- and inter-regionally.

Goals:

- ▷ To show the effect of birth region of the impostor and enrollee on false match rates.
- ▷ To determine whether some algorithms give better impostor distribution stability.

Methods:

- ▷ For the visa images, NIST defined 10 regions: Sub-Saharan Africa, South Asia, Polynesia, North Africa, Middle East, Europe, East Asia, Central and South America, Central Asia, and the Caribbean.
- ▷ For the visa images, NIST mapped each country of birth to a region. There is some arbitrariness to this. For example, Egypt could reasonably be assigned to the Middle East instead of North Africa. An alternative methodology could, for example, assign the Philippines to *both* Polynesia and East Asia.
- ▷ FMR is computed for cases where all face images of impostors born in region r_2 are compared with enrolled face images of persons born in region r_1.

$$\text{FMR}(r_1, r_2, T) = \frac{\sum_{i=1}^{N_{r_1,r_2}} H(s_i - T)}{N_{r_1,r_2}} \tag{5}$$

  where the same threshold, T, is used in all cells, and H is the unit step function. The threshold is set to give FMR(T) = 0.001 over the entire set of visa image impostor comparisons.
- ▷ This analysis is then repeated by country-pair, but only for those country pairs where both have at least 1000 images available. The countries[1] appear in the axes of graphs that follow.
- ▷ The mean number of impostor scores in any cross-region bin is 33 million. The smallest number of impostor scores in any bin is 135000, for Central Asia - North Africa. While these counts are large enough to support reasonable significance, the number of individual faces is much smaller, on the order of $N^{0.5}$.
- ▷ The numbers of impostor scores in any cross-country bin is shown in Figure 905.

Results: Subsequent figures show heatmaps that use color to represent the base-10 logarithm of the false match rate. Red colors indicate high (bad) false match rates. Dark colors indicate benign false match rates. There are two series of graphs corresponding to aggregated geographical regions, and to countries. The notable observations are:

- ▷ The on-diagonal elements correspond to within-region impostors. FMR is generally above the nominal value of FMR = 0.001. Particularly there is usually higher FMR in, Sub-Saharan Africa, South Asia, and the Caribbean. Europe and Central Asia, on the other hand, usually give FMR closer to the nominal value.
- ▷ The off-diagonal elements correspond to across-region impostors. The highest FMR is produced between the Caribbean and Sub-Saharan Africa.
- ▷ Algorithms vary.

[1] These are Argentina, Australia, Brazil, Chile, China, Costa Rica, Cuba, Czech Republic, Dominican Republic, Ecuador, Egypt, El Salvador, Germany, Ghana, Great Britain, Greece, Guatamala, Haiti, Hong Kong, Honduras, Indonesia, India, Israel, Jamaica, Japan, Kenya, Korea, Lebanon, Mexico, Malaysia, Nepal, Nigeria, Peru, Philippines, Pakistan, Poland, Romania, Russia, South Africa, Saudi Arabia, Thailand, Trinidad, Turkey, Taiwan, Ukraine, Venezuela, and Vietnam.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

▷ We computed the same quantities for a global FMR = 0.0001. The effects are similar.

Caveats:

▷ The effects of variable impostor rates on one-to-many identification systems may well differ from what's implied by these one-to-one verification results. Two reasons for this are a) the enrollment galleries are usually imbalanced across countries of birth, age and sex; b) one-to-many identification algorithms often implement techniques aimed at stabilizing the impostor distribution. Further research is necessary.

▷ In principle, the effects seen in this subsection could be due to differences in the image capture process. We consider this unlikely since the effects are maintained across geography - e.g. Caribbean vs. Africa, or Japan vs. China.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

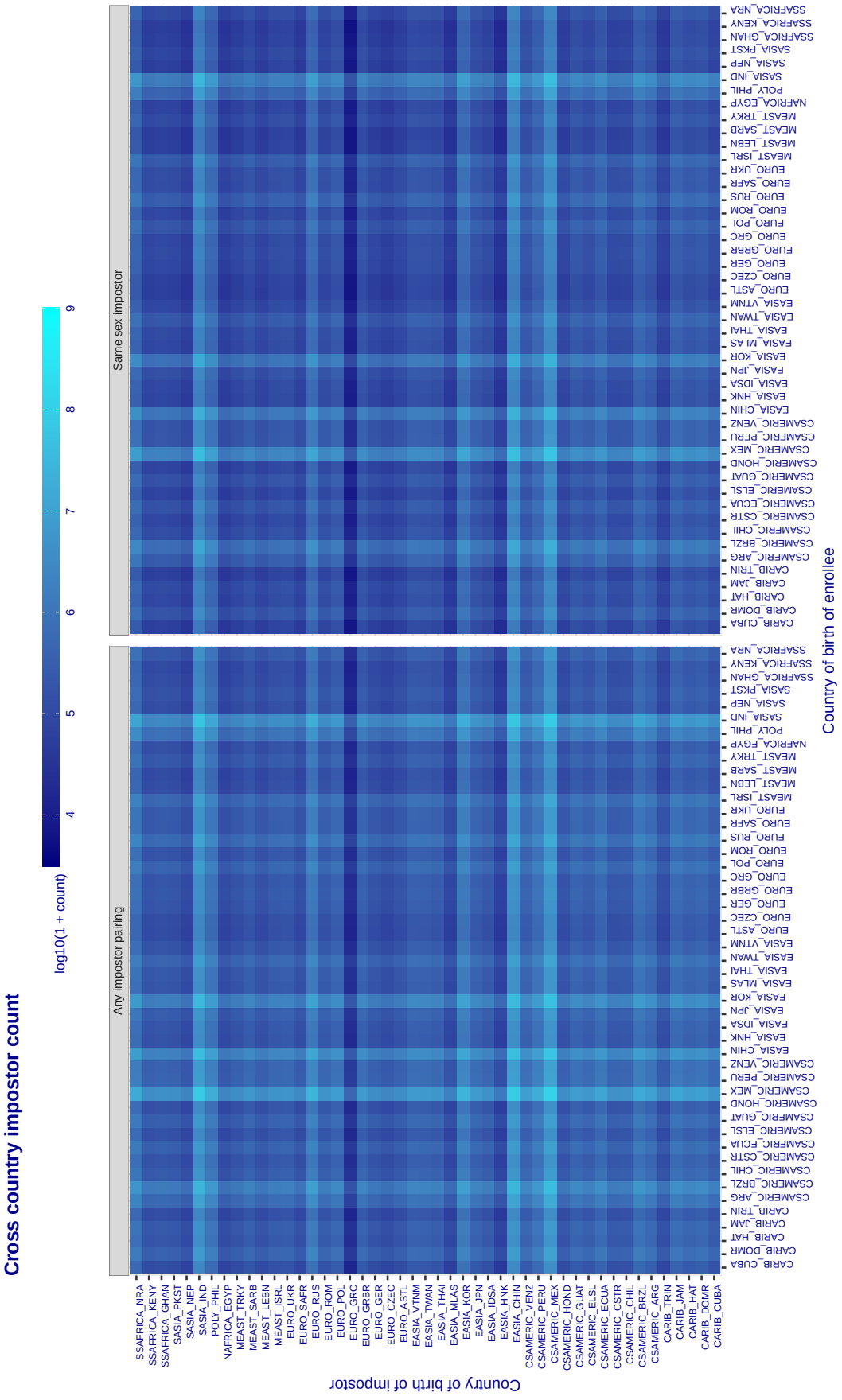


Figure 905: For visa images, the heatmap shows the count of impostor comparisons of faces from different individuals who were born in the given country pair. The FMR heatmaps themselves appear in the 1:1 report cards, for example, this one.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

3.6.2 Effect of age on impostors

Background: This section shows the effect of age on the impostor distribution. The ideal behaviour is that the age of the enrollee and the impostor would not affect impostor scores. This would support FMR stability over sub-populations.

Goals:

- ▷ To show the effect of relative ages of the impostor and enrollee on false match rates.
- ▷ To determine whether some algorithms have better impostor distribution stability.

Methods:

- ▷ Define 14 age group bins, spanning 0 to over 100 years old.
- ▷ Compute FMR over all impostor comparisons for which the subjects in the enrollee and impostor images have ages in two bins.
- ▷ Compute FMR over all impostor comparisons for which the subjects are additionally of the same sex, and born in the same geographic region.

Results:

The notable aspects are:

- ▷ Diagonal dominance: Impostors are more likely to be matched against their same age group.
- ▷ Same sex and same region impostors are more successful. On the diagonal, an impostor is more likely to succeed by posing as someone of the same sex. If $\Delta \log_{10}$ FMR $= 0.2$, then same-sex same-region FMR exceeds the all-pairs FMR by factor of $10^{0.2} = 1.6$.
- ▷ Young children impostors give elevated FMR against young children. Older adult impostor give elevated FMR against older adults. These effects are quite large, for example if $\Delta \log_{10}$ FMR $= 1.0$ larger than a 32 year old, then these groups have higher FMR by a factor of $10^1 = 10$. This would imply an FMR above 0.01 for a nominal (global) FMR = 0.001.
- ▷ Algorithms vary.
- ▷ We computed the same quantities for a global FMR = 0.0001. The effects are similar.

Note the calculations in this section include impostors paired across all countries of birth.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"

Accuracy Terms + Definitions

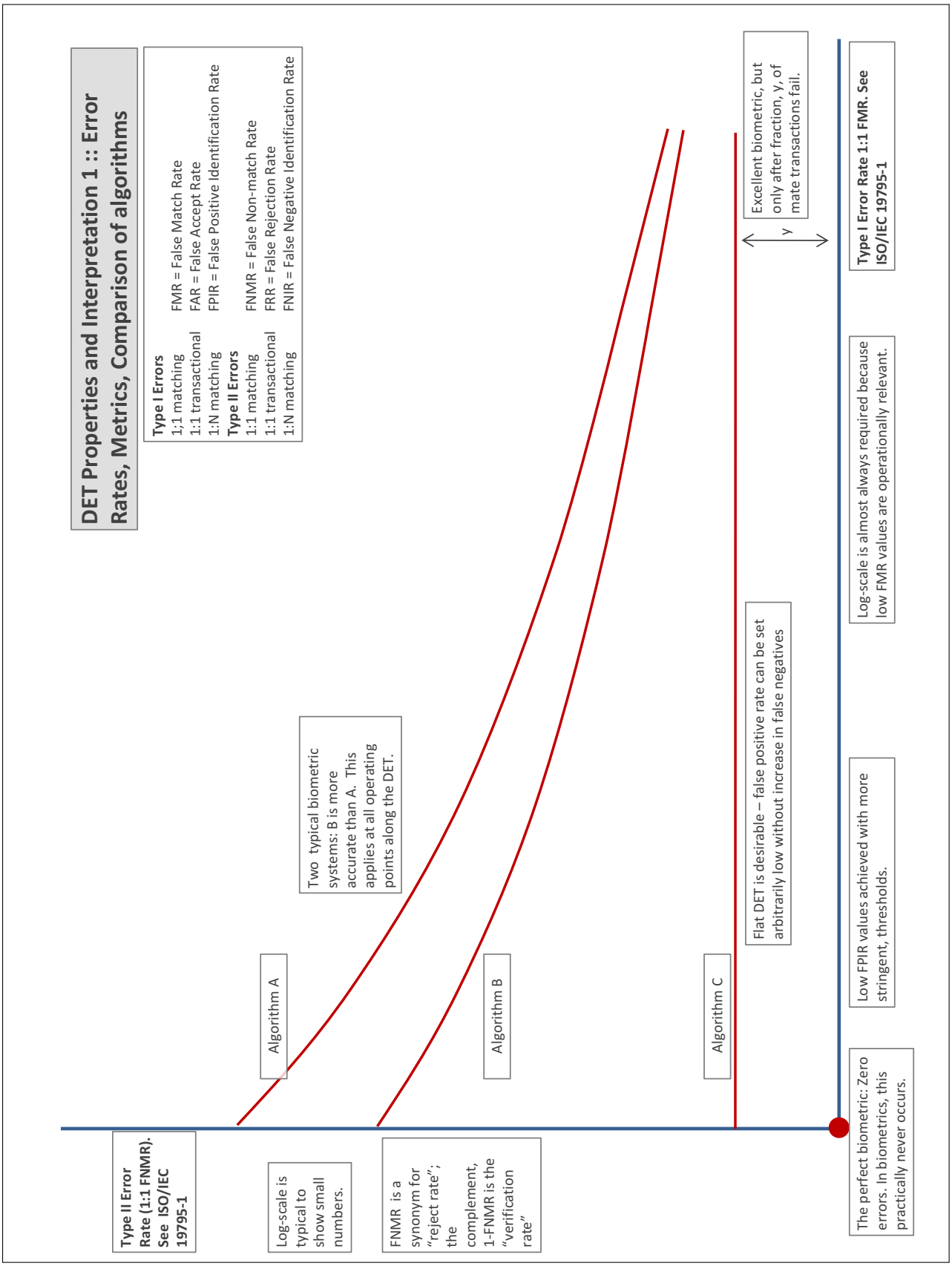
In biometrics, Type II errors occur when two samples of one person do not match – this is called a **false negative**. Correspondingly, Type I errors occur when samples from two persons do match – this is called a **false positive**. Matches are declared by a biometric system when the native comparison score from the recognition algorithm meets some **threshold**. Comparison scores can be either **similarity scores**, in which case higher values indicate that the samples are more likely to come from the same person, or **dissimilarity scores**, in which case higher values indicate different people. Similarity scores are traditionally computed by **fingerprint** and **face** recognition algorithms, while dissimilarities are used in **iris recognition**. In some cases, the dissimilarity score is a distance; this applies only when **metric** properties are obeyed. In any case, scores can be either **mate** scores, coming from a comparison of one person's samples, or **nonmate** scores, coming from comparison of different persons' samples. The words **genuine** or **authentic** are synonyms for mate, and the word **impostor** is used a synonym for nonmate. The words mate and nonmate are traditionally used in identification applications (such as law enforcement search, or background checks) while genuine and impostor are used in verification applications (such as access control).

A **error tradeoff** characteristic represents the tradeoff between Type II and Type I classification errors. For verification this plots false non-match rate (FNMR) vs. false match rate (FMR) parametrically with T.

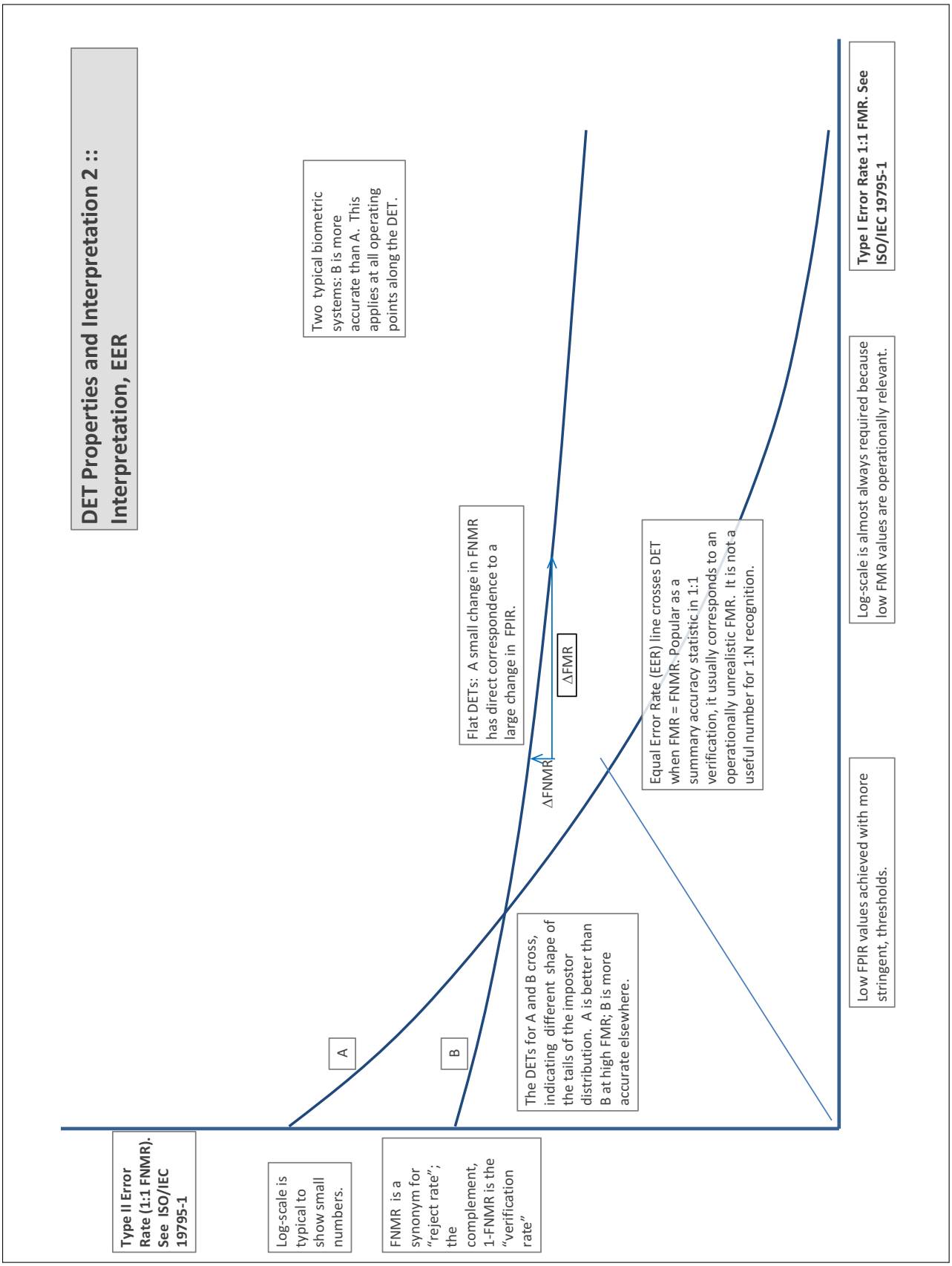
The error tradeoff plots are often called **detection error tradeoff (DET)** characteristics or **receiver operating characteristic** (ROC). These serve the same function but differ, for example, in plotting the complement of an error rate (e.g. TMR = 1 – FNMR) and in transforming the axes most commonly using logarithms, to show multiple decades of FMR. More rarely, the function might be the inverse Gaussian function.

More detail and generality is provided in formal biometrics testing standards, see the various parts of ISO/IEC 19795 Biometrics Testing and Reporting. More terms, including and beyond those to do with accuracy, see ISO/IEC 2382-37 Information technology -- Vocabulary -- Part 37: Harmonized biometric vocabulary

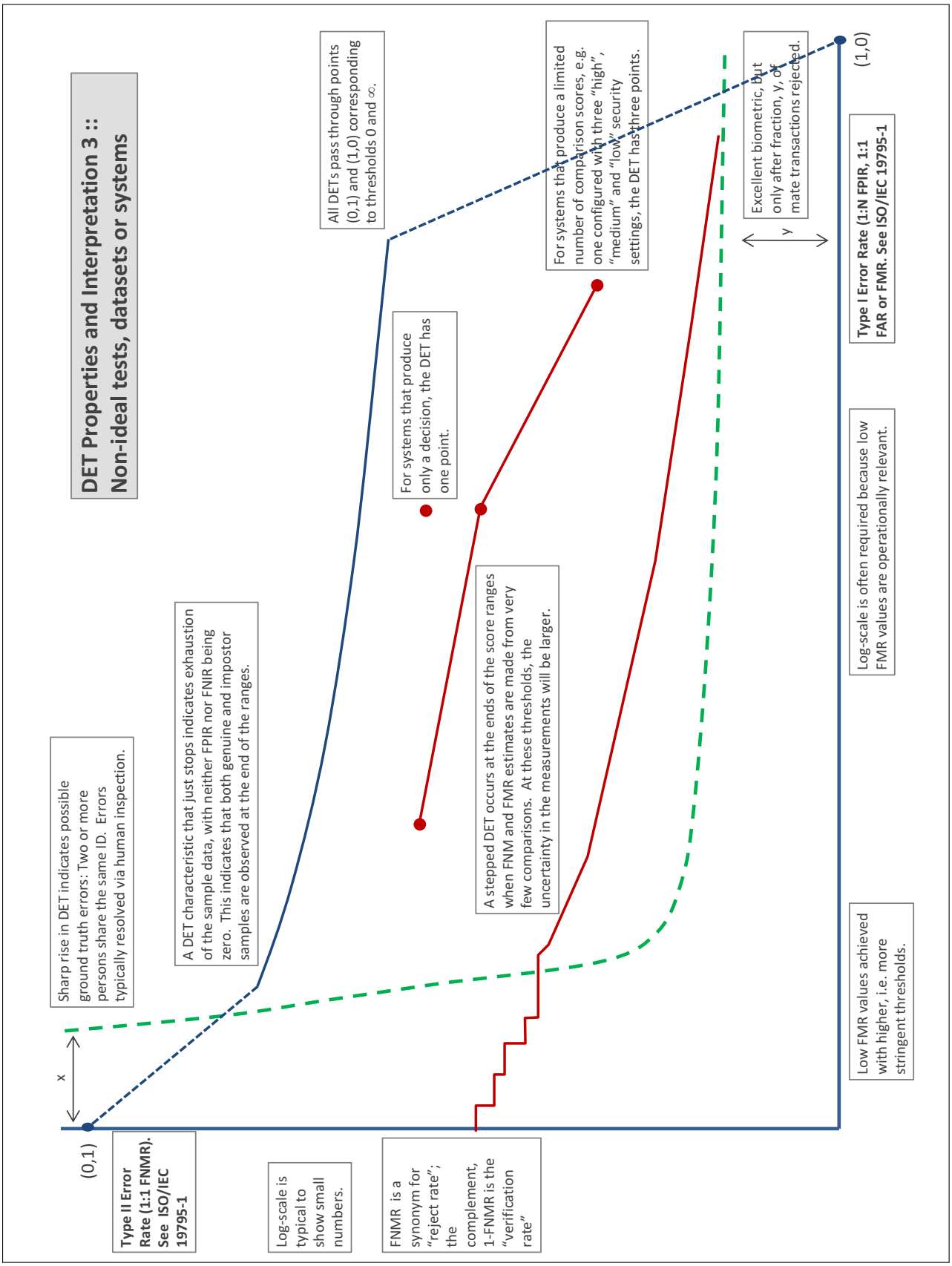
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



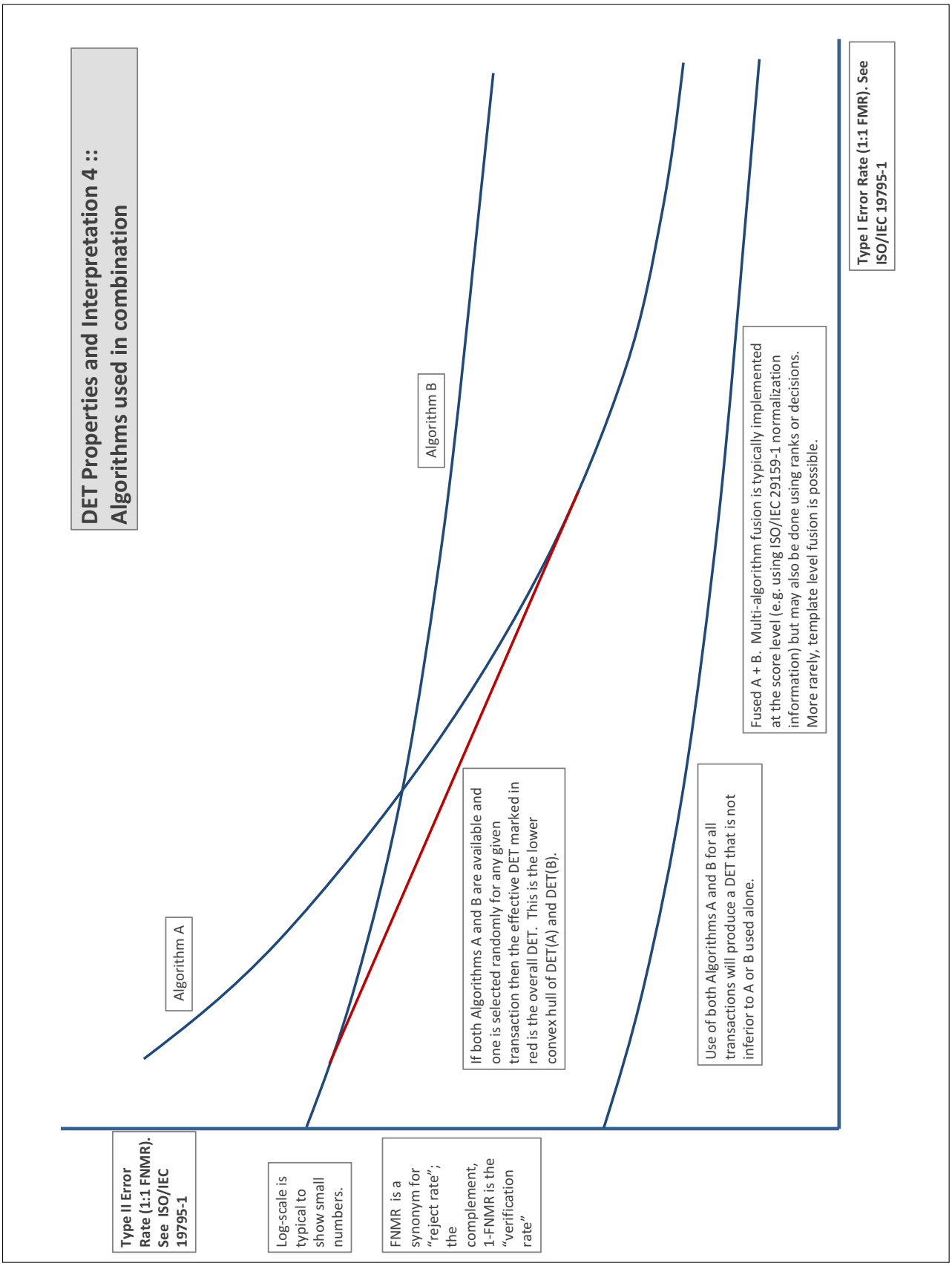
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



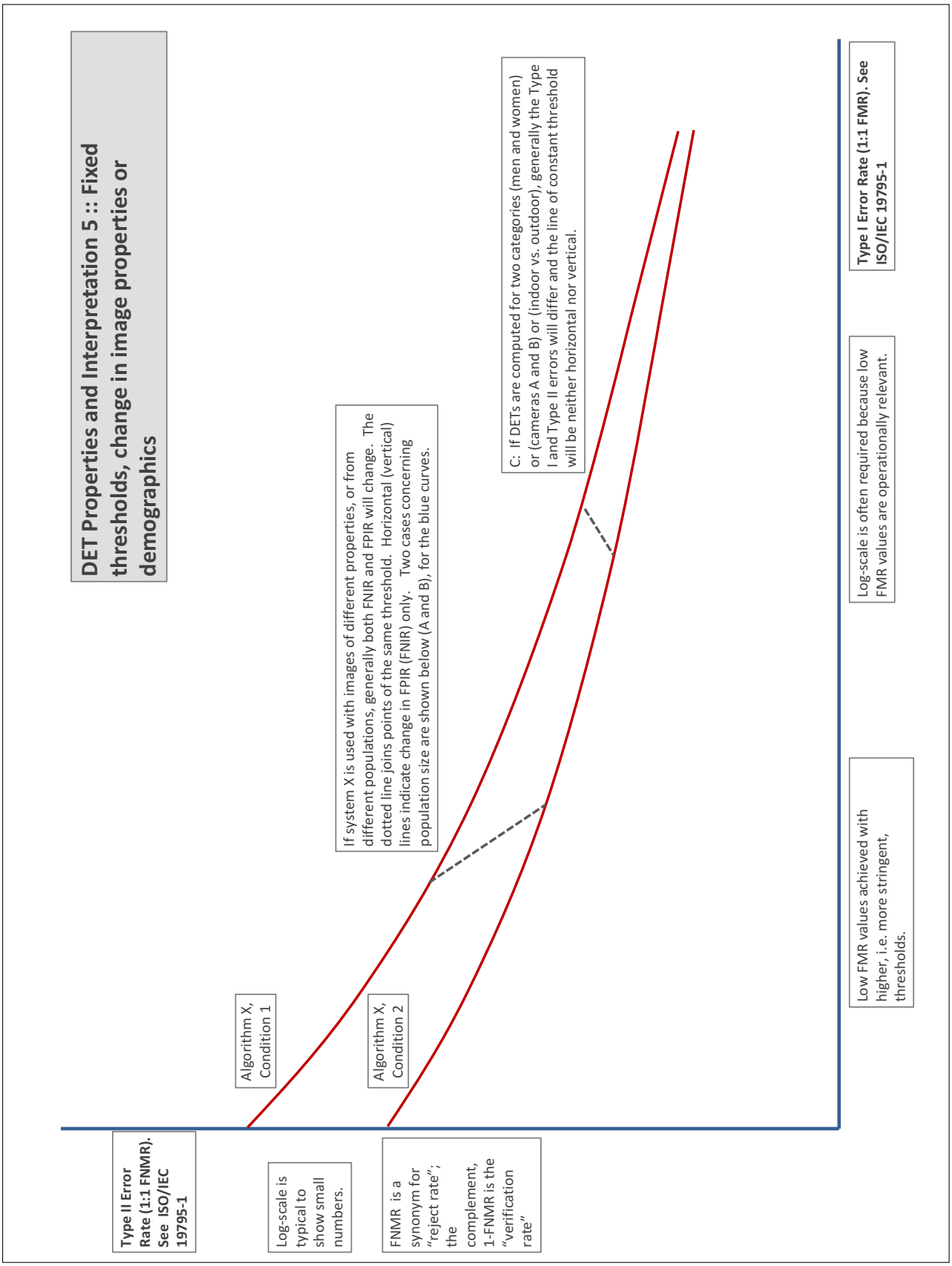
FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
FMR(T) "False match rate"



FNMR(T) "False non-match rate"
FMR(T) "False match rate"

References

[1] P. Jonathon Phillips, Amy N. Yates, Ying Hu, Carina A. Hahn, Eilidh Noyes, Kelsey Jackson, Jacqueline G. Cavazos, Géraldine Jeckeln, Rajeev Ranjan, Swami Sankaranarayanan, Jun-Cheng Chen, Carlos D. Castillo, Rama Chellappa, David White, and Alice J. O'Toole. Face recognition accuracy of forensic examiners, superrecognizers, and face recognition algorithms. *Proceedings of the National Academy of Sciences*, 115(24):6171–6176, 2018.

FNMR(T) "False non-match rate"
FMR(T) "False match rate"